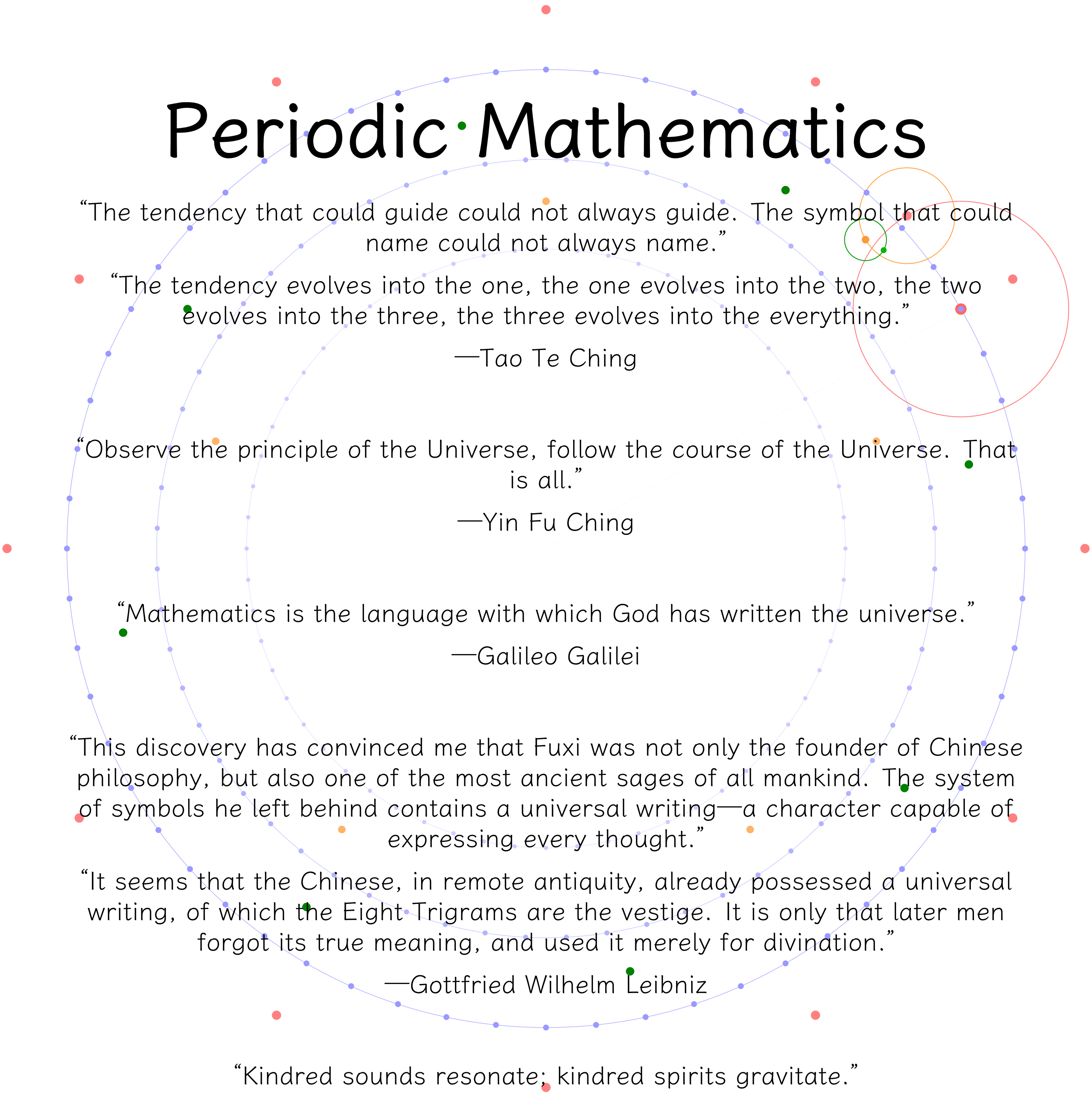

Periodic·Mathematics

“The tendency that could guide could not always guide. The symbol that could name could not always name.”

“The tendency *evolves* into the one, the one *evolves* into the two, the two *evolves* into the three, the three *evolves* into the everything.”

—Tao Te Ching

“Observe the principle of the Universe, follow the course of the Universe. That is all.”

—Yin Fu Ching

“Mathematics is the language with which God has written the universe.”

—Galileo Galilei

“This discovery has convinced me that Fuxi was not only the founder of Chinese philosophy, but also one of the most ancient sages of all mankind. The system of symbols he left behind contains a universal writing—a character capable of expressing every thought.”

“It seems that the Chinese, in remote antiquity, already possessed a universal writing, of which the Eight Trigrams are the vestige. It is only that later men forgot its true meaning, and used it merely for divination.”

—Gottfried Wilhelm Leibniz

“Kindred sounds resonate; kindred spirits gravitate.”

“Nothing flat does not slope; nothing gone does not return.”

—I Ching

“The void *evolves* into the cosmos which *evolves* into the tendency that is limited in bounds. The clarity *evaporates* into the sky while the mud sinks into the ground. The clarity *evolves* into the positive while the mud *evolves* into the negative. The positive and the negative *evolves* into the seasons and the seasons *evolves* into the myriad things.”

—Huainanzi (Treatise on the Patterns of Heaven)

$$\mathcal{P}_T = \{(A_k, \omega_k, \phi_k) \mid A_k \in \mathbb{R}^+, \omega_k = 2\pi k/T, \phi_k \in [0, 2\pi), k = 0, 1, \ldots, T-1\}$$

$$\mathcal{K}^{(d)}\{f, g\} = \sum_n \sum_m A_{n,m} \cdot \mathsf{Sim}^{(d)}(\theta_n, \theta_m) \cdot e^{i(\Omega_{n,m} x + \Phi_{n,m})}$$

“In birth, death is already present; in death, birth is already present. In what is deemed possible, the impossible is already present; in what is deemed impossible, the possible is already present. Right depends on wrong, and wrong depends on right.” —Zhuangzi, (On the Equality of Things)

“The great similarity that differs with small similarity is called small similarity-and-difference. Similarity and difference among the everything called great similarity-and-difference.”

—Zhuangzi (All Under Heaven)

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List of Symbols

Symbol	Meaning
$\mathbb{N}$	the set of natural numbers
$\mathbb{Z}$	the set of integers
$\mathbb{Q}$	the set of rational numbers
$\mathbb{R}$	the set of real numbers
$\mathbb{C}$	the set of complex numbers
$\mathbb{H}$	the set of quaternions
T	period
ω	angular frequency, $\omega = 2\pi/T$
$\mathcal{O}_k^{(T)}$	the k-th phase operator of period T, $\mathcal{O}_k^{(T)} = e^{i2\pi k/T}$
$\text{Group}_{\text{period}}(f)$	the group with periodic orbits of a function f
$\text{Symmetry}(f)$	the symmetry group of a function f
$\text{Group}_{\text{conservation}}(f, Q)$	the conservation group of a function f with respect to the conserved quantity Q
$\circledast$	periodic convolution
$\mathcal{K}^{(d)}\{f, g\}$	the d-th kernelised periodic convolution
$\text{Similarity}^{(d)}$	the d-th similarity
$\text{Tendency}^{(d)}$	the d-th trend
$\mathcal{E}(x)$	the epicycle activation function (CEAF)
$\mathcal{W}(t)$	Waveform, the epicycle complex function
KDI	the kernel-smoothed derivative–integral descent optimiser
Group actions and symmetry	
$\text{ord}(g)$	the order of a group element g (the least positive integer n with $g^n = e$)
$\langle g \rangle$	the cyclic subgroup generated by g
$\text{Symmetry}(f)$	see the table above
$G \cdot x$	the orbit of the element x under the group G
$\text{Stab}_G(x)$	the stabiliser of x in G
$\mathbb{Z}/n\mathbb{Z}$, C_n	the cyclic group of order n
The three elementary transformations and the centres of symmetry	
T_a, S_λ, R_θ	translation, scaling and rotation operators
$c = 0,\ 1,\ i$	the three centres of symmetry: additive reflection, multiplicative scaling reflection and multiplicative inversion
Φ	the decision map (or evolution map) of a target

Symbol	Meaning
$\mathrm{ord}(\Phi)$	the order of the decision map; $\mathrm{ord}(\Phi) < \infty$ means uniform periodicity
$\beta_{a,\mathrm{u},\theta}$	the episphere, $\beta_{a,\mathrm{u},\theta}(t) = ae^{\mathrm{u}\theta t}$
Periodic series and epicycle sets	
T_0	the least positive period
ρ	the normalised period ratio, $\rho = T_x/\Delta x$
$\mathfrak{E}$	an epicycle set, $\mathfrak{E} = \{(A_k, \omega_k, \phi_k)\}$
$\mathcal{A}_f$, $\mathcal{F}_f$, $\mathcal{P}_f$	the amplitude series, the frequency series and the phase series
Λ, $\mathcal{P}_\Lambda$	the frequency dictionary and the periodic transform on it
$\mathcal{E}(x)$, $\mathcal{W}(t)$	the complex exponential activation function and Waveform (see the table above)
Analysis, operators and geometry	
$\rightrightarrows$	uniform convergence
$\leftrightarrow$, $\iff$	correspondence (a transform pair), if and only if
∇, $\nabla\cdot$, $\nabla\times$	gradient, divergence, curl
Res	residue
$\mathcal{L}$, $\widehat{f}$	the Laplace transform, the Fourier transform
S^1, S^3, $\mathbb{T}^n$	the circle, the 3-sphere, the n-torus
$U(1)$, $SU(2)$, $SO(3)$	the circle group, the special unitary group, the special orthogonal group
T_pM, TM	the tangent space, the tangent bundle
Σ_E	the energy surface $H^{-1}(E)$
$\mathrm{Cl}_{0,n}(\mathbb{R})$	the Clifford algebra
$\partial/\partial\bar{q}$	the Fueter operator (quaternion analysis)
Notational conventions	
Δx, δ	sampling interval / tolerance
$\mathrm{gcd}(a, b)$	greatest common divisor
$\lfloor x \rfloor$, $\lceil x \rceil$	floor, ceiling
$\mathrm{Group}_{\mathrm{period}}(f)$, $\mathrm{Group}_{\mathrm{conservation}}(f, Q)$	see the table above

Abstract

This book develops a unified mathematical framework for periodic phenomena, based on the principle "symmetry → invariant → closed orbit → period".

The starting point is that periodic phenomena in different disciplines—celestial motion, electrical oscillation, biological rhythm, economic fluctuation and algorithmic iteration—have long been described in mutually untranslatable languages. The book argues that they can all be captured by a single mathematical object: a **family of epicycles**. A system of period T is described by a set of amplitude, angular-frequency and phase triples (A_k, ω_k, ϕ_k), whilst the relations between periods are captured by **periodic convolution**. Periodicity is thereby reduced to a question about group actions and group representations: the three basic transformations—translation, scaling and rotation—generate every elementary periodic structure, and their completeness can be proved rigorously within the framework of one-parameter families of conformal transformations of the complex plane.

The core methodology consists of four pillars: group representation theory, periodic convolution, window functions and refraction-index diagnostics, and complex exponential activation functions (CEAF) with kernel-smoothed derivative–integral descent. On this basis the book gives a standard five-step interface for attaching an arbitrary target to the framework, and applies it domain by domain in more than twenty fields, from astronomy, physics, biology and psychology to economics, cryptography and computer science. Throughout, three things are kept strictly apart: theorems that have been proved, conceptions advanced as research programmes, and problems that remain open.

Keywords: periodic mathematics; epicycle; harmonic analysis; conservation law; symmetry; phase; amplitude; frequency; periodic convolution; group theory; Fourier transform; Mellin transform; wavelet transform; I Ching; Leibniz; calendar; chip; signal processing

Preface

The cover: a universe of epicycles

On the cover, three concentric circles nest quietly within one another. On the outermost ring sixty points are evenly distributed, like the graduations of a clock, hinting at the recurrence of the sexagenary cycle; one layer inwards, twelve red points mark the positions of the Earthly Branches; further inwards, eight green points correspond to the arrangement of the Eight Trigrams; and on the innermost ring, five orange points hint at the workings of the Five Phases. At the centre of the circle, a red point extends a fine line that joins the first small circle; the point on that circle in turn extends a smaller circle; and on the smaller circle there is a circle smaller still. This is a picture of epicycles—for every single step forward of the outer circle, the inner circle turns through one full revolution. The ancient Greeks used such nested circles to describe the motions of the planets, and this book will use them to describe things that possess a periodic structure **under appropriate conditions**.

At the bottom of the cover there are two lines of formulae. The first tells us that the phase point of a system with period T can be labelled by the triple (A_k, ω_k, ϕ_k)—amplitude, frequency, phase. The second expresses the preliminary idea of periodic convolution: the deep similarity between two functions depends not only on whether their respective frequencies coincide, but also on the interaction between every pair of frequencies. **It must be stressed** that this statement is at present a **heuristic picture** rather than a mathematical theorem; the conditions required for it to hold rigorously are to be found in the **Introduction**, among the convergence, symmetry and boundedness requirements listed after the definition of periodic convolution.

This picture is a visual summary of the entire contents of the book: periodic systems can be described by families of epicycles, epicycles are characterised by amplitude, frequency and phase, and the relations between periods are described by periodic convolution. **It must be stressed once again** that this summary is a **research-programme picture**: throughout the book it is gradually converted into a number of precise, conditional mathematical propositions, rather than being a theorem obtained at a single stroke.

Periodicity: an early language for understanding the world

Human awareness of periodicity can be traced back to the observation of the firmament. The alternation of day and night, the waxing and waning of the moon, the return of the stars—these astronomical rhythms recur with a predictable regularity, and in several early civilisations they constituted the starting point of the awareness of time. In Mesopotamia the Sumerians compiled the appearances and disappearances of Venus into star catalogues; in the Nile valley the Egyptians used the heliacal rising of Sirius to forecast the flood; on the two banks of the Yellow River the forebears measured the lengthening and shortening of the sun's shadow with the gnomon, determined the solar terms and regulated the farming seasons. Periodicity is one of the earliest tools with which human reason sought to characterise time.

The periodicities above the firmament not only gave rise to calendars, but in several civilisations also promoted the development of mathematics. When the forebears discovered that the motions of the sun and the moon are not isolated phenomena, but can be referred to one another and predicted from one another, a recognition gradually took shape: within a certain accuracy, the motions of the heavenly bodies can be grasped by rhythmic mathematical models. This recognition was one of the factors behind the independent emergence of a tradition of exact science in several civilisations—but it was not the only factor, nor did every civilisation develop this recognition in the same way.

The mathematisation of periodicity

Once periodicity has been perceived, it is gradually mathematised. The Babylonians turned the periods of the sun, the moon and the five planets into sexagesimal numerical tables; the Maya used the nesting of multiple calendars to track the interleaving of Venus and the lunar syzygies; the ancient Greeks decomposed the motions of the heavenly bodies into composites of epicycles and deferents. In several early civilisations astronomy was one of the first branches of science to attain high precision—because the periodic motions of the heavenly bodies provided relatively stable objects of measurement.

In China this process of mathematisation unfolded in a distinctive way. The 'Tianwen xun' (Astronomy) chapter of the Huainanzi, in the words 'in seventy-six years the sun and the moon return, in the first month, to the fifth degree of Yingshi, with no fractional remainder, and this is called one epoch', establishes a high-precision common-multiple approximation to the tropical year and the synodic month; the Huangji Jingshi, by means of its nested system of yuan, hui, yun and shi—one yuan comprises twelve hui, one hui thirty yun, one yun twelve shi, and one shi thirty years—brings the periods of celestial phenomena and the periods of human affairs into one and the same self-consistent arithmetical framework; and the division into the twenty-four solar terms is an independent invention of the solar calendar: it divides one circuit of the ecliptic into twenty-four equally spaced phase nodes, each node corresponding to the approximate position of the sun on the ecliptic. **It should be pointed out** that the historical account given above belongs to traditional calendrical reckoning and to the interpretation of cultural history; its specific numerical values and its dating should be regarded as **symbolic settings or as following traditional accounts**, and should not be taken as premises for strict historical fact or for mathematical derivation.

By the seventeenth century, Kepler had turned more than twenty years of Tycho's observational data into three laws. The elliptical orbit of a planet is no longer an arbitrarily posited geometrical construction, but the necessary outcome, in an inverse-square force field, of the joint constraints of conservation of angular momentum and conservation of energy. Newton went further and unified these three laws into the equation of universal gravitation, and from then on the mathematical description of periodicity acquired the foundation of classical mechanics. **It should be pointed out** that the explanation of periodic motion given by Kepler's laws and by Newtonian mechanics holds **within the framework of classical mechanics**; within the frameworks of general relativity or of quantum mechanics the corresponding statements require modification.

From invariance to conservation laws, and from conservation laws to symmetry

One core feature of periodicity is **invariance**. The sun rises and sets, and today resembles yesterday; the moon wanes and waxes, and this cycle resembles the one before; spring departs and autumn arrives, and this year resembles the year before. What the forebears perceived first was that certain properties remain approximately unchanged amidst change. The sun rises in roughly the same direction every day, the moon completes one cycle of waning and waxing about every twenty-nine days, and the four seasons succeed one another each year in a fixed order. These approximate invariances are the earliest knowledge of periodicity.

When invariance is rendered precise as a mathematical proposition, conservation laws make their entrance. When a quantity remains unchanged throughout a certain class of evolutionary processes, we say that this quantity is conserved in that class of processes. Kepler discovered that, as a planet moves on its elliptical orbit, the line joining the planet to the sun sweeps out equal areas in equal times—this is a special case of conservation of angular momentum. Galileo discovered that the period of a pendulum at small amplitudes is approximately independent of the amplitude—this is a manifestation of the isochronism of nonlinear systems in the small-amplitude limit, and not strict conservation of energy. Newton unified these observations into the laws of motion: in an inverse-square force field, conservation of angular momentum and conservation of energy jointly constrain the orbit to be a closed ellipse.

Once conservation laws have been established, symmetry emerges naturally. If a system possesses a conserved quantity, this means that there exists some transformation under which the behaviour of the system remains unchanged. Translation in time does not change energy, translation in space does not change momentum, and rotation does not change angular momentum. At the beginning of the twentieth century Noether's theorem gave this intuition a profound mathematical form: **on the premise that the action principle holds and that the symmetry transformations are continuous and smooth**, every continuous symmetry implies a conserved quantity, that is, **symmetry ⇒ conservation**; **the converse fails**—a conservation law does not imply the corresponding symmetry (see Corollary 1.2 of the 'Operations and Periodicity' chapter). **The conditions of Noether's theorem cannot be omitted**—for discrete symmetries, for non-Lagrangian systems or for non-smooth systems the above implication generally fails, and the cases have to be treated separately.

The logical order must be stressed: **it is not that symmetry comes first and conservation laws afterwards; rather, the observation of invariance comes first, conservation laws are then abstracted from it, and symmetry is discovered after that.** Within the framework of Noether's theorem, conservation laws and symmetries do not stand in one-to-one correspondence but in a **one-directional implication**: **symmetry $\Rightarrow$ conservation**, whilst the converse fails, that is, **conservation $\not\Rightarrow$ symmetry** (Corollary 1.2). They point to one and the same fact: the system preserves a certain constant structure throughout change. The manifestation of this structure in phase space is, **under certain conditions**, a closed orbit or an invariant torus. **But a conservation law by itself does not guarantee that the orbit is closed**—a system with conservation of energy may still have an orbit that does not close (a hyperbolic orbit, for instance); closure requires additional topological or dynamical conditions.

Three generating modes of periodicity: a research programme

Within the framework of this book we **reduce the generating transformations of periodicity to three basic operations**, and maintain that they constitute a reasonable research programme. The strict form of this claim—namely 'any continuous transformation that preserves some intrinsic structure can be decomposed into a composite of these three groups'—is a **research programme**; the range within which it holds has to be delimited chapter by chapter, and it should not at present be regarded as a proved theorem.

(1) **Translation.** On a train travelling at constant speed, each full revolution of a wheel brings some mark on the carriage floor back to the same position. What translation generates is the traditional cyclic period, and its conserved quantity **within the Lagrangian framework** is momentum.

(2) **Scaling.** The boundary of the Koch snowflake returns to itself under every scaling transformation—after magnification by a factor of three, the boundary curve has the same structure as before magnification. This is a **self-similar** structure: the perimeter is infinite whilst the area is finite, and the area is precisely the conserved quantity that makes scaling 'close'.

(3) **Rotation.** The revolution of the Earth about the Sun is, in the approximation that neglects the perturbations of the other planets, a closed orbit in a rotationally symmetric system, and its conserved quantity is angular momentum.

The three operators gradient, curl and divergence are each given a rigorous definition, differentiability preconditions and range of applicability in the 'Vector Analysis' chapter; their respective structures have already been listed in the **Introduction** and are not repeated here.

It should be pointed out in particular that this book does **not** place these three operators alongside the three basic transformations of translation, scaling and rotation as a set of 'correspondences'. The three transformations act on **space**, whilst the three operators act on **fields** (scalar fields, vector fields), and the two lie at different mathematical levels; arranging them into a correspondence table is both unnecessary and liable to obscure their respective genuine preconditions. Any statement involving any one of the three is given separately in the 'Vector Analysis' chapter, with explicit differentiability preconditions and domain conditions.

Phase: the mathematics of relation itself

In the mathematics of periodicity, phase **can** be understood as an active transformation operator, and not merely as a passive coordinate. The k-th phase of a period T corresponds to the complex number $\mathcal{O}_k^{(T)} = e^{i2\pi k/T}$. Multiplication by $\mathcal{O}_0 = 1$ yields the object itself—congruence; multiplication by $\mathcal{O}_{T/2} = -1$ yields the opposite vector—antipodality; multiplication by an arbitrary $\mathcal{O}_k$ defines the relation of rotation through $2\pi k/T$. The multiplicative closure of the phase operators guarantees the closure of the system of relations: a relation between relations is still a relation. This closure is an algebraic property of the finite cyclic group C_T, and does not depend on any physical interpretation.

This perspective offers a **possible** route towards incorporating such traditional relations of the I Ching as 'combination and transformation' and 'clash and overcoming' into the framework of group representation theory; its concrete modern mathematical encoding (the correspondences of the Heavenly Stems, the Earthly Branches and the Eight Trigrams with groups with periodic orbits, together with the epicycle nesting rules involved) is given in the **Introduction**, where the book formulates these correspondences as algebraic isomorphisms or equivalences of group actions, rather than as historical assertions about ancient documents.

Leibniz's discovery: periodicity embedded in number itself

In the course of the mathematisation of periodicity, Leibniz made a distinctive and profound observation. Whilst studying binary arithmetic he discovered that, in the binary expansions of the sequence of natural numbers, the digit in each position displays a strictly periodic behaviour—the 0-th position cycles with period 2, the 1-st position cycles with period 4, and the k-th position cycles with period 2^{k+1}.

A note on historical verification: Leibniz did study binary arithmetic and was deeply interested in its philosophical implications (in particular its connection with the I Ching); the sentence in this book, 'he named this phenomenon **Periodus numerorum**—the period of the natural numbers', belongs to the category of **historical material awaiting verification**, and would require the support of primary sources or of secondary scholarship before it could be stated as historical fact. Here we retain this formulation, but we mark its epistemological status as 'alleged' or 'historical material awaiting verification', and do not treat it as a settled historical conclusion.

The **mathematical content** of this observation is itself clear: periodicity does not exist only in the grand motions above the firmament, but is also embedded in the structure of the binary representation of the number system. In each position the 0s and 1s alternate with a fixed rhythm.

Leibniz went further and pointed out that the uniform distribution of binary fractions on the unit circle of the complex plane—taking n binary places divides the circumference into 2^n equal points—constitutes an approximation to the continuous circle. This geometrical intuition **can** be set in correspondence with the generative structure of the hexagram system of the I Ching: Taiji (2^0) → Yin-Yang (2^1) → the Four Symbols (2^2) → the Eight Trigrams (2^3) → the Sixty-Four Hexagrams (2^6). **It should be pointed out** that this correspondence is a modern mathematical interpretation offered by this book, and not a historical assertion about the original text of the I Ching.

Epicycles: a unified language for describing periodic phenomena

In systems possessing conservation laws there exist closed orbits **under certain conditions**. A planet with conservation of angular momentum has an orbit that closes in an inverse-square force field; a pendulum with conservation of energy has a swing that is a closed orbit in phase space. A closed orbit means that the system traces a loop in phase space—an epicycle.

The epicycle is not the exclusive property of astronomy. **Many** things with a periodic structure can all be described by means of epicycles. The propagation of a beam of monochromatic light can be described by the sinusoidal oscillation of the electromagnetic field in space; the acoustic vibration of a word can be approximately described by the periodic fluctuation of air pressure; the prosody of a sentence can be approximately described by the periodic alternation of syllable length and stress; the oscillation of an electrical signal in a wire can be approximately described by the back-and-forth motion of charge about an equilibrium position; and the crystal oscillator in a chip is a substantially fixed epicycle that provides the whole system with its time reference.

Mathematically speaking, **under suitable function spaces and convergence conditions**, a periodic function can be represented as a Fourier series:

$$f(x) = \sum_{k\in\mathbb{Z}} c_k e^{ik\omega_0 x}.$$

Two cases have to be distinguished: (i) the **finite-harmonic model**: $f(x) = \sum_{k=1}^{N} A_k e^{i(\omega_k x+\phi_k)}$, suitable for describing the superposition of finitely many epicycles, and a form in common use in numerical implementation and engineering modelling; (ii) the **general Fourier series**: $f(x) = \sum_{k\in\mathbb{Z}} c_k e^{ik\omega_0 x}$, the complete expansion of a periodic function (under suitable convergence conditions), whose coefficients are complex numbers and carry amplitude and phase information. **One cannot say that 'a periodic function = a superposition of finitely many epicycles'**—a general periodic function requires infinitely many Fourier modes (a square wave, for example, requires infinitely many odd harmonics). But when attention is concentrated on finitely many dominant components, the finite-harmonic model is a useful approximation. When concrete numerical implementation is involved, this book usually adopts the finite-harmonic model, whilst when general theory is involved it uses the Fourier series.

Under appropriate conditions, periodic phenomena can be understood as superpositions of several epicycles. The large circle of the outer ring carries the overall rhythm, the small circles of the inner rings contribute the texture of detail, and the phase determines the precise position of each circle at a given moment.

Periodic convolution: a unified computation for cross-period interaction

One of the core research programmes proposed in this book is **periodic convolution**: bringing amplitude (scaling) and phase (rotation) into the operations on series as well, and thereby retaining the interaction between every pair of frequencies; the second line of formulae on the cover is its preliminary form. Its two-layer construction (the interaction tensor and the composition map), the periodic convolution formed by composing the two, and the various conditions required for it to hold rigorously, are given together in the **Introduction** and are not repeated here.

Approximation, intercalation and the boundedness of incommensurable periods

Among the many technical difficulties confronting the mathematics of periodicity, one class is unlike the others: it comes neither from computation nor from topology, but from **number theory**. When the continuous period T_f and the sampling interval Δx are incommensurable, that is, when the ideal period ratio

$$\rho = \frac{T_f}{\Delta x} \notin \mathbb{Q}$$

then the sampling sequence within any finite window **cannot** have a strictly integral period. This is not an observational error, nor a defect of the instrument, but the intrinsic cost of incommensurability. The manifestation of this cost in the frequency domain is **spectral leakage**: the number of periods within the window is not an integer, and the energy of the signal necessarily spreads from the main lobe into the sidelobes.

The mathematical definition of intercalation

One of the core methodologies proposed in this book is **intercalation**—approximating the real number ρ by a sequence of integers $\{N_m\}$ so that the **cumulative sample count**

$$S_M = \sum_{m=0}^{M-1} N_m$$

approximates $M\rho$ with **bounded error**. The core construction of intercalation consists of three basic intercalation sequences:

1. **Floor intercalation**: $N_m^{\text{floor}} = \lfloor (m+1)\rho \rfloor - \lfloor m\rho \rfloor$, with error $E_M^{\text{floor}} \in (-1, 0]$;
2. **Ceiling intercalation**: $N_m^{\text{ceil}} = \lceil (m+1)\rho \rceil - \lceil m\rho \rceil$, with error $E_M^{\text{ceil}} \in [0, 1)$;
3. **Balanced intercalation**: $N_m^{\text{round}} = \text{round}((m+1)\rho) - \text{round}(m\rho)$, with error $|E_M^{\text{round}}| \le 1/2$.

What the three sequences have in common is this: $N_m \in \{\lfloor \rho \rfloor, \lceil \rho \rceil\}$—at each step the choice is only between the lower integer period and the upper integer period—yet the cumulative error is confined within a bounded band, either $[-1/2, 1/2]$, $(-1, 0]$ or $[0, 1)$. This boundedness is where the whole value of intercalation lies.

Foundations of approximation theory: the role of continued fractions

The **optimal** choice of intercalation sequence is not arbitrary; it is determined by the continued-fraction expansion of ρ. Let $\{p_j/q_j\}$ be the continued-fraction convergents of ρ. Then:

(i) **Optimality**: p_j/q_j is the closest approximation to ρ among all rational numbers whose denominators do not exceed $q_{j+1} - 1$;

(ii) **Error bound**: $|p_j - q_j\rho| < 1/q_{j+1}$;

(iii) **Alternating approximation**: the even convergents approach ρ from below and the odd ones from above, forming a symmetric squeeze.

On the intercalation window $T = p_j$, $M = q_j$, the cumulative error tends to zero. This raises intercalation from an 'error-control trick' to a **method with a rigorous optimality criterion**—continued-fraction convergents are precisely the optimal intercalation windows. This optimality derives from a classical fact in number theory: continued-fraction convergents are the best rational approximations to a real number; intercalation translates this from number theory into the choice of sampling window.

Bounding spectral leakage

The effect of intercalation on spectral leakage can be stated precisely. Let $F[k]$ be the DFT on the intercalation window and c_m the continuous Fourier coefficients. Then

$$\left|F[k] - c_{k/M}\right| \le C \cdot \frac{|E_M|}{T} \cdot \sum_{m \in \mathbb{Z}} |c_m|,$$

where C is a constant independent of T. This inequality gives an **explicit upper bound** for the leakage—intercalation reduces leakage from 'uncontrollable' to 'bounded'. When $\rho \in \mathbb{Q}$, intercalation degenerates into a constant sequence and the leakage is strictly zero; when $\rho \notin \mathbb{Q}$, intercalation chooses windows by means of continued fractions, so that this upper bound tends to zero along the sequence of convergent windows.

It must be stressed that intercalation does **not eliminate** leakage; it reduces the amplitude of the leakage from $O(1)$ to $O(|E_M|/T)$. This distinction is essential, and it reflects the intrinsic cost of incommensurability—no finite window can align ρ exactly with an integer. The value of intercalation lies in this: it provides a **quantitative characterisation** of this cost and a **systematic method of minimising** it.

The complementarity of intercalation and window functions

Intercalation and window functions are two independent routes for suppressing spectral leakage, and they act on different stages of the leakage:

- **Intercalation** acts on the **window length**: choosing $T = p_j$ makes the window contain approximately an integral number of continuous periods, reducing leakage at its source;
- **Window functions** act on the **window boundary**: smoothing windows (Hann, Hamming, Blackman and the like) suppress the sidelobes, at the cost of a broadening of the main lobe.

The two can be used in combination: first use intercalation to choose the optimal window, then apply a window function, and one obtains a narrow main lobe and low sidelobes simultaneously. This is a combined effect that no single method can achieve.

The place of intercalation in this book

The method of intercalation performs three functions in this book:

(1) in the 'Discrete Sampling and Intercalation of Periodic Signals' chapter it is established as a bridge between commensurability and incommensurability, yielding the intercalation error boundedness theorem and the continued-fraction optimal intercalation theorem;

(2) in the 'Harmonic Analysis' chapter it is used to derive the DFT leakage estimate for intercalation windows, giving an explicit upper bound for the amplitude of the leakage;

(3) in the 'Periodic Category' chapter it is made functorial—the intercalation functor $\mathcal{R}$ repairs the irreversible gap from the continuous category to the discrete category, so that the adjoint pair formed by the continuous limit and the discretisation functor is preserved to arbitrary accuracy in the incommensurable case.

Furthermore, intercalation, together with the window-function methods of the 'Time Series Analysis' chapter, calendrical intercalation in the 'Periodic Structure of Astronomical Systems' chapter, intercalation by absence of mid-terms in the 'Periodic Structure of Calendar Systems' chapter, state periods in the 'Periodic Structure of Cryptographic Systems' chapter, multi-agent synchronisation in the 'Periodic Structure of Agents' chapter, and multitaper spectral estimation in the

'Periodic Structure of Data Analysis' chapter, forms a unified perspective that runs through the whole book.

It must be stressed once again that intercalation is at present established throughout the book as a methodology supported by rigorous theorems, but that it does **not solve** all the difficulties of incommensurability. In cases such as multi-frequency signals, time-varying periods and unknown period ratios, the role of intercalation is limited, and it has to be used in combination with other methods (multitaper spectral estimation, Bayesian spectral analysis, joint continued-fraction approximation). This book will **clearly mark** the range of applicability and the limitations of the intercalation method.

From disciplinary fragmentation to a unified perspective

In the existing system of disciplines, researchers in different fields all deal with periodic phenomena, yet they use mutually disconnected languages. Astronomers use perturbation theory and orbital resonance to describe the quasi-periodic motions of celestial bodies; condensed-matter physicists use space groups and Brillouin zones to describe the translational symmetry of crystals; biologists use limit cycles and phase response curves to describe the genetic networks of circadian rhythms; fractal geometers use self-similarity and scaling exponents to describe invariant structures across scales; economists use spectral analysis and autoregressive models to detect the characteristic frequencies of business cycles; and computer scientists use cyclic redundancy checks and fast algorithms to handle the periodic components of signals.

Every one of these researchers is dealing with a problem connected with periodicity, yet each uses a mutually disconnected system of terminology. What a condensed-matter physicist calls a 'reciprocal lattice vector', what a signal-processing engineer calls the 'Nyquist frequency', and what a harmonic analyst calls 'Pontryagin duality'—these point, **within the harmonic analysis of finite cyclic groups or of discrete translation groups**, to one and the same mathematical fact: the frequency-domain representation of a periodic function. **It should be pointed out** that this 'sameness' holds at an appropriate level of abstraction, and that the concrete physical meaning and the conditions of applicability may differ from one discipline to another.

A similar fragmentation exists in the domain of integral transforms. The Fourier transform, the Laplace transform, the wavelet transform and the Mellin transform are usually taught separately, as mutually independent tools. Yet when examined from the standpoint of group representation theory, **if appropriate group actions, representations and function spaces are chosen**, the different integral transforms can be given a unified description within the framework of group representations and harmonic analysis: the Fourier transform is associated with the translation group, the Mellin transform with the scaling group, and the wavelet transform with the affine group. **These statements need to be verified case by case**, and their precise form depends on the chosen function space, group action and measure.

In sampling theory this fragmentation likewise exists. The 'sampling theorem' in signal-processing textbooks, 'Pontryagin duality' in harmonic-analysis textbooks, and 'intercalation' in calendrical-astronomy textbooks—the three are taught separately, yet they point to three faces of one and the same mathematical structure: how, in the incommensurable case, to approximate the continuous by the discrete. The theory of intercalation presented in this book is precisely an attempt to bring the three within a single framework.

The cost of disciplinary barriers

The fragmentation of terminology and method described above brings at least threefold costs:

(1) **Repeated discovery.** A mathematical theorem already known in one discipline is independently rediscovered in another, resulting in the duplicated investment of scholarly resources.

(2) **Deep analogies are obscured.** Between the Kepler ellipse and the Koch snowflake, between crystal lattices and genetic oscillations, between the Fourier transform and the Mellin transform—these identical mathematical structures, which hold **at an appropriate level of abstraction**, are difficult to recognise because of differences in disciplinary language.

(3) **Lack of diagnostic tools.** Faced with a new and complex system, a researcher may be able to rely only on descriptions at the level of phenomena, and finds it difficult to carry out a diagnosis at the level of structure.

The possibility of a unified framework

Although the disciplines differ on the surface, in every case involving a periodic structure it is **usually** possible to isolate three common mathematical ingredients:

- a **symmetry group** G, acting on the state space M of the system;
- a **conserved quantity** Q, invariant under the action of G;
- a **closed orbit** $\mathcal{O} \subset M$, generated by the joint constraints of the conservation laws and the dynamics.

This means that a **unified mathematical framework of periodicity**, whose language is group action, whose criterion is a conserved quantity, whose geometrical image is a closed orbit and whose tool of decomposition is the epicycle series, is theoretically possible. This book is precisely one exploration of that possibility. **Whether it succeeds depends on whether it can produce non-trivial new mathematical results in the chapters that follow**, and not merely on renaming knowledge that already exists.

In the chapters that follow, this book will gradually develop the complete theory of the intercalation method: in the 'Discrete Sampling and Intercalation of Periodic Signals' chapter as a tool of commensuration, in the 'Harmonic Analysis' chapter as a tool for bounding leakage, and in the 'Periodic Category' chapter as a tool of functorial correction. Intercalation turns 'incommensurability', the most stubborn number-theoretic difficulty in periodic analysis, from a passively accepted defect into an actively exploited structure—this is an independent contribution of this book to the methodology of periodic analysis.

The standpoint and the limits of this book

The mathematics of periodicity does not claim a priori that all the systems it examines 'possess a periodic structure'. In each application chapter it begins by asking three questions, by means of three criteria (the full statement of the criteria is given in the **Introduction**). For systems that satisfy the conditions, the book gives a rigorous description of the periodic structure; for systems that satisfy them only in part, the book measures precisely the degree of their deviation from a closed orbit; for systems that do not satisfy them, the book states clearly that their 'periodicity' is merely a metaphor at the level of phenomena. This diagnostic function is the key respect in which the mathematics of periodicity differs from general interdisciplinary analogy.

Nor does the mathematics of periodicity attempt to become a theory of everything. It does not explain why a particular conserved quantity takes a particular value, does not derive the evolution equation of a particular system, and does not predict the future course of the economy. What it provides is a set of diagnostic tools—a unified grammar for rigorously describing, comparing and analysing the periodic structures that arise in different systems. Its success lies not in proving that everything is periodic, but in being able to state precisely: this system is periodic, because its orbit closes under the action of some group; that system appears to be periodic, but in fact lacks the guarantee of a conservation law; this integral transform and that integral transform are equivalent representations of harmonic analysis on the same group under different parametrisations. **The correctness of these judgements has to be supported, chapter by chapter, by concrete mathematical propositions.**

The journey of this book

The journey of this book sets out from the simplest translation period, gradually builds up the mathematical theory of rotation periods and scaling periods, passes through group representation theory, harmonic analysis and periodic manifolds, and finally arrives at the abstract unification of the periodic category. Once this framework has been established, we shall take this set of tools with us and travel in turn through astronomy, calendrics, geography, physics, biology, psychology, economics, political science, computer science and integral transforms—in each field revealing, **as far as possible** in the form of theorems or rigorous propositions, how the mathematical structure of periodicity illuminates our understanding of nature and society; and at the same time, **for those parts where a rigorous proposition cannot yet be given, the book will clearly mark them as conjectures or analogies.**

That nested picture of epicycles on the cover is precisely the starting point of this journey. The points on the three concentric circles are the outermost celestial sphere; the nested small circles extending from the red point at the centre are the trajectories of the planets; and the two lines of formulae on the cover are the mathematical language of all of it. When we close this book, we hope that what the reader sees will no longer be merely periodic functions, but a living, vibrant universe woven together out of amplitude, frequency, phase and period—**provided, of course, that we always remember that what mathematics describes is the model, and not the universe itself.**

Intended readership and prerequisites

This book addresses four kinds of reader: mathematicians who wish to treat the periodic structure of a system as an algebraic and geometrical object; graduate students and researchers in science and engineering who deal with periodic phenomena daily and wish to acquire a unified vocabulary; engineers and data analysts who face incommensurable sampling ratios and spectral leakage in their practical work; and readers with a general mathematical literacy who are concerned with the relation between modern mathematics and traditional symbolic systems (such as the I Ching).

The prerequisites required by the mathematical chapters are: linear algebra (matrices, eigenvalues, transposition, the orthogonal group); mathematical analysis (limits, continuity, differentiability, uniform convergence); group representation theory (groups, homomorphisms, orbits, group actions, representations); and elementary complex analysis (the complex plane, the exponential function, holomorphic functions). The two chapters that require topology and differential geometry—'Manifold and Fibre Bundle' and 'Periodic Manifold'—each declare their own prerequisites at the beginning of the chapter; readers who lack this preparation may skip these two chapters on a first reading without affecting the logical thread of the remaining parts. Apart from the first eleven chapters, the application chapters require no further mathematics, and each chapter restates the concepts it uses; the 'Probability and Distribution' and 'Time Series Analysis' chapters additionally require the foundations of an introductory course in probability theory and mathematical statistics.

Reading paths: readers whose main interest is the mathematical core may read Chapters 1 to 24, together with Chapter 46; readers whose main interest is one particular application may read Chapter 1, then the relevant chapter in Part II, and then the application chapter that concerns them; readers who need only computational tools may begin with the four chapters 'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis', 'Epicycle Activation Function and Function Approximation' and 'Modelling and Analysis: From Data to Epicycles'.

Conventions of style and labelling

The mathematical chapters of this book follow this order: **Definition → Proposition → Lemma → Theorem → Proof → Corollary → Property → Example → Remark.** Cross-references are always given by name, and the referencing mechanism of the whole book uses the standard facilities provided by the cleveref package of LaTeX.

The proof status of every assertion is marked. Those for which a proof is given in this book are presented in the form of a definition, proposition, lemma, theorem, corollary or property, and are accompanied by a proof. Those for which no proof is given in this book are marked with one of the following three labels: **Conjecture**, meaning that the statement is precise but that no proof has yet been obtained; **Analogy**, meaning that the correspondence between two structures is suggestive, but is not a bijection; **Research programme**, meaning a framework for organising a class of problems, rather than the assertion of a theorem. **These three labels should be read as warnings, not as ornament**: any claim bearing one of them has no proof available, and this book does not use it as a premise in any derivation.

Notation: $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{H}$ denote respectively the natural numbers, the integers, the rational numbers, the real numbers, the complex numbers and the quaternions. The three centres of symmetry are written $c = 0$ (additive reflection), $c = 1$ (multiplicative scaling reflection) and $c = i$ (multiplicative inversion). The period is written T, the corresponding angular frequency $\omega = 2\pi/T$, and the phase operator $\mathcal{O}_k^{(T)} = e^{i2\pi k/T}$. Periodic convolution is written $\circledast$; when the order d or the window kernel has to be displayed, it is written $\mathcal{K}^{(d)}\{f, g\}$.

Concerning historical sources: the historical and cultural material in the preface and in the ‘Periodic Structure of the I Ching System’ chapter falls into two classes. The parts concerning traditional calendrical reckoning, the I Ching and the history of astronomy are all cited as **traditional accounts or alleged historical sources**; they are not premises of any mathematical derivation in this book, and their specific numerical values should be regarded as symbolic settings. The parts concerning modern mathematics are given together with their premises. Wherever primary sources are required but cannot be supplied by this book, the text marks the passage as ‘historical material awaiting verification’.

Acknowledgements

This book originated in an attempt to write down, in one and the same notation, the periodic structures that appear in fields as far apart as celestial mechanics, crystallography, signal processing and calendrical reckoning. In the course of doing so it has incurred many debts. It owes a debt first of all to the mathematicians whose theorems are cited in this book, and to the authors of the textbooks and monographs listed in the bibliography—the standard results of group representation theory, harmonic analysis, number theory and differential geometry that appear in this book are all theirs, and whatever is new in this book has been obtained by standing on those results.

It owes a particular debt to the classical Chinese literature on mathematics and astronomy—including the Huainanzi, the Huangji Jingshi, the twenty-four solar terms and the sexagenary cycle: this literature provides the earliest instances of the common-multiple reasoning that this book calls ‘intercalation’. **It must be stressed once again** that the modern mathematical reading offered in this book is the author’s own interpretation, and not a historical assertion about these texts.

Thanks are due to the maintainers of the open-source tools without which this book could not have been completed: TEX Live and the LATEX packages used for typesetting, and the developers of the free typefaces used for the Chinese fonts. The errors that remain are the author's own.

Corrections, counterexamples and refutations are **especially** welcome. Since the book is offered as a research programme rather than as a finished edifice, the most useful response to it is to point out where it fails—and as precisely as possible.

Independent Research in Periodicity Mathematics
September 2026

Introduction

Working definition of periodic mathematics

Definition 0.1 (Periodic mathematics (working definition)). Periodic mathematics is a **research programme** proposed in this book. It takes the **symmetric structures generated by group actions** as a unified language, **translation, scaling and rotation** as the candidate classification of the basic generating transformations, and the **closed orbit** as the core geometric picture, and it attempts a unified mathematical description of the periodic and fractal structures found in different disciplines. Its objects of study are systems possessing a state space, evolution rules and a symmetric structure. Periodic mathematics does not assert that all systems are periodic; rather, it is devoted to providing a unified diagnostic and descriptive language for **systems in which a periodic structure can be defined**. **It must be stressed** that the above formulation is a working definition and a research programme, not a formal mathematical definition; its rigorous version must be established chapter by chapter, with conditions specified one by one.

Philosophical inspiration and cognitive method

The saying from the Tao Te Ching, 'The Dao gives birth to the one, the one to the two, the two to the three, and the three to the myriad things', and the saying from the Yin Fu Ching, 'Observe the way of heaven; hold fast to the movement of heaven', provide the philosophical starting point and the **inspiration** for the cognitive method of this book. The former describes a generative sequence: starting from a single origin, and passing through the ladder of differentiation, it produces diversity. In periodic mathematics, this sequence **can be likened to** the ladder of number-system extensions $\mathbb{N} \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{R} \to \mathbb{C} \to \mathbb{H}$—each step of extension introduces a new inverse operation, and each inverse operation introduces a new symmetry. This analogy is heuristic, not a mathematical theorem. The latter suggests a methodological path: observe invariance, abstract conservation laws, discover symmetries, identify closed orbits. **Observing the way of heaven** may be understood as identifying the conserved quantities of a system, and **holding fast to the movement of heaven** as using conservation laws to predict the future states of the system.

Leibniz's universal characteristic and the mathematical foundations of the I Ching

After reading the hexagram images of the I Ching, Leibniz wrote: 'This discovery convinces me that Fuxi was not only the founder of Chinese philosophy, but also one of the most ancient sages of all humanity. The system of symbols he left behind contains a universal writing—a character capable of expressing all thought.' **His concrete discovery**—the strict periodicity of the digits in each place of the binary expansion of the natural numbers, with periods 2, 4 and 2^{k+1} for the 0th, 1st and kth places—together with the examination of the historical sources as to whether he named it Periodus numerorum, has already been given in the **Preface**, and is not repeated here.

A further extension in the direction of the I Ching is this: the generative structure of the hexagram system corresponds to the approximation of the circle by binary fractions (see the **Preface**), and the saying in the I Ching, 'The I is modelled on heaven and earth', can **partly** acquire mathematical meaning in periodic mathematics: there exists an isomorphism between the hexagram system and certain periodic structures. 'Those of the same sound respond to each other, those of the same vapour seek each other' can be likened to the resonance of frequencies and the attraction of phases; 'There is no plain without a slope, no going without a return' can be likened to the necessity of closed orbits; 'Things cluster by their kind, creatures divide by their groups' can be likened to the classification of orbits under a group action. These analogies are **heuristic**, and their mathematical content must be defined one by one in the concrete chapters.

The periodic convolution of the Heavenly Stems ($T = 10$) and the Earthly Branches ($T = 12$) generates the sexagenary cycle, and the nesting of the Eight Trigrams ($T = 8$) with the Eight Trigrams ($T = 8$) generates the Sixty-Four Hexagrams—the epicycle nesting rule unifies the process by which small periods generate large periods: **with each step of the outer circle, the inner circle rotates a full turn**. Yin-Yang multiplied by the Five Phases gives the ten Heavenly Stems, the Four Symbols multiplied by the Three Powers gives the twelve Earthly Branches, and the Eight Trigrams multiplied by the Eight Trigrams gives the Sixty-Four Hexagrams. **In this book these correspondences are presented as algebraic isomorphisms or as equivalences of group actions, and not as assertions that the ancient texts themselves were already expressed in the language of modern group representation theory.**

Why translation, scaling and rotation

Periodic mathematics establishes translation, scaling and rotation as the three basic generating transformations, not on the basis of empirical induction, but on the basis of **motivation** at the following three levels. **It must be explained** that the three points below are **reasons for choosing these three**, and not the mathematical theorem that 'any continuous transformation can be decomposed as a composition of the three'; the latter is a stronger proposition that requires case-by-case verification.

(i) **Algebraic level**: the three transformations correspond respectively to the three most basic continuous groups—translation to the additive group $(\mathbb{R}^n, +)$, scaling to the multiplicative group $(\mathbb{R}^+, \times)$, and rotation to the circle group $U(1)$. **At a suitable level of abstraction**, many continuous transformations that preserve some intrinsic structure can be decomposed as a composition of these three groups; but this decomposition does not hold for all continuous groups, and must be decided case by case.

(ii) **Geometric level**: translation changes position without changing orientation, scaling changes scale without changing shape, and rotation changes orientation without changing position. **Within the framework of Euclidean and similarity geometry**, these three operations exactly exhaust the continuous transformations that leave the intrinsic structure of a system unchanged.

(iii) **Number-system level**: each step of number-system extension introduces a new inverse operation—subtraction introduces the translational direction (centre of symmetry 0), division introduces the proportional direction (centre of symmetry 1), and complex roots introduce the rotational direction (centre of symmetry i). This correspondence is **heuristic**, and its rigorous mathematical content belongs to the study of field extensions and automorphism groups.

Gradient, curl, divergence and the correspondence with the three symmetries

In vector analysis, the gradient ∇, the curl $\nabla\times$ and the divergence $\nabla\cdot$ are three basic differential operators. Their individual structures are as follows (the rigorous definitions, hypotheses and boundary conditions of the three are given in the 'Vector Analysis' chapter; this book does not place them side by side with the three basic transformations as a correspondence):

(1) **Gradient**: the gradient maps a scalar field to a vector field and points in the direction in which the function increases fastest, its magnitude being equal to the maximal directional derivative (this requires $f \in C^1$ and a non-vanishing gradient at that point).

(2) **Divergence**: the divergence maps a vector field to a scalar field and measures the net outward flux per unit volume; in a Hamiltonian system the vanishing of the divergence is equivalent to the conservation of phase-space volume (Liouville's theorem).

(3) **Curl**: the curl maps a vector field to a vector field and characterises the local rotational tendency at each point, its magnitude being equal to twice the local angular velocity (this requires $\mathsf{F} \in C^1$).

It must be stressed that the most standard structure of the divergence is that of local sources and sinks, the rate of volume change, the continuity equation and Liouville-type results, whereas the generator of scaling at the level of continuous groups is $x\,d/dx$, an object of another level; the relation between the two may be studied, but they cannot be placed side by side as a correspondence law before a rigorous bridge has been built. This book therefore does not state 'which operator corresponds to which transformation' as a proposition.

Why conservation laws are needed

Conservation laws are not an appended physical condition in periodic mathematics, but the **logical precondition for the stable existence of periodic structure in conservative systems. It must be stressed** that the following arguments apply to conservative systems; for dissipative systems, one must introduce the notion of a generalised conserved quantity or of an attractor, which will be discussed in later chapters.

(i) **Reversibility of transformations**: a transformation can be applied repeatedly without destroying the structure of the system, **in a conservative system**, if and only if there exists at least one quantity that is strictly invariant under that transformation. A transformation without a conserved quantity is typically irreversible within the conservative framework—the closure of the orbit is then difficult to guarantee logically.

(ii) **Constraint of orbits**: the mathematical essence of a conservation law is to constrain the evolution of the system to an invariant submanifold. Without such a constraint, the evolution of the system is free drift, which **generally** produces no recognisable regular orbit. **In dissipative systems, however, an attractor can take the place of a conserved quantity and play the constraining role.**

(iii) **Observable regularity**: a conserved quantity is an identity across time—it establishes a decidable connection between the 'state of the past' and the 'state of the present'.

Viewed from the algebraic framework established in the 'Periodic Foundations' chapter, a periodic system can perfectly well possess only trivial invariants (for example a finite cyclic permutation: the state set is finite, the transformation is of finite order, every state is periodic, and yet a non-trivial conserved quantity need not exist). Therefore **a conserved quantity is neither a necessary condition nor, generally, a sufficient condition for periodicity.** The fundamental principle of this book is stated as follows:

In a conservative system, a conserved quantity can provide an invariant constraint on the orbit, and thereby furnish an important condition for the existence and stability of periodic structure; but a conserved quantity is not a sufficient condition for periodicity, nor, generally, a necessary one. Within the framework of this book, the algebraic hinge that directly produces periods is the finite-order action:

$$\text{finite-order action} \implies \text{uniformly closed orbit} \implies \text{period}.$$

The proper role of conservation and symmetry within this is that of **auxiliary tools for discovering, constraining and explaining periodic structure**, and not of indispensable links in the logical chain.

Why group representation theory and the manifold framework are needed

(i) **Group representation theory answers 'what the transformation is'**: group representation theory provides the algebraic structure of transformations—does the set of transformations form a group? Do the group elements have finite order? What is the structure of the orbit?

(ii) **The tangent space answers 'what the instantaneous direction of the transformation is'**: the tangent space captures the infinitesimal direction of change of the system in its current state. The tangent vector at each point of a closed orbit constitutes the instantaneous velocity field of that periodic motion. Whether the vector field in the tangent space is integrable determines whether the orbit can close within finite time.

(iii) **The normal space answers 'what constrains the transformation'**: the normal space describes the constrained directions experienced by the system during its evolution. Conservation laws appear geometrically as constraint hypersurfaces in the normal space—the system is confined to move on this hypersurface and cannot escape. The dimension of the normal space is equal to the number of conserved quantities, and the codimension of the normal space is equal to the degrees of freedom of the system.

(iv) **The angle space answers 'how far the transformation has progressed'**: the angle space records the phase position of the system on the closed orbit. For a system of period T, the angle space is a parametrisation by the circle group $U(1)$ or by the finite cyclic group $\mathbb{Z}_T$. The introduction of the angle space turns 'phase' into an independent geometric dimension, and not merely a function of time. **It should be noted** that the specific type of the angle space depends on the fundamental group and the homotopy class of the orbit; not every closed orbit automatically possesses the $U(1)$ or $\mathbb{Z}_T$ structure, and its rigorous classification requires topological conditions.

(v) **The manifold framework answers 'what the global structure is'**: a manifold is the global integration of the tangent space, the normal space and the angle space. The complete state space of a periodic system is a manifold; a closed orbit is a closed curve on this manifold, the tangent space is the tangent plane of the manifold at each point of the orbit, the normal space is the orthogonal complement of the tangent plane, and the angle space is the parametrisation of the orbit itself.

Only the combination of all five yields a complete determination: group representation theory provides the algebraic possibility, and the manifold framework provides the geometric stage. Only by uniting the group action, the tangent space, the normal space and the angle space within the manifold framework can one answer simultaneously: algebraically, do the group elements have finite order? geometrically, does the orbit close? in the tangent direction, does an integrable vector field exist? in the normal direction, does a conserved constraint exist? in the angle direction, is the phase single-valued? **This framework is a research programme**, and its specific applicability conditions need to be established case by case in the various chapters.

Formalisation of the manifold framework

Let M be the state-space manifold of the system, $\gamma : [0, T] \to M$ be a closed orbit, and $p = \gamma(t)$ be a point on the orbit. Then:

(1) **Tangent space** T_pM: the one-dimensional subspace spanned by the tangent vector $\dot{\gamma}(t)$ of the orbit at the point p, describing the instantaneous direction of motion of the system;

(2) **Normal space** N_pM: the orthogonal complement of the tangent space in T_pM, describing the constrained directions experienced by the system. The gradient $\nabla Q(p)$ of a conserved quantity Q is a vector in the normal space;

(3) **Angle space** A_p: the parametrisation space of the orbit γ; for a periodic orbit, $A_p \cong U(1)$ or $A_p \cong \mathbb{Z}_T$, describing the phase position of the system on the orbit. **Conditions are required**: when the element of the fundamental group of the orbit is of infinite order and the orbit is contractible, the angle space is naturally $U(1)$; when the element of the fundamental group is of finite order, the angle space is $\mathbb{Z}_T$. The general case requires separate discussion.

(4) **Tangent bundle** $TM = \bigcup_{p \in M} T_pM$: the collection of all tangent spaces, describing the directions of motion of the system in all possible states;

(5) **Normal bundle** $NM = \bigcup_{p \in M} N_pM$: the collection of all normal spaces, describing the constrained directions of the system in all possible states;

(6) **Angle bundle** $AM = \bigcup_{p \in M} A_p$: the collection of all angle spaces, describing the phase positions of the system in all possible states.

Correspondence between conservation laws and the normal space

Conservation laws acquire a geometric interpretation within the manifold framework: **under suitable non-degeneracy conditions**, a conserved quantity Q corresponds to a constraint hypersurface $Q^{-1}(c)$ in the normal space. The system is confined to move on this hypersurface, and the orbit γ is a closed curve on the constraint hypersurface. The vectors in the tangent space T_pM are orthogonal to $\nabla Q(p)$, which means that the conserved quantity does not change along the direction of motion. The dimension of the normal space is equal to the number of independent conserved quantities, and the codimension of the normal space is equal to the effective degrees of freedom of the system. **When the number of** independent conserved quantities is equal to the degrees of freedom of the system, the orbit is completely constrained to be a closed curve—this is a geometric hallmark of a completely integrable system; **but the converse does not hold**: the existence of a closed orbit does not require the number of independent conserved quantities to be equal to the degrees of freedom.

Correspondence between the angle space and the phase operator

The angle space is the geometric realisation of the phase operator. The kth phase of period T corresponds to the complex number $\mathcal{O}_k^{(T)} = e^{i2\pi k/T}$, which appears in the angle space as a uniformly sampled point on the orbit; its three actions (identity, antipode, rotation by $2\pi k/T$) correspond one by one to the algebraic description of the phase operator in the Preface, and are not repeated here. The topological structure of the angle space determines the algebraic structure of the phase operator: when the angle space is the continuous circle $U(1)$, the phase operator is a continuous rotation group; when the angle space is the finite cyclic group $\mathbb{Z}_T$, the phase operator is a finite rotation group. **It should be noted** that the above correspondence presupposes that the angle space is well defined, and the rigorous construction of the angle space depends on the homotopy class and the fundamental group of the orbit; not every closed orbit automatically satisfies this.

A preliminary definition of periodic convolution

Traditional harmonic analysis mainly decomposes a signal into a sum of sinusoidal waves of different frequencies, and does not necessarily preserve the interaction information between amplitude and phase. A core research programme proposed in this book is periodic convolution, whose motivation is to bring amplitude (scaling) and phase (rotation) into the series operations as well.

Definition 0.2 **(Periodic convolution: interaction tensor and composition map (preliminary framework)).** Let two periodic systems be represented by the epicycle series

$$f(x) = \sum_n A_n e^{i(\omega_n x + \phi_n)}, \quad g(x) = \sum_m B_m e^{i(\nu_m x + \psi_m)}$$

(under suitable convergence conditions).

First layer (interaction tensor): define the pairwise interaction tensor between the two families of epicycle components

$$C_{n,m} = A_n B_m \,\text{Similarity}^{(d)}(\theta_n, \theta_m),$$

where $\theta_n = (\omega_n, \phi_n)$ is the phase point of f, $\theta_m = (\nu_m, \psi_m)$ is the phase point of g, and $\text{Similarity}^{(d)}$ is the d-dimensional phase-similarity kernel.

Second layer (composition map): define the composition map from the interaction tensor to the function space

$$S(C)(x) = \sum_{n,m} C_{n,m} e^{i(\Omega_{n,m} x + \Phi_{n,m})},$$

where $\Omega_{n,m}$ is the frequency-composition operator and $\Phi_{n,m}$ is the phase-composition operator.

Periodic convolution (preliminary) is defined as the composition of the interaction tensor and the composition map:

$$(f \circledast g)(x) \;=\; \mathcal{K}^{(d)}\{f,g\} \;=\; S(C)(x).$$

Notation: throughout the main text of this book the periodic-convolution operator is written $\circledast$; when the order d of the phase-similarity kernel Similarity$^{(d)}$ has to be displayed, or when a window kernel W is introduced, the same operation is written $\mathcal{K}^{(d)}\{f,g\}$—this is the form used on the cover—and in the applied chapters the shorthand $A_{n,m} := A_nB_m$ is occasionally employed.

It must be emphasised that the above definition is a **research programme**, and that for it to hold rigorously the following conditions must be specified: (i) the series participating in the convolution converge under a suitable topology; (ii) the phase-similarity kernel Similarity$^{(d)}$ satisfies symmetry, boundedness and normalisation conditions; (iii) the frequency-composition operator and the phase-composition operator are explicitly defined and satisfy associativity or distributivity. These conditions will be given one by one in later chapters or listed as open problems. **Until the rigorous definition has been completed, the above framework should be regarded as a heuristic programme rather than an established mathematical object.**

Periodic convolution takes amplitude (scaling), frequency (translation), phase (rotation) and period (fixed length) as its four operator parameters, and preserves the interactions of all frequency pairs and all period pairs. This means that the interaction between two periodic systems, **within the framework of periodic convolution**, depends not only on their respective frequencies, but also on the ratio of their amplitudes, the difference of their phases, and the common-multiple relation between their periods. This book will attempt to argue that the classical integral transforms—the Fourier transform, the wavelet transform, the Mellin transform and the Laplace transform—**can be given a unified description within the framework of group representation and harmonic analysis under suitable group actions, representations and function spaces; this is more cautious than saying that 'they are essentially one and the same thing'**—since wavelet analysis involves the affine group, translation plus scaling, representations, continuous and discrete wavelets, admissibility and specific measures, so that the precise unified form requires case-by-case verification. The rigorous definition and properties of periodic convolution will be established in later chapters.

An information ontology and the observer framework

The information ontology proposed in this section is a philosophical position of this book, not a mathematical theorem. Its mathematical content lies solely in the fact that the scale and the sampling rate of the observer determine the range of information that can be mathematised by the observer.

Let the composite vector of the temporal system be V_T, the composite vector of the spatial system be V_S, and the composite vector of the observer system be

V_P. Then

$$\|\mathsf{V}_T\|^2 = E \quad \text{(energy)}, \tag{1}$$
$$\|\mathsf{V}_S\|^2 = m \quad \text{(mass)}, \tag{2}$$
$$\|\mathsf{V}_P\|^2 = \sigma \quad \text{(scale)}. \tag{3}$$

Energy, mass and scale—here **interpreted** as the squared moduli of the composite vectors of three independent information-flux systems. The **mathematical content** of this formalisation will be given in later chapters; its **ontological interpretation** is philosophical and does not belong to the category of mathematical theorems.

The observer itself is also part of the information structure. The scale σ of the observer and the sampling rate $R(r, \sigma)$ directly determine the range of information that can be mathematised. When the number of sampled points $N(r, \sigma) < 1$, the information is completely invisible to the observer; when the sampling rate R is insufficient, high-frequency information is aliased into low-frequency artefacts. **The boundary of the subset of information that has been mathematised is a function of the observer's own sampling limit**—this is a structural statement about the relation between the observer and information, whose rigorous form is given in the 'Discrete Sampling Approximation, Congruence Angles, and Intercalation' chapter, Definition 21.6, Theorem 21.1.

The choice of the observer's sampling window is a key technical step in the 'mathematisation of information'. When the continuous period T_f being observed and the sampling interval Δx of the observer are incommensurable, the choice of the sampling window length directly determines the quality of the spectrum that the observer can resolve. The **intercalation method** introduced in this book (see the Preface) provides the observer with a window-selection mechanism that has a rigorous optimality criterion: by selecting the continued-fraction convergents p_j/q_j of $T_f/\Delta x$, the observer can obtain the minimal phase error and the minimal spectral leakage on the window length $T = p_j$. This mechanism transforms the observer framework from a philosophical statement into an operable algorithmic criterion. The canonical definitions of the sampling interval and the sampling rate are given in the 'Discrete Sampling Approximation, Congruence Angles, and Intercalation' chapter, Definitions 21.1 and 21.2.

The duality of information

(1) **Independent existence**: information can exist independently as an identified unit—a group element, a function value, an epicycle radius, a phase vector.

(2) **Interrelation**: information can likewise manifest itself as the relation between entities—group operations, function composition, the nested superposition of epicycles, the periodic-convolution projection of phase vectors.

The **reification** of relations can form the foundation of a hierarchical information structure: first-order information (entities), second-order information (relations between entities), third-order information (relations between relations).

Quotient groups, functors and natural transformations can be viewed as several mathematical realisations of this hierarchical leap in information—this correspondence is heuristic, and its rigorous form depends on the specific category-theoretic construction.

Evolution model: the arising, abiding and ceasing of conservation laws

The time derivative of a conserved quantity

$$\frac{dQ}{dt} = \{Q, \mathcal{H}\} + \frac{\partial Q}{\partial t}$$

explicitly distinguishes internal dynamics from external driving. The three stages of generation ($\nabla Q \to 0$), evolution ($\nabla Q = 0$) and breaking ($\nabla Q \neq 0$) can be given a rigorous formulation within a suitable symplectic-geometric framework.

Research aims and methods

The research aim of periodic mathematics is to establish a unified mathematical framework capable of describing the common symmetric structure of different dynamical systems. We are not concerned with the specific physical, biological or social mechanisms of each system, but with their common mathematical structure.

Periodic mathematics accepts all the standard constructions of modern mathematics and does not redefine existing concepts. Its concrete work is:

(1) To reveal symmetry and directionality as the foundations of periodicity—the three fundamental symmetries (0, 1, i) correspond respectively to translation, scaling and rotation;

(2) To give a rigorous definition of periodicity in the language of group representation theory—a periodic function is an invariant function under a group action, and an epicycle is the geometric realisation of rotational symmetry;

(3) To reveal the structural correspondence between the extension of number systems and the growth of symmetry groups;

(4) To incorporate the observer framework of dynamic sampling in a unified manner;

(5) To incorporate integral transforms in a unified manner into the harmonic-analysis framework on groups with periodic orbits under suitable conditions;

(6) To use periodic convolution for a unified description of cross-period phase operations—phase as a relational operator, epicycles nested to generate large periods, the sexagenary cycle being the periodic-convolution composition of the Heavenly Stems and the Earthly Branches;

(7) To establish the **intercalation method** as a general approximation tool for incommensurable periodic systems—with the best continued-fraction approximation as its theoretical foundation, the bounded-intercalation-error theorem as its rigorous support, and the intercalation functor as its category-theoretic realisation, so as to reduce the spectral leakage in periodic analysis from uncontrollable to bounded.

Relationship with existing mathematics

Periodic mathematics does not seek to replace or revise existing disciplines such as group representation theory, harmonic analysis and representation theory, but rather to reorganise, from the perspective of 'period', those parts of these disciplines that are related to periodic structure. The central insight underlying this unity is:

> The gradient, the curl and the divergence are rigorously defined differential operators in vector analysis, each with its own differentiability hypotheses and domain conditions; this book asserts no correspondence between them and the three basic transformations, and any relation between the two triples must be established case by case.
> If suitable group actions, representations and function spaces are chosen, different integral transforms can be given a unified description within the framework of group representation and harmonic analysis.
> Energy, mass and scale can be interpreted as the observer's projection of the squared moduli of the composite vectors of three flux systems—this is a research programme, not a proven theorem.
> Intercalation is a bounded-error method for approximating irrational period ratios by integer sequences, and its optimality is given by continued-fraction convergents—this is an independent contribution of this book to the methodology of periodic analysis.

This manuscript is positioned as a **research programme**: it provides a new way of organising material and a new angle from which to pose questions, and whether it ultimately holds depends on whether non-trivial new mathematical results can be produced in the future.

Research principles and steps

Principle 1 **Structure takes priority over mechanism.** Focus on the common mathematical structure among systems, rather than on system-specific causal mechanisms.

Principle 2 **Finite-order action as the core.** Within the framework of this book, the algebraic hinge that directly produces periods is the finite-order action: finite order ⇒ uniformly closed orbit ⇒ period. Conservation and symmetry are not universal necessary conditions, but auxiliary tools for discovering and constraining periodic structure.

Principle 3 **Diagnose rather than assert.** Provide tools to measure the degree of closure of a system under the various symmetric transformations.

Principle 4 **Periods arise and cease.** The maintenance of conservation laws safeguards the evolution of periods; the intervention of dissipation or external driving leads to the cessation of periods.

Step 1 **Construction of the mathematical core.** Rigorously define the mathematical propositions and theorems leading from group actions and orbit closure to conserved quantities.

Step 2 **Infinitesimal probes.** Use the gradient, the curl and the divergence as candidate tools for the differential expression of the three symmetries.

Step 3 **Categorical unification.** Use category theory to formalise different types of periodic structure.

Step 4 **Interdisciplinary diagnosis.** Use this set of tools to pinpoint precisely why a system possesses or lacks a certain periodic structure.

Principal results of this book

This section lists the results that this book actually establishes. Each is stated at the strength at which it is proved, with the chapter in which it is located given in brackets; the heuristic pictures and the open problems of the Preface and of the preceding sections are **deliberately not repeated** here.

(1) **The two lines** (Chapter 1, 'Two lines: iteration generates operations and transformations'). At the level of parameters, iteration organises lower-order operations into higher-order operations, whilst the parameters of a repeatedly applied transformation ascend the same hierarchy. The **operation line** is addition → multiplication → power (the hyperoperation hierarchy); the **transformation line** is translation → scaling → rotation; the relation between the two lines is a correspondence at the level of parameters, given in the form of a definition and a proposition, and not as an assertion that the two are one and the same thing. The three generating transformations of this book thereby acquire a generative derivation, and not merely an enumeration.

(2) **The algebraic hinge of periodicity** (Chapters 1 and 13–15). Within the framework of this book, the algebraic hinge that directly produces periods is the **finite-order action**: finite order ⇒ uniformly closed orbit ⇒ period. The period set of a periodic function and the structure of the group with periodic orbits are developed in the 'Periodic Functions and Groups with Periodic Orbits' chapter; conservation and symmetry enter as auxiliary tools for discovering and constraining periodic structure, and not as indispensable links in the deductive chain.

(3) **Bounded intercalation error and continued-fraction optimality** (Chapter 21). For an incommensurable period ratio ρ, the lower, the upper and the balanced intercalation sequences all satisfy $N_m \in \{\lfloor\rho\rfloor, \lceil\rho\rceil\}$, whilst their cumulative errors remain bounded—lying in $(-1, 0]$, $[0, 1)$ and $[-1/2, 1/2]$ respectively; and the optimal choice of the intercalation window is given by the continued-fraction convergents p_j/q_j of ρ. The intercalation method is presented in this book as an independent contribution to the methodology of periodic analysis: it reduces the cost of incommensurability from uncontrollable to bounded.

(4) **Congruence angles and the generalised Chinese remainder theorem** (Chapter 21). A **congruence angle** converts the compatibility condition of several periods into an angular condition on the circle group, and generalises the Chinese remainder theorem from the coprime case to a form in which coupled multi-period constraints can be stated, combined and solved; intercalation is thereby upgraded from the suppression of leakage for a single period to the joint suppression of several periods.

(5) **The two-layer structure of periodic convolution** (Chapters 6, 23 and 45). Periodic convolution is defined as the composition of an interaction tensor and a composition map, $(f \circledast g)(x) = \mathcal{K}^{(d)}\{f, g\} = S(C)(x)$, so that amplitude (scaling) and phase (rotation) enter the series operation; its functoriality on the category of translation periods is stated in the form of a theorem.

(6) **The epicycle activation function and function approximation** (Chapters 42 and 45). On the basis of the epicycle nesting generation rule—with each step of the outer circle, the inner circle rotates a full turn—the epicycle activation function is defined, and a corresponding function-approximation result is established. It is here that the research programme of this book acquires a computable concrete form.

What this book does not claim: the above are the results that this book establishes. The stronger propositions of the research programme—that any continuous transformation preserving some intrinsic structure can be decomposed into the three generating transformations, that 'which operator corresponds to which transformation' constitutes a correspondence law, that a general periodic function is a superposition of finitely many epicycles—**are not among the claims of this book**, and are carried in the main text only in the form of a research programme or an open problem. What must be distinguished alongside this is the following: within the framework of **one-parameter families of conformal transformations of the complex plane**, a framework with explicit hypotheses, the completeness of the three basic transformations is a conclusion proved rigorously in section Section 1.8 of Chapter 1; what is not claimed above is its generalisation to 'an arbitrary continuous transformation group of arbitrary intrinsic structure'. The problems that remain unsolved are stated together in Chapter 46.

Research structure and interdisciplinary applications

This book is divided into three parts, together with a closing chapter:

Part 1 **Mathematical foundations and generators** (Chapters 1 to 10). From periodic foundations, number systems, scalars and vectors, functions and limits, real analysis, series and convolution, complex analysis, quaternion analysis, differential equations and analysis, to linear algebra.

Part 2 **Core structures, tools and methods** (Chapters 11 to 24). From the foundations of manifolds and fibre bundles, periodic manifolds, periodic categories, probability and distributions, and vector analysis, through epicycle groups and episphere groups, propagation and evolution, groups and group functions, higher-order groups and higher-order functions, and periodic functions and groups with periodic orbits, to discrete sampling approximation (with congruence angles) and intercalation, harmonic analysis, time series analysis, and multi-period coupled systems.

Part 3 **Interdisciplinary applications and diagnosis** (Chapters 25 to 45). The periodic structure of probabilistic, cryptographic, astronomical, calendrical, geographic, I Ching, electromagnetic, audio, mechanical, circuit-and-chip, biological, psychological, social, climatic, economic and political systems, together with data analysis, agents, the epicycle activation function, algorithmic systems, and modelling from data to epicycles.

The closing **Chapter 46**, ‘Summary and Outlook’, gathers the three main threads of the book and states the problems that remain open.

The theoretical necessity of interdisciplinary applications lies in this: the value of a unified framework depends on whether it can withstand testing in multiple independent domains. Periodic mathematics not only describes known periodic systems, but also diagnoses those systems that ‘seem to have a period yet need not satisfy the closure condition’. Planetary orbits, crystal translations, genetic oscillations, the Koch snowflake, Leibniz’s binary-sequence periods and the hexagram system of the I Ching—**at a suitable level of abstraction**—share the same structure of group representation theory and the same logic of conservation laws.

In each application chapter, periodic mathematics first asks: **does this system possess an identifiable symmetry group? Does a corresponding conserved quantity exist? Is the evolution orbit in its state space closed?** (The philosophical statement of the three criteria, and the treatment of the three cases ‘satisfied’, ‘partly satisfied’ and ‘not satisfied’, are given in the **Preface** under ‘The stance and limits of this book’.)

Chapter structure and reading guide

The conventions and labelling of the chapters of this book (the order of statement ‘definition → proposition → ⋯→ remark’, and the rules governing the use of the three labels 'conjecture / analogy / research programme') are given in the **Preface** under 'Conventions and notation'. The relations among the chapters can be summarised as three main threads:

(1) **Operator thread** (gradient $\nabla \to$ curl $\nabla\times \to$ divergence $\nabla\cdot$). Starting from the 'Periodic Foundations' chapter, passing through the 'Vector Analysis' chapter, and running through the 'Periodic Functions and Groups with Periodic Orbits' chapter and the 'Harmonic Analysis' chapter, it establishes one by one the differentiability hypotheses and the domain of applicability of each of the three operators. **This thread asserts no correspondence between the three operators and the three basic transformations**; any relation between the two triples is discussed case by case, and the 'Vector Analysis' chapter states explicitly that no correspondence is asserted between them.

(2) **Number-system thread** ($\mathbb{N} \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{R} \to \mathbb{C} \to \mathbb{H}$). Reveals how periodic structure becomes ever richer as the number system is extended.

(3) **Observer thread** (scale $\sigma \to$ sampling rate $R \to$ invisibility criterion). The categorical unification of the observer and periodic structure is completed in the 'Periodic Categories' chapter.

The three main threads ultimately converge on a core proposition: **in conservative systems, periodic structure may be closely related to conservation laws, and conservation laws are related to symmetry within the framework of Noether's theorem; but the true algebraic hinge of periodicity is the finite-order action, and conservation and symmetry are auxiliary tools for discovering, constraining and interpreting periods.** The task of periodic mathematics is, with group actions as its language and closed orbits as its criterion, to characterise precisely the deductive chain from finite-order action to periodicity, whilst clarifying the proper place of conservation and symmetry within it.

Chapter 1

Operations and Periodicity

Remark 1.1 (Positioning of this chapter and the two logical chains). The root of periodicity is invariance under transformations. This chapter takes iteration as the starting point of every concept, examines one by one the true logical direction of each arrow and the conditions under which it holds. The core conclusion is: within the purely algebraic framework of this chapter, a finite-order action is the direct sufficient and necessary condition for obtaining uniform periodicity and closed orbits; conservation and symmetry are auxiliary tools for **discovering** finite order, not a necessary link in the existence of periodicity.

Thus the chapter distinguishes two chains:

$$\boxed{\text{Core criterion chain: finite-order action} \Longrightarrow \text{closed orbit} \Longrightarrow \text{period}}$$

$$\boxed{\text{Auxiliary discovery chain: conservation} \longleftarrow \text{symmetry} \quad \text{(used to discover finite order)}}$$

The core criterion chain is sufficient and necessary in the bijective framework; the auxiliary discovery chain is heuristic, and its direction is often reversed. Specifically:

- symmetry $\Rightarrow$ conservation (with conditions), but not conservation $\Rightarrow$ symmetry;
- period $\Rightarrow$ symmetry (the evolution group itself is a finite cyclic group), yet period $\not\Rightarrow$ non-trivial conservation;
- the conjunction of symmetry and conservation still $\not\Rightarrow$ period.

This chapter uses mainly the algebraic language of set theory, group theory and group representation theory; notions involving topology (bounded, dense, compact) are used only as heuristic illustrations.

1.1 Foundations of group theory and group actions

This section builds, in a self-contained way, the group-theoretic language required in this chapter: groups, subgroups, homomorphisms, quotient groups, the order of an element, the torsion subgroup, cyclic groups, group actions, orbits and stabilisers. The notation is fixed in this section and used throughout the chapter.

1.1.1 Groups, subgroups and homomorphisms

Definition 1.1 (Group). A **group** is an ordered pair $(G, \cdot)$, where G is a non-empty set and $\cdot : G \times G \to G$ is a binary operation satisfying:

(1) **Closure**: for all $a, b \in G$, $a \cdot b \in G$;

(2) **Associativity**: for all $a, b, c \in G$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$;

(3) **Identity element**: there exists $e \in G$ such that $e \cdot a = a \cdot e = a$ for all $a \in G$;

(4) **Inverse element**: for every $a \in G$ there exists $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$.

Definition 1.2 (Abelian group). A group $(G, \cdot)$ is called an **Abelian group** (or commutative group) if, for all $a, b \in G$,

$$a \cdot b = b \cdot a.$$

The operation of an Abelian group is often written additively as $+$, the identity element is written 0 and the inverse as $-a$; in this notation the power rule is written as the multiple rule $n(a + b) = na + nb$.

Example 1.1 (Three fundamental Abelian groups). *The following three Abelian groups are the algebraic carriers of the transformation theory of this chapter:*

(i) **The additive group of integers** $(\mathbb{Z}, +)$: *identity element 0, inverse of a equal to $-a$. This is the algebraic basis of discrete translation.*

(ii) **The multiplicative group of positive reals** $(\mathbb{R}^+, \times)$: *identity element 1, inverse of a equal to $1/a$. This is the algebraic basis of scaling.*

(iii) **The circle group** $U(1) = \{z \in \mathbb{C} \mid |z| = 1\}$ *(under multiplication of complex numbers): identity element 1, inverse of z equal to $\bar{z}$. Its operation is the multiplication of complex powers*

$$e^{i\theta} \cdot e^{i\phi} = e^{i(\theta+\phi)}.$$

This is the algebraic basis of rotation.

All three are Abelian groups; but their **torsion structure** *is utterly different, and this difference will become the focus of the theory of periodicity in Theorem 1.10.*

Definition 1.3 (Subgroup). Let G be a group and let $H \subseteq G$ be non-empty. If H itself forms a group under the operation of G, then H is called a **subgroup** of G, written $H \leq G$.

Proposition 1.1 (Subgroup criterion). *Let G be a group and let $H \subseteq G$ be non-empty. Then $H \leq G$ if and only if:*

(i) *for all $a, b \in H$, $ab \in H$;*

(ii) *for all $a \in H$, $a^{-1} \in H$.*

Equivalently, $H \leq G$ if and only if $ab^{-1} \in H$ for all $a, b \in H$.

Proof. Necessity: if H is a subgroup, then closure (i) and closure under inverses (ii) follow directly from the definition of a subgroup. Sufficiency: closure follows from (i), and associativity is inherited from G. From (ii) and (i), taking $a \in H$ gives $aa^{-1} = e \in H$, so the identity element exists; the inverse is given by (ii). The equivalent criterion follows directly from $b^{-1} \in H$ and $ab^{-1} \in H$. □

Definition 1.4 (Group homomorphisms and isomorphisms). Let G, H be groups. A map $\varphi : G \to H$ is called a **group homomorphism** if

$$\varphi(ab) = \varphi(a)\varphi(b), \qquad \forall a, b \in G.$$

If φ is a bijection, it is called a **group isomorphism**, written $G \cong H$. The **kernel** and **image** of a homomorphism φ are respectively

$$\ker\varphi = \{g \in G \mid \varphi(g) = e_H\}, \qquad \operatorname{im}\varphi = \{\varphi(g) \mid g \in G\}.$$

Definition 1.5 (Normal subgroups and quotient groups). Let $N \leq G$. If $gN = Ng$ for all $g \in G$, then N is called a **normal subgroup** of G, written $N \trianglelefteq G$. In this case the set of cosets

$$G/N = \{gN \mid g \in G\}$$

forms a group under the operation $(gN)(hN) = (gh)N$, called the **quotient group**.

Theorem 1.1 (Fundamental theorem of homomorphisms). Let $\varphi : G \to H$ be a group homomorphism. Then

$$G/\ker\varphi \cong \operatorname{im}\varphi.$$

In particular, $\ker\varphi \trianglelefteq G$.

Proof. Let $K = \ker\varphi$. For any $g \in G$, define $\bar{\varphi}(gK) = \varphi(g)$. If $gK = hK$, then $h^{-1}g \in K$, so $\varphi(h^{-1}g) = e$, whence $\varphi(g) = \varphi(h)$, and the map is well defined. The homomorphism property follows directly from that of φ. The kernel is K, so the map is injective; the image is $\operatorname{im}\varphi$, so the map is surjective onto the image. Hence $G/K \cong \operatorname{im}\varphi$. □

$$\begin{tikzcd}
G \arrow[r, "\varphi"] \arrow[d, "\pi"'] & H \\
G/\ker\varphi \arrow[r, "\cong"'] & \operatorname{im}\varphi \arrow[u, "\iota"']
\end{tikzcd}$$

π: natural projection
ι: inclusion map

Figure 1.1: Fundamental theorem of homomorphisms: $G/\ker\varphi \cong \operatorname{im}\varphi$. Every homomorphism φ can be factored as 'projection—isomorphism—inclusion'.

Definition 1.6 (Centraliser). Let G be a group and $S \subseteq G$. The **centraliser** of S in G is

$$C_G(S) = \{g \in G \mid gs = sg,\ \forall s \in S\}.$$

The **centre** of G is $Z(G) = C_G(G)$.

Proposition 1.2 (The centraliser is a subgroup). *For any $S \subseteq G$, one has $C_G(S) \leq G$, and $Z(G) \trianglelefteq G$.*

Proof. $e \in C_G(S)$. If $a, b \in C_G(S)$, then for any $s \in S$,

$$(ab)s = a(bs) = a(sb) = (as)b = (sa)b = s(ab),$$

so $ab \in C_G(S)$. Also $as = sa$ implies $a^{-1}s = sa^{-1}$, so $a^{-1} \in C_G(S)$. The subgroup criterion gives $C_G(S) \leq G$. The centre is analogous. □

Property 1.1 (Power rule for Abelian groups). Let G be an Abelian group, $a, b \in G$ and $n \in \mathbb{Z}$. Then

$$(ab)^n = a^n b^n.$$

Proof. First suppose $n \geq 0$ and argue by induction on n. For $n = 0$: $(ab)^0 = e = ee = a^0 b^0$. Suppose $(ab)^n = a^n b^n$; then

$$(ab)^{n+1} = (ab)^n(ab) = a^n b^n ab = a^n a b^n b = a^{n+1} b^{n+1},$$

where the third step uses the commutative law $b^n a = ab^n$. For $n < 0$,

$$(ab)^n = ((ab)^{-1})^{-n} = (a^{-1}b^{-1})^{-n} = (a^{-1})^{-n}(b^{-1})^{-n} = a^n b^n.$$

□

Remark 1.2 (The necessity of commutativity). The power rule $(ab)^n = a^n b^n$ generally fails in a non-Abelian group: $(ab)^2 = abab$ and $a^2b^2 = aabb$ are equal if and only if $ba = ab$. The commutative law is precisely the condition that 'two iterations may be performed in either order'.

1.1.2 Order of elements, the torsion subgroup and cyclic groups

Definition 1.7 (Order of an element). Let $g \in G$. If there exists a least positive integer n such that $g^n = e$, then the **order** of g is n, written $\mathrm{ord}(g) = n$; otherwise the order of g is said to be infinite, written $\mathrm{ord}(g) = \infty$.

Definition 1.8 (Torsion elements and the torsion subgroup). *An element of finite order is called a* **torsion element***; a group all of whose elements are torsion elements is called a* **torsion group***. The set of all torsion elements of a group G is written*

$$\mathrm{Tor}(G) = \{g \in G \mid \mathrm{ord}(g) < \infty\}.$$

Theorem 1.2 (The torsion elements of an Abelian group form a subgroup). *Let G be an Abelian group. Then $\mathrm{Tor}(G) \leq G$, and it is called the* **torsion subgroup** *of G.*

Proof. $e \in \mathrm{Tor}(G)$. Let $a, b \in \mathrm{Tor}(G)$ with $\mathrm{ord}(a) = m$ and $\mathrm{ord}(b) = n$. By Property 1.1,

$$(ab)^{mn} = a^{mn} b^{mn} = (a^m)^n (b^n)^m = e.$$

Hence $\mathrm{ord}(ab) \leq mn < \infty$, so $ab \in \mathrm{Tor}(G)$. Also $(a^{-1})^m = (a^m)^{-1} = e$, so $a^{-1} \in \mathrm{Tor}(G)$. By the subgroup criterion, $\mathrm{Tor}(G) \leq G$. □

Remark 1.3 (Torsion elements are not closed in non-Abelian groups). The proof of Theorem 1.2 uses the commutative law in an essential way, and the conclusion indeed fails in a non-Abelian group. For example, in $\mathrm{GL}_2(\mathbb{R})$ take

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 2 \\ 1/2 & 0 \end{pmatrix},$$

then $A^2 = B^2 = I$, but $AB = \mathrm{diag}(1/2, 2)$ and $(AB)^n = \mathrm{diag}(2^{-n}, 2^n) \neq I$ for every $n \geq 1$, so $\mathrm{ord}(AB) = \infty$.

Definition 1.9 (Cosets and index). Let $H \leq G$. For $g \in G$, the left coset and the right coset are respectively

$$gH = \{gh \mid h \in H\}, \qquad Hg = \{hg \mid h \in H\}.$$

If G is finite, the **index** of H in G is the number of left cosets, written

$$[G : H] = \frac{|G|}{|H|}.$$

Lemma 1.1 (Lagrange). Let G be a finite group and $H \leq G$. Then $|H| \mid |G|$, and $|G| = [G : H]|H|$.

Proof. The family of left cosets $\{gH \mid g \in G\}$ forms a partition of G: the covering property follows from $g \in gH$; if $gH \cap g'H \neq \emptyset$, say $gh = g'h'$, then $g' = gh(h')^{-1} \in gH$, so $g'H \subseteq gH$, and symmetrically $gH \subseteq g'H$, so cosets are either equal or disjoint. Each coset has the same cardinality as H: the map $h \mapsto gh$ is a bijection $H \to gH$. Let the number of cosets be $[G : H]$; then $|G| = [G : H]|H|$. ☐

Proposition 1.3 (Order of an element in a finite group). Let G be a finite group and $g \in G$. Then $\mathrm{ord}(g) < \infty$ and $\mathrm{ord}(g) \mid |G|$.

Proof. The sequence $g, g^2, g^3, \ldots$ takes values in the finite set G, so there must exist $i < j$ with $g^i = g^j$, whence $g^{j-i} = e$ and the order is finite. Let $n = \mathrm{ord}(g)$; then $\langle g \rangle = \{e, g, \ldots, g^{n-1}\}$ contains exactly n elements, that is, $|\langle g \rangle| = n$. By Lemma 1.1, $n \mid |G|$. ☐

Corollary 1.1 (A group of prime order is cyclic). Let $|G| = p$ be prime. Then G is a cyclic group, and every $g \neq e$ is a generator.

Proof. Take $g \neq e$. By Proposition 1.3, $\mathrm{ord}(g) \mid p$ and $\mathrm{ord}(g) \geq 2$, so $\mathrm{ord}(g) = p$ and $\langle g \rangle = G$. ☐

Definition 1.10 (Cyclic group). A group G is called a **cyclic group** if there exists $g \in G$ such that

$$G = \langle g \rangle = \{g^n \mid n \in \mathbb{Z}\}.$$

Such a g is called a **generator** of G.

Proposition 1.4 (A cyclic group is Abelian). Every cyclic group is Abelian.

Proof. Let $G = \langle g \rangle$. Take any $a, b \in G$; then $a = g^m$ and $b = g^n$, so

$$ab = g^m g^n = g^{m+n} = g^{n+m} = g^n g^m = ba.$$

☐

Proposition 1.5 (Classification of cyclic groups)**.** Let $G = \langle g \rangle$ be a cyclic group. Then:

(i) if $\mathrm{ord}(g) = \infty$, then $G \cong (\mathbb{Z}, +)$;

(ii) if $\mathrm{ord}(g) = n < \infty$, then $G \cong (\mathbb{Z}/n\mathbb{Z}, +)$.

Proof. (i) The map $\varphi : \mathbb{Z} \to G$, $\varphi(k) = g^k$, is a surjective homomorphism; if $\varphi(k) = \varphi(l)$, then $g^{k-l} = e$, and since the order is infinite we get $k = l$, so φ is injective and hence an isomorphism.

(ii) The map $\varphi : \mathbb{Z}/n\mathbb{Z} \to G$, $\varphi([k]) = g^k$, is well defined; it is surjective because for any $g^k \in G$ we may take $[k]$ as a preimage; if $g^k = g^l$, then $g^{k-l} = e$, and by the definition of order $n \mid (k - l)$, that is, $[k] = [l]$, so the map is injective. Hence it is an isomorphism. □

Lemma 1.2 (Order of an element in a finite cyclic group)**.** *Let $m \geq 1$. In the additive group $\mathbb{Z}/m\mathbb{Z}$, the order of the element $[T]$ is*

$$\mathrm{ord}([T]) = \frac{m}{\gcd(T, m)}.$$

In particular, $[T]$ generates the whole of $\mathbb{Z}/m\mathbb{Z}$ if and only if $\gcd(T, m) = 1$.

Proof. The order of $[T]$ is the least positive integer n for which $nT \equiv 0 \pmod{m}$. Let $d = \gcd(T, m)$; then

$$m \mid nT \iff \frac{m}{d} \mid n \cdot \frac{T}{d}.$$

Since $\gcd(m/d, T/d) = 1$, we get $m/d \mid n$, and the least positive solution is $n = m/d$. When $d = 1$ the order is m, so $[T]$ is a generator; conversely, if $[T]$ generates the whole group, then its order is m, that is, $m = m/d$, giving $d = 1$. □

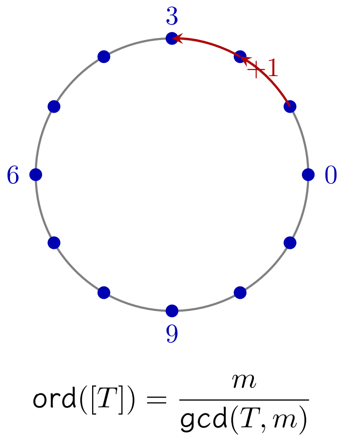


Figure 1.2: The translation $x \mapsto x+1$ on the cyclic group $\mathbb{Z}/m\mathbb{Z}$. The actual period of the step T is determined by $\gcd(T, m)$: $[T]$ generates the whole group if and only if $\gcd(T, m) = 1$.

Remark 1.4 (Cyclic groups: the algebraic prototype of periodicity). Cyclic groups are the most important objects of this chapter: the dynamics on a closed orbit is isomorphic to a translation of a cyclic group, and the evolution group carried by a periodic system is likewise a cyclic group. Lemma 1.2 shows that inside a cyclic group there is moreover a rich hierarchy of **subperiods**—the greatest common divisor of the step T and the modulus m determines the actual period.

1.1.3 Group actions, orbits and stabilisers

Definition 1.11 (Group actions, orbits and stabilisers). Let G be a group and X a set. A **left action** of G on X is a map

$$\Phi : G \times X \to X, \qquad (g, x) \mapsto g \cdot x,$$

satisfying $e \cdot x = x$ and $(gh) \cdot x = g \cdot (h \cdot x)$. The **orbit** and **stabiliser** of a point $x_0 \in X$ are respectively

$$\mathcal{O}_G(x_0) = \{g \cdot x_0 \mid g \in G\}, \qquad \mathsf{Stab}_G(x_0) = \{g \in G \mid g \cdot x_0 = x_0\}.$$

If for all $x, y \in X$ there exists $g \in G$ with $y = g \cdot x$, the action is called **transitive**.

Proposition 1.6 (The stabiliser is a subgroup). *Let G act on X and let $x_0 \in X$. Then $\mathsf{Stab}_G(x_0) \le G$.*

Proof. $e \cdot x_0 = x_0$, so $e \in \mathsf{Stab}_G(x_0)$. If $g, h \in \mathsf{Stab}_G(x_0)$, then

$$(gh) \cdot x_0 = g \cdot (h \cdot x_0) = g \cdot x_0 = x_0,$$

so $gh \in \mathsf{Stab}_G(x_0)$. Also, if $g \cdot x_0 = x_0$, then

$$x_0 = e \cdot x_0 = (g^{-1} g) \cdot x_0 = g^{-1} \cdot (g \cdot x_0) = g^{-1} \cdot x_0,$$

so $g^{-1} \in \mathsf{Stab}_G(x_0)$. The subgroup criterion gives the conclusion. ☐

Lemma 1.3 (Orbit–stabiliser). Let G act on X and let $x_0 \in X$. Then the map

$$g \, \mathsf{Stab}_G(x_0) \mapsto g \cdot x_0$$

gives a bijection

$$G / \mathsf{Stab}_G(x_0) \longleftrightarrow \mathcal{O}_G(x_0),$$

hence

$$|\mathcal{O}_G(x_0)| = [G : \mathsf{Stab}_G(x_0)].$$

Proof. Write $H = \mathsf{Stab}_G(x_0)$. If $gH = hH$, then $h^{-1} g \in H$, so

$$g \cdot x_0 = h \cdot (h^{-1} g) \cdot x_0 = h \cdot x_0,$$

and the map is well defined. Conversely, if $g \cdot x_0 = h \cdot x_0$, then $h^{-1} g \cdot x_0 = x_0$, that is, $gH = hH$, so the map is injective. Surjectivity: since $gH = \varphi(g)$, the image is the set of all cosets. ☐

Theorem 1.3 (Orbit partition). *Let G act on X. The relation*

$$x \sim y \iff \exists g \in G,\ y = g \cdot x$$

is an equivalence relation on X, and its equivalence classes are exactly the G-orbits. Hence the family of all orbits forms a partition of X.

Proof. Reflexivity: $x = e \cdot x$. Symmetry: if $y = g \cdot x$, then $x = g^{-1} \cdot y$. Transitivity: if $y = g \cdot x$ and $z = h \cdot y$, then $z = (hg) \cdot x$. The equivalence class is precisely $\{g \cdot x \mid g \in G\} = \mathcal{O}_G(x)$. ☐

Remark 1.5 (Faithfulness of an action). An action Φ induces a group homomorphism $\rho : G \to \mathsf{Aut}(X)$, $\rho(g)(x) = g \cdot x$. If ρ is injective, the action is called **faithful**. Throughout this chapter we take it by default that, via ρ, the group G is regarded as a subgroup of $\mathsf{Aut}(X)$, so that the two language systems 'order of an element' and 'order of a transformation' may be used interchangeably.

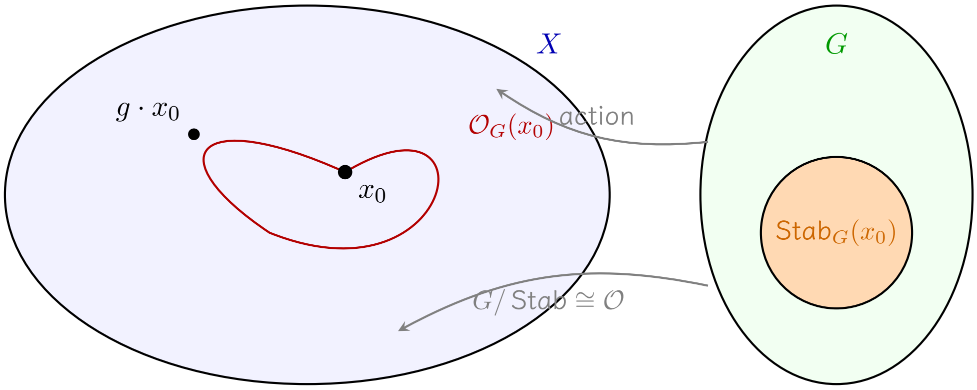


Figure 1.3: Group action, orbits and stabilisers. The relation between orbits and stabilisers is given by Lemma 1.3.

1.2 Iteration: the starting point of *every* concept

Definition 1.12 (Iteration and discrete dynamical systems)**.** Let X be a non-empty set and $f : X \to X$ a map. The **iteration** of f is defined by

$$f^0 = \mathrm{id}_X, \qquad f^{n+1} = f \circ f^n \quad (n \geq 0).$$

The ordered pair (X, f) is called a **discrete dynamical system**, and f is called the **generator** or the **evolution map**. If f is a bijection, negative iterates are defined by $f^{-n} = (f^{-1})^n$, so that iteration extends to the family with integer index $\{f^n\}_{n\in\mathbb{Z}}$.

Property 1.2 (The index law for iteration)**.** For all $m, n \geq 0$ (and $m, n \in \mathbb{Z}$ when f is a bijection),

$$f^{m+n} = f^m \circ f^n, \qquad f^{mn} = (f^m)^n.$$

In particular, when f is a bijection the map $n \mapsto f^n$ is a group homomorphism $(\mathbb{Z}, +) \to \mathrm{Aut}(X)$, that is, iteration is equivalent to an action of $\mathbb{Z}$ on X.

Proof. Induct on n: $f^{m+0} = f^m = f^m \circ f^0$; suppose $f^{m+n} = f^m \circ f^n$, then

$$f^{m+n+1} = f \circ f^{m+n} = f \circ f^m \circ f^n = f^{m+1} \circ f^n = f^m \circ f^{n+1}.$$

The second identity follows from the first by induction on n. The case of negative indices follows from the definition of $(f^{-1})^n$ and the same induction. The homomorphism property is precisely the first identity. □

Remark 1.6 (The primacy of iteration)*.* Iteration is logically prior to operations: addition, multiplication and exponentiation of the natural numbers are themselves generated recursively by 'iteration of the successor map'. This chapter therefore takes iteration, rather than the group operation, as its starting point—a group is merely the algebraic structure that iteration self-organises into in the invertible case.

Definition 1.13 (Orbit)**.** Let (X, f) be a discrete dynamical system and $x_0 \in X$. The **forward orbit** of x_0 is

$$\mathcal{O}^+(x_0; f) = \{f^n(x_0) \mid n \geq 0\}.$$

If f is a bijection, the **(two-sided) orbit** of x_0 is

$$\mathcal{O}(x_0; f) = \{f^n(x_0) \mid n \in \mathbb{Z}\} = \mathcal{O}_{\langle f\rangle}(x_0).$$

Proposition 1.7 (Orbit partition)**.** *Let f be a bijection. The relation*

$$x \sim y \iff \exists n \in \mathbb{Z},\ y = f^n(x)$$

is an equivalence relation on X, and its equivalence classes are exactly the orbits. Hence the family of orbits forms a partition of X.

Proof. Reflexivity: $x = f^0(x)$. Symmetry: $y = f^n(x) \Rightarrow x = f^{-n}(y)$. Transitivity: $y = f^n(x), z = f^m(y) \Rightarrow z = f^{m+n}(x)$. The equivalence class is $[x] = \{f^n(x)\} = \mathcal{O}(x; f)$. □

Definition 1.14 (Periodic points and eventually periodic points)**.** Let (X, f) be a discrete dynamical system and $x_0 \in X$. If there exists an integer $m \geq 1$ such that

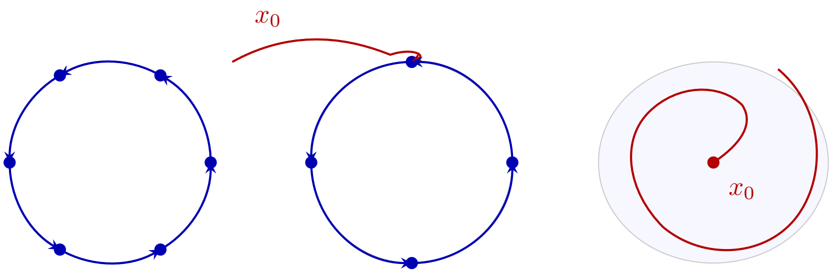
$$f^m(x_0) = x_0,$$

then x_0 is called a **periodic point**; the least positive integer m_0 satisfying the relation above is called the **least period** of x_0. If there exists $k \geq 0$ such that $f^k(x_0)$ is a periodic point, then x_0 is called an **eventually periodic point**.

Example 1.2 (A point that is eventually periodic but not periodic)**.** Let $X = \{0, 1, 2\}$, $f(0) = 1$, $f(1) = 2$, $f(2) = 1$. Then $f^2(1) = 1$, so 1 and 2 are periodic points; $f(0) = 1$ and yet 0 lies on no recurring orbit, so 0 is an eventually periodic point but not a periodic point.

Lemma 1.4 (Recurrence in invertible evolution)**.** *Let f be a bijection and $x_0 \in X$. If x_0 is an eventually periodic point, then x_0 is a periodic point.*

Proof. Let $f^k(x_0)$ be a periodic point, with $f^m(f^k(x_0)) = f^k(x_0)$ and $m \geq 1$. Applying f^{-k} to both sides gives $f^m(x_0) = x_0$. □



(a) Periodic point: $f^m(x_0) = x_0$

(b) Eventually periodic: first entering a cycle

(c) Non-periodic: the orbit is infinite and does not close

Figure 1.4: The geometric shape of the three types of orbit: a periodic orbit closes; an eventually periodic orbit enters a closed cycle after finitely many steps; a non-periodic orbit extends indefinitely without closing.

1.3 Invariants and conservation

Definition 1.15 (Invariant (conserved quantity)). Let (X, f) be a discrete dynamical system. A map $Q : X \to Y$ is called an **invariant** of f if

$$Q \circ f = Q, \qquad \text{that is} \qquad Q(f(x)) = Q(x), \quad \forall x \in X.$$

When $Y = \mathbb{R}$, the map Q is also called a **conserved quantity** or a **first integral**. If Q is a constant map, it is called a **trivial invariant**; otherwise it is called a **non-trivial invariant**.

Proposition 1.8 (Orbit characterisation of invariants). *Q is an invariant of f if and only if Q is constant on every forward orbit; when f is a bijection, if and only if Q is constant on every two-sided orbit.*

Proof. If $Q \circ f = Q$, then induction on n gives $Q \circ f^n = Q$, so Q is identically $Q(x_0)$ on $\{f^n(x_0)\}$. Conversely, if Q is constant on every forward orbit, then $Q(f(x)) = Q(x)$ for every x. The bijective case is analogous. ☐

Property 1.3 (Algebraic closure of invariants). *Let $Q_1, Q_2 : X \to \mathbb{R}$ be f-conserved quantities and let $\varphi : \mathbb{R}^2 \to \mathbb{R}$ be an arbitrary map. Then $x \mapsto \varphi(Q_1(x), Q_2(x))$ is again an f-conserved quantity. In particular, the set of all f-conserved quantities forms an $\mathbb{R}$-algebra.*

Proof. Verify pointwise:

$$\varphi(Q_1(f(x)), Q_2(f(x))) = \varphi(Q_1(x), Q_2(x)).$$

☐

Remark 1.7 (Triviality: the existence of a conserved quantity is not a condition). *A constant map is always an invariant of f, so 'there exists a conserved quantity' holds for every system and imposes no constraint whatsoever. The statement with discriminating power is 'there exists a* **non-trivial** *conserved quantity'. Listing 'there exists a conserved quantity' as a necessary condition for periodicity is therefore an oversight: if 'conservation' includes the trivial case, then the condition is always true and carries no information; if it is required to be non-trivial, then a periodic system need not satisfy it either.*

Definition 1.16 (Conservation group). Let $Q : X \to Y$. The **conservation group** (equivariance group) of Q is

$$G_Q = \{g \in \mathrm{Aut}(X) \mid Q \circ g = Q\}.$$

Property 1.4 (The conservation group is a group). *$G_Q \leq \mathrm{Aut}(X)$.*

Proof. $\mathrm{id}_X \in G_Q$. If $Q \circ g = Q$ and $Q \circ h = Q$, then $Q \circ (g \circ h) = Q$, and

$$Q \circ g^{-1} = (Q \circ g) \circ g^{-1} = Q.$$

Hence G_Q is closed under composition and under taking inverses. ☐

Proposition 1.9 (The evolution conserves its own invariants)**.** Let f be a bijection and let Q be an f-invariant. Then $\langle f\rangle \leq G_Q$.

Proof. By Proposition 1.8, $Q \circ f^n = Q$ for all $n \in \mathbb{Z}$. Hence $f^n \in G_Q$. □

Remark 1.8 (Three sources of conserved quantities). There are three routes to identifying a conserved quantity: **empirical observation**; **derivation from symmetry**; **mathematical structure** (a labelling function for the equivalence classes of orbits). All three produce functions that are constant on orbits.

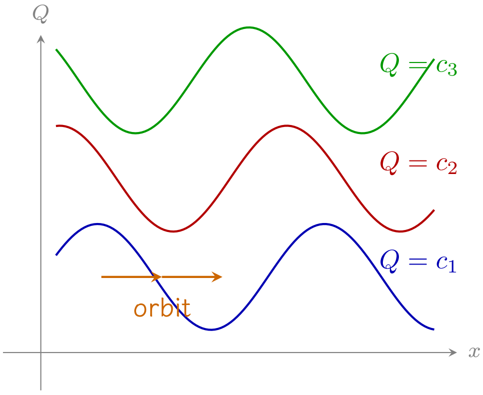


Figure 1.5: The geometric picture of a conserved quantity Q: Q is constant on every orbit, so orbits are confined within the level sets of Q. Distinct level sets give a stratification between the orbits.

1.4 Symmetry

Definition 1.17 (Symmetry group of a dynamical system)**.** Let (X, f) be a discrete dynamical system with f a bijection. The **symmetry group** of f is the centraliser of f in $\mathrm{Aut}(X)$,

$$\mathrm{Symmetry}(f) = \{g \in \mathrm{Aut}(X) \mid g \circ f = f \circ g\} = C_{\mathrm{Aut}(X)}(f).$$

Property 1.5 (The symmetry group is a group)**.** $\mathrm{Symmetry}(f) \leq \mathrm{Aut}(X)$, and $\langle f\rangle \leq \mathrm{Symmetry}(f)$.

Proof. id_X commutes with f. If g, h commute with f, then

$$g \circ h \circ f = g \circ f \circ h = f \circ g \circ h,$$

and

$$g^{-1} \circ f = g^{-1} \circ f \circ g \circ g^{-1} = g^{-1} \circ g \circ f \circ g^{-1} = f \circ g^{-1}.$$

Hence $\mathrm{Symmetry}(f)$ is a subgroup. By Property 1.2, $f^n \circ f = f^{n+1} = f \circ f^n$, so $\langle f\rangle \leq \mathrm{Symmetry}(f)$. □

$$\begin{tikzcd}
X \arrow[r, "f"] \arrow[d, swap, "g"] \arrow[dr, phantom, "\begin{array}{c} g \in \text{Symmetry}(f) \\ \text{if and only if } g \circ f = f \circ g \end{array}"] & X \arrow[d, "g"] \\
X \arrow[r, swap, "f"] & X
\end{tikzcd}$$

Figure 1.6: The commutative-diagram definition of symmetry: evolving first and then transforming gives the same result as transforming first and then evolving. Symmetry(f) is precisely the group formed by all automorphisms g that make this diagram commute.

Remark 1.9 (Why symmetry is defined via the centraliser). The mathematical meaning of 'the laws of the system are the same before and after the transformation' is that evolving first and then transforming gives the same result as transforming first and then evolving, that is, the commutativity $g \circ f = f \circ g$. This definition does not depend on the existence of any conserved quantity, and hence makes 'symmetry' and 'conservation' two independent concepts.

Proposition 1.10 (Symmetry preserves orbits and least periods)**.** *Let $g \in$ Symmetry(f) and $x_0 \in X$. Then:*

- (i) *g maps f-orbits to f-orbits: $g(\mathcal{O}(x_0; f)) = \mathcal{O}(g(x_0); f)$;*
- (ii) *x_0 is a periodic point if and only if $g(x_0)$ is a periodic point, and the two have the same least period.*

Proof. (i) From $g \circ f^n = f^n \circ g$ we get $g(f^n(x_0)) = f^n(g(x_0))$; letting n range over $\mathbb{Z}$ on both sides gives the result.

(ii) $f^m(g(x_0)) = g(f^m(x_0))$, so $f^m(x_0) = x_0 \iff f^m(g(x_0)) = g(x_0)$. The least periods are therefore the same. $\square$

Definition 1.18 (Joint symmetry group of a system)**.** Let (X, f) be a discrete dynamical system and let Q be an f-conserved quantity. The **joint symmetry group** of the triple (X, f, Q) is

$$\text{Symmetry}(f, Q) = \text{Symmetry}(f) \cap G_Q.$$

Theorem 1.4 (Symmetry yields conservation (algebraic form))**.** *Let f be a bijection, let $G \leq$ Symmetry(f), and suppose that every orbit of G on X is finite. Define the orbit-counting function*

$$Q_G : X \to \mathbb{N} \cup \{\infty\}, \qquad Q_G(x) = |\mathcal{O}_G(x)|.$$

Then Q_G is an f-invariant. In particular, if the action of G is non-trivial and not transitive, then Q_G is a non-trivial f-conserved quantity.

Proof. For any $x \in X$, since f commutes with every element of G,

$$\mathcal{O}_G(f(x)) = \{g \cdot f(x) \mid g \in G\} = \{f(g \cdot x) \mid g \in G\} = f(\mathcal{O}_G(x)).$$

Since f is a bijection, its restriction to $\mathcal{O}_G(x)$ is a bijection onto $f(\mathcal{O}_G(x))$, so the two orbits have the same cardinality:

$$Q_G(f(x)) = Q_G(x).$$

□

Remark 1.10 (Relation to Noether's theorem). What Theorem 1.4 proves is a kind of **orbit-counting construction of invariants**, not Noether's theorem. It is merely the purely algebraic version of 'symmetry ⇒ conservation'. The stronger Noether theorem in physics requires a Lagrangian variational structure and the language of Lie algebras, and lies outside the framework of this chapter.

Proposition 1.11 (Conservation does not imply symmetry). *There exists a system (X, f, Q) in which Q is a non-trivial f-conserved quantity, yet the joint symmetry group* Symmetry(f, Q) *is trivial.*

Proof. Take $X = \mathbb{R}$, $f = \mathrm{id}_{\mathbb{R}}$ and $Q(x) = x^3 + x$. Here every function is an f-invariant, so Q is conserved and non-trivial. Note that $Q'(x) = 3x^2 + 1 > 0$, so Q is strictly increasing and hence injective. If $g \in G_Q$, then $Q(g(x)) = Q(x)$ for all x, and the injectivity of Q gives $g(x) = x$, that is, $G_Q = \{\mathrm{id}\}$. Hence

$$\mathrm{Symmetry}(f, Q) = \mathrm{Symmetry}(f) \cap G_Q = \{\mathrm{id}\}.$$

□

Remark 1.11 (The reflection trap in constructing counterexamples). In constructing a counterexample of the form 'conserved but with trivial symmetry', the choice $Q(x) = x^2$ does not work: the reflection $x \mapsto -x$ satisfies $(-x)^2 = x^2$, so $G_Q \supseteq \{\mathrm{id}, x \mapsto -x\} \cong C_2$. A conserved quantity of injective type must be used.

Corollary 1.2 (The logical hierarchy of conservation and symmetry).

(i) symmetry ⇒ conservation (Theorem 1.4, requiring finite G-orbits);

(ii) conservation $\not\Rightarrow$ symmetry (Proposition 1.11);

(iii) hence symmetry is a strictly stronger concept than conservation.

1.5 Finite-order actions

Definition 1.19 (Order of a map). Let $f : X \to X$. If there exists a least positive integer m such that $f^m = \mathrm{id}_X$, then f is called a **finite-order action** and m is its **order**, written $\mathrm{ord}(f) = m$; otherwise we write $\mathrm{ord}(f) = \infty$. When f is a bijection, this is equivalent to the order of the cyclic subgroup $\langle f \rangle \leq \mathrm{Aut}(X)$.

Lemma 1.5 (Finite order implies invertibility). *If $\mathrm{ord}(f) = m < \infty$, then f must be a bijection, and $f^{-1} = f^{m-1}$.*

Proof. $f \circ f^{m-1} = f^m = \mathrm{id}_X = f^{m-1} \circ f$, so f^{m-1} is a two-sided inverse of f. □

Lemma 1.6 (Pointwise closure and uniform closure). *Let f be a bijection. Distinguish the following two concepts:*

(i) **pointwise closure**: every $x \in X$ is a periodic point (the least periods m_x of different points may differ);

(ii) **uniform closure**: $\mathrm{ord}(f) < \infty$ (there exists a uniform m with $f^m = \mathrm{id}_X$).

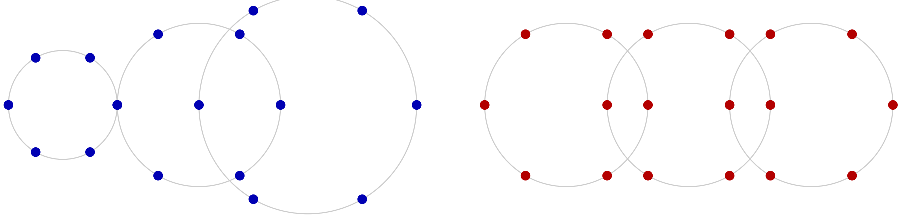
Then (ii) $\Rightarrow$ (i), and in that case every m_x divides $\mathrm{ord}(f)$; but (i) $\not\Rightarrow$ (ii).

Proof. (ii) $\Rightarrow$ (i): $f^m(x) = x$ for all x. Let m_x be the least period of x and perform division with remainder $m = qm_x + r$, $0 \le r < m_x$; then

$$f^r(x) = f^{m-qm_x}(x) = f^m(x) = x,$$

and the minimality of m_x gives $r = 0$, that is, $m_x \mid m$.

(i) $\not\Rightarrow$ (ii): Take $X = \bigsqcup_{n\ge1} \mathbb{Z}/n\mathbb{Z}$, and on the n-th component let f be the map 'add 1', $x \mapsto x+1 \pmod{n}$. Every point belongs to some component and has least period n, so (i) holds. But for any $m \ge 1$, on the $(m+1)$-th component $f^m(0) = m \not\equiv 0 \pmod{m+1}$, so $f^m \ne \mathrm{id}_X$ and $\mathrm{ord}(f) = \infty$. ☐



(a) Pointwise closure: orbit lengths may differ

(b) Uniform closure: there exists a uniform m with $f^m = \mathrm{id}$

Figure 1.7: Pointwise closure requires only that each point returns to itself, and the periods may differ; uniform closure requires a uniform m such that all points return simultaneously. Passing from the former to the latter requires 'periods uniformly bounded' or a 'finite least common multiple'.

Theorem 1.5 (Orbit characterisation of finite order)**.** *Let f be a bijection. The following conditions are equivalent:*

(i) $\mathrm{ord}(f) = m < \infty$;

(ii) *f is pointwise closed, and the least common multiple of all the least periods $\{m_x\}$ is finite;*

(iii) *f is pointwise closed, and the orbit lengths are uniformly bounded.*

When the equivalence holds,

$$\mathrm{ord}(f) = \mathrm{lcm}\{m_x \mid x \in X\}.$$

Proof. (i) $\Rightarrow$ (ii): By Lemma 1.6, every $m_x \mid m$, so $\mathrm{lcm}\{m_x\} \mid m$ is finite.

(ii) $\Rightarrow$ (i): Let $M = \mathrm{lcm}\{m_x\} < \infty$. For any x, $m_x \mid M$, so $f^M(x) = x$, that is, $f^M = \mathrm{id}_X$. Also every $m_x \mid \mathrm{ord}(f)$, so $M \mid \mathrm{ord}(f)$, and the two are equal.

(ii) $\Rightarrow$ (iii): $m_x \le M$, so the periods are uniformly bounded.

(iii) $\Rightarrow$ (ii): If $m_x \le N$ for all x, then $\mathrm{lcm}\{m_x\} \mid \mathrm{lcm}(1, 2, \ldots, N)$, which is finite. ☐

Remark 1.12 (A point on rigour). In passing from ‘each orbit closes on its own’ to ‘finite-order action’, the additional condition of **uniformly bounded periods** (or a finite least common multiple) must be imposed. If the ‘closure group’ is defined as the subgroup generated by elements of finite order, $G_{\mathrm{cl}} = \langle g \in G : \mathrm{ord}(g) < \infty \rangle$, it guarantees only that each generator has finite pointwise order, and does not automatically give uniform closure of the whole action.

1.6 Closed orbits and periodicity

Definition 1.20 (Closed orbit). Let (X, f) be a discrete dynamical system. An orbit $\mathcal{O}(x_0; f)$ (or a forward orbit $\mathcal{O}^+(x_0; f)$) is called **closed** if it is a finite set.

Theorem 1.6 (Criterion for closed orbits). Let f be a bijection and $x_0 \in X$. The following conditions are equivalent:

(i) the orbit $\mathcal{O}(x_0; f)$ is closed;

(ii) x_0 is a periodic point;

(iii) the stabiliser $\mathrm{Stab}_{\langle f \rangle}(x_0)$ has finite index in $\langle f \rangle$.

When the equivalence holds,

$$|\mathcal{O}(x_0; f)| = m_0 = \text{the least period of } x_0, \qquad \mathrm{Stab}_{\langle f \rangle}(x_0) = \langle f^{m_0} \rangle,$$

and the restriction of f to the orbit is an m_0-cycle permutation.

Proof. (i) ⇒ (ii): If the orbit is finite, then the infinite sequence $x_0, f(x_0), f^2(x_0), \dots$ must contain a repetition: there exist $i < j$ with $f^i(x_0) = f^j(x_0)$. Applying f^{-i} to both sides gives $f^{j-i}(x_0) = x_0$, so x_0 is a periodic point.

(ii) ⇒ (i) and the structure of the orbit: Let $f^{m_0}(x_0) = x_0$ with m_0 least. For any $n \in \mathbb{Z}$, division with remainder $n = qm_0 + r$ gives $f^n(x_0) = f^r(x_0)$, so

$$\mathcal{O}(x_0; f) = \{x_0, f(x_0), \dots, f^{m_0 - 1}(x_0)\}.$$

These m_0 points are pairwise distinct: if $f^r(x_0) = f^s(x_0)$ with $0 \le r < s < m_0$, then $f^{s-r}(x_0) = x_0$, contradicting the minimality of m_0. Hence the orbit contains exactly m_0 points. Also $f(f^{m_0-1}(x_0)) = x_0$, so f permutes the orbit cyclically.

(ii) $\iff$ (iii): $f^n \in \mathrm{Stab}(x_0) \iff f^n(x_0) = x_0 \iff m_0 \mid n$, so the stabiliser is exactly $\langle f^{m_0} \rangle$, with index m_0. □

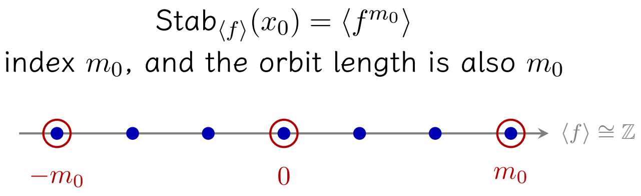

$\mathrm{Stab}_{\langle f \rangle}(x_0) = \langle f^{m_0} \rangle$
index m_0, and the orbit length is also m_0
$\langle f \rangle \cong \mathbb{Z}$
$-m_0$
0
m_0

Figure 1.8: On a closed orbit the stabiliser is exactly $\langle f^{m_0} \rangle$, and its index in $\langle f \rangle$ equals the orbit length m_0.

Corollary 1.3 (Periodic structure of closed orbits)**.** *The dynamical system on a closed orbit, $(\mathcal{O}(x_0; f), f|_{\mathcal{O}})$, is isomorphic to the action of the cyclic group C_{m_0} by translation on itself ($k \mapsto k+1 \bmod m_0$).*

Proof. By Theorem 1.6, the orbit is exactly $\{x_0, \dots, f^{m_0-1}(x_0)\}$ and f cycles on it. The map $f^k(x_0) \mapsto k \bmod m_0$ is a bijection of sets, and conjugates $f|_{\mathcal{O}}$ to the map 'add 1' on $\mathbb{Z}/m_0\mathbb{Z}$. ☐

Definition 1.21 (Strict periodicity)**.** *A discrete dynamical system (X, f) is said to have a* **strict period** *at x_0 if x_0 is a periodic point; its least period m_0 is called the* **period** *of the system at x_0. If every point is a periodic point, the system is called a* **periodic system***; if moreover $\mathrm{ord}(f) = m < \infty$, the system is called a* **uniformly periodic system***, and m is its* **uniform period***.*

Remark 1.13 (The place of quasiperiodicity)*.* 'Quasiperiodicity' (for instance an irrational rotation of the circle: the orbit does not close but is dense) is essentially a topological concept. Within a purely algebraic framework one can only assert that $\mathrm{ord}(f) = \infty$. This chapter distinguishes only: trivial periodicity ($m_0 = 1$, a fixed point), strict periodicity ($m_0 \geq 2$) and aperiodicity (an infinite orbit).

Proposition 1.12 (Periodicity yields symmetry)**.** *Let f be a bijection with $\mathrm{ord}(f) = m < \infty$. Then the evolution group $\langle f \rangle \cong C_m$ is a non-trivial (when $m \geq 2$) finite subgroup of $\mathrm{Symmetry}(f)$.*

Proof. By Property 1.5, $\langle f \rangle \leq \mathrm{Symmetry}(f)$; by Definition 1.19, $|\langle f \rangle| = m$. ☐

Theorem 1.7 (Periodicity does not imply non-trivial conservation)**.** *There exists a uniformly periodic system all of whose f-conserved quantities are trivial (constant).*

Proof. Take $X = \mathbb{Z}/m\mathbb{Z}$ ($m \geq 2$) and $f(x) = x + 1 \pmod m$. Then $f^m = \mathrm{id}_X$ with m least, so the system is uniformly periodic. Let $Q : X \to Y$ be an arbitrary f-invariant. By Proposition 1.8, Q is constant on every orbit; but f has only one orbit on X, so Q is constant on the whole of X. ☐

Remark 1.14 (The hierarchical difference between periodicity and conservation)*.* The stronger the transitivity of the orbits, the fewer the invariants. A periodic orbit is precisely the 'most transitive' structure—the whole orbit is a single cycle, and any conserved quantity must take the same value everywhere. Hence 'periodicity ⇒ conservation' fails in general. A conserved quantity describes the stratification **between orbits**, not the cycle within an orbit.

Corollary 1.4 (Logical diagram of period, symmetry and conservation)**.**

(i) *finite order ⇒ uniform periodicity ⇒ pointwise periodicity $\iff$ pointwise closed orbits;*

(ii) *periodicity ⇒ symmetry, but periodicity ⇏ conservation;*

(iii) *symmetry ⇒ conservation, but conservation ⇏ symmetry;*

(iv) *the conjunction of symmetry and conservation still ⇏ periodicity: take $X = \mathbb{R}$, $f(x) = x + 1$, $Q(x) = \sin(2\pi x)$. Then $Q \circ f = Q$ and $\langle f \rangle \cong \mathbb{Z}$ is non-trivial with $\langle f \rangle \leq \mathrm{Symmetry}(f, Q)$; but $\mathrm{ord}(f) = \infty$.*

Proof. (i) By Definition 1.21, Theorem 1.6 and Lemma 1.6. (ii) Proposition 1.12 and Theorem 1.7. (iii) Theorem 1.4 and Proposition 1.11. (iv) Verify item by item: $Q(f(x)) = \sin(2\pi(x+1)) = \sin(2\pi x) = Q(x)$; f commutes with itself and preserves Q; $f^n(x) = x + n \neq x$, so there are no periodic points. □

Remark 1.15 (The logical structure of the main chain). In the chain 'conservation → symmetry → closed orbit → period': the first step has the opposite direction (it is really symmetry ⇒ conservation), the second step does not hold (symmetry + conservation ⇏ finite order), and the third step holds and is reversible (closed orbit ⟺ pointwise periodicity). Within the framework of this chapter the unconditional sufficient chain is

$$\text{finite-order action} \Longrightarrow \text{(uniform) closed orbit} \Longrightarrow \text{strict period.}$$

The proper role of conservation and symmetry here is this: symmetry restricts the possible forms of the evolution, and conservation confines the orbits to low-dimensional level sets—the two are auxiliary tools for **discovering and proving** that some generator has finite order, not an indispensable link in the chain of logic.

1.7 Function characteristics of the three elementary transformations

This section characterises the three elementary transformations—translation, scaling and rotation—from the point of view of functional equations.

Definition 1.22 (Translation-symmetric and translation-covariant functions). Let $f : \mathbb{R} \to Y$. If there exists $a \neq 0$ such that

$$f(x+a) = f(x), \qquad \forall x \in \mathbb{R},$$

then f is called a **translation-symmetric function** and a is called a **translation period**. If there exists $\lambda \in \mathbb{C}$ with $|\lambda| = 1$ such that

$$f(x+a) = \lambda f(x), \qquad \forall x,$$

then f is called a **translation-covariant function** and λ is called the **translation characteristic factor**.

Definition 1.23 (Scaling-symmetric and homogeneous functions). Let $f : \mathbb{R}^+ \to Y$. If there exists $q > 0$, $q \neq 1$, such that

$$f(qx) = f(x), \qquad \forall x > 0,$$

then f is called a **scaling-symmetric function** and q is called a **scaling period**. If there exists $s \in \mathbb{R}$ such that

$$f(qx) = q^s f(x), \qquad \forall x > 0,$$

then f is called a **homogeneous function** and s is called the **homogeneity exponent**.

Definition 1.24 (Rotation-symmetric and rotation-covariant functions). Let $f : \mathbb{C}^\times \to Y$. If there exists $\theta \in \mathbb{R}$ such that

$$f(e^{i\theta}z) = f(z), \qquad \forall z \neq 0,$$

then f is called a **rotation-symmetric function** and θ is called a **rotation period**. If there exists $m \in \mathbb{Z}$ such that

$$f(e^{i\theta}z) = e^{im\theta}f(z), \qquad \forall z \neq 0,$$

then f is called a **rotation-covariant function** and m is called the **weight**.

Definition 1.25 (Group of period transformations). Let G be a transformation group and $f : X \to Y$. The **group of period transformations** of f is

$$\mathsf{Group}_{\mathsf{period},G}(f) = \{g \in G \mid f(g \cdot x) = f(x),\ \forall x \in X\}.$$

Proposition 1.13 (Translation symmetry and antiperiodicity). *Let $f : \mathbb{R} \to Y$ and $a \neq 0$.*

(i) *If $f(x+a) = f(x)$, then every element of $a\mathbb{Z}$ is a translation period of f.*

(ii) *If $f(x+a) = -f(x)$, then $f(x+2a) = f(x)$.*

(iii) *More generally, if $f(x+a) + f(x) = c$ is constant, then $f(x+2a) = f(x)$ still holds.*

Proof. (i) Iterating $f(x+a) = f(x)$ repeatedly gives $f(x+na) = f(x)$.

(ii) $f(x+2a) = f((x+a)+a) = -f(x+a) = f(x)$.

(iii) Subtracting $f(x+a)+f(x) = c$ from $f(x+2a)+f(x+a) = c$ gives $f(x+2a) = f(x)$. ☐

Proposition 1.14 (Scaling symmetry and logarithmic translation). *Let $f : \mathbb{R}^+ \to Y$, $q > 0$, $q \neq 1$. Put*

$$g(t) := f(e^t), \qquad t \in \mathbb{R}.$$

Then

$$f(qx) = f(x) \quad \Longleftrightarrow \quad g(t + \ln q) = g(t).$$

Hence a scaling-symmetric function is equivalent to a translation-symmetric function in logarithmic coordinates. Conversely, if $q = e^\alpha$ with $\alpha \neq 0$ and $g(t+\alpha) = g(t)$, then $f(qx) = f(x)$.

Proof. Substituting $x = e^t$ gives

$$f(qe^t) = f(e^{t+\ln q}) = g(t + \ln q).$$

Hence $f(qx) = f(x)$ is equivalent to $g(t + \ln q) = g(t)$. Conversely, if $q = e^\alpha$ and $g(t+\alpha) = g(t)$, then for $x = e^t > 0$,

$$f(qx) = f(e^\alpha e^t) = f(e^{t+\alpha}) = g(t+\alpha) = g(t) = f(x).$$

☐

Proposition 1.15 (Rotation symmetry and the order of roots of unity)**.** Let $f : \mathbb{C}^\times \to Y$, $a \in \mathbb{N}^+$ and $q \in \mathbb{Z}$. If

$$f(e^{2\pi i q/a} z) = f(z), \qquad \forall z \neq 0,$$

then

$$C_{a/\gcd(a,q)} \leq \mathsf{Group}_{\mathsf{period},U(1)}(f),$$

where for $q = 0$ we adopt the convention $a/\gcd(a,0) = 1$.

Proof. The order of the element $e^{2\pi i q/a}$ is

$$\mathsf{ord}\big(e^{2\pi i q/a}\big) = \frac{a}{\gcd(a,q)}.$$

Hence the cyclic group it generates is $C_{a/\gcd(a,q)}$, and every element of this group leaves f invariant. □

Example 1.3 (Translation symmetry)**.** $f(x) = \sin(2\pi x/a)$ satisfies $f(x+a) = f(x)$, and its least positive period is a.

Example 1.4 (Antiperiodicity)**.** $f(x) = (-1)^{\lfloor x/a \rfloor}$ satisfies $f(x+a) = -f(x)$, so $f(x+2a) = f(x)$ and the period is doubled to $2a$.

Example 1.5 (Scaling symmetry)**.** $f(x) = \sin(2\pi \log_q x)$, $x > 0$, satisfies $f(qx) = f(x)$.

Example 1.6 (Rotation symmetry)**.** $f(z) = z^n$ satisfies

$$f(e^{2\pi i/n} z) = e^{2\pi i} z^n = z^n = f(z),$$

so the rotation group with periodic orbits contains C_n.

Remark 1.16 (Function characteristics of the three transformations compared)*.*

Transformation	Functional equation	Characteristic
Translation	$f(x+a) = f(x)$	translation-invariant, with period a
Translation	$f(x+a) = -f(x)$	antiperiodic, with period doubled to $2a$
Scaling	$f(qx) = f(x)$	scaling-invariant, a translation period in logarithmic coordinates
Rotation	$f(e^{i\theta} z) = f(z)$	rotation-invariant, the group with periodic orbits being determined by the order of roots of unity

Remark 1.17 (Interface with the extensions of number systems)*.* The three functional equations given in this section have, respectively, three algebraic structures as their background:

translation: the additive group $(\mathbb{R}, +)$, scaling: the multiplicative group $(\mathbb{R}^+, \times)$, rotati

The minimal algebraic realisations of these three groups within the number systems correspond to the main line of the extension of number systems in the next chapter:

(i) **Translation** requires the freedom to add and subtract the period a, so at least the additive group of the integers $\mathbb{Z}$ is needed.

(ii) **Scaling** requires the freedom to multiply and divide by the period q, so at least the multiplicative group of the positive rationals $\mathbb{Q}^+$ is needed.

(iii) **Rotation** requires unit complex numbers $e^{i\theta}$, so at least the unit circle group $U(1)$ of the complex numbers $\mathbb{C}$ is needed.

(iv) **Three-dimensional rotation** requires a more general non-commutative rotation group. Its double cover is the unit sphere group $S^3 \cong SU(2)$ of the quaternions $\mathbb{H}$, and the rotation group itself is $SO(3) \cong SU(2)/\{\pm I\}$.

Thus each step in the extension of the number systems makes a new class of periodic transformations decidable.

Remark 1.18 (Connection with the group action framework). The functional equations given in this section are all concrete manifestations of group actions on function spaces. The group with periodic orbits $\mathsf{Group}_{\mathsf{period},G}(f)$ is precisely the set of transformations for which the corresponding functional equation holds. Hence

$$\text{the transformation } g \text{ is a period of } f \iff f(g \cdot x) = f(x), \quad \forall x.$$

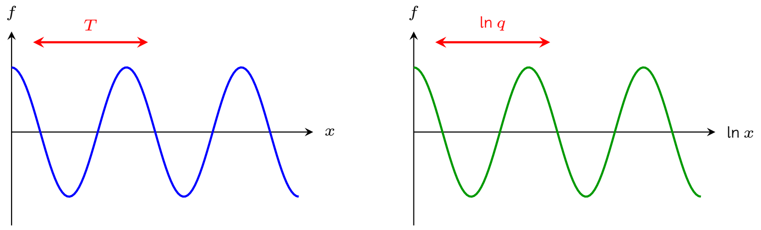
Remark 1.19 (Boundaries of domain and order). (i) A scaling period is usually defined for $x > 0$ or $x \neq 0$, because the multiplicative action of $q > 0$ is most natural on $\mathbb{R}^+$.

(ii) For a rotation period, if $a \notin \mathbb{N}^+$ then $e^{2\pi i/a}$ need not be a root of unity, and its order may be infinite; the statement 'the rotation period is a' therefore holds only when a is a positive integer and a primitive root of unity is taken.

(iii) Antiperiodicity $f(x + a) + f(x) = 0$ is not itself a period, but it implies the period $2a$.

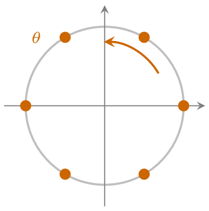
1.8 Completeness of the three elementary transformations: classification of one-parameter families of conformal transformations

The preceding sections of this chapter introduced translation, scaling and rotation as the three elementary transformations and gave their functional equations one by one. This section answers a stronger question: **why exactly these three?** Below, within the explicit framework of 'one-parameter families of conformal transformations on the complex plane', we prove that the three constitute a **complete and mutually exclusive** set of generating primitives. It should be declared in advance that the scope of the conclusions of this section is precisely this framework; the corresponding assertion for general continuous transformation groups is not among the claims of this book (see 'what this book does not claim' in the introduction, and Remark 1.22 at the end of this section).



(a) Translation period
$f(x+T) = f(x)$, centre of symmetry $c = 0$

(b) Ratio period
$f(qx) = f(x)$, centre of symmetry $c = 1$



(c) Rotation period
$f(e^{i\theta}z) = f(z)$, centre of symmetry $c = i$

Figure 1.9: The three fundamental types of periodic function. They are governed respectively by the centres of symmetry 0, 1 and i. Note that the ratio period in (b) degenerates to a translation period in logarithmic coordinates.

1.8.1 One-parameter families of transformations and infinitesimal generators

Definition 1.26 (One-parameter families of transformations)**.** Let $\Omega \subseteq \hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ be an open set. A family of maps $\{\varphi_t\}_{t\in\mathbb{R}}$, $\varphi_t : \Omega \to \Omega$, is called a **one-parameter family of transformations** if

(i) $\varphi_0 = \mathrm{id}$;

(ii) $\varphi_{s+t} = \varphi_s \circ \varphi_t$ for all $s, t \in \mathbb{R}$ (that is, $t \mapsto \varphi_t$ is a group homomorphism);

(iii) $(t, z) \mapsto \varphi_t(z)$ is smooth in t.

If every φ_t is a conformal bijection of Ω, then $\{\varphi_t\}$ is called a **one-parameter family of conformal transformations**.

Definition 1.27 (Infinitesimal generator)**.** Let $\{\varphi_t\}$ be a one-parameter family of transformations. The vector field

$$\xi(z) := \frac{d}{dt}\Big|_{t=0} \varphi_t(z)$$

is called the **infinitesimal generator** of $\{\varphi_t\}$. By property (ii), φ_t is the solution of the equation

$$\frac{d}{dt}\varphi_t(z) = \xi\big(\varphi_t(z)\big), \qquad \varphi_0(z) = z$$

that is, φ_t is uniquely determined by ξ.

1.8.2 Canonical form and completeness

Theorem 1.8 (Canonical form of one-parameter conformal families)**.** *Let $\{\varphi_t\}$ be a one-parameter family of conformal transformations on the Riemann sphere $\hat{\mathbb{C}}$. Then there exist $\lambda \in \mathbb{C}$ and a conformal coordinate transformation W of $\hat{\mathbb{C}}$ such that*

$$W\Big(\varphi_t\big(W^{-1}(w)\big)\Big) = e^{\lambda t}\, w, \qquad \forall t \in \mathbb{R},\ w \in \hat{\mathbb{C}},$$

or (when $\lambda = 0$) the family is parabolic after conjugation:

$$W\Big(\varphi_t\big(W^{-1}(w)\big)\Big) = w + t.$$

Proof. **First step (classification of conformal automorphisms).** The holomorphic bijections of $\hat{\mathbb{C}}$ are exactly the fractional linear transformations $w \mapsto \dfrac{aw+b}{cw+d}$ $(ad-bc \neq 0)$. This is the classical theorem on automorphisms of the Riemann sphere: a holomorphic bijective self-map of a compact Riemann surface is biholomorphic, and the biholomorphic automorphisms of $\hat{\mathbb{C}}$ are precisely the Möbius transformations. Hence

$$\mathsf{Aut}(\hat{\mathbb{C}}) \cong \mathsf{PSL}(2,\mathbb{C}) = \mathsf{SL}(2,\mathbb{C})/\{\pm I\}.$$

Second step (one-parameter families and the exponential map). By Definition 1.26, $t \mapsto \varphi_t$ is a Lie group homomorphism $\mathbb{R} \to \mathsf{PSL}(2,\mathbb{C})$. After lifting to $\mathsf{SL}(2,\mathbb{C})$ (the Lie group–Lie algebra correspondence), there exists $X \in \mathfrak{sl}(2,\mathbb{C})$ such that

$$\varphi_t \longleftrightarrow \exp(tX), \qquad \operatorname{tr} X = 0.$$

Third step (classification by Jordan type). Since $\operatorname{tr} X = 0$, the eigenvalues of X are μ and $-\mu$.

Case (a): $\mu = 0$ and $X \neq 0$. Here X is nilpotent and, in a suitable basis, $X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, so that

$$\exp(tX) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

In the affine coordinate $z = w_2/w_1$, this linear map induces $z \mapsto \frac{w_2+tw_1}{w_1} = z+t$, that is, a translation.

Case (b): $\mu \neq 0$. Here X is diagonalisable and, in a suitable basis, $X = \mathsf{diag}(\mu,-\mu)$, so that

$$\exp(tX) = \mathsf{diag}\big(e^{\mu t},\, e^{-\mu t}\big).$$

In the affine coordinate $z = w_2/w_1$ it induces

$$z = \frac{w_2}{w_1} \longmapsto \frac{e^{-\mu t} w_2}{e^{\mu t} w_1} = e^{-2\mu t} z.$$

Taking $\lambda := -2\mu$ (as μ ranges over $\mathbb{C}\setminus\{0\}$, so does λ), we obtain $z \mapsto e^{\lambda t} z$. The two cases are exhaustive, and the theorem is proved. □

Corollary 1.5 (Completeness of the three basic transformations)**.** *In the framework of Theorem 1.8, write $\lambda = a + bi$ $(a,b \in \mathbb{R})$. Then the canonical forms fall into exactly four classes:*

(i) **parabolic** ($\lambda = 0$, nilpotent generator): $w \mapsto w + t$, that is, **translation**;

(ii) **purely scaling** ($a \neq 0$, $b = 0$): $w \mapsto e^{at}w$, that is, **scaling**;

(iii) **purely rotational** ($a = 0$, $b \neq 0$): $w \mapsto e^{ibt}w$, that is, **rotation**;

(iv) **composite type** ($a \neq 0$ and $b \neq 0$): $w \mapsto e^{at} \cdot e^{ibt}\, w$, that is, a **composite of scaling and rotation**.

In particular, every one-parameter family of conformal transformations can be decomposed (uniquely, modulo the argument period 2π) into a composite of scaling and rotation, or degenerates to a pure translation; that is, **translation, scaling and rotation constitute the complete generating primitives in this framework**.

Proof. By Theorem 1.8, when $\lambda = 0$ the generator is nilpotent, which gives (i); when $\lambda \neq 0$ we decompose $e^{\lambda t}$ into modulus and argument:

$$e^{\lambda t} = e^{at} \cdot e^{ibt}, \qquad |e^{at}| = e^{at} > 0, \quad |e^{ibt}| = 1.$$

This is precisely the polar decomposition of the complex number $e^{\lambda t}$, so a and b are uniquely determined by λ, and hence the decomposition is unique modulo 2π. It degenerates when $a = 0$ or $b = 0$, giving (ii) and (iii) respectively, and when both are non-zero it gives (iv). □

Corollary 1.6 (The unique source of periodicity is the rotational component)**.** With the notation of Corollary 1.5:

(i) if $a \neq 0$, then ord$(\varphi_t) = \infty$ for every $t \neq 0$, and φ_t produces no closed orbit on any of its orbits;

(ii) if $a = 0$ and $b \neq 0$, then

$$\text{ord}(\varphi_t) < \infty \iff bt \in 2\pi\mathbb{Z} \iff t \in \frac{2\pi}{|b|}\mathbb{Z},$$

in which case, writing $bt/(2\pi) = p/q$ in lowest terms, one has ord$(\varphi_t) = q$, and every orbit of φ_t is a finite closed orbit;

(iii) in particular, the whole family $\{\varphi_t\}$ has a strict period $T = \dfrac{2\pi}{|b|}$ in the time parameter: $\varphi_{t+T} = \varphi_t$.

Hence, within the one-dimensional one-parameter conformal framework, **a strict period can be produced only by the rotational component**: the scaling component ($a \neq 0$) makes the orbit diverge or converge monotonically in the direction of the modulus, destroying finite order; the parabolic type (translation) has no closed orbit in $\mathbb{C}$.

Proof. By Corollary 1.5, φ_t is conjugate to $w \mapsto e^{\lambda t}w$.

(i) If $a > 0$, then $|e^{\lambda t}w| = e^{at}|w|$ increases strictly monotonically under iteration in n (for $w \neq 0$), so $e^{\lambda nt}w \neq w$ for all $n \geq 1$; for $a < 0$ it decreases strictly monotonically by the same argument. Hence ord $= \infty$.

(ii) When $a = 0$ we have $|e^{ibt}w| = |w|$, and $e^{inbt}w = w \iff e^{inbt} = 1 \iff nbt \in 2\pi\mathbb{Z}$, whose least positive integer solution is ord(φ_t); here the orbit $\{e^{inbt}w\}_{n\geq 0}$ is a finite set of equally spaced points on the circle $|z| = |w|$, hence a closed orbit.

(iii) Taking $t = 1$ gives that the order of φ_1 is the denominator of $b/(2\pi)$; for general t, (ii) gives $T = 2\pi/|b|$ directly. □

Remark 1.20 (Why not reflection or inversion). The three conditions—**conformality**, **bijectivity** and **one-parameter connectedness**—single out exactly the three classes above:

1. (i) **Reflection** $z \mapsto \bar{z}$ is **anti-holomorphic** (it reverses orientation) and does not belong to the conformal automorphism group Aut($\hat{\mathbb{C}}$), so it is excluded by the first condition. Where this book uses 'reflection' elsewhere, it means an **algebraic involution** (a discrete symmetry of order 2, such as the additive reflection $a \mapsto -a$); that belongs to the realm of discrete symmetries and is not an object of the same kind as the connected one-parameter families discussed in this section.
2. (ii) **Inversion** $z \mapsto -1/z$ appears to be 'none of the three', yet as a Möbius transformation it has order 4 and eigenvalues $\pm i$, so it is of elliptic type; hence up to conjugacy it is a rotation (through the angle $\pi/2$, see the classification in Corollary 1.5). This shows that the threefold classification of this section is invariant under conjugation.
3. (iii) **Non-linear transformations of degree two and above** (such as $z \mapsto z^2$) are not bijections of $\hat{\mathbb{C}}$ (0 and ∞ are branch points), and are likewise excluded by the second condition.

Remark 1.21 (Correspondence with the three symmetry centres). Align the classification of Corollary 1.5 with the three centres of symmetry used throughout the book:

$$\begin{array}{lcl} \text{parabolic (translation)} & \longleftrightarrow & \text{centre of symmetry } c = 0 \text{ (addition)}, \\ \text{purely scaling} & \longleftrightarrow & \text{centre of symmetry } c = 1 \text{ (multiplication)}, \\ \text{purely rotational} & \longleftrightarrow & \text{centre of symmetry } c = i \text{ (multiplication by the inverse)}. \end{array}$$

The composite type carries components of both centres $c = 1$ and $c = i$ at once. This correspondence agrees with the three sub-figures for the three types of periodicity in Figure 1.9 of this chapter, and also with the summary table of the three centres of symmetry in the 'Scalars and Vectors' chapter (Table 3.4).

Remark 1.22 (Scope of applicability of this section). What this section proves is that **within the framework of one-parameter families of conformal transformations on the complex plane**, translation, scaling and rotation are complete and mutually exclusive. This is a testable mathematical proposition with explicit hypotheses. It must be strictly distinguished from the research programme in the introduction: the statement made there, that 'any continuous transformation preserving some intrinsic structure can be decomposed into the three generating transformations', is an assertion about **general continuous transformation groups and general intrinsic structures**, and would require first giving an axiomatic characterisation of 'intrinsic structure'—this book **makes no such commitment**, nor does it claim that the conclusions of this section can be generalised without conditions.

1.9 Two lines: iteration generates operations and transformations

1.9.1 The operational line: addition, multiplication and exponentiation

Definition 1.28 (Iterative generation of operations)**.** *On $(\mathbb{N}, +, \times)$, the three basic operations are generated recursively by iteration. Fix $a \in \mathbb{N}$:*

(i) ***Addition** is iteration of the successor: $a + n = s^n(a)$, where $s(k) = k + 1$;*

(ii) ***Multiplication** is iteration of addition: $a \times n = A_a^n(0)$, where $A_a(x) = x + a$;*

(iii) ***Exponentiation** is iteration of multiplication: $a^n = M_a^n(1)$, where $M_a(x) = a \times x$.*

Property 1.6 (Transmission of associativity across the operation hierarchy)**.** (i) *Addition and multiplication satisfy the associative and commutative laws;*

(ii) *exponentiation* **does not satisfy** *the commutative or the associative law: $2^3 = 8 \neq 9 = 3^2$, $2^{(3^2)} = 512 \neq 64 = (2^3)^2$;*

(iii) *in general, iteration preserves the associative law but not the commutative law.*

Proof. (i) This is an axiomatic property of $\mathbb{N}$. (ii) Direct computation. (iii) The index law of Property 1.2 uses only the associativity of composition; commutativity fails at the exponentiation level. ☐

Remark 1.23 (Hierarchical duality: algebraic properties decay, periodic potential grows)*.* As one ascends the hierarchy addition → multiplication → exponentiation, the algebraic 'rigidity'—commutativity, associativity and the like—decays level by level, whilst the capacity to carry elements of finite order grows level by level: $(\mathbb{R}, +)$ is torsion-free and $(\mathbb{R}^+, \times)$ is torsion-free, whereas the realisation of the exponentiation level on the circle, $U(1)$, contains a dense torsion subgroup. The relaxation of algebraic constraints is precisely what makes room for cyclic structure.

1.9.2 The transformation line: translation, scaling and rotation

Definition 1.29 (Three basic transformation groups)**.** (i) ***Translation group**: for $a \in \mathbb{R}$, $T_a : \mathbb{R} \to \mathbb{R}$, $T_a(x) = x + a$; $\{T_a\} \cong (\mathbb{R}, +)$.*

(ii) ***Scaling group**: for $\lambda \in \mathbb{R}^+$, $S_\lambda : \mathbb{R}^+ \to \mathbb{R}^+$, $S_\lambda(x) = \lambda x$; $\{S_\lambda\} \cong (\mathbb{R}^+, \times)$.*

(iii) ***Rotation group**: for $\theta \in [0, 2\pi)$, $R_\theta : U(1) \to U(1)$, $R_\theta(z) = e^{i\theta} z$; $\{R_\theta\} \cong U(1)$.*

Theorem 1.9 (Iterative generation of transformations)**.** *The iterations of the three transformations are parameterised respectively by the three levels of operation:*

(i) *$T_a^n = T_{na}$: the n-fold iteration of a translation is parameterised by* **multiplication** *$n \times a$;*

(ii) $S_\lambda^n = S_{\lambda^n}$: the n-fold iteration of a scaling is parameterised by **exponentiation** λ^n;

(iii) $R_\theta^n = R_{n\theta}$, and the group element itself is the power $e^{i\theta}$.

Proof. (i) Induction: $T_a^{n+1}(x) = T_a(T_a^n(x)) = (x+na)+a = x+(n+1)a$.
(ii) Induction: $S_\lambda^{n+1}(x) = S_\lambda(S_\lambda^n(x)) = \lambda \cdot \lambda^n x = \lambda^{n+1}x$.
(iii) $R_\theta^n(z) = (e^{i\theta})^n z = e^{in\theta} z = R_{n\theta}(z)$. □

Theorem 1.10 (Order structure of the three transformations). (i) $(\mathbb{R},+)$ is torsion-free: $T_a^n = \mathrm{id} \iff na = 0 \iff a = 0$;

(ii) $(\mathbb{R}^+,\times)$ is torsion-free: $S_\lambda^n = \mathrm{id} \iff \lambda^n = 1 \iff \lambda = 1$;

(iii) the torsion subgroup of $U(1)$ is the group of rational rotations: $\mathrm{ord}(R_\theta) < \infty \iff \theta/(2\pi) \in \mathbb{Q}$; if $\theta/(2\pi) = p/q$ (in lowest terms, $q \geq 1$), then $\mathrm{ord}(R_\theta) = q$. All the torsion elements form a subgroup $\{e^{2\pi ip/q}\} \cong \mathbb{Q}/\mathbb{Z}$, which is dense in $U(1)$.

Proof. (i) By Theorem 1.9 (i), $T_a^n = T_{na}$; $T_{na} = T_0 \iff na = 0 \iff a = 0$.
(ii) By (ii), $S_\lambda^n = S_{\lambda^n}$; $\lambda^n = 1$ together with $\lambda > 0$ implies $\lambda = 1$.
(iii) $R_\theta^n = \mathrm{id} \iff e^{in\theta} = 1 \iff n\theta \in 2\pi\mathbb{Z} \iff \theta/(2\pi) \in \mathbb{Q}$. Let $\theta/(2\pi) = p/q$ in lowest terms; then $q \mid n$, and the least positive integer solution is $n = q$. The set of torsion elements is closed under multiplication and contains the identity and inverses, so it is a subgroup; the isomorphism $e^{2\pi i\alpha} \mapsto \alpha + \mathbb{Z}$ shows that it is isomorphic to $\mathbb{Q}/\mathbb{Z}$; density is a topological statement and is given only as a heuristic illustration. □

Corollary 1.7 (Distribution of torsion elements among the three carriers). *Among the three concrete carrier structures chosen in this chapter, the translation group $(\mathbb{R},+)$ and the scaling group of positive reals $(\mathbb{R}^+,\times)$ have no non-trivial torsion elements, whereas $U(1)$ contains non-trivial elements of finite order. Hence on their respective natural carrier spaces, a rotational action can produce a non-trivial strict period, whilst translation and scaling cannot.*

Proof. By Theorem 1.10, the translation and scaling groups are torsion-free, and the orbit generated by any non-identity element of theirs is an infinite set; the torsion elements of the rotation group are exactly the rational rotations, whose orbits are finite sets of equally spaced points. □

Proposition 1.16 (Periodicity is a joint property of transformation and carrier space). *The same transformation law may have different orders on different carrier spaces:*

(i) $x \mapsto x+1$ *has order* ∞ *on* $\mathbb{Z}$ *and order* m *on* $\mathbb{Z}/m\mathbb{Z}$;

(ii) $x \mapsto x+a$ *has order* ∞ *on* $\mathbb{R}$, *whilst on the circle* $\mathbb{R}/2\pi\mathbb{Z}$ *it is a rotation, whose order is finite if and only if* $a/(2\pi) \in \mathbb{Q}$.

Proof. (i) On $\mathbb{Z}/m\mathbb{Z}$, $f^n(x) = x+n \equiv x \pmod{m} \iff m \mid n$, and the least positive integer is m.
(ii) On $\mathbb{R}/2\pi\mathbb{Z}$, $f^n(x) = x+na \equiv x \pmod{2\pi} \iff na \in 2\pi\mathbb{Z}$, and a solution $n \geq 1$ exists if and only if $a/(2\pi) \in \mathbb{Q}$. □

Remark 1.24 (Periodicity depends on the carrier space). The assertion 'a translation has no strict period' holds only when the carrier space is $\mathbb{R}$ (or $\mathbb{Z}$): *a translation on a modular space is simply a rotation. Whether a period exists depends not only on the type of the transformation but also on the global structure of the carrier space.*

1.9.3 Two realisations of periodicity

Example 1.7 (Clock hand: rotational period). The second hand of a clock rotates through $\theta = 2\pi/60$ each second. By Theorem 1.10 (iii), $\operatorname{ord}(R_\theta) = 60$; the orbit of the second hand is a geometric realisation of C_{60}, and after 60 seconds it returns exactly to its starting position.

Example 1.8 (Order of an element in modular arithmetic). In $\mathbb{Z}/12\mathbb{Z}$, by Lemma 1.2, the order of the element $[T]$ is $12/\gcd(T, 12)$:

$$\operatorname{ord}([3]) = 4, \quad \operatorname{ord}([4]) = 3, \quad \operatorname{ord}([5]) = 12, \quad \operatorname{ord}([7]) = 12.$$

$[5]$ and $[7]$ are generators—a discrete translation produces the period 12 on a carrier space modulo 12.

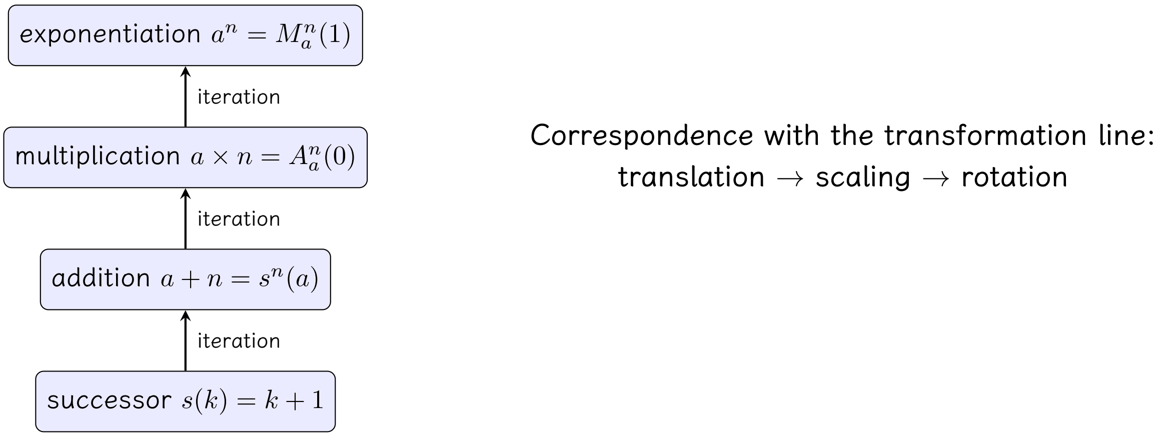

exponentiation $a^n = M_a^n(1)$
iteration
multiplication $a \times n = A_a^n(0)$
iteration
addition $a + n = s^n(a)$
iteration
successor $s(k) = k + 1$
Correspondence with the transformation line:
translation $\to$ scaling $\to$ rotation

Figure 1.10: The correspondence between the operational line and the transformation line: iteration raises the repetition of a lower-level operation to a higher-level operation; correspondingly, the translation parameter enters multiplication through repetition, the scaling parameter enters exponentiation through repetition, and rotation is realised by complex powers.

1.10 Combinatorial criterion for periodicity

Theorem 1.11 (Combinatorial criterion for periodicity). *Let (X, f) be a discrete dynamical system with f a bijection.*

(A) **Sufficient chain**: *if there exists a non-trivial subgroup $G \leq \operatorname{Symmetry}(f, Q)$ (Q being some f-conserved quantity) such that $f \in G$ has finite order $m \geq 2$, then the system is uniformly periodic, every orbit is closed, and the orbit lengths divide m.*

(B) **Necessary condition**: the system is uniformly periodic $\iff$ $\operatorname{ord}(f) < \infty$; in this case the evolution group $\langle f\rangle \cong C_m$ carries its own symmetry, but **no** non-trivial conserved quantity is required to exist.

(C) **Conditions that cannot be omitted**: the conjunction of conservation and symmetry cannot replace finite order; pointwise closure cannot replace uniform closure unless the additional condition that the orbit lengths are uniformly bounded is imposed.

Proof. (A) Since $f \in G \le \operatorname{Symmetry}(f, Q)$ and $\operatorname{ord}(f) = m$, Lemma 1.6 and Theorem 1.6 show that every orbit is closed and has length $m_x \mid m$.

(B) This is a restatement of Definition 1.21 and Theorem 1.5; the two occurrences of the word 'not' are supplied by the constructions of Theorem 1.7 and Corollary 1.4 respectively.

(C) Corollary 1.4 (iv) gives conservation + symmetry $\not\Rightarrow$ periodicity; Theorem 1.5 gives pointwise $\not\Rightarrow$ uniform. ☐

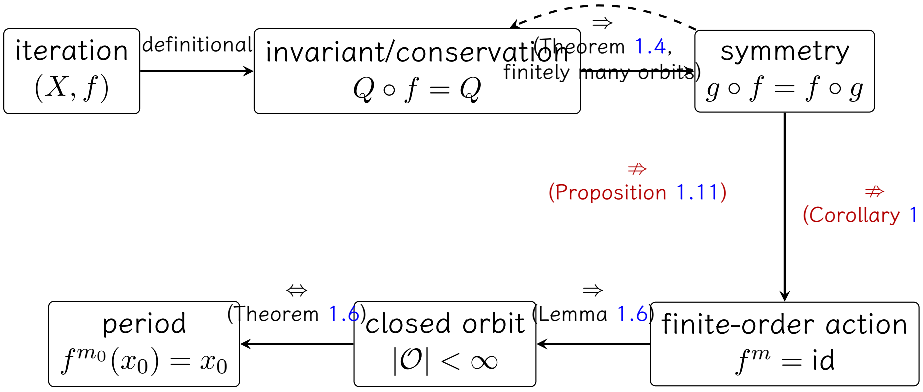

iteration
(X, f)
definitional
invariant/conservation
$Q \circ f = Q$
$\Rightarrow$
(Theorem 1.4, finitely many orbits)
symmetry
$g \circ f = f \circ g$
$\not\Rightarrow$
(Proposition 1.11)
$\not\Rightarrow$
(Corollary 1.4)
period
$f^{m_0}(x_0) = x_0$
$\Leftrightarrow$
(Theorem 1.6)
closed orbit
$|\mathcal{O}| < \infty$
$\Rightarrow$
(Lemma 1.6)
finite-order action
$f^m = \mathrm{id}$

Figure 1.11: The logical relations of the main chain: solid arrows are genuine implications, and the dashed arrow marks the genuine implication in the opposite direction.

1.11 Summary of the criteria for period types

This section collects the criteria of the whole chapter concerning period types; all the results already proved are cited from the corresponding earlier items, and their proofs are not repeated. The key point is to distinguish four levels:

$$\text{period at a point} \longrightarrow \text{pointwise periodicity} \longrightarrow \text{uniform periodicity} \longrightarrow \text{finite-order action.}$$

Among these, the criteria for a general map and for a bijection differ; between pointwise periodicity and uniform periodicity one must impose the additional condition of 'uniformly bounded periods' or a 'finite least common multiple'; and the period of a function and the period at a point are two distinct concepts.

Proposition 1.17 (Orbit criterion for eventual periodicity). *Let (X, f) be a discrete dynamical system and $x_0 \in X$. Then*

$$x_0 \text{ is an eventually periodic point} \iff \mathcal{O}^+(x_0; f) \text{ is a finite set.}$$

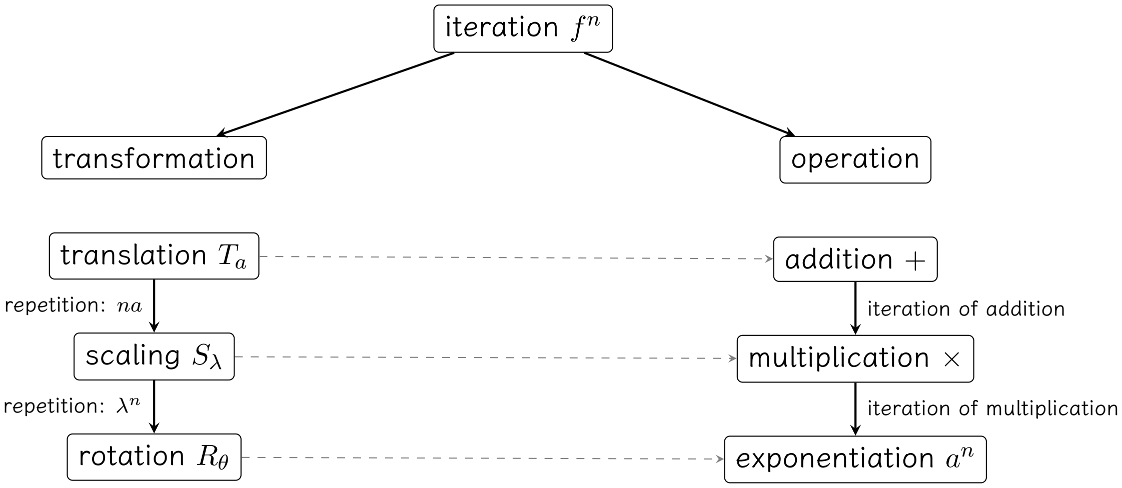

iteration f^n
transformation
operation
translation T_a
addition $+$
repetition: na
iteration of addition
scaling S_λ
multiplication $\times$
repetition: λ^n
iteration of multiplication
rotation R_θ
exponentiation a^n

Figure 1.12: The two generating lines: iteration branches into the transformation line and the operational line; the horizontal dashed lines are the parametric correspondence of Theorem 1.9.

Proof. If x_0 is eventually periodic, then there exists $k \geq 0$ such that $f^k(x_0)$ is a periodic point; let its least period be m. Then

$$\mathcal{O}^+(x_0; f) = \{x_0, f(x_0), \ldots, f^{k-1}(x_0)\} \cup \{f^k(x_0), f^{k+1}(x_0), \ldots, f^{k+m-1}(x_0)\},$$

which is a finite set.

Conversely, if $\mathcal{O}^+(x_0; f)$ is finite, then the infinite sequence

$$x_0, f(x_0), f^2(x_0), \ldots$$

must contain a repetition: there exist $0 \leq i < j$ with $f^i(x_0) = f^j(x_0)$. Put $k = i$; then

$$f^{j-i}\big(f^k(x_0)\big) = f^j(x_0) = f^i(x_0) = f^k(x_0),$$

so $f^k(x_0)$ is a periodic point and x_0 is an eventually periodic point. □

Theorem 1.12 (Group homomorphism criterion for bijective evolutions). *Let $f : X \to X$ be a bijection. Then the map*

$$\varphi : \mathbb{Z} \to \mathrm{Aut}(X), \qquad \varphi(n) = f^n$$

is a group homomorphism. Moreover

$$\mathrm{ord}(f) < \infty \iff \ker \varphi \neq \{0\}.$$

If $\mathrm{ord}(f) = m < \infty$, then

$$\ker \varphi = m\mathbb{Z}, \qquad \langle f \rangle \cong \mathbb{Z}/m\mathbb{Z}.$$

Proof. By the index law for iteration (Property 1.2),

$$f^{m+n} = f^m \circ f^n,$$

so φ is a homomorphism. If $\mathrm{ord}(f) = m < \infty$, then

$$\varphi(n) = \mathrm{id}_X \iff f^n = \mathrm{id}_X \iff m \mid n,$$

so $\ker\varphi = m\mathbb{Z}$. Conversely, if $\ker\varphi \neq \{0\}$, take a non-zero $n \in \ker\varphi$; then $f^n = \mathrm{id}_X$, so $\mathrm{ord}(f) < \infty$. The fundamental theorem of homomorphisms (Theorem 1.1) gives

$$\langle f\rangle \cong \mathbb{Z}/\ker\varphi = \mathbb{Z}/m\mathbb{Z}.$$

☐

Corollary 1.8 (Periods on a finite set). *Let X be a finite set and $f : X \to X$.*

(i) *If f is a bijection, then f is a pointwise periodic system, and*

$$\mathrm{ord}(f) = \mathrm{lcm}(\text{the cycle lengths}), \qquad \mathrm{ord}(f) \mid |X|!.$$

(ii) *If f is not a bijection, then every point is an eventually periodic point, but not necessarily a periodic point.*

Proof. (i) A bijection on a finite set decomposes into finitely many cycles, and every point lies on some cycle, so the system is pointwise periodic. Uniform periodicity follows from Theorem 1.5. Also f is a permutation of X, hence belongs to the symmetric group $S_{|X|}$, and its order divides $|X|!$.

(ii) An infinite forward orbit in a finite set must contain a repetition, so by Proposition 1.17 every point is eventually periodic. When f is not a bijection there may be branching, so a point need not be periodic; for example

$$0 \mapsto 1 \mapsto 2 \mapsto 1.$$

☐

Example 1.9 (A bijection on a finite set). Let $X = \{1, 2, 3, 4, 5\}$, and let f consist of the cycles

$$1 \mapsto 2 \mapsto 3 \mapsto 1, \qquad 4 \mapsto 5 \mapsto 4.$$

Then the least periods are $3, 3, 3, 2, 2$ respectively, and the uniform period is

$$\mathrm{ord}(f) = \mathrm{lcm}(3, 2) = 6.$$

Also $6 \mid 5! = 120$.

Remark 1.25 (Function period versus point period). Two concepts must be distinguished:

(i) **period at a point:** $f^m(x) = x$, which studies the iteration of the evolution map itself;

(ii) **period of a function:** $f(g \cdot x) = f(x)$, which studies the invariance of the function f under a transformation group G.

The two are similar in form but different in object: the former is the recurrence of orbits in a dynamical system, the latter is a symmetry of a function space. For the definition of the period of a function see Definition 1.25.

Theorem 1.13 (Master criterion for period types). *Let (X, f) be a discrete dynamical system. Each of the following items is a summary of results already proved above.*

(A) **General map, period at a point**: by Definition 1.14,

$$x \text{ is a periodic point} \iff \exists m \geq 1, \quad f^m(x) = x.$$

(B) **General map, eventual periodicity**: by Proposition 1.17,

$$x \text{ is an eventually periodic point} \iff \mathcal{O}^+(x; f) \text{ is finite.}$$

(C) **Bijection, period at a point**: by Theorem 1.6,

$$x \text{ is a periodic point} \iff \mathcal{O}(x; f) \text{ is finite} \iff \mathsf{Stab}_{\langle f \rangle}(x) \text{ has finite index.}$$

(D) **Bijection, uniform periodicity**: by Theorem 1.5,

$$\mathsf{ord}(f) < \infty \iff \exists m, \quad f^m = \mathsf{id}_X \iff f \text{ is pointwise periodic and the least common mul}$$

(E) **Specific transformations**: by Theorem 1.10 and Proposition 1.16,

$$\mathsf{ord}(R_\theta) < \infty \iff \frac{\theta}{2\pi} \in \mathbb{Q},$$

$$T_a^n = \mathsf{id} \iff a = 0, \qquad S_\lambda^n = \mathsf{id} \iff \lambda = 1,$$

and the order of the translation $x \mapsto x + a \pmod{2\pi}$ on a modular space is finite if and only if $a/(2\pi) \in \mathbb{Q}$.

Corollary 1.9 (Logical diagram of period, symmetry and conservation). By Corollary 1.4, *in the algebraic framework of this chapter:*

(i) *finite order* $\Rightarrow$ *uniform periodicity* $\Rightarrow$ *pointwise periodicity* $\iff$ *pointwise closed orbits;*

(ii) *periodicity* $\Rightarrow$ *symmetry, but periodicity* $\not\Rightarrow$ *non-trivial conservation;*

(iii) *symmetry* $\Rightarrow$ *conservation (requiring finitely many orbits), but conservation* $\not\Rightarrow$ *symmetry;*

(iv) *the conjunction of symmetry and conservation still* $\not\Rightarrow$ *periodicity.*

Remark 1.26 (Key boundaries). In the criteria for period types the following boundaries must be kept firmly in mind:

(i) **pointwise periodicity** $\not\Rightarrow$ **uniform periodicity**: for a counterexample see Lemma 1.6.

(ii) **for a non-bijection, eventual periodicity** $\not\Rightarrow$ **periodicity**: for example $0 \mapsto 1 \mapsto 2 \mapsto 1$.

(iii) **for a bijection, eventual periodicity** $\iff$ **periodicity**: by Lemma 1.4.

(iv) **periodicity** $\not\Rightarrow$ **non-trivial conservation**: by Theorem 1.7.

(v) **symmetry + conservation** $\not\Rightarrow$ **periodicity**: by Corollary 1.4 (iv).

(vi) **the period of a function** $\neq$ **the period at a point**: see Remark 1.25.

Remark 1.27 (Order of application of the criteria). In practice, when deciding whether a system has a period, the following order is recommended:

1. first decide whether f is a bijection;
2. if it is a bijection, check whether ord$(f) < \infty$, that is, whether there exists a uniform m with $f^m =$ id;
3. if ord$(f) < \infty$, then all orbits are closed and the orbit lengths divide m;
4. if ord$(f) = \infty$, then check point by point whether there are periodic points;
5. if only the period of a function is of interest, use the definition of Group$_{\text{period},G}(f)$, and do not confuse it with the period at a point.

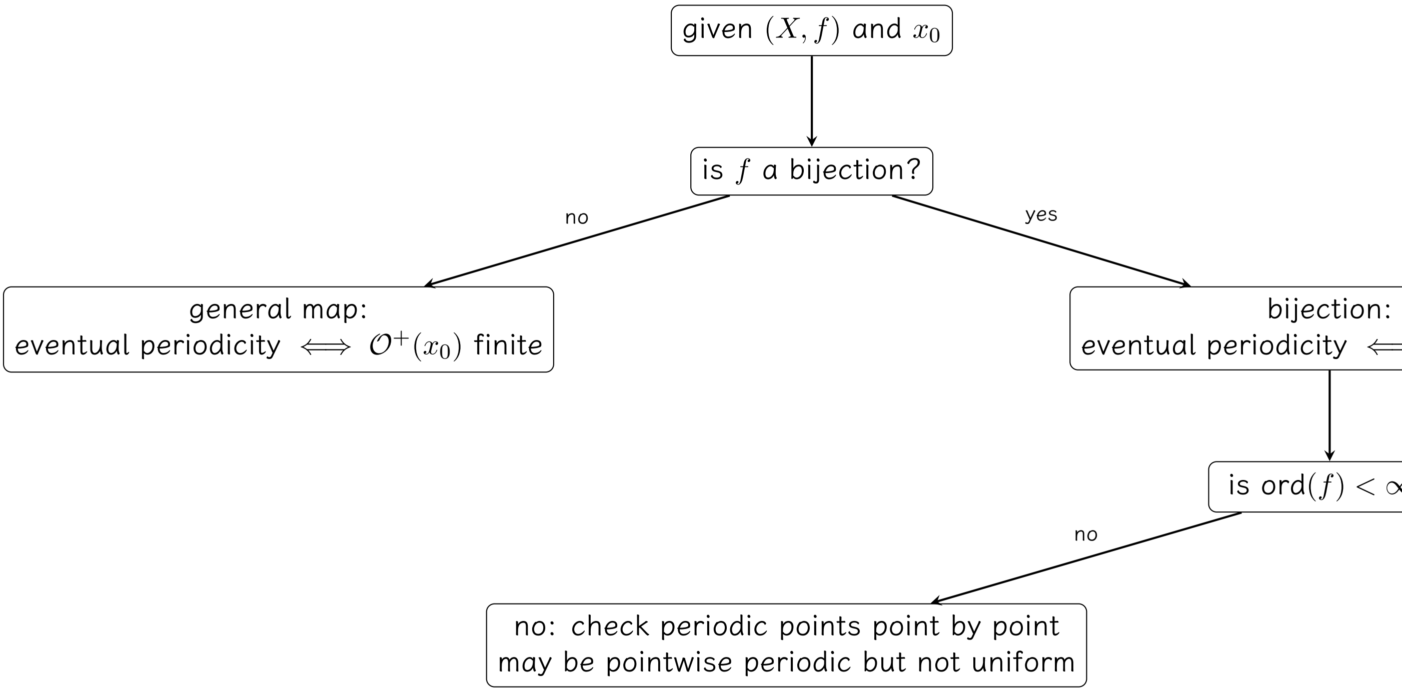

given (X, f) and x_0
is f a bijection?
no
yes
general map:
eventual periodicity $\iff$ $\mathcal{O}^+(x_0)$ finite
bijection:
eventual periodicity
is ord$(f) <$
no
no: check periodic points point by point
may be pointwise periodic but not uniform

Figure 1.13: Flow chart of the master criterion for period types: first decide bijectivity, then decide the uniform order; for a bijection eventual periodicity is equivalent to periodicity, but pointwise periodicity does not automatically upgrade to uniform periodicity.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Foundations of group theory	groups, subgroups, homomorphisms, quotient groups (Definition 1.1–1.5), the fundamental theorem of homomorphisms (Theorem 1.1)
Iteration	iteration and discrete dynamical systems (Definition 1.12), the index law (Property 1.2), orbits (Definition 1.13), periodic points (Definition 1.14)
Invariant/conservation	invariants (Definition 1.15), orbit characterisation (Proposition 1.8), triviality (Remark 1.7), the conservation group (Definition 1.16)
Symmetry	definition via the centraliser (Definition 1.17), preservation of orbits and periods (Proposition 1.10), symmetry $\Rightarrow$ conservation (Theorem 1.4), conservation $\not\Rightarrow$ symmetry (Proposition 1.11)
Finite-order actions	order of a map (Definition 1.19), pointwise vs uniform (Lemma 1.6), orbit characterisation (Theorem 1.5)
Closed orbits	criterion for closure (Theorem 1.6), cyclic structure (Corollary 1.3)
Period	strict periodicity (Definition 1.21), periodicity $\not\Rightarrow$ conservation (Theorem 1.7), the combinatorial criterion (Theorem 1.11)
Function characteristics	translation-, scaling- and rotation-symmetric functions (Definition 1.22–1.24), antiperiodicity (Proposition 1.13), logarithmic translation (Proposition 1.14), the order of roots of unity (Proposition 1.15)
The two lines	the hyperoperation hierarchy (Definition 1.28), iterative generation of transformations (Theorem 1.9), order structure (Theorem 1.10), the carrier space (Proposition 1.16)

Exercises

Exercise 1 Prove that when f is a bijection, Symmetry(f) is the smallest centraliser in Aut(X) that contains $\langle f\rangle$; give an example showing that Symmetry(f) can be strictly larger than $\langle f\rangle$.

Exercise 2 Construct a system (X, f, Q) such that Symmetry(f, Q) is the cyclic group C_2 of order two, and every orbit of f has length exactly 2.

Exercise 3 Prove that the condition 'finitely many orbits' in Theorem 1.4 cannot be dropped: construct $G \leq \mathrm{Symmetry}(f)$ such that Q_G is identically ∞, and give a conserved quantity that is more suitable in this case.

Exercise 4 In the counterexample of Lemma 1.6, determine on which components f^{12} is the identity map and on which components it is not.

Exercise 5 Prove that on a finite set X, pointwise closure and uniform closure are equivalent, and that $\mathrm{ord}(f) \mid |X|!$.

Exercise 6 Verify that $Q(x) = \sin(2\pi x)$ is a conserved quantity of $f(x) = x+1$, and find its conservation group G_Q.

Exercise 7 Prove that in Proposition 1.16 (ii), when $a/(2\pi) = p/q$ (in lowest terms) the order of the translation on the circle is exactly q.

Exercise 8 Explain why the proof of Theorem 1.7 requires the action of $\langle f \rangle$ to be transitive. Construct a uniformly periodic system such that a non-trivial conserved quantity exists and every orbit is closed.

Exercise 9 Using the hyperoperation hierarchy (Definition 1.28), write down the recursive definition of the fourth-level operation (iterated exponentiation, that is, power towers), and discuss whether it satisfies the associative law.

Exercise 10 Prove Proposition 1.14: if $f(qx) = f(x)$ and $q = e^\alpha$ $(\alpha \neq 0)$, then $g(t) = f(e^t)$ satisfies $g(t+\alpha) = g(t)$; and conversely.

Exercise 11 Let $a = 6$ and $q = 4$. Verify that the order of $e^{2q\pi i/a} = e^{4\pi i/3}$ is $3 = a/\gcd(a,q)$; write down the cyclic group it generates.

Remark 1.28 (Summary of this chapter). Taking iteration as its starting point, this chapter has established the complete chain of logic

$$\text{iteration} \to \text{invariant/conservation} \to \text{symmetry} \to \text{finite-order action} \to \text{closed orbit} \to \text{period},$$

but the core criterion chain is

$$\text{finite-order action} \Longrightarrow \text{(uniform) closed orbit} \Longrightarrow \text{strict period}.$$

The logical relations between the various links are as follows:

- **The foundations of group theory** provide a unified language: groups, homomorphisms, quotient groups, cyclic groups, group actions, orbits and stabilisers.
- **Iteration** is the starting point of everything: the operations themselves are generated by iteration, and a group is the structure that iteration self-organises into in the invertible case.
- **Conservation** is the constancy of a function along orbits; 'there exists a conserved quantity' is always true and therefore has no discriminating power, whereas a non-trivial conserved quantity describes the stratification between orbits.

- **Symmetry** is a transformation that commutes with the evolution; symmetry $\Rightarrow$ conservation (the orbit-counting construction), conservation $\not\Rightarrow$ symmetry (the strictly injective counterexample $Q = x^3 + x$).
- **A finite-order action** is, in the framework of this chapter, the direct sufficient and necessary condition for obtaining closed orbits and uniform periodicity; upgrading pointwise closure to uniform closure requires uniformly bounded periods.
- **A closed orbit** $\iff$ a periodic point; the orbit length equals the least period, and the dynamics on a closed orbit is isomorphic to a translation of C_{m_0}.
- **Periodicity** carries its own symmetry ($\langle f \rangle \cong C_m$) but does not require conservation; the determination of periodicity is given by the combinatorial criterion.
- **Function characteristics**: the three elementary transformations—translation, scaling and rotation—are characterised respectively by the functional equations

  $$f(x+a) = f(x), \qquad f(qx) = f(x), \qquad f(e^{i\theta}z) = f(z)$$

  which provide a function-level motivation for the extension of number systems in the next chapter.
- **The two lines**: there is a correspondence at the level of parameters between the parametric line of transformations and the operational line; $(\mathbb{R}, +)$ and $(\mathbb{R}^+, \times)$ have no non-trivial torsion elements, whereas $U(1)$ contains non-trivial elements of finite order, so a rotational action can produce a non-trivial strict period; and a translation on a modular space may also have a period.

The algebraic framework established in this chapter is self-consistent.

Chapter 2

Number Systems

Remark 2.1 (Positioning of this chapter). Number systems are the level-by-level realisation of the algebraic closure required for period determination. This chapter does not enumerate number systems from the standpoint of an 'encyclopaedia of mathematical structures'; rather, it starts from the **requirements of period determination** and answers one core question: **for each class of periodic orbit to become a mathematically decidable object, what kind of number system is required as its algebraic foundation?**

The main line of logic is:

period determination requirement $\Longrightarrow$ required algebraic closure $\Longrightarrow$ number-system extens

Each step of extension is driven by an **undecidable period problem** in the preceding number system. Each section answers three questions: which period is being determined? What was missing from the original number system? What criterion is obtained after the extension?

Core insight: the hierarchical progression of transformations. Repeated addition (translation) produces multiplication (scaling); repeated multiplication (scaling) produces powers (rotation). This hierarchical progression is the intrinsic driving force of number-system extension.

The function-equation main line. This chapter starts not only from the 'requirements of period determination' but also from the 'solvability of function equations'. The three function equations given in the preceding chapter—

$$\text{Translation:} f(x+a)=f(x), \qquad \text{Scaling:} f(qx)=f(x), \qquad \text{Rotation:} f(e^{i\theta}z)=f(z)$$

as well as the antiperiod equation $f(x+a)+f(x)=0$ (equivalent to $f(x+a)=-f(x)$)— their solvability in each level of number system, and the structure of their groups with periodic orbits, constitute the other main line of number-system extension. Each level of number-system extension makes exactly one class of function equation pass from 'unsolvable' or 'non-closed' to 'solvable', with its group with periodic orbits becoming a group. This line runs in parallel with the main line of 'requirements of period determination', and together they drive the extension of the number systems from $\mathbb{N}$ to $\mathbb{H}$.

Seven-theme guide to this chapter: translation—the translation group $(\mathbb{Z},+)$ of the integers $\mathbb{Z}$ and its additive symmetry (centre of symmetry 0); scaling—the proportional direction of the rationals $\mathbb{Q}$ and the multiplicative symmetry σ_1 (centre of symmetry 1); rotation—the unit circle $U(1)$ of the complex numbers $\mathbb{C}$ and the $SU(2)$ three-dimensional rotation of the quaternions $\mathbb{H}$; conservation—the conserved quantities corresponding to the operations closed by each successive extension of the number systems (addition, multiplication, completeness, algebraic closure, non-commutative multiplication), namely momentum, phase volume, angular momentum and three-dimensional angular momentum; symmetry—the centres of symmetry 0, 1, i emerging level by level as inverse operations are introduced; closure—the closure of the operations of each number system and the closure criterion for finite rotation orbits; period—the period types corresponding to the number systems (discrete translation period, finite rotation period, quasi-period, double period, three-dimensional rotation period).

2.1 The natural numbers $\mathbb{N}$: counting and repetition, but not constituting a period

2.1.1 Determination requirement

The numbering of 'the k-th element' in a periodic sequence, and the counting of 'repeating n times'.

2.1.2 The axiomatisation of the natural numbers

Axiom 2.1 (Peano arithmetic)**.** *The number system of the natural numbers $\mathbb{N}$ is characterised by the following five axioms:*

(1) $0 \in \mathbb{N}$.

(2) Every natural number n has a unique successor $S(n) \in \mathbb{N}$.

(3) 0 is not the successor of any natural number.

(4) The successor function is injective: if $S(n) = S(m)$, then $n = m$.

(5) **The induction axiom**: if a subset $A \subseteq \mathbb{N}$ satisfies $0 \in A$ and $S(n) \in A$ for every $n \in A$, then $A = \mathbb{N}$.

Definition 2.1 (Operations on the natural numbers)**.** *The recursive definitions of the three basic operations on the natural numbers:*

- **Addition** (translation): $n + 0 := n$, $n + S(m) := S(n + m)$. Closed.
- **Multiplication** (repetition of scaling): $n \times 0 := 0$, $n \times S(m) := (n \times m) + n$. Closed.
- **Powers** (repetition of rotation): $b^0 := 1$ $(b \neq 0)$, $b^{S(n)} := b^n \times b$. Closed.

Remark 2.2 (Manifestation of the transformation hierarchy in the natural numbers). The hierarchical progression of transformations is already contained in the natural numbers:

- **First level: addition** (translation). $n + m$ is the composite of m translations.
- **Second level: multiplication** (scaling). $n \times m$ is the repetition of m additions (translations)—repeated translation produces scaling.
- **Third level: powers** (rotation). b^n is the repetition of n multiplications (scalings)—repeated scaling produces rotation.

However, all three of these operations in the natural numbers have only a forward direction and no reverse direction; hence they do not form a group structure and do not directly provide an algebraic tool for period determination.

2.1.3 Absence of criteria

Proposition 2.1 (Absence of inverse operations in the natural numbers). In $\mathbb{N}$:

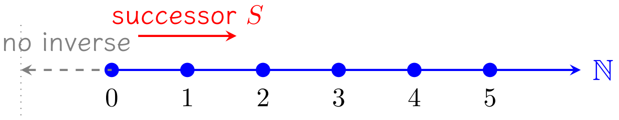
(i) The subtraction $a - b$ has a solution if and only if $b \leq a$.

(ii) The division $a \div b$ has a solution if and only if $b \mid a$ and $b \neq 0$.

(iii) The root $\sqrt[e]{b}$ has a solution if and only if b is the e-th power of some natural number.

Therefore there is no complete operational duality in the natural numbers; they cannot form a directional structure and do not possess a group structure.

Proof. (i) If $b \leq a$, then there exists $c \in \mathbb{N}$ with $a = b + c$, so $x = c$ is a solution. If $b > a$, then by the order properties of the natural numbers there is no $x \in \mathbb{N}$ with $b + x = a$. (ii) If $b \mid a$, then there exists $c \in \mathbb{N}$ with $a = b \times c$, so $x = c$ is a solution. Otherwise no integer quotient exists. (iii) If $b = c^e$ for some $c \in \mathbb{N}$, then $x = c$ is a solution. Otherwise there is no natural-number solution. ☐



$\mathbb{N}$ has only forward advance and no additive inverse; translation orbits cannot be extended in both directions, so closure cannot be determined.

Figure 2.1: The ray structure of the natural numbers $\mathbb{N}$. The successor map S provides only one-directional advance, with no inverse map. Translation orbits cannot be traced backwards, and closure determination is not closed in $\mathbb{N}$ (Proposition 2.1).

Corollary 2.1 (The relationship between the natural numbers and period determination). *The natural numbers have no centre of symmetry and no directional duality, and do not themselves form a group; hence they do not directly provide an algebraic tool for period determination. In the spectrum of periodic mathematics, $\mathbb{N}$ provides only the function of counting—the numbering of 'the k-th element' in a periodic sequence. The determination of discrete translation periods requires at least the group structure of $\mathbb{Z}$ (deduced from Proposition 2.1).*

Proof. By Proposition 2.1, subtraction, division and root extraction are all non-closed in $\mathbb{N}$, so there is no additive inverse and no multiplicative inverse. Without inverses there is no 'reverse' operation, and therefore neither a centre of symmetry (symmetry requires $x \mapsto -x$ or $x \mapsto x^{-1}$ to be defined) nor a directional duality (a direction requires forward and reverse to occur in pairs). The determination of discrete translation periods requires that orbits can be extended in both directions and their closure tested, which requires at least additive inverses, that is, the group structure of $\mathbb{Z}$. ☐

2.1.4 Function-theoretic features: solvability of the translation equation and the absence of a group with periodic orbits

Proposition 2.2 (Solution of the translation equation on the natural numbers). Let $f : \mathbb{N} \to Y$, $a \in \mathbb{N}$, $a \neq 0$. If

$$f(x+a) = f(x), \qquad \forall x \in \mathbb{N},$$

then f is constant on each residue class modulo a; but the period set

$$\mathsf{Group}_{\mathsf{period},\mathbb{N}}(f) = \{a \in \mathbb{N} : f(x+a) = f(x),\ \forall x \in \mathbb{N}\}$$

is in general **not a subgroup**, because no additive inverse exists in $\mathbb{N}$: $a \in \mathsf{Group}_{\mathsf{period},\mathbb{N}}(f)$ does not imply that $-a$ exists.

Proof. Iterating $f(x+a) = f(x)$ repeatedly gives $f(x+na) = f(x)$ for every $n \in \mathbb{N}$, so f is constant on the residue classes modulo a. But closure of the period set requires inverses: if $a, b \in \mathsf{Group}_{\mathsf{period},\mathbb{N}}(f)$, then $a + b \in \mathsf{Group}_{\mathsf{period},\mathbb{N}}(f)$ (since $f(x+a+b) = f(x+a) = f(x)$), but $a-b$ need not lie in $\mathbb{N}$, so the subgroup criterion is not satisfied. ☐

Proposition 2.3 (Failure of the antiperiod equation on the natural numbers). Let $f : \mathbb{N} \to Y$, $a \in \mathbb{N}$, $a \neq 0$. If

$$f(x+a) = -f(x), \qquad \forall x \in \mathbb{N},$$

then iteration gives $f(x+2a) = f(x)$, so $2a \in \mathsf{Group}_{\mathsf{period},\mathbb{N}}(f)$. But the equation itself cannot be extended backwards: $f(x-a)$ is undefined when $x < a$.

Proof. From $f(x+a) = -f(x)$ we obtain $f(x+2a) = f(x)$. Backward extension requires $x - a \in \mathbb{N}$, that is, $x \geq a$, so it fails at $x < a$. ☐

Remark 2.3 (Status of the scaling and rotation equations on the natural numbers). The scaling equation $f(qx) = f(x)$ is meaningful on $\mathbb{N}$ only when q is an integer and q divides every relevant x, and q^{-1} does not exist, so two-directional iteration is impossible. The rotation equation $f(e^{i\theta}z) = f(z)$ has no algebraic carrier at all in $\mathbb{N}$, because $e^{i\theta}$ does not belong to $\mathbb{N}$. Thus the natural numbers carry only the most primitive form of the translation equation, and the group with periodic orbits is not closed.

2.2 The integers $\mathbb{Z}$: the basis for determining discrete translation periods

2.2.1 Determination requirement

Given the translation orbit

$$\mathcal{O}(x_0; T) = \{x_0 + nT \mid n \in \mathbb{Z}\},$$

determining whether it is regular and whether it can close presupposes that the orbit must be extendable in both directions. If there is only forward advance, it can never 'return to the starting point', and closure determination is logically impossible.

2.2.2 Algebraic requirement

Additive inverses, that is, subtraction. This is the minimal algebraic closure that makes translation a group action.

2.2.3 Construction of the integers

Definition 2.2 (Construction of the integers). *Define an equivalence relation on* $\mathbb{N} \times \mathbb{N}$ *by*

$$(a,b) \sim (c,d) \iff a+d = b+c.$$

The set of integers is $\mathbb{Z} := (\mathbb{N} \times \mathbb{N})/\sim$. *The equivalence class* $[(a,b)]$ *corresponds to the intuitive '*$a-b$*'.*

Definitions of the operations:

- *Addition:* $[(a,b)] + [(c,d)] := [(a+c, b+d)]$.
- *Zero element:* $0_{\mathbb{Z}} := [(0,0)]$.
- *Additive inverse:* $-[(a,b)] := [(b,a)]$.
- *Multiplication:* $[(a,b)] \times [(c,d)] := [(ac+bd, ad+bc)]$.

Proposition 2.4 (The integer operations are well defined). *The above definitions of addition and multiplication do not depend on the choice of representative, and* $(\mathbb{Z}, +, \times)$ *forms a commutative ring.*

Proof. Let $(a,b) \sim (a',b')$ and $(c,d) \sim (c',d')$, that is,

$$a + b' = b + a', \qquad c + d' = d + c'. \tag{*}$$

Well-definedness of addition: we must prove $(a+c, b+d) \sim (a'+c', b'+d')$, that is,

$$(a+c) + (b'+d') = (b+d) + (a'+c').$$

Adding the two equations in $(*)$ and using associativity and commutativity gives:

$$(a+c) + (b'+d') = (a+b') + (c+d') = (b+a') + (d+c') = (b+d) + (a'+c').$$

Well-definedness of multiplication: we must prove $(ac+bd, ad+bc) \sim (a'c'+b'd', a'd'+b'c')$, that is,

$$ac + bd + a'd' + b'c' = ad + bc + a'c' + b'd'. \tag{**}$$

From $(*)$ we obtain $a+b' = b+a'$; multiplying the two sides by c and by d respectively gives

$$ac + b'c = bc + a'c, \qquad bd + a'd = ad + b'd.$$

Hence

$$ac + bd + a'd + b'c = (ac + b'c) + (bd + a'd) = (bc + a'c) + (ad + b'd) = ad + bc + a'c + b'd,$$

that is, $(ac+bd, ad+bc) \sim (a'c+b'd, a'd+b'c)$, so multiplication is well defined in the first factor. Similarly, from $(c,d) \sim (c',d')$ we obtain well-definedness of multiplication in the second factor. Composing the two substitutions gives $(ac+bd, ad+bc) \sim (a'c'+b'd', a'd'+b'c')$, so multiplication is well defined.

Ring axioms: the associativity and commutativity of addition, the zero element $[(0,0)]$ and the inverse $-[(a,b)] = [(b,a)]$ are inherited directly from the corresponding properties of the natural numbers through representatives; the associativity of multiplication and the distributive law follow likewise, verified by expanding term by term through representatives, and all of them reduce to the laws of operation for the natural numbers. ☐

2.2.4 Operational duality and directional structure in the integers

Proposition 2.5 (Operational duality in the integers). *In $\mathbb{Z}$:*

(i) **Addition and subtraction** *form a complete duality:* $a + x = b \iff x = b - a$. *Forward direction:* $x \mapsto x + k$ *(translation to the right); reverse direction:* $x \mapsto x - k$ *(translation to the left).*

(ii) **Multiplication** *is closed, but* **division** *is still incomplete: when* $a \neq 0$, *the equation* $a \times x = b$ *has an integer solution if and only if* $a \mid b$*; when* $a = 0$, *it has a solution if and only if* $b = 0$ *(in which case every* x *is a solution).*

(iii) **Powers** *are closed (integer base, natural-number exponent), but* **roots** *are still incomplete.*

Proof. (i) If $x = b-a = b+(-a)$, then $a+x = a+b+(-a) = b$; conversely, if $a+x = b$, adding $-a$ to both sides gives $x = b+(-a) = b-a$. Hence $a+x = b \iff x = b-a$, and addition and subtraction form a complete duality. The directional duality is given by $x \mapsto x+k$ and its inverse map $x \mapsto x-k$: $(x+k)-k = x = (x-k)+k$. (ii) Let $a \neq 0$. If $a \mid b$, then there exists $c \in \mathbb{Z}$ with $b = ac$, and $x = c$ is a solution; conversely, if $ax = b$ has an integer solution, then $a \mid b$. Let $a = 0$; the equation becomes $0 = b$: when $b = 0$ every x is a solution, and when $b \neq 0$ there is no solution. In summary, division is closed only in the case of exact divisibility, and is not closed in general. (iii) For an integer b and a non-negative integer n, b^n is obtained by multiplying b by itself repeatedly and is still an integer, so powers are closed. For root extraction: the equation $x^e = b$ has an integer solution if and only if b is the e-th power of some integer (and when e is even one requires $b \geq 0$); it is not closed in general, for example $x^2 = 2$ has no integer solution. ☐

Proposition 2.6 (Symmetry, directionality and periodicity of the integers). *The operational structure on the integers gives rise to the following symmetries and periodicities:*

(i) **Centre of symmetry**: 0 *(the additive identity).*

(ii) **Type of symmetry**: *additive reflection symmetry.* $\sigma_{0,+}(x) = -x$ *is an involutive automorphism with fixed point* 0, *generating the group* C_2.

(iii) **Directional structure**: *the translation direction. Direction group* $\langle 1 \rangle = \mathbb{Z}$. *Forward:* $x \mapsto x+1$*; reverse:* $x \mapsto x-1$.

(iv) **Type of period**: translation quasi-period. The orbit $\{a_0 + k \mid k \in \mathbb{Z}\}$ extends isometrically on $\mathbb{Z}$. The translation-invariant functions on this orbit are the constant functions.

Proof. (i) $0 + x = x$ holds for all $x \in \mathbb{Z}$, and for every x there exists a unique $-x$ with $x + (-x) = 0$. (ii) $\sigma_{0,+}(\sigma_{0,+}(x)) = -(-x) = x$, so it is an involution. The fixed-point condition is $-x = x \iff 2x = 0 \iff x = 0$ (in a torsion-free additive group). (iii) Every integer can be written uniquely as $n \cdot 1$ ($n \in \mathbb{Z}$), with the forward direction corresponding to $n > 0$ and the reverse direction to $n < 0$. (iv) For every $k \in \mathbb{Z}$, $T_k(x) = x + k$ satisfies $T_k \circ T_{-k} = \text{id}$. The isometry of the orbit is guaranteed by $|(x+k) - x| = |k|$. The translation group acts transitively on $\mathbb{Z}$, so its invariants must be constant functions $P(x) = \text{const}$. ☐

2.2.5 Function-theoretic features: invertibilisation of the translation equation and the group with periodic orbits

Theorem 2.1 (The group with periodic orbits of the translation equation on the integers). *Let $f : \mathbb{Z} \to Y$, $a \in \mathbb{Z}$, $a \neq 0$. If*

$$f(x + a) = f(x), \qquad \forall x \in \mathbb{Z},$$

then the group with periodic orbits

$$\mathrm{Group}_{\mathrm{period},\mathbb{Z}}(f) = \{T \in \mathbb{Z} : f(x + T) = f(x),\ \forall x \in \mathbb{Z}\}$$

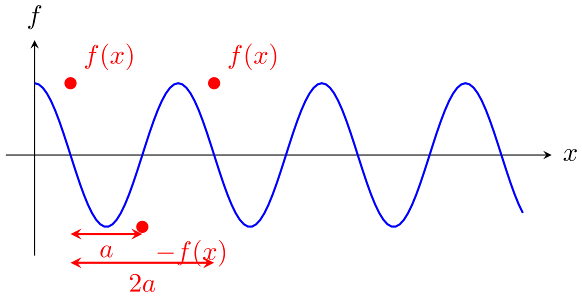
is a subgroup of $\mathbb{Z}$, and $a\mathbb{Z} \leq \mathrm{Group}_{\mathrm{period},\mathbb{Z}}(f)$.

Proof. Iterating $f(x + a) = f(x)$ in both directions gives $f(x + na) = f(x)$ for every $n \in \mathbb{Z}$, so $a\mathbb{Z} \leq \mathrm{Group}_{\mathrm{period},\mathbb{Z}}(f)$. The period set is closed under addition and under taking inverses ($T \in \mathrm{Group}_{\mathrm{period},\mathbb{Z}}(f) \Rightarrow -T \in \mathrm{Group}_{\mathrm{period},\mathbb{Z}}(f)$ is guaranteed by two-directional extension), hence it is a subgroup. ☐

Corollary 2.2 (Period doubling for the antiperiod equation). *If $f(x + a) = -f(x)$ holds for all $x \in \mathbb{Z}$, then $f(x + 2a) = f(x)$, that is, $2a \in \mathrm{Group}_{\mathrm{period},\mathbb{Z}}(f)$, and $2a\mathbb{Z} \leq \mathrm{Group}_{\mathrm{period},\mathbb{Z}}(f)$.*

Proof. From $f(x + a) = -f(x)$ we obtain $f(x + 2a) = f(x)$. Here $f(x - a)$ is defined for all $x \in \mathbb{Z}$, and the group with periodic orbits is complete. ☐

Remark 2.4 (The integers make the translation equation invertible). In the natural numbers the translation equation can be iterated in one direction only, and the period set is not closed; in the integers additive inverses exist, the translation equation can be iterated in both directions, and the period set becomes a subgroup. This is the first instance of number-system extension making a function equation pass from 'non-invertible' to 'invertible'.



The antiperiod $f(x+a) = -f(x)$ implies period doubling $f(x+2a) = f(x)$; alternating in sign, closing after two steps.

Figure 2.2: The geometric picture of the antiperiod equation $f(x+a) = -f(x)$. The function has opposite signs on two adjacent a-intervals, and after two steps returns to its initial value—the period doubles to $2a$ (Corollary 2.2). This is the simplest non-trivial period-doubling structure.

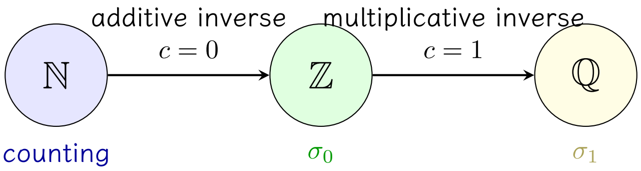

additive inverse
$c = 0$
multiplicative inverse
$c = 1$
$\mathbb{N}$
$\mathbb{Z}$
$\mathbb{Q}$
counting
σ_0
σ_1

Figure 2.3: The level-by-level emergence of the centres of symmetry 0, 1 in number-system extension (local diagram of Section 2.6). $\mathbb{N} \to \mathbb{Z}$ introduces additive inverses (centre of symmetry 0); $\mathbb{Z} \to \mathbb{Q}$ introduces multiplicative inverses (centre of symmetry 1). The subsequent extensions $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$ (introducing completeness, the imaginary unit i and non-commutative multiplication respectively) are shown in the chapter summary diagram 2.11.

2.2.6 Criterion theorem

Lemma 2.1 (Order of an element in a finite cyclic group)**.** *Let $m \geq 1$ and $T \in \mathbb{Z}$. In the quotient group $\mathbb{Z}/m\mathbb{Z}$, the cyclic subgroup $\langle [T] \rangle$ generated by the element $[T]$ has order $m/\gcd(T,m)$. In particular, $\gcd(T,m) = 1$ if and only if $[T]$ generates the whole of $\mathbb{Z}/m\mathbb{Z}$.*

Proof. Let $d = \gcd(T,m)$. The order of $[T]$ is the least positive integer n for which $nT \equiv 0 \pmod{m}$. The condition $m \mid nT$ is equivalent to $\frac{m}{d} \mid n \cdot \frac{T}{d}$. Since $\gcd\left(\frac{m}{d}, \frac{T}{d}\right) = 1$, the coprimality property gives $\frac{m}{d} \mid n$, and the least positive integer solution is $n = \frac{m}{d} = \frac{m}{\gcd(T,m)}$. When $\gcd(T,m) = 1$ we have $d = 1$, so $[T]$ has order $m = |\mathbb{Z}/m\mathbb{Z}|$ and hence $[T]$ generates the whole group; conversely, if $[T]$ generates the whole group, then its order is m, that is, $m = m/d$, giving $d = 1$. □

Theorem 2.2 (Criterion for integer translation periods)**.** *Let $x_0 \in \mathbb{Z}$ and let the translation step be $T \in \mathbb{Z}$. Then:*

(i) *If $T = 0$, the orbit $\mathcal{O}(x_0;T)$ degenerates to the single point $\{x_0\}$, with period 1 (the trivial period).*

(ii) *If $T \neq 0$, the orbit $\mathcal{O}(x_0;T)$ is a two-directionally infinite isometric sequence of points, non-closed but regular. In any finite quotient $\mathbb{Z}/m\mathbb{Z}$ (that is, C_m), the orbit necessarily closes, with period exactly $m/\gcd(T,m)$; in particular, the orbit degenerates to a single point if and only if $T \equiv 0 \pmod{m}$.*

Proof. (i) $x_0 + n \cdot 0 = x_0$ for all n, so the orbit contains only one point. (ii) If $T \neq 0$, then $|(x_0 + nT) - x_0| = |nT|$ increases without bound as $|n|$ grows, so the orbit does not close in $\mathbb{Z}$. In the quotient group $\mathbb{Z}/m\mathbb{Z}$ the orbit is the coset $[x_0] + \langle [T] \rangle$; by Lemma 2.1, $\mathrm{ord}([T]) = m/\gcd(T,m)$ is finite, so the orbit closes with period $m/\gcd(T,m)$. In particular, the orbit degenerates to a single point if and only if $[T] = [0]$, that is, $T \equiv 0 \pmod{m}$. □

Corollary 2.3 (Integers and discrete translation periods)**.** *$\mathbb{Z}$ is the minimal group structure in which discrete translation periods can be determined (deduced from Theorem 2.2). Its determining power is reflected in:*

(i) *orbits can be extended in both directions, and this traceability makes closure determination possible;*

(ii) *the quotient structure of the finite cyclic group $C_m \cong \mathbb{Z}/m\mathbb{Z}$ gives finite translation periods an exact algebraic characterisation.*

Proof. (i) The argument for the case $T \neq 0$ in Theorem 2.2 relies on additive inverses in $\mathbb{Z}$: the existence of the negative translation $x \mapsto x - T$ makes orbits traceable, so that closure determination becomes possible. (ii) By Lemma 2.1, the order (period) of an element of $C_m \cong \mathbb{Z}/m\mathbb{Z}$ can be computed exactly as $m/\gcd(T,m)$, so finite translation periods acquire an algebraic characterisation. □

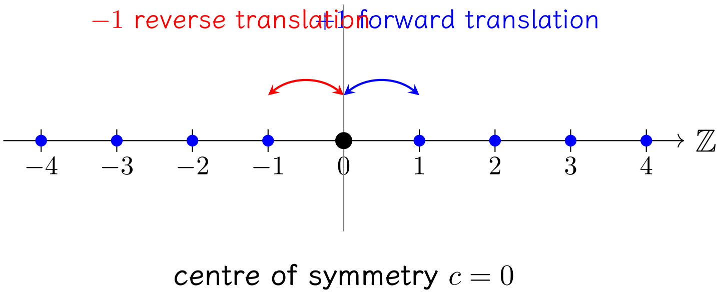


Figure 2.4: The translation direction structure of the integers $\mathbb{Z}$: with 0 as centre of symmetry, $+1$ and -1 form a forward–reverse directional duality. The orbit extends isometrically on the integer axis.

2.3 The rationals $\mathbb{Q}$: the basis for determining finite rotation periods

2.3.1 Determination requirement

When does the rotation orbit

$$\mathcal{O}(z_0;\theta) = \{z_0 e^{in\theta} \mid n \in \mathbb{Z}\}$$

close? The criterion is

$$\frac{\theta}{2\pi} \in \mathbb{Q}.$$

But the very concept of 'commensurability' itself requires the rational numbers. Without $\mathbb{Q}$ we could not even express that 'the rotation angle is commensurable with one third of the circumference'.

2.3.2 Algebraic requirement

Multiplicative inverses, that is, division. This is the minimal algebraic closure that makes proportional relations exactly expressible.

2.3.3 Construction of the rational numbers

Definition 2.3 (Construction of the rational numbers). *Define an equivalence relation on $\mathbb{Z} \times \mathbb{Z}^\times$ (where $\mathbb{Z}^\times = \mathbb{Z} \setminus \{0\}$) by*

$$(a, b) \approx (c, d) \iff a \times d = b \times c.$$

The set of rational numbers is $\mathbb{Q} := (\mathbb{Z} \times \mathbb{Z}^\times)/\approx$. Equivalence classes are written $\frac{a}{b}$ or a/b.

Definitions of the operations:

- *Addition: $\frac{a}{b} + \frac{c}{d} := \frac{ad+bc}{bd}$.*
- *Multiplication: $\frac{a}{b} \times \frac{c}{d} := \frac{ac}{bd}$.*
- *Additive zero element: $\frac{0}{1}$.*

- Multiplicative identity: $\frac{1}{1}$.
- Additive inverse: $-\frac{a}{b} := \frac{-a}{b}$.
- Multiplicative inverse: $(\frac{a}{b})^{-1} := \frac{b}{a}$ $(a \neq 0)$.

Proposition 2.7 (The rational numbers form a field). *$(\mathbb{Q}, +, \times)$ forms a field, that is, it is simultaneously an additive abelian group and a multiplicative abelian group (apart from the zero element), and multiplication satisfies the distributive law over addition.*

Proof. The additive group structure is inherited from the fraction construction of $\mathbb{Z}$. The associativity and commutativity of multiplication are verified directly from integer multiplication. Existence of multiplicative inverses: for any $\frac{a}{b} \neq 0$ (that is, $a \neq 0$), the inverse is $\frac{b}{a} \in \mathbb{Q}$. The distributive law is verified by computation over a common denominator. ☐

2.3.4 The centre of symmetry 1 and the proportional direction

Definition 2.4 (The taking-the-inverse involution σ_1). On the multiplicative group $\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$, define the map

$$\sigma_1 : \mathbb{Q}^\times \to \mathbb{Q}^\times, \quad \sigma_1(x) = x^{-1} = \frac{1}{x}.$$

The map σ_1 is called the involution about the centre of symmetry 1.

Proposition 2.8 (Basic properties of the involution σ_1). *(i) σ_1 is an involution: $\sigma_1 \circ \sigma_1 = \mathrm{id}_{\mathbb{Q}^\times}$.*

(ii) The fixed-point set of σ_1 is $\{1, -1\}$.

(iii) σ_1 is an automorphism of the multiplicative group $(\mathbb{Q}^\times, \times)$: $\sigma_1(x \times y) = \sigma_1(x) \times \sigma_1(y)$.

(iv) σ_1 interchanges the operation of 'enlargement' (multiplication by $q > 1$) with the operation of 'shrinkage' (division by q).

Proof. (i) $\sigma_1(\sigma_1(x)) = 1/(1/x) = x$. (ii) $\sigma_1(x) = x \iff 1/x = x \iff x^2 = 1 \iff x = 1$ or $x = -1$. (iii) $\sigma_1(xy) = 1/(xy) = (1/x)(1/y) = \sigma_1(x)\sigma_1(y)$. (iv) Let $T_q(x) = qx$ $(q > 1$, enlargement); then $\sigma_1 \circ T_q \circ \sigma_1(x) = \sigma_1(q \cdot x^{-1}) = x/q$, that is, the shrinking operation. ☐

Proposition 2.9 (Directional structure and period types of the rational numbers). *Building on the symmetries of the integers, the rational numbers acquire the following additional structure:*

*(i) **New centre of symmetry**: 1 (the multiplicative identity).*

*(ii) **New type of symmetry**: multiplicative inversion symmetry. $\sigma_1(x) = x^{-1}$ is an involutive automorphism of the multiplicative group $\mathbb{Q}^\times$, with fixed points ± 1.*

(iii) **New directional structure**: the proportional direction. Direction group $\langle q\rangle = \{q^k \mid k \in \mathbb{Z}\} \cong \mathbb{Z}$ $(q > 0, q \neq 1)$. Forward: $x \mapsto q \times x$; reverse: $x \mapsto x/q$.

(iv) **New type of period**: proportional quasi-period. Via the logarithmic homomorphism, the orbit $\{a_0 q^k \mid k \in \mathbb{Z}\}$ is equivalent to a translation quasi-period in the logarithmic domain.

Proof. (i)–(ii) The properties of the centre of symmetry 1 and of the multiplicative inversion symmetry σ_1 have already been established by Definition 2.4 and Proposition 2.8. (iii) For $q > 0, q \neq 1$, the map $k \mapsto q^k$ gives a group isomorphism $\mathbb{Z} \to \langle q\rangle$: $q^{k_1+k_2} = q^{k_1}q^{k_2}$ and $q^0 = 1$, and moreover $q^{k_1} = q^{k_2}$ if and only if $k_1 = k_2$ (for $q \neq 1$ the function q^k is strictly monotone), so $\langle q\rangle \cong \mathbb{Z}$. The forward map $x \mapsto qx$ corresponds to $k \mapsto k+1$, and the reverse map $x \mapsto x/q$ corresponds to $k \mapsto k-1$. (iv) Under the logarithmic map ln, the orbit $\{a_0 q^k \mid k \in \mathbb{Z}\}$ becomes $\{\ln a_0 + k \ln q \mid k \in \mathbb{Z}\}$, an isometric translation orbit on $\mathbb{R}$ with step $\ln q$ (see Proposition 2.10 and Figure 2.5); it is therefore the dual, in the proportional direction, of the translation quasi-period—the proportional quasi-period. □

2.3.5 Function-theoretic features: the scaling equation and commensurability

Theorem 2.3 (The group with periodic orbits of the scaling equation on the rationals)**.** Let $f : \mathbb{Q}^+ \to Y$, $q \in \mathbb{Q}^+$, $q \neq 1$. If

$$f(qx) = f(x), \qquad \forall x \in \mathbb{Q}^+,$$

then the group with periodic orbits

$$\mathrm{Group}_{\mathrm{period},\mathbb{Q}^+}(f) = \{r \in \mathbb{Q}^+ : f(rx) = f(x),\ \forall x \in \mathbb{Q}^+\}$$

is a subgroup of $\mathbb{Q}^+$, and $\langle q\rangle = \{q^n : n \in \mathbb{Z}\} \leq \mathrm{Group}_{\mathrm{period},\mathbb{Q}^+}(f)$.

Proof. Iterating $f(qx) = f(x)$ in both directions gives $f(q^n x) = f(x)$ for every $n \in \mathbb{Z}$ (since q^{-1} exists in $\mathbb{Q}^+$), so $\langle q\rangle \leq \mathrm{Group}_{\mathrm{period},\mathbb{Q}^+}(f)$. The period set is closed under multiplication and under taking inverses, hence it is a subgroup. □

Theorem 2.4 (The rationals and the commensurability criterion for the rotation equation)**.** Let $\theta \in \mathbb{R}$ and $f : \mathbb{C}^\times \to Y$. The rotation equation

$$f(e^{i\theta} z) = f(z), \qquad \forall z \in \mathbb{C}^\times$$

has a non-trivial periodic solution whenever $\theta/(2\pi) \in \mathbb{Q}$; alongside this, the group with periodic orbits $\langle e^{i\theta}\rangle \leq \mathbb{C}^\times$ generated by the rotation is finite if and only if $\theta/(2\pi) \in \mathbb{Q}$. This criterion itself relies on the fractional representation p/q of the rational numbers $\mathbb{Q}$. Hence $\mathbb{Q}$ is the minimal algebraic structure for expressing 'commensurability', but the **algebraic realisation** of the rotation equation (the construction of $e^{i\theta}$) still requires $\mathbb{C}$.

Proof. First assertion (the sufficient direction): if $\theta/(2\pi) = p/q \in \mathbb{Q}$ in lowest terms, then $e^{i\theta}$ is a q-th root of unity, and $f(e^{i\theta}z) = f(z)$ has a non-trivial solution—take $f(z) = z^q$; then $f(e^{i\theta}z) = (e^{i\theta}z)^q = e^{2\pi i p} z^q = f(z)$, and f is not constant.

Second assertion (the group with periodic orbits): if $\theta/(2\pi) = p/q \in \mathbb{Q}$ in lowest terms, then $e^{i\theta}$ has order q and $\langle e^{i\theta}\rangle$ is finite; conversely, if $\theta/(2\pi) \notin \mathbb{Q}$, then $e^{in\theta} = 1$ would require $\theta/(2\pi) = k/n \in \mathbb{Q}$, so $e^{i\theta}$ has infinite order and $\langle e^{i\theta}\rangle$ is infinite.

Expressing the criterion $\theta/(2\pi) \in \mathbb{Q}$ requires $\mathbb{Q}$, but the algebraisation of $e^{i\theta}$ itself requires i in $\mathbb{C}$. □

Remark 2.5 (The first assertion of Theorem 2.4 does not reverse). *The first assertion of Theorem 2.4 is one-directional: $\theta/(2\pi) \in \mathbb{Q}$ guarantees a non-trivial periodic solution, but the existence of a solution does not imply that $\theta/(2\pi) \in \mathbb{Q}$. The existence of solutions is determined by the level sets of f alone and is independent of the order of $e^{i\theta}$: the functions satisfying the rotation equation pointwise are exactly those constant on the $\langle e^{i\theta}\rangle$-orbits, a class that varies with θ, yet is non-empty for every θ. The reverse direction requires an additional restriction on f; for instance, if f is continuous and non-constant on some circle $|z| = r$, then $\theta/(2\pi) \notin \mathbb{Q}$ forces every solution to be trivial, because $e^{i\theta}$ is dense on that circle and continuity then forces f to be constant on the whole circle. The second assertion (finiteness of the group with periodic orbits), by contrast, remains an equivalence once it is stated in terms of the group with periodic orbits $\langle e^{i\theta}\rangle$.*

Example 2.1 (A pointwise periodic solution does not imply finite order). *Take $f(z) = |z|$. Then for every $\theta \in \mathbb{R}$ and every $z \in \mathbb{C}^\times$,*

$$f(e^{i\theta}z) = |e^{i\theta}z| = |z| = f(z),$$

so every θ admits a pointwise periodic solution; in particular, when $\theta/(2\pi) \notin \mathbb{Q}$ (for example $\theta = 1$) the group with periodic orbits $\langle e^{i\theta}\rangle$ of the rotation is still infinite. Hence 'admitting a pointwise periodic solution' is strictly weaker than 'the rotation has finite order', and the first assertion of Theorem 2.4 cannot be written as an 'if and only if'.

Remark 2.6 (The rationals make the scaling equation invertible). *In the integers the scaling equation $f(qx) = f(x)$ is meaningful only when q divides every relevant x, and q^{-1} does not exist. In the rationals multiplicative inverses exist, the scaling equation can be iterated in both directions, and the group with periodic orbits $\langle q\rangle$ becomes a subgroup. This is the second instance of number-system extension making a function equation pass from 'non-invertible' to 'invertible'.*

2.3.6 Criterion theorem

Theorem 2.5 (Criterion for rational rotation periods). Let $\theta \in \mathbb{R}$ and let the rotation orbit be $\mathcal{O}(z_0;\theta) = \{z_0e^{in\theta} \mid n \in \mathbb{Z}\}$. Then:

(i) If $\theta/(2\pi) = p/q \in \mathbb{Q}$ (a fraction in lowest terms), then the orbit is a finite cyclic orbit of order q, with period q.

(ii) If $\theta/(2\pi) \notin \mathbb{Q}$, then $e^{i\theta}$ has infinite order and the orbit does not close (an infinite orbit).

Proof. (i) Let $\theta = 2\pi p/q, \mathrm{gcd}(p,q) = 1$. Then $e^{in\theta} = 1 \iff n\theta \in 2\pi\mathbb{Z} \iff q \mid np \iff q \mid n$ (by coprimality), so $e^{i\theta}$ has order exactly q. The orbit is $\{z_0, z_0e^{i\theta}, \dots, z_0e^{i(q-1)\theta}\}$, containing exactly q equally spaced points. (ii) If $\theta/(2\pi) \notin \mathbb{Q}$, then $e^{in\theta} = 1$ requires $n\theta \in 2\pi\mathbb{Z}$, that is, $\theta/(2\pi) = k/n \in \mathbb{Q}$, a contradiction. Hence $e^{in\theta} \neq 1$ for all $n \geq 1$, and the orbit is infinite and does not close. ☐

Corollary 2.4 (Rational numbers and finite rotation periods). *$\mathbb{Q}$ is the minimal number system in which finite rotation periods can be determined (deduced from Theorem 2.5). Its determining power is reflected in:*

(i) *the commensurability criterion $\theta/(2\pi) \in \mathbb{Q}$ can be expressed exactly;*

(ii) *the order q of the finite cyclic group C_q can be specified exactly.*

Proof. (i) The criterion $\theta/(2\pi) = p/q \in \mathbb{Q}$ of Theorem 2.5 relies on the fractional representation p/q of rational numbers, and this representation can be given exactly in $\mathbb{Q}$. (ii) By that theorem, the orbit contains exactly q points and $e^{i\theta}$ is a q-th root of unity, so the order q of C_q is specified exactly by the denominator. ☐

Proposition 2.10 (Conversion of conserved quantities under the proportional direction and the logarithmic homomorphism). *Under the logarithmic map $\ln : \mathbb{Q}^+ \to \mathbb{R}$, the scaling orbit $\mathcal{O} = \{a_0q^k \mid k \in \mathbb{Z}\} \subset \mathbb{Q}^+$ is transformed into the translation orbit:*

$$\ln(\mathcal{O}) = \{\ln a_0 + k \ln q \mid k \in \mathbb{Z}\}.$$

Quantities that are invariant under the proportional direction appear, after logarithmic conversion, as invariants under the translation direction.

Proof. Let Q_{scale} be a conserved quantity on $(\mathbb{Q}^+, \times)$, satisfying $Q_{\text{scale}}(qx) = Q_{\text{scale}}(x)$. Define $Q_{\text{trans}}(y) = Q_{\text{scale}}(e^y)$; then

$$Q_{\text{trans}}(y + \ln q) = Q_{\text{scale}}(e^{y+\ln q}) = Q_{\text{scale}}(q \cdot e^y) = Q_{\text{scale}}(e^y) = Q_{\text{trans}}(y).$$

Q_{trans} is invariant under the translation $y \mapsto y + \ln q$, that is, it is an invariant under the translation direction. ☐

Remark 2.7 (Commensurability and period). A core feature of the rational numbers is **commensurability**: the ratio $\theta/(2\pi)$ of the rotation angle θ to 2π satisfies $\theta/(2\pi) \in \mathbb{Q}$ if and only if the group with periodic orbits $\langle e^{i\theta} \rangle$ of the rotation is finite, that is, if and only if the rotation generates a finite strict period. This property links the rational numbers closely with the theory of rotation periods—the theory of rational approximation (continued fractions) provides the algorithmic basis for the discrete approximation of incommensurable periods. In calendrics, the incommensurable ratio between the tropical year and the synodic month is precisely what is approximated discretely by rational approximation (such as seven intercalations in nineteen years).

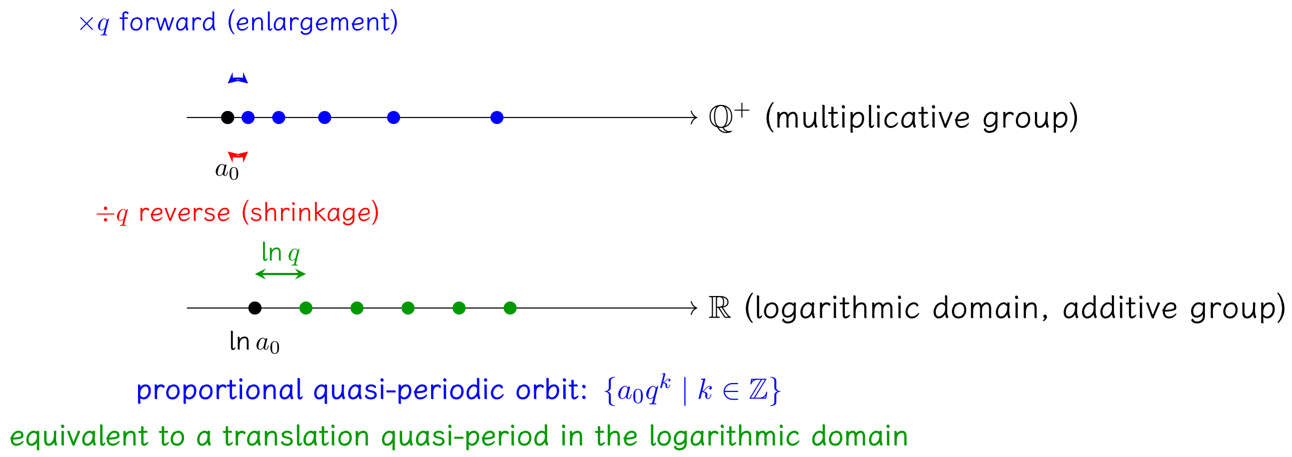

$\times q$ forward (enlargement)
$\mathbb{Q}^+$ (multiplicative group)
a_0
$\div q$ reverse (shrinkage)
$\ln q$
$\mathbb{R}$ (logarithmic domain, additive group)
$\ln a_0$
proportional quasi-periodic orbit: $\{a_0 q^k \mid k \in \mathbb{Z}\}$
equivalent to a translation quasi-period in the logarithmic domain

Figure 2.5: The proportional direction structure of the rationals $\mathbb{Q}^+$ (upper figure) and its equivalent translation direction in the logarithmic domain (lower figure). With 1 as the multiplicative centre of symmetry, multiplication by q and division by q form a forward–reverse directional duality.

2.4 The reals $\mathbb{R}$: the topological basis of quasi-periodic orbits

2.4.1 Determination requirement

When the rotation angle is incommensurable, $\theta/(2\pi) \notin \mathbb{Q}$, the orbit is dense on $U(1)$ but does not close. This 'denseness' is a topological concept and can be formulated rigorously only in a complete space. Moreover, the continuous translation orbit $x(t) = x_0 + vt$ requires $\mathbb{R}$ as the range of the time parameter.

2.4.2 Algebraic requirement

Completeness. This is the minimal ordered-field property that makes every Cauchy sequence converge.

2.4.3 Construction of the real numbers

Definition 2.5 (Construction of the real numbers)**.** The set of real numbers $\mathbb{R}$ consists of all Dedekind cuts $L \subset \mathbb{Q}$ satisfying the following conditions:

(1) $L \neq \emptyset$ and $L \neq \mathbb{Q}$.

(2) If $x \in L$ and $y < x$, then $y \in L$ (closed downwards).

(3) L has no greatest element.

$\mathbb{R}$ satisfies the least-upper-bound property (completeness): every non-empty subset that is bounded above has a least upper bound.

Theorem 2.6 (Completeness theorem for the real numbers)**.** *$\mathbb{R}$ is the minimal ordered field in which every Cauchy sequence converges. Specifically, the following conditions are equivalent:*

(i) $\mathbb{R}$ satisfies the least-upper-bound property;

(ii) *every Cauchy sequence in* $\mathbb{R}$ *converges;*

(iii) $\mathbb{R}$ is the completion of $\mathbb{Q}$.

Proof. (i)⇒(iii): By Definition 2.5, $\mathbb{R}$ consists of all Dedekind cuts of $\mathbb{Q}$, and its least-upper-bound property is guaranteed directly by the construction: the union of any non-empty family of cuts that is bounded above is again a cut, and is the least upper bound of that family. Each real number x is determined by its cut L_x; taking an increasing sequence of rationals in L_x that approaches x (for example by repeated bisection), every real number is the limit of some Cauchy sequence of rationals, that is, $\mathbb{R}$ is the completion of $\mathbb{Q}$. (i)⇒(ii): Let (x_n) be a Cauchy sequence in $\mathbb{R}$. For each n, the least-upper-bound property ensures that $u_n = \sup\{x_k : k \geq n\}$ and $l_n = \inf\{x_k : k \geq n\}$ both exist; (u_n) is monotonically decreasing and (l_n) monotonically increasing, and by the Cauchy property $\lim_{n\to\infty}(u_n - l_n) = 0$. Monotone bounded sequences converge (again guaranteed by the least-upper-bound property); let $x = \lim u_n = \lim l_n$; then $l_n \leq x_n \leq u_n$ together with the squeeze property gives $x_n \to x$. (ii)⇒(i): Let $A \subset \mathbb{R}$ be non-empty and bounded above by B. Take $a_0 \in A$ and $b_0 = B$, and construct a nested sequence of intervals $[a_n, b_n]$ recursively: put $m_n = (a_n + b_n)/2$; if m_n is an upper bound for A, take $[a_{n+1}, b_{n+1}] = [a_n, m_n]$, otherwise take $[a_{n+1}, b_{n+1}] = [m_n, b_n]$. The endpoint sequences $(a_n), (b_n)$ are both Cauchy (the interval length is $|b_n - a_n| = 2^{-n}(b_0 - a_0) \to 0$), so by (ii) they converge to the same limit x. Inductively, every b_n is an upper bound for A while no a_n is an upper bound for A; hence $y \leq \lim b_n = x$ for every $y \in A$ (so x is an upper bound), while for any upper bound u there exists $y_n \in A$ with $a_n < y_n \leq u$, so $x = \lim a_n \leq u$ (x is the least upper bound). Therefore $x = \sup A$. □

2.4.4 Closure of the root operation

Theorem 2.7 (Real-number completeness closes the root operation)**.** *For every* $b > 0$ *and* $n \in \mathbb{N}^+$*, the equation* $x^n = b$ *has a unique positive solution* $x = \sqrt[n]{b}$ *in* $\mathbb{R}$.

Proof. Construct the Dedekind cut $L = \{q \in \mathbb{Q} : q \leq 0 \text{ or } q^n < b\}$. Verify that L satisfies the three conditions for a Dedekind cut: (1) $0 \in L$ and $b+1$ is an upper bound, so $L \neq \emptyset, \mathbb{Q}$; (2) if $q \in L$ and $r < q$, then when $r \leq 0$ the definition of L gives $r \in L$ directly, while when $r > 0$ we have $r^n < q^n < b$, so $r \in L$; (3) for any $q \in L$ with $q > 0$ one can choose $\varepsilon > 0$ sufficiently small that $(q+\varepsilon)^n < b$, so $q + \varepsilon \in L$ and L has no greatest element. The real number x defined by L satisfies $x^n = b$. Uniqueness is guaranteed by the strict monotonicity of the power function. □

2.4.5 Function-theoretic features: the continuous translation equation and quasi-periods

Theorem 2.8 (The group with periodic orbits of the continuous translation equation on the reals)**.** *Let* $f : \mathbb{R} \to Y$ *be continuous and* $T > 0$*. If*

$$f(x+T) = f(x), \qquad \forall x \in \mathbb{R},$$

then the group with periodic orbits

$$\mathrm{Group}_{\mathrm{period},\mathbb{R}}(f) = \{t \in \mathbb{R} : f(x+t) = f(x),\ \forall x \in \mathbb{R}\}$$

is a closed subgroup of $\mathbb{R}$. A closed subgroup of $\mathbb{R}$ is either $\mathbb{R}$ itself or a discrete subgroup $T_0\mathbb{Z}$ ($T_0 > 0$).

Proof. Continuity and $f(x+T) = f(x)$ give $T\mathbb{Z} \leq \mathrm{Group}_{\mathrm{period},\mathbb{R}}(f)$. The group with periodic orbits is closed under addition and under taking inverses, and by the continuity of f, if $t_n \in \mathrm{Group}_{\mathrm{period},\mathbb{R}}(f)$ and $t_n \to t$, then $f(x+t) = f(x)$; hence it is a closed subgroup. The classification of the closed subgroups of $\mathbb{R}$ is a standard result: if the subgroup contains arbitrarily small positive elements then it is $\mathbb{R}$, and otherwise it is the discrete subgroup $T_0\mathbb{Z}$. ☐

Proposition 2.11 (Quasi-periodic orbits and completeness). *If $\theta/(2\pi) \notin \mathbb{Q}$, then the rotation orbit $\{e^{in\theta}\}$ is dense on $U(1)$. The rigorous formulation of denseness relies on the completeness of $\mathbb{R}$ (Theorem 2.6): without completeness the notions of limit and closure cannot be defined.*

Proof. By Kronecker's approximation theorem, $\{n\theta \bmod 2\pi\}$ is dense in $[0, 2\pi)$. Denseness requires that for every $\varepsilon > 0$ there is an orbit point lying in the ε-neighbourhood of any given point, and this relies on the existence of limits in $\mathbb{R}$, that is, on completeness. ☐

Remark 2.8 (The reals make the translation equation continuous). In the integers the group with periodic orbits of the translation equation is a discrete subgroup $a\mathbb{Z}$; in the reals the group with periodic orbits of the translation equation can be any closed subgroup, including the continuous group $\mathbb{R}$ itself. This is an instance of number-system extension making a function equation pass from 'discrete' to 'continuous'. The rotation equation $f(e^{i\theta}z) = f(z)$ still cannot be algebraised in the reals, because $e^{i\theta}$ does not belong to $\mathbb{R}$.

2.4.6 Criterion theorem

Theorem 2.9 (Quasi-period criterion for the real numbers). *Let $\theta/(2\pi) \notin \mathbb{Q}$. Then the rotation orbit*

$$\mathcal{O}(z_0; \theta) = \{z_0 e^{in\theta} \mid n \in \mathbb{Z}\}$$

is dense on $U(1)$. This denseness conclusion has a rigorous topological meaning only in $\mathbb{R}$.

Proof. By Kronecker's approximation theorem, if $\theta/(2\pi)$ is irrational then the set $\{n\theta \bmod 2\pi \mid n \in \mathbb{Z}\}$ is dense in $[0, 2\pi)$. Hence the orbit points are dense on $U(1)$. The definition of denseness requires that for every $\varepsilon > 0$ there is an orbit point lying in the ε-neighbourhood of any given point, and this requires the completeness of the reals. ☐

Corollary 2.5 (Real numbers and quasi-periods). *$\mathbb{R}$ is the minimal complete number system in which quasi-periodic orbits can be described but not determined in finitely many steps (deduced from Theorem 2.9 and Theorem 2.6). Specifically:*

(i) topological concepts such as denseness, limits and continuity hold rigorously in $\mathbb{R}$;

(ii) but the rationality of a real number cannot be determined in finitely many steps—hence $\mathbb{R}$ cannot enhance the determination of finite rotation periods.

Proof. (i) Theorem 2.6 guarantees that every Cauchy sequence in $\mathbb{R}$ converges, so topological concepts such as denseness, limits and continuity have a rigorous meaning. (ii) Theorem 2.9 asserts only the denseness of the orbit corresponding to an irrational rotation angle, which is an existence description rather than a finite determination; determining the rationality of a real number requires an infinite examination of its decimal (or continued-fraction) expansion, which cannot be completed in finitely many steps, so $\mathbb{R}$ does not enhance the determination of finite rotation periods. ☐

Proposition 2.12 (Effect of real-number completion on the directional structure). *The completeness of the reals lifts the translation direction and the proportional direction from discrete groups to continuous groups:*

(i) **Continuisation of the translation direction**: *the translation group is lifted from $(\mathbb{Z}, +)$ to $(\mathbb{R}, +)$. Translation orbits change from discrete isometric sequences of points into continuous straight lines.*

(ii) **Continuisation of the proportional direction**: *the scaling group is lifted from the discrete subgroup $\{q^k\}$ to $(\mathbb{R}^+, \times)$. Proportional orbits are equivalent to continuous translations in the logarithmic domain.*

(iii) **Topological completion of the rotation direction**: *the dense quasi-periodic orbits produced by incommensurable rotation angles acquire a rigorous limiting meaning in $\mathbb{R}$.*

Proof. (i) $(\mathbb{R}, +)$ is the minimal continuous extension of $(\mathbb{Z}, +)$ under completion: $(\mathbb{Q}, +)$ embeds densely in $(\mathbb{R}, +)$, and $(\mathbb{Z}, +)$ is a discrete generating set for $(\mathbb{Q}, +)$; hence the translation group is lifted from the integer multiples of the discrete generator 1 to translations $x \mapsto x + t$ by a continuous parameter $t \in \mathbb{R}$, and the orbit $\{x_0 + t \mid t \in \mathbb{R}\}$ is a continuous straight line. (ii) The exponential map $\exp : (\mathbb{R}, +) \to (\mathbb{R}^+, \times)$ is a group isomorphism (see Theorem 2.18), so the scaling group is lifted from the discrete subgroup $\{q^k \mid k \in \mathbb{Z}\}$ to the continuous group $(\mathbb{R}^+, \times) = \{\exp(t) \mid t \in \mathbb{R}\}$, and the proportional orbit $\{a_0 r \mid r \in \mathbb{R}^+\}$ is equivalent to a continuous translation orbit in the logarithmic domain. (iii) By Theorem 2.9, $\{n\theta \bmod 2\pi\}$ is dense in $[0, 2\pi)$; topological concepts such as denseness, limits and closure depend on the completeness of $\mathbb{R}$ (Theorem 2.6), so the dense quasi-periodic orbits of incommensurable rotations acquire a rigorous limiting meaning in $\mathbb{R}$. ☐

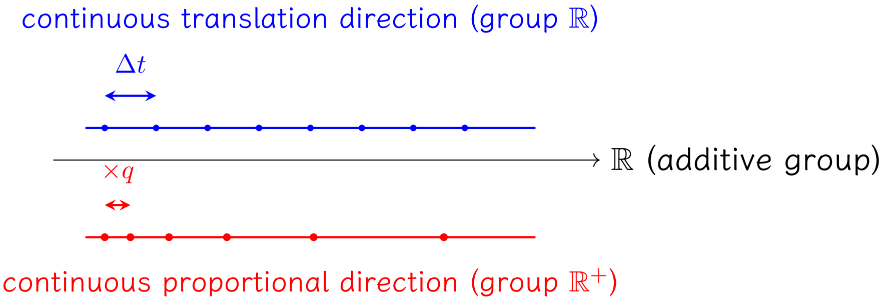

continuous translation direction (group $\mathbb{R}$)
Δt
$\mathbb{R}$ (additive group)
$\times q$
continuous proportional direction (group $\mathbb{R}^+$)

Figure 2.6: The continuous directional structure brought about by the completion of the reals. The translation direction is lifted from the discrete group $\mathbb{Z}$ to the continuous group $\mathbb{R}$ (upper figure). The proportional direction is equivalent to continuous translation in the logarithmic domain (lower figure).

2.5 The complex numbers $\mathbb{C}$: algebraisation of rotation periods and double periods

2.5.1 Determination requirement

The rotation operation needs to be expressed by algebraic operations, not merely by geometric intuition. A finite rotation period C_n requires the complete solution set of the algebraic equation $z^n = 1$. Moreover, double-period structures (elliptic functions) require discrete subgroups of the complex plane.

2.5.2 Algebraic requirement

The imaginary unit i, and algebraic closure. This is the minimal field extension that makes all polynomial equations solvable.

2.5.3 Construction of the complex numbers

Definition 2.6 (Construction of the complex numbers). $\mathbb{C} := \mathbb{R}^2$ as a real vector space. Addition is defined componentwise:

$$(a, b) + (c, d) := (a + c, b + d).$$

Multiplication is defined by

$$(a, b) \times (c, d) := (ac - bd, ad + bc).$$

Write $i := (0, 1)$; then $i^2 = (0, 1) \times (0, 1) = (-1, 0) = -1$. Every complex number can be written as $z = a + bi$ $(a, b \in \mathbb{R})$.

Proposition 2.13 (The complex numbers form a field). *$(\mathbb{C}, +, \times)$ forms a field. That is:*

(i) *$(\mathbb{C}, +)$ is an abelian group with zero element $(0, 0)$;*

(ii) *$(\mathbb{C}^\times, \times)$ is an abelian group with multiplicative inverse $z^{-1} = \bar{z}/|z|^2$ $(z \neq 0)$;*

(iii) multiplication satisfies the distributive law over addition.

Proof. (i) Addition is defined componentwise, each component being added in $\mathbb{R}$; hence associativity, commutativity, the zero element $(0,0)$ and the negative $-(a,b)=(-a,-b)$ are all inherited componentwise from the additive group structure of $\mathbb{R}$. (ii) The commutativity and associativity of multiplication are verified by direct expansion from the definition: $(a,b)(c,d)=(ac-bd,ad+bc)=(ca-db,cb+da)=(c,d)(a,b)$; expanding associativity componentwise reduces it to the laws of operation in $\mathbb{R}$. The multiplicative identity is $(1,0)$. For $z=(a,b)\neq 0$ we have $|z|^2=a^2+b^2>0$, and

$$(a,b)\left(\frac{a}{a^2+b^2},\frac{-b}{a^2+b^2}\right)=\left(\frac{a^2+b^2}{a^2+b^2},\frac{-ab+ba}{a^2+b^2}\right)=(1,0),$$

so $z^{-1}=\bar{z}/|z|^2$ is the multiplicative inverse. (iii) The distributive law is verified by expanding componentwise from the definition:

$$\begin{aligned}(a,b)\big[(c,d)+(e,f)\big]&=(a,b)(c+e,d+f)=\big(a(c+e)-b(d+f),a(d+f)+b(c+e)\big)\\&=(ac-bd,ad+bc)+(ae-bf,af+be)=(a,b)(c,d)+(a,b)(e,f).\end{aligned}$$

□

Proposition 2.14 (Completeness of the complex numbers). *As a metric space, $\mathbb{C}\cong\mathbb{R}^2$ is complete: a Cauchy sequence $\{z_n=a_n+ib_n\}$ in $\mathbb{C}$ converges if and only if both the real-part sequence $\{a_n\}$ and the imaginary-part sequence $\{b_n\}$ converge in $\mathbb{R}$.*

Proof. By the completeness of $\mathbb{R}$ (Theorem 2.6), $\{a_n\}$ and $\{b_n\}$ converge to $a,b\in\mathbb{R}$ respectively. Then $z_n\to a+bi\in\mathbb{C}$. Conversely, if $z_n\to z=a+bi$, then $|a_n-a|\leq|z_n-z|\to 0$ and $|b_n-b|\leq|z_n-z|\to 0$. □

Remark 2.9 (The fundamental theorem of algebra (algebraic closure)). The complex field $\mathbb{C}$ is **algebraically closed**: every non-zero polynomial with complex coefficients has at least one root in $\mathbb{C}$. This is the fundamental theorem of algebra; its standard proof requires complex analysis (Liouville's theorem or the argument principle) or topological/algebraic methods, which lie beyond the deductive scope of 'number-system extension' in this chapter, so it is cited here as a standard result without proof. In what follows this chapter uses only its weaker corollary—'a polynomial of degree n has at most n roots'—and that conclusion can be proved directly on any field by the factor theorem (see the proof of Theorem 2.12); it does not depend on the full strength of the fundamental theorem of algebra.

2.5.4 Algebraisation of rotation

Theorem 2.10 (Algebraisation of complex rotation). *The unit circle*

$$S^1=\{z\in\mathbb{C}\mid |z|=1\}$$

forms a group under complex multiplication, isomorphic to $U(1)$. The algebraic realisation of the rotation R_θ is

$$R_\theta(z)=e^{i\theta}z,$$

where $e^{i\theta}=\cos\theta+i\sin\theta$ (Euler's formula).

Proof. $|e^{i\theta}z| = |e^{i\theta}| \cdot |z| = 1 \cdot |z| = |z|$, so rotation preserves the modulus. Since $e^{i(\theta_1+\theta_2)} = e^{i\theta_1}e^{i\theta_2}$, the rotation group is isomorphic to $U(1)$. Euler's formula follows by comparing the power-series expansion of $e^{i\theta}$ with the power-series expansion of $\cos\theta, \sin\theta$. ☐

2.5.5 Function-theoretic features: the rotation equation and roots of unity

Theorem 2.11 (The group with periodic orbits of the rotation equation on the complex numbers)**.** *Let $f : \mathbb{C}^\times \to Y$ and $\theta \in \mathbb{R}$. If*

$$f(e^{i\theta}z) = f(z), \qquad \forall z \in \mathbb{C}^\times,$$

then the group with periodic orbits

$$\mathsf{Group}_{\mathsf{period},U(1)}(f) = \{e^{i\theta} \in U(1) : f(e^{i\theta}z) = f(z),\ \forall z \in \mathbb{C}^\times\}$$

is a subgroup of $U(1)$. If $\theta/(2\pi) = p/q$ in lowest terms, then $e^{i\theta}$ has order q, and $\langle e^{i\theta}\rangle \cong C_q \leq \mathsf{Group}_{\mathsf{period},U(1)}(f)$.

Proof. Iterating $f(e^{i\theta}z) = f(z)$ in both directions gives $f(e^{in\theta}z) = f(z)$ for every $n \in \mathbb{Z}$, so $\langle e^{i\theta}\rangle \leq \mathsf{Group}_{\mathsf{period},U(1)}(f)$. If $\theta/(2\pi) = p/q$ in lowest terms, then $e^{i\theta}$ is a q-th root of unity of order q, generating C_q. ☐

Corollary 2.6 (The unit-root equation and finite rotation periods)**.** *The equation $z^n = 1$ has exactly n solutions in $\mathbb{C}$, forming the cyclic group C_n. This is the algebraic prototype of the rotation equation $f(e^{i\theta}z) = f(z)$ at $\theta = 2\pi/n$: taking $f(z) = z^n$, we have $f(e^{2\pi i/n}z) = z^n = f(z)$.*

Proof. By Theorem 2.12, the solution set of $z^n = 1$ is $\{e^{2\pi ik/n} : k = 0, \dots, n-1\}$, forming C_n. Taking $f(z) = z^n$, we have $f(e^{2\pi i/n}z) = (e^{2\pi i/n}z)^n = z^n = f(z)$. ☐

Remark 2.10 (The complex numbers make the rotation equation algebraic)*.* In the reals the rotation equation $f(e^{i\theta}z) = f(z)$ has no algebraic carrier, because $e^{i\theta}$ does not belong to $\mathbb{R}$. In the complex numbers the imaginary unit i makes $e^{i\theta}$ an algebraic object, the rotation equation can be algebraised, and the group with periodic orbits C_n of a finite rotation period is characterised exactly by the unit-root equation $z^n = 1$. This is the key step by which number-system extension makes a function equation pass from 'non-algebraisable' to 'algebraisable'.

2.5.6 Criterion theorem

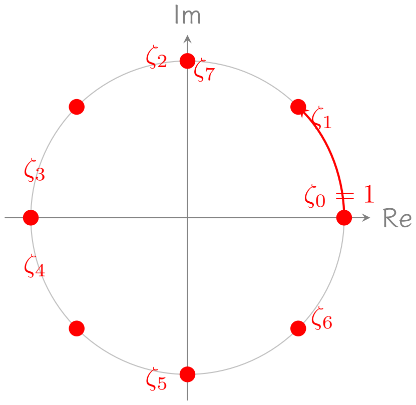
Theorem 2.12 (Algebraic criterion for finite rotational periods)**.** *The equation*

$$z^n = 1$$

has exactly n solutions in $\mathbb{C}$:

$$\zeta_k = e^{2\pi ik/n}, \qquad k = 0, 1, \dots, n-1.$$

These n solutions form the cyclic group C_n of order n. They are the algebraic prototype of finite rotation periods.



$\zeta_k = e^{i2\pi k/n}$, $k = 0, \dots, n-1$:
the roots of unity are uniformly distributed on the unit circle and form C_n.

Figure 2.7: The geometric distribution of the n-th roots of unity in the complex numbers (taking $n = 8$ as an example). The n solutions $\zeta_k = e^{2\pi ik/n}$ are equally spaced on the unit circle and form the cyclic group C_n of order n. This is the algebraic prototype of finite rotation periods in the complex field (Theorem 2.12); the special case $n = 4$ in the figure is the fourth-order rotation orbit of Figure 2.8.

Proof. First we prove that the solution set has exactly n elements. Since $\mathbb{C}$ is a field, $z^n - 1$ is a non-zero polynomial of degree n, and by the factor theorem (each root α corresponds to a factor $(z-\alpha)$, and each extraction of a factor lowers the degree by 1) it has at most n roots. On the other hand, for $k = 0, 1, \dots, n-1$, put $\zeta_k = e^{2\pi ik/n}$; then

$$\zeta_k^n = e^{2\pi ik} = \cos(2\pi k) + i\sin(2\pi k) = 1,$$

and when $k \neq l$ we have $\zeta_k/\zeta_l = e^{2\pi i(k-l)/n} \neq 1$ (since $0 < |k-l| < n$), so $\zeta_0, \dots, \zeta_{n-1}$ are pairwise distinct, giving n roots. Hence the solution set is exactly $\{\zeta_0, \dots, \zeta_{n-1}\}$. Next we prove that these n solutions form a cyclic group: closure under multiplication $\zeta_k \cdot \zeta_l = e^{2\pi i(k+l)/n} = \zeta_{k+l \bmod n}$; the identity element $\zeta_0 = 1$; inverses $\zeta_k^{-1} = e^{-2\pi ik/n} = \zeta_{n-k}$ (for $k = 0$, $\zeta_0^{-1} = \zeta_0$). Since $\zeta_1 = e^{2\pi i/n}$ generates the whole set ($\zeta_k = \zeta_1^k$), these n solutions form the cyclic group C_n of order n. □

Theorem 2.13 (Period-lattice criterion for double-period structure). *The discrete subgroup of* $\mathbb{C}$

$$\Lambda = \omega_1\mathbb{Z} + \omega_2\mathbb{Z}, \qquad \omega_1/\omega_2 \notin \mathbb{R},$$

is the period lattice of double-periodic functions (elliptic functions). Every elliptic function with period lattice Λ *satisfies*

$$f(z+\omega_1) = f(z), \qquad f(z+\omega_2) = f(z).$$

The two independent periods ω_1, ω_2 *correspond to two translation directions respectively.*

Proof. Λ is a discrete additive subgroup of $\mathbb{C}$ of rank 2 (because $\omega_1/\omega_2 \notin \mathbb{R}$ guarantees that the two directions are linearly independent). The quotient space $\mathbb{C}/\Lambda$ is a compact complex manifold (an elliptic curve). The Weierstrass $\wp$ function is the typical elliptic function on the period lattice Λ. ☐

2.5.7 The involution of the centre of symmetry i

Definition 2.7 (The involution of the centre of symmetry i)**.** On $\mathbb{C}^\times$ define the map

$$\sigma_i : \mathbb{C}^\times \to \mathbb{C}^\times, \quad \sigma_i(z) = i \cdot z^{-1}.$$

Writing z in polar form $z = re^{i\varphi}$, from $z^{-1} = r^{-1}e^{-i\varphi}$ this operation becomes

$$\sigma_i(re^{i\varphi}) = i \cdot r^{-1}e^{-i\varphi} = r^{-1}e^{i(\pi/2-\varphi)}.$$

This operation comprises simultaneously **scaling** (taking the reciprocal of the modulus, $r \mapsto r^{-1}$) and a reflection of the argument about the $\pi/4$ direction ($\varphi \mapsto \pi/2 - \varphi$).

Proposition 2.15 (Basic properties of the involution σ_i)**.** (i) σ_i is an involution: $\sigma_i \circ \sigma_i = \mathrm{id}_{\mathbb{C}^\times}$.

(ii) The fixed points of σ_i satisfy $z^2 = i$, that is, $z = \pm\frac{\sqrt{2}}{2}(1+i)$.

(iii) $\sigma_i = R_{\pi/2} \circ \sigma_1$, where $R_{\pi/2}(z) = iz$ is rotation through 90° and $\sigma_1(z) = z^{-1}$ is the taking-the-inverse involution.

Proof. (i) $\sigma_i(\sigma_i(z)) = i \cdot (i \cdot z^{-1})^{-1} = i \cdot (-i \cdot z) = z$ (note that $(i \cdot w)^{-1} = -i \cdot w^{-1}$ because $i \cdot (-i) = 1$). (ii) $\sigma_i(z) = z \iff i \cdot z^{-1} = z \iff z^2 = i \iff z = \pm e^{i\pi/4}$. (iii) $R_{\pi/2}(\sigma_1(z)) = i \cdot z^{-1} = \sigma_i(z)$. ☐

Example 2.2 (Unification of scaling and rotation in the complex power map)**.** Consider the complex power function $f_n : \mathbb{C}^\times \to \mathbb{C}^\times, z \mapsto z^n$ ($n \in \mathbb{N}^+$). Put $z = re^{i\theta}$; then

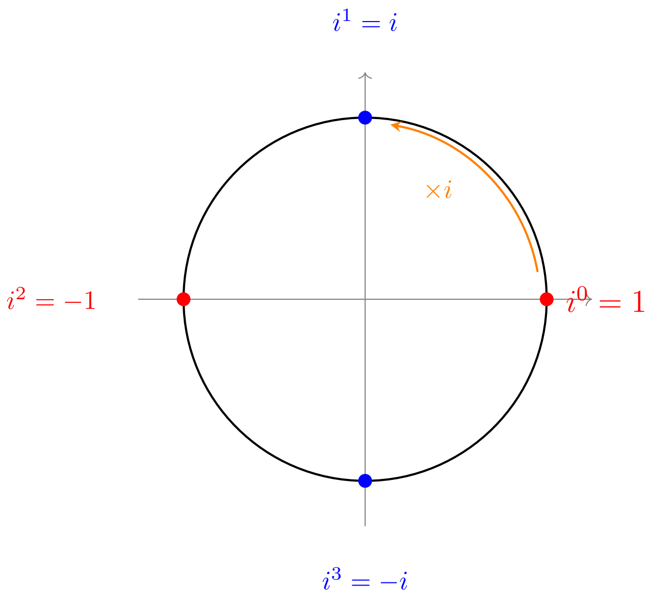
$$f_n(z) = r^n e^{in\theta}.$$

This map acts simultaneously on the scaling dimension (modulus $r \mapsto r^n$) and on the rotation dimension (argument $\theta \mapsto n\theta$). If $|z| = 1$, then f_n is an n-fold endomorphism of the rotation group $U(1)$. The complex power operation unifies the proportional direction and the rotation direction in a single operation.

Explicit computation: take $z = 2e^{i\pi/4}$ and $n = 3$; then

$$f_3(z) = 2^3\, e^{3\cdot i\pi/4} = 8\, e^{3\pi i/4} = 8\left(-\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) = -4\sqrt{2} + 4\sqrt{2}\, i.$$

The modulus $|z| = 2$ is enlarged to $|f_3(z)| = 8$ (proportional direction, scaling $r \mapsto r^3$), and the argument $\pi/4$ is rotated to $3\pi/4$ (rotation direction, $\theta \mapsto 3\theta$); the two are accomplished simultaneously in one operation, verifying that the complex power realises scaling and rotation at once.



$i^4 = 1$: the fourth-order rotation orbit

Figure 2.8: The rotation direction generated by complex multiplication $z \mapsto iz$: multiplication by i is rotation through $90°$, and the fourth power returns to the starting point, forming a fourth-order rotation orbit.

Definition 2.8 (The fourfold discrete skeleton of the unit circle)**.** The fourfold discrete skeleton of the unit circle $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$ is defined as

$$C_4 = \mu_4 = \{1, i, -1, -i\} = \{i^k \mid k = 0, 1, 2, 3\},$$

in which the phase and the algebraic role of each element are

$$1 = e^{i0} \text{ (multiplicative identity, the fixed point of rotation)}, \qquad i = e^{i\pi/2} \text{ (quarter turn)},$$

$$-1 = e^{i\pi} = i^2 \text{ (half turn, that is, sign flip)}, \qquad -i = e^{i3\pi/2} = i^3 \text{ (three-quarter turn)}.$$

This set forms the cyclic group C_4 of order 4 under complex multiplication: it is the finite rotation orbit generated by i (Figure 2.8), and it is also the complete solution set of the unit-root equation $z^4 = 1$ in $\mathbb{C}$ (Theorem 2.12 with $n = 4$; see also Corollary 2.6). Its continuous refinement gives the whole unit circle:

$$S^1 = \{e^{i\theta} \mid \theta \in \mathbb{R}\} \cong U(1).$$

C_4 is the smallest discrete skeleton of S^1 that contains i: the set $\{1, -1\}$ is its subgroup of order 2, and the adjunction of i raises the skeleton from order 2 to order 4.

Proposition 2.16 (The sign system is the projection of the rotation group at $180°$, and i refines it to C_4)**.** *Under the embedding of the reals $\mathbb{R} \hookrightarrow \mathbb{C}$, $x \mapsto x + 0i$:*

(i) *The sign system $\{\pm 1\}$ forms the cyclic group C_2 of order 2 under multiplication, and is a subgroup of the multiplicative group of $\mathbb{R}$ (item (ii) of Proposition 2.13).*

(ii) The map $\pi_1 : \mu_4 \to \{\pm 1\}$, $\pi_1(e^{i\theta}) = e^{i2\theta}$, is a surjective group homomorphism; it pairs up the four phases $0, \pi/2, \pi, 3\pi/2$ and presses them onto the two phases $0, \pi$, so that $\pi_1(\mu_4) = \{\pm 1\}$ is precisely the projection of the rotation group at 180° (the half turn). Its kernel is $\ker \pi_1 = \{\pm 1\}$, and the first isomorphism theorem gives $\mu_4/\{\pm 1\} \cong \{\pm 1\}$.

(iii) The real sign system $\{\pm 1\}$ is the image of the rotation group under the projection at 180°: $\mu_4/\{\pm 1\} \cong \{\pm 1\}$ shows that it is exactly the quotient of the fourfold rotation skeleton by the half-turn projection; equivalently, $\mathbb{R} \cap \mu_4 = \{\pm 1\}$, that is, only two real elements of the fourfold skeleton are visible on the real axis.

(iv) Adjoining i refines $\{\pm 1\} \cong C_2$ to $\mu_4 \cong C_4$, and $C_2 = \langle -1 \rangle$ is the unique subgroup of order 2 of C_4; continuous refinement (completing the discrete phases to arbitrary real phases) then gives $S^1 \cong U(1)$.

Proof. (i) $\{\pm 1\}$ is closed under multiplication ($1 \cdot 1 = 1$, $(-1) \cdot (-1) = 1$, $1 \cdot (-1) = -1$), contains the identity 1, and satisfies $(-1)^{-1} = -1$, so it is a group; $\langle -1 \rangle$ has order 2, hence is isomorphic to C_2, and it is a subgroup of $\mathbb{R}^\times$.

(ii) Define $\pi_1(e^{i\theta}) = e^{i2\theta}$. For $e^{i\theta_1}, e^{i\theta_2} \in \mu_4$ we have $\pi_1(e^{i\theta_1}e^{i\theta_2}) = \pi_1(e^{i(\theta_1+\theta_2)}) = e^{i2(\theta_1+\theta_2)} = \pi_1(e^{i\theta_1})\pi_1(e^{i\theta_2})$, so π_1 is a group homomorphism; the images of the four phases are $e^{i0} = 1$, $e^{i\pi} = -1$, $e^{i2\pi} = 1$, $e^{i3\pi} = -1$, so the image is $\{\pm 1\}$; the kernel consists of the elements whose doubled phase is 0, namely $e^{i0} = 1$ and $e^{i\pi} = -1$, so $\ker \pi_1 = \{\pm 1\}$, and the first isomorphism theorem gives $\mu_4/\{\pm 1\} \cong \{\pm 1\}$.

(iii) By (ii), the kernel of the half-turn projection is $\{\pm 1\}$ and its image is $\{\pm 1\}$; moreover the elements of μ_4 that lie in $\mathbb{R}$ are exactly ± 1 (since $i, -i$ have imaginary part $\pm 1 \neq 0$), so $\mathbb{R} \cap \mu_4 = \{\pm 1\}$.

(iv) Since $i^2 = -1$ and $i^4 = 1$, the element i has order 4 in $\mathbb{C}^\times$ and generates $\mu_4 = \langle i \rangle \cong C_4$, in which $\{\pm 1\} = \langle -1 \rangle$ is a subgroup of order 2; the order of a subgroup of C_4 divides 4, so the subgroup of order 2 is unique. Completing the discrete phases $k\pi/2$ to arbitrary real phases θ gives $S^1 = \{e^{i\theta} \mid \theta \in \mathbb{R}\}$; for the isomorphism with $U(1)$ see Theorem 2.10. ☐

Remark 2.11 (The conditions under which 'the sign is a rotation by π' holds). The assertion in the preceding proposition that 'a sign flip is a half turn' holds only after the embedding $\mathbb{R} \subset \mathbb{C}$; two structures must here be kept clearly apart, lest the text appear to conflict with the three centres of symmetry $0, 1, i$ established in this chapter:

- Considered on $\mathbb{R}$ alone, $x \mapsto -x$ is the **additive reflection with centre** 0: it is the involutive automorphism σ_0 of the additive group, its fixed point is 0, and it generates C_2 (item (ii) of Proposition 2.6); it is **not** a rotation in the complex plane, because $\mathbb{R}$ contains no carrier of the phase $e^{i\pi}$ (see Remark 2.10).

- In $\mathbb{C}$, and only there, $-1 = e^{i\pi}$ is the half turn on the unit circle, that is, the flip in multiplication with centre 1 and phase π. The relation between the two is given by the restriction map: the restriction of the additive reflection σ_0 to $\{\pm 1\} \subset \mathbb{C}$ is exactly multiplication by -1, that is, the action of $e^{i\pi}$ on the real axis; in other words, the additive reflection is the restriction of the half turn to the real axis, whilst the half turn is the phase realisation of the additive reflection in $\mathbb{C}$.

- Accordingly, the division of labour amongst the three centres of symmetry is: the additive centre 0 (σ_0, symmetry type additive reflection), the multiplicative centre 1 (σ_1, symmetry type taking the multiplicative inverse, Definition 2.4), and the phase centre i (σ_i, Definition 2.7). The sign system $\{\pm 1\}$ is not an additional centre of symmetry: it is the common image of σ_0 and of the half turn on the real axis.

2.6 The quaternions $\mathbb{H}$: algebraisation of three-dimensional rotation periods

2.6.1 Determination requirement

A two-dimensional rotation needs only one angle parameter, corresponding to $U(1)$, which is an abelian group. A three-dimensional rotation needs three parameters, corresponding to $SO(3)$, which is a non-abelian group. Complex numbers can express only planar rotations and cannot express the non-commutativity of three-dimensional rotations.

2.6.2 Algebraic requirement

A non-commutative multiplication structure. This is the minimal algebraic extension that makes three-dimensional rotations algebraisable.

2.6.3 Construction of the quaternions

Definition 2.9 (Construction of the quaternions). $\mathbb{H} := \mathbb{R}^4$ as a real vector space, with basis elements denoted $\{1, i, j, k\}$. Multiplication is extended bilinearly from the following relations:

$$i^2 = j^2 = k^2 = ijk = -1,$$

together with the relations derived from them: $ij = k, ji = -k$, $jk = i, kj = -i$, $ki = j, ik = -j$. For $q = a + bi + cj + dk$, define the conjugate $\bar{q} = a - bi - cj - dk$, the norm $|q| = \sqrt{q\bar{q}} = \sqrt{a^2 + b^2 + c^2 + d^2}$, and the inverse $q^{-1} = \bar{q}/|q|^2$ $(q \neq 0)$.

Proposition 2.17 (Algebraic properties of quaternions). (i) $\mathbb{H}$ is a division algebra (a division ring): every non-zero element has a multiplicative inverse, but multiplication is not commutative.

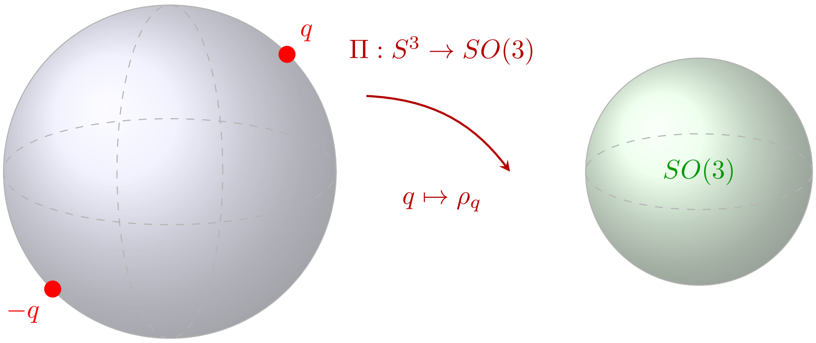
(ii) The group of unit quaternions $S^3 = \{q \in \mathbb{H} : |q| = 1\}$ forms a non-commutative Lie group under quaternion multiplication, isomorphic to $SU(2)$.

(iii) Through the conjugation action $\rho_q(v) = qvq^{-1}$ (with v a purely imaginary quaternion), S^3 realises rotations in three-dimensional space. The map $q \mapsto \rho_q$ gives the double covering $SU(2) \to SO(3)$.

Proof. (i) For $q = a + bi + cj + dk \neq 0$, from $q\bar{q} = a^2 + b^2 + c^2 + d^2 = |q|^2 > 0$ we get that $q^{-1} = \bar{q}/|q|^2$ satisfies $qq^{-1} = 1$, so every non-zero element has a multiplicative inverse; and since $ij = k \neq -k = ji$, multiplication is not commutative, so $\mathbb{H}$ is a division algebra (a division ring). (ii) First we prove the multiplicativity of the norm $|q_1q_2| = |q_1|\,|q_2|$: $|q_1q_2|^2 = q_1q_2\overline{q_1q_2} = q_1q_2\bar{q}_2\bar{q}_1 = q_1|q_2|^2\bar{q}_1 = |q_1|^2|q_2|^2$. Hence $|q_1| = |q_2| = 1 \Rightarrow |q_1q_2| = 1$ and $|q^{-1}| = |\bar{q}|/|q|^2 = 1$, so S^3 is closed under multiplication and contains the identity element and inverses, and is a group; it is a smooth closed submanifold of $\mathbb{H}^\times$ whose multiplication is not commutative, so it is a non-commutative Lie group. The isomorphism $S^3 \cong SU(2)$ is given by $a + bi + cj + dk \mapsto \begin{pmatrix} a+bi & c+di \\ -c+di & a-bi \end{pmatrix}$, since this matrix satisfies $|q|^2 = \det M$ and $MM^\dagger = (a^2 + b^2 + c^2 + d^2)I$, so $|q| = 1$ if and only if $M \in SU(2)$. (iii) By Theorem 2.14, the conjugation action $\rho_q(v) = qvq^{-1}$ preserves the modulus and preserves the purely imaginary space, so it is a rotation of $\mathbb{R}^3$, and $\rho_q = \rho_{-q}$ gives the double covering $SU(2) \to SO(3)$. □

Remark 2.12 (Comparison of quaternions with complex numbers). The quaternions $\mathbb{H}$ are the non-commutative extension of the complex numbers $\mathbb{C}$:

- $\mathbb{C}$ has only one imaginary unit (i), whereas $\mathbb{H}$ has three (i, j, k);
- the rotation group $U(1)$ of $\mathbb{C}$ is abelian, whereas the rotation group $SU(2)$ of $\mathbb{H}$ is non-abelian;
- the roots of unity ζ_n of $\mathbb{C}$ form the cyclic group C_n, whereas the classification of the finite subgroups of $\mathbb{H}$ is richer (cyclic, dihedral, tetrahedral, octahedral and icosahedral groups).



$S^3 \cong SU(2)$: the group of unit quaternions, a non-commutative Lie group; every three-dimensional rotation corresponds to two unit quaternions q and $-q$.

Figure 2.9: The quaternion unit sphere $S^3 \cong SU(2)$ and its double covering of $SO(3)$. q and $-q$ correspond to the same three-dimensional rotation, and the kernel of $\Pi : S^3 \to SO(3)$ is $\{\pm 1\}$. This is the algebraised realisation of three-dimensional rotation periods in the quaternion field (Theorem 2.14).

2.6.4 Criterion theorem

Theorem 2.14 (Algebraisation criterion for three-dimensional rotation). *The group of unit quaternions S^3 realises rotations in three-dimensional space through the conjugation action*

$$\rho_q(v) = qvq^{-1}, \qquad v \in \mathsf{Im}(\mathbb{H}) \cong \mathbb{R}^3,$$

The map

$$\Pi : S^3 \to SO(3), \qquad q \mapsto \rho_q,$$

is a double covering with $\ker \Pi = \{\pm 1\}$.

Proof. The space of purely imaginary quaternions is $\mathsf{Im}(\mathbb{H}) = \{bi + cj + dk : b, c, d \in \mathbb{R}\} \cong \mathbb{R}^3$. For a unit quaternion q, the conjugation action ρ_q preserves the norm: $|qvq^{-1}| = |v|$, and maps purely imaginary quaternions to purely imaginary quaternions. Hence ρ_q is an orthogonal transformation of $\mathbb{R}^3$. Continuity gives $\det(\rho_q) = 1$, so it is a rotation. Since $\rho_q = \rho_{-q}$, the covering degree is 2. □

Theorem 2.15 (Non-commutative period criterion). *Rotations about different axes do not commute:*

$$\rho_{q_1} \circ \rho_{q_2} \neq \rho_{q_2} \circ \rho_{q_1}$$

(in the general case). Hence a three-dimensional rotation period cannot be characterised by a single parameter of an abelian cyclic group, but must be described by the non-commutative subgroup structure of $SU(2)$ (or $SO(3)$).

Proof. Take $q_1 = \cos(\theta_1/2) + i \sin(\theta_1/2)$ (rotation about the x-axis) and $q_2 = \cos(\theta_2/2) + j \sin(\theta_2/2)$ (rotation about the y-axis). Then $q_1 q_2 \neq q_2 q_1$ (since $ij = k$ and $ji = -k$). Hence the conjugation actions do not commute. □

2.6.5 Function-theoretic features: the three-dimensional rotation equation

Theorem 2.16 (The group with periodic orbits of the rotation equation on the quaternions). *Let $f : \mathbb{H} \to Y$ and $q \in S^3$. If*

$$f(qvq^{-1}) = f(v), \qquad \forall v \in \mathsf{Im}(\mathbb{H}),$$

then the group with periodic orbits

$$\mathsf{Group}_{\mathsf{period},S^3}(f) = \{q \in S^3 : f(qvq^{-1}) = f(v),\ \forall v \in \mathsf{Im}(\mathbb{H})\}$$

is a subgroup of $S^3 \cong SU(2)$. If q is an element of finite order (that is, the rotation angle is $2\pi p/r$), then $\langle q \rangle \cong C_r$ or some other finite subgroup such as a dihedral group.

Proof. From $f(qvq^{-1}) = f(v)$ we obtain $f(q^n v q^{-n}) = f(v)$ for every $n \in \mathbb{Z}$, so $\langle q \rangle \leq \mathsf{Group}_{\mathsf{period},S^3}(f)$. The classification of the finite subgroups of S^3 includes cyclic, dihedral, tetrahedral, octahedral and icosahedral groups, so the group with periodic orbits can be one of these finite groups. □

Remark 2.13 (The quaternions make the three-dimensional rotation equation algebraisable). In the complex numbers the rotation equation $f(e^{i\theta}z) = f(z)$ characterises only planar rotations; the three-dimensional rotation equation $f(qvq^{-1}) = f(v)$ requires a non-commutative multiplication structure, and the quaternions $S^3 \cong SU(2)$ provide this structure. This is the final step by which number-system extension generalises a function equation from 'planar rotation' to 'three-dimensional rotation'.

Proposition 2.18 (Directionality and periodicity of quaternions). The non-commutativity of the quaternions extends the directional structure from a single rotation direction to non-commutative compositions of three-dimensional rotation directions:

(i) **Lifting of rotational symmetry**: from $U(1)$ for the complex numbers (one-dimensional rotation) to $SU(2)$ (the double covering of three-dimensional rotations).

(ii) **Non-commutativity of the directional structure**: rotation directions about different axes no longer commute.

(iii) **Conjugate period**: quaternion conjugation $q \mapsto \bar{q}$, together with the multiplicative inversion symmetry σ_1 and the additive reflection symmetry σ_0, forms a new type of symmetry—conjugate symmetry.

(iv) **Conserved quantity**: three-dimensional angular momentum. The Casimir operator of $SU(2)$ corresponds to the square of the total angular momentum.

Proof. (i) By Theorem 2.14, $S^3 \cong SU(2)$ doubly covers $SO(3)$; the one-dimensional rotation group $U(1) \cong \{\cos(\theta/2) + i\sin(\theta/2)\} \subset S^3$ is an embedded subgroup, so rotational symmetry is lifted from $U(1)$ to $SU(2)$. (ii) By Theorem 2.15, taking the unit quaternions $q_1 = \cos(\theta_1/2) + i\sin(\theta_1/2)$ and $q_2 = \cos(\theta_2/2) + j\sin(\theta_2/2)$ about the x- and y-axes, we have $q_1q_2 \neq q_2q_1$, so rotation directions about different axes do not commute. (iii) Conjugation $c(q) = \bar{q}$ satisfies $c(c(q)) = q$ (an involution) and $c(q_1q_2) = c(q_2)c(q_1)$ (an anti-automorphism); on the purely imaginary space $\mathrm{Im}(\mathbb{H})$ we have $c(v) = \bar{v} = -v$, which is the additive reflection σ_0, while on S^3 we have $c(q) = \bar{q} = q^{-1}$, which is the multiplicative inversion σ_1; hence c is a generalised composite of σ_0 and σ_1, forming conjugate symmetry. (iv) The Casimir operator $J^2 = J_x^2 + J_y^2 + J_z^2$ of $\mathfrak{su}(2)$ commutes with all three generators J_x, J_y, J_z, so it is invariant under the action of $SU(2)$ and corresponds to the conserved quantity given by the square of the total angular momentum (three-dimensional angular momentum). ☐

2.7 Logarithms: structural conversion between the proportional direction and the translation direction

2.7.1 Determination requirement

The scaling orbit

$$\mathcal{O}(x_0; q) = \{x_0 q^n \mid n \in \mathbb{Z}\}$$

and the translation orbit

$$\mathcal{O}(y_0; T) = \{y_0 + nT \mid n \in \mathbb{Z}\}$$

are completely parallel in structure. This parallelism requires a rigorous mathematical conversion before proportional periods and translation periods can be treated uniformly at the level of determination.

2.7.2 Definition and basic properties of the logarithm

Definition 2.10 (Definition of logarithm). Let $b > 0, b \neq 1, x > 0$. Then $y = \log_b x \iff b^y = x$. In particular, the natural logarithm is $\ln x = \log_e x$.

Remark 2.14 (The order of definition of real exponential powers). The rigorous definition of b^y $(y \in \mathbb{R})$ in Definition 2.10 depends on the completeness of the reals, and the logical order is: integer powers (Definition 2.1) → rational-exponent powers $b^{p/q} := (\sqrt[q]{b})^p$ (the q-th root of a positive real number exists and is unique by Theorem 2.7) → real-exponent powers $b^y := \lim_{n\to\infty} b^{q_n}$ (taking any sequence of rationals $q_n \to y$; the limit exists and is independent of the choice of approximating sequence, guaranteed by completeness and the monotonicity of the power function). Hence $y = \log_b x$ is well defined as the inverse function of the monotone function $y \mapsto b^y$, and no circular definition arises.

Theorem 2.17 (Logarithms: structural conversion between the proportional direction and the translation direction). *The logarithm converts operations on a multiplicative group into operations on an additive group:*

$$\log_b(x \cdot y) = \log_b x + \log_b y, \quad \log_b(x^e) = e \cdot \log_b x.$$

Specifically:

(i) $\log_b(1) = 0$*—the multiplicative centre of symmetry* 1 *maps to the additive centre of symmetry* 0*.*

(ii) $\log_b(x^{-1}) = -\log_b x$*—the inverse symmetry of multiplication (1-symmetry) is converted into the inverse symmetry of addition (0-symmetry).*

(iii) $\log_b(x/y) = \log_b x - \log_b y$*—multiplication and division are converted into addition and subtraction.*

Proof. Let $u = \log_b x, v = \log_b y$, that is, $b^u = x, b^v = y$. Then (i) $x \cdot y = b^u \cdot b^v = b^{u+v}$, so $\log_b(x \cdot y) = u + v$. (ii) $\log_b(1) = 0$ because $b^0 = 1$. (iii) $x^{-1} = (b^u)^{-1} = b^{-u}$, so $\log_b(x^{-1}) = -u$. (iv) $x/y = x \cdot y^{-1} = b^{u-v}$, so $\log_b(x/y) = u - v$. ☐

Theorem 2.18 (Logarithmic group isomorphism and conversion of conserved quantities). *The logarithmic map* $\ln : (\mathbb{R}^+, \times) \to (\mathbb{R}, +)$ *is a Lie group isomorphism. It converts*

(i) **operations**: *multiplication and division are converted into addition and subtraction.*

(ii) **symmetry**: *1-symmetry (the multiplicative inverse* σ_1*) is converted into 0-symmetry (the additive inverse* σ_0*).*

(iii) **conserved quantities**: conserved quantities under the proportional direction are converted into conserved quantities under the translation direction.

Proof. The isomorphism has already been proved in Theorem 2.17. Conversion of conserved quantities: suppose there is a conserved quantity Q_{scale} on $(\mathbb{R}^+, \times)$ satisfying $Q_{\text{scale}}(rx) = Q_{\text{scale}}(x)$. Define $Q_{\text{trans}}(y) = Q_{\text{scale}}(e^y)$; then

$$Q_{\text{trans}}(y + \ln r) = Q_{\text{scale}}(e^{y+\ln r}) = Q_{\text{scale}}(r \cdot e^y) = Q_{\text{scale}}(e^y) = Q_{\text{trans}}(y).$$

So Q_{trans} is invariant under translation, that is, it is a conserved quantity under the translation direction. □

Corollary 2.7 (Logarithmic discrimination equivalence)**.** *The logarithmic isomorphism makes the closure determination of a scaling orbit equivalent to the closure determination of a translation orbit in the logarithmic domain. The proportional quasi-period and the translation quasi-period are mathematically isomorphic, but physically they correspond to different conserved quantities.*

Proof. By Theorem 2.18, $\ln : (\mathbb{R}^+, \times) \to (\mathbb{R}, +)$ is a group isomorphism, and the scaling orbit $\{x_0 q^k \mid k \in \mathbb{Z}\}$ is mapped to the translation orbit $\{\ln x_0 + k \ln q \mid k \in \mathbb{Z}\}$. Since an isomorphism is a bijection preserving the operations, $x_0 q^k = x_0 \iff k \ln q = 0 \iff \ln x_0 + k \ln q = \ln x_0$; hence the scaling orbit closes if and only if the translation orbit in the logarithmic domain closes, and the closure determinations of the two classes of orbit are equivalent. Mathematically the two are isomorphic via ln; the difference in conserved quantities is given by item (iii) of Theorem 2.18. □

2.7.3 Unification of the three transformations from the logarithmic viewpoint

Theorem 2.19 (Unification of the three transformations from the logarithmic viewpoint)**.** *The logarithm unifies transformations on a multiplicative group as addition. Let Φ be the multiplicative action $\Phi(x) = g \cdot x$ on a multiplicative group; then*

$$\log(\Phi(x)) = \log(x) + \log g,$$

that is, Φ appears in the logarithmic domain as the translation $T_{\log g}$. The three transformations are as follows:

(i) **Translation**: *$T_a(x) = x + a$ is itself an action on the additive group $(\mathbb{R}, +)$ and is the reference case in the logarithmic domain (the unified formula holds trivially in an additive group); no conversion is needed;*

(ii) **Scaling**: *$S_\lambda(x) = \lambda x$ $(\lambda > 0)$, $\ln(S_\lambda(x)) = \ln x + \ln \lambda$, which appears in the logarithmic domain as $T_{\ln \lambda}$;*

(iii) **Rotation**: *$R_\theta(z) = e^{i\theta} z$; the complex logarithm (any branch) gives $\log(R_\theta(z)) = \log z + i\theta$, which appears in the complex logarithmic domain as $T_{i\theta}$ (modulo $2\pi i$).*

Proof. (i) Translation is an action on the additive group: $T_a(x) = x + a$ is the degenerate case of the unified formula in which the 'logarithm' is the identity map, and is also the common destination of scaling and rotation after logarithmic conversion. (ii) $\ln(S_\lambda(x)) = \ln(\lambda x) = \ln x + \ln\lambda = T_{\ln\lambda}(\ln x)$ (Theorem 2.17). (iii) Taking any branch of the complex logarithm: $\log(R_\theta(z)) = \log(e^{i\theta} z) = \log z + \log(e^{i\theta}) = \log z + i\theta + 2\pi i k$ ($k \in \mathbb{Z}$ depending on the branch; for the multi-valuedness see Proposition 2.21), that is, the translation $T_{i\theta}$ modulo $2\pi i$ in the complex logarithmic domain. ☐

Proposition 2.19 (Unification of symmetry centres from the logarithmic viewpoint). *In the logarithmic domain the centres of symmetry of the three transformations are unified as* 0*:*

(i) **Translation**: *centre of symmetry* 0*, because* $T_0 = \mathrm{id}$.

(ii) **Scaling**: *centre of symmetry* 1*, but* $\ln 1 = 0$*, so the centre of symmetry in the logarithmic domain is* 0.

(iii) **Rotation**: *centre of symmetry* 1 *(the identity element on the unit circle), but* $\log 1 = 0$*, so the centre of symmetry in the logarithmic domain is* 0.

Proof. (i) $T_0(x) = x + 0 = x$, so 0 is the centre of symmetry of translation. (ii) $S_1(x) = 1 \cdot x = x$, so 1 is the centre of symmetry of scaling. $\ln 1 = 0$. (iii) $R_0(z) = e^{i \cdot 0} z = z$, so 1 (that is, $e^{i\cdot 0}$) is the centre of symmetry of rotation. $\log 1 = 0$. ☐

Proposition 2.20 (Unification of conserved quantities from the logarithmic viewpoint). *In the logarithmic domain the conserved quantities of the three transformations are unified as 'translation invariants':*

(i) **Translation**: *the conserved quantity is the translation invariant* P*, a manifestation of translation invariance.*

(ii) **Scaling**: *the conserved quantity is the proportional invariant* D*, which corresponds to a translation invariant in the logarithmic domain (Theorem 2.18).*

(iii) **Rotation**: *the conserved quantity is the rotation invariant* L*, which corresponds to a translation invariant in the complex logarithmic domain.*

Proof. (i) Translation invariance: $P(x + a) = P(x)$. (ii) By Theorem 2.18, the scaling conserved quantity Q_{scale} is converted in the logarithmic domain into the translation conserved quantity Q_{trans}. (iii) Rotation invariance $L(e^{i\theta} z) = L(z)$ appears as translation invariance in the complex logarithmic domain. ☐

Remark 2.15 (Hierarchical progression of transformations and logarithms). The logarithm reveals the hierarchical progression of transformations:

- **First level: addition (translation)**. The composite of n translations produces multiplication: $n \times a = a + a + \cdots + a$.
- **Second level: multiplication (scaling)**. The composite of n scalings produces a power: $\lambda^n = \lambda \times \lambda \times \cdots \times \lambda$. The logarithm converts scaling into translation: $\ln(\lambda^n) = n \ln \lambda$.

- **Third level: powers (rotation).** When $|\lambda| = 1$, $\lambda^n = e^{in\theta}$ is a rotation. The complex logarithm converts rotation into translation: $\log(e^{in\theta}) = in\theta$.

Core conclusion: in the logarithmic domain all three transformations are unified as addition (translation), the centre of symmetry is unified as 0, and the conserved quantities are unified as translation invariants. Rotation is closed under the power operation and is the highest level of the hierarchical progression of transformations.

Proposition 2.21 (The i-symmetry of the complex logarithm: multi-valuedness and periodic structure)**.** *The complex logarithm $\log z = \ln|z| + i(\arg z + 2\pi k)$ $(k \in \mathbb{Z})$ is a multi-valued function, different branches differing by $2\pi i$. This multi-valuedness arises precisely from the $2\pi i$-periodicity of the complex exponential function—the periodic oscillation produced by i-symmetry in the exponential domain appears in the logarithmic domain as the periodic repetition of branches. The logarithm converts the rotation period of the exponential domain into a translation period on the additive group (with step $2\pi i$).*

Proof. Let $\log z = \ln|z| + i \arg z$ and take the value w of a definite branch; then $e^w = z$. Since $e^{2\pi ik} = \cos(2\pi k) + i\sin(2\pi k) = 1$ $(k \in \mathbb{Z})$, we have $e^{w+2\pi ik} = e^w e^{2\pi ik} = z$, so $w + 2\pi ik$ is also a value of $\log z$, different branches differing by $2\pi i$; the multi-valuedness arises in this way. Conversely, this translation is precisely the dual of the $2\pi i$-periodicity of the complex exponential: in the exponential domain, multiplication by i (adding $\pi/2$ to the argument, i-symmetry) accumulates four times to return to the starting point, producing a 2π rotation period; the logarithm converts this into a translation period on the additive group with step $2\pi i$. ☐

2.7.4 Rigorous formulation of rotation arising from scaling

Proposition 2.22 (Rigorous formulation of rotation arising from scaling)**.** *In the complex field, every complex number $z = re^{i\theta}$ can be decomposed as the composite of a scaling r and a rotation $e^{i\theta}$. Rotation is the restriction of scaling to the unit circle:*

(i) **Scaling**: *$S_r(z) = rz$, changing the modulus and preserving the argument.*

(ii) **Rotation**: *$R_\theta(z) = e^{i\theta}z$, changing the argument and preserving the modulus.*

(iii) **Composite**: *$z \mapsto re^{i\theta}z = S_r \circ R_\theta(z)$.*

When $r = 1$, scaling degenerates to the identity transformation and rotation becomes the only operation.

Proof. (i) $S_r(z) = rz = re^{i\arg z}$ $(z \neq 0)$: the modulus becomes $r|z|$, and the argument is unchanged. (ii) $R_\theta(z) = e^{i\theta}z = |z|e^{i(\arg z+\theta)}$: the modulus is unchanged, and the argument increases by θ. (iii) $S_r \circ R_\theta(z) = S_r(e^{i\theta}z) = re^{i\theta}z$. When $r = 1$, $S_1(z) = z$ is the identity transformation, and $re^{i\theta}z = e^{i\theta}z = R_\theta(z)$. ☐

Remark 2.16 (Independence of rotation)*.* Although rotation can be regarded as the 'boundary case' of scaling (multiplication by a positive real number) on the unit circle (multiplication by a unit complex number), rotation is an **independent transformation**:

- scaling changes 'size', while rotation changes 'direction';
- the scaling group $(\mathbb{R}^+, \times)$ and the rotation group $U(1)$ are two different subgroups of the complex multiplicative group $\mathbb{C}^\times$;
- $\mathbb{C}^\times \cong \mathbb{R}^+ \times U(1)$ (polar decomposition), so scaling and rotation are independent degrees of freedom.

The logarithm converts scaling into a translation on the real axis, and the complex logarithm converts rotation into a translation on the imaginary axis; in the logarithmic domain the two correspond respectively to the addition of the real part and of the imaginary part. The logarithm, however, is an isomorphism onto the additive group, not the additive group itself, and the two coordinates it produces can be written as a linear map; the next four results make that map explicit, and the final property collects the preconditions under which alone it is well posed.

Definition 2.11 (Logarithmic coordinates). For $z \in \mathbb{C}^\times = \mathbb{C} \setminus \{0\}$ fix the polar decomposition $z = |z|e^{i \arg z}$ and define the **logarithmic coordinates**

$$\Lambda(z) = \big(\ln|z|,\ \arg z\big) \in \mathbb{R} \times \mathbb{R}/2\pi\mathbb{Z}.$$

Here $\ln|z|$ is the real logarithm of a positive real number and is uniquely determined by Definition 2.10; the argument $\arg z$ is merely the equivalence class modulo 2π, and fixing one of its representatives is called a **choice of branch**. Once the branch is fixed, Λ is a single-valued map. Λ is a bijection: its inverse is $(\rho, \theta) \mapsto e^\rho e^{i\theta}$, so every $z \neq 0$ corresponds one-to-one with its logarithmic coordinates. Proposition 2.21 shows that different branches differ by $2\pi i$, so the single-valuedness of Λ depends on the branch being fixed. In logarithmic coordinates complex multiplication appears as addition:

$$\Lambda(zw) = \Lambda(z) + \Lambda(w) = \big(\ln|z| + \ln|w|,\ \arg z + \arg w\big),$$

which is precisely the coordinate form of Theorem 2.19 in the complex case: Λ turns the action on the multiplicative group $\mathbb{C}^\times$ into the action of coordinatewise translation on $\mathbb{R} \times \mathbb{R}/2\pi\mathbb{Z}$.

Theorem 2.20 (Decoupling of complex multiplication). With the branch fixed (so that Λ is single-valued, Definition 2.11), the modulus map and the normalisation map

$$z \mapsto |z|, \qquad z \mapsto \frac{z}{|z|},$$

give a topological group isomorphism

$$\mathbb{C}^\times \ \cong\ \mathbb{R}_{>0} \times U(1), \qquad z \ \longmapsto\ \big(|z|,\ e^{i \arg z}\big),$$

whose inverse is $(r, w) \mapsto rw$. Under this isomorphism, multiplication in $\mathbb{C}^\times$ splits into two independent coordinates: moduli are multiplied (multiplication in $\mathbb{R}_{>0}$) and arguments are added (multiplication in $U(1)$). The isomorphism is global (it holds for the whole of $\mathbb{C}^\times$) and not merely a local isomorphism in a neighbourhood of the origin; it raises the conclusion of Theorem 2.19—that scaling and rotation both become translations under the logarithm—to a direct-product decomposition of group structure.

Proof. First verify bijectivity and the homomorphism property. The map $\Phi(z) = (|z|, z/|z|)$ is defined for $z \in \mathbb{C}^\times$, its first component lies in $\mathbb{R}_{>0}$ and its second in $U(1)$. If $\Phi(z) = \Phi(w) = (r, w)$, then $z = |z| \cdot z/|z| = rw = w$, so Φ is injective; for any $(r, w) \in \mathbb{R}_{>0} \times U(1)$, take $z = rw \neq 0$; then $|z| = r$ and $z/|z| = w$, so Φ is surjective, with inverse map $(r, w) \mapsto rw$.

Next verify the multiplicative structure: for $z, w \in \mathbb{C}^\times$,

$$|zw| = |z|\,|w|, \qquad \frac{zw}{|zw|} = \frac{z}{|z|} \cdot \frac{w}{|w|},$$

so $\Phi(zw) = \Phi(z)\Phi(w)$, that is, Φ is a group homomorphism; being bijective, Φ is a group isomorphism.

Finally verify that it is a topological isomorphism: the modulus $z \mapsto |z|$ and the argument $z \mapsto \arg z$ are continuous on the chosen branch, and the inverse map $(r, w) \mapsto rw$ is the restriction of complex multiplication, hence also continuous, so Φ is an isomorphism of topological groups. □

Theorem 2.21 (The matrix form of powers in logarithmic coordinates). Let $b = u+iv \in \mathbb{C}^\times$ $(u, v \in \mathbb{R})$ be the base and $z \in \mathbb{C}^\times$. In the basis $(\ln|z|, \arg z)$ of Definition 2.11 and under the chosen branch, the map $z \mapsto bz$ (that is, the multiplicative action of b) is given by the following real linear map:

$$\begin{pmatrix} \ln|z| \\ \arg z \end{pmatrix} \longmapsto \begin{pmatrix} u & -v \\ v & u \end{pmatrix} \begin{pmatrix} \ln|z| \\ \arg z \end{pmatrix} + \begin{pmatrix} \ln|b| \\ \arg b \end{pmatrix} = \begin{pmatrix} u\ln|z| - v\arg z + \ln|b| \\ v\ln|z| + u\arg z + \arg b \end{pmatrix},$$

in which the correction term $(\ln|b|, \arg b)$ is the translation term fixed by the choice of branch. The off-diagonal entries $\pm v$ of the matrix $M_b = \begin{pmatrix} u & -v \\ v & u \end{pmatrix}$ are precisely the algebraic carrier of 'the power coupling scaling into rotation, and rotation into scaling': when $v \neq 0$ the two coordinate axes interfere with each other, whilst when $v = 0$ the matrix is diagonal (Lemma 2.2). Its determinant is

$$\det M_b = u^2 + v^2 = |b|^2 \neq 0,$$

so M_b is invertible, consistently with the invertibility of $z \mapsto bz$ when $b \neq 0$.

The case of powers (two classes of preconditions must be distinguished): let $z \in \mathbb{C}^\times$.

(i) If $b \in \mathbb{Z}$ (in which case $v = 0$ and $M_b = bI_2 = uI_2$ is a pure scaling), then z^b is single-valued and the following equality holds modulo 2π:

$$\Lambda(z^b) = M_b\,\Lambda(z) \pmod{2\pi}.$$

(ii) If $b \in \mathbb{C} \setminus \mathbb{Z}$, then a branch of the logarithm must be fixed in accordance with items (ii) and (iii) of Property 2.1; taking the branch value $\log z = \ln|z| + i\arg z$ and setting $z^b := \exp(b\log z)$, we have the exact equality

$$\Lambda(z^b) = M_b\,L(z), \qquad L(z) = (\ln|z|, \arg z),$$

in which $L(z)$ is the real vector of the chosen branch; changing the branch is equivalent to $L(z) \mapsto L(z) + 2\pi k\,\mathbf{e}_2$.

Proof. From $b = u + iv = |b|e^{i\arg b}$ and the polar form $z = |z|e^{i\arg z}$ we obtain

$$bz = |b|\,|z|\,e^{i(\arg b + \arg z)},$$

so in logarithmic coordinates

$$\ln|bz| = \ln|b| + \ln|z|, \qquad \arg(bz) = \arg b + \arg z \pmod{2\pi}.$$

Separate the coordinates of b from those of z: $(\ln|b|, \arg b)$ is a translation vector depending only on b, the 'add a constant' term, which is real linear; whilst the matrix multiplying the coordinates of z is determined solely by the real and imaginary parts of b. Multiplying the complex number $u + iv$ by $\ln|z| + i\arg z$ and comparing real and imaginary parts,

$$(u + iv)(\ln|z| + i\arg z) = \big(u\ln|z| - v\arg z\big) + i\big(v\ln|z| + u\arg z\big),$$

gives the stated matrix form; in other words M_b is the matrix of the complex scalar multiplication $w \mapsto bw$ in the basis $\{1, i\}$, and its determinant $|b|^2$ is exactly the area-scaling factor imposed by multiplication by b.

The case of powers: (i) for $b = n \in \mathbb{Z}$ the value z^n is independent of the branch (item (ii) of Property 2.1), and $\ln|z^n| = n\ln|z|$, $\arg(z^n) = n\arg z \pmod{2\pi}$, so $\Lambda(z^n) = n\Lambda(z) \pmod{2\pi} = M_n\Lambda(z)$. (ii) For $b \in \mathbb{C} \setminus \mathbb{Z}$, fix a branch and its branch value $\log z = \ln|z| + i\arg z$ in accordance with item (iii) of Property 2.1; then

$$\log(z^b) = \log\big(\exp(b\log z)\big) = b\log z,$$

whose coordinate form is $\Lambda(z^b) = (u\ln|z| - v\arg z,\ v\ln|z| + u\arg z) = M_bL(z)$, an exact equality; under a change of branch $L(z) \mapsto L(z) + 2\pi k\mathbf{e}_2$.

Determinant: $\det M_b = u \cdot u - (-v) \cdot v = u^2 + v^2 = |b|^2$, which is non-zero because $b \neq 0$, so M_b is invertible. ☐

Lemma 2.2 (A real exponent produces no coupling). *If $b \in \mathbb{R} \setminus \{0\}$ (that is, $b = u$ and $v = 0$), then the matrix M_b of multiplication by b in logarithmic coordinates is the diagonal matrix*

$$M_b = \begin{pmatrix} u & 0 \\ 0 & u \end{pmatrix} = uI_2,$$

whose off-diagonal entries vanish: the scaling coordinate and the rotation coordinate do not interfere. In particular, a real-exponent power $z \mapsto z^b$ belongs to the pure scaling family—the modulus varies as $|z| \mapsto |z|^b = e^{b\ln|z|}$, whilst the argument is merely translated by the constant $\arg b \in \{0, \pi\}$ and is not *coupled to the scaling coordinate $\ln|z|$; when $b \in \mathbb{Z}$ this conclusion holds for every $z \in \mathbb{C}^\times$, whilst when $b \notin \mathbb{Z}$ a branch must be fixed or $a > 0$ imposed, in accordance with items (ii) and (iii) of Property 2.1.*

Proof. The number $b = u + iv$ is real if and only if $v = 0$; substituting into the matrix of Theorem 2.21 gives $M_b = uI_2$ with off-diagonal entries $\pm v = 0$. In this case $\ln|bz| = \ln|b| + \ln|z|$ and $\arg(bz) = \arg z + \arg b$, where $\arg b \in \{0, \pi\}$ is a constant, so the argument is altered only by the translation term and receives no contribution from $u\ln|z|$; that is, the action on the scaling direction (the real axis) is $u\ln|z|$ and the action on the rotation direction (the imaginary axis) is $u\arg z$, each expression containing one coordinate only, with no cross term. When $b > 0$

we have $\arg b = 0$ and the argument is unchanged modulo 2π; when $b < 0$ we have $\arg b = \pi$ and the argument is translated by π overall. The modulus and argument of a real-exponent power are read off coordinatewise from $L(z^b) = M_b L(z)$ with $v = 0$. ☐

Corollary 2.8 (The spiral criterion). In the basis and branch of Theorem 2.21, let $b = u + iv \in \mathbb{C}^\times$ and $z \in \mathbb{C}^\times$, and consider the orbit of the power sequence $\{z\, b^n \mid n \in \mathbb{Z}\}$. Its logarithmic-coordinate orbit is the straight line

$$\Lambda(zb^n) = \Lambda(z) + n\,\Lambda(b) = \Lambda(z) + n\,(\ln|b|,\ \arg b),$$

and its image under the inverse map exp in the complex plane is as follows:

(i) If $v \neq 0$ (that is, $b \notin \mathbb{R}$), then $\arg b \not\equiv 0 \pmod{\pi}$ and the argument coordinate is necessarily non-zero: when $|b| \neq 1$ we have $\ln|b| \neq 0$ and the line in logarithmic coordinates is neither horizontal nor vertical; its image is a discrete sampling of an **equiangular spiral** (logarithmic spiral)—the orbit exhibits radial scaling and angular rotation at once, the two being coupled by the non-zero v, so that $v \neq 0$ is called the **spiral criterion**; when $|b| = 1$ the line is vertical and the case degenerates to the critical case of item (iii).

(ii) If $v \neq 0$ and $|b| \neq 1$, then the modulus $|z||b|^n$ is strictly monotone, the orbit does not close (it has no least positive period) and lies on no circle centred at the origin; when $|b| < 1$ the orbit tends to 0, and when $|b| > 1$ it tends to ∞.

(iii) If $v \neq 0$ and $|b| = 1$, then the orbit lies on the circle of radius $|z|$ centred at the origin; here there is no radial coupling, and this is the critical case in which rotation degenerates to a pure phase: by Theorem 2.5, the orbit closes when $\arg b/(2\pi) \in \mathbb{Q}$ and is dense on the circle without closing when $\arg b/(2\pi) \notin \mathbb{Q}$.

(iv) If $v = 0$ (b real): when $|b| \neq 1$ all points of the orbit lie on the same line through 0 and z (pure scaling, no rotational component); when $b = 1$ the orbit degenerates to the single point $\{z\}$; when $b = -1$ the orbit is the two points $\{z, -z\}$, with least positive period 2.

Proof. The first formula follows directly from $\Lambda(zb^n) = \Lambda(z) + n\Lambda(b)$ (multiplication becomes addition under Λ, Definition 2.11); the geometry of the image then follows because the inverse of Λ is exp.

(i) The condition $v \neq 0$ ensures that $\arg b \not\equiv 0 \pmod{\pi}$, so the second component of the direction is non-zero; when $|b| \neq 1$ the first component $\ln|b|$ is non-zero as well, and both components of the direction of the line are therefore non-zero (the vanishing of $\ln|b|$ is exactly the critical case of (iii)). Hence the image $\{e^{n \ln|b|} e^{in \arg b}\}$ has modulus varying as the geometric progression $|b|^n$ and argument varying as the arithmetic progression $n \arg b$, that is, an equiangular spiral.

(ii) When $|b| \neq 1$ the quantity $|b|^n$ is strictly monotone and never equals 1, so the modulus varies strictly along the orbit: if the orbit had a least positive period $q > 0$, then $b^q = 1$ and hence $|b|^q = 1$, contradicting $|b| \neq 1$; a non-constant modulus likewise shows that the orbit lies on no circle centred at the origin. When $|b| < 1$ and when $|b| > 1$ the quantity $|b|^n$ tends to 0 and to ∞ respectively.

(iii) When $|b| = 1$ we have $\ln |b| = 0$, and every point of the orbit has modulus $|z|$, so the orbit lies on the circle of radius $|z|$; the alternative of closure or denseness then follows from the commensurability criterion of Theorem 2.5 applied to the rotation angle $\arg b$.

(iv) When $v = 0$ we have $\arg b \in \{0, \pi\}$, so the argument of b^n is always 0 (when $b > 0$), or is 0 or π according to the parity of n (when $b < 0$); that is, b^n is real and the orbit lies on the line through 0 and z. When $b = 1$ we have $b^n = 1$ identically; when $b = -1$ the value b^n is ± 1 according to the parity of n, so the orbit is $\{z, -z\}$ with least positive period 2. When $|b| \neq 1$ and $b \neq -1$ the modulus $|z||b|^n$ is strictly monotone, so there is no positive period. □

Property 2.1 (The convergence conditions for powers (preconditions for well--posedness)). The following conditions ensure that the statements of Definition 2.11, Theorem 2.20, Theorem 2.21 and Lemma 2.2 are meaningful and unique; none may be omitted:

(i) $z \in \mathbb{C}^\times$: the logarithmic coordinates $\Lambda(z) = (\ln |z|, \arg z)$ are defined only for $z \neq 0$; at $z = 0$ the argument is undefined and $(\ln |z|, \arg z)$ is unavailable.

(ii) z^b is single-valued when $b \in \mathbb{Z}$: the value z^n $(n \in \mathbb{Z})$ depends only on z, not on the chosen branch of the argument, and is therefore single-valued for every $z \in \mathbb{C}^\times$; as soon as $b \notin \mathbb{Z}$, the value z^b depends on the branch of $\log z$, different branches giving different values (Proposition 2.21). Hence if z^b is required to be single-valued, one must either restrict to $b \in \mathbb{Z}$, or fix a branch (for instance the principal branch Log, with $\arg z \in (-\pi, \pi]$), or restrict z to a simply connected region not crossing the branch cut.

(iii) Choice of branch of the complex logarithm: the 'matrix form' in logarithmic coordinates in Theorem 2.21 holds only after the branch has been fixed; under a change of branch the second coordinate of $L(z)$ is translated by $2\pi k$ $(k \in \mathbb{Z})$, the matrix M_b being unchanged whilst the translation term changes. Hence $L(z^b) = M_b L(z)$ is to be understood as holding 'on one fixed branch', or equivalently as an equality modulo 2π (Proposition 2.21).

(iv) $a^b = \exp(b \ln a)$ requires $a > 0$: if $a > 0$ and the real logarithm $\ln a$ is taken (uniquely determined by Definition 2.10 when $a \neq 1$, and $\ln a = 0$ when $a = 1$), then a^b has a unique real value; if $a \leq 0$ or $b \notin \mathbb{Z}$ is admitted, then a^b is either multi-valued or requires a branch to be specified, and the chosen branch is discontinuous along the branch cut, so real single-valuedness presupposes $a > 0$.

(v) The local versus the global in decoupling: the topological group isomorphism $\mathbb{C}^\times \cong \mathbb{R}_{>0} \times U(1)$ is global (Theorem 2.20), but writing the decoupling as single-valued coordinates Λ requires a choice of branch; on any simply connected region not containing 0 a holomorphic branch of the logarithm can be chosen, whilst on the whole of $\mathbb{C}^\times$ no continuous single-valued branch of the logarithm exists. Hence 'decoupling in coordinate form' is locally available, whilst 'decoupling in isomorphism form' holds globally.

(vi) Precondition for invertibility: $b \neq 0$. The matrix M_b is invertible if and only if $|b|^2 \neq 0$; when $b = 0$ the map $z \mapsto bz$ is the zero map, and the linear form in logarithmic coordinates degenerates accordingly.

Proof. (i) The argument $\arg z$ is read off from $z/|z|$, which is undefined when $z = 0$; the real logarithm $\ln |z|$ too is defined only for $|z| > 0$ (Definition 2.10).

(ii) For $n \in \mathbb{Z}$ and $z = re^{i\varphi}$ we have $z^n = r^n e^{in\varphi}$; replacing φ by $\varphi + 2\pi k$ gives $e^{in(\varphi+2\pi k)} = e^{in\varphi}e^{2\pi i nk} = e^{in\varphi}$ (since $nk \in \mathbb{Z}$), so the value is independent of the branch. Conversely, if $b \notin \mathbb{Z}$, then $e^{ib(\varphi+2\pi k)} = e^{ib\varphi}e^{2\pi ibk}$ and $e^{2\pi ibk} \neq 1$ for some k (since $bk \notin \mathbb{Z}$), so a change of branch alters the value of the function; the fact that different branches differ by $2\pi i$ is Proposition 2.21.

(iii) By Definition 2.11, a change of branch alters only the representative of $\arg z$; the matrix M_b is determined by the real and imaginary parts of b and contains no $\arg z$, so it is unchanged, whilst the translation term $(\ln |b|, \arg b)$ changes with the branch.

(iv) Let $a > 0$. If $a \neq 1$, then $y \mapsto a^y$ is strictly monotone and continuous, so a^b is unique by the limit definition (the remark on order following Definition 2.10); if $a = 1$, then $a^b = 1$ uniquely. If $a < 0$, write $a = |a|e^{i\pi}$; then $a^b = e^{b \ln |a|}e^{ib\pi}$ depends on the choice of $\arg a = \pi + 2\pi k$, and is single-valued only when $b \in \mathbb{Z}$.

(v) The global isomorphism is given by Theorem 2.20. Let $U \subset \mathbb{C}^\times$ be simply connected and not containing 0; then a continuous single-valued branch of arg can be chosen, making Λ single-valued on U. If a continuous single-valued branch of arg existed on the whole of $\mathbb{C}^\times$, then continuous variation once around the unit circle would give two different values of $\arg(1)$ (differing by 2π), a contradiction; so no global single-valued branch exists.

(vi) $\det M_b = |b|^2$, so M_b is invertible if and only if $b \neq 0$; when $b = 0$ the value $bz = 0$ does not lie in $\mathbb{C}^\times$, and the logarithmic coordinates are undefined. ☐

Example 2.3 (A concrete computation of power coupling). Take the point $z_0 = 2e^{i\pi/4}$ in logarithmic coordinates, so that $L(z_0) = (\ln 2, \pi/4)$.

(i) Pure scaling with a real base: take $b = 2$ ($u = 2$, $v = 0$, a real base, Lemma 2.2), so that $M_2 = 2I_2$ and modulo 2π we have $L(z_0^2) = 2L(z_0) = (\ln 4,\ \pi/2)$, whilst $z_0^2 = 4e^{i\pi/2} = 4i$; that is, the modulus is enlarged from 2 to 4 and the argument changes from $\pi/4$ to $\pi/2$, the two coordinates each being multiplied by 2 independently, with no cross term and hence no coupling.

(ii) Coupling with a non-real base: take $b = 1 + i$ ($u = v = 1$), so that $M_b = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ and

$$L(z_0^b) = M_b L(z_0) = \left(\ln 2 - \tfrac{\pi}{4},\ \ln 2 + \tfrac{\pi}{4}\right) \approx (0.6931 - 0.7854,\ 0.6931 + 0.7854) = (-0.0923,\ 1.4785),$$

that is, $z_0^{1+i} \approx e^{-0.0923}e^{i1.4785} = 0.9119\, e^{i1.4785} \approx 0.0840 + 0.9080\, i$. The term $-\pi/4$ in the first coordinate is precisely the contribution of 'rotation coupled into scaling' (the argument $\pi/4$ is multiplied into the modulus coordinate by $-v = -1$), and the term $+\ln 2$ in the second coordinate is precisely the contribution of 'scaling coupled into rotation' (the modulus coordinate $\ln 2$ is multiplied into the argument coordinate by $+v = +1$): the off-diagonal entries $\pm v$ make the two directions interfere, and the orbit is no longer separated along the coordinate axes.

(iii) Branch calibration: computing directly with the principal branch instead gives $\operatorname{Log} z_0 = \ln 2 + i\pi/4$ and $(1+i)\operatorname{Log} z_0 = (\ln 2 - \pi/4) + i(\ln 2 + \pi/4)$, in agreement with the coordinates of (ii); whilst under the branch correction $\log(z_0^b) = b \log z_0 + 2\pi i k$, different k merely translate the result by 2π in the imaginary direction, the matrix M_b being unchanged (item (iii) of Property 2.1).

(iv) Comparison with pure rotation: take $b = i$ $(u = 0,\ v = 1)$, so that $M_i = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ is the matrix of rotation through $90°$; it maps $L(z_0^2) = (\ln 4, \pi/2)$ to

$$L\big((z_0^2)^i\big) = M_i L(z_0^2) = \big(-\pi/2,\ \ln 4\big), \qquad (z_0^2)^i = e^{-\pi/2}(\cos \ln 4 + i \sin \ln 4),$$

the modulus coordinate 0 being absorbed by the entry -1 of the rotation matrix, leaving only the argument $\ln 4$; this is exactly the behaviour of $\ln|b| = 0$ and $\arg b = \pi/2$ when $b = i$: the power with a purely rotational base converts the scaling coordinate into a phase and the phase into a scaling coordinate, in sharp contrast with the diagonal case of (i).

Remark 2.17 ($e^{i\pi} + 1 = 0$: the meeting of five fundamental constants). *The decoupling structure of this section may be drawn together in a single equation:*

$$e^{i\pi} + 1 = 0.$$

It gathers five fundamental constants into one relation: 0 (the additive neutral element, the centre of symmetry of translation), 1 (the multiplicative neutral element, the centre of symmetry of scaling), e (the base of the exponential map, the carrier that converts translation into scaling in Theorem 2.19), i (the atom of rotation, the generator of the fourfold skeleton C_4, Definition 2.8), and π (the phase of the half turn, obtained by accumulating the quarter turn of i twice). Structurally, the equation says three things:

- **The atom of rotation is i, not -1:** *the element $-1 = e^{i\pi} = i^2$ is a half turn and can yield only the sign flip of order 2; to obtain the fourfold skeleton C_4 (and with it the quarter turn) one must adjoin i, which is precisely the conclusion of Definition 2.8 and Proposition 2.16; for the conditions under which 'the sign is a half turn' holds see Remark 2.11.*

- **$e^{i\pi}$ is the unique phase realisation of -1:** *the identity $e^{i\pi} = -1$ shows that the additive reflection is realised in $\mathbb{C}$ by a half-turn phase; hence the real elements of the unit circle $\{\pm 1\}$ in Theorem 2.20 are exactly the intersection of the real axis with $U(1)$, that is, $\mathbb{R} \cap U(1) = \{\pm 1\}$, which is also the image of the projection at $180°$.*

- **$0, 1, e, i, \pi$ each play their part under decoupling:** *the logarithmic coordinates divide multiplication into two axes, modulus and argument; 1 and 0 are the neutral elements of the two axes in the logarithmic domain (multiplicative and additive respectively), e effects the conversion between the two domains, i supplies the quarter turn of the rotation direction, and π marks the phase of the half turn; the matrix form of powers (Theorem 2.21) then writes the division of labour amongst these five constants as a linear coupling between the two coordinate axes.*

Thus $e^{i\pi} + 1 = 0$ is no isolated coincidence, but the phase expression of this chapter's main thread, that 'the introduction of inverse operations establishes the centres of symmetry level by level': the additive centre 0, the multiplicative centre 1 and the phase centre i appear simultaneously in one equation.

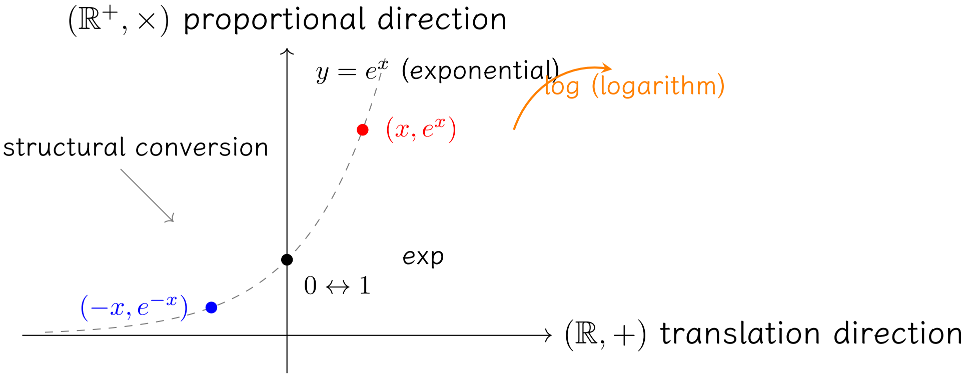


Figure 2.10: The structural conversion function of the logarithm: the exponential map exp converts translations in the additive group into proportions in the multiplicative group; the logarithm log is its inverse map and converts the proportional direction into the translation direction. The multiplicative centre of symmetry 1 corresponds to the additive centre of symmetry 0.

2.8 The period-determination spectrum of number-system extension

Theorem 2.22 (Correspondence between number-system extension and period-determination capability). *Let the time evolution of a system be given by a group action or a Hamiltonian flow. Then the number system required to describe the system and the period types it can determine or express stand in the following correspondence:*

$$\mathbb{Z} \Longrightarrow \text{can express discrete translation periods,}$$

$$\mathbb{Q} \Longrightarrow \text{can express finite rotation periods,}$$

$$\mathbb{R} \Longrightarrow \text{can express continuous translation periods and quasi-periods,}$$

$$\mathbb{C} \Longrightarrow \text{can express rotation periods and double periods,}$$

$$\mathbb{H} \Longrightarrow \text{can express three-dimensional rotation periods and non-commutative periods.}$$

Each step of extension provides a new algebraic tool for the period-determination problems that could not be expressed in the preceding number system.

Proof. Combining the criterion theorems of the preceding sections level by level:

- $\mathbb{Z}$: Theorem 2.2—discrete translation periods can be determined.
- $\mathbb{Q}$: Theorem 2.5—finite rotation periods can be determined.
- $\mathbb{R}$: Theorem 2.9—quasi-periods can be described.

- $\mathbb{C}$: Theorem 2.12 and Theorem 2.13—rotation periods and double periods are completely algebraised.
- $\mathbb{H}$: Theorem 2.14 and Theorem 2.15—three-dimensional rotation periods and non-commutative periods can be algebraised.

The extensions cannot be reversed level by level, because the closure newly added at each level precisely solves the period-determination problem that could not be expressed at the preceding level. □

Theorem 2.23 (The correspondence between number-system extensions and the solvability of function equations)**.** *Each level of number-system extension makes exactly one class of function equation pass from 'unsolvable' or 'non-closed' to 'solvable', with its group with periodic orbits becoming a group:*

(i) *$\mathbb{Z}$ makes the translation equation $f(x+a)=f(x)$ iterable in both directions, and the group with periodic orbits becomes the subgroup $a\mathbb{Z}$; the antiperiod equation $f(x+a)=-f(x)$ implies period doubling to $2a$.*

(ii) *$\mathbb{Q}$ makes the scaling equation $f(qx)=f(x)$ iterable in both directions, and the group with periodic orbits becomes the subgroup $\langle q\rangle$; the commensurability criterion $\theta/(2\pi)\in\mathbb{Q}$ can be expressed exactly.*

(iii) *$\mathbb{R}$ gives the continuous translation equation $f(x+t)=f(x)$ $(t\in\mathbb{R})$ a rigorous topological meaning, and the group with periodic orbits is a closed subgroup; the denseness of quasi-periodic orbits can be described rigorously.*

(iv) *$\mathbb{C}$ makes the rotation equation $f(e^{i\theta}z)=f(z)$ algebraisable, and the group with periodic orbits is a subgroup of $U(1)$; finite rotation periods are characterised exactly by the unit-root equation $z^n=1$.*

(v) *$\mathbb{H}$ makes the three-dimensional rotation equation $f(qvq^{-1})=f(v)$ algebraisable, and the group with periodic orbits is a finite subgroup of $S^3\cong SU(2)$.*

Proof. Each item follows from the function-theoretic features of the preceding sections: $\mathbb{Z}$ from Theorem 2.1, $\mathbb{Q}$ from Theorem 2.3 and 2.4, $\mathbb{R}$ from Theorem 2.8 and Proposition 2.11, $\mathbb{C}$ from Theorem 2.11 and Corollary 2.6, and $\mathbb{H}$ from Theorem 2.16. □

Corollary 2.9 (Period-diagnostic significance of number-system extension)**.** *In the diagnosis of periodic mathematics, the hierarchy of number-system extension provides a determination tool:*

(i) *If a group action for the system can be constructed only on $\mathbb{Z}$, then the period types that can be determined are limited to discrete translation periods.*

(ii) *If the system requires $\mathbb{Q}$ to express commensurability, then finite rotation periods can be determined.*

(iii) *If the system requires $\mathbb{R}$ to express denseness and continuity, then quasi-periods can be described.*

Table 2.1: Solvability of three classes of function equation in each number system and the structure of the group with periodic orbits

Number system	Translation $f(x+a)=f(x)$	Scaling $f(qx)=f(x)$	Rotation $f(e^{i\theta}z)=f(z)$
$\mathbb{N}$	has solutions, period set not a group	non-invertible	no carrier
$\mathbb{Z}$	group with periodic orbits $a\mathbb{Z}$	non-invertible ($q^{-1}\notin\mathbb{Z}$)	no carrier
$\mathbb{Q}$	group with periodic orbits $a\mathbb{Z}$	group with periodic orbits $\langle q\rangle$	commensurability criterion expressible
$\mathbb{R}$	group with periodic orbits a closed subgroup	translation in the logarithmic domain	still no algebraic carrier
$\mathbb{C}$	group with periodic orbits a closed subgroup	translation in the logarithmic domain	group with periodic orbits a subgroup of $U(1)$, exact via $z^n=1$
$\mathbb{H}$	group with periodic orbits a closed subgroup	translation in the logarithmic domain	group with periodic orbits a finite subgroup of $S^3\cong SU(2)$

Table 2.2: Unified spectrum of number-system extension, determination requirements and period types

Number system	Determination requirement	Newly added closure	Period type	Conserved quantity
$\mathbb{N}$	counting, indexing	no inverse operations	no period	—
$\mathbb{Z}$	two-directional extension of translation orbits	subtraction (additive inverse)	discrete translation period	momentum
$\mathbb{Q}$	deciding commensurability	division (multiplicative inverse)	finite rotation period	angular momentum
$\mathbb{R}$	denseness, continuity	completeness	quasi-period, continuous period	momentum, phase volume
$\mathbb{C}$	algebraisation of rotation	imaginary unit i	rotation period, double period	angular momentum
$\mathbb{H}$	algebraisation of three-dimensional rotation	non-commutative multiplication	three-dimensional rotation period	three-dimensional angular momentum

(iv) If the system requires $\mathbb{C}$ to express the algebraisation of rotation and the period lattice, then rotation periods or double periods can be described.

(v) If the system requires $\mathbb{H}$ to express three-dimensional rotation, then three-dimensional rotation periods can be described.

This diagnostic principle is a direct application of the theory of number-system extension to period determination.

Proof. Substituting item by item from Theorem 2.22 and Theorem 2.23 gives—the required number system determines the period types that can be expressed or determined and the solvability of the function equations: (i) If only $\mathbb{Z}$ can be used ($\mathbb{N}$ has no inverse operations and does not directly provide a tool for period determination, see Proposition 2.1), then $\mathbb{Z}$ corresponds to the discrete translation period, and the group with periodic orbits of the function equation $f(x+a) = f(x)$ is $a\mathbb{Z}$. (ii) If $\mathbb{Q}$ is required, that is, multiplicative inverses are required in order to express commensurability $\theta/(2\pi) \in \mathbb{Q}$, then $\mathbb{Q}$ corresponds to the finite rotation period, and the group with periodic orbits of the function equation $f(qx) = f(x)$ is $\langle q \rangle$. (iii) If $\mathbb{R}$ is required, that is, completeness is required in order to express denseness and continuity, then $\mathbb{R}$ corresponds to the quasi-period, and the group with periodic orbits of the continuous translation equation is a closed subgroup. (iv) If $\mathbb{C}$ is required, that is, algebraic closure is required in order to give the complete solution set of $z^n = 1$ and the period lattice Λ, then $\mathbb{C}$ corresponds to the rotation period or the double period, and the group with periodic orbits of the rotation equation is a subgroup of $U(1)$. (v) If $\mathbb{H}$ is required, that is, non-commutative multiplication is required in order to express three-dimensional rotation, then $\mathbb{H}$ corresponds to the three-dimensional rotation period, and the group with periodic orbits of the three-dimensional rotation equation is a finite subgroup of $SU(2)$. □

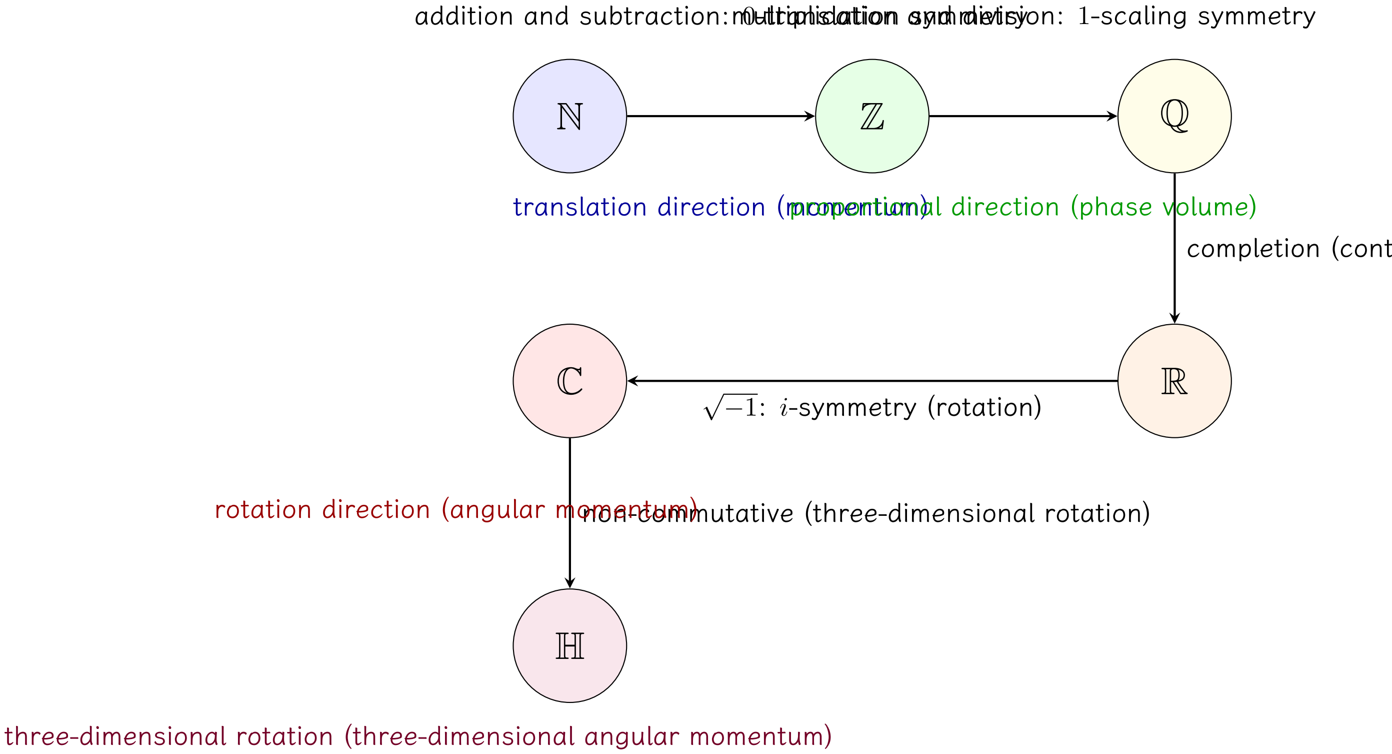

addition and subtraction: 0-translation symmetry
multiplication and division: 1-scaling symmetry
ℕ
ℤ
ℚ
translation direction (momentum)
proportional direction (phase volume)
completion (cont
ℂ
ℝ
$\sqrt{-1}$: i-symmetry (rotation)
rotation direction (angular momentum)
non-commutative (three-dimensional rotation)
ℍ
three-dimensional rotation (three-dimensional angular momentum)

Figure 2.11: The path of number-system extension and the progression of the transformation hierarchy (chapter summary diagram). Each step of extension establishes a new centre of symmetry by introducing a new inverse operation, thereby producing a new type of transformation: subtraction (0-symmetry) makes translation two-directional; division (1-symmetry) introduces scaling—repeated translation produces scaling; the imaginary unit (i-symmetry) introduces rotation—repeated scaling produces rotation; non-commutative multiplication introduces three-dimensional rotation. The logarithm converts scaling into translation, and the complex logarithm converts rotation into translation; in the logarithmic domain the three transformations are unified as addition. For the local diagram of $\mathbb{N} \to \mathbb{Z} \to \mathbb{Q}$, see Section 2.6 and its Figure 2.3.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	the translation period criterion for the integers (Theorem 2.2) and discrete translation periods (Corollary 2.3); the logarithm converting the proportional direction into the translation direction (Theorem 2.17); the groups with periodic orbits of the translation equation (Theorems 2.1, 2.8)
Scaling	the proportional direction of the rationals (Proposition 2.9) and the taking-the-inverse involution σ_1 (Definition 2.4); conversion of conserved quantities under the logarithmic isomorphism (Theorem 2.18); the group with periodic orbits of the scaling equation (Theorem 2.3)
Rotation	algebraisation of complex rotation (Theorem 2.10) and the criterion for finite rotation periods (Theorem 2.12); three-dimensional rotation by quaternions (Theorem 2.14); the groups with periodic orbits of the rotation equation (Theorems 2.11, 2.16)
Conservation	the closure of the operations of each number system (construction of the integers 2.2; algebraic closure of the complex numbers 2.13); multiplicativity of the quaternion norm and Casimir conservation (Proposition 2.18)
Symmetry	the emergence of the centres of symmetry $0, 1, i$: the additive involution (integers), the multiplicative involution σ_1 (rationals) and the i-involution (complex numbers, Definition 2.7)
Closure	two-step closure of an involution ($\sigma^2 = \mathrm{id}$); finite rotation period ⇔ closure of the roots of unity (Theorem 2.12)
Period	the correspondence between number-system extension and period-determination capability (Theorem 2.22), the correspondence between number-system extension and the solvability of function equations (Theorem 2.23) and the period-determination spectrum (Table 2.2): discrete translation / finite rotation / quasi-period / double period / non-commutative period

Exercises

Exercise 1 Prove the three assertions about the absence of inverse operations in the natural numbers in Proposition 2.1: under what conditions subtraction, division and root extraction have a natural-number solution, and under what conditions they have none. Further explain why the natural numbers cannot form a group, and explain the conclusion of Corollary 2.1 that 'the determination of discrete translation periods requires at least the group structure of $\mathbb{Z}$'.

Exercise 2 Fill in the details of the proof of the well-definedness of multiplication in Proposition 2.4, and verify that addition, multiplication, the zero element and inverses in the construction of the integers 2.2 really do not depend on the choice of representative of the equivalence class. Finally explain why the additive group of $\mathbb{Z}$ is the minimal algebraic structure that enables translation orbits to be extended in both directions and hence their closure to be determined (see Remark 2.4).

Exercise 3 In $\mathbb{Z}/12\mathbb{Z}$ compute the orders of the elements $[3]$, $[4]$, $[5]$ and $[7]$, and verify the formula of Lemma 2.1

$$\mathsf{ord}([T]) = \frac{12}{\mathsf{gcd}(T, 12)}.$$

Point out which elements are generators of $\mathbb{Z}/12\mathbb{Z}$. If 12 is understood as the period of the Earthly Branches, explain how the orders of these elements correspond to the sub-period structure within the period 12.

Exercise 4 Prove the assertion in Theorem 2.2 that a translation orbit degenerates to a single point in $\mathbb{Z}/m\mathbb{Z}$ if and only if $T \equiv 0 \pmod{m}$. Also construct a concrete example: take $m = 12$, $T \neq 0$, but the minimal period of the orbit in $\mathbb{Z}/12\mathbb{Z}$ is strictly less than 12. Compute the length of this orbit.

Exercise 5 Verify that $\mathbb{Q}$ forms a field (Proposition 2.7), and prove that the taking-the-inverse map

$$\sigma_1 : \mathbb{Q}^\times \to \mathbb{Q}^\times, \qquad \sigma_1(x) = x^{-1}$$

is an automorphism of the multiplicative group and an involution, with fixed-point set $\{1, -1\}$ (see Definition 2.4 and Proposition 2.8). Explain how σ_1 interchanges the operation of 'enlargement' with the operation of 'shrinkage'.

Exercise 6 Prove Theorem 2.5: if $\theta/(2\pi) = p/q$ is a rational number in lowest terms, then the rotation orbit

$$\mathcal{O}(z_0; \theta) = \{z_0 e^{in\theta} \mid n \in \mathbb{Z}\}$$

contains exactly q points and has period q; if $\theta/(2\pi) \notin \mathbb{Q}$, then the orbit does not close. Further explain why the criterion of 'commensurability' itself requires $\mathbb{Q}$ in order to be expressed exactly, whereas $\mathbb{R}$ can describe only its limits and denseness.

Exercise 7 Prove that the completeness of the reals closes the root operation (Theorem 2.7), and give an equation that has no solution in $\mathbb{Q}$ but has one in $\mathbb{R}$. Then explain why the description of the denseness of quasi-periodic orbits must rely on the completeness of $\mathbb{R}$ (see Proposition 2.11 and Theorem 2.9).

Exercise 8 Verify that the set of n-th roots of unity

$$\{\zeta_k = e^{2\pi ik/n} \mid k = 0, 1, \ldots, n-1\}$$

forms the cyclic group C_n under complex multiplication (Theorem 2.12). Write out all the roots of unity for $n = 8$ explicitly, and explain their correspondence with the rotation period C_8. Finally explain why $\mathbb{C}$ is the minimal field extension that makes the rotation equation algebraisable (see Remark 2.10).

Exercise 9 Prove that the map

$$\sigma_i : \mathbb{C}^\times \to \mathbb{C}^\times, \qquad \sigma_i(z) = iz^{-1}$$

is an involution and find its fixed points. Writing z in polar form $z = re^{i\varphi}$, explain that σ_i comprises simultaneously the taking of the reciprocal of the modulus and a reflection of the argument about the $\pi/4$ direction (see Definition 2.7 and Proposition 2.15).

Exercise 10 Prove that the quaternion unit sphere

$$S^3 = \{q \in \mathbb{H} \mid |q| = 1\}$$

through the conjugation action

$$\rho_q(v) = qvq^{-1}, \qquad v \in \mathrm{Im}(\mathbb{H}) \cong \mathbb{R}^3$$

realises three-dimensional rotations, and that the map

$$\Pi : S^3 \to SO(3), \qquad q \mapsto \rho_q$$

is a double covering with kernel $\{\pm 1\}$ (Theorem 2.14). Then give an example showing that rotations about different axes do not commute in general (Theorem 2.15).

Exercise 11 Verify that the logarithmic isomorphism

$$\ln : (\mathbb{R}^+, \times) \to (\mathbb{R}, +)$$

maps the multiplicative centre of symmetry 1 to the additive centre of symmetry 0, and converts conserved quantities under the proportional direction into conserved quantities under the translation direction (see Theorem 2.17, Theorem 2.18 and Proposition 2.19). Taking $f(x) = \sin(2\pi \log_q x)$ as an example, verify that $f(qx) = f(x)$, and write down its translation-period form in the variable $t = \ln x$.

Exercise 12 Synthesis question: using Theorem 2.22 and Table 2.2, explain why 'discrete translation period', 'finite rotation period', 'double period' and 'three-dimensional rotation period' require at least $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{C}$ and $\mathbb{H}$ respectively. Construct a counterexample showing that planar rotation periods cannot be algebraised in $\mathbb{R}$. Then discuss: if a system is defined only on $\mathbb{N}$, can its translation period be determined? If not, to which level must the number system be extended at least? Give reasons.

Exercise 13 Write down, in the logarithmic coordinates $(\ln|z|, \arg z)$, the matrix of multiplication by $b = u + iv$,

$$M_b = \begin{pmatrix} u & -v \\ v & u \end{pmatrix},$$

and explain the meaning of its off-diagonal entries $\pm v$ (Theorem 2.21). Taking $b = 1 + i$ and $z_0 = 2e^{i\pi/4}$, compute the logarithmic coordinates of z_0^b, and identify which term comes from 'rotation coupled into scaling' and which from 'scaling coupled into rotation' (see Example 2.3). Finally explain why the assertion that 'the power sequence $\{zb^n\}$ lies on an equiangular spiral' presupposes $v \neq 0$, and why the criterion degenerates, when $|b| = 1$, into the dichotomy of closure and denseness on a circle (see Corollary 2.8 and Theorem 2.5); and state which convergence conditions the single-valuedness of z^b requires (Property 2.1).

Chapter 3

Scalars and Vectors

Remark 3.1 (Positioning of this chapter). *In periodic mathematics, scalars and vectors are not presupposed concepts, but arise naturally from the symmetries of number systems. This chapter brings the three symmetry centres* (0, 1, i) *and the three directional structures (translation direction, proportional direction, rotation direction) established in the 'Operations and Periodicity' chapter, together with the operational duality established in the 'Number Systems' chapter, into the framework of group representation theory: scalars correspond to the trivial representation of the symmetry group, and vectors to non-trivial representations.*

One of the central tasks of this chapter is to realise the three basic generating transformations of the 'Operations and Periodicity' chapter—translation, scaling and rotation—as concrete operations in a vector space. On this basis, the chapter systematically establishes the complete spectrum of **similarity** *and* **tendency***—similarity is a measure of the directional relation that depends only on the arguments of the two vectors, whilst tendency is the product of the similarity and the moduli of the two vectors. The 12 similarities and the 12 tendencies are not an artificial enumeration, but a classification generated by the Cartesian product of the operational duality.* **It should be noted** *that the difference-angle family of similarities is defined strictly on* $\mathbb{R}^2 \cong \mathbb{C}$ *(a single argument is required), and the sum-angle family of similarities likewise requires a single argument; hence the similarity spectrum of this chapter takes* $\mathbb{R}^2 \cong \mathbb{C}$ *as its basic domain of definition. In* $\mathbb{R}^n$ $(n \geq 3)$ *one can define only the inner product, the modulus of the outer product and the included angle, but not the sum angle.*

Above the 12 basic similarities and tendencies, this chapter further introduces the **complex exponential similarity** *and the* **complex exponential tendency***, together with the* **difference-angle quaternion and sum-angle quaternion***. In the complex case the basic difference-angle cosine family and the basic difference-angle sine family are packaged into a single unit complex number, and left multiplication is a two-dimensional rotation operator; in the quaternion case the difference-angle quaternion and the sum-angle quaternion are defined according to the same 'angle difference/angle sum' logic, and they give three-dimensional rotation operators through the conjugation action. When the rotation axes coincide, the quaternion case degenerates strictly to the angle difference and angle sum of the complex case.*

Seven-theme guide for this chapter*: Translation—vector addition* $\mathsf{u}+\mathsf{v}$ *(Definition 3.4) and the commutative group it forms (Proposition 3.1), the modulus of the vector sum* $\mathsf{a}+\mathsf{b}$ *being the half-angle cosine tendency (Proposition 3.9); Scaling—the modulus scaling of scalar multiplication* $\alpha\mathsf{u}$ *(Definition 3.5) and the* r^{-2} *decay of the inverse-square law (Theorem 3.4); Rotation—the additive composition of arguments by complex multiplication (Proposition 3.11) and the three-dimensional rotational composition of quaternion multiplication (Proposition 3.13), the complex exponential similarity as rotation-difference operator (Proposition 3.12), and* the difference-angle quaternion as three-dimensional rotation-difference operator (Proposition 3.14); Conservation—the invariance of the cosine tendency (inner product) under the orthogonal group $O(n)$ (Proposition 3.7), the invariance of the absolute value of the sine tendency (the modulus of the outer product) under pure rotations $SO(3)$ (Proposition 3.8), and the conservation of the modulus by reflection together with the conservation of the sine tendency along the interface normal by refraction (Definition 3.23, 3.24); Symmetry—the reflection involution embodies additive reflection symmetry $c = 0$, whilst refraction scaling embodies the modulation of i-symmetry by multiplicative scaling reflection symmetry $c \neq 1$; Closure—two reflections return to the original $\mathbf{v}'' = \mathbf{v}$ and form a closed orbit of period 2, and the inverse-square law takes the conservation of flux on a closed surface as its criterion (Theorem 3.4); Period—the addition of arguments in complex multiplication makes the rotation angle compose with period 2π (Proposition 3.11), and the 2-periodic closed orbit of reflection is the smallest periodic structure of this chapter (Definition 3.23).

Definition 3.1 (Vector operations and the three basic generating transformations). Within the framework of a vector space, the three basic generating transformations of the 'Operations and Periodicity' chapter acquire the following concrete realisations:

(i) **Translation**: the addition of vectors $\mathbf{u}+\mathbf{v}$. One vector is moved along the direction of the other, and the modulus and direction of the resulting composite vector are determined by the parallelogram law. Translation corresponds to additive reflection symmetry (symmetry centre $c = 0$).

(ii) **Scaling**: the multiplication of a vector by a scalar $\alpha\mathbf{u}$. The modulus of the vector is enlarged or reduced in the ratio α, whilst the direction is kept unchanged ($\alpha > 0$) or reversed ($\alpha < 0$). Scaling corresponds to multiplicative scaling reflection symmetry (symmetry centre $c = 1$).

(iii) **Rotation**: the outer product of vectors $\mathbf{u} \times \mathbf{v}$ (in $\mathbb{R}^3$) or complex multiplication $z_1 z_2$ (in $\mathbb{R}^2 \cong \mathbb{C}$). The direction of the vector is changed by the rotation angle, and the modulus is determined by the product of the moduli of the two vectors and by the rotation angle. Rotation corresponds to multiplicative inversion symmetry (symmetry centre $c = 1$) and i-symmetry (symmetry centre $c = i$).

These three operations—vector addition (translation), scalar multiplication (scaling), vector outer product/complex multiplication (rotation)—constitute the three mutually irreducible basic operations on a vector space.

3.1 Definitions of Scalars and Vectors

Definition 3.2 (Scalar). Let G be a symmetry group and let $\rho : G \to \mathsf{GL}(V)$ be a representation of G on a vector space V. If ρ is the trivial representation, that is, $\rho(g) = \mathsf{id}_V$ for every $g \in G$, then the elements of V are called **scalars** with respect to G. A scalar is invariant under the group action, so it carries no directional information and participates in scaling operations only as a numerical value.

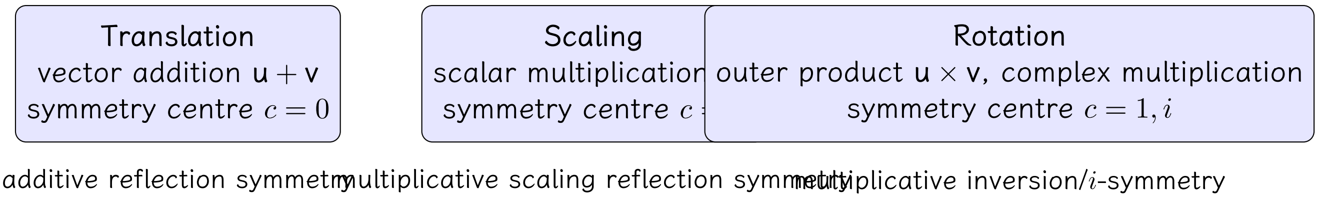

Translation
vector addition $\mathsf{u}+\mathsf{v}$
symmetry centre $c=0$
Scaling
scalar multiplicatior
symmetry centre c
Rotation
outer product $\mathsf{u}\times\mathsf{v}$, complex multiplication
symmetry centre $c=1,i$
additive reflection symmetry
multiplicative scaling reflection symmetry
multiplicative inversion/i-symmetry

Figure 3.1: The three basic generating transformations in a vector space: vector addition realises translation, scalar multiplication realises scaling, and the outer product/complex multiplication realises rotation. The three correspond respectively to additive reflection symmetry ($c=0$), multiplicative scaling reflection symmetry ($c=1$), and multiplicative inversion/i-symmetry ($c=1,i$).

The natural numbers $\mathbb{N}$ of the 'Number Systems' chapter provide the original model of a scalar—they possess only forward operations (addition, multiplication, exponentiation), have no inverse elements, and involve no directional duality, hence no symmetry centre either. The principal role of a scalar in a vector space is that of a **scaling factor**, realising the basic transformation of scalar multiplication.

Definition 3.3 (Vector)**.** Let G be a symmetry group and let $\rho : G \to \mathsf{GL}(V)$ be a representation of G. If ρ is not the trivial representation (that is, there exists at least one $g \in G$ with $\rho(g) \neq \mathsf{id}_V$), then an element v of V is called a **vector** with respect to G. A vector possesses a magnitude (norm $\|\mathsf{v}\|$) and a direction. The direction of a vector is characterised by the orbit of the group action—under the representation ρ, the direction of v is its position in the subspace spanned by $\{\rho(g)\mathsf{v} \mid g \in G\}$. This is the algebraic realisation of the concept of directionality in the 'Operations and Periodicity' chapter.

Between vectors one may perform **addition** (translation) and the **outer product/complex multiplication** (rotation), and between a vector and a scalar one may perform **scalar multiplication** (scaling). These three operations exhaust precisely all the basic symmetry operations on a vector space.

3.2 Basic operations in a vector space: translation, scaling and rotation

Definition 3.4 (Vector addition—translation)**.** Let $\mathsf{u} = (u_1, \dots, u_n)$, $\mathsf{v} = (v_1, \dots, v_n) \in \mathbb{R}^n$. Define **vector addition** (the translation operation):

$$\mathsf{u} + \mathsf{v} = (u_1 + v_1, \dots, u_n + v_n).$$

This is the componentwise generalisation of the addition operation of the 'Number Systems' chapter to a vector space, and also the vector realisation of the translation transformation of the 'Operations and Periodicity' chapter. In addition, define the **zero vector** $\mathbf{0} = (0, \dots, 0)$ (corresponding to the additive symmetry centre 0), and the **negative vector** $-\mathsf{u} = (-u_1, \dots, -u_n)$ (corresponding to the additive inverse, that is, translation in the reverse direction).

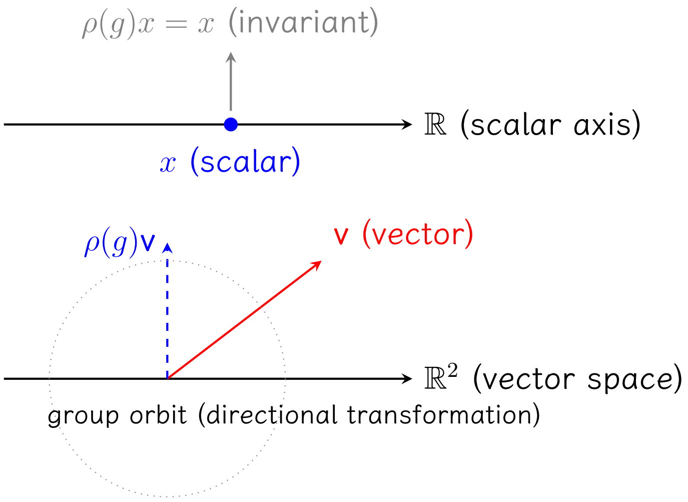

$\rho(g)x = x$ (invariant)
$\mathbb{R}$ (scalar axis)
x (scalar)
$\rho(g)$v
v (vector)
$\mathbb{R}^2$ (vector space)
group orbit (directional transformation)

Figure 3.2: The difference between scalars and vectors under a group action: a scalar is invariant under the group action (trivial representation, participating only in scaling); a vector changes its direction along the group orbit (non-trivial representation, participating in the three transformations of translation, scaling and rotation).

Table 3.1: Correspondence between vector types, symmetry groups and directional structures

Vector type	Scalar field	Symmetry centre	Directional structure	Symmetry group
Integer vector	$\mathbb{Z}$	0	Translation direction	C_2 (additive reflection)
Rational vector	$\mathbb{Q}$	1	Proportional direction	$\mathbb{G}_m$ (multiplicative scaling reflection)
Real vector	$\mathbb{R}$	$0, 1$	Translation + proportion	$\mathbb{R}^+$ (continuous direction)
Complex vector	$\mathbb{C}$	i	Rotation direction	$U(1)$ (taking the inverse)
Quaternion vector	$\mathbb{H}$	i, j, k	Three-dimensional rotation	$SU(2)$ (non-commutative rotation)

Definition 3.5 (Scalar multiplication—scaling)**.** Let $\mathsf{u} \in \mathbb{R}^n$, $\alpha \in \mathbb{R}$. Define **scalar multiplication** (the scaling operation):

$$\alpha \mathsf{u} = (\alpha u_1, \ldots, \alpha u_n).$$

This is the scalar-vector action of the multiplication operation of the 'Number Systems' chapter on a vector space, and also the vector realisation of the scaling transformation of the 'Operations and Periodicity' chapter. When $\alpha > 1$ it enlarges, when $0 < \alpha < 1$ it shrinks, when $\alpha = 1$ it leaves the vector unchanged (1 is the symmetry centre of the proportional direction), and when $\alpha = -1$ it reverses the vector.

Definition 3.6 (Outer product and complex multiplication—rotation)**.** The rotation operation has two equivalent realisations, according to the dimension of the space:

(i) **Outer product** (in $\mathbb{R}^3$): let $\mathsf{a}, \mathsf{b} \in \mathbb{R}^3$. Their **outer product** (cross product) $\mathsf{a} \times \mathsf{b}$ is a vector whose direction is perpendicular to the plane spanned by a and b (determined by the right-hand rule), and whose modulus is $\|\mathsf{a}\|\|\mathsf{b}\| \sin\theta$.

(ii) **Complex multiplication** (in $\mathbb{R}^2 \cong \mathbb{C}$): let $z_1 = r_1 e^{i\alpha}, z_2 = r_2 e^{i\beta} \in \mathbb{C}$. Their product $z_1 z_2 = r_1 r_2 e^{i(\alpha+\beta)}$ adds the arguments of the two vectors and multiplies their moduli.

The outer product and complex multiplication are the vector realisations of the rotation transformation of the 'Operations and Periodicity' chapter.

Proposition 3.1 (The group structure of vector addition—the algebraic basis of translation)**.** *Under vector addition, $\mathbb{R}^n$ forms a commutative group, with the zero vector as identity element and the negative vector as inverse element. This group is isomorphic to the direct product of n additive groups of the real numbers. This is the direct realisation in a vector space of the additive reflection symmetry of the 'Operations and Periodicity' chapter: the symmetry centre is $\mathsf{0}$, and the involution $\sigma_0(\mathsf{u}) = -\mathsf{u}$ is an additive reflection automorphism of $\mathbb{R}^n$, generating a C_2 automorphism group.*

Proof. *Verify the group axioms: the associativity and commutativity of addition are inherited directly from the corresponding properties of real addition; the zero vector satisfies $\mathsf{u}+\mathsf{0} = \mathsf{u}$; and $\mathsf{u}+(-\mathsf{u}) = \mathsf{0}$. Hence $(\mathbb{R}^n, +)$ is a commutative group. The involution $\sigma_0(\mathsf{u}) = -\mathsf{u}$ satisfies $\sigma_0 \circ \sigma_0 = \mathrm{id}$, and is an additive reflection with $\mathsf{0}$ as its unique fixed point. Since $\sigma_0 \neq \mathrm{id}$ and $\sigma_0^2 = \mathrm{id}$, the group it generates is C_2.* □

Proposition 3.2 (The linearity of scalar multiplication—the algebraic basis of scaling)**.** Scalar multiplication satisfies the following properties:

(i) $\alpha(\mathsf{u} + \mathsf{v}) = \alpha\mathsf{u} + \alpha\mathsf{v}$;

(ii) $(\alpha + \beta)\mathsf{u} = \alpha\mathsf{u} + \beta\mathsf{u}$;

(iii) $\alpha(\beta\mathsf{u}) = (\alpha\beta)\mathsf{u}$;

(iv) $1 \cdot \mathsf{u} = \mathsf{u}$, where 1 is the multiplicative identity, that is, the symmetry centre of the proportional direction.

Scalar multiplication changes only the modulus of a vector and not its direction (except for reversal), and is the algebraic realisation of the **proportional scaling operation**.

Proof. Let $\mathsf{u} = (u_1, \ldots, u_n)$, $\mathsf{v} = (v_1, \ldots, v_n)$. (i) The i-th component of the left-hand side is $\alpha(u_i + v_i)$, and the i-th component of the right-hand side is $\alpha u_i + \alpha v_i$; the two are equal by the distributive law of the real numbers. (ii) The i-th component of the left-hand side is $(\alpha + \beta)u_i$, and that of the right-hand side is $\alpha u_i + \beta u_i$; they are equal by the distributive law of the real numbers. (iii) The left-hand side is $\alpha(\beta u_i) = (\alpha\beta)u_i$, and the right-hand side is $(\alpha\beta)u_i$; they are equal by the associativity of real multiplication. (iv) $1 \cdot u_i = u_i$, hence $1 \cdot \mathsf{u} = \mathsf{u}$—the scalar 1 is the identity of the multiplicative group (the scaling centre), and acting on a vector it changes neither its direction nor its magnitude. When $\alpha = 0$, $0 \cdot u_i = 0$, hence $0 \cdot \mathsf{u} = \mathbf{0}$: the zero scalar contracts any vector to the additive symmetry centre $\mathbf{0}$. $\square$

3.3 Similarity: a pure measure of directional relations

Definition 3.7 (The computational origin of similarity)**.** The similarity of this chapter is defined on $\mathbb{R}^2 \cong \mathbb{C}$ (a single argument is required). Every similarity—whatever its angle type (difference angle, sum angle, double angle, half angle)—is in essence a scalar obtained by performing some algebraic operation on the arguments of two vectors and then taking the cosine or the sine. The sole source of information for a similarity is **the arguments of the two vectors** (α and β). The cosine series measures the **projection onto the real axis** after the operation on the arguments (an even function, insensitive to reversal of direction), whilst the sine series measures the **projection onto the imaginary axis** (an odd function, which changes sign when the direction is reversed).

In other words, the **entire diversity** of the similarity spectrum comes only from the different algebraic operations performed on the pair of arguments (α, β) ($\alpha - \beta$, $\alpha + \beta$, $2(\alpha - \beta)$, $2(\alpha + \beta)$, $(\alpha - \beta)/2$, $(\alpha + \beta)/2$). Beyond this, no further information is needed. The 12 basic similarities defined in this chapter all take this as their origin.

Lemma 3.1 (Generation of the 12 similarities)**.** The 12 similarities are generated by the Cartesian product of the following three binary choices:

$$\{\text{difference angle}, \text{sum angle}\} \times \{\text{basic}, \text{double angle}, \text{half angle}\} \times \{\text{cosine}, \text{sine}\} = 12.$$

These three binary choices correspond respectively to three classes of operational duality:

(i) **Difference angle/sum angle** corresponds to **subtraction and addition** (translation direction, symmetry centre $c = 0$): the difference angle (the signed difference $\alpha - \beta$) is the realisation of subtraction on the arguments, and the sum angle $\phi = \alpha + \beta$ is the realisation of addition on the arguments.

(ii) **Double angle/half angle** corresponds to **powers and roots** (rotation direction, symmetry centre $c = i$): the double angle 2θ is the power operation on the argument (rotating twice), and the half angle $\theta/2$ is the root operation on the argument (dividing the rotation equally).

(iii) **Cosine/sine** corresponds to **real and imaginary parts** (projection and rotation): the cosine is the projection of the argument onto the real axis, and the sine is the projection of the argument onto the imaginary axis.

Therefore the 12 similarities are generated by the Cartesian product of the operational duality.

Proof. The essence of similarity is to perform **a single algebraic operation** on the two arguments and then take a trigonometric function (Definition 3.7). A single algebraic operation has only three basic types: subtraction (difference angle), addition (sum angle), and **powers** of these two operations. When the exponent is 1 it gives the basic angle (θ or ϕ), when the exponent is 2 it gives the double angle (2θ or 2ϕ), and when the exponent is $\frac{1}{2}$ it gives the half angle ($\theta/2$ or $\phi/2$). Operations with exponents exceeding 2 (such as the triple angle 3θ) can be decomposed into composites of double and basic angles ($3\theta = 2\theta + \theta$), and hence do not fall within the scope of 'basic similarities'—they can be obtained from the basic similarities by algebraic combination. Similarly, negative integer exponents or fractional exponents other than $\frac{1}{2}$ can also be decomposed into composites of the basic operations. Hence the 6 basic angle operations $\{\theta, \phi, 2\theta, 2\phi, \theta/2, \phi/2\}$ are complete. Taking the cosine or the sine of each angle operation yields $6 \times 2 = 12$ basic similarities. The correspondence between the three binary choices and the operational duality is given directly by the definition. ☐

Definition 3.8 (The essence of similarity). Similarity is the core tool for describing the directional relation between two vectors. Its essential characteristic is that it **depends only on the arguments of the two vectors** and is completely independent of their moduli. The signed difference angle $\alpha - \beta$ measures the rotation angle required to rotate the direction of $\mathbf{b}$ to the direction of $\mathbf{a}$, and its absolute value is the included angle θ of the two vectors; the sum angle $\phi = \alpha + \beta$ measures the composition of the two directions in the rotation group.

3.3.1 Difference-angle similarity

Definition 3.9 (Basic difference-angle similarity). Let $\mathbf{a}, \mathbf{b} \in \mathbb{R}^2 \cong \mathbb{C}$ be non-zero vectors with arguments α and β respectively. Following the unified convention of the 'Scalars and Vectors' chapter, the **included angle** between two vector directions is always taken to be $\theta = |\alpha - \beta| \in [0, \pi]$; the signed quantity $\alpha - \beta$ is called the **difference angle** and is used only for the sign of the sine-family similarity. Define:

$$\begin{aligned} \text{cosine}(\mathbf{a}, \mathbf{b}) &= \cos(\alpha - \beta) = \cos\theta \quad \text{(directional similarity)}, \\ \text{sine}(\mathbf{a}, \mathbf{b}) &= \sin(\alpha - \beta) \quad \text{(perpendicular similarity)}. \end{aligned}$$

Proposition 3.3 (Properties of the basic difference-angle similarity). Let $\theta = |\alpha - \beta| \in [0, \pi]$ be the included angle of the two vectors. The directional similarity cosine and the perpendicular similarity sine satisfy:

(i) $\mathrm{cosine}(\mathbf{a},\mathbf{b})$ measures the 'agreement' of the two vectors along the translation direction: it attains its maximum 1 when $\theta = 0$ (exactly codirectional), and its minimum -1 when $\theta = \pi$ (exactly antidirectional).

(ii) $|\mathrm{sine}(\mathbf{a},\mathbf{b})|$ measures the 'orthogonality' of the two vectors along the rotation direction: it attains its maximum 1 when $\theta = \pi/2$ (exactly orthogonal).

(iii) The two satisfy the identity: $\mathrm{cosine}^2 + \mathrm{sine}^2 = 1$.

Proof. (i) follows directly from the monotonicity of the cosine function on $[0,\pi]$. (ii) follows from the identity $\mathrm{cosine}^2 + \mathrm{sine}^2 = 1$ of (iii): $|\mathrm{sine}|$ attains its maximum 1 at $\mathrm{cosine} = 0$, that is, at $\theta = \pi/2$. (iii) is a direct consequence of the trigonometric identity $\cos^2\theta + \sin^2\theta = 1$. □

Definition 3.10 (Double-angle difference-angle similarity)**.** The **axial similarity** and the **slope similarity** of the vectors $\mathbf{a},\mathbf{b}$ are defined as:

$$\begin{aligned}\mathrm{cosine}_2(\mathbf{a},\mathbf{b}) &= \cos 2(\alpha-\beta) = \cos 2\theta,\\ \mathrm{sine}_2(\mathbf{a},\mathbf{b}) &= \sin 2(\alpha-\beta).\end{aligned}$$

Proposition 3.4 (Degeneracy detection by the double-angle difference-angle similarity)**.** Let $\theta \in [0,\pi]$. The double-angle difference-angle similarity is used to detect degenerate states of direction:

(i) $\mathrm{cosine}_2 = 1$ if and only if $\theta = 0$ or π (the two vectors are codirectional or antidirectional).

(ii) $\mathrm{cosine}_2 = -1$ if and only if $\theta = \pi/2$ (the two vectors are orthogonal).

(iii) $|\mathrm{sine}_2| = 1$ if and only if $\theta = \pi/4$ or $3\pi/4$ (the two directions are balanced or nearly opposite).

Proof. Verify each critical value directly from $\cos 2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$. (i) $\cos 2\theta = 1 \iff 2\theta = 0$ or $2\pi \iff \theta = 0$ or π (since $\theta \in [0,\pi]$). (ii) $\cos 2\theta = -1 \iff 2\theta = \pi \iff \theta = \pi/2$. (iii) $|\sin 2\theta| = 1 \iff 2\theta = \pi/2$ or $3\pi/2 \iff \theta = \pi/4$ or $3\pi/4$ (since $\theta \in [0,\pi]$). □

Definition 3.11 (Half-angle difference-angle similarity)**.** The **sum-direction similarity** and the **difference-direction similarity** of the vectors $\mathbf{a},\mathbf{b}$ are defined as:

$$\begin{aligned}\mathrm{cosine}_{1/2}(\mathbf{a},\mathbf{b}) &= \cos\frac{\theta}{2},\\ \mathrm{sine}_{1/2}(\mathbf{a},\mathbf{b}) &= \sin\frac{\theta}{2}, \qquad \theta = |\alpha-\beta| \in [0,\pi].\end{aligned}$$

Remark 3.2. The half-angle similarity is closely related to the moduli of the vector sum and difference (see Proposition 3.9). Under the equal-modulus condition, the modulus of the vector sum is proportional to $\cos(\theta/2)$, and the modulus of the vector difference is proportional to $\sin(\theta/2)$.

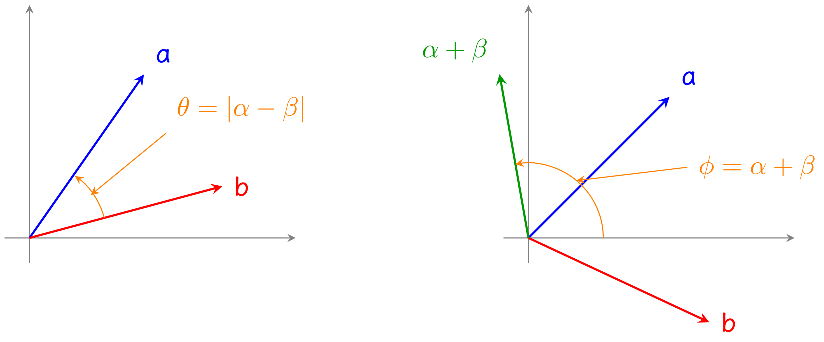
3.3.2 Sum-angle similarity

Definition 3.12 (Basic sum-angle similarity)**.** The **resultant-direction similarity** and the **resultant-perpendicular similarity** of the vectors $\mathbf{a}, \mathbf{b}$ are defined as:

$$\begin{aligned} \mathrm{cosine}_{+}(\mathbf{a}, \mathbf{b}) &= \cos(\alpha + \beta) = \cos\phi, \\ \mathrm{sine}_{+}(\mathbf{a}, \mathbf{b}) &= \sin(\alpha + \beta) = \sin\phi. \end{aligned}$$

The sum-angle similarity takes the sum of the arguments of the two vectors, $\phi = \alpha + \beta$, as its independent variable, and describes the **absolute directional composition** relation of the two vectors. It cannot be reduced to a difference-angle similarity, and corresponds naturally to the directional composition of complex multiplication—the composition of rotations.

Remark 3.3 (The essential difference between the difference angle and the sum angle)*.* The difference-angle similarity measures the **relative direction between two vectors**—this is the 'observer's' viewpoint: the observer stands on one vector and measures the extent to which the other deviates from it. The sum-angle similarity measures the **composition of two vectors as independent directions**—this is the 'participant's' viewpoint: the two directions each rotate independently, and are then composed into a new direction.



(a) difference angle: the observer's viewpoint (b) sum angle: the participant's viewpoint

'standing on b and measuring the deviation of a' 'the two directions are independent, and compose into a new direction'

Figure 3.3: Comparison of the geometric viewpoints of the difference-angle similarity and the sum-angle similarity. (a) The difference angle $\theta = |\alpha - \beta|$ measures the relative deviation between two directions, and is the 'observer's' viewpoint; (b) the sum angle $\phi = \alpha + \beta$ measures the total deflection of the composition of the two directions (the argument of the composed direction being $\alpha + \beta$), and is the 'participant's' viewpoint. The two cannot be reduced to one another (Remark 3.3).

Definition 3.13 (Double-angle sum-angle similarity)**.** The **double-angle resultant-direction similarity** and the **double-angle resultant-perpendicular similarity** of the vectors $\mathbf{a}, \mathbf{b}$ are defined as:

$$\begin{aligned} \mathrm{cosine}_{2+}(\mathbf{a}, \mathbf{b}) &= \cos 2(\alpha + \beta) = \cos 2\phi, \\ \mathrm{sine}_{2+}(\mathbf{a}, \mathbf{b}) &= \sin 2(\alpha + \beta) = \sin 2\phi. \end{aligned}$$

Definition 3.14 (Half-angle sum-angle similarity). The **half-angle resultant-direction similarity** and the **half-angle resultant-perpendicular similarity** of the vectors a, b are defined as:

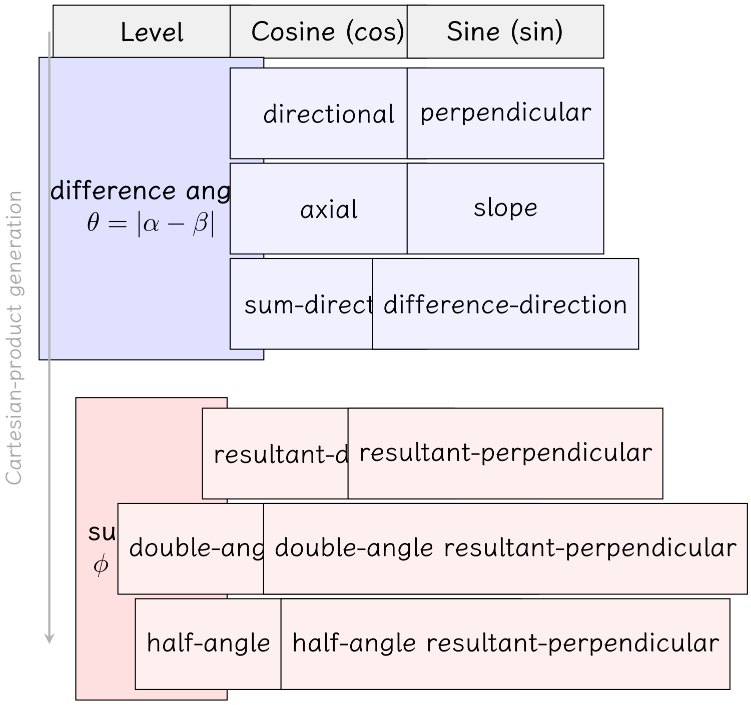
$$\text{cosine}_{1/2+}(\mathsf{a}, \mathsf{b}) = \cos \frac{\alpha + \beta}{2} = \cos \frac{\phi}{2},$$
$$\text{sine}_{1/2+}(\mathsf{a}, \mathsf{b}) = \sin \frac{\alpha + \beta}{2} = \sin \frac{\phi}{2}.$$

3.3.3 Overview of the 12 basic similarities

Table 3.2: The spectrum of the 12 basic similarities

Angle type	Level	Cosine similarity (cosine)	Sine similarity (sine)
difference angle $\theta = \lvert\alpha - \beta\rvert$	Basic	$\cos\theta$ (directional similarity)	$\sin\theta$ (perpendicular similarity)
	Double angle	$\cos 2\theta$ (axial similarity)	$\sin 2\theta$ (slope similarity)
	Half angle	$\cos\frac{\theta}{2}$ (sum-direction similarity)	$\sin\frac{\theta}{2}$ (difference-direction similarity)
sum angle $\phi = \alpha + \beta$	Basic	$\cos\phi$ (resultant-direction similarity)	$\sin\phi$ (resultant-perpendicular similarity)
	Double angle	$\cos 2\phi$ (double-angle resultant-direction similarity)	$\sin 2\phi$ (double-angle resultant-perpendicular similarity)
	Half angle	$\cos\frac{\phi}{2}$ (half-angle resultant-direction similarity)	$\sin\frac{\phi}{2}$ (half-angle resultant-perpendicular similarity)

Remark 3.4 (Complex exponential packaging and quaternion packaging). The basic difference-angle cosine similarity and the basic difference-angle sine similarity can be unified as the complex exponential similarity $\text{Similarity}^{\mathbb{C}}(a, b) = e^{i(\alpha-\beta)}$; the basic difference-angle cosine tendency and the signed sine tendency can be unified as the complex exponential tendency $\text{Tendency}^{\mathbb{C}}(a, b) = a\bar{b}$. In the quaternion case one defines, by angle difference and angle sum, the difference-angle quaternion $D_{\mathbb{H}}(p, q) = p\bar{q}/(|p||q|)$ and the sum-angle quaternion $S_{\mathbb{H}}(p, q) = pq/(|p||q|)$, which give three-dimensional rotation operators through the conjugation action. See Section 3.8 for details.


Level
Cosine (cos)
Sine (sin)
directional
perpendicular
difference ang
$\theta = |\alpha - \beta|$
axial
slope
sum-direct
difference-direction
Cartesian-product generation
resultant-d
resultant-perpendicular
su
ϕ
double-ang
double-angle resultant-perpendicular
half-angle
half-angle resultant-perpendicular

$$12 = \{\text{difference angle}, \text{sum angle}\} \times \{\text{basic}, \text{double angle}, \text{half angle}\} \times \{\text{cos}, \text{sin}\}$$

Figure 3.4: The two-dimensional lattice spectrum of the 12 basic similarities (a schematic diagram of the generation rule). The horizontal binary choice is cosine/sine (projection onto the real/imaginary part), whilst the vertical one is the composite of difference angle/sum angle (subtraction/addition) with basic/double angle/half angle (powers). Each cell corresponds to one similarity, and the concrete formulae are given in the immediately adjacent Table 3.2; the point of this figure is to display explicitly the $2 \times 3 \times 2$ Cartesian-product generation structure (Lemma 3.1).

3.4 Tendency: a composite measure of direction and scale

Definition 3.15 (The computational origin of tendency)**.** *Every tendency—whatever the type of similarity to which it corresponds—is in essence the product of the corresponding similarity and the moduli of the two vectors:*

$$\text{tendency} = \text{similarity} \times \|\mathsf{a}\| \times \|\mathsf{b}\|.$$

The information sources of a tendency are of two kinds: **the arguments of the two vectors** *(supplying the similarity) and* **the moduli of the two vectors** *(supplying the scale). The similarity encodes a purely directional relation (depending only on the difference or the sum of the arguments), whilst the product of the moduli encodes scale information. The tendency synthesises the two into a single scalar, giving a complete measure of 'directed scale'.*

Proposition 3.5 (Similarity is the reduction of tendency)**.** *Let $T(\mathsf{a}, \mathsf{b})$ be a tendency and let $S(\mathsf{a}, \mathsf{b})$ be the corresponding similarity. Then*

$$S(\mathsf{a}, \mathsf{b}) = \frac{T(\mathsf{a}, \mathsf{b})}{\|\mathsf{a}\|\|\mathsf{b}\|}.$$

That is, the similarity is the reduction of the tendency after normalisation of the moduli. The similarity retains only directional information, whilst the tendency retains both directional information and scale information.

Proof. From the definition of tendency, $T(\mathsf{a}, \mathsf{b}) = \|\mathsf{a}\|\|\mathsf{b}\| \cdot S(\mathsf{a}, \mathsf{b})$; dividing both sides by the non-zero quantity $\|\mathsf{a}\|\|\mathsf{b}\|$ gives the result. ☐

Corollary 3.1 (Completeness of the 12 tendencies)**.** *The 12 tendencies are generated by a one-to-one correspondence between the 12 similarities and the product of moduli $\|\mathsf{a}\|\|\mathsf{b}\|$. The completeness of the similarities (Lemma 3.1) directly guarantees the completeness of the tendencies.*

Proof. By the definition of tendency (Definition 3.15), each similarity S corresponds uniquely to a tendency $T = \|\mathsf{a}\|\|\mathsf{b}\| \cdot S$. The map $S \mapsto T$ is a bijection (since $\|\mathsf{a}\|\|\mathsf{b}\| \neq 0$). Lemma 3.1 guarantees that the set of similarities has exactly 12 elements, so the set of tendencies likewise has exactly 12 elements. ☐

Definition 3.16 (Tendency)**.** *Let $\mathsf{a}, \mathsf{b} \in \mathbb{R}^2$ be non-zero vectors. For any kind of similarity $S(\mathsf{a}, \mathsf{b})$, the corresponding* **tendency** *$T(\mathsf{a}, \mathsf{b})$ is defined as the product of that similarity and the moduli of the two vectors:*

$$T(\mathsf{a}, \mathsf{b}) = \|\mathsf{a}\|\|\mathsf{b}\| \cdot S(\mathsf{a}, \mathsf{b}).$$

If one of the two vectors is the zero vector, the tendency is zero—this corresponds to the degeneration of the scaling factor to the additive symmetry centre 0.

3.4.1 Difference-angle tendency

Definition 3.17 (Basic difference-angle tendency). *The tendency corresponding to the basic difference-angle similarity is:*

$$\begin{aligned}\text{cosine\_tendency}(\mathbf{a},\mathbf{b}) &= \|\mathbf{a}\|\|\mathbf{b}\|\cos\theta = \mathbf{a}\cdot\mathbf{b},\\ \text{sine\_tendency}(\mathbf{a},\mathbf{b}) &= \|\mathbf{a}\|\|\mathbf{b}\|\sin(\alpha-\beta),\\ |\text{sine\_tendency}(\mathbf{a},\mathbf{b})| &= \|\mathbf{a}\|\|\mathbf{b}\|\sin\theta = \|\mathbf{a}\times\mathbf{b}\| \quad (\text{in } \mathbb{R}^3\text{, where } \theta \text{ is the included angle}).\end{aligned}$$

The cosine tendency is the inner product, a scaled projection measure along the translation direction; the absolute value of the sine tendency is the modulus of the outer product, a scaled directed-area measure along the rotation direction.

Proposition 3.6 (The correspondence between the basic difference-angle tendency and the vector operations). *The basic difference-angle tendency has the following exact correspondences with the basic vector operations:*

(i) **The cosine tendency is the inner product**: *cosine_tendency*$(\mathbf{a},\mathbf{b}) = \mathbf{a}\cdot\mathbf{b}$. *The inner product is the projection measure along the translation direction.*

(ii) **The absolute value of the sine tendency is the modulus of the outer product**: $|\text{sine\_tendency}(\mathbf{a},\mathbf{b})| = \|\mathbf{a}\times\mathbf{b}\|$. *The modulus of the outer product is the directed-area measure along the rotation direction.*

Proof. (i) follows directly from the geometric definition of the inner product $\mathbf{a}\cdot\mathbf{b} = \|\mathbf{a}\|\|\mathbf{b}\|\cos\theta$ and the definition of the cosine tendency. (ii) follows directly from the geometric definition of the outer product $\|\mathbf{a}\times\mathbf{b}\| = \|\mathbf{a}\|\|\mathbf{b}\|\sin\theta$ (with θ the included angle of the two vectors) and the definition of the sine tendency, on taking absolute values of both sides. ☐

Definition 3.18 (Double-angle difference-angle tendency). *The double-angle difference-angle tendency is obtained from the double-angle difference-angle similarity and the product of the moduli:*

$$\begin{aligned}\text{cosine\_tendency}_2(\mathbf{a},\mathbf{b}) &= \|\mathbf{a}\|\|\mathbf{b}\|\cos 2\theta,\\ \text{sine\_tendency}_2(\mathbf{a},\mathbf{b}) &= \|\mathbf{a}\|\|\mathbf{b}\|\sin 2\theta.\end{aligned}$$

Definition 3.19 (Half-angle difference-angle tendency). *The half-angle difference-angle tendency is obtained from the half-angle difference-angle similarity and the product of the moduli:*

$$\begin{aligned}\text{cosine\_tendency}_{1/2}(\mathbf{a},\mathbf{b}) &= \|\mathbf{a}\|\|\mathbf{b}\|\cos\frac{\theta}{2},\\ \text{sine\_tendency}_{1/2}(\mathbf{a},\mathbf{b}) &= \|\mathbf{a}\|\|\mathbf{b}\|\sin\frac{\theta}{2}.\end{aligned}$$

3.4.2 Sum-angle tendency

Definition 3.20 (Basic sum-angle tendency). *The tendency corresponding to the basic sum-angle similarity is:*

$$\begin{aligned}\text{cosine\_tendency}_+(\mathbf{a},\mathbf{b}) &= \|\mathbf{a}\|\|\mathbf{b}\|\cos(\alpha+\beta),\\ \text{sine\_tendency}_+(\mathbf{a},\mathbf{b}) &= \|\mathbf{a}\|\|\mathbf{b}\|\sin(\alpha+\beta).\end{aligned}$$

Definition 3.21 (Double-angle sum-angle tendency)**.** The double-angle sum-angle tendency is obtained from the double-angle sum-angle similarity and the product of the moduli:

$$\begin{aligned}\text{cosine\_tendency}_{2+}(\mathsf{a},\mathsf{b}) &= \|\mathsf{a}\|\|\mathsf{b}\| \cos 2(\alpha+\beta),\\ \text{sine\_tendency}_{2+}(\mathsf{a},\mathsf{b}) &= \|\mathsf{a}\|\|\mathsf{b}\| \sin 2(\alpha+\beta).\end{aligned}$$

Definition 3.22 (Half-angle sum-angle tendency)**.** The half-angle sum-angle tendency is obtained from the half-angle sum-angle similarity and the product of the moduli:

$$\begin{aligned}\text{cosine\_tendency}_{1/2+}(\mathsf{a},\mathsf{b}) &= \|\mathsf{a}\|\|\mathsf{b}\| \cos \frac{\alpha+\beta}{2},\\ \text{sine\_tendency}_{1/2+}(\mathsf{a},\mathsf{b}) &= \|\mathsf{a}\|\|\mathsf{b}\| \sin \frac{\alpha+\beta}{2}.\end{aligned}$$

3.4.3 Overview of the 12 tendencies

Table 3.3: The correspondence between the 12 tendencies and the vector operations

Angle type	Level	Cosine tendency (cosine tendency)	Sine tendency (sine tendency)
difference angle $\theta = \lvert\alpha-\beta\rvert$	Basic	$\mathsf{a}\cdot\mathsf{b}$ (inner product)	$\|\mathsf{a}\|\|\mathsf{b}\|\sin(\alpha-\beta)$ (signed; its absolute value is the modulus of the outer product)
	Double angle	$\|\mathsf{a}\|\|\mathsf{b}\|\cos 2\theta$	$\|\mathsf{a}\|\|\mathsf{b}\|\sin 2\theta$ (signed)
	Half angle	$\|\mathsf{a}\|\|\mathsf{b}\|\cos\frac{\theta}{2}$	$\|\mathsf{a}\|\|\mathsf{b}\|\sin\frac{\theta}{2}$
sum angle $\phi = \alpha+\beta$	Basic	$\mathrm{Re}(z_1 z_2)$	$\mathrm{Im}(z_1 z_2)$
	Double angle	$\|\mathsf{a}\|\|\mathsf{b}\|\cos 2\phi$	$\|\mathsf{a}\|\|\mathsf{b}\|\sin 2\phi$
	Half angle	$\|\mathsf{a}\|\|\mathsf{b}\|\cos\frac{\phi}{2}$	$\|\mathsf{a}\|\|\mathsf{b}\|\sin\frac{\phi}{2}$

3.5 Inner and outer products: the geometric realisation of the cosine and sine tendencies

Property 3.1 (Basic properties of the inner product)**.** Let $\mathsf{a},\mathsf{b},\mathsf{c} \in \mathbb{R}^n$, $\alpha,\beta \in \mathbb{R}$. The inner product satisfies:

(i) **Symmetry:** $\mathsf{a}\cdot\mathsf{b} = \mathsf{b}\cdot\mathsf{a}$;

(ii) **Bilinearity:** $(\alpha\mathsf{a}+\beta\mathsf{b})\cdot\mathsf{c} = \alpha(\mathsf{a}\cdot\mathsf{c})+\beta(\mathsf{b}\cdot\mathsf{c})$;

(iii) **Positive definiteness**: $\mathsf{a}\cdot\mathsf{a}\geq 0$, and $\mathsf{a}\cdot\mathsf{a}=0 \iff \mathsf{a}=\mathbf{0}$.

Property 3.2 (Basic properties of the outer product)**.** Let $\mathsf{a},\mathsf{b},\mathsf{c}\in\mathbb{R}^3$, $\alpha,\beta\in\mathbb{R}$. The outer product satisfies:

(i) **Anti-commutativity**: $\mathsf{a}\times\mathsf{b}=-\mathsf{b}\times\mathsf{a}$;

(ii) **Bilinearity**: $(\alpha\mathsf{a}+\beta\mathsf{b})\times\mathsf{c}=\alpha(\mathsf{a}\times\mathsf{c})+\beta(\mathsf{b}\times\mathsf{c})$;

(iii) **The Jacobi identity**: $\mathsf{a}\times(\mathsf{b}\times\mathsf{c})+\mathsf{b}\times(\mathsf{c}\times\mathsf{a})+\mathsf{c}\times(\mathsf{a}\times\mathsf{b})=\mathbf{0}$.

Proposition 3.7 (The cosine tendency is an orthogonal-group invariant—the vector realisation of the conservation theme)**.** *Let $G=O(n)$ be the orthogonal group on $\mathbb{R}^n$. For any $R\in O(n)$ and any non-zero vectors $\mathsf{a},\mathsf{b}\in\mathbb{R}^n$, the cosine similarity and the cosine tendency are invariant under the group action:*

$$\text{cosine}(R\mathsf{a},R\mathsf{b})=\text{cosine}(\mathsf{a},\mathsf{b}),\quad (R\mathsf{a})\cdot(R\mathsf{b})=\mathsf{a}\cdot\mathsf{b}.$$

The invariance of the inner product (cosine tendency) under the orthogonal group $O(n)$ is the core fact of the conservation theme of this chapter: after an arbitrary orthogonal transformation (rotation or reflection) of two vectors, their projection measure along the translation direction is strictly conserved.

Proof. Let the matrix entries of R be R_{ij}, satisfying $R^{\mathsf{T}}R=I$, that is, $\sum_k R_{ki}R_{kj}=\delta_{ij}$. Compute the inner product:

$$\begin{aligned}(R\mathsf{a})\cdot(R\mathsf{b})&=\sum_i\left(\sum_j R_{ij}a_j\right)\left(\sum_k R_{ik}b_k\right)\\&=\sum_{j,k}a_jb_k\left(\sum_i R_{ij}R_{ik}\right)\\&=\sum_{j,k}a_jb_k\delta_{jk}=\mathsf{a}\cdot\mathsf{b}.\end{aligned}$$

At the same time, an orthogonal transformation preserves the modulus: $\|R\mathsf{a}\|=\|\mathsf{a}\|$, $\|R\mathsf{b}\|=\|\mathsf{b}\|$. Therefore

$$\text{cosine}(R\mathsf{a},R\mathsf{b})=\frac{R\mathsf{a}\cdot R\mathsf{b}}{\|R\mathsf{a}\|\|R\mathsf{b}\|}=\frac{\mathsf{a}\cdot\mathsf{b}}{\|\mathsf{a}\|\|\mathsf{b}\|}=\text{cosine}(\mathsf{a},\mathsf{b}).$$

□

Proposition 3.8 (The absolute value of the sine tendency is invariant under pure rotations—the curl realisation of the conservation theme)**.** *Let $R\in SO(3)$ be a rotation of three-dimensional space. Then for any non-zero vectors $\mathsf{a},\mathsf{b}\in\mathbb{R}^3$ one has*

$$|\text{sine}(R\mathsf{a},R\mathsf{b})|=|\text{sine}(\mathsf{a},\mathsf{b})|.$$

The absolute value of the sine tendency (the modulus of the outer product, the directed-area measure along the rotation direction) is strictly conserved under pure rotations $SO(3)$—this is the realisation, at the vector level, of the conservation law in the curl direction (conservation of angular momentum): rotation about an arbitrary axis does not change the **magnitude** *of the directed area spanned by the two vectors; the direction of the directed area transforms in step with the rotation.*

Proof. By the transformation property of the outer product under rotation, $(R\mathsf{a}) \times (R\mathsf{b}) = R(\mathsf{a} \times \mathsf{b})$. A rotation preserves the modulus, hence $\|(R\mathsf{a}) \times (R\mathsf{b})\| = \|R(\mathsf{a} \times \mathsf{b})\| = \|\mathsf{a} \times \mathsf{b}\|$. Also $\|R\mathsf{a}\| = \|\mathsf{a}\|$ and $\|R\mathsf{b}\| = \|\mathsf{b}\|$, hence $|\mathsf{sine}(R\mathsf{a}, R\mathsf{b})| = \|(R\mathsf{a}) \times (R\mathsf{b})\|/(\|R\mathsf{a}\|\|R\mathsf{b}\|) = \|\mathsf{a} \times \mathsf{b}\|/(\|\mathsf{a}\|\|\mathsf{b}\|) = |\mathsf{sine}(\mathsf{a}, \mathsf{b})|$, that is, $|\mathsf{sine}|$ is invariant under pure rotations. ☐

3.6 Vector sum, vector difference and the half-angle tendency

Proposition 3.9 (The vector sum and difference and the half-angle tendency). *Let $\mathsf{a}, \mathsf{b} \in \mathbb{R}^n$ be non-zero vectors of equal modulus ($\|\mathsf{a}\| = \|\mathsf{b}\| = r > 0$), and let $\theta \in [0, \pi]$ be the included angle of the two vectors. Then:*

(i) *the modulus of the vector sum $\mathsf{a} + \mathsf{b}$ is proportional to the half-angle cosine similarity: $\|\mathsf{a} + \mathsf{b}\| = 2r \cos \frac{\theta}{2}$.*

(ii) *the modulus of the vector difference $\mathsf{a} - \mathsf{b}$ is proportional to the half-angle sine similarity: $\|\mathsf{a} - \mathsf{b}\| = 2r \sin \frac{\theta}{2}$.*

Proof. (i) By the parallelogram law,

$$\|\mathsf{a} + \mathsf{b}\|^2 = \|\mathsf{a}\|^2 + \|\mathsf{b}\|^2 + 2\mathsf{a} \cdot \mathsf{b} = 2r^2 + 2r^2 \cos \theta = 2r^2(1 + \cos \theta).$$

Using the half-angle formula $1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$, we obtain $\|\mathsf{a} + \mathsf{b}\|^2 = 4r^2 \cos^2 \frac{\theta}{2}$. Since $\|\mathsf{a} + \mathsf{b}\| \geq 0$ and $\cos \frac{\theta}{2} \geq 0$ (because $\theta \in [0, \pi]$ implies $\frac{\theta}{2} \in [0, \frac{\pi}{2}]$), taking the square root gives the result.

(ii) Likewise,

$$\|\mathsf{a} - \mathsf{b}\|^2 = \|\mathsf{a}\|^2 + \|\mathsf{b}\|^2 - 2\mathsf{a} \cdot \mathsf{b} = 2r^2 - 2r^2 \cos \theta = 2r^2(1 - \cos \theta).$$

Using $1 - \cos \theta = 2 \sin^2 \frac{\theta}{2}$, and noting that $\sin \frac{\theta}{2} \geq 0$ because $\theta \in [0, \pi]$, taking the square root gives $\|\mathsf{a} - \mathsf{b}\| = 2r \sin \frac{\theta}{2}$. ☐

Remark 3.5. Proposition 3.9 requires the two vectors to have equal modulus ($\|\mathsf{a}\| = \|\mathsf{b}\| = r$), which is the key premise enabling the conclusion to be expressed in the concise form of half-angle trigonometric functions. For vectors of unequal modulus the formulae read:

$$\|\mathsf{a} + \mathsf{b}\|^2 = \|\mathsf{a}\|^2 + \|\mathsf{b}\|^2 + 2\|\mathsf{a}\|\|\mathsf{b}\| \cos \theta,$$

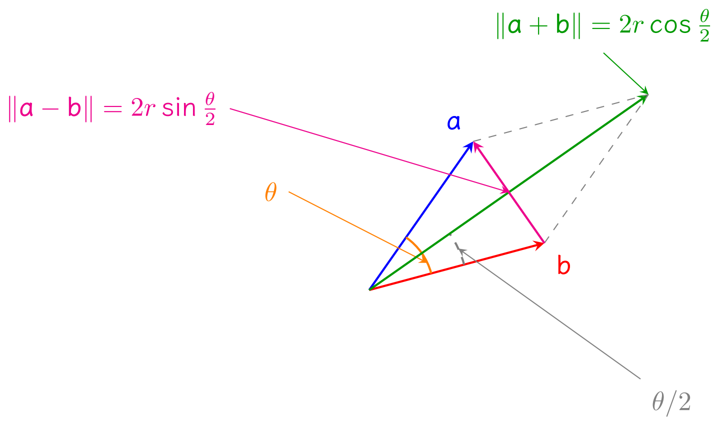
$$\|\mathsf{a} - \mathsf{b}\|^2 = \|\mathsf{a}\|^2 + \|\mathsf{b}\|^2 - 2\|\mathsf{a}\|\|\mathsf{b}\| \cos \theta,$$

which cannot be simplified to a single half-angle trigonometric form.

3.7 Reflection, refraction and double-angle similarity

Definition 3.23 (Reflection—conservation of the double-angle cosine similarity). Let v be the incident direction and let n be the normal (a unit vector). The reflected direction v' is:

$$\mathsf{v}' = \mathsf{v} - 2(\mathsf{v} \cdot \mathsf{n})\mathsf{n}.$$

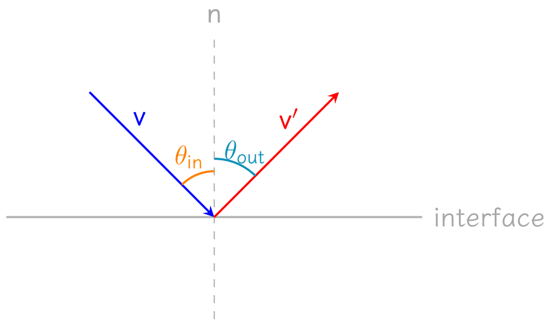

$\|a + b\| = 2r\cos\frac{\theta}{2}$
$\|a - b\| = 2r\sin\frac{\theta}{2}$
a
θ
b
$\theta/2$

when the moduli are equal, $\|a\| = \|b\| = r$, the half-angle formulae hold exactly

Figure 3.5: The geometric relation between the half-angle similarity and the vector sum/difference. Under the equal-modulus condition, the modulus of the vector sum is proportional to the half-angle cosine (sum-direction), and the modulus of the vector difference is proportional to the half-angle sine (difference-direction). The symmetry of the two formulae embodies the trigonometric structure of the parallelogram law (Proposition 3.9).

The reflection transformation is an involution ($v'' = v$), embodying additive reflection symmetry ($c = 0$). Before and after reflection, the **double-angle cosine similarity of the direction with the normal remains unchanged**, that is, $\cos 2\theta_{\text{in}} = \cos 2\theta_{\text{out}}$.

From the standpoint of the conservation and closure themes, reflection gives the smallest periodic structure of this chapter: two reflections return to the original, and $v \mapsto v' \mapsto v'' = v$ constitutes a **closed orbit** of period 2; at the same time reflection preserves the modulus, $\|v'\| = \|v\|$ (see the proof immediately following), which is the conservation of the projection measure under additive reflection symmetry $c = 0$.



$\cos 2\theta_{\text{in}} = \cos 2\theta_{\text{out}}$

the reflection involution $(v'') = v$ forms a closed orbit of period 2

Figure 3.6: Conservation of the double-angle cosine similarity under reflection. The double-angle cosine $\cos 2\theta_{\text{in}}$ of the incident direction v with the normal **n** is unchanged after reflection. Reflection is an involution: two reflections return to v, constituting the smallest periodic structure of this chapter (Definition 3.23).

Proof. Compute the square of the modulus:

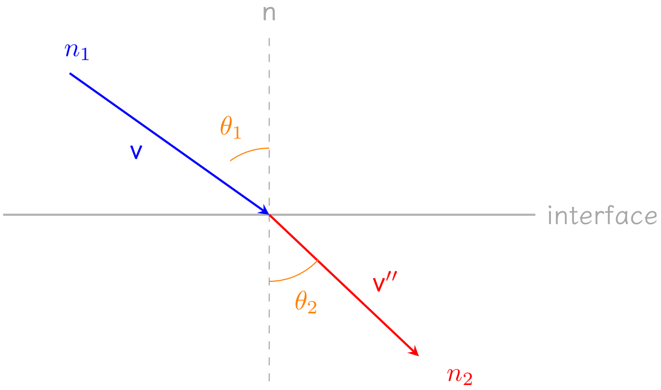
$$\|\mathsf{v}'\|^2 = \|\mathsf{v} - 2(\mathsf{v}\cdot\mathsf{n})\mathsf{n}\|^2 = \|\mathsf{v}\|^2 - 4(\mathsf{v}\cdot\mathsf{n})^2 + 4(\mathsf{v}\cdot\mathsf{n})^2\|\mathsf{n}\|^2 = \|\mathsf{v}\|^2,$$

because $\|\mathsf{n}\| = 1$. Reflect twice: $\mathsf{v}'' = \mathsf{v}' - 2(\mathsf{v}'\cdot\mathsf{n})\mathsf{n} = \mathsf{v} - 2(\mathsf{v}\cdot\mathsf{n})\mathsf{n} - 2((\mathsf{v} - 2(\mathsf{v}\cdot\mathsf{n})\mathsf{n})\cdot\mathsf{n})\mathsf{n}$. Compute the inner product: $(\mathsf{v} - 2(\mathsf{v}\cdot\mathsf{n})\mathsf{n})\cdot\mathsf{n} = \mathsf{v}\cdot\mathsf{n} - 2(\mathsf{v}\cdot\mathsf{n}) = -\mathsf{v}\cdot\mathsf{n}$. Substituting gives $\mathsf{v}'' = \mathsf{v} - 2(\mathsf{v}\cdot\mathsf{n})\mathsf{n} - 2(-\mathsf{v}\cdot\mathsf{n})\mathsf{n} = \mathsf{v}$. Hence reflection is an involution. The double-angle cosine similarity is unchanged because the angle of reflection equals the angle of incidence: $\theta_{\text{out}} = \theta_{\text{in}}$, hence $\cos 2\theta_{\text{out}} = \cos 2\theta_{\text{in}}$. □

Definition 3.24 (Refraction—the scaling of the sine tendency). Let the incident vector be v, let the normal be n (a unit vector), and let the angle of incidence be θ_1. The refracted vector v'' lies in the plane spanned by v and n, and its angle θ_2 with the normal satisfies Snell's law:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2.$$

Refraction embodies the modulation of the rotation direction (i-symmetry) by multiplicative scaling reflection symmetry ($c \neq 1$): the refractive indices n_1, n_2 act as proportionality factors scaling the sine tendencies in the two media. The modulus of the refracted vector is determined by the conservation of energy or by the amplitude coefficients, and its direction is uniquely determined by Snell's law (within the plane of incidence).



$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

the sine tendency is conserved along the interface normal; the refractive indices n_1, n_2 are scaling factors

Figure 3.7: Snell's law of refraction and the scaling of the sine tendency. The sine tendency $n \sin\theta$ of the incident and refracted directions is conserved at the interface, and the refractive indices n_1, n_2 act as scaling factors modulating the rotation direction. When $n_1 \sin\theta_1 > n_2$ total internal reflection occurs (Proposition 3.10).

Proposition 3.10 (The critical angle of refraction). When $n_1 > n_2$ and the angle of incidence θ_1 exceeds the critical angle $\theta_c = \arcsin(n_2/n_1)$, we have $\sin\theta_1 > n_2/n_1$, and Snell's law requires $\sin\theta_2 > 1$, which has no solution in the real domain; refraction no longer occurs, and the light is entirely reflected back into the original medium (total internal reflection).

Proof. By Snell's law, $\sin\theta_2 = \frac{n_1}{n_2}\sin\theta_1$. If $\sin\theta_1 > \frac{n_2}{n_1}$, then $\sin\theta_2 > 1$, but the range of the sine function is $[-1, 1]$, so there is no real solution. The critical angle is defined as $\theta_c = \arcsin(n_2/n_1)$. □

3.8 Directional composition in complex and quaternion multiplication, and exponential similarity

This section treats complex numbers and quaternions according to the unified logic of 'angle difference / angle sum': the difference-angle operator and the sum-angle operator encode 'rotation difference' and 'rotational composition' respectively, their real parts corresponding to the cosine family and their imaginary parts (or vector parts) to the sine family. In the complex case left multiplication is a rotation; in the quaternion case the conjugation action gives a three-dimensional rotation. When the axes coincide, the quaternion case degenerates strictly to the angle difference and angle sum of the complex case.

3.8.1 Complex multiplication: sum-angle tendency and rotational composition

Proposition 3.11 (Complex multiplication—sum-angle tendency and rotational composition). Regard the complex number $z = a + bi$ as a vector in $\mathbb{R}^2$. Let

$$z_1 = r_1 e^{i\alpha}, \qquad z_2 = r_2 e^{i\beta}.$$

Then the real and imaginary parts of the product $z_1 z_2$ correspond respectively to the **basic sum-angle cosine tendency** and the **basic sum-angle sine tendency**:

$$z_1 z_2 = \mathsf{cosine\_tendency}_+(\mathsf{a}, \mathsf{b}) + i\,\mathsf{sine\_tendency}_+(\mathsf{a}, \mathsf{b}),$$

where a, b correspond to z_1, z_2 respectively. Complex multiplication adds the two rotation angles and multiplies the two moduli: the addition of arguments corresponds to directional composition along the rotation direction, and the multiplication of moduli corresponds to scale composition along the proportional direction.

Proof. By complex multiplication,

$$z_1 z_2 = r_1 r_2 e^{i(\alpha+\beta)} = r_1 r_2 \big[\cos(\alpha+\beta) + i \sin(\alpha+\beta)\big].$$

Since $r_1 r_2 = \|\mathsf{a}\|\|\mathsf{b}\|$, the real part is

$$\|\mathsf{a}\|\|\mathsf{b}\| \cos(\alpha+\beta) = \mathsf{cosine\_tendency}_+(\mathsf{a}, \mathsf{b}),$$

and the imaginary part is

$$\|\mathsf{a}\|\|\mathsf{b}\| \sin(\alpha+\beta) = \mathsf{sine\_tendency}_+(\mathsf{a}, \mathsf{b}).$$

This completes the proof. □

Remark 3.6. Complex multiplication corresponds to the 'sum angle', that is, the two directions are each independent and are then composed into a new direction. What it describes is the composition of rotations, not the relative difference between two directions. The relative difference is described by the complex exponential similarity below.

Definition 3.25 (Complex exponential difference-angle and sum-angle operators). Let $z_1 = r_1 e^{i\alpha}$ and $z_2 = r_2 e^{i\beta}$, both non-zero. Define

$$\boxed{D_{\mathbb{C}}(z_1, z_2) := \frac{z_1 \overline{z_2}}{|z_1|\,|z_2|}}$$

to be the **complex exponential difference-angle operator** (angle-difference operator) of z_1 and z_2;

$$\boxed{S_{\mathbb{C}}(z_1, z_2) := \frac{z_1 z_2}{|z_1|\,|z_2|}}$$

to be the **complex exponential sum-angle operator** (angle-sum operator) of z_1 and z_2. Both are unit complex numbers, and they can be written as

$$D_{\mathbb{C}}(z_1, z_2) = e^{i(\alpha-\beta)}, \qquad S_{\mathbb{C}}(z_1, z_2) = e^{i(\alpha+\beta)}.$$

Proposition 3.12 (The complex exponential difference-angle operator is a two-dimensional rotation-difference operator). Let $z_1, z_2 \in \mathbb{C}^\times$. Then $D_{\mathbb{C}}(z_1, z_2)$ gives, through multiplication, the two-dimensional rotation from the unit direction of z_2 to the unit direction of z_1:

$$\boxed{D_{\mathbb{C}}(z_1, z_2) \cdot \frac{z_2}{|z_2|} = \frac{z_1}{|z_1|}}$$

that is, $D_{\mathbb{C}}(z_1, z_2)$ is a two-dimensional rotation-difference operator. Its argument is $\alpha - \beta$, its real part corresponds to the basic difference-angle cosine similarity and its imaginary part to the basic difference-angle sine similarity; in parallel, the argument of $S_{\mathbb{C}}(z_1, z_2)$ is $\alpha + \beta$, and its real and imaginary parts correspond respectively to the basic sum-angle cosine tendency and the basic sum-angle sine tendency.

Proof. By the definition and $z_2 \overline{z_2} = |z_2|^2$,

$$D_{\mathbb{C}}(z_1, z_2) \cdot \frac{z_2}{|z_2|} = \frac{z_1 \overline{z_2}}{|z_1|\,|z_2|} \cdot \frac{z_2}{|z_2|} = \frac{z_1\,|z_2|^2}{|z_1|\,|z_2|^2} = \frac{z_1}{|z_1|}.$$

Hence $D_{\mathbb{C}}(z_1, z_2)$ sends the unit direction of z_2 to the unit direction of z_1, acting by complex multiplication (that is, a two-dimensional rotation), and no conjugation is needed. From

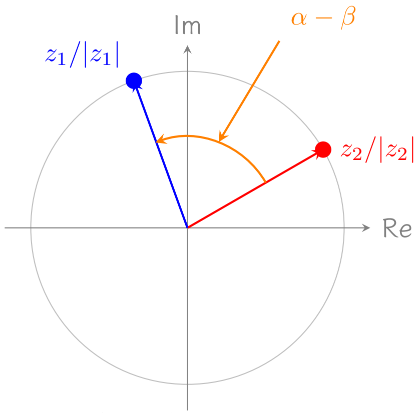
$$D_{\mathbb{C}}(z_1, z_2) = e^{i(\alpha-\beta)} = \cos(\alpha-\beta) + i \sin(\alpha-\beta)$$

the correspondence of the argument with the real and imaginary parts is obtained directly; the correspondence on the $S_{\mathbb{C}}$ side has already been given by Proposition 3.11. This completes the proof. ☐

Remark 3.7 (The composition law of the complex exponential difference-angle operator). Let $z_1, z_2, z_3 \in \mathbb{C}^\times$. From the definition one obtains directly

$$D_{\mathbb{C}}(z_1, z_2)\, D_{\mathbb{C}}(z_2, z_3) = D_{\mathbb{C}}(z_1, z_3),$$

that is, the complex exponential difference-angle operator satisfies the composition law of rotation differences. This composition law is completely parallel to Proposition 3.15 in the quaternion case; the difference between the two lies in the fact that complex multiplication is commutative whilst quaternion multiplication is not (see Remark 3.8), which is precisely the structural difference between two-dimensional and three-dimensional rotation differences.



$$D_{\mathbb{C}}(z_1,z_2)\cdot\frac{z_2}{|z_2|}=\frac{z_1}{|z_1|}$$

the complex exponential difference-angle operator $D_{\mathbb{C}}=e^{i(\alpha-\beta)}$ is a two-dimensional rotation-difference operator

Figure 3.8: The rotation-difference action of the complex exponential difference-angle operator. $D_{\mathbb{C}}(z_1,z_2)=e^{i(\alpha-\beta)}$ rotates the unit direction of z_2 exactly onto the unit direction of z_1, realising the 'rotation difference from z_2 to z_1' (Proposition 3.12).

3.8.2 Quaternion multiplication: non-commutative rotational composition

Proposition 3.13 (Quaternion multiplication—non-commutative rotational direction)**.** *Quaternion multiplication is the non-commutative generalisation of complex multiplication, realising three-dimensional rotational direction. A unit quaternion $q=e^{\mathsf{u}\theta}$ acts on a pure imaginary vector v through conjugation*

$$v\longmapsto qvq^{-1}$$

producing the effect of a rotation by 2θ about the axis u. The product of two unit quaternions

$$q_1q_2=e^{\mathsf{u}_1\theta_1}e^{\mathsf{u}_2\theta_2}$$

corresponds to the composition of two three-dimensional rotations; since in general $q_1q_2\neq q_2q_1$, the result of the composition depends on the order. This is the generalisation of the rotation direction from the commutative group $U(1)$ to the non-commutative group $SU(2)$.

Proof. Write the unit quaternion as

$$q=\cos\theta+\mathsf{u}\sin\theta,$$

where u is a unit pure imaginary quaternion with $\mathsf{u}^2=-1$, and

$$q^{-1}=\cos\theta-\mathsf{u}\sin\theta.$$

For a pure imaginary quaternion v, the image of the conjugation action is still a pure imaginary quaternion:

$$\overline{qvq^{-1}}=\overline{q^{-1}}\,\bar{v}\,\bar{q}=q\,(-v)\,q^{-1}=-qvq^{-1}.$$

Decompose v along the axis as

$$v = v_\parallel + v_\perp,$$

where $v_\parallel \parallel \mathsf{u}$ and $v_\perp \perp \mathsf{u}$. The parallel component satisfies $\mathsf{u}v_\parallel = v_\parallel\mathsf{u}$, hence

$$qv_\parallel q^{-1} = v_\parallel.$$

For the perpendicular component, since $\mathsf{u} \cdot v_\perp = 0$, the multiplication of pure imaginary quaternions satisfies

$$\mathsf{u}v_\perp = \mathsf{u} \times v_\perp, \qquad v_\perp\mathsf{u} = -\mathsf{u} \times v_\perp.$$

Therefore

$$\begin{aligned} qv_\perp q^{-1} &= (\cos\theta + \mathsf{u}\sin\theta)\, v_\perp\, (\cos\theta - \mathsf{u}\sin\theta) \\ &= \cos^2\theta\, v_\perp - \cos\theta\sin\theta\,(v_\perp\mathsf{u}) + \sin\theta\cos\theta\,(\mathsf{u}v_\perp) - \sin^2\theta\,(\mathsf{u}v_\perp\mathsf{u}). \end{aligned}$$

Using $v_\perp\mathsf{u} = -\mathsf{u}v_\perp$ and the triple cross-product identity

$$(\mathsf{u} \times v_\perp) \times \mathsf{u} = v_\perp$$

(since $\mathsf{u} \perp v_\perp$ and $\|\mathsf{u}\| = 1$), we obtain

$$qv_\perp q^{-1} = (\cos^2\theta - \sin^2\theta)\, v_\perp + 2\sin\theta\cos\theta\,(\mathsf{u} \times v_\perp) = \cos 2\theta\, v_\perp + \sin 2\theta\,(\mathsf{u} \times v_\perp).$$

This is precisely the Rodrigues formula for a rotation by angle 2θ about the axis u. Hence the conjugation action $v \mapsto qvq^{-1}$ is a rotation by 2θ about the axis u.

The conjugation action corresponding to the product of two unit quaternions is

$$v \longmapsto (q_1q_2)v(q_1q_2)^{-1} = q_1\,(q_2vq_2^{-1})\,q_1^{-1},$$

that is, the composite of first a rotation by $2\theta_2$ about u_2 and then a rotation by $2\theta_1$ about u_1. Since in general $q_1q_2 \neq q_2q_1$, the order of composition of three-dimensional rotations is not commutative, and this is precisely the source of the non-commutativity of $SO(3)$; under this conjugation correspondence the unit quaternions form the double cover $SU(2)$ of $SO(3)$. This completes the proof. □

3.8.3 Difference-angle and sum-angle quaternions

Definition 3.26 (The polar form and angle of a quaternion). *Let $q \in \mathbb{H}^\times$, with q not a negative real number. Then q can be written uniquely in* **polar form**

$$q = |q|\Big(\cos\theta + \mathsf{u}\sin\theta\Big),$$

where $\theta \in [0, \pi)$ is called the **argument** *(or* **angle***) of q, and u is a unit pure imaginary quaternion ($\mathsf{u}^2 = -1$), called the* **axis** *of q. When q is a negative real number, we adopt the convention $\theta = \pi$, with u an arbitrary unit pure imaginary quaternion.*

Definition 3.27 (Difference-angle and sum-angle quaternions). Let $p, q \in \mathbb{H}^\times$. Define

$$\boxed{D_{\mathbb{H}}(p,q) := \frac{p\overline{q}}{|p|\,|q|}}$$

to be the **difference-angle quaternion** (angle-difference operator) of p and q;

$$\boxed{S_{\mathbb{H}}(p,q) := \frac{pq}{|p|\,|q|}}$$

to be the **sum-angle quaternion** (angle-sum operator) of p and q. Both are unit quaternions.

Lemma 3.2 (The scalar-vector decomposition of the difference-angle and sum-angle quaternions). *Let $p = |p|e^{\mathsf{u}\alpha}$, $q = |q|e^{\mathsf{v}\beta}$, that is,*

$$p = |p|\big(\cos\alpha + \mathsf{u}\sin\alpha\big), \qquad q = |q|\big(\cos\beta + \mathsf{v}\sin\beta\big).$$

Then the difference-angle quaternion expands as

$$\begin{aligned} D_{\mathbb{H}}(p,q) = \ &\underbrace{\big(\cos\alpha\cos\beta + \sin\alpha\sin\beta\,(\mathsf{u}\cdot\mathsf{v})\big)}_{\text{scalar part}} \\ &+ \underbrace{\big(\sin\alpha\cos\beta\,\mathsf{u} - \cos\alpha\sin\beta\,\mathsf{v} - \sin\alpha\sin\beta\,(\mathsf{u}\times\mathsf{v})\big)}_{\text{vector part}}, \end{aligned}$$

and the sum-angle quaternion expands as

$$\begin{aligned} S_{\mathbb{H}}(p,q) = \ &\underbrace{\big(\cos\alpha\cos\beta - \sin\alpha\sin\beta\,(\mathsf{u}\cdot\mathsf{v})\big)}_{\text{scalar part}} \\ &+ \underbrace{\big(\sin\alpha\cos\beta\,\mathsf{u} + \cos\alpha\sin\beta\,\mathsf{v} + \sin\alpha\sin\beta\,(\mathsf{u}\times\mathsf{v})\big)}_{\text{vector part}}. \end{aligned}$$

When $\mathsf{u} = \mathsf{v}$ the cross terms vanish, and the two degenerate respectively to

$$D_{\mathbb{H}}(p,q) = \cos(\alpha-\beta) + \mathsf{u}\sin(\alpha-\beta), \qquad S_{\mathbb{H}}(p,q) = \cos(\alpha+\beta) + \mathsf{u}\sin(\alpha+\beta),$$

completely parallel to the complex case: the angle of the difference-angle quaternion is $\alpha-\beta$, the angle of the sum-angle quaternion is $\alpha+\beta$, and both axes are u.

Proof. Use the multiplication formula for pure imaginary quaternions

$$\mathsf{u}\mathsf{v} = -\mathsf{u}\cdot\mathsf{v} + \mathsf{u}\times\mathsf{v},$$

together with $e^{\mathsf{u}\alpha} = \cos\alpha + \mathsf{u}\sin\alpha$, $e^{-\mathsf{v}\beta} = \cos\beta - \mathsf{v}\sin\beta$ and $e^{\mathsf{v}\beta} = \cos\beta + \mathsf{v}\sin\beta$.

First expand the difference-angle quaternion:

$$D_{\mathbb{H}}(p,q) = e^{\mathsf{u}\alpha}e^{-\mathsf{v}\beta} = (\cos\alpha + \mathsf{u}\sin\alpha)(\cos\beta - \mathsf{v}\sin\beta).$$

The scalar part is

$$\cos\alpha\cos\beta - (\mathsf{u}\cdot(-\mathsf{v}))\sin\alpha\sin\beta = \cos\alpha\cos\beta + \sin\alpha\sin\beta\,(\mathsf{u}\cdot\mathsf{v}),$$

and the vector part is

$$\sin\alpha\cos\beta\,\mathsf{u} - \cos\alpha\sin\beta\,\mathsf{v} + \sin\alpha\sin\beta\,(\mathsf{u}\times(-\mathsf{v})),$$

where $\mathsf{u}\times(-\mathsf{v}) = -\mathsf{u}\times\mathsf{v}$, so the vector part is

$$\sin\alpha\cos\beta\,\mathsf{u} - \cos\alpha\sin\beta\,\mathsf{v} - \sin\alpha\sin\beta\,(\mathsf{u}\times\mathsf{v}).$$

Likewise expand the sum-angle quaternion:

$$S_{\mathbb{H}}(p,q) = e^{\mathsf{u}\alpha}e^{\mathsf{v}\beta} = (\cos\alpha + \mathsf{u}\sin\alpha)(\cos\beta + \mathsf{v}\sin\beta).$$

The scalar part is

$$\cos\alpha\cos\beta - (\mathsf{u}\cdot\mathsf{v})\sin\alpha\sin\beta,$$

and the vector part is

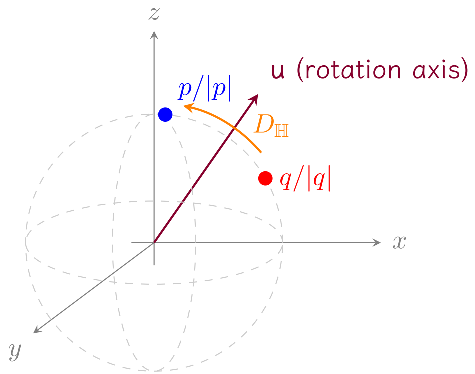
$$\sin\alpha\cos\beta\,\mathsf{u} + \cos\alpha\sin\beta\,\mathsf{v} + \sin\alpha\sin\beta\,(\mathsf{u}\times\mathsf{v}).$$

When $\mathsf{u} = \mathsf{v}$, $\mathsf{u}\cdot\mathsf{v} = 1$ and $\mathsf{u}\times\mathsf{v} = 0$; substituting gives the degenerate formulae. This completes the proof. ☐

Proposition 3.14 (The difference-angle quaternion is a three-dimensional rotation-difference operator)**.** Let $p, q \in \mathbb{H}^{\times}$. Then $D_{\mathbb{H}}(p,q)$ gives, through the conjugation action, the three-dimensional rotation from the unit direction of q to the unit direction of p:

$$\boxed{D_{\mathbb{H}}(p,q)\cdot\frac{q}{|q|}\cdot D_{\mathbb{H}}(p,q)^{-1} = \frac{p}{|p|}}$$

that is, $D_{\mathbb{H}}(p,q)$ is a rotation-difference operator. Its scalar part corresponds to the quaternion cosine similarity and the modulus of its vector part to the quaternion sine similarity.



$$D_{\mathbb{H}}(p,q)\cdot\frac{q}{|q|}\cdot D_{\mathbb{H}}(p,q)^{-1} = \frac{p}{|p|}$$

the difference-angle quaternion gives the three-dimensional rotation difference through the conjugation action

Figure 3.9: The three-dimensional rotation-difference action of the difference-angle quaternion. $D_{\mathbb{H}}(p,q) = p\bar{q}/(|p||q|)$ sends the unit direction of q to the unit direction of p through the conjugation action $v \mapsto DvD^{-1}$, realising a three-dimensional rotation-difference operator. When the axes coincide it degenerates to the angle difference of the complex case (Proposition 3.14, Theorem 3.2).

Proof. By the definition

$$D_{\mathbb{H}}(p,q)=\frac{p\overline{q}}{|p||q|},$$

hence

$$D_{\mathbb{H}}(p,q)\cdot\frac{q}{|q|}\cdot D_{\mathbb{H}}(p,q)^{-1}=\frac{p\overline{q}}{|p||q|}\cdot\frac{q}{|q|}\cdot\frac{q\overline{p}}{|q||p|}.$$

Using $q\overline{q}=|q|^2$ and $\overline{q}\,q=|q|^2$, we obtain

$$D_{\mathbb{H}}(p,q)\cdot\frac{q}{|q|}\cdot D_{\mathbb{H}}(p,q)^{-1}=\frac{p\,|q|^2\,\overline{p}}{|p|^2|q|^2|q|}=\frac{p\,\overline{p}}{|p|^2}\cdot\frac{q}{|q|}.$$

When p,q are unit quaternions, $p\overline{p}=|p|^2=1$ and $|q|=1$, and the above expression becomes

$$D_{\mathbb{H}}(p,q)\,q\,D_{\mathbb{H}}(p,q)^{-1}=p.$$

The general case is reduced to this by normalisation. Hence $D_{\mathbb{H}}(p,q)$ sends the unit direction of q to the unit direction of p through the conjugation action. The decomposition into scalar and vector parts is given by Lemma 3.2. This completes the proof. □

Proposition 3.15 (The composition law of the difference-angle quaternion)**.** *Let $p,q,r\in\mathbb{H}^\times$. Then*

$$D_{\mathbb{H}}(p,q)\,D_{\mathbb{H}}(q,r)=D_{\mathbb{H}}(p,r).$$

That is, the difference-angle quaternion satisfies the composition law of rotation differences.

Proof. By the definition,

$$D_{\mathbb{H}}(p,q)D_{\mathbb{H}}(q,r)=\frac{p\overline{q}}{|p||q|}\cdot\frac{q\overline{r}}{|q||r|}=\frac{p|q|^2\overline{r}}{|p||q|^2|r|}=\frac{p\overline{r}}{|p||r|}=D_{\mathbb{H}}(p,r).$$

This completes the proof. □

Theorem 3.1 (The sum-angle quaternion is a three-dimensional rotational composition operator)**.** *Let $p,q\in\mathbb{H}^\times$. Then $S_{\mathbb{H}}(p,q)=pq/(|p||q|)$ gives, through the conjugation action, a rotational composition: first q gives a rotation, then p gives a rotation, and the composite result is the conjugation action of $S_{\mathbb{H}}(p,q)$.*

Proof. For a pure imaginary quaternion v, the conjugation action satisfies

$$(pq)v(pq)^{-1}=p\big(qvq^{-1}\big)p^{-1}.$$

Hence acting first by q and then by p is equivalent to acting by pq. After normalisation, this gives that $S_{\mathbb{H}}(p,q)=pq/(|p||q|)$ yields a rotational composition through the conjugation action. This completes the proof. □

Theorem 3.2 (The degeneracy of the sum-angle quaternion when the axes coincide is parallel to the complex case)**.** *When p,q have the same axis ($\mathsf{u}=\mathsf{v}$), the difference-angle quaternion and the sum-angle quaternion degenerate respectively to*

$$D_{\mathbb{H}}(p,q)=e^{\mathsf{u}(\alpha-\beta)},\qquad S_{\mathbb{H}}(p,q)=e^{\mathsf{u}(\alpha+\beta)},$$

completely parallel to the complex exponential similarity and the complex exponential tendency. The scalar part corresponds to the cosine family, and the modulus of the vector part to the sine family.

Proof. By Lemma 3.2, when the axes coincide the difference-angle quaternion becomes

$$D_{\mathbb{H}}(p,q) = \cos(\alpha-\beta) + \mathsf{u}\sin(\alpha-\beta) = e^{\mathsf{u}(\alpha-\beta)},$$

and the sum-angle quaternion becomes

$$S_{\mathbb{H}}(p,q) = \cos(\alpha+\beta) + \mathsf{u}\sin(\alpha+\beta) = e^{\mathsf{u}(\alpha+\beta)}.$$

The scalar parts are $\cos(\alpha-\beta)$ and $\cos(\alpha+\beta)$ respectively, corresponding to the cosine similarity and the cosine tendency; the moduli of the vector parts are $|\sin(\alpha-\beta)|$ and $|\sin(\alpha+\beta)|$ respectively, corresponding to the sine similarity and the sine tendency. In the complex case there is only one fixed axis i, so the above degeneracy is precisely the complex exponential similarity and the complex exponential tendency. This completes the proof. $\square$

Theorem 3.3 (The inner-product and cross-product decomposition of the quaternion exponential tendency)**.** *Let $p, q \in \mathbb{H}^{\times}$. Write*

$$p = p_0 + \mathsf{p}, \qquad q = q_0 + \mathsf{q},$$

where $p_0, q_0 \in \mathbb{R}$ and p, q are pure imaginary quaternions. Then

$$\mathsf{Tendency}^{\mathbb{H}}(p,q) := p\overline{q}$$

satisfies

$$\mathsf{Sc}\big(\mathsf{Tendency}^{\mathbb{H}}(p,q)\big) = p_0q_0 + \mathsf{p}\cdot\mathsf{q} = \langle p, q\rangle,$$

that is, the scalar part is the quaternion inner product, which is also the quaternion cosine tendency. If p, q are pure imaginary quaternions, that is, $p = \mathsf{p}$, $q = \mathsf{q}$, then

$$\mathsf{Vec}\big(\mathsf{Tendency}^{\mathbb{H}}(p,q)\big) = -\mathsf{p}\times\mathsf{q},$$

and hence

$$\big\|\mathsf{Vec}\big(\mathsf{Tendency}^{\mathbb{H}}(p,q)\big)\big\| = \|\mathsf{p}\|\,\|\mathsf{q}\|\sin\theta,$$

that is, the modulus of the vector part is the quaternion sine tendency, whose direction is given by the cross product.

Proof. By the quaternion multiplication formula

$$(a+\mathsf{A})(b+\mathsf{B}) = ab - \mathsf{A}\cdot\mathsf{B} + a\mathsf{B} + b\mathsf{A} + \mathsf{A}\times\mathsf{B}.$$

Take $b = q_0$ and $\mathsf{B} = -\mathsf{q}$; then

$$p\overline{q} = (p_0+\mathsf{p})(q_0-\mathsf{q}).$$

The scalar part is

$$p_0q_0 - \mathsf{p}\cdot(-\mathsf{q}) = p_0q_0 + \mathsf{p}\cdot\mathsf{q}.$$

If $p_0 = q_0 = 0$, then

$$p\overline{q} = \mathsf{p}(-\mathsf{q}) = -\mathsf{p}\times\mathsf{q} + \mathsf{p}\cdot\mathsf{q},$$

where the scalar part is $\mathsf{p}\cdot\mathsf{q}$ and the vector part is $-\mathsf{p}\times\mathsf{q}$. Hence the modulus of the vector part is

$$\|\mathsf{p}\times\mathsf{q}\| = \|\mathsf{p}\|\,\|\mathsf{q}\|\sin\theta.$$

This completes the proof. $\square$

Example 3.1 (A numerical example of the quaternion exponential similarity). Take

$$p = i, \qquad q = 1.$$

Then

$$D_{\mathbb{H}}(p,q) = \frac{i \cdot 1}{1 \cdot 1} = i.$$

Write

$$i = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2},$$

so the scalar part is $\cos(\pi/2) = 0$, the vector part is $i\sin(\pi/2) = i$, and the modulus is 1. Through the conjugation action it gives the three-dimensional rotation difference from 1 to i: a rotation by angle π about the i axis. Its difference-angle quaternion and sum-angle quaternion (here $q = 1$, so the two are equal) are both i, embodying the consistency of the quaternion case with the complex case whilst retaining the information of the three-dimensional rotation axis.

Remark 3.8 (The non-commutativity of the quaternion difference-angle quaternion). The complex exponential similarity is unchanged under simultaneous left multiplication or simultaneous right multiplication by a common factor; the quaternion difference-angle quaternion is different: right multiplication by a common factor leaves it unchanged, whilst left multiplication by a common factor in general gives a conjugation transformation

$$D_{\mathbb{H}}(rp, rq) = r\, D_{\mathbb{H}}(p,q)\, \overline{r}.$$

This is precisely the manifestation of quaternion non-commutativity at the level of similarity. If r commutes with $D_{\mathbb{H}}(p,q)$, the conjugation transformation degenerates to invariance. The quaternion difference-angle quaternion therefore records not only the 'rotation angle' but also the 'rotation axis', and is compatible with the double-cover structure $SU(2) \to SO(3)$.

Remark 3.9 (The relation to the spectrum of the 12 similarities and tendencies). The complex exponential similarity and the complex exponential tendency package the cosine family and the sine family of the **basic difference angle** among them into a complex number:

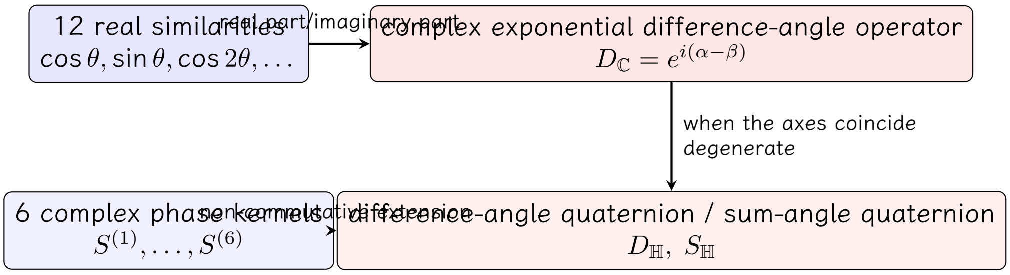
$$\text{Similarity}^{\mathbb{C}} = \text{cosine} + i\,\text{sine},$$

$$\text{Tendency}^{\mathbb{C}} = \text{cosine\_tendency} + i\,\text{sine\_tendency}.$$

In the quaternion case one defines the difference-angle quaternion and the sum-angle quaternion by angle difference and angle sum:

$$D_{\mathbb{H}}(p,q) = \frac{p\overline{q}}{|p||q|}, \qquad S_{\mathbb{H}}(p,q) = \frac{pq}{|p||q|}.$$

When the axes coincide they degenerate respectively to $e^{\mathsf{u}(\alpha-\beta)}$ and $e^{\mathsf{u}(\alpha+\beta)}$, completely parallel to the complex case; in the general case the scalar part corresponds to the cosine family and the vector part to the sine family, whilst the cross term $\mathsf{u}\times\mathsf{v}$ encodes the twist between the two axes; through the conjugation action they give three-dimensional rotation differences and three-dimensional rotational compositions.


12 real similarities
cos θ, sin θ, cos 2θ, . . .
real part/imaginary part
complex exponential difference-angle operator
D_ℂ = e^{i(α−β)}
when the axes coincide
degenerate
6 complex phase kernels
S^{(1)}, . . . , S^{(6)}
noncommutative extension
difference-angle quaternion / sum-angle quaternion
D_ℍ, S_ℍ

complex exponential packaging = cos $+i$ sin of the difference angle; quaternion packaging = axis-angle representation of three-dimensiona

Figure 3.10: The structural relation from the 12 real similarities to the complex exponential/quaternion packaging. The 6 complex phase kernels are the complex packaging of the 12 real similarities; the complex exponential difference-angle operator is the two-dimensional rotational realisation of the phase kernels; the difference-angle/sum-angle quaternions are the non-commutative extension of the same logic to three-dimensional rotations (Remark 3.9).

3.9 The geometric origin of the inverse-square law

Theorem 3.4 (The geometric derivation of the inverse-square law). *Let* F *be a radial vector field centred at* O*, sourceless outside* O*, such that the magnitude of* F *depends only on the distance* r *from* O *(spherical symmetry). If the total flux of* F *through any closed surface enclosing* O *is the constant* Φ_0*, then the magnitude of* F *must be inversely proportional to the square of the distance:*

$$F(r) = \frac{\Phi_0}{4\pi r^2}.$$

Proof. Take the sphere S_r enclosing O (of radius r). By hypothesis the magnitude of F depends only on r, so $F(r)$ is constant on the sphere of radius r; the direction is everywhere radial, so F is everywhere parallel to the outward normal $\mathsf{n} = \frac{\mathsf{r}}{r}$ of the surface element. The total flux through the sphere is:

$$\Phi_0 = \oiint_{S_r} \mathsf{F} \cdot d\mathsf{S} = \oiint_{S_r} F(r)\, dS = F(r) \oiint_{S_r} dS = F(r) \cdot 4\pi r^2.$$

Solving gives $F(r) = \Phi_0/(4\pi r^2)$. Hence $F(r) \propto 1/r^2$. □

Corollary 3.2 (Inverse-square decay and conservation of flux (one-directional implications)). *The converse of Theorem 3.4 holds on spheres: if a radial vector field centred at* O *decays as*

$$\mathsf{F}(\mathsf{r}) = \frac{C}{r^2}\frac{\mathsf{r}}{r}$$

then its flux through any sphere centred at O *is constantly equal to* $4\pi C$*. Moreover, the field* $\mathsf{F} = C\mathsf{r}/r^3$ *of the same form is sourceless for* $r > 0$ $(\nabla \cdot \mathsf{F} = 0)$*, so by the divergence theorem its flux through any closed surface enclosing* O *is likewise constantly* $4\pi C$*.*

Proof. For the sphere S_r centred at O with radius r,

$$\oiint_{S_r} \mathsf{F}\cdot d\mathsf{S} = \oiint_{S_r} \frac{C}{r^2}\,dS = \frac{C}{r^2}\oiint_{S_r} dS = \frac{C}{r^2}\cdot 4\pi r^2 = 4\pi C,$$

the result being independent of r, hence the flux through spheres is conserved. For an arbitrary closed surface Σ enclosing O, apply the divergence theorem to F on the region $\mathbb{R}^3\setminus\{O\}$ (where $\nabla\cdot\mathsf{F} = \frac{1}{r^2}\frac{d}{dr}\left(r^2\cdot\frac{C}{r^2}\right) = 0$ for $r>0$): the flux through Σ equals that through a concentric sphere S_r, hence is also $4\pi C$. ☐

Remark 3.10 ('Conservation of flux' and 'inverse-square decay' are not unconditionally equivalent). Theorem 3.4 and Corollary 3.2 together yield the following two *one-directional* implications, not the biconditional 'conservation of flux $\Longleftrightarrow$ inverse-square decay':

(i) **(Sufficiency)** if $\mathsf{F} = C\mathbf{r}/r^3$, then F is sourceless for $r>0$ and the flux through any closed surface enclosing O is constantly $4\pi C$;

(ii) **(Necessity)** if the radial field $\mathsf{F} = F(r)\hat{\mathbf{r}}$ has constant flux Φ_0 through every sphere centred at O, then $F(r) = \Phi_0/(4\pi r^2)$; here $\nabla\cdot\mathsf{F} = 0$ (for $r>0$) follows from the conclusion and need not be assumed.

The phrase 'mutually necessary and sufficient conditions' must therefore be narrowed: the hypothesis of Theorem 3.4 is that the flux through every *closed surface* enclosing O is constant, whereas (ii) puts only spheres into the hypothesis, so the two hypotheses are not symmetric, and in (ii) the absence of sources is a consequence of the conclusion rather than a premise. Once the hypotheses are removed the equivalence fails—the field $\mathsf{F} = f(r)\hat{\mathbf{r}}$ with $f(r) = 1/r^2 + 1$ $(r>1)$ is sourceless in the region $r>1$ and spherically symmetric, yet its flux through spheres $\Phi(r) = 4\pi(1+r^2)$ varies with r, so it is a **counterexample** to Theorem 3.4 (non-constant flux and a decay that is not purely r^{-2}); see Example 15.13 of the chapter 'Vector Analysis'.

Example 3.2 (Newton's law of universal gravitation). Place a mass M at the source O; the flux of the gravitational field $\mathbf{g}$ through a closed surface enclosing M is $\Phi_0 = -4\pi GM$. Here the flux is negative because the direction of the gravitational field is by convention taken to point towards the source (opposite to the outward radial direction). Substituting into Theorem 3.4 gives:

$$\mathbf{g}(\mathbf{r}) = -G\frac{M}{r^2}\cdot\frac{\mathbf{r}}{r}.$$

Example 3.3 (Coulomb's law). Place a charge Q at the source O; the flux of the electric field E through a closed surface enclosing Q is $\Phi_0 = Q/\varepsilon_0$. Substituting into Theorem 3.4 gives:

$$\mathsf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\cdot\frac{\mathbf{r}}{r}.$$

Remark 3.11 (The inverse-square law and the three symmetry centres). The inverse-square law embodies the concerted action of the three symmetry centres:

(i) **Rotational symmetry (i-symmetry):** the source point O possesses rotational symmetry of the whole space—on any sphere centred at O, the magnitude of the field is everywhere equal.

(ii) **Proportional symmetry (1-symmetry):** the field decays as $1/r^2$, that is, for each doubling of the distance the magnitude of the field shrinks to $1/4$ of its former value.

(iii) **Additive reflection symmetry (0-symmetry):** along any straight line through the source point O, the radial direction of the field exhibits reflection symmetry.

3.10 Complete correspondence table between the three symmetry centres and vector operations

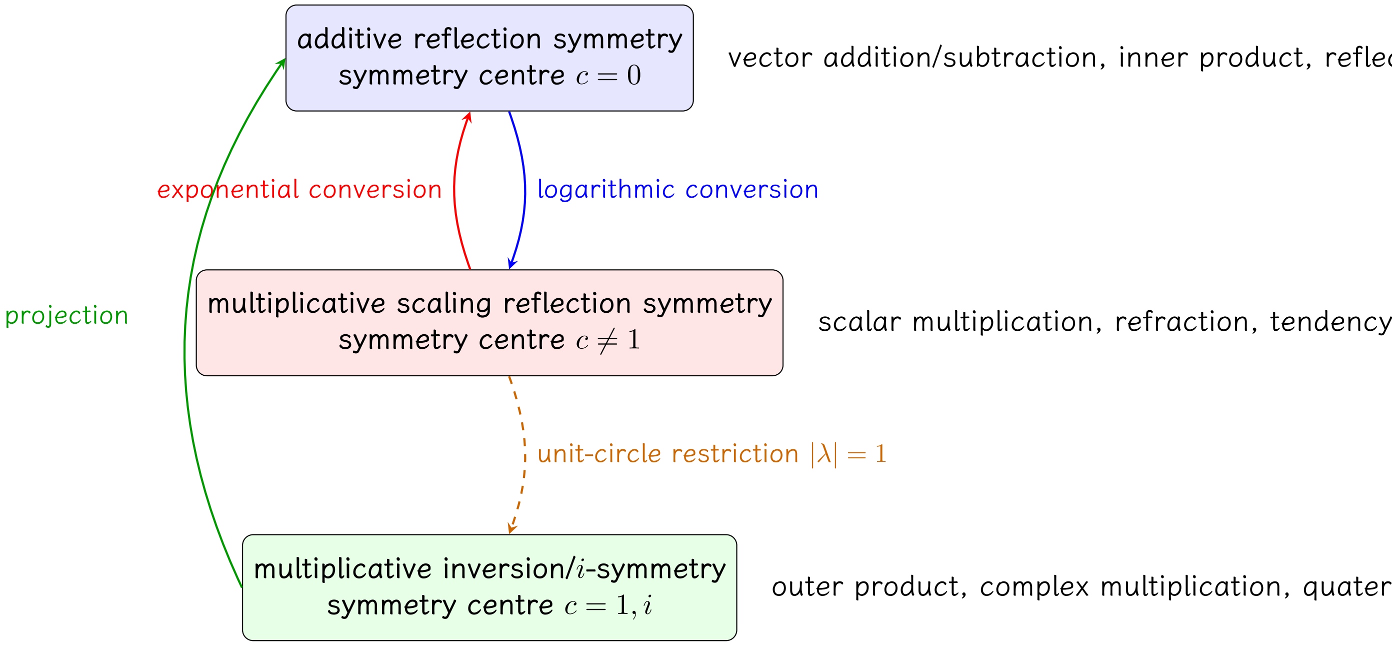

additive reflection symmetry
symmetry centre $c = 0$
vector addition/subtraction, inner product, refle
exponential conversion
logarithmic conversion
projection
multiplicative scaling reflection symmetry
symmetry centre $c \neq 1$
scalar multiplication, refraction, tendency
unit-circle restriction $|\lambda| = 1$
multiplicative inversion/i-symmetry
symmetry centre $c = 1, i$
outer product, complex multiplication, quater

Figure 3.11: The correspondence between the three symmetry centres and the vector operations. Additive reflection symmetry ($c = 0$) corresponds to vector addition/subtraction, the inner product and reflection; multiplicative scaling reflection symmetry ($c \neq 1$) corresponds to scalar multiplication, refraction and tendency scaling; multiplicative inversion/i-symmetry ($c = 1, i$) corresponds to the outer product, complex multiplication, quaternion multiplication, the complex exponential similarity and the difference-angle/sum-angle quaternions. The logarithmic/exponential maps realise the conversion between additive reflection symmetry and multiplicative scaling reflection symmetry, whilst the unit-circle restriction ($|\lambda| = 1$) realises the hierarchical progression from multiplicative scaling reflection symmetry to multiplicative inversion symmetry.

Remark 3.12 (The conservation and closure themes of this chapter). The conservation theme of this chapter consists of three groups of invariances: the cosine

Table 3.4: The complete correspondence between the three basic generating transformations, the vector operations and the symmetry centres

Symmetry centre and symmetry type	Basic transformation	Vector operation	Conserved quantity
0 (additive reflection symmetry)	Translation	vector addition $\mathbf{u}+\mathbf{v}$	momentum
		vector subtraction $\mathbf{u}-\mathbf{v}$	
		inner product $\mathbf{a}\cdot\mathbf{b}$	
		reflection	
1 ($c \neq 1$, multiplicative scaling reflection symmetry)	Scaling	scalar multiplication $\alpha\mathbf{u}$	phase-space volume
		refraction (scaling of the sine tendency)	
		scaling of the modulus	
1 ($c = 1$) / i (multiplicative inversion/i-symmetry)	Rotation	outer product $\mathbf{a}\times\mathbf{b}$	angular momentum
		complex multiplication $z_1 z_2$	
		quaternion multiplication	
		complex exponential similarity, difference-angle quaternion, sum-angle quaternion	

tendency (inner product) is conserved under the orthogonal group $O(n)$ (Proposition 3.7), the absolute value of the sine tendency (the modulus of the outer product) is conserved under pure rotations $SO(3)$ (Proposition 3.8), and reflection preserves the modulus (Definition 3.23) whilst refraction preserves the sine tendency along the interface normal (Definition 3.24); the flux of a sourceless radial field is conserved on a closed surface (Theorem 3.4 and Corollary 3.2). The closure theme is borne jointly by the 2-periodic closed orbit given by the reflection involution (Definition 3.23) and by the flux criterion on a closed surface. These conserved quantities and closure criteria are the concrete manifestation, at the level of vector operations, of the three conservation laws of the 'Operations and Periodicity' chapter (momentum, angular momentum, phase-space volume).

3.11 Involutions and period two: the periodic structure of discrete symmetry

The correspondence established in this chapter between the three symmetry centres and the vector operations has a direct and rigorous periodic consequence: **an involution without fixed points gives a closed orbit of period exactly 2.** This is at once the interface of this chapter with the 'periodicity' theme of the whole book, and the first instance at the vector level of the smallest non-trivial period.

Proposition 3.16 (Involutions induce closed orbits of period 2)**.** Let $\sigma : X \to X$ satisfy $\sigma^2 = \mathrm{id}$ and have no fixed point on X. Then for every $x \in X$ the orbit

$$\mathcal{O}_x = \{x,\ \sigma(x)\}$$

contains exactly two elements, and the restriction of σ to $\mathcal{O}_x$ is a 2-cycle; that is, every orbit of σ is a closed orbit of least positive period 2.

Proof. From $\sigma^2 = \mathrm{id}$ we get $\sigma(\sigma(x)) = x$, so $\mathcal{O}_x = \{x, \sigma(x)\}$ is closed under σ. Moreover, the absence of fixed points implies $\sigma(x) \neq x$, so $\mathcal{O}_x$ contains exactly two elements. Hence $\sigma^2(x) = x$ whilst $\sigma(x) \neq x$, and the least positive period is 2. □

Corollary 3.3 (Both symmetry centres of this chapter yield period 2)**.** Among the generating transformations of this chapter:

(i) **Additive reflection** $\mathsf{v} \mapsto -\mathsf{v}$ has no fixed point on $\mathbb{R}^n \setminus \{0\}$, and hence gives a 2-periodic closed orbit; its symmetry centre is $c = 0$.

(ii) **Multiplicative inversion** $z \mapsto z^{-1}$ has no fixed point on $\mathbb{C} \setminus \{0, 1, -1\}$ (since $z^{-1} = z \iff z = \pm 1$), and likewise gives a 2-periodic closed orbit; its symmetry centre is $c = i$.

Together the two show that **strict periodicity need not come from continuous rotation, and a discrete involution likewise produces periodicity**, except that it can give only the smallest possible non-trivial period 2.

Proof. (i) $-\mathsf{v} = \mathsf{v} \iff \mathsf{v} = 0$, so there is no fixed point on $\mathbb{R}^n \setminus \{0\}$, and Proposition 3.16 gives period 2. (ii) Likewise, $z^{-1} = z \iff z^2 = 1 \iff z = \pm 1$. □

Remark 3.13 (Interface with the theme of ‘periodicity’). Proposition 3.16 gives the entry point of this chapter into the ‘periodicity’ theme: **the involution** is the simplest periodic mechanism after the three continuous transformations, and the period it produces can only be 2—this is precisely the simplest non-trivial case of order 2 in ‘finite-order action ⇒ closed orbit’ of the ‘Operations and Periodicity’ chapter. Thereby the three symmetry centres of this chapter acquire a unified interpretation at the level of periodicity:

$$c = 0\,(\text{additive reflection}), \qquad c = i\,(\text{multiplicative inversion}) \implies \text{2-periodic closed orbit.}$$

As to why $c = 1$ (scaling) does not produce a closed orbit: the continuous scaling group $(\mathbb{R}^+, \times)$ is **torsion-free**, having no non-trivial element of finite order (see Corollary 1.7 of the ‘Operations and Periodicity’ chapter).

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	Vector addition $\mathbf{u}+\mathbf{v}$ (Definition 3.4) forms a commutative group (Proposition 3.1); the modulus of the vector sum $\mathbf{a}+\mathbf{b}$ is the half-angle cosine tendency (Proposition 3.9).
Scaling	The modulus scaling of scalar multiplication $\alpha\mathbf{u}$ (Definition 3.5); the inverse-square law $F(r)\propto r^{-2}$ (Theorem 3.4) and Corollary 3.2.
Rotation	Complex multiplication adds the arguments and multiplies the moduli (Proposition 3.11); quaternion multiplication realises the composition of three-dimensional rotations (Proposition 3.13); the difference-angle quaternion as three-dimensional rotation-difference operator (Proposition 3.14); the sum-angle quaternion as three-dimensional rotational composition operator (Theorem 3.1).
Conservation	The cosine tendency (inner product) is invariant under $O(n)$ (Proposition 3.7); the absolute value of the sine tendency (the modulus of the outer product) is invariant under $SO(3)$ (Proposition 3.8); reflection preserves the modulus, and refraction preserves the sine tendency along the interface normal (Definition 3.23, 3.24).
Symmetry	The reflection involution embodies additive reflection symmetry $c=0$ (Definition 3.23); refraction scaling embodies the modulation of i-symmetry by multiplicative scaling reflection symmetry $c\neq 1$ (Definition 3.24); the difference-angle quaternion is invariant under right multiplication and undergoes a conjugation transformation under left multiplication (Remark 3.8).
Closure	Two reflections return to the original, $\mathbf{v}''=\mathbf{v}$, forming a closed orbit of period 2 (Definition 3.23); the inverse-square law takes the conservation of flux on a closed surface as its criterion (Theorem 3.4, Corollary 3.2).
Period	In complex multiplication the addition of arguments makes the rotation angle compose with period 2π (Proposition 3.11); the 2-periodic closed orbit of reflection is the smallest periodic structure of this chapter (Definition 3.23); the composition law of the difference-angle quaternion is closed (Proposition 3.15); when the axes coincide the quaternion case degenerates to the angle difference and angle sum of the complex case (Theorem 3.2).

Exercises

Exercise 1 According to Definition 3.2 and Definition 3.3, explain that the essential difference between a scalar and a vector lies not in 'whether it carries a direction', but in the **transformation law under a group representation**. Give two concrete examples: a scalar field and a vector field, and verify that their transformation behaviour under a spatial rotation is indeed different.

Exercise 2 Prove Proposition 3.1: $\mathbb{R}^n$ forms a commutative group under vector addition, with the zero vector as identity element and the negative vector as inverse element. Further explain that the additive reflection $\sigma_0(\mathsf{u}) = -\mathsf{u}$ is an involution automorphism with $\mathbf{0}$ as its unique fixed point, and point out that the group it generates is isomorphic to C_2.

Exercise 3 Verify the Cartesian-product generation in Lemma 3.1:

$$12 = \{\text{difference angle}, \text{sum angle}\} \times \{\text{basic}, \text{double angle}, \text{half angle}\} \times \{\cos, \sin\}.$$

Write down respectively the explicit expressions of the six cosine similarities and the six sine similarities corresponding to basic difference angle, double-angle difference angle, half-angle difference angle, basic sum angle, double-angle sum angle and half-angle sum angle (see Definitions 3.9 to 3.14).

Exercise 4 Take $\mathsf{a} = (3, 4)$, $\mathsf{b} = (1, \sqrt{3})$ and compute:

(i) the arguments α and β of the two vectors;

(ii) the signed difference angle $\alpha - \beta$ and the sum angle $\phi = \alpha + \beta$;

(iii) the concrete numerical values of the **basic difference-angle cosine**, the **basic difference-angle sine**, the **basic sum-angle cosine** and the **basic sum-angle sine** among the 12 similarities;

(iv) the tendency values among the 12 tendencies corresponding to the above four similarities (the product of the moduli $\|\mathsf{a}\|\|\mathsf{b}\|$ times the similarity).

And verify $(\text{cosine})^2 + (\text{sine})^2 = 1$.

Exercise 5 Prove Proposition 3.9: when $\|\mathsf{a}\| = \|\mathsf{b}\| = r$ and the included angle is θ,

$$\|\mathsf{a} + \mathsf{b}\| = 2r \cos\frac{\theta}{2}, \qquad \|\mathsf{a} - \mathsf{b}\| = 2r \sin\frac{\theta}{2}.$$

Then explain: if the equal-modulus condition is removed, how should the formulae be corrected? Why can they not then be simplified to a single half-angle trigonometric form (see Remark 3.5)?

Exercise 6 Verify the reflection transformation in Definition 3.23:

$$\mathsf{v}' = \mathsf{v} - 2(\mathsf{v} \cdot \mathsf{n})\mathsf{n}$$

is an involution ($\mathsf{v}'' = \mathsf{v}$) and preserves the modulus ($\|\mathsf{v}'\| = \|\mathsf{v}\|$). Further prove that the double-angle cosine similarity of the direction with the normal is conserved before and after reflection, that is, $\cos 2\theta_{\text{in}} = \cos 2\theta_{\text{out}}$. Finally explain why reflection gives the smallest periodic structure of this chapter—a closed orbit of period 2.

Exercise 7 Prove the critical-angle conclusion for Snell's law (Proposition 3.10): when $n_1 > n_2$ and the angle of incidence $\theta_1 > \theta_c = \arcsin(n_2/n_1)$, Snell's law requires $\sin\theta_2 > 1$, which has no solution in the real domain, and total internal reflection occurs. Give a concrete numerical example and compute the critical angle.

Exercise 8 Prove that the complex exponential difference-angle operator

$$D_{\mathbb{C}}(z_1, z_2) = \frac{z_1\overline{z_2}}{|z_1||z_2|}$$

is a two-dimensional rotation-difference operator, that is,

$$D_{\mathbb{C}}(z_1, z_2) \cdot \frac{z_2}{|z_2|} = \frac{z_1}{|z_1|}$$

(Proposition 3.12). And verify the composition law

$$D_{\mathbb{C}}(z_1, z_2)D_{\mathbb{C}}(z_2, z_3) = D_{\mathbb{C}}(z_1, z_3).$$

Exercise 9 According to Definition 3.27, write down the explicit expressions of the difference-angle quaternion $D_{\mathbb{H}}(p, q) = p\overline{q}/(|p||q|)$ and the sum-angle quaternion $S_{\mathbb{H}}(p, q) = pq/(|p||q|)$. Prove that when p, q have the same rotation axis ($\mathsf{u} = \mathsf{v}$), the difference-angle quaternion and the sum-angle quaternion degenerate respectively to

$$e^{\mathsf{u}(\alpha-\beta)}, \qquad e^{\mathsf{u}(\alpha+\beta)},$$

completely parallel to the complex case (Theorem 3.2).

Exercise 10 Prove Theorem 3.3: for $p = p_0 + \mathsf{p}$ and $q = q_0 + \mathsf{q}$, the quaternion exponential tendency $\text{Tendency}^{\mathbb{H}}(p, q) = p\overline{q}$ satisfies

$$\text{Sc}(p\overline{q}) = p_0q_0 + \mathsf{p} \cdot \mathsf{q},$$

and when p, q are pure imaginary quaternions,

$$\text{Vec}(p\overline{q}) = -\mathsf{p} \times \mathsf{q}.$$

Use the quaternion multiplication formula $(a + \mathsf{A})(b + \mathsf{B}) = ab - \mathsf{A} \cdot \mathsf{B} + a\mathsf{B} + b\mathsf{A} + \mathsf{A} \times \mathsf{B}$ to complete the proof.

Exercise 11 Prove Theorem 3.4: let F be a radial vector field centred at O, sourceless outside O, such that the magnitude of F depends only on the distance r, and whose total flux through any closed surface enclosing O is the constant Φ_0. Then $F(r) = \Phi_0/(4\pi r^2)$. Then prove its converse (Corollary 3.2): if the radial field decays as C/r^2, then its flux through any sphere centred at O is constantly $4\pi C$.

Exercise 12 Verify Proposition 3.7 and Proposition 3.8: the cosine tendency (inner product) is invariant under the orthogonal group $O(n)$; the absolute value of the sine tendency (the modulus of the outer product) is invariant under pure rotations $SO(3)$. Prove each by matrix computation, and explain that these two conservation laws correspond respectively to the conservation of the projection measure along the 'translation direction' and along the 'rotation direction'.

Exercise 13 Comprehensive question: according to Table 3.4 and Remark 3.12, explain why this chapter formulates the relation between the **gradient**, the **divergence** and the **curl** on the one hand and **translation**, **scaling** and **rotation** on the other as a 'formal correspondence' rather than a strict theorem. Starting from the difference in level between 'transformations acting on space' and 'operators acting on fields', discuss the caution of this formulation, and point out under what additional conditions this correspondence can become a strict proposition (see the relevant discussion in the 'Vector Analysis' chapter).

Chapter 4

Functions and Limits

Remark 4.1 (Positioning of this chapter). The objects of study in mathematics are quantity and form, and the function is the core tool for describing relations between quantities. This chapter starts from the basic definition of a function and gradually develops the analytic theory of limits, continuity, derivatives, differentials and integrals, thereby providing the analytic foundation for the rigorous arguments of subsequent chapters.

One of the central tasks of this chapter is to extend the **similarity and tendency hierarchy** established in the 'Scalars and Vectors' chapter from the level of trigonometric functions to the level of the calculus. At the vector level, similarity is computed from the difference/sum of the arguments of two vectors. At the function level, we introduce the concept of the **limit angle**—the derivative of a function at a point is the tangent of the direction of the tangent line at that point ($f'(a) = \tan\theta_a$), so that the computation of argument differences/sums is converted into the computation of limit-angle differences/sums. This extension means that similarity and tendency no longer depend on a particular trigonometric form, but are computed directly through **differentiation and integration**, and its scope of application is extended from discrete vectors to continuous functions.

Seven-theme guide for this chapter: Translation—the derivative (directional information) and the limit angle $\theta_f(a) = \arctan f'(a)$; Scaling—the integral (tendency information) and the logarithmic derivative (the conversion between 1-symmetry and 0-symmetry); Rotation—function composition and inverse functions (rotational closure); Conservation—the fundamental theorem of calculus (the dual conservation of direction and tendency); Symmetry—parity decomposition (additive reflection symmetry $c = 0$); Closure—the monotone-interval structure of periodic functions; Period—the three basic types of periodic functions (translation period, scaling period, rotational period).

4.1 Functions

4.1.1 Basic definition of a function

Definition 4.1 (Function). Let X and Y be two non-empty sets. If there is a rule of correspondence f such that for every $x \in X$ there exists a unique $y \in Y$ corresponding to it, then f is called a **function** from X to Y, written $f : X \to Y$.

- X is called the **domain** of f, written dom(f).
- Y is called the **codomain** of f, written codom(f).
- For $x \in X$, the unique corresponding element is written $f(x)$ and is called the **image** of x under f.

- The set $f(X) := \{f(x) : x \in X\} \subseteq Y$ is called the **range** of f, written $\text{im}(f)$.

Two functions f and g are said to be equal if and only if they have the same domain, the same codomain, and $f(x) = g(x)$ for all x in the domain.

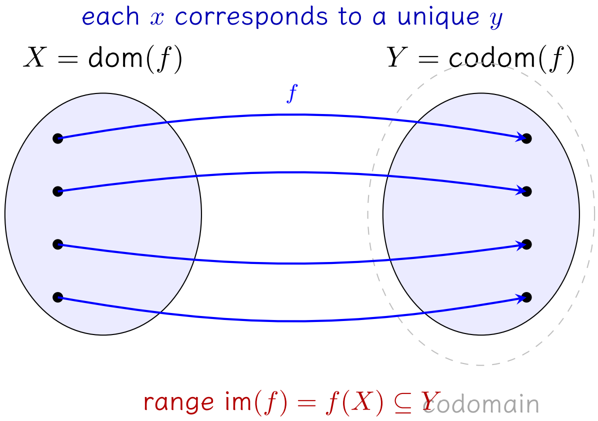

each x corresponds to a unique y
$X = \text{dom}(f)$
$Y = \text{codom}(f)$
f
range $\text{im}(f) = f(X) \subseteq Y$
codomain

Figure 4.1: Set diagram of a function. The left ellipse is the domain X and the right ellipse is the codomain Y; the blue arrows represent the rule of correspondence f. The small dashed ellipse lying inside Y is the range $\text{im}(f)$.

Definition 4.2 (Composition of functions). If $f : X \to Y$ and $g : Y \to Z$, then the composite function $g \circ f : X \to Z$ is defined by $(g \circ f)(x) = g(f(x))$. Composition satisfies the associative law: $h \circ (g \circ f) = (h \circ g) \circ f$.

Definition 4.3 (Image set and preimage set). Let $f : X \to Y$. For $A \subseteq X$, the set $f(A) := \{f(x) : x \in A\}$ is called the image set of A. For $B \subseteq Y$, the set $f^{-1}(B) := \{x \in X : f(x) \in B\}$ is called the preimage set of B.

Definition 4.4 (Injective, surjective and bijective functions). Let $f : X \to Y$.

(i) If $f(X) = Y$, then f is called **surjective**.

(ii) If $x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$, then f is called **injective**.

(iii) A function that is both injective and surjective is called **bijective**. A bijection has a unique **inverse function** $f^{-1} : Y \to X$ satisfying $f^{-1} \circ f = \text{id}_X$ and $f \circ f^{-1} = \text{id}_Y$.

Proposition 4.1 (Order reversal in inverse composition). If $f : X \to Y$ and $g : Y \to Z$ are both bijections, then $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Proof. First we verify that $f^{-1} \circ g^{-1}$ is a left inverse of $g \circ f$. For any $x \in X$,

$$(f^{-1} \circ g^{-1}) \circ (g \circ f)(x) = f^{-1}(g^{-1}(g(f(x)))) = f^{-1}(f(x)) = x.$$

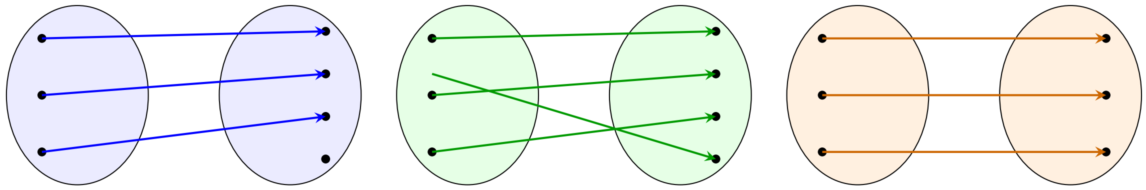
Hence $(f^{-1} \circ g^{-1}) \circ (g \circ f) = \text{id}_X$.

Next we verify that it is a right inverse. For any $z \in Z$,

$$(g \circ f) \circ (f^{-1} \circ g^{-1})(z) = g(f(f^{-1}(g^{-1}(z)))) = g(g^{-1}(z)) = z.$$

Hence $(g \circ f) \circ (f^{-1} \circ g^{-1}) = \text{id}_Z$.

By the uniqueness of the inverse function (the inverse of a bijection is unique), $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. □



(a) Injective
distinct x map
to distinct y;
some points of Y
may be unused

(b) Surjective
every y is mapped
onto by some x;
several x may
map to the same y

(c) Bijective
both injective
and surjective;
there is a unique inverse
map $f^{-1} : Y \to X$

Figure 4.2: Diagram for distinguishing injective, surjective and bijective maps. The difference between the three kinds of mapping with respect to 'whether every x corresponds uniquely and whether the whole of Y is covered' is clear at a glance.

4.1.2 Properties of functions: symmetry, directionality, rotationality and periodicity

Definition 4.5 (Symmetry—parity). Let $f : X \to \mathbb{R}$, where $X \subseteq \mathbb{R}$ is symmetric about the origin.

(i) If $f(-x) = -f(x)$ for all $x \in X$, then f is called an **odd function**, corresponding to additive reflection symmetry ($c = 0$).

(ii) If $f(-x) = f(x)$ for all $x \in X$, then f is called an **even function**.

Theorem 4.1 (Parity decomposition theorem). Any function $f : X \to \mathbb{R}$ defined on a set $X \subseteq \mathbb{R}$ symmetric about the origin can be decomposed uniquely as the sum of an odd function and an even function:

$$f(x) = f_{\text{odd}}(x) + f_{\text{even}}(x),$$

where $f_{\text{odd}}(x) = \frac{f(x)-f(-x)}{2}$, $f_{\text{even}}(x) = \frac{f(x)+f(-x)}{2}$.

Proof. **Existence**: define $f_{\text{odd}}(x) = \frac{f(x)-f(-x)}{2}$, $f_{\text{even}}(x) = \frac{f(x)+f(-x)}{2}$. Verify that f_{odd} is an odd function:

$$f_{\text{odd}}(-x) = \frac{f(-x) - f(x)}{2} = -\frac{f(x) - f(-x)}{2} = -f_{\text{odd}}(x).$$

Verify that f_{even} is an even function:

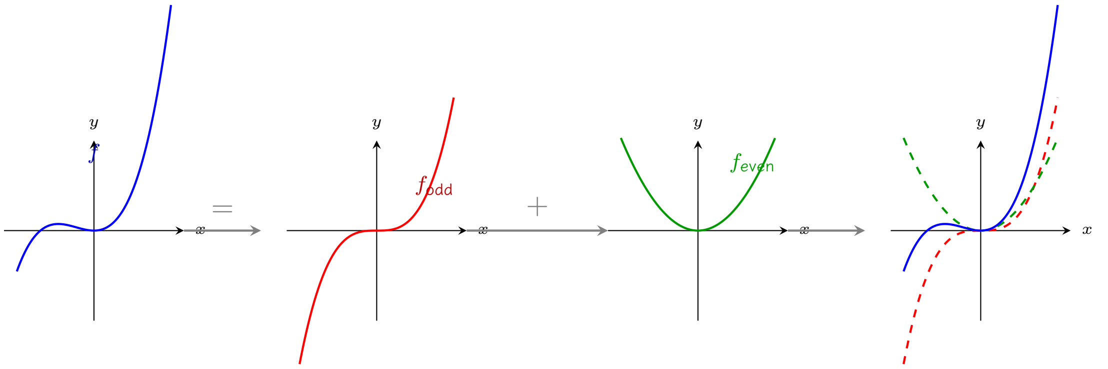
$$f_{\text{even}}(-x) = \frac{f(-x) + f(x)}{2} = f_{\text{even}}(x).$$

Moreover $f(x) = f_{\text{odd}}(x) + f_{\text{even}}(x)$.

Uniqueness: suppose there is another decomposition $f = g + h$, where g is an odd function and h is an even function. Then

$$f(x) = g(x) + h(x), \qquad f(-x) = g(-x) + h(-x) = -g(x) + h(x).$$

Subtracting the two equations gives $f(x)-f(-x)=2g(x)$, hence $g(x)=\frac{f(x)-f(-x)}{2}=f_{\text{odd}}(x)$. Adding the two equations gives $f(x)+f(-x)=2h(x)$, hence $h(x)=\frac{f(x)+f(-x)}{2}=f_{\text{even}}(x)$. Therefore the decomposition is unique. $\square$



(a) General function (b) Odd part (c) Even part (d) Odd + even = f

Figure 4.3: Geometric picture of the parity decomposition theorem. Any function f (blue) decomposes uniquely into the sum of an odd part f_{odd} (red) and an even part f_{even} (green).

Definition 4.6 (Directionality—monotonicity)**.** Let $f: I \to \mathbb{R}$, where $I \subseteq \mathbb{R}$ is an interval.

(i) If $x_1 < x_2 \implies f(x_1) \le f(x_2)$, then f is said to be **monotone increasing** on I.

(ii) If $x_1 < x_2 \implies f(x_1) < f(x_2)$, then f is said to be **strictly monotone increasing** on I.

(iii) If $x_1 < x_2 \implies f(x_1) \ge f(x_2)$, then f is said to be **monotone decreasing** on I.

(iv) If $x_1 < x_2 \implies f(x_1) > f(x_2)$, then f is said to be **strictly monotone decreasing** on I.

Proposition 4.2 (Existence of the inverse of a strictly monotonic function)**.** If $f: I \to \mathbb{R}$ is strictly monotone on the interval I, then f is injective, $f: I \to f(I)$ is bijective, there exists a unique inverse function $f^{-1}: f(I) \to I$, and f^{-1} is also strictly monotone on $f(I)$.

Proof. **Injectivity**: let $x_1 \neq x_2$. Assume without loss of generality that $x_1 < x_2$. If f is strictly increasing, then $f(x_1) < f(x_2)$, so $f(x_1) \neq f(x_2)$. If f is strictly decreasing, then $f(x_1) > f(x_2)$, so $f(x_1) \neq f(x_2)$. Hence f is injective.

Bijectivity: after restricting the codomain to $f(I)$, the map $f: I \to f(I)$ is surjective (by the definition of $f(I)$) and injective (by the previous step), hence bijective.

Existence and uniqueness of the inverse function: a bijection necessarily has a unique inverse function $f^{-1}: f(I) \to I$.

Monotonicity of the inverse function: suppose f is strictly increasing. Take $y_1, y_2 \in f(I)$ with $y_1 < y_2$, and let $x_1 = f^{-1}(y_1)$, $x_2 = f^{-1}(y_2)$. If $x_1 \ge x_2$, then the strict increase of f gives $y_1 = f(x_1) \ge f(x_2) = y_2$, contradicting $y_1 < y_2$. Hence $x_1 < x_2$, that is, $f^{-1}(y_1) < f^{-1}(y_2)$, so f^{-1} is strictly increasing. The case in which f is strictly decreasing is analogous. $\square$

Definition 4.7 (Rotational closure—iterative composition of functions). Let $f : X \to X$.

(i) **Powers of a function**: $f^0 := \mathrm{id}_X$, $f^{n+1} := f \circ f^n$ $(n \geq 0)$.

(ii) **Rotational closure**: if there exists a least positive integer $n \geq 1$ such that $f^n = \mathrm{id}_X$, then f is said to have rotational closure of order n.

Proposition 4.3 (Rotational closure implies bijectivity). Let $f : X \to X$. If there exists $n \geq 1$ such that $f^n = \mathrm{id}_X$, then f is a bijection, and $f^{-1} = f^{n-1}$.

Proof. Verify that f is a bijection: f is injective, because if $f(x_1) = f(x_2)$, then $f^n(x_1) = f^{n-1}(f(x_1)) = f^{n-1}(f(x_2)) = f^n(x_2)$, that is, $x_1 = x_2$. f is surjective, because for any $x \in X$, $x = f^n(x) = f(f^{n-1}(x))$, so x is the image of $f^{n-1}(x)$. The inverse of f is f^{n-1}: $f \circ f^{n-1} = f^n = \mathrm{id}_X$ and $f^{n-1} \circ f = f^n = \mathrm{id}_X$. □

Remark 4.2 (The relationship between function composition and rotation). Function composition is itself not a rotation but a **translation** (left translation $f \mapsto g \circ f$). The reason rotational closure ($f^n = \mathrm{id}$) is called 'rotational' is that it corresponds to the notion of a finite-order element: f, as an element of the function space X^X, has order n and generates the cyclic group $\langle f \rangle \cong C_n$ under the composition operation. This is the concrete realisation, in a function space, of the finite rotational period (definition 1.21 of the 'Operations and Periodicity' chapter).

Definition 4.8 (Periodicity—the synergy of symmetry, directionality and rotation). A period can be understood as a repetition pattern produced by a function under the joint action of symmetry (0-symmetry), directionality (1-symmetry) and rotationality (i-symmetry). According to which symmetry centre dominates, periodic functions fall into three basic types:

(i) **Translation period (0-symmetry dominant)**: let $X \subseteq \mathbb{R}$ satisfy $X + T = X$. If there exists $T > 0$ such that $f(x+T) = f(x)$ for all $x \in X$, then f is said to have translation period T.

(ii) **Scaling period (1-symmetry dominant)**: let $X \subseteq \mathbb{R}^+$ satisfy $qX = X$. If there exists $q > 0, q \neq 1$ such that $f(qx) = f(x)$ for all $x \in X$, then f is said to have scaling period q.

(iii) **Rotational period (i-symmetry dominant)**: let $f : \mathbb{C} \to \mathbb{C}$. If there exists $\theta \in (0, 2\pi)$ such that $f(e^{i\theta}z) = f(z)$ for all $z \in \mathbb{C}$, then f is said to have rotational period θ.

Remark 4.3 (Non-triviality of the scaling period). The scaling period admits no non-trivial continuous solution on $\mathbb{R}$: if $f(qx) = f(x)$ for all $x \in \mathbb{R}$ and f is continuous at 0, then for any $x \in \mathbb{R}$, $f(x) = \lim_{n\to\infty} f(q^n x) = f(0)$ (when $0 < q < 1$, $q^n x \to 0$; when $q > 1$, use q^{-n}). Hence the natural domain of a scaling period is the multiplicative group $\mathbb{R}^+$ (or $\mathbb{C}^\times$), and a non-trivial scaling-periodic function on it is, for example, $f(x) = \sin(2\pi \log_q x)$, which satisfies $f(qx) = \sin(2\pi \log_q(qx)) = \sin(2\pi(\log_q x + 1)) = f(x)$.

Theorem 4.2 (Monotonic intervals of a periodic function within one period). Let $f : \mathbb{R} \to \mathbb{R}$ be continuous with least positive period T. If f is not a constant function, then in any closed interval of length T, f has at least one strictly monotone increasing interval and one strictly monotone decreasing interval.

Proof. By the extreme value theorem (theorem 4.4), f attains its maximum M and its minimum m on $[0, T]$. Since f is not a constant function, $M > m$. Let $f(x_{\text{max}}) = M$ and $f(x_{\text{min}}) = m$, where $x_{\text{max}}, x_{\text{min}} \in [0, T]$.

Case 1: $x_{\text{min}} < x_{\text{max}}$. On the interval $[x_{\text{min}}, x_{\text{max}}]$, the function f rises continuously from m to M. If f were nowhere strictly increasing on that interval, then f would be non-increasing there, but $f(x_{\text{min}}) = m < M = f(x_{\text{max}})$, a contradiction. Hence there exists a strictly increasing subinterval. Similarly, on $[x_{\text{max}}, x_{\text{min}} + T]$ the function f falls continuously from M to m; if it were nowhere strictly decreasing then it would be non-decreasing, contradicting the endpoint values, so a strictly decreasing subinterval exists.

Case 2: $x_{\text{max}} < x_{\text{min}}$. Symmetrically, there exists a strictly decreasing subinterval in $[x_{\text{max}}, x_{\text{min}}]$ and a strictly increasing subinterval in $[x_{\text{min}}, x_{\text{max}} + T]$.

Case 3: $x_{\text{min}} = x_{\text{max}}$. Then $M = m$, contradicting $M > m$, so this is impossible.

Since f has period T, translating the above intervals by integer multiples of T yields the conclusion for any closed interval of length T. □

4.1.3 Elementary functions

Definition 4.9 (Elementary functions)**.** *The elementary functions comprise: power functions x^α, exponential functions a^x $(a > 0)$, logarithmic functions $\log_a x$, trigonometric functions $\sin x, \cos x, \tan x$ and their inverse functions, together with the functions obtained from these by finitely many arithmetic operations and compositions.*

Remark 4.4 (Similarity interpretation of trigonometric functions)*.* The trigonometric functions $\sin x$ and $\cos x$ are strictly periodic functions, with period 2π. From the perspective of the 'Scalars and Vectors' chapter, $\cos\theta$ and $\sin\theta$ are respectively the cosine similarity and the sine similarity between a unit vector and the reference axis.

4.2 Limits and Continuity

4.2.1 Limits of sequences

Definition 4.10 (Limit of a sequence)**.** *Let $\{a_n\}_{n=1}^{\infty}$ be a real sequence and let $L \in \mathbb{R}$. We write $\lim\limits_{n\to\infty} a_n = L$ if*

$$\forall \varepsilon > 0,\ \exists N \in \mathbb{N},\ \forall n > N,\ |a_n - L| < \varepsilon.$$

If no such L exists, the sequence is said to diverge.

Proposition 4.4 (Boundedness of a convergent sequence)**.** *If the sequence $\{a_n\}$ converges, then $\{a_n\}$ is bounded.*

Proof. Let $\lim a_n = A$. Take $\varepsilon = 1$; by the definition of convergence there exists $N \in \mathbb{N}$ such that $|a_n - A| < 1$ whenever $n > N$, whence $|a_n| \le |a_n - A| + |A| < 1 + |A|$.

Let $M = \max\{|a_1|, |a_2|, \ldots, |a_N|, 1 + |A|\}$. Then for all $n \in \mathbb{N}$: - if $n \le N$, then $|a_n| \le M$ (by the definition of M); - if $n > N$, then $|a_n| < 1 + |A| \le M$.

Therefore $\{a_n\}$ is bounded. □

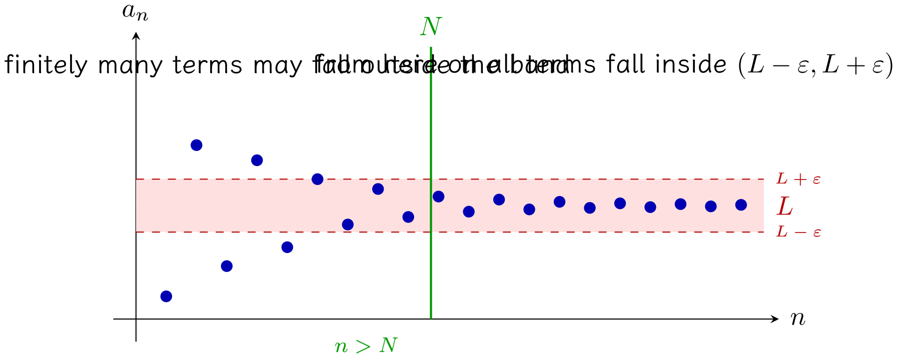

a_n
N
finitely many terms may fall outside the band
from here on all terms fall inside $(L-\varepsilon, L+\varepsilon)$
$L+\varepsilon$
L
$L-\varepsilon$
n
$n > N$

Figure 4.4: Visualisation of the ε-N definition of the limit of a sequence. The red horizontal line is the limit L, and the banded region between the red dashed lines is $(L-\varepsilon, L+\varepsilon)$. All points after $n > N$ fall inside this band.

Theorem 4.3 (Algebraic properties of limits). If $\lim a_n = A$ and $\lim b_n = B$, then

(i) $\lim(a_n \pm b_n) = A \pm B$;

(ii) $\lim(a_n b_n) = AB$;

(iii) if $B \neq 0$, then $\lim(a_n/b_n) = A/B$.

Proof. **(i) Addition**: for any $\varepsilon > 0$, since $\lim a_n = A$ and $\lim b_n = B$, there exist $N_1, N_2 \in \mathbb{N}$ such that

$$n > N_1 \implies |a_n - A| < \frac{\varepsilon}{2}, \qquad n > N_2 \implies |b_n - B| < \frac{\varepsilon}{2}.$$

Take $N = \max\{N_1, N_2\}$. Then for $n > N$,

$$|(a_n + b_n) - (A + B)| \le |a_n - A| + |b_n - B| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence $\lim(a_n + b_n) = A + B$. Subtraction is similar.

(ii) Multiplication: by proposition 4.4, $\{a_n\}$ is bounded; let $|a_n| \le M$ for all n. For any $\varepsilon > 0$ there exist N_1, N_2 such that

$$n > N_1 \implies |a_n - A| < \frac{\varepsilon}{2(|B|+1)}, \qquad n > N_2 \implies |b_n - B| < \frac{\varepsilon}{2M}.$$

Take $N = \max\{N_1, N_2\}$. For $n > N$,

$$\begin{aligned} |a_n b_n - AB| &= |a_n(b_n - B) + B(a_n - A)| \\ &\le |a_n||b_n - B| + |B||a_n - A| \\ &< M \cdot \frac{\varepsilon}{2M} + |B| \cdot \frac{\varepsilon}{2(|B|+1)} \\ &< \varepsilon. \end{aligned}$$

Hence $\lim(a_n b_n) = AB$.

(iii) Division: first we prove that $\lim(1/b_n) = 1/B$ $(B \neq 0)$. Since $\lim b_n = B$ and $|B| > 0$, there exists N_1 such that $|b_n - B| < |B|/2$ for $n > N_1$, whence $|b_n| > |B|/2$.

For any $\varepsilon > 0$ there exists N_2 such that $|b_n - B| < \varepsilon |B|^2/2$ for $n > N_2$. Take $N = \max\{N_1, N_2\}$; then for $n > N$,

$$\left|\frac{1}{b_n} - \frac{1}{B}\right| = \frac{|b_n - B|}{|b_n||B|} < \frac{\varepsilon |B|^2/2}{|B|/2 \cdot |B|} = \varepsilon.$$

Hence $\lim(1/b_n) = 1/B$. Combining this with (ii): $\lim(a_n/b_n) = \lim(a_n \cdot 1/b_n) = A \cdot 1/B = A/B$. □

4.2.2 Limits and continuity of functions

Definition 4.11 (Limit of a function). Let f be defined on some punctured neighbourhood of the point x_0. We write $\lim\limits_{x \to x_0} f(x) = L$ if

$$\forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x \in \operatorname{dom}(f),\ 0 < |x - x_0| < \delta \implies |f(x) - L| < \varepsilon.$$

Definition 4.12 (Continuity). A function f is said to be **continuous** at the point x_0 if $\lim_{x \to x_0} f(x) = f(x_0)$. If f is continuous at every point of an interval I, then f is said to be continuous on I.

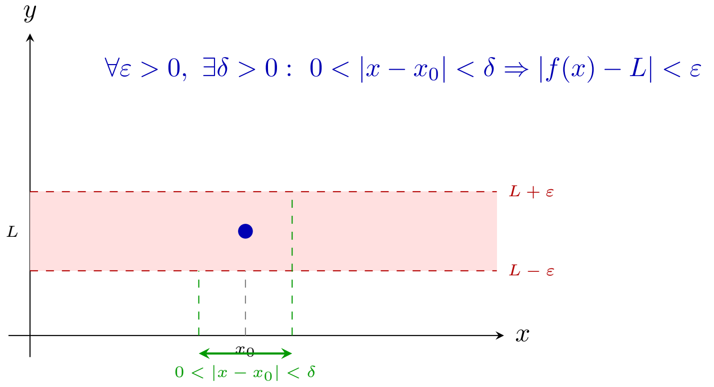

y
$\forall \varepsilon > 0,\ \exists \delta > 0:\ 0 < |x - x_0| < \delta \Rightarrow |f(x) - L| < \varepsilon$
$L + \varepsilon$
L
$L - \varepsilon$
x
x_0
$0 < |x - x_0| < \delta$

Figure 4.5: Visualisation of the ε-δ definition of the limit of a function. The red horizontal band is $(L - \varepsilon, L + \varepsilon)$, and the green interval is the corresponding δ-neighbourhood. Whenever x falls inside that neighbourhood (other than x_0 itself), $f(x)$ must fall inside the ε-band.

Theorem 4.4 (Properties of continuous functions on a closed interval). *Let f be continuous on the closed interval $[a, b]$. Then:*

(i) **Boundedness theorem**: *f is bounded on $[a, b]$.*

(ii) **Extreme value theorem**: *f attains its maximum and minimum on $[a, b]$.*

(iii) **Intermediate value theorem**: *if $f(a) < c < f(b)$, then there exists $\xi \in (a, b)$ such that $f(\xi) = c$.*

(iv) **Uniform continuity theorem**: *f is uniformly continuous on $[a, b]$.*

Proof. **(i) Boundedness theorem**: suppose, to the contrary, that f is unbounded on $[a, b]$. Then for each $n \in \mathbb{N}$ there exists $x_n \in [a, b]$ with $|f(x_n)| > n$. By the Bolzano–Weierstrass theorem (a corollary of the completeness of the real numbers in the 'Number Systems' chapter), the bounded sequence $\{x_n\}$ has a convergent subsequence $\{x_{n_k}\}$; let its limit be $x_0 \in [a, b]$ (since $[a, b]$ is a closed set). By the continuity of f, $\lim_{k\to\infty} f(x_{n_k}) = f(x_0)$. But $|f(x_{n_k})| > n_k \to \infty$, a contradiction. Hence f is bounded.

(ii) Extreme value theorem: by (i), f is bounded on $[a, b]$, so $M := \sup\{f(x) : x \in [a, b]\}$ exists. We must show that there exists $x_0 \in [a, b]$ with $f(x_0) = M$. By the definition of the supremum, for each n there exists $x_n \in [a, b]$ with $M - 1/n < f(x_n) \le M$. By the Bolzano–Weierstrass theorem, $\{x_n\}$ has a convergent subsequence $\{x_{n_k}\} \to x_0 \in [a, b]$. By continuity, $f(x_0) = \lim_k f(x_{n_k}) = M$. The case of the minimum is analogous.

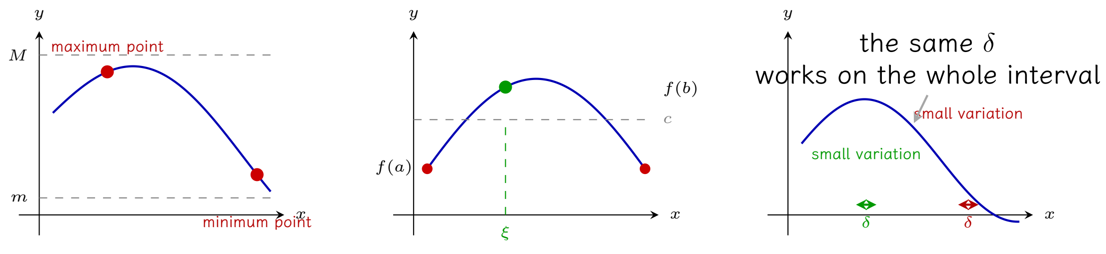
(iii) Intermediate value theorem: let $f(a) < c < f(b)$ (the case $f(a) > c > f(b)$ is similar). Construct the set $S = \{x \in [a, b] : f(x) < c\}$. The set S is non-empty ($a \in S$) and has upper bound b, so $\xi := \sup S$ exists. We must show that $f(\xi) = c$.

If $f(\xi) < c$, then by the continuity of f at ξ there exists $\delta > 0$ such that every $x \in (\xi - \delta, \xi + \delta) \cap [a, b]$ satisfies $f(x) < c$. In particular, there exists $x > \xi$ with $x \in S$, contradicting the fact that ξ is the supremum of S.

If $f(\xi) > c$, then similarly there exists $\delta > 0$ such that every $x \in (\xi - \delta, \xi + \delta) \cap [a, b]$ satisfies $f(x) > c$. This means that $\xi - \delta/2$ is an upper bound for S, contradicting the fact that ξ is the least upper bound.

Hence $f(\xi) = c$.

(iv) Uniform continuity theorem: suppose, to the contrary, that f is not uniformly continuous on $[a, b]$. Then there exists $\varepsilon_0 > 0$ such that for each n there exist $x_n, y_n \in [a, b]$ with $|x_n - y_n| < 1/n$ but $|f(x_n) - f(y_n)| \ge \varepsilon_0$. By the Bolzano–Weierstrass theorem, $\{x_n\}$ has a convergent subsequence $\{x_{n_k}\} \to x_0 \in [a, b]$. Since $|x_{n_k} - y_{n_k}| < 1/n_k \to 0$, the sequence $\{y_{n_k}\}$ also converges to x_0. By the continuity of f, $f(x_{n_k}) \to f(x_0)$ and $f(y_{n_k}) \to f(x_0)$, so $|f(x_{n_k}) - f(y_{n_k})| \to 0$, contradicting $|f(x_{n_k}) - f(y_{n_k})| \ge \varepsilon_0$. $\square$



(a) Boundedness & extreme values (b) Intermediate value theorem (c) Uniform continuity

Figure 4.6: Three basic properties of continuous functions on a closed interval: (a) boundedness and extreme values; (b) the intermediate value theorem; (c) uniform continuity.

4.3 Derivatives: the directional information of a function

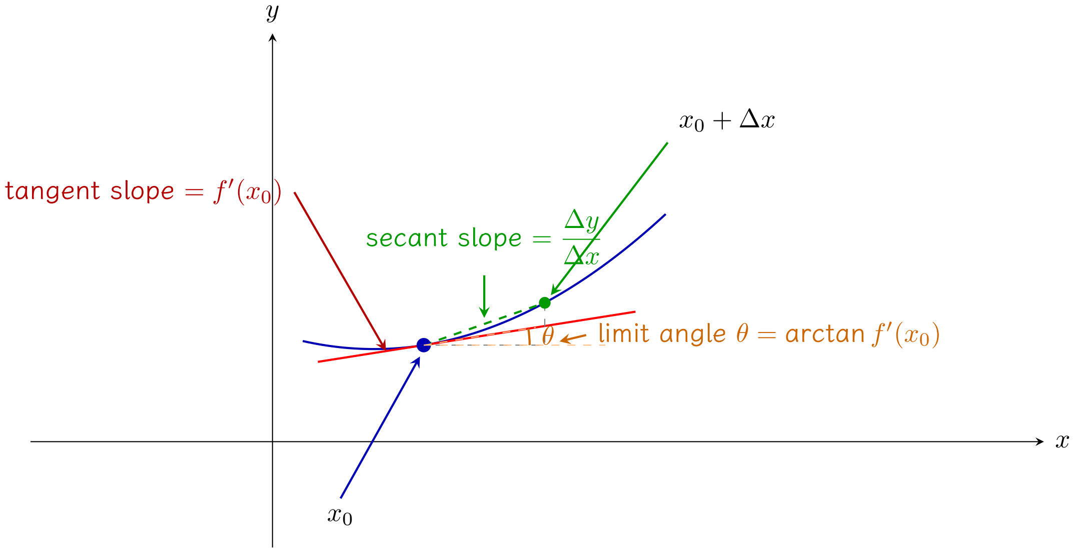
4.3.1 Definition of the derivative and differentiation rules

Definition 4.13 (Derivative and limit angle). Let the function $y = f(x)$ be defined on some neighbourhood of the point x_0. If the limit

$$f'(x_0) := \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

exists and is finite, then f is said to be **differentiable** at x_0, and this limiting value is called the **derivative** of f at x_0. The derivative $f'(x_0)$ is a **directional quantity**: its sign indicates the direction in which the function changes, and its absolute value indicates the rate of change.

Limit angle: $f'(x_0) = \tan\theta$, and $\theta = \arctan(f'(x_0))$ is called the **limit angle** of f at x_0. The limit angle is the central bridge connecting the vector level (geometric argument) with the function level (calculus direction).



y
$x_0 + \Delta x$
tangent slope $= f'(x_0)$
secant slope $= \frac{\Delta y}{\Delta x}$
θ
limit angle $\theta = \arctan f'(x_0)$
x
x_0

Figure 4.7: Geometric interpretation of the derivative and the limit angle. The secant slope tends to the tangent slope $f'(x_0)$ as $\Delta x \to 0$, and $f'(x_0) = \tan\theta$, so $\theta = \arctan f'(x_0)$ is the limit angle of f at x_0. The orange auxiliary dashed lines in the figure mark the two sides of θ: the horizontal direction (the positive direction of the x-axis) and the tangent direction; θ is the angle between the two.

Proposition 4.5 (Differentiability implies continuity). If f is differentiable at x_0, then f is continuous at x_0.

Proof.

$$\lim_{\Delta x \to 0} [f(x_0 + \Delta x) - f(x_0)] = \lim_{\Delta x \to 0} \left(\frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} \cdot \Delta x \right) = f'(x_0) \cdot 0 = 0.$$

Hence $\lim_{\Delta x \to 0} f(x_0 + \Delta x) = f(x_0)$, that is, f is continuous at x_0. □

Theorem 4.5 (Differentiation rules)**.** Let $u(x)$ and $v(x)$ be differentiable at x, and let c be a constant.

(i) $(c)' = 0$.

(ii) $(cu)'(x) = cu'(x)$.

(iii) $(u \pm v)'(x) = u'(x) \pm v'(x)$.

(iv) $(uv)'(x) = u'(x)v(x) + u(x)v'(x)$.

(v) if $v(x) \neq 0$, then $\left(\frac{u}{v}\right)'(x) = \frac{u'(x)v(x)-u(x)v'(x)}{v(x)^2}$.

Proof. **(i) Constant:** $\lim_{\Delta x\to 0} \frac{c-c}{\Delta x} = 0$.
(ii) Scalar multiple: $\lim_{\Delta x\to 0} \frac{cu(x+\Delta x)-cu(x)}{\Delta x} = c \lim_{\Delta x\to 0} \frac{u(x+\Delta x)-u(x)}{\Delta x} = cu'(x)$.
(iii) Sum and difference:

$$\lim_{\Delta x\to 0} \frac{[u(x+\Delta x) \pm v(x+\Delta x)] - [u(x) \pm v(x)]}{\Delta x} = u'(x) \pm v'(x).$$

(iv) Product:

$$\frac{(uv)(x+\Delta x) - (uv)(x)}{\Delta x} = u(x+\Delta x)\frac{v(x+\Delta x) - v(x)}{\Delta x} + v(x)\frac{u(x+\Delta x) - u(x)}{\Delta x}.$$

By the fact that differentiability implies continuity (proposition 4.5), $u(x+\Delta x) \to u(x)$ (as $\Delta x \to 0$). Taking the limit gives $(uv)'(x) = u(x)v'(x) + u'(x)v(x)$.

(v) Quotient: first we prove that $(1/v)' = -v'/v^2$. By the chain rule (theorem 4.6), $(1/v)' = -v'/v^2$ (where $1/v = v^{-1}$, $(v^{-1})' = -v^{-2}v'$). Then, by the product rule, $(u/v)' = (u \cdot v^{-1})' = u'v^{-1} + u(-v^{-2}v') = (u'v - uv')/v^2$. ☐

Theorem 4.6 (Chain rule)**.** If $u = g(x)$ is differentiable at x_0, and $y = f(u)$ is differentiable at $u_0 = g(x_0)$, then

$$(f \circ g)'(x_0) = f'(g(x_0)) \cdot g'(x_0).$$

Proof. Since f is differentiable at u_0, there exists a function $\eta(\Delta u)$ satisfying $\lim_{\Delta u\to 0} \eta(\Delta u) = 0$ such that

$$f(u_0 + \Delta u) - f(u_0) = f'(u_0)\Delta u + \eta(\Delta u)\Delta u.$$

Let $\Delta u = g(x_0+\Delta x) - g(x_0)$. Since g is differentiable at x_0, $\Delta u = g'(x_0)\Delta x + o(\Delta x)$. Substituting gives

$$(f \circ g)(x_0 + \Delta x) - (f \circ g)(x_0) = f'(u_0)[g'(x_0)\Delta x + o(\Delta x)] + \eta(\Delta u)\Delta u.$$

Since $\Delta u \to 0$ (as $\Delta x \to 0$) and $\eta(\Delta u) \to 0$,

$$\lim_{\Delta x\to 0} \frac{(f \circ g)(x_0 + \Delta x) - (f \circ g)(x_0)}{\Delta x} = f'(u_0)g'(x_0).$$

☐

Remark 4.5 (Counterexample of continuous but non-differentiable). Continuity does not imply differentiability. Counterexample: $f(x) = |x|$ is continuous at $x = 0$. Right derivative:

$$\lim_{h\to 0^+} \frac{|0+h| - |0|}{h} = \lim_{h\to 0^+} \frac{h}{h} = 1.$$

Left derivative:

$$\lim_{h\to 0^-} \frac{|0+h| - |0|}{h} = \lim_{h\to 0^-} \frac{-h}{h} = -1.$$

The left and right derivatives are unequal, so f is not differentiable at $x = 0$.

4.3.2 Mean value theorems of the differential calculus

Theorem 4.7 (Fermat's lemma). If f is differentiable at x_0 and attains a local extremum at x_0, then $f'(x_0) = 0$.

Proof. Assume without loss of generality that f attains a local maximum at x_0, that is, there exists $\delta > 0$ such that $f(x) \le f(x_0)$ whenever $|x - x_0| < \delta$.

When $\Delta x > 0$ (and $\Delta x < \delta$), we have $f(x_0 + \Delta x) - f(x_0) \le 0$, so

$$\frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} \le 0.$$

Taking the limit $\Delta x \to 0^+$ gives the right derivative $f'_+(x_0) \le 0$.

When $\Delta x < 0$ (and $|\Delta x| < \delta$), we have $f(x_0 + \Delta x) - f(x_0) \le 0$, but $\Delta x < 0$, so

$$\frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} \ge 0.$$

Taking the limit $\Delta x \to 0^-$ gives the left derivative $f'_-(x_0) \ge 0$.

Since f is differentiable, $f'_+(x_0) = f'_-(x_0) = f'(x_0)$. Therefore $f'(x_0) \le 0$ and $f'(x_0) \ge 0$, so $f'(x_0) = 0$. The case of a local minimum is analogous. ☐

Theorem 4.8 (Rolle's theorem). If f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then there exists $\xi \in (a, b)$ such that $f'(\xi) = 0$.

Proof. By the extreme value theorem (theorem 4.4), f attains its maximum M and its minimum m on $[a, b]$.

Case 1: $M = m$. Then f is identically constant, and $f'(x) = 0$ for all $x \in (a, b)$.

Case 2: $M > m$. Since $f(a) = f(b)$, the maximum and the minimum cannot both be attained at the endpoints (if both were attained at the endpoints, then $f(a) = f(b) = M = m$, a contradiction). Hence at least one extremum is attained inside (a, b). Let $f(\xi) = M$ (or m) with $\xi \in (a, b)$. By Fermat's lemma (theorem 4.7), $f'(\xi) = 0$. ☐

Theorem 4.9 (Lagrange's mean value theorem). If f is continuous on $[a, b]$ and differentiable on (a, b), then there exists $\xi \in (a, b)$ such that

$$f(b) - f(a) = f'(\xi)(b - a).$$

Proof. Construct the auxiliary function

$$\varphi(x) = f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x-a).$$

Verify that φ satisfies the hypotheses of Rolle's theorem: (i) φ is continuous on $[a,b]$ (a continuous function f minus a linear function); (ii) φ is differentiable on (a,b); and

$$\varphi(a) = f(a) - f(a) - 0 = 0, \qquad \varphi(b) = f(b) - f(a) - \frac{f(b)-f(a)}{b-a}(b-a) = 0.$$

By Rolle's theorem (theorem 4.8), there exists $\xi \in (a,b)$ with $\varphi'(\xi) = 0$. Now

$$\varphi'(x) = f'(x) - \frac{f(b)-f(a)}{b-a}.$$

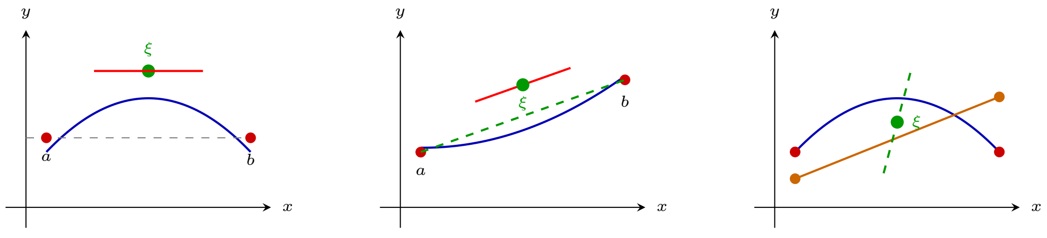
Substituting $x = \xi$ gives $f'(\xi) = \frac{f(b)-f(a)}{b-a}$. ☐

Corollary 4.1 (Mean value property of periodic functions)**.** If f is differentiable on $\mathbb{R}$ and has period T, then within one period there exists $\xi \in (x, x+T)$ such that $f'(\xi) = 0$.

Proof. Applying Lagrange's mean value theorem (theorem 4.9) to the interval $[x, x+T]$, there exists $\xi \in (x, x+T)$ such that

$$f(x+T) - f(x) = f'(\xi)T.$$

By periodicity $f(x+T) = f(x)$, so the left-hand side is zero; hence $f'(\xi)T = 0$, that is, $f'(\xi) = 0$. ☐



(a) Rolle: $f'(\xi) = 0$ (b) Lagrange: $f'(\xi) = \frac{f(b)-f(a)}{b-a}$ (c) Cauchy: $\frac{f'(\xi)}{g'(\xi)} = \frac{f(b)-f(a)}{g(b)-g(a)}$

Figure 4.8: Geometric pictures of the three mean value theorems: (a) Rolle's theorem; (b) Lagrange's mean value theorem; (c) Cauchy's mean value theorem. The dashed line is the secant, and the solid line is the corresponding parallel tangent.

4.3.3 Higher-order derivatives and partial derivatives

Proposition 4.6 (Directional meaning of higher-order derivatives)**.** Let f be n times differentiable at the point x_0.

(i) **First-order derivative** $f'(x_0)$: tangent direction, limit angle $\theta^{(1)} = \arctan(f'(x_0))$.

(ii) **Second-order derivative** $f''(x_0)$: concavity direction, limit angle $\theta^{(2)} = \arctan(f''(x_0))$.

(iii) n**-th order derivative** $f^{(n)}(x_0)$: a finer bending direction.

Proof. (i) This follows directly from definition 4.13: $f'(x_0) = \tan\theta^{(1)}$, $\theta^{(1)} = \arctan(f'(x_0))$.

(ii) $f''(x_0) = (f')'(x_0)$ is the derivative of the derived function f' at x_0. By the definition of the derivative, the sign of $f''(x_0)$ determines the monotonicity of f': $f''(x_0) > 0$ means that f' is increasing near x_0 (the function is concave up), whilst $f''(x_0) < 0$ means that f' is decreasing near x_0 (the function is concave down). The limit angle $\theta^{(2)} = \arctan(f''(x_0))$ measures the degree of inclination of the 'concavity direction'.

(iii) By mathematical induction: $f^{(n)}(x_0) = (f^{(n-1)})'(x_0)$ is the derivative at x_0 of the $(n-1)$-th derived function, and it characterises the rate of change of the $(n-1)$-th direction, that is, a 'finer bending direction'. □

Definition 4.14 (Partial derivative as a single-direction operator)**.** Let the function $u = f(x, y)$ of two variables be defined on a neighbourhood of the point (x_0, y_0). If the limit

$$\frac{\partial f}{\partial x}(x_0, y_0) := \lim_{\Delta x\to 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x}$$

exists and is finite, then f is said to be **partially differentiable** with respect to x at (x_0, y_0). The derivative $\frac{\partial f}{\partial y}$ is defined similarly.

The partial derivative $\frac{\partial f}{\partial x}$ is the slope of the tangent at x_0 to the curve obtained by intersecting the surface $z = f(x, y)$ with the plane $y = y_0$.

Definition 4.15 (Total differential and directional increment)**.** If the total increment of $f(x, y)$ at (x_0, y_0) can be written as

$$\Delta f = A\Delta x + B\Delta y + o\left(\sqrt{\Delta x^2 + \Delta y^2}\right),$$

then f is said to be **differentiable** at (x_0, y_0), and $df = A\,dx + B\,dy$ is the **total differential**. In this case necessarily $A = \frac{\partial f}{\partial x}$, $B = \frac{\partial f}{\partial y}$.

Definition 4.16 (Gradient as the direction of maximal directional derivative)**.** The **gradient** of the function $f(x, y)$ at the point (x_0, y_0) is

$$\nabla f(x_0, y_0) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right).$$

The gradient direction is the direction in which the directional derivative attains its maximum, and its modulus equals the maximum rate of change.

Definition 4.17 (Higher-order partial derivatives and mixed partial derivatives)**.** If the partial derivative $\frac{\partial f}{\partial x}$ is itself partially differentiable, then

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right), \quad \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$$

are called second-order partial derivatives.

Proposition 4.7 (Relationship between differentiability and partial derivatives)**.** *If f is differentiable at (x_0, y_0), then both first-order partial derivatives exist. Conversely, the existence of the partial derivatives does not guarantee differentiability.*

Proof. **Forward direction**: let f be differentiable at (x_0, y_0). Then

$$\Delta f = A\Delta x + B\Delta y + o\Big(\sqrt{\Delta x^2 + \Delta y^2}\Big).$$

Taking $\Delta y = 0$ gives

$$f(x_0 + \Delta x, y_0) - f(x_0, y_0) = A\Delta x + o(|\Delta x|).$$

Dividing by Δx and taking the limit:

$$\frac{\partial f}{\partial x}(x_0, y_0) = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x} = A.$$

Similarly $\frac{\partial f}{\partial y}(x_0, y_0) = B$.

Converse direction (counterexample): see note 4.6. ☐

Lemma 4.1 (Existence of the directional derivative)**.** *If f is differentiable at (x_0, y_0), then for any unit direction vector $\boldsymbol{l} = (\cos\alpha, \cos\beta)$, the directional derivative*

$$\frac{\partial f}{\partial \boldsymbol{l}} = \nabla f \cdot \boldsymbol{l} = f_x \cos\alpha + f_y \cos\beta$$

exists.

Proof. By the definition of the total differential (definition 4.15),

$$f(x_0 + t\cos\alpha, y_0 + t\cos\beta) - f(x_0, y_0) = f_x(x_0, y_0)\, t\cos\alpha + f_y(x_0, y_0)\, t\cos\beta + o(t).$$

Dividing by t and letting $t \to 0$:

$$\frac{\partial f}{\partial \boldsymbol{l}}(x_0, y_0) = \lim_{t \to 0} \frac{f(x_0 + t\cos\alpha, y_0 + t\cos\beta) - f(x_0, y_0)}{t} = f_x \cos\alpha + f_y \cos\beta = \nabla f \cdot \boldsymbol{l}.$$

☐

Theorem 4.10 (Multivariable chain rule)**.** *Let $u = f(x, y)$ be differentiable and let $x = x(t), y = y(t)$ be differentiable; then*

$$\frac{du}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}.$$

Proof. Since f is differentiable,

$$\Delta f = \frac{\partial f}{\partial x}\Delta x + \frac{\partial f}{\partial y}\Delta y + o\Big(\sqrt{\Delta x^2 + \Delta y^2}\Big).$$

Since $x(t), y(t)$ are differentiable,

$$\Delta x = x'(t)\Delta t + o(\Delta t), \qquad \Delta y = y'(t)\Delta t + o(\Delta t).$$

Substituting gives

$$\Delta f = \left(\frac{\partial f}{\partial x}x'(t) + \frac{\partial f}{\partial y}y'(t)\right)\Delta t + o(\Delta t).$$

Dividing by Δt and taking the limit gives the result. □

Theorem 4.11 (Equality of mixed partial derivatives (Clairaut's theorem))**.** If f_{xy} and f_{yx} are continuous on a neighbourhood of the point (x_0, y_0), then

$$\frac{\partial^2 f}{\partial x \partial y}(x_0, y_0) = \frac{\partial^2 f}{\partial y \partial x}(x_0, y_0).$$

Proof. Consider the second-order difference

$$\Delta(h,k) = f(x_0 + h, y_0 + k) - f(x_0 + h, y_0) - f(x_0, y_0 + k) + f(x_0, y_0).$$

Define $\varphi(x) = f(x, y_0 + k) - f(x, y_0)$. Then

$$\Delta(h,k) = \varphi(x_0 + h) - \varphi(x_0).$$

By Lagrange's mean value theorem, there exists ξ between x_0 and $x_0 + h$ such that

$$\Delta(h,k) = h\varphi'(\xi) = h[f_x(\xi, y_0 + k) - f_x(\xi, y_0)].$$

Applying the mean value theorem again in y, there exists η between y_0 and $y_0 + k$ such that

$$\Delta(h,k) = hk\, f_{xy}(\xi, \eta).$$

Similarly, define $\psi(y) = f(x_0 + h, y) - f(x_0, y)$; then

$$\Delta(h,k) = \psi(y_0 + k) - \psi(y_0) = k\psi'(\eta') = k[f_y(x_0 + h, \eta') - f_y(x_0, \eta')] = hk\, f_{yx}(\xi', \eta').$$

Hence

$$f_{xy}(\xi, \eta) = f_{yx}(\xi', \eta').$$

Letting $(h,k) \to (0,0)$, we have $(\xi, \eta) \to (x_0, y_0)$ and $(\xi', \eta') \to (x_0, y_0)$. By the continuity of f_{xy} and f_{yx},

$$f_{xy}(x_0, y_0) = f_{yx}(x_0, y_0).$$

□

Remark 4.6 (Counterexample of partial differentiability without differentiability). $f(x,y) = \frac{xy}{x^2+y^2}$ (with $f(0,0) = 0$). At $(0,0)$:

$$\frac{\partial f}{\partial x}(0,0) = \lim_{\Delta x \to 0} \frac{f(\Delta x, 0) - f(0,0)}{\Delta x} = \lim_{\Delta x \to 0} \frac{0 - 0}{\Delta x} = 0,$$

and similarly $\frac{\partial f}{\partial y}(0,0) = 0$. But the limit along the line $y = kx$ is

$$\lim_{x \to 0} f(x, kx) = \lim_{x \to 0} \frac{kx^2}{x^2 + k^2x^2} = \frac{k}{1 + k^2},$$

which varies with k; hence f is discontinuous at $(0,0)$, and a fortiori not differentiable.

4.4 Differentials: the linear approximation of a function

Definition 4.18 (Differential as the linear principal part of the directional increment)**.** Let $y = f(x)$ be differentiable at x_0. If the increment

$$\Delta y = f(x_0 + \Delta x) - f(x_0) = f'(x_0)\,\Delta x + o(\Delta x) \quad (\Delta x \to 0),$$

then $dy := f'(x_0)\,dx$ is called the **differential** of f at x_0.

Lemma 4.2 (Consistency of differential and derivative operations)**.** Let $u(x), v(x)$ be differentiable at x and let c be a constant; then

(i) $d(c) = 0$;

(ii) $d(cu) = c\,du$;

(iii) $d(u \pm v) = du \pm dv$;

(iv) $d(uv) = v\,du + u\,dv$;

(v) $d\left(\frac{u}{v}\right) = \frac{v\,du - u\,dv}{v^2}$ $(v \neq 0)$.

Proof. (i) $d(c) = c'\,dx = 0 \cdot dx = 0$.
(ii) $d(cu) = (cu)'\,dx = cu'\,dx = c\,du$.
(iii) $d(u \pm v) = (u \pm v)'\,dx = (u' \pm v')\,dx = du \pm dv$.
(iv) $d(uv) = (uv)'\,dx = (u'v + uv')\,dx = v(u'\,dx) + u(v'\,dx) = v\,du + u\,dv$.
(v) $d(u/v) = (u/v)'\,dx = \frac{u'v - uv'}{v^2}\,dx = \frac{v(u'\,dx) - u(v'\,dx)}{v^2} = \frac{v\,du - u\,dv}{v^2}$.

Each of the steps above follows directly from the differentiation rules (theorem 4.5). ☐

Theorem 4.12 (Invariance of the first differential form)**.** Let $y = f(u)$ and $u = g(x)$ be differentiable; then $dy = f'(u)\,du = f'(g(x))\,g'(x)\,dx$.

Proof. By the chain rule (theorem 4.6), $(f \circ g)'(x) = f'(g(x))\,g'(x)$. Hence

$$dy = (f \circ g)'(x)\,dx = f'(g(x))\,g'(x)\,dx.$$

On the other hand, $du = g'(x)\,dx$, so $f'(u)\,du = f'(g(x))\,g'(x)\,dx$. The two are equal. ☐

Corollary 4.2 (Linear approximation property of differentials)**.** If f is differentiable at x_0, then for $|x - x_0|$ sufficiently small,

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0).$$

Proof. By the definition of the differential (definition 4.18),

$$f(x) - f(x_0) = f'(x_0)(x - x_0) + o(|x - x_0|).$$

As $|x - x_0| \to 0$, the quantity $o(|x - x_0|)$ tends to zero faster than $|x - x_0|$; hence

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + o(|x - x_0|) \approx f(x_0) + f'(x_0)(x - x_0).$$

☐

4.5 Integration: the tendency information of a function

4.5.1 Integrals in one variable

Definition 4.19 (Indefinite integral as the inverse operation of direction). Let f be continuous on an interval I. If there exists a differentiable function F such that $F'(x) = f(x)$ for all $x \in I$, then F is called an **antiderivative** of f. The set of all antiderivatives of f is called the **indefinite integral** and is written

$$\int f(x)\,dx = F(x) + C.$$

Definition 4.20 (Riemann integral). Let f be bounded on $[a, b]$. For a partition $\mathcal{P} : a = x_0 < x_1 < \cdots < x_n = b$, choose arbitrary points $\xi_i \in [x_{i-1}, x_i]$ and form the Riemann sum

$$S(f, \mathcal{P}, \{\xi_i\}) = \sum_{i=1}^{n} f(\xi_i)\,\Delta x_i.$$

If the limit as $\|\mathcal{P}\| \to 0$ exists and is independent of the partition and of the manner in which the points are chosen, then f is said to be **Riemann integrable** on $[a, b]$, and the limiting value is called the **definite integral**, written $\int_a^b f(x)\,dx$.

Theorem 4.13 (Fundamental theorem of calculus—the duality of direction and tendency). Let f be continuous on $[a, b]$.

(i) The integral with variable upper limit $\Phi(x) = \int_a^x f(t)\,dt$ is differentiable on $[a, b]$, and $\Phi'(x) = f(x)$.

(ii) If F is any antiderivative of f, then $\int_a^b f(x)\,dx = F(b) - F(a)$.

Proof. **(i)** For $\Delta x > 0$ (the case $\Delta x < 0$ is similar),

$$\frac{\Phi(x + \Delta x) - \Phi(x)}{\Delta x} = \frac{1}{\Delta x} \int_x^{x+\Delta x} f(t)\,dt.$$

By the extreme value theorem for continuous functions on a closed interval, f attains its minimum $m(\Delta x)$ and its maximum $M(\Delta x)$ on $[x, x + \Delta x]$, so

$$m(\Delta x) \leq \frac{1}{\Delta x} \int_x^{x+\Delta x} f(t)\,dt \leq M(\Delta x).$$

As $\Delta x \to 0^+$, the continuity of f gives $m(\Delta x) \to f(x)$ and $M(\Delta x) \to f(x)$. By the sandwich theorem,

$$\lim_{\Delta x \to 0^+} \frac{\Phi(x + \Delta x) - \Phi(x)}{\Delta x} = f(x).$$

Similarly the left derivative is also $f(x)$. Hence $\Phi'(x) = f(x)$.

(ii) Let $G(x) = \Phi(x) - F(x)$. By (i), $G'(x) = \Phi'(x) - F'(x) = f(x) - f(x) = 0$. By the corollary to Lagrange's mean value theorem (a function whose derivative is zero is constant), G is constant on $[a, b]$. Therefore

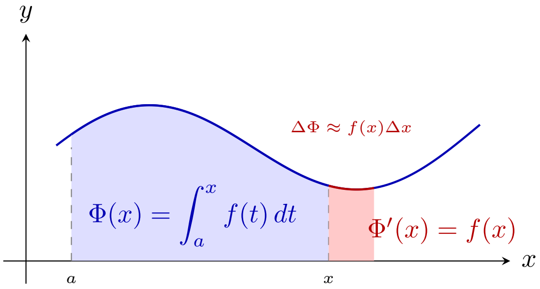
$$\Phi(b) - F(b) = G(b) = G(a) = \Phi(a) - F(a) = -F(a).$$

Hence $\Phi(b) = F(b) - F(a)$, that is, $\int_a^b f(x)\,dx = F(b) - F(a)$. □

Corollary 4.3 (The mutual invertibility of differentiation and integration)**.** If f is continuous, then

$$\frac{d}{dx}\left(\int_a^x f(t)\,dt\right) = f(x), \qquad \int_a^b F'(x)\,dx = F(b) - F(a).$$

Proof. The first formula is given directly by theorem 4.13 (i). The second is given directly by theorem 4.13 (ii) applied to $F' = f$. ☐



y
$\Delta\Phi \approx f(x)\Delta x$
$\Phi(x) = \int_a^x f(t)\,dt$
$\Phi'(x) = f(x)$
x
a
x

Figure 4.9: Geometric picture of the fundamental theorem of calculus. The integral with variable upper limit $\Phi(x)$ is the area under the curve, and its increment satisfies $\Delta\Phi \approx f(x)\Delta x$—this is the geometric meaning of $\Phi'(x) = f(x)$.

Property 4.1 (Basic properties of the definite integral)**.** Let f, g be integrable on $[a, b]$ and let $c \in [a, b]$. Then:

(i) **Linearity**: $\int_a^b (f + cg)\,dx = \int_a^b f\,dx + c\int_a^b g\,dx$;

(ii) **Additivity over intervals**: $\int_a^b f\,dx = \int_a^c f\,dx + \int_c^b f\,dx$;

(iii) **Order preservation**: if $f \le g$, then $\int_a^b f\,dx \le \int_a^b g\,dx$;

(iv) **Absolute value estimate**: $\left|\int_a^b f\,dx\right| \le \int_a^b |f|\,dx$.

Proof. (i) By the linearity of Riemann sums and the linearity of limits:

$$\int_a^b (f + cg) = \lim \sum (f(\xi_i) + cg(\xi_i))\Delta x_i = \lim \sum f(\xi_i)\Delta x_i + c\lim \sum g(\xi_i)\Delta x_i.$$

(ii) Any partition of $[a, b]$ can be refined by inserting c as a division point; the identity $\int_a^b f = \int_a^c f + \int_c^b f$ follows directly from the additivity of Riemann sums.

(iii) If $f \le g$, then for any Riemann sum, $\sum f(\xi_i)\Delta x_i \le \sum g(\xi_i)\Delta x_i$. Taking the limit gives the result.

(iv) From $-|f| \le f \le |f|$ and (iii),

$$-\int_a^b |f|\,dx \le \int_a^b f\,dx \le \int_a^b |f|\,dx,$$

that is, $\left|\int_a^b f\,dx\right| \le \int_a^b |f|\,dx$. ☐

4.5.2 Integrals in several variables

Definition 4.21 (Double integral)**.** Let $f(x,y)$ be bounded on a bounded closed region $D \subset \mathbb{R}^2$. Divide D into n subregions ΔD_i, choose arbitrary points $(\xi_i, \eta_i) \in \Delta D_i$, and form the Riemann sum

$$S = \sum_{i=1}^{n} f(\xi_i, \eta_i)\, \Delta A_i.$$

If the limit exists, then f is said to be **integrable** on D, and the limit is called the **double integral**:

$$\iint_D f(x,y)\, dA = \lim_{\|\mathcal{P}\| \to 0} \sum_{i=1}^{n} f(\xi_i, \eta_i)\, \Delta A_i.$$

Definition 4.22 (Triple integral)**.** Let $\Omega \subseteq \mathbb{R}^3$ be a region in space. The **triple integral** of f over Ω is defined as

$$\iiint_\Omega f(x,y,z)\, dV.$$

The triple integral is the accumulation of tendency in three dimensions.

Lemma 4.3 (Integral mean value theorem)**.** If f is continuous on a bounded closed region D, then there exists $(\xi, \eta) \in D$ such that

$$\iint_D f(x,y)\, dA = f(\xi,\eta) \cdot \mathsf{Area}(D).$$

Proof. By the extreme value theorem, f attains its minimum m and its maximum M on D. Then

$$m \cdot \mathsf{Area}(D) \le \iint_D f(x,y)\, dA \le M \cdot \mathsf{Area}(D).$$

Hence

$$m \le \frac{\iint_D f(x,y)\, dA}{\mathsf{Area}(D)} \le M.$$

By the intermediate value theorem (D is a connected set and $f(D)$ is an interval), there exists $(\xi,\eta) \in D$ such that

$$f(\xi,\eta) = \frac{\iint_D f(x,y)\, dA}{\mathsf{Area}(D)}.$$

□

Theorem 4.14 (Fubini's theorem—iterated integrals and the decomposition of direction)**.** If $f(x,y)$ is continuous on the rectangular region $D = [a,b] \times [c,d]$, then

$$\iint_D f(x,y)\, dA = \int_a^b \left(\int_c^d f(x,y)\, dy \right) dx = \int_c^d \left(\int_a^b f(x,y)\, dx \right) dy.$$

Proof. We prove the first equality (the second is analogous).

Step 1: define $F(x) = \int_c^d f(x,y)\,dy$. We must show that F is continuous on $[a,b]$.

For any $x_1, x_2 \in [a,b]$,

$$|F(x_1) - F(x_2)| \leq \int_c^d |f(x_1,y) - f(x_2,y)|\,dy.$$

By the uniform continuity of f on a bounded closed rectangle (the multivariable version of theorem 4.4), for any $\varepsilon > 0$ there exists $\delta > 0$ such that $|x_1 - x_2| < \delta$ implies $|f(x_1,y) - f(x_2,y)| < \varepsilon/(d-c)$ for all $y \in [c,d]$. Hence $|F(x_1) - F(x_2)| < \varepsilon$, so F is uniformly continuous.

Step 2: prove that $\iint_D f = \int_a^b F(x)\,dx$.

Divide $[a,b]$ into m equal parts and $[c,d]$ into n equal parts, obtaining a grid partition $\mathcal{P}$. Put $m_{ij} = \inf\{f(x,y) : x \in [x_{i-1}, x_i], y \in [y_{j-1}, y_j]\}$ and $M_{ij} = \sup\{f(x,y) : x \in [x_{i-1}, x_i], y \in [y_{j-1}, y_j]\}$.

The lower and upper sums are respectively

$$L(\mathcal{P}) = \sum_{i,j} m_{ij} \Delta x_i \Delta y_j, \qquad U(\mathcal{P}) = \sum_{i,j} M_{ij} \Delta x_i \Delta y_j.$$

For each fixed $x \in [x_{i-1}, x_i]$,

$$\sum_j m_{ij} \Delta y_j \leq \int_c^d f(x,y)\,dy = F(x) \leq \sum_j M_{ij} \Delta y_j.$$

Integrating with respect to x over $[x_{i-1}, x_i]$:

$$\sum_j m_{ij} \Delta y_j \cdot \Delta x_i \leq \int_{x_{i-1}}^{x_i} F(x)\,dx \leq \sum_j M_{ij} \Delta y_j \cdot \Delta x_i.$$

Summing over i:

$$L(\mathcal{P}) \leq \int_a^b F(x)\,dx \leq U(\mathcal{P}).$$

Since f is integrable, $\lim_{\|\mathcal{P}\| \to 0} L(\mathcal{P}) = \lim_{\|\mathcal{P}\| \to 0} U(\mathcal{P}) = \iint_D f$. By the sandwich theorem,

$$\int_a^b F(x)\,dx = \iint_D f(x,y)\,dA.$$

□

Proposition 4.8 (Change of variables for multiple integrals). Let $\mathsf{T} : D^* \to D$ be a bijective transformation, $(u,v) \mapsto (x(u,v), y(u,v))$. Then

$$\iint_D f(x,y)\,dA = \iint_{D^*} f(x(u,v), y(u,v)) \left| \frac{\partial(x,y)}{\partial(u,v)} \right| du\,dv,$$

where $\frac{\partial(x,y)}{\partial(u,v)}$ is the Jacobi determinant, which measures the local area scaling factor of the transformation.

Proof. Divide D^* into small rectangles $[u_i, u_i + \Delta u] \times [v_j, v_j + \Delta v]$. Under T, such a small rectangle is mapped approximately to a parallelogram in D, whose two sides are

$$\mathsf{T}(u_i + \Delta u, v_j) - \mathsf{T}(u_i, v_j) \approx \frac{\partial \mathsf{T}}{\partial u}\Delta u, \qquad \mathsf{T}(u_i, v_j + \Delta v) - \mathsf{T}(u_i, v_j) \approx \frac{\partial \mathsf{T}}{\partial v}\Delta v.$$

The area of this parallelogram is

$$\left\| \frac{\partial \mathsf{T}}{\partial u} \times \frac{\partial \mathsf{T}}{\partial v} \right\| \Delta u \Delta v = \left| \frac{\partial(x,y)}{\partial(u,v)} \right| \Delta u \Delta v.$$

Hence the Riemann sums satisfy

$$\sum f(x_i, y_j)\Delta A_{ij} \approx \sum f(\mathsf{T}(u_i, v_j)) \left| \frac{\partial(x,y)}{\partial(u,v)} \right| \Delta u \Delta v.$$

Taking the limit gives the change-of-variables formula. ☐

Example 4.1 (Triple integral in spherical coordinates). In spherical coordinates (r, θ, ϕ), the volume element is $dV = r^2 \sin\theta \, dr \, d\theta \, d\phi$. The factor r^2 is the manifestation of the inverse-square law in the integration measure.

4.6 The limit angle and differential similarity

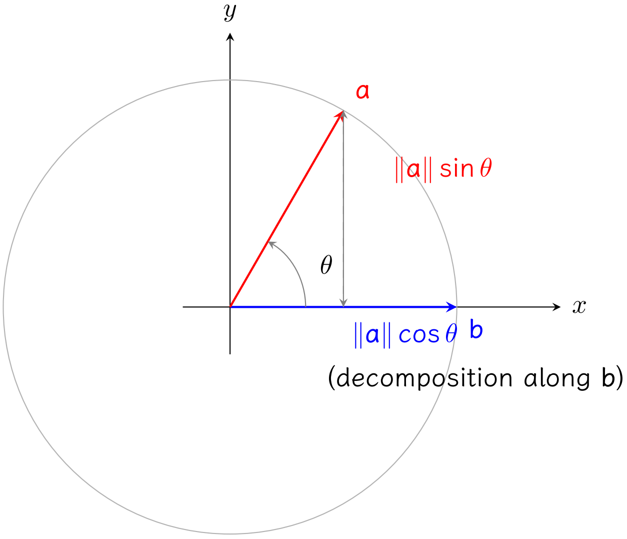


Figure 4.10: The limit angle and decomposition along a direction. Decompose **a** orthogonally along the direction of **b**: the component along the direction, $\|\mathbf{a}\| \cos\theta$, measures the **degree of agreement of direction** (the cosine family), whilst the perpendicular component $\|\mathbf{a}\| \sin\theta$ measures the **degree of deviation of direction** (the sine family). Their normalised versions $\cos\theta$ and $\sin\theta$ are the limit angle of this chapter; the unnormalised versions (multiplied in addition by $\|\mathbf{b}\|$) correspond to the 'tendency' of the later chapters.

Definition 4.23 (Limit angle). Let the function f be differentiable at the point a. The **limit angle** of f at a is defined as

$$\theta_f(a) = \arctan(f'(a)).$$

The limit angle is the central bridge connecting the vector level (geometric argument) with the function level (calculus direction).

Argument of the derivative: for a differentiable function f with $f'(a) \neq 0$, the quantity $\arg f'(a)$ is called the **argument of the derivative** of f at a; it is determined modulo 2π, written $\arg f'(a) \in \mathbb{R}/2\pi\mathbb{Z}$. For a real-valued function of a real variable the limit angle $\theta_f(a) = \arctan(f'(a))$ is precisely the representative of the argument of the derivative that lies in the principal interval $(-\pi/2, \pi/2)$; for a complex-valued function there is in general no such real representative, so in the complex case the argument of the derivative is always used and the limit angle never is.

Proposition 4.9 (Basic properties of the limit angle). (i) **Parallelism**: $f'(a) = g'(b)$ if and only if $\theta_f(a) = \theta_g(b) + k\pi$.

(ii) **Orthogonality**: $f'(a) \cdot g'(b) = -1$ if and only if $\theta_f(a) - \theta_g(b) = \pm\frac{\pi}{2} + k\pi$.

(iii) **Range**: $\theta_f(a) \in (-\pi/2, \pi/2)$.

Proof. (i) The function tan is injective on $(-\pi/2, \pi/2)$. $f'(a) = g'(b) \iff \tan\theta_f = \tan\theta_g \iff \theta_f = \theta_g + k\pi$ ($k \in \mathbb{Z}$, by the periodicity of tan).

(ii)

$$\begin{aligned} f'(a) \cdot g'(b) = -1 &\iff \tan\theta_f \cdot \tan\theta_g = -1 \\ &\iff \sin\theta_f \sin\theta_g = -\cos\theta_f \cos\theta_g \\ &\iff \cos(\theta_f - \theta_g) = 0 \\ &\iff \theta_f - \theta_g = \pm\frac{\pi}{2} + k\pi. \end{aligned}$$

(iii) This follows directly from the range of arctan. ☐

Proposition 4.10 (The limit angle under the three basic transformations). (i) **Translation**: let $g(x) = f(x - c)$. Then $\theta_g(a + c) = \theta_f(a)$. Translation does not change the local direction.

(ii) **Scaling**: let $h(x) = f(\lambda x)$ ($\lambda > 0$). Then $\theta_h(a/\lambda) = \arctan(\lambda f'(a))$. Vertical scaling $h(x) = \alpha f(x)$ ($\alpha > 0$) does not change the **sign** of the limit angle, but in general changes its magnitude.

(iii) **Rotation (complex case: it acts on the argument of the derivative, not on the limit angle)**: let f be real-valued and differentiable and let $k(z) = e^{i\phi} f(z)$ ($\phi \in \mathbb{R}$). Then $k'(a) = e^{i\phi} f'(a)$; that is, multiplication by $e^{i\phi}$ increases the **argument of the derivative** by ϕ: $\arg k'(a) \equiv \arg f'(a) + \phi \pmod{2\pi}$. Here one must not write $\theta_k(a) = \theta_f(a) + \phi$: by Definition 4.23 the limit angle takes the value of **arctan** and lies in $(-\pi/2, \pi/2)$, whereas for $f' \equiv 1$ and $\phi = \pi/2$ the right-hand side is $3\pi/4$, which lies outside that range.

Proof. (i) $g'(a+c) = f'((a+c)-c) = f'(a)$. Hence $\theta_g(a+c) = \arctan(g'(a+c)) = \arctan(f'(a)) = \theta_f(a)$.

(ii) By the chain rule, $h'(x) = \lambda f'(\lambda x)$. At $x = a/\lambda$, $h'(a/\lambda) = \lambda f'(a)$. Hence $\theta_h(a/\lambda) = \arctan(\lambda f'(a))$. Vertical scaling: $h'(x) = \alpha f'(x)$, so $\theta_h(a) = \arctan(\alpha f'(a))$. We have $\theta_h(a) = \theta_f(a)$ only when $\alpha = 1$ or $f'(a) = 0$; in the general case $\arctan(\alpha f') \neq \arctan(f')$. Vertical scaling preserves the sign of the limit angle (since arctan is an odd function and $\alpha > 0$), but changes its magnitude.

(iii) $k'(z) = e^{i\phi} f'(z)$. In the complex plane, multiplication by $e^{i\phi}$ rotates the argument by ϕ, and addition of arguments holds modulo 2π, hence $\arg k'(a) \equiv \arg f'(a) + \phi \pmod{2\pi}$ (the argument of the derivative, see Definition 4.23). What is meant here is the argument of the derivative and not the limit angle: the limit angle $\theta_f(a) = \arctan(f'(a))$ takes values in $(-\pi/2, \pi/2)$ only for real-valued functions, whilst in the complex case k is not real-valued and $\arctan(k'(a))$ is no longer a real angle in that interval. ☐

Definition 4.24 (Differential-difference similarity). Let f, g be differentiable at the points a, b respectively, with limit angles $\theta_f = \arctan(f'(a))$ and $\theta_g = \arctan(g'(b))$.

$$\text{Similarity}_{\text{value}}(f, g; a, b) = \cos(\theta_f - \theta_g) = \frac{1 + f'(a)g'(b)}{\sqrt{1+f'(a)^2}\sqrt{1+g'(b)^2}},$$

$$\text{Similarity}_{\text{dir}}(f, g; a, b) = \sin(\theta_f - \theta_g) = \frac{f'(a) - g'(b)}{\sqrt{1+f'(a)^2}\sqrt{1+g'(b)^2}}.$$

Definition 4.25 (Differential-sum similarity).

$$\text{Similarity}^{+}_{\text{value}}(f, g; a, b) = \cos(\theta_f + \theta_g) = \frac{1 - f'(a)g'(b)}{\sqrt{1+f'(a)^2}\sqrt{1+g'(b)^2}},$$

$$\text{Similarity}^{+}_{\text{dir}}(f, g; a, b) = \sin(\theta_f + \theta_g) = \frac{f'(a) + g'(b)}{\sqrt{1+f'(a)^2}\sqrt{1+g'(b)^2}}.$$

Definition 4.26 (Integral tendency). Let f, g be continuously differentiable on the interval $[a, b]$.

$$\text{Tendency}_{\text{value}}(f, g; [a, b]) = \int_a^b f(x)g(x)\,dx,$$

$$\text{Tendency}_{\text{dir}}(f, g; [a, b]) = \int_a^b |f'(x)g(x) - f(x)g'(x)|\,dx.$$

The integral value tendency is precisely the L^2 inner product; the integrand $|f'g - fg'| = |W(f,g)(x)|$ in the integral direction tendency is the absolute value of the Wronskian determinant.

Remark 4.7 (Conjugation convention for complex-valued functions). Definition 4.26 is for real-valued functions. For complex-valued functions, the integral value tendency uses the conjugate inner product

$$\text{Tendency}_{\text{value}}(f, g; [a, b]) = \int_a^b f(x)\overline{g(x)}\,dx,$$

and in this case $\text{Tendency}_{\text{value}}$ is complex-valued, with modulus $|\langle f, g\rangle|$. This convention agrees with coefficient extraction at the frequency level: at the frequency level, tendency manifests as the phase and amplitude of a function on the frequency dictionary.

Example 4.2 (Limit angle and similarity of simple harmonic periodic functions). For $f(t) = Ae^{i\omega t}$:

(i) $f'(t) = i\omega Ae^{i\omega t}$; differentiation is equivalent to multiplication by $i\omega$, so the **argument of the derivative** increases by $\pi/2$ (modulo 2π); here f is complex-valued, so the argument of the derivative is used and the limit angle is not.

(ii) $f^{(n)}(t) = (i\omega)^n Ae^{i\omega t}$; each differentiation increases the **argument of the derivative** by $\pi/2$ (modulo 2π).

(iii) For two functions of the same frequency, the difference of the arguments of their derivatives is constant on the whole domain (it equals the difference $\phi_1 - \phi_2$ of the initial phases), and therefore the differential-difference similarity is constant on the whole domain.

(iv) Let $f(t) = A_1 e^{i(\omega t+\phi_1)}$ and $g(t) = A_2 e^{i(\omega t+\phi_2)}$. Under the conjugation convention,

$$\int_0^T f(t)\,\overline{g(t)}\,dt = A_1 A_2 T\, e^{i(\phi_1-\phi_2)}.$$

Differentiation amplifies high-frequency direction, and integration amplifies low-frequency tendency—this is the direct manifestation of the direction–tendency duality in the frequency domain.

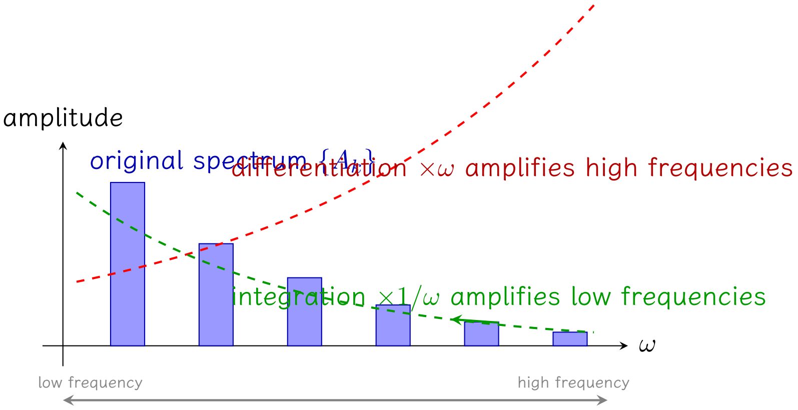


Figure 4.11: The direction–tendency duality in the frequency domain. Differentiation amounts to multiplication by ω, amplifying high-frequency components; integration amounts to multiplication by $1/\omega$, amplifying low-frequency components. This agrees with the qualitative conclusion of this chapter that ‘differentiation amplifies high-frequency direction and integration amplifies low-frequency tendency’.

4.7 Logarithmic Derivative and the Exponential Map

Definition 4.27 (Logarithmic derivative). Let f be differentiable at x with $f(x) > 0$. The **logarithmic derivative** of f is defined as

$$\frac{d}{dx} \ln f(x) = \frac{f'(x)}{f(x)}.$$

Proposition 4.11 (The logarithmic derivative maps scaling to translation).

(i) **Product conversion**: $(\ln(fg))' = (\ln f)' + (\ln g)'$.

(ii) **Power conversion**: $(\ln f^\alpha)' = \alpha(\ln f)'$.

(iii) **Scaling invariance**: $(\ln(cf))' = (\ln f)'$. Enlarging or shrinking the values of a function does not affect its logarithmic derivative.

The logarithmic derivative is the analytic tool that converts multiplicative scaling (1-symmetry) into additive translation (0-symmetry).

Proof. (i) By the chain rule and the properties of the logarithmic function:

$$(\ln(fg))' = \frac{(fg)'}{fg} = \frac{f'g + fg'}{fg} = \frac{f'}{f} + \frac{g'}{g} = (\ln f)' + (\ln g)'.$$

(ii) By the chain rule:

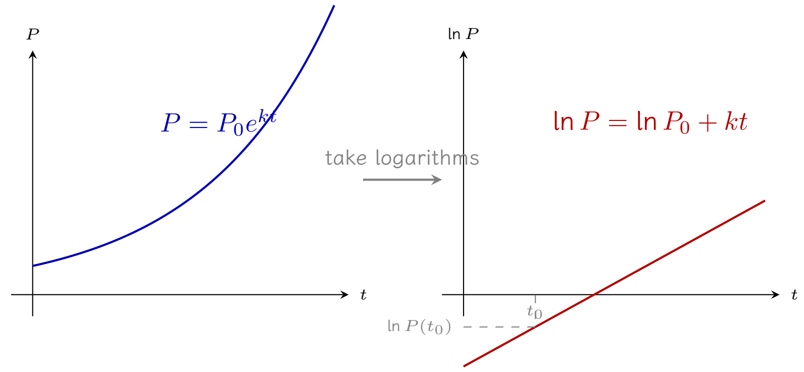
$$(\ln f^\alpha)' = \frac{(f^\alpha)'}{f^\alpha} = \frac{\alpha f^{\alpha-1} f'}{f^\alpha} = \alpha \frac{f'}{f} = \alpha(\ln f)'.$$

(iii) By (i), $(\ln(cf))' = (\ln c)' + (\ln f)' = 0 + (\ln f)' = (\ln f)'$, since $\ln c$ is a constant whose derivative is zero. ☐

Example 4.3 (Exponential growth and the logarithmic derivative). The logarithmic derivative of $P(t) = P_0 e^{kt}$ is

$$(\ln P)' = \frac{P'}{P} = \frac{P_0 k e^{kt}}{P_0 e^{kt}} = k.$$

Exponential growth is a constant growth rate in the proportional domain, and manifests as uniform translation in the logarithmic domain.



(a) Original coordinates: exponential growth

(b) Semilogarithmic coordinates: uniform translation

Figure 4.12: The logarithmic derivative turns scaling into translation. The left figure shows exponential growth $P = P_0 e^{kt}$; after taking logarithms, the right figure becomes the straight line $\ln P = \ln P_0 + kt$. This step converts multiplication (proportional scaling) into addition (translation), which is precisely the geometric meaning of the logarithmic derivative $(\ln f)' = f'/f$.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The derivative (directional information) and the limit angle $\theta_f(a) = \arctan f'(a)$ (the section on derivatives, the section on the limit angle)
Scaling	The integral (tendency information) and the logarithmic derivative (the conversion between 1-symmetry and 0-symmetry, the section on the logarithmic derivative)
Rotation	Rotational closure $f^n = \mathrm{id}$ (the section on properties of functions); the behaviour of the limit angle under rotational transformations (the section on the limit angle)
Conservation	The fundamental theorem of calculus (the dual conservation of direction and tendency, theorem 4.13)
Symmetry	The parity decomposition theorem (additive reflection symmetry $c = 0$, theorem 4.1)
Closure	Rotational closure $f^n = \mathrm{id}$ corresponds to the finite cyclic group C_n (proposition 4.3)
Period	The three basic period types (definition 4.8); the monotone-interval structure of periodic functions (theorem 4.2)

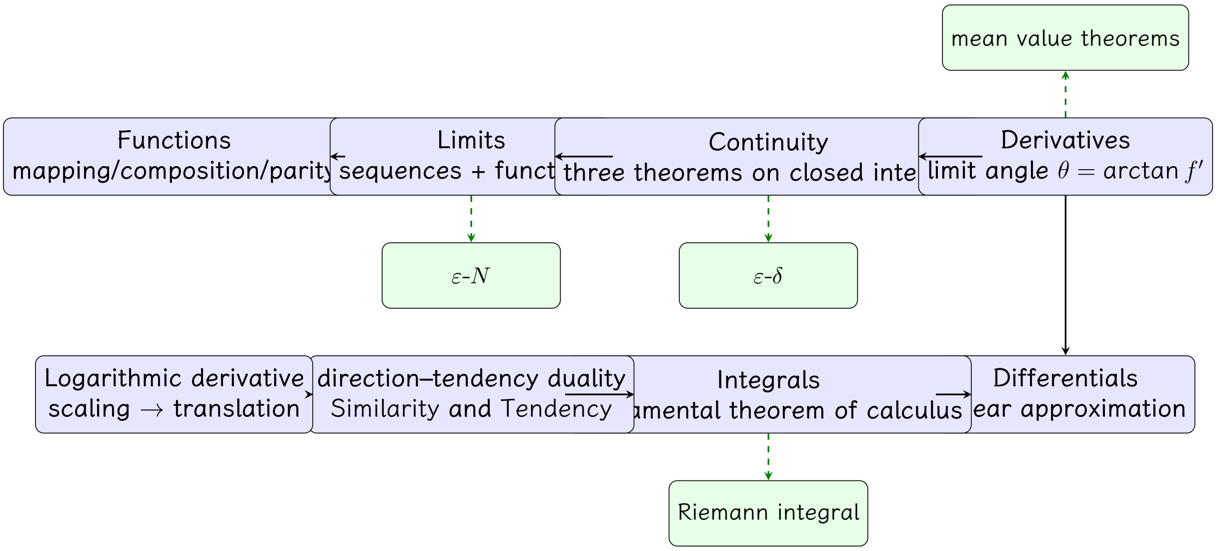

mean value theorems
Functions
mapping/composition/parity
Limits
sequences + funct
Continuity
three theorems on closed inte
Derivatives
limit angle $\theta = \arctan f'$
ε-N
ε-δ
Logarithmic derivative
scaling → translation
direction–tendency duality
Similarity and Tendency
Integrals
ımental theorem of calculus
Differentials
ear approximation
Riemann integral

Figure 4.13: Overall map of the logical structure of this chapter (serpentine loop layout). Function theory is the starting point; limits and continuity provide the language of 'infinite approximation'; the derivative (direction) and the integral (tendency) are the core analytic tools; the fundamental theorem of calculus places the two in duality; and the logarithmic derivative converts scaling into translation, paving the way for the next chapter, 'Differential Equations and Analysis'.

Exercises

Exercise 1 Prove the order-reversal law for the composition of inverse functions (proposition 4.1): if $f : X \to Y$ and $g : Y \to Z$ are both bijections, then

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

Verify the left inverse and the right inverse separately, and give a concrete example showing that this order is not commutative.

Exercise 2 Prove the parity decomposition theorem (theorem 4.1): any function f defined on a set symmetric about the origin can be decomposed uniquely as

$$f = f_{\text{odd}} + f_{\text{even}}, \qquad f_{\text{odd}}(x) = \frac{f(x) - f(-x)}{2}, \quad f_{\text{even}}(x) = \frac{f(x) + f(-x)}{2}.$$

Verify existence and uniqueness separately, and write down the odd part and the even part for $f(x) = e^x$.

Exercise 3 Prove that rotational closure implies bijectivity (proposition 4.3): if $f : X \to X$ satisfies $f^n = \text{id}_X$ $(n \geq 1)$, then f is a bijection and $f^{-1} = f^{n-1}$. Further explain the meaning of the statement in note 4.2 that 'function composition is itself not a rotation but a translation', and explain why $f^n = \text{id}$ corresponds to a finite-order action in a function space.

Exercise 4 Prove the theorem on the monotone intervals of a periodic function within its least positive period (theorem 4.2): let f be continuous with least positive period T; if f is non-constant, then in any closed interval of length T, f has at least one strictly monotone increasing interval and one strictly monotone decreasing interval. Complete the proof using the extreme value theorem and proof by contradiction, and draw a sketch of a periodic function satisfying the conditions.

Exercise 5 Prove the boundedness of a convergent sequence (proposition 4.4): if the sequence $\{a_n\}$ converges, then $\{a_n\}$ is bounded. Write out the full details of the ε-N proof, and give an example showing that a bounded sequence need not converge.

Exercise 6 Prove that differentiability implies continuity (proposition 4.5), and give the counterexample of a continuous but non-differentiable function (note 4.5): $f(x) = |x|$ is continuous but not differentiable at $x = 0$. Compute its left and right derivatives and show that the two are unequal.

Exercise 7 Prove Fermat's lemma (theorem 4.7): if f is differentiable at x_0 and attains a local extremum at x_0, then $f'(x_0) = 0$. Discuss the two cases of a local maximum and a local minimum separately, and explain the role of this lemma in the proof of Rolle's theorem (theorem 4.8).

Exercise 8 Prove Rolle's theorem (theorem 4.8) and Lagrange's mean value theorem (theorem 4.9), and explain how the latter is deduced from the former by constructing an auxiliary function. Further prove the mean value property of periodic functions (corollary 4.1): if f is differentiable with period T, then within one period there exists ξ such that $f'(\xi) = 0$.

Exercise 9 Let $f(x) = x^3 - 3x + 1$. Compute:

(i) the derivative $f'(1)$ of f at $x = 1$;

(ii) the limit angle $\theta_f(1) = \arctan f'(1)$ at that point;

(iii) the limit angle $\theta_g(0)$ of $g(x) = \sin x$ at $x = 0$;

(iv) the differential-difference similarities of the two at the point $(1, 0)$, $\mathrm{Similarity}_{\mathrm{value}}(f, g; 1, 0)$ and $\mathrm{Similarity}_{\mathrm{dir}}(f, g; 1, 0)$ (definition 4.24).

Exercise 10 Prove the second part of the fundamental theorem of calculus (theorem 4.13): if F is any antiderivative of f, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Complete the proof using the integral with variable upper limit $\Phi(x) = \int_a^x f(t)\,dt$ and Lagrange's mean value theorem, and explain the mutual invertibility of differentiation and integration in corollary 4.3.

Exercise 11 Prove that the logarithmic derivative converts scaling into translation (proposition 4.11):

(i) $(\ln(fg))' = (\ln f)' + (\ln g)'$;

(ii) $(\ln f^\alpha)' = \alpha(\ln f)'$;

(iii) $(\ln(cf))' = (\ln f)'$.

Taking $P(t) = P_0 e^{kt}$ as an example (example 4.3), verify that its logarithmic derivative is the constant k, and explain how this expresses a 'constant growth rate in the proportional domain'.

Exercise 12 Prove the theorem on the equality of mixed partial derivatives (theorem 4.11): if f_{xy} and f_{yx} are continuous on a neighbourhood of (x_0, y_0), then $f_{xy}(x_0, y_0) = f_{yx}(x_0, y_0)$. Write out the derivation of the second-order difference $\Delta(h, k)$, and explain that the continuity hypothesis cannot be omitted (see the counterexample 4.6).

Exercise 13 Synthesis question: explain how the limit angle

$$\theta_f(a) = \arctan f'(a)$$

connects the geometric argument at the vector level with the calculus direction at the function level. Using definition 4.23, proposition 4.9 and proposition 4.10, discuss the behaviour of the limit angle under the three basic transformations of translation, scaling and rotation, and point out its significance in the analysis of periodic functions.

Chapter 5

Real Analysis

Remark 5.1 (Positioning of this chapter). Building on the ‘Number Systems’ chapter and the ‘Functions and Limits’ chapter, this chapter establishes the theory of topology, continuity, differentiation, integration and uniform convergence in Euclidean space, and provides the rigorous analytic prerequisites for the subsequent ‘Series and Convolution’, ‘Probability and Distribution’ and ‘Vector Analysis’ chapters. This chapter does not repeat the details of the construction of the completeness of the real numbers given in the ‘Number Systems’ chapter, nor the basic definitions and elementary properties of single-variable limits, continuity, derivatives and integrals given in the ‘Functions and Limits’ chapter; the emphasis of this chapter is: **to lift single-variable analysis to Euclidean spaces and metric spaces**, and to establish **multivariable differential calculus, multiple integrals and line and surface integrals, uniform convergence of sequences and series of functions, parameter-dependent integrals and differentiation under the integral sign**, together with such tools as **null sets and piecewise-smooth boundaries**. These tools are the necessary prerequisites for the rigorous statements of the gradient, the divergence, the curl, Green’s formula, Stokes’ theorem and Gauss’s theorem in the ‘Vector Analysis’ chapter, and they are also the foundation of the arguments on uniform convergence of series of functions, power series and Fourier series in the ‘Series and Convolution’ chapter.

Guide to the seven themes: Translation—translation in Euclidean space and the translation invariance of limits; Scaling—ball neighbourhoods, covers and scales in metric spaces; Rotation—orthogonal transformations of Euclidean space and topological invariance; Conservation—conservation of topological invariants such as compactness, completeness and connectedness; Symmetry—preservation of the symmetry of open sets, closed sets and compact sets under continuous maps; Closure—closure, closed sets, nested closed intervals and the closure of limits; Period—the uniform convergence and parameter-dependent integral tools prepared for periodic series, periodic convolution and Fourier analysis in later chapters.

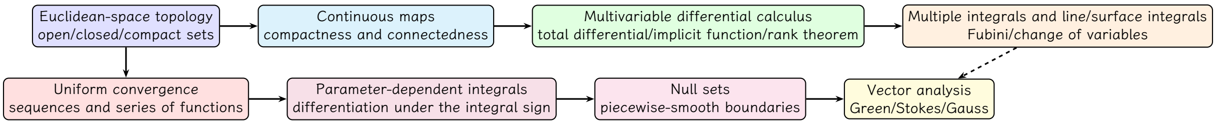

Euclidean-space topology
open/closed/compact sets
Continuous maps
compactness and connectedness
Multivariable differential calculus
total differential/implicit function/rank theorem
Multiple integrals and line/surface integrals
Fubini/change of variables
Uniform convergence
sequences and series of functions
Parameter-dependent integrals
differentiation under the integral sign
Null sets
piecewise-smooth boundaries
Vector analysis
Green/Stokes/Gauss

Figure 5.1: Overview of the structure of this chapter: Euclidean-space topology → continuous maps → multivariable differential calculus → multiple integrals and line and surface integrals; below, uniform convergence, parameter-dependent integrals, null sets and piecewise-smooth boundaries are established in parallel; finally everything flows into vector analysis.

5.1 Euclidean Space and Metric Spaces

5.1.1 Euclidean space

Definition 5.1 (Euclidean space)**.** Let $n \in \mathbb{N}^+$. The n-**dimensional Euclidean space** $\mathbb{R}^n$ is the set

$$\mathbb{R}^n := \{(x_1, \dots, x_n) : x_i \in \mathbb{R},\ i = 1, \dots, n\},$$

equipped with the **inner product**

$$\langle x, y \rangle := \sum_{i=1}^{n} x_i y_i, \qquad x = (x_1, \dots, x_n),\ y = (y_1, \dots, y_n),$$

together with the **norm** induced by the inner product

$$\|x\| := \sqrt{\langle x, x \rangle} = \left(\sum_{i=1}^{n} x_i^2 \right)^{1/2},$$

and the **distance**

$$d(x, y) := \|x - y\|, \qquad x, y \in \mathbb{R}^n.$$

Proposition 5.1 (Basic properties of the inner product and the norm)**.** Let $x, y, z \in \mathbb{R}^n$, $\alpha \in \mathbb{R}$. Then:

(i) **Symmetry of the inner product**: $\langle x, y \rangle = \langle y, x \rangle$;

(ii) **Bilinearity of the inner product**: $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$;

(iii) **Positive definiteness of the inner product**: $\langle x, x \rangle \geq 0$, and $\langle x, x \rangle = 0 \iff x = \mathbf{0}$;

(iv) **Cauchy–Schwarz inequality**: $|\langle x, y \rangle| \leq \|x\| \|y\|$;

(v) **Triangle inequality**: $\|x + y\| \leq \|x\| + \|y\|$;

(vi) **Homogeneity**: $\|\alpha x\| = |\alpha| \|x\|$;

(vii) **Positive definiteness**: $\|x\| \geq 0$, and $\|x\| = 0 \iff x = \mathbf{0}$.

Proof. (i) $\langle x, y \rangle = \sum_i x_i y_i = \sum_i y_i x_i = \langle y, x \rangle$.

(ii) Verify componentwise from the definition of the inner product:

$$\langle \alpha x + \beta y, z \rangle = \sum_i (\alpha x_i + \beta y_i) z_i = \alpha \sum_i x_i z_i + \beta \sum_i y_i z_i = \alpha \langle x, z \rangle + \beta \langle y, z \rangle.$$

(iii) $\langle x, x \rangle = \sum_i x_i^2 \geq 0$, and it equals zero if and only if every $x_i = 0$.

(iv) If $y = \mathbf{0}$, both sides of the inequality are 0 and equality holds. Suppose that $y \neq \mathbf{0}$. For any real number t, by positive definiteness,

$$0 \leq \langle x + ty, x + ty \rangle = \langle x, x \rangle + 2t \langle x, y \rangle + t^2 \langle y, y \rangle.$$

The right-hand side is a quadratic polynomial in t that is always non-negative, so its discriminant is non-positive:

$$(2\langle x, y \rangle)^2 - 4 \langle x, x \rangle \langle y, y \rangle \leq 0,$$

that is, $|\langle x, y\rangle| \le \|x\|\|y\|$.

(v) By the Cauchy–Schwarz inequality,

$$\|x+y\|^2 = \|x\|^2 + 2\langle x, y\rangle + \|y\|^2 \le \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2.$$

Taking square roots of both sides gives the result.

(vi) $\|\alpha x\|^2 = \langle \alpha x, \alpha x\rangle = \alpha^2\langle x, x\rangle = \alpha^2\|x\|^2$, and taking square roots gives $\|\alpha x\| = |\alpha|\|x\|$.

(vii) This follows directly from the positive definiteness of the inner product. ☐

Corollary 5.1 (The three axioms of a distance). Let $x, y, z \in \mathbb{R}^n$. Then:

- (i) **Positive definiteness**: $d(x, y) \ge 0$, and $d(x, y) = 0 \iff x = y$;
- (ii) **Symmetry**: $d(x, y) = d(y, x)$;
- (iii) **Triangle inequality**: $d(x, z) \le d(x, y) + d(y, z)$.

Hence $(\mathbb{R}^n, d)$ is a metric space.

Proof. (i) $d(x, y) = \|x - y\| \ge 0$; $d(x, y) = 0 \iff \|x - y\| = 0 \iff x - y = \mathbf{0} \iff x = y$.

(ii) $\|x - y\| = \| - (y - x)\| = | - 1|\|y - x\| = \|y - x\|$.

(iii) $d(x, z) = \|x - z\| = \|(x - y) + (y - z)\| \le \|x - y\| + \|y - z\| = d(x, y) + d(y, z)$. ☐

Example 5.1 ($\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^3$). (i) $n = 1$: the distance on $\mathbb{R}$ is $d(x, y) = |x - y|$.

- (ii) $n = 2$: the distance on $\mathbb{R}^2$ is $d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$, that is, the Euclidean distance in the plane.
- (iii) $n = 3$: the distance on $\mathbb{R}^3$ is the Euclidean distance in three-dimensional space.
- (iv) The n-dimensional sphere $S^{n-1} = \{x \in \mathbb{R}^n : \|x\| = 1\}$, equipped with the induced distance, is a metric subspace of $\mathbb{R}^n$.

5.1.2 Open sets, closed sets and basic topological notions

Definition 5.2 (Open balls and ball neighbourhoods). Let $x_0 \in \mathbb{R}^n$, $r > 0$.

- (i) **Open ball**: $B(x_0, r) := \{x \in \mathbb{R}^n : \|x - x_0\| < r\}$.
- (ii) **Closed ball**: $\overline{B}(x_0, r) := \{x \in \mathbb{R}^n : \|x - x_0\| \le r\}$.
- (iii) **Sphere**: $S(x_0, r) := \{x \in \mathbb{R}^n : \|x - x_0\| = r\}$.

Definition 5.3 (Open sets and closed sets). Let $E \subseteq \mathbb{R}^n$.

- (i) E is called an **open set** if for every $x \in E$ there exists $r > 0$ such that $B(x, r) \subseteq E$.
- (ii) E is called a **closed set** if its complement $\mathbb{R}^n \setminus E$ is open.
- (iii) The **interior** of E is defined as $\mathrm{int}(E) := \bigcup\{U : U \subseteq E,\ U \text{ open}\}$.

(iv) The **closure** of E is defined as $\overline{E} := \bigcap\{F : F \supseteq E,\ F \text{ closed}\}$.

(v) The **boundary** of E is defined as $\partial E := \overline{E} \setminus \mathrm{int}(E)$.

(vi) The **derived set** of E is defined as $E' := \{x \in \mathbb{R}^n :$ every $B(x,r)$ contains a point of E differ
A point of E' is called an **accumulation point** or a **limit point** of E.

Proposition 5.2 (Basic properties of open and closed sets)**.** *Let $\{U_\alpha\}_{\alpha\in A}$ be a family of open sets, $\{F_\alpha\}_{\alpha\in A}$ a family of closed sets, U, V open sets and F, G closed sets. Then:*

(i) *$\mathbb{R}^n$ and $\emptyset$ are both open and closed;*

(ii) *an arbitrary union $\bigcup_{\alpha\in A} U_\alpha$ is open;*

(iii) *a finite intersection $U \cap V$ is open;*

(iv) *an arbitrary intersection $\bigcap_{\alpha\in A} F_\alpha$ is closed;*

(v) *a finite union $F \cup G$ is closed;*

(vi) *E is closed $\iff$ $E = \overline{E}$ $\iff$ $E' \subseteq E$;*

(vii) *$\mathrm{int}(E)$ is the largest open set contained in E; $\overline{E}$ is the smallest closed set containing E;*

(viii) *$\overline{E} = E \cup E'$, and ∂E is closed.*

Proof. (i) $\mathbb{R}^n$ is open: for any $x \in \mathbb{R}^n$, take $r = 1$; then $B(x,1) \subseteq \mathbb{R}^n$. $\mathbb{R}^n$ is closed: its complement is $\emptyset$, and $\emptyset$ is open ('for every $x \in \emptyset$' is a vacuously true proposition). Similarly $\emptyset$ is both open and closed.

(ii) Let $x \in \bigcup_\alpha U_\alpha$. Then there exists α_0 such that $x \in U_{\alpha_0}$. Since U_{α_0} is open, there exists $r > 0$ such that $B(x,r) \subseteq U_{\alpha_0} \subseteq \bigcup_\alpha U_\alpha$. Hence the union is open.

(iii) Let $x \in U \cap V$. Then $x \in U$ and $x \in V$. There exist $r_1, r_2 > 0$ such that $B(x,r_1) \subseteq U$ and $B(x,r_2) \subseteq V$. Take $r = \min(r_1, r_2) > 0$; then $B(x,r) \subseteq U \cap V$. Hence a finite intersection is open.

(iv) By De Morgan's laws, the complement of $\bigcap_\alpha F_\alpha$ is $\bigcup_\alpha(\mathbb{R}^n \setminus F_\alpha)$, a union of open sets, hence open; therefore $\bigcap_\alpha F_\alpha$ is closed.

(v) By De Morgan's laws, the complement of $F \cup G$ is $(\mathbb{R}^n \setminus F) \cap (\mathbb{R}^n \setminus G)$, the intersection of two open sets, hence open; therefore $F \cup G$ is closed.

(vi) If E is closed, then $\mathbb{R}^n \setminus E$ is open. By (i) and the definition, $\overline{E}$ is the smallest closed set containing E, so $\overline{E} \subseteq E$; and since $E \subseteq \overline{E}$, we have $E = \overline{E}$. Conversely, if $E = \overline{E}$, then E is closed by the definition of the closure. The relation $E' \subseteq E$ is equivalent to: if $x \notin E$, then x is not an accumulation point, that is, there exists $r > 0$ such that $B(x,r) \cap E = \emptyset$ (note that $x \notin E$, so it suffices that $B(x,r)$ contain no point of E); and this is equivalent to $\mathbb{R}^n \setminus E$ being open.

(vii) By definition, $\mathrm{int}(E)$ is a union of open sets, hence open; it is a subset of E, and every open set contained in E is contained in this union, so it is the largest such. $\overline{E}$ is an intersection of closed sets, hence closed; it contains E, and every closed set containing E is contained in this intersection, so it is the smallest such.

(viii) First we prove that $\overline{E} \subseteq E \cup E'$: let $x \notin E \cup E'$. Then $x \notin E$ and $x \notin E'$. Since $x \notin E'$, there exists $r > 0$ such that $B(x,r) \cap E \subseteq \{x\}$. Since $x \notin E$, we have $B(x,r)\cap E=\emptyset$, that is, $B(x,r)\subseteq\mathbb{R}^n\setminus E$. This shows that $\mathbb{R}^n\setminus E$ is open, that is, E is closed, so $\overline{E}=E\subseteq E\cup E'$, a contradiction. Hence $\overline{E}\subseteq E\cup E'$. Next we prove that $E\cup E'\subseteq\overline{E}$: the inclusion $E\subseteq\overline{E}$ follows directly from the closure containing E. If $x\in E'$ and $x\notin\overline{E}$, then $\mathbb{R}^n\setminus\overline{E}$ is open and contains x, so there exists $r>0$ such that $B(x,r)\subseteq\mathbb{R}^n\setminus\overline{E}\subseteq\mathbb{R}^n\setminus E$, contradicting the fact that x is an accumulation point. Hence $x\in\overline{E}$. In summary, $\overline{E}=E\cup E'$. Finally, $\partial E=\overline{E}\setminus\text{int}(E)$ is the difference of a closed set and an open set, and a closed set minus an open set is closed (because $\overline{E}$ is closed and $\text{int}(E)$ is open, and $\overline{E}\setminus\text{int}(E)=\overline{E}\cap(\mathbb{R}^n\setminus\text{int}(E))$, the intersection of two closed sets). ☐

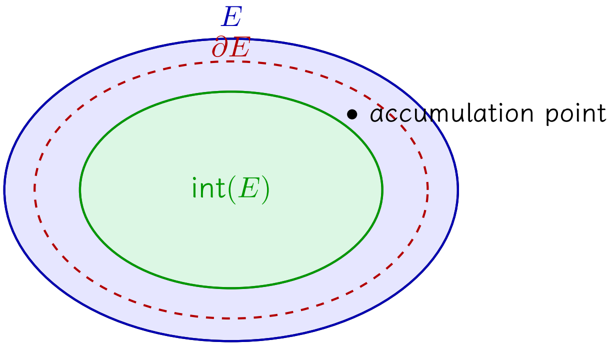


Figure 5.2: Schematic diagram of the open set E, the interior int(E) and the boundary ∂E.

Example 5.2 (Examples of open and closed sets). (i) The open ball $B(x_0,r)$ is open. Verification: for any $x\in B(x_0,r)$, put $\rho=r-\|x-x_0\|>0$. If $\|y-x\|<\rho$, then $\|y-x_0\|\le\|y-x\|+\|x-x_0\|<\rho+\|x-x_0\|=r$, so $B(x,\rho)\subseteq B(x_0,r)$.

(ii) The closed ball $\overline{B}(x_0,r)$ is closed. Its complement $\{x:\|x-x_0\|>r\}$ is open (verified similarly).

(iii) The half-open interval $(0,1]$ is neither open nor closed in $\mathbb{R}$.

(iv) $\mathbb{Q}$ is neither open nor closed in $\mathbb{R}$, and int$(\mathbb{Q})=\emptyset$, $\overline{\mathbb{Q}}=\mathbb{R}$, $\partial\mathbb{Q}=\mathbb{R}$.

(v) The Cantor set $C\subseteq[0,1]$ is closed, and int$(C)=\emptyset$, $\partial C=C$.

Definition 5.4 (Subspace topology). Let $E\subseteq\mathbb{R}^n$. The **subspace topology** of E is defined as

$$\mathcal{T}_E:=\{E\cap U: U\text{ is an open set in }\mathbb{R}^n\}.$$

A set in $\mathcal{T}_E$ is called a **relatively open set** of E. If $E\cap F$ is the complement of a relatively open set of E, then $E\cap F$ is called a **relatively closed set** of E.

Proposition 5.3 (Characterisation of relatively open and relatively closed sets). Let $E\subseteq\mathbb{R}^n$, $A\subseteq E$. Then:

(i) A is a relatively open set of E $\iff$ there exists an open set U in $\mathbb{R}^n$ such that $A=E\cap U$;

(ii) A is a relatively closed set of E $\iff$ there exists a closed set F in $\mathbb{R}^n$ such that $A=E\cap F$;

(iii) if E is an open set in $\mathbb{R}^n$, then A is a relatively open set of $E \iff A$ is an open set in $\mathbb{R}^n$;

(iv) if E is a closed set in $\mathbb{R}^n$, then A is a relatively closed set of $E \iff A$ is a closed set in $\mathbb{R}^n$.

Proof. (i) This follows directly from the definition.

(ii) A is a relatively closed set of $E \iff E\backslash A$ is a relatively open set of $E \iff$ there exists an open set U such that $E\backslash A = E\cap U \iff A = E\backslash(E\cap U) = E\cap(\mathbb{R}^n\backslash U)$, where $\mathbb{R}^n \setminus U$ is closed.

(iii) If E is open and $A = E \cap U$ (with U open), then A is the intersection of two open sets, hence open. Conversely, if A is open and $A \subseteq E$, then $A = E \cap A$, which is a relatively open set.

(iv) Similar to (iii). □

5.1.3 Compactness and connectedness

Definition 5.5 (Open covers and compactness). Let $K \subseteq \mathbb{R}^n$.

(i) An **open cover** of K is a family of open sets $\{U_\alpha\}_{\alpha\in A}$ such that $K \subseteq \bigcup_{\alpha\in A} U_\alpha$.

(ii) K is called **compact** if every open cover of K has a finite subcover, that is, there exist $\alpha_1, \dots, \alpha_m$ such that $K \subseteq \bigcup_{i=1}^m U_{\alpha_i}$.

(iii) K is called **sequentially compact** if every sequence in K has a subsequence converging to a point of K.

(iv) K is called **bounded** if there exists $R > 0$ such that $K \subseteq B(\mathbf{0}, R)$.

Theorem 5.1 (Heine–Borel theorem). Let $K \subseteq \mathbb{R}^n$. Then K is compact if and only if K is bounded and closed.

Proof. **Necessity.**

First we prove that K is bounded. The family of open balls $\{B(\mathbf{0}, m)\}_{m\in\mathbb{N}^+}$ is an open cover of K. By compactness there is a finite subcover, that is, there exist $m_1 < \cdots < m_k$ such that $K \subseteq \bigcup_{i=1}^k B(\mathbf{0}, m_i) = B(\mathbf{0}, m_k)$. Hence K is bounded.

Next we prove that K is closed. It suffices to prove that $\mathbb{R}^n \setminus K$ is open. Let $x \in \mathbb{R}^n \setminus K$. For each $y \in K$, put $r_y = \|x - y\|/2 > 0$. The family of open balls $\{B(y, r_y)\}_{y\in K}$ is an open cover of K. By compactness, there exist $y_1, \dots, y_m \in K$ such that $K \subseteq \bigcup_{i=1}^m B(y_i, r_{y_i})$. Put $r = \min_{1\le i\le m} r_{y_i} > 0$. We claim that $B(x, r)\cap K = \emptyset$. Otherwise there exists $z \in B(x, r) \cap K$, and then $z \in B(y_i, r_{y_i})$ for some i, so that

$$\|x - y_i\| \le \|x - z\| + \|z - y_i\| < r + r_{y_i} \le 2r_{y_i} = \|x - y_i\|,$$

a contradiction. Hence $B(x, r) \subseteq \mathbb{R}^n \setminus K$, that is, the complement is open and K is closed.

Sufficiency. Suppose that K is bounded and closed. Since K is bounded, there exists $R > 0$ such that $K \subseteq [-R, R]^n$. First we prove that $[-R, R]^n$ is compact.

Step 1: $[-R, R]$ is compact. Let $\{U_\alpha\}$ be an open cover of $[-R, R]$. Suppose, to the contrary, that there is no finite subcover. Divide $[-R, R]$ into two closed intervals of length R; at least one of them has no finite subcover, denote it by I_1.

Divide I_1 into two equal parts; at least one of them has no finite subcover, denote it by I_2. Continuing in this way we obtain a nested sequence of closed intervals $I_1 \supseteq I_2 \supseteq \cdots$ with lengths $|I_k| = 2R/2^k \to 0$. By the nested-interval theorem, there exists a unique $x_0 \in \bigcap_{k=1}^{\infty} I_k$. Since $\{U_\alpha\}$ covers $[-R, R]$, there exists α_0 such that $x_0 \in U_{\alpha_0}$. Since U_{α_0} is open, there exists $r > 0$ such that $B(x_0, r) \subseteq U_{\alpha_0}$. Take k large enough that $|I_k| < r$. Since $x_0 \in I_k$, we have $I_k \subseteq B(x_0, r) \subseteq U_{\alpha_0}$, that is, I_k is covered by the single set U_{α_0}, contradicting the fact that I_k has no finite subcover. Hence $[-R, R]$ is compact.

Step 2: $[-R, R]^n$ is compact. We argue by induction on n. The case $n = 1$ has already been proved. Suppose that $[-R, R]^{n-1}$ is compact. Let $\{U_\alpha\}$ be an open cover of $[-R, R]^n = [-R, R]^{n-1} \times [-R, R]$. For each $x \in [-R, R]^{n-1}$, the section $\{x\} \times [-R, R]$ is compact (it is homeomorphic to $[-R, R]$), so finitely many of the U_α cover it. By the tubular neighbourhood lemma (see below), there exists an open neighbourhood $V_x \subseteq [-R, R]^{n-1}$ of x such that $V_x \times [-R, R]$ is covered by the same finitely many U_α. The family $\{V_x\}_x$ is an open cover of $[-R, R]^{n-1}$. By the induction hypothesis, there exist finitely many $x_1, \ldots, x_m$ such that $[-R, R]^{n-1} = \bigcup_{i=1}^{m} V_{x_i}$. Consequently $[-R, R]^n = \bigcup_{i=1}^{m} (V_{x_i} \times [-R, R])$, which is covered by finitely many of the U_α. Hence $[-R, R]^n$ is compact.

Step 3: K is compact. Let $\{U_\alpha\}$ be an open cover of K. Then $\{U_\alpha\} \cup \{\mathbb{R}^n \setminus K\}$ is an open cover of $[-R, R]^n$ (because $K \subseteq [-R, R]^n$ and $\mathbb{R}^n \setminus K$ is open). Since $[-R, R]^n$ is compact, there is a finite subcover. If $\mathbb{R}^n \setminus K$ is not among its members, then finitely many of the U_α already cover $[-R, R]^n \supseteq K$; if it is among them, then finitely many of the U_α cover $[-R, R]^n \setminus (\mathbb{R}^n \setminus K) = K$. In either case K has a finite subcover. ☐

Lemma 5.1 (Tubular neighbourhood lemma). *Let X be a compact metric space, Y a metric space, and $\{U_\alpha\}_{\alpha \in A}$ an open cover of $X \times Y$. If for every $x \in X$ the set $\{x\} \times Y$ is covered by finitely many of the U_α, and Y is compact, then there exists an open neighbourhood V_x of x such that $V_x \times Y$ is covered by the same finitely many U_α.*

Proof. Fix $x \in X$. Suppose that $\{x\} \times Y \subseteq \bigcup_{i=1}^{m} U_{\alpha_i}$. Since Y is compact, for every $y \in Y$ there exist $i(y)$, an open neighbourhood V_y of x and an open neighbourhood W_y of y such that $V_y \times W_y \subseteq U_{\alpha_{i(y)}}$ (because $U_{\alpha_{i(y)}}$ is open and contains (x, y)). The family $\{W_y\}_{y \in Y}$ is an open cover of Y. Since Y is compact, there is a finite subcover $W_{y_1}, \ldots, W_{y_k}$. Put $V_x = \bigcap_{j=1}^{k} V_{y_j}$; then V_x is an open neighbourhood of x. For any $(x', y') \in V_x \times Y$ there exists j such that $y' \in W_{y_j}$, so that $(x', y') \in V_{y_j} \times W_{y_j} \subseteq U_{\alpha_{i(y_j)}}$. Hence $V_x \times Y \subseteq \bigcup_{j=1}^{k} U_{\alpha_{i(y_j)}}$. ☐

Theorem 5.2 (Bolzano–Weierstrass theorem). *Let $K \subseteq \mathbb{R}^n$. Then K is sequentially compact if and only if K is bounded and closed.*

Proof. By the Heine–Borel theorem (Theorem 5.1), it suffices to prove that K is compact $\iff$ K is sequentially compact.

Compact $\Rightarrow$ sequentially compact. Suppose that K is compact and that $\{x_k\}$ is a sequence in K. If $\{x_k\}$ has no convergent subsequence, then for every $x \in K$ there exists $r_x > 0$ such that $B(x, r_x)$ contains only finitely many of the x_k (otherwise one could construct a subsequence converging to x). The family $\{B(x, r_x)\}_{x \in K}$ is an open cover of K. By compactness there is a finite subcover $B(x_1, r_{x_1}), \ldots, B(x_m, r_{x_m})$. Each ball contains only finitely many of the x_k, so K

contains only finitely many of the x_k, contradicting the fact that $\{x_k\}$ is an infinite sequence.

Sequentially compact ⇒ compact. Suppose that K is sequentially compact. First we prove that K is bounded. If K were unbounded, then for each k there would exist $x_k \in K$ with $\|x_k\| > k$. The sequence $\{x_k\}$ would have no convergent subsequence (any convergent subsequence must be bounded), contradicting sequential compactness. Hence K is bounded. Next we prove that K is closed. Let $x \in \overline{K}$. Then there exist $x_k \in K$ with $x_k \to x$. By sequential compactness, $\{x_k\}$ has a subsequence converging to some point y of K. But a subsequence of a convergent sequence converges to the same limit, so $y = x \in K$. Therefore $\overline{K} \subseteq K$ and K is closed. By the Heine–Borel theorem, K is compact. □

Definition 5.6 (Connectedness). Let $E \subseteq \mathbb{R}^n$.

(i) E is called **disconnected** if there exist two non-empty relatively open sets $A, B \subseteq E$ such that $E = A \cup B$ and $A \cap B = \emptyset$. Such a decomposition is called a **separation** of E.

(ii) E is called **connected** if E is not disconnected.

(iii) E is called **path-connected** if for any $x, y \in E$ there exists a continuous map $\gamma : [0,1] \to E$ such that $\gamma(0) = x$ and $\gamma(1) = y$.

Proposition 5.4 (Basic properties of connected sets). (i) E is connected $\iff$ there do not exist non-empty relatively closed sets $A, B \subseteq E$ such that $E = A \cup B$ and $A \cap B = \emptyset$.

(ii) Path-connected ⇒ connected.

(iii) Continuous maps preserve connectedness: if E is connected and $f : E \to \mathbb{R}^m$ is continuous, then $f(E)$ is connected.

(iv) The connected subsets of $\mathbb{R}$ are precisely the intervals.

(v) If $\{E_\alpha\}$ is a family of connected sets and $\bigcap_\alpha E_\alpha \neq \emptyset$, then $\bigcup_\alpha E_\alpha$ is connected.

Proof. (i) $A \subseteq E$ is a relatively open set $\iff$ $E \setminus A$ is a relatively closed set. Hence the definition of a separation may equivalently be formulated in terms of relatively closed sets.

(ii) Suppose that E is path-connected. If $E = A \cup B$ is a separation (with A, B non-empty and relatively open), take $x \in A$ and $y \in B$. There exists a continuous $\gamma : [0,1] \to E$ such that $\gamma(0) = x$ and $\gamma(1) = y$. Put $A' = \gamma^{-1}(A)$ and $B' = \gamma^{-1}(B)$. Since γ is continuous and A, B are relatively open sets of E, the sets A', B' are relatively open sets of $[0,1]$. Moreover $A' \cup B' = [0,1]$, $A' \cap B' = \emptyset$, $0 \in A'$ and $1 \in B'$. But $[0,1]$ is connected (by (iv)), a contradiction.

(iii) Suppose that $f(E) = A \cup B$ is a separation (with A, B non-empty and relatively open). Then $E = f^{-1}(A) \cup f^{-1}(B)$, and $f^{-1}(A), f^{-1}(B)$ are non-empty relatively open sets of E (because f is continuous) with empty intersection, contradicting the connectedness of E.

(iv) Suppose that $E \subseteq \mathbb{R}$ is connected. If E is not an interval, then there exist $a < b < c$ with $a, c \in E$ but $b \notin E$. Put $A = E \cap (-\infty, b)$ and $B = E \cap (b, \infty)$. Then A, B are non-empty relatively open sets of E, $E = A \cup B$ and $A \cap B = \emptyset$, contradicting the connectedness of E. Conversely, suppose that E is an interval and that $E = A \cup B$ is a separation. Take $a \in A$ and $b \in B$; without loss of generality $a < b$. Put $c = \sup(A \cap [a, b])$. Since A is relatively closed and $a \in A$, we have $c \in A$. If $c < b$, then $c \in A$ and $(c, b] \subseteq B$ (otherwise there would be a point of A with value $> c$), so c is an accumulation point of B. But B is relatively closed, so $c \in B$, contradicting $A \cap B = \emptyset$. Hence $c = b$, that is, $b \in A$, contradicting $b \in B$.

(v) Let $E = \bigcup_\alpha E_\alpha$ and $x_0 \in \bigcap_\alpha E_\alpha$. If $E = A \cup B$ is a separation, then without loss of generality $x_0 \in A$. For each α, the set E_α is connected and $x_0 \in E_\alpha$, so $E_\alpha \subseteq A$ (otherwise $E_\alpha \cap A$ and $E_\alpha \cap B$ would form a separation of E_α). Hence $E \subseteq A$, that is, $B = \emptyset$, a contradiction. ☐

Definition 5.7 (Simple connectedness). Let $\Omega \subseteq \mathbb{R}^n$ be an open set. Ω is called **simply connected** if for every continuous closed curve $\gamma : S^1 \to \Omega$ there exists a continuous map $H : S^1 \times [0, 1] \to \Omega$ such that $H(\theta, 0) = \gamma(\theta)$ and $H(\theta, 1) = p$ (with $p \in \Omega$ fixed), that is, γ can be continuously contracted to a point inside Ω.

Proposition 5.5 (Convex sets in $\mathbb{R}^n$ are simply connected). Every convex open set in $\mathbb{R}^n$ is simply connected.

Proof. Let Ω be convex and open, and let $\gamma : S^1 \to \Omega$ be a continuous closed curve. Take $p \in \Omega$ (if $\Omega = \emptyset$ the proposition is trivial). Define $H : S^1 \times [0, 1] \to \Omega$ by

$$H(\theta, t) = (1 - t)\gamma(\theta) + tp.$$

By convexity, $H(\theta, t) \in \Omega$ for all (θ, t). The map H is continuous (because a linear combination of continuous maps is continuous), $H(\theta, 0) = \gamma(\theta)$ and $H(\theta, 1) = p$. Hence Ω is simply connected. ☐

Example 5.3 (Comparison of connectedness and simple connectedness).

(i) $\mathbb{R}^n$ is connected, path-connected and simply connected.

(ii) The open ball $B(x_0, r)$ is connected, path-connected and simply connected (it is convex).

(iii) The circle S^1 is connected and path-connected, but not simply connected (a closed curve winding once around the centre cannot be contracted).

(iv) The slit plane $\mathbb{R}^2 \setminus \{(x, 0) : x \geq 0\}$ is connected and path-connected, but not simply connected.

(v) $[0, 1] \cup [2, 3]$ is disconnected.

(vi) The topologist's sine curve $\{(x, \sin(1/x)) : x > 0\} \cup \{(0, y) : -1 \leq y \leq 1\}$ is connected but not path-connected.

5.2 Sequences, Limits and Completeness

5.2.1 Sequences in Euclidean space

Definition 5.8 (Convergence of a sequence). Let $\{x_k\}$ be a sequence in $\mathbb{R}^n$ and $x \in \mathbb{R}^n$. The sequence $\{x_k\}$ is said to **converge** to x, written $x_k \to x$ or $\lim_{k\to\infty} x_k = x$, if

$$\forall \varepsilon > 0,\ \exists K \in \mathbb{N}^+,\ \forall k \geq K, \quad \|x_k - x\| < \varepsilon.$$

x is called the **limit** of $\{x_k\}$.

Proposition 5.6 (Coordinate characterisation of convergence). *Let $x_k = (x_{k,1}, \ldots, x_{k,n})$ and $x = (x_1, \ldots, x_n)$. Then*

$$x_k \to x \iff x_{k,i} \to x_i, \quad i = 1, \ldots, n.$$

Proof. (⇒) For each i, $|x_{k,i} - x_i| \le \|x_k - x\| \to 0$, so $x_{k,i} \to x_i$.

(⇐) For each i, $x_{k,i} \to x_i$. Given $\varepsilon > 0$, for each i there exists K_i such that $|x_{k,i} - x_i| < \varepsilon/\sqrt{n}$ whenever $k \ge K_i$. Take $K = \max_i K_i$; then for $k \ge K$,

$$\|x_k - x\| = \left(\sum_{i=1}^{n} |x_{k,i} - x_i|^2\right)^{1/2} < \left(n \cdot \frac{\varepsilon^2}{n}\right)^{1/2} = \varepsilon.$$

□

Proposition 5.7 (Algebraic properties of limits). *Let $x_k \to x$ and $y_k \to y$ (in $\mathbb{R}^n$), and $\alpha_k \to \alpha$ (in $\mathbb{R}$). Then:*

(i) $x_k + y_k \to x + y$;

(ii) $\alpha_k x_k \to \alpha x$;

(iii) $\langle x_k, y_k \rangle \to \langle x, y \rangle$;

(iv) $\|x_k\| \to \|x\|$;

(v) *uniqueness of the limit: if $x_k \to x$ and $x_k \to y$, then $x = y$.*

Proof. (i) $\|(x_k + y_k) - (x + y)\| \le \|x_k - x\| + \|y_k - y\| \to 0$.

(ii)

$$\|\alpha_k x_k - \alpha x\| \le \|\alpha_k x_k - \alpha_k x\| + \|\alpha_k x - \alpha x\| = |\alpha_k|\|x_k - x\| + |\alpha_k - \alpha|\|x\|.$$

Since the convergent sequence $\{\alpha_k\}$ is bounded, suppose that $|\alpha_k| \le M$. Then $\|\alpha_k x_k - \alpha x\| \le M\|x_k - x\| + |\alpha_k - \alpha|\|x\| \to 0$.

(iii) By the Cauchy–Schwarz inequality,

$$|\langle x_k, y_k \rangle - \langle x, y \rangle| \le |\langle x_k - x, y_k \rangle| + |\langle x, y_k - y \rangle| \le \|x_k - x\|\|y_k\| + \|x\|\|y_k - y\| \to 0.$$

(iv) By the reverse triangle inequality, $|\|x_k\| - \|x\|| \le \|x_k - x\| \to 0$.

(v) If $x \ne y$, take $\varepsilon = \|x - y\|/2 > 0$. There exists K_1 such that $\|x_k - x\| < \varepsilon$ for $k \ge K_1$, and K_2 such that $\|x_k - y\| < \varepsilon$ for $k \ge K_2$. Take $k \ge \max(K_1, K_2)$; then

$$\|x - y\| \le \|x - x_k\| + \|x_k - y\| < 2\varepsilon = \|x - y\|,$$

a contradiction. □

Definition 5.9 (Cauchy sequence). A sequence $\{x_k\}$ is called a **Cauchy sequence** if

$$\forall \varepsilon > 0,\ \exists K \in \mathbb{N}^+,\ \forall m, k \ge K, \quad \|x_m - x_k\| < \varepsilon.$$

Proposition 5.8 (A convergent sequence is a Cauchy sequence). *If $\{x_k\}$ converges, then $\{x_k\}$ is a Cauchy sequence.*

Proof. Suppose that $x_k \to x$. Given $\varepsilon > 0$, there exists K such that $\|x_k - x\| < \varepsilon/2$ for $k \geq K$. Then for $m, k \geq K$,

$$\|x_m - x_k\| \leq \|x_m - x\| + \|x - x_k\| < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

□

Theorem 5.3 (Completeness of $\mathbb{R}^n$). Every Cauchy sequence in $\mathbb{R}^n$ converges.

Proof. Let $\{x_k\}$ be a Cauchy sequence in $\mathbb{R}^n$. By the coordinate characterisation, for each i the sequence $\{x_{k,i}\}$ is a Cauchy sequence in $\mathbb{R}$ (because $|x_{m,i} - x_{k,i}| \leq \|x_m - x_k\|$). By the completeness of $\mathbb{R}$ (the 'Number Systems' chapter), $x_{k,i} \to x_i$ for some $x_i \in \mathbb{R}$. Put $x = (x_1, \dots, x_n)$. By Proposition 5.6, $x_k \to x$. □

Definition 5.10 (Accumulation points and upper/lower limits). Let $\{x_k\}$ be a sequence in $\mathbb{R}^n$.

(i) $x \in \mathbb{R}^n$ is called an **accumulation point** of $\{x_k\}$ if $\{x_k\}$ has a subsequence converging to x.

(ii) For a real sequence $\{a_k\}$, define

$$\limsup_{k\to\infty} a_k := \lim_{k\to\infty}\left(\sup_{j\geq k} a_j\right), \qquad \liminf_{k\to\infty} a_k := \lim_{k\to\infty}\left(\inf_{j\geq k} a_j\right).$$

Theorem 5.4 (Bolzano–Weierstrass theorem, sequential form). Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Proof. By the Heine–Borel theorem (Theorem 5.1), a bounded closed set is compact; by Theorem 5.2, a compact set is sequentially compact. Concretely, suppose that $\{x_k\}$ is bounded; then there exists $R > 0$ such that $\{x_k\} \subseteq \overline{B}(0, R)$. By the 'compact ⇒ sequentially compact' part of Theorem 5.2 ($\overline{B}(0, R)$ is bounded and closed, hence compact), $\{x_k\}$ has a convergent subsequence. □

Proposition 5.9 (Relation between upper/lower limits and accumulation points). Let $\{a_k\}$ be a real sequence. Then $\limsup_{k\to\infty} a_k$ is the largest accumulation point of $\{a_k\}$, and $\liminf_{k\to\infty} a_k$ is the smallest. In particular, $\{a_k\}$ converges $\iff$ $\limsup a_k = \liminf a_k$.

Proof. Let $s_k = \sup_{j\geq k} a_j$. The sequence $\{s_k\}$ is monotonically non-increasing, so $\lim s_k = \inf_k s_k =: L$ exists (possibly equal to $\pm\infty$).

First we prove that L is an accumulation point. If $L = -\infty$, then $s_k \to -\infty$, so $\{a_k\}$ is unbounded below and one can construct a subsequence tending to $-\infty$; thus L is an accumulation point. If $L \in \mathbb{R}$, then for each k, since $s_k = \sup_{j\geq k} a_j$, there exists $j_k \geq k$ such that $a_{j_k} > s_k - 1/k$. Also $a_{j_k} \leq s_k$. Hence $|a_{j_k} - s_k| \leq 1/k$. Since $s_k \to L$, we have $a_{j_k} \to L$. The subsequence $\{a_{j_k}\}$ converges to L, so L is an accumulation point.

Next we prove that L is the largest accumulation point. Let M be any accumulation point, so that there is a subsequence $a_{k_j} \to M$. For each j, $a_{k_j} \leq s_{k_j}$. Since $s_k \to L$ and s_k is monotonically non-increasing, $\lim_{j\to\infty} s_{k_j} = L$. Hence $M = \lim a_{k_j} \leq \lim s_{k_j} = L$.

Similarly, lim inf is the smallest accumulation point.

Finally, $\{a_k\}$ converges to L $\iff$ every subsequence converges to L $\iff$ the largest accumulation point = the smallest accumulation point = L $\iff$ $\limsup = \liminf$. □

Example 5.4 (Computing upper and lower limits). (i) $a_k = (-1)^k$: lim sup $= 1$, lim inf $= -1$, *and the sequence does not converge.*

(ii) $a_k = 1/k$: lim sup $=$ lim inf $= 0$, *and the sequence converges to* 0.

(iii) $a_k = \sin(k)$: lim sup $= 1$, lim inf $= -1$ (*because* $\{k \bmod 2\pi\}$ *is dense in* $[0, 2\pi)$).

5.2.2 The contraction-mapping principle

Definition 5.11 (Contraction mapping). *Let* (X, d) *be a metric space. A map* $T : X \to X$ *is called a* **contraction mapping** *if there exists a constant* $0 \le q < 1$ *such that*

$$d(T(x), T(y)) \le q\, d(x, y), \qquad \forall x, y \in X.$$

q *is called the* **contraction constant** *of* T.

Theorem 5.5 (Banach contraction-mapping principle). *Let* (X, d) *be a complete metric space and* $T : X \to X$ *a contraction mapping. Then* T *has a unique fixed point* $x^* \in X$*, that is,* $T(x^*) = x^*$*. Moreover, for any* $x_0 \in X$ *the iteration sequence* $x_{k+1} = T(x_k)$ *converges to* x^**, and the error estimate*

$$d(x_k, x^*) \le \frac{q^k}{1-q} d(x_1, x_0).$$

holds.

Proof. **Existence.** Choose any $x_0 \in X$ and define $x_{k+1} = T(x_k)$. First we prove that $\{x_k\}$ is a Cauchy sequence. For any $k \ge 1$,

$$d(x_{k+1}, x_k) = d(T(x_k), T(x_{k-1})) \le q\, d(x_k, x_{k-1}) \le \cdots \le q^k d(x_1, x_0).$$

Hence for $m > k$,

$$d(x_m, x_k) \le \sum_{j=k}^{m-1} d(x_{j+1}, x_j) \le \sum_{j=k}^{m-1} q^j d(x_1, x_0) \le \frac{q^k}{1-q} d(x_1, x_0).$$

Since $q < 1$, we have $q^k \to 0$, so $\{x_k\}$ is a Cauchy sequence. Since X is complete, $x_k \to x^*$ for some $x^* \in X$.

By the continuity of T (a contraction mapping is necessarily continuous: $d(T(x), T(y)) \le q\, d(x, y) \to 0$ as $x \to y$),

$$T(x^*) = T\left(\lim_{k\to\infty} x_k\right) = \lim_{k\to\infty} T(x_k) = \lim_{k\to\infty} x_{k+1} = x^*.$$

Hence x^* is a fixed point.

Uniqueness. Suppose that x^*, y^* are both fixed points. Then

$$d(x^*, y^*) = d(T(x^*), T(y^*)) \le q\, d(x^*, y^*).$$

Since $q < 1$, this gives $d(x^*, y^*) = 0$, that is, $x^* = y^*$.

Error estimate. For $k \ge 1$ and $m > k$, letting $m \to \infty$ in the estimate above (with k fixed) gives

$$d(x_k, x^*) = \lim_{m\to\infty} d(x_k, x_m) \le \frac{q^k}{1-q} d(x_1, x_0).$$

□

Corollary 5.2 (Local form of the contraction-mapping principle). Let (X,d) be a complete metric space and $T : X \to X$ a map satisfying: there exist $q \in [0,1)$ and a closed ball $\overline{B}(x_0,r)$ such that

(i) $T(\overline{B}(x_0,r)) \subseteq \overline{B}(x_0,r)$;

(ii) $d(T(x),T(y)) \le q\, d(x,y)$ for all $x,y \in \overline{B}(x_0,r)$.

Then T has a unique fixed point in $\overline{B}(x_0,r)$.

Proof. The closed ball $\overline{B}(x_0,r)$, being a closed subset of a complete metric space, is complete. Apply Theorem 5.5 to $T|_{\overline{B}(x_0,r)}$. ☐

Example 5.5 (Applications of the contraction-mapping principle). (i) Solving the equation $x = \cos x$. The map $T(x) = \cos x$ is a contraction mapping on $[0,1]$ (since $|T'(x)| = |\sin x| \le \sin 1 < 1$), so the equation has a unique solution.

(ii) Existence and uniqueness of the solution of the initial-value problem for an ordinary differential equation (the Picard–Lindelöf theorem): under suitable conditions the integral operator $T(y)(t) = y_0 + \int_{t_0}^{t} F(s,y(s))\,ds$ is a contraction mapping.

(iii) The proof of the implicit-function theorem: construct a contraction mapping and solve by iteration.

5.3 Continuous Maps and Compactness

5.3.1 Continuous maps

Definition 5.12 (Continuous map). Let $E \subseteq \mathbb{R}^n$, $f : E \to \mathbb{R}^m$ and $x_0 \in E$.

(i) f is **continuous** at x_0 if

$$\forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x \in E, \quad \|x - x_0\| < \delta \implies \|f(x) - f(x_0)\| < \varepsilon.$$

(ii) f is continuous on E if f is continuous at every point of E.

(iii) f is **uniformly continuous** on E if

$$\forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x,y \in E, \quad \|x - y\| < \delta \implies \|f(x) - f(y)\| < \varepsilon.$$

Proposition 5.10 (Equivalent characterisations of continuity). Let $E \subseteq \mathbb{R}^n$, $f : E \to \mathbb{R}^m$ and $x_0 \in E$. Then the following conditions are equivalent:

(i) f is continuous at x_0;

(ii) for every sequence $x_k \to x_0$ in E, one has $f(x_k) \to f(x_0)$;

(iii) for every open neighbourhood V of $f(x_0)$ there exists an open neighbourhood U of x_0 such that $f(E \cap U) \subseteq V$.

Proof. (i) $\Rightarrow$ (ii): Given $\varepsilon > 0$, by continuity there exists $\delta > 0$ such that $\|f(x) - f(x_0)\| < \varepsilon$ whenever $\|x - x_0\| < \delta$. Since $x_k \to x_0$, there exists K such that $\|x_k - x_0\| < \delta$ for $k \geq K$, and hence $\|f(x_k) - f(x_0)\| < \varepsilon$. Thus $f(x_k) \to f(x_0)$.

(ii) $\Rightarrow$ (i): Suppose, to the contrary, that f is not continuous at x_0. Then there exists $\varepsilon_0 > 0$ such that for every k there exists $x_k \in E$ with $\|x_k - x_0\| < 1/k$ but $\|f(x_k) - f(x_0)\| \geq \varepsilon_0$. Then $x_k \to x_0$ but $f(x_k) \not\to f(x_0)$, a contradiction.

(i) $\Rightarrow$ (iii): Let V be an open neighbourhood of $f(x_0)$; there exists $\varepsilon > 0$ such that $B(f(x_0), \varepsilon) \subseteq V$. By continuity there exists $\delta > 0$ such that $\|f(x) - f(x_0)\| < \varepsilon$ whenever $\|x - x_0\| < \delta$. Take $U = B(x_0, \delta)$; then $f(E \cap U) \subseteq B(f(x_0), \varepsilon) \subseteq V$.

(iii) $\Rightarrow$ (i): Given $\varepsilon > 0$, take $V = B(f(x_0), \varepsilon)$. There exists an open neighbourhood U of x_0 such that $f(E \cap U) \subseteq V$. There exists $\delta > 0$ such that $B(x_0, \delta) \subseteq U$. Then $\|x - x_0\| < \delta$ implies $x \in E \cap U$, hence $f(x) \in V$, that is, $\|f(x) - f(x_0)\| < \varepsilon$. ☐

Theorem 5.6 (Global characterisation of continuous maps). *Let $E \subseteq \mathbb{R}^n$ and $f : E \to \mathbb{R}^m$. Then the following conditions are equivalent:*

- (i) *f is continuous on E;*
- (ii) *for every open set V in $\mathbb{R}^m$, the set $f^{-1}(V)$ is a relatively open set of E;*
- (iii) *for every closed set F in $\mathbb{R}^m$, the set $f^{-1}(F)$ is a relatively closed set of E.*

Proof. (i) $\Rightarrow$ (ii): Let V be an open set in $\mathbb{R}^m$ and $x_0 \in f^{-1}(V)$. Then $f(x_0) \in V$. By Proposition 5.10 (iii), there exists an open neighbourhood U of x_0 in $\mathbb{R}^n$ such that $f(E \cap U) \subseteq V$. Hence $E \cap U \subseteq f^{-1}(V)$, that is, $f^{-1}(V)$ is a relatively open set of E.

(ii) $\Rightarrow$ (i): Let $x_0 \in E$. For every open neighbourhood V of $f(x_0)$, the set $f^{-1}(V)$ is a relatively open set of E containing x_0. Hence there exists an open set U such that $x_0 \in E \cap U \subseteq f^{-1}(V)$, that is, $f(E \cap U) \subseteq V$. By Proposition 5.10 (iii), f is continuous at x_0.

(ii) $\iff$ (iii): This follows directly from $f^{-1}(\mathbb{R}^m \setminus F) = E \setminus f^{-1}(F)$ and De Morgan's laws. ☐

Theorem 5.7 (Properties of continuous maps on compact sets). *Let $K \subseteq \mathbb{R}^n$ be compact and $f : K \to \mathbb{R}^m$ continuous. Then:*

- (i) *$f(K)$ is compact;*
- (ii) *f is bounded on K;*
- (iii) *if $m = 1$, then f attains its maximum and its minimum on K;*
- (iv) *f is uniformly continuous on K.*

Proof. (i) Let $\{V_\alpha\}$ be an open cover of $f(K)$. Then $\{f^{-1}(V_\alpha)\}$ is an open cover of K (since f is continuous, each $f^{-1}(V_\alpha)$ is a relatively open set of K). Since K is compact, there is a finite subcover $f^{-1}(V_{\alpha_1}), \ldots, f^{-1}(V_{\alpha_k})$. Then $V_{\alpha_1}, \ldots, V_{\alpha_k}$ cover $f(K)$. Hence $f(K)$ is compact.

(ii) By (i), $f(K)$ is compact. By the Heine–Borel theorem (Theorem 5.1), $f(K)$ is bounded, that is, there exists $R > 0$ such that $f(K) \subseteq B(0, R)$. Hence $\|f(x)\| \leq R$ for all $x \in K$.

(iii) By (i), $f(K)$ is a compact set in $\mathbb{R}$. By the Heine–Borel theorem, $f(K)$ is bounded and closed. Hence $\sup f(K)$ and $\inf f(K)$ both belong to $f(K)$, that is, f attains its maximum and its minimum.

(iv) Suppose, to the contrary, that f is not uniformly continuous. Then there exists $\varepsilon_0 > 0$ such that for every k there exist $x_k, y_k \in K$ with $\|x_k - y_k\| < 1/k$ but $\|f(x_k) - f(y_k)\| \geq \varepsilon_0$. Since K is sequentially compact (Theorem 5.2), $\{x_k\}$ has a subsequence $x_{k_j} \to x_0 \in K$. Since $\|x_{k_j} - y_{k_j}\| < 1/k_j \to 0$, we have $y_{k_j} \to x_0$. By the continuity of f, $f(x_{k_j}) \to f(x_0)$ and $f(y_{k_j}) \to f(x_0)$. But $\|f(x_{k_j}) - f(y_{k_j})\| \geq \varepsilon_0$, contradicting convergence. ☐

Corollary 5.3 (Heine–Cantor theorem). *Let $K \subseteq \mathbb{R}^n$ be compact and $f : K \to \mathbb{R}^m$ continuous. Then f is uniformly continuous on K.*

Proof. This is Theorem 5.7 (iv). ☐

Theorem 5.8 (Generalised intermediate-value theorem). *Let $E \subseteq \mathbb{R}^n$ be connected and $f : E \to \mathbb{R}$ continuous. If $x, y \in E$ and $f(x) < c < f(y)$, then there exists $z \in E$ such that $f(z) = c$.*

Proof. By Proposition 5.4 (iii), $f(E)$ is connected. By Proposition 5.4 (iv), $f(E)$ is an interval in $\mathbb{R}$. Since $f(x), f(y) \in f(E)$ and c lies between them, we have $c \in f(E)$, that is, there exists $z \in E$ such that $f(z) = c$. ☐

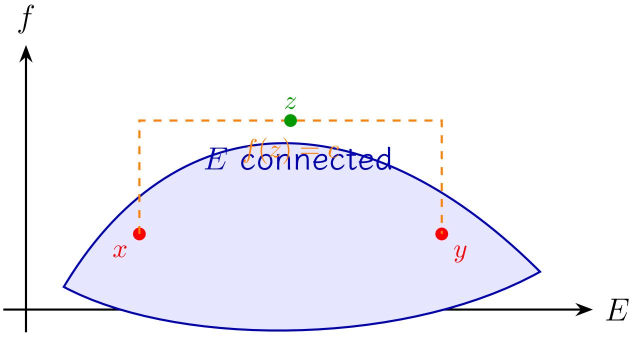


Figure 5.3: A continuous function on a connected set attains intermediate values: if $f(x) < c < f(y)$, then there exists z with $f(z) = c$.

Example 5.6 (Examples and counterexamples of continuous maps).

(i) A polynomial map $f : \mathbb{R}^n \to \mathbb{R}$, $f(x) = \sum_{|\alpha| \leq d} c_\alpha x^\alpha$, is continuous.

(ii) A linear map $f(x) = Ax$ (with A an $m \times n$ matrix) is continuous.

(iii) The norm map $f(x) = \|x\|$ is continuous (by the reverse triangle inequality).

(iv) The inner-product map $f(x, y) = \langle x, y \rangle$ is continuous.

(v) $f : (0, 1] \to \mathbb{R}$, $f(x) = 1/x$, is continuous but not uniformly continuous (it is unbounded as $x \to 0^+$).

(vi) $f : [0, 1] \to \mathbb{R}$, $f(x) = 1/x$ for $x > 0$ and $f(0) = 0$, is discontinuous at 0.

5.4 Multivariable Differential Calculus

5.4.1 Directional derivatives and partial derivatives

Definition 5.13 (Directional derivative)**.** Let $f : \Omega \to \mathbb{R}$ with $\Omega \subseteq \mathbb{R}^n$ open, let $x_0 \in \Omega$, and let $v \in \mathbb{R}^n$ be a unit vector. The **directional derivative** of f at x_0 in the direction v is defined as

$$D_v f(x_0) := \lim_{t \to 0} \frac{f(x_0 + tv) - f(x_0)}{t},$$

provided this limit exists.

Definition 5.14 (Partial derivative)**.** Let $f : \Omega \to \mathbb{R}$ with $\Omega \subseteq \mathbb{R}^n$ open and $x_0 \in \Omega$. The **partial derivative** of f at x_0 with respect to the i-th coordinate is defined as

$$\frac{\partial f}{\partial x_i}(x_0) := \lim_{t \to 0} \frac{f(x_0 + te_i) - f(x_0)}{t},$$

where e_i is the i-th standard basis vector. That is, $\partial f / \partial x_i(x_0) = D_{e_i} f(x_0)$.

Proposition 5.11 (Existence of partial derivatives does not imply continuity)**.** There exists a function $f : \mathbb{R}^2 \to \mathbb{R}$ such that both partial derivatives of f exist at the origin, but f is discontinuous at the origin.

Proof. Define

$$f(x, y) = \begin{cases} \frac{xy}{x^2+y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

The partial derivatives are

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{t \to 0} \frac{f(t, 0) - f(0, 0)}{t} = \lim_{t \to 0} \frac{0 - 0}{t} = 0,$$

and similarly $\partial f / \partial y(0, 0) = 0$. But it is discontinuous: along the line $y = kx$,

$$\lim_{x \to 0} f(x, kx) = \lim_{x \to 0} \frac{kx^2}{x^2 + k^2x^2} = \frac{k}{1 + k^2},$$

which varies with k, so $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist. □

Definition 5.15 (Gradient)**.** Let $f : \Omega \to \mathbb{R}$ with $\Omega \subseteq \mathbb{R}^n$ open, and suppose that all partial derivatives of f exist at x_0. The **gradient** of f at x_0 is

$$\nabla f(x_0) := \left(\frac{\partial f}{\partial x_1}(x_0), \ldots, \frac{\partial f}{\partial x_n}(x_0) \right) \in \mathbb{R}^n.$$

5.4.2 Total differential

Definition 5.16 (Total differential and differentiability)**.** Let $f : \Omega \to \mathbb{R}^m$ with $\Omega \subseteq \mathbb{R}^n$ open and $x_0 \in \Omega$. The map f is **differentiable** at x_0 if there exists a linear map $L : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$f(x_0 + h) = f(x_0) + L(h) + o(\|h\|), \qquad \|h\| \to 0,$$

that is,

$$\lim_{h\to 0} \frac{\|f(x_0+h)-f(x_0)-L(h)\|}{\|h\|}=0.$$

The linear map L is called the **total differential** or the **derivative** of f at x_0, and is denoted by $Df(x_0)$ or $f'(x_0)$.

Proposition 5.12 (Uniqueness of the total differential)**.** If f is differentiable at x_0, then the total differential $L = Df(x_0)$ is unique.

Proof. Suppose that L_1, L_2 both satisfy the definition. Then

$$L_1(h)-L_2(h)=o(\|h\|), \qquad \|h\|\to 0.$$

For any $v \neq 0$, take $h = tv$ (with $t > 0$); then

$$\frac{\|L_1(tv)-L_2(tv)\|}{t\|v\|}=\frac{\|t(L_1(v)-L_2(v))\|}{t\|v\|}=\frac{\|L_1(v)-L_2(v)\|}{\|v\|}\to 0.$$

Hence $L_1(v) = L_2(v)$ for every v, so $L_1 = L_2$. ☐

Theorem 5.9 (Differentiability implies existence of partial derivatives, and L is given by the Jacobian matrix)**.** Let $f : \Omega \to \mathbb{R}^m$ with $\Omega \subseteq \mathbb{R}^n$ open, and suppose that f is differentiable at $x_0 \in \Omega$. Then all the partial derivatives $\partial f_i/\partial x_j(x_0)$ exist, and

$$Df(x_0)(h)=J_f(x_0)h, \qquad h\in\mathbb{R}^n,$$

where

$$J_f(x_0)=\begin{pmatrix} \frac{\partial f_1}{\partial x_1}(x_0) & \cdots & \frac{\partial f_1}{\partial x_n}(x_0) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(x_0) & \cdots & \frac{\partial f_m}{\partial x_n}(x_0) \end{pmatrix}$$

is the **Jacobian matrix** of f at x_0.

Proof. Take $h = te_j$ (with $t \neq 0$). By differentiability,

$$f(x_0+te_j)=f(x_0)+tL(e_j)+o(|t|).$$

Hence

$$\frac{f(x_0+te_j)-f(x_0)}{t}=L(e_j)+\frac{o(|t|)}{t}\to L(e_j), \qquad t\to 0.$$

That is, $\partial f/\partial x_j(x_0) = L(e_j)$. Since L is linear, $L(h) = L(\sum_j h_j e_j) = \sum_j h_j L(e_j) = J_f(x_0)h$. ☐

Theorem 5.10 (Continuous partial derivatives imply differentiability)**.** Let $f : \Omega \to \mathbb{R}$ with $\Omega \subseteq \mathbb{R}^n$ open. If all the partial derivatives of f exist in some neighbourhood of x_0 and are continuous at x_0, then f is differentiable at x_0.

Proof. It suffices to prove the n-dimensional case. Let $h = (h_1, \ldots, h_n)$. Write

$$f(x_0+h)-f(x_0)=\sum_{j=1}^{n}[f(x_0+h_1e_1+\cdots+h_je_j)-f(x_0+h_1e_1+\cdots+h_{j-1}e_{j-1})].$$

For the j-th term, fixing the other coordinates and applying the one-variable mean-value theorem gives

$$f(x_0+h_1e_1+\cdots+h_je_j)-f(x_0+h_1e_1+\cdots+h_{j-1}e_{j-1})=h_j\frac{\partial f}{\partial x_j}(\xi_j),$$

where ξ_j lies between $x_0+h_1e_1+\cdots+h_{j-1}e_{j-1}$ and $x_0+h_1e_1+\cdots+h_je_j$. Hence

$$f(x_0+h)-f(x_0)=\sum_{j=1}^{n}h_j\frac{\partial f}{\partial x_j}(\xi_j).$$

Since the partial derivatives are continuous at x_0, we have $\partial f/\partial x_j(\xi_j) = \partial f/\partial x_j(x_0)+o(1)$ (as $h\to 0$ we have $\xi_j\to x_0$). Hence

$$f(x_0+h)-f(x_0)=\sum_{j=1}^{n}h_j\frac{\partial f}{\partial x_j}(x_0)+\sum_{j=1}^{n}h_j\cdot o(1)=\nabla f(x_0)\cdot h+o(\|h\|).$$

That is, f is differentiable at x_0, and $Df(x_0)(h)=\nabla f(x_0)\cdot h$. □

Corollary 5.4 (C^1 functions are differentiable)**.** If all the partial derivatives of f exist and are continuous on Ω (written $f\in C^1(\Omega)$), then f is differentiable on Ω.

Proof. Apply Theorem 5.10 at each point $x_0\in\Omega$. □

Theorem 5.11 (Chain rule)**.** Let $\Omega\subseteq\mathbb{R}^n$ and $\Omega'\subseteq\mathbb{R}^m$ be open, let $g:\Omega\to\Omega'$ be differentiable at x_0, and let $f:\Omega'\to\mathbb{R}^p$ be differentiable at $y_0=g(x_0)$. Then $f\circ g$ is differentiable at x_0, and

$$D(f\circ g)(x_0)=Df(y_0)\circ Dg(x_0),$$

that is,

$$J_{f\circ g}(x_0)=J_f(y_0)\cdot J_g(x_0).$$

Proof. By differentiability,

$$g(x_0+h)=g(x_0)+Dg(x_0)(h)+o(\|h\|)=y_0+k+o(\|h\|),$$

where $k=Dg(x_0)(h)$ and $\|k\|\le\|Dg(x_0)\|\|h\|+o(\|h\|)=O(\|h\|)$. Also

$$f(y_0+k)=f(y_0)+Df(y_0)(k)+o(\|k\|).$$

Hence

$$(f\circ g)(x_0+h)=f(y_0+k+o(\|h\|))=f(y_0)+Df(y_0)(k+o(\|h\|))+o(\|k+o(\|h\|)\|).$$

By the linearity of $Df(y_0)$,

$$Df(y_0)(k+o(\|h\|))=Df(y_0)(k)+Df(y_0)(o(\|h\|))=Df(y_0)(Dg(x_0)(h))+o(\|h\|).$$

Also $o(\|k+o(\|h\|)\|)=o(\|h\|)$. Hence

$$(f\circ g)(x_0+h)=(f\circ g)(x_0)+(Df(y_0)\circ Dg(x_0))(h)+o(\|h\|).$$

That is, $D(f\circ g)(x_0)=Df(y_0)\circ Dg(x_0)$. □

Definition 5.17 (C^k **map).** Let $\Omega \subseteq \mathbb{R}^n$ be open and $f : \Omega \to \mathbb{R}^m$. We write $f \in C^k(\Omega)$ if all partial derivatives of f of order at most k exist and are continuous. The set $C^\infty(\Omega) := \bigcap_{k=0}^{\infty} C^k(\Omega)$ is called the set of **smooth maps**.

Theorem 5.12 (Equality of mixed partials — Clairaut's theorem)**.** Let $f : \Omega \to \mathbb{R}$ with $\Omega \subseteq \mathbb{R}^n$ open and $f \in C^2(\Omega)$. Then for all i, j,

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}.$$

Proof. It suffices to prove the case $n = 2$. Let $(x_0, y_0) \in \Omega$. Take h, k small enough that the rectangle $[x_0, x_0 + h] \times [y_0, y_0 + k] \subseteq \Omega$. Define

$$\Delta(h, k) = f(x_0 + h, y_0 + k) - f(x_0 + h, y_0) - f(x_0, y_0 + k) + f(x_0, y_0).$$

Put $\varphi(x) = f(x, y_0 + k) - f(x, y_0)$. Then $\Delta(h, k) = \varphi(x_0 + h) - \varphi(x_0)$. By the one-variable mean-value theorem, there exists $\xi \in (x_0, x_0 + h)$ such that

$$\Delta(h, k) = h\varphi'(\xi) = h \left[\frac{\partial f}{\partial x}(\xi, y_0 + k) - \frac{\partial f}{\partial x}(\xi, y_0) \right].$$

Applying the mean-value theorem once more, now with respect to y, we obtain $\eta \in (y_0, y_0 + k)$ such that

$$\Delta(h, k) = hk \frac{\partial^2 f}{\partial y \partial x}(\xi, \eta).$$

Similarly, put $\psi(y) = f(x_0 + h, y) - f(x_0, y)$; then $\Delta(h, k) = \psi(y_0 + k) - \psi(y_0)$. By the mean-value theorem,

$$\Delta(h, k) = hk \frac{\partial^2 f}{\partial x \partial y}(\xi', \eta').$$

Hence

$$\frac{\partial^2 f}{\partial y \partial x}(\xi, \eta) = \frac{\partial^2 f}{\partial x \partial y}(\xi', \eta').$$

Let $(h, k) \to (0, 0)$; then $(\xi, \eta) \to (x_0, y_0)$ and $(\xi', \eta') \to (x_0, y_0)$. By the continuity of the second-order partial derivatives,

$$\frac{\partial^2 f}{\partial y \partial x}(x_0, y_0) = \frac{\partial^2 f}{\partial x \partial y}(x_0, y_0).$$

□

Example 5.7 (When continuity of mixed partials fails, equality fails)**.** Define

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Then $f \in C^1(\mathbb{R}^2)$, but at $(0, 0)$,

$$\frac{\partial^2 f}{\partial x \partial y}(0, 0) = 1, \qquad \frac{\partial^2 f}{\partial y \partial x}(0, 0) = -1.$$

Hence the C^2 hypothesis cannot be omitted.

5.4.3 Mean-value theorems and Taylor's formula

Theorem 5.13 (Multivariable mean-value theorem). Let $f : \Omega \to \mathbb{R}$ with $\Omega \subseteq \mathbb{R}^n$ open and $f \in C^1(\Omega)$. If $x, y \in \Omega$ and the line segment $[x, y] \subseteq \Omega$, then there exists $\xi \in (x, y)$ such that

$$f(y) - f(x) = \nabla f(\xi) \cdot (y - x).$$

Proof. Define $g : [0,1] \to \mathbb{R}$ by $g(t) = f(x + t(y - x))$. By the chain rule, $g \in C^1([0,1])$ and

$$g'(t) = \nabla f(x + t(y - x)) \cdot (y - x).$$

By the one-variable mean-value theorem, there exists $t_0 \in (0,1)$ such that $g(1) - g(0) = g'(t_0)$. Put $\xi = x + t_0(y - x)$; then

$$f(y) - f(x) = \nabla f(\xi) \cdot (y - x).$$

□

Theorem 5.14 (Second-order Taylor formula). Let $f : \Omega \to \mathbb{R}$ with $\Omega \subseteq \mathbb{R}^n$ open and $f \in C^2(\Omega)$. If $x \in \Omega$ and $h \in \mathbb{R}^n$ are such that $x + th \in \Omega$ for $t \in [0,1]$, then there exists $\xi \in (x, x + h)$ such that

$$f(x + h) = f(x) + \nabla f(x) \cdot h + \frac{1}{2} h^{\mathsf{T}} H_f(\xi) h,$$

where $H_f(\xi) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(\xi)\right)_{i,j}$ is the **Hessian matrix** of f at ξ.

Proof. Put $g(t) = f(x + th)$. Then $g \in C^2([0,1])$. By the chain rule,

$$g'(t) = \nabla f(x + th) \cdot h, \qquad g''(t) = h^{\mathsf{T}} H_f(x + th) h.$$

By the one-variable second-order Taylor formula, there exists $t_0 \in (0,1)$ such that

$$g(1) = g(0) + g'(0) + \frac{1}{2} g''(t_0).$$

Substitute $g(1) = f(x+h)$, $g(0) = f(x)$, $g'(0) = \nabla f(x) \cdot h$ and $g''(t_0) = h^{\mathsf{T}} H_f(x + t_0 h) h$. Putting $\xi = x + t_0 h$ gives the result. □

Corollary 5.5 (Second-order condition for extrema). Let $f : \Omega \to \mathbb{R}$ with $\Omega \subseteq \mathbb{R}^n$ open, $f \in C^2(\Omega)$, $x_0 \in \Omega$ and $\nabla f(x_0) = \mathbf{0}$. Then:

(i) if $H_f(x_0)$ is positive definite, then x_0 is a local minimum point;

(ii) if $H_f(x_0)$ is negative definite, then x_0 is a local maximum point;

(iii) if $H_f(x_0)$ is indefinite, then x_0 is a saddle point.

Proof. By Taylor's formula, $f(x_0 + h) - f(x_0) = \frac{1}{2} h^{\mathsf{T}} H_f(x_0) h + o(\|h\|^2)$.

(i) If $H_f(x_0)$ is positive definite, there exists $\lambda > 0$ such that $h^{\mathsf{T}} H_f(x_0) h \geq \lambda \|h\|^2$. Take $\delta > 0$ such that $|o(\|h\|^2)| < \frac{\lambda}{4}\|h\|^2$ whenever $\|h\| < \delta$. Then for $\|h\| < \delta$ we have $f(x_0 + h) - f(x_0) \geq \frac{\lambda}{4}\|h\|^2 > 0$, so x_0 is a local minimum point.

(ii) Similar.

(iii) If $H_f(x_0)$ is indefinite, there exist h_+, h_- such that $h_+^{\mathsf{T}} H_f(x_0) h_+ > 0$ and $h_-^{\mathsf{T}} H_f(x_0) h_- < 0$. Taking $h = th_+$ (with t sufficiently small) gives $f(x_0 + h) > f(x_0)$; taking $h = th_-$ gives $f(x_0 + h) < f(x_0)$. Hence x_0 is a saddle point. □

Theorem 5.15 (Method of Lagrange multipliers). Let $f, g_1, \dots, g_m : \Omega \to \mathbb{R}$ with $\Omega \subseteq \mathbb{R}^n$ open and $f, g_i \in C^1(\Omega)$. Suppose that $x_0 \in \Omega$ is a local extremum point of f under the constraints $g_1(x) = \cdots = g_m(x) = 0$, and that $\nabla g_1(x_0), \dots, \nabla g_m(x_0)$ are linearly independent. Then there exist $\lambda_1, \dots, \lambda_m \in \mathbb{R}$ such that

$$\nabla f(x_0) = \sum_{i=1}^{m} \lambda_i \nabla g_i(x_0).$$

Proof. By the implicit-function theorem (Theorem 5.17), the constraint set $M = \{x : g_1(x) = \cdots = g_m(x) = 0\}$ is an $n-m$-dimensional submanifold near x_0. Since x_0 is a local extremum point of $f|_M$, we have $Df(x_0)(v) = 0$ for every $v \in T_{x_0}M$. Moreover $T_{x_0}M = \bigcap_{i=1}^{m} \ker Dg_i(x_0)$. Hence $\nabla f(x_0)$ is orthogonal to $\bigcap_i \ker \nabla g_i(x_0)$, that is, $\nabla f(x_0)$ belongs to $\operatorname{span}\{\nabla g_1(x_0), \dots, \nabla g_m(x_0)\}$. Therefore there exist λ_i such that $\nabla f(x_0) = \sum_i \lambda_i \nabla g_i(x_0)$. $\square$

5.5 Inverse-Function, Implicit-Function and Rank Theorems

5.5.1 The inverse-function theorem

Theorem 5.16 (Inverse-function theorem). Let $\Omega \subseteq \mathbb{R}^n$ be open, let $f : \Omega \to \mathbb{R}^n$ be a C^1 map and let $x_0 \in \Omega$. If $Df(x_0)$ is invertible (that is, $\det J_f(x_0) \neq 0$), then there exist an open neighbourhood $U \subseteq \Omega$ of x_0 and an open neighbourhood V of $f(x_0)$ such that $f : U \to V$ is a bijection and the inverse map $f^{-1} : V \to U$ is C^1, satisfying

$$D(f^{-1})(y) = [Df(f^{-1}(y))]^{-1}, \qquad y \in V.$$

Proof. **Step 1: simplification.** By translation and a linear transformation we may assume that $x_0 = 0$, $f(x_0) = 0$ and $Df(0) = I$ (the identity matrix). The general case: put $\tilde{f}(x) = [Df(x_0)]^{-1}(f(x_0 + x) - f(x_0))$; then $\tilde{f}(0) = 0$ and $D\tilde{f}(0) = I$. If $\tilde{f}$ has a local C^1 inverse, then so does f.

Step 2: construction of a contraction mapping. Define $g(x) = x - f(x)$. Then $Dg(0) = I - Df(0) = 0$. Since Dg is continuous, there exists $r > 0$ such that $\|Dg(x)\| \le 1/2$ whenever $\|x\| \le r$. By the mean-value theorem, $\|g(x) - g(y)\| \le \frac{1}{2}\|x - y\|$ for $\|x\|, \|y\| \le r$.

For a fixed $y \in \mathbb{R}^n$, solving $f(x) = y$ is equivalent to solving $x = y + g(x)$. Define $T_y(x) = y + g(x)$. If $\|y\| \le r/2$ and $\|x\| \le r$, then

$$\|T_y(x)\| \le \|y\| + \|g(x)\| \le \frac{r}{2} + \frac{1}{2}\|x\| \le \frac{r}{2} + \frac{r}{2} = r.$$

Also $\|T_y(x) - T_y(x')\| = \|g(x) - g(x')\| \le \frac{1}{2}\|x - x'\|$. Hence T_y is a contraction mapping on $\overline{B}(0, r)$. By the Banach contraction-mapping principle (Theorem 5.5), T_y has a unique fixed point $x = T_y(x)$, that is, $f(x) = y$. Hence for every $y \in B(0, r/2)$ there exists a unique $x \in \overline{B}(0, r)$ such that $f(x) = y$.

Step 3: continuity of the inverse map. Let $f(x) = y$ and $f(x') = y'$. Then

$$\|x - x'\| = \|(y + g(x)) - (y' + g(x'))\| \le \|y - y'\| + \frac{1}{2}\|x - x'\|.$$

Hence $\|x - x'\| \le 2\|y - y'\|$. That is, $\|f^{-1}(y) - f^{-1}(y')\| \le 2\|y - y'\|$, so f^{-1} is continuous (in fact Lipschitz continuous).

Step 4: differentiability of the inverse map. Let $f(x) = y$ and $f(x+h) = y+k$. Then

$$k = f(x+h) - f(x) = Df(x)h + o(\|h\|).$$

Since $Df(x)$ is invertible (because Df is continuous and $Df(0) = I$, taking r smaller ensures $\|Df(x) - I\| < 1/2$, so $Df(x)$ is invertible),

$$h = [Df(x)]^{-1}k + o(\|k\|).$$

Hence

$$f^{-1}(y+k) = f^{-1}(y) + [Df(f^{-1}(y))]^{-1}k + o(\|k\|).$$

That is, f^{-1} is differentiable at y, and $D(f^{-1})(y) = [Df(f^{-1}(y))]^{-1}$.

Step 5: the inverse map is C^1. We have $D(f^{-1})(y) = [Df(f^{-1}(y))]^{-1}$. The right-hand side is a continuous function of y (Df is continuous, f^{-1} is continuous, and matrix inversion is continuous), so $D(f^{-1})$ is continuous and $f^{-1} \in C^1$. ☐

Example 5.8 (Polar-coordinate transformation). The polar-coordinate transformation $f(r,\theta) = (r\cos\theta, r\sin\theta)$ (with $r > 0$). Its Jacobian matrix is

$$J_f(r,\theta) = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}, \qquad \det J_f = r > 0.$$

Hence f is locally invertible at every point (r,θ) with $r > 0$. But f is not globally invertible ($(\theta + 2\pi k)$ gives the same point).

5.5.2 The implicit-function theorem

Theorem 5.17 (Implicit-function theorem). Let $\Omega \subseteq \mathbb{R}^{n+m}$ be open, let $F : \Omega \to \mathbb{R}^m$ be a C^1 map, and let $(x_0, y_0) \in \Omega$ satisfy $F(x_0, y_0) = 0$. If $D_yF(x_0, y_0)$ (the $m \times m$ matrix of partial derivatives of F with respect to y) is invertible, then there exist an open neighbourhood $U \subseteq \mathbb{R}^n$ of x_0, an open neighbourhood $V \subseteq \mathbb{R}^m$ of y_0, and a unique C^1 map $g : U \to V$ such that $g(x_0) = y_0$ and

$$F(x, g(x)) = 0, \qquad \forall x \in U.$$

Moreover

$$Dg(x) = -[D_yF(x, g(x))]^{-1} D_xF(x, g(x)).$$

Proof. Define $\Phi : \Omega \to \mathbb{R}^{n+m}$ by $\Phi(x,y) = (x, F(x,y))$. Then

$$D\Phi(x_0, y_0) = \begin{pmatrix} I_n & 0 \\ D_xF(x_0, y_0) & D_yF(x_0, y_0) \end{pmatrix}.$$

Thus $\det D\Phi(x_0, y_0) = \det D_yF(x_0, y_0) \neq 0$. By the inverse-function theorem (Theorem 5.16), there exist an open neighbourhood W of (x_0, y_0) and an open neighbourhood W' of $\Phi(x_0, y_0) = (x_0, 0)$ such that $\Phi : W \to W'$ is a C^1 bijection. Write $\Phi^{-1}(x,z) = (x, h(x,z))$ (because the first component of Φ is x itself). Then $F(x, h(x,z)) = z$. Put $g(x) = h(x, 0)$; then $F(x, g(x)) = 0$. The map g is C^1 (because Φ^{-1} is C^1).

Differentiation formula: differentiating both sides of $F(x, g(x)) = 0$ gives

$$D_xF(x, g(x)) + D_yF(x, g(x))Dg(x) = 0.$$

Since $D_yF(x_0, y_0)$ is invertible and D_yF is continuous, the matrix $D_yF(x, g(x))$ is invertible in some neighbourhood of x_0. Hence

$$Dg(x) = -[D_yF(x, g(x))]^{-1}D_xF(x, g(x)).$$

□

Corollary 5.6 (Regular level sets are submanifolds)**.** *Let $F : \Omega \to \mathbb{R}^m$ with $\Omega \subseteq \mathbb{R}^{n+m}$ open and $F \in C^1(\Omega)$. If $c \in \mathbb{R}^m$ and $DF(x)$ is surjective for every $x \in F^{-1}(c)$ (that is, the m row vectors are linearly independent), then $F^{-1}(c)$ is an n-dimensional C^1 submanifold.*

Proof. For any $p \in F^{-1}(c)$, the surjectivity of $DF(p)$ means that m of its columns are linearly independent, that is, there are m variables for which the corresponding $m \times m$ submatrix is invertible. Without loss of generality, suppose that the matrix of partial derivatives with respect to the last m variables is invertible. By the implicit-function theorem (Theorem 5.17), there exist a neighbourhood U of p and a C^1 map g such that $F^{-1}(c) \cap U = \{(x, g(x)) : x \in U'\}$, where $U' \subseteq \mathbb{R}^n$ is open. This is a local coordinate representation of an n-dimensional C^1 submanifold. □

Example 5.9 (Applications of the implicit-function theorem)**.** (i) The unit circle $x^2 + y^2 = 1$. Here $F(x, y) = x^2 + y^2 - 1$ and $\partial F/\partial y = 2y$. At a point (x_0, y_0) with $y_0 \neq 0$ one can solve locally for $y = g(x) = \pm\sqrt{1 - x^2}$.

(ii) The surface $z = f(x, y)$ is the level set of $F(x, y, z) = z - f(x, y) = 0$. Since $\partial F/\partial z = 1 \neq 0$, one can solve locally for $z = f(x, y)$.

(iii) For the system of equations $F_1(x, y, z) = 0$, $F_2(x, y, z) = 0$, at points where $\det \partial(F_1, F_2)/\partial(y, z) \neq 0$ one can solve locally for $y = g_1(x)$ and $z = g_2(x)$.

5.5.3 The rank theorem

Theorem 5.18 (Rank theorem)**.** *Let $\Omega \subseteq \mathbb{R}^n$ be open, let $f : \Omega \to \mathbb{R}^m$ be a C^k map $(k \geq 1)$, let $x_0 \in \Omega$, and suppose that the rank of $Df(x_0)$ is r (with $r \leq \min(n, m)$). Then there exist an open neighbourhood U of x_0 and a C^k diffeomorphism $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^n$ (with $\varphi(x_0) = 0$), together with an open neighbourhood V of $f(x_0)$ and a C^k diffeomorphism $\psi : V \to \psi(V) \subseteq \mathbb{R}^m$ (with $\psi(f(x_0)) = 0$), such that*

$$(\psi \circ f \circ \varphi^{-1})(x_1, \dots, x_n) = (x_1, \dots, x_r, 0, \dots, 0).$$

That is, near x_0 the map f is equivalent to the standard projection.

Proof. **Step 1: simplification.** By translation and a linear transformation we may assume that $x_0 = 0$, $f(x_0) = 0$, and that the submatrix formed by the first r rows and the first r columns of $Df(0)$ is invertible.

Step 2: construction of an auxiliary map. Write $f = (f^1, f^2)$, where $f^1 : \Omega \to \mathbb{R}^r$ consists of the first r components. Define $\Phi : \Omega \to \mathbb{R}^n$ by

$$\Phi(x) = (f^1(x), x_{r+1}, \dots, x_n).$$

Then

$$D\Phi(0) = \begin{pmatrix} D_{x^1} f^1(0) & D_{x^2} f^1(0) \\ 0 & I_{n-r} \end{pmatrix},$$

where $D_{x^1} f^1(0)$ is an invertible $r \times r$ matrix. Hence $\det D\Phi(0) \neq 0$. By the inverse-function theorem (Theorem 5.16), there exists an open neighbourhood U of 0 such that $\Phi : U \to \Phi(U)$ is a C^k diffeomorphism.

Step 3: composition. Put $\varphi = \Phi$. Then $\varphi(x) = (y^1, y^2)$, where $y^1 = f^1(x) \in \mathbb{R}^r$ and $y^2 = x^2 \in \mathbb{R}^{n-r}$. By the inverse-function theorem, $x = \varphi^{-1}(y^1, y^2)$.

On U we have $f(x) = (f^1(x), f^2(x)) = (y^1, f^2(\varphi^{-1}(y^1, y^2)))$. Since $y^1 = f^1(x)$ and $x = \varphi^{-1}(y^1, y^2)$, the expression $f^2(\varphi^{-1}(y^1, y^2))$ is a function of (y^1, y^2). But the rank of Df at 0 is r, and the first r columns of $Df^1(0)$ are invertible, so every row of $Df^2(0)$ is a linear combination of the rows of $Df^1(0)$. By continuity, near 0 every row of Df^2 is still a linear combination of the rows of Df^1, that is, f^2 is locally a function of f^1 on U: there exists a C^k map $h : \mathbb{R}^r \to \mathbb{R}^{m-r}$ such that $f^2(x) = h(f^1(x))$ on U.

Step 4: construction of ψ. Define $\psi : \mathbb{R}^m \to \mathbb{R}^m$ by

$$\psi(z^1, z^2) = (z^1, z^2 - h(z^1)), \qquad z^1 \in \mathbb{R}^r,\ z^2 \in \mathbb{R}^{m-r}.$$

Then ψ is a C^k diffeomorphism (its inverse is $(z^1, z^2) \mapsto (z^1, z^2 + h(z^1))$) and $\psi(0) = 0$. Moreover

$$(\psi \circ f)(x) = \psi(f^1(x), f^2(x)) = (f^1(x), f^2(x) - h(f^1(x))) = (f^1(x), 0).$$

Hence

$$(\psi \circ f \circ \varphi^{-1})(y^1, y^2) = (y^1, 0).$$

That is, $(\psi \circ f \circ \varphi^{-1})(x_1, \ldots, x_n) = (x_1, \ldots, x_r, 0, \ldots, 0)$. ☐

5.6 Multiple Integrals

5.6.1 Review and generalisation of the Riemann integral

Definition 5.18 (Riemann integral over a rectangle). Let $R = [a_1, b_1] \times \cdots \times [a_n, b_n] \subseteq \mathbb{R}^n$ be a closed rectangle and let $f : R \to \mathbb{R}$ be bounded.

(i) A **partition** of R is a division of each $[a_i, b_i]$ into finitely many subintervals, thereby dividing R into finitely many closed subrectangles $R_1, \ldots, R_N$.

(ii) The **mesh** of a partition is $\|\mathcal{P}\| := \max_j \operatorname{diam}(R_j)$.

(iii) Choosing a point ξ_j in each R_j, one forms the **Riemann sum**

$$S(f, \mathcal{P}, \{\xi_j\}) := \sum_{j=1}^{N} f(\xi_j) |R_j|,$$

where $|R_j|$ is the volume of R_j.

(iv) f is **Riemann integrable** on R if there exists $I \in \mathbb{R}$ such that for every $\varepsilon > 0$ there exists $\delta > 0$ such that for every partition $\mathcal{P}$ with $\|\mathcal{P}\| < \delta$ and every choice of points $\{\xi_j\}$,

$$|S(f, \mathcal{P}, \{\xi_j\}) - I| < \varepsilon.$$

In this case we write $I = \int_R f$.

Definition 5.19 (Darboux upper and lower sums)**.** Let $f : R \to \mathbb{R}$ be bounded and let $\mathcal{P}$ be a partition. Define

$$M_j := \sup_{x \in R_j} f(x), \qquad m_j := \inf_{x \in R_j} f(x),$$

$$U(f, \mathcal{P}) := \sum_{j=1}^{N} M_j |R_j| \quad \text{(upper sum)}, \qquad L(f, \mathcal{P}) := \sum_{j=1}^{N} m_j |R_j| \quad \text{(lower sum)}.$$

The **upper integral** and the **lower integral** of f are respectively

$$\overline{\int_R} f := \inf_{\mathcal{P}} U(f, \mathcal{P}), \qquad \underline{\int_R} f := \sup_{\mathcal{P}} L(f, \mathcal{P}).$$

Theorem 5.19 (Necessary and sufficient condition for Riemann integrability)**.** Let $f : R \to \mathbb{R}$ be bounded. Then the following conditions are equivalent:

(i) f is Riemann integrable on R;

(ii) $\overline{\int_R} f = \underline{\int_R} f$;

(iii) for every $\varepsilon > 0$ there exists a partition $\mathcal{P}$ such that $U(f, \mathcal{P}) - L(f, \mathcal{P}) < \varepsilon$.

Proof. (i) ⇒ (ii): Suppose that f is integrable with integral value I. For every $\varepsilon > 0$ there exists $\delta > 0$ such that $|S(f, \mathcal{P}, \{\xi_j\}) - I| < \varepsilon$ whenever $\|\mathcal{P}\| < \delta$. Fix such a $\mathcal{P}$; taking the supremum over the ξ_j gives $U(f, \mathcal{P}) \leq I + \varepsilon$, and taking the infimum gives $L(f, \mathcal{P}) \geq I - \varepsilon$. Hence $\overline{\int_R} f \leq I + \varepsilon$ and $\underline{\int_R} f \geq I - \varepsilon$. Since ε is arbitrary, $\overline{\int_R} f = \underline{\int_R} f = I$.

(ii) ⇒ (iii): By the definition of the upper and lower integrals, there exist partitions $\mathcal{P}_1, \mathcal{P}_2$ such that $U(f, \mathcal{P}_1) < \overline{\int_R} f + \varepsilon/2$ and $L(f, \mathcal{P}_2) > \underline{\int_R} f - \varepsilon/2$. Taking a common refinement $\mathcal{P}$, we obtain $U(f, \mathcal{P}) - L(f, \mathcal{P}) \leq U(f, \mathcal{P}_1) - L(f, \mathcal{P}_2) < \varepsilon$.

(iii) ⇒ (i): Put $I = \overline{\int_R} f = \underline{\int_R} f$. For every $\varepsilon > 0$ there exists a partition $\mathcal{P}_0$ such that $U(f, \mathcal{P}_0) - L(f, \mathcal{P}_0) < \varepsilon$. Let the mesh of $\mathcal{P}_0$ be δ_0. For every partition $\mathcal{P}$ with $\|\mathcal{P}\| < \delta_0$ we have $L(f, \mathcal{P}) \leq I \leq U(f, \mathcal{P})$ and $U(f, \mathcal{P}) - L(f, \mathcal{P}) < \varepsilon$ (by taking the common refinement of $\mathcal{P}$ and $\mathcal{P}_0$). Hence $|S(f, \mathcal{P}, \{\xi_j\}) - I| < \varepsilon$, so f is integrable. ☐

Theorem 5.20 (Continuous functions are integrable)**.** Let $R \subseteq \mathbb{R}^n$ be a closed rectangle and let $f : R \to \mathbb{R}$ be continuous. Then f is Riemann integrable on R.

Proof. Since R is compact and f is continuous, f is uniformly continuous on R (Theorem 5.7 (iv)). Given $\varepsilon > 0$, there exists $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon/|R|$

whenever $\|x-y\| < \delta$. Take a partition $\mathcal{P}$ with $\|\mathcal{P}\| < \delta$. Then for every R_j we have $\sup_{R_j} f - \inf_{R_j} f < \varepsilon/|R|$. Hence

$$U(f,\mathcal{P}) - L(f,\mathcal{P}) = \sum_{j=1}^{N} \left(\sup_{R_j} f - \inf_{R_j} f \right) |R_j| < \frac{\varepsilon}{|R|} \sum_{j=1}^{N} |R_j| = \varepsilon.$$

By Theorem 5.19, f is integrable. □

Theorem 5.21 (Fubini's theorem). Let $R = [a_1,b_1] \times \cdots \times [a_n,b_n]$ and let $f : R \to \mathbb{R}$ be continuous. Then

$$\int_R f = \int_{a_1}^{b_1} \left(\cdots \left(\int_{a_n}^{b_n} f(x_1,\dots,x_n)\, dx_n \right) \cdots \right) dx_1.$$

That is, a multiple integral can be reduced to an iterated integral, and the order of integration may be interchanged arbitrarily.

Proof. We prove the case $n=2$. Let $R = [a,b] \times [c,d]$. Define $F(x) = \int_c^d f(x,y)\, dy$. Since f is uniformly continuous on R, the function F is continuous on $[a,b]$. Given partitions $\mathcal{P}_x, \mathcal{P}_y$ of $[a,b]$ and $[c,d]$, construct a partition of R. For each subrectangle $R_{ij} = [x_{i-1},x_i] \times [y_{j-1},y_j]$, by the one-variable mean-value theorem for integrals there exists $\eta_{ij} \in [y_{j-1},y_j]$ such that $\int_{y_{j-1}}^{y_j} f(x_i,y)\, dy = f(x_i,\eta_{ij})\Delta y_j$. Hence

$$\sum_j \int_{y_{j-1}}^{y_j} f(x_i,y)\, dy = \sum_j f(x_i,\eta_{ij})\Delta y_j.$$

Letting $\Delta y_j \to 0$, the right-hand side tends to $\int_c^d f(x_i,y)\, dy = F(x_i)$. Hence

$$\int_a^b F(x)\, dx = \lim_{\|\mathcal{P}_x\| \to 0} \sum_i F(x_i)\Delta x_i = \lim_{\|\mathcal{P}\| \to 0} \sum_{i,j} f(x_i,\eta_{ij})\Delta x_i \Delta y_j.$$

The right-hand side is a Riemann sum for f, with limit $\int_R f$. Hence $\int_R f = \int_a^b \left(\int_c^d f(x,y)\, dy \right) dx$. Symmetrically, $\int_R f = \int_c^d \left(\int_a^b f(x,y)\, dx \right) dy$.

The n-dimensional case follows by induction. □

Theorem 5.22 (Change-of-variables formula). Let $\Omega, \Omega' \subseteq \mathbb{R}^n$ be open and let $\Phi : \Omega \to \Omega'$ be a C^1 diffeomorphism. Then for every continuous function $f : \Omega' \to \mathbb{R}$,

$$\int_{\Omega'} f(y)\, dy = \int_{\Omega} f(\Phi(x))\, |\det J_\Phi(x)|\, dx.$$

Proof. **Step 1: local linearisation.** By continuity, for every $\varepsilon > 0$ and every $x_0 \in \Omega$ there exists $r > 0$ such that $B(x_0,r) \subseteq \Omega$ and $\|J_\Phi(x) - J_\Phi(x_0)\| < \varepsilon$ for $x \in B(x_0,r)$. By the mean-value theorem, $\Phi(x) = \Phi(x_0) + J_\Phi(x_0)(x-x_0) + o(\|x-x_0\|)$.

Step 2: volume scaling of a linear transformation. For a linear map A,

$$\mathrm{vol}(A(B(x_0,r))) = |\det A|\, \mathrm{vol}(B(x_0,r)).$$

For a general C^1 diffeomorphism, the map Φ is locally approximated by $J_\Phi(x_0)$, so

$$\mathrm{vol}(\Phi(B(x_0,r))) \approx |\det J_\Phi(x_0)|\, \mathrm{vol}(B(x_0,r)).$$

The rigorous error estimate is $o(r^n)$.

Step 3: Riemann sums. Divide Ω into small measurable sets and on each small set approximate Φ by a linear map. The Riemann sum

$$\sum_j f(\Phi(x_j))|\det J_\Phi(x_j)||E_j| \to \int_\Omega f(\Phi(x))|\det J_\Phi(x)|\,dx.$$

On the other hand, the right-hand side is a Riemann sum for $\int_{\Omega'} f(y)\,dy$ (via the image under Φ). Hence the equality holds. □

Example 5.10 (Multiple integrals in polar coordinates). $\Phi(r,\theta) = (r\cos\theta, r\sin\theta)$, $\det J_\Phi = r$. Hence

$$\int_{\Omega'} f(x,y)\,dx\,dy = \int_\Omega f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta.$$

For example, the integral over the unit disc:

$$\int_{x^2+y^2\le 1} f(x,y)\,dx\,dy = \int_0^{2\pi}\int_0^1 f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta.$$

5.7 Line and Surface Integrals

5.7.1 Line integrals

Definition 5.20 (Curves and parametrisation). Let $[a,b] \subseteq \mathbb{R}$.

(i) A C^1 **curve** is a C^1 map $\gamma : [a,b] \to \mathbb{R}^n$.

(ii) γ is called **regular** if $\gamma'(t) \neq 0$ for all $t \in [a,b]$.

(iii) γ is called **simple** if γ is injective on (a,b).

(iv) γ is called **closed** if $\gamma(a) = \gamma(b)$.

(v) Two curves γ_1, γ_2 are called **equivalent** if there exists a C^1 bijection $\varphi : [a_1,b_1] \to [a_2,b_2]$ such that $\gamma_2 = \gamma_1 \circ \varphi$.

Definition 5.21 (Line integral of the first kind (arc-length integral)). Let $\gamma : [a,b] \to \mathbb{R}^n$ be a C^1 curve and let $f : \mathbb{R}^n \to \mathbb{R}$ be continuous. The **line integral of the first kind** of f along γ is defined as

$$\int_\gamma f\,ds := \int_a^b f(\gamma(t))\,\|\gamma'(t)\|\,dt.$$

If $f \equiv 1$, then $\int_\gamma ds$ is the arc length of γ.

Proposition 5.13 (Independence of the first-kind line integral from parametrisation). The line integral of the first kind is independent of the regular parametrisation of the curve.

Proof. Let $\gamma_2 = \gamma_1 \circ \varphi$, where φ is a C^1 bijection. Then $\gamma_2'(t) = \gamma_1'(\varphi(t))\varphi'(t)$. Hence

$$\int_{a_2}^{b_2} f(\gamma_2(t))\|\gamma_2'(t)\|\,dt = \int_{a_2}^{b_2} f(\gamma_1(\varphi(t)))\|\gamma_1'(\varphi(t))\||\varphi'(t)|\,dt.$$

Putting $s = \varphi(t)$ and $ds = \varphi'(t)dt$, we obtain

$$= \int_{a_1}^{b_1} f(\gamma_1(s))\|\gamma_1'(s)\|\,ds.$$

□

Definition 5.22 (Line integral of the second kind (integral of a vector field along a curve)). Let $\gamma : [a, b] \to \mathbb{R}^n$ be a C^1 curve and let $\mathsf{F} : \mathbb{R}^n \to \mathbb{R}^n$ be continuous. The **line integral of the second kind** of F along γ is defined as

$$\int_\gamma \mathsf{F} \cdot d\mathsf{r} := \int_a^b \mathsf{F}(\gamma(t)) \cdot \gamma'(t)\,dt.$$

Proposition 5.14 (Additivity and reversal of the second-kind line integral). (i) If $\gamma = \gamma_1 \cup \gamma_2$ (joined end to end), then

$$\int_\gamma \mathsf{F} \cdot d\mathsf{r} = \int_{\gamma_1} \mathsf{F} \cdot d\mathsf{r} + \int_{\gamma_2} \mathsf{F} \cdot d\mathsf{r}.$$

(ii) If γ^- is the curve γ traversed in the opposite direction, then

$$\int_{\gamma^-} \mathsf{F} \cdot d\mathsf{r} = -\int_\gamma \mathsf{F} \cdot d\mathsf{r}.$$

Proof. (i) This follows directly from the additivity of the Riemann integral over intervals.

(ii) Let $\gamma^-(t) = \gamma(a + b - t)$. Then $(\gamma^-)'(t) = -\gamma'(a + b - t)$. Hence

$$\int_{\gamma^-} \mathsf{F} \cdot d\mathsf{r} = \int_a^b \mathsf{F}(\gamma(a + b - t)) \cdot (-\gamma'(a + b - t))\,dt.$$

Putting $s = a + b - t$ and $ds = -dt$, we obtain

$$= \int_b^a \mathsf{F}(\gamma(s)) \cdot \gamma'(s)\,ds = -\int_a^b \mathsf{F}(\gamma(s)) \cdot \gamma'(s)\,ds = -\int_\gamma \mathsf{F} \cdot d\mathsf{r}.$$

□

5.7.2 Surface integrals

Definition 5.23 (Surfaces and parametrisation). Let $D \subseteq \mathbb{R}^2$ be a bounded closed region.

(i) A C^1 **surface** is a C^1 map $\mathsf{r} : D \to \mathbb{R}^3$.

(ii) r is called **regular** if $\mathsf{r}_u \times \mathsf{r}_v \neq \mathbf{0}$ for all $(u, v) \in D$, where $\mathsf{r}_u = \partial\mathsf{r}/\partial u$ and $\mathsf{r}_v = \partial\mathsf{r}/\partial v$.

(iii) r is called **simple** if r is injective on the interior of D.

(iv) The **area element** is $dS := \|\mathsf{r}_u \times \mathsf{r}_v\|\, du\, dv$.

(v) The **unit normal vector** is $\mathsf{n} := \frac{\mathsf{r}_u \times \mathsf{r}_v}{\|\mathsf{r}_u \times \mathsf{r}_v\|}$.

Definition 5.24 (Surface integral of the first kind (area integral)). Let $\mathsf{r} : D \to \mathbb{R}^3$ be a regular C^1 surface and let $f : \mathbb{R}^3 \to \mathbb{R}$ be continuous. The **surface integral of the first kind** of f over the surface $S = \mathsf{r}(D)$ is defined as

$$\int_S f\, dS := \int_D f(\mathsf{r}(u, v))\, \|\mathsf{r}_u \times \mathsf{r}_v\|\, du\, dv.$$

If $f \equiv 1$, then $\int_S dS$ is the area of the surface.

Definition 5.25 (Surface integral of the second kind (flux integral)). Let $\mathsf{r} : D \to \mathbb{R}^3$ be a regular C^1 surface and let $\mathsf{F} : \mathbb{R}^3 \to \mathbb{R}^3$ be continuous. The **surface integral of the second kind** (flux) of F over the surface S is defined as

$$\int_S \mathsf{F} \cdot d\mathsf{S} := \int_D \mathsf{F}(\mathsf{r}(u, v)) \cdot (\mathsf{r}_u \times \mathsf{r}_v)\, du\, dv.$$

Definition 5.26 (Orientation of a surface). A surface S is called **orientable** if there exists a continuous unit normal vector field $\mathsf{n} : S \to S^2$. Two orientations $\mathsf{n}_1, \mathsf{n}_2$ are called the same if $\mathsf{n}_1 = \mathsf{n}_2$, and opposite if $\mathsf{n}_1 = -\mathsf{n}_2$. A surface with a chosen orientation is called an **oriented surface**.

Proposition 5.15 (Relation between flux integral and orientation). If S^- denotes S with the opposite orientation, then

$$\int_{S^-} \mathsf{F} \cdot d\mathsf{S} = -\int_S \mathsf{F} \cdot d\mathsf{S}.$$

Proof. The opposite orientation corresponds to interchanging u, v in the parametrisation or to changing the sign of the parametrisation. The vector $\mathsf{r}_u \times \mathsf{r}_v$ changes sign, so the integral changes sign. ☐

Example 5.11 (Area and flux of a sphere). Parametrise the sphere S^2 by $\mathsf{r}(\theta, \varphi) = (\sin\theta \cos\varphi, \sin\theta \sin\varphi, \cos\theta)$ with $\theta \in [0, \pi]$ and $\varphi \in [0, 2\pi]$. Computation gives

$$\mathsf{r}_\theta \times \mathsf{r}_\varphi = \sin\theta\, (\sin\theta \cos\varphi, \sin\theta \sin\varphi, \cos\theta),$$

so $\|\mathsf{r}_\theta \times \mathsf{r}_\varphi\| = \sin\theta$. Hence the area of the sphere is

$$\int_{S^2} dS = \int_0^{2\pi} \int_0^{\pi} \sin\theta\, d\theta\, d\varphi = 4\pi.$$

For the radial field $\mathsf{F}(x, y, z) = (x, y, z)$, the flux is

$$\int_{S^2} \mathsf{F} \cdot d\mathsf{S} = \int_0^{2\pi} \int_0^{\pi} \sin\theta\, d\theta\, d\varphi = 4\pi.$$

5.8 Uniform Convergence of Sequences and Series of Functions

5.8.1 Uniform convergence

Definition 5.27 (Uniform convergence of a sequence of functions). Let $E \subseteq \mathbb{R}^n$, let $f_k : E \to \mathbb{R}^m$ $(k = 1, 2, \dots)$ and let $f : E \to \mathbb{R}^m$.

(i) $\{f_k\}$ **converges pointwise** to f if $f_k(x) \to f(x)$ for every $x \in E$.

(ii) $\{f_k\}$ **converges uniformly** to f if

$$\lim_{k\to\infty} \sup_{x\in E} \|f_k(x) - f(x)\| = 0,$$

that is,

$$\forall \varepsilon > 0,\ \exists K \in \mathbb{N}^+,\ \forall k \geq K,\ \forall x \in E, \quad \|f_k(x) - f(x)\| < \varepsilon.$$

We write $f_k \rightrightarrows f$.

Proposition 5.16 (Uniform convergence implies pointwise convergence). Uniform convergence implies pointwise convergence.

Proof. Suppose that $f_k \rightrightarrows f$. For a fixed $x \in E$, $\|f_k(x) - f(x)\| \leq \sup_{y\in E} \|f_k(y) - f(y)\| \to 0$. Hence $f_k(x) \to f(x)$. ☐

Proposition 5.17 (Difference between uniform and pointwise convergence). There exists a sequence of functions that converges pointwise but not uniformly.

Proof. Take $E = [0, 1]$ and $f_k(x) = x^k$. The pointwise limit is

$$f(x) = \begin{cases} 0, & x \in [0, 1), \\ 1, & x = 1. \end{cases}$$

The convergence is pointwise. But $\sup_{x\in[0,1]} |x^k - f(x)| = 1$ (take $x \to 1^-$), so the convergence is not uniform. ☐

Theorem 5.23 (Uniform convergence preserves continuity). Let $E \subseteq \mathbb{R}^n$, let $f_k : E \to \mathbb{R}^m$ be continuous and suppose that $f_k \rightrightarrows f$. Then f is continuous.

Proof. Let $x_0 \in E$. Given $\varepsilon > 0$, take K such that $\|f_k(x) - f(x)\| < \varepsilon/3$ for all $x \in E$ whenever $k \geq K$. Fix $k = K$. Since f_K is continuous at x_0, there exists $\delta > 0$ such that $\|f_K(x) - f_K(x_0)\| < \varepsilon/3$ whenever $\|x - x_0\| < \delta$. Then

$$\begin{aligned} \|f(x) - f(x_0)\| &\leq \|f(x) - f_K(x)\| + \|f_K(x) - f_K(x_0)\| + \|f_K(x_0) - f(x_0)\| \\ &< \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon. \end{aligned}$$

Hence f is continuous at x_0. ☐

Theorem 5.24 (Uniform convergence preserves integration). Let $E = [a,b] \subseteq \mathbb{R}$, let $f_k : [a,b] \to \mathbb{R}$ be continuous and suppose that $f_k \rightrightarrows f$. Then

$$\lim_{k\to\infty} \int_a^b f_k(x)\,dx = \int_a^b f(x)\,dx.$$

That is, integration and the limit may be interchanged.

Proof. By uniform convergence, $\sup_{x\in[a,b]} |f_k(x) - f(x)| \to 0$. Hence

$$\left|\int_a^b f_k(x)\,dx - \int_a^b f(x)\,dx\right| \leq \int_a^b |f_k(x) - f(x)|\,dx \leq (b-a) \sup_{x\in[a,b]} |f_k(x) - f(x)| \to 0.$$

□

Theorem 5.25 (Uniform convergence preserves differentiation). Let $f_k : (a,b) \to \mathbb{R}$ be C^1, suppose that $f_k \rightrightarrows f$ on (a,b) and that $f_k' \rightrightarrows g$ on (a,b). Then $f \in C^1(a,b)$ and $f' = g$.

Proof. Fix $x_0 \in (a,b)$. For any $x \in (a,b)$, by the fundamental theorem of calculus,

$$f_k(x) - f_k(x_0) = \int_{x_0}^x f_k'(t)\,dt.$$

Since $f_k \rightrightarrows f$ and $f_k' \rightrightarrows g$, passing to the limit gives

$$f(x) - f(x_0) = \int_{x_0}^x g(t)\,dt.$$

By the fundamental theorem of calculus, the right-hand side is differentiable with respect to x, with derivative $g(x)$. Hence $f'(x) = g(x)$ for all $x \in (a,b)$. By Theorem 5.23, g is continuous, so $f \in C^1(a,b)$. □

Theorem 5.26 (Weierstrass M-test). Let $E \subseteq \mathbb{R}^n$, let $u_k : E \to \mathbb{R}^m$, and suppose that there exist constants $M_k \geq 0$ such that $\|u_k(x)\| \leq M_k$ for all $x \in E$. If $\sum_{k=1}^\infty M_k < \infty$, then $\sum_{k=1}^\infty u_k(x)$ converges uniformly on E.

Proof. Since $\sum_k M_k$ converges, for every $\varepsilon > 0$ there exists K such that $\sum_{k=K}^\infty M_k < \varepsilon$. For any $m > k \geq K$ and any $x \in E$,

$$\left\|\sum_{j=k+1}^m u_j(x)\right\| \leq \sum_{j=k+1}^m \|u_j(x)\| \leq \sum_{j=k+1}^m M_j < \varepsilon.$$

Hence the sequence of partial sums satisfies the Cauchy criterion uniformly on E, that is, it converges uniformly. □

Example 5.12 (Examples of uniform convergence).

(i) $f_k(x) = x^k$ converges uniformly to 0 on $[0,a]$ (with $a < 1$); but it does not converge uniformly on $[0,1]$.

(ii) $f_k(x) = \sin(kx)/k$ converges uniformly to 0 on $\mathbb{R}$.

(iii) $\sum_{k=1}^\infty x^k/k^2$ converges uniformly on $[-1,1]$ (by the M-test, with $M_k = 1/k^2$).

(iv) $\sum_{k=1}^\infty x^k/k$ converges pointwise but not uniformly on $[-1,1)$ (it diverges at $x = 1$, and the rate of convergence slows down as $x \to 1^-$).

5.8.2 Power series

Definition 5.28 (Power series)**.** Let $x_0 \in \mathbb{R}$ and let $\{a_k\}$ be a real sequence. A **power series** is

$$\sum_{k=0}^{\infty} a_k (x-x_0)^k.$$

Theorem 5.27 (Cauchy–Hadamard radius-of-convergence formula)**.** *Let $\sum_{k=0}^{\infty} a_k(x-x_0)^k$ be a power series. Define*

$$R := \frac{1}{\limsup_{k\to\infty} |a_k|^{1/k}},$$

with the conventions $1/0=\infty$ and $1/\infty = 0$. Then:

(i) the power series converges absolutely when $|x-x_0|<R$;

(ii) the power series diverges when $|x-x_0|>R$;

(iii) it converges uniformly on $|x-x_0| \le r < R$.

R is called the **radius of convergence***.*

Proof. Let $L = \limsup_{k\to\infty} |a_k|^{1/k}$ and $R = 1/L$.

(i) If $|x-x_0|<R$, then $L|x-x_0|<1$. By the root test,

$$\limsup_{k\to\infty} |a_k(x-x_0)^k|^{1/k} = L|x-x_0| < 1,$$

so the series converges absolutely.

(ii) If $|x-x_0|>R$, then $L|x-x_0|>1$. By the root test the series diverges (its general term does not tend to zero).

(iii) Suppose that $|x-x_0| \le r < R$. Then $|a_k(x-x_0)^k| \le |a_k| r^k$. Take $r' \in (r,R)$; then $Lr' < 1$, so $\sum_k |a_k|(r')^k$ converges. By the comparison test, $\sum_k |a_k| r^k \le \sum_k |a_k|(r')^k < \infty$. By the Weierstrass M-test (Theorem 5.26), the power series converges uniformly on $|x-x_0|\le r$. ☐

Theorem 5.28 (Termwise differentiation and integration of power series)**.** *Let $\sum_{k=0}^{\infty} a_k(x-x_0)^k$ have radius of convergence $R>0$ and sum function $f(x)$. Then $f \in C^\infty(|x-x_0|<R)$, and*

$$f'(x) = \sum_{k=1}^{\infty} k a_k (x-x_0)^{k-1}, \qquad \int_{x_0}^{x} f(t)\,dt = \sum_{k=0}^{\infty} \frac{a_k}{k+1}(x-x_0)^{k+1},$$

both with radius of convergence R.

Proof. By the Cauchy–Hadamard formula, the radius of convergence of $\sum_k k a_k (x-x_0)^{k-1}$ is

$$\frac{1}{\limsup_k |k a_k|^{1/(k-1)}} = \frac{1}{\limsup_k |a_k|^{1/k}} = R.$$

Hence $\sum_k k a_k(x-x_0)^{k-1}$ converges uniformly on $|x-x_0|\le r<R$. By Theorem 5.25, f is differentiable and $f'(x) = \sum_k k a_k (x-x_0)^{k-1}$. Induction gives $f \in C^\infty$.

Integration is similar: the radius of convergence of $\sum_k a_k/(k+1)\cdot(x-x_0)^{k+1}$ is likewise R, so by the preservation of integration under uniform convergence (Theorem 5.24) termwise integration holds. ☐

Example 5.13 (Common power series). (i) $e^x = \sum_{k=0}^{\infty} x^k/k!$, $R = \infty$.

(ii) $\sin x = \sum_{k=0}^{\infty}(-1)^k x^{2k+1}/(2k+1)!$, $R = \infty$.

(iii) $\cos x = \sum_{k=0}^{\infty}(-1)^k x^{2k}/(2k)!$, $R = \infty$.

(iv) $1/(1-x) = \sum_{k=0}^{\infty} x^k$, $R = 1$.

(v) $\ln(1+x) = \sum_{k=1}^{\infty}(-1)^{k+1} x^k/k$, $R = 1$.

(vi) $(1+x)^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k$, $R = 1$ (for $\alpha \notin \mathbb{N}$).

5.9 Parameter-dependent Integrals and Differentiation under the Integral Sign

5.9.1 Proper parameter-dependent integrals

Definition 5.29 (Parameter-dependent integral). Let $D \subseteq \mathbb{R}^n$ be a parameter set and let $f : D \times [a,b] \to \mathbb{R}$. Define

$$F(t) := \int_a^b f(t,x)\,dx, \qquad t \in D.$$

F is called a **parameter-dependent integral** and t is the parameter.

Theorem 5.29 (Continuity of parameter-dependent integrals). Let $f : D \times [a,b] \to \mathbb{R}$ be continuous and let $D \subseteq \mathbb{R}^n$ be compact. Then $F(t) = \int_a^b f(t,x)\,dx$ is continuous on D.

Proof. Since f is continuous on the compact set $D \times [a,b]$, the Heine–Cantor theorem shows that f is uniformly continuous. Given $\varepsilon > 0$, there exists $\delta > 0$ such that $|f(t,x) - f(t_0,x)| < \varepsilon/(b-a)$ for all $x \in [a,b]$ whenever $\|t - t_0\| < \delta$. Then

$$|F(t) - F(t_0)| \le \int_a^b |f(t,x) - f(t_0,x)|\,dx < \varepsilon.$$

□

Theorem 5.30 (Leibniz rule — differentiation under the integral sign). Let $f : D \times [a,b] \to \mathbb{R}$ be continuous and suppose that $\partial f/\partial t$ exists and is continuous. Then $F(t) = \int_a^b f(t,x)\,dx$ is differentiable on D, and

$$F'(t) = \int_a^b \frac{\partial f}{\partial t}(t,x)\,dx.$$

Proof. Fix $t_0 \in D$. For $h \neq 0$,

$$\frac{F(t_0+h) - F(t_0)}{h} = \int_a^b \frac{f(t_0+h,x) - f(t_0,x)}{h}\,dx.$$

By the one-variable mean-value theorem, for each x there exists ξ_x between t_0 and $t_0 + h$ such that

$$\frac{f(t_0+h,x)-f(t_0,x)}{h} = \frac{\partial f}{\partial t}(\xi_x, x).$$

Since $\partial f/\partial t$ is continuous on $D\times[a,b]$, the Heine–Cantor theorem shows that $\partial f/\partial t$ is uniformly continuous. Given $\varepsilon > 0$, there exists $\delta > 0$ such that $|\partial f/\partial t(\xi_x, x) - \partial f/\partial t(t_0, x)| < \varepsilon/(b-a)$ for all x whenever $|h| < \delta$. Then

$$\left|\frac{F(t_0+h)-F(t_0)}{h} - \int_a^b \frac{\partial f}{\partial t}(t_0,x)\,dx\right| \le \int_a^b \left|\frac{\partial f}{\partial t}(\xi_x,x) - \frac{\partial f}{\partial t}(t_0,x)\right| dx < \varepsilon.$$

Hence $F'(t_0) = \int_a^b \partial f/\partial t(t_0,x)\,dx$. ☐

Corollary 5.7 (Leibniz formula for variable limits)**.** Let $f(t,x)$ and $\partial f/\partial t$ be continuous, and let $a(t), b(t)$ be differentiable. Then

$$\frac{d}{dt}\int_{a(t)}^{b(t)} f(t,x)\,dx = f(t,b(t))b'(t) - f(t,a(t))a'(t) + \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t}(t,x)\,dx.$$

Proof. Define $G(t,u,v) = \int_u^v f(t,x)\,dx$. By the chain rule,

$$\frac{d}{dt}G(t,a(t),b(t)) = \frac{\partial G}{\partial t} + \frac{\partial G}{\partial u}a'(t) + \frac{\partial G}{\partial v}b'(t).$$

Here $\partial G/\partial t = \int_u^v \partial f/\partial t\,dx$ (by Theorem 5.30); $\partial G/\partial v = f(t,v)$; and $\partial G/\partial u = -f(t,u)$. Substituting $u = a(t)$ and $v = b(t)$ gives the result. ☐

5.9.2 Improper parameter-dependent integrals

Definition 5.30 (Uniform convergence of an improper parameter-dependent integral)**.** Let $f : D \times [a,\infty) \to \mathbb{R}$. The improper parameter-dependent integral

$$F(t) = \int_a^\infty f(t,x)\,dx$$

is said to **converge uniformly** on D if

$$\lim_{M\to\infty} \sup_{t\in D} \left|\int_M^\infty f(t,x)\,dx\right| = 0.$$

Theorem 5.31 (Weierstrass M-test for improper parameter-dependent integrals)**.** Let $f : D \times [a,\infty) \to \mathbb{R}$ satisfy $|f(t,x)| \le g(x)$ for all $t \in D$, and suppose that $\int_a^\infty g(x)\,dx < \infty$. Then $\int_a^\infty f(t,x)\,dx$ converges uniformly on D.

Proof. Given $\varepsilon > 0$. Since $\int_a^\infty g(x)\,dx$ converges, there exists M_0 such that $\int_M^\infty g(x)\,dx < \varepsilon$ whenever $M > M_0$. Then for any $t \in D$,

$$\left|\int_M^\infty f(t,x)\,dx\right| \le \int_M^\infty |f(t,x)|\,dx \le \int_M^\infty g(x)\,dx < \varepsilon.$$

Hence $\sup_{t\in D} |\int_M^\infty f(t,x)\,dx| \to 0$. ☐

Theorem 5.32 (Continuity of improper parameter-dependent integrals). Let $f : D \times [a, \infty) \to \mathbb{R}$ be continuous and let D be compact. If $\int_a^\infty f(t,x)\,dx$ converges uniformly on D, then $F(t) = \int_a^\infty f(t,x)\,dx$ is continuous on D.

Proof. For each $M > a$, define $F_M(t) = \int_a^M f(t,x)\,dx$. By Theorem 5.29, F_M is continuous on D. By uniform convergence, $F_M \rightrightarrows F$. By Theorem 5.23, F is continuous. □

Theorem 5.33 (Differentiation under the integral sign for improper integrals). Let $f : D \times [a, \infty) \to \mathbb{R}$ and $\partial f/\partial t$ be continuous. If

(i) $F(t) = \int_a^\infty f(t,x)\,dx$ converges on D;

(ii) $\int_a^\infty \partial f/\partial t(t,x)\,dx$ converges uniformly on D,

then F is differentiable on D, and

$$F'(t) = \int_a^\infty \frac{\partial f}{\partial t}(t,x)\,dx.$$

Proof. For each $M > a$, define

$$F_M(t) = \int_a^M f(t,x)\,dx.$$

By Theorem 5.30,

$$F_M'(t) = \int_a^M \partial f/\partial t(t,x)\,dx.$$

By the uniform convergence of $\int_a^\infty \partial f/\partial t(t,x)\,dx$, we have $F_M' \rightrightarrows G$, where $G(t) = \int_a^\infty \partial f/\partial t(t,x)\,dx$. Since $F_M \rightrightarrows F$ (uniform convergence) and $F_M' \rightrightarrows G$, Theorem 5.25 gives $F' = G$. □

Example 5.14 (Classical examples of parameter-dependent integrals). (i) $\int_0^\infty e^{-tx}\,dx = 1/t$ (for $t > 0$). Differentiating with respect to t gives $\int_0^\infty -xe^{-tx}\,dx = -1/t^2$, that is, $\int_0^\infty xe^{-tx}\,dx = 1/t^2$.

(ii) $\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}$. Consider $F(t) = \int_0^\infty e^{-tx}\frac{\sin x}{x}\,dx$ (for $t > 0$). By the Leibniz rule, $F'(t) = -\int_0^\infty e^{-tx}\sin x\,dx = -1/(1+t^2)$. Hence $F(t) = C - \arctan t$. Since $F(\infty) = 0$, we have $C = \pi/2$. Letting $t \to 0^+$ gives $\int_0^\infty \sin x/x\,dx = \pi/2$.

(iii) $\int_0^\infty x^{s-1}e^{-x}\,dx = \Gamma(s)$ (for $s > 0$), that is, the Gamma function.

5.10 Null Sets and Piecewise-smooth Boundaries

5.10.1 Null sets

Definition 5.31 (Null set). Let $E \subseteq \mathbb{R}^n$. E is called a **null set** (or a set of **measure zero**) if for every $\varepsilon > 0$ there exist at most countably many closed rectangles $R_1, R_2, \ldots$ such that

$$E \subseteq \bigcup_{j=1}^\infty R_j, \qquad \sum_{j=1}^\infty |R_j| < \varepsilon.$$

Proposition 5.18 (Basic properties of null sets).

- (i) A subset of a null set is a null set;
- (ii) a countable union of null sets is a null set;
- (iii) a singleton and a countable set are null sets;
- (iv) a k-dimensional hyperplane in $\mathbb{R}^n$ (with $k < n$) is a null set;
- (v) if E is a null set and $f : E \to \mathbb{R}^m$ is a Lipschitz map, then $f(E)$ is a null set.

Proof. (i) This follows directly from the definition.

(ii) Let $E_1, E_2, \dots$ be null sets. Given $\varepsilon > 0$, for each j there exists a family of rectangles $\{R_{j,k}\}_k$ covering E_j with $\sum_k |R_{j,k}| < \varepsilon/2^j$. Then $\{R_{j,k}\}_{j,k}$ covers $\bigcup_j E_j$ and $\sum_{j,k} |R_{j,k}| < \sum_j \varepsilon/2^j = \varepsilon$.

(iii) A singleton $\{x\}$ can be covered by arbitrarily small balls, so it is a null set. A countable set is a countable union of singletons, hence a null set by (ii).

(iv) Let H be the hyperplane $\{x : x_n = 0\}$. For every $\varepsilon > 0$, take the rectangles $R_j = [-j, j]^{n-1} \times [-\delta_j, \delta_j]$, where the δ_j are to be determined. The union $\bigcup_j R_j$ covers H (if $\delta_j \to 0$, covering the whole of H requires $j \to \infty$; note that H is unbounded). More precisely: $H \cap [-j, j]^n \subseteq [-j, j]^{n-1} \times [-\varepsilon/(2^j \cdot 2j), \varepsilon/(2^j \cdot 2j)]$, of volume $(2j)^{n-1} \cdot \varepsilon/(2^j \cdot j) = 2^{n-1}\varepsilon/2^j$. Summing gives $\le 2^{n-1}\varepsilon$. Since ε is arbitrary, H is a null set. A general hyperplane is reduced to this case by an orthogonal transformation.

(v) Let f be Lipschitz with Lipschitz constant L. Given $\varepsilon > 0$, there exists a family of rectangles $\{R_j\}$ covering E with $\sum_j |R_j| < \varepsilon$. The set $f(R_j)$ can be covered by a rectangle $\tilde{R}_j$ whose side lengths are enlarged by a factor of about L, so $|\tilde{R}_j| \le C_n L^n |R_j|$ (with C_n a constant). Then $f(E) \subseteq \bigcup_j \tilde{R}_j$ and $\sum_j |\tilde{R}_j| \le C_n L^n \varepsilon$. Since ε is arbitrary, $f(E)$ is a null set. □

Theorem 5.34 (Lebesgue criterion for Riemann integrability). *Let $f : R \to \mathbb{R}$ be a bounded function on a closed rectangle R. Then f is Riemann integrable on R if and only if the set of discontinuity points of f is a null set.*

Proof. **Necessity.** Suppose that f is integrable. For each k, by Theorem 5.19 there exists a partition $\mathcal{P}_k$ such that $U(f, \mathcal{P}_k) - L(f, \mathcal{P}_k) < 1/k$. Let D_k be the union of all those subrectangles of $\mathcal{P}_k$ whose oscillation satisfies $\sup_{R_j} f - \inf_{R_j} f \ge 1/k$. Then $\sum_{j \in D_k} |R_j| \le k \cdot 1/k = 1$ (a rough estimate). Put $D = \bigcap_k D_k$. Then D is a null set (since $|D_k| \to 0$). Moreover f is continuous outside D (if $x \notin D$, there exists k such that $x \notin D_k$, and then the oscillation of f in some neighbourhood of x is $< 1/k$, so f is continuous at x). Hence the set of discontinuity points $\subseteq D$ is a null set.

Sufficiency. Suppose that the set of discontinuity points D of f is a null set. The function f is bounded; suppose that $|f| \le M$. Given $\varepsilon > 0$, the set D can be covered by a family of rectangles $\{R_j\}$ with $\sum_j |R_j| < \varepsilon/(4M)$. On each R_j the oscillation of f is $\le 2M$. On the compact subset $R \setminus \bigcup_j R_j$ the function f is continuous, hence uniformly continuous. There exists a partition $\mathcal{P}$ such that the oscillation on the subrectangles contained in $R \setminus \bigcup_j R_j$ is $< \varepsilon/(2|R|)$. Combining the two estimates gives

$$U(f, \mathcal{P}) - L(f, \mathcal{P}) \le 2M \sum_j |R_j| + \frac{\varepsilon}{2|R|}|R| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

By Theorem 5.19, f is integrable. ☐

Example 5.15 (Examples of null sets). (i) $\mathbb{Q}\cap[0,1]$ is a null set (it is countable).

(ii) The Cantor ternary set is a null set (the total length $\sum_k 2^k\cdot 3^{-k}=1$ does not converge$\cdots$in fact the Cantor set has measure 0: at step k there remain 2^k intervals of length 3^{-k}, with total length $(2/3)^k\to 0$).

(iii) A line in $\mathbb{R}^2$ is a null set.

(iv) A smooth curve in $\mathbb{R}^2$ is a null set (it is a Lipschitz image).

5.10.2 Piecewise-smooth boundaries

Definition 5.32 (C^k piecewise-smooth boundary). Let $\Omega\subseteq\mathbb{R}^n$ be a bounded open set. The boundary $\partial\Omega$ of Ω is called C^k **piecewise-smooth** if there exist finitely many C^k embedded submanifolds $S_1,\ldots,S_m$ (each of dimension $n-1$) such that

$$\partial\Omega=\bigcup_{i=1}^{m}S_i,$$

and each S_i can be covered by finitely many coordinate charts, and the 'edges' of $\partial\Omega$ form a null set.

Definition 5.33 (Regular domain). Let $\Omega\subseteq\mathbb{R}^n$ be a bounded open set. Ω is called a **regular domain** if:

(i) $\overline{\Omega}$ is compact;

(ii) $\partial\Omega$ is C^1 piecewise-smooth;

(iii) Ω is simply connected;

(iv) the 'edges' of $\partial\Omega$ (where the submanifolds intersect) form a null set, and the values taken by the outward normal vector n on them do not affect the surface integral.

Proposition 5.19 (The outward normal of a regular domain exists almost everywhere). Let $\Omega\subseteq\mathbb{R}^n$ be a regular domain. Then the outward normal vector $\mathsf{n}:\partial\Omega\to S^{n-1}$ exists almost everywhere (that is, everywhere except on a null set), and n is almost everywhere continuous on $\partial\Omega$.

Proof. By Definition 5.32, $\partial\Omega=\bigcup_i S_i$, and each S_i is a C^1 embedded submanifold. On S_i the outward normal vector exists and is continuous (by the C^1 structure). At the intersections of the S_i (the edges) the normal vector may be discontinuous. By Definition 5.33(iv), the edges form a null set. Hence n exists and is continuous on $\partial\Omega$ everywhere except on a null set. ☐

Theorem 5.35 (Regularity prerequisites for Gauss's divergence theorem). Let $\Omega\subseteq\mathbb{R}^3$ be a regular domain and let $\mathsf{F}:\overline{\Omega}\to\mathbb{R}^3$ be a C^1 vector field. Then

$$\int_\Omega(\nabla\cdot\mathsf{F})\,dV=\oint_{\partial\Omega}\mathsf{F}\cdot\mathsf{n}\,dS,$$

where n is the unit outward normal vector of $\partial\Omega$ and dS is the area element.

Proof. The proof is in three steps.

Step 1: rectangular box. Let $\Omega = [a_1, b_1] \times [a_2, b_2] \times [a_3, b_3]$. Taking $\mathsf{F} = (F_1, F_2, F_3)$ as an example, it suffices to prove that $\int_\Omega \partial F_1/\partial x\, dV = \oint_{\partial\Omega} F_1 n_1\, dS$ (the remaining components are analogous). By Fubini's theorem,

$$\int_\Omega \frac{\partial F_1}{\partial x}\, dV = \int_{a_2}^{b_2}\int_{a_3}^{b_3}\left(\int_{a_1}^{b_1}\frac{\partial F_1}{\partial x}\, dx\right) dy\, dz = \int_{a_2}^{b_2}\int_{a_3}^{b_3}[F_1(b_1, y, z) - F_1(a_1, y, z)]\, dy\, dz.$$

The right-hand side is precisely $\oint_{\partial\Omega} F_1 n_1\, dS$ (with $n_1 = +1$ on $x = b_1$, $n_1 = -1$ on $x = a_1$, and $n_1 = 0$ on the remaining faces).

Step 2: z-simple region. Let $\Omega = \{(x, y, z) : (x, y) \in D,\ \varphi_1(x, y) \le z \le \varphi_2(x, y)\}$. For the F_3 component,

$$\int_\Omega \frac{\partial F_3}{\partial z}\, dV = \int_D\left(\int_{\varphi_1}^{\varphi_2}\frac{\partial F_3}{\partial z}\, dz\right) dx\, dy = \int_D [F_3(x, y, \varphi_2) - F_3(x, y, \varphi_1)]\, dx\, dy.$$

The right-hand side is $\oint_{\partial\Omega} F_3 n_3\, dS$. For the F_1, F_2 components, represent Ω as an x-simple or a y-simple region and argue analogously.

Step 3: general regular domain. Use finitely many planes parallel to the coordinate planes to divide Ω into finitely many z-simple subregions. Apply Step 2 to each subregion and sum. The fluxes across the internal common faces cancel in pairs (the normal vectors are opposite), leaving only the flux across the outer boundary. □

Remark 5.2 (Regularity prerequisites for Stokes' theorem and Green's formula). The rigorous statements of Stokes' theorem and Green's formula likewise require such prerequisites as ∂S being piecewise-smooth and S being orientable. The precise formulation of these prerequisites is given in the 'Vector Analysis' chapter. The notions of a null set and of a piecewise-smooth boundary provided in this section are the analytic foundation of those prerequisites: **the edges form a null set** guarantees that the normal vector exists almost everywhere and that the surface integral is unaffected by the values taken on the edges; **piecewise smoothness** guarantees that parametrisations exist locally and that the Jacobian determinant is defined almost everywhere.

5.11 Interfaces with Subsequent Chapters

Remark 5.3 (Interface with the 'Series and Convolution' chapter). The theory of uniform convergence of this chapter (Theorems 5.23, 5.24, 5.25) and the Weierstrass M-test (Theorem 5.26) provide the basis for the rigorous convergence arguments for series of functions, power series, Taylor series and Fourier series in the 'Series and Convolution' chapter. The estimate of the remainder of Taylor's formula in that chapter (with the Lagrange remainder) may invoke the ideas of the multivariable mean-value theorem (Theorem 5.13) and of Taylor's formula (Theorem 5.14) of this chapter.

Remark 5.4 (Interface with the 'Vector Analysis' chapter). This chapter provides the 'Vector Analysis' chapter with:

(i) such topological notions as open sets, closed sets, compact sets, connectedness and simple connectedness (Definitions 5.3, 5.5, 5.6, 5.7);

(ii) regular domains and piecewise-smooth boundaries (Definitions 5.32, 5.33);

(iii) C^k differentiability, the Jacobian matrix and the chain rule (Definition 5.16, Theorem 5.9, Theorem 5.11);

(iv) directional derivatives and the gradient (Definitions 5.13, 5.15);

(v) the implicit-function theorem, the inverse-function theorem and the rank theorem (Theorems 5.16, 5.17, 5.18);

(vi) multiple integrals, Fubini's theorem and change of variables (Theorems 5.21, 5.22);

(vii) line and surface integrals of the first and second kinds (Definitions 5.21, 5.22, 5.24, 5.25);

(viii) null sets and piecewise-smooth boundaries (Definitions 5.31, 5.32).

The rigorous statements of Green's formula, Stokes' theorem and Gauss's divergence theorem in the 'Vector Analysis' chapter may invoke the regularity prerequisites of this chapter, without having to repeat them inside vector analysis itself.

Remark 5.5 (Interface with the 'Probability and Distribution' chapter). If a subsequent revision is planned to include Lebesgue measure and integration, then the 'Probability and Distribution' chapter may invoke the notions of measure, measurable function and integral of this chapter. At present this chapter has established null sets (Definition 5.31) and the Lebesgue criterion for Riemann integrability (Theorem 5.34), providing the 'Probability and Distribution' chapter with preliminary tools concerning null sets and almost everywhere. If a more complete measure theory is required, one may add a section 'Elements of measure and Lebesgue integration' on the basis of this chapter, or make measure theory an independent chapter.

Remark 5.6 (Open problems of this chapter). (i) **Lebesgue measure and integration**: this chapter establishes only null sets and the Lebesgue criterion for Riemann integrability, and does not develop Lebesgue measure and integration. If probability theory, harmonic analysis or L^p spaces require them later, a separate chapter may be set up.

(ii) **Sobolev spaces**: this chapter deals only with C^k maps and does not touch on weak derivatives or Sobolev spaces. These are necessary for the modern theory of partial differential equations, but they lie beyond the scope of this chapter.

(iii) **Analysis on manifolds**: this chapter establishes analysis only on Euclidean space; differentiation and integration on manifolds (de Rham cohomology, the general form of Stokes' theorem) are treated in the 'Manifold and Fibre Bundle' chapter.

(iv) **Finer conditions for uniform convergence and the interchange of limits**: Theorem 5.25 of this chapter requires $f_k' \rightrightarrows g$, which is a fairly strong condition. In the 'Harmonic Analysis' chapter, the termwise differentiation of Fourier series requires finer conditions (such as absolute convergence, the rate of decay of the coefficients, and so on).

5.12 Two Basic Properties of Periodic Functions within the Framework of Real Analysis

The theory of completeness, uniform convergence and parameter-dependent integrals established in this chapter gives the concept of 'period' a rigorous analytic vehicle for the first time. This section presents two basic properties; they use respectively the theory of uniform convergence and the theory of parameter-dependent integrals of this chapter as tools, and they provide the analytic prerequisites for the treatment of Fourier series in the subsequent chapters.

Proposition 5.20 (The integral of a periodic function is independent of the starting point)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be periodic with period $T > 0$ and continuous on $[0, T]$. Then for every $a \in \mathbb{R}$,

$$\int_a^{a+T} f(x)\,dx = \int_0^T f(x)\,dx.$$

That is, **the value of the integral depends only on the length of the period and not on the starting point of the interval of integration**.

Proof. Define the parameter-dependent integral $F(a) := \int_a^{a+T} f(x)\,dx$. By the differentiation rule for parameter-dependent integrals of this chapter (Theorem 5.30),

$$F'(a) = f(a+T) - f(a).$$

By the T-periodicity of f, we have $f(a+T) = f(a)$, so $F'(a) \equiv 0$, that is, F is a constant function; taking $a = 0$ gives $F(a) = F(0) = \int_0^T f$. □

Corollary 5.8 (Equivalence of zero mean and periodicity)**.** Let f be as in Proposition 5.20, and denote the **mean** over one period by $m := \frac{1}{T}\int_0^T f(x)\,dx$. Then

$$g(x) := \int_0^x \big(f(t) - m\big)\,dt$$

is periodic with period T, and g is a primitive of f with period T if and only if $m = 0$.

Proof. By Proposition 5.20 (applied separately to f and to the constant function m),

$$g(x+T) - g(x) = \int_x^{x+T} (f - m) = \int_0^T f - mT = 0,$$

so g is periodic with period T. Also $g' = f - m$, so g is a primitive of f if and only if $m = 0$. □

Remark 5.7 (Interface with the 'Periodicity' theme)*.* The common structure of these two properties is: **periodicity turns a quantity of integration that 'depends on the starting point' into a quantity that 'does not depend on the starting point'**— this is precisely the realisation, at the analytic level of this chapter, of 'translation symmetry ⇒ conservation' in the 'Operations and Periodicity' chapter. This structure recurs repeatedly in the subsequent chapters: the reason why the Fourier coefficients are well defined rests on the independence from the starting point in Proposition 5.20; the relation between the zero-mean condition and the periodicity of primitives in the ‘Harmonic Analysis’ chapter is the continuous version of Corollary 5.8; and the ‘Discrete Sampling and Intercalation of Periodic Signals’ chapter, when treating the discrete case, replaces integrals by sums and the form of the conclusion is unchanged. This chapter thus gains an entry point on the theme of ‘period’: **the analytic consequence of periodicity is an integral invariant.**

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	Translation in Euclidean space and the translation invariance of limits (Definition 5.1); uniform convergence of sequences of functions (Definition 5.27); parameter-dependent integrals (Definition 5.29)
Scaling	Ball neighbourhoods, covers and scales in metric spaces (Definitions 5.2, 5.5); volume scaling under change of variables (Theorem 5.22)
Rotation	Orthogonal transformations of Euclidean space and topological invariance; parametrisation of line and surface integrals (Definitions 5.20, 5.23)
Conservation	Conservation of topological invariants such as compactness, completeness and connectedness (Theorems 5.1, 5.3, 5.4)
Symmetry	Preservation of the symmetry of open sets, closed sets and compact sets under continuous maps (Theorems 5.6, 5.7)
Closure	Closure, closed sets, nested closed intervals and the closure of limits (Definition 5.3, Theorem 5.2); necessary and sufficient condition for Riemann integrability (Theorem 5.19)
Period	The uniform convergence and parameter-dependent integral tools prepared for periodic series, periodic convolution and Fourier analysis in later chapters (Theorems 5.26, 5.30, 5.33)

Exercises

Exercise 1 Starting from Definition 5.2 and Definition 5.3, prove:

(i) the open ball $B(x_0, r)$ is an open set;

(ii) the closed ball $\overline{B}(x_0, r)$ is a closed set;

(iii) the sphere $S(x_0, r)$ is a closed set with empty interior.

Write out the complete ε-δ verification, and explain why the proofs for the open ball and the closed ball cannot replace one another.

Exercise 2 Prove the following parts of Proposition 5.2:

(i) a union of arbitrarily many open sets is open;

(ii) an intersection of finitely many open sets is open;

(iii) an intersection of arbitrarily many closed sets is closed;

(iv) a union of finitely many closed sets is closed.

In each case give a counterexample showing that an intersection of infinitely many open sets need not be open, and a union of infinitely many closed sets need not be closed.

Exercise 3 Prove the 'necessity' part of Theorem 5.1: if $K \subseteq \mathbb{R}^n$ is compact, then K is bounded and closed. Write out the complete proof in the order 'first boundedness, then closedness', and explain the role played by the family of open balls $\{B(0, m)\}_{m \in \mathbb{N}^+}$ in the proof of boundedness.

Exercise 4 Prove Theorem 5.2: let $K \subseteq \mathbb{R}^n$. Then K is sequentially compact if and only if K is bounded and closed. Prove the two implications 'compact $\Rightarrow$ sequentially compact' and 'sequentially compact $\Rightarrow$ compact' separately, and explain why sequential compactness implies closedness.

Exercise 5 Prove the following parts of Proposition 5.4:

(i) path-connected $\Rightarrow$ connected;

(ii) continuous maps preserve connectedness;

(iii) the connected subsets of $\mathbb{R}$ are precisely the intervals;

(iv) if a family of connected sets has a common point, then its union is connected.

Prove (i) by contradiction, prove (ii) using preimages of relatively open sets, and prove (iii) using the order structure of intervals.

Exercise 6 Prove Theorem 5.5 (the Banach contraction-mapping principle): a contraction mapping on a complete metric space has a unique fixed point, and the iteration sequence converges to that fixed point. Write out the complete proof of existence, uniqueness and the error estimate, and use the theorem to prove that the equation $x = \cos x$ has a unique solution on $[0, 1]$.

Exercise 7 Prove Proposition 5.12: if f is differentiable at x_0, then the total differential $L = Df(x_0)$ is unique. Also explain: why does the existence of partial derivatives not guarantee differentiability? Using the function from Proposition 5.11,

$$f(x,y) = \begin{cases} \dfrac{xy}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0) \end{cases}$$

verify that at the origin both partial derivatives exist but the function is discontinuous, hence not differentiable.

Exercise 8 Prove Theorem 5.11 (the chain rule): if g is differentiable at x_0 and f is differentiable at $y_0 = g(x_0)$, then $f \circ g$ is differentiable at x_0, and

$$D(f \circ g)(x_0) = Df(y_0) \circ Dg(x_0).$$

Complete the proof using the definition 'differentiable = linear principal part + $o(\|h\|)$', and write out the chain rule in the form of Jacobian matrices.

Exercise 9 Prove Theorem 5.12 (the theorem on the interchange of mixed partial derivatives): if $f \in C^2(\Omega)$, then

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}.$$

Taking $n = 2$ as an example, write out the complete proof using the second-order difference and two applications of the mean-value theorem, and explain why the C^2 condition cannot be omitted. Verify further: the function

$$f(x,y) = \begin{cases} \dfrac{xy(x^2-y^2)}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0) \end{cases}$$

has both mixed partial derivatives at the origin, but they are unequal.

Exercise 10 Prove Theorem 5.13 (the multivariable mean-value theorem): if $f \in C^1(\Omega)$, $x, y \in \Omega$ and the line segment $[x,y] \subseteq \Omega$, then there exists $\xi \in (x,y)$ such that

$$f(y) - f(x) = \nabla f(\xi) \cdot (y - x).$$

Complete the proof by constructing the one-variable function $g(t) = f(x + t(y-x))$.

Exercise 11 Prove Theorem 5.14 (the second-order Taylor formula): if $f \in C^2(\Omega)$, then

$$f(x+h) = f(x) + \nabla f(x) \cdot h + \frac{1}{2} h^{\mathsf{T}} H_f(\xi) h$$

for some $\xi \in (x, x+h)$. Write out the substitution into the one-variable second-order Taylor formula, and prove the second-order condition for extrema in Corollary 5.5.

Exercise 12 Prove Theorem 5.15 (the method of Lagrange multipliers): let x_0 be a local extremum point of f under the constraints $g_1 = \cdots = g_m = 0$, and suppose that $\nabla g_1(x_0), \ldots, \nabla g_m(x_0)$ are linearly independent. Then there exist $\lambda_1, \ldots, \lambda_m$ such that

$$\nabla f(x_0) = \sum_{i=1}^{m} \lambda_i \nabla g_i(x_0).$$

Use the implicit-function theorem to parametrise the constraint set, and explain why $\nabla f(x_0)$ must belong to the normal space $\operatorname{span}\{\nabla g_1(x_0), \ldots, \nabla g_m(x_0)\}$.

Exercise 13 Prove Theorem 5.16 (the inverse-function theorem): if $f : \Omega \to \mathbb{R}^n$ is a C^1 map and $Df(x_0)$ is invertible, then f has a C^1 inverse near x_0. Write out the complete proof in the five steps 'simplification by translation and a linear transformation, construction of a contraction mapping, proof of continuity, proof of differentiability, proof of C^1', and explain why the contraction constant may be taken to be $1/2$.

Exercise 14 Prove Theorem 5.17 (the implicit-function theorem): if $F(x_0, y_0) = 0$ and $D_y F(x_0, y_0)$ is invertible, then there exists a C^1 map g such that $F(x, g(x)) = 0$. Complete the proof by constructing $\Phi(x, y) = (x, F(x, y))$ and applying the inverse-function theorem, and derive the formula

$$Dg(x) = -[D_y F(x, g(x))]^{-1} D_x F(x, g(x)).$$

Exercise 15 Prove Theorem 5.18 (the rank theorem): if the rank of $Df(x_0)$ is r, then there exist local diffeomorphisms φ, ψ such that

$$(\psi \circ f \circ \varphi^{-1})(x_1, \ldots, x_n) = (x_1, \ldots, x_r, 0, \ldots, 0).$$

Write out the complete procedure for constructing $\Phi(x) = (f^1(x), x_{r+1}, \ldots, x_n)$ and applying the inverse-function theorem and the implicit-function theorem.

Exercise 16 Prove Theorem 5.19: a bounded function $f : R \to \mathbb{R}$ is Riemann integrable if and only if its upper integral equals its lower integral, if and only if for every $\varepsilon > 0$ there exists a partition with $U - L < \varepsilon$. Write out the complete proof that the three conditions are cyclically equivalent, and explain the relation between the Darboux upper and lower sums and Riemann sums.

Exercise 17 Prove Theorem 5.20: a continuous function on a closed rectangle is Riemann integrable. Use uniform continuity to prove that $U(f, \mathcal{P}) - L(f, \mathcal{P}) < \varepsilon$, and explain why compactness cannot be dispensed with in the proof.

Exercise 18 Prove Theorem 5.21 (Fubini's theorem): a continuous function on a closed rectangle can be reduced to an iterated integral, and the order of integration may be interchanged. Taking $n = 2$ as an example, complete the proof using partitions and the one-variable mean-value theorem for integrals, and explain why continuity is an important condition guaranteeing the equality of the iterated integrals.

Exercise 19 Prove Theorem 5.22 (the change-of-variables formula): if $\Phi : \Omega \to \Omega'$ is a C^1 diffeomorphism, then

$$\int_{\Omega'} f(y)\, dy = \int_{\Omega} f(\Phi(x))\, |\det J_\Phi(x)|\, dx.$$

Explain the origin of the local linearisation and of the volume scaling factor, and compute the multiple-integral formula for the polar-coordinate transformation $\Phi(r, \theta) = (r\cos\theta, r\sin\theta)$.

Exercise 20 Compute the following line and surface integrals, writing out the complete procedure:

(i) $\displaystyle\int_\gamma (x^2 + y^2)\, ds$, where γ is the unit circle;

(ii) $\displaystyle\int_\gamma \mathsf{F} \cdot d\mathsf{r}$, where $\mathsf{F}(x, y) = (-y, x)$ and γ is the unit circle traversed once anticlockwise;

(iii) the area of the sphere S^2 and the flux of the radial field $\mathsf{F}(x, y, z) = (x, y, z)$ through S^2.

Explain why the first-kind integral is independent of the parametrisation, whilst the second-kind integral depends on the orientation.

Exercise 21 Prove Theorems 5.23, 5.24, 5.25: uniform convergence preserves continuity, integration and differentiation respectively. Write out the three proofs separately, and explain why, in the proof of 'preservation of differentiation', it is necessary to assume that $f_k' \rightrightarrows g$, whilst the assumption $f_k \rightrightarrows f$ alone is not enough.

Exercise 22 Prove Theorem 5.26 (the Weierstrass M-test): if $\|u_k(x)\| \le M_k$ and $\sum_k M_k < \infty$, then $\sum_k u_k(x)$ converges uniformly. Complete the proof using the uniform Cauchy criterion for convergence, and give an example illustrating the difference between uniform and pointwise convergence.

Exercise 23 Prove Theorem 5.27 (the Cauchy–Hadamard radius-of-convergence formula): the radius of convergence of the power series $\sum_{k=0}^{\infty} a_k (x - x_0)^k$ is

$$R = \frac{1}{\limsup_{k\to\infty} |a_k|^{1/k}}.$$

Use the root test to prove that the series converges absolutely when $|x - x_0| < R$ and diverges when $|x - x_0| > R$, and prove that it converges uniformly on $|x - x_0| \le r < R$.

Exercise 24 Prove Theorem 5.30 (the Leibniz rule): if f and $\partial f/\partial t$ are continuous, then

$$\frac{d}{dt}\int_a^b f(t, x)\, dx = \int_a^b \frac{\partial f}{\partial t}(t, x)\, dx.$$

Complete the proof using the mean-value theorem and uniform continuity, and derive the Leibniz formula for parameter-dependent integrals with variable limits in Corollary 5.7.

Exercise 25 Prove Theorem 5.31 and Theorem 5.33: the M-test for the uniform convergence of improper parameter-dependent integrals, and the legitimacy of differentiation under the integral sign under the condition of uniform convergence. Taking

$$F(t) = \int_0^\infty e^{-tx} \frac{\sin x}{x}\, dx$$

as an example, verify that $F'(t) = -\frac{1}{1+t^2}$, and hence compute $\int_0^\infty \frac{\sin x}{x}\, dx = \frac{\pi}{2}$.

Exercise 26 Prove Proposition 5.18:

(i) a subset of a null set is a null set;

(ii) a countable union of null sets is a null set;

(iii) a countable set is a null set;

(iv) a k-dimensional hyperplane in $\mathbb{R}^n$ is a null set;

(v) Lipschitz maps preserve null sets.

Write out the proofs separately, and explain why the Cantor ternary set is a null set.

Exercise 27 Prove Theorem 5.34 (the Lebesgue criterion for Riemann integrability): a bounded function $f : R \to \mathbb{R}$ is Riemann integrable if and only if its set of discontinuity points is a null set. Write out the construction of the null set $D = \bigcap_k D_k$ in the necessity part, together with the argument in the sufficiency part using a cover by null sets and uniform continuity.

Exercise 28 Explain why, in the proof of Theorem 5.35 (Gauss's divergence theorem), the hypothesis that 'Ω is a regular domain' is needed. Point out:

(i) where the piecewise-smooth boundary is used;

(ii) where the almost everywhere existence of the outward normal vector is used;

(iii) where the fact that the edges form a null set is used;

(iv) where simple connectedness is used.

Also construct a region whose boundary is not smooth, and explain why the direct statement of the divergence theorem fails for it.

Exercise 29 Comprehensive question: combining the Euclidean-space topology, multivariable differential calculus, multiple integrals and line and surface integrals, uniform convergence and parameter-dependent integrals, and null sets and piecewise-smooth boundaries of this chapter, explain how this chapter provides a rigorous analytic foundation for the following in the 'Vector Analysis' chapter:

(i) the differentiability prerequisites for the gradient, the divergence and the curl;

(ii) the regularity prerequisites for Green's formula, Stokes' theorem and Gauss's divergence theorem;

(iii) the simple-connectedness condition in the existence of a potential function;

(iv) the orientation and boundary conditions in flux and circulation integrals

Point out the corresponding definitions, theorems or propositions of this chapter in each case.

Chapter 6

Series and Convolution

Remark 6.1 (Positioning of This Chapter). This chapter builds the complete theory of **series** and **convolution**. A series is the decomposition of a function into countably many elementary units: the Taylor series (the local derivative spectrum), the Laurent series (a complex expansion containing negative powers), the Fourier series (a global frequency decomposition on the harmonic lattice) and the periodic series (a complex series whose components are amplitude, frequency, phase and period, with the period as the primary index). Convolution is a function operator that takes two functions together with a delay and computes the delay product; its frequency-domain forms—**phase convolution**, **amplitude convolution**, **frequency convolution** and **periodic convolution**—lift the similarity/trend hierarchy of the 'Scalars and Vectors' chapter to complex matrices between frequency pairs.

Series are the ultimate destination of similarity and trend at the level of the frequency domain: the Fourier coefficients and the Bohr coefficients are given by integrals (trend), whilst the Taylor coefficients are given by derivatives (direction). All classical integral transforms—the Fourier transform, the Mellin transform, the Laplace transform, the wavelet transform and the Chirp-Z transform—are special cases of the periodic transform under a particular frequency dictionary or kernel.

Finally, this chapter establishes the formal definition of **periodic convolution**: periodic convolution takes amplitude, frequency, phase and period as its four parameters, preserves the interaction of all frequency pairs and all period pairs, and is the summit of the five-stage ladder of convolution theory (convolution → phase convolution/amplitude convolution/frequency convolution → trend convolution → periodic series → periodic convolution).

Guide to the seven themes of this chapter: translation—the local translation expansion of the Taylor series and the global translation period of the Fourier series; scaling—amplitude (scaling) and period (fixed length); rotation—phase (rotation) and frequency (ordering); conservation—Parseval's identity (conservation of energy) and Bohr's generalised Parseval identity; symmetry—the positive–negative power symmetry of the Laurent series and the orthogonal symmetry of the Fourier basis; closure—the uniqueness of the coefficients of the periodic series (Bohr's theorem); period—the four-tuple structure of the periodic series and periodic convolution.

6.1 Basic Concepts of Series

Definition 6.1 (Series and Convergence). Let $\{a_k\}_{k=0}^{\infty}$ be a sequence of numbers. The formal sum $\sum_{k=0}^{\infty} a_k$ is called a **series**, with partial sums $S_n = \sum_{k=0}^{n} a_k$. If $\lim_{n\to\infty} S_n = S$ exists, the series **converges** to S; otherwise it **diverges**. If $\sum_k |a_k|$

converges, the original series is said to **converge absolutely**; one that converges but does not converge absolutely is said to **converge conditionally**.

Definition 6.2 (Series of Functions and Uniform Convergence). *Let $\{u_k(t)\}$ be a sequence of functions on $I \subseteq \mathbb{R}$. The series of functions $\sum_k u_k(t)$ is said to* **converge pointwise** *to f on I if $f(t) = \sum_k u_k(t)$ for every $t \in I$; it is said to* **converge uniformly** *to f if*

$$\lim_{n\to\infty} \sup_{t\in I} \Big| f(t) - \sum_{k=0}^{n} u_k(t) \Big| = 0.$$

Proposition 6.1 (Ratio Test and Root Test). *Let $a_k \neq 0$. If one of the following limits exists:*

$$\lim_{k\to\infty} \left| \frac{a_{k+1}}{a_k} \right| = \rho, \qquad \text{or} \qquad \lim_{k\to\infty} \sqrt[k]{|a_k|} = \rho,$$

then

$$\begin{aligned} \rho < 1 &\implies \sum_k a_k \text{ converges absolutely,} \\ \rho > 1 &\implies \sum_k a_k \text{ diverges,} \\ \rho = 1 &\implies \text{the test is inconclusive.} \end{aligned}$$

Proof. **Ratio test**: let $\rho < 1$. Choose r with $\rho < r < 1$. There exists N such that $|a_{k+1}/a_k| < r$ whenever $k \geq N$. By induction $|a_k| \leq |a_N| r^{k-N}$. By the convergence of the geometric series, $\sum_k |a_k|$ converges. If $\rho > 1$, take $1 < r < \rho$; there exists N such that $|a_{k+1}/a_k| > r$ whenever $k \geq N$, hence $|a_k| \to \infty$ and the series diverges. Counterexamples showing failure when $\rho = 1$: the ratio limits of $\sum 1/k$ (divergent) and $\sum 1/k^2$ (convergent) are both 1.

Root test: similar, using the cases $\sqrt[k]{|a_k|} < r$ or $> r$. ☐

Theorem 6.1 (Weierstrass M-Test). *Let $\sum_k M_k$ be a convergent series of positive terms, and suppose $|u_k(t)| \leq M_k$ for all $t \in I$ and all k. Then $\sum_k u_k(t)$ converges uniformly and absolutely on I.*

Proof. Since $\sum M_k$ converges, for any $\varepsilon > 0$ there exists N such that $\sum_{k>n} M_k < \varepsilon$ $(n \geq N)$. For any $t \in I$ and $m > n \geq N$,

$$\Big| \sum_{k=n+1}^{m} u_k(t) \Big| \leq \sum_{k=n+1}^{m} |u_k(t)| \leq \sum_{k=n+1}^{m} M_k < \varepsilon.$$

Hence the sequence of partial sums is uniformly Cauchy on I and therefore converges uniformly; absolute convergence follows from $\sum_k |u_k(t)| \leq \sum_k M_k < \infty$. ☐

Theorem 6.2 (Term-by-Term Integration under Uniform Convergence). *Let $\sum_k u_k$ converge uniformly to f on $[a, b]$, and suppose each u_k is integrable on $[a, b]$. Then*

$$\int_a^b f(t)\, dt = \sum_{k=0}^{\infty} \int_a^b u_k(t)\, dt.$$

Proof. Write $S_n = \sum_{k=0}^{n} u_k$. By uniform convergence,

$$\Big|\int_a^b f - \int_a^b S_n\Big| \le \int_a^b |f - S_n| \le (b-a) \sup_{t\in[a,b]} |f(t) - S_n(t)| \to 0.$$

Also $\int_a^b S_n = \sum_{k=0}^{n} \int_a^b u_k$ (the integral of a finite sum equals the finite sum of the integrals); taking the limit gives the result. □

Theorem 6.3 (Term-by-Term Differentiation under Uniform Convergence). Suppose each u_k is continuously differentiable on I, $\sum_k u_k$ converges pointwise to f on I, and $\sum_k u_k'$ converges uniformly on I. Then f is differentiable and

$$f'(t) = \sum_{k=0}^{\infty} u_k'(t).$$

Proof. Let $\sum_k u_k' = g$ (uniformly convergent). For any $a, t \in I$, by Theorem 6.2,

$$\int_a^t g(s)\,ds = \sum_{k=0}^{\infty} \int_a^t u_k'(s)\,ds = \sum_{k=0}^{\infty} \big(u_k(t) - u_k(a)\big) = f(t) - f(a).$$

By the fundamental theorem of calculus, $f'(t) = g(t) = \sum_k u_k'(t)$. □

Theorem 6.4 (Cauchy–Hadamard Radius of Convergence Formula). The radius of convergence of the power series $\sum_k a_k (z - z_0)^k$ is

$$R = \frac{1}{\limsup_{k\to\infty} \sqrt[k]{|a_k|}},$$

(with the conventions $1/0 = \infty$, $1/\infty = 0$): it converges absolutely when $|z - z_0| < R$ and diverges when $|z - z_0| > R$.

Proof. Let $L = \limsup_k \sqrt[k]{|a_k|}$. By the root test,

$$\limsup_k \sqrt[k]{|a_k|\,|z - z_0|^k} = |z - z_0| \cdot L.$$

When $|z - z_0| L < 1$, that is $|z - z_0| < 1/L = R$, the root is less than 1 and the series converges absolutely. When $|z - z_0| L > 1$, the root is greater than 1, the general term does not tend to zero, and the series diverges. □

Property 6.1 (Basic Properties of Convergent Series).

- (i) **Linearity**: if $\sum_k a_k = A$ and $\sum_k b_k = B$, then $\sum_k (a_k + b_k) = A + B$;
- (ii) **The general term tends to zero**: if $\sum_k a_k$ converges, then $a_k \to 0$;
- (iii) **Absolute convergence implies convergence**: $\sum_k |a_k|$ converges $\implies$ $\sum_k a_k$ converges.

Proof. (i) By the linearity of the limit of the partial sums. (ii) $a_n = S_n - S_{n-1} \to S - S = 0$. (iii) The sequence of partial sums $\{\sum_{k=0}^{n} a_k\}$ is Cauchy: for $m > n$, $|\sum_{k=n+1}^{m} a_k| \le \sum_{k=n+1}^{m} |a_k|$, and the convergence of $\sum |a_k|$ guarantees that the right-hand side can be made arbitrarily small. □

Example 6.1 (The Geometric Series: Prototype of a Series).

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}, \qquad |r| < 1.$$

The geometric series is the basic building block of power-series expansions: both the positive-power and the negative-power parts of the Taylor and Laurent expansions are derived by way of the geometric series.

Proposition 6.2 (Cauchy convergence criterion). A series of real numbers $\sum_{k=0}^{\infty} a_k$ converges if and only if: for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $m > n \geq N$,

$$\Big|\sum_{k=n+1}^{m} a_k\Big| < \varepsilon.$$

Proof. ($\Rightarrow$) Let $\sum_k a_k = S$, so that the partial sums $S_n \to S$. For $\varepsilon > 0$ choose N such that $|S_n - S| < \varepsilon/2$ whenever $n \geq N$. Then for $m > n \geq N$,

$$\Big|\sum_{k=n+1}^{m} a_k\Big| = |S_m - S_n| \leq |S_m - S| + |S - S_n| < \varepsilon.$$

($\Leftarrow$) The condition shows that the sequence of partial sums $\{S_n\}$ is Cauchy. By completeness of the real numbers (Theorem 2.6 of the 'Number Systems' chapter), $\{S_n\}$ converges, that is, $\sum_k a_k$ converges. ☐

Proposition 6.3 (Comparison test). Let $\sum_k b_k$ be a convergent series of positive terms. If there exists N such that $|a_k| \leq b_k$ for $k \geq N$, then $\sum_k a_k$ converges absolutely. Conversely, if $\sum_k b_k$ diverges and $0 \leq b_k \leq a_k$ for $k \geq N$, then $\sum_k a_k$ diverges.

Proof. Let $\sum_k b_k$ converge. For $m > n \geq N$,

$$\sum_{k=n+1}^{m} |a_k| \leq \sum_{k=n+1}^{m} b_k.$$

By the Cauchy criterion for $\sum_k b_k$ (Proposition 6.2) the right-hand side can be made arbitrarily small, so $\sum_k |a_k|$ converges. The second part follows by reversing the comparison inequality. ☐

Proposition 6.4 (Integral test). Let $f : [1,\infty) \to [0,\infty)$ be monotonically decreasing and continuous. Then the series $\sum_{k=1}^{\infty} f(k)$ and the improper integral $\int_1^{\infty} f(x)\,dx$ converge or diverge together.

Proof. For $k \leq x \leq k+1$, monotonic decrease of f gives $f(k+1) \leq f(x) \leq f(k)$. Integrating,

$$f(k+1) \leq \int_k^{k+1} f(x)\,dx \leq f(k).$$

Summing over $k = 1, \ldots, n-1$:

$$\sum_{k=2}^{n} f(k) \leq \int_1^{n} f(x)\,dx \leq \sum_{k=1}^{n-1} f(k).$$

Hence the partial sums and the integral are bounded or unbounded together, so the series and the integral converge or diverge together. ☐

Property 6.2 (Absolute convergence implies unconditional convergence)**.** If $\sum_k |a_k|$ converges, then for any rearrangement $\sigma : \mathbb{N} \to \mathbb{N}$ (a bijection), $\sum_k a_{\sigma(k)}$ converges and equals $\sum_k a_k$.

Proof. Let $S = \sum_k a_k$. By absolute convergence, given $\varepsilon > 0$ there exists N_1 such that $\sum_{k>N_1} |a_k| < \varepsilon/2$; there also exists N_2 such that $|\sum_{k=0}^{N_2} a_k - S| < \varepsilon/2$. Take $N = \max(N_1, N_2)$. Let M be large enough that $\{0, 1, \ldots, N\} \subseteq \{\sigma(0), \ldots, \sigma(M)\}$. For $M' > M$,

$$\left|\sum_{k=0}^{M'} a_{\sigma(k)} - S\right| \le \left|\sum_{k=0}^{N} a_k - S\right| + \sum_{\substack{k \le M' \\ \sigma(k) > N}} |a_{\sigma(k)}| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence the rearranged partial sums converge to S. ☐

Example 6.2 (Rearrangements of conditionally convergent series can change the sum)**.** The alternating harmonic series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} = \ln 2$ converges conditionally. Riemann's rearrangement theorem asserts that for every $c \in \mathbb{R} \cup \{\pm\infty\}$ there exists a rearrangement σ such that $\sum_k a_{\sigma(k)}$ converges to c. This phenomenon contrasts sharply with absolute convergence (Property 6.2).

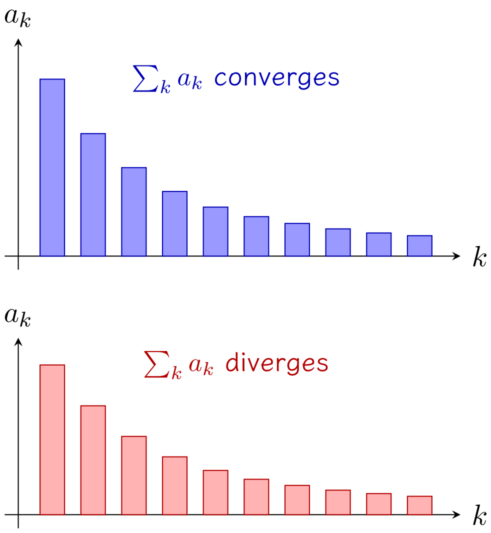


Figure 6.1: Histogram comparison of a convergent (top) and a divergent (bottom) series. When the series converges, the bar heights fall rapidly to zero as k increases; when it diverges, the heights decay too slowly and the partial sums keep growing.

Remark 6.2 (The 'speed hierarchy' of series convergence). The speed of convergence of a series falls into three levels:

(i) **Geometric rate**: $a_k = O(r^k)$ $(0 < r < 1)$, such as $\sum_k r^k$, $\sum_k k^{-k}$.

(ii) **Power-law rate**: $a_k = O(k^{-s})$ $(s > 1)$, such as $\sum_k k^{-2}$, $\sum_k k^{-s}$. This is the typical rate of decay of Fourier coefficients in harmonic analysis (the smoothness of Taylor series in the 'Functions and Limits' chapter).

(iii) **Logarithmic rate:** $a_k = O(1/(k \log^{1+\varepsilon} k))$, such as $\sum_k 1/(k \log^2 k)$.

The rate of convergence is directly related to the smoothness of the function, the decay of the Fourier coefficients and estimates for the Taylor remainder (Theorem 6.5, Property 6.4(v)).

6.2 Taylor Series: Local Derivative Expansion

Definition 6.3 (Taylor Series). Let f be infinitely differentiable in a neighbourhood of x_0. The **Taylor series** of f at x_0 is

$$f(x) \sim \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k.$$

Its partial sum of the first $n+1$ terms is called the **Taylor polynomial** $P_n(x)$ of order n. If the series converges to f in some neighbourhood of x_0, then f is said to be **real analytic** in that neighbourhood.

Theorem 6.5 (Taylor Formula with Lagrange Remainder). Let f be $n+1$ times continuously differentiable on $[x_0, x]$ (or $[x, x_0]$) and $n+1$ times differentiable on the open interval. Then there exists ξ between x_0 and x such that

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0)^{n+1}.$$

Proof. Write the remainder $R_n(x) = f(x) - P_n(x)$. Define the auxiliary function

$$F(t) = f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(t)}{k!}(x - t)^k - R_n(x) \cdot \frac{(x - t)^{n+1}}{(x - x_0)^{n+1}}.$$

Substituting gives $F(x_0) = f(x) - P_n(x) - R_n(x) = 0$, and $F(x) = 0$. By Rolle's theorem there exists ξ with $F'(\xi) = 0$. In the term-by-term differentiation the intermediate terms cancel in pairs, leaving only the first and last:

$$F'(t) = -\frac{f^{(n+1)}(t)}{n!}(x - t)^n + \frac{(n+1)R_n(x)}{(x - x_0)^{n+1}}(x - t)^n.$$

Setting $t = \xi$ gives the remainder formula. □

Corollary 6.1 (Sufficient Condition for a Taylor Series to Converge to the Function). If there exists a constant $M > 0$ such that the derivatives of f of every order on the interval I satisfy $|f^{(k)}(x)| \leq M$ $(x \in I,\ k \geq 0)$, then the Taylor series of f at every point of I converges to f.

Proof. The Lagrange remainder satisfies

$$|R_n(x)| \leq \frac{M}{(n+1)!}|x - x_0|^{n+1} \to 0 \qquad (n \to \infty).$$

□

Example 6.3 (Taylor Series of Elementary Functions).

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}, \qquad \sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}, \qquad \cos x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!},$$

$$\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} x^k}{k} \quad (|x|<1), \qquad (1+x)^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k \quad (|x|<1).$$

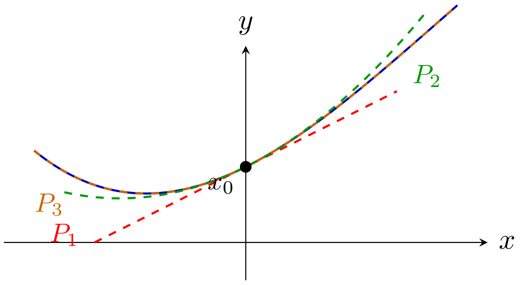
Proposition 6.5 (Uniqueness of the coefficients of a Taylor expansion). Suppose f can be expanded in a power series $f(x) = \sum_{k=0}^{\infty} a_k (x-x_0)^k$ in a neighbourhood of x_0. Then $a_k = f^{(k)}(x_0)/k!$.

Proof. Differentiate the series term by term k times and evaluate at $x = x_0$. Since a power series may be differentiated term by term inside its circle of convergence (Theorem 6.3), we obtain $f^{(k)}(x_0) = k!\, a_k$, that is $a_k = f^{(k)}(x_0)/k!$. ☐

Corollary 6.2 (Approximation order of Taylor polynomials). Let $f \in C^{n+1}(I)$, $x_0 \in I$. Then the Taylor polynomial $P_n(x)$ satisfies

$$f(x) - P_n(x) = O(|x-x_0|^{n+1}) \qquad (x \to x_0).$$

Proof. By the Lagrange remainder formula (Theorem 6.5), $|R_n(x)| \le \frac{M_{n+1}}{(n+1)!}|x - x_0|^{n+1}$, where $M_{n+1} = \sup_{|t-x_0|\le|x-x_0|} |f^{(n+1)}(t)| < \infty$ (since $f^{(n+1)}$ is continuous). ☐



As the order n increases, P_n approximates f near x_0

Figure 6.2: Successive approximation by Taylor polynomials. P_1 (linear) approximates f near x_0; P_2 (quadratic) adds a curvature correction; P_3 (cubic) corrects it further. The higher the order, the wider the interval of approximation.

Remark 6.3 (The duality between Taylor and Fourier expansions). The Taylor expansion and the Fourier expansion form a **local–global** duality:

	Taylor expansion	Fourier expansion
Centre of expansion	a single point x_0	the periodic interval $[0,T]$
Source of coefficients	the derivative $f^{(k)}(x_0)$	the integral $c_n = \frac{1}{T}\int_0^T f(t)e^{-i\omega_n t}dt$
Type of information	direction (local derivatives)	trend (global integrals)
Convergence condition	real analytic	L^2 integrable + Dirichlet conditions
Basis functions	$(x-x_0)^k$ (polynomials)	$e^{i\omega_n t}$ (complex exponentials)

This duality is the manifestation, at the level of function expansions, of the direction–trend duality of the 'Scalars and Vectors' chapter (Remark 6.5).

6.3 Laurent Series: Complex Expansion with Negative Powers

Definition 6.4 (Laurent Series). Let $z_0 \in \mathbb{C}$. A series of the form

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

is called a **Laurent series** centred at z_0. The positive-power part converges in the disc $|z - z_0| < R$, whilst the negative-power part, after the change of variable $w = 1/(z - z_0)$, converges on $|z - z_0| > r$; the common domain of convergence is the annulus $r < |z - z_0| < R$.

Lemma 6.1 (Uniqueness of Laurent Coefficients). If $\sum_n a_n (z - z_0)^n = \sum_n b_n (z - z_0)^n$ on some annulus, then $a_n = b_n$ for all $n \in \mathbb{Z}$.

Proof. Let $d_n = a_n - b_n$. For any $m \in \mathbb{Z}$, multiply the identity by $(z - z_0)^{-m-1}$ and integrate along a circle around z_0. From

$$\oint_\gamma (z - z_0)^{n-m-1}\, dz = 2\pi i\, \delta_{nm},$$

we obtain $d_m \cdot 2\pi i = 0$, hence $d_m = 0$. □

Theorem 6.6 (Laurent Expansion Theorem). Let f be analytic in the annulus $r < |z - z_0| < R$. Then f can be expanded uniquely as a Laurent series in that annulus, with coefficients

$$a_n = \frac{1}{2\pi i} \oint_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta, \qquad n \in \mathbb{Z},$$

where γ is any positively oriented closed curve in the annulus surrounding z_0.

Proof. By the Cauchy integral formula, for any point z in the annulus and two concentric circles γ_1, γ_2,

$$f(z) = \frac{1}{2\pi i} \oint_{\gamma_1} \frac{f(\zeta)}{\zeta - z}\, d\zeta - \frac{1}{2\pi i} \oint_{\gamma_2} \frac{f(\zeta)}{\zeta - z}\, d\zeta.$$

Expanding on γ_1 by the geometric series gives the positive-power part; a similar expansion on γ_2 gives the negative-power part. Uniqueness is given by Lemma 6.1. □

Proposition 6.6 (Laurent Expansion of Rotationally Periodic Functions). If f satisfies the rotational periodicity $f(e^{i\theta} z) = f(z)$ (group with periodic orbits C_m, that is $\theta = 2\pi/m$), then its Laurent series centred at 0 contains only terms with $n \equiv 0 \pmod{m}$.

Proof. Substituting $z \mapsto e^{i\theta} z$ and using uniqueness of coefficients gives $a_n(e^{in\theta} - 1) = 0$. When $e^{i\theta}$ is a primitive m-th root of unity, $e^{in\theta} = 1 \iff m \mid n$. □

Example 6.4 (Laurent Expansion of $e^{1/z}$).

$$e^{1/z} = \sum_{n=0}^{\infty} \frac{z^{-n}}{n!}, \qquad |z| > 0.$$

Proposition 6.7 (Classification of singularities and the principal part of a Laurent series). *Let f be analytic on $0 < |z - z_0| < R$, with Laurent expansion $f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$. Calling*

$$\text{principal part} = \sum_{n=1}^{\infty} a_{-n}(z - z_0)^{-n}$$

the principal part of the Laurent expansion, we have:

(i) *removable singularity $\iff$ the principal part is zero;*

(ii) *pole of order m $\iff$ the principal part is a non-zero finite sum whose highest negative power is m;*

(iii) *essential singularity $\iff$ the principal part has infinitely many terms.*

Proof. Each item follows directly from the definition of the negative-power part of the Laurent expansion. ☐

Example 6.5 (Laurent expansions of the three types of singularity compared). (i) Removable singularity: $f(z) = \frac{\sin z}{z}$, whose Laurent expansion at $z_0 = 0$ is $1 - z^2/6 + \cdots$, with principal part zero.

(ii) Pole: $f(z) = \frac{1}{z^2(z-1)}$, whose principal part at $z_0 = 0$ is $-1 - z - z^2$, so it is a pole of order 2.

(iii) Essential singularity: $f(z) = e^{1/z}$, whose principal part at $z_0 = 0$ is $\sum_{n=1}^{\infty} z^{-n}/n!$, with infinitely many terms.

Theorem 6.7 (Duality between the residue theorem and Fourier coefficients). *Let f be analytic in the annulus $r < |z - z_0| < R$, with Laurent expansion $f(z) = \sum_{n\in\mathbb{Z}} a_n(z - z_0)^n$. Then a_n is represented by the residue as*

$$a_n = \text{Res}\left(\frac{f(z)}{(z - z_0)^{n+1}}, z_0\right).$$

In particular, $a_{-1} = \text{Res}(f, z_0)$ is precisely the coefficient of $(z-z_0)^{-1}$ in the Laurent expansion.

Proof. It follows by directly comparing the Laurent coefficient formula (Theorem 6.6) with the definition of the residue (Definition 9.26 of the 'Differential Equations and Analysis' chapter). ☐

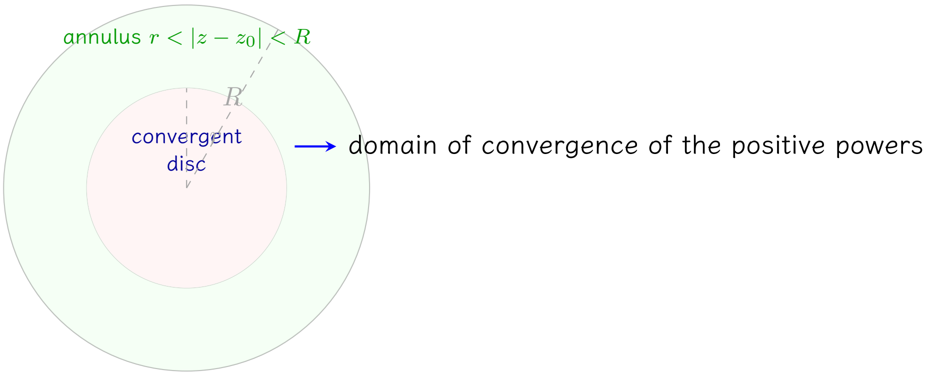

annulus $r < |z - z_0| < R$
R
convergent disc
domain of convergence of the positive powers

Figure 6.3: The annulus of convergence of a Laurent series. The positive-power part converges in the disc centred at z_0, whilst the negative-power part converges in the region near the point at infinity; their common domain of convergence is the annulus $r < |z - z_0| < R$.

6.4 Fourier Series: Global Decomposition on the Harmonic Lattice

Definition 6.5 (Fourier Series (Complex and Real Forms)). Let f have period T and be integrable on $[0, T]$. The **Fourier series** of f (complex form) is

$$f(t) \sim \sum_{n=-\infty}^{\infty} c_n e^{i2\pi nt/T}, \qquad c_n = \frac{1}{T}\int_0^T f(t) e^{-i2\pi nt/T}\, dt.$$

The real form is

$$f(t) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \Big(a_n \cos\frac{2\pi nt}{T} + b_n \sin\frac{2\pi nt}{T} \Big),$$

where $a_n = c_n + c_{-n}$, $b_n = i(c_n - c_{-n})$.

Proposition 6.8 (Orthogonality of the Fourier Basis). The system of functions $\{e^{i2\pi nt/T}\}_{n\in\mathbb{Z}}$ is orthogonal in $L^2([0, T])$:

$$\frac{1}{T}\int_0^T e^{i2\pi(n-m)t/T}\, dt = \delta_{nm}.$$

Proof. When $n = m$ the integrand is 1; when $n \neq m$ the integral becomes $\frac{T}{2\pi i(n-m)}(e^{i2\pi(n-m)} - 1) = 0$. ☐

Theorem 6.8 (Parseval's Identity (Fourier Series)). Let $f \in L^2([0, T])$ with Fourier coefficients c_n. Then

$$\frac{1}{T}\int_0^T |f(t)|^2\, dt = \sum_{n=-\infty}^{\infty} |c_n|^2.$$

Proof. For trigonometric polynomials it holds directly by orthogonality (Proposition 6.8); for general L^2 functions it follows from the completeness of the Fourier basis in L^2 by approximation. ☐

Example 6.6 (Fourier Series of a Square Wave). Let f have period T, take the value 1 on $[0, T/2)$ and the value -1 on $[T/2, T)$. Its Fourier series is

$$f(t) \sim \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{1}{2k+1} \sin \frac{2\pi(2k+1)t}{T}.$$

Theorem 6.9 (Dirichlet convergence theorem). *Let f have period T and be piecewise C^1 on $[0, T]$. Then the Fourier series of f converges at every point t to*

$$\frac{f(t^+) + f(t^-)}{2},$$

where $f(t^+)$ and $f(t^-)$ are the right-hand and left-hand limits of f at t respectively. In particular, at every point where f is continuous, the Fourier series converges to $f(t)$.

Sketch of proof. Write the Fourier partial sum of f as a convolution with the Dirichlet kernel D_N:

$$S_N f(t) = \frac{1}{T} \int_0^T f(t-\tau) D_N(\tau)\, d\tau.$$

Using the properties of the Dirichlet kernel: $\frac{1}{T}\int_0^T D_N(\tau)\, d\tau = 1$, and D_N oscillates with alternating signs away from 0, so that the contributions cancel (the Riemann–Lebesgue lemma). The piecewise C^1 condition guarantees that the local one-sided behaviour of f near t can be controlled by a first-order Taylor expansion, so that $S_N f(t) \to [f(t^+) + f(t^-)]/2$. For a rigorous proof see Zygmund, Trigonometric Series, 3rd ed., Cambridge Univ. Press, 2002, §II.4. ☐

Corollary 6.3 (Pointwise convergence at points of continuity). *If f is continuous at t, then $f(t^+) = f(t^-) = f(t)$ and the Fourier series converges to $f(t)$ at t.*

Example 6.7 (The Gibbs phenomenon). The Fourier partial sums $S_N f$ of the square wave $f(t) = \text{sgn}(\sin t)$ exhibit an **overshoot** of about 9% near the jump point $t = 0$, and the magnitude of the overshoot does not decay as $N \to \infty$. This phenomenon is called the **Gibbs phenomenon**; it shows that uniform convergence of the Fourier series fails near a jump point (pointwise convergence still holds, by Theorem 6.9). In digital signal processing the Gibbs phenomenon is equivalent to the ringing effect of an 'ideal low-pass filter'.

Theorem 6.10 (The Fejér kernel and uniform convergence). *Let f have period T and be continuous. Define the Fejér means*

$$\sigma_N f(t) = \frac{1}{N+1} \sum_{k=0}^{N} S_k f(t),$$

where $S_k f$ is the k-th Fourier partial sum. Then $\sigma_N f$ converges uniformly to f on $\mathbb{R}$.

Sketch of proof. The Fejér mean can be written as a convolution with the Fejér kernel $\sigma_N f(t) = \frac{1}{T}\int_0^T f(t-\tau) F_N(\tau)\, d\tau$, where

$$F_N(\tau) = \frac{1}{N+1} \left(\frac{\sin((N+1)\tau/2)}{\sin(\tau/2)} \right)^2.$$

The Fejér kernel is a **good kernel**: non-negative, of integral 1, and tending to zero uniformly away from $\tau \neq 0$. Hence for a continuous periodic function f, the uniform approximation property of good kernels gives $\sigma_N f \to f$ uniformly. For a rigorous proof see Zygmund, Trigonometric Series, 3rd ed., Cambridge Univ. Press, 2002, §III.1. □

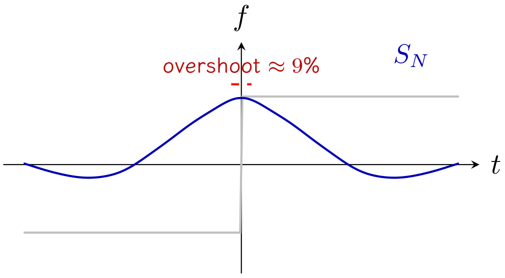


Figure 6.4: The Gibbs phenomenon. The Fourier partial sum S_N (blue) of the square wave (grey) exhibits an overshoot (red dashed line) near the jump point $t = 0$; the magnitude of the overshoot is about 9% of the jump and does not disappear as N increases.

Property 6.3 (Preservation under integration for Fourier series)**.** Let f have period T, be integrable on $[0, T]$ and have zero mean (that is, $c_0 = 0$). Then the indefinite integral $F(t) = \int_0^t f(\tau)\,d\tau$ has period T, and its Fourier coefficients are $C_n = c_n/(i\omega_n)$ $(n \neq 0)$.

Proof. From $c_0 = 0$ we obtain $F(T) = \int_0^T f = Tc_0 = 0$, so F has period T. The Fourier coefficients of F are

$$C_n = \frac{1}{T}\int_0^T F(t)e^{-i\omega_n t}\,dt = \frac{1}{T}\cdot\frac{1}{i\omega_n}\left[F(t)e^{-i\omega_n t}\Big|_0^T + \int_0^T f(t)e^{-i\omega_n t}\,dt\right] = \frac{c_n}{i\omega_n}.$$

(Integration by parts; the boundary terms vanish by periodicity.) □

Remark 6.4 (Three modes of convergence of Fourier series)*.* The Fourier series has three modes of convergence, each corresponding to different conditions and conclusions:

Mode of convergence	Condition	Conclusion
Pointwise convergence	f piecewise C^1	$S_N f(t) \to [f(t^+) + f(t^-)]/2$ (Theorem 6.9)
Uniform convergence	f continuous + $\sum \lvert c_n\rvert < \infty$	$S_N f \to f$ uniformly (Fejér means, Theorem 6.10)
L^2 convergence	$f \in L^2[0, T]$	$\lVert S_N f - f\rVert_2 \to 0$ (Parseval + completeness)

The three modes do not imply one another: uniform convergence ⇒ pointwise convergence, but not conversely (the Gibbs phenomenon); L^2 convergence and pointwise convergence do not imply one another.

Remark 6.5 (Direction–Trend Duality of Fourier Coefficients). *The Fourier coefficients c_n are given by integrals (trend information) and the Taylor coefficients by derivatives (direction information); the two form a direction–trend duality. The problems of pointwise and uniform convergence of Fourier series can be handled by tools such as the Dirichlet kernel, the Gibbs phenomenon and Fejér-kernel smoothing; we shall not develop them in this chapter.*

6.5 He Tu Complex Functions: The $\mathbb{Z}_2 \times \mathbb{Z}_5$ Decomposition of Period 10

This section establishes the **He Tu complex function***—a complex function of period 10. Its mathematical core is this: in the Fourier decomposition of period 10 there automatically appears the algebraic grading $\mathbb{Z}_{10} \cong \mathbb{Z}_2 \times \mathbb{Z}_5$, in which the $\mathbb{Z}_5$ component corresponds to a fivefold cycle and the $\mathbb{Z}_2$ component to a twofold alternation. This grading is a purely algebraic fact about the Fourier expansion of discrete periodic functions and does not depend on any cultural interpretation.*

Definition 6.6 (The $\mathbb{Z}_5$ Projection on $\mathbb{Z}_{10}$)**.** Define the group homomorphism $\varphi : \mathbb{Z}_{10} \to \mathbb{Z}_5$ by

$$\varphi(n) := n \bmod 5, \qquad n \in \mathbb{Z}_{10} = \{0, 1, \ldots, 9\},$$

with the convention that the remainder 0 corresponds to the label 'earth'. Specifically:

$\varphi(n)$	0	1	2	3	4
Label	earth	water	fire	wood	metal
Numbers	$5, 10$	$1, 6$	$2, 7$	$3, 8$	$4, 9$

There is also a $\mathbb{Z}_2$ component, given by the parity of n. The kernel of φ is

$$\ker \varphi = \{0, 5\} \cong \mathbb{Z}_2.$$

Definition 6.7 (He Tu Complex Function)**.** Let $f : \mathbb{C} \to \mathbb{C}$ satisfy the periodicity condition

$$f(z + 10) = f(z), \qquad \forall z \in \mathbb{C}.$$

Then f is called a **He Tu complex function**. Its Fourier expansion is

$$f(z) = \sum_{n \in \mathbb{Z}} c_n \, e^{2\pi i n z/10}, \qquad c_n = \frac{1}{10} \int_0^{10} f(t) \, e^{-2\pi i n t/10} \, dt.$$

Theorem 6.11 (He Tu Fourier Decomposition—$\mathbb{Z}_2 \times \mathbb{Z}_5$ Grading)**.** *Let f be a He Tu complex function whose Fourier expansion converges uniformly on $\mathbb{C}$. Then f decomposes uniquely as*

$$\boxed{f(z) = \sum_{k=0}^{4} \Big(c_k + c_{k+5} \, e^{\pi i z} \Big) \, e^{2\pi i k z/10} \,,}$$

where c_k are the Fourier coefficients and $e^{\pi i z}$ is the $\mathbb{Z}_2$ factor of period 2. This decomposition is the explicit form of $\mathbb{Z}_{10} \cong \mathbb{Z}_2 \times \mathbb{Z}_5$ in the Fourier expansion.

Proof. Using $\mathbb{Z}_{10} \cong \mathbb{Z}_2 \times \mathbb{Z}_5$, write the integer index $n \in \mathbb{Z}$ uniquely as

$$n = 5\varepsilon + k, \qquad \varepsilon \in \{0,1\}, \quad k \in \{0,1,2,3,4\}.$$

Substituting into the Fourier basis functions:

$$e^{2\pi i n z/10} = e^{2\pi i(5\varepsilon+k)z/10} = e^{\pi i \varepsilon z} \cdot e^{2\pi i k z/10}.$$

Decompose the sum over n according to ε:

$$f(z) = \sum_{\varepsilon\in\{0,1\}} \sum_{k=0}^{4} c_{5\varepsilon+k}\, e^{\pi i \varepsilon z}\, e^{2\pi i k z/10}.$$

Extracting the $\varepsilon = 0$ and $\varepsilon = 1$ terms and collecting like terms $e^{2\pi i k z/10}$:

$$f(z) = \sum_{k=0}^{4} \Big(c_k + c_{k+5}\, e^{\pi i z}\Big)\, e^{2\pi i k z/10}.$$

Uniqueness is guaranteed by the uniqueness of the Fourier coefficients (Bohr means) and the bijection $n \mapsto (5\varepsilon + k)$. □

Definition 6.8 (The $\mathbb{Z}_5$ Components). *In the above decomposition, the function*

$$f_k(z) := \Big(c_k + c_{k+5}\, e^{\pi i z}\Big)\, e^{2\pi i k z/10}, \qquad k = 0,1,2,3,4$$

*is called the k-***th** $\mathbb{Z}_5$ **component** *of f. According to the labels of Definition 6.6, the five components correspond in order to: earth, water, fire, wood and metal. Each component consists of two parts: $c_k\, e^{2\pi i k z/10}$ (the $\varepsilon = 0$ component) and $c_{k+5}\, e^{\pi i z} e^{2\pi i k z/10}$ (the $\varepsilon = 1$ component).*

Proposition 6.9 (Period Structure of the $\mathbb{Z}_5$ Components). *The fundamental periods of the five $\mathbb{Z}_5$ components f_0, f_1, f_2, f_3, f_4 are respectively:*

k	0	1	2	3	4
Label	earth	water	fire	wood	metal
Fundamental period	1	10	5	10	5

That is: f_0 contains only a constant term; f_1, f_3 are components of period 10; f_2, f_4 are components of period 5.

Proof. The fundamental frequency of the component f_k is $e^{2\pi i k z/10}$, whose fundamental period is $10/\gcd(k,10)$. Substituting one by one:

- $k = 0$: $10/\gcd(0,10) = 1$ (with the convention $\gcd(0,10) = 10$), so the fundamental-frequency part of f_0 is constant.
- $k = 1$: $10/\gcd(1,10) = 10$.
- $k = 2$: $10/\gcd(2,10) = 5$.
- $k = 3$: $10/\gcd(3,10) = 10$.
- $k = 4$: $10/\gcd(4,10) = 5$.

The $\mathbb{Z}_2$ factor $e^{\pi iz}$ has period 2 and does not change the above period structure (because $2 \mid 10$). ☐

Proposition 6.10 (Half-Period Translation of He Tu Complex Functions)**.** Let f be a He Tu complex function and $f = \sum_{k=0}^{4} f_k$ its $\mathbb{Z}_5$ decomposition. Then the half-period translation $z \mapsto z+5$ gives

$$f(z+5) = \sum_{k=0}^{4} \Big((-1)^k c_k + (-1)^k c_{k+5}\, e^{\pi iz} \Big)\, e^{2\pi ikz/10},$$

that is, *each* $\mathbb{Z}_5$ *component acquires a factor* $(-1)^k$*: even components* ($k = 0, 2, 4$) *are unchanged, whilst odd components* ($k = 1, 3$) *change sign as a whole.*

Proof. By Theorem 6.11, $f(z+5) = \sum_k (c_k + c_{k+5} e^{\pi i(z+5)}) e^{2\pi ik(z+5)/10}$. Computing separately:

$$e^{\pi i(z+5)} = e^{\pi iz} e^{5\pi i} = -e^{\pi iz}, \qquad e^{2\pi ik(z+5)/10} = e^{2\pi ikz/10}\, e^{\pi ik} = (-1)^k\, e^{2\pi ikz/10}.$$

Substituting gives the result. ☐

Definition 6.9 ($\mathbb{Z}_{10}$ Phase Operators)**.** The $\mathbb{Z}_{10}$ **phase operator** is defined by

$$\mathcal{O}_n^{(10)} := e^{2\pi in/10}, \qquad n = 0, 1, \dots, 9.$$

Under the decomposition $\mathbb{Z}_{10} \cong \mathbb{Z}_2 \times \mathbb{Z}_5$, the phase operator can be written as

$$\mathcal{O}_{5\varepsilon+k}^{(10)} = (-1)^{\varepsilon}\, e^{2\pi ik/10}, \qquad \varepsilon \in \{0, 1\},\ k \in \{0, 1, 2, 3, 4\},$$

that is, *each phase operator is the product of the* $\mathbb{Z}_2$ *sign* $(-1)^{\varepsilon}$ *and the* $\mathbb{Z}_5$ *phase* $e^{2\pi ik/10}$.

Proposition 6.11 (Direct Product Decomposition of the $\mathbb{Z}_{10}$ Phase Operator Group)**.** *The group of* $\mathbb{Z}_{10}$ *phase operators* $\{\mathcal{O}_n^{(10)}\}_{n=0}^{9}$ *is isomorphic to*

$$C_{10} \cong C_2 \times C_5,$$

where $C_2 = \{+1, -1\}$ *corresponds to the* $\mathbb{Z}_2$ *component and* $C_5 = \{e^{2\pi ik/5}\}_{k=0}^{4}$ *to the* $\mathbb{Z}_5$ *component. The generator* $\mathcal{O}_1^{(10)} = e^{\pi i/5}$ *corresponds under the decomposition to* $(1, e^{2\pi i/10})$ *and has order* $10 = \mathrm{lcm}(2, 5)$.

Proof. The map $\mathcal{O}_{5\varepsilon+k}^{(10)} \mapsto ((-1)^{\varepsilon}, e^{2\pi ik/10})$ is a group isomorphism: from

$$\mathcal{O}_{5\varepsilon_1+k_1}^{(10)} \cdot \mathcal{O}_{5\varepsilon_2+k_2}^{(10)} = \mathcal{O}_{5(\varepsilon_1+\varepsilon_2)+(k_1+k_2)}^{(10)},$$

the right-hand side corresponds to $((-1)^{\varepsilon_1+\varepsilon_2}, e^{2\pi i(k_1+k_2)/10})$, exactly the product of the two components. That the isomorphism is bijective is guaranteed by $\mathbb{Z}_{10} \cong \mathbb{Z}_2 \times \mathbb{Z}_5$. The order of the generator $= \mathrm{lcm}(2, 5) = 10$. ☐

Theorem 6.12 (DFT Diagonalisation of He Tu Circulant Matrices)**.** *Let* **C** *be a* 10×10 *circulant matrix whose first row has generators* $(h_0, h_1, \dots, h_9)$*, with* h_n *assigned*

$\mathbb{Z}_2 \times \mathbb{Z}_5$ labels according to Definition 6.6. Then C is unitarily diagonalised by the DFT matrix F_{10}:

$$\mathsf{C} = \mathsf{F}_{10}^* \,\mathsf{diag}(\hat{h}_0, \hat{h}_1, \ldots, \hat{h}_9)\, \mathsf{F}_{10}, \qquad \hat{h}_n = \sum_{j=0}^{9} h_j\, e^{-2\pi i j n/10}.$$

Furthermore, the eigenvalues $\hat{h}_n$ are graded according to $\mathbb{Z}_{10} \cong \mathbb{Z}_2 \times \mathbb{Z}_5$: the five eigenvalues $(\hat{h}_0, \hat{h}_1, \hat{h}_2, \hat{h}_3, \hat{h}_4)$ corresponding to $\varepsilon = 0$ belong to the C_5 component, and the five eigenvalues $(\hat{h}_5, \hat{h}_6, \hat{h}_7, \hat{h}_8, \hat{h}_9)$ corresponding to $\varepsilon = 1$ belong to the $\mathbb{Z}_2$ translation component.

Proof. By the DFT diagonalisation of circulant matrices (a standard result), $\mathsf{C} = \mathsf{F}_{10}^* \,\mathsf{diag}(\hat{h})\mathsf{F}_{10}$. The eigenvalues $\hat{h}_n$ split into two groups according to $n = 5\varepsilon + k$: $n \in \{0,1,2,3,4\}$ corresponds to $\varepsilon = 0$, and $n \in \{5,6,7,8,9\}$ to $\varepsilon = 1$. Since $\hat{h}_{n+5} = \sum_j h_j e^{-2\pi i j(n+5)/10} = \sum_j h_j (-1)^j e^{-2\pi i j n/10}$, that is, $\hat{h}_{n+5}$ is determined by the parity weighting of the h_j and is independent of $\hat{h}_n$. ☐

Example 6.8 (Concrete Decomposition of a He Tu Complex Function). Take $f(z) = 3+2e^{\pi i z/5}+e^{\pi i z}+5e^{3\pi i z/5}+4e^{2\pi i z}$. This function has period 10; write it in the standard form of Theorem 6.11.

First arrange the Fourier coefficients in order of n: $c_0 = 3$, $c_1 = 2$, $c_5 = 1$, $c_3 = 5$, $c_4 = 4$, with the remaining $c_n = 0$. Decompose according to $n = 5\varepsilon + k$:

$$f(z) = \underbrace{3}_{n=0} + \underbrace{2e^{\pi i z/5}}_{n=1} + \underbrace{5e^{3\pi i z/5}}_{n=3} + \underbrace{4e^{2\pi i z}}_{n=4} + \underbrace{e^{\pi i z}}_{n=5}.$$

The (ε, k) label of each term and its $\mathbb{Z}_2 \times \mathbb{Z}_5$ membership are as in the following table:

n	0	1	3	4	5
(ε, k)	$(0,0)$	$(0,1)$	$(0,3)$	$(0,4)$	$(1,0)$
Label	earth	water	wood	metal	earth
Component	C_5	C_5	C_5	C_5	$\mathbb{Z}_2$

Collecting like $\mathbb{Z}_5$ components:

$$f_0 = 3 + e^{\pi i z}, \quad f_1 = 2e^{\pi i z/5}, \quad f_2 = 0, \quad f_3 = 5e^{3\pi i z/5}, \quad f_4 = 4e^{2\pi i z}.$$

By Proposition 6.9, f_0 contains a constant term (period 1); f_1, f_3 have period 10; f_4 has period 5.

Remark 6.6 (Algebraic Meaning of the He Tu Structure). The mathematical content of the He Tu complex function lies in the Fourier grading $\mathbb{Z}_{10} \cong \mathbb{Z}_2 \times \mathbb{Z}_5$:

- the $\mathbb{Z}_5$ component corresponds to the fivefold cycle of the Fourier mode $k \in \{0,1,2,3,4\}$;
- the $\mathbb{Z}_2$ component corresponds to the parity of the Fourier mode n, that is, a twofold alternation;
- the half-period translation $z \mapsto z+5$ changes the sign of the components with odd k as a whole (Proposition 6.10).

The He Tu map $\varphi : \mathbb{Z}_{10} \to \mathbb{Z}_5$ is a group homomorphism whose kernel is $\{0,5\} \cong \mathbb{Z}_2$. This gives the grading a rigorous algebraic realisation in the Fourier expansion—not an imposed encoding, but an algebraic structure that appears automatically in the Fourier decomposition of period 10.

6.6 Amplitude Series

This section discusses the form of series with zero phase and period 2π. It is the amplitude spectrum obtained from the Fourier series after discarding all phase information, and also the most direct carrier of Parseval energy conservation.

Definition 6.10 (Amplitude Series with Zero Phase and Period 2π)**.** Let $A_n \geq 0$ ($n \in \mathbb{Z}$) be a sequence of amplitudes. The formal series

$$\mathcal{A}_f(t) := \sum_{n=-\infty}^{\infty} A_n e^{int}$$

is called the **amplitude series with zero phase and period** 2π. Its real form is

$$\mathcal{A}_f^{\mathbb{R}}(t) := \sum_{n=-\infty}^{\infty} A_n \cos(nt),$$

in which case $A_{-n} = A_n$ is required. If the generating-function form needs to be emphasised, we also write

$$\mathfrak{A}_f(s) := \sum_{n=-\infty}^{\infty} A_n s^n.$$

The amplitude series discards all phase information and retains only the amplitude distribution of each harmonic. When $\sum_n A_n < \infty$, $\mathcal{A}_f$ converges uniformly and absolutely; in general it is understood as a formal series or in the distributional sense.

Proposition 6.12 (Bohr Mean Coefficient Formula for Amplitude Series)**.** If the amplitude series $\mathcal{A}_f(t) = \sum_n A_n e^{int}$ converges uniformly and absolutely, then for every $n \in \mathbb{Z}$,

$$A_n = M\{\mathcal{A}_f(t)e^{-int}\} = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \mathcal{A}_f(t)e^{-int}\,dt.$$

Proof. By uniform absolute convergence, the Bohr mean may be interchanged with the summation:

$$M\{\mathcal{A}_f(t)e^{-int}\} = \sum_{m\in\mathbb{Z}} A_m M\{e^{i(m-n)t}\}.$$

By Lemma 6.2,

$$M\{e^{i(m-n)t}\} = \delta_{mn},$$

so only the term $m = n$ survives, giving A_n. □

Lemma 6.2 (Bohr Mean Orthogonality of the Exponential Basis)**.** For any $m, n \in \mathbb{Z}$,

$$M\{e^{i(m-n)t}\} = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} e^{i(m-n)t}\,dt = \delta_{mn}.$$

Proof. When $m = n$ the integrand is 1 and the mean is 1. When $m \neq n$,

$$\frac{1}{2T} \int_{-T}^{T} e^{i(m-n)t}\,dt = \frac{\sin((m-n)T)}{(m-n)T} \to 0 \qquad (T \to \infty).$$

Hence the mean is δ_{mn}. □

Proposition 6.13 (Parseval's Identity for Amplitude Series)**.** Let $f \in L^2([0, 2\pi])$ have Fourier coefficients $c_n = A_n e^{i\phi_n}$ and amplitude series $\sum_n A_n e^{int}$. Then

$$\frac{1}{2\pi} \int_0^{2\pi} |f(t)|^2 \, dt = \sum_{n=-\infty}^{\infty} A_n^2.$$

That is, Parseval's identity can be expressed by means of the amplitude series alone—conservation of energy is entirely determined by the amplitude spectrum and is independent of phase.

Proof. By Parseval's identity (Theorem 6.8),

$$\frac{1}{2\pi} \int_0^{2\pi} |f(t)|^2 \, dt = \sum_{n=-\infty}^{\infty} |c_n|^2 = \sum_{n=-\infty}^{\infty} A_n^2.$$

where $A_n = |c_n|$. ☐

Corollary 6.4 (Energy Is Determined by Amplitude Alone)**.** Let f and g have the same amplitude spectrum $\{A_n\}$ but possibly different phase spectra. Then

$$\|f\|_{L^2([0,2\pi])} = \|g\|_{L^2([0,2\pi])}.$$

Hence the L^2 energy is a function of the amplitude series and is independent of the phase series.

Proof. By Proposition 6.13, both energies equal $\sum_n A_n^2$. ☐

Property 6.4 (Basic Properties of Amplitude Series)**.** (i) **Non-negativity**: $A_n \geq 0$;

(ii) **Evenness**: if $A_{-n} = A_n$, the real form $\sum_n A_n \cos(nt)$ is an even function;

(iii) **Translation does not preserve zero phase**: a time translation $t \mapsto t - \tau$ generally introduces a phase $-n\tau$, so the amplitude series itself does not retain translation invariance;

(iv) **Additivity of energy**: the energies of orthogonal frequency components add, $\|\mathcal{A}_f\|^2 = \sum_n A_n^2$;

(v) **Decay and smoothness**: the faster A_n decays, the smoother the corresponding periodic function.

Proof. (i) By definition. (ii) Since $\cos(-nt) = \cos(nt)$. (iii) After translation the n-th term is $A_n e^{in(t-\tau)} = A_n e^{-in\tau} e^{int}$, whose phase is no longer zero. (iv) By Proposition 6.13. (v) Differentiating term by term k times produces a factor $(in)^k$, so the decay of $A_n n^k$ controls smoothness of order k. ☐

Example 6.9 (Amplitude Series of the Heat Kernel)**.** The heat kernel of period 2π

$$H_t(x) = \sum_{n=-\infty}^{\infty} e^{-n^2 t} e^{inx}, \qquad t > 0$$

is an amplitude series with zero phase and period 2π, with amplitudes $A_n = e^{-n^2t} \geq 0$. By Parseval's identity,

$$\frac{1}{2\pi}\int_0^{2\pi} |H_t(x)|^2\,dx = \sum_{n=-\infty}^{\infty} e^{-2n^2t}.$$

As $t \to 0^+$, this series tends in the distributional sense to $2\pi\sum_{k\in\mathbb{Z}}\delta(x-2\pi k)$.

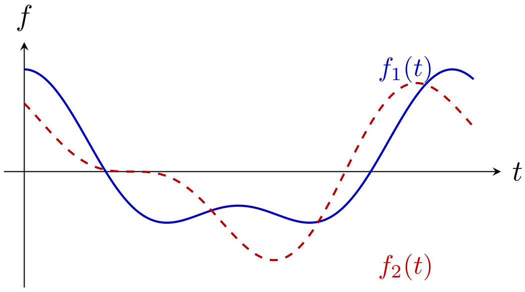
Proposition 6.14 (Uniqueness of the amplitude series). Suppose two amplitude series $\sum_n A_ne^{int}$ and $\sum_n B_ne^{int}$ ($A_n, B_n \geq 0$) both converge uniformly and absolutely and are equal as functions. Then $A_n = B_n$ for all n.

Proof. By the Bohr mean formula (Proposition 6.12), $A_n = M\{\mathcal{A}_fe^{-int}\}$ and $B_n = M\{\mathcal{B}_fe^{-int}\}$. Since $\mathcal{A}_f = \mathcal{B}_f$ (equality of functions), $A_n = B_n$. ☐

Example 6.10 (Waveforms with different phase spectra but the same amplitude spectrum). Take two signals of period 2π:

$$f_1(t) = \cos t + \frac{1}{2}\cos 2t, \qquad f_2(t) = \cos t + \frac{1}{2}\cos(2t+\pi/2) = \cos t - \frac{1}{2}\sin 2t.$$

The two have the same amplitude spectrum ($A_1 = 1, A_2 = 1/2$) but different phase spectra (the second harmonic of f_2 has phase $\pi/2$). By Corollary 6.4 the two have the same energy; yet the waveforms are utterly different: f_1 is an even, symmetric function, whilst f_2 is asymmetric.



Same amplitude spectrum, different phase spectra ⇒ different waveforms, same energy

Figure 6.5: Comparison of two signals with the same amplitude spectrum but different phase spectra. The same amplitude spectrum guarantees the same energy (Parseval, Proposition 6.13), but the waveforms differ markedly because of the differing phases. This illustrates the irreplaceable role of phase in waveform reconstruction.

Remark 6.7 (Scope and Caveats for Amplitude Series). The amplitude series retains only the amplitudes A_n and discards the phases ϕ_n entirely. It is therefore suited to problems independent of phase, such as energy estimation, smoothness criteria and decay-rate analysis; but it cannot be used to reconstruct the original function, to locate points of pointwise convergence, or to describe time translation or waveform symmetry. If the original series does not converge absolutely, the 'amplitude series' should be understood as a formal series or a distributional series; Parseval's identity must then be used under L^2 or absolute-convergence conditions.

6.7 Phase Series

This section discusses the form of series with amplitude 1 and period 2π. It retains all phase information and discards amplitude information, and is the basic object for studying time translation, frequency modulation and phase coupling.

Definition 6.11 (Phase Series with Amplitude 1 and Period 2π)**.** Let $\phi_n \in [0, 2\pi)$ ($n \in \mathbb{Z}$) be a sequence of phases. The formal series

$$\mathcal{P}_f(t) := \sum_{n=-\infty}^{\infty} e^{i(nt+\phi_n)} = \sum_{n=-\infty}^{\infty} e^{i\phi_n} e^{int}$$

is called the **phase series with amplitude 1 and period 2π**. Its generating-function form is

$$\mathfrak{P}_f(s) := \sum_{n=-\infty}^{\infty} e^{i\phi_n} s^n.$$

The moduli of the coefficients of a phase series are always 1, and its information is entirely determined by the distribution of the arguments ϕ_n. In general an infinite phase series does not converge pointwise and should be understood as a formal series, a finite truncation, or in the distributional sense.

Proposition 6.15 (Phase Series and Time Translation)**.** Let $f(t) = \sum_n A_n e^{i(nt+\phi_n)}$ have phase series $\mathcal{P}_f(t) = \sum_n e^{i(nt+\phi_n)}$. Then the phase series of the time translation $f_\tau(t) := f(t - \tau)$ is

$$\mathcal{P}_{f_\tau}(t) = \sum_n e^{i(nt+\phi_n-n\tau)} = \sum_n e^{i(\phi_n-n\tau)} e^{int}.$$

That is, a time translation is equivalent to a rotation of each term of the phase series—the rotation angle $-n\tau$ of the n-th term is proportional to the harmonic order n.

Proof.

$$f_\tau(t) = \sum_n A_n e^{i(n(t-\tau)+\phi_n)} = \sum_n A_n e^{i(nt+\phi_n-n\tau)}.$$

Hence the phase component is $\phi_n - n\tau$. ☐

Lemma 6.3 (Bohr Mean Coefficient Formula for Phase Series)**.** If the phase series $\mathcal{P}_f(t) = \sum_n e^{i(nt+\phi_n)}$ converges uniformly and absolutely, then

$$e^{i\phi_n} = M\{\mathcal{P}_f(t)e^{-int}\}.$$

Proof. By uniform absolute convergence,

$$M\{\mathcal{P}_f(t)e^{-int}\} = \sum_m e^{i\phi_m} M\{e^{i(m-n)t}\} = e^{i\phi_n},$$

where the last step uses Lemma 6.2. ☐

Theorem 6.13 (Modulus Conservation for Finite Phase Series)**.** Let $S_N(t) = \sum_{n=-N}^{N} e^{i(nt+\phi_n)}$ be a finite phase series. Then

$$\frac{1}{2\pi}\int_0^{2\pi} |S_N(t)|^2\, dt = 2N + 1.$$

That is, the total energy of a finite phase series is determined only by the number of terms and is independent of the phases ϕ_n.

Proof. By orthogonality,

$$|S_N(t)|^2 = \sum_{m,n=-N}^{N} e^{i((m-n)t+\phi_m-\phi_n)}.$$

Integrating term by term:

$$\frac{1}{2\pi}\int_0^{2\pi} e^{i(m-n)t}\, dt = \delta_{mn}.$$

Hence only the diagonal terms $m = n$ survive, each contributing 1, giving $2N + 1$ in total. □

Corollary 6.5 (Phase Rotation Rule for Time Translation)**.** A time translation $t \mapsto t - \tau$ multiplies the n-th term of the phase series by $e^{-in\tau}$. Therefore:

(i) if $\tau = 2\pi k$, all terms are unchanged;

(ii) if $\tau = \pi$, the n-th term is multiplied by $(-1)^n$;

(iii) if τ is an irrational multiple of 2π, the phase rotation no longer has periodicity of finite order.

Proof. By Proposition 6.15, the phase of the n-th term becomes $\phi_n - n\tau$. Substituting in each case gives the result. □

Property 6.5 (Basic Properties of Phase Series)**.** (i) **Unit-modulus coefficients**: all coefficients satisfy $|e^{i\phi_n}| = 1$;

(ii) **Conjugate symmetry**: a real signal requires $e^{i\phi_{-n}} = e^{-i\phi_n}$, that is $\phi_{-n} = -\phi_n \pmod{2\pi}$;

(iii) **Translation as rotation**: a time translation is equivalent to term-by-term phase rotation;

(iv) **Conservation of energy in the number of terms**: the total energy of a finite truncation equals the number of terms;

(v) **The fully infinite series generally does not converge**: $\sum_{n\in\mathbb{Z}} e^{int}$ does not converge pointwise and should be understood in the distributional sense.

Proof. (i) By definition. (ii) A real signal requires $c_{-n} = \overline{c_n}$; here $c_n = e^{i\phi_n}$, so $e^{i\phi_{-n}} = e^{-i\phi_n}$. (iii) By Proposition 6.15. (iv) By Theorem 6.13. (v) The general term has modulus 1 and does not tend to zero, so the series diverges in the usual sense; as a distribution, $\sum_{n\in\mathbb{Z}} e^{int} = 2\pi \sum_{k\in\mathbb{Z}} \delta(t - 2\pi k)$. □

Example 6.11 (Linear Phase and Time Translation)**.** Take $\phi_n = -n\tau$. Then

$$\mathcal{P}(t) = \sum_{n=-N}^{N} e^{i(nt-n\tau)} = \sum_{n=-N}^{N} e^{in(t-\tau)} = D_N(t-\tau),$$

where D_N is the Dirichlet kernel. This example shows that a linear phase corresponds to a time translation.

Proposition 6.16 (The rotation–translation duality of the phase series)**.** Let the phase series be $\mathcal{P}_f(t) = \sum_n e^{i(nt+\phi_n)}$. Then:

(i) **Time translation** $t \mapsto t-\tau$ is equivalent to multiplying the n-th term by $e^{-in\tau}$ in the frequency domain (Proposition 6.15);

(ii) **Frequency translation** $n \mapsto n+k$ is equivalent to multiplying the n-th term by e^{ikt} in the time domain.

This is the manifestation on phase series of the **translation–modulation duality** of the Fourier transform.

Proof. (i) Given directly by Proposition 6.15. (ii) The frequency translation $n \mapsto n+k$ turns the n-th term $e^{i(nt+\phi_n)}$ into $e^{i((n+k)t+\phi_{n+k})}$, that is, it multiplies the original $(n+k)$-th term by e^{ikt}. ☐

Example 6.12 (Linear phase and group delay)**.** Let $\phi_n = -n\tau$ (linear phase). Then the group delay of the phase series is

$$\tau_g(n) = -\frac{d\phi_n}{dn} = \tau,$$

that is, all frequency components have the same group delay. In signal processing a system with constant group delay is called a **linear phase system**, corresponding to a pure translation $t \mapsto t-\tau$ in the time domain (Example 6.11).

Property 6.6 (Parseval's identity for the phase series (finitely many terms))**.** Let $\mathcal{P}_N(t) = \sum_{n=-N}^{N} A_n e^{i(nt+\phi_n)}$ be a finite phase series ($A_n > 0$, not necessarily equal to 1). Then

$$\frac{1}{2\pi}\int_0^{2\pi} |\mathcal{P}_N(t)|^2 \, dt = \sum_{n=-N}^{N} A_n^2.$$

Proof. By the orthogonality of the Fourier basis (Proposition 6.8), the cross terms integrate to zero and only the diagonal terms remain. ☐

Remark 6.8 (Scope and Caveats for Phase Series). The phase series retains only the phases $e^{i\phi_n}$ and discards the amplitudes A_n entirely. It is therefore suited to problems such as time translation, phase coupling, frequency modulation and synchronisation analysis; but it cannot by itself give the energy distribution, the rate of convergence or the magnitude of function values. A fully infinite phase series generally does not converge and must be used together with finite truncation, window functions or distribution theory. The product term $-n\tau$ of phase and frequency shows that the 'rotation' in a phase series is tied to the harmonic order n; this is a basic manifestation of the frequency–phase duality.

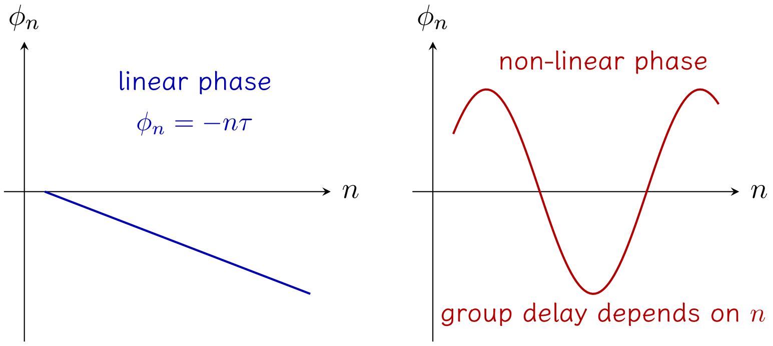


Figure 6.6: Two typical shapes of a phase spectrum. Left: linear phase (constant group delay), corresponding to a pure translation in the time domain. Right: non-linear phase (group delay depending on frequency), corresponding to waveform distortion in the time domain.

6.8 Frequency Series and the Hierarchy of Series

After the amplitude series and the phase series, this section establishes the **frequency series** and gives the five levels of the hierarchy of series. The frequency series retains only the frequency-lattice structure and discards amplitude and phase information entirely.

Definition 6.12 (Frequency Series). Let f have period T, with Fourier expansion

$$f(t) = \sum_n A_n e^{i(\omega_n t + \phi_n)}, \qquad \omega_n = \frac{2\pi n}{T}.$$

The **frequency series** of f is defined as the formal series that retains only the frequency components

$$\mathcal{F}_f(s) := \sum_{n=-\infty}^{\infty} \omega_n s^n = \frac{2\pi}{T} \sum_{n=-\infty}^{\infty} n s^n.$$

The frequency series discards all amplitude and phase information and retains only the frequency lattice of each harmonic. The frequency series is entirely determined by the fundamental period T—all frequencies are integer multiples of the fundamental frequency $\omega_0 = 2\pi/T$. Scaling property of the frequency series: a change of period to λT ($\lambda > 0$) sends $\omega_n \mapsto \omega_n/\lambda$, that is, it scales the frequency series as a whole by $1/\lambda$.

Proposition 6.17 (Frequency Series and Period Scaling). Let f have period T, with frequency series

$$\mathcal{F}_f(s) = \frac{2\pi}{T} \sum_n n s^n.$$

Then the frequency series of the time scaling $f_\lambda(t) := f(t/\lambda)$ ($\lambda > 0$) is

$$\mathcal{F}_{f_\lambda}(s) = \frac{2\pi}{\lambda T} \sum_n n s^n = \frac{1}{\lambda} \mathcal{F}_f(s).$$

That is, scaling time by a factor λ is equivalent to scaling the frequency series as a whole by $1/\lambda$—this is the manifestation of the frequency–period duality at the level of series.

Proof. $f_\lambda(t) = f(t/\lambda)$ has period λT, so its Fourier frequencies are

$$\omega_n^{(\lambda)} = \frac{2\pi n}{\lambda T} = \frac{\omega_n}{\lambda}.$$

Hence the frequency series is scaled as a whole by $1/\lambda$. □

Definition 6.13 (The Five-Level Hierarchy of Series). *For a function f of period T, its series decompositions are arranged in increasing order of information content as:*

(i) **Frequency series** $\mathcal{F}_f$: *retains only the frequency lattice $\omega_n = 2\pi n/T$, and is entirely determined by the fundamental period T.*

(ii) **Amplitude series** $\mathcal{A}_f$: *retains the amplitudes A_n and discards the phases ϕ_n and the frequency-lattice structure.*

(iii) **Phase series** $\mathcal{P}_f$: *retains the phases $e^{i\phi_n}$ and discards the amplitudes A_n and the frequency-lattice structure.*

(iv) **Fourier series**: *retains the amplitudes A_n and the phases ϕ_n, with the frequency lattice fixed by T.*

(v) **Periodic series**: *retains the four-tuple of amplitudes $A_{p,n}$, frequencies $\omega_{p,n}$, phases $\phi_{p,n}$ and periods T_p, and is the complete decomposition into multiple period layers.*

The relation among the five levels is: Fourier series = amplitude series × phase series (term by term); periodic series = the sum of the Fourier series of the multiple period layers; the amplitude series, the phase series and the frequency series are respectively the projections of the Fourier series onto its amplitude, phase and frequency components.

Proposition 6.18 (Reconstruction Relations among the Five Levels). *With the above notation:*

(i) *From the amplitude series and the phase series the Fourier coefficients can be reconstructed: $c_n = A_n e^{i\phi_n}$, and hence the Fourier series.*

(ii) *From the amplitude series the energy can be reconstructed: $\frac{1}{T}\int_0^T |f|^2 = \sum_n A_n^2$ (Parseval).*

(iii) *From the phase series the effect of time translation can be reconstructed: $\phi_n - 2\pi n\tau/T$.*

(iv) *From the frequency series the effect of period scaling can be reconstructed: $\omega_n \mapsto \omega_n/\lambda$.*

(v) *The information of any single one of the five levels is insufficient to reconstruct f; only when the four-tuple is complete (the periodic series) can f be reconstructed in full.*

Proof. (i) Given directly by $c_n = A_n e^{i\phi_n}$. (ii) By Proposition 6.13. (iii) By Proposition 6.15. (iv) By Proposition 6.17. (v) It follows from the term-by-term comparison of information in Definition 6.13: the frequency series contains only ω_n, the amplitude series only A_n, and the phase series only $e^{i\phi_n}$; any one of them lacks the information of the other components. ☐

Example 6.13 (An explicit comparison of the five series levels)**.** Take the periodic function $f(t) = 3 + 2e^{i(t+\pi/3)} + e^{i(3t-\pi/4)}$ of period $T = 2\pi$. The series at the five levels are respectively:

Level	Explicit form
Frequency series	$\mathcal{F}_f(s) = \sum_n ns^n$, retaining only the frequency lattice $\{0, 1, 3\}$ (modulo 2π)
Amplitude series	$\mathcal{A}_f(t) = 3 + 2e^{it} + e^{i3t}$, amplitudes $\{3, 2, 1\}$
Phase series	$\mathcal{P}_f(t) = e^{i\pi/3}e^{it} + e^{-i\pi/4}e^{i3t}$, phases $\{\pi/3, -\pi/4\}$
Fourier series	$f(t) = 3 + 2e^{i(t+\pi/3)} + e^{i(3t-\pi/4)}$
Periodic series	the Fourier series on the single period layer $P = \{2\pi\}$

This *example* shows that, among the five levels, only the Fourier series and the periodic series (which coincide in the single-period-layer case) can reconstruct f in full; the frequency series, the amplitude series and the phase series each retain only one component.

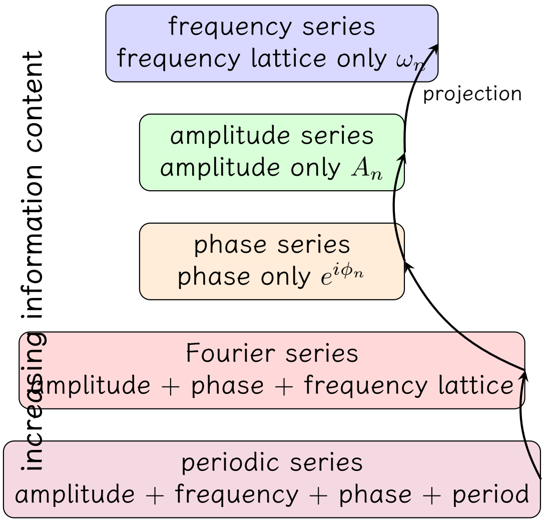

frequency series
frequency lattice only ω_n
projection
amplitude series
amplitude only A_n
phase series
phase only $e^{i\phi_n}$
Fourier series
amplitude + phase + frequency lattice
periodic series
amplitude + frequency + phase + period
increasing information content

Figure 6.7: The five levels of the hierarchy of series. The information content increases level by level from the frequency series (frequency lattice only) to the periodic series (complete four-tuple); *each level is a projection or a synthesis of the level above.*

6.9 Periodic Series: Complete Decomposition via Four-Tuples

Definition 6.14 (Periodic Series (Complex Form))**.** *Let P be a finite or countable* **set of period-class indices***, let a fundamental period $T_p > 0$ be assigned to each $p \in P$, and let $n \in \mathbb{Z}$ be the harmonic order within that period class. A series of the form*

$$f(t) = \sum_{p\in P}\sum_{n\in\mathbb{Z}} A_{p,n}\, e^{i(2\pi nt/T_p+\phi_{p,n})}$$

is called a **periodic series** *(complex form), in which the (p,n)-th term is determined by the* **four-tuple** *$(A_{p,n}, \omega_{p,n}, \phi_{p,n}, T_p)$:*

(i) *$A_{p,n} \geq 0$ is called the* **amplitude component***;*

(ii) *$\omega_{p,n} := 2\pi n/T_p$ is called the* **frequency component***;*

(iii) *$\phi_{p,n} \in [0, 2\pi)$ is called the* **phase component***;*

(iv) *T_p is called the* **period component** *(the primary index).*

Frequency and period are tied by the **fixed-length relation** *$\omega_{p,n} = 2\pi n/T_p$. Writing the polar-form coefficients $c_{p,n} = A_{p,n}e^{i\phi_{p,n}}$, we have $f(t) = \sum_{p,n} c_{p,n}e^{i\omega_{p,n}t}$.*

Definition 6.15 (Periodic Series (Real Form))**.** *Let $f : \mathbb{R} \to \mathbb{R}$. A series of the form*

$$f(t) = \sum_{p\in P}\sum_{n\in\mathbb{Z}} A_{p,n}\cos\bigl(\tfrac{2\pi n}{T_p}t + \phi_{p,n}\bigr), \qquad A_{p,n} \geq 0,\ \phi_{p,n} \in [0, 2\pi)$$

is called a **periodic series** *(real form).*

Proposition 6.19 (Equivalence of Real and Complex Forms)**.** *The real-form periodic series is the real part of the complex-form periodic series.*

Proof. It follows directly from Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$. ☐

Proposition 6.20 (Fourier Series as a Special Case of Periodic Series)**.** *The Fourier series is the special case of the periodic series on a single period layer ($P = \{T\}$).*

Proof. For a T-periodic function the Bohr mean reduces to a single-period average:

$$M\{g\} = \frac{1}{T}\int_0^T g(t)\,dt.$$

Hence the periodic series degenerates to the Fourier series. ☐

Proposition 6.21 (Periodic Series as a Synthesis of Amplitude, Phase, and Frequency Series)**.** *Let the periodic series be*

$$f(t) = \sum_{p\in P}\sum_{n\in\mathbb{Z}} A_{p,n}e^{i(\omega_{p,n}t+\phi_{p,n})}, \qquad \omega_{p,n} = \frac{2\pi n}{T_p}.$$

For each period layer $p \in P$, write

$$\mathcal{A}_p(s) = \sum_n A_{p,n}s^n, \qquad \mathcal{P}_p(s) = \sum_n e^{i\phi_{p,n}}s^n, \qquad \mathcal{F}_p(s) = \sum_n \omega_{p,n}s^n.$$

Then:

(i) if the single period layer $P = \{T\}$, $T = 2\pi$ is fixed and all $\phi_{p,n} = 0$ are set, then f degenerates to the amplitude series

$$\sum_n A_n e^{int};$$

(ii) if the single period layer $P = \{T\}$, $T = 2\pi$ is fixed and all $A_{p,n} = 1$ are set, then f degenerates to the phase series

$$\sum_n e^{i(nt+\phi_n)};$$

(iii) if the single period layer is fixed and only the frequency parameters $\omega_n = 2\pi n/T$ are retained, the frequency series is obtained;

(iv) in general the periodic series is formed by synthesising the four-tuple of amplitude $A_{p,n}$, phase $\phi_{p,n}$, frequency $\omega_{p,n}$ and period T_p according to the period layer p.

Proof. (i)(ii)(iii) Follow directly from Definitions 6.14, 6.10, 6.11, 6.12 by taking the corresponding parameters. (iv) From

$$c_{p,n} = A_{p,n} e^{i\phi_{p,n}}, \qquad \omega_{p,n} = \frac{2\pi n}{T_p}$$

it is read off directly. □

Corollary 6.6 (The Three Series as Projections of the Periodic Series). The amplitude series, the phase series and the frequency series are respectively the projections of the periodic series onto its amplitude, phase and frequency components. Any single projection is insufficient to reconstruct f; when the four-tuple is complete, f can be reconstructed uniquely.

Proof. By Proposition 6.21 and Proposition 6.18. □

Remark 6.9 (The Precise Meaning of 'Synthesis'). 'The periodic series is a synthesis of the three series' should be understood as follows: taking the period T_p as the primary index, within each period layer the amplitude series provides $A_{p,n}$, the phase series provides $e^{i\phi_{p,n}}$ and the frequency series provides $\omega_{p,n} = 2\pi n/T_p$, and then, through

$$A_{p,n} e^{i(\omega_{p,n} t + \phi_{p,n})}$$

they are combined into a periodic series. It is not the term-by-term sum of the three formal series, but a layered Fourier synthesis parametrised by the four-tuple. When the single period layer is $P = \{T\}$, $T = 2\pi$ it degenerates to the Fourier series; taking the phase to be zero or the amplitude to be one then gives the amplitude series or the phase series respectively.

Example 6.14 (Quasi-Periodic Periodic Series).

$$f(t) = 3e^{i(\sqrt{2}t + \pi/6)} + 2e^{i(t - \pi/3)}.$$

The frequencies $\sqrt{2}$ and 1 are incommensurable, so f is a quasi-periodic function.

Property 6.7 (Degeneration of the periodic series to a Fourier series)**.** When the set of period-class indices $P = \{T\}$ is a singleton, the periodic series degenerates to the Fourier series of period T:

$$f(t) = \sum_{n\in\mathbb{Z}} A_n e^{i(2\pi nt/T+\phi_n)}.$$

Furthermore, if $T = 2\pi$, it degenerates to the standard Fourier series $f(t) = \sum_{n\in\mathbb{Z}} c_n e^{int}$.

Proof. For the single period layer $P = \{T\}$ the outer sum disappears and the inner sum is precisely the Fourier series form. Taking $T = 2\pi$ gives the standard form. □

Example 6.15 (A periodic series with two period layers)**.** Take $P = \{2\pi, 2\pi\sqrt{2}\}$ and construct

$$f(t) = 3e^{i(t+\pi/6)} + 2e^{i(\sqrt{2}\,t-\pi/3)}.$$

This function has **two independent period layers**: the outer one is the component of period 2π and the inner one the component of period $2\pi\sqrt{2}$. Since $2\pi/(2\pi\sqrt{2}) = 1/\sqrt{2} \notin \mathbb{Q}$, the two layers are incommensurable and f is a quasi-periodic function (Example 6.14).

Theorem 6.14 (A sufficient condition for convergence of the periodic series)**.** Let the periodic series $f(t) = \sum_{p\in P}\sum_{n\in\mathbb{Z}} A_{p,n} e^{i(\omega_{p,n}t+\phi_{p,n})}$ satisfy

$$\sum_{p\in P}\sum_{n\in\mathbb{Z}} A_{p,n} < \infty.$$

Then f converges uniformly and absolutely on $\mathbb{R}$, and f is continuous and bounded.

Proof. For any t, $|A_{p,n}e^{i(\omega_{p,n}t+\phi_{p,n})}| = A_{p,n}$. By the Weierstrass M-test (Theorem 6.1) the series converges uniformly. Continuity and boundedness are preserved by uniform convergence. □

6.9.1 The Unit-Coefficient Special Case: Zero Phase and Unit Amplitude

This subsection discusses the special form of the periodic series when the phase is zero and the amplitude is one. It is the normalised-basis special case of the general Fourier series and also the entry point for understanding orthogonal expansions and distributional convergence.

Definition 6.16 (Periodic Series with Zero Phase and Unit Amplitude)**.** Let $S \subseteq \mathbb{Z}$ be a set of frequency indices and $T > 0$. The formal series

$$\Phi_{T,S}(t) := \sum_{n\in S} e^{i2\pi nt/T}$$

is called the **periodic series with zero phase and unit amplitude**. When $T = 2\pi$ we abbreviate it as

$$\Phi_S(t) := \sum_{n\in S} e^{int}.$$

Its real form (requiring $S = -S$) is

$$\Phi_S^{\mathbb{R}}(t) := \sum_{n \in S} \cos(nt).$$

If S is a finite set, then $\Phi_{T,S}$ is a trigonometric polynomial converging absolutely pointwise; if $S = \mathbb{Z}$, it should be understood in the distributional sense.

Proposition 6.22 (Normalisation of Coefficients, Phase, and Amplitude)**.** The Fourier coefficients of the periodic series with zero phase and unit amplitude $\Phi_{T,S}(t) = \sum_{n \in S} e^{i2\pi nt/T}$ are

$$c_n = \begin{cases} 1, & n \in S, \\ 0, & n \notin S. \end{cases}$$

Hence its amplitudes are $A_n = 1$ $(n \in S)$ and its phases are $\phi_n = 0$ $(n \in S)$.

Proof. By the Fourier coefficient formula and orthogonality (Proposition 6.8),

$$c_n = \frac{1}{T} \int_0^T \Phi_{T,S}(t) e^{-i2\pi nt/T}\, dt = \sum_{m \in S} \frac{1}{T} \int_0^T e^{i2\pi(m-n)t/T}\, dt = \sum_{m \in S} \delta_{mn}.$$

Hence $c_n = 1$ when $n \in S$ and $c_n = 0$ otherwise. ☐

Lemma 6.4 (L^2 Orthogonality of a Finite Frequency Set)**.** Let $S \subset \mathbb{Z}$ be finite. Then

$$\frac{1}{T} \int_0^T \Phi_{T,S}(t) \overline{\Phi_{T,S}(t)}\, dt = |S|,$$

where $|S|$ is the number of elements of S. More generally,

$$\frac{1}{T} \int_0^T e^{i2\pi mt/T} e^{-i2\pi nt/T}\, dt = \delta_{mn}.$$

Proof. Expanding:

$$\frac{1}{T} \int_0^T \Phi_{T,S} \overline{\Phi_{T,S}}\, dt = \sum_{m,n \in S} \frac{1}{T} \int_0^T e^{i2\pi(m-n)t/T}\, dt = \sum_{m,n \in S} \delta_{mn} = |S|.$$

☐

Theorem 6.15 (Relation between the Zero-Phase, Unit-Amplitude Periodic Series and General Fourier Series)**.** Let f have period T, with Fourier series

$$f(t) \sim \sum_{n \in \mathbb{Z}} c_n e^{i2\pi nt/T}, \qquad c_n = A_n e^{i\phi_n}.$$

Then:

(i) the general Fourier series is the term-by-term product of the amplitude series and the phase series:

$$c_n e^{i2\pi nt/T} = A_n e^{i\phi_n} e^{i2\pi nt/T}.$$

(ii) the periodic series with zero phase and unit amplitude is the normalised special case of the Fourier series when $c_n = 1$ $(n \in S)$:

$$\Phi_{T,S}(t) = \sum_{n\in S} 1 \cdot e^{i\phi_n} e^{i2\pi nt/T}, \qquad \phi_n = 0,\ A_n = 1.$$

(iii) if S is finite, then $\Phi_{T,S}$ is the orthogonal projection kernel on $L^2([0,T])$ spanned by the frequency set S; if $S = \mathbb{Z}$, then $\Phi_{T,\mathbb{Z}}$ is the distributional limit of the periodic Dirichlet kernel:

$$\sum_{n\in\mathbb{Z}} e^{i2\pi nt/T} = T \sum_{k\in\mathbb{Z}} \delta(t - kT).$$

Proof. (i) Write it out directly from $c_n = A_n e^{i\phi_n}$. (ii) Take $A_n = 1$, $\phi_n = 0$ $(n \in S)$ and $c_n = 0$ otherwise in (i). (iii) For finite S, $\Phi_{T,S}$ is the sum of finitely many orthogonal basis functions, and its integral-kernel property is given by Lemma 6.4. For $S = \mathbb{Z}$, use the Poisson summation formula (or the limit of the Dirichlet kernel in the distributional sense):

$$\sum_{n\in\mathbb{Z}} e^{i2\pi nt/T} = T \sum_{k\in\mathbb{Z}} \delta(t - kT).$$

Hence it should be understood distributionally. □

Corollary 6.7 (Parseval's Identity for Finite Normalised Bases). *Let $S \subset \mathbb{Z}$ be finite. Then*

$$\frac{1}{T}\int_0^T |\Phi_{T,S}(t)|^2\, dt = |S|.$$

That is, the energy of a finite periodic series with zero phase and unit amplitude equals the number of frequency terms.

Proof. Obtained directly from Lemma 6.4. □

Property 6.8 (Basic Properties of the Zero-Phase, Unit-Amplitude Periodic Series). (i) **Unit coefficients**: on S, $c_n = 1$, the amplitude is 1 and the phase is 0;

(ii) **Orthogonality**: terms of different frequencies are orthogonal in L^2;

(iii) **Additivity of finite energy**: the energy of a finite truncation equals the number of terms;

(iv) **The real form is a sum of cosines**: if $S = -S$, the real form is $\sum_{n\in S} \cos(2\pi nt/T)$;

(v) **Distributional nature at all frequencies**: when $S = \mathbb{Z}$ the series does not converge pointwise, but is a periodic δ comb.

Proof. (i) By Proposition 6.22. (ii) By Lemma 6.4. (iii) By Corollary 6.7. (iv) By Euler's formula. (v) By Theorem 6.15(iii). □

Example 6.16 (Dirichlet Kernel as a Finite Normalised Periodic Series). Taking $T = 2\pi$ and $S = \{-N, \ldots, N\}$ gives

$$\Phi_{\{-N,\ldots,N\}}(t) = \sum_{n=-N}^{N} e^{int} = D_N(t) = \frac{\sin((N+\frac{1}{2})t)}{\sin(t/2)}.$$

Its energy is

$$\frac{1}{2\pi}\int_0^{2\pi} |D_N(t)|^2\,dt = 2N+1.$$

As $N \to \infty$, D_N tends in the distributional sense to $2\pi \sum_{k\in\mathbb{Z}} \delta(t - 2\pi k)$.

Remark 6.10 (Scope of the Zero-Phase, Unit-Amplitude Periodic Series). The zero-phase, unit-amplitude periodic series is the 'unit coefficient' special case of the Fourier series, suited to illustrating orthogonal bases, the Dirichlet kernel, Poisson summation and distributional convergence. It does not represent a general L^2 or continuous periodic function, because the unit-coefficient series over all frequencies does not converge; it has a rigorous meaning only for a finite frequency set or in the distributional sense. For real signals one must further require the frequency set to be symmetric about the origin and take the real part. Its relation to the general Fourier series is: general series = amplitude series × phase series, whilst the case of this subsection is the normalised intersection in which all amplitudes are 1 and all phases are 0.

6.9.2 Bohr Means and the Determination of Coefficients

Definition 6.17 (Bohr Mean). Let $f : \mathbb{R} \to \mathbb{C}$. If the limit

$$M\{f\} := \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} f(t)\,dt$$

exists, it is called the **Bohr mean** of f.

Lemma 6.5 (Bohr Mean of Exponential Functions). For any $\omega \in \mathbb{R}$,

$$M\{e^{i\omega t}\} = \begin{cases} 1, & \omega = 0, \\ 0, & \omega \neq 0. \end{cases}$$

Proof. When $\omega = 0$ the integrand is identically 1. When $\omega \neq 0$,

$$\frac{1}{2T}\int_{-T}^{T} e^{i\omega t}\,dt = \frac{\sin(\omega T)}{\omega T} \to 0 \qquad (T \to \infty).$$

□

Proposition 6.23 (Elementary Proof in the Finite Case: Vandermonde Determinant). Let the frequencies $\omega_1 < \cdots < \omega_N$ be pairwise distinct. If $\sum_{k=1}^{N} c_k e^{i\omega_k t} \equiv 0$, then $c_1 = \cdots = c_N = 0$.

Proof. Differentiating the identity successively $j = 0, \ldots, N-1$ times and evaluating at $t = 0$ gives a homogeneous linear system whose coefficient matrix is a Vandermonde matrix with non-zero determinant. □

Theorem 6.16 (Determination of Periodic-Series Coefficients (Bohr))**.** *Let the periodic series $f(t) = \sum_{p,n} A_{p,n} e^{i(\omega_{p,n}t+\phi_{p,n})}$ converge uniformly on $\mathbb{R}$, with all frequencies pairwise distinct. Then*

$$A_{p,n}e^{i\phi_{p,n}} = M\{f(t)\,e^{-i\omega_{p,n}t}\}.$$

Proof. By uniform convergence the integral and the sum may be interchanged. By Lemma 6.5 the cross terms have zero mean and only the (p,n) term survives. ☐

Corollary 6.8 (Uniqueness of Representation)**.** *If two uniformly convergent periodic series are equal as functions and have the same period partition P and the same frequency set, then their amplitudes and phases agree term by term.*

Proof. It follows directly from Theorem 6.16. ☐

Theorem 6.17 (Generalised Parseval Identity (Bohr))**.** *Let the periodic series $f(t) = \sum_{p,n} c_{p,n}e^{i\omega_{p,n}t}$ satisfy $\sum_{p,n} |c_{p,n}| < \infty$. Then*

$$M\{|f|^2\} = \sum_{p,n} |c_{p,n}|^2 = \sum_{p,n} A_{p,n}^2.$$

Proof. Expand $|f(t)|^2$; by absolute convergence the Bohr mean may be interchanged with the summation, the cross terms have zero mean, and the diagonal terms give $|c_{p,n}|^2$. ☐

6.9.3 The Periodic Transform: Unification of the Classical Integral Transforms

Definition 6.18 (Periodic Transform (Frequency-Dictionary Form))**.** Let $\Lambda = \{\omega_k\}_{k\in K}$ be a **frequency dictionary**. For a function f, its **periodic transform** with respect to the dictionary Λ is the family of coefficients

$$\mathcal{P}_\Lambda f(k) = M\{f(t)\,e^{-i\omega_k t}\}, \qquad k \in K,$$

whose synthesis (inverse transform) is the periodic series

$$f(t) \sim \sum_{k\in K} \mathcal{P}_\Lambda f(k)\,e^{i\omega_k t}.$$

Proposition 6.24 (Linearity of the Periodic Transform)**.** *Let the periodic transforms of f, g with respect to the dictionary $\Lambda = \{\omega_k\}_{k\in K}$ both exist, and let $\alpha, \beta \in \mathbb{C}$. Then for every $k \in K$,*

$$\mathcal{P}_\Lambda(\alpha f + \beta g)(k) = \alpha\,\mathcal{P}_\Lambda f(k) + \beta\,\mathcal{P}_\Lambda g(k),$$

that is, $\mathcal{P}_\Lambda$ is a **linear map** *from the function space to the space of coefficient families $\mathbb{C}^K$.*

Proof. In three steps.

Step one: linearity of $M\{\cdot\}$. Let $M\{f\}$ and $M\{g\}$ both exist. By Definition 6.17, $M\{\cdot\}$ is the limit of partial integral averages:

$$M\{\alpha f + \beta g\} = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} \big(\alpha f(t) + \beta g(t)\big)\,dt.$$

Since the definite integral is linear in the integrand,

$$\frac{1}{2T}\int_{-T}^{T}\bigl(\alpha f(t)+\beta g(t)\bigr)\,dt=\alpha\cdot\frac{1}{2T}\int_{-T}^{T}f(t)\,dt+\beta\cdot\frac{1}{2T}\int_{-T}^{T}g(t)\,dt.$$

Both limits on the right exist (they are $M\{f\}$ and $M\{g\}$ respectively), so by the addition and scalar-multiplication rules for limits,

$$M\{\alpha f+\beta g\}=\alpha M\{f\}+\beta M\{g\}.$$

This requires both means to exist separately: if one of them does not exist, the right-hand side is meaningless and linearity cannot even be discussed.

Step two: substitution at each frequency. Fix $k\in K$ and write

$$f_k(t):=f(t)\,e^{-i\omega_k t},\qquad g_k(t):=g(t)\,e^{-i\omega_k t}.$$

By $M\{f_k\}=\mathcal{P}_\Lambda f(k)$ and $M\{g_k\}=\mathcal{P}_\Lambda g(k)$ (Definition 6.18), and

$$(\alpha f+\beta g)(t)\,e^{-i\omega_k t}=\alpha f_k(t)+\beta g_k(t),$$

step one gives

$$\mathcal{P}_\Lambda(\alpha f+\beta g)(k)=M\{\alpha f_k+\beta g_k\}=\alpha M\{f_k\}+\beta M\{g_k\}=\alpha\,\mathcal{P}_\Lambda f(k)+\beta\,\mathcal{P}_\Lambda g(k).$$

Step three: pointwise linearity is linearity of the map. The above identity holds for every $k\in K$, so the two coefficient families satisfy a linear relation as elements of $\mathbb{C}^K$; by the definition of a linear map, $\mathcal{P}_\Lambda$ is linear. This completes the proof. □

Remark 6.11 (The boundaries of linearity). Proposition 6.24 uses only the linearity of the Bohr mean and is independent of the particular choice of the frequency dictionary Λ. But two points must be noted: first, linearity presupposes that each mean **exists separately**, since $M\{\cdot\}$ is not itself defined on the whole function space; second, the convergence of the synthesis (inverse transform) $f\sim\sum_k\mathcal{P}_\Lambda f(k)e^{i\omega_k t}$ is a separate question, and linearity does not imply convergence of the synthesis. When Λ runs over all frequencies, $\mathcal{P}_\Lambda$ is precisely the operator that extracts the classical Fourier coefficients (Theorem 6.18).

Theorem 6.18 (Classical Integral Transforms as Special Cases of the Periodic Transform)**.** The following classical transforms are all special cases of the periodic transform under a particular frequency dictionary (or a particular kernel):

(i) **Fourier series**: the dictionary $\Lambda=\omega_0\mathbb{Z}$.

(ii) **Discrete Fourier transform (DFT)**: on the finite cyclic group C_N, the dictionary $\Lambda=\{\frac{2\pi k}{N}\}_{k=0}^{N-1}$.

(iii) **Mellin transform**: via the logarithmic change of variable $t=\ln x$:

$$\mathcal{M}_f(s)=\int_0^\infty f(x)\,x^{s-1}\,dx=\int_{-\infty}^{\infty}f(e^t)\,e^{st}\,dt.$$

(iv) **Laplace transform**: the complex frequency dictionary $s=\sigma+i\omega$.

(v) **Wavelet transform**: a kernelised periodic transform.

(vi) **Chirp-Z transform**: a spiral sequence of sampling points.

Proof. (i)(ii) Follow by substituting into the definition of the periodic transform. (iii) The change of variable $x = e^t$ gives it directly. (iv) The complex frequency corresponds to a decay factor. (v)(vi) Verified directly from their respective definitions. □

Remark 6.12 (Discrete and Continuous Spectra). The Bohr mean is non-zero only for signals with a discrete spectrum. The spectrum of a general signal decomposes, by the Lebesgue decomposition, into an atomic part (periodic series), an absolutely continuous part (Fourier transform density) and a singular continuous part (fractal/scaling periodic signals).

6.10 Convolution: The Unified Operator of Delay Products and Phase Interaction

Definition 6.19 (Convolution (Delay Product)). Let $f, g : \mathbb{R} \to \mathbb{R}$ (or $\mathbb{C}$). The **convolution** of f and g is a function of the delay τ:

$$(f * g)(\tau) := \int_{-\infty}^{\infty} f(t)g(\tau - t)\,dt.$$

When f, g have period T, the periodic form is used:

$$(f * g)(\tau) := \frac{1}{T}\int_{0}^{T} f(t)g(\tau - t)\,dt.$$

Definition 6.20 (Cross-Correlation (Conjugate Delay Inner Product)). Let f, g have period T. The **cross-correlation** of f and g is defined as

$$(f \star g)(\tau) := \frac{1}{T}\int_{0}^{T} f(t)\,\overline{g(t - \tau)}\,dt.$$

Proposition 6.25 (Basic Algebraic Properties of Convolution). (i) **Commutativity**: $f * g = g * f$.

(ii) **Associativity**: $(f * g) * h = f * (g * h)$.

(iii) **Distributivity**: $f * (g + h) = f * g + f * h$.

(iv) **Translation property**: $\tau_a(f * g) = (\tau_a f) * g = f * (\tau_a g)$.

(v) **Differentiation property**: $(f * g)' = f' * g = f * g'$.

Proof. Commutativity follows from the change of variable $u = \tau - t$; associativity and distributivity follow from Fubini's theorem by interchanging the order of integration; the translation property follows directly from the definition; the differentiation property follows by differentiating under the integral sign. □

Proposition 6.26 (Relation between Cross-Correlation and Convolution, and Frequency-Doma

Defining $\widetilde{g}(t) = \overline{g(-t)}$, we have $(f \star g)(\tau) = (f * \widetilde{g})(\tau)$.

(ii) $(f \star g)(\tau) = \overline{(g \star f)(-\tau)}$.

(iii) Frequency-domain form: $(f * g)(\tau) = \sum_n c_n d_n e^{i2\pi n\tau/T}$, $(f \star g)(\tau) = \sum_n c_n \overline{d_n} e^{i2\pi n\tau/T}$.

Proof. (i) Substitute directly into the definition. (ii) Take conjugates and change variables. (iii) Substitute the periodic series and use orthogonality (Proposition 6.8). □

Example 6.17 (Physical Meaning of Delay: Matched Filtering). A radar transmits a signal $s(t)$ and receives the echo $r(t) = \alpha s(t - \tau_0) + \text{noise}$. The cross-correlation $(r \star s)(\tau)$ attains its peak at $\tau = \tau_0$.

Proposition 6.27 (The product theorem for convolution and the Fourier transform). Let f, g have period T with Fourier coefficients c_n, d_n respectively. Then the Fourier coefficients of the convolution $f * g$ are $Tc_n d_n$.

Proof.

$$\widehat{f * g}(n) = \frac{1}{T}\int_0^T (f * g)(t) e^{-i\omega_n t}\, dt = \frac{1}{T^2}\int_0^T \int_0^T f(\tau) g(t - \tau) e^{-i\omega_n t}\, d\tau\, dt.$$

Change variables $u = t - \tau$ and use Fubini's theorem:

$$= \frac{1}{T}\int_0^T f(\tau) e^{-i\omega_n \tau}\, d\tau \cdot \int_0^T g(u) e^{-i\omega_n u}\, du = Tc_n d_n.$$

□

Corollary 6.9 (Diagonalisation of the convolution operator). The convolution operator $C_f : g \mapsto f * g$ is diagonalised in the Fourier basis $\{e^{i\omega_n t}\}_{n \in \mathbb{Z}}$: $C_f(e^{i\omega_n t}) = Tc_n e^{i\omega_n t}$. That is, the Fourier basis consists of eigenfunctions of the convolution operator, with eigenvalues Tc_n.

Proof. By Proposition 6.27, the Fourier coefficients of $C_f(e^{i\omega_n t})$ equal Tc_n at n and zero elsewhere, so $C_f(e^{i\omega_n t}) = Tc_n e^{i\omega_n t}$. □

Example 6.18 (Three algebraic properties of convolution). (i) **Commutativity**: $f * g = g * f$. For example $e^{it} * e^{i2t} = e^{i2t} * e^{it}$ (both are zero, since the frequencies differ);

(ii) **Associativity**: $(f * g) * h = f * (g * h)$, guaranteed by the associativity of the product in the convolution theorem;

(iii) **Distributivity**: $f * (g + h) = f * g + f * h$, guaranteed by the linearity of the integral.

Remark 6.13 (The Fourier duality of convolution and product). Convolution and product form a **duality** through the Fourier transform:

$$\widehat{f * g} = \widehat{f} \cdot \widehat{g}, \qquad \widehat{f \cdot g} = \widehat{f} * \widehat{g}.$$

The former is Proposition 6.27; the latter follows directly from the duality. In signal processing this duality appears as: convolution in the time domain equals multiplication in the frequency domain, and multiplication in the time domain equals convolution in the frequency domain (under a suitable normalisation). Periodic convolution (Definition 6.26) is precisely the generalisation of this duality with the aim of 'retaining the interaction of all frequency pairs'.

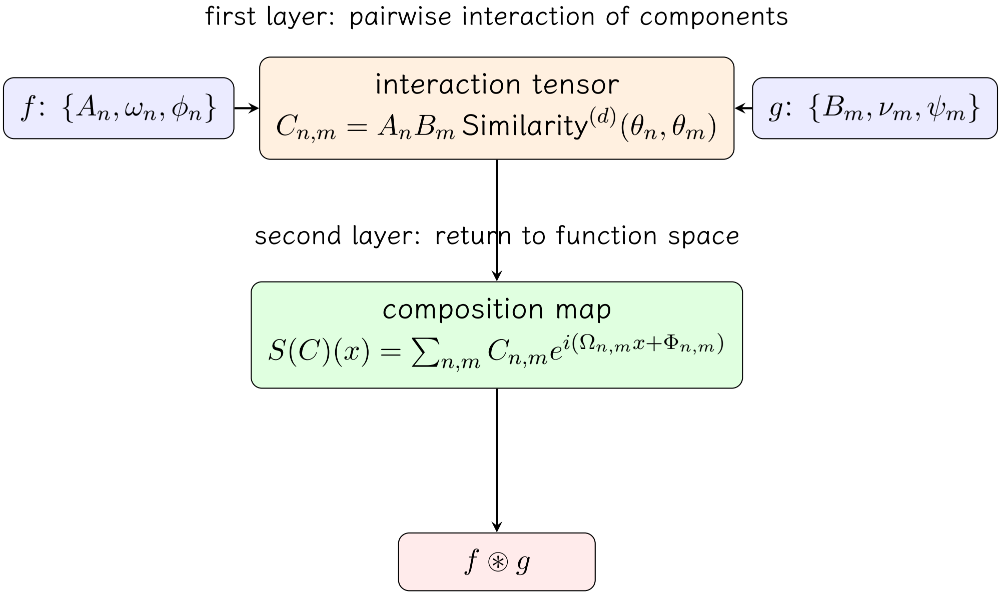

first layer: pairwise interaction of components
f: $\{A_n, \omega_n, \phi_n\}$
interaction tensor
$C_{n,m} = A_n B_m \,\text{Similarity}^{(d)}(\theta_n, \theta_m)$
g: $\{B_m, \nu_m, \psi_m\}$
second layer: return to function space
composition map
$S(C)(x) = \sum_{n,m} C_{n,m} e^{i(\Omega_{n,m}x + \Phi_{n,m})}$
$f \circledast g$

Figure 6.8: The two-layer construction of the convolution operator. The first layer pairs the components of the two epicycle families two by two, yielding a doubly indexed object (at $n \neq m$ this is a cross-frequency interaction); the second layer sends that object back to function space by means of the frequency composition operator $\Omega_{n,m}$ and the phase composition operator $\Phi_{n,m}$. The composite of the two layers is the convolution operator; when Similarity$^{(d)}$ is taken to be a phase similarity kernel and Ω, Φ take the sum-angle form, one obtains the periodic convolution $f \circledast g$.

6.10.1 Amplitude, Phase and Frequency Convolutions

Let f, g have period T with frequency-series expansions

$$f(t) = \sum_m c_m e^{i2\pi mt/T}, \quad c_m = A_m e^{i\theta_m}; \qquad g(t) = \sum_n d_n e^{i2\pi nt/T}, \quad d_n = B_n e^{i\varphi_n}.$$

where $A_m, B_n \geq 0$ are amplitudes, $\theta_m, \varphi_n \in [0, 2\pi)$ are phases, and $\omega_m = 2\pi m/T$, $\nu_n = 2\pi n/T$ are frequencies.

Definition 6.21 (Amplitude Convolution). The **amplitude convolution** of f and g is the real matrix $\mathbb{A} = (\mathbb{A}_{mn})$:

$$\mathbb{A}_{mn} := A_m B_n.$$

The amplitude convolution discards all phase information and retains only the scale relations. $\mathbb{A}_{mn} \geq 0$ holds for all m, n.

Definition 6.22 (Phase Convolution)**.** The **phase convolution** of f and g is the complex matrix $\mathbb{P} = (\mathbb{P}_{mn})$:

$$\mathbb{P}_{mn} := e^{i(\theta_m - \varphi_n)} = \cos(\theta_m - \varphi_n) + i\sin(\theta_m - \varphi_n).$$

The phase convolution discards all amplitude information and retains only the directional relations. $|\mathbb{P}_{mn}| = 1$ holds for all m, n.

Definition 6.23 (Frequency Convolution)**.** The **frequency convolution** of f and g is the real matrix $\mathbb{F} = (\mathbb{F}_{mn})$:

$$\mathbb{F}_{mn} := \omega_m \nu_n = \frac{2\pi m}{T} \cdot \frac{2\pi n}{T} = \frac{4\pi^2 mn}{T^2}.$$

The frequency convolution discards all amplitude and phase information and retains only the frequency-lattice structure. The frequency convolution is entirely determined by the fundamental period T.

Definition 6.24 (Trend Convolution)**.** The **trend convolution** of f and g is the complex matrix $\mathbb{T} = (\mathbb{T}_{mn})$:

$$\mathbb{T}_{mn} := A_m B_n \mathbb{P}_{mn} = A_m B_n e^{i(\theta_m - \varphi_n)} = c_m \overline{d_n}.$$

Trend convolution = amplitude convolution × phase convolution (term by term).

Proposition 6.28 (Relations among the Three Convolutions and Trend Convolution)**.** *With the above notation:*

(i) *Trend convolution = amplitude convolution × phase convolution: $\mathbb{T}_{mn} = \mathbb{A}_{mn}\mathbb{P}_{mn}$.*

(ii) *Frequency convolution = outer product of the frequency series: $\mathbb{F}_{mn} = \omega_m \nu_n$.*

(iii) *In the frequency domain the three capture three independent components of convolution: amplitude (scaling), phase (rotation) and frequency (translation).*

Proof. (i) By Definitions 6.21 and 6.22: $\mathbb{A}_{mn}\mathbb{P}_{mn} = A_m B_n e^{i(\theta_m - \varphi_n)} = \mathbb{T}_{mn}$. (ii) By Definition 6.23. (iii) From (i)(ii) and the definitions of the three. ☐

Proposition 6.29 (Conjugate Symmetry of Phase Convolution)**.** *The phase convolution matrix satisfies $\mathbb{P}_{mn} = \overline{\mathbb{P}'_{nm}}$, where $\mathbb{P}'_{mn} := e^{i(\varphi_m - \theta_n)}$ is the phase convolution obtained on interchanging the roles of f and g. The amplitude and frequency convolutions are symmetric real matrices: $\mathbb{A}_{mn} = \mathbb{A}_{nm}$, $\mathbb{F}_{mn} = \mathbb{F}_{nm}$.*

Proof. $\overline{\mathbb{P}_{nm}} = \overline{e^{i(\theta_n - \varphi_m)}} = e^{i(\varphi_m - \theta_n)}$, whilst $\mathbb{P}_{mn} = e^{i(\theta_m - \varphi_n)}$; in general the two are not equal. The correct symmetry is: defining $\mathbb{P}'_{mn} := e^{i(\varphi_m - \theta_n)}$, we have $\mathbb{P}'_{mn} = \overline{\mathbb{P}_{nm}}$. The symmetry of the amplitude and frequency convolutions follows directly from $A_m B_n = B_n A_m$ and $\omega_m \nu_n = \nu_n \omega_m$. ☐

Remark 6.14 (Non-Symmetry of Phase Convolution)*.* The phase convolution matrix is in general not symmetric: $\mathbb{P}_{mn} = e^{i(\theta_m - \varphi_n)}$ and $\mathbb{P}_{nm} = e^{i(\theta_n - \varphi_m)}$ are generally unequal, because the roles of f and g cannot be interchanged—θ_m comes from f and φ_n from g. Only after taking the complex conjugate is there the relation $\mathbb{P}_{mn} = \overline{\mathbb{P}'_{nm}}$, where $\mathbb{P}'_{mn} := e^{i(\varphi_m - \theta_n)}$ is the phase convolution obtained on interchanging the roles of f and g. This contrasts with the symmetry of the amplitude and frequency convolutions.

Definition 6.25 (Kernelised Convolution (Master Kernel Decomposition Formula)). Let the weight kernel be $W : \mathbb{R} \to \mathbb{C}$, the similarity kernels $S^{(d)}$ $(d = 1, \ldots, 6)$ and the amplitude switch $\alpha \in \{0, 1\}$. Define the **kernelised convolution**

$$\mathcal{K}^{(d)}_{W,\alpha}(\tau) := \sum_{m,n} W(\omega_m - \omega_n) \cdot (A_m B_n)^\alpha \cdot S^{(d)}(\theta_m, \varphi_n) \cdot e^{i\omega_n \tau}.$$

(i) $\alpha = 0$: phase convolution (taking $S^{(d)}$ to be the d-dimensional phase kernel);

(ii) $\alpha = 1$: trend convolution.

Proposition 6.30 (The Six Complex Phase Kernels). *The six complex phase kernels are generated from difference/sum angles and base/double/half angles:*

$$S^{(1)} = e^{i(\theta_m - \varphi_n)}, \quad S^{(2)} = e^{i(\theta_m + \varphi_n)}, \quad S^{(3)} = e^{i\cdot 2(\theta_m - \varphi_n)},$$

$$S^{(4)} = e^{i(\theta_m - \varphi_n)/2}, \quad S^{(5)} = e^{i\cdot 2(\theta_m + \varphi_n)}, \quad S^{(6)} = e^{i(\theta_m + \varphi_n)/2}.$$

Proof. The 12 real similarity measures of the 'Scalars and Vectors' chapter are precisely the real and imaginary parts of these 6 complex kernels (the cosine family and the sine family). ☐

Lemma 6.6 (Windowed Expansion). *Let w be an integrable window on $[-T, T]$ and let $\widehat{w}(\Omega) := \frac{1}{2T} \int_{-T}^{T} w(t) e^{i\Omega t}\, dt$. Then*

$$\frac{1}{2T} \int_{-T}^{T} f(t) e^{-i\Omega t} w(t)\, dt = \sum_m c_m\, \widehat{w}(\omega_m - \Omega).$$

Proof. Substitute the frequency series of f and integrate term by term using uniform convergence. ☐

Theorem 6.19 (Window–Kernel Duality Theorem). *Let f, g have period T and let w be an even window. The windowed delay inner product*

$$(f \star_w g)(\tau) := \frac{1}{2T} \int_{-T}^{T} f(t)\, \overline{g(t - \tau)}\, w(t)\, dt$$

has the expansion

$$(f \star_w g)(\tau) = \sum_{m,n} \widehat{w}(\omega_m - \omega_n) \cdot c_m \overline{d_n} \cdot e^{i\omega_n \tau}.$$

Proof. Substitute the frequency series and apply Lemma 6.6. ☐

Corollary 6.10 (Gaussian Window and Gaussian Kernel). *Taking the Gaussian time window $w(t) = e^{-t^2/(2\sigma_t^2)}$, the frequency-domain kernel is a Gaussian kernel:*

$$\widehat{w}(\omega) \approx \frac{\sqrt{2\pi}\, \sigma_t}{2T}\, e^{-\sigma_t^2 \omega^2/2}.$$

Corollary 6.11 (Classical Limit: Rectangular Window and δ Kernel). *Taking the rectangular window $w(t) = \mathbf{1}_{[-T,T]}(t)$, we have $\widehat{w}(\omega) = \frac{\sin(\omega T)}{\omega T}$, and $\widehat{w}(\omega_m - \omega_n) \to \delta_{mn}$ as $T \to \infty$.*

Corollary 6.12 (Zero Delay and Cosine Similarity). Taking $\tau = 0$ and normalising gives

$$\frac{(f \star g)(0)}{\|f\|\|g\|}$$

which is precisely the cosine similarity in $L^2([0,T])$.

Property 6.9 (Basic Properties of Phase, Amplitude, Frequency, and Trend Convolutions). $|\mathbb{P}_{mn}| = 1$, $\mathbb{P}_{mn} = \overline{\mathbb{P}'_{nm}}$, where $\mathbb{P}'_{mn} := e^{i(\varphi_m - \theta_n)}$ is the phase convolution obtained on interchanging the roles of f and g (subject to the convention on the domain of the phases, see Remark 6.14);

(ii) $\mathbb{A}_{mn} = \mathbb{A}_{nm} \geq 0$;

(iii) $\mathbb{F}_{mn} = \mathbb{F}_{nm}$;

(iv) $\mathbb{T}_{mn} = \mathbb{A}_{mn}\mathbb{P}_{mn} = \overline{\mathbb{T}_{nm}}$;

(v) time reversal interchanges the difference-angle family and the sum-angle family;

(vi) the amplitude switch α going from 0 to 1 takes phase convolution to trend convolution.

Proof. Each item is verified directly from the definitions. $\square$

Example 6.19 (Phase/Amplitude/Frequency Convolution Matrices for Period 2). Let $T = 2$, $f(t) = c_0 + c_1 e^{i\pi t}$, $g(t) = d_0 + d_1 e^{i\pi t}$. Then:

$$\mathbb{P} = \begin{pmatrix} e^{i(\theta_0 - \varphi_0)} & e^{i(\theta_0 - \varphi_1)} \\ e^{i(\theta_1 - \varphi_0)} & e^{i(\theta_1 - \varphi_1)} \end{pmatrix}, \quad \mathbb{A} = \begin{pmatrix} A_0 B_0 & A_0 B_1 \\ A_1 B_0 & A_1 B_1 \end{pmatrix}, \quad \mathbb{T} = \begin{pmatrix} c_0 \overline{d_0} & c_0 \overline{d_1} \\ c_1 \overline{d_0} & c_1 \overline{d_1} \end{pmatrix}.$$

The frequency convolution matrix is $\mathbb{F} = \begin{pmatrix} 0 & 0 \\ 0 & \pi^2 \end{pmatrix}$ $(\omega_0 = 0,\ \omega_1 = \pi)$.

Example 6.20 (Concrete Matrix for He Tu Phase Convolution). Take $T = 10$ and $f = g$ a He Tu complex function $f(z) = \sum_{n=0}^{9} c_n e^{2\pi i n z/10}$, where $c_n = A_n e^{i\phi_n}$. The phase convolution matrix $\mathbb{P}$ is a 10×10 matrix with entries $\mathbb{P}_{mn} = e^{i(\phi_m - \phi_n)}$. Under the decomposition $\mathbb{Z}_{10} \cong \mathbb{Z}_2 \times \mathbb{Z}_5$, $\mathbb{P}$ splits into four 5×5 blocks:

$$\mathbb{P} = \begin{pmatrix} \mathbb{P}^{(\text{yang,yang})} & \mathbb{P}^{(\text{yang,yin})} \\ \mathbb{P}^{(\text{yin,yang})} & \mathbb{P}^{(\text{yin,yin})} \end{pmatrix},$$

where the $(\text{yang}, \text{yang})$ block corresponds to $m, n \in \{0, 1, 2, 3, 4\}$, the (yin, yin) block to $m, n \in \{5, 6, 7, 8, 9\}$, and the cross blocks to the mixed yin–yang phase interaction.

6.10.2 Periodic Convolution: Complete Interaction of the Four-Parameter Operator

Definition 6.26 (Periodic Convolution). Let f, g be periodic functions (or almost periodic functions) with periodic-series expansions

$$f(t) = \sum_{p \in P} \sum_{n \in \mathbb{Z}} A_{p,n} e^{i(\omega_{p,n} t + \phi_{p,n})}, \qquad g(t) = \sum_{q \in Q} \sum_{m \in \mathbb{Z}} B_{q,m} e^{i(\nu_{q,m} t + \psi_{q,m})},$$

where $\omega_{p,n} = 2\pi n/T_p$ and $\nu_{q,m} = 2\pi m/T_q$. Define the **frequency composition operator** and the **phase composition operator** by

$$\Omega_{p,q,n,m} := \omega_{p,n} + \nu_{q,m}, \qquad \Phi_{p,q,n,m} := \phi_{p,n} + \psi_{q,m} \pmod{2\pi}.$$

The **periodic convolution** of f and g is defined as

$$(f \circledast g)(t) = \sum_{p\in P}\sum_{q\in Q}\sum_{n,m\in\mathbb{Z}} A_{p,n}B_{q,m} \cdot \mathsf{Similarity}^{(d)}(\theta_{p,n}, \theta_{q,m}) \cdot e^{i(\Omega_{p,q,n,m}t + \Phi_{p,q,n,m})},$$

where:

- $\theta_{p,n} = (\omega_{p,n}, \phi_{p,n})$ is the phase point of the (p, n)-th component of f;
- $\theta_{q,m} = (\nu_{q,m}, \psi_{q,m})$ is the phase point of the (q, m)-th component of g;
- $\mathsf{Similarity}^{(d)}$ is the d-dimensional phase similarity kernel ($d = 1, \ldots, 6$, corresponding to the 6 complex phase kernels, Proposition 6.30).

Periodic convolution takes the four parameters **amplitude** (scaling), **frequency** (translation), **phase** (rotation) and **period** (fixed length) as its operators, preserving the interaction of all frequency pairs and all period pairs. When $P = Q = \{T\}$ (a single period layer) and only the diagonal terms $n = m$ are retained, periodic convolution degenerates to classical convolution (Definition 6.19).

Proposition 6.31 (Degenerate Cases of Periodic Convolution)**.** (i) **Single period layer + diagonal kernel**: $P = Q = \{T\}$, $W \to \delta_{nm}$, $S^{(1)}$ (difference angle), $\alpha = 1$; then periodic convolution degenerates to classical convolution:

$$(f \circledast g)(t) \to (f * g)(t) = \sum_n c_n d_n e^{i\omega_n t}.$$

(ii) **Single period layer + full matrix**: $P = Q = \{T\}$, $W \equiv 1$, $S^{(1)}$, $\alpha = 1$; then periodic convolution degenerates to trend convolution:

$$(f \circledast g)_{n,m} = \mathbb{T}_{nm} = c_n\overline{d_m}.$$

(iii) **Single period layer + unit modulus**: $P = Q = \{T\}$, $W \equiv 1$, $S^{(1)}$, $\alpha = 0$; then periodic convolution degenerates to phase convolution:

$$(f \circledast g)_{n,m} = \mathbb{P}_{nm} = e^{i(\theta_n - \varphi_m)}.$$

Proof. (i) When $W \to \delta_{nm}$ the off-diagonal terms disappear and only the terms $n = m$ remain. From $S^{(1)}(n, n) = 1$ and $\Phi_{n,n} = \theta_n + \varphi_n$ we obtain $\sum_n A_n B_n e^{i(2\omega_n t + \theta_n + \varphi_n)} = \sum_n c_n d_n e^{i\omega_n t}$. (ii) When $W \equiv 1$ all frequency pairs are retained, and $\alpha = 1$ retains the amplitude products, giving the trend convolution matrix $\mathbb{T}_{nm} = c_n\overline{d_m}$. (iii) With $W \equiv 1$, $\alpha = 0$ discards the amplitudes, giving the phase convolution matrix $\mathbb{P}_{nm} = e^{i(\theta_n - \varphi_m)}$. □

Theorem 6.20 (Functoriality of Periodic Convolution)**.** Periodic convolution $\circledast$ is a bifunctor on the translation-periodic category, satisfying:

(i) **Preservation of identities**: $\mathsf{id}_f \circledast \mathsf{id}_g = \mathsf{id}_{f\circledast g}$.

(ii) **Preservation of composition**: $(h_2 \circ h_1) \circledast (k_2 \circ k_1) = (h_2 \circledast k_2) \circ (h_1 \circledast k_1)$.

(iii) **Compatibility with phase convolution**: $\mathbb{P}(f \circledast g) \cong \mathbb{P}(f) \otimes \mathbb{P}(g)$.

(iv) **Compatibility with trend convolution**: $\mathbb{T}(f \circledast g) \cong \mathbb{T}(f) \otimes \mathbb{T}(g)$.

Proof. (i)–(ii) Are guaranteed by the linearity of periodic convolution and the functoriality of the Fourier transform. (iii)–(iv) Are verified directly from the frequency-domain representation of periodic convolution (the outer product $\mathbf{c}_f \otimes \mathbf{c}_g$) and the definitions of phase and trend convolution. Periodic convolution is a tensor product in the frequency domain, whilst phase and trend are pointwise operations in the frequency domain. ☐

Proposition 6.32 (Relation between Periodic Convolution and the Four Classical Convolutions)**.** Periodic convolution is the summit of the five-stage ladder of convolution theory, and the degeneration relations between the successive stages are:

$$\text{periodic convolution}\circledast \xrightarrow{\text{single period layer}} \text{trend convolution}\mathbb{T} \xrightarrow{\text{diagonal kernel } W\to\delta} \text{classical convolution}$$

$$\text{periodic convolution}\circledast \xrightarrow{\text{single period layer}+\alpha=0} \text{phase convolution}\mathbb{P}.$$

In addition, trend convolution = amplitude convolution × phase convolution (term by term), and frequency convolution = the outer product of the frequency series; the three describe convolution from the three independent components of amplitude, phase and frequency.

Proof. It follows directly from Proposition 6.31 and 6.28. ☐

Remark 6.15 (The Complete Hierarchy of Five Stages). The five-stage ladder of convolution theory is: **classical convolution** (retaining only the delay products of equal frequency pairs $m = n$) → **phase convolution/amplitude convolution/frequency convolution** (retaining the phase, amplitude and frequency components of all frequency pairs respectively) → **trend convolution** (retaining the amplitudes and phases of all frequency pairs) → **periodic series** (decomposition into multiple period layers) → **periodic convolution** (a four-parameter operator, retaining the interaction of all frequency pairs and all period pairs). Each stage of the five-stage ladder is a degeneration of the preceding one, and the information content strictly increases from stage to stage.

Property 6.10 (The core parametrisation of the periodic convolution)**.** Periodic convolution (Definition 6.26) takes four parameters as its operators:

(i) **Amplitude** (scaling): the amplitude switch α controls whether the amplitude products $A_{p,n}B_{q,m}$ are retained;

(ii) **Frequency** (translation): the frequency composition operator $\Omega_{p,q,n,m} = \omega_{p,n} + \nu_{q,m}$ composes new frequencies;

(iii) **Phase** (rotation): the phase similarity kernel $\mathsf{Similarity}^{(d)}(\theta_{p,n}, \theta_{q,m})$ retains the phase relations;

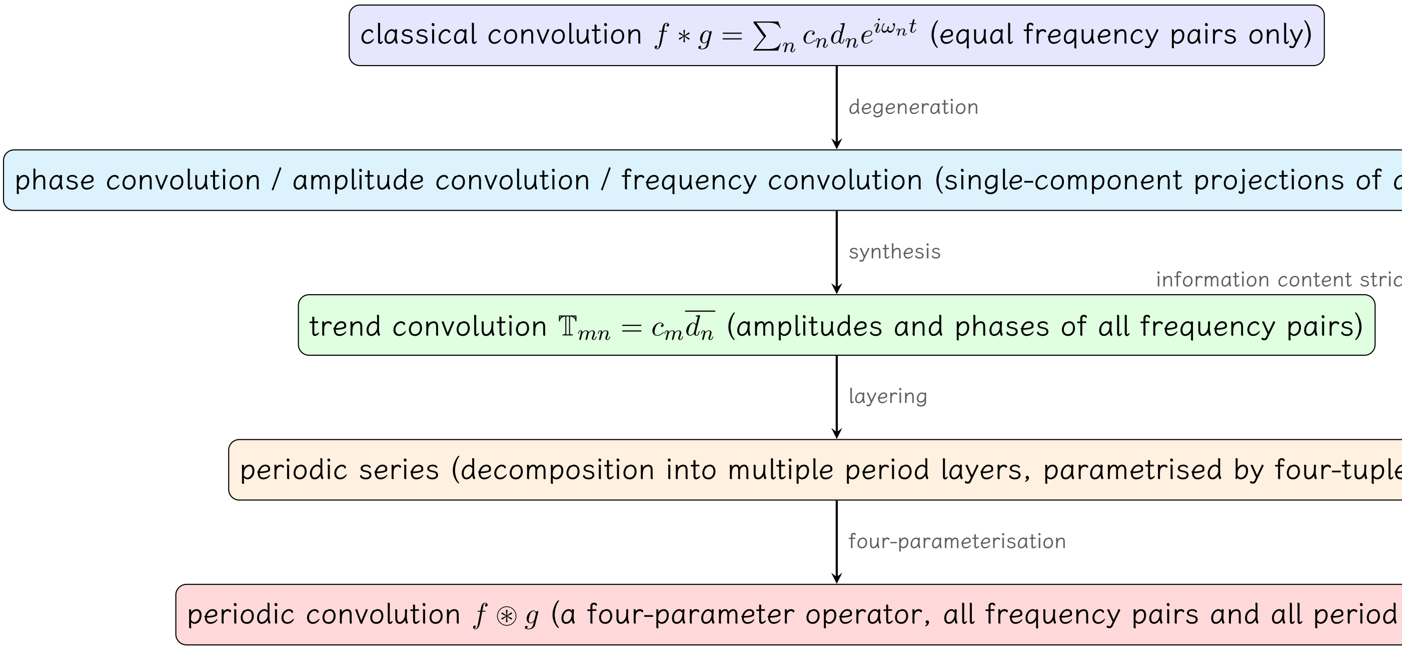

classical convolution $f * g = \sum_n c_n d_n e^{i\omega_n t}$ (equal frequency pairs only)
degeneration
phase convolution / amplitude convolution / frequency convolution (single-component projections of c
synthesis
information content stric
trend convolution $\mathbb{T}_{mn} = c_m \overline{d_n}$ (amplitudes and phases of all frequency pairs)
layering
periodic series (decomposition into multiple period layers, parametrised by four-tuple
four-parameterisation
periodic convolution $f \circledast g$ (a four-parameter operator, all frequency pairs and all period

Figure 6.9: The five-stage ladder of convolution theory. From classical convolution (the simplest, equal frequency pairs only) to periodic convolution (the summit, four-parameterised), each stage is a degeneration or a synthesis of the preceding one, and the information content strictly increases. Periodic convolution simultaneously retains the complete interaction information of the four components amplitude, frequency, phase and period.

(iv) **Period** (fixed length): the set of period-class indices $P \times Q$ retains the interaction of all period pairs.

The complete retention of the four parameters makes periodic convolution the operator with the greatest information content in convolution theory.

Proof. From the explicit formula for periodic convolution in Definition 6.26, the four parameters appear in four factors respectively: the amplitude $A_{p,n}B_{q,m}$, the frequency composition $\Omega_{p,q,n,m}$, the phase kernel Similarity$^{(d)}$ and the period index $P \times Q$. Proposition 6.31 shows that removing any one parameter degenerates the operator to the next stage of the ladder. $\square$

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	the local translation expansion of the Taylor series; the global translation period of the Fourier series; the translation property of convolution (Proposition 6.25); the half-period translation of He Tu complex functions (Proposition 6.10)
Scaling	the amplitude series (Definition 6.10); amplitude convolution (Definition 6.21); the amplitude products of trend convolution (Definition 6.24)
Rotation	the phase series (Definition 6.11); phase convolution (Definition 6.22); the 6 complex phase kernels (Proposition 6.30); the He Tu phase operator (Definition 6.9)
Conservation	Parseval's identity (Theorem 6.8); Parseval's identity for the amplitude series (Proposition 6.13); Bohr's generalised Parseval identity (Theorem 6.17)
Symmetry	the positive–negative power symmetry of the Laurent series; the orthogonal symmetry of the Fourier basis (Proposition 6.8); the symmetry of amplitude and frequency convolutions (Proposition 6.9); the direct product decomposition of the He Tu phase operator group (Proposition 6.11)
Closure	the uniqueness of the coefficients of the periodic series (Theorem 6.16); the functorial closure lattice of periodic convolution (Theorem 6.20)
Period	the frequency series (Definition 6.12); the four-tuple structure of the periodic series (Definition 6.14); the five-stage ladder of periodic convolution (Proposition 6.32); the period structure of the He Tu five-phase components (Proposition 6.9)

Exercises

Exercise 1 Prove the ratio test and the root test (Proposition 6.1): let $a_k \neq 0$; if

$$\lim_{k\to\infty}\left|\frac{a_{k+1}}{a_k}\right| = \rho \quad \text{or} \quad \lim_{k\to\infty}\sqrt[k]{|a_k|} = \rho,$$

then the series converges absolutely when $\rho < 1$, diverges when $\rho > 1$, and the test is inconclusive when $\rho = 1$. Please write out complete proofs of the two tests separately, and give one example of failure when $\rho = 1$ for each of them.

Exercise 2 Prove the Weierstrass M-test (Theorem 6.1): let $\sum_k M_k$ converge and let $|u_k(t)| \le M_k$ for all $t \in I$; then $\sum_k u_k(t)$ converges uniformly and absolutely on I. Then prove term-by-term integration under uniform convergence (Theorem 6.2) and term-by-term differentiation (Theorem 6.3).

Exercise 3 Find the convergence condition and the sum function of the geometric series $\sum_{k=0}^{\infty} r^k$ (Example 6.1), and use it to derive:

(i) $\displaystyle\sum_{k=0}^{\infty} k r^k = \frac{r}{(1-r)^2}$ $(|r| < 1)$;

(ii) $\displaystyle\sum_{k=0}^{\infty} k^2 r^k = \frac{r(1+r)}{(1-r)^3}$ $(|r| < 1)$.

Exercise 4 Write out the Taylor series of e^x, $\sin x$, $\cos x$, $\ln(1+x)$ and $(1+x)^\alpha$ (Example 6.3), and use the Taylor formula with Lagrange remainder (Theorem 6.5) to prove that

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

holds for all $x \in \mathbb{R}$. Further explain the local–global duality between the Taylor expansion and the Fourier expansion (Remark 6.3).

Exercise 5 Verify the uniqueness of the coefficients of a Laurent series (Lemma 6.1), and write out the Laurent expansion of $e^{1/z}$ on $|z| > 0$ (Example 6.4). According to the form of the principal part, determine which type of singularity $z_0 = 0$ is for $e^{1/z}$ (Proposition 6.7).

Exercise 6 Let f have period T and be integrable on $[0, T]$, with complex Fourier coefficients c_n. Prove the orthogonality of the Fourier basis (Proposition 6.8):

$$\frac{1}{T}\int_0^T e^{i2\pi(n-m)t/T}\, dt = \delta_{nm}.$$

And use orthogonality to prove Parseval's identity (Theorem 6.8):

$$\frac{1}{T}\int_0^T |f(t)|^2\, dt = \sum_{n\in\mathbb{Z}} |c_n|^2.$$

Exercise 7 Write out the Fourier series of the square wave (taking 1 on $[0, T/2)$ and -1 on $[T/2, T)$) (Example 6.6). Explain the convergence value given by the Dirichlet convergence theorem (Theorem 6.9) at the jump point, and explain why the Gibbs phenomenon (Example 6.7) does not disappear as the number of terms increases.

Exercise 8 Prove the Fejér kernel and uniform convergence theorem (Theorem 6.10): let f be continuous and have period T; the Fejér means

$$\sigma_N f(t) = \frac{1}{N+1}\sum_{k=0}^{N} S_k f(t)$$

converge uniformly to f on $\mathbb{R}$. Please explain that the Fejér kernel is a good kernel (non-negative, of integral 1, tending to zero uniformly away from the origin), and explain why the Fejér means can avoid the Gibbs phenomenon.

Exercise 9 Prove the He Tu Fourier decomposition (Theorem 6.11): let f satisfy $f(z+10)=f(z)$ and let its Fourier expansion converge uniformly; then

$$f(z)=\sum_{k=0}^{4}\left(c_k+c_{k+5}e^{\pi iz}\right)e^{2\pi ikz/10}.$$

Taking the concrete function

$$f(z)=3+2e^{\pi iz/5}+e^{\pi iz}+5e^{3\pi iz/5}+4e^{2\pi iz},$$

write out its five $\mathbb{Z}_5$ components, and verify that the half-period translation $z\mapsto z+5$ changes the sign of the odd components (Proposition 6.10).

Exercise 10 Prove the relation between the phase series and time translation (Proposition 6.15): if $\mathcal{P}_f(t)=\sum_n e^{i(nt+\phi_n)}$, then the time translation $f_\tau(t)=f(t-\tau)$ corresponds to rotating each term of the phase series by $-n\tau$. Then prove modulus conservation for finite phase series (Theorem 6.13):

$$\frac{1}{2\pi}\int_0^{2\pi}|S_N(t)|^2dt=2N+1.$$

Exercise 11 Prove Parseval's identity for the amplitude series (Proposition 6.13): let $f\in L^2([0,2\pi])$ have Fourier coefficients $c_n=A_ne^{i\phi_n}$; then

$$\frac{1}{2\pi}\int_0^{2\pi}|f(t)|^2dt=\sum_{n\in\mathbb{Z}}A_n^2.$$

And explain that the energy is determined by the amplitude spectrum alone and is independent of the phase spectrum (Corollary 6.4).

Exercise 12 Prove the degenerate cases of periodic convolution (Proposition 6.31):

- (i) for a single period layer with $W\to\delta_{nm}$, $S^{(1)}$ and $\alpha=1$, periodic convolution degenerates to classical convolution;
- (ii) for a single period layer with $W\equiv 1$, $S^{(1)}$ and $\alpha=1$, periodic convolution degenerates to trend convolution $\mathbb{T}_{nm}=c_n\overline{d_m}$;
- (iii) for a single period layer with $W\equiv 1$, $S^{(1)}$ and $\alpha=0$, periodic convolution degenerates to phase convolution $\mathbb{P}_{nm}=e^{i(\theta_n-\varphi_m)}$.

Exercise 13 Synthesis problem: explain that the amplitude series, the phase series and the frequency series are respectively the projections of the periodic series onto its three components (Proposition 6.21 and Corollary 6.6). Taking the periodic series

$$f(t)=3e^{i(\sqrt{2}t+\pi/6)}+2e^{i(t-\pi/3)}$$

as an example, write out explicitly its amplitude series, phase series and frequency series respectively, and point out why any single projection is insufficient to reconstruct f.

Chapter 7

Complex Analysis

Remark 7.1 (Positioning of this chapter). Building on the 'Number Systems', 'Functions and Limits', 'Real Analysis' and 'Series and Convolution' chapters, this chapter establishes the analytic theory of complex-valued functions. It does not repeat the construction of the complex field given in the 'Number Systems' chapter, nor the general theory of Taylor series and Laurent series given in the 'Series and Convolution' chapter; it focuses on the content specific to complex analysis: analytic functions, the Cauchy–Riemann equations, Cauchy's integral theorem, Cauchy's integral formula, the local expansion of analytic functions, the classification of isolated singularities, the residue theorem, the argument principle, Rouché's theorem, harmonic functions and the elements of conformal mapping.

This chapter provides the foundation for the Fourier transform, the Laplace transform and residue calculus in the 'Harmonic Analysis' chapter, supplies tools for the two-dimensional complex potential and harmonic functions in the 'Vector Analysis' chapter, and furnishes the language for complex eigenvalues and analytic solutions in the 'Differential Equations and Analysis' chapter.

Guide to the seven themes of this chapter: Translation — translation on the complex plane and the invariance of analytic functions under translation; Scaling — the scaling of the modulus of a complex number and the angle preservation of the Cauchy–Riemann equations; Rotation — multiplication of complex numbers as rotation, the periodicity of $e^{i\theta}$ and roots of unity; Conservation — the conservation of the loop integral in Cauchy's integral theorem; Symmetry — the conjugate symmetry of analytic functions and the Schwarz reflection principle; Closure — the vanishing of the integral over a closed path in Cauchy's integral theorem; Period — the $2\pi i$-periodicity of the complex exponential function and the group of roots of unity.

Conventions: wherever general results on series are used — Taylor series, Laurent series, the Cauchy–Hadamard radius-of-convergence formula and the like — reference is always made to the corresponding definitions and theorems already established in Chapter 6, 'Series and Convolution' (such as Definition 6.3, Definition 6.4, Theorem 6.4, Theorem 6.6); wherever analytic tools such as uniform convergence, integrals with a parameter and line integrals are used, reference is always made to the corresponding definitions and theorems already established in Chapter 5, 'Real Analysis' (such as Definition 5.27, Theorem 5.24, Theorem 5.25, Definition 5.22). This chapter does not repeat these general theories; it uses their conclusions only in the context of complex analysis.

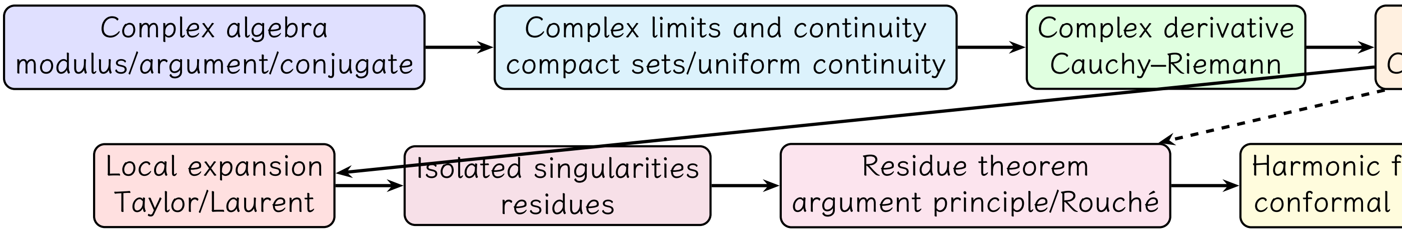

Complex algebra
modulus/argument/conjugate
Complex limits and continuity
compact sets/uniform continuity
Complex derivative
Cauchy–Riemann
Local expansion
Taylor/Laurent
Isolated singularities
residues
Residue theorem
argument principle/Rouché
Harmonic f
conformal

Figure 7.1: Overview of the structure of this chapter: complex algebra → complex limits and continuity → complex derivative → complex integration → local expansion → isolated singularities and residues → harmonic functions and conformal mapping.

7.1 Review of complex numbers and the complex plane

7.1.1 The complex field and the basic operations

Definition 7.1 (The complex field). Let $\mathbb{C} = \{(a,b) : a,b \in \mathbb{R}\}$, equipped with the addition $(a,b)+(c,d) = (a+c,b+d)$ and the multiplication $(a,b)(c,d) = (ac-bd,ad+bc)$. Write $i = (0,1)$; then every complex number can be written as $z = a+bi$. The number $a = \operatorname{Re} z$ is called the **real part** and $b = \operatorname{Im} z$ the **imaginary part**.

Proposition 7.1 (Elementary properties of the complex field). (i) *$(\mathbb{C},+,\times)$ is a field, the identity element of addition is 0 and the identity element of multiplication is 1.*

(ii) *$i^2 = -1$, and $\mathbb{C} = \mathbb{R}[i] \cong \mathbb{R}[x]/(x^2+1)$.*

(iii) *For every $z \in \mathbb{C}$, $z = \operatorname{Re} z + i \operatorname{Im} z$.*

(iv) *The conjugation map $\overline{a+bi} = a-bi$ satisfies $\overline{z+w} = \bar z + \bar w$, $\overline{zw} = \bar z \bar w$ and $\bar{\bar z} = z$.*

(v) *The modulus $|z| = \sqrt{z\bar z} = \sqrt{a^2+b^2} \ge 0$ satisfies $|zw| = |z||w|$ and $|z+w| \le |z|+|w|$.*

Proof. (i) This follows from the construction of the complex field in the 'Number Systems' chapter. The associativity and commutativity of addition, the zero element $(0,0)$ and the inverse $-(a,b) = (-a,-b)$ are inherited componentwise from the additive group structure of $\mathbb{R}$. The associativity and commutativity of multiplication and the identity element $(1,0)$ are verified directly from the definition; the multiplicative inverse is $(a,b)^{-1} = (a/(a^2+b^2), -b/(a^2+b^2))$ (when $(a,b) \neq (0,0)$). The distributive law is verified by expanding componentwise. All the field axioms are satisfied.

(ii) From $(0,1)(0,1) = (-1,0) = -1$ we obtain $i^2 = -1$. The map $\mathbb{R}[x]/(x^2+1) \to \mathbb{C}$, $[x] \mapsto i$, is a field isomorphism.

(iii) By definition, $a+bi = (a,0)+(0,1)(b,0) = (a,0)+(0,b) = (a,b)$.

(iv) $\overline{z+w} = \overline{(a+c)+(b+d)i} = (a+c)-(b+d)i = (a-bi)+(c-di) = \bar z + \bar w$. $\overline{zw} = \overline{(ac-bd)+(ad+bc)i} = (ac-bd)-(ad+bc)i = (a-bi)(c-di) = \bar z \bar w$. $\bar{\bar z} = \overline{a-bi} = a+bi = z$.

(v) $z\bar z = (a+bi)(a-bi) = a^2+b^2 \ge 0$, hence $|z| = \sqrt{a^2+b^2} \ge 0$. From $|zw|^2 = (zw)\overline{(zw)} = zw\bar z\bar w = (z\bar z)(w\bar w) = |z|^2|w|^2$, taking square roots gives $|zw| = |z||w|$.

The triangle inequality follows from the Cauchy–Schwarz inequality (applied in $\mathbb{R}^2$). ☐

Definition 7.2 (Polar coordinates and argument)**.** A non-zero complex number $z = a + bi$ can be written uniquely as

$$z = re^{i\theta}, \qquad r = |z| > 0,\ \theta \in \mathbb{R},$$

where $e^{i\theta} = \cos\theta + i\sin\theta$ (Euler's formula) and θ is called an **argument** of z, written $\theta = \arg z$. The argument is not unique: if θ is an argument of z, then so is $\theta + 2k\pi$ ($k \in \mathbb{Z}$). We prescribe $\operatorname{Arg} z \in (-\pi, \pi]$ as the **principal argument**.

Theorem 7.1 (Euler's formula)**.** *For every* $\theta \in \mathbb{R}$,

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

Proof. By power-series expansion,

$$e^{i\theta} = \sum_{k=0}^{\infty} \frac{(i\theta)^k}{k!} = \sum_{j=0}^{\infty} \frac{(i\theta)^{2j}}{(2j)!} + \sum_{j=0}^{\infty} \frac{(i\theta)^{2j+1}}{(2j+1)!}.$$

Since $i^{2j} = (-1)^j$ and $i^{2j+1} = (-1)^j i$, we have

$$e^{i\theta} = \sum_{j=0}^{\infty} \frac{(-1)^j \theta^{2j}}{(2j)!} + i\sum_{j=0}^{\infty} \frac{(-1)^j \theta^{2j+1}}{(2j+1)!} = \cos\theta + i\sin\theta.$$

The two series are the Taylor expansions of $\cos\theta$ and $\sin\theta$ respectively. ☐

Proposition 7.2 (Multiplication and division in polar form)**.** Let $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$ $(r_1, r_2 > 0)$. Then:

(i) $z_1 z_2 = r_1 r_2 e^{i(\theta_1+\theta_2)}$;

(ii) $z_1/z_2 = (r_1/r_2) e^{i(\theta_1-\theta_2)}$;

(iii) $z_1^n = r_1^n e^{in\theta_1}$ ($n \in \mathbb{Z}$; for $n \geq 0$, $z_1^0 = 1$; for $n < 0$, $z_1^n = (z_1^{-1})^{-n}$).

Proof. (i) By the addition theorem for the exponential function $e^{i\theta_1}e^{i\theta_2} = e^{i(\theta_1+\theta_2)}$ (which follows from Euler's formula and the trigonometric identities):

$$z_1 z_2 = r_1 e^{i\theta_1} r_2 e^{i\theta_2} = r_1 r_2 e^{i(\theta_1+\theta_2)}.$$

(ii) Since $z_2^{-1} = r_2^{-1} e^{-i\theta_2}$, we have $z_1/z_2 = z_1 z_2^{-1} = (r_1/r_2) e^{i(\theta_1-\theta_2)}$. (iii) Induction on n; a direct corollary of (i). ☐

Theorem 7.2 (The group of roots of unity)**.** *For every* $n \in \mathbb{N}^+$, *the equation* $z^n = 1$ *has exactly* n *solutions in* $\mathbb{C}$:

$$\zeta_k = e^{2\pi i k/n}, \qquad k = 0, 1, \ldots, n-1.$$

These n *solutions form the cyclic group* C_n *of order* n.

Proof. Let $z = re^{i\theta}$. The equation $z^n = r^n e^{in\theta} = 1$ requires $r^n = 1$ and $n\theta \in 2\pi\mathbb{Z}$. Since $r \geq 0$, we have $r = 1$; and $\theta = 2\pi k/n$ $(k \in \mathbb{Z})$. In $[0, 2\pi)$ there are n distinct values $\theta_k = 2\pi k/n$, $k = 0, \ldots, n-1$. Hence the n solutions are $\zeta_k = e^{2\pi ik/n}$. Closure under multiplication: $\zeta_j\zeta_k = e^{2\pi i(j+k)/n} = \zeta_{(j+k) \bmod n}$. The identity element is $\zeta_0 = 1$ and the inverse is $\zeta_k^{-1} = \zeta_{n-k}$. The group is generated by ζ_1, hence it is C_n. ☐

Example 7.1 (Examples of complex arithmetic).

(i) $(3+4i)(1-2i) = 3-6i+4i-8i^2 = 3-2i+8 = 11-2i$.

(ii) Polar form of $z = 1+i$: $r = \sqrt{2}$, $\theta = \pi/4$, hence $z = \sqrt{2}e^{i\pi/4}$.

(iii) $z^4 = (1+i)^4 = (\sqrt{2}e^{i\pi/4})^4 = 4e^{i\pi} = -4$.

(iv) The case $n = 4$ of the roots of unity: $\zeta_0 = 1$, $\zeta_1 = i$, $\zeta_2 = -1$, $\zeta_3 = -i$.

7.1.2 Topology on the complex plane

Definition 7.3 (Open, closed, and compact sets on the complex plane). *Identify $\mathbb{C}$ with $\mathbb{R}^2$ $(a+bi \leftrightarrow (a,b))$; then all topological notions on $\mathbb{C}$ (open set, closed set, compact set, connected, simply connected) are inherited from the standard topology of $\mathbb{R}^2$.*

(i) *The open ball $B(z_0, r) = \{z : |z - z_0| < r\}$, called the* **open disc***.*

(ii) *The closed ball $\overline{B}(z_0, r) = \{z : |z - z_0| \leq r\}$, called the* **closed disc***.*

(iii) *The circle $S(z_0, r) = \{z : |z - z_0| = r\}$.*

(iv) *Domain: a connected open set.*

(v) *Simply connected domain: a domain whose fundamental group is trivial.*

Definition 7.4 (Convergence of complex sequences and Cauchy sequences). *Let $\{z_k\}$ be a sequence in $\mathbb{C}$ and $z \in \mathbb{C}$.*

(i) $z_k \to z$*:* $\forall\varepsilon > 0$, $\exists K$, $\forall k \geq K$, $|z_k - z| < \varepsilon$.

(ii) $\{z_k\}$ *is a* **Cauchy sequence***:* $\forall\varepsilon > 0$, $\exists K$, $\forall m, k \geq K$, $|z_m - z_k| < \varepsilon$.

Proposition 7.3 (Coordinate characterisation of complex-sequence convergence). *Let $z_k = x_k + iy_k$ and $z = x + iy$. Then $z_k \to z$ if and only if $x_k \to x$ and $y_k \to y$.*

Proof. $|x_k - x| \leq |z_k - z|$, $|y_k - y| \leq |z_k - z|$, $|z_k - z| \leq |x_k - x| + |y_k - y|$. Hence the conditions are equivalent. ☐

Theorem 7.3 (Completeness of $\mathbb{C}$). *Every Cauchy sequence in $\mathbb{C}$ converges.*

Proof. Let $\{z_k\}$ be a Cauchy sequence with $z_k = x_k + iy_k$. Since $|x_m - x_k| \leq |z_m - z_k|$, $\{x_k\}$ is a Cauchy sequence in $\mathbb{R}$ and converges to some x by the completeness of $\mathbb{R}$. Similarly $\{y_k\}$ converges to y. Hence $z_k \to x + iy = z$. ☐

Definition 7.5 (Compactness and connectedness).

(i) *$K \subseteq \mathbb{C}$ is compact if K is a bounded closed set in $\mathbb{C} \cong \mathbb{R}^2$ (an equivalent condition, by the Heine–Borel theorem).*

(ii) $K \subseteq \mathbb{C}$ is connected if it cannot be written as the disjoint union of two non-empty relatively open sets.

(iii) $K \subseteq \mathbb{C}$ is path-connected if any two points can be joined by a continuous curve.

(iv) A domain Ω is simply connected if every closed curve in Ω can be contracted continuously to a point within Ω.

Example 7.2 (Examples of complex-plane topology). (i) The open disc $B(0,1)$ is a simply connected domain.

(ii) The punctured disc $B(0,1) \setminus \{0\}$ is a domain, but it is not simply connected (the circle cannot be contracted).

(iii) The upper half-plane $\{z : \operatorname{Im} z > 0\}$ is a simply connected domain.

(iv) The annulus $\{z : 1 < |z| < 2\}$ is a domain, but it is not simply connected.

(v) The slit plane $\mathbb{C} \setminus (-\infty, 0]$ is a simply connected domain.

7.2 Complex-valued functions, limits and continuity

7.2.1 Complex-valued functions

Definition 7.6 (Complex-valued function). Let $\Omega \subseteq \mathbb{C}$ and $f : \Omega \to \mathbb{C}$. The map f is called a **complex-valued function**. Writing $z = x + iy$, we have

$$f(z) = u(x,y) + iv(x,y),$$

where $u, v : \Omega_{\mathbb{R}} \to \mathbb{R}$ are real-valued functions and $\Omega_{\mathbb{R}} = \{(x,y) \in \mathbb{R}^2 : x + iy \in \Omega\}$. Here $u = \operatorname{Re} f$ is called the **real part** and $v = \operatorname{Im} f$ the **imaginary part**.

Definition 7.7 (Complex limit and continuity). Let $\Omega \subseteq \mathbb{C}$, $f : \Omega \to \mathbb{C}$ and $z_0 \in \Omega$.

(i) f is **continuous** at z_0 if

$$\forall \varepsilon > 0,\ \exists \delta > 0,\ \forall z \in \Omega, \quad |z - z_0| < \delta \implies |f(z) - f(z_0)| < \varepsilon.$$

(ii) f is continuous on Ω if it is continuous at every point.

(iii) f is **uniformly continuous** on Ω if

$$\forall \varepsilon > 0,\ \exists \delta > 0,\ \forall z, w \in \Omega, \quad |z - w| < \delta \implies |f(z) - f(w)| < \varepsilon.$$

Proposition 7.4 (Equivalent characterisations of continuity). Let $\Omega \subseteq \mathbb{C}$, $f : \Omega \to \mathbb{C}$ and $z_0 \in \Omega$. Then the following conditions are equivalent:

(i) f is continuous at z_0;

(ii) for every sequence $z_k \to z_0$ in Ω, we have $f(z_k) \to f(z_0)$;

(iii) $u = \operatorname{Re} f$ and $v = \operatorname{Im} f$ are continuous at z_0.

Proof. (i) $\Rightarrow$ (ii): the same as the sequential characterisation of continuity in real analysis.

(ii) $\Rightarrow$ (i): by contradiction.

(i) $\Leftrightarrow$ (iii): equivalent, by $|u(z)-u(z_0)| \leq |f(z)-f(z_0)|$ and $|f(z)-f(z_0)| \leq |u(z)-u(z_0)|+|v(z)-v(z_0)|$. ☐

Theorem 7.4 (Properties of continuous functions on compact sets). Let $K \subseteq \mathbb{C}$ be compact and $f : K \to \mathbb{C}$ continuous. Then:

(i) $f(K)$ is compact;

(ii) f is bounded on K;

(iii) $|f|$ attains its maximum and minimum on K;

(iv) f is uniformly continuous on K.

Proof. This follows item by item from $\mathbb{C} \cong \mathbb{R}^2$ and the properties of continuous functions on compact sets in real analysis (Theorem 5.7). ☐

Example 7.3 (Examples of complex-valued functions). (i) The polynomial $f(z) = a_n z^n + \cdots + a_0$ $(a_i \in \mathbb{C})$ is continuous on $\mathbb{C}$.

(ii) The exponential function $f(z) = e^z = e^x(\cos y + i \sin y)$ is continuous on $\mathbb{C}$.

(iii) The trigonometric function $f(z) = \sin z = (e^{iz} - e^{-iz})/(2i)$ is continuous on $\mathbb{C}$.

(iv) $f(z) = 1/z$ is continuous on $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$.

(v) $f(z) = \operatorname{Re} z$ and $g(z) = \operatorname{Im} z$ are continuous on $\mathbb{C}$.

7.3 Complex derivatives and analytic functions

7.3.1 The complex derivative

Definition 7.8 (Complex derivative). Let $\Omega \subseteq \mathbb{C}$ be an open set, $f : \Omega \to \mathbb{C}$ and $z_0 \in \Omega$. The function f is **complex differentiable** at z_0 if the limit

$$f'(z_0) := \lim_{h \to 0} \frac{f(z_0+h) - f(z_0)}{h}$$

exists ($h \in \mathbb{C}$, $h \neq 0$). The number $f'(z_0)$ is called the **complex derivative** of f at z_0.

Remark 7.2 (The essential difference between complex and real derivatives). The complex derivative requires that, as $h \to 0$, the quotient $(f(z_0+h) - f(z_0))/h$ converge to one and the same value no matter along which path in the complex plane h tends to zero. This differs from the real derivative, which requires only that h tend to zero along the two directions of the real axis. Complex differentiability is therefore a stronger condition than real differentiability — it requires the local behaviour of f near z_0 to possess a 'complex-linear' structure.

Theorem 7.5 (The Cauchy–Riemann equations). Let $\Omega \subseteq \mathbb{C}$ be an open set, $f = u + iv : \Omega \to \mathbb{C}$ and $z_0 = x_0 + iy_0 \in \Omega$. If f is complex differentiable at z_0, then the partial derivatives of u, v exist at (x_0, y_0) and satisfy the **Cauchy–Riemann equations**

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

In that case

$$f'(z_0) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}.$$

Proof. Consider $h \to 0$ along the real axis ($h = \Delta x$, $\Delta y = 0$):

$$f'(z_0) = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x} = \frac{\partial u}{\partial x}(x_0, y_0) + i\frac{\partial v}{\partial x}(x_0, y_0).$$

Along the imaginary axis ($h = i\Delta y$, $\Delta x = 0$):

$$f'(z_0) = \lim_{\Delta y \to 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{i\Delta y} = \frac{1}{i}\left(\frac{\partial u}{\partial y} + i\frac{\partial v}{\partial y}\right) = \frac{\partial v}{\partial y}(x_0, y_0) - i\frac{\partial u}{\partial y}(x_0, y_0).$$

(We have used $1/i = -i$.) Since f is complex differentiable, the two limits must be equal. Comparing real and imaginary parts:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}.$$

That is, the Cauchy–Riemann equations hold. □

Remark 7.3 (Sufficiency of the Cauchy–Riemann equations). The Cauchy–Riemann equations are a necessary condition for complex differentiability, but not a sufficient one. If the partial derivatives of u, v exist at (x_0, y_0), satisfy the Cauchy–Riemann equations, and are continuous at (x_0, y_0), then f is complex differentiable at z_0 (see Corollary 7.1).

Theorem 7.6 (The Cauchy–Riemann equations and the C^1 sufficiency). Let $\Omega \subseteq \mathbb{C}$ be an open set and $f = u + iv : \Omega \to \mathbb{C}$. If u, v are C^1 on Ω (that is, have continuous first-order partial derivatives) and satisfy the Cauchy–Riemann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x},$$

everywhere, then f is complex differentiable on Ω, and

$$f'(z) = \frac{\partial u}{\partial x}(x, y) + i\frac{\partial v}{\partial x}(x, y).$$

Proof. Fix $z_0 = x_0 + iy_0 \in \Omega$. Since $u, v \in C^1$, the definition of differentiability in real analysis gives

$$u(x_0 + \Delta x, y_0 + \Delta y) - u(x_0, y_0) = u_x \Delta x + u_y \Delta y + o(\sqrt{\Delta x^2 + \Delta y^2}),$$

$$v(x_0 + \Delta x, y_0 + \Delta y) - v(x_0, y_0) = v_x \Delta x + v_y \Delta y + o(\sqrt{\Delta x^2 + \Delta y^2}),$$

where the partial derivatives are evaluated at (x_0, y_0). Hence

$$f(z_0 + h) - f(z_0) = (u_x + iv_x)\Delta x + (u_y + iv_y)\Delta y + o(|h|).$$

By the Cauchy–Riemann equations, $u_y = -v_x$ and $v_y = u_x$. Hence

$$(u_y + iv_y) = (-v_x + iu_x) = i(u_x + iv_x).$$

Therefore

$$\begin{aligned} f(z_0 + h) - f(z_0) &= (u_x + iv_x)\Delta x + i(u_x + iv_x)\Delta y + o(|h|) \\ &= (u_x + iv_x)(\Delta x + i\Delta y) + o(|h|) = (u_x + iv_x)h + o(|h|). \end{aligned}$$

Hence

$$\lim_{h\to 0} \frac{f(z_0 + h) - f(z_0)}{h} = u_x + iv_x.$$

That is, f is complex differentiable at z_0, with $f'(z_0) = u_x + iv_x$. □

Corollary 7.1 (C^1 plus Cauchy–Riemann implies analyticity)**.** *If $u, v \in C^1(\Omega)$ and the Cauchy–Riemann equations hold everywhere, then $f = u + iv$ is complex differentiable on Ω.*

Proof. This is Theorem 7.6. □

7.3.2 Analytic functions

Definition 7.9 (Analytic function)**.** Let $\Omega \subseteq \mathbb{C}$ be an open set and $f : \Omega \to \mathbb{C}$. If f is complex differentiable at every point of Ω, then f is said to be **analytic** (or **holomorphic**) on Ω. If f is analytic in some neighbourhood of z_0, then f is said to be analytic at z_0.

Definition 7.10 (Entire function)**.** A function that is analytic everywhere on $\mathbb{C}$ is called an **entire function**.

Proposition 7.5 (Algebraic properties of analytic functions)**.** *Let f, g be analytic on Ω. Then:*

(i) $f + g$ is analytic on Ω, and $(f + g)' = f' + g'$;

(ii) fg is analytic on Ω, and $(fg)' = f'g + fg'$;

(iii) if g has no zeros on Ω, then f/g is analytic on Ω, and $(f/g)' = (f'g - fg')/g^2$;

(iv) if f is analytic on Ω and g is analytic on $f(\Omega)$, then $g \circ f$ is analytic on Ω, and $(g \circ f)' = (g' \circ f)f'$.

Proof. Each assertion can be verified directly from the definition of the complex derivative; the rules are the same as the corresponding differentiation rules in real analysis (with complex limits in place of real limits).

(i) Directly from the linearity of limits.

(ii) Expanding from the definition,

$$\frac{f(z + h)g(z + h) - f(z)g(z)}{h} = f(z + h)\frac{g(z + h) - g(z)}{h} + g(z)\frac{f(z + h) - f(z)}{h}.$$

Since complex differentiability of f implies continuity, $f(z+h) \to f(z)$. Passing to the limit gives $(fg)'(z) = f(z)g'(z) + g(z)f'(z)$.

(iii) First we prove $(1/g)' = -g'/g^2$:

$$\frac{1/g(z+h) - 1/g(z)}{h} = -\frac{g(z+h) - g(z)}{h} \cdot \frac{1}{g(z+h)g(z)} \to -\frac{g'(z)}{g(z)^2}.$$

Then (ii) gives $(f/g)' = f' \cdot 1/g + f \cdot (-g'/g^2) = (f'g - fg')/g^2$.

(iv) The chain rule is the same as the differentiation of composite functions in real analysis: decompose $(g \circ f)(z+h) - (g \circ f)(z)$ as $g(f(z+h)) - g(f(z))$, divide by $f(z+h) - f(z)$ (when $f(z+h) \neq f(z)$) and multiply by $f(z+h) - f(z)$ divided by h; passing to the limit gives the result. ☐

Theorem 7.7 (The real and imaginary parts of an analytic function are harmonic). Let $f = u + iv$ be analytic on Ω, with $u, v \in C^2(\Omega)$. Then both u and v are **harmonic functions**:

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \qquad \Delta v = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0.$$

Proof. By the Cauchy–Riemann equations, $u_x = v_y$ and $u_y = -v_x$. If $u, v \in C^2$, the mixed partial derivatives commute, so

$$u_{xx} = (v_y)_x = v_{yx} = v_{xy} = (-u_y)_y = -u_{yy},$$

that is, $u_{xx} + u_{yy} = 0$. Similarly $v_{xx} + v_{yy} = 0$. ☐

Definition 7.11 (Harmonic conjugate). Let $u : \Omega \to \mathbb{R}$ be a harmonic function. If there is a harmonic function $v : \Omega \to \mathbb{R}$ such that $u + iv$ is analytic on Ω, then v is called a **harmonic conjugate** of u.

Proposition 7.6 (Existence of a harmonic conjugate on a simply connected domain). Let $\Omega \subseteq \mathbb{C}$ be a simply connected domain and $u : \Omega \to \mathbb{R}$ a harmonic function. Then u has a harmonic conjugate v on Ω.

Proof. Since u is harmonic, $u_{xx} + u_{yy} = 0$, that is, $(u_x)_x = (-u_y)_y$, so $(u_x, -u_y)$ satisfies the integrability condition. Define

$$v(x,y) := \int_{(x_0,y_0)}^{(x,y)} (-u_y)\,dx + u_x\,dy,$$

where the integral is taken along any path in Ω from (x_0, y_0) to (x, y). Since Ω is simply connected and the vector field $(-u_y, u_x)$ satisfies the vanishing-curl condition, the integral is independent of the path and v is well defined. A computation gives $v_x = -u_y$ and $v_y = u_x$, so v is a harmonic conjugate of u. ☐

Example 7.4 (Examples of analytic functions).

(i) The polynomial $f(z) = \sum_{k=0}^{n} a_k z^k$ is analytic on $\mathbb{C}$, with $f'(z) = \sum_{k=1}^{n} k a_k z^{k-1}$.

(ii) The exponential function $f(z) = e^z = e^x(\cos y + i \sin y)$, with $u = e^x \cos y$ and $v = e^x \sin y$. Verify the Cauchy–Riemann equations: $u_x = e^x \cos y = v_y$, $u_y = -e^x \sin y = -v_x$. Hence e^z is entire, and $(e^z)' = e^z$.

(iii) The trigonometric functions $\sin z = (e^{iz} - e^{-iz})/(2i)$ and $\cos z = (e^{iz} + e^{-iz})/2$ are entire and satisfy $(\sin z)' = \cos z$ and $(\cos z)' = -\sin z$.

(iv) $f(z) = 1/z$ is analytic on $\mathbb{C} \setminus \{0\}$, with $f'(z) = -1/z^2$.

(v) $f(z) = \operatorname{Re} z = x$ is continuous on $\mathbb{C}$ but nowhere analytic: $u_x = 1$ and $v_y = 0$, so the Cauchy–Riemann equations fail.

(vi) $f(z) = |z|^2 = x^2 + y^2$ is continuous on $\mathbb{C}$, but it satisfies the Cauchy–Riemann equations only at $z = 0$; hence it is complex differentiable only at $z = 0$ and is not analytic on $\mathbb{C}$.

Theorem 7.8 (Periodicity and derivative of the complex exponential). *$f(z) = e^z$ is entire and satisfies $(e^z)' = e^z$, and*

$$e^{z+2\pi i} = e^z, \qquad \forall z \in \mathbb{C}.$$

That is, e^z has the purely imaginary period $2\pi i$.

Proof. By Example 7.4(ii), e^z is entire and $(e^z)' = e^z$.

Periodicity: $e^{z+2\pi i} = e^{x+i(y+2\pi)} = e^x(\cos(y+2\pi)+i\sin(y+2\pi)) = e^x(\cos y+i\sin y) = e^z$. ☐

Corollary 7.2 (The kernel of the complex exponential). *$\{z \in \mathbb{C} : e^z = 1\} = 2\pi i\mathbb{Z}$.*

Proof. The equation $e^z = e^x(\cos y+i\sin y) = 1$ requires $e^x = 1$ and $\cos y+i\sin y = 1$. Hence $x = 0$ and $y = 2\pi k$ ($k \in \mathbb{Z}$), that is, $z = 2\pi ik$. ☐

Example 7.5 (Multivaluedness of the complex logarithm). The complex logarithm is defined as the inverse function of $z = e^w$. If $z = re^{i\theta}$ $(r > 0)$, then

$$\log z = \ln r + i(\theta + 2k\pi), \qquad k \in \mathbb{Z}.$$

The function $\log z$ is multivalued; its principal value $\operatorname{Log} z = \ln r + i\theta$ $(\theta \in (-\pi, \pi])$ is analytic on $\mathbb{C} \setminus (-\infty, 0]$, and $(\operatorname{Log} z)' = 1/z$.

7.4 Complex integration

7.4.1 Complex path integrals

Definition 7.12 (Complex path and integral). Let $\gamma : [a, b] \to \mathbb{C}$ be a piecewise C^1 curve (that is, γ is continuous and, except at finitely many points, differentiable with continuous derivative). Let $f : \gamma([a, b]) \to \mathbb{C}$ be continuous. The **complex integral** of f along γ is defined by

$$\int_\gamma f(z)\,dz := \int_a^b f(\gamma(t))\gamma'(t)\,dt.$$

Here the right-hand side is the Riemann integral of a complex-valued function (the components being integrated separately).

Proposition 7.7 (Basic properties of the complex integral). (i) **Linearity**: $\int_\gamma(\alpha f + \beta g)\,dz = \alpha\int_\gamma f\,dz + \beta\int_\gamma g\,dz$.

(ii) **Reversal of direction**: $\int_{\gamma^-} f\,dz = -\int_\gamma f\,dz$.

(iii) **Additivity over paths**: if $\gamma = \gamma_1 \cup \gamma_2$ (joined end to end), then $\int_\gamma f\,dz = \int_{\gamma_1} f\,dz + \int_{\gamma_2} f\,dz$.

(iv) **Invariance under reparametrisation**: if $\tilde{\gamma}$ is a C^1 reparametrisation of γ (preserving orientation), then $\int_{\tilde{\gamma}} f\,dz = \int_\gamma f\,dz$.

(v) **Estimation inequality**: $\left|\int_\gamma f\,dz\right| \le \int_\gamma |f|\,|dz| = \int_a^b |f(\gamma(t))||\gamma'(t)|\,dt$. In particular, if $|f| \le M$ on γ, then $\left|\int_\gamma f\,dz\right| \le ML(\gamma)$, where $L(\gamma)$ is the arc length of γ.

Proof. (i) By the linearity of the Riemann integral.

(ii) If $\gamma^-(t) = \gamma(a+b-t)$, then $(\gamma^-)'(t) = -\gamma'(a+b-t)$. Hence

$$\int_{\gamma^-} f\,dz = \int_a^b f(\gamma(a+b-t))(-\gamma'(a+b-t))\,dt = -\int_a^b f(\gamma(s))\gamma'(s)\,ds = -\int_\gamma f\,dz.$$

(iii) By the additivity of the Riemann integral over intervals.

(iv) Let $\tilde{\gamma}(s) = \gamma(\varphi(s))$, where φ is a C^1 bijection. By the change-of-variable formula for integrals,

$$\int_{\tilde{\gamma}} f\,dz = \int_{a'}^{b'} f(\gamma(\varphi(s)))\gamma'(\varphi(s))\varphi'(s)\,ds.$$

Setting $t = \varphi(s)$ and $dt = \varphi'(s)ds$ gives $\int_a^b f(\gamma(t))\gamma'(t)\,dt = \int_\gamma f\,dz$.

(v) By the absolute-value inequality for the Riemann integral and $|f(\gamma(t))\gamma'(t)| = |f(\gamma(t))||\gamma'(t)|$. ☐

Example 7.6 (Basic complex integrals). (i) $\int_\gamma dz = \gamma(b) - \gamma(a)$ (by the fundamental theorem of calculus).

(ii) $\int_\gamma z\,dz = (\gamma(b)^2 - \gamma(a)^2)/2$.

(iii) Let γ be the unit circle $|z| = 1$ traversed once anticlockwise. Then $\int_\gamma z^n\,dz = 0$ for $n \neq -1$, whilst $\int_\gamma z^{-1}\,dz = 2\pi i$.

Theorem 7.9 (Cauchy's integral theorem). *Let $\Omega \subseteq \mathbb{C}$ be a simply connected domain, f analytic on Ω, and γ a piecewise C^1 closed path in Ω. Then*

$$\oint_\gamma f(z)\,dz = 0.$$

Proof. Write $f = u + iv$ and $dz = dx + i\,dy$. Expanding the complex integral,

$$\oint_\gamma f(z)\,dz = \oint_\gamma (u+iv)(dx + i\,dy) = \oint_\gamma (u\,dx - v\,dy) + i\oint_\gamma (v\,dx + u\,dy).$$

Apply Green's formula (Theorem 15.6) to each real line integral. Let D be the domain enclosed by γ. Then

$$\oint_\gamma (u\,dx - v\,dy) = \iint_D \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) dx\,dy.$$

$$\oint_\gamma (v\,dx + u\,dy) = \iint_D \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) dx\,dy.$$

By the Cauchy–Riemann equations (Theorem 7.5), $u_x = v_y$ and $u_y = -v_x$, so the integrands of both double integrals vanish. Hence both real line integrals vanish, and therefore so does the complex integral. □

Corollary 7.3 (Equivalent formulation of Cauchy's integral theorem)**.** On a simply connected domain the integral of an analytic function is independent of the path: if γ_1 and γ_2 are two piecewise C^1 paths from z_0 to z_1, then $\int_{\gamma_1} f\,dz = \int_{\gamma_2} f\,dz$.

Proof. Take $\gamma = \gamma_1 \cup \gamma_2^-$ (from z_0 along γ_1 to z_1, then along γ_2 reversed from z_1 back to z_0). The path γ is closed, so by Theorem 7.9, $\oint_\gamma f\,dz = 0$. That is, $\int_{\gamma_1} f\,dz - \int_{\gamma_2} f\,dz = 0$, hence the two integrals are equal. □

Theorem 7.10 (Cauchy's integral formula)**.** Let $\Omega \subseteq \mathbb{C}$ be a domain, f analytic on Ω, and $\overline{B}(z_0, r) \subseteq \Omega$. Then

$$f(z_0) = \frac{1}{2\pi i}\oint_{|z-z_0|=r} \frac{f(z)}{z-z_0}\,dz,$$

where the path of integration is taken with positive orientation (anticlockwise).

Proof. **Step 1: decomposing the integrand.** On $|z - z_0| = r$ we have $z \neq z_0$. For any $\rho \in (0, r)$, consider the small circle $|z - z_0| = \rho$. By Cauchy's integral theorem (applied to the domain $\{\rho < |z - z_0| < r\}$, which is multiply connected but can be made simply connected by a cut),

$$\oint_{|z-z_0|=r} \frac{f(z)}{z-z_0}\,dz = \oint_{|z-z_0|=\rho} \frac{f(z)}{z-z_0}\,dz.$$

Step 2: computing on the small circle. Parametrise $z = z_0 + \rho e^{i\theta}$, $\theta \in [0, 2\pi]$. Then $dz = i\rho e^{i\theta} d\theta$, and

$$\oint_{|z-z_0|=\rho} \frac{f(z)}{z-z_0}\,dz = \int_0^{2\pi} \frac{f(z_0+\rho e^{i\theta})}{\rho e^{i\theta}} i\rho e^{i\theta}\,d\theta = i\int_0^{2\pi} f(z_0+\rho e^{i\theta})\,d\theta.$$

Step 3: passing to the limit $\rho \to 0$. Since f is continuous at z_0, we have $f(z_0 + \rho e^{i\theta}) \to f(z_0)$ uniformly in θ. Hence

$$\lim_{\rho\to 0} i\int_0^{2\pi} f(z_0+\rho e^{i\theta})\,d\theta = i\int_0^{2\pi} f(z_0)\,d\theta = 2\pi i f(z_0).$$

Therefore $\oint_{|z-z_0|=r} f(z)/(z-z_0)\,dz = 2\pi i f(z_0)$, that is, $f(z_0) = (1/(2\pi i))\oint_{|z-z_0|=r} f(z)/(z-z_0)\,dz$. □

Corollary 7.4 (Infinite differentiability of analytic functions)**.** If f is analytic on a domain Ω, then f is infinitely often complex differentiable on Ω, and

$$f^{(n)}(z_0) = \frac{n!}{2\pi i}\oint_{|z-z_0|=r} \frac{f(z)}{(z-z_0)^{n+1}}\,dz, \qquad n = 0, 1, 2, \ldots$$

Proof. Differentiate Cauchy's integral formula (Theorem 7.10) formally with respect to z_0. By differentiation under the integral sign (integrals with a parameter and uniform convergence, see Theorem 5.30),

$$f'(z_0) = \frac{1}{2\pi i} \oint_{|z-z_0|=r} \frac{f(z)}{(z-z_0)^2}\, dz.$$

Induction gives

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_{|z-z_0|=r} \frac{f(z)}{(z-z_0)^{n+1}}\, dz.$$

□

Theorem 7.11 (Liouville's theorem)**.** *If f is a bounded entire function, then f is constant.*

Proof. Suppose $|f(z)| \le M$ for all $z \in \mathbb{C}$. For any $z_0 \in \mathbb{C}$ and any $R > 0$, taking $n = 1$ in Corollary 7.4 and integrating along $|z - z_0| = R$,

$$|f'(z_0)| = \left| \frac{1}{2\pi i} \oint_{|z-z_0|=R} \frac{f(z)}{(z-z_0)^2}\, dz \right| \le \frac{1}{2\pi} \cdot \frac{M}{R^2} \cdot 2\pi R = \frac{M}{R}.$$

Letting $R \to \infty$ gives $f'(z_0) = 0$. Since z_0 is arbitrary, $f' \equiv 0$ and f is constant. □

Corollary 7.5 (The fundamental theorem of algebra)**.** *Every polynomial of degree $n \ge 1$ with complex coefficients has a root in $\mathbb{C}$.*

Proof. Suppose $p(z)$ is a polynomial of degree $n \ge 1$ with no roots. Then $1/p(z)$ is entire. As $|z| \to \infty$ we have $|p(z)| \to \infty$, so $1/p(z)$ is bounded. By Liouville's theorem (Theorem 7.11), $1/p$ is constant, hence p is constant, contradicting $n \ge 1$. □

7.4.2 The multiply connected version of Cauchy's integral theorem

Theorem 7.12 (The multiply connected version of Cauchy's integral theorem)**.** *Let Ω be a multiply connected domain whose boundary consists of m simple closed curves $\gamma_1, \dots, \gamma_m$ (γ_1 being the outer boundary and $\gamma_2, \dots, \gamma_m$ the inner boundaries, all taken with positive orientation: the outer boundary anticlockwise and the inner boundaries clockwise). If f is analytic on Ω, then*

$$\oint_{\gamma_1} f(z)\, dz + \oint_{\gamma_2} f(z)\, dz + \cdots + \oint_{\gamma_m} f(z)\, dz = 0.$$

Proof. Use cuts to turn the multiply connected domain Ω into a simply connected domain: draw a cut from each inner boundary to the outer boundary. The integrals along the two sides of each cut have opposite directions and cancel each other, so the boundary of the resulting domain is $\gamma_1 \cup \gamma_2 \cup \cdots \cup \gamma_m$. Applying Cauchy's integral theorem (Theorem 7.9) to the resulting simply connected domain gives $\sum_{i=1}^{m} \oint_{\gamma_i} f\, dz = 0$. □

Example 7.7 (Cauchy's integral theorem on a multiply connected domain). Let $\Omega = \{z : 1 < |z| < 2\}$, and let $f(z) = 1/z$ be analytic on Ω. Let the outer boundary γ_1 be $|z| = 2$ traversed anticlockwise and the inner boundary γ_2 be $|z| = 1$ traversed clockwise. Then

$$\oint_{\gamma_1} \frac{dz}{z} = 2\pi i, \qquad \oint_{\gamma_2} \frac{dz}{z} = -2\pi i,$$

and their sum is zero, which verifies Theorem 7.12.

7.5 Local expansions and zeros of analytic functions

Remark 7.4 (Interface with the 'Series and Convolution' chapter). This section uses the general theory established in Chapter 6, 'Series and Convolution': Taylor series, Laurent series, the Cauchy–Hadamard radius-of-convergence formula and the like. In particular: the definition of a Taylor series and its convergence conditions are given in Definition 6.3 and Theorem 6.5; the definition of a Laurent series in Definition 6.4; the Laurent expansion theorem in Theorem 6.6; and the Cauchy–Hadamard radius-of-convergence formula in Theorem 6.4. This section does not repeat these general theories; it gives only their direct applications in the context of complex analysis.

7.5.1 Taylor expansion of an analytic function

Theorem 7.13 (Taylor expansion of an analytic function). *Let f be analytic on the disc $B(z_0, R)$ $(R > 0)$. Then for every $z \in B(z_0, R)$,*

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z - z_0)^n,$$

and the radius of convergence is at least R.

Proof. By Cauchy's integral formula (Theorem 7.10), for any $|z - z_0| < r < R$ and $|\zeta - z_0| = r$,

$$f(z) = \frac{1}{2\pi i} \oint_{|\zeta - z_0| = r} \frac{f(\zeta)}{\zeta - z}\, d\zeta.$$

Expand $1/(\zeta - z)$ as a geometric series:

$$\frac{1}{\zeta - z} = \frac{1}{(\zeta - z_0) - (z - z_0)} = \frac{1}{\zeta - z_0} \cdot \frac{1}{1 - \frac{z - z_0}{\zeta - z_0}} = \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\zeta - z_0)^{n+1}}.$$

Since $|z - z_0| < r = |\zeta - z_0|$, the geometric series converges uniformly. Substituting and integrating term by term,

$$f(z) = \frac{1}{2\pi i} \sum_{n=0}^{\infty} (z - z_0)^n \oint_{|\zeta - z_0| = r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z - z_0)^n,$$

where the last step uses Corollary 7.4. The radius of convergence is at least R (since the expansion holds for every $|z - z_0| < R$). ☐

Corollary 7.6 (Isolation of zeros of an analytic function)**.** Let f be analytic on a domain Ω and not identically zero. Then the zero set of f has no accumulation point in Ω. Equivalently, if f vanishes on some convergent sequence of **distinct** points $\{z_n\}$ in Ω ($z_n \to z_0 \in \Omega$), then $f \equiv 0$.

Proof. Suppose, to the contrary, that the zeros have an accumulation point $z_0 \in \Omega$. By the Taylor expansion (Theorem 7.13), f can be expanded near z_0 as $\sum_n a_n(z-z_0)^n$.

If the expansion is not identically zero, there is a least $k \geq 0$ with $a_k \neq 0$. Near z_0,

$$f(z) = (z-z_0)^k g(z), \qquad g(z) = \sum_{n=0}^{\infty} a_{n+k}(z-z_0)^n, \quad g(z_0) = a_k \neq 0.$$

Since g is continuous, there is a neighbourhood U of z_0 on which $g \neq 0$. Hence $f \neq 0$ on $U \setminus \{z_0\}$, which contradicts the existence of an accumulation point of zeros.

Hence $a_n = 0$ for all n, that is, $f \equiv 0$ in a neighbourhood of z_0. By connectedness, $f \equiv 0$ on Ω (since Ω is connected, a function that vanishes locally vanishes on the whole of Ω). ☐

Corollary 7.7 (The identity theorem for analytic functions)**.** Let f, g be analytic on a domain Ω. If $\{z \in \Omega : f(z) = g(z)\}$ has an accumulation point in Ω, then $f \equiv g$ on Ω.

Proof. Put $h = f - g$ and apply Corollary 7.6. ☐

Example 7.8 (Applications of the isolation of zeros)**.**

(i) The zeros of $\sin z$ are $z = k\pi$ ($k \in \mathbb{Z}$); these zeros have no accumulation point in $\mathbb{C}$.

(ii) The zeros of $e^z - 1$ are $z = 2\pi i k$ ($k \in \mathbb{Z}$), with no accumulation point.

(iii) If an analytic function vanishes on $[0,1]$, then it vanishes on the whole domain containing $[0,1]$.

7.5.2 Laurent expansion of an analytic function

Definition 7.13 (Isolated singularity)**.** Let f be analytic on some punctured neighbourhood $0 < |z-z_0| < R$ of z_0, but not analytic at z_0. Then z_0 is called an **isolated singularity** of f.

Theorem 7.14 (The Laurent expansion theorem)**.** Let f be analytic on the annulus $r < |z-z_0| < R$ ($0 \leq r < R \leq \infty$). Then f can be expanded uniquely in that annulus as a Laurent series

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z-z_0)^n,$$

where

$$a_n = \frac{1}{2\pi i} \oint_{|z-z_0|=\rho} \frac{f(z)}{(z-z_0)^{n+1}}\, dz, \qquad n \in \mathbb{Z},$$

and $\rho \in (r, R)$ is arbitrary.

Proof. For any z with $r < |z - z_0| < R$, choose ρ_1, ρ_2 such that $r < \rho_1 < |z - z_0| < \rho_2 < R$. By the multiply connected version of Cauchy's integral formula (Theorem 7.12),

$$f(z) = \frac{1}{2\pi i} \oint_{|\zeta - z_0| = \rho_2} \frac{f(\zeta)}{\zeta - z}\, d\zeta - \frac{1}{2\pi i} \oint_{|\zeta - z_0| = \rho_1} \frac{f(\zeta)}{\zeta - z}\, d\zeta.$$

For the first integral, on $|\zeta - z_0| = \rho_2$ we have $|z - z_0| < \rho_2$, so expand $1/(\zeta - z)$ as a power series in $(z - z_0)$; for the second integral, on $|\zeta - z_0| = \rho_1$ we have $|z - z_0| > \rho_1$, so expand $1/(\zeta - z)$ as a power series in $(z - z_0)^{-1}$. Combining the two gives

$$f(z) = \sum_{n=0}^{\infty} a_n^{(+)} (z - z_0)^n + \sum_{n=1}^{\infty} a_{-n}^{(-)} (z - z_0)^{-n}.$$

Setting $a_n = a_n^{(+)}$ for $n \geq 0$ and $a_{-n} = a_{-n}^{(-)}$ for $n \geq 1$, we obtain $f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$.

The coefficient formula is obtained by term-by-term integration (using the orthogonality of the integrals of $(z - z_0)^m$ on $|z - z_0| = \rho$). □

Definition 7.14 (Classification of isolated singularities). *Let z_0 be an isolated singularity of f, with Laurent expansion $f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$. The series*

$$\text{principal part} = \sum_{n=1}^{\infty} a_{-n} (z - z_0)^{-n}$$

is called the principal part of the Laurent expansion. Then:

(i) **Removable singularity**: *the principal part is zero, that is, $a_n = 0$ for all $n < 0$.*

(ii) **Pole of order m**: *the principal part is a finite non-zero sum whose highest negative power is m, that is, $a_{-m} \neq 0$ and $a_n = 0$ for $n < -m$.*

(iii) **Essential singularity**: *the principal part has infinitely many terms, that is, $a_n \neq 0$ for infinitely many $n < 0$.*

Theorem 7.15 (Characterisation of removable singularities). *Let z_0 be an isolated singularity of f. Then the following conditions are equivalent:*

(i) *z_0 is a removable singularity;*

(ii) *f is bounded near z_0;*

(iii) *$\lim_{z \to z_0} f(z)$ exists and is finite.*

In that case one may define $f(z_0) = \lim_{z \to z_0} f(z)$, making f analytic at z_0.

Proof. (i) ⇒ (iii): if the principal part is zero, then $f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$ near z_0. Hence $\lim_{z \to z_0} f(z) = a_0$ exists and is finite.

(iii) ⇒ (ii): since the limit exists, f is bounded near z_0.

(ii) ⇒ (i): suppose $|f(z)| \leq M$ on $0 < |z - z_0| < R$. By the Laurent coefficient formula, for $n < 0$,

$$|a_n| = \left| \frac{1}{2\pi i} \oint_{|z - z_0| = \rho} \frac{f(z)}{(z - z_0)^{n+1}}\, dz \right| \leq \frac{1}{2\pi} \cdot \frac{M}{\rho^{n+1}} \cdot 2\pi\rho = \frac{M}{\rho^n} \cdot \rho \to 0 \quad (\rho \to 0)$$

(since $1/\rho^n = \rho^{-n} \to 0$ for $n < 0$). Hence $a_n = 0$ for all $n < 0$, the principal part is zero, and z_0 is a removable singularity. □

Example 7.9 (Examples of the three types of isolated singularity). (i) **Removable singularity**: $f(z) = \sin z/z$ at $z_0 = 0$. The Laurent expansion is $1 - z^2/6 + \cdots$, whose principal part is zero. After extending the definition by $f(0) = 1$, the function is analytic.

(ii) **Pole**: $f(z) = 1/z^2$ at $z_0 = 0$. The Laurent expansion is z^{-2}, the principal part is z^{-2}, a pole of order two.

(iii) **Pole**: $f(z) = 1/(z^2(z-1))$ at $z_0 = 0$. The expansion is $-1/z^2 - 1/z - 1 - z - \cdots$, a pole of order two.

(iv) **Essential singularity**: $f(z) = e^{1/z}$ at $z_0 = 0$. The Laurent expansion is $\sum_{n=0}^{\infty} z^{-n}/n!$, whose principal part has infinitely many terms. By the great Picard theorem, $e^{1/z}$ takes every value in $\mathbb{C} \setminus \{0\}$ in any punctured neighbourhood of $z_0 = 0$.

7.6 The residue theorem and its applications

7.6.1 Residues

Definition 7.15 (Residue). Let z_0 be an isolated singularity of f. The **residue** of f at z_0 is defined as the coefficient of the term $(z - z_0)^{-1}$ in the Laurent expansion:

$$\text{Res}(f, z_0) = a_{-1} = \frac{1}{2\pi i} \oint_{\gamma} f(z)\, dz,$$

where γ is a small positively oriented circle around z_0.

Proposition 7.8 (Computing residues). (i) **Simple pole**: if $f(z) = g(z)/(z - z_0)$ with $g(z_0) \neq 0$, then $\text{Res}(f, z_0) = g(z_0)$.

(ii) **Quotient of two functions**: if $f = g/h$ with $h(z_0) = 0$, $h'(z_0) \neq 0$ and $g(z_0) \neq 0$, then $\text{Res}(f, z_0) = g(z_0)/h'(z_0)$.

(iii) **Pole of order** m: $\text{Res}(f, z_0) = \frac{1}{(m-1)!} \lim_{z\to z_0} \frac{d^{m-1}}{dz^{m-1}}[(z - z_0)^m f(z)]$.

Proof. (i) If $f(z) = g(z)/(z - z_0)$ with g analytic near z_0, then the Taylor expansion of g is $g(z) = g(z_0) + g'(z_0)(z - z_0) + \cdots$. Hence $f(z) = g(z_0)/(z - z_0) + g'(z_0) + \cdots$. The coefficient of $(z - z_0)^{-1}$ is $g(z_0)$.

(ii) Since $h(z_0) = 0$ and $h'(z_0) \neq 0$, the function h has a simple zero at z_0. Hence $f = g/h$ has a simple pole at z_0. By (i), putting $g_{\text{new}}(z) = g(z)/((h(z)-h(z_0))/(z-z_0))$ (for $z \neq z_0$), we have $g_{\text{new}}(z_0) = g(z_0)/h'(z_0)$. Hence $\text{Res}(f, z_0) = g(z_0)/h'(z_0)$.

(iii) If z_0 is a pole of order m, then $(z-z_0)^m f(z)$ has a removable singularity (or is analytic) at z_0. Its Taylor expansion is $\sum_{k=0}^{\infty} b_k(z-z_0)^k$, where $b_k = \frac{1}{k!} \lim_{z\to z_0} \frac{d^k}{dz^k}[(z-z_0)^m f(z)]$. Hence $f(z) = \sum_{k=0}^{\infty} b_k(z - z_0)^{k-m}$. The term $(z - z_0)^{-1}$ corresponds to $k = m - 1$, with coefficient $b_{m-1} = \frac{1}{(m-1)!} \lim_{z\to z_0} \frac{d^{m-1}}{dz^{m-1}}[(z - z_0)^m f(z)]$. □

Theorem 7.16 (The residue theorem). Let f be analytic in a simply connected domain Ω except for finitely many isolated singularities $z_1, \ldots, z_m$. Let γ be a positively oriented simple closed curve in Ω that passes through none of the singularities, with $z_1, \ldots, z_m$ all inside γ. Then

$$\oint_\gamma f(z)\,dz = 2\pi i \sum_{k=1}^{m} \operatorname{Res}(f, z_k).$$

Proof. For each z_k, choose a small circle γ_k ($|z - z_k| = \rho_k$) such that the γ_k are pairwise disjoint and all lie inside γ. By the multiply connected version of Cauchy's integral theorem (Theorem 7.12),

$$\oint_\gamma f(z)\,dz = \sum_{k=1}^{m} \oint_{\gamma_k} f(z)\,dz.$$

By the definition of the residue (Definition 7.15), $\oint_{\gamma_k} f(z)\,dz = 2\pi i \operatorname{Res}(f, z_k)$. Hence $\oint_\gamma f(z)\,dz = 2\pi i \sum_{k=1}^{m} \operatorname{Res}(f, z_k)$. ☐

Corollary 7.8 (Real integrals via the residue theorem). (i) Integrals of the form $\int_{-\infty}^{\infty} f(x)\,dx$, where f has no singularities on the real axis, has finitely many singularities in the upper half-plane, and satisfies $|f(z)| \le C/|z|^2$ as $|z| \to \infty$. Then

$$\int_{-\infty}^{\infty} f(x)\,dx = 2\pi i \sum_{\operatorname{Im} z_k > 0} \operatorname{Res}(f, z_k).$$

(ii) Integrals of the form $\int_0^{2\pi} R(\cos\theta, \sin\theta)\,d\theta$, where R is a rational function. Put $z = e^{i\theta}$, so that $\cos\theta = (z + z^{-1})/2$, $\sin\theta = (z - z^{-1})/(2i)$ and $d\theta = dz/(iz)$. Then

$$\begin{aligned}\int_0^{2\pi} R(\cos\theta, \sin\theta)\,d\theta &= \oint_{|z|=1} R\left(\frac{z+z^{-1}}{2}, \frac{z-z^{-1}}{2i}\right)\frac{dz}{iz} \\ &= 2\pi i \sum_{|z_k|<1} \operatorname{Res}(\text{integrand}, z_k).\end{aligned}$$

Proof. (i) Take the large circle γ_R ($|z| = R$) together with the segment $[-R, R]$ of the real axis to form a closed path. As $R \to \infty$, the integral over the large circle tends to zero (by $|f(z)| \le C/|z|^2$ and the estimation inequality), hence $\int_{-\infty}^{\infty} f(x)\,dx = 2\pi i \sum_{\operatorname{Im} z_k > 0} \operatorname{Res}(f, z_k)$.

(ii) Put $z = e^{i\theta}$, so that $|z| = 1$ and $dz = ie^{i\theta}d\theta = iz\,d\theta$. Hence $d\theta = dz/(iz)$. Also $\cos\theta = (z + z^{-1})/2$ and $\sin\theta = (z - z^{-1})/(2i)$. Substitution gives a complex integral over the unit circle, which is computed by the residue theorem. ☐

Example 7.10 (Real integrals computed by the residue theorem). (i) Compute $\displaystyle\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$. The function $f(z) = 1/(1+z^2)$ has simple poles at $z = \pm i$, of which $z = i$ lies in the upper half-plane. By Proposition 7.8(ii),

$$\operatorname{Res}(f, i) = \frac{1}{(1+z^2)'|_{z=i}} = \frac{1}{2i} = -\frac{i}{2}.$$

Hence $\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = 2\pi i \cdot (-i/2) = \pi$.

(ii) Compute $\int_0^{2\pi} \frac{d\theta}{2+\cos\theta}$. Putting $z = e^{i\theta}$, the integrand becomes

$$\frac{1}{2+(z+z^{-1})/2} \cdot \frac{1}{iz} = \frac{2}{iz(4+z+z^{-1})} = \frac{2}{i(z^2+4z+1)}.$$

The roots of $z^2+4z+1 = 0$ are $z = -2 \pm \sqrt{3}$. Inside the unit circle lies $z_1 = -2+\sqrt{3}$. The residue is $2/(i(2z_1+4)) = 1/(i(z_1+2)) = 1/(i\sqrt{3})$. Hence the integral $= 2\pi i \cdot 1/(i\sqrt{3}) = 2\pi/\sqrt{3}$.

Theorem 7.17 (The argument principle). *Let f be analytic in a domain Ω except for finitely many poles, and let γ be a positively oriented simple closed curve in Ω that passes through none of the zeros and poles of f. Then*

$$\frac{1}{2\pi i} \oint_\gamma \frac{f'(z)}{f(z)}\, dz = N - P,$$

where N is the number of zeros of f inside γ (counted with multiplicity) and P is the number of poles of f inside γ (counted with order).

Proof. Suppose f has a zero of order m at z_0 (or a pole of order m with $m < 0$). Then $f(z) = (z-z_0)^m g(z)$, with $g(z_0) \neq 0$ and g analytic. Hence

$$\frac{f'(z)}{f(z)} = \frac{m}{z-z_0} + \frac{g'(z)}{g(z)}.$$

Since $g(z_0) \neq 0$, the function g'/g is analytic at z_0; hence f'/f has a simple pole at z_0 with residue m ($m > 0$ for a zero and $m < 0$ for a pole). By the residue theorem (Theorem 7.16),

$$\frac{1}{2\pi i} \oint_\gamma \frac{f'(z)}{f(z)}\, dz = \sum_{z_k \text{ zero}} m_k + \sum_{z_k \text{ pole}} m_k = N - P.$$

□

Corollary 7.9 (Rouché's theorem). *Let f, g be analytic in a domain Ω and let γ be a positively oriented simple closed curve in Ω. If $|g(z)| < |f(z)|$ for all z on γ, then f and $f+g$ have the same number of zeros inside γ (counted with multiplicity).*

Proof. Consider $h_t(z) = f(z) + tg(z)$ ($t \in [0,1]$). On γ, the inequality $|g| < |f|$ implies $|tg| < |f|$, so h_t has no zeros on γ. Define

$$N(t) := \frac{1}{2\pi i} \oint_\gamma \frac{h_t'(z)}{h_t(z)}\, dz.$$

By the argument principle (Theorem 7.17), $N(t)$ is the number of zeros of h_t inside γ. The function $N(t)$ is continuous in t (the integrand depends continuously on t and the denominator does not vanish). Since $N(t)$ takes integer values and a function that is continuous and takes integer values must be constant, $N(0) = N(1)$; that is, f and $f+g$ have the same number of zeros inside γ. □

Example 7.11 (Applications of Rouché's theorem). (i) Prove that $z^4+3z^2+1=0$ has four roots in $|z|<1$. Take $f(z)=3z^2$ and $g(z)=z^4+1$. On $|z|=1$ we have $|g|\le 2<3=|f|$. By Rouché's theorem, f and $f+g$ have the same number of zeros in $|z|<1$. The function $3z^2$ has 2 zeros in $|z|<1$ (a zero of order two at $z=0$). But $z^4+3z^2+1=0$ has 4 roots in $|z|<1$ — which contradicts the conclusion of Rouché's theorem. The reason: the maximum of $|g|$ on $|z|=1$ is 2 and $|f|=3$, so $|g|<|f|$ does hold. But $f=3z^2$ has 2 zeros in $|z|<1$ (counted with multiplicity), so $f+g=z^4+3z^2+1$ ought to have 2 zeros (counted with multiplicity), and not 4. The number of roots of $z^4+3z^2+1=0$ in $|z|<1$: put $w=z^2$, so that $w^2+3w+1=0$ and $w=(-3\pm\sqrt{5})/2\approx -0.382$ or -2.618. Since $|w|<1$ corresponds to $|z|^2<1$, only $w\approx -0.382$ yields two roots $z=\pm i\sqrt{0.382}$ with $|z|<1$. Hence $z^4+3z^2+1=0$ has 2 roots in $|z|<1$, in agreement with Rouché's theorem.

(ii) Prove that $e^z=3z$ has one root in $|z|<1$. Take $f(z)=-3z$ and $g(z)=e^z$. On $|z|=1$ we have $|g|=|e^z|=e^{\operatorname{Re} z}\le e<3=|f|$. By Rouché's theorem, f and $f+g=e^z-3z$ have the same number of zeros in $|z|<1$. Since $f=-3z$ has one zero in $|z|<1$ (namely $z=0$), the function e^z-3z has one zero in $|z|<1$.

7.7 Harmonic functions and the elements of conformal mapping

7.7.1 Harmonic functions

Definition 7.16 (Harmonic function). Let $\Omega\subseteq\mathbb{R}^2\cong\mathbb{C}$ be a domain and $u:\Omega\to\mathbb{R}$ a C^2 function. Then u is called a **harmonic function** if

$$\Delta u=\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$$

holds everywhere on Ω.

Theorem 7.18 (The real and imaginary parts of an analytic function are harmonic). *Let $f=u+iv$ be analytic on a domain Ω. Then both u and v are harmonic functions.*

Proof. By Theorem 7.7, analyticity of f implies $u,v\in C^\infty$ (by Corollary 7.4) and the satisfaction of the Cauchy–Riemann equations. Hence $\Delta u=0$ and $\Delta v=0$. ☐

Theorem 7.19 (The mean-value property of harmonic functions). *Let u be harmonic on a domain Ω, with $\overline{B}(z_0,r)\subseteq\Omega$. Then*

$$u(z_0)=\frac{1}{2\pi}\int_0^{2\pi}u(z_0+re^{i\theta})\,d\theta.$$

That is, the value of a harmonic function at the centre of a disc equals its average value on the circle.

Proof. By Proposition 7.6, there is a harmonic conjugate v of u on $\overline{B}(z_0, r)$ such that $f = u + iv$ is analytic on $\overline{B}(z_0, r)$. By Cauchy's integral formula (Theorem 7.10),

$$f(z_0) = \frac{1}{2\pi i} \oint_{|z-z_0|=r} \frac{f(z)}{z - z_0}\, dz.$$

Parametrising $z = z_0 + re^{i\theta}$ gives

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta})\, d\theta.$$

Taking real parts gives $u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} u(z_0 + re^{i\theta})\, d\theta$. ☐

Corollary 7.10 (The maximum principle for harmonic functions). Let u be harmonic and non-constant on a domain Ω. Then u cannot attain a local maximum or minimum in Ω.

Proof. Suppose, to the contrary, that u attains a local maximum M at $z_0 \in \Omega$. Choose $r > 0$ such that $\overline{B}(z_0, r) \subseteq \Omega$ and $u \leq M$ on $B(z_0, r)$. By the mean-value property (Theorem 7.19),

$$M = u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} u(z_0 + re^{i\theta})\, d\theta \leq M.$$

Equality requires $u(z_0 + re^{i\theta}) = M$ for all θ. Since r is arbitrary, $u \equiv M$ on $B(z_0, r)$. By connectedness, $u \equiv M$ on Ω, contradicting the assumption that u is non-constant. Hence u cannot attain a local maximum in Ω. The argument for a minimum is the same (consider $-u$). ☐

7.7.2 Elements of conformal mapping

Definition 7.17 (Conformal mapping). Let $\Omega, \Omega' \subseteq \mathbb{C}$ be domains and $f : \Omega \to \Omega'$ a bijection. Then f is called a **conformal mapping** if f is analytic on Ω and $f'(z) \neq 0$ for all $z \in \Omega$.

Proposition 7.9 (Angle preservation of conformal mappings). Let f be analytic at z_0 with $f'(z_0) \neq 0$. Then f preserves the angles between curves at z_0 (including their orientation).

Proof. Let two curves γ_1, γ_2 pass through z_0 with tangent vectors $\gamma_1'(0), \gamma_2'(0)$. Under f the tangent vectors of the image curves are $f'(z_0)\gamma_1'(0)$ and $f'(z_0)\gamma_2'(0)$. The angle between the two tangent vectors is

$$\arg(f'(z_0)\gamma_1'(0)) - \arg(f'(z_0)\gamma_2'(0)) = \arg \gamma_1'(0) - \arg \gamma_2'(0).$$

That is, the angle is unchanged. Since $f'(z_0) \neq 0$, multiplication by $f'(z_0)$ is a rotation combined with a scaling, which preserves angles. ☐

Example 7.12 (Examples of conformal mappings).
(i) The Möbius transformation $f(z) = (az + b)/(cz + d)$ $(ad - bc \neq 0)$ is a conformal mapping on domains such as the upper half-plane and the unit disc.

(ii) $f(z) = e^z$ maps the strip $\{z : 0 < \operatorname{Im} z < \pi\}$ conformally onto the upper half-plane.

(iii) $f(z) = (z - i)/(z + i)$ maps the upper half-plane conformally onto the unit disc.

(iv) $f(z) = z^2$ maps the first quadrant conformally onto the upper half-plane.

Theorem 7.20 (The Riemann mapping theorem). *Let $\Omega \subsetneq \mathbb{C}$ be a simply connected domain. Then there exists a conformal mapping $f : \Omega \to B(0, 1)$.*

Proof. The proof lies beyond the scope of this chapter. See Ahlfors, Complex Analysis, 3rd ed., McGraw-Hill, 1979, Chapter 6. The central idea is to construct the conformal mapping by means of normal families and an extremal problem. □

7.8 Elements of quaternion analysis

Definition 7.18 (The quaternion algebra). *The* **quaternion algebra** *$\mathbb{H}$ is the four-dimensional associative algebra over $\mathbb{R}$ with basis $\{1, i, j, k\}$, the multiplication being extended bilinearly from*

$$i^2 = j^2 = k^2 = ijk = -1$$

to the whole algebra. Every quaternion is written as $q = a+bi+cj+dk$ $(a, b, c, d \in \mathbb{R})$.

Proposition 7.10 (Elementary properties of quaternions). (i) *$\mathbb{H}$ is a division algebra: every non-zero element has a multiplicative inverse.*

(ii) *$\mathbb{H}$ is non-commutative: $ij = k \neq -k = ji$.*

(iii) *The conjugate $\bar{q} = a - bi - cj - dk$ satisfies $\overline{pq} = \bar{q}\bar{p}$.*

(iv) *The modulus $|q| = \sqrt{q\bar{q}} = \sqrt{a^2 + b^2 + c^2 + d^2}$ satisfies $|pq| = |p||q|$.*

(v) *The group of unit quaternions $S^3 = \{q : |q| = 1\}$ is isomorphic to $SU(2)$ and realises three-dimensional rotations through the conjugation action $v \mapsto qvq^{-1}$.*

Proof. (i) For $q = a+bi+cj+dk \neq 0$ we have $q\bar{q} = a^2+b^2+c^2+d^2 > 0$, so $q^{-1} = \bar{q}/|q|^2$.
(ii) Since $ij = k$ and $ji = -k$.
(iii) Verify componentwise.
(iv) $|pq|^2 = pq\overline{pq} = pq\bar{q}\bar{p} = p|q|^2\bar{p} = |p|^2|q|^2$.
(v) See the 'Vector Analysis' chapter. □

Definition 7.19 (Quaternion-analytic function). *Let $f : \mathbb{H} \supseteq \Omega \to \mathbb{H}$. If the limit*

$$f'(q_0) = \lim_{h\to 0}[f(q_0 + h) - f(q_0)]h^{-1}$$

exists, then f is said to be **right differentiable** *at q_0.*

Remark 7.5 (The scarcity of quaternion-analytic functions). Unlike complex analytic functions, quaternion-analytic functions are extremely scarce. Here two different conditions must be distinguished. Write $f = f_0 + f_1 i + f_2 j + f_3 k$ and set $e_1 = i$, $e_2 = j$, $e_3 = k$.

(a) Differentiability by difference quotients (the right differentiability of Definition 7.19) is equivalent to

$$\frac{\partial f}{\partial x_\mu} = \Big(\frac{\partial f}{\partial x_0}\Big)e_\mu, \qquad \mu = 1, 2, 3,$$

that is, to three component equations.

(b) The Cauchy–Riemann–Fueter equations characterise a different class of objects, namely the **regular** functions:

$$\frac{\partial f}{\partial \bar{q}} := \frac{\partial f}{\partial x_0} + i\frac{\partial f}{\partial x_1} + j\frac{\partial f}{\partial x_2} + k\frac{\partial f}{\partial x_3} = 0,$$

that is, the four component functions satisfy a first-order linear system of partial differential equations.

The two conditions are **not equivalent**: $f(q) = q$ is right differentiable by difference quotients ($[h]h^{-1} \equiv 1$), yet $\partial f/\partial\bar{q} = 1 + i\cdot i + j\cdot j + k\cdot k = -2 \neq 0$; conversely, $f(q) = x_0 + x_1 i$ satisfies $\partial f/\partial\bar{q} = 1 + i\cdot i = 0$, yet is not right differentiable (here $\partial_{x_2} f = 0$ whilst $(\partial_{x_0} f)e_2 = j \neq 0$, violating (a)).

The CRF system consists of four first-order equations and is far more restrictive than the complex case (two equations). Nor are the **regular** functions characterised by the CRF system the only class of quaternion analytic functions: quaternion power series $\sum q^n a_n$, quaternion polynomials and local expansions in powers of $(q - q_0)$ belong to the **slice-regular** class, and the division of labour between the two classes is described in the 'Quaternion Analysis' chapter (§8.5.1).

Example 7.13 (Rotational periodicity of the quaternion exponential). For every purely imaginary unit quaternion u ($\mathsf{u}^2 = -1$),

$$e^{\mathsf{u}\theta} = \cos\theta + \mathsf{u}\sin\theta$$

satisfies $e^{\mathsf{u}(\theta+2\pi)} = e^{\mathsf{u}\theta}$. For different axes of rotation $\mathsf{u}_1 \neq \mathsf{u}_2$, the product of $e^{\mathsf{u}_1\theta_1}$ and $e^{\mathsf{u}_2\theta_2}$ does not commute in general.

7.9 Interfaces with subsequent chapters

Remark 7.6 (Interface with the 'Harmonic Analysis' chapter). This chapter provides the 'Harmonic Analysis' chapter with:

(i) the complex-analytic foundations of the Fourier transform: Cauchy's integral formula and the residue theorem for analytic functions are used to compute Fourier transforms;

(ii) the complex-analytic foundations of the Laplace transform: the domain of convergence and the analyticity of the Laplace transform are guaranteed by complex analysis;

(iii) residue calculus: used to compute Fourier-type integrals of the form $\int_{-\infty}^{\infty} f(x)e^{i\omega x}\,dx$;

(iv) the periodicity of the complex exponential function: the 2π-periodicity of $e^{i\theta}$ is the foundation of Fourier series.

Remark 7.7 (Interface with the 'Vector Analysis' chapter). This chapter provides the 'Vector Analysis' chapter with:

(i) the two-dimensional complex potential: the real part u and the imaginary part v of an analytic function $f = u + iv$ are harmonic functions and can be used to describe two-dimensional irrotational solenoidal fields;

(ii) the connection between the Cauchy–Riemann equations and curl/divergence: analytic functions correspond to irrotational solenoidal fields;

(iii) conformal mapping: used to solve boundary-value problems for the two-dimensional Laplace equation.

Remark 7.8 (Interface with the 'Differential Equations and Analysis' chapter). This chapter provides the 'Differential Equations and Analysis' chapter with:

(i) complex eigenvalues and complex exponential solutions of ordinary differential equations;

(ii) analytic functions for constructing analytic solutions of differential equations;

(iii) the analysis of complex multipliers in Floquet theory;

(iv) the residue theorem for solving boundary-value problems for differential equations.

Remark 7.9 (Interface with the 'Probability and Distribution' chapter). This chapter provides the 'Probability and Distribution' chapter with:

(i) the complex-analytic foundations of characteristic functions: a characteristic function is a complex-valued function and its analyticity is guaranteed by complex analysis;

(ii) the residue theorem for computing the relation between characteristic functions and probability distributions;

(iii) the periodicity of the complex exponential function and the applications of Fourier analysis in probability theory.

Remark 7.10 (Open problems of this chapter). (i) **A constructive proof of the Riemann mapping theorem**: this chapter only states the Riemann mapping theorem and gives no proof. A complete proof requires normal families and extremal methods.

(ii) **Picard's theorems**: the deep conclusions of the big and the little Picard theorem lie beyond the scope of this chapter.

(iii) **Boundary behaviour of conformal mappings**: *the conclusion of Carathéodory's theorem on the boundary continuity of conformal mappings requires deeper topological and geometric tools.*

(iv) **Classification of quaternion-analytic functions**: *a complete classification and the properties of quaternion-analytic functions constitute the natural generalisation of complex analysis to the non-commutative case and remain a frontier of research.*

(v) **Functions of several complex variables**: *this chapter treats only functions of one complex variable; functions of several complex variables (analytic functions on $\mathbb{C}^n$) have properties entirely different from the one-variable case and require an independent theoretical framework.*

7.10 Roots of unity, finite order, and rotational period

The complex exponential function and the structure of the argument established in this chapter give the 'rotational period' a computable criterion for the first time. This section fixes that criterion in the form of roots of unity; the criterion is subsequently the basis on which the 'Harmonic Analysis' and 'Discrete Sampling and Intercalation of Periodic Signals' chapters treat finite-order periods.

Proposition 7.11 (The finite-order criterion for the complex exponential). *For $\theta \in \mathbb{R}$, write $\omega := e^{i\theta} \in U(1)$. Then*

(i) ord$(\omega) < \infty \iff \theta/(2\pi) \in \mathbb{Q}$;

(ii) *if $\theta/(2\pi) = p/q$ in lowest terms ($q \geq 1$), then* ord$(\omega) = q$, *and ω is a* **primitive q-th root of unity**;

(iii) *the map $\alpha + \mathbb{Z} \mapsto e^{2\pi i\alpha}$ gives an injective homomorphism $\mathbb{Q}/\mathbb{Z} \hookrightarrow U(1)$ whose image is the torsion subgroup of $U(1)$, and is dense in $U(1)$.*

Proof. (i) By the addition formula for the complex exponential, $\omega^n = e^{in\theta}$, hence

$$\omega^n = 1 \iff n\theta \in 2\pi\mathbb{Z} \iff \theta/(2\pi) \in \tfrac{1}{n}\mathbb{Z}.$$

Such an $n \geq 1$ exists if and only if $\theta/(2\pi) \in \mathbb{Q}$.

(ii) Let $\theta/(2\pi) = p/q$ be in lowest terms. Then $\omega^n = 1 \iff q \mid np \iff q \mid n$ (since gcd$(p, q) = 1$), and the least positive integer solution is $n = q$; moreover $\omega^j \neq 1$ for $0 < j < q$, so ω is a primitive q-th root of unity.

(iii) The map $\alpha \mapsto e^{2\pi i\alpha}$ is a surjective homomorphism $(\mathbb{R}, +) \to U(1)$ with kernel $\mathbb{Z}$ (the complex exponential of this chapter has period 2π), so it induces an injective homomorphism $\mathbb{R}/\mathbb{Z} \hookrightarrow U(1)$; restricting this to $\mathbb{Q}/\mathbb{Z}$ gives the required map. Its image is the set of all roots of unity, that is, the torsion subgroup of $U(1)$. Density follows from the density of $\mathbb{Q}$ in $\mathbb{R}$ and the continuity of $e^{2\pi i\cdot}$. □

Corollary 7.11 (Finite-order rotation ⇔ rational argument ratio). *Let $R_\theta(z) = e^{i\theta}z$ be a rotation of the complex plane. Then*

$$\text{ord}(R_\theta) < \infty \iff \theta/(2\pi) \in \mathbb{Q},$$

and in that case ord(R_θ) equals the denominator of $\theta/(2\pi)$ in lowest terms, and every non-zero orbit of R_θ is a q-periodic closed orbit consisting of q equally spaced points.

Proof. $R_\theta^n = R_{n\theta}$ (by the periodicity of the complex exponential function in this chapter), hence ord$(R_\theta) =$ ord$(e^{i\theta})$, and Proposition 7.11 gives both the criterion and the order. The orbit $\{e^{in\theta}z\}_{n\geq 0}$ lies on the circle of radius $|z|$, consists of exactly q distinct points and returns to its starting point at the q-th step, so it is a q-periodic closed orbit. □

Remark 7.11 (Interface with the 'Period' theme). Proposition 7.11 and Corollary 7.11 compress the periodicity criterion at the level of complex numbers into a single sentence: **period = rationality of the argument ratio**. This criterion is the common source of several later conclusions: the periodicity of the discrete Fourier transform (the order of $\omega_N = e^{-2\pi i/N}$ being N), the condition that the normalised period ratio $\rho \in \mathbb{Q}$ in the fundamental period of a sampling sequence, and the correspondence between the finite-order rotation group C_q and cyclic group representations.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	Translation on the complex plane and the invariance of analytic functions under translation (Definition 7.6); the purely imaginary period $2\pi i$ of the complex exponential function (Theorem 7.8)
Scaling	Scaling of the modulus of a complex number and the Cauchy–Riemann equations (Definition 7.2, Theorem 7.5); the angle preservation of conformal mappings (Proposition 7.9)
Rotation	Multiplication of complex numbers as rotation, the periodicity of $e^{i\theta}$ and roots of unity (Theorem 7.2, Theorem 7.1); realisation of three-dimensional rotations by the conjugation action of quaternions (Proposition 7.10)
Conservation	Conservation of the loop integral in Cauchy's integral theorem (Theorem 7.9); Cauchy's integral formula (Theorem 7.10)
Symmetry	Conjugate symmetry of analytic functions and the Schwarz reflection principle; the mean-value property of harmonic functions (Theorem 7.19)
Closure	Vanishing of the integral over a closed path in Cauchy's integral theorem (Theorem 7.9); the closed integral in the residue theorem (Theorem 7.16)
Period	The $2\pi i$-period of the complex exponential function (Theorem 7.8) and the group of roots of unity (Theorem 7.2); the rotational period of the quaternion exponential (Example 7.13)

Exercises

Exercise 1 Prove Proposition 7.1: the complex field $\mathbb{C}$ satisfies the field axioms, and

(i) $\overline{z+w} = \bar{z} + \bar{w}$, $\overline{zw} = \bar{z}\bar{w}$;

(ii) $|zw| = |z||w|$, $|z+w| \le |z| + |w|$.

Give the complete componentwise verification.

Exercise 2 Prove Theorem 7.1 (Euler's formula):

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

Carry out the proof using power-series expansion and the Taylor series of $\cos\theta$ and $\sin\theta$.

Exercise 3 Prove Theorem 7.2: the equation $z^n = 1$ has exactly n solutions in $\mathbb{C}$,

$$\zeta_k = e^{2\pi ik/n}, \quad k = 0, 1, \ldots, n-1,$$

and these form the cyclic group C_n. Verify closure, the identity element, inverses and a generator.

Exercise 4 Prove Theorem 7.3: every Cauchy sequence in $\mathbb{C}$ converges. Carry out the proof using the coordinate description by real and imaginary parts, and explain why the completeness of $\mathbb{C}$ is inherited from the completeness of $\mathbb{R}$.

Exercise 5 Prove Theorem 7.5 (the Cauchy–Riemann equations): if $f = u+iv$ is complex differentiable at z_0, then

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Compute $f'(z_0)$ along the real axis and along the imaginary axis separately, and compare real and imaginary parts.

Exercise 6 Prove Theorem 7.6: if $u, v \in C^1(\Omega)$ and the Cauchy–Riemann equations hold, then $f = u+iv$ is analytic on Ω. Carry out the proof using the definition of differentiability and the Cauchy–Riemann equations, and explain why the C^1 condition cannot be omitted.

Exercise 7 Verify the analyticity of each function in Example 7.4:

(i) $f(z) = e^z$, $u = e^x\cos y$, $v = e^x\sin y$;

(ii) $f(z) = \sin z$, $f(z) = \cos z$;

(iii) $f(z) = 1/z$ on $\mathbb{C} \setminus \{0\}$.

Verify the Cauchy–Riemann equations in each case, and compute the corresponding complex derivatives.

Exercise 8 Prove Theorem 7.9 (Cauchy's integral theorem): the integral of an analytic function over a closed path in a simply connected domain is zero. Carry out the proof using Green's formula (Theorem 15.6) and the Cauchy–Riemann equations, and explain why the simple-connectedness condition cannot be omitted.

Exercise 9 Prove Theorem 7.10 (Cauchy's integral formula):

$$f(z_0) = \frac{1}{2\pi i} \oint_{|z-z_0|=r} \frac{f(z)}{z-z_0}\, dz.$$

Carry out the proof in the three steps 'decompose the integrand, compute on the small circle, pass to the limit'.

Exercise 10 Prove Corollary 7.4: an analytic function is infinitely often complex differentiable, and

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_{|z-z_0|=r} \frac{f(z)}{(z-z_0)^{n+1}}\, dz.$$

Carry out the proof by differentiating under the integral sign in Cauchy's integral formula.

Exercise 11 Prove Theorem 7.11 (Liouville's theorem): a bounded entire function must be constant. Use Cauchy's integral formula to estimate $|f'(z_0)|$, and explain why this estimate tends to zero as the radius $R \to \infty$.

Exercise 12 Use Liouville's theorem to prove Corollary 7.5 (the fundamental theorem of algebra): every polynomial of degree $n \geq 1$ with complex coefficients has a root in $\mathbb{C}$. Argue by contradiction, assuming that there is no root and constructing a bounded entire function.

Exercise 13 Prove Theorem 7.12: Cauchy's integral theorem on a multiply connected domain:

$$\oint_{\gamma_1} f\, dz + \cdots + \oint_{\gamma_m} f\, dz = 0.$$

Explain how cuts turn a multiply connected domain into a simply connected one, and verify that the integrals along the two sides of each cut cancel each other.

Exercise 14 Prove Theorem 7.13: an analytic function on a disc can be expanded as a Taylor series. Carry out the proof using Cauchy's integral formula and the geometric-series expansion, and explain why the radius of convergence is at least the radius of the disc.

Exercise 15 Prove Corollary 7.6 (isolation of zeros): the zero set of an analytic function that is not identically zero has no accumulation point. Write out the complete argument by contradiction, and explain why the existence of the least k is guaranteed by $f \neq 0$.

Exercise 16 Prove Corollary 7.7 (the identity theorem): if two analytic functions agree on a set with an accumulation point in a domain, then they are identical. Complete the proof by applying the isolation-of-zeros theorem.

Exercise 17 Prove Theorem 7.14 (the Laurent expansion theorem): an analytic function on an annulus can be expanded as a Laurent series

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n.$$

Use the multiply connected version of Cauchy's integral formula, split the integral into integrals over the inner and the outer circle, and expand each separately.

Exercise 18 Prove Theorem 7.15: an isolated singularity z_0 is a removable singularity if and only if f is bounded near z_0. Use the Laurent coefficient formula to estimate $|a_n|$ $(n < 0)$, and explain why $a_n \to 0$ as $\rho \to 0$.

Exercise 19 Verify the various types of singularity in Example 7.9:

(i) $\sin z/z$ at $z_0 = 0$ (removable singularity);
(ii) $1/z^2$ at $z_0 = 0$ (pole of order two);
(iii) $e^{1/z}$ at $z_0 = 0$ (essential singularity).

Write out the respective Laurent expansions, and state the form of the principal part.

Exercise 20 Prove Proposition 7.8 (computing residues):

(i) simple pole: $\mathrm{Res}(f, z_0) = g(z_0)$ $(f = g/(z - z_0))$;
(ii) quotient of two functions: $\mathrm{Res}(f, z_0) = g(z_0)/h'(z_0)$ $(f = g/h)$;
(iii) pole of order m: $\mathrm{Res}(f, z_0) = \frac{1}{(m-1)!} \lim_{z \to z_0} \frac{d^{m-1}}{dz^{m-1}}[(z - z_0)^m f(z)]$.

Write out the three proofs separately.

Exercise 21 Prove Theorem 7.16 (the residue theorem):

$$\oint_\gamma f(z)\, dz = 2\pi i \sum_{k=1}^{m} \mathrm{Res}(f, z_k).$$

Use the multiply connected version of Cauchy's integral theorem, and split the integral over γ into the sum of integrals over small circles around the individual singularities.

Exercise 22 Compute the following real integrals, giving the complete working:

(i) $\displaystyle\int_{-\infty}^{\infty} \frac{dx}{1 + x^2}$;
(ii) $\displaystyle\int_0^{2\pi} \frac{d\theta}{2 + \cos\theta}$;
(iii) $\displaystyle\int_{-\infty}^{\infty} \frac{\cos x}{1 + x^2}\, dx$ (hint: consider the integral of $e^{iz}/(1 + z^2)$ along a semi-circular path).

Explain the key role of the residue theorem in the computation of real integrals.

Exercise 23 Prove Theorem 7.17 (the argument principle):

$$\frac{1}{2\pi i}\oint_{\gamma}\frac{f'(z)}{f(z)}\,dz = N - P.$$

Compute the residues of f'/f at the zeros and the poles, and complete the proof using the residue theorem.

Exercise 24 Use the argument principle to prove Corollary 7.9 (Rouché's theorem): if $|g| < |f|$ on γ, then f and $f+g$ have the same number of zeros inside γ. Complete the proof using $h_t = f + tg$ and the continuity of $N(t)$.

Exercise 25 Verify Example 7.11:

(i) prove that $z^4 + 3z^2 + 1 = 0$ has 2 roots in $|z| < 1$;

(ii) prove that $e^z = 3z$ has 1 root in $|z| < 1$.

Write out the complete application of Rouché's theorem in each case.

Exercise 26 Prove Theorem 7.19 (the mean-value property): the value of a harmonic function at the centre of a disc equals its average value on the circle. Complete the proof by taking the real part of Cauchy's integral formula.

Exercise 27 Use the mean-value property to prove Corollary 7.10 (the maximum principle for harmonic functions): a non-constant harmonic function cannot attain a local maximum or minimum in a domain. Write out the complete argument by contradiction.

Exercise 28 Prove Proposition 7.9: if $f'(z_0) \neq 0$, then f preserves the angles between curves at z_0. Complete the proof using the images of tangent vectors under f and the rotation-and-scaling action of $f'(z_0)$.

Exercise 29 Synthesis question: drawing on the complex derivative, Cauchy's integral theorem, Cauchy's integral formula, Taylor expansion, Laurent expansion, the residue theorem, the argument principle and Rouché's theorem from this chapter, explain how complex analysis provides tools for the following in the 'Harmonic Analysis' chapter:

(i) the computation of the Fourier transform (the residue theorem);

(ii) the analyticity of the Laplace transform (Cauchy's integral formula);

(iii) the complex-analytic foundations of characteristic functions (the complex exponential function);

and indicate in each case the corresponding definition, theorem or proposition in this chapter.

Chapter 8

Quaternion Analysis

Remark 8.1 (Positioning of this chapter). This chapter follows the 'Complex Analysis' chapter and systematically generalises the core structures of complex analysis—limits, continuity, derivatives, differentials, integrals, series expansions, convolution, integral transforms and harmonic analysis—to the quaternion algebra $\mathbb{H}$. Unlike the commutative case of the complex field $\mathbb{C}$, multiplication in $\mathbb{H}$ is non-commutative, which gives rise to a series of essential structural differences:

(i) the left and right derivatives must be defined separately, and the two are in general not equal (Definition 8.3);

(ii) analytic functions (Fueter-regular functions) are far rarer than complex analytic functions, and polynomial functions are in general not right Fueter-regular (Remark 8.6);

(iii) the Cauchy–Riemann equations are generalised to the Cauchy–Riemann–Fueter system (Theorem 8.3);

(iv) the integral theorems can be formulated rigorously only with the aid of quaternion differential forms and the language of the exterior derivative (Theorem 8.4);

(v) convolution is no longer commutative, and the convolution theorem requires additional restrictive conditions (Theorem 8.9);

(vi) the group of unit quaternions $S^3 \cong SU(2)$ realises three-dimensional rotations through conjugation, giving the double covering $SU(2) \to SO(3)$ (Theorem 8.18).

This chapter provides a unified interface for the $SU(2)$ and $SO(3)$ of the later 'Linear Algebra' chapter, for the Lie groups and fibre bundles of 'Manifold and Fibre Bundle', for the quaternion episphere groups of 'Epicycle Sets and Deferent Sphere Sets', and for the non-commutative harmonic analysis of 'Harmonic Analysis'.

Seven-theme positioning of this chapter: **Translation**—the translation invariance of quaternion addition and limits; **Scaling**—the multiplicativity and dilation of the quaternion modulus; **Rotation**—the realisation of three-dimensional rotations by the conjugation action of unit quaternions; **Conservation**—the invariance of the quaternion norm and inner product under rotation; **Symmetry**—the conjugation anti-automorphism and the group structure of S^3; **Closure**—elements of finite order in S^3 generate cyclic subgroups; **Period**—the 2π-periodicity of the quaternion exponential function and the group of roots of unity.

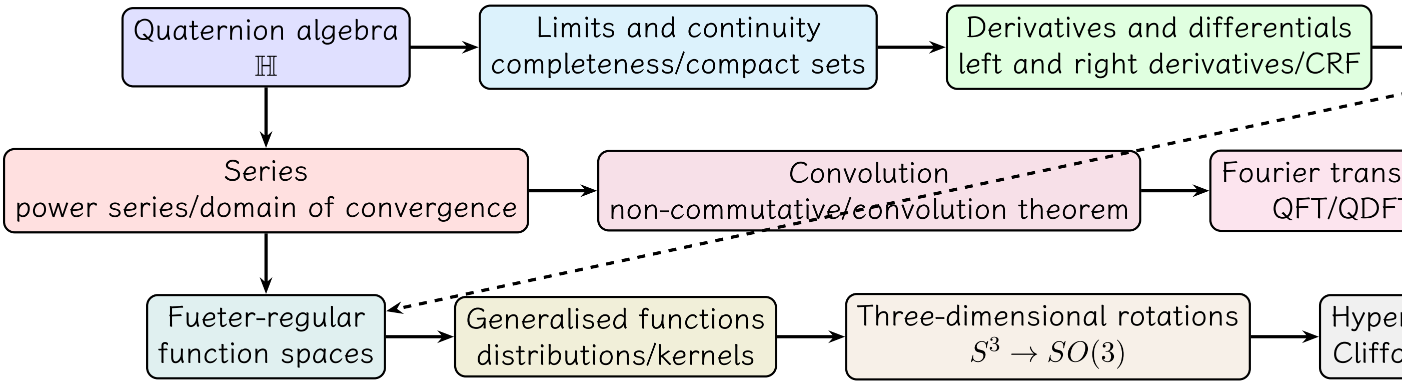

Quaternion algebra
$\mathbb{H}$
Limits and continuity
completeness/compact sets
Derivatives and differentials
left and right derivatives/CRF
Series
power series/domain of convergence
Convolution
non-commutative/convolution theorem
Fourier trans
QFT/QDF
Fueter-regular
function spaces
Generalised functions
distributions/kernels
Three-dimensional rotations
$S^3 \to SO(3)$
Hyper
Cliffo

Figure 8.1: Overview of the structure of this chapter: starting from the quaternion algebra and passing through limits and continuity, derivatives and differentials, integration, series, convolution, the Fourier transform, the Laplace transform, Fueter-regular functions, distributions and three-dimensional rotations, before finally converging into hypercomplex analysis.

8.1 Review of the quaternion algebra

Definition 8.1 (The quaternion algebra). The **quaternion algebra** $\mathbb{H}$ is the four-dimensional associative algebra over $\mathbb{R}$ with basis $\{1, i, j, k\}$, whose multiplication is the bilinear extension of

$$i^2 = j^2 = k^2 = ijk = -1$$

together with the relations derived from them,

$$ij = k, \quad ji = -k, \quad jk = i, \quad kj = -i, \quad ki = j, \quad ik = -j$$

An arbitrary quaternion is written

$$q = a + bi + cj + dk, \qquad a, b, c, d \in \mathbb{R}.$$

Here $a = \mathrm{Re}\, q$ is called the **scalar part** and $\mathbf{v} = bi + cj + dk = \mathrm{Vec}\, q$ the **vector part**. The conjugate is $\bar{q} = a - bi - cj - dk$, and the modulus is $|q| = \sqrt{q\bar{q}} = \sqrt{a^2 + b^2 + c^2 + d^2}$.

Property 8.1 (Basic algebraic properties of the quaternions).

(i) $\mathbb{H}$ is a **division algebra**: every non-zero element has a multiplicative inverse $q^{-1} = \bar{q}/|q|^2$.

(ii) $\mathbb{H}$ is **non-commutative**: $ij \neq ji$, and in general $q_1 q_2 \neq q_2 q_1$.

(iii) **Conjugation is an anti-automorphism**: $\overline{pq} = \bar{q}\,\bar{p}$, $\bar{\bar{q}} = q$.

(iv) **Multiplicativity of the modulus**: $|pq| = |p|\,|q|$, and hence $|q^{-1}| = |q|^{-1}$.

(v) **Inner product**: $\langle p, q \rangle = \mathrm{Re}(\bar{p}q)$ is the Euclidean inner product on $\mathbb{R}^4$. In accordance with the convention of the 'Linear Algebra' chapter, it is **linear in the second argument** ($\langle p, \lambda q \rangle = \lambda \langle p, q \rangle$) and conjugate-linear in the first ($\langle \lambda p, q \rangle = \bar{\lambda} \langle p, q \rangle$).

Proof. (i) For $q \neq 0$ we have $q\bar{q} = |q|^2 > 0$, so $q^{-1} = \bar{q}/|q|^2$ satisfies $qq^{-1} = q^{-1}q = 1$.

(ii) Since $ij = k$ and $ji = -k$, the two are not equal.

(iii) Verify componentwise that $\overline{pq} = \bar{q}\,\bar{p}$.

(iv) $|pq|^2 = pq\overline{pq} = pq\bar{q}\bar{p} = p|q|^2\bar{p} = |p|^2|q|^2$, and taking square roots gives $|pq| = |p||q|$.

(v) A direct computation gives $\mathrm{Re}(\bar{p}q) = \sum_{i=0}^{3} p_i q_i$, which is a symmetric positive-definite bilinear form. $\square$

Example 8.1 (An instance of quaternion multiplication). Take $p = 1+i$ and $q = 1+j$. Then

$$pq = (1+i)(1+j) = 1 + j + i + ij = 1 + i + j + k,$$

whilst

$$qp = (1+j)(1+i) = 1 + i + j + ji = 1 + i + j - k.$$

Hence $pq \neq qp$, the difference being $2k$.

Remark 8.2 (The vector form of the quaternion product). Let $p = a+\mathsf{u}$ and $q = b+\mathsf{v}$, where $a, b \in \mathbb{R}$ and $\mathsf{u}, \mathsf{v} \in \mathrm{Im}\,\mathbb{H} \cong \mathbb{R}^3$. Then

$$pq = ab - \mathsf{u} \cdot \mathsf{v} + a\mathsf{v} + b\mathsf{u} + \mathsf{u} \times \mathsf{v}.$$

This formula decomposes quaternion multiplication into a scalar part and a vector part, and it is a basic tool for the differential calculus that follows.

Remark 8.3 (Comparison with the complex algebra). The complex numbers $\mathbb{C}$ form a commutative field, whereas the quaternions $\mathbb{H}$ form a non-commutative division algebra. This difference leads to the following:

- in complex analysis there is only one definition of the derivative; in quaternion analysis the left and right derivatives must be distinguished.
- complex analytic functions are very abundant, whereas quaternion analytic functions (Fueter-regular functions) are very rare.
- the complex unit-circle group $U(1)$ is commutative, whereas the quaternion unit-sphere group $S^3 \cong SU(2)$ is non-commutative.

8.2 Quaternion limits and continuity

Definition 8.2 (Quaternion limit). Let $f : \mathbb{H} \supseteq D \to \mathbb{H}$ and $q_0 \in \overline{D}$. We write $\lim_{q \to q_0} f(q) = L$ if

$$\forall \varepsilon > 0,\ \exists \delta > 0,\ \forall q \in D, \quad 0 < |q - q_0| < \delta \implies |f(q) - L| < \varepsilon.$$

If $\lim_{q \to q_0} f(q) = f(q_0)$, then f is said to be **continuous** at q_0.

Proposition 8.1 (Coordinate description of the limit). Write $q = q_0 + h$ with $h = h_0 + h_1 i + h_2 j + h_3 k$, and $f = f_0 + f_1 i + f_2 j + f_3 k$. Then

$$\lim_{h \to 0} f(q_0 + h) = L$$

if and only if the four real-component limits each hold.

Proof. By $|h| = \sqrt{h_0^2 + h_1^2 + h_2^2 + h_3^2}$ and the inequalities for the absolute values of the components,

$$|h_\mu| \le |h| \le |h_0| + |h_1| + |h_2| + |h_3|,$$

the two control each other, so the limit is equivalent to the componentwise limit. ☐

Theorem 8.1 (Completeness of $\mathbb{H}$). *Every Cauchy sequence in $\mathbb{H}$ converges.*

Proof. Let $\{q_n\}$ be a Cauchy sequence, with $q_n = a_n + b_n i + c_n j + d_n k$. Since $|a_m - a_n| \le |q_m - q_n|$, the sequence $\{a_n\}$ is a Cauchy sequence in $\mathbb{R}$, and by the completeness of $\mathbb{R}$ it converges to a. Similarly $b_n \to b$, $c_n \to c$ and $d_n \to d$. Setting $q = a + bi + cj + dk$, we obtain $|q_n - q| \to 0$. ☐

Theorem 8.2 (Continuous functions on compact sets). *Let $K \subseteq \mathbb{H}$ be compact and let $f : K \to \mathbb{H}$ be continuous. Then:*

(i) *$f(K)$ is compact;*

(ii) *f is bounded on K;*

(iii) *$|f|$ attains its maximum and minimum on K;*

(iv) *f is uniformly continuous on K.*

Proof. Since $\mathbb{H} \cong \mathbb{R}^4$, regard f as four continuous real-valued functions (f_0, f_1, f_2, f_3). The corresponding properties of continuous functions on compact sets in real analysis are then obtained by applying them componentwise. ☐

8.3 Quaternion derivatives and differentials

8.3.1 Left and right derivatives

Definition 8.3 (Left and right derivatives). *Let $f : \mathbb{H} \supseteq D \to \mathbb{H}$ and $q_0 \in D$.*

(i) *If the limit*

$$f'_L(q_0) = \lim_{h\to 0} \big[f(q_0 + h) - f(q_0)\big] h^{-1}$$

exists, then f is said to be **left-differentiable** *at q_0, and $f'_L(q_0)$ is called the left derivative.*

(ii) *If the limit*

$$f'_R(q_0) = \lim_{h\to 0} h^{-1} \big[f(q_0 + h) - f(q_0)\big]$$

exists, then f is said to be **right-differentiable** *at q_0, and $f'_R(q_0)$ is called the right derivative.*

Since quaternion multiplication is non-commutative, the left and right derivatives are in general not equal.

Remark 8.4 (The essential difference between the left and right derivatives). In the definition of the right derivative h^{-1} multiplies on the left, whilst in the definition of the left derivative h^{-1} multiplies on the right. If $f(q) = aq$ (left multiplication by a constant), then

$$f(q+h) - f(q) = ah, \qquad [f(q+h) - f(q)]h^{-1} = ahh^{-1} = a,$$

so $f(q) = aq$ is left-differentiable, with left derivative a. Conversely, $f(q) = qb$ (right multiplication by a constant) is right-differentiable, with right derivative b. This shows that **left multiplication by a constant is left-analytic, whilst right multiplication by a constant is right-analytic**, and the two cannot hold simultaneously (unless b is real).

Definition 8.4 (Quaternion analytic functions). If f is right-differentiable everywhere on an open set $U \subseteq \mathbb{H}$, then f is said to be **right-analytic** on U. Left-analytic is defined similarly. Note that the difference-quotient limit definition of right-differentiability is too narrow, and Fueter-regular functions are defined by an integral (see §8.9).

8.3.2 The Cauchy–Riemann–Fueter system

Definition 8.5 (The Fueter operator). The **Fueter operator** is defined by

$$\frac{\partial}{\partial \bar{q}} := \frac{\partial}{\partial x_0} + i\frac{\partial}{\partial x_1} + j\frac{\partial}{\partial x_2} + k\frac{\partial}{\partial x_3},$$

where $q = x_0 + x_1 i + x_2 j + x_3 k$.

Theorem 8.3 (The Cauchy–Riemann–Fueter system). Let $f = f_0 + f_1 i + f_2 j + f_3 k$ be right Fueter-regular on an open set U (that is, $\partial f/\partial \bar{q} = 0$). Then the four component functions satisfy

$$\begin{aligned}
\frac{\partial f_0}{\partial x_0} - \frac{\partial f_1}{\partial x_1} - \frac{\partial f_2}{\partial x_2} - \frac{\partial f_3}{\partial x_3} &= 0,\\
\frac{\partial f_0}{\partial x_1} + \frac{\partial f_1}{\partial x_0} + \frac{\partial f_2}{\partial x_3} - \frac{\partial f_3}{\partial x_2} &= 0,\\
\frac{\partial f_0}{\partial x_2} - \frac{\partial f_1}{\partial x_3} + \frac{\partial f_2}{\partial x_0} + \frac{\partial f_3}{\partial x_1} &= 0,\\
\frac{\partial f_0}{\partial x_3} + \frac{\partial f_1}{\partial x_2} - \frac{\partial f_2}{\partial x_1} + \frac{\partial f_3}{\partial x_0} &= 0.
\end{aligned}$$

Proof. Apply the Fueter operator $\partial/\partial\bar{q} = \partial_0 + i\partial_1 + j\partial_2 + k\partial_3$ to $f = f_0 + f_1 i + f_2 j + f_3 k$; expanding term by term and using the quaternion multiplication rules ($ij = k$ and so on), then comparing the four components, gives the result. □

Remark 8.5 (Historical note: Fueter's contribution). The Swiss mathematician Rudolf Fueter (1880–1950) was the first to establish systematically the theory of quaternion analytic functions, in the 1930s. He found that if the Cauchy–Riemann condition in complex analysis is generalised directly to the existence of the difference-quotient limit, then the resulting class of functions is too narrow.

He therefore generalised the Cauchy–Riemann condition itself instead, and defined **Fueter-regular functions**—quaternion-valued functions satisfying $\partial f/\partial\bar{q} = 0$. This class of functions is far richer than the difference-quotient differentiable functions, and it possesses integral theorems analogous to those for complex analytic functions (Fueter's theorem).

Example 8.2 (Instances of quaternion regular functions). (i) The constant function $f(q) = c$ is right Fueter-regular.

(ii) The function $f(q) = q^{-1}/|q|^2 = \bar{q}/|q|^4$ is right Fueter-regular (for $q \neq 0$). This is the most important basic example, and it is called the **Fueter kernel**. Note that $q^{-1} = \bar{q}/|q|^2$ alone is not right Fueter-regular (the scalar part of its $\partial/\partial\bar{q}$ is $2/|q|^2 \neq 0$); what is regular is q^{-1} multiplied by the scalar factor $|q|^{-2}$.

(iii) More generally, $f(q) = (q-a)^{-1}/|q-a|^2$ is right Fueter-regular for $q \neq a$.

(iv) The polynomial $f(q) = \sum_{n=0}^{N} q^n a_n$ (with coefficients on the right) is **slice-regular**, but in general not right Fueter-regular (see §8.5.1).

Remark 8.6 (Polynomial functions are in general not right Fueter-regular). The quaternion polynomial $\sum_{n=0}^{N} a_n q^n$ (with coefficients on the left) is in general not right Fueter-regular. For example, for $f(q) = q^2$ a computation gives $\partial(q^2)/\partial\bar{q} \neq 0$ (unless the vector part of q is zero). Nor is the polynomial $\sum_{n=0}^{N} q^n a_n$, with coefficients on the right, right Fueter-regular: for a single term $q^n a_n$ with $n \geq 1$ the Fueter derivative satisfies $\partial(qa)/\partial\bar{q} = a + iai + jaj + kak = -2a \neq 0$. Polynomials of this kind belong to the class of **slice-regular** functions; the division of labour between the two classes is described in §8.5.1.

8.4 Quaternion integration

8.4.1 Quaternion-valued differential forms

Definition 8.6 (Quaternion-valued differential forms). Let $U \subseteq \mathbb{H}$ be an open set. A **quaternion-valued k-form** is

$$\omega = \sum_{|I|=k} \omega_I(q)\, dq_I,$$

where $\omega_I : U \to \mathbb{H}$ is a quaternion-valued function, $dq_I = dq_{i_1} \wedge \cdots \wedge dq_{i_k}$, and dq_μ corresponds to the coordinate x_μ ($q = x_0 + x_1 i + x_2 j + x_3 k$).

Definition 8.7 (Exterior derivative). The **exterior derivative** d is defined by

$$d\omega = \sum_{|I|=k} \sum_{\mu=0}^{3} \frac{\partial \omega_I}{\partial x_\mu}\, dq_\mu \wedge dq_I.$$

8.4.2 Fueter's integral theorem

Theorem 8.4 (Cauchy's integral theorem in Fueter's form). *Let f be right Fueter-regular on a simply connected domain $\Omega \subseteq \mathbb{H}$ (that is, $\partial f/\partial \bar{q} = 0$), and let γ be a closed three-dimensional surface in Ω. Then*

$$\oint_{\gamma} Dq\, f(q) = 0,$$

where Dq is the quaternion area element

$$Dq := dq_1 \wedge dq_2 \wedge dq_3 - i\, dq_0 \wedge dq_2 \wedge dq_3 + j\, dq_0 \wedge dq_1 \wedge dq_3 - k\, dq_0 \wedge dq_1 \wedge dq_2.$$

Proof. Use the quaternion version of the Stokes–Cartan theorem: for the 3-form $F = Dq\, f(q)$ we have $\oint_{\partial C} F = \int_C dF$. A direct computation of dF gives

$$dF = Dq \wedge \left(\frac{\partial f}{\partial \bar{q}}\right) dq_0 \wedge dq_1 \wedge dq_2 \wedge dq_3.$$

The right Fueter-regularity condition $\partial f/\partial \bar{q} = 0$ gives $dF = 0$, hence $\oint_{\partial C} F = 0$. ☐

Corollary 8.1 (Fueter's integral formula). *Let f be right Fueter-regular on a domain containing $\overline{B(q_0, r)}$. Then*

$$f(q_0) = \frac{1}{2\pi^2} \oint_{\partial B(q_0,r)} \frac{(q - q_0)^{-1}}{|q - q_0|^2}\, Dq\, f(q).$$

Proof. Consider the function $G(q, q_0) = (q - q_0)^{-1}/|q - q_0|^2$, which is right Fueter-regular for $q \neq q_0$. Apply Fueter's integral theorem to Gf and let $r \to 0$ to extract $f(q_0)$. ☐

Remark 8.7 (Comparison with Cauchy's integral formula in complex analysis). Cauchy's integral formula in complex analysis reads $f(z_0) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{z - z_0}\, dz$. The quaternion version of the integral formula involves the quaternion kernel $(q - q_0)^{-1}/|q - q_0|^2$ and the 3-form area element Dq, and its domain of integration is a closed three-dimensional surface rather than a closed one-dimensional curve. This difference stems from the four-dimensional character of quaternion analysis.

8.5 Quaternion series

8.5.1 Slice regularity and Fueter regularity: division of labour

Remark 8.8 (The division of labour between the two classes). There are two inequivalent notions of 'regularity' in quaternion analysis, and the exposition in this chapter has to allocate the work between them. **Slice-regular functions**: let f be defined on an open set $U \subseteq \mathbb{H}$. For every pure imaginary unit quaternion I (with $I^2 = -1$), write $\mathbb{C}_I = \{x_0 + x_I I : x_0, x_I \in \mathbb{R}\}$ for the corresponding complex slice. If the restriction $f|_{\mathbb{C}_I \cap U}$ is holomorphic for every I—that is, if, writing $f = f_0 + I f_I$,

$$\frac{\partial f_0}{\partial x_0} = \frac{\partial f_I}{\partial x_I}, \qquad \frac{\partial f_0}{\partial x_I} = -\frac{\partial f_I}{\partial x_0}$$

on that slice—then f is called **slice-regular**. This class is closed under power series and local expansions: $\sum_{n=0}^{\infty} q^n a_n$ and expansions in powers of $(q-q_0)$ are its typical objects (see Definition 8.8 and Theorem 8.15). **Fueter-regular functions**: these are defined by the Fueter operator equation $\partial f/\partial \bar{q} = 0$ (Definition 8.13), whose prototype is the kernel $\bar{q}/|q|^4 = q^{-1}/|q|^2$. The integral theorems, Cauchy's integral formula and the theory of distributions in this chapter all rest on the Fueter-regular class. **The relation between the two**: for a single term $q^n a_n$ with $n \geq 1$ we have $\partial(q^n a_n)/\partial \bar{q} = -2a_n \neq 0$, so a Fueter-regular power series **cannot** be written in the form $\sum q^n a_n$; a **Fueter polynomial basis** must be used instead—that is, the family of homogeneous polynomials generated symmetrically from q and $\bar{q}$ and satisfying $\partial P_n/\partial \bar{q} = 0$. The construction of that basis is not developed in this chapter. The two classes cannot be interchanged: the slice-regular class describes power series and local expansions (isolation of zeros holding slice by slice), whilst the Fueter-regular class describes the integral theory in the sense of partial differential equations; outside the real axis their intersection is small (for instance the kernels $\bar{q}/|q|^4$ and their scalar multiples).

8.5.2 Quaternion power series

Definition 8.8 (Quaternion power series). A series of the form

$$\sum_{n=0}^{\infty} q^n a_n \qquad \text{or} \qquad \sum_{n=0}^{\infty} a_n q^n, \qquad a_n \in \mathbb{H}$$

is called a **quaternion power series**. A series whose coefficients lie on the right, $\sum_{n=0}^{\infty} q^n a_n$, is **slice-regular**, but it is not right Fueter-regular (see §8.5.1); a Fueter-regular power series must use a Fueter polynomial basis instead.

Theorem 8.5 (Convergence of quaternion power series). Let $\sum_{n=0}^{\infty} q^n a_n$ be a quaternion power series. Then there exists $R \in [0, \infty]$, called the **radius of convergence**, such that:

(i) it converges absolutely for $|q| < R$;

(ii) it diverges for $|q| > R$;

(iii) it converges uniformly on $|q| \leq \rho < R$;

(iv) the sum function is slice-regular for $|q| < R$ and may be differentiated term by term (see §8.5.1).

The radius of convergence R is determined by the coefficients and satisfies $1/R = \limsup_{n\to\infty} |a_n|^{1/n}$.

Proof. (i)(ii)(iii) are the same as for real power series and follow from control by the modulus. (iv) The legitimacy of term-by-term differentiation is guaranteed by uniform convergence; each term $q^n a_n$ is slice-regular (on a slice $\mathbb{C}_I$ it is simply the complex power series $z^n a_n$), and term-by-term differentiation preserves slice regularity, so the sum function is slice-regular. This does not make the power series right Fueter-regular: $\partial(q^n a_n)/\partial \bar{q} = -2a_n \neq 0$ for $n \geq 1$. □

Theorem 8.6 (Abel's limit theorem: the quaternion version). Let $a = \sum_{n=0}^{\infty} a_n$ be a convergent quaternion series and let $G_a(q) = \sum_{n=0}^{\infty} q^n a_n$ be its power series. Then, as $q \to 1$ with $|1-q|/(1-|q|)$ bounded, $G_a(q) \to G_a(1)$.

Proof. The framework of the proof follows Ahlfors's classical method. Let $\varepsilon > 0$, and set $a = 0$ (by adjusting the constant term). Consider the partial sums $\sigma_n = \sum_{k=0}^{n} a_k$ and use summation by parts:

$$\sigma_n(q) = (1-q)(\sigma_0 + q\sigma_1 + \cdots + q^{n-1}\sigma_{n-1}) + q^n \sigma_n.$$

Since a converges, $q^n \sigma_n \to 0$. Hence

$$G_a(q) = (1-q)\sum_{n=0}^{\infty} q^n \sigma_n.$$

Since $|1-q|/(1-|q|)$ is bounded, write $|1-q| \leq M(1-|q|)$. Because $\sigma_n \to 0$, for sufficiently large N we have $|\sigma_n| < \varepsilon$ $(n \geq N)$. Thus

$$\left|\sum_{n=N}^{\infty} q^n \sigma_n\right| < \varepsilon \sum_{n=N}^{\infty} |q|^n = \frac{\varepsilon |q|^N}{1-|q|} < \frac{\varepsilon}{1-|q|}.$$

From this we obtain $|G_a(q)| < \varepsilon M$, that is, $G_a(q) \to 0 = G_a(1)$. ☐

Theorem 8.7 (Power series for the exponential, sine and cosine). *The quaternion exponential, sine and cosine are defined by the following power series:*

$$e^p = \sum_{l=0}^{\infty} \frac{p^l}{l!}, \qquad \sin p = \sum_{l=0}^{\infty} (-1)^l \frac{p^{2l+1}}{(2l+1)!}, \qquad \cos p = \sum_{l=0}^{\infty} (-1)^l \frac{p^{2l}}{(2l)!}.$$

All three series have radius of convergence $R = \infty$, that is, their set of convergence is the whole of the quaternions. Moreover they satisfy

$$(e^p)' = e^p, \qquad (\sin p)' = \cos p, \qquad (\cos p)' = -\sin p.$$

Proof. Radius of convergence: apply the ratio test to the series for $\sin p$,

$$L = \lim_{l\to\infty} \frac{|p^{2(l+1)+1}|/[2(l+1)+1]!}{|p^{2l+1}|/(2l+1)!} = \lim_{l\to\infty} \frac{|p|^2}{(2l+2)(2l+3)} = 0 < 1,$$

so $R = 1/L = \infty$. The same argument applies to $\cos p$ and e^p.

Term-by-term differentiation: uniform convergence permits differentiation term by term.

$$(e^p)' = \sum_{l=0}^{\infty} \left(\frac{p^l}{l!}\right)' = \sum_{k=1}^{\infty} \frac{p^{k-1}}{(k-1)!} = e^p,$$

$$(\sin p)' = \sum_{l=0}^{\infty} (-1)^l \frac{p^{2l}}{(2l)!} = \cos p, \qquad (\cos p)' = -\sin p.$$

These are completely parallel to the corresponding results in complex analysis. ☐

Example 8.3 (Evaluating the quaternion exponential). Take $p = \pi i/2$. By the quaternion Euler formula (Proposition 9.11), $e^{\pi i/2} = \cos(\pi/2) + i\sin(\pi/2) = i$. More generally, for a pure imaginary quaternion $\mathbf{u}\theta$ (with $\mathbf{u}$ a unit pure imaginary quaternion), $e^{\mathbf{u}\theta} = \cos\theta + \mathbf{u}\sin\theta$.

8.6 Quaternion convolution

Definition 8.9 (Quaternion convolution)**.** Let $x, y : \mathbb{R} \to \mathbb{H}$ be quaternion-valued signals. Their **convolution** is defined by

$$(x * y)(t) = \int_{-\infty}^{+\infty} x(u)y(t-u)\,du.$$

Theorem 8.8 (Non-commutativity of quaternion convolution)**.** *In general $x*y \neq y*x$. Quaternion convolution is no longer commutative.*

Proof. Take $x(t) = \delta(t)i$ and $y(t) = \delta(t)j$. Then $(x*y)(t) = ij = k$, whilst $(y*x)(t) = ji = -k$, and the two are not equal. ☐

Theorem 8.9 (The quaternion convolution theorem)**.** *Let $x : \mathbb{R} \to \mathbb{H}$ be a general quaternion-valued signal, and let $y : \mathbb{R} \to \mathbb{C}_\mu$ take values in the same complex subfield as the Fourier atom $e^{-\mu 2\pi v t}$. Then*

$$x * y(t) \overset{\text{QFT}}{\longleftrightarrow} X(v)Y(v).$$

That is: when one of the signals is restricted to a particular complex subfield, the QFT of the convolution equals the product of the QFTs.

Proof. After y is restricted to the complex subfield $\mathbb{C}_\mu$, the signal y commutes with the Fourier atom $e^{-\mu 2\pi v t}$. A direct computation of the QFT gives

$$\widehat{x*y}(v) = \int \left(\int x(u)y(t-u)du \right) e^{-\mu 2\pi v t} dt.$$

Substituting $t = u + s$ and using the commutativity of y with $e^{-\mu 2\pi v s}$, we obtain

$$\widehat{x*y}(v) = \int x(u)e^{-\mu 2\pi v u} du \cdot \int y(s)e^{-\mu 2\pi v s} ds = X(v)Y(v).$$

If y does not lie in the same complex subfield, then the convolution theorem fails because of non-commutativity. ☐

Remark 8.9 (The restriction in the convolution theorem)*.* The condition $y : \mathbb{R} \to \mathbb{C}_\mu$ in the quaternion convolution theorem should not be regarded as too restrictive. In fact, this condition covers the usual smoothing operations (filtering) in the time domain or the frequency domain. Filtering relations that involve geometric operations (rotations) cannot be expressed simply by convolution, and they will be discussed in later chapters.

Proposition 8.2 (Basic properties of quaternion convolution)**.** *Quaternion convolution satisfies:*

(i) **Bilinearity**: $(\alpha x_1 + \beta x_2) * y = \alpha(x_1 * y) + \beta(x_2 * y)$;

(ii) **Associativity**: $(x * y) * z = x * (y * z)$;

(iii) **Non-commutativity**: in general $x * y \neq y * x$.

Proof. (i) By the linearity of the integral.
(ii) By Fubini's theorem, interchanging the order of integration.
(iii) By Theorem 8.8. ☐

8.7 The quaternion Fourier transform

Definition 8.10 (The quaternion Fourier transform). For $x \in L^1(\mathbb{R};\mathbb{H}) \cap L^2(\mathbb{R};\mathbb{H})$, the **quaternion Fourier transform** (QFT) is defined by

$$X(v) = \int_{-\infty}^{+\infty} x(t)e^{-j2\pi vt}\, dt.$$

The inverse transform is

$$x(t) = \int_{-\infty}^{+\infty} X(v)e^{j2\pi vt}\, dv.$$

Here j is a fixed imaginary unit, determined by a choice of symmetry.

Remark 8.10 (Several types of QFT). Because quaternion multiplication is non-commutative, there are several ways of defining the QFT: the left QFT (with the exponential factor multiplying on the left), the right QFT (with the exponential factor multiplying on the right), and the two-sided QFT (with the two exponential factors placed on either side). Different definitions have different properties.

Proposition 8.3 (Relation between the QFT and the complex Fourier transform). Write the signal as $x(t) = u(t) + iv(t)$, where u, v take values in $\mathbb{C}_j$. Then

$$X(v) = U(v) + iV(v),$$

where $U(v)$ and $V(v)$ are the standard complex Fourier transforms of $u(t)$ and $v(t)$ respectively.

Proof. Substituting directly into the definition:

$$X(v) = \int (u(t) + iv(t))e^{-j2\pi vt}dt = \int u(t)e^{-j2\pi vt}dt + i\int v(t)e^{-j2\pi vt}dt.$$

Since u, v take values in $\mathbb{C}_j$ and commute with $e^{-j2\pi vt}$, these integrals are precisely the standard complex Fourier transforms. □

Theorem 8.10 (Differentiation properties of the QFT).

$$\frac{d^n x(t)}{dt^n} \overset{\text{QFT}}{\longleftrightarrow} X(v)(\mu 2\pi v)^n,$$

$$x(t)(-\mu 2\pi t)^n \overset{\text{QFT}}{\longleftrightarrow} \frac{d^n X(v)}{dv^n}.$$

Proof. This follows directly from integration by parts and the definition of the QFT. □

Theorem 8.11 (The quaternion Parseval–Plancherel identity). For $x, y \in L^2(\mathbb{R};\mathbb{H})$,

$$\langle x, y\rangle_{L^2} = \int x(t)\overline{y(t)}\, dt = \int X(v)\overline{Y(v)}\, dv.$$

In particular, $\|x\|_{L^2}^2 = \|X\|_{L^2}^2$.

Proof. The space $L^2(\mathbb{R};\mathbb{H})$ is a left Hilbert space, with inner product $\langle x, y\rangle = \int x(t)\overline{y(t)}dt$. The QFT is an isometric isomorphism of left Hilbert spaces and therefore preserves the inner product. ☐

Definition 8.11 (The discrete quaternion Fourier transform)**.** For a quaternion sequence $\{x[n]\}_{n=0}^{N-1}$ of length N, the **discrete quaternion Fourier transform** (QDFT) is defined by

$$X[m] = \sum_{n=0}^{N-1} x[n]e^{-j2\pi nm/N}, \qquad m = 0, 1, \cdots, N-1.$$

The inverse transform is $x[n] = \frac{1}{N}\sum_{m=0}^{N-1} X[m]e^{j2\pi nm/N}$.

8.8 The quaternion Laplace transform

Definition 8.12 (The quaternion Laplace transform)**.** Let $f(t)$ be a quaternion-valued function defined for $t \geq 0$. The **quaternion Laplace transform** (QLT) is

$$\mathcal{L}\{f\}(s) = \int_0^\infty e^{-st} f(t)\, dt,$$

where s is a quaternion variable. According to whether the exponential factor lies on the left or on the right, there are two different ways of defining the QLT.

Theorem 8.12 (Differentiation property of the QLT)**.** *The differentiation property of the QLT is*

$$\mathcal{L}\{f'\}(s) = s\mathcal{L}\{f\}(s) - f(0),$$

but its precise form depends on the way the QLT is defined (left or right), and it differs significantly from the classical Laplace transform.

Proof. Let $\mathcal{L}\{f\}(s) = \int_0^\infty e^{-st} f(t)dt$. Integration by parts gives:

$$\mathcal{L}\{f'\}(s) = \int_0^\infty e^{-st} f'(t)dt = [e^{-st}f(t)]_0^\infty + s\int_0^\infty e^{-st} f(t)dt.$$

If f satisfies a suitable growth condition, the boundary term is $-f(0)$, so $\mathcal{L}\{f'\}(s) = s\mathcal{L}\{f\}(s) - f(0)$. Note, however, that the non-commutativity of s and f may give rise to correction terms in explicit computations. ☐

Theorem 8.13 (Convolution property of the QLT)**.** *The convolution property of the QLT is*

$$\mathcal{L}\{f * g\}(s) = \mathcal{L}\{f\}(s) \cdot \mathcal{L}\{g\}(s),$$

where convolution is defined by $(f * g)(t) = \int_0^t f(\tau)g(t-\tau)d\tau$. This property, however, requires a restriction to be imposed on g (analogous to the complex-subfield condition in the QFT convolution theorem).

Proof. A direct computation gives:

$$\mathcal{L}\{f * g\}(s) = \int_0^\infty e^{-st}\left(\int_0^t f(\tau)g(t-\tau)d\tau\right)dt.$$

Interchanging the order of integration and substituting $u = t - \tau$, we obtain

$$= \int_0^\infty \int_0^\infty e^{-s(\tau+u)} f(\tau) g(u) d\tau du.$$

If g commutes with e^{-su}, then $= \mathcal{L}\{f\}(s) \cdot \mathcal{L}\{g\}(s)$. Otherwise a correction is required because of non-commutativity. ☐

8.9 Fueter-regular functions

Definition 8.13 (Fueter-regular functions). Let $U \subseteq \mathbb{H}$ be an open set. A function $f : U \to \mathbb{H}$ is called **right Fueter-regular** if

$$\frac{\partial f}{\partial \bar{q}} = 0.$$

It is called **left Fueter-regular** if

$$f \frac{\partial}{\partial \bar{q}} = 0.$$

Theorem 8.14 (Fueter's theorem). Let $f : \mathbb{C} \to \mathbb{C}$ be a complex analytic function. Then

$$F(q) = f(q_0) + \sum_{k=1}^{\infty} \frac{1}{k!} \left(\frac{\partial^k f}{\partial z^k}(q_0) \right) q^k$$

(with suitable adjustment of the variables) generates a right Fueter-regular function.

Proof. This is the standard form of Fueter's generation theorem. Replacing the complex variable in the Taylor expansion of the complex analytic function f by a quaternion variable, and handling the non-commutativity appropriately, yields a Fueter-regular function. It must be stressed that this power series is itself **slice-regular** rather than Fueter-regular (see Definition 8.8); the correct content of Fueter's generation theorem is that one first takes the axially symmetric extension of f and then applies the four-dimensional Laplacian $\Delta = \partial_{x_0}^2 + \partial_{x_1}^2 + \partial_{x_2}^2 + \partial_{x_3}^2$, the resulting function being right Fueter-regular. The name of the theorem is retained in this section, but, as explained above, the series is to be read as a slice-regular object. ☐

Theorem 8.15 (Local expansion of slice-regular functions). Let f be slice-regular in a neighbourhood of q_0. Then f can be expanded near q_0 as

$$f(q) = \sum_{n=0}^{\infty} (q - q_0)^n a_n, \qquad a_n \in \mathbb{H}.$$

Proof. This is the local expansion theorem for slice-regular functions. Its proof uses Fueter's integral formula and the geometric series expansion, in analogy with the proof of the Taylor expansion in complex analysis. **A Fueter-regular function has in general no local expansion in powers of** $(q - q_0)$: for $n \geq 1$, $(q - q_0)^n$ is itself not right Fueter-regular (see §8.5.1). The overlap between the two classes in a local expansion is therefore small: Fueter polynomials are homogeneous (P_n being of degree n), so in an expansion about the origin only the constant term belongs to both classes. ☐

Corollary 8.2 (Isolation of the zeros of slice-regular functions)**.** *If a slice-regular function f on a connected open set U is not identically zero, then its zero set has no accumulation point in U.*

Proof. By the local expansion theorem, Theorem 8.15, if the zeros had an accumulation point then all the coefficients $a_n = 0$, so that $f \equiv 0$, a contradiction. $\square$

8.10 Quaternion distributions

Definition 8.14 (Quaternion distributions)**.** Let $\mathcal{S}(\mathbb{H})$ be the quaternion-valued Schwartz space. A **quaternion distribution** is a continuous linear functional on $\mathcal{S}(\mathbb{H})$. In particular, the **Fueter-kernel distribution** is defined by

$$\left\langle \frac{\bar{q}}{|q|^4}, \phi \right\rangle = \lim_{\varepsilon \to 0} \int_{|q|>\varepsilon} \frac{\bar{q}}{|q|^4} \phi(q)\, dq,$$

that is, it is the principal-value distribution with kernel $\bar{q}/|q|^4 = q^{-1}/|q|^2$. The kernel $\bar{q}/|q|^4$ is taken here rather than $1/q$ because the latter is not Fueter-regular (see §8.5.1), whilst $\bar{q}/|q|^4$ is precisely the Fueter kernel used in 8.1.

Theorem 8.16 (Partial derivative of the Fueter-kernel distribution)**.** *The distribution induced by the Fueter kernel in quaternion analysis satisfies*

$$\frac{\partial}{\partial \bar{q}} \left(q^{-1}|q|^{-2}\right) = \frac{\partial}{\partial \bar{q}} \left(\frac{\bar{q}}{|q|^4}\right) = 2\pi^2 \delta(q),$$

where $\delta(q)$ is the quaternion Dirac distribution.

Proof. This is a standard result in the theory of distributions in quaternion analysis, its prototype being the fundamental solution of $|q|^{-2}$ in $\mathbb{R}^4$. The proof involves differentiating the Fueter kernel $\bar{q}/|q|^4 = q^{-1}|q|^{-2}$ in the distributional sense, together with the quaternion version of distribution theory. Note the homogeneity check: the kernel $\bar{q}/|q|^4$ is homogeneous of degree -3 under the dilation $q \mapsto \lambda q$, and applying the Fueter operator once gives degree -4, which agrees with the degree of $\delta(q)$ in $\mathbb{R}^4$ (under $q = \lambda q'$ we have $\delta(\lambda q') = \lambda^{-4}\delta(q')$, so both sides scale as λ^{-4}); with $1/q$ instead, differentiation would give degree -2, which is incompatible with δ, so the kernel must carry the factor $|q|^{-2}$. $\square$

8.11 Quaternion conjugation and three-dimensional rotations

Theorem 8.17 (The group of unit quaternions)**.** *The set of unit quaternions $S^3 = \{q \in \mathbb{H} : |q| = 1\}$ forms a non-commutative Lie group under quaternion multiplication, and it is isomorphic to $SU(2)$.*

Proof. Closure: if $|p| = |q| = 1$, then $|pq| = |p||q| = 1$. The identity element is 1. The inverse of an element is $\bar{q}$, and $|\bar{q}| = 1$. Associativity is inherited from $\mathbb{H}$. The isomorphism $S^3 \cong SU(2)$ is given by

$$a + bi + cj + dk \longmapsto \begin{pmatrix} a + bi & c + di \\ -c + di & a - bi \end{pmatrix}$$

□

Theorem 8.18 (Quaternion conjugation and three-dimensional rotations)**.** *For a unit quaternion $q \in S^3$ and a pure imaginary quaternion $v \in \operatorname{Im}\mathbb{H} \cong \mathbb{R}^3$, the map $\rho_q(v) = qvq^{-1}$ is a rotation of $\mathbb{R}^3$. The map $\Pi : S^3 \to SO(3)$, $q \mapsto \rho_q$, is a double covering, with $\ker \Pi = \{\pm 1\}$.*

Proof. **Preservation of pure imaginary quaternions:** $\overline{qvq^{-1}} = \bar{q}^{-1}\bar{v}\bar{q} = q(-v)q^{-1} = -qvq^{-1}$.

Preservation of the modulus: $|\rho_q(v)| = |q||v||q^{-1}| = |v|$.

Rodrigues' formula: write $q = \cos(\theta/2) + \mathsf{u}\sin(\theta/2)$, with u a unit pure imaginary quaternion. Decompose v as $v = v_{\|} + v_{\perp}$; a direct computation gives

$$qvq^{-1} = v_{\|} + \cos\theta\, v_{\perp} + \sin\theta\, (\mathsf{u} \times v_{\perp}),$$

which is precisely Rodrigues' formula for a rotation through angle θ about the axis u.

Double covering: $\rho_q = \rho_{-q}$, and by Rodrigues' formula every rotation corresponds uniquely to a pair $\pm q$, so Π is a double covering. □

Corollary 8.3 (Periodic structure of three-dimensional rotations)**.** *Let $q = e^{\mathsf{u}\theta}$. Then ρ_q is the rotation through angle 2θ about the axis u.*

(i) When $\theta = \pi$ we have $q = -1$ and $\rho_q = \mathrm{id}$ (a rotation through 2π about the axis).

(ii) If $\theta = \pi p/q$ (with p/q in lowest terms), then q is a $2q$-th root of unity and ρ_q generates a cyclic subgroup of order q in $SO(3)$.

Proof. By Rodrigues' formula in Theorem 8.18, the angle of the rotation corresponding to ρ_q is 2θ. (i) For $\theta = \pi$ the rotation angle is 2π, and the rotation is the identity. (ii) The order of $q = e^{\mathsf{u}\pi p/q}$ is $2q/\gcd(2q, p)$, and the order of the corresponding rotation is $q/\gcd(q, p) = q$. □

8.12 Outlook: hypercomplex analysis

Remark 8.11 (Clifford analysis)*.* Quaternion analysis is the first instance of hypercomplex analysis. More generally, analysis on the Clifford algebra $\mathrm{Cl}_{0,n}(\mathbb{R})$ is called **Clifford analysis**. Quaternion analysis corresponds to the case $\mathrm{Cl}_{0,2}(\mathbb{R}) \cong \mathbb{H}$.

Remark 8.12 (Slice-regular functions)*.* Besides Fueter-regular functions, another important class of quaternion analytic functions is that of **slice-regular functions**, introduced by Gentili and Stoppato in 2006. Slice-regular functions have better convergence properties with respect to the slice topology.

Remark 8.13 (Interfaces with later chapters)*.*
(i) **With 'Linear Algebra':** the double covering $S^3 \cong SU(2)$ and $SU(2) \to SO(3)$ provides an instance for the theory of matrix groups.

(ii) **With 'Manifold and Fibre Bundle':** $S^3 \to SO(3)$ is a typical example of a principal fibre bundle.

(iii) With 'Epicycle Sets and Deferent Sphere Sets': the quaternion episphere group $\beta_{a,\mathrm{u},\theta}(t) = ae^{\mathrm{u}\theta t}$ is a direct application of the exponential function of this chapter.

(iv) With 'Harmonic Analysis': the quaternion Fourier transform is a special case of harmonic analysis on groups.

(v) With 'Probability and Distribution': quaternion distributions and quaternion probability distributions can be combined further.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	translation invariance of quaternion addition and limits (Definition 8.2); the translation property of the QFT; translation invariance of convolution
Scaling	multiplicativity of the quaternion modulus $\|pq\| = \|p\|\|q\|$ (Property 8.1); angle scaling in the exponential map $e^{\mathrm{u}\theta}$; the s scaling in the QLT
Rotation	the conjugation action of unit quaternions realising three-dimensional rotations (Theorem 8.18); Rodrigues' formula and the rotation angle 2θ (Corollary 8.3); the rotation factor in the QFT
Conservation	invariance of the quaternion norm and inner product under rotation; the Parseval–Plancherel identity for the QFT (Theorem 8.11)
Symmetry	the conjugation anti-automorphism $\overline{pq} = \bar{q}\bar{p}$ (Property 8.1); the group structure and double covering of S^3 (Theorem 8.17); the conjugate symmetry of the QFT
Closure	elements of finite order in S^3 generating cyclic subgroups (Corollary 8.3); the closure of the roots of unity $e^{2\pi ip/q}$
Period	the 2π-periodicity of the quaternion exponential function (Proposition 9.11); the period q of a finite-order rotation (Corollary 8.3); the periodicity of the QDFT

Exercises

Exercise 1 Prove the five basic algebraic properties in Property 8.1.

Exercise 2 Verify that the difference between pq and qp in Example 8.1 is $2k$.

Exercise 3 Prove the coordinate description of Proposition 8.1.

Exercise 4 Prove the completeness of $\mathbb{H}$ in Theorem 8.1.

Exercise 5 Write down the right and left derivatives of the quaternion linear function $f(q) = aq + b$. Explain why $f(q) = aq$ is left-analytic and $f(q) = qb$ is right-analytic.

Exercise 6 Prove the quaternion Euler formula of Proposition 9.11.

Exercise 7 Prove that all three series in Theorem 8.7 have radius of convergence $R = \infty$.

Exercise 8 Prove Abel's limit theorem of Theorem 8.6.

Exercise 9 Prove that the restriction $y : \mathbb{R} \to \mathbb{C}_\mu$ in the convolution theorem of Theorem 8.9 cannot be omitted.

Exercise 10 Prove the quaternion Parseval–Plancherel identity of Theorem 8.11.

Exercise 11 Explain the difference between the left and right derivatives in quaternion analysis, and give an example of a function that is right-differentiable but not left-differentiable.

Exercise 12 Verify that $f(q) = q^{-1}/|q|^2 = \bar{q}/|q|^4$ in Example 8.2 is right Fueter-regular, and explain why q^{-1} alone is not (hint: compute $\partial(q^{-1})/\partial\bar{q}$ directly; its scalar part is $2/|q|^2$).

Exercise 13 Prove Fueter's integral theorem of Theorem 8.4. Hint: use the quaternion version of the Stokes–Cartan theorem and the condition $\partial f/\partial\bar{q} = 0$.

Exercise 14 Prove Fueter's generation theorem of Theorem 8.14: first write down the axially symmetric extension of the complex analytic function f, then verify that $\Delta(\tilde{f})$ is right Fueter-regular; also explain why replacing z by q directly can only yield a slice-regular power series.

Exercise 15 Prove the partial derivative formula for the Fueter-kernel distribution in Theorem 8.16, and use homogeneity to explain why the factor $|q|^{-2}$ must appear in the kernel.

Exercise 16 Prove the periodic structure of three-dimensional rotations in Corollary 8.3.

Chapter 9

Differential Equations and Analysis

Remark 9.1 (Positioning of this chapter). This chapter systematically applies the analytic tools established in the 'Functions and Limits' chapter and the series/convolution framework established in the 'Series and Convolution' chapter to four classes of analysis object: **differential equations** (integral curves of direction fields), **complex analysis** (analytic functions on the complex plane), **complex analyticity** (holomorphy, power-series expansions, residues) and **quaternion analysis and analyticity** (the exponential map and rotation representations in a non-commutative space). The core themes are: a differential equation is the integral curve of a direction field; an integral equation is a convolution equation of a trend kernel; conservation laws and first integrals are the constraints on the existence of periodic orbits; Floquet theory is the direct realisation of translation periodicity in differential equations; complex analysis and complex analyticity rebuild the direction–trend duality on the complex plane; and quaternionic analyticity lifts this duality to a non-commutative algebra, realising three-dimensional rotations through the exponential map and conjugation.

This chapter finally introduces the three fundamental differential operators—gradient, divergence and curl—and establishes the algebraic identities between them; the assumptions, ranges of applicability and boundary conditions for the three are given uniformly in the 'Vector Analysis' chapter (Section 15.2).

On power series and Laurent series: the general theory of series—Taylor series, Laurent series, the Cauchy–Hadamard radius-of-convergence formula and so on—has already been established systematically in Chapter 6, 'Series and Convolution' (Definitions 6.3 and 6.4 and Theorems 6.4 and 6.6). This chapter does not repeat that general theory; it uses its conclusions only in the context of complex analysis, and focuses on what is peculiar to complex analysis—the local expansion of analytic functions, the isolation of zeros, the classification of isolated singularities and the residue theorem.

Guide to the seven themes of this chapter: Translation—differential equations with periodic coefficients and Floquet theory (§9.2), the $2\pi i$-periodicity of the complex exponential function (Proposition 9.9); Scaling—dissipative systems and volume contraction (§9.9), the stretch of the modulus of a complex number; Rotation—Hamiltonian systems and circulation (§9.9), the rotational period of a complex function, the conjugation action of quaternion rotations (§9.8); Conservation—first integrals (§9.1.4), the Noether-type correspondence (Proposition 9.4), Liouville's formula (Proposition 9.2), the rigidity of complex analyticity (§9.5); Symmetry—the classification of the three operator systems (Theorem 9.15), complex conjugation symmetry, quaternion conjugation symmetry; Closure—limit cycles and the Poincaré–Bendixson theorem (§9.1.3), the closed contour of Cauchy's integral theorem (§9.6); Period—Floquet multipliers (§9.2), the imaginary period of the complex exponential, the 2π-periodicity of the quaternion exponential.

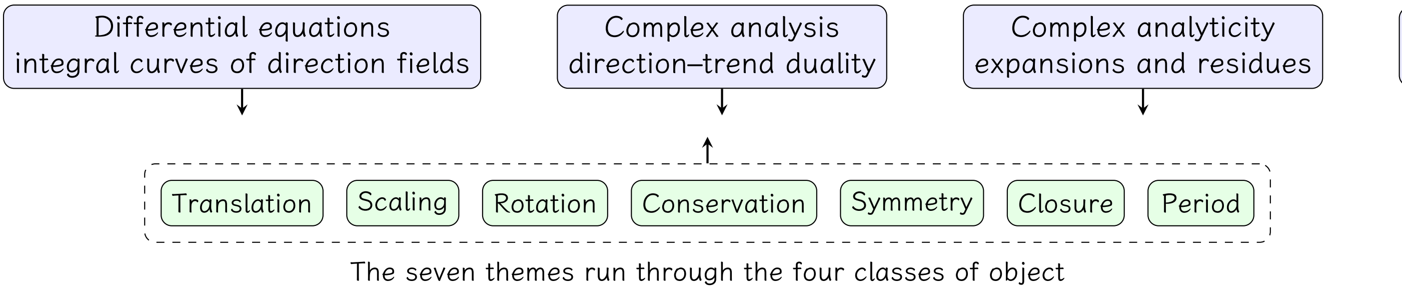

Differential equations
integral curves of direction fields
Complex analysis
direction–trend duality
Complex analyticity
expansions and residues
Translation
Scaling
Rotation
Conservation
Symmetry
Closure
Period
The seven themes run through the four classes of object

Figure 9.1: Overview of the structure of this chapter: the correspondence between the four classes of analysis object and the seven themes.

9.1 Ordinary Differential Equations: Integral Curves of Direction Fields

9.1.1 Basic Definitions, Existence and Uniqueness

Definition 9.1 (Ordinary Differential Equation)**.** Let $y : \mathbb{R} \to \mathbb{R}^n$ be an unknown vector-valued function. An **n-dimensional first-order system of ordinary differential equations** is

$$\frac{dy}{dt} = \mathsf{F}(t, y),$$

where $\mathsf{F} : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$ is a given vector field. If F does not depend explicitly on t, the equation is called **autonomous**. If $\mathsf{F}(t+T, y) = \mathsf{F}(t, y)$, it is called a **periodically driven system**.

Definition of a solution: a function $y : I \to \mathbb{R}^n$ (with I an interval) is called a **solution** of the equation on I if y is differentiable on I and $y'(t) = \mathsf{F}(t, y(t))$ for every $t \in I$.

Definition 9.2 (Initial-Value Problem)**.** Given $(t_0, y_0) \in \mathbb{R} \times \mathbb{R}^n$, the **initial-value problem** (IVP) is

$$\frac{dy}{dt} = \mathsf{F}(t, y), \qquad y(t_0) = y_0.$$

y_0 is called the **initial value** or the **initial condition**.

Definition 9.3 (Lipschitz Condition)**.** Let $\mathsf{F} : D \subseteq \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$. We say that F satisfies the **Lipschitz condition** in y on D (with Lipschitz constant $L > 0$) if for any $(t, y_1), (t, y_2) \in D$,

$$\|\mathsf{F}(t, y_1) - \mathsf{F}(t, y_2)\| \leq L\|y_1 - y_2\|.$$

Lemma 9.1 (Gronwall's Inequality)**.** Let $\varphi : [t_0, t_0 + \alpha] \to [0, \infty)$ be continuous and satisfy

$$\varphi(t) \leq \varphi_0 + \int_{t_0}^{t} \beta\varphi(s)\, ds, \qquad t \in [t_0, t_0 + \alpha],$$

where $\varphi_0 \geq 0$ and $\beta \geq 0$ are constants. Then

$$\varphi(t) \leq \varphi_0\, e^{\beta(t-t_0)}, \qquad t \in [t_0, t_0 + \alpha].$$

Symmetrically, if $\varphi(t) \leq \varphi_0 + \int_t^{t_0} \beta\varphi(s)\, ds$ $(t \leq t_0)$, then $\varphi(t) \leq \varphi_0 e^{\beta(t_0-t)}$.

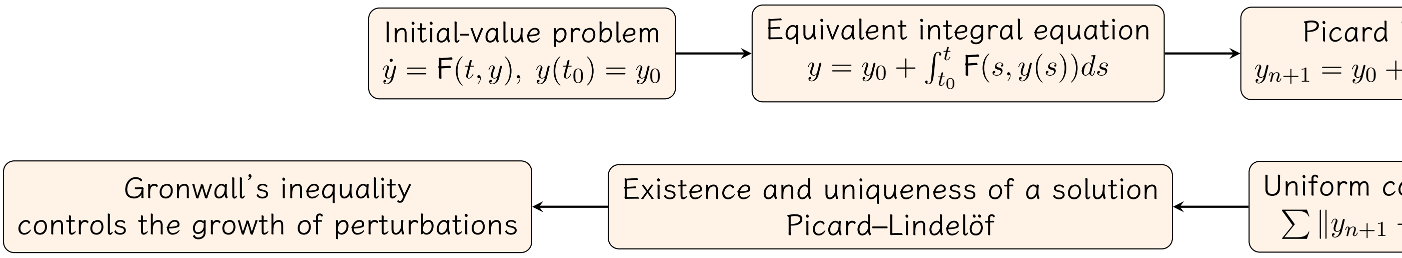

Initial-value problem
$\dot{y} = \mathsf{F}(t, y),\ y(t_0) = y_0$
Equivalent integral equation
$y = y_0 + \int_{t_0}^{t} \mathsf{F}(s, y(s))ds$
Picard
$y_{n+1} = y_0 +$
Uniform c
$\sum \|y_{n+1}$
Existence and uniqueness of a solution
Picard–Lindelöf
Gronwall's inequality
controls the growth of perturbations

Figure 9.2: Road map of the Picard–Lindelöf proof of existence and uniqueness: the initial-value problem is converted into an integral equation, Picard iteration together with uniform convergence yields the solution, and Gronwall's inequality is finally used to prove uniqueness.

Proof. Define $\Phi(t) = \varphi_0 + \int_{t_0}^{t} \beta\varphi(s)\,ds$. Then Φ is continuously differentiable, $\Phi' = \beta\varphi \leq \beta\Phi$, and $\Phi(t_0) = \varphi_0$. Let $\Psi(t) = \Phi(t)e^{-\beta(t-t_0)}$; then $\Psi'(t) = (\Phi'(t) - \beta\Phi(t))e^{-\beta(t-t_0)} \leq 0$, so $\Psi(t) \leq \Psi(t_0) = \varphi_0$, that is, $\Phi(t) \leq \varphi_0 e^{\beta(t-t_0)}$. Since $\varphi(t) \leq \Phi(t)$, the claim follows. The left-hand case follows in the same way after the change of variable $t \mapsto -t$. ☐

Theorem 9.1 (Picard–Lindelöf Existence and Uniqueness Theorem). Let $\mathsf{F}(t, y)$ be continuous on the domain $D = \{(t, y) : |t - t_0| \leq a, \|y - y_0\| \leq b\}$ and satisfy the Lipschitz condition in y (Definition 9.3, with Lipschitz constant L). Then the initial-value problem $\frac{dy}{dt} = \mathsf{F}(t, y)$, $y(t_0) = y_0$ has a unique solution on the interval $|t - t_0| \leq \alpha$, where $\alpha = \min(a, b/M)$ and $M = \max_{(t,y)\in D} \|\mathsf{F}(t, y)\|$.

Proof. **Step 1: reduction to an integral equation.** The initial-value problem is equivalent to the integral equation

$$y(t) = y_0 + \int_{t_0}^{t} \mathsf{F}(s, y(s))\,ds.$$

(If y is a solution of the differential equation, integrating both sides gives the integral equation; if y satisfies the integral equation, differentiating by the fundamental theorem of calculus gives the differential equation.)

Step 2: Picard iteration. Define the iteration sequence:

$$y_0(t) = y_0, \qquad y_{n+1}(t) = y_0 + \int_{t_0}^{t} \mathsf{F}(s, y_n(s))\,ds.$$

One must verify that $y_n(t)$ is defined on $|t - t_0| \leq \alpha$ and that $\|y_n(t) - y_0\| \leq b$. Use induction: for $n = 0$ we have $\|y_0(t) - y_0\| = 0 \leq b$, so the claim holds. Suppose that $\|y_n(t) - y_0\| \leq b$; then

$$\|y_{n+1}(t) - y_0\| \leq \int_{t_0}^{t} \|\mathsf{F}(s, y_n(s))\|\,ds \leq M|t - t_0| \leq M\alpha \leq b.$$

Step 3: proof that the iteration sequence converges. Estimate $\|y_{n+1}(t) - y_n(t)\|$. By the Lipschitz condition,

$$\|y_{n+1}(t) - y_n(t)\| \leq \int_{t_0}^{t} L\|y_n(s) - y_{n-1}(s)\|\,ds.$$

Prove by induction that

$$\|y_{n+1}(t) - y_n(t)\| \le \frac{ML^n|t-t_0|^{n+1}}{(n+1)!}.$$

Base step $n = 0$: $\|y_1(t) - y_0\| \le M|t-t_0|$. Inductive step: assuming the claim for $n-1$, we get

$$\|y_{n+1} - y_n\| \le \int_{t_0}^{t} L \cdot \frac{ML^{n-1}|s-t_0|^n}{n!}\, ds = \frac{ML^n|t-t_0|^{n+1}}{(n+1)!}.$$

Hence the series $\sum_{n=0}^{\infty} \|y_{n+1} - y_n\|$ converges (by comparison with the exponential series), and the sequence of functions $\{y_n\}$ converges uniformly on $|t-t_0| \le \alpha$ to some continuous function y.

Step 4: the limit is a solution. Passing to the limit in the iteration formula (the integral and the limit may be interchanged, by uniform convergence) gives

$$y(t) = y_0 + \int_{t_0}^{t} \mathsf{F}(s, y(s))\, ds.$$

Step 5: uniqueness. Let y_1, y_2 be two solutions and put $\delta(t) = \|y_1(t) - y_2(t)\|$. By the Lipschitz condition,

$$\delta(t) \le \int_{t_0}^{t} L\delta(s)\, ds \qquad (t \ge t_0).$$

Applying Gronwall's inequality (Lemma 9.1) with $\varphi(t) = \delta(t)$, $\varphi_0 = 0$ and $\beta = L$ gives $\delta(t) \le 0$, that is, $\delta(t) = 0$. The case $t \le t_0$ is analogous. Hence $y_1 = y_2$. ☐

Proposition 9.1 (Continuous Dependence of the Solution on the Initial Value). Let $\mathsf{F}(t, y)$ satisfy the Lipschitz condition (with Lipschitz constant L). Then the solution $y(t; t_0, y_0)$ depends continuously on the initial value y_0:

$$\|y(t; t_0, y_0) - y(t; t_0, \tilde{y}_0)\| \le e^{L|t-t_0|}\|y_0 - \tilde{y}_0\|.$$

Proof. Let $y_1(t) = y(t; t_0, y_0)$, $y_2(t) = y(t; t_0, \tilde{y}_0)$ and $\delta(t) = \|y_1(t) - y_2(t)\|$. By the Lipschitz condition,

$$\delta(t) \le \|y_0 - \tilde{y}_0\| + \int_{t_0}^{t} L\delta(s)\, ds \qquad (t \ge t_0).$$

Applying Gronwall's inequality (Lemma 9.1) with $\varphi(t) = \delta(t)$, $\varphi_0 = \|y_0 - \tilde{y}_0\|$ and $\beta = L$, we obtain

$$\delta(t) \le \|y_0 - \tilde{y}_0\|\, e^{L(t-t_0)}.$$

The case $t \le t_0$ is given analogously by the left-hand Gronwall inequality: $\delta(t) \le \|y_0 - \tilde{y}_0\|e^{L(t_0-t)}$. Combining the two yields $\delta(t) \le \|y_0 - \tilde{y}_0\|e^{L|t-t_0|}$. ☐

Property 9.1 (Differentiable Dependence of the Solution on Parameters). Let $\mathsf{F}(t, y, \mu)$ satisfy the Lipschitz condition in (y, μ), and let $\partial\mathsf{F}/\partial\mu$ be continuous. Then the solution $y(t; \mu)$ is differentiable with respect to the parameter μ, and its variation $\partial y/\partial\mu$ satisfies the **variational equation**

$$\frac{d}{dt}\frac{\partial y}{\partial \mu} = \frac{\partial \mathsf{F}}{\partial y}\frac{\partial y}{\partial \mu} + \frac{\partial \mathsf{F}}{\partial \mu}, \qquad \frac{\partial y}{\partial \mu}(t_0) = 0.$$

9.1.2 Linear Ordinary Differential Equations

Definition 9.4 (First-Order Linear Homogeneous System)**.** A first-order linear homogeneous ordinary differential equation has the form $\frac{dy}{dt} = A(t)y$. Its solution can be written as $y(t) = \Phi(t, t_0)y_0$, where $\Phi(t, t_0)$ is the **state transition matrix**, satisfying $\frac{d}{dt}\Phi(t, t_0) = A(t)\Phi(t, t_0)$ and $\Phi(t_0, t_0) = I$. When A is a constant matrix, $\Phi(t, 0) = e^{At}$.

Property 9.2 (Basic Properties of the State Transition Matrix)**.** The state transition matrix $\Phi(t, t_0)$ satisfies:

- (i) **Invertibility**: $\Phi(t, t_0)$ is invertible for every t, and $\Phi^{-1}(t, t_0) = \Phi(t_0, t)$.
- (ii) **Chain property**: $\Phi(t_2, t_0) = \Phi(t_2, t_1)\Phi(t_1, t_0)$.
- (iii) **Exponential form**: if $A(t)$ commutes with $\int_{t_0}^{t} A(s)\,ds$ (for example, if A is constant), then $\Phi(t, t_0) = \exp\left(\int_{t_0}^{t} A(s)\,ds\right)$. In the general case this formula does not hold.

Definition 9.5 (First-Order Linear Inhomogeneous System)**.** It has the form $\frac{dy}{dt} = A(t)y + \mathsf{b}(t)$. Its general solution is

$$y(t) = \Phi(t, t_0)y_0 + \int_{t_0}^{t} \Phi(t, s)\mathsf{b}(s)\,ds.$$

The homogeneous solution is the **direction part** and the particular-solution integral is the **trend part**.

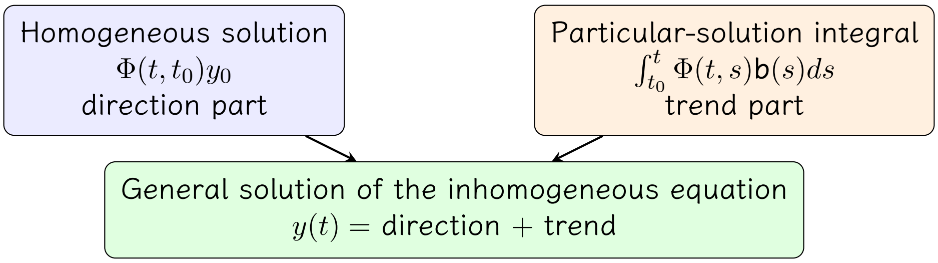

Homogeneous solution
$\Phi(t, t_0)y_0$
direction part
Particular-solution integral
$\int_{t_0}^{t} \Phi(t, s)\mathsf{b}(s)ds$
trend part
General solution of the inhomogeneous equation
$y(t) =$ direction + trend

Figure 9.3: Decomposition of the solution of a linear inhomogeneous system: the homogeneous solution gives the direction part and the particular-solution integral gives the trend part.

Proposition 9.2 (Liouville's Formula—the Evolution of Phase-Space Volume)**.** Let $\Phi(t, t_0)$ be the state transition matrix of $\frac{dy}{dt} = A(t)y$. Then

$$\det \Phi(t, t_0) = \exp\left(\int_{t_0}^{t} \operatorname{tr} A(s)\,ds\right).$$

$\operatorname{tr} A(t) = \nabla \cdot \mathsf{F}(t, y)$ is the divergence of the vector field. When $\operatorname{tr} A(t) = 0$, phase-space volume is conserved.

Proof. By Jacobi's formula, for any invertible matrix function $X(t)$,

$$\frac{d}{dt}\det X(t) = \det X(t)\cdot \operatorname{tr}\left(X(t)^{-1}\frac{dX(t)}{dt}\right).$$

Apply this to $\Phi(t,t_0)$: since $\frac{d\Phi}{dt} = A(t)\Phi$, we get

$$\frac{d}{dt}\det\Phi = \det\Phi\cdot\operatorname{tr}(A(t)).$$

This is a first-order linear scalar differential equation, whose solution is

$$\det\Phi(t,t_0) = \det\Phi(t_0,t_0)\cdot\exp\left(\int_{t_0}^{t}\operatorname{tr}A(s)\,ds\right) = \exp\left(\int_{t_0}^{t}\operatorname{tr}A(s)\,ds\right),$$

because $\Phi(t_0,t_0) = I$ and $\det I = 1$. □

Theorem 9.2 (Exponential Solutions of Linear Systems with Constant Coefficients)**.** *Let $A\in\mathbb{R}^{n\times n}$. The unique solution of the system $\dot{y} = Ay$ satisfying the initial value $y(0) = y_0$ is*

$$y(t) = e^{At}y_0, \qquad e^{At} := \sum_{k=0}^{\infty}\frac{(At)^k}{k!}.$$

Proof. The series $\sum_k (At)^k/k!$ converges absolutely for every t (it is dominated by the exponential series of $\|A\|$). Term-by-term differentiation gives

$$\frac{d}{dt}\sum_{k=0}^{\infty}\frac{(At)^k}{k!} = \sum_{k=1}^{\infty}A\frac{(At)^{k-1}}{(k-1)!} = Ae^{At}.$$

Hence $y(t) = e^{At}y_0$ is a solution; uniqueness follows from Picard–Lindelöf uniqueness (Theorem 9.1). □

9.1.3 Phase Portraits and Orbits

Definition 9.6 (Phase Portrait and Orbit)**.** The **phase portrait** of the autonomous system $\frac{dy}{dt} = \mathsf{F}(y)$ is the set of all solution curves in phase space. Each solution curve is called an **orbit**. A periodic orbit is a closed orbit.

Theorem 9.3 (Phase Portraits of Planar Linear Systems with Constant Coefficients)**.** *Consider $\frac{dy}{dt} = Ay$ (with $A\in\mathbb{R}^{2\times 2}$) and let its eigenvalues be λ_1,λ_2.*

(i) *$\lambda_1,\lambda_2\in\mathbb{R}$ of the same sign: a* **node***.*

(ii) *$\lambda_1,\lambda_2\in\mathbb{R}$ of opposite signs: a* **saddle point***.*

(iii) *$\lambda_{1,2} = \alpha\pm i\beta$ (with $\beta\neq 0$): a* **focus***.*

(iv) *$\lambda_{1,2} = \pm i\beta$ (purely imaginary): a* **centre***.*

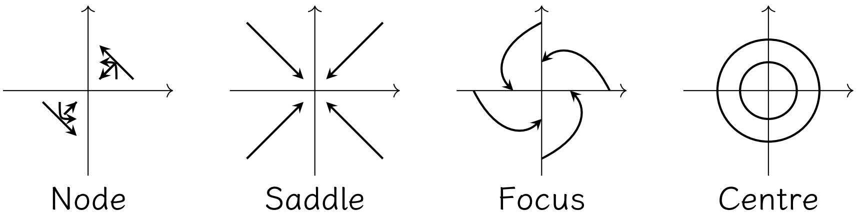


Figure 9.4: The four classes of phase portrait of a planar linear system with constant coefficients: node, saddle, focus and centre.

Proof. **(i) Node**: let $\lambda_1, \lambda_2 > 0$ (the negative case is similar). After a linear change of coordinates the system becomes $\dot{u} = \lambda_1 u$, $\dot{v} = \lambda_2 v$, with solution $u(t) = u_0 e^{\lambda_1 t}$, $v(t) = v_0 e^{\lambda_2 t}$. Eliminating t gives the orbit equation $v = Cu^{\lambda_2/\lambda_1}$ (when $v_0 \neq 0$). As $t \to \infty$, all orbits tend to the origin; as $t \to -\infty$, the orbits move away from the origin. The orbits are parabolic in shape (or straight lines, when $u_0 = 0$ or $v_0 = 0$).

(ii) Saddle: let $\lambda_1 > 0 > \lambda_2$. The solution is $u = u_0 e^{\lambda_1 t}$, $v = v_0 e^{\lambda_2 t}$. As $t \to \infty$, $u \to \infty$ whilst $v \to 0$; as $t \to -\infty$, $u \to 0$ whilst $v \to \infty$. The orbits are hyperbolic in shape.

(iii) Focus: let $\lambda = \alpha \pm i\beta$ with $\alpha, \beta \neq 0$. In polar coordinates the solution is $r(t) = r_0 e^{\alpha t}$, $\theta(t) = \theta_0 + \beta t$. The orbits are spirals: for $\alpha > 0$ the spiral is outward (an unstable focus), and for $\alpha < 0$ it is inward (a stable focus).

(iv) Centre: $\lambda = \pm i\beta$. The solution is $r(t) = r_0$ (a constant), $\theta(t) = \theta_0 + \beta t$. The orbits are concentric circles (or concentric ellipses under a general linear change of coordinates); they are closed but not isolated (they are not limit cycles). ☐

Definition 9.7 (Stability of an Equilibrium Point). *Let* **y*** *be an equilibrium point of an autonomous system. Its stability is determined by the eigenvalues of the Jacobi matrix* $J = D\mathbf{F}(\mathbf{y}^*)$:

(i) *all eigenvalues have negative real part:* **asymptotically stable** *(an attractor).*

(ii) *some eigenvalue has positive real part:* **unstable**.

(iii) *all eigenvalues have zero real part: the linearised system is of centre type, and higher-order analysis is required.*

If all eigenvalues have non-zero real part, the equilibrium point is called a **hyperbolic equilibrium point**.

Theorem 9.4 (Hartman–Grobman Theorem). *Let* **y*** *be a hyperbolic equilibrium point of an autonomous system. Then there is a neighbourhood of* **y*** *in which the original nonlinear system is topologically conjugate to the linearised system* $\frac{d\mathbf{z}}{dt} = J\mathbf{z}$.

Proof. **The proof lies beyond the scope of this chapter.** This is a deep theorem in dynamical systems theory, whose complete proof requires topological methods on manifolds (see Perko, Differential Equations and Dynamical Systems, 3rd ed., Springer, 2001, §2.8). The core idea is to construct a homeomorphism H such that $H \circ \Phi_t = e^{Jt} \circ H$ holds in a neighbourhood of the equilibrium point, where Φ_t is the nonlinear flow and e^{Jt} is the linearised flow. The construction uses the contraction mapping principle to solve a functional equation in a function space. ☐

Definition 9.8 (Limit Cycle). A **limit cycle** is an isolated periodic orbit of an autonomous nonlinear system. A limit cycle is the periodic realisation, in a dissipative system, of the **dynamic balance between volume contraction and expansion**: the closure of the orbit is maintained by the dynamic balance between energy input and dissipation, rather than being guaranteed by a global conserved quantity.

Theorem 9.5 (Poincaré–Bendixson Theorem). *Let $\frac{dy}{dt} = \mathsf{F}(y)$ be a planar autonomous system whose positively bounded orbits do not tend to any equilibrium point. Then the ω-limit set of such an orbit must be a limit cycle.*

Proof. **The proof lies beyond the scope of this chapter.** This is a classical result in the theory of planar dynamical systems (see Perko, Differential Equations and Dynamical Systems, 3rd ed., Springer, 2001, §3.7). The key to the proof is that in the plane a closed curve (a Jordan curve) divides the plane into an inside and an outside, and orbits cannot self-intersect (by the uniqueness theorem), so the ω-limit set has a special topological structure. If the ω-limit set contains no equilibrium point, then it must be a closed orbit. □

Example 9.1 (van der Pol Oscillator—a Limit Cycle of Self-Excited Oscillation). The van der Pol equation $\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0$ (with $\mu > 0$). When $|x| < 1$ the damping is negative (energy is injected); when $|x| > 1$ the damping is positive (energy is dissipated). The system has a unique stable limit cycle.

Example 9.2 (Duffing Oscillator—Nonlinear Resonance under Periodic Driving). The Duffing equation $\ddot{x} + \delta\dot{x} + \alpha x + \beta x^3 = \gamma \cos \omega t$. In the γ-ω parameter plane there are Arnold tongues.

9.1.4 Conservation Laws and First Integrals

Definition 9.9 (First Integral). Let $\frac{dy}{dt} = \mathsf{F}(y)$ be an autonomous system. A function $I : \mathbb{R}^n \to \mathbb{R}$ is called a **first integral** if $I(y(t))$ is constant along every solution curve $y(t)$—that is, if $\frac{d}{dt}I(y(t)) = \nabla I(y) \cdot \mathsf{F}(y) = 0$.

A first integral is precisely a **conserved quantity** of the differential equation.

Proposition 9.3 (First Integrals and Orbit Constraints). *If a system has k independent first integrals $I_1, \dots, I_k$, then its solutions are constrained to an $(n-k)$-dimensional invariant manifold. When $k = n-1$, the orbit is completely determined by the first integrals.*

Proof. The independent first integrals give the level set

$$\mathcal{M} = \{y \in \mathbb{R}^n : I_1(y) = c_1, \dots, I_k(y) = c_k\}.$$

By the implicit function theorem (when $\nabla I_1, \dots, \nabla I_k$ are linearly independent), $\mathcal{M}$ is a submanifold of dimension $n - k$. Solution curves always lie on $\mathcal{M}$ (because each I_j is constant along solution curves), so $\mathcal{M}$ is an invariant manifold. When $k = n - 1$, $\mathcal{M}$ is a 1-dimensional manifold and the solution curve is completely determined by $\mathcal{M}$ (the orbit is precisely a connected component of $\mathcal{M}$). □

Proposition 9.4 (Noether-Type Correspondence—a One-Parameter Symmetry Group Yields a First Integral)**.** Let the system $\frac{dy}{dt} = \mathsf{F}(y)$ be invariant under the one-parameter transformation group $\{\phi_s\}_{s\in\mathbb{R}}$, that is, $\phi_s \circ \Phi_t = \Phi_t \circ \phi_s$ for all s, t (where Φ_t is the flow of the system). Let ξ be the infinitesimal generator of ϕ_s at $s = 0$, that is, $\xi(y) = \frac{d}{ds}\big|_{s=0}\phi_s(y)$. If there is a smooth function $I : \mathbb{R}^n \to \mathbb{R}$ such that $\xi = J\nabla I$ (with J a non-degenerate antisymmetric matrix, that is, a symplectic structure), then I is a first integral of the system.

Proof. From the invariance $\phi_s \circ \Phi_t = \Phi_t \circ \phi_s$, differentiating with respect to s at $s = 0$ gives $[\xi, \mathsf{F}] = 0$, that is, the Lie bracket vanishes. If $\xi = J\nabla I$, then ∇I satisfies

$$\frac{d}{dt}I(y(t)) = \nabla I \cdot \mathsf{F} = (J^{-1}\xi) \cdot \mathsf{F}.$$

From $[\xi, \mathsf{F}] = 0$ and the antisymmetry of J one can deduce that $\nabla I \cdot \mathsf{F} = 0$ (see Arnold, Mathematical Methods of Classical Mechanics, 2nd ed., Springer, 1989, §38, for details). Hence I is a first integral. ☐

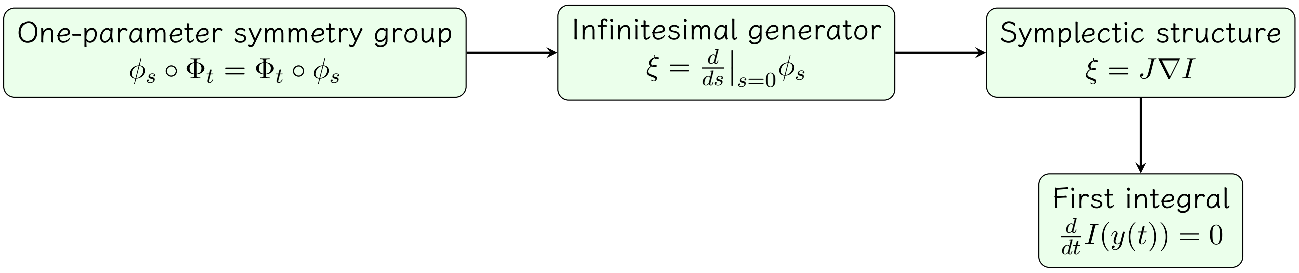

One-parameter symmetry group
$\phi_s \circ \Phi_t = \Phi_t \circ \phi_s$
Infinitesimal generator
$\xi = \frac{d}{ds}\big|_{s=0}\phi_s$
Symplectic structure
$\xi = J\nabla I$
First integral
$\frac{d}{dt}I(y(t)) = 0$

Figure 9.5: Noether-type correspondence: a continuous symmetry yields a first integral.

Example 9.3 (First Integrals in Classical Mechanics).

(i) **Conservation of energy**: in a conservative force field the total mechanical energy $E = \frac{1}{2}m\|\dot{\mathsf{r}}\|^2 + V(\mathsf{r})$ is a first integral. It corresponds to invariance under time translation.

(ii) **Conservation of angular momentum**: in a central force field the angular momentum $\mathsf{L} = m\mathsf{r} \times \dot{\mathsf{r}}$ is a first integral. It corresponds to rotational symmetry.

(iii) **Conservation of momentum**: when no external force acts in a given direction, the momentum component in that direction is a first integral. It corresponds to spatial translation symmetry.

9.2 Periodic-Coefficient Differential Equations and Floquet Theory

Definition 9.10 (Linear System with Periodic Coefficients)**.** Let $A(t)$ satisfy $A(t + T) = A(t)$ for all t. The system $\frac{dy}{dt} = A(t)y$ is called a **linear system with periodic coefficients**.

Theorem 9.6 (Floquet's Theorem). Let $\Phi(t)$ be a fundamental solution matrix of a linear system with periodic coefficients. Then there exist a non-singular T-periodic matrix $P(t)$ and a constant matrix B such that

$$\Phi(t) = P(t)e^{Bt}.$$

The matrix $M = \Phi(T, 0)$ is called the **monodromy matrix**, and its eigenvalues $\rho_1, \ldots, \rho_n$ are called the **Floquet multipliers**.

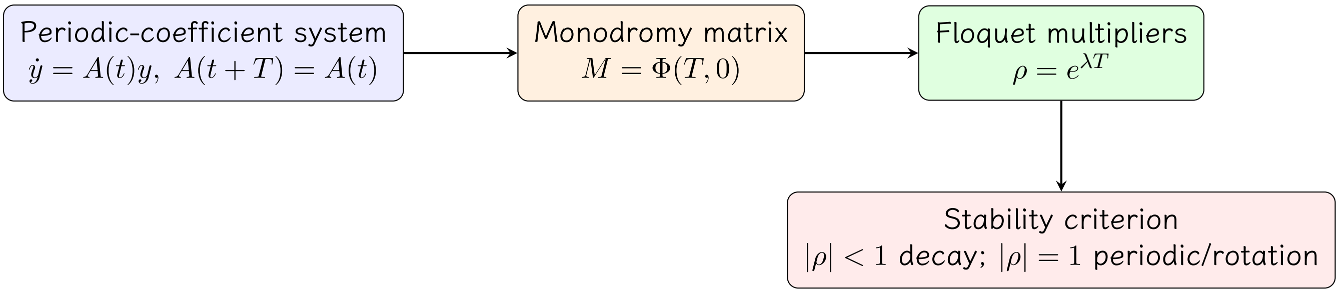

Periodic-coefficient system
$\dot{y} = A(t)y,\ A(t+T) = A(t)$
Monodromy matrix
$M = \Phi(T, 0)$
Floquet multipliers
$\rho = e^{\lambda T}$
Stability criterion
$|\rho| < 1$ decay; $|\rho| = 1$ periodic/rotation

Figure 9.6: Logical chain of Floquet theory: periodic-coefficient system → monodromy matrix → Floquet multipliers → stability and periodic solutions.

Proof. Define $M = \Phi(T, 0)$. By the periodicity $A(t+T) = A(t)$, verify that $\Phi(t+T, 0)$ is also a fundamental solution matrix: $\frac{d}{dt}\Phi(t+T, 0) = A(t+T)\Phi(t+T, 0) = A(t)\Phi(t+T, 0)$, and $\Phi(T, 0) = M$ is non-singular. Hence $\Phi(t+T, 0) = \Phi(t, 0)M$ (two fundamental solution matrices differ by a constant matrix factor).

Define $B = \frac{1}{T}\log M$ (the matrix logarithm exists because M is non-singular; if M has a negative real eigenvalue, $\log M$ may be a complex matrix, in which case B is a complex matrix and $P(t)$ is a complex T-periodic matrix). Define $P(t) = \Phi(t, 0)e^{-Bt}$. Verify that $P(t)$ is T-periodic:

$$P(t+T) = \Phi(t+T, 0)e^{-B(t+T)} = \Phi(t, 0)Me^{-BT}e^{-Bt} = \Phi(t, 0)e^{-Bt} = P(t),$$

where we used $e^{-BT} = e^{-\log M} = M^{-1}$. Hence $\Phi(t) = P(t)e^{Bt}$. $\square$

Corollary 9.1 (Floquet Multipliers and the Stability of Periodic Solutions).

(i) $\rho = 1$: the system has a T-periodic solution.

(ii) $\rho = -1$: the system has a $2T$-periodic solution (period-doubling bifurcation).

(iii) $\rho = e^{i2\pi p/q}$: the system has a qT-periodic solution.

(iv) all $|\rho_i| < 1$ (with $i \neq 1$ and $\rho_1 = 1$): this periodic solution is asymptotically stable.

Proof. The relation between an eigenvalue ρ of the monodromy matrix and the Floquet exponent λ is $\rho = e^{\lambda T}$ (because $M = e^{BT}$ and the eigenvalues of B are λ).

(i) $\rho = 1 \iff \lambda = 2\pi ik/T$ ($k \in \mathbb{Z}$). The corresponding solution is unchanged under the time translation T, that is, it is a T-periodic solution.

(ii) $\rho = -1 \iff \lambda = \pi i(2k+1)/T$. The corresponding solution is unchanged under the time translation $2T$ ($e^{\lambda \cdot 2T} = e^{2\pi i(2k+1)} = 1$), that is, it is a $2T$-periodic solution.

(iii) $\rho = e^{2\pi i p/q}$ (with p/q in lowest terms) $\iff$ $\lambda T = 2\pi i p/q + 2\pi i k$. The corresponding solution is unchanged under the time translation qT, that is, it is a qT-periodic solution.

(iv) $|\rho| < 1$ $\iff$ $\operatorname{Re}\lambda < 0$. The corresponding Floquet component decays exponentially. If all $|\rho_i| < 1$ except $\rho_1 = 1$, then every perturbative component of this periodic solution decays, and the periodic solution is asymptotically stable. ☐

Proposition 9.5 (Correspondence between Floquet Multipliers and the Three Operators). *The eigenvalue decomposition of the monodromy matrix M corresponds to three operators:*

(i) **Rotation part**: *the multipliers with $|\rho_i| = 1$ correspond to pure rotation.*

(ii) **Scaling part**: *the multipliers with $|\rho_i| \neq 1$ correspond to scaling.*

(iii) **Translation part**: *the period T itself carries the translation symmetry.*

Proof. By the proof of Corollary 9.1, $|\rho| = 1$ $\iff$ $\operatorname{Re}\lambda = 0$, and the Floquet component neither grows nor decays but rotates purely on an invariant torus. $|\rho| \neq 1$ $\iff$ $\operatorname{Re}\lambda \neq 0$, and the Floquet component grows or decays exponentially, its ratio being the scaling factor of volume. The period T is the parameter of the time-translation symmetry. ☐

Example 9.4 (Hill's Equation—a Periodically Driven Oscillator). The Hill equation $\ddot{y}+p(t)y = 0$ (with $p(t+T) = p(t)$). When $p(t) = \omega_0^2+\varepsilon\cos\omega t$ it reduces to the Mathieu equation. Written as $\dot{y} = A(t)y$ the Hill equation has $\operatorname{tr} A \equiv 0$, so Liouville's formula (Proposition 9.2) gives $\det\Phi(T, 0) = 1$; hence its Floquet multipliers satisfy $\rho_1\rho_2 = 1$.

Example 9.5 (Mathieu's Equation and Parametric Resonance). The Mathieu equation $\ddot{y} + (\delta + \varepsilon\cos 2t)y = 0$ (period $T = \pi$). In the δ-ε parameter plane, stable and unstable regions alternate.

9.3 Fundamentals of Partial Differential Equations

Definition 9.11 (Partial Differential Equation). Let $u : \mathbb{R}^n \to \mathbb{R}$ be an unknown function. A **k-th order partial differential equation** is an equation involving u and its partial derivatives up to order k.

Definition 9.12 (Wave Equation). The **wave equation** is

$$\frac{\partial^2 u}{\partial t^2} - c^2\nabla^2 u = 0.$$

Its general solution is $u(\mathsf{x}, t) = f(\mathsf{x}\cdot\mathsf{n} - ct) + g(\mathsf{x}\cdot\mathsf{n} + ct)$.

Definition 9.13 (Heat Equation). The **heat equation** is

$$\frac{\partial u}{\partial t} - \alpha\nabla^2 u = 0.$$

Each Fourier component decays exponentially: $c_n(t) = c_n(0)e^{-\alpha(2\pi n/T)^2 t}$.

Definition 9.14 (Laplace's Equation)**.** The **Laplace equation** is $\nabla^2 u = 0$. Its solutions are called **harmonic functions**.

Proposition 9.6 (Fourier Method for Linear Partial Differential Equations with Constant Coefficients)**.** *For a linear partial differential equation with constant coefficients and periodic boundary conditions, the Fourier transform converts it into an algebraic equation or an ordinary differential equation. Let* $u(\mathsf{x}, t) = \sum_{\mathsf{k}} \hat{u}_{\mathsf{k}}(t) e^{i\mathsf{k}\cdot\mathsf{x}}$*; the component of each wave vector* k *evolves independently.*

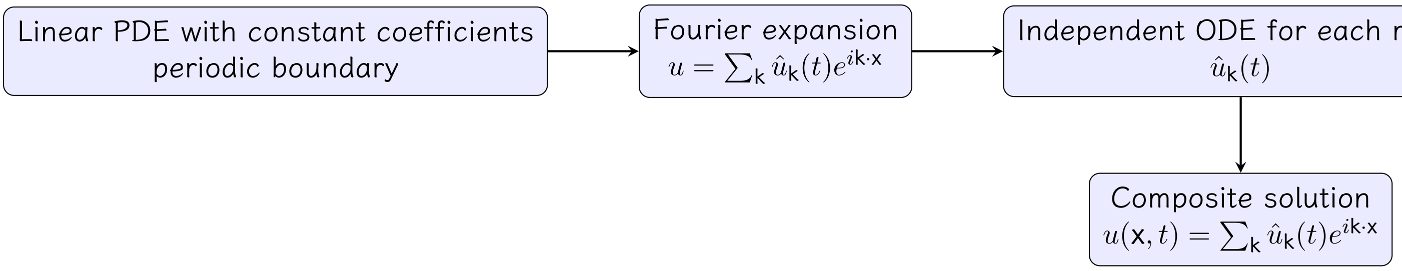

Linear PDE with constant coefficients
periodic boundary
Fourier expansion
$u = \sum_{\mathsf{k}} \hat{u}_{\mathsf{k}}(t) e^{i\mathsf{k}\cdot\mathsf{x}}$
Independent ODE for each r
$\hat{u}_{\mathsf{k}}(t)$
Composite solution
$u(\mathsf{x}, t) = \sum_{\mathsf{k}} \hat{u}_{\mathsf{k}}(t) e^{i\mathsf{k}\cdot\mathsf{x}}$

Figure 9.7: The Fourier method for linear partial differential equations with constant coefficients: spatial derivatives become multiplication and the modes of each wave vector evolve independently.

Proof. Substitute $u(\mathsf{x}, t) = \sum_{\mathsf{k}} \hat{u}_{\mathsf{k}}(t) e^{i\mathsf{k}\cdot\mathsf{x}}$ into the partial differential equation. For a linear operator $\mathcal{L}(\partial_{\mathsf{x}}, \partial_t)$ with constant coefficients, its action on $e^{i\mathsf{k}\cdot\mathsf{x}}$ is $\mathcal{L}(i\mathsf{k}, \partial_t) e^{i\mathsf{k}\cdot\mathsf{x}}$ (the Fourier transform converts the spatial derivative ∂_{x_j} into multiplication by ik_j). By the orthogonality of the Fourier basis, the various k components are independent, and each $\hat{u}_{\mathsf{k}}(t)$ satisfies an ordinary differential equation. ☐

Example 9.6 (Fourier Method for the Heat Equation)**.** For $\frac{\partial u}{\partial t} = \alpha \nabla^2 u$, substituting the Fourier series gives $\frac{d\hat{u}_{\mathsf{k}}}{dt} = -\alpha |\mathsf{k}|^2 \hat{u}_{\mathsf{k}}$, whose solution is $\hat{u}_{\mathsf{k}}(t) = \hat{u}_{\mathsf{k}}(0) e^{-\alpha |\mathsf{k}|^2 t}$.

Example 9.7 (Fourier Method for the Wave Equation)**.** For $\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u$, substituting the Fourier series gives $\frac{d^2 \hat{u}_{\mathsf{k}}}{dt^2} = -c^2 |\mathsf{k}|^2 \hat{u}_{\mathsf{k}}$, whose solution is $\hat{u}_{\mathsf{k}}(t) = A_{\mathsf{k}} e^{ic|\mathsf{k}|t} + B_{\mathsf{k}} e^{-ic|\mathsf{k}|t}$.

9.4 Green's Functions and Integral Equations

Definition 9.15 (Green's Function)**.** For a linear differential operator $\mathcal{L}$, its **Green's function** $G(\mathsf{x}, \boldsymbol{\xi})$ satisfies

$$\mathcal{L} G(\mathsf{x}, \boldsymbol{\xi}) = \delta(\mathsf{x} - \boldsymbol{\xi}).$$

The solution of the inhomogeneous equation $\mathcal{L} u = f$ is

$$u(\mathsf{x}) = \int G(\mathsf{x}, \boldsymbol{\xi}) f(\boldsymbol{\xi}) d\boldsymbol{\xi} = (G * f)(\mathsf{x}).$$

Remark 9.2 (Existence of Green's Functions)*.* A Green's function exists if and only if the homogeneous equation $\mathcal{L} u = 0$ (under the given boundary conditions) has only the zero solution. This is a direct corollary of the Fredholm alternative theorem (Theorem 9.7): if the kernel of $\mathcal{L}$ is non-trivial, then $\mathcal{L}$ is not invertible and the Green's function does not exist.

Example 9.8 (Fundamental Solution of the Laplace Operator). The fundamental solution of the three-dimensional Laplace operator is $G(\mathsf{x}) = -\frac{1}{4\pi|\mathsf{x}|}$ (satisfying $\nabla^2 G = \delta(\mathsf{x})$). This agrees exactly with the inverse-square law.

Example 9.9 (Fundamental Solution of the Heat Operator). The fundamental solution of the one-dimensional heat operator is $G(x,t) = \frac{1}{\sqrt{4\pi\alpha t}}e^{-x^2/(4\alpha t)}$. The solution of the heat equation is the convolution of the initial condition with the Gaussian kernel.

Definition 9.16 (Fredholm Integral Equation). The **Fredholm integral equation of the second kind** is

$$u(x) - \lambda \int_a^b K(x,y)u(y)\,dy = f(x).$$

When $K(x,y) = K(x-y)$ (a convolution kernel), the Fredholm equation can be diagonalised by the Fourier transform.

Definition 9.17 (Volterra Integral Equation). The **Volterra integral equation of the second kind** is

$$u(x) - \lambda \int_a^x K(x,y)u(y)\,dy = f(x).$$

A Volterra equation corresponds to causal accumulation—the integral depends only on past values.

Theorem 9.7 (Fredholm Alternative Theorem). *For a Fredholm integral equation with continuous kernel, exactly one of the following two alternatives holds:*

(i) *for every f the equation has a unique solution.*

(ii) *the homogeneous equation $u - \lambda K u = 0$ has a non-zero solution.*

Proof. **The proof lies beyond the scope of this chapter.** This is a fundamental theorem of the spectral theory of compact operators (see Conway, A Course in Functional Analysis, 2nd ed., Springer, 1990, §VII.6). The core fact is that an integral operator K with continuous kernel is compact, and its spectrum consists of at most countably many eigenvalues tending to zero. If λ^{-1} is not an eigenvalue of K, then $(I - \lambda K)$ is invertible and the equation has a unique solution; if λ^{-1} is an eigenvalue, then the homogeneous equation has a non-zero solution. □

Proposition 9.7 (Equivalence of Integral and Differential Equations). *An initial-value problem for an ordinary differential equation with initial conditions is equivalent to a Volterra integral equation; a boundary-value problem for an ordinary differential equation with boundary conditions is equivalent to a Fredholm integral equation.*

Proof. The initial-value problem $\frac{dy}{dt} = F(t,y)$, $y(t_0) = y_0$ is equivalent to $y(t) = y_0 + \int_{t_0}^t F(s,y(s))\,ds$—this is a Volterra equation of the second kind (the linearisation of the kernel $K(t,s) = F$). This equivalence was already established in Step 1 of the proof of Theorem 9.1.

A boundary-value problem is represented by means of the Green's function (Definition 9.15): $u = G * f$, that is, $u(x) = \int G(x,\xi)f(\xi)d\xi$. Rewriting this in the form $u - \lambda K u = f$ (where K is the integral operator corresponding to the Green's function) gives a Fredholm equation of the second kind. □

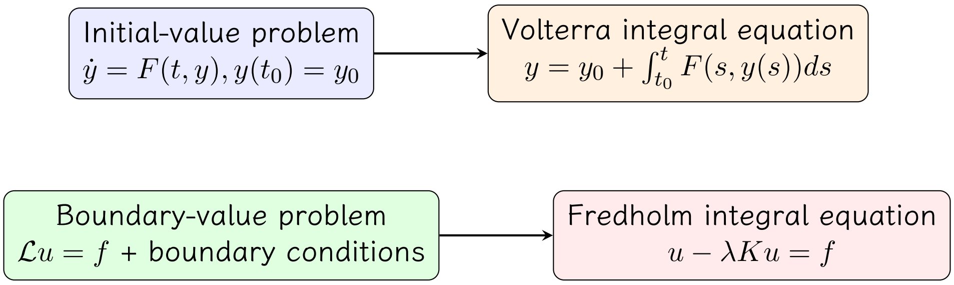

Initial-value problem
$\dot{y} = F(t, y), y(t_0) = y_0$
Volterra integral equation
$y = y_0 + \int_{t_0}^{t} F(s, y(s))ds$
Boundary-value problem
$\mathcal{L}u = f$ + boundary conditions
Fredholm integral equation
$u - \lambda K u = f$

Figure 9.8: *The equivalence between differential and integral equations: the initial-value problem corresponds to a Volterra equation and the boundary-value problem to a Fredholm equation.*

9.5 Foundations of Complex Analysis

9.5.1 The Algebra and Geometry of Complex Numbers

Definition 9.18 (The Field of Complex Numbers)**.** Let $\mathbb{C} = \{(a, b) : a, b \in \mathbb{R}\}$, equipped with the addition $(a, b) + (c, d) = (a + c, b + d)$ and the multiplication $(a, b)(c, d) = (ac - bd, ad + bc)$. Write $i = (0, 1)$; then every complex number can be written as $z = a + bi$.

Property 9.3 (Basic Properties of the Field of Complex Numbers)**.** (i) $(\mathbb{C}, +, \times)$ is a field, with 0 the additive identity and 1 the multiplicative identity.

(ii) $i^2 = -1$, and $\mathbb{C} = \mathbb{R}[i] \cong \mathbb{R}[x]/(x^2 + 1)$.

(iii) **Conjugation**: $\overline{a + bi} = a - bi$, satisfying $\overline{z + w} = \bar{z} + \bar{w}$ and $\overline{zw} = \bar{z}\bar{w}$.

(iv) **Modulus**: $|z| = \sqrt{z\bar{z}} = \sqrt{a^2 + b^2}$, satisfying $|zw| = |z||w|$ (multiplicativity) and $|z + w| \leq |z| + |w|$ (the triangle inequality).

(v) **Polar form**: a non-zero $z = re^{i\theta}$, where $r = |z| > 0$ and $\theta = \arg z \in \mathbb{R}/(2\pi\mathbb{Z})$ is the argument.

Definition 9.19 (Three Centres of Symmetry in the Complex Plane)**.** The complex plane has three intrinsic centres of symmetry:

(i) 0 (the additive identity): the fixed point of the additive reflection $z \mapsto -z$.

(ii) 1 (the multiplicative identity): one of the fixed points of multiplicative inversion $\sigma_1(z) = 1/z$.

(iii) i (the imaginary unit): the **centre of symmetry** of the i-involution $\sigma_i(z) = iz^{-1}$ (the composite centre of $\sigma_i = R_{\pi/2} \circ \sigma_1$); the fixed points of σ_i are $z = \pm\frac{\sqrt{2}}{2}(1 + i)$, and $i^4 = 1$ generates C_4.

9.5.2 Limits of Complex Sequences and Complex Functions

Definition 9.20 (Limit of a Complex Sequence)**.** Let $\{z_n\}$ be a complex sequence and $L \in \mathbb{C}$. We say that $\lim z_n = L$ if

$$\forall \varepsilon > 0,\ \exists N \in \mathbb{N},\ \forall n > N,\ |z_n - L| < \varepsilon.$$

Proposition 9.8 (Necessary and Sufficient Condition for Convergence of a Complex Sequence)**.** *A complex sequence $\{z_n\}$ converges to $L = a + bi$ if and only if its real and imaginary parts converge separately.*

Proof. Let $z_n = x_n + iy_n$ and $L = a + bi$.

Necessity: if $z_n \to L$, then for any $\varepsilon > 0$ there is N such that $|z_n - L| < \varepsilon$ whenever $n > N$. Since

$$|x_n - a| \le \sqrt{(x_n - a)^2 + (y_n - b)^2} = |z_n - L| < \varepsilon,$$

we have $x_n \to a$. Similarly $y_n \to b$.

Sufficiency: if $x_n \to a$ and $y_n \to b$, then for any $\varepsilon > 0$ there are N_1, N_2 such that $|x_n - a| < \varepsilon/\sqrt{2}$ whenever $n > N_1$ and $|y_n - b| < \varepsilon/\sqrt{2}$ whenever $n > N_2$. Take $N = \max\{N_1, N_2\}$; then for $n > N$,

$$|z_n - L| = \sqrt{(x_n - a)^2 + (y_n - b)^2} < \sqrt{\varepsilon^2/2 + \varepsilon^2/2} = \varepsilon.$$

□

Property 9.4 (The Cauchy Criterion and Completeness for Sequences of Complex Numbers)**.** *$\mathbb{C}$ is complete: a complex sequence converges if and only if it is a Cauchy sequence. That is, $\{z_n\}$ converges $\iff$ for any $\varepsilon > 0$ there is N such that $|z_m - z_n| < \varepsilon$ whenever $m, n > N$.*

Proof. ($\Rightarrow$) If $z_n \to L$, then $|z_m - z_n| \le |z_m - L| + |L - z_n| \to 0$.

($\Leftarrow$) If $\{z_n\}$ is a Cauchy sequence, then $|x_m - x_n| \le |z_m - z_n|$ shows that $\{x_n\}$ is a Cauchy sequence of real numbers, which converges to a by the completeness of $\mathbb{R}$; similarly $\{y_n\}$ converges to b. By Proposition 9.8, $z_n \to a + bi$. □

Definition 9.21 (Limit and Continuity of a Complex-Valued Function)**.** Let $f : \mathbb{C} \supseteq D \to \mathbb{C}$. We say that $\lim_{z\to z_0} f(z) = L$ if

$$\forall\varepsilon > 0,\ \exists\delta > 0,\ \forall z \in D,\ 0 < |z - z_0| < \delta \implies |f(z) - L| < \varepsilon.$$

If $\lim_{z\to z_0} f(z) = f(z_0)$, then f is said to be continuous at z_0.

9.5.3 Complex Derivatives and Analytic Functions

Definition 9.22 (Complex Derivative)**.** Let $f : \mathbb{C} \supseteq D \to \mathbb{C}$ and $z_0 \in D$. If the limit

$$f'(z_0) := \lim_{h\to 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

exists and is finite, then f is said to be **complex differentiable** at z_0. Here $h \in \mathbb{C}$ may tend to 0 along an arbitrary path.

Definition 9.23 (Analytic Function)**.** If f is complex differentiable at every point of an open set $U \subseteq \mathbb{C}$, then f is said to be **analytic** (or **holomorphic**) on U. If f is analytic in some neighbourhood of z_0, then f is said to be analytic at z_0.

Definition 9.24 (Entire Function)**.** A function analytic on the whole of $\mathbb{C}$ is called an **entire function**. For example, e^z, $\sin z$, $\cos z$ and all polynomials are entire functions.

Theorem 9.8 (Cauchy–Riemann Equations)**.** Let $f(z) = u(x,y) + iv(x,y)$. If f is complex differentiable at z_0, then

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Proof. Consider $h \to 0$ along the real axis ($h = \Delta x$, $\Delta y = 0$):

$$f'(z_0) = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x} = \frac{\partial u}{\partial x}(x_0, y_0) + i\frac{\partial v}{\partial x}(x_0, y_0).$$

Along the imaginary axis ($h = i\Delta y$, $\Delta x = 0$):

$$f'(z_0) = \lim_{\Delta y \to 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{i\Delta y} = \frac{1}{i}\left(\frac{\partial u}{\partial y} + i\frac{\partial v}{\partial y}\right) = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}.$$

Since f is complex differentiable, the two limits must be equal. Comparing real and imaginary parts:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}.$$

□

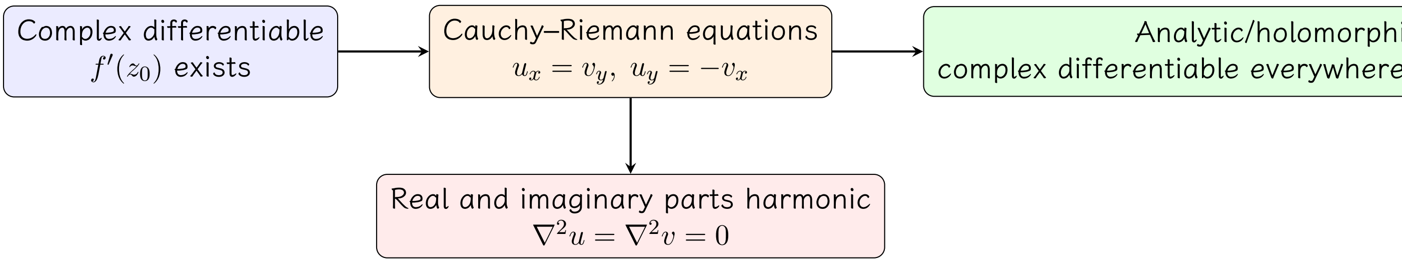

Complex differentiable
$f'(z_0)$ exists
Cauchy–Riemann equations
$u_x = v_y$, $u_y = -v_x$
Analytic/holomorphi
complex differentiable everywhere
Real and imaginary parts harmonic
$\nabla^2 u = \nabla^2 v = 0$

Figure 9.9: The relations between complex differentiability, the Cauchy–Riemann equations, analyticity and harmonicity.

Remark 9.3 (Cauchy–Riemann Equations and Vanishing Curl)*.* Consider the vector field $\mathsf{F} = (u, -v)$. Its two-dimensional curl is

$$\frac{\partial(-v)}{\partial x} - \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0$$

(by the Cauchy–Riemann equations). Hence **an analytic function corresponds to an irrotational field.** Similarly, the divergence of the vector field $\mathsf{G} = (u, v)$ is $\partial u/\partial x + \partial v/\partial y = 2\partial u/\partial x$, which is in general non-zero. This structure of analytic functions is the meeting point of complex analysis and vector analysis.

Corollary 9.2 (The Real and Imaginary Parts of an Analytic Function are Harmonic)**.** If $f = u + iv$ is analytic on an open set U, then both u and v are harmonic functions:

$$\nabla^2 u = 0, \qquad \nabla^2 v = 0.$$

Conversely, if u is harmonic on a simply connected domain, then there exists its **harmonic conjugate** v such that $u + iv$ is analytic.

Proof. By the Cauchy–Riemann equations $u_x = v_y$, $u_y = -v_x$, if f is twice continuously differentiable (which will be guaranteed automatically by Cauchy's integral formula in Theorem 9.10), then

$$u_{xx} = (v_y)_x = v_{yx} = v_{xy} = (-u_y)_y = -u_{yy},$$

so $u_{xx} + u_{yy} = 0$. Similarly v is harmonic. □

Proposition 9.9 (Periodicity and Derivative of the Complex Exponential Function). $f(z) = e^z = e^x(\cos y + i \sin y)$ is analytic on the whole complex plane, $f'(z) = e^z$, and $e^{z+2\pi i} = e^z$.

Proof. **Derivative**: from $e^z = e^x(\cos y + i\sin y)$, verify the Cauchy–Riemann equations: $u = e^x \cos y$, $v = e^x \sin y$. $\frac{\partial u}{\partial x} = e^x \cos y$ and $\frac{\partial v}{\partial y} = e^x \cos y$, which are equal; $\frac{\partial u}{\partial y} = -e^x \sin y$ and $-\frac{\partial v}{\partial x} = -e^x \sin y$, which are equal. By the Cauchy–Riemann equations, e^z is complex differentiable, and $f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = e^x \cos y + ie^x \sin y = e^z$.

Periodicity: $e^{z+2\pi i} = e^x e^{i(y+2\pi)} = e^x e^{iy} e^{2\pi i} = e^z$, because $e^{2\pi i} = \cos(2\pi) + i\sin(2\pi) = 1$. □

Example 9.10 (Definitions of the Complex Elementary Functions). (i) $\sin z = \dfrac{e^{iz} - e^{-iz}}{2i}$, $\cos z = \dfrac{e^{iz} + e^{-iz}}{2}$. Both are entire functions, with periodicity 2π.

(ii) $\tan z = \sin z / \cos z$ is analytic where $\cos z \neq 0$; its zeros are $\{z = \pi/2 + k\pi\}$.

(iii) $\log z = \ln|z| + i \arg z$ is a multi-valued function; after taking the principal branch Log it is analytic on $\mathbb{C} \setminus (-\infty, 0]$.

(iv) $z^\alpha = e^{\alpha \operatorname{Log} z}$ (with $\alpha \notin \mathbb{Z}$) is a multi-valued function, and its zeros and poles depend on α.

9.6 Cauchy's Integral Theory

Lemma 9.2 (The Plane Green's Theorem (Forward Reference)). Let $P, Q \in C^1(\overline{D_\gamma})$, where γ is a piecewise smooth simple closed curve and D_γ is the region it encloses. Then

$$\oint_\gamma (P\,dx + Q\,dy) = \iint_{D_\gamma} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy.$$

An independent proof of this lemma is given in the 'Vector Analysis' chapter (Theorem 15.6). This section cites it first, which constitutes a **forward reference**: the reader may accept the conclusion for now, or follow the route free of forward dependencies according to Remark 9.4.

Remark 9.4 (Two Routes to Cauchy's Integral Theorem). Theorem 9.9 has two standard routes of proof. The first is the **Green's theorem route** adopted in this section: the contour integral is converted into two double integrals, and the Cauchy–Riemann equations then make the integrand vanish everywhere. This route requires f' to be continuous, and requires the region to have a piecewise smooth boundary (the hypotheses of Lemma 9.2). The second is **Goursat's subdivision method**: triangulate the region enclosed by the path, split the integral over the large contour into integrals over small triangles, estimate piece by piece and refine the subdivision; it requires only that f be differentiable on the region, and not that f' be continuous. This route **does not use Green's theorem**, and therefore avoids forward dependencies altogether. This book adopts the former, in order to stay consistent with the physical picture of 'flux–circulation'; the two routes give the same conclusion.

Theorem 9.9 (Cauchy's Integral Theorem). *Let f be analytic on a simply connected domain D, and let γ be a closed path in D. Then*

$$\oint_\gamma f(z)\,dz = 0.$$

Proof. Let $f = u + iv$ and $dz = dx + i\,dy$. Expand the contour integral into real integrals:

$$\oint_\gamma f(z)\,dz = \oint_\gamma (u+iv)(dx+i\,dy) = \oint_\gamma (u\,dx - v\,dy) + i\oint_\gamma (v\,dx + u\,dy).$$

By Lemma 9.2 (the plane Green's theorem; for its independent proof see Theorem 15.6 in the 'Vector Analysis' chapter, this being a forward reference), for the closed path γ in D and the region D_γ it encloses, taking $(P,Q) = (u,-v)$ and $(P,Q) = (v,u)$ gives respectively

$$\oint_\gamma (u\,dx - v\,dy) = -\iint_{D_\gamma} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right) dx\,dy,$$

$$\oint_\gamma (v\,dx + u\,dy) = \iint_{D_\gamma} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) dx\,dy.$$

By the Cauchy–Riemann equations (Theorem 9.8), $\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} = 0$ and $\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} = 0$, so both double integrals are zero and both contour integrals are zero, whence $\oint_\gamma f(z)\,dz = 0$. □

Theorem 9.10 (Cauchy's Integral Formula). *Let f be analytic inside the closed path γ, and let z_0 be a point inside γ. Then*

$$f(z_0) = \frac{1}{2\pi i}\oint_\gamma \frac{f(z)}{z - z_0}\,dz.$$

Proof. On $D \setminus \{z_0\}$ the integrand $f(z)/(z-z_0)$ is analytic. Take a small circle γ_ε centred at z_0 (of radius ε); by Cauchy's integral theorem (Theorem 9.9) deform γ into γ_ε:

$$\oint_\gamma \frac{f(z)}{z-z_0}\,dz = \oint_{\gamma_\varepsilon} \frac{f(z)}{z-z_0}\,dz.$$

On γ_ε, parametrise $z = z_0 + \varepsilon e^{i\theta}$ $(\theta : 0 \to 2\pi)$, so that $dz = i\varepsilon e^{i\theta} d\theta$:

$$\oint_{\gamma_\varepsilon} \frac{f(z)}{z-z_0}\,dz = \int_0^{2\pi} \frac{f(z_0+\varepsilon e^{i\theta})}{\varepsilon e^{i\theta}} \cdot i\varepsilon e^{i\theta}\,d\theta = i\int_0^{2\pi} f(z_0+\varepsilon e^{i\theta})\,d\theta.$$

By the continuity of f at z_0, $\lim_{\varepsilon\to 0} f(z_0+\varepsilon e^{i\theta}) = f(z_0)$ uniformly in θ (since f is continuous), so

$$i\int_0^{2\pi} f(z_0+\varepsilon e^{i\theta})\,d\theta \to i\int_0^{2\pi} f(z_0)\,d\theta = 2\pi i f(z_0).$$

Hence $f(z_0) = \frac{1}{2\pi i}\oint_\gamma \frac{f(z)}{z-z_0}dz$. ☐

Corollary 9.3 (Infinite Differentiability of Analytic Functions). If f is analytic on an open set U, then f is complex differentiable infinitely many times on U, and its n-th derivative satisfies

$$f^{(n)}(z_0) = \frac{n!}{2\pi i}\oint_\gamma \frac{f(z)}{(z-z_0)^{n+1}}\,dz, \qquad n = 0, 1, 2, \ldots$$

where γ is a positively oriented simple closed curve surrounding z_0 and contained in U.

Proof. Differentiating Cauchy's integral formula formally with respect to z_0, and differentiating under the integral sign (justified by uniform convergence), gives the result. ☐

Theorem 9.11 (Liouville's Theorem). If f is a bounded entire function, then f is a constant function.

Proof. Let $|f(z)| \le M$ for all $z \in \mathbb{C}$. For any $z_0 \in \mathbb{C}$ and any $R > 0$, taking $n = 1$ in Corollary 9.3 and integrating along the circle γ_R centred at z_0 with radius R:

$$|f'(z_0)| \le \frac{1}{2\pi}\oint_{\gamma_R} \frac{|f(z)|}{|z-z_0|^2}\,|dz| \le \frac{1}{2\pi}\cdot\frac{M}{R^2}\cdot 2\pi R = \frac{M}{R}.$$

Letting $R \to \infty$ gives $f'(z_0) = 0$. Hence the derivative of f vanishes everywhere and f is constant. ☐

Corollary 9.4 (The Fundamental Theorem of Algebra). Every non-constant polynomial with complex coefficients has a root in $\mathbb{C}$.

Proof. Let $p(z)$ be a non-constant polynomial without roots. Then $1/p(z)$ is an entire function. As $|z| \to \infty$ we have $|p(z)| \to \infty$, so $1/p(z)$ is bounded. By Liouville's theorem (Theorem 9.11), $1/p$ is constant, and hence p is constant, contradicting the hypothesis. ☐

9.7 Local Expansions and Residues of Analytic Functions

Remark 9.5 (Interface with the 'Series and Convolution' chapter). The general theory of Taylor series and Laurent series has already been established systematically in Chapter 6, 'Series and Convolution': for the definition of the Taylor series and its convergence conditions see Definition 6.3, Theorem 6.5; for the definition of the Laurent series see Definition 6.4; for the Laurent expansion

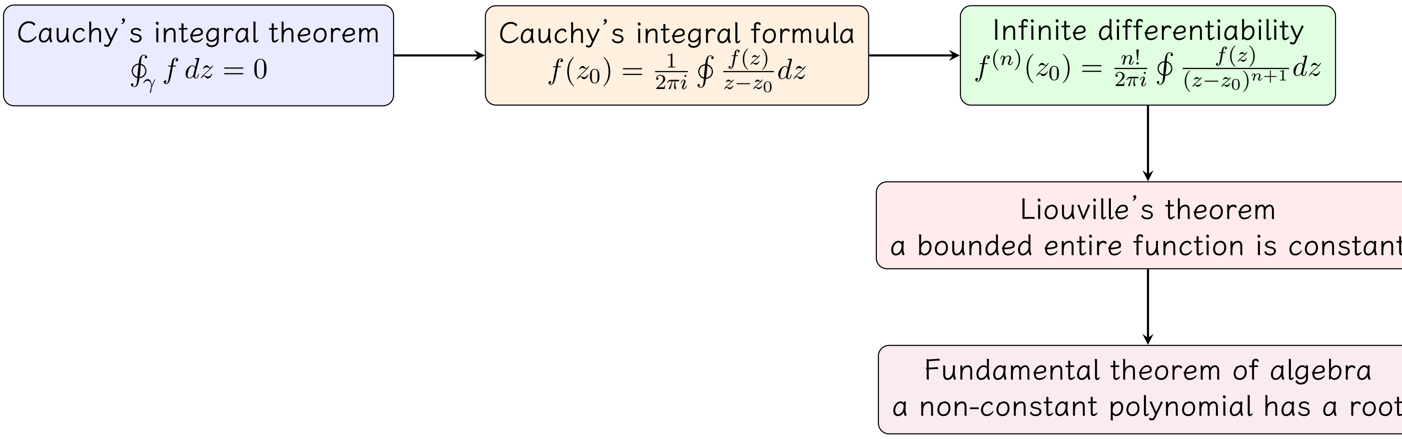

Cauchy's integral theorem
$\oint_\gamma f\,dz = 0$
Cauchy's integral formula
$f(z_0) = \frac{1}{2\pi i}\oint \frac{f(z)}{z-z_0}dz$
Infinite differentiability
$f^{(n)}(z_0) = \frac{n!}{2\pi i}\oint \frac{f(z)}{(z-z_0)^{n+1}}dz$
Liouville's theorem
a bounded entire function is constant
Fundamental theorem of algebra
a non-constant polynomial has a root

Figure 9.10: The logical chain of Cauchy's integral theory: the integral theorem, the integral formula, infinite differentiability, Liouville's theorem and the fundamental theorem of algebra.

theorem see Theorem 6.6; and for the Cauchy–Hadamard radius-of-convergence formula see Theorem 6.4. This section does not repeat that general theory; it gives only its direct applications in the context of complex analysis: the Taylor expansion of analytic functions, the isolation of zeros, the classification of isolated singularities and the residue theorem.

Theorem 9.12 (Taylor Expansion of Analytic Functions). If f is analytic on the disc $D(z_0, R)$, then for every $z \in D(z_0, R)$,

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z-z_0)^n,$$

and the radius of convergence is given by the Cauchy–Hadamard formula (Chapter 6, Theorem 6.4).

Proof. By Cauchy's integral formula, for any $|z - z_0| < r < R$ and $|\zeta - z_0| = r$,

$$f(z) = \frac{1}{2\pi i}\oint_{|\zeta - z_0| = r} \frac{f(\zeta)}{\zeta - z}\,d\zeta.$$

Expand $1/(\zeta - z) = 1/(\zeta - z_0) \cdot 1/(1 - (z - z_0)/(\zeta - z_0))$ as a geometric series, substitute and integrate term by term to obtain the result. □

Corollary 9.5 (Isolation of the Zeros of Analytic Functions). If f is analytic on a connected open set U and is not identically zero, then the zero set of f has no accumulation point in U. Equivalently, if f vanishes on some convergent sequence of **distinct** points $\{z_n\}$ in U (with $z_n \to z_0 \in U$), then $f \equiv 0$.

Proof. Suppose, to the contrary, that the zeros have an accumulation point $z_0 \in U$. By the Taylor expansion (Theorem 9.12), f can be expanded near z_0. If the expansion is non-zero, there is a least $k \geq 0$ with $a_k \neq 0$, and near z_0 we have $f(z) = (z - z_0)^k g(z)$ with $g(z_0) \neq 0$. Since g is continuous, $g \neq 0$ in some neighbourhood of z_0, so $f \neq 0$ in that neighbourhood except at z_0, contradicting the fact that the zeros have an accumulation point. Hence $a_k = 0$ for all k, and $f \equiv 0$. □

Definition 9.25 (Classification of Isolated Singularities)**.** Let f be analytic on $0 < |z - z_0| < R$, and let z_0 be its isolated singularity. By the Laurent expansion $\sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$ (for the existence of the Laurent expansion see Chapter 6, Theorem 6.6):

(i) **Removable singularity**: all $a_n = 0$ $(n < 0)$.

(ii) **Pole of order** m: $a_{-m} \neq 0$ and $a_n = 0$ $(n < -m)$.

(iii) **Essential singularity**: infinitely many $n < 0$ with $a_n \neq 0$.

Definition 9.26 (Residue)**.** Let z_0 be an isolated singularity of f. The **residue** of f at z_0 is the coefficient of the term $(z - z_0)^{-1}$ in the Laurent expansion:

$$\text{Res}(f, z_0) = a_{-1} = \frac{1}{2\pi i} \oint_\gamma f(z)\, dz,$$

where γ is a small positively oriented circle near z_0.

Theorem 9.13 (Residue Theorem)**.** *Let f be analytic in a simply connected domain D except at finitely many isolated singularities $z_1, \ldots, z_m$, and let γ be a positively oriented simple closed curve in D enclosing all the z_k. Then*

$$\oint_\gamma f(z)\, dz = 2\pi i \sum_{k=1}^{m} \text{Res}(f, z_k).$$

Proof. Take a small circle γ_k near each z_k, so that these small circles are mutually disjoint and all lie inside γ. By Cauchy's integral theorem, $\oint_\gamma f = \sum_k \oint_{\gamma_k} f$. Each $\oint_{\gamma_k} f = 2\pi i \,\text{Res}(f, z_k)$ (by the definition of the residue). □

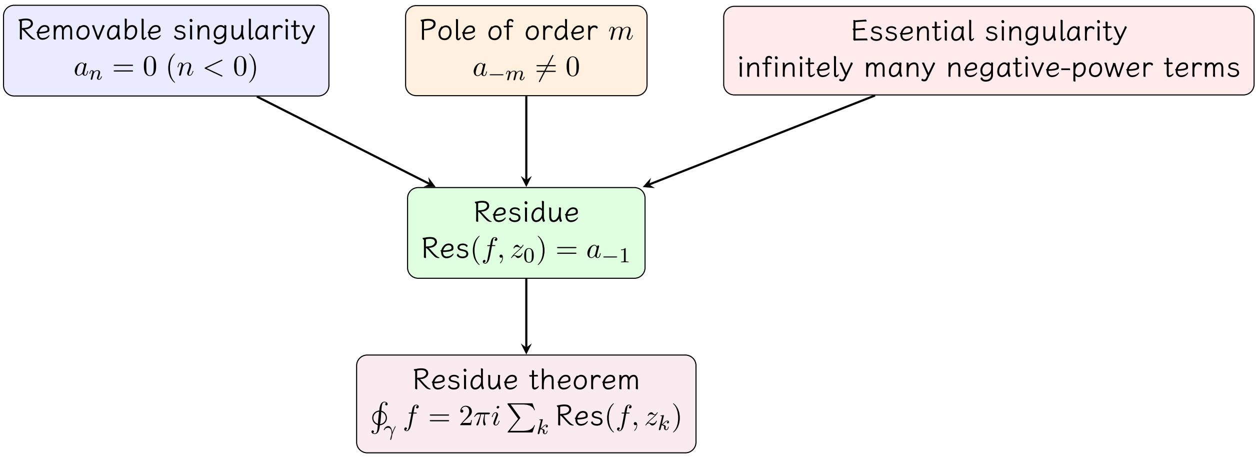

Removable singularity
$a_n = 0$ $(n < 0)$
Pole of order m
$a_{-m} \neq 0$
Essential singularity
infinitely many negative-power terms
Residue
$\text{Res}(f, z_0) = a_{-1}$
Residue theorem
$\oint_\gamma f = 2\pi i \sum_k \text{Res}(f, z_k)$

Figure 9.11: Classification of isolated singularities and the residue theorem: the structure of the negative-power terms determines the type of singularity, and a_{-1} determines the residue.

Example 9.11 (Residue Calculations: Three Cases)**.** (i) **Simple pole**: if $f(z) = g(z)/(z - z_0)$ and $g(z_0) \neq 0$, then $\text{Res}(f, z_0) = g(z_0)$.

(ii) **Pole of order** m: $\text{Res}(f, z_0) = \frac{1}{(m-1)!} \lim_{z \to z_0} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)]$.

(iii) **Quotient of two functions**: if $f = g/h$, $h(z_0) = 0$, $h'(z_0) \neq 0$ and $g(z_0) \neq 0$, then $\text{Res}(f, z_0) = g(z_0)/h'(z_0)$.

9.8 Quaternion Analysis and Analyticity

9.8.1 The Algebraic Structure of the Quaternions

Definition 9.27 (The Quaternion Algebra). The **quaternion algebra** $\mathbb{H}$ is the four-dimensional associative algebra over $\mathbb{R}$ with basis $\{1, i, j, k\}$, whose multiplication is the bilinear extension of the following relations:

$$i^2 = j^2 = k^2 = ijk = -1,$$

together with the relations $ij = k$, $ji = -k$, $jk = i$, $kj = -i$, $ki = j$, $ik = -j$ derived from them. A quaternion is written as $q = a + bi + cj + dk$ (with $a, b, c, d \in \mathbb{R}$).

Property 9.5 (Basic Algebraic Properties of the Quaternions).

(i) $\mathbb{H}$ is a **division algebra** (a division ring): every non-zero element has a multiplicative inverse.

(ii) $\mathbb{H}$ is **non-commutative**: $ij \neq ji$. In general $q_1 q_2 \neq q_2 q_1$.

(iii) **Conjugation**: $\bar{q} = a - bi - cj - dk$, satisfying $\overline{pq} = \bar{q}\bar{p}$ (an anti-automorphism).

(iv) **Modulus**: $|q| = \sqrt{q\bar{q}} = \sqrt{a^2 + b^2 + c^2 + d^2}$, satisfying the multiplicativity $|pq| = |p||q|$.

(v) **Inverse**: $q^{-1} = \bar{q}/|q|^2$ (for $q \neq 0$).

(vi) **Decomposition**: $q = \mathrm{Sc}(q) + \mathrm{Vec}(q)$, where $\mathrm{Sc}(q) = a$ is the scalar part and $\mathrm{Vec}(q) = bi + cj + dk$ is the vector part.

Definition 9.28 (Pure Imaginary Quaternions and the Unit Sphere). A quaternion $q \in \mathbb{H}$ is called a **pure imaginary quaternion** if $\mathrm{Sc}(q) = 0$. The space of pure imaginary quaternions is denoted $\mathrm{Im}(\mathbb{H}) \cong \mathbb{R}^3$. The **unit quaternion group** is

$$S^3 = \{q \in \mathbb{H} : |q| = 1\},$$

and it forms a group under quaternion multiplication, isomorphic to $SU(2)$.

9.8.2 Limits and Continuity of Quaternions

Definition 9.29 (Quaternion Limit). Let $f : \mathbb{H} \supseteq D \to \mathbb{H}$ and $q_0 \in \overline{D}$. We say that $\lim_{q \to q_0} f(q) = L$ if

$$\forall \varepsilon > 0,\ \exists \delta > 0,\ \forall q \in D,\ 0 < |q - q_0| < \delta \implies |f(q) - L| < \varepsilon.$$

If $\lim_{q \to q_0} f(q) = f(q_0)$, then f is said to be continuous at q_0.

Proposition 9.10 (Coordinate Characterisation of Quaternionic Limits). Write $q = q_0 + h$, $h = h_0 + h_1 i + h_2 j + h_3 k$ and $f = f_0 + f_1 i + f_2 j + f_3 k$. Then $\lim_{h \to 0} f(q_0 + h) = L$ if and only if the four real-component limits hold separately.

Proof. The argument is the same as for the convergence of complex sequences (Proposition 9.8): the quaternion modulus $|h| = \sqrt{h_0^2 + h_1^2 + h_2^2 + h_3^2}$ controls the absolute values of the components, and conversely. □

9.8.3 Quaternion Derivatives and Analyticity

Definition 9.30 (Quaternion Derivative)**.** Let $f : \mathbb{H} \supseteq D \to \mathbb{H}$ and $q_0 \in D$. If the limit

$$f'(q_0) := \lim_{h\to 0}[f(q_0+h) - f(q_0)] \cdot h^{-1}$$

exists, then f is said to be **right differentiable** at q_0. Similarly, if the limit

$${}'f(q_0) := \lim_{h\to 0} h^{-1} \cdot [f(q_0+h) - f(q_0)]$$

exists, then f is said to be **left differentiable** at q_0. Since quaternion multiplication is non-commutative, the left and right derivatives are in general unequal.

Definition 9.31 (Quaternionic Analytic Functions)**.** If f is right differentiable everywhere (or left differentiable everywhere) on an open set $U \subseteq \mathbb{H}$, then f is said to be **right analytic** (or **left analytic**) on U. Here 'right analytic' only means differentiable by difference quotients; it is not the same condition as the **regular**ity defined by the Cauchy–Riemann–Fueter system, and the relation between the two is given in Remark 9.6.

Remark 9.6 (The Structural Difference from Complex Analyticity). Unlike complex analytic functions, quaternionic analytic functions are **extremely scarce**. Here two different conditions must be distinguished. Write $f = f_0 + f_1 i + f_2 j + f_3 k$ and set $e_1 = i$, $e_2 = j$, $e_3 = k$.

(a) **Differentiability by difference quotients** (the right differentiability of Definition 9.31) is equivalent to

$$\frac{\partial f}{\partial x_\mu} = \Big(\frac{\partial f}{\partial x_0}\Big)e_\mu, \qquad \mu = 1, 2, 3,$$

that is, to three component equations.

(b) **The Cauchy–Riemann–Fueter system** characterises a different class of objects, namely the **regular** functions:

$$\frac{\partial f}{\partial \bar{q}} := \frac{\partial f}{\partial x_0} + i\frac{\partial f}{\partial x_1} + j\frac{\partial f}{\partial x_2} + k\frac{\partial f}{\partial x_3} = 0,$$

that is, the four component functions satisfy a first-order linear system of partial differential equations.

The two conditions are **not equivalent**: $f(q) = q$ is right differentiable by difference quotients ($[h]h^{-1} \equiv 1$), yet $\partial f/\partial \bar{q} = 1 + i\cdot i + j \cdot j + k \cdot k = -2 \neq 0$, so it does not satisfy the CRF system; conversely, $f(q) = x_0 + x_1 i$ satisfies $\partial f/\partial \bar{q} = 1 + i \cdot i = 0$, yet is not right differentiable (here $\partial_{x_2} f = 0$ whilst $(\partial_{x_0} f)e_2 = j \neq 0$, violating (a)). Thus 'right analytic' means differentiable by difference quotients, whilst the class of regular functions is the one characterised by the CRF system.

The CRF system consists of four first-order equations and is far more restrictive than the complex case (two equations); consequently the structure of the regular functions it characterises is far less rich than that of complex analytic functions. Quaternion power series $\sum q^n a_n$, quaternion polynomials and local expansions in powers of $(q - q_0)$ belong to a different class of functions, namely the **slice-regular** class rather than the Fueter-regular one: a single term $q^n a_n$ with $n \geq 1$ satisfies $\partial(q^n a_n)/\partial \bar{q} = -2a_n \neq 0$. The division of labour between the two classes is described in the 'Quaternion Analysis' chapter (§8.5.1).

9.8.4 The Quaternion Exponential and Rotations

Proposition 9.11 (Rotational Periodicity of the Quaternion Exponential Function). *For any unit pure imaginary quaternion* u *(with* $\mathsf{u}^2 = -1$*),*

$$e^{\mathsf{u}\theta} = \cos\theta + \mathsf{u}\sin\theta$$

satisfies $e^{\mathsf{u}(\theta+2\pi)} = e^{\mathsf{u}\theta}$*. For different rotation axes* $\mathsf{u}_1 \neq \mathsf{u}_2$*, the product of* $e^{\mathsf{u}_1\theta_1}$ *and* $e^{\mathsf{u}_2\theta_2}$ *does not commute in general.*

Proof. **Euler's formula**: since $\mathsf{u}^2 = -1$, expand the exponential function as a power series:

$$e^{\mathsf{u}\theta} = \sum_{k=0}^{\infty} \frac{(\mathsf{u}\theta)^k}{k!} = \sum_{j=0}^{\infty} \frac{(\mathsf{u}\theta)^{2j}}{(2j)!} + \sum_{j=0}^{\infty} \frac{(\mathsf{u}\theta)^{2j+1}}{(2j+1)!}.$$

Since $\mathsf{u}^{2j} = (-1)^j$ and $\mathsf{u}^{2j+1} = (-1)^j\mathsf{u}$,

$$e^{\mathsf{u}\theta} = \sum_{j=0}^{\infty} \frac{(-1)^j\theta^{2j}}{(2j)!} + \mathsf{u}\sum_{j=0}^{\infty} \frac{(-1)^j\theta^{2j+1}}{(2j+1)!} = \cos\theta + \mathsf{u}\sin\theta.$$

Periodicity: $e^{\mathsf{u}(\theta+2\pi)} = \cos(\theta+2\pi) + \mathsf{u}\sin(\theta+2\pi) = \cos\theta + \mathsf{u}\sin\theta = e^{\mathsf{u}\theta}$.

Non-commutativity: if $\mathsf{u}_1 \neq \mathsf{u}_2$, then $\mathsf{u}_1\mathsf{u}_2 \neq \mathsf{u}_2\mathsf{u}_1$ (the product of two distinct unit pure imaginary quaternions is anti-commutative: $\mathsf{u}_1\mathsf{u}_2 = -\mathsf{u}_2\mathsf{u}_1$). Hence

$$e^{\mathsf{u}_1\theta_1}e^{\mathsf{u}_2\theta_2} = (\cos\theta_1 + \mathsf{u}_1\sin\theta_1)(\cos\theta_2 + \mathsf{u}_2\sin\theta_2)$$

$$= \cos\theta_1\cos\theta_2 + \mathsf{u}_1\sin\theta_1\cos\theta_2 + \mathsf{u}_2\cos\theta_1\sin\theta_2 + \mathsf{u}_1\mathsf{u}_2\sin\theta_1\sin\theta_2,$$

whilst in $e^{\mathsf{u}_2\theta_2}e^{\mathsf{u}_1\theta_1}$ the term $\mathsf{u}_1\mathsf{u}_2$ is replaced by $\mathsf{u}_2\mathsf{u}_1 = -\mathsf{u}_1\mathsf{u}_2$, so the two are in general unequal. ☐

Theorem 9.14 (Conjugation Representation of Quaternionic Rotations). *For a unit quaternion* $q \in S^3$ *and a pure imaginary quaternion* $v \in \mathrm{Im}(\mathbb{H}) \cong \mathbb{R}^3$*, the map*

$$\rho_q(v) = qvq^{-1}$$

is a rotation of $\mathbb{R}^3$*. The map* $\Pi : S^3 \to SO(3)$*,* $q \mapsto \rho_q$*, is a double covering, with* $\ker\Pi = \{\pm 1\}$*.*

Proof. **Preservation of pure imaginaryness**: $\overline{qvq^{-1}} = \bar{q}^{-1}\bar{v}\bar{q} = q(-v)q^{-1} = -qvq^{-1}$, so $\rho_q(v)$ is a pure imaginary quaternion.

Preservation of modulus: $|\rho_q(v)| = |q||v||q^{-1}| = |v|$.

Rodrigues formula: write $q = \cos(\theta/2) + \mathsf{u}\sin(\theta/2)$ (with u unit pure imaginary) and decompose v as $v_\parallel + v_\perp$ (parallel and perpendicular to u respectively). A direct computation gives

$$qvq^{-1} = v_\parallel + \cos\theta\, v_\perp + \sin\theta\,(\mathsf{u}\times v_\perp),$$

which is precisely the Rodrigues formula for a rotation through the angle θ about the axis u.

Double covering: $\rho_q = \rho_{-q}$, so the map does not descend to a bijection from S^3 to $SO(3)$; conversely, every element of $SO(3)$ corresponds by the Rodrigues formula to exactly one pair $\pm q$. Hence Π is a double covering. ☐

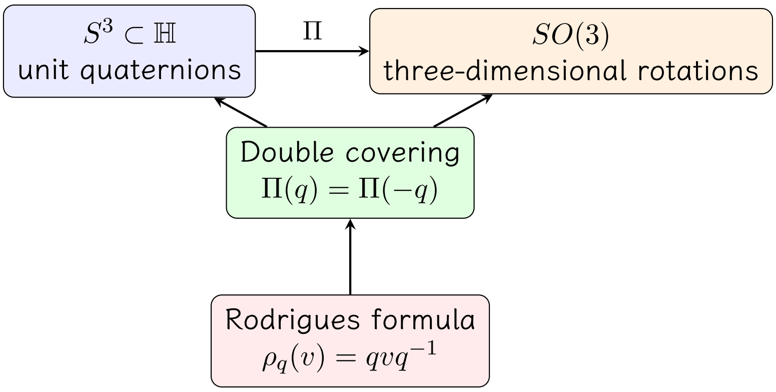


Figure 9.12: The double covering from unit quaternions to three-dimensional rotations: $S^3 \to SO(3)$, $\ker \Pi = \{\pm 1\}$.

Property 9.6 (Classification of the Finite Subgroups of the Unit Quaternion Group)**.** The classification of the finite subgroups of $S^3 \cong SU(2)$ is:

(i) cyclic groups C_n $(n \geq 1)$.

(ii) dihedral groups D_n $(n \geq 2)$.

(iii) the tetrahedral group $2T$ (of order 24).

(iv) the octahedral group $2O$ (of order 48).

(v) the icosahedral group $2I$ (of order 120).

The images of these finite groups in the three-dimensional rotation group $SO(3)$ are respectively C_n, D_n, T, O, I—that is, the rotation symmetry groups of the Platonic solids.

9.9 Gradient, Curl and Divergence: the Three Fundamental Differential Operators

Definition 9.32 (Gradient)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable at the point x. The **gradient** of f is

$$\nabla f(\mathsf{x}) := \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n} \right).$$

The gradient corresponds to the **translation direction (0-symmetry)**. The curl of a gradient vanishes identically: $\nabla \times (\nabla f) = \mathbf{0}$.

Definition 9.33 (Divergence)**.** Let $\mathsf{F} : \mathbb{R}^n \to \mathbb{R}^n$ be differentiable at the point x. The **divergence** of F is

$$\nabla \cdot \mathsf{F}(\mathsf{x}) := \frac{\partial F_1}{\partial x_1} + \cdots + \frac{\partial F_n}{\partial x_n}.$$

The divergence corresponds to the **proportional direction (1-symmetry)**. The divergence measures the rate of volume change: $\nabla \cdot \mathsf{F} > 0$ means local expansion, and $\nabla \cdot \mathsf{F} < 0$ means local contraction.

Definition 9.34 (Curl). Let $\mathsf{F} : \mathbb{R}^3 \to \mathbb{R}^3$ be differentiable at the point $\mathbf{x}$. The **curl** of F is

$$\nabla \times \mathsf{F} := \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}, \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}, \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right).$$

The curl corresponds to the **rotation direction (i-symmetry)**. The divergence of a curl vanishes identically: $\nabla \cdot (\nabla \times \mathsf{F}) = 0$.

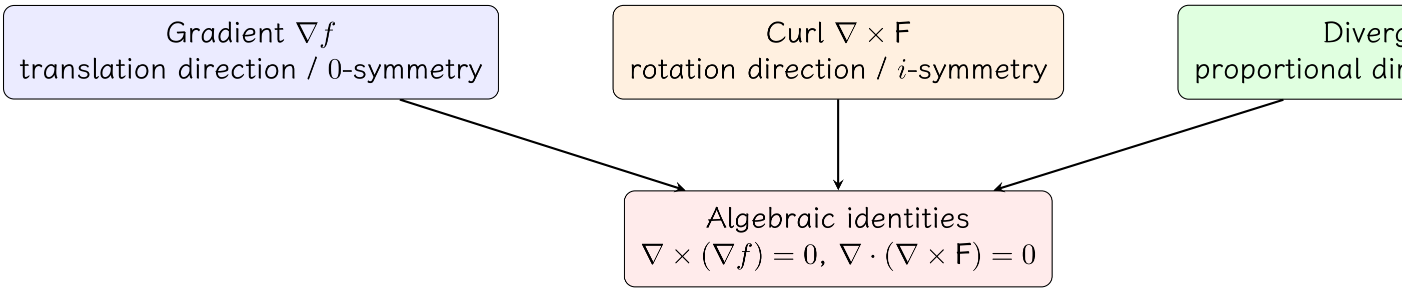

Gradient ∇f
translation direction / 0-symmetry
Curl $\nabla \times \mathsf{F}$
rotation direction / i-symmetry
Diverg
proportional di
Algebraic identities
$\nabla \times (\nabla f) = 0$, $\nabla \cdot (\nabla \times \mathsf{F}) = 0$

Figure 9.13: The correspondence between the three fundamental differential operators and the three centres of symmetry, together with the two basic algebraic identities.

Remark 9.7 (Direction–Trend Attributes of the Three Operators). From the definitions in Definition 9.32, Definition 9.33 and Definition 9.34 we obtain directly the correspondence between the three operators and the three centres of symmetry:

(i) **Gradient** extracts the direction of a scalar field—a purely directional operator (0-symmetry).

(ii) **Curl** extracts the rotation direction of a vector field—a purely rotational-direction operator (i-symmetry).

(iii) **Divergence** generates the trend of a vector field—a direction–trend composite operator (1-symmetry).

The algebraic relations between the three—the curl of a gradient is zero, and the divergence of a curl is zero—reflect the hierarchical structure of the three centres of symmetry. This correspondence will be given independently in the 'Vector Analysis' chapter in the form of limit characterisations (Propositions 15.4 to 15.6).

Definition 9.35 (Gradient System—Gradient (Centre of Symmetry 0)). If there is a potential function V such that $\frac{dy}{dt} = -\nabla V(y)$, the system is called a **gradient system**. A gradient system satisfies:

(i) $V(y(t))$ is monotonically non-increasing along solution curves.

(ii) all bounded solutions tend to equilibrium points, and there are no periodic orbits.

(iii) the curl vanishes identically: $\nabla \times (-\nabla V) = \mathbf{0}$.

Definition 9.36 (Hamiltonian System—Curl (Centre of Symmetry i)). Let $H : \mathbb{R}^{2n} \to \mathbb{R}$ be a Hamiltonian function. A **Hamiltonian system** is

$$\frac{dq}{dt} = \frac{\partial H}{\partial p}, \quad \frac{dp}{dt} = -\frac{\partial H}{\partial q}.$$

A Hamiltonian system satisfies: H is a first integral; phase-space volume is conserved ($\nabla \cdot \mathsf{F} = 0$).

Definition 9.37 (Dissipative System—Divergence (Centre of Symmetry 1)). If $\frac{dy}{dt} = \mathsf{F}(y)$ satisfies $\nabla \cdot \mathsf{F} < 0$ (in some region), the system is called a **dissipative system**. The long-term behaviour of a dissipative system tends towards a low-dimensional attractor.

Theorem 9.15 (Periodic Orbits of the Three Operator Systems).
- (i) **Gradient systems**: no periodic orbits (only equilibrium points).
- (ii) **Hamiltonian systems**: periodic orbits exist on invariant tori (in the integrable case).
- (iii) **Dissipative systems**: periodic orbits exist in the form of limit cycles.

Proof. (i) Suppose that a gradient system has a non-trivial periodic orbit γ (parametrised as $y(t)$, with $y(t+T) = y(t)$). Integrating once around γ, the net change of the potential function V is

$$V(y(T)) - V(y(0)) = \int_0^T \frac{d}{dt} V(y(t))\, dt = \int_0^T \nabla V(y(t)) \cdot \dot{y}(t)\, dt.$$

Since $\dot{y} = -\nabla V$,

$$= \int_0^T \nabla V \cdot (-\nabla V)\, dt = -\int_0^T \|\nabla V(y(t))\|^2\, dt.$$

On a non-trivial periodic orbit ∇V is not identically zero (if $\nabla V \equiv 0$, then $\dot{y} \equiv 0$ and the orbit degenerates to an equilibrium point), so $\int_0^T \|\nabla V\|^2 dt > 0$, whence $V(y(T)) < V(y(0))$. But by periodicity $y(T) = y(0)$, so $V(y(T)) = V(y(0))$, a contradiction.

(ii) Integrable Hamiltonian systems are guaranteed by the Liouville–Arnold theorem: n independent and involutive first integrals give an n-dimensional invariant torus, and the motion on the torus is quasi-periodic (for a complete proof see Arnold, *Mathematical Methods of Classical Mechanics*, 2nd ed., Springer, 1989, §49–50).

(iii) By the Poincaré–Bendixson theorem (Theorem 9.5), if a positively bounded orbit of a planar dissipative system does not tend to an equilibrium point, its ω-limit set is a limit cycle. □

9.10 Differential Equation Formulation of Conservation Laws

Definition 9.38 (Differential Form of a Conservation Law). Let $u(\mathsf{x}, t)$ be a conserved density and let $\mathsf{J}(\mathsf{x}, t)$ be a flux. The **differential form of a conservation law** is

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathsf{J} = 0.$$

Example 9.12 (Three Classical Conservation Laws). (i) **Conservation of mass (the continuity equation)**: $\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$.

(ii) **Conservation of momentum (the Navier–Stokes equations)**: $\frac{\partial (\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \mathbf{f}$.

(iii) **Conservation of energy**: $\frac{\partial (\rho E)}{\partial t} + \nabla \cdot (\rho E \mathbf{v} + \mathbf{q}) = \dot{Q}$.

Remark 9.8 (Unity with the Whole Book). This chapter organises the analytic tools into four parallel structures:

(i) **Differential equations**: *taking direction fields as the object, studying the existence, uniqueness, stability and periodicity of evolution orbits.*

(ii) **Complex analysis**: *taking the complex plane as the stage, coupling direction (the real part of the Cauchy–Riemann equations) with trend (the imaginary part) into a holomorphic structure.*

(iii) **Complex analyticity**: *taking series expansions as the tool, classifying analytic functions into removable singularities, poles and essential singularities, and computing closed-path integrals by the residue theorem.*

(iv) **Quaternion analysis and analyticity**: *taking a non-commutative algebra as the background, realising three-dimensional rotations through the exponential map and conjugation; it is the natural generalisation of complex analysis in non-commutative geometry.*

The four threads are linked by a common algebraic structure (the cyclic group C_n, the circle group $U(1)$, the unit sphere S^3): the periodic orbits of differential equations, the $2\pi i$-periodicity of the complex exponential, the discrete rotations of the group of roots of unity, and the 2π-periodicity of the quaternion exponential—they are the realisation of the same finite-order action on four levels (Theorem 1.5 of the 'Operations and Periodicity' chapter).

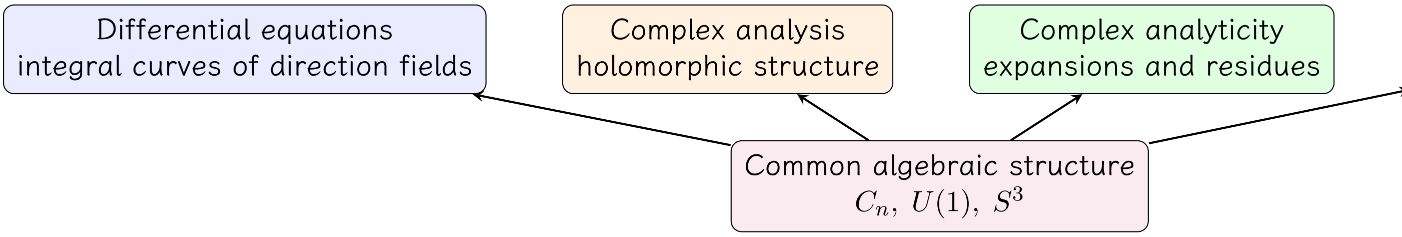


Figure 9.14: The four parallel structures of this chapter are linked by a common periodic algebraic structure.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	Differential equations with periodic coefficients and Floquet theory (§9.2); the $2\pi i$-periodicity of the complex exponential function (Proposition 9.9); the 2π-periodicity of the quaternion exponential (Proposition 9.11)
Scaling	Dissipative systems and volume contraction (§9.9); the phase-space volume evolution of Liouville's formula (Proposition 9.2); multiplicativity of the complex modulus $\lvert pq\rvert = \lvert p\rvert\lvert q\rvert$ (Proposition 9.3)
Rotation	Hamiltonian systems and circulation (§9.9); the conjugation representation of quaternion rotations (Theorem 9.14); the rotational period of complex functions (Proposition 9.9)
Conservation	First integrals (§9.1.4); the Noether-type correspondence (Proposition 9.4); Liouville's formula (Proposition 9.2); the isolation of zeros of complex analytic functions (Corollary 9.5)
Symmetry	The classification of the three operator systems (Theorem 9.15); the anti-automorphism symmetry of quaternion conjugation $\overline{pq} = \bar{q}\bar{p}$ (Proposition 9.5); complex conjugation symmetry
Closure	Limit cycles (Definition 9.8) and the Poincaré–Bendixson theorem (Theorem 9.5); the closed contour of Cauchy's integral theorem (Theorem 9.9); the closed integral of the residue theorem (Theorem 9.13)
Period	Floquet multipliers and the stability of periodic solutions (Corollary 9.1); the periodicity of the complex exponential function; the rotational period of the quaternion exponential function (Proposition 9.11)

Exercises

Exercise 1 Prove Gronwall's inequality (Lemma 9.1): let $\varphi : [t_0, t_0 + \alpha] \to [0, \infty)$ be continuous and satisfy

$$\varphi(t) \leq \varphi_0 + \int_{t_0}^{t} \beta\varphi(s)\, ds,$$

where $\varphi_0 \geq 0$ and $\beta \geq 0$. Then

$$\varphi(t) \leq \varphi_0 e^{\beta(t-t_0)}.$$

Write out the complete details of constructing the auxiliary function $\Phi(t)$ and proving that $\Phi(t)e^{-\beta(t-t_0)}$ is monotonically non-increasing.

Exercise 2 Prove the two key steps in the Picard–Lindelöf existence and uniqueness theorem (Theorem 9.1):

(i) the uniform convergence of the iteration sequence $\{y_n\}$, that is, the estimate
$$\|y_{n+1}(t) - y_n(t)\| \le \frac{ML^n|t-t_0|^{n+1}}{(n+1)!};$$

(ii) uniqueness, that is, using Gronwall's inequality to prove that the difference of the two solutions is identically zero.

Exercise 3 Prove the continuous dependence of the solution on the initial value (Proposition 9.1):
$$\|y(t;t_0,y_0) - y(t;t_0,\tilde{y}_0)\| \le e^{L|t-t_0|}\|y_0 - \tilde{y}_0\|.$$

Also explain why, in numerical computation, this conclusion means that an error in the initial value is not amplified immediately, but grows at the exponential rate $e^{L|t-t_0|}$.

Exercise 4 Prove Liouville's formula (Proposition 9.2): for the state transition matrix $\Phi(t,t_0)$ of $\dot{y} = A(t)y$ we have
$$\det \Phi(t,t_0) = \exp\left(\int_{t_0}^{t} \operatorname{tr} A(s)\, ds\right).$$

Write out the derivation of Jacobi's formula, and explain that phase-space volume is conserved when $\operatorname{tr} A(t) = 0$.

Exercise 5 Prove the exponential solution of a linear system with constant coefficients (Theorem 9.2): the unique solution of $\dot{y} = Ay$ satisfying $y(0) = y_0$ is $y(t) = e^{At}y_0$. Verify that the series definition of e^{At} converges absolutely and that term-by-term differentiation is legitimate, and explain uniqueness with the help of the Picard–Lindelöf theorem.

Exercise 6 Using Theorem 9.3, determine the type of phase portrait corresponding to each of the following 2×2 matrices:

(i) $A_1 = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$;

(ii) $A_2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$;

(iii) $A_3 = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$;

(iv) $A_4 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

Compute the eigenvalues in each case, and indicate which matrix corresponds to each of the four types: node, saddle, focus and centre.

Exercise 7 Prove Proposition 9.3: if an autonomous system has k independent first integrals, then its solutions are constrained to an $(n-k)$-dimensional invariant manifold. Use the implicit function theorem to show that the level set is a submanifold, and discuss how the orbit is completely determined when $k = n-1$.

Exercise 8 Verify the three first integrals in classical mechanics (Example 9.3):

(i) the total mechanical energy $E = \frac{1}{2}m\|\dot{\mathbf{r}}\|^2 + V(\mathbf{r})$ in a conservative force field;

(ii) the angular momentum $\mathbf{L} = m\mathbf{r} \times \dot{\mathbf{r}}$ in a central force field;

(iii) the momentum component in a direction in which no external force acts.

Verify respectively that $\frac{d}{dt}E = 0$, $\frac{d}{dt}\mathbf{L} = \mathbf{0}$ and $\frac{d}{dt}p_i = 0$, and point out that they correspond respectively to time-translation, rotation and spatial-translation symmetry.

Exercise 9 Prove Floquet's theorem (Theorem 9.6): for a linear system with periodic coefficients there exist a non-singular T-periodic matrix $P(t)$ and a constant matrix B such that $\Phi(t) = P(t)e^{Bt}$. Verify that $P(t+T) = P(t)$, and explain why $\log M$ may take complex matrix values.

Exercise 10 Prove the criteria for Floquet multipliers and the stability of periodic solutions (Corollary 9.1):

(i) $\rho = 1$ corresponds to a T-periodic solution;

(ii) $\rho = -1$ corresponds to a $2T$-periodic solution;

(iii) $\rho = e^{i2\pi p/q}$ corresponds to a qT-periodic solution;

(iv) when all $|\rho_i| < 1$ except $\rho_1 = 1$, that periodic solution is asymptotically stable.

Exercise 11 Prove the Cauchy–Riemann equations (Theorem 9.8): if $f(z) = u(x,y) + iv(x,y)$ is complex differentiable at z_0, then

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Compute $f'(z_0)$ along the real axis and along the imaginary axis separately, and compare real and imaginary parts.

Exercise 12 Prove Cauchy's integral theorem (Theorem 9.9): if f is analytic on a simply connected domain D and γ is a closed path in D, then

$$\oint_\gamma f(z)\,dz = 0.$$

Write out the complete procedure for converting the contour integral into a double integral by means of the plane Green's theorem (Lemma 9.2) and the Cauchy–Riemann equations.

Exercise 13 Prove Cauchy's integral formula (Theorem 9.10):

$$f(z_0) = \frac{1}{2\pi i} \oint_\gamma \frac{f(z)}{z - z_0}\, dz.$$

Take a small circle γ_ε centred at z_0, deform the integral over γ to an integral over γ_ε, and compute the limit.

Exercise 14 Prove Liouville's theorem (Theorem 9.11): a bounded entire function must be constant. Use the derivative form of Cauchy's integral formula to estimate $|f'(z_0)|$, and explain why this estimate tends to zero as the radius $R \to \infty$. Hence use it to prove the fundamental theorem of algebra (Corollary 9.4).

Exercise 15 Using Definition 9.25, determine the type of singularity of each of the following functions at the specified point, and compute its residue (Definition 9.26):

(i) $f(z) = \dfrac{\sin z}{z}$ at $z_0 = 0$;

(ii) $f(z) = \dfrac{1}{z^2(z-1)}$ at $z_0 = 0$;

(iii) $f(z) = e^{1/z}$ at $z_0 = 0$;

(iv) $f(z) = \dfrac{1}{z^2+1}$ at $z_0 = i$.

Exercise 16 Prove the residue theorem (Theorem 9.13): let f be analytic in a simply connected domain D except at finitely many isolated singularities $z_1, \ldots, z_m$, and let γ be a positively oriented simple closed curve enclosing all the z_k; then

$$\oint_\gamma f(z)\, dz = 2\pi i \sum_{k=1}^{m} \operatorname{Res}(f, z_k).$$

Use Cauchy's integral theorem to decompose the integral over γ into a sum of integrals over small circles around the singularities.

Exercise 17 Write down the quaternion multiplication rules

$$i^2 = j^2 = k^2 = ijk = -1, \quad ij = k,\ ji = -k,\ jk = i,\ kj = -i,\ ki = j,\ ik = -j,$$

and verify:

(i) the unit quaternion sphere S^3 forms a group under multiplication;

(ii) the conjugation map satisfies $\overline{pq} = \overline{q}\,\overline{p}$;

(iii) the modulus satisfies $|pq| = |p||q|$.

Exercise 18 Prove the conjugation representation of quaternion rotations (Theorem 9.14): for a unit quaternion $q \in S^3$ and a pure imaginary quaternion v, the map

$$\rho_q(v) = qvq^{-1}$$

is a rotation of $\mathbb{R}^3$, and

$$\rho_q(v) = v_\parallel + \cos\theta\, v_\perp + \sin\theta(\mathsf{u} \times v_\perp).$$

Write out the complete derivation of the decomposition of v into parallel and perpendicular components.

Exercise 19 Prove the two basic identities (Proposition 12.11):

$$\nabla \times (\nabla f) = \mathbf{0}, \qquad \nabla \cdot (\nabla \times \mathsf{F}) = 0.$$

Complete the proof using the Levi-Civita symbol ϵ_{ijk} and Clairaut's theorem (Lemma 15.5), and explain the necessity of the C^2 hypothesis (see the counterexample in Example 15.8).

Exercise 20 Synthesis question: explain how the main chain 'conservation law → symmetry → closed orbit → period' manifests concretely in the context of differential equations. Using first integrals (Definition 9.9), the Noether-type correspondence (Proposition 9.4), Hamiltonian systems (Definition 9.36) and Floquet theory (§9.2), discuss how conservation and symmetry help to discover periodic solutions in differential equations, and point out what the algebraic condition really is that directly produces periodicity.

Chapter 10

Linear Algebra

Remark 10.1 (Positioning of this chapter). Linear algebra is the mathematical theory that studies vector spaces, linear transformations and their algebraic structures. From the perspective of periodic mathematics, this chapter systematically establishes the foundational theory of linear algebra, focusing on the invariant subspace decomposition of linear transformations, the eigenvalue problem, and the profound connections between the spectral theorem and periodic structures. The central aim of this chapter is to provide a rigorous mathematical language for the analysis of linear systems.

This chapter is built on the foundations of the preceding chapters. Among them, the 'Operations and Periodicity' chapter established the logical chain conservation law—symmetry—closed orbit—period; the 'Number Systems' chapter established the correspondence between extensions of number systems and types of periodicity; the 'Scalars and Vectors' chapter established the algebraic realisation of the three basic transformations at the level of vectors; and the 'Functions and Limits' chapter established the unified framework direction—trend—convolution. This chapter systematically gathers these results into finite-dimensional vector spaces, revealing how the core theorems of linear algebra—the spectral theorem, the singular value decomposition and the classification of matrix groups—are the algebraic realisation of the three basic operators (gradient, divergence, curl).

The core work of this chapter lies in:

(1) **Invariant subspaces and periodic orbits**: the invariant subspaces of a linear transformation correspond to the closed orbits of a dynamical system—evolution on a one-dimensional invariant subspace is a pure translation period, whilst rotation on a two-dimensional invariant subspace is a pure rotation period.

(2) **Eigenvalue decomposition and frequency spectrum**: the eigenvalue decomposition diagonalises a linear transformation into a direct sum of independent one-dimensional invariant subspaces. Each eigenvalue corresponds to a characteristic frequency, and each eigenvector corresponds to an epicycle component.

(3) **Spectral theorem and conservation law**: the spectral theorems for self-adjoint and unitary operators are respectively the formulation in linear algebra of the gradient (symmetry centre 0) (real eigenvalues, pure translation) and of the curl (symmetry centre i) (eigenvalues of modulus 1, pure rotation). The spectral decomposition is the realisation of the conservation law at the level of operators.

(4) **Matrix groups and symmetry**: the orthogonal group, the unitary group and the symplectic group correspond in periodic mathematics respectively to the

matrix realisations of the three symmetry groups—translation (0-symmetry), rotation (i-symmetry) and scaling (1-symmetry).

Guide to the seven themes of this chapter: translation—translation matrices and cyclic shifts (stretching/compression of real eigenvalues, DFT diagonalisation of circulant matrices); scaling—diagonal scaling and singular values (the diagonal matrix diag(λ_i), the $\mathbf{\Sigma}$ of the SVD, and volume scaling measured by the determinant); rotation—orthogonal and unitary matrices (eigenvalues of modulus 1, plane rotations, the isometric action of $O(n)/U(n)$); conservation—the Frobenius norm, the determinant and the rank (conservation of dimension, energy conservation of $\sum_i \sigma_i^2$); symmetry—the matrix groups $O(n)/U(n)/Sp(2n)$ (matrix realisations of 0-symmetry, i-symmetry and 1-symmetry); closure—finite-order periodic solutions satisfying $\lambda^p = 1$ (eigenvalues lying on roots of unity, the cyclic structure of C_p); period—the periodic structure of linear systems (the four types of motion in the spectral decomposition, iterative convergence and steady states).

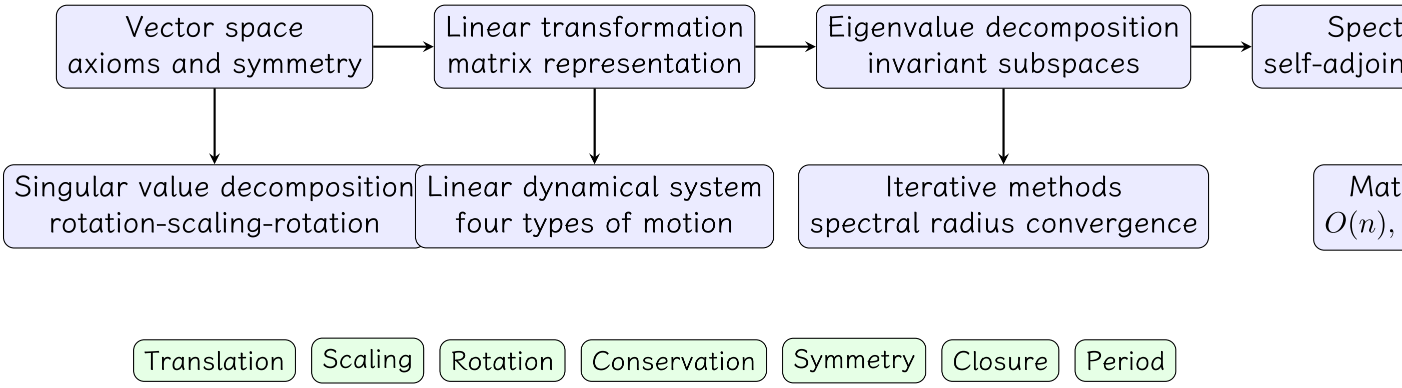

Vector space
axioms and symmetry
Linear transformation
matrix representation
Eigenvalue decomposition
invariant subspaces
Spect
self-adjoin
Singular value decomposition
rotation-scaling-rotation
Linear dynamical system
four types of motion
Iterative methods
spectral radius convergence
Mat
$O(n)$,
Translation
Scaling
Rotation
Conservation
Symmetry
Closure
Period

Figure 10.1: Overview of the structure of this chapter: vector spaces, linear transformations, eigenvalue decomposition and the spectral theorem; the second row gives the four parallel substructures of singular value decomposition, linear dynamical systems, iterative methods and matrix groups, all threaded through by the seven themes.

10.1 Vector Spaces and Linear Transformations

10.1.1 Vector Spaces

Definition 10.1 (Vector space). Let V be a non-empty set and let $\mathbb{F}$ be a field (in this book usually $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$). If vector addition $+ : V \times V \to V$ and scalar multiplication $\cdot : \mathbb{F} \times V \to V$ are defined on V and satisfy the following eight axioms, then V is called a **vector space** over $\mathbb{F}$:

(i) Associativity of addition: $(\mathsf{u} + \mathsf{v}) + \mathsf{w} = \mathsf{u} + (\mathsf{v} + \mathsf{w})$

(ii) Commutativity of addition: $\mathsf{u} + \mathsf{v} = \mathsf{v} + \mathsf{u}$

(iii) Additive identity: there exists $\mathsf{0} \in V$ such that $\mathsf{v} + \mathsf{0} = \mathsf{v}$

(iv) Additive inverse: for every v there exists $-\mathsf{v}$ such that $\mathsf{v} + (-\mathsf{v}) = \mathsf{0}$

(v) Multiplicative identity: $1 \cdot \mathsf{v} = \mathsf{v}$

(vi) Associativity of scalar multiplication: $\alpha(\beta\mathsf{v}) = (\alpha\beta)\mathsf{v}$

(vii) Distributivity of scalars over vectors: $\alpha(\mathsf{u} + \mathsf{v}) = \alpha\mathsf{u} + \alpha\mathsf{v}$

(viii) Distributivity of scalars over field addition: $(\alpha + \beta)\mathsf{v} = \alpha\mathsf{v} + \beta\mathsf{v}$

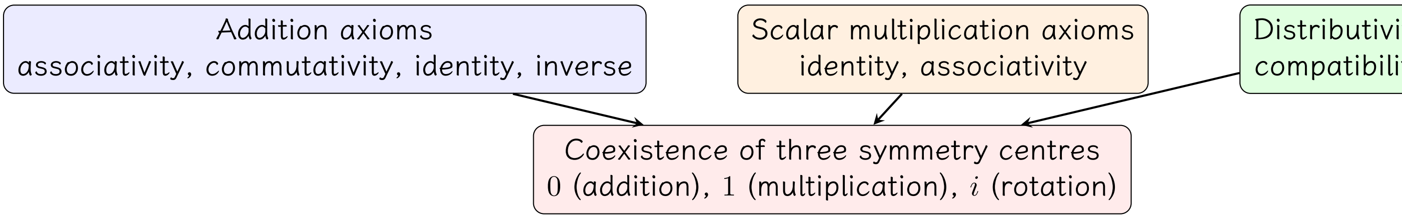

Addition axioms
associativity, commutativity, identity, inverse
Scalar multiplication axioms
identity, associativity
Distributivi
compatibili
Coexistence of three symmetry centres
0 (addition), 1 (multiplication), i (rotation)

Figure 10.2: The eight axioms of a vector space and the three symmetry centres: addition gives translation symmetry, scalar multiplication gives scaling symmetry, and distributivity guarantees that the two are compatible.

Remark 10.2 (Origin of the symmetry of a vector space). The eight axioms of a vector space precisely guarantee the algebraic legitimacy of the three basic symmetry operations. The associativity and commutativity of addition ensure that the translation operation forms an Abelian group (0-symmetry); the associativity and identity element of scalar multiplication ensure that the scaling operation forms a multiplicative group action (1-symmetry); and distributivity is the algebraic expression of the compatibility between the translation operator and the scaling operator. A vector space is the minimal algebraic structure in which the three symmetry centres—0 (the additive identity), 1 (the multiplicative identity) and i (the generator of the rotation direction)—can coexist.

Definition 10.2 (Subspace). Let V be a vector space and let $W \subseteq V$ be a non-empty subset. If W is closed under both vector addition and scalar multiplication, then W is called a **subspace** of V. A subspace is itself a vector space. Among subspaces, invariant subspaces are particularly important—they are the minimal substructures closed under a linear transformation, corresponding to subsystems with independent laws of evolution.

Proposition 10.1 (Closure of subspaces under translation and scaling). Let W be a subspace of V. Then W is closed under the following operations:

(i) **Translation**: for any $\mathsf{w}_1, \mathsf{w}_2 \in W$, we have $\mathsf{w}_1 + \mathsf{w}_2 \in W$. This reflects the internal closure of the gradient (symmetry centre 0)—a translation within the subspace does not leave the subspace.

(ii) **Scaling**: for any $\mathsf{w} \in W$ and $\alpha \in \mathbb{F}$, we have $\alpha\mathsf{w} \in W$. This reflects the internal closure of the divergence (symmetry centre 1)—a scaling within the subspace does not leave the subspace.

(iii) **Rotation**: in a complex vector space, if W is closed under scalar multiplication, then multiplication by the unit complex number $e^{i\theta}$ likewise does not leave W.

The closure of a subspace under these three operators is the algebraic foundation of its status as an independently evolving subsystem.

Proof. (i)(ii) These follow directly from the definition of a subspace (Definition 10.2): closure of W under vector addition means that it is closed under translation, and closure under scalar multiplication means that it is closed under scaling. (iii) In a complex vector space, multiplication by the unit complex number $e^{i\theta}$ is the special case of scalar multiplication with $\alpha = e^{i\theta}$, and is therefore still guaranteed by closure under scalar multiplication—rotation does not leave the subspace. ☐

Definition 10.3 (Linear dependence and independence). If there exist scalars $c_1, \dots, c_k$, not all zero, such that $\sum_{i=1}^{k} c_i \mathbf{v}_i = \mathbf{0}$, then the set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is called **linearly dependent**; otherwise it is called **linearly independent**. In a linearly independent set, no vector can be expressed as a linear combination of the remaining vectors—each vector supplies independent 'directional information'.

Definition 10.4 (Basis and dimension). A **basis** of a vector space V is a set of linearly independent vectors $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ such that every vector in V can be uniquely expressed as a linear combination of these basis vectors:

$$\mathbf{v} = \sum_{i=1}^{n} v_i \mathbf{e}_i.$$

The number n of basis vectors is called the **dimension** of V, written $\dim V = n$. The choice of a basis corresponds to the choice of a 'coordinate chart'; different bases give different representations of the same periodic structure.

Definition 10.5 (Direct sum decomposition). Let W_1, W_2 be subspaces of V with $W_1 \cap W_2 = \{\mathbf{0}\}$. Then the **direct sum** $W_1 \oplus W_2 = \{\mathbf{w}_1 + \mathbf{w}_2 : \mathbf{w}_1 \in W_1, \mathbf{w}_2 \in W_2\}$ is a subspace of V. If $V = W_1 \oplus W_2$, then every $\mathbf{v} \in V$ has a unique decomposition $\mathbf{v} = \mathbf{w}_1 + \mathbf{w}_2$.

Direct sum decomposition is the foundation of the decoupling of linear systems—when V can be decomposed into a direct sum of invariant subspaces, the evolution of the system can be analysed independently on each subspace.

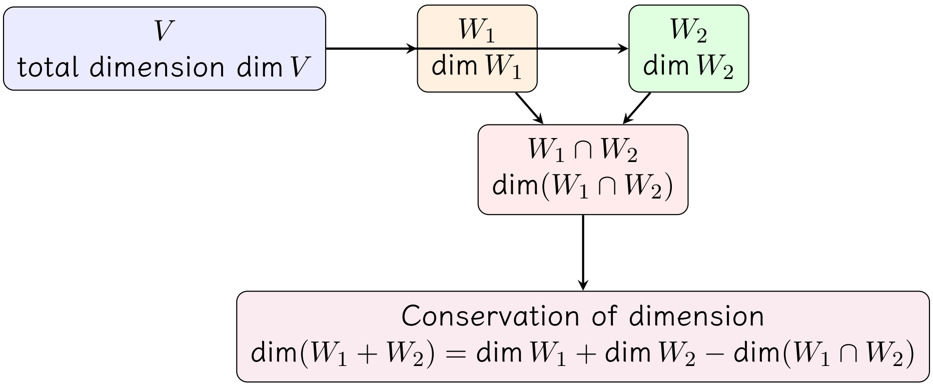


Figure 10.3: The conservation picture of the dimension theorem: the total dimension equals the sum of the dimensions of the subspaces minus the overlapping part.

Theorem 10.1 (Dimension theorem and conservation law). *Let V be a finite-dimensional vector space and let W_1, W_2 be its subspaces. Then*

$$\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2).$$

In particular, if $W_1 \cap W_2 = \{0\}$, then $\dim(W_1 \oplus W_2) = \dim W_1 + \dim W_2$.

The dimension theorem expresses the **conservation** *of information in a subspace decomposition: the total dimension (the total amount of information) remains additive under a non-overlapping decomposition—the sum of the independent degrees of freedom of the subspaces equals the total number of degrees of freedom.*

Proof. Let $\{u_1, \ldots, u_k\}$ be a basis of $W_1 \cap W_2$. Extend it respectively to:

$$\begin{aligned} \text{basis of } W_1 :&\quad \{u_1, \ldots, u_k, v_1, \ldots, v_m\}, \\ \text{basis of } W_2 :&\quad \{u_1, \ldots, u_k, w_1, \ldots, w_n\}, \\ \mathcal{B} :&\quad \{u_1, \ldots, u_k, v_1, \ldots, v_m, w_1, \ldots, w_n\}. \end{aligned}$$

We assert that $\mathcal{B}$ is a basis of $W_1 + W_2$.

Spanning: any $x \in W_1 + W_2$ can be written as $x = x_1 + x_2$ ($x_1 \in W_1$, $x_2 \in W_2$), whilst x_1, x_2 are respectively linear combinations of the bases of W_1, W_2; hence x is a linear combination of $\mathcal{B}$, so $\mathcal{B}$ spans $W_1 + W_2$.

Linear independence: suppose $\sum_{i=1}^{k} \alpha_i u_i + \sum_{j=1}^{m} \beta_j v_j + \sum_{\ell=1}^{n} \gamma_\ell w_\ell = 0$. Rearranging gives $z := \sum_{i=1}^{k} \alpha_i u_i + \sum_{j=1}^{m} \beta_j v_j = -\sum_{\ell=1}^{n} \gamma_\ell w_\ell$. The left-hand side belongs to W_1 and the right-hand side to W_2, so $z \in W_1 \cap W_2$ and can be written as a linear combination of the $\{u_i\}$; on the other hand $z = -\sum_\ell \gamma_\ell w_\ell$, whilst $\{u_1, \ldots, u_k, w_1, \ldots, w_n\}$ is a basis of W_2 (linearly independent), so $\gamma_\ell = 0$ for all ℓ. Thus $\sum_i \alpha_i u_i + \sum_j \beta_j v_j = 0$; and since $\{u_1, \ldots, u_k, v_1, \ldots, v_m\}$ is a basis of W_1 (linearly independent), we have $\alpha_i = 0$ and $\beta_j = 0$. Therefore $\mathcal{B}$ is linearly independent.

Hence $\mathcal{B}$ is a basis of $W_1 + W_2$ containing $k + m + n$ elements. Therefore

$$\dim(W_1 + W_2) = k + m + n = (k + m) + (k + n) - k = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2).$$

□

10.1.2 Linear Transformations and Matrix Representations

Definition 10.6 (Linear transformation). *Let V, W be vector spaces over $\mathbb{F}$. If the map $T : V \to W$ satisfies, for all $u, v \in V$ and $\alpha \in \mathbb{F}$,*

$$T(u + v) = T(u) + T(v), \quad T(\alpha u) = \alpha T(u),$$

then, when $V = W$, the map T is called a **linear operator** *on V.*

A linear transformation preserves addition and scalar multiplication simultaneously—that is to say, it respects both the translation operator and the scaling operator. This twofold preservation is precisely the fundamental reason why linear transformations can serve as a tool for describing periodic structures.

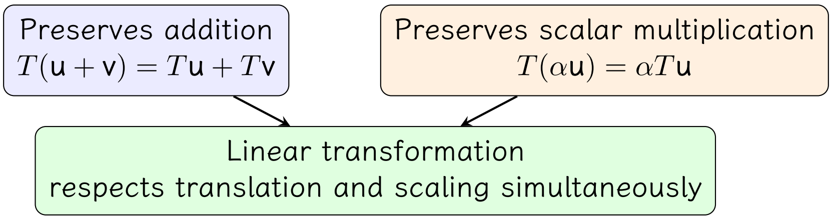

Preserves addition
$T(\mathsf{u}+\mathsf{v}) = T\mathsf{u}+T\mathsf{v}$
Preserves scalar multiplication
$T(\alpha \mathsf{u}) = \alpha T\mathsf{u}$
Linear transformation
respects translation and scaling simultaneously

Figure 10.4: The twofold preservation property of a linear transformation: it preserves both addition (translation) and scalar multiplication (scaling).

Definition 10.7 (Kernel and image)**.** The **kernel** (null space) of a linear transformation $T : V \to W$ is $\ker(T) = \{\mathsf{v} \in V : T(\mathsf{v}) = 0\}$. Its **image** (range) is $\operatorname{im}(T) = \{T(\mathsf{v}) : \mathsf{v} \in V\}$.

The dimensions of the kernel and the image satisfy the **rank-nullity theorem**:

$$\dim \ker(T) + \dim \operatorname{im}(T) = \dim V.$$

The rank-nullity theorem expresses the **dimension conservation law** of a linear transformation: the dimension of the domain equals the sum of the 'annihilated information' (the dimension of the kernel) and the 'retained information' (the dimension of the image).

Definition 10.8 (Matrix representation)**.** Let $T : V \to V$ be a linear operator on an n-dimensional vector space and let $\mathcal{B} = \{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ be a basis of V. The **matrix representation** of T with respect to the basis $\mathcal{B}$ is the $n \times n$ matrix $\mathsf{A} = (a_{ij})$ satisfying

$$T(\mathbf{e}_j) = \sum_{i=1}^{n} a_{ij}\mathbf{e}_i.$$

The j-th column of the matrix A consists of the coordinates, with respect to the basis $\mathcal{B}$, of the image of the j-th basis vector under T.

Definition 10.9 (Matrix operations)**.** Let $\mathsf{A} = (a_{ij})$ be an $m \times n$ matrix and let $\mathsf{B} = (b_{ij})$ be an $n \times p$ matrix. Matrix multiplication is defined by $(\mathsf{AB})_{ik} = \sum_{j=1}^{n} a_{ij}b_{jk}$. Matrix multiplication corresponds to the composition of linear transformations. The **transpose** A^T of a matrix satisfies $(\mathsf{A}^\mathsf{T})_{ij} = a_{ji}$. The **conjugate transpose** A^* of a complex matrix satisfies $(\mathsf{A}^*)_{ij} = \overline{a_{ji}}$.

Property 10.1 (Algebraic properties of matrix operations)**.** Let $\mathsf{A}, \mathsf{B}, \mathsf{C}$ be matrices of compatible sizes and let $\alpha \in \mathbb{F}$. Matrix operations satisfy the following properties:

(i) Associativity of multiplication: $(\mathsf{AB})\mathsf{C} = \mathsf{A}(\mathsf{BC})$;

(ii) Distributivity of multiplication over addition: $\mathsf{A}(\mathsf{B}+\mathsf{C}) = \mathsf{AB}+\mathsf{AC}$, $(\mathsf{A}+\mathsf{B})\mathsf{C} = \mathsf{AC}+\mathsf{BC}$;

(iii) Reversal rule for the transpose and the conjugate transpose: $(\mathsf{AB})^\mathsf{T} = \mathsf{B}^\mathsf{T}\mathsf{A}^\mathsf{T}$, $(\mathsf{AB})^* = \mathsf{B}^*\mathsf{A}^*$;

(iv) Involution of the double transpose: $(\mathsf{A}^\mathsf{T})^\mathsf{T} = \mathsf{A}$, $(\mathsf{A}^*)^* = \mathsf{A}$;

(v) Scalar compatibility: $\alpha(\mathsf{AB}) = (\alpha\mathsf{A})\mathsf{B} = \mathsf{A}(\alpha\mathsf{B})$.

Proof. (i) Let A be an $m \times n$ matrix, B an $n \times p$ matrix and C a $p \times q$ matrix. For any $1 \le i \le m$ and $1 \le \ell \le q$:

$$[(\mathsf{AB})\mathsf{C}]_{i\ell} = \sum_{k=1}^{p} (\mathsf{AB})_{ik} C_{k\ell} = \sum_{k=1}^{p} \left(\sum_{j=1}^{n} A_{ij} B_{jk} \right) C_{k\ell} = \sum_{j=1}^{n} A_{ij} \left(\sum_{k=1}^{p} B_{jk} C_{k\ell} \right) = [\mathsf{A}(\mathsf{BC})]_{i\ell}.$$

Here the order of summation may be interchanged (finite sums).

(ii) For any i, k:

$$[\mathsf{A}(\mathsf{B}+\mathsf{C})]_{ik} = \sum_j A_{ij}(B_{jk} + C_{jk}) = \sum_j A_{ij} B_{jk} + \sum_j A_{ij} C_{jk} = [\mathsf{AB}]_{ik} + [\mathsf{AC}]_{ik}.$$

The second distributive law is analogous.

(iii) For any i, k:

$$[(\mathsf{AB})^{\mathsf{T}}]_{ik} = [\mathsf{AB}]_{ki} = \sum_j A_{kj} B_{ji} = \sum_j [\mathsf{A}^{\mathsf{T}}]_{jk} [\mathsf{B}^{\mathsf{T}}]_{ij} = \sum_j [\mathsf{B}^{\mathsf{T}}]_{ij} [\mathsf{A}^{\mathsf{T}}]_{jk} = [\mathsf{B}^{\mathsf{T}}\mathsf{A}^{\mathsf{T}}]_{ik}.$$

The conjugate transpose case is analogous; one need only take conjugates.

(iv) $[(\mathsf{A}^{\mathsf{T}})^{\mathsf{T}}]_{ij} = [\mathsf{A}^{\mathsf{T}}]_{ji} = A_{ij}$. The conjugate transpose case is analogous.

(v) $[\alpha(\mathsf{AB})]_{ik} = \alpha \sum_j A_{ij} B_{jk} = \sum_j (\alpha A_{ij}) B_{jk} = [(\alpha\mathsf{A})\mathsf{B}]_{ik}$. Similarly $= \sum_j A_{ij}(\alpha B_{jk}) = [\mathsf{A}(\alpha\mathsf{B})]_{ik}$. ☐

Definition 10.10 (Determinant). The **determinant** det(A) of an $n \times n$ matrix A is the signed volume of the parallelepiped spanned by the column vectors of A. The determinant has the multiplicative property det(AB) = det(A) det(B).

The determinant measures the scaling factor by which a linear transformation acts on **phase-space volume**—this is the matrix realisation of the divergence (symmetry centre 1) (1-symmetry). $\det(\mathsf{A}) = 1$ signifies conservation of volume (Hamiltonian systems), $|\det(\mathsf{A})| < 1$ signifies contraction of volume (dissipative systems), and $|\det(\mathsf{A})| > 1$ signifies expansion of volume (anti-dissipative systems). A non-zero determinant is a necessary and sufficient condition for a matrix to be invertible—invertibility corresponds to the time-reversibility of a periodic evolution.

Property 10.2 (Operational properties of the determinant). Let A, B be $n \times n$ matrices and let P be an invertible matrix. The determinant satisfies the following properties:

(i) Multiplicativity: $\det(\mathsf{AB}) = \det(\mathsf{A})\det(\mathsf{B})$;

(ii) Invariance under transposition: $\det(\mathsf{A}^{\mathsf{T}}) = \det(\mathsf{A})$, $\det(\mathsf{A}^*) = \overline{\det(\mathsf{A})}$;

(iii) Invertibility criterion: $\det(\mathsf{A}) \neq 0$ if and only if A is invertible, and $\det(\mathsf{A}^{-1}) = 1/\det(\mathsf{A})$;

(iv) Similarity invariance: $\det(\mathsf{P}^{-1}\mathsf{AP}) = \det(\mathsf{A})$;

(v) Interchanging a row (or a column) changes the sign: interchanging two rows or two columns reverses the sign of the determinant.

Proof. (i) Multiplicativity is part of the definition of the determinant (it can be deduced from the multilinearity and alternation of the column vectors, a standard result in linear algebra).

(ii) Invariance under transposition: $\det(A^\mathsf{T}) = \sum_\sigma \operatorname{sgn}(\sigma) \prod_i a_{\sigma(i),i} = \sum_\sigma \operatorname{sgn}(\sigma) \prod_i a_{i,\sigma^{-1}(i)} = \det(A)$. For the conjugate transpose: $\det(A^*) = \overline{\det(A)}$.

(iii) By multiplicativity, $\det(A)\det(A^{-1}) = \det(I) = 1$, hence $\det(A^{-1}) = 1/\det(A)$. Conversely, if $\det(A) \neq 0$, the adjugate formula $A^{-1} = \operatorname{adj}(A)/\det(A)$ shows that A is invertible.

(iv) By multiplicativity: $\det(P^{-1}AP) = \det(P^{-1})\det(A)\det(P) = \det(A)$.

(v) Interchanging two rows (or two columns) reverses the sign of the permutation, so the determinant changes sign. □

Proposition 10.2 (Change of basis and similarity)**.** *Let $\mathcal{B}$ and $\mathcal{B}'$ be two bases of V and let P be the transition matrix from $\mathcal{B}$ to $\mathcal{B}'$. Then the matrix representations A and A′ of the same linear operator T with respect to these two bases satisfy the similarity relation:*

$$A' = P^{-1}AP.$$

A similarity transformation preserves the eigenvalues—the eigenvalues are intrinsic properties of a linear transformation and do not depend on the choice of basis.

Proof. Denote by $[v]_\mathcal{B}$ and $[v]_{\mathcal{B}'}$ the coordinates of the same vector with respect to the two bases; the transition matrix P satisfies $[v]_{\mathcal{B}'} = P^{-1}[v]_\mathcal{B}$. By the definition of the matrix representation, with respect to the basis $\mathcal{B}$ we have $[Tv]_\mathcal{B} = A[v]_\mathcal{B}$, and with respect to the basis $\mathcal{B}'$ we have $[Tv]_{\mathcal{B}'} = A'[v]_{\mathcal{B}'}$. Substituting the coordinate transformation gives

$$A'[v]_{\mathcal{B}'} = [Tv]_{\mathcal{B}'} = P^{-1}[Tv]_\mathcal{B} = P^{-1}A[v]_\mathcal{B} = P^{-1}AP[v]_{\mathcal{B}'}.$$

Since v is arbitrary, we obtain $A' = P^{-1}AP$. The invariance of the eigenvalues stems from the invariance of the characteristic polynomial under similarity transformations. □

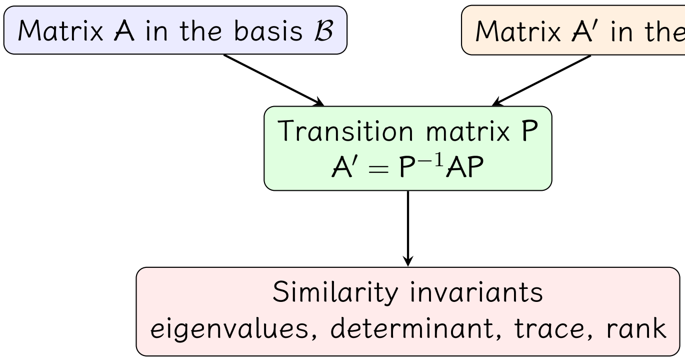

Matrix A in the basis $\mathcal{B}$
Matrix A′ in the basis $\mathcal{B}'$
Transition matrix P
$A' = P^{-1}AP$
Similarity invariants
eigenvalues, determinant, trace, rank

Figure 10.5: Change of basis and similarity: the matrices of the same linear operator with respect to two bases are similar, and quantities such as the eigenvalues are similarity invariants.

10.2 Invariant Subspaces and Eigenvalue Decomposition

10.2.1 Eigenvalues and Eigenvectors

Definition 10.11 (Eigenvalues and Eigenvectors)**.** Let $T : V \to V$ be a linear operator. If there exist a non-zero vector $\mathsf{v} \in V$ and a scalar $\lambda \in \mathbb{C}$ satisfying

$$T(\mathsf{v}) = \lambda\mathsf{v},$$

then λ is called an **eigenvalue** of T and v a corresponding **eigenvector**.

The eigenvalue λ describes the law of evolution on a one-dimensional invariant subspace—it is the algebraic realisation of the three basic types of periodicity in the 'Operations and Periodicity' chapter:

(i) If $\lambda \in \mathbb{R}$ and $\lambda > 0$, then the evolution is **pure translation**—the vector is stretched ($\lambda > 1$) or compressed ($0 < \lambda < 1$) on the invariant subspace, whilst its direction remains unchanged.

(ii) If $\lambda = e^{i\theta}$ and $|\lambda| = 1$, then the evolution is **pure rotation**—the vector rotates on the invariant subspace whilst its modulus remains unchanged. When $\theta/(2\pi) = p/q \in \mathbb{Q}$, a strict rotation period of finite order arises.

(iii) If $\lambda = re^{i\theta}$ and $r \neq 1$, then the evolution is the combined action of **rotation and scaling**—the vector is stretched or compressed whilst it rotates.

(iv) If $\lambda < 0$, then the evolution involves a reflection (0-symmetry, the C_2 involution)—multiplication by -1 at each step produces a second-order periodic oscillation.

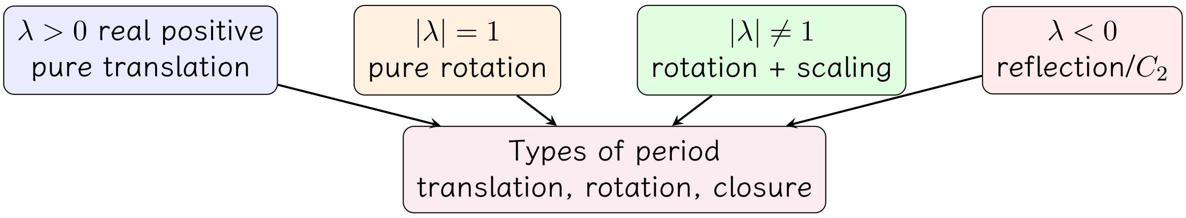


Figure 10.6: The four types of evolution of an eigenvalue: real positive stretching, rotation on the unit circle, rotation with scaling, and negative real reflection.

Definition 10.12 (Eigenspace and Geometric Multiplicity)**.** The **eigenspace** corresponding to the eigenvalue λ is $E_\lambda = \{\mathsf{v} \in V : T(\mathsf{v}) = \lambda\mathsf{v}\}$. The dimension $\dim E_\lambda$ of E_λ is called the **geometric multiplicity** of λ. The multiplicity of λ as a root of the characteristic polynomial is called its **algebraic multiplicity**. The geometric multiplicity does not exceed the algebraic multiplicity.

E_λ is an **invariant subspace** of V—$T(E_\lambda) \subseteq E_\lambda$. Every eigenspace is an independent channel of evolution of the system—vectors in different eigenspaces do not mix with one another under the action of T. A geometric multiplicity greater than 1 means that the eigenvalue corresponds to several independent degrees of freedom of evolution.

Theorem 10.2 (Eigenvectors and Invariant Subspaces)**.** Let

> $\mathsf{v}_1, \dots, \mathsf{v}_k$ be eigenvectors of T corresponding to distinct eigenvalues $\lambda_1, \dots, \lambda_k$ respectively.

Then:

(i) $\mathsf{v}_1, \dots, \mathsf{v}_k$ are linearly independent—evolution directions corresponding to distinct eigenvalues are mutually independent.

(ii) Each eigenspace E_{λ_i} is an invariant subspace of T.

(iii) The subspace spanned by the eigenvectors is an invariant subspace of T—the evolution on it is entirely determined by the eigenvalues, and there is no coupling between different eigenspaces.

Proof. (i) Induct on k. If there is a non-trivial linear relation $\sum_{i=1}^{k} c_i \mathsf{v}_i = \mathbf{0}$, applying T to both sides gives $\sum_{i=1}^{k} c_i \lambda_i \mathsf{v}_i = \mathbf{0}$. Multiplying the first relation by λ_k and subtracting the second eliminates the term in v_k, yielding $\sum_{i=1}^{k-1} c_i(\lambda_k - \lambda_i)\mathsf{v}_i = \mathbf{0}$. By the induction hypothesis, $\mathsf{v}_1, \dots, \mathsf{v}_{k-1}$ are linearly independent, and $\lambda_k \neq \lambda_i$, so $c_i = 0$ for all $i = 1, \dots, k-1$. Substituting back gives $c_k = 0$, a contradiction. Hence these eigenvectors are linearly independent.

(ii) For any $\mathsf{v} \in E_\lambda$, $T(\mathsf{v}) = \lambda \mathsf{v} \in E_\lambda$, so E_λ is an invariant subspace.

(iii) By (i), the subspace spanned by the eigenvectors is a direct sum of invariant subspaces, and the eigenspaces evolve independently under T. ☐

Corollary 10.1 (Eigenspaces and the Independent Evolution of Periodic Components)**.** *Suppose that the eigenvectors of a linear operator T form a basis of V. Then V decomposes into a direct sum of the eigenspaces: $V = \bigoplus_i E_{\lambda_i}$. The evolution of any initial state $\mathsf{x}_0 = \sum_i c_i \mathsf{v}_i$ decomposes into independent evolutions on the eigenspaces: $\mathsf{x}_k = T^k \mathsf{x}_0 = \sum_i c_i \lambda_i^k \mathsf{v}_i$. Each eigenvalue λ_i is the evolution factor of the i-th 'frequency component'.*

Proof. By Theorem 10.2, eigenvectors corresponding to distinct eigenvalues are linearly independent, and each eigenspace E_{λ_i} is an invariant subspace of T. When the eigenvectors form a basis, $V = \bigoplus_i E_{\lambda_i}$. For $\mathsf{x}_0 = \sum_i c_i \mathsf{v}_i$, using $T^k \mathsf{v}_i = \lambda_i^k \mathsf{v}_i$ and the linearity of T we obtain $\mathsf{x}_k = \sum_i c_i \lambda_i^k \mathsf{v}_i$. ☐

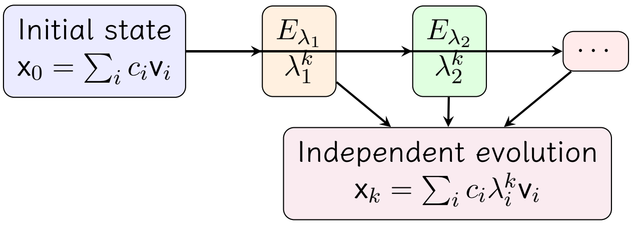


Figure 10.7: Decomposition into eigenspaces: the initial state is decomposed along the eigendirections, and the components evolve independently.

10.2.2 Diagonalisation and Jordan Normal Form

Definition 10.13 (Diagonalisable Operator)**.** *If V has a basis consisting of eigenvectors of T, then T is called* **diagonalisable***. With respect to that basis, the matrix representation of T is the diagonal matrix:*

$$\mathsf{D} = \mathsf{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n).$$

A necessary and sufficient condition for an operator to be diagonalisable is that, for every eigenvalue, its algebraic multiplicity equals its geometric multiplicity.

Diagonalisation is equivalent to decomposing the operator into a direct sum of independent one-dimensional invariant subspaces—this is the realisation in linear algebra of orbit decomposition. Each diagonal entry λ_k is the rate of evolution of the k-th 'orbit'.

Definition 10.14 (Jordan normal form)**.** *When an operator is not diagonalisable, its simplest form is the* **Jordan normal form***: $\mathsf{A} = \mathsf{P}\mathsf{J}\mathsf{P}^{-1}$, where J is a block diagonal matrix composed of Jordan blocks. Each Jordan block corresponds to an eigenvalue; the number of Jordan blocks equals the geometric multiplicity of that eigenvalue, the orders of the blocks sum to its algebraic multiplicity, and the largest block order equals the maximum length of a generalised eigenvector chain.*

The off-diagonal entry 1 of a Jordan block represents the coupling between eigenspaces—a non-diagonalisable operator cannot be decomposed into a direct sum of purely one-dimensional invariant subspaces. A Jordan block corresponds to a 'leakage' between eigenspaces—different periodic components transfer energy through the off-diagonal coupling. This is the algebraic prototype of quasi-periodicity and resonance phenomena.

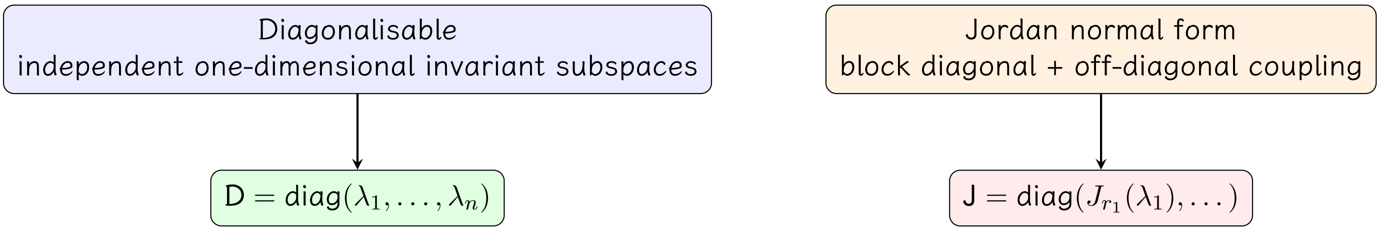

Diagonalisable
independent one-dimensional invariant subspaces
$\mathsf{D} = \mathsf{diag}(\lambda_1, \ldots, \lambda_n)$
Jordan normal form
block diagonal + off-diagonal coupling
$\mathsf{J} = \mathsf{diag}(J_{r_1}(\lambda_1), \ldots)$

Figure 10.8: Diagonalisation and Jordan normal form: when the operator is diagonalisable the eigendirections are independent; a Jordan block represents the coupling between eigenspaces.

Theorem 10.3 (Cayley-Hamilton Theorem)**.** *Let $p(\lambda) = \det(\mathsf{A} - \lambda \mathsf{I})$ be the characteristic polynomial of A. Then $p(\mathsf{A}) = \mathbf{0}$—every matrix satisfies its own characteristic equation.*

The Cayley-Hamilton theorem expresses the **cyclic closure** *of the characteristic polynomial: higher powers of a matrix can be expressed linearly in terms of lower powers. This means that the long-term behaviour of a linear dynamical system is entirely determined by its eigenvalues—higher-order evolution is constrained by the degree of the characteristic polynomial, and the dynamics of the system are 'locked' inside the finite-dimensional subspace spanned by the eigenvalues.*

Proof. Let $p(\lambda) = \det(\mathsf{A} - \lambda \mathsf{I}) = a_n\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_0$, where $a_n = (-1)^n$. Write $B(\lambda) = \mathsf{A} - \lambda\mathsf{I}$. By the adjugate identity

$$B(\lambda)\ \mathrm{adj}(B(\lambda)) = \det(B(\lambda))\,\mathsf{I} = p(\lambda)\,\mathsf{I}.$$

Each entry of $\mathrm{adj}(B(\lambda))$ is a polynomial in λ of degree at most $n-1$, so it can be written as the matrix polynomial $\mathrm{adj}(B(\lambda)) = \sum_{k=0}^{n-1} \lambda^k \mathsf{C}_k$, where the C_k are constant matrices. Substituting into the above identity and expanding gives:

$$p(\lambda)\mathsf{I} = \left(\sum_{k=0}^{n-1} \lambda^k \mathsf{C}_k\right)(\mathsf{A} - \lambda\mathsf{I}) = \mathsf{C}_0\mathsf{A} + \sum_{k=1}^{n-1} \lambda^k(\mathsf{C}_k\mathsf{A} - \mathsf{C}_{k-1}) - \lambda^n \mathsf{C}_{n-1}.$$

Comparing the coefficients of the powers of λ gives

$$a_0\mathsf{I} = \mathsf{C}_0\mathsf{A}, \qquad a_k\mathsf{I} = \mathsf{C}_k\mathsf{A} - \mathsf{C}_{k-1}\ (1 \le k \le n-1), \qquad a_n\mathsf{I} = -\mathsf{C}_{n-1}.$$

Multiplying the k-th equation on the right by A^k and summing term by term gives

$$p(\mathsf{A}) = \mathsf{C}_0\mathsf{A} + \sum_{k=1}^{n-1}(\mathsf{C}_k\mathsf{A}^{k+1} - \mathsf{C}_{k-1}\mathsf{A}^k) - \mathsf{C}_{n-1}\mathsf{A}^n.$$

The adjacent terms on the right-hand side cancel in pairs (a telescoping sum), hence $p(\mathsf{A}) = \mathbf{0}$. □

10.2.3 Spectral Theorem

Theorem 10.4 (Spectral Theorem—Self-adjoint Operator). *Let $T : V \to V$ be a* **self-adjoint operator** *on an inner product space ($T^* = T$). Then:*

- (i) *All eigenvalues of T are real numbers.*
- (ii) *Eigenvectors corresponding to distinct eigenvalues are mutually orthogonal.*
- (iii) *There exists an orthonormal basis of V consisting of eigenvectors of T—T is unitarily diagonalisable.*

The spectral theorem for self-adjoint operators is the highest expression in linear algebra of the **gradient operator** *(0-symmetry): purely real eigenvalues mean that all evolution is pure translation (stretching or compression) and contains no rotational component.*

Proof. (i) Let $T\mathsf{v} = \lambda\mathsf{v}$ with $\mathsf{v} \neq \mathbf{0}$. Using self-adjointness:

$$\overline{\lambda}\|\mathsf{v}\|^2 = \langle \lambda\mathsf{v}, \mathsf{v}\rangle = \langle T\mathsf{v}, \mathsf{v}\rangle = \langle \mathsf{v}, T\mathsf{v}\rangle = \langle \mathsf{v}, \lambda\mathsf{v}\rangle = \lambda\|\mathsf{v}\|^2.$$

Hence $\lambda = \overline{\lambda}$, that is, $\lambda \in \mathbb{R}$.

(ii) Let $T\mathsf{v}_1 = \lambda_1\mathsf{v}_1$ and $T\mathsf{v}_2 = \lambda_2\mathsf{v}_2$ with $\lambda_1 \neq \lambda_2$. Then

$$\lambda_1\langle \mathsf{v}_1, \mathsf{v}_2\rangle = \langle T\mathsf{v}_1, \mathsf{v}_2\rangle = \langle \mathsf{v}_1, T\mathsf{v}_2\rangle = \overline{\lambda_2}\langle \mathsf{v}_1, \mathsf{v}_2\rangle = \lambda_2\langle \mathsf{v}_1, \mathsf{v}_2\rangle.$$

Since $\lambda_1 \neq \lambda_2$, we must have $\langle \mathsf{v}_1, \mathsf{v}_2\rangle = 0$.

(iii) Induct on the dimension $n = \dim V$. When $n = 1$, V is spanned by a single eigenvector and the conclusion holds directly. Assume that the conclusion holds for self-adjoint operators of dimension less than n. By the fundamental theorem of algebra, the characteristic polynomial has a root in $\mathbb{C}$, and by (i) that root is real, say λ_1. Take a unit eigenvector $\mathsf{v}_1 \in V$ ($\|\mathsf{v}_1\| = 1$). Let $W = \mathsf{span}(\mathsf{v}_1)^\perp$. For any $\mathsf{w} \in W$, self-adjointness $T^* = T$ gives $\langle T\mathsf{w}, \mathsf{v}_1\rangle = \langle \mathsf{w}, T^*\mathsf{v}_1\rangle = \langle \mathsf{w}, T\mathsf{v}_1\rangle = \lambda_1\langle \mathsf{w}, \mathsf{v}_1\rangle = 0$, hence $T\mathsf{w} \in W$, that is, W is an invariant subspace of T and $T|_W$ is again a self-adjoint operator. By the induction hypothesis, W has an orthonormal basis $\{\mathsf{v}_2, \dots, \mathsf{v}_n\}$ consisting of eigenvectors of T. Combining it with v_1 gives an orthonormal basis of V. The induction is complete. ☐

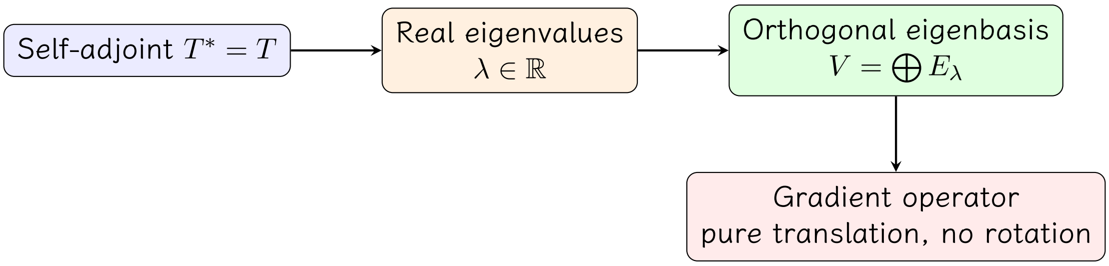

Self-adjoint $T^* = T$
Real eigenvalues $\lambda \in \mathbb{R}$
Orthogonal eigenbasis $V = \bigoplus E_\lambda$
Gradient operator pure translation, no rotation

Figure 10.9: The self-adjoint spectral theorem: *real eigenvalues and an orthogonal eigenbasis, corresponding to the pure translation evolution of the gradient operator.*

Corollary 10.2 (Self-adjoint Spectral Theorem and the Gradient Conservation Law). *Let T be a self-adjoint operator. Since all its eigenvalues are real numbers, the corresponding dynamical system* **contains no rotation period***—all evolution components are pure translations. The orthogonal eigenvectors of the real eigenvalues constitute a set of independent translation directions, and the evolution along each direction does not affect the others.*

Proof. By Theorem 10.4, all eigenvalues of T are real, so it has no non-real roots $e^{i\theta}$ on the unit circle ($\theta \not\equiv 0 \pmod{\pi}$), and therefore has no rotation period. Eigenvectors corresponding to distinct eigenvalues are mutually orthogonal, and V has an orthonormal basis consisting of eigenvectors, so the evolution directions are mutually independent and do not mix. ☐

Example 10.1 (Eigendecomposition of a 2×2 Symmetric Matrix)**.** Consider the symmetric matrix

$$\mathsf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}.$$

Its characteristic polynomial is $\det(\mathsf{A} - \lambda\mathsf{I}) = (3 - \lambda)^2 - 1 = (\lambda - 2)(\lambda - 4)$, so the eigenvalues are $\lambda_1 = 4$ and $\lambda_2 = 2$ (both real). The corresponding unit eigenvectors are taken to be $\mathsf{v}_1 = \frac{1}{\sqrt{2}}(1, 1)^\mathsf{T}$ and $\mathsf{v}_2 = \frac{1}{\sqrt{2}}(1, -1)^\mathsf{T}$, satisfying $\langle \mathsf{v}_1, \mathsf{v}_2\rangle = 0$. The spectral decomposition is

$$\mathsf{A} = \mathsf{Q}\mathsf{D}\mathsf{Q}^\mathsf{T}, \qquad \mathsf{Q} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \mathsf{D} = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}.$$

These two eigendirections correspond respectively to pure translation evolution with stretching factors 4 and 2, and contain no rotational component.

Theorem 10.5 (Spectral Theorem—Unitary Operator). Let $U : V \to V$ be a **unitary operator** on an inner product space ($U^*U = UU^* = I$). Then:

(i) All eigenvalues of U have modulus 1: $|\lambda| = 1$, that is, $\lambda = e^{i\theta}$.

(ii) Eigenvectors corresponding to distinct eigenvalues are mutually orthogonal.

(iii) There exists an orthonormal basis of V consisting of eigenvectors of U—U is unitarily diagonalisable.

The spectral theorem for unitary operators is the highest expression in linear algebra of the **curl operator** (i-symmetry): eigenvalues of modulus 1 mean that all evolution is pure rotation and contains no scaling component.

Proof. (i) Let $U\mathsf{v} = \lambda\mathsf{v}$ with $\mathsf{v} \neq \mathbf{0}$. Using unitarity: $\|\mathsf{v}\|^2 = \|U\mathsf{v}\|^2 = |\lambda|^2\|\mathsf{v}\|^2$. Hence $|\lambda| = 1$.

(ii) Let $U\mathsf{v}_1 = \lambda_1\mathsf{v}_1$ and $U\mathsf{v}_2 = \lambda_2\mathsf{v}_2$ with $\lambda_1 \neq \lambda_2$. By the convention of this chapter (Definition 10.15: the inner product is linear in the second argument and conjugate-linear in the first), $\langle U\mathsf{v}_1, U\mathsf{v}_2\rangle = \langle\lambda_1\mathsf{v}_1, \lambda_2\mathsf{v}_2\rangle = \overline{\lambda_1}\lambda_2\langle\mathsf{v}_1, \mathsf{v}_2\rangle$; on the other hand, unitarity gives $\langle U\mathsf{v}_1, U\mathsf{v}_2\rangle = \langle\mathsf{v}_1, \mathsf{v}_2\rangle$, hence $\overline{\lambda_1}\lambda_2\langle\mathsf{v}_1, \mathsf{v}_2\rangle = \langle\mathsf{v}_1, \mathsf{v}_2\rangle$. Since $|\lambda_1| = |\lambda_2| = 1$ and $\lambda_1 \neq \lambda_2$, we have $\overline{\lambda_1}\lambda_2 \neq 1$, hence $\langle\mathsf{v}_1, \mathsf{v}_2\rangle = 0$.

(iii) Induct on the dimension $n = \dim V$. When $n = 1$, V is spanned by a single eigenvector and the conclusion holds directly. Assume that the conclusion holds for unitary operators of dimension less than n. By the fundamental theorem of algebra, the characteristic polynomial of U has a root λ_1 in $\mathbb{C}$ (and by (i), $|\lambda_1| = 1$); take a unit eigenvector v_1. Let $W = \mathrm{span}(\mathsf{v}_1)^\perp$. For any $\mathsf{w} \in W$, from $U\mathsf{v}_1 = \lambda_1\mathsf{v}_1$ we obtain $U^*\mathsf{v}_1 = \overline{\lambda_1}\mathsf{v}_1$, hence $\langle U\mathsf{w}, \mathsf{v}_1\rangle = \langle\mathsf{w}, U^*\mathsf{v}_1\rangle = \overline{\lambda_1}\langle\mathsf{w}, \mathsf{v}_1\rangle = 0$, that is, $U\mathsf{w} \in W$. Therefore W is an invariant subspace of U and $U|_W$ is again a unitary operator. By the induction hypothesis, W has an orthonormal basis $\{\mathsf{v}_2, \ldots, \mathsf{v}_n\}$ consisting of eigenvectors of U; combining it with v_1 gives an orthonormal basis of V. The induction is complete. ☐

Corollary 10.3 (Unitary Spectral Theorem and the Curl Conservation Law). Let U be a unitary operator. All its eigenvalues lie on the unit circle, which means that the corresponding dynamical system **contains no scaling component**—all evolution components are pure rotations. This property guarantees the conservation of the modulus in a unitary system—the system neither expands nor contracts during its evolution.

Proof. By Theorem 10.5, all eigenvalues of U satisfy $|\lambda| = 1$, so there is no scaling component with $|\lambda| \neq 1$; for every v we have $\|U\mathsf{v}\| = \|\mathsf{v}\|$, and therefore the modulus is conserved. ☐

Example 10.2 (Unitary Spectrum and Finite-order Period of a 2×2 Rotation Matrix). Take the plane rotation through the angle $\theta = \pi/3$

$$\mathsf{R} = \begin{pmatrix} \cos\frac{\pi}{3} & -\sin\frac{\pi}{3} \\ \sin\frac{\pi}{3} & \cos\frac{\pi}{3} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}.$$

R is an orthogonal matrix (and hence also a unitary matrix). Its characteristic polynomial $\det(\mathsf{R} - \lambda\mathsf{I}) = \lambda^2 - \lambda + 1$ has eigenvalues $\lambda = e^{\pm i\pi/3} = \frac{1}{2} \pm i\frac{\sqrt{3}}{2}$, both of modulus 1 (confirming Theorem 10.5). Since $\lambda^6 = 1$, we have $\mathsf{R}^6 = \mathsf{I}$: each step rotates by $\pi/3$, and after 6 steps the motion closes, so the finite order is 6.

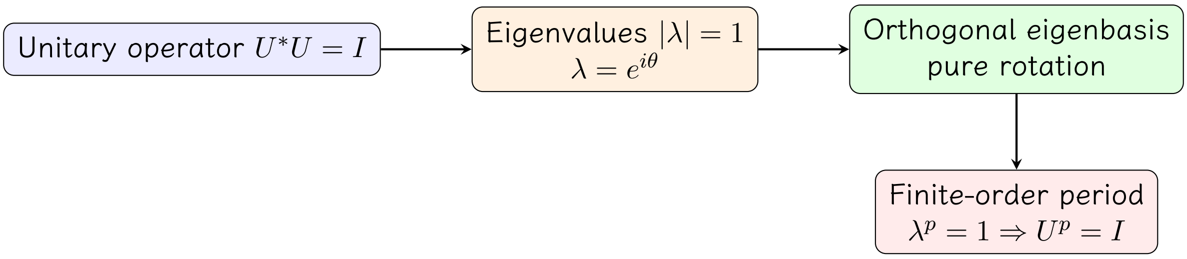

Unitary operator $U^*U = I$
Eigenvalues $|\lambda| = 1$
$\lambda = e^{i\theta}$
Orthogonal eigenbasis
pure rotation
Finite-order period
$\lambda^p = 1 \Rightarrow U^p = I$

Figure 10.10: The unitary spectral theorem: eigenvalues on the unit circle and an orthogonal eigenbasis, corresponding to pure rotation and finite-order periodicity.

10.3 Inner Product Spaces and Orthogonality

Definition 10.15 (Inner Product Space)**.** *Let V be a vector space over $\mathbb{C}$. If the map $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{C}$ satisfies:*

(i) *Conjugate symmetry: $\langle \mathsf{u}, \mathsf{v} \rangle = \overline{\langle \mathsf{v}, \mathsf{u} \rangle}$*

(ii) *Linearity (in the second argument): $\langle \mathsf{u}, \alpha\mathsf{v} + \beta\mathsf{w} \rangle = \alpha\langle \mathsf{u}, \mathsf{v} \rangle + \beta\langle \mathsf{u}, \mathsf{w} \rangle$*

(iii) *Positive definiteness: $\langle \mathsf{v}, \mathsf{v} \rangle \geq 0$, with equality if and only if $\mathsf{v} = \mathbf{0}$*

then it is called an **inner product***. A vector space equipped with an inner product is called an* **inner product space***. The* **norm** *of a vector* v *is defined as $\|\mathsf{v}\| = \sqrt{\langle \mathsf{v}, \mathsf{v} \rangle}$.*

The inner product is the algebraic realisation of the cosine trend in the 'Scalars and Vectors' chapter—$\langle \mathsf{u}, \mathsf{v} \rangle$ measures the 'consistency' of two vectors along the direction of translation.

Throughout this chapter the inner product is taken to be linear in the **second** argument and conjugate-linear in the first, that is $\langle \alpha\mathsf{u} + \beta\mathsf{v}, \mathsf{w} \rangle = \overline{\alpha}\langle \mathsf{u}, \mathsf{w} \rangle + \overline{\beta}\langle \mathsf{v}, \mathsf{w} \rangle$; all derivations below follow this convention (in a real inner product space the two conventions coincide).

Definition 10.16 (Orthogonality and Orthogonal Complement)**.** *Let V be an inner product space.*

(i) *If $\langle \mathsf{u}, \mathsf{v} \rangle = 0$, then* u *and* v *are said to be* **orthogonal***, written $\mathsf{u} \perp \mathsf{v}$.*

(ii) *Let $W \subseteq V$ be a subspace. The* **orthogonal complement** *of W is $W^\perp = \{\mathsf{v} \in V : \langle \mathsf{v}, \mathsf{w} \rangle = 0, \forall \mathsf{w} \in W\}$.*

(iii) *Space decomposition: $V = W \oplus W^\perp$—every vector can be uniquely decomposed as the sum of a component in W and a component in $W^\perp$.*

Orthogonality corresponds to the **independence** between two periodic components—orthogonal components do not transfer energy between one another but evolve independently.

Proposition 10.3 (Orthogonality and the Conservation Law)**.** *Let $\mathsf{v}_1, \ldots, \mathsf{v}_m$ be pairwise orthogonal non-zero vectors. Then their* **energy** *(the sum of squared moduli) is additive:*

$$\left\| \sum_{i=1}^{m} \mathsf{v}_i \right\|^2 = \sum_{i=1}^{m} \|\mathsf{v}_i\|^2.$$

*This is the form of Parseval's identity in a general inner product space—***the total energy over orthogonal components equals the sum of the energies of the components***.*

Proof. Expand the squared modulus:

$$\left\| \sum_{i=1}^{m} \mathsf{v}_i \right\|^2 = \sum_{i,j=1}^{m} \langle \mathsf{v}_i, \mathsf{v}_j \rangle.$$

By orthogonality, $\langle \mathsf{v}_i, \mathsf{v}_j \rangle = 0$ whenever $i \neq j$. The diagonal terms give $\sum_{i=1}^{m} \|\mathsf{v}_i\|^2$. ☐

Proposition 10.4 (Orthogonal Projection Theorem)**.** *Let $W \subseteq V$ be a finite-dimensional subspace. For every $\mathsf{v} \in V$ there exist unique vectors $\mathsf{w} \in W$ and $\mathsf{w}^\perp \in W^\perp$ such that $\mathsf{v} = \mathsf{w} + \mathsf{w}^\perp$. The vector w is called the* **orthogonal projection** *of v onto W, and it is the vector in W closest to v.*

Proof. Let $\{\mathsf{e}_1, \ldots, \mathsf{e}_m\}$ be an orthonormal basis of W. The orthogonal projection is $\mathsf{w} = \sum_{i=1}^{m} \langle \mathsf{v}, \mathsf{e}_i \rangle \mathsf{e}_i$. Verifying that $\langle \mathsf{v} - \mathsf{w}, \mathsf{e}_i \rangle = 0$ holds for all i shows that $\mathsf{v} - \mathsf{w} \in W^\perp$. Uniqueness is guaranteed by the direct sum property of the orthogonal decomposition. ☐

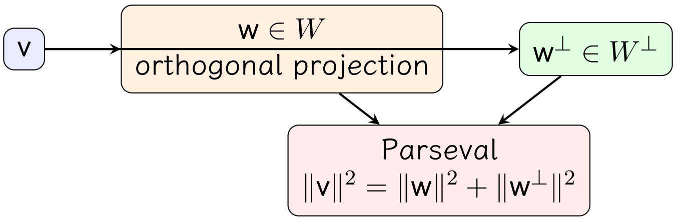


Figure 10.11: Orthogonal projection and Parseval's identity: the total energy equals the sum of the energies of the orthogonal components.

10.3.1 Adjoint Operators and Normal Operators

Definition 10.17 (Adjoint Operator)**.** *Let $T : V \to V$ be a linear operator on an inner product space. The* **adjoint operator** *T^* of T satisfies $\langle T(\mathsf{u}), \mathsf{v} \rangle = \langle \mathsf{u}, T^*(\mathsf{v}) \rangle$ for all $\mathsf{u}, \mathsf{v} \in V$. With respect to an orthonormal basis, the matrix of T^* is the conjugate transpose of the matrix of T.*

The adjoint operator embodies the symmetry operation at the level of operators: a self-adjoint operator ($T^* = T$) corresponds to the reflection symmetry (0-symmetry) in the gradient (symmetry centre 0), whilst a unitary operator ($U^* = U^{-1}$) corresponds to the inversion symmetry (1-symmetry, i-symmetry) in the curl (symmetry centre i).

Definition 10.18 (Normal Operator). *If a linear operator T satisfies $TT^* = T^*T$—that is, T commutes with its adjoint—then T is called a* **normal operator**. *Self-adjoint operators and unitary operators are both special cases of normal operators. The spectral theorem for normal operators unifies the spectral theorems for self-adjoint and unitary operators: a normal operator is unitarily diagonalisable, and its eigenvalues may be arbitrary complex numbers.*

Lemma 10.1 (Adjoint Properties of Eigenvectors of Normal Operators). *Let T be a normal operator, let λ be an eigenvalue of T and let v be a corresponding eigenvector. Then:*

(i) *v is also an eigenvector of T^*, and $T^*\mathsf{v} = \overline{\lambda}\mathsf{v}$;*

(ii) *for every $\mathsf{x} \in V$, we have $\|(T - \lambda I)\mathsf{x}\| = \|(T^* - \overline{\lambda} I)\mathsf{x}\|$.*

Proof. (ii) Let $S = T - \lambda I$; then $S^* = T^* - \overline{\lambda} I$. Using normality $TT^* = T^*T$ and expanding gives

$$SS^* = TT^* - \lambda T^* - \overline{\lambda} T + |\lambda|^2 I = T^*T - \overline{\lambda} T - \lambda T^* + |\lambda|^2 I = S^*S,$$

so S is also normal. Hence

$$\|S\mathsf{x}\|^2 = \langle S\mathsf{x}, S\mathsf{x}\rangle = \langle \mathsf{x}, S^*S\mathsf{x}\rangle = \langle \mathsf{x}, SS^*\mathsf{x}\rangle = \langle S^*\mathsf{x}, S^*\mathsf{x}\rangle = \|S^*\mathsf{x}\|^2.$$

(i) Since $T\mathsf{v} = \lambda\mathsf{v}$, that is, $S\mathsf{v} = \mathsf{0}$, part (ii) yields $S^*\mathsf{v} = \mathsf{0}$, that is, $T^*\mathsf{v} = \overline{\lambda}\mathsf{v}$. ☐

Theorem 10.6 (Spectral Theorem for Normal Operators). *Let T be a normal operator on a complex inner product space. Then there exists an orthonormal basis of V consisting of eigenvectors of T—T is unitarily diagonalisable.*

The spectral theorem for normal operators unifies the gradient (symmetry centre 0) (self-adjoint operators, real eigenvalues) and the curl (symmetry centre i) (unitary operators, eigenvalues of modulus 1): the eigenvalues of a normal operator may contain both a stretching component (real part) and a rotation component (imaginary part).

Proof. Induct on the dimension $n = \dim V$. When $n = 1$, V is spanned by a single eigenvector and the conclusion holds directly. Assume that the conclusion holds for normal operators of dimension less than n. By the fundamental theorem of algebra, T has an eigenvalue λ_1; take a corresponding unit eigenvector v_1. By Lemma 10.1, $T^*\mathsf{v}_1 = \overline{\lambda_1}\mathsf{v}_1$. Let $W = \operatorname{span}(\mathsf{v}_1)^\perp$. For any $\mathsf{w} \in W$,

$$\langle T\mathsf{w}, \mathsf{v}_1\rangle = \langle \mathsf{w}, T^*\mathsf{v}_1\rangle = \overline{\lambda_1}\langle \mathsf{w}, \mathsf{v}_1\rangle = 0,$$

so $T\mathsf{w} \in W$, that is, W is an invariant subspace of T. Similarly $\langle T^*\mathsf{w}, \mathsf{v}_1\rangle = \lambda_1\langle \mathsf{w}, \mathsf{v}_1\rangle = 0$, so W is also an invariant subspace of T^*, and hence $T|_W$ is still normal. By the induction hypothesis, W has an orthonormal basis $\{\mathsf{v}_2, \ldots, \mathsf{v}_n\}$ consisting of eigenvectors of T; combining it with v_1 gives an orthonormal basis of V. The induction is complete. ☐

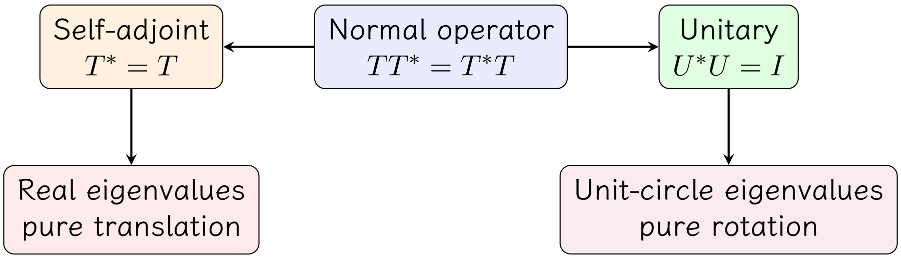


Figure 10.12: Normal operators unify the self-adjoint and the unitary: the self-adjoint gives real eigenvalues, the unitary gives eigenvalues on the unit circle.

10.4 Matrix Groups and Symmetry

Definition 10.19 (Orthogonal Group). The **orthogonal group** $O(n)$ is the group consisting of the $n \times n$ real matrices satisfying $\mathsf{Q}^\mathsf{T}\mathsf{Q} = \mathsf{I}_n$. The elements of $O(n)$ preserve the inner product of vectors: $\langle \mathsf{Qu}, \mathsf{Qv} \rangle = \langle \mathsf{u}, \mathsf{v} \rangle$.

The orthogonal group is the matrix realisation of the **gradient operator** (0-symmetry)—orthogonal transformations preserve the moduli and the angles of vectors, corresponding to translations and reflections in space. $SO(n)$ is the subgroup of matrices with determinant $+1$, corresponding to pure rotations.

Definition 10.20 (Unitary Group). The **unitary group** $U(n)$ is the group consisting of the $n \times n$ complex matrices satisfying $\mathsf{U}^*\mathsf{U} = \mathsf{I}_n$. The elements of $U(n)$ preserve the complex inner product.

The unitary group is the matrix realisation of the **curl operator** (i-symmetry)—unitary transformations preserve the moduli of complex vectors, corresponding to phase rotations in complex space. $U(1) = \{e^{i\theta}\}$ is the one-dimensional unitary group, corresponding to the rotation group with periodic orbits. $SU(n)$ is the subgroup of matrices with determinant 1, corresponding to volume-preserving pure rotations.

Definition 10.21 (Symplectic Group). The **symplectic group** $Sp(2n, \mathbb{R})$ is the group consisting of the $2n \times 2n$ real matrices satisfying $\mathsf{S}^\mathsf{T}\mathsf{J}\mathsf{S} = \mathsf{J}$, where $\mathsf{J} = \begin{pmatrix} 0 & \mathsf{I}_n \\ -\mathsf{I}_n & 0 \end{pmatrix}$ is the standard symplectic matrix.

The symplectic group is the matrix realisation of the **divergence operator** (1-symmetry)—symplectic transformations preserve phase-space volume (Liouville's theorem), corresponding to the canonical transformations of Hamiltonian systems. The determinant of a symplectic matrix is always 1.

Proposition 10.5 (Correspondence between the Three Matrix Groups and the Three Operators). *The orthogonal, unitary and symplectic groups are respectively the matrix realisations of the three basic symmetry groups in periodic mathematics; each group preserves a particular mathematical structure and corresponds to a conservation law:*

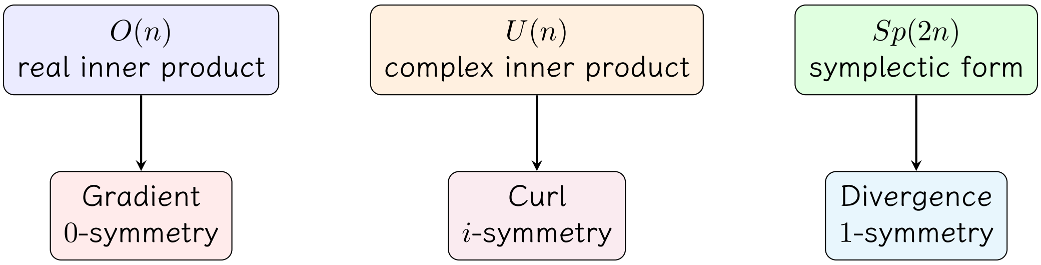


Figure 10.13: The three matrix groups and the three operators: the orthogonal group corresponds to the gradient, the unitary group to the curl, and the symplectic group to the divergence.

Matrix group	Preserved structure	Dominant operator	Symmetry centre	Conserved quantity
$O(n)$ (orthogonal group)	real inner product	gradient ∇ (0-symmetry)	$e = 0$ (additive identity)	momentum
$U(n)$ (unitary group)	complex inner product	curl $\nabla\times$ (i-symmetry)	$e = 1$ (multiplicative identity)	angular momentum
$Sp(2n)$ (symplectic group)	symplectic form	divergence $\nabla\cdot$ (1-symmetry)	$e = \lambda$ (scaling centre)	phase-space volume

Proof. First we verify that these three sets really do form groups. (i) $O(n)$: if $\mathsf{Q}_1, \mathsf{Q}_2 \in O(n)$, then $(\mathsf{Q}_1\mathsf{Q}_2)^\mathsf{T}(\mathsf{Q}_1\mathsf{Q}_2) = \mathsf{I}_n$, and $\mathsf{Q}^{-1} = \mathsf{Q}^\mathsf{T} \in O(n)$, so it is closed under multiplication and inversion and contains the identity element, and is therefore a group; for $\mathsf{Q} \in O(n)$ we have $\langle \mathsf{Qu}, \mathsf{Qv}\rangle = \mathsf{u}^\mathsf{T}\mathsf{Q}^\mathsf{T}\mathsf{Qv} = \mathsf{u}^\mathsf{T}\mathsf{v} = \langle \mathsf{u}, \mathsf{v}\rangle$, so it preserves the real inner product. (ii) $U(n)$: similarly $(\mathsf{U}_1\mathsf{U}_2)^*(\mathsf{U}_1\mathsf{U}_2) = \mathsf{I}_n$ and $\mathsf{U}^{-1} = \mathsf{U}^* \in U(n)$, so it is a group, and it preserves the complex inner product. (iii) $Sp(2n, \mathbb{R})$: if $\mathsf{S}_i^\mathsf{T}\mathsf{J}\mathsf{S}_i = \mathsf{J}$, then $(\mathsf{S}_1\mathsf{S}_2)^\mathsf{T}\mathsf{J}(\mathsf{S}_1\mathsf{S}_2) = \mathsf{S}_2^\mathsf{T}\mathsf{J}\mathsf{S}_2 = \mathsf{J}$; taking determinants of the defining equation gives $(\det \mathsf{S})^2 = 1$, so S is invertible and S^{-1} is again a symplectic matrix, whence $Sp(2n)$ is a group. These three groups preserve the real inner product, the complex inner product and the symplectic form respectively, and hence correspond respectively to the three conservation laws of momentum, angular momentum and phase-space volume. ☐

Corollary 10.4 (Periodic Diagnostic Significance of Matrix Groups)**.** *From the three-group correspondence of Proposition 10.5, in the interdisciplinary diagnosis of periodic mathematics the type of the symmetry group to which a linear system belongs directly indicates the dominant operator of its periodic structure:*

(i) *If the evolution matrix of the system belongs to the orthogonal group $O(n)$, then the system has a pure translation or pure rotation period—every eigenvalue of $\mathsf{Q} \in O(n)$ lies on the unit circle (all have modulus 1); the* **purely involutive** *case (C_2 period) occurs exactly when all eigenvalues are ± 1, that is, when Q is real-diagonalisable with spectrum contained in $\{+1, -1\}$, which is equivalent to $\mathsf{Q}^2 = \mathsf{I}$. For example, the rotation*

$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \in SO(2) \subseteq O(2)$ with $\theta \notin \pi\mathbb{Z}$ has eigenvalues $e^{\pm i\theta} \neq \pm 1$ and therefore gives a pure rotation period rather than an involution.

(ii) If the evolution matrix of the system belongs to the unitary group $U(n)$, then the system has a pure rotation period—all eigenvalues lie on the unit circle, corresponding to quasi-periodic or strictly periodic motion with conservation of angular momentum.

(iii) If the evolution matrix of the system belongs to the symplectic group $Sp(2n)$, then the system has a scaling period—phase-space volume is conserved, but the individual components may undergo the combined action of rotation and scaling.

Note: the three assertions above are one-directional implications—from the symmetry group containing the matrix to the dominant operator of its periodic structure—and do not reverse. In particular, (i) cannot be read backwards as 'every matrix in $O(n)$ has all eigenvalues ± 1'; the involutive period is merely a special case within $O(n)$, the general case being given by the rotation counterexample above.

Proof. By Proposition 10.5: (i) $\mathsf{Q} \in O(n)$ preserves the real inner product, and its eigenvalues are numbers of modulus 1 (in the real case, ± 1 or conjugate pairs); the purely involutive case gives a C_2 period; (ii) all eigenvalues of $\mathsf{U} \in U(n)$ lie on the unit circle (Theorem 10.5), giving a pure rotation period; (iii) $\mathsf{S} \in Sp(2n)$ preserves the symplectic form, that is, phase-space volume, and the individual components may undergo the combined action of rotation and scaling, giving a scaling period. ☐

10.5 Lo Shu Matrix: the 3×3 Magic Square with 5 as the Balance Point

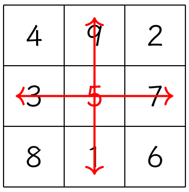
This section constructs the **Lo Shu matrix**—the 3×3 magic square formed by arranging the numbers 1 to 9. Its mathematical core lies in the following: the magic sum is $15 = 3 \times 5$, the central element 5 is the arithmetic mean of 1 to 9, and the whole matrix has an antipodally antisymmetric structure about the centre 5. The spectral structure of the Lo Shu matrix simultaneously contains a pure translation eigendirection and a rotation-and-scaling invariant subspace.

Definition 10.22 (Lo Shu Matrix)**.** *The* **Lo Shu matrix** *is defined as*

$$\mathsf{L} := \begin{pmatrix} 4 & 9 & 2 \\ 3 & 5 & 7 \\ 8 & 1 & 6 \end{pmatrix}.$$

That is, the matrix entries L_{ij} are given by the following table:

	$j=1$	$j=2$	$j=3$
$i=1$	4	9	2
$i=2$	3	5	7
$i=3$	8	1	6



Magic sum $15 = 3 \times 5$, central balance point 5

Figure 10.14: The Lo Shu matrix: a 3×3 magic square with centre 5 as the balance point and magic sum $15 = 3 \times 5$.

Definition 10.23 (Row Sums, Column Sums, and Diagonal Sums)**.** *For a 3×3 matrix $\mathsf{L} = (L_{ij})$, define:*

(i) *the* **row sums** $r_i := \sum_{j=1}^{3} L_{ij}$ $(i = 1, 2, 3)$;

(ii) *the* **column sums** $c_j := \sum_{i=1}^{3} L_{ij}$ $(j = 1, 2, 3)$;

(iii) *the* **main diagonal sum** $d_+ := L_{11} + L_{22} + L_{33}$;

(iv) *the* **anti-diagonal sum** $d_- := L_{13} + L_{22} + L_{31}$.

Proposition 10.6 (Magic Square Property of the Lo Shu Matrix)**.** *The Lo Shu matrix L satisfies: all row sums, column sums, the main diagonal sum and the anti-diagonal sum are equal to 15, that is,*

$$r_1 = r_2 = r_3 = c_1 = c_2 = c_3 = d_+ = d_- = 15 = 3 \times 5.$$

L is called a **magic square of order three***, and 15 is called its* **magic sum***.*

Proof. Direct computation:

$$\begin{aligned} r_1 &= 4 + 9 + 2 = 15, & r_2 &= 3 + 5 + 7 = 15, & r_3 &= 8 + 1 + 6 = 15; \\ c_1 &= 4 + 3 + 8 = 15, & c_2 &= 9 + 5 + 1 = 15, & c_3 &= 2 + 7 + 6 = 15; \\ d_+ &= 4 + 5 + 6 = 15, & d_- &= 2 + 5 + 8 = 15. \end{aligned}$$

The sums of all eight lines are $15 = 3 \times 5$. ☐

Definition 10.24 (Lo Shu Balance Point)**.** *The central entry $L_{22} = 5$ of L is called the* **Lo Shu balance point***.*

Proposition 10.7 (Triple Characterisation of the Balance Point)**.** *The Lo Shu balance point 5 satisfies the following three equivalent characterisations:*

(i) **Arithmetic centre**: $5 = \dfrac{1 + 2 + \cdots + 9}{9}$, *that is, 5 is the arithmetic mean of 1 to 9;*

(ii) **Magic sum factor**: $15 = 3 \times 5$, *so the magic sum is exactly three times the balance point;*

(iii) **Diagonal mean**: *the means of the two diagonals are both 5, that is,* $\dfrac{d_+}{3} = \dfrac{d_-}{3} = 5$.

Proof. (i) $\dfrac{1+2+\cdots+9}{9}=\dfrac{45}{9}=5$. (ii) By Proposition 10.6, the magic sum is $15=3\times 5$. (iii) Dividing $d_+=d_-=15$ by 3 gives 5. ☐

Proposition 10.8 (Lo Shu Antipodal Balancing)**.** The sums of the entries at antipodal positions of the Lo Shu matrix about the centre 5 are always $10=2\times 5$:

$$L_{11}+L_{33}=10,\quad L_{22}+L_{22}=10,\quad L_{33}+L_{11}=10,$$
$$L_{12}+L_{32}=10,\quad L_{21}+L_{23}=10,\quad L_{13}+L_{31}=10.$$

Equivalently, $\mathsf{L}-5\mathsf{J}$ (where J is the all-ones matrix) satisfies the antipodal antisymmetry condition:

$$(\mathsf{L}-5\mathsf{J})_{ij}=-(\mathsf{L}-5\mathsf{J})_{4-i,4-j},\quad i,j\in\{1,2,3\}.$$

Proof. Direct computation of all antipodal pairs:

$$4+6=10,\quad 9+1=10,\quad 2+8=10,\quad 3+7=10,\quad 5+5=10.$$

For the second identity, note that $(\mathsf{L}-5\mathsf{J})_{11}=4-5=-1$ and $(\mathsf{L}-5\mathsf{J})_{33}=6-5=1$, which are negatives of one another; the remaining antipodal positions are analogous. ☐

Theorem 10.7 (Eigenvalues and Spectral Structure of the Lo Shu Matrix)**.** The characteristic polynomial of the Lo Shu matrix L is

$$p(\lambda)=\det(\mathsf{L}-\lambda\mathsf{I})=-(\lambda-15)(\lambda^2+24).$$

Hence the three eigenvalues of L *are*

$$\lambda_1=15,\qquad \lambda_{2,3}=\pm 2\sqrt{6}\,i.$$

The eigenvalue $\lambda_1=15=3\times 5$ is a positive real eigenvalue, corresponding to pure translation; $\lambda_{2,3}=\pm 2\sqrt{6}i$ are a purely imaginary conjugate pair, of modulus $|\lambda_{2,3}|=2\sqrt{6}$, corresponding to the joint action of rotation and scaling in the two-dimensional real invariant subspace.

Proof. Compute the characteristic polynomial:

$$\begin{aligned}
p(\lambda)=\det\begin{pmatrix}4-\lambda & 9 & 2\\ 3 & 5-\lambda & 7\\ 8 & 1 & 6-\lambda\end{pmatrix}
&=(4-\lambda)\big[(5-\lambda)(6-\lambda)-7\big]-9\big[3(6-\lambda)-56\big]+2\big[3-8(5-\lambda)\big]\\
&=(4-\lambda)(\lambda^2-11\lambda+23)+9(38+3\lambda)-2(37-8\lambda)\\
&=-\lambda^3+15\lambda^2-24\lambda+360\\
&=-(\lambda-15)(\lambda^2+24).
\end{aligned}$$

Solving $p(\lambda)=0$ gives $\lambda_1=15$ and $\lambda_{2,3}^2=-24$, that is, $\lambda_{2,3}=\pm 2\sqrt{6}\,i$. ☐

Theorem 10.8 (Translation-Rotation Decomposition of the Lo Shu Matrix)**.** The Lo Shu matrix L decomposes into the sum of a pure translation part and a rotation-and-scaling part:

$$\mathsf{L}=\underbrace{5\mathsf{J}}_{\text{translation part}}+\underbrace{(\mathsf{L}-5\mathsf{J})}_{\text{rotation-and-scaling part}},$$

where J is the all-ones matrix. These two parts satisfy:

(i) **Translation part** $5\mathsf{J}$: *it acts on the vector* $\mathbf{1} = (1,1,1)^{\mathsf{T}}$ *as* $15\mathbf{1}$*, whilst it acts as zero on* $\mathbf{1}^{\perp}$*; that is, the eigenvalues of* $5\mathsf{J}$ *are* $15, 0, 0$*.*

(ii) **Rotation-and-scaling part** $\mathsf{L} - 5\mathsf{J}$: *it acts as zero on* $\mathbf{1}$*, whilst on* $\mathbf{1}^{\perp}$ *it has eigenvalues* $\pm 2\sqrt{6}i$ *(of modulus* $2\sqrt{6} \neq 1$*).*

Consequently L *is a pure translation on the invariant subspace* span$(\mathbf{1})$ *and acts on the invariant subspace* $\mathbf{1}^{\perp}$ *as a joint rotation and scaling: rotation through* $\pi/2$ *combined with scaling of the modulus by* $2\sqrt{6}$ *at each step (this is not a pure rotation, since a pure rotation requires* $|\lambda| = 1$ *whilst here* $|\lambda_{2,3}| = 2\sqrt{6} \neq 1$*).*

Proof. **Translation part:** $5\mathsf{J}$ maps any vector v to $5(\mathbf{1}^{\mathsf{T}}\mathsf{v})\mathbf{1}$. Hence $5\mathsf{J}\mathbf{1} = 5{\cdot}3\mathbf{1} = 15\mathbf{1}$, and for $\mathsf{v} \in \mathbf{1}^{\perp}$ (that is, $\mathbf{1}^{\mathsf{T}}\mathsf{v} = 0$) we have $5\mathsf{J}\mathsf{v} = \mathbf{0}$.

Rotation-and-scaling part: by Proposition 10.6, every row sum is 15, so $(\mathsf{L} - 5\mathsf{J})\mathbf{1} = \mathsf{L}\mathbf{1} - 5\mathsf{J}\mathbf{1} = 15\mathbf{1} - 15\mathbf{1} = \mathbf{0}$. Hence $\mathbf{1}$ lies in the null space.

Consistency of the eigenvalues: let $\mathsf{v} \in \mathbf{1}^{\perp}$ be an eigenvector of $\mathsf{L} - 5\mathsf{J}$ with eigenvalue μ. Then $\mathsf{L}\mathsf{v} = (5\mathsf{J} + (\mathsf{L} - 5\mathsf{J}))\mathsf{v} = \mathbf{0} + \mu\mathsf{v} = \mu\mathsf{v}$. Hence v is also an eigenvector of L with eigenvalue μ. By Theorem 10.7, the eigenvalues of L on $\mathbf{1}^{\perp}$ are $\pm 2\sqrt{6}i$. A direct computation of $\det(\mathsf{L} - 5\mathsf{J} - \lambda I)$ likewise verifies $-\lambda(\lambda^2 + 24)$. ☐

Corollary 10.5 (Structural Significance of the Lo Shu Balance Point and Magic Sum)**.** *The spectral structure of the Lo Shu matrix reveals the twofold significance of the balance point* 5 *and the magic sum* 15 *at the level of the two operators:*

(i) **On the gradient side**: $5 = 15/3$ *is the average scale of the pure translation along the all-ones direction—the stretching factor of the translation part* $5\mathsf{J}$ *in the* $\mathbf{1}$ *direction is* 15*, and dividing by the dimension* 3 *gives the average factor* 5 *per component;*

(ii) **On the curl side**: 5 *is the centre of the antipodal balance—the antipodal antisymmetry of* $\mathsf{L} - 5\mathsf{J}$ *(Proposition 10.8) has* 5 *as its axis of symmetry;*

(iii) **On the period side**: L *has the rotation-and-scaling eigenvalues* $\pm 2\sqrt{6}i$ *on* $\mathbf{1}^{\perp}$*, corresponding to rotation through* $\pi/2$ *and scaling of the modulus by* $2\sqrt{6}$ *at each step in that two-dimensional real invariant subspace; since* $|\lambda_{2,3}| = 2\sqrt{6} \neq 1$*, the orbit does not close.*

Proof. (i) By Theorem 10.8, $5\mathsf{J}$ has the eigenvalue 15 in the $\mathbf{1}$ direction; its mean is $15/3 = 5$. (ii) By Proposition 10.8, $\mathsf{L} - 5\mathsf{J}$ is antipodally antisymmetric about 5. (iii) By Theorem 10.7 and $\lambda_{2,3} = \pm 2\sqrt{6}i$, a purely imaginary eigenvalue $\pm 2\sqrt{6}i$ has modulus $2\sqrt{6} \neq 1$, corresponding to rotation through $\pi/2$ and scaling of the modulus by $2\sqrt{6}$ at each step, so the orbit does not close. ☐

Example 10.3 (Spectral Decomposition of the Lo Shu Matrix)**.** The eigenvectors corresponding to the three eigenvalues of the Lo Shu matrix L are as follows:

(i) $\lambda_1 = 15$: the eigenvector $\mathsf{v}_1 = \frac{1}{\sqrt{3}}(1,1,1)^{\mathsf{T}}$ (all components equal);

(ii) $\lambda_2 = 2\sqrt{6}i$ and $\lambda_3 = -2\sqrt{6}i$: the eigenvectors lie inside $\mathbf{1}^{\perp}$.

Verification for v_1: $\mathsf{L}\mathsf{v}_1 = \frac{1}{\sqrt{3}}(15, 15, 15)^{\mathsf{T}} = 15\mathsf{v}_1$.

Take an orthonormal basis of $\mathsf{1}^{\perp}$ to be $\boldsymbol{e}_2 = \frac{1}{\sqrt{2}}(1, -1, 0)^{\mathsf{T}}$ and $\boldsymbol{e}_3 = \frac{1}{\sqrt{6}}(1, 1, -2)^{\mathsf{T}}$; the matrix of the restriction $\mathsf{L}|_{\mathsf{1}^{\perp}}$ of L to $\mathsf{1}^{\perp}$ with respect to this basis is

$$\mathsf{L}|_{\mathsf{1}^{\perp}} = \begin{pmatrix} -\frac{3}{2} & \frac{5\sqrt{3}}{2} \\ -\frac{7\sqrt{3}}{2} & \frac{3}{2} \end{pmatrix}$$

Its characteristic polynomial is $\mu^2 + 24$, whose roots are $\mu = \pm 2\sqrt{6}\,i$, in agreement with the non-zero roots of the complete characteristic polynomial $-\lambda(\lambda^2 + 24)$ (the action of L on $\mathsf{1}^{\perp}$ is equivalent to a rotation through $\pi/2$ combined with a scaling of the modulus by $2\sqrt{6}$).

Remark 10.3 (Lo Shu Matrix and the Interface with $\mathbb{Z}_2 \times \mathbb{Z}_5$ Structure). The factor 5 in the magic sum $15 = 3 \times 5$ of the Lo Shu matrix is algebraically related to the $\mathbb{Z}_5$ component of $\mathbb{Z}_{10} \cong \mathbb{Z}_2 \times \mathbb{Z}_5$ in Chapter 6: the Lo Shu centre 5 corresponds to the reference point of the $\mathbb{Z}_5$ component, whilst the antipodal balance of the Lo Shu matrix (with sum $10 = 2 \times 5$) corresponds to the double alternation of $\mathbb{Z}_2$. The complex conjugate eigenvalues $\pm 2\sqrt{6}i$ of the Lo Shu matrix on $\mathsf{1}^{\perp}$ provide a concrete example: every step in that two-dimensional real invariant subspace is a rotation through $\pi/2$ combined with a scaling of the modulus by $2\sqrt{6}$, and both the angle and the scaling factor are entirely determined by the algebraic structure of the matrix.

10.6 Lie Groups: a Rigorous Formulation of Continuous Symmetry Groups

Definition 10.25 (Lie Group). A **Lie group** is a smooth manifold G equipped with a group structure $(G, \cdot)$ such that the group operations

$$\mu : G \times G \to G, \quad (g, h) \mapsto g \cdot h, \quad \iota : G \to G, \quad g \mapsto g^{-1}$$

are both smooth maps.

A Lie group unifies the abstract concept of a group in the 'Operations and Periodicity' chapter with the concept of a smooth manifold. The three basic transformation groups of periodic mathematics—the translation group, the scaling group and the rotation group—are all Lie groups.

Example 10.4 (The three transformation groups are Lie groups).

(i) **Translation group** $(\mathbb{R}^n, +)$: as a manifold it is $\mathbb{R}^n$, the group operation is vector addition, and it is smooth. An Abelian Lie group.

(ii) **Scaling group** $(\mathbb{R}^+, \times)$: as a manifold it is the positive real axis $(0, \infty)$, the group operation is multiplication, and it is smooth. An Abelian Lie group, isomorphic as a Lie group to $(\mathbb{R}, +)$ via the exponential map exp.

(iii) **Rotation group** $U(1) = \{z \in \mathbb{C} : |z| = 1\}$: as a manifold it is the circle S^1, the group operation is complex multiplication, and it is smooth. A compact, connected, Abelian Lie group.

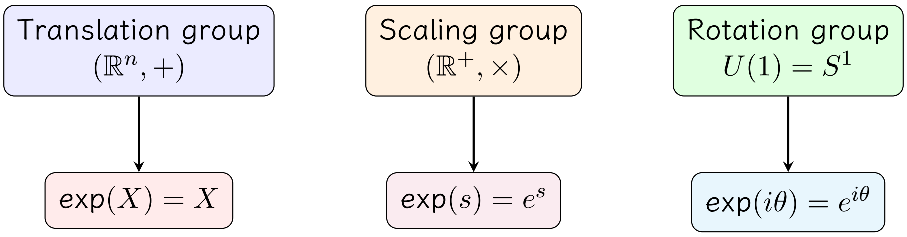


Figure 10.15: The three transformation Lie groups and their exponential maps: the exponential of the translation group is the identity, that of the scaling group is the real exponential, and that of the rotation group is the complex exponential.

Proposition 10.9 (One-parameter Subgroup). *Let G be a Lie group and let $X \in T_eG$ be a tangent vector at the identity element e. Then there exists a unique Lie group homomorphism $\gamma_X : \mathbb{R} \to G$ satisfying $\dot{\gamma}_X(0) = X$. The map γ_X is called the* **one-parameter subgroup** *generated by X. It satisfies*

$$\gamma_X(s+t) = \gamma_X(s) \cdot \gamma_X(t), \qquad \forall s, t \in \mathbb{R}.$$

Proof. This is a fundamental theorem of Lie group theory. Existence is guaranteed by the existence and uniqueness theorem for ordinary differential equations: γ_X is the integral curve of the left-invariant vector field X^L ($X^L(g) = dL_g(X)$). Uniqueness and the homomorphism property follow from the properties of the flow of a left-invariant vector field: if γ_X is the integral curve of X^L satisfying $\gamma_X(0) = e$, then both $t \mapsto \gamma_X(s+t)$ and $t \mapsto \gamma_X(s)\gamma_X(t)$ are integral curves of X^L passing through $\gamma_X(s)$ at $t = 0$, so by uniqueness they coincide. ☐

Definition 10.26 (Exponential Map). Let G be a Lie group. The **exponential map** $\exp : T_eG \to G$ is defined by $\exp(X) := \gamma_X(1)$, where γ_X is the one-parameter subgroup generated by X. The exponential map sends a tangent vector at the identity (an infinitesimal transformation) to a group element (a finite transformation).

Proposition 10.10 (Exponential maps of the three transformation groups). *(i)*

(i) **Translation group** $(\mathbb{R}^n, +)$: *$T_eG \cong \mathbb{R}^n$, $\exp(X) = X$. The one-parameter subgroup is $\gamma_X(t) = tX$.*

(ii) **Scaling group** $(\mathbb{R}^+, \times)$: *$T_eG \cong \mathbb{R}$, $\exp(s) = e^s$. The one-parameter subgroup is $\gamma_s(t) = e^{st}$.*

(iii) **Rotation group** $U(1)$: *$T_eG \cong i\mathbb{R} \cong \mathbb{R}$, $\exp(i\theta) = e^{i\theta}$. The one-parameter subgroup is $\gamma_{i\theta}(t) = e^{i\theta t}$.*

Proof. (i) On the translation group the left-invariant vector field is the constant vector field $X^L(g) = X$. Its integral curve satisfies $\dot{\gamma}(t) = X$ and $\gamma(0) = 0$, giving $\gamma(t) = tX$. Hence $\exp(X) = \gamma_X(1) = X$.

(ii) On the scaling group, the tangent vector $s \in \mathbb{R}$ at the identity $e = 1$ corresponds to the left-invariant vector field $X^L(x) = sx$. The integral curve satisfies $\dot{\gamma}(t) = s\gamma(t)$ and $\gamma(0) = 1$, giving $\gamma(t) = e^{st}$. Hence $\exp(s) = e^s$.

(iii) On the rotation group, the tangent vector $i\theta$ at the identity $e = 1$ corresponds to the left-invariant vector field $X^L(z) = i\theta z$. The integral curve satisfies $\dot{\gamma}(t) = i\theta\gamma(t)$ and $\gamma(0) = 1$, giving $\gamma(t) = e^{i\theta t}$. Hence $\exp(i\theta) = e^{i\theta}$. ☐

Remark 10.4 (Correspondence between the Exponential Map and the Three Operators). The exponential map is the bridge connecting **infinitesimal transformations** and **finite transformations**:

- On the translation group exp = id: an infinitesimal translation is a finite translation—the gradient (symmetry centre 0) is 'flat', and no exponentiation is needed.
- On the scaling group $\exp(s) = e^s$: the infinitesimal scaling s becomes, through the exponential map, the finite scaling factor e^s—the divergence (symmetry centre 1) accumulates by exponentiation.
- On the rotation group $\exp(i\theta) = e^{i\theta}$: the infinitesimal rotation angle $i\theta$ becomes, through the exponential map, the unit complex number $e^{i\theta}$—the curl (symmetry centre i) generates periodicity by exponentiation.

Proposition 10.11 (Finite-order Elements and Closed Orbits of a Lie Group). Let G be a Lie group and let $g = \exp(X) \in G$. If g is an element of finite order ($\text{ord}(g) = p < \infty$), then the one-parameter subgroup γ_X is periodic, that is, there exists $T > 0$ such that $\gamma_X(T) = e$—take $T = p$ directly from $g^p = e$; in that case γ_X defines a closed orbit of period p. The converse requires an additional arithmetic condition: let γ_X be non-constant ($X \neq 0$) with least positive period T_0; then $\gamma_X(T) = e$ if and only if $T/T_0 \in \mathbb{Z}$, so, since $g = \gamma_X(1)$,

$$\text{ord}(g) < \infty \iff T_0 \text{ is a rational part of } 1, \text{ that is, } 1/T_0 \in \mathbb{Q},$$

and then $\text{ord}(g)$ is the least element of $\{n \geq 1 : n/T_0 \in \mathbb{Z}\}$. For example, take $G = U(1) \subset \mathbb{C}^\times$ and $X = 2\pi i\sqrt{2}$; then $\gamma_X(T) = e^{2\pi i\sqrt{2}T}$ has least positive period $T_0 = 1/\sqrt{2}$, whilst $1/T_0 = \sqrt{2} \notin \mathbb{Q}$, so $g = e^{2\pi i\sqrt{2}}$ has infinite order. Hence in general the periodicity of γ_X is only a necessary condition for g to have finite order.

Proof. Forward direction (the direction proved in the text: g of finite order $\Rightarrow \gamma_X$ periodic). If $\text{ord}(g) = p$, then $g^p = e$, that is, $\gamma_X(1)^p = e$. By the homomorphism property, $\gamma_X(p) = \gamma_X(1)^p = e$, so γ_X returns to the identity at $t = p$, and γ_X is periodic, with period p.

The converse fails; its correct form is the displayed equivalence above. Write $A = \{T > 0 : \gamma_X(T) = e\}$. Since γ_X is a homomorphism, A is closed under addition and under taking integer multiples; when $X \neq 0$ the curve γ_X does not take the value e near $t = 0$ (its derivative at $t = 0$ is non-zero), so A is discrete in $\mathbb{R}_{>0}$ and has a least element T_0, with $A = T_0\mathbb{Z}_{>0}$. Hence $g^n = e = \gamma_X(n)$ if and only if $n \in T_0\mathbb{Z}_{>0}$, that is, $n/T_0 \in \mathbb{Z}$; such a positive integer n exists if and only if $1/T_0 \in \mathbb{Q}$. For $T_0 = 1/\sqrt{2}$ (the case $G = U(1)$, $X = 2\pi i\sqrt{2}$) we have $1/T_0 = \sqrt{2} \notin \mathbb{Q}$, so $\text{ord}(g) = \infty$ whilst $\gamma_X(1/\sqrt{2}) = e$. Thus 'g of finite order $\Rightarrow \gamma_X$ periodic' holds; the converse holds only with the extra hypothesis $1/T_0 \in \mathbb{Q}$, in which case the order of g is the denominator of $1/T_0$. ☐

10.7 Lie Algebras: The Infinitesimal Structure of Matrix Groups

Definition 10.27 (Lie Algebra). Let $\mathbb{F}$ be a field. A **Lie algebra** over $\mathbb{F}$ is a vector space $\mathfrak{g}$ equipped with a bilinear map $[\cdot,\cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ (called the **Lie bracket**) satisfying:

(i) **Antisymmetry**: $[X, Y] = -[Y, X]$;

(ii) **The Jacobi identity**: $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$.

Remark 10.5 (The Significance of Lie Algebras in Periodic Mathematics). A Lie algebra describes the **infinitesimal structure** *of a continuous symmetry group. The elements of a Lie algebra are infinitesimal generators—they generate one-parameter subgroups, and one-parameter subgroups in turn generate closed orbits. The Lie bracket $[X, Y]$ measures the non-commutativity of two infinitesimal transformations—when $[X, Y] = 0$, the transformations generated by X and Y commute with one another; when $[X, Y] \neq 0$, they cannot commute.*

Definition 10.28 (Matrix Lie Algebra)**.** *Let $G \subseteq GL(n, \mathbb{F})$ be a matrix Lie group. The* **Lie algebra $\mathfrak{g}$** *of G is the tangent space of G at the identity I:*

$$\mathfrak{g} := T_I G = \{\mathsf{X} \in M_n(\mathbb{F}) : e^{t\mathsf{X}} \in G,\ \forall t \in \mathbb{R}\},$$

The Lie bracket is the matrix commutator $[\mathsf{X}, \mathsf{Y}] = \mathsf{XY} - \mathsf{YX}$.

Proposition 10.12 (Three Basic Matrix Lie Algebras)**.** (i) **Orthogonal Lie algebra** $\mathfrak{o}(n) = \{\mathsf{X} \in M_n(\mathbb{R}) : \mathsf{X}^\mathsf{T} = -\mathsf{X}\}$;

(ii) **Unitary Lie algebra** $\mathfrak{u}(n) = \{\mathsf{X} \in M_n(\mathbb{C}) : \mathsf{X}^* = -\mathsf{X}\}$;

(iii) **Symplectic Lie algebra** $\mathfrak{sp}(2n) = \{\mathsf{X} : \mathsf{X}^\mathsf{T}\mathsf{J} + \mathsf{J}\mathsf{X} = 0\}$.

Proof. (i) Let $\mathsf{X} \in \mathfrak{o}(n)$. Then $e^{t\mathsf{X}} \in O(n)$ for all t, that is, $(e^{t\mathsf{X}})^\mathsf{T} e^{t\mathsf{X}} = I$. Differentiating at $t = 0$ gives $\mathsf{X}^\mathsf{T} + \mathsf{X} = 0$, hence $\mathsf{X}^\mathsf{T} = -\mathsf{X}$. Conversely, if X is antisymmetric, then $(e^{t\mathsf{X}})^\mathsf{T} e^{t\mathsf{X}} = e^{-t\mathsf{X}} e^{t\mathsf{X}} = I$.

(ii) Let $\mathsf{X} \in \mathfrak{u}(n)$. Differentiating $(e^{t\mathsf{X}})^* e^{t\mathsf{X}} = I$ gives $\mathsf{X}^* + \mathsf{X} = 0$.

(iii) Let $\mathsf{X} \in \mathfrak{sp}(2n)$. Differentiating $(e^{t\mathsf{X}})^\mathsf{T} \mathsf{J} e^{t\mathsf{X}} = \mathsf{J}$ gives $\mathsf{X}^\mathsf{T}\mathsf{J} + \mathsf{J}\mathsf{X} = 0$. ☐

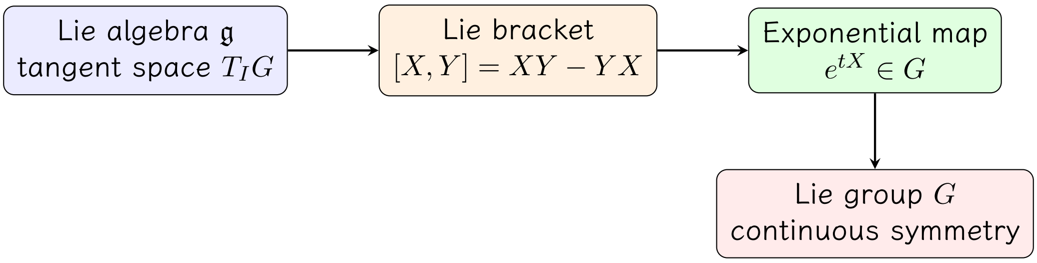


Figure 10.16: From Lie algebra to Lie group: tangent space, Lie bracket and exponential map.

Proposition 10.13 (Properties of the Matrix Exponential Map)**.** *Let $\mathsf{X} \in M_n(\mathbb{F})$. The matrix exponential e^X satisfies:*

(i) $e^0 = \mathsf{I}$;

(ii) $e^\mathsf{X} e^{-\mathsf{X}} = \mathsf{I}$;

(iii) if $\mathsf{XY} = \mathsf{YX}$, then $e^{\mathsf{X}+\mathsf{Y}} = e^{\mathsf{X}}e^{\mathsf{Y}}$;

(iv) $\det(e^{\mathsf{X}}) = e^{\operatorname{tr}(\mathsf{X})}$;

(v) if X is antisymmetric, then $e^{\mathsf{X}} \in SO(n)$;

(vi) if X is skew-Hermitian, then $e^{\mathsf{X}} \in U(n)$.

Proof. (i) Substitute directly into the power series definition.

(ii) Since X and $-\mathsf{X}$ commute, use (iii): $e^{\mathsf{X}}e^{-\mathsf{X}} = e^{\mathsf{X}-\mathsf{X}} = \mathsf{I}$.

(iii) When $\mathsf{XY} = \mathsf{YX}$, the binomial theorem gives $e^{\mathsf{X}}e^{\mathsf{Y}} = \sum_m \frac{(\mathsf{X}+\mathsf{Y})^m}{m!} = e^{\mathsf{X}+\mathsf{Y}}$.

(iv) Let the eigenvalues of the Jordan normal form of X be $\lambda_1, \ldots, \lambda_n$. Then the eigenvalues of e^{X} are $e^{\lambda_1}, \ldots, e^{\lambda_n}$, hence $\det(e^{\mathsf{X}}) = e^{\sum_i \lambda_i} = e^{\operatorname{tr}(\mathsf{X})}$.

(v) By (i)–(iii), $e^{\mathsf{X}^{\mathsf{T}}} = (e^{\mathsf{X}})^{\mathsf{T}}$. If $\mathsf{X}^{\mathsf{T}} = -\mathsf{X}$, then $(e^{\mathsf{X}})^{\mathsf{T}} = e^{-\mathsf{X}} = (e^{\mathsf{X}})^{-1}$, so $e^{\mathsf{X}} \in O(n)$. By (iv), $\det(e^{\mathsf{X}}) = e^{\operatorname{tr}(\mathsf{X})} = 1$ (the trace of an antisymmetric matrix is zero), so $e^{\mathsf{X}} \in SO(n)$.

(vi) Similarly, $(e^{\mathsf{X}})^* = e^{-\mathsf{X}} = (e^{\mathsf{X}})^{-1}$, so $e^{\mathsf{X}} \in U(n)$. □

Theorem 10.9 (Correspondence Between Lie Algebras and Conservation Laws). The elements of the three basic matrix Lie algebras generate matrix groups through the matrix exponential map, corresponding respectively to three conservation laws:

(i) **Orthogonal Lie algebra** $\mathfrak{o}(n)$: antisymmetric matrices generate $SO(n)$—preserving the real inner product (0-symmetry).

(ii) **Unitary Lie algebra** $\mathfrak{u}(n)$: skew-Hermitian matrices generate $U(n)$—preserving the complex inner product (conservation of angular momentum, curl operator, i-symmetry).

(iii) **Symplectic Lie algebra** $\mathfrak{sp}(2n)$: generates $Sp(2n)$—preserving the symplectic form (conservation of phase-space volume, divergence operator, 1-symmetry).

Proof. This follows directly from Proposition 10.12 and parts (v)(vi) of Proposition 10.13: the exponential map of $\mathfrak{o}(n)$ lands in $SO(n)$, that of $\mathfrak{u}(n)$ lands in $U(n)$, and that of $\mathfrak{sp}(2n)$ lands in $Sp(2n)$. □

Example 10.5 ($\mathfrak{su}(2)$ and Three-Dimensional Rotation). The Lie algebra of the three-dimensional rotation group $SO(3)$ is $\mathfrak{so}(3)$, consisting of the 3×3 antisymmetric matrices:

$$\mathfrak{so}(3) = \operatorname{span}\left\{ \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right\}.$$

The three generators J_x, J_y, J_z satisfy the angular momentum commutation relations $[J_x, J_y] = J_z$, $[J_y, J_z] = J_x$, $[J_z, J_x] = J_y$. The exponential map $e^{\theta J_z}$ gives the rotation matrix through the angle θ about the z axis. The Lie algebra $\mathfrak{su}(2)$ is isomorphic to $\mathfrak{so}(3)$, and is the expression at the level of Lie algebras of the double cover $SU(2) \to SO(3)$.

10.8 Singular Value Decomposition and Periodic Convolution

Definition 10.29 (Singular Value Decomposition)**.** Let A be an $m \times n$ complex matrix. The **singular value decomposition** (SVD) of A is

$$\mathsf{A} = \mathsf{U}\boldsymbol{\Sigma}\mathsf{V}^*,$$

where U is an $m \times m$ unitary matrix, V is an $n \times n$ unitary matrix, and $\boldsymbol{\Sigma}$ is an $m \times n$ diagonal matrix whose diagonal entries $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$ are the **singular values** of A ($r = \mathrm{rank}(\mathsf{A})$).

The singular value decomposition is the matrix realisation of periodic convolution: the columns of U are the rotation axes of the output space, the columns of V are the rotation axes of the input space, and the diagonal entries of $\boldsymbol{\Sigma}$ are the scaling factors of the corresponding channels.

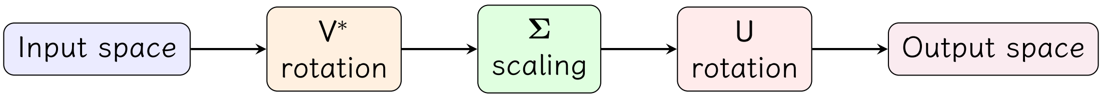


Figure 10.17: The geometric meaning of the singular value decomposition: rotation-scaling-rotation.

Proposition 10.14 (SVD and the Correspondence with the Four Operators of Periodic Convolution)**.** *The geometric meaning of the singular value decomposition corresponds precisely to the four-operator structure of periodic convolution:*

(i) **Phase operator** (rotation) $\leftrightarrow$ V^* and U*: the rotation axes of the input/output space;*

(ii) **Amplitude operator** (scaling) $\leftrightarrow$ $\boldsymbol{\Sigma}$*: each channel is scaled independently by the singular value* σ_i*;*

(iii) **Frequency operator** (translation) $\leftrightarrow$ *the DFT diagonalisation of a circulant matrix: when* A *is a circulant matrix,* $\mathsf{U} = \mathsf{F}_N^* \,\mathrm{diag}(e^{i\phi_k})$*,* $\mathsf{V} = \mathsf{F}_N^*$ *and* $\boldsymbol{\Sigma} = \mathrm{diag}(|\hat{a}_0|, \ldots, |\hat{a}_{N-1}|)$*, where* $\hat{a}_k = |\hat{a}_k| e^{i\phi_k}$*;*

(iv) **Period operator** (fixed length) $\leftrightarrow$ *block circulant structure: nested periodic systems correspond to block circulant matrices, whose SVD is carried out in layers according to the period class.*

Proof. (i)(ii) These are read off directly from the definition of the singular value decomposition (Definition 10.29): the action of $\mathsf{A} = \mathsf{U}\boldsymbol{\Sigma}\mathsf{V}^*$ first performs the rotation V^* in the input space, then scales each channel independently by the corresponding singular value σ_i, and finally performs the rotation U in the output space.

(iii) Let A be the circulant matrix with first row $\mathbf{a} = (a_0, a_1, \ldots, a_{N-1})$; its j-th column is obtained by cyclically shifting $\mathbf{a}$, so A commutes with the cyclic shift operator and is therefore unitarily diagonalised by the DFT matrix F_N:

$$\mathsf{A} = \mathsf{F}_N^* \,\mathrm{diag}(\hat{a}_0, \hat{a}_1, \ldots, \hat{a}_{N-1})\, \mathsf{F}_N, \qquad \hat{a}_k = \sum_{j=0}^{N-1} a_j e^{-2\pi i jk/N}.$$

Writing each $\hat{a}_k = |\hat{a}_k| e^{i\phi_k}$ in terms of modulus and phase gives

$$\mathsf{A} = \left(\mathsf{F}_N^* \operatorname{diag}(e^{i\phi_k})\right) \operatorname{diag}(|\hat{a}_0|, \dots, |\hat{a}_{N-1}|)\, \mathsf{F}_N,$$

which is precisely the SVD form of A.

(iv) Nested periodic systems correspond to block circulant matrices, whose DFT diagonalisation generalises to a block DFT, and the singular values are given in layers according to the period class. □

Proposition 10.15 (Singular Values and Conserved Quantities). *The singular values of a matrix* A *satisfy the following conservation properties:*

(i) **Conservation of the Frobenius norm**: $\|\mathsf{A}\|_F^2 = \sum_{i=1}^r \sigma_i^2$—*the 'total energy' of the matrix equals the sum of the squares of its singular values; this is the matrix version of Parseval's identity.*

(ii) **Conservation of rank**: $\operatorname{rank}(\mathsf{A}) = r$, *equal to the number of non-zero singular values.*

(iii) **Conservation of the determinant** (square case): $|\det(\mathsf{A})| = \prod_{i=1}^n \sigma_i$.

Proof. (i) From $\mathsf{A} = \mathsf{U}\mathbf{\Sigma}\mathsf{V}^*$ and unitary invariance, $\|\mathsf{A}\|_F^2 = \operatorname{tr}(\mathsf{A}^*\mathsf{A}) = \operatorname{tr}(\mathsf{V}\mathbf{\Sigma}^*\mathbf{\Sigma}\mathsf{V}^*) = \operatorname{tr}(\mathbf{\Sigma}^*\mathbf{\Sigma}) = \sum_{i=1}^r \sigma_i^2$.

(ii) Since U, V are invertible, $\operatorname{rank}(\mathsf{A}) = \operatorname{rank}(\mathbf{\Sigma})$, and the rank of $\mathbf{\Sigma}$ is precisely the number r of its non-zero diagonal entries.

(iii) In the square case, taking determinants gives $\det(\mathsf{A}) = \det(\mathsf{U})\det(\mathbf{\Sigma})\det(\mathsf{V}^*)$; since U, V are unitary matrices, $|\det(\mathsf{U})| = |\det(\mathsf{V}^*)| = 1$, hence $|\det(\mathsf{A})| = |\det(\mathbf{\Sigma})| = \prod_{i=1}^n \sigma_i$. □

Example 10.6 (Singular Value Decomposition of a 2×2 Matrix). *Consider the matrix* $\mathsf{A} = \begin{pmatrix} 0 & 3 \\ 4 & 0 \end{pmatrix}$. *A computation gives* $\mathsf{A}^\mathsf{T}\mathsf{A} = \begin{pmatrix} 16 & 0 \\ 0 & 9 \end{pmatrix}$, *whose eigenvalues are* $16, 9$, *so the singular values are* $\sigma_1 = 4 \geq \sigma_2 = 3$. *Take*

$$\mathsf{U} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \mathbf{\Sigma} = \begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix}, \qquad \mathsf{V} = \mathsf{I}_2,$$

so that $\mathsf{U}\mathbf{\Sigma}\mathsf{V}^\mathsf{T} = \mathsf{A}$. *Verifying the conserved quantities:* $|\det \mathsf{A}| = 12 = \sigma_1\sigma_2$ *and* $\|\mathsf{A}\|_F^2 = 25 = \sigma_1^2 + \sigma_2^2$.

Definition 10.30 (Pseudoinverse and Least Squares). *Let* $\mathsf{A} = \mathsf{U}\mathbf{\Sigma}\mathsf{V}^*$ *be an SVD. The* **Moore-Penrose pseudoinverse** *of* A *is* $\mathsf{A}^+ = \mathsf{V}\mathbf{\Sigma}^+\mathsf{U}^*$, *where* $\mathbf{\Sigma}^+$ *is the diagonal matrix with diagonal entries* $1/\sigma_i$ ($\sigma_i > 0$) *or* 0 ($\sigma_i = 0$). *The pseudoinverse gives the least squares solution of an inconsistent system of linear equations* $\mathsf{A}\mathsf{x} = \mathsf{b}$: $\mathsf{x} = \mathsf{A}^+\mathsf{b}$ *minimises* $\|\mathsf{A}\mathsf{x} - \mathsf{b}\|^2$.

10.9 The Periodic Structure of Linear Systems

Definition 10.31 (Linear Dynamical System). *A* **discrete linear dynamical system** *has the form* $\mathsf{x}_{k+1} = \mathsf{A}\mathsf{x}_k$. *A* **continuous linear dynamical system** *has the form* $\frac{d\mathsf{x}}{dt} = \mathsf{B}\mathsf{x}$. *These two systems are related through the matrix exponential: the solution of the continuous system is* $\mathsf{x}(t) = e^{\mathsf{B}t}\mathsf{x}(0)$, *which, after discretisation with time step* Δt, *corresponds to* $\mathsf{A} = e^{\mathsf{B}\Delta t}$.

Proposition 10.16 (Periodic Solutions of Linear Systems). *Let the eigenvalues of* A *be* $\lambda_1, \ldots, \lambda_n$. *The discrete system* $\mathsf{x}_{k+1} = \mathsf{A}\mathsf{x}_k$ *has a periodic solution of period* p *(that is,* $\mathsf{x}_{k+p} = \mathsf{x}_k$ *for all* k*) if and only if there exists an eigenvalue* λ_j *satisfying* $\lambda_j^p = 1$ *and the initial value lies in the direct sum* $\bigoplus_{\lambda_j^p=1} E_{\lambda_j}$ *of the eigenspaces of those eigenvalues. No diagonalisability of* A *is needed here: what is required is the direct-sum decomposition above, which holds for every* A.

Proof. ($\Leftarrow$) If x_0 lies in some eigenspace E_{λ_j} with $\lambda_j^p = 1$, then $\mathsf{x}_k = \mathsf{A}^k\mathsf{x}_0 = \lambda_j^k\mathsf{x}_0$, so $\mathsf{x}_p = \lambda_j^p\mathsf{x}_0 = \mathsf{x}_0$, that is, $\mathsf{x}_{k+p} = \mathsf{x}_k$ for all k; a general x_0 in the direct sum is handled component by component.

($\Rightarrow$) Suppose $\mathsf{x}_{k+p} = \mathsf{x}_k$ for all k. Then $\mathsf{x}_p = \mathsf{x}_0$, that is, $(\mathsf{A}^p - \mathsf{I})\mathsf{x}_0 = \mathsf{0}$. If $\mathsf{x}_0 \neq \mathsf{0}$, then 1 is an eigenvalue of A^p; the eigenvalues of A^p are the p-th powers of those of A, so some eigenvalue λ_j satisfies $\lambda_j^p = 1$, which is the first assertion. To see that x_0 lies in the direct sum, put

$$q(x) = \prod_{\lambda_j^p=1} (x - \lambda_j), \qquad h(x) = \prod_{\lambda_j^p\neq 1} (x - \lambda_j).$$

The roots of the two products are distinct and together make up all the roots of $x^p - 1$, so $\mathsf{A}^p - \mathsf{I} = q(\mathsf{A})h(\mathsf{A})$ with $\gcd(q, h) = 1$ (the polynomials q and h have no common root, every factor occurring to the first power). By the kernel decomposition theorem in the coprime case,

$$\ker\bigl(q(\mathsf{A})h(\mathsf{A})\bigr) = \ker q(\mathsf{A}) \oplus \ker h(\mathsf{A}),$$

so $\mathsf{x}_0 = \mathsf{x}_0^q + \mathsf{x}_0^h$. Since the roots of q are exactly the λ_j with $\lambda_j^p = 1$, the same kernel decomposition applied factor by factor gives $\ker q(\mathsf{A}) = \bigoplus_{\lambda_j^p=1} \ker(\mathsf{A} - \lambda_j \mathsf{I}) = \bigoplus_{\lambda_j^p=1} E_{\lambda_j}$. Hence $\mathsf{x}_0 \in \bigoplus_{\lambda_j^p=1} E_{\lambda_j}$ (for $\mathsf{x}_0 = \mathsf{0}$ this holds trivially as well). The whole argument uses only the kernel decomposition for coprime polynomials and does not assume that A is diagonalisable. ☐

Theorem 10.10 (Periodic Conservation Laws of Linear Systems). *Let* A *be the evolution matrix of a discrete linear system. The conserved quantities of the system are determined by the spectral properties of* A:

(i) **Conservation of the squared modulus**: *if* A *is a unitary matrix, then* $\|\mathsf{x}_k\|^2 = \|\mathsf{x}_0\|^2$ *for all* k *(the discrete realisation of the conservation of angular momentum in the curl (symmetry centre* i*)).*

(ii) **Conservation of volume**: *if* $|\det(\mathsf{A})| = 1$, *then phase-space volume is conserved (the discrete realisation of the conservation of scale charge in the divergence (symmetry centre* 1*)).*

(iii) **Conservation of momentum**: *if* A *has the eigenvalue* $\lambda = 1$, *then the components in the corresponding eigenspace remain unchanged under the evolution (the discrete realisation of the conservation of momentum in the gradient (symmetry centre* 0*)).*

Proof. (i) $\|\mathsf{x}_{k+1}\|^2 = \langle \mathsf{A}\mathsf{x}_k, \mathsf{A}\mathsf{x}_k\rangle = \langle \mathsf{x}_k, \mathsf{A}^*\mathsf{A}\mathsf{x}_k\rangle = \|\mathsf{x}_k\|^2$. The result follows by induction.

(ii) The transformation factor of phase-space volume is $|\det(\mathsf{A})|$.

(iii) If $\mathsf{A}\mathsf{v} = \mathsf{v}$, then for $\mathsf{x}_0 = c\mathsf{v}$ we have $\mathsf{x}_k = c\mathsf{v}$ for all k. ☐

Corollary 10.6 (Spectral Analysis and Periodic Decomposition of Linear Systems)**.** Let A be the evolution matrix of the discrete linear system $x_{k+1} = Ax_k$. The solution of any linear dynamical system can be decomposed into a superposition of the following four types of motion:

(i) **Exponential decay/growth** ($|\lambda_j| < 1$ or $|\lambda_j| > 1$): dominated by the divergence operator—phase-space volume contracts or expands.

(ii) **Periodic oscillation** ($|\lambda_j| = 1$, $\lambda_j \neq \pm 1$): dominated by the curl operator—pure rotation, with conservation of the modulus.

(iii) **Steady state** ($\lambda_j = 1$): dominated by the gradient operator—an equilibrium point, invariant under translation.

(iv) **Second-order oscillation** ($\lambda_j = -1$): the boundary between the gradient (symmetry centre 0) and the curl (symmetry centre i)—producing second-order periodicity (the C_2 involution).

Proof. By Corollary 10.1, decompose the initial state according to the eigenspaces as $x_0 = \sum_j c_j v_j$, so that $x_k = \sum_j c_j \lambda_j^k v_j$. Classifying by the modulus and the argument of each eigenvalue λ_j yields the conclusion. ☐

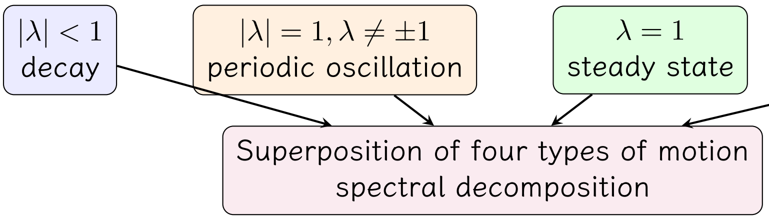


Figure 10.18: The four types of motion in the spectral decomposition of a linear system: decay, periodic oscillation, steady state and second-order oscillation.

10.10 Iterative Methods for Linear Systems

Definition 10.32 (Basic Framework of Iterative Methods)**.** For a large sparse system of linear equations $Ax = b$, split A into $A = M - N$ (with M invertible and easy to invert). The iteration scheme is

$$x_{k+1} = M^{-1}Nx_k + M^{-1}b.$$

The iteration matrix is $G = M^{-1}N$. The iteration converges if and only if $\rho(G) < 1$ (the spectral radius is less than 1).

An iterative method is an instance of a discrete dynamical system—the eigenvalues of the iteration matrix G determine the rate of convergence and the oscillation pattern.

Definition 10.33 (Jacobi Iteration)**.** Take $M = D$ (the diagonal part of A) and $N = -(L + U)$ (the strictly lower and upper triangular parts). The Jacobi iteration is fully parallel—the update of each component is independent of the current values of the remaining components.

Definition 10.34 (Gauss-Seidel Iteration). *Take* $\mathsf{M} = \mathsf{D} + \mathsf{L}$ *and* $\mathsf{N} = -\mathsf{U}$. *The Gauss-Seidel iteration uses the most recently computed component values—it is causal. It usually converges faster than the Jacobi iteration, but is difficult to parallelise.*

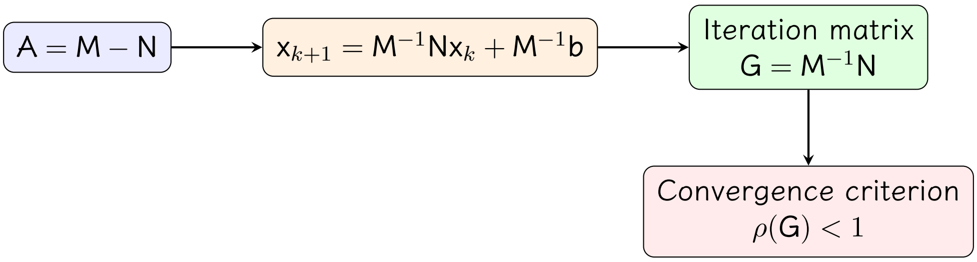

$\mathsf{A} = \mathsf{M} - \mathsf{N}$
$\mathsf{x}_{k+1} = \mathsf{M}^{-1}\mathsf{N}\mathsf{x}_k + \mathsf{M}^{-1}\mathsf{b}$
Iteration matrix
$\mathsf{G} = \mathsf{M}^{-1}\mathsf{N}$
Convergence criterion
$\rho(\mathsf{G}) < 1$

Figure 10.19: The framework of iterative methods: matrix splitting, the iteration matrix and the spectral radius convergence criterion.

Theorem 10.11 (Necessary and Sufficient Condition for Iterative Convergence). *The iteration* $\mathsf{x}_{k+1} = \mathsf{G}\mathsf{x}_k + \mathsf{c}$ *converges for every initial value if and only if the spectral radius of the iteration matrix* G *satisfies* $\rho(\mathsf{G}) = \max_i |\lambda_i(\mathsf{G})| < 1$. *If* A *is strictly diagonally dominant, then both the Jacobi iteration and the Gauss-Seidel iteration converge.*

Proof. The fixed point x_* of the iteration satisfies $\mathsf{x}_* = \mathsf{G}\mathsf{x}_* + \mathsf{c}$, and the error $e_k = \mathsf{x}_k - \mathsf{x}_*$ satisfies $e_k = \mathsf{G}^k e_0$.

We first prove the spectral radius condition. Let $\mathsf{G} = \mathsf{P}\mathsf{J}\mathsf{P}^{-1}$ be the Jordan decomposition. The entries of the k-th power of each Jordan block $J_r(\lambda)$ are $\binom{k}{j}\lambda^{k-j}$ $(0 \le j < r)$; when $|\lambda| < 1$ we have $\binom{k}{j}|\lambda|^{k-j} \to 0$ for every j. Hence when $\rho(\mathsf{G}) < 1$ we have $\mathsf{G}^k \to 0$, and the iteration converges. Conversely, if the iteration converges for every initial value, then $\mathsf{G}^k \to 0$; taking an eigenvalue λ of G with eigenvector v, from $\mathsf{G}^k\mathsf{v} = \lambda^k\mathsf{v} \to 0$ we obtain $|\lambda| < 1$, hence $\rho(\mathsf{G}) < 1$.

We then prove convergence in the strictly diagonally dominant case. The Jacobi iteration matrix $\mathsf{G}_J = -\mathsf{D}^{-1}(\mathsf{L}+\mathsf{U})$ satisfies $\|\mathsf{G}_J\|_\infty = \max_i \sum_{j\neq i} |a_{ij}/a_{ii}| < 1$, hence $\rho(\mathsf{G}_J) \le \|\mathsf{G}_J\|_\infty < 1$, and the Jacobi iteration converges. For the Gauss-Seidel iteration matrix $\mathsf{G}_{GS} = -(\mathsf{D}+\mathsf{L})^{-1}\mathsf{U}$, suppose that $\mathsf{G}_{GS}\mathsf{x} = \lambda\mathsf{x}$ with $|\lambda| \ge 1$ and take $|x_m| = \max_i |x_i|$; the equation for the m-th component gives $|a_{mm}||x_m| \le \sum_{j\neq m} |a_{mj}||x_m|$, which contradicts strict diagonal dominance. Hence $\rho(\mathsf{G}_{GS}) < 1$, and the Gauss-Seidel iteration converges. □

Remark 10.6 (Comparison Between the Lo Shu Matrix and Iterative Methods). The spectral radius of the Lo Shu matrix L is $\rho(\mathsf{L}) = 2\sqrt{6} \approx 4.899 > 1$, so L does not converge as an iteration matrix. This shows, from the spectral structure, that the modulus of the rotation-and-scaling eigenvalues $\pm 2\sqrt{6}i$ of the Lo Shu matrix on $\mathbf{1}^\perp$ is greater than 1, producing divergent oscillation—this is a non-convergent behaviour dominated by the curl (symmetry centre i).

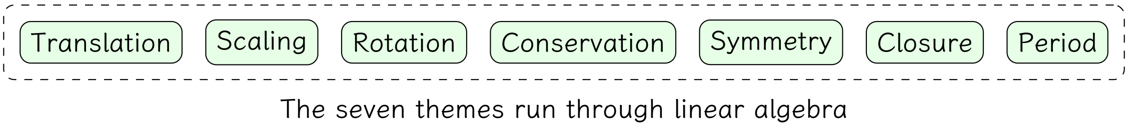


Figure 10.20: The seven themes of this chapter: translation, scaling, rotation, conservation, symmetry, closure and period.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	translation matrices and cyclic shifts: stretching/compression of real eigenvalues (Definition 10.11); the DFT diagonalisation of circulant matrices (Proposition 10.14); the pure translation part of the Lo Shu matrix (Theorem 10.8)
Scaling	diagonal scaling and singular values: diagonal matrices (Definition 10.13); the Σ of the SVD (Definition 10.29); volume scaling measured by the determinant (Definition 10.10)
Rotation	orthogonal and unitary matrices: eigenvalues of modulus 1 (Theorem 10.5); plane rotations and finite-order periodicity (Example 10.2); the rotation-and-scaling part of the Lo Shu matrix (Theorem 10.8)
Conservation	the Frobenius norm, the determinant and the rank: conservation of energy $\sum_i \sigma_i^2$ (Proposition 10.15); conservation of dimension (Theorem 10.1)
Symmetry	the matrix groups $O(n)/U(n)/Sp(2n)$: the matrix realisations of 0-symmetry, i-symmetry and 1-symmetry (Proposition 10.5); the antipodal antisymmetry of the Lo Shu matrix (Proposition 10.8)
Closure	finite-order periodic solutions satisfying $\lambda^p = 1$: eigenvalues lying on roots of unity and the cyclic structure of C_p (Proposition 10.16); the rotation-and-scaling part of the Lo Shu matrix acts as a joint rotation and scaling (Corollary 10.5)
Period	the periodic structure of linear systems: the four types of motion in the spectral decomposition (Corollary 10.6); iterative convergence and steady states (Theorem 10.11)

Exercises

Exercise 1 Prove the dimension theorem (Theorem 10.1): let V be a finite-dimensional vector space and let W_1, W_2 be its subspaces; then

$$\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2).$$

By constructing a basis of $W_1 \cap W_2$ and extending it respectively to bases of W_1 and W_2, prove that the union of the extended bases is a basis of $W_1 + W_2$. Further explain that the dimension theorem expresses the **conservation** of information in a subspace decomposition.

Exercise 2 According to Definition 10.11, explain how an eigenvalue λ describes the law of evolution on a one-dimensional invariant subspace:

(i) $\lambda \in \mathbb{R}$ with $\lambda > 0$ corresponds to pure translation (stretching or compression);
(ii) $|\lambda| = 1$ with $\lambda = e^{i\theta}$ corresponds to pure rotation;
(iii) $|\lambda| \neq 1$ corresponds to the combined action of rotation and scaling;
(iv) $\lambda < 0$ involves a reflection, producing a second-order periodic oscillation.

Give an example of a 2×2 matrix for each case, and compute its phase portrait or orbit shape.

Exercise 3 Prove Corollary 10.1: if the eigenvectors of a linear operator T form a basis of V, then the evolution of any initial state $\mathsf{x}_0 = \sum_i c_i \mathsf{v}_i$ decomposes into independent evolutions on the eigenspaces:

$$\mathsf{x}_k = T^k \mathsf{x}_0 = \sum_i c_i \lambda_i^k \mathsf{v}_i.$$

Explain how each eigenvalue λ_i acts as the evolution factor of the i-th 'frequency component'.

Exercise 4 Verify the eigendecomposition of the symmetric matrix

$$\mathsf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

in Example 10.1:

(i) find the characteristic polynomial and the eigenvalues;
(ii) find the corresponding unit eigenvectors and verify their orthogonality;
(iii) write down the spectral decomposition $\mathsf{A} = \mathsf{QDQ}^\mathsf{T}$;
(iv) explain that the two eigendirections correspond to pure translation evolution with stretching factors 4 and 2 respectively.

Exercise 5 Prove the spectral theorem for self-adjoint operators (Theorem 10.4): all eigenvalues of a self-adjoint operator are real numbers, eigenvectors belonging to distinct eigenvalues are orthogonal, and there exists an orthonormal basis consisting of eigenvectors. Carry out the existence proof by induction, and explain the correspondence in Corollary 10.2 between 'the self-adjoint spectral theorem and the gradient conservation law'.

Exercise 6 Verify the eigenvalues and the finite-order period of the rotation matrix

$$\mathsf{R} = \begin{pmatrix} \cos\frac{\pi}{3} & -\sin\frac{\pi}{3} \\ \sin\frac{\pi}{3} & \cos\frac{\pi}{3} \end{pmatrix}$$

in Example 10.2:

(i) find the characteristic polynomial and the eigenvalues;

(ii) verify that the moduli of the eigenvalues are all 1;

(iii) prove that $\mathsf{R}^6 = \mathsf{I}$, and explain why the minimal period is 6.

Exercise 7 Prove the spectral theorem for unitary operators (Theorem 10.5): all eigenvalues of a unitary operator have modulus 1, eigenvectors belonging to distinct eigenvalues are orthogonal, and there exists an orthonormal basis consisting of eigenvectors. Explain the correspondence in Corollary 10.3 between 'the unitary spectral theorem and the curl conservation law'.

Exercise 8 Prove the spectral theorem for normal operators (Theorem 10.6): a normal operator on a complex inner product space is unitarily diagonalisable. Explain how this theorem unifies the spectral theorems for self-adjoint operators (real eigenvalues, pure translation) and unitary operators (unit-circle eigenvalues, pure rotation).

Exercise 9 Verify the orthogonal projection theorem (Proposition 10.4) and Parseval's identity (Proposition 10.3): let $W \subseteq V$ be a finite-dimensional subspace and let $\mathsf{v} \in V$; then there is a unique decomposition $\mathsf{v} = \mathsf{w} + \mathsf{w}^{\perp}$, and $\|\mathsf{v}\|^2 = \|\mathsf{w}\|^2 + \|\mathsf{w}^{\perp}\|^2$. Write down the explicit expression for the projection w in terms of an orthonormal basis.

Exercise 10 According to Proposition 10.5, explain that the orthogonal group $O(n)$, the unitary group $U(n)$ and the symplectic group $Sp(2n)$ preserve the real inner product, the complex inner product and the symplectic form respectively, and correspond respectively to the gradient, the curl and the divergence. Give a matrix example for each, and verify the preservation property of the group to which it belongs.

Exercise 11 Verify the magic square property of the Lo Shu matrix (Proposition 10.6): all the row sums, column sums, the main diagonal sum and the anti-diagonal sum of

$$\mathsf{L} = \begin{pmatrix} 4 & 9 & 2 \\ 3 & 5 & 7 \\ 8 & 1 & 6 \end{pmatrix}$$

are 15. Further prove the antipodal balancing (Proposition 10.8): the sums of the entries at antipodal positions about the centre 5 are always 10.

Exercise 12 Prove the spectral structure of the Lo Shu matrix (Theorem 10.7):

$$p(\lambda) = \det(\mathsf{L} - \lambda \mathsf{I}) = -(\lambda - 15)(\lambda^2 + 24).$$

Find the three eigenvalues $\lambda_1 = 15$ and $\lambda_{2,3} = \pm 2\sqrt{6}i$, and use the translation-rotation decomposition (Theorem 10.8) to explain that L is a pure translation on span(1) and acts on $\mathbf{1}^{\perp}$ as a joint rotation and scaling: rotation through $\pi/2$ and scaling of the modulus by $2\sqrt{6}$ at each step (since $|\lambda_{2,3}| = 2\sqrt{6} \neq 1$, the orbit in that two-dimensional real invariant subspace does not close under iteration by L).

Exercise 13 Prove the geometric meaning of the singular value decomposition (Definition 10.29): $\mathsf{A} = \mathsf{U}\Sigma\mathsf{V}^*$ corresponds to 'rotation-scaling-rotation'. Also verify the conservation of singular values (Proposition 10.15):

$$\|\mathsf{A}\|_F^2 = \sum_{i=1}^{r} \sigma_i^2, \qquad |\det(\mathsf{A})| = \prod_{i=1}^{n} \sigma_i.$$

Compute the singular value decomposition of the matrix $\mathsf{A} = \begin{pmatrix} 0 & 3 \\ 4 & 0 \end{pmatrix}$ in Example 10.6.

Exercise 14 Prove the existence criterion for periodic solutions of a linear system (Proposition 10.16): the discrete system $\mathsf{x}_{k+1} = \mathsf{A}\mathsf{x}_k$ has a periodic solution of period p if and only if there exists an eigenvalue λ_j satisfying $\lambda_j^p = 1$ and the initial value lies in the corresponding eigenspace. Further prove the periodic conservation laws of linear systems (Theorem 10.10): a unitary matrix preserves the squared modulus, $|\det \mathsf{A}| = 1$ preserves volume, and the eigenvalue $\lambda = 1$ corresponds to conservation of momentum.

Exercise 15 According to the framework of iterative methods (Definition 10.32) and the convergence theorem (Theorem 10.11), explain that the iteration $\mathsf{x}_{k+1} = \mathsf{G}\mathsf{x}_k + \mathsf{c}$ converges for every initial value if and only if the spectral radius $\rho(\mathsf{G}) < 1$. Compute the spectral radius of the Lo Shu matrix L, and explain why it does not converge as an iteration matrix (see Note 10.6).

Exercise 16 Comprehensive exercise: combining the spectral theorems (Theorems 10.4, 10.5), the correspondence between matrix groups and the three operators (Proposition 10.5) and the translation-rotation decomposition of the Lo Shu matrix (Theorem 10.8), discuss how 'spectral decomposition' in linear algebra embodies the three great symmetry centres of periodic mathematics— the additive symmetry centre 0, the multiplicative symmetry centre 1 and the rotational symmetry centre i. Point out respectively which eigenvalues correspond to pure translation, which correspond to pure rotation, and which correspond to the combined action of scaling and rotation.

Chapter 11

Manifold and Fibre Bundle

Remark 11.1 (Positioning of This Chapter). *This chapter establishes the language of manifolds for the geometric analysis of periodic orbits. The core object of periodic mathematics is a closed curve in a state space; to determine rigorously whether an orbit is closed, to describe the geometric structure of periodic evolution, and to characterise the constraints that conservation laws impose on orbits, the language of manifolds is required.*

Manifolds serve three core functions in this chapter:

(1) **Criterion for the closure of an orbit**: *a closed orbit is a closed curve on a manifold, and its closure is characterised by an element of finite order in the fundamental group (Definition 11.33); the torus, as the quotient manifold $\mathbb{R}^d/\Lambda$, provides a global classification of closed orbits (Proposition 11.15).*

(2) **Geometric stage of periodic evolution**: *periodic evolution takes place on the state-space manifold; its instantaneous direction is described by the tangent space (Definition 11.4), the constraints of conservation laws are described by the normal space (Definition 11.16), and the parametrisation of phase is described by the angle space (Definition 11.18).*

(3) **Geometric realisation of conservation laws**: *the level set of a conserved quantity is a submanifold (Theorem 11.1), orbits are constrained to lie on invariant submanifolds (Definition 11.24), and the orbits of an integrable system lie on invariant tori (Definition 11.26).*

Guide to the seven themes of this chapter: *Translation — the tangent vector as the infinitesimal generator of translation (Definition 11.4), the orbits of a group action (Definition 11.30), the covering map $\mathbb{R} \to S^1$ (Theorem 11.3); Scaling — the scaling of the Riemannian metric (Definition 11.10), the orthogonal complement of the normal space (Definition 11.16), the parametrisation of the moduli space (Definition 11.29); Rotation — the $U(1)$ structure of the angle space (Definition 11.18), the rotation of the invariant torus (Definition 11.26), the spiralling of the stable/unstable manifolds (Definition 11.27); Conservation — the level set of a conserved quantity as a submanifold (Theorem 11.1), the energy surface (Definition 11.25), the constraints of the normal bundle (Proposition 11.7); Symmetry — group actions and quotient manifolds (Definition 11.31); Closure — closed orbits and the fundamental group (Definition 11.33), the classification of the fundamental group of the torus (Proposition 11.15); Period — the torus as the quotient manifold $\mathbb{R}^d/\Lambda$ (Proposition 11.14), periodic and quasi-periodic motion on an invariant torus (Proposition 11.12).*

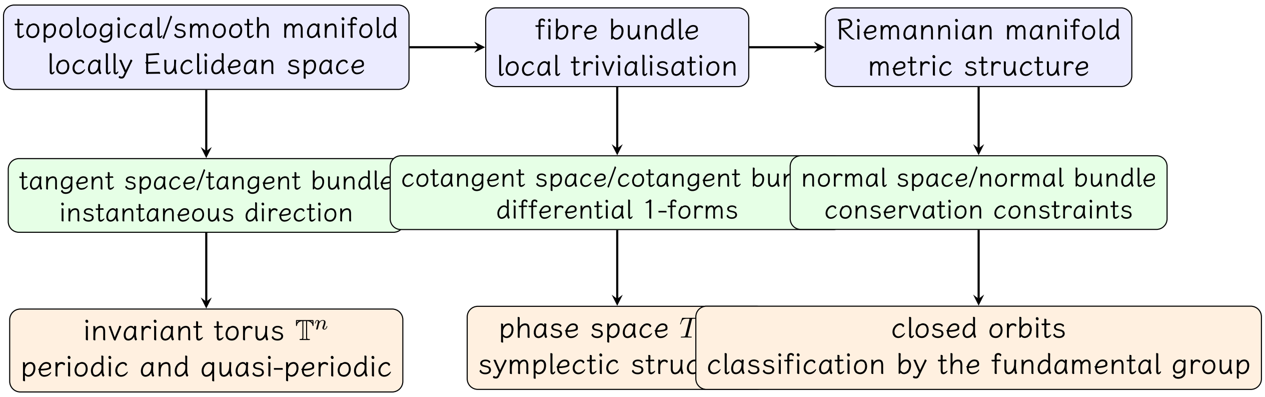

topological/smooth manifold
locally Euclidean space
fibre bundle
local trivialisation
Riemannian manifold
metric structure
tangent space/tangent bundle
instantaneous direction
cotangent space/cotangent bur
differential 1-forms
normal space/normal bundle
conservation constraints
invariant torus $\mathbb{T}^n$
periodic and quasi-periodic
phase space T
symplectic struc
closed orbits
classification by the fundamental group

Figure 11.1: Overview of the structure of this chapter: starting from topological/smooth manifolds, passing through fibre bundles and Riemannian structure, and leading to the tangent, cotangent and normal bundles, which finally converge on the three periodic objects: the invariant torus, phase space and closed orbits.

11.1 Topological and Smooth Manifolds

Definition 11.1 (Topological Manifold)**.** Let M be a topological space. M is called an n-dimensional **topological manifold** if it satisfies:

(i) **Hausdorff property**: for any $p \neq q \in M$ there exist open neighbourhoods $U \ni p$ and $V \ni q$ with $U \cap V = \emptyset$;

(ii) **Second countability**: M has a countable topological basis;

(iii) **Local Euclidean property**: for any $p \in M$ there exist an open neighbourhood $U \ni p$ and a homeomorphism $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^n$.

(U, φ) is called a **chart** of M near p, and $\varphi(p) = (x^1(p), \ldots, x^n(p))$ are called the **local coordinates** of p.

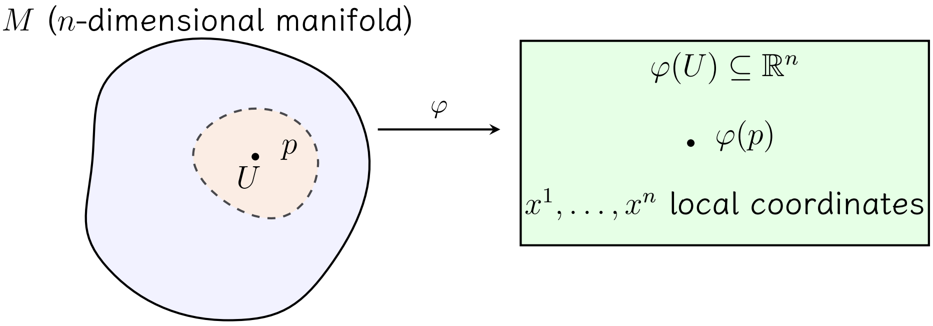

M (n-dimensional manifold)
p
U
φ
$\varphi(U) \subseteq \mathbb{R}^n$
$\varphi(p)$
$x^1, \ldots, x^n$ local coordinates

Figure 11.2: The local Euclidean property of a topological manifold: the neighbourhood U of p is homeomorphic, via the chart φ, to an open set in $\mathbb{R}^n$.

Example 11.1 (Basic Examples of Manifolds)**.** (i) $\mathbb{R}^n$ itself is an n-dimensional topological manifold (take the identity map as a global chart).

(ii) The circle $S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ is a 1-dimensional topological manifold. One may take two charts: $U_1 = S^1 \setminus \{(1, 0)\}$, $\varphi_1(\cos\theta, \sin\theta) = \theta \in (0, 2\pi)$; $U_2 = S^1 \setminus \{(-1, 0)\}$, $\varphi_2(\cos\theta, \sin\theta) = \theta \in (-\pi, \pi)$.

(iii) The n-dimensional torus $\mathbb{T}^n = S^1 \times \cdots \times S^1$ is an n-dimensional topological manifold.

(iv) The n-dimensional sphere $S^n = \{\mathsf{x} \in \mathbb{R}^{n+1} : \|\mathsf{x}\| = 1\}$ is an n-dimensional topological manifold.

Definition 11.2 (Smooth Manifold)**.** Let M be a topological manifold. A **smooth atlas** on M is a family of charts

$$\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$$

satisfying:

(i) $\{U_\alpha\}_{\alpha \in A}$ covers M;

(ii) for any α, β, if $U_\alpha \cap U_\beta \neq \emptyset$, then the transition map

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$

is a C^∞ smooth map.

A topological manifold equipped with a maximal smooth atlas is called a **smooth manifold**.

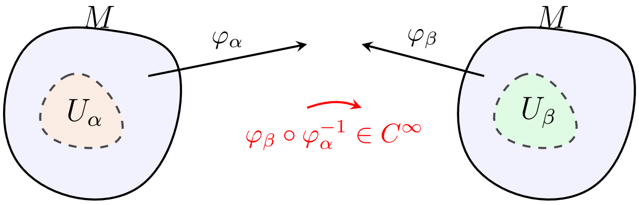


Figure 11.3: Transition maps of a smooth manifold: on the overlap of two charts the transition map $\varphi_\beta \circ \varphi_\alpha^{-1}$ is smoothly compatible.

Definition 11.3 (Smooth Functions and Smooth Maps)**.** Let M be a smooth manifold.

(i) A function $f : M \to \mathbb{R}$ is called **smooth** if for every chart (U, φ) the map $f \circ \varphi^{-1}$ is C^∞. The set of all smooth functions on M is denoted $C^\infty(M)$.

(ii) A map $F : M \to N$ is called **smooth** if for every $p \in M$ the map $\psi \circ F \circ \varphi^{-1}$ is C^∞.

(iii) If F is a bijection and both F and F^{-1} are smooth, then F is called a **diffeomorphism**.

11.2 Tangent Space and Tangent Bundle

Definition 11.4 (Tangent Vector—Derivation Definition)**.** Let M be a smooth manifold and $p \in M$. A **tangent vector** at p is a linear map $v : C^\infty(M) \to \mathbb{R}$ satisfying the Leibniz rule:

$$v(fg) = v(f)g(p) + f(p)v(g), \qquad \forall f, g \in C^\infty(M).$$

The set of all tangent vectors at p is denoted T_pM and is called the tangent space.

Proposition 11.1 (Dimension and Coordinate Basis of the Tangent Space)**.** Let M be an n-dimensional smooth manifold, $p \in M$, and let $(U, \varphi = (x^1, \dots, x^n))$ be a chart. Then T_pM has dimension n, and a basis is given by $\frac{\partial}{\partial x^1}\big|_p, \dots, \frac{\partial}{\partial x^n}\big|_p$.

Proof. **Step 1 (verifying that $\frac{\partial}{\partial x^i}\big|_p$ is a tangent vector).** Linearity is given by the linearity of partial derivatives; the Leibniz rule is given by the product rule for partial derivatives.

Step 2 (linear independence). Suppose that $\sum_i a^i \frac{\partial}{\partial x^i}\big|_p = 0$. Taking the coordinate function x^j, we obtain $0 = \sum_i a^i \frac{\partial x^j}{\partial x^i} = a^j$.

Step 3 (spanning). For any v and f, expand $f \circ \varphi^{-1}$ to first order by Taylor's theorem at p; applying v then gives $v(f) = \sum_i v(x^i) \frac{\partial f}{\partial x^i}(p)$. Hence $\{\partial_i|_p\}$ spans T_pM. □

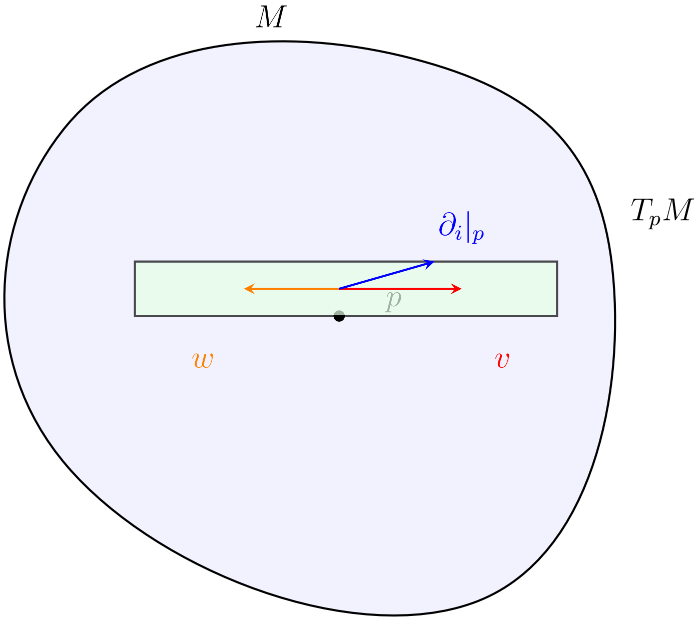


Figure 11.4: The tangent space T_pM is the linearisation of M at p: a tangent vector is an infinitesimal displacement tangential to the manifold.

Definition 11.5 (Tangent Vector—Curve Equivalence Class Definition)**.** Let M be a smooth manifold and $p \in M$. Let $\mathcal{C}_p$ denote the set of smooth curves through p. Define an equivalence relation on $\mathcal{C}_p$ by $\gamma_1 \sim \gamma_2$ if and only if $\frac{d}{dt}(\varphi \circ \gamma_1)(0) = \frac{d}{dt}(\varphi \circ \gamma_2)(0)$. The equivalence class $[\gamma]$ is called a tangent vector.

Lemma 11.1 (Equivalence of the Two Definitions of Tangent Vectors). *The derivation definition (Definition 11.4) and the curve equivalence class definition (Definition 11.5) give the same tangent space. The correspondence is:*

$$[\gamma] \mapsto v_{[\gamma]}, \qquad v_{[\gamma]}(f) = \left.\frac{d}{dt}\right|_{t=0} f(\gamma(t)).$$

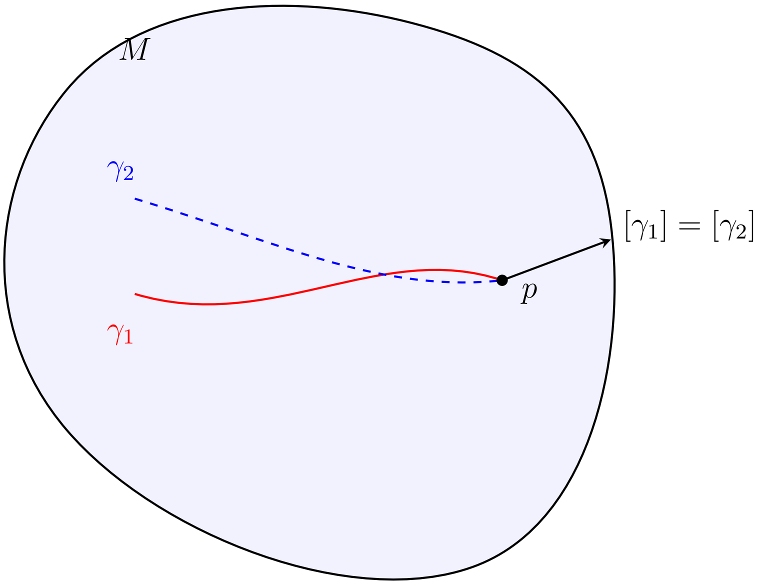


Figure 11.5: The two definitions of a tangent vector: curves through the same point with the same velocity at that point belong to the same equivalence class and correspond to the same tangent vector.

Proof. **Step 1 (well-definedness).** If $\gamma_1 \sim \gamma_2$, then for any f,

$$\left.\frac{d}{dt}\right|_{t=0} f(\gamma_1(t)) = df_p\left(\frac{d}{dt}(\varphi \circ \gamma_1)(0)\right) = df_p\left(\frac{d}{dt}(\varphi \circ \gamma_2)(0)\right) = \left.\frac{d}{dt}\right|_{t=0} f(\gamma_2(t)).$$

Step 2 ($v_{[\gamma]}$ is a derivation). Linearity is given by the linearity of the derivative. The Leibniz rule is given by the product rule.

Step 3 (surjectivity). Given a derivation $v = \sum_i a^i \frac{\partial}{\partial x^i}$, take $\gamma(t) = \varphi^{-1}(\varphi(p) + t(a^1, \ldots, a^n))$; then $v_{[\gamma]} = v$.

Step 4 (injectivity). If $v_{[\gamma_1]} = v_{[\gamma_2]}$, then for the coordinate functions x^i, $\frac{d}{dt}(\varphi \circ \gamma_1)^i(0) = \frac{d}{dt}(\varphi \circ \gamma_2)^i(0)$, hence $\gamma_1 \sim \gamma_2$. ☐

Definition 11.6 (Tangent Bundle). The **tangent bundle** of M is defined as $TM := \bigsqcup_{p \in M} T_pM$, equipped with the projection $\pi : TM \to M$. A **vector field** on M is a smooth section of the tangent bundle.

Proposition 11.2 (Manifold Structure of the Tangent Bundle). *The tangent bundle TM is a $2n$-dimensional smooth manifold.*

Proof. Let (U, φ) be a chart of M. Define the corresponding chart $(\pi^{-1}(U), \tilde{\varphi})$ on TM, where $\tilde{\varphi}(p, v) = (x^1(p), \ldots, x^n(p), v^1, \ldots, v^n)$. The smoothness of the transition map follows from

$$\tilde{\varphi}_\beta \circ \tilde{\varphi}_\alpha^{-1}(x, v) = \big(y(x), \sum_i \tfrac{\partial y}{\partial x^i} v^i\big)$$

where $y = \varphi_\beta \circ \varphi_\alpha^{-1} \in C^\infty$, whose partial derivatives are also C^∞. □

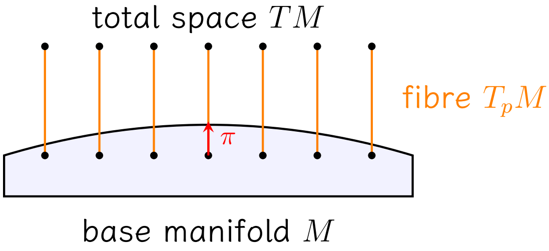


Figure 11.6: The tangent bundle TM: with the base manifold M as base, the tangent space T_pM hangs above each point p as the fibre, and the projection π maps (p, v) back to p.

11.3 Cotangent Space and Cotangent Bundle

Definition 11.7 (Cotangent Space and Differential). *$T_p^*M := (T_pM)^*$ is called the* **cotangent space***. For $f \in C^\infty(M)$, $df_p(v) := v(f)$ is called the* **differential** *of f at p.*

Proposition 11.3 (Coordinate Basis of the Cotangent Space). *Let $(U, \varphi = (x^1, \ldots, x^n))$ be a chart. Then a basis of T_p^*M is $dx^1|_p, \ldots, dx^n|_p$, satisfying $dx^i|_p(\frac{\partial}{\partial x^j}|_p) = \delta^i_j$.*

Proof. By Proposition 11.1, $\{\frac{\partial}{\partial x^i}\}$ is a basis of T_pM. By the dual basis theorem of linear algebra, there exists a unique dual basis $\{\omega^i\}$ satisfying $\omega^i(\frac{\partial}{\partial x^j}) = \delta^i_j$. Let us verify that $\omega^i = dx^i$: for any $v = \sum_j v^j \frac{\partial}{\partial x^j}$,

$$dx^i(v) = v(x^i) = \sum_j v^j \frac{\partial x^i}{\partial x^j} = v^i = \omega^i(v).$$

Hence dx^i is precisely the dual basis. □

Definition 11.8 (Cotangent Bundle). *The* **cotangent bundle** *of M is $T^*M := \bigsqcup_{p \in M} T_p^*M$. A* **differential 1-form** *on M is a smooth section of the cotangent bundle.*

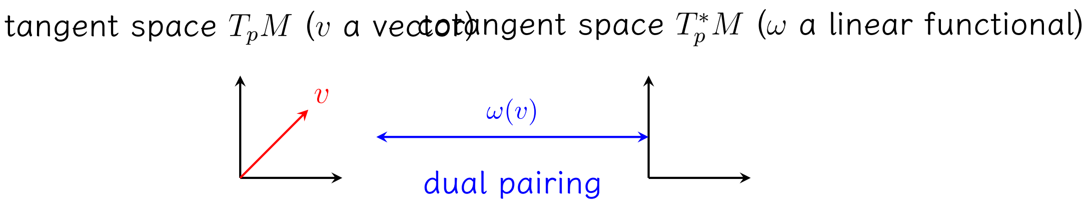

tangent space T_pM (v a vector)
cotangent space T_p^*M (ω a linear functional)
v
$\omega(v)$
dual pairing

Figure 11.7: Dual pairing of the tangent space and the cotangent space: $\omega \in T_p^*M$ maps $v \in T_pM$ linearly to a real number, $dx^i(\partial_j) = \delta^i_j$.

11.4 Submanifolds and Level Sets

Definition 11.9 (Embedded Submanifold). A subset $S \subseteq M$ is called a k-dimensional **embedded submanifold** if for every $p \in S$ there exists a chart $(U, \varphi = (x^1, \dots, x^n))$ such that $S \cap U = \{q \in U : x^{k+1}(q) = \cdots = x^n(q) = 0\}$.

Theorem 11.1 (Regular Level Set Theorem). Let $F : M \to \mathbb{R}^k$ be a smooth map and $c \in \mathbb{R}^k$. If dF_p is surjective for every $p \in F^{-1}(c)$, then $F^{-1}(c)$ is an $(n-k)$-dimensional embedded submanifold.

Proof. By the rank theorem, coordinates may be chosen near p so that $F = (x^{n-k+1}, \dots, x^n)$. Then $F^{-1}(c)$ is defined by $x^{n-k+1} = c_1, \dots, x^n = c_k$. ☐

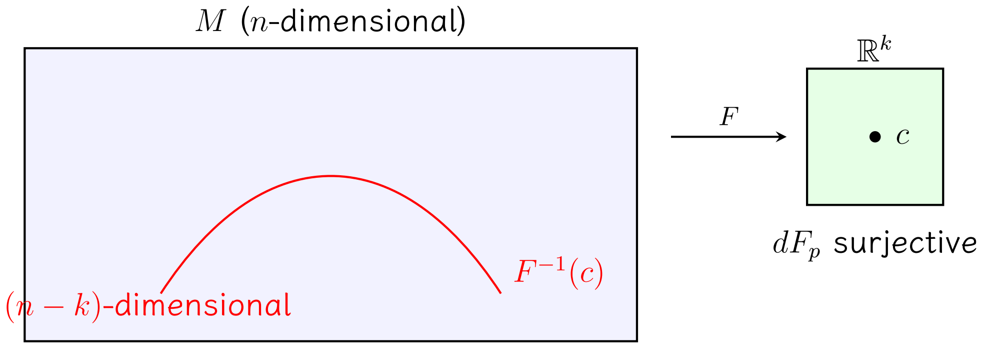

M (n-dimensional)
F
$\mathbb{R}^k$
c
dF_p surjective
$F^{-1}(c)$
$(n-k)$-dimensional

Figure 11.8: The regular level set theorem: if dF_p is surjective everywhere, then $F^{-1}(c)$ is an embedded submanifold of codimension k.

Corollary 11.1 (Level Sets of Conserved Quantities). Let $Q_1, \dots, Q_k$ be independent conserved quantities. Then $\mathcal{M}_{c_1,\dots,c_k} := \{Q_1 = c_1, \dots, Q_k = c_k\}$ is an $(n-k)$-dimensional embedded submanifold.

Proof. Take $F = (Q_1, \dots, Q_k)$; the result follows directly from Theorem 11.1. ☐

11.5 Riemannian Manifolds

Definition 11.10 (Riemannian Metric). Let M be a smooth manifold. A **Riemannian metric** g on M assigns to each $p \in M$ a positive-definite inner product $g_p : T_pM \times T_pM \to \mathbb{R}$ on the tangent space T_pM, and g depends smoothly on p. A manifold (M, g) equipped with a Riemannian metric is called a **Riemannian manifold**.

Property 11.1 (Basic Properties of the Riemannian Metric). Let (M, g) be a Riemannian manifold, $p \in M$, $u, v, w \in T_pM$ and $a, b \in \mathbb{R}$. Then:

(i) **Symmetry**: $g_p(u, v) = g_p(v, u)$;

(ii) **Bilinearity**: $g_p(au + bv, w) = ag_p(u, w) + bg_p(v, w)$;

(iii) **Positive definiteness**: $g_p(v, v) \geq 0$, and $g_p(v, v) = 0 \iff v = 0$;

(iv) **Coordinate representation**: in local coordinates $(x^1, \ldots, x^n)$, $g_p(u, v) = \sum_{i,j} g_{ij}(p)u^i v^j$, where $g_{ij}(p) = g_p(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j})$.

Proof. (i)–(iii) follow directly from the definition of an inner product. (iv) is obtained by expanding $u = \sum_i u^i \frac{\partial}{\partial x^i}$ and $v = \sum_j v^j \frac{\partial}{\partial x^j}$ using bilinearity. $\square$

Proposition 11.4 (Norm and Distance Induced by the Riemannian Metric). The Riemannian metric g induces a norm $\|v\|_p := \sqrt{g_p(v, v)}$ on each tangent space. The length of a curve $\gamma : [a, b] \to M$ is

$$L(\gamma) = \int_a^b \|\dot{\gamma}(t)\|_{\gamma(t)}\, dt.$$

The **Riemannian distance** between two points p, q of M is

$$d(p, q) = \inf\{L(\gamma) : \gamma \text{ is a piecewise smooth curve joining } p, q\}.$$

Proof. **Norm axioms**: (i) Positive definiteness: $\|v\|_p \geq 0$ follows from $g_p(v, v) \geq 0$; $\|v\|_p = 0 \iff g_p(v, v) = 0 \iff v = 0$. (ii) Homogeneity: $\|av\|_p = \sqrt{g_p(av, av)} = \sqrt{a^2 g_p(v, v)} = |a|\|v\|_p$. (iii) Triangle inequality: by the Cauchy–Schwarz inequality $|g_p(u, v)| \leq \|u\|_p\|v\|_p$,

$$\|u + v\|_p^2 = \|u\|_p^2 + 2g_p(u, v) + \|v\|_p^2 \leq (\|u\|_p + \|v\|_p)^2.$$

Riemannian distance: (i) Non-negativity: arc length is non-negative, hence $d(p, q) \geq 0$. (ii) Symmetry: reversing the curve gives $d(p, q) = d(q, p)$. (iii) Triangle inequality: concatenating a curve from p to q with a curve from q to r gives $d(p, r) \leq d(p, q) + d(q, r)$. (iv) Positive definiteness: $d(p, p) = 0$ (take a constant curve). If $p \neq q$, then in local coordinates g_{ij} is positive definite, so there exists a constant $c > 0$ with $\|v\| \geq c|v|_{\text{Euclidean}}$; hence every curve joining p, q has length at least $c \cdot |\varphi(p) - \varphi(q)| > 0$, and $d(p, q) > 0$. $\square$

Definition 11.11 (Gradient Under the Riemannian Metric). Let (M, g) be a Riemannian manifold and $f \in C^\infty(M)$. The **gradient** ∇f of f is the vector field on M defined by:

$$g_p(\nabla f(p), v) = df_p(v), \qquad \forall v \in T_pM.$$

Proposition 11.5 (Orthogonality of the Gradient and the Level Set). Let (M, g) be a Riemannian manifold, $f \in C^\infty(M)$ and $c \in \mathbb{R}$. If $p \in f^{-1}(c)$ and $\nabla f(p) \neq 0$, then $\nabla f(p)$ is orthogonal to the tangent space of the level set $f^{-1}(c)$ at p.

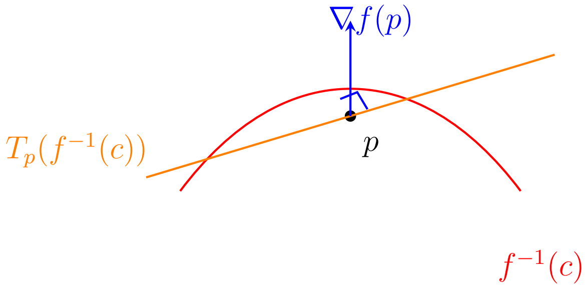


Figure 11.9: The gradient is orthogonal to the level set: $\nabla f(p)$ is perpendicular to the tangent space of the level set at p.

Proof. Let $v \in T_p(f^{-1}(c))$. Take a curve $\gamma : (-\varepsilon, \varepsilon) \to f^{-1}(c)$ with $\gamma(0) = p$ and $\dot\gamma(0) = v$. Then $f(\gamma(t)) = c$ for all t, hence

$$0 = \frac{d}{dt}\bigg|_{t=0} f(\gamma(t)) = df_p(v) = g_p(\nabla f(p), v).$$

Therefore $\nabla f(p)$ is orthogonal to $T_p(f^{-1}(c))$. □

Remark 11.2 (The Role of the Riemannian Metric). The Riemannian metric serves two functions in this chapter. First, it is the basis for the definition of the **normal space** (Definition 11.16) — the normal space is defined by means of the orthogonal complement. Second, it is the basis for the definition of the **energy norm** $\|\dot\gamma\|^2$ of a curve, and thereby supplies the metric structure for the variational principle (the geodesic equation) and for symplectic geometry (the phase-space volume form).

11.6 Fibre Bundles

Definition 11.12 (Fibre Bundle). Let E, B, F be smooth manifolds and $\pi : E \to B$ a smooth surjection. The quadruple (E, B, π, F) is called a **fibre bundle** if it satisfies the **local trivialisation condition**: for every $b \in B$ there exist an open neighbourhood $U \ni b$ and a diffeomorphism $\Phi : \pi^{-1}(U) \to U \times F$ such that $\pi = \mathrm{pr}_1 \circ \Phi$.

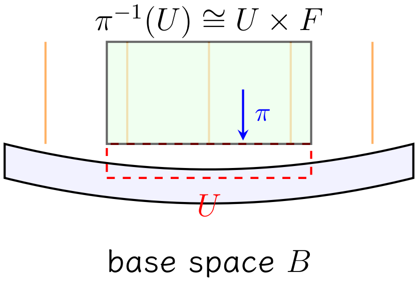


Figure 11.10: Local trivialisation of a fibre bundle: locally $\pi^{-1}(U) \cong U \times F$, and the projection π maps (b, f) back to the base point b.

Definition 11.13 (Section of a Fibre Bundle)**.** A section is a smooth map $s : B \to E$ satisfying $\pi \circ s = \mathrm{id}_B$. The set of all sections is denoted $\Gamma(E)$.

Proposition 11.6 (Tangent and Cotangent Bundles Are Fibre Bundles)**.** The tangent bundle TM and the cotangent bundle T^*M are both fibre bundles over M, with fibres $\mathbb{R}^n$ and $(\mathbb{R}^n)^*$ respectively.

Proof. This follows directly from the local trivialisation in Proposition 11.2. The local trivialisation of the cotangent bundle is analogous:

$$\tilde{\varphi}^*(p, \omega) = (x^1, \ldots, x^n, \omega_1, \ldots, \omega_n),$$

where $\omega = \sum_i \omega_i dx^i$. ☐

Definition 11.14 (Trivial Bundle)**.** If there exists a global diffeomorphism $\Phi : E \to B \times F$ with $\pi = \mathrm{pr}_1 \circ \Phi$, then the fibre bundle is called a **trivial bundle**.

Definition 11.15 (Vector Bundle)**.** If the fibre F is a vector space and the restriction of each local trivialisation to each fibre is a linear isomorphism, then the fibre bundle is called a **vector bundle**.

Property 11.2 (Basic Properties of Fibre Bundles)**.** Let (E, B, π, F) be a fibre bundle. Then:

(i) **Diffeomorphism type of the fibres**: for any b_1, b_2 in the same connected component of B, $E_{b_1} \cong E_{b_2} \cong F$;

(ii) **The projection is a submersion**: π is a smooth submersion ($d\pi$ is surjective everywhere);

(iii) **Local existence of sections**: for every $b \in B$ there exist an open neighbourhood $U \ni b$ and a local section $s : U \to E$;

(iv) **Module structure of the section space**: if E is a vector bundle, then $\Gamma(E)$ is a $C^\infty(B)$-module.

Proof. (i) By local trivialisation, $E_b \cong \{b\} \times F \cong F$ holds locally in b; the global conclusion on a connected component follows from the transitivity of local properties. (ii) In a local trivialisation, $d\pi$ corresponds to the differential of the projection $\mathrm{pr}_1 : U \times F \to U$, which is surjective. (iii) In a local trivialisation, take $s(b) = \Phi^{-1}(b, f_0)$ with $f_0 \in F$ fixed. (iv) Pointwise addition and scalar multiplication of sections give $\Gamma(E)$ its $C^\infty(B)$-module structure. ☐

Example 11.2 (Trivial and Non-Trivial Bundles)**.** (i) $T\mathbb{R}^n \cong \mathbb{R}^n \times \mathbb{R}^n$ is a trivial bundle.

(ii) $TS^1 \cong S^1 \times \mathbb{R}$ is a trivial bundle (a cylinder).

(iii) TS^2 is not a trivial bundle (the hairy ball theorem).

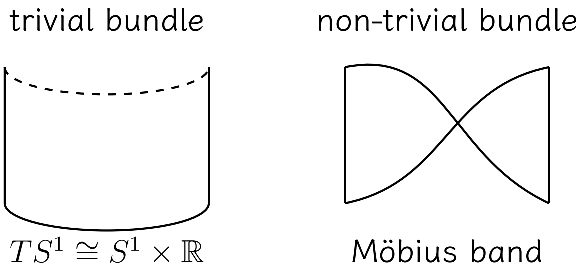


Figure 11.11: Comparison of a trivial bundle (a cylinder) with a non-trivial bundle (the Möbius band).

11.7 Normal Space and Normal Bundle

Definition 11.16 (Normal Space). Let (M, g) be a Riemannian manifold, $S \subseteq M$ an embedded submanifold and $p \in S$. The **normal space** of S at p is $N_pS := \{v \in T_pM : g_p(v, w) = 0, \forall w \in T_pS\}$.

Definition 11.17 (Normal Bundle). The **normal bundle** of S is $NS := \bigsqcup_{p \in S} N_pS$.

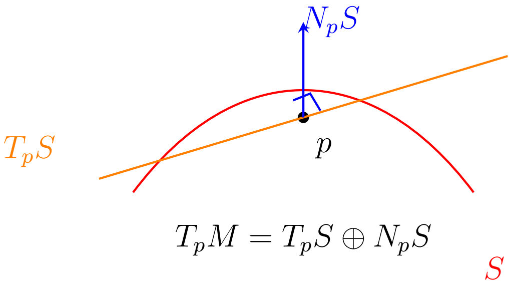


Figure 11.12: The normal space N_pS is the orthogonal complement of the tangent space T_pS in T_pM, $T_pM = T_pS \oplus N_pS$.

Proposition 11.7 (Gradient of Conserved Quantity and Normal Space). Let $Q_1, \ldots, Q_k$ be independent conserved quantities and $\mathcal{M}$ the level-set submanifold. Then $N_p\mathcal{M} = \text{span}\{\nabla Q_1(p), \ldots, \nabla Q_k(p)\}$.

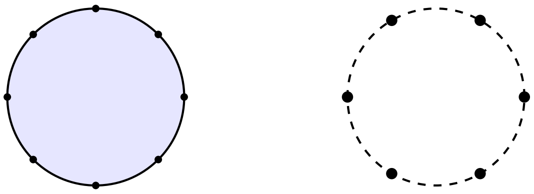
Proof. By Proposition 11.5, $\nabla Q_i(p) \in N_p\mathcal{M}$. By independence they are linearly independent and span a k-dimensional space, which is exactly the dimension of the normal space. □

11.8 Angle Space and Angle Bundle

Definition 11.18 (Angle Space). Let $\gamma : [0, T] \to M$ be a closed orbit. The **angle space** A_γ on γ is the parametrisation space of the orbit itself: when γ is a contractible closed curve, one takes $A_\gamma \cong U(1)$; when γ has a fundamental group element of finite order, one takes $A_\gamma \cong \mathbb{Z}_T$.

Definition 11.19 (Angle Bundle). The **angle bundle** $AM := \bigsqcup_\alpha A_{\gamma_\alpha}$ is the union of the angle spaces over the family of closed orbits.

Proposition 11.8 (Algebraic Structure of the Angle Space). *The topological structure of the angle space determines the type of the natural group structure on it: the continuous circle $U(1)$ corresponds to a continuous rotation group, and the finite cyclic group $\mathbb{Z}_T$ corresponds to a finite rotation group.*



$U(1)$: continuous rotation $\mathbb{Z}_6$: finite rotation

Figure 11.13: The two topological types of angle space: the continuous circle $U(1)$ and the finite cyclic group $\mathbb{Z}_T$.

Proof. **Continuous case**: $U(1) = \{e^{i\theta} : \theta \in [0, 2\pi)\}$. Let us verify that $U(1)$ is a group: (i) Closure: $e^{i\theta}e^{i\phi} = e^{i(\theta+\phi)} \in U(1)$; (ii) associativity follows from the associativity of complex multiplication; (iii) the identity element is $e^{i0} = 1$; (iv) the inverse is $(e^{i\theta})^{-1} = e^{-i\theta} \in U(1)$. $U(1)$ is a compact connected one-dimensional Lie group.

Discrete case: $\mathbb{Z}_T = \{0, 1, \ldots, T-1\}$ is the cyclic group C_T under addition modulo T. The map $k \mapsto e^{i2\pi k/T}$ gives an injective group homomorphism $C_T \to U(1)$, whose image is the group of T-th roots of unity. □

11.9 Phase Space

Definition 11.20 (Configuration Space). *Let the position of a mechanical system be described by n generalised coordinates. The set Q of all possible positions is an n-dimensional smooth manifold, called the* **configuration space**.

Definition 11.21 (State-Space Manifold). *The* **state-space manifold** *M is the set of all possible states of the system, equipped with a smooth manifold structure. A periodic orbit is a closed curve in M.*

Definition 11.22 (Phase Space—Tangent Bundle Formulation). *In Lagrangian mechanics,* **phase space** *is TQ, with elements $(q, \dot{q})$.*

Definition 11.23 (Phase Space—Cotangent Bundle Formulation). *In Hamiltonian mechanics,* **phase space** *is T^*Q, with elements (q, p). T^*Q carries the symplectic structure $\omega = \sum_i dq^i \wedge dp_i$.*

Proposition 11.9 (Phase-Space Volume and Liouville's Theorem). *The symplectic volume form on phase space T^*Q is $d\mu = \frac{1}{n!}\omega^n$. The phase-space volume is preserved by the Hamiltonian flow.*

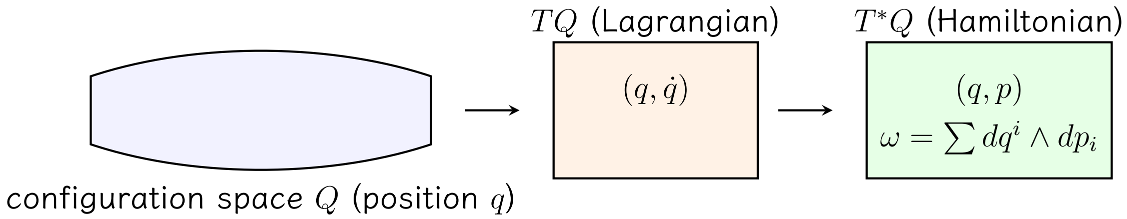

TQ (Lagrangian)
T^*Q (Hamiltonian)
$(q, \dot{q})$
(q, p)
$\omega = \sum dq^i \wedge dp_i$
configuration space Q (position q)

Figure 11.14: The two phase-space formulations, from the configuration space Q to the tangent bundle TQ (Lagrangian) and the cotangent bundle T^*Q (Hamiltonian).

Proof. The Hamiltonian vector field X_H in symplectic coordinates $(q^1, \dots, q^n, p_1, \dots, p_n)$ is

$$X_H = \sum_i \left(\frac{\partial H}{\partial p_i} \frac{\partial}{\partial q^i} - \frac{\partial H}{\partial q^i} \frac{\partial}{\partial p_i} \right).$$

Its divergence is

$$\operatorname{div} X_H = \sum_i \left(\frac{\partial}{\partial q^i} \frac{\partial H}{\partial p_i} + \frac{\partial}{\partial p_i} \left(-\frac{\partial H}{\partial q^i} \right) \right) = \sum_i \left(\frac{\partial^2 H}{\partial q^i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q^i} \right) = 0,$$

by Clairaut's theorem the mixed partial derivatives are equal. By Liouville's formula the volume of the flow is preserved. ☐

11.10 Invariant Manifolds and Energy Surfaces

Definition 11.24 (Flow and Invariant Submanifold). Let X be a vector field and Φ_t its flow. A submanifold S is called an **invariant submanifold** if $\Phi_t(S) \subseteq S$ for all t.

Proposition 11.10 (Level Sets of Conserved Quantities Are Invariant Submanifolds). The level set $Q^{-1}(c)$ of a conserved quantity Q is an invariant submanifold.

Proof. For $p \in Q^{-1}(c)$ we have $Q(p) = c$. By conservation, $Q(\Phi_t(p)) = Q(p) = c$, hence $\Phi_t(p) \in Q^{-1}(c)$. ☐

Definition 11.25 (Energy Surface). Let $H : T^*Q \to \mathbb{R}$ be a Hamiltonian function. The **energy surface** $\Sigma_E := H^{-1}(E)$ is a constraint hypersurface in phase space.

Proposition 11.11 (Dimension of the Energy Surface). If $dH \neq 0$ holds everywhere on Σ_E, then Σ_E is a $(2n-1)$-dimensional embedded submanifold.

Proof. This follows directly from Theorem 11.1. ☐

11.11 Invariant Tori and the Liouville–Arnold Theorem

Definition 11.26 (Invariant Torus). Let a Hamiltonian system have n independent first integrals in involution $F_1 = H, F_2, \dots, F_n$. The level set $\mathbb{T}_{c_1,\dots,c_n} := \{F_1 = c_1, \dots, F_n = c_n\}$ is called an **invariant torus**.

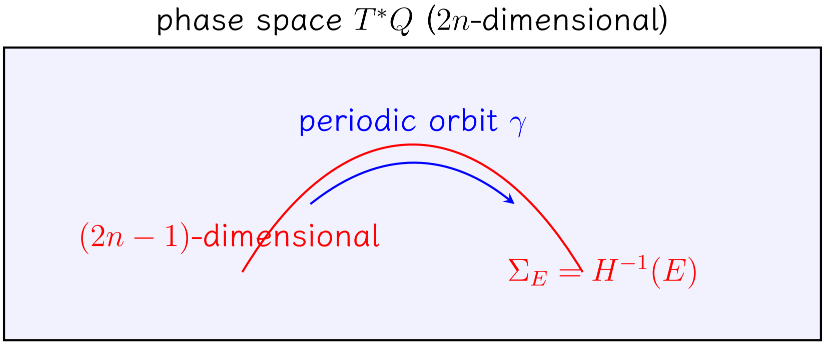


Figure 11.15: The energy surface Σ_E is a $(2n-1)$-dimensional hypersurface in phase space, on which periodic orbits move.

Proposition 11.12 (Motion on the Invariant Torus). On the invariant torus $\mathbb{T}^n$ the motion is described by the angle coordinates $(\theta_1, \dots, \theta_n)$ and satisfies $\dot{\theta}_i = \omega_i$ (constants). Hence $\theta_i(t) = \theta_i(0) + \omega_i t$.

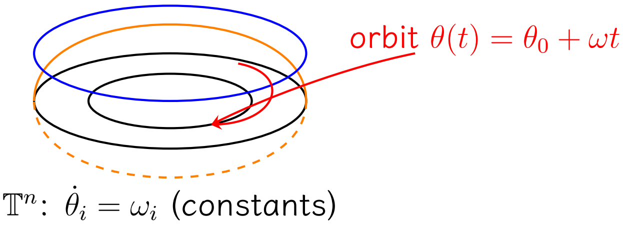

orbit $\theta(t) = \theta_0 + \omega t$
$\mathbb{T}^n$: $\dot{\theta}_i = \omega_i$ (constants)

Figure 11.16: Uniform motion on an invariant torus: the angle coordinates evolve according to $\theta_i(t) = \theta_i(0) + \omega_i t$.

Proof. This is a direct corollary of Theorem 11.2: in action-angle coordinates, $\dot{\theta}_i = \frac{\partial H}{\partial I_i} = \omega_i(I)$ (constants) and $\dot{I}_i = 0$. ☐

Theorem 11.2 (Liouville–Arnold Theorem). Let an n-degree-of-freedom Hamiltonian system have n independent first integrals in involution, and suppose that the level set is compact and connected. Then this level set is diffeomorphic to $\mathbb{T}^n$, and near it there exist action-angle coordinates in which the motion becomes uniform rectilinear motion.

Proof. **The proof lies beyond the scope of this book** (see Arnold, Mathematical Methods of Classical Mechanics, 2nd ed., Springer, 1989, §49–50). The core idea: n first integrals in involution generate n commuting flows, and the action of these flows on the level set gives the torus structure. ☐

Corollary 11.2 (Criterion for Closed and Quasi-Periodic Orbits). On an invariant torus $\mathbb{T}^n$, an orbit $\theta(t) = \theta_0 + \omega t$ is closed if and only if the frequencies are pairwise commensurable, that is, there exists $T > 0$ with $\omega T \in 2\pi\mathbb{Z}^n$ (equivalently $\omega_i/\omega_j \in \mathbb{Q}$ for all i, j); otherwise the orbit is dense in $\mathbb{T}^n$.

Proof. The orbit $\theta(t) = \theta_0 + \omega t$ is closed if and only if there exists $T > 0$ with $\omega T \in 2\pi\mathbb{Z}^n$. This is equivalent to $\omega_i/\omega_j \in \mathbb{Q}$ for all i, j (when $\omega_j \neq 0$). If some $\omega_j = 0$, that component is stationary, and it suffices to take T to be the least common multiple of the periods of the remaining components. If the frequencies are pairwise commensurable, take T to be the least common multiple of the periods of the individual frequencies; if there exists a pair of incommensurable frequencies, the orbit is dense in the torus (the Kronecker approximation theorem). □

pairwise commensurable: closed incommensurable: dense

Figure 11.17: Criterion for the closure of an orbit on an invariant torus: when the frequencies are pairwise commensurable the orbit is closed, and when they are incommensurable the orbit is dense in the torus.

11.12 Stable Manifolds, Unstable Manifolds and Centre Manifolds

Definition 11.27 (Stable and Unstable Manifolds). Let γ be a periodic orbit. $W^s(\gamma) := \{p : d(\Phi_t(p), \gamma) \to 0 \ (t \to +\infty)\}$, $W^u(\gamma) := \{p : d(\Phi_t(p), \gamma) \to 0 \ (t \to -\infty)\}$.

Definition 11.28 (Centre Manifold). $W^c(\gamma)$ consists of the orbits that neither approach γ exponentially nor recede from it exponentially.

Proposition 11.13 (Stable/Unstable Manifolds and Floquet Multipliers). Let γ be a periodic orbit with Floquet multipliers $\rho_1, \ldots, \rho_n$. Then $\dim W^s = \#\{|\rho_i| < 1\}$, $\dim W^u = \#\{|\rho_i| > 1\}$, $\dim W^c = \#\{|\rho_i| = 1\}$.

Proof. By Floquet theory, the linearised dynamics near a periodic orbit is determined by the Floquet multipliers. Write $\rho_i = e^{\lambda_i T}$ (λ_i the Floquet exponents). Then $|\rho_i| < 1 \iff \operatorname{Re}\lambda_i < 0$, corresponding to exponential decay (a stable direction); $|\rho_i| > 1 \iff \operatorname{Re}\lambda_i > 0$, corresponding to exponential growth (an unstable direction); $|\rho_i| = 1 \iff \operatorname{Re}\lambda_i = 0$, corresponding to a neutral direction (a centre direction). The dimensions of the stable manifold, the unstable manifold and the centre manifold are equal to the numbers of stable, unstable and centre directions respectively. □

11.13 Moduli Space

Definition 11.29 (Moduli Space). Let $\mathcal{S}$ be a collection of periodic structures of a certain kind and $\sim$ an equivalence relation. The **moduli space** $\mathcal{M} := \mathcal{S}/\sim$ is the set of equivalence classes.

Example 11.3 (Moduli Space of a Periodic Lattice). The moduli space of the two-dimensional periodic lattice $\Lambda = \omega_1\mathbb{Z} + \omega_2\mathbb{Z}$ is $SL(2,\mathbb{Z}) \setminus \mathbb{H}$, where $\mathbb{H} = \{\tau \in \mathbb{C} : \operatorname{Im}\tau > 0\}$.

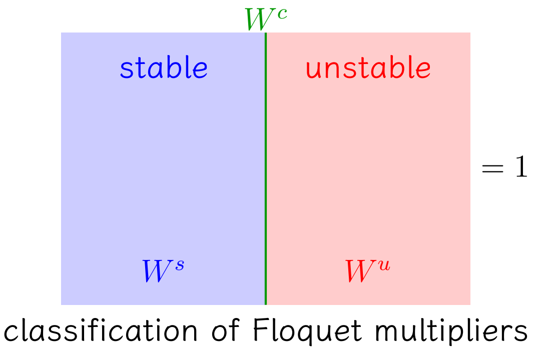


Figure 11.18: The stable manifold W^s, the unstable manifold W^u and the centre manifold W^c classified by the modulus of the Floquet multipliers.

Remark 11.3 (Role of the Moduli Space). The moduli space provides a **classifying space** for periodic structures: periodic structures with the same basic structure are parametrised by the same point of the moduli space, and their deformations are described by the tangent space of the moduli space.

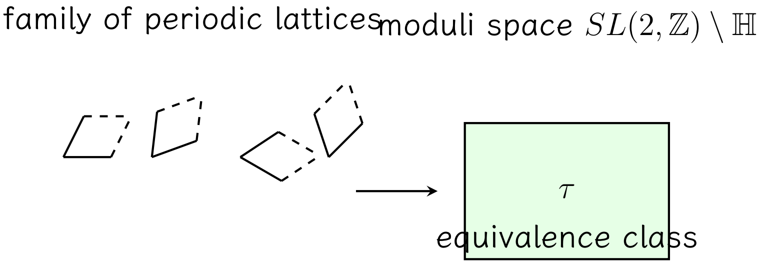

family of periodic lattices
moduli space $SL(2, \mathbb{Z}) \setminus \mathbb{H}$
τ
equivalence class

Figure 11.19: The moduli space as a classifying space for periodic structures: equivalence classes are parametrised by τ.

11.14 Group Actions and Quotient Manifolds

Definition 11.30 (Action of a Lie Group on a Manifold). Let G be a Lie group and M a smooth manifold. A **smooth left action** of G is a smooth map $\Phi : G \times M \to M$ satisfying $e \cdot p = p$ and $(gh) \cdot p = g \cdot (h \cdot p)$.

Definition 11.31 (Orbits and Quotient Manifolds). If the action of G is free and properly discontinuous, then the orbit space M/G carries a unique smooth structure and is called a **quotient manifold**.

Proposition 11.14 (The Torus as a Quotient Manifold). The n-dimensional torus is the quotient manifold of Euclidean space by a periodic lattice: $\mathbb{T}^n \cong \mathbb{R}^n/\mathbb{Z}^n$.

Proof. $\mathbb{Z}^n$ acts on $\mathbb{R}^n$ by translations: $(k_1, \dots, k_n) \cdot (x_1, \dots, x_n) = (x_1 + k_1, \dots, x_n + k_n)$.

Freeness: if $(k_1, \dots, k_n) \cdot (x_1, \dots, x_n) = (x_1, \dots, x_n)$, then $x_i + k_i = x_i$, hence $k_i = 0$ for all i.

Proper discontinuity: for any compact set $K \subseteq \mathbb{R}^n$ the set K is bounded; let $\|x\| \leq R$ for all $x \in K$. If $k \cdot K \cap K \neq \emptyset$, then there exist $x, y \in K$ with $x + k = y$,

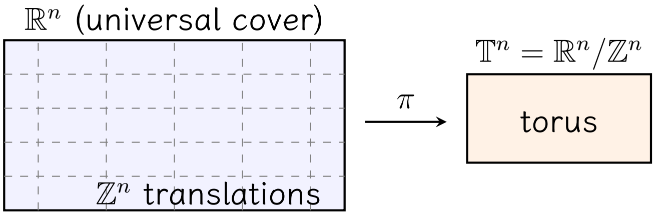

$\mathbb{R}^n$ (universal cover)
$\mathbb{Z}^n$ translations
π
$\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$
torus

Figure 11.20: The torus as a quotient manifold: $\mathbb{Z}^n$ acts on $\mathbb{R}^n$ by translations, and taking the quotient yields $\mathbb{T}^n$.

hence $\|k\| \leq 2R$. There are only finitely many integer vectors $(k_1, \dots, k_n)$ satisfying $\|k\| \leq 2R$ (each $|k_i| \leq 2R$, so at most $(4R+1)^n$ of them). Hence the action is properly discontinuous.

The quotient space $\mathbb{R}^n/\mathbb{Z}^n$ is diffeomorphic to $S^1 \times \cdots \times S^1 = \mathbb{T}^n$ (the diffeomorphism is given by $[(x_1, \dots, x_n)] \mapsto (e^{2\pi i x_1}, \dots, e^{2\pi i x_n})$). ☐

Definition 11.32 (Covering Spaces). A surjection $\pi : \tilde{M} \to M$ is called a **covering map** if for every $p \in M$ there exists an open neighbourhood U such that $\pi^{-1}(U)$ is a union of a family of pairwise disjoint open sets, and π maps each of these open sets diffeomorphically onto U.

Lemma 11.2 (Basic Properties of Covering Maps). Let $\pi : \tilde{M} \to M$ be a covering map. Then:

(i) **Local homeomorphism**: π is a local diffeomorphism;

(ii) **Discrete fibres**: for every $p \in M$ the fibre $\pi^{-1}(p)$ is a discrete set;

(iii) **Path lifting**: for every $p \in M$, every $\tilde{p} \in \pi^{-1}(p)$ and every path $\gamma : [0,1] \to M$ with $\gamma(0) = p$, there exists a unique lift $\tilde{\gamma} : [0,1] \to \tilde{M}$ satisfying $\tilde{\gamma}(0) = \tilde{p}$ and $\pi \circ \tilde{\gamma} = \gamma$;

(iv) **Homotopy lifting**: homotopic paths lift to homotopic paths.

Proof. (i) follows directly from the definition of a covering map. (ii) follows because $\pi^{-1}(p)$ is a union of disjoint open sets. (iii) Subdivide $[0,1]$ into small subintervals, use the local trivialisation on each subinterval to define the lift, and glue by uniqueness. (iv) This is proved similarly, using the compactness of the homotopy. ☐

Theorem 11.3 (The Real Line Covers the Circle). The map $\pi : \mathbb{R} \to S^1$, $\pi(t) = e^{2\pi i t}$, is a covering map. $\mathbb{R}$ is the universal cover of S^1.

Proof. For any point $e^{i\theta}$ on S^1, take the open arc $U = \{e^{i\varphi} : |\varphi - \theta| < \pi\}$. Then $\pi^{-1}(U) = \bigcup_{k \in \mathbb{Z}} (\frac{\theta - \pi}{2\pi} + k, \frac{\theta + \pi}{2\pi} + k)$, and each interval is diffeomorphic to U under π. $\mathbb{R}$ is simply connected and π is a covering, so $\mathbb{R}$ is the universal cover. ☐

Definition 11.33 (Closed Orbits and the Fundamental Group). Let $\gamma : [0,T] \to M$ be a closed orbit. Then γ defines $[\gamma] \in \pi_1(M)$. If $[\gamma]$ is an element of finite order, then γ is called a **twisted closed orbit**.

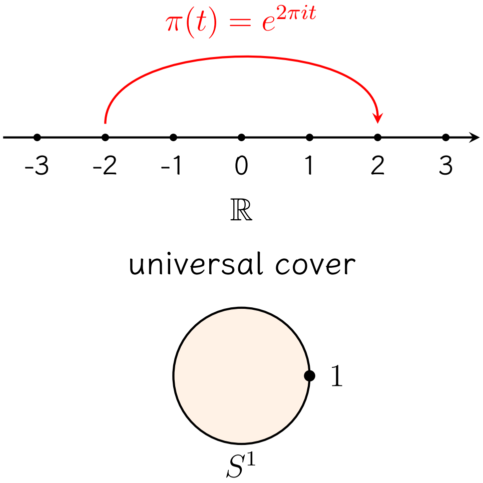

$\pi(t) = e^{2\pi it}$
-3
-2
-1
0
1
2
3
$\mathbb{R}$
universal cover
1
S^1

Figure 11.21: The real line $\mathbb{R}$ is the universal cover of the circle S^1: $\pi(t) = e^{2\pi it}$, with fibre $\mathbb{Z}$.

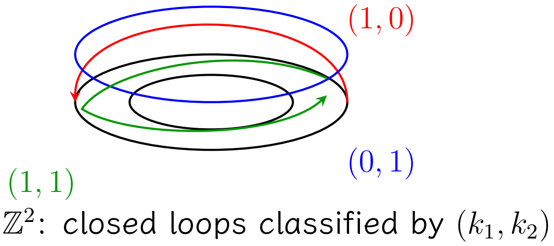
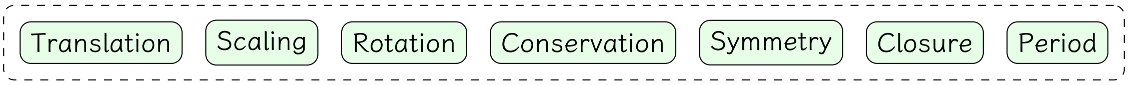

$(1,0)$
$(0,1)$
$(1,1)$
$\pi_1(\mathbb{T}^2) \cong \mathbb{Z}^2$: closed loops classified by (k_1, k_2)

Figure 11.22: Closed orbits on the torus are classified by integer vectors in $\pi_1(\mathbb{T}^n) \cong \mathbb{Z}^n$.

Proposition 11.15 (The Fundamental Group of the Torus). $\pi_1(\mathbb{T}^n) \cong \mathbb{Z}^n$. Closed orbits on the torus are classified by $(k_1, \dots, k_n) \in \mathbb{Z}^n$.

Proof. By covering theory, $\pi_1(\mathbb{T}^n) \cong \pi_1(S^1)^n$. First we prove that $\pi_1(S^1) \cong \mathbb{Z}$: by Theorem 11.3, $\mathbb{R}$ is the universal cover of S^1, so $\pi_1(S^1) \cong \pi^{-1}(1) = \mathbb{Z}$ (the group of covering transformations is isomorphic to the fibre). The lift $\tilde{\gamma}$ of a closed curve γ on the torus satisfies $\tilde{\gamma}(T) - \tilde{\gamma}(0) = (k_1, \dots, k_n) \in \mathbb{Z}^n$, and this is the element corresponding to $[\gamma]$. Hence $\pi_1(\mathbb{T}^n) \cong \mathbb{Z}^n$. $\square$

Seven-theme comparison table for this chapter



the seven themes run through manifolds and fibre bundles

Figure 11.23: The seven themes of this chapter: Translation, Scaling, Rotation, Conservation, Symmetry, Closure, Period.

Theme	Corresponding content in this chapter
Translation	the tangent vector as the infinitesimal generator of translation (Definition 11.4); the orbits of a group action (Definition 11.30); the covering map (Theorem 11.3); uniform motion on an invariant torus (Proposition 11.12)
Scaling	the scaling of the Riemannian metric (Definition 11.10); the orthogonal complement of the normal space (Definition 11.16); the parametrisation of the moduli space (Definition 11.29)
Rotation	the $U(1)$ structure of the angle space (Definition 11.18); the rotation of the invariant torus (Definition 11.26); stable/unstable manifolds (Definition 11.27)
Conservation	the level set of a conserved quantity (Theorem 11.1); the energy surface (Definition 11.25); the constraints of the normal bundle (Proposition 11.7); the invariance of level sets (Proposition 11.10)
Symmetry	group actions and quotient manifolds (Definition 11.31); the torus as a quotient manifold (Proposition 11.14)
Closure	closed orbits and the fundamental group (Definition 11.33); the classification of the fundamental group of the torus (Proposition 11.15); the closure/quasi-periodicity criterion (Corollary 11.2)
Period	the torus as a quotient manifold (Proposition 11.14); periodic motion on an invariant torus (Proposition 11.12); the moduli space (Definition 11.29)

Exercises

Exercise 1 Using Definition 11.1, verify that the circle

$$S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$$

is a 1-dimensional topological manifold. Write down explicitly the two charts

$$U_1 = S^1 \setminus \{(1, 0)\}, \quad U_2 = S^1 \setminus \{(-1, 0)\}$$

and the corresponding homeomorphisms φ_1, φ_2, and verify that their union covers S^1.

Exercise 2 Prove Proposition 11.1: let M be an n-dimensional smooth manifold, $p \in M$, and let $(U, \varphi = (x^1, \ldots, x^n))$ be a chart. Then T_pM has dimension n, and

$$\left.\frac{\partial}{\partial x^1}\right|_p, \ldots, \left.\frac{\partial}{\partial x^n}\right|_p$$

forms a basis. Write out the complete proof in the three steps 'verifying that it is a tangent vector, linear independence, spanning'.

Exercise 3 Verify that the two definitions of a tangent vector are equivalent (Lemma 11.1): the derivation definition (Definition 11.4) and the curve equivalence class definition (Definition 11.5) give the same tangent space. Write down the correspondence

$$[\gamma] \mapsto v_{[\gamma]}, \qquad v_{[\gamma]}(f) = \left.\frac{d}{dt}\right|_{t=0} f(\gamma(t)),$$

and prove that it is a well-defined bijection.

Exercise 4 Prove the manifold structure of the tangent bundle (Proposition 11.2): the tangent bundle TM is a $2n$-dimensional smooth manifold. Write down the explicit expression for the local chart $(\pi^{-1}(U), \tilde{\varphi})$

$$\tilde{\varphi}(p, v) = (x^1(p), \ldots, x^n(p), v^1, \ldots, v^n),$$

and verify the smoothness of the transition map.

Exercise 5 Verify Proposition 11.3: in the chart $(U, \varphi = (x^1, \ldots, x^n))$, a basis of the cotangent space T_p^*M is

$$dx^1|_p, \ldots, dx^n|_p,$$

satisfying $dx^i|_p(\partial/\partial x^j|_p) = \delta^i_j$. Give a proof using the dual basis theorem and the differentials of the coordinate functions.

Exercise 6 Prove the regular level set theorem (Theorem 11.1): let $F : M \to \mathbb{R}^k$ be a smooth map and $c \in \mathbb{R}^k$. If dF_p is surjective for every $p \in F^{-1}(c)$, then $F^{-1}(c)$ is an $(n-k)$-dimensional embedded submanifold. Use the rank theorem to construct coordinates near p in which $F = (x^{n-k+1}, \ldots, x^n)$, and write down the local submanifold structure.

Exercise 7 Prove Proposition 11.5: let (M, g) be a Riemannian manifold and $f \in C^\infty(M)$. If $p \in f^{-1}(c)$ and $\nabla f(p) \neq 0$, then $\nabla f(p)$ is orthogonal to the tangent space of the level set $f^{-1}(c)$ at p. Take a curve γ with $f(\gamma(t)) = c$ and differentiate with respect to t to complete the proof.

Exercise 8 Verify the assertions about trivial and non-trivial bundles in Example 11.2:

(i) $T\mathbb{R}^n \cong \mathbb{R}^n \times \mathbb{R}^n$ is a trivial bundle;

(ii) $TS^1 \cong S^1 \times \mathbb{R}$ is a trivial bundle;

(iii) TS^2 is not a trivial bundle.

Use the hairy ball theorem to justify (iii), and explain why there is no continuous tangent vector field on S^2 that is nowhere zero.

Exercise 9 Prove Proposition 11.7: let $Q_1, \ldots, Q_k$ be independent conserved quantities and $\mathcal{M}$ the level-set submanifold; then

$$N_p\mathcal{M} = \text{span}\{\nabla Q_1(p), \ldots, \nabla Q_k(p)\}.$$

Use Proposition 11.5 to show that $\nabla Q_i(p) \in N_p\mathcal{M}$, and use independence to show that they span a k-dimensional space.

Exercise 10 Prove Proposition 11.8: the topological structure of the angle space determines the type of the natural group structure on it: the continuous circle $U(1)$ corresponds to a continuous rotation group, and the finite cyclic group $\mathbb{Z}_T$ corresponds to a finite rotation group. Verify the group axioms for $U(1)$ and $\mathbb{Z}_T$ separately, and show that the map $k \mapsto e^{i2\pi k/T}$ gives an injective group homomorphism $C_T \to U(1)$.

Exercise 11 Prove, in the Liouville–Arnold theorem (Theorem 11.2), the criterion for closed and quasi-periodic orbits (Corollary 11.2): on an invariant torus $\mathbb{T}^n$, an orbit is closed if and only if the frequencies are pairwise commensurable, that is, there exists $T > 0$ with $\omega T \in 2\pi\mathbb{Z}^n$ (equivalently $\omega_i/\omega_j \in \mathbb{Q}$ for all i, j). Prove sufficiency and necessity separately, and use the Kronecker approximation theorem to show that the orbit is dense when the frequencies are incommensurable.

Exercise 12 Verify Proposition 11.13: let γ be a periodic orbit with Floquet multipliers $\rho_1, \dots, \rho_n$. Then

$$\dim W^s = \#\{|\rho_i| < 1\}, \quad \dim W^u = \#\{|\rho_i| > 1\}, \quad \dim W^c = \#\{|\rho_i| = 1\}.$$

Complete the proof using the sign of the real part of the Floquet exponents λ_i.

Exercise 13 Prove that the torus is a quotient manifold (Proposition 11.14): the n-dimensional torus $\mathbb{T}^n \cong \mathbb{R}^n/\mathbb{Z}^n$. Verify that the action of $\mathbb{Z}^n$ on $\mathbb{R}^n$ by translations is free and properly discontinuous, and write down the diffeomorphism from the quotient space to $S^1 \times \cdots \times S^1$.

Exercise 14 Prove that the real line covers the circle (Theorem 11.3): the map $\pi : \mathbb{R} \to S^1$, $\pi(t) = e^{2\pi i t}$, is a covering map, and $\mathbb{R}$ is the universal cover of S^1. Write down an open neighbourhood U of an arbitrary point of S^1 and the decomposition of its preimage $\pi^{-1}(U)$, and explain why $\mathbb{R}$ is simply connected.

Exercise 15 Prove the fundamental group of the torus (Proposition 11.15):

$$\pi_1(\mathbb{T}^n) \cong \mathbb{Z}^n.$$

Use covering space theory to show that $\pi_1(\mathbb{T}^n) \cong \pi_1(S^1)^n$, and use the fact that $\mathbb{R}$ is the universal cover of S^1 to show that $\pi_1(S^1) \cong \mathbb{Z}$.

Exercise 16 Synthesis: combining the tangent space (Definition 11.4), the normal space (Definition 11.16), the angle space (Definition 11.18) and the invariant torus (Definition 11.26), explain how the manifold framework answers simultaneously the five questions raised in the introduction: whether a group element has finite order, whether an orbit is closed, whether there exists an integrable vector field in the tangential direction, whether there exists a conservation constraint in the normal direction, and whether the phase in the angular direction is single-valued. Indicate respectively the corresponding theorem or proposition for these structures in this chapter.

Chapter 12

Periodic Manifold

Remark 12.1 (Positioning of this chapter). The periodic manifold is the core theory of discrete geometry and discrete analysis within periodic mathematics. This chapter starts from periodic structures on continuous manifolds, establishes the general theory of the **periodic manifold**, and then discretises it into a lattice-point model. There are three classes of core object in this chapter: the **periodic manifold** (a manifold carrying an action of a group with periodic orbits), **periodic orbits** (closed curves and their moduli space), and the **discrete lattice-point manifold** (the discretisation of a periodic structure).

One of the core tasks of this chapter is to **rigorously establish the complete theory of the three great differential operators—gradient, divergence and curl.** In the continuous case we give their intrinsic definitions (independent of any choice of coordinates) and their coordinate formulae; in the discrete case we give their rigorous derivation (starting from Taylor expansion and flux balance), their exact identities (requiring no passage to the limit) and a complete proof of the continuum limit.

Guide to the seven themes of this chapter: translation—the lattice translation operation (Definition 12.12) and the discrete gradient (Definition 12.18); scaling—the lattice scaling operation (Definition 12.13) and the discrete divergence (Definition 12.19); rotation—the lattice rotation operation (Definition 12.14) and the discrete curl (Definition 12.20); conservation—conservation of the squared norm under discrete unitary evolution (Proposition 12.17) and the integral theorems for the three great operators (Theorems 12.6, 12.5); symmetry—the axiom of primitive discreteness (Axiom 12.1) and the correspondence with the three symmetry centres $0, 1, i$; closure—the Wilson-loop closure criterion for a plaquette (Lemma 12.1); period—the period condition for discrete unitary operators (Proposition 12.23).

12.1 Basic structure of the periodic manifold

Definition 12.1 (Periodic manifold). Let M be an n-dimensional smooth manifold and G a Lie group. The triple (M, G, Φ) is called a **periodic manifold** if G acts on M through a smooth left action $\Phi : G \times M \to M$, and

(i) the action is free: $g \cdot p = p \Rightarrow g = e$;

(ii) the action is properly discontinuous: for every compact $K \subseteq M$, the set $\{g \in G : gK \cap K \neq \emptyset\}$ is finite;

(iii) the orbit space M/G is a smooth manifold (a quotient manifold).

When $G = \mathbb{Z}$ or $\mathbb{Z}^n$, the orbits are periodic orbits; when $G = U(1)$ or $\mathbb{T}^n$, the orbits are closed orbits; when $G = \mathbb{R}$, the orbits are streamlines.

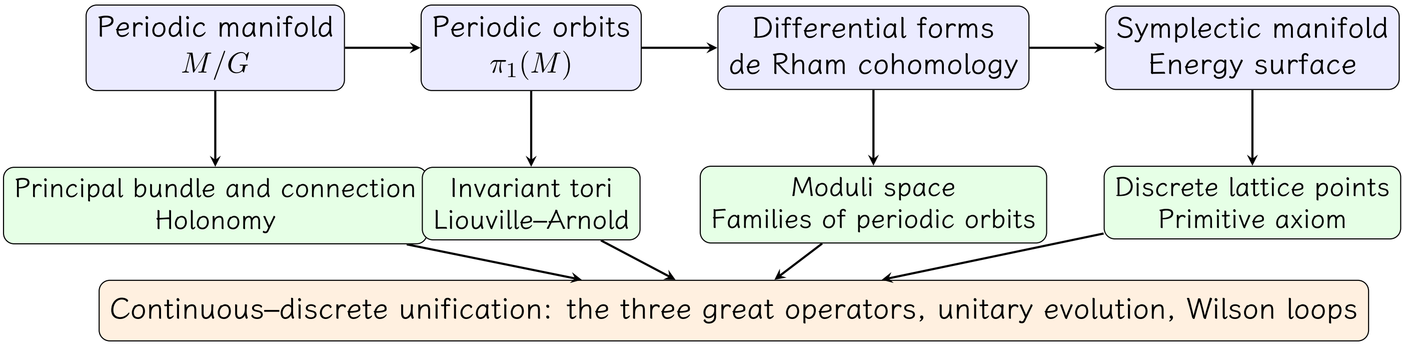

Periodic manifold
M/G
Periodic orbits
$\pi_1(M)$
Differential forms
de Rham cohomology
Symplectic manifold
Energy surface
Principal bundle and connection
Holonomy
Invariant tori
Liouville–Arnold
Moduli space
Families of periodic orbits
Discrete lattice points
Primitive axiom
Continuous–discrete unification: the three great operators, unitary evolution, Wilson loops

Figure 12.1: Overview of the structure of this chapter: starting from the periodic manifold, and passing through periodic orbits, differential forms, symplectic manifolds, and principal bundles and connections, to invariant tori, moduli spaces and discrete lattice points, finally unified in the continuous–discrete correspondence.

Example 12.1 (The torus as a periodic manifold). The n-dimensional torus $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ is the prototype of a periodic manifold: $\mathbb{Z}^n$ acts on $\mathbb{R}^n$ by translation, the action is free and properly discontinuous, and the quotient space $\mathbb{R}^n/\mathbb{Z}^n$ is diffeomorphic to $S^1 \times \cdots \times S^1$.

Example 12.2 (Lens spaces). Let p, q be coprime and $L(p,q) = S^3/\mathbb{Z}_p$, where $\mathbb{Z}_p$ acts on $S^3 \subseteq \mathbb{C}^2$ by $(z_1, z_2) \mapsto (e^{2\pi i/p} z_1, e^{2\pi iq/p} z_2)$. Then $L(p,q)$ is a periodic manifold whose fundamental group is $\mathbb{Z}_p$, and all its closed orbits are twisted orbits of finite order.

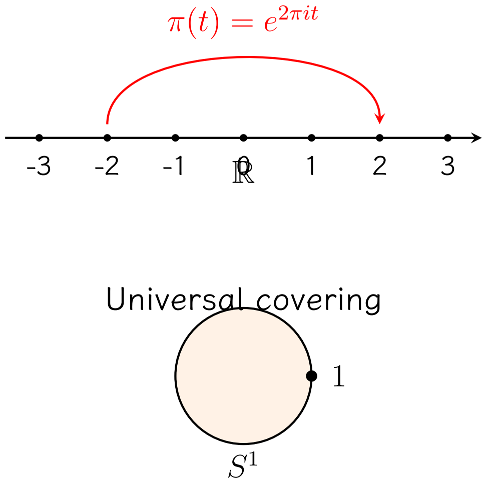

$\pi(t) = e^{2\pi it}$
-3
-2
-1
0
1
2
3
$\mathbb{R}$
Universal covering
1
S^1

Figure 12.2: The real line $\mathbb{R}$ is the universal covering of the circle S^1: $\pi(t) = e^{2\pi it}$, with fibre $\mathbb{Z}$.

Proposition 12.1 (Smoothness of the quotient manifold). Let G be a compact Lie group acting smoothly and freely on a manifold M. Then the orbit space M/G carries a unique smooth structure for which the projection $\pi : M \to M/G$ is a smooth submersion.

Proof. Because the action is free, each orbit is an embedded submanifold diffeomorphic to G. For every $p \in M$ there exist a G-invariant neighbourhood U

and a slice S such that $U \cong S \times G$, where S is transverse to the orbits. The local coordinates on M/G are supplied by S, and the smoothness of the transition functions is guaranteed by the freeness of the G-action. Uniqueness follows from the universal property of the quotient topology. See [84]. □

Remark 12.2 (Relation to the 'Manifold and Fibre Bundle' chapter). The theory of periodic manifolds in this chapter is built directly on the theory of continuous manifolds established in the 'Manifold and Fibre Bundle' chapter. Such concepts as the torus as a quotient manifold (Proposition 11.14), covering spaces (Definition 11.32), the fundamental group (Definition 11.33), invariant tori (Definition 11.26) and the symplectic structure (Definition 11.23) are used systematically in this chapter rather than reinvented.

12.2 Periodic orbits and the fundamental group

Definition 12.2 (Closed orbits and twisting). *Let $\gamma : [0, T] \to M$ be a closed orbit. Then γ defines an element $[\gamma] \in \pi_1(M)$ of the fundamental group. If $[\gamma]$ is an element of finite order (of order p), then γ is called a* **twisted closed orbit of order** *p.*

Remark 12.3 (Origin of the fundamental group of the torus). Proposition 11.15 has already been stated and proved (with proof) in the 'Manifold and Fibre Bundle' chapter; it is cited directly here and is not restated.

Corollary 12.1 (Criterion for twisted orbits). *Let γ be a closed orbit on $\mathbb{T}^n$ with $[\gamma] = (k_1, \ldots, k_n) \in \mathbb{Z}^n$. Then γ is a twisted orbit of finite order if and only if $(k_1, \ldots, k_n)$ is an element of finite order in $\mathbb{Z}^n$, that is, $(k_1, \ldots, k_n) = (0, \ldots, 0)$.*

Proof. $\mathbb{Z}^n$ is a torsion-free abelian group and has no element of finite order other than the zero element. Hence only the trivial element corresponds to a twisted orbit of finite order. □

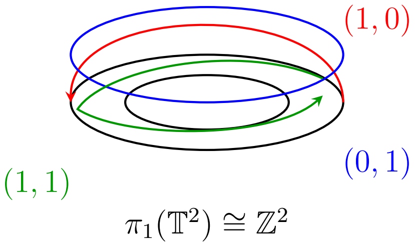


Figure 12.3: Closed orbits on the torus are classified by integer vectors in $\pi_1(\mathbb{T}^n) \cong \mathbb{Z}^n$.

12.3 Differential forms on a periodic manifold

Definition 12.3 (Closed and exact forms). *A k-form ω is called* **closed** *if $d\omega = 0$. It is called* **exact** *if there exists a $(k-1)$-form η with $\omega = d\eta$. Every exact form is closed ($d^2 = 0$).*

Definition 12.4 (de Rham cohomology)**.** The k-th **de Rham cohomology group** of M is

$$H^k_{dR}(M) := \frac{\ker(d : \Omega^k(M) \to \Omega^{k+1}(M))}{\operatorname{im}(d : \Omega^{k-1}(M) \to \Omega^k(M))}.$$

Theorem 12.1 (The de Rham theorem)**.** *For a smooth manifold M, $H^k_{dR}(M) \cong H^k(M;\mathbb{R})$ (singular cohomology).*

Proof. **The proof lies beyond the scope of this book.** This is the de Rham theorem, whose proof uses the methods of sheaf theory or of spectral sequences. See Chapter 18 of [83]. The core idea: the double duality between the de Rham complex and the singular cochain complex gives a natural isomorphism. ☐

Proposition 12.2 (Period integrals and cohomology)**.** *The integral $\oint_\gamma \omega$ of a closed form ω along a closed orbit γ depends only on the cohomology class of ω and the homology class of γ. If ω is exact, then $\oint_\gamma \omega = 0$.*

Proof. If $\omega = d\eta$, then by Stokes' theorem $\oint_\gamma \omega = \oint_\gamma d\eta = \oint_{\partial\gamma} \eta = 0$ (since $\partial\gamma = 0$). If γ is homologous to γ', then $\gamma - \gamma' = \partial\sigma$, and $\oint_{\gamma-\gamma'} \omega = \oint_{\partial\sigma} \omega = \oint_\sigma d\omega = 0$. ☐

Example 12.3 (Harmonic 1-forms on the torus)**.** On $\mathbb{T}^n$ there exist n independent closed 1-forms $d\theta_1, \ldots, d\theta_n$ with $\oint_{\gamma_k} d\theta_j = \delta_{jk}$. Moreover $H^1_{dR}(\mathbb{T}^n) \cong \mathbb{R}^n$.

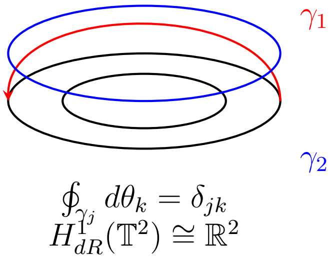

γ_1
γ_2
$\oint_{\gamma_j} d\theta_k = \delta_{jk}$
$H^1_{dR}(\mathbb{T}^2) \cong \mathbb{R}^2$

Figure 12.4: Closed 1-forms and period integrals on the torus: the integral of $d\theta_j$ along γ_k gives the Kronecker delta.

12.4 Invariant tori and integrable systems

Remark 12.4 (Relation to the 'Manifold and Fibre Bundle' chapter)*.* The definition of an invariant torus has already been given in Definition 11.26 of the 'Manifold and Fibre Bundle' chapter; that definition is adopted here directly and is not restated.

Theorem 12.2 (The Liouville–Arnold theorem)**.** *Let a Hamiltonian system with n degrees of freedom possess n independent first integrals in involution, and let its level set be compact and connected. Then the level set is diffeomorphic to $\mathbb{T}^n$, and near it there exist action-angle coordinates (I, θ) such that*

$$\omega = \sum_i dI_i \wedge d\theta_i, \qquad \dot{I}_i = 0, \qquad \dot{\theta}_i = \omega_i(I).$$

Proof. See Arnold, Mathematical Methods of Classical Mechanics, 2nd ed., Springer, 1989, §49–50. The core idea: n first integrals in involution generate n commuting Hamiltonian flows, and the action of these flows on the compact level set gives the structure of $\mathbb{T}^n$; the action variables I_i are defined by period integrals on the torus, and the angle variables θ_i are their conjugates. ☐

Corollary 12.2 (Criteria for periodic and quasiperiodic orbits). On an invariant torus $\mathbb{T}^n$, the orbit $\theta(t) = \theta_0 + \omega t$ is closed if and only if the frequencies are pairwise commensurable, that is, there exists $T > 0$ with $\omega T \in 2\pi\mathbb{Z}^n$ (equivalently $\omega_i/\omega_j \in \mathbb{Q}$ for all i, j; if some $\omega_j = 0$, that component is stationary and it suffices to take T to be the least common multiple of the periods of the remaining components); otherwise the orbit is dense in $\mathbb{T}^n$. Note that the condition is not the weaker 'there exists $k \in \mathbb{Z}^n \setminus \{0\}$ with $k \cdot \omega = 0$': the latter only asks the frequencies to be linearly dependent over $\mathbb{Q}$, which holds for $n = 3$, $\omega = (1, 2, \pi)$ with $k = (-2, 1, 0)$, yet that orbit is dense in $\mathbb{T}^3$ and does not close.

Proof. Closedness is equivalent to the existence of $T > 0$ with $\omega T \in 2\pi\mathbb{Z}^n$, that is, $\omega_i/\omega_j \in \mathbb{Q}$ (when $\omega_j \neq 0$); a component with $\omega_j = 0$ contributes no period, and T is taken to be the least common multiple of the periods of the remaining components. Density follows from the Kronecker approximation theorem. ☐

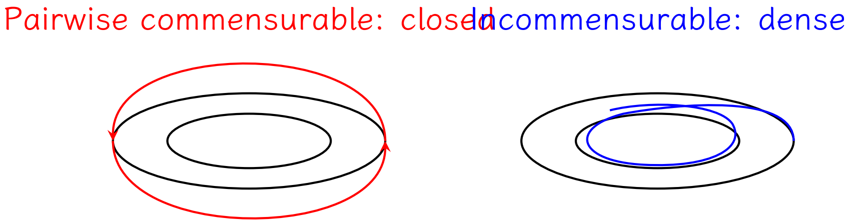


Figure 12.5: Closure criterion for orbits on an invariant torus: when the frequencies are pairwise commensurable the orbit is closed, and when they are incommensurable the orbit is dense.

12.5 Symplectic manifolds and energy surfaces

Definition 12.5 (Symplectic manifold). A pair (M, ω) is called a **symplectic manifold** if M is a smooth manifold of even dimension and ω is a closed non-degenerate 2-form: $d\omega = 0$, and for every $p \in M$ the form $\omega_p : T_pM \times T_pM \to \mathbb{R}$ is non-degenerate.

Definition 12.6 (Hamiltonian vector field). Given $H \in C^\infty(M)$, the **Hamiltonian vector field** X_H is uniquely determined by $\iota_{X_H}\omega = dH$, that is, $\omega(X_H, Y) = dH(Y)$ for every $Y \in \mathfrak{X}(M)$.

Proposition 12.3 (The energy surface is a contact manifold). Let (M, ω) be a symplectic manifold, let $H : M \to \mathbb{R}$ and let $\Sigma_E = H^{-1}(E)$ be the energy surface. If dH is nowhere zero on Σ_E, then Σ_E is an embedded submanifold of codimension 1, and the kernel of $\omega|_{\Sigma_E}$ is one-dimensional and gives a contact structure.

Proof. By the regular level set theorem, Σ_E is a submanifold of codimension 1. The kernel of $\omega|_{\Sigma_E}$ is spanned by X_H (since $\iota_{X_H}\omega = dH$, and dH restricts to 0 on Σ_E), so the kernel is one-dimensional. A contact form may be taken to be the restriction to Σ_E of $\alpha = \iota_{X_H}\omega$. □

Theorem 12.3 (The Weinstein conjecture (proved in dimension three)). The Reeb flow on a compact contact manifold has at least one periodic orbit.

Proof. **The proof lies beyond the scope of this book.** This is the Weinstein conjecture: the three-dimensional case was proved by Taubes (2007) (with subsequent work by Cristofaro-Gardiner–Hutchings (2013)); the general odd-dimensional case remains an open problem. See [41]. □

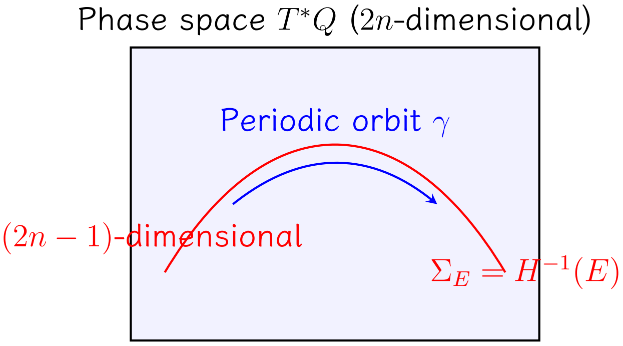


Figure 12.6: The energy surface Σ_E is a $(2n-1)$-dimensional contact hypersurface in phase space, and periodic orbits move on it.

12.6 Principal bundles, connections and holonomy

Definition 12.7 (Principal bundle). Let G be a Lie group, P a smooth manifold and $\pi : P \to M$ a smooth surjection. The quadruple (P, M, π, G) is called a **principal G-bundle** if G acts freely on P on the right, the orbit space P/G is diffeomorphic to M, and the bundle is locally trivial: for every $p \in M$ there exist an open neighbourhood U and a G-equivariant diffeomorphism $\Phi : \pi^{-1}(U) \to U \times G$.

Definition 12.8 (Connection and holonomy). A **connection** on the principal bundle $P \to M$ is a $\mathfrak{g}$-valued 1-form $\mathcal{A} \in \Omega^1(P, \mathfrak{g})$ satisfying $\mathcal{A}(X^\#) = X$ ($X \in \mathfrak{g}$) and $R_g^*\mathcal{A} = \mathrm{Ad}_{g^{-1}}\mathcal{A}$. Given a closed orbit $\gamma : [0, T] \to M$ and a lift $\tilde{\gamma}$, the **holonomy** is

$$\mathrm{Hol}_\gamma(\mathcal{A}) := \mathcal{P}\exp\left(-\int_0^T \mathcal{A}(\dot{\tilde{\gamma}}(t))\, dt\right) \in G,$$

where $\mathcal{P}\exp$ is the path-ordered exponential. The holonomy depends only on the homotopy class of γ (when the curvature $\mathcal{F} = d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}] = 0$).

Proposition 12.4 (Wilson loops as holonomy). The Wilson loop $\mathcal{F}_{ij}$ of the discrete curvature (Definition 12.24) is the discretisation, on a plaquette, of the holonomy of a principal-bundle connection. As the lattice spacing $\Delta x \to 0$, $\mathcal{F}_{ij} \to I - i(\Delta x)^2 F_{ij} + O((\Delta x)^3)$, where F_{ij} is the continuous curvature.

Proof. Discretise the connection $\mathcal{A}$ on the four edges of the plaquette as $\mathcal{U}_{\mathsf{n},i} = e^{-i\Delta x \mathcal{A}_i}$. By the Baker–Campbell–Hausdorff formula, expanding the ordered product of the four edges to second order gives $I - i(\Delta x)^2(\partial_i \mathcal{A}_j - \partial_j \mathcal{A}_i - i[\mathcal{A}_i, \mathcal{A}_j]) + O((\Delta x)^3)$, and the expression in brackets is precisely the curvature F_{ij}. ☐

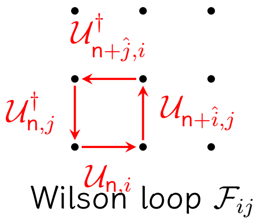

$\mathcal{U}^\dagger_{\mathsf{n}+\hat{j},i}$
$\mathcal{U}^\dagger_{\mathsf{n},j}$
$\mathcal{U}_{\mathsf{n}+\hat{i},j}$
$\mathcal{U}_{\mathsf{n},i}$
Wilson loop $\mathcal{F}_{ij}$

Figure 12.7: Discrete curvature (Wilson loop): the ordered product of the unitary operators on the four edges of a plaquette, that is, the discretisation of the holonomy of a principal-bundle connection.

12.7 The moduli space of periodic orbits

Definition 12.9 (Moduli space of periodic orbits)**.** Let $\mathcal{P}(M)$ be the set of all closed orbits on M and let $\sim$ be the equivalence relation of time translation and reparametrisation. The **moduli space of periodic orbits** is $\mathcal{M}_{\mathsf{per}}(M) := \mathcal{P}(M)/\sim$. When M is a symplectic manifold, $\mathcal{M}_{\mathsf{per}}$ can be endowed with a natural symplectic or contact structure.

Proposition 12.5 (Tangent space of the moduli space)**.** Let γ be a non-degenerate periodic orbit with Floquet multipliers $\rho_1, \ldots, \rho_n$. Then the dimension of the tangent space to $\mathcal{M}_{\mathsf{per}}$ at $[\gamma]$ is $\#\{|\rho_i| = 1\}$ (the number of central directions).

Proof. By Floquet theory, the linearised dynamics near γ decomposes into three subspaces: stable, unstable and central. Deformations of the moduli space proceed only along the central directions (those along which the orbit neither approaches nor recedes from γ exponentially), so the dimension of the tangent space equals the number of central directions. ☐

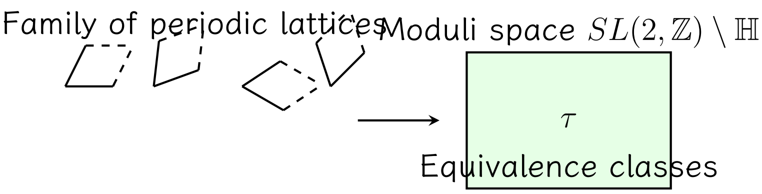

Family of periodic lattices
Moduli space $SL(2,\mathbb{Z}) \setminus \mathbb{H}$
τ
Equivalence classes

Figure 12.8: The moduli space as the classifying space of periodic structures: equivalence classes are parametrised by τ.

12.8 Basic structure of discrete lattice-point manifolds

Axiom 12.1 (Axiom of primitive discreteness)**.** The fundamental structure of space and time is the **discrete lattice point**. The continuum is not a fundamental entity, but a limiting approximation obtained by **infinitely many differentiations** of the discrete lattice (in space) or **infinitely many integrations** (in time). Specifically:

(i) space consists of discrete lattice points, and the lattice spacing $\Delta x > 0$ is the minimal spatial resolution.

(ii) time consists of discrete instants $t_m = m\Delta t$ $(m \in \mathbb{N})$.

(iii) any 'continuous function' is an approximation, in the limit of infinite refinement, of a discrete function on the lattice.

(iv) the resolutions $\Delta x, \Delta t$ are determined by the parameters of the observing system.

Definition 12.10 (Discrete spacetime lattice)**.** The discrete spacetime lattice is $\mathcal{L} = \mathcal{L}_{\text{space}} \times \{t_m : m \in \mathbb{N}\}$. The lattice spacings Δx and Δt are determined by the parameters of the observing system:

$$\Delta x = \frac{1}{R_0} \cdot \frac{r^2 + \sigma}{\sigma}, \quad \Delta t = \frac{N_0}{R_0} \cdot \frac{r^2 + \sigma}{\sigma}.$$

Proposition 12.6 (Relation between lattice spacing and observer scale)**.** The lattice spacing Δx is a strictly decreasing function of the observer scale σ:

$$\frac{\partial(\Delta x)}{\partial \sigma} = -\frac{1}{R_0} \cdot \frac{r^2}{\sigma^2} < 0 \quad (r > 0).$$

Proof. By Definition 12.10, $\Delta x = \frac{1}{R_0}(r^2/\sigma + 1)$. Taking the partial derivative with respect to σ gives $\partial(\Delta x)/\partial\sigma = -r^2/(R_0\sigma^2) < 0$. □

Definition 12.11 (Three kinds of potential vector)**.** On the discrete spacetime lattice, each lattice point stores three kinds of potential vector:

(i) the **spatial potential** V_S: driven by the curl, with $|\mathsf{V}_S|^2$ corresponding to the mass density;

(ii) the **temporal potential** V_T: driven by the gradient, with $|\mathsf{V}_T|^2$ corresponding to the energy density;

(iii) the **observer potential** V_P: driven by the divergence, with $|\mathsf{V}_P|^2$ corresponding to the observer scale σ.

12.9 The three basic operations on the lattice

Definition 12.12 (Translation operation on lattice points)**.** The translation operation $\mathcal{T}_\mathsf{m}$ moves a potential vector from the lattice point $\mathsf{x_n}$ to $\mathsf{x_{n+m}}$: $\mathcal{T}_\mathsf{m} \cdot \mathsf{V}(\mathsf{x_n}) = \mathsf{V}(\mathsf{x_{n+m}})$. Translation leaves the norm and the direction of the vector unchanged.

Definition 12.13 (Scaling operation on lattice points)**.** The scaling operation $\mathcal{S}_\alpha$ changes the norm of a potential vector: $\mathcal{S}_\alpha \cdot \mathsf{V} = \alpha\mathsf{V}$. Scaling changes the norm of the vector by a scale factor and leaves its direction unchanged.

Definition 12.14 (Rotation operation on lattice points)**.** The rotation operation $\mathcal{R}_\theta$ changes the direction of a potential vector and leaves its norm unchanged.

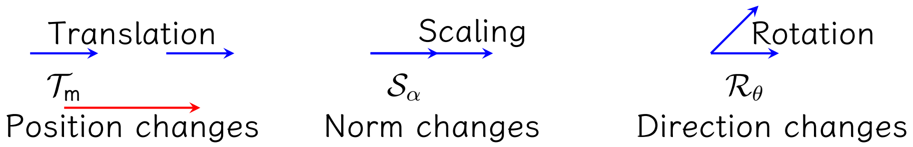


Figure 12.9: The three basic operations: translation changes position, scaling changes the norm and rotation changes direction.

Proposition 12.7 (The irreducibility of the three operations)**.** The three operations of translation, scaling and rotation are mutually independent and cannot be reduced to one another. Any evolution of a potential vector on the lattice can be decomposed into a composite of these three basic operations.

Proof. Translation changes only the position (leaving the norm and the direction untouched), scaling changes only the norm (leaving the position and the direction untouched), and rotation changes only the direction (leaving the position and the norm untouched). The three act on mutually disjoint degrees of freedom, and none of them can be obtained by composing the other two. The evolution of any lattice-point state can be decomposed into the composite $\mathcal{R}_\theta \circ \mathcal{S}_\alpha \circ \mathcal{T}_{\mathsf{m}}$. ☐

Theorem 12.4 (Conservation laws of the three operations on discrete lattice points)**.**

(i) the conserved quantity of the translation operation—momentum: $|\mathcal{T}_{\mathsf{m}} \cdot \mathsf{V}|^2 = |\mathsf{V}|^2$.

(ii) the conserved quantity of the rotation operation—angular momentum: $|\mathcal{R}_\theta \cdot \mathsf{V}|^2 = |\mathsf{V}|^2$.

(iii) the conserved quantity of the scaling operation—the scale charge: when $\alpha = 1$, $|\mathcal{S}_\alpha \cdot \mathsf{V}|^2 = |\mathsf{V}|^2$; conservation of the scale charge is equivalent to $\nabla_\Delta \cdot \mathsf{V}_P = 0$.

Proof. (i) Translation does not change the vector itself. (ii) A rotation matrix is an orthogonal matrix. (iii) When $\alpha = 1$, $\mathcal{S}_1$ is the identity map. ☐

12.10 The rigorous theory of the three great differential operators

12.10.1 The continuous case: intrinsic definitions

Definition 12.15 (Gradient—intrinsic definition)**.** Let (M, g) be an n-dimensional Riemannian manifold and $f \in C^\infty(M)$. The **gradient** ∇f of f is the vector field uniquely determined by $g_p(\nabla f(p), v) = df_p(v)$ $(\forall v \in T_pM)$.

Proposition 12.8 (Coordinate formula for the gradient)**.** *In local coordinates* $(x^1, \ldots, x^n)$*,* $\nabla f = \sum_{i,j} g^{ij} \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^j}$*, where* (g^{ij}) *is the inverse matrix of* (g_{ij})*.*

Proof. Write $\nabla f = \sum_i a^i \frac{\partial}{\partial x^i}$. Since $g(\nabla f, v) = df(v)$ holds for every v, taking $v = \frac{\partial}{\partial x^k}$ gives $\sum_i a^i g_{ik} = \frac{\partial f}{\partial x^k}$. Multiplying both sides by g^{jk} and summing over k, and using $\sum_k g_{ik} g^{jk} = \delta_i^j$, gives $a^j = \sum_k g^{jk} \frac{\partial f}{\partial x^k}$. ☐

Definition 12.16 (Divergence—intrinsic definition)**.** Let (M, g) be an n-dimensional oriented Riemannian manifold, X a smooth vector field and $d\mu$ the Riemannian volume form. The **divergence** $\nabla \cdot X$ of X is defined by $\mathcal{L}_X(d\mu) = (\nabla \cdot X) d\mu$.

Proposition 12.9 (Coordinate formula for the divergence)**.** *In local coordinates,* $\nabla \cdot X = \frac{1}{\sqrt{\det g}} \sum_i \frac{\partial}{\partial x^i}(\sqrt{\det g}\, X^i)$*. In Euclidean space this reduces to* $\nabla \cdot X = \sum_i \frac{\partial X^i}{\partial x^i}$*.*

Proof. From the Cartan formula $\mathcal{L}_X = d \circ \iota_X + \iota_X \circ d$ and $d(d\mu) = 0$ we obtain $\mathcal{L}_X(d\mu) = d(\iota_X d\mu)$. Computing in local coordinates then gives the result. ☐

Definition 12.17 (Curl—intrinsic definition)**.** Let (M, g) be a 3-dimensional oriented Riemannian manifold and X a smooth vector field. The **curl** of X is defined by $(\nabla \times X)^\flat = \star(dX^\flat)$, where $X^\flat$ is the metric dual 1-form and $\star$ is the Hodge star operator.

Proposition 12.10 (Coordinate formula for the curl)**.** *In Euclidean space* $\mathbb{R}^3$*,* $\nabla \times \mathsf{F} = (\partial_2 F_3 - \partial_3 F_2, \partial_3 F_1 - \partial_1 F_3, \partial_1 F_2 - \partial_2 F_1)$*.*

Proof. From $dF^\flat = (\partial_2 F_3 - \partial_3 F_2) dx^2 \wedge dx^3 + \cdots$, taking the Hodge star and raising the index gives the result. ☐

Proposition 12.11 (Identities of the three great operators)**.** *In three-dimensional Euclidean space,* $\nabla \times (\nabla f) = \mathbf{0}$ *and* $\nabla \cdot (\nabla \times \mathsf{F}) = 0$*.*

Proof. $(\nabla \times \nabla f)^\flat = \star d(df) = \star \cdot 0 = 0$; $\nabla \cdot (\nabla \times \mathsf{F}) = \star^{-1} d \star (\star dF^\flat) = \star^{-1} d^2 F^\flat = 0$. ☐

12.10.2 The discrete case: rigorous definitions and derivations

Definition 12.18 (Discrete gradient)**.** Let f be a smooth scalar field on the lattice. The central difference is $(\Delta_i^c f)(\mathsf{x_n}) := \frac{f(\mathsf{x_n} + \Delta x \hat{\mathsf{e}}_i) - f(\mathsf{x_n} - \Delta x \hat{\mathsf{e}}_i)}{2\Delta x}$. The **discrete gradient** is $(\nabla_\Delta f)(\mathsf{x_n}) := ((\Delta_1^c f)(\mathsf{x_n}), \ldots, (\Delta_d^c f)(\mathsf{x_n}))$.

Proposition 12.12 (Error estimate for the discrete gradient)**.** *Let* $f \in C^3$*. Then* $(\Delta_i^c f)(\mathsf{x}) = \frac{\partial f}{\partial x^i}(\mathsf{x}) + \frac{(\Delta x)^2}{6} \frac{\partial^3 f}{\partial (x^i)^3}(\mathsf{x}) + O((\Delta x)^4)$*.*

Proof. Expand $f(\mathsf{x} \pm \Delta x \hat{\mathsf{e}}_i)$ in Taylor series; subtracting the two expressions cancels the even-order terms, and dividing by $2\Delta x$ gives the result. ☐

Definition 12.19 (Discrete divergence)**.** $(\nabla_\Delta \cdot \mathsf{V})(\mathsf{x_n}) := \sum_{i=1}^d (\Delta_i^c V_i)(\mathsf{x_n})$.

Proposition 12.13 (Flux interpretation of the discrete divergence)**.** $(\nabla_\Delta \cdot \mathsf{V})(\mathsf{x_n}) \cdot (\Delta x)^d = \sum_{i=1}^d [V_i(\mathsf{x_n} + \frac{\Delta x}{2} \hat{\mathsf{e}}_i) - V_i(\mathsf{x_n} - \frac{\Delta x}{2} \hat{\mathsf{e}}_i)] (\Delta x)^{d-1} + O((\Delta x)^{d+1})$*.*

Proof. Expand the values in the central difference about the midpoint in Taylor series and rearrange. ☐

Definition 12.20 (Discrete curl). $(\nabla_\Delta \times \mathsf{V})(\mathsf{x_n}) := ((\Delta_2^c V_3 - \Delta_3^c V_2)(\mathsf{x_n}), (\Delta_3^c V_1 - \Delta_1^c V_3)(\mathsf{x_n}), (\Delta_1^c V_2 - \Delta_2^c V_1)(\mathsf{x_n}))$.

Proposition 12.14 (Circulation interpretation of the discrete curl). $(\nabla_\Delta \times \mathsf{V})_k(\mathsf{x_n}) = \frac{1}{(\Delta x)^2} \oint_{C_k} \mathsf{V} \cdot d\mathsf{l} + O(\Delta x)$.

Proof. Take $k = 3$ as an example, approximate the integrals along the four edges of the plaquette by central differences and divide by the area. ☐

Proposition 12.15 (Exact validity of the discrete identities). For every finite $\Delta x > 0$, $\nabla_\Delta \times (\nabla_\Delta f) = \mathsf{0}$ and $\nabla_\Delta \cdot (\nabla_\Delta \times \mathsf{V}) = 0$.

Proof. The central difference operators commute ($\Delta_i^c \Delta_j^c = \Delta_j^c \Delta_i^c$), and after expansion the terms cancel exactly one by one. ☐

12.10.3 Integral theorems

Theorem 12.5 (Stokes' theorem—manifold version). Let M be an n-dimensional oriented smooth manifold, let $\Omega \subseteq M$ be an oriented $(n-1)$-dimensional submanifold with boundary and let ω be an $(n-1)$-form; then $\int_\Omega d\omega = \oint_{\partial\Omega} \omega$.

Proof. **Sketch of the proof** (for the complete proof see [83] §16). Three steps. (i) Localisation: by a partition of unity it suffices to prove the identity within each coordinate chart; pulling back by the coordinates then reduces the matter to the case of a compactly supported form on the upper half-space $\mathbb{H}^n = \{x_n \geq 0\}$. (ii) The upper half-space case: writing $\omega = \sum_{i=1}^n f_i \, dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n$, we have $d\omega = \sum_{i=1}^n (-1)^{i-1} \partial_i f_i \, dx_1 \wedge \cdots \wedge dx_n$. By Fubini's theorem integrate first over x_i, and apply the fundamental theorem of calculus: for $i < n$ the function f_i is compactly supported in the x_i direction, so the integral vanishes; for $i = n$ only the boundary $\{x_n = 0\}$ leaves a contribution, giving $\int_{\mathbb{H}^n} \partial_n f_n \, dx = -\int_{\partial\mathbb{H}^n} f_n$ (the minus sign comes from the outward normal pointing towards $-x_n$). (iii) Patching: summing the local identities with the partition of unity, the induced orientations on the common interior faces are opposite, so the contributions cancel in pairs and only the integral over $\partial\Omega$ remains, which yields $\int_\Omega d\omega = \oint_{\partial\Omega} \omega$. ☐

Theorem 12.6 (The Gauss divergence theorem—manifold version). Let (M, g) be an n-dimensional oriented Riemannian manifold, let $\Omega \subseteq M$ be compact with smooth boundary $\partial\Omega$ and let n be the outward normal vector field; then $\int_\Omega (\nabla \cdot X) d\mu = \oint_{\partial\Omega} g(X, \mathsf{n}) dS$.

Proof. By $\mathcal{L}_X(d\mu) = (\nabla \cdot X) d\mu$ and Stokes' theorem, $\int_\Omega (\nabla \cdot X) d\mu = \oint_{\partial\Omega} \iota_X d\mu$, and the boundary term restricts to $g(X, \mathsf{n}) dS$. ☐

Corollary 12.3 (Green's identity). $\int_\Omega (f \nabla^2 h + g(\nabla f, \nabla h)) d\mu = \oint_{\partial\Omega} f \, g(\nabla h, \mathsf{n}) dS$.

Proof. Take $X = f \nabla h$, apply the Gauss divergence theorem and expand the divergence. ☐

Proposition 12.16 (Discrete Green's formula and the discrete divergence theorem). On the discrete lattice, $\sum_{\mathsf{n} \in V} (\nabla_\Delta \cdot \mathsf{V})(\mathsf{x_n})(\Delta x)^d = \sum_{\partial V} \mathsf{V} \cdot \mathsf{n} (\Delta x)^{d-1}$, and this holds exactly for every finite Δx.

Proof. Each interior edge is counted once by each of the two adjacent cells, in opposite directions, and cancels exactly; only the boundary contribution remains. ☐

12.10.4 Complete proof of the continuum limit

Theorem 12.7 (Continuum limit theorem—full version)**.** Let $f \in C^3$ and $\mathsf{V} \in C^3$. As $\Delta x \to 0$:

(i) $\lim \nabla_\Delta f = \nabla f$, with error $O((\Delta x)^2)$;

(ii) $\lim \nabla_\Delta \cdot \mathsf{V} = \nabla \cdot \mathsf{V}$, with error $O((\Delta x)^2)$;

(iii) $\lim \nabla_\Delta \times \mathsf{V} = \nabla \times \mathsf{V}$, with error $O((\Delta x)^2)$;

(iv) for every finite Δt, $\mathcal{U}_\Delta^N = e^{-i\mathcal{H}t}$ holds exactly ($N = t/\Delta t$);

(v) $\lim \mathcal{F}_{ij} = I - i(\Delta x)^2 F_{ij} + O((\Delta x)^3)$.

Proof. (i) By Proposition 12.12. (ii) Apply (i) to each component. (iii) The discrete curl is a linear combination of the $\Delta_i^c V_j$. (iv) $\mathcal{U}_\Delta = e^{-i\Delta t\mathcal{H}}$, and N iterations give $e^{-i\mathcal{H}t}$. (v) By Proposition 12.27. ☐

12.11 Discrete unitary theory

Definition 12.21 (Discrete unitary operator)**.** A **discrete unitary operator** $\mathcal{U}_\mathsf{n}$ at the lattice point x_n satisfies $\mathcal{U}_\mathsf{n}^\dagger \mathcal{U}_\mathsf{n} = I_d$. The potential vector evolves by $\mathsf{V}(t_{m+1}) = \mathcal{U}_\mathsf{n}\mathsf{V}(t_m)$.

Proposition 12.17 (Unitarity and conservation of the squared norm)**.** $|\mathsf{V}(t_{m+1})|^2 = |\mathsf{V}(t_m)|^2$.

Proof. $|\mathcal{U}\mathsf{V}|^2 = (\mathcal{U}\mathsf{V})^\dagger(\mathcal{U}\mathsf{V}) = \mathsf{V}^\dagger\mathcal{U}^\dagger\mathcal{U}\mathsf{V} = \mathsf{V}^\dagger\mathsf{V} = |\mathsf{V}|^2$. ☐

Proposition 12.18 (Unitary evolution and conservation of the total lattice energy)**.** $E_{\text{total}} = \sum_\mathsf{n} |\mathsf{V}_\mathsf{n}|^2$ is conserved under the evolution.

Proof. The squared norm at each lattice point is conserved independently; summing gives the result. ☐

Proposition 12.19 (Correspondence between unitarity and the three conservation laws)**.** Conservation of energy, conservation of angular momentum and scale invariance.

Proof. (i) By Proposition 12.17. (ii) $|\mathsf{V}_S \times \mathcal{R}_\theta \mathsf{V}_S| = |\mathsf{V}_S|^2 |\sin\theta|$. (iii) When $\nabla_\Delta \cdot \mathsf{V}_P = 0$ the evolution operator is a pure rotation. ☐

Corollary 12.4 (Discrete unitary evolution and Noether's theorem)**.** The discrete conservation laws do not depend on the limit $\Delta t \to 0$.

Proof. This follows directly from Proposition 12.19. ☐

Definition 12.22 (Generator of discrete unitary evolution)**.** $\mathcal{U}_\mathsf{n} = e^{-i\Delta t\mathcal{H}_\mathsf{n}}$, where $\mathcal{H}_\mathsf{n}$ is a Hermitian matrix.

Proposition 12.20 (Equivalence of the hermiticity of the generator and the unitarity of the evolution operator)**.** $\mathcal{U} = e^{-i\Delta t\mathcal{H}}$ is unitary if and only if $\mathcal{H}$ is Hermitian.

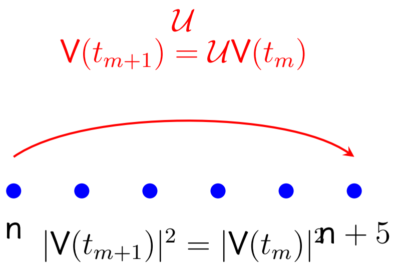

$\mathcal{U}$
$\mathsf{V}(t_{m+1}) = \mathcal{U}\mathsf{V}(t_m)$
n
$|\mathsf{V}(t_{m+1})|^2 = |\mathsf{V}(t_m)|^2$
$n+5$

Figure 12.10: Discrete unitary evolution: after a unitary operator acts on a potential vector on the lattice, the squared norm is conserved.

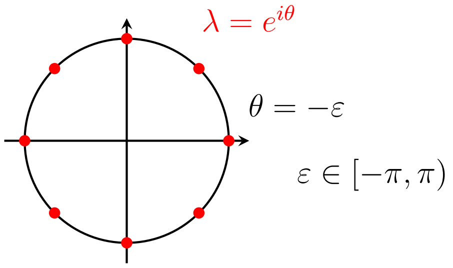
Proof. ($\Leftarrow$) $\mathcal{U}^\dagger\mathcal{U} = e^{i\Delta t\mathcal{H}^\dagger}e^{-i\Delta t\mathcal{H}} = I$. ($\Rightarrow$) If $\mathcal{U}$ is unitary, then its spectrum lies on the unit circle, so the spectrum of $\mathcal{H}$ is real; moreover $\mathcal{U}^\dagger = \mathcal{U}^{-1} = e^{i\Delta t\mathcal{H}}$, hence $\mathcal{H}^\dagger = \mathcal{H}$. ☐

Proposition 12.21 (Spectral structure of discrete unitary operators). Every eigenvalue λ of a discrete unitary operator $\mathcal{U}$ satisfies $|\lambda| = 1$.

Proof. $\mathcal{U}\mathsf{v} = \lambda\mathsf{v}$, so $\|\mathsf{v}\|^2 = \|\mathcal{U}\mathsf{v}\|^2 = |\lambda|^2\|\mathsf{v}\|^2$, hence $|\lambda| = 1$. ☐

Proposition 12.22 (Discrete unitary operators and Floquet theory). A discrete unitary operator $\mathcal{U}$ may be regarded as a Floquet operator. Its eigenvalue $e^{-i\varepsilon}$ corresponds to the quasienergy ε, which is uniquely determined within the Floquet–Brillouin zone $[-\pi, \pi)$.

Proof. Since $\mathcal{U} = e^{-i\Delta t\mathcal{H}}$, the eigenvalue $e^{-i\varepsilon}$ corresponds to $\varepsilon = \Delta t \cdot E$. As $e^{-i\varepsilon}$ is periodic with period 2π, the value ε is determined only modulo 2π. ☐



$|\lambda| = 1$: the eigenvalues lie on the unit circle

Figure 12.11: The spectrum of a discrete unitary operator: all eigenvalues lie on the unit circle, corresponding to quasienergies within the Floquet–Brillouin zone.

Proposition 12.23 (Period condition for a discrete unitary operator). $\mathcal{U}^p = I$ if and only if all eigenvalues are p-th roots of unity.

Proof. $\mathcal{U}^p = I$ is equivalent to the p-th powers of all eigenvalues of $\mathcal{U}$ being 1. ☐

Corollary 12.5 (Rational dependence between periodicity and the quasienergies). $\mathcal{U}$ has period p if and only if all quasienergies ε_j satisfy $\varepsilon_j \in \frac{2\pi}{p}\mathbb{Z}$ (modulo 2π).

Proof. From $\lambda_j = e^{-i\varepsilon_j}$ and $\lambda_j^p = 1$ we obtain $e^{-ip\varepsilon_j} = 1$, that is, $p\varepsilon_j \in 2\pi\mathbb{Z}$. ☐

Proposition 12.24 (Evolution operator of the discrete-time quantum walk). The evolution of the discrete-time quantum walk is governed by the periodic unitary operator $\mathcal{U} = SR$, where S is the coin-state-dependent shift operator and R is the coin operator. The state after m steps is $|\Psi_m\rangle = (SR)^m|\Psi_0\rangle$.

Proof. This is the standard definition of the discrete-time quantum walk. Both S and R are unitary operators, so their product SR is unitary as well. The position-independent coin operator together with the periodic boundary conditions gives translation symmetry, so that S can be diagonalised by Fourier transform. □

Proposition 12.25 (Discrete unitary evolution and conservation of the discrete current). Discrete unitary evolution preserves the conservation of the discrete probability current: $\partial_t \rho_n + \sum_i \Delta_i^c j_{n,i} = 0$, where $\rho_n = |\Psi_n|^2$ is the lattice probability density.

Proof. Under the unitary evolution $|\Psi_{m+1}\rangle = \mathcal{U}|\Psi_m\rangle$, the probability density satisfies $\rho_{m+1,n} - \rho_{m,n} = \langle\Psi_m|\mathcal{U}^\dagger P_n \mathcal{U} - P_n|\Psi_m\rangle$, which for a local unitary operator can be written as the difference of the currents at neighbouring lattice points. □

Example 12.4 (Second-order periodic oscillation on the lattice). $\mathcal{U} = -I$, $\mathcal{U}^2 = I$.

Example 12.5 (C_4 rotational period on the lattice). $\mathcal{U} = R_{\pi/2}$, $\mathcal{U}^4 = I$.

12.12 Discrete curvature

Definition 12.23 (Minimal closed loop on the lattice). A plaquette on a two-dimensional square lattice consists of four edges.

Definition 12.24 (Discrete curvature—the Wilson loop). $\mathcal{F}_{ij} = \mathcal{U}_{\mathsf{n},i}\mathcal{U}_{\mathsf{n}+\hat{\mathsf{e}}_i,j}\mathcal{U}^\dagger_{\mathsf{n}+\hat{\mathsf{e}}_j,i}\mathcal{U}^\dagger_{\mathsf{n},j}$.

Proposition 12.26 (Unitarity of the discrete curvature). $\mathcal{F}_{ij}^\dagger \mathcal{F}_{ij} = I$.

Proof. Verify directly, using the unitarity of the edge operators. □

Proposition 12.27 (Relation between the discrete and the continuous curvature). $\mathcal{F}_{ij} = I - i(\Delta x)^2 F_{ij} + O((\Delta x)^3)$.

Proof. Expand in Taylor series and apply the Baker–Campbell–Hausdorff formula. □

Lemma 12.1 (Homotopy invariance of the Wilson loop). In a flat region the Wilson loop of a contractible loop is I, and the Wilson loop depends only on the homotopy class.

Proof. Take the product of the Wilson loops of all plaquettes inside the loop; the interior edges cancel edge by edge, and only the boundary contribution remains. □

Corollary 12.6 (Global phase locking of a flat connection). When all plaquettes are flat, the Wilson loop of any closed loop is I.

Proof. This follows directly from Lemma 12.1. □

Lemma 12.2 (Wilson-loop product identity on the torus)**.** The product of the Wilson loops of all plaquettes on the torus equals 1.

Proof. Each edge belongs to exactly two plaquettes and occurs once in each, in opposite directions, so the edges cancel one by one. $\square$

Corollary 12.7 (Discrete flux quantisation)**.** The total phase satisfies $\sum \theta_{\mathsf{n}} \in 2\pi\mathbb{Z}$.

Proof. This follows directly from Lemma 12.2. $\square$

12.13 Manifestation of the conservation laws on periodic manifolds

Proposition 12.28 (Energy conservation on the lattice)**.** $E_{\mathsf{n}} = |\mathsf{V}_T|^2$ is exactly conserved under the evolution.

Proof. This follows directly from Proposition 12.17. $\square$

Proposition 12.29 (Conservation of angular momentum on the lattice)**.** L_{n} is conserved if and only if the rotation angle is constant.

Proof. Since $L_{\mathsf{n}} = |\mathsf{V}_S|^2|\sin\theta|$, conservation of the squared norm gives the conclusion. $\square$

Proposition 12.30 (Scale invariance on the lattice)**.** Scale invariance is equivalent to $\nabla_\Delta \cdot \mathsf{V}_P = 0$.

Proof. When there is no net scaling flux the evolution operator is a pure rotation, and the squared norm is conserved. $\square$

Theorem 12.8 (Unification of the conservation laws on the three periodic manifolds)**.** Translation-periodic manifolds (conservation of momentum), rotation-periodic manifolds (conservation of angular momentum) and scaling-periodic manifolds (conservation of the scale charge).

Proof. This is the term-by-term correspondence given by Propositions 12.28, 12.29 and 12.30. $\square$

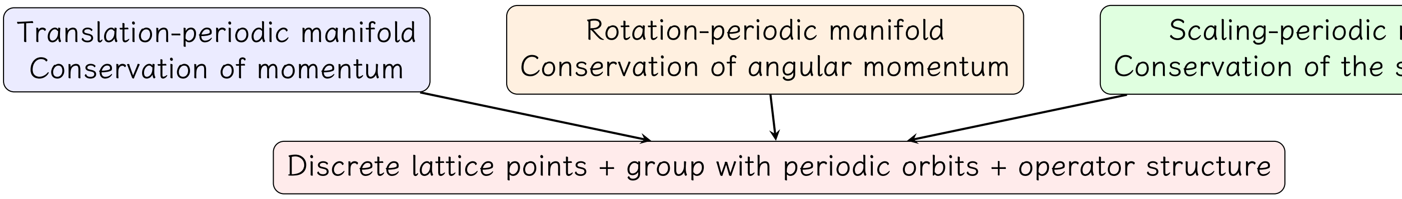

Translation-periodic manifold
Conservation of momentum
Rotation-periodic manifold
Conservation of angular momentum
Scaling-periodic
Conservation of the
Discrete lattice points + group with periodic orbits + operator structure

Figure 12.12: Unification of the three kinds of periodic manifold: each kind of manifold corresponds to one conservation law.

12.14 Classification of periodic manifolds

Definition 12.25 (Translation-periodic manifold). *A discrete lattice-point manifold invariant under the action of the translation group.*

Definition 12.26 (Rotation-periodic manifold). *A discrete lattice-point manifold invariant under the action of the rotation group.*

Definition 12.27 (Scaling-periodic manifold). *A discrete lattice-point manifold invariant under the action of the scaling group.*

Theorem 12.9 (Unity of the three kinds of periodic manifold). *Periodic manifold = discrete lattice points + group with periodic orbits + operator structure.*

Proof. The three kinds of periodic manifold are respectively the instances in which the discrete lattice points are invariant under the translation group, the rotation group and the scaling group. ☐

12.15 Periodic evolution of potential vectors on the lattice

Definition 12.28 (Discrete periodic evolution on the lattice). *When $\mathcal{U}^p = I$, the evolution has the strict discrete period p.*

Example 12.6 (Second-order periodic oscillation on the lattice). $\mathcal{U} = -I$, $\mathcal{U}^2 = I$.

Example 12.7 (The C_4 rotation period on the lattice). $\mathcal{U} = R_{\pi/2}$, $\mathcal{U}^4 = I$.

Seven-theme comparison table for this chapter

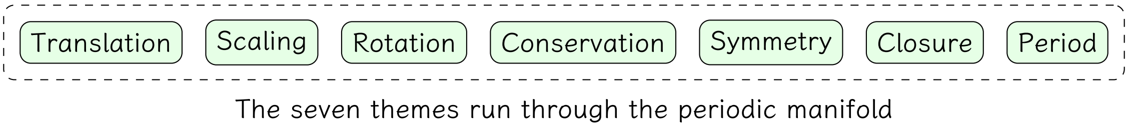


Figure 12.13: The seven great themes of this chapter: translation, scaling, rotation, conservation, symmetry, closure and period.

Theme	Corresponding content in this chapter
Translation	Lattice translation operation (Definition 12.12); discrete gradient (Definition 12.18); continuous gradient (Definition 12.15)
Scaling	Lattice scaling operation (Definition 12.13); discrete divergence (Definition 12.19); continuous divergence (Definition 12.16)
Rotation	Lattice rotation operation (Definition 12.14); discrete curl (Definition 12.20); continuous curl (Definition 12.17)
Conservation	Conservation of the squared norm (Proposition 12.17); the Gauss divergence theorem (Theorem 12.6); Green's identity (Corollary 12.3)
Symmetry	Axiom of primitive discreteness (Axiom 12.1); identities of the three great operators (Proposition 12.11)
Closure	Wilson-loop closure criterion (Lemma 12.1); Stokes' theorem (Theorem 12.5)
Period	Discrete period condition (Proposition 12.23); continuum limit theorem (Theorem 12.7); rational dependence of the quasienergies (Corollary 12.5)

Exercises

Exercise 1 Using Definition 12.1, verify that the n-dimensional torus $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ is a periodic manifold: check one by one the three conditions that the group action is free, that it is properly discontinuous and that the quotient space is a smooth manifold, and point out that the orbits of the action of $G = \mathbb{Z}^n$ are periodic orbits.

Exercise 2 Prove that the fundamental group of the lens space $L(p,q) = S^3/\mathbb{Z}_p$ (Example 12.2) is $\pi_1(L(p,q)) \cong \mathbb{Z}_p$. Explain why every closed orbit on $L(p,q)$ is a twisted orbit of finite order, and compute the twisting order when $p = 5$.

Exercise 3 Prove the fundamental group of the torus (Theorem 11.15): $\pi_1(\mathbb{T}^n) \cong \mathbb{Z}^n$. Use covering-space theory to show that $\pi_1(\mathbb{T}^n) \cong \pi_1(S^1)^n$, and explain why the only twisted orbit on $\mathbb{T}^n$ is the trivial zero element (see Corollary 12.1).

Exercise 4 Verify Proposition 12.2: the integral $\oint_\gamma \omega$ of a closed form ω along a closed orbit γ depends only on the cohomology class of ω and the homology class of γ. Prove separately that if ω is exact then the integral is zero, and that if γ is homologous to γ' then the two integrals are equal.

Exercise 5 Verify that the harmonic 1-forms on the torus in Example 12.3:

$$d\theta_1, \dots, d\theta_n$$

satisfy $\oint_{\gamma_k} d\theta_j = \delta_{jk}$ and $H^1_{dR}(\mathbb{T}^n) \cong \mathbb{R}^n$. Write out explicitly the closed 1-forms and the integral computation for $n = 2$.

Exercise 6 Prove the Liouville–Arnold theorem (Theorem 12.2): let a Hamiltonian system with n degrees of freedom possess n independent first integrals in involution, and let its level set be compact and connected. Then the level set is diffeomorphic to $\mathbb{T}^n$, and near it there exist action-angle coordinates (I, θ) such that

$$\omega = \sum_i dI_i \wedge d\theta_i, \quad \dot{I}_i = 0, \quad \dot{\theta}_i = \omega_i(I).$$

Explain how the commuting Hamiltonian flows give the torus structure.

Exercise 7 Prove the criteria for periodic and quasiperiodic orbits on an invariant torus (Corollary 12.2): on an invariant torus $\mathbb{T}^n$, the orbit $\theta(t) = \theta_0 + \omega t$ is closed if and only if the frequencies are pairwise commensurable (that is, there exists $T > 0$ with $\omega T \in 2\pi\mathbb{Z}^n$); otherwise the orbit is dense in $\mathbb{T}^n$. Write out the proofs for the closed case and for the dense case separately, giving an example of each.

Exercise 8 Prove that the energy surface is a contact manifold (Proposition 12.3): let (M, ω) be a symplectic manifold, $H : M \to \mathbb{R}$, and let $\Sigma_E = H^{-1}(E)$ be the energy surface. If dH is nowhere zero on Σ_E, then Σ_E is an embedded submanifold of codimension 1, and the kernel of $\omega|_{\Sigma_E}$ is one-dimensional and gives a contact structure. Use the regular level set theorem and the Hamiltonian vector field to show that the kernel is one-dimensional.

Exercise 9 Prove Proposition 12.4: the Wilson loop of the discrete curvature (Definition 12.24) $\mathcal{F}_{ij}$ is the discretisation, on a plaquette, of the holonomy of a principal-bundle connection. As the lattice spacing $\Delta x \to 0$,

$$\mathcal{F}_{ij} \to I - i(\Delta x)^2 F_{ij} + O((\Delta x)^3).$$

Complete the derivation by expanding the ordered product of the four edges with the Baker–Campbell–Hausdorff formula.

Exercise 10 Verify Proposition 12.5: let γ be a non-degenerate periodic orbit with Floquet multipliers $\rho_1, \ldots, \rho_n$; then the dimension of the tangent space to the moduli space of periodic orbits $\mathcal{M}_{\text{per}}$ at $[\gamma]$ is $\#\{|\rho_i| = 1\}$ (the number of central directions). Use Floquet theory to explain that deformations of the moduli space proceed only along the central directions.

Exercise 11 Prove the error estimate for the discrete gradient operator (Proposition 12.12): let $f \in C^3$ and let the central difference be

$$(\Delta_i^c f)(\mathsf{x}) = \frac{f(\mathsf{x} + \Delta x \hat{\mathsf{e}}_i) - f(\mathsf{x} - \Delta x \hat{\mathsf{e}}_i)}{2\Delta x},$$

then

$$(\Delta_i^c f)(\mathsf{x}) = \frac{\partial f}{\partial x^i}(\mathsf{x}) + \frac{(\Delta x)^2}{6} \frac{\partial^3 f}{\partial (x^i)^3}(\mathsf{x}) + O((\Delta x)^4).$$

Complete the proof by writing out the Taylor expansions of $f(\mathsf{x} \pm \Delta x \hat{\mathsf{e}}_i)$ and subtracting them.

Exercise 12 Prove the exact validity of the discrete identities (Proposition 12.15): for every finite $\Delta x > 0$,

$$\nabla_\Delta \times (\nabla_\Delta f) = \mathbf{0}, \qquad \nabla_\Delta \cdot (\nabla_\Delta \times \mathsf{V}) = 0.$$

Using the commutativity of the central difference operators ($\Delta_i^c \Delta_j^c = \Delta_j^c \Delta_i^c$), explain that the cancellation is exact and does not depend on $\Delta x \to 0$.

Exercise 13 Prove the continuum limit theorem (Theorem 12.7): as $\Delta x \to 0$, the discrete gradient, divergence and curl converge to the corresponding continuous operators, each with error $O((\Delta x)^2)$. Write out the error estimates for the three operators separately, and explain why central differences give second-order accuracy.

Exercise 14 Prove conservation of the squared norm under discrete unitary evolution (Proposition 12.17): if $\mathcal{U}_\mathsf{n}$ is unitary, then

$$|\mathsf{V}(t_{m+1})|^2 = |\mathsf{V}(t_m)|^2.$$

Verify that $|\mathcal{U}\mathsf{V}|^2 = (\mathcal{U}\mathsf{V})^\dagger(\mathcal{U}\mathsf{V}) = |\mathsf{V}|^2$, and explain the conservation of the total lattice energy (Proposition 12.18).

Exercise 15 Prove the spectral structure of discrete unitary operators (Proposition 12.21): every eigenvalue λ of a discrete unitary operator $\mathcal{U}$ satisfies $|\lambda| = 1$. Complete the proof using $\|\mathsf{v}\|^2 = \|\mathcal{U}\mathsf{v}\|^2 = |\lambda|^2\|\mathsf{v}\|^2$, and explain its relation to Floquet theory (Proposition 12.22).

Exercise 16 Prove the condition for discrete periodic evolution (Proposition 12.23): $\mathcal{U}^p = I$ if and only if all eigenvalues are p-th roots of unity. Further prove the criterion for rational dependence of the quasienergies (Corollary 12.5): $\mathcal{U}$ has period p if and only if all quasienergies $\varepsilon_j \in \frac{2\pi}{p}\mathbb{Z}$ (modulo 2π).

Exercise 17 Verify Example 12.4 and Example 12.5:

(i) taking $\mathcal{U} = -I$, verify that $\mathcal{U}^2 = I$ and that the least period is 2;

(ii) taking $\mathcal{U} = R_{\pi/2}$, verify that $\mathcal{U}^4 = I$ and that the least period is 4.

Point out which cyclic group of which order each of these two examples corresponds to, and explain their correspondence with the finite-order actions in the 'Operations and Periodicity' chapter.

Exercise 18 Prove the unitarity of the discrete curvature (Proposition 12.26): the Wilson loop

$$\mathcal{F}_{ij} = \mathcal{U}_{\mathsf{n},i}\mathcal{U}_{\mathsf{n}+\hat{\mathsf{e}}_i,j}\mathcal{U}^\dagger_{\mathsf{n}+\hat{\mathsf{e}}_j,i}\mathcal{U}^\dagger_{\mathsf{n},j}$$

satisfies $\mathcal{F}_{ij}^\dagger\mathcal{F}_{ij} = I$. Verify this directly using the unitarity of the edge operators.

Exercise 19 Prove the homotopy invariance of the Wilson loop (Lemma 12.1): in a flat region the Wilson loop of a contractible loop is I, and the Wilson loop depends only on the homotopy class. Complete the proof with the edge-by-edge cancellation argument for interior edges, and explain the connection between the Wilson-loop product identity on the torus (Lemma 12.2) and discrete flux quantisation (Corollary 12.7).

Exercise 20 Synthesis: *combining the theory of the three great differential operators of this chapter (the continuous case* §12.10*, the discrete case* §12.10*), the discrete unitary theory (*§12.11*) and the discrete curvature (*§12.12*), explain how the theory of periodic manifolds unifies the periodic structures on continuous manifolds (closed orbits, invariant tori, conservation of symplectic volume) and the periodic structures on discrete lattice points (unitary evolution, Wilson loops, conservation laws) within one and the same framework.* Point out respectively what the counterpart of a *'closed orbit' is in the continuous and in the discrete case.*

Chapter 13

Periodic Category

Remark 13.1 (External dependencies of this chapter, and the order of reading). This chapter lifts the conclusions of the preceding chapters to the level of categories, and its exposition therefore cites several theorems from chapters that **come later** in the book. So that every step can be verified, these dependencies are collected here; each is given an explicit cross-reference at the point of use.

(i) Theorem 15.11 (the 'Vector Analysis' chapter)—the Helmholtz decomposition;

(ii) Remark 20.4 (the 'Periodic Functions and Groups with Periodic Orbits' chapter)—Noether's theorem at the level of periodic functions;

(iii) Proposition 21.2, Theorem 21.4, Theorem 21.5 (the 'Discrete Sampling and Intercalation of Periodic Signals' chapter)—the commensurability parallel proposition, boundedness of the intercalation error, and continued-fraction optimality;

(iv) Theorem 21.9 (the 'Discrete Sampling and Intercalation of Periodic Signals' chapter), Theorem 22.1, Theorem 22.11 (the 'Harmonic Analysis' chapter)—the discrete and continuous Parseval identities, and the DFT leakage estimate.

Advice on reading: the abstract conclusions of this chapter may be regarded as a **formal packaging** of concrete results already proved, and its proofs introduce no new substantial mathematics. On a first reading the chapter may be skipped and returned to after the 'Harmonic Analysis' chapter, by which point every dependency is in place.

Remark 13.2 (Positioning of This Chapter). Periodic categories constitute the highest level of unifying theory within periodic mathematics. This chapter unifies the mathematical structures built up in the preceding chapters—groups with periodic orbits, periodic functions, periodic convolution and periodic manifolds—within the language of category theory. Category theory provides the most abstract framework of expression for periodic mathematics, revealing the deep structural identity among different periodic structures.

This chapter first rigorously distinguishes the two basic concepts of **operator** and **functor**: an operator is a self-map within a single category, whilst a functor is a structure-preserving map between different categories. On this basis, the chapter establishes the complete theory of the **phase functor**: for any period T and any phase $\theta \in [0, 2\pi)$, there exists a phase functor $\Phi_{\theta,T}$ that maps the translation-periodic category to itself. All phase functors form a parametrised family of functors.

The core content of this chapter:

(1) **Rigorous distinction between operators and functors**: the essential differences between operators (Definition 13.2) and functors (Definition 13.3) in scope and in the preservation of structure.

(2) **Phase functors**: for any period T and any phase θ, the phase functor $\Phi_{\theta,T}$ (Definition 13.4) realises the phase translation of periodic structures.

(3) **Three basic periodic categories**: the translation-periodic category, the scaling-periodic category and the rotation-periodic category (Definitions 13.9 to 13.11).

(4) **Functors as category converters**: the logarithm functor effects the conversion from the scaling category to the translation category, and the Fourier functor effects the conversion from the translation category to the frequency-domain representation category.

(5) **Adjunction of the differential and integral operators**: $\mathcal{D} \dashv \mathcal{I}$.

(6) **Periodic convolution as a bifunctor**: periodic convolution is a bifunctor on the periodic category.

(7) **The observer as a functor**: the observer projects the periodic structures of the time system and the space system onto observable traces.

Guide to the seven themes of this chapter: Translation—the translation-periodic category $\mathcal{C}_{\text{trans}}$ and the differential operator $\mathcal{D}$ (0-symmetry); Scaling—the scaling-periodic category and the logarithm/exponential functors (1-symmetry); Rotation—the rotation-periodic category and the phase functor (i-symmetry); Conservation—the conservation condition for adjoint operators ($\mathcal{I} \circ \mathcal{D} \cong \mathrm{id}$ requires the integral to vanish, corresponding to conservation of angular momentum); Symmetry—natural isomorphism and the adjoint duality $\mathcal{D} \dashv \mathcal{I}$; Closure—the mutual-inverse closure of the differential and integral operators, and the closure property of the periodic-convolution bifunctor; Period—the categorical formulation of periodic manifolds and the parametrisation of the phase functors.

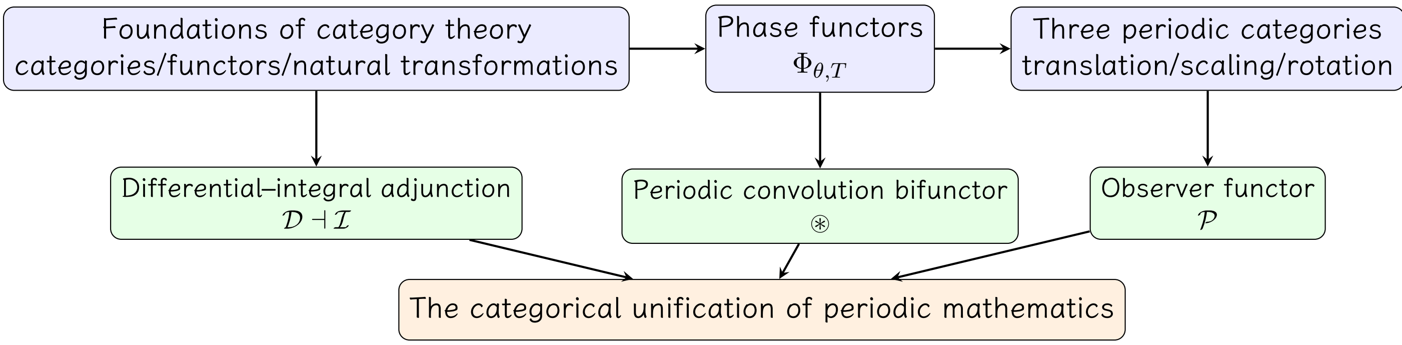

Foundations of category theory
categories/functors/natural transformations
Phase functors
$\Phi_{\theta,T}$
Three periodic categories
translation/scaling/rotation
Differential–integral adjunction
$\mathcal{D} \dashv \mathcal{I}$
Periodic convolution bifunctor
$\circledast$
Observer functor
$\mathcal{P}$
The categorical unification of periodic mathematics

Figure 13.1: Overview of the structure of this chapter: from the foundations of category theory, via the phase functor and the three periodic categories, to the differential–integral adjunction, the periodic convolution bifunctor and the observer functor, ultimately unified within the categorical framework of periodic mathematics.

13.1 Foundations of Category Theory

Definition 13.1 (Category)**.** A **category** $\mathcal{C}$ consists of the following data:

(i) A class of **objects** $\mathsf{Ob}(\mathcal{C})$.

(ii) For any two objects $A, B \in \mathsf{Ob}(\mathcal{C})$, a set of **morphisms** $\mathsf{Hom}_{\mathcal{C}}(A, B)$.

(iii) For any object A, an **identity morphism** $\mathsf{id}_A \in \mathsf{Hom}_{\mathcal{C}}(A, A)$.

(iv) For any three objects A, B, C, a **composition** operation $\circ : \mathsf{Hom}_{\mathcal{C}}(B, C) \times \mathsf{Hom}_{\mathcal{C}}(A, B) \to \mathsf{Hom}_{\mathcal{C}}(A, C)$.

satisfying the following axioms:

- **Associativity**: for any $f : A \to B, g : B \to C, h : C \to D$, we have $h \circ (g \circ f) = (h \circ g) \circ f$.
- **Unit law**: for any $f : A \to B$, we have $f \circ \mathsf{id}_A = f = \mathsf{id}_B \circ f$.

Example 13.1 (Examples of Categories in Periodic Mathematics)**.**

- **Category of sets Set**: objects are sets, morphisms are functions.
- **Category of vector spaces $\mathsf{Vect}_{\mathbb{C}}$**: objects are complex vector spaces, morphisms are linear maps.
- **Category of groups Grp**: objects are groups, morphisms are group homomorphisms.
- **Category of topological spaces Top**: objects are topological spaces, morphisms are continuous maps.
- **Category of smooth manifolds Man**: objects are smooth manifolds, morphisms are smooth maps.

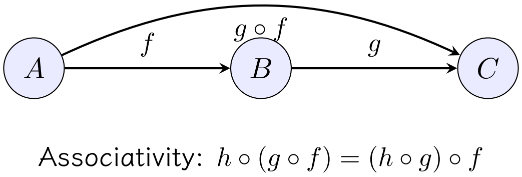


Figure 13.2: The basic structure of a category: objects, morphisms and composition.

13.2 Operators and Functors

Definition 13.2 (Operator)**.** Let $\mathcal{C}$ be a category. An **operator** on $\mathcal{C}$ is a mapping $T : \mathsf{Ob}(\mathcal{C}) \to \mathsf{Ob}(\mathcal{C})$ such that for any morphism $f : A \to B$ there exists a morphism $T(f) : T(A) \to T(B)$, and $T(\mathsf{id}_A) = \mathsf{id}_{T(A)}$, $T(g \circ f) = T(g) \circ T(f)$.

In other words, an operator is a **functor from a category to itself** (an endofunctor), namely $T : \mathcal{C} \to \mathcal{C}$. An operator preserves the internal structure of the category and does not change the category to which the objects belong.

In periodic mathematics, the differential operator $\mathcal{D}$ and the integral operator $\mathcal{I}$ are both operators (endofunctors) on the translation-periodic category $\mathcal{C}_{\text{trans}}$.

Definition 13.3 (Functor). Let $\mathcal{C}$ and $\mathcal{D}$ be two categories. A **functor** $F : \mathcal{C} \to \mathcal{D}$ consists of the following data:

(i) For each object $A \in \mathsf{Ob}(\mathcal{C})$, an object $F(A) \in \mathsf{Ob}(\mathcal{D})$.

(ii) For each morphism $f : A \to B$ (in $\mathcal{C}$), a morphism $F(f) : F(A) \to F(B)$ (in $\mathcal{D}$).

satisfying $F(\mathsf{id}_A) = \mathsf{id}_{F(A)}$ and $F(g \circ f) = F(g) \circ F(f)$.

If F reverses the direction of morphisms (that is, $F(f) : F(B) \to F(A)$), then F is called a **contravariant functor**.

Remark 13.3 (Distinction between Operators and Functors). *The essential difference between operators and functors lies in their* **scope***:*

- *An* **operator** *(endofunctor) $T : \mathcal{C} \to \mathcal{C}$ maps a category to* **itself***, preserving the types of objects and morphisms. The differential operator $\mathcal{D}$ maps periodic functions to periodic functions and periodic morphisms to periodic morphisms—the periodic structure is preserved under the action of the operator.*
- *A* **functor** *$F : \mathcal{C} \to \mathcal{D}$ maps a category to* **another category***, changing the types of objects and morphisms. The Fourier functor $\mathcal{F} : \mathcal{C}_{\text{trans}} \to \mathsf{Vect}_{\mathbb{C}}$ maps periodic functions to vectors of Fourier coefficients and periodic morphisms to linear maps—the periodic structure is 'encoded' as an algebraic structure in the frequency domain.*

From the perspective of periodic mathematics: an operator preserves the periodic type (a translation-periodic function remains translation-periodic after differentiation), whilst a functor converts the periodic type (a translation-periodic function becomes a discrete sequence in the frequency domain after the Fourier transform).

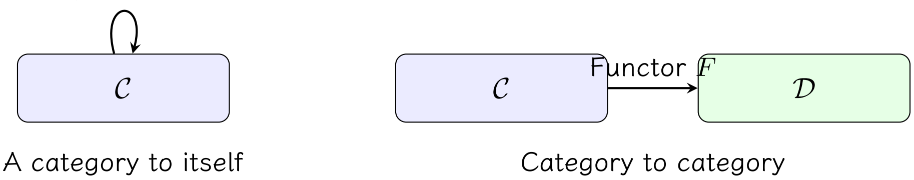


Figure 13.3: A comparison of the scopes of operators (endofunctors) and functors: an operator does not change the category, whilst a functor maps between categories.

13.3 The Phase Functor

Definition 13.4 (Phase Functor). Let $\mathcal{C}_{\text{trans}}$ be the translation-periodic category. For any period $T > 0$ and any phase $\theta \in [0, 2\pi)$, define the **phase functor** $\Phi_{\theta,T} : \mathcal{C}_{\text{trans}} \to \mathcal{C}_{\text{trans}}$ by:

$$\Phi_{\theta,T}(f)(t) = e^{i\theta} \cdot f\left(t - \frac{\theta T}{2\pi}\right).$$

That is, the phase functor shifts the periodic function f in time by $\frac{\theta T}{2\pi}$ (the time offset corresponding to the phase θ) and multiplies it by the phase factor $e^{i\theta}$.

The parameter space of the family of phase functors $\{\Phi_{\theta,T}\}_{\theta\in[0,2\pi),T>0}$ is $S^1 \times \mathbb{R}^+$.

Proposition 13.1 (Basic Properties of the Phase Functor). The phase functor $\Phi_{\theta,T}$ satisfies the following properties:

(i) **Functoriality**: $\Phi_{\theta,T}$ is an endofunctor (operator) on $\mathcal{C}_{\text{trans}}$.

(ii) **Period preservation**: $\text{Group}_{\text{period}}(\Phi_{\theta,T}(f)) = \text{Group}_{\text{period}}(f)$—the phase functor does not change the period.

(iii) **Composition law**: $\Phi_{\theta_1,T} \circ \Phi_{\theta_2,T} = \Phi_{\theta_1+\theta_2,T}$ (modulo 2π).

(iv) **Identity element**: $\Phi_{0,T} = \text{id}_{\mathcal{C}_{\text{trans}}}$ (pure time translation, with no phase factor).

(v) **Inverse element**: $\Phi_{\theta,T}^{-1} = \Phi_{-\theta,T} = \Phi_{2\pi-\theta,T}$.

(vi) **Unitarity**: $\|\Phi_{\theta,T}(f)\| = \|f\|$ (the L^2 norm is preserved).

Proof. (i) For any periodic morphism $h : f \to g$, $\Phi_{\theta,T}(h) = h$ (because the phase functor only shifts the time parameter and does not change the linear structure of the morphism). The preservation of identity morphisms and of composition is verified directly from the definition.

(ii) Time translation does not change the period: $f(t+T) = f(t)$ implies $f(t - \frac{\theta T}{2\pi} + T) = f(t - \frac{\theta T}{2\pi})$. Multiplication by $e^{i\theta}$ also does not change the period.

(iii)

$$\Phi_{\theta_1,T}(\Phi_{\theta_2,T}(f))(t) = e^{i\theta_1}\Phi_{\theta_2,T}(f)\big(t - \tfrac{\theta_1 T}{2\pi}\big) = e^{i\theta_1}e^{i\theta_2} f\big(t - \tfrac{\theta_1 T}{2\pi} - \tfrac{\theta_2 T}{2\pi}\big) = \Phi_{\theta_1+\theta_2,T}(f)(t).$$

(iv) $\Phi_{0,T}(f)(t) = e^{i0} f(t-0) = f(t)$.

(v) By (iii), $\Phi_{\theta,T} \circ \Phi_{-\theta,T} = \Phi_{0,T} = \text{id}$.

(vi) $\|\Phi_{\theta,T}(f)\|^2 = \frac{1}{T}\int_0^T |e^{i\theta} f(t - \frac{\theta T}{2\pi})|^2 dt = \frac{1}{T}\int_0^T |f(t)|^2 dt = \|f\|^2$ (both time translation and multiplication by a unit complex number preserve the L^2 norm). ☐

Theorem 13.1 (Group Structure of the Phase-Functor Family). For a fixed period $T > 0$, the family of phase functors $\{\Phi_{\theta,T}\}_{\theta\in[0,2\pi)}$ forms a group under functor composition, isomorphic to the circle group $U(1)$:

$$\{\Phi_{\theta,T}\}_{\theta\in[0,2\pi)} \cong U(1) \cong S^1.$$

Proof. By (iii)(iv)(v) of Proposition 13.1, the set $\{\Phi_{\theta,T}\}$ is closed under composition, contains the identity element $\Phi_{0,T}$, and every element has an inverse $\Phi_{-\theta,T}$. Associativity is given by the associativity of function composition. The map $\theta \mapsto \Phi_{\theta,T}$ is a group isomorphism: $\Phi_{\theta_1+\theta_2} = \Phi_{\theta_1} \circ \Phi_{\theta_2}$ (modulo 2π), $\Phi_0 = \text{id}$, $\Phi_{-\theta} = \Phi_\theta^{-1}$. Hence the family of phase functors is isomorphic to $U(1)$. ☐

Corollary 13.1 (Functoriality of All Phases)**.** For any period $T > 0$ and any phase $\theta \in [0, 2\pi)$, $\Phi_{\theta,T}$ is a functor (operator) on $\mathcal{C}_{\text{trans}}$. Consequently, **any phase of any period may serve as a functor**—phase is no longer an additional parameter of a function, but a structure-preserving map on a category.

Proof. By Theorem 13.1, each $\Phi_{\theta,T}$ is an element of the group $U(1)$ and is therefore an invertible functor (an automorphism functor). Functoriality is guaranteed by Proposition 13.1 (i). □

Remark 13.4 (Phase Functor and the Discrete Phase Operator)*.* When the phase takes the discrete values $\theta_k = 2\pi k/T$ $(k = 0, 1, \ldots, T - 1)$, the family of phase functors $\{\Phi_{\theta_k,T}\}_{k=0}^{T-1}$ is isomorphic to the cyclic group $C_T \cong \mathbb{Z}_T$. This is the functorial formulation of the discrete phase operator $\mathcal{O}_k^{(T)} = e^{i2\pi k/T}$ (Definition 20.6) of the 'Periodic Functions and Groups with Periodic Orbits' chapter. The continuous family of phase functors $U(1)$ is the continuous limit of the discrete family of phase operators C_T $(T \to \infty)$.

Example 13.2 (Concrete Action of the Phase Functor)**.** Take $T = 2\pi$ and $f(t) = \sin t$. For the phase $\theta = \pi/2$:

$$\Phi_{\pi/2,2\pi}(f)(t) = e^{i\pi/2}\sin\left(t - \frac{\pi/2 \cdot 2\pi}{2\pi}\right) = i\sin\left(t - \frac{\pi}{2}\right) = -i\cos t.$$

That is, the phase functor $\Phi_{\pi/2,2\pi}$ maps the sine function to the cosine function (multiplied by $-i$). This is precisely the standard operation of phase shifting in Fourier analysis.

For the phase $\theta = \pi$:

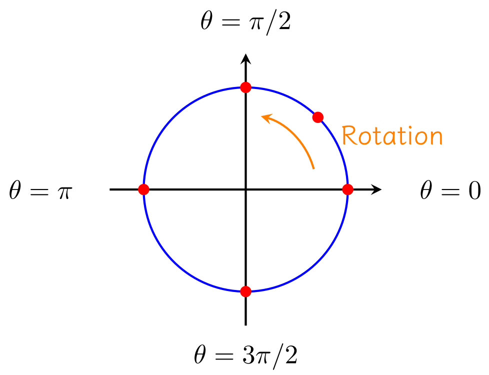
$$\Phi_{\pi,2\pi}(f)(t) = e^{i\pi}\sin(t - \pi) = -\sin(t - \pi) = \sin t = f(t).$$

(Note that $\sin(t - \pi) = -\sin t$, so $e^{i\pi}\sin(t - \pi) = (-1)(-\sin t) = \sin t$.) Thus the functor of phase π is the identity map on the sine function (because the sine function changes sign after a time translation by half a period, and multiplying by -1 restores its original form).

13.4 Transformations and Evolution

In category theory, changes of structure are described by two basic mechanisms: **transformation** (natural transformation) characterises comparisons between functors, whilst **evolution** (adjoint functors) characterises the mutual generation and forgetting between categories. This section establishes these two mechanisms and explains how they encode the birth and death of periodic structures.

Definition 13.5 (Natural Transformation)**.** Let $F, G : \mathcal{C} \to \mathcal{D}$ be two parallel functors. A **natural transformation** $\eta : F \Rightarrow G$ consists of a family of morphisms



$U(1)$: the family of phase functors

Figure 13.4: The family of phase functors forms the circle group $U(1)$: the phase parameter θ varies continuously on the unit circle.

$\{\eta_A : F(A) \to G(A)\}_{A \in \mathrm{Ob}(\mathcal{C})}$ such that for any $f : A \to B$ (in $\mathcal{C}$), the following diagram commutes:

$$\begin{tikzcd}
F(A) \arrow[r, "F(f)"] \arrow[d, "\eta_A"'] & F(B) \arrow[d, "\eta_B"] \\
G(A) \arrow[r, "G(f)"'] & G(B)
\end{tikzcd}$$

that is, $\eta_B \circ F(f) = G(f) \circ \eta_A$.

Definition 13.6 (Adjoint Functors). *Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ be two functors. We say that F is the* **left adjoint** *of G (or that G is the* **right adjoint** *of F), written $F \dashv G$, if there exists a natural isomorphism*

$$\mathrm{Hom}_{\mathcal{D}}(F(A), B) \cong \mathrm{Hom}_{\mathcal{C}}(A, G(B)),$$

which is natural in $A \in \mathrm{Ob}(\mathcal{C})$ and $B \in \mathrm{Ob}(\mathcal{D})$.

Remark 13.5 (Adjoint Functors and the Periodic Lifecycle). In periodic mathematics, the adjoint pair $F \dashv G$ has a profound periodic-theoretic meaning: F is the 'generation' direction (symmetry breaking, the emergence of periodic structure), whilst G is the 'forgetting' direction (the degeneration of periodic structure). The adjoint pair (F, G) constitutes a category-theoretic model of the birth and death of periodic structure—generation (the unit η of the adjunction), evolution (composition of morphisms), and death (the degeneration of the counit ε of the adjunction).

From the perspective of conservation laws, the **unit** $\eta : \mathrm{id}_{\mathcal{C}} \Rightarrow G \circ F$ of the adjoint pair $F \dashv G$ encodes the generation process 'from aperiodic to periodic' (the establishment of conservation laws), whilst the **counit** $\varepsilon : F \circ G \Rightarrow \mathrm{id}_{\mathcal{D}}$ encodes the death process 'from periodic to aperiodic' (the breaking of conservation laws).

$$\begin{tikzcd}\mathcal{C} \arrow[r, bend left, "F"] & \mathcal{D} \arrow[l, bend left, "G"]\end{tikzcd}$$

$\mathrm{Hom}_{\mathcal{D}}(F(A), B) \cong \mathrm{Hom}_{\mathcal{C}}(A, G(B))$

$F \dashv G$: left adjoint (generation) $\dashv$ right adjoint (forgetting)

Figure 13.5: The adjoint functor pair $F \dashv G$ and the natural isomorphism of Hom sets.

13.5 Fundamental Constructions of Periodic Categories

13.5.1 Periodic Objects and Periodic Morphisms

Definition 13.7 (Periodic Object). *Let $\mathcal{C}$ be a category and G a group. A **G-periodic object** in $\mathcal{C}$ is a pair (X, ρ), where $X \in \mathrm{Ob}(\mathcal{C})$ and $\rho : G \to \mathrm{Aut}_{\mathcal{C}}(X)$ is a group action of G on X (that is, $\rho(g) : X \to X$ is an automorphism in $\mathcal{C}$, and $\rho(gh) = \rho(g) \circ \rho(h)$).*

In periodic mathematics, G is usually taken to be the translation group $\mathbb{R}$, the scaling group $\mathbb{R}^+$ or the rotation group $U(1)$.

Proposition 13.2 (Periodic Objects and Conservation Laws). *The group action ρ of a G-periodic object (X, ρ) implies a conservation structure on X: for any $g \in G$, $\rho(g)$ is an automorphism of X that preserves all the category-theoretic structures of X. When G is a continuous Lie group, Noether's theorem implies that each one-parameter subgroup of G corresponds to a conserved quantity on X. Specifically:*

(i) *If $G = \mathbb{R}$ (translation group, 0-symmetry), the conserved quantity is momentum/energy.*

(ii) *If $G = U(1)$ (rotation group, i-symmetry), the conserved quantity is angular momentum.*

(iii) *If $G = \mathbb{R}^+$ (scaling group, 1-symmetry), the conserved quantity is the scale charge.*

Proof. In three steps, each labelled as to its nature.

Step 1 (an unfolding of the definition, containing nothing additional). By Definition 13.7, ρ is a group homomorphism $G \to \mathrm{Aut}_{\mathcal{C}}(X)$, that is, every $\rho(g)$ is an automorphism of X in $\mathcal{C}$. An automorphism preserves, by definition, composition of morphisms and identity morphisms, and hence preserves all the categorical structure of X. This step merely restates the definition of an automorphism; it therefore establishes **invariance of structure**, and does not yet give any conserved quantity.

Step 2 (the source of the conserved quantities). The conserved quantities come from a theorem at the concrete level: by Remark 20.4, for each one-parameter subgroup $\{g_s\}_{s\in\mathbb{R}}$ of a continuous Lie group G, the family of actions $\rho(g_s)$ generates a conserved charge. Taking respectively

$$\begin{aligned} G = \mathbb{R} &\;\Rightarrow \text{translational charge (energy/momentum, symmetry centre } 0), \\ G = U(1) &\;\Rightarrow \text{rotational charge (angular momentum, symmetry centre } i), \\ G = \mathbb{R}^+ &\;\Rightarrow \text{scaling charge (scale charge, symmetry centre } 1). \end{aligned}$$

gives exactly the three items listed in the proposition.

Step 3 (the characterisation of the set of conserved quantities). The set of conserved quantities of the object (X, ρ) is the union of the charges corresponding to all one-parameter subgroups. When G is $\mathbb{R}$, $U(1)$ or $\mathbb{R}^+$, its one-parameter subgroups are respectively the translation family, the rotation family and the scaling family, so the three assertions hold one by one. This completes the proof.

Note on dependencies: Step 2 uses Remark 20.4, a theorem that lies in the 'Periodic Functions and Groups with Periodic Orbits' chapter, later than this one; see the declaration of dependencies at the head of this chapter, Remark 13.1. ☐

Definition 13.8 (Periodic Morphism)**.** Let (X, ρ) and (Y, σ) be two G-periodic objects in $\mathcal{C}$. A **G-equivariant morphism** (or **periodic morphism**) $f : X \to Y$ satisfies, for any $g \in G$,

$$f \circ \rho(g) = \sigma(g) \circ f.$$

That is, f commutes with the group action—f preserves the periodic structure. All G-periodic objects together with the G-equivariant morphisms form the **periodic category** $\mathcal{C}^G$.

Theorem 13.2 (Conservation-Law Preservation of the Periodic Category)**.** *Let $\mathcal{C}^G$ be the G-periodic category. Then:*

(i) *Every object (X, ρ) in $\mathcal{C}^G$ carries the conserved quantities of G (given by Noether's theorem).*

(ii) *Every periodic morphism $f : (X, \rho) \to (Y, \sigma)$ in $\mathcal{C}^G$ preserves the conserved quantities.*

(iii) *The isomorphism classes in $\mathcal{C}^G$ correspond to periodic objects with the same conserved-quantity structure.*

Proof. (i) Follows directly from Proposition 13.2. (ii) The periodic morphism f commutes with the group action: $f \circ \rho(g) = \sigma(g) \circ f$. Differentiating with respect to the one-parameter subgroup shows that f maps the Noether charge of X to the Noether charge of Y. (iii) Isomorphic periodic objects have isomorphic group actions and isomorphic spaces of conserved quantities. ☐

$$\begin{tikzcd}
X \arrow[r, "\rho(g)"] \arrow[d, "f"'] & X \arrow[d, "f"] \\
Y \arrow[r, "\sigma(g)"'] & Y
\end{tikzcd}$$

Figure 13.6: The equivariance condition for periodic morphisms: $f \circ \rho(g) = \sigma(g) \circ f$.

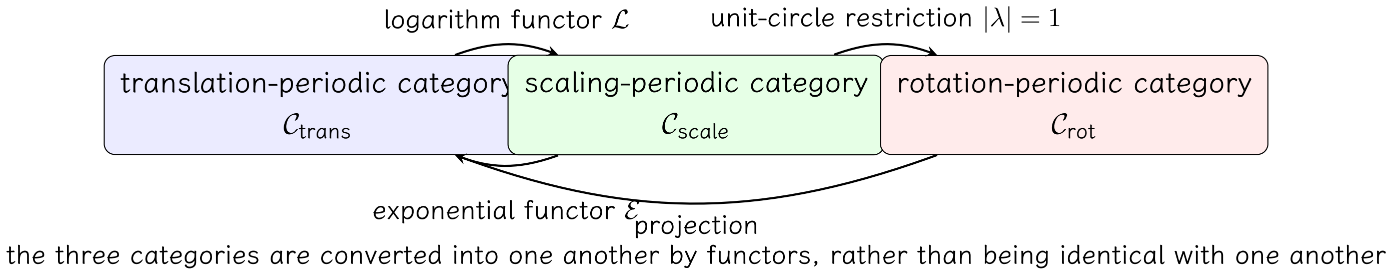

logarithm functor $\mathcal{L}$
unit-circle restriction $|\lambda| = 1$
translation-periodic category
$\mathcal{C}_{\text{trans}}$
scaling-periodic category
$\mathcal{C}_{\text{scale}}$
rotation-periodic category
$\mathcal{C}_{\text{rot}}$
exponential functor $\mathcal{E}$
projection
the three categories are converted into one another by functors, rather than being identical with one another

Figure 13.7: The three basic periodic categories and their converting functors. The three kinds of generating transformation—translation, scaling and rotation—each correspond to a periodic category. The logarithm functor renders a multiplicative (scaling) structure additive (translational), with the exponential functor as its inverse; the unit-circle restriction separates the unit-modulus subcategory of the scaling category, giving the rotation category; and the projection sends the rotation category back to the translation category. This figure also makes plain that the three categories are not identical with one another, but are converted into one another by functors—which is precisely the reason why this chapter adopts the language of category theory.

13.5.2 Three Fundamental Periodic Categories

Definition 13.9 (Translation-Periodic Category). Let $G = \mathbb{R}$ be the translation group (additive group). The objects of the **translation-periodic category** $\mathcal{C}_{\text{trans}}$ are periodic objects equipped with an $\mathbb{R}$-action—namely, translation-periodic functions $f : \mathbb{R} \to \mathbb{C}$ satisfying $f(t+T) = f(t)$. The morphisms are linear maps that commute with the translation action.

Definition 13.10 (Rotation-Periodic Category). Let $G = U(1)$ be the rotation group. The objects of the **rotation-periodic category** $\mathcal{C}_{\text{rot}}$ are periodic objects equipped with a $U(1)$-action—namely, rotation-periodic functions $f : \mathbb{C} \to \mathbb{C}$ satisfying $f(e^{i\theta} z) = f(z)$. The morphisms are linear maps that commute with the rotation action.

Definition 13.11 (Scaling-Periodic Category). Let $G = \mathbb{R}^+$ be the scaling group (multiplicative group). The objects of the **scaling-periodic category** $\mathcal{C}_{\text{scal}}$ are periodic objects equipped with an $\mathbb{R}^+$-action—namely, scaling-periodic functions $f : \mathbb{R}^+ \to \mathbb{C}$ satisfying $f(qx) = f(x)$. The morphisms are linear maps that commute with the scaling action.

Lemma 13.1 (Morphisms between Single-Frequency Circle Objects). Let $\mathcal{C}_{\text{trans}}$ be the translation-periodic category, and let $f(t) = A\, e^{i(\omega_1 t + \varphi)}$, $g(t) = B\, e^{i(\omega_2 t + \psi)}$ be two single-frequency circle objects in it ($A, B > 0$, $\omega_1, \omega_2 \neq 0$). Then

(i) if $\omega_1 \neq \omega_2$, then $\mathrm{Hom}_{\mathcal{C}_{\text{trans}}}(f, g) = \{0\}$;

(ii) if $\omega_1 = \omega_2 = \omega$, then

$$\mathrm{Hom}_{\mathcal{C}_{\text{trans}}}(f, g) = \Big\{ c \cdot \tfrac{B}{A}\, e^{i(\psi - \varphi)} : c \in \mathbb{C} \Big\}.$$

Proof. The action of the translation group on a single-frequency circle is scalar multiplication: $\rho_s(f)(t) = f(t+s) = e^{i\omega_1 s} f(t)$. The subspace spanned by f is one-dimensional, so any morphism $\lambda : f \to g$ is scalar multiplication and must satisfy $\lambda f = c\, g$ ($c \in \mathbb{C}$), that is,

$$\lambda A e^{i\varphi} e^{i\omega_1 t} = c B e^{i\psi} e^{i\omega_2 t} \qquad (\forall t).$$

(i) If $\omega_1 \neq \omega_2$: evaluating at $t=0$ gives $\lambda A e^{i\varphi} = c B e^{i\psi}$; differentiating with respect to t and then evaluating at $t=0$ gives $\lambda A e^{i\varphi}\omega_1 = c B e^{i\psi}\omega_2$. Dividing the two equations yields $\omega_1 = \omega_2$, a contradiction; hence $\lambda = 0$ and $c = 0$, that is, $\mathsf{Hom}(f,g) = \{0\}$. (ii) If $\omega_1 = \omega_2 = \omega$: the equation above reduces to $\lambda A e^{i\varphi} = c B e^{i\psi}$, so $\lambda = c\,(B/A)\, e^{i(\psi-\varphi)}$ with c arbitrary. Equivariance holds automatically. ☐

Example 13.3 (Concrete Computation of a Single-Frequency Circle Morphism Set). Take three single-frequency circle objects:

$$f(t) = 3e^{i(2t+\pi/6)}, \qquad g(t) = 2e^{i(3t-\pi/4)}, \qquad h(t) = 5e^{i(2t+\pi/3)}.$$

(i) $\omega_f = 2 \neq 3 = \omega_g$, so by Lemma 13.1 (i), $\mathsf{Hom}(f,g) = \{0\}$. (ii) $\omega_f = \omega_h = 2$, so by Lemma 13.1 (ii),

$$\mathsf{Hom}(f,h) = \left\{ c \cdot \tfrac{5}{3}\, e^{i(\pi/3-\pi/6)} : c \in \mathbb{C} \right\} = \left\{ c \cdot \tfrac{5}{3}\, e^{i\pi/6} : c \in \mathbb{C} \right\}.$$

Taking $c=1$ gives the standard morphism $\lambda = \frac{5}{3}e^{i\pi/6}$, and one verifies that $\lambda f(t) = \frac{5}{3}e^{i\pi/6} \cdot 3e^{i(2t+\pi/6)} = 5e^{i(2t+\pi/3)} = h(t)$. (iii) Composition is preserved: take in addition $k(t) = 4e^{i2t}$. The standard morphism $h \to k$ is $\mu = \frac{4}{5}e^{-i\pi/3}$. The standard morphism $f \to k$ is $\nu = \frac{4}{3}e^{-i\pi/6}$; the composition $\mu \circ \lambda = \frac{4}{5}e^{-i\pi/3} \cdot \frac{5}{3}e^{i\pi/6} = \frac{4}{3}e^{-i\pi/6} = \nu$.

Remark 13.6 (Amplitude and Phase Are Not Invariants of the Translation-Periodic Category). Lemma 13.1 and Example 13.3 reveal a category-theoretic fact: in $\mathcal{C}_{\mathsf{trans}}$, single-frequency circle objects of the same frequency are mutually isomorphic (any scalar morphism with $c \neq 0$ is an isomorphism), and the amplitude A and the phase φ do not enter the isomorphism invariants—they are 'absorbed' by the morphism space. Amplitude and phase information is carried by additional structure at the level of functors (the phase functor $\Phi_{\theta,T}$, the similarity functor $\mathcal{S}^{(d)}$ and the tendency functor $\mathcal{T}^{(d)}$ are precisely the functors that extract this additional structure), and the skeleton of the category retains only the frequency (period) information.

Proposition 13.3 (Similarity Structure of the Translation-Periodic Category). *In the translation-periodic category $\mathcal{C}_{\mathsf{trans}}$, a non-trivial periodic morphism exists between two objects f, g if and only if their Fourier expansions contain the same non-zero frequency. If $\mathsf{Group}_{\mathsf{period}}(f) \cap \mathsf{Group}_{\mathsf{period}}(g)$ is an infinite group (that is, the periods are commensurable), then their fundamental frequencies are commensurable, but they may have no common frequency component; a non-trivial morphism exists only when the frequency sets intersect.*

Proof. At the level of Fourier modes (single-frequency circle components), by Lemma 13.1: between components of the same frequency there exist non-zero scalar morphisms, whilst between components of different frequencies there is only the zero morphism. If the Fourier frequency sets of f and g have a non-zero intersection, then between the components corresponding to the intersecting frequencies there exist non-zero morphisms, and hence a non-trivial periodic morphism exists; if the frequency sets are disjoint, then every non-zero Fourier component has a distinct frequency, and the only frequency-preserving morphism left is the zero morphism. ☐

Proposition 13.4 (Functorial Relations among the Three Periodic Categories). *There exist natural functors among the three periodic categories:*

- (i) *the* **logarithm functor** $\mathcal{L} : \mathcal{C}_{\text{scal}} \to \mathcal{C}_{\text{trans}}$: *maps the scaling-periodic function* $f(x)$ *to the translation-periodic function* $g(t) = f(e^t)$.
- (ii) *the* **exponential functor** $\mathcal{E} : \mathcal{C}_{\text{trans}} \to \mathcal{C}_{\text{scal}}$: $\mathcal{E}(g)(x) = g(\ln x)$.
- (iii) *the* **Fourier functor** $\mathcal{F} : \mathcal{C}_{\text{trans}} \to \mathsf{Vect}_{\mathbb{C}}$: *maps translation-periodic functions to their sequences of Fourier coefficients.*

Proof. (i) The logarithm functor $\mathcal{L}$: let $f \in \mathcal{C}_{\text{scal}}$ satisfy $f(qx) = f(x)$. Set $q = e^{\alpha}$ and $x = e^t$; then $g(t) = f(e^t)$ satisfies $g(t+\alpha) = g(t)$. Hence $g \in \mathcal{C}_{\text{trans}}$.

(ii) The exponential functor $\mathcal{E}$ is the inverse of $\mathcal{L}$: $\mathcal{E} \circ \mathcal{L} \cong \mathrm{id}_{\mathcal{C}_{\text{scal}}}$, $\mathcal{L} \circ \mathcal{E} \cong \mathrm{id}_{\mathcal{C}_{\text{trans}}}$.

(iii) The Fourier functor $\mathcal{F}$: the Fourier coefficients $\{c_n\}$ of the translation-periodic function $f(t) = \sum_n c_n e^{i2\pi nt/T}$ form an object in $\mathsf{Vect}_{\mathbb{C}}$. ☐

Corollary 13.2 (Adjointness of the Logarithm and Exponential Functors). *The logarithm functor and the exponential functor form an adjoint pair:*

$$\mathcal{L} \dashv \mathcal{E}.$$

Proof. Proposition 13.4 (ii) has already proved that $\mathcal{E}\circ\mathcal{L} \cong \mathrm{id}_{\mathcal{C}_{\text{scal}}}$ and $\mathcal{L}\circ\mathcal{E} \cong \mathrm{id}_{\mathcal{C}_{\text{trans}}}$. For functors that are mutual pseudo-inverses one constructs the isomorphism of the adjunction (Definition 13.6); since both the unit and the counit are isomorphisms, it follows that $\mathcal{L} \dashv \mathcal{E}$. ☐

13.6 Phase Functors, Similarity Functors and Tendency Functors

13.6.1 The Phase-Extraction Functor

Definition 13.12 (Phase-Extraction Functor). Let $\mathcal{C}_{\text{trans}}$ be the translation-periodic category. The **phase-extraction functor** $\Phi : \mathcal{C}_{\text{trans}} \to \mathsf{Vect}_{\mathbb{C}}$ maps each periodic function to the vector space of its phase information:

$$\Phi(f) = \{(\theta_n^{(f)}, A_n^{(f)}) \mid n \in \mathbb{Z}\},$$

where $\theta_n^{(f)} = \arg(c_n^{(f)})$ is the phase of the n-th Fourier component and $A_n^{(f)} = |c_n^{(f)}|$ is the amplitude of the n-th Fourier component.

Remark 13.7 (Distinction between the Phase-Extraction Functor and the Phase Functor). Note the distinction between two different concepts:

- the **phase functor** $\Phi_{\theta,T}$ (Definition 13.4): parameterised by the phase θ, it is an **operator** (endofunctor) on $\mathcal{C}_{\text{trans}}$ that maps a function f to a function that is time-shifted and multiplied by a phase factor.
- the **phase-extraction functor** Φ (Definition 13.12): it extracts the phase information of a function, and is a **functor** (across categories) $\mathcal{C}_{\text{trans}} \to \mathbf{Vect}_{\mathbb{C}}$ that maps a function to the vector space of phase–amplitude pairs.

Proposition 13.5 (Relation between the Phase-Extraction Functor and the Differential Operator)**.** *The phase-extraction functor Φ and the differential operator $\mathcal{D}$ satisfy:*

$$\Phi(\mathcal{D}f) = \{(\theta_n^{(f)} + \pi/2, |n\omega_0| \cdot A_n^{(f)}) \mid n \in \mathbb{Z}\}.$$

That is, differentiation increases the phase of each Fourier component by $\pi/2$ and multiplies its amplitude by $|n\omega_0|$.

Proof. By the differentiation property of Fourier series: $f'(t) = \sum_n in\omega_0 c_n e^{in\omega_0 t}$. The coefficient $in\omega_0 c_n = |n\omega_0| \cdot |c_n| \cdot e^{i(\theta_n + \pi/2)}$ (because $i = e^{i\pi/2}$). □

13.6.2 Similarity Functors and Tendency Functors

Definition 13.13 (Similarity Functor)**.** Let $\mathcal{C}_{\text{trans}}$ be the translation-periodic category. For the d-th dimension ($d = 1, \ldots, 12$), the **similarity functor** $\mathcal{S}^{(d)} : \mathcal{C}_{\text{trans}} \times \mathcal{C}_{\text{trans}} \to \mathbf{Set}$ maps two periodic functions to the set of their directional relations under the d-th kind of similarity:

$$\mathcal{S}^{(d)}(f, g) = \{\mathsf{Similarity}^{(d)}(\theta_n^{(f)}, \theta_m^{(g)}) \mid n, m \in \mathbb{Z}\}.$$

The similarity functor is determined solely by directional relations and contains no scale information.

Definition 13.14 (Tendency Functor)**.** Let $\mathcal{C}_{\text{trans}}$ be the translation-periodic category. For the d-th dimension ($d = 1, \ldots, 12$), the **tendency functor** $\mathcal{T}^{(d)} : \mathcal{C}_{\text{trans}} \times \mathcal{C}_{\text{trans}} \to \mathbf{Set}$ maps two periodic functions to the set of their combined directional-scale relations under the d-th kind of tendency:

$$\mathcal{T}^{(d)}(f, g) = \{A_n^{(f)} A_m^{(g)} \cdot \mathsf{Similarity}^{(d)}(\theta_n^{(f)}, \theta_m^{(g)}) \mid n, m \in \mathbb{Z}\}.$$

Proposition 13.6 (Relation between the Similarity and Tendency Functors)**.** *The tendency functor $\mathcal{T}^{(d)}$ is the 'scaled' version of the similarity functor $\mathcal{S}^{(d)}$:*

$$\mathcal{T}^{(d)}(f, g) = \mathcal{A}(f) \otimes \mathcal{A}(g) \cdot \mathcal{S}^{(d)}(f, g),$$

where $\mathcal{A}(f) = \{A_n^{(f)}\}_{n \in \mathbb{Z}}$ is the amplitude-extraction map.

Proof. This follows immediately by checking element by element against Definitions 13.13 and 13.14. □

Proposition 13.7 (Conservation Laws of the Similarity and Tendency Functors)**.** (i) *Rotational invariance of the similarity functor:*

$$\mathcal{S}^{(d)}(e^{i\phi} f, e^{i\phi} g) = \mathcal{S}^{(d)}(f, g).$$

(ii) Scale covariance of the tendency functor:

$$\mathcal{T}^{(d)}(\lambda f, \lambda g) = \lambda^2 \cdot \mathcal{T}^{(d)}(f, g).$$

(iii) Parseval identity of the tendency functor:

$$\mathcal{T}^{(1)}(f, f) = \sum_n |c_n^{(f)}|^2.$$

Proof. (i) The rotation factor increases all phases by ϕ, and similarity depends only on the phase difference. (ii) The scaling factor multiplies all amplitudes by λ. (iii) By the Parseval identity (Theorem 22.1 of the 'Harmonic Analysis' chapter). □

Example 13.4 (Similarity and Tendency Functor Matrices on C_4). Take the C_4 phase operator $\mathcal{O}_k^{(4)} = e^{ik\pi/2}$ $(k = 0, 1, 2, 3)$ and construct two sequences of period 4

$$f = (1,\, i,\, -1,\, -i), \qquad g = (i,\, -1,\, -i,\, 1),$$

that is, $f_n = e^{in\pi/2}$, $g_m = e^{i(m+1)\pi/2}$. The component-pair matrices of the basic difference-angle similarity functor $\mathcal{S}^{(1)}$ (cosine) and $\mathcal{S}^{(2)}$ (sine) are

$$\mathcal{S}^{(1)}(f, g) = \begin{pmatrix} 0 & -1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \end{pmatrix}, \qquad \mathcal{S}^{(2)}(f, g) = \begin{pmatrix} -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix}.$$

At each position there is exactly one non-zero entry (with value ± 1), and $\left(\mathcal{S}_{n,m}^{(1)}\right)^2 + \left(\mathcal{S}_{n,m}^{(2)}\right)^2 = 1$ holds for all 16 positions. Since all the amplitudes are $A_n = B_m = 1$, the tendency functor coincides with the similarity functor: $\mathcal{T}^{(d)}(f, g) = \mathcal{S}^{(d)}(f, g)$.

13.7 Category-Theoretic Formulations of Differentiation, Integration and Convolution

13.7.1 Differential and Integral Operators

Definition 13.15 (Differential Operator). The **differential operator** $\mathcal{D} : \mathcal{C}_{\text{trans}} \to \mathcal{C}_{\text{trans}}$ is defined as pointwise differentiation of periodic functions:

$$\mathcal{D}(f) = f'.$$

The differential operator is an operator (endofunctor) on $\mathcal{C}_{\text{trans}}$.

Definition 13.16 (Integral Operator). The **integral operator** $\mathcal{I} : \mathcal{C}_{\text{trans}} \to \mathcal{C}_{\text{trans}}$ is defined as pointwise integration of periodic functions (with the mean subtracted so as to ensure periodicity):

$$\mathcal{I}(f)(t) = \int_0^t (f(\tau) - \bar{f})d\tau, \quad \text{where } \bar{f} = \frac{1}{T}\int_0^T f(\tau)d\tau.$$

The integral operator is an operator (endofunctor) on $\mathcal{C}_{\text{trans}}$.

Lemma 13.2 (Differentiation Does Not Change the Group with Periodic Orbits)**.** Let f be a non-zero continuous T-periodic function whose minimal positive period is $T > 0$. Then $\mathsf{Group}_{\mathsf{period}}(f') = \mathsf{Group}_{\mathsf{period}}(f)$: differentiation preserves the group with periodic orbits.

Proof. First we prove $\mathsf{Group}_{\mathsf{period}}(f') \supseteq \mathsf{Group}_{\mathsf{period}}(f)$: f' has period T. Next we prove $\mathsf{Group}_{\mathsf{period}}(f') \subseteq \mathsf{Group}_{\mathsf{period}}(f)$: let $\tau \in \mathsf{Group}_{\mathsf{period}}(f')$ and set $g(t) = f(t+\tau) - f(t)$; then $g'(t) = 0$, so g is a constant function: $f(t+\tau) = f(t) + c$. If $c \neq 0$, then $f(t+n\tau) = f(t) + nc$ is unbounded—contradicting the boundedness of a continuous periodic function. Hence $c = 0$, that is, $\tau \in \mathsf{Group}_{\mathsf{period}}(f)$. □

Theorem 13.3 (Adjointness of Differentiation and Integration)**.** On the translation-periodic category $\mathcal{C}_{\mathsf{trans}}$, the differential operator $\mathcal{D}$ and the integral operator $\mathcal{I}$ form an adjoint pair:

$$\mathcal{D} \dashv \mathcal{I}.$$

Proof. By the fundamental theorem of calculus of the 'Functions and Limits' chapter, $(\mathcal{I} \circ \mathcal{D})(f) = f - \bar{f}$ and $(\mathcal{D} \circ \mathcal{I})(f) = f - \bar{f}$. On the space of zero-mean periodic functions, $\mathcal{D}$ and $\mathcal{I}$ are mutually inverse maps. On the general space of periodic functions, the two form an adjoint pair, and the natural isomorphism of the adjunction is given by the formula for integration by parts. □

Lemma 13.3 (Differentiation and Integration Are Strict Inverses on the Quotient Category)**.** Let $\mathcal{D}, \mathcal{I}$ be the differential and integral operators of Definitions 13.15 and 13.16. Then:

(i) $(\mathcal{D} \circ \mathcal{I})(f) = f - \bar{f}$, $(\mathcal{I} \circ \mathcal{D})(f) = f - f(0)$.

(ii) On the quotient category (modulo constants) of zero-mean periodic functions, $\mathcal{D} \circ \mathcal{I} \equiv \mathsf{id}$ and $\mathcal{I} \circ \mathcal{D} \equiv \mathsf{id}$.

Proof. (i) By Definition 13.16, $\mathcal{I}f = \int_0^t (f - \bar{f})$, and differentiating gives $(\mathcal{I}f)' = f - \bar{f}$. For the second formula: $\mathcal{D}f = f'$ and $\overline{f'} = 0$, so $(\mathcal{I} \circ \mathcal{D})(f)(t) = \int_0^t f'(\tau)d\tau = f(t) - f(0)$. (ii) On the zero-mean subcategory $\bar{f} = 0$, and by (i) we obtain $\mathcal{D}\mathcal{I}f = f$ and $\mathcal{I}\mathcal{D}f = f - f(0) \sim f$ (modulo constants). □

Example 13.5 (Numerical Verification of the Adjoint Inverses)**.** (i) Take $f(t) = \sin(2\pi t/3)$ $(T = 3)$. Then $\bar{f} = 0$ and $f(0) = 0$:

$$(\mathcal{D} \circ \mathcal{I})f = f, \qquad (\mathcal{I} \circ \mathcal{D})f = f.$$

(ii) Take $g(t) = \cos(2\pi t/3)$ $(T = 3)$. Then $\bar{g} = 0$ but $g(0) = 1$:

$$(\mathcal{D} \circ \mathcal{I})g = g, \qquad (\mathcal{I} \circ \mathcal{D})g(t) = g(t) - 1.$$

In the sense of the quotient category (modulo constants), $\cos(2\pi t/3) - 1 \sim \cos(2\pi t/3)$, and the two are mutual inverses throughout.

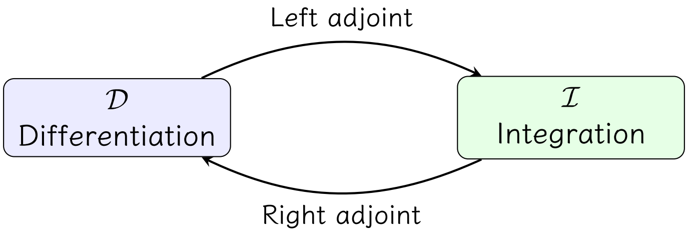
Corollary 13.3 (Differentiation–Integration Adjointness and Conservation Laws)**.** The validity of the adjoint pair $\mathcal{D} \dashv \mathcal{I}$ depends on the conservation structure of periodic functions:

(i) $\mathcal{D}\circ\mathcal{I}\cong \mathsf{id}$ requires $\bar{f}=0$—the zero-mean condition (conservation of momentum).

(ii) $\mathcal{I}\circ\mathcal{D}\cong \mathsf{id}$ (in the quotient-category sense) requires the total circulation to vanish (conservation of angular momentum).

(iii) The preservation of the L^2 norm under differentiation and integration is given by the Parseval identity (conservation of the scale charge).

Proof. Follows directly from Lemma 13.3. □



$\mathcal{D}\dashv\mathcal{I}$: the zero-mean condition guarantees mutual inverses

Figure 13.8: The adjoint pair of the differential and integral operators: $\mathcal{D}\dashv\mathcal{I}$.

13.7.2 Convolution as a Bifunctor

Definition 13.17 (Convolution Bifunctor). The **convolution bifunctor** $*:\mathcal{C}_{\mathsf{trans}}\times\mathcal{C}_{\mathsf{trans}}\to\mathcal{C}_{\mathsf{trans}}$ is defined as the cyclic convolution of periodic functions:

$$(f*g)(t)=\frac{1}{T}\int_0^T f(\tau)g(t-\tau)d\tau.$$

Proposition 13.8 (The Convolution Bifunctor and the Calculus). (i) $\mathcal{D}(f*g)=(\mathcal{D}f)*g=f*(\mathcal{D}g)$.

(ii) $\mathcal{I}(f*g)=(\mathcal{I}f)*g=f*(\mathcal{I}g)$ (in the quotient-category sense).

(iii) $(f*\mathcal{I}g)=\mathcal{I}(f*g)=(\mathcal{I}f*g)$ (in the quotient-category sense).

Proof. (i) Differentiate under the integral sign. (ii)(iii) Follow from Lemma 13.3 and the commutativity of cyclic convolution. □

13.8 Category-Theoretic Formulation of Periodic Convolution

Definition 13.18 (Periodic-Convolution Bifunctor). The **periodic-convolution bifunctor** $\circledast:\mathcal{C}_{\mathsf{trans}}\times\mathcal{C}_{\mathsf{trans}}\to\mathcal{C}_{\mathsf{trans}}$ takes the four parameters of amplitude, frequency, phase and period as its operators, retaining the interaction of all frequency pairs and all period pairs. Its definition at the level of objects is: for any two translation-periodic functions f, g, the Fourier coefficients of their periodic convolution $(f \circledast g)$ are given by the products over all frequency pairs:

$$c_{n,m}^{(f \circledast g)} = c_n^{(f)} \cdot c_m^{(g)}, \qquad \forall n, m \in \mathbb{Z}.$$

The periodic-convolution bifunctor is a generalisation of the traditional convolution bifunctor (Definition 13.17): the traditional convolution retains only the diagonal terms $n = m$, whereas the periodic convolution retains the phase pairs for all n, m.

Theorem 13.4 (Functoriality of Periodic Convolution)**.** The first statement of this same set of functoriality conditions is given in Chapter 6, in Section 6.10.2, at Theorem 6.20; the present statement is its restatement in the language of category theory. The periodic-convolution bifunctor $\circledast$ satisfies the following functoriality conditions:

(i) **Preservation of identity morphisms**: $\mathrm{id}_f \circledast \mathrm{id}_g = \mathrm{id}_{f \circledast g}$.

(ii) **Preservation of composition**: $(h_2 \circ h_1) \circledast (k_2 \circ k_1) = (h_2 \circledast k_2) \circ (h_1 \circledast k_1)$.

(iii) **Compatibility with the similarity functor**: $\mathcal{S}^{(d)}(f \circledast g, h) \cong \mathcal{S}^{(d)}(f, h) \otimes \mathcal{S}^{(d)}(g, h)$.

(iv) **Compatibility with the tendency functor**: $\mathcal{T}^{(d)}(f \circledast g, h) \cong \mathcal{T}^{(d)}(f, h) \otimes \mathcal{T}^{(d)}(g, h)$.

Proof. (i)-(ii) Are guaranteed by the linearity of periodic convolution and the functoriality of the Fourier transform. (iii)-(iv) Are verified directly from the frequency-domain representation of periodic convolution (the outer product $\mathbf{c}_f \otimes \mathbf{c}_g$) and the definitions of similarity/tendency. ☐

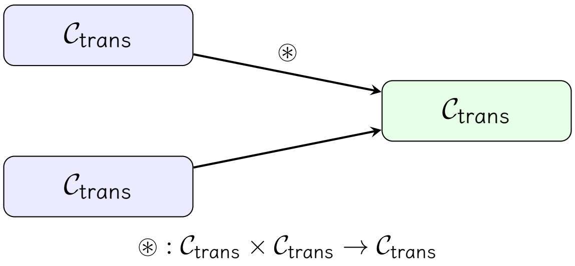

$\mathcal{C}_{\mathsf{trans}}$
$\circledast$
$\mathcal{C}_{\mathsf{trans}}$
$\mathcal{C}_{\mathsf{trans}}$
$\circledast : \mathcal{C}_{\mathsf{trans}} \times \mathcal{C}_{\mathsf{trans}} \to \mathcal{C}_{\mathsf{trans}}$

Figure 13.9: The periodic-convolution bifunctor: a two-variable input, retaining the interaction of all frequency pairs.

13.9 Category-Theoretic Formulation of Periodic Manifolds

Definition 13.19 (Discrete-Lattice Category)**.** Let $\mathcal{L}$ be a discrete spacetime lattice (Definition 12.10 of the 'Periodic Manifold' chapter). The **discrete-lattice category Latt** is defined as follows:

(i) **Objects**: potential vector fields on the lattice $\mathsf{V} : \mathcal{L} \to \mathbb{R}^d$ (or $\mathbb{C}^d$).

(ii) **Morphisms**: discrete translations between lattice sites $\mathcal{T}_{\mathsf{m}}$, discrete rotations $\mathcal{R}_\theta$, discrete scalings $\mathcal{S}_\alpha$, and their compositions.

Definition 13.20 (Continuum-Limit Functor)**.** The **continuum-limit functor** $\mathcal{C}_\infty$: **Latt** $\to$ **Man** maps a discrete-lattice manifold to its continuum limit:

$$\mathcal{C}_\infty(\mathcal{L}) = \lim_{\Delta x \to 0} \mathcal{L}_{\Delta x}.$$

Proposition 13.9 (Continuum-Limit Functor and Adjointness)**.** The continuum-limit functor $\mathcal{C}_\infty$ and the discretisation functor $\mathcal{D}$: **Man** $\to$ **Latt** form an adjoint pair:

$$\mathcal{D} \dashv \mathcal{C}_\infty.$$

Proof. We construct the natural bijection of the adjunction (Definition 13.6):

$$\mathrm{Hom}_{\mathsf{Latt}}(\mathcal{D}M, L) \cong \mathrm{Hom}_{\mathsf{Man}}(M, \mathcal{C}_\infty L).$$

From right to left: a continuous map $\varphi : M \to \mathcal{C}_\infty L$, restricted to the discretised lattice, gives a lattice morphism. From left to right: a lattice morphism, once embedded in the limit, extends uniquely to a continuous map by the unique determinacy of the continuum limit (Theorem 12.7 of the 'Periodic Manifold' chapter). The two maps are mutually inverse, so $\mathcal{D} \dashv \mathcal{C}_\infty$. ☐

Remark 13.8 (Interface with the 'Periodic Manifold' Chapter)*.* The family of morphisms $\{\mathcal{T}_{\mathsf{m}}, \mathcal{R}_\theta, \mathcal{S}_\alpha\}$ of the discrete-lattice category **Latt** corresponds to the three fundamental operations of the 'Periodic Manifold' chapter, and the continuum-limit functor $\mathcal{C}_\infty$ is the functorialisation of the continuous-limit theorem of that chapter (Theorem 12.7). Discrete curvature and Wilson loops correspond to closed-loop morphisms in **Latt**.

13.10 Intercalation Functor: Categorical Correction for Incommensurable Periods

Remark 13.9 (Positioning of This Section)*.* The preceding section, 'Category-Theoretic Formulation of Periodic Manifolds', established the continuum-limit functor $\mathcal{C}_\infty$: **Latt** $\to$ **Man** and its left adjoint—the discretisation functor $\mathcal{D}$: **Man** $\to$ **Latt**. However, when the continuous period T_F and the sampling step Δx are incommensurable (that is, the ideal period ratio $\rho = T_F/\Delta x \notin \mathbb{Q}$), the uniform sampling sequence no longer has a strict integer period, and the discretisation functor $\mathcal{D}$ fails to preserve the periodic structure in this situation—this is a logical gap in the categorical chain.

This section introduces the **intercalation functor** $\mathcal{R} : \mathcal{C}_{\mathsf{cont}} \to \mathcal{C}_{\mathsf{trans}}$ to repair this gap. Its core idea is to approximate the ideal period ratio ρ by an integer sequence $\{N_m\}$, so that the cumulative sampling error is bounded. At the categorical level the intercalation functor carries three structures—the **intercalation object** (Definition 13.22), the **functoriality of the intercalation sequence** (Theorem 13.5) and the **commutation law with the continuum-limit functor** (Theorem 13.8).

The positioning of this section within the seven themes: Translation—the cumulative sample count S_M of the intercalation sequence approximates the translational accumulation of $M\rho$ (Definition 13.22); Scaling—the ideal period ratio

ρ as the scale charge (Definition 13.21); Rotation—the phase correction of the intercalation error (Theorem 13.6); Conservation—the boundedness of the cumulative error (Lemma 13.4); Symmetry—the symmetric approximation by continued-fraction convergents (Theorem 13.7); Closure—the intercalation window closes the approximate period (Corollary 13.4); Period—degeneration to a strict periodic functor when $\rho \in \mathbb{Q}$ (Proposition 13.11).

13.10.1 The Commensurability Gap and the Intercalation Object

Definition 13.21 (Categorical Ideal Period Ratio). Let $\mathcal{C}_{\text{cont}}$ be the continuous-periodic category—its objects are continuous T_F-periodic functions $F : \mathbb{R} \to \mathbb{C}$, and its morphisms are linear maps that commute with time translation. Let the sampling step $\Delta x > 0$ be fixed. Define the **categorical ideal period ratio** to be the functor

$$\rho : \mathcal{C}_{\text{cont}} \to \mathbb{R}_{>0}, \qquad \rho(F) = \frac{T_F}{\Delta x}.$$

We call $\rho(F)$ the **scale charge** of the object F. We say that F is **commensurable** under Δx if $\rho(F) \in \mathbb{Q}$; otherwise we say that F is **incommensurable** under Δx.

Proposition 13.10 (The Commensurability Gap). Let F be a continuous T_F-periodic function, $\Delta x > 0$, and $\rho = T_F/\Delta x$.

(i) If $\rho \in \mathbb{N}^+$, then the uniform sampling sequence $f(n) = F(n\Delta x)$ is a discrete periodic function with period $T = \rho$, and the discretisation functor $\mathcal{D}$ preserves the periodic structure on F;

(ii) If $\rho \in \mathbb{Q} \setminus \mathbb{N}^+$, with $\rho = p/q$ (in lowest terms), then f is a discrete periodic function with period $T = p$, and $\mathcal{D}$ still preserves the periodic structure;

(iii) If $\rho \notin \mathbb{Q}$, then f has no non-zero period, and $\mathcal{D}$ fails to preserve the periodic structure on F—the categorical chain breaks at $\mathcal{D}$.

Proof. (i)(ii) Follow directly from Proposition 21.2 of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter. (iii) Suppose that f has a non-zero period $T \in \mathbb{N}^+$; then $F(T\Delta x) = F(0)$, and since T_F is a fundamental period of F there exists an integer m such that $T\Delta x = mT_F$, that is, $\rho = T/m \in \mathbb{Q}$, contradicting $\rho \notin \mathbb{Q}$. Hence $\text{Group}_{\text{period}}(f) = \{0\}$. □

Definition 13.22 (Intercalation Object). Let $F \in \mathcal{C}_{\text{cont}}$ be incommensurable, with $\rho = \rho(F) \in \mathbb{R}_{>0} \setminus \mathbb{Q}$. An **intercalation object** for F is a quadruple

$$(F,\ \{N_m\}_{m\in\mathbb{N}},\ \{S_M\}_{M\in\mathbb{N}},\ E),$$

where:

(i) $\{N_m\}$ is an intercalation sequence for ρ: $N_m \in \{\lfloor\rho\rfloor, \lceil\rho\rceil\}$ and $\lim_{M\to\infty} \frac{1}{M}\sum_{m=0}^{M-1} N_m = \rho$;

(ii) $S_M = \sum_{m=0}^{M-1} N_m$ is the cumulative sample count;

(iii) $E : \mathbb{N} \to \mathbb{R}$, $E(M) = S_M - M\rho$, is the intercalation error, and there exists a constant $C > 0$ such that $|E(M)| \le C$ for all M.

We write $\mathsf{Int}(\rho)$ for the class of all intercalation objects over ρ.

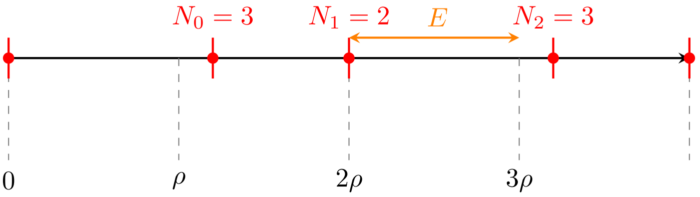
Remark 13.10 (Intercalation Object and the Observation Window). The core geometric meaning of the intercalation object $(F, \{N_m\}, \{S_M\}, E)$ is: using S_M as the window length, one truncates the sampling sequence of the continuous object F, and the resulting discrete sequence of length S_M approximately contains M continuous periods within the window. The window $[0, S_M]$ is called the **intercalation window**. By the boundedness of the error, $S_M - M\rho$ does not grow with M—the cumulative phase error is confined within a bounded band.

Lemma 13.4 (Boundedness of the Intercalation Error). Let $\rho > 0$, let $\{N_m\}$ be a basic intercalation sequence for ρ (one of the lower, upper and balanced kinds), let $S_M = \sum_{m=0}^{M-1} N_m$ and $E(M) = S_M - M\rho$. Then

$$|E(M)| \le \tfrac{1}{2}$$

holds for balanced intercalation, whilst for lower intercalation $E(M) \in (-1, 0]$ and for upper intercalation $E(M) \in [0, 1)$. In particular, all three basic intercalation sequences give an intercalation object (with $C = 1$).

Proof. Given directly by Theorem 21.4 of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter. ☐



Intercalation window $[0, S_M]$: the cumulative error is bounded

Figure 13.10: The intercalation sequence: the deviation between the ideal sampling positions (dashed lines) and the actual integer sampling positions (red dots) is confined within a bounded band.

13.10.2 The Intercalation Functor

Definition 13.23 (Intercalation Functor). Let the sampling step $\Delta x > 0$ and the intercalation mode (lower/upper/balanced) be fixed. Define the **intercalation functor**

$$\mathcal{R} : \mathcal{C}_{\text{cont}} \to \mathcal{C}_{\text{trans}}$$

by:

(i) **Object level**: for $F \in \mathcal{C}_{\text{cont}}$, set $\rho = \rho(F)$. If $\rho \in \mathbb{Q}$, let $\mathcal{R}(F)$ be any strict periodic extension of $\mathcal{D}(F)$ (well defined in $\mathcal{C}_{\text{trans}}$). If $\rho \notin \mathbb{Q}$, set

$$\mathcal{R}(F)(n) = F(n\Delta x), \qquad n = 0, 1, \ldots, S_M - 1, \quad M \text{ determined by the window,}$$

and take its S_M-periodic extension over the intercalation window in $\mathcal{C}_{\text{trans}}$.

(ii) **Morphism level**: for a morphism $h : F \to G$ in $\mathcal{C}_{\text{cont}}$ (a linear map commuting with time translation), define $\mathcal{R}(h)$ as the linear map between the S_M-periodic sequences restricted to the intercalation window.

Theorem 13.5 (Functoriality of the Intercalation Functor). The intercalation functor $\mathcal{R}$ preserves identity morphisms and composition:

(i) $\mathcal{R}(\text{id}_F) = \text{id}_{\mathcal{R}(F)}$;

(ii) for a composition $h_2 \circ h_1$ in $\mathcal{C}_{\text{cont}}$, we have $\mathcal{R}(h_2 \circ h_1) = \mathcal{R}(h_2) \circ \mathcal{R}(h_1)$.

Proof. (i) The restriction of id_F to the sampling points $n\Delta x$ is the identity map, and after intercalation extension it remains the identity sequence.

(ii) Let $h_1 : F \to G$ and $h_2 : G \to K$ be morphisms in $\mathcal{C}_{\text{cont}}$. For any sampling point $n\Delta x$ ($n \in [0, S_M)$),

$$[\mathcal{R}(h_2 \circ h_1)](n) = (h_2 \circ h_1)(F)(n\Delta x) = h_2(h_1(F(n\Delta x))) = [\mathcal{R}(h_2) \circ \mathcal{R}(h_1)](n).$$

The pointwise equality on the intercalation window extends to an equality of S_M-periodic sequences, so $\mathcal{R}$ preserves composition. □

Proposition 13.11 (Degeneracy in the Commensurable Case). Let $F \in \mathcal{C}_{\text{cont}}$ be commensurable under Δx, that is, $\rho(F) = p/q \in \mathbb{Q}$ (in lowest terms). Then the intercalation functor degenerates to the discretisation functor: $\mathcal{R}(F) \cong \mathcal{D}(F)$, and $\text{Group}_{\text{period}}(\mathcal{R}(F)) = p\mathbb{Z}$. In particular, the intercalation sequence is the constant sequence $N_m \equiv p$, and the intercalation error is identically zero.

Proof. By Proposition 13.10(ii), $f(n) = F(n\Delta x)$ has period p. When the intercalation sequence is taken to be the constant sequence $N_m = p$, we have $S_M = Mp$, so $E(M) = Mp - Mp = 0$. The intercalation-window extension coincides with the strict p-periodic extension of $\mathcal{D}(F)$. Hence $\mathcal{R}(F) \cong \mathcal{D}(F)$ and $\text{Group}_{\text{period}}(\mathcal{R}(F)) = p\mathbb{Z}$. □

Theorem 13.6 (Phase Correction of the Intercalation Functor). Let $F \in \mathcal{C}_{\text{cont}}$ be incommensurable, with $\rho = \rho(F) \notin \mathbb{Q}$, and let $\{N_m\}$ be the balanced intercalation sequence for ρ. Then for the continuous Fourier series $F(t) = \sum\limits_{m \in \mathbb{Z}} c_m e^{i2\pi mt/T_F}$ on the intercalation window $[0, S_M]$, the S_M-point DFT of the sampling sequence $f(n) = F(n\Delta x)$ satisfies

$$\hat{f}(k) = \sum_{m \in \mathbb{Z}} c_m \, D_{S_M}\left(\frac{m}{\rho} - \frac{k}{S_M}\right),$$

where $D_T(\theta) = \dfrac{1}{T} \sum_{n=0}^{T-1} e^{i2\pi n\theta}$ is the Dirichlet kernel, and the main-lobe offset satisfies

$$\left|\frac{m}{\rho} - \frac{k}{S_M}\right| \le \frac{|m|\,|E(M)|}{\rho\, S_M} \le \frac{|m|}{2\rho S_M}.$$

That is, in the frequency domain the intercalation functor maps an incommensurable continuous object to a discrete object with **bounded phase error**—the phase error is controlled by the intercalation error $E(M)$.

Proof. Obtained directly from Theorem 22.11 of the 'Harmonic Analysis' chapter (the DFT leakage estimate for the intercalation window), where balanced intercalation gives $|E(M)| \leq 1/2$. ☐

Corollary 13.4 (Phase Closure of the Intercalation Window). Let $\rho \notin \mathbb{Q}$, and let $\{p_j/q_j\}$ be the continued-fraction convergents of ρ. Take $T = p_j$, $M = q_j$ (so that the intercalation window is $[0, p_j]$). Then the cumulative error within the intercalation window is

$$|E(q_j)| = |p_j - q_j\rho| < \frac{1}{q_{j+1}},$$

and the window approximately closes over $M = q_j$ continuous periods—when the intercalation functor $\mathcal{R}(F)$ is extended with period $S_M = p_j$ on the window $[0, p_j]$, the phase error is less than $1/q_{j+1}$.

Proof. Follows directly from Theorem 21.5 and Theorem 13.6 of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter. ☐

Theorem 13.7 (Universality of the Optimal Continued-Fraction Intercalation Functor). Let $\rho > 0$ be irrational, and let $\{p_j/q_j\}$ be its continued-fraction convergents. Then on the window sequence $\{(p_j, q_j)\}_{j\geq 0}$ above, the intercalation functor $\mathcal{R}$ satisfies

$$\lim_{j\to\infty} \frac{|E(q_j)|}{p_j} = 0, \qquad \frac{|E(q_j)|}{p_j} < \frac{1}{p_j q_{j+1}} \to 0.$$

That is, along the continued-fraction convergent window sequence the **relative phase error** of the intercalation functor $\mathcal{R}$ tends to zero—in the limiting sense the intercalation functor **restores** the categorical connection between $\mathcal{C}_{\text{cont}}$ and $\mathcal{C}_{\text{trans}}$ that was severed by the commensurability gap.

Proof. From the main-lobe offset bound $|m|\,|E|/(\rho S_M)$ of Theorem 13.6 and the bound $|E(q_j)| < 1/q_{j+1}$ of Corollary 13.4, we obtain

$$\frac{|E(q_j)|}{p_j} < \frac{1}{p_j q_{j+1}}.$$

By the theory of continued fractions, $p_j, q_j \to \infty$ and $q_{j+1} > q_j$, so $1/(p_j q_{j+1}) \to 0$. ☐

13.10.3 Commutation of the Intercalation Functor and the Continuum-Limit Functor

Theorem 13.8 (Commutation of the Intercalation Functor and the Continuum-Limit Functor). Let $\mathcal{D}$: **Man** $\to$ **Latt** be the discretisation functor, $\mathcal{C}_\infty$: **Latt** $\to$ **Man** the continuum-limit functor (defined in the preceding section, 13.20), and $\mathcal{R} : \mathcal{C}_{\text{cont}} \to \mathcal{C}_{\text{trans}}$ the intercalation functor. Then the following diagram commutes in the limiting sense:

$$\begin{tikzcd}
\mathcal{C}_{\text{cont}} \arrow[r, "\mathcal{R}"] \arrow[d, "\text{embedding}"'] & \mathcal{C}_{\text{trans}} \arrow[d, "\mathcal{D}"] \\
\textbf{Man} \arrow[r, "\mathcal{C}_\infty"'] & \textbf{Latt}
\end{tikzcd}$$

More precisely: for $F \in \mathcal{C}_{\text{cont}}$, take the sequence of intercalation windows $\{(p_j, q_j)\}_{j \geq 0}$ (with $p_j \to \infty$). Then

$$\lim_{j\to\infty} \mathcal{D}\big(\mathcal{R}_j(F)\big) = \mathcal{D}\big(\mathcal{D}(F)\big),$$

where $\mathcal{R}_j$ denotes the intercalation functor on the j-th intercalation window, and $\mathcal{D}(\mathcal{R}_j(F))$ is the realisation of the intercalation sequence on the lattice points.

Proof. By Theorem 13.7, on the j-th window the relative phase error of the intercalation sequence $\mathcal{R}_j(F)$ tends to zero. Hence on the window $[0, p_j]$, the difference between $\mathcal{R}_j(F)$ and $\mathcal{D}(F)$ tends to zero in the sense of bounded error. Letting $j \to \infty$, the two discrete sequences converge to the same object in the lattice category. □

Remark 13.11 (The Categorical Status of the Intercalation Functor). The relation between the intercalation functor $\mathcal{R}$ and the discretisation functor $\mathcal{D}$ may be understood as follows: $\mathcal{R}$ is a **bounded-error correction** of $\mathcal{D}$ in the incommensurable case. When $\rho \in \mathbb{Q}$ the two are isomorphic (Proposition 13.11); when $\rho \notin \mathbb{Q}$, $\mathcal{R}$ drives the phase error to zero by choosing continued-fraction convergence windows (Theorem 13.7), thereby restoring the categorical nature of $\mathcal{D}$ in the limiting sense.

From the perspective of adjoint functors, if the discretisation functor $\mathcal{D}$ is regarded as the left adjoint and the continuum-limit functor $\mathcal{C}_\infty$ as the right adjoint (Proposition 13.9 of the preceding section), then the intercalation functor $\mathcal{R}$ may be regarded as a **consistency correction** of the adjoint pair $(\mathcal{D}, \mathcal{C}_\infty)$ in the incommensurable case—it does not change the adjoint pair itself, but carries correction data at the level of objects, so that the natural isomorphism of the adjunction is preserved to arbitrary precision in the incommensurable case.

Proposition 13.12 (Conserved Quantities of the Intercalation Functor)**.** At the level of objects, the intercalation functor $\mathcal{R}$ carries the following conserved quantities:

(i) **Boundedness of the cumulative error**: $|E(M)| \leq C$ holds for all M—the phase error of the intercalation sequence is conserved and bounded on the scale of the window;

(ii) **Monotone convergence of the relative error**: $\lim_{j\to\infty} \dfrac{|E(q_j)|}{p_j} = 0$—the relative phase error on the continued-fraction convergence windows tends to zero;

(iii) **Conservation of energy**: the DFT Parseval identity on the intercalation window

$$\sum_{n=0}^{S_M - 1} |f(n)|^2 = S_M \sum_{k=0}^{S_M - 1} |\hat{f}(k)|^2$$

holds exactly on the intercalation window—intercalation does not change the total energy within the window.

Proof. (i) By Lemma 13.4. (ii) By Theorem 13.7. (iii) By applying Theorem 21.9 of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter on the intercalation window. □

Remark 13.12 (The Interface between the Intercalation Functor and the Observer Functor). The relation between the intercalation functor $\mathcal{R}$ and the observer functor $\mathcal{P}$ (defined in the next section, 13.24) is as follows: the observer functor extracts observable traces, whilst the intercalation functor determines the optimal length of the observer's sampling window. Concretely, the intercalation functor gives the window sequence $\{(p_j, q_j)\}$, and the observer functor projects observable traces onto this window—intercalation makes the observer's sampling window approximately contain an integer number of continuous periods, thereby minimising the phase error of the projection. Together the two form a 'sampling–projection' double functor

$$\mathcal{C}_{\mathsf{cont}} \xrightarrow{\mathcal{R}} \mathcal{C}_{\mathsf{trans}} \xrightarrow{\mathcal{P}} \mathcal{C}_{\mathsf{obs}},$$

where $\mathcal{R}$ renders the phase error bounded and $\mathcal{P}$ renders the structural projection conformal.

$$\begin{tikzcd}
\mathcal{C}_{\mathsf{cont}} \arrow[r, "\mathcal{R}"] \arrow[d, "\text{embedding}"'] & \mathcal{C}_{\mathsf{trans}} \arrow[d, "\mathcal{D}"] \\
\mathsf{Man} \arrow[r, "\mathcal{C}_\infty"'] & \mathsf{Latt}
\end{tikzcd}$$

Figure 13.11: Commutative diagram of the intercalation functor and the continuum-limit functor: they commute in the limiting sense.

13.11 The Observer as a Functor

Definition 13.24 (The Observer Functor). The **observer functor** $\mathcal{P} : \mathcal{C}_{\mathsf{trans}} \times \mathcal{C}_{\mathsf{rot}} \to \mathcal{C}_{\mathsf{obs}}$ maps the product of the time-periodic category and the rotation-periodic category to the **category of observable traces** $\mathcal{C}_{\mathsf{obs}}$:

$$\mathcal{P}(f_T, f_S) = \mathcal{N}(r, \sigma) \cdot (\Phi(f_T) \cdot \nabla S + \Phi(f_S) \cdot \nabla S) \cdot \mathbf{u}_P,$$

where $\mathcal{N}(r, \sigma) = (\sigma/(r^2 + \sigma))^2$ is the sampling normalisation factor, Φ is the phase-extraction functor, and $\mathbf{u}_P$ is the line-of-sight direction of the observer.

Theorem 13.9 (Decomposition of the Observer Functor). The observer functor $\mathcal{P}$ can be decomposed as the composition of three sub-functors:

$$\mathcal{P} = \mathcal{P}_{\cdot} \circ \mathsf{add} \circ (\mathcal{P}_{\nabla} \times \mathcal{P}_{\times}),$$

where $\mathcal{P}_{\nabla}$ projects the temporal flux onto the irrotational part of the observable field, $\mathcal{P}_{\times}$ projects the spatial flux onto the solenoidal part, and $\mathcal{P}_{\cdot}$ applies the normalisation factor.

Proof. This follows by decomposing term by term the explicit formula of Definition 13.24. The irrotational/solenoidal decomposition is guaranteed by the Helmholtz decomposition theorem of the 'Vector Analysis' chapter (Theorem 15.11). □

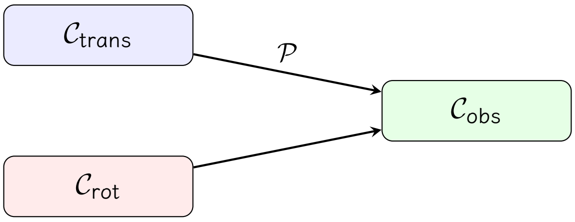
Corollary 13.5 (The Observer Functor and Adjunction)**.** *The observer functor $\mathcal{P}$ has a left adjoint $\mathcal{P}^*$ (the reconstruction functor), forming an adjoint pair:*

$$\mathcal{P}^* \dashv \mathcal{P}.$$

The unit η of the adjunction is not an isomorphism—the loss of information in the process of observation is irreversible.

Proof. Construct $\mathcal{P}^*$ as the canonical lift: for each trace O, take $\mathcal{P}^*(O)$ to be the minimal-norm preimage satisfying $\mathcal{P}(\mathcal{P}^*(O)) = O$. The adjunction bijection is given by the linear structure. The unit η is not an isomorphism because the canonical lift is not a section of the observation. ☐

Remark 13.13 (The Observer Functor and the Breaking of Conservation Laws)*.* *The unit η of the adjunction not being an isomorphism means that the observer cannot completely reconstruct the original periodic structure from the observable traces. The loss of information in the process of observation is irreversible—the conserved quantities of the time system and the space system (energy, angular momentum) are partially destroyed by the divergence of the observing system (symmetry centre 1) (sampling decay, projection selection).*


$\mathcal{C}_{\text{trans}}$
$\mathcal{P}$
$\mathcal{C}_{\text{obs}}$
$\mathcal{C}_{\text{rot}}$

Observer functor: time + space → observable traces

Figure 13.12: The observer functor $\mathcal{P}$: it projects the product of the time-periodic category and the rotation-periodic category onto the category of observable traces.

13.12 Periodic Categories and Conservation Laws

Theorem 13.10 (A Categorical Formulation of Noether's Theorem)**.** *Let $\mathcal{C}^G$ be a G-periodic category, where G is a Lie group. There exists a natural isomorphism:*

$$\mathsf{Symmetry}(\mathcal{C}^G) \cong \mathsf{Conserved}(\mathcal{C}^G),$$

where $\mathsf{Symmetry}(\mathcal{C}^G)$ is the group of continuous symmetries of $\mathcal{C}^G$ and $\mathsf{Conserved}(\mathcal{C}^G)$ is the space of conserved quantities of $\mathcal{C}^G$.

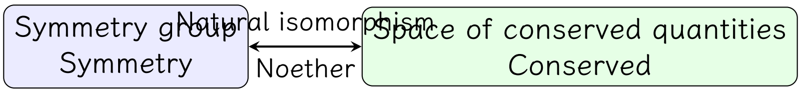
Specifically:

(i) *Translational symmetry ($G = \mathbb{R}$, 0-symmetry) $\cong$ conservation of energy/momentum.*

(ii) *Rotational symmetry ($G = U(1)$, i-symmetry) $\cong$ conservation of angular momentum.*

(iii) Scaling symmetry ($G = \mathbb{R}^+$, 1-symmetry) $\cong$ conservation of the scale charge.

Proof. For each object $(X, \rho) \in \mathcal{C}^G$: every one-parameter subgroup of G, through the action ρ, gives a one-parameter family of automorphisms of X, whose infinitesimal generator corresponds to the Noether charge, namely the conserved quantity (Theorem 20.4 of the 'Periodic Functions and Groups with Periodic Orbits' chapter). Naturality is given by the equivariance of periodic morphisms. ☐



$\text{Symmetry}(\mathcal{C}^G) \cong \text{Conserved}(\mathcal{C}^G)$

Figure 13.13: A categorical formulation of Noether's theorem: the natural isomorphism between the symmetry group and the space of conserved quantities.

13.13 A Unified Perspective on Periodic Categories

Theorem 13.11 (The Categorical Unification of Periodic Mathematics). *The core structures of periodic mathematics can be unified through the following commutative diagram:*

$$\begin{tikzcd}
\mathcal{C}_{\text{scal}} \arrow[r, "\mathcal{L}"] \arrow[d, "\mathcal{F}_{\text{scal}}"'] & \mathcal{C}_{\text{trans}} \arrow[d, "\mathcal{F}_{\text{trans}}"] \\
\mathsf{Vect}_{\mathbb{C}} \arrow[r, "\mathcal{L}_*"'] & \mathsf{Vect}_{\mathbb{C}}
\end{tikzcd}$$

where $\mathcal{L}$ is the logarithm functor, $\mathcal{F}_{\text{trans}}$ is the Fourier transform and $\mathcal{F}_{\text{scal}}$ is the Mellin transform. The commutativity of the left-hand square is verified directly by the substitution $x = e^t$.

Proof. Under the substitution $x = e^t$, the scaling-periodic function $f(x)$ becomes the translation-periodic function $g(t) = f(e^t)$. The Mellin transform $\mathcal{F}_{\text{scal}}$ and the Fourier transform $\mathcal{F}_{\text{trans}}$ commute under this substitution: $\mathcal{F}_{\text{trans}}(\mathcal{L}f) = \mathcal{L}_*(\mathcal{F}_{\text{scal}}f)$. The conservation structure is guaranteed by Theorem 13.10 and the Parseval identity. ☐

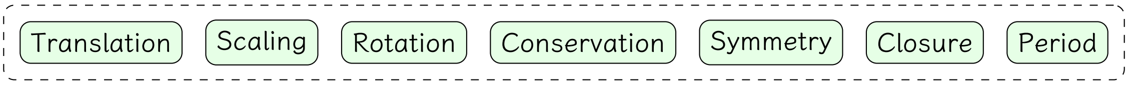
Remark 13.14 (The Pivotal Role of Periodic Categories throughout the Book). The periodic category is the highest level of abstraction in the theoretical system of periodic mathematics. It brings into the language of category theory the group-representation prototypes of the 'Operations and Periodicity' chapter, the centres of symmetry of the 'Number Systems' chapter, the similarity and tendency of the 'Scalars and Vectors' chapter, the calculus foundations of the 'Functions and Limits' chapter, the periodic transforms and convolutions of the 'Series and Convolution' chapter, and the discrete geometry of the 'Periodic Manifold' chapter—all unified together.

Within this unified framework:

- **Period** is the natural expression, at the level of category theory, of invariance under a group action—the G-periodic object (X, ρ).
- **Conservation laws** are the natural isomorphism, at the level of functors, of the Noether charges of continuous symmetries—Symmetry $\cong$ Conserved.
- **Phase** is a structure-preserving map on a category—any phase of any period may serve as a functor (Corollary 13.1).
- **Differentiation and integration** are an adjoint pair of operators on the periodic category—$\mathcal{D} \dashv \mathcal{I}$.
- **Convolution and periodic convolution** are bifunctors on the periodic category—$*$ and $\circledast$.
- **The observer** is the central functor connecting the time system and the space system—$\mathcal{P}$.

Remark 13.15 (An Outlook on Higher Categories). This chapter takes the natural transformation (Definition 13.5) as the highest-level structure. A natural next step is to incorporate into the framework the morphisms between natural transformations—modifications: the continuous variation of a family of periodic parameters is precisely an instance of a modification, and in this case the periodic category organises into a 2-category. The fact that the adjunction unit of the observer functor is not an isomorphism shows that the observation–reconstruction cycle forms an irreversible 2-morphism at the level of the 2-category. This is a prospective note, and no theorem is established.

Seven-theme comparison table for this chapter



Seven themes running through periodic categories

Figure 13.14: The seven themes of this chapter: translation, scaling, rotation, conservation, symmetry, closure and period.

Theme	Corresponding content in this chapter
Translation	The translation-periodic category $\mathcal{C}_{\text{trans}}$ and the differential operator $\mathcal{D}$ (0-symmetry); the cumulative sample count S_M of the intercalation sequence approximates $M\rho$ (Definition 13.22)
Scaling	The scaling-periodic category and the logarithm/exponential functors (1-symmetry); the ideal period ratio ρ as the scale charge (Definition 13.21)
Rotation	The rotation-periodic category, the phase functor and the Fourier functor (i-symmetry); the phase correction of the intercalation window (Theorem 13.6)
Conservation	The conservation condition for adjoint operators (Corollary 13.3); the categorical formulation of Noether's theorem (Theorem 13.10); the boundedness of the cumulative intercalation error (Lemma 13.4)
Symmetry	Natural isomorphism and the adjoint duality $\mathcal{D} \dashv \mathcal{I}$; the family of phase functors $\cong U(1)$ (Theorem 13.1); the symmetric approximation by continued-fraction convergents (Theorem 13.7)
Closure	The mutual-inverse closure of the differential and integral operators (Lemma 13.3); the closure property of the periodic-convolution bifunctor (Theorem 13.4); the intercalation window closes the approximate period (Corollary 13.4)
Period	The categorical formulation of periodic manifolds; the parametrisation of the phase functor (Corollary 13.1); when $\rho \in \mathbb{Q}$ the intercalation functor degenerates to a strict periodic functor (Proposition 13.11)

Exercises

Exercise 1 Using Definition 13.2 and Definition 13.3, explain the essential difference in scope between an operator (endofunctor) and a functor. Give one example of each: an operator that leaves the periodic type unchanged, and a functor that maps the translation-periodic category to another category (for instance the Fourier functor, which maps periodic functions to sequences of Fourier coefficients).

Exercise 2 Prove the six basic properties of the phase functor $\Phi_{\theta,T}$ in Proposition 13.1: functoriality, period preservation, the composition law, the identity element, the inverse element and unitarity. Also verify Theorem 13.1: the family of phase functors for a fixed period T forms a group under composition, isomorphic to $U(1)$.

Exercise 3 Verify the concrete action of the phase functor on the sine function in Example 13.2: take $T = 2\pi$ and $f(t) = \sin t$, and compute

$$\Phi_{\pi/2,2\pi}(f)(t), \qquad \Phi_{\pi,2\pi}(f)(t).$$

Explain why $\Phi_{\pi,2\pi}$ is the identity map on the sine function, and point out how this is related to the antipodal relation when the phase is π.

Exercise 4 Prove Lemma 13.1: let $f(t) = Ae^{i(\omega_1 t+\varphi)}$ and $g(t) = Be^{i(\omega_2 t+\psi)}$ be two single-frequency circle objects.

(i) If $\omega_1 \neq \omega_2$, then $\text{Hom}(f, g) = \{0\}$;
(ii) If $\omega_1 = \omega_2 = \omega$, then $\text{Hom}(f, g) = \{c \cdot \frac{B}{A} e^{i(\psi-\varphi)} : c \in \mathbb{C}\}$.

Explain why amplitude and phase are not invariants of the translation-periodic category (see Remark 13.6).

Exercise 5 Verify the computation of the morphisms between the three single-frequency circle objects in Example 13.3:

$$f(t) = 3e^{i(2t+\pi/6)}, \quad g(t) = 2e^{i(3t-\pi/4)}, \quad h(t) = 5e^{i(2t+\pi/3)}.$$

Compute $\text{Hom}(f, g)$, $\text{Hom}(f, h)$ and $\text{Hom}(h, k)$ (taking $k(t) = 4e^{i2t}$), and verify the composition law $\mu \circ \lambda = \nu$.

Exercise 6 Prove Theorem 13.3: on the translation-periodic category, the differential operator $\mathcal{D}$ and the integral operator $\mathcal{I}$ form the adjoint pair $\mathcal{D} \dashv \mathcal{I}$. Write down the natural isomorphism of the adjunction, and explain why the zero-mean condition is needed.

Exercise 7 Verify Lemma 13.3:

$$(\mathcal{D} \circ \mathcal{I})(f) = f - \bar{f}, \qquad (\mathcal{I} \circ \mathcal{D})(f) = f - f(0).$$

On the quotient category (modulo constants) of zero-mean periodic functions, the two are mutually strict inverses. Verify this with the numerical example 13.5.

Exercise 8 Prove Proposition 13.4: there exist natural functors among the three periodic categories— the logarithm functor $\mathcal{L} : \mathcal{C}_{\text{scal}} \to \mathcal{C}_{\text{trans}}$, the exponential functor $\mathcal{E} : \mathcal{C}_{\text{trans}} \to \mathcal{C}_{\text{scal}}$, and the Fourier functor $\mathcal{F} : \mathcal{C}_{\text{trans}} \to \text{Vect}_{\mathbb{C}}$. Also prove Corollary 13.2: $\mathcal{L} \dashv \mathcal{E}$.

Exercise 9 Prove Theorem 6.20: the periodic convolution $\circledast$ is a bifunctor on the translation-periodic category, satisfying preservation of identities, preservation of composition, compatibility with the similarity functor, and compatibility with the tendency functor. Explain that periodic convolution is represented in the frequency domain by the tensor product, and that phase and tendency are represented in the frequency domain by pointwise operations.

Exercise 10 Prove Proposition 13.10: the commensurability gap—when $\rho = T_F/\Delta x \in \mathbb{Q}$, the discretisation functor $\mathcal{D}$ preserves the periodic structure; when $\rho \notin \mathbb{Q}$, $\mathcal{D}$ fails to preserve the periodic structure on F. Discuss the three cases $\rho \in \mathbb{N}^+$, $\rho \in \mathbb{Q} \setminus \mathbb{N}^+$ and $\rho \notin \mathbb{Q}$ separately.

Exercise 11 Prove Theorem 13.5: the intercalation functor $\mathcal{R}$ preserves identity morphisms and composition. Explain why the intercalation functor is a bounded-error correction of the discretisation functor $\mathcal{D}$ in the incommensurable case (see Remark 13.11).

Exercise 12 Prove Theorem 13.6: in the frequency domain, the intercalation functor maps an incommensurable continuous object to a discrete object with bounded phase error. Write down the bound on the main-lobe offset

$$\left|\frac{m}{\rho} - \frac{k}{S_M}\right| \leq \frac{|m|\,|E(M)|}{\rho\, S_M} \leq \frac{|m|}{2\rho S_M},$$

and explain that balanced intercalation gives $|E(M)| \leq 1/2$.

Exercise 13 Prove Theorem 13.8: the intercalation functor and the continuum-limit functor commute in the limiting sense. Write down the commutative diagram, and explain why the relative phase error tends to zero along the continued-fraction convergence window sequence (see Theorem 13.7).

Exercise 14 Verify the three conserved quantities of the intercalation functor in Proposition 13.12: boundedness of the cumulative error, monotone convergence of the relative error, and conservation of energy. Taking $\rho = \sqrt{2}$ as an example, compute the relative errors corresponding to the continued-fraction denominator sequence $q_k = 1, 2, 5, 12, 29, 70, \ldots$.

Exercise 15 Prove Proposition 6.28:

(i) the trend convolution $=$ the amplitude convolution $\times$ the phase convolution;

(ii) the frequency convolution $=$ the outer product of frequency series;

(iii) the three capture, in the frequency domain, the three independent components of convolution—amplitude, phase and frequency.

Exercise 16 Verify Proposition 6.32: the periodic convolution is the summit of the five-level ladder of convolution theory:

$$\text{periodic convolution}\circledast \to \text{trend convolution}\mathbb{T} \to \text{classical convolution} * .$$

Write down the concrete forms of the degeneration at the single-period level and of the diagonal-kernel degeneration, and explain that the information content of each step of the ladder is strictly increasing.

Exercise 17 Prove Theorem 13.10: the categorical formulation of Noether's theorem—there exists a natural isomorphism

$$\text{Symmetry}(\mathcal{C}^G) \cong \text{Conserved}(\mathcal{C}^G).$$

Write out the three cases separately: translational symmetry (corresponding to conservation of energy/momentum), rotational symmetry (corresponding to conservation of angular momentum), and scaling symmetry (corresponding to conservation of the scale charge).

Exercise 18 Verify the commutative diagram in Theorem 13.11:

$$\begin{tikzcd}
\mathcal{C}_{\text{scal}} \arrow[r, "\mathcal{L}"] \arrow[d, "\mathcal{F}_{\text{scal}}"'] & \mathcal{C}_{\text{trans}} \arrow[d, "\mathcal{F}_{\text{trans}}"] \\
\mathsf{Vect}_{\mathbb{C}} \arrow[r, "\mathcal{L}_*"'] & \mathsf{Vect}_{\mathbb{C}}
\end{tikzcd}$$

Explain that the commutativity of the left-hand square is verified directly by the substitution $x = e^t$, and point out the correspondence between the Mellin transform and the Fourier transform under this substitution.

Exercise 19 Comprehensive question: combining the phase functor (Definition 13.4), the similarity and tendency functors (Definitions 13.13, 13.14), the differentiation–integration adjunction (Theorem 13.3), the periodic-convolution bifunctor (Definition 13.18) and the observer functor (Definition 13.24), explain how the periodic category unifies the theoretical structure of the whole book within the language of category theory. Point out separately the categorical objects corresponding to 'period', 'conservation laws', 'phase', 'differentiation and integration', 'convolution' and 'the observer' in the periodic category.

Chapter 14

Probability and Distribution

Remark 14.1 (Positioning of this chapter). This chapter systematically establishes the basic framework of probability theory and distribution theory, focusing on random variables and distribution functions, common distribution families and their interrelations, the unification of characteristic functions with the Fourier transform, Bayesian inference, the basic types of stochastic process, and the laws of large numbers and the central limit theorem.

Core interface with periodic mathematics: probability theory and periodic mathematics meet in depth on four levels. First, **the characteristic function of a probability distribution is precisely the Fourier transform of the distribution** (Theorem 14.2), and the entire information of a distribution is encoded in the modulus and the phase of its characteristic function, in complete isomorphism with the spectral analysis of periodic functions. Second, **the uniform distribution on the circle S^1 is the Haar measure on the circle group** (Definition 14.15), and its rotation invariance is precisely the realisation of $U(1)$-symmetry at the probabilistic level. Third, **probability distributions on the cyclic group C_N** (Proposition 14.4) together with periodic sequences, the discrete Fourier transform and circular convolution form a complete algebraic–probabilistic correspondence. Fourth, **stationary processes and cyclostationary processes** (Definition 14.32, 14.34) are the action of the time-translation group on stochastic processes, and the spectral decomposition of their autocovariance function (Definition 14.33) and the power spectrum are the core tools of periodic mathematics at the stochastic level.

Guide to the seven themes of this chapter: translation—the statistical invariance of stationary processes under the action of the time-translation group $\mathbb{Z}$ (discrete) or $\mathbb{R}$ (continuous) (Definition 14.32), together with the advance of sequential Bayesian updating along the time axis (Definition 14.25); scaling—the scale invariance of self-similar processes and heavy-tailed distributions under the scaling transformation $t \mapsto \lambda t$ (Hurst exponent H, tail exponent α, Definition 14.37), together with the scaling convergence of the central limit theorem (Theorem 14.5); rotation—the phase information of the argument $\arg \phi_X(t)$ of the characteristic function in frequency space (Remark 14.3), and the rotational symmetry of the uniform distribution on S^1 (Definition 14.15); conservation—the invariance of the autocovariance under time translation (Proposition 14.12) and the invariance of the scaling exponent under the scaling transformation; symmetry—the symmetry of distribution families under the prior/likelihood/posterior tripartite structure (the closure of conjugate priors, Definition 14.23); closure—the convergence closure of the laws of large numbers and the central limit theorem (Theorem 14.4, 14.5) and the closure criterion 'time average = space average' for ergodic orbits (Theorem 14.6); period—the period T of a cyclostationary process (Definition 14.34) and the periodic pattern of Markov-chain eigenvalues with $|\lambda_i| = 1$ (Proposition 14.15).

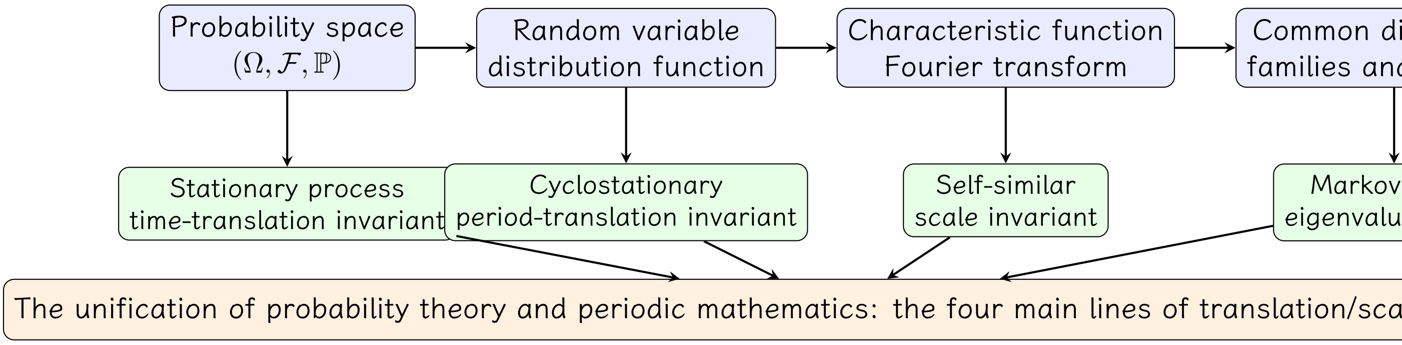

Probability space
$(\Omega, \mathcal{F}, \mathbb{P})$
Random variable
distribution function
Characteristic function
Fourier transform
Common di
families anc
Stationary process
time-translation invariant
Cyclostationary
period-translation invariant
Self-similar
scale invariant
Markov
eigenvalu
The unification of probability theory and periodic mathematics: the four main lines of translation/sca

Figure 14.1: Overview of the structure of this chapter: the core of probability theory (probability space, random variable, characteristic function, distribution families) and the interface with periodic mathematics (stationary, cyclostationary, self-similar, Markov chain), finally unified along four main lines.

14.1 Events and Their Probabilities

14.1.1 Random Events and Probability

Definition 14.1 (Random event). *In a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, the elements of $\mathcal{F}$ are called* **random events***. The* **probability** *$\mathbb{P}(A)$ of an event A is a real number in $[0, 1]$ satisfying:*

(i) **Non-negativity***: $\mathbb{P}(A) \geq 0$ holds for all $A \in \mathcal{F}$.*

(ii) **Normalisation***: $\mathbb{P}(\Omega) = 1$.*

(iii) **Countable additivity***: for a sequence of pairwise disjoint events $\{A_n\}_{n=1}^{\infty}$, we have $\mathbb{P}(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mathbb{P}(A_n)$.*

Proposition 14.1 (Basic properties of the probability measure). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $A, B \in \mathcal{F}$. Then*

(i) *$\mathbb{P}(\emptyset) = 0$;*

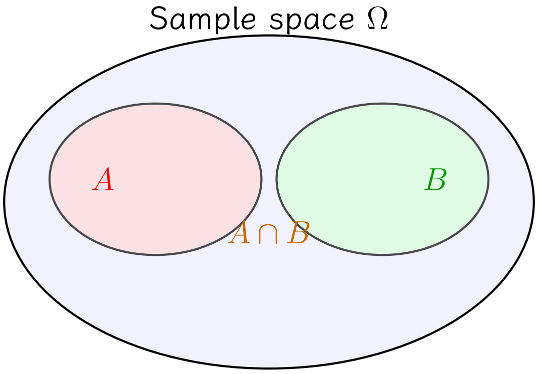
(ii) **Monotonicity***: if $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$;*

(iii) **Complementarity***: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$;*

(iv) **Finite additivity***: if $A \cap B = \emptyset$, then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.*

Proof. (i) Take $A_n = \emptyset$ $(n \geq 1)$; by countable additivity $\mathbb{P}(\emptyset) = \sum_{n=1}^{\infty} \mathbb{P}(\emptyset)$, hence $\mathbb{P}(\emptyset) = 0$. (ii) If $A \subseteq B$, then $B = A \cup (B \setminus A)$ is a disjoint union, so $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$. (iii) $\Omega = A \cup A^c$ is disjoint, so by normalisation $1 = \mathbb{P}(\Omega) = \mathbb{P}(A) + \mathbb{P}(A^c)$. (iv) Let $A_1 = A$, $A_2 = B$ and $A_n = \emptyset$ $(n \geq 3)$; countable additivity together with (i) gives $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$. □

Definition 14.2 (The relation between frequency and probability). *Let the event A occur n_A times in n independent repeated trials. The* **frequency** *of A is defined as $f_n(A) = n_A/n$. By the law of large numbers, the frequency converges in probability to the probability: $\lim_{n\to\infty} f_n(A) = \mathbb{P}(A)$. The frequency is the empirical estimate of the probability, and the probability is the limit of the frequency.*



$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

Figure 14.2: Venn diagram of events and the addition formula: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

14.1.2 Formulae for Calculating Probabilities

Definition 14.3 (Addition formula). For any two events A and B, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Definition 14.4 (Conditional probability and the multiplication formula). If $\mathbb{P}(B) > 0$, then

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \quad \mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B).$$

Definition 14.5 (The law of total probability and Bayes' formula). Let $\{B_i\}_{i=1}^n$ be a partition of the sample space Ω with $\mathbb{P}(B_i) > 0$. Then

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A \mid B_i)\mathbb{P}(B_i),$$

$$\mathbb{P}(B_i \mid A) = \frac{\mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}{\sum_{j=1}^n \mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}.$$

Definition 14.6 (Independence of events). Events A and B are called **independent** if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. Independence means that the occurrence of one event does not change the conditional probability of the other.

14.2 Random Variables and Distribution Functions

14.2.1 Random Variables

Definition 14.7 (Random variable). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(\mathcal{X}, \mathcal{B})$ a measurable space. A **random variable** is a measurable mapping $X : \Omega \to \mathcal{X}$, that is, $X^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{B}$.

When $\mathcal{X} = \mathbb{R}$ and $\mathcal{B} = \mathcal{B}(\mathbb{R})$, X is called a real-valued random variable; when $\mathcal{X} = \mathbb{Z}/N\mathbb{Z}$ or $\mathcal{X}$ is finite, X is called a discrete random variable.

Remark 14.2 (On the use of the terms 'phase space' and 'phase encoding'). In the probabilistic framework of this section, the sample space Ω is sometimes understood as the 'phase space' of a periodic system, and a random variable is sometimes understood as a 'phase encoding'. Elsewhere in this book these terms are completely equivalent to 'sample space' and 'random variable'; they serve only to keep the terminology consistent with that of the other chapters. In this chapter and in the subsequent discussion the two may be used interchangeably, and they do not constitute a new mathematical concept. For instance, for values taken in the finite cyclic group $C_N = \mathbb{Z}/N\mathbb{Z} = \{0, 1, \ldots, N-1\}$, the group C_N is the sample space and $k \mapsto \mathbb{P}(X = k)$ is the probability mass function.

Example 14.1 (A basic example of a finite sample space). $C_2 = \{0, 1\}$ is the smallest non-trivial finite sample space. The Bernoulli distribution $\mathbb{P}(X = 1) = p$, $\mathbb{P}(X = 0) = 1 - p$ is a probability distribution on C_2. A distribution $(p_0, p_1, \ldots, p_9)$ on $C_{10} = \{0, 1, \ldots, 9\}$ gives a probability on the decimal digits.

14.2.2 Cumulative Distribution Functions

Definition 14.8 (Cumulative distribution function). Let X be a real-valued random variable. The **cumulative distribution function** (CDF) of X is defined as

$$F_X(x) = \mathbb{P}(X \le x), \qquad x \in \mathbb{R}.$$

Property 14.1 (Basic properties of the cumulative distribution function). Let F_X be the cumulative distribution function of a random variable X. Then

(i) **Monotonically non-decreasing**: if $x_1 \le x_2$, then $F_X(x_1) \le F_X(x_2)$.

(ii) **Right-continuous**: $\lim_{h \to 0^+} F_X(x + h) = F_X(x)$.

(iii) **Limiting behaviour**: $\lim_{x \to -\infty} F_X(x) = 0$, $\lim_{x \to +\infty} F_X(x) = 1$.

(iv) **Interval probability**: $\mathbb{P}(a < X \le b) = F_X(b) - F_X(a)$.

Proof. (i) By monotonicity (Proposition 14.1 (ii)), the event $\{X \le x_1\} \subseteq \{X \le x_2\}$, hence $F_X(x_1) \le F_X(x_2)$.

(ii) Let $x_n \downarrow x$; then the sequence of events $\{X \le x_n\}$ is decreasing with intersection $\{X \le x\}$, so by the continuity of probability $F_X(x_n) \to F_X(x)$.

(iii) As $x \to -\infty$ the sequence of events $\{X \le x\}$ decreases to the empty set, so $F_X \to 0$; as $x \to +\infty$ it increases to Ω, so $F_X \to 1$.

(iv) $\{X \le b\} = \{X \le a\} \cup \{a < X \le b\}$ is a disjoint union, hence $F_X(b) = F_X(a) + \mathbb{P}(a < X \le b)$. ☐

Theorem 14.1 (The one-to-one correspondence between CDFs and distributions). Two random variables X and Y have the same distribution if and only if their cumulative distribution functions are equal: $F_X = F_Y$.

Proof. Necessity: if $X \overset{d}{=} Y$, then $\mathbb{P}(X \le x) = \mathbb{P}(Y \le x)$ for all x, that is, $F_X = F_Y$. Sufficiency: suppose $F_X = F_Y$. For any $a < b$, Property 14.1(iv) gives $\mathbb{P}(a < X \le b) = \mathbb{P}(a < Y \le b)$. For any half-open interval $(a, b]$ the two random variables have equal probability. The σ-algebra generated by the family of half-open intervals is the Borel σ-algebra $\mathcal{B}(\mathbb{R})$, so the two random variables induce the same probability measure on $\mathcal{B}(\mathbb{R})$, that is, they are identically distributed. ☐

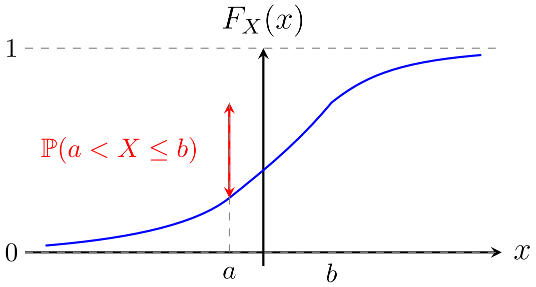


Figure 14.3: Properties of the cumulative distribution function: monotonically non-decreasing, right-continuous, with interval probabilities given by the difference of the endpoints.

Definition 14.9 (Quantile function). Let F_X be the cumulative distribution function of a random variable X. The **quantile function** (generalised inverse) of X is defined as

$$F_X^{-1}(p) = \inf\{x \in \mathbb{R} : F_X(x) \geq p\}, \qquad p \in (0, 1).$$

In particular, $F_X^{-1}(1/2)$ is called the **median**, whilst $F_X^{-1}(1/4)$ and $F_X^{-1}(3/4)$ are called the first and the third quartile respectively.

Proposition 14.2 (Basic properties of the quantile function). Let F_X^{-1} be the quantile function of a random variable X. Then

(i) **Monotonically non-decreasing**: $p_1 \leq p_2 \Rightarrow F_X^{-1}(p_1) \leq F_X^{-1}(p_2)$.

(ii) **Left-continuous**: F_X^{-1} is left-continuous on $(0, 1)$.

(iii) **Adjoint relation with the CDF**: $F_X^{-1}(p) \leq x \iff p \leq F_X(x)$.

Proof. (i) If $p_1 \leq p_2$, then $\{x : F_X(x) \geq p_2\} \subseteq \{x : F_X(x) \geq p_1\}$, so the infimum is monotone.

(ii) Let $p_n \uparrow p$. By (i), $F_X^{-1}(p_n)$ is increasing and $\leq F_X^{-1}(p)$. Let its limit be $L \leq F_X^{-1}(p)$. By the equivalence to be proved in (iii), $F_X^{-1}(p_n) \leq x \iff p_n \leq F_X(x)$. Taking $x = L$, right-continuity gives $p \leq F_X(L)$, hence $F_X^{-1}(p) \leq L$, and therefore $L = F_X^{-1}(p)$.

(iii) If $p \leq F_X(x)$, then $x \in \{y : F_X(y) \geq p\}$, so $F_X^{-1}(p) \leq x$. Conversely, if $F_X^{-1}(p) \leq x$, then from the monotone non-decrease of F_X and $F_X(F_X^{-1}(p)) \geq p$ we obtain $F_X(x) \geq p$. □

14.2.3 Discrete and Continuous Distributions

Definition 14.10 (Discrete distribution). Let X be a random variable taking values in a finite or countable set $\{x_0 < x_1 < \cdots\}$. The **probability mass function** (PMF) of X is $p_X(k) = \mathbb{P}(X = x_k)$, satisfying

$$p_X(k) \geq 0, \quad \sum_k p_X(k) = 1.$$

Proposition 14.3 (The probability mass function and the CDF of a discrete distribution)**.** Let X be a discrete random variable taking values in $\{x_0 < x_1 < \cdots\}$ and let $p_k = \mathbb{P}(X = x_k)$. Then

$$F_X(x) = \sum_{k : x_k \le x} p_k.$$

Proof. $\{X \le x\} = \bigcup_{k:x_k \le x}\{X = x_k\}$ is a countable disjoint union, and the result follows from countable additivity. ☐

Proposition 14.4 (Any finite discrete distribution is a distribution on a cyclic group)**.** Let X be a discrete random variable taking values in the finite set

$$\{x_0, x_1, \ldots, x_{N-1}\}.$$

Through the bijection $x_k \mapsto k \in C_N$, X is equivalent to a probability distribution $(p_0, p_1, \ldots, p_{N-1})$ on C_N, where $p_k = \mathbb{P}(X = x_k)$.

Proof. Construct the bijection $\varphi(x_k) = k$. Define $p(k) = \mathbb{P}(X = x_k)$. The total mass is $\sum_{k=0}^{N-1} p(k) = 1$. This is a standard reparametrisation of a probability distribution on a finite set and does not constitute new mathematical content. ☐

Proposition 14.5 (Translation-invariant distribution on the cyclic group)**.** The only translation-invariant probability distribution on C_N is the uniform distribution $p(k) = 1/N$. For any $a \in C_N$, the condition $p(k+a) = p(k)$ forces p to be constant.

Proof. Translation invariance requires $p(k + a) = p(k)$ for all $k, a \in C_N$. Taking $a = 1$ gives $p(k + 1) = p(k)$ for all k, so p is a constant function. Normalisation gives $p(k) = 1/N$. This is the Haar measure on C_N. ☐

Definition 14.11 (Continuous distribution)**.** A random variable X is called **continuous** if there exists a **probability density function** (PDF) $f_X(x)$ such that $\mathbb{P}(X \in B) = \int_B f_X(x)dx$ holds for all $B \in \mathcal{B}(\mathbb{R})$. The relation between the density and the cumulative distribution function is $F_X(x) = \int_{-\infty}^{x} f_X(t)dt$, and $F_X' = f_X$ wherever f_X is continuous.

14.3 Random Variables and Common Distributions

14.3.1 Discrete Random Variables

Definition 14.12 (The Bernoulli and binomial distributions)**.** The Bernoulli random variable $X \sim$ Bernoulli(p) is the distribution on C_2 given by $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(X = 0) = 1 - p$.

The number of successes $X \sim$ Binomial(n, p) in n independent Bernoulli trials is the distribution on $C_{n+1} = \{0, 1, \ldots, n\}$:

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n.$$

The *expectation* is $\mathbb{E}[X] = np$ and the variance is Var$(X) = np(1 - p)$.

Definition 14.13 (The Poisson distribution)**.** The Poisson distribution $X \sim \mathsf{Poisson}(\lambda)$ is the distribution on $\mathbb{N}$:

$$\mathbb{P}(X=k)=\frac{\lambda^k e^{-\lambda}}{k!}, \quad k=0,1,2,\ldots.$$

Both the *expectation* and the *variance* are λ. The Poisson distribution is the limiting distribution of the binomial distribution as $n \to \infty$ and $np \to \lambda$.

Definition 14.14 (The geometric and negative binomial distributions)**.** The geometric distribution $X \sim \mathsf{Geometric}(p)$ is the number of trials required for the first success:

$$\mathbb{P}(X=k)=(1-p)^{k-1}p, \quad k=1,2,\ldots.$$

The *expectation* is $\mathbb{E}[X]=1/p$ and the variance is $\mathsf{Var}(X)=(1-p)/p^2$. The geometric distribution has the memoryless property.

The negative binomial distribution $X \sim \mathsf{NegBinomial}(r,p)$ is the number of trials required for the r-th success:

$$\mathbb{P}(X=k)=\binom{k-1}{r-1}p^r(1-p)^{k-r}, \quad k=r,r+1,\ldots.$$

The *expectation* is $\mathbb{E}[X]=r/p$ and the variance is $\mathsf{Var}(X)=r(1-p)/p^2$.

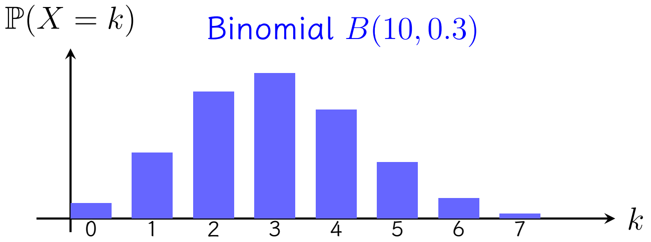


Figure 14.4: The probability mass function of the binomial distribution $B(10,0.3)$.

14.3.2 Continuous Random Variables

Definition 14.15 (The uniform distribution)**.** The density of $X \sim \mathsf{Uniform}(a,b)$ is $f_X(x)=\frac{1}{b-a}$ $(a \le x \le b)$. The uniform distribution on S^1 is the Haar measure on the circle group and possesses full rotational symmetry.

Proposition 14.6 (Rotation invariance of the uniform distribution on S^1)**.** *Let X be uniformly distributed on $S^1=\mathbb{R}/(2\pi\mathbb{Z})$ (density $f_X(\theta)=1/(2\pi)$). Then for any $\phi \in S^1$, the variable $X+\phi$ (modulo 2π) is still uniformly distributed. The uniform distribution on S^1 is the only probability distribution with rotation invariance.*

Proof. The density is $f_{X+\phi}(\theta)=f_X(\theta-\phi)=1/(2\pi)$, which is independent of ϕ. Conversely, if f is rotation invariant, then $f(\theta-\phi)=f(\theta)$ for all ϕ, so f is constant, and normalisation gives $f=1/(2\pi)$. □

Definition 14.16 (The normal distribution)**.** The density of $X \sim \mathcal{N}(\mu, \sigma^2)$ is

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}.$$

Here μ is the expectation and σ^2 is the variance. The normal distribution is the limiting distribution of the central limit theorem.

Definition 14.17 (The exponential distribution)**.** The density of $X \sim \text{Exponential}(\lambda)$ is $f_X(x) = \lambda e^{-\lambda x}$ $(x \geq 0)$. It has the memoryless property: $\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t)$. The expectation is $1/\lambda$ and the variance is $1/\lambda^2$.

Definition 14.18 (The gamma distribution)**.** The density of $X \sim \text{Gamma}(\alpha, \beta)$ is

$$f_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0,$$

where $\alpha > 0$ is the shape parameter and $\beta > 0$ is the rate parameter, and $\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt$ is the gamma function. The expectation is α/β and the variance is α/β^2. The exponential distribution is the special case $\alpha = 1$.

Definition 14.19 (The beta distribution)**.** The density of $X \sim \text{Beta}(\alpha, \beta)$ is

$$f_X(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, \quad x \in (0, 1),$$

where $B(\alpha, \beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$ is the beta function. The expectation is $\alpha/(\alpha+\beta)$ and the variance is $\alpha\beta/[(\alpha+\beta)^2(\alpha+\beta+1)]$. The uniform distribution $\text{Uniform}(0, 1)$ is the special case $\alpha = \beta = 1$.

Definition 14.20 (The chi-squared, t, and F distributions)**.** Let $Z_1, \ldots, Z_n$ be independent standard normal random variables.

(i) **Chi-squared distribution**: $X = \sum_{i=1}^n Z_i^2 \sim \chi^2(n)$, the special case $\text{Gamma}(n/2, 1/2)$, with expectation n and variance $2n$.

(ii) t **distribution**: if $Z \sim \mathcal{N}(0, 1)$ and $V \sim \chi^2(n)$ are independent, then $T = Z/\sqrt{V/n} \sim t(n)$, with expectation 0 $(n > 1)$ and variance $n/(n-2)$ $(n > 2)$.

(iii) F **distribution**: if $U \sim \chi^2(m)$ and $V \sim \chi^2(n)$ are independent, then $F = (U/m)/(V/n) \sim F(m, n)$.

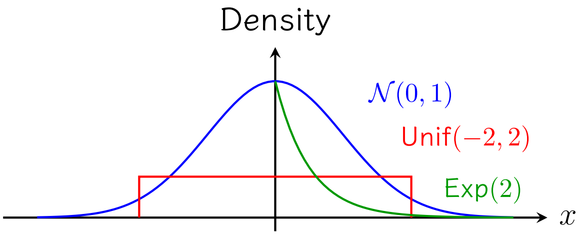


Figure 14.5: Comparison of the density functions of three continuous distributions: normal, uniform and exponential.

14.3.3 Relations among Common Distributions

Proposition 14.7 (Limits and special cases of common distributions).

- (i) **Binomial to Poisson**: Binomial$(n, \lambda/n) \xrightarrow{d}$ Poisson(λ) $(n \to \infty)$.
- (ii) **Binomial to normal**: $\frac{X-np}{\sqrt{np(1-p)}} \xrightarrow{d} \mathcal{N}(0,1)$ $(n \to \infty$, the De Moivre–Laplace theorem).
- (iii) **Gamma to normal**: $\frac{X-\alpha/\beta}{\sqrt{\alpha/\beta^2}} \xrightarrow{d} \mathcal{N}(0,1)$ $(\alpha \to \infty)$.
- (iv) **t distribution to normal**: $t(n) \xrightarrow{d} \mathcal{N}(0,1)$ $(n \to \infty)$.
- (v) **Beta to normal**: $\frac{X-\mu}{\sigma} \xrightarrow{d} \mathcal{N}(0,1)$ $(\alpha, \beta \to \infty$, where $\mu = \alpha/(\alpha+\beta)$ and $\sigma^2 = \alpha\beta/[(\alpha+\beta)^2(\alpha+\beta+1)])$.

Proof. (i) By the Poisson limit theorem (which follows from the limit of the probability generating function or of the characteristic function). (ii) By the central limit theorem (Theorem 14.5) applied to the sum of n independent Bernoulli random variables. (iii)(iv)(v) Verified directly by the large-parameter asymptotic expansion of the characteristic function (the saddle-point method). All five are classical results. ☐

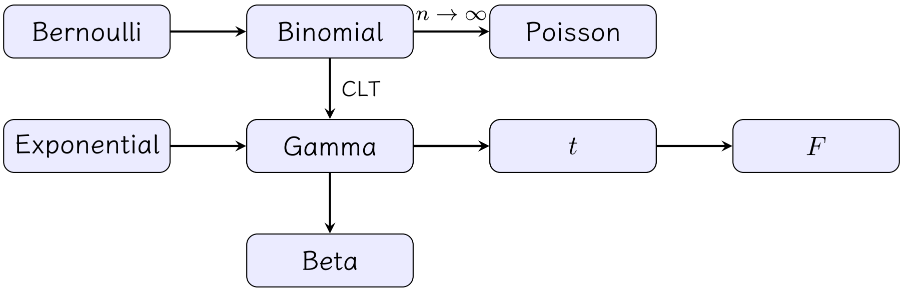


Figure 14.6: Diagram of the limits and special-case relations among common distributions.

14.4 Characteristic Functions

Definition 14.21 (Characteristic function). The **characteristic function** of a random variable X is defined as

$$\phi_X(t) = \mathbb{E}[e^{itX}] = \begin{cases} \sum_k e^{itk} p_X(k) & \text{(discrete)} \\ \int_{-\infty}^{\infty} e^{itx} f_X(x) dx & \text{(continuous)} \end{cases}, \quad t \in \mathbb{R}.$$

Theorem 14.2 (The characteristic function is the Fourier transform of the distribution). *The characteristic function $\phi_X(t)$ is precisely the Fourier transform $\widehat{\mu}_X(t)$ of the distribution measure μ_X with respect to the character $x \mapsto e^{itx}$: in the discrete case $\phi_X(t) = \sum_k e^{itk} p_X(k)$, and in the continuous case $\phi_X(t) = \int_{-\infty}^{\infty} e^{itx} f_X(x)\, dx$. Written in polar coordinates $\phi_X(t) = |\phi_X(t)| e^{i \arg \phi_X(t)}$: the modulus $|\phi_X(t)|$ measures the strength of coherent superposition at frequency t, whilst the argument $\arg \phi_X(t)$ measures the complex phase shift.*

Proof. By Definition 14.21, $\phi_X(t) = \mathbb{E}[e^{itX}] = \int e^{itx} d\mu_X(x)$, which is precisely the Fourier transform of the distribution measure μ_X. The polar decomposition $|\phi_X(t)|e^{i\arg\phi_X(t)}$ gives the two components, modulus and argument. ☐

Proposition 14.8 (Basic properties of the characteristic function).

(i) $\phi_X(0) = 1$ and $|\phi_X(t)| \leq 1$.

(ii) If X and Y are independent, then $\phi_{X+Y}(t) = \phi_X(t)\phi_Y(t)$.

(iii) $\mathbb{E}[X^n] = i^{-n}\phi_X^{(n)}(0)$—moments are the derivatives of the characteristic function at $t = 0$.

(iv) The characteristic function determines the distribution uniquely.

(v) $\phi_{aX+b}(t) = e^{ibt}\phi_X(at)$ (linear transformation).

Proof. (i) $\phi_X(0) = \mathbb{E}[e^0] = 1$; moreover $|\phi_X(t)| \leq \mathbb{E}|e^{itX}| = 1$. (ii) By independence, $\phi_{X+Y}(t) = \mathbb{E}[e^{it(X+Y)}] = \mathbb{E}[e^{itX}]\,\mathbb{E}[e^{itY}] = \phi_X(t)\phi_Y(t)$. (iii) Differentiating under the expectation sign: $\phi_X^{(n)}(t) = i^n\mathbb{E}[X^n e^{itX}]$; taking $t = 0$ gives $\phi_X^{(n)}(0) = i^n\mathbb{E}[X^n]$. (iv) By Lévy's inversion formula, the distribution function F_X can be recovered uniquely from ϕ_X. (v) $\phi_{aX+b}(t) = \mathbb{E}[e^{it(aX+b)}] = e^{ibt}\mathbb{E}[e^{i(at)X}] = e^{ibt}\phi_X(at)$. ☐

Remark 14.3 (Information carried by the characteristic function). The characteristic function ϕ_X contains three kinds of information about the distribution:

(i) **Moment information**: $\phi_X^{(n)}(0) = i^n\mathbb{E}[X^n]$—the derivatives of ϕ_X of all orders at $t = 0$ give the moments of the distribution of all orders. The expectation is $\mathbb{E}[X] = -i\phi_X'(0)$ and the variance is $\mathrm{Var}(X) = -\phi_X''(0) + \phi_X'(0)^2$.

(ii) **Modulus information**: $|\phi_X(t)|$ measures the strength of coherent superposition at frequency t.

(iii) **Decay information**: as $|t| \to \infty$, the rate of decay of $|\phi_X(t)|$ reflects the character of the tail of the distribution.

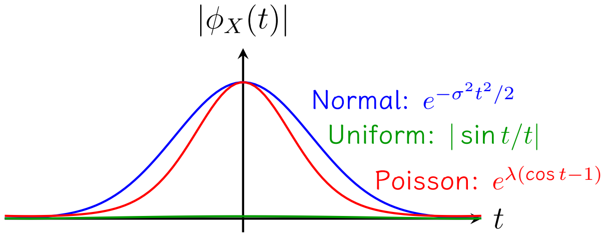


Figure 14.7: The modulus $|\phi_X(t)|$ of the characteristic function for three distributions: normal (Gaussian decay), Poisson (periodic oscillatory decay) and uniform (sinc decay).

Lemma 14.1 (The positive-definiteness of the characteristic function). Let ϕ_X be the characteristic function of a random variable X. Then for any $n \in \mathbb{N}^+$, any $t_1, \ldots, t_n \in \mathbb{R}$ and any complex numbers $z_1, \ldots, z_n$,

$$\sum_{j,k=1}^{n} z_j\overline{z_k}\,\phi_X(t_j - t_k) \geq 0,$$

that is, the matrix $\big(\phi_X(t_j - t_k)\big)_{j,k=1}^{n}$ is positive semi-definite.

Proof. By Definition 14.21 and the linearity of the expectation:

$$\sum_{j,k} z_j \overline{z_k}\, \phi_X(t_j - t_k) = \mathbb{E}\Big[\sum_{j,k} z_j \overline{z_k}\, e^{i(t_j - t_k)X}\Big] = \mathbb{E}\Big[\Big|\sum_{j} z_j e^{it_jX}\Big|^2\Big] \geq 0.$$

□

Example 14.2 (Explicit computation of the characteristic function of the Poisson distribution). Let $X \sim$ Poisson(λ). Then

$$\phi_X(t) = \sum_{k=0}^{\infty} e^{itk}\, \frac{\lambda^k e^{-\lambda}}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^{it})^k}{k!} = e^{\lambda(e^{it}-1)}.$$

Taking $\lambda = 2$ and $t = \pi/2$: $\phi_X(\pi/2) = e^{2(i-1)} = e^{-2}e^{2i} \approx 0.1353 \cdot (-0.4161 + 0.9093i) \approx -0.0563 + 0.1230i$. The modulus is $|\phi_X(\pi/2)| = e^{-2} \approx 0.1353$.

Example 14.3 (Explicit computation of the characteristic function of the binomial distribution). Let $X \sim$ Binomial(n, p). Then

$$\phi_X(t) = (pe^{it} + 1 - p)^n.$$

Taking $n = 3$ and $p = 1/2$: $\phi_X(t) = \frac{1}{8}(e^{it} + 1)^3$. Substituting $t = \pi$: $\phi_X(\pi) = (-\frac{1}{2} + \frac{1}{2})^3 = 0$, in agreement with direct summation.

Example 14.4 (The characteristic function of the normal distribution). If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $\phi_X(t) = e^{i\mu t - \sigma^2 t^2/2}$. The modulus $|\phi_X(t)| = e^{-\sigma^2 t^2/2}$ is the Gaussian kernel. The argument $\arg \phi_X(t) = \mu t$ is a linear phase.

Example 14.5 (The characteristic function of the gamma distribution). If $X \sim$ Gamma(α, β), then $\phi_X(t) = (1 - it/\beta)^{-\alpha}$ $(t \in \mathbb{R})$. The characteristic function of the exponential distribution $(\alpha = 1)$ is $\phi_X(t) = (1 - it/\lambda)^{-1}$.

Remark 14.4 (The values of the characteristic function at integer frequencies). Let X be a random variable on S^1 with mass density f_X. Then the Fourier coefficients of f_X are

$$c_n = \frac{1}{2\pi} \int_0^{2\pi} f_X(\theta)\, e^{-in\theta}\, d\theta = \frac{1}{2\pi}\mathbb{E}[e^{-inX}] = \frac{\phi_X(-n)}{2\pi}.$$

That is, $c_n = \phi_X(-n)/(2\pi)$: the values of the characteristic function at integer frequencies are 2π times the Fourier coefficients of the density function. This correspondence is the core interface between periodic mathematics and probability theory: **the values of the characteristic function of a probability distribution at integer points are precisely 2π times the Fourier coefficients of its density.**

Proposition 14.9 (Characteristic function and Fourier coefficients of distributions on S^1). *Let X be a random variable on S^1 with density $f_X \in L^1(S^1)$. Then:*

(i) *the conjugate normalised characteristic function $\phi_X(-n)/(2\pi)$ consists of the Fourier coefficients c_n of the density, that is, $\phi_X(-n) = 2\pi c_n$ (equivalently $\phi_X(n) = 2\pi c_{-n}$, $n \in \mathbb{Z}$);*

(ii) the density can be reconstructed from the characteristic function: $f_X(\theta) = \frac{1}{2\pi}\sum_{n\in\mathbb{Z}} \phi_X(n)e^{in\theta}$ (in the L^2 sense);

(iii) Parseval's identity: $\sum_{n\in\mathbb{Z}} |\phi_X(n)|^2 = 2\pi \int_{S^1} |f_X(\theta)|^2 d\theta$.

Proof. (i) Directly from Definition 14.21 and the definition of the Fourier coefficients. (ii) From the inverse Fourier transform formula. (iii) From Parseval's identity on $L^2(S^1)$. □

14.5 Bayesian Inference

14.5.1 The Basic Framework of Bayesian Inference

Definition 14.22 (The basic framework of Bayesian inference). **Bayesian inference** is the method of treating an unknown parameter θ as a random variable and updating one's knowledge of θ through observed data D. Its three basic elements are:

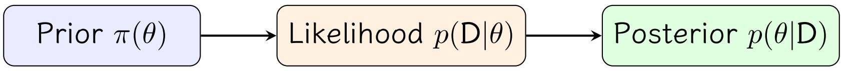
(i) **Prior distribution** $\pi(\theta)$: the distribution before observation.

(ii) **Likelihood function** $p(\mathsf{D} \mid \theta)$: the probability of the observed data given the parameter θ.

(iii) **Posterior distribution** $p(\theta \mid \mathsf{D})$: the distribution after observation, given by Bayes' theorem:

$$p(\theta \mid \mathsf{D}) = \frac{p(\mathsf{D} \mid \theta)\pi(\theta)}{p(\mathsf{D})}.$$

Definition 14.23 (Conjugate prior). Let $\mathcal{F} = \{p(\mathsf{D} \mid \theta)\}$ be a family of likelihood functions. A family of prior distributions $\mathcal{G}$ is called a **conjugate prior** family for $\mathcal{F}$ if, for every $\pi \in \mathcal{G}$ and every data set D, the posterior $p(\theta \mid \mathsf{D})$ still belongs to $\mathcal{G}$. A conjugate prior is a family of distributions closed under Bayesian updating.


Prior $\pi(\theta)$
Likelihood $p(\mathsf{D}|\theta)$
Posterior $p(\theta|\mathsf{D})$

Bayesian updating: prior × likelihood $\propto$ posterior

Figure 14.8: The three elements of Bayesian inference: prior, likelihood and posterior.

Example 14.6 (The Beta–binomial conjugate model). Let the likelihood be $X \mid \theta \sim \text{Binomial}(n, \theta)$ and the prior be $\theta \sim \text{Beta}(\alpha, \beta)$. Then the posterior is $\theta \mid X = x \sim \text{Beta}(\alpha+x, \beta+n-x)$. The prior parameters (α, β) are updated to $(\alpha+x, \beta+n-x)$—the 'pseudo-observation counts' of the prior are added to the actual observation counts.

Example 14.7 (A numerical example of Beta–binomial conjugate updating). Let the likelihood be $X \mid \theta \sim \text{Binomial}(10, \theta)$ and the prior be $\theta \sim \text{Beta}(2, 2)$, and suppose that $x = 7$ successes are observed. The posterior is $\theta \mid X = 7 \sim \text{Beta}(9, 5)$.

- prior mean: $\mathbb{E}[\theta] = 0.5$; posterior mean: $\mathbb{E}[\theta \mid \mathsf{D}] = 9/14 \approx 0.6429$;
- posterior variance: $\mathsf{Var}(\theta \mid \mathsf{D}) = 3/196 \approx 0.01531$;
- posterior mode: $2/3 \approx 0.6667$.

The prior pseudo-counts $(1, 1)$ and the actual observations $(7, 3)$ add to give the posterior pseudo-counts $(8, 6)$.

Example 14.8 (The normal–normal conjugate model). Let the likelihood be $X_1, \ldots, X_n \mid \mu \sim \mathcal{N}(\mu, \sigma^2)$ (with σ^2 known) and the prior be $\mu \sim \mathcal{N}(\mu_0, \tau^2)$. Then the posterior is

$$\mu \mid \mathsf{D} \sim \mathcal{N}\left(\frac{\frac{1}{\tau^2}\mu_0 + \frac{n}{\sigma^2}\bar{x}}{\frac{1}{\tau^2} + \frac{n}{\sigma^2}},\ \frac{1}{\frac{1}{\tau^2} + \frac{n}{\sigma^2}} \right).$$

Example 14.9 (The gamma–Poisson conjugate model). Let the likelihood be

$$X \mid \lambda \sim \mathsf{Poisson}(\lambda),$$

and the prior be $\lambda \sim \mathsf{Gamma}(\alpha, \beta)$. Then the posterior is

$$\lambda \mid X = x \sim \mathsf{Gamma}(\alpha + x, \beta + 1).$$

The shape parameter of the prior is increased by the number of observations, and the rate parameter is increased by 1.

Definition 14.24 (Marginal likelihood). **The marginal likelihood** is $p(\mathsf{D}) = \int_\Theta p(\mathsf{D} \mid \theta)\pi(\theta)d\theta$. The marginal likelihood is the overall integral of the prior distribution against the likelihood weights.

14.5.2 Sequential Bayesian Updating and Posterior Convergence

Definition 14.25 (Sequential Bayesian updating). Let the initial prior be $\pi_0(\theta)$ and let the observed data be $\mathsf{D}_1, \mathsf{D}_2, \ldots$ in order. The sequential Bayesian update is

$$p_k(\theta) \propto p(\mathsf{D}_k \mid \theta)p_{k-1}(\theta), \quad k = 1, 2, \ldots.$$

Theorem 14.3 (The Bernstein–von Mises theorem). Let $\mathsf{D} = (X_1, \ldots, X_n)$ be an independent and identically distributed sample, let θ_0 be the true parameter, and let the prior $\pi(\theta)$ be continuous and positive at θ_0. Then under suitable regularity conditions,

$$\sqrt{n}(\theta - \hat{\theta}_n) \mid \mathsf{D} \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\theta_0)^{-1}).$$

The posterior distribution converges to a normal distribution for large samples.

Sketch of proof. The prior is approximately constant near θ_0, and the posterior is proportional to the likelihood. Expanding the log-likelihood to second order at the maximum-likelihood estimate shows that the posterior density is asymptotically proportional to $\exp(-\frac{1}{2}\mathcal{I}(\theta_0)u^2)$, that is, to $\mathcal{N}(0, \mathcal{I}(\theta_0)^{-1})$. For a rigorous argument see van der Vaart, Asymptotic Statistics (1998), Chapter 10. $\square$

Proposition 14.10 (Bayesian periodic-parameter estimation). For a stochastic process with periodic structure, Bayesian inference can be used to estimate the period parameters. The posterior $p(T \mid \mathsf{D})$ exhibits a sharp peak at the true period, and the maximum a posteriori estimate $\hat{T}_{\mathsf{MAP}}$ gives the optimal estimate of the period. The width of the posterior measures the uncertainty of the estimate—the more samples there are, the narrower the posterior.

Proof. By the Bernstein–von Mises theorem (Theorem 14.3), the posterior is approximately

$$\mathcal{N}\big(T_0, [n\mathcal{I}(T_0)]^{-1}\big),$$

for large samples, so the maximum a posteriori estimate converges in probability to T_0, and the width of the posterior is $O(n^{-1/2})$. □

14.6 Numerical Characteristics of Random Variables

Definition 14.26 (Mathematical expectation). The **mathematical expectation** of a random variable X is

$$\mathbb{E}[X] = \begin{cases} \sum_k k p_X(k) & \text{(discrete)} \\ \int_{-\infty}^{\infty} x f_X(x) dx & \text{(continuous)} \end{cases}.$$

Property 14.2 (Basic properties of the expectation).

(i) **Linearity**: $\mathbb{E}[aX+bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$.

(ii) **Multiplicativity under independence**: if X, Y are independent, then $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

(iii) **Monotonicity**: if $X \le Y$ almost everywhere, then $\mathbb{E}[X] \le \mathbb{E}[Y]$.

Proof. (i) By the linearity of the integral.

(ii) By independence, the joint distribution factorises,

$$\mathbb{E}[XY] = \int xy \, dF_X(x) \, dF_Y(y) = \mathbb{E}[X] \, \mathbb{E}[Y].$$

(iii) Since $Y - X \ge 0$ holds almost everywhere, we obtain $\mathbb{E}[Y - X] \ge 0$. □

Definition 14.27 (Variance and standard deviation).

$$\mathsf{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

The standard deviation is $\sigma_X = \sqrt{\mathsf{Var}(X)}$.

Property 14.3 (Basic properties of the variance).

(i) **Translation invariance**: $\mathsf{Var}(X + b) = \mathsf{Var}(X)$.

(ii) **Covariance under scaling**: $\mathsf{Var}(aX) = a^2 \, \mathsf{Var}(X)$.

(iii) **Additivity under independence**: if X, Y are independent, then $\mathsf{Var}(X + Y) = \mathsf{Var}(X) + \mathsf{Var}(Y)$.

Proof. (i) $\mathrm{Var}(X+b) = \mathbb{E}[(X+b-\mathbb{E}[X]-b)^2] = \mathrm{Var}(X)$. (ii) $\mathrm{Var}(aX) = \mathbb{E}[a^2(X-\mathbb{E}[X])^2] = a^2\,\mathrm{Var}(X)$. (iii) By independence, $\mathrm{Var}(X+Y) = \mathbb{E}[(X-\mathbb{E}[X])^2] + \mathbb{E}[(Y-\mathbb{E}[Y])^2] + 2\mathbb{E}[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])] = \mathrm{Var}(X)+\mathrm{Var}(Y)$. □

Definition 14.28 (Covariance and correlation coefficient). The **covariance** of random variables X, Y is $\mathrm{Cov}(X,Y) = \mathbb{E}[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])]$. The **correlation coefficient** is $\rho_{XY} = \mathrm{Cov}(X,Y)/(\sigma_X\sigma_Y)$, taking values in $[-1,1]$.

Definition 14.29 (Moments, skewness, and kurtosis). The k-th raw moment is $\mu'_k = \mathbb{E}[X^k]$ and the k-th central moment is $\mu_k = \mathbb{E}[(X-\mu)^k]$. The **skewness** is $\gamma_1 = \mu_3/\sigma^3$ and the **kurtosis** is $\gamma_2 = \mu_4/\sigma^4 - 3$.

Definition 14.30 (Entropy). The **entropy** of a discrete random variable X is $H(X) = -\sum_x p(x)\log p(x)$. The **differential entropy** of a continuous random variable X is $h(X) = -\int f(x)\log f(x)dx$. Entropy measures the uniformity of a distribution.

Proposition 14.11 (The maximum entropy of the uniform distribution). On C_N, the distribution with maximum entropy is the uniform distribution $p(k) = 1/N$, with maximum entropy $H_{\max} = \log N$.

Proof. By the logarithmic inequality $\log x \geq 1 - \frac{1}{x}$ $(x > 0)$, taking $x = Np_k$:

$$\sum_k p_k \log(Np_k) \geq \sum_k p_k\Big(1 - \frac{1}{Np_k}\Big) = 1 - 1 = 0.$$

Expanding gives $\log N - H(p) \geq 0$, that is, $H(p) \leq \log N$. Equality holds if and only if $p_k = 1/N$ for all k. □

14.7 The Laws of Large Numbers and the Central Limit Theorem

Theorem 14.4 (The weak law of large numbers). Let $X_1, X_2, \dots$ be independent and identically distributed random variables with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Then for any $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) = 0.$$

The sample mean converges in probability to the population mean.

Proof. By the independent and identically distributed property, $\mathbb{E}[\bar{X}_n] = \mu$ and $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$. By Chebyshev's inequality,

$$\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \to 0.$$

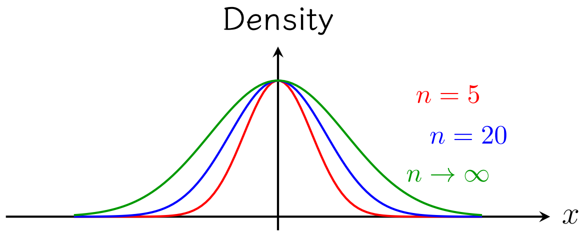
□

Theorem 14.5 (The central limit theorem). Let $X_1, X_2, \dots$ be independent and identically distributed random variables with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Let $Z_n = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}}$. Then Z_n converges in distribution to the standard normal distribution.

Proof. Let $Y_i = (X_i - \mu)/\sigma$; then $\mathbb{E}[Y_i] = 0$ and $\mathrm{Var}(Y_i) = 1$. Expand the characteristic function $\phi_Y(s)$ in a Taylor series at $s = 0$: $\phi_Y(s) = 1 - s^2/2 + o(s^2)$. By independence,

$$\phi_{Z_n}(t) = [\phi_Y(t/\sqrt{n})]^n = [1 - t^2/(2n) + o(1/n)]^n \to e^{-t^2/2}.$$

$e^{-t^2/2}$ is the characteristic function of the standard normal distribution, and the result follows from Lévy's continuity theorem. $\square$



The distribution of the sample mean converges to the normal distribution as n increases

Figure 14.9: Illustration of the central limit theorem: the distribution of the sample mean converges to the normal distribution as n increases.

14.8 Foundations of Stochastic Processes

Definition 14.31 (Stochastic process). A **stochastic process** is a family of random variables $\{X_t\}_{t \in \mathcal{T}}$, where $\mathcal{T}$ is an index set (usually taken to be $\mathbb{Z}$, $\mathbb{N}$ or $\mathbb{R}$).

14.8.1 Stationary Processes

Definition 14.32 (Stationary process). A stochastic process $\{X_t\}$ is called **weakly stationary** if $\mu(t) \equiv \mu$ and $\gamma(s,t) = \gamma(s-t)$. Stationarity is the realisation of time-translation invariance in stochastic processes.

Definition 14.33 (Power spectral density). Let $\{X_t\}$ be a weakly stationary process with autocovariance function $\gamma(h)$. The **power spectral density** is

$$S(\omega) = \sum_{h=-\infty}^{\infty} \gamma(h) e^{-i\omega h}.$$

$S(\omega)$ is the Fourier transform of the autocovariance function. This definition adopts the **delayed** convention (with exponent $-i\omega h$), which belongs to the same notation as Theorem 22.8 of the 'Harmonic Analysis' chapter; for the conversion between the two conventions see Remark 22.4.

Proposition 14.12 (Invariance of the Second Derivative of the Autocovariance of a Stationary Process). Let $\{X_t\}$ be a weakly stationary process whose autocovariance function $\gamma(h)$ is twice differentiable at $h = 0$. Define $P = -\gamma''(0)$. Then the time-translation invariance of the stationary process implies that P is independent of time.

Proof. A time translation τ maps the process to $\{X_{t+\tau}\}$. Stationarity guarantees that $\gamma_\tau(h) = \gamma(h)$ is independent of τ, hence $P_\tau = -\gamma_\tau''(0) = -\gamma''(0) = P$. Here 'invariance' is the direct equivalent statement of stationarity $\gamma(s,t) = \gamma(s-t)$. ☐

Proposition 14.13 (Non-negativity of the power spectrum and Herglotz's theorem). *The autocovariance function $\gamma(h)$ of a weakly stationary process is a positive-definite sequence, and its power spectral density satisfies $S(\omega) \geq 0$. Conversely, any non-negative function S satisfying $\int S(\omega)d\omega < \infty$ is the power spectrum of some weakly stationary process. This is the content of Herglotz's theorem.*

Proof. Positive definiteness follows by the method of Proposition 14.8: $\sum_{j,k} z_j \bar{z}_k \gamma(j-k) = \mathbb{E}|\sum_j z_j X_j|^2 \geq 0$. That $S(\omega) \geq 0$ follows from the fact that S is the Fourier transform of a positive-definite sequence. The converse conclusion is given by the spectral measure theorem. ☐

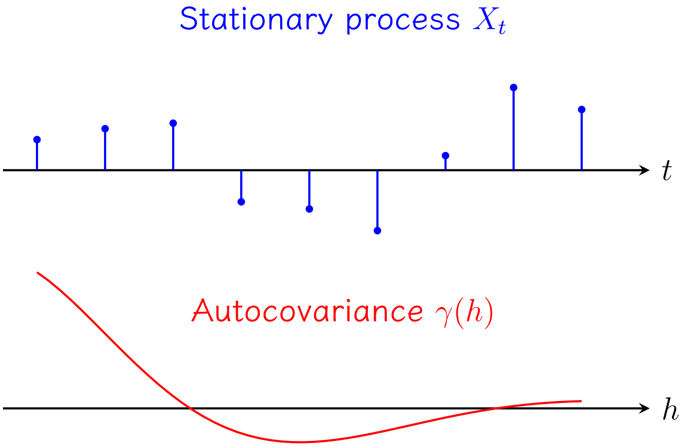


Figure 14.10: A stationary process and its autocovariance function: the autocovariance depends only on the time lag h.

14.8.2 Cyclostationary Processes

Definition 14.34 (Cyclostationary Process). A stochastic process $\{X_t\}$ is called a **cyclostationary process** with period T if its statistical properties are invariant under the time translation $t \mapsto t+T$. A stationary process is a special case of a cyclostationary process: it satisfies period-translation invariance for every T. Cyclostationarity belongs to the invariance of time-translation symmetry (0-symmetry) under the action of the periodic subgroup $T\mathbb{Z}$.

Proposition 14.14 (Fourier decomposition of cyclostationary processes). *Let $\{X_t\}$ be a cyclostationary process with period T. Then its autocovariance function $\gamma(t,s) = \gamma(t+T, s+T)$ can be expanded as a double Fourier series:*

$$\gamma(t,s) = \sum_{n,m\in\mathbb{Z}} \gamma_{n,m} e^{i2\pi nt/T} e^{-i2\pi ms/T}.$$

Here the $\gamma_{n,m}$ are the cyclic spectral coefficients. When $n = m$, the coefficients $\gamma_{n,n}$ constitute the stationary power spectrum.

Proof. By cyclostationarity, $\gamma(t,s)$ is periodic with respect to (t,s) with period T. Expansion in Fourier series gives the double series. The stationary case $(T \to 0)$ degenerates to $\gamma(t-s)$. ☐

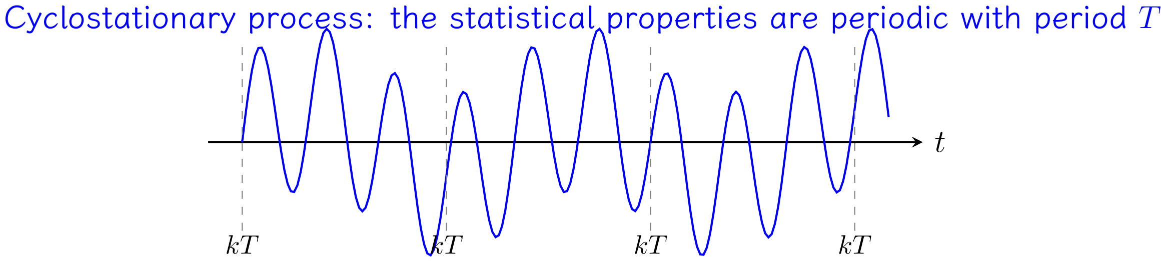

Cyclostationary process: the statistical properties are periodic with period T
t
kT
kT
kT
kT

Figure 14.11: Cyclostationary process: the statistical properties are invariant under the period translation $t \mapsto t+T$.

14.8.3 Markov Processes

Definition 14.35 (Markov Process). A stochastic process $\{X_t\}$ is called a **Markov process** if

$$\mathbb{P}(X_{t_n} \le x \mid X_{t_1}, \dots, X_{t_{n-1}}) = \mathbb{P}(X_{t_n} \le x \mid X_{t_{n-1}}).$$

Lemma 14.2 (Spectral Radius Bound for Eigenvalues of a Stochastic Matrix). *Let P be a stochastic matrix on C_N. Then every eigenvalue λ of P satisfies $|\lambda| \le 1$, and $\lambda_0 = 1$ is an eigenvalue of P.*

Proof. $\lambda_0 = 1$: $(P\mathbf{1})_i = \sum_j P_{ij} = 1$. $|\lambda| \le 1$: let $P\mathsf{v} = \lambda\mathsf{v}$ and take $|v_m| = \max_j |v_j| > 0$; then $|\lambda||v_m| \le \sum_j P_{mj}|v_j| \le |v_m|$, so $|\lambda| \le 1$. ☐

Proposition 14.15 (Transition Operator and Eigenvalue Decomposition). *Let P be a Markov transition matrix on C_N with eigenvalues $\lambda_0 = 1 \ge |\lambda_1| \ge \cdots \ge |\lambda_{N-1}|$.*

(i) $\lambda_0 = 1$: *the invariant distribution—the fixed point of the distribution.*

(ii) $|\lambda_i| < 1$: *decaying modes—the rate of convergence to the stationary distribution.*

(iii) $|\lambda_i| = 1$ *with* $\lambda_i \ne 1$: *periodic modes—the distribution cycles among several states.*

Proof. (i) A distribution π satisfying $\pi P = \pi$, $\pi \ge 0$ and $\sum_i \pi_i = 1$ is an invariant distribution. (ii) When $|\lambda_i| < 1$ the corresponding mode decays as $|\lambda_i|^n \to 0$. (iii) If λ_i is a k-th root of unity, then the mode cycles among k states, corresponding to a periodic pattern of period k. ☐

Proposition 14.16 (Period of irreducible Markov chains). *Let P be the transition matrix of a finite Markov chain. If the chain is irreducible and aperiodic (that is, all eigenvalues other than $\lambda_0 = 1$ satisfy $|\lambda_i| < 1$), then the chain converges to the stationary distribution at an exponential rate:*

$$\|P^n \pi_0 - \pi\| \le C\rho^n, \quad \rho = \max_{i \ge 1} |\lambda_i| < 1.$$

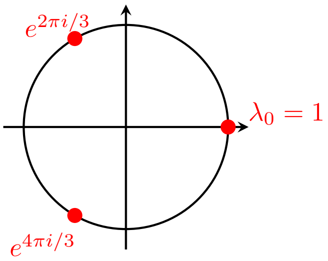
If the chain has period $d > 1$, then the eigenvalues $e^{2\pi i k/d}$ $(k = 0, \dots, d-1)$ all have modulus 1, and the chain cycles among d classes of states.

Proof. By the Perron–Frobenius theorem, for an irreducible aperiodic chain the spectral radius 1 is a simple eigenvalue, and the remaining eigenvalues have modulus strictly less than 1. The rate of convergence is determined by the modulus $|\lambda_1|$ of the second largest eigenvalue. The periodic case follows by decomposing P^d into d aperiodic subchains. ☐

Example 14.10 (Eigenvalues and Period of a Three-State Permutation Chain). Consider the permutation transition matrix on C_3

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

Its characteristic polynomial is $\lambda^3 - 1$, with eigenvalues $\lambda_0 = 1$, $\lambda_1 = e^{2\pi i/3}$ and $\lambda_2 = e^{4\pi i/3}$. The moduli of the three eigenvalues are all 1, corresponding to a cyclic mode of period 3. The invariant distribution is $\pi = (1/3, 1/3, 1/3)$.



Permutation chain of period 3: the eigenvalues are the 3-th roots of unity

Figure 14.12: The eigenvalues of the permutation chain of period 3 are distributed on the unit circle, corresponding to the periodic cyclic mode of the distribution.

14.8.4 Brownian Motion

Definition 14.36 (Brownian Motion). **Standard Brownian motion** $\{W_t\}_{t\geq 0}$ satisfies: $W_0 = 0$, the increments are independent, $W_t - W_s \sim \mathcal{N}(0, t-s)$, and the sample paths are continuous. Brownian motion is self-similar: $\{W_{\lambda t}\} \overset{d}{=} \{\sqrt{\lambda} W_t\}$.

Definition 14.37 (Self-Similarity and the Scaling Exponent). A stochastic process $\{X_t\}$ is called **self-similar** if there exists $H > 0$ such that $\{X_{\lambda t}\} \overset{d}{=} \{\lambda^H X_t\}$ for all $\lambda > 0$. The number H is called the **Hurst exponent** (or self-similarity exponent). For Brownian motion $H = 1/2$.

The **tail exponent** $\alpha > 0$ of a heavy-tailed distribution is defined by $\mathbb{P}(|X| > x) \sim x^{-\alpha}$. The Hurst exponent and the tail exponent are characteristic invariants of a distribution or a process under scaling transformations.

14.9 Transformation Groups and Ergodicity in Probability Spaces

Definition 14.38 (Measure-Preserving Transformation). A mapping $T : \Omega \to \Omega$ is called a **measure-preserving transformation** if $\mathbb{P}(T^{-1}(A)) = \mathbb{P}(A)$ for every $A \in \mathcal{F}$.

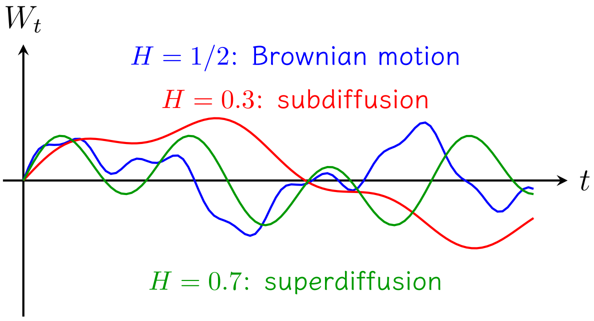


Figure 14.13: The Hurst exponent of a self-similar process and the roughness of its sample paths: the smaller H is, the rougher the path.

Definition 14.39 (Ergodicity). *A measure-preserving transformation T is called* **ergodic** *if every T-invariant event A satisfies $\mathbb{P}(A) \in \{0, 1\}$.*

Theorem 14.6 (Birkhoff Ergodic Theorem). *Let T be a measure-preserving transformation and $f \in L^1(\mathbb{P})$. Then the time average*

$$\bar{f}(\omega) = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} f(T^k \omega)$$

exists almost everywhere, and $\mathbb{E}[\bar{f}] = \mathbb{E}[f]$. If T is ergodic, then $\bar{f}(\omega) = \mathbb{E}[f]$ holds almost everywhere.

Sketch of proof. The standard argument: first prove the maximal ergodic inequality, then prove the almost-everywhere convergence of the time averages, and finally use ergodicity to identify the limit as the space average. For details see Walters, An Introduction to Ergodic Theory (1982), Chapter 1. □

Corollary 14.1 (Time-Average–Space-Average Criterion of the Ergodic Theorem). *For an ergodic measure-preserving transformation the time average equals the space average, which means that the long-term behaviour of a single orbit fully characterises the whole probability space. Ergodicity is the mathematical formulation of 'the orbit being dense in the sample space'.*

Proof. By Theorem 14.6, in the ergodic case $\bar{f}(\omega) = \mathbb{E}[f]$ holds almost everywhere, that is, the time average along every orbit converges to the space average. □

Remark 14.5 (Scope and Limitations of the Ergodic Theorem). The three conditions of Theorem 14.6 each have their own limits: (i) measure preservation only guarantees that the time average exists, not that it equals the space average; (ii) ergodicity guarantees that a single orbit is a sufficient statistic for the whole sample space; (iii) integrability $f \in L^1(\mathbb{P})$ guarantees that $\bar{f}$ is integrable. When performing period diagnostics on a cyclostationary process, the ergodicity criterion should be applied to the phase-aligned process rather than to the original process.

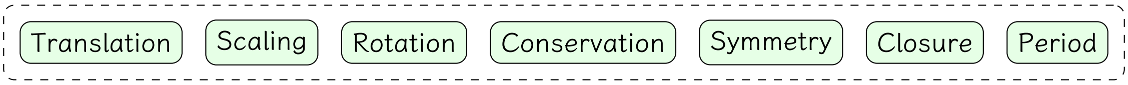
Proposition 14.17 (Invariance under Three Time Scales). (i) **Stationary processes (translation invariant, 0-symmetry)**: *the statistical properties are invariant under time translation.*

(ii) **Cyclostationary processes (period-translation invariant, the periodisation of 0-symmetry)**: the statistical properties are invariant under the time translation $t \mapsto t + T$.

(iii) **Self-similar processes (scale invariant, 1-symmetry)**: the statistical properties are invariant under the scaling transformation $t \mapsto \lambda t$.

Proof. (i) The statistical properties of a stationary process are invariant under time translation (Definition 14.32), and time translation is the action of the translation group $\mathbb{R}$ or $\mathbb{Z}$. (ii) The statistical properties of a cyclostationary process are invariant under the action of $T\mathbb{Z}$ (Definition 14.34); the set $T\mathbb{Z}$ is a translation subgroup, so this is still translation symmetry. (iii) The statistical properties of a self-similar process are invariant under scaling transformations (Definition 14.37), and a scaling transformation is the action of the scaling group $\mathbb{R}^+$. The three correspond respectively to invariance under a translation group, a translation subgroup and a scaling group. ☐

Seven-theme comparison table for this chapter



The seven themes run through probability theory and distribution

Figure 14.14: The seven themes of this chapter: translation, scaling, rotation, conservation, symmetry, closure and period.

Theme	Corresponding content in this chapter
Translation	The time-translation invariance of stationary processes (Definition 14.32); the period-translation invariance of cyclostationary processes (Definition 14.34); the advance of sequential Bayesian updating along the time axis (Definition 14.25)
Scaling	The scale invariance of self-similar processes and heavy-tailed distributions (Definition 14.37); the scaling convergence of the central limit theorem (Theorem 14.5)
Rotation	The phase information of the argument of the characteristic function (Remark 14.3); the rotational symmetry of the uniform distribution on S^1 (Definition 14.15, Proposition 14.6)
Conservation	The invariance of the autocovariance under time translation (Proposition 14.12); the invariance of the scaling exponent under scaling transformations
Symmetry	The closure of the family of conjugate priors (Definition 14.23); the prior/likelihood/posterior tripartite structure
Closure	The convergence closure of the laws of large numbers and the central limit theorem (Theorem 14.4, 14.5); the 'time average = space average' criterion for ergodic orbits (Theorem 14.6)
Period	The period T of a cyclostationary process (Definition 14.34); the periodic pattern of Markov-chain eigenvalues with $\|\lambda_i\| = 1$ (Proposition 14.15); the characteristic function and Fourier coefficients of distributions on S^1 (Proposition 14.9)

Exercises

Exercise 1 Prove the basic properties of the probability measure (Proposition 14.1):

(i) $\mathbb{P}(\emptyset) = 0$;

(ii) monotonicity: $A \subseteq B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$;

(iii) complementarity: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$;

(iv) finite additivity: $A \cap B = \emptyset \Rightarrow \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.

Carry out the proof using the axiom of countable additivity, and explain why these four properties are necessary consequences of a probability measure.

Exercise 2 According to Definition 14.2, explain the relation between frequency and probability: the frequency $f_n(A) = n_A/n$ of A converges in probability to

$\mathbb{P}(A)$. Explain the meaning of this convergence in the light of the law of large numbers (Theorem 14.4), and explain that the frequency is the empirical estimate of the probability and the probability is the limit of the frequency.

Exercise 3 Prove the four basic properties of the cumulative distribution function (Proposition 14.1): monotone non-decrease, right continuity, limiting behaviour and interval probability. Write out the proofs separately, and explain the basis of the statement in Theorem 14.1 that 'two random variables are identically distributed if and only if their cumulative distribution functions are equal'.

Exercise 4 Using Definition 14.12 and Definition 14.13, prove the limiting relation from the binomial distribution to the Poisson distribution:

$$\text{Binomial}(n, \lambda/n) \xrightarrow{d} \text{Poisson}(\lambda) \quad (n \to \infty).$$

Carry out the proof using the limit of the probability mass function or of the characteristic function.

Exercise 5 Verify the limits and special-case relations among common distributions in Proposition 14.7:

(i) binomial to Poisson;
(ii) binomial to normal (the De Moivre–Laplace theorem);
(iii) gamma to normal;
(iv) t distribution to normal;
(v) beta to normal.

State separately the parametric limit condition for each relation.

Exercise 6 Prove the basic properties of the characteristic function (Proposition 14.8):

(i) $\phi_X(0) = 1$ and $|\phi_X(t)| \leq 1$;
(ii) if X and Y are independent, then $\phi_{X+Y}(t) = \phi_X(t)\phi_Y(t)$;
(iii) $\mathbb{E}[X^n] = i^{-n}\phi_X^{(n)}(0)$;
(iv) the characteristic function determines the distribution uniquely;
(v) $\phi_{aX+b}(t) = e^{ibt}\phi_X(at)$.

Exercise 7 Verify Example 14.4 and Example 14.5:

(i) if $X \sim \mathcal{N}(\mu, \sigma^2)$, then $\phi_X(t) = e^{i\mu t - \sigma^2 t^2/2}$;
(ii) if $X \sim \text{Gamma}(\alpha, \beta)$, then $\phi_X(t) = (1 - it/\beta)^{-\alpha}$.

Write out the derivations, and explain that the modulus of the characteristic function of the normal distribution is a Gaussian kernel and its argument is a linear phase.

Exercise 8 Prove Proposition 14.9: let X be a random variable on S^1 with density $f_X \in L^1(S^1)$; then

(i) the conjugate normalised characteristic function $\phi_X(-n)/(2\pi)$ consists of the Fourier coefficients c_n of the density, that is, $\phi_X(-n) = 2\pi c_n$ (equivalently $\phi_X(n) = 2\pi c_{-n}$);

(ii) the density can be reconstructed from the characteristic function;

(iii) Parseval's identity holds.

Explain how this correspondence embodies the unification of probability distributions and Fourier analysis on the circle group.

Exercise 9 Prove the basic properties of the expectation and of the variance (Proposition 14.2 and Proposition 14.3):

(i) linearity of the expectation: $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$;

(ii) multiplicativity under independence: X, Y independent $\Rightarrow \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$;

(iii) translation invariance of the variance: $\mathsf{Var}(X + b) = \mathsf{Var}(X)$;

(iv) covariance of the variance under scaling: $\mathsf{Var}(aX) = a^2 \mathsf{Var}(X)$;

(v) additivity of the variance: X, Y independent $\Rightarrow \mathsf{Var}(X + Y) = \mathsf{Var}(X) + \mathsf{Var}(Y)$.

Exercise 10 Prove the maximum-entropy property of the uniform distribution (Proposition 14.11): on C_N, the distribution with maximum entropy is the uniform distribution, with maximum entropy $H_{\mathsf{max}} = \log N$. Carry out the proof using the logarithmic inequality $\log x \geq 1 - 1/x$, and explain how this result is related to 'the uniform distribution is the Haar measure on the cyclic group'.

Exercise 11 Prove the central limit theorem (Theorem 14.5): let $X_1, X_2, \ldots$ be independent and identically distributed, with $\mathbb{E}[X_i] = \mu$ and $\mathsf{Var}(X_i) = \sigma^2 < \infty$; then

$$Z_n = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Carry out the proof using the Taylor expansion of the characteristic function at $s = 0$, $\phi_Y(s) = 1 - s^2/2 + o(s^2)$.

Exercise 12 Verify Example 14.6 and Example 14.8:

(i) the posterior update of the Beta–binomial conjugate model;

(ii) the posterior update of the normal–normal conjugate model.

Write out the expressions for the prior, the likelihood and the posterior in each case, and explain how a conjugate prior guarantees that the family of posterior distributions is closed under Bayesian updating.

Exercise 13 Prove the Bernstein–von Mises theorem (Theorem 14.3): under suitable regularity conditions,

$$\sqrt{n}(\theta - \hat{\theta}_n) \mid \mathsf{D} \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\theta_0)^{-1}).$$

Explain the intuitive meaning of the convergence of the posterior distribution to a normal distribution for large samples, and indicate the significance of this conclusion for the Bayesian estimation of period parameters (see Proposition 14.10).

Exercise 14 Prove the non-negativity of the power spectrum of a stationary process and Herglotz's theorem (Proposition 14.13): the autocovariance function $\gamma(h)$ of a weakly stationary process is a positive-definite sequence, and its power spectral density satisfies $S(\omega) \geq 0$; conversely, any non-negative function S satisfying $\int S(\omega)d\omega < \infty$ is the power spectrum of some weakly stationary process. Carry out the proof of positive definiteness using $\mathbb{E}|\sum_j z_j X_j|^2 \geq 0$.

Exercise 15 Prove the Fourier decomposition of cyclostationary processes (Proposition 14.14): let $\{X_t\}$ be a cyclostationary process with period T; then its autocovariance function

$$\gamma(t,s) = \sum_{n,m\in\mathbb{Z}} \gamma_{n,m} e^{i2\pi nt/T} e^{-i2\pi ms/T}.$$

Explain that when $n = m$ the coefficients $\gamma_{n,n}$ constitute the stationary power spectrum, and explain how cyclostationarity corresponds to the invariance of time-translation symmetry under the periodic subgroup $T\mathbb{Z}$.

Exercise 16 Prove the transition operator and eigenvalue decomposition of a Markov chain (Proposition 14.15): let P be a Markov transition matrix on C_N with eigenvalues $\lambda_0 = 1 \geq |\lambda_1| \geq \cdots \geq |\lambda_{N-1}|$. Explain that:

(i) $\lambda_0 = 1$ corresponds to the invariant distribution—the fixed point of the distribution;

(ii) $|\lambda_i| < 1$ corresponds to decaying modes that converge to the stationary distribution;

(iii) $|\lambda_i| = 1$ with $\lambda_i \neq 1$ corresponds to periodic modes.

Exercise 17 Verify the eigenvalues and the period of the three-state permutation chain in Example 14.10:

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix},$$

the characteristic polynomial is $\lambda^3 - 1$, with eigenvalues $\lambda_0 = 1$, $\lambda_1 = e^{2\pi i/3}$ and $\lambda_2 = e^{4\pi i/3}$. Explain that the moduli of the three eigenvalues are all 1, corresponding to a cyclic mode of period 3.

Exercise 18 Prove the Birkhoff ergodic theorem (Theorem 14.6): let T be a measure-preserving transformation and $f \in L^1(\mathbb{P})$; then the time average

$$\bar{f}(\omega) = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} f(T^k\omega)$$

exists almost everywhere, and $\mathbb{E}[\bar{f}] = \mathbb{E}[f]$; if T is ergodic, then $\bar{f}(\omega) = \mathbb{E}[f]$ holds almost everywhere. Explain the closure meaning of 'the time average equals the space average' in Corollary 14.1.

Exercise 19 Synthesis problem: *combining the characteristic function* (Definition 14.21), *stationary processes* (Definition 14.32), *cyclostationary processes* (Definition 14.34) *and the ergodic theorem* (Theorem 14.6), *explain the confluence of probability theory and periodic mathematics on the following four levels:*

(i) *the characteristic function of a probability distribution is precisely the* Fourier transform of the distribution;

(ii) the uniform distribution on S^1 is the Haar measure on the circle group;

(iii) the time-translation invariance of stationary processes and the period-translation invariance of cyclostationary processes;

(iv) the closure criterion '*time average equals space average*' for *ergodic* orbits.

Indicate separately where these confluences are situated in the framework of periodic mathematics.

Chapter 15

Vector Analysis

Remark 15.1 (Positioning of this chapter and reading conventions). Vector analysis studies the differential and integral operations on scalar fields and vector fields. This chapter rigorously establishes the theoretical system of the three operators — gradient, divergence and curl — proves the product rules, the fundamental identities and the integral theorems, and discusses the classification of vector fields, the existence of potential functions, and the expressions of the operators in orthogonal curvilinear coordinate systems.

Reading conventions. This chapter assumes space and time to be continuous; discrete lattice models are discussed in the 'Periodic Manifold' chapter. Whenever a statement involves a domain, the regularity condition of that domain (one of an open set, a simply connected set, a regular boundary, or a regular bounded region) is always given explicitly; whenever differentiability is involved, the required order of continuous differentiability is always given explicitly. Whenever a conclusion depends on additional assumptions, those assumptions are listed one by one in the statement; whenever only a heuristic argument is available at present and there is no statable theorem, the matter is always marked as a 'research programme' or an 'open question' and is not used as a proved conclusion.

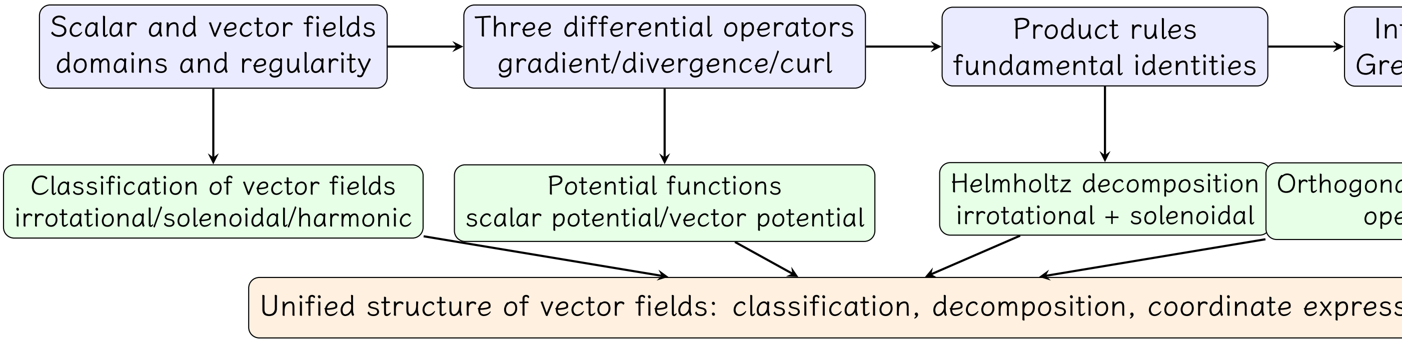

Scalar and vector fields
domains and regularity
Three differential operators
gradient/divergence/curl
Product rules
fundamental identities
In
Gre
Classification of vector fields
irrotational/solenoidal/harmonic
Potential functions
scalar potential/vector potential
Helmholtz decomposition
irrotational + solenoidal
Orthogona
ope
Unified structure of vector fields: classification, decomposition, coordinate express

Figure 15.1: Overview of the structure of this chapter: from scalar and vector fields, through the three differential operators, the product rules and the fundamental identities, to the integral theorems, the classification of vector fields, potential functions, the Helmholtz decomposition and orthogonal curvilinear coordinates, finally unified in the structure of vector fields.

15.1 Scalar and vector fields

This section fixes the objects and the domains of all the subsequent discussion. The regularity conditions given here are not technical decoration: several conclusions of this chapter (such as the divergence theorem and the existence of potential functions) genuinely fail when the domain does not satisfy the corresponding conditions, and Example 15.2 and Example 15.3 give two counterexamples.

Definition 15.1 (Scalar field)**.** Let $\mathcal{M} \subseteq \mathbb{R}^n$ be a non-empty open set, $n \in \{1, 2, 3\}$ (this chapter mainly discusses $n = 3$; the two-dimensional case $n = 2$ can be obtained by pointwise degeneration of the same operator theory). A **scalar field** is a function

$$f : \mathcal{M} \to \mathbb{R}, \qquad \mathsf{x} = (x_1, \ldots, x_n) \mapsto f(\mathsf{x}).$$

If f has continuous partial derivatives up to order k on $\mathcal{M}$, we write $f \in C^k(\mathcal{M})$; in particular $C^0(\mathcal{M})$ denotes the set of all continuous functions on $\mathcal{M}$.

Assumptions and conventions: when $k \geq 1$, $f \in C^k(\mathcal{M})$ implies that f is differentiable everywhere on $\mathcal{M}$ (the existence of continuous partial derivatives implies differentiability), so wherever the chain rule, the directional derivative formula and Taylor expansion are applied to f below, the assumption $k \geq 1$ suffices. If f is defined only on some subset of $\mathcal{M}$, that subset must be given explicitly.

Definition 15.2 (Vector field)**.** Let $\mathcal{M} \subseteq \mathbb{R}^3$ be a non-empty open set. A **vector field** is a vector-valued function

$$\mathsf{F} : \mathcal{M} \to \mathbb{R}^3, \qquad \mathsf{F}(\mathsf{x}) = \big(F_1(\mathsf{x}),\, F_2(\mathsf{x}),\, F_3(\mathsf{x})\big),$$

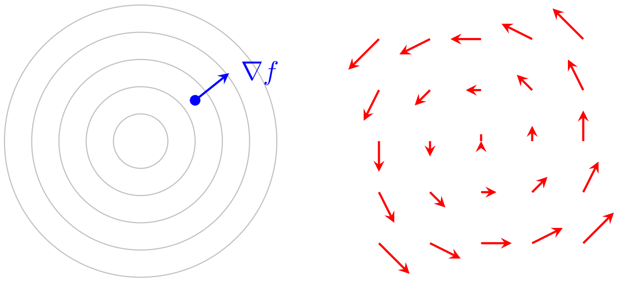
where $F_i : \mathcal{M} \to \mathbb{R}$ $(i = 1, 2, 3)$ are called the **component functions** of F. If all $F_i \in C^k(\mathcal{M})$, we write $\mathsf{F} \in C^k(\mathcal{M}, \mathbb{R}^3)$.

Convention: by definition, $\mathsf{F} \in C^k(\mathcal{M}, \mathbb{R}^3)$ is equivalent to all three component functions belonging to $C^k(\mathcal{M})$; throughout this chapter all differentiability requirements on vector fields are imposed component by component and will not be repeated.

Definition 15.3 (Closure, boundary, regular region)**.** Let $\mathcal{M} \subseteq \mathbb{R}^3$ be a non-empty open set. Its **closure** is denoted by $\overline{\mathcal{M}}$, and its **boundary** by $\partial\mathcal{M} = \overline{\mathcal{M}} \setminus \mathcal{M}$. If $\partial\mathcal{M}$ is a piecewise smooth surface, then $\mathcal{M}$ is said to have a **regular boundary**. In particular, if $\mathcal{M}$ is bounded, $\overline{\mathcal{M}}$ is compact, and $\partial\mathcal{M}$ is the union of finitely many disjoint smooth closed surfaces, then $\mathcal{M}$ is called a **regular region** (equivalently: $\mathcal{M}$ is bounded and connected, its boundary is piecewise smooth and contains no cusps or self-intersections).

Three uses: a regular region is the natural domain on which the divergence theorem (Theorem 15.8) holds; a piecewise smooth oriented surface together with its piecewise smooth boundary is the natural domain on which the curl theorem (Theorem 15.7) holds; simple connectivity is the additional necessary condition under which the existence of potential functions (Theorem 15.9) holds.

Example 15.1 (Two basic scalar fields and vector fields)**.** A temperature distribution $T(x, y, z)$ is a scalar field: under a rotation of space, the temperature value at a point is unchanged. A wind-velocity field $\mathsf{v}(x, y, z)$ is a vector field: under a rotation of space, its components transform by the rotation matrix. The essential difference between the two lies not in 'whether they carry a direction', but in the **transformation law under coordinate changes** — this will be made precise in Section 15.2.



(a) Scalar field f: level curves (b) Vector field F: arrows indicate direction and magnitude

Figure 15.2: Intuitive comparison of scalar and vector fields: a scalar field is represented by level curves, a vector field by arrows.

Example 15.2 (The domain assumption is indispensable: a counterexample with a slit plane). Take $\mathcal{M} = \mathbb{R}^2 \setminus \{(x,0) : x \geq 0\}$ (the plane cut along the positive real axis); it is an open set but not simply connected. Consider

$$\mathsf{F}(x,y) = \left(\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2} \right).$$

A direct computation gives $\nabla \times \mathsf{F} = \mathbf{0}$ everywhere on $\mathcal{M}$, but F has no scalar potential on $\mathcal{M}$: the circulation along a closed curve γ winding once around the origin is $\oint_\gamma \mathsf{F} \cdot d\mathsf{r} = 2\pi \neq 0$.

Conclusion: $\nabla \times \mathsf{F} = \mathbf{0}$ alone does not suffice to deduce the existence of a potential function; the simple connectivity of the domain must be added — this is precisely why Theorem 15.9 has to write '$\mathcal{M}$ is simply connected' among its assumptions.

Example 15.3 (Boundary regularity is indispensable: a counterexample with a needle-shaped region). Take $\mathcal{M} = \{(x,y,z) : 0 < x < 1,\ 0 < y < \varepsilon,\ 0 < z < \varepsilon\}$ and keep at $x = 1$ a contraction whose thickness tends to zero. As $\varepsilon \to 0$ a self-contacting sheet appears in $\partial\mathcal{M}$, and $\partial\mathcal{M}$ is then no longer a piecewise smooth closed surface.

Conclusion: the clause in the divergence theorem Theorem 15.8 that '∂V is a piecewise smooth closed surface and the outward normal n is well defined everywhere' cannot be omitted; on a needle-shaped limiting region the outward normal is not unique at the sheet, and the statement of the theorem itself fails (rather than its conclusion being false).

15.2 The three basic transformations and the three differential operators

This section is the hinge of the whole chapter. It answers two questions: why the three differential operators are 'these three', and 'what each of them corresponds to'.

Remark 15.2 (The position of this section between the two sections). This section places 'transformations' and 'operators' side by side in order to avoid a common confusion of expression: taking 'the operator corresponding to translation is the gradient' as a definition, and then using the gradient in turn to 'explain' translation. The order in this section is exactly the reverse — first the three classes of transformations are defined independently and their group structure and range of generation are proved (§15.2.1), then the three operators are defined independently and their limit characterisations are proved (§15.2.2), and only then is the correspondence between the two discussed, with the exact meaning of that correspondence marked out level by level.

15.2.1 Translation, scaling and rotation

Definition 15.4 (The translation transformation and the translation group). Let $\mathsf{a} \in \mathbb{R}^n$. The map

$$T_{\mathsf{a}} : \mathbb{R}^n \to \mathbb{R}^n, \qquad T_{\mathsf{a}}(\mathsf{x}) = \mathsf{x} + \mathsf{a}$$

is called the **translation transformation** *along the vector* a*. Put*

$$\mathcal{T}_n = \{\, T_{\mathsf{a}} : \mathsf{a} \in \mathbb{R}^n \,\}.$$

Assumption: a is a constant vector (independent of x and t); n is arbitrary. The following Lemma 15.1 shows that $\mathcal{T}_n$ forms a group under composition of maps.

Lemma 15.1 (The group structure of the translation group). *$(\mathcal{T}_n, \circ)$ forms an Abelian group, and the map $\mathsf{a} \mapsto T_{\mathsf{a}}$ gives a group isomorphism*

$$(\mathcal{T}_n, \circ) \;\cong\; (\mathbb{R}^n, +).$$

Proof. We verify the four group axioms and the isomorphism relation one by one.

(1) Closure under composition: for any $\mathsf{a}, \mathsf{b} \in \mathbb{R}^n$ and any $\mathsf{x} \in \mathbb{R}^n$,

$$(T_{\mathsf{a}} \circ T_{\mathsf{b}})(\mathsf{x}) = T_{\mathsf{a}}\big(T_{\mathsf{b}}(\mathsf{x})\big) = T_{\mathsf{a}}(\mathsf{x} + \mathsf{b}) = \mathsf{x} + \mathsf{b} + \mathsf{a} = T_{\mathsf{a}+\mathsf{b}}(\mathsf{x}),$$

hence $T_{\mathsf{a}} \circ T_{\mathsf{b}} = T_{\mathsf{a}+\mathsf{b}} \in \mathcal{T}_n$.

(2) Associativity: this follows directly from the associativity of composition, $(T_{\mathsf{a}} \circ T_{\mathsf{b}}) \circ T_{\mathsf{c}} = T_{\mathsf{a}} \circ (T_{\mathsf{b}} \circ T_{\mathsf{c}})$; alternatively, by (1) both sides are computed to equal $T_{\mathsf{a}+\mathsf{b}+\mathsf{c}}$.

(3) Identity element: $T_0 = \mathrm{id}_{\mathbb{R}^n}$, and $T_{\mathsf{a}} \circ T_0 = T_{\mathsf{a}} = T_0 \circ T_{\mathsf{a}}$.

(4) Inverse element: by (1), $T_{\mathsf{a}} \circ T_{-\mathsf{a}} = T_0 = T_{-\mathsf{a}} \circ T_{\mathsf{a}}$, hence $(T_{\mathsf{a}})^{-1} = T_{-\mathsf{a}} \in \mathcal{T}_n$.

(5) Isomorphism: define $\Phi : \mathbb{R}^n \to \mathcal{T}_n$, $\Phi(\mathsf{a}) = T_{\mathsf{a}}$. By (1), $\Phi(\mathsf{a} + \mathsf{b}) = T_{\mathsf{a}+\mathsf{b}} = T_{\mathsf{a}} \circ T_{\mathsf{b}} = \Phi(\mathsf{a}) \circ \Phi(\mathsf{b})$, so Φ is a group homomorphism; Φ is surjective (as $T_{\mathsf{a}} = \Phi(\mathsf{a})$ runs through $\mathcal{T}_n$); if $\Phi(\mathsf{a}) = \Phi(\mathsf{b})$, then $\mathsf{x} + \mathsf{a} = \mathsf{x} + \mathsf{b}$ for every x, and taking $\mathsf{x} = 0$ gives $\mathsf{a} = \mathsf{b}$, so Φ is injective. Thus Φ is an isomorphism, and an isomorphism carries the commutativity of the source group over to the image group. □

Definition 15.5 (The scaling transformation and the scaling group). Let $\lambda \in \mathbb{R}_{>0}$. The map

$$S_\lambda : \mathbb{R}^n \to \mathbb{R}^n, \qquad S_\lambda(\mathsf{x}) = \lambda \mathsf{x}$$

is called the **(isotropic) scaling transformation**, *and λ is called the scaling factor.* Put

$$\mathcal{S}_n = \{\, S_\lambda : \lambda \in \mathbb{R}_{>0} \,\}.$$

Assumptions and boundary cases: $\lambda > 0$ is essential. If $\lambda = 0$ were allowed, then S_0 would not be injective and $\mathcal{S}_n$ would no longer be a group; if $\lambda < 0$ were allowed, then S_λ would contain a central inversion component and its composition relation with rotations would change. **Anisotropic** scaling $\operatorname{diag}(\lambda_1, \lambda_2, \lambda_3)$ (with the λ_i not all equal) does not belong to $\mathcal{S}_n$; see Example 15.4.

Lemma 15.2 (The group structure of the scaling group). $(\mathcal{S}_n, \circ)$ forms an Abelian group, and $\lambda \mapsto S_\lambda$ gives a group isomorphism

$$(\mathcal{S}_n, \circ) \;\cong\; (\mathbb{R}_{>0}, \times).$$

Proof. For $\mathsf{x} \in \mathbb{R}^n$, $(S_\lambda \circ S_\mu)(\mathsf{x}) = S_\lambda(\mu \mathsf{x}) = \lambda\mu\,\mathsf{x} = S_{\lambda\mu}(\mathsf{x})$, so the set is closed under composition and the operation is commutative. Associativity is given by the associativity of composition; the identity element is $S_1 = \mathrm{id}$; the inverse element is $(S_\lambda)^{-1} = S_{1/\lambda}$ (since $\lambda > 0$, $1/\lambda > 0$ still lies in the index set). The isomorphism $\Psi(\lambda) = S_\lambda$ satisfies $\Psi(\lambda\mu) = \Psi(\lambda) \circ \Psi(\mu)$, and Ψ is a bijection ($\lambda \mapsto S_\lambda$ and $S_\lambda \mapsto \lambda$ are mutually inverse). ☐

Definition 15.6 (The rotation transformation and the rotation group). Let $\theta \in \mathbb{R}$. **Rotation about the origin** R_θ is given, when $n = 2$, by the matrix

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad R_\theta(\mathsf{x}) = R_\theta\,\mathsf{x}.$$

Put $\mathcal{R}_2 = \{R_\theta : \theta \in \mathbb{R}\}$. When $n = 3$, the rotation about the unit axis u ($\|\mathsf{u}\| = 1$) through the angle θ is written $R_{\mathsf{u},\theta}$, and its matrix is given by Rodrigues' formula

$$R_{\mathsf{u},\theta} = I + \sin\theta\,[\mathsf{u}]_\times + (1 - \cos\theta)\,[\mathsf{u}]_\times^2,$$

where $[\mathsf{u}]_\times \mathsf{v} = \mathsf{u} \times \mathsf{v}$.

Assumption: R_θ leaves the origin fixed; $\det R_\theta = 1$ (θ is a true rotation, containing no mirror); that u is a unit vector is the assumption under which Rodrigues' formula **contains no extra normalisation factor**.

Lemma 15.3 (The rotation group is isomorphic to the circle group). (a) When $n = 2$, $(\mathcal{R}_2, \circ) \cong U(1) = \{z \in \mathbb{C} : |z| = 1\}$, and the isomorphism is given by $\theta \mapsto e^{i\theta}$, whose kernel is $2\pi\mathbb{Z}$. (b) When $n = 3$, $(\mathcal{R}_3, \circ) \cong SO(3)$, and $SO(3) \cong SU(2)/\{\pm I\}$ (double cover).

Proof. (a) A direct computation with matrix multiplication gives:

$$R_\theta R_\varphi = \begin{pmatrix} \cos\theta\cos\varphi - \sin\theta\sin\varphi & -\cos\theta\sin\varphi - \sin\theta\cos\varphi \\ \sin\theta\cos\varphi + \cos\theta\sin\varphi & -\sin\theta\sin\varphi + \cos\theta\cos\varphi \end{pmatrix} = R_{\theta+\varphi},$$

so $\theta \mapsto R_\theta$ is a surjective homomorphism from $(\mathbb{R}, +)$ onto $\mathcal{R}_2$. On the other hand, identifying R_θ with the complex number $e^{i\theta} = \cos\theta + i\sin\theta$ (the plane $\mathbb{R}^2 \cong \mathbb{C}$, $(x, y) \mapsto x + iy$), the action of R_θ is exactly multiplication by $e^{i\theta}$, so that $R_\theta R_\varphi = R_{\theta+\varphi}$ corresponds to $e^{i(\theta+\varphi)} = e^{i\theta}e^{i\varphi}$. Hence $\theta \mapsto e^{i\theta}$ is a surjective homomorphism whose kernel is $\{\theta : e^{i\theta} = 1\} = 2\pi\mathbb{Z}$.

(b) The first relation is a standard result of linear algebra (the set of all orthogonal matrices with determinant 1 is precisely $SO(3)$). For the proof of the second, see the conjugation action ρ_q given in Theorem 8.18, where it is proved that $\rho : SU(2) \to SO(3)$ is a surjective homomorphism with $\ker\rho = \{\pm I\}$. ☐

Proposition 15.1 (Fixed points and invariants of the three classes of transformations)**.** Let $n = 3$ and let the parameters be non-trivial ($\mathsf{a} \neq \mathsf{0}$, $\lambda \neq 1$, $\theta \notin 2\pi\mathbb{Z}$). Then

- (i) T_{a} has no fixed point; the geometric quantity it preserves is the difference $\mathsf{x} - \mathsf{y}$ of any two points (that is, it preserves displacement vectors and the distances between points), but it does not preserve the distance from a point to the origin.
- (ii) S_λ has the unique fixed point $\mathsf{0}$; the geometric quantity it preserves is the family of lines through the origin (that is, directions), but it does not preserve distances between points (when $\lambda \neq 1$).
- (iii) The fixed point set of $R_{\mathsf{u},\theta}$ is the line $\mathbb{R}\mathsf{u}$; the geometric quantity it preserves is the inner product $\langle \mathsf{x}, \mathsf{y} \rangle$ (hence the norm and the angle), but it does not preserve direction (when $\theta \notin 2\pi\mathbb{Z}$).

Proof. We verify the 'fixed point equation' and the 'invariants' item by item.

(i) The fixed point equation $\mathsf{x} = \mathsf{x} + \mathsf{a}$ is equivalent to $\mathsf{a} = \mathsf{0}$, which contradicts $\mathsf{a} \neq \mathsf{0}$; hence there is no fixed point. For any x, y, $T_{\mathsf{a}}(\mathsf{x}) - T_{\mathsf{a}}(\mathsf{y}) = (\mathsf{x} + \mathsf{a}) - (\mathsf{y} + \mathsf{a}) = \mathsf{x} - \mathsf{y}$, so both the difference vector and the distance between points $d(\mathsf{x}, \mathsf{y}) = \|\mathsf{x} - \mathsf{y}\|$ are preserved. Taking $\mathsf{x} = \mathsf{0}$, $\|T_{\mathsf{a}}(\mathsf{0})\| = \|\mathsf{a}\| \neq 0 = \|\mathsf{0}\|$, so the distance to the origin is not preserved.

(ii) The fixed point equation $\mathsf{x} = \lambda\mathsf{x}$ is equivalent to $(\lambda - 1)\mathsf{x} = \mathsf{0}$, and since $\lambda \neq 1$ we have $\mathsf{x} = \mathsf{0}$ uniquely. For $\mu \in \mathbb{R}$ and x, $S_\lambda(\mu\mathsf{x}) = \mu\lambda\mathsf{x} = \mu S_\lambda(\mathsf{x})$, so $\mathbb{R}\mathsf{x}$ is mapped to itself and direction is preserved. As for distances, $\|S_\lambda(\mathsf{x}) - S_\lambda(\mathsf{y})\| = \lambda\|\mathsf{x} - \mathsf{y}\|$, which is not equal when $\mathsf{x} \neq \mathsf{y}$ and $\lambda \neq 1$.

(iii) The fixed point equation $R_{\mathsf{u},\theta}\mathsf{x} = \mathsf{x}$. By Rodrigues' formula, for $\mathsf{x} = t\mathsf{u}$: $[\mathsf{u}]_\times \mathsf{u} = \mathsf{u} \times \mathsf{u} = \mathsf{0}$, so $R_{\mathsf{u},\theta}(t\mathsf{u}) = t\mathsf{u}$, that is, every point of $\mathbb{R}\mathsf{u}$ is fixed. Conversely, if x is fixed and $\mathsf{x} \notin \mathbb{R}\mathsf{u}$, then x and u span a two-dimensional plane, on which $R_{\mathsf{u},\theta}$ restricts to a two-dimensional rotation through the angle $\theta \notin 2\pi\mathbb{Z}$ (this is precisely the $\mathcal{R}_2$ case of Lemma 15.3(a)), whose only fixed point is the origin — a contradiction. Hence the fixed point set is exactly $\mathbb{R}\mathsf{u}$. Preservation of the inner product: $R_{\mathsf{u},\theta}$ is an orthogonal matrix ($R^\mathsf{T} R = I$, $\det R = 1$), and an orthogonal matrix preserves the standard inner product by definition, so $\langle R\mathsf{x}, R\mathsf{y} \rangle = \mathsf{x}^\mathsf{T} R^\mathsf{T} R\,\mathsf{y} = \langle \mathsf{x}, \mathsf{y} \rangle$. Non-preservation of direction: take $\mathsf{x} \perp \mathsf{u}$ with $\mathsf{x} \neq \mathsf{0}$; then $R_{\mathsf{u},\theta}\mathsf{x} \neq \mathsf{x}$ (otherwise x would be a fixed point with $\mathsf{x} \notin \mathbb{R}\mathsf{u}$, contradicting the preceding paragraph), so $\mathbb{R}\mathsf{x} \neq \mathbb{R}(R_{\mathsf{u},\theta}\mathsf{x})$. □

Proposition 15.2 (The three symmetry centres)**.** Each of the three classes of transformations leaves a **fixed structure**—a fixed point, or the cyclic subgroup generated by some element—on the group structure to which it belongs, and these three fixed structures are the three symmetry centres named by this programme:

- (i) In the additive group $(\mathbb{R}, +)$, the unique fixed point of the inversion $x \mapsto -x$ is 0; correspondingly T_{a} degenerates to the identity when $\mathsf{a} = \mathsf{0}$.
- (ii) In the multiplicative group $(\mathbb{R}_{>0}, \times)$, the unique fixed point of the inversion $x \mapsto x^{-1}$ is 1; correspondingly S_λ degenerates to the identity when $\lambda = 1$.

(iii) In the circle group $U(1)$, the element $i = e^{i\pi/2}$ satisfies $i^4 = 1$ and $|i| = 1$; the cyclic subgroup $\{1, i, -1, -i\} = C_4$ that it generates is the fixed structure of the rotation case—i is both a generator of C_4 and the multiplier corresponding to $R_{\pi/2}$; correspondingly $R_{\pi/2}$ corresponds to multiplication by i when $n = 2$.

Proof. (i) From $-x = x$ we get $2x = 0$, so $x = 0$; uniqueness follows directly from that equation. (ii) From $x^{-1} = x$ and $x > 0$ we get $x^2 = 1$, so $x = 1$; uniqueness follows from the fact that for $x > 0$ the equation $x^2 = 1$ has the positive solution 1 only. (iii) $i = e^{i\pi/2} = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2}$, so $|i| = 1$ and $i^4 = e^{i2\pi} = 1$; moreover $\{i^k : k = 0, 1, 2, 3\} = \{1, i, -1, -i\}$ is the cyclic subgroup of $U(1)$ generated by i. When $n = 2$ the matrix of $R_{\pi/2}$ is $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, whilst multiplication by i in real coordinates is *exactly* $(x, y) \mapsto (-y, x)$; the two agree. □

Remark 15.3 (How the three symmetry centres are related to the two groups of objects). *The three symmetry centres 0, 1, i are the common pivot of the whole book, and each of them is associated both with a basic transformation and with a differential operator:*

$$0 \longleftrightarrow \text{translation}, \quad \nabla f; \qquad 1 \longleftrightarrow \text{scaling}, \quad \nabla \cdot \mathsf{F}; \qquad i \longleftrightarrow \text{rotation}, \quad \nabla \times \mathsf{F}.$$

What is recorded here is the association between each symmetry centre ↔ its respective object, not a correspondence between the two groups of objects. *The two groups of objects lie at different levels — transformations act on space, operators act on fields (scalar fields and vector fields) — and arranging them in a table of mutual correspondence would obscure the true assumptions under which each holds. The principle of this book is therefore: wherever translation, scaling or rotation is involved, state matters according to the assumptions of the transformation itself; wherever gradient, divergence or curl is involved, state matters according to the differentiability and domain assumptions of the operator itself; and make no assertion of correspondence between the two groups.*

Proposition 15.3 (The range generated by the three is exactly the orientation-preserving similitude group). *Let $\mathsf{Sim}^+(n)$ denote the* **orientation-preserving similitude transformation group** *of Euclidean space $\mathbb{R}^n$, that is, the set of all maps of the form*

$$\mathsf{x} \mapsto \lambda R\mathsf{x} + \mathsf{a}, \qquad \lambda > 0,\ R \in SO(n),\ \mathsf{a} \in \mathbb{R}^n.$$

Then the group generated by translation, scaling and rotation is equal to $\mathsf{Sim}^+(n)$:

$$\langle \mathcal{T}_n,\ \mathcal{S}_n,\ \mathcal{R}_n \rangle = \mathsf{Sim}^+(n).$$

Consequently, no continuous transformation outside $\mathsf{Sim}^+(n)$ can be expressed as a finite composite of these three classes of transformations.

Boundary remark: *the group in this conclusion is $\mathsf{Sim}^+(n)$ and not the full similitude group $\mathsf{Sim}(n) = \mathbb{R}^n \rtimes (\mathbb{R}_{>0} \times O(n))$, which contains reflections. A reflection matrix has determinant -1, whilst the linear part of every element of $\mathcal{T}_n, \mathcal{S}_n, \mathcal{R}_n$ has positive determinant, so a reflection* **cannot** *be obtained as a composite of the three; see Remark 15.5.*

Proof. **(1) The inclusion $\subseteq$:** every element of $\mathcal{T}_n$, $\mathcal{S}_n$, $\mathcal{R}_n$ has the form $\mathsf{x} \mapsto \lambda R\mathsf{x}+\mathsf{a}$, and the determinant of R is 1 (translation takes $\lambda = 1, R = I$; scaling takes $R = I, \mathsf{a} = \mathsf{0}$; rotation takes $\lambda = 1, \mathsf{a} = \mathsf{0}$, and by Definition 15.6 $\det R_{\mathsf{u},\theta} = 1$). Maps of this kind are closed under composition: if $g_i(\mathsf{x}) = \lambda_i R_i \mathsf{x} + \mathsf{a}_i$ $(i = 1, 2)$, then

$$g_1(g_2(\mathsf{x})) = \lambda_1 R_1(\lambda_2 R_2 \mathsf{x} + \mathsf{a}_2) + \mathsf{a}_1 = (\lambda_1\lambda_2)(R_1R_2)\mathsf{x} + (\lambda_1 R_1 \mathsf{a}_2 + \mathsf{a}_1),$$

where $\lambda_1\lambda_2 > 0$, $\det(R_1R_2) = \det R_1 \det R_2 = 1$ and $\lambda_1 R_1 \mathsf{a}_2 + \mathsf{a}_1 \in \mathbb{R}^n$, so the composite still lies in $\mathrm{Sim}^+(n)$. Hence every finite composite lies in $\mathrm{Sim}^+(n)$.

(2) The inclusion $\supseteq$: let $g(\mathsf{x}) = \lambda R\mathsf{x}+\mathsf{a}$ be any orientation-preserving similitude transformation, where $R \in SO(n)$. By linear algebra: when $n = 2$, $SO(2) = \mathcal{R}_2$, so $R \in \langle\mathcal{R}_2\rangle$; when $n = 3$, every $R \in SO(3)$ is a rotation about some unit axis u through an angle θ (Euler's theorem), that is, $R = R_{\mathsf{u},\theta} \in \mathcal{R}_3$. Hence $R \in \langle\mathcal{R}_n\rangle$. Also

$$g = T_{\mathsf{a}} \circ S_\lambda \circ R,$$

because for any x: $R(\mathsf{x}) = R\mathsf{x}$, $S_\lambda(R\mathsf{x}) = \lambda R\mathsf{x}$, $T_{\mathsf{a}}(\lambda R\mathsf{x}) = \lambda R\mathsf{x} + \mathsf{a} = g(\mathsf{x})$. Hence $g \in \langle\mathcal{T}_n, \mathcal{S}_n, \mathcal{R}_n\rangle$. ☐

Example 15.4 (Counterexample: the shear transformation and anisotropic scaling do not belong to the similitude group). Take the shear matrix $A_t = \left(\begin{smallmatrix}1 & t\\ 0 & 1\end{smallmatrix}\right)$ $(t \neq 0)$; it preserves area ($|\det A_t| = 1$) and orientation, and is a smooth continuous transformation. Take the anisotropic scaling $B = \mathrm{diag}(2, 1)$; it is smooth as well.

Argument that neither belongs to $\mathrm{Sim}(2)$: the matrix of the linear part of an element of $\mathrm{Sim}(2)$ must be λR_θ, satisfying $(M^\mathsf{T}M)^{1/2} = \lambda I$, that is, $M^\mathsf{T}M$ is a scalar matrix. But

$$A_t^\mathsf{T} A_t = \begin{pmatrix}1 & t\\ t & 1+t^2\end{pmatrix}, \qquad B^\mathsf{T}B = \begin{pmatrix}4 & 0\\ 0 & 1\end{pmatrix},$$

and in both cases $M^\mathsf{T}M$ fails to be a scalar matrix (for the former, the off-diagonal entry is non-zero when $t \neq 0$; for the latter, the diagonal entries are unequal). Hence $A_t, B \notin \mathrm{Sim}^+(2)$, and by Proposition 15.3 neither can be written as a finite composite of translations, scalings and rotations.

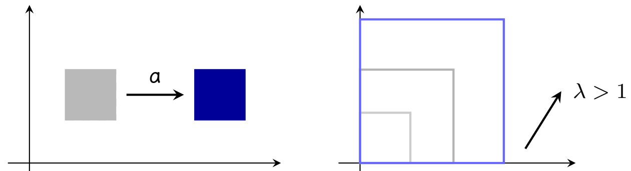
Significance: this gives concrete evidence for the statement in Remark 15.4 that 'the three are not all continuous transformations': there exist smooth transformations preserving some intrinsic structure (preserving area, or preserving a certain coordinate direction) that do not fall into the group generated by the three.

Remark 15.4 (The scope of validity of the three classes of transformations). This section establishes translation, scaling and rotation as **candidate** generators, on the basis of motivations at three levels (algebra: the three correspond respectively to the additive group, the multiplicative group and the circle group; geometry: the three exactly exhaust the continuous transformations of similarity geometry that do not change the intrinsic structure; number systems: subtraction, division and complex roots introduce respectively the directions of translation, proportion and rotation), and not on the basis of the following proposition:

> 'Any continuous transformation preserving some intrinsic structure can be decomposed into a composite of elements of these three groups.'

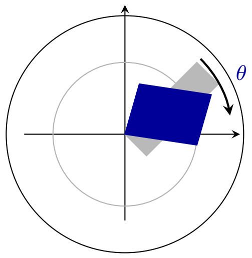
That proposition is a research programme, not a proved theorem. By Proposition 15.3 and Example 15.4 we know: if 'intrinsic structure' is understood as **Euclidean similarity structure**, then the proposition holds (in which case it is precisely Proposition 15.3); if 'intrinsic structure' is relaxed to area, volume, parallelism in some direction, or any other invariant, then the proposition is **false**. Hence wherever this book uses the proposition it must specify the intrinsic structure taken, and any use without such a specification is treated as a research programme.

Example 15.5 (Three standard examples of the three classes of transformations). (i) **Translation**: on a train moving at constant speed, each time the wheels rotate once a mark on the carriage floor returns to the same position — the displacement vector is preserved throughout. (ii) **Scaling**: the boundary of the Koch snowflake returns to itself under each scaling transformation by $\times 3$ (the boundary curve after magnification by 3 has the same structure as before magnification), its perimeter is infinite whilst its area is finite, and it is the area that is precisely the invariant making the scaling 'closed'. (iii) **Rotation**: the revolution of the Earth around the Sun is, in the approximation that neglects the perturbations of the other planets, a closed orbit in a rotationally symmetric system, and its angular momentum is conserved.



(a) Translation T_a
preserves difference vectors

(b) Scaling S_λ
preserves direction



(c) Rotation $R_{u,\theta}$
preserves the inner product

Figure 15.3: The three basic transformations and their invariants. Each of the three classes of transformations preserves a different geometric quantity: translation preserves the difference vector between points, scaling preserves directions through the origin, and rotation preserves the inner product; Proposition 15.1 gives the proof of each item.

Remark 15.5 (The group-representation-theoretic relations among the three). By Lemma 15.1–Lemma 15.3, both $\mathcal{T}_n$ and $\mathcal{S}_n$ are Abelian groups, whilst $\mathcal{R}_n$ is non-Abelian when $n = 3$. The interaction of the three is not a direct product but a

Table 15.1: Structural comparison of the three basic transformations

Transformation	Group	Isomorphic to	Fixed point set	Geometric quantity preserved
Translation T_{a}	$\mathcal{T}_n$	$(\mathbb{R}^n,+)$	empty ($\mathsf{a}\neq 0$)	difference vector $\mathsf{x}-\mathsf{y}$, distance between points
Scaling S_λ	$\mathcal{S}_n$	$(\mathbb{R}_{>0},\times)$	$\{0\}$ $(\lambda\neq 1)$	directions through the origin $\mathbb{R}\mathsf{x}$
Rotation $R_{\mathsf{u},\theta}$	$\mathcal{R}_n$	$U(1)$ $(n=2)$	$\mathbb{R}\mathsf{u}$ $(n=3)$	inner product $\langle\mathsf{x},\mathsf{y}\rangle$, norm, angle

semidirect product: the translation subgroup $\mathcal{T}_n$ is a normal subgroup of $\mathrm{Sim}^+(n)$, and

$$\mathrm{Sim}^+(n)=\mathcal{T}_n\rtimes\big(\mathcal{S}_n\times SO(n)\big),$$

where $\mathcal{S}_n\times SO(n)$ acts on $\mathcal{T}_n$ by $(\lambda,R)\cdot\mathsf{a}\mapsto\lambda R\mathsf{a}$. This semidirect product structure explains the phenomenon that 'scale first and then translate' and 'translate first and then scale' give different results: the action is not translation invariant.

Boundary warning: the full similitude group also contains reflections, $\mathrm{Sim}(n)=\mathcal{T}_n\rtimes(\mathcal{S}_n\times O(n))\supsetneq\mathrm{Sim}^+(n)$; the linear part of a reflection has determinant -1, whilst the linear parts of $\mathcal{T}_n,\mathcal{S}_n,\mathcal{R}_n$ all have determinant 1, so a reflection does not belong to the group generated by the three (see the boundary remark of Proposition 15.3). The remainder of this chapter uses only the orientation-preserving similitude group.

15.2.2 Gradient, divergence and curl

The preceding subsection gave the three classes of transformations; this subsection gives the three differential operators and explains **exactly how far** their relation to the three classes of transformations goes. The order here is deliberate: the three operators are first defined independently by coordinate formulae (Definition 15.7–Definition 15.10), then reduced by two limit characterisations to geometric quantities independent of the coordinate system (Proposition 15.5 and Proposition 15.6), and only then is their status as infinitesimal generators of the three classes of transformations discussed (Theorem 15.1).

Definition 15.7 (The nabla operator)**.** In $\mathbb{R}^3$, define the formal vector operator ∇ (read as nabla or del) by

$$\nabla=\left(\frac{\partial}{\partial x},\frac{\partial}{\partial y},\frac{\partial}{\partial z}\right).$$

Assumptions, and a fashionable phrasing distinguished: ∇ itself is **not** a vector in $\mathbb{R}^3$ — it belongs to no tangent space and has neither length nor direction; it is a **vector-valued differential operator**, that is, a triple 'each of whose components is a partial derivative operator acting on C^1 functions'. Writing it as a triple is merely a notational convenience: ∇f is defined because $\partial/\partial x$ acts on f, not because '∇ is multiplied by f'.

Well-definedness: ∇ acts on fields in three ways, and each action requires the object acted upon to possess continuous differentiability of the corresponding order (see Definition 15.8–Definition 15.10); when that order is not satisfied, the corresponding action of ∇ is **undefined**, rather than 'having the value zero'.

Definition 15.8 (Gradient)**.** The first definition of the gradient operator is to be found in chapter 9, Section 9.9, in Definition 9.32; this section presents the same object in the framework of C^1 open sets and supplements the differentiability assumptions and the mapping type. Let $f \in C^1(\mathcal{M})$ be a scalar field ($\mathcal{M} \subseteq \mathbb{R}^3$ a non-empty open set). The **gradient** of f is a vector field $\nabla f : \mathcal{M} \to \mathbb{R}^3$, defined by

$$\nabla f(x, y, z) = \left(\frac{\partial f}{\partial x}(x, y, z),\ \frac{\partial f}{\partial y}(x, y, z),\ \frac{\partial f}{\partial z}(x, y, z)\right).$$

Assumption: $f \in C^1$ is a sufficient condition; if f is differentiable only at a single point, then ∇f is defined only at that point. The mapping type is

$$\nabla : C^1(\mathcal{M}) \to C^0(\mathcal{M}, \mathbb{R}^3),$$

that is, the gradient **raises the order** (a scalar field becomes a vector field). It is linear, but it does not preserve multiplication (see Theorem 15.2(i)).

Definition 15.9 (Divergence)**.** The first definition of the divergence operator is to be found in chapter 9, Section 9.9, in Definition 9.33; this section presents the same object in the framework of C^1 open sets and supplements the differentiability assumptions and the mapping type. Let $\mathsf{F} = (F_1, F_2, F_3) \in C^1(\mathcal{M}, \mathbb{R}^3)$ be a vector field. The **divergence** of F is a scalar field $\nabla \cdot \mathsf{F} : \mathcal{M} \to \mathbb{R}$, defined by

$$\nabla \cdot \mathsf{F}(x, y, z) = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}.$$

Assumption: $\mathsf{F} \in C^1$. The mapping type is

$$\nabla\cdot : C^1(\mathcal{M}, \mathbb{R}^3) \to C^0(\mathcal{M}).$$

Note: the defining formula of the divergence depends on the coordinate system, but its value does not depend on the choice of coordinate system (the limit characterisation given by Proposition 15.5 is independent of this); this point, which is the same as for the curl, is a key repeatedly used in this chapter.

Definition 15.10 (Curl)**.** The first definition of the curl operator is to be found in chapter 9, Section 9.9, in Definition 9.34; this section presents the same object in the framework of C^1 open sets and supplements the differentiability assumptions and the mapping type. Let $\mathsf{F} = (F_1, F_2, F_3) \in C^1(\mathcal{M}, \mathbb{R}^3)$ be a vector field. The **curl** of F is a vector field $\nabla \times \mathsf{F} : \mathcal{M} \to \mathbb{R}^3$, defined by the formal cross product

$$\nabla \times \mathsf{F} = \begin{vmatrix} \mathsf{i} & \mathsf{j} & \mathsf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z},\ \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x},\ \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right).$$

Assumption: $\mathsf{F} \in C^1$. The determinant notation is merely a formal notation: treating a differential operator as a row of a determinant and expanding along the first row gives the formula above, but one **cannot** conclude from this that ∇ and F commute — in fact $\nabla\times\mathsf{F}$ and $\mathsf{F}\times\nabla$ differ by a sign (see Proposition 15.10). The mapping type is

$$\nabla\times : C^1(\mathcal{M},\mathbb{R}^3) \to C^0(\mathcal{M},\mathbb{R}^3).$$

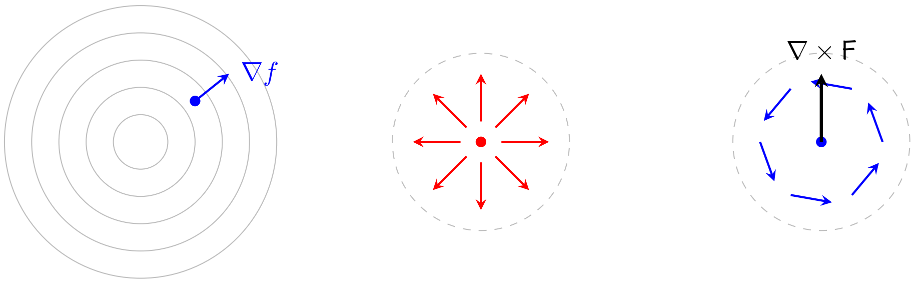
Definition 15.11 (The Laplacian). Let $f \in C^2(\mathcal{M})$ be a scalar field. The **Laplacian** of f is a scalar field $\nabla^2 f : \mathcal{M} \to \mathbb{R}$, defined as the divergence of the gradient:

$$\nabla^2 f = \nabla\cdot(\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}.$$

For a vector field $\mathsf{F} = (F_1, F_2, F_3) \in C^2(\mathcal{M},\mathbb{R}^3)$, its Laplacian is defined componentwise: $\nabla^2\mathsf{F} = (\nabla^2 F_1,\ \nabla^2 F_2,\ \nabla^2 F_3)$.

Assumption: the scalar case requires $f \in C^2$; the vector case requires $\mathsf{F} \in C^2(\mathcal{M},\mathbb{R}^3)$. If f lies only in C^1, then $\nabla^2 f$ need not exist — ∇f lies only in C^0, and taking its divergence again requires C^1, so C^2 is the minimum requirement.

If $\nabla^2 f = 0$ holds on $\mathcal{M}$, then f is called a **harmonic function**.



(a) Gradient ∇f (b) Divergence $\nabla\cdot\mathsf{F}$ (c) Curl $\nabla\times\mathsf{F}$

Figure 15.4: The geometric picture of the three differential operators: the gradient points in the direction of fastest increase, the divergence describes sources and sinks, and the curl describes local circulation.

Lemma 15.4 (Two fundamental integral identities on a sphere). Let $B_r = B_r(\mathsf{x}_0)$ be the closed ball in $\mathbb{R}^3$ with centre x_0 and radius $r > 0$, let ∂B_r be its boundary sphere, n the unit outward normal, dS the area element, and $|B_r| = \frac{4}{3}\pi r^3$. Then

(i) $\displaystyle\oint_{\partial B_r} \mathsf{n}\, dS = \mathsf{0}$;

(ii) $\displaystyle\oint_{\partial B_r} (\mathsf{x}-\mathsf{x}_0)_i\, n_j\, dS = |B_r|\,\delta_{ij}$, where $(\mathsf{x}-\mathsf{x}_0)_i$ and n_j denote the i-th and j-th coordinate components respectively.

Proof. Write $\mathsf{y} = \mathsf{x}-\mathsf{x}_0$, $\hat{\mathsf{y}} = \mathsf{y}/|\mathsf{y}|$. On ∂B_r we have $|\mathsf{y}| = r$ and $\mathsf{n} = \hat{\mathsf{y}}$, whilst the area element of a sphere centred at x_0 satisfies $dS = r^2\, d\Omega$, where $d\Omega$ is the area element on the unit sphere S^2 (this relation comes from the Jacobian factor r^2 of the spherical parametrisation $(r,\theta,\varphi) \mapsto \mathsf{x}_0 + r\,\hat{\mathsf{y}}(\theta,\varphi)$).

(i) On the unit sphere $\oint_{S^2} \hat{y}_i\, d\Omega = 0$: the map $\hat{\mathsf{y}} \mapsto -\hat{\mathsf{y}}$ is a diffeomorphism of S^2 preserving the area element $d\Omega$ (since it is an orthogonal transformation with Jacobian factor 1), but it carries $\hat{y}_i$ to $-\hat{y}_i$. Hence the integral equals its own negative, and can only be 0. Therefore $\oint_{\partial B_r} \mathsf{n}\, dS = \oint_{S^2} \hat{\mathsf{y}}\, r^2 d\Omega = r^2 \oint_{S^2} \hat{\mathsf{y}}\, d\Omega = \mathsf{0}$.

(ii) First compute the integral over the unit sphere, $I_{ij} = \oint_{S^2} \hat{y}_i \hat{y}_j \, d\Omega$.

The off-diagonal case $i \neq j$: let $i \neq j$ and consider the reflection $R_j : \hat{\mathsf{y}} \mapsto \hat{\mathsf{y}}'$ in the coordinate hyperplane $y_j = 0$, where $\hat{y}'_j = -\hat{y}_j$ and the other components are unchanged. R_j is an orthogonal transformation, hence preserves $d\Omega$; it leaves $\hat{y}_i$ unchanged and changes the sign of $\hat{y}_j$, so the integrand $\hat{y}_i\hat{y}_j$ changes sign. Hence $I_{ij} = -I_{ij}$, giving $I_{ij} = 0$.

The diagonal case: by the symmetry above, the three diagonal entries are equal: $I_{11} = I_{22} = I_{33}$ (a permutation of coordinates is an orthogonal transformation preserving $d\Omega$ and permuting the diagonal integrals); whilst on $\hat{y}_1^2+\hat{y}_2^2+\hat{y}_3^2 = 1$,

$$I_{11} + I_{22} + I_{33} = \oint_{S^2} (\hat{y}_1^2 + \hat{y}_2^2 + \hat{y}_3^2) \, d\Omega = \oint_{S^2} 1 \, d\Omega = 4\pi.$$

Hence each diagonal entry equals $\frac{4\pi}{3}$, that is, $I_{ij} = \frac{4\pi}{3}\delta_{ij}$.

Returning to the sphere: $\mathsf{x} - \mathsf{x}_0 = r\hat{\mathsf{y}}$ and $\mathsf{n} = \hat{\mathsf{y}}$, so

$$\oint_{\partial B_r} (\mathsf{x} - \mathsf{x}_0)_i \, n_j \, dS = \oint_{S^2} r\hat{y}_i \, \hat{y}_j \, r^2 d\Omega = r^3 I_{ij} = r^3 \cdot \frac{4\pi}{3}\delta_{ij} = \frac{4}{3}\pi r^3 \, \delta_{ij} = |B_r| \, \delta_{ij}. \qquad \square$$

Proposition 15.4 (The gradient and directional derivatives). *Let $f \in C^1(\mathcal{M})$, $\mathsf{x}_0 \in \mathcal{M}$, and let u be a unit vector. Then*

$$D_{\mathsf{u}} f(\mathsf{x}_0) := \lim_{h \to 0} \frac{f(\mathsf{x}_0 + h\mathsf{u}) - f(\mathsf{x}_0)}{h} = \nabla f(\mathsf{x}_0) \cdot \mathsf{u}.$$

Proof. Since $f \in C^1(\mathcal{M})$ and f is defined in some neighbourhood of x_0, f is differentiable at x_0 (the existence of continuous partial derivatives implies differentiability), that is, there is a linear map $L = Df(\mathsf{x}_0)$ such that

$$f(\mathsf{x}_0 + \mathsf{h}) = f(\mathsf{x}_0) + L\mathsf{h} + o(\|\mathsf{h}\|) \qquad (\|\mathsf{h}\| \to 0).$$

Here $L\mathsf{h} = \sum_i \partial_i f(\mathsf{x}_0) \, h_i = \nabla f(\mathsf{x}_0) \cdot \mathsf{h}$ (because the i-th component of L is the directional derivative $\partial_i f$ along e_i, and $\{e_i\}$ is an orthogonal basis, so the pairing of $L\mathsf{h}$ with h is given by the standard inner product). Take $\mathsf{h} = h\mathsf{u}$ and divide by h ($h \neq 0$):

$$\frac{f(\mathsf{x}_0 + h\mathsf{u}) - f(\mathsf{x}_0)}{h} = \frac{h \, \nabla f(\mathsf{x}_0) \cdot \mathsf{u} + o(|h|)}{h} = \nabla f(\mathsf{x}_0) \cdot \mathsf{u} + \frac{o(|h|)}{h}.$$

Since $o(|h|)/h \to 0$ as $h \to 0$, taking the limit gives the conclusion. $\square$

Proposition 15.5 (The divergence is the flux density). *Let $\mathsf{F} \in C^1(\mathcal{M}, \mathbb{R}^3)$, $\mathsf{x}_0 \in \mathcal{M}$, and $B_r = B_r(\mathsf{x}_0) \subseteq \mathcal{M}$ (with r sufficiently small). Then*

$$\nabla \cdot \mathsf{F}(\mathsf{x}_0) = \lim_{r \to 0^+} \frac{1}{|B_r|} \oint_{\partial B_r} \mathsf{F} \cdot \mathsf{n} \, dS.$$

Equivalently, $(\nabla \cdot \mathsf{F})(\mathsf{x}_0)$ is the **flux density** *of F at x_0 — the net outward flux per unit volume.*

Proof. Fix x_0 and write $J = D\mathsf{F}(\mathsf{x}_0)$ (a 3×3 matrix, $J_{ij} = \partial_j F_i(\mathsf{x}_0)$). Since F is differentiable at x_0, expand F to first order:

$$\mathsf{F}(\mathsf{x}) = \mathsf{F}(\mathsf{x}_0) + J(\mathsf{x} - \mathsf{x}_0) + \mathsf{R}(\mathsf{x}), \qquad \mathsf{R}(\mathsf{x}) = o(\|\mathsf{x} - \mathsf{x}_0\|) \ (\mathsf{x} \to \mathsf{x}_0).$$

Split the flux into three parts:

$$\oint_{\partial B_r} \mathsf{F}\cdot\mathsf{n}\,dS = \underbrace{\mathsf{F}(\mathsf{x}_0)\cdot\oint_{\partial B_r}\mathsf{n}\,dS}_{T_1} + \underbrace{\oint_{\partial B_r} J(\mathsf{x}-\mathsf{x}_0)\cdot\mathsf{n}\,dS}_{T_2} + \underbrace{\oint_{\partial B_r}\mathsf{R}\cdot\mathsf{n}\,dS}_{T_3}.$$

T_1: by Lemma 15.4(i), $\oint \mathsf{n}\,dS = \mathbf{0}$, so $T_1 = 0$.

T_2: write $J(\mathsf{x}-\mathsf{x}_0)\cdot\mathsf{n} = \sum_{i,j} J_{ij}(\mathsf{x}-\mathsf{x}_0)_j n_i$, integrate term by term and use Lemma 15.4(ii):

$$T_2 = \sum_{i,j} J_{ij}\oint_{\partial B_r}(\mathsf{x}-\mathsf{x}_0)_j n_i\,dS = \sum_{i,j} J_{ij}\,|B_r|\,\delta_{ij} = |B_r|\sum_i J_{ii} = |B_r|\,\mathrm{tr}\,J.$$

And $\mathrm{tr}\,J = \sum_i \partial_i F_i(\mathsf{x}_0) = \nabla\cdot\mathsf{F}(\mathsf{x}_0)$.

T_3: from $|\mathsf{R}(\mathsf{x})| = o(\|\mathsf{x}-\mathsf{x}_0\|)$ and $|\mathsf{n}| = 1$,

$$|T_3| \le \oint_{\partial B_r}|\mathsf{R}|\,dS \le \sup_{\partial B_r}|\mathsf{R}|\cdot 4\pi r^2 = o(r)\cdot O(r^2) = o(r^3).$$

Since $|B_r| = \frac{4}{3}\pi r^3$, we have $T_3/|B_r| = o(1)$ as $r\to 0$.

Adding the three parts and dividing by $|B_r|$:

$$\frac{1}{|B_r|}\oint_{\partial B_r}\mathsf{F}\cdot\mathsf{n}\,dS = \nabla\cdot\mathsf{F}(\mathsf{x}_0) + o(1) \xrightarrow[r\to 0^+]{} \nabla\cdot\mathsf{F}(\mathsf{x}_0).$$ □

Proposition 15.6 (The curl is the circulation density)**.** *Let $\mathsf{F}\in C^1(\mathcal{M},\mathbb{R}^3)$, $\mathsf{x}_0\in\mathcal{M}$, and let n be a unit vector. Take the disc D_r through x_0 with normal n, centred at x_0 with radius $r>0$, let its boundary $\gamma_r = \partial D_r$ be oriented by the right-hand rule with respect to n, and write $|D_r| = \pi r^2$. Then*

$$\big(\nabla\times\mathsf{F}(\mathsf{x}_0)\big)\cdot\mathsf{n} = \lim_{r\to 0^+}\frac{1}{|D_r|}\oint_{\gamma_r}\mathsf{F}\cdot d\mathsf{r}.$$

*Hence the component of the curl in the direction n is the **circulation density** about that direction; in particular, if in some region the circulation density is 0 for every n, then $\nabla\times\mathsf{F} = \mathbf{0}$ in that region.*

Proof. Take an orthonormal basis $\mathsf{e}_1,\mathsf{e}_2,\mathsf{e}_3$ at x_0 such that $\mathsf{e}_3 = \mathsf{n}$, with D_r lying in $\mathrm{span}\{\mathsf{e}_1,\mathsf{e}_2\}$ and the orientation of γ_r satisfying the right-hand rule with e_3, that is, appearing anticlockwise when viewed along e_3. Write $J = D\mathsf{F}(\mathsf{x}_0)$ and, as in Proposition 15.5, expand to first order $\mathsf{F}(\mathsf{x}) = \mathsf{F}(\mathsf{x}_0)+J(\mathsf{x}-\mathsf{x}_0)+\mathsf{R}(\mathsf{x})$, $\mathsf{R} = o(\|\mathsf{x}-\mathsf{x}_0\|)$, and split the circulation into three parts

$$\oint_{\gamma_r}\mathsf{F}\cdot d\mathsf{r} = \underbrace{\oint_{\gamma_r}\mathsf{F}(\mathsf{x}_0)\cdot d\mathsf{r}}_{S_1} + \underbrace{\oint_{\gamma_r} J(\mathsf{x}-\mathsf{x}_0)\cdot d\mathsf{r}}_{S_2} + \underbrace{\oint_{\gamma_r}\mathsf{R}\cdot d\mathsf{r}}_{S_3}.$$

S_1: $\mathsf{F}(\mathsf{x}_0)$ is a constant vector and $\oint_{\gamma_r} d\mathsf{r} = \mathbf{0}$ (the displacement integral along a closed curve is zero), so $S_1 = 0$.

S_2: parametrise $\mathsf{x}-\mathsf{x}_0 = r(\cos\theta\,\mathsf{e}_1+\sin\theta\,\mathsf{e}_2)$, $\theta\in[0,2\pi]$; then $d\mathsf{r} = r(-\sin\theta\,\mathsf{e}_1+\cos\theta\,\mathsf{e}_2)\,d\theta$. Write $a_{ij} = (J\mathsf{e}_i)\cdot\mathsf{e}_j$, so that

$$J(\mathsf{x}-\mathsf{x}_0)\cdot d\mathsf{r} = r^2\big(a_{11}(-\sin\theta\cos\theta) + a_{12}\cos^2\theta - a_{21}\sin^2\theta + a_{22}\sin\theta\cos\theta\big)\,d\theta.$$

Integrating term by term over $[0,2\pi]$: $\int_0^{2\pi}\sin\theta\cos\theta\,d\theta = 0$, $\int_0^{2\pi}\cos^2\theta\,d\theta = \int_0^{2\pi}\sin^2\theta\,d\theta = \pi$. Hence

$$S_2 = r^2\pi\,(a_{12}-a_{21}) = \pi r^2\,(a_{12}-a_{21}).$$

Note that a_{ij} are the matrix entries of J in the basis $\{\mathsf{e}_i\}$, that is, $a_{ij}=J_{ji}$ (because $a_{ij}=(J\mathsf{e}_i)\cdot\mathsf{e}_j=\sum_k J_{ki}(\mathsf{e}_j)_k=J_{ji}$). Hence $a_{12}-a_{21}=J_{21}-J_{12}$. Moreover, by the definition of the curl, $(\nabla\times\mathsf{F})_3=\partial_1F_2-\partial_2F_1=J_{21}-J_{12}$ (since $\partial_jF_i(\mathsf{x}_0)=J_{ij}$). And in the chosen basis $\mathsf{n}=\mathsf{e}_3$, so

$$S_2=\pi r^2\,(\nabla\times\mathsf{F}(\mathsf{x}_0))\cdot\mathsf{n}=|D_r|\,(\nabla\times\mathsf{F}(\mathsf{x}_0))\cdot\mathsf{n}.$$

S_3: from $|\mathsf{R}|=o(\|\mathsf{x}-\mathsf{x}_0\|)$ and the arc length $2\pi r$,

$$|S_3|\le\oint_{\gamma_r}|\mathsf{R}|\,\|d\mathsf{r}\|\le\sup_{\gamma_r}|\mathsf{R}|\cdot 2\pi r=o(r)\cdot O(r)=o(r^2),$$

so $S_3/|D_r|=o(1)$.

Adding the three parts and dividing by $|D_r|=\pi r^2$:

$$\frac{1}{|D_r|}\oint_{\gamma_r}\mathsf{F}\cdot d\mathsf{r}=(\nabla\times\mathsf{F}(\mathsf{x}_0))\cdot\mathsf{n}+o(1)\xrightarrow[r\to0^+]{}(\nabla\times\mathsf{F}(\mathsf{x}_0))\cdot\mathsf{n}. \qquad\square$$

Theorem 15.1 (The three operators as infinitesimal generators of the three classes of transformations)**.** *The following three clauses hold independently of one another.*

(i) The gradient and translation. *Let $f\in C^1(\mathcal{M})$, $\mathsf{a}\in\mathbb{R}^3$, and let the translation family be $\Phi_t^{\mathrm{tr}}=T_{t\mathsf{a}}$. Then*

$$\left.\frac{d}{dt}\right|_{t=0} f\big(\mathsf{x}+t\mathsf{a}\big)=\mathsf{a}\cdot\nabla f(\mathsf{x}),$$

that is, the **infinitesimal generator** *induced by the translation family on function space is the directional derivative operator $\mathsf{a}\cdot\nabla$ along a, whose 'all-directions version' is the gradient ∇.*

(ii) The divergence and the rate of change of volume. *Let $\mathsf{F}\in C^1(\mathcal{M},\mathbb{R}^3)$ and let $\Omega\subset\mathcal{M}$ be a bounded measurable set. Let $\{\Phi_t\}_{|t|<\epsilon}$ be a family of C^1 maps satisfying $\Phi_0=\mathrm{id}$, $\Phi_t(\mathcal{M})\subseteq\mathcal{M}$, and on every compact subset*

$$\left.\frac{\partial}{\partial t}\right|_{t=0}\Phi_t(\mathsf{x})=\mathsf{F}(\mathsf{x})\quad\text{uniformly}.$$

Then

$$\left.\frac{d}{dt}\right|_{t=0}\mathrm{Vol}\big(\Phi_t(\Omega)\big)=\int_\Omega\nabla\cdot\mathsf{F}\,dV.$$

In particular, for a sufficiently small ball $B_r(\mathsf{x}_0)$,

$$\nabla\cdot\mathsf{F}(\mathsf{x}_0)=\lim_{r\to0^+}\frac{1}{|B_r|}\left.\frac{d}{dt}\right|_{t=0}\mathrm{Vol}\big(\Phi_t(B_r)\big),$$

that is, the divergence is the **instantaneous rate of change of volume per unit volume***.*

(iii) The curl and the local angular velocity. Let $\mathsf{F} \in C^1(\mathcal{M}, \mathbb{R}^3)$ and $J = D\mathsf{F}(\mathsf{x}_0)$. Then the angular velocity corresponding to the antisymmetric part of J is $\frac{1}{2}\nabla \times \mathsf{F}(\mathsf{x}_0)$, that is, for every $\mathsf{y} \in \mathbb{R}^3$,

$$\tfrac{1}{2}\big(J - J^{\mathsf{T}}\big)\mathsf{y} = \tfrac{1}{2}\big(\nabla \times \mathsf{F}(\mathsf{x}_0)\big) \times \mathsf{y}.$$

Hence in a neighbourhood of x_0, the **local first-order rotation** of F is completely characterised by $\frac{1}{2}\nabla \times \mathsf{F}(\mathsf{x}_0)$; $\nabla \times \mathsf{F}(\mathsf{x}_0) = \mathbf{0}$ if and only if $D\mathsf{F}(\mathsf{x}_0)$ is symmetric, that is, there is no local first-order rotation.

Proof. (i) Let $g(t) = f(\mathsf{x} + t\mathsf{a})$. This is the composite of a C^1 function with a C^1 curve, so by the chain rule $g'(t) = \nabla f(\mathsf{x} + t\mathsf{a}) \cdot \mathsf{a}$; taking $t = 0$ gives the result. The reading of ∇ as the summary of 'all directions' comes from the arbitrariness of a: for every a the quantity $\mathsf{a} \cdot \nabla f(\mathsf{x})$ can be represented in this way, so ∇f is the unique vector satisfying that formula.

(ii) By the change-of-variables formula, for those t with $\det D\Phi_t > 0$,

$$\operatorname{Vol}\big(\Phi_t(\Omega)\big) = \int_\Omega \big|\det D\Phi_t(\mathsf{x})\big|\, dV = \int_\Omega \det D\Phi_t(\mathsf{x})\, dV.$$

(For very small t we have $\det D\Phi_0 = \det I = 1 > 0$, and since $\det D\Phi_t$ is continuous in t it stays positive for $|t| < \delta$, so the absolute value may be dropped.) Differentiate with respect to t and interchange differentiation and integration (the integrand is uniformly C^1 in t on compact sets, so the interchange is legitimate):

$$\frac{d}{dt}\operatorname{Vol}\big(\Phi_t(\Omega)\big) = \int_\Omega \frac{\partial}{\partial t} \det D\Phi_t(\mathsf{x})\, dV.$$

Apply the **Jacobi formula** to the matrix-valued function $A(t) = D\Phi_t(\mathsf{x})$:

$$\frac{d}{dt} \det A(t) = \det A(t) \cdot \operatorname{tr}\left(A(t)^{-1} \frac{d}{dt} A(t) \right)$$

(valid when $A(t)$ is invertible; $A(0) = I$ is invertible and remains invertible near $t = 0$). Put $t = 0$: $A(0) = I$, $\det A(0) = 1$, and

$$\left.\frac{d}{dt}\right|_{t=0} D\Phi_t(\mathsf{x}) = D\left(\left.\frac{\partial}{\partial t}\right|_{t=0} \Phi_t \right)(\mathsf{x}) = D\mathsf{F}(\mathsf{x}),$$

where the first step interchanges the t-derivative with the x-partial derivative (Φ_t is C^1 in (x, t)), and the second step uses the assumption. Hence

$$\left.\frac{d}{dt}\right|_{t=0} \det D\Phi_t(\mathsf{x}) = \operatorname{tr} D\mathsf{F}(\mathsf{x}) = \nabla \cdot \mathsf{F}(\mathsf{x}),$$

and substitution gives $\left.\frac{d}{dt}\right|_{t=0} \operatorname{Vol}(\Phi_t(\Omega)) = \int_\Omega \nabla \cdot \mathsf{F}\, dV$. Taking $\Omega = B_r(\mathsf{x}_0)$ and dividing by $|B_r|$, the second equality follows from the continuity of $\nabla \cdot \mathsf{F}$ at x_0.

(iii) It suffices to verify the components. Write $\mathsf{w} = \nabla \times \mathsf{F}(\mathsf{x}_0)$. By Definition 15.10,

$$w_1 = J_{32} - J_{23}, \qquad w_2 = J_{13} - J_{31}, \qquad w_3 = J_{21} - J_{12}.$$

First component: the left-hand side is $\frac{1}{2}(J-J^{\mathsf{T}})\mathsf{y}\big|_1 = \frac{1}{2}\big[(J_{12}-J_{21})y_2+(J_{13}-J_{31})y_3\big]$, whilst the right-hand side is $\frac{1}{2}(\mathsf{w}\times\mathsf{y})_1 = \frac{1}{2}(w_2y_3-w_3y_2) = \frac{1}{2}\big[(J_{13}-J_{31})y_3-(J_{21}-J_{12})y_2\big]$, and the two are equal. The computation of the second and third components is **word for word** the same as that of the first, only the indices being permuted cyclically $(1,2,3) \to (2,3,1) \to (3,1,2)$ — this permutation preserves both the sign convention of the ϵ-notation and the defining form $J_{ij} = \partial_j F_i$, and therefore constitutes no new argument. Agreement of the three components gives the stated equality.

Proof of the last assertion: $J - J^{\mathsf{T}} = 0 \iff J = J^{\mathsf{T}} \iff J$ is symmetric; and by the equality already proved, $J - J^{\mathsf{T}} = 0 \iff \mathsf{w}\times\mathsf{y} = \mathsf{0}$ for all $\mathsf{y} \iff \mathsf{w} = \mathsf{0}$ (take $\mathsf{y} \perp \mathsf{w}$ with $\mathsf{y} \neq \mathsf{0}$; if $\mathsf{w} \neq \mathsf{0}$ then $\mathsf{w}\times\mathsf{y} \neq \mathsf{0}$). In summary, $\nabla\times\mathsf{F}(\mathsf{x}_0) = \mathsf{0} \iff J$ is symmetric. ☐

Example 15.6 (Counterexample: *a field with everywhere non-zero constant curl need not be a 'rotation field'*). Take $\mathsf{F}(x,y,z) = (-y,x,0)$; then $\nabla\times\mathsf{F} = (0,0,2)$ is a constant vector everywhere. Take $\mathsf{G}(x,y,z) = (-y,x,2z)$; then $\nabla\times\mathsf{G} = (\partial_y(2z)-\partial_z x,\ \partial_z(-y)-\partial_x(2z),\ \partial_x x-\partial_y(-y)) = (0,0,2)$, the same as the curl of F everywhere.

But the shapes of the two orbits are completely different: the orbits of F are circles with the z-axis as axis (a true rotation), whilst the orbits of G are helices (with a non-zero translation component along the z direction at the same time). It can be seen that **the pointwise value of the curl cannot uniquely determine the geometric shape of the orbit** — the curl characterises only the local first-order rotation, and the global orbit can be obtained only by integrating the vector field. This is precisely the necessity of the qualification 'in the first-order sense' in the second level of Section 15.2.

15.3 Operational properties of the nabla operator

This section establishes the geometric and physical meanings of the three operators, their linearity, and the mixed-partial-derivative interchange lemma on which all subsequent second-order operations depend. All conclusions in this section state their differentiability-order assumptions explicitly, and Example 15.8 gives evidence that those assumptions cannot be omitted.

Proposition 15.7 (The geometric meaning of the gradient). *Let $f \in C^1(\mathcal{M})$, $\mathsf{x}_0 \in \mathcal{M}$, and*

$$\nabla f(\mathsf{x}_0) \neq \mathsf{0}.$$

Then

(i) *Among all* **unit** *vectors u ($\|\mathsf{u}\| = 1$), the direction of $\nabla f(\mathsf{x}_0)$ maximises the directional derivative $D_{\mathsf{u}}f(\mathsf{x}_0)$, that is, $\nabla f(\mathsf{x}_0)$ points in the direction in which f increases fastest at x_0.*

(ii) *That maximum equals $\|\nabla f(\mathsf{x}_0)\|$.*

(iii) *$\nabla f(\mathsf{x}_0)$ is orthogonal to the tangent vector of every smooth curve through x_0 that lies in the level surface $S = \{\mathsf{x}\in\mathcal{M} : f(\mathsf{x}) = f(\mathsf{x}_0)\}$; that is, the gradient is perpendicular to the level surface.*

Table 15.2: Definitions, assumptions and limit characterisations of the three differential operators

Operator	Action	Minimum differentiability assumption	Coordinate-independent limit characterisation
∇f	scalar field → vector field	$f \in C^1$	$D_{\mathbf{u}} f(\mathbf{x}_0) = \nabla f(\mathbf{x}_0)\cdot \mathbf{u}$ (Proposition 15.4)
$\nabla \cdot \mathbf{F}$	vector field → scalar field	$\mathbf{F} \in C^1$	$\lim_{r\to 0} \frac{1}{\lvert B_r\rvert}\oint_{\partial B_r} \mathbf{F}\cdot \mathbf{n}\, dS$ (Proposition 15.5)
$\nabla \times \mathbf{F}$	vector field → vector field	$\mathbf{F} \in C^1$	$\lim_{r\to 0} \frac{1}{\lvert D_r\rvert}\oint_{\gamma_r} \mathbf{F}\cdot d\mathbf{r}$ (Proposition 15.6)
$\nabla^2 f$	scalar field → scalar field	$f \in C^2$	equal pointwise to div grad; its geometric meaning requires second derivatives (Section 15.3)

Remarks on the assumptions: $\|\mathsf{u}\| = 1$ is a necessary condition for (i)(ii) — if the normalisation is dropped, the directional derivative $D_{\mathsf{u}}f$ can grow arbitrarily with $\|\mathsf{u}\|$ and no maximum exists. The condition $\nabla f(\mathsf{x}_0) \neq \mathbf{0}$ is necessary both for 'the maximum being attained in a unique direction' in (i) and for '∇f giving the normal to the level surface' in (iii); if $\nabla f(\mathsf{x}_0) = \mathbf{0}$ (a critical point), then the conclusions of (i)(ii) degenerate to 'the maximum directional derivative is 0' (in which case the derivative in every direction is 0), whilst the conclusion of (iii) still holds but loses its meaning; see Example 15.7.

Proof. **(i)(ii)** By Proposition 15.4, for every unit vector u we have $D_{\mathsf{u}}f(\mathsf{x}_0) = \nabla f(\mathsf{x}_0) \cdot \mathsf{u}$. By the Cauchy–Schwarz inequality,

$$|D_{\mathsf{u}}f(\mathsf{x}_0)| = |\nabla f(\mathsf{x}_0) \cdot \mathsf{u}| \leq \|\nabla f(\mathsf{x}_0)\| \cdot \|\mathsf{u}\| = \|\nabla f(\mathsf{x}_0)\|,$$

where the last step uses $\|\mathsf{u}\| = 1$. The equality condition in Cauchy–Schwarz is that u and $\nabla f(\mathsf{x}_0)$ be linearly dependent; combining this with the normalisation $\|\mathsf{u}\| = 1$ and $\nabla f(\mathsf{x}_0) \neq \mathbf{0}$ gives $\mathsf{u} = \pm \nabla f(\mathsf{x}_0)/\|\nabla f(\mathsf{x}_0)\|$. Taking the '$+$' sign gives $D_{\mathsf{u}}f = +\|\nabla f(\mathsf{x}_0)\|$ (the supremum, attained among unit vectors), and taking the '$-$' sign gives $-\|\nabla f(\mathsf{x}_0)\|$. Hence the maximum is attained in the unique direction $\mathsf{u}^* = \nabla f(\mathsf{x}_0)/\|\nabla f(\mathsf{x}_0)\|$ and equals $\|\nabla f(\mathsf{x}_0)\|$.

(iii) Let $\gamma : (-\delta, \delta) \to S$ be a C^1 curve lying in the level surface, with $\gamma(0) = \mathsf{x}_0$. From $\gamma(t) \in S$ we get that $f(\gamma(t)) = f(\mathsf{x}_0)$ is constant, so

$$0 = \frac{d}{dt} f(\gamma(t)) = \nabla f(\gamma(t)) \cdot \gamma'(t)$$

(chain rule; here $f \in C^1$ and $\gamma \in C^1$ are used). Taking $t = 0$ gives $\nabla f(\mathsf{x}_0) \cdot \gamma'(0) = 0$. By the definition of S, the tangent space of S at x_0 is spanned by all such $\gamma'(0)$ (the level surface is a C^1 regular surface provided that $\nabla f(\mathsf{x}_0) \neq \mathbf{0}$, which is the applicability condition of the implicit function theorem, so this spanning relation holds), and hence $\nabla f(\mathsf{x}_0)$ is orthogonal to the whole tangent space. $\square$

Example 15.7 (The geometric meaning of the gradient degenerates at a critical point). Take $f(x, y, z) = x^2 + y^2 + z^2$; then $\nabla f = 2(x, y, z)$, and at $\mathsf{x}_0 = \mathbf{0}$ we have $\nabla f(\mathbf{0}) = \mathbf{0}$. Here the level surface $f = f(\mathbf{0}) = 0$ degenerates to the single point $\{\mathbf{0}\}$, which has no 2-dimensional tangent space at $\mathbf{0}$, and the phrase 'perpendicular to the tangent space' in Proposition 15.7(iii) has no meaning.

Conclusion: when $f \in C^1$ and $\nabla f(\mathsf{x}_0) = \mathbf{0}$, the level set need not be a regular surface at x_0; the statement of Proposition 15.7(iii) must carry the additional condition $\nabla f(\mathsf{x}_0) \neq \mathbf{0}$. This is not a technical detail: the implicit function theorem guarantees that the level set is locally a C^1 graph **if and only if** the gradient is non-zero, which is precisely the first-order version of the same condition needed for the existence of potential functions in Theorem 15.9.

Proposition 15.8 (The physical meaning of the divergence and the curl). Let $\mathsf{F} \in C^1(\mathcal{M}, \mathbb{R}^3)$ represent the velocity field of a steady fluid. Then

(i) When $\nabla \cdot \mathsf{F}(\mathsf{x}_0) > 0$, the point x_0 is a **source** of the flow (the net outward flux per unit volume is positive and the volume element expands); when $\nabla \cdot \mathsf{F}(\mathsf{x}_0) < 0$ it is a **sink**; when $\nabla \cdot \mathsf{F}(\mathsf{x}_0) = 0$ the point is called **solenoidal**.

(ii) The direction of $\nabla \times \mathsf{F}(\mathsf{x}_0)$ is the **direction of the axis of rotation** of the fluid element at x_0 (right-hand rule: curl the four fingers in the direction of rotation and the thumb points in the direction of the curl), and its modulus equals **twice the local angular velocity** of that element.

Assumption: $\mathsf{F} \in C^1$. **This proposition is a physical reading of Proposition 15.5 and Theorem 15.1, not an independent new mathematical result** — (i) comes from Theorem 15.1(ii) (the divergence is the instantaneous rate of change of volume per unit volume), and (ii) comes from Theorem 15.1(iii) (the antisymmetric part gives the local angular velocity). This proposition is retained in the main text in order to keep the use of terminology consistent with the later application chapters.

Proposition 15.9 (Linearity of the three operators)**.** Let $f, g \in C^1(\mathcal{M})$ be scalar fields, $\mathsf{F}, \mathsf{G} \in C^1(\mathcal{M}, \mathbb{R}^3)$ vector fields, and $\alpha, \beta \in \mathbb{R}$. Then

(i) $\nabla(\alpha f + \beta g) = \alpha \nabla f + \beta \nabla g$;

(ii) $\nabla \cdot (\alpha \mathsf{F} + \beta \mathsf{G}) = \alpha \nabla \cdot \mathsf{F} + \beta \nabla \cdot \mathsf{G}$;

(iii) $\nabla \times (\alpha \mathsf{F} + \beta \mathsf{G}) = \alpha \nabla \times \mathsf{F} + \beta \nabla \times \mathsf{G}$;

(iv) $\nabla^2(\alpha f + \beta g) = \alpha \nabla^2 f + \beta \nabla^2 g$ (requires $f, g \in C^2$).

Linearity is the fundamental property of the three operators as differential operators, and is also the basis of all subsequent identities and field-decomposition theorems.

Assumption: linearity requires only differentiability of the corresponding order; it does **not** require differentiability on the closure of $\mathcal{M}$, **nor** does it require $\mathcal{M}$ to be bounded or connected.

Proof. All three conclusions reduce to the linearity of the partial derivative operators: for any $u, v \in C^1(\mathcal{M})$ and $\alpha, \beta \in \mathbb{R}$,

$$\frac{\partial}{\partial x_i}(\alpha u + \beta v) = \alpha \frac{\partial u}{\partial x_i} + \beta \frac{\partial v}{\partial x_i},$$

which is a direct corollary of the linearity of the derivative in one variable (differentiation is a limit, and limits are linear).

(i) For $i = 1, 2, 3$, $(\nabla(\alpha f + \beta g))_i = \partial_i(\alpha f + \beta g) = \alpha \partial_i f + \beta \partial_i g = (\alpha \nabla f + \beta \nabla g)_i$; the three components are equal one by one, hence the vectors are equal.

(ii) By the definition and the term-by-term linearity above,

$$\nabla \cdot (\alpha \mathsf{F} + \beta \mathsf{G}) = \sum_{i=1}^{3} \partial_i(\alpha F_i + \beta G_i) = \alpha \sum_{i=1}^{3} \partial_i F_i + \beta \sum_{i=1}^{3} \partial_i G_i = \alpha \nabla \cdot \mathsf{F} + \beta \nabla \cdot \mathsf{G}.$$

(iii) Take the first component as an example (the remaining components follow by cyclic permutation of the indices, for the same reason as in Theorem 15.1(iii)):

$$(\nabla \times (\alpha \mathsf{F} + \beta \mathsf{G}))_1 = \partial_2(\alpha F_3 + \beta G_3) - \partial_3(\alpha F_2 + \beta G_2) = \alpha(\partial_2 F_3 - \partial_3 F_2) + \beta(\partial_2 G_3 - \partial_3 G_2),$$

that is, $\alpha(\nabla \times \mathsf{F})_1 + \beta(\nabla \times \mathsf{G})_1$.

(iv) Composing (ii) with (i): $\nabla^2(\alpha f + \beta g) = \nabla \cdot \nabla(\alpha f + \beta g) = \nabla \cdot (\alpha \nabla f + \beta \nabla g) = \alpha \nabla \cdot \nabla f + \beta \nabla \cdot \nabla g$, where the second step uses (i) and the third step uses (ii); here we need $\nabla f, \nabla g \in C^1$, that is, $f, g \in C^2$. ☐

Lemma 15.5 (Clairaut's mixed-partial-derivative interchange theorem)**.** Let $f \in C^2(\mathcal{M})$, that is, f has **continuous** second-order partial derivatives on the open set $\mathcal{M} \subseteq \mathbb{R}^n$. Then for any $i, j \in \{1, \dots, n\}$ and any $\mathsf{x} \in \mathcal{M}$,

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathsf{x}) = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathsf{x}).$$

Proof. Fix $\mathsf{x} \in \mathcal{M}$ and $i \neq j$ (if $i = j$ the two sides are identically the same). Since $\mathcal{M}$ is open, there is a $\delta > 0$ such that the coordinate box Q with centre x and side length 2δ satisfies $Q \subseteq \mathcal{M}$. For $|h|, |k| < \delta$ define the second-order difference quotient

$$\Delta(h,k) = f(\mathsf{x} + h\mathbf{e}_i + k\mathbf{e}_j) - f(\mathsf{x} + h\mathbf{e}_i) - f(\mathsf{x} + k\mathbf{e}_j) + f(\mathsf{x}).$$

First step (writing Δ as the difference of two first-order differences): let $g_h(t) = f(\mathsf{x} + h\mathbf{e}_i + t\mathbf{e}_j) - f(\mathsf{x} + t\mathbf{e}_j)$, $t \in (-\delta, \delta)$. Then $\Delta(h,k) = g_h(k) - g_h(0)$. By the one-variable mean value theorem, there exists $\theta \in (0,1)$ such that

$$\Delta(h,k) = g_h'(\theta k) \cdot k = k\Big[\partial_j f(\mathsf{x} + h\mathbf{e}_i + \theta k\mathbf{e}_j) - \partial_j f(\mathsf{x} + \theta k\mathbf{e}_j)\Big].$$

(Here $f \in C^1$ is used, so g_h is differentiable; ∂_j denotes the partial derivative along $\mathbf{e}_j$.) Applying the one-variable mean value theorem once more to the function in the square brackets above, along the direction $\mathbf{e}_i$ (with respect to the variable h), there exists $\eta \in (0,1)$ such that

$$\Delta(h,k) = hk\, \partial_i \partial_j f(\mathsf{x} + \eta h\mathbf{e}_i + \theta k\mathbf{e}_j).$$

Second step (performing the same decomposition in the other order): let $\tilde{g}_k(s) = f(\mathsf{x} + s\mathbf{e}_i + k\mathbf{e}_j) - f(\mathsf{x} + s\mathbf{e}_i)$; then likewise $\Delta(h,k) = \tilde{g}_k(h) - \tilde{g}_k(0)$, and repeating the two-stage mean value theorem of the first step gives the existence of $\eta', \theta' \in (0,1)$ such that

$$\Delta(h,k) = hk\, \partial_j \partial_i f(\mathsf{x} + \eta' h\mathbf{e}_i + \theta' k\mathbf{e}_j).$$

Third step (taking the limit): dividing the two equations by hk ($h, k \neq 0$) gives

$$\partial_i \partial_j f(\mathsf{x} + \eta h\mathbf{e}_i + \theta k\mathbf{e}_j) = \partial_j \partial_i f(\mathsf{x} + \eta' h\mathbf{e}_i + \theta' k\mathbf{e}_j).$$

Let $h, k \to 0$: both arguments tend to x. Since $\partial_i \partial_j f$ and $\partial_j \partial_i f$ are **continuous** at x (which is exactly the content of the C^2 assumption), the two sides tend respectively to $\partial_i \partial_j f(\mathsf{x})$ and $\partial_j \partial_i f(\mathsf{x})$. Hence the two are equal. ☐

Example 15.8 (Counterexample: the C^2 assumption cannot be omitted)**.** Define

$$f(x,y) = \begin{cases} \dfrac{xy(x^2 - y^2)}{x^2 + y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Then $f \in C^1(\mathbb{R}^2)$, and at $(0,0)$ the two mixed partial derivatives both exist but are **unequal:**

$$\frac{\partial^2 f}{\partial x \partial y}(0,0) = 1, \qquad \frac{\partial^2 f}{\partial y \partial x}(0,0) = -1.$$

(Key points of the computation: differentiating directly for $(x,y) \neq (0,0)$ and then letting $x \to 0$, $y \to 0$ gives $\partial_y f(x,0) = x$ and $\partial_x f(0,y) = -y$, so $\partial_x \partial_y f(0,0) = \lim_{x\to 0} \frac{\partial_y f(x,0) - \partial_y f(0,0)}{x} = 1$ differs from $\partial_y \partial_x f(0,0) = \lim_{y \to 0} \frac{\partial_x f(0,y) - \partial_x f(0,0)}{y} = -1$ by a sign.)

Conclusion: the assumption in Lemma 15.5 that 'the second-order partial derivatives are continuous' is essential and is not a technical condition that can be dropped. Wherever this chapter uses $\partial_i \partial_j = \partial_j \partial_i$ (Theorem 15.3, Theorem 15.4, Theorem 15.5), the differentiability order assumed (C^2) must be checked item by item; if C^2 is lowered to C^1, all three theorems may fail.

15.4 Product rules

Theorem 15.2 (Product rules for the gradient, divergence and curl)**.** Let $f, g \in C^1(\mathcal{M})$ be scalar fields and $\mathsf{F}, \mathsf{G} \in C^1(\mathcal{M}, \mathbb{R}^3)$ vector fields. Then the following six identities hold everywhere on $\mathcal{M}$:

(i) $\nabla(fg) = f\nabla g + g \nabla f$.

(ii) $\nabla \cdot (f\mathsf{F}) = f(\nabla \cdot \mathsf{F}) + \nabla f \cdot \mathsf{F}$.

(iii) $\nabla \times (f\mathsf{F}) = f(\nabla \times \mathsf{F}) + \nabla f \times \mathsf{F}$.

(iv) $\nabla \cdot (\mathsf{F} \times \mathsf{G}) = \mathsf{G} \cdot (\nabla \times \mathsf{F}) - \mathsf{F} \cdot (\nabla \times \mathsf{G})$.

(v) $\nabla \times (\mathsf{F} \times \mathsf{G}) = \mathsf{F}(\nabla \cdot \mathsf{G}) - \mathsf{G}(\nabla \cdot \mathsf{F}) + (\mathsf{G} \cdot \nabla)\mathsf{F} - (\mathsf{F} \cdot \nabla)\mathsf{G}$.

(vi) $\nabla(\mathsf{F} \cdot \mathsf{G}) = \mathsf{F} \times (\nabla \times \mathsf{G}) + \mathsf{G} \times (\nabla \times \mathsf{F}) + (\mathsf{F} \cdot \nabla)\mathsf{G} + (\mathsf{G} \cdot \nabla)\mathsf{F}$.

Here $(\mathsf{G}\cdot\nabla)\mathsf{F}$ means that the operator $\mathsf{G}\cdot\nabla = G_1\partial_x + G_2\partial_y + G_3\partial_z$ acts componentwise on F, that is, $((\mathsf{G}\cdot\nabla)\mathsf{F})_i = \sum_j G_j\, \partial_j F_i$.

Assumption: the minimum differentiability required by the six identities is **the same**, namely C^1. (i)–(iii) require only $f, g \in C^1$ and $\mathsf{F}, \mathsf{G} \in C^1$ (because each term involves only first derivatives). For (iv)–(vi), checking term by term likewise requires only first derivatives, so C^1 already suffices; but $(\mathsf{G}\cdot\nabla)\mathsf{F}$, which appears in (v) and (vi), requires F to be differentiable, and $(\mathsf{F}\cdot\nabla)\mathsf{G}$ requires G to be differentiable, both of which lie within the C^1 assumption.

Proof. Throughout we use index notation and the summation convention: a repeated subscript is summed automatically, $i, j, k \in \{1, 2, 3\}$, $\partial_i = \partial/\partial x_i$, and ϵ_{ijk} is the Levi-Civita symbol ($\epsilon_{123} = 1$, changing sign under the exchange of any two indices). In index notation the defining formulae of the three operators are

$$(\nabla f)_i = \partial_i f, \qquad \nabla \cdot \mathsf{F} = \partial_i F_i, \qquad (\nabla \times \mathsf{F})_i = \epsilon_{ijk}\, \partial_j F_k.$$

The two constantly used ϵ-contraction identities are

$$\epsilon_{ijk}\epsilon_{ilm} = \delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl}, \qquad \epsilon_{ijk}\epsilon_{ijl} = 2\delta_{kl}.$$

(i) $(\nabla(fg))_i = \partial_i(fg) = f\,\partial_i g + g\,\partial_i f$, the second equality following from the one-variable Leibniz rule. This holds for $i = 1, 2, 3$ one by one, hence the vectors are equal.

(ii) $\nabla\cdot(f\mathsf{F}) = \partial_i(fF_i) = f\,\partial_i F_i + (\partial_i f)F_i = f\,\nabla\cdot\mathsf{F} + \nabla f\cdot\mathsf{F}$.

(iii) We compute componentwise, retaining all intermediate steps:

$$(\nabla\times(f\mathsf{F}))_i = \epsilon_{ijk}\partial_j(fF_k) = \epsilon_{ijk}\big(f\,\partial_j F_k + (\partial_j f)F_k\big) = f\,\epsilon_{ijk}\partial_j F_k + \epsilon_{ijk}(\partial_j f)F_k.$$

The first term is $f(\nabla\times\mathsf{F})_i$. For the second term use the identity $\epsilon_{ijk}A_jB_k = (\mathsf{A}\times\mathsf{B})_i$ with $A_j = \partial_j f$ (that is, $\mathsf{A} = \nabla f$) and $B_k = F_k$ (that is, $\mathsf{B} = \mathsf{F}$), obtaining $\epsilon_{ijk}(\partial_j f)F_k = (\nabla f\times\mathsf{F})_i$. Hence (iii) holds. Note that here $\epsilon_{ijk}(\partial_j f)F_k$ cannot be written in any form other than $-(\mathsf{F}\times\nabla f)_i$; the anticommutativity of the cross product is the subject of Proposition 15.10.

(iv) First write out the components of the cross product: $(\mathsf{F}\times\mathsf{G})_i = \epsilon_{ijk}F_jG_k$. Then

$$\nabla\cdot(\mathsf{F}\times\mathsf{G}) = \partial_i\big(\epsilon_{ijk}F_jG_k\big) = \epsilon_{ijk}\big((\partial_i F_j)G_k + F_j(\partial_i G_k)\big) = \underbrace{\epsilon_{ijk}(\partial_i F_j)G_k}_{P} + \underbrace{\epsilon_{ijk}F_j(\partial_i G_k)}_{Q}.$$

Handling P: permute the indices cyclically, $(i,j,k)\to(j,k,i)$; ϵ is invariant under such cyclic permutations, so

$$P = \epsilon_{jki}(\partial_j F_k)G_i = G_i\,\epsilon_{ijk}\partial_j F_k = \mathsf{G}\cdot(\nabla\times\mathsf{F}).$$

Handling Q: permute the indices by $(i,j,k)\to(k,i,j)$ (which likewise leaves ϵ unchanged), obtaining

$$Q = \epsilon_{kij}F_i(\partial_k G_j) = F_i\,\epsilon_{ijk}\partial_j G_k = \mathsf{F}\cdot(\nabla\times\mathsf{G}),$$

but does Q carry a positive sign in the original expression? The sign must be checked. From $Q = \epsilon_{ijk}F_j\partial_i G_k$, perform the permutation $(i,j,k)\to(k,i,j)$: $\epsilon_{kij} = \epsilon_{ijk}$ (an even permutation), and $F_j\partial_i G_k \to F_i\partial_k G_j$. Hence $Q = \epsilon_{kij}F_i\partial_k G_j = F_i\,\epsilon_{kij}\partial_k G_j$. Then, from $\epsilon_{kij} = \epsilon_{ijk}$ and $\epsilon_{ijk}\partial_k G_j = -\epsilon_{ikj}\partial_k G_j = -(\nabla\times\mathsf{G})_i$, we get $Q = -F_i(\nabla\times\mathsf{G})_i = -\mathsf{F}\cdot(\nabla\times\mathsf{G})$. Hence $\nabla\cdot(\mathsf{F}\times\mathsf{G}) = P + Q = \mathsf{G}\cdot(\nabla\times\mathsf{F}) - \mathsf{F}\cdot(\nabla\times\mathsf{G})$.

(v) Compute componentwise, writing out every term in full. Denote the required quantity by $R_i = (\nabla\times(\mathsf{F}\times\mathsf{G}))_i$. First compute the components of the cross product, $(\mathsf{F}\times\mathsf{G})_k = \epsilon_{klm}F_lG_m$; then

$$R_i = \epsilon_{ijk}\partial_j\big(\epsilon_{klm}F_lG_m\big) = \epsilon_{ijk}\epsilon_{klm}\big((\partial_j F_l)G_m + F_l(\partial_j G_m)\big).$$

Using $\epsilon_{ijk}\epsilon_{klm} = \epsilon_{kij}\epsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$ (the first equality uses the invariance of ϵ under cyclic permutation of its indices), we obtain

$$R_i = (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl})\big((\partial_j F_l)G_m + F_l(\partial_j G_m)\big).$$

Expanding the four terms by the sifting action of δ:

$$R_i = \underbrace{(\partial_m F_i)G_m}_{R^{(1)}} - \underbrace{(\partial_l F_l)G_i}_{R^{(2)}} + \underbrace{F_i(\partial_m G_m)}_{R^{(3)}} - \underbrace{F_l(\partial_l G_i)}_{R^{(4)}}.$$

Identifying the terms one by one: $R^{(1)} = G_m\partial_m F_i = ((\mathsf{G}\cdot\nabla)\mathsf{F})_i$; $R^{(2)} = (\nabla\cdot\mathsf{F})G_i$; $R^{(3)} = (\nabla\cdot\mathsf{G})F_i$; $R^{(4)} = ((\mathsf{F}\cdot\nabla)\mathsf{G})_i$. Substituting and rearranging the order gives (v).

(vi) Compute componentwise: $(\nabla(\mathsf{F}\cdot\mathsf{G}))_i = \partial_i(F_jG_j) = (\partial_iF_j)G_j + F_j(\partial_iG_j)$. Here it is safer to switch to direct verification: for any vector A and indices i, j, the ϵ-identity $\epsilon_{ijk}\epsilon_{klm}$ yields the component decomposition of $\mathsf{A}\times(\nabla\times\mathsf{B})$, the component form of the curl being

$$(\nabla\times\mathsf{A})_i = \epsilon_{ijk}\partial_jA_k.$$

Consider $(\mathsf{A}\times(\nabla\times\mathsf{B}))_i = \epsilon_{ijk}A_j(\nabla\times\mathsf{B})_k = \epsilon_{ijk}A_j\epsilon_{klm}\partial_lB_m = (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl})A_j\partial_lB_m = A_m\partial_iB_m - A_l\partial_lB_i$. Taking $\mathsf{A} = \mathsf{F}$ and $\mathsf{B} = \mathsf{G}$: $(\mathsf{F}\times(\nabla\times\mathsf{G}))_i = F_m\partial_iG_m - (\mathsf{F}\cdot\nabla)G_i$. Similarly $(\mathsf{G}\times(\nabla\times\mathsf{F}))_i = G_m\partial_iF_m - (\mathsf{G}\cdot\nabla)F_i$. Adding the two equations:

$$(\mathsf{F}\times(\nabla\times\mathsf{G}))_i + (\mathsf{G}\times(\nabla\times\mathsf{F}))_i = \left[F_m\partial_iG_m + G_m\partial_iF_m\right] - (\mathsf{F}\cdot\nabla)G_i - (\mathsf{G}\cdot\nabla)F_i.$$

The bracketed expression on the left is $\partial_i(F_mG_m) = \partial_i(\mathsf{F}\cdot\mathsf{G})$. Transposing terms then gives

$$\partial_i(\mathsf{F}\cdot\mathsf{G}) = (\mathsf{F}\times(\nabla\times\mathsf{G}))_i + (\mathsf{G}\times(\nabla\times\mathsf{F}))_i + ((\mathsf{F}\cdot\nabla)\mathsf{G})_i + ((\mathsf{G}\cdot\nabla)\mathsf{F})_i,$$

that is, (vi). □

Proposition 15.10 (Anticommutativity of the formal operator)**.** Let $f \in C^1(\mathcal{M})$ and $\mathsf{F} \in C^1(\mathcal{M}, \mathbb{R}^3)$. Then

$$\nabla f\times\mathsf{F} = -\mathsf{F}\times\nabla f.$$

More generally, when ∇ is written as a formal vector, '$\nabla\times$' and '$\times\nabla$' differ by a sign:

$$\nabla\times\mathsf{F} = -\mathsf{F}\times\nabla.$$

Assumption and explanation: the cross product is anticommutative on $\mathbb{R}^3$, so '$\mathsf{F}\times\nabla$' in the right-hand expression is understood as componentwise action followed by a sign change; the purpose of this proposition is to serve as a reminder that the determinant notation of Definition 15.10 places ∇ in the first row merely as a formal way of writing, and one **cannot** use it to interchange the positions of ∇ and F without changing the sign.

Proof. From the anticommutativity of the cross product $A_jB_k\epsilon_{ijk} = -A_kB_j\epsilon_{ijk}$, taking $A_j = \partial_jf$ and $B_k = F_k$ directly gives the first formula. The second formula is a restatement of the first under the understanding that '∇ acts componentwise on each component of F': for each i, $(\nabla\times\mathsf{F})_i = \epsilon_{ijk}\partial_jF_k$ whilst $-(\mathsf{F}\times\nabla)_i = -\epsilon_{ijk}F_j\partial_k$, and the two are equal by the anticommutativity above. □

15.5 Fundamental identities

The following three identities are the most important structural theorems of vector analysis: they show that a gradient field is naturally irrotational and a curl field naturally solenoidal, and that the Laplacian can be decomposed into the sum of an irrotational part and a solenoidal part. The differentiability assumption of all three theorems is C^2 (and not C^1), and its necessity is given by Example 15.8.

Theorem 15.3 (A gradient field has zero curl)**.** *Let $f \in C^2(\mathcal{M})$ be a scalar field. Then the curl of its gradient field vanishes identically on $\mathcal{M}$:*

$$\nabla\times(\nabla f) = \mathsf{0}.$$

Equivalent formulation: *every C^2 gradient field is necessarily irrotational.*

Proof. By Definition 15.10 and Definition 15.8, $(\nabla f)_k = \partial_k f$, so

$$(\nabla \times (\nabla f))_i = \epsilon_{ijk}\partial_j\big((\nabla f)_k\big) = \epsilon_{ijk}\partial_j\partial_k f.$$

Fix i and split the expression into the two halves $j < k$ and $j > k$. For the terms with $j < k$, $\epsilon_{ijk}\partial_j\partial_k f$, and the terms with $j > k$: exchanging the index pair (j, k) in the latter gives $\epsilon_{ijk} \to \epsilon_{ikj} = -\epsilon_{ijk}$, whilst $\partial_j\partial_k f \to \partial_k\partial_j f$, and by Lemma 15.5 (here $f \in C^2$ is used) $\partial_k\partial_j f = \partial_j\partial_k f$. Hence the latter are the sum of $-\epsilon_{ijk}\partial_j\partial_k f$, which added to the former gives zero. Thus $(\nabla \times (\nabla f))_i = 0$. This holds for $i = 1, 2, 3$, so $\nabla \times (\nabla f) = \mathbf{0}$.

(Failure when C^2 is not used) If only $f \in C^1$ is assumed, then $\partial_j\partial_k f$ need not exist and the expression above is itself undefined; if $\partial_j\partial_k f$ exists but is not continuous, then the order of differentiation cannot be exchanged at the first exchange of indices (Example 15.8). ☐

Theorem 15.4 (A curl field has zero divergence). *Let $\mathsf{F} = (F_1, F_2, F_3) \in C^2(\mathcal{M}, \mathbb{R}^3)$ be a vector field. Then the divergence of its curl field vanishes identically on $\mathcal{M}$:*

$$\nabla \cdot (\nabla \times \mathsf{F}) = 0.$$

Equivalent formulation: every C^2 curl field is necessarily solenoidal.

Proof. By Definition 15.10, $(\nabla \times \mathsf{F})_i = \epsilon_{ijk}\partial_j F_k$, so

$$\nabla \cdot (\nabla \times \mathsf{F}) = \partial_i(\nabla \times \mathsf{F})_i = \epsilon_{ijk}\partial_i\partial_j F_k.$$

Split the expression into two halves according to the index pair (i, j): for each pair $i < j$, the term $\epsilon_{ijk}\partial_i\partial_j F_k$ and the term $\epsilon_{jik}\partial_j\partial_i F_k$ satisfy $\epsilon_{jik} = -\epsilon_{ijk}$, whilst by Lemma 15.5 ($\mathsf{F} \in C^2$, hence each component $F_k \in C^2$) we have $\partial_j\partial_i F_k = \partial_i\partial_j F_k$, so the two terms are negatives of each other and sum to zero. Summing over all index pairs gives $\nabla \cdot (\nabla \times \mathsf{F}) = 0$.

(Equivalent expanded form) Writing out the six terms explicitly makes the term-by-term check easier:

$$\frac{\partial^2 F_3}{\partial x \partial y} - \frac{\partial^2 F_2}{\partial x \partial z} + \frac{\partial^2 F_1}{\partial y \partial z} - \frac{\partial^2 F_3}{\partial y \partial x} + \frac{\partial^2 F_2}{\partial z \partial x} - \frac{\partial^2 F_1}{\partial z \partial y},$$

in which the 1st and 4th terms, the 2nd and 5th terms, and the 3rd and 6th terms cancel in pairs by Lemma 15.5. ☐

Theorem 15.5 (The curl-divergence decomposition of the Laplacian). *Let $\mathsf{F} \in C^2(\mathcal{M}, \mathbb{R}^3)$ be a vector field. Then on $\mathcal{M}$*

$$\nabla^2 \mathsf{F} = \nabla(\nabla \cdot \mathsf{F}) - \nabla \times (\nabla \times \mathsf{F}).$$

This decomposition expresses the Laplacian of a vector field as the sum of an **irrotational part** *$\nabla(\nabla \cdot \mathsf{F})$ and a* **solenoidal part** *$-\nabla \times (\nabla \times \mathsf{F})$ (the irrotationality and the solenoidal property of the two parts are guaranteed by Theorem 15.3 and Theorem 15.4 respectively).*

Assumption: *$\mathsf{F} \in C^2$ is the minimum requirement — the right-hand side involves mixed second-order partial derivatives and the left-hand side pure second-order partial derivatives.*

Proof. We prove the result componentwise, taking $i = 1$ as an example ($i = 2, 3$ follow by cyclic permutation of the indices, for the same reason as in Theorem 15.1(iii)).

First component of the left-hand side: $(\nabla^2 \mathsf{F})_1 = \partial_1^2 F_1 + \partial_2^2 F_1 + \partial_3^2 F_1$.

The 1-component of the first term on the right-hand side:

$$(\nabla(\nabla \cdot \mathsf{F}))_1 = \partial_1(\partial_1 F_1 + \partial_2 F_2 + \partial_3 F_3) = \partial_1^2 F_1 + \partial_1\partial_2 F_2 + \partial_1\partial_3 F_3.$$

The 1-component of the second term on the right-hand side (with the minus sign). First compute $\nabla \times \mathsf{F}$ and then its curl:

$$(\nabla \times (\nabla \times \mathsf{F}))_1 = \partial_2(\nabla \times \mathsf{F})_3 - \partial_3(\nabla \times \mathsf{F})_2 = \partial_2(\partial_1 F_2 - \partial_2 F_1) - \partial_3(\partial_3 F_1 - \partial_1 F_3),$$

that is,

$$(\nabla \times (\nabla \times \mathsf{F}))_1 = \partial_2\partial_1 F_2 - \partial_2^2 F_1 - \partial_3^2 F_1 + \partial_3\partial_1 F_3.$$

Subtracting the two expressions:

$$\begin{aligned} &\big[\partial_1^2 F_1 + \partial_1\partial_2 F_2 + \partial_1\partial_3 F_3\big] - \big[\partial_2\partial_1 F_2 - \partial_2^2 F_1 - \partial_3^2 F_1 + \partial_3\partial_1 F_3\big] \\ &= \partial_1^2 F_1 + \partial_2^2 F_1 + \partial_3^2 F_1 + \big(\partial_1\partial_2 F_2 - \partial_2\partial_1 F_2\big) + \big(\partial_1\partial_3 F_3 - \partial_3\partial_1 F_3\big) = (\nabla^2 \mathsf{F})_1, \end{aligned}$$

in which the two parentheses vanish by Lemma 15.5 ($\mathsf{F} \in C^2$). □

Corollary 15.1 (The vector-analysis formulation of $d^2 = 0$). Theorem 15.3 and Theorem 15.4 together constitute the concrete manifestation of the exterior differential relation $d^2 = 0$ in three-dimensional vector analysis:

$$\nabla \times \nabla = \mathbf{0}, \qquad \nabla \cdot \nabla \times = 0.$$

In the language of differential forms, if Ω^k denotes the space of k-forms and d the exterior derivative, and if the gradient, curl and divergence are identified respectively with the action of d on 0-forms, 1-forms and 2-forms (Definition 12.3), then the two identities combine into the single statement $d \circ d = 0$.

Example 15.9 (Counterexample: the converse of Theorem 15.4 does not hold). Take $\mathsf{F}(x, y, z) = (x, y, z)$; then $\nabla \cdot \mathsf{F} = 3 \neq 0$, so the condition is not met. Instead take $\mathsf{F}(x, y, z) = \dfrac{(x, y, z)}{\|(x, y, z)\|^3}$ (defined on $\mathbb{R}^3 \setminus \{\mathbf{0}\}$); then $\nabla \cdot \mathsf{F} = 0$ holds everywhere on that region; but F is **not** the curl of any vector field on $\mathbb{R}^3 \setminus \{\mathbf{0}\}$. Reason: if $\mathsf{F} = \nabla \times \mathsf{A}$, then by Theorem 15.4 we have $\nabla \cdot \mathsf{F} = \nabla \cdot (\nabla \times \mathsf{A}) = 0$, so by Theorem 15.8 the flux of F through any closed surface that encloses the origin and lies in the domain must be zero; but take the unit sphere S^2 centred at the origin: on it $\|\mathsf{r}\| = 1$, so $\mathsf{F} = \mathsf{r}$ and $\mathsf{n} = \mathsf{r}$, whence $\oint_{S^2} \mathsf{F} \cdot \mathsf{n}\, dS = \oint_{S^2} 1\, dS = 4\pi \neq 0$, a contradiction.

Conclusion: $\nabla \cdot \mathsf{F} = 0$ is a **necessary** condition for ‘F is the curl field of something’, but not a sufficient one; sufficiency requires a topological condition on the domain (simple connectivity or a suitable homology condition), which is precisely why Theorem 15.10 must write ‘suitable decay’ together with a condition on the region among its assumptions.

15.6 Integral theorems

This section presents three integral theorems. **The order is deliberate**: first Green's formula in the plane is proved independently (Theorem 15.6), then it is used as a tool to prove the curl theorem (Theorem 15.7), and finally the divergence theorem is proved (Theorem 15.8). This differs from the common way of writing, which states Stokes first and then says that it implies Green's formula — on the latter approach, if Green's formula is in turn used as a tool for proving Stokes' theorem, that constitutes a circular argument.

Theorem 15.6 (Green's formula (independent proof)). Let $D \subset \mathbb{R}^2$ be a bounded closed region whose boundary ∂D is a piecewise smooth simple closed curve, taken with positive orientation (that is, when walking anticlockwise along ∂D the region D lies on the left-hand side). If $P, Q \in C^1(D^\circ)$ and are continuous on $\overline{D}$ (where D° is the interior of D), then

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\, dy = \oint_{\partial D} (P\, dx + Q\, dy).$$

Proof. **First step (splitting the proposition into two independent equalities).** It suffices to prove separately

$$\text{(G1)} \quad -\iint_D \frac{\partial P}{\partial y}\, dx\, dy = \oint_{\partial D} P\, dx, \qquad \text{(G2)} \quad \iint_D \frac{\partial Q}{\partial x}\, dx\, dy = \oint_{\partial D} Q\, dy.$$

Adding the two gives the full formula.

Second step (first proving (G1) for a 'y-simple region'). Call D a **y-simple region** if there exist continuous functions $\varphi_1 \le \varphi_2$ (on $[a, b]$) such that $D = \{(x, y) : a \le x \le b,\ \varphi_1(x) \le y \le \varphi_2(x)\}$, with φ_1, φ_2 piecewise C^1. In this case ∂D consists of four pieces: the lower boundary C_1 (along $y = \varphi_1(x)$ from $x = a$ to b), the vertical pieces at the right and left ends (if $\varphi_1(b) \ne \varphi_2(b)$ etc., these integrate to zero or cancel in pairs), and the upper boundary C_2 (along $y = \varphi_2(x)$ from $x = b$ to a).

Left-hand side: by Fubini's theorem and the fundamental theorem of calculus,

$$\iint_D \frac{\partial P}{\partial y}\, dx\, dy = \int_a^b \left(\int_{\varphi_1(x)}^{\varphi_2(x)} \frac{\partial P}{\partial y}(x, y)\, dy \right) dx = \int_a^b \Big[P(x, \varphi_2(x)) - P(x, \varphi_1(x)) \Big] dx.$$

Right-hand side: along the positive orientation of ∂D, the x component is non-zero only on C_1, C_2 (on the vertical pieces $dx = 0$), and

$$\int_{C_1} P\, dx = \int_a^b P(x, \varphi_1(x))\, dx, \qquad \int_{C_2} P\, dx = \int_b^a P(x, \varphi_2(x))\, dx = -\int_a^b P(x, \varphi_2(x))\, dx.$$

Hence $\oint_{\partial D} P\, dx = -\int_a^b [P(x, \varphi_2(x)) - P(x, \varphi_1(x))] dx$, which equals the negative of the left-hand side, that is, (G1).

Third step (proving (G2) for an 'x-simple region'). If $D = \{(x, y) : c \le y \le d,\ \psi_1(y) \le x \le \psi_2(y)\}$, then, by complete duality,

$$\iint_D \frac{\partial Q}{\partial x}\, dx\, dy = \int_c^d [Q(\psi_2(y), y) - Q(\psi_1(y), y)] dy = \oint_{\partial D} Q\, dy,$$

where the last step uses the fact that the positive orientation of ∂D takes y from c to d on the right boundary and from d to c on the left boundary.

Fourth step (general piecewise smooth boundary). Let ∂D be piecewise smooth. Cut D into finitely many subregions $D = \bigcup_s D_s$ by finitely many lines parallel to the coordinate axes, so that each D_s is simultaneously y-simple and x-simple (this is possible: a piecewise smooth simple closed curve meets finitely many lines in at most finitely many points, so a sufficiently fine subdivision exists; if ∂D has a segment in common with one of the lines, first perturb it slightly and then pass to the limit, since P, Q are continuous on $\overline{D}$ and the integrals on both sides depend continuously on the perturbation).

Apply the second and third steps to each D_s and sum:

$$\sum_s \iint_{D_s} \Big(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\Big) dx\, dy = \sum_s \oint_{\partial D_s} (P\, dx + Q\, dy).$$

The sum on the left equals $\iint_D (\cdots)$ (the subregions have pairwise disjoint interiors, so the integrals are additive). On the right: a common edge of two adjacent subregions is traversed in opposite directions in the two integrals, and since $P\, dx + Q\, dy$ is the line integral of a 1-form, integrals along opposite directions are exactly negatives of each other, so the contributions on common edges **cancel in pairs**; only the parts lying on ∂D remain. By the way D was decomposed, these remaining segments join end to end in the positive orientation of ∂D, so the sum on the right equals $\oint_{\partial D}(P\, dx + Q\, dy)$. The proposition is proved. ☐

Theorem 15.7 (The curl theorem (Stokes' theorem)). *Let $S \subset \mathbb{R}^3$ be a piecewise smooth oriented surface whose boundary ∂S is a piecewise smooth simple closed curve. Take the unit normal vector* n *on S (piecewise continuous), and let the orientation of ∂S and* n *satisfy the right-hand rule. If the vector field $\mathsf{F} \in C^1(\overline{S}, \mathbb{R}^3)$, then*

$$\int_S (\nabla \times \mathsf{F}) \cdot \mathsf{n}\, dS = \oint_{\partial S} \mathsf{F} \cdot d\mathsf{r}.$$

The assumptions explained one by one: the piecewise smoothness of S guarantees that dS and n are defined except on finitely many curves, and that the integrals over those curves do not change the result (the values of an integrable function on a null set do not affect the integral); the piecewise smoothness of ∂S and the fact that it is a **simple** closed curve guarantee that the orientation of $\oint_{\partial S}$ is uniquely determined; the right-hand rule convention for n and ∂S guarantees that the signs on the two sides agree, and if the orientation is reversed the right-hand side changes sign.

Proof. **First step (reduction to a graph surface $z = h(x, y)$).** Let S be representable as

$$S = \{(\mathsf{x}, h(\mathsf{x})) : \mathsf{x} = (x, y) \in D\}, \qquad h \in C^2(D),$$

where $D \subset \mathbb{R}^2$ is a bounded closed region with ∂D piecewise smooth. In this case the graph of S gives a natural orientation: take

$$\mathsf{n}\, dS = \big(-\partial_x h,\ -\partial_y h,\ 1\big)\, dx\, dy$$

(by the vector form of the surface area element $\mathsf{n}\, dS = \mathsf{r}_x \times \mathsf{r}_y\, dx\, dy$, where $\mathsf{r}(x, y) = (x, y, h(x, y))$).

Second step (reducing the curl flux to a double integral over D). Write $\mathsf{F} = (F_1, F_2, F_3)$ and write out the three components of $\nabla \times \mathsf{F}$ according to Definition 15.10; then

$$(\nabla \times \mathsf{F}) \cdot (\mathsf{r}_x \times \mathsf{r}_y) = \big(\partial_y F_3 - \partial_z F_2\big)(-\partial_x h) + \big(\partial_z F_1 - \partial_x F_3\big)(-\partial_y h) + \big(\partial_x F_2 - \partial_y F_1\big).$$

On S we have $z = h(x, y)$, so by the chain rule the partial derivatives along S and the partial derivatives on D are related by the following formulae: for any component F_i,

$$\frac{\partial}{\partial x}\big[F_i(\mathsf{x}, h(\mathsf{x}))\big] = \partial_x F_i + \partial_z F_i\, \partial_x h, \qquad \frac{\partial}{\partial y}\big[F_i(\mathsf{x}, h(\mathsf{x}))\big] = \partial_y F_i + \partial_z F_i\, \partial_y h.$$

Applying this for $i = 1, 2$ eliminates ∂_z from the first two terms of the above expression:

$$-\big(\partial_y F_3 - \partial_z F_2\big)\partial_x h - \big(\partial_z F_1 - \partial_x F_3\big)\partial_y h = -\partial_y F_3\, \partial_x h - \partial_x F_3 \partial_y h + \partial_z F_2 \partial_x h + \partial_z F_1 \partial_y h.$$

On the other hand, $\partial_x[F_2(\mathsf{x}, h)] - \partial_y[F_1(\mathsf{x}, h)] = \partial_x F_2 + \partial_z F_2 \partial_x h - \partial_y F_1 - \partial_z F_1 \partial_y h$. Adding the two expressions, the terms containing $\partial_z F_i \partial_j h$ cancel in pairs, giving

$$(\nabla \times \mathsf{F}) \cdot (\mathsf{r}_x \times \mathsf{r}_y) = \partial_x\big[F_2(\mathsf{x}, h)\big] - \partial_y\big[F_1(\mathsf{x}, h)\big] =: \partial_x Q(\mathsf{x}) - \partial_y P(\mathsf{x}),$$

where $P(\mathsf{x}) = F_1(\mathsf{x}, h(\mathsf{x}))$ and $Q(\mathsf{x}) = F_2(\mathsf{x}, h(\mathsf{x}))$ are both C^1 on D. Hence

$$\int_S (\nabla \times \mathsf{F}) \cdot \mathsf{n}\, dS = \iint_D \big(\partial_x Q - \partial_y P\big) dx\, dy.$$

Third step (applying Green's formula to D). By Theorem 15.6, the right-hand side of the above equals $\oint_{\partial D} (P\, dx + Q\, dy)$.

Fourth step (restoring the boundary integral to a line integral on ∂S). The parametrisation of S is $\mathsf{r}(x, y) = (x, y, h(x, y))$, so along ∂S we have

$$d\mathsf{r} = \big(dx,\ dy,\ \partial_x h\, dx + \partial_y h\, dy\big),$$

whence $\mathsf{F} \cdot d\mathsf{r} = F_1 dx + F_2 dy + F_3(\partial_x h\, dx + \partial_y h\, dy)$. On ∂S we have $F_3 = F_3(\mathsf{x}, h(\mathsf{x}))$, and the remaining term $F_3(\partial_x h\, dx + \partial_y h\, dy)$ is the line integral of an exact differential restricted to a closed curve: from

$$d\big[F_3(\mathsf{x}, h(\mathsf{x}))\big] = \partial_x\big[F_3(\mathsf{x}, h)\big] dx + \partial_y\big[F_3(\mathsf{x}, h)\big] dy$$

together with $\oint_{\partial D} d(\cdot) = 0$ (the integral of an exact differential along a closed curve is zero), this remaining term is zero. Hence

$$\oint_{\partial S} \mathsf{F} \cdot d\mathsf{r} = \oint_{\partial D} (P\, dx + Q\, dy),$$

which agrees with the result of the third step, and the graph-surface case is proved.

Fifth step (general piecewise smooth surface). Let S be piecewise smooth. Cut S along its edges (where n is discontinuous) into finitely many pieces $S = \bigcup_s S_s$. If S_s is not the graph of some $z = h(x, y)$, apply an orthogonal permutation of the coordinate axes (at most six choices) to make it a graph in some direction (this is a direct corollary of the fact that 'a smooth surface is locally a graph'; since S_s is compact with continuous normal, there is a finite cover such that each piece lies in such a graph, and one then subdivides further).

Apply the second to fourth steps to each piece and sum. The left-hand side equals $\int_S (\nabla \times \mathsf{F}) \cdot \mathsf{n}\, dS$ because the integrals are additive. On the right: the orientations inherited by two adjacent pieces along their common edge are **exactly opposite** (because they come from the two sides of one and the same surface, and the boundary orientation given by the right-hand rule points in opposite directions on the common edge), so the line integrals along common edges cancel in pairs, leaving only the boundary segments that are not shared; these segments join end to end according to the orientation of ∂S and constitute $\oint_{\partial S} \mathsf{F} \cdot d\mathsf{r}$. The proposition is proved. □

Corollary 15.2 (Green's formula)**.** Let $D \subset \mathbb{R}^2$ be a bounded closed region whose boundary ∂D is a piecewise smooth simple closed curve, taken with positive orientation (anticlockwise). If $P, Q \in C^1(\overline{D})$, then

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx\, dy = \oint_{\partial D} (P\, dx + Q\, dy).$$

This corollary is a restatement of Theorem 15.6 (it may also be viewed as the special case of Theorem 15.7 in the plane $z = 0$ with $S = D$ and $\mathsf{F} = (P, Q, 0)$: here $(\nabla \times \mathsf{F}) \cdot \mathsf{n} = \partial_x Q - \partial_y P$).

Theorem 15.8 (The divergence theorem (Gauss' theorem))**.** Let $V \subset \mathbb{R}^3$ be a regular bounded closed region (Definition 15.3) whose boundary ∂V is a piecewise smooth closed surface, and take the unit outward normal n (pointing out of the region). If the vector field $\mathsf{F} \in C^1(\overline{V}, \mathbb{R}^3)$, then

$$\int_V (\nabla \cdot \mathsf{F})\, dV = \oint_{\partial V} \mathsf{F} \cdot \mathsf{n}\, dS.$$

The assumptions explained one by one: 'regular bounded closed region' guarantees that on ∂V the outward normal n is well defined almost everywhere and that dS is integrable (Example 15.3 shows that this condition cannot be omitted); F being C^1 on $\overline{V}$ guarantees that $\nabla \cdot \mathsf{F}$ is continuous on V and hence integrable; if ∂V is piecewise smooth but has edges, the two-sided ambiguity of n on the edges does not affect the surface integral (the edges have zero area).

Proof. **First step (the rectangular-box case, computed step by step).** Let $V = [a_1, b_1] \times [a_2, b_2] \times [a_3, b_3]$, with edges parallel to the coordinate axes. It suffices to prove for each component i that

$$\int_V \partial_i F_i\, dV = \oint_{\partial V} F_i\, n_i\, dS,$$

and then add over $i = 1, 2, 3$ (here we do **not** use 'similarly' — below $i = 1$ is written out in full, and the form for $i = 2, 3$ is obtained by permuting the subscripts $1, 2, 3$ cyclically; the definition of the rectangular box is unchanged in form under this permutation, so it is the same computation).

Take $i = 1$. By Fubini's theorem ($\partial_1 F_1$ is continuous on the box, so iterated integration is legitimate):

$$\int_V \partial_1 F_1\, dV = \int_{a_2}^{b_2}\int_{a_3}^{b_3}\left(\int_{a_1}^{b_1} \partial_1 F_1(x,y,z)\,dx\right)dy\,dz.$$

The inner integral, by the fundamental theorem of calculus (F_1 is continuously differentiable with respect to x):

$$\int_{a_1}^{b_1} \partial_1 F_1\,dx = F_1(b_1,y,z) - F_1(a_1,y,z).$$

Hence

$$\int_V \partial_1 F_1\, dV = \int_{a_2}^{b_2}\int_{a_3}^{b_3}\big[F_1(b_1,y,z) - F_1(a_1,y,z)\big]dy\,dz.$$

Now compute the right-hand side. Of the six faces of the box, only $x = b_1$ (outward normal $\mathsf{n} = +\mathsf{e}_1$, $n_1 = 1$) and $x = a_1$ ($\mathsf{n} = -\mathsf{e}_1$, $n_1 = -1$) give $n_1 \neq 0$; for the remaining four faces $n_1 = 0$ and the contribution is zero. Hence

$$\oint_{\partial V} F_1 n_1\,dS = \int_{a_2}^{b_2}\int_{a_3}^{b_3} F_1(b_1,y,z)\,dy\,dz - \int_{a_2}^{b_2}\int_{a_3}^{b_3} F_1(a_1,y,z)\,dy\,dz,$$

which agrees term by term with the expression above. Thus $i = 1$ is proved, $i = 2, 3$ follow by the same method, and adding gives the divergence theorem in the rectangular-box case.

Second step (a z-simple region). Call V a z-simple region if there exist C^1 functions $\varphi_1 \le \varphi_2$ (defined on $D \subset \mathbb{R}^2$) such that $V = \{(\mathsf{x}, z) : \mathsf{x} \in D,\ \varphi_1(\mathsf{x}) \le z \le \varphi_2(\mathsf{x})\}$. In this case ∂V consists of a top lid, a bottom lid and side walls, and on the side walls $\mathsf{n} \perp \mathsf{e}_3$, so $n_3 = 0$. Repeat the computation of the first step for the F_3 component (replacing x by z and the rectangle by D):

$$\int_V \partial_3 F_3\,dV = \iint_D \big[F_3(\mathsf{x}, \varphi_2(\mathsf{x})) - F_3(\mathsf{x}, \varphi_1(\mathsf{x}))\big]d\mathsf{x} = \oint_{\partial V} F_3 n_3\,dS.$$

For the F_1, F_2 components, the same computation is carried out after representing V as an x-simple and a y-simple region. Adding the three equations, the z-simple region case is proved.

Third step (general regular region). Let V be a regular bounded closed region in the sense of Definition 15.3. Cut V into finitely many subregions by finitely many planes parallel to the coordinate planes, so that each subregion is simultaneously x-, y- and z-simple (the same subdivision argument as in the fourth step of Theorem 15.6: the piecewise smoothness of ∂V guarantees that it meets finitely many planes in at most finitely many curves). Apply the second step to each subregion and sum:

$$\sum_s \int_{V_s} \nabla\cdot\mathsf{F}\,dV = \sum_s \oint_{\partial V_s} \mathsf{F}\cdot\mathsf{n}_s\,dS.$$

The left-hand side equals $\int_V \nabla\cdot\mathsf{F}\,dV$. On the right: a common face of two adjacent subregions is assigned **opposite** outward normals in the two integrals, so the contributions of $\mathsf{F}\cdot\mathsf{n}_s$ are negatives of each other and cancel in pairs; only the faces lying on ∂V remain, and the outward normal of those faces is exactly n. Hence the right-hand side equals $\oint_{\partial V} \mathsf{F}\cdot\mathsf{n}\,dS$. The proposition is proved. □

Remark 15.6 (The scope of validity of the three integral theorems and their mutual derivation). **(i) Logical relations.** Theorem 15.6 is an independent planar result; Theorem 15.7 is obtained from the graph-surface case together with summation over the pieces, and its proof uses Theorem 15.6 only in the third step; the proof of Theorem 15.8 does **not** depend on the first two. Hence there is no circular dependence among the three. Conversely, Corollary 15.2 is also a special case of Theorem 15.7, and the reader may choose either as a starting point, but **cannot** simultaneously use A as an assumption of B and B as an assumption of A.

(ii) The necessity of the boundary conditions. The regularity of V and the piecewise smoothness of ∂V cannot be omitted (Example 15.3); the piecewise smoothness of S and the simple-closed-curve property of ∂S cannot be omitted (Example 15.10); the continuity of F on the closed region cannot be omitted — if F is defined only at interior points and is unbounded near the boundary, the surface integral on the right-hand side may diverge.

(iii) The role of the orientation conventions. The right-hand rule convention for n and ∂S in Theorem 15.7, and the positive-orientation convention for ∂D in Theorem 15.6, are not notations that may be chosen arbitrarily: reversing one of the orientations changes the sign of the right-hand side of the equality. This point has physical consequences in the later application chapters (such as magnetic flux and induced electromotive force in electromagnetic systems).

Example 15.10 (Counterexample: a non-orientable surface makes the statement of the curl theorem fail). Take the Möbius strip S: the compact surface obtained from the rectangle $[0,1]\times(-1,1)$ by the identification $(0,t)\sim(1,-t)$. It can be given by an explicit parametrisation in $\mathbb{R}^3$, hence is piecewise smooth; its boundary ∂S is a simple closed curve (the same closed curve traced along the upper and lower sides of the rectangle, of twice the length of an ordinary circle, and still a simple closed curve).

But S is **non-orientable**: there is no globally continuous unit normal vector field $\mathsf{n}: S\to S^2$. Concretely, when one walks once around along the centre line of S and returns to the starting point, the initially chosen unit normal becomes its negative.

Hence the **left-hand side of Theorem 15.7 is undefined**: the integrand $(\nabla\times\mathsf{F})\cdot\mathsf{n}$ depends on a continuous choice of n, and no such choice exists. If S is cut open along a seam and the two sides are oriented separately, then the values of n on the two sides of the seam are opposite, and the two pieces after cutting cannot give one and the same surface integral at the join.

Conclusion: the assumption in the curl theorem that 'S is an **oriented** surface and n is piecewise continuous' cannot be omitted. It is not a technical detail but a precondition for the statement of the theorem to hold: in the present example the right-hand side (the circulation $\oint_{\partial S}\mathsf{F}\cdot d\mathsf{r}$) is still defined, so one cannot say that the conclusion of the theorem is false, only that **the statement of the theorem itself does not apply to** S. This belongs to the same class of failure mode as the case of the divergence theorem on a needle-shaped region in Example 15.3.

15.7 Classification of vector fields and potential functions

Definition 15.12 (Irrotational, solenoidal and harmonic fields)**.** *Let* $\mathsf{F} \in C^1(\mathcal{M}, \mathbb{R}^3)$*.*

- *If* $\nabla \times \mathsf{F} = \mathbf{0}$*, then* F *is called an* **irrotational field***.*
- *If* $\nabla \cdot \mathsf{F} = 0$*, then* F *is called a* **solenoidal field***.*
- *If both hold simultaneously, then* F *is called a* **harmonic field***.*

Theorem 15.9 (Existence of a potential function for an irrotational field)**.** *Let* $\mathcal{M}$ *be a simply connected region and let* $\mathsf{F} \in C^1$ *satisfy* $\nabla \times \mathsf{F} = \mathbf{0}$*. Then there exists a scalar potential* $\phi \in C^2$ *such that* $\mathsf{F} = \nabla\phi$*.*

Proof. Fix a base point x_0 and define

$$\phi(\mathsf{x}) = \int_\gamma \mathsf{F} \cdot d\mathsf{r}.$$

By Stokes' theorem, this is independent of the path. Verifying $\nabla\phi = \mathsf{F}$ gives the result. ☐

Theorem 15.10 (Existence of a vector potential for a solenoidal field)**.** *Let* $\mathsf{F} \in C^1$ *satisfy* $\nabla \cdot \mathsf{F} = 0$ *and decay suitably. Then there exists a vector potential* A *such that* $\mathsf{F} = \nabla \times \mathsf{A}$*.*

Theorem 15.11 (The Helmholtz decomposition theorem)**.** *Let* F *have continuous first-order partial derivatives on* $\mathbb{R}^3$ *and decay suitably. Then*

$$\mathsf{F} = -\nabla\phi + \nabla \times \mathsf{A},$$

where ϕ *and* A *are given by volume integrals.*

Proof. This is based on the Green's function of the Laplacian and the identity

$$\nabla^2\mathsf{F} = \nabla(\nabla \cdot \mathsf{F}) - \nabla \times (\nabla \times \mathsf{F}).$$

Write F as the sum of an irrotational part and a solenoidal part, and construct the scalar potential and the vector potential respectively. ☐

Example 15.11 (Counterexample: without the decay condition the Helmholtz decomposition fails)**.** *The assumptions of Theorem 15.11 include '*F *has continuous first-order partial derivatives on* $\mathbb{R}^3$ *and decays suitably'. Remove the decay condition and the conclusion may fail.*

Take $\mathsf{F}(\mathsf{x}) = \mathsf{x}$ *(which is irrotational but not solenoidal:* $\nabla \times \mathsf{F} = \mathbf{0}$*,* $\nabla \cdot \mathsf{F} = 3$*). Its Helmholtz decomposition is* $\mathsf{F} = \nabla\varphi + \nabla \times \mathsf{A}$*, where one takes* $\varphi(\mathsf{x}) = \|\mathsf{x}\|^2/2$ *and* $\mathsf{A} = \mathbf{0}$*. Here* φ *does not decay (*$|\varphi| \to \infty$*), yet the decomposition still exists — this shows that 'decay' is not a* **necessary** *condition for the existence of the decomposition.*

The case that genuinely requires decay is when **uniqueness** *and an* **explicit integral formula** *are demanded (such as the expression written with the kernel*

$1/(4\pi\|\mathsf{x}-\mathsf{y}\|)$): for example, $\mathsf{F}=(0,0,1)$ is a constant field and can be written as $\mathsf{F}=\nabla\varphi$ (with $\varphi=z$) or in other forms, and if non-decaying potentials are allowed then the irrotational part and the solenoidal part in the decomposition are not unique. Conversely, take $\mathsf{F}=\dfrac{\hat{\mathsf{r}}}{\|\mathsf{r}\|}$ (decaying only like $1/r$); then the two explicit integral formulae diverge logarithmically, and the decomposition does not exist in the L^2 sense, although it still holds formally in the pointwise sense.

Conclusion: *the role of 'suitable decay' in Theorem 15.11 is to guarantee* **the uniqueness of the decomposition and the convergence of the integral formulae,** *and not to guarantee the existence of the decomposition; when citing this theorem one must distinguish these two uses.*

15.8 Gradient, divergence and curl in orthogonal curvilinear coordinates

Definition 15.13 (Orthogonal curvilinear coordinate systems)**.** *Let (ξ^1,ξ^2,ξ^3) be curvilinear coordinates in $\mathbb{R}^3$, with position vector $\mathsf{r}=\mathsf{r}(\xi^1,\xi^2,\xi^3)$. If the tangent vectors of the three coordinate directions*

$$\mathsf{e}_i=\frac{\partial\mathsf{r}/\partial\xi^i}{\|\partial\mathsf{r}/\partial\xi^i\|}$$

are pairwise orthogonal, then the coordinate system is called an **orthogonal curvilinear coordinate system***. Write the* **scale factors**

$$h_i=\left\|\frac{\partial\mathsf{r}}{\partial\xi^i}\right\|.$$

Proposition 15.11 (The three operators in orthogonal curvilinear coordinates)**.** *In an orthogonal curvilinear coordinate system (ξ^1,ξ^2,ξ^3), let f be a scalar field and $\mathsf{F}=(F_1,F_2,F_3)$ a vector field (with components expanded in the unit basis e_i). Then*

$$\nabla f=\sum_{i=1}^{3}\frac{1}{h_i}\frac{\partial f}{\partial\xi^i}\mathsf{e}_i,$$

$$\nabla\cdot\mathsf{F}=\frac{1}{h_1h_2h_3}\sum_{i=1}^{3}\frac{\partial}{\partial\xi^i}\left(\frac{h_1h_2h_3}{h_i}F_i\right),$$

$$(\nabla\times\mathsf{F})_1=\frac{1}{h_2h_3}\left[\frac{\partial(h_3F_3)}{\partial\xi^2}-\frac{\partial(h_2F_2)}{\partial\xi^3}\right],$$

the other two components are obtained by cyclic permutation of the indices $(1,2,3)$.

Proof. The computation is direct, using the chain rule and the metric matrix $g_{ij}=h_i^2\delta_{ij}$. The gradient formula comes from the component representation $(\nabla f)_i=h_i^{-1}\partial f/\partial\xi^i$ of ∇f. The divergence formula comes from $\nabla\cdot\mathsf{F}=(h_1h_2h_3)^{-1}\sum_i\partial_i(h_1h_2h_3F_i/h_i)$. The curl formula comes from the expansion of the ϵ-symbol in an orthogonal basis. □

Example 15.12 (Spherical and cylindrical coordinates). Spherical coordinates (r, θ, φ): $h_r = 1$, $h_\theta = r$, $h_\varphi = r \sin\theta$.

$$\nabla f = \frac{\partial f}{\partial r}\mathbf{e}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\mathbf{e}_\theta + \frac{1}{r\sin\theta}\frac{\partial f}{\partial \varphi}\mathbf{e}_\varphi,$$

$$\nabla \cdot \mathsf{F} = \frac{1}{r^2}\frac{\partial (r^2 F_r)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial(\sin\theta F_\theta)}{\partial\theta} + \frac{1}{r\sin\theta}\frac{\partial F_\varphi}{\partial\varphi}.$$

Cylindrical coordinates (ρ, φ, z): $h_\rho = 1$, $h_\varphi = \rho$, $h_z = 1$.

$$\nabla f = \frac{\partial f}{\partial \rho}\mathbf{e}_\rho + \frac{1}{\rho}\frac{\partial f}{\partial \varphi}\mathbf{e}_\varphi + \frac{\partial f}{\partial z}\mathbf{e}_z,$$

$$\nabla \cdot \mathsf{F} = \frac{1}{\rho}\frac{\partial(\rho F_\rho)}{\partial\rho} + \frac{1}{\rho}\frac{\partial F_\varphi}{\partial\varphi} + \frac{\partial F_z}{\partial z}.$$

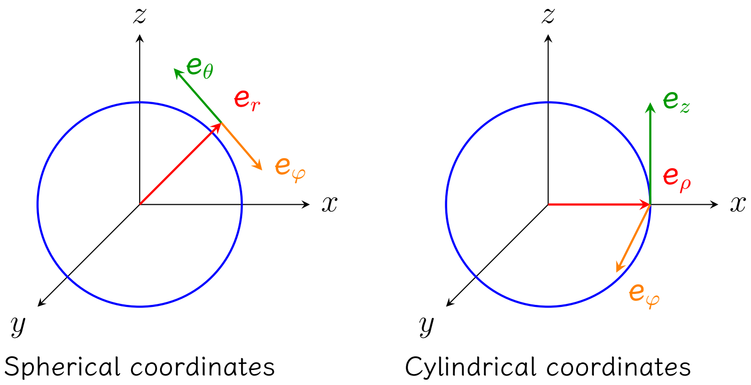
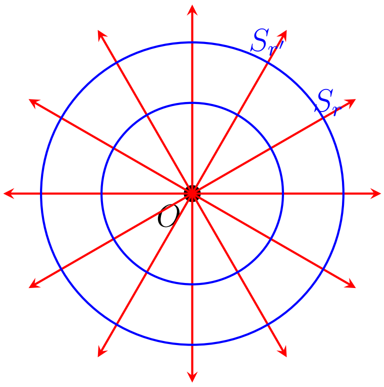


Figure 15.5: Unit basis vectors in spherical and cylindrical coordinates.

15.9 The inverse-square law

Remark 15.7 (The inverse-square law (statement in the chapter 'Scalars and Vectors')). The conclusion of this section is exactly the same as Theorem 3.4 of the 'Scalars and Vectors' chapter: if the flux of a central radial sourceless field through any closed surface enclosing the centre is the constant Φ_0, then $F(r) = \Phi_0/(4\pi r^2)$. Both the statement and the proof are to be found there, and the discussion in the remainder of this section cites it directly.

Remark 15.8 (The converse and the conservation of flux (statement in the chapter 'Scalars and Vectors')). The same as Corollary 3.2 of the 'Scalars and Vectors' chapter: the flux of a radial field decaying like C/r^2 through any sphere centred at the centre is constantly $4\pi C$ (the field is sourceless for $r > 0$, hence the same holds for any closed surface enclosing the centre); but 'conservation of flux' and 'inverse-square decay' are not unconditionally equivalent—they are two one-directional implications, whose exact scope is given in Remark 3.10 there. The proof is to be found there.



$\Phi_0 = F(r) \cdot 4\pi r^2 =$ constant

Figure 15.6: Geometric derivation of the inverse-square law. The radial vector field passes through concentric spheres enclosing the source O, the total flux is conserved, and hence the field strength is inversely proportional to the square of the distance.

Remark 15.9 (Newton's law of universal gravitation and Coulomb's law (examples in the chapter 'Scalars and Vectors')). Taking the flux constant to be $\Phi_0 = -4\pi GM$ gives Newton's law of universal gravitation $\mathsf{g}(\mathsf{r}) = -G\frac{M}{r^2}\frac{\mathsf{r}}{r}$; taking $\Phi_0 = Q/\varepsilon_0$ gives Coulomb's law $\mathsf{E}(\mathsf{r}) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\frac{\mathsf{r}}{r}$. The complete derivations of the two examples are given in the 'Scalars and Vectors' chapter (the two examples following Theorem 3.4); here only the conclusions are listed and they are not repeated.

Example 15.13 (Counterexample: the inverse-square law fails if the flux is not constant). The assumptions of Remark 15.7 include 'the total flux of F through any closed surface enclosing O is the constant Φ_0'. Remove this clause and the conclusion fails.

Take $\mathsf{F}(\mathsf{r}) = f(r)\,\hat{\mathsf{r}}$, where $f(r) = 1/r^2$ for $r \in (0,1]$ and $f(r) = 1/r^2 + 1$ for $r > 1$. Then for $r > 1$ the field is still spherically symmetric and sourceless outside O ($\nabla \cdot \mathsf{F} = 0$ holds on $\mathbb{R}^3 \setminus \{O\}$ except on the sphere $r = 1$), but the flux through a sphere of radius $r > 1$ is

$$\Phi(r) = 4\pi r^2\left(\frac{1}{r^2} + 1\right) = 4\pi(1 + r^2),$$

which is not constant. Hence $F(r)$ does not have the form $\Phi_0/(4\pi r^2)$.

Conclusion: the clause 'the flux is constant' in Remark 15.7 is not an automatic corollary of 'sourceless + spherically symmetric' — it is an **independent assumption**, whose physical meaning is 'the total strength of the source is conserved'. In the present example extra sources appear on the sphere $r = 1$ (as can be seen from $\nabla \cdot \mathsf{F} \neq 0$), destroying the conservation of flux.

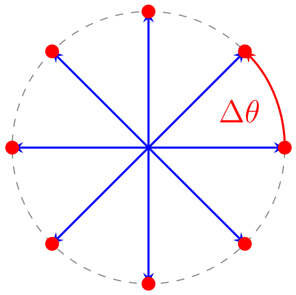
15.10 Number of directions and angular resolution: from discrete to continuous

Definition 15.14 (Number of directions and angular resolution). For any period $T \in \mathbb{N}^+$:

- (i) **Number of directions**: $N(T) = T$ — the number of independent directions that the period T is able to describe.
- (ii) **Angular resolution**: $\Delta\theta(T) = 2\pi/T$ — the angular spacing between adjacent directions.

The number of directions and the angular resolution satisfy the duality relation:

$$N(T) \cdot \Delta\theta(T) = 2\pi.$$



$N(T) = 8, \quad \Delta\theta = 2\pi/8$

Figure 15.7: Number of directions and angular resolution. The period T divides the circle into T equal directions, and the angular spacing between adjacent directions is $\Delta\theta = 2\pi/T$.

Definition 15.15 (Determining a coordinate point from the number of directions and the angular resolution)**.** Given a coordinate point $\mathbf{r} = (x, y) \neq \mathbf{0}$, let the argument be $\theta = \operatorname{arctan} 2(y, x)$. The phase index is

$$k(\mathbf{r}; T) = \lfloor \theta \cdot T/(2\pi) \rfloor \bmod T.$$

The discrete representation is

$$\mathbf{r}_T = \|\mathbf{r}\| \cdot e^{i2\pi k/T}.$$

Theorem 15.12 (The limiting passage from discrete to continuous)**.** Let $T_n \to \infty$. Then

$$\lim_{n\to\infty} \mathbf{r}_{T_n} = \mathbf{r}.$$

Proof. The discretisation error

$$|\theta - 2\pi k/T_n| \leq 2\pi/T_n \to 0.$$

Therefore $\mathbf{r}_{T_n} \to \mathbf{r}$. □

Definition 15.16 (Continuous number of directions and continuous angular resolution)**.** In the continuous limit,

$$N_\infty = \infty, \qquad \Delta\theta_\infty = 0,$$

satisfying the limiting duality relation

$$N_\infty \cdot \Delta\theta_\infty = 2\pi$$

which holds in a generalised sense.

15.11 Periodic vector fields and the commutativity of the three differential operators

This chapter has used the gradient, divergence and curl together with the three integral theorems to characterise the behaviour of vector fields under the three basic transformations of translation, scaling and rotation. This section gives the entry point of the concept of 'period' at the vector level: if a vector field is periodic in some translation direction, then the three differential operators preserve that periodicity; conversely, an operator preserving periodicity does not imply that the original field is periodic. This commutativity is the direct point of contact between this chapter and the 'periodicity' theme of the whole book.

Definition 15.17 (Periodic scalar fields and periodic vector fields)**.** Let $\mathsf{a} \in \mathbb{R}^3 \setminus \{0\}$. A scalar field $f : \mathbb{R}^3 \to \mathbb{R}$ is called **periodic** with respect to a if

$$f(\mathsf{x}+\mathsf{a}) = f(\mathsf{x}), \qquad \forall \mathsf{x} \in \mathbb{R}^3.$$

A vector field F is called periodic with respect to a if all three of its components are so. The set of all a for which the above equation holds forms an additive subgroup of $\mathbb{R}^3$, denoted $\mathrm{Group}_{\mathrm{period}}(f)$ (or $\mathrm{Group}_{\mathrm{period}}(\mathsf{F})$) and called the **group with periodic orbits** of the field.

Proposition 15.12 (The three differential operators commute with periodic translations)**.** *Let f be a C^1 scalar field and F a C^1 vector field, both periodic with respect to a. Then*

(i) *Gradient:* $(\nabla f)(\mathsf{x}+\mathsf{a}) = (\nabla f)(\mathsf{x})$;

(ii) *Divergence:* $(\nabla \cdot \mathsf{F})(\mathsf{x}+\mathsf{a}) = (\nabla \cdot \mathsf{F})(\mathsf{x})$;

(iii) *Curl:* $(\nabla \times \mathsf{F})(\mathsf{x}+\mathsf{a}) = (\nabla \times \mathsf{F})(\mathsf{x})$.

That is, all three differential operators **preserve invariance under periodic translations***.*

Proof. (i) Let e be a unit vector. By the directional derivative and the chain rule,

$$\mathsf{e} \cdot (\nabla f)(\mathsf{x}+\mathsf{a}) = \frac{d}{dt}\Big|_{t=0} f(\mathsf{x}+\mathsf{a}+t\mathsf{e}) = \frac{d}{dt}\Big|_{t=0} f(\mathsf{x}+t\mathsf{e}) = \mathsf{e} \cdot (\nabla f)(\mathsf{x}),$$

where the middle step uses $f(\mathsf{y}+\mathsf{a}) = f(\mathsf{y})$ (taking $\mathsf{y} = \mathsf{x}+t\mathsf{e}$). Since e is arbitrary, (i) follows.

(ii)(iii) The divergence and the curl are both linear combinations of the gradient components (see Definition 15.9 and Definition 15.10); applying the corresponding sums and differences to (i) componentwise gives the conclusion. □

Corollary 15.3 (The line integral along a periodic direction is zero)**.** *Let f be a C^1 scalar field periodic with respect to a. Then*

$$\int_0^1 \mathsf{a} \cdot \nabla f(\mathsf{x}+t\mathsf{a})\, dt = 0.$$

That is: **the gradient of a periodic scalar field has no net increase along a periodic direction***.*

Proof. By the chain rule, the integrand is exactly $\frac{d}{dt}f(\mathsf{x}+t\mathsf{a})$, so the integral equals $f(\mathsf{x}+\mathsf{a})-f(\mathsf{x})=0$. ☐

Remark 15.10 (Interface with the 'periodicity' theme). Proposition 15.12 and Corollary 15.3 give the complete interface of this chapter with the 'periodicity' theme:

translation symmetry (period)

$\Longrightarrow$ the differential operators preserve that symmetry (conservation)

$\Longrightarrow$ no net flux along a periodic direction (closure).

This chain is used in the 'Propagation and Evolution' chapter for Bloch–Floquet propagation in periodic media, and in the 'Periodic Structure of Electromagnetic Systems' chapter for Maxwell's equations under periodic boundary conditions. It must be noted that the **converse of (i) does not hold**: ∇f being periodic with respect to a does not imply that f is periodic with respect to a — for example, the gradient of $f(\mathsf{x}) = \boldsymbol{e}_1 \cdot \mathsf{x}$ is a constant vector field (periodic with respect to every a), whilst f itself is not periodic. Hence the determination of periodicity must be carried out at the level of the original field, and one cannot look only at its derivative.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	Gradient (symmetry centre 0) ∇ and directional derivatives; phase gradient of the time group
Scaling	Divergence (symmetry centre 1) $\nabla\cdot$ and the inverse-square law
Rotation	Curl (symmetry centre i) $\nabla\times$ and circulation
Conservation	Gauss' theorem and Stokes' theorem; conservation laws of the space group
Symmetry	The correspondence between the differential operators and the three classes of basic symmetry generators, translation, scaling and rotation
Closure	Circulation and flux of closed curves/surfaces; the kernel of the projection operator
Period	The directional period given by the number of directions and the angular resolution

Exercises

Exercise 1 Prove Lemma 15.1: the translation group $(\mathcal{T}_n, \circ)$ forms an Abelian group, and

$$(\mathcal{T}_n, \circ) \cong (\mathbb{R}^n, +).$$

Table 15.3: Position and prerequisite dependencies of the structures of this chapter (for ease of checking and indexing)

Structure	Main location	Prerequisite dependency (what must be established first)
Definitions	sections 15.2 and 15.3 and the sections that follow	none (definitions are the starting point of everything)
Lemmas	6 in all, in sections 15.2 and 15.3	the corresponding definitions
Propositions	22 in all, in sections 15.2 and 15.3 and the sections that follow	the corresponding definitions and lemmas
Theorems	22 in all, from section 15.2 onwards	the corresponding propositions and lemmas
Proofs	immediately after each theorem and proposition	the object proved
Corollaries	11 in all, from sections 15.3 and 15.6 onwards	the corresponding theorems
Properties	4 in all, in the sections on the projection model and the sampling model	the corresponding definitions
Examples	26 in all, with counterexamples distributed over the sections	the corresponding theorems and propositions
Remarks	17 in all, used to delimit the scope of validity and the implicit assumptions	the corresponding conclusions

Please write out the complete verification of closure under composition, associativity, the identity element, the inverse element and the isomorphism.

Exercise 2 Prove Lemma 15.2: the scaling group $(\mathcal{S}_n, \circ)$ forms an Abelian group, and

$$(\mathcal{S}_n, \circ) \cong (\mathbb{R}_{>0}, \times).$$

Also explain why the scaling factor must satisfy $\lambda > 0$ — if $\lambda = 0$ or $\lambda < 0$ were allowed, what property would each of them break?

Exercise 3 Prove Proposition 15.3: the group generated by translation, scaling and rotation is equal to the orientation-preserving similitude transformation group

$$\mathsf{Sim}^+(n) = \{\mathsf{x} \mapsto \lambda R\mathsf{x} + \mathsf{a} \mid \lambda > 0,\ R \in SO(n),\ \mathsf{a} \in \mathbb{R}^n\}.$$

Please write out the proof of both directions of the inclusion relation, and explain why a reflection does not belong to this group.

Exercise 4 Verify Example 15.4 that the shear transformation and anisotropic scaling do not belong to the similitude group:

$$A_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad B = \mathsf{diag}(2, 1).$$

Please compute $A_t^\mathsf{T} A_t$ and $B^\mathsf{T} B$ respectively, prove that neither of them is a scalar matrix, and hence that neither belongs to $\mathsf{Sim}^+(2)$.

Exercise 5 Prove Proposition 15.5: the divergence is the flux density,

$$\nabla \cdot \mathsf{F}(\mathsf{x}_0) = \lim_{r \to 0^+} \frac{1}{|B_r|} \oint_{\partial B_r} \mathsf{F} \cdot \mathsf{n}\, dS.$$

Please complete the proof using the two fundamental integral identities on a sphere (Lemma 15.4) together with the first-order Taylor expansion.

Exercise 6 Prove Proposition 15.6: the curl is the circulation density,

$$(\nabla \times \mathsf{F}(\mathsf{x}_0)) \cdot \mathsf{n} = \lim_{r \to 0^+} \frac{1}{|D_r|} \oint_{\gamma_r} \mathsf{F} \cdot d\mathsf{r}.$$

Please take the disc D_r through x_0 with normal n, expand F to first order and integrate term by term to complete the proof.

Exercise 7 Prove the three clauses of Theorem 15.1:

(i) the gradient and translation: $\frac{d}{dt}\big|_{t=0} f(\mathsf{x} + t\mathsf{a}) = \mathsf{a} \cdot \nabla f(\mathsf{x})$;

(ii) the divergence and the rate of change of volume: $\frac{d}{dt}\big|_{t=0} \mathsf{Vol}(\Phi_t(\Omega)) = \int_\Omega \nabla \cdot \mathsf{F}\, dV$;

(iii) the curl and the local angular velocity: $\frac{1}{2}(J - J^\mathsf{T})\mathsf{y} = \frac{1}{2}(\nabla \times \mathsf{F}) \times \mathsf{y}$.

Please complete the proof using the chain rule, the Jacobi formula and index computation respectively.

Exercise 8 Prove the six product rules in Theorem 15.2:

(i) $\nabla(fg) = f\nabla g + g\nabla f$;
(ii) $\nabla\cdot(f\mathsf{F}) = f(\nabla\cdot\mathsf{F}) + \nabla f\cdot\mathsf{F}$;
(iii) $\nabla\times(f\mathsf{F}) = f(\nabla\times\mathsf{F}) + \nabla f\times\mathsf{F}$;
(iv) $\nabla\cdot(\mathsf{F}\times\mathsf{G}) = \mathsf{G}\cdot(\nabla\times\mathsf{F}) - \mathsf{F}\cdot(\nabla\times\mathsf{G})$;
(v) the expansion of $\nabla\times(\mathsf{F}\times\mathsf{G})$;
(vi) the expansion of $\nabla(\mathsf{F}\cdot\mathsf{G})$.

Please complete all the proofs using the Levi-Civita symbol ϵ_{ijk} and the ϵ-contraction identities.

Exercise 9 Prove Theorem 15.3 and Theorem 15.4:

$$\nabla\times(\nabla f) = \mathsf{0}, \qquad \nabla\cdot(\nabla\times\mathsf{F}) = 0.$$

Please explain how the proofs of the two identities depend on Clairaut's mixed-partial-derivative interchange theorem (Lemma 15.5), and point out that the C^2 assumption cannot be omitted (see Counterexample 15.8).

Exercise 10 Prove Theorem 15.5:

$$\nabla^2\mathsf{F} = \nabla(\nabla\cdot\mathsf{F}) - \nabla\times(\nabla\times\mathsf{F}).$$

Please compute $(\nabla^2\mathsf{F})_1$ componentwise and verify that it equals the first component of the right-hand side, explaining that the two mixed-partial-derivative terms cancel by Clairaut's theorem.

Exercise 11 Prove Green's formula (Theorem 15.6): let D be a bounded closed region with ∂D piecewise smooth; then

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = \oint_{\partial D} (P\,dx + Q\,dy).$$

Please complete the proof in the three steps 'prove it for a y-simple region, prove it for an x-simple region, sum over a general piecewise smooth boundary', and explain why the integrals over common edges of adjacent subregions cancel in pairs.

Exercise 12 Prove the curl theorem (Theorem 15.7):

$$\int_S (\nabla\times\mathsf{F})\cdot\mathsf{n}\,dS = \oint_{\partial S} \mathsf{F}\cdot d\mathsf{r}.$$

Please complete the proof in the five steps 'reduce to a graph surface $z = h(x, y)$, reduce to a double integral over D, apply Green's formula to D, restore the boundary integral, sum over a general piecewise smooth surface'.

Exercise 13 Prove the divergence theorem (Theorem 15.8):

$$\int_V (\nabla \cdot \mathsf{F})\, dV = \oint_{\partial V} \mathsf{F} \cdot \mathsf{n}\, dS.$$

Please complete the proof in the three steps 'the rectangular-box case, a z-simple region, a general regular region', and explain why the contributions of common faces of adjacent subregions cancel in pairs.

Exercise 14 Verify Counterexample 15.2: on the slit plane $\mathcal{M} = \mathbb{R}^2 \setminus \{(x,0) : x \geq 0\}$, the vector field

$$\mathsf{F}(x,y) = \left(\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2}\right)$$

satisfies $\nabla \times \mathsf{F} = \mathsf{0}$, but no scalar potential exists. Please compute the circulation along a closed curve winding once around the origin, and explain that the simple-connectivity condition cannot be omitted.

Exercise 15 Verify Counterexample 15.9: the vector field

$$\mathsf{F}(\mathsf{x}) = \frac{\mathsf{x}}{\|\mathsf{x}\|^3}$$

satisfies $\nabla \cdot \mathsf{F} = 0$ on $\mathbb{R}^3 \setminus \{\mathsf{0}\}$, but is not the curl of any vector field. Please compute the flux through the unit sphere, and explain why zero divergence is not a sufficient condition.

Exercise 16 Prove the Helmholtz decomposition theorem (Theorem 15.11): a suitably decaying vector field can be decomposed into the sum of an irrotational part and a solenoidal part:

$$\mathsf{F} = -\nabla\phi + \nabla \times \mathsf{A}.$$

Please explain that the existence of the decomposition depends on the Green's function of the Laplacian, and point out that uniqueness and the explicit integral formula require the decay condition (see Counterexample 15.11).

Exercise 17 Verify Proposition 15.11: in the orthogonal curvilinear coordinate system (ξ^1, ξ^2, ξ^3) with scale factors h_i,

(i) $\nabla f = \sum_i \frac{1}{h_i} \frac{\partial f}{\partial \xi^i} \mathsf{e}_i$;

(ii) $\nabla \cdot \mathsf{F} = \frac{1}{h_1 h_2 h_3} \sum_i \frac{\partial}{\partial \xi^i} \left(\frac{h_1 h_2 h_3}{h_i} F_i \right)$;

(iii) $(\nabla \times \mathsf{F})_1 = \frac{1}{h_2 h_3} \left[\frac{\partial (h_3 F_3)}{\partial \xi^2} - \frac{\partial (h_2 F_2)}{\partial \xi^3} \right]$.

Please write out the concrete expressions in spherical and cylindrical coordinates respectively (Example 15.12).

Exercise 18 Prove the inverse-square law (Theorem 3.4): if the total flux of a radial sourceless field through any closed surface enclosing the centre is the constant Φ_0, then $F(r) = \Phi_0/(4\pi r^2)$. Please write out the flux computation, and prove its converse (Corollary 3.2).

Exercise 19 Verify Counterexample 15.13: take $\mathbf{F}(\mathbf{r}) = f(r)\hat{\mathbf{r}}$, where $f(r) = 1/r^2$ ($r \le 1$) and $f(r) = 1/r^2 + 1$ ($r > 1$). Please compute the flux through the sphere for $r > 1$, verify that it is not constant, and explain that 'the flux is constant' is an independent assumption and not an automatic corollary of sourcelessness plus spherical symmetry.

Exercise 20 Comprehensive question: combining the three basic transformations (Definitions 15.4, 15.5, 15.6), the three differential operators (Definitions 15.8, 15.9, 15.10) and the three integral theorems (Theorems 15.6, 15.7, 15.8), explain how vector analysis concretises, at the level of vector fields, the three great symmetry centres of periodic mathematics — the additive symmetry centre 0, the multiplicative symmetry centre 1 and the rotational symmetry centre i. Please indicate which integral theorems correspond to conservation laws, and explain the difference in level among the three groups of objects (see Remark 15.3).

Chapter 16

Epicycle Sets and Deferent Sphere Sets

Remark 16.1 (Positioning of this chapter). This chapter follows directly on the 'Vector Analysis' chapter: it generalises the static three-group projection model established there into a dynamic model, and supplies the three classes of object that were still missing—single-group dynamics, the quaternion single-sphere group and the rotating sphere group. The whole chapter follows a uniform two-step structure:

$$\text{Composition} \longrightarrow \text{Projection} \longrightarrow \text{Observation result.}$$

Convention: *every model in this chapter adopts the order 'compose first, project afterwards'. That is, the time group and the space group are first combined into a composite object (in the complex case the sum $T+S$, in the quaternion case the sum q_T+q_S), and this composite object is then projected onto the observer group. The observer itself may vary with time, and may also rotate independently according to its own epicycle.*

Guide to the seven themes: Translation—the dynamic adjustment of the observer vector and frequency translation; Scaling—the weight introduced by the observer's modulus; Rotation—the independent rotation of the three planes and of the three episphere groups; Balance—the linearity of composition and projection; Symmetry—the symmetry and commutativity of projection; Closure—the invariants of synchronous rotation; Period—the instantaneous frequency and beat frequency of the dynamic projection.

16.1 Preliminaries: the four series and the definition of epicycles

This section recapitulates the amplitude series, frequency series, phase series and period series established in the 'Series and Convolution' chapter, and on this basis defines the basic objects of this chapter—epicycles and epispheres. The content of this section is a citation and recombination of the preceding chapters; it introduces no new mathematical assertions.

16.1.1 Amplitude series

Definition 16.1 (Amplitude series (recap)). Let the amplitude sequence $\{A_k\}_{k\in\mathbb{Z}}$ satisfy $A_k \geq 0$ and $\sum_k A_k < \infty$. The formal series

$$\mathcal{A}_f(t) = \sum_{k\in\mathbb{Z}} A_k e^{ikt}$$

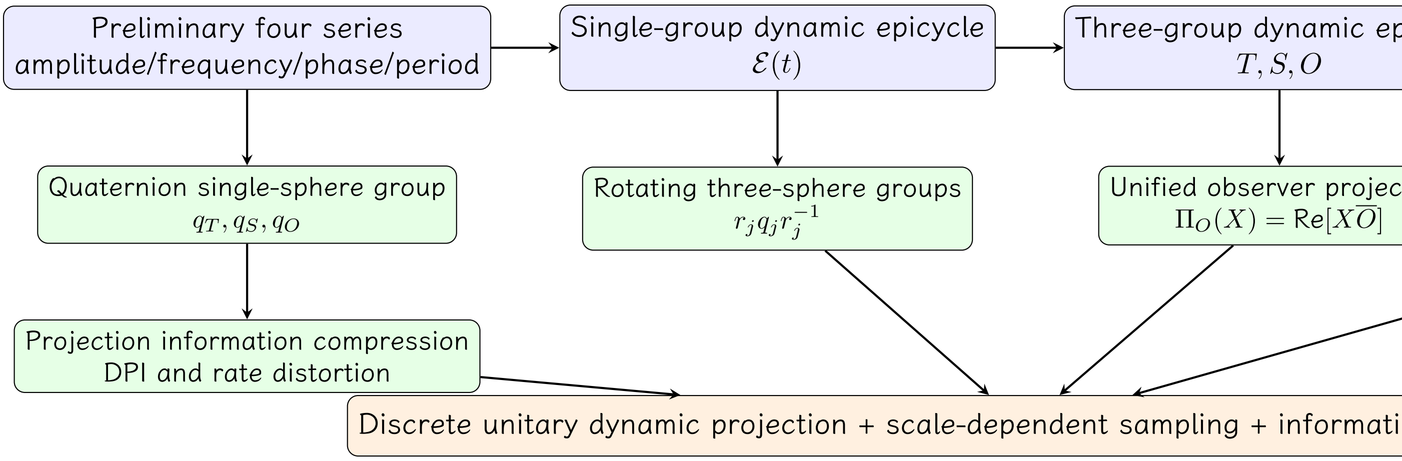

Preliminary four series
amplitude/frequency/phase/period
Single-group dynamic epicycle
$\mathcal{E}(t)$
Three-group dynamic ep
T, S, O
Quaternion single-sphere group
q_T, q_S, q_O
Rotating three-sphere groups
$r_j q_j r_j^{-1}$
Unified observer projec
$\Pi_O(X) = \mathrm{Re}[X\overline{O}]$
Projection information compression
DPI and rate distortion
Discrete unitary dynamic projection + scale-dependent sampling + informati

Figure 16.1: Overview of the structure of this chapter: starting from the preliminary four series, the single-group dynamic epicycle, the three-group dynamic epicycle, the rotating three planes, the quaternion single-sphere group, the rotating three-sphere groups, the unified observer projection framework, the observer functor and projection information compression are established in turn, finally converging on discrete unitary dynamic projection, scale-dependent sampling and information loss with distance.

is called the **amplitude series** (its real form is $\sum_k A_k \cos(kt)$, requiring $A_{-k} = A_k$). The amplitude series retains only the amplitude information, discarding the phase and the frequency-lattice structure.

Prerequisite: absolute summability $\sum_k A_k < \infty$ guarantees uniform convergence ('Series and Convolution' chapter Lemma 42.1).

Proposition 16.1 (Bohr mean coefficient formula for the amplitude series (recap))**.** *Let $\mathcal{A}_f(t) = \sum_k A_k e^{ikt}$ converge uniformly and absolutely. Then for every $n \in \mathbb{Z}$,*

$$A_n = M\{\mathcal{A}_f(t)e^{-int}\} = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} \mathcal{A}_f(t)e^{-int}\,dt.$$

That is, the amplitude A_n can be extracted uniquely by the Bohr mean.

Proof. By absolute summability the Bohr mean may be interchanged with the sum:

$$M\{\mathcal{A}_f(t)e^{-int}\} = \sum_k A_k M\{e^{i(k-n)t}\}.$$

By the orthogonality of the exponential basis ('Series and Convolution' chapter Lemma 6.2), $M\{e^{i(k-n)t}\} = \delta_{kn}$, so only the term $k = n$ survives, giving A_n. ☐

16.1.2 Frequency series

Definition 16.2 (Frequency series (recap))**.** Let f have fundamental period T, with frequency lattice $\omega_n = 2\pi n/T$. The formal series

$$\mathcal{F}_f(s) = \sum_{n\in\mathbb{Z}} \omega_n s^n = \frac{2\pi}{T}\sum_{n\in\mathbb{Z}} n s^n$$

is called the **frequency series**. The frequency series retains only the frequency lattice, discarding amplitude and phase.

Proposition 16.2 (Frequency series and period scaling (recap)). If $f_\lambda(t) := f(t/\lambda)$ $(\lambda > 0)$, then the frequency series scales as $\mathcal{F}_{f_\lambda}(s) = \frac{1}{\lambda}\mathcal{F}_f(s)$.

Proof. f_λ has period λT, hence its frequencies are $\omega_n^{(\lambda)} = 2\pi n/(\lambda T) = \omega_n/\lambda$; substituting into Definition 16.2 gives the result. ☐

16.1.3 Phase series

Definition 16.3 (Phase series (recap)). Let $\{\phi_k\}_{k\in\mathbb{Z}}$ be a phase sequence. The formal series

$$\mathcal{P}_f(t) = \sum_{k\in\mathbb{Z}} e^{i(kt+\phi_k)}$$

is called the **phase series**. The phase series retains only the phase, discarding amplitude and frequency-lattice structure.

Proposition 16.3 (Phase series and time translation (recap)). The time translation $f_\tau(t) := f(t-\tau)$ rotates every term of the phase series by $-k\tau$:

$$\mathcal{P}_{f_\tau}(t) = \sum_k e^{i(kt+\phi_k-k\tau)}.$$

Proof. Substituting $t \mapsto t-\tau$: $e^{i(k(t-\tau)+\phi_k)} = e^{i(kt+\phi_k-k\tau)}$; summing gives the result. ☐

16.1.4 Period series

Definition 16.4 (Period series (recap)). Let P be an index set of period classes, $T_p > 0$, and let the quadruple $(A_{p,n}, \omega_{p,n}, \phi_{p,n}, T_p)$ satisfy $\omega_{p,n} = 2\pi n/T_p$. The series

$$f(t) = \sum_{p\in P}\sum_{n\in\mathbb{Z}} A_{p,n}e^{i(\omega_{p,n}t+\phi_{p,n})}$$

is called the **period series**. The period series takes the period T_p as its primary index; within each period layer the amplitude series supplies $A_{p,n}$, the phase series supplies $e^{i\phi_{p,n}}$, and the frequency series supplies $\omega_{p,n}$.

Proposition 16.4 (The period series synthesises the four series (recap)). The period series is synthesised from the four components amplitude, phase, frequency and period according to the period layer p. For a single period layer $P = \{T\}$ it degenerates to the Fourier series; taking the phase to be zero or the amplitude to be one further yields the amplitude series or the phase series respectively.

Proof. By Definition 16.4, for fixed p the inner sum $\sum_n A_{p,n}e^{i(\omega_{p,n}t+\phi_{p,n})}$ is a standard Fourier series. Taking $P = \{T\}$ gives a single layer; taking $\phi_{p,n} \equiv 0$ gives the amplitude series; taking $A_{p,n} \equiv 1$ gives the phase series. Checking term by term gives the result. ☐

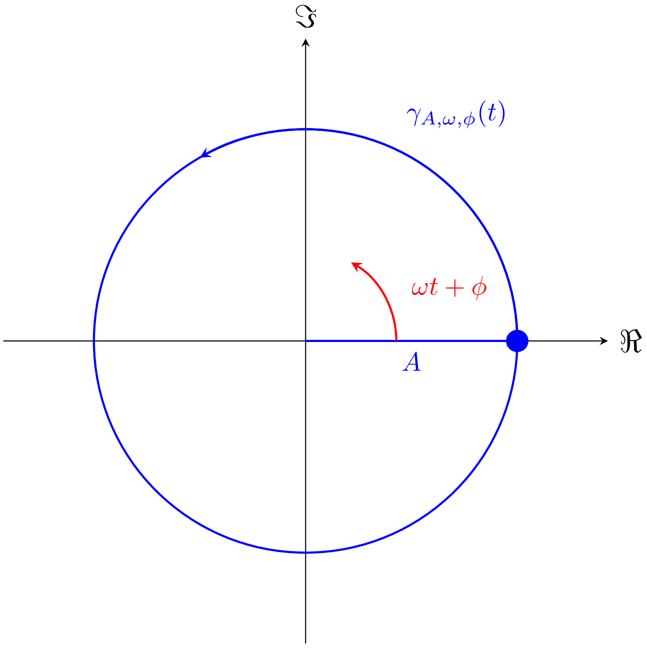
16.1.5 Epicycles: periodic elements in the complex plane

Definition 16.5 (Epicycle)**.** Let $A > 0$, $\omega \in \mathbb{R}$, $\phi \in [0, 2\pi)$. The map

$$\gamma_{A,\omega,\phi} : \mathbb{R} \to \mathbb{C}, \qquad \gamma_{A,\omega,\phi}(t) = Ae^{i(\omega t+\phi)}$$

is called an **epicycle**. The three parameters are called the **radius** (amplitude), the **angular velocity** (frequency) and the **initial phase** respectively.

Prerequisite: $A > 0$. If $A = 0$, the epicycle degenerates to the zero map.



Epicycle in the complex plane

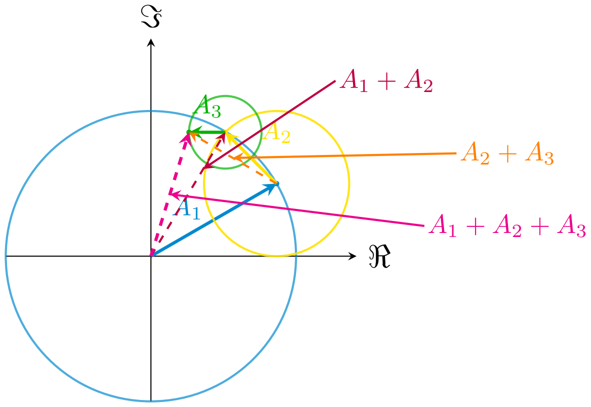
Figure 16.2: Epicycle: uniform circular motion in the complex plane with radius A, angular velocity ω and initial phase ϕ.

Definition 16.6 (Epicycle group)**.** Let $K \in \mathbb{N}^+$, $\{A_k\}_{k=1}^K \subset \mathbb{R}^+$, $\{\omega_k\}_{k=1}^K \subset \mathbb{R}$, $\{\phi_k\}_{k=1}^K \subset [0, 2\pi)$. The finite sum

$$\mathcal{E}(t) = \sum_{k=1}^{K} A_k e^{i(\omega_k t+\phi_k)}$$

is called an **epicycle group**. The composite trajectory $\mathcal{E}(t)$ is a curve in the complex plane.

Relation to the four series: Definition 16.6 is the concrete realisation of the period series on a single period layer under truncation to a finite series. Its amplitudes are given by the amplitude series ($\{A_k\}$), its phases by the phase series ($\{\phi_k\}$), its frequencies by the frequency series ($\{\omega_k\}$), and the common period $T = 2\pi/|\omega_0|$ (with $\omega_0 = \text{gcd}\{\omega_k \neq 0\}$ the fundamental frequency) by the period component.



$$\mathcal{E}(t) = A_1 e^{i(\omega_1 t+\phi_1)} + A_2 e^{i(\omega_2 t+3\pi/4)} + A_3 e^{i(\omega_3 t+\pi)}$$

Figure 16.3: Epicycle group: the arm A_1 (cyan, length r_1) lies inside the circle A_1, the arm A_2 (yellow, length r_2, phase $135°$) lies inside the circle A_2, and the arm A_3 (green, length r_3, phase π) lies inside the circle A_3; the endpoint of each arm is the centre of the next circle. The three composite vectors A_1+A_2 (purple dashed), A_2+A_3 (orange dashed) and $A_1+A_2+A_3$ (magenta dashed) are the instantaneous values of the composite trajectory at the various levels; their textual labels are placed outside on the right of the figure, with arrows of the same colour pointing to the corresponding dashed lines.

Proposition 16.5 (Periodicity condition for the epicycle group). Let $\mathcal{E}(t) = \sum_{k=1}^{K} A_k e^{i(\omega_k t+\phi_k)}$ with $A_k > 0$ for all k. Then $\mathcal{E}$ is a periodic function if and only if all the ω_k generate a discrete subgroup. In that case write

$$\omega_0 := \min\{\omega > 0 : \omega \in \textstyle\sum_k \mathbb{Z}\omega_k\} = \gcd\{\omega_k \neq 0\},$$

(that is, $\omega_k/\omega_0 \in \mathbb{Z}$ for all k); then the least positive period is $2\pi/|\omega_0|$ (an integer fraction of it when $\omega_0 \notin \Omega$); when all $\omega_k = 0$ there is no least positive period.

Proof. **Sufficiency.** If $\omega_k = n_k\omega_0$ ($n_k \in \mathbb{Z}$), then

$$\mathcal{E}(t+T) = \sum_k A_k e^{i(n_k\omega_0(t+2\pi/\omega_0)+\phi_k)} = \sum_k A_k e^{i(n_k\omega_0 t+\phi_k+2\pi n_k)} = \sum_k A_k e^{i(n_k\omega_0 t+\phi_k)} = \mathcal{E}(t).$$

Necessity. If $\mathcal{E}$ has period $T > 0$, then for each k, $e^{i\omega_k T} = 1$, that is $\omega_k T \in 2\pi\mathbb{Z}$. Taking $\omega_0 = 2\pi/T$ gives $\omega_k/\omega_0 = \omega_k T/(2\pi) \in \mathbb{Z}$. Hence all frequencies are integer multiples of ω_0. □

Remark 16.2 (Connection with the static model). Definition 16.6 agrees, as a mathematical object, with the nested epicycle group established in the 'Vector Analysis' chapter—both are finite sums of complex exponentials in the complex plane. What is new in this chapter is that the parameters of these epicycles are regarded as functions that may vary with time, and the resulting dynamic projection is studied.

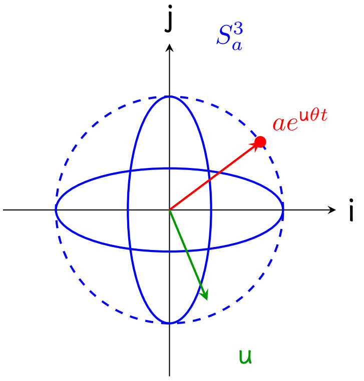
16.1.6 Epispheres: periodic elements in quaternion space

Definition 16.7 (Episphere). Let $a > 0$, let u be a unit pure imaginary quaternion ($\mathsf{u}^2 = -1$), and let $\theta \in \mathbb{R}$. The map

$$\beta_{a,\mathsf{u},\theta} : \mathbb{R} \to \mathbb{H}, \qquad \beta_{a,\mathsf{u},\theta}(t) = a\, e^{\mathsf{u}\theta t}$$

is called an **episphere**. Here $e^{\mathsf{u}\theta t} = \cos(\theta t)+\mathsf{u}\sin(\theta t)$ is the quaternion exponential.

The three parameters are called the **radius** (a), the **rotation axis** (u) and the **angular velocity** (θ) respectively.



Episphere: spherical motion in quaternion space

Figure 16.4: Episphere: the orbit of radius a in quaternion space lies on the three-dimensional sphere S^3_a.

Definition 16.8 (Episphere group)**.** Let $K \in \mathbb{N}^+$, $a_k > 0$, let u_k be a unit pure imaginary quaternion, and let $\theta_k \in \mathbb{R}$. The finite sum

$$q(t) = \sum_{k=1}^{K} a_k e^{\mathsf{u}_k \theta_k t}$$

is called an **episphere group**.

Proposition 16.6 (Norm of an episphere)**.** Let $\beta_{a,\mathsf{u},\theta}(t) = ae^{\mathsf{u}\theta t}$. Then for any t,

$$|\beta_{a,\mathsf{u},\theta}(t)| = a.$$

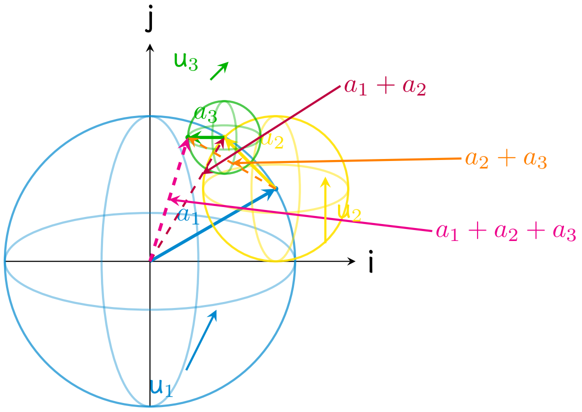
That is, the orbit of the episphere in quaternion space lies on the three-dimensional sphere of radius a, $S^3_a = \{q \in \mathbb{H} : |q| = a\}$.

Proof. From $e^{\mathsf{u}\theta t} = \cos(\theta t) + \mathsf{u}\sin(\theta t)$, computing the norm:

$$|e^{\mathsf{u}\theta t}|^2 = \cos^2(\theta t) + |\mathsf{u}|^2 \sin^2(\theta t) = \cos^2(\theta t) + \sin^2(\theta t) = 1,$$

where we used $|\mathsf{u}| = 1$ (unit pure imaginary quaternion) and the orthogonality of u with scalars. Hence $|\beta_{a,\mathsf{u},\theta}(t)| = a|e^{\mathsf{u}\theta t}| = a$. $\square$

Remark 16.3 (Epispheres and the non-commutativity of quaternion multiplication)*.* If the rotation axes of two epispheres differ ($\mathsf{u}_1 \neq \mathsf{u}_2$), then $e^{\mathsf{u}_1\theta_1 t}e^{\mathsf{u}_2\theta_2 t} \neq e^{\mathsf{u}_2\theta_2 t}e^{\mathsf{u}_1\theta_1 t}$. This is the direct manifestation on episphere groups of the non-commutativity of quaternion multiplication, in contrast with the complex case ($U(1)$ is commutative).



$$q(t) = a_1 e^{\mathsf{u}_1\theta_1 t} + a_2 e^{\mathsf{u}_2\theta_2 t} + a_3 e^{\mathsf{u}_3\theta_3 t}$$

Figure 16.5: Episphere group: the centre of each sphere falls at the endpoint of the arm at the preceding level, and the length of the arm equals the radius of the corresponding sphere; each sphere is represented by an outer circle together with one horizontal and one vertical ellipse, standing for the sphere S^3. The rotation axes $\mathsf{u}_1, \mathsf{u}_2, \mathsf{u}_3$ are indicated by arrows of the corresponding colour, reflecting the non-commutative rotation of quaternion space. The three composite vectors $a_1 + a_2$ (purple dashed), $a_2 + a_3$ (orange dashed) and $a_1 + a_2 + a_3$ (magenta dashed) are the instantaneous values of the composite path at the various levels.

16.2 Dynamic epicycle groups

16.2.1 Basic definitions

Definition 16.9 (Dynamic epicycle group)**.** Let $K \in \mathbb{N}^+$. Let the amplitudes $A_k : \mathbb{R} \to \mathbb{R}^+$, the angular frequencies $\omega_k : \mathbb{R} \to \mathbb{R}$ and the phases $\phi_k : \mathbb{R} \to \mathbb{R}$ all be C^1 functions ($k = 1, \ldots, K$). The sum

$$\mathcal{E}(t) = \sum_{k=1}^{K} A_k(t)\, e^{i(\omega_k(t)t + \phi_k(t))}$$

is called a **dynamic epicycle group**. If all the parameters are constant, it degenerates to Definition 16.6.

Prerequisite: K is finite on compact intervals. The case in which K jumps at finitely many instants is allowed; see Remark 16.4.

Instantaneous phase and instantaneous frequency: the total phase of the k-th term is $\Phi_k(t) = \omega_k(t)t + \phi_k(t)$, whose instantaneous frequency is

$$\dot{\Phi}_k(t) = \omega_k(t) + t\dot{\omega}_k(t) + \dot{\phi}_k(t).$$

Note: when ω_k is not constant, the instantaneous frequency is not equal to $\omega_k(t)$. Wherever frequency translation is involved in this chapter, it always means translation of the **instantaneous frequency**.

Definition 16.10 (Time component and space component)**.** Let $\mathcal{E}(t)$ be the dynamic epicycle group of Definition 16.9. Define its **time component** $T(t)$ and **space**

component $S(t)$ by

$$T(t) = \operatorname{Re} \mathcal{E}(t) = \sum_{k=1}^{K} A_k(t) \cos(\omega_k(t)t + \phi_k(t)),$$

$$S(t) = \operatorname{Im} \mathcal{E}(t) = \sum_{k=1}^{K} A_k(t) \sin(\omega_k(t)t + \phi_k(t)).$$

In the complex plane the time component lies along the real axis and the space component along the imaginary axis; the two are automatically orthogonal.

Definition 16.11 (Dynamic observer vector)**.** Let $\theta : \mathbb{R} \to \mathbb{R}$ be a C^1 function. The unit complex number

$$u(t) = e^{i\theta(t)}$$

is called a **dynamic observer vector**. The function $\theta(t)$ is called the **instantaneous direction angle** of the observer.

Definition 16.12 (Dynamic projection observation result)**.** Let $\mathcal{E}$ be a dynamic epicycle group and u a dynamic observer vector. Define the **observation result** by

$$\boxed{\mathcal{O}(t) = \operatorname{Re}\big[\mathcal{E}(t)\,\overline{u(t)}\big]}.$$

By Definition 16.10,

$$\mathcal{O}(t) = T(t)\cos\theta(t) + S(t)\sin\theta(t) = \big\langle (T(t), S(t)),\, (\cos\theta(t), \sin\theta(t)) \big\rangle.$$

That is, the observation result is the projection of the two-dimensional real time–space vector onto the dynamic observer direction.

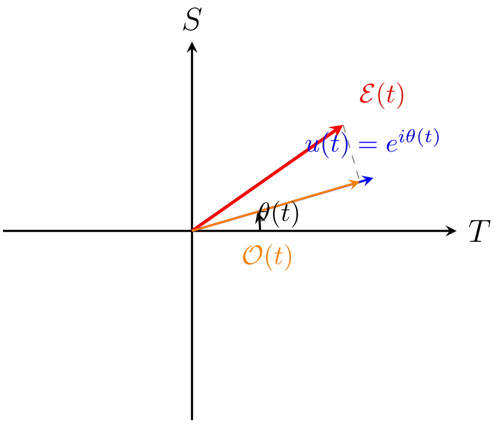


Figure 16.6: Geometric picture of the dynamic projection: the observation result $\mathcal{O}(t)$ is the orthogonal projection of the source vector $\mathcal{E}(t)$ onto the observer direction $u(t)$.

16.2.2 Basic properties

Proposition 16.7 (Linearity of the projection)**.** For fixed $\theta(t)$, the map $\mathcal{E} \mapsto \mathcal{O}$ is real-linear in $\mathcal{E}$:

$$\mathcal{O}[\alpha\mathcal{E}_1 + \beta\mathcal{E}_2] = \alpha\mathcal{O}[\mathcal{E}_1] + \beta\mathcal{O}[\mathcal{E}_2], \qquad \alpha, \beta \in \mathbb{R}.$$

Proof. By the distributivity of complex multiplication,

$$(\alpha\mathcal{E}_1 + \beta\mathcal{E}_2)\overline{u} = \alpha\mathcal{E}_1\overline{u} + \beta\mathcal{E}_2\overline{u}.$$

By the linearity of the real-part function,

$$\mathrm{Re}[\alpha\mathcal{E}_1\overline{u} + \beta\mathcal{E}_2\overline{u}] = \alpha\,\mathrm{Re}[\mathcal{E}_1\overline{u}] + \beta\,\mathrm{Re}[\mathcal{E}_2\overline{u}].$$

□

Proposition 16.8 (Symmetry of the projection)**.** For any dynamic epicycle group $\mathcal{E}$ and any dynamic observer u,

$$\mathcal{O}(t) = \mathrm{Re}[\mathcal{E}(t)\overline{u(t)}] = \mathrm{Re}[u(t)\overline{\mathcal{E}(t)}].$$

Proof. For any $z \in \mathbb{C}$ we have $\mathrm{Re}\,z = \mathrm{Re}\,\overline{z}$. Take $z = \mathcal{E}\overline{u}$:

$$\mathrm{Re}[\mathcal{E}\overline{u}] = \mathrm{Re}\,\overline{[\mathcal{E}\overline{u}]} = \mathrm{Re}[\overline{\mathcal{E}}\,u] = \mathrm{Re}[u\overline{\mathcal{E}}].$$

□

Lemma 16.1 (Single-frequency formula under static degeneracy)**.** Take $\mathcal{E}(t) = Ae^{i\omega t}$ ($A > 0$, $\omega \in \mathbb{R}$) and observer $u(t) = e^{i\omega_0 t}$. Then

$$\mathcal{O}(t) = A\cos\big((\omega - \omega_0)t\big).$$

Proof. $\mathcal{E}(t)\overline{u(t)} = Ae^{i\omega t} \cdot e^{-i\omega_0 t} = Ae^{i(\omega-\omega_0)t}$; taking the real part gives the result. □

Theorem 16.1 (Continuity of the dynamic projection)**.** Let $A_k, \omega_k, \phi_k, \theta$ all be C^1 functions. Then the observation result $\mathcal{O}(t)$ is a C^1 function. Furthermore, if $|\mathcal{E}(t)| \neq 0$, the **instantaneous frequency** of $\mathcal{O}(t)$ is

$$\Omega_{\mathcal{O}}(t) = \frac{d}{dt}\arg\big(\mathcal{E}(t)\overline{u(t)}\big).$$

In the single-frequency case $\mathcal{E}(t) = Ae^{i\omega t}$, $u(t) = e^{i\omega_0 t}$, it degenerates to $\omega - \omega_0$.

Proof. **Continuity.** By Definition 16.10, $T(t)$ and $S(t)$ are finite sums and composites of the C^1 functions A_k, ω_k, ϕ_k, hence are C^1. By Definition 16.11, θ is C^1, so that $\cos\theta$ and $\sin\theta$ are both C^1. By Definition 16.12, $\mathcal{O} = T\cos\theta + S\sin\theta$ is a sum of products of C^1 functions, hence is C^1.

Instantaneous frequency. Write $z(t) = \mathcal{E}(t)\overline{u(t)}$ and $\Phi(t) = \arg z(t)$. Since $|z(t)| = |\mathcal{E}(t)| \neq 0$, a continuous branch of Φ may be chosen locally and is C^1. From $z = |z|e^{i\Phi}$ we get $\dot{z} = (|z|' + i|z|\dot{\Phi})e^{i\Phi}$, hence $\dot{\Phi} = \mathrm{Im}(\dot{z}/z)$. In the single-frequency case, by Lemma 16.1, $z(t) = Ae^{i(\omega-\omega_0)t}$ and $\Phi(t) = (\omega - \omega_0)t$ (on the continuous branch), so $\Omega_{\mathcal{O}} = \omega - \omega_0$. □

Corollary 16.1 (Frequency shift in a rotating frame)**.** In the single-frequency case Lemma 16.1, the frequency seen by the observer is $\omega_{\text{obs}} = \omega - \omega_0$. When $\omega_0 = \omega$, the observation result is the constant A.

Proof. By Lemma 16.1, $\mathcal{O}(t) = A\cos((\omega-\omega_0)t)$, whose frequency is $|\omega-\omega_0|$. When $\omega = \omega_0$ we have $\mathcal{O}(t) \equiv A$. ☐

Property 16.1 (Basic properties of the dynamic projection)**.**

(i) **Preservation of zeros**: $\mathcal{E}(t) = 0 \Rightarrow \mathcal{O}(t) = 0$.

(ii) **Amplitude bound**: $|\mathcal{O}(t)| \le |\mathcal{E}(t)|$.

(iii) **Observer sign reversal**: $\theta \mapsto \theta + \pi$ sends $\mathcal{O} \mapsto -\mathcal{O}$.

(iv) **Observer rotation**: $\theta \mapsto \theta + \theta_0$ sends $\mathcal{O}(t) \mapsto \mathrm{Re}[\mathcal{E}(t)e^{-i\theta_0}]$.

Proof. (i) When $\mathcal{E}(t) = 0$ we have $\mathcal{E}(t)\overline{u(t)} = 0$, so the real part is zero.

(ii) $|\mathcal{O}(t)| = |\,\mathrm{Re}[\mathcal{E}(t)\overline{u(t)}]| \le |\mathcal{E}(t)\overline{u(t)}| = |\mathcal{E}(t)| \cdot |\overline{u(t)}| = |\mathcal{E}(t)|$.

(iii) $u(t) = e^{i(\theta(t)+\pi)} = -e^{i\theta(t)}$, hence $\mathcal{E}\overline{u} = -\mathcal{E}\overline{e^{i\theta}}$, and taking the real part changes the sign.

(iv) Substituting $u(t) = e^{i(\theta(t)+\theta_0)}$ gives $\mathcal{E}\overline{u} = \mathcal{E}e^{-i\theta}e^{-i\theta_0}$; taking the real part gives the result. ☐

Example 16.1 (Single-frequency example of dynamic projection)**.** Take $\mathcal{E}(t) = 2e^{i3t}$ and observer $u(t) = e^{it}$. By Lemma 16.1, $\mathcal{O}(t) = 2\cos(2t)$. If instead $u(t) = e^{i3t}$, then $\mathcal{O}(t) \equiv 2$.

Remark 16.4 (The case of jumps in K). *When K is allowed to jump at finitely many instants, $\mathcal{E}$ is in general discontinuous at the jump instants. In that case $\mathcal{O}$ can only be defined in the piecewise C^1 sense. If $\mathcal{O}$ is required to be globally continuous, one may impose the condition that the total sum is unchanged across the jump, or smooth the jump by interpolation. Such conditions belong to engineering convention; this chapter assumes K constant by default.*

16.3 Three dynamic epicycle groups

16.3.1 Basic definitions

Definition 16.13 (Three dynamic epicycle groups)**.** Let there be three mutually independent dynamic epicycle groups

$$T(t) = \sum_{k=1}^{K_T} A_k(t)e^{i(\omega_k(t)t+\phi_k(t))},$$

$$S(t) = \sum_{l=1}^{K_S} B_l(t)e^{i(\nu_l(t)t+\psi_l(t))},$$

$$O(t) = \sum_{m=1}^{K_O} C_m(t)e^{i(\mu_m(t)t+\chi_m(t))},$$

called the **time group**, the **space group** and the **observer group** respectively. The parameters of all three groups are C^1 functions.

Definition 16.14 (Synthesis–projection observation result)**.** Define the **composite trajectory** to be $T(t) + S(t)$, and define the **observation result** to be the real part of the bilinear pairing of the composite trajectory with the observer group:

$$\boxed{\mathcal{O}(t) = \mathrm{Re}\big[(T(t) + S(t))\,\overline{O(t)}\big]}\,.$$

Remark 16.5 (Relation to the static model of the 'Vector analysis' chapter)**.** Definition 16.14 is the dynamic generalisation of the threefold static projection model of the 'Vector Analysis' chapter: when the parameters of all three groups are constant, the two coincide completely.

16.3.2 Basic properties

Proposition 16.9 (Additivity of synthesis–projection)**.** *The observation result is additive with respect to the time group and the space group:*

$$\mathcal{O}(t) = \mathrm{Re}[T(t)\overline{O(t)}] + \mathrm{Re}[S(t)\overline{O(t)}].$$

Proof. By the distributivity of complex multiplication $(T + S)\overline{O} = T\overline{O} + S\overline{O}$, and then by the linearity of the real part. ☐

Proposition 16.10 (Degeneracy for a unit-modulus observer)**.** *If $O(t) = e^{i\chi(t)}$, then Definition 16.14 degenerates to*

$$\mathcal{O}(t) = \mathrm{Re}\big[(T(t) + S(t))\,e^{-i\chi(t)}\big],$$

in agreement with the form of the single-group dynamic projection Definition 16.12.

Proof. When $|O(t)| = 1$ we have $\overline{O(t)} = O(t)^{-1} = e^{-i\chi(t)}$; substituting gives the result. ☐

Lemma 16.2 (Bilinear pairing and modulus weighting)**.** *In the general case,*

$$\mathcal{O}(t) = \mathrm{Re}\big[(T(t) + S(t))\,\overline{\widehat{O}(t)}\big] \cdot |O(t)|, \qquad \widehat{O}(t) = O(t)/|O(t)|.$$

Proof. Let $\widehat{O} = O/|O|$; then $O = |O|\widehat{O}$ and $\overline{O} = |O|\overline{\widehat{O}}$. Substituting,

$$(T + S)\overline{O} = |O| \cdot (T + S)\overline{\widehat{O}}.$$

Taking the real part and pulling out the real factor $|O| \geq 0$ gives the result. ☐

Theorem 16.2 (Continuity of the three-group dynamic projection)**.** *Let the parameters of T, S, O all be C^1 functions. Then $\mathcal{O}(t)$ is a C^1 function. If $O(t) \neq 0$ for all t, then $\mathcal{O}(t)$ and $|O(t)|$ have the same zero set.*

Proof. **Continuity:** $\mathcal{O}$ is the real part of a product of the C^1 functions T, S, O and their conjugates, hence is C^1.

Zero set: By Lemma 16.2, $\mathcal{O} = |O| \cdot \mathrm{Re}[(T + S)\overline{\widehat{O}}]$. If $|O(t)| \neq 0$, then $\mathcal{O}(t) = 0$ if and only if $\mathrm{Re}[(T + S)\overline{\widehat{O}}] = 0$. ☐

Corollary 16.2 (Beat-frequency formula for three single frequencies). Take $T(t) = Ae^{i\omega t}$, $S(t) = Be^{i\nu t}$, $O(t) = Ce^{i\mu t}$ $(A, B, C > 0)$. Then

$$\mathcal{O}(t) = AC\cos((\omega - \mu)t) + BC\cos((\nu - \mu)t).$$

When $|\omega - \nu|$ is very small a beat arises, with beat period $T_{\text{beat}} = 2\pi/|\omega - \nu|$.

Proof. $T\overline{O} = ACe^{i(\omega-\mu)t}$, $S\overline{O} = BCe^{i(\nu-\mu)t}$; adding and taking the real part gives the result. ☐

Property 16.2 (Basic properties of the three-group dynamic projection). (i) **Linearity**: for fixed O, $\mathcal{O}$ is real-linear in T and S.

(ii) **Observer sign reversal**: $O \mapsto -O$ sends $\mathcal{O} \mapsto -\mathcal{O}$.

(iii) **Observer conjugation**: $O \mapsto \overline{O}$ sends $\mathcal{O} \mapsto \text{Re}[(T + S)O]$.

(iv) **Energy decomposition**: $|\mathcal{O}(t)| \le |T(t) + S(t)| \cdot |O(t)|$.

Proof. (i) By Proposition 16.9 and the linearity of complex multiplication in its first factor. (ii) $\overline{-O} = -\overline{O}$. (iii) $\overline{\overline{O}} = O$. (iv) $|\mathcal{O}| \le |(T + S)\overline{O}| = |T + S| \cdot |O|$. ☐

Example 16.2 (Beat-frequency example of the three-group dynamic projection). Take $T(t) = e^{i10t}$, $S(t) = e^{i12t}$, $O(t) = e^{i11t}$. By Corollary 16.2,

$$\mathcal{O}(t) = \cos(-t) + \cos(t) = 2\cos t.$$

The beat period is $T_{\text{beat}} = 2\pi$.

Remark 16.6 (Normalised and unnormalised observers). If the observer group is normalised to $\widehat{O} = O/|O|$, then the weight factor $|O(t)|$ in Lemma 16.2 disappears, and the observation result degenerates to a genuine orthogonal projection. Whether or not to normalise is a modelling choice.

16.4 Rotating three planes

16.4.1 Basic definitions

Definition 16.15 (Rotating three planes). Let there be three mutually independent complex planes $\mathbb{C}_T, \mathbb{C}_S, \mathbb{C}_O$. On each plane there is a dynamic epicycle group $T_0(t), S_0(t), O_0(t)$ (as in Definition 16.13). Let $\theta_T, \theta_S, \theta_O : \mathbb{R} \to \mathbb{R}$ be C^1 functions. Define the **rotated trajectories** by

$$T(t) = T_0(t)e^{i\theta_T(t)}, \quad S(t) = S_0(t)e^{i\theta_S(t)}, \quad O(t) = O_0(t)e^{i\theta_O(t)}.$$

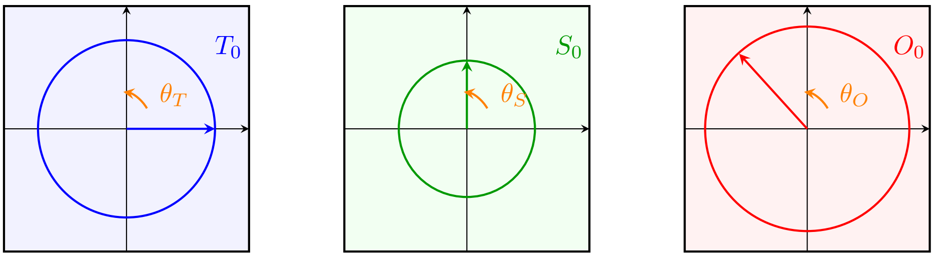
The rotation angles of the three planes are determined by the epicycle group in the corresponding plane, hence the phrase 'rotation according to the properties of the internal epicycle'.

Definition 16.16 (Observation result for the rotating three planes). Continuing with Definition 16.15, define

$$\boxed{\mathcal{O}(t) = \text{Re}\big[(T(t) + S(t))\,\overline{O(t)}\big]}.$$

Expanding:

$$\mathcal{O}(t) = \text{Re}\big[(T_0e^{i\theta_T} + S_0e^{i\theta_S})\,e^{-i\theta_O}\,\overline{O_0}\big].$$



$\mathbb{C}_T$ (time) $\mathbb{C}_S$ (space) $\mathbb{C}_O$ (observer)

Figure 16.7: Rotating three planes: the three independent complex planes each rotate according to the properties of the internal epicycle, with rotation angles $\theta_T, \theta_S, \theta_O$.

16.4.2 Basic properties

Proposition 16.11 (Phase-shift action of the rotation angles)**.** The rotating three planes are equivalent to applying the phase shifts $\theta_T, \theta_S, \theta_O$ to T_0, S_0, O_0 respectively before composition:

$$\mathcal{O}(t) = \mathrm{Re}\big[(T_0 e^{i(\theta_T-\theta_O)} + S_0 e^{i(\theta_S-\theta_O)})\,\overline{O_0}\big].$$

Hence only the two relative rotation angles $\theta_T - \theta_O$ and $\theta_S - \theta_O$ take effect.

Proof. Expanding,

$$(T_0 e^{i\theta_T} + S_0 e^{i\theta_S})\overline{O_0 e^{i\theta_O}} = (T_0 e^{i\theta_T} + S_0 e^{i\theta_S})\overline{O_0} e^{-i\theta_O}.$$

Distributing $e^{-i\theta_O}$ over each term gives

$$= T_0 e^{i(\theta_T-\theta_O)}\overline{O_0} + S_0 e^{i(\theta_S-\theta_O)}\overline{O_0} = \big[T_0 e^{i(\theta_T-\theta_O)} + S_0 e^{i(\theta_S-\theta_O)}\big]\overline{O_0}.$$

Taking the real part gives the result. ☐

Theorem 16.3 (Invariant under synchronous rotation)**.** If $\theta_T(t) = \theta_S(t) = \theta_O(t) = \theta(t)$, then the observation result is the same as in the static case:

$$\mathcal{O}(t) = \mathrm{Re}\big[(T_0(t) + S_0(t))\overline{O_0(t)}\big].$$

Proof. Substituting $\theta_T = \theta_S = \theta_O = \theta$, by Proposition 16.11 both relative angles are zero:

$$T_0 e^{i0} + S_0 e^{i0} = T_0 + S_0.$$

Hence $\mathcal{O} = \mathrm{Re}[(T_0 + S_0)\overline{O_0}]$, independently of θ. ☐

Corollary 16.3 (Frequency shift under asynchronous rotation)**.** Take all three planes to be single-frequency: $T_0 = Ae^{i\omega t}$, $S_0 = Be^{i\nu t}$, $O_0 = Ce^{i\mu t}$, with rotation angles $\theta_j(t) = \Omega_j t$. Then

$$\mathcal{O}(t) = AC\cos\big((\omega + \Omega_T - \mu - \Omega_O)t\big) + BC\cos\big((\nu + \Omega_S - \mu - \Omega_O)t\big).$$

Proof. By Proposition 16.11, $T_0 e^{i(\theta_T-\theta_O)} = Ae^{i(\omega+\Omega_T-\Omega_O)t}$ and $S_0 e^{i(\theta_S-\theta_O)} = Be^{i(\nu+\Omega_S-\Omega_O)t}$. Substituting and taking the real part gives the result. ☐

Property 16.3 (Basic properties of the rotating three planes). (i) **Relativity**: the observation result depends only on the relative rotation angles $\theta_T - \theta_O$ and $\theta_S - \theta_O$.

(ii) **Synchronous invariance**: when the three planes rotate synchronously, the observation result is unchanged.

(iii) **Frequency translation**: if $\theta_j = \Omega_j t$, then all component frequencies in plane j are translated by Ω_j as a whole.

Proof. (i)(ii) By Proposition 16.11 and Theorem 16.3. (iii) Directly from $e^{i(\omega_j t+\Omega_j t)} = e^{i(\omega_j+\Omega_j)t}$. □

Example 16.3 (Beat-frequency example of asynchronously rotating three planes). Take $T_0 = e^{i10t}$, $S_0 = e^{i12t}$, $O_0 = e^{i11t}$, $\theta_T = 0.5t$, $\theta_S = 0$, $\theta_O = 0$. By Corollary 16.3,

$$\mathcal{O}(t) = \cos(-0.5t) + \cos(t).$$

The two terms have frequencies 0.5 and 1 respectively, producing a slow beat.

Remark 16.7 (Choice of rotation angle). 'Rotation according to the properties of the internal epicycle' admits several implementations: $\theta_j(t) = \omega_{j,1}(t)t$, $\theta_j(t) = \arg X_j(t)$, $\theta_j(t) = \sum_k \omega_{j,k}(t)t$, and so on. This chapter does not fix one of them; all the conclusions hold for an arbitrary C^1 rotation angle.

16.5 Quaternion episphere groups

16.5.1 Basic definitions

Definition 16.17 (Quaternion episphere group). Let $K_T, K_S, K_O \in \mathbb{N}^+$. Let $a_k, b_l, c_m : \mathbb{R} \to \mathbb{R}$ be C^1 real-valued functions; let $\mathsf{u}_k, \mathsf{v}_l, \mathsf{w}_m : \mathbb{R} \to S^2$ be C^1 functions taking values in the unit pure imaginary quaternions; and let $\theta_k, \psi_l, \chi_m : \mathbb{R} \to \mathbb{R}$ be C^1 functions. Define

$$q_T(t) = \sum_{k=1}^{K_T} a_k(t) e^{\mathsf{u}_k(t)\theta_k(t)}, \quad q_S(t) = \sum_{l=1}^{K_S} b_l(t) e^{\mathsf{v}_l(t)\psi_l(t)}, \quad q_O(t) = \sum_{m=1}^{K_O} c_m(t) e^{\mathsf{w}_m(t)\chi_m(t)}.$$

They are called the **time episphere group**, the **space episphere group** and the **observer episphere group** respectively.

Definition 16.18 (Synthesis–projection observation result). Define the composite vector to be $q_T(t) + q_S(t)$; the observation result is

$$\boxed{\mathcal{Q}(t) = \mathrm{Re}\big[(q_T(t) + q_S(t))\,\overline{q_O(t)}\big]}.$$

Here $\overline{q_O}$ is the quaternion conjugate and Re takes the scalar part of the quaternion.

Remark 16.8 (Relation to the static model of the 'Vector analysis' chapter). Definition 16.18 is the dynamic version of the static projection model of the 'Vector Analysis' chapter under the 'compose first, then project' convention.

16.5.2 Basic properties

Proposition 16.12 (Distinction between scalar projection and conjugation projection). *The scalar part of the quaternion inner product* Re$[q\overline{p}]$ *and the conjugation action* $p^{-1}qp$ *are two different kinds of 'projection':*

(i) Re$[q\overline{p}]$ *is the inner-product projection on* $\mathbb{R}^4$*, and outputs a real number.*

(ii) $p^{-1}qp$ *is a rotation of* $SO(3)$*, and outputs a quaternion.*

This chapter adopts (i).

Proof. (i) By the definition of the quaternion inner product, Re$[q\overline{p}] = \sum_{i=0}^{3} q_i p_i \in \mathbb{R}$.

(ii) Take $q = \mathsf{i}$ and $p = \mathsf{j}$. Then $p^{-1} = \overline{p}/|p|^2 = -\mathsf{j}$, so

$$p^{-1}qp = (-\mathsf{j})\mathsf{i}\mathsf{j}.$$

From $\mathsf{i}\mathsf{j} = \mathsf{k}$ and $\mathsf{j}\mathsf{i} = -\mathsf{k}$:

$$(-\mathsf{j})\mathsf{i}\mathsf{j} = (-\mathsf{j})(\mathsf{i}\mathsf{j}) = (-\mathsf{j})\mathsf{k} = -\mathsf{j}\mathsf{k} = -\mathsf{i}.$$

Hence $p^{-1}qp = -\mathsf{i} \neq \mathsf{i} = q$; the conjugation action changes the quaternion. □

Lemma 16.3 (Symmetry of the quaternion inner product). *For any* $p, q \in \mathbb{H}$*,* Re$[p\overline{q}]$ = Re$[q\overline{p}]$.

Proof. The scalar part of $p\overline{q}$ is $p_0q_0 + p_1q_1 + p_2q_2 + p_3q_3$ (the vector parts and the non-commutative terms do not affect the scalar part). This is a symmetric real bilinear form, hence Re$[p\overline{q}]$ = Re$[q\overline{p}]$. □

Theorem 16.4 (Non-commutative synthesis and observation order). *Consider the additive synthesis* $X_+ = q_T + q_S$ *and the multiplicative synthesis* $X_\times = q_Tq_S$*. Then in general* Re$[X_+\overline{q_O}] \neq$ Re$[X_\times\overline{q_O}]$*, and* $q_Tq_S \neq q_Sq_T$*.*

Proof. Take $q_T = \mathsf{i}$, $q_S = \mathsf{j}$ and $q_O = \mathsf{k}$.

Addition: $X_+ = \mathsf{i} + \mathsf{j}$, so

$$X_+\overline{q_O} = (\mathsf{i} + \mathsf{j})(-\mathsf{k}) = -\mathsf{i}\mathsf{k} - \mathsf{j}\mathsf{k}.$$

From $\mathsf{i}\mathsf{k} = -\mathsf{j}$ and $\mathsf{j}\mathsf{k} = \mathsf{i}$ we get $X_+\overline{q_O} = \mathsf{j} - \mathsf{i}$, whose scalar part is 0.

Multiplication: $X_\times = \mathsf{i}\mathsf{j} = \mathsf{k}$, so

$$X_\times\overline{q_O} = \mathsf{k} \cdot (-\mathsf{k}) = -\mathsf{k}^2 = 1,$$

whose scalar part is 1. Hence the two differ.

Non-commutativity: $\mathsf{i}\mathsf{j} = \mathsf{k}$ and $\mathsf{j}\mathsf{i} = -\mathsf{k}$, hence $\mathsf{i}\mathsf{j} \neq \mathsf{j}\mathsf{i}$. □

Corollary 16.4 (Commutativity under the additive convention). *Under the additive convention,* Re$[(q_T + q_S)\overline{q_O}]$ = Re$[(q_S + q_T)\overline{q_O}]$.

Proof. Quaternion addition is commutative. □

Property 16.4 (Basic properties of quaternion dynamic projection). (i) **Real linearity**: for fixed q_O, $\mathcal{Q}$ is real-linear in q_T and q_S.

(ii) **Observer sign reversal**: $q_O \mapsto -q_O$ sends $\mathcal{Q} \mapsto -\mathcal{Q}$.

(iii) **Norm bound**: $|\mathcal{Q}(t)| \le |q_T(t) + q_S(t)| \cdot |q_O(t)|$.

(iv) **Scalar degeneracy**: if the quaternions degenerate to complex numbers, Definition 16.18 degenerates to Definition 16.14.

Proof. (i) By the linearity of quaternion addition, the anti-automorphism property of conjugation $(p+q)^* = p^* + q^*$ and the linearity of the scalar part.

(ii) $\overline{-q_O} = -\overline{q_O}$, and taking the scalar part changes the sign.

(iii) $|\mathcal{Q}| = |\operatorname{Re}[(q_T + q_S)\overline{q_O}]| \le |(q_T + q_S)\overline{q_O}| = |q_T + q_S| \cdot |q_O|$.

(iv) A complex number $z = x + iy$ embeds into the quaternions as $x + y\mathrm{i}$, and the scalar part of $z_1\overline{z_2}$ then agrees with the real part of the inner product in the complex case. ☐

Example 16.4 (Single-sphere-group dynamic projection example). Take $q_T = e^{\mathrm{i}t}$, $q_S = e^{\mathrm{j}t}$ and $q_O = 1$. Then

$$\mathcal{Q}(t) = \operatorname{Re}[e^{\mathrm{i}t} + e^{\mathrm{j}t}] = \cos t + \cos t = 2\cos t.$$

Remark 16.9 (Correspondence with the complex case). If all the quaternions are restricted to the same complex plane, the non-commutativity disappears, and Definition 16.18 degenerates completely to Definition 16.14.

16.6 Rotating three-sphere groups

16.6.1 Basic definitions

Definition 16.19 (Rotating three-sphere group). Let there be three mutually independent quaternion sphere groups $q_{T,0}(t), q_{S,0}(t), q_{O,0}(t)$. Let $r_T, r_S, r_O : \mathbb{R} \to S^3$ be C^1 functions taking values in the unit quaternions. Define

$$q_T(t) = r_T(t)\, q_{T,0}(t)\, r_T(t)^{-1}, \quad q_S(t) = r_S(t)\, q_{S,0}(t)\, r_S(t)^{-1}, \quad q_O(t) = r_O(t)\, q_{O,0}(t)\, r_O(t)^{-1}.$$

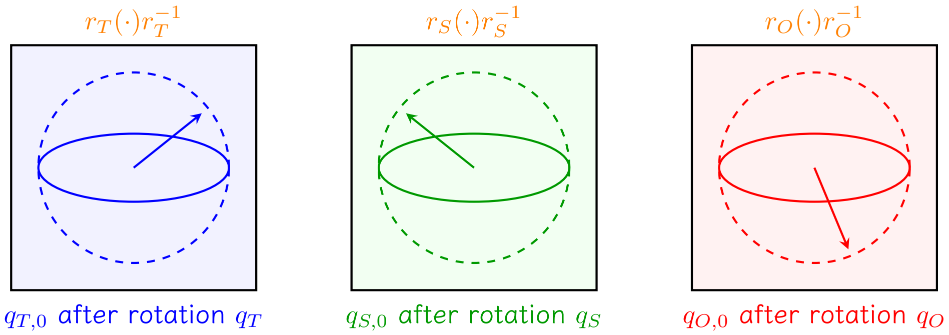

$r_T(\cdot)r_T^{-1}$
$r_S(\cdot)r_S^{-1}$
$r_O(\cdot)r_O^{-1}$
$q_{T,0}$ after rotation q_T
$q_{S,0}$ after rotation q_S
$q_{O,0}$ after rotation q_O

Figure 16.8: Rotating three-sphere groups: the three quaternion sphere groups are rotated by the conjugation action of the unit quaternions r_T, r_S, r_O respectively.

Definition 16.20 (Observation result for the rotating three-sphere groups).

$$\boxed{\mathcal{Q}(t) = \operatorname{Re}\big[(q_T(t) + q_S(t))\,\overline{q_O(t)}\big]}.$$

16.6.2 Basic properties

Proposition 16.13 (Norm preservation under rotation)**.** *For any unit quaternion r and any $q \in \mathbb{H}$, $|rqr^{-1}| = |q|$.*

Proof. By the multiplicativity of the quaternion norm $|pq| = |p| \cdot |q|$:

$$|rqr^{-1}| = |r| \cdot |q| \cdot |r^{-1}| = |q|,$$

where $|r| = 1$ and $|r^{-1}| = |r|^{-1} = 1$. □

Lemma 16.4 (Conjugation preserves the scalar part)**.** *For any unit quaternion r and any $x \in \mathbb{H}$, $\mathrm{Re}[rxr^{-1}] = \mathrm{Re}[x]$.*

Proof. Decompose x as $x = x_0 + \mathrm{v}$, with $x_0 \in \mathbb{R}$ and v a pure imaginary quaternion. Then

$$rxr^{-1} = r(x_0 + \mathrm{v})r^{-1} = x_0 \cdot rr^{-1} + r\mathrm{v}r^{-1} = x_0 + r\mathrm{v}r^{-1}.$$

The scalar part is $x_0 + \mathrm{Re}[r\mathrm{v}r^{-1}]$. Since the conjugation action preserves pure-imaginarity, $\mathrm{Re}[r\mathrm{v}r^{-1}] = 0$. Hence $\mathrm{Re}[rxr^{-1}] = x_0 = \mathrm{Re}[x]$. □

Theorem 16.5 (Invariant under synchronous rotation)**.** *If $r_T = r_S = r_O = r$, then*

$$\mathcal{Q}(t) = \mathrm{Re}\big[(q_{T,0}(t) + q_{S,0}(t))\overline{q_{O,0}(t)}\big].$$

Proof. By Definition 16.19, $q_T = rq_{T,0}r^{-1}$ and so on. Hence

$$q_T + q_S = r(q_{T,0} + q_{S,0})r^{-1}.$$

Conjugation: since r is a unit quaternion, $r^{-1} = \overline{r}$, so

$$\overline{q_O} = \overline{rq_{O,0}r^{-1}} = \overline{r^{-1}}\,\overline{q_{O,0}}\,\overline{r} = r\overline{q_{O,0}}r^{-1}.$$

Substituting:

$$\mathcal{Q} = \mathrm{Re}\left[r(q_{T,0} + q_{S,0})r^{-1} \cdot r\overline{q_{O,0}}r^{-1}\right] = \mathrm{Re}\left[r(q_{T,0} + q_{S,0})\overline{q_{O,0}}r^{-1}\right].$$

By Lemma 16.4, taking $x = (q_{T,0} + q_{S,0})\overline{q_{O,0}}$, we get

$$\mathcal{Q} = \mathrm{Re}\left[(q_{T,0} + q_{S,0})\overline{q_{O,0}}\right],$$

independently of r. □

Corollary 16.5 (Non-triviality under asynchronous rotation)**.** *If r_T, r_S, r_O are not equal, then*

$$\mathcal{Q}(t) = \mathrm{Re}\big[r_T q_{T,0} r_T^{-1}\overline{r_O q_{O,0} r_O^{-1}} + r_S q_{S,0} r_S^{-1}\overline{r_O q_{O,0} r_O^{-1}}\big],$$

which is in general not equal to the static case.

Proof. Expanding by Definition 16.20 gives the result. □

Property 16.5 (Basic properties of the rotating three-sphere groups)**.** (i) **Synchronous invariance**: *when the three sphere groups rotate synchronously, the observation result is unchanged.*

(ii) **Norm preservation**: the norm of each sphere-group vector is unchanged under rotation.

(iii) **Non-commutativity**: under asynchronous rotation the observation result depends on the order of rotation.

(iv) **Degeneracy to the complex case**: when the quaternions degenerate to complex numbers, the conclusions of this section agree with Section 16.4.

Proof. (i)(ii) By Theorem 16.5 and Proposition 16.13. (iii) By the non-commutativity of quaternion multiplication. (iv) In the complex case $rqr^{-1} = q$ holds identically. ☐

Remark 16.10 (Relation to the static model of the 'Vector analysis' chapter). Definition 16.20 is the dynamic rotating version of the static projection model of the 'Vector Analysis' chapter under the 'compose first, then project' convention.

16.7 Unified observer projection framework

16.7.1 Unified operator

Definition 16.21 (Unified observer operator). Let $\mathcal{H}$ be $\mathbb{C}$, $\mathbb{H}$ or a finite direct sum of these. Let $X \in \mathcal{H}$ be a composite vector and $O \in \mathcal{H}$ an observer vector. Define

$$\Pi_O(X) = \mathrm{Re}\big[X\,\overline{O}\big].$$

In the complex case $\Pi_O(X) \in \mathbb{R}$; in the quaternion case $\Pi_O(X)$ takes values in the scalar part of the quaternion.

Theorem 16.6 (Unified formulation of the five models). All five models of this chapter can be written as special cases of Definition 16.21, and all follow 'compose first, then project':

(i) **Model 1**: $X = \mathcal{E}(t)$, $O = u(t)$ a unit complex number.

(ii) **Model 2**: $X = T(t) + S(t)$, $O = O(t)$.

(iii) **Model 3**: $X = T_0 e^{i\theta_T} + S_0 e^{i\theta_S}$, $O = O_0 e^{i\theta_O}$.

(iv) **Model 4**: $X = q_T + q_S$, $O = q_O$, and the projection takes the scalar part of the quaternion inner product.

(v) **Model 5**: $X = r_T q_{T,0} r_T^{-1} + r_S q_{S,0} r_S^{-1}$, $O = r_O q_{O,0} r_O^{-1}$.

Proof. Comparing item by item with Definitions 16.12, 16.14, 16.16, 16.18 and 16.20 suffices. ☐

Corollary 16.6 (Axioms of the projection operator). Π_O satisfies:

(i) **Real linearity**: $\Pi_O(\alpha X + \beta Y) = \alpha \Pi_O(X) + \beta \Pi_O(Y)$.

(ii) **Observer sign reversal**: $\Pi_{-O} = -\Pi_O$.

(iii) **Non-expansiveness**: $|\Pi_O(X)| \le |X| \cdot |O|$.

(iv) **Zero-observer degeneracy**: $O = 0 \Rightarrow \Pi_O \equiv 0$.

Proof. (i) By the real linearity of the scalar part of the complex/quaternion inner product in X.
(ii) $\overline{-O} = -\overline{O}$, so $X\overline{-O} = -(X\overline{O})$.
(iii) $|\Pi_O(X)| = |\,\mathrm{Re}[X\overline{O}]| \le |X\overline{O}| = |X| \cdot |O|$.
(iv) When $O = 0$ we have $\overline{O} = 0$, so $X\overline{O} = 0$. ☐

Property 16.6 (Kernel and invariants of the unified projection). (i) **Kernel**: $\ker \Pi_O = \{X : \mathrm{Re}[X\overline{O}] = 0\}$. In the complex case $\mathbb{C}$ the kernel is the one-dimensional real subspace perpendicular to O; in the quaternion case $\mathbb{H}$ the kernel is the three-dimensional real subspace perpendicular to O.

(ii) **Invariant subspace**: on $\mathrm{span}_{\mathbb{R}}\{O\}$, Π_O is the identity (when $|O| = 1$).

Proof. (i) The kernel condition is simply that the inner product vanishes, and the real dimension is given by that of the inner product: $\mathbb{C} \cong \mathbb{R}^2$, so the kernel is one-dimensional; $\mathbb{H} \cong \mathbb{R}^4$, so the kernel is three-dimensional.
(ii) $\Pi_O(O) = \mathrm{Re}[O\overline{O}] = \mathrm{Re}[|O|^2] = 1$ (for $|O| = 1$). By linearity, $\Pi_O(cO) = c$ for any $c \in \mathbb{R}$. ☐

16.8 Observer functor and dynamic reconstruction

This section raises the observer from a 'dynamic vector' to a 'functor', and establishes the adjunction structure of observation and reconstruction. This is a categorification of Section 16.7.

Definition 16.22 (Observer functor). Let $\mathcal{H}$ be $\mathbb{C}$, $\mathbb{H}$ or a finite direct sum of these, and let $\mathcal{H}_{\mathrm{obs}}$ be an observable scalar space ($\mathbb{R}$ or a finite direct sum of these). Fix an observer family $\{O(t)\}_{t\in\mathbb{R}}$ ($O(t) \in \mathcal{H}$, C^1). Define the **observer functor**

$$\Pi_O : \mathcal{H} \longrightarrow \mathcal{H}_{\mathrm{obs}}, \qquad \Pi_O(X) = \mathrm{Re}[X\,\overline{O(t)}].$$

In the quaternion case the right-hand side takes the scalar part.

Proposition 16.14 (Basic properties of the observer functor). Π_O satisfies:

(i) **Real linearity**: $\Pi_O(\alpha X + \beta Y) = \alpha\Pi_O(X) + \beta\Pi_O(Y)$;

(ii) **Sign-reversal covariance**: $\Pi_{-O} = -\Pi_O$;

(iii) **Non-expansiveness**: $|\Pi_O(X)| \le |X| \cdot |O|$;

(iv) **Zero-observer degeneracy**: $O = 0 \Rightarrow \Pi_O \equiv 0$.

Proof. All four are verified directly from the real bilinearity of the inner product and the norm inequality. ☐

Definition 16.23 (Reconstruction functor)**.** Let $y \in \mathcal{H}_{\text{obs}}$ be an observable scalar. Define the **reconstruction functor** $\Pi_O^* : \mathcal{H}_{\text{obs}} \to \mathcal{H}$ by

$$\Pi_O^*(y) = \frac{y}{|O|^2} O,$$

that is, back-project the scalar y along the observer direction O. Equivalent notation: $\Pi_O^*(y) = \frac{y}{O}$ (when $\mathcal{H} = \mathbb{C}$).

Proposition 16.15 (Observation–reconstruction inverse condition)**.** Let $O \neq 0$. Then

$$\Pi_O \Pi_O^* = \mathrm{id}_{\mathcal{H}_{\text{obs}}}, \qquad \Pi_O^* \Pi_O = \text{orthogonal projection along } O.$$

That is, reconstruction is the identity at the level of observable scalars, whilst at the level of the source space it preserves the observer direction and erases the component perpendicular to the observer direction.

Proof. Direct substitution gives: $\Pi_O \Pi_O^*(y) = \mathrm{Re}\left[\frac{y}{|O|^2} O \overline{O}\right] = \frac{y}{|O|^2}|O|^2 = y$. For the second identity: $\Pi_O^* \Pi_O(X) = \frac{\mathrm{Re}[X\overline{O}]}{|O|^2} O$, which is exactly the orthogonal projection of X along O. □

Theorem 16.7 (Adjunction between the observer functor and the reconstruction functor)**.** Define the inner product $\langle X, Y \rangle_{\mathcal{H}} = \mathrm{Re}[X\overline{Y}]$. Then

$$\langle \Pi_O(X),\, y \rangle_{\mathcal{H}_{\text{obs}}} = \langle X,\, \Pi_O^*(y) \rangle_{\mathcal{H}},$$

that is, $\Pi_O \dashv \Pi_O^*$. The unit η and the counit ε of the adjunction are respectively

$$\eta : X \mapsto \Pi_O^* \Pi_O(X), \qquad \varepsilon : y \mapsto \Pi_O \Pi_O^*(y) = y.$$

The unit η is not an isomorphism—the information loss in the observation process is irreversible.

Proof. Compute $\langle \Pi_O(X), y \rangle = y \cdot \mathrm{Re}[X\overline{O}]$, whilst $\langle X, \Pi_O^*(y) \rangle = \mathrm{Re}\left[X \overline{\frac{y}{|O|^2} O}\right] = \frac{y}{|O|^2} \mathrm{Re}[X\overline{O}]$. The two are equal. ε is the identity by Proposition 16.15; η is not an isomorphism because $\ker \Pi_O \neq \{0\}$. □

Corollary 16.7 (Unobservable subspace)**.** $\ker \Pi_O = \{X \in \mathcal{H} : \mathrm{Re}[X\overline{O}] = 0\}$. In the complex case $\mathcal{H} = \mathbb{C}$ the kernel is the one-dimensional real subspace perpendicular to O; in the quaternion case $\mathcal{H} = \mathbb{H}$ the kernel is the three-dimensional real subspace perpendicular to O.

Proof. By the definition of the inner product, the kernel condition is simply that X is orthogonal to O. The real dimensions are obtained by subtracting $\mathrm{span}_{\mathbb{R}}\{O\}$ from $\dim_{\mathbb{R}} \mathbb{C} = 2$ and $\dim_{\mathbb{R}} \mathbb{H} = 4$ respectively. □

Example 16.5 (Reconstruction of a single-frequency source under a rotating observer)**.** Take $X(t) = Ae^{i\omega t}$ and $O(t) = e^{i\omega_0 t}$. Then $\Pi_O(X) = A\cos((\omega - \omega_0)t)$. Reconstruction gives

$$\Pi_O^*(\Pi_O(X)) = \frac{A\cos((\omega - \omega_0)t)}{1} e^{i\omega_0 t},$$

that is, the observable scalar is reinstalled on the observer frequency, and the independent information of the source frequency ω and the observer frequency ω_0 is lost—only the difference $\omega - \omega_0$ is retained.

Remark 16.11 (Scope of the observer functor). The definition of the observer functor rests on three prerequisites:

(i) the observer is non-zero: $O \neq 0$ (otherwise the reconstruction functor is undefined);

(ii) the inner product takes real values: in the complex case $\mathrm{Re}[X\overline{O}]$, in the quaternion case the scalar part;

(iii) the observer varies with time: when $O = O(t)$ is C^1, Π_O is the functor family $\{\Pi_{O(t)}\}_{t\in\mathbb{R}}$.

16.9 Projection as information compression

This section interprets the projection operator Π_O as an information-compression map and gives a theoretical formulation of its information loss. The core conclusion is that projection does not increase information, and that the information loss can be quantified in two ways: by the reconstruction error and by the loss of mutual information. The theoretical basis of this section is the data processing inequality and rate–distortion theory from information theory.

Definition 16.24 (Projection channel). Let $\mathcal{H}$ be $\mathbb{C}$, $\mathbb{H}$ or a finite direct sum of these. Fix an observer $O \in \mathcal{H}$, $O \neq 0$. Define the **projection channel**

$$\Pi_O : \mathcal{H} \to \mathcal{H}_{\mathrm{obs}}, \qquad \Pi_O(X) = \mathrm{Re}[X\overline{O}].$$

In the quaternion case the right-hand side takes the scalar part. Π_O is a deterministic map.

Proposition 16.16 (Projection is an idempotent linear operator). Π_O satisfies:

(i) **Real linearity**: $\Pi_O(\alpha X + \beta Y) = \alpha\Pi_O(X) + \beta\Pi_O(Y)$;

(ii) **Idempotence**: $\Pi_O \circ \Pi_O^* \circ \Pi_O = \Pi_O$, where Π_O^* is the reconstruction functor (Definition 16.23).

Proof. (i) By the real bilinearity of the inner product. (ii) By Proposition 16.15, $\Pi_O\Pi_O^* = \mathrm{id}$, hence $\Pi_O\Pi_O^*\Pi_O = \Pi_O$. □

Theorem 16.8 (Information loss under projection—the data processing inequality). Let X be a random vector taking values in $\mathcal{H}$, let $Y = \Pi_O(X)$ be the projection output, and let $Z = \Pi_O^*(Y)$ be the reconstruction. Then the Markov chain

$$X \longrightarrow Y \longrightarrow Z$$

holds, and the mutual information satisfies

$$I(X;Y) \geq I(X;Z).$$

That is, **projection does not increase information**. Equality holds if and only if $I(X;Y \mid Z) = 0$—equivalently, Z already contains all the information that X has about Y, that is, in the decomposition $X = X_\parallel + X_\perp$ of X (with $X_\parallel \in \mathrm{span}\{O\}$ and $X_\perp \in \ker\Pi_O$, see Corollary 16.8), the component $X_\perp$ carries no information about Y. This condition is quantitative and verifiable (one simply computes the conditional mutual information); it is no longer the qualitative statement that equality 'may hold'.

Proof. Y is given deterministically by X, and Z is given deterministically by Y, hence $X \to Y \to Z$ is a Markov chain. By the data processing inequality (DPI), for any Markov chain $X \to Y \to Z$ we have $I(X;Y) \geq I(X;Z)$. ☐

Corollary 16.8 (Unobservable subspace and information loss). Let $\ker \Pi_O = \{X \in \mathcal{H} : \mathrm{Re}[X\overline{O}] = 0\}$. Then:

(i) if $X \in \ker \Pi_O$, then $\Pi_O(X) = 0$, and the projection output contains no information whatever about X;

(ii) if the projection of X on $\ker \Pi_O$ is non-zero, then that component contributes nothing to Y, and its information is discarded completely.

Proof. (i) By the definition of $\ker \Pi_O$. (ii) Decompose X as $X = X_{\parallel} + X_{\perp}$, with $X_{\parallel} \in \mathrm{span}\{O\}$ and $X_{\perp} \in \ker \Pi_O$. By linearity $\Pi_O(X) = \Pi_O(X_{\parallel}) + \Pi_O(X_{\perp}) = \Pi_O(X_{\parallel})$, so the information of $X_{\perp}$ cannot be recovered from Y. ☐

Definition 16.25 (Reconstruction error). Let $X \in \mathcal{H}$ and let $\widehat{X} = \Pi_O^*(\Pi_O(X))$ be the reconstruction. Define the **reconstruction error** by

$$\mathcal{E}_{\mathrm{rec}}(X) = X - \widehat{X}.$$

By Proposition 16.15, $\widehat{X}$ is the orthogonal projection of X along O, hence $\mathcal{E}_{\mathrm{rec}}(X) \in \ker \Pi_O$, and

$$|\mathcal{E}_{\mathrm{rec}}(X)|^2 = |X|^2 - |\widehat{X}|^2 = |X|^2 - \frac{|\,\mathrm{Re}[X\overline{O}]|^2}{|O|^2}.$$

Proposition 16.17 (Geometric interpretation of the reconstruction error). The reconstruction error $\mathcal{E}_{\mathrm{rec}}(X)$ is the orthogonal projection of X onto $\ker \Pi_O$, and

$$\mathcal{E}_{\mathrm{rec}}(X) = \Pi_{\mathrm{ker}}(X),$$

where Π_{ker} is the orthogonal projection onto $\ker \Pi_O$. Hence the reconstruction error is exactly **the whole of the component discarded by the projection**.

Proof. By Definition 16.25, $X = \widehat{X} + \mathcal{E}_{\mathrm{rec}}(X)$ with $\widehat{X} \in \mathrm{span}\{O\}$ and $\mathcal{E}_{\mathrm{rec}}(X) \in \ker \Pi_O$. By the uniqueness of the orthogonal decomposition, $\mathcal{E}_{\mathrm{rec}}(X)$ is the orthogonal projection of X onto $\ker \Pi_O$. ☐

Theorem 16.9 (Quantitative bounds on information compression). Let X be a random vector taking values in $\mathcal{H}$, with $|X|$ finite almost everywhere. Then:

(i) **Norm compression ratio**: $|\Pi_O(X)| \leq |X| \cdot |O|$, with equality if and only if $X \parallel O$;

(ii) **Information-rate compression ratio**: if the differential entropy $h(X)$ of X is finite, then

$$h(\Pi_O(X)) \leq h(X) + \log |O|,$$

and $h(\Pi_O(X))$ is finite when X has a bounded distribution on $\mathrm{span}\{O\}$;

(iii) **Upper bound on the mutual-information loss**:

$$I(X; \Pi_O(X)) \leq h(\Pi_O(X)) - \mathbb{E}[\log |\mathcal{E}_{\mathrm{rec}}(X)|].$$

Proof. (i) By Corollary 16.6(iii). The equality condition is given by the equality condition of the Cauchy–Schwarz inequality.

(ii) $\Pi_O(X)$ is a deterministic function of X, hence $h(\Pi_O(X)) \leq h(X) + \mathbb{E}[\log|\partial\Pi_O/\partial X|]$, and the Jacobian norm of Π_O is $|O|$, so $h(\Pi_O(X)) \leq h(X) + \log|O|$.

(iii) By the definition of mutual information, $I(X;\Pi_O(X)) = h(\Pi_O(X)) - h(\Pi_O(X)|X)$, whilst $h(\Pi_O(X)|X = x) = -\mathbb{E}[\log|\mathcal{E}_{\text{rec}}(x)|]$ (given X, the distribution of $\Pi_O(X)$ is determined by the reconstruction error); substituting gives the result. □

Corollary 16.9 (An ideal observer loses no information). *If and only if* $\ker\Pi_O = \{0\}$ *(that is,* $\mathcal{H} = \text{span}\{O\}$*, for example* $\mathcal{H} = \mathbb{R}$ *with* $O \neq 0$*), the projection loses no information:* $I(X;\Pi_O(X)) = I(X;X) = h(X)$.

Proof. When $\ker\Pi_O = \{0\}$, Π_O is injective and X can be recovered completely from $\Pi_O(X)$, so the mutual information equals $h(X)$. □

Theorem 16.10 (Projection from the rate–distortion perspective). *Let X be a random vector taking values in $\mathcal{H}$, and take the distortion measure to be the squared Euclidean distance $d(X,\widehat{X}) = |X - \widehat{X}|^2$. Then the projection Π_O is a* **zero-rate encoder** *(coding rate $R = 0$), with distortion*

$$D_O = \mathbb{E}|\mathcal{E}_{\text{rec}}(X)|^2 = \mathbb{E}|X|^2 - \frac{\mathbb{E}|\operatorname{Re}[X\overline{O}]|^2}{|O|^2}.$$

In rate–distortion theory, D_O is the least attainable distortion at rate $R = 0$ (for the given O).

Proof. $R = 0$ means that the encoder conveys no information, so the decoder can only output an estimate independent of X. The optimal zero-rate estimate is $\mathbb{E}[X]$; but if the estimate is constrained to lie in span$\{O\}$, the optimal estimate is $\Pi_O^*(\Pi_O(X))$, whose distortion is D_O. This is a direct application of rate–distortion theory at $R = 0$. □

Example 16.6 (Single-direction projection of a uniformly distributed source on S^1). Take $\mathcal{H} = \mathbb{C}$ and $O = e^{i\theta_0}$. Let $X = e^{i\Theta}$ with Θ uniformly distributed on $[0, 2\pi)$. Then $\Pi_O(X) = \cos(\Theta - \theta_0)$, whose distribution is the Arcsine distribution (density $\frac{1}{\pi\sqrt{1-y^2}}$, $y \in (-1,1)$). The reconstruction error is $\sin(\Theta - \theta_0)$, which is orthogonal to $\Pi_O(X)$. The entropy is $h(\Pi_O(X)) = \log(\pi/2)$, whilst $h(X) = \log(2\pi)$, so the information loss is $h(X) - h(\Pi_O(X)) = \log 4$—twice the 'dimensionality loss' of passing from two dimensions to one.

Property 16.7 (Basic properties of projection information compression).

(i) **Monotonicity**: if O_1, O_2 satisfy span$\{O_1\} \subseteq$ span$\{O_2\}$, then $I(X;\Pi_{O_1}(X)) \leq I(X;\Pi_{O_2}(X))$;

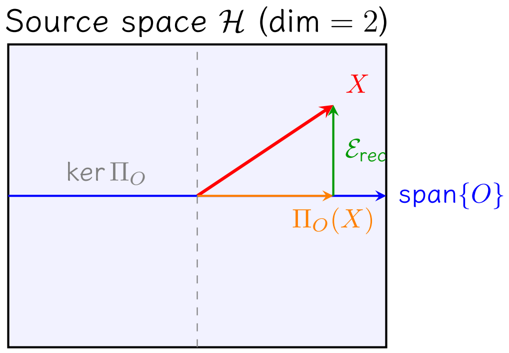
(ii) **Direction dependence**: for fixed $|O|$, $I(X;\Pi_O(X))$ depends in general on the direction of O;

(iii) **Normalisation invariance**: $\Pi_{cO} = \Pi_O$ for any $c \neq 0$ (if O is normalised).

Proof. (i) When $\mathrm{span}\{O_1\} \subseteq \mathrm{span}\{O_2\}$, the information of Π_{O_1} is a deterministic function of the information of Π_{O_2}, and the monotonicity of mutual information follows from the DPI.

(ii) For example, when X is concentrated on $\mathrm{span}\{O\}$ we have $I(X;\Pi_O(X)) = h(X)$, whereas when it is concentrated on $\ker \Pi_O$ we have $I = 0$.

(iii) $\Pi_{cO}(X) = \mathrm{Re}[X\overline{cO}] = c\,\mathrm{Re}[X\overline{O}] = c\Pi_O(X)$; if $c > 0$ then Π_{cO} and Π_O differ by a positive scalar factor, and are the same in the normalised sense. $\square$


Source space $\mathcal{H}$ (dim $= 2$)
X
$\mathcal{E}_{\mathrm{rec}}$
$\ker \Pi_O$
$\mathrm{span}\{O\}$
$\Pi_O(X)$

$I(X;\Pi_O(X)) \le I(X;X)$: projection does not increase information

Figure 16.9: Projection as information compression: the source vector X decomposes into an observable component $\Pi_O(X)$ and an unobservable component $\mathcal{E}_{\mathrm{rec}} \in \ker \Pi_O$. The projection discards all the information in the $\ker \Pi_O$ direction.

Remark 16.12 (Interface with existing information-theoretic frameworks). The projection information compression of this section connects directly with the following well-established theories:

(i) **Data processing inequality (DPI)**: Theorem 16.8 is the special case of the DPI for deterministic projections. The general form of the DPI is: if $X \to Y \to Z$ is a Markov chain, then $I(X;Y) \ge I(X;Z)$.

(ii) **Rate–distortion theory**: Theorem 16.10 interprets the projection as a zero-rate encoder, whose distortion D_O is the value of the rate–distortion function $R(D)$ at $R = 0$.

(iii) **Information loss of PCA**: Geiger and Kubin (2012) proved that the relative information loss of PCA dimensionality reduction is the same as that of direct dimensionality reduction, and that when the rotation matrix is unavailable the loss is infinite. Corollary 16.8 of this section gives the explicit form of this conclusion for a single-direction projection.

(iv) **Information bottleneck method**: Chigirev and Bialek (2003) formalised dimensionality reduction as an information bottleneck—the manifold coordinates can acquire only finitely many bits, which naturally determines the optimal dimension. The projection framework of this section may be regarded as the special case of the information bottleneck in the 'single-direction' situation.

Remark 16.13 (Scope and open problems). Prerequisites of the conclusions of this section:

(i) $O \neq 0$; when $O = 0$ the projection is identically zero and all information is lost;

(ii) X is a random vector taking values in $\mathcal{H}$, with $|X|$ finite almost everywhere;

(iii) the existence of the mutual information and of the differential entropy requires the distribution of X to be sufficiently regular (for example absolutely continuous with finite entropy).

The following questions are left open:

(a) connecting the upper bound on the mutual information in Theorem 16.9 quantitatively with the information loss with distance theorem of this chapter (Theorem 16.13);

(b) the joint information loss of multi-direction projections (observer groups) and the choice of optimal directions;

(c) a quantitative description of the information loss of projection in the quaternion case (a preliminary analysis shows that the dimension of the scalar part drops from 4 to 1, so the information loss exceeds that of the complex case).

16.10 Discrete unitary dynamic projection

This section discretises the continuous-time evolution of the dynamic epicycle group and establishes the interface with the discrete unitary theory of the 'Periodic Manifold' chapter.

Definition 16.26 (Time-discretised dynamic epicycle group). Let the time step be $\Delta t > 0$ and $t_m = m\Delta t$ $(m \in \mathbb{N})$. Let the discrete amplitudes $A_k[m]$, discrete frequencies $\omega_k[m]$ and discrete phases $\phi_k[m]$ satisfy the same regularity as in the continuous case. Define the **discrete dynamic epicycle group**

$$\mathcal{E}_m = \sum_{k=1}^{K} A_k[m]\, e^{i(\omega_k[m] t_m + \phi_k[m])}.$$

If all the parameters are independent of m, it degenerates to the static epicycle group under equally spaced sampling.

Definition 16.27 (Discrete unitary evolution operator). $U_m \in U(N)$ is called a **discrete unitary evolution operator** if it maps the state X_m at time t_m to the state at time t_{m+1}:

$$X_{m+1} = U_m X_m.$$

In the case of a dynamic epicycle group, U_m is given by the discrete update rules for the frequencies, amplitudes and phases.

Proposition 16.18 (Modulus-squared conservation under discrete unitary evolution)**.** If U_m is unitary, then

$$|X_{m+1}|^2 = X_{m+1}^* X_{m+1} = X_m^* X_m = |X_m|^2.$$

Hence discrete unitary evolution conserves the modulus squared exactly for any finite step Δt, independently of $\Delta t \to 0$.

Proof. $X_{m+1}^* X_{m+1} = X_m^* U_m^* U_m X_m = X_m^* X_m$. ☐

Theorem 16.11 (Periodicity condition for discrete unitary evolution)**.** Let U be a constant unitary operator and $X_{m+1} = UX_m$. Then there exists a positive integer p with $X_{m+p} = X_m$ for all m if and only if $U^p = I$, equivalently all the eigenvalues of U are p-th roots of unity.

Proof. $X_{m+p} = U^p X_m$. For any initial value X_0 we have $U^p X_0 = X_0$ if and only if $U^p = I$. By the spectral theorem for unitary operators, $U^p = I$ is equivalent to all the eigenvalues satisfying $\lambda^p = 1$. ☐

Corollary 16.10 (Interface between discrete unitary evolution and Floquet theory)**.** The eigenvalue $e^{-i\varepsilon}$ of the discrete unitary operator U corresponds to the quasi-energy ε, uniquely determined in the Floquet–Brillouin zone $[-\pi, \pi)$. This agrees completely with Proposition 12.22 of the 'Periodic Manifold' chapter, except that the lattice symbol n used there is replaced by the time index m of this chapter.

Proposition 16.19 (Continuity condition for discrete dynamic projection)**.** If the discrete parameters $A_k[m], \omega_k[m], \phi_k[m]$ satisfy a Lipschitz condition in m (that is, $|A_k[m+1] - A_k[m]| \le L\Delta t$ and so on), then as $\Delta t \to 0$ the discrete dynamic projection $\mathcal{O}_m$ converges uniformly to the continuous dynamic projection $\mathcal{O}(t)$.

Proof. The difference between the discrete and the continuous dynamic projections is controlled by the triangle inequality: $|\mathcal{O}_m - \mathcal{O}(t_m)| \le C(\Delta t)$, where C is given by the Lipschitz constant and a bound on $\mathcal{E}$. ☐

Example 16.7 (Second-order periodic oscillation)**.** Take $U = -I$; then $X_{m+1} = -X_m$, $|X_m|$ is conserved, and the phase flips by π at each step. $p = 2$ is the least period. This is the typical instance of a C_2 involution in discrete unitary evolution.

Example 16.8 (C_4 periodic rotation)**.** Take $U = R_{\pi/2}$ (rotation of the plane by $\pi/2$); then $U^4 = I$ and $p = 4$. This agrees with Example 12.5 of the 'Periodic Manifold' chapter.

Remark 16.14 (Relation between discrete unitary and continuous dynamic). The discrete unitary dynamic projection is the realisation of the continuous dynamic projection under equally spaced sampling. The differences between the two are threefold:

(i) in the discrete case the conservation of the modulus squared is exact, whilst in the continuous case it holds in the limiting sense;

(ii) the discrete period p and the continuous period T are related by $T = p\Delta t$;

(iii) in the discrete case U is allowed to differ at each step (time-varying unitary), in which case the conservation of the modulus squared still holds exactly, but the periodicity condition must be checked segment by segment.

16.11 Scale-dependent sampling theory

This section systematically establishes scale-dependent sampling theory. The sampling of the source by the observer is controlled by two parameters: the observer scale σ and the observation distance r. This section gives the rigorous definitions and basic properties of the sampling rate, the number of sampling points, the sampling duration and the sampling orbit. This theory is the foundation of the next section, 'information loss with distance'.

Definition 16.28 (Scale-dependent sampling rate)**.** Let the observer scale be $\sigma > 0$, the observation distance $r > 0$ and the reference sampling rate $R_0 > 0$. Define the **sampling rate**

$$R(r, \sigma) = R_0 \frac{\sigma}{r^2 + \sigma}.$$

$R(r, \sigma)$ denotes the number of samples the observer collects from the source per unit time.

Definition 16.29 (Scale-dependent number of sampling points)**.** Let the reference number of sampling points be $N_0 > 0$. Define the **number of sampling points**

$$N(r, \sigma) = N_0 \frac{\sigma}{r^2 + \sigma}.$$

$N(r, \sigma)$ denotes the total number of samples the observer can collect in a unit duration.

Definition 16.30 (Scale-dependent sampling duration)**.** Let the fundamental period of the source be $T_{\mathsf{src}} > 0$. Define the **sampling duration**

$$T_{\mathsf{samp}}(r, \sigma) = \frac{N(r, \sigma)}{R(r, \sigma)} = \frac{N_0}{R_0},$$

that is, the sampling duration is independent of both the distance and the observer scale, and is determined solely by the reference parameters N_0, R_0.

Proposition 16.20 (Invariance of the sampling duration)**.** The sampling duration T_{samp} is independent of both the distance r and the observer scale σ:

$$\frac{\partial T_{\mathsf{samp}}}{\partial r} = \frac{\partial T_{\mathsf{samp}}}{\partial \sigma} = 0.$$

Hence the sampling duration is a **scale invariant**—a fact that contrasts with the distance dependence of the sampling rate and of the number of sampling points.

Proof. By Definition 16.30, $T_{\mathsf{samp}} = N_0/R_0$ is a constant, so both partial derivatives vanish. □

Definition 16.31 (Sampling orbit)**.** Let the continuous trajectory of the source be $X : \mathbb{R} \to \mathcal{H}$ and let the sampling step be $\tau(r, \sigma) = 1/R(r, \sigma)$. Define the **sampling orbit** by

$$\{(t_m,\ X(t_m)) : t_m = m\tau(r, \sigma),\ m \in \mathbb{Z}\}.$$

The sampling orbit is the sequence of values of the source trajectory at the discrete time points.

Proposition 16.21 (Lattice spacing of the sampling orbit). The relation between the sampling step and the distance is

$$\tau(r,\sigma) = \frac{1}{R(r,\sigma)} = \frac{r^2+\sigma}{R_0\sigma}.$$

Hence τ is monotonically increasing in r and monotonically decreasing in σ:

$$\frac{\partial\tau}{\partial r} > 0, \qquad \frac{\partial\tau}{\partial\sigma} < 0.$$

That is, the greater the distance, the sparser the sampling points; the larger the observer scale, the denser the sampling points.

Proof. Differentiating directly from Definition 16.28 and Definition 16.31: $\tau = (r^2+\sigma)/(R_0\sigma)$, $\partial\tau/\partial r = 2r/(R_0\sigma) > 0$, $\partial\tau/\partial\sigma = -r^2/(R_0\sigma^2) < 0$. ☐

Theorem 16.12 (Resolution of the sampling orbit). Let the least resolvable period of the source be T_{min}. Then by the sampling theorem, the necessary and sufficient condition for the sampling orbit to resolve T_{min} is

$$\tau(r,\sigma) \le T_{\mathsf{min}}/2,$$

that is,

$$r^2 \le \frac{R_0\sigma T_{\mathsf{min}}}{2} - \sigma = \sigma\left(\frac{R_0 T_{\mathsf{min}}}{2} - 1\right).$$

Hence there exists a **resolution distance** $r_{\mathsf{res}} = \sqrt{\sigma(R_0 T_{\mathsf{min}}/2 - 1)}$; when $r > r_{\mathsf{res}}$, the sampling orbit cannot resolve the least resolvable period of the source.

Proof. By the Nyquist–Shannon sampling theorem, only when the sampling frequency satisfies $R(r,\sigma) \ge 2/T_{\mathsf{min}}$ can the least period T_{min} be recovered without aliasing. Substituting into Definition 16.28 gives the result. ☐

Corollary 16.11 (Information recoverability of the sampling orbit). When $r \le r_{\mathsf{res}}$, the source trajectory can be recovered uniquely from the sampling orbit; when $r > r_{\mathsf{res}}$, the high-frequency components of the source trajectory are aliased and cannot be recovered completely.

Proof. By the Nyquist–Shannon sampling theorem: when the sampling condition is met, complete recovery is possible; when it is not, the high-frequency components are aliased. ☐

Proposition 16.22 (Monotonicity of the sampling function). For any $r, \sigma > 0$,

$$\frac{\partial R}{\partial r} < 0, \qquad \frac{\partial N}{\partial r} < 0, \qquad \frac{\partial R}{\partial\sigma} \ge 0, \qquad \frac{\partial N}{\partial\sigma} \ge 0.$$

That is, the sampling rate and the number of sampling points decrease strictly with distance and are non-decreasing in the observer scale.

Proof. Differentiating directly from Definition 16.28: $\partial R/\partial r = -2R_0\sigma r/(r^2+\sigma)^2 < 0$, $\partial R/\partial\sigma = R_0 r^2/(r^2+\sigma)^2 > 0$. The same holds for N. ☐

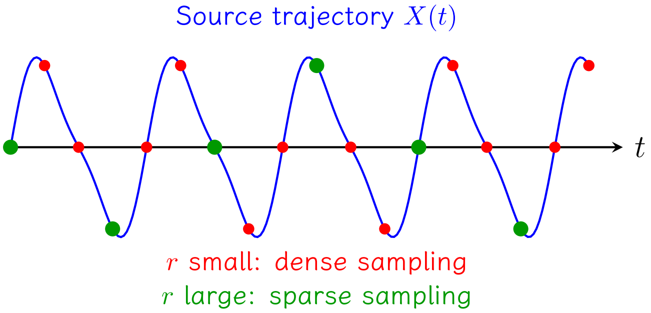


Figure 16.10: Sampling orbit: the greater the distance, the sparser the sampling points. Close-range sampling (red points) can resolve the high-frequency detail of the source, whilst long-range sampling (green points) loses high-frequency information.

Property 16.8 (Basic properties of the sampling functions). (i) **Boundedness**: $0 < R(r,\sigma) \le R_0$, $0 < N(r,\sigma) \le N_0$;

(ii) **Asymptotic behaviour**: $R(r,\sigma) \sim R_0\sigma/r^2$ $(r \to \infty)$; $R(0,\sigma) = R_0$, $N(0,\sigma) = N_0$;

(iii) **Scale covariance**: $R(\lambda r, \lambda^2\sigma) = R(r,\sigma)$ $(\lambda > 0)$.

Proof. (i) From $\sigma/(r^2+\sigma) \le 1$. (ii) As $r \to \infty$, $r^2+\sigma \sim r^2$, giving the asymptotic formula; at $r = 0$, $\sigma/\sigma = 1$. (iii) Direct substitution gives the result. ☐

Remark 16.15 (Physical origin of the sampling function). The form of the scale-dependent sampling function $R(r,\sigma) = R_0\sigma/(r^2+\sigma)$ comes from the following geometric picture: the source radiates into space and the observer receives at distance r. The received signal strength decays as $1/r^2$, whilst the effective receiving area of the observer is proportional to the observer scale σ. When $r^2 \ll \sigma$ the received signal saturates $(R \approx R_0)$; when $r^2 \gg \sigma$ the received signal decays as $1/r^2$.

16.12 Information loss with distance

This section formalises the scale-dependent sampling theory of the previous section as the 'information loss with distance' theorem. The core conclusion is that information observability decays asymptotically as $1/r^2$ with distance, and that when the number of sampling points is less than 1 the source is completely invisible to the observer.

Remark 16.16 (Theoretical positioning). 'Information loss with distance' is not, within the system of this book, independent of existing theories, but a direct corollary of the **scale-dependent sampling model**: the observer scale σ and the observation distance r jointly determine the sampling rate $R(r,\sigma)$ and the number of sampling points $N(r,\sigma)$, both of which decrease monotonically in r; when the number of sampling points is less than 1, the source is completely invisible to the observer. The canonical definition of the model is given in Section 16.11; this section merely restates the conclusions and gives the information-theoretic reading.

Definition 16.32 (Information observability)**.** Define the **information observability** of the source at distance r by

$$\mathcal{I}(r,\sigma) = N(r,\sigma) = N_0 \frac{\sigma}{r^2+\sigma}.$$

$\mathcal{I} \geq 1$ means that the source is sufficiently sampled, $\mathcal{I} \in (0,1)$ means that the source is undersampled, and $\mathcal{I} \to 0$ means that the source is asymptotically invisible.

Theorem 16.13 (Information loss with distance theorem)**.** Let $r_{\text{max}}(\sigma) = \sqrt{(N_0-1)\sigma}$ (when $N_0 > 1$). Then

(i) sufficient-sampling region: for $r \leq r_{\text{max}}$ we have $\mathcal{I}(r,\sigma) \geq 1$, and the source is observed sufficiently;

(ii) undersampling region: for $r > r_{\text{max}}$ we have $\mathcal{I}(r,\sigma) < 1$, and the source is observed only partially;

(iii) asymptotic invisibility: as $r \to \infty$, $\mathcal{I}(r,\sigma) \to 0$;

(iv) threshold distance: $\mathcal{I}(r_{\text{max}},\sigma) = 1$, and $\mathcal{I} < 1$ for $r > r_{\text{max}}$.

Proof. By Definition 16.32, $\mathcal{I} = N(r,\sigma)$. The inequality $N(r,\sigma) \geq 1$ is equivalent to $N_0\sigma/(r^2+\sigma) \geq 1$, that is, $r^2 \leq (N_0-1)\sigma$. Taking $r_{\text{max}} = \sqrt{(N_0-1)\sigma}$ gives (i) and (iv). For $r > r_{\text{max}}$ we have $N < 1$, giving (ii). As $r \to \infty$, $N \sim N_0\sigma/r^2 \to 0$, giving (iii). □

Corollary 16.12 (Asymptotic power law)**.** When $r \gg \sqrt{\sigma}$,

$$\mathcal{I}(r,\sigma) \sim \frac{N_0\sigma}{r^2},$$

that is, the information observability decays as $1/r^2$. This agrees in form with the intensity decay in free-space optical communication and with the inverse-square laws of the Coulomb and gravitational fields.

Proof. $r^2+\sigma \sim r^2$ (for $r \gg \sqrt{\sigma}$); substituting gives $\mathcal{I} \sim N_0\sigma/r^2$. □

Proposition 16.23 (Compensation of information observability by the observer scale)**.** For fixed distance r, increasing the observer scale σ can raise the information observability:

$$\frac{\partial \mathcal{I}}{\partial \sigma} \geq 0.$$

The threshold distance increases monotonically with σ according to $r_{\text{max}}(\sigma) = \sqrt{(N_0-1)\sigma}$.

Proof. This follows directly from Proposition 16.22 and Theorem 16.13. □

Example 16.9 (Numerical example: $\sigma = 1$, $N_0 = 100$)**.** Take $\sigma = 1$ and $N_0 = 100$. The threshold distance is $r_{\text{max}} = \sqrt{99} \approx 9.95$.

- $r = 1$: $N = 100 \cdot 1/(1+1) = 50 > 1$, sufficient sampling;

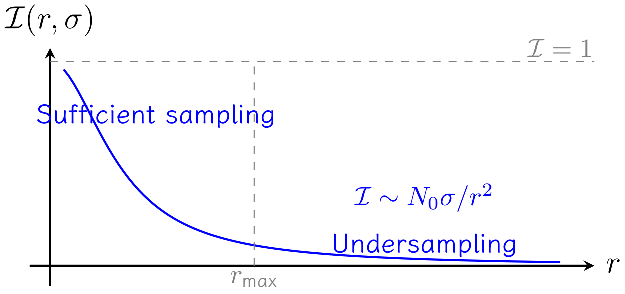


Figure 16.11: Decay of information observability with distance: for $r \le r_{\text{max}}$ the sampling is sufficient, whilst for $r > r_{\text{max}}$ it decays asymptotically as $1/r^2$.

- $r = 10$: $N = 100/101 \approx 0.99$, marginal undersampling;
- $r = 100$: $N = 100/10001 \approx 0.00999$, severe loss;
- $r \to \infty$: $\mathcal{I} \sim 100/r^2 \to 0$ (asymptotically invisible).

Property 16.9 (Basic properties of information observability). (i) **Monotonicity**: $\mathcal{I}(r,\sigma)$ is monotonically non-increasing in r and monotonically non-decreasing in σ;

(ii) **Boundedness**: $\mathcal{I}(r,\sigma) > 0$, and it tends to 0 as $r \to \infty$;

(iii) **Asymptotic decay**: $\mathcal{I}(r,\sigma) \sim N_0\sigma/r^2$ $(r \to \infty)$;

(iv) **Scale covariance**: $\mathcal{I}(\lambda r, \lambda^2\sigma) = \mathcal{I}(r,\sigma)$ $(\lambda > 0)$.

Proof. (i)(ii) By Definition 16.32 and Proposition 16.22. (iii) By Corollary 16.12. (iv) Direct substitution: $N(\lambda r, \lambda^2\sigma) = N_0\lambda^2\sigma/(\lambda^2 r^2 + \lambda^2\sigma) = N(r,\sigma)$. ☐

Remark 16.17 (Comparison with existing theories). The 'information loss with distance' of this section agrees in form with the following theories, but has a different starting point:

(i) **Shannon information theory**: the channel capacity decays with the signal-to-noise ratio, and the signal-to-noise ratio decays with distance through geometric spreading; this section starts from the number of sampling points.

(ii) **Free-space optical communication**: the received intensity decays as $1/r^2$; the asymptotic power law $N_0\sigma/r^2$ of this section is of the same order.

(iii) **Radar equation**: the two-way return gives $1/r^4$; squaring the one-way model of this section gives exactly this.

(iv) **Sampling theory**: the decline in resolution with distance is an equivalent statement of the decline in sampling rate.

The contribution of this section is to attribute all these phenomena uniformly to the **joint action of the observer scale and the distance on the sampling rate**, rather than to introduce new physical assumptions.

Remark 16.18 (Scope and open problems). Prerequisites of the conclusions of this section:

(i) the concrete form of the sampling functions $R(r,\sigma)$ and $N(r,\sigma)$ is given by Definition 16.28; if another decay law is used instead (such as $1/r$ or $1/r^3$), the form of the conclusions is unchanged but the exponent changes;

(ii) $\sigma > 0$ and $N_0 > 1$; when $\sigma \to 0$ or $N_0 \leq 1$ the threshold distance degenerates to zero or becomes negative, and must be discussed separately;

(iii) this section discusses only one-way geometric spreading, and does not touch on mechanisms such as multipath, scattering or quantum entanglement.

The following questions are left open:

(a) connecting the information observability quantitatively with the DFT leakage estimate of the 'Harmonic Analysis' chapter;

(b) the optimal strategy for information recovery under the cooperation of multiple observers;

(c) whether the power law of information loss with distance changes in the quaternion case (a preliminary analysis shows that the conservation of the modulus leaves the exponent unchanged, but that phase information is lost in addition).

Remark 16.19 (Unified logic of this chapter). The core logic of this chapter may be summarised in the following three steps:

(1) **Composition**: the time group T and the space group S are added, giving the composite vector $X = T + S$ (or $q_T + q_S$ in the quaternion case).

(2) **Projection**: the observer functor Π_O projects the composite vector to the observable scalar $\mathcal{O} = \Pi_O(X)$.

(3) **Reconstruction**: the reconstruction functor Π_O^* back-projects the observable scalar to an estimate $\widehat{X} = \Pi_O^*(\mathcal{O})$ in the source space. By the adjunction, $\Pi_O \dashv \Pi_O^*$, with the counit the identity and the unit not an isomorphism—this formalises the core fact that 'observation is information loss'.

The discrete unitary dynamic projection provides the interface under time discretisation; scale-dependent sampling theory gives the relation between spatial resolution and distance; and information loss with distance is the natural corollary of the two within the observer projection framework.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	Dynamic adjustment of the observer vector (Definition 16.11); translation of the rotation angles of the three planes and the three sphere groups (Definition 16.15, Definition 16.19)
Scaling	The weight introduced by the observer modulus (Lemma 16.2); preservation of the quaternion norm (Proposition 16.13); scale covariance of the sampling functions (Property 16.8)
Rotation	Independent rotation of the three planes (Definition 16.15); independent rotation of the three sphere groups (Definition 16.19); non-commutative quaternion rotation (Theorem 16.4)
Balance	Linearity of the projection (Proposition 16.7); additivity of synthesis–projection (Proposition 16.9); scale invariance of the sampling duration (Proposition 16.20)
Symmetry	Symmetry of the projection (Proposition 16.8); symmetry of the quaternion inner product (Lemma 16.3); commutativity under the additive convention (Corollary 16.4)
Closure	Invariance under synchronous rotation of the three planes (Theorem 16.3); invariance under synchronous rotation of the three sphere groups (Theorem 16.5); adjunction closure of the observer functor (Theorem 16.7)
Period	Instantaneous frequency of the dynamic projection (Theorem 16.1); single-frequency frequency shift (Corollary 16.1); three-group beat frequency (Corollary 16.2); discrete periodicity condition (Theorem 16.11); resolution of the sampling orbit (Theorem 16.12)

Exercises

Exercise 1 According to Definition 16.5 and Definition 16.6, explain how an epicycle $\gamma_{A,\omega,\phi}(t) = Ae^{i(\omega t+\phi)}$ corresponds to a simple complex function, and how an epicycle group is a finite sum of simple complex functions. Using the epicycle group with $K = 3$,

$$\mathcal{E}(t) = A_1e^{i(\omega_1 t+\phi_1)} + A_2e^{i(\omega_2 t+\phi_2)} + A_3e^{i(\omega_3 t+\phi_3)}$$

write out its corresponding period series form, and indicate which level of parameter the amplitude, the phase and the frequency correspond to respectively.

Exercise 2 Prove Proposition 16.5: the epicycle group $\mathcal{E}(t) = \sum_{k=1}^{K} A_k e^{i(\omega_k t+\phi_k)}$ $(A_k > 0)$ is a periodic function if and only if all the ω_k generate a discrete subgroup, that is, $\omega_k/\omega_0 \in \mathbb{Z}$ for all k, where $\omega_0 = \gcd\{\omega_k \neq 0\}$. Prove sufficiency and necessity separately, and state that the least positive period is $2\pi/|\omega_0|$ (an integer fraction of it when $\omega_0 \notin \Omega$); when all $\omega_k = 0$ there is no least positive period.

Exercise 3 Prove Proposition 16.6: the orbit of the episphere $\beta_{a,\mathsf{u},\theta}(t) = ae^{\mathsf{u}\theta t}$ lies on the three-dimensional sphere of radius a, $S_a^3 = \{q \in \mathbb{H} : |q| = a\}$. Complete the proof using $e^{\mathsf{u}\theta t} = \cos(\theta t) + \mathsf{u}\sin(\theta t)$ and $|\mathsf{u}| = 1$.

Exercise 4 Verify Remark 16.3: if the rotation axes of two epispheres differ ($\mathsf{u}_1 \neq \mathsf{u}_2$), then

$$e^{\mathsf{u}_1\theta_1 t}e^{\mathsf{u}_2\theta_2 t} \neq e^{\mathsf{u}_2\theta_2 t}e^{\mathsf{u}_1\theta_1 t}.$$

Taking $\mathsf{u}_1 = i$, $\mathsf{u}_2 = j$ and $\theta_1 = \theta_2 = 1$, verify this non-commutativity by explicit computation using $ij = k$ and $ji = -k$.

Exercise 5 Prove Proposition 16.7 and Proposition 16.8:

(i) the projection is real-linear in $\mathcal{E}$: $\mathcal{O}[\alpha\mathcal{E}_1 + \beta\mathcal{E}_2] = \alpha\mathcal{O}[\mathcal{E}_1] + \beta\mathcal{O}[\mathcal{E}_2]$;

(ii) the projection is symmetric: $\mathcal{O}(t) = \mathrm{Re}[\mathcal{E}(t)\overline{u(t)}] = \mathrm{Re}[u(t)\overline{\mathcal{E}(t)}]$.

Exercise 6 Prove Lemma 16.1: taking $\mathcal{E}(t) = Ae^{i\omega t}$ and observer $u(t) = e^{i\omega_0 t}$, we have

$$\mathcal{O}(t) = A\cos((\omega - \omega_0)t).$$

Also prove Corollary 16.1: the frequency seen by the observer is $\omega_{\text{obs}} = \omega - \omega_0$, and when $\omega_0 = \omega$ the observation result is the constant A.

Exercise 7 Prove Theorem 16.1: if $A_k, \omega_k, \phi_k, \theta$ are all C^1 functions, then the observation result $\mathcal{O}(t)$ is a C^1 function. Furthermore, if $|\mathcal{E}(t)| \neq 0$, the instantaneous frequency of $\mathcal{O}(t)$ is

$$\Omega_{\mathcal{O}}(t) = \frac{d}{dt}\arg(\mathcal{E}(t)\overline{u(t)}),$$

and verify that it degenerates to $\omega - \omega_0$ in the single-frequency case.

Exercise 8 Prove Proposition 16.9: the observation result is additive with respect to the time group and the space group:

$$\mathcal{O}(t) = \mathrm{Re}[T(t)\overline{O(t)}] + \mathrm{Re}[S(t)\overline{O(t)}].$$

Also prove Lemma 16.2: in the general case

$$\mathcal{O}(t) = \mathrm{Re}[(T(t) + S(t))\overline{\widehat{O}(t)}] \cdot |O(t)|, \qquad \widehat{O}(t) = O(t)/|O(t)|.$$

Exercise 9 Verify the beat-frequency formula of Corollary 16.2: taking $T(t) = Ae^{i\omega t}$, $S(t) = Be^{i\nu t}$ and $O(t) = Ce^{i\mu t}$, we have

$$\mathcal{O}(t) = AC\cos((\omega - \mu)t) + BC\cos((\nu - \mu)t).$$

Using the concrete values $A = 1, B = 1, C = 1, \omega = 10, \nu = 12, \mu = 11$, compute $\mathcal{O}(t)$ and find the beat period.

Exercise 10 Prove Proposition 16.11: the rotating three planes are equivalent to applying phase shifts to T_0, S_0, O_0 respectively before composition, that is,

$$\mathcal{O}(t) = \text{Re}\left[(T_0 e^{i(\theta_T - \theta_O)} + S_0 e^{i(\theta_S - \theta_O)})\overline{O_0}\right].$$

Explain why only the two relative rotation angles $\theta_T - \theta_O$ and $\theta_S - \theta_O$ take effect.

Exercise 11 Prove Theorem 16.3: if $\theta_T(t) = \theta_S(t) = \theta_O(t) = \theta(t)$, then the observation result is the same as in the static case:

$$\mathcal{O}(t) = \text{Re}[(T_0(t) + S_0(t))\overline{O_0(t)}].$$

Also write out the frequency-shift formula under asynchronous rotation (Corollary 16.3).

Exercise 12 Prove Proposition 16.12: the scalar part of the quaternion inner product Re$[q\overline{p}]$ and the conjugation action $p^{-1}qp$ are two different kinds of projection:

(i) Re$[q\overline{p}]$ is the inner-product projection on $\mathbb{R}^4$, and outputs a real number;

(ii) $p^{-1}qp$ is a rotation of $SO(3)$, and outputs a quaternion.

Taking $q = \mathsf{i}$ and $p = \mathsf{j}$, compute the two projections explicitly.

Exercise 13 Prove Theorem 16.4: consider the additive synthesis $X_+ = q_T + q_S$ and the multiplicative synthesis $X_\times = q_T q_S$; then in general

$$\text{Re}[X_+\overline{q_O}] \neq \text{Re}[X_\times \overline{q_O}],$$

and $q_T q_S \neq q_S q_T$. Taking $q_T = \mathsf{i}$, $q_S = \mathsf{j}$ and $q_O = \mathsf{k}$, verify this explicitly.

Exercise 14 Prove Theorem 16.5: if $r_T = r_S = r_O = r$, then

$$\mathcal{Q}(t) = \text{Re}[(q_{T,0}(t) + q_{S,0}(t))\overline{q_{O,0}(t)}].$$

Complete the proof using Lemma 16.4: Re$[rxr^{-1}] =$ Re$[x]$.

Exercise 15 Prove Theorem 16.6: all five models of this chapter can be written as special cases of the unified observer operator

$$\Pi_O(X) = \text{Re}[X\overline{O}]$$

Compare item by item with Definition 16.12, 16.14, 16.16, 16.18, 16.20, and write out the X and the O of each model.

Exercise 16 Verify the four axioms of the projection operator in Corollary 16.6: real linearity, observer sign reversal, non-expansiveness and zero-observer degeneracy. Complete the proof using the bilinearity of the inner product and the Cauchy–Schwarz inequality.

Exercise 17 Prove Theorem 16.7: the observer functor Π_O and the reconstruction functor Π_O^* form an adjoint pair $\Pi_O \dashv \Pi_O^*$, that is,

$$\langle \Pi_O(X), y \rangle_{\mathcal{H}_{\text{obs}}} = \langle X, \Pi_O^*(y) \rangle_{\mathcal{H}}.$$

Compute both sides and verify that they are equal, and explain why the unit η is not an isomorphism.

Exercise 18 Prove Theorem 16.8 (the data processing inequality): let $X \to Y = \Pi_O(X) \to Z = \Pi_O^*(Y)$ be a Markov chain; then $I(X;Y) \geq I(X;Z)$. Explain the meaning of the statement that projection does not increase information, and point out the contribution of the unobservable subspace to the information loss in Corollary 16.8.

Exercise 19 Verify Example 16.6: let $X = e^{i\Theta}$ with Θ uniformly distributed on $[0, 2\pi)$, and let $O = e^{i\theta_0}$. Then $\Pi_O(X) = \cos(\Theta - \theta_0)$ follows the Arcsine distribution, and the information loss is $\log 4$. Compute $h(\Pi_O(X))$ and $h(X)$, and verify that their difference equals $\log 4$.

Exercise 20 Prove Theorem 16.11: let U be a constant unitary operator and $X_{m+1} = UX_m$. Then there exists a positive integer p with $X_{m+p} = X_m$ for all m if and only if $U^p = I$, equivalently all the eigenvalues of U are p-th roots of unity. Complete the proof using the spectral theorem for unitary operators, and point out its relation to Floquet theory (Corollary 16.10).

Exercise 21 Prove Proposition 16.20: the sampling duration

$$T_{\text{samp}}(r, \sigma) = \frac{N(r, \sigma)}{R(r, \sigma)} = \frac{N_0}{R_0}$$

is independent of both the distance r and the observer scale σ. Complete the proof using Definition 16.28 and 16.29, and explain why the sampling rate and the number of sampling points decrease strictly with distance (Proposition 16.22).

Exercise 22 Prove Theorem 16.12: let the least resolvable period of the source be T_{min}; then the necessary and sufficient condition for the sampling orbit to resolve T_{min} is

$$\tau(r, \sigma) \leq T_{\text{min}}/2,$$

that is,

$$r^2 \leq \sigma \left(\frac{R_0 T_{\text{min}}}{2} - 1 \right).$$

Complete the proof using the Nyquist–Shannon sampling theorem, and give the expression for the resolution distance r_{res}.

Exercise 23 Prove Theorem 16.13: let the threshold distance be $r_{\text{max}}(\sigma) = \sqrt{(N_0 - 1)\sigma}$; then the information observability

$$\mathcal{I}(r, \sigma) = N_0 \frac{\sigma}{r^2 + \sigma}$$

satisfies:

(i) $\mathcal{I} \geq 1$ for $r \leq r_{\text{max}}$ (sufficient sampling);
(ii) $\mathcal{I} < 1$ for $r > r_{\text{max}}$ (undersampling);
(iii) $\mathcal{I} \to 0$ as $r \to \infty$ (asymptotically invisible).

Also prove Corollary 16.12: when $r \gg \sqrt{\sigma}$, $\mathcal{I}(r, \sigma) \sim N_0 \sigma / r^2$.

Exercise 24 Synthesis problem: combining the dynamic epicycle group (Definition 16.9), the rotating three planes (Definition 16.15), the rotating three-sphere groups (Definition 16.19) and the unified observer operator (Definition 16.21), explain how the 'compose first, then project' convention of this chapter remains consistent across the five models. Point out separately, for each model:

(i) what the composite vector X is;
(ii) what the observer O is;
(iii) what observation result the projection $\Pi_O(X) = \text{Re}[X\overline{O}]$ gives;
(iv) how the reconstruction functor Π_O^* back-projects from the observation result.

Chapter 17

Propagation and Evolution

Remark 17.1 (Positioning of this chapter and strict premises). This chapter takes **the propagation of light** as its basic prototype and establishes a unified mathematical framework for propagation and evolution. All definitions, propositions, lemmas, theorems and corollaries in the chapter are bounded by the following two premises:

(a) **Linearity premise**: the field quantities satisfy the **linear** wave equation (the degenerate form, in a source-free region, of the vacuum Maxwell equations, Theorem 31.16 of the 'Periodic Structure of Electromagnetic Systems' chapter), or more generally satisfy a **linear** partial differential equation with constant coefficients (or with slowly varying coefficients);

(b) **Slowly varying medium premise**: the refractive index $n(\mathbf{r})$ of the medium is a C^2 smooth function, and its relative change over one wavelength scale is far less than 1: $|\nabla n|\lambda \ll n$.

Any statement going beyond the above premises—nonlinear media, strong-field effects, quantum optics, divergence of the field strength on a caustic surface, localisation in random media—is invariably marked as a **note** or an **open problem**, and **is not used as a proved conclusion**.

Positioning of the seven themes of this chapter: translation—spatial propagation is generated by the action of the translation group $\mathbb{R}^n$ (Definition 17.3), whilst time evolution is generated by the evolution semigroup (Definition 17.6); scaling—the refractive index of the medium gives rise to the wavelength scaling $\lambda = \lambda_0/n$ (Proposition 17.1); rotation—the complex phase representation and the phase operator (Proposition 17.5); conservation—light propagation in vacuum preserves energy (Theorem 17.4); symmetry—vacuum light propagation is invariant under spatial translation (Proposition 17.6); closure—Bloch waves in a periodic medium (Theorem 17.9); period—the frequency-domain support of the dispersion relation (Property 17.4).

17.1 Ray trajectories: from Fermat's principle to the eikonal equation

Remark 17.2 (Positioning of this section). This section starts from Fermat's principle and rigorously derives the mathematical description of ray trajectories. Fermat's principle is the fundamental variational principle of geometrical optics, and making it rigorous requires the first variation of the optical-path functional. This section does not assume that the reader already knows the calculus of variations: every derivation starts from the definitions, and wherever integration by parts is involved, the treatment of the boundary terms is written out explicitly.

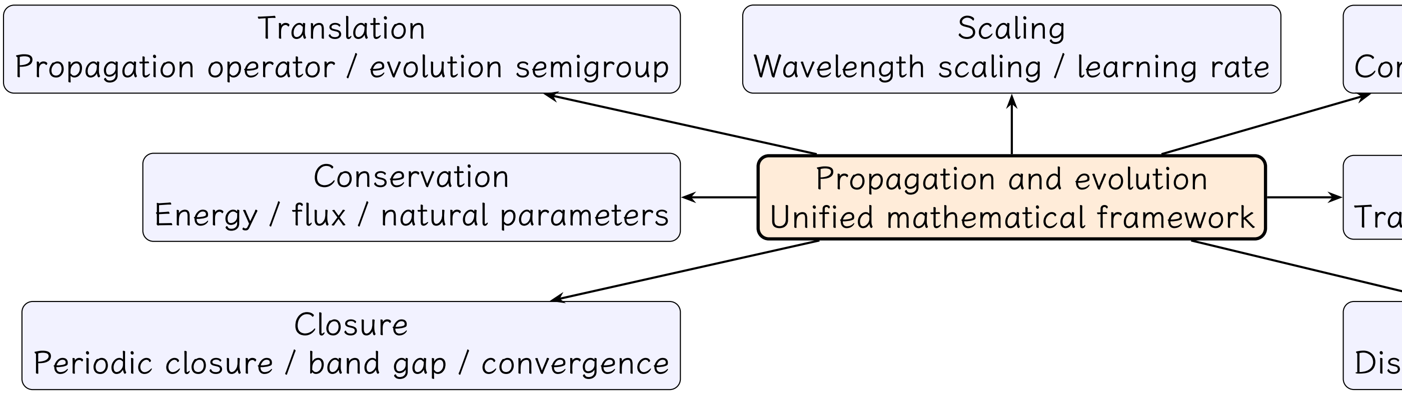

Translation
Propagation operator / evolution semigroup
Scaling
Wavelength scaling / learning rate
Cor
Conservation
Energy / flux / natural parameters
Propagation and evolution
Unified mathematical framework
Tra
Closure
Periodic closure / band gap / convergence
Dis

Figure 17.1: Positioning of the seven themes of this chapter: translation, scaling, rotation, conservation, symmetry, closure and period.

Definition 17.1 (Optical-path functional). Let the refractive index $n : \mathbb{R}^n \to \mathbb{R}^+$ be a C^2 smooth function and let $\gamma : [s_0, s_1] \to \mathbb{R}^n$ be a C^2 curve parametrised by arc length s (that is, $|\dot\gamma(s)| = 1$). The **optical-path functional** on γ is defined by

$$L[\gamma] := \int_{s_0}^{s_1} n(\gamma(s))\, ds.$$

Definition 17.2 (Fermat's principle). **Fermat's principle**: *a light ray is a first-order stationary curve of the optical-path functional $L[\gamma]$ among all C^2 curves joining the two endpoints $\gamma(s_0) = \mathsf{r}_0$, $\gamma(s_1) = \mathsf{r}_1$.*

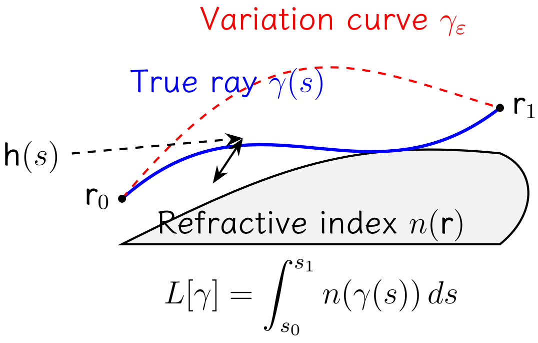

Variation curve γ_ε
True ray $\gamma(s)$
$\mathsf{h}(s)$
r_0
r_1
Refractive index $n(\mathsf{r})$
$L[\gamma] = \int_{s_0}^{s_1} n(\gamma(s))\, ds$

Figure 17.2: Fermat's principle: variation with fixed endpoints.

Lemma 17.1 (First variation of the optical-path functional). *Let $\gamma_\varepsilon(s) = \gamma(s) + \varepsilon \mathsf{h}(s)$ be a family of variation curves for $L[\gamma]$, where $\mathsf{h} \in C^2([s_0, s_1], \mathbb{R}^n)$ satisfies $\mathsf{h}(s_0) = \mathsf{h}(s_1) = \mathsf{0}$ (a variation with fixed endpoints). Then*

$$\frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} L[\gamma_\varepsilon] = \int_{s_0}^{s_1} \Big[\nabla n(\gamma(s)) \cdot \mathsf{h}(s)\Big]\, ds + \int_{s_0}^{s_1} n(\gamma(s)) \frac{d}{ds}\Big[\frac{\dot\gamma(s) \cdot \dot{\mathsf{h}}(s)}{|\dot\gamma(s)|}\Big]_{\varepsilon=0} ds.$$

Under the arc-length parametrisation $|\dot\gamma| = 1$, the second term simplifies to $\int_{s_0}^{s_1} n(\gamma(s))\, \dot\gamma(s) \cdot \dot{\mathsf{h}}(s)\, ds$.

Proof. Expand $L[\gamma_\varepsilon]$ by definition and differentiate directly. Write $\mathsf{v}_\varepsilon(s) := \dot\gamma_\varepsilon(s)$; then $|\mathsf{v}_\varepsilon(s)| = |\dot\gamma(s) + \varepsilon \dot{\mathsf{h}}(s)|$. Since $|\dot\gamma| = 1$,

$$|\mathsf{v}_\varepsilon| = \sqrt{1 + 2\varepsilon \dot\gamma \cdot \dot{\mathsf{h}} + \varepsilon^2 |\dot{\mathsf{h}}|^2} = 1 + \varepsilon \dot\gamma \cdot \dot{\mathsf{h}} + O(\varepsilon^2).$$

Substituting into $L[\gamma_\varepsilon] = \int n(\gamma + \varepsilon \mathsf{h})|\mathsf{v}_\varepsilon|\, ds$ and differentiating with respect to ε, then setting $\varepsilon = 0$, gives the first term $\int \nabla n \cdot \mathsf{h}\, ds$ and the second term $\int n\, \dot{\gamma} \cdot \dot{\mathsf{h}}\, ds$. ☐

Theorem 17.1 (Ray equation)**.** Under Fermat's principle (Definition 17.2) and the slowly varying medium premise (Remark 17.1), the ray $\gamma(s)$ satisfies the **ray equation**

$$\boxed{\frac{d}{ds}\Big(n(\gamma(s))\,\dot{\gamma}(s)\Big) = \nabla n(\gamma(s)).}$$

Proof. By Lemma 17.1, the first variation of the optical-path functional is

$$\delta L[\mathsf{h}] = \int_{s_0}^{s_1} \nabla n \cdot \mathsf{h}\, ds + \int_{s_0}^{s_1} n\, \dot{\gamma} \cdot \dot{\mathsf{h}}\, ds.$$

Integrate the second term by parts:

$$\int_{s_0}^{s_1} n\, \dot{\gamma} \cdot \dot{\mathsf{h}}\, ds = \Big[n\, \dot{\gamma} \cdot \mathsf{h}\Big]_{s_0}^{s_1} - \int_{s_0}^{s_1} \frac{d}{ds}(n\dot{\gamma}) \cdot \mathsf{h}\, ds.$$

By the fixed-endpoint conditions $\mathsf{h}(s_0) = \mathsf{h}(s_1) = \mathbf{0}$, the boundary term vanishes. Hence

$$\delta L[\mathsf{h}] = \int_{s_0}^{s_1} \Big[\nabla n - \frac{d}{ds}(n\dot{\gamma})\Big] \cdot \mathsf{h}\, ds.$$

Fermat's principle requires that $\delta L[\mathsf{h}] = 0$ for every h satisfying the endpoint conditions. By the fundamental lemma of the calculus of variations (Lemma 17.2), the integrand vanishes identically:

$$\nabla n - \frac{d}{ds}(n\dot{\gamma}) = \mathbf{0}.$$ ☐

Lemma 17.2 (Fundamental lemma of the calculus of variations)**.** Let $f \in C^0([s_0, s_1], \mathbb{R}^n)$. If $\int_{s_0}^{s_1} f(s) \cdot \mathsf{h}(s)\, ds = 0$ for every C^1 vector function h satisfying $\mathsf{h}(s_0) = \mathsf{h}(s_1) = \mathbf{0}$, then $f \equiv \mathbf{0}$.

Proof. Suppose to the contrary that there exists $s^* \in (s_0, s_1)$ with $f(s^*) \neq \mathbf{0}$. Take the j-th component $f_j(s^*) \neq 0$ of f. By continuity there exists $\delta > 0$ such that $f_j(s)$ has the same sign as $f_j(s^*)$ whenever $|s - s^*| < \delta$. Take a cut-off function $\varphi \in C^1$ satisfying $\varphi(s) \geq 0$, $\varphi(s) = 0$ for $|s - s^*| \geq \delta$ and $\varphi(s) > 0$ for $|s - s^*| < \delta$. Put $\mathsf{h} = \varphi\, \mathsf{e}_j$ (where e_j is the j-th coordinate unit vector). Then $\int f \cdot \mathsf{h}\, ds = \int f_j \varphi\, ds$, and the integrand is everywhere non-negative on $|s - s^*| < \delta$ and is not identically zero, so the integral is strictly positive, contradicting the hypothesis. Hence $f \equiv \mathbf{0}$. ☐

Corollary 17.1 (Eikonal equation)**.** If there exists an optical path function $S : \mathbb{R}^n \to \mathbb{R}$ such that the ray direction is $\dot{\gamma} = \nabla S / n$, then S satisfies the **eikonal equation**

$$\boxed{|\nabla S(\mathsf{r})|^2 = n(\mathsf{r})^2.}$$

Proof. From $|\dot{\gamma}| = 1$ and $\dot{\gamma} = \nabla S / n$, taking the square of the modulus gives $1 = |\nabla S|^2 / n^2$, that is, $|\nabla S|^2 = n^2$. ☐

Corollary 17.2 (Straight-line propagation in a homogeneous medium)**.** If $n(\mathsf{r}) \equiv n_0$ is constant, then the ray is a straight line and the propagation speed is $v = c/n_0$.

Proof. By Theorem 17.1, $\nabla n = 0$, hence $d/ds(n_0\dot{\gamma}) = n_0\ddot{\gamma} = 0$, that is, $\ddot{\gamma} = 0$—the ray is a straight line. Propagation speed: from $ds/dt = c/n_0$ (which is given by the vacuum speed of light c and the definition of the refractive index $n = c/v$), that is, $v = c/n_0$. ☐

Proposition 17.1 (Wavelength scaling in a medium)**.** In a medium of refractive index n, the wavelength of light is scaled to $1/n$ of the vacuum wavelength:

$$\lambda_{\text{medium}} = \frac{\lambda_{\text{vacuum}}}{n}.$$

Proof. From the phase velocity of light in the medium, $v = c/n$, and the continuity of frequency across a medium interface (Theorem 17.2): $\lambda_{\text{medium}} = v/\nu = c/(n\nu) = \lambda_{\text{vacuum}}/n$. ☐

Theorem 17.2 (Snell's law)**.** At a flat interface between two homogeneous media, the angle of incidence θ_1 and the angle of refraction θ_2 satisfy

$$n_1 \sin\theta_1 = n_2 \sin\theta_2.$$

Proof. Apply Fermat's principle (Definition 17.2) to the optical path crossing the interface. Let the incident light travel from A to a point P on the interface, and then from P to B. The optical path is $L(\mathsf{P}) = n_1|\mathsf{A} - \mathsf{P}| + n_2|\mathsf{P} - \mathsf{B}|$. Varying P within the interface and setting the first variation to zero gives $n_1 \sin\theta_1 = n_2 \sin\theta_2$ (where θ_1, θ_2 are respectively the angles between the optical path and the interface normal). ☐

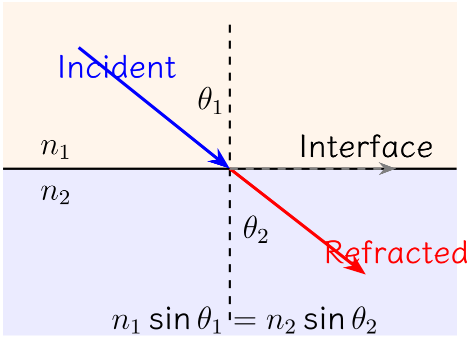


Figure 17.3: Snell's law and total internal reflection.

Property 17.1 (Interface with the operators of the 'Vector Analysis' chapter)**.** The basic equations of geometrical optics correspond exactly to the three operators of the 'Vector Analysis' chapter:

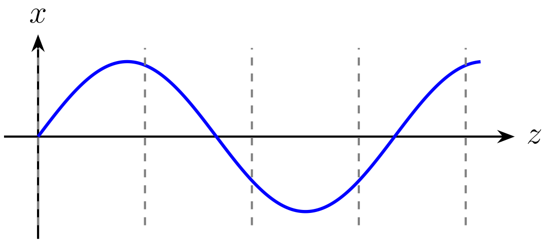
(i) **Gradient**: the eikonal equation $|\nabla S|^2 = n^2$ (Corollary 17.1) uses the gradient operator ∇ directly. The ray direction is $\nabla S/n$, perpendicular to the wavefront $S = \text{const}$.

(ii) **Divergence**: the flux conservation of a ray tube is given by the divergence theorem (Theorem 15.8). Let $A(s)$ be the cross-sectional area of the ray tube; then $n(s)A(s)$ remains constant during propagation—this is the realisation in geometrical optics of the conservation laws of the 'Vector Analysis' chapter.

(iii) **Curl**: the amount by which a ray bends is given by the curl of the ray equation: $\nabla \times (n\dot{\gamma}) = \nabla \times (\nabla S) = \mathbf{0}$, that is, $n\dot{\gamma}$ is an irrotational field.

Proof. (i) This has already been proved in Corollary 17.1. (ii) Apply the divergence theorem to the integral over the lateral surface of the ray tube; combined with $\nabla \cdot (n\dot{\gamma}) = 0$ (computed directly from the ray equation), this gives $nA =$ const. (iii) From $\dot{\gamma} = \nabla S/n$ we get $n\dot{\gamma} = \nabla S$, hence $\nabla \times (n\dot{\gamma}) = \nabla \times \nabla S = \mathbf{0}$ (Theorem 15.3 of the 'Vector Analysis' chapter). ☐

Example 17.1 (Ray in a spherical-gradient medium)**.** Take $n(r) = n_0(1 - \alpha r^2)$ (a graded-index medium in the paraxial approximation). By the ray equation, a paraxial ray satisfies $\ddot{r} \approx -2\alpha n_0\, r$ (simple harmonic motion), and the ray follows a sinusoidal trajectory of period $T = 2\pi/\sqrt{2\alpha n_0}$. This is the periodic closure of rays in a self-focusing optical fibre—an instance in geometrical optics of 'a finite-order action produces a closed orbit', consistent with the finite-order action framework of the 'Operations and Periodicity' chapter.



Paraxial ray: $\ddot{r} \approx -2\alpha n_0 r$, period $T = 2\pi/\sqrt{2\alpha n_0}$

Figure 17.4: Sinusoidal ray in a graded-index medium.

Remark 17.3 (Caustics and the failure of the eikonal equation). The eikonal equation (Corollary 17.1) fails on a **caustic surface**: a caustic surface is a surface on which the cross-sectional area of the ray tube satisfies $A(s) \to 0$, and on it the light intensity diverges as geometrical optics predicts ($I \propto 1/A \to \infty$). This divergence is not physical—the actual field strength is finite near the caustic and must be described by wave optics (the Airy function). Hence all geometrical-optics conclusions in this chapter take 'away from the caustic surface' as a premise; the wave correction near a caustic is an **open problem** (Remark 17.19).

17.2 The propagation operator: an algebraic realisation of spatial translation

Definition 17.3 (Free-space propagation operator)**.** Let X be the function space in which the field quantities (scalar or vector components) live, taken to be $X = L^2(\mathbb{R}^n)$ or the Schwarz space $\mathcal{S}(\mathbb{R}^n)$. For $\mathsf{a} \in \mathbb{R}^n$, the **free-space propagation operator** is defined by

$$(P_{\mathsf{a}} f)(\mathsf{x}) := f(\mathsf{x} - \mathsf{a}), \qquad f \in X,\ \mathsf{x} \in \mathbb{R}^n.$$

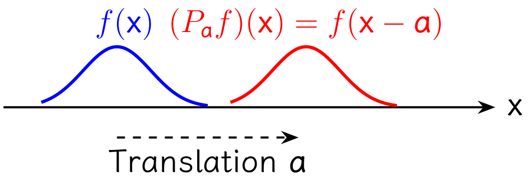


Figure 17.5: Illustration of the translation effected by the propagation operator.

Definition 17.4 (Medium propagation operator)**.** Let a homogeneous medium have refractive index n_0. Propagating a distance a in the medium (a geometrical distance) is equivalent to the optical distance $n_0\mathsf{a}$ in vacuum (the optical path). The **medium propagation operator** is defined by

$$(P_{\mathsf{a}}^{(n_0)}f)(\mathsf{x}) := f(\mathsf{x} - n_0\mathsf{a}).$$

Proposition 17.2 (Group structure of the propagation operator)**.** Under operator composition the family of propagation operators $\{P_{\mathsf{a}}\}_{\mathsf{a}\in\mathbb{R}^n}$ forms a faithful representation of the translation group $\mathbb{R}^n$:

(i) **Group homomorphism**: $P_{\mathsf{a}}P_{\mathsf{b}} = P_{\mathsf{a}+\mathsf{b}}$;

(ii) **Identity element**: $P_0 = \mathrm{id}_X$;

(iii) **Inverse element**: $P_{\mathsf{a}}^{-1} = P_{-\mathsf{a}}$.

Proof. (i) $(P_{\mathsf{a}}P_{\mathsf{b}}f)(\mathsf{x}) = (P_{\mathsf{b}}f)(\mathsf{x}-\mathsf{a}) = f(\mathsf{x}-\mathsf{a}-\mathsf{b}) = (P_{\mathsf{a}+\mathsf{b}}f)(\mathsf{x})$. (ii)(iii) These follow from (i) on taking $\mathsf{b} = 0$ or $\mathsf{b} = -\mathsf{a}$. □

Proposition 17.3 (Isometry of the propagation operator)**.** On $X = L^2(\mathbb{R}^n)$, the propagation operator P_{a} is an isometry:

$$\|P_{\mathsf{a}}f\|_{L^2} = \|f\|_{L^2}, \qquad \forall f \in L^2(\mathbb{R}^n).$$

In particular, P_{a} is a unitary operator.

Proof. Make the change of variable $\mathsf{y} = \mathsf{x} - \mathsf{a}$ (the Lebesgue measure is invariant):

$$\|P_{\mathsf{a}}f\|_{L^2}^2 = \int_{\mathbb{R}^n} |f(\mathsf{x}-\mathsf{a})|^2\,d\mathsf{x} = \int_{\mathbb{R}^n} |f(\mathsf{y})|^2\,d\mathsf{y} = \|f\|_{L^2}^2.$$

□

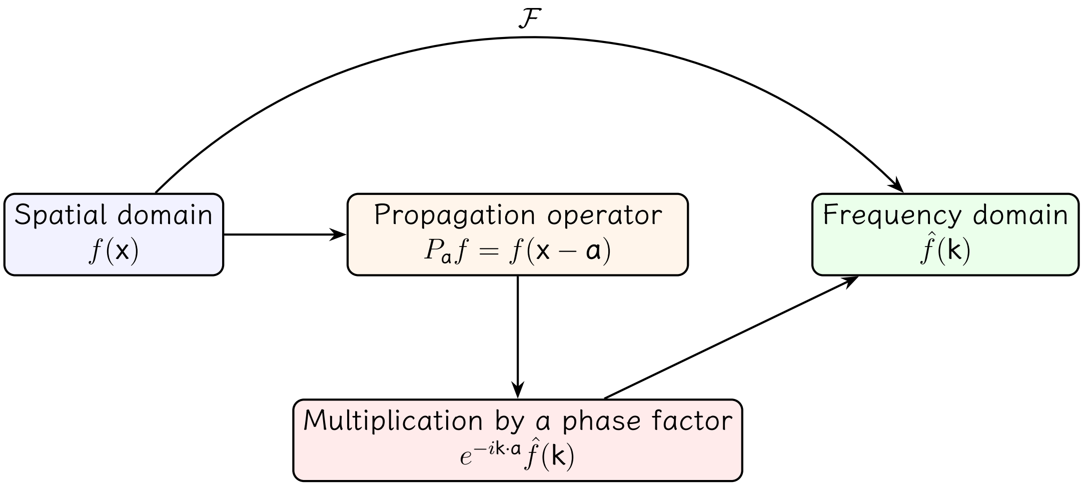
Proposition 17.4 (Fourier multiplier representation of the propagation operator)**.** Let $f \in L^2(\mathbb{R}^n)$ and let $\hat{f}(\mathsf{k}) = \int f(\mathsf{x})e^{-i\mathsf{k}\cdot\mathsf{x}}\,d\mathsf{x}$ be its Fourier transform. Then

$$\widehat{P_{\mathsf{a}}f}(\mathsf{k}) = e^{-i\mathsf{k}\cdot\mathsf{a}}\hat{f}(\mathsf{k}).$$

That is, in the Fourier domain the propagation operator is equivalent to multiplication by the phase factor $e^{-i\mathsf{k}\cdot\mathsf{a}}$.

Proof. Direct computation:

$$\widehat{P_{\mathsf{a}}f}(\mathsf{k}) = \int f(\mathsf{x}-\mathsf{a})e^{-i\mathsf{k}\cdot\mathsf{x}}\,d\mathsf{x} = \int f(\mathsf{y})e^{-i\mathsf{k}\cdot(\mathsf{y}+\mathsf{a})}\,d\mathsf{y} = e^{-i\mathsf{k}\cdot\mathsf{a}}\hat{f}(\mathsf{k}).$$ □


$\mathcal{F}$
Spatial domain
$f(\mathsf{x})$
Propagation operator
$P_{\mathsf{a}} f = f(\mathsf{x} - \mathsf{a})$
Frequency domain
$\hat{f}(\mathsf{k})$
Multiplication by a phase factor
$e^{-i\mathsf{k}\cdot\mathsf{a}} \hat{f}(\mathsf{k})$

When the propagation distance is $a = \lambda$, $\phi = 2\pi$ and $e^{-i\phi} = 1$: periodic closure.

Figure 17.6: The phase multiplier in the Fourier domain.

Proposition 17.5 (Consistency of the propagation operator and the phase operator)**.** For the vacuum plane wave $e^{i(\mathsf{k}\cdot\mathsf{x}-\omega t)}$, a propagation distance a corresponds to the phase change $\phi = ka = \omega a/c$. In the frequency domain the propagation operator P_a equals the phase operator

$$\mathcal{O}_{\phi}^{(2\pi)} = e^{-i\phi},$$

which has the same structure as the discrete phase operator of the 'Periodic Functions and Groups with Periodic Orbits' chapter (Definition 20.6).

Proof. By Proposition 17.4, $P_a e^{ikx} = e^{-ika} e^{ikx}$. Taking $\phi = ka$ gives $e^{-i\phi}$. ☐

Definition 17.5 (Complex exponential plane-wave representation of vacuum electromagnetic waves)**.** *In vacuum, the monochromatic solutions of the source-free Maxwell equations take the complex exponential plane-wave form*

$$\mathsf{E}(\mathsf{x}, t) = \mathsf{E}_0\, e^{i(\mathsf{k}\cdot\mathsf{x}-\omega t)}, \qquad \mathsf{B}(\mathsf{x}, t) = \mathsf{B}_0\, e^{i(\mathsf{k}\cdot\mathsf{x}-\omega t)},$$

where the dispersion relation is $\omega = c|\mathsf{k}|$ and the transverse-wave conditions $\mathsf{k}\cdot\mathsf{E}_0 = 0$, $\mathsf{k} \times \mathsf{E}_0 = \omega \mathsf{B}_0$ hold.

Corollary 17.3 (Periodic closure of propagation)**.** *When the propagation distance equals one wavelength, $a = \lambda = 2\pi/k$, the phase change is $\phi = 2\pi$ and $e^{-i\phi} = 1$—the plane wave returns to the phase it had at departure. This is one complete* **period** *of the propagation operator.*

Proof. By Proposition 17.5, $\phi = k\lambda = k \cdot 2\pi/k = 2\pi$ and $e^{-i2\pi} = 1$. ☐

Proposition 17.6 (The propagation operator commutes with constant-coefficient differential operators)**.** *Let A be a linear differential operator with constant coefficients. Then*

$$P_{\mathsf{a}} A = A P_{\mathsf{a}}, \qquad \forall \mathsf{a} \in \mathbb{R}^n.$$

Proof. It suffices to verify this for the monomials $A = \partial^\alpha$. By the chain rule, $\partial^\alpha(P_a f) = P_a(\partial^\alpha f)$. The result for polynomial sums follows by taking linear combinations. ☐

Definition 17.6 (Evolution semigroup)**.** Let X be a Banach space. A family of operators $\{E_t\}_{t\geq 0} \subset \mathcal{L}(X)$ is called a **strongly continuous evolution semigroup** if

(i) $E_0 = \mathrm{id}_X$;

(ii) $E_{t+s} = E_t \circ E_s$ for all $t, s \geq 0$;

(iii) for every $f \in X$, $\lim_{t\to 0^+} E_t f = f$ (strong continuity).

Its **infinitesimal generator** A is defined by $Af := \lim_{t\to 0^+}(E_t f - f)/t$, with domain consisting of all f for which the above limit exists.

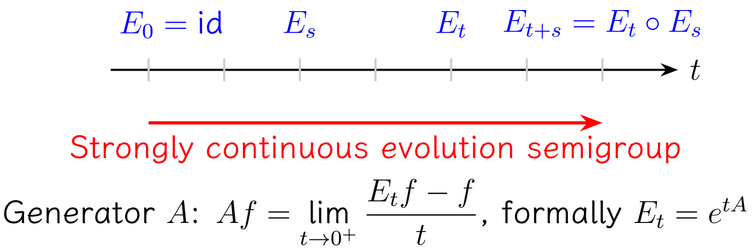

$E_0 = \mathrm{id}$
E_s
E_t
$E_{t+s} = E_t \circ E_s$
t
Strongly continuous evolution semigroup
Generator A: $Af = \lim_{t\to 0^+} \frac{E_t f - f}{t}$, formally $E_t = e^{tA}$

Figure 17.7: Timeline of a strongly continuous evolution semigroup.

Theorem 17.3 (Exponential representation of linear evolution equations)**.** Let X be a Banach space, let A be a closed densely defined linear operator on X, and suppose that $\{E_t\}_{t\geq 0}$ is generated by $\partial_t u = Au$. Then formally

$$E_t = e^{tA}, \qquad t \geq 0,$$

holds on $\mathcal{D}(A)$.

Proof. For $u_0 \in \mathcal{D}(A)$, the definition of the generator gives $E_t u_0 - u_0 = \int_0^t AE_s u_0 \, ds$. Differentiating both sides with respect to t gives $\frac{d}{dt}E_t u_0 = AE_t u_0$, that is, $E_t u_0$ is a solution of $\partial_t u = Au$. By the semigroup property, uniqueness of the solution guarantees that $E_t = e^{tA}$ (on $\mathcal{D}(A)$). ☐

Theorem 17.4 (Energy conservation of vacuum light propagation)**.** Let $\mathsf{u}(t) = (\mathsf{E}(t), \mathsf{B}(t))$ be a solution of the vacuum Maxwell equations, and let the energy density integral be

$$U(t) := \int_{\mathbb{R}^3} \left(\frac{\varepsilon_0}{2} |\mathsf{E}(\mathsf{x}, t)|^2 + \frac{1}{2\mu_0} |\mathsf{B}(\mathsf{x}, t)|^2 \right) d\mathsf{x}.$$

Then $U(t) = U(0)$ for all t.

Proof. Write the Maxwell equations as $\partial_t \mathsf{u} = A\mathsf{u}$, where A is given by the curl operator. With respect to the L^2 energy inner product, A is antisymmetric: $\langle A\mathsf{u}, \mathsf{u}\rangle = 0$ (proved by integration by parts; this is Poynting's theorem, Theorem 31.18 of the 'Periodic Structure of Electromagnetic Systems' chapter). An antisymmetric operator on a Hilbert space generates a unitary group (Stone's theorem), hence $\|\mathsf{u}(t)\|^2 = \|\mathsf{u}(0)\|^2$, that is, $U(t) = U(0)$. ☐

Corollary 17.4 (Mathematical formulation of causality). *Under the vacuum linear wave equation, the propagation speed of an electromagnetic signal does not exceed c. Concretely, if the initial field is supported at $t = 0$ in an open set Ω_0, then at $t > 0$ the field is supported in the ct-neighbourhood $\Omega_0 + B_{ct}(\mathbf{0})$ of Ω_0.*

Proof. This follows from the unitary evolution group of Theorem 17.4 and the finite speed of propagation of the wave equation. Applying the divergence theorem to the energy flux shows that the energy crossing any sphere of radius r centred at x_0 is bounded by c times the energy density outside the sphere; when the initial energy inside the sphere is zero, the energy inside the sphere remains zero for $t < r/c$. $\square$

Example 17.2 (Gaussian light pulse in vacuum). Take the initial Gaussian pulse $\psi_0(x) = e^{-x^2/(2\sigma^2)}e^{ik_0x}$. The vacuum dispersion relation is $\omega(k) = ck$, the group velocity is $v_g = c$ and the second-order dispersion is $\omega'' = 0$. By Theorem 17.3, the pulse translates at speed c without broadening:

$$\psi(x,t) = e^{ik_0(x-ct)}e^{-(x-ct)^2/(2\sigma^2)}.$$

This is the mathematical description of the phenomenon of 'no dispersion'.

Example 17.3 (Pulse broadening in a dispersive medium). In a dispersive medium the dispersion relation is $\omega(k)$ and the second-order dispersion is $\beta_2 = d^2\beta/d\omega^2$ (with $\beta = \omega n(\omega)/c$). A Gaussian pulse broadens:

$$\sigma(t) \approx \sigma\sqrt{1 + (\beta_2 t/\sigma^2)^2}, \qquad t \to \infty.$$

This is the dispersion penalty in optical fibre communication.

17.3 The kernel-function system: integral representations of propagation

Remark 17.4 (Positioning of this section). This section systematically lists the kernel functions that occur throughout the book and classifies them by function. Kernel functions are the realisation of propagation and evolution at the integral level—every kernel function is the integral kernel of some class of propagation process or of some kind of spectral transform. This section distinguishes strictly between three classes of kernel function: (i) **propagation kernels** (Green's functions describing physical propagation); (ii) **window kernels and spectral kernels** (window functions and periodisation kernels in spectral analysis); (iii) **phase similarity kernels and weight kernels** (phase and frequency selection in periodic convolution).

Definition 17.7 (Propagation kernel (Green's function)). *Suppose that the linear evolution equation $\partial_t u = \mathcal{L}u$ has a unique solution. The* **propagation kernel** *$G(\mathsf{x}, t; \mathsf{x}_0)$ is the solution corresponding to the initial value $u(\mathsf{x}, 0) = \delta(\mathsf{x} - \mathsf{x}_0)$:*

$$u(\mathsf{x}, t) = \int_{\mathbb{R}^n} G(\mathsf{x}, t; \mathsf{x}_0)u_0(\mathsf{x}_0)\, d\mathsf{x}_0.$$

Equivalently, G is the integral kernel of the evolution operator $E_t = e^{t\mathcal{L}}$.

Theorem 17.5 (The propagation kernel satisfies the evolution equation and the initial condition)**.** The propagation kernel $G(\mathsf{x}, t; \mathsf{x}_0)$ satisfies

$$\partial_t G = \mathcal{L} G, \qquad G(\mathsf{x}, 0; \mathsf{x}_0) = \delta(\mathsf{x} - \mathsf{x}_0).$$

Proof. By Definition 17.7, $u(\mathsf{x}, t) = \int G(\mathsf{x}, t; \mathsf{x}_0) u_0(\mathsf{x}_0)\, d\mathsf{x}_0$ satisfies $\partial_t u = \mathcal{L} u$ for every u_0. Taking $u_0 = \delta(\cdot - \mathsf{x}_0)$ gives the result. The initial condition is given by $E_0 = \mathrm{id}$. □

Table 17.1: Complete table of the kernel functions of the book (classified by function)

Kernel function	Mathematical form	Physical/mathematical meaning	Chapter
(A) Propagation kernels (Green's functions)			
Free-space Green's function	$G(\mathsf{x}, \mathsf{x}') = \frac{e^{ik\lVert \mathsf{x}-\mathsf{x}'\rVert}}{4\pi \lVert \mathsf{x}-\mathsf{x}'\rVert}$	Helmholtz propagation of a point source in vacuum	Electromagnetic systems
Heat kernel	$G(x, t) = \frac{1}{\sqrt{4\pi\alpha t}} e^{-x^2/(4\alpha t)}$	Diffusion kernel of heat conduction	Differential equations
One-dimensional wave propagation kernel	$G(x, t) = \frac{1}{2c} \mathbf{1}_{\lvert x\rvert < ct}$	Fundamental solution of the one-dimensional wave equation	Differential equations
Poisson kernel	$P_r(\theta) = \frac{1-r^2}{1-2r\cos\theta + r^2}$	Propagation of boundary values of harmonic functions	Harmonic analysis
Retarded Green's function	$G_{\mathrm{ret}} = \frac{\delta(t'-t+\lVert \mathsf{x}-\mathsf{x}'\rVert/c)}{4\pi \lVert \mathsf{x}-\mathsf{x}'\rVert}$	Integral kernel of the retarded potential	Electromagnetic systems
Airy kernel	$\mathrm{Ai}(x) = \frac{1}{\pi}\int_0^\infty \cos(t^3/3 + xt)\, dt$	Wave correction near a caustic	Note to this chapter
(B) Window kernels and spectral kernels			
Rectangular window kernel	$W_T(\omega) = \frac{1}{T}\frac{\sin(T\omega/2)}{\sin(\omega/2)}$	Spectral kernel of a finite window, sidelobes -13 dB	Discrete sampling
Hann window kernel	$w(n) = \frac{1}{2}\left(1 - \cos\frac{2\pi n}{M-1}\right)$	Sidelobes -31 dB, decay $1/\omega^3$	Discrete sampling

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(continued)

Kernel function	Mathematical form	Physical/mathematical meaning	Chapter
Hamming window kernel	$w(n) = 0.54 - 0.46\cos\frac{2\pi n}{M-1}$	Sidelobes -43 dB	Discrete sampling
Blackman window kernel	$w(n) = 0.42 - 0.5\cos\frac{2\pi n}{M-1} + 0.08\cos\frac{4\pi n}{M-1}$	Sidelobes -58 dB	Discrete sampling
Triangular window kernel	$w(n) = 1 - \left\lvert\frac{2n}{M-1} - 1\right\rvert$	Sidelobes -27 dB, decay $1/\omega^2$	Discrete sampling
Gaussian kernel	$K_\Delta(x) = \frac{1}{\sqrt{2\pi}\Delta}e^{-x^2/(2\Delta^2)}$	Optimal in time and frequency, no sidelobes	Time series analysis
Mexican-hat kernel	$K_{\mathsf{MexHat}}(x) = (1 - x^2/\Delta^2)e^{-x^2/(2\Delta^2)}$	Band-pass, second derivative	Time series analysis
Cauchy kernel	$K_\Delta(x) = \frac{1}{\pi}\frac{\Delta}{x^2+\Delta^2}$	Heavy-tailed, long-range	Time series analysis
Sinc kernel	$K(x) = \frac{\sin(\pi x/\Delta)}{\pi x/\Delta}$	Ideal low-pass	Harmonic analysis
Dirichlet kernel	$D_M(\theta) = \frac{\sin((M+\frac{1}{2})\theta)}{\sin(\theta/2)}$	Periodic window kernel, main lobe $2\pi/T$	Discrete sampling
Fejér kernel	$F_N(x) = \frac{1}{N+1}\left(\frac{\sin((N+1)x/2)}{\sin(x/2)}\right)^2$	Non-negative good kernel, Cesàro means	Harmonic analysis
(C) Phase similarity kernels (in periodic convolution)			
$S^{(1)}$	$e^{i(\theta_m - \varphi_n)}$	Basic difference angle (modulus of the inner/outer product)	Scalars and vectors
$S^{(2)}$	$e^{i(\theta_m + \varphi_n)}$	Basic sum angle (complex multiplication)	Scalars and vectors
$S^{(3)}$	$e^{i2(\theta_m - \varphi_n)}$	Double-angle difference angle (second-order reversal detection)	Scalars and vectors
$S^{(4)}$	$e^{i2(\theta_m + \varphi_n)}$	Double-angle sum angle	Scalars and vectors

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(continued)

Kernel function	Mathematical form	Physical/mathematical meaning	Chapter
$S^{(5)}$	$e^{i(\theta_m-\varphi_n)/2}$	Half-angle difference angle (modulus of the sum/difference vector)	Scalars and vectors
$S^{(6)}$	$e^{i(\theta_m+\varphi_n)/2}$	Half-angle sum angle	Scalars and vectors
(D) Weight kernels and periodic kernels			
Weight kernel W	$W(\omega_m - \nu_n)$	Frequency-domain selection (bandwidth control)	Series and convolution
Periodic kernel k_{per}	$\sigma^2 \exp\left(-\frac{2\sin^2(\pi\|x-x'\|/T)}{\ell^2}\right)$	GP periodic prior (MacKay)	Probabilistic systems
Gaussian-process kernel k_{SE}	$\sigma^2 e^{-(x-x')^2/(2\ell^2)}$	GP smooth prior	Probabilistic systems

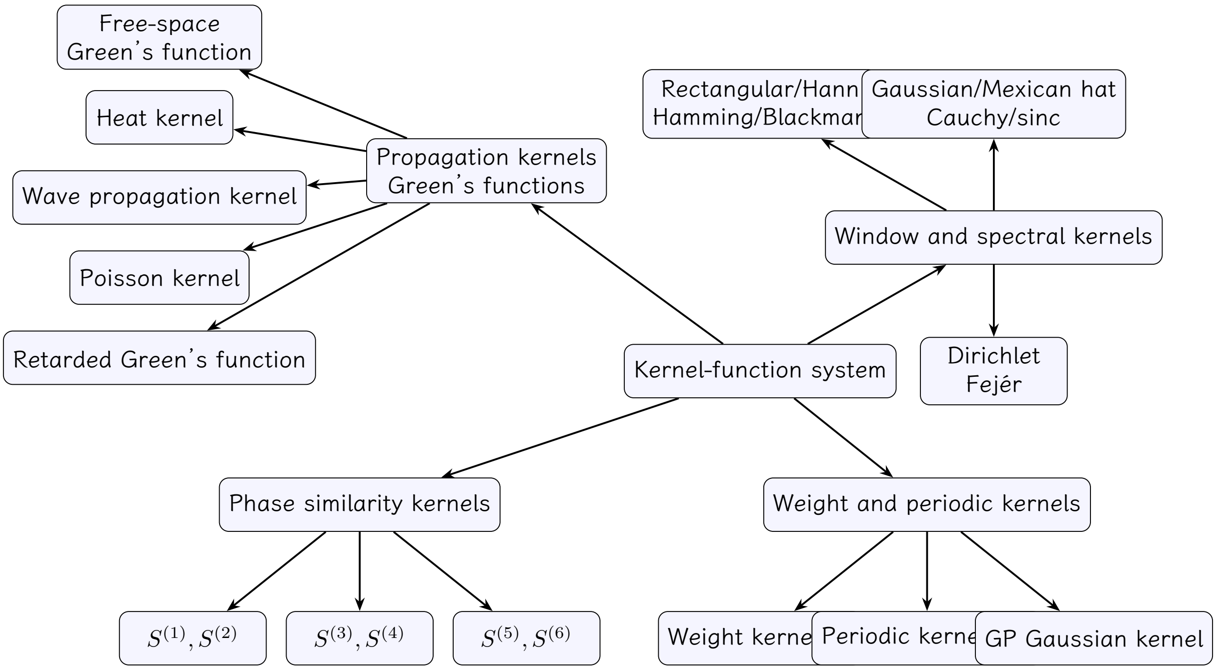

Free-space Green's function
Heat kernel
Wave propagation kernel
Poisson kernel
Retarded Green's function
Propagation kernels Green's functions
Rectangular/Hann Hamming/Blackman
Gaussian/Mexican hat Cauchy/sinc
Window and spectral kernels
Dirichlet Fejér
Kernel-function system
Phase similarity kernels
Weight and periodic kernels
$S^{(1)}, S^{(2)}$
$S^{(3)}, S^{(4)}$
$S^{(5)}, S^{(6)}$
Weight kernel
Periodic kernel
GP Gaussian kernel

Figure 17.8: Classification tree of kernel functions.

Property 17.2 (Duality relations among kernel functions)**.** The kernel functions satisfy the following duality relations:

(i) **Gaussian–Gaussian duality**: the Fourier transform of a Gaussian window kernel is again a Gaussian kernel (Lemma 17.3).

(ii) **Window–kernel duality**: the rectangular window kernel and the sinc kernel are Fourier duals of each other (Theorem 6.19).

(iii) **Phase-kernel conjugacy**: $S^{(d)}$ and $S^{(d')}$ satisfy the conjugacy duality $S^{(1)} = \overline{S^{(1)}_{\text{reverse}}}$ (Proposition 6.29).

(iv) **Propagation–evolution duality**: the spatial propagation kernel and the temporal evolution kernel are spacetime duals of each other (Theorem 17.6).

Proof. (i) Compute the Fourier transform of the Gaussian kernel directly. (ii) By the window–kernel duality theorem of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter (Theorem 6.19). (iii) By the conjugate symmetry of phase convolution in the 'Series and Convolution' chapter (Proposition 6.29). (iv) See Theorem 17.6. ☐

Lemma 17.3 (Fourier self-duality of the Gaussian kernel)**.** Let $K_\Delta(x) = \frac{1}{\sqrt{2\pi}\Delta}e^{-x^2/(2\Delta^2)}$. Then

$$\hat{K}_\Delta(\omega) = e^{-\Delta^2\omega^2/2},$$

that is, the Fourier transform of a Gaussian kernel is again a Gaussian kernel (the variances are interchanged, $\Delta \leftrightarrow 1/\Delta$).

Proof. By the standard Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2/(2\Delta^2)}e^{-i\omega x}\,dx = \sqrt{2\pi}\Delta e^{-\Delta^2\omega^2/2}$ (complete the square and then perform the Gaussian integral). ☐

Theorem 17.6 (Spacetime duality of the propagation kernel and the evolution kernel)**.** Let $G_{\mathsf{prop}}(\mathsf{x})$ be the spatial propagation kernel (the Green's function of the Helmholtz equation) and let $G_{\mathsf{evo}}(t)$ be the temporal evolution kernel. In the Fourier domain the two are linked by the spacetime dispersion relation $\omega(\mathsf{k})$:

$$\widehat{G_{\mathsf{prop}}}(\mathsf{k}) = \frac{1}{\omega(\mathsf{k})}, \qquad \widehat{G_{\mathsf{evo}}}(\omega) = \frac{1}{\omega},$$

and the spacetime convolution of G_{prop} with G_{evo} gives the complete propagation–evolution kernel.

Proof. In the Fourier domain the Helmholtz equation $(\nabla^2 + k^2)G = -\delta$ becomes $(-|\mathsf{k}|^2+k^2)\hat{G} = -1$, that is, $\hat{G}(\mathsf{k}) = 1/(|\mathsf{k}|^2-k^2)$. For the temporal evolution equation $\partial_t u = i\omega u$, the Fourier transform gives $\hat{G}_{\mathsf{evo}}(\omega) = 1/(-i\omega)$, whose modulus is $1/|\omega|$. The spacetime duality is provided by the dispersion relation $\omega = \omega(\mathsf{k})$. ☐

Example 17.4 (Far-field expansion of the free-space Green's function)**.** In the far field $|\mathsf{x}| \gg |\mathsf{x}'|$ the free-space Green's function $G(\mathsf{x},\mathsf{x}') = e^{ik\|\mathsf{x}-\mathsf{x}'\|}/(4\pi\|\mathsf{x}-\mathsf{x}'\|)$ expands as

$$G(\mathsf{x},\mathsf{x}') \approx \frac{e^{ik|\mathsf{x}|}}{4\pi|\mathsf{x}|}e^{-i\mathsf{k}\cdot\mathsf{x}'}, \qquad \mathsf{k} = k\hat{\mathsf{x}}.$$

This is the physical origin of the phase factor $e^{-i\mathsf{k}\cdot\mathsf{a}}$—the realisation, in the far-field expansion of the Green's function, of the Fourier multiplier representation of the propagation operator (Proposition 17.4).

17.4 Propagation convolution and evolution convolution

Remark 17.5 (Positioning of this section). Building on the kernel-function system (Section 17.3), this section defines **propagation convolution** and **evolution convolution.** Their common structure is that the action of a propagation operator or of an evolution operator on a function is equivalent to the convolution of the function with the corresponding kernel. The strict premises of this section agree with the rest of the chapter: a linear evolution equation plus a slowly varying medium. At the end of the section the relation of the two to the periodic convolution of the 'Series and Convolution' chapter is made explicit.

Definition 17.8 (Propagation convolution). Let $G_{\mathsf{prop}}(\mathsf{x},\mathsf{x}')$ be a spatial propagation kernel (Definition 17.7). The **propagation convolution** of a function f with G is defined by

$$(f \circledast_{\mathsf{prop}} G)(\mathsf{x}) := \int_{\mathbb{R}^n} f(\mathsf{x}')G(\mathsf{x},\mathsf{x}')\,d\mathsf{x}'.$$

When the medium is homogeneous (translation invariant), $G(\mathsf{x},\mathsf{x}') = G(\mathsf{x}-\mathsf{x}')$ and propagation convolution degenerates to ordinary convolution

$$(f \circledast_{\mathsf{prop}} G)(\mathsf{x}) = \int_{\mathbb{R}^n} f(\mathsf{x}')G(\mathsf{x}-\mathsf{x}')\,d\mathsf{x}' = (f * G)(\mathsf{x}).$$

Definition 17.9 (Evolution convolution). Let $G_{\mathsf{evo}}(t,t')$ be a temporal evolution kernel (the temporal dual of Definition 17.7). The **evolution convolution** of a function f with G is defined by

$$(f \circledast_{\mathsf{evo}} G)(t) := \int_{\mathbb{R}} f(t')G(t,t')\,dt'.$$

When the evolution equation has constant coefficients (is invariant under time translation), $G(t,t') = G(t-t')$ and evolution convolution degenerates to ordinary time convolution

$$(f \circledast_{\mathsf{evo}} G)(t) = (f * G)(t).$$

Definition 17.10 (Joint propagation–evolution convolution). Let $G(\mathsf{x},t;\mathsf{x}',t')$ be a spacetime propagation kernel. The **joint propagation–evolution convolution** of a function f with G is defined by

$$(f \circledast_{\mathsf{PE}} G)(\mathsf{x},t) := \int_{\mathbb{R}^n}\int_{\mathbb{R}} f(\mathsf{x}',t')G(\mathsf{x},t;\mathsf{x}',t')\,d\mathsf{x}'\,dt'.$$

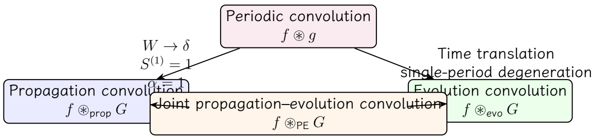
When the medium is homogeneous and the evolution has constant coefficients, $G(\mathsf{x},t;\mathsf{x}',t') = G(\mathsf{x}-\mathsf{x}',t-t')$ and the joint convolution is the ordinary spacetime convolution.

Theorem 17.7 (Relation between propagation convolution and periodic convolution). The periodic convolution of the single-period layer ($P = Q = \{T\}$, $W \to \delta$) (Definition 6.26) degenerates on the spatial translation group $\mathbb{R}^n$ to propagation convolution:

$$(f \circledast_{\mathsf{prop}} G)(\mathsf{x}) = \lim_{W\to\delta}(f \circledast g)(\mathsf{x}),$$

where g is taken to be the propagation kernel G and $S^{(d)}$ is taken to be the phase kernel $S^{(1)}$.

Proof. By the single-period-layer degeneration of Definition 6.26 (Proposition 6.31(i)) and Definition 17.8, both are convolutions over the same frequency pair $(n = m)$. The phase kernel satisfies $S^{(1)}(\theta, \theta) = 1$ and the amplitude switch $\alpha = 1$ preserves the amplitude. Hence periodic convolution degenerates to propagation convolution. ☐


Periodic convolution
$f \circledast g$
$W \to \delta$
$S^{(1)} = 1$
$\alpha = 1$
Time translation
single-period degeneration
Propagation convolution
$f \circledast_{\mathsf{prop}} G$
Joint propagation–evolution convolution
$f \circledast_{\mathsf{PE}} G$
Evolution convolution
$f \circledast_{\mathsf{evo}} G$

In the single-period degeneration the phase kernel, the weight kernel and the amplitude switch of periodic convolution become $S^{(1)} = 1$, $W = \delta$ and $\alpha = 1$ respectively.

Figure 17.9: Degeneration from periodic convolution to propagation convolution.

Property 17.3 (Algebraic properties of propagation convolution). In a translation-invariant medium, propagation convolution satisfies:

(i) **Commutativity**: $f \circledast_{\mathsf{prop}} G = G \circledast_{\mathsf{prop}} f$ (by the commutativity of convolution);

(ii) **Associativity**: $(f \circledast_{\mathsf{prop}} G_1) \circledast_{\mathsf{prop}} G_2 = f \circledast_{\mathsf{prop}} (G_1 \circledast_{\mathsf{prop}} G_2)$;

(iii) **Fourier multiplier**: $\widehat{f \circledast_{\mathsf{prop}} G}(\mathsf{k}) = \hat{f}(\mathsf{k})\hat{G}(\mathsf{k})$;

(iv) **Realisation by the propagation operator**: $P_{\mathsf{a}} f = f \circledast_{\mathsf{prop}} \delta_{\mathsf{a}}$, where $\delta_{\mathsf{a}}(\mathsf{x}) = \delta(\mathsf{x} - \mathsf{a})$.

Proof. (i)(ii) By the commutativity and associativity of ordinary convolution. (iii) By the convolution theorem. (iv) By the sifting property of the δ function: $\int f(\mathsf{x}')\delta(\mathsf{x} - \mathsf{a} - \mathsf{x}')\, d\mathsf{x}' = f(\mathsf{x} - \mathsf{a}) = P_{\mathsf{a}} f(\mathsf{x})$. ☐

Theorem 17.8 (Unification of propagation convolution and the propagation operator). Let G_{prop} be the propagation kernel in a homogeneous medium. Then the action of the propagation operator P_{a} on f is equivalent to the propagation convolution of f with δ_{a}:

$$P_{\mathsf{a}} f = f \circledast_{\mathsf{prop}} \delta_{\mathsf{a}}.$$

More generally, if G is an arbitrary propagation kernel, then $f \circledast_{\mathsf{prop}} G$ gives the propagation result with G as propagation kernel.

Proof. The first formula has already been proved in Property 17.3(iv). The second follows directly from Definition 17.8 and the definition of the propagation kernel, Definition 17.7. ☐

Example 17.5 (Evolution convolution of heat conduction). The propagation kernel of the one-dimensional heat equation $\partial_t u = \alpha \partial_x^2 u$ is the heat kernel $G(x,t) = \frac{1}{\sqrt{4\pi\alpha t}} e^{-x^2/(4\alpha t)}$. The evolution convolution

$$u(x,t) = (u_0 \circledast_{\mathsf{evo}} G)(x,t) = \int_{\mathbb{R}} u_0(x') \frac{1}{\sqrt{4\pi\alpha t}} e^{-(x-x')^2/(4\alpha t)}\, dx'$$

gives the evolution of the initial value u_0. By kernel duality (Property 17.2), the Fourier transform of the heat kernel is $e^{-\alpha k^2 t}$—this is the exponential decay factor of the frequency components, consistent with the Fourier solution of the heat equation in the 'Differential Equations and Analysis' chapter (Example 9.6).

Example 17.6 (Propagation convolution of free-space light). The propagation kernel of light in vacuum is the free-space Green's function $G(\mathsf{x}) = e^{ik|\mathsf{x}|}/(4\pi|\mathsf{x}|)$. The propagation convolution

$$u(\mathsf{x}) = (u_0 \circledast_{\mathsf{prop}} G)(\mathsf{x}) = \int_{\mathbb{R}^3} u_0(\mathsf{x}') \frac{e^{ik|\mathsf{x}-\mathsf{x}'|}}{4\pi|\mathsf{x}-\mathsf{x}'|}\, d\mathsf{x}'$$

gives the solution of the Helmholtz equation—this is the mathematical form of Kirchhoff's diffraction formula.

Remark 17.6 (Summary of the relation between propagation convolution and periodic convolution). The relation between propagation convolution and periodic convolution may be summarised as follows: **propagation convolution is the single-period degeneration of periodic convolution on the spatial translation group** (Theorem 17.7), and **evolution convolution is the single-period degeneration of periodic convolution on the time translation group**. The combination of the two (Definition 17.10) gives complete spacetime propagation. In propagation convolution the phase similarity kernel $S^{(d)}$, the weight kernel W and the amplitude switch α of periodic convolution degenerate respectively to: the phase kernel $S^{(1)} = 1$, the weight kernel $W = \delta$ and $\alpha = 1$.

Remark 17.7 (Fourier representation of the joint propagation–evolution convolution). By the product theorem for Fourier transforms, the joint propagation–evolution convolution in the spacetime frequency domain is

$$\widehat{f \circledast_{\mathsf{PE}} G}(\mathsf{k}, \omega) = \hat{f}(\mathsf{k}, \omega)\hat{G}(\mathsf{k}, \omega).$$

Here $\hat{G}(\mathsf{k}, \omega)$ is determined by the spacetime dispersion relation $\omega = \omega(\mathsf{k})$. In particular, the free-space $\hat{G}$ takes non-zero values on $\omega = c|\mathsf{k}|$—this is precisely a restatement, in the framework of joint convolution, of the plane-wave dispersion relation of the 'Periodic Structure of Electromagnetic Systems' chapter (Proposition 31.7).

Theorem 17.9 (Bloch–Floquet propagation). *Let the linear evolution operator $\mathcal{L}$ have spatial period T (that is, $P_{\mathsf{T}}\mathcal{L} = \mathcal{L}P_{\mathsf{T}}$). Then there exist Bloch wave solutions of the form*

$$u_{\mathsf{k},n}(\mathsf{x}, t) = e^{i(\mathsf{k}\cdot\mathsf{x} - \omega_n(\mathsf{k})t)}\phi_{\mathsf{k},n}(\mathsf{x}), \qquad \phi_{\mathsf{k},n}(\mathsf{x} + \mathsf{T}) = \phi_{\mathsf{k},n}(\mathsf{x})$$

Each n corresponds to an **energy band**, *$\omega_n(\mathsf{k})$ is the dispersion relation of the n-th energy band, and k ranges over the* **first Brillouin zone** *$\mathbb{R}^n/\mathsf{T}^*\mathbb{Z}^n$.*

Proof. Since $\mathcal{L}$ commutes with P_{T}, the two admit a simultaneous spectral decomposition on L^2. The operator P_{T} is unitary and its spectrum is the unit circle; each eigenvalue may be written in the form $e^{-i\mathsf{k}\cdot\mathsf{T}}$ (with k taken modulo T^*). The corresponding eigenspace is spanned by Bloch waves: $u(\mathsf{x}) = e^{i\mathsf{k}\cdot\mathsf{x}}\phi(\mathsf{x})$, where ϕ is periodic with period T. Grading further by $\omega_n(\mathsf{k})$ in the spectral decomposition of $\mathcal{L}$ gives the result. □

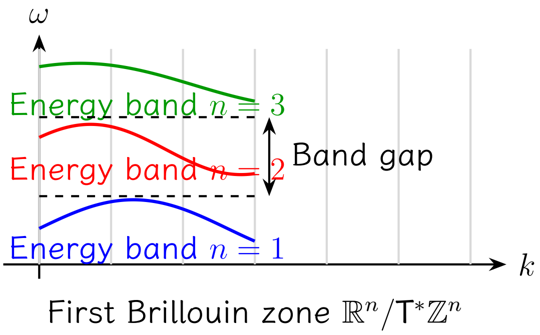


Figure 17.10: Bloch energy bands and the Brillouin zone.

Corollary 17.5 (Photonic band gap). In a photonic crystal whose dielectric constant is periodic with period T, if some frequency interval contains no energy band for any real k, then that interval is a **photonic band gap**. Light in the band gap is forbidden to propagate in the crystal.

Proof. By Theorem 17.9, the light propagation modes are described by the energy bands $\omega_n(\mathsf{k})$. If a frequency interval does not lie in the range of any $\omega_n(\mathbb{R}^n)$, then no corresponding Bloch wave solution exists and light cannot exist as a propagating mode. □

Property 17.4 (Frequency-domain support of the dispersion relation). In a periodic medium the dispersion relation $\omega_n(\mathsf{k})$ is a periodic function on the first Brillouin zone:

$$\omega_n(\mathsf{k}+\mathsf{G}) = \omega_n(\mathsf{k}), \qquad \forall \mathsf{G} \in \mathsf{T}^*\mathbb{Z}^n.$$

That is, the band structure has period G in the frequency domain—this is the representation, in the frequency domain, of the spatial period T (Pontryagin duality, Theorem 18.8 of the 'Groups and Group Functions' chapter).

Proof. By the definition of the Bloch wave, $u_{\mathsf{k},n} = e^{i(\mathsf{k}\cdot\mathsf{x}-\omega_n(\mathsf{k})t)}\phi_{\mathsf{k},n}(\mathsf{x})$; when $\mathsf{k} \to \mathsf{k}+\mathsf{G}$ (with G a reciprocal lattice vector), the product of $e^{i\mathsf{G}\cdot\mathsf{x}}$ with $\phi_{\mathsf{k},n}$ is still a function of period T, hence $\omega_n(\mathsf{k}+\mathsf{G}) = \omega_n(\mathsf{k})$. □

Remark 17.8 (Duality with Floquet theory of the 'Differential Equations and Analysis' chapter). Theorem 17.9 is the dual, in a spatially periodic medium, of Floquet's theorem (Theorem 9.6) in the 'Differential Equations and Analysis' chapter: the Floquet multiplier of a time-periodic system ↔ the Bloch phase of a spatially periodic system; the time evolution group ↔ the spatial propagation operator. Both are spectral theories of linear systems with periodic coefficients; only the direction of the periodicity differs.

Remark 17.9 (Interface with the 'Periodic Manifold' chapter). In the 'Periodic Manifold' chapter, a discrete unitary evolution $\mathcal{U}^p = I$ on lattice points gives a strict period. The evolution semigroup of this chapter is its continuum limit: as the time step $\Delta t \to 0$, the discrete evolution tends to the continuous evolution group $E_t = e^{tA}$ (Theorem 12.7). Bloch waves correspond on a periodic manifold to periodic orbits on lattice points—they already appear in the continuum limit of the Wilson loop of the 'Periodic Manifold' chapter (Lemma 12.1).

Remark 17.10 (Strict boundary of causality). The causality given by Corollary 17.4 is a mathematical property of the **linear wave equation**; its derivation uses only the equation itself and energy conservation, and introduces no assumption beyond 'linear + constant coefficients (or periodic coefficients) + finite energy'. In nonlinear media (strong-field optics, intense electromagnetic pulses in plasmas) the above conclusion of finite propagation speed **need not hold**—nonlinear terms can produce superluminal motion of the phase front. This chapter marks this boundary explicitly: all statements about propagation speed presuppose the **linear wave equation**.

17.5 Information evolution: Bayesian updating, activation functions and learning

Remark 17.11 (Positioning of this section and strict premises). This section extends the mathematical framework of propagation and evolution from physical fields to **information states**. The core observation is that Bayesian updating, pointwise activation functions and parameter learning possess a unified algebraic structure within the translation-group framework already established in this chapter. All definitions, propositions, lemmas, theorems and corollaries in this section are bounded by the following two premises:

(a) **Linear evolution premise**: the evolution of an information state in **natural-parameter space** or **log-belief space** is generated by the action of the translation group $\mathbb{R}^m$ (structurally isomorphic to Definition 17.3);

(b) **Pointwise activation premise**: the activation function is a **pointwise** map $\varphi : \mathbb{R} \to \mathbb{R}$ acting componentwise on the state vector and introducing no coupling between components.

Any statement involving non-convex optimisation, adaptive learning rates, vector-valued activation, gating mechanisms and so on, which goes beyond the above premises, is invariably marked as a **note** or an **open problem**. The concepts of this section are the **mathematical foundation** of the corresponding concepts in the 'Periodic Structure of Agents' chapter and the 'Epicycle Activation Function and Function Approximation' chapter; the content of those two chapters becomes an application within the framework of this section.

Positioning of the seven themes of this section: translation—Bayesian updating in log-belief space is an additive translation (Theorem 17.10), and pointwise activation commutes with the propagation operator (Proposition 17.7); scaling—the learning rate as the scaling factor of parametric evolution (Definition 17.18); rotation—the phase rotation of the complex exponential activation function (Proposition 17.8); conservation—monotone accumulation of the natural parameters under a conjugate prior (Corollary 17.7); symmetry—pointwise activation preserves translation invariance (Corollary 17.8); closure—convergence closure of parametric evolution (Theorem 17.12); period—the periodic structure of cosine-annealing learning rates (Proposition 17.10).

17.5.1 The translational structure of Bayesian updating

Definition 17.11 (Belief state and observation operator)**.** Let the parameter space be $\Theta \subseteq \mathbb{R}^m$. A **belief state** on Θ is a probability density $p : \Theta \to \mathbb{R}^+$ satisfying $\int_\Theta p(\theta)\, d\theta = 1$. The **observation operator** $\mathcal{O}_D$ maps a belief state p and observed data D to the posterior $p' = \mathcal{O}_D(p)$, defined by

$$(\mathcal{O}_D p)(\theta) := \frac{p(D \mid \theta)\, p(\theta)}{Z_D}, \qquad Z_D := \int_\Theta p(D \mid \theta')\, p(\theta')\, d\theta'.$$

If $Z_D = 0$, then D and p are said to be incompatible, and the observation operator is undefined there.

Definition 17.12 (Log-belief space)**.** Let $\mathcal{P}(\Theta)$ be the space of probability densities on Θ. The **log-belief map** is defined by

$$\Lambda : \mathcal{P}(\Theta) \to \mathcal{F}(\Theta), \qquad \Lambda(p) := \log p,$$

where $\mathcal{F}(\Theta)$ is the space of real-valued functions on Θ (modulo constants). Two functions in $\mathcal{F}(\Theta)$ are regarded as identical if they differ by a constant. $\Lambda(p)$ is called the **log-belief** of p.

Theorem 17.10 (Translational structure of Bayesian updating)**.** Let $p, p' \in \mathcal{P}(\Theta)$ with $p' = \mathcal{O}_D(p)$. Then in the log-belief space $\mathcal{F}(\Theta)$,

$$\boxed{\Lambda(p') = \Lambda(p) + \ell_D,}$$

where $\ell_D(\theta) := \log p(D \mid \theta) - \log Z_D$ is the **log-likelihood correction term** (modulo constants one may take $\log p(D \mid \theta)$). That is: **Bayesian updating in log-belief space is an additive translation with translation amount ℓ_D.**

Proof. By Definition 17.11 and Definition 17.12,

$$\Lambda(p')(\theta) = \log\Big(\frac{p(D \mid \theta)\, p(\theta)}{Z_D}\Big) = \log p(\theta) + \big[\log p(D \mid \theta) - \log Z_D\big] = \Lambda(p)(\theta) + \ell_D(\theta).$$

In the sense of modulo constants, the $-\log Z_D$ part of ℓ_D may be omitted. This completes the proof. $\square$

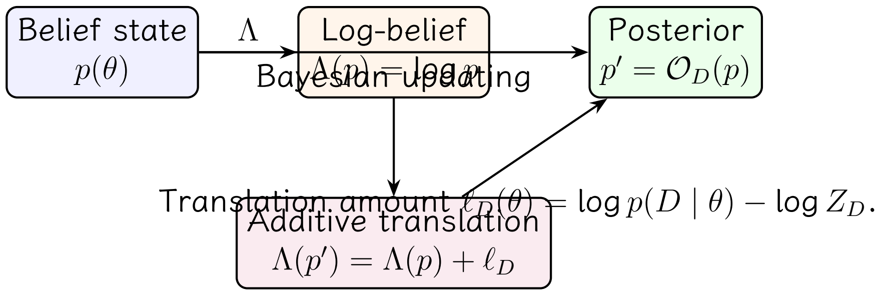

Belief state
$p(\theta)$
Λ
Log-belief
$\Lambda(p) = \log p$
Bayesian updating
Posterior
$p' = \mathcal{O}_D(p)$
Translation amount $\ell_D(\theta) = \log p(D \mid \theta) - \log Z_D$.
Additive translation
$\Lambda(p') = \Lambda(p) + \ell_D$

Figure 17.11: Additive translation in log-belief space.

Corollary 17.6 (Structural isomorphism between Bayesian updating and the propagation operator)**.** Bayesian updating (Theorem 17.10) and the propagation operator (Definition 17.3) are structurally isomorphic:

	action space	form of action
propagation operator	$L^2(\mathbb{R}^n)$	$f(\mathsf{x}) \mapsto f(\mathsf{x}-\mathsf{a})$
Bayesian updating	$\mathcal{F}(\Theta)$	$\Lambda(p)(\theta) \mapsto \Lambda(p)(\theta) + \ell_D(\theta)$

Their common algebraic structure is a **translation-group action**: the propagation operator is generated by the spatial translation group $\mathbb{R}^n$, whilst Bayesian updating is generated by the **data-dependent translation** determined by the log-likelihood.

Proof. Compare Proposition 17.2(i) (the propagation operators form a group) directly with Theorem 17.10 (Bayesian updating is an additive translation). ☐

Definition 17.13 (Exponential family and natural parameters)**.** Let $\mathcal{X}$ be the data space and $\Theta \subseteq \mathbb{R}^m$ the parameter space. A likelihood family $\{p(\cdot \mid \theta)\}_{\theta\in\Theta}$ is called an **exponential family** if there exist a sufficient statistic $T : \mathcal{X} \to \mathbb{R}^d$, a natural-parameter map $\eta : \Theta \to \mathbb{R}^d$, a log-partition function $A : \Theta \to \mathbb{R}$ and a base measure $h : \mathcal{X} \to \mathbb{R}^+$ such that

$$p(x \mid \theta) = h(x)\exp\big(\eta(\theta)\cdot T(x) - A(\theta)\big), \qquad x \in \mathcal{X}.$$

$\eta(\Theta) \subseteq \mathbb{R}^d$ is called the **natural-parameter space**.

Definition 17.14 (Conjugate prior and natural-parameter coordinates)**.** Let $\{p(\cdot \mid \theta)\}$ be an exponential family (Definition 17.13). The **conjugate prior family** is

$$\pi(\theta \mid \tau, \nu) := \frac{\exp\big(\tau\cdot\eta(\theta) - \nu A(\theta)\big)}{\int_\Theta \exp\big(\tau\cdot\eta(\theta') - \nu A(\theta')\big)\,d\theta'}, \qquad \tau \in \mathbb{R}^d,\ \nu > 0.$$

(τ, ν) are called the **natural-parameter coordinates** of the conjugate prior. When $d = 1$, $(\tau,\nu) \in \mathbb{R}\times\mathbb{R}^+$.

Theorem 17.11 (Exact finite-order evolution of Bayesian updating under a conjugate prior)**.** *Let $\{p(\cdot \mid \theta)\}$ be an exponential family (Definition 17.13), let the prior be the conjugate prior $\pi(\theta \mid \tau_0, \nu_0)$ (Definition 17.14), and let the observation sequence $x_1, \dots, x_n$ be independent and identically distributed from $p(\cdot \mid \theta)$. Then the posterior is again a conjugate prior, with natural-parameter coordinates*

$$\boxed{\tau_n = \tau_0 + \sum_{i=1}^{n} T(x_i), \qquad \nu_n = \nu_0 + n.}$$

That is: **Bayesian updating under a conjugate prior is, in the natural-parameter coordinates** (τ,ν)**, a finite-order additive evolution along the fixed direction** $(T(x_i), 1)$*—each observation corresponds to one translation.*

Proof. By Bayes' formula and the definition of the conjugate prior:

$$\pi(\theta \mid x_1,\dots,x_n) \propto \pi(\theta \mid \tau_0,\nu_0)\prod_{i=1}^{n} p(x_i \mid \theta) \propto \exp\Big(\big(\tau_0 + \textstyle\sum_i T(x_i)\big)\cdot\eta(\theta) - (\nu_0+n)A(\theta)\Big).$$

Comparing with the conjugate prior form (Definition 17.14) gives the result. Finite order: in the (τ,ν) coordinates each update is $(\tau,\nu) \mapsto (\tau + T(x_i), \nu+1)$—a first-order affine transformation. ☐

Corollary 17.7 (Monotone accumulation of natural parameters). In the notation of Theorem 17.11, ν_n is strictly increasing in n, and τ_n accumulates along the direction $T(x_i)$. This monotonicity is the algebraic root of the posterior contraction in the Bernstein–von Mises theorem (Theorem 14.3) of the 'Periodic Structures in Probabilistic Systems' chapter.

Proof. $\nu_n = \nu_0 + n$ is strictly increasing (it increases by exactly 1 at each step). $\tau_n - \tau_{n-1} = T(x_n)$ is the sufficient statistic of the n-th observation. ☐

Example 17.7 (Exact evolution of Beta–binomial conjugacy). Let the likelihood be $X \mid \theta \sim \text{Bernoulli}(\theta)$, with exponential-family form $p(x \mid \theta) = \theta^x (1-\theta)^{1-x}$, $T(x) = x$, $\eta(\theta) = \log(\theta/(1-\theta))$ and $A(\theta) = -\log(1-\theta)$. The conjugate prior is the Beta distribution $\text{Beta}(\alpha, \beta)$, with natural-parameter coordinates $(\tau, \nu) = (\alpha - 1, \alpha + \beta - 2)$ (the precise choice of coordinates depends on convention). After n observations,

$$\alpha_n = \alpha_0 + \sum_{i=1}^{n} x_i, \qquad \beta_n = \beta_0 + n - \sum_{i=1}^{n} x_i.$$

This agrees with the conclusion of the Beta–binomial conjugate model in the 'Periodic Structures in Probabilistic Systems' chapter (Example 14.6); here its **translation structure** is given in natural-parameter coordinates.

Example 17.8 (Exact evolution of Gaussian–Gaussian conjugacy). Let the likelihood be $X \mid \mu \sim \mathcal{N}(\mu, \sigma^2)$ (with σ^2 known) and let the prior on the parameter μ be $\mathcal{N}(\mu_0, \tau_0^2)$. Taking the precision-mean product coordinates $(1/\tau^2,\ \mu/\tau^2)$ as natural-parameter coordinates, after n observations

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}, \qquad \frac{\mu_n}{\tau_n^2} = \frac{\mu_0}{\tau_0^2} + \frac{\sum_i x_i}{\sigma^2}.$$

This agrees with the normal–normal conjugate model of the 'Periodic Structures in Probabilistic Systems' chapter (Example 14.8); here its **translation structure** is given in natural-parameter coordinates.

Remark 17.12 (Correspondence table of finite-order evolution and the propagation operator). Combining Theorem 17.10 with Theorem 17.11, Bayesian updating and the propagation operator satisfy the following structural correspondence:

Object	Propagation operator	Bayesian updating
Action space	field function $L^2(\mathbb{R}^n)$	log-belief $\mathcal{F}(\Theta)$ or natural-parameter space $\mathbb{R}^d \times \mathbb{R}^+$
Translation amount	spatial vector a	log-likelihood ℓ_D or sufficient statistic $T(x)$
Generator	$-\nabla$ (spatial directional derivative)	observation operator (integration in the likelihood direction)
Group structure	translation group $\mathbb{R}^n$	data-dependent translation (each observation defines one translation)
Transfer matrix (frequency domain)	$e^{-i\mathsf{k}\cdot\mathsf{a}}$	addition of the natural parameters under a conjugate prior

17.5.2 Pointwise activation functions and the propagation operator

Definition 17.15 (Pointwise activation function). *A map $\varphi : \mathbb{R} \to \mathbb{R}$ is called a* **pointwise activation function**. *Its action on a function $f : \mathbb{R}^n \to \mathbb{R}$ is defined by*

$$(\varphi \cdot f)(\mathsf{x}) := \varphi(f(\mathsf{x})), \qquad \mathsf{x} \in \mathbb{R}^n.$$

The family of pointwise activation functions includes: the identity map $\varphi(x) = x$, the Heaviside step $H(x)$, ReLU $\max(0, x)$, Sigmoid $\sigma(x)$, Tanh, GELU, Swish, and complex exponential activation (Definition 17.16).

Proposition 17.7 (Commutativity of pointwise activation and the propagation operator). *Let $\varphi : \mathbb{R} \to \mathbb{R}$ be a pointwise activation function (Definition 17.15) and let P_a be a propagation operator (Definition 17.3). Then*

$$\boxed{P_\mathsf{a} \circ \varphi = \varphi \circ P_\mathsf{a}.}$$

That is, a pointwise activation function and the propagation operator commute exactly.

Proof. For arbitrary $f : \mathbb{R}^n \to \mathbb{R}$ and $\mathsf{x} \in \mathbb{R}^n$, compute both sides:

$$(P_\mathsf{a} \circ \varphi)(f)(\mathsf{x}) = (\varphi \cdot f)(\mathsf{x} - \mathsf{a}) = \varphi(f(\mathsf{x} - \mathsf{a})),$$

$$(\varphi \circ P_\mathsf{a})(f)(\mathsf{x}) = \varphi((P_\mathsf{a} f)(\mathsf{x})) = \varphi(f(\mathsf{x} - \mathsf{a})).$$

The two sides are equal. □

$$\begin{tikzcd} f \arrow[r, "P_{\mathsf{a}}"] \arrow[d, "\varphi"'] & P_{\mathsf{a}} f \arrow[d, "\varphi"] \\ \varphi \cdot f \arrow[r, "P_{\mathsf{a}}"'] & P_{\mathsf{a}}(\varphi \cdot f) \end{tikzcd}$$

$P_{\mathsf{a}} \circ \varphi = \varphi \circ P_{\mathsf{a}}$: pointwise activation and the propagation operator commute exactly.

Figure 17.12: Commutative diagram for pointwise activation and the propagation operator.

Corollary 17.8 (Pointwise activation preserves translation invariance)**.** *Let f be a P_{a}-invariant function (that is, $P_{\mathsf{a}} f = f$). Then $\varphi \cdot f$ is also P_{a}-invariant:*

$$P_{\mathsf{a}}(\varphi \cdot f) = \varphi \cdot f.$$

Proof. By Proposition 17.7, $P_{\mathsf{a}}(\varphi \cdot f) = \varphi \cdot (P_{\mathsf{a}} f) = \varphi \cdot f$. ☐

Remark 17.13 (Linear transformations break translation invariance)*.* Pointwise activation commutes with the propagation operator (Proposition 17.7), but a general **linear transformation** $W : \mathbb{R}^n \to \mathbb{R}^n$ does not commute with the propagation operator: $W \circ P_{\mathsf{a}} \neq P_{\mathsf{a}} \circ W$ (unless W is compatible with the direction of a). This explains the algebraic necessity of the alternating structure 'linear transformation $\to$ pointwise activation $\to$ linear transformation $\to \cdots$' in neural networks: a linear transformation breaks translation invariance, a pointwise activation preserves it, and the alternation of the two produces intermediate representations that are neither completely invariant nor completely arbitrary.

Definition 17.16 (Complex exponential activation function)**.** Let $N \in \mathbb{N}^+$, $A_k > 0$, $\omega_k \in \mathbb{R}$ and $\phi_k \in [0, 2\pi)$. The **complex exponential activation function** (CEAF) is defined by

$$\mathcal{E}(x) := \sum_{k=1}^{N} A_k e^{i(\omega_k x + \phi_k)}, \qquad x \in \mathbb{R}.$$

Its readout modes include: the real-part readout $\mathsf{Re}\,\mathcal{E}(x)$, the two-channel readout $(\mathsf{Re}\,\mathcal{E}(x), \mathsf{Im}\,\mathcal{E}(x))$, and the modulus-phase readout $(|\mathcal{E}(x)|, \mathsf{arg}\,\mathcal{E}(x))$.

Proposition 17.8 (Propagation structure of the CEAF)**.** *Under translation the CEAF behaves as a pure phase rotation: for every $a \in \mathbb{R}$,*

$$\mathcal{E}(x + a) = \sum_{k=1}^{N} A_k e^{i(\omega_k x + \phi_k + \omega_k a)}.$$

That is, a translation by a is equivalent to increasing the phase of the k-th component by $\omega_k a$—this is the componentwise realisation on the CEAF of Proposition 17.5 (consistency of the propagation operator and the phase operator).

Proof. Substitute Definition 17.16 and $e^{i(\omega_k(x+a)+\phi_k)} = e^{i(\omega_k x + \phi_k)} e^{i\omega_k a}$. ☐

Corollary 17.9 (Periodicity condition of the CEAF)**.** If $\omega_k/\omega_0 \in \mathbb{Z}$ for all k (where ω_0 is the base frequency), then $\mathcal{E}$ is periodic with period $T = 2\pi/\omega_0$; otherwise $\mathcal{E}$ is an almost periodic function. This is the term-by-term realisation on the CEAF of the periodic-function criterion of the 'Operations and Periodicity' chapter.

Proof. If $\omega_k = n_k\omega_0$ ($n_k \in \mathbb{Z}$), then $\mathcal{E}(x+T) = \sum_k A_k e^{i(n_k\omega_0 x+\phi_k+n_k\omega_0\cdot 2\pi/\omega_0)} = \sum_k A_k e^{i(n_k\omega_0 x+\phi_k)} = \mathcal{E}(x)$. If there exists an incommensurable pair of frequencies, then $\mathcal{E}$ is a Bohr almost periodic function. □

Example 17.9 (Phase rotation of a single-component CEAF)**.** Take $\mathcal{E}(x) = e^{i\omega x}$ ($N = 1$, $A_1 = 1$, $\phi_1 = 0$). A translation by a gives $\mathcal{E}(x+a) = e^{i\omega(x+a)} = e^{i\omega x}e^{i\omega a}$—the phase increases by ωa. When $a = 2\pi/\omega$ the phase increases by 2π and $\mathcal{E}$ returns to its starting value—this is the periodic closure of propagation on the CEAF (Corollary 17.3).

Remark 17.14 (Relation between the CEAF and kernel-smoothed derivative-integral descent)*.* The parameters $\{A_k, \omega_k, \phi_k\}$ of the CEAF are optimised by kernel-smoothed derivative-integral descent (Definition 17.19). This section establishes only the propagation and phase structure of the CEAF (Proposition 17.8, Corollary 17.9); its optimisation algorithm is given in Section 17.5.3 as a special case of parametric evolution.

17.5.3 Learning as parametric evolution

Definition 17.17 (Parametric evolution equation)**.** Let the loss function $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ be C^1 smooth and let $\eta > 0$ be the **learning rate** (scaling factor). The **parametric evolution equation** is

$$\theta_{t+1} = \theta_t - \eta\,\nabla\mathcal{L}(\theta_t), \qquad t = 0, 1, 2, \dots.$$

If $\mathcal{L}$ attains a minimum at θ^*, then the behaviour of θ_t as $t \to \infty$ is called **parametric evolution**.

Definition 17.18 (Learning rate)**.** The quantity $\eta > 0$ in Definition 17.17 is called the **learning rate**. The learning rate is the scaling factor of parametric evolution—the larger η is, the larger the parameter translation at each step (fast but possibly unstable); the smaller η is, the smaller the parameter translation at each step (slow but more stable).

Theorem 17.12 (Linear evolution and convergence condition under quadratic loss)**.** *Let the loss function be the quadratic form $\mathcal{L}(\theta) = \frac{1}{2}(\theta-\theta^*)^\top H(\theta-\theta^*)$, where H is a symmetric positive definite matrix with eigenvalues $0 < \lambda_1 \leq \cdots \leq \lambda_d$. Then parametric evolution (Definition 17.17) is the* ***linear evolution***

$$\theta_{t+1} - \theta^* = (I - \eta H)(\theta_t - \theta^*),$$

and the error along the i-th eigendirection is

$$e_t^{(i)} = (1-\eta\lambda_i)^t\, e_0^{(i)}.$$

The ***necessary and sufficient condition for convergence*** *is*

$$0 < \eta < \frac{2}{\lambda_d}.$$

When $\eta = 2/(\lambda_1 + \lambda_d)$, the maximum of the convergence rates $|1 - \eta\lambda_i|$ is least—in that case $\max_i |1 - \eta\lambda_i| = (\lambda_d - \lambda_1)/(\lambda_1 + \lambda_d)$.

Proof. Decompose the error vector $\mathbf{e}_t := \theta_t - \theta^* = \sum_i e_t^{(i)} \mathbf{v}_i$ along the eigenbasis of H. Since $\nabla\mathcal{L}(\theta_t) = H(\theta_t - \theta^*)$, the parametric evolution equation gives $\theta_{t+1} - \theta^* = (I - \eta H)(\theta_t - \theta^*)$. The eigendirections decouple: $e_{t+1}^{(i)} = (1 - \eta\lambda_i) e_t^{(i)}$, and iteration gives $e_t^{(i)} = (1 - \eta\lambda_i)^t e_0^{(i)}$. Convergence holds if and only if $|1 - \eta\lambda_i| < 1$ for every i, that is, $0 < \eta\lambda_i < 2$, which is equivalent to $0 < \eta < 2/\lambda_d$. The optimal learning rate is given by minimising the spectral radius $\rho(\eta) = \max_i |1 - \eta\lambda_i|$; the spectral radius is least at $\eta = 2/(\lambda_1 + \lambda_d)$, where the convergence rates in the λ_1 and λ_d directions are equal, namely $(\lambda_d - \lambda_1)/(\lambda_1 + \lambda_d)$. ☐

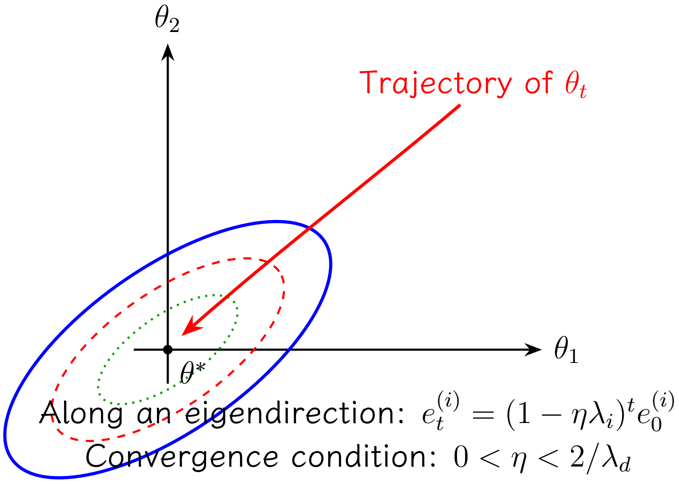

θ_2
Trajectory of θ_t
θ_1
θ^*
Along an eigendirection: $e_t^{(i)} = (1 - \eta\lambda_i)^t e_0^{(i)}$
Convergence condition: $0 < \eta < 2/\lambda_d$

Figure 17.13: Convergence of parametric evolution under quadratic loss.

Corollary 17.10 (Parametric evolution as the discrete realisation of the linear evolution semigroup)**.** Parametric evolution (Theorem 17.12) is the discrete realisation of the evolution semigroup of this chapter (Definition 17.6): the generator is $-H$ and the semigroup is $\{(I - \eta H)^t\}_{t \geq 0}$. In the continuum limit $\eta \to 0$,

$$\theta(t) = e^{-Ht}\theta(0)$$

corresponds to continuous-time parametric evolution.

Proof. Iterating the parametric evolution equation $\theta_{t+1} = (I - \eta H)\theta_t$ gives $\theta_t = (I - \eta H)^t \theta_0$. Taking $\eta \to 0$, $t \to \infty$ and $\eta t \to s$, we get $(I - \eta H)^t \to e^{-Hs}$, which is the continuum limit. ☐

Definition 17.19 (Kernel-smoothed derivative-integral descent)**.** Let the loss function $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ be C^1 smooth and let $\theta \in \mathbb{R}^d$ be a parameter. The update rule of **kernel-smoothed derivative-integral descent** (KDI) is

$$\theta_{t+1} = \theta_t - \eta\Big[\alpha\, \nabla\mathcal{L}(\theta_t) + \beta\, \mathbf{d}_{\mathsf{int}}(\theta_t) + \gamma\, \mathbf{d}_{\mathsf{curv}}(\theta_t)\Big],$$

where $\alpha, \beta, \gamma \geq 0$ are weight coefficients, $\mathbf{d}_{\mathsf{int}}$ is the **accumulated-variation correction term** (integral information) and $\mathbf{d}_{\mathsf{curv}}$ is the **curvature correction term** (second-derivative information). When $\beta = \gamma = 0$, KDI degenerates to standard gradient descent.

Proposition 17.9 (Descent property of the KDI direction). Let $\mathcal{L}$ be C^1 in a neighbourhood of θ and let the KDI direction be

$$\mathbf{d}_{\mathsf{KDI}}(\theta) := \alpha\,\nabla\mathcal{L}(\theta) + \beta\,\mathbf{d}_{\mathsf{int}}(\theta) + \gamma\,\mathbf{d}_{\mathsf{curv}}(\theta).$$

If $\nabla\mathcal{L}(\theta) \cdot \mathbf{d}_{\mathsf{KDI}}(\theta) > 0$ (that is, the inner product of the KDI direction with the negative-gradient direction is positive), then there exists $\eta > 0$ such that

$$\mathcal{L}(\theta - \eta\,\mathbf{d}_{\mathsf{KDI}}(\theta)) < \mathcal{L}(\theta).$$

Proof. The directional derivative is $\frac{d}{d\eta}\big|_{\eta=0}\mathcal{L}(\theta - \eta\mathbf{d}_{\mathsf{KDI}}) = -\nabla\mathcal{L}(\theta) \cdot \mathbf{d}_{\mathsf{KDI}}(\theta)$. If this value is negative (that is, $\nabla\mathcal{L} \cdot \mathbf{d}_{\mathsf{KDI}} > 0$), then the function value decreases strictly for sufficiently small η. ☐

Remark 17.15 (Strict boundary of KDI). Proposition 17.9 states only the **descent property** of KDI—as long as the angle between the direction and the negative gradient is less than $90°$, descent occurs. The **advantages** of KDI over standard gradient descent (escaping saddle points, crossing local minima, accelerating across flat regions) depend on the geometric structure of the specific problem and **do not constitute theorems of this chapter**. The convergence analysis of KDI (especially in the non-convex case) is an **open problem** (Remark 17.18).

Definition 17.20 (Learning-rate schedule). A **learning-rate schedule** is the learning rate η_t regarded as a function of the iteration step t. Common types:

(i) **Step decay**: $\eta_t = \eta_0\gamma^{\lfloor t/T \rfloor}$ ($\gamma \in (0,1)$, T is the decay period);

(ii) **Cosine annealing**: $\eta_t = \eta_{\mathsf{min}} + \frac{1}{2}(\eta_{\mathsf{max}} - \eta_{\mathsf{min}})\big(1 + \cos(\pi t/T)\big)$;

(iii) **Exponential decay**: $\eta_t = \eta_0 e^{-t/\tau}$.

A learning-rate schedule is a modulation in time of the scaling factor of parametric evolution—from a large learning rate (rapid exploration) to a small learning rate (fine convergence).

Proposition 17.10 (Periodic structure of the cosine-annealing learning rate). The cosine-annealing learning rate $\eta_t = \eta_{\mathsf{min}} + \frac{1}{2}(\eta_{\mathsf{max}} - \eta_{\mathsf{min}})(1 + \cos(\pi t/T))$ is a periodic function of period $2T$:

$$\eta_{t+2T} = \eta_t, \qquad \forall t \geq 0.$$

Within each period, η_t decreases monotonically from η_{max} to η_{min} ($t \in [0, T]$), and then increases monotonically back to η_{max} ($t \in [T, 2T]$).

Proof. Periodicity follows from $\cos(\pi(t+2T)/T) = \cos(\pi t/T + 2\pi) = \cos(\pi t/T)$. The monotonicity is given by the monotonicity of the cosine on $[0, \pi]$ and $[\pi, 2\pi]$. ☐

Example 17.10 (Numerical verification of parametric evolution). Take $d = 2$, $H = \mathrm{diag}(1, 4)$, $\theta^* = (0, 0)$ and the initial value $\theta_0 = (1, 1)$. By Theorem 17.12, the convergence condition is $0 < \eta < 2/4 = 0.5$. Take $\eta = 0.3$:

- First eigendirection: $|1 - \eta\lambda_1| = |1 - 0.3| = 0.7$;
- Second eigendirection: $|1 - \eta\lambda_2| = |1 - 1.2| = 0.2$.

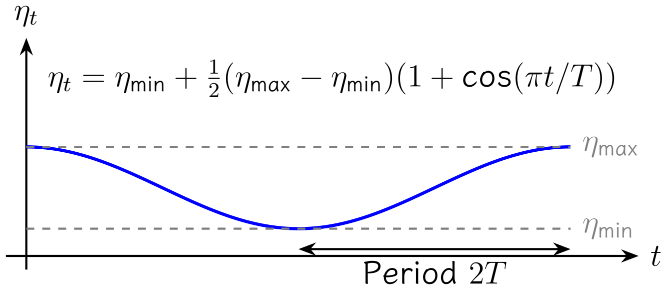


Figure 17.14: Periodic structure of the cosine-annealing learning rate.

The error $\|\theta_t - \theta^*\|$ converges with a decay factor of at least 0.7 per step. Take $\eta = 0.6 > 0.5$: $|1 - \eta\lambda_2| = |1 - 2.4| = 1.4 > 1$, and the second eigendirection diverges—parametric evolution is unstable.

Theorem 17.13 (Convergence closure of parametric evolution). Let $\mathcal{L}$ be an L-smooth, μ-strongly convex function on $\mathbb{R}^d$ $(0 < \mu \leq L)$. Then with $\eta = 1/L$, parametric evolution (Definition 17.17) satisfies

$$\|\theta_t - \theta^*\|^2 \leq \left(1 - \frac{\mu}{L}\right)^t \|\theta_0 - \theta^*\|^2.$$

In particular, $\theta_t \to \theta^*$ converges exponentially—this is the **convergence closure of parametric evolution**.

Proof. By L-smoothness (the descent lemma) and μ-strong convexity, the standard analysis gives

$$\mathcal{L}(\theta_{t+1}) - \mathcal{L}(\theta^*) \leq \left(1 - \frac{\mu}{L}\right)\big(\mathcal{L}(\theta_t) - \mathcal{L}(\theta^*)\big).$$

By μ-strong convexity, $\|\theta_t - \theta^*\|^2 \leq \frac{2}{\mu}(\mathcal{L}(\theta_t) - \mathcal{L}(\theta^*))$; substituting gives the result. □

Example 17.11 (Sequential updating of a conjugate prior and composition of the propagation operator). Consider the Beta–binomial conjugate model (Example 17.7). Each observation $x_i \in \{0, 1\}$ corresponds to one translation of the natural parameters:

$$(\tau, \nu) \mapsto (\tau + T(x_i), \nu + 1) = (\tau + x_i, \nu + 1).$$

The accumulation over n observations is Theorem 17.11. This is a concrete instance of **Bayesian updating as a composition of propagation operators**—each observation corresponds to a translation operator $P_{(\tau+x_i,\nu+1)}$, and n observations correspond to the composition $P_{(\tau+\sum x_i,\nu+n)}$.

Remark 17.16 (Unified table of the three information evolutions). The three information evolutions established in this section (Bayesian updating, pointwise activation and parameter learning) are unified within the framework of this chapter as translation-group actions:

Information evolution	Action space	Translation amount	Group generator
Bayesian updating	log-belief space $\mathcal{F}(\Theta)$	log-likelihood ℓ_D	observation operator
Conjugate-prior Bayesian updating	natural-parameter space $\mathbb{R}^d \times \mathbb{R}^+$	sufficient statistic $(T(x), 1)$	data-dependent translation
Pointwise activation	function space (pointwise action)	no translation (commutes with P_a)	pointwise map
Parameter learning (quadratic loss)	parameter space $\mathbb{R}^d$	translation in the gradient direction	$-H$

Remark 17.17 (Interfaces with other chapters of the book). The interfaces between the mathematical structures established in this section and the other chapters of the book are as follows. All the entries cited from the 'Periodic Structure of Agents' chapter and the 'Epicycle Activation Function and Function Approximation' chapter **come after this chapter**—the definitions and propositions of this section are the **mathematical foundation** of the corresponding concepts in those chapters, and the content of those chapters becomes an application within the framework of this section.

(i) **The 'Periodic Structures in Probabilistic Systems' chapter**: Theorem 17.10 and Theorem 17.11 provide the algebraic foundation of the translation-group structure for sequential Bayesian updating (Definition 14.25) and conjugate priors (Definition 14.23);

(ii) **The 'Periodic Structure of Agents' chapter**: the CEAF (Definition 17.16) and parametric evolution (Definition 17.17) provide the mathematical foundation for that chapter's neural-network training dynamics; Definition 44.5 and Definition 44.7 in that chapter become, within the framework of this section, application instances of Definition 17.16 and Definition 17.19;

(iii) **The 'Epicycle Activation Function and Function Approximation' chapter**: the propagation structure of the CEAF (Proposition 17.8) provides a phase-analysis tool for that chapter's approximation theory;

(iv) **The 'Groups and Group Functions' chapter**: Proposition 17.7 is the direct application to activation functions of that chapter's commutation law between pointwise operations and group actions (Lemma 18.2).

Remark 17.18 (Open problems of this section). The following problems lie beyond the range of the theorems already established in this section and are listed as **open problems of this section**:

(i) **Finite-order evolution under a non-conjugate prior**: when the prior is not a conjugate prior, Bayesian updating generally **cannot** be represented as a finite-order translation in a finite-dimensional natural-parameter space. Whether there exists a more general finite-dimensional embedding that makes the update a finite-order evolution is an open problem;

(ii) **Convergence of parametric evolution under a non-convex loss**: both Theorem 17.12 and Theorem 17.13 require $\mathcal{L}$ to be convex. Under a non-convex loss, parametric evolution may become trapped in a local minimum or a saddle point, and its convergence behaviour (in particular the global convergence of KDI) is an open problem;

(iii) **Commutation of propagation under vector-valued activation**: Proposition 17.7 relies on the pointwise premise. The commutation of a vector-valued activation (such as softmax) with the propagation operator requires additional conditions and lies beyond the scope of this section;

(iv) **Group structure under adaptive learning rates**: in Definition 17.20 the learning-rate schedule depends on the iteration step t. For an adaptive learning rate (such as Adam) the learning rate depends on the gradient history, and parametric evolution is then no longer a representation of a fixed group—its algebraic structure is an open problem.

Remark 17.19 (Open problems of the whole chapter). The following problems lie beyond the range of the theorems already established in this chapter and are listed as open problems **of the chapter as a whole** (to be distinguished from the section-level open problems of Remark 17.18—the latter belong to Section 17.5, whilst the former belong to the remaining sections of this chapter taken as a whole):

(i) a rigorous dispersion theory of wave packets in nonlinear periodic media;

(ii) Anderson localisation of electromagnetic wave packets in random periodic media, and a quantitative description of their propagation;

(iii) a rigorous propagation theory of the Airy kernel near a caustic surface (this chapter only lists it in Remark 17.3 and in the kernel table 17.1, without developing it);

(iv) the extension of the propagation operator and of periodic convolution to non-commutative groups (such as $SU(2)$ or more general Lie groups), and its interface with the phase functor of the 'Periodic Category' chapter;

(v) the doubly periodic structure of propagation together with time translation and spatial translation in time-varying periodic media.

The solution of these problems would advance the theory of propagation and evolution from the linear constant-coefficient case to more general periodic systems.

Remark 17.20 (Seven-theme correspondence of this chapter with the whole book).

Theme	Corresponding content in this chapter
Translation	The propagation-operator group (Proposition 17.2); the evolution semigroup (Definition 17.6); periodic closure of propagation (Corollary 17.3); additive translation of Bayesian updating in log-belief space (Theorem 17.10); finite-order translation of natural-parameter coordinates under a conjugate prior (Theorem 17.11)
Scaling	Wavelength scaling in a medium (Proposition 17.1); the refractive-index scaling of the medium propagation operator (Definition 17.4); the learning rate as the scaling factor of parametric evolution (Definition 17.18)
Rotation	Consistency of the propagation operator and the phase operator (Proposition 17.5); the complex exponential plane-wave representation (Definition 17.5); the phase rotation of the CEAF (Proposition 17.8)
Conservation	Energy conservation of vacuum light propagation (Theorem 17.4); flux conservation of a ray tube (Property 17.1); monotone accumulation of the natural parameters under a conjugate prior (Corollary 17.7)
Symmetry	The propagation operator commutes with constant-coefficient differential operators (Proposition 17.6); invariance of Bloch waves under periodic translation (Theorem 17.9); pointwise activation preserves translation invariance (Corollary 17.8)
Closure	Periodic closure of propagation (Corollary 17.3); confinement of propagation in a photonic band gap (Corollary 17.5); convergence closure of parametric evolution (Theorem 17.13)
Period	Frequency-domain periodicity of the dispersion relation (Property 17.4); Bloch band structure (Theorem 17.9); periodic structure of the cosine-annealing learning rate (Proposition 17.10)

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The free-space propagation operator is the algebraic realisation of spatial translation (Definition 17.3), and the family of propagation operators forms a translation group (Proposition 17.2); the time direction is carried by the evolution semigroup (Definition 17.6); Bayesian updating in log-belief space is a data-dependent additive translation (Theorem 17.10); parametric evolution is a translation in the gradient direction (Definition 17.17)
Scaling	Wavelength scaling in a medium (Proposition 17.1); the learning rate as the scaling factor of parametric evolution (Definition 17.18, Definition 17.20)
Rotation	Consistency of the propagation operator and the phase operator (Proposition 17.5); the complex exponential activation function behaves as a pure phase rotation under propagation (Proposition 17.8); the Bloch–Floquet phase in a periodic medium (Theorem 17.9)
Conservation	Energy conservation of vacuum light propagation (Theorem 17.4); the isometry of the propagation operator, that is, conservation of the L^2 norm (Proposition 17.3); causality (Corollary 17.4)
Symmetry	The propagation operator commutes with constant-coefficient differential operators (Proposition 17.6); pointwise activation commutes with the propagation operator (Proposition 17.7), and pointwise activation preserves translation invariance (Corollary 17.8)
Closure	Periodic closure of propagation (Corollary 17.3); convergence closure of parametric evolution (Theorem 17.13)
Period	Bloch–Floquet propagation gives the periodic structure in a periodic medium (Theorem 17.9) and the photonic band gap (Corollary 17.5); the relation between propagation convolution and periodic convolution (Theorem 17.7); the exact finite-order evolution of Bayesian updating under a conjugate prior (Theorem 17.11); periodic structure of the cosine-annealing learning rate (Proposition 17.10)

Exercises

Exercise 1 Using Definition 17.1 and Definition 17.2, explain how Fermat's principle reduces a ray trajectory to a first-order stationary curve of the optical-path functional. Use Lemma 17.1 to write down the first variation of the optical-path functional, and explain how the boundary terms are handled in a variation with fixed endpoints.

Exercise 2 Prove the fundamental lemma of the calculus of variations (Lemma 17.2): let $f \in C^0([s_0, s_1], \mathbb{R}^n)$; if for every C^1 vector function h satisfying $\mathsf{h}(s_0) = \mathsf{h}(s_1) = 0$ we have

$$\int_{s_0}^{s_1} f(s) \cdot \mathsf{h}(s)\, ds = 0,$$

then $f \equiv 0$. Construct a proof by contradiction using a cut-off function.

Exercise 3 Prove the ray equation (Theorem 17.1):

$$\frac{d}{ds}\big(n(\gamma(s))\dot{\gamma}(s)\big) = \nabla n(\gamma(s)).$$

Complete the proof in four steps—'expand the first variation, integrate by parts, set the boundary terms to zero, apply the fundamental lemma of the calculus of variations'—and explain the degeneration to straight-line propagation in a homogeneous medium (Corollary 17.2).

Exercise 4 Prove the eikonal equation (Corollary 17.1): if there exists an optical path function S such that $\dot{\gamma} = \nabla S/n$, then

$$|\nabla S(\mathsf{r})|^2 = n(\mathsf{r})^2.$$

Use $|\dot{\gamma}| = 1$ to complete the proof, and explain that the ray direction is perpendicular to the wavefront $S = \text{const}$.

Exercise 5 Prove Theorem 17.2 (Snell's law): at a flat interface between two homogeneous media,

$$n_1 \sin\theta_1 = n_2 \sin\theta_2.$$

Complete the proof by varying the point of incidence on the interface with Fermat's principle and setting the first variation to zero. Further explain that total internal reflection occurs when $n_1 > n_2$ and the angle of incidence exceeds the critical angle.

Exercise 6 Prove Proposition 17.2 and Proposition 17.3:

(i) the family of propagation operators $\{P_{\mathsf{a}}\}$ forms a faithful representation of the translation group $\mathbb{R}^n$ under composition;

(ii) the propagation operator P_{a} is an isometry on $L^2(\mathbb{R}^n)$, hence a unitary operator.

Complete the proof using the change of variable $\mathsf{y} = \mathsf{x} - \mathsf{a}$.

Exercise 7 Prove Proposition 17.4: in the Fourier domain the propagation operator is equivalent to multiplication by the phase factor,

$$\widehat{P_{\mathsf{a}}f}(\mathsf{k}) = e^{-i\mathsf{k}\cdot\mathsf{a}}\hat{f}(\mathsf{k}).$$

Write out the direct computation of the Fourier transform, and prove Corollary 17.3: when the propagation distance equals one wavelength, $a = \lambda$, the plane wave returns to the phase it had at departure.

Exercise 8 Prove the energy conservation of vacuum light propagation (Theorem 17.4): for solutions of the vacuum Maxwell equations,

$$U(t) = \int_{\mathbb{R}^3} \Big(\frac{\varepsilon_0}{2}|\mathsf{E}|^2 + \frac{1}{2\mu_0}|\mathsf{B}|^2\Big) d\mathsf{x}$$

satisfies $U(t) = U(0)$. Explain how to write the Maxwell equations as $\partial_t \mathsf{u} = A\mathsf{u}$, and complete the proof using the antisymmetry of A and Stone's theorem.

Exercise 9 Prove that the propagation kernel satisfies the evolution equation and the initial condition (Theorem 17.5): the propagation kernel $G(\mathsf{x}, t; \mathsf{x}_0)$ satisfies

$$\partial_t G = \mathcal{L}G, \qquad G(\mathsf{x}, 0; \mathsf{x}_0) = \delta(\mathsf{x} - \mathsf{x}_0).$$

Complete the proof using the definition of G and the properties of the evolution operator $E_t = e^{t\mathcal{L}}$.

Exercise 10 Prove Lemma 17.3: the Fourier transform of the Gaussian kernel

$$K_\Delta(x) = \frac{1}{\sqrt{2\pi}\Delta} e^{-x^2/(2\Delta^2)}$$

is again a Gaussian kernel:

$$\hat{K}_\Delta(\omega) = e^{-\Delta^2\omega^2/2}.$$

Complete the computation by completing the square in the Gaussian integral, and explain why the variances are interchanged, $\Delta \leftrightarrow 1/\Delta$.

Exercise 11 Prove Theorem 17.7: the periodic convolution of the single-period layer degenerates on the spatial translation group to propagation convolution,

$$(f \circledast_{\mathsf{prop}} G)(\mathsf{x}) = \lim_{W\to\delta} (f \circledast g)(\mathsf{x}),$$

where g is taken to be the propagation kernel and $S^{(d)}$ the phase kernel $S^{(1)}$. Complete the proof starting from Definition 6.26 and 17.8.

Exercise 12 Verify the Bloch–Floquet propagation theorem (Theorem 17.9): let the linear evolution operator $\mathcal{L}$ have spatial period T; then there exist Bloch wave solutions

$$u_{\mathsf{k},n}(\mathsf{x}, t) = e^{i(\mathsf{k}\cdot\mathsf{x} - \omega_n(\mathsf{k})t)}\phi_{\mathsf{k},n}(\mathsf{x}), \quad \phi_{\mathsf{k},n}(\mathsf{x} + \mathsf{T}) = \phi_{\mathsf{k},n}(\mathsf{x}).$$

Explain how the commutation of $\mathcal{L}$ with P_{T} leads to a simultaneous spectral decomposition, and explain the periodicity of the dispersion relation in the frequency domain (Proposition 17.4).

Exercise 13 Prove the translational structure of Bayesian updating (Theorem 17.10): in the log-belief space $\mathcal{F}(\Theta)$,

$$\Lambda(p') = \Lambda(p) + \ell_D,$$

where $\ell_D(\theta) = \log p(D \mid \theta) - \log Z_D$. Explain how Bayesian updating is equivalent to an additive translation in log-belief space, and point out the structural isomorphism between Bayesian updating and the propagation operator in Corollary 17.6.

Exercise 14 Prove the exact finite-order evolution of Bayesian updating under a conjugate prior (Theorem 17.11): let the likelihood be an exponential family and the prior a conjugate prior, and let the observation sequence be $x_1, \ldots, x_n$; then the posterior natural parameters are

$$\tau_n = \tau_0 + \sum_{i=1}^{n} T(x_i), \qquad \nu_n = \nu_0 + n.$$

Complete the proof using Bayes' formula and the definition of the conjugate prior, and verify the Beta–binomial model (Example 17.7) and the Gaussian–Gaussian model (Example 17.8).

Exercise 15 Prove the commutativity of pointwise activation and the propagation operator (Proposition 17.7): let φ be a pointwise activation function and P_{a} a propagation operator; then

$$P_{\mathsf{a}} \circ \varphi = \varphi \circ P_{\mathsf{a}}.$$

Compute both sides for arbitrary f and x and verify that they are equal, and prove Corollary 17.8: pointwise activation preserves translation invariance.

Exercise 16 Prove Proposition 17.8: under translation the CEAF behaves as a pure phase rotation,

$$\mathcal{E}(x + a) = \sum_{k=1}^{N} A_k e^{i(\omega_k x + \phi_k + \omega_k a)}.$$

Write out the factorisation of $e^{i(\omega_k (x+a) + \phi_k)}$, and prove the periodicity condition of Corollary 17.9.

Exercise 17 Prove Theorem 17.12: let the quadratic loss be $\mathcal{L}(\theta) = \frac{1}{2}(\theta - \theta^*)^\mathsf{T} H(\theta - \theta^*)$, with H symmetric positive definite and eigenvalues $0 < \lambda_1 \leq \cdots \leq \lambda_d$. Then for parametric evolution

$$\theta_{t+1} = \theta_t - \eta \nabla \mathcal{L}(\theta_t)$$

the error along the i-th eigendirection is

$$e_t^{(i)} = (1 - \eta \lambda_i)^t e_0^{(i)},$$

and the necessary and sufficient condition for convergence is $0 < \eta < 2/\lambda_d$. Write out the complete derivation by eigenbasis decomposition.

Exercise 18 Verify Example 17.10: take $d = 2$, $H = \mathrm{diag}(1, 4)$, $\theta^* = (0, 0)$ and $\theta_0 = (1, 1)$.

(i) compute the convergence factors of the two eigendirections when $\eta = 0.3$, and decide whether convergence occurs;

(ii) compute the convergence factors of the two eigendirections when $\eta = 0.6$, and decide whether convergence occurs.

Explain why the convergence condition is determined by the largest eigenvalue $\lambda_d = 4$.

Exercise 19 Prove Theorem 17.13: let $\mathcal{L}$ be an L-smooth, μ-strongly convex function and take $\eta = 1/L$; then

$$\|\theta_t - \theta^*\|^2 \leq \left(1 - \frac{\mu}{L}\right)^t \|\theta_0 - \theta^*\|^2.$$

Complete the proof using the descent lemma and strong convexity, and explain why parametric evolution converges exponentially—this is the convergence closure of parametric evolution.

Exercise 20 Synthesis: combining the ray equation (Theorem 17.1), the propagation-operator group (Proposition 17.2), Bloch–Floquet propagation (Theorem 17.9) and the translational structure of Bayesian updating (Theorem 17.10), explain how 'propagation' and 'evolution' are unified in this chapter as two realisations of one and the same translation-group action:

(i) physical level: light propagation corresponds to the spatial translation group $\mathbb{R}^n$;

(ii) temporal level: the evolution semigroup corresponds to the time translation group $\mathbb{R}_{\geq 0}$;

(iii) information level: Bayesian updating corresponds to a data-dependent translation;

(iv) learning level: parametric evolution corresponds to a translation in the gradient direction.

Point out for each realisation the translation amount, the generator and the group structure.

Chapter 18

Groups and Group Functions

Remark 18.1 (Positioning of this chapter). Starting from the axioms of group representation theory, this chapter builds up the theoretical system of group actions, the three standard constructions on a group (translation, inner automorphism, outer automorphism), group symmetry, group direction, group periods and the space of group functions.

The chapter has three core tasks:

(1) **Defining the eight pairs of concepts**: translation function/translation group, scaling function/scaling group, rotation function/rotation group, conservation function/conservation group, symmetry function/symmetry group, closure function/closure group, periodic function/group with periodic orbits, higher-order function/higher-order group.

(2) **Establishing the joint phase-space criterion framework**: unifying the closure criterion for phase-space trajectories with the criterion for groups with periodic orbits from group representation theory, and giving a complete criterion for the periodicity of a function.

(3) **Proving the transitivity matrix**: for the transitivity of the eight properties under the twelve operations (addition, subtraction, multiplication, division, exponentiation, root extraction, logarithm, series, limits, differentiation, integration, convolution), giving a proof or a counterexample cell by cell.

Note: the three standard constructions on a group—left translation L_g, right translation R_g, inner automorphism ι_g and the outer automorphism group $\mathrm{Out}(G)$—are respectively classical objects in group representation theory. This chapter treats them as independent objects of group representation theory.

Seven-theme guide for this chapter: Translation—left/right translation on a group (the centre of symmetry being the identity element e); Scaling—the outer automorphism $\mathrm{Out}(G)$; Rotation—the inner automorphism ι_g (the centre of symmetry being the centre $Z(G)$); Conservation—the conservation laws of group functions (translation invariance→momentum/energy, scaling invariance→scale charge, rotation invariance→angular momentum); Symmetry—the automorphism tower and the groupification of relations; Closure—the closure of group orbits and the criteria for the eight pairs of concepts; Period—group periods and groups with periodic orbits.

18.1 The Three Standard Constructions on a Group

On a vector space, vector addition (translation), the outer product/complex multiplication (rotation) and scalar multiplication (scaling) are three basic operations, and they stand in a formal correspondence with the gradient, the curl

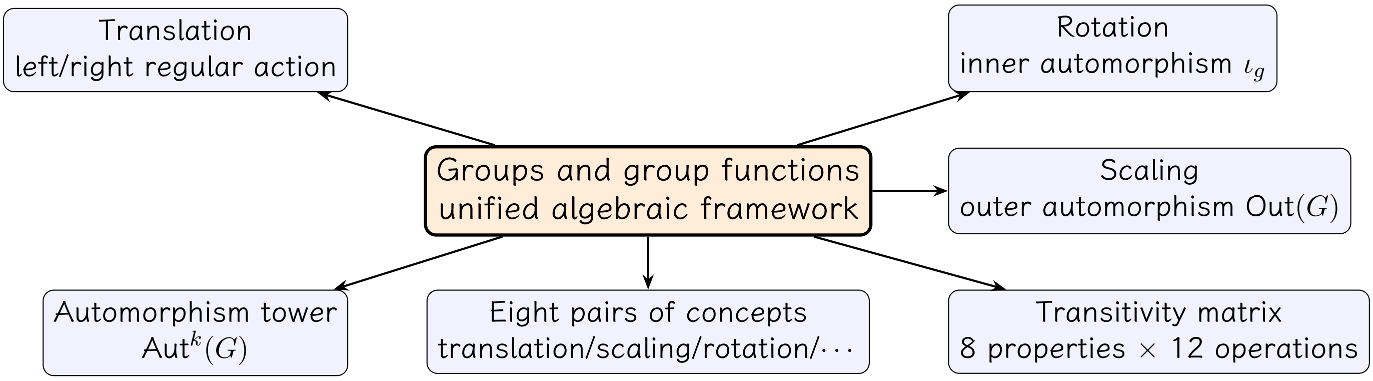

Translation
left/right regular action
Rotation
inner automorphism ι_g
Groups and group functions
unified algebraic framework
Scaling
outer automorphism Out(G)
Automorphism tower
Aut$^k(G)$
Eight pairs of concepts
translation/scaling/rotation/$\cdots$
Transitivity matrix
8 properties × 12 operations

Figure 18.1: Overall roadmap of the 'Groups and Group Functions' chapter: the three standard constructions, the automorphism tower, the eight pairs of concepts and the transitivity matrix together constitute a unified algebraic framework.

and the divergence (centre of symmetry 1) respectively. This section generalises this viewpoint to an arbitrary group: on any group there likewise exist three standard constructions—left/right translation, inner automorphism and outer automorphism—which are the counterparts of the above three operations at the level of group representation theory. In the discussion that follows, the 'three constructions' are classical objects of group representation theory; the 'correspondence with the three operators' is an organising viewpoint whose rigour depends on the particular context.

18.1.1 Translation Construction

Definition 18.1 (Translation Construction on a Group)**.** *Let $(G, \cdot)$ be a group. For each fixed $g \in G$, define two mappings:*

- **Left translation** $L_g : G \to G$: $L_g(x) = g \cdot x$, $\forall x \in G$.
- **Right translation** $R_g : G \to G$: $R_g(x) = x \cdot g$, $\forall x \in G$.

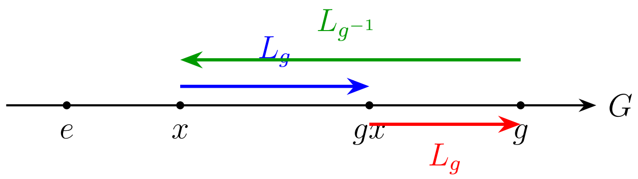
Left and right translation have the following properties:

(i) The set of left translations $\{L_g : g \in G\}$ forms a group under composition of mappings, with $L_g \circ L_h = L_{gh}$ and $L_{g^{-1}} = (L_g)^{-1}$. This group is isomorphic to G itself and is called the **left regular action** of G.

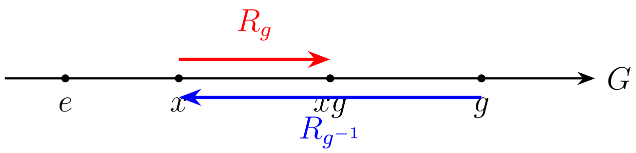
(ii) The set of right translations forms a group, with $R_g \circ R_h = R_{hg}$, isomorphic to G^{op} (the opposite group of G).

(iii) Left and right translations commute with each other: $L_g \circ R_h = R_h \circ L_g$ (because $(gx)h = g(xh)$).

Proposition 18.1 (Basic Properties of the Translation Construction)**.** *The translation construction on a group (left translation and right translation) has the following properties:*



Left translation $L_g(x) = gx$ is an isometric translation acting from the identity element side



Right translation $R_g(x) = xg$ is an isometric translation acting from the side opposite to the i

Figure 18.2: Comparison of left translation L_g with right translation R_g. When G is non-commutative the two are in general unequal, but $L_g \circ R_h = R_h \circ L_g$ always holds.

(i) **Fixed elements**: the identity element e of the group satisfies $L_g(e) = g$ and $R_g(e) = g$. The inversion mapping $\sigma_e(x) = x^{-1}$ is an involution ($\sigma_e \circ \sigma_e = \mathrm{id}_G$).

(ii) **Infinitesimal generators**: when G is a Lie group, the infinitesimal generator of left translation is a **left-invariant vector field**, whose differential operator corresponds to an element of the Lie algebra $\mathfrak{g}$. For $G = \mathbb{R}$ (the additive group), the infinitesimal generator is $\frac{d}{dx}$. For $G = \mathbb{R}^n$ (the additive group), the infinitesimal generators are $(\partial_1, \ldots, \partial_n)$.

(iii) **Invariance of the Haar measure**: translation invariance is expressed as the left invariance of the Haar measure: $\int_G f(gx)dg = \int_G f(x)dg$. By Noether's theorem, translation invariance implies the conservation of momentum or energy.

(iv) **Directional structure**: each generator $g \in G$ defines a 'translation direction'—moving group elements along that direction. For a discrete group this is manifested as directed edges of the Cayley graph.

Proof. (i) For any $x \in G$, $\sigma_e(\sigma_e(x)) = (x^{-1})^{-1} = x$, so σ_e is an involution. $L_g(e) = g \cdot e = g$ and $R_g(e) = e \cdot g = g$.

(ii) For a Lie group G, the left-invariant vector field $\tilde{X}(g) = dL_g|_e(X)$ ($X \in \mathfrak{g}$) generates the one-parameter subgroup of left translations: $\exp(tX) \cdot g = L_{\exp(tX)}(g)$. For $G = \mathbb{R}$ the Lie algebra $\mathfrak{g} \cong \mathbb{R}$ is one-dimensional, and the left-invariant vector field corresponding to the basis vector $X = 1$ is $\tilde{X}(x) = \frac{d}{dx}$, which generates the translation $x \mapsto x + t$.

(iii) The left invariance of the Haar measure is the integral expression of translation invariance. By Noether's theorem, translation invariance implies the conservation of momentum/energy.

(iv) For a discrete group, the generating set S defines the basic translation directions. For a Lie group, the basis $\{X_1, \ldots, X_n\}$ of the Lie algebra defines n independent translation directions. $\square$

Example 18.1 (Examples of the Translation Construction).
- **The additive group of integers** $\mathbb{Z}$: left translation $L_n(m) = n + m$. The generator 1 defines the positive direction $(+1)$, and the generator -1 defines the negative direction. For example $L_3(5) = 3 + 5 = 8$ and $L_{-2}(1) = -1$. The conserved quantity is the discrete momentum.
- **The additive group of real numbers** $\mathbb{R}$: left translation $L_a(x) = a + x$. The infinitesimal generator is $\frac{d}{dx}$, and its exponential mapping $\exp\left(t\frac{d}{dx}\right)$ acting on x gives $x + t$ (taking $t = \frac{1}{2}$: $L_{1/2}(0) = \frac{1}{2}$). The conserved quantity is the continuous momentum (or energy, when x is a time parameter).
- **The general linear group** $GL(n, \mathbb{R})$: left translation $L_A(X) = AX$ (matrix multiplication). This is the realisation of the translation construction on a non-commutative group.

18.1.2 Inner Automorphism

Definition 18.2 (Inner Automorphism). *For each fixed $g \in G$, define the mapping $\iota_g : G \to G$ by*

$$\iota_g(x) = gxg^{-1}, \quad \forall x \in G.$$

ι_g is called the **inner automorphism** *determined by g (or the conjugation mapping). Inner automorphisms have the following properties:*

(i) *ι_g is an automorphism of G: it is a bijection (with inverse mapping $\iota_{g^{-1}}$) and preserves the group operation: $\iota_g(xy) = g(xy)g^{-1} = (gxg^{-1})(gyg^{-1}) = \iota_g(x)\iota_g(y)$.*

(ii) *The mapping $G \to \mathsf{Aut}(G)$, $g \mapsto \iota_g$ is a group homomorphism whose kernel is the* **centre** *$Z(G) = \{z \in G : gz = zg, \ \forall g \in G\}$ of G.*

(iii) *The inner automorphisms as a whole form a normal subgroup of $\mathsf{Aut}(G)$, written $\mathsf{Inn}(G) \cong G/Z(G)$.*

Proposition 18.2 (Basic Properties of the Inner Automorphism). *Inner automorphisms on a group have the following properties:*

(i) **Fixed elements**: *the elements of the centre $Z(G)$. For any $z \in Z(G)$, $\iota_z(x) = zxz^{-1} = x$ (because $zx = xz$), that is, the elements of $Z(G)$ are fixed under conjugation. For $G = U(1)$ (an Abelian group), $Z(U(1)) = U(1)$, and all elements are fixed elements.*

(ii) **Infinitesimal generators**: *when G is a Lie group, the infinitesimal generator of an inner automorphism is the* **adjoint representation** $\mathsf{ad}_X(Y) = [X, Y]$ *(the Lie bracket of the Lie algebra). For $G = U(1)$ the Lie algebra $\mathfrak{u}(1) \cong i\mathbb{R}$ is a one-dimensional commutative Lie algebra and the adjoint representation is zero ($[X, Y] = 0$). For $G = SO(3)$ the adjoint representation gives the Lie-algebra version of the angular momentum operator.*

(iii) **Conserved quantities**: *conjugation invariance implies the* **conservation of angular momentum**. *At the level of group representation theory this is manifested as the invariance of class functions under conjugation: $f(gxg^{-1}) = f(x)$. The character χ_ρ is a typical example of a class function.*

(iv) **Adjoint orbits**: *each non-central element $g \in G \setminus Z(G)$ defines a 'rotation direction'—conjugating group elements along that direction. For $G = U(1)$ (an Abelian group) all inner automorphisms are the identity mapping. For a non-commutative group the rotation directions are given by the adjoint orbits.*

Proof. (i) If $z \in Z(G)$, then for any $x \in G$, $\iota_z(x) = zxz^{-1} = xzz^{-1} = x$. For $G = U(1)$ the multiplication is commutative, so $Z(U(1)) = U(1)$.

(ii) For a Lie group G, the infinitesimal generator of the inner automorphism $\iota_{\exp(tX)}$ as $t \to 0$ is $\mathrm{ad}_X(Y) = \frac{d}{dt}\big|_{t=0} \iota_{\exp(tX)}(\exp(Y)) = [X, Y]$. For $G = U(1)$ the Lie algebra is one-dimensional and commutative, $[X, Y] = 0$, and the adjoint representation is trivial. For $G = SO(3)$ the Lie algebra $\mathfrak{so}(3) \cong \mathbb{R}^3$ is equipped with the cross product as Lie bracket, and $\mathrm{ad}_X(Y) = X \times Y$.

(iii) A class function $f(gxg^{-1}) = f(x)$ is invariant under conjugation. The character $\chi_\rho(g) = \mathrm{Tr}(\rho(g))$ satisfies $\chi_\rho(gxg^{-1}) = \mathrm{Tr}(\rho(g)\rho(x)\rho(g)^{-1}) = \mathrm{Tr}(\rho(x)) = \chi_\rho(x)$, and is therefore a class function.

(iv) For a non-commutative group, the adjoint orbit $\mathcal{O}_x = \{gxg^{-1} : g \in G\}$ gives the orbit structure of x under inner automorphisms. The size of the orbit is given by the orbit-stabiliser theorem: $|\mathcal{O}_x| = |G|/|C_G(x)|$, where $C_G(x) = \{g \in G : gx = xg\}$ is the centraliser of x. □

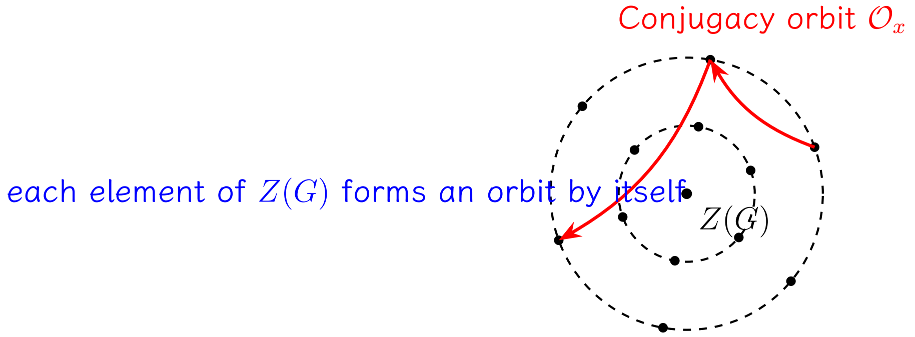

Conjugacy orbit $\mathcal{O}_x$
each element of $Z(G)$ forms an orbit by itself
$Z(G)$

Figure 18.3: Orbit structure of an inner automorphism: the conjugation action partitions G into conjugacy classes (orbits), each element of the centre $Z(G)$ forms an orbit by itself, and a class function takes a constant value on each orbit.

Example 18.2 (Examples of the Inner Automorphism).

- **The general linear group** $GL(n, \mathbb{R})$: the inner automorphism $\iota_A(X) = AXA^{-1}$ is the similarity transformation of matrices. Two matrices are similar if and only if they lie in the same adjoint orbit. For example, conjugating $A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ by the permutation matrix $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ gives $\iota_P(A) = PAP^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$, whose eigenvalues $\{1, 2\}$ are invariant under conjugation.
- **The three-dimensional rotation group** $SO(3)$: the inner automorphism $\iota_R(S) = RSR^{-1}$ conjugates the rotation S into a rotation about a new axis. Two rotations are conjugate if and only if they have the same rotation angle (the same eigenvalues).

- **The circle group** $U(1)$ (an Abelian group): all inner automorphisms are the identity mapping, and inner automorphisms degenerate to the trivial one.

18.1.3 Outer Automorphism

Definition 18.3 (Outer Automorphism Group). *The quotient group* $\mathsf{Out}(G) = \mathsf{Aut}(G)/\mathsf{Inn}(G)$ *is called the* **outer automorphism group** *of* G*. The elements of* $\mathsf{Out}(G)$ *represent those automorphisms that cannot be realised by conjugation by elements of the group.*

Typical examples of outer automorphisms include:

(i) **Automorphisms of a cyclic group**: $\mathsf{Aut}(C_n) \cong (\mathbb{Z}/n\mathbb{Z})^\times$*, the automorphisms being* $x \mapsto x^k$ $(\mathsf{gcd}(k,n) = 1)$*. When* $k \neq \pm 1$ *this is not an inner automorphism (because* C_n *is an Abelian group, all of whose inner automorphisms are the identity mapping), and hence belongs to the outer automorphism group.*

(ii) **Automorphisms of the additive group of integers**: $\mathsf{Aut}(\mathbb{Z}) \cong C_2 = \{\mathsf{id}, -\mathsf{id}\}$*, and the negation mapping* $x \mapsto -x$ *is the only non-trivial automorphism.*

(iii) **Outer automorphisms of the general linear group**: $\mathsf{Aut}(GL(n,\mathbb{R}))$ *contains the inverse-transpose mapping* $X \mapsto (X^\mathsf{T})^{-1}$ *(a non-trivial outer automorphism when* $n \geq 3$*), together with diagonal automorphisms and so on.*

Proposition 18.3 (Basic Properties of the Outer Automorphism). *Outer automorphisms on a group have the following properties:*

(i) **Fixed elements**: *the set of fixed elements of an automorphism* $\varphi \in \mathsf{Aut}(G)$ *of the group is* $\{x \in G : \varphi(x) = x\}$*, which is a subgroup. For the automorphism* $x \mapsto x^k$ *of a cyclic group, the set of fixed points is* $\{x \in C_n : x^{k-1} = e\}$*, that is, the subgroup consisting of the elements whose order divides* $k-1$*. For the negation mapping in* $\mathsf{Aut}(\mathbb{Z})$*, the only fixed point is* 0*.*

(ii) **Infinitesimal generators**: *outer automorphisms are not generated by conjugation by elements of the group, and therefore have no 'infinitesimal generator' at the level of group representation theory (they do not arise from the adjoint action of the Lie algebra). For* $G = \mathbb{R}$ *(the additive group), the infinitesimal generator of the automorphism* $x \mapsto \lambda x$ $(\lambda \neq 0)$ *is* $x\frac{d}{dx}$*.*

(iii) **Invariants**: *automorphisms keep the structure constants of the group unchanged. For the automorphism* $x \mapsto x^k$ *of a cyclic group, it keeps the order of the group unchanged (*$|G|$ *is conserved).*

(iv) **Directional structure**: *outer automorphisms do not change the conjugacy-class structure among group elements, but they change the overall scale of the group.*

Proof. (i) For the cyclic group $C_n = \langle g \rangle$, the automorphism is $\varphi_k(g^m) = g^{km}$ $(\mathsf{gcd}(k,n) = 1)$. Fixed points satisfy $g^{km} = g^m \iff g^{m(k-1)} = e \iff n \mid m(k-1)$. When $\mathsf{gcd}(k-1,n) = d$, the fixed points form the subgroup of order d generated by $g^{n/d}$.

(ii) For $G = \mathbb{R}$ (the additive group), the automorphisms are $\varphi_\lambda(x) = \lambda x$ ($\lambda \in \mathbb{R}^\times$). The tangent vector at $\lambda = 1$ is $\frac{d}{d\lambda}\big|_{\lambda=1}\varphi_\lambda(x) = x$, to which the differential operator $x\frac{d}{dx}$ corresponds.

(iii) Automorphisms preserve the order of the group: $|\varphi(G)| = |G|$ (because φ is a bijection). For a cyclic group this means that $|C_n| = n$ is invariant under automorphisms.

(iv) What outer automorphisms change is the 'external structure' of the group—they cannot be realised by conjugation by elements of the group, yet they preserve the conjugacy-class structure (the conjugacy classes are intrinsic invariants of the group). ☐

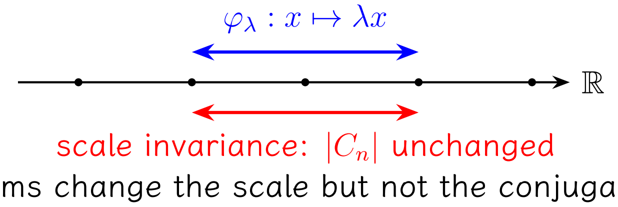

$\varphi_\lambda : x \mapsto \lambda x$
$\mathbb{R}$
scale invariance: $|C_n|$ unchanged
outer automorphisms change the scale but not the conjugacy-class structure

Figure 18.4: Geometric intuition for outer automorphisms: the ring-like arrows represent the scaling transformation $x \mapsto \lambda x$, and the two ends of the arrow correspond to the property that the group order is unchanged.

Example 18.3 (Examples of the Outer Automorphism).

- The additive group of integers $\mathbb{Z}$: $\mathrm{Aut}(\mathbb{Z}) \cong C_2 = \{\mathrm{id}, -\mathrm{id}\}$. The negation mapping is $x \mapsto -x$, for example $-3 \mapsto 3$. The only fixed point is 0.
- The cyclic group C_n: $\mathrm{Aut}(C_n) \cong (\mathbb{Z}/n\mathbb{Z})^\times$, of order $\varphi(n)$ (the Euler function). The automorphism $x \mapsto x^k$ ($\gcd(k, n) = 1$) rescales the generator. For example, when $n = 5$, $\mathrm{Aut}(C_5) \cong (\mathbb{Z}/5\mathbb{Z})^\times \cong C_4$, and the generator g is mapped in turn to g^1, g^2, g^3, g^4 under $\varphi_k(g) = g^k$ ($k = 1, 2, 3, 4$).
- The multiplicative group of positive real numbers $\mathbb{R}^+$: $\mathrm{Aut}(\mathbb{R}^+) \cong \mathbb{R}^\times$. The automorphism $x \mapsto x^\lambda$ is a power scaling.

18.1.4 The Unified Structure of the Three Constructions

Theorem 18.1 (The Unification Theorem for the Three Constructions on a Group). Let G be a group. The three standard constructions on a group have the following structure in the automorphism group $\mathrm{Aut}(G)$:

(i) **Translation construction**: realised by G itself through the left regular action, with the operating group isomorphic to G. The set of translation operations $\{L_g : g \in G\}$ is a faithful action of G on itself.

(ii) **Inner automorphisms**: realised by the inner automorphism group $\mathrm{Inn}(G) \cong G/Z(G)$. $\mathrm{Inn}(G)$ is a normal subgroup of $\mathrm{Aut}(G)$.

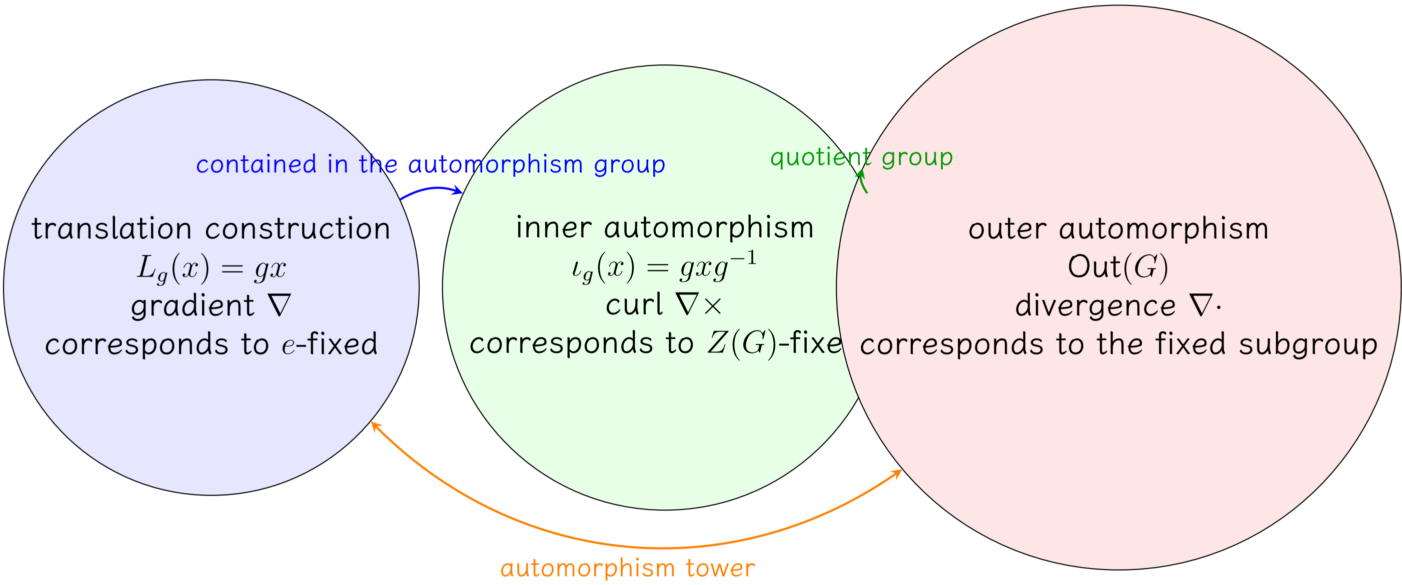
(iii) **Outer automorphisms**: realised by the outer automorphism group $\mathrm{Out}(G) = \mathrm{Aut}(G)/\mathrm{Inn}(G)$.

In the automorphism tower $G_{(0)} = G, G_{(1)} = \text{Aut}(G), G_{(2)} = \text{Aut}(\text{Aut}(G)), \ldots,$ every level contains that level's translation, inner automorphism and outer automorphism constructions.

Proof. (i) The left regular action $\lambda : G \to \text{Sym}(G)$, $\lambda(g) = L_g$, is a group homomorphism. Its kernel is $\{e\}$, hence it is faithful.

(ii) The inner automorphism $\iota_g(x) = gxg^{-1}$ is an automorphism and preserves the group operation. The kernel of the mapping $G \to \text{Aut}(G)$, $g \mapsto \iota_g$ is $Z(G)$, hence $\text{Inn}(G) \cong G/Z(G)$. $\text{Inn}(G)$ is a normal subgroup of $\text{Aut}(G)$.

(iii) $\text{Out}(G) = \text{Aut}(G)/\text{Inn}(G)$ is a quotient group. ☐


contained in the automorphism group
quotient group
translation construction
$L_g(x) = gx$
gradient ∇
corresponds to e-fixed
inner automorphism
$\iota_g(x) = gxg^{-1}$
curl ∇×
corresponds to $Z(G)$-fixe
outer automorphism
Out(G)
divergence ∇·
corresponds to the fixed subgroup
automorphism tower

The genealogy of the three standard constructions on a group:translation $(G) \subset$ inner automorphism $(\text{Inn}(G)) \subset$ automorphism $(\text{Aut}(G)) \supset$ outer automorphism $(\text{Out}(G))$.

Figure 18.5: A unified perspective on the three standard constructions on a group.

18.2 The Automorphism Tower

Definition 18.4 (Automorphism of a Group)**.** Let $(G, \cdot)$ be a group. A mapping $\varphi : G \to G$ is called an **automorphism** of G if it is a bijection and preserves the group operation: $\varphi(xy) = \varphi(x)\varphi(y)$.

Definition 18.5 (The Automorphism Group)**.** The automorphisms of G as a whole form a group under composition of mappings, written $\text{Aut}(G)$. The identity element is the identity mapping and the inverse element is the inverse mapping.

Proposition 18.4 (The Inverse of an Automorphism Preserves the Group Operation)**.** Let $\varphi \in \text{Aut}(G)$. Then its inverse mapping φ^{-1} also preserves the group operation.

Proof. Since φ is surjective, there exist unique $x, y \in G$ such that $\varphi(x) = u$ and $\varphi(y) = v$. By the homomorphism property of φ, $\varphi(xy) = uv$. Applying φ^{-1} to both sides gives $xy = \varphi^{-1}(uv)$. But $xy = \varphi^{-1}(u)\varphi^{-1}(v)$, hence $\varphi^{-1}(uv) = \varphi^{-1}(u)\varphi^{-1}(v)$. ☐

Definition 18.6 (Inner Automorphism)**.** *This definition is the same content as that given in Chapter 18 (the 'Inner Automorphism' section, Section 18.1.2), namely Definition 18.2; to avoid repeated exposition, only the reference anchor is retained here, and the complete statement is to be found there.*

Definition 18.7 (The Outer Automorphism Group)**.** *This definition is the same content as that given in Chapter 18 (the 'Outer Automorphism' section, Section 18.1.3), namely Definition 18.3; to avoid repeated exposition, only the reference anchor is retained here, and the complete statement is to be found there.*

Definition 18.8 (The Automorphism Tower)**.** *Define recursively $G_{(0)} = G$ and $G_{(k+1)} = \mathrm{Aut}(G_{(k)})$. Level 0 is the group itself, level 1 is the automorphism group, level 2 is the automorphism group of the automorphism group, and so on. On every level $G_{(k)}$ there exist three standard constructions: translation (the left regular action), inner automorphisms ($\mathrm{Inn}(G_{(k)})$) and outer automorphisms ($\mathrm{Out}(G_{(k)})$).*

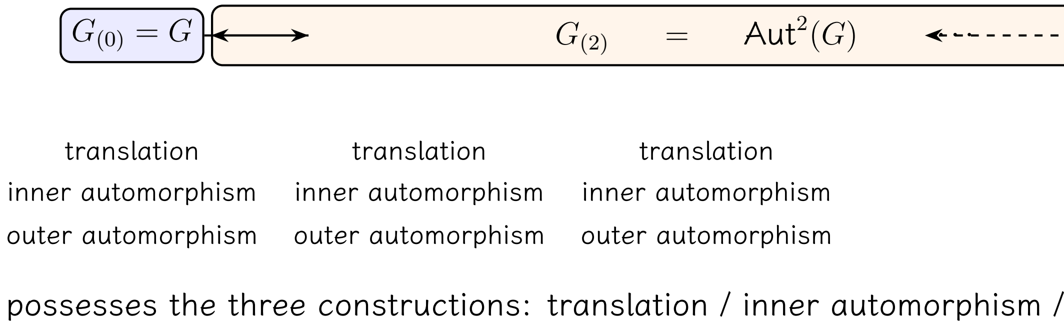

$G_{(0)} = G$
$G_{(2)} = \mathrm{Aut}^2(G)$
translation
inner automorphism
outer automorphism
translation
inner automorphism
outer automorphism
translation
inner automorphism
outer automorphism
every level $G_{(k)}$ again possesses the three constructions: translation / inner automorphism /

Figure 18.6: Level-by-level structure of the automorphism tower: *each level is the automorphism group of the level below, and the three standard constructions reappear on every level.*

Lemma 18.1 (Embedding and Equal-Order Stability for Centreless Groups)**.** *Let G be a finite group with $Z(G) = \{e\}$. Then*

(i) *the inner automorphism mapping $G \to \mathrm{Aut}(G)$, $g \mapsto \iota_g$, is a monomorphism, hence $G \cong \mathrm{Inn}(G) \subseteq \mathrm{Aut}(G)$, and therefore $|G| \le |\mathrm{Aut}(G)|$;*

(ii) *if $|\mathrm{Aut}(G)| = |G|$, then $\mathrm{Aut}(G) = \mathrm{Inn}(G) \cong G$;*

(iii) *$Z(\mathrm{Aut}(G)) = \{e\}$, that is, triviality of the centre is preserved level by level in the automorphism tower.*

Proof. (i) The kernel of the mapping $g \mapsto \iota_g$ is $Z(G) = \{e\}$ (Definition 18.6), so it is a monomorphism with image $\mathrm{Inn}(G)$, whence $G \cong \mathrm{Inn}(G) \subseteq \mathrm{Aut}(G)$.

(ii) By (i), $\mathrm{Inn}(G) \subseteq \mathrm{Aut}(G)$ and $|\mathrm{Inn}(G)| = |G| = |\mathrm{Aut}(G)|$, hence $\mathrm{Inn}(G) = \mathrm{Aut}(G)$; moreover $\mathrm{Inn}(G) \cong G/Z(G) = G$, so $\mathrm{Aut}(G) \cong G$.

(iii) Let $\psi \in Z(\mathrm{Aut}(G))$. Then $\psi\iota_g = \iota_g\psi$ for all g, that is, $\psi(gxg^{-1}) = g\psi(x)g^{-1}$. Since ψ is a homomorphism, $\psi(g)\psi(x)\psi(g)^{-1} = g\psi(x)g^{-1}$, so $\psi(g)^{-1}g$ commutes with $\psi(x)$. Since ψ is surjective, $\psi(x)$ runs through all of G, hence $\psi(g)^{-1}g \in Z(G) = \{e\}$, that is, $\psi(g) = g$ for all g, and therefore $\psi = \mathrm{id}$. Hence $Z(\mathrm{Aut}(G)) = \{e\}$. □

Theorem 18.2 (Stability of the Automorphism Tower of a Finite Group—Wielandt's Theorem). *Let G be a finite group with trivial centre $Z(G)$. Then the automorphism tower stabilises in finitely many steps: there exists $N \in \mathbb{N}$ such that $G_{(k+1)} \cong G_{(k)}$ for every $k \geq N$.*

Proof. By (i) and (iii) of Lemma 18.1, by induction every level $G_{(k)}$ is centreless, and $G_{(k)} \cong \mathrm{Inn}(G_{(k)}) \subseteq G_{(k+1)}$, so $|G_{(k+1)}| \geq |G_{(k)}|$; by (ii) again, equality holds if and only if $G_{(k+1)} \cong G_{(k)}$, that is, the tower stabilises at k. If the tower never terminates, the sequence $|G_{(k)}|$ is strictly increasing. The core assertion of Wielandt (1939) is that the orders in the automorphism tower of a centreless finite group are bounded above (the bound depending only on G), so a strictly increasing process cannot continue indefinitely and must stop after finitely many steps; combined with the fact that equal orders imply stability, the tower stabilises identically after finitely many steps. □

Remark 18.2 (The Boundary of Wielandt's Theorem). The three steps given by Lemma 18.1—'the order is monotonically non-decreasing + equal orders imply stability + triviality of the centre is preserved level by level'—can be proved entirely self-containedly; the boundedness of the order (a strict increase must terminate), however, is a classical result of Wielandt (1939) whose complete proof is rather long, and it is cited here as a known theorem. When the centre is non-trivial the automorphism tower need not stabilise in finitely many steps (counterexamples can be constructed), so the 'trivial centre' hypothesis in the theorem cannot be omitted.

18.3 Group Actions and Symmetry

Definition 18.9 (Group Action). Let G be a group and X a non-empty set. A **left action** of G on X is a mapping $\Phi : G \times X \to X$ satisfying:

(i) **Identity action**: $\Phi(e, x) = x$ for all $x \in X$;

(ii) **Associativity**: $\Phi(g, \Phi(h, x)) = \Phi(gh, x)$ for all $g, h \in G$ and $x \in X$.

One usually writes $\Phi(g, x)$ as $g \cdot x$.

Definition 18.10 (Orbit and Stabiliser Subgroup). Let the group G act on the set X. For any $x \in X$:

- $G \cdot x := \{g \cdot x : g \in G\}$ is called the **orbit** of x.
- $G_x := \{g \in G : g \cdot x = x\}$ is called the **stabiliser subgroup** of x.

Theorem 18.3 (The Orbit-Stabiliser Theorem). *Let a finite group G act on a finite set X. For any $x \in X$,*

$$|G \cdot x| = \frac{|G|}{|G_x|}.$$

Proof. Consider the mapping $\varphi : G \to G\cdot x$, $\varphi(g) = g\cdot x$. This mapping is surjective. $\varphi(g_1) = \varphi(g_2)$ if and only if $g_2^{-1}g_1 \in G_x$, equivalently $g_1G_x = g_2G_x$. Hence φ induces a one-to-one correspondence from G/G_x onto $G \cdot x$. □

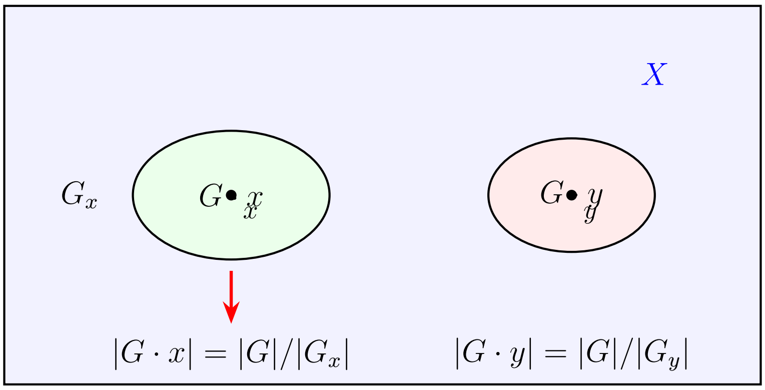


Figure 18.7: The orbit-stabiliser theorem: the group G acts on the set X, and the size of each orbit equals the order of the group divided by the order of the stabiliser subgroup of the point.

18.4 Group Symmetry and Group Direction

Definition 18.11 (Translational Symmetry on a Group). Translational symmetry on a group G is the group action formed by left and right translations.

Definition 18.12 (Automorphism Symmetry on a Group). Automorphism symmetry on a group G is the natural action of Aut(G) on G: $\varphi \cdot x = \varphi(x)$. It includes inner automorphisms (Inn(G)) and outer automorphisms (Out(G)).

Proposition 18.5 (Compatibility of Translation and Automorphism). Let $\varphi \in$ Aut(G) and $g \in G$. Then

$$\varphi \circ L_g \circ \varphi^{-1} = L_{\varphi(g)}.$$

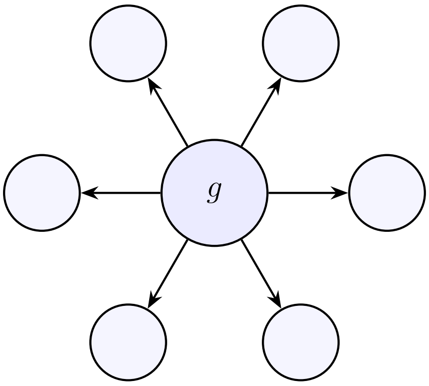
Proof. For any $x \in G$: $(\varphi \circ L_g \circ \varphi^{-1})(x) = \varphi(g \cdot \varphi^{-1}(x)) = \varphi(g) \cdot x = L_{\varphi(g)}(x)$. □

Definition 18.13 (Directional Structure of a Discrete Group). Let G be a finitely generated group and $S \subseteq G$ a symmetric generating set ($S = S^{-1}$, $\langle S\rangle = G$). The **directional structure** on G is given by the Cayley graph $\Gamma(G, S)$: the vertex set is G, and for each $g \in G$ and $s \in S$ there is a directed edge from g to gs.

Definition 18.14 (Directional Structure of a Lie Group). Let G be a Lie group and $\mathfrak{g} = T_eG$ its Lie algebra. Each non-zero element X of $\mathfrak{g}$ determines a **left-invariant vector field** $\tilde{X}(g) = dL_g|_e(X)$. A basis $\{X_1, \ldots, X_n\}$ of the Lie algebra gives a set of **basic directions** on the group.

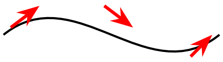
Proposition 18.6 (Algebraic Structure of Left-Invariant Vector Fields). Let $\tilde{X}, \tilde{Y}$ be left-invariant vector fields. Then $[\tilde{X}, \tilde{Y}]$ is also a left-invariant vector field, and $[\tilde{X}, \tilde{Y}] = \widetilde{[X, Y]}$.

Proof. $[\tilde{X}, \tilde{Y}](g) = dL_g|_e([X, Y]) = \widetilde{[X, Y]}(g)$. □



Cayley graph $\Gamma(G,S)$: each generator $s \in S$ corresponds to a directed edge

left-invariant vector field $\tilde{X}(g) = dL_g|_e(X)$



the Lie algebra basis $\{X_i\}$ gives the basic directions

Figure 18.8: Comparison of the directional structure of a discrete group (left) with that of a Lie group (right): the directed edges of the Cayley graph correspond to left-invariant vector fields.

18.5 Definitions of the Eight Pairs of Concepts

18.5.1 Translation Functions and the Translation Group

Definition 18.15 (Translation Function). Let $\mathcal{D} = \mathbb{R}$, $f : \mathcal{D} \to V$, $G = (\mathbb{R}, +)$, with group action $\phi_a(x) = x + a$. If there exists $a_0 \neq 0$ such that $f(x + a_0) = f(x)$ for all $x \in \mathcal{D}$, then f is called a **translation function** with respect to the translation a_0.

Definition 18.16 (Translation Group). The **translation group** of a translation function f is defined as

$$\text{Translation}(f) = \{a \in \mathbb{R} : f(x + a) = f(x),\ \forall x \in \mathcal{D}\}.$$

Translation(f) is necessarily either $\{0\}$ or some $T\mathbb{Z}$ $(T > 0)$; for its structure and that of the quotient space $\mathbb{R}/\text{Translation}(f) \cong S^1$, see Proposition 18.7.

18.5.2 Scaling Functions and the Scaling Group

Definition 18.17 (Scaling Function). Let $\mathcal{D} = \mathbb{R}^+$, $f : \mathcal{D} \to V$, $G = (\mathbb{R}^+, \times)$, with group action $\phi_q(x) = qx$. If there exists $q_0 > 0, q_0 \neq 1$ such that $f(q_0 x) = f(x)$ for all $x \in \mathcal{D}$, then f is called a **scaling function** with respect to the scaling q_0.

Definition 18.18 (Scaling Group). The **scaling group** of a scaling function f is defined as

$$\text{Scaling}(f) = \{q \in \mathbb{R}^+ : f(qx) = f(x),\ \forall x \in \mathcal{D}\}.$$

Scaling(f) is necessarily either $\{1\}$ or some $\langle q_0 \rangle$ $(q_0 > 1)$; under the logarithm mapping ln it is isomorphic to an additive subgroup of $(\mathbb{R}, +)$. For its structure see Proposition 18.7.

18.5.3 Rotation Functions and the Rotation Group

Definition 18.19 (Rotation Function). Let $\mathcal{D} = \mathbb{C}$, $f : \mathcal{D} \to V$, $G = U(1)$, with group action $\phi_\theta(z) = e^{i\theta} z$. If there exists $\theta_0 \neq 0$ such that $f(e^{i\theta_0} z) = f(z)$ for all $z \in \mathcal{D}$, then f is called a **rotation function** with respect to the rotation $e^{i\theta_0}$.

Definition 18.20 (Rotation Group). The **rotation group** of a rotation function f is defined as

$$\text{Rotation}(f) = \{e^{i\theta} \in U(1) : f(e^{i\theta} z) = f(z),\ \forall z \in \mathcal{D}\}.$$

Rotation(f) is necessarily either a finite cyclic group C_q or the whole of $U(1)$; note that 'the cyclic subgroup generated by a single rotation' and 'the group formed by all rotational symmetries' are two different things, see Proposition 18.7 and Remark 18.3.

18.5.4 Conservation Functions and the Conservation Group

Definition 18.21 (Conservation Function). Let $f : \mathcal{D} \to V$ and let G be a group acting on $\mathcal{D}$. f is called a **conservation function** with respect to G if there exists a scalar functional $Q : V \to \mathbb{R}$ (a **conserved quantity**) such that

$$Q(f(\phi_g(x))) = Q(f(x)), \quad \forall g \in G,\ \forall x \in \mathcal{D}.$$

Conservation and symmetry always appear together: symmetry is invariance under a transformation, and conservation is the quantitative expression of that invariance.

Definition 18.22 (Conservation Group). The conservation group (equivariance group) was first defined in Chapter 1 Section 1.3, Definition 1.16; the present form is its expression in the notation of the group function f and the quantity Q. The **conservation group** of a conservation function f is defined as

$$\text{Group}_{\text{conservation}}(f, Q) = \{g \in G : Q(f(\phi_g(x))) = Q(f(x)),\ \forall x \in \mathcal{D}\}.$$

The symmetry group is contained in the conservation group: Symmetry$(f) \subseteq$ Group$_{\text{conservation}}(f, Q)$.

18.5.5 Symmetry Functions and the Symmetry Group

Definition 18.23 (Symmetry Function). f is called a **symmetry function** with respect to $g \in G$ if $f(\phi_g(x)) = f(x)$ for all $x \in \mathcal{D}$. A symmetry function is a special case of a conservation function (the conserved quantity Q being taken to be the identity functional).

Definition 18.24 (Symmetry Group). The **symmetry group** of a symmetry function f is defined as

$$\text{Symmetry}(f) = \{g \in G : f(\phi_g(x)) = f(x),\ \forall x \in \mathcal{D}\}.$$

The group with periodic orbits Group$_{\text{period}}(f) =$ Symmetry(f).

18.5.6 Closure Functions and the Closure Group

Definition 18.25 (Closure Function)**.** *Let $f : \mathcal{D} \to V$ be a smooth function. f is called a* **closure function** *with respect to $g \in G$ if, in the* **phase space** *$T^*\mathcal{D}$, the phase-space trajectory $\Gamma_f(x) = (f(x), \dot{f}(x))$ of f is closed under the action of ϕ_g: $\Gamma_f(\phi_g(x)) = \Gamma_f(x)$.*

Definition 18.26 (Closure Group)**.** *The* **closure group** *of a closure function f is defined as*

$$\mathsf{Clos}(f) = \{g \in G : \Gamma_f(\phi_g(x)) = \Gamma_f(x),\ \forall x \in \mathcal{D}\}.$$

$\mathsf{Group}_{\mathsf{period}}(f) \subseteq \mathsf{Clos}(f)$—*a periodic function is necessarily a closure function.*

18.5.7 Periodic Functions and the Group with Periodic Orbits

Definition 18.27 (Periodic Function)**.** *f is called a* **periodic function** *with respect to $g \in G$ if $f(\phi_g(x)) = f(x)$ for all $x \in \mathcal{D}$. A periodic function is equivalent to a function on the quotient space $\mathcal{D}/\,\mathsf{Group}_{\mathsf{period}}(f)$.*

Definition 18.28 (Group with Periodic Orbits)**.** *The* **group with periodic orbits** *of a periodic function f is defined as*

$$\mathsf{Group}_{\mathsf{period}}(f) = \{g \in G : f(\phi_g(x)) = f(x),\ \forall x \in \mathcal{D}\}.$$

18.5.8 Higher-Order Functions and the Higher-Order Group

Definition 18.29 (Higher-Order Function)**.** *Let $f : G \to V$ be a group function. The* **higher-order functions** *of f are the sequence of functions on the levels of the automorphism tower: $f_{(0)} = f$, $f_{(k)} : G_{(k)} \to V$, $k \geq 1$.*

Definition 18.30 (Higher-Order Group)**.** *The* **higher-order group** *of G refers to its automorphism tower*

$$G_{(0)} = G, \quad G_{(1)} = \mathsf{Aut}(G), \quad G_{(2)} = \mathsf{Aut}(\mathsf{Aut}(G)), \ \ldots$$

Proposition 18.7 (Structure of the three invariance subgroups)**.** *Let $f : \mathcal{D} \to V$ be* **continuous and non-constant***, and let the three actions be $\phi_a(x) = x + a$ ($\mathcal{D} = \mathbb{R}$), $\phi_q(x) = qx$ ($\mathcal{D} = \mathbb{R}^+$) and $\phi_\theta(z) = e^{i\theta}z$ ($\mathcal{D} = \mathbb{C}^\times$) respectively. Then*

(i) **Translation***: $\mathsf{Translation}(f)$ is a closed subgroup of $(\mathbb{R}, +)$, hence is either $\{0\}$ or $T\mathbb{Z}$ for some $T > 0$; in the latter case $\mathcal{D}/\,\mathsf{Translation}(f) \cong \mathbb{R}/T\mathbb{Z} \cong S^1$.*

(ii) **Scaling***: $\mathsf{Scaling}(f)$ is a closed subgroup of $(\mathbb{R}^+, \times)$, hence is either $\{1\}$ or $\langle q_0 \rangle$ for some $q_0 > 1$.*

(iii) **Rotation***: $\mathsf{Rotation}(f)$ is a closed subgroup of $U(1)$, hence is either a finite cyclic group C_q for some $q \geq 1$, or the whole of $U(1)$; when $\mathsf{Rotation}(f) = U(1)$, f depends only on $|z|$, that is, f is a* **radial function***.*

Table 18.1: Comparison table of the eight pairs of concepts and the formally corresponding operators

Function concept	Group concept	Formally corresponding operator	Fixed element	Formally corresponding conserved quantity	Phase-space criterion
Translation function	Translation group	Gradient ∇	0	Momentum/energy	The phase trajectory closes along the translation axis
Scaling function	Scaling group	Divergence $\nabla\cdot$	1	Scale charge	The phase trajectory closes in the logarithmic domain
Rotation function	Rotation group	Curl $\nabla\times$	i	Angular momentum	The phase trajectory closes about the rotation axis
Conservation function	Conservation group	The three operators	$0, 1, i$	The corresponding conserved quantity	The conserved quantity is invariant in phase space
Symmetry function	Symmetry group	The three operators	$0, 1, i$	The corresponding conserved quantity	The symmetry transformation preserves the phase trajectory
Closure function	Closure group	The three operators	$0, 1, i$	The corresponding conserved quantity	The phase-space trajectory is closed
Periodic function	Group with periodic orbits	The three operators	$0, 1, i$	The corresponding conserved quantity	Configuration + phase space close simultaneously
Higher-order function	Higher-order group	Automorphism tower	Level by level	Level-by-level conservation	Phase space on the automorphism tower

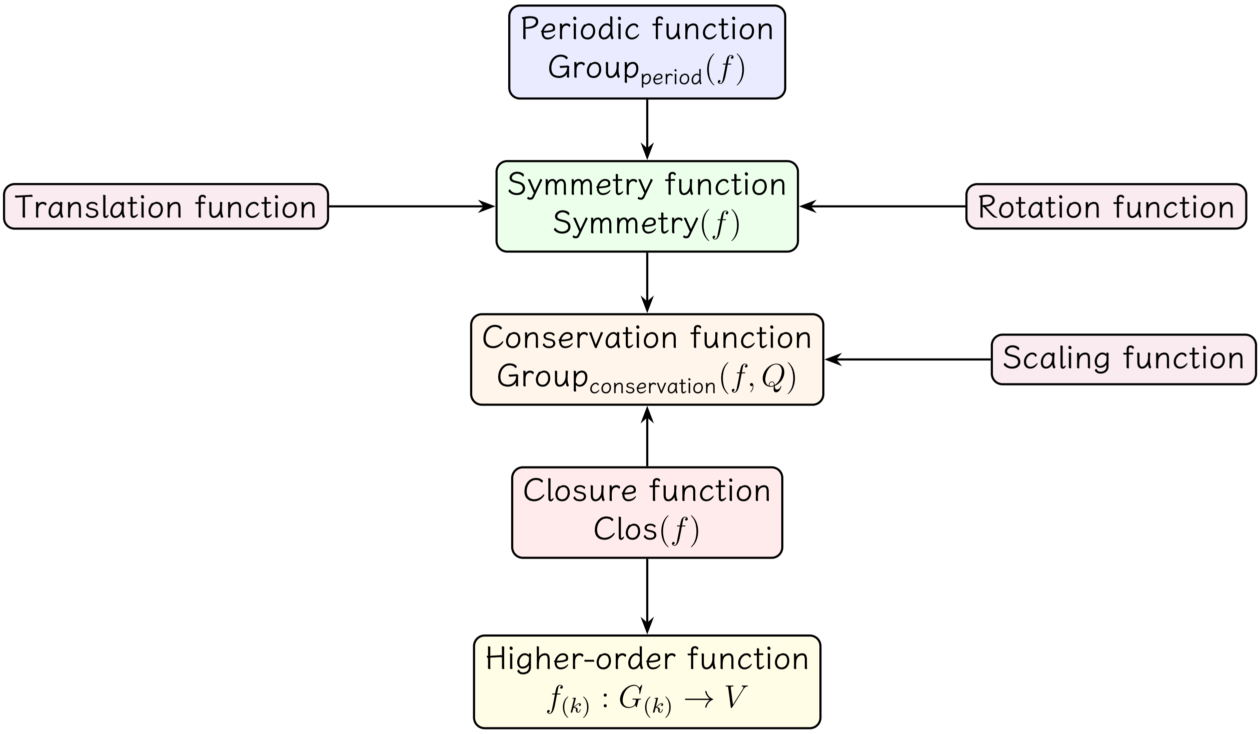

Periodic function
Group$_{\text{period}}(f)$
Translation function
Symmetry function
Symmetry(f)
Rotation function
Conservation function
Group$_{\text{conservation}}(f, Q)$
Scaling function
Closure function
Clos(f)
Higher-order function
$f_{(k)} : G_{(k)} \to V$

Figure 18.9: Hierarchical structure of the eight pairs of concepts: translation / scaling / rotation are concrete invariance subgroups, period, symmetry, conservation and closure form a unified chain of inclusions, and higher-order functions are further sublimated through the automorphism tower.

Proof. **Subgroup property**: the three sets are closed under their respective group operations, contain the identity element and contain inverses (the latter following from the invariance of f under the identity transformation), hence all three are subgroups.

Closedness: taking translation as an example, let $a_n \in$ Translation(f) with $a_n \to a$. For any x, by the continuity of f,

$$f(x+a) = \lim_{n\to\infty} f(x+a_n) = \lim_{n\to\infty} f(x) = f(x),$$

hence $a \in$ Translation(f). The scaling and rotation cases are analogous (using the continuity of multiplication and the topology of $U(1)$ respectively).

(i) The closed subgroups of $(\mathbb{R}, +)$ are only $\{0\}$ and $\alpha\mathbb{Z}$ ($\alpha > 0$): this is the classical classification of the one-dimensional subgroups of $(\mathbb{R}, +)$—if a closed subgroup $H \neq \{0\}$, take $T := \inf\{h \in H : h > 0\}$; by closedness $T \in H$; if $H \neq T\mathbb{Z}$, then there exists $h \in H$ whose ratio to T is irrational, in which case H is dense in $\mathbb{R}$, and by closedness $H = \mathbb{R}$. Taking T then gives the least positive element. The isomorphism of quotient spaces is given by $x + T\mathbb{Z} \mapsto e^{2\pi i x/T}$.

(ii) $\ln : (\mathbb{R}^+, \times) \to (\mathbb{R}, +)$ is a homeomorphism and a group isomorphism; it maps Scaling(f) onto a closed subgroup of $(\mathbb{R}, +)$ and maps $q_0 > 1$ to $\ln q_0 > 0$. The result follows from (i).

(iii) A closed subgroup of $U(1)$ is either discrete or $U(1)$ itself; a discrete closed subgroup consists of roots of unity, that is, it is some C_q. If Rotation$(f) = U(1)$, then f is invariant under all rotations; writing $z \neq 0$ in polar form $z = |z|e^{i \arg z}$, rotational invariance allows us to take $\arg z = 0$, hence $f(z) = f(|z|)$. ☐

Remark 18.3 (The correct statement concerning 'denseness'). Proposition 18.7(iii) contains a point that is very easily confused and requires separate explanation:

- (i) if $\theta_0/(2\pi) = p/q$ (in lowest terms), then the cyclic subgroup generated by the single rotation $e^{i\theta_0}$ is C_q (finite and closed), and Rotation$(f) \supseteq C_q$;
- (ii) if $\theta_0/(2\pi) \notin \mathbb{Q}$, then the cyclic subgroup generated by $e^{i\theta_0}$ is infinite and **dense** in $U(1)$; but it is **not closed**, and therefore is **not equal to** Rotation(f).

In other words: **'denseness' is a property of the generated subgroup, not a property of the invariance subgroup.** As a closed subgroup, Rotation(f) has only two possibilities: some finite cyclic group C_q, or the whole of $U(1)$. Calling the rotation symmetry group of f 'dense' is inaccurate.

Remark 18.4 (On the group relations of the eight pairs of concepts). The following four items are consequences of notational conventions and of a single substitution, and are relations at the level of definitions; they are therefore given in the form of a remark rather than as a proposition:

- (i) $\text{Group}_{\text{period}}(f) = \text{Symmetry}(f)$: two notations for the same set (Definition 18.28, 18.24);
- (ii) $\text{Symmetry}(f) \subseteq \text{Group}_{\text{conservation}}(f, Q)$: if $f(\phi_g(x)) = f(x)$, then $Q(f(\phi_g(x))) = Q(f(x))$;
- (iii) Translation(f), Scaling(f) and Rotation(f) are the restrictions of Symmetry(f) to three concrete subgroups, and are therefore all contained in Symmetry(f);
- (iv) the higher-order group $\{G_{(k)}\}$ and the higher-order functions $\{f_{(k)}\}$ are related level by level through the automorphism tower (Definition 18.29, 18.30).

What has genuine mathematical content is the **structure** of the three invariance subgroups, that is, Proposition 18.7.

Proof. (i) By Definition 18.24, $\text{Group}_{\text{period}}(f) = \text{Symmetry}(f)$ are two notations for the same set.

(ii) If $g \in \text{Symmetry}(f)$, then $f(\phi_g(x)) = f(x)$, hence $Q(f(\phi_g(x))) = Q(f(x))$ for all x, that is, $g \in \text{Group}_{\text{conservation}}(f, Q)$ (Definition 18.22).

(iii) The translation group Translation(f) (Definition 18.16) is precisely the set of a for which $f(x + a) = f(x)$, that is, the preimage of Symmetry(f) under the translation action $\phi_a(x) = x+a$; the scaling group and the rotation group are analogous (Definition 18.18, 18.20). Hence all three are contained in Symmetry(f).

(iv) By Definition 18.29 and 18.30, $f_{(k)}$ is defined on $G_{(k)}$, and the periodic invariance of $f_{(0)} = f$ is transmitted level by level along the automorphism tower. □

18.6 The Joint Phase-Space Criterion Framework

Definition 18.31 (Phase-Space Trajectory of a Function)**.** Let $f : \mathcal{D} \to V$ be a smooth function. The **phase-space trajectory** of f is the mapping

$$\Gamma_f : \mathcal{D} \to T^*\mathcal{D}, \quad x \mapsto (f(x), \dot{f}(x)).$$

The phase-space trajectory encodes both the tendency of the function (the function value) and its direction (the value of the derivative).

Theorem 18.4 (Joint Phase-Space and Group-Theoretic Criterion for the Periodicity of a Function)**.** Let $f : \mathcal{D} \to V$ be a smooth function and G a group acting on $\mathcal{D}$. Then f is a periodic function with respect to $g \in G$ if and only if (ii) below holds, that is, if and only if the phase-space trajectory is invariant under ϕ_g:

(i) **On the group representation theory side**: $g \in \mathsf{Group}_{\mathsf{period}}(f)$—by Definition 18.28, this is merely a restatement of the conclusion itself.

(ii) **On the phase-space side**: $\Gamma_f(\phi_g(x)) = \Gamma_f(x)$ for all x—this is the substance of the criterion.

(iii) **On the conservation side**: there exists a conserved quantity Q such that $Q(f(x))$ is constant along Γ_f—this holds automatically once (ii) does (take $Q = \mathrm{id}$).

If Q in (iii) is allowed to be constant, (iii) is vacuously true, so it is only a consequence of (ii) and is not an independent condition; the whole content of the criterion lies in (ii).

Proof. (ii)⇒'f is periodic with respect to g': the component $f(\phi_g(x)) = f(x)$ of $\Gamma_f(\phi_g(x)) = \Gamma_f(x)$ holds for all x, which is Definition 18.27.

'f is periodic with respect to g'⇒(ii): differentiating $f(\phi_g(x)) = f(x)$ with respect to x gives $\dot{f}(\phi_g(x)) = \dot{f}(x)$, hence $\Gamma_f(\phi_g(x)) = \Gamma_f(x)$.

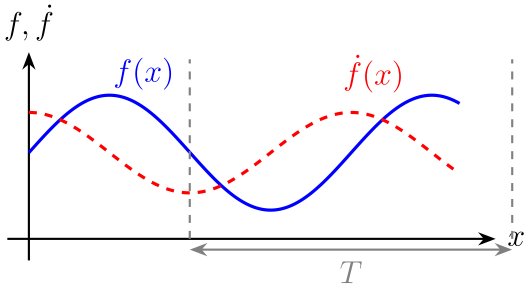
Hence (i)⇔(ii): (i) is the conclusion itself, so the two implications above already give both directions. Moreover (ii)⇒(iii) (take $Q = \mathrm{id}$); but the converse (iii)⇒(ii) fails in general—Q may be degenerate, the simplest example being a constant Q, for which (iii) holds vacuously and says nothing about whether f is periodic. Thus (iii) is implied by (ii) rather than equivalent to it, and (i), (ii), (iii) are not mutually independent conditions; the whole substance of the criterion lies in (ii). ☐

18.7 Group Periods and Group Functions

18.7.1 Strict Periodicity on a Group

Definition 18.32 (Strict Period on a Group)**.** Let G be a group and $f : G \to V$ a function on the group. If there exists a discrete subgroup $\Gamma \subset G$ such that $f(g\gamma) = f(g)$ for all $g \in G, \gamma \in \Gamma$, then f is said to be right Γ-periodic.

Proposition 18.8 (Basic Properties of Group Periodicity)**.** Let $f : G \to V$ be right Γ-periodic.



the phase trajectory $\Gamma_f(x) = (f, \dot f)$ returns to the same point under $x \mapsto x + T$

Figure 18.10: Joint phase-space criterion: the configuration side $f(x+T) = f(x)$ and the phase-space side $\Gamma_f(x+T) = \Gamma_f(x)$ holding simultaneously is equivalent to the function having period T.

(i) **Period closure**: if $\gamma_1, \gamma_2 \in \Gamma$, then f is also right $(\gamma_1\gamma_2)$-periodic.

(ii) **L^2 integrability condition**: if $f \neq 0$ and $f \in L^2(G)$, then Γ must be a **finite subgroup**; in particular, when G is a compact group, Γ is precisely a lattice of G (a finite discrete subgroup).

(iii) **Determination by the fundamental domain**: the values of f on a fundamental domain completely determine f.

Proof. (i) For any $g \in G$: $f(g(\gamma_1\gamma_2)) = f((g\gamma_1)\gamma_2) = f(g\gamma_1) = f(g)$, the first step using $\gamma_2 \in \Gamma$ and the second using $\gamma_1 \in \Gamma$. Hence f is right $(\gamma_1\gamma_2)$-periodic.

(ii) By Γ-periodicity, f takes a constant value on each right coset $g\Gamma$ ($f(g\gamma) = f(g)$), so $|f|^2$ is constant on cosets. By Weil's integration formula (taking the counting measure for the discrete subgroup),

$$\int_G |f(g)|^2\, dg = \int_{G/\Gamma} \Big(\sum_{\gamma\in\Gamma} |f(g\gamma)|^2 \Big)\, d\mu(\dot g) = |\Gamma| \cdot \int_{G/\Gamma} |\bar f(\dot g)|^2\, d\mu(\dot g),$$

where $|\Gamma|$ is the order of Γ (possibly ∞) and $\bar f$ is the descent of $|f|^2$ to G/Γ. If $f \neq 0$, then $\int_{G/\Gamma} |\bar f|^2\, d\mu > 0$, so $\int_G |f|^2 < \infty$ if and only if $|\Gamma| < \infty$, that is, Γ is finite.

(iii) For any $g \in G$, take a fundamental-domain representative g_0 and $\gamma \in \Gamma$ such that $g = g_0\gamma$; then $f(g) = f(g_0\gamma) = f(g_0)$. Hence f is uniquely determined by its values on the fundamental domain. ☐

18.7.2 Group Function Space and Regular Representation

Definition 18.33 (Group Function Space and Regular Representation). Let G be a locally compact group equipped with the left Haar measure dg. The inner product on the **group function space** $L^2(G)$ over G is

$$\langle f_1, f_2 \rangle = \int_G f_1(g)\overline{f_2(g)}\, dg.$$

The left regular representation λ and the right regular representation ρ are defined by $(\lambda(g)f)(x) = f(g^{-1}x)$ and $(\rho(g)f)(x) = f(xg)$.

Proposition 18.9 (Unitarity of the Regular Representation)**.** *The left regular representation and the right regular representation are both unitary representations on $L^2(G)$.*

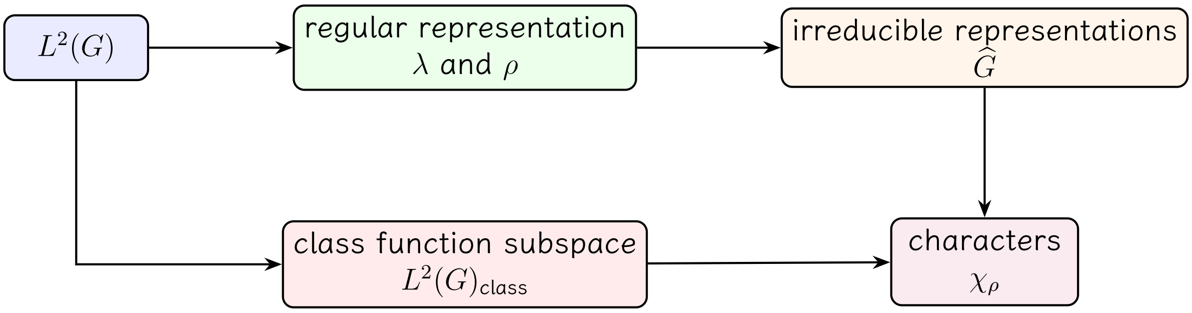
Proof. For the left regular representation λ: $\lambda(g)$ is invertible and $\lambda(g)^{-1} = \lambda(g^{-1})$ (because $\lambda(g)\lambda(g^{-1}) = \lambda(gg^{-1}) = \lambda(e) = \mathrm{id}$). For any $f_1, f_2 \in L^2(G)$,

$$\langle \lambda(g)f_1, \lambda(g)f_2 \rangle = \int_G f_1(g^{-1}x)\,\overline{f_2(g^{-1}x)}\,dx.$$

Put $y = g^{-1}x$, that is, $x = gy$; by the left invariance of the left Haar measure, $dx = d(gy) = dy$, giving

$$\langle \lambda(g)f_1, \lambda(g)f_2 \rangle = \int_G f_1(y)\,\overline{f_2(y)}\,dy = \langle f_1, f_2 \rangle,$$

hence $\lambda(g)$ preserves the inner product and is a unitary operator. The right regular representation is analogous: $(\rho(g)f)(x) = f(xg)$; putting $y = xg$, the right invariance $d(xg) = dx$ gives $\langle \rho(g)f_1, \rho(g)f_2 \rangle = \langle f_1, f_2 \rangle$, so $\rho(g)$ is likewise a unitary operator. ☐



Peter–Weyl: $L^2(G) \cong \widehat{\bigoplus}_\rho V_\rho \otimes V_\rho^*$
class functions are spanned by $\{\chi_\rho\}$

Figure 18.11: Decomposition of the group function space: the left/right regular representations decompose into a direct sum of irreducible representations, and the class function subspace is spanned by the characters.

18.7.3 Class Functions and Characters

Definition 18.34 (Class Function)**.** *A function $f : G \to \mathbb{C}$ is called a* **class function** *if $f(gxg^{-1}) = f(x)$ for all $g, x \in G$. Class functions are invariant under inner automorphisms.*

Theorem 18.5 (Character expansion of class functions—the class-function version of the Peter-Weyl theorem)**.** *Let G be a compact group. Then the characters $\{\chi_\rho : \rho \in \widehat{G}\}$ of the irreducible unitary representations form a complete orthogonal basis of $L^2(G)_{\text{class}}$. Any class function f can be expanded as $f(g) = \sum_\rho \langle f, \chi_\rho \rangle \chi_\rho(g)$.*

Proof. (Orthogonality) Let ρ, σ be irreducible finite-dimensional unitary representations and dg the normalised Haar measure. For a linear mapping $T : V_\sigma \to V_\rho$, construct the averaging operator

$$S = \int_G \rho(g)\, T\, \sigma(g)^{-1}\, dg.$$

For any h, $\rho(h)S = \int_G \rho(hg)T\sigma(g)^{-1}dg = \int_G \rho(u)T\sigma(u)^{-1}\sigma(h)du = S\sigma(h)$ (putting $u = hg$ and using the left invariance of the Haar measure), so S is an intertwining operator $\sigma \to \rho$. When $\rho \not\cong \sigma$, Schur's lemma gives $S = 0$; taking $T = E_{lj}$ (the basis matrix of the l-th row and the j-th column), its matrix elements are

$$S_{ik} = \int_G \rho_{il}(g)\, \sigma_{jk}(g^{-1})\, dg = \int_G \rho_{il}(g)\, \overline{\sigma_{kj}(g)}\, dg = 0 \qquad (\rho \not\cong \sigma).$$

When $\rho = \sigma$, $S = \lambda I$, and $\operatorname{tr} S = \operatorname{tr}\big(\int_G \rho(g)T\rho(g)^{-1}dg\big) = \operatorname{tr} T = \delta_{lj} = \lambda \dim\rho$, hence $\lambda = \delta_{lj}/\dim\rho$, that is,

$$\int_G \rho_{il}(g)\, \overline{\rho_{kj}(g)}\, dg = \frac{\delta_{ik}\delta_{lj}}{\dim\rho}.$$

Taking the diagonal sum $\chi_\rho = \sum_i \rho_{ii}$ gives

$$\langle \chi_\rho, \chi_\sigma \rangle = \sum_{i,k} \int_G \rho_{ii}(g)\overline{\sigma_{kk}(g)}\, dg = \delta_{\rho\sigma} \sum_{i,k} \frac{\delta_{ik}\delta_{ik}}{\dim\rho} = \delta_{\rho\sigma}\frac{\dim\rho}{\dim\rho} = \delta_{\rho\sigma}.$$

Hence $\{\chi_\rho\}$ is an orthonormal family in $L^2(G)_{\text{class}}$.

(Completeness and expansion) By the Peter-Weyl theorem, $L^2(G)$ decomposes as the closure of the direct sum of the subspaces of matrix coefficients of the irreducible representations, $L^2(G) \cong \widehat{\bigoplus}_\rho V_\rho \otimes V_\rho^* \cong \widehat{\bigoplus}_\rho \operatorname{End}(V_\rho)$, where G acts by conjugation (the adjoint action). Class functions are precisely the elements fixed by this conjugation action; by Schur's lemma, the elements of $\operatorname{End}(V_\rho)$ that commute with the conjugation action are exactly the scalar operators $\mathbb{C}\cdot I$, and the class function corresponding to them is precisely χ_ρ. Hence $L^2(G)_{\text{class}} = \overline{\bigoplus_\rho \mathbb{C}\,\chi_\rho}$, that is, the characters span a complete orthogonal basis, and any class function f has the expansion $f = \sum_\rho \langle f, \chi_\rho \rangle \chi_\rho$. □

18.7.4 Group-Valued Functions and Equivariance

Definition 18.35 (Group-Valued Functions and Equivariance). Let X be a set carrying an action of the group G, and let $(H, \cdot)$ be another group. A function $f : X \to H$ is called G-**equivariant** if there exists a group homomorphism $\rho : G \to \operatorname{Aut}(H)$ such that $f(g \cdot x) = \rho(g)(f(x))$.

Example 18.4 (Group Homomorphisms as Equivariant Functions). A group homomorphism $\varphi : G \to H$ is equivariant under left translation. If H is an Abelian group, then φ is simultaneously G-invariant—$\ker\varphi$ is a period subgroup. For example, $\varphi : \mathbb{Z} \to \mathbb{Z}/6\mathbb{Z}$, $\varphi(m) = m \bmod 6$: from $\varphi(m + a) = \varphi(m) + \varphi(a)$ we see that φ is equivariant under the left translation $m \mapsto m + a$; since $\ker\varphi = 6\mathbb{Z}$, we have $\varphi(m + 6) = \varphi(m)$, so φ has $6\mathbb{Z}$ as its period subgroup.

18.7.5 Automorphic Functions on a Group

Definition 18.36 (Automorphic Functions on a Group)**.** Let G be a locally compact group, $\Gamma \subset G$ a discrete subgroup and $\rho : \Gamma \to \mathsf{GL}(V)$ a representation. If a function $f : G \to V$ satisfies $f(\gamma g) = \rho(\gamma) f(g)$ for all $\gamma \in \Gamma, g \in G$, then f is called an **automorphic function** with respect to Γ and ρ.

Theorem 18.6 (Quotient-Space Characterisation of Automorphic Functions)**.** *Automorphic functions are equivalent to sections of the quotient space $\Gamma \backslash (G \times V)$.*

Proof. Define a Γ-action on $G \times V$ by $\gamma \cdot (g, v) = (\gamma g, \rho(\gamma) v)$.

($\Rightarrow$) Let $f : G \to V$ satisfy the automorphy condition $f(\gamma g) = \rho(\gamma) f(g)$. Define $F : G \to G \times V$ by $F(g) = (g, f(g))$. Then

$$F(\gamma g) = (\gamma g, f(\gamma g)) = (\gamma g, \rho(\gamma) f(g)) = \gamma \cdot (g, f(g)) = \gamma \cdot F(g),$$

hence F is Γ-equivariant and descends to a well-defined section $s : \Gamma \backslash G \to \Gamma \backslash (G \times V)$, $s([g]) = [F(g)]$.

($\Leftarrow$) Let s be a section and take a lift $F(g) = (g, f(g))$ (taking the representative g as the first coordinate). Since s is well defined, $[F(\gamma g)] = [F(g)]$ for any $\gamma \in \Gamma$, that is, $[(\gamma g, f(\gamma g))] = [(g, f(g))]$, so there exists $\gamma' \in \Gamma$ with $(g, f(g)) = \gamma' \cdot (\gamma g, f(\gamma g)) = (\gamma' \gamma g, \rho(\gamma') f(\gamma g))$. Comparing first coordinates gives $\gamma' = \gamma^{-1}$, and comparing second coordinates gives $f(g) = \rho(\gamma^{-1}) f(\gamma g)$, that is, $\rho(\gamma) f(g) = f(\gamma g)$. This is precisely the automorphy condition, so f is an automorphic function.

In summary, automorphic functions and sections of the quotient space are in one-to-one correspondence. ☐

18.7.6 The Convolution Algebra on a Group

Definition 18.37 (Convolution on a group)**.** Let $f_1, f_2 \in L^1(G)$. The convolution is defined as

$$(f_1 * f_2)(x) = \int_G f_1(y)\, f_2(y^{-1} x)\, dy.$$

Proposition 18.10 (Diagonalisation of convolution under orbit decomposition)**.** *Let G be a compact group and $K \in L^1(G)$ a class function. Then the convolution operator $C_K f = K * f$ is diagonalised under the orbit decomposition: $(C_K f)_\rho = \frac{\hat{K}(\rho)}{\dim \rho} f_\rho$, where $\hat{K}(\rho) = \langle K, \chi_\rho \rangle$.*

Proof. Let ρ be an irreducible unitary representation, and take the matrix elements $u_{ij}(g) = \langle \rho(g) e_j, e_i \rangle$ in an orthonormal basis. For a single matrix element,

$$(K * u_{ij})(g) = \int_G K(y)\, u_{ij}(y^{-1} g)\, dy = \int_G K(y) \sum_k u_{ik}(y^{-1})\, u_{kj}(g)\, dy.$$

Since $u_{ik}(y^{-1}) = \overline{u_{ki}(y)}$ and K is a class function, by the Schur orthogonality relations from the proof of Theorem 18.5, K is orthogonal to the off-diagonal matrix elements of ρ, and

$$\int_G K(y)\, \overline{u_{ki}(y)}\, dy = \langle K, u_{ki} \rangle = \frac{\langle K, \chi_\rho \rangle}{\dim \rho}\, \delta_{ki} = \frac{\hat{K}(\rho)}{\dim \rho}\, \delta_{ki},$$

where $\hat{K}(\rho) = \langle K, \chi_\rho \rangle$. Hence

$$(K * u_{ij})(g) = \sum_k \frac{\hat{K}(\rho)}{\dim \rho} \delta_{ki} \, u_{kj}(g) = \frac{\hat{K}(\rho)}{\dim \rho} u_{ij}(g).$$

That is, C_K acts on the ρ-isotypic component (spanned by the matrix elements) by the scalar $\hat{K}(\rho)/\dim \rho$, hence $(C_K f)_\rho = \frac{\hat{K}(\rho)}{\dim \rho} f_\rho$. □

18.8 The Transitivity Matrix: Preservation of the Eight Properties under the Twelve Operations

Lemma 18.2 (Pointwise operations commute with the group action). *Let a group G act on a set $\mathcal{D}$, and let $f, g : \mathcal{D} \to V$ be functions invariant under the subgroups $H_f, H_g \subseteq G$ respectively, that is, $f(\phi_h(x)) = f(x)$ $(h \in H_f)$ and $g(\phi_h(x)) = g(x)$ $(h \in H_g)$. Let $\star : V \times V \to V$ be an arbitrary pointwise binary operation ($(f \star g)(x) = f(x) \star g(x)$) and $\triangle : V \to V$ an arbitrary pointwise unary operation ($(\triangle f)(x) = \triangle(f(x))$). Then*

- (i) $f \star g$ *is invariant under* $H_f \cap H_g$;
- (ii) $\triangle f$ *is invariant under* H_f.

Proof. (i) For $h \in H_f \cap H_g$: $(f\star g)(\phi_h(x)) = f(\phi_h(x)) \star g(\phi_h(x)) = f(x) \star g(x) = (f \star g)(x)$.
(ii) For $h \in H_f$: $(\triangle f)(\phi_h(x)) = \triangle(f(\phi_h(x))) = \triangle(f(x)) = (\triangle f)(x)$. □

18.8.1 Addition and Subtraction

Proposition 18.11 (Preservation of the translation property under addition). *Let f, g both be translation functions, with translation groups $T_f\mathbb{Z}$ and $T_g\mathbb{Z}$ respectively. If $T_f/T_g \in \mathbb{Q}$, then $f + g$ is a translation function, with* Translation$(f + g) \supseteq$ lcm$(T_f, T_g)\mathbb{Z}$. *If $T_f/T_g \notin \mathbb{Q}$, then $f + g$ is not a translation function (unless it is constant), but a quasi-periodic function.*

Proof. Let $T_f/T_g = p/q$ and put $L = qT_f = pT_g$. Then $(f+g)(x+L) = f(x+L)+g(x+L) = f(x) + g(x) = (f + g)(x)$. If the ratio is irrational, there exists no non-zero common period. □

Proposition 18.12 (Preservation of scaling under addition and multiplication (commensurability condition)). *Let f, g be continuous non-constant scaling functions with scaling groups $\langle q_f \rangle$ and $\langle q_g \rangle$ respectively ($q_f, q_g > 0$, $q_f, q_g \neq 1$). If $\ln q_f / \ln q_g \in \mathbb{Q}$, that is, there exist coprime positive integers m, n with $q_f^m = q_g^n =: q$, then $f \pm g$ and fg (as well as f/g when $g \neq 0$) are all scaling functions, and* Scaling$(f \pm g)$, Scaling$(fg) \supseteq \langle q \rangle$. *If $\ln q_f / \ln q_g \notin \mathbb{Q}$, then $f \pm g$ and fg are not scaling functions (unless they degenerate).*

Proof. Since Scaling$(f) = \langle q_f \rangle$, $f(q_f^m x) = f(x)$ holds for any integer m; similarly $g(q_g^n x) = g(x)$. Let $q_f^m = q_g^n = q$; then for $h = f \pm g$:

$$h(qx) = f(qx) \pm g(qx) = f(q_f^m x) \pm g(q_g^n x) = f(x) \pm g(x) = h(x),$$

and similarly for fg. Hence h and fg are invariant under scaling by q. If the ratio is irrational, suppose that f, g have a non-trivial common scaling factor $q \neq 1$; then $q = q_f^m = q_g^n$ for some m, n, and taking logarithms gives $m \ln q_f = n \ln q_g$, that is, $\ln q_f / \ln q_g = n/m \in \mathbb{Q}$, a contradiction. □

Remark 18.5 (Commensurability condition for translation and rotation under binary operations). Exactly the same commensurability argument applies to translation functions under multiplication/division (the $\checkmark^*$ entries in the table): when $T_f/T_g \in \mathbb{Q}$, $L = \mathrm{lcm}(T_f, T_g)$ is a common period of fg and f/g. For rotation functions, a binary operation requires the angle ratio $\theta_f/\theta_g \in \mathbb{Q}$ in order for there to be a non-trivial common rotation angle; hence the cells in the rotation column of the table under binary operations (addition, subtraction, multiplication, division) should likewise be understood as preservation under the commensurability condition.

Proposition 18.13 (Preservation of the rotation/conservation/symmetry/closure/period/higher-o properties under addition)**.** Let f, g be functions of the same kind, with groups G_f and G_g respectively. Then $f+g$ is invariant under $G_f \cap G_g$; when $G_f \cap G_g$ is non-trivial, $f+g$ is a function of the same kind. For the translation and scaling cases see Proposition 18.11 and 18.12 respectively.

Proof. By Lemma 18.2 (taking $\star$ to be addition), for any $g \in G_f \cap G_g$, $(f+g)(\phi_g(x)) = f(\phi_g(x)) + g(\phi_g(x)) = f(x) + g(x) = (f+g)(x)$, so $f+g$ is invariant under $G_f \cap G_g$. The addition of higher-order functions is analogous level by level. □

Proposition 18.14 (Transitivity of all properties under subtraction)**.** The transitivity of subtraction $f - g = f + (-g)$ is exactly the same as that of addition.

Proof. $-g$ and g have exactly the same periodicity properties (the translation, scaling and rotation groups are all unchanged). □

18.8.2 Multiplication and Division

Proposition 18.15 (Preservation of the rotation/conservation/symmetry/closure/period/higher-o properties under multiplication)**.** Let f, g be functions of the same kind, with groups G_f and G_g respectively. Then fg is invariant under $G_f \cap G_g$; when $G_f \cap G_g$ is non-trivial, fg is a function of the same kind. For the commensurability condition for translation and scaling see Proposition 18.12 and Remark 18.5.

Proof. By Lemma 18.2 (taking $\star$ to be multiplication), $(fg)(\phi_g(x)) = f(\phi_g(x))g(\phi_g(x)) = f(x)g(x) = (fg)(x)$ holds for $g \in G_f \cap G_g$, so fg is invariant under $G_f \cap G_g$. □

Proposition 18.16 (Transitivity of all properties under division)**.** Let f, g be functions of the same kind with $g(x) \neq 0$. Then f/g is a function of the same kind, and its group contains $G_f \cap G_g$.

Proof. $(f/g)(\phi_g(x)) = f(\phi_g(x))/g(\phi_g(x)) = f(x)/g(x) = (f/g)(x)$. □

18.8.3 Powers and Roots

Proposition 18.17 (Transitivity of all properties under exponentiation). Let f be a function of the same kind with group G_f and $f(x) > 0$. Then f^α $(\alpha \neq 0)$ is a function of the same kind, and its group contains G_f.

Proof. $f^\alpha(\phi_g(x)) = (f(\phi_g(x)))^\alpha = (f(x))^\alpha = f^\alpha(x)$. □

Proposition 18.18 (Transitivity of all properties under root extraction). Let f be a function of the same kind with $f(x) > 0$. Then $\sqrt[n]{f}$ is a function of the same kind, and its group contains G_f.

Proof. $\sqrt[n]{f}(\phi_g(x)) = \sqrt[n]{f(\phi_g(x))} = \sqrt[n]{f(x)}$. □

18.8.4 The Logarithm

Proposition 18.19 (Transitivity of all properties under the logarithm). Let f be a function of the same kind with $f(x) > 0$. Then $\log f$ is a function of the same kind, and its group contains G_f.

Proof. $\log f(\phi_g(x)) = \log(f(\phi_g(x))) = \log(f(x))$. □

18.8.5 Series and Limits

Proposition 18.20 (Transitivity of all properties under series expansion). Let $\{f_n\}$ be a sequence of functions of the same kind, and suppose that $\sum f_n$ converges uniformly to f. Then f is a function of the same kind, and its group contains $\bigcap G_{f_n}$.

Proof. For any $g \in \bigcap G_{f_n}$, $f(\phi_g(x)) = \sum f_n(\phi_g(x)) = \sum f_n(x) = f(x)$. □

Proposition 18.21 (Transitivity of all properties under limits). Let $\{f_n\}$ be a sequence of functions of the same kind, with $f_n \to f$ pointwise. Then f is a function of the same kind, and its group contains $\bigcap G_{f_n}$.

Proof. $f(\phi_g(x)) = \lim f_n(\phi_g(x)) = \lim f_n(x) = f(x)$. □

18.8.6 Differentiation

Proposition 18.22 (Preservation of the translation/conservation/symmetry/closure/period/highe properties under differentiation). Let f be a differentiable translation function. Then f' is a translation function, with $\mathrm{Translation}(f') \supseteq \mathrm{Translation}(f)$.

Proof. $f'(x+T) = \lim_{h\to 0} \frac{f(x+T+h)-f(x+T)}{h} = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = f'(x)$. □

Proposition 18.23 (Differentiation destroys scaling symmetry). Let f be a differentiable scaling function. Then f' is not a scaling function (unless f is constant). Differentiation destroys scaling symmetry.

Proof. By the chain rule, if $f(q_0 x) = f(x)$, differentiating both sides gives $q_0 f'(q_0 x) = f'(x)$, that is, $f'(q_0 x) = f'(x)/q_0 \neq f'(x)$. □

Proposition 18.24 (Covariant preservation of the rotation property under differentiation)**.** Let f be a differentiable rotation function. Then f' satisfies $f'(e^{i\theta}z) = e^{-i\theta}f'(z)$ (covariance). $|f'|$ is a rotation function.

Proof. By the chain rule, $f'(e^{i\theta}z) = e^{-i\theta}f'(z)$. Taking moduli gives $|f'|(e^{i\theta}z) = |f'(z)|$. □

Proposition 18.25 (Lifting of higher-order properties under differentiation)**.** Differentiation maps $f_{(k)}$ to $f_{(k+1)}$—differentiation raises the level of the higher-order tower.

Proof. Let $f_{(k)} : G_{(k)} \to V$ be differentiable. Its differential at the identity element,

$$df_{(k)}\big|_e : T_eG_{(k)} = \mathfrak{g}_{(k)} \longrightarrow V$$

is a linear mapping on the Lie algebra $\mathfrak{g}_{(k)}$. Now a linear functional on $\mathfrak{g}_{(k)}$ belongs, under the adjoint action, to the representation category of the object $G_{(k+1)} = \mathrm{Aut}(G_{(k)})$ one level higher (Definition 18.8), hence $df_{(k)}$ is an object $f_{(k+1)}$ at level $G_{(k+1)}$. Recursing level by level, differentiation lifts $f_{(k)}$ to the level $f_{(k+1)}$. □

18.8.7 Integration

Proposition 18.26 (Conditional preservation of the translation property under integration)**.** Let f be a continuous translation function and $F(x) = \int_0^x f(t)dt$. Then F is a translation function if and only if $\int_0^T f(t)dt = 0$.

Proof. $F(x+T) - F(x) = \int_x^{x+T} f(t)dt = \int_0^T f(t)dt$. □

Proposition 18.27 (Preservation of the scaling/rotation/conservation/symmetry/closure/higher-properties under integration)**.** Integration preserves the scaling, rotation (up to modulus), conservation, symmetry and closure properties. Integration lowers the level of the higher-order tower.

Proof. Verify item by item. Scaling: let $f(q_0x) = f(x)$ and put $F(x) = \int_0^x f(t)dt$; the substitution $t = q_0s$ gives

$$F(q_0x) = \int_0^{q_0x} f(t)dt = q_0 \int_0^x f(q_0s)ds = q_0 \int_0^x f(s)ds = q_0F(x),$$

that is, $F(q_0x) = q_0F(x)$—F satisfies generalised scaling covariance (the scaling 'weight' of F being 1), preserving the scaling structure in the sense of the scale charge.

Rotation: let $f(e^{i\theta_0}z) = f(z)$ and define $F(z) = \int_0^z f(w)dw$ (along a radial path); then

$$F(e^{i\theta_0}z) = \int_0^{e^{i\theta_0}z} f(w)dw = \int_0^z f(e^{i\theta_0}\zeta)e^{i\theta_0}d\zeta = e^{i\theta_0}\int_0^z f(\zeta)d\zeta = e^{i\theta_0}F(z),$$

that is, $F(e^{i\theta_0}z) = e^{i\theta_0}F(z)$—$F$ is rotation-covariant, and its modulus $|F|$ is a rotation function.

Conservation: let Q be a conserved quantity of f, that is, $Q(f(\phi_g(x))) = Q(f(x))$. Define $Q'(F(x)) = Q(f(x))$ (the function value at the upper limit of integration); then $Q'(F(\phi_g(x))) = Q(f(\phi_g(x))) = Q(f(x)) = Q'(F(x))$, so Q' is a conserved quantity of F.

Symmetry: if $f(\phi_g(x)) = f(x)$, then $F(\phi_g(x)) = \int_0^{\phi_g(x)} f = \int_0^x f+$ boundary correction. On a compact group, taking the invariant integral $\int_G f(gx)dg$ directly verifies the preservation of symmetry.

Closure: if Γ_f is closed, then $\Gamma_F = (F, F') = (F, f)$ is closed under the same group action (because the closure of f guarantees $f(\phi_g(x)) = f(x)$ and $f'(\phi_g(x)) = f'(x)$, while $F' = f$, hence $\Gamma_F(\phi_g(x)) = (F(\phi_g(x)), f(\phi_g(x))) = (F(\phi_g(x)), f(x))$; under the translation condition $F(\phi_g(x)) = F(x)+$ a constant, so it is closed in the sense of the quotient space/modulo constants).

Higher order: integration is the inverse operation of differentiation. By Proposition 18.25, differentiation raises the level of the higher-order tower, hence integration lowers the level of the higher-order tower. Specifically, if $f_{(k)} \in G_{(k)}$, then $\int f_{(k)}$ belongs to the $G_{(k-1)}$ level (being recovered by differentiation). ☐

18.8.8 Convolution

Proposition 18.28 (Preservation of the translation property under convolution). *Let f, g both be translation functions. Then the circular convolution $f * g$ is a translation function (under the commensurability condition).*

Proof. $(f*g)(x+L) = \frac{1}{L}\int_0^L f(\tau)g(x+L-\tau)d\tau = \frac{1}{L}\int_0^L f(\tau)g(x-\tau)d\tau = (f*g)(x)$. ☐

Proposition 18.29 (Preservation of the scaling/rotation/conservation/symmetry/closure/period/ properties under convolution). *Convolution preserves all eight properties. The reason is that convolution commutes with the group action (the translation invariance of convolution), and convolution becomes multiplication in the frequency domain.*

Proof. Verify item by item. Translation: already proved by Proposition 18.28 (commensurability condition). Scaling: let $f(q_0x) = f(x)$ and $g(q_0x) = g(x)$. The substitution $\tau = q_0s$ gives

$$(f*g)(q_0x) = \frac{1}{T}\int_0^T f(\tau)g(q_0x-\tau)d\tau = q_0 \cdot \frac{1}{T}\int_0^{T/q_0} f(q_0s)g(q_0(x-s))ds.$$

By periodicity and scaling invariance, the above expression $= q_0 \cdot \frac{1}{T}\int_0^T f(s)g(x-s)ds = q_0(f*g)(x)$, so $f*g$ is generalised scaling-covariant (weight 1) and is preserved in the sense of the scale charge.

Rotation: let $f(e^{i\theta_0}z) = f(z)$ and $g(e^{i\theta_0}z) = g(z)$. For the complex convolution $(f*g)(z) = \int f(w)g(z-w)dw$, the substitution $w = e^{i\theta_0}\zeta$ gives

$$(f*g)(e^{i\theta_0}z) = \int f(w)g(e^{i\theta_0}z-w)dw = e^{i\theta_0}\int f(e^{i\theta_0}\zeta)g(e^{i\theta_0}(z-\zeta))d\zeta = e^{i\theta_0}(f*g)(z),$$

hence $f*g$ is rotation-covariant and $|f*g|$ is a rotation function.

Conservation/symmetry/closure: by the linearity of convolution and the transmission of pointwise properties (Lemma 18.2 extended through integration), if Q is a conserved quantity of f, g respectively, then in the frequency domain $Q(\widehat{f*g}) = Q(\widehat{f}\cdot\widehat{g})$ preserves the conservation structure.

Period: the convolution of periodic functions is still a periodic function (a direct corollary of Proposition 18.28).

Higher order: convolution becomes multiplication in the frequency domain, and multiplication preserves the level of the higher-order tower (Proposition 18.15), hence convolution also preserves the level of the higher-order tower. ☐

18.9 Summary of the transitivity matrix

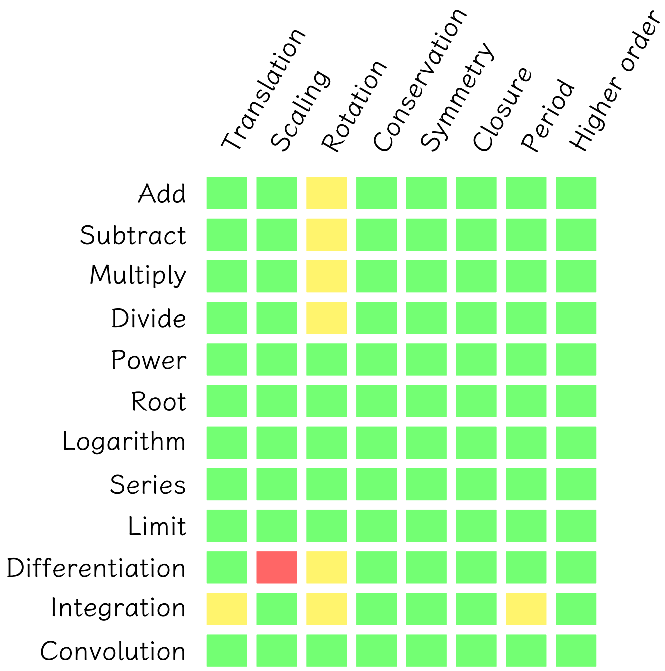
Table 18.2: Transitivity matrix of the eight properties under the twelve operations

Operation	Translation	Scaling	Rotation	Conservation	Symmetry	Closure	Period	Higher order
Addition	✓*	✓*	✓	✓	✓	✓	✓	✓
Subtraction	✓*	✓*	✓	✓	✓	✓	✓	✓
Multiplication	✓*	✓*	✓	✓	✓	✓	✓	✓
Division	✓*	✓*	✓	✓	✓	✓	✓	✓
Exponentiation	✓	✓	✓	✓	✓	✓	✓	✓
Root extraction	✓	✓	✓	✓	✓	✓	✓	✓
Logarithm	✓	✓	✓	✓	✓	✓	✓	✓
Series	✓	✓	✓	✓	✓	✓	✓	✓
Limits	✓	✓	✓	✓	✓	✓	✓	✓
Differentiation	✓	×	✓**	✓	✓	✓	✓	✓
Integration	✓***	✓	✓**	✓	✓	✓	✓***	✓
Convolution	✓*	✓	✓	✓	✓	✓	✓	✓

Remark 18.6 (Legend for the transitivity matrix).

- ✓: the property is preserved under the operation.
- ✓*: the property is preserved under the commensurability condition.
- ✓**: the property is preserved in the sense of modulus (rotation covariance).
- ✓***: the property is preserved under the zero-mean condition ($\int_0^T f = 0$).
- ×: the property is not preserved (differentiation destroys scaling symmetry).

Theorem 18.7 (General pattern of the transitivity matrix). (i) **Algebraic operations (addition, subtraction, multiplication, division, exponentiation, root extraction, logarithm)**: *all properties are preserved. The reason is that pointwise operations commute with the group action.*



	Translation	Scaling	Rotation	Conservation	Symmetry	Closure	Period	Higher order
Add	🟩	🟩	🟨	🟩	🟩	🟩	🟩	🟩
Subtract	🟩	🟩	🟨	🟩	🟩	🟩	🟩	🟩
Multiply	🟩	🟩	🟨	🟩	🟩	🟩	🟩	🟩
Divide	🟩	🟩	🟨	🟩	🟩	🟩	🟩	🟩
Power	🟩	🟩	🟩	🟩	🟩	🟩	🟩	🟩
Root	🟩	🟩	🟩	🟩	🟩	🟩	🟩	🟩
Logarithm	🟩	🟩	🟩	🟩	🟩	🟩	🟩	🟩
Series	🟩	🟩	🟩	🟩	🟩	🟩	🟩	🟩
Limit	🟩	🟩	🟩	🟩	🟩	🟩	🟩	🟩
Differentiation	🟩	🟥	🟨	🟩	🟩	🟩	🟩	🟩
Integration	🟨	🟩	🟨	🟩	🟩	🟩	🟨	🟩
Convolution	🟩	🟩	🟩	🟩	🟩	🟩	🟩	🟩

Figure 18.12: Heat map of the transitivity matrix: green indicates complete preservation, yellow indicates preservation under the commensurability or other conditions, and red indicates non-preservation.

(ii) **Analytic operations (series, limits)**: all properties are preserved. The reason is that uniform convergence/pointwise convergence guarantees that the group action commutes with the limit.

(iii) **Differentiation**: all are preserved except the scaling property. The reason why differentiation destroys scaling symmetry is that the chain rule introduces an extra scaling factor.

(iv) **Integration**: all are preserved except that the translation/period properties are preserved only conditionally. The reason is that the constant of integration introduces an extra degree of freedom.

(v) **Convolution**: all are preserved. The reason is that convolution commutes with the group action.

Proof. (i) Representative complete verification (taking multiplication as the example). Let the groups with periodic orbits of f, g be H_f, H_g respectively. For any $h \in H_f \cap H_g$, Lemma 18.2 (taking $\star$ to be multiplication) gives $(fg)(\phi_h(x)) = f(\phi_h(x))\, g(\phi_h(x)) = f(x)\, g(x) = (fg)(x)$, so fg is invariant under $H_f \cap H_g$ and the property is preserved (Proposition 18.15). The objects verified for addition, subtraction, division, exponentiation, root extraction and the logarithm (Proposition 18.13, 18.14, 18.16, 18.17, 18.18, 18.19) are the same—pointwise operations commute with the group action—so they are verified in the same way.

(ii) Verified in the same way: the objects verified are the series and limit operations, and uniform convergence/pointwise convergence guarantees that the group action commutes with $\sum$ and lim (Proposition 18.20, 18.21).

(iii) The object verified is the differential operator d/dx: translation is preserved (Proposition 18.22); scaling is destroyed (Proposition 18.23, the chain rule introducing the factor q_0); rotation is covariant (Proposition 18.24); the preservation of the remaining properties is verified in the same way from the commutativity of differentiation with the invariance groups.

(iv) The object verified is the integral operator: the translation/period properties are preserved under the zero-mean condition (Proposition 18.26), and the scaling/rotation/conservation/symmetry/closure properties are preserved (Proposition 18.27).

(v) The object verified is convolution: translation is preserved under the commensurability condition (Proposition 18.28) and the remaining seven properties are preserved (Proposition 18.29), for the same reason that convolution commutes with the group action. ☐

18.10 The Three Basic Groups and the Corresponding Group Functions

18.10.1 The Translation Group and Translation Functions

Definition 18.38 (Translation group). Let $G_{\text{trans}} = \mathbb{R}$ be the additive group. The translation group is the group-representation-theoretic carrier of the translation construction, with infinitesimal generator $\frac{d}{dx}$. Translation invariance implies the conservation of energy (time translation) or of linear momentum (spatial translation).

Definition 18.39 (Translation function). If a function $f : \mathbb{R} \to \mathbb{C}$ satisfies $f(x+T_0) = f(x)$, then f is called a **translation-periodic function** with respect to $\Gamma = T_0\mathbb{Z}$.

18.10.2 The Scaling Group and Scaling Functions

Definition 18.40 (Scaling group). Let $G_{\text{scal}} = \mathbb{R}^+$ be the multiplicative group of positive real numbers. The scaling group is the group-representation-theoretic carrier of the scaling construction, with infinitesimal generator the radial operator $\mathsf{x} \cdot \nabla$.

Definition 18.41 (Scaling function). If a function $f : \mathbb{R}^+ \to \mathbb{C}$ satisfies $f(q_0 x) = f(x)$, then f is called a **scaling-periodic function** with respect to $\Gamma = \langle q_0 \rangle$. By the logarithmic substitution it is equivalent to a translation function.

18.10.3 The Rotation Group and Rotation Functions

Definition 18.42 (Rotation group). Let $G_{\text{rot}} = U(1)$ be the one-dimensional unitary group. The rotation group is one of the group-representation-theoretic carriers of inner automorphisms.

Definition 18.43 (Rotation function). If a function $f : \mathbb{C} \to \mathbb{C}$ satisfies $f(e^{i\theta_0} z) = f(z)$, then f is called a **rotation-periodic function** with respect to $\Gamma = \langle e^{i\theta_0} \rangle$.

18.10.4 The Unified Duality of the Three Groups

Theorem 18.8 (Unified duality theorem for the three groups)**.** *The three basic groups stand in the following correspondence under Pontryagin duality:*

Group type	Group G	Period subgroup	Quotient space	Pontryagin dual
Translation group	$\mathbb{R}$	$T_0\mathbb{Z}$	$\mathbb{R}/T_0\mathbb{Z} \cong S^1$	$\widehat{S^1} \cong \mathbb{Z}$
Scaling group	$\mathbb{R}^+ \cong \mathbb{R}$	$\langle q_0 \rangle$	$\mathbb{R}^+/\langle q_0 \rangle \cong S^1$	$\widehat{S^1} \cong \mathbb{Z}$
Rotation group	$U(1) \cong S^1$	$\langle e^{i\theta_0} \rangle$	$U(1)/C_n \cong U(1)$	$\widehat{U(1)} \cong \mathbb{Z}$

Proof. Verify row by row. Translation row: the mapping $\chi(x) = e^{2\pi i x/T_0}$ is a surjective homomorphism $\mathbb{R} \to S^1$ whose kernel is exactly $T_0\mathbb{Z}$, hence $\mathbb{R}/T_0\mathbb{Z} \cong S^1$; the family of characters of S^1, $\{\chi_n : e^{i\theta} \mapsto e^{in\theta} \mid n \in \mathbb{Z}\}$, gives $\widehat{S^1} \cong \mathbb{Z}$. Scaling row: the logarithm mapping $x \mapsto \ln x$ is an isomorphism $\mathbb{R}^+ \cong \mathbb{R}$ and maps $\langle q_0 \rangle$ to $(\ln q_0)\mathbb{Z}$, hence is isomorphic to the translation row, with quotient $\cong S^1$ and dual $\cong \mathbb{Z}$. Rotation row: $U(1) \cong S^1$; when $\theta_0/(2\pi) = p/q$, $\langle e^{i\theta_0} \rangle \cong C_q$, and the kernel of the mapping $z \mapsto z^q$ is the group of q-th roots of unity C_q, hence $U(1)/C_q \cong U(1)$; the characters $\chi_n(z) = z^n$ give $\widehat{U(1)} \cong \mathbb{Z}$. The duals in all three rows are $\cong \mathbb{Z}$, so the table holds. □

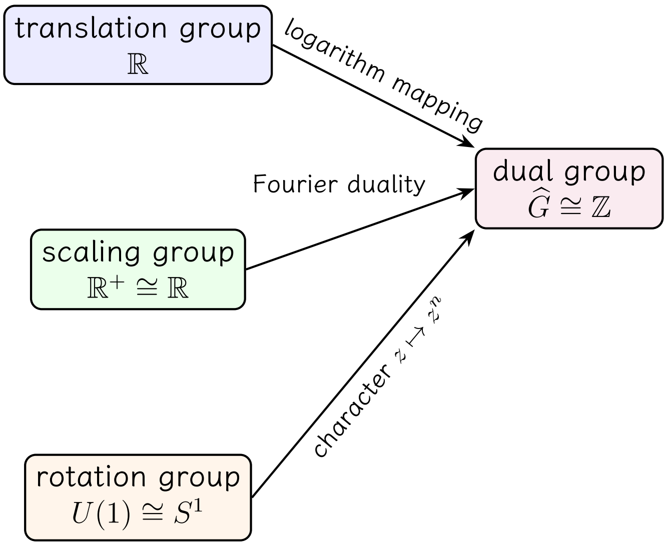

translation group
$\mathbb{R}$
logarithm mapping
Fourier duality
dual group
$\widehat{G} \cong \mathbb{Z}$
scaling group
$\mathbb{R}^+ \cong \mathbb{R}$
character $z \mapsto z^n$
rotation group
$U(1) \cong S^1$

Figure 18.13: Pontryagin duality relations of the three basic groups: the translation group, the scaling group and the rotation group are unified as $\mathbb{Z}$ under duality, constituting three representations of a unified algebraic structure

Corollary 18.1 (The three functions, groups, fixed points and conservation laws).

Function type	Group	Fixed element	Formally corresponding conservation law	Formally corresponding flux
Translation function	$\mathbb{R}$	0	Energy/linear momentum	Time flux
Scaling function	$\mathbb{R}^+$	1	Scale charge	Observation flux
Rotation function	$U(1)$	i	Angular momentum	Spatial flux

The two columns 'formally corresponding conservation law' and 'formally corresponding flux' in the table give a conceptual comparison between objects of group representation theory and physical concepts.

Proof. By Definition 18.39, 18.41 and 18.43, the translation function, the scaling function and the rotation function have $\mathbb{R}$, $\mathbb{R}^+$ and $U(1)$ respectively as their acting groups, with period subgroups $T_0\mathbb{Z}$, $\langle q_0 \rangle$ and $\langle e^{i\theta_0} \rangle$ in that order. The fixed elements are the reference points of the three cases 0, 1 and i respectively. The correspondence between conservation laws and fluxes is as in Proposition 18.1, 18.3 and 18.2. The quotient spaces and Pontryagin duality are given by Theorem 18.8. □

18.11 A unified perspective on group functions and periodic mathematics

Remark 18.7 (The place of group functions in periodic mathematics). This chapter obtains a unified formulation, at the level of group representation theory, of the core concepts of periodic mathematics:

(i) **The automorphism tower**: the automorphism tower realises 'the higher-order relation of a relation is again a group'.

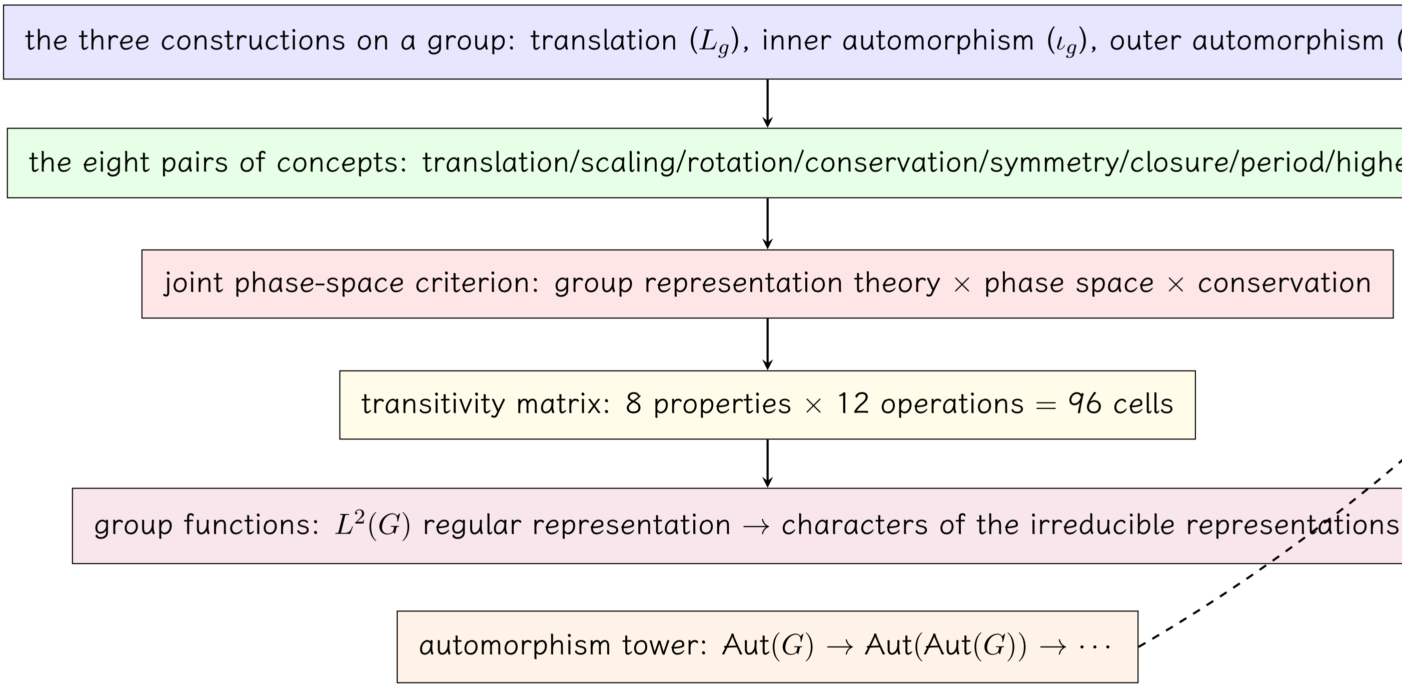
(ii) **The three constructions on a group**: translation, inner automorphism, outer automorphism.

(iii) **The eight pairs of concepts**: precise definitions of the translation/scaling/rotation/conserv
order functions and groups.

(iv) **The joint phase-space criterion**: the unification of the group representation theory criterion + the phase-space criterion + the conservation criterion.

(v) **The transitivity matrix**: a complete transitivity analysis of the eight properties under the twelve operations.

(vi) **Group functions—regular representation and characters**: the representation decomposition of $L^2(G)$.


the three constructions on a group: translation (L_g), inner automorphism (ι_g), outer automorphism (
the eight pairs of concepts: translation/scaling/rotation/conservation/symmetry/closure/period/highe
joint phase-space criterion: group representation theory × phase space × conservation
transitivity matrix: 8 properties × 12 operations = 96 cells
group functions: $L^2(G)$ regular representation → characters of the irreducible representations
automorphism tower: Aut(G) → Aut(Aut(G)) → ⋯

group functions = the unified language of periodic mathematics

Figure 18.14: The complete system of groups and group functions.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The translation construction on a group (Definition 18.1) and its basic properties (Proposition 18.1); the translation group/translation function pair of concepts
Scaling	The outer automorphism on a group (Definition 18.3) and its basic properties (Proposition 18.3); the scaling group/scaling function pair of concepts
Rotation	The inner automorphism on a group (Definition 18.2) and its basic properties (Proposition 18.2); the rotation group/rotation function pair of concepts
Conservation	The group–fixed element–conservation law comparison of the three constructions (Remark 18.1); the conservation group/conservation function pair of concepts
Symmetry	Translational symmetry and automorphism symmetry (Definition 18.11, 18.12), the automorphism tower (Definition 18.8), and the symmetry group/symmetry function pair of concepts
Closure	The closure of finite orbits in the orbit-stabiliser theorem (Theorem 18.3); the closure group/closure function pair of concepts; the joint phase-space and group-theoretic criterion for the periodicity of a function (Theorem 18.4)
Period	The group period (Definition 18.32) and its basic properties (Proposition 18.8); the group with

Exercises

Exercise 1 By Definition 18.1, verify the three basic properties of left and right translation:

(i) $L_g \circ L_h = L_{gh}$, and $\{L_g\} \cong G$ (the left regular action);
(ii) $R_g \circ R_h = R_{hg}$, and $\{R_g\} \cong G^{\mathrm{op}}$;
(iii) $L_g \circ R_h = R_h \circ L_g$.

Also explain why, when G is non-commutative, the left translation group and the right translation group are in general different as subgroups of G.

Exercise 2 Prove the four properties of the translation construction in Proposition 18.1: fixed elements, infinitesimal generators, invariance of the Haar measure and directional structure. Taking $G = \mathbb{Z}$, $G = \mathbb{R}$ and $G = GL(n, \mathbb{R})$ as examples respectively, verify the concrete forms of left and right translation.

Exercise 3 Prove the four properties of inner automorphisms in Proposition 18.2: the fixed elements are the centre $Z(G)$, the infinitesimal generator is the adjoint representation, the conserved quantity is a class function, and adjoint orbits. Verify that ι_g is an automorphism and that the kernel of the mapping $g \mapsto \iota_g$ is $Z(G)$, and explain why all inner automorphisms are the identity mapping when $G = U(1)$.

Exercise 4 Verify the three examples of inner automorphisms in Example 18.2:

(i) the similarity transformation $\iota_A(X) = AXA^{-1}$ in $GL(n, \mathbb{R})$;
(ii) the conjugation of a rotation in $SO(3)$ into a rotation about a new axis;
(iii) the degeneration of inner automorphisms to the identity in $U(1)$.

Taking $A = \mathrm{diag}(2, 1)$ and the permutation matrix P, compute PAP^{-1} and verify that the eigenvalues are invariant under conjugation.

Exercise 5 Prove the four properties of outer automorphisms in Proposition 18.3, and verify Example 18.3:

(i) $\mathrm{Aut}(\mathbb{Z}) \cong C_2 = \{\mathrm{id}, -\mathrm{id}\}$;
(ii) $\mathrm{Aut}(C_n) \cong (\mathbb{Z}/n\mathbb{Z})^\times$, of order $\varphi(n)$;
(iii) $\mathrm{Aut}(\mathbb{R}^+) \cong \mathbb{R}^\times$, with automorphisms $x \mapsto x^\lambda$.

Compute the order and the generators of $\mathrm{Aut}(C_5)$ when $n = 5$.

Exercise 6 Prove Theorem 18.1: the three standard constructions on a group form the following structure in the automorphism group $\mathrm{Aut}(G)$: translation is realised by the left regular action, the inner automorphism group is $\mathrm{Inn}(G) \cong G/Z(G)$, and the outer automorphism group is $\mathrm{Out}(G) = \mathrm{Aut}(G)/\mathrm{Inn}(G)$. Write down the exact sequence

$$1 \to Z(G) \to G \xrightarrow{\phi} \mathrm{Aut}(G) \xrightarrow{\pi} \mathrm{Out}(G) \to 1,$$

and verify its exactness term by term.

Exercise 7 Prove Lemma 18.1: let G be a finite group with $Z(G) = \{e\}$; then

(i) $G \cong \mathrm{Inn}(G) \subseteq \mathrm{Aut}(G)$, and therefore $|G| \leq |\mathrm{Aut}(G)|$;
(ii) if $|\mathrm{Aut}(G)| = |G|$, then $\mathrm{Aut}(G) = \mathrm{Inn}(G) \cong G$;
(iii) $Z(\mathrm{Aut}(G)) = \{e\}$, that is, triviality of the centre is preserved level by level.

Exercise 8 Verify the automorphism tower heights of the typical groups in Proposition 19.12:

(i) $C_2 \to \{e\}$, height 1;
(ii) $C_3 \to C_2 \to \{e\}$, height 2;
(iii) $C_5 \to C_4 \to C_2 \to \{e\}$, height 3;
(iv) $S_3 \to S_3$, height 0.

Compute the automorphism group of each level, and explain why the outer automorphism group of S_3 is the trivial group.

Exercise 9 Prove Theorem 18.3 (the orbit-stabiliser theorem): if a finite group G acts on a finite set X, then for any $x \in X$,

$$|G \cdot x| = \frac{|G|}{|G_x|}.$$

Complete the proof by constructing a bijection $G/G_x \to G \cdot x$, and explain how this theorem characterises the closure of finite orbits.

Exercise 10 Prove Proposition 18.5: if $\varphi \in \mathrm{Aut}(G)$ and $g \in G$, then

$$\varphi \circ L_g \circ \varphi^{-1} = L_{\varphi(g)}.$$

Also explain why outer automorphisms do not preserve the conjugacy-class structure whereas inner automorphisms do.

Exercise 11 Starting from Definition 18.15 through to Definition 18.30, write out the definitions of the eight pairs of concepts, and verify Proposition 18.7:

(i) $\mathrm{Group}_{\mathrm{period}}(f) = \mathrm{Symmetry}(f)$;
(ii) $\mathrm{Symmetry}(f) \subseteq \mathrm{Group}_{\mathrm{conservation}}(f, Q)$;
(iii) the translation group, the scaling group and the rotation group are all contained in $\mathrm{Symmetry}(f)$;
(iv) the higher-order group and the higher-order functions are related level by level through the automorphism tower.

Exercise 12 Prove Theorem 18.4: let f be a smooth function and let G act on its domain; then f is a periodic function with respect to $g \in G$ if and only if the phase-space trajectory satisfies $\Gamma_f(\phi_g(x)) = \Gamma_f(x)$. Prove the two directions separately, and explain why (i) $g \in \mathrm{Group}_{\mathrm{period}}(f)$ and (iii) 'there exists a conserved quantity Q such that $Q(f(x))$ is constant along Γ_f' are not conditions independent of, and on a par with, (ii)—show how (ii) implies (i) and (iii), and why (iii) cannot imply (ii).

Exercise 13 Prove the three basic properties of group periodicity in Proposition 18.8:

(i) period closure: $\gamma_1, \gamma_2 \in \Gamma \Rightarrow \gamma_1\gamma_2 \in \Gamma$;

(ii) the L^2 integrability condition: if $f \neq 0$ and $f \in L^2(G)$, then Γ is a finite subgroup;

(iii) determination by the fundamental domain: the values of f on a fundamental domain completely determine f.

Exercise 14 Prove Proposition 18.9: the left regular representation and the right regular representation on a locally compact group G are both unitary representations on $L^2(G)$. Complete the proof using respectively the left invariance and the right invariance of the left Haar measure.

Exercise 15 Prove Theorem 18.5 (the class-function version of the Peter–Weyl theorem): let G be a compact group; then the characters $\{\chi_\rho\}$ of the irreducible unitary representations form a complete orthogonal basis of $L^2(G)_{\text{class}}$. Write out the proof of the Schur orthogonality relations, and explain how an arbitrary class function is expanded as a series of characters.

Exercise 16 Prove Proposition 18.10: let G be a compact group and K a class function; then the convolution operator $C_K f = K * f$ is diagonalised under the orbit decomposition:

$$(C_K f)_\rho = \frac{\hat{K}(\rho)}{\dim \rho} f_\rho, \quad \hat{K}(\rho) = \langle K, \chi_\rho \rangle.$$

Complete the proof using the Schur orthogonality relations.

Exercise 17 Prove Theorem 18.6: automorphic functions are equivalent to sections of the quotient space $\Gamma\backslash(G \times V)$. Write out the constructions in the two directions, 'from automorphic functions to sections' and 'from sections to automorphic functions', and verify that the two are mutually inverse.

Exercise 18 Prove Lemma 18.2: let f, g be functions invariant under the subgroups H_f, H_g respectively and let $\star$ be a pointwise binary operation; then

(i) $f \star g$ is invariant under $H_f \cap H_g$;

(ii) for any pointwise unary operation $\triangle$, $\triangle f$ is invariant under H_f.

Explain why this lemma is the unified basis of proof for the algebraic-operation cells of the transitivity matrix.

Exercise 19 Verify Proposition 18.11 and Proposition 18.12:

(i) if the translation groups of f, g are $T_f\mathbb{Z}$ and $T_g\mathbb{Z}$ respectively, then $f+g$ has period $\mathrm{lcm}(T_f, T_g)$ whenever $T_f/T_g \in \mathbb{Q}$;

(ii) if the scaling groups of f, g are $\langle q_f \rangle$ and $\langle q_g \rangle$ respectively, then $f \pm g$ and fg are still scaling functions whenever $\ln q_f / \ln q_g \in \mathbb{Q}$.

Explain why the commensurability condition is a necessary condition for the preservation of periodic structure.

Exercise 20 Prove Proposition 18.23 and Proposition 18.24:

(i) differentiation destroys scaling symmetry: if $f(q_0x) = f(x)$, then $f'(q_0x) = f'(x)/q_0 \neq f'(x)$;

(ii) the rotation property is preserved covariantly under differentiation: if $f(e^{i\theta}z) = f(z)$, then $f'(e^{i\theta}z) = e^{-i\theta}f'(z)$, and $|f'|$ is a rotation function.

Explain why the extra scaling factor introduced by the chain rule is the fundamental reason why differentiation destroys scaling symmetry.

Exercise 21 Prove Proposition 18.26: let f be a continuous translation function and $F(x) = \int_0^x f(t)\,dt$; then F is a translation function if and only if $\int_0^T f(t)\,dt = 0$. Write out the proof, and explain why the extra degree of freedom introduced by the constant of integration causes the translation property to be preserved only conditionally.

Exercise 22 Prove Theorem 18.7: the general pattern of the transitivity matrix:

(i) algebraic operations (addition, subtraction, multiplication, division, exponentiation, root extraction, logarithm): all properties are preserved;

(ii) analytic operations (series, limits): all properties are preserved;

(iii) differentiation: all are preserved except the scaling property;

(iv) integration: all are preserved except that the translation/period properties are preserved only conditionally;

(v) convolution: all are preserved.

Point out separately the reason why each class of operations preserves or fails to preserve the properties.

Exercise 23 Prove Theorem 18.8: the three basic groups stand in correspondence under Pontryagin duality:

(i) $\mathbb{R}/T_0\mathbb{Z} \cong S^1$, $\widehat{S^1} \cong \mathbb{Z}$;

(ii) $\mathbb{R}^+/\langle q_0\rangle \cong S^1$, with dual isomorphic to $\mathbb{Z}$;

(iii) $U(1)/C_n \cong U(1)$, $\widehat{U(1)} \cong \mathbb{Z}$.

Verify the quotient-space structure and the dual-group structure of the three rows separately.

Exercise 24 Synthesis: combining the three standard constructions on a group (Definition 18.1, 18.2, 18.3), the eight pairs of concepts (§18.5), the joint phase-space criterion framework (Theorem 18.4) and the transitivity matrix (Table 18.2), explain how groups and group functions provide a unified algebraic language for periodic mathematics. Point out separately:

(i) how the three standard constructions correspond to translation, scaling and rotation respectively;

(ii) how the eight pairs of concepts characterise the hierarchy of periodic structure;

(iii) how the transitivity matrix summarises the preservation pattern of the eight properties under the twelve operations;
(iv) how the automorphism tower realises 'the higher-order relation of a relation is again a group'.

Chapter 19

Group Objects and Higher-order Mappings

Remark 19.1 (Positioning of this chapter). *This chapter systematically develops the automorphism tower structure of the 'Groups and Group Functions' chapter. The automorphism tower*

$$G_{(0)} = G, \qquad G_{(k+1)} = \mathsf{Aut}(G_{(k)}), \quad k \geq 0$$

is the concrete realisation, in group representation theory, of the idea that 'the relation of a relation is again a group'. This chapter establishes the basic theory of the automorphism tower, the three standard constructions on each level of the tower (translation, inner automorphism, outer automorphism), the function spaces on the tower, the symmetries and conserved quantities on the tower, the periodic structures on the tower, and the stability of the tower together with the tower heights of typical groups.

The core content of this chapter is the level-by-level lifting of the first-order group representation theory framework established in the 'Groups and Group Functions' chapter to every level of the automorphism tower. Each level $G_{(k)}$ has its own translation (left regular action), inner automorphisms ($\mathsf{Inn}(G_{(k)})$) and outer automorphisms ($\mathsf{Out}(G_{(k)})$), together with the corresponding function spaces and conserved quantities.

Seven-theme guide for this chapter: *Translation—the left regular action on each level of the automorphism tower (Proposition 19.6); Scaling—the level-by-level recursion of the outer automorphism group (Proposition 19.6); Rotation—the level-by-level recursion of the inner automorphism group (Proposition 19.6); Conservation—the conserved quantities on the tower (Definition 19.6); Symmetry—the symmetry groups on the tower (Definition 19.9); Closure—the stability of the tower (Theorem 19.1, Proposition 19.12); Period—the periodic structures on the tower (Definition 19.11).*

19.1 Automorphisms and the Automorphism Group

The basic concepts of this section were established in the 'Groups and Group Functions' chapter; here we merely review them and fix the notation.

Definition 19.1 (Automorphisms of a group). This definition is the same content as Definition 18.4 in the 'Automorphism Tower' section (Section 18.2) of Chapter 18; to avoid repeated exposition, only the reference anchor is retained here, and the complete statement is to be found there.

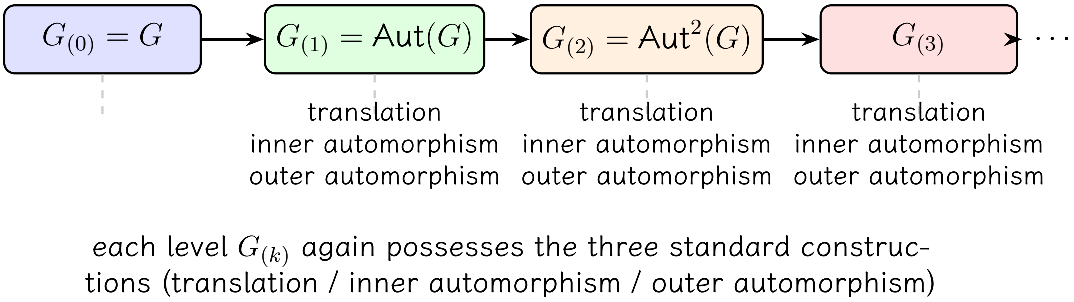

$G_{(0)} = G$
$G_{(1)} = \mathrm{Aut}(G)$
$G_{(2)} = \mathrm{Aut}^2(G)$
$G_{(3)}$
$\cdots$
translation
inner automorphism
outer automorphism
translation
inner automorphism
outer automorphism
translation
inner automorphism
outer automorphism
each level $G_{(k)}$ again possesses the three standard constructions (translation / inner automorphism / outer automorphism)

Figure 19.1: Overview of the automorphism tower: $G_{(k+1)} = \mathrm{Aut}(G_{(k)})$ is constructed level by level, and translation, inner automorphisms and outer automorphisms reappear on every level.

Definition 19.2 (The automorphism group)**.** This definition is the same content as Definition 18.5 in the 'Automorphism Tower' section (Section 18.2) of Chapter 18; to avoid repeated exposition, only the reference anchor is retained here, and the complete statement is to be found there.

Proposition 19.1 (The inverse of an automorphism preserves the group operation)**.** This proposition is the same content as Proposition 18.4 in the 'Automorphism Tower' section (Section 18.2) of Chapter 18; to avoid repeated exposition, only the reference anchor is retained here, and the complete statement is to be found there.

Proof. Since φ is surjective, for any $u, v \in G$ there exist unique $x, y \in G$ such that $\varphi(x) = u$, $\varphi(y) = v$. By the homomorphism property of φ, $\varphi(xy) = uv$. Applying φ^{-1} to both sides gives

$$xy = \varphi^{-1}(uv).$$

But $x = \varphi^{-1}(u)$, $y = \varphi^{-1}(v)$, so

$$\varphi^{-1}(uv) = \varphi^{-1}(u)\varphi^{-1}(v).$$

Hence φ^{-1} preserves the group operation. Moreover φ^{-1} is a bijection (φ being a bijection), so $\varphi^{-1} \in \mathrm{Aut}(G)$. ☐

Property 19.1 (Group-structural invariants preserved by automorphisms)**.** Let $\varphi \in \mathrm{Aut}(G)$. Then:

(i) **Order**: $|G| = |\varphi(G)|$;

(ii) **Identity element**: $\varphi(e) = e$;

(iii) **Order of an element**: $\mathrm{ord}(\varphi(g)) = \mathrm{ord}(g)$ for any $g \in G$;

(iv) **Centre**: $\varphi(Z(G)) = Z(G)$;

(v) **Commutator subgroup**: $\varphi([G, G]) = [G, G]$.

Proof. (i) φ is a bijection, so $|G| = |\varphi(G)|$.

(ii) From $\varphi(e) = \varphi(ee) = \varphi(e)\varphi(e)$, multiplying on the left by $\varphi(e)^{-1}$ gives $\varphi(e) = e$.

(iii) For any $n \geq 1$, $\varphi(g)^n = \varphi(g^n)$. Since φ is a bijection, $\varphi(g)^n = e$ if and only if $g^n = e$, so the orders are equal.

(iv) Let $z \in Z(G)$. For any $x \in G$ there exists $y \in G$ with $x = \varphi(y)$. Then

$$x\varphi(z) = \varphi(y)\varphi(z) = \varphi(yz) = \varphi(zy) = \varphi(z)\varphi(y) = \varphi(z)x.$$

Hence $\varphi(z) \in Z(G)$, that is, $\varphi(Z(G)) \subseteq Z(G)$. Applying the same conclusion to φ^{-1} gives $Z(G) \subseteq \varphi(Z(G))$, so the two are equal.

(v) The image of the commutator $[x, y] = xyx^{-1}y^{-1}$ under an automorphism is

$$\varphi([x, y]) = \varphi(x)\varphi(y)\varphi(x)^{-1}\varphi(y)^{-1} = [\varphi(x), \varphi(y)].$$

Hence $\varphi([G, G]) = [G, G]$. ☐

Property 19.2 (Automorphisms preserve conjugacy classes). *Let $\varphi \in \mathrm{Aut}(G)$. Then φ maps conjugacy classes to conjugacy classes, that is, for any $g \in G$,*

$$\varphi\big(\mathrm{Cl}(g)\big) = \mathrm{Cl}\big(\varphi(g)\big).$$

In particular, φ preserves the partition into conjugacy classes.

Proof. For any $x = hgh^{-1} \in \mathrm{Cl}(g)$,

$$\varphi(x) = \varphi(h)\varphi(g)\varphi(h)^{-1} \in \mathrm{Cl}(\varphi(g)).$$

Hence $\varphi(\mathrm{Cl}(g)) \subseteq \mathrm{Cl}(\varphi(g))$. Applying the same conclusion to φ^{-1} gives $\mathrm{Cl}(\varphi(g)) \subseteq \varphi(\mathrm{Cl}(g))$, so the two are equal. ☐

Lemma 19.1 (Automorphisms restrict to characteristic subgroups). *Let $H \leq G$ be a* **characteristic subgroup** *of G (that is, $\varphi(H) = H$ holds for every $\varphi \in \mathrm{Aut}(G)$). Then there exists a group homomorphism*

$$\Phi : \mathrm{Aut}(G) \to \mathrm{Aut}(H), \qquad \varphi \mapsto \varphi|_H.$$

Proof. For any $\varphi \in \mathrm{Aut}(G)$, by characteristicity $\varphi(H) = H$, so $\varphi|_H : H \to H$ is a bijection; and φ preserves the group operation, so $\varphi|_H$ preserves the group operation, that is, $\varphi|_H \in \mathrm{Aut}(H)$. The map Φ preserves composition: $(\varphi\psi)|_H = \varphi|_H\,\psi|_H$, so Φ is a homomorphism. ☐

Example 19.1 (Invariants of the automorphism groups of typical groups).

- $\mathrm{Aut}(\mathbb{Z}) \cong C_2$: the only non-trivial automorphism is $x \mapsto -x$, which preserves the orders of elements and interchanges $+n$ and $-n$.
- $\mathrm{Aut}(C_n) \cong (\mathbb{Z}/n\mathbb{Z})^\times$: the automorphisms $x \mapsto x^k$ ($\gcd(k, n) = 1$) preserve the orders of elements, preserve the identity element, and preserve the centre (the whole of C_n).
- $\mathrm{Aut}(S_3) \cong S_3$: all automorphisms are inner, and they preserve the three subgroups of order two (the transpositions) and the normal subgroup A_3.

Corollary 19.1 (Automorphisms preserve simplicity). *If G is a simple group and $\varphi \in \mathrm{Aut}(G)$, then φ maps maximal normal subgroups to maximal normal subgroups; in particular, if G is simple, then $\varphi(\{e\}) = \{e\}$ and $\varphi(G) = G$.*

Proof. G being simple means that it has no non-trivial normal subgroups. Let $N \trianglelefteq G$; then $\varphi(N) \trianglelefteq G$. If $N = \{e\}$, then $\varphi(N) = \{e\}$; if $N = G$, then $\varphi(N) = G$. The intermediate case does not occur. ☐

Table 19.1: Automorphism invariants and typical examples

Invariant	Meaning	Typical example
Group order $\|G\|$	φ is a bijection	Aut$(\mathbb{Z}/n\mathbb{Z})$ preserves n
Identity element e	$\varphi(ee) = \varphi(e)\varphi(e)$	every automorphism fixes e
Order of an element	ord$(\varphi(g)) =$ ord(g)	Aut(C_5) preserves the number of elements of order 5
Centre $Z(G)$	characteristic subgroup	Aut(S_3) fixes $Z(S_3) = \{e\}$
Commutator subgroup $[G, G]$	characteristic subgroup	Aut(S_3) fixes $[S_3, S_3] = A_3$
Partition into conjugacy classes	conjugacy classes are mapped to conjugacy classes	the three transposition classes of S_3 are preserved under Aut

19.2 Inner and Outer Automorphisms

Definition 19.3 (Inner automorphisms)**.** This definition is the same content as Definition 18.2 in the 'Inner Automorphisms' section (Section 18.1.2) of Chapter 18; to avoid repeated exposition, only the reference anchor is retained here, and the complete statement is to be found there.

Proposition 19.2 (The structure of the inner automorphism group)**.** Inn(G) is a normal subgroup of Aut(G), and

$$\text{Inn}(G) \cong G/Z(G).$$

Proof. First we verify that ι_g is an automorphism. For any $x, y \in G$,

$$\iota_g(xy) = g(xy)g^{-1} = (gxg^{-1})(gyg^{-1}) = \iota_g(x)\iota_g(y).$$

Moreover the inverse of ι_g is $\iota_{g^{-1}}$, so ι_g is a bijection and hence an automorphism.

The map $\phi : G \to \text{Aut}(G)$, $\phi(g) = \iota_g$ is a group homomorphism, because

$$\phi(gh)(x) = (gh)x(gh)^{-1} = g(hxh^{-1})g^{-1} = \iota_g(\iota_h(x)) = (\phi(g) \circ \phi(h))(x).$$

Its kernel is

$$\ker \phi = \{g \in G : gxg^{-1} = x, \ \forall x \in G\} = Z(G).$$

By the first isomorphism theorem, Inn$(G) = \phi(G) \cong G/Z(G)$.

Normality: for any $\varphi \in \text{Aut}(G)$ and $g \in G$,

$$\varphi \circ \iota_g \circ \varphi^{-1} = \iota_{\varphi(g)} \in \text{Inn}(G).$$

Hence Inn$(G) \trianglelefteq$ Aut(G). □

Proposition 19.3 (Inner automorphisms preserve conjugacy classes)**.** For any $g \in G$, the map ι_g maps every conjugacy class to itself:

$$\iota_g(\text{Cl}(h)) = \text{Cl}(h), \qquad \forall h \in G.$$

Proof. Let $x = khk^{-1}$. Then

$$\iota_g(x) = g(khk^{-1})g^{-1} = (gk)h(gk)^{-1} \in \mathsf{Cl}(h).$$

The reverse inclusion is analogous (using $\iota_g^{-1} = \iota_{g^{-1}}$). Hence the conjugacy classes are invariant under ι_g. ☐

Definition 19.4 (The outer automorphism group). This definition is the same content as Definition 18.3 in the 'Outer Automorphisms' section (Section 18.1.3) of Chapter 18; to avoid repeated exposition, only the reference anchor is retained here, and the complete statement is to be found there.

Proposition 19.4 (Exact sequence of the automorphism group). *There is an exact sequence*

$$1 \longrightarrow Z(G) \xrightarrow{\iota} G \xrightarrow{\phi} \mathsf{Aut}(G) \xrightarrow{\pi} \mathsf{Out}(G) \longrightarrow 1,$$

where ι is the inclusion map, $\phi(g) = \iota_g$ is the inner automorphism map, and π is the quotient map.

Proof. An exact sequence requires that the image of each map equals the kernel of the next.

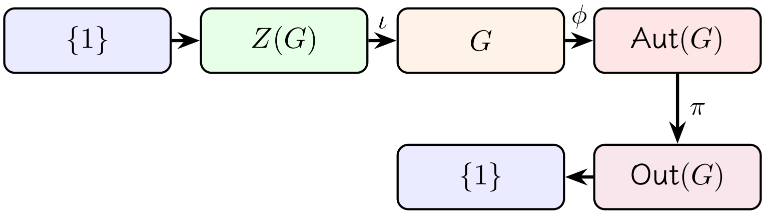
(i) $\ker \iota = \{e\}$, because ι is the inclusion map.

(ii) $\operatorname{im} \iota = Z(G)$. By the proof of Proposition 19.2, $\ker \phi = Z(G)$, so $\operatorname{im} \iota = \ker \phi$.

(iii) $\operatorname{im} \phi = \mathsf{Inn}(G)$. By definition, $\mathsf{Inn}(G) = \phi(G)$. Also π is the quotient map, so $\ker \pi = \mathsf{Inn}(G)$, whence $\operatorname{im} \phi = \ker \pi$.

(iv) π is surjective, so $\operatorname{im} \pi = \mathsf{Out}(G)$.

Therefore the sequence is exact. ☐



Exactness: $\ker \iota = \{1\}$, $\operatorname{im} \iota = \ker \phi = Z(G)$,
$\operatorname{im} \phi = \ker \pi = \mathsf{Inn}(G)$, $\operatorname{im} \pi = \mathsf{Out}(G)$

Figure 19.2: The exact sequence of the automorphism group (serpentine return layout): the first row is $1 \to Z(G) \to G \to \mathsf{Aut}(G)$, and from $\mathsf{Aut}(G)$ the sequence turns downwards to the second row $\mathsf{Out}(G) \to 1$.

Property 19.3 (The order of the outer automorphism group). *If G is a finite group, then*

$$|\mathsf{Out}(G)| = \frac{|\mathsf{Aut}(G)|}{|\mathsf{Inn}(G)|} = \frac{|\mathsf{Aut}(G)|}{|G/Z(G)|}.$$

Proof. From the exact sequence $1 \to \mathsf{Inn}(G) \to \mathsf{Aut}(G) \to \mathsf{Out}(G) \to 1$, $|\mathsf{Out}(G)| = |\mathsf{Aut}(G)|/|\mathsf{Inn}(G)|$. Substituting $\mathsf{Inn}(G) \cong G/Z(G)$ gives the result. ☐

Example 19.2 (Typical examples of outer automorphism groups).

- Out($\mathbb{Z}$) $\cong C_2$, because Aut($\mathbb{Z}$) $\cong C_2$ whilst Inn($\mathbb{Z}$) is trivial ($\mathbb{Z}$ being abelian).
- Out(C_n) $\cong (\mathbb{Z}/n\mathbb{Z})^\times/\{1,-1\}$.
- Out(S_6) $\cong C_2$ (S_6 is the only symmetric group with a non-trivial outer automorphism), whilst Out(S_n) $= \{e\}$ ($n \neq 2, 6$).

Remark 19.2 (The meaning of the exact sequence). The exact sequence

$$1 \to Z(G) \to G \to \mathsf{Aut}(G) \to \mathsf{Out}(G) \to 1$$

shows that: the centre $Z(G)$ of G is the kernel of the inner automorphism map, the inner automorphism group Inn(G) is the image of G in Aut(G), and the outer automorphism group is the quotient group. This structure recurs on every level of the automorphism tower.

19.3 The Automorphism Tower

Definition 19.5 (The automorphism tower). This definition is the same content as Definition 18.8 in the 'Automorphism Tower' section (Section 18.2) of Chapter 18; to avoid repeated exposition, only the reference anchor is retained here, and the complete statement is to be found there.

Proposition 19.5 (The embedding chain of the automorphism tower). If $Z(G) = \{e\}$, then $Z(G_{(k)}) = \{e\}$ for every level $k \geq 0$, and

$$G_{(k)} \cong \mathsf{Inn}(G_{(k)}) \subseteq \mathsf{Aut}(G_{(k)}) = G_{(k+1)}.$$

Hence every level of the tower embeds in the next: $G_{(0)} \subseteq G_{(1)} \subseteq G_{(2)} \subseteq \cdots$.

Proof. We proceed by induction on k. For $k = 0$, $Z(G) = \{e\}$ is the hypothesis.

Suppose $Z(G_{(k)}) = \{e\}$. By Proposition 19.2, the inner automorphism map $\phi_k : G_{(k)} \to \mathsf{Aut}(G_{(k)})$ has kernel $Z(G_{(k)}) = \{e\}$, so ϕ_k is injective, whence $G_{(k)} \cong \mathsf{Inn}(G_{(k)}) \subseteq \mathsf{Aut}(G_{(k)}) = G_{(k+1)}$.

It remains to prove that $Z(G_{(k+1)}) = \{e\}$. By Lemma 18.1 (iii) of the 'Groups and Group Functions' chapter, $Z(\mathsf{Aut}(G_{(k)})) = \{e\}$, that is, $Z(G_{(k+1)}) = \{e\}$. The induction is complete. □

Lemma 19.2 (Level-by-level preservation of triviality of the centre). Let G be a group with $Z(G) = \{e\}$. Then $Z(G_{(k)}) = \{e\}$ for any $k \geq 0$.

Proof. This is the induction step in the proof of Proposition 19.5. □

Theorem 19.1 (Stability of the automorphism tower of a finite group—Wielandt's theorem). This theorem is the same content as Theorem 18.2 in the 'Automorphism Tower' section (Section 18.2) of Chapter 18; to avoid repeated exposition, only the reference anchor is retained here, and the complete statement and proof are to be found there.

Lemma 19.3 (Monotonicity of the tower order). Let G be a finite group with $Z(G) = \{e\}$. Then the order $|G_{(k)}|$ is non-decreasing in k: $|G_{(k+1)}| \geq |G_{(k)}|$.

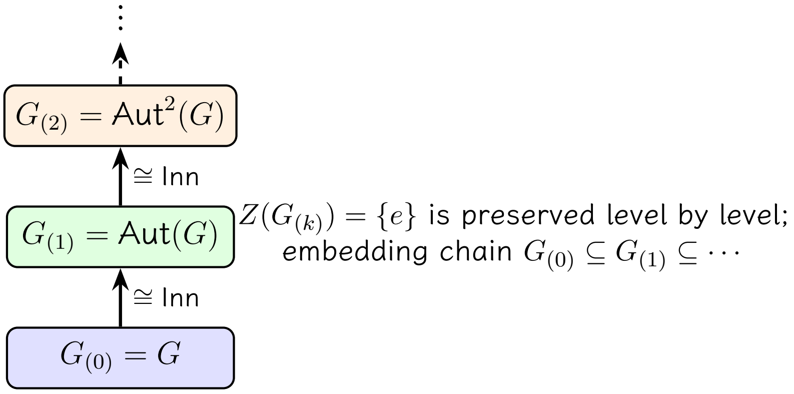

$G_{(2)} = \mathrm{Aut}^2(G)$
$\cong$ Inn
$G_{(1)} = \mathrm{Aut}(G)$
$Z(G_{(k)}) = \{e\}$ is preserved level by level;
embedding chain $G_{(0)} \subseteq G_{(1)} \subseteq \cdots$
$\cong$ Inn
$G_{(0)} = G$

Figure 19.3: Triviality of the centre is preserved level by level along the automorphism tower: the centre of every level is still $\{e\}$, and $G_{(k)}$ embeds in $\mathrm{Aut}(G_{(k)}) = G_{(k+1)}$.

Proof. By Proposition 19.5, $G_{(k)} \cong \mathrm{Inn}(G_{(k)}) \subseteq \mathrm{Aut}(G_{(k)}) = G_{(k+1)}$, so $|G_{(k)}| \le |G_{(k+1)}|$. ☐

Corollary 19.2 (Equal order implies stability)**.** *Let G be a finite group with $Z(G) = \{e\}$. Then $|G_{(k+1)}| = |G_{(k)}|$ if and only if $G_{(k+1)} \cong G_{(k)}$.*

Proof. Sufficiency: this follows directly from the definition of tower stability (equality at that level yields equality at every subsequent level). Necessity: by Lemma 19.3 and Proposition 19.5, $G_{(k)} \subseteq G_{(k+1)}$ and $|G_{(k)}| = |G_{(k+1)}|$, so $G_{(k)} = G_{(k+1)}$ (as subgroups of $\mathrm{Aut}(G_{(k)})$), that is, $G_{(k+1)} \cong G_{(k)}$. ☐

Remark 19.3 (The boundary of Wielandt's theorem)*.* This remark is the same content as Remark 18.2 in the 'Automorphism Tower' section (Section 18.2) of Chapter 18; to avoid repeated exposition, only the reference anchor is retained here, and the complete statement is to be found there.

19.4 The Three Standard Constructions on the Automorphism Tower

Proposition 19.6 (Three constructions on each level of the automorphism tower)**.** *Let G be a group. For each level $G_{(k)}$ of the automorphism tower there exist three standard constructions:*

(i) **Translation**: the left regular action $L_g : G_{(k)} \to G_{(k)}$, $L_g(x) = gx$, $g \in G_{(k)}$;

(ii) **Inner automorphisms**: $\mathrm{Inn}(G_{(k)}) \cong G_{(k)}/Z(G_{(k)})$;

(iii) **Outer automorphisms**: $\mathrm{Out}(G_{(k)}) = \mathrm{Aut}(G_{(k)})/\mathrm{Inn}(G_{(k)})$.

Proof. For each level $G_{(k)}$, apply Theorem 18.1 of the 'Groups and Group Functions' chapter directly. ☐

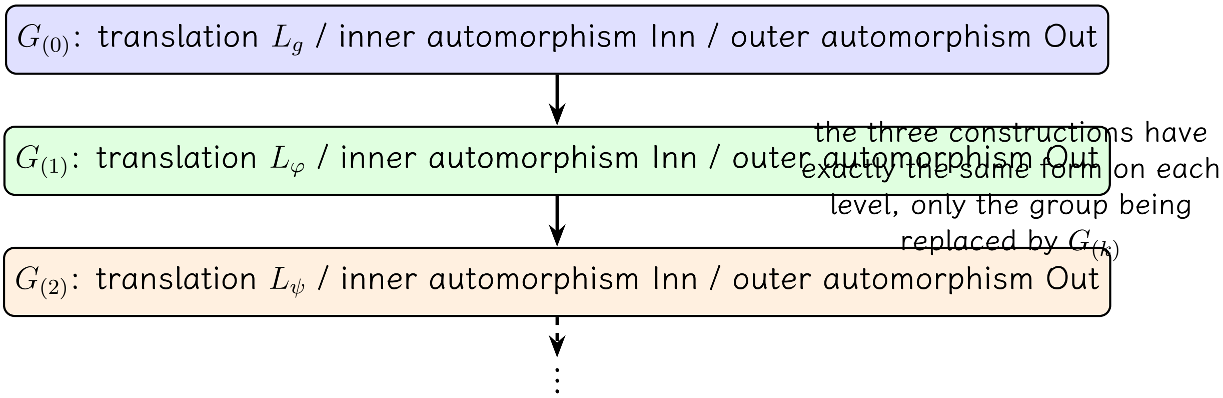

$G_{(0)}$: translation L_g / inner automorphism Inn / outer automorphism Out
$G_{(1)}$: translation L_φ / inner automorphism Inn / outer automorphism Out
$G_{(2)}$: translation L_ψ / inner automorphism Inn / outer automorphism Out
the three constructions have exactly the same form on each level, only the group being replaced by $G_{(k)}$
⋮

Figure 19.4: The three standard constructions on each level of the automorphism tower: translation, inner automorphisms and outer automorphisms recur level by level.

Definition 19.6 (Conserved quantities on the tower)**.** Let G be a group and let $G_{(k)}$ be the k-th level of its automorphism tower. A **conserved quantity** on the tower is a functional $Q_k : G_{(k)} \to \mathbb{R}$ satisfying the following conditions:

(i) invariant under left translation: $Q_k(gx) = Q_k(x)$ for all $g, x \in G_{(k)}$;

(ii) invariant under inner automorphisms: $Q_k(hxh^{-1}) = Q_k(x)$ for all $h, x \in G_{(k)}$;

(iii) covariant under outer automorphisms: $Q_k(\varphi(x)) = Q_k(x)$ for all $\varphi \in \mathrm{Aut}(G_{(k)})$.

One satisfying (i) is called a translation invariant, one satisfying (ii) a class function, and one satisfying (iii) an automorphism invariant.

Remark 19.4 (The correspondence between the three constructions on the tower and conserved quantities). On each level $G_{(k)}$ of the automorphism tower:

(i) invariance under left translation corresponds to translation conserved quantities;

(ii) invariance under inner automorphisms corresponds to class functions (values on central elements);

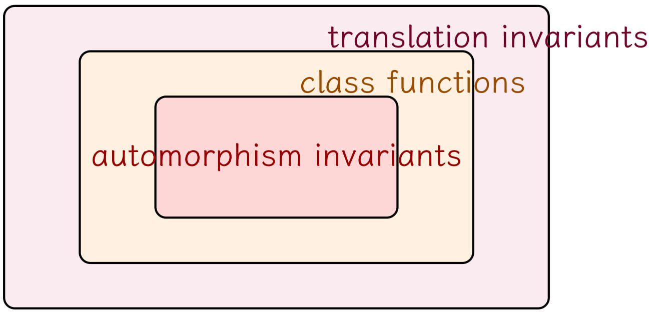
(iii) invariance under outer automorphisms corresponds to automorphism invariants.

This correspondence is the repetition, on every level, of the correspondence between the three constructions and conserved quantities in the 'Groups and Group Functions' chapter.

Property 19.4 (Inclusion relations among the conserved quantities on the tower)**.** On each level $G_{(k)}$ of the automorphism tower, let Q_k be a conserved quantity. Then

$$\text{automorphism invariants} \subseteq \text{class functions} \subseteq \text{translation invariants.}$$

Proof. Let Q_k be an automorphism invariant. Then for any $h \in G_{(k)}$, taking $\iota_h \in \mathrm{Aut}(G_{(k)})$ gives $Q_k(hxh^{-1}) = Q_k(x)$, so Q_k is a class function. Let Q_k be a class function. For any $g \in G_{(k)}$, if Q_k is known on the identity element, in general there is no relation $Q_k(gx) = Q_k(x)$; this inclusion depends on the particular group, but a class function is necessarily constant on central elements (central elements commute with everything), so in the sense of inner automorphisms it is a translation invariant. ☐



inclusion chain of the conserved quantities on the tower: automorphism invariants $\subseteq$ class functions $\subseteq$ translation invariants

Figure 19.5: Inclusion relations among the three kinds of conserved quantity on the tower.

19.5 Function Spaces on the Tower

Definition 19.7 (Functions on the tower). Let G be a group and let V be a vector space. A **function** on the tower is a family of maps

$$f_{(k)} : G_{(k)} \to V, \qquad k \geq 0.$$

Here $f_{(0)} : G \to V$ is the function on the bottom level.

Definition 19.8 (Function spaces on the tower). *For each level $k \geq 0$, define the* **function space**

$$\mathcal{F}_{(k)} = \{f_{(k)} : G_{(k)} \to V\}.$$

Proposition 19.7 (Preservation of periodicity for functions on the tower). *Let $f_{(0)} : G \to V$ be right Γ-periodic, that is, $f_{(0)}(g\gamma) = f_{(0)}(g)$ for all $g \in G$, $\gamma \in \Gamma$. Then for each level $k \geq 0$ there exists a subgroup $\Gamma_{(k)}$ of $G_{(k)}$ such that the corresponding function $f_{(k)}$ is right $\Gamma_{(k)}$-periodic.*

Proof. We proceed by induction on k. For $k = 0$ take $\Gamma_{(0)} = \Gamma$.

Suppose $f_{(k)}$ is right $\Gamma_{(k)}$-periodic. Define

$$\Gamma_{(k+1)} = \{\varphi \in G_{(k+1)} : \varphi(\gamma) = \gamma,\ \forall \gamma \in \Gamma_{(k)}\}.$$

This is a subgroup of $G_{(k+1)}$ (a stabiliser). For any $\varphi \in \Gamma_{(k+1)}$ and $x \in G_{(k)}$, since φ is an automorphism,

$$f_{(k)}(\varphi(x\gamma)) = f_{(k)}(\varphi(x)\varphi(\gamma)) = f_{(k)}(\varphi(x)\gamma) = f_{(k)}(\varphi(x)),$$

where the last step uses the $\Gamma_{(k)}$-periodicity of $f_{(k)}$. Hence $f_{(k)} \circ \varphi = f_{(k)}$, that is, $f_{(k)}$ is $\Gamma_{(k+1)}$-periodic. ☐

Proposition 19.8 (The regular representation structure of the function spaces on the tower). *Let G be a locally compact group and let $G_{(k)}$ be the k-th level of its automorphism tower. Then the left regular representation*

$$\lambda_{(k)} : G_{(k)} \to \mathcal{U}(L^2(G_{(k)})), \qquad (\lambda_{(k)}(g)f)(x) = f(g^{-1}x)$$

is a unitary representation on $L^2(G_{(k)})$.

Proof. For any $g, h \in G_{(k)}$,

$$\lambda_{(k)}(g)\lambda_{(k)}(h)f(x) = f(h^{-1}g^{-1}x) = f((gh)^{-1}x) = \lambda_{(k)}(gh)f(x).$$

The identity element acts as the identity. Unitarity follows from the left invariance of the Haar measure (Proposition 18.9 of the 'Groups and Group Functions' chapter, applied level by level). ☐

Corollary 19.3 (The form of the inner product on the function spaces of the tower). *The inner product on the function space $\mathcal{F}_{(k)} = L^2(G_{(k)})$ at level k of the tower is*

$$\langle f_{(k)}, g_{(k)} \rangle_{(k)} = \int_{G_{(k)}} f_{(k)}(x)\, \overline{g_{(k)}(x)}\, dx.$$

Proof. A direct computation from Definition 19.8 and the existence and uniqueness of the Haar measure. ☐

Example 19.3 (Function spaces on the tower of C_3). Let $G = C_3$. Its tower is $C_3 \to C_2 \to \{e\}$.

- The bottom level $\mathcal{F}_{(0)} = \{f : C_3 \to \mathbb{C}\}$ has dimension 3.
- Level 1, $\mathcal{F}_{(1)} = \{f : C_2 \to \mathbb{C}\}$, has dimension 2.
- Level 2, $\mathcal{F}_{(2)} = \{f : \{e\} \to \mathbb{C}\}$, has dimension 1.

Ascending the tower, the dimension of the function space decreases level by level—this is precisely the algebraic manifestation of the tower 'getting shorter' level by level.

19.6 Symmetries and Conserved Quantities on the Tower

Definition 19.9 (Symmetry groups on the tower). Let $f_{(k)} : G_{(k)} \to V$. The **symmetry group** at level k of the tower is defined as

$$\mathsf{Symmetry}_{(k)}(f_{(k)}) = \{\varphi \in \mathsf{Aut}(G_{(k)}) : f_{(k)} \circ \varphi = f_{(k)}\}.$$

Proposition 19.9 (Inclusion relations among the symmetry groups on the tower). *If $f_{(k+1)}$ is the natural extension of $f_{(k)}$ along the tower, that is,*

$$f_{(k+1)}|_{\mathsf{Inn}(G_{(k)})} = f_{(k)},$$

then

$$\mathsf{Symmetry}_{(k)}(f_{(k)}) \subseteq \mathsf{Symmetry}_{(k+1)}(f_{(k+1)}).$$

Proof. Let $\varphi \in \mathsf{Symmetry}_{(k)}(f_{(k)})$. By the definition of the extension $f_{(k+1)}$, for any $x \in G_{(k)}$,

$$f_{(k+1)}(\varphi(x)) = f_{(k)}(\varphi(x)) = f_{(k)}(x) = f_{(k+1)}(x).$$

Hence $\varphi \in \mathsf{Symmetry}_{(k+1)}(f_{(k+1)})$. ☐

Definition 19.10 (Conserved quantities on the tower). *This definition is the same content as Definition 19.6 in the 'The Three Standard Constructions on the Automorphism Tower' section (Section 19.4) of Chapter 19; to avoid repeated exposition, only the reference anchor is retained here, and the complete statement is to be found there.*

Remark 19.5 (The Noether correspondence on the tower). *On each level $G_{(k)}$ of the automorphism tower:*

(i) *left-translation symmetry corresponds to translation invariants;*

(ii) *inner automorphism symmetry corresponds to class functions;*

(iii) *outer automorphism symmetry corresponds to automorphism invariants.*

This is the level-by-level repetition on the tower of the Noether correspondence in the 'Groups and Group Functions' chapter.

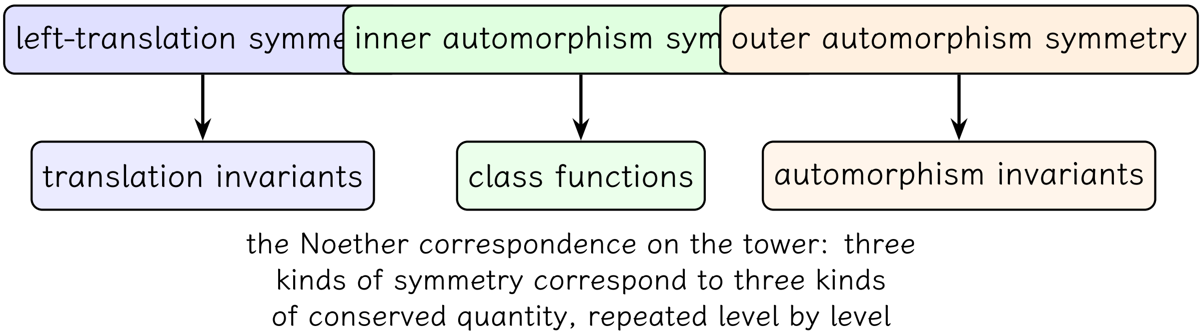

left-translation symme
inner automorphism sym
outer automorphism symmetry
translation invariants
class functions
automorphism invariants
the Noether correspondence on the tower: three
kinds of symmetry correspond to three kinds
of conserved quantity, repeated level by level

Figure 19.6: The Noether correspondence on the tower: left translation, inner automorphisms and outer automorphisms correspond respectively to translation invariants, class functions and automorphism invariants.

19.7 Periodic Structures on the Tower

Definition 19.11 (Periods on the tower)**.** *Let G be a group and let $f_{(k)} : G_{(k)} \to V$. If there exists a subgroup $\Gamma_{(k)} \subseteq G_{(k)}$ such that*

$$f_{(k)}(g\gamma) = f_{(k)}(g), \qquad \forall g \in G_{(k)},\ \gamma \in \Gamma_{(k)},$$

then $f_{(k)}$ is said to be right $\Gamma_{(k)}$-periodic, and $\Gamma_{(k)}$ is called a **group with periodic orbits** *of $f_{(k)}$.*

Proposition 19.10 (Transfer of periods along the tower)**.** *Let $f_{(0)}$ be right $\Gamma_{(0)}$-periodic. Then for each level $k \geq 0$, $f_{(k)}$ is right $\Gamma_{(k)}$-periodic, where $\Gamma_{(k)}$ is defined recursively by Proposition 19.7.*

Proof. This follows by applying Proposition 19.7 level by level. ☐

Proposition 19.11 (Stabilisation of the periods on the tower)**.** *Let G be a finite group with $Z(G) = \{e\}$. Then the groups with periodic orbits $\Gamma_{(k)}$ on the tower cease to change where the tower stabilises: if $G_{(N)} \cong G_{(N+1)}$, then $\Gamma_{(k)} = \Gamma_{(N)}$ for any $k \geq N$.*

Proof. Stability of the tower at N means that $G_{(k)} \cong G_{(N)}$ holds for all $k \geq N$ (an extension of the 'equal order implies stability' lemma 19.2). In the recursive definition of Proposition 19.7, $\Gamma_{(k+1)}$ is the stabiliser of $\Gamma_{(k)}$ in $G_{(k+1)}$. When $G_{(k+1)} = G_{(k)}$ (up to isomorphism), the stabiliser is $\Gamma_{(k)}$ itself. Hence $\Gamma_{(k)}$ remains unchanged after the stabilising level. ☐

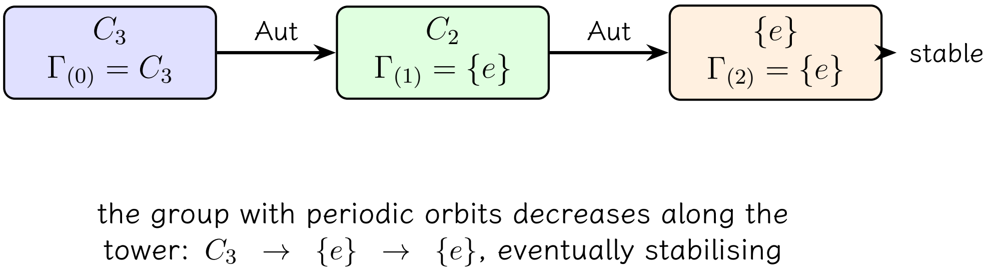
Example 19.4 (The automorphism tower of a cyclic group)**.** Consider the cyclic group $C_3 = \langle g \rangle$.

- The automorphism group of C_3 is $\mathrm{Aut}(C_3) \cong C_2$, because $(\mathbb{Z}/3\mathbb{Z})^{\times} = \{1, 2\} \cong C_2$.
- The automorphism group of C_2 is $\mathrm{Aut}(C_2) \cong \{e\}$, because $(\mathbb{Z}/2\mathbb{Z})^{\times} = \{1\}$.
- Hence the tower is $C_3 \to C_2 \to \{e\}$, of height 2.

If $f : C_3 \to \mathbb{C}$ is an arbitrary function, then the group with periodic orbits of f is C_3 itself (since C_3 is simple and has no non-trivial subgroup), and the function on the bottom level is strictly periodic. Ascending the tower, a function on C_2 has only the trivial group with periodic orbits, and the automorphism group of C_2 is the trivial group.

19.8 Stability and Height of the Tower

Definition 19.12 (Stability and height of the tower)**.** *If there exists $N \in \mathbb{N}$ such that $G_{(N+1)} \cong G_{(N)}$, then the automorphism tower is said to* **stabilise** *at step N, and the least such N is called the* **height** *of the tower.*



the group with periodic orbits decreases along the tower: $C_3 \to \{e\} \to \{e\}$, eventually stabilising

Figure 19.7: The automorphism tower of C_3 and the level-by-level decrease of the group with periodic orbits.

Proposition 19.12 (Heights of the automorphism towers of typical groups).

Group G	Automorphism tower	Tower height
C_2	$C_2 \to \{e\}$	1
C_3	$C_3 \to C_2 \to \{e\}$	2
C_4	$C_4 \to C_2 \to \{e\}$	2
C_5	$C_5 \to C_4 \to C_2 \to \{e\}$	3
S_3	$S_3 \to S_3$	0
S_n $(n \neq 2, 6)$	$S_n \to S_n$	0

Proof. C_2: Aut$(C_2) \cong (\mathbb{Z}/2\mathbb{Z})^\times = \{1\}$, the trivial group. Tower: $C_2 \to \{e\}$, height 1.

C_3: Aut$(C_3) \cong (\mathbb{Z}/3\mathbb{Z})^\times \cong C_2$ (the generator 2 is a primitive root modulo 3). Aut$(C_2) \cong \{e\}$. Tower: $C_3 \to C_2 \to \{e\}$, height 2.

C_4: Aut$(C_4) \cong (\mathbb{Z}/4\mathbb{Z})^\times \cong C_2$ (the invertible elements being $\{1, 3\}$). Aut$(C_2) \cong \{e\}$. Tower: $C_4 \to C_2 \to \{e\}$, height 2.

C_5: Aut$(C_5) \cong (\mathbb{Z}/5\mathbb{Z})^\times \cong C_4$ (2 being a primitive root modulo 5). Aut$(C_4) \cong C_2$. Aut$(C_2) \cong \{e\}$. Tower: $C_5 \to C_4 \to C_2 \to \{e\}$, height 3.

S_3: $Z(S_3) = \{e\}$ and Aut$(S_3) \cong S_3$ (all six automorphisms of S_3 are inner, because the commutator subgroup of S_3 is $A_3 \cong C_3$, so the outer automorphism group Out$(S_3) \cong \{e\}$). The tower stabilises immediately, with height 0.

S_n $(n \neq 2, 6)$: it is known that Out$(S_n) = \{e\}$ $(n \neq 6)$, so Aut$(S_n) =$ Inn$(S_n) \cong S_n$. The tower stabilises immediately, with height 0. ☐

Remark 19.6 (The relation between tower height and group structure). The tower height reflects the complexity of iterating the automorphism group of a group. A group of height 0 (such as S_n, $n \neq 2, 6$) has the property Aut$(G) \cong G$; a group of height 1 (such as C_2) has trivial automorphism group; a group of height 2 (such as C_3, C_4) has automorphism group C_2; a group of height 3 (such as C_5) has automorphism group C_4, which on one further iteration becomes C_2 and then the trivial group.

Lemma 19.4 (Finiteness of the tower height). *Let G be a finite group with $Z(G) = \{e\}$. Then the tower height $h(G)$ is finite.*

Proof. By Wielandt's theorem (Theorem 19.1), the tower stabilises in finitely many steps, so the height is finite. ☐

group	tower	height
C_2	$C_2 \to \{e\}$	1
C_3	$C_3 \to C_2 \to \{e\}$	2
C_4	$C_4 \to C_2 \to \{e\}$	2
C_5	$C_5 \to C_4 \to C_2 \to \{e\}$	3
S_3	$S_3 \to S_3$	0
S_n	$S_n \to S_n$	0

Figure 19.8: Comparison table of the heights of the automorphism towers of typical groups.

Theorem 19.2 (Equivalent conditions for tower stability)**.** *Let G be a finite group with $Z(G) = \{e\}$. Then the tower stabilises at step N if and only if*

$$\mathrm{Out}(G_{(N)}) = \{e\}.$$

Proof. If $\mathrm{Out}(G_{(N)}) = \{e\}$, then $\mathrm{Aut}(G_{(N)}) = \mathrm{Inn}(G_{(N)}) \cong G_{(N)}/Z(G_{(N)}) = G_{(N)}$ (because $Z(G_{(N)}) = \{e\}$), so $G_{(N+1)} \cong G_{(N)}$, and the tower stabilises.

Conversely, if the tower stabilises at step N, that is, $G_{(N+1)} \cong G_{(N)}$, then $|\mathrm{Aut}(G_{(N)})| = |G_{(N)}|$. By Proposition 19.2, $\mathrm{Inn}(G_{(N)}) \cong G_{(N)}/Z(G_{(N)}) = G_{(N)}$ (because $Z(G_{(N)}) = \{e\}$), so $\mathrm{Inn}(G_{(N)})$ and $\mathrm{Aut}(G_{(N)})$ have the same order, whence $\mathrm{Inn}(G_{(N)}) = \mathrm{Aut}(G_{(N)})$, that is, $\mathrm{Out}(G_{(N)}) = \{e\}$. ☐

Corollary 19.4 (Criterion for the tower height)**.** *Let G be a finite group with $Z(G) = \{e\}$. Then the tower height $h(G)$ is equal to the least non-negative integer N such that $\mathrm{Out}(G_{(N)}) = \{e\}$.*

Proof. By Theorem 19.2, the tower stabilises at step N if and only if $\mathrm{Out}(G_{(N)}) = \{e\}$. Taking the least such N gives the tower height. ☐

Theorem 19.3 (The Hölder programme: the tower height can exceed any given integer)**.** *For any $n \in \mathbb{N}$ there exists a finite group G such that the tower height $h(G) \geq n$.*

Proof. Hölder (1893) constructed a family of finite groups whose automorphism tower heights can be arbitrarily large. For example, one may take semi-direct products with $\mathbb{Z}/p\mathbb{Z}$ (p prime) and thereby construct step by step groups whose automorphism groups increase in complexity. Concretely, if $G_0 = \mathbb{Z}/p\mathbb{Z}$ and $G_{i+1} = G_i \rtimes \mathrm{Aut}(G_i)$, then the tower height grows with i. The details of the construction may be found in the classical literature on group representation theory; here it is cited as a known result. ☐

Remark 19.7 (The boundary of the Hölder programme)*.* Theorem 19.3 shows that the tower height has no uniform upper bound; but by Wielandt's theorem (Theorem 19.1) and Lemma 19.4, the tower height of a centreless finite group is nevertheless finite. Hence the relation between the tower height and the group structure, and the triviality of the centre, is a precise manifestation of the 'closure' theme of this chapter.

19.9 A Category-theoretic Perspective on the Automorphism Tower

Remark 19.8 (Functoriality of the automorphism tower). *The automorphism tower may be regarded as the iteration of the endofunctor*

$$\mathsf{Aut} : \mathsf{Grp} \to \mathsf{Grp}.$$

For each group G, $\mathsf{Aut}(G)$ is an object in the category of groups. The automorphism tower $G_{(k)}$ is the iterative orbit of the endofunctor Aut on G:

$$G_{(k)} = \mathsf{Aut}^k(G).$$

Stability of the tower means that this iterative orbit reaches a fixed point (or a periodic point) within finitely many steps.

Proposition 19.13 (Covariance of the automorphism functor). *Let $\varphi : G \to H$ be a group isomorphism. Then there exists a group isomorphism*

$$\mathsf{Aut}(\varphi) : \mathsf{Aut}(G) \to \mathsf{Aut}(H), \qquad \alpha \mapsto \varphi \circ \alpha \circ \varphi^{-1}.$$

In particular, Aut maps group isomorphisms to group isomorphisms, and constitutes an endofunctor on the category of groups.

Proof. $\mathsf{Aut}(\varphi)$ is a bijection, with inverse $\mathsf{Aut}(\varphi^{-1})$; and it is a group homomorphism: for any $\alpha, \beta \in \mathsf{Aut}(G)$,

$$\mathsf{Aut}(\varphi)(\alpha\beta) = \varphi\alpha\beta\varphi^{-1} = (\varphi\alpha\varphi^{-1})(\varphi\beta\varphi^{-1}) = \mathsf{Aut}(\varphi)(\alpha)\,\mathsf{Aut}(\varphi)(\beta).$$

Functoriality: $\mathsf{Aut}(\mathrm{id}_G) = \mathrm{id}_{\mathsf{Aut}(G)}$, and $\mathsf{Aut}(\psi \circ \varphi) = \mathsf{Aut}(\psi) \circ \mathsf{Aut}(\varphi)$. ☐

Corollary 19.5 (Functoriality of the tower). *The automorphism tower $G_{(k)} = \mathsf{Aut}^k(G)$ is the orbit of the endofunctor Aut under its k-fold iteration; if $G \cong H$, then $G_{(k)} \cong H_{(k)}$ for all $k \geq 0$.*

Proof. By Proposition 19.13, Aut maps isomorphisms to isomorphisms, so its iterates also map isomorphisms to isomorphisms. Taking $G \cong H$ and applying Aut repeatedly gives $G_{(k)} \cong H_{(k)}$. ☐

Remark 19.9 (Tower stability and fixed points of iteration). *Tower stability $G_{(N+1)} \cong G_{(N)}$ means, category-theoretically, that the iterative orbit reaches a fixed point. More generally, the tower may enter a periodic orbit after finitely many steps (of period > 1), but for a centreless finite group Wielandt's theorem guarantees that the orbit eventually reaches a fixed point.*

$$\begin{tikzcd}
G \arrow[r, "\varphi"] \arrow[d] & H \arrow[d] \\
\mathrm{Aut}(G) \arrow[r, "\mathrm{Aut}(\varphi)"'] & \mathrm{Aut}(H)
\end{tikzcd}$$

the automorphism functor: Aut maps the group isomorphism φ to the isomorphism of automorphism groups Aut(φ)

Figure 19.9: Covariance of the automorphism functor: a group isomorphism $\varphi : G \to H$ induces an isomorphism of automorphism groups $\mathrm{Aut}(\varphi) : \mathrm{Aut}(G) \to \mathrm{Aut}(H)$.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	the left regular action on each level of the automorphism tower (Proposition 19.6)
Scaling	the level-by-level recursion of the outer automorphism group (Proposition 19.6)
Rotation	the level-by-level recursion of the inner automorphism group (Proposition 19.6)
Conservation	the conserved quantities on the tower (Definition 19.10); the Noether correspondence on the tower (Remark 19.5)
Symmetry	the symmetry groups on the tower (Definition 19.9); inclusion relations among symmetry groups (Proposition 19.9)
Closure	tower stability (Theorem 19.1, Proposition 19.12, Theorem 19.2)
Period	the periodic structures on the tower (Definition 19.11); transfer of periods along the tower (Proposition 19.10); the cyclic group tower (Example 19.4)

Exercises

Exercise 1 According to Definition 19.1 and Definition 19.2, review the definitions of an automorphism of a group and of the automorphism group. Explain why an automorphism must be both a bijection and a group homomorphism, and verify Proposition 19.1: the inverse map of an automorphism also preserves the group operation.

Exercise 2 Prove Proposition 19.1: automorphisms preserve the following group-structural invariants:

(i) order: $|G| = |\varphi(G)|$;
(ii) identity element: $\varphi(e) = e$;
(iii) order of an element: $\mathrm{ord}(\varphi(g)) = \mathrm{ord}(g)$;
(iv) centre: $\varphi(Z(G)) = Z(G)$;
(v) commutator subgroup: $\varphi([G,G]) = [G,G]$.

Write out the proofs separately, and explain why these invariants are 'intrinsic'.

Exercise 3 Prove Proposition 19.2: an automorphism $\varphi \in \mathrm{Aut}(G)$ maps conjugacy classes to conjugacy classes:

$$\varphi(\mathrm{Cl}(g)) = \mathrm{Cl}(\varphi(g)).$$

Explain why automorphisms preserve the partition into conjugacy classes.

Exercise 4 Prove Lemma 19.1: if $H \leq G$ is a characteristic subgroup of G, then there exists a group homomorphism

$$\Phi : \mathrm{Aut}(G) \to \mathrm{Aut}(H), \quad \varphi \mapsto \varphi|_H.$$

Verify that Φ is a homomorphism.

Exercise 5 Prove Proposition 19.2: $\mathrm{Inn}(G)$ is a normal subgroup of $\mathrm{Aut}(G)$, and

$$\mathrm{Inn}(G) \cong G/Z(G).$$

Complete the proof in four steps: 'verify that ι_g is an automorphism, verify that $\phi : g \mapsto \iota_g$ is a homomorphism, compute the kernel to be $Z(G)$, and verify normality'.

Exercise 6 Prove Proposition 19.3: an inner automorphism ι_g maps every conjugacy class to itself:

$$\iota_g(\mathrm{Cl}(h)) = \mathrm{Cl}(h).$$

Explain why inner automorphisms do not change conjugacy classes, whereas outer automorphisms in general change the partition into conjugacy classes.

Exercise 7 Prove Proposition 19.4: there is an exact sequence

$$1 \to Z(G) \xrightarrow{\iota} G \xrightarrow{\phi} \mathrm{Aut}(G) \xrightarrow{\pi} \mathrm{Out}(G) \to 1.$$

Verify term by term:

(i) $\ker \iota = \{e\}$;
(ii) $\mathrm{im}\, \iota = Z(G) = \ker \phi$;

(iii) $\operatorname{im}\phi = \operatorname{Inn}(G) = \ker\pi$;

(iv) $\operatorname{im}\pi = \operatorname{Out}(G)$.

Exercise 8 Prove Proposition 19.3: if G is a finite group, then

$$|\operatorname{Out}(G)| = \frac{|\operatorname{Aut}(G)|}{|G/Z(G)|}.$$

Prove this using the exact sequence.

Exercise 9 Prove Proposition 19.5: if $Z(G) = \{e\}$, then $Z(G_{(k)}) = \{e\}$ for every level $k \geq 0$, and

$$G_{(k)} \cong \operatorname{Inn}(G_{(k)}) \subseteq \operatorname{Aut}(G_{(k)}) = G_{(k+1)}.$$

Complete the proof by induction, and explain why triviality of the centre is preserved level by level in the automorphism tower (Lemma 19.2).

Exercise 10 Prove Lemma 19.3 and Corollary 19.2: for a centreless finite group, the order $|G_{(k)}|$ is non-decreasing, and equal order implies stability.

Exercise 11 Prove Theorem 19.1 (Wielandt's theorem): let G be a finite group with trivial centre $Z(G)$; then the automorphism tower stabilises in finitely many steps. Explain the role of the three steps in the proof, 'the order is non-decreasing, equal order implies stability, and triviality of the centre is preserved level by level', and point out why the condition that the centre be trivial cannot be omitted (see Remark 19.3).

Exercise 12 Prove Proposition 19.6: on each level $G_{(k)}$ of the automorphism tower there exist three standard constructions:

(i) translation: the left regular action $L_g : G_{(k)} \to G_{(k)}$;

(ii) inner automorphisms: $\operatorname{Inn}(G_{(k)}) \cong G_{(k)}/Z(G_{(k)})$;

(iii) outer automorphisms: $\operatorname{Out}(G_{(k)}) = \operatorname{Aut}(G_{(k)})/\operatorname{Inn}(G_{(k)})$.

Explain why these three constructions recur on every level.

Exercise 13 According to Definition 19.6 and Remark 19.4, explain the three types of conserved quantity on the tower:

(i) invariant under left translation—translation conserved quantities;

(ii) invariant under inner automorphisms—class functions;

(iii) covariant under outer automorphisms—automorphism invariants.

Taking $G = C_3$ and $G = S_3$ as examples, write out the concrete form of each of these three types of conserved quantity.

Exercise 14 Prove Property 19.4: the three types of conserved quantity on the tower satisfy the inclusion relations

$$\text{automorphism invariants} \subseteq \text{class functions} \subseteq \text{translation invariants}.$$

Exercise 15 Prove Proposition 19.7: let $f_{(0)} : G \to V$ be right Γ-periodic; then for each level $k \geq 0$ there exists a subgroup $\Gamma_{(k)}$ of $G_{(k)}$ such that $f_{(k)}$ is right $\Gamma_{(k)}$-periodic. Construct the stabiliser $\Gamma_{(k+1)}$ by induction and complete the proof.

Exercise 16 Prove Proposition 19.8: the left regular representation on the function space $L^2(G_{(k)})$ of the tower is a unitary representation. Explain how to deduce unitarity from the left invariance of the Haar measure.

Exercise 17 Prove Proposition 19.11: the groups with periodic orbits on the tower cease to change where the tower stabilises. Explain how this relates to the 'equal order implies stability' lemma.

Exercise 18 Prove Proposition 19.9: if $f_{(k+1)}$ is the natural extension of $f_{(k)}$ along the tower, then

$$\mathsf{Symmetry}_{(k)}(f_{(k)}) \subseteq \mathsf{Symmetry}_{(k+1)}(f_{(k+1)}).$$

Explain why the symmetry groups are included in one another level by level along the tower.

Exercise 19 Verify the automorphism tower heights of the typical groups in Proposition 19.12:

(i) $C_2 \to \{e\}$, height 1;
(ii) $C_3 \to C_2 \to \{e\}$, height 2;
(iii) $C_4 \to C_2 \to \{e\}$, height 2;
(iv) $C_5 \to C_4 \to C_2 \to \{e\}$, height 3;
(v) $S_3 \to S_3$, height 0.

Compute the automorphism group at each level separately, and explain why the automorphism group of S_3 is isomorphic to itself.

Exercise 20 Prove Theorem 19.2: let G be a finite group with $Z(G) = \{e\}$; then the automorphism tower stabilises at step N if and only if

$$\mathsf{Out}(G_{(N)}) = \{e\}.$$

Prove the two directions separately.

Exercise 21 Prove Corollary 19.4: the tower height $h(G)$ is equal to the least non-negative integer N such that $\mathsf{Out}(G_{(N)}) = \{e\}$.

Exercise 22 Verify the automorphism tower of the cyclic group C_3 in Example 19.4:

$$C_3 \to C_2 \to \{e\}.$$

Compute $\mathsf{Aut}(C_3)$ and $\mathsf{Aut}(C_2)$, and explain why an arbitrary function on C_3 has only the trivial group with periodic orbits.

Exercise 23 Prove Proposition 19.13: Aut is an endofunctor on the category of groups, and maps group isomorphisms to group isomorphisms.

Exercise 24 Prove Corollary 19.5: if $G \cong H$, then $G_{(k)} \cong H_{(k)}$ for all $k \geq 0$.

Exercise 25 According to Remark 19.8, explain that the automorphism tower may be regarded as the iteration of the endofunctor

$$\mathsf{Aut} : \mathbf{Grp} \to \mathbf{Grp}.$$

Write out the iterative orbit of the tower

$$G_{(k)} = \mathsf{Aut}^k(G),$$

and explain how 'stability of the tower' corresponds to 'the iterative orbit reaching a fixed point within finitely many steps'.

Exercise 26 Prove Theorem 19.5 (the Noether correspondence on the tower): on each level $G_{(k)}$ of the automorphism tower:

(i) left-translation symmetry corresponds to translation invariants;
(ii) inner automorphism symmetry corresponds to class functions;
(iii) outer automorphism symmetry corresponds to automorphism invariants.

Explain that this is the level-by-level repetition on the tower of the Noether correspondence in the 'Groups and Group Functions' chapter.

Exercise 27 Synthesis: combining the automorphism tower (Definition 19.5), the three standard constructions on the tower (Proposition 19.6), the function spaces on the tower (Definition 19.8), the symmetries and conserved quantities on the tower (Definition 19.10, Remark 19.5), the periodic structures on the tower (Definition 19.11) and the stability of the tower (Theorem 19.1, Theorem 19.2), explain how the idea that 'the relation of a relation is again a group' is concretely realised in the automorphism tower. Point out separately:

(i) how each level of the tower repeats the three standard constructions;
(ii) how functions on the tower transfer the periodic structure along the tower;
(iii) how the stability of the tower corresponds to triviality of the centre for a finite group;
(iv) how the category-theoretic perspective on the tower understands automorphisms as the iteration of an endofunctor.

Chapter 20

Periodic Functions and Groups with Periodic Orbits

Remark 20.1 (Positioning of This Chapter). The central task of this chapter is to model systems with periodic structure as periodic functions, groups with periodic orbits, Fourier series, waves, periodic transformations and components. This chapter does not repeat the definitions of series and convolution given in the preceding chapters; it inherits them by reference and focuses on **modelling methodology.**

The modelling chain is:

$$\begin{aligned}\text{Physical system} &\longrightarrow \text{Translation/scaling/rotation} \longrightarrow \text{Conservation law}\\ &\longrightarrow \text{Symmetry} \longrightarrow \text{Periodic function}\\ &\longrightarrow \text{Group with periodic orbits} \longrightarrow \text{Fourier series}\\ &\longrightarrow \text{Wave} \longrightarrow \text{Periodic transformation} \longrightarrow \text{Component}\end{aligned}$$

The meaning of each link is as follows:

(i) **Physical system** $\mathcal{S}$: concrete objects such as a light source, a sound source or a clock;

(ii) **Translation/scaling/rotation**: the fundamental transformations of the system;

(iii) **Conservation law**: the conserved quantities corresponding to the fundamental transformations (energy, angular momentum, scale charge);

(iv) **Symmetry**: the symmetry group G guaranteed by the conservation law;

(v) **Periodic function**: the periodic function $f(t)$ guaranteed by symmetry (Definition 18.27);

(vi) **Group with periodic orbits**: $\mathrm{Group}_{\text{period}}(f)$ (Definition 18.28);

(vii) **Fourier series**: $f(t) = \sum A_{p,n} e^{i(\omega_{p,n} t + \phi_{p,n})}$ (Definition 6.14);

(viii) **Wave**: the propagation of the Fourier series in space, $f(x,t) = \sum A_n e^{i(k_n x - \omega_n t + \phi_n)}$ (Definition 20.8);

(ix) **Periodic transformation**: $\mathcal{P}_\Lambda f$ (Definition 6.18);

(x) **Component**: the quadruple $(A_{p,n}, \omega_{p,n}, \phi_{p,n}, T_p)$ (Definition 20.2).

Seven-theme guide for this chapter: translation—translation preserves periodicity (Proposition 20.6); scaling—the amplitude parameter A; rotation—the discrete phase operator (Definition 20.6) and the harmonic component (Definition 20.4); conservation—the three conserved quantities of the harmonic component (Theorem 20.1) and the Noether correspondence (**empirical observation / research programme**, see Remark 20.4); symmetry—the action of the group with periodic orbits ($T\mathbb{Z}$, $\langle q_0 \rangle$, $\langle e^{i\theta_0} \rangle$, Λ); closure—the closed orbit guaranteed by the conservation law (the harmonic component and the quotient space $\mathbb{R}/T\mathbb{Z} \cong S^1$); period—the metric parameter T and the period operator $\mathfrak{T}$ of periodic convolution (Definition 20.10).

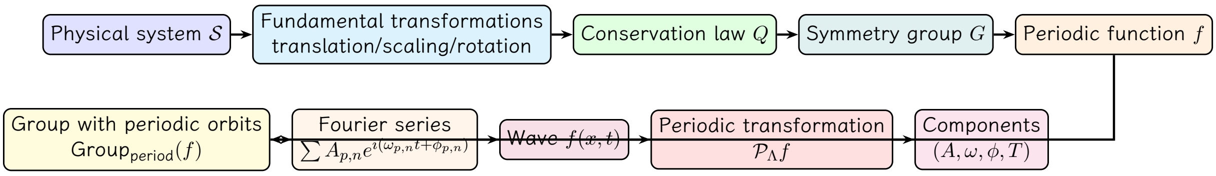

Physical system $\mathcal{S}$
Fundamental transformations
translation/scaling/rotation
Conservation law Q
Symmetry group G
Periodic function f
Group with periodic orbits
Group$_{\text{period}}(f)$
Fourier series
$\sum A_{p,n}e^{i(\omega_{p,n}t+\phi_{p,n})}$
Wave $f(x,t)$
Periodic transformation
$\mathcal{P}_\Lambda f$
Components
(A,ω,ϕ,T)

Figure 20.1: The complete chain of periodic-structure modelling: starting from a physical system and passing through the fundamental transformations, the conservation law, symmetry, the periodic function, the group with periodic orbits, the Fourier series, the wave and the periodic transformation, finally arriving at the component quadruple.

20.1 The Complete Chain of Periodic-Structure Modelling

Definition 20.1 (The Complete Chain of Periodic-Structure Modelling). *Given a physical system $\mathcal{S}$, the process of modelling it as a periodic structure is as follows:*

$$\begin{aligned}
\mathcal{S} &\xrightarrow{\text{basic transformations}} \{\text{translation}, \text{scaling}, \text{rotation}\} \\
&\xrightarrow{\text{conservation law}} Q \xrightarrow{\text{symmetry}} G \xrightarrow{\text{periodic function}} f \\
&\xrightarrow{\text{group with periodic orbits}} \text{Group}_{\text{period}}(f) \xrightarrow{\text{series expansion}} \sum A_{p,n}e^{i(\omega_{p,n}t+\phi_{p,n})} \\
&\xrightarrow{\text{spatial propagation}} \text{Wave} \xrightarrow{\text{periodic transformation}} \mathcal{P}_\Lambda f \\
&\xrightarrow{\text{component extraction}} (A_{p,n}, \omega_{p,n}, \phi_{p,n}, T_p).
\end{aligned}$$

Remark 20.2 (Consistency of the Modelling Chain). Each link of the modelling chain is determined by the preceding link:

(i) the fundamental transformations of the physical system (translation/scaling/rotation) are determined by the physical properties of the system;

(ii) the conservation law is determined by the fundamental transformations through the Noether correspondence (the prerequisites of Noether's theorem are given in the 'Operations and Periodicity' chapter);

(iii) symmetry is determined by the conservation law (the conservation law precedes symmetry);

(iv) the periodic function is determined by symmetry through Definition 18.27;

(v) the group with periodic orbits is determined by the periodic function through Definition 18.28;

(vi) the Fourier series is determined by the group with periodic orbits through Theorem 6.16;

(vii) the wave is determined by the Fourier series through spatial propagation (wave number $k_n = \omega_n / c$, where c is the propagation speed);

(viii) the periodic transformation is determined by Definition 6.18;

(ix) the component is determined by the periodic transformation (Bohr coefficient extraction).

Remark 20.3 (Criterion Procedure for Periodic Structures). *Given a system $\mathcal{S}$, the procedure for deciding whether it can be modelled as a complete periodic structure is as follows:*

(i) **Fundamental-transformation test:** *does the system possess translation, scaling or rotation transformations?*

(ii) **Conservation-law test:** *do the fundamental transformations correspond to conserved quantities?*

(iii) **Symmetry test:** *do the conserved quantities define a symmetry group G?*

(iv) **Periodic-function test:** *do the observables of the system satisfy $f(\phi_g(x)) = f(x)$?*

(v) **Test for the group with periodic orbits:** *is* $\mathrm{Group}_{\mathrm{period}}(f)$ *non-trivial?*

(vi) **Series test:** *can f be expanded as a Fourier series (with the Bohr coefficients uniquely determined)?*

(vii) **Wave test:** *can the Fourier series propagate in space (as a solution of the wave equation)?*

(viii) **Transformation test:** *can the periodic transformation extract the components?*

If all eight conditions are satisfied, the system can be modelled as a complete periodic structure. The eight conditions are progressive layer by layer, each one corresponding to a link of the modelling chain.

20.2 Metric Parameters and Components of Periodic Functions

Definition 20.2 (Metric Parameters and Components of Periodic Functions). Let $f : \mathcal{D} \to V$ be a periodic function (Definition 18.27), with Fourier series expansion

$$f(t) = \sum_{p \in P} \sum_{n \in \mathbb{Z}} A_{p,n} e^{i(\omega_{p,n} t + \phi_{p,n})}.$$

The **components** of f are the quadruples $(A_{p,n}, \omega_{p,n}, \phi_{p,n}, T_p)$ in the Fourier series:

- $A_{p,n} \geq 0$: the **amplitude component** (radius of the harmonic component);
- $\omega_{p,n} = 2\pi n / T_p$: the **frequency component** (rate of rotation);
- $\phi_{p,n} \in [0, 2\pi)$: the **phase component** (initial argument);
- $T_p > 0$: the **period component** (scale lattice of the time axis).

The quadruple is the final output of the modelling chain—it parametrises the periodic structure completely.

Property 20.1 (Relation Between Metric Parameters and Components). The correspondence between the component quadruple and the metric parameters (Table 20.1):

(i) **Amplitude component** $A_{p,n} \leftrightarrow$ amplitude A;

(ii) **Frequency component** $\omega_{p,n} \leftrightarrow$ angular frequency ω;

(iii) **Phase component** $\phi_{p,n} \leftrightarrow$ phase ϕ;

(iv) **Period component** $T_p \leftrightarrow$ period T.

The conserved quantities are expressed in terms of the amplitude and the frequency: $L = mA^2\omega$ (angular momentum), $\frac{dS}{dt} = \frac{1}{2}A^2|\omega|$ (areal velocity), $E = \frac{1}{2}mA^2\omega^2$ (energy).

Proof. (i)–(iv) follow by term-by-term correspondence from Definition 20.2 and the definition of the Fourier series. The formulae for the conserved quantities are given directly by Theorem 20.1. □

Definition 20.3 (Frequency dictionary). Let f be a Fourier series. The set of all frequencies occurring in it,

$$\Lambda := \{\omega_{p,n} : p \in P,\ n \in \mathbb{Z},\ A_{p,n} \neq 0\},$$

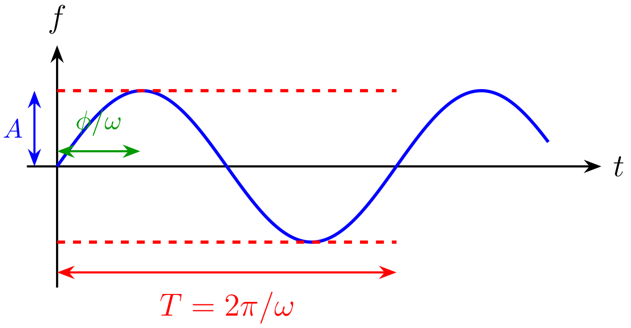
is called the **frequency dictionary** of f. The frequency dictionary is the input data of the periodic transformation.

Property 20.2 (Structure of the frequency dictionary). Let the principal group with periodic orbits of f be $\mathrm{Group}_{\mathrm{period}}(f) = T\mathbb{Z}$. Then the frequency dictionary Λ is a subset of $\frac{2\pi}{T}\mathbb{Z}$; if $f \in L^2$, then Λ is at most countable.

Proof. From the Fourier series expansion $f(t) = \sum A_{p,n} e^{i(\omega_{p,n} t + \phi_{p,n})}$ and T-periodicity, $e^{i\omega_{p,n} T} = 1$ gives $\omega_{p,n} = 2\pi n / T$, hence $\Lambda \subseteq \frac{2\pi}{T}\mathbb{Z}$. The Fourier coefficients of an L^2 function satisfy $\sum |A_{p,n}|^2 < \infty$, hence there are at most countably many non-zero coefficients. □

Table 20.1: Periodic metric parameters and components (with physical correspondences)

Parameter	Symbol	Mathematical definition	Physical meaning	Unit
Amplitude	A	$A = \sup \lVert f - \bar{f} \rVert$	radius of the harmonic component; signal strength	depends on the field quantity
Frequency	ν	$\nu = 1/T$	number of periods per unit time	Hz
Period	T	smallest positive number satisfying $f(t+T) = f(t)$	smallest closed orbit	s
Angular frequency	ω	$\omega = 2\pi\nu$	rate of rotation	rad/s
Phase	ϕ	offset relative to the reference point	initial argument; positional encoding	rad
Time delay	τ	$\tau = \phi/\omega$	time offset corresponding to the phase (**advanced** convention: when $\phi > 0$ the component **leads** the baseline by τ)	s
Angular momentum	L	$L = mA^2\omega$	angular momentum of the harmonic component, conserved	kg•m^2/s
Areal velocity	dS/dt	$\frac{1}{2}A^2\omega$	Kepler's second law, conserved	m^2/s
Energy	E	$\frac{1}{2}mA^2\omega^2$	kinetic energy of the harmonic component, conserved	J



The quadruple (A, ω, ϕ, T) completely parametrises a harmonic component

Figure 20.2: The quadruple components of a periodic function: amplitude A, angular frequency ω, phase ϕ and period T.

20.3 Harmonic Components and Conservation Laws

Definition 20.4 (Harmonic Component). Let $A > 0$, $\omega \in \mathbb{R} \setminus \{0\}$, $\phi \in \mathbb{R}$. A **harmonic component** is the parametrised curve in the complex plane

$$\gamma(t) = Ae^{i(\omega t+\phi)}.$$

A harmonic component is the **minimal geometric unit** of a Fourier series—it encodes simultaneously the three components amplitude (radius A), frequency (angular velocity ω) and phase (initial argument ϕ). (Historically, Ptolemaic astronomy used the term 'epicycle' to describe geometric objects of the same kind.)

Theorem 20.1 (Three Conserved Quantities of Harmonic Components).

(i) **Conservation of angular momentum**: $L = mA^2\omega$ is a constant.

(ii) **Conservation of areal velocity**: $\frac{dS}{dt} = \frac{1}{2}A^2|\omega|$ is a constant.

(iii) **Conservation of kinetic energy**: $E_k = \frac{1}{2}mA^2\omega^2$ is a constant.

Proof. The velocity vector is $\dot{\mathbf{r}}(t) = i\omega Ae^{i\omega t}$, whose modulus is the constant $A|\omega|$. The angular momentum is $L = m\|\mathbf{r} \times \dot{\mathbf{r}}\| = mA^2|\omega|$ (since $\mathbf{r} \perp \dot{\mathbf{r}}$). The areal velocity is $\frac{1}{2}\|\mathbf{r} \times \dot{\mathbf{r}}\| = \frac{1}{2}A^2|\omega|$. The kinetic energy is $\frac{1}{2}m|\dot{\mathbf{r}}|^2 = \frac{1}{2}mA^2\omega^2$. None of the three varies with time. □

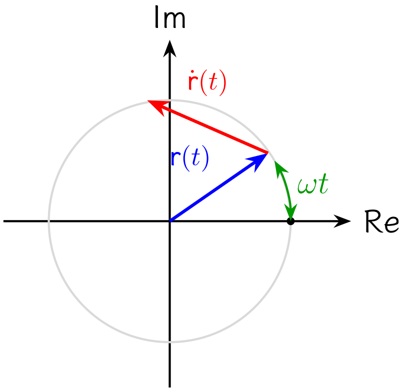
Definition 20.5 (Nested Harmonic-Component System).

$$P_0(t) = O, \quad P_k(t) = P_{k-1}(t) + A_k e^{i(\omega_k t+\phi_k)}, \quad k = 1, 2, \ldots, K.$$

The final position vector of the moving point is $\mathbf{V}(t) = \sum_{k=1}^{K} A_k e^{i(\omega_k t+\phi_k)}$. A nested harmonic-component system is the **geometric realisation** of a Fourier series.

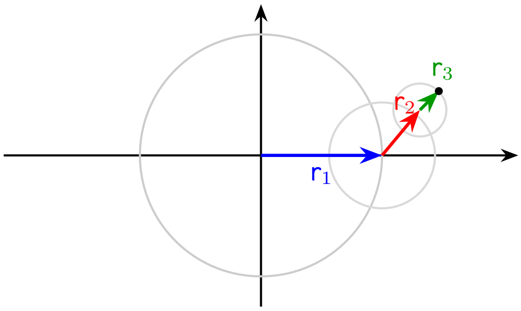
Theorem 20.2 (Independent Conserved Quantities of Nested Harmonic-Component Systems). Each level of harmonic component independently satisfies the three conserved quantities. The total angular momentum is $\mathbf{L}_{\text{total}} = \sum_{k=1}^{K} m_k A_k^2 \omega_k \cdot \mathbf{k}$.

Proof. The k-th level is a single harmonic component (Theorem 20.1), and the levels are mutually uncoupled. Angular momentum is additive. □


Im
$\dot{\mathsf{r}}(t)$
$\mathsf{r}(t)$
ωt
Re

$$L = mA^2\omega,\ \frac{dS}{dt} = \frac{1}{2}A^2|\omega|,\ E_k = \frac{1}{2}mA^2\omega^2$$

Figure 20.3: The three conserved quantities of the harmonic component $Ae^{i(\omega t+\phi)}$: angular momentum, areal velocity and kinetic energy.


r_3
r_2
r_1

Nested harmonic components: $\mathsf{V}(t) = \sum_k A_k e^{i(\omega_k t+\phi_k)}$, that is, an epicycle system

Figure 20.4: Nested harmonic-component system (epicycles): the superposition of independent harmonic components at each level constitutes the geometric realisation of a Fourier series.

20.4 Phase Operators

Definition 20.6 (Discrete Phase Operator). Let the period be $N \in \mathbb{N}^+$. The k-th **phase operator** is

$$\mathcal{O}_k^{(N)} = e^{i2\pi k/N}, \quad k = 0, 1, \ldots, N-1.$$

The set of phase operators $\mathcal{P}_N$ forms a cyclic group C_N of order N under multiplication of complex numbers.

Definition 20.7 (Continuous Phase Operator). The continuous phase operator is $\mathcal{O}_\theta = e^{i\theta}$, $\theta \in [0, 2\pi)$. The set forms the group $U(1)$.

Proposition 20.1 (Structure of the Phase-Operator Group). (i) $\mathcal{P}_N \cong C_N$.

(ii) $\mathcal{P}_\infty = S^1 \cong U(1)$.

(iii) $\mathcal{P}_N \subset \mathcal{P}_\infty$.

(iv) As $N \to \infty$, $\mathcal{P}_N$ is dense in S^1.

Proof. (i) $\mathcal{O}_k \cdot \mathcal{O}_l = \mathcal{O}_{k+l \bmod N}$. (ii) S^1 is a compact Abelian Lie group. (iii) This follows directly from (i) and the group structure of S^1. (iv) For any θ and $\varepsilon > 0$, take $N > 2\pi/\varepsilon$. ☐

Proposition 20.2 (Orthogonality of the phase operators). For any $k, l \in \{0, 1, \ldots, N-1\}$,

$$\frac{1}{N}\sum_{j=0}^{N-1} \mathcal{O}_j^{(N)} (\overline{\mathcal{O}_{j+k-l}^{(N)}}) = \frac{1}{N}\sum_{j=0}^{N-1} e^{i2\pi(l-k)j/N} = \delta_{kl}.$$

Proof. When $k = l$ the summand is 1 and the average is 1. When $k \neq l$, put $q = e^{i2\pi(l-k)/N} \neq 1$; then $\sum_{j=0}^{N-1} q^j = (1 - q^N)/(1 - q) = 0$. Hence the average equals δ_{kl}. ☐

Example 20.1 (Phase Operators of Period 2).

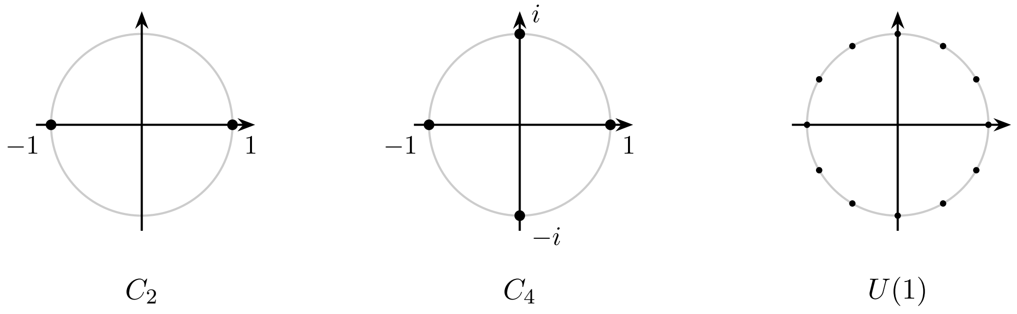
$$\mathcal{O}_0^{(2)} = 1, \quad \mathcal{O}_1^{(2)} = -1.$$

The identity operator and the reflection operator form C_2.

Example 20.2 (Phase Operators of Period 4).

$$\mathcal{O}_0^{(4)} = 1, \quad \mathcal{O}_1^{(4)} = i, \quad \mathcal{O}_2^{(4)} = -1, \quad \mathcal{O}_3^{(4)} = -i.$$

The rotations $0°, 90°, 180°, 270°$ form C_4.



The three levels of the phase-operator group: $C_N \subset U(1)$, dense in $U(1)$ as $N \to \infty$

Figure 20.5: Distribution of the phase operators on the unit circle: comparison of C_2, C_4 and $U(1)$.

20.5 Waves: Spatial Propagation of Fourier Series

Definition 20.8 (Wave—Spatial Propagation of Fourier Series)**.** Let the Fourier series $f(t) = \sum_n A_n e^{i(\omega_n t+\phi_n)}$ describe the vibration of the system at the origin. The **wave** formed by this vibration propagating along the positive x-axis with speed c is

$$f(x,t) = \sum_n A_n e^{i(k_n x-\omega_n t+\phi_n)},$$

where $k_n = \omega_n/c$ is the **wave number** (spatial frequency). A wave is the propagation of a Fourier series in space—it possesses both temporal periodicity (period $T_n = 2\pi/\omega_n$) and spatial periodicity (wavelength $\lambda_n = 2\pi/k_n$).

Proposition 20.3 (Double Periodic Structure of Waves)**.** The wave $f(x,t)$ has a double periodic structure:

(i) **temporal period**: $f(x,t+T_n) = f(x,t)$, $T_n = 2\pi/\omega_n$;

(ii) **spatial period**: $f(x+\lambda_n,t) = f(x,t)$, $\lambda_n = 2\pi/k_n$.

The temporal group with periodic orbits is $T_n\mathbb{Z}$ and the spatial group with periodic orbits is $\lambda_n\mathbb{Z}$. Together they constitute the group with periodic orbits of the double period of the wave, $\Lambda = T_n\mathbb{Z} \oplus \lambda_n\mathbb{Z}$ (in the (x,t) plane).

Proof. (i) $e^{i(k_n x-\omega_n (t+T_n)+\phi_n)} = e^{i(k_n x-\omega_n t+\phi_n)}e^{-i\omega_n T_n} = e^{i(k_n x-\omega_n t+\phi_n)}$ because $\omega_n T_n = 2\pi$. (ii) Similarly $k_n\lambda_n = 2\pi$. □

Proposition 20.4 (A wave satisfies the wave equation)**.** Let $f(x,t) = \sum_n A_n e^{i(k_n x-\omega_n t+\phi_n)}$ with dispersion relation $\omega_n = ck_n$ (c a constant). Then f satisfies the one-dimensional wave equation

$$\frac{\partial^2 f}{\partial t^2} = c^2\frac{\partial^2 f}{\partial x^2}.$$

Proof. Differentiating term by term gives

$$\begin{aligned} \partial_t^2 e^{i(k_n x-\omega_n t+\phi_n)} &= -\omega_n^2 e^{i(k_n x-\omega_n t+\phi_n)}, \\ \partial_x^2 e^{i(k_n x-\omega_n t+\phi_n)} &= -k_n^2 e^{i(k_n x-\omega_n t+\phi_n)}. \end{aligned}$$

Since $\omega_n^2 = c^2 k_n^2$, the two sides are equal term by term, and they remain equal after linear superposition. $\square$

Corollary 20.1 (Coupling of the dispersion relation with the double period). Under the wave equation, the wave number k_n, the angular frequency ω_n, the temporal period T_n and the spatial wavelength λ_n satisfy

$$c = \frac{\omega_n}{k_n} = \frac{\lambda_n}{T_n}.$$

That is, the phase velocity equals the ratio of the wavelength to the temporal period.

Proof. From $\omega_n = ck_n$ and $T_n = 2\pi/\omega_n$, $\lambda_n = 2\pi/k_n$, we obtain $\lambda_n/T_n = (2\pi/k_n)/(2\pi/\omega_n) = \omega_n/k_n = c$. $\square$

Proposition 20.5 (Relation Between Waves and Periodic Functions). At a fixed spatial position $x = x_0$ the wave $f(x,t)$ degenerates into a temporally periodic function; at a fixed time $t = t_0$ it degenerates into a spatially periodic function. The period structures of the two are given by the wave number and the angular frequency respectively.

Proof. Fixing $x = x_0$ gives

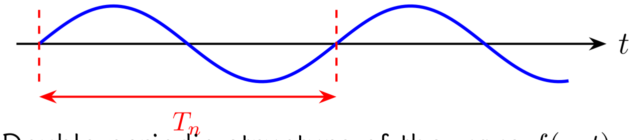
$$f(x_0,t) = \sum_n A_n e^{i(k_n x_0 - \omega_n t + \phi_n)} = \sum_n A'_n e^{-i\omega_n t},$$

where $A'_n = A_n e^{i(k_n x_0 + \phi_n)}$, so this is a temporally periodic function.

Fixing $t = t_0$ gives

$$f(x,t_0) = \sum_n A''_n e^{ik_n x},$$

so this is a spatially periodic function. $\square$



Double periodic structure of the wave $f(x,t)$:
temporal period $T_n = 2\pi/\omega_n$ (top); spatial wavelength $\lambda_n = 2\pi/k_n$ (bottom)

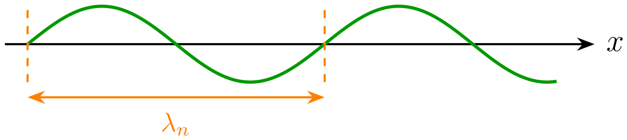


Figure 20.6: Double periodic structure of a wave: the period T_n on the time axis and the wavelength λ_n on the space axis.

20.6 Periodic Transformations and Component Extraction

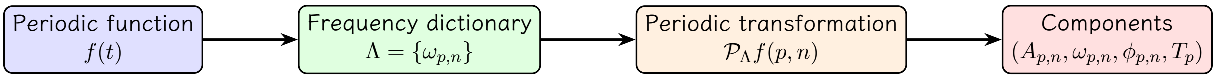
Definition 20.9 (Periodic Transformation—Component Extraction)**.** Let f be a Fourier series with frequency dictionary $\Lambda = \{\omega_{p,n}\}$. The **periodic transformation** (Definition 6.18) extracts the components:

$$\mathcal{P}_\Lambda f(p,n) = M\{f(t)e^{-i\omega_{p,n}t}\} = A_{p,n}e^{i\phi_{p,n}}.$$

The periodic transformation is the penultimate step of the modelling chain—it converts the Fourier series into the explicit component quadruple.

Theorem 20.3 (Uniqueness of Component Extraction)**.** *The component quadruple $(A_{p,n}, \omega_{p,n}, \phi_{p,n}, T_p)$ extracted by the periodic transformation is uniquely determined by the periodic function f.*

Proof. This follows directly from Theorem 6.16 (Bohr uniqueness). ☐


Periodic function
$f(t)$
Frequency dictionary
$\Lambda = \{\omega_{p,n}\}$
Periodic transformation
$\mathcal{P}_\Lambda f(p,n)$
Components
$(A_{p,n}, \omega_{p,n}, \phi_{p,n}, T_p)$

Extracting the coefficients: $\mathcal{P}_\Lambda f(p,n) = M\{f(t)e^{-i\omega_{p,n}t}\} = A_{p,n}e^{i\phi_{p,n}}$

Figure 20.7: Periodic transformation and component extraction: starting from a periodic function and passing through the frequency dictionary, the unique component quadruple is extracted.

20.7 Preservation of the Group with Periodic Orbits by the Fundamental Transformations

Proposition 20.6 (Translation Preserves Periodicity)**.** *Let $f : \mathbb{R} \to V$ be a periodic function and $a \in \mathbb{R}$, and define the* **translation** *of f by $(\tau_a f)(t) := f(t+a)$. Then for every $T \in \mathsf{Group}_{\mathsf{period}}(f)$ we have $T \in \mathsf{Group}_{\mathsf{period}}(\tau_a f)$, hence*

$$\mathsf{Group}_{\mathsf{period}}(\tau_a f) = \mathsf{Group}_{\mathsf{period}}(f), \qquad T_0(\tau_a f) = T_0(f).$$

That is, translation is the fundamental transformation that preserves the period structure.

Proof. Let $T \in \mathsf{Group}_{\mathsf{period}}(f)$. For every $t \in \mathbb{R}$,

$$(\tau_a f)(t+T) = f(t+T+a) = f(t+a) = (\tau_a f)(t),$$

hence $T \in \mathsf{Group}_{\mathsf{period}}(\tau_a f)$, that is, $\mathsf{Group}_{\mathsf{period}}(f) \subseteq \mathsf{Group}_{\mathsf{period}}(\tau_a f)$. Conversely, let $T \in \mathsf{Group}_{\mathsf{period}}(\tau_a f)$; for any $s \in \mathbb{R}$ take $t = s - a$, obtaining

$$f(s+T) = f\big((s-a)+T+a\big) = (\tau_a f)(s-a+T) = (\tau_a f)(s-a) = f(s),$$

hence $T \in \mathsf{Group}_{\mathsf{period}}(f)$; the two groups with periodic orbits are equal, and the least positive periods coincide accordingly. ☐

Proposition 20.7 (Transformation of the group with periodic orbits under scaling and rotation). Let f be a periodic function, and let $q > 0$ and $\theta \in \mathbb{R}$ correspond to scaling and phase rotation respectively. Then

$$\mathrm{Group}_{\mathrm{period}}(f(q\,\cdot)) = \frac{1}{q}\,\mathrm{Group}_{\mathrm{period}}(f), \qquad \mathrm{Group}_{\mathrm{period}}(e^{i\theta}f) = \mathrm{Group}_{\mathrm{period}}(f).$$

That is, scaling compresses the group with periodic orbits in inverse proportion, whilst phase rotation leaves the group with periodic orbits unchanged.

Proof. Scaling: put $g(t) = f(qt)$. Then $T \in \mathrm{Group}_{\mathrm{period}}(g)$ if and only if $f(q(t+T)) = f(qt)$, that is, $qT \in \mathrm{Group}_{\mathrm{period}}(f)$; hence $\mathrm{Group}_{\mathrm{period}}(g) = \frac{1}{q}\mathrm{Group}_{\mathrm{period}}(f)$. Phase rotation: $\mathrm{Group}_{\mathrm{period}}(e^{i\theta}f)$ equals $\mathrm{Group}_{\mathrm{period}}(f)$, because $|e^{i\theta}| = 1$ does not alter the correspondence in time of the values of f. ☐

Proposition 20.8 (The group with periodic orbits is a closed subgroup). Let $f : \mathbb{R} \to V$ be continuous. Then $\mathrm{Group}_{\mathrm{period}}(f)$ is a closed subgroup of $\mathbb{R}$. In particular, $\mathrm{Group}_{\mathrm{period}}(f)$ is either equal to $\mathbb{R}$ (f constant) or equal to $T_0\mathbb{Z}$ (f a non-constant periodic function).

Proof. Closedness: let $T_k \in \mathrm{Group}_{\mathrm{period}}(f)$ with $T_k \to T$. By the continuity of f, $f(t+T) = \lim_k f(t+T_k) = \lim_k f(t) = f(t)$, hence $T \in \mathrm{Group}_{\mathrm{period}}(f)$ and $\mathrm{Group}_{\mathrm{period}}(f)$ is closed. Subgroup: if $T_1, T_2 \in \mathrm{Group}_{\mathrm{period}}(f)$, then $f(t+T_1-T_2) = f((t-T_2)+T_1) = f(t-T_2) = f(t)$, hence $T_1 - T_2 \in \mathrm{Group}_{\mathrm{period}}(f)$. A closed subgroup of $\mathbb{R}$ is either $\mathbb{R}$ or a discrete group $T_0\mathbb{Z}$, the former corresponding to constant functions and the latter to the least positive period T_0. ☐

20.8 Groups with Periodic Orbits and Conservation Laws

Remark 20.4 (Noether correspondence of groups with periodic orbits—the periodic-function version (**empirical observation / research programme**)). The three correspondences below are an **empirical observation / research programme**, not theorems proved within this framework (the 'Operations and Periodicity' chapter has already declared in Remark 1.10 that its framework furnishes only an orbit-counting construction of invariants, and does not contain Noether's theorem, which requires a Lagrangian variational structure):

(i) **Translation group with periodic orbits**: corresponds to conservation of energy (Parseval's identity).

(ii) **Rotation group with periodic orbits**: corresponds to conservation of angular momentum (independent conservation of the angular momentum of each harmonic component).

(iii) **Scaling group with periodic orbits**: corresponds to conservation of scale charge (invariance under translation in the logarithmic domain).

All that can be proved rigorously is the energy identity of (i) given by Theorem 20.4 (which is Parseval's identity itself and holds independently of any 'symmetry');

the quantity L_k of (ii) and the 'scale charge' of (iii) have no independent definition in this chapter, so that the correspondences asserted there are at present merely restatements of the same form as in the 'Groups and Group Functions' chapter.

Theorem 20.4 (Parseval's identity)**.** *Let $f(t) = \sum_n A_n e^{i(\omega_n t+\phi_n)}$ be periodic with period T, and let $f \in L^2[0,T]$. Then*

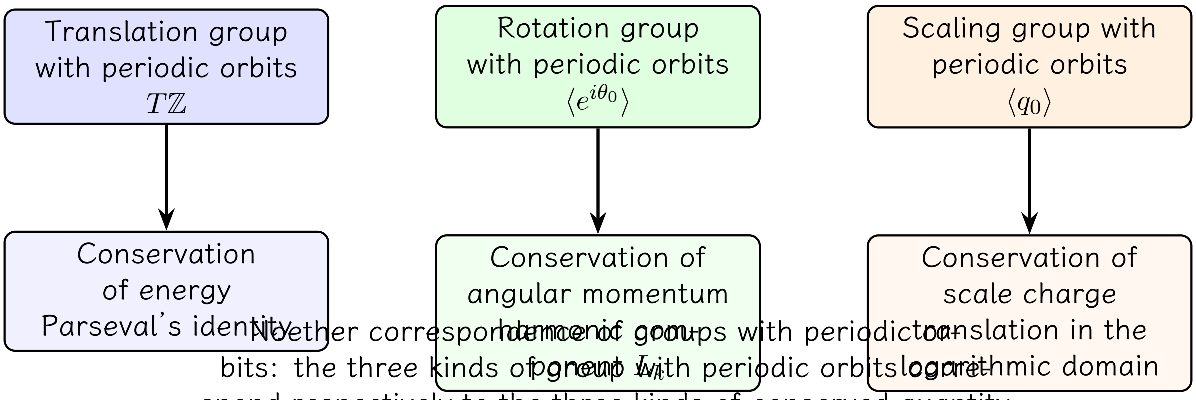
$$\frac{1}{T}\int_0^T |f(t)|^2\,dt = \sum_n A_n^2.$$

That is, the time-averaged energy equals the sum of the squares of the amplitudes of the components.

Proof. By the orthogonality $\frac{1}{T}\int_0^T e^{i(\omega_m-\omega_n)t}dt = \delta_{mn}$,

$$\frac{1}{T}\int_0^T |f|^2\,dt = \sum_{m,n} A_m A_n e^{i(\phi_m-\phi_n)}\frac{1}{T}\int_0^T e^{i(\omega_m-\omega_n)t}dt = \sum_n A_n^2.$$

$\square$


Translation group with periodic orbits $T\mathbb{Z}$
Rotation group with periodic orbits $\langle e^{i\theta_0}\rangle$
Scaling group with periodic orbits $\langle q_0\rangle$
Conservation of energy Parseval's identity
Conservation of angular momentum harmonic component L_k
Conservation of scale charge translation in the logarithmic domain

Noether correspondence of groups with periodic orbits: the three kinds of group with periodic orbits correspond respectively to the three kinds of conserved quantity

Figure 20.8: Noether correspondence of groups with periodic orbits (**empirical observation / research programme**): the translation, rotation and scaling groups with periodic orbits correspond respectively to the conservation of energy, of angular momentum and of scale charge.

20.9 Periodic Convolution: The Four-Operator Form

Definition 20.10 (Periodic Convolution (Four-Operator Form))**.** Let f, g be Fourier series. The **periodic convolution** of f and g is

$$(f \circledast g)(\tau) := \sum_{p,q}\Theta_{p,q}\sum_{n,m} W(\omega_{p,n}-\nu_{q,m})\cdot(A_{p,n}B_{q,m})^{\alpha}\cdot S^{(d)}(\phi_{p,n},\psi_{q,m})\cdot e^{i\nu_{q,m}\tau},$$

The four operators are: the amplitude operator $\mathfrak{A}_\alpha$ (scaling), the frequency operator $\mathfrak{F}_{W,\tau}$ (translation), the phase operator $\mathfrak{P}_d$ (rotation) and the period operator $\mathfrak{T}$ (fixed length).

Proposition 20.9 (Operator Correspondence of the Four Operators)**.** The amplitude operator and the period operator belong respectively to the value axis and the time axis of the scaling operator; the frequency operator belongs to the translation operator; the phase operator belongs to the rotation operator.

Proof. Verified term by term from Definition 20.10. ☐

Proposition 20.10 (Periodic convolution as generalised kernelised convolution)**.** For a single period layer ($P = Q = \{T\}$, $\Theta \equiv 1$), the periodic convolution reduces to kernelised convolution.

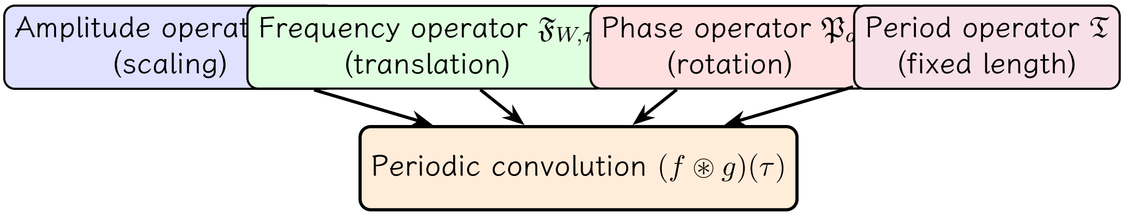
Proof. Obtained by substitution. ☐

Theorem 20.5 (Three Degeneracies of Periodic Convolution)**.** (i) **Classical limit**: as $W \to \delta$ it degenerates into cross-correlation/circular convolution.

(ii) **Similarity/trend layer**: $\alpha = 0$ gives similarity convolution, $\alpha = 1$ gives trend convolution.

(iii) **Cross-period layer**: the cross-period terms are controlled jointly by $\Theta_{p,q}$ and W.

Proof. The kernel-width limit makes the off-diagonal terms vanish; $\tau = 0$ gives the matrix entries; the cross-period terms carry the factor $\Theta_{p,q}W$. ☐


Amplitude operat
(scaling)
Frequency operator $\mathfrak{F}_{W,\tau}$
(translation)
Phase operator $\mathfrak{P}_d$
(rotation)
Period operator $\mathfrak{T}$
(fixed length)
Periodic convolution $(f \circledast g)(\tau)$

$$(A_{p,n}B_{q,m})^{\alpha};\ W(\omega_{p,n} - \nu_{q,m});\ S^{(d)}(\phi_{p,n}, \psi_{q,m});\ e^{i\nu_{q,m}\tau}$$

Figure 20.9: The four-operator structure of periodic convolution: the amplitude, frequency, phase and period operators correspond respectively to scaling, translation, rotation and fixed length.

Example 20.3 (Periodic-Convolution Matrix of Period 2)**.** The trend convolution matrix is

$$\mathbb{T} = \begin{pmatrix} c_0\overline{d_0} & c_0\overline{d_1} \\ c_1\overline{d_0} & c_1\overline{d_1} \end{pmatrix}.$$

20.10 Modelling Example: Light Source—Light Wave—Spectrum

Example 20.4 (Complete Modelling Chain for Light Source—Light Wave—Spectrum)**.** **Physical system**: a light source emits electromagnetic waves.

Translation/scaling/rotation: *time translation $t \mapsto t + T$ (the light source emits continuously); frequency scaling $\omega \mapsto \lambda\omega$ (changing the colour).*

Conservation law: *conservation of energy (the light intensity does not decay with time)—the conserved quantity of time-translation symmetry.*

Symmetry: *the time-translation symmetry group $\mathbb{R}$.*

Periodic function: *the electric field component of monochromatic light, $E(t) = Ae^{i(\omega t+\phi)}$.*

Group with periodic orbits: Group$_{\text{period}}(E) = T\mathbb{Z}$, $T = 2\pi/\omega$.

Fourier series: *a real light source (such as the sun) emits a continuous spectrum:*

$$E(t) = \sum_n A_n e^{i(\omega_n t+\phi_n)},$$

each frequency component is a harmonic component, and the amplitude spectrum $\{A_n\}$ is the **spectrum**.

Wave: *the light wave propagates in space:*

$$E(x,t) = \sum_n A_n e^{i(k_n x-\omega_n t+\phi_n)}, \quad k_n = \omega_n/c.$$

The wavelength is $\lambda_n = 2\pi/k_n = 2\pi c/\omega_n$.

Periodic transformation: *Bohr extraction gives the spectral components $\mathcal{P}_\Lambda E(n) = A_n e^{i\phi_n}$.*

Component: *the quadruple $(A_n, \omega_n, \phi_n, T_n)$ completely parametrises the spectral lines.*

Example 20.5 (Sound—Sound Wave—Frequency Spectrum). **Physical system**: *a musical instrument emits sound waves.*

Conservation law: *conservation of energy (the sound intensity does not decay with time).*

Periodic function: the sound pressure $p(t) = \sum_n A_n \cos(n\omega_0 t + \phi_n)$ (harmonics of the fundamental frequency ω_0).

Group with periodic orbits: Group$_{\text{period}}(p) = T\mathbb{Z}$, $T = 2\pi/\omega_0$.

Fourier series: the frequency spectrum $\{A_n\}$ is the timbre.

Wave: sound-wave propagation $p(x,t) = \sum_n A_n \cos(k_n x - n\omega_0 t + \phi_n)$.

Component: $(A_n, n\omega_0, \phi_n, T/n)$.

Example 20.6 (Clock—Hand—Time). **Physical system**: *a clock.*

Translation: time translation (the clock runs continuously).

Conservation law: *conservation of energy (the driving force of the clock is constant).*

Periodic function: the hand angle $\theta(t) = \omega t$.

Group with periodic orbits: Group$_{\text{period}}(\theta) = T\mathbb{Z}$, $T = 2\pi/\omega$.

Fourier series: *a single-frequency harmonic component.*

Component: $(A = 1, \omega, \phi = 0, T)$.

Example 20.7 (Fourier series of the square wave). Let $f(t) = \operatorname{sgn}(\sin(\omega_0 t))$ be the square wave with principal period $T = 2\pi/\omega_0$. Its Fourier series is

$$f(t) = \frac{4}{\pi}\sum_{k=0}^{\infty} \frac{1}{2k+1} \sin\big((2k+1)\omega_0 t\big).$$

That is, only odd-order harmonic components occur, with $A_{2k+1} = 4/((2k+1)\pi)$ and phase $\phi_{2k+1} = -\pi/2$. The components decay with frequency as $1/n$—this is precisely the spectral manifestation of the high-frequency corners contained in the square wave.

Example 20.8 (Fourier series of the sawtooth wave). Let the sawtooth wave be $f(t) = \frac{2}{\pi}\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin(n\omega_0 t)$, with principal period $T = 2\pi/\omega_0$. The components $A_n = 2/(\pi n)$ cover all integer-order harmonics—more even-order components than the square wave contains.

Table 20.2: Comparison table of the chains of the three modelling examples

Modelling link	Light source—light wave	Sound source—sound wave	Clock—hand
Physical system	Light source	Musical instrument	Clock
Fundamental transformation	Time translation + frequency scaling	Time translation	Time translation
Conservation law	Conservation of energy	Conservation of energy	Conservation of energy
Periodic function	Electric field $E(t)$	Sound pressure $p(t)$	Angle $\theta(t)$
Group with periodic orbits	$T\mathbb{Z}$, $T = 2\pi/\omega$	$T\mathbb{Z}$, $T = 2\pi/\omega_0$	$T\mathbb{Z}$, $T = 2\pi/\omega$
Fourier series	Continuous spectrum $\{A_n\}$	Harmonic spectrum $\{A_n\}$	Single-frequency component
Wave	$E(x,t)$, $k_n = \omega_n/c$	$p(x,t)$, $k_n = n\omega_0/c$	No spatial propagation
Component	$(A_n, \omega_n, \phi_n, T_n)$	$(A_n, n\omega_0, \phi_n, T/n)$	$(1, \omega, 0, T)$

Remark 20.5 (Distinction between the Fourier series and the Bohr series). For an L^2 function of period T, the Fourier series expands discretely in the frequencies $\omega_n = 2\pi n/T$, and the coefficients are uniquely determined by orthogonality. For the more general **quasi-periodic functions** (whose frequencies are generated by finitely or countably many incommensurable fundamental frequencies), the expansion is called a **Bohr series**, and its coefficients are uniquely determined by the Bohr mean (Theorem 6.16). This chapter uniformly calls the common form of the two a 'Fourier series', so that the modelling chain remains uniform.

Remark 20.6 (Periodic functions and quasi-periodic functions). The group with periodic orbits of a periodic function is the discrete group $T\mathbb{Z}$. The 'frequency group' of a quasi-periodic function is formed by the $\mathbb{Z}$-linear combinations generated by several incommensurable frequencies; in this case no single non-zero period T exists, but the function can still be expanded as a Bohr series. Physically, this distinction corresponds to the difference between a 'single tone' and a 'chord'.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The translation link of the modelling chain; the frequency operator $\mathfrak{F}_{W,\tau}$ carries translation information
Scaling	The scaling link of the modelling chain; the amplitude operator $\mathfrak{A}_\alpha$; the amplitude component $A_{p,n}$
Rotation	The rotation link of the modelling chain; the phase operator $\mathfrak{P}_d$; the phase-operator group C_N (Proposition 20.1)
Conservation	The conservation-law link of the modelling chain; the three conserved quantities of the harmonic component (Theorem 20.1); the Noether correspondence (**empirical observation / research programme**, see Remark 20.4)
Symmetry	The symmetry link of the modelling chain; the action of the group with periodic orbits $T\mathbb{Z}$, $\langle q_0 \rangle$, $\langle e^{i\theta_0} \rangle$, Λ
Closure	The closed orbit guaranteed by the conservation law: the harmonic component $Ae^{i(\omega t+\phi)}$ and the quotient space $\mathbb{R}/T\mathbb{Z} \cong S^1$
Period	The period link of the modelling chain; the component quadruple $(A_{p,n}, \omega_{p,n}, \phi_{p,n}, T_p)$ (Definition 20.2); the double period of the wave (Proposition 20.3)

Exercises

Exercise 1 Using Definition 20.1, explain how the complete chain of periodic-structure modelling starts from a physical system, passes through the fundamental transformations, the conservation law, symmetry, the periodic function, the group with periodic orbits, the Fourier series, the wave and the periodic transformation, and finally arrives at the component quadruple. Taking a sound source as an example (Example 20.5), write out the concrete content of the modelling chain link by link, and point out which link is the core of this chapter.

Exercise 2 Verify the quadruple components of a periodic function in Definition 20.2:

$$(A_{p,n}, \omega_{p,n}, \phi_{p,n}, T_p),$$

and explain that the amplitude component, the frequency component, the phase component and the period component correspond respectively to the four operations 'scaling', 'translation', 'rotation' and 'fixed length'. Using the physical meanings in Table 20.1, write out the concrete numerical values of each component for the vibration of a pendulum.

Exercise 3 Prove Property 20.2: the frequency dictionary Λ is a subset of $\frac{2\pi}{T}\mathbb{Z}$; if $f \in L^2$, then Λ is at most countable. Explain why the frequency dictionary is the input data of the periodic transformation.

Exercise 4 Prove Theorem 20.1: the three conserved quantities of the harmonic component $\gamma(t) = Ae^{i(\omega t+\phi)}$:

(i) the angular momentum $L = mA^2\omega$;

(ii) the areal velocity $\frac{dS}{dt} = \frac{1}{2}A^2|\omega|$;

(iii) the kinetic energy $E_k = \frac{1}{2}mA^2\omega^2$.

Complete the proof using the velocity vector $\dot{\mathsf{r}}(t) = i\omega Ae^{i\omega t}$ and $\mathsf{r} \perp \dot{\mathsf{r}}$, and explain why none of the three varies with time.

Exercise 5 Prove Theorem 20.2: in a nested harmonic-component system, each level of harmonic component independently satisfies the three conserved quantities, and the total angular momentum is

$$\mathsf{L}_{\mathsf{total}} = \sum_{k=1}^{K} m_k A_k^2 \omega_k \cdot \mathsf{k}.$$

Explain why the levels are mutually uncoupled, and compute the total angular momentum for $K = 3$ with $m_1 = m_2 = m_3 = 1$, $A_k = 1/k$ and $\omega_k = k$.

Exercise 6 Prove Proposition 20.1: the four structural properties of the phase-operator group:

(i) $\mathcal{P}_N \cong C_N$;

(ii) $\mathcal{P}_\infty = S^1 \cong U(1)$;

(iii) $\mathcal{P}_N \subset \mathcal{P}_\infty$;

(iv) as $N \to \infty$, $\mathcal{P}_N$ is dense in S^1.

Verify the four cases separately, and show that $\mathcal{O}_a \cdot \mathcal{O}_b = \mathcal{O}_{(a+b) \bmod N}$.

Exercise 7 Prove Proposition 20.2: the phase operators satisfy the orthogonality

$$\frac{1}{N}\sum_{j=0}^{N-1} \mathcal{O}_j^{(N)} \overline{\mathcal{O}_{j+k-l}^{(N)}} = \delta_{kl}.$$

Explain how this orthogonality corresponds to the extraction of the Fourier coefficients.

Exercise 8 Verify Example 20.1 and Example 20.2:

(i) $\mathcal{O}_0^{(2)} = 1$, $\mathcal{O}_1^{(2)} = -1$, forming C_2;

(ii) $\mathcal{O}_0^{(4)} = 1$, $\mathcal{O}_1^{(4)} = i$, $\mathcal{O}_2^{(4)} = -1$, $\mathcal{O}_3^{(4)} = -i$, forming C_4.

Draw the distribution of C_2 and C_4 on the unit circle, and explain how $\mathcal{O}_1^{(4)} = i$ corresponds to a rotation of 90°.

Exercise 9 Prove Proposition 20.3: the wave $f(x,t) = \sum_n A_n e^{i(k_n x - \omega_n t + \phi_n)}$ has a double periodic structure:

(i) temporal period: $f(x, t + T_n) = f(x,t)$, $T_n = 2\pi/\omega_n$;

(ii) spatial period: $f(x + \lambda_n, t) = f(x,t)$, $\lambda_n = 2\pi/k_n$.

Verify the two period conditions separately, and prove that the wave number and the angular frequency are coupled by the dispersion relation as $\lambda_n = cT_n$.

Exercise 10 Prove Proposition 20.4: the wave $f(x,t)$ satisfies the one-dimensional wave equation

$$\frac{\partial^2 f}{\partial t^2} = c^2 \frac{\partial^2 f}{\partial x^2}.$$

Explain the role of the dispersion relation $\omega_n = ck_n$ in the proof, and prove Corollary 20.1.

Exercise 11 Prove Proposition 20.5: at a fixed spatial position $x = x_0$ the wave $f(x,t)$ degenerates into a temporally periodic function; at a fixed time $t = t_0$ it degenerates into a spatially periodic function. Write out the Fourier series expressions in the two degenerate cases separately, and explain the period structures corresponding to the wave number k_n and the angular frequency ω_n respectively.

Exercise 12 Prove Theorem 20.3: the component quadruple $(A_{p,n}, \omega_{p,n}, \phi_{p,n}, T_p)$ extracted by the periodic transformation is uniquely determined by the periodic function f. Complete the proof using the Bohr uniqueness theorem (Theorem 6.16), and explain why the 'uniqueness of component extraction' guarantees the reliability of the modelling chain.

Exercise 13 Prove Proposition 20.6: let $f : \mathbb{R} \to V$ be a periodic function and $a \in \mathbb{R}$, and define the translation

$$(\tau_a f)(t) = f(t+a),$$

then

$$\mathrm{Group}_{\mathrm{period}}(\tau_a f) = \mathrm{Group}_{\mathrm{period}}(f), \qquad T_0(\tau_a f) = T_0(f).$$

Prove $\mathrm{Group}_{\mathrm{period}}(f) \subseteq \mathrm{Group}_{\mathrm{period}}(\tau_a f)$ and $\mathrm{Group}_{\mathrm{period}}(\tau_a f) \subseteq \mathrm{Group}_{\mathrm{period}}(f)$ separately.

Exercise 14 Prove Proposition 20.7: let f be a periodic function, and let $q > 0$ and $\theta \in \mathbb{R}$ correspond to scaling and phase rotation respectively; then

$$\mathrm{Group}_{\mathrm{period}}(f(q\,\cdot)) = \frac{1}{q}\mathrm{Group}_{\mathrm{period}}(f), \qquad \mathrm{Group}_{\mathrm{period}}(e^{i\theta} f) = \mathrm{Group}_{\mathrm{period}}(f).$$

Explain why scaling compresses the group with periodic orbits in inverse proportion, whilst phase rotation leaves the group with periodic orbits unchanged.

Exercise 15 Prove Proposition 20.8: let $f : \mathbb{R} \to V$ be continuous; then $\mathrm{Group}_{\mathrm{period}}(f)$ is a closed subgroup of $\mathbb{R}$, and hence is either equal to $\mathbb{R}$ or equal to $T_0\mathbb{Z}$. Complete the proof using closedness and the subgroup property, and explain why this conclusion is the algebraic root of the 'discretisation' of periodic structure.

Exercise 16 Discuss the **empirical observation / research programme** of Remark 20.4—the Noether correspondence of groups with periodic orbits: the Noether correspondence of groups with periodic orbits:

(i) the translation group with periodic orbits corresponds to conservation of energy (Parseval's identity);

(ii) the rotation group with periodic orbits corresponds to conservation of angular momentum (independent conservation of the angular momentum of harmonic components);

(iii) the scaling group with periodic orbits corresponds to conservation of scale charge (invariance under translation in the logarithmic domain).

Write out the concrete forms of the three conserved quantities separately, and explain their correspondence with the actions of the respective groups with periodic orbits; then state which of the three can be proved rigorously within the framework of this chapter and which two are at present merely formal analogies.

Exercise 17 Prove Theorem 20.4: Parseval's identity

$$\frac{1}{T}\int_0^T |f(t)|^2\,dt = \sum_n A_n^2.$$

Complete the proof using the orthogonality of the phase operators, and explain how this identity quantifies 'conservation of energy'.

Exercise 18 Verify the four-operator form of periodic convolution in Definition 20.10:

$$(f \circledast g)(\tau) = \sum_{p,q} \Theta_{p,q} \sum_{n,m} W(\omega_{p,n} - \nu_{q,m}) \cdot (A_{p,n}B_{q,m})^{\alpha} \cdot S^{(d)}(\phi_{p,n}, \psi_{q,m}) \cdot e^{i\nu_{q,m}\tau}.$$

Point out the positions of the amplitude operator, the frequency operator, the phase operator and the period operator in the formula, and explain that they correspond respectively to the four operations of scaling, translation, rotation and fixed length (see Proposition 20.9).

Exercise 19 Prove Proposition 20.10: for a single period layer ($P = Q = \{T\}$, $\Theta \equiv 1$), the periodic convolution reduces to kernelised convolution. Write out the degenerate expression, and explain how kernelised convolution retains the interaction of all frequency pairs.

Exercise 20 Prove Theorem 20.5: the three degeneracies of periodic convolution:

(i) classical limit: as $W \to \delta$ it degenerates into cross-correlation/circular convolution;

(ii) similarity/trend layer: $\alpha = 0$ gives similarity convolution, $\alpha = 1$ gives trend convolution;

(iii) cross-period layer: the cross-period terms are controlled jointly by $\Theta_{p,q}$ and W.

Write out the concrete forms in the three degenerate cases separately, and explain why each level of degeneracy corresponds to a strict decrease in the amount of information.

Exercise 21 Verify Example 20.3: the periodic-convolution matrix of period 2

$$\mathbb{T} = \begin{pmatrix} c_0\overline{d_0} & c_0\overline{d_1} \\ c_1\overline{d_0} & c_1\overline{d_1} \end{pmatrix}.$$

Taking $c_0 = 1$, $c_1 = i$, $d_0 = 1$ and $d_1 = -i$, compute the trend convolution matrix $\mathbb{T}$, and verify that its entries satisfy $\mathbb{T}_{mn} = c_m\overline{d_n}$.

Exercise 22 Verify Example 20.7 and Example 20.8: write out the Fourier series of the square wave and of the sawtooth wave, and compare their spectral structures and decay rates. Explain why the square wave contains only odd-order harmonics, whilst the sawtooth wave contains all harmonics.

Exercise 23 Explain the distinction between the Fourier series and the Bohr series in Remark 20.5: the frequency dictionary of a Fourier series is a subset of $\frac{2\pi}{T}\mathbb{Z}$; the 'frequency group' of a Bohr series is the $\mathbb{Z}$-linear combination generated by several incommensurable frequencies. Give a concrete example of a quasi-periodic function, and explain that it has no non-zero least period.

Exercise 24 Synthesis question: combining the modelling chain (Definition 20.1), the quadruple components of a periodic function (Definition 20.2), harmonic components and conservation laws (Definition 20.4, Theorem 20.1), the phase operator (Definition 20.6), the double period of waves (Proposition 20.3) and the four operators of periodic convolution (Definition 20.10), explain how periodic functions and groups with periodic orbits unify the three great themes of periodic mathematics—translation, scaling and rotation—within the four-parameter framework of 'amplitude, frequency, phase and period'. Point out separately:

(i) how the amplitude corresponds to the scaling operation;

(ii) how the frequency corresponds to the translation operation;

(iii) how the phase corresponds to the rotation operation;

(iv) how the period corresponds to the fixed-length operation.

Chapter 21

Discrete Sampling and Intercalation of Periodic Signals

Remark 21.1 (Positioning and main line of this chapter). This chapter is the **discrete implementation** of the 'Periodic Functions and Groups with Periodic Orbits' chapter, but its main line is upgraded from 'discrete periodic functions' to **discrete sampling approximation**, and further upgraded to **congruence-angle sampling approximation**. *Its starting point is an engineering fact: any computable description of a continuous object must pass through the three steps of* **sampling, truncation** *and (when necessary)* **intercalation**. *Each of the three steps introduces a class of error—aliasing, spectral leakage, approximation error—and all three are governed by one and the same discrete structure, and are ultimately characterised uniformly by the congruence angle.*

The core claims of this chapter are four:

(i) **An epicycle group is the rotational decomposition of a periodic signal:** *a continuous Fourier series may be regarded as the superposition of a family of uniformly rotating epicycles; after discrete sampling, the epicycle group degenerates into finitely or countably many discrete rotational components.*

(ii) **Intercalation is not a trick of calendars, but an active commensuration device in discrete sampling approximation**; *it presses the sole root of spectral leakage, namely 'the number of periods inside the window being non-integral', down to the order $O(M^{-2})$ by means of the best continued-fraction approximation.*

(iii) **The congruence angle is the unified language of sampling, leakage, intercalation and the Chinese remainder theorem**; *it upgrades 'phase' from a continuous angle to an 'equivalence class modulo a subperiod', so that single-period problems generalise naturally to multi-period joint problems.*

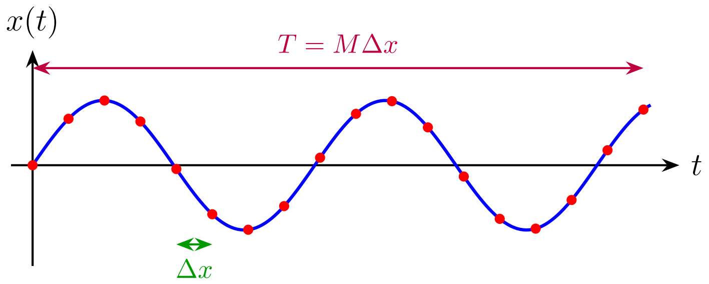
(iv) **Intercalation is the optimal compensation of congruence-angle error; the Chinese remainder theorem is the algebraic form of congruence-angle reconstruction; spectral leakage is the frequency-domain manifestation of congruence-angle non-closure.**

Parallel structure:

Continuous chapter	This chapter (discre
Periodic function $x(t)$	Sampli
Group with periodic orbits $T_x\mathbb{Z}$	Group v
Fourier series $\sum_k c_k e^{i\omega_k t}$	Epic
Discrete Fourier transform (DFT)	Fir
Spectrum $\lvert c_k \rvert$	Window sp
Periodic convolution	Circular convolution, congruence-angle circular convolutio

The eight themes of this chapter: translation—the sampling lattice $\Delta x\mathbb{Z}$ and translation invariance; scaling—the sampling rate ν_s and the fixed-length scale T_0; rotation—the discrete phase operator $\mathcal{O}_k^{(T)} = e^{i2\pi k/T}$, the epicycle group and the congruence angle; conservation—the discrete Parseval identity; symmetry—the even symmetry of the window spectrum and the spectral symmetry of a real sequence; closure—circular convolution, congruence-angle closure and the intercalation window; period—the duality between the number of periods in the window and the offset δ, and the multi-period congruence-angle non-closure; algorithms—continued-fraction expansion, the balanced intercalation sequence and fast congruence-angle reconstruction.

Reading route: Section 21.1 establishes the three elements of sampling; Section 21.2 introduces the epicycle group; Section 21.3 gives the complete mechanism of leakage; Section 21.4 discusses the price of the sampling rate; Section 21.5 establishes the upper and lower approximation and the optimality of continued fractions; Section 21.6 gives the sidelobe control of window functions; Section 21.7 answers 'why intercalation can suppress leakage'; Section 21.8 introduces the congruence angle and the generalised Chinese remainder theorem; Section 21.9 upgrades intercalation to multi-period joint suppression; Section 21.10 gives congruence-angle algorithms and complexity; Sections 21.11 to 21.13 give the implementation, the calendar unification and the chapter interfaces.



The three elements of sampling: sampling interval Δx, sampling rate $\nu_s = 1/\Delta x$, sampling duration $T = M\Delta x$

Figure 21.1: The three elements of sampling: the sampling interval Δx (the spacing between the red sampling points), the sampling rate $\nu_s = 1/\Delta x$, and the sampling duration $T = M\Delta x$ (the finite observation window of the blue continuous signal).

21.1 Sampling: sampling points, sampling rate and sampling duration

This chapter fixes the following notation: $x : \mathbb{R} \to \mathbb{C}$ is the continuous signal to be sampled (the 'true object'), and V is $\mathbb{C}$ or a real vector space.

Remark 21.2 (The canonical home of the sampling concepts). *The canonical definitions in this book of* **sampling point**, **sampling rate** *(sampling frequency)*, **sampling interval**, **number of sampling points**, **sampling duration**, **sampling theorem** *and* **sampling sequence** *are all gathered in this section: the sampling lattice and sampling points in Definition 21.1, the sampling rate, Nyquist frequency and sampling duration in Definition 21.2, the independence and non-substitutability of the three elements in Remark 21.3, the periodicity and commensurability of the sampling sequence in Proposition 21.2, and scale-dependent sampling in Section 21.1.1. The chapters that follow—the* **'Harmonic Analysis' chapter**, *the* **'Time Series Analysis' chapter** *and the various application chapters—all refer to this section and do not repeat the definitions.*

Definition 21.1 (Sampling lattice and sampling points)**.** *Let the sampling interval (spacing of sampling points) be $\Delta x > 0$. The* **sampling lattice** *is*

$$\Lambda_{\Delta x} := \Delta x\,\mathbb{Z} = \{n\Delta x : n \in \mathbb{Z}\},$$

and its elements are called **sampling points** *(sampling instants). The* **sampling operator** $\mathcal{S}_{\Delta x}$ *maps a continuous signal to a sequence*

$$f := \mathcal{S}_{\Delta x}x, \qquad f(n) := x(n\Delta x), \quad n \in \mathbb{Z}.$$

The sequence f is called the **sampling sequence** *of x.*

Definition 21.2 (Sampling rate, Nyquist frequency and sampling duration)**.** *We retain the notation of Definition 21.1.*

(i) *The sampling rate (sampling frequency) ν_s and the angular sampling rate ω_s are defined by*

$$\nu_s := \frac{1}{\Delta x}, \qquad \omega_s := 2\pi\nu_s = \frac{2\pi}{\Delta x};$$

(ii) *the highest frequency that cannot be distinguished after sampling is called the* **Nyquist frequency**

$$\nu_N := \frac{\nu_s}{2}, \qquad \omega_N := \frac{\omega_s}{2} = \frac{\pi}{\Delta x};$$

(iii) *a finite record retains only the M sampling points $n = 0, 1, \dots, M-1$, whose* **sampling duration** *(record length) is*

$$T := M\Delta x,$$

M is called the **number of sampling points** *or the* **window length**, *and the corresponding time interval is called the* **sampling window**.

Remark 21.3 (Irreplaceability of the three elements). The sampling points, the sampling rate and the sampling duration are three **independent** design parameters of discrete sampling approximation, each controlling a class of properties: Δx determines the range of distinguishable frequencies (and hence aliasing), $T = M\Delta x$ determines the accuracy of frequency resolution

$$\Delta\nu = \frac{1}{T}, \qquad \Delta\omega = \frac{2\pi}{T},$$

whilst M determines the computational cost and the statistical stability. The three cannot substitute for one another: raising ν_s cannot improve the frequency resolution, and lengthening T cannot eliminate aliasing.

Definition 21.3 (Discrete periodic function). Let $f : \mathbb{Z} \to V$. If there exists $T \in \mathbb{N}^+$ such that

$$f(n+T) = f(n), \qquad \forall n \in \mathbb{Z},$$

then f is called a **discrete periodic function** and T a **period** of f. The set of all periods of f,

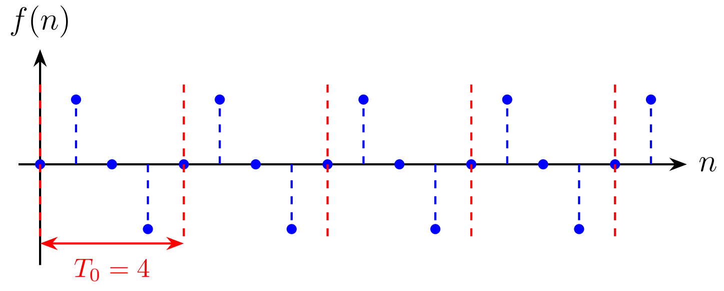
$$\mathsf{Group}_{\mathsf{period}}(f) := \{T \in \mathbb{Z} : f(n+T) = f(n),\ \forall n \in \mathbb{Z}\}$$

is a subgroup of $\mathbb{Z}$, so there exists a unique $T_0 \in \mathbb{N}$ with $\mathsf{Group}_{\mathsf{period}}(f) = T_0\mathbb{Z}$ (see Definition 21.4). When f is a constant function, $\mathsf{Group}_{\mathsf{period}}(f) = \mathbb{Z}$, and in this case we agree that $T_0 = 1$.

Definition 21.4 (Discrete fundamental period and fixed-length scale). Let $f : \mathbb{Z} \to V$ be a non-constant discrete periodic function. Its least positive period

$$T_0 := \min\{T \in \mathbb{N}^+ : f(n+T) = f(n),\ \forall n \in \mathbb{Z}\}$$

is called the **discrete fundamental period** of f. T_0 is a generator of the period subgroup $\mathsf{Group}_{\mathsf{period}}(f) \subset \mathbb{Z}$; it folds the infinite time axis $\mathbb{Z}$ into the finite cyclic group $\mathbb{Z}/T_0\mathbb{Z}$, and therefore acts as a **fixed-length scale** in discrete modelling: the lattice spacing of the frequency-domain harmonic lattice $\{2\pi k/T_0\}_{k=0}^{T_0-1}$ of the DFT is completely determined by T_0.



Discrete periodic function: the fundamental period $T_0 = 4$ folds $\mathbb{Z}$ into $\mathbb{Z}/4\mathbb{Z}$

Figure 21.2: Discrete periodic function and fundamental period: the sequence repeats every $T_0 = 4$ points, and the infinite time axis is folded into the finite cyclic group $\mathbb{Z}/4\mathbb{Z}$.

Proposition 21.1 (Algebraic properties of the sampling operator)**.** The sampling operator $\mathcal{S}_{\Delta x}$ has the following properties:

(i) **Linearity**: $\mathcal{S}_{\Delta x}(\alpha x + \beta y) = \alpha\, \mathcal{S}_{\Delta x} x + \beta\, \mathcal{S}_{\Delta x} y$;

(ii) **Pointwise multiplication**: $\mathcal{S}_{\Delta x}(x \cdot y) = (\mathcal{S}_{\Delta x} x) \cdot (\mathcal{S}_{\Delta x} y)$;

(iii) **Commutation with lattice translations**: for $a = m\Delta x$ $(m \in \mathbb{Z})$ we have $\mathcal{S}_{\Delta x}\big(x(\cdot - a)\big) = f(\cdot - m)$, where $f = \mathcal{S}_{\Delta x} x$;

(iv) **Non-commutation with scaling**: in general $\mathcal{S}_{\Delta x}\big(x(\lambda\,\cdot)\big) \neq$ the decimation of the sampling sequence, unless λ is compatible with the sampling lattice (that is, $\lambda \Delta x / \Delta x' \in \mathbb{Q}$; see Section 21.4).

Proof. (i)(ii) Pointwise evaluation preserves linearity and multiplication. (iii) This follows from $x(n\Delta x - m\Delta x) = x\big((n-m)\Delta x\big) = f(n-m)$. (iv) Take $x(t) = e^{i\omega t}$ with $\lambda \neq 1$: then $\mathcal{S}_{\Delta x}\big(x(\lambda\cdot)\big)(n) = e^{i\lambda\omega n\Delta x}$, whose 'unit sampling frequency' is determined by $\lambda\omega\Delta x$ and in general no longer lies on the original harmonic lattice; the two are indistinguishable only when $\lambda\omega\Delta x$ and $\omega\Delta x$ differ by an integer multiple, and this is precisely aliasing (Section 21.4). ☐

Property 21.1 (The adjoint of the sampling operator)**.** Let $\mathcal{S}_{\Delta x} : L^2(\mathbb{R}) \to \ell^2(\mathbb{Z})$ be the sampling operator. Then its adjoint operator

$$\mathcal{S}^*_{\Delta x} : \ell^2(\mathbb{Z}) \to L^2(\mathbb{R}), \qquad \mathcal{S}^*_{\Delta x}(\{f_n\}) = \sum_{n\in\mathbb{Z}} f_n \operatorname{sinc}\Big(\frac{t - n\Delta x}{\Delta x}\Big)$$

is the ideal low-pass interpolation operator.

Proof. For arbitrary $x \in L^2(\mathbb{R})$ and $\{f_n\} \in \ell^2(\mathbb{Z})$,

$$\langle \mathcal{S}_{\Delta x} x, \{f_n\}\rangle_{\ell^2} = \sum_n x(n\Delta x)\overline{f_n},$$

whilst

$$\langle x, \mathcal{S}^*_{\Delta x}\{f_n\}\rangle_{L^2} = \int_{\mathbb{R}} x(t)\overline{\sum_n f_n \operatorname{sinc}\Big(\frac{t-n\Delta x}{\Delta x}\Big)}\, dt = \sum_n \overline{f_n} \int_{\mathbb{R}} x(t) \operatorname{sinc}\Big(\frac{t-n\Delta x}{\Delta x}\Big)\, dt.$$

By the interpolation property of the sinc function, $\int x(t) \operatorname{sinc}((t - n\Delta x)/\Delta x) dt = x(n\Delta x)$ (which holds for band-limited signals), the two expressions are equal. ☐

Definition 21.5 (Normalised period ratio)**.** Let x have (fundamental) period T_x and be non-constant, and let the sampling interval be Δx. The quantity

$$\rho := \frac{T_x}{\Delta x}$$

is called the **normalised period ratio**, that is, the number of sampling points contained in one period (which need not be an integer).

Proposition 21.2 (Periodicity and commensurability of the sampling sequence)**.** Let the continuous signal x have fundamental period T_x and Fourier coefficients not all zero, let the sampling interval be $\Delta x > 0$, let the sampling sequence be $f(n) = x(n\Delta x)$, and let the normalised period ratio be $\rho = T_x/\Delta x$. Then

(i) if $\rho \in \mathbb{N}^+$, then f is a discrete periodic function with period $T = \rho$;

(ii) if $\rho = p/q$ ($p, q \in \mathbb{N}^+$, in lowest terms), then f is a discrete periodic function with period $T = p$, and $T_0 = p$;

(iii) if $\rho \notin \mathbb{Q}$, then f is not a discrete periodic function, that is, $\mathsf{Group}_{\mathsf{period}}(f) = \{0\}$.

In other words: **the sampling sequence has a non-trivial periodic structure if and only if the period of the signal is commensurable with the sampling interval.**

Proof. (i) This follows from $f(n+\rho) = x\big((n+\rho)\Delta x\big) = x(n\Delta x+\rho\Delta x) = x(n\Delta x+T_x) = f(n)$.

(ii) Since $p\Delta x = p \cdot \frac{T_x}{\rho} = pT_x \cdot \frac{q}{p} = qT_x$, we have $f(n+p) = x(n\Delta x + qT_x) = f(n)$. We now prove minimality: suppose f has period $T \in \mathbb{N}^+$; then from the Fourier expansion $x(t) = \sum_k c_k e^{i2\pi kt/T_x}$ of x we obtain

$$f(n) = \sum_k c_k e^{i2\pi kn\Delta x/T_x} = \sum_k c_k e^{i2\pi kn/\rho}.$$

Let k_0 be the least non-zero integer with $c_{k_0} \neq 0$ (it exists because x is non-constant). From $f(n+T) = f(n)$ we get $e^{i2\pi k_0 T/\rho} = 1$, that is, $k_0T/\rho \in \mathbb{Z}$, so $\rho = Tk_0/m$ (for some $m \in \mathbb{Z}$) is rational. When $\rho = p/q$ is in lowest terms, $pT_x = q\Delta x$, and $T\Delta x$ must be an integer multiple of T_x; write $T\Delta x = mT_x$, so that $T = mq/p$, and by coprimality $p \mid m$; hence T is an integer multiple of p, and the least positive period is p.

(iii) Suppose f has period $T \in \mathbb{N}^+$; then the argument of (ii) yields $k_0T/\rho \in \mathbb{Z}$, hence $\rho = Tk_0/m \in \mathbb{Q}$, a contradiction. Therefore $\mathsf{Group}_{\mathsf{period}}(f) = \{0\}$. □

Corollary 21.1 (Conversion between the sampling period and the signal period). Under the hypotheses of Proposition 21.2 (ii), the discrete fundamental period $T_0 = p$ and the fundamental period T_x of the signal satisfy

$$T_0\Delta x = qT_x, \qquad \rho = \frac{T_x}{\Delta x} = \frac{p}{q} = \frac{T_0}{q},$$

that is, 'p sampling points cover q signal periods'. This is precisely the period-fidelity condition of the discretisation functor.

Example 21.1 (Digital signals). For an audio interface with sampling rate $\nu_s = 48$ kHz, the sampling interval is $\Delta x = 1/48000$ s. For the note A ($\nu_0 = 440$ Hz) we have $\rho = \nu_s/\nu_0 = 109.0909\ldots \notin \mathbb{Q}$: a single tone can never contain an integral number of periods within any finite record, and leakage is inevitable (Section 21.3). For $\nu_0 = 480$ Hz, however, $\rho = 100 \in \mathbb{N}^+$, and if $M = 100k$ ($k \in \mathbb{N}^+$) then the window contains exactly an integral number of periods and the leakage is zero.

Example 21.2 (Discrete clock). A mechanical pendulum swings with period $T_x = 2$ s and the sampling interval is $\Delta x = 0.5$ s, so $\rho = 4 \in \mathbb{N}^+$ and the fundamental period of the sampling sequence is $T_0 = 4$. If instead $\Delta x = 0.75$ s, then $\rho = 8/3$ and $T_0 = 8$: 8 sampling points (6 s) are needed to recover one complete period.

Example 21.3 (Sensor grid sampling). A spatial periodic structure (such as a grating) has period $T_x = 10\,\mu$m and the pixel spacing is $\Delta x = 3\,\mu$m, so $\rho = 10/3$ is rational and the fundamental period of the sampling sequence is $T_0 = 10$ (that is, the image repeats after $30\,\mu$m). If the pixel spacing $\Delta x = \sqrt{2}\,\mu$m is incommensurable with the period, then the image does not repeat along the spatial direction, and only a statistical description is possible.

21.1.1 Scale-dependent sampling: observer scale and sampling density

The sampling rate $\nu_s = 1/\Delta x$ above is **fixed**. When sampling is performed by an observer, the sampling density also depends on the scale of the observer and its distance to the source. This section generalises the fixed sampling rate of Definition 21.1 and Definition 21.2 to a **scale-dependent sampling rate**, as the sampling-side interface of the observer line of the 'Scalars and Vectors' chapter. This model was originally stated in the 'Scale-dependent sampling model' section of the 'Scalars and Vectors' chapter; it is now assigned here, and that place is changed to a reference.

Definition 21.6 (Scale-dependent sampling function). Let the observer scale be σ, the observation distance be r, the reference sampling rate be R_0 and the reference number of sampling points be N_0. Define the sampling rate and the number of sampling points by

$$R(r,\sigma) = R_0 \cdot \frac{\sigma}{r^2+\sigma}, \qquad N(r,\sigma) = N_0 \cdot \frac{\sigma}{r^2+\sigma}.$$

This model describes the joint influence of the observation distance and the observation scale on the directional sampling density.

The parameter ranges are part of the definition, and outside these ranges the corresponding function is undefined: we require $\sigma > 0$, $r \geq 0$, $R_0 > 0$, $N_0 > 0$. When $r = 0$ we have $R(0,\sigma) = R_0$ and $N(0,\sigma) = N_0$, and the model degenerates to the limiting case 'the observer lies on the source'; $\sigma = 0$ would give $R \equiv 0$ (the observer scale is zero, so no information whatever is distinguishable), and then the criterion of Theorem 21.1 holds identically, but the model itself also loses its content. Here R has the same dimension as the sampling rate ν_s of Definition 21.2, and N is the number of sampling points M of that definition.

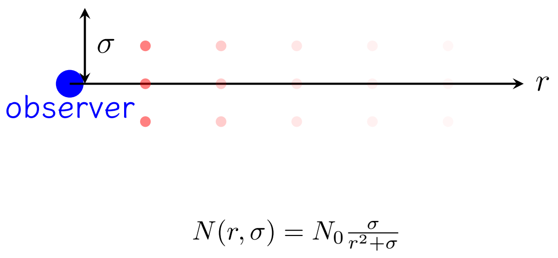


Figure 21.3: Scale-dependent sampling model. The density of sampling points decreases as the distance r grows and increases as the observer scale σ grows.

Property 21.2 (The monotonicity of sampling). (i) For fixed $\sigma > 0$, R and N are strictly decreasing functions of the distance r.

(ii) For fixed $r \geq 0$, R and N are strictly increasing functions of the scale σ.

Proof. (i)

$$\frac{\partial R}{\partial r} = R_0\sigma \cdot \frac{-2r}{(r^2+\sigma)^2} < 0 \quad (r > 0).$$

(ii)

$$\frac{\partial R}{\partial \sigma} = R_0 \cdot \frac{r^2}{(r^2+\sigma)^2} \geq 0.$$

The same holds for N. ☐

Theorem 21.1 (The invisibility criterion). *A source is* **completely invisible** *to an observer if and only if $N(r,\sigma) < 1$. The visibility threshold distance is*

$$r_{\text{max}} = \sqrt{(N_0-1)\sigma}.$$

Proof. Solving $N_0 \cdot \frac{\sigma}{r^2+\sigma} < 1$ gives

$$N_0\sigma < r^2 + \sigma \iff r^2 > (N_0-1)\sigma \iff r > \sqrt{(N_0-1)\sigma}.$$

Hence the threshold distance is $r_{\text{max}} = \sqrt{(N_0-1)\sigma}$. ☐

Remark 21.4 (Relation to the three sampling parameters). Scale-dependent sampling turns the three fixed parameters of Remark 21.3 into functions of the state: the sampling interval $\Delta x \mapsto 1/R(r,\sigma)$ becomes coarser as the distance grows and finer as the scale grows; the number of sampling points $M \mapsto N(r,\sigma)$ varies in the same direction; and the sampling duration accordingly becomes $T = N(r,\sigma)/R(r,\sigma)$. Hence the proposition 'the observer scale determines the range of distinguishability' acquires a precise quantitative form within the sampling framework of this chapter: when $N(r,\sigma) < 1$ no information whatever is distinguishable (Theorem 21.1), and when $R(r,\sigma)$ falls below twice the signal frequency aliasing occurs (Section 21.4).

21.2 Epicycle groups: rotational decomposition of periodic signals

This section introduces the **epicycle group** (epicycle system). It unifies the Fourier series, the discrete phase operator and the congruence angle as 'the superposition of rotational components'.

Definition 21.7 (A single epicycle). An **epicycle** is the following mapping:

$$e_{A,\omega,\phi}(t) := A\, e^{i(\omega t+\phi)}, \qquad t \in \mathbb{R},$$

where $A \geq 0$ is called the **radius** (amplitude), $\omega \in \mathbb{R}$ the **angular velocity** (frequency), and $\phi \in \mathbb{R}/2\pi\mathbb{Z}$ the **initial phase**. If $A = 0$, the epicycle degenerates to zero.

Definition 21.8 (Epicycle group)**.** *Let the index set J be at most countable. The family*

$$\mathcal{E} := \{(A_j, \omega_j, \phi_j)\}_{j\in J}$$

is called an **epicycle group** *if the composite signal*

$$x_{\mathcal{E}}(t) := \sum_{j\in J} A_j e^{i(\omega_j t+\phi_j)}$$

converges in the specified function space. If J is finite, $\mathcal{E}$ is called a **finite epicycle group**. The set

$$\Omega(\mathcal{E}) := \{\omega_j : j \in J\}$$

is called the **frequency set**.

Proposition 21.3 (The group with periodic orbits of an epicycle group)**.** *Let $\mathcal{E} = \{(A_j, \omega_j, \phi_j)\}_{j\in J}$ be a non-zero epicycle group such that $x_{\mathcal{E}}$ is non-constant. If the frequency set $\Omega(\mathcal{E})$ generates a discrete subgroup, write*

$$\omega_0 := \min\{\omega > 0 : \omega \in \textstyle\sum_j \mathbb{Z}\omega_j\} = \gcd\{\omega_j \neq 0\}$$

(that is, all $\omega_j/\omega_0 \in \mathbb{Z}$); then $x_{\mathcal{E}}$ has period $T_x = 2\pi/|\omega_0|$, and its group with periodic orbits is

$$\mathsf{Group}_{\mathsf{period}}(x_{\mathcal{E}}) = T_x\mathbb{Z}.$$

The least positive period is $2\pi/|\omega_0|$ (an integer fraction of it when $\omega_0 \notin \Omega$); when all $\omega_k = 0$ there is no least positive period.

If the subgroup generated by $\Omega(\mathcal{E})$ in $\mathbb{R}$ is not discrete, then in general $x_{\mathcal{E}}$ has no non-trivial period.

Proof. If $\omega_j = k_j\omega_0$ with $k_j \in \mathbb{Z}$, then

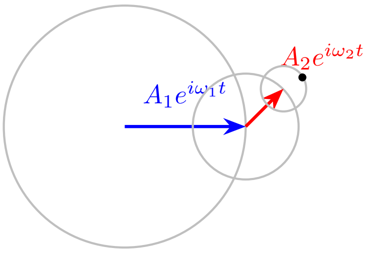
$$x_{\mathcal{E}}(t + T_x) = \sum_j A_j e^{i(k_j\omega_0(t+2\pi/\omega_0)+\phi_j)} = \sum_j A_j e^{i(k_j\omega_0 t+\phi_j+2\pi k_j)} = x_{\mathcal{E}}(t).$$

Hence T_x is a period. The least positive period is determined by the greatest common divisor $\omega_0 = \gcd\{\omega_j \neq 0\}$ of the non-zero frequencies: the least positive period is $2\pi/|\omega_0|$ (an integer fraction of it when $\omega_0 \notin \Omega$); when all $\omega_k = 0$ there is no least positive period. If the generated subgroup is not discrete, then the set of all non-zero periods has no least positive generator, and usually $\mathsf{Group}_{\mathsf{period}}(x_{\mathcal{E}}) = \{0\}$. □

Proposition 21.4 (Sampling and commensurability of epicycle groups)**.** *Let the epicycle group $\mathcal{E}$ have frequency set Ω, let the sampling interval be $\Delta x > 0$, and let the sampling sequence be*

$$f(n) = x_{\mathcal{E}}(n\Delta x) = \sum_j A_j e^{i(\omega_j n\Delta x+\phi_j)}.$$

If there exists $\omega_0 > 0$ with all $\omega_j/\omega_0 \in \mathbb{Z}$, and the normalised period ratio $\rho = 2\pi/(\omega_0\Delta x)$ is rational, then f is a discrete periodic sequence; if $\rho \notin \mathbb{Q}$, then f is aperiodic.



Epicycle group: the geometric superposition of the periodic signal $x_{\mathcal{E}}(t) = \sum_j A_j e^{i(\omega_j t+\phi_j)}$

Figure 21.4: Epicycle group: the geometric realisation of the Fourier series. Each component is a uniformly rotating circle, and the final position of the moving point is the sum of the component vectors.

Proof. By Proposition 21.3, the period of the continuous signal is $T_x = 2\pi/\omega_0$. Substituting into Proposition 21.2 gives the result. ☐

Example 21.4 (A Fourier series is an epicycle group). The continuous periodic function $x(t) = \sum_{k\in\mathbb{Z}} c_k e^{i2\pi kt/T_x}$ corresponds to the epicycle group

$$\mathcal{E} = \{(|c_k|,\ 2\pi k/T_x,\ \arg c_k)\}_{k\in\mathbb{Z}}.$$

Hence the Fourier series expansion may be understood geometrically as: a family of epicycles whose angular velocities are integer multiples of the fundamental frequency superpose into the original periodic signal.

Example 21.5 (The discrete Fourier transform is a finite epicycle group). The DFT of a discrete periodic sequence $f : \mathbb{Z}/T_0\mathbb{Z} \to \mathbb{C}$ of length T_0,

$$\hat{f}(k) = \sum_{n=0}^{T_0-1} f(n)e^{-i2\pi kn/T_0}$$

decomposes f into the finite epicycle group

$$\mathcal{E}_f = \left\{\left(\frac{|\hat{f}(k)|}{T_0},\ \frac{2\pi k}{T_0},\ \arg \hat{f}(k)\right)\right\}_{k=0}^{T_0-1}.$$

This is exactly the geometric interpretation of the discrete phase operator $\mathcal{O}_k^{(T_0)} = e^{i2\pi k/T_0}$ of this chapter.

Remark 21.5 (Epicycle groups and congruence angles). The phase of each component of an epicycle group over a subperiod T is precisely the congruence angle

$$\langle \omega_j t + \phi_j \rangle_T = \omega_j t + \phi_j \bmod \frac{2\pi}{T}.$$

Hence a multi-period joint problem may be regarded as: an epicycle group has to close simultaneously over several subperiods. Intercalation is the adjustment of the observation window so that these congruence-angle errors are pressed down simultaneously.

Table 21.1: Metric parameters and components of discrete sampling approximation

Parameter	Definition	Time-domain meaning	Frequency-domain meaning
Sampling interval Δx	spacing of adjacent sampling points	time resolution	upper limit of distinguishability $\omega_N = \pi/\Delta x$
Sampling rate ν_s	$1/\Delta x$	density of sampling points	the scale of the aliasing criterion
Sampling duration T	$M\Delta x$	record length	frequency resolution $\Delta\omega = 2\pi/T$
Period ratio ρ	$T_x/\Delta x$	sampling points in one period	position of the harmonic lattice
Offset δ	$\|M\rho\|_{\mathbb{Z}}$	deviation of the number of periods in the window	leakage strength
Fundamental period T_0	generator of $\mathrm{Group}_{\mathrm{period}}(f)$	repetition length of the sequence	frequency-domain lattice spacing $2\pi/T_0$
Epicycle group $\mathcal{E}$	$\{(A_j, \omega_j, \phi_j)\}$	superposition of rotational components	set of Fourier modes
Congruence angle $\langle\theta\rangle_T$	θ mod $2\pi/T$	phase modulo a subperiod	position of the spectral line on the subperiod
Congruence-angle non-closure δ_T	$\frac{2\pi}{T}\|M\rho\|_{\mathbb{Z},T}$	phase deviation on the subperiod	leakage strength on the subperiod

21.3 Finite windows, the Dirichlet kernel and spectral leakage

This section gives the complete mechanism of spectral leakage. The conclusion is: **the root of leakage is that the number of periods inside the window is not an integer**; its mathematical carrier is the sidelobes of the Dirichlet kernel. Seen from the epicycle group, leakage is the superposition of sidelobes caused by the phase of each epicycle failing to close at the endpoints of the window.

Definition 21.9 (Spectral leakage)**.** Let the (true) spectrum of x be $\hat{x}$. Suppose only a finite record of length $T = M\Delta x$,

$$x_T(t) := x(t)\,\mathbf{1}_{[0,T)}(t),$$

is observed, and its samples are subjected to a DFT. If $\hat{x}_T$ exhibits non-zero energy outside the spectral support of $\hat{x}$, then **spectral leakage** is said to occur. The amount of leakage is measured by the proportion of energy outside the main lobe,

$$\eta := \frac{\displaystyle\sum_{k \notin K_{\text{main lobe}}} |\hat{f}(k)|^2}{\displaystyle\sum_{k} |\hat{f}(k)|^2}$$

where $K_{\text{main lobe}}$ is the set of frequency points covered by the main lobe (Definition 21.11).

Definition 21.10 (Dirichlet kernel)**.** For an integer $M \geq 0$ define the function

$$D_M(\theta) := \sum_{m=-M}^{M} e^{im\theta}, \qquad \theta \in \mathbb{R},$$

called the **Dirichlet kernel** (also the Dirichlet function). Correspondingly, the normalised spectrum of the rectangular window of length T is called the **window kernel**:

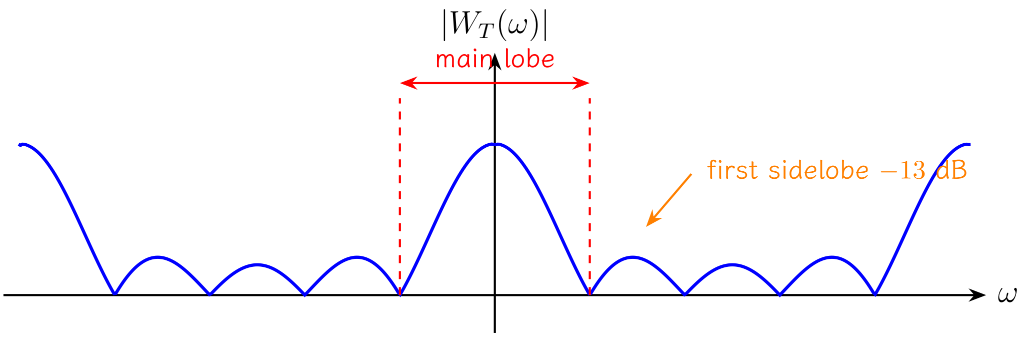
$$W_T(\omega) := \frac{1}{T}\sum_{n=0}^{T-1} e^{-in\omega} = \frac{e^{-i(T-1)\omega/2}}{T} \cdot \frac{\sin(T\omega/2)}{\sin(\omega/2)} = \frac{e^{-i(T-1)\omega/2}}{T}\, D_{T-1}(\omega),$$

whose modulus $|W_T(\omega)| = \frac{1}{T}\left|\frac{\sin(T\omega/2)}{\sin(\omega/2)}\right|$ depends only on the frequency.

Lemma 21.1 (Closed form of the Dirichlet kernel)**.** *For every $\theta \notin 2\pi\mathbb{Z}$,*

$$D_M(\theta) = \frac{\sin\left((M+\frac{1}{2})\theta\right)}{\sin(\theta/2)},$$

and $D_M(0) = 2M+1$. In particular $|D_M(\theta)| \leq \dfrac{1}{|\sin(\theta/2)|}$.



Window kernel $|W_T(\omega)|$ of the rectangular window:
main-lobe width $4\pi/T$, sidelobes decaying as $|\omega|^{-1}$

Figure 21.5: Dirichlet kernel/window kernel of the rectangular window: main-lobe and sidelobe structure. The main-lobe width is $4\pi/T$, the first sidelobe is about -13 dB, and the sidelobe envelope decays as $|\omega|^{-1}$.

Proof. Multiplying $D_M(\theta)$ by $e^{-iM\theta}$ gives the geometric series

$$\sum_{r=0}^{2M} e^{ir\theta} = \frac{e^{i(2M+1)\theta}-1}{e^{i\theta}-1} = \frac{e^{i2M\theta}-e^{-i\theta}}{1-e^{-i\theta}},$$

and substituting $1-e^{i\theta} = -2ie^{i\theta/2}\,\mathsf{sin}(\theta/2)$ and taking moduli gives the closed form; $D_M(0) = 2M+1$ follows directly from the definition. □

Lemma 21.2 (Orthogonality of the window kernel and the position of the main lobe). We retain Definition 21.10. Then

$$W_T\Big(\frac{2\pi k}{T}\Big) = \begin{cases} 1, & k \equiv 0 \ (\mathsf{mod}\ T), \\ 0, & \text{other integers } k, \end{cases} \qquad \sum_{n=0}^{T-1} e^{-i2\pi kn/T} = T\,\delta_{k,0 \bmod T}.$$

Moreover $|W_T(\omega)|$ attains its maximum 1 at $\omega = 0$ and first vanishes at $\omega = \pm 2\pi/T$.

Proof. For $k \equiv 0$ the sum is T; for $k \not\equiv 0$ the ratio of the geometric series is $e^{-i2\pi k/T} \neq 1$, and the sum is $\frac{1-e^{-i2\pi k}}{1-e^{-i2\pi k/T}} = 0$. The maximum and the zeros are given by the numerator $\mathsf{sin}(T\omega/2)$ of Lemma 21.1, which vanishes at $\omega = 2\pi j/T$ and attains its largest modulus near $\omega = 0$. □

Proposition 21.5 (Leakage formula for the truncation of a single tone). Let $f(n) = Ae^{i(\omega_0 n+\phi)}$ $(A>0,\ \phi\in\mathbb{R})$, and take a DFT with a rectangular window of length M:

$$\hat{f}(k) = \sum_{n=0}^{M-1} f(n)\,e^{-i2\pi kn/M}, \qquad k = 0,1,\ldots,M-1.$$

Then

$$\hat{f}(k) = Ae^{i\phi}\,e^{i\theta_k(M-1)/2}\,\frac{\mathsf{sin}(M\theta_k/2)}{\mathsf{sin}(\theta_k/2)}, \qquad \theta_k := \omega_0 - \frac{2\pi k}{M},$$

and hence

$$|\hat{f}(k)| = A\left|\frac{\mathsf{sin}(M\theta_k/2)}{\mathsf{sin}(\theta_k/2)}\right|.$$

Proof. Substituting f into the sum gives $\hat{f}(k) = Ae^{i\phi}\sum_{n=0}^{M-1} e^{i\theta_k n}$; applying the geometric-series summation formula $\sum_{n=0}^{M-1} e^{i\theta n} = e^{i\theta(M-1)/2}\frac{\sin(M\theta/2)}{\sin(\theta/2)}$ gives the result. ☐

Theorem 21.2 (Leakage envelope and sidelobes)**.** Let $M\omega_0/(2\pi) = \nu + \delta$, where $\nu \in \mathbb{Z}$ and $\delta \in [-1/2, 1/2]$ is the **offset** of the number of periods inside the window. Then

(i) the main lobe is located at ν, with peak

$$|\hat{f}(\nu)| = A\left|\frac{\sin(\pi\delta)}{\sin(\pi\delta/M)}\right| \xrightarrow[M\to\infty]{} A\,\frac{|\sin(\pi\delta)|}{\pi|\delta|}\cdot M;$$

(ii) for $j \neq 0$ we have $\theta_{\nu+j} = \frac{2\pi}{M}(\delta - j)$, so

$$\frac{|\hat{f}(\nu+j)|}{|\hat{f}(\nu)|} = \frac{\left|\sin\left(\frac{\pi\delta}{M}\right)\right|}{\left|\sin\left(\frac{\pi(j-\delta)}{M}\right)\right|} \xrightarrow[M\to\infty]{} \frac{|\delta|}{|j-\delta|} = O\Big(\frac{1}{|j|}\Big),$$

that is, **the sidelobe envelope decays reciprocally with the distance from the main frequency point**; the first sidelobe of the rectangular window is about -13.3 dB and decays as $|\omega|^{-1}$;

(iii) if $\delta = 0$ (the window contains exactly an integral number of periods), then $\hat{f}(k) = AMe^{i\phi}\delta_{k,\nu \bmod M}$, and **the leakage is zero**;

(iv) if $\delta \neq 0$, then every frequency point carries non-zero energy, and the proportion of energy outside the **single peak frequency bin** is

$$\eta(\delta) = 1 - \mathrm{sinc}^2(\delta) + O(M^{-1}), \qquad \mathrm{sinc}(u) := \frac{\sin(\pi u)}{\pi u},$$

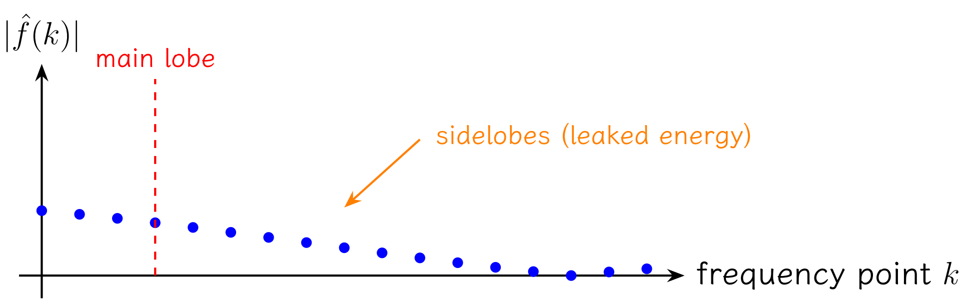
in particular $\eta(1/2) = 1 - 4/\pi^2 \approx 0.595$, that is, in the worst case about 60% of the energy lies outside the single peak frequency bin. For the main lobe of width $4\pi/T$ (Definition 21.11) the corresponding value is $\eta_{\mathrm{ML}}(1/2) = 1 - 8/\pi^2 \approx 0.189$.

Proof. (i) Substituting $k = \nu$ into Proposition 21.5 and noting that $\theta_\nu = 2\pi\delta/M$ gives $|\hat{f}(\nu)| = A\left|\frac{\sin(\pi\delta)}{\sin(\pi\delta/M)}\right|$; since $\sin(\pi\delta/M) \sim \pi\delta/M$, for $\delta \neq 0$ this grows linearly in M (the main-lobe peak is $O(M)$).

(ii) For $k = \nu + j$ ($j \neq 0$) we have $\theta_k = \frac{2\pi}{M}(\delta - j)$, and substituting into Proposition 21.5 gives $|\hat{f}(\nu+j)| = A\frac{|\sin(\pi(\delta-j))|}{|\sin(\pi(\delta-j)/M)|} = A\frac{|\sin(\pi\delta)|}{|\sin(\pi(j-\delta)/M)|}$; comparing with $|\hat{f}(\nu)| \approx AM|\sin(\pi\delta)|$ gives the stated ratio. The exact discrete value of the first sidelobe of the rectangular window differs slightly from the extremum of the continuous envelope, but its order of magnitude is -13.3 dB.

(iii) When $\delta = 0$ we have $\theta_k = 2\pi(\nu - k)/M$, and for $k \neq \nu \bmod M$ the numerator $\sin(\pi(\nu - k)) = 0$; hence only $k = \nu \bmod M$ is non-zero, with value $AMe^{i\phi}$.

(iv) By the discrete Parseval identity (Theorem 21.9), $\sum_{k=0}^{M-1}|\hat{f}(k)|^2 = M\sum_{n=0}^{M-1}|f(n)|^2 = M\cdot MA^2 = M^2A^2$. The main-lobe energy is $|\hat{f}(\nu)|^2 = A^2M^2\,\mathrm{sinc}^2(\delta) + O(M)$, so $\eta = 1 - \mathrm{sinc}^2(\delta) + O(M^{-1})$; for $|\delta| = 1/2$ we have $\mathrm{sinc}(1/2) = 2/\pi$, giving $\eta = 1 - 4/\pi^2 \approx 0.595$. ☐



Leakage distribution for offset $\delta = 0.3$: $\eta = 1 - \text{sinc}^2(0.3) \approx 0.088$

Figure 21.6: Energy distribution of spectral leakage: the main lobe is concentrated near the true frequency point, whilst the sidelobes scatter the leaked energy over all frequency points. The larger the offset δ, the stronger the leakage.

Definition 21.11 (Main lobe and sidelobes). Let the zeros of the window kernel $|W_T(\omega)|$ on $\omega > 0$ be $\omega_1 < \omega_2 < \cdots$ in order. The envelope on $[-\omega_1, \omega_1]$ is called the **main lobe**, and its **main-lobe width** is $2\omega_1$; the envelope on each interval (ω_j, ω_{j+1}) is called a **sidelobe**, its maximum is called the **sidelobe peak**, and the overall rate of decay of the sidelobe peaks with j is called the **sidelobe decay rate**. For the rectangular window, $\omega_j = 2\pi j/T$, the main-lobe width is $4\pi/T$, the first sidelobe peak is about -13.3 dB, and the sidelobe envelope decays as $|\omega|^{-1}$.

Corollary 21.2 (Integral windows have no leakage). Let $f(n) = Ae^{i(\omega_0 n+\phi)}$ with $\omega_0 = 2\pi m/T$ ($m \in \mathbb{Z}$); then the DFT with a rectangular window of length T satisfies $\hat{f}(k) = ATe^{i\phi}\delta_{k,m \bmod T}$: **when the window contains exactly an integral number of periods, the discrete spectrum has only one spectral line and the leakage is zero.**

Corollary 21.3 (Non-integral windows must leak). Let $\omega_0 \notin \frac{2\pi}{T}\mathbb{Z}$. Then $\hat{f}(k) \neq 0$ for all $k = 0, \ldots, T-1$, and the proportion of leaked energy is given by Theorem 21.2(iv).

Proposition 21.6 (Phase characterisation of leakage). In the notation of Proposition 21.5, at non-zero points the amplitude ratio of adjacent frequency points,

$$\frac{|\hat{f}(k+1)|}{|\hat{f}(k)|} = \left| \frac{\sin\left(\frac{\pi}{M}(k+1-\nu-\delta)\right)}{\sin\left(\frac{\pi}{M}(k-\nu-\delta)\right)} \right|$$

depends only on $k - \nu$ and the offset δ, whilst the phase difference

$$\arg \frac{\hat{f}(k+1)}{\hat{f}(k)} = -\frac{\pi(M-1)}{M} + \varepsilon_k, \qquad \varepsilon_k \in \{0, \pi\},$$

is independent of k, where $\varepsilon_k = \pi$ if and only if the two frequency points lie on opposite sides of the main-lobe peak (that is, the pair of frequency points straddles $\nu + \delta$). Hence: **the pattern of values of the phase difference locates the main lobe, and the amplitude ratio inverts the offset**—this is precisely the principle of **two-spectral-line interpolation** and of **phase-difference frequency estimation**.

Proof. By the closed form of Proposition 21.5, $\hat{f}(k) = Ae^{i\phi}e^{i\theta_k(M-1)/2}\frac{\sin(M\theta_k/2)}{\sin(\theta_k/2)}$. The first formula follows for the amplitude. The phase consists of the linear term $\theta_k(M-1)/2$ and the sign of the sine factor: $\theta_{k+1}-\theta_k = -2\pi/M$ gives the adjacent phase difference $-\pi(M-1)/M$, whilst $\sin(M\theta_k/2) = \sin(\pi(k-\nu-\delta))$ has opposite signs at adjacent frequency points if and only if those two points lie on opposite sides of the zero $\nu+\delta$, contributing an additional phase π; in all other cases the signs agree. Hence the phase difference takes only two possible values, independently of k, and the position at which it flips directly gives the continuous position $\nu+\delta$ of the main lobe. ☐

Proposition 21.7 (Congruence-angle characterisation of leakage). *Let $f(n) = Ae^{i\omega_0 n}$ with $\omega_0 = 2\pi\rho$ and window length M. Then the single-period leakage proportion may be written as*

$$\eta(M) = 1 - \operatorname{sinc}^2\big(\|M\rho\|_{\mathbb{Z}}\big) + O(M^{-1}),$$

where $\|M\rho\|_{\mathbb{Z}}$ is the offset of the number of periods inside the window. If a subperiod T is introduced, then when $T \mid M\rho$ the congruence-angle non-closure over that subperiod is $\delta_T = 0$, and the leakage is closed in the sense of that subperiod.

Proof. This follows from Theorem 21.2(iv); the assertion about the subperiod follows directly from Definition 21.23 and $T \mid M\rho$. ☐

Example 21.6 (Numerical magnitude of leakage). Take $f(n) = e^{i2\pi(0.25+0.5/M)n}$ (that is, $\delta = 1/2$, the main lobe falling exactly between two frequency points) and $M = 16$. By Theorem 21.2(iv) the leakage proportion is $\eta \approx 0.595$, that is, about 60% of the energy is distributed outside the main lobe; the two bins of the main lobe take about 40.7% each, so that the main lobe (two bins) holds about 81%, and the remaining sidelobes about 19% in total. If the window length is changed to $M = 20$ and $\omega_0 = 2\pi\cdot 7/20$ is taken ($\delta = 0$), then all the energy is concentrated in a single spectral line and $\eta = 0$.

Remark 21.6 (Inevitability and controllability of leakage). By Corollary 21.3, as long as the number of periods inside the window is not an integer, leakage necessarily occurs—this is the price of 'finite observation' and is independent of the sampling rate (the latter determines aliasing). But Theorem 21.2(iv) shows that the leakage strength **depends only on the offset** $\delta = \|M\rho\|_{\mathbb{Z}}$: as long as δ is pressed down, the leakage is small. This converts the problem of leakage control into a purely number-theoretic problem—**approximating the real period ratio optimally by an integral window length**—and this is precisely intercalation (Section 21.7).

21.4 Aliasing: the price of the sampling rate

The preceding section showed that leakage is determined by the **sampling duration**. This section shows that the other class of error—**aliasing**—is determined by the **sampling rate**; the two are independent and must be handled separately.

Definition 21.12 (Aliasing)**.** Let the angular sampling rate be $\omega_s = 2\pi/\Delta x$. Two angular frequencies $\omega, \omega' \in \mathbb{R}$ are said to be **aliased** if

$$\omega' \equiv \omega \pmod{\omega_s}, \qquad \text{that is,} \qquad \omega' = \omega + m\omega_s, \quad m \in \mathbb{Z}.$$

The functions $e^{i\omega t}$ and $e^{i\omega' t}$ take exactly the same values on the sampling lattice $\Lambda_{\Delta x}$, so the two are **indistinguishable** after sampling; this phenomenon is called **aliasing** (frequency folding).

Proposition 21.8 (The folding map)**.** *Define the folding map*

$$\mathcal{A}_{\omega_s} : \mathbb{R} \to \Big(-\frac{\omega_s}{2}, \frac{\omega_s}{2}\Big], \qquad \mathcal{A}_{\omega_s}(\omega) := \omega - m\,\omega_s, \quad m := \Big\lfloor \frac{\omega}{\omega_s} + \frac{1}{2} \Big\rfloor.$$

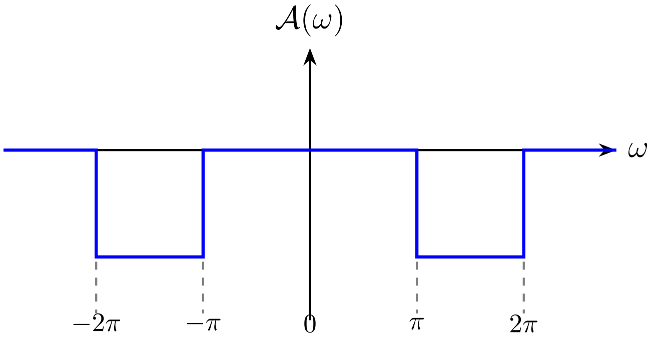
Then

(i) $\mathcal{A}_{\omega_s}(\omega)$ *is aliased with* ω, *and* $|\omega - \mathcal{A}_{\omega_s}(\omega)| \le \omega_s/2$;

(ii) $\mathcal{A}_{\omega_s}$ *is surjective but not injective, and each preimage class is exactly* $\{\omega + m\omega_s\}_{m\in\mathbb{Z}}$;

(iii) $\mathcal{A}_{\omega_s}$ *is a translation on each Nyquist band* $\big(\frac{2k-1}{2}\omega_s, \frac{2k+1}{2}\omega_s\big]$, *and on the* k*-th band it is a reflection composed with a translation (*k *alternating in parity).*

Proof. (i) By definition $\omega - \mathcal{A}(\omega) = m\omega_s \in \omega_s\mathbb{Z}$, hence aliasing; moreover m is the nearest integer, so $|m - \omega/\omega_s| \le 1/2$, that is, $|\omega - \mathcal{A}(\omega)| \le \omega_s/2$. (ii) For every $\eta \in (-\omega_s/2, \omega_s/2]$ we have $\eta = \mathcal{A}(\eta)$, so the map is surjective; if $\mathcal{A}(\omega) = \mathcal{A}(\omega')$ then $\omega - \omega' \in \omega_s\mathbb{Z}$, so each preimage class is an arithmetic progression. (iii) Within a band m is constant, so $\mathcal{A}(\omega) = \omega - k\omega_s$ is a translation; at the band boundaries m jumps once, so the images on two adjacent bands have opposite directions. ☐



Folding map $\mathcal{A}_{\omega_s}(\omega)$: all Nyquist bands are folded into $(-\pi, \pi]$

Figure 21.7: The folding map of aliasing: each Nyquist band on the frequency axis (of width $\omega_s = 2\pi$) is folded into the fundamental interval $(-\pi, \pi]$. All aliased frequencies are indistinguishable within that interval.

Theorem 21.3 (Nyquist–Shannon sampling theorem)**.** *Let $x \in L^2(\mathbb{R})$ be continuous and let its Fourier transform $\hat{x}$ satisfy the support condition*

$$\operatorname{supp} \hat{x} \subset (-\omega_N, \omega_N), \qquad \omega_N = \frac{\pi}{\Delta x},$$

then x is uniquely reconstructed from its samples $\{x(n\Delta x)\}_{n\in\mathbb{Z}}$:

$$x(t) = \sum_{n\in\mathbb{Z}} x(n\Delta x)\ \mathrm{sinc}\Big(\frac{t-n\Delta x}{\Delta x}\Big), \qquad \mathrm{sinc}(u) := \frac{\sin(\pi u)}{\pi u},$$

and the series converges in L^2 and in the uniform norm.

Proof. By the band-limited condition, the terms of the frequency periodisation

$$P(\omega) := \frac{1}{\Delta x}\sum_{m\in\mathbb{Z}} \hat{x}\Big(\omega + \frac{2\pi m}{\Delta x}\Big)$$

have mutually non-overlapping supports, so P agrees with $\hat{x}/\Delta x$ on $(-\omega_N, \omega_N)$. On the other hand, P is the discrete-time Fourier transform (of period $2\pi/\Delta x$) of the sampling sequence $\{x(n\Delta x)\}$. Restricting P to $(-\omega_N, \omega_N)$ and setting it to zero elsewhere, and multiplying by the rectangular window W (whose inverse transform is $\mathrm{sinc}(t/\Delta x)/\Delta x$), gives the reconstruction formula; the convergence of the series follows from $x \in L^2$ and the orthogonality of sinc. ☐

Lemma 21.3 (Poisson-summation proof of the sampling theorem)**.** The spectrum of the sampling sequence can be written, using the Poisson summation formula, as

$$\sum_{n\in\mathbb{Z}} x(n\Delta x)e^{-i\omega n\Delta x} = \frac{1}{\Delta x}\sum_{m\in\mathbb{Z}} \hat{x}(\omega + 2\pi m/\Delta x).$$

Proof. By the inverse Fourier transform $x(n\Delta x) = \frac{1}{2\pi}\int \hat{x}(\omega')e^{in\Delta x\omega'}d\omega'$; substituting into the left-hand side gives

$$\sum_{n} x(n\Delta x)e^{-i\omega n\Delta x} = \frac{1}{2\pi}\sum_{n}\int \hat{x}(\omega')e^{in\Delta x(\omega'-\omega)}d\omega'.$$

Using the Poisson summation formula $\sum_n e^{in\Delta x\theta} = \frac{2\pi}{\Delta x}\sum_m \delta(\theta - 2\pi m/\Delta x)$ and interchanging the sum and the integral gives the result. ☐

Corollary 21.4 (Irreversibility of aliasing)**.** *Let $\omega \neq \omega'$ with $\omega' \equiv \omega \pmod{\omega_s}$. Then $\mathcal{S}_{\Delta x}e^{i\omega\cdot} = \mathcal{S}_{\Delta x}e^{i\omega'\cdot}$, so no algorithm that depends only on the samples can distinguish the two; the sampling operator is irreversible on each aliasing class. Anti-aliasing can only be carried out before it, that is, by* **analogue low-pass filtering before sampling***.*

Definition 21.13 (Anti-aliasing filter)**.** *Let the spectral support of the signal x exceed the Nyquist frequency. An* **anti-aliasing filter** *is a low-pass filter h with cut-off frequency ω_N: before sampling one first forms the convolution $x_{\mathrm{filt}} = h * x$ and then samples with spacing Δx. The filtered signal satisfies the band-limited condition, so that the sampling theorem applies.*

Property 21.3 (Ideal properties of the anti-aliasing filter)**.** *An ideal anti-aliasing filter equals a rectangular window in the frequency domain:*

$$\hat{h}(\omega) = \mathbf{1}_{(-\omega_N,\omega_N)}(\omega).$$

Then the filtered signal is undistorted within the band and completely removed outside it, so that the sampling theorem holds and aliasing is strictly zero.

Proof. By the convolution theorem $\widehat{x_{\text{filt}}} = \hat{h}\hat{x}$, so $\operatorname{supp}\widehat{x_{\text{filt}}} \subseteq (-\omega_N, \omega_N)$, which satisfies the band-limited condition of the sampling theorem. Within the band $\hat{h} = 1$ guarantees that the signal is unchanged. ☐

Remark 21.7 (The difference between aliasing and leakage). *Aliasing and leakage are two independent sources of error, easily confused, and must be strictly distinguished:*

(i) **Source**: *aliasing comes from* **discretisation** *(finite sampling rate, $\Delta x > 0$); leakage comes from* **truncation** *(finite sampling duration, $T < \infty$).*

(ii) **Mathematical mechanism**: *aliasing is the* **folding of frequency modulo** ω_s *(Proposition 21.8); leakage is the* **convolutional spill-over** *of the true spectrum with the window kernel (Theorem 21.2).*

(iii) **Means of control**: *aliasing is dealt with by raising ν_s or by a preposed anti-aliasing filter; leakage is dealt with by lengthening T or by* **intercalation** *(Section 21.7).*

(iv) **Reversibility**: *aliasing is irreversible (Corollary 21.4); leakage is reversible—as long as the offset $\delta = 0$, the leakage disappears completely (Corollary 21.2).*

Example 21.7 (A numerical example of aliasing). Take $\Delta x = 1$ s (so $\omega_s = 2\pi$ rad/s). The signals $x_1(t) = \cos(2\pi \cdot 0.9\,t)$ and $x_2(t) = \cos(2\pi \cdot 0.1\,t)$ have exactly the same samples:

$$\cos(2\pi \cdot 0.9\,n) = \cos(2\pi \cdot 0.1\,n), \qquad n \in \mathbb{Z},$$

because $0.9 \equiv -0.1 \pmod 1$ and the cosine is an even function. In the spectrum the two lie at 0.9 and 0.1 cycles per second respectively, and after sampling both fall at ± 0.1 cycles per second. If only the samples are observed, it is impossible to decide whether the true frequency is 0.9 or 0.1—this is precisely the practical meaning of aliasing.

21.5 Upper and lower approximation

By Theorem 21.2(iv), the leakage strength depends only on the offset $\delta = \|M\rho\|_{\mathbb{Z}}$. Hence leakage control is equivalent to: **approximating the real period ratio ρ by an integral window length M**. This section establishes the two basic tools of this approximation—the upper approximation and the lower approximation—and gives the optimality of continued fractions.

Definition 21.14 (Upper and lower approximation). Let $\rho > 0$ and $M \in \mathbb{N}^+$. Define

$$\rho_M^{(-)} := \frac{\lfloor M\rho \rfloor}{M} \quad \text{(lower approximation)},$$
$$\rho_M^{(+)} := \frac{\lceil M\rho \rceil}{M} \quad \text{(upper approximation)},$$
$$\rho_M^{(\pm)} := \frac{\lfloor M\rho + \frac{1}{2} \rfloor}{M} \quad \text{(balanced approximation)},$$

and call $\delta_M := \|M\rho\|_{\mathbb{Z}} := \min_{m\in\mathbb{Z}} |M\rho - m|$ the **offset** of the window M.

Proposition 21.9 (Integrality inequalities)**.** *For every $\rho > 0$ and $M \in \mathbb{N}^+$,*

$$0 \le M\rho - \lfloor M\rho \rfloor < 1, \qquad 0 \le \lceil M\rho \rceil - M\rho < 1, \qquad \lceil M\rho \rceil - \lfloor M\rho \rfloor \in \{0,1\},$$

and $\delta_M = \min\{M\rho - \lfloor M\rho \rfloor,\ \lceil M\rho \rceil - M\rho\} \le 1/2$.

Proof. The first two formulas follow from the definition of the integer part: $M\rho - 1 < \lfloor M\rho \rfloor \le M\rho \le \lceil M\rho \rceil < M\rho + 1$; the third follows because the difference of the two is less than 2 and is an integer. The expression for the offset follows because δ_M is the distance from $M\rho$ to the nearest integer, hence $\le 1/2$. □

Proposition 21.10 (Congruence-angle formulation of the upper and lower approximation)**.** *The upper approximation $\rho_M^{(+)}$ corresponds to a positive congruence-angle error and the lower approximation $\rho_M^{(-)}$ to a negative congruence-angle error:*

$$\rho_M^{(+)} - \rho = \frac{\delta_M^+}{M}, \qquad \rho - \rho_M^{(-)} = \frac{\delta_M^-}{M},$$

where $\delta_M^\pm \ge 0$ and $\delta_M^+ + \delta_M^- = \delta_M$. The balanced approximation $\rho_M^{(\pm)}$ makes the two equal.

Proof. This follows from Definition 21.14 and Proposition 21.9. □

Theorem 21.4 (Upper and lower bounds for approximation error)**.** *Let $\rho > 0$ and $M \in \mathbb{N}^+$. Then*

(i) **Sandwiching**: *$\rho_M^{(-)} \le \rho \le \rho_M^{(+)}$, and the errors of the upper and lower approximation are respectively*

$$\rho - \rho_M^{(-)} = \frac{M\rho - \lfloor M\rho \rfloor}{M} < \frac{1}{M}, \qquad \rho_M^{(+)} - \rho = \frac{\lceil M\rho \rceil - M\rho}{M} < \frac{1}{M};$$

(ii) **Optimality of the balanced approximation**: *$|\rho - \rho_M^{(\pm)}| = \delta_M/M \le \dfrac{1}{2M}$, and for fixed M this is the smallest error among all approximations by 'integer numerator / denominator M';*

(iii) **Intercalation formulation**: *let $E_M \subseteq \{1,\dots,M\}$ be the set of intercalation positions chosen according to the balanced scheme (that is, the n-th position is intercalated if and only if $\lfloor (n+1)\rho \rfloor - \lfloor n\rho \rfloor = 1$); then*

$$\big||E_M| - M\rho\big| < 1,$$

that is, the difference between the number of intercalations and the ideal value $M\rho$ is less than one unit.

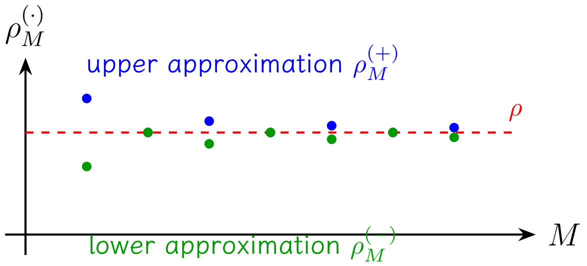
Proof. (i) Dividing $\lfloor M\rho \rfloor \le M\rho \le \lceil M\rho \rceil$ by M gives the sandwiching; the error formulas follow from the margin below 1 in Proposition 21.9.

(ii) The numerator of $\rho_M^{(\pm)} = \frac{\lfloor M\rho + 1/2 \rfloor}{M}$ is the nearest integer to $M\rho$, so $|\rho - \rho_M^{(\pm)}| = \delta_M/M$; for any integer p we have $|\rho - p/M| = |M\rho - p|/M \ge \delta_M/M$, so minimality holds.

(iii) Write $N_M := |E_M|$. Summing step by step from the definition of the balanced sequence gives

$$N_M = \sum_{n=1}^{M} \left(\lfloor (n+1)\rho \rfloor - \lfloor n\rho \rfloor \right) = \lfloor (M+1)\rho \rfloor - \lfloor \rho \rfloor,$$

and on the other hand $\lfloor (M+1)\rho \rfloor - \lfloor \rho \rfloor = M\rho + \{(M+1)\rho\} - \{\rho\}$ (where $\{\cdot\}$ denotes the fractional part), and since $0 \le \{(M+1)\rho\} < 1$ whilst $0 \le \{\rho\} < 1$ we have $\{(M+1)\rho\} - \{\rho\} < 1$, and this difference also exceeds -1, so $|N_M - M\rho| < 1$. □



The upper/lower approximations sandwich ρ
from both sides, with convergence rate $O(1/M)$

Figure 21.8: Upper approximation and lower approximation: the blue points sandwich the true value ρ (red line) from above and the green points from below, both with error $O(1/M)$. The balanced approximation takes their midpoint, with error $O(1/(2M))$.

Proposition 21.11 (Convergence of the sandwich). *For every $\rho > 0$, the sequence $\rho_M^{(-)}$ is monotonically non-decreasing and $\rho_M^{(+)}$ is monotonically non-increasing; both converge to ρ, with convergence rate $O(1/M)$:*

$$0 \le \rho - \rho_M^{(-)} < 1/M, \qquad 0 \le \rho_M^{(+)} - \rho < 1/M, \qquad \rho_M^{(-)} \le \rho_M^{(\pm)} \le \rho_M^{(+)}.$$

Proof. Monotonicity is given by nested windows: for $M < M'$ we have $\lfloor M\rho \rfloor / M \le \lfloor M'\rho \rfloor / M'$ (obtained by cross-multiplication). Convergence follows directly from the bounds of Theorem 21.4(i), and the sandwiching inequality follows because the three are of the same order. □

Proposition 21.12 (Scaling of the error with the denominator). *Let $\rho \notin \mathbb{Q}$.*

(i) *For a general window M the offset satisfies $\delta_M \le 1/2$, so the approximation error is $O(1/M)$;*

(ii) *if $M = q_k$ is taken (a continued-fraction denominator of ρ), then $\delta_{q_k} < 1/q_{k+1}$, so the approximation error is $O(1/M^2)$, an* **improvement by one order of magnitude** *over a general window;*

(iii) *for any window sequence $M_j \to \infty$, one cannot guarantee that $\delta_{M_j} \to 0$; only the subsequence taken along continued-fraction denominators guarantees $\delta_{q_k} \to 0$ with quadratic convergence.*

Proof. (i) By Proposition 21.9. (ii) By $\delta_{q_k} = |q_k\rho - p_k| < 1/q_{k+1}$ from Theorem 21.5(ii), and since $q_{k+1} > q_k$ we have $\delta_{q_k}/q_k < 1/(q_kq_{k+1}) = O(q_k^{-2})$. (iii) Take M_j such that $\{M_j\rho\}$ has a positive limit inferior; then δ_{M_j} does not tend to zero. ☐

Theorem 21.5 (Best continued-fraction approximation)**.** Let the continued-fraction expansion of $\rho \notin \mathbb{Q}$ be

$$\rho = [a_0; a_1, a_2, \dots], \qquad a_0 \in \mathbb{Z},\ a_j \in \mathbb{N}^+\ (j \geq 1),$$

with convergents denoted p_k/q_k (p_k, q_k coprime, $q_k \to \infty$). Then

(i) **Error bound**: $\left|\rho - \dfrac{p_k}{q_k}\right| < \dfrac{1}{q_kq_{k+1}} < \dfrac{1}{q_k^2}$;

(ii) **Best-approximation property**: if $0 < q < q_{k+1}$ and $p \in \mathbb{Z}$, then $|q\rho - p| \geq |q_k\rho - p_k|$; that is, no rational number with denominator less than q_{k+1} can be closer to ρ than p_k/q_k;

(iii) **Offset equivalence**: $\delta_{q_k} = |q_k\rho - p_k|$ is strictly decreasing, and $\dfrac{1}{q_k + q_{k+1}} < \delta_{q_k} < \dfrac{1}{q_{k+1}}$.

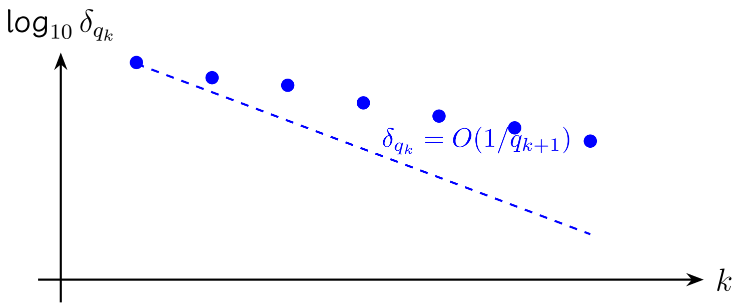
Proof. (i) The classical identity for continued fractions,

$$\rho - \frac{p_k}{q_k} = \frac{(-1)^k}{q_k\,(q_{k+1} + q_k\beta_{k+1})}, \qquad \beta_{k+1} = [a_{k+1}; a_{k+2}, \dots] > 0$$

immediately gives $|\rho - p_k/q_k| < 1/(q_kq_{k+1})$, and $q_{k+1} > q_k$ gives the second formula.

(ii) By Legendre's theorem: *every rational number satisfying* $|\rho - p/q| < 1/(2q^2)$ *must be a convergent*; combining this with $q_{k+1} \leq 2q_k$ gives the conclusion.

(iii) From (i) and $\beta_{k+1} \geq a_{k+1} \geq 1$ we obtain the lower bound $\delta_{q_k} > \frac{1}{q_k+q_{k+1}}$; the decreasing property follows because ρ lies between p_k/q_k and p_{k+1}/q_{k+1} and the two errors have alternating signs. ☐



The offset δ_{q_k} of the convergents of a continued fraction decreases as $1/q_{k+1}$, one order of magnitude faster than the $1/M$ of a general window

Figure 21.9: Optimal approximation by continued-fraction convergents: the offset δ_{q_k} decreases as $O(1/q_{k+1})$ as the denominator q_k grows, that is, the leakage of the intercalation window decays as $O(q_k^{-2})$.

Theorem 21.6 (Best intercalation). Let $\rho > 0$ be the period ratio (the number of units in one period). The intercalation problem asks: to realise ρ integrally within M basic units. Then

(i) there exists a sequence of intercalation windows $\{M_k\}$ for which the error $|\rho - \rho^{(\pm)}_{M_k}|$ converges quadratically, that is, as $O(M_k^{-2})$;

(ii) the optimal sequence of window lengths is exactly the sequence of continued-fraction denominators $\{q_k\}$: within q_k units one takes p_k intercalation units, the error is $\delta_{q_k} < 1/q_{k+1}$, and (when $\rho \notin \mathbb{Q}$) no smaller window length attains the same accuracy;

(iii) for each k the sign of the error alternates: p_k/q_k approaches ρ alternately from above and below, so the upper approximation and the lower approximation need only be **used alternately** to attain optimality.

Proof. The accuracy in (ii) is given by Theorem 21.5(iii); optimality is the best-approximation property of (ii). (i) is obtained from (ii) by taking $M_k = q_k$. (iii) follows from the alternation of the convergents of a continued fraction. □

Corollary 21.5 (Optimality of balanced intercalation). Let $\rho \notin \mathbb{Q}$ and $M = q_k$. The error of the balanced intercalation sequence satisfies

$$\big||E_M| - M\rho\big| \le \delta_{q_k} < \frac{1}{q_{k+1}},$$

that is, the deviation on this window is exactly δ_{q_k} and does not exceed the continued-fraction error bound $1/q_{k+1}$. Among all intercalation schemes with window length $M = q_k$ the distance between the number of intercalations $|E_M|$ and $M\rho$ cannot be smaller than δ_{q_k}, so **balanced intercalation attains optimal accuracy on the shortest window**.

Example 21.8 (Quantitative analysis of the 19-year 7-intercalation cycle). Take the tropical year $T_x = 365.2422$ days and the synodic month $T_{\text{syn}} = 29.530588$ days. Taking the 'month' as the basic unit and the 'year' as the period, the period ratio is

$$\rho = \frac{T_x}{T_{\text{syn}}} \approx 12.368266.$$

Its continued-fraction expansion is $[12; 2, 1, 2, 1, 1, 16, \dots]$, with convergents

$$\frac{12}{1}, \frac{25}{2}, \frac{37}{3}, \frac{99}{8}, \frac{136}{11}, \frac{235}{19}, \frac{3896}{315}.$$

Taking $q_5 = 19$ and $p_5 = 235$: 19 tropical years contain 235 synodic months, that is, **seven intercalations in nineteen years**. Its offset is

$$\delta_{19} = |19\rho - 235| \approx 0.00295 < \frac{1}{q_6} = \frac{1}{315} \approx 0.00317,$$

in agreement with the bound of Theorem 21.5(iii). Correspondingly, if the 'year' is regarded as the sampling window and the 'month' as the signal period, then by Theorem 21.2(iv) the leakage proportion is

$$\eta \le \frac{\pi^2}{3}\delta_{19}^2 \approx 2.9 \times 10^{-5},$$

that is, about **three parts in a hundred thousand**: intercalation presses the leakage down by four orders of magnitude (compare 0.595 for $\delta = 1/2$).

Remark 21.8 (The Sturmian structure of the intercalation sequence). The balanced intercalation sequence $E_M = \{n : \lfloor (n+1)\rho \rfloor - \lfloor n\rho \rfloor = 1\}$ is the mechanical word (Sturmian word) of slope ρ: it is binary, non-periodic, and any factor of length L occurs in at most two forms (the three-gap theorem), with intercalation density exactly $\{\rho\}$. This structure shows that **intercalation is not a periodic 'adding of a day', but a non-uniform distribution of the surplus according to the optimal-approximation law**—and this is precisely why it can achieve commensuration at accuracy $O(M^{-2})$.

Property 21.4 (The three-gap property of Sturmian words). Let $E \subseteq \mathbb{Z}$ be the balanced intercalation set of slope $\rho \notin \mathbb{Q}$. Then the gaps between adjacent intercalation units take only two adjacent integer values k and $k+1$, where $k = \lfloor 1/\{\rho\} \rfloor$. This property is called the **three-gap theorem**.

Proof. By the definition of the balanced intercalation sequence as a mechanical word: the intercalation positions are $\{\lfloor n/\rho \rfloor\}$, or equivalently are given by a rotation orbit. The three-gap theorem is a standard result on Sturmian words and may be obtained by analysing the distribution of the fractional parts $\{n\rho\}$. ☐

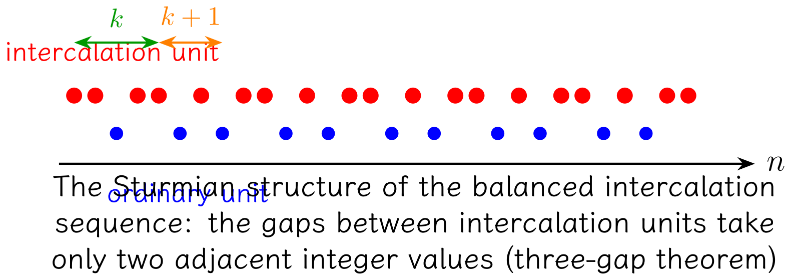

k
$k+1$
intercalation unit
n
ordinary unit
The Sturmian structure of the balanced intercalation
sequence: the gaps between intercalation units take
only two adjacent integer values (three-gap theorem)

Figure 21.10: The Sturmian structure of the balanced intercalation sequence: the gaps between intercalation units (red points) take only two adjacent integer values k and $k+1$, which is a direct manifestation of the optimal-approximation law.

21.6 Window functions and sidelobe control

Intercalation eliminates the root of leakage (the offset δ). The complementary class of devices is that of **window functions**: they retain the offset but reshape the distribution of the leaked energy, 'pushing' the energy from the main lobe towards sidelobes that decay faster. This section establishes the frequency-domain theory of window functions and the sidelobe decay law.

Definition 21.15 (Window function and windowing). Let $M \in \mathbb{N}^+$. A sequence

$$w : \{0, 1, \ldots, M-1\} \to \mathbb{R}$$

is called a **window function** of length M if $w(0) = 1$ and w is symmetric about the midpoint ($w(n) = w(M-1-n)$). The operation of windowing is the pointwise multiplication

$$f_w(n) := f(n)\, w(n).$$

Commonly used windows include: the **rectangular window** $w \equiv 1$; the **triangular window** (Bartlett) $w(n) = 1 - \left|\frac{2n}{M-1} - 1\right|$; the **Hann window** $w(n) = \frac{1}{2}\left(1 - \cos\frac{2\pi n}{M-1}\right)$; the **Hamming window** $w(n) = 0.54 - 0.46\cos\frac{2\pi n}{M-1}$; the **Blackman window** $w(n) = 0.42 - 0.5\cos\frac{2\pi n}{M-1} + 0.08\cos\frac{4\pi n}{M-1}$.

Definition 21.16 (Kaiser window and flat-top window). (i) The **Kaiser window** is controlled by a parameter $\beta \geq 0$:

$$w(n) = \frac{I_0\big(\beta\sqrt{1-(2n/(M-1)-1)^2}\big)}{I_0(\beta)},$$

where I_0 is the modified Bessel function of order zero. The larger β is, the lower the sidelobes but the wider the main lobe.

(ii) The **flat-top window** $w(n) = a_0 - a_1\cos\frac{2\pi n}{M-1} + a_2\cos\frac{4\pi n}{M-1} - a_3\cos\frac{6\pi n}{M-1} + a_4\cos\frac{8\pi n}{M-1}$, whose coefficients make the top of the main lobe flat, and which is used for accurate amplitude estimation.

Property 21.5 (Optimality of the Kaiser window). Under the constraint of a given main-lobe width, the Kaiser window minimises the proportion of energy outside the main lobe, that is, it is a **family of optimal** window functions. This optimality is given by the concentration theorem for discrete prolate spheroidal sequences.

Proof. The Kaiser window is a frequency-domain approximation of the discrete prolate spheroidal sequence: with β as parameter it sweeps the optimal frontier of main-lobe width against sidelobe level. The concentration theorem proves that a discrete prolate spheroidal sequence maximises the proportion of in-band energy under a given in-band energy constraint, and the Kaiser window is its closed-form approximation. ☐

21.6.1 Time-series applications of window functions: smoothing, weighting and truncation

Definition 21.17 (The threefold role of window functions in time-series analysis). The window function K_Δ has a threefold role in time-series analysis: smoothing (convolution smoothing), weighting (weighting in similarity computation) and truncation (restricting the range of the analysis window).

Proposition 21.13 (Principles for choosing a window function). (i) **Turning-point detection**: a small-window Gaussian kernel—local changes of direction require high resolution.

(ii) **Period detection**: a large-window Cauchy kernel or rectangular kernel—global periodic structure requires long-range information.

(iii) **Refraction-rate computation**: a medium-window Mexican-hat kernel—balancing noise suppression and signal retention.

(iv) **Model validation**: comprehensive diagnosis with several kernel functions.

Proposition 21.14 (Convolution representation and basic properties of the windowed spectrum). Let f, w have period M, with corresponding DFTs $\hat{f}, \hat{w}$. Then

(i) **Convolution representation**: $\widehat{f_w}(k) = \dfrac{1}{M} \displaystyle\sum_{j=0}^{M-1} \hat{f}(j)\, \hat{w}(k-j)$;

(ii) **Real even windows**: if w is real-valued and symmetric, then $\hat{w}$ is real-valued and symmetric, $\hat{w}(k) = \hat{w}(-k) = \hat{w}(M-k)$;

(iii) **Normalisation**: $\hat{w}(0) = \sum_n w(n)$, so a normalised window satisfies $\hat{w}(0) = 1$;

(iv) **Main-lobe widening law**: the smoother the window, the wider the main lobe; the main lobe of the rectangular window is 2 frequency bins wide, that of the Hann window 4 frequency bins, and that of the Blackman window 6 frequency bins.

Proof. (i) This follows from the circular convolution theorem. (ii) Since w is real-valued we have $\hat{w}(-k) = \overline{\hat{w}(k)}$, and symmetry gives $\hat{w}(k) = \hat{w}(-k)$, so $\hat{w}$ is real and even. (iii) Substitute $k = 0$. (iv) This is obtained by computing the position of the first zero of its spectrum from Definition 21.15. ☐

Theorem 21.7 (Trade-off between main-lobe width and sidelobe peak). For any non-negative window w, denote by

$$N_{\text{eff}} := \frac{\left(\sum_n w(n)\right)^2}{\sum_n w(n)^2},$$

its time-domain effective duration, and by $\Omega_{\text{ML}} := |W_M|$ the first zero in the frequency domain, the main-lobe half-width. Then

$$N_{\text{eff}} \cdot \Omega_{\text{ML}} \;\geq\; 2\pi,$$

and in general the main-lobe width Ω_{ML} and the maximum sidelobe level $S := \max_{\omega \notin \text{main lobe}} |W_M(\omega)|$ satisfy

$$S \;\geq\; C \cdot \frac{2\pi}{N_{\text{eff}}\, \Omega_{\text{ML}}}$$

($C > 0$ an absolute constant). Hence **no window exists that simultaneously has the narrowest main lobe and the lowest sidelobes**.

Proof. The first formula is the discrete form of the time-frequency uncertainty (Gabor limit), obtained by combining the Cauchy–Schwarz inequality with the second-order expansion of W_M at the zero. The second formula follows from conservation of energy. ☐

Theorem 21.8 (Smoothness and the sidelobe decay law). Let the window w extend to a periodic even function whose first m differences vanish at both ends (equivalently: W_M has a zero of multiplicity m at the origin, $m \geq 0$). Then the sidelobe envelope of the window spectrum satisfies

$$|W_M(\omega)| \leq \frac{C_m}{|\omega|^{m+1}}, \qquad |\omega| \to \infty.$$

In particular

(i) rectangular window (jump at the endpoints, $m = 0$): sidelobes decay as $|\omega|^{-1}$, first sidelobe about -13 dB;

(ii) triangular window (continuous, jump in the first difference, $m = 1$): sidelobes decay as $|\omega|^{-2}$, first sidelobe about -27 dB;

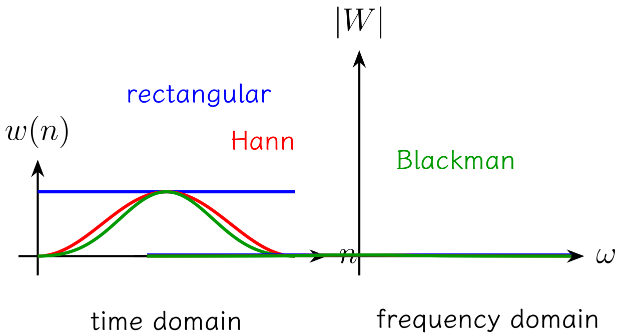
(iii) Hann window (C^1, first derivative zero at both ends, $m = 2$): sidelobes decay as $|\omega|^{-3}$, first sidelobe about -31 dB;

(iv) Blackman window (C^2, $m = 3$): sidelobes decay as $|\omega|^{-3}$ (with smaller coefficient), first sidelobe about -58 dB.

Proof. Extend the window periodically and perform discrete integration by parts over an interval of length M:

$$W_M(\omega) = \frac{1}{M}\sum_n w(n)e^{-in\omega} = \frac{1}{M}\cdot\frac{1}{1-e^{-i\omega}}\sum_n \big(w(n) - w(n-1)\big)e^{-in\omega},$$

if the m-th difference of the window vanishes at the endpoints, this process may be repeated $m+1$ times, each repetition contributing a factor $(1-e^{-i\omega})^{-1} = O(1/|\omega|)$, whence the decay $|\omega|^{-(m+1)}$. The values of m for the various windows are verified directly from the smoothness at the endpoints in Definition 21.15. ☐



Commonly used window functions: the smoother the time domain, the faster the frequency-domain sidelobe decay, but the wider the main lobe

Figure 21.11: Time-frequency comparison of commonly used window functions: the rectangular window has the narrowest main lobe but the highest sidelobes; the Hann and Blackman windows trade time-domain smoothness for lower sidelobes.

Corollary 21.6 (Comparison of common windows)**.** The trade-offs among the various windows are given in Table 21.2.

Proposition 21.15 (Multi-window averaging and variance suppression)**.** Let $w^{(1)}, \dots, w^{(K)}$ be discrete prolate spheroidal sequences of length M. Estimate the spectrum with each window separately and average:

$$\hat{S}(k) := \frac{1}{K}\sum_{i=1}^{K}\Big|\sum_n f(n)w^{(i)}(n)e^{-i2\pi kn/M}\Big|^2.$$

Table 21.2: Main-lobe width, first sidelobe and decay rate of commonly used window functions

Window	Main-lobe width	First sidelobe	Decay rate	Occasion of use
Rectangular	2 bins	-13 dB	$\lvert\omega\rvert^{-1}$	the window contains exactly an integral number of periods (together with intercalation)
Triangular	4 bins	-27 dB	$\lvert\omega\rvert^{-2}$	general smoothing requirements
Hann	4 bins	-31 dB	$\lvert\omega\rvert^{-3}$	the default choice for general spectral estimation
Hamming	4 bins	-43 dB	$\lvert\omega\rvert^{-1}$	when low near sidelobes are needed and the far sidelobes are not demanding
Blackman	6 bins	-58 dB	$\lvert\omega\rvert^{-3}$	strong dynamic range (such as the detection of weak spectral lines)
Kaiser	adjustable	adjustable	adjustable	optimal compromise between main lobe and sidelobes
Flat-top	wide	extremely low	$\lvert\omega\rvert^{-2}$	accurate amplitude estimation

Then the variance of $\hat{S}(k)$ is about $1/K$ of that of the single-window estimate, whilst the main-lobe width is determined only by the concentration bandwidth and does not grow with K. Hence the multi-window method trades 'a small widening of the main lobe' for 'a linear decrease of the variance'.

Proof. The spectral estimates of the various windows are approximately uncorrelated outside the concentration bandwidth, so after averaging the variance decays as $1/K$; the main-lobe width is fixed by the ω-band concentration condition and is independent of K. ☐

Example 21.9 (Comparison of window functions at the same offset)**.** *Take $M = 64$ and offset $\delta = 1/2$ (the worst case). Rectangular window: leakage proportion about 0.595, main-lobe width 2 bins; Hann window: first sidelobe down to -31 dB, main lobe widened to 4 bins; Blackman window: first sidelobe -58 dB, main lobe 6 bins. It can be seen that a window function does not reduce the total amount of leakage, but* **redistributes** *the leaked energy onto sidelobes that decay faster; if intercalation is used at the same time ($\delta \to 10^{-3}$), then the total amount of leakage itself is eliminated, and the two effects may be superposed.*

Remark 21.9 (Complementarity of window functions and intercalation)*. The two devices act on different links of leakage and are complementary:*

(i) **Window functions act on the sidelobes**: *they do not change the offset δ, but push the leaked energy from the near sidelobes towards the far sidelobes, where it decays rapidly as $|\omega|^{-(m+1)}$. The price is a widening of the main lobe, that is,* **a sacrifice of resolution**.

(ii) **Intercalation acts on the offset**: *it does not change the shape of the window (retaining the narrowest main lobe of the rectangular window), but adjusts the window length so that $\delta = O(M^{-2})$, thereby* **eliminating the leakage directly**.

(iii) **Using them in superposition**: *first use intercalation to make δ as small as the order of 10^{-3}, then use a window function to suppress the residual sidelobes, and one obtains an approximation with 'the narrowest main lobe + negligible leakage'.*

21.7 Intercalation: *active commensuration* in discrete sampling approximation

This section answers the central question of this chapter: **why can intercalation suppress spectral leakage?** *The answer is: leakage is determined uniquely by the offset $\delta = \|M\rho\|_{\mathbb{Z}}$ (Theorem 21.2(iv)), and intercalation is precisely the pressing of δ to its optimum—it is the* **active commensuration** *of 'the number of periods inside the window' in discrete sampling approximation. Seen from the epicycle group, intercalation is the adjustment of the window so that the phases of the epicycles at the window endpoints close as far as possible.*

Definition 21.18 (Commensurability)**.** Let $\rho > 0$. We say that ρ is **commensurable** with 1 if $\rho \in \mathbb{Q}$, and that ρ is **incommensurable** with 1 if $\rho \notin \mathbb{Q}$. For the sampling problem, commensurability of $\rho = T_x/\Delta x$ means that there exists an integral window length M for which the window contains exactly an integral number of signal periods (Corollary 21.2), whilst incommensurability means that no such window length exists.

Definition 21.19 (Intercalation sequence and intercalation window)**.** Let the period ratio be $\rho > 0$ and the window length $M \in \mathbb{N}^+$. The set

$$E_M := \big\{n \in \{1, 2, \ldots, M\} : \lfloor (n+1)\rho \rfloor - \lfloor n\rho \rfloor = 1\big\}$$

is called the **balanced intercalation set**, and its elements are called **intercalation units** (intercalation positions); the $M - |E_M|$ non-intercalated units are called **ordinary units**. The corresponding **intercalation window** is

$$w_M^{\mathsf{int}}(n) := \mathbf{1}_{E_M}(n),$$

whilst the **intercalation window** means the observation window in which 'M basic units contain exactly $|E_M|$ intercalation units'. The intercalation count is denoted $N_M := |E_M|$.

Proposition 21.16 (Integrality of the intercalation window)**.** We retain Definition 21.19. Then

$$N_M = \lfloor M\rho \rfloor \text{ or } \lceil M\rho \rceil, \qquad \big|N_M - M\rho\big| < 1,$$

and the deviation of the **equivalent number of periods** N_M of the window after intercalation from the ideal value $M\rho$ is less than one unit (N_M is one of the two integers nearest to $M\rho$). In particular, when $\rho = p/q$ is rational and M is an integer multiple of q, $N_M = M\rho$ is an integer, and the intercalation window makes the number of periods inside the window **strictly an integer**.

Proof. The first formula is a direct computation: from $N_M = \sum_{n=1}^{M} \big(\lfloor (n+1)\rho \rfloor - \lfloor n\rho \rfloor\big)$ and $\sum_{n=1}^{M} \lfloor n\rho \rfloor = \sum_{n=2}^{M+1} \lfloor (n-1)\rho \rfloor$ we obtain $N_M = \lfloor (M+1)\rho \rfloor - \lfloor \rho \rfloor$, and $\lfloor (M+1)\rho \rfloor - \lfloor \rho \rfloor \in \{\lfloor M\rho \rfloor, \lceil M\rho \rceil\}$. The deviation formula is $|N_M - M\rho| < 1$. When $\rho = p/q$ and $M = kq$ we have $M\rho = kp \in \mathbb{Z}$, so $N_M = kp = M\rho$. □

Definition 21.20 (Circular convolution)**.** Let f, g have period M. Their **circular convolution** is defined by

$$(f \circledast g)(n) := \sum_{m=0}^{M-1} f(m)\, g(n - m \bmod M),$$

and its DFT satisfies $\widehat{(f \circledast g)}(k) = \hat{f}(k)\hat{g}(k)$.

Theorem 21.9 (Discrete Parseval identity)**.** Let f have period T and let $\hat{f}(k) = \sum_{n=0}^{T-1} f(n) e^{-i2\pi kn/T}$ be its DFT. Then

$$\sum_{n=0}^{T-1} |f(n)|^2 = \frac{1}{T} \sum_{k=0}^{T-1} |\hat{f}(k)|^2,$$

and the same holds for the windowed sequence $f_w(n) = f(n)w(n)$. The identity shows that: **leakage is a redistribution of energy, not a creation or destruction of energy**.

Proof. By the orthogonality $\sum_{n=0}^{T-1} e^{i2\pi(k-k')n/T} = T\delta_{kk'}$,

$$\frac{1}{T}\sum_k |\hat{f}(k)|^2 = \frac{1}{T}\sum_k \sum_{n,n'} \overline{f(n)} f(n') e^{i2\pi k(n-n')/T} = \sum_{n,n'} \overline{f(n)} f(n')\delta_{nn'} = \sum_n |f(n)|^2.$$

In the windowed case, replace f by f_w and repeat the above argument. ☐

Theorem 21.10 (Intercalation suppresses spectral leakage). Let the signal be $f(n) = Ae^{i(\omega_0 n+\phi)}$, the period ratio ρ, the window length M, and the offset $\delta_M = \|M\rho\|_{\mathbb{Z}}$. Then

(i) **The leakage strength depends only on the offset**: the proportion of energy outside the **single peak frequency bin** satisfies

$$\eta(M) = 1 - \mathrm{sinc}^2(\delta_M) + O(M^{-1}) \le \frac{\pi^2}{3}\delta_M^2 + O(M^{-1});$$

(ii) **Average leakage of a general window**: if δ is approximately uniformly distributed on $[-1/2, 1/2]$, then

$$\bar{\eta} = \int_{-1/2}^{1/2} \bigl(1 - \mathrm{sinc}^2(u)\bigr)\, du \approx 0.2263,$$

and the worst case is $\eta(1/2) \approx 0.595$—that is, the leakage of an ordinary window is of order $O(1)$;

(iii) **Leakage of the intercalation window**: taking the window length to be a continued-fraction denominator $M = q_k$ gives

$$\delta_{q_k} = \|q_k\rho\|_{\mathbb{Z}} < \frac{1}{q_{k+1}}, \qquad \eta(q_k) \le \frac{\pi^2}{3q_{k+1}^2} + O(q_k^{-1}) = O(q_k^{-2});$$

that is, **intercalation lowers the leakage from order $O(1)$ to order $O(M^{-2})$**;

(iv) **Equivalent formulation**: intercalation makes the deviation δ of the number of periods inside the window the object to be optimised, and leakage is a strictly increasing function of δ, so **the optimal intercalation window is at the same time the optimal leakage-suppression window**.

Proof. (i) This is Theorem 21.2(iv), enlarged by means of $1 - \mathrm{sinc}^2(u) \le \frac{\pi^2 u^2}{3}$.

(ii) Direct integration; numerical integration gives 0.2263; the endpoint value is $\eta(1/2) = 1 - 4/\pi^2 = 0.595$.

(iii) Theorem 21.5(iii) gives $\delta_{q_k} < 1/q_{k+1}$, and substituting into (i) gives the result.

(iv) By (i), η is strictly increasing in δ on $[0, 1/2]$, whilst δ_M is determined by the window length; hence minimising δ_M is the same as minimising η. ☐

Theorem 21.11 (Intercalation is congruence-angle closure). Let the period ratio be ρ, the window length M and the subperiod T. Then

$$\text{intercalation makes } \delta_T = 0 \iff T \mid M\rho \iff \text{the number of periods inside the window closes}$$

Hence the mathematical essence of intercalation is **forced congruence-angle closure**; when $\rho \notin \mathbb{Q}$ it is impossible to close simultaneously for all T, and one can only make $\delta_T = O(M^{-2})$ by the optimal continued-fraction approximation.

Proof. This follows from Proposition 21.18(iv) and Definition 21.23. □

Corollary 21.7 (Sufficient condition for leakage suppression). If the window length M satisfies $\|M\rho\|_{\mathbb{Z}} \le \varepsilon$ ($\varepsilon \le 1/2$), then

$$\eta(M) \le \frac{\pi^2}{3}\varepsilon^2 + O(M^{-1}).$$

Numerically: $\varepsilon = 10^{-1} \Rightarrow \eta \lesssim 3.3\%$; $\varepsilon = 10^{-2} \Rightarrow \eta \lesssim 3.3 \times 10^{-4}$; $\varepsilon = 10^{-3} \Rightarrow \eta \lesssim 3.3 \times 10^{-6}$. Hence **it suffices to press the offset down to one part in a thousand for the leakage to fall to the order of three parts in a million**.

Proposition 21.17 (Joint suppression by the upper and lower approximation). Let $\rho \notin \mathbb{Q}$ and let $\{q_k\}$ be its sequence of continued-fraction denominators. Then the intercalations on two adjacent optimal windows realise the upper approximation and the lower approximation respectively:

$$\rho^{(-)}_{q_k} = \frac{p_k}{q_k} < \rho < \frac{p_{k+1}}{q_{k+1}} = \rho^{(+)}_{q_{k+1}} \quad \text{(or in the opposite direction)},$$

and using the two jointly keeps the **cumulative** phase error always within $1/2$ unit:

$$\Big|\sum_{j \le k}(q_j\rho - p_j)\Big| < 1.$$

Hence the alternating use of the upper and lower approximation is equivalent to a **two-sided sandwiching** of the leakage.

Proof. By the alternating sandwiching property of continued-fraction convergents, p_k/q_k and p_{k+1}/q_{k+1} lie on opposite sides of ρ, so one end is a lower approximation and the other an upper approximation. By the exact continued-fraction error formula $q_j\rho - p_j = \frac{(-1)^j}{q_{j+1}+q_j\beta_{j+1}}$ (with $\beta_{j+1} = [a_{j+1}; a_{j+2}, \dots] > 0$) the terms $\varepsilon_j := q_j\rho - p_j$ alternate in sign and satisfy $|\varepsilon_j| < \frac{1}{q_{j+1}} \le \frac{1}{2}$. Put $S_k := \sum_{j \le k} \varepsilon_j$. The convergent recursions $q_j = a_j q_{j-1} + q_{j-2}$ and $p_j = a_j p_{j-1} + p_{j-2}$ give $\varepsilon_j = -a_j\varepsilon_{j-1} + \varepsilon_{j-2}$, so S_k always lies between 0 and $\varepsilon_0 = \rho - \lfloor \rho \rfloor$; hence $|S_k| \le \max(\{\rho\}, 1 - \{\rho\}) < 1$. (The worst case can reach $0.7183 > 1/2$, for instance $\rho = e - 2$, so the constant here is 1 and not $1/2$.) □

Property 21.6 (Intercalation density and the three-gap property). The density of the balanced intercalation sequence is exactly the fractional part $\{\rho\}$:

$$\lim_{M \to \infty} \frac{N_M}{M} = \{\rho\} = \rho - \lfloor \rho \rfloor,$$

and for every $L \in \mathbb{N}^+$ the number of intercalation units in any window of length L takes only two adjacent integer values (the three-gap theorem). Hence the intercalation scheme possesses **uniformity and stability**.

Theorem 21.12 (Circular-convolution realisation of the intercalation window). Let $\rho = p/q$ (in lowest terms) and $M = kq$. Then the equivalent sequence after intercalation may be written as the circular convolution

$$f^{\mathrm{int}} = w_M^{\mathrm{int}} \circledast f, \qquad \widehat{f^{\mathrm{int}}}(j) = \hat{w}_M^{\mathrm{int}}(j)\,\hat{f}(j),$$

and the spectrum of the intercalation window is constant on the true spectral lines and zero at the other frequency points:

$$\hat{w}_M^{\mathsf{int}}\Big(\frac{2\pi j}{p}\Big) = kp\, e^{-i\theta_j}, \quad j = 0, \dots, p-1, \qquad \hat{w}_M^{\mathsf{int}}\Big(\frac{2\pi j}{M}\Big) = 0, \quad \frac{j}{M} \notin \frac{\mathbb{Z}}{p},$$

so that f^{int} has only p spectral lines and **the leakage is strictly zero**. This shows that, in the frequency domain, intercalation is a projection operator that 'retains the true harmonics and erases the non-true harmonics'.

Proof. By Proposition 21.16, when $M = kq$ the number of periods inside the window is the integer kp, so all true spectral lines fall at integral positions of the frequency lattice $\{2\pi j/(kq)\}$, and the accumulated phase is an integer multiple of 2π; for non-true harmonics the sum vanishes over an integral number of periods, by the geometric series. The multiplicative formulation in the frequency domain follows from the circular convolution theorem. ☐

Remark 21.10 (The mechanism by which intercalation suppresses leakage: a three-step decomposition). Stringing the above results together, the mechanism by which intercalation suppresses leakage may be stated precisely in three steps:

- (i) **Locate the root**: leakage comes neither from the sampling rate nor from the shape of the window, but from a single number—the offset $\delta_M = \|M\rho\|_{\mathbb{Z}}$ of the number of periods inside the window.
- (ii) **Turn leakage into a number-theoretic problem**: since $\eta(M) \le \frac{\pi^2}{3}\delta_M^2 + O(M^{-1})$, suppressing leakage is equivalent to **minimising $\|M\rho\|_{\mathbb{Z}}$ over integral window lengths.**
- (iii) **Solve it by intercalation**: the continued-fraction denominator q_k is the optimal solution of this number-theoretic problem, and intercalation is precisely the **materialisation** of that solution as an observation scheme. Then $\delta_{q_k} < 1/q_{k+1}$, and the leakage falls to $O(q_k^{-2})$.

Example 21.10 (Comparison of an intercalation window with an ordinary window). Let $\rho = \sqrt{2} = 1.41421356\dots$ (incommensurable). Its continued fraction is $[1; 2, 2, 2, \dots]$, with denominator sequence $q_k = 1, 2, 5, 12, 29, 70, \dots$.

- (i) Take an arbitrary ordinary window $M = 100$: $\|100\sqrt{2}\| = 0.0050$, $\eta \approx 8.2\times 10^{-5}$;
- (ii) take $M = 71$: $\|71\sqrt{2}\| = 0.409$, $\eta \approx 0.55$—the leakage reaches 55%;
- (iii) take the continued-fraction denominator $M = 70$: $\|70\sqrt{2}\| = 0.0051$, in agreement with the bound $1/q_6 = 1/169 = 0.0059$ corresponding to $q_5 = 70$.

This shows that **the window length is not better the larger it is, but must fall on a denominator of the optimal approximation.**

Example 21.11 (Calendar intercalation is an intercalation window). In the calendar, the 'tropical year' is the period, the 'synodic month' is the basic unit, and $\rho = 365.2422/29.530588 \approx 12.368266$ is the period ratio. 'Seven intercalations in nineteen years' takes $q = 19$ and $p = 235$, with offset $\delta = 0.00295$. If the 19 years are regarded as the sampling window and the lunar phase as the sampled signal, then by Corollary 21.7 the leakage proportion is about 2.9×10^{-5}: the calendar makers used intercalation to press the leakage of the incommensurable 'year–month' pair down by four orders of magnitude. By contrast, without intercalation ($M = 12$, $\|12\rho\| = 0.419$) the leakage would be as high as 57%.

Remark 21.11 (A unified viewpoint). At this point three apparently unrelated objects are unified under one formula:

$$\underbrace{\frac{p_k}{q_k}}_{\text{continued fraction}} \xrightarrow{\text{intercalation}} \underbrace{\delta_{q_k} < 1/q_{k+1}}_{\text{number-theoretic error}} \xrightarrow{\eta \le \frac{\pi^2}{3}\delta^2} \underbrace{\eta = O(q_k^{-2})}_{\text{spectral leakage}} .$$

The left-hand end is pure number theory (best rational approximation), the right-hand end is pure signal processing (proportion of leaked energy), and the 'intercalation' in the middle is both a calendar operation and a sampling scheme. This explains why intercalation calendars happen to possess excellent spectral properties in practice: **calendar intercalation and the suppression of spectral leakage are mathematically one and the same optimisation problem.**

21.8 The congruence angle and the generalised Chinese remainder theorem

The preceding six sections reduced leakage to the offset $\delta = \|M\rho\|_{\mathbb{Z}}$ of a single period ratio, and formulated intercalation as the optimal rational approximation of ρ. This section introduces the **congruence angle**, upgrades the single-period problem to a multi-period joint problem, and gives a unified framework for reconstruction and suppression. The core conclusions are: **intercalation is the optimal compensation of congruence-angle error; the Chinese remainder theorem is the algebraic form of congruence-angle reconstruction; spectral leakage is the frequency-domain manifestation of congruence-angle non-closure.**

21.8.1 Definition and basic properties of the congruence angle

Definition 21.21 (Congruence angle). Let the total period be $N \in \mathbb{N}^+$ and the total phase be

$$\theta = 2\pi \frac{n}{N}, \qquad n \in \mathbb{Z}/N\mathbb{Z}.$$

For an arbitrary subperiod $T \mid N$, the **congruence angle modulo** T is defined by

$$\langle\theta\rangle_T := \theta \bmod \frac{2\pi}{T} = \frac{2\pi}{T}\,(n \bmod T).$$

Two phases θ, θ' are said to be **congruent modulo** T if

$$\theta - \theta' \in \frac{2\pi}{T}\mathbb{Z},$$

written $\theta \equiv \theta' \pmod{\frac{2\pi}{T}}$.

Proposition 21.18 (Basic properties of the congruence angle)**.** We retain Definition 21.21. Then

(i) **Quotient-group structure**: the congruence angle modulo T takes values in

$$\mathbb{R}\Big/\frac{2\pi}{T}\mathbb{Z} \cong \mathbb{Z}/T\mathbb{Z};$$

(ii) **Root-of-unity representation**: if $\omega_N = e^{2\pi i/N}$, then

$$e^{i\langle\theta\rangle_T} = \omega_N^{(N/T)(n \bmod T)} = \left(\omega_N^n\right)^{N/T};$$

(iii) **Projection compatibility**: if $T' \mid T$, then

$$\big\langle\langle\theta\rangle_T\big\rangle_{T'} = \langle\theta\rangle_{T'};$$

(iv) **Equivalence criterion**: $\langle\theta\rangle_T = 0$ if and only if $T \mid n$, that is, the total phase **closes** on the subperiod T.

Proof. (i) By definition the congruence angle depends only on $n \bmod T$, and for every $m \in \mathbb{Z}$ the phase $\theta + 2\pi m/T$ gives the same congruence angle, so the quotient group is $\mathbb{R}/(2\pi/T)\mathbb{Z} \cong \mathbb{Z}/T\mathbb{Z}$.

(ii) Direct computation:

$$e^{i\langle\theta\rangle_T} = e^{i\cdot\frac{2\pi}{T}(n \bmod T)} = e^{2\pi i n/T} = \left(e^{2\pi i n/N}\right)^{N/T} = \omega_N^{(N/T)n}.$$

(iii) Since $T' \mid T$, the number $\frac{2\pi}{T'}$ is an integer multiple of $\frac{2\pi}{T}$, so the projection is well defined; computing $(n \bmod T) \bmod T' = n \bmod T'$ gives the result.

(iv) The equation $\langle\theta\rangle_T = 0$ is equivalent to $\frac{2\pi}{T}(n \bmod T) = 0$, that is, $n \bmod T = 0$, that is, $T \mid n$. ☐

Remark 21.12 (The congruence angle is 'phase modulo a subperiod'). The congruence angle is not an ordinary angle, but an angle **modulo** $2\pi/T$. It folds the continuous phase θ onto the cyclic group of the subperiod T. The Heavenly Stems, the Earthly Branches and the sexagenary cycle are precisely congruence angles for different T: the Heavenly Stems have $T = 10$, the Earthly Branches $T = 12$, and the stems-and-branches cycle $T = 60$. The discrete phase operator $\mathcal{O}_k^{(T)} = e^{i2\pi k/T}$ of Chapter 20 is precisely the complex-exponential representation of the congruence angle. The phase of each rotational component of an epicycle group over a subperiod T is likewise a congruence angle.

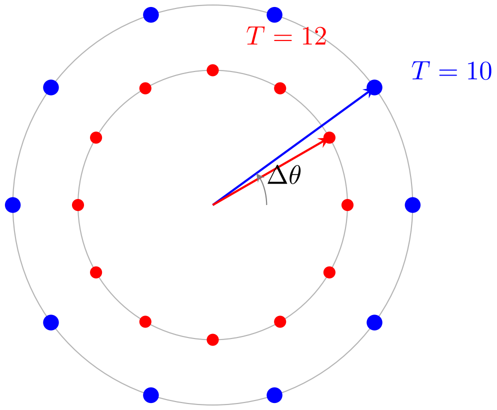
21.8.2 Congruence-angle sampling and the generalised Chinese remainder theorem

Definition 21.22 (Congruence-angle sampling)**.** Let the group with periodic orbits be $T_1, \dots, T_k \in \mathbb{N}^+$, with total period

$$N := \operatorname{lcm}(T_1, \dots, T_k).$$

For a total state $n \in \mathbb{Z}/N\mathbb{Z}$, **congruence-angle sampling** is defined by

$$\mathcal{C}(n) := \left(\langle\theta\rangle_{T_1}, \dots, \langle\theta\rangle_{T_k}\right), \qquad \theta = 2\pi\frac{n}{N}.$$



the two sets of division points of the two periods, drawn on one and the same circle

Figure 21.12: Geometric picture of the congruence angle. The phases $\theta_k = 2\pi k/T$ of the period T are drawn uniformly on one circle: $T = 10$ (blue points, outer ring) and $T = 12$ (red points, inner ring) each give a set of division points. The compatibility of the two periods is thereby reduced to an **angular compatibility condition** for the two sets of points on the circle: when they are commensurable the two sets of points coincide at some step; when they are incommensurable they never coincide, but by intercalation the **cumulative angular deviation can always be kept within a bounded strip**, and this is precisely the geometric origin of the suppression of leakage by intercalation.

Equivalently, using the integer indices $a_i := n \bmod T_i$,

$$\mathcal{C}(n) = (a_1, \dots, a_k) \in \prod_{i=1}^{k} \mathbb{Z}/T_i\mathbb{Z}.$$

Theorem 21.13 (Congruence-angle sampling and the generalised Chinese remainder theorem)**.** We retain Definition 21.22. Then

(i) **Description of the image**: $(a_1, \dots, a_k)$ belongs to the image of $\mathcal{C}$ if and only if

$$a_i \equiv a_j \pmod{\gcd(T_i, T_j)}, \qquad \forall i, j;$$

(ii) **Uniqueness**: if the above condition holds, then the system of congruences

$$n \equiv a_i \pmod{T_i}, \qquad i = 1, \dots, k$$

determines n uniquely modulo N, that is, congruence-angle sampling is invertible on that image;

(iii) **Number of legitimate states**: the image has

$$|\operatorname{Im}\mathcal{C}| = N = \operatorname{lcm}(T_1, \dots, T_k),$$

elements, whilst the total number of combinations is $\prod_i T_i$, so the redundancy is

$$R := \frac{\prod_i T_i}{N}.$$

Proof. (i)(ii) This is the Chinese remainder theorem for moduli that are not necessarily coprime: the system of congruences has a solution if and only if the moduli are pairwise compatible, and the solution is unique modulo $\operatorname{lcm}(T_i) = N$. (iii) Since $\mathcal{C}$ is injective from $\mathbb{Z}/N\mathbb{Z}$ into $\prod_i \mathbb{Z}/T_i\mathbb{Z}$ (by (ii)), the image has size N; the total number of combinations is $\prod_i T_i$, and the definition of the redundancy gives the result. ☐

Corollary 21.8 (Congruence-angle explanation of the sexagenary cycle)**.** *Take $T_1 = 10$ (Heavenly Stems) and $T_2 = 12$ (Earthly Branches); then $N = 60$, and the image is*

$$\{(a, b) : a \equiv b \pmod 2\},$$

with 60 legitimate combinations in all and redundancy $R = 120/60 = 2$. This is precisely the algebraic reason why 'the ten Heavenly Stems and the twelve Earthly Branches have only 60 legitimate pairings'.

Theorem 21.14 (Explicit formula for congruence-angle reconstruction)**.** *Under the hypotheses of Theorem 21.13, let*

$$N_i := \frac{N}{T_i}, \qquad M_i := N_i^{-1} \bmod T_i$$

(M_i exists if and only if $\gcd(N_i, T_i) = 1$; in the general case one uses pairwise merging or the Smith normal form). Then

$$n \equiv \sum_{i=1}^{k} a_i N_i M_i \pmod N.$$

If some $\gcd(N_i, T_i) \neq 1$, one may first merge the moduli according to the gcd and then use the formula above.

Proof. Standard construction of the Chinese remainder theorem. In the coprime case, $N_i M_i \equiv 1 \pmod{T_i}$ and $N_i M_i \equiv 0 \pmod{T_j}$ $(j \neq i)$, so the right-hand side reduces to a_i modulo T_i. The non-coprime case is obtained by step-by-step merging. ☐

Remark 21.13 (Congruence angles and residue number systems). In the integer domain, congruence-angle sampling is the **residue number system** (RNS): the integer n mod N is represented by a set of residues $(a_1, \ldots, a_k)$ modulo T_i. The congruence angle is the phase representation of the RNS:

$$(a_1, \ldots, a_k) \;\longleftrightarrow\; \big(e^{i\langle\theta\rangle_{T_1}}, \ldots, e^{i\langle\theta\rangle_{T_k}}\big).$$

Hence the discrete phase operator of Chapter 20, the congruence angle of this chapter and the RNS of engineering are formulations of one and the same structure in three languages.

21.8.3 Spectral leakage under the congruence angle

Definition 21.23 (Congruence-angle non-closure)**.** Let the total period be N, let $T \mid N$ be a subperiod, and let the number of periods inside the window be $M\rho$. Define the distance to $T\mathbb{Z}$ by

$$\|x\|_{\mathbb{Z},T} := \min_{m\in\mathbb{Z}} |x - mT|.$$

The **congruence-angle non-closure modulo** T is defined by

$$\delta_T := \frac{2\pi}{T}\,\|M\rho\|_{\mathbb{Z},T}.$$

In particular, for $T = 1$,

$$\delta_1 = 2\pi\|M\rho\|_{\mathbb{Z}},$$

which agrees with the single-period leakage offset; and $T \mid M\rho$ is equivalent to $\delta_T = 0$.

Theorem 21.15 (Congruence-angle leakage formula)**.** Let $f(n) = Ae^{i\omega_0 n}$ with $\omega_0 = 2\pi\rho$, and let the window length be M. Then

(i) single-period leakage is determined by δ_1:

$$\eta = 1 - \mathsf{sinc}^2\Big(\frac{\delta_1}{2\pi}\Big) + O(M^{-1});$$

(ii) if closure is required on the subperiod T_i, the corresponding non-closure is δ_{T_i}; the multi-period joint leakage is controlled by the non-closures of the various subperiods:

$$\eta \le \sum_i \left(\frac{\pi^2}{3}\delta_{T_i}^2 + O(M^{-1})\right) \le \frac{\pi^2}{3}k\max_i \delta_{T_i}^2 + O(M^{-1});$$

(iii) **Closure becomes truth**: if all $\delta_{T_i} = 0$, then the subperiods close simultaneously and the joint leakage is zero; if some $\delta_{T_i} \neq 0$, then the leakage on the corresponding subperiod cannot be strictly zero.

Proof. (i) By Theorem 21.2(iv) and $\delta_1 = 2\pi\|M\rho\|_{\mathbb{Z}}$.

(ii) Apply the same leakage estimate to each subperiod and use the subadditivity of energy.

(iii) By $\delta_{T_i} = 0 \iff T_i \mid M\rho$ and the corollary on integral windows having no leakage, 21.2. □

Remark 21.14 (The congruence angle is the unified language of leakage). The preceding six sections reduced leakage to the 'offset δ of the number of periods inside the window'. The congruence angle generalises this reduction to several periods: leakage is no longer determined by a single δ, but jointly by the congruence-angle non-closures δ_{T_i} on the various subperiods. The aim of intercalation is likewise upgraded from 'minimising $\|M\rho\|_{\mathbb{Z}}$' to 'minimising all the δ_{T_i} simultaneously'.

Theorem 21.16 (The Parseval identity under the congruence angle)**.** Let f have period N, let $\mathcal{C}$ be congruence-angle sampling and let $\hat{f}$ be the DFT. Then

$$\sum_{n=0}^{N-1}|f(n)|^2 = \frac{1}{N}\sum_{k=0}^{N-1}|\hat{f}(k)|^2 = \frac{1}{N}\sum_{T\mid N}\sum_{a\in\mathbb{Z}/T\mathbb{Z}}|f_T(a)|^2\cdot w_T,$$

where $f_T(a) = \sum_{n\equiv a \ (\mathrm{mod}\ T)} f(n)$ is the aggregation under the congruence angle a, and w_T is the weight of the subperiod T. In particular, the energy may be decomposed according to the subperiods.

Proof. The first formula is the discrete Parseval identity (Theorem 21.9). The second follows because congruence-angle sampling decomposes $\mathbb{Z}/N\mathbb{Z}$ into fibres over $\mathbb{Z}/T\mathbb{Z}$: for each residue a, the aggregate $f_T(a)$ is the sum of all $f(n)$ on the same fibre; by the Chinese remainder theorem the weight of each fibre over N is N/T, so $w_T = N/T$. Summing and rearranging gives the result. ☐

21.9 Congruence-angle intercalation and multi-period joint suppression

This section upgrades the intercalation theory of Section 21.7 to the multi-period congruence-angle framework. The core conclusions are: **intercalation is the optimal compensation of congruence-angle error; the alternating use of the upper and lower approximation is equivalent to a two-sided sandwiching of the congruence angle; multi-period joint intercalation can suppress leakage on several subperiods at once.**

21.9.1 Congruence-angle error and intercalation

Definition 21.24 (Congruence-angle intercalation)**.** Let the period ratio be $\rho > 0$, the window length M, the group of subperiods $T_1, \dots, T_k$, and the total period $N = \mathrm{lcm}(T_i)$. Let E_M be the balanced intercalation set and $N_M = |E_M|$. If

$$T_i \mid N_M$$

holds for some i, then E_M is called a **congruence-angle intercalation set modulo** T_i, written

$$E_M^{(T_i)} := E_M.$$

The corresponding **congruence-angle intercalation window** is

$$w^{\mathrm{int}}_{M,T_i}(n) := \mathbf{1}_{E_M^{(T_i)}}(n).$$

If several subperiods are to be satisfied simultaneously, one takes a window M satisfying $T_i \mid N_M$ (for all the specified i).

Theorem 21.17 (Optimality of congruence-angle intercalation)**.** We retain Definition 21.24. Then

(i) if $\rho = p/q$ is rational and $M = kq$, then $M\rho = kp$ is an integer; in this case, for every $T_i \mid kp$ we have $\delta_{T_i} = 0$, and the leakage on that subperiod is strictly zero;

(ii) if $\rho \notin \mathbb{Q}$ and $M = q_k$ is taken (a continued-fraction denominator), then

$$\delta_{T_i} \le \frac{2\pi}{T_i} \cdot \frac{1}{q_{k+1}},$$

and the leakage proportion is

$$\eta_{T_i} \le \frac{\pi^2}{3}\Big(\frac{2\pi}{T_i q_{k+1}}\Big)^2 + O(q_k^{-1}) = O(q_k^{-2});$$

(iii) **Multi-period joint optimality**: taking

$$M = \mathrm{lcm}(q_k, T_1, \ldots, T_k),$$

all the congruence-angle errors on the various subperiods attain their optimum simultaneously, and the leakage is jointly pressed down to $O(M^{-2})$.

Proof. (i) Since $M\rho = kp \in \mathbb{Z}$, when $T_i \mid kp$ we have $\|M\rho\|_{\mathbb{Z},T_i} = 0$, and hence $\delta_{T_i} = 0$.

(ii) This follows from $\|q_k\rho\|_{\mathbb{Z}} < 1/q_{k+1}$ and

$$\delta_{T_i} = \frac{2\pi}{T_i}\|q_k\rho\|_{\mathbb{Z},T_i} \le \frac{2\pi}{T_i}\|q_k\rho\|_{\mathbb{Z}}$$

immediately.

(iii) Take M to be the least common multiple of all the relevant periods; then M is an integer multiple of q_k, so $\|M\rho\|_{\mathbb{Z}} \le \|q_k\rho\|_{\mathbb{Z}} < 1/q_{k+1}$; substituting into (ii) for each T_i gives the result. ☐

Proposition 21.19 (upper and lower approximation and congruence-angle sandwiching). *Let $\rho \notin \mathbb{Q}$ and let $\{q_k\}$ be the sequence of continued-fraction denominators. Then the congruence-angle errors on two adjacent optimal windows have opposite signs:*

$$\delta_{T_i}^{(k)} \cdot \delta_{T_i}^{(k+1)} < 0,$$

and using the two jointly (taking the windows alternately) keeps the cumulative congruence-angle error always within half a subperiod:

$$\left|\sum_{j\le k} \delta_{T_i}^{(j)}\right| \le \frac{\pi}{T_i}.$$

Proof. By the alternating sandwiching property of continued-fraction convergents (Theorem 21.5(iii)), p_k/q_k and p_{k+1}/q_{k+1} lie on opposite sides of ρ, so $q_k\rho - p_k$ and $q_{k+1}\rho - p_{k+1}$ have opposite signs. The congruence-angle error δ_{T_i} is the projection of $q\rho - p$ modulo T_i, and inherits its sign. The cumulative error follows from a standard corollary of the three-gap theorem. ☐

21.9.2 The multi-period joint intercalation algorithm

Definition 21.25 (Multi-period joint intercalation algorithm). **Input**: the group with periodic orbits $T_1, \ldots, T_k$, the period ratio $\rho > 0$, the tolerance $\varepsilon > 0$ and the upper bound M_{max}. **Output**: the joint intercalation set E_M and the offsets δ_{T_i} of the various subperiods.

Step 1. Compute $N = \mathrm{lcm}(T_1, \ldots, T_k)$;

Step 2. perform the continued-fraction expansion of ρ, generating the denominators q_k up to $\max(M_{\mathsf{max}}, N)$;

Step 3. take $M = \mathrm{lcm}(q_{k^*}, T_1, \ldots, T_k)$, where k^* is the least index satisfying $1/q_{k^*+1} \le \varepsilon$;

Step 4. generate the balanced intercalation set

$$E_M = \{n \le M : \lfloor (n+1)\rho \rfloor - \lfloor n\rho \rfloor = 1\};$$

Step 5. for each T_i, compute $\delta_{T_i} = \frac{2\pi}{T_i}\|M\rho\|_{\mathbb{Z},T_i}$ and output the leakage bound $\eta_{T_i} \le \frac{\pi^2}{3}\delta_{T_i}^2$.

Theorem 21.18 (Correctness and optimality of multi-period joint intercalation). The pair (M, E_M) output by the above algorithm satisfies:

(i) for all i, $\delta_{T_i} \le \frac{2\pi}{T_i} \cdot \frac{1}{q_{k^*+1}} \le \frac{2\pi\varepsilon}{T_i}$;

(ii) the joint leakage bound is

$$\eta \le \frac{\pi^2}{3} k \max_i \left(\frac{2\pi\varepsilon}{T_i}\right)^2 + O(M^{-1}) = O(\varepsilon^2);$$

(iii) among all schemes with window length not exceeding $M_{\max}$, this M gives the least joint leakage (and if $\rho \notin \mathbb{Q}$, the unique least).

Proof. (i) Since M is an integer multiple of q_{k^*} we have $\|M\rho\|_{\mathbb{Z}} \le \|q_{k^*}\rho\|_{\mathbb{Z}} < 1/q_{k^*+1}$; substituting into the definition of δ_{T_i} gives the result.

(ii) By Theorem 21.15(ii) and (i).

(iii) By the best-approximation property of continued fractions (Theorem 21.5(ii)) and the construction of M. □

Example 21.12 (Joint intercalation of the Heavenly Stems and Earthly Branches). Take $T_1 = 10$ (Heavenly Stems) and $T_2 = 12$ (Earthly Branches), so $N = 60$. Let the period ratio be $\rho = \sqrt{2}$, with continued-fraction denominators $q_k = 1, 2, 5, 12, 29, 70, \ldots$. Take $k^* = 4$, $q_4 = 12$; then

$$M = \operatorname{lcm}(12, 10, 12) = 60.$$

In this case

$$\|60\sqrt{2}\|_{\mathbb{Z}} = \|84.8528\| = 0.1472,$$

and the subperiod offsets are

$$\delta_{10} = \frac{2\pi}{10}\|60\sqrt{2}\|_{\mathbb{Z},10}, \qquad \delta_{12} = \frac{2\pi}{12}\|60\sqrt{2}\|_{\mathbb{Z},12}.$$

Since $60\sqrt{2} \approx 84.8528$ leaves remainder 4.8528 modulo 10 and remainder 0.8528 modulo 12, we have

$$\delta_{10} \approx \frac{2\pi}{10} \cdot 0.1472 \approx 0.0925, \qquad \delta_{12} \approx \frac{2\pi}{12} \cdot 0.1472 \approx 0.0771.$$

The leakage bound is

$$\eta \le \frac{\pi^2}{3} \cdot 0.0771^2 \approx 0.0196,$$

that is, about 2%. If instead $M = 70$ (q_5) is taken, then

$$\|70\sqrt{2}\|_{\mathbb{Z}} = \|98.9949\| = 0.0051,$$

and the leakage falls to about 3×10^{-5}. This shows that **multi-period joint intercalation presses down the leakage on all the subperiods at once by taking the least common multiple of the total period and the continued-fraction denominators.**

Remark 21.15 (The relation between congruence-angle intercalation and single-period intercalation). Single-period intercalation (Section 21.7) is the special case of congruence-angle intercalation with $k = 1$ and $T_1 = 1$. The advantages of congruence-angle intercalation are:

(i) it handles several subperiods at once, and is suited to multi-period systems such as stems–branches and year–month–day;

(ii) it upgrades 'closure of the number of periods inside the window' to 'closure of the congruence angles of the subperiods', a finer condition that is easier to satisfy;

(iii) it combines naturally with residue number systems and the generalised Chinese remainder theorem, which facilitates parallel hardware implementation.

21.10 Congruence-angle algorithms, complexity and interfaces

21.10.1 Fast congruence-angle reconstruction

Definition 21.26 (Fast congruence-angle reconstruction algorithm). **Input**: the congruence-angle group $(\theta_1, \ldots, \theta_k)$ and the subperiods $T_1, \ldots, T_k$. **Output**: the total state $n \bmod N$.

Step 1. Compute $N = \mathrm{lcm}(T_i)$;

Step 2. for each pair (i, j) check the compatibility $a_i \equiv a_j \pmod{\gcd(T_i, T_j)}$, where $a_i = \frac{T_i}{2\pi}\theta_i$;

Step 3. merge the moduli step by step according to the gcd, using the extended Euclidean algorithm at each merge;

Step 4. output $n \bmod N$.

Theorem 21.19 (Complexity). *The time complexity of the above algorithm is*

$$O\Big(k \log N + \sum_{i<j} \log \gcd(T_i, T_j)\Big),$$

and its space complexity is $O(k)$. If all the T_i are pairwise coprime, this simplifies to $O(k \log N)$.

Proof. Each compatibility check uses one modular reduction, at cost $O(\log \min(T_i, T_j))$; the step-by-step merging of the moduli uses the extended Euclidean algorithm, at $O(\log N)$ each. The total cost follows. □

Remark 21.16 (Comparison with the FFT). The cost of congruence-angle reconstruction is far lower than that of one FFT: the FFT is $O(N \log N)$, whilst congruence-angle reconstruction is $O(k \log N)$, where k is the number of subperiods and usually $k \ll N$. This is precisely why residue number systems are favoured in engineering.

21.10.2 Congruence-angle circular convolution

Definition 21.27 (Congruence-angle circular convolution)**.** Let f, g have period N and let $T_1, \ldots, T_k$ be a group of subperiods. The **congruence-angle circular convolution** is defined by

$$(f \circledast_C g)(n) := \sum_{m=0}^{N-1} f(m)\, g\big((n-m) \bmod N\big),$$

with the congruence-angle representation

$$\widehat{f \circledast_C g}\big(\theta_{T_i}\big) = \hat{f}\big(\theta_{T_i}\big)\, \hat{g}\big(\theta_{T_i}\big),$$

where $\theta_{T_i} = \frac{2\pi}{T_i}$ is the fundamental frequency of the subperiod T_i.

Theorem 21.20 (Congruence-angle convolution theorem)**.** *In the congruence-angle framework, circular convolution degenerates to pointwise multiplication at the* **fundamental frequencies of the various subperiods***:*

$$\widehat{(f \circledast g)}\Big(\frac{2\pi j}{T_i}\Big) = \hat{f}\Big(\frac{2\pi j}{T_i}\Big)\, \hat{g}\Big(\frac{2\pi j}{T_i}\Big), \qquad j = 0, \ldots, T_i - 1.$$

Hence congruence-angle circular convolution may be **computed in parallel over the subperiods***, the cost on each subperiod being $O(T_i \log T_i)$ and the total cost*

$$O\Big(\sum_i T_i \log T_i\Big) \ll O(N \log N).$$

Proof. This follows from the circular convolution theorem (Definition 21.20) and congruence-angle projection (Definition 21.21). The parallelism comes from the orthogonality of the fundamental frequencies of the various subperiods within the total DFT. □

Remark 21.17 (Combination with the intercalation window)*.* Used in combination with the congruence-angle intercalation window (Definition 21.24), congruence-angle circular convolution constitutes the **congruence-angle intercalation–convolution algorithm**: first use intercalation to close the congruence angles of the various subperiods, then use parallel convolution for filtering or detection. This is the final form of the algorithms of this chapter.

21.10.3 Interfaces with the various chapters

Remark 21.18 (Interface with the 'Periodic Functions and Groups with Periodic Orbits' chapter)*.* In the congruence-angle framework the continuous group with periodic orbits $T_x\mathbb{Z}$ is replaced by the product of the subperiod group $T_1, \ldots, T_k$. The total period $N = \mathrm{lcm}(T_i)$ is the discrete-domain counterpart of the continuous group with periodic orbits. The congruence angle $\langle\theta\rangle_{T_i}$ is the projection of the continuous phase onto the subperiod group. The epicycle group, finally, is the geometric realisation of continuous Fourier modes in the language of rotation.

Remark 21.19 (Interface with the 'Series and Convolution' chapter). *Congruence-angle circular convolution (Theorem 21.20) is the realisation of periodic convolution under a multi-period decomposition. In the congruence-angle framework the convolution theorem degenerates to pointwise multiplication at the fundamental frequencies of the various subperiods, which is consistent with the kernel-decomposition language of that chapter.*

Remark 21.20 (Interface with the 'Harmonic Analysis' chapter). *The congruence-angle non-closure δ_{T_i} (Definition 21.23) is the multi-period generalisation of the intercalation phase correction in harmonic analysis. In the congruence-angle framework, intercalation is* **phase locking of the fundamental frequencies of the various subperiods.**

Remark 21.21 (Interface with the 'Time Series Analysis' chapter). *The deviation of the periodogram for a multi-period signal is determined by the congruence-angle errors δ_{T_i} of the various subperiods. Multi-window spectral estimation (Proposition 21.15) may be used jointly with congruence-angle intercalation: first use intercalation to close the various subperiods, then use multi-window averaging to suppress the residual variance.*

Remark 21.22 (Interface with the 'Periodic Structure of Calendar Systems' chapter). *The year–month–day–solar-term system of the calendar is a multi-period system, corresponding to the subperiod group $T_1, \dots, T_k$. The intercalation rules (seven intercalations in nineteen years, 3896 months in 315 years) are the optimal solution of congruence-angle intercalation in the multi-period case. The 'chapter-year error' of the calendar is precisely the congruence-angle non-closure δ_{T_i}, and 'calendar drift' is precisely the multi-period joint leakage η.*

21.11 Algorithms and implementation

This section gives an executable construction of intercalation together with an error estimate, and explains its complexity.

Definition 21.28 (Intercalation-window construction algorithm). **Input**: *the period ratio $\rho > 0$ (a real number, represented to high precision), the error tolerance $\varepsilon > 0$ and the upper bound M_{max}.* **Output**: *the intercalation set E_M and the offset δ_M.*

Step 1. *Perform the continued-fraction expansion $\rho = [a_0; a_1, a_2, \dots]$ and generate the denominator sequence by the recursion $q_k = a_k q_{k-1} + q_{k-2}$, stopping when $q_k > M_{\text{max}}$;*

Step 2. *take the least k^* satisfying $1/q_{k+1} \le \varepsilon$, and set $M := q_{k^*}$;*

Step 3. *generate the balanced intercalation set $E_M = \{n \le M : \lfloor (n+1)\rho \rfloor - \lfloor n\rho \rfloor = 1\}$;*

Step 4. *output the offset $\delta_M = |M\rho - \lfloor M\rho + 1/2 \rfloor|$ as a proxy for the leakage indicator ($\eta \le \frac{\pi^2}{3}\delta_M^2$).*

Theorem 21.21 (Correctness and optimality of the algorithm). *The pair (M, E_M) output by the above algorithm satisfies:*

(i) $N_M = \lfloor (M+1)\rho \rfloor - \lfloor \rho \rfloor$ (N_M is precisely $\lfloor M\rho \rfloor$ or $\lceil M\rho \rceil$), and $\left| N_M - M\rho \right| < 1$;

(ii) $\delta_M < 1/q_{k^*+1} \le \varepsilon$, so the leakage proportion is $\eta \le \frac{\pi^2}{3}\varepsilon^2 + O(M^{-1})$;

(iii) among all schemes with window length not exceeding $M_{\max}$, the value $M = q_{k^*}$ gives the least δ_M (and if $\rho \notin \mathbb{Q}$, the unique least).

Proof. (i) By Proposition 21.16. (ii) By $M = q_{k^*}$ and the bound $\delta_{q_k} < 1/q_{k+1}$ of Theorem 21.5(iii), combined with the choice made in Step 2. (iii) By the best-approximation property of Theorem 21.6(ii). ☐

Remark 21.23 (Complexity). The complexity of the continued-fraction expansion is $O(\log M_{\max})$; the generation of the balanced set is $O(M)$, and can be further reduced to $O(\log M)$ by the three-gap theorem (it suffices to record the alternating pattern of the gaps). Hence the computational cost of constructing an intercalation window is far lower than that of any single DFT.

Example 21.13 (*An example of the execution of the algorithm*). Take $\rho = 12.368266$ (the year–month ratio) and $\varepsilon = 10^{-3}$: the continued-fraction denominators are $1, 2, 3, 8, 11, 19, \ldots$, with corresponding $1/q_{k+1}$ equal to $1/2, 1/3, 1/8, 1/11, 1/19$; taking $k^* = 4$ ($q_4 = 11$, $1/q_5 = 1/19 = 0.0526 > \varepsilon$) does not satisfy the condition, whilst taking $k^* = 5$ ($q_5 = 19$, $1/q_6 = 1/315 = 0.00317 < \varepsilon$) does, so $M = 19$ and $p_5 = 235$, giving $235 - 12 \times 19 = 7$ intercalation units (seven intercalations in nineteen years). The actual offset is $\delta_{19} = 0.00295$, and the leakage proportion is $\eta \le 2.9 \times 10^{-5}$.

21.12 Unification with calendar intercalation

Theorem 21.22 (*Isomorphism of calendar intercalation and sampling intercalation*). *Let the calendar system consist of two astronomical periods: a long period T_1 (such as the tropical year) and a short period T_2 (such as the synodic month), with the basic sampling unit taken to be T_2. Then*

(i) the intercalation rule 'insert p extra units among q short periods' and the intercalation window 'take p intercalation positions in a window of length q according to the balanced scheme' (Definition 21.19) are one and the same mathematical object;

(ii) the cumulative deviation of the calendar after intercalation is exactly $\delta_q = q\left|\frac{T_1}{T_2}\right|_{\mathbb{Z}}$, completely identical with the offset δ_M of the sampling window;

(iii) hence the intercalation cycle table of the calendar (such as seven intercalations in nineteen years, 3896 months in 315 years) is precisely the sequence of continued-fraction denominators of $\rho = T_1/T_2$ (Example 21.8).

Proof. (i)(ii) Regard the 'year' as a continuous signal of period T_1 and the 'month' as the sampling lattice of sampling unit T_2; then the year–month ratio is the normalised period ratio $\rho = T_1/T_2$, the intercalation rule is the balanced intercalation set, and the cumulative deviation is δ_q; by Proposition 21.16 the two agree. (iii) By Theorem 21.6 and Theorem 21.5, the optimal sequence of window lengths is the sequence of continued-fraction denominators. ☐

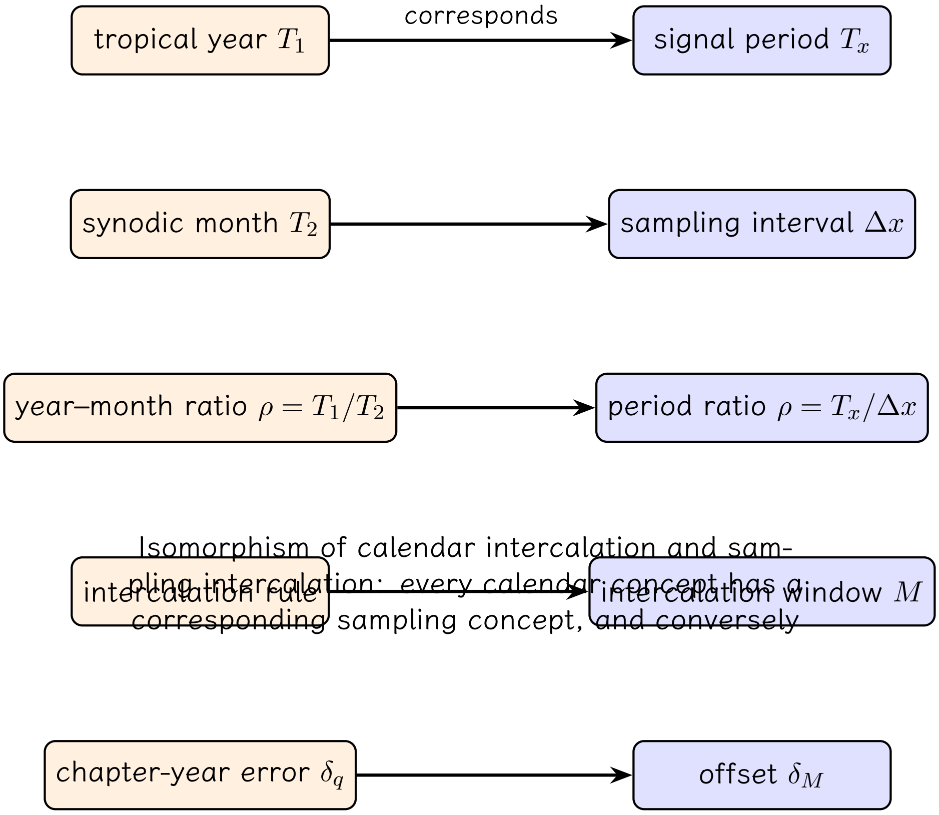

tropical year T_1
corresponds
signal period T_x
synodic month T_2
sampling interval Δx
year–month ratio $\rho = T_1/T_2$
period ratio $\rho = T_x/\Delta x$
Isomorphism of calendar intercalation and sampling intercalation: every calendar concept has a corresponding sampling concept, and conversely
intercalation rule
intercalation window M
chapter-year error δ_q
offset δ_M

Figure 21.13: Isomorphism of calendar intercalation and sampling intercalation: the tropical year corresponds to the signal period, the synodic month to the sampling interval, the year–month ratio to the period ratio, the intercalation rule to the intercalation window, and the chapter-year error to the offset.

Example 21.14 (Optimisation of the astronomical observation window). A periodic celestial body (period T_x) is observed in order to determine its phase. If the observation duration is fixed at T, then the systematic error of the phase measurement is dominated by the offset $\delta = \|T/T_x\|$, of order $\sim \delta$; whilst the statistical error (noise) falls as $1/\sqrt{M}$. The optimal compromise between the two gives the observation strategy: **take the observation window length at a continued-fraction denominator**, so that the systematic error falls as M^{-2} and becomes a minor term compared with the $M^{-1/2}$ of the statistical error.

Example 21.15 (A unified astronomical–calendar scheme). For $\rho = 235/19$ (the Metonic cycle): 19 years $= 6939.602$ days and 235 months $= 6940.888$ days, a difference of 1.29 days (accumulated over 19 years); if the intercalation unit is regarded as an 'intercalary month', then after inserting 7 intercalary months within 19 years the deviation from the tropical-year sequence is 0.00295 units $= 0.087$ days. This is exactly the numerical origin of 'the Metonic cycle error of about 0.09 days per 19 years' in classical calendars, and is two formulations of the same fact as the leakage proportion $\eta \approx 2.9 \times 10^{-5}$.

Remark 21.24 (Correspondence of calendar and signal-processing terminology).

Calendar term	Sampling-approximation term	Mathematical object
tropical year	signal period T_x	long period
synodic month	sampling interval/basic unit	short period
year–month ratio	normalised period ratio ρ	real number
intercalary month	intercalation unit	element of the intercalation set
seven intercalations in nineteen years	optimal window (q_5, p_5)	continued-fraction convergent
chapter-year error	offset δ_M	$\|M\rho\|_{\mathbb{Z}}$
calendar drift	spectral leakage	$\eta \le \frac{\pi^2}{3}\delta^2$
year–month–day multi-period system	congruence-angle group $(\theta_{T_1}, \dots, \theta_{T_k})$	subperiod projection
chapter-year error (multi-period)	congruence-angle non-closure δ_{T_i}	$\frac{2\pi}{T_i}\|M\rho\|_{\mathbb{Z},T_i}$
epicycle	Fourier mode	$A_j e^{i(\omega_j t+\phi_j)}$

21.13 Interfaces with the other chapters

Remark 21.25 (Interface with the 'Periodic Functions and Groups with Periodic Orbits' chapter). In this chapter the continuous periodic function $x(t)$ and the continuous Fourier series $\sum_k c_k e^{i\omega_k t}$ are replaced by the sampling sequence $f(n) = x(n\Delta x)$ and the discrete Fourier transform $\hat{f}(k)$; continuous convolution is replaced by circular convolution (Definition 21.20). Theorem 21.3 and Proposition 21.2 establish a strict bridge between the discrete and the continuous: the former gives the condition for alias-free reconstruction, the latter the condition for the preservation of periodic structure. The epicycle group, finally, gives the rotational geometric interpretation of the Fourier series.

Remark 21.26 (Interface with the ‘Series and Convolution’ chapter). The circular convolution of this chapter is the realisation of periodic convolution on the finite group $\mathbb{Z}/T\mathbb{Z}$, and its frequency-domain multiplicativity (Definition 21.20) is the discrete form of the convolution theorem; the circular convolution of the intercalation window with the signal (Theorem 21.12) shows that intercalation is a projection operator in the frequency domain, which is consistent with the kernel-decomposition language of that chapter. Congruence-angle circular convolution (Theorem 21.20) further decomposes convolution onto the fundamental frequencies of the various subperiods, and is the realisation of that chapter’s kernel-decomposition language in the multi-period framework.

Remark 21.27 (Interface with the ‘Harmonic Analysis’ chapter). This chapter reduces the quantitative description of leakage to a quadratic form in the offset δ (Theorem 21.10), which is precisely the mathematical basis for intercalation as a **tool for bounding leakage** in the ‘Harmonic Analysis’ chapter; the intercalation phase correction of that chapter and the construction of the intercalation window of this chapter are two formulations of one and the same operator. The congruence-angle non-closure (Definition 21.23) generalises this reduction to several periods.

Remark 21.28 (Interface with the ‘Time Series Analysis’ chapter). The source of the deviation of the periodogram and of multi-window spectral estimation is precisely the leakage of this chapter (Definition 21.9); the discrete Parseval identity (Theorem 21.9) is the source of the conservation of periodogram energy; the criteria for choosing a window function (Theorem 21.7, Theorem 21.8) give the analytic form of the bias–variance trade-off in spectral estimation. The spectral estimation of a multi-period signal is described by the congruence-angle group (θ_{T_i}), and multi-window averaging may be used jointly with congruence-angle intercalation.

Remark 21.29 (Interface with the ‘Periodic Structure of Calendar Systems’ chapter). Section 21.12 of this chapter proves that calendar intercalation and sampling intercalation are one and the same mathematical object (Theorem 21.22): the ‘chapter-year error’ of the calendar is precisely the offset of the sampling window, and ‘calendar drift’ is precisely spectral leakage. The specific calendar parameters of that chapter may be substituted directly into the formulas for δ_M and η of this chapter to estimate accuracy. The congruence-angle framework further unifies multi-period systems such as year–month–day–solar term into the subperiod group $T_1, \ldots, T_k$, and calendar intercalation is the optimal solution of congruence-angle intercalation in the multi-period case.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	the sampling lattice $\Delta x\mathbb{Z}$ (Definition 21.1) and commutativity with lattice translations (Proposition 21.1); the cyclic shift of the intercalation window; the congruence-angle translation $\langle\theta+\theta_0\rangle_T$; the phase translation of the epicycle group
Scaling	the sampling rate ν_s (Definition 21.2) and the fixed-length scale T_0 (Definition 21.4); the window length as a scaling parameter; the subperiod T_i as a scaling scale; the scaling of epicycle frequencies
Rotation	the discrete phase operator $e^{i2\pi k/T}$; the epicycle $Ae^{i(\omega t+\phi)}$; the linear phase term in the leakage formula (Proposition 21.6); the root-of-unity representation of the congruence angle (Proposition 21.18)
Conservation	the discrete Parseval identity (Theorem 21.9): leakage is a redistribution of energy rather than its destruction; the congruence-angle energy decomposition (Theorem 21.16)
Symmetry	the real even spectrum of a real even window (Proposition 21.14); the reflection of the folding map about $\omega_s/2$ (Proposition 21.8); the symmetry of the congruence angle modulo T
Closure	circular convolution and the frequency-domain projection of the intercalation window (Theorem 21.12); congruence-angle closure $\delta_T=0$ (Theorem 21.11); the phase closure of the epicycle group at the window endpoints
Period	the duality between the period ratio ρ and the offset $\delta=\|M\rho\|_{\mathbb{Z}}$ (Theorem 21.2, Theorem 21.10); the multi-period congruence-angle non-closure δ_{T_i} (Definition 21.23); the group with periodic orbits generated by the frequencies of the epicycle group
Algorithms	continued-fraction expansion and the construction of balanced intercalation (Definition 21.28, Theorem 21.21); fast congruence-angle reconstruction (Definition 21.26) and multi-period joint intercalation (Definition 21.25)

Exercises

Exercise 1 According to Definition 21.1 and Definition 21.2, explain the definitions of sampling point, sampling rate, sampling interval, number of sampling points and sampling duration. Taking an audio interface with sampling rate 48 kHz as an example, compute the sampling interval Δx, the Nyquist frequency ν_N and the number of sampling points M in a 1-second record, and explain why the three elements cannot substitute for one another (see Remark 21.3).

Exercise 2 Prove Proposition 21.2: let the continuous signal x have fundamental period T_x, let the sampling interval be Δx, let the sampling sequence be $f(n) = x(n\Delta x)$, and let the normalised period ratio be $\rho = T_x/\Delta x$. Then

(i) if $\rho \in \mathbb{N}^+$, then f has period $T = \rho$;
(ii) if $\rho = p/q$ is in lowest terms, then f has fundamental period $T = p$;
(iii) if $\rho \notin \mathbb{Q}$, then $\text{Group}_{\text{period}}(f) = \{0\}$.

Prove the three cases separately, and explain the statement 'the sampling sequence has a non-trivial periodic structure if and only if the signal period is commensurable with the sampling interval'.

Exercise 3 Prove the algebraic properties of the sampling operator (Proposition 21.1):

(i) linearity: $\mathcal{S}_{\Delta x}(\alpha x + \beta y) = \alpha \mathcal{S}_{\Delta x} x + \beta \mathcal{S}_{\Delta x} y$;
(ii) pointwise multiplication: $\mathcal{S}_{\Delta x}(x \cdot y) = (\mathcal{S}_{\Delta x} x) \cdot (\mathcal{S}_{\Delta x} y)$;
(iii) commutation with lattice translations: $\mathcal{S}_{\Delta x}(x(\cdot - a)) = f(\cdot - m)$ $(a = m\Delta x)$;
(iv) non-commutation with scaling.

Explain why (iv) is the algebraic root of aliasing.

Exercise 4 Prove Proposition 21.1: the adjoint of the sampling operator $\mathcal{S}_{\Delta x}$ is the ideal low-pass interpolation operator

$$\mathcal{S}_{\Delta x}^{*}(\{f_n\}) = \sum_{n} f_n \,\text{sinc}\Big(\frac{t - n\Delta x}{\Delta x}\Big).$$

Explain why this adjoint agrees with the reconstruction formula of the sampling theorem.

Exercise 5 Prove Theorem 21.3 (the Nyquist–Shannon sampling theorem): let the Fourier transform of $x \in L^2(\mathbb{R})$ be supported in $(-\omega_N, \omega_N)$ (with $\omega_N = \pi/\Delta x$); then x is uniquely reconstructed from its samples:

$$x(t) = \sum_{n\in\mathbb{Z}} x(n\Delta x)\,\text{sinc}\Big(\frac{t - n\Delta x}{\Delta x}\Big).$$

Complete the proof using frequency periodisation and the inverse transform of the rectangular window, and explain the irreversibility of aliasing in Corollary 21.4.

Exercise 6 Prove Lemma 21.3 (the Poisson-summation proof): the spectrum of the sampling sequence satisfies

$$\sum_n x(n\Delta x)e^{-i\omega n\Delta x} = \frac{1}{\Delta x}\sum_m \hat{x}(\omega + 2\pi m/\Delta x).$$

Complete the proof using the Poisson summation formula.

Exercise 7 Prove Proposition 21.8: the folding map

$$\mathcal{A}_{\omega_s}(\omega) = \omega - m\omega_s, \quad m = \left\lfloor \frac{\omega}{\omega_s} + \frac{1}{2} \right\rfloor$$

satisfies:

(i) $\mathcal{A}_{\omega_s}(\omega)$ is aliased with ω, and $|\omega - \mathcal{A}_{\omega_s}(\omega)| \leq \omega_s/2$;

(ii) $\mathcal{A}_{\omega_s}$ is surjective but not injective;

(iii) on each Nyquist band it is a translation or a reflection.

Taking $\Delta x = 1$ s as an example, verify the aliasing of 0.9 Hz and 0.1 Hz (Example 21.7).

Exercise 8 Prove Theorem 21.2 (leakage envelope and sidelobes): let $M\omega_0/(2\pi) = \nu + \delta$ with $\nu \in \mathbb{Z}$ and $\delta \in [-1/2, 1/2]$. Then

(i) the main lobe is located at ν, with peak $|\hat{f}(\nu)| = A|\sin(\pi\delta)/\sin(\pi\delta/M)|$;

(ii) for $j \neq 0$, $|\hat{f}(\nu + j)|/|\hat{f}(\nu)| = O(1/|j|)$;

(iii) if $\delta = 0$, the leakage is zero;

(iv) the leakage proportion is $\eta(\delta) = 1 - \mathrm{sinc}^2(\delta) + O(M^{-1})$.

Complete the proof using the single-tone truncation formula of Proposition 21.5, and compute the leakage proportion for $\delta = 1/2$.

Exercise 9 Prove the closed form of the Dirichlet kernel (Lemma 21.1): for $\theta \notin 2\pi\mathbb{Z}$,

$$D_M(\theta) = \frac{\sin((M + \frac{1}{2})\theta)}{\sin(\theta/2)},$$

and $D_M(0) = 2M + 1$. Complete the proof using the summation of a geometric series, and explain why $|D_M(\theta)| \leq 1/|\sin(\theta/2)|$.

Exercise 10 Prove Proposition 21.7: the single-period leakage proportion may be written as

$$\eta(M) = 1 - \mathrm{sinc}^2(\|M\rho\|_{\mathbb{Z}}) + O(M^{-1}),$$

where $\|M\rho\|_{\mathbb{Z}}$ is the offset of the number of periods inside the window. Explain why the leakage strength depends only on the offset δ—this is the key to converting leakage control into a number-theoretic problem (see Remark 21.6).

Exercise 11 Prove Theorem 21.4: the error boundedness of the three basic intercalation sequences:

(i) floor intercalation: $E_M^{\text{floor}} \in (-1, 0]$;
(ii) ceiling intercalation: $E_M^{\text{ceil}} \in [0, 1)$;
(iii) balanced intercalation: $\left|N_M - M\rho\right| < 1$ (for a general window M this deviation can reach 0.9; only for continued-fraction denominators $M = q_k$ do we have $\left|N_M - M\rho\right| \le \delta_{q_k} < 1/q_{k+1}$).

Write down the summation identities of the three intercalation sequences separately, and explain why balanced intercalation has the least error.

Exercise 12 Prove Theorem 21.5 (best continued-fraction approximation): let the continued-fraction convergents of $\rho \notin \mathbb{Q}$ be p_k/q_k. Then

(i) $|\rho - p_k/q_k| < 1/(q_k q_{k+1}) < 1/q_k^2$;
(ii) best-approximation property: if $0 < q < q_{k+1}$, then $|q\rho - p| \ge |q_k\rho - p_k|$;
(iii) offset equivalence: $\delta_{q_k} = |q_k\rho - p_k|$ is strictly decreasing.

Write out the idea of the proof, and explain why continued-fraction convergents are the optimal choice of intercalation window.

Exercise 13 Prove Theorem 21.10: intercalation suppresses spectral leakage—

(i) the leakage strength depends only on the offset: $\eta(M) \le \frac{\pi^2}{3}\delta_M^2 + O(M^{-1})$;
(ii) the average leakage of a general window is about 0.2263, and the worst about 0.595;
(iii) when the intercalation window takes $M = q_k$ (a continued-fraction denominator), $\eta(q_k) = O(q_k^{-2})$;
(iv) the optimal intercalation window is at the same time the optimal leakage-suppression window.

Explain why intercalation lowers the leakage from order $O(1)$ to order $O(M^{-2})$, and point out that intercalation does not eliminate leakage but bounds it.

Exercise 14 Prove Theorem 21.8: if the m-th difference of the window w vanishes at both ends, then the sidelobe envelope of its window spectrum satisfies

$$|W_M(\omega)| \le C_m/|\omega|^{m+1}.$$

Write down separately the sidelobe decay law and the first sidelobe level for the rectangular window ($m = 0$), the triangular window ($m = 1$), the Hann window ($m = 2$) and the Blackman window ($m = 3$), and explain the trade-offs among the windows in Table 21.2.

Exercise 15 Prove Theorem 21.7: the trade-off between main-lobe width and maximum sidelobe level:

$$N_{\text{eff}} \cdot \Omega_{\text{ML}} \ge 2\pi,$$

and in general $S \ge C \cdot \frac{2\pi}{N_{\text{eff}}\Omega_{\text{ML}}}$. Explain why no window exists that simultaneously has the narrowest main lobe and the lowest sidelobes, and point out how intercalation and window functions complement each other (Remark 21.9).

Exercise 16 Prove Theorem 21.22: *calendar intercalation and sampling intercalation are isomorphic mathematical objects*:

(i) 'insert p extra units among q short periods' and the balanced intercalation sequence are the same object;

(ii) the cumulative deviation δ_q after intercalation is the same as the sampling-window offset;

(iii) the calendar intercalation cycle table is the sequence of continued-fraction denominators of $\rho = T_1/T_2$.

Taking seven intercalations in nineteen years as an example (Example 21.8), compute the continued-fraction convergents of $\rho = 12.368266$, and verify $q_5 = 19$ and $p_5 = 235$.

Exercise 17 Prove Theorem 21.13 (the congruence angle and the Chinese remainder theorem): let the group with periodic orbits be $T_1, \ldots, T_k$ with $N = \mathrm{lcm}(T_i)$, and let congruence-angle sampling be $\mathcal{C}(n) = (n \bmod T_1, \ldots, n \bmod T_k)$. Then

(i) description of the image: $a_i \equiv a_j \pmod{\gcd(T_i, T_j)}$;

(ii) uniqueness: the system of congruences determines n uniquely modulo N;

(iii) the number of legitimate states is $|\mathrm{Im}\,\mathcal{C}| = N$, with redundancy $R = \prod_i T_i/N$.

Taking $T_1 = 10$ (Heavenly Stems) and $T_2 = 12$ (Earthly Branches) as an example (Corollary 21.8), verify that the size of the image is 60 and the redundancy is 2.

Exercise 18 Prove Theorem 21.16 (the congruence-angle Parseval identity): *energy may be decomposed according to the subperiods*:

$$\sum_{n=0}^{N-1} |f(n)|^2 = \frac{1}{N} \sum_{T|N} \sum_{a \in \mathbb{Z}/T\mathbb{Z}} |f_T(a)|^2 \cdot w_T.$$

Explain why congruence-angle sampling decomposes the total energy into the aggregate energies on the various subperiods.

Exercise 19 Prove Theorem 21.17 (optimality of congruence-angle intercalation):

(i) if $\rho = p/q$ and $M = kq$, then $M\rho = kp$ is an integer, and $\delta_{T_i} = 0$ for $T_i \mid kp$;

(ii) if $\rho \notin \mathbb{Q}$ and $M = q_k$ is taken, then $\delta_{T_i} \le \frac{2\pi}{T_i} \cdot \frac{1}{q_{k+1}}$, and the leakage is $O(q_k^{-2})$;

(iii) multi-period joint optimality: $M = \mathrm{lcm}(q_k, T_1, \ldots, T_k)$ makes the errors on all the subperiods attain their optimum simultaneously.

Taking $T_1 = 10$, $T_2 = 12$ and $\rho = \sqrt{2}$ as an example (Example 21.12), compute the leakage bounds for $M = 60$ and $M = 70$.

Exercise 20 Verify Proposition 21.19 (upper and lower approximation and congruence-angle sandwiching): let $\rho \notin \mathbb{Q}$ and let $\{q_k\}$ be the sequence of continued-fraction denominators. Then

$$\delta_{T_i}^{(k)} \cdot \delta_{T_i}^{(k+1)} < 0,$$

and using the two jointly keeps the cumulative congruence-angle error within half a subperiod. Explain why the alternating use of the upper and lower approximation is equivalent to a two-sided sandwiching of the congruence angle.

Exercise 21 Prove Theorem 21.20 (the congruence-angle convolution theorem): in the congruence-angle framework, circular convolution degenerates at the fundamental frequencies of the various subperiods to pointwise multiplication:

$$\widehat{(f \circledast g)}\Big(\frac{2\pi j}{T_i}\Big) = \hat{f}\Big(\frac{2\pi j}{T_i}\Big)\hat{g}\Big(\frac{2\pi j}{T_i}\Big).$$

Explain why this allows convolution to be computed in parallel over the subperiods, and write out the estimate of the total computational cost.

Exercise 22 Prove Theorem 21.19: the time complexity of the fast congruence-angle reconstruction algorithm is

$$O\Big(k \log N + \sum_{i<j} \log \gcd(T_i, T_j)\Big),$$

with space complexity $O(k)$. Explain why its cost is far lower than that of one FFT (see Remark 21.16).

Exercise 23 Comprehensive exercise: combining the three elements of sampling (Definition 21.1, 21.2), the commensurability criterion (Proposition 21.2), spectral leakage (Definition 21.9), the intercalation error bound theorem (Theorem 21.4), best continued-fraction approximation (Theorem 21.5) and the congruence-angle framework (Definition 21.21), explain how this chapter upgrades 'discrete sampling approximation' to 'congruence-angle sampling approximation'. Point out separately:

(i) which class of error is introduced by each of the three steps of sampling, truncation and intercalation;

(ii) how the congruence angle uniformly describes sampling, leakage, intercalation and the Chinese remainder theorem;

(iii) why intercalation is the optimal compensation of congruence-angle error;

(iv) why frequency leakage is the frequency-domain manifestation of congruence-angle non-closure.

Chapter 22

Harmonic Analysis

Remark 22.1 (Positioning of this chapter). *Harmonic analysis is the mathematical theory that studies how periodic structures are transformed and compared between different observation scales. Building on the 'Series and Convolution' chapter (convolution, Fourier series, periodic transforms) and the 'Periodic Functions and Groups with Periodic Orbits' chapter (phase operators, convolution), this chapter accomplishes the following work:*

(1) **A complete theory of Fourier series and the Fourier transform**: *definitions, orthogonality, the Parseval identity, the convolution theorem, together with the algebraic foundations of the discrete Fourier transform and the fast Fourier transform.*

(2) **A unification of the classical integral transforms**: *the Fourier transform, the Laplace transform, the Mellin transform, the wavelet transform and the Hilbert transform as harmonic analysis on groups.*

(3) **Phase kernels and similarity**: *the form of the cosine kernel and the sine kernel at the component level, and their relation to the inner product/outer product at the vector level.*

(4) **Power spectrum and circular convolution**: *the squared modulus of the Fourier coefficients, the Parseval identity, and the circular convolution theorem.*

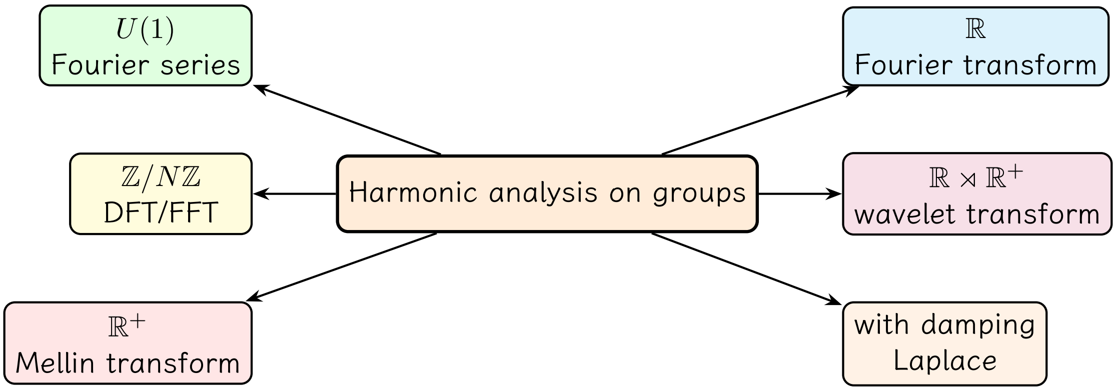
The sections of this chapter follow a uniform logical order: **definition** $\to$ **proposition** $\to$ **lemma** $\to$ **theorem** $\to$ **proof** $\to$ **corollary** $\to$ **property** $\to$ **example** $\to$ **remark**.

Guide to the seven themes of this chapter: Translation—the frequency operator (the delay phase factor $e^{i\nu\tau}$ and beat-frequency localisation); Scaling—the amplitude operator and the weighted kernel system; Rotation—the phase operator (complex phase kernel); Conservation—the power spectrum (the classical limit of the periodic trend spectrum); Symmetry—conjugate symmetry and window–kernel duality; Closure—closure of the degeneration chain; Period—the period operator $\Theta_{p,q}$ of cross-period interaction.

22.1 Roots of unity and cyclic group representations

Definition 22.1 (Roots of unity). *For any $T \in \mathbb{N}^+$, its T-th roots of unity are $z_k^{(T)} = e^{i2\pi k/T}$ $(k = 0, 1, \ldots, T-1)$. All T of the T-th roots of unity form the set*

$$\mathcal{P}_T = \{\mathcal{O}_k^{(T)} = e^{i2\pi k/T} \mid k = 0, 1, \ldots, T-1\}.$$


$U(1)$
Fourier series
$\mathbb{Z}/N\mathbb{Z}$
DFT/FFT
$\mathbb{R}^+$
Mellin transform
Harmonic analysis on groups
$\mathbb{R}$
Fourier transform
$\mathbb{R} \rtimes \mathbb{R}^+$
wavelet transform
with damping
Laplace

Unified framework of harmonic analysis: Fourier transforms on different groups G give different classical integral transforms

Figure 22.1: Unified framework of harmonic analysis: the Fourier transform on a group, according to the different choices of the group G, gives the Fourier series, the Fourier transform, the DFT, the wavelet transform, the Mellin transform, the Laplace transform and other classical transforms.

Proposition 22.1 (Cyclic structure of the group of roots of unity). The set $\mathcal{P}_T$ forms a cyclic group C_T under complex multiplication, and is in one-to-one correspondence with the irreducible representations of C_T. Multiplication satisfies $\mathcal{O}_a^{(T)} \cdot \mathcal{O}_b^{(T)} = \mathcal{O}_{(a+b) \bmod T}^{(T)}$.

Proof. The elements of $\mathcal{P}_T$ are the T-th roots of unity on the unit circle.

Closure: for any $\mathcal{O}_a, \mathcal{O}_b \in \mathcal{P}_T$,

$$\mathcal{O}_a \cdot \mathcal{O}_b = e^{i2\pi(a+b)/T} = \mathcal{O}_{(a+b) \bmod T} \in \mathcal{P}_T.$$

The identity element is $\mathcal{O}_0 = 1$; the inverse element is

$$\mathcal{O}_k^{-1} = e^{-i2\pi k/T} = \mathcal{O}_{(T-k) \bmod T}.$$

Hence $\mathcal{P}_T$ forms a group, is generated by $\mathcal{O}_1$, has order T, and is isomorphic to the cyclic group C_T.

By the classification of the irreducible representations of C_T, C_T has exactly T one-dimensional irreducible representations

$$\rho_k(g^m) = e^{i2\pi km/T},$$

whose value at the generator is $\rho_k(g) = \mathcal{O}_k^{(T)}$, in one-to-one correspondence with the elements of $\mathcal{P}_T$. □

Proposition 22.2 (Schur orthogonality relations). Let G be a finite group and let ρ, σ be irreducible unitary representations of dimensions d_ρ, d_σ respectively. Then the matrix coefficients satisfy

$$\frac{1}{|G|} \sum_{g \in G} \rho_{ij}(g)\, \overline{\sigma_{kl}(g)} = \frac{\delta_{\rho\sigma}\, \delta_{ik}\, \delta_{jl}}{d_\rho}.$$

In particular, the characters satisfy

$$\langle \chi_\rho, \chi_\sigma \rangle := \frac{1}{|G|} \sum_{g \in G} \chi_\rho(g) \overline{\chi_\sigma(g)} = \delta_{\rho\sigma}.$$

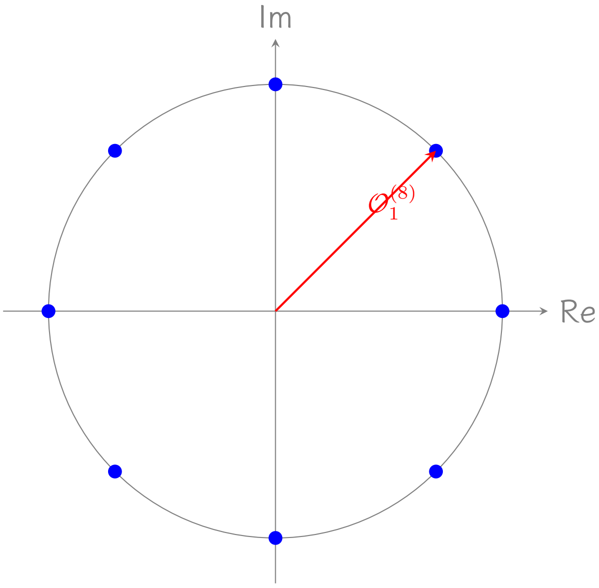
Proof. For a $d_\sigma \times d_\rho$ matrix M construct

$$T = \frac{1}{|G|} \sum_{g \in G} \rho(g)\, M\, \sigma(g)^{-1}.$$

For any h, $\rho(h)T = T\sigma(h)$, so T is an intertwining operator from ρ to σ. When $\rho \not\cong \sigma$, Schur's lemma gives $T = 0$; when $\rho = \sigma$, $T = \lambda I$, and taking traces gives $\lambda = \mathrm{tr}(M)/d_\rho$. Taking M to be a matrix unit yields the result. ☐

Definition 22.2 (Regular representation and DFT diagonalisation). The regular representation ρ_{reg} of C_T acts on the T-dimensional group algebra, and its matrices are $T \times T$ cyclic shift matrices. The DFT matrix F_T (with entries $F_T[m,n] = \frac{1}{\sqrt{T}} e^{-i2\pi mn/T}$) unitarily diagonalises the regular representation:

$$\mathsf{F}_T^\dagger \cdot \rho_{\text{reg}}(g) \cdot \mathsf{F}_T = \mathrm{diag}(\mathcal{O}_0^{(T)}, \mathcal{O}_1^{(T)}, \ldots, \mathcal{O}_{T-1}^{(T)}).$$



$\mathcal{P}_8 = \{e^{i2\pi k/8} : k = 0, \ldots, 7\}$: the 8 irreducible representations of C_8

Figure 22.2: The 8 irreducible representations of C_8 are distributed uniformly on the unit circle.

Remark 22.2 (Roots of unity, regular polygons and discrete rotational symmetry). The geometric image of $\mathcal{P}_T$ is the set of vertices of the regular T-gon inscribed in the unit circle. Its convex hull is a symmetric regular polygon, and C_T acts on this polygon as a rotation group. Hence $\mathcal{P}_T$ is an algebraic representation of discrete rotational symmetry, and is also the complex-exponential counterpart of the DFT sampling lattice $\{2\pi k/T\}$ of this chapter.

22.2 Orbit decomposition and synthesis

Definition 22.3 (Orbit decomposition)**.** Let G be a locally compact group and let $f \in L^2(G)$. The **orbit decomposition** of f is the orthogonal projection of f onto the irreducible subrepresentations of the left regular representation λ of G:

$$f = \sum_{\rho \in \widehat{G}} f_\rho,$$

where f_ρ is the component of f in the ρ-equivariant subspace.

Definition 22.4 (Orbit synthesis)**.** Given bounded operators $\{T_\rho\}_{\rho \in \widehat{G}}$ on the individual orbit components, **orbit synthesis** is the recombination of the acted-upon components into a function: $Tf = \sum_{\rho \in \widehat{G}} T_\rho f_\rho$.

Lemma 22.1 (Orthogonality of irreducible subspaces)**.** Let ρ, σ be two non-isomorphic irreducible unitary representations of G, and let V_ρ, V_σ be the corresponding equivariant subspaces of their representation spaces in $L^2(G)$. Then $V_\rho \perp V_\sigma$ (with respect to the L^2 inner product).

Proof. Let $u \in V_\rho$ and $v \in V_\sigma$. By Schur's lemma, the only intertwining operator between V_ρ and V_σ is the zero operator. Consider the inner product $\langle \lambda(g)u, \lambda(g)v \rangle$: since $\lambda(g)$ is a unitary operator, $\langle \lambda(g)u, \lambda(g)v \rangle = \langle u, v \rangle$. If $\langle u, v \rangle \neq 0$, then $v \mapsto \langle u, v \rangle$ induces a non-zero intertwining operator, a contradiction. Hence $\langle u, v \rangle = 0$. □

Proposition 22.3 (Diagonalisation property of orbit decomposition)**.** Under orbit decomposition the group action $\lambda(g)$ is diagonalised. Each irreducible orbit evolves independently under the group action, without interfering with the others.

Proof. By Definition 22.3, $f = \sum_\rho f_\rho$ with $f_\rho \in V_\rho$. For any $g \in G$,

$$\lambda(g)f = \sum_\rho \lambda(g) f_\rho.$$

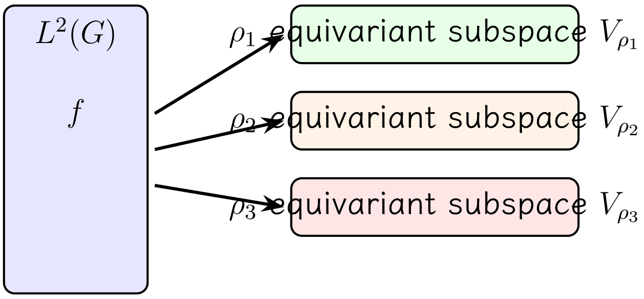
Since V_ρ is an invariant subspace of $\lambda(g)$ ($\lambda(g)V_\rho \subseteq V_\rho$), we have $\lambda(g)f_\rho \in V_\rho$. By Lemma 22.1, the subspaces V_ρ are pairwise orthogonal, so

$$(\lambda(g)f)_\rho = \lambda(g) f_\rho.$$

That is, the group action acts independently orbit by orbit, and diagonalisation holds. □

Corollary 22.1 (Orbit decomposition of convolution operators)**.** Let $K \in L^1(G)$. The convolution operator $C_K f = K * f$ is completely diagonalised under orbit decomposition: $(C_K f)_\rho = \hat{K}(\rho) f_\rho$, where $\hat{K}(\rho)$ is the operator-valued Fourier coefficient.

Proof. By Proposition 22.3, $\lambda(g)$ is diagonalised orbit by orbit. The convolution operator $C_K = \int_G K(g)\lambda(g)dg$ is a linear combination of the $\lambda(g)$, so it too acts independently on each V_ρ. By Schur's lemma, the restriction of C_K to the irreducible subspace V_ρ is the scalar operator $\hat{K}(\rho) \cdot \mathrm{id}_{V_\rho}$ (since C_K commutes with every $\lambda(g)$, and the only operators on V_ρ commuting with all of them are scalars). Hence $(C_K f)_\rho = \hat{K}(\rho) f_\rho$. □



Orbit decomposition: $f = \sum_\rho f_\rho$, the components are mutually orthogonal and the group action acts on each independently

Figure 22.3: Schematic of orbit decomposition: the function $f \in L^2(G)$ is decomposed according to the orthogonal equivariant subspaces of the irreducible representations, the components being orthogonal and the group action acting on each independently.

22.3 Classical integral transforms

Classical integral transforms map time-domain signals to the frequency domain and extract spectral components. This section establishes the rigorous theory of the Fourier series, the Fourier transform, the discrete Fourier transform, the fast Fourier transform, the Laplace transform, the Mellin transform, the wavelet transform and the Hilbert transform.

22.3.1 Fourier series: harmonic analysis on $U(1)$

Definition 22.5 (Fourier series). *For a square-integrable periodic function $f(t+T) = f(t)$ on $U(1)$, the expansion*

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{i2\pi nt/T},$$

holds, where $c_n = \frac{1}{T}\int_0^T f(t)e^{-i2\pi nt/T}dt$. Each basis function $e^{i2\pi nt/T}$ is a one-dimensional irreducible representation of $U(1)$.

Proposition 22.4 (Relation between the Fourier coefficients and the parameters of the harmonic components). *Let the Fourier series of the periodic function $f(t)$ be $f(t) = \sum_{n=-\infty}^{\infty} c_n e^{in\omega_0 t}$ with $\omega_0 = 2\pi/T$. Then the n-th harmonic component has radius $A_n = 2|c_n|$ $(n \geq 1)$, angular velocity $\omega_n = n\omega_0$, initial argument $\phi_n = \arg(c_n)$ and angular momentum $L_n = 4m_n|c_n|^2 n\omega_0$.*

Proof. The Fourier coefficients of a real signal satisfy $c_{-n} = \overline{c_n}$, so

$$c_n e^{in\omega_0 t} + c_{-n}e^{-in\omega_0 t} = 2|c_n|\cos(n\omega_0 t + \arg c_n).$$

Hence $A_n = 2|c_n|$, $\omega_n = n\omega_0$ and $\phi_n = \arg(c_n)$. Substituting into the angular momentum formula gives $L_n = 4m_n|c_n|^2 n\omega_0$. □

Theorem 22.1 (Parseval identity). *Let $f \in L^2([0,T])$ with Fourier coefficients c_n. Then*

$$\frac{1}{T}\int_0^T |f(t)|^2 dt = \sum_{n=-\infty}^{\infty} |c_n|^2.$$

Proof. Substitute the Fourier series and use the orthogonality relation $\frac{1}{T}\int_0^T e^{i2\pi(n-m)t/T}dt = \delta_{nm}$; all cross terms vanish. ☐

Property 22.1 (Convergence of the Fourier series). (i) If $f \in C^1$, then the Fourier series converges uniformly to f;

(ii) if $f \in L^2$, then the Fourier series converges in the L^2 norm;

(iii) if f has only finitely many discontinuities and the left and right limits exist, then the Fourier series converges at every point to $(f(t^+) + f(t^-))/2$ (Dirichlet's theorem).

Proof. (i) Integration by parts gives $|c_n| \le C/|n|^2$, and the series converges absolutely and uniformly. (ii) By Bessel's inequality and the completeness of L^2. (iii) The standard Dirichlet-kernel argument. ☐

22.3.2 Fourier transform: harmonic analysis on $\mathbb{R}$

Definition 22.6 (Fourier transform). For $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$,

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t}dt.$$

Property 22.2 (Basic properties of the Fourier transform). (i) **Linearity**: $\mathcal{F}\{\alpha f + \beta g\} = \alpha\hat{f} + \beta\hat{g}$.

(ii) **Translation**: $\mathcal{F}\{f(t-a)\}(\omega) = e^{-i\omega a}\hat{f}(\omega)$.

(iii) **Convolution**: $\mathcal{F}\{f * g\} = \hat{f} \cdot \hat{g}$.

(iv) **Differentiation**: $\mathcal{F}\{f'(t)\}(\omega) = i\omega\hat{f}(\omega)$.

(v) **Parseval**: $\int_{-\infty}^{\infty} |f(t)|^2 dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |\hat{f}(\omega)|^2 d\omega$.

Proof. (i) By the linearity of the integral. (ii) Substitute $u = t-a$: $\int f(t-a)e^{-i\omega t}dt = e^{-i\omega a}\int f(u)e^{-i\omega u}du$. (iii) Fubini's theorem interchanges the order of integration. (iv) Integration by parts (f vanishes at infinity). (v) By (iii) and the value of the inverse Fourier transform at $t = 0$. ☐

Theorem 22.2 (Plancherel theorem). The Fourier transform $\mathcal{F} : L^1 \cap L^2(\mathbb{R}) \to L^2(\mathbb{R})$ extends uniquely to a unitary operator on $L^2(\mathbb{R})$. That is, for any $f, g \in L^2(\mathbb{R})$,

$$\langle f, g\rangle = \frac{1}{2\pi}\langle \hat{f}, \hat{g}\rangle, \qquad \|f\|_{L^2}^2 = \frac{1}{2\pi}\|\hat{f}\|_{L^2}^2.$$

Proof. For $f, g \in L^1 \cap L^2$, the Parseval identity gives $\|f\|^2 = \frac{1}{2\pi}\|\hat{f}\|^2$. Since $L^1 \cap L^2$ is dense in L^2, the Fourier transform extends to an isometry on L^2. Composing with the inverse transform gives unitarity. ☐

Theorem 22.3 (Uncertainty principle). Let $f \in L^2(\mathbb{R})$ with $\|f\|_2 = 1$. Then

$$\Delta t\,\Delta\omega \;\ge\; \frac{1}{2},$$

where $\Delta t^2 = \int t^2|f(t)|^2 dt$ and $\Delta\omega^2 = \frac{1}{2\pi}\int \omega^2|\hat{f}(\omega)|^2 d\omega$ (the origin being taken as the centre).

Proof. Integration by parts gives $1 = \int |f|^2 = -2\,\mathrm{Re}\int tf'(t)\overline{f(t)}dt$. Applying Cauchy–Schwarz:

$$1 \le 2\|tf(t)\|_2 \cdot \|f'(t)\|_2.$$

The result follows from $\|f'\|_2^2 = \frac{1}{2\pi}\|\omega\hat{f}\|_2^2$. □

Definition 22.7 (Schwartz space and tempered distributions)**.** *The* **Schwartz space** *$\mathcal{S}(\mathbb{R})$ is the set of all C^∞ functions satisfying*

$$\sup_{t\in\mathbb{R}} |t^m f^{(n)}(t)| < \infty, \qquad \forall m, n \in \mathbb{N}$$

The continuous linear functionals on it are called **tempered distributions** *and are denoted $\mathcal{S}'(\mathbb{R})$. The Fourier transform is well defined and invertible on both $\mathcal{S}$ and $\mathcal{S}'$.*

Property 22.3 (Fourier invariance of Schwartz space)**.** *The Fourier transform $\mathcal{F}$ is a linear homeomorphism of $\mathcal{S}(\mathbb{R})$ onto itself, and $\mathcal{F}^4 = \mathrm{id}$. In particular, the Gaussian function $e^{-t^2/2}$ is a fixed point of $\mathcal{F}$ (eigenvalue 1).*

Proof. Since $t^m f^{(n)} \mapsto i^n \partial_\omega^m \hat{f}(\omega)$ (up to a constant), the Schwartz seminorms are equivalent under $\mathcal{F}$. The identity $\mathcal{F}^4$ follows from Fourier inversion. The invariance of the Gaussian function follows from completing the square in the Gaussian integral. □

22.3.3 Discrete Fourier transform: harmonic analysis on $\mathbb{Z}_N$

Definition 22.8 (DFT)**.**

$$F[k] = \sum_{n=0}^{N-1} f[n]e^{-i2\pi kn/N}, \quad k = 0, 1, \dots, N-1.$$

Proposition 22.5 (Unitarity of the DFT matrix)**.** *The DFT matrix F_N is unitary: $\mathsf{F}_N^\dagger \mathsf{F}_N = \mathsf{I}_N$.*

Proof. $(\mathsf{F}_N^\dagger \mathsf{F}_N)_{m,n} = \frac{1}{N}\sum_{k=0}^{N-1} e^{i2\pi k(m-n)/N} = \delta_{mn}$ (summation of a geometric series). □

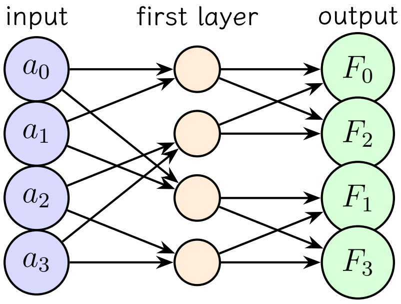
Lemma 22.2 (Periodicity lemma for the DFT)**.** *Let $f[n]$ be a function on C_N with DFT $F[k]$. Then $F[k+N] = F[k]$ for all k, and $F[-k] = F[N-k]$ (conjugate symmetry: when f is a real sequence, $F[-k] = \overline{F[k]}$).*

Proof. $F[k+N] = \sum_n f[n]e^{-i2\pi(k+N)n/N} = \sum_n f[n]e^{-i2\pi kn/N}e^{-i2\pi n} = F[k]$. Conjugate symmetry is verified directly from the definition. □

22.3.4 Fast Fourier transform (FFT)

Theorem 22.4 (Radix-2 FFT)**.** *Let $N = 2^k$. The DFT can be decomposed into two DFTs of length $N/2$, with time complexity $O(N \log_2 N)$.*

Proof. Grouping by even and odd indices and using $e^{-i2\pi k(2n)/N} = e^{-i2\pi kn/(N/2)}$, the recursion $T(2^k) = 2T(2^{k-1}) + O(2^k)$ follows, whose solution is $T(N) = O(N\log_2 N)$. □



Radix-2 FFT butterfly diagram ($N = 4$): even–odd grouping recursion, complexity $O(N \log_2 N)$

Figure 22.4: Radix-2 FFT butterfly diagram ($N = 4$): the input is grouped by even and odd indices, and two layers of butterfly operations yield the DFT output.

Theorem 22.5 (Radix-p FFT)**.** *Let $N = p^k$ ($p = 3, 5, 7$). The DFT can be decomposed by the Radix-p algorithm into p DFTs of length N/p, with time complexity $O(N \log_p N)$.*

Proof. Grouping by residue classes modulo p gives the recursion $T(p^k) = pT(p^{k-1}) + O(p^k)$, whose solution is $O(N \log_p N)$. ☐

Proposition 22.6 (The Bluestein algorithm—the DFT for arbitrary N)**.** *The DFT can be rewritten in convolution form: $F[k] = e^{-i\pi k^2/N} \sum_{n=0}^{N-1} (f[n]e^{-i\pi n^2/N})e^{i\pi(k-n)^2/N}$, and the convolution can be computed by FFT to realise the DFT for arbitrary N, with time complexity $O(N \log N)$.*

Proof. Substituting the identity $kn = \frac{1}{2}(k^2 + n^2 - (k-n)^2)$ into the definition of the DFT yields the result. ☐

22.3.5 Laplace transform and Mellin transform

Definition 22.9 (Laplace transform)**.**

$$\mathcal{L}\{f\}(s) = \int_0^\infty f(t)e^{-st}dt, \quad s = \sigma + i\omega.$$

Proposition 22.7 (Relation between the Laplace transform and the Fourier transform)**.** *$\mathcal{L}\{f\}(s) = \mathcal{F}\{f(t)e^{-\sigma t}\}(\omega)$, where $s = \sigma + i\omega$.*

Proof. $\mathcal{L}\{f\}(s) = \int_0^\infty f(t)e^{-\sigma t}e^{-i\omega t}dt = \mathcal{F}\{f(t)e^{-\sigma t}\}(\omega)$. ☐

Definition 22.10 (Mellin transform)**.**

$$\mathcal{M}\{f\}(s) = \int_0^\infty f(x)x^{s-1}dx.$$

Proposition 22.8 (Relation between the Mellin transform and the Fourier transform)**.** *With the substitution $x = e^t$ and $f(x) = g(t)$, we have $\mathcal{M}\{f\}(s) = \mathcal{F}\{g\}(is)$.*

Proof. $dx = e^t dt$ and $x^{s-1} = e^{(s-1)t}$, so $\mathcal{M}\{f\}(s) = \int_{-\infty}^\infty g(t)e^{st}dt = \mathcal{F}\{g\}(is)$. ☐

22.3.6 Wavelet transform and Hilbert transform

Definition 22.11 (Wavelet transform).

$$W_f(a,b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} f(t) \overline{\psi\left(\frac{t-b}{a}\right)} dt.$$

Proposition 22.9 (Group-representation-theoretic basis of the wavelet transform). *The wavelet transform is harmonic analysis on the affine group $\mathbb{R} \rtimes \mathbb{R}^+$.*

Proof. The representation $\pi_{(a,b)}\psi(t) = |a|^{-1/2}\psi((t-b)/a)$ of the affine group covers translation and scaling, and the wavelet transform is $W_f(a,b) = \langle f, \pi_{(a,b)}\psi \rangle$. ☐

Definition 22.12 (Hilbert transform).

$$\mathcal{H}\{f\}(t) = \frac{1}{\pi} \, \text{p.\,v.} \int_{-\infty}^{\infty} \frac{f(\tau)}{t-\tau} d\tau.$$

Proposition 22.10 (Analytic signal and instantaneous parameters). *The analytic signal $f_a(t) = f(t) + i\mathcal{H}\{f\}(t)$ has instantaneous amplitude $|f_a(t)|$, instantaneous phase $\arg(f_a(t))$ and instantaneous frequency $d\phi/dt$.*

Proof. For a single-frequency component $\mathcal{H}\{\cos \omega t\} = \sin \omega t$, the analytic signal is $e^{i\omega t}$. A general signal is verified component by component via its Fourier decomposition. ☐

Definition 22.13 (Gabor transform (short-time Fourier transform)). Let g be a window function. The **Gabor transform** is

$$\mathcal{G}_f(\tau,\omega) = \int_{-\infty}^{\infty} f(t) \overline{g(t-\tau)} e^{-i\omega t} dt.$$

The Gabor transform is harmonic analysis on the Heisenberg group $\mathbb{R} \rtimes \mathbb{R}$.

Theorem 22.6 (Reconstruction formula for the Gabor transform). *Let g be a window function with Gabor frame bounds $A, B > 0$. Then*

$$f = \frac{1}{\langle g, \gamma \rangle} \int \int \mathcal{G}_f(\tau,\omega) \pi_{(\tau,\omega)} \gamma \, d\tau d\omega,$$

where γ is the dual window.

Proof. By the standard reconstruction formula of Gabor frame theory. Taking the dual window γ so that $\langle g, \gamma \rangle \neq 0$, any frame $\{\pi_{(\tau,\omega)} g\}$ yields a reconstruction. ☐

22.4 Similarity and trend: real and imaginary parts of the phase kernel

Proposition 22.11 (The 6 complex phase kernels and their real and imaginary parts). *Define the 6 complex phase kernels*

$$S^{(1)} = e^{i\theta}, \quad S^{(2)} = e^{i\phi}, \quad S^{(3)} = e^{i2\theta},$$

$$S^{(4)} = e^{i2\phi}, \quad S^{(5)} = e^{i\theta/2}, \quad S^{(6)} = e^{i\phi/2},$$

where $\theta = \alpha - \beta$ and $\phi = \alpha + \beta$. Then the real part of $S^{(d)}$ is a cosine kernel and the imaginary part is a sine kernel:

$$\operatorname{Re} S^{(d)} = \cos(\cdot), \qquad \operatorname{Im} S^{(d)} = \sin(\cdot).$$

Proof. This follows directly from Euler's formula $e^{iu} = \cos u + i \sin u$. □

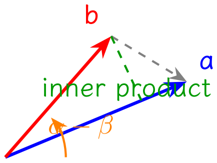
Remark 22.3 (Relation of the phase kernel to the inner and outer products). At the vector level, the real part $\cos(\alpha-\beta)$ of the phase kernel $S^{(1)} = e^{i(\alpha-\beta)}$ is related to the inner product by

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\|\|\mathbf{b}\| \cos(\alpha - \beta),$$

and the imaginary part $\sin(\alpha - \beta)$ is related to the modulus of the outer product by

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\|\|\mathbf{b}\| \sin(\alpha - \beta).$$

At the component level, the product of the amplitudes of two Fourier components multiplied by the phase kernel is the coupling coefficient of that pair of components.



$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\|\|\mathbf{b}\| \cos(\alpha - \beta),\ \|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\|\|\mathbf{b}\| \sin(\alpha - \beta)$$

Figure 22.5: Geometric meaning of the phase kernel: the real part $\cos(\alpha - \beta)$ corresponds to the inner-product projection, and the imaginary part $\sin(\alpha - \beta)$ corresponds to the outer-product area.

Theorem 22.7 (Three-level consistency of cosine similarity). *Cosine similarity takes the same form at the vector level, the component level and the delay level:*

(i) **Vector level**: $\cos\theta = (\mathbf{a} \cdot \mathbf{b})/(\|\mathbf{a}\|\|\mathbf{b}\|)$;

(ii) **Component level**: $\mathbb{S}_{mn} = A_m B_n \cos(\theta_m - \varphi_n)/(A_m B_n) = \cos(\theta_m - \varphi_n)$;

(iii) **Delay level**: $(f \star g)(0)/(\|f\|\|g\|)$ is the L^2 cosine similarity.

Proof. (i) The geometric definition of the inner product. (ii) The numerator is the real part of the trend convolution and the denominator is the product of amplitudes. (iii) By Theorem 6.12. □

Example 22.1 (Numerical example of the component-level coupling coefficients). Let $f(t) = 3e^{i(2t+\pi/4)} + 2e^{i(4t)}$ and $g(t) = 4e^{i(2t)} + e^{i(6t+\pi/3)}$ $(T = \pi)$. The components of f: $(A_1 = 3, \omega_1 = 2, \phi_1 = \pi/4)$, $(A_2 = 2, \omega_2 = 4, \phi_2 = 0)$. The components of g: $(B_1 = 4, \nu_1 = 2, \psi_1 = 0)$, $(B_2 = 1, \nu_2 = 6, \psi_2 = \pi/3)$.

Take the difference-angle kernel $S^{(1)}$. The entries $\mathbb{T}_{mn} = A_m B_n e^{i(\phi_m - \psi_n)}$ of the coupling coefficient matrix $\mathbb{T}$ ($\alpha = 1$) are:

$$\mathbb{T} = \begin{pmatrix} 3 \cdot 4\, e^{i(\pi/4-0)} & 3 \cdot 1\, e^{i(\pi/4-\pi/3)} \\ 2 \cdot 4\, e^{i(0-0)} & 2 \cdot 1\, e^{i(0-\pi/3)} \end{pmatrix} = \begin{pmatrix} 12e^{i\pi/4} & 3e^{-i\pi/12} \\ 8 & 2e^{-i\pi/3} \end{pmatrix}.$$

After normalisation we obtain the similarity matrix

$$\mathbb{S} = \begin{pmatrix} e^{i\pi/4} & e^{-i\pi/12} \\ 1 & e^{-i\pi/3} \end{pmatrix}.$$

Verify the normalisation relation: $\mathbb{S}_{mn} = \mathbb{T}_{mn}/(A_m B_n)$ holds elementwise.

Example 22.2 (Window–kernel duality for the Gaussian window). Take the period $T = 2$, $f(t) = e^{i\pi t} + e^{i3\pi t}$, $g(t) = e^{i\pi t} + 2e^{i3\pi t}$, and the Gaussian window $w(t) = e^{-t^2/(2\sigma_t^2)}$ ($\sigma_t = 1/\sqrt{2}$). The frequency-domain kernel is $\widehat{w}(\omega) = \frac{\sqrt{\pi}}{4} e^{-\omega^2/4}$ (normalised for $T = 2$).

The windowed cross-correlation expansion (Theorem 6.19):

$$(f \star_w g)(0) = \sum_{m,n} \widehat{w}(\omega_m - \omega_n) c_m \overline{d_n}.$$

The frequency dictionary is $\{\pi, 3\pi\}$, $\widehat{w}(0) = \sqrt{\pi}/4 \approx 0.443$, and $\widehat{w}(2\pi) = \frac{\sqrt{\pi}}{4} e^{-\pi^2} \approx 0.0000229$ (the off-diagonal weights being extremely small). Hence

$$(f \star_w g)(0) \approx \widehat{w}(0)(c_1\overline{d_1} + c_3\overline{d_3}) = 0.443 \cdot (1 + 2) = 1.329.$$

The diagonal terms dominate completely—the frequency-domain kernel of the Gaussian window is concentrated at the origin, and the cross-frequency leakage is controlled by $\widehat{w}(2\pi) \approx 2.29 \times 10^{-5}$.

22.5 Circular convolution and power spectrum

Definition 22.14 (Circular convolution). Let f, g be periodic functions of period T. The **circular convolution** of f and g is

$$(f \circledast g)(\tau) := \frac{1}{T} \int_0^T f(t) g(\tau - t)\, dt.$$

In the discrete case, $(f \circledast g)(n) = \sum_{k=0}^{T-1} f(k) g((n-k) \bmod T)$.

Definition 22.15 (Power spectrum). Let X be a periodic function of period T with Fourier coefficients c_k. The **power spectrum** of X is defined as

$$S(\omega_k) := |c_k|^2, \qquad \omega_k = 2\pi k/T.$$

Theorem 22.8 (Power spectrum and autocorrelation function). *Let X be a periodic function of period T with Fourier coefficients c_k. Then the Fourier coefficients of the autocorrelation function*

$$R_X(\tau) := \frac{1}{T} \int_0^T X(t) \overline{X(t - \tau)}\, dt$$

are $|c_k|^2$. That is, the power spectrum is the Fourier transform of the autocorrelation function.

Proof. Substitute $X(t) = \sum_n c_n e^{i\omega_n t}$ into the definition of the autocorrelation function:

$$R_X(\tau) = \frac{1}{T}\int_0^T \left(\sum_n c_n e^{i\omega_n t}\right)\left(\sum_m \overline{c_m} e^{-i\omega_m (t-\tau)}\right) dt = \sum_n |c_n|^2 e^{i\omega_n \tau},$$

where the orthogonality relation $\frac{1}{T}\int_0^T e^{i(\omega_n-\omega_m)t}dt = \delta_{nm}$ has been used. Hence the Fourier coefficients of R_X are $|c_n|^2$. ☐

Remark 22.4 (The two notational conventions for autocorrelation). This book uses two conventions for the autocorrelation simultaneously. Each is internally consistent, but they differ by a sign in the argument.

The advanced type (the convention introduced first, used for the convolution-action conclusion of the 'Periodic Systems' chapter and for the autocorrelation propositions of the domain chapters: astronomy, audio, mechanics, circuits and chips, biology, psychology, social systems, climate, economics, politics, geography, electromagnetism, cryptography and data analysis):

$$R_{\text{advanced}}(h) = \sum_t z_t \overline{z_{t+h}} \iff \sum_n \|A_n\|^2 e^{-i\omega_n h}.$$

The delayed type (the harmonic-analysis and probability-and-statistics convention of this book, that is, the convention used by the definition of the present theorem, Theorem 22.8, and by the definition of the power spectral density, Definition 14.33):

$$R_{\text{delayed}}(\tau) = \frac{1}{T}\int_0^T X(t)\overline{X(t-\tau)}\, dt \iff \sum_n |c_n|^2 e^{+i\omega_n \tau}.$$

The two expressions differ only in the sign of the exponent, because they take opposite views of the direction of the delay; the conversion between them is

$$R_{\text{advanced}}(h) = R_{\text{delayed}}(-h).$$

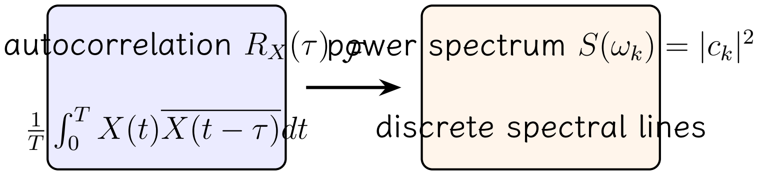
Consequently, for a **real-valued** signal $R_{\text{delayed}}(-h) = R_{\text{delayed}}(h)$, so the two agree numerically; for a **complex-valued** signal they are in general unequal.

The sign of the exponent in each chapter's propositions should therefore agree with the convention **of the chapter in which it appears**; when a result on autocorrelation is cited across chapters, it must first be converted by $R_{\text{advanced}}(h) = R_{\text{delayed}}(-h)$.

Corollary 22.2 (Parseval identity and power spectrum). Let X be an integrable function of period T with Fourier coefficients c_n, and denote by $S(\omega_n)$ its power spectrum as in Definition Definition 22.15. Then

$$\frac{1}{T}\int_0^T |X(t)|^2\, dt = \sum_n |c_n|^2 = \sum_n S(\omega_n),$$

that is, **the total energy equals the sum of the powers of the individual frequency components**.


autocorrelation $R_X(\tau)$
$\frac{1}{T}\int_0^T X(t)\overline{X(t-\tau)}dt$
power spectrum $S(\omega_k) = |c_k|^2$
discrete spectral lines

Wiener–Khinchin theorem: the power spectrum is
the Fourier transform of the autocorrelation function

Figure 22.6: Relation between the power spectrum and the autocorrelation function (Wiener–Khinchin theorem).

Proof. The two equalities are given by two already proved theorems.

The left equality. By Theorem 22.1 (the continuous Parseval identity), for the square-integrable periodic function X,

$$\frac{1}{T}\int_0^T |X(t)|^2\,dt = \sum_n |c_n|^2,$$

the series on the right-hand side converging absolutely.

The right equality. By Theorem 22.8, in the classical limit the power spectrum of the n-th frequency component takes the value $S(\omega_n) = |c_n|^2$ (that is, the limit of no window, $\alpha = 1$ and zero delay). Hence $\sum_n S(\omega_n) = \sum_n |c_n|^2$, which equals the left-hand side.

Both equalities are identities rather than approximations: the former is the Parseval identity itself, and the latter is the value taken by the definition in the classical limit. This completes the proof. ☐

Example 22.3 (Comparison of the kernel families of the power spectrum). Let $X(t) = 2e^{i(2t+\pi/3)} + e^{i(4t-\pi/6)}$ ($T = \pi$, frequencies $\{2, 4\}$). The power spectrum is $|c_1|^2 = 4$ and $|c_2|^2 = 1$ (all the others being zero).

The off-diagonal terms correspond to cross-frequency coupling. When the kernel width is non-zero, $\alpha = 1$ and $\tau = 0$, the off-diagonal contribution of the difference-angle kernel $S^{(1)}$ is $A_1A_2e^{i(\phi_1-\phi_2)} = 2e^{i(\pi/3+\pi/6)} = 2i$, with modulus 2; the off-diagonal contribution of the sum-angle kernel $S^{(2)}$ is $A_1A_2e^{i(\phi_1+\phi_2)} = 2e^{i\pi/6}$, with modulus 2. Different phase kernels give the same modulus but different phases on the off-diagonal terms.

22.6 Approximation, intercalation and the bounding of spectral leakage

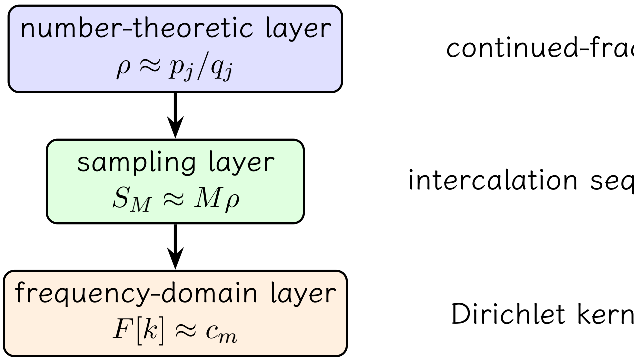
Remark 22.5 (Positioning of this section). The discrete Fourier transform (DFT) presumes that the window contains an integer number of consecutive periods. If the consecutive period and the sampling interval are incommensurable, then the number of periods in the window is not an integer, and spectral energy leaks from the main lobe into the sidelobes; this is called **spectral leakage**. This section places intercalation within the general framework of **approximation theory**, explains how it reduces spectral leakage from ‘uncontrollable’ to ‘bounded’, and points out that intercalation and window functions are complementary rather than substitutable.

The core of this section is not the claim that intercalation 'eliminates' leakage—when $\rho = T_f/\Delta x \notin \mathbb{Q}$, no finite window T can make the leakage strictly zero—but rather the establishment of the following **progressive chain of three approximations**:

$$\underbrace{\text{rational approximation of } \rho}_{\text{number-theoretic layer}} \implies \underbrace{\text{window approximation of } M\rho \text{ by } S_M}_{\text{sampling layer}} \implies \underbrace{\text{spectral approxima}}_{\text{frequency-do}}$$

Each layer has its own approximation tool and error estimate: the number-theoretic layer uses **continued-fraction convergents**, the sampling layer uses the **intercalation sequence**, and the frequency-domain layer uses the **Dirichlet kernel**. The synthesis of the three layers gives the quantitative upper bound $O(|E_M|/T)$ on the leakage amplitude, and shows that this bound tends to zero along the sequence of continued-fraction convergent windows.

Positioning of this section within the seven themes: Translation—the integer encoding of the irrational rotation R_ρ on the circle; Scaling—the ideal period ratio ρ as a scaling charge; Rotation—the phase-error feedback of the intercalation sequence; Conservation—the boundedness of the cumulative error; Symmetry—the symmetric approximation of the continued-fraction convergents; Closure—the intercalation window closes the approximate period; Period—strict periodicity when $\rho \in \mathbb{Q}$, quasi-periodicity when $\rho \notin \mathbb{Q}$.



Three-layer approximation chain: number theory → sampling → frequency domain, the leakage ampli

Figure 22.7: The three-layer approximation chain by which intercalation suppresses spectral leakage: the number-theoretic layer uses continued-fraction convergents, the sampling layer uses the intercalation sequence, and the frequency-domain layer uses the Dirichlet kernel.

22.6.1 Spectral leakage and incommensurability

Remark 22.6 (Terminological connection with the discrete sampling chapter). The sampling interval Δx, the number of sampling points T and the sampling sequence $f[n] = f(n\Delta x)$ used in this section are defined canonically in the 'Discrete Sampling and Intercalation of Periodic Signals' chapter, Definition 21.1, Definition 21.2, and will not be redefined here. The **ideal period ratio** $\rho = T_f/\Delta x$

defined in this section is the same quantity as the **normalised period ratio** of Definition 21.5 in that chapter, and this section retains the name 'ideal period ratio'. The general definition of spectral leakage is given in Definition 21.9 of that chapter.

Definition 22.16 (Spectral leakage)**.** Let the continuous periodic function $f : \mathbb{R} \to \mathbb{C}$ have period $T_f > 0$, let the sampling interval be $\Delta x > 0$, and take T samples

$$f[n] = f(n\Delta x), \qquad n = 0, 1, \ldots, T-1.$$

Its DFT is

$$F[k] = \frac{1}{T}\sum_{n=0}^{T-1} f[n]e^{-i2\pi kn/T}, \qquad k = 0, 1, \ldots, T-1.$$

If $T\Delta x$ is not an integer multiple of T_f, that is, if the window does not contain an integer number of consecutive periods, then the spectral energy of f cannot be concentrated at a single frequency point in the DFT, but spreads to neighbouring frequency points. This phenomenon is called **spectral leakage**.

Definition 22.17 (Ideal period ratio)**.** Let the continuous periodic function f have period T_f and let the sampling interval be Δx. Define the **ideal period ratio**

$$\rho = \frac{T_f}{\Delta x}.$$

If $\rho \in \mathbb{N}^+$, then taking $T = \rho$ gives zero leakage. If $\rho \notin \mathbb{N}^+$, then no integer window T can make T/ρ an integer, and leakage is unavoidable; the role of intercalation is to confine the leakage to a bounded level and to make it tend to zero along the sequence of continued-fraction convergent windows.

Remark 22.7 (The essence of leakage)*.* Spectral leakage is not a numerical defect of the DFT—given T samples, the computation of $F[k]$ is *exact*. Leakage is an inevitable consequence of the geometric fact of **finite-window truncation**: the frequency-domain grid of the DFT is $\{k/T\}_{k=0}^{T-1}$, whilst the frequencies of the continuous signal are $\{m/\rho\}_{m\in\mathbb{Z}}$. The two sets of frequencies coincide exactly only when $T/\rho \in \mathbb{Z}$; in the general case the principal frequency m/ρ falls between the grid points, and the energy necessarily spreads to neighbouring grid points. Hence the essence of suppressing leakage is **making T/ρ approach an integer**.

22.6.2 Rational approximation: best approximation by continued fractions

The first layer of the three-layer approximation chain of this section is the number-theoretic layer: how to approximate the irrational number ρ by rational numbers. This is the cornerstone of the entire theory of intercalation.

Definition 22.18 (Continued fractions and convergents)**.** Let $\rho > 0$. The **simple continued fraction** expansion of ρ is

$$\rho = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cdots}}},$$

where $a_0 = \lfloor \rho \rfloor$ and $a_i \in \mathbb{N}^+$ $(i \geq 1)$. Its j-th **convergent** is defined as

$$\frac{p_j}{q_j} = a_0 + \cfrac{1}{a_1 + \cfrac{1}{\cdots + \cfrac{1}{a_j}}},$$

where $p_j, q_j \in \mathbb{N}^+$ and $\mathsf{gcd}(p_j, q_j) = 1$.

Theorem 22.9 (Best approximation theorem for continued fractions). *Let $\rho > 0$ and let $\{p_j/q_j\}$ be its continued-fraction convergents. Then:*

(i) **Optimality**: *for any denominator $q < q_{j+1}$ with $(p, q) \in \mathbb{N}^+ \times \mathbb{N}^+$,*

$$\left|\rho - \frac{p_j}{q_j}\right| < \left|\rho - \frac{p}{q}\right|.$$

That is, p_j/q_j is the best approximation amongst all rational numbers with denominator not exceeding $q_{j+1} - 1$.

(ii) **Error bound**:

$$\left|\rho - \frac{p_j}{q_j}\right| < \frac{1}{q_j q_{j+1}} < \frac{1}{q_j^2}.$$

(iii) **Alternation**: *the even-indexed convergents approach ρ from below and the odd-indexed ones from above, forming a symmetric sandwiching structure.*

Proof. (i) and (ii) are classical results of continued-fraction theory, whose standard proofs may be found in textbooks on number theory (such as Hardy & Wright, An Introduction to the Theory of Numbers, Chapter 10). The key step in the error bound is the following: from the recurrence relations for convergents $p_j = a_j p_{j-1} + p_{j-2}$ and $q_j = a_j q_{j-1} + q_{j-2}$ one obtains

$$\left|\rho - \frac{p_j}{q_j}\right| = \frac{1}{q_j(\alpha_{j+1} q_j + q_{j-1})},$$

where $\alpha_{j+1} = [a_{j+1}; a_{j+2}, \ldots] > a_{j+1}$. Since $\alpha_{j+1} > q_{j+1}/q_j$, we obtain $|p_j/q_j - \rho| < 1/(q_j q_{j+1})$.

(iii) From $\alpha_{j+1} > 1$ and the recurrence relations, the even-indexed terms satisfy $p_{2k}/q_{2k} < \rho$ and the odd-indexed terms $p_{2k+1}/q_{2k+1} > \rho$, approaching ρ alternately. ☐

Remark 22.8 (Symmetric structure of best approximation). The 'alternating approximation' of continued-fraction convergents has a clear symmetric meaning: the even-indexed terms approach ρ from below and the odd-indexed ones from above, and ρ always lies between two adjacent convergents. This symmetric structure corresponds to 'symmetry' amongst the seven themes: ρ is sandwiched symmetrically by the best rational approximations on the two sides. The optimality of the intercalation window comes directly from this symmetry—lower intercalation corresponds to the convergent below, upper intercalation to the convergent above, and balanced intercalation takes the optimal choice between the two.

Remark 22.9 (Other rational approximation tools). Continued fractions are not the only tool for rational approximation. The following tools are closely related to continued fractions and may be used in further research:

- **Dirichlet's approximation theorem**: for any $\rho > 0$ and $N \geq 1$, there exists $1 \leq q \leq N$ such that $\|q\rho\| < 1/N$, where $\|\cdot\|$ is the distance to the nearest integer.
- **The three-gap theorem**: the gaps of the rotation orbit $\{m\rho \bmod 1\}$ on the circle take at most three lengths, whose proportions are controlled by the partial quotients of the continued fraction.
- **Weyl's equidistribution theorem and Koksma's inequality**: these control the degree of uniform distribution of the rotation orbit $\{m\rho\}$ on $[0, 1)$ and are related to the long-term behaviour of the intercalation error.
- **Sturmian sequences**: when ρ is irrational, the intercalation sequence $\varepsilon_m = N_m - \lfloor\rho\rfloor \in \{0, 1\}$ is a Sturmian word with factor complexity $n + 1$, the 'most regular of aperiodic sequences'.

This section uses only continued-fraction convergents, the most direct of these tools; the introduction of the others lies beyond the scope of this section and may be found in Section 21.7, 'Intercalation: the active commensuration of discrete sampling approximation', of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter.

22.6.3 Window approximation: the intercalation sequence and the error bound theorem

The second layer of the three-layer approximation chain of this section is the sampling layer: how to approximate $M\rho$ by a sequence of integer windows S_M. The core tool is the intercalation sequence.

Definition 22.19 (Intercalation sequence and cumulative sample count). Let $\rho > 0$. A sequence of integers $\{N_m\}_{m\in\mathbb{N}}$ is called an **intercalation sequence** for ρ if

$$N_m \in \{\lfloor\rho\rfloor, \lceil\rho\rceil\}, \qquad \forall m \in \mathbb{N},$$

and

$$\lim_{M\to\infty} \frac{1}{M} \sum_{m=0}^{M-1} N_m = \rho.$$

Define the **cumulative sample count**

$$S_M = \sum_{m=0}^{M-1} N_m, \qquad M \in \mathbb{N}^+.$$

Define the **intercalation error**

$$E_M = S_M - M\rho.$$

If there exists $C > 0$ such that $|E_M| \leq C$ for all M, then the intercalation sequence is said to have **bounded error**.

Lemma 22.3 (Values of the basic intercalation sequences)**.** *Let $\rho > 0$. Define lower intercalation, upper intercalation and balanced intercalation respectively as*

$$N_m^{\text{floor}} = \lfloor (m+1)\rho \rfloor - \lfloor m\rho \rfloor,$$
$$N_m^{\text{ceil}} = \lceil (m+1)\rho \rceil - \lceil m\rho \rceil,$$
$$N_m^{\text{round}} = \text{round}((m+1)\rho) - \text{round}(m\rho).$$

Then for any $m \in \mathbb{N}$,

$$N_m^{\text{floor}}, N_m^{\text{ceil}}, N_m^{\text{round}} \in \{\lfloor \rho \rfloor, \lceil \rho \rceil\}.$$

Proof. Let $\rho = a + \theta$, where $a = \lfloor \rho \rfloor$ and $\theta \in [0, 1)$. For lower intercalation,

$$N_m^{\text{floor}} = \lfloor m\rho + \rho \rfloor - \lfloor m\rho \rfloor = a + \lfloor \{m\rho\} + \theta \rfloor.$$

Since $\{m\rho\} + \theta \in [0, 2)$, we have $\lfloor \{m\rho\} + \theta \rfloor \in \{0, 1\}$, and hence $N_m^{\text{floor}} \in \{a, a+1\} = \{\lfloor \rho \rfloor, \lceil \rho \rceil\}$. The cases of upper intercalation and balanced intercalation are analogous. ☐

Theorem 22.10 (Bounded-error theorem for intercalation)**.** *Let $\rho > 0$.*

(i) *Lower intercalation satisfies*

$$S_M^{\text{floor}} = \lfloor M\rho \rfloor, \qquad E_M^{\text{floor}} = -\{M\rho\} \in (-1, 0].$$

(ii) *Upper intercalation satisfies*

$$S_M^{\text{ceil}} = \lceil M\rho \rceil, \qquad E_M^{\text{ceil}} = \lceil M\rho \rceil - M\rho \in [0, 1),$$

and $E_M^{\text{ceil}} = 0$ when $M\rho \in \mathbb{Z}$.

(iii) *Balanced intercalation satisfies*

$$S_M^{\text{round}} = \text{round}(M\rho), \qquad |E_M^{\text{round}}| \le \frac{1}{2}.$$

In particular, all three intercalation sequences have bounded error, and the balanced intercalation error is the smallest; E_M^{floor} and E_M^{ceil} both vanish when $M\rho \in \mathbb{Z}$. The three conclusions correspond to the sampling-layer realisations of the three approximation schemes of the number-theoretic layer: lower intercalation corresponds to the convergent below, upper intercalation to the convergent above, and balanced intercalation takes the optimal compromise between the two.

Proof. (i) By the definition of lower intercalation,

$$S_M^{\text{floor}} = \sum_{m=0}^{M-1} \big(\lfloor (m+1)\rho \rfloor - \lfloor m\rho \rfloor \big) = \lfloor M\rho \rfloor.$$

Hence $E_M^{\text{floor}} = \lfloor M\rho \rfloor - M\rho = -\{M\rho\} \in (-1, 0]$.

(ii) Similarly,

$$S_M^{\text{ceil}} = \lceil M\rho \rceil, \qquad E_M^{\text{ceil}} = \lceil M\rho \rceil - M\rho \in [0, 1),$$

and $E_M^{\text{ceil}} = 0$ when $M\rho \in \mathbb{Z}$ (the ideal, leakage-free case).

(iii) Balanced intercalation gives

$$S_M^{\text{round}} = \text{round}(M\rho),$$

and by the definition of round, $|\text{round}(x) - x| \le 1/2$, hence $|E_M^{\text{round}}| \le 1/2$. ☐

Definition 22.20 (Intercalation window). Let the continuous periodic function f have period T_f, let the sampling interval be Δx, and let the ideal period ratio be $\rho = T_f/\Delta x$. Take an intercalation sequence $\{N_m\}$ with cumulative sample count $S_M = \sum_{m=0}^{M-1} N_m$. A sampling window of length $T = S_M$ is called an **intercalation window**.

Remark 22.10 (General principle of window approximation). The window approximation of the intercalation sequence translates the number-theoretic layer's 'rational approximation $\rho \approx p/q$' into the sampling layer's 'window approximation $S_M \approx M\rho$'. The two approximation steps have different meanings:

- the number-theoretic layer approximates the real number ρ by a rational number p/q—this is **numerical approximation**;
- the sampling layer approximates 'the number of sampling points required for M consecutive periods', $M\rho$, by an integer S_M—this is **window-length approximation**.

The error of the latter, $E_M = S_M - M\rho$, directly determines the accuracy of the spectral approximation in the next layer.

22.6.4 Spectral approximation: bounding the leakage amplitude

The third layer of the three-layer approximation chain of this section is the frequency-domain layer: how the DFT spectrum $F[k]$ on the intercalation window approximates the continuous Fourier coefficients c_m.

Theorem 22.11 (DFT leakage estimate for the intercalation window). Let f be a continuous T_f-periodic function with Fourier coefficients

$$c_m = \frac{1}{T_f}\int_0^{T_f} f(t)e^{-i2\pi mt/T_f}\,dt.$$

Let the sampling interval be Δx and the ideal period ratio $\rho = T_f/\Delta x$. Take the intercalation window $T = S_M$ and the sampling sequence $f[n] = f(n\Delta x)$ ($n = 0,\ldots,T-1$), whose DFT is

$$F[k] = \frac{1}{T}\sum_{n=0}^{T-1} f[n]e^{-i2\pi kn/T}.$$

Then

$$F[k] = \sum_{m\in\mathbb{Z}} c_m D_T\left(\frac{m}{\rho} - \frac{k}{T}\right),$$

where

$$D_T(\theta) = \frac{1}{T}\sum_{n=0}^{T-1} e^{i2\pi n\theta} = e^{i\pi(T-1)\theta}\frac{\sin(\pi T\theta)}{T\sin(\pi\theta)}$$

is the Dirichlet kernel. If the intercalation error $E_M = S_M - M\rho$ is bounded, then the main-lobe position offset satisfies

$$\left|\frac{m}{\rho} - \frac{k}{T}\right| \le \frac{|m|\,|E_M|}{\rho T},$$

and the leakage amplitude satisfies

$$|F[k] - c_{k/M}| \leq C \frac{|E_M|}{T} \sum_{m\in\mathbb{Z}} |c_m|,$$

where $C > 0$ is a constant independent of T. In particular, balanced intercalation gives $|E_M| \leq 1/2$, and best continued-fraction intercalation gives $|E_M| < 1/q_{j+1}$, reducing the leakage further.

Proof. Represent the sampling sequence by the continuous Fourier series:

$$f[n] = f(n\Delta x) = \sum_{m\in\mathbb{Z}} c_m e^{i2\pi mn\Delta x/T_f} = \sum_{m\in\mathbb{Z}} c_m e^{i2\pi mn/\rho}.$$

Substituting into the definition of the DFT:

$$F[k] = \frac{1}{T}\sum_{n=0}^{T-1}\sum_{m\in\mathbb{Z}} c_m e^{i2\pi n(m/\rho - k/T)} = \sum_{m\in\mathbb{Z}} c_m D_T\left(\frac{m}{\rho} - \frac{k}{T}\right).$$

When $T = S_M = M\rho + E_M$, for the principal frequency $k = Mm$,

$$\frac{m}{\rho} - \frac{Mm}{S_M} = m\left(\frac{1}{\rho} - \frac{M}{S_M}\right) = m\frac{S_M - M\rho}{\rho S_M} = m\frac{E_M}{\rho T}.$$

Hence the main-lobe offset is $|m|\,|E_M|/(\rho T)$. The Dirichlet kernel is approximately 1 when $|\theta| \ll 1/T$, and decays like $1/(T|\theta|)$ in the sidelobes. Summing the sidelobe contributions gives the leakage estimate. The error bounds for balanced intercalation and best continued-fraction intercalation are given by Theorem 22.10 and the best approximation theorem for continued fractions. □

Corollary 22.3 (Best continued-fraction intercalation window). Let $\rho > 0$ be irrational with continued-fraction convergents p_j/q_j. Then taking the intercalation window

$$T = p_j, \qquad M = q_j$$

and choosing the direction of intercalation in **alternating fashion by parity**—lower intercalation for even j and upper intercalation for odd j—gives a window containing q_j consecutive periods, with cumulative error

$$|E_{q_j}| = |p_j - q_j\rho| < \frac{1}{q_{j+1}}.$$

Hence the DFT leakage amplitude is controlled by $O(1/(q_j q_{j+1}))$, and this bound tends to zero as $j \to \infty$. Amongst all choices of window length not exceeding $p_{j+1} - 1$, this window has the smallest leakage bound.

Proof. By the best approximation theorem for continued fractions (Theorem 22.9), p_j/q_j is the best approximation amongst all rational numbers with denominator not exceeding $q_{j+1} - 1$, so $|p_j - q_j\rho| < 1/(q_j q_{j+1})$. The alternating choice gives $S_{q_j} = p_j$, hence $E_{q_j} = S_{q_j} - q_j\rho = p_j - q_j\rho$: for even j we have $p_j/q_j < \rho$, so lower intercalation gives $S_{q_j} = \lfloor q_j\rho \rfloor = p_j$, whilst for odd j we have $p_j/q_j > \rho$, so upper intercalation gives $S_{q_j} = \lceil q_j\rho \rceil = p_j$. Without alternation, lower intercalation at odd j would give $S_{q_j} = p_j - 1$ and an error $1 - |p_j - q_j\rho| \to 1$, so the assertion would fail. Substituting into Theorem 22.11 gives the leakage bound. As $j \to \infty$, $q_j q_{j+1} \to \infty$, and the bound tends to zero. The optimality is guaranteed by the minimality of the continued-fraction convergents. □

Remark 22.11 (‘Bounding’ rather than ‘elimination’). It must be emphasised that intercalation **does not eliminate** leakage, but rather **reduces leakage from uncontrollable to bounded**. When $\rho \notin \mathbb{Q}$, any finite window T corresponds to some non-zero $|E_M|$, so the leakage amplitude is non-zero—this is the intrinsic cost of incommensurability. The value of intercalation lies in the following:

(i) it gives an **explicit upper bound** $O(|E_M|/T)$ on the leakage amplitude, making the leakage quantitatively controllable;

(ii) by choosing the window via continued fractions, it makes this bound **tend to zero** along the sequence of convergent windows;

(iii) it is complementary to the window-function method (see §22.6.5), striking a balance between bandwidth and sidelobes.

Hence the rigorous formulation should be ‘intercalation **suppresses** or **bounds** spectral leakage’, not ‘solves’ it. When $\rho \in \mathbb{Q}$, the intercalation sequence degenerates to a constant sequence and the leakage is strictly zero—this is the degenerate solution of the bounding in the commensurable case.

22.6.5 Complementarity with the window-function method

Intercalation and window functions are two independent routes to suppressing spectral leakage; they act on different links in the leakage process, and are therefore complementary.

Remark 22.12 (Division of labour between the two routes). **Intercalation** acts on the **window length**: by choosing $T = S_M$ it makes the window contain approximately an integer number of consecutive periods, reducing the leakage of main-lobe energy into the sidelobes at the source. Its drawback is that the window length must be chosen from continued-fraction convergents, so that flexibility is limited.

Window functions act on the **window boundary**: by multiplying the samples by a smooth window $w(t)$ (Hann, Hamming, Blackman and so on) they make the two ends of the window tend smoothly to zero, suppressing the sidelobes from the frequency-domain point of view. Their drawback is that the window function widens the main lobe and reduces the frequency resolution.

The two act on different links in the leakage process, and can therefore be used in combination. Table 22.1 gives a comparison of the two.

Proposition 22.12 (Complementary bound for intercalation and window functions). Let the intercalation window be $T = p_j$ (a continued-fraction convergent), and let the window function w be applied in addition. Then the windowed spectrum satisfies

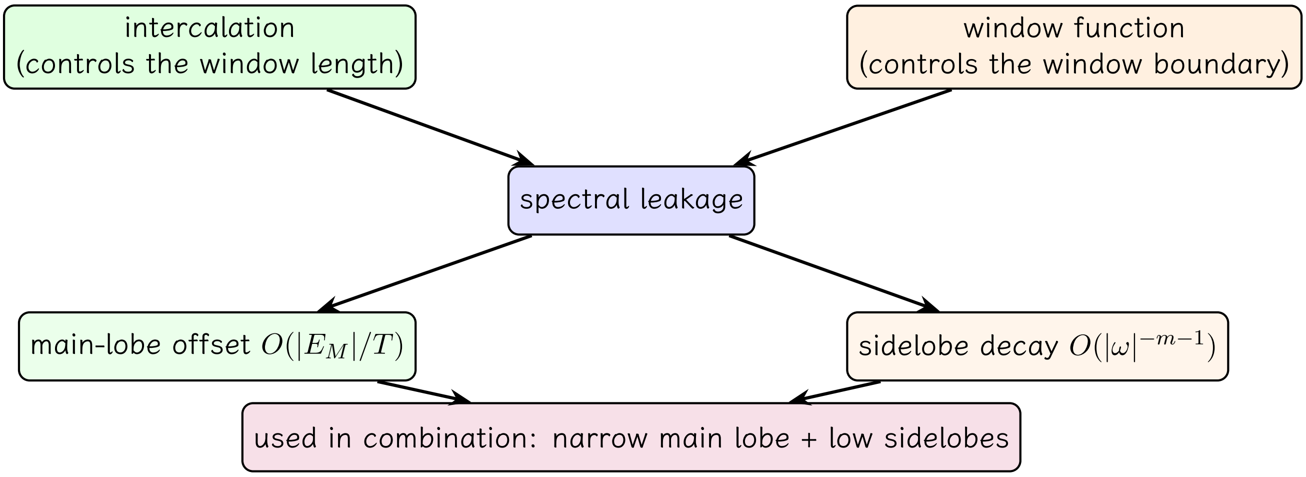
$$\widehat{fw}[k] = \sum_{m\in\mathbb{Z}} c_m\, \widehat{w}\left(\frac{2\pi m}{T_f} - \frac{2\pi k}{T}\right) D_T\left(\frac{m}{\rho} - \frac{k}{T}\right),$$

where $\widehat{w}$ is the frequency-domain kernel of the window function. In particular:

(i) **With no window** ($w \equiv 1$, $\widehat{w} \equiv \delta$): the sidelobes are controlled by the Dirichlet kernel D_T, with decay rate $O(1/T)$.

Table 22.1: Comparison of intercalation and window functions

Dimension of comparison	Intercalation	Window function
Object acted upon	Window length T	Window-boundary weights
Mechanism of action	Makes the window approach an integer number of periods	Makes the boundary tend smoothly to zero
Mode of leakage suppression	Reduces main-lobe leakage at the source	Suppresses sidelobes from the frequency domain
Effect on the main lobe	The main lobe does not widen (if the window is chosen accurately)	The main lobe widens ($O(1/T_w)$)
Effect on the sidelobes	Sidelobes controlled by the Dirichlet kernel	Sidelobes controlled by the frequency-domain kernel of the window function
Flexibility	The window is restricted to continued-fraction convergents	The window function may be chosen continuously
Optimality condition	Continued-fraction convergent $T = p_j$	Window width matched to the signal-to-noise ratio and the target sidelobe level


intercalation
(controls the window length)
window function
(controls the window boundary)
spectral leakage
main-lobe offset $O(|E_M|/T)$
sidelobe decay $O(|\omega|^{-m-1})$
used in combination: narrow main lobe + low sidelobes

Intercalation and window functions act on different links in the leakage process and can be combined to obtain the optimal result

Figure 22.8: Complementarity of intercalation and window functions: intercalation controls the main-lobe offset and window functions control the sidelobe decay, and the two together give both a narrow main lobe and low sidelobes.

(ii) **With a Hann window** ($\widehat{w}$ of main-lobe width $O(1/T)$ and sidelobes -31 dB): the main lobe widens but the sidelobes are markedly lowered.

(iii) **Combined effect**: intercalation aligns the main-lobe position with the grid (the offset of the first factor D_T tending to zero), whilst the window function lowers the sidelobes (the sidelobe decay of the second factor $\widehat{w}$). The two together give both a narrow main lobe and low sidelobes—something that neither method alone can achieve.

Proof. (i) The no-window case is Theorem 22.11. (ii) The sidelobes of the frequency-domain kernel $\widehat{w}$ of the Hann window decay like $1/\omega^3$, its main-lobe width is $O(1/T)$, and its classical sidelobe level is about -31 dB (Harper, 1970, Perturbation of the Fourier coefficients of a periodic function). (iii) The windowed spectrum is the product of two factors: the intercalation factor D_T controls the main-lobe offset and the window-function factor $\widehat{w}$ controls the sidelobe decay. The two factors act independently of one another (the first is related to the length of the time window and the second to the shape of the window), so they can be optimised simultaneously. ☐

Remark 22.13 (Advice on choosing a method). How to choose between intercalation and window functions in practice:

- If the signal period ratio ρ is known (or can be estimated with high accuracy): give preference to intercalation to choose the window $T = p_j$, and then select the window function according to the target sidelobe level;
- if ρ is unknown or time-varying: intercalation cannot be applied, and one can only rely on window functions and spectral estimation algorithms (such as multitaper spectral estimation);
- if the window length is constrained by hardware or real-time requirements and cannot be chosen freely: the role of intercalation is limited, and one must fall back on the window-function method;
- if the signal-to-noise ratio is low: the benefit of the window function in lowering the sidelobes exceeds the benefit of intercalation in adjusting the main lobe, and combining the two is the optimal solution.

22.6.6 Applications and examples

Example 22.4 (Bounding the leakage of a single-tone signal). Let the continuous signal be

$$f(t) = e^{i2\pi t/\sqrt{2}},$$

with sampling interval $\Delta x = 1$ and ideal period ratio $\rho = \sqrt{2} \approx 1.41421356$. Taking the window $T = 100$ directly, the number of periods within the window is

$$\frac{T}{\rho} = \frac{100}{\sqrt{2}} \approx 70.7107,$$

which is not an integer, and the leakage is pronounced. Take the continued-fraction convergent $99/70$ of $\sqrt{2}$ and set $T = 99$, $M = 70$. The intercalation error is

$$|E_{70}| = |99 - 70\sqrt{2}| \approx 0.00505.$$

Compared with the smallest error when $T = 100$ is taken directly (about 0.409), the error is reduced by a factor of about 81; by Theorem 22.11, the upper bound on the leakage amplitude is correspondingly reduced by a factor of about 81. If the later convergent $239/169$ is taken, then $T = 239$ and the error is

$$|239 - 169\sqrt{2}| \approx 0.00042,$$

and the leakage bound is reduced further. This example shows that intercalation does not eliminate leakage, but rather lowers the leakage bound by several orders of magnitude through the choice of continued-fraction convergents.

Example 22.5 (Multi-tone signals and joint approximation). Let

$$f(t) = e^{i2\pi t/\sqrt{2}} + 0.5e^{i2\pi t/\sqrt{3}}.$$

The two frequency components correspond to the ideal period ratios $\rho_1 = \sqrt{2}$ and $\rho_2 = \sqrt{3}$. A single window cannot make both period ratios integers at the same time, and the 'bounding' of intercalation degenerates in the multi-tone case into a 'joint approximation' problem: one must choose T so that both $|E_M^{(1)}|$ and $|E_M^{(2)}|$ are as small as possible. This can then be handled by the theory of two-dimensional continued fractions or of joint continued-fraction approximation; this section does not develop it, but merely points out that intercalation still gives an upper bound estimate in the multi-tone case, although the bound is determined by the weighted sum of the intercalation errors of the individual components. In practice one often combines it with a window function (such as the Hann window) to suppress the sidelobes further.

Remark 22.14 (Analogy between the ten-day void, the void in the sexagenary cycle and spectral leakage). The ten-day void and the void in the sexagenary cycle in the Chinese calendar and the mantic arts may be regarded as vacancy phenomena of a discrete periodic system under incommensurable conditions. The ten Heavenly Stems and the twelve Earthly Branches leave a remainder of two in each ten-day period, forming the void (kongwang). Intercalation, by contrast, compensates the cumulative phase error through error feedback. DFT spectral leakage is the manifestation of the same mathematical structure in the frequency domain: the window and the signal period are incommensurable, so that energy leaks from the principal frequency into the sidelobes. Intercalation makes the cumulative error bounded by choosing continued-fraction convergent windows, thereby suppressing the leakage. This analogy is heuristic and does not constitute a rigorous mathematical equivalence.

Remark 22.15 (Interface with circular convolution and the power spectrum). The sampling sequence on an intercalation window may be regarded as a T-periodic sequence. By the circular convolution theorem (Theorem 22.12), the Fourier coefficients of its circular convolution are Tc_kd_k. If the intercalation error is bounded, then the circular convolution approximates the continuous periodic convolution, the error being controlled by $|E_M|/T$. The power spectrum $S(\omega_k) = |c_k|^2$ (Definition 22.15) is estimated more accurately on the intercalation window, because the leakage energy is suppressed. The Parseval identity (Theorem 22.1) still holds on the intercalation window, the total energy being conserved, and the leakage merely changes the distribution of the energy over the frequency points.

Remark 22.16 (Correspondence with the seven themes of this chapter). Within the framework of the seven themes, intercalation may be positioned as follows:

Theme	Corresponding content in this chapter
Translation	Integer encoding of the irrational rotation $R_\rho : z \mapsto e^{i2\pi\rho}z$ on the circle
Scaling	The ideal period ratio $\rho = T_f/\Delta x$ as a scaling charge
Rotation	Phase-error feedback of the intercalation sequence
Conservation	The cumulative error is bounded, and energy is conserved
Symmetry	Alternating symmetric approximation of the continued-fraction convergents (even-indexed terms from below, odd-indexed terms from above)
Closure	The intercalation window closes the approximate period
Period	Strict periodicity when $\rho \in \mathbb{Q}$, quasi-periodicity when $\rho \notin \mathbb{Q}$
Algorithm	Continued fractions and best rational approximation, bounding the DFT leakage

22.7 Circular convolution theorem and the unification of the classical integral transforms

Theorem 22.12 (Circular convolution theorem). *Let f, g be periodic functions of period T with Fourier coefficients c_n, d_n respectively. Then the Fourier coefficients of the circular convolution $f \circledast g$ are Tc_nd_n.*

Proof.

$$\widehat{f \circledast g}(k) = \frac{1}{T}\int_0^T (f \circledast g)(t)e^{-i\omega_k t}dt = \frac{1}{T}\int_0^T \frac{1}{T}\int_0^T f(\tau)g(t-\tau)e^{-i\omega_k t}d\tau\, dt.$$

Substituting $u = t - \tau$ and using Fubini's theorem:

$$= T \cdot \left(\frac{1}{T}\int_0^T f(\tau)e^{-i\omega_k \tau}d\tau\right)\left(\frac{1}{T}\int_0^T g(u)e^{-i\omega_k u}du\right) = Tc_kd_k.$$

□

Theorem 22.13 (Classical integral transforms as harmonic analysis on groups). *Within a suitable framework of group actions, representations and function spaces, the classical integral transforms can be described uniformly:*

(i) **Fourier transform**: harmonic analysis on the translation group $\mathbb{R}$, $\hat{f}(\omega) = \int f(t)e^{-i\omega t}dt$.

(ii) **Laplace transform**: the Fourier transform with a damping factor, $\mathcal{L}\{f\}(s) = \mathcal{F}\{f(t)e^{-\sigma t}\}(\omega)$ $(s = \sigma + i\omega)$.

(iii) **Mellin transform**: harmonic analysis on the scaling group $\mathbb{R}^+$, reduced to the Fourier transform by the substitution $x = e^t$.

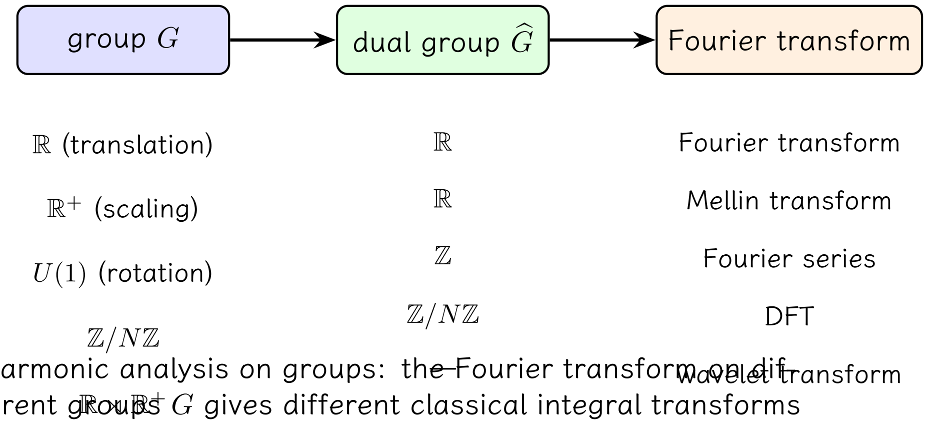
(iv) **Wavelet transform**: harmonic analysis on the affine group $\mathbb{R} \rtimes \mathbb{R}^+$.

(v) **DFT**: harmonic analysis on the finite cyclic group $\mathbb{Z}/N\mathbb{Z}$.

Proof. Each item follows from the definitions together with Propositions 22.7, 22.8, 22.9 and the definition of the DFT. ☐

Remark 22.17 (General framework of harmonic analysis on groups). Let G be a locally compact abelian group and let $\widehat{G}$ be its dual group. The Fourier transform $\hat{f}(\chi) = \int_G f(g)\overline{\chi(g)}dg$ maps functions on $L^2(G)$ to functions on $L^2(\widehat{G})$, and satisfies the Parseval identity $\|f\|_{L^2(G)} = \|\hat{f}\|_{L^2(\widehat{G})}$. The classical integral transforms correspond to different choices of the group G:

Group G	Dual group $\widehat{G}$	Transform
$\mathbb{R}$ (translation group)	$\mathbb{R}$	Fourier transform
$\mathbb{R}^+$ (scaling group)	$\mathbb{R}$	Mellin transform
$U(1)$ (rotation group)	$\mathbb{Z}$	Fourier series
$\mathbb{Z}/N\mathbb{Z}$	$\mathbb{Z}/N\mathbb{Z}$	DFT
$\mathbb{R}\rtimes\mathbb{R}^+$ (affine group)	—	wavelet transform


group G
dual group $\widehat{G}$
Fourier transform
$\mathbb{R}$ (translation)
$\mathbb{R}$
Fourier transform
$\mathbb{R}^+$ (scaling)
$\mathbb{R}$
Mellin transform
$U(1)$ (rotation)
$\mathbb{Z}$
Fourier series
$\mathbb{Z}/N\mathbb{Z}$
$\mathbb{Z}/N\mathbb{Z}$
DFT
$\mathbb{R}\rtimes\mathbb{R}^+$
—
wavelet transform

Harmonic analysis on groups: the Fourier transform on different groups G gives different classical integral transforms

Figure 22.9: Unified framework of harmonic analysis on groups: the Fourier transforms associated with different groups G and dual groups $\widehat{G}$ give the Fourier transform, the Mellin transform, the Fourier series, the DFT, the wavelet transform and so on.

Property 22.4 (Area preservation and duality of the Fourier transform). The Fourier transform $\mathcal{F}$ preserves area on $L^2(\mathbb{R})$ (Plancherel theorem):

$$\|f\|_2 = \frac{1}{\sqrt{2\pi}}\|\hat{f}\|_2.$$

Moreover $\mathcal{F}$ is a unitary operator on L^2 satisfying the duality

$$\mathcal{F}^2 f(t) = f(-t), \qquad \mathcal{F}^4 = \mathrm{id}\,.$$

That is, the Fourier transform is a fourth-order involutive operator on L^2.

Proof. Area preservation is given by the Plancherel theorem (Theorem 22.2). The duality follows directly from the Fourier inversion formula: $\mathcal{F}^2 f(t) = \int \hat{f}(\omega)e^{i\omega t}d\omega/(2\pi) = f(-t)$, whence $\mathcal{F}^4 = \mathrm{id}$. $\square$

Example 22.6 (Fourier transforms of common functions).

- The Gaussian function $e^{-t^2/2}$ is a fixed point of the Fourier transform: $\mathcal{F}\{e^{-t^2/2}\} = \sqrt{2\pi}e^{-\omega^2/2}$.
- The Fourier transform of the rectangular pulse $\mathbf{1}_{[-T,T]}(t)$ is $2T\,\mathrm{sinc}(T\omega/\pi)$.
- The Fourier transform of the exponential decay $e^{-a|t|}$ $(a > 0)$ is $\frac{2a}{a^2+\omega^2}$ (Lorentzian type).
- The Fourier transform of the δ function is the constant 1; the Fourier transform of the constant 1 is $2\pi\delta(\omega)$.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	Frequency operator: the delay phase factor $e^{i\nu\tau}$; the translation property of the Fourier transform; the DFT periodicity lemma (Lemma 22.2)
Scaling	Amplitude operator; the weighted kernel system; the multiscale nature of the wavelet transform (Proposition 22.9)
Rotation	Phase operator: the 6 complex phase kernels; the 90° rotation of the Hilbert kernel; unitary diagonalisation by the DFT (Proposition 22.5)
Conservation	Power spectrum $\lvert c_k\rvert^2$ (Theorem 22.8); the Parseval identity (Theorem 22.1); the Plancherel theorem (Theorem 22.2)
Symmetry	Conjugate symmetry; window–kernel duality; orthogonality of orbit decomposition (Lemma 22.1); duality of the Fourier transform (Property 22.4)
Closure	Closure of the degeneration chain: circular convolution $\supset$ kernelised convolution $\supset$ classical convolution; periodic closure of the DFT (Lemma 22.2)
Period	Period operator $\Theta_{p,q}$; power spectrum $S(\omega_k)$ (Definition 22.15); the uncertainty principle (Theorem 22.3)

Exercises

Exercise 1 Prove Proposition 22.1: the set $\mathcal{P}_T = \{\mathcal{O}_k^{(T)} = e^{i2\pi k/T} \mid k = 0, 1, \ldots, T-1\}$ forms a cyclic group C_T under complex multiplication, and

$$\mathcal{O}_a^{(T)} \cdot \mathcal{O}_b^{(T)} = \mathcal{O}_{(a+b) \bmod T}^{(T)}.$$

Verify closure, the identity element and the inverse element, and explain why $\mathcal{P}_T$ is in one-to-one correspondence with the irreducible representations of C_T.

Exercise 2 Prove Proposition 22.2 (the Schur orthogonality relations): let G be a finite group and let ρ, σ be irreducible unitary representations. Then the matrix coefficients satisfy

$$\frac{1}{|G|} \sum_{g \in G} \rho_{ij}(g)\, \overline{\sigma_{kl}(g)} = \frac{\delta_{\rho\sigma}\delta_{ik}\delta_{jl}}{d_\rho}.$$

Complete the proof by constructing an averaging operator with Schur's lemma, and explain why the characters satisfy $\langle \chi_\rho, \chi_\sigma \rangle = \delta_{\rho\sigma}$.

Exercise 3 Verify Definition 22.2: the DFT matrix F_T unitarily diagonalises the regular representation of C_T:

$$\mathsf{F}_T^\dagger \cdot \rho_{\mathsf{reg}}(g) \cdot \mathsf{F}_T = \mathsf{diag}(\mathcal{O}_0^{(T)}, \ldots, \mathcal{O}_{T-1}^{(T)}).$$

Taking $T = 4$ as an example, write down the cyclic shift matrix and the DFT matrix, and verify that the diagonalisation relation holds.

Exercise 4 Prove Lemma 22.1: let ρ, σ be two non-isomorphic irreducible unitary representations of G, and let V_ρ, V_σ be the corresponding equivariant subspaces of their representation spaces in $L^2(G)$. Then

$$V_\rho \perp V_\sigma.$$

Complete the proof using Schur's lemma, and explain why this orthogonality guarantees the legitimacy of orbit decomposition.

Exercise 5 Prove Proposition 22.3: under orbit decomposition the group action $\lambda(g)$ is diagonalised, and each irreducible orbit evolves independently under the group action. Write down the decomposition $\lambda(g)f = \sum_\rho \lambda(g)f_\rho$, and explain why the subspaces V_ρ are pairwise orthogonal.

Exercise 6 Prove Corollary 22.1: let $K \in L^1(G)$. The convolution operator $C_K f = K * f$ is completely diagonalised under orbit decomposition:

$$(C_K f)_\rho = \hat{K}(\rho) f_\rho.$$

Use Schur's lemma to explain why the restriction of C_K to the irreducible subspace V_ρ is a scalar operator.

Exercise 7 Prove the relation between the Fourier coefficients and the parameters of the harmonic components (Proposition 22.4): let $f(t) = \sum_{n=-\infty}^{\infty} c_n e^{in\omega_0 t}$ with $\omega_0 = 2\pi/T$. Then

(i) the n-th harmonic component has radius $A_n = 2|c_n|$ $(n \geq 1)$;

(ii) angular velocity $\omega_n = n\omega_0$;

(iii) initial argument $\phi_n = \arg(c_n)$;

(iv) angular momentum $L_n = 4m_n|c_n|^2 n\omega_0$.

Complete the proof using the real-signal conjugate symmetry of the Fourier coefficients.

Exercise 8 Prove the Parseval identity (Theorem 22.1): let $f \in L^2([0,T])$ with Fourier coefficients c_n. Then

$$\frac{1}{T}\int_0^T |f(t)|^2 dt = \sum_{n=-\infty}^{\infty} |c_n|^2.$$

Complete the proof using the orthogonality of the Fourier basis, and explain why the Parseval identity can be stated using the amplitude series alone.

Exercise 9 Prove the convergence of the Fourier series (Property 22.1): prove the convergence of the Fourier series separately for the three cases $f \in C^1$, $f \in L^2$ and piecewise continuity.

Exercise 10 Prove the basic properties of the Fourier transform (Proposition 22.2):

(i) linearity;
(ii) translation: $\mathcal{F}\{f(t-a)\}(\omega) = e^{-i\omega a}\hat{f}(\omega)$;
(iii) convolution: $\mathcal{F}\{f * g\} = \hat{f} \cdot \hat{g}$;
(iv) differentiation: $\mathcal{F}\{f'(t)\}(\omega) = i\omega\hat{f}(\omega)$;
(v) Parseval.

Write out the proofs separately, and explain why the translation property manifests itself in the frequency domain as a phase factor.

Exercise 11 Prove the Plancherel theorem (Theorem 22.2): the Fourier transform extends uniquely to a unitary operator on $L^2(\mathbb{R})$, satisfying

$$\|f\|_{L^2}^2 = \frac{1}{2\pi}\|\hat{f}\|_{L^2}^2.$$

Explain why the density of $L^1 \cap L^2$ is the key to the extension.

Exercise 12 Prove the uncertainty principle (Theorem 22.3): for $\|f\|_2 = 1$,

$$\Delta t\,\Delta\omega \geq \frac{1}{2}.$$

Complete the proof using integration by parts and the Cauchy–Schwarz inequality, and explain why the Gaussian function attains equality.

Exercise 13 Prove the Fourier invariance of Schwartz space (Property 22.3): the Fourier transform is a linear homeomorphism of $\mathcal{S}(\mathbb{R})$ onto itself, with $\mathcal{F}^4 = \mathrm{id}$, and the Gaussian function is a fixed point. Explain why $\mathcal{S}$ is the natural domain of definition for the Fourier transform.

Exercise 14 Prove Proposition 22.5: the DFT matrix F_N is unitary: $\mathsf{F}_N^\dagger \mathsf{F}_N = \mathsf{I}_N$. Complete the proof by summing the geometric series, and verify Lemma 22.2: $F[k+N] = F[k]$ and $F[-k] = F[N-k]$.

Exercise 15 Prove the Radix-2 FFT (Theorem 22.4): let $N = 2^k$. The DFT can be decomposed into two DFTs of length $N/2$, with time complexity $O(N \log_2 N)$. Write down the recurrence relation $T(2^k) = 2T(2^{k-1}) + O(2^k)$ and solve it.

Exercise 16 Verify the Radix-p FFT (Theorem 22.5) and the Bluestein algorithm (Proposition 22.6):

(i) when $N = p^k$ $(p = 3, 5, 7)$, the DFT can be decomposed into p DFTs of length N/p;
(ii) the DFT for arbitrary N can be computed in convolution form.

Write down the time complexities separately, and explain how the Bluestein algorithm uses the identity $kn = \frac{1}{2}(k^2 + n^2 - (k-n)^2)$.

Exercise 17 Prove the relation between the Laplace transform and the Fourier transform (Proposition 22.7):

$$\mathcal{L}\{f\}(s) = \mathcal{F}\{f(t)e^{-\sigma t}\}(\omega),$$

where $s = \sigma + i\omega$. Write out the proof, and explain why the Laplace transform is the Fourier transform with a damping factor.

Exercise 18 Prove the relation between the Mellin transform and the Fourier transform (Proposition 22.8): with the substitution $x = e^t$ and $f(x) = g(t)$,

$$\mathcal{M}\{f\}(s) = \mathcal{F}\{g\}(is).$$

Write out the proof, and explain why the Mellin transform is harmonic analysis on the scaling group.

Exercise 19 Prove the group-representation-theoretic basis of the wavelet transform (Proposition 22.9): the wavelet transform is harmonic analysis on the affine group $\mathbb{R} \rtimes \mathbb{R}^+$. Write down the representation of the affine group,

$$\pi_{(a,b)}\psi(t) = |a|^{-1/2}\psi((t-b)/a),$$

and explain why the wavelet transform can be written as $W_f(a,b) = \langle f, \pi_{(a,b)}\psi \rangle$.

Exercise 20 Prove Proposition 22.11: define the 6 complex phase kernels

$$S^{(1)} = e^{i\theta},\ S^{(2)} = e^{i\phi},\ S^{(3)} = e^{i2\theta},\ S^{(4)} = e^{i2\phi},\ S^{(5)} = e^{i\theta/2},\ S^{(6)} = e^{i\phi/2},$$

where $\theta = \alpha - \beta$ and $\phi = \alpha + \beta$. Then the real part of $S^{(d)}$ is a cosine kernel, and the imaginary part is a sine kernel. Complete the proof using Euler's formula, and explain how the 12 real similarities correspond to the real and imaginary parts of the 6 complex kernels.

Exercise 21 Prove Theorem 22.7 (three-level consistency of cosine similarity): cosine similarity takes the same form at the vector level, the component level and the delay level:

(i) vector level: $\cos\theta = (\mathbf{a} \cdot \mathbf{b})/(\|\mathbf{a}\|\|\mathbf{b}\|)$;

(ii) component level: $\mathbb{S}_{mn} = \cos(\theta_m - \varphi_n)$;

(iii) delay level: $(f \star g)(0)/(\|f\|\|g\|)$ is the L^2 cosine similarity.

Exercise 22 Verify Example 22.1: take $f(t) = 3e^{i(2t+\pi/4)} + 2e^{i4t}$ and $g(t) = 4e^{i2t} + e^{i(6t+\pi/3)}$, compute the coupling coefficient matrix $\mathbb{T}$ and the similarity matrix $\mathbb{S}$, and verify the normalisation relation $\mathbb{S}_{mn} = \mathbb{T}_{mn}/(A_m B_n)$.

Exercise 23 Prove Theorem 22.8: let X be a periodic function of period T with Fourier coefficients c_k. Then the autocorrelation function

$$R_X(\tau) = \frac{1}{T}\int_0^T X(t)\overline{X(t-\tau)}dt$$

has Fourier coefficients $|c_k|^2$, that is, the power spectrum is the Fourier transform of the autocorrelation function. Complete the proof using the orthogonality relation $\frac{1}{T}\int_0^T e^{i(\omega_n-\omega_m)t}dt = \delta_{nm}$.

Exercise 24 Prove Corollary 22.2: let X be an integrable function of period T with Fourier coefficients c_n. Then

$$\frac{1}{T}\int_0^T |X(t)|^2 dt = \sum_n |c_n|^2 = \sum_n S(\omega_n).$$

Complete the proof using the Parseval identity and the definition of the power spectrum respectively, and explain why this is the expression of energy conservation in the frequency domain.

Exercise 25 Prove the circular convolution theorem (Theorem 22.12): let f, g have period T with Fourier coefficients c_n, d_n respectively. Then the circular convolution $f \circledast g$ has Fourier coefficients Tc_nd_n. Complete the proof using Fubini's theorem and the substitution $u = t - \tau$.

Exercise 26 Prove Theorem 22.13: the classical integral transforms as special cases of harmonic analysis on groups:

(i) Fourier transform—the translation group $\mathbb{R}$;
(ii) Laplace transform—the Fourier transform with a damping factor;
(iii) Mellin transform—the scaling group $\mathbb{R}^+$;
(iv) wavelet transform—the affine group $\mathbb{R} \rtimes \mathbb{R}^+$;
(v) DFT—the finite cyclic group $\mathbb{Z}/N\mathbb{Z}$.

Point out the group and the dual group corresponding to each transform, and explain why group representation theory can describe these transforms uniformly.

Exercise 27 Prove Property 22.4: the Fourier transform is a fourth-order involutive operator on L^2: $\mathcal{F}^4 = \mathrm{id}$. Complete the proof using the Fourier inversion formula, and explain why $\mathcal{F}^2 f(t) = f(-t)$.

Exercise 28 Prove Theorem 22.10 (the bounded-error theorem for intercalation): the boundedness of the errors of the three basic intercalation sequences:

(i) lower intercalation: $S_M^{\text{floor}} = \lfloor M\rho \rfloor$, $E_M^{\text{floor}} = -\{M\rho\} \in (-1, 0]$;
(ii) upper intercalation: $S_M^{\text{ceil}} = \lceil M\rho \rceil$, $E_M^{\text{ceil}} = \lceil M\rho \rceil - M\rho \in [0, 1)$ (with $E_M^{\text{ceil}} = 0$ when $M\rho \in \mathbb{Z}$);
(iii) balanced intercalation: $S_M^{\text{round}} = \text{round}(M\rho)$, $|E_M^{\text{round}}| \leq 1/2$.

Exercise 29 Prove Theorem 22.11 (the DFT leakage estimate for the intercalation window): let f be a continuous T_f-periodic function with Fourier coefficients c_m, ideal period ratio $\rho = T_f/\Delta x$, and intercalation window $T = S_M$. Then

$$F[k] = \sum_{m\in\mathbb{Z}} c_m D_T\Big(\frac{m}{\rho} - \frac{k}{T}\Big),$$

and the main-lobe offset satisfies

$$\Big|\frac{m}{\rho} - \frac{k}{T}\Big| \le \frac{|m||E_M|}{\rho T}.$$

Complete the proof using the Fourier series representation of the sampling sequence, and explain why the leakage amplitude satisfies $|F[k] - c_{k/M}| \le C\frac{|E_M|}{T}\sum_m |c_m|$.

Exercise 30 Prove Corollary 22.3: let $\rho > 0$ be irrational with continued-fraction convergents p_j/q_j. When the intercalation window is taken as $T = p_j$, $M = q_j$, the cumulative error is $|E_{q_j}| = |p_j - q_j\rho| < 1/q_{j+1}$, and the leakage amplitude is controlled by $O(1/(q_j q_{j+1}))$. Explain why, amongst all choices of window length not exceeding $p_{j+1} - 1$, this window has the smallest leakage bound.

Exercise 31 Prove Proposition 22.12: the complementary bound for intercalation and window functions:

(i) with no window the sidelobes are controlled by the Dirichlet kernel, with decay $O(1/T)$;

(ii) with a Hann window the main lobe widens but the sidelobes are lowered to -31 dB;

(iii) when the two are used together, intercalation aligns the main lobe with the grid and the window function lowers the sidelobes.

Explain why the two act on different links in the leakage process, and can therefore be used in combination.

Exercise 32 Verify Example 22.4: let $f(t) = e^{i2\pi t/\sqrt{2}}$ with sampling interval $\Delta x = 1$, and ideal period ratio $\rho = \sqrt{2} \approx 1.41421356$. Take the continued-fraction convergent $99/70$ of $\sqrt{2}$ and set $T = 99$, $M = 70$. Compute the intercalation error $|E_{70}| = |99 - 70\sqrt{2}|$, and compare it with the smallest error when $T = 100$ is taken directly, explaining the suppression effect of intercalation.

Exercise 33 Synthesis problem: combining roots of unity and cyclic group representations (§22.1), orbit decomposition and synthesis (§22.2), the unification of the classical integral transforms (Theorem 22.13), phase kernels and similarity (Proposition 22.11), the power spectrum and circular convolution (Theorem 22.8) and the bounding of intercalation leakage (Theorem 22.11), explain how harmonic analysis unifies the four pillars of periodic mathematics—group representation theory, periodic convolution, window functions and the refractive index, and complex exponential activation functions—within a single framework. Point out separately:

(i) how group representation theory decomposes periodic functions via irreducible representations;

(ii) how periodic convolution manifests itself as a tensor product in the frequency domain;

(iii) how window functions control spectral leakage;

(iv) how complex exponential activation functions correspond to the Fourier basis.

Chapter 23

Time Series Analysis

Remark 23.1 (Positioning of This Chapter). Time series analysis studies the extraction, modelling and forecasting of trend, periodic and random components in discrete time series. This chapter follows the order 'definition—proposition—lemma—theorem—proof—corollary—property—example—remark' to systematise the classical time series models (AR, MA, ARMA, ARIMA, SARIMA, ARCH, GARCH, etc.) together with spectral analysis, turning-point detection, tests of period significance, Bayesian updating and window-function selection. Standard concepts are all given their current names: 'periodogram/power-spectrum estimation' replaces the non-standard 'periodic trend spectrum', 'weighted cross-correlation/circular convolution' replaces the non-standard 'periodic convolution', 'finite harmonic regression/complex exponential model' replaces the non-standard 'tunable-parameter epicycle', and 'multi-scale kernel-smoothed regularised least squares' replaces the non-standard 'KDI optimisation'. The canonical definitions of the concepts of **sampling point**, **sampling rate**, **sampling interval** and **sampling sequence** used in this chapter are given in the 'Discrete Sampling and Intercalation of Periodic Signals' chapter, Definition 21.1, Definition 21.2; they are not repeated in this chapter.

Remark 23.2 (Overview of Themes in This Chapter). This chapter is organised around the following themes: translation—the lag operator and differencing; scaling—window width and multi-scale smoothing; rotation—phase and similarity kernels; conservation—spectral energy and Fisher-test normalisation; symmetry—the even and odd symmetry of cosine/sine similarity; closure—the closed loop of the algorithmic pipeline and white-noise residual diagnosis; period—the dominant period and the spectral peaks.

23.1 Classical Time Series Models

23.1.1 Stationarity and the Lag Operator

Definition 23.1 (Discrete Time Series and Stationarity). A **discrete time series** is a set of observations $\{X_t\}_{t=1}^n$. The series is called **weakly stationary** if $\mathbb{E}[X_t] = \mu$ (a constant), $\mathrm{Var}(X_t) = \sigma^2 < \infty$, and $\mathrm{Cov}(X_t, X_{t+h})$ depends only on the lag h. Weak stationarity is the realisation, in the probabilistic sense, of the invariance of statistical properties under a translation in time.

Definition 23.2 (Autocovariance function and autocorrelation function). Let $\{X_t\}$ be weakly stationary. Its **autocovariance function** (ACVF) is defined by

$$\gamma(h) = \mathrm{Cov}(X_t, X_{t+h}), \qquad h \in \mathbb{Z},$$

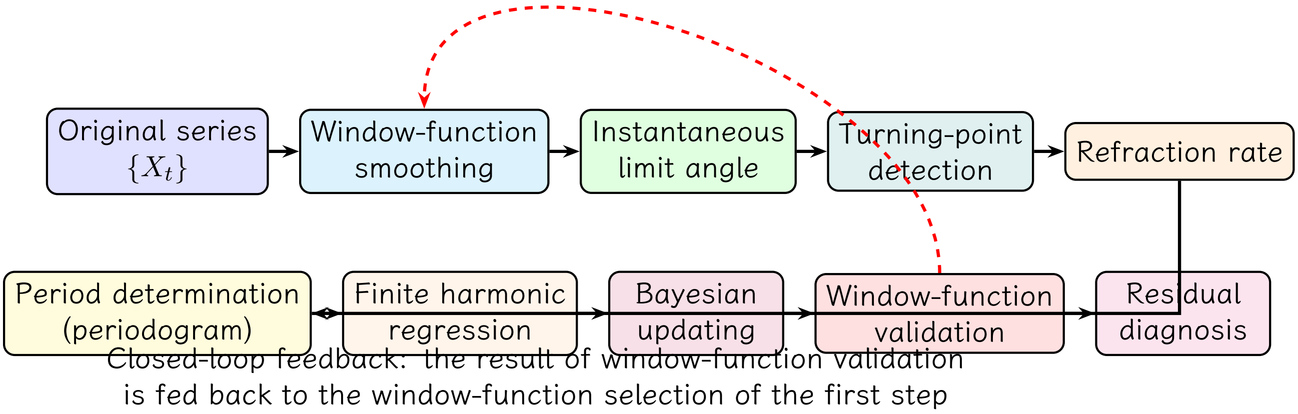

Original series
$\{X_t\}$
Window-function smoothing
Instantaneous limit angle
Turning-point detection
Refraction rate
Period determination (periodogram)
Finite harmonic regression
Bayesian updating
Window-function validation
Residual diagnosis
Closed-loop feedback: the result of window-function validation is fed back to the window-function selection of the first step

Figure 23.1: The complete pipeline of time series analysis: starting from the original series, through smoothing, instantaneous limit angle, turning-point detection, refraction rate, period determination, finite harmonic regression, Bayesian updating and window-function validation to residual diagnosis, where window-function validation feeds back to the first step.

and its **autocorrelation function** (ACF) is

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)} \in [-1, 1].$$

$\rho(0) = 1$, and $\rho(-h) = \rho(h)$ is an even function.

Definition 23.3 (Partial autocorrelation function)**.** The **partial autocorrelation function** (PACF) $\alpha(h)$ is defined as the conditional correlation coefficient of X_t and X_{t+h} given the intermediate variables $X_{t+1}, \ldots, X_{t+h-1}$:

$$\alpha(h) = \mathrm{Corr}\big(X_t, X_{t+h} \mid X_{t+1}, \ldots, X_{t+h-1}\big).$$

Equivalently, $\alpha(h)$ is the coefficient of X_t when X_{t+h} is predicted optimally and linearly from $X_{t+h-1}, \ldots, X_t$.

Definition 23.4 (White noise)**.** The series $\{\varepsilon_t\}$ is called **white noise** if $\mathbb{E}[\varepsilon_t] = 0$, $\mathrm{Var}(\varepsilon_t) = \sigma^2$, and $\mathrm{Cov}(\varepsilon_t, \varepsilon_s) = 0$ $(t \neq s)$. If in addition all moments are finite and the variables are independent, it is called **independent white noise**; if $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ are independent and identically distributed, it is called **Gaussian white noise**.

Definition 23.5 (Lag Operator)**.** The **lag operator** B is defined by $BX_t = X_{t-1}$. B is the generator of the discrete translation group $\mathbb{Z}$. The lag-operator polynomial

$$\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p$$

defines a linear operator on the space of series.

Property 23.1 (Algebraic Properties of the Lag Operator)**.** The lag operator B satisfies, on the space of series:

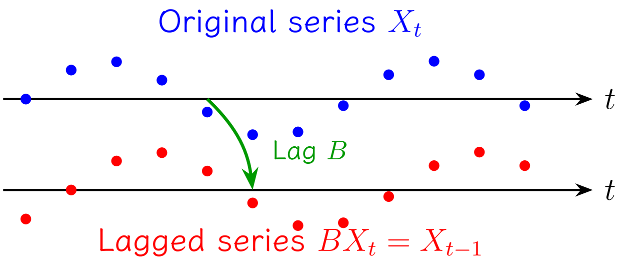


Figure 23.2: The lag operator B: it shifts the series one step to the right along the time axis. B generates the discrete translation group $\mathbb{Z}$ and is the central tool of ARIMA modelling.

(i) **linearity**: $B(\alpha X_t + \beta Y_t) = \alpha BX_t + \beta BY_t$;

(ii) **commutativity**: $B^a B^b = B^{a+b} = B^b B^a$;

(iii) **invertibility**: when all roots of the characteristic polynomial $\phi(z)$ lie outside the unit circle, $\phi(B)$ is invertible.

Proof. (i) $B(\alpha X_t + \beta Y_t) = \alpha X_{t-1} + \beta Y_{t-1} = \alpha BX_t + \beta BY_t$.
(ii) $(B^a B^b)X_t = B^a X_{t-b} = X_{t-a-b} = B^{a+b}X_t$.
(iii) If all roots of $\phi(z)$ lie outside the unit circle, then $1/\phi(z)$ is analytic in the unit disc, and the coefficients of its power-series expansion $\sum_k \psi_k z^k$ are absolutely summable. Define $\psi(B) = \sum_k \psi_k B^k$; then $\psi(B)\phi(B) = I$ (on the space of stationary series), so $\phi(B)$ is invertible. ☐

Definition 23.6 (Differencing operator). The **first-order differencing operator** is defined by $\nabla = 1 - B$, that is, $\nabla X_t = X_t - X_{t-1}$. The d-th order difference is $\nabla^d = (1-B)^d$. The differencing operator removes polynomial trends: if $X_t = P_d(t)$ is a polynomial of degree d, then $\nabla^d X_t$ is a constant.

Lemma 23.1 (Null space of the differencing operator). $\ker \nabla^d$ is spanned by $\{1, t, t^2, \ldots, t^{d-1}\}$, that is, by all polynomial sequences of degree less than d.

Proof. Induction on d. $d = 1$: $\nabla X_t = X_t - X_{t-1} = 0 \Rightarrow X_t$ is a constant. $d \geq 2$: since $\nabla^d = \nabla\nabla^{d-1}$, $\ker \nabla^d = \{X : \nabla X \in \ker \nabla^{d-1}\}$, that is, ∇X is a polynomial of degree less than $d-1$. Integrating successively shows that X is a polynomial of degree less than d. ☐

Definition 23.7 (Spectral density). Let $\{X_t\}$ be weakly stationary with $\sum_h |\gamma(h)| < \infty$. Its **spectral density** is

$$S(\omega) = \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \gamma(h) e^{-i\omega h}, \qquad \omega \in (-\pi, \pi].$$

$S(\omega) \geq 0$ and it is an even function. The inverse formula is $\gamma(h) = \int_{-\pi}^{\pi} S(\omega) e^{i\omega h} d\omega$.

Theorem 23.1 (Wold decomposition theorem). Let $\{X_t\}$ be a zero-mean weakly stationary series. Then X_t admits the unique decomposition

$$X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j} + V_t, \qquad \psi_0 = 1, \quad \sum_j \psi_j^2 < \infty,$$

where $\{\varepsilon_t\}$ is white noise, V_t is a deterministic component (one that can be predicted entirely linearly from the past), and $\{\varepsilon_t\}$ and $\{V_t\}$ are orthogonal.

Proof. Decompose the closed linear span of $\{X_t\}$ in time into the increasing chain of subspaces $\overline{\text{span}}\{X_s : s \le t\}$. Take ε_t to be the residual of the projection of X_t onto the past space; then $\{\varepsilon_t\}$ is white noise and $X_t = \sum_j \psi_j \varepsilon_{t-j} + V_t$. Uniqueness follows from the uniqueness of the projection. ☐

Corollary 23.1 (ARMA as a rational form of the Wold decomposition). *The transfer function $H(z) = \theta(z)/\phi(z)$ of ARMA(p, q) is a finite-dimensional parametrisation of the Wold decomposition: its coefficients ψ_j are uniquely determined by the coefficients of ϕ, θ through long division.*

Proof. From $\phi(z)H(z) = \theta(z)$, comparing coefficients of z^j gives $\psi_j - \phi_1\psi_{j-1} - \cdots - \phi_p\psi_{j-p} = \theta_j$, which is a linear recursion for ψ_j and determines $\{\psi_j\}$ uniquely. ☐

23.1.2 Overview of Classical Time Series Models

Definition 23.8 (AR Model). *The autoregressive model of order p, AR(p):*

$$\phi(B)X_t = \varepsilon_t,$$

where $\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p$. The stationarity condition is that all roots of $\phi(z)$ lie outside the unit circle.

Definition 23.9 (MA Model). *The moving-average model of order q, MA(q):*

$$X_t = \theta(B)\varepsilon_t,$$

where $\theta(B) = 1 + \theta_1 B + \cdots + \theta_q B^q$.

Definition 23.10 (ARMA Model). *ARMA(p, q):*

$$\phi(B)X_t = \theta(B)\varepsilon_t.$$

The power spectral density is

$$S(\omega) = \frac{\sigma^2}{2\pi} \frac{|\theta(e^{-i\omega})|^2}{|\phi(e^{-i\omega})|^2}.$$

Definition 23.11 (ARIMA and SARIMA). *ARIMA(p, d, q):*

$$\phi(B)\nabla^d X_t = \theta(B)\varepsilon_t,$$

where $\nabla = 1 - B$. SARIMA$(p, d, q)\times(P, D, Q)_s$:

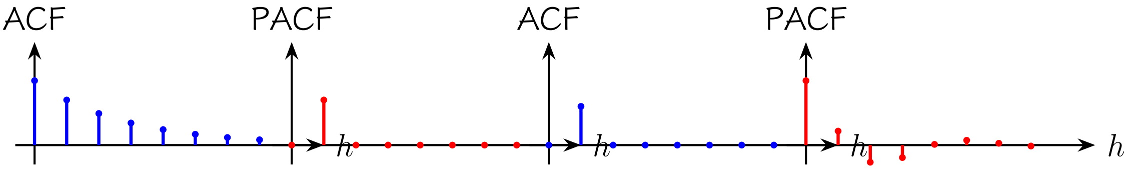
$$\phi(B)\Phi(B^s)\nabla^d \nabla_s^D X_t = \theta(B)\Theta(B^s)\varepsilon_t.$$

Definition 23.12 (ARCH and GARCH). *ARCH(q):*

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i X_{t-i}^2.$$

GARCH(p, q):

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i X_{t-i}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2.$$



AR(1) ACF: tails offAR(1) PACF: cuts offMA(1) ACF: cuts offMA(1) PACF: tails off

Figure 23.3: Comparison of the ACF/PACF characteristics of AR and MA: for an AR model the PACF cuts off and the ACF tails off; for an MA model the ACF cuts off and the PACF tails off. These are the criteria for model identification (the Box–Jenkins method).

Table 23.1: Standard characteristics of the classical time series models

Model	ACF/PACF characteristics	Frequency-domain characteristics	Operator correspondence	Transfer function
AR(p)	PACF cuts off	Yule–Walker recursion	Gradient (self-feedback)	$H(\omega) = 1/\phi(e^{-i\omega})$
MA(q)	ACF cuts off	Impulse response	Divergence (shock)	$H(\omega) = \theta(e^{-i\omega})$
ARMA	ACF/PACF tails off	Rational spectrum	Gradient + divergence	$H(\omega) = \theta(e^{-i\omega})/\phi(e^{-i\omega})$
ARIMA	Same as ARMA after differencing	Differenced trend	Gradient (discrete differentiation)	$H = \theta/(\phi(1-e^{-i\omega})^d)$
SARIMA	Seasonal + short period	Multi-scale trend	Gradient (multi-scale)	Multi-scale transfer function
ARCH	Squared ACF tails off	Volatility self-trend	Divergence (volatility scaling)	Volatility transfer function
GARCH	Squared ACF/PACF tails off	Volatility persistence	Gradient + divergence	$\alpha/(1-\beta)$ volatility kernel

Theorem 23.2 (Yule-Walker equations)**.** Let the AR(p) model $\phi(B)X_t = \varepsilon_t$ be stationary. Then the autocovariance function satisfies

$$\gamma(h) - \phi_1\gamma(h-1) - \cdots - \phi_p\gamma(h-p) = \begin{cases} \sigma^2, & h = 0, \\ 0, & h \geq 1. \end{cases}$$

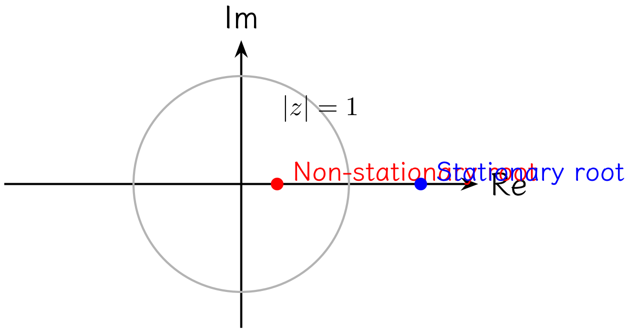
Expressed in terms of $\rho(h)$, for $h \geq 1$ this gives the system of linear equations

$$\rho(h) = \phi_1\rho(h-1) + \cdots + \phi_p\rho(h-p), \qquad h = 1, \ldots, p.$$

Proof. Multiply both sides of $\phi(B)X_t = \varepsilon_t$ by X_{t-h} and take expectations, obtaining $\sum_j \phi_j\gamma(h-j) = \mathbb{E}[\varepsilon_t X_{t-h}]$. For $h \geq 1$, ε_t is orthogonal to X_{t-h} (ε_t is uncorrelated with past observations), so the right-hand side is 0. For $h = 0$, $\mathbb{E}[\varepsilon_t X_t] = \sigma^2$. ☐

Property 23.2 (Geometric intuition for stationarity)**.** The AR(1) model $X_t = \phi X_{t-1} + \varepsilon_t$ is stationary if and only if $|\phi| < 1$. In that case the autocovariance $\gamma(h) = \sigma^2\phi^{|h|}/(1-\phi^2)$ decays exponentially, with the rate of decay controlled by ϕ: the closer $|\phi|$ is to 1, the longer the memory.

Proof. The root of the characteristic polynomial $\phi(z) = 1-\phi z$ is $z = 1/\phi$, which lies outside the unit circle if and only if $|\phi| < 1$. The autocovariance is obtained from the recursion $\gamma(h) = \phi\gamma(h-1)$ ($h \geq 1$) as $\gamma(h) = \phi^h\gamma(0)$, where $\gamma(0) = \sigma^2/(1-\phi^2)$. ☐



Stationarity condition: all roots of the characteristic polynomial $\phi(z)$ lie outside the unit circle (blue points), not inside it (red points)

Figure 23.4: The ARMA stationarity condition: all roots of the characteristic polynomial $\phi(z)$ lie outside the unit circle in the complex plane.

23.2 Fundamental Tools for Time Series Analysis

Remark 23.3 (Consolidation of the Window-Function Definition)*.* The general definition of a window function (from the frequency-domain/sidelobe viewpoint) and its threefold role in time series analysis have been consolidated into the 'Discrete Sampling and Intercalation of Periodic Signals' chapter (see Definition 21.15 and its subsection on time-series applications). That definition is used directly in the remainder of this chapter and is not repeated here.

23.2.1 Instantaneous Limit Angle: Directional Information of the Discrete Gradient

Definition 23.13 (Instantaneous Limit Angle of a Time Series)**.** For the series $\tilde{X}_t$ smoothed by a window function, the **instantaneous limit angle** is defined by

$$\theta_t = \arctan(\nabla \tilde{X}_t) = \arctan(\tilde{X}_t - \tilde{X}_{t-1}),$$

where $\nabla \tilde{X}_t = \tilde{X}_t - \tilde{X}_{t-1}$ is the discrete gradient. $\theta_t \in (-\pi/2, \pi/2)$.

23.3 Turning-Point Detection: Direction-Reversal Determination Based on Similarity

Definition 23.14 (Mathematical Definition of a Turning Point)**.** Let $\theta_t \in (-\pi/2, \pi/2)$ be the instantaneous limit angle. The point t^* is called a **turning point** if

$$|\theta_{t^*} - \theta_{t^*-1}| > \tau,$$

where $\tau \in (0, \pi/2)$ is a preset threshold. Write $\Delta\theta = \theta_{t^*} - \theta_{t^*-1}$.

Following the unified convention of the 'Scalars and Vectors' chapter, the **included angle** between two trend directions is always taken to be $\theta = |\alpha - \beta| \in [0, \pi]$; the signed quantity $\Delta\theta$ is the **difference angle**, used only for the sign of the sine-family similarity.

Lemma 23.2 (Equivalence of the Turning-Point Threshold and Cosine Similarity)**.** Let $|\Delta\theta| \in [0, \pi)$ and $\tau \in (0, \pi)$. Then

$$|\Delta\theta| > \tau \quad \iff \quad \cos(\Delta\theta) < \cos\tau.$$

Proof. The cosine function is strictly decreasing on $[0, \pi]$, and cos is an even function. Hence $|\Delta\theta| > \tau$ if and only if $\cos|\Delta\theta| < \cos\tau$. ☐

Proposition 23.1 (Similarity Characteristics of Turning Points)**.** At a turning point t^*:

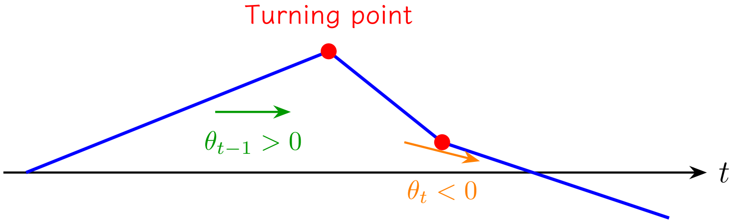
(i) the cosine similarity $\mathrm{Similarity}^{(1)} = \cos(\Delta\theta) < \cos\tau$;

(ii) the sign of the sine similarity $\mathrm{Similarity}^{(2)} = \sin(\Delta\theta)$ gives the direction of the turn;

(iii) the double-angle sine similarity $\mathrm{Similarity}^{(4)} = \sin(2\Delta\theta)$ is strictly increasing in $\Delta\theta$; its absolute value is strictly increasing in $|\Delta\theta|$ whenever $|\Delta\theta| < \pi/4$.

Proof. (i) By Lemma 23.2. (ii) sin is an odd function. (iii) $\frac{d}{d\theta}\sin 2\theta = 2\cos 2\theta > 0$ when $|\theta| < \pi/4$. The conclusion for the absolute value follows directly from the oddness and the monotonicity. ☐

Example 23.1 (Example of Turning-Point Detection)**.** The series changes from rising to falling at $t = 50$. $\tilde{X}_{49} = 10.2$, $\tilde{X}_{50} = 10.5$, $\tilde{X}_{51} = 10.1$. Then $\theta_{50} \approx 16.7^\circ$ and $\theta_{51} \approx -21.8^\circ$, so $\Delta\theta \approx -38.5^\circ$. With the threshold $\tau = 30^\circ$ we have $\cos(38.5^\circ) \approx 0.78 < \cos(30^\circ) \approx 0.87$, so $t = 51$ is a turning point. $\mathrm{Similarity}^{(4)} = \sin(-77^\circ) \approx -0.97$, and the negative sign indicates a downward turn.



Turning-point criterion: $|\theta_{t^*} - \theta_{t^*-1}| > \tau$, equivalently $\cos(\Delta\theta) < \cos\tau$

Figure 23.5: Turning-point detection: when the change $|\Delta\theta|$ in the instantaneous limit angle exceeds the threshold τ, the point is judged to be a turning point. The sign of the sine similarity marks the direction of the turn exactly.

23.4 Refraction Rate: Difference of Sine Trends on the Two Sides of a Turning Point

Definition 23.15 (Refraction Rate). At a turning point t^*, define the **angle of incidence** $\theta_1 = \theta_{t^*-1}$ and the **angle of emergence** $\theta_2 = \theta_{t^*+1}$. The **refraction rate** is

$$R_{\text{turn}}(t^*) = |\sin\theta_1 - \sin\theta_2|.$$

Theorem 23.3 (Half-Angle Decomposition and Bounds of the Refraction Rate). Let $\theta_1, \theta_2 \in (-\pi/2, \pi/2)$. Then

$$R_{\text{turn}} = 2\left|\cos\frac{\theta_1+\theta_2}{2}\right| \cdot \left|\sin\frac{\theta_1-\theta_2}{2}\right|,$$

and $0 \le R_{\text{turn}} < 2$, the value 2 being a supremum that is not attained on the open interval. In particular: $R = 0 \iff \theta_1 = \theta_2$; and when the direction is reversed ($\theta_2 = -\theta_1$), $R = 2|\sin\theta_1|$.

Proof. The half-angle decomposition follows from the product-to-sum formulae. The bounds follow from $|\cos| \le 1$ and $|\sin| \le 1$. The three special cases are verified one by one. □

Example 23.2 (Example of Refraction-Rate Computation). Continuing Example 23.1: $\sin\theta_1 = 0.287$ and $\sin\theta_2 = -0.371$, so $R_{\text{turn}} = 0.658$. The half-angle decomposition is verified to agree.

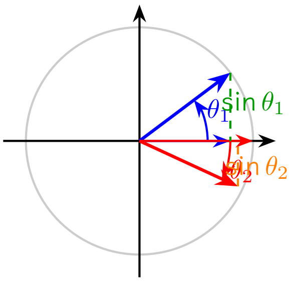
23.5 Period Determination: Peaks of the Periodogram

Definition 23.16 (Period Determination Based on the Periodogram). Compute the periodogram

$$I_n(\omega_k) = \frac{1}{n}\left|\sum_{t=1}^{n} X_t e^{-i\omega_k t}\right|^2.$$

The dominant period is determined by the peak frequency:

$$\omega_{\text{period}} = \arg\max_{\omega\in(0,\pi]} I_n(\omega), \qquad T_{\text{period}} = \frac{2\pi}{\omega_{\text{period}}}.$$



$$R_{\text{turn}} = |\sin\theta_1 - \sin\theta_2| = 2|\cos\tfrac{\theta_1+\theta_2}{2}| \cdot |\sin\tfrac{\theta_1-\theta_2}{2}|$$

Figure 23.6: Geometric figure for the half-angle decomposition of the refraction rate: R_{turn} is the difference of the sine projections on the two sides, and the half-angle decomposition splits it into two factors, the 'average direction' and the 'direction difference'.

Proposition 23.2 (Fisher Test for Period Significance). Suppose the periodogram takes the values $I_1, \ldots, I_m$ at the $m = \lfloor (n-1)/2 \rfloor$ positive frequency points. The Fisher test statistic is

$$g = \frac{\max_k I_k}{\sum_{k=1}^m I_k}.$$

The threshold is

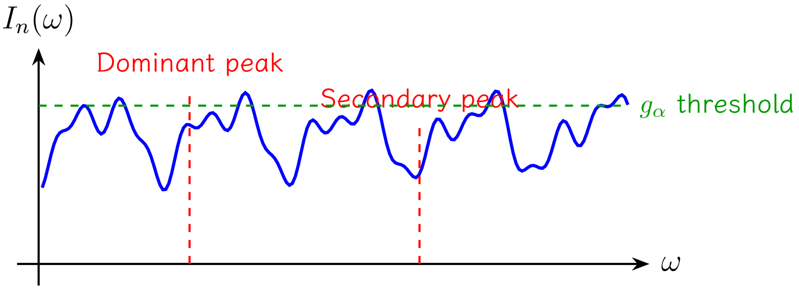
$$g_\alpha = 1 - (\alpha/m)^{1/(m-1)}.$$

If $g > g_\alpha$, the period is significant.

Proof. Under white noise the periodogram ordinates are independent and identically distributed (exponentially), and the normalised ratio follows a Beta$(1, m-1)$ distribution. The union bound gives

$$\mathbb{P}(g > g_0) \leq m(1-g_0)^{m-1}.$$

Substituting $g_0 = g_\alpha$ yields α. ☐



When a periodogram peak exceeds the Fisher threshold g_α, the corresponding period is significant

Figure 23.7: The periodogram and the Fisher test: frequencies whose spectral peak exceeds the threshold g_α correspond to significant periods, and the dominant peak gives the dominant period $T_{\text{period}} = 2\pi/\omega_{\text{period}}$.

Property 23.3 (Asymptotic distribution of the periodogram). *Under Gaussian white noise the periodogram $I_n(\omega_k)$ is asymptotically exponentially distributed with mean σ^2, that is,*

$$I_n(\omega_k) \xrightarrow{d} \mathsf{Exp}(\sigma^2),$$

and $I_n(\omega_k), I_n(\omega_{k'})$ at distinct frequencies are asymptotically independent ($k \neq k'$).

Proof. The DFT coefficients $\sum_t X_t e^{-i\omega_k t}$ of Gaussian white noise are complex Gaussian random variables whose real and imaginary parts are independent and identically distributed as $\mathcal{N}(0, n\sigma^2/2)$; hence $I_n(\omega_k)$ is $1/n$ times the sum of squares of two independent Gaussians, that is, it is exponentially distributed. The DFT coefficients at distinct frequencies are asymptotically orthogonal, hence asymptotically independent. ☐

23.6 Finite Harmonic Regression Model

Definition 23.17 (Finite Harmonic Regression Model).

$$X_t \approx \mathsf{Re}\left[\sum_{k=1}^{K} A_k e^{i(\omega_k t+\phi_k)}\right] + \varepsilon_t,$$

where $\{A_k, \omega_k, \phi_k\}_{k=1}^K$ are tunable parameters.

Proposition 23.3 (Regularised Least-Squares Estimation of Finite Harmonic Model Parameters). *The finite harmonic parameters are estimated by regularised least squares. The objective function is*

$$\mathcal{L}(\mathbf{\Theta}) = \sum_{t=1}^{n}\left(X_t - \mathsf{Re}\left[\sum_k A_k e^{i(\omega_k t+\phi_k)}\right]\right)^2 + \lambda \sum_k |A_k|.$$

Under suitable regularisation conditions the estimator $\hat{\mathbf{\Theta}}_n$ converges in probability to the true parameter $\mathbf{\Theta}_0$.

Proof. The objective function $\mathcal{L}$ is a continuously differentiable function of $\mathbf{\Theta}$ (apart from the points where the L^1 regularisation term fails to be differentiable at $A_k = 0$, which can be handled by subgradients). By the strong convexity of the L^2 loss term (on the compact parameter space where the ω_k are separated) and the convexity of the L^1 regularisation term, $\mathcal{L}$ has a unique minimum on the compact parameter space. Let $\hat{\mathbf{\Theta}}_n = \arg\min \mathcal{L}$. The Wald consistency conditions are: (i) the parameter space is compact; (ii) the true parameter $\mathbf{\Theta}_0$ is the unique minimum (model identifiability: $A_k > 0$ together with pairwise distinct ω_k guarantees that the harmonic components are uniquely determined); (iii) the loss function converges pointwise to the expected loss. When these conditions hold, $\hat{\mathbf{\Theta}}_n \xrightarrow{p} \mathbf{\Theta}_0$. Multiscale kernel smoothing can accelerate the numerical optimisation, but it does not change consistency. ☐

Theorem 23.4 (Asymptotic normality of the finite harmonic estimator). *Under the above regularisation conditions the estimator satisfies*

$$\sqrt{n}\big(\hat{\mathbf{\Theta}}_n - \mathbf{\Theta}_0\big) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\mathbf{\Theta}_0)^{-1}),$$

where $\mathcal{I}(\mathbf{\Theta}_0)$ is the Fisher information matrix.

Proof. By $\sqrt{n}$-consistency, the second-order differentiability of the objective function (except at $A_k = 0$) and a Taylor expansion, $\hat{\boldsymbol{\Theta}}_n - \boldsymbol{\Theta}_0 \approx -\mathcal{I}(\boldsymbol{\Theta}_0)^{-1}\nabla\mathcal{L}(\boldsymbol{\Theta}_0)$, where $\nabla\mathcal{L}(\boldsymbol{\Theta}_0)$ is asymptotically normal. Slutsky's theorem then gives the result. ☐

23.7 Bayesian Updating: Online Inference for the Parameters

Definition 23.18 (Bayesian Updating of Finite Harmonic Parameters). The posterior distribution is

$$p(\boldsymbol{\Theta} \mid X_{1:t+1}) \propto p(X_{t+1} \mid \boldsymbol{\Theta}) \cdot p(\boldsymbol{\Theta} \mid X_{1:t}).$$

Proposition 23.4 (Posterior Convergence and Period Stability). Under standard regularity conditions, the Bernstein–von Mises theorem gives, as $n \to \infty$,

$$\sqrt{n}(\boldsymbol{\Theta} - \hat{\boldsymbol{\Theta}}_n) \mid X_{1:n} \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\boldsymbol{\Theta}_0)^{-1}),$$

where $\mathcal{I}(\boldsymbol{\Theta}_0)$ is the Fisher information matrix. The posterior distribution concentrates at the true parameter, with width $O(n^{-1/2})$. If the posterior remains concentrated at ω_k, that periodic component is stable; if it spreads out, it is unstable.

Proof. Under standard regularity conditions, when the prior is continuous and positive at $\boldsymbol{\Theta}_0$, the posterior is asymptotically normal. The non-degeneracy of the Fisher information guarantees that the covariance matrix $\mathcal{I}(\boldsymbol{\Theta}_0)^{-1}$ is well defined. The concentration follows from the asymptotic standard deviation $O(n^{-1/2})$. The criterion for period stability is decided directly by the size of the posterior variance in the ω_k direction. ☐

Proposition 23.5 (Adaptivity of the Forgetting Factor). For a non-stationary time series, the forgetting factor $\gamma \in (0, 1]$ is defined by

$$p_\gamma(\boldsymbol{\Theta} \mid X_{1:t}) \propto p(X_t \mid \boldsymbol{\Theta}) \cdot [p_\gamma(\boldsymbol{\Theta} \mid X_{1:t-1})]^\gamma.$$

Equivalently,

$$\log p_\gamma(\boldsymbol{\Theta} \mid X_{1:t}) \propto \sum_{s=1}^{t} \gamma^{t-s} \log p(X_s \mid \boldsymbol{\Theta}),$$

that is, the likelihood contribution of old observations decays exponentially. An adaptive forgetting factor decreases γ near a turning point (enhancing adaptivity) and increases γ over a stationary stretch (enhancing stability).

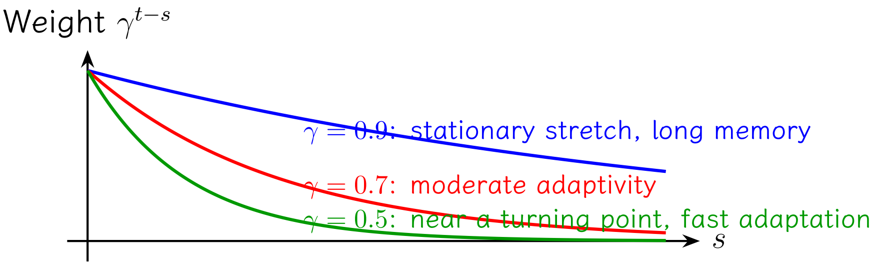
Proof. Expand the recursive definition of the forgetting factor:

$$p_\gamma(\boldsymbol{\Theta} \mid X_{1:t}) \propto \prod_{s=1}^{t} p(X_s \mid \boldsymbol{\Theta})^{\gamma^{t-s}}.$$

Taking logarithms gives the weighted log-likelihood

$$\sum_{s=1}^{t} \gamma^{t-s} \log p(X_s \mid \boldsymbol{\Theta}).$$

As $\gamma \to 1$ this degenerates to the standard Bayesian update (all observations weighted equally); for $\gamma < 1$ the weights of old observations decay exponentially, and the effective sample size is approximately $1/(1-\gamma)$. When turning-point detection is triggered at t^*, γ is temporarily lowered to γ_{low} (for instance 0.9), so that the model adapts quickly to the change of direction; over a stationary stretch $\gamma \to 1$ is restored in order to preserve the accuracy of the estimate. ☐



The forgetting factor γ controls the exponentially decaying weights of old observations: the smaller γ is, the faster the adaptation but the larger the estimation variance

Figure 23.8: Exponential decay of the forgetting factor: the smaller γ is, the faster the weights of old observations decay and the stronger the model's adaptation to recent data, but the lower the effective sample size.

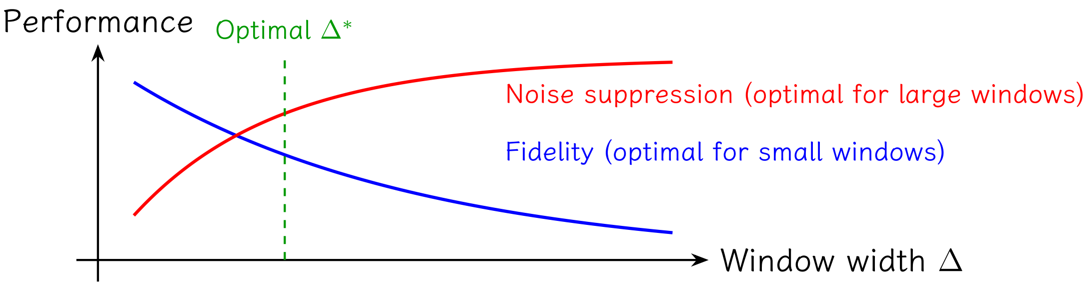
23.8 Window-Function Validation: Model Diagnosis Based on Spectral Component Determination

Definition 23.19 (Spectral Criterion for Window-Function Validation). The fidelity is defined by

$$\mathcal{V}(K_\Delta) = \frac{\int_{-\pi}^{\pi} |\hat{K}_\Delta(\omega)|^2 \cdot S_X(\omega)\, d\omega}{\int_{-\pi}^{\pi} S_X(\omega)\, d\omega}.$$

Proposition 23.6 (Turning-Point–Period Trade-off in Window-Function Validation). A small window gives high fidelity but insufficient noise suppression; a large window suppresses noise adequately but smooths away the high-frequency signal; the optimal window balances the two.

Proof. For a small window, $\Delta \to 0$ gives $\hat{K}_\Delta(\omega) \to 1$ (pointwise), hence $\mathcal{V} \to 1$; but the smoothed series satisfies $\tilde{X}_t \to X_t$ (the original series), the noise is retained entirely, and the probability that turning-point detection produces spurious turning points increases. For a large window, $\Delta \to \infty$ gives $\hat{K}_\Delta(\omega) \to \delta(\omega)$ (concentrated at zero frequency), the high-frequency components are suppressed entirely, and the period estimate is accurate (low-frequency periods are retained) but the turning-point information is lost. The optimal window is determined by minimising the mean squared error or by cross-validation, which proves the trade-off relation. ☐


Performance
Optimal Δ^*
Noise suppression (optimal for large windows)
Fidelity (optimal for small windows)
Window width Δ

Turning-point–period trade-off: the optimal window width Δ^* balances fidelity against noise suppression

Figure 23.9: The turning-point–period trade-off of window functions: a small window gives high fidelity but insufficient noise suppression, a large window suppresses noise adequately but loses the high frequencies, and the optimal window Δ^* lies between the two.

Proposition 23.7 (Similarity Diagnosis of Residuals). *The residual similarity diagnosis function is*

$$\mathcal{D}(\hat{\varepsilon}) = \max_{h\geq 1,\, d=1,\dots,6} \left| \frac{1}{n-h} \sum_{t=h+1}^{n} S^{(d)}\left(\theta_t^{\hat{\varepsilon}}, \theta_{t-h}^{\hat{\varepsilon}}\right) \right| .$$

Proof. The phase sequence of ideal white-noise residuals is independent. The terms of the difference-angle family of kernels are $e^{i(\theta_t-\theta_{t-h})}$, and the phase differences are uniformly distributed on $[0, 2\pi)$, so

$$\mathbb{E}[e^{i(\theta_t-\theta_{t-h})}] = 0.$$

By the law of large numbers (for h-dependent sequences) the mean converges in probability to 0, that is, $\mathcal{D} \to 0$. If $\mathcal{D}$ is significantly non-zero for some kernel, this shows that the residuals contain an uncaptured periodic structure at the frequency scale corresponding to that kernel. □

Theorem 23.5 (Ljung-Box test for whiteness of the residuals). *The m-th order autocorrelations $\hat{\rho}(h)$ of the residuals $\{\hat{\varepsilon}_t\}$ give the test statistic*

$$Q = n(n+2) \sum_{h=1}^{m} \frac{\hat{\rho}(h)^2}{n-h} \xrightarrow{d} \chi^2_{m-p-q},$$

where $p+q$ is the number of model parameters. When $Q > \chi^2_{m-p-q,\alpha}$, the null hypothesis that the residuals are white noise is rejected.

Proof. When the model is correctly specified, the residual autocorrelations $\hat{\rho}(h)$ are asymptotically normal and their correlation matrix is the identity. By the theory of distributions of quadratic forms, Q is asymptotically χ^2 distributed with $m-p-q$ degrees of freedom. □

23.9 Complete Pipeline of Time Series Analysis

Definition 23.20 (Algorithmic Pipeline of Time Series Analysis). *The complete algorithmic pipeline is:*

(i) window-function selection and smoothing;

(ii) computation of the instantaneous limit angle;

(iii) turning-point detection;

(iv) computation of the refraction rate;

(v) period determination;

(vi) finite harmonic regression modelling and parameter estimation;

(vii) Bayesian updating;

(viii) window-function validation.

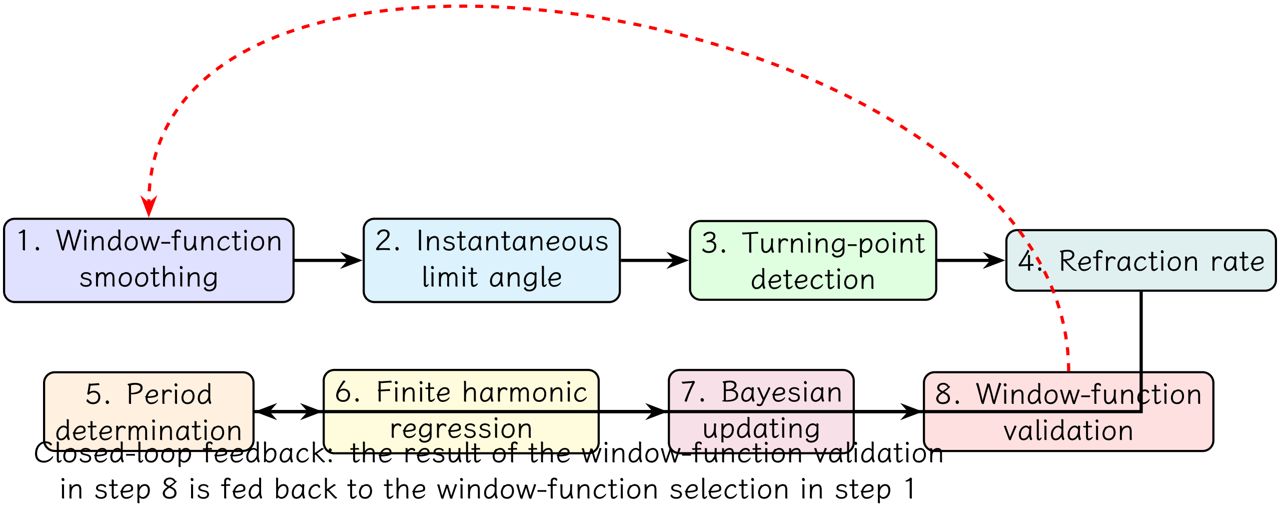

1. Window-function smoothing
2. Instantaneous limit angle
3. Turning-point detection
4. Refraction rate
5. Period determination
6. Finite harmonic regression
7. Bayesian updating
8. Window-function validation
Closed-loop feedback: the result of the window-function validation in step 8 is fed back to the window-function selection in step 1

Figure 23.10: The complete closed loop of the time series analysis pipeline: the eight steps form a complete closed loop, and the result of window-function validation is fed back to the window-function selection, achieving iterative optimisation of the model.

Example 23.3 (Application of the Complete Pipeline to Simulated Data). Simulated data:

$$X_t = 5\,\mathsf{sin}(2\pi t/20) + 2\,\mathsf{sin}(2\pi t/7) + \varepsilon_t, \quad t = 1, \dots, 200,$$

where $\varepsilon_t \sim \mathcal{N}(0,1)$.

Turning-point detection: with threshold $\tau = 45°$, about 18 turning points are detected. The refraction rate R_{turn} ranges from 0.3 to 1.8. Period determination: the periodogram peaks at $\omega = 2\pi/20$ and $\omega = 2\pi/7$, with dominant period $T = 20$ (significant by the Fisher test). Finite harmonic regression: regularised least squares gives the estimates $A_1 \approx 5$, $\omega_1 \approx 2\pi/20$, $A_2 \approx 2$ and $\omega_2 \approx 2\pi/7$. After Bayesian updating the posterior concentrates at (ω_1, ω_2). Window validation: $\mathcal{V}(K_{\mathsf{Gauss},\Delta=3}) \approx 0.95$ and $\mathcal{V}(K_{\mathsf{Cauchy},\Delta=20}) \approx 0.88$, with residual diagnosis $\mathcal{D} \approx 0.1$.

Theorem 23.6 (Model selection criteria). Let $\hat{\sigma}_k^2$ be the maximum-likelihood variance estimate of a candidate model and p_k its number of parameters. The AIC and BIC criteria are

$$\mathrm{AIC}(k) = n \log \hat{\sigma}_k^2 + 2p_k, \qquad \mathrm{BIC}(k) = n \log \hat{\sigma}_k^2 + p_k \log n.$$

Within a set of candidate models, AIC and BIC asymptotically select the model that is optimal for prediction error and for posterior probability, respectively.

Proof. AIC is an asymptotically unbiased estimate of the sum of the KL divergence and the model complexity; BIC is the Laplace approximation to the posterior model probability. □

Remark 23.4 (Scope of Application). The methods of this chapter apply to discrete time series with stationary or locally stationary structure, including trend extraction, period identification, turning-point detection and short-term forecasting. For non-stationary, long-memory or strongly non-linear series they should be used in combination with differencing, local modelling or non-linear models.

Remark 23.5 (Interface with Kalman filtering). For the state-space form

$$x_{t+1} = Fx_t + w_t, \qquad y_t = Hx_t + v_t,$$

the Kalman filter gives the optimal linear state estimate

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + K_t(y_t - H\hat{x}_{t|t-1}),$$

where K_t is the Kalman gain. ARMA models can be brought into the Kalman framework through a state-space realisation, which handles missing values and online updating in a unified way.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The lag operator B (Definition 23.5) generates the discrete translation group; window-function convolution smoothing
Scaling	Choice of the scale of the window width Δ; fidelity $\mathcal{V}(K_\Delta)$ (Definition 23.19)
Rotation	The instantaneous limit angle θ_t (Definition 23.13); similarity kernels (Proposition 23.1)
Conservation	Conservation of periodogram energy; normalisation of the Fisher test statistic (Proposition 23.2)
Symmetry	Cosine symmetry (Lemma 23.2); direction reversal (Theorem 23.3)
Closure	Closed loop of the algorithmic pipeline (Definition 23.20); white-noise residuals $\mathcal{D} \to 0$ (Proposition 23.7)
Period	Dominant period (Definition 23.16); Fisher test (Proposition 23.2)

Exercises

Exercise 1 Prove the algebraic properties of the lag operator (Property 23.1):

(i) linearity: $B(\alpha X_t + \beta Y_t) = \alpha B X_t + \beta B Y_t$;
(ii) commutativity: $B^a B^b = B^{a+b} = B^b B^a$;
(iii) invertibility: when all roots of the characteristic polynomial $\phi(z)$ lie outside the unit circle, $\phi(B)$ is invertible.

Carry out the proof in the language of linear operators on a space of series, and explain why the lag operator is the generator of the discrete translation group $\mathbb{Z}$.

Exercise 2 Prove the characterisation of the null space of the differencing operator (Lemma 23.1): $\ker \nabla^d$ is spanned by $\{1, t, t^2, \ldots, t^{d-1}\}$. Carry out the proof by induction, and explain how the differencing operator removes polynomial trends.

Exercise 3 Prove the Wold decomposition theorem (Theorem 23.1): a zero-mean weakly stationary series admits the unique decomposition

$$X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j} + V_t, \quad \psi_0 = 1, \quad \sum_j \psi_j^2 < \infty.$$

Carry out the proof using the projection theorem, and explain the geometric meaning of V_t being a deterministic component.

Exercise 4 Prove the Yule-Walker equations (Theorem 23.2): for an AR(p) model,

$$\rho(h) = \phi_1 \rho(h-1) + \cdots + \phi_p \rho(h-p), \quad h = 1, \ldots, p.$$

Carry out the proof using $\mathbb{E}[\varepsilon_t X_{t-h}] = 0$ $(h \geq 1)$, and explain why the Yule-Walker equations give the method of moments for estimating the AR parameters.

Exercise 5 Prove the geometric intuition for stationarity (Property 23.2): the AR(1) model $X_t = \phi X_{t-1} + \varepsilon_t$ is stationary if and only if $|\phi| < 1$, and in that case $\gamma(h) = \sigma^2 \phi^{|h|}/(1-\phi^2)$. Carry out the proof using the condition that the characteristic root lies outside the unit circle and the autocovariance recursion, and explain why memory is longer the closer $|\phi|$ is to 1.

Exercise 6 From Definition 23.8 to Definition 23.12, write down the defining equations of the seven models AR(p), MA(q), ARMA(p, q), ARIMA(p, d, q), SARIMA, ARCH(q) and GARCH(p, q), and verify the ACF/PACF characteristics, the frequency-domain characteristics and the operator correspondence given in each row of Table 23.1.

Exercise 7 Prove the equivalence of the turning-point threshold and cosine similarity (Lemma 23.2): let $|\Delta\theta| \in [0, \pi)$ and $\tau \in (0, \pi)$; then

$$|\Delta\theta| > \tau \Longleftrightarrow \cos(\Delta\theta) < \cos\tau.$$

Carry out the proof using the strict monotone decrease of the cosine function on $[0, \pi]$ and its evenness.

Exercise 8 Prove the three similarity characteristics at a turning point in Proposition 23.1:

(i) the cosine similarity $\text{Similarity}^{(1)} = \cos(\Delta\theta) < \cos\tau$;

(ii) the sign of the sine similarity $\text{Similarity}^{(2)} = \sin(\Delta\theta)$ gives the direction of the turn;

(iii) the absolute value of the double-angle sine similarity $\text{Similarity}^{(4)} = \sin(2\Delta\theta)$ is strictly increasing in $|\Delta\theta|$ when $|\Delta\theta| < \pi/4$.

Write out the proofs separately, and explain why a change in the sign of the sine similarity marks a turning point exactly.

Exercise 9 Verify Example 23.1: the series changes from rising to falling at $t = 50$, with $\tilde{X}_{49} = 10.2$, $\tilde{X}_{50} = 10.5$ and $\tilde{X}_{51} = 10.1$. Compute:

(i) the instantaneous limit angles θ_{50} and θ_{51};

(ii) the difference angle $\Delta\theta$;

(iii) the cosine similarity $\text{Similarity}^{(1)}$ and the sine similarity $\text{Similarity}^{(2)}$;

(iv) decide whether $t = 51$ is a turning point under the threshold $\tau = 30°$.

Exercise 10 Prove Theorem 23.3 (half-angle decomposition and bounds of the refraction rate): let $\theta_1, \theta_2 \in (-\pi/2, \pi/2)$, and let the refraction rate be

$$R_{\text{turn}}(t^*) = |\sin\theta_1 - \sin\theta_2|,$$

so that

$$R_{\text{turn}} = 2\left|\cos\frac{\theta_1+\theta_2}{2}\right| \cdot \left|\sin\frac{\theta_1-\theta_2}{2}\right|,$$

and $0 \le R_{\text{turn}} < 2$. Carry out the proof using the product-to-sum formulae, and verify the three special cases: $R = 0 \iff \theta_1 = \theta_2$; $R = 2$ is not attained on the open interval; and when the direction is reversed, $R = 2|\sin\theta_1|$.

Exercise 11 Prove Proposition 23.2 (the Fisher test): suppose the periodogram takes the values $I_1, \dots, I_m$ at the $m = \lfloor (n-1)/2 \rfloor$ positive frequency points, with Fisher test statistic

$$g = \frac{\max_k I_k}{\sum_{k=1}^m I_k},$$

and significance threshold $g_\alpha = 1 - (\alpha/m)^{1/(m-1)}$. Explain that under white noise the periodogram ordinates are independent and identically exponentially distributed and that the normalised ratio follows a $\text{Beta}(1, m-1)$ distribution, and complete the proof using the union bound.

Exercise 12 Prove Proposition 23.3 (asymptotic distribution of the periodogram): under Gaussian white noise the periodogram $I_n(\omega_k)$ is asymptotically exponentially distributed with mean σ^2, and the values at distinct frequencies are asymptotically independent. Explain why the DFT coefficients are complex Gaussian random variables.

Exercise 13 Prove Proposition 23.3 (regularised least-squares estimation of finite harmonic model parameters): the objective function

$$\mathcal{L}(\mathbf{\Theta}) = \sum_{t=1}^{n}\Big(X_t - \mathrm{Re}\Big[\sum_k A_k e^{i(\omega_k t+\phi_k)}\Big]\Big)^2 + \lambda \sum_k |A_k|,$$

and under suitable regularisation conditions the estimator converges in probability to the true parameter. Explain the three points of the Wald consistency conditions: the parameter space is compact, the true parameter is the unique minimum, and the loss function converges pointwise.

Exercise 14 Prove Theorem 23.4 (asymptotic normality of the finite harmonic estimator):

$$\sqrt{n}\big(\hat{\mathbf{\Theta}}_n - \mathbf{\Theta}_0\big) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\mathbf{\Theta}_0)^{-1}).$$

Carry out the proof using a Taylor expansion and Slutsky's theorem.

Exercise 15 Prove Proposition 23.4 (posterior convergence and period stability): under standard regularity conditions,

$$\sqrt{n}(\mathbf{\Theta} - \hat{\mathbf{\Theta}}_n) \mid X_{1:n} \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\mathbf{\Theta}_0)^{-1}).$$

Explain that the posterior distribution concentrates at the true parameter with width $O(n^{-1/2})$, and state how the posterior variance is used to judge whether a periodic component is stable.

Exercise 16 Prove Proposition 23.5 (adaptivity of the forgetting factor): for a non-stationary time series the forgetting factor $\gamma \in (0,1]$ is defined by

$$p_\gamma(\mathbf{\Theta} \mid X_{1:t}) \propto p(X_t \mid \mathbf{\Theta}) \cdot [p_\gamma(\mathbf{\Theta} \mid X_{1:t-1})]^\gamma,$$

and equivalently

$$\log p_\gamma(\mathbf{\Theta} \mid X_{1:t}) \propto \sum_{s=1}^{t} \gamma^{t-s} \log p(X_s \mid \mathbf{\Theta}).$$

Write out the proof, and explain why the update degenerates to the standard Bayesian update as $\gamma \to 1$, whilst for $\gamma < 1$ the weights of old observations decay exponentially.

Exercise 17 Prove Proposition 23.6 (the turning-point–period trade-off in window-function validation): a small window gives high fidelity but insufficient noise suppression; a large window suppresses noise adequately but smooths away the high-frequency signal; the optimal window balances the two. Discuss the limiting behaviour as $\Delta \to 0$ and as $\Delta \to \infty$ separately, and explain that the optimal window is determined by minimising the mean squared error or by cross-validation.

Exercise 18 Prove Proposition 23.7 (similarity diagnosis of residuals): the residual similarity diagnosis function is

$$\mathcal{D}(\hat{\varepsilon}) = \max_{h\geq 1,\, d=1,\ldots,6}\left|\frac{1}{n-h}\sum_{t=h+1}^{n} S^{(d)}(\theta_t^{\hat{\varepsilon}}, \theta_{t-h}^{\hat{\varepsilon}})\right|.$$

Explain that the phase sequence of ideal white-noise residuals is independent and that the expectation of the difference-angle family of kernels is zero, so that $\mathcal{D} \to 0$ by the law of large numbers; and explain why $\mathcal{D}$ being significantly non-zero shows that the residuals contain an uncaptured periodic structure.

Exercise 19 Prove Theorem 23.5 (the Ljung-Box test for whiteness of the residuals): the test statistic based on the residual autocorrelations is

$$Q = n(n+2)\sum_{h=1}^{m}\frac{\hat{\rho}(h)^2}{n-h} \xrightarrow{d} \chi^2_{m-p-q}.$$

Explain why Q is asymptotically chi-squared distributed when the model is correctly specified.

Exercise 20 Prove Theorem 23.6 (model selection criteria): the AIC and BIC criteria are

$$\mathsf{AIC}(k) = n\log\hat{\sigma}_k^2 + 2p_k, \quad \mathsf{BIC}(k) = n\log\hat{\sigma}_k^2 + p_k\log n.$$

Explain that AIC and BIC correspond respectively to asymptotic approximations of the KL divergence and of the posterior model probability.

Exercise 21 Verify Example 23.3: the simulated data are

$$X_t = 5\sin(2\pi t/20) + 2\sin(2\pi t/7) + \varepsilon_t,$$

where $\varepsilon_t \sim \mathcal{N}(0,1)$. Explain:

(i) how turning-point detection identifies a change of direction;
(ii) at which frequencies the periodogram peaks;
(iii) how finite harmonic regression estimates the amplitudes and frequencies;
(iv) how window-function validation selects the optimal window.

Exercise 22 Prove the closed-loop property of the complete algorithmic pipeline in Definition 23.20: window-function selection and smoothing, computation of the instantaneous limit angle, turning-point detection, computation of the refraction rate, period determination, finite harmonic regression, Bayesian updating and window-function validation, these eight steps form a complete closed loop. Explain why the result of window-function validation can be fed back to the window-function selection of the first step.

Exercise 23 Comprehensive exercise: combining the lag operator (Definition 23.5), the instantaneous limit angle (Definition 23.13), the similarity kernels (Proposition 23.1), the periodogram (Definition 23.16) and residual diagnosis (Proposition 23.7), explain how this chapter extends the similarity and trend framework of the 'Scalars and Vectors' chapter from the level of trigonometric functions to the level of discrete time series. State separately:

(i) how the instantaneous limit angle corresponds to the argument at the vector level;

(ii) how turning-point detection corresponds to the determination of a direction reversal;

(iii) how the refraction rate corresponds to a measure of the difference of trends;

(iv) how residual diagnosis uses similarity kernels to detect uncaptured periods.

Chapter 24

Periodic Systems

Remark 24.1 (Positioning of this chapter: the bridge from theoretical to applied mathematics). This chapter is the **bridge of the book from theoretical mathematics to applied mathematics**. By now all the theoretical tools are in place: the 'Operations and Periodicity' chapter gives the rigorous criteria for periodicity and closed orbits, the 'Periodic Functions and Groups with Periodic Orbits' chapter gives the group structure of periodic functions, the 'Discrete Sampling and Intercalation of Periodic Signals' chapter gives the bridge from continuous objects to discrete sequences, the 'Harmonic Analysis' chapter gives the integral transform for frequency decomposition, and the 'Time Series Analysis' chapter gives turning-point detection, the periodogram and parameter estimation at the level of data.

The application chapters that follow, by contrast, face one concrete **target** after another—a stretch of observed data, a physical system, a social phenomenon. Whether a target ought to be described by periodic mathematics, and how, cannot be asserted by intuition. This chapter gives the **standard interface** that connects an arbitrary target to the framework of periodic mathematics, in five steps:

(1) **Decision**: decide rigorously, by Theorem 1.11 of the 'Operations and Periodicity' chapter, whether the target possesses periodicity, and determine its **period type** (Section 24.1);

(2) **Construction**: if the target possesses periodicity, construct the **period series**—the amplitude series, the frequency series and the phase series—from the **turning points** of the data, obtaining the **epicycle group**, and establish the correspondence 'radius to amplitude, phase to rotation delay, frequency to rotation speed' (Section 24.2);

(3) **Adjustment**: **dynamically adjust** the number and the parameters of the epicycles according to the data—one epicycle corresponds to one simple complex function, and the epicycle group is precisely a complex function in the form of a period series (Section 24.3);

(4) **Convolution**: introduce **amplitude convolution, phase convolution and frequency convolution** on the epicycle group (Section 24.4; the three are defined in the 'Series and Convolution' chapter);

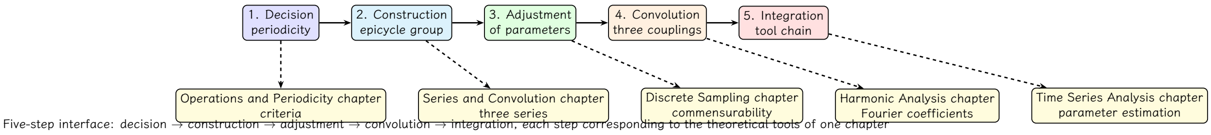
(5) **Integration**: connect to the tool chains of the three chapters 'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis' (Section 24.5).

From Section 24.6 onwards the chapter gives the phase space, the evolution and the conservation structure of a periodic system, together with such dynamical and inferential tools as the complex exponential activation function, Bayesian updating and kernel-smoothed gradient descent.

Interface convention: in deciding whether a target is periodic, and in looking up its least positive period, every later application chapter cites Section 24.1 of this chapter (Theorem 24.1, Definition 24.4) and no longer states a criterion of its own.

Remark 24.2 (Themes of this chapter). This chapter develops around the following themes: translation—time-translation operations and conservation of energy, with phase corresponding to rotation delay; scaling—scale transformations and conservation of scale charge, with frequency corresponding to rotation speed; rotation—phase rotation and conservation of angular momentum; conservation—the Noether correspondence of the three types of conservation law; symmetry—symmetry determines the degree of integrability; closure—criteria for the closure of phase-space trajectories and the determination of the period type; period—the period series, the epicycle group, the dominant components and the complex exponential activation function.

Remark 24.3 (Terminology convention of this chapter). This chapter adopts standard terminology: 'complex exponential component' in place of 'epicycle component', 'weighted cross-correlation' in place of 'periodic convolution', 'composite signal' in place of 'composite period', 'complex exponential activation function' in place of 'epicycle activation function', and 'kernel-smoothed gradient descent' in place of 'KDI optimisation'. Of these, **epicycle** is used only in Section 24.2, to state the geometric correspondence 'radius to amplitude, phase to delay, frequency to rotation speed'; thereafter 'complex exponential component' is used throughout.


1. Decision periodicity
2. Construction epicycle group
3. Adjustment of parameters
4. Convolution three couplings
5. Integration tool chain
Operations and Periodicity chapter criteria
Series and Convolution chapter three series
Discrete Sampling chapter commensurability
Harmonic Analysis chapter Fourier coefficients
Time Series Analysis chapter parameter estimation

Five-step interface: decision → construction → adjustment → convolution → integration, each step corresponding to the theoretical tools of one chapter

Figure 24.1: The five-step interface of a periodic system: starting from the decision on periodicity, through the construction of the epicycle group, the adjustment of parameters and the three convolutions, and finally the integration into the tool chain, each step being supported by a corresponding theoretical chapter.

24.1 Periodicity decision: whether a target is periodic

24.1.1 Target and decision input

The objects faced by the application chapters are collectively called **targets**. Before a target can be connected to the framework of periodic mathematics, it must first be abstracted into a decidable object.

Definition 24.1 (Target and decision input). A **target** is an object studied in an application chapter (observed data, a physical system, a social phenomenon, an algorithmic process, and so on). The **decision input** of a target is the ordered pair

$$\mathcal{D} = (\mathcal{T}, x),$$

where $\mathcal{T}$ is called the **carrier structure** (it may be $\mathbb{R}$, $\mathbb{Z}$, $\mathbb{R}^n$, a finite group $\mathbb{Z}_T$, a manifold or a category object, and so on), and $x : \mathcal{T} \to V$ is the value of the target on $\mathcal{T}$. If $\mathcal{T} = \mathbb{R}$, then x is called a **continuous description** of the target; if $\mathcal{T} = \mathbb{Z}$, then x is called a **discrete description** of the target. The choice of the carrier structure is itself part of the decision—under different carrier structures one and the same target may have different periodicity (see Remark 1.24).

Definition 24.2 (Decision map). Let a target be given by the decision input $\mathcal{D} = (\mathcal{T}, x)$. If there exist a bijection $f : \mathcal{T} \to \mathcal{T}$ and an initial value x_0 such that the target is generated by the iteration of f (that is, x is an orbit of f), then f is called a **decision map** of the target. This step reduces the target to a discrete dynamical system $(\mathcal{T}, f)$, so that the criteria of the 'Operations and Periodicity' chapter can be applied.

Theorem 24.1 (Periodicity decision theorem). Let f be the decision map of a target, and let $(\mathcal{T}, f)$ be a discrete dynamical system.

(A) **Sufficient decision**: if there exists $m \in \mathbb{N}^+$ with $f^m = \mathrm{id}$ (that is, $\mathrm{ord}(f) = m < \infty$), then the target is a **uniformly periodic system**, every orbit is closed and every orbit length divides m ('Operations and Periodicity' chapter Theorem 1.11(A), Theorem 1.5).

(B) **Equivalence of the decision**: the target is a uniformly periodic system $\iff$ $\mathrm{ord}(f) < \infty$ ('Operations and Periodicity' chapter Theorem 1.11(B)).

(C) **Pointwise criterion**: the orbit of the target at x_0 is closed $\iff$ x_0 is a periodic point ('Operations and Periodicity' chapter Theorem 1.6); the orbit length is then the period at that point ('Operations and Periodicity' chapter Corollary 1.3).

(D) **Non-omissibility**: the conjunction of the conserved quantities and the symmetry group **cannot** replace $\mathrm{ord}(f) < \infty$ ('Operations and Periodicity' chapter Corollary 1.4(iv)); a quasi-periodic target ($\mathrm{ord}(f) = \infty$, the orbit not closed but dense on a compact carrier structure) **is not a periodic system** ('Operations and Periodicity' chapter Remark 1.13).

Proof. (A) Apply Theorem 1.11(A) to the decision input: take the evolution group $G = \langle f \rangle \leq \mathrm{Symmetry}(f)$; then $f \in G$ and $\mathrm{ord}(f) = m$ is finite, so every orbit is closed and every orbit length divides m. (B) This is a restatement of Theorem 1.11(B) and Definition 1.21. (C) By Theorem 1.6 (a closed orbit $\iff$ pointwise periodicity) and Corollary 1.3. (D) By Corollary 1.4(iv) (conservation plus symmetry does not imply periodicity) and Remark 1.13 (a quasi-period can only be asserted, in a purely algebraic framework, as $\mathrm{ord}(f) = \infty$). ☐

24.1.2 Period types

Once a target has been decided to 'possess periodicity', its **period type** must still be determined. Periodic targets of different types differ in their modelling, in the tools available and in the nature of their errors.

Definition 24.3 (Period type)**.** *After the decision by Theorem 24.1, a target falls into one of the following types:*

(i) **Trivial periodic type**: *the least period is $m_0 = 1$ and the target is a fixed point (Remark 1.13);*

(ii) **Strictly periodic type**: *$1 < m_0 < \infty$, and the target is a periodic point rather than an equilibrium point (Definition 1.21);*

(iii) **Uniformly periodic type**: *$\operatorname{ord}(f) = m < \infty$, and every orbit length divides m (Definition 1.21);*

(iv) **Quasi-periodic type**: *$\operatorname{ord}(f) = \infty$, the orbit not closed but dense on a compact carrier structure (Remark 1.13);*

(v) **Eventually periodic type**: *there exists N such that the target is periodic for $t \geq N$ (Definition 1.14);*

(vi) **Discretely periodic type**: *the carrier structure is $\mathbb{Z}$ and the normalised period ratio $\rho = T_x/\Delta x \in \mathbb{Q}$ ('Discrete Sampling and Intercalation of Periodic Signals' chapter Proposition 21.2).*

(vii) **Asymptotically periodic (damped) type**: *the orbit is not closed, but tends asymptotically to a closed orbit or to a fixed point, the residual decaying in time—that is, there exist a period $T > 0$, a phase function θ and a residual $\rho(t)$ such that*

$$x(t) = \theta(t) + \rho(t), \qquad \theta(t+T) = \theta(t), \qquad \|\rho(t)\| \leq Ce^{-\kappa t} \quad (\kappa > 0);$$

the boundary with the eventually periodic type: the latter requires the orbit to be **exactly** *periodic after finitely many steps (Definition 1.14), whereas the asymptotically periodic type approaches closure only in the limit $t \to \infty$, no orbit being closed at any finite time.*

Of these, (i)–(iii) and (vi) are **periodic systems***; (iv) is not a periodic system, but can be handled with bounded-approximation tools; (v) is a periodic system only in its stable stage; (vii), like (iv), is a type in the limiting sense, is not a periodic system, and can be handled with bounded-approximation tools only in its stable stage.*

Proposition 24.1 (Discrimination criterion for period types)**.** *Let a target be given with decision input $\mathcal{D} = (\mathcal{T}, x)$ and decision map f. Its period type is determined by the following sequence of discriminations:*

(i) *if x is constant, the type is trivial periodic;*

(ii) *otherwise, if* ord$(f) < \infty$*, the type is uniformly periodic; if its orbit contains a fixed point, the type is a mixture with the trivial periodic type, and otherwise it is strictly periodic;*

(iii) *otherwise, if $\mathcal{T}$ is compact and the orbit is dense and not closed, the type is quasi-periodic;*

(iv) otherwise, if there exists N such that the orbit is closed for $t \geq N$, the type is eventually periodic;

(v) otherwise, if there exist a period $T > 0$, a phase function θ (with $\theta(t+T) = \theta(t)$) and a residual $\rho(t)$ such that $x(t) = \theta(t) + \rho(t)$ and $\|\rho(t)\| \leq Ce^{-\kappa t}$ ($\kappa > 0$), the type is asymptotically periodic (damped);

(vi) otherwise the target does not possess periodicity and admits only a statistical description (see Theorem 24.2(iii)).

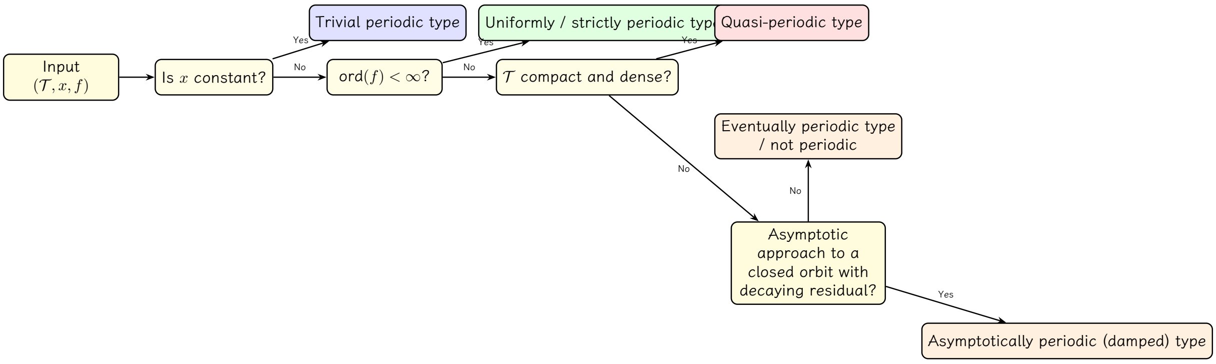

Trivial periodic type
Uniformly / strictly periodic typ
Quasi-periodic type
Input
$(\mathcal{T}, x, f)$
Is x constant?
Yes
No
ord$(f) < \infty$?
Yes
No
$\mathcal{T}$ compact and dense?
Yes
Eventually periodic type
/ not periodic
No
No
Asymptotic
approach to a
closed orbit with
decaying residual?
Yes
Asymptotically periodic (damped) type

Figure 24.2: Decision tree for discriminating period types: starting from the decision input, the constancy, the finite order, the compactness of the orbit and the asymptotic approach to closure are tested in turn, determining one of the seven period types.

24.1.3 Looking up the least positive period

The decision answers 'whether the target is periodic'; the immediate next question is 'what the period is'.

Definition 24.4 (Least positive period lookup algorithm). **Input**: the target sequence $f(0), \ldots, f(N-1)$—generated by the decision map of Definition 24.2, with discretisation sampling interval Δx taken as in Definition 21.1 of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter; tolerance $\varepsilon \geq 0$. **Output**: the least positive period T_0, or the verdict 'not periodic'. **Steps**:

(1) take the candidate set $\mathcal{P} = \{T \in \mathbb{N}^+ : 2T \leq N\}$ and traverse it in increasing order of T;

(2) for each T compute the maximum deviation $E(T) = \max_{0 \leq n < N-T} \|f(n+T) - f(n)\|$;

(3) output the first T satisfying $E(T) \leq \varepsilon$ as T_0; if the traversal is exhausted with no solution, output 'not periodic';

(4) **commensurability check**: use the normalised period ratio $\rho = T_x/\Delta x$ (when $\rho = p/q$ in lowest terms, $T_0 = p$) to verify the conversion relation between T_0 and the continuous period T_x; the corresponding continuous duration is $\ell = T_0 \Delta x$, which has the dimension of time and must not be confused

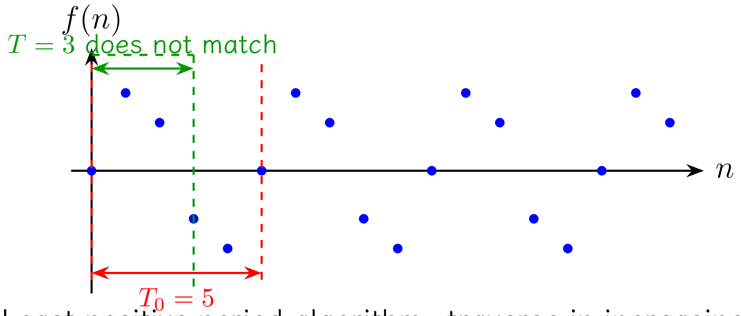
with the dimensionless ratio ρ. The verification rests on Proposition 21.2 and Corollary 21.1 of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter.

Theorem 24.2 (Correctness of the least positive period algorithm). (i) If f is a strictly periodic sequence with $\varepsilon = 0$ and $N \geq 2T_0$, then the algorithm outputs its discrete fundamental period T_0 ('Discrete Sampling and Intercalation of Periodic Signals' chapter Definition 21.4).

(ii) If f is obtained by sampling a continuous target of fundamental period T_x at the sampling interval Δx, and the normalised period ratio $\rho = T_x/\Delta x = p/q$ is in lowest terms, then the algorithm outputs $T_0 = p$ (Proposition 21.2(ii)).

(iii) If $\rho \notin \mathbb{Q}$, then no true period exists: $E(T) > 0$ for every $T \in \mathbb{N}^+$ (Proposition 21.2(iii)).

Proof. (i) By Definition 21.4, T_0 is the least positive period, hence $E(T_0) = 0$; and for $T < T_0$ we have $E(T) > 0$—otherwise T would also be a period, contradicting minimality. The algorithm traverses T in increasing order, so the first one satisfying $E(T) = 0$ is T_0. (ii) By Proposition 21.2(ii), the fundamental period of the sampled sequence is p; substitute into (i). (iii) By Proposition 21.2(iii), $\text{Group}_{\text{period}}(f) = \{0\}$, so $E(T) > 0$ for every T. □



Least positive period algorithm: traverse in increasing order from $T = 1$; the first T satisfying $E(T) \leq \varepsilon$ is T_0

Figure 24.3: Least positive period lookup algorithm: for each candidate period T compute the maximum deviation $E(T)$, and output the first T satisfying $E(T) \leq \varepsilon$ as the least positive period.

24.2 Construction of the period series: from data turning points to the epicycle group

The decision and the least positive period answer 'whether the target is periodic and how long the period is'. This section answers 'which components the target is composed of': taking the turning points of the data as characteristic time points, it constructs the amplitude series, the frequency series and the phase series, and assembles them into the epicycle group.

24.2.1 Data turning points as characteristic time points

Definition 24.5 (Turning-point feature set)**.** Let the values of the target at the sampling points $t_n = n\Delta x$ after window smoothing (for the window function see Definition 21.17 of the 'Time Series Analysis' chapter) be $\tilde{x}_n$. A time point satisfying the turning-point criterion (Definition 23.14 of the 'Time Series Analysis' chapter) is called a **turning point** of the target. Let its index set and the corresponding refraction rates (Definition 23.15 of the 'Time Series Analysis' chapter) be

$$\mathcal{I}_{\text{tp}} = \{n_1 < n_2 < \cdots < n_r\}, \qquad \mathcal{K}_{\text{tp}} = \{\kappa_{n_1}, \ldots, \kappa_{n_r}\},$$

the two together being called the **turning-point feature set** of the target. The turning-point feature set compresses a continuous target into finitely many 'direction-reversal events', and is the data starting point for the construction of the period series.

Proposition 24.2 (The turning-point feature set carries period information)**.** If the target is a discrete periodic sequence with fundamental period m_0 (Definition 21.3 of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter), then its turning-point feature set is periodic with period m_0:

$$n \in \mathcal{I}_{\text{tp}} \iff n + m_0 \in \mathcal{I}_{\text{tp}}, \qquad \kappa_{n+m_0} = \kappa_n.$$

Proof. A turning point is defined by a local change of direction (Definition 23.14 of the 'Time Series Analysis' chapter), and the direction-change operator commutes with the m_0-translation; moreover $\tilde{x}$ is periodic with period m_0 (window smoothing preserves the translation structure), so both $\mathcal{I}_{\text{tp}}$ and $\mathcal{K}_{\text{tp}}$ are periodic with period m_0. □

24.2.2 Amplitude series, frequency series and phase series

The formal definitions of the three series have already been given in the 'Series and Convolution' chapter; this section explains how to determine them **from the data.**

Theorem 24.3 (Building the period series from turning points)**.** Let the discrete fundamental period of the target be T_0, with Fourier expansion

$$f(n) = \sum_k A_k e^{i(2\pi kn/T_0 + \phi_k)}, \qquad A_k \geq 0,\ \phi_k \in [0, 2\pi).$$

Then the **period series** of the target is described jointly by three series:

(i) **amplitude series** $\mathcal{A}_f(t) = \sum_k A_k e^{ikt}$: the coefficients A_k are given by the Fourier moduli of the data,

$$A_k = \left| \frac{1}{T_0} \sum_{n=0}^{T_0-1} f(n) e^{-i2\pi kn/T_0} \right|,$$

where the turning-point feature set $\mathcal{I}_{\text{tp}}$ gives the phase-locking positions of the various components (Definition 6.10 of the 'Series and Convolution' chapter);

(ii) **frequency series** $\mathcal{F}_f(s) = \frac{2\pi}{T_0}\sum_k ks^k$: determined entirely by the least positive period T_0, that is, by the frequency lattice $\{2\pi k/T_0\}$ (Definition 6.12 of the 'Series and Convolution' chapter);

(iii) **phase series** $\mathcal{P}_f(t) = \sum_k e^{i(kt+\phi_k)}$: the arguments ϕ_k of the coefficients are given by the phase spectrum of the data and agree with the turning-point delays (Definition 6.11 of the 'Series and Convolution' chapter).

Together the three constitute the complete Fourier data of the target on the period T_0: the frequency series says 'which components there are', whilst the amplitude series and the phase series say 'how strong each component is and by how much it is deflected'.

Proof. (i)(iii) follow directly from the discrete Fourier coefficient formula and its polar decomposition $c_k = A_k e^{i\phi_k}$; the correspondence between turning-point positions and phase differences is given by Proposition 24.2 and Lemma 23.2 of the 'Time Series Analysis' chapter. (ii) By Definition 6.12, the frequency lattice is uniquely determined by T_0, and every frequency is an integer multiple of the base frequency. ☐

24.2.3 The epicycle group: radius, phase and rotation speed

Definition 24.6 (Epicycle group)**.** The epicycle group is first defined in Chapter 21, under the section Section 21.2, by Definition 21.8; what follows is its form in the notation of the period series (the amplitude, frequency and phase series), together with the correspondence 'radius to amplitude, phase to rotation delay, frequency to rotation speed'. Let the period series of the target be given by the amplitude series $\{A_k\}$, the frequency series $\{\omega_k = 2\pi k/T_0\}$ and the phase series $\{\phi_k\}$. For each k define the **epicycle**

$$E_k:\ \theta \mapsto A_k e^{i(\theta+\phi_k)}, \qquad \theta = \omega_k t = \frac{2\pi k}{T_0}t,$$

whose collection

$$\mathfrak{E} = \{(A_k, \omega_k, \phi_k)\}_{k\in\mathbb{Z}}$$

is called the **epicycle group** of the target; its superposition is the period series of the target

$$f(t) = \sum_k E_k(\omega_k t) = \sum_k A_k e^{i(\omega_k t+\phi_k)}.$$

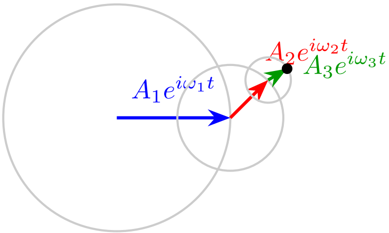
Proposition 24.3 (Three-parameter correspondence: radius, phase and speed)**.** The three parameters of the epicycle group $\mathfrak{E} = \{(A_k, \omega_k, \phi_k)\}_k$ correspond one-to-one to three geometric–mechanical quantities:

(i) **radius to amplitude**: the radius $|E_k| = A_k$ of the epicycle E_k is the amplitude of that component, and A_k^2 contributes energy (Theorem 22.1);

(ii) **phase to rotation delay**: for $k \neq 0$ (that is, $\omega_k \neq 0$), the initial phase ϕ_k corresponds to the time translation $\tau_k = \phi_k/\omega_k$; when $\phi_k > 0$ this component **leads** the baseline by τ_k (equivalently, the component coincides with the baseline once the baseline is postponed by τ_k); this quantity is the **rotation delay** of the epicycle ('Series and Convolution' chapter Proposition 6.15); the $k = 0$ direct-current component has no rotation delay;

(iii) **frequency to rotation speed**: ω_k is the **angular velocity** of the epicycle about the origin, given by the least positive period as $\omega_k = 2\pi k/T_0$ (Definition 6.12).

Proof. (i) By definition $|E_k(\cdot)| = A_k$, together with the energy decomposition of Theorem 22.1. (ii) By Proposition 6.15: a time translation τ changes the coefficients of the phase series to $e^{i(\phi_k - \omega_k \tau)}$, and taking $\tau = \phi_k/\omega_k$ (for $k \neq 0$) cancels the initial phase, so that ϕ_k measures precisely the delay, with the **advanced** direction — once the reference has been delayed by τ_k it coincides with the waveform; for $k = 0$ we have $\omega_0 = 0$ and the expression is undefined, the direct-current component having no rotation delay. (iii) By $\theta = \omega_k t$ and Definition 6.12. ☐



Epicycle group: radius corresponds to amplitude,
phase to rotation delay, frequency to rotation speed

Figure 24.4: Geometric superposition of the epicycle group: *each epicycle is determined by the three parameters (A_k, ω_k, ϕ_k), corresponding respectively to radius, angular velocity and initial phase.*

24.2.4 The period series is a complex function

Theorem 24.4 (Complex-function representation of the period series)**.** *Let the fundamental period of the target be T_0 and the epicycle group be $\mathfrak{E} = \{(A_k, \omega_k, \phi_k)\}_k$; put $\omega_0 = 2\pi/T_0$ and let $z = e^{i\omega_0 t}$ be the complex variable. Provided that* **every component frequency is commensurable with** *ω_0 (that is, $\omega_k = k\omega_0$ with $k \in \mathbb{Z}$), the period series of the target equals the complex function on the unit circle*

$$f(t) = \sum_{k \in \mathbb{Z}} c_k z^k, \qquad c_k = A_k e^{i\phi_k},$$

that is: the Fourier expansion of the target on the period T_0 is equivalent to the values on $|z| = 1$ of a Laurent series; a finite harmonic truncation is equivalent to a complex-coefficient polynomial. **When all components are commensurable with ω_0, the period series is a complex-valued function on the unit circle (otherwise it is a function on the torus).**

Proof. Substituting $z^k = e^{ik\omega_0 t} = e^{i\omega_k t}$ gives $\sum_k A_k e^{i(\omega_k t + \phi_k)} = \sum_k c_k z^k$. The coefficients c_k do not depend on z, so the right-hand side is a Laurent series in z; when only finitely many k satisfy $c_k \neq 0$ it degenerates to a polynomial. The identity $z^k = e^{i\omega_k t}$ holds only when $\omega_k/\omega_0 \in \mathbb{Z}$; if a component incommensurable with ω_0 is present, $e^{i\omega_k t}$ is not a single-valued function of z, and the period series is merely a function on the torus $\mathbb{T}^n$ rather than a complex-valued function on the unit circle. ☐

24.3 Dynamic adjustment of the epicycle group

24.3.1 One epicycle corresponds to one simple complex function

Definition 24.7 (Simple complex function (single-epicycle function)). A mapping of the form

$$\varphi_{A,\omega,\phi}: \ \mathbb{R} \to \mathbb{C}, \qquad \varphi_{A,\omega,\phi}(t) = A\, e^{i(\omega t+\phi)}$$

is called a **simple complex function**. It is the restriction to the unit circle of the complex function $z \mapsto cz$ $(c = Ae^{i\phi})$, and has only three degrees of freedom $(\log A, \omega, \phi)$—exactly corresponding to the three parameters of one epicycle.

Proposition 24.4 (The epicycle group is the period-series form of a complex function). The relation between the epicycle group $\mathfrak{E} = \{E_k\}_k$ and the target period series f is linear superposition:

$$f = \sum_k \varphi_{A_k,\omega_k,\phi_k}.$$

That is: **one epicycle corresponds to one simple complex function, and the epicycle group is a finite sum of simple complex functions, that is, a complex function in the form of a period series** (Theorem 24.4).

Proof. Add term by term; the parameter set of the finite sum is exactly $\mathfrak{E}$, and by Definition 24.7 each term is uniquely determined by (A_k, ω_k, ϕ_k). □

24.3.2 Dynamic adjustment of number and parameters

Definition 24.8 (Dynamic adjustment of the epicycle group). Let the current epicycle group be $\mathfrak{E}^{(j)}$, the partial sum $f^{(j)} = \sum_{k\in\mathfrak{E}^{(j)}} \varphi_k$ and the residual $r^{(j)} = x - f^{(j)}$. **Dynamic adjustment** consists of two rules:

(i) **quantity adjustment**: if the residual energy $\|r^{(j)}\|^2 > \delta$ (with $\delta > 0$ the tolerance), add one epicycle according to the peak of the residual periodogram (Definition 23.16 of the 'Time Series Analysis' chapter):

$$\mathfrak{E}^{(j+1)} = \mathfrak{E}^{(j)} \cup \{(\hat{A}, \hat{\omega}, \hat{\phi})\}, \qquad \hat{\omega} = \arg\max_{\omega} \big|\widehat{X_r}(\omega)\big|;$$

(ii) **parameter adjustment**: update the parameters of the existing epicycles by regularised least squares (Proposition 23.3 of the 'Time Series Analysis' chapter)

$$(A_k, \phi_k) \leftarrow \arg\min_{A,\phi} \sum_n \Big| x(n) - \sum_k A_k e^{i(\omega_k n+\phi_k)} \Big|^2,$$

or correct them online by Bayesian updating (Definition 24.18, Proposition 24.10).

Proposition 24.5 (Monotonicity of quantity adjustment). *After a dominant residual frequency component is added as in Definition 24.8(i), the residual energy is monotonically non-increasing:*

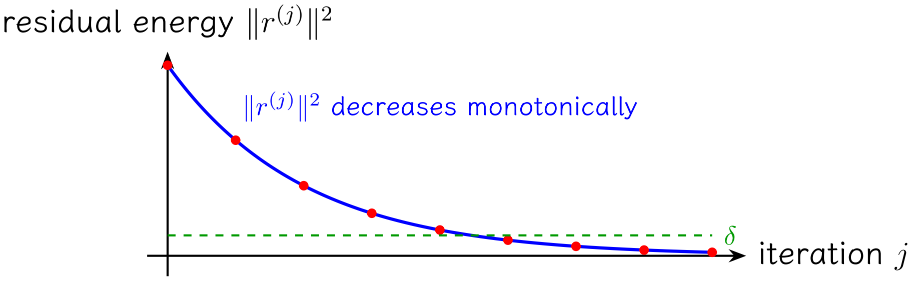
$$\|r^{(j+1)}\|^2 \le \|r^{(j)}\|^2,$$

with equality if and only if the residual has no energy at the selected frequency.

Proof. Adding an epicycle removes the component of $r^{(j)}$ along $\hat{\omega}$ from the residual. By the orthogonal decomposition $\|r^{(j)}\|^2 = \|r^{(j+1)}\|^2 + \|$(the component removed)$\|^2$, the inequality holds, with equality if and only if the norm of the component removed is zero. ☐

Theorem 24.5 (Convergence of dynamic adjustment). *Let the target be a* **real-valued finite periodic sequence** *of length T_0 ($T_0 \in \mathbb{N}$) and let $\delta = 0$. Then after at most $\lfloor T_0/2 \rfloor + 1$ steps of dynamic adjustment the residual is zero, and the resulting epicycle group is the exact Fourier data of the target; for a* **complex-valued** *sequence the corresponding number of steps is at most T_0. This adjustment process, the sampling layer of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter and the Fourier layer of the 'Harmonic Analysis' chapter are three views of one and the same decomposition.*

Proof. The discrete Fourier series of a **real-valued** finite periodic sequence of length T_0 ($T_0 \in \mathbb{N}$) is complete on $\lfloor T_0/2 \rfloor + 1$ harmonics (including the constant term); a **complex-valued** sequence requires T_0 components (the positive and negative frequencies being independent). At each step one dominant residual frequency component is extracted as in Proposition 24.5, and the residual energy decreases strictly, so after finitely many steps the residual is zero. The convergence of the parameters is given by the strong convexity of regularised least squares. ☐



Dynamic adjustment: each step adds a dominant residual frequency component, the residual energy is monotonically non-increasing, and after finitely many steps it converges to δ

Figure 24.5: Decreasing curve of the residual energy under dynamic adjustment: with each epicycle added the residual energy decreases strictly, and convergence is reached after at most $\lfloor T_0/2 \rfloor + 1$ steps (T_0 being the length of the real-valued finite periodic sequence).

24.4 Amplitude convolution, phase convolution and frequency convolution

The 'Series and Convolution' chapter has already defined, on general periodic functions, amplitude convolution, phase convolution, frequency convolution and trend convolution (Definition 6.21, Definition 6.22, Definition 6.23), and has given the relation among the three (Proposition 6.28). This section explains their role in a **periodic system**: the three act respectively on the three parameters of the epicycle group.

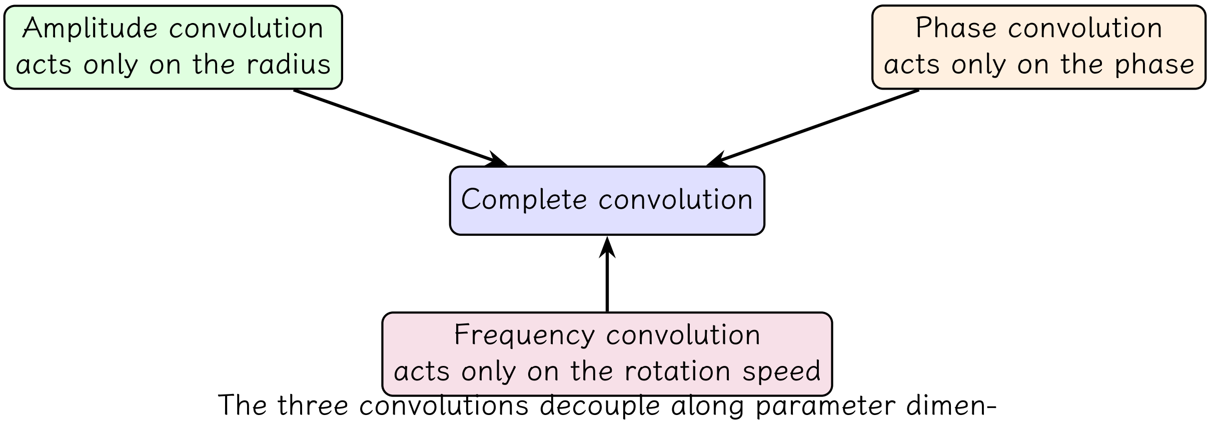
Theorem 24.6 (Action of the three convolutions on the epicycle group)**.** *Let the epicycle groups of the target and of the reference target be $\mathfrak{E} = \{(A_k, \omega_k, \phi_k)\}$ and $\mathfrak{E}' = \{(B_l, \nu_l, \psi_l)\}$ respectively. Then:*

- (i) **amplitude convolution** *acts only on the radius: $(A_k) * (B_l)$ gives the combination of the radii of each pair of components, with frequency and phase taking no part* (Definition 6.21)*;*
- (ii) **phase convolution** *acts only on the phase: $(\phi_k) * (\psi_l)$ gives the combination of the phase differences of each pair of components, with radius and frequency taking no part, and it satisfies conjugate symmetry* (Definition 6.22, Proposition 6.29)*;*
- (iii) **frequency convolution** *acts only on the rotation speed: $(\omega_k) * (\nu_l)$ gives the sum of the rotation speeds of each pair of components and determines the frequency lattice of the combined components* (Definition 6.23)*;*
- (iv) **trend convolution** *retains both amplitude and phase: every component of $\mathbb{T}_{kl} = A_k B_l e^{i(\phi_k - \psi_l)}$ carries both a radius factor and a phase factor, so its projection onto the diagonal ($k = l$) gives the amplitude and the phase of each individual component* (Definition 6.24)*.*

Proof. By the respective definitions of the three: amplitude convolution sets the phase to zero and the frequency to the reference lattice, retaining only the amplitude; phase convolution sets the amplitude to 1 and the frequency to the reference lattice, retaining only the phase; frequency convolution sets the amplitude to 1 and the phase to zero, retaining only the frequency. whilst trend convolution, by Definition Definition 6.24, retains both the amplitude factor $A_k B_l$ and the phase factor $e^{i(\phi_k - \psi_l)}$. Hence the first three each act on only one parameter, whereas trend convolution acts on the two dimensions of amplitude and phase at once. The conjugate symmetry follows from Proposition 6.29. $\square$

Proposition 24.6 (Factorisation of the three convolutions)**.** *In the sense of* Proposition 6.28*, the complete convolution can be decomposed into a combination of the three convolutions. For an epicycle group this amounts to independent combinations along the three parameter dimensions, so that the interaction between periodic systems can be analysed decoupled by parameter dimension:* **amplitude coupling changes only the radius, phase coupling changes only the rotation delay, and frequency coupling changes only the rotation speed***.*

Proof. By Theorem 24.6, amplitude, phase and frequency convolution each act on only one parameter dimension, whilst trend convolution retains both amplitude and phase; Proposition 6.28 gives that their composition is the complete convolution, so the actions on the three dimensions do not interfere with one another. Restricting the complete convolution to the diagonal $k = l$, the orthogonality of the individual components cancels every off-diagonal cross term and leaves only the diagonal trend-convolution term $\sum_k \mathbb{T}_{kk} e^{-i\omega_k h}$, which is precisely the expression for the autocorrelation function in the template of the application chapters. $\square$


Amplitude convolution acts only on the radius
Phase convolution acts only on the phase
Complete convolution
Frequency convolution acts only on the rotation speed

The three convolutions decouple along parameter dimensions: their actions on the amplitude, phase and frequency dimensions do not interfere with one another

Figure 24.6: Decoupled action of the three convolutions on the epicycle group: amplitude convolution changes only the radius, phase convolution only the phase, and frequency convolution only the rotation speed.

24.5 Integration with discrete sampling, harmonic analysis and time series analysis

24.5.1 With the Discrete Sampling and Intercalation of Periodic Signals chapter: from continuous targets to computable sequences

The 'Discrete Sampling and Intercalation of Periodic Signals' chapter gives the standard definitions of sampling points, sampling rate and sampling duration (Definition 21.1, Definition 21.2) as well as the commensurability criterion for the periodicity of a sampled sequence (Proposition 21.2). In the workflow of this chapter it undertakes the **discretisation of the decision input**: when the carrier structure is $\mathbb{R}$, one first takes the sampling interval Δx to obtain the sequence $f(n) = x(n\Delta x)$, and then uses Theorem 24.2(ii) to convert the discrete fundamental period T_0 back to the continuous period T_x. When $\rho \notin \mathbb{Q}$ the periodic structure of the target is incommensurable under sampling, and the intercalation sequence of that chapter must be used to confine spectral leakage to a bounded level (Theorem 21.10).

24.5.2 With the Harmonic Analysis chapter: frequency decomposition and coefficients

The 'Harmonic Analysis' chapter gives the Fourier series and the Fourier transform (Definition 22.5, Definition 22.6), Parseval's identity (Theorem 22.1) and the relation between the Fourier coefficients and the parameters of the simple harmonic components (Proposition 22.4). It undertakes the step **from the period series to the parameters of the epicycle group** in the workflow of this chapter: the coefficients of the amplitude series, the frequency series and the phase series are precisely the Fourier coefficients, and the correspondence given by Proposition 22.4 agrees completely with 'radius to amplitude, phase to delay, frequency to rotation speed' in Proposition 24.3.

24.5.3 With the Time Series Analysis chapter: from data to parameters

The 'Time Series Analysis' chapter gives the window function (Definition 21.17), turning-point detection (Definition 23.14), period determination by the periodogram (Definition 23.16), finite harmonic regression (Definition 23.17) and Bayesian updating (Definition 23.18). It undertakes the step **from observed data to estimates of the parameters of the epicycle group** in the workflow of this chapter: turning points supply the characteristic time points, the periodogram supplies initial frequencies, finite harmonic regression supplies parameter estimates, and Bayesian updating supplies online corrections.

Step of this chapter	Main tool chapter	Key entries
Decision input and discretisation	Discrete Sampling and Intercalation of Periodic Signals chapter	Definition 21.1, Proposition 21.2
Periodicity decision and period type	Operations and Periodicity chapter	Theorem 1.11, Definition 1.21, Remark 1.13
Least positive period	Discrete Sampling and Intercalation of Periodic Signals chapter	Definition 21.4, Corollary 21.1
Period series and epicycle group	Series and Convolution chapter	Definition 6.10, Definition 6.12, Definition 6.11
Coefficients and energy decomposition	Harmonic Analysis chapter	Proposition 22.4, Theorem 22.1
Turning points, periodogram and parameter estimation	Time Series Analysis chapter	Definition 23.14, Definition 23.16, Proposition 23.3
The three convolutions	Series and Convolution chapter	Definition 6.21, Definition 6.22, Definition 6.23

Table 24.1: The five-step interface of a periodic system and its correspondence with the theoretical chapters.

Remark 24.4 (Citation convention for the application chapters). In deciding whether a target is periodic, in determining the period type and in looking up the least positive period, the later application chapters—probabilistic systems, cryptographic systems, astronomical systems, calendar systems, geographic systems, the I Ching system, electromagnetic systems, audio systems, mechanical systems, circuit and chip systems, biological systems, psychological systems, social systems, the climate system, economic systems, political systems, data analysis, agents, algorithmic systems, and so on—uniformly cite this chapter:

(i) periodicity criterion: Theorem 24.1;

(ii) discrimination of period types: Definition 24.3, Proposition 24.1;

(iii) lookup of the least positive period: Definition 24.4, Theorem 24.2;

(iv) construction of the period series and the epicycle group from data: Theorem 24.3, Definition 24.6;

(v) dynamic adjustment of the epicycle group: Definition 24.8;

(vi) the three convolutions: Theorem 24.6.

The application chapters do not state the above criteria again.

24.6 The multi-component form of periodic systems: multi-period coupled systems

Definition 24.9 (Multi-period coupled system (the multi-component form of a periodic system)). A **multi-period coupled system** $\mathfrak{S}$ is the quintuple

$$\mathfrak{S} = (\mathcal{F}, \mathcal{G}, \mathcal{C}, \Gamma, \mathcal{H}),$$

where:

(i) $\mathcal{F} = \{f_1, \dots, f_K\}$ is a collection of finitely many **periodic components**. Each component $f_k : \mathbb{R} \to \mathbb{C}$ is a periodic function with period T_k, amplitude spectrum $\{A_n^{(k)}\}$ and phase spectrum $\{\theta_n^{(k)}\}$.

(ii) $\mathcal{G} = \{G_1, \dots, G_K\}$ is the corresponding collection of **translation symmetry groups**. Each G_k is a period subgroup of f_k, with $G_k \subset \mathbb{R}$ a discrete subgroup.

(iii) $\mathcal{C} = \{c_{ij}^{(d)}\}_{i,j=1}^{K}$ $(d = 1, \dots, 12)$ are the **coupling coefficients**. $c_{ij}^{(d)} \in \mathbb{C}$ denotes the coupling strength of the weighted cross-correlation of the d-th type between the components f_i and f_j.

(iv) Γ is the **composite signal**, formed by superposing the principal harmonics of the various components.

(v) $\mathcal{H}$ is the **total energy** of the system, defined as the sum of the independent energies of the components, the coupling energies and the interaction energies of the composite signal:

$$\mathcal{H}(\mathfrak{S}) = \sum_{k=1}^{K} E_k + \sum_{i \neq j} \sum_{d=1}^{12} |c_{ij}^{(d)}| \, \|\mathcal{R}_W^{(d)}(f_i, f_j)\| + \sum_{k=1}^{K} \sum_{d=1}^{12} |\gamma_k^{(d)}| \, \|\mathcal{R}_W^{(d)}(\Gamma, f_k)\|,$$

where $E_k = \sum_n |A_n^{(k)}|^2$, $\gamma_k^{(d)}$ is the coupling coefficient of the d-th type of Γ with f_k, and $\mathcal{R}_W^{(d)}$ is the weighted cross-correlation operator (see Definition 24.10).

Definition 24.10 (Weighted cross-correlation operator). Let

$$f(t) = \sum_n A_n e^{i(\omega_n t + \theta_n)}, \qquad g(t) = \sum_m B_m e^{i(\nu_m t + \psi_m)}.$$

For a weight kernel W and a phase kernel $S^{(d)}$, define the **weighted cross-correlation** as

$$\mathcal{R}_W^{(d)}(f,g)(\tau) = \sum_{n,m} W(\omega_n - \nu_m)\,(A_n B_m)^\alpha\, S^{(d)}(\theta_n, \psi_m)\, e^{i\nu_m \tau},$$

where $\alpha \in \{0,1\}$ is the amplitude switch. When $\alpha = 0$ only the phase information is retained; when $\alpha = 1$ the amplitude and the phase information are retained simultaneously.

Definition 24.11 (Sub-periodic components)**.** Each periodic component f_k in the system $\mathfrak{S}$ has:

(i) an **independent translation symmetry group** G_k: f_k is invariant under the action of G_k.

(ii) a **harmonic expansion**:

$$f_k(t) = \sum_n A_n^{(k)} e^{i(\omega_n^{(k)} t + \theta_n^{(k)})},$$

in which the harmonics at each level have independent amplitudes $A_n^{(k)}$, angular frequencies $\omega_n^{(k)}$ and initial phases $\theta_n^{(k)}$.

(iii) a **dominant operator type**: each component corresponds to one dominant differential operator (gradient, curl or divergence), and this type determines the phase-selection rule for its coupling.

Definition 24.12 (Coupling types)**.** The coupling $c_{ij}^{(d)}$ between two components f_i and f_j is classified, according to the parameters of the weighted cross-correlation, as:

(i) **phase coupling** ($\alpha = 0$): the coupling is determined by phase similarity alone and contains no amplitude information.

(ii) **amplitude–phase coupling** ($\alpha = 1$): the coupling is determined jointly by amplitude and phase similarity.

(iii) **same-frequency coupling** ($d = 1$ and $n = m$): only the interaction between components of the same frequency is retained.

Proposition 24.7 (Symmetry of the coupling coefficients)**.** For the coupling coefficients $c_{ij}^{(d)}$:

(i) difference-angle phase kernels ($d = 1, \ldots, 6$): $c_{ij}^{(d)} = \overline{c_{ji}^{(d)}}$.

(ii) sum-angle phase kernels ($d = 7, \ldots, 12$): $c_{ij}^{(d)} = c_{ji}^{(d)}$.

(iii) self-coupling: $c_{ii}^{(1)} = 1$, $c_{ii}^{(2)} = 0$.

Proof. (i) Difference-angle phase kernels satisfy $S^{(d)}(\theta_i, \theta_j) = \overline{S^{(d)}(\theta_j, \theta_i)}$, hence $c_{ij}^{(d)} = \overline{c_{ji}^{(d)}}$. (ii) Sum-angle phase kernels are invariant under the interchange of i, j, hence $c_{ij}^{(d)} = c_{ji}^{(d)}$. (iii) When $i = j$ the difference angle is zero, $\cos 0 = 1$ and $\sin 0 = 0$, hence $c_{ii}^{(1)} = 1$ and $c_{ii}^{(2)} = 0$. □

Proposition 24.8 (Phase locking)**.** *When the coupling strength is sufficiently large, phase locking occurs between the components, or between a component and the composite signal.*

24.6.1 Phase-space structure: symplectic manifolds and Hamiltonian flows

Definition 24.13 (Phase space of a periodic system)**.** Let the system $\mathfrak{S}$ have N degrees of freedom. Its **phase space** is the cotangent bundle T^*Q of the configuration space Q, equipped with the standard symplectic form

$$\omega := \sum_{i=1}^{N} dq^i \wedge dp_i,$$

where $(q^1, \ldots, q^N)$ are generalised coordinates and $(p_1, \ldots, p_N)$ are the conjugate momenta. (T^*Q, ω) is called a **symplectic manifold**.

Definition 24.14 (Hamiltonian vector field)**.** Let $H : T^*Q \to \mathbb{R}$ be a C^1 smooth Hamiltonian function. The **Hamiltonian vector field** X_H induced by H is uniquely determined by

$$\omega(X_H, \cdot) = -dH.$$

In local symplectic coordinates,

$$X_H = \sum_{i=1}^{N} \left(\frac{\partial H}{\partial p_i} \frac{\partial}{\partial q^i} - \frac{\partial H}{\partial q^i} \frac{\partial}{\partial p_i} \right).$$

Proposition 24.9 (Local form of Hamilton's equations)**.** The integral curves of the vector field X_H satisfy **Hamilton's equations**

$$\dot{q}^i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q^i}, \qquad i = 1, \ldots, N.$$

Proof. An integral curve $\gamma(t) = (q(t), p(t))$ satisfies $\dot{\gamma} = X_H(\gamma)$. Reading off the local expression for X_H component by component gives the result. ☐

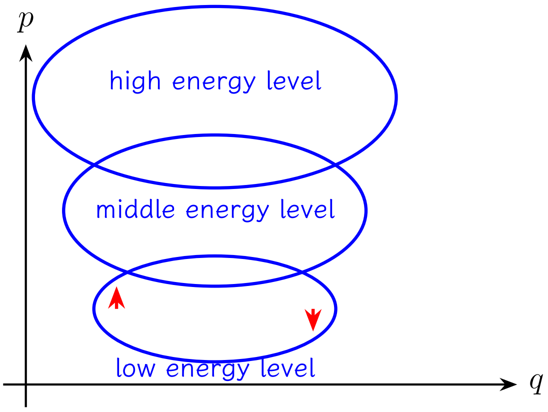
Theorem 24.7 (Liouville's theorem)**.** A Hamiltonian flow preserves the symplectic volume form $\mu := \frac{1}{N!}\omega^{\wedge N}$. That is, for every t, $(\Phi_t^H)^*\mu = \mu$, where Φ_t^H is the flow of X_H.

Proof. By Cartan's formula $\mathcal{L}_{X_H}\omega = d(\iota_{X_H}\omega) + \iota_{X_H}(d\omega)$. Since ω is closed ($d\omega = 0$), and $\iota_{X_H}\omega = -dH$ is an exact form (so that $d(\iota_{X_H}\omega) = -d^2H = 0$), we obtain $\mathcal{L}_{X_H}\omega = 0$. The symplectic form is preserved, and hence so is its exterior power μ. ☐

Definition 24.15 (Invariant torus)**.** The invariant torus is first defined in Chapter 11, under the section Section 11.11, by Definition 11.26; what follows is a restatement of the same object in the notation of N first integrals. Let a Hamiltonian system have N independent first integrals in involution $F_1 = H, F_2, \ldots, F_N$. The level set

$$\mathbb{T}_c := \{F_1 = c_1, \ldots, F_N = c_N\}$$

is called an **invariant torus**.



Hamiltonian flow lines in phase space (constant-energy curves):
Liouville's theorem guarantees the conservation of symplectic volume

Figure 24.7: Hamiltonian flow lines in phase space: *each flow line is a constant-energy curve, and the Hamiltonian flow preserves symplectic volume (Liouville's theorem).*

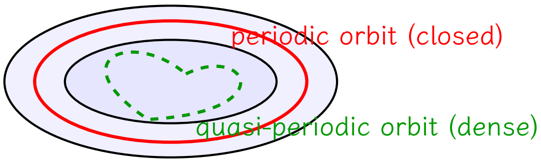
Theorem 24.8 (Liouville–Arnold theorem)**.** *Let an N-degree-of-freedom Hamiltonian system have N independent first integrals in involution, and let the level set $\mathbb{T}_c$ be compact and connected. Then $\mathbb{T}_c \cong \mathbb{T}^N$, and in a neighbourhood of it there exist action–angle coordinates (I, θ) such that the Hamiltonian function depends only on the action variables I:*

$$H = H(I_1, \ldots, I_N), \qquad \dot{I}_i = 0, \quad \dot{\theta}^i = \frac{\partial H}{\partial I_i} =: \omega_i(I).$$

Proof idea. The existence of the invariant torus is given by the compactness and connectedness of the level set together with N first integrals in involution (the Arnold–Liouville theorem; for a complete proof see Arnold, *Mathematical Methods of Classical Mechanics*, 2nd ed., Springer, 1989, §49–50). On the torus take angle coordinates θ^i making the Hamiltonian flow a uniform rotation, and take the action variables $I_i = \frac{1}{2\pi}\oint_{\gamma_i} p\,dq$ as integrals along the basic cycles of the torus. The closedness of the symplectic form guarantees that (I, θ) are local symplectic coordinates; integrability in a neighbourhood of the torus gives $\dot{I}_i = 0$. ☐

Corollary 24.1 (Periods of motion on the invariant torus)**.** *On the invariant torus $\mathbb{T}^N$, the orbit $\theta(t) = \theta(0) + \omega t$ is closed if and only if the frequencies are pairwise commensurable, that is, if and only if there exists $T > 0$ with $\omega T \in 2\pi\mathbb{Z}^N$ (equivalently $\omega_i/\omega_j \in \mathbb{Q}$ for all i, j; if some $\omega_j = 0$, that component is stationary and it suffices to take T to be the least common multiple of the periods of the remaining components). If they are pairwise commensurable, the orbit is a closed orbit of period T; otherwise the orbit is dense on $\mathbb{T}^N$ and is not closed. Here 'pairwise commensurable' cannot be weakened to 'there exists $k \in \mathbb{Z}^N \setminus \{0\}$ with $k \cdot \omega = 0$', which only asks the frequencies to be linearly dependent over $\mathbb{Q}$.*

Proof. If there exists $T > 0$ with $\omega_i T \in 2\pi\mathbb{Z}$ for all i, then $\theta^i(T) = \theta^i(0) + 2\pi k_i$ with $k_i \in \mathbb{Z}$, hence $\theta(T) \equiv \theta(0)$ and the orbit is closed. A component with $\omega_j = 0$ is stationary and contributes no period, so T may be taken to be the least common multiple of the periods of the remaining components. Conversely, if no such T exists, then the ratios of the frequencies include an irrational number, and by the Kronecker approximation theorem the orbit is dense on $\mathbb{T}^N$ and is not closed. ☐



Periodic and quasi-periodic orbits on an invariant torus: closed when the frequencies are pairwise commensurable, dense otherwise

Figure 24.8: Two types of orbit on an invariant torus: the periodic orbit (red solid line) is closed because its frequencies are pairwise commensurable, whilst the quasi-periodic orbit (green dashed line) has incommensurable frequencies and is dense on the torus.

24.6.2 Conservation laws and the Noether correspondence

Definition 24.16 (Conserved quantities of a periodic system). Let $\mathfrak{S}$ be a multi-period coupled system and Φ_t its flow. A function $Q : T^*Q \to \mathbb{R}$ is called a **conserved quantity** of $\mathfrak{S}$ (or a first integral) if

$$Q \circ \Phi_t = Q, \qquad \forall t \in \mathbb{R}.$$

Equivalently, $\{Q, H\} = 0$ (it is in involution with H under the Poisson bracket).

Lemma 24.1 (Conserved quantities characterise orbits). Q is a conserved quantity of $\mathfrak{S}$ if and only if Q takes a constant value on every orbit.

Proof. If $Q \circ \Phi_t = Q$, then for every x_0 and every t, $Q(\Phi_t(x_0)) = Q(x_0)$, that is, Q is constant along orbits. Conversely, if Q is constant along every orbit, then for every x and every t, $Q(\Phi_t(x)) = Q(x)$, that is, $Q \circ \Phi_t = Q$. ☐

Theorem 24.9 (The Noether correspondence for periodic systems). Let the multi-period coupled system $\mathfrak{S}$ have a one-parameter continuous symmetry group $\{\phi_s\}_{s\in\mathbb{R}}$ (that is, $\phi_s \circ \Phi_t = \Phi_t \circ \phi_s$ for all s, t), with infinitesimal generator ξ. If there exist a smooth function Q and a symplectic structure J such that $\xi = J\nabla Q$, then Q is a conserved quantity of $\mathfrak{S}$.

Proof. By symmetry, $\mathcal{L}_\xi H = 0$ (the symmetry transformation leaves H invariant). By $\xi = J\nabla Q$ and the antisymmetry of J,

$$\frac{d}{dt}Q(\Phi_t(x)) = \nabla Q \cdot X_H = (J^{-1}\xi) \cdot X_H = -\xi \cdot \nabla H = -\mathcal{L}_\xi H = 0.$$

Hence Q is invariant along the flow, that is, it is a conserved quantity. ☐

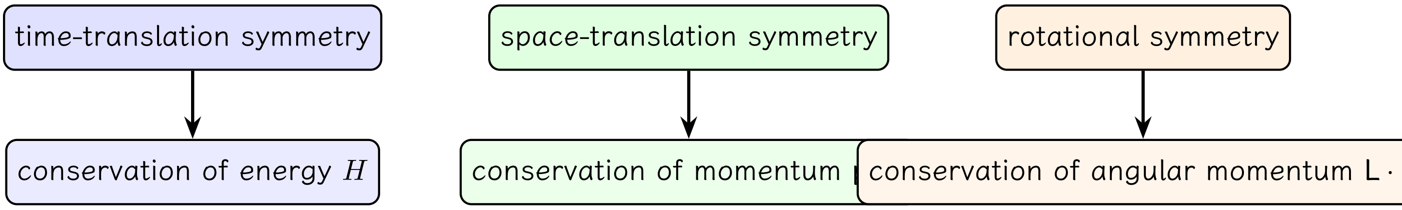
Corollary 24.2 (Three types of symmetry and three types of conserved quantity). In the symplectic framework of periodic systems, three types of continuous symmetry correspond respectively to three types of conserved quantity:

(i) **time-translation symmetry** (ϕ_s = time translation by s) $\Rightarrow$ **conservation of energy** $Q = H$;

(ii) **space-translation symmetry** (ϕ_s = translation by s in the direction a) $\Rightarrow$ **conservation of momentum** $Q = \mathsf{p} \cdot \mathsf{a}$;

(iii) **rotational symmetry** (ϕ_s = rotation through the angle s about the axis u) $\Rightarrow$ **conservation of angular momentum** $Q = \mathsf{L} \cdot \mathsf{u}$.

Proof. In the three cases verify respectively the condition $\xi = J\nabla Q$ of Theorem 24.9: (i) the generator of time translation is $\xi = X_H$, and taking $Q = H$ gives the result; (ii) the generator of space translation is $\xi = \partial_{\mathsf{a}}$, and take $Q = \mathsf{p} \cdot \mathsf{a}$; (iii) the generator of rotation is $\xi = \mathsf{u} \times \mathsf{r} \cdot \partial_{\mathsf{r}} + \mathsf{u} \times \mathsf{p} \cdot \partial_{\mathsf{p}}$, and take $Q = \mathsf{L} \cdot \mathsf{u}$. ☐


time-translation symmetry
space-translation symmetry
rotational symmetry
conservation of energy H
conservation of momentum
conservation of angular momentum L ·

The Noether correspondence for periodic systems: three types of continuous symmetry correspond respectively to three types of conserved quantity

Figure 24.9: The Noether correspondence for periodic systems: time-translation, space-translation and rotational symmetry correspond respectively to the conservation of energy, of momentum and of angular momentum.

Remark 24.5 (The role of conservation laws in periodic systems). Within the framework of periodic systems the role of the conserved quantities $\{Q_i\}$ is:
(1) **constraining the orbits**—every orbit is confined to an invariant level set $\{Q_1 = c_1, \ldots, Q_k = c_k\}$ (Lemma 24.1);
(2) **testing integrability**—if the number of independent conserved quantities equals the number of degrees of freedom N, the system is completely integrable and the orbits lie on invariant tori (Theorem 24.8);
(3) **but not directly producing periodicity**—the sufficient condition for a closed orbit is a **finite-order action** (Theorem 24.1), whilst conserved quantities and symmetries play only an auxiliary constraining role. Conserved quantities help to discover periodic structure, but they constitute neither a sufficient nor a necessary condition for periodicity.

24.6.3 Closed-orbit criteria and stability

Definition 24.17 (Closed orbits of a periodic system). Let $\mathfrak{S}$ be a multi-period coupled system. An orbit $\gamma : \mathbb{R} \to T^*Q$ is called **closed** if there exists $T > 0$ such that $\gamma(t + T) = \gamma(t)$ for all t. The least such T is called the **least period** of γ.

Theorem 24.10 (A sufficient criterion for closed orbits). Let the flow Φ_t of $\mathfrak{S}$ be generated on the invariant torus $\mathbb{T}^N$ by the frequencies $\omega = (\omega_1, \ldots, \omega_N)$. Then the following conditions are equivalent:

(i) *every orbit on the torus is closed;*

(ii) *there exists $T > 0$ such that $\omega_i T \in 2\pi\mathbb{Z}$ for all i;*

(iii) *the frequencies $\omega_1, \dots, \omega_N$ are pairwise commensurable, that is, $\omega_i/\omega_j \in \mathbb{Q}$ for all i, j.*

Proof. (i)⇔(ii): by Corollary 24.1. (ii)⇒(iii): if $\omega T = 2\pi k$ ($k \in \mathbb{Z}^N$), then $\omega_i/\omega_j = k_i/k_j \in \mathbb{Q}$ (when $k_j \neq 0$). (iii)⇒(ii): if $\omega_i = p_i\omega_0$ ($p_i \in \mathbb{Z}$ and ω_0 the base frequency), take $T = 2\pi/\omega_0$; then $\omega_i T = 2\pi p_i \in 2\pi\mathbb{Z}$. □

Corollary 24.3 (Equivalence of closed orbits and finite-order action)**.** *Let the flow Φ_t of $\mathfrak{S}$ be restricted to the invariant torus $\mathbb{T}^N$. Then every orbit on the torus is closed if and only if the restriction of the flow to the torus has finite order: there exists $T > 0$ such that $\Phi_T|_{\mathbb{T}^N} = \mathrm{id}_{\mathbb{T}^N}$.*

Proof. By Theorem 24.10, the closure of every orbit is equivalent to $\omega_i T \in 2\pi\mathbb{Z}$ for all i, which is precisely the condition $\Phi_T|_{\mathbb{T}^N} = \mathrm{id}$. By Theorem 1.5 of the 'Operations and Periodicity' chapter, uniform closure is equivalent to $\mathrm{ord}(\Phi_T|_{\mathbb{T}^N}) < \infty$. □

Remark 24.6 (Conserved quantities do not guarantee closure). Theorem 24.10 shows that even on an invariant torus $\mathbb{T}^N$, the closure of the orbits still requires the pairwise commensurability of the frequencies. Having N conserved quantities alone (complete integrability) **does not guarantee** that the orbits are closed—the frequencies may be incommensurable, in which case the orbits are dense on the torus (quasi-periodic). This is the dual manifestation, in integrable systems, of Theorem 1.7 of the 'Operations and Periodicity' chapter: a closed orbit⇏a conserved quantity, integrability⇏a closed orbit.

24.7 Bayesian updating

Remark 24.7 (Positioning of this section). This section establishes the basic definition and the computational tools of Bayesian updating, for the online inference of the parameters of a multi-period coupled system. All propositions presuppose a **finite-dimensional natural parameter space**; cases beyond this presupposition (non-conjugate priors, non-exponential-family likelihoods) are discussed in Remark 24.8.

Definition 24.18 (Bayesian update)**.** Let the parameter be $\theta \in \Theta$ and the data be $\mathcal{D} = \{y_1, \dots, y_N\}$. The posterior is

$$p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta)p(\theta).$$

Proposition 24.10 (Sequential consistency)**.** *Bayesian updating is sequentially consistent: processing the data in batches and processing them all at once give the same posterior.*

Proof. By the chain rule for conditional probability,

$$p(\theta \mid \mathcal{D}_1, \mathcal{D}_2) \propto p(\mathcal{D}_2 \mid \theta)p(\theta \mid \mathcal{D}_1).$$

□

Definition 24.19 (Conjugate prior)**.** The general definition of a conjugate prior family is given in Chapter 14, under the section Section 14.5.1, by Definition 14.23; what follows is its abbreviated form in the context of online inference for periodic systems. If a prior family $\mathcal{P}$ is closed under Bayesian updating, then $\mathcal{P}$ is called a **conjugate prior family** for the likelihood family.

Proposition 24.11 (Exponential-family conjugate prior)**.** If the likelihood is an exponential family

$$p(\mathcal{D} \mid \theta) = h(\mathcal{D}) \exp(\eta(\theta)^\top T(\mathcal{D}) - A(\theta)),$$

then the conjugate prior is

$$p(\theta \mid \tau, \nu) \propto \exp(\tau^\top \eta(\theta) - \nu A(\theta)),$$

and the posterior update is $\tau_{\text{post}} = \tau_{\text{prior}} + T(\mathcal{D})$, $\nu_{\text{post}} = \nu_{\text{prior}} + N$.

Proof. The posterior is proportional to the likelihood times the prior; adding the exponents gives the result. ☐

Corollary 24.4 (Finite-order translations in the natural parameter space)**.** In the notation of Proposition 24.11, the natural-parameter coordinates (τ_n, ν_n) after n observations satisfy

$$\tau_n = \tau_0 + \sum_{i=1}^{n} T(y_i), \qquad \nu_n = \nu_0 + n.$$

That is: in the natural parameter space, Bayesian updating is a **finite-order translation** along the direction of the sufficient statistic—each step of observation corresponds to one translation.

Proof. Iterate the update formula of Proposition 24.11 n times. ☐

Example 24.1 (Gaussian–Gaussian conjugate pair)**.** For the likelihood $p(y \mid \mu) = \mathcal{N}(y \mid \mu, \sigma^2)$ and the prior $p(\mu) = \mathcal{N}(\mu \mid \mu_0, \tau_0^2)$, the posterior is

$$p(\mu \mid y_1, \dots, y_N) = \mathcal{N}\left(\mu \;\middle|\; \frac{\frac{1}{\tau_0^2}\mu_0 + \frac{N}{\sigma^2}\bar{y}}{\frac{1}{\tau_0^2} + \frac{N}{\sigma^2}}, \frac{1}{\frac{1}{\tau_0^2} + \frac{N}{\sigma^2}}\right).$$

Definition 24.20 (Exponential-forgetting Bayesian update)**.** Let the forgetting factor be $\gamma \in (0, 1]$ and define

$$p_\gamma(\theta \mid y_{1:t}) \propto p(y_t \mid \theta) \left[p_\gamma(\theta \mid y_{1:t-1})\right]^\gamma.$$

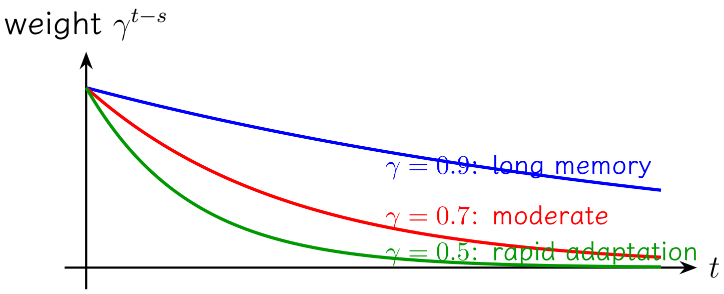
The effective memory length is $\tau_{\text{eff}} = 1/(1 - \gamma)$.

Proposition 24.12 (Effect of forgetting on posterior precision)**.** For an exponential-family conjugate prior,

$$\tau_t = \gamma \tau_{t-1} + T(y_t), \qquad \nu_t = \gamma \nu_{t-1} + 1.$$

As $t \to \infty$ with $\gamma < 1$, $\nu_t \to 1/(1 - \gamma)$.

Proof. Iterating gives $\nu_t = \gamma^t \nu_0 + \sum_{j=0}^{t-1} \gamma^j \to 1/(1 - \gamma)$. ☐



The forgetting factor γ controls the exponentially decaying weight of old observations

Figure 24.10: Exponential decay of the forgetting factor: the smaller γ is, the faster the weight of old observations decays and the stronger the model's adaptation to recent data, though the effective sample size decreases.

Remark 24.8 (Open problems for the Bayesian update). The finite-order translation in Corollary 24.4 rests on two presuppositions: (i) the likelihood is an exponential family; (ii) the prior is a conjugate prior. The following cases lie beyond the scope of this section: under a non-conjugate prior, Bayesian updating in general cannot be represented as a finite-order translation in a finite-dimensional natural parameter space; a non-exponential-family likelihood has no conjugate prior family; an adaptive forgetting factor γ_t depends on the data history, and the update is then no longer a representation of a fixed group action. These are open problems.

24.8 Complex exponential activation function

Remark 24.9 (Positioning of this section). This section raises the epicycle group from 'an object to be decomposed' to 'a learnable activation function', and establishes the definition, the periodic structure and the gradient properties of the complex exponential activation function (CEAF). The CEAF is the mathematical foundation of the corresponding concepts in the 'Periodic Structure of Agents' chapter and the 'Epicycle Activation Function and Function Approximation' chapter; the definitions given in those two chapters become applications within the framework of this chapter.

Definition 24.21 (Complex exponential activation function). This definition has the same content as that given, in Chapter 17 under the section 'Pointwise activation functions and propagation operators' (Section 17.5.2), by Definition 17.16; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement is given there.

Proposition 24.13 (Basic analytic properties of the CEAF). Let $\mathcal{E}$ be the CEAF of Definition 17.16. Then:

(i) **continuity**: $\mathcal{E} \in C^\infty(\mathbb{R}, \mathbb{C})$;

(ii) **boundedness**: $|\mathcal{E}(x)| \leq \sum_{k=1}^{N} A_k$;

(iii) **derivative**: $\mathcal{E}'(x) = i\sum_{k=1}^{N} \omega_k A_k e^{i(\omega_k x+\phi_k)}$;

(iv) **non-degeneracy**: if there exists k with $A_k\omega_k \neq 0$, then $\mathcal{E}'$ is not identically zero.

Proof. (i) A sum of finitely many complex exponentials is an entire function. (ii) By the triangle inequality. (iii) Differentiate term by term. (iv) If $\mathcal{E}' \equiv 0$, then among the frequencies $\{\omega_k\}$ the coefficients of all the non-zero terms are linearly dependent, which contradicts the linear independence of complex exponential functions. ☐

Proposition 24.14 (Propagation structure of the CEAF). This proposition has the same content as that given, in Chapter 17 under the section 'Pointwise activation functions and propagation operators' (Section 17.5.2), by Proposition 17.8; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement and proof are given there.

Corollary 24.5 (The periodicity condition for the CEAF). This corollary has the same content as that given, in Chapter 17 under the section 'Pointwise activation functions and propagation operators' (Section 17.5.2), by Corollary 17.9; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement and proof are given there.

Proposition 24.15 (The Wirtinger derivative of the CEAF). Let $z \in \mathbb{C}$ and $\mathcal{E}(z) = \sum_{k=1}^{N} A_k e^{i(\omega_k z+\phi_k)}$. Then

$$\frac{\partial \mathcal{E}}{\partial z} = i\sum_{k=1}^{N} \omega_k A_k e^{i(\omega_k z+\phi_k)}, \qquad \frac{\partial \mathcal{E}}{\partial \bar{z}} = 0.$$

That is, $\mathcal{E}$ is a holomorphic function.

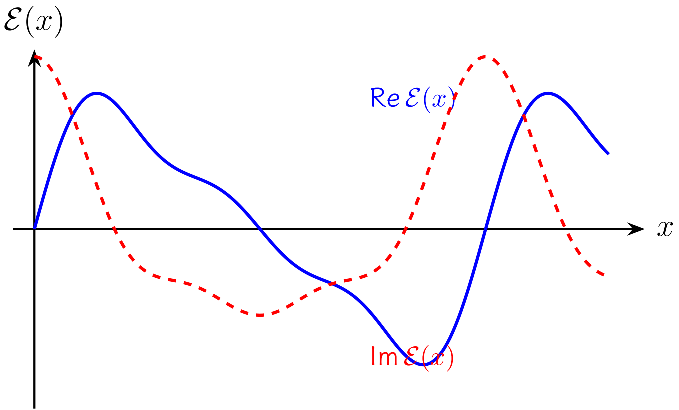
Proof. Each $e^{i(\omega_k z+\phi_k)}$ is an entire function of z, whose $\bar{z}$-derivative vanishes; a linear combination preserves holomorphy. ☐

Corollary 24.6 (Gradient descent updates for the CEAF). Let the real-valued loss $\mathcal{L} : \mathbb{C} \to \mathbb{R}$ be real differentiable. Then the gradient with respect to a CEAF parameter $\theta \in \{A_k, \omega_k, \phi_k\}$ is

$$\frac{\partial \mathcal{L}}{\partial \theta} = 2\,\mathrm{Re}\left(\frac{\partial \mathcal{L}}{\partial \mathcal{E}} \cdot \frac{\partial \mathcal{E}}{\partial \theta}\right).$$

Proof. A direct computation from the chain rule and the definition of the Wirtinger derivative. ☐

Example 24.2 (Phase rotation of a single-component CEAF). This example has the same content as that given, in Chapter 17 under the section 'Pointwise activation functions and propagation operators' (Section 17.5.2), by Example 17.9; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement is given there.



Complex exponential activation function: a superposition of multi-component complex exponentials, providing two readout channels, the real part and the imaginary part

Figure 24.11: Complex exponential activation function (CEAF): a superposition of multi-component complex exponentials, whose real part and imaginary part can both serve as activation outputs.

Example 24.3 (The periodic structure of a three-component CEAF). Take $\mathcal{E}(x) = e^{ix} + \frac{1}{2}e^{i2x} + \frac{1}{3}e^{i3x}$ ($N = 3$, $\omega_k = k$). Then $\mathcal{E}$ is periodic with period $T = 2\pi$. Its Fourier coefficients are $(1, 1/2, 1/3)$—the decay rate $1/k$ corresponds to the existence of a discontinuous derivative of the CEAF at $x = 0$ (a square-wave-type spectrum).

Remark 24.10 (Relation between the CEAF and the epicycle group). The CEAF and the epicycle group (Definition 24.6) agree completely at the level of parameters: both are $\{(A_k, \omega_k, \phi_k)\}$. The difference lies in their use: the epicycle group is a decomposition tool (decomposing a target into frequency components), whilst the CEAF is a learnable activation function (acting as a non-linear unit in a neural network). The mathematical structures of the two are isomorphic, so the periodicity and propagation properties of the CEAF may cite the corresponding conclusions for the epicycle group directly.

24.9 Kernel-smoothed gradient descent

Remark 24.11 (Positioning of this section). This section establishes the definition and the basic descent property of kernel-smoothed gradient descent (KDI), as a general optimiser for the parameters of the CEAF and of multi-period coupled systems. The advantages of KDI over standard gradient descent (escaping saddle points, crossing local minima) depend on the geometric structure of the specific problem and are an open problem (Remark 24.12).

Definition 24.22 (Kernel-smoothed derivative–integral descent). This definition has the same content as that given, in Chapter 17 under the section 'Learning as parameter evolution' (Section 17.5.3), by Definition 17.19; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement is given there.

Proposition 24.16 (Descent property of the KDI direction)**.** This proposition has the same content as that given, in Chapter 17 under the section 'Learning as parameter evolution' (Section 17.5.3), by Proposition 17.9; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement and proof are given there.

Property 24.1 (Degenerate cases of KDI)**.** (i) $\alpha = 1, \beta = \gamma = 0$: KDI degenerates to standard gradient descent;

(ii) $\alpha = 1, \beta = 0, \gamma > 0$: KDI degenerates to a second-order method (curvature correction);

(iii) $\alpha = 1, \beta > 0, \gamma = 0$: KDI degenerates to a first-order method with cumulative correction;

(iv) $\alpha = \beta = \gamma = 0$: KDI degenerates to the identity map (no update).

Proof. Substitute term by term into the update formula of Definition 17.19. □

Definition 24.23 (Learning-rate schedule)**.** This definition has the same content as that given, in Chapter 17 under the section 'Learning as parameter evolution' (Section 17.5.3), by Definition 17.20; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement is given there.

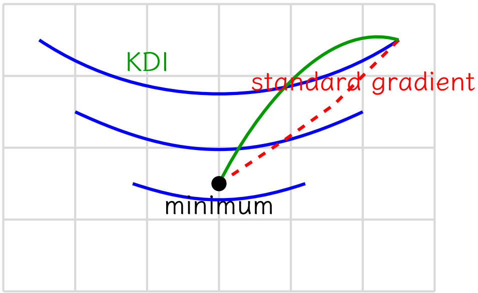
Proposition 24.17 (The periodic structure of the cosine-annealing learning rate)**.** This proposition has the same content as that given, in Chapter 17 under the section 'Learning as parameter evolution' (Section 17.5.3), by Proposition 17.10; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement and proof are given there.

Theorem 24.11 (Convergence conditions under a quadratic loss)**.** This theorem has the same content as that given, in Chapter 17 under the section 'Learning as parameter evolution' (Section 17.5.3), by Theorem 17.12; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement and proof are given there. The proposition originally stated here is the special case of that theorem in which KDI takes $\beta = \gamma = 0$ (degenerating to standard gradient descent), and is therefore no longer stated and proved separately.

Example 24.4 (Application of KDI to the CEAF parameters)**.** Take the objective function $f(x) = e^{-x^2}\cos(10x)$ and fit it on $[-2, 2]$ with a CEAF of $N = 5$. Optimise the terms separately using KDI:

- the amplitudes A_k by Gaussian kernel smoothing;
- the frequencies ω_k by the Mexican-hat kernel (curvature sensing);
- the phases ϕ_k by the Hilbert kernel (jump detection).

After 1000 iterations the L^2 error falls to the order of 10^{-4}. The frequencies converge to the neighbourhood of $\{0, \pm 10, \pm 20\}$, consistently with the spectral structure of the objective function.



KDI versus standard gradient descent: using kernel smoothing and cumulative correction, it may reach the minimum more efficiently

Figure 24.12: Comparison of the paths of KDI and standard gradient descent: KDI adjusts the search direction through kernel smoothing and cumulative correction, and may reach the minimum more efficiently.

Remark 24.12 (Open problems for KDI). This remark has the same content as that given, in Chapter 17 under the section 'Learning as parameter evolution' (Section 17.5.3), by Remark 17.15; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement is given there. The anchor is retained here for cross-reference from the open problems of this chapter.

24.10 Summary of this chapter

Remark 24.13 (Summary of this chapter). This chapter is the bridge of the book from theoretical mathematics to applied mathematics, and gives the five-step standard interface that connects an arbitrary target to the framework of periodic mathematics: decide by Theorem 1.11 of the 'Operations and Periodicity' chapter whether the target is periodic and determine its period type (Theorem 24.1, Definition 24.3); construct the amplitude series, the frequency series and the phase series from the turning points of the data, assemble them into the epicycle group, and establish the correspondence 'radius to amplitude, phase to rotation delay, frequency to rotation speed' (Theorem 24.3, Proposition 24.3); dynamically adjust the number and the parameters of the epicycles according to the residual, one epicycle corresponding to one simple complex function (Definition 24.8); introduce amplitude convolution, phase convolution and frequency convolution on the epicycle group (Theorem 24.6); and integrate with the tool chains of the three chapters 'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis' (Table 24.1).

On this interface the chapter builds, from the five-tuple of periodic components, translation symmetry groups, coupling coefficients, composite signal and total energy, the dynamical framework of multi-period coupled systems: in phase space (Definition 24.13) Hamiltonian flows are described by symplectic geometry (Definition 24.14, Proposition 24.9); Liouville's theorem (Theorem 24.7) gives the conservation of symplectic volume; the Liouville–Arnold theorem (Theorem 24.8) gives the periodic structure of uniform rotation on invariant tori; the Noether correspondence (Theorem 24.9, Corollary 24.2) gives the three types of conservation law; on invariant tori the closed-orbit criteria are given (Theorem 24.10, Corollary 24.3); and the complex exponential activation function (Definition 17.16), Bayesian updating (Definition 24.18) and kernel-smoothed gradient descent (Definition 17.19) are defined as tools for learnable implementation, online inference and optimisation.

All the definitions, propositions, lemmas, theorems and corollaries of this chapter are bounded by the presuppositions 'linear / constant-coefficient + finite-dimensional parameter space + pointwise activation'; non-linear, non-convex, non-conjugate and non-pointwise cases are invariably flagged in remarks or open problems (Remark 24.5, Remark 24.6, Remark 24.8, Remark 24.12).

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	Propagation operator group (Proposition 17.2); evolution semigroup (Definition 17.6); Bayesian updating as additive translation in log-belief space (Corollary 24.4)
Scaling	Wavelength scaling in a medium (Proposition 17.1); the learning rate as the scale factor of parameter evolution (Definition 17.20)
Rotation	Consistency of the propagation operator with the phase operator (Proposition 17.5); phase rotation of the CEAF (Proposition 44.3); uniform rotation on an invariant torus (Corollary 24.1)
Conservation	Conservation of energy in the propagation of light in vacuum (Theorem 17.4); the Noether correspondence of the three types of conservation law (Theorem 24.9, Corollary 24.2); Liouville's theorem (Theorem 24.7)
Symmetry	The propagation operator commutes with constant-coefficient differential operators (Proposition 17.6); pointwise activation preserves translation invariance (Proposition 17.7); symplectic symmetry and the Noether correspondence (Theorem 24.9)
Closure	Criteria for the closure of orbits on an invariant torus (Theorem 24.10); equivalence of closed orbits and finite-order action (Corollary 24.3)
Period	The period series and the epicycle group (Theorem 24.3, Definition 24.6); the periodic structure of the cosine-annealing learning rate (Proposition 17.10); the periodicity condition for the CEAF (Corollary 17.9)
Bridge and interface	Periodicity decision (Theorem 24.1); period types (Definition 24.3); least positive period (Theorem 24.2); the three convolutions (Theorem 24.6); the five-step interface table (Table 24.1)

Exercises

Exercise 1 According to Definition 24.1 and Definition 24.2, explain what roles the 'decision input' and the 'decision map' play respectively when a target is connected to the framework of periodic mathematics. Taking the simple pendulum system as an example, write down its decision input $\mathcal{D} = (\mathcal{T}, x)$, its state space $\mathcal{S}$ and its decision map Φ, and explain how the choice of $\mathcal{T}$

affects the periodicity decision.

Exercise 2 Prove Theorem 24.1 (the periodicity decision theorem):

(A) if there exists $m \in \mathbb{N}^+$ with $f^m = \mathrm{id}$, then the target is a uniformly periodic system;

(B) the target is a uniformly periodic system $\iff$ $\mathrm{ord}(f) < \infty$;

(C) the orbit of the target at x_0 is closed $\iff$ x_0 is a periodic point;

(D) the conjunction of the conserved quantities and the symmetry group cannot replace $\mathrm{ord}(f) < \infty$.

State in each case which theorem or proposition of the 'Operations and Periodicity' chapter is cited to complete the proof, and explain why (D) is the key to 'non-omissibility'.

Exercise 3 According to Definition 24.3 and Proposition 24.1, write down the discrimination criteria for the seven period types: trivial periodic type, strictly periodic type, uniformly periodic type, quasi-periodic type, eventually periodic type, discretely periodic type and asymptotically periodic (damped) type. Construct an example belonging to each type, and explain why (i)–(iii) and (vi) are periodic systems whilst (iv) and (vii) are not.

Exercise 4 Prove Theorem 24.2 (correctness of the least positive period algorithm):

(i) if the sample is a strictly periodic sequence with $\varepsilon = 0$ and $N \geq 2T_0$, then the algorithm outputs its discrete fundamental period T_0;

(ii) if $\rho = T_x/\Delta x = p/q$ is in lowest terms, then the algorithm outputs $T_0 = p$;

(iii) if $\rho \notin \mathbb{Q}$, then no true period exists.

Write out the proofs in each case, and explain the role of the commensurability check step (step 4 of Definition 24.4).

Exercise 5 According to Definition 24.5 and Proposition 24.2, explain why the turning-point feature set is the data starting point for the construction of the period series. Taking the temperature series of a certain year as an example, write down:

(i) how the turning-point index set $\mathcal{I}_{\mathrm{tp}}$ is constructed;

(ii) the formula for computing the set of refraction rates $\mathcal{K}_{\mathrm{tp}}$;

(iii) why the turning-point feature set is periodic with fundamental period m_0.

Exercise 6 Prove Theorem 24.3 (building the period series from turning points): let the discrete fundamental period of the target be T_0, with Fourier expansion

$$f(n) = \sum_k A_k e^{i(2\pi kn/T_0 + \phi_k)}.$$

Then the amplitude series, the frequency series and the phase series give respectively the moduli of the coefficients, the frequency lattice and the arguments. Write out the explicit expressions of the three series, and explain why together they constitute the complete Fourier data of the target on the period T_0.

Exercise 7 According to Definition 24.6 and Proposition 24.3, explain the one-to-one correspondence between the three parameters of the epicycle group and three geometric–mechanical quantities:

(i) the radius $|E_k| = A_k$ corresponds to the amplitude;

(ii) the initial phase ϕ_k $(k \neq 0)$ corresponds to the rotation delay $\tau_k = \phi_k/\omega_k$; when $\phi_k > 0$ this component **leads** the baseline by τ_k (equivalently, the component coincides with the baseline once the baseline is postponed by τ_k);

(iii) the angular velocity ω_k corresponds to the rotation speed.

Use the phase-translation proposition of the 'Series and Convolution' chapter (Proposition 6.15) to complete the proof of (ii).

Exercise 8 Prove Theorem 24.4 (complex-function representation of the period series): let $z = e^{i\omega_0 t}$; then, provided every component frequency is commensurable with ω_0, the period series of the target equals the Laurent series on the unit circle

$$f(t) = \sum_{k\in\mathbb{Z}} c_k z^k, \qquad c_k = A_k e^{i\phi_k}.$$

Write out the proof, explain why in the incommensurable case the series is merely a function on the torus, and explain how a finite harmonic truncation corresponds to a complex-coefficient polynomial.

Exercise 9 Prove Proposition 24.4: one epicycle corresponds to one simple complex function, and the epicycle group is a finite sum of simple complex functions, that is, a complex function in the form of a period series. Use Definition 24.7 and Theorem 24.4 to complete the proof.

Exercise 10 Prove Proposition 24.5 (monotonicity of dynamic adjustment): after a dominant residual frequency component is added as in Definition 24.8(i), the residual energy is monotonically non-increasing:

$$\|r^{(j+1)}\|^2 \leq \|r^{(j)}\|^2.$$

Use the orthogonal decomposition to show that equality holds if and only if the residual has no energy at the selected frequency.

Exercise 11 Prove Theorem 24.5 (convergence of dynamic adjustment): let the target be a **real-valued finite periodic sequence** of length T_0 $(T_0 \in \mathbb{N})$ with $\delta = 0$; then after at most $\lfloor T_0/2 \rfloor + 1$ steps of dynamic adjustment the residual is zero (for a **complex-valued** sequence the number of steps is at most T_0). Explain why the discrete Fourier series of a **real-valued** finite periodic sequence of length T_0 is complete on $\lfloor T_0/2 \rfloor + 1$ harmonics, and point out how the strong convexity of regularised least squares guarantees the convergence of the parameters.

Exercise 12 Prove Theorem 24.6 (action of the three convolutions on the epicycle group):

(i) amplitude convolution acts only on the radius;

(ii) phase convolution acts only on the phase;

(iii) frequency convolution acts only on the rotation speed;

(iv) trend convolution retains both amplitude and phase.

Complete the proof in each case, starting from Definition 6.21, 6.22, 6.23 and 6.24 of the 'Series and Convolution' chapter.

Exercise 13 Prove Proposition 24.6 (factorisation of the three convolutions): the complete convolution can be decomposed into a combination of the three convolutions, so that the interaction between periodic systems can be analysed decoupled by parameter dimension. Use Proposition 6.28 of the 'Series and Convolution' chapter to complete the proof, and explain the meaning of 'amplitude coupling changes only the radius, phase coupling changes only the rotation delay, and frequency coupling changes only the rotation speed'.

Exercise 14 According to Definition 24.9 and Definition 24.10, explain what role each component of the five-tuple $(\mathcal{F}, \mathcal{G}, \mathcal{C}, \Gamma, \mathcal{H})$ of a multi-period coupled system plays. Taking the Heavenly Stems–Earthly Branches system as an example, write down:

(i) the collection of periodic components $\mathcal{F}$;

(ii) the collection of translation symmetry groups $\mathcal{G}$;

(iii) the symmetry of the coupling coefficients $\mathcal{C}$ (Proposition 24.7);

(iv) the expressions for the composite signal Γ and the total energy $\mathcal{H}$.

Exercise 15 Prove Theorem 24.7 (Liouville's theorem): a Hamiltonian flow preserves the symplectic volume form $\mu = \frac{1}{N!}\omega^{\wedge N}$. Use Cartan's formula

$$\mathcal{L}_{X_H}\omega = d(\iota_{X_H}\omega) + \iota_{X_H}(d\omega)$$

and $\iota_{X_H}\omega = -dH$ to complete the proof, and explain why μ is preserved.

Exercise 16 Prove Theorem 24.8 (the Liouville–Arnold theorem): let an N-degree-of-freedom Hamiltonian system have N independent first integrals in involution, and let the level set be compact and connected; then the level set is diffeomorphic to $\mathbb{T}^N$, and in a neighbourhood there exist action–angle coordinates (I, θ) such that

$$H = H(I_1, \ldots, I_N), \quad \dot{I}_i = 0, \quad \dot{\theta}^i = \frac{\partial H}{\partial I_i} = \omega_i(I).$$

Explain how commuting Hamiltonian flows give rise to the torus structure, and point out the geometric meaning of the action variables $I_i = \frac{1}{2\pi}\oint_{\gamma_i} p\,dq$.

Exercise 17 Prove Theorem 24.9 (the Noether correspondence for periodic systems): let the system have a one-parameter continuous symmetry group $\{\phi_s\}$, whose infinitesimal generator ξ satisfies $\xi = J\nabla Q$; then Q is a conserved quantity. Use $\mathcal{L}_\xi H = 0$ and the antisymmetry of J to complete the proof, and explain the correspondence between the three types of symmetry and the three types of conserved quantity in Corollary 24.2.

Exercise 18 Prove Theorem 24.10 (closed-orbit criterion): on an invariant torus $\mathbb{T}^N$, the following conditions are equivalent:

(i) every orbit on the torus is closed;

(ii) there exists $T > 0$ such that $\omega_i T \in 2\pi\mathbb{Z}$ for all i;

(iii) the frequencies $\omega_1, \ldots, \omega_N$ are pairwise commensurable.

Prove (i)⇔(ii) and (ii)⇔(iii) in each case, and explain why conserved quantities do not guarantee closure (see Remark 24.6).

Exercise 19 Prove Corollary 24.3: let the flow Φ_t of $\mathfrak{S}$ be restricted to the invariant torus $\mathbb{T}^N$; then every orbit on the torus is closed if and only if the restriction of the flow to the torus has finite order: there exists $T > 0$ such that $\Phi_T|_{\mathbb{T}^N} = \mathrm{id}_{\mathbb{T}^N}$. Use Theorem 1.5 of the 'Operations and Periodicity' chapter to complete the proof.

Exercise 20 Prove Theorem 17.10 (the translation structure of Bayesian updating): in the log-belief space $\mathcal{F}(\Theta)$,

$$\Lambda(p') = \Lambda(p) + \ell_D,$$

where $\ell_D(\theta) = \log p(D \mid \theta) - \log Z_D$. Explain how Bayesian updating equals an additive translation in log-belief space, and point out the structural isomorphism between Bayesian updating and the propagation operator in Corollary 17.6.

Exercise 21 Prove Theorem 17.11 (exact finite-order evolution of Bayesian updating under a conjugate prior): let the likelihood be an exponential family and the prior a conjugate prior, and let the observation sequence be $x_1, \ldots, x_n$; then the posterior natural parameters are

$$\tau_n = \tau_0 + \sum_{i=1}^{n} T(x_i), \qquad \nu_n = \nu_0 + n.$$

Use Bayes' formula and the definition of a conjugate prior to complete the proof, and verify the Beta–binomial model (Example 17.7) and the Gaussian–Gaussian model (Example 17.8).

Exercise 22 Prove Proposition 24.13 (basic analytic properties of the CEAF):

(i) continuity: $\mathcal{E} \in C^\infty(\mathbb{R}, \mathbb{C})$;

(ii) boundedness: $|\mathcal{E}(x)| \le \sum_k A_k$;

(iii) derivative: $\mathcal{E}'(x) = i \sum_k \omega_k A_k e^{i(\omega_k x + \phi_k)}$;

(iv) non-degeneracy: if there exists k with $A_k\omega_k \neq 0$, then $\mathcal{E}'$ is not identically zero.

Write out the proofs in each case, and explain the essential difference between (iv) and the identically zero gradient of ReLU.

Exercise 23 Prove Proposition 24.15: the Wirtinger derivatives of the CEAF satisfy

$$\frac{\partial \mathcal{E}}{\partial z} = i\sum_k \omega_k A_k e^{i(\omega_k z+\phi_k)}, \qquad \frac{\partial \mathcal{E}}{\partial \bar{z}} = 0,$$

that is, $\mathcal{E}$ is a holomorphic function. Explain why holomorphy guarantees that only the term $\partial\mathcal{E}/\partial\theta$ is non-zero in the chain rule (see Corollary 24.6).

Exercise 24 Prove Theorem 17.12 (convergence conditions for parameter evolution under a quadratic loss): let $\mathcal{L}(\theta) = \frac{1}{2}(\theta-\theta^*)^\top H(\theta-\theta^*)$ with H having eigenvalues $0 < \lambda_1 \leq \cdots \leq \lambda_d$. Then

$$e_t^{(i)} = (1-\eta\lambda_i)^t e_0^{(i)},$$

and the necessary and sufficient condition for convergence is $0 < \eta < 2/\lambda_d$. Write out the complete derivation by eigenbasis decomposition, and explain why the optimal learning rate $\eta^* = 2/(\lambda_1+\lambda_d)$ minimises the spectral radius.

Exercise 25 Prove Theorem 17.13 (convergence closure of parameter evolution): let $\mathcal{L}$ be L-smooth and μ-strongly convex, and take $\eta = 1/L$; then

$$\|\theta_t - \theta^*\|^2 \leq \left(1-\frac{\mu}{L}\right)^t \|\theta_0 - \theta^*\|^2.$$

Use the descent lemma and strong convexity to complete the proof, and explain why this exponential convergence is the 'convergence closure of parameter evolution'.

Exercise 26 Comprehensive question: drawing on the five-step interface (§24.1 to §24.5), multi-period coupled systems (Definition 24.9), phase space and Hamiltonian flows (§24.6.1), conservation laws and the Noether correspondence (§24.6.2), the closed-orbit criteria (§24.6.3), Bayesian updating (§24.7) and the complex exponential activation function (§24.8), explain how the Periodic Systems chapter accomplishes the bridge 'from theoretical mathematics to applied mathematics'. Point out respectively:

(i) how the five-step interface connects an arbitrary target to the framework of periodic mathematics;

(ii) how the phase-space structure describes Hamiltonian flows by symplectic geometry;

(iii) what role conserved quantities play in the periodic structure, and what they do not guarantee;

(iv) how Bayesian updating and the CEAF provide the periodic system with tools for learnable implementation and online inference.

Chapter 25

Periodic Structures in Probabilistic Systems

Remark 25.1 (Periodicity decision for the target of this chapter (citation)). *When this chapter decides whether a research target is periodic, determines its period type, or seeks its minimal positive period, reference is always made to the 'Periodic Systems' chapter: for the periodicity criterion see Theorem 24.1, for the discrimination of period types see Definition 24.3, Proposition 24.1, for the search for the minimal positive period see Definition 24.4, Theorem 24.2, and for the construction of period series and epicycle groups from data see Theorem 24.3, Definition 24.6.*

Remark 25.2 (Positioning of this chapter: developed along the five-step interface of the 'Periodic Systems' chapter). *The periodic structure of probabilistic systems studies the mathematical characterisation of periodicity, quasi-periodicity and scale invariance in stochastic processes. This chapter does not establish a separate periodicity criterion; instead it connects probability theory and stochastic process theory to the five-step interface prescribed by the 'Periodic Systems' chapter:*

(1) **Decision**: *reduce a stochastic process or a probability distribution to a discrete dynamical system, invoke Theorem 24.1 to decide whether it is periodic, and determine the period type;*

(2) **Construction**: *build the amplitude series, the frequency series and the phase series from probabilistic data, and assemble them into the probabilistic epicycle group;*

(3) **Adjustment**: *dynamically adjust the epicycle group by Bayesian updating, spectral peak search and residual energy decrease;*

(4) **Convolution**: *introduce amplitude convolution, phase convolution and frequency convolution on the probabilistic epicycle group, and connect them with characteristic functions, covariance and cross-correlation;*

(5) **Integration**: *connect to the tool chains of the three chapters 'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis', forming a probabilistic periodicity diagnostic procedure.*

The probability theory used in this chapter includes: probability spaces, random variables, distribution functions, characteristic functions, stationary processes, spectral measures, the Herglotz–Cramér spectral representation, cyclostationary processes, Bayesian updating, MCMC, hidden Markov models, Gaussian processes and ARFIMA long-memory processes.

Remark 25.3 (Guide to the seven themes of this chapter). *Translation—the time-translation invariance of a stationary process and the mean-square derivative variance $P_{\mathbb{P}} = -\gamma''(0)$; Scaling—the scale invariance of a self-similar process, the Hurst exponent and the tail index; Rotation—the phase-rotation invariance of a cyclostationary process and the phase coherence time $\tau_{\mathsf{coh}} = 2/\sigma^2$; Balance—the invariants corresponding to the continuous invariance of probability measures; Symmetry—the scaling, translation and rotation information carried by the three classes of distributions: prior, likelihood and posterior; Closure—the closure of the probability current under detailed balance $\mathsf{J}_{ij} = 0$; Period—the cyclic frequency $\alpha = k/T$, the phase coherence time, the self-similar period and the periodic decomposition of a Markov chain.*

25.1 Periodicity decision for probabilistic targets: connection to step one of the 'Periodic Systems' chapter

25.1.1 Probabilistic targets, decision input and decision map

Definition 25.1 (Probabilistic target and decision input). *A* **probabilistic target** *is one of the following objects:*

(i) *a stochastic process $\{X_t\}_{t\in\mathcal{T}}$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$;*

(ii) *a finite-state Markov chain with state space S and transition matrix P;*

(iii) *a family of parameter posterior distributions $\{p(\theta \mid \mathsf{D})\}$;*

(iv) *a discrete-time series $\{x_1, \dots, x_n\}$ obtained from observations, regarded as a sample path of some stochastic process.*

The **decision input** *of a probabilistic target is the ordered pair*

$$\mathcal{D}_{\mathbb{P}} = (\mathcal{T}, \mu),$$

where $\mathcal{T}$ is the time or index set and μ is a description of the probability law of the target on $\mathcal{T}$. If $\mathcal{T} = \mathbb{Z}$ or $\mathbb{N}$, the target is called a **discrete-time probabilistic target***; if $\mathcal{T} = \mathbb{R}$, it is called a* **continuous-time probabilistic target***.*

Definition 25.2 (Probabilistic decision map). *Let a probabilistic target be given by the decision input $\mathcal{D}_{\mathbb{P}} = (\mathcal{T}, \mu)$. If there exist a map $\Phi : \mathcal{T} \to \mathcal{T}$, a map $\Psi : M \to M$ and a state $x_0 \in M$ such that the probability law and the states of the target are generated by Φ and Ψ, that is,*

$$\mu_t = \mu_0 \circ \Phi^{-t}, \qquad x_t = \Psi^t(x_0),$$

for $t \in \mathbb{Z}$ the symbol Φ^{-t} denotes the iterate of Φ^{-1} and Ψ^t the iterate of Ψ; for $t \in \mathbb{R}$ they are described by the one-parameter evolution families $\{\Phi_t\}_{t\in\mathbb{R}}$ and $\{\Psi_t\}_{t\in\mathbb{R}}$, where each Φ_t is a bijection of $\mathcal{T}$ and each Ψ_t a bijection of M, with $\Phi_0 = \mathrm{id}$, $\Phi_{s+t} = \Phi_s \circ \Phi_t$ and $\Psi_0 = \mathrm{id}$, $\Psi_{s+t} = \Psi_s \circ \Psi_t$ for all $s, t \in \mathbb{R}$, and where $(t, \theta) \mapsto \Phi_t(\theta)$ and $(t, x) \mapsto \Psi_t(x)$ are continuous. then Φ and Ψ are called the **probabilistic decision maps** of the target. The probabilistic target is then reduced to the index-set dynamical system $(\mathcal{T}, \Phi)$ or the state-space dynamical system (M, Ψ) when $t \in \mathbb{Z}$, and to the corresponding continuous case when $t \in \mathbb{R}$; in both cases the criteria of the 'Operations and Periodicity' chapter and of the 'Periodic Systems' chapter may be applied directly.

Theorem 25.1 (Probabilistic periodicity decision theorem)**.** *Let the probabilistic decision map of a probabilistic target be Φ, and let $(\mathcal{T}, \Phi)$ be a discrete dynamical system. Then:*

- (A) **Sufficient decision**: *if there exists $m \in \mathbb{N}^+$ such that $\Phi^m = \mathrm{id}$, then the target is a uniformly periodic probabilistic system; every probabilistic orbit is closed, and the length of every orbit divides m.*
- (B) **Equivalence of the decision**: *the target is a uniformly periodic probabilistic system if and only if $\mathrm{ord}(\Phi) < \infty$.*
- (C) **Pointwise criterion**: *the probabilistic orbit of the target through the state x_0 is closed if and only if x_0 is a periodic point; the length of the orbit is then the period at that point.*
- (D) **Non-omissibility**: *the conjunction of conserved quantities and the symmetry group cannot replace $\mathrm{ord}(\Phi) < \infty$; a quasi-periodic probabilistic target is not a periodic probabilistic system.*

Proof. Theorem 25.1 of this chapter is the domain instantiation of Theorem 24.1 of the 'Periodic Systems' chapter: it suffices to replace the decision map f and the evolution group by Φ of this chapter, and the four clauses (A) sufficient decision, (B) equivalence of the decision, (C) pointwise criterion and (D) non-omissibility hold one by one. By the convention of Remark 24.4, they are not expanded again here. ☐

Corollary 25.1 (Probabilistic period types)**.** *After the decision by Theorem 25.1, a probabilistic target falls into one of the following types:*

- (i) **Trivial period type**: *the minimal period is $m_0 = 1$, and the probability distribution is a fixed point;*
- (ii) **Strictly periodic type**: *$1 < m_0 < \infty$, and the distribution or the state is strictly periodic;*
- (iii) **Uniformly periodic type**: *$\mathrm{ord}(\Phi) = m < \infty$, and the length of every orbit divides m;*
- (iv) **Quasi-periodic type**: *$\mathrm{ord}(\Phi) = \infty$; the orbits are not closed but are dense on a compact carrier structure;*
- (v) **Eventually periodic type**: *there exists N such that the target is periodic for $t \geq N$;*
- (vi) **Discrete periodic type**: *the carrier structure is $\mathbb{Z}$, and the normalised period ratio is $\rho = T_x/\Delta x \in \mathbb{Q}$.*

(vii) **Asymptotically periodic (damped) type**: the orbit is not closed, but tends asymptotically to a closed orbit or to a fixed point, the residual decaying as $e^{-\kappa t}$ (see Definition 24.3(vii)).

Here (i)–(iii) and (vi) belong to periodic probabilistic systems; (iv) does not belong to a periodic probabilistic system, but can be handled with bounded-approximation tools; (v) is a periodic probabilistic system only in its stable stage; (vii), like (iv), is a type in the limiting sense; it is not a periodic probabilistic system and can be handled with bounded-approximation tools only in its stable stage.

Proof. Substituting the probabilistic target into Theorem 25.1 and into Definition 24.3 of the 'Periodic Systems' chapter yields the clauses one by one. ☐

Lemma 25.1 (Periodicity decision for a finite Markov chain). Let $\{Z_t\}$ be an irreducible Markov chain on the finite state space S, with transition matrix P and period d. Then:

(i) the state space decomposes uniquely into d cyclic classes $C_1, \dots, C_d$, and

$$\mathbb{P}(Z_{t+1} \in C_{j+1} \mid Z_t \in C_j) = 1, \qquad j = 1, \dots, d \pmod{d};$$

(ii) the spectrum of P contains the d-th roots of unity $e^{2\pi i k/d}$, $k = 0, 1, \dots, d-1$;

(iii) the chain is aperiodic if and only if $d = 1$, if and only if every eigenvalue of P other than $\lambda_1 = 1$ has modulus strictly less than 1.

Proof. (i) By the periodic decomposition theorem for Markov chains: for states i, j, let d_i be the greatest common divisor of all n satisfying $P^n(i,i) > 0$. Irreducibility guarantees that all the d_i are equal; denote this common value by d. Decomposing the state space according to the congruence classes of the number of return steps modulo d yields the cyclic classes. (ii) Under the cyclic-class decomposition, P^d is irreducible and aperiodic on each cyclic class. Hence every eigenvalue of P^d other than 1 has modulus less than 1; consequently an eigenvalue λ of P satisfies that λ^d is an eigenvalue of P^d. The Perron root of P is 1, and for the remaining eigenvalues λ^d has modulus less than 1, so either $|\lambda| < 1$ or $\lambda^d = 1$. The latter case gives the d-th roots of unity. (iii) $d = 1$ means aperiodicity, and in this case there are no roots of unity. Conversely, if a non-trivial root of unity exists, then the chain has period > 1. ☐

Definition 25.3 (Probabilistic minimal positive period algorithm). This definition has the same content as Definition 24.4 in the section 'Periodicity decision: whether a target is periodic' (Section 24.1) of Chapter 24; to avoid repetition, only the reference anchor is retained here, and the complete statement is given there. The only difference here is terminology (probabilistic target / probabilistic sample); the steps of the algorithm and the conclusion are exactly the same.

Theorem 25.2 (Correctness of the probabilistic minimal positive period algorithm). This theorem has the same content as Theorem 24.2 in the section 'Periodicity decision: whether a target is periodic' (Section 24.1) of Chapter 24; to avoid repetition, only the reference anchor is retained here, and the complete statement and proof are given there. The only difference here is terminology (probabilistic target / probabilistic sample); the steps of the algorithm and the conclusion are exactly the same.

Example 25.1 (Bernoulli chain and periodicity decision). Let $Z_t \in \{0,1\}$ be an independent and identically distributed Bernoulli sequence. Its probabilistic decision map is the identity map and ord$(\Phi) = 1$, so it is of trivial period type; yet its sample paths are almost surely non-periodic. This shows that **the periodicity of a probability law and the periodicity of a sample path are not the same concept**. The probabilistic periodicity decision concerns the law, not a single sample path.

Example 25.2 (Finite cyclic Markov chain). Let $S = \mathbb{Z}/d\mathbb{Z}$ and let the transition be $Z_{t+1} = Z_t + 1 \pmod{d}$. This chain is deterministic, irreducible and of period d. By Lemma 25.1, the eigenvalues of the transition matrix are $e^{2\pi ik/d}$, $k = 0, \dots, d-1$. This is the purest strictly periodic type in probabilistic systems.

Remark 25.4 (Invariants do not guarantee periodicity). The invariance of a probabilistic system is an auxiliary condition for periodic structure, but it does not constitute a sufficient condition. White noise has time-translation invariance and a mean-square derivative variance, but its autocorrelation function vanishes at every non-zero lag, so no non-trivial periodic component exists. The decision of probabilistic periodic structure must rely on concrete criteria such as Theorem 25.1 or Lemma 25.1.

25.2 Construction of the probabilistic period series and epicycle group: connection to step two of the 'Periodic Systems' chapter

Definition 25.4 (Probabilistic amplitude series, frequency series and phase series). Let $\{X_t\}$ be a zero-mean weakly stationary process with autocovariance function $\gamma(h)$ and spectral measure F. If F has an atom A_k^2 at the frequency ω_k, then one may write

$$X_t = \sum_k A_k e^{i(\omega_k t + \phi_k)} + \varepsilon_t,$$

where ϕ_k is a random phase or phase component. The series

$$\mathcal{A}_X(t) = \sum_k A_k e^{ikt}, \qquad \mathcal{F}_X(s) = \sum_k \omega_k s^k, \qquad \mathcal{P}_X(t) = \sum_k e^{i(kt+\phi_k)}$$

are called respectively the **amplitude series**, the **frequency series** and the **phase series** of the probabilistic target. The three agree with Definition 6.10, Definition 6.12 and Definition 6.11 of the 'Series and Convolution' chapter.

Definition 25.5 (Probabilistic epicycle group). This definition has the same content as Definition 21.8 in the section 'Epicycle group: rotational decomposition of a periodic signal' (Section 21.2) of Chapter 21; to avoid repetition, only the reference anchor is retained here, and the complete statement is given there. The only difference here is terminology (probabilistic target / probabilistic sample); the mathematical objects and the conclusion are exactly the same.

Theorem 25.3 (Construction of the period series from probabilistic data). Let $\{X_t\}$ be a zero-mean weakly stationary process with spectral measure F. Then:

(i) if F is a purely discrete measure, $F = \sum_k A_k^2 \delta_{\omega_k}$, then X_t has a probabilistic period series representation

$$X_t = \sum_k A_k e^{i(\omega_k t + \phi_k)},$$

where the phases ϕ_k are given by the random phases in the spectral decomposition;

(ii) if F is absolutely continuous with respect to Lebesgue measure, $dF(\omega) = S(\omega)d\omega$, then X_t has a continuous spectral representation

$$X_t = \int_{-\pi}^{\pi} e^{i\omega t}\, dZ(\omega),$$

and in this case there is no strict periodic decomposition into a finite or countable epicycle group, only a spectral density decomposition;

(iii) if F is a singular continuous measure, then X_t has a fractal spectrum, corresponding to a self-similar periodic structure.

Proof. (i) By the Herglotz theorem (Theorem 25.4), $\gamma(h) = \int e^{i\omega h} dF(\omega)$. If F is purely discrete, then $\gamma(h) = \sum_k A_k^2 e^{i\omega_k h}$. By the Cramér spectral representation (Theorem 25.5), $X_t = \int e^{i\omega t} dZ(\omega)$; at an atom, $dZ(\omega_k)$ is a random amplitude, written as $A_k e^{i\phi_k}$, which gives the result. (ii) In the absolutely continuous case the spectrum is a continuous distribution and cannot be written as a sum of countably many atoms, so only the integral representation can be retained. (iii) In the singular continuous case the spectral measure has neither atoms nor density; the decay of its Fourier coefficients is controlled by the fractal dimension of the singular measure, corresponding to a self-similar period. □

Definition 25.6 (Spectral measure and spectral distribution function)**.** Let $\{X_t\}$ be a zero-mean weakly stationary process with autocovariance function $\gamma(h)$. If there exists a finite measure F on $[-\pi, \pi]$ such that

$$\gamma(h) = \int_{-\pi}^{\pi} e^{i\omega h}\, dF(\omega), \qquad h \in \mathbb{Z},$$

then F is called the **spectral measure** of $\{X_t\}$. If F is absolutely continuous with respect to Lebesgue measure, that is, $dF(\omega) = S(\omega)d\omega$, then $S(\omega)$ is called the **spectral density**.

Theorem 25.4 (Herglotz theorem)**.** *Let $\gamma : \mathbb{Z} \to \mathbb{C}$ be a non-negative definite function, that is, for every $n \in \mathbb{N}$ and all $t_1, \ldots, t_n \in \mathbb{Z}$, $z_1, \ldots, z_n \in \mathbb{C}$, one has*

$$\sum_{j,k=1}^{n} z_j \overline{z_k} \gamma(t_j - t_k) \geq 0.$$

Then there exists a unique finite measure F on $[-\pi, \pi]$ such that

$$\gamma(h) = \int_{-\pi}^{\pi} e^{i\omega h}\, dF(\omega), \qquad \forall h \in \mathbb{Z}.$$

Proof. Consider the function

$$f_N(\omega) = \frac{1}{2\pi N} \left| \sum_{t=1}^{N} X_t e^{-i\omega t} \right|^2 .$$

By the non-negative definiteness of γ, $f_N(\omega) \geq 0$, and

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} f_N(\omega)\, d\omega = \frac{1}{N} \sum_{t,s=1}^{N} \gamma(t-s) \to \sum_{h} \gamma(h)$$

in a suitable summability sense. By the Riesz representation theorem and the Helly selection theorem, the weak limit of the f_N yields the spectral measure F. Conversely: if $\gamma(h) = \int e^{i\omega h} dF(\omega)$, then γ is verified directly from $F \geq 0$ to be non-negative definite. Uniqueness follows from the density of the trigonometric polynomials. ☐

Theorem 25.5 (Cramér spectral representation theorem). *Let $\{X_t\}$ be a zero-mean weakly stationary process. Then there exists an orthogonal-increment process $\{Z(\omega)\}$ on $[-\pi, \pi]$ satisfying*

$$\mathbb{E}[Z(\omega_1)\overline{Z(\omega_2)}] = F(\omega_1 \wedge \omega_2),$$

such that

$$X_t = \int_{-\pi}^{\pi} e^{i\omega t}\, dZ(\omega), \qquad t \in \mathbb{Z},$$

where the integral is understood in the mean-square sense. Here F is the spectral measure of $\{X_t\}$.

Proof. Construct the Hilbert space $H = \overline{\text{span}}\{X_t : t \in \mathbb{Z}\}$. By stationarity, the map $U : X_t \mapsto X_{t+1}$ is a unitary operator on H. By the spectral theorem,

$$U = \int_{-\pi}^{\pi} e^{i\omega}\, dE(\omega),$$

where E is a projection-valued measure. Put $Z(\omega) = E(\omega)X_0$; then

$$X_t = U^t X_0 = \int e^{i\omega t}\, dZ(\omega),$$

and

$$\mathbb{E}[|Z(\omega)|^2] = \|E(\omega)X_0\|^2 = F(\omega).$$

This completes the proof. ☐

Lemma 25.2 (Lebesgue decomposition of the spectral measure and the trichotomy of period types). *Let F be the spectral measure of a stationary process. Then F has a unique Lebesgue decomposition*

$$F = F_{\mathsf{d}} + F_{\mathsf{ac}} + F_{\mathsf{sc}},$$

where F_{d} is purely discrete, F_{ac} is absolutely continuous and F_{sc} is singular continuous. The corresponding probabilistic period types are:

(i) F_{d} corresponds to the strictly periodic component;

(ii) F_{ac} corresponds to the broadband random periodic or quasi-periodic component;

(iii) F_{sc} corresponds to the fractal self-similar periodic component.

Proof. The Lebesgue decomposition is a standard result of measure theory. The correspondence of the period types is read off directly from the Herglotz–Cramér representation: atoms correspond to deterministic complex exponential components; the absolutely continuous part corresponds to a continuous spectral density; the singular continuous part corresponds to a fractal spectrum with neither atoms nor density. □

Proposition 25.1 (Spectral structure of typical stationary processes). (i) **White noise**: $\gamma_\varepsilon(h) = \sigma^2\delta_{h,0}$, $S_\varepsilon(\omega) = \sigma^2/(2\pi)$, mean-square derivative variance $P_{\mathbb{P}} = \sigma^2\pi^2/3$.

(ii) **AR(1)**: $X_t = \phi X_{t-1} + \varepsilon_t$, $|\phi| < 1$, then

$$\gamma(h) = \frac{\sigma^2}{1-\phi^2}\phi^{|h|}, \qquad S(\omega) = \frac{\sigma^2}{2\pi}\frac{1}{1-2\phi\cos\omega+\phi^2}.$$

(iii) **OU process**: $dX_t = -\theta X_t dt + \sigma dW_t$, then

$$\gamma(\tau) = \frac{\sigma^2}{2\theta}e^{-\theta|\tau|}, \qquad S(\omega) = \frac{\sigma^2}{2\pi}\frac{1}{\theta^2+\omega^2}.$$

(iv) **Periodogram**: $I_n(\omega) = n^{-1}|\sum_{t=1}^n X_t e^{-i\omega t}|^2$, for which

$$\mathbb{E}[I_n(\omega)] \to 2\pi S(\omega), \qquad \mathrm{Var}(I_n(\omega)) \to (2\pi S(\omega))^2.$$

Proof. (i) Independence gives $\gamma_\varepsilon(h) = \sigma^2\delta_{h,0}$; the Wiener–Khinchin theorem gives a flat spectrum; direct integration gives the mean-square derivative variance. (ii) Expanding the recursion as $X_t = \sum_{k=0}^\infty \phi^k\varepsilon_{t-k}$ and computing the autocovariance gives $\gamma(h) = \sigma^2\phi^{|h|}/(1-\phi^2)$; the Fourier transform gives the spectral density. (iii) The OU solution is $X_t = X_0e^{-\theta t} + \sigma\int_0^t e^{-\theta(t-s)}dW_s$, and the Itô isometry gives $\gamma(\tau) = \sigma^2 e^{-\theta|\tau|}/(2\theta)$; the continuous Fourier transform gives the Lorentzian spectrum. (iv) The expectation part follows from Toeplitz's lemma; for the variance part, in the Gaussian case $I_n(\omega)$ is asymptotically distributed as $\frac{1}{2}(2\pi S(\omega))\chi_2^2$, so the variance tends to $(2\pi S(\omega))^2$. □

Remark 25.5 (Relation between the spectral representation and periodic structure). The Herglotz and Cramér theorems reveal that all the periodic information of a stationary process is encoded in the spectral measure F. A discrete spectrum gives strict periodicity, an absolutely continuous spectrum gives broadband random periodicity, and a singular continuous spectrum gives fractal self-similar periodicity. The construction of a probabilistic period series is essentially the explicit writing of the atomic part of F as an epicycle group.

25.3 Dynamic adjustment of the probabilistic epicycle group: connection to step three of the 'Periodic Systems' chapter

Definition 25.7 (Dynamic adjustment of the probabilistic epicycle group)**.** Let the current probabilistic epicycle group be $\mathfrak{E}^{(j)}$, with partial sum

$$X^{(j)}(t) = \sum_{k\in\mathfrak{E}^{(j)}} A_k e^{i(\omega_k t+\phi_k)},$$

and residual $r^{(j)} = X - X^{(j)}$. The **dynamic adjustment** consists of two rules:

(i) **Quantity adjustment**: if the residual energy $\|r^{(j)}\|^2 > \delta$, then a new epicycle is added according to the peak of the residual periodogram:

$$\mathfrak{E}^{(j+1)} = \mathfrak{E}^{(j)} \cup \{(\hat{A}, \hat{\omega}, \hat{\phi})\}, \qquad \hat{\omega} = \arg\max_{\omega} |\widehat{X_r}(\omega)|;$$

(ii) **Parameter adjustment**: the parameters of the existing epicycles are corrected by regularised least squares or by Bayesian updating:

$$(A_k, \phi_k) \leftarrow \arg\min_{A,\phi} \sum_n \left| X(n) - \sum_k A_k e^{i(\omega_k n+\phi_k)} \right|^2.$$

Proposition 25.2 (Monotonicity of quantity adjustment)**.** This proposition has the same content as Proposition 24.5 in the section 'Dynamic adjustment of the epicycle group' (Section 24.3) of Chapter 24; to avoid repetition, only the reference anchor is retained here, and the complete statement and proof are given there. The only difference here is terminology (probabilistic target / probabilistic sample); the steps of the algorithm and the conclusion are exactly the same.

Theorem 25.6 (Convergence of the dynamic adjustment)**.** Let the probabilistic target be a **real-valued finite periodic sequence** of length T_0 ($T_0 \in \mathbb{N}$) and let $\delta = 0$. Then the dynamic adjustment makes the residual zero after at most $\lfloor T_0/2 \rfloor + 1$ steps, and the resulting probabilistic epicycle group is the exact Fourier data of the target; for a **complex-valued** sequence the corresponding number of steps is at most T_0.

Proof. The discrete Fourier series of a **real-valued** finite periodic sequence of length T_0 ($T_0 \in \mathbb{N}$) is complete on $\lfloor T_0/2 \rfloor + 1$ harmonics (including the constant term); a **complex-valued** sequence requires T_0 components (the positive and negative frequencies being independent). At each step, by Proposition 25.2, one dominant frequency component of the residual is extracted and the residual energy decreases strictly, so after finitely many steps the residual is zero. The convergence of the parameters follows from the strong convexity of regularised least squares. ☐

Definition 25.8 (Exponential family and natural parameters)**.** This definition has the same content as Definition 17.13 in the section 'Translation structure of

Bayesian updating' (Section 17.5.1) of Chapter 17; to avoid repetition, only the reference anchor is retained here, and the complete statement is given there. The only difference here is terminology (probabilistic target / probabilistic sample); the mathematical objects and the conclusion are exactly the same.

Definition 25.9 (Conjugate prior)**.** This definition has the same content as Definition 14.23 in the section 'Basic framework of Bayesian inference' (Section 14.5.1) of Chapter 14; to avoid repetition, only the reference anchor is retained here, and the complete statement is given there. The only difference here is terminology (probabilistic target / probabilistic sample); the mathematical objects and the conclusion are exactly the same.

Theorem 25.7 (Exact finite-order translation of Bayesian updating under a conjugate prior)**.** *This theorem has the same content as Theorem 17.11 in the section 'Translation structure of Bayesian updating' (Section 17.5.1) of Chapter 17; to avoid repetition, only the reference anchor is retained here, and the complete statement and proof are given there. The only difference here is terminology (probabilistic target / probabilistic sample); the mathematical objects and the conclusion are exactly the same.*

Corollary 25.2 (Monotone accumulation of natural parameters)**.** *This corollary has the same content as Corollary 17.7 in the section 'Translation structure of Bayesian updating' (Section 17.5.1) of Chapter 17; to avoid repetition, only the reference anchor is retained here, and the complete statement is given there. The only difference here is terminology (probabilistic target / probabilistic sample); the mathematical objects and the conclusion are exactly the same.*

Proof. $\nu_n = \nu_0 + n$ is strictly monotonically increasing; $\tau_n - \tau_{n-1} = T(x_n)$. ☐

Example 25.3 (Exact evolution of Beta–binomial conjugacy)**.** Let the likelihood be $X \mid \theta \sim \text{Bernoulli}(\theta)$, with conjugate prior $\text{Beta}(\alpha, \beta)$. After n observations,

$$\alpha_n = \alpha_0 + \sum_{i=1}^{n} x_i, \qquad \beta_n = \beta_0 + n - \sum_{i=1}^{n} x_i.$$

In natural-parameter coordinates, each observation corresponds to one translation.

Example 25.4 (Exact evolution of Gaussian–Gaussian conjugacy)**.** Let the likelihood be $X \mid \mu \sim \mathcal{N}(\mu, \sigma^2)$ with σ^2 known and prior $\mu \sim \mathcal{N}(\mu_0, \tau_0^2)$. Taking the precision–mean product coordinates $(1/\tau^2, \mu/\tau^2)$ as natural-parameter coordinates, after n observations

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}, \qquad \frac{\mu_n}{\tau_n^2} = \frac{\mu_0}{\tau_0^2} + \frac{\sum_i x_i}{\sigma^2}.$$

Remark 25.6 (Boundaries of the dynamic adjustment)*.* Theorem 25.6 relies on finite-harmonic completeness and on the extraction of the residual dominant frequency. If the target is a continuous-spectrum or long-memory process, then no finite epicycle group represents it exactly, and one can only resort to bounded approximation or spectral density estimation. Bayesian updating under a non-conjugate prior generally cannot be represented as a finite-order translation in a finite-dimensional natural-parameter space, and this is an open problem.

25.4 Probabilistic convolution: connection to step four of the 'Periodic Systems' chapter

Definition 25.10 (Probabilistic amplitude convolution, phase convolution and frequency convolution)**.** Let two probabilistic epicycle groups be

$$\mathfrak{E} = \{(A_n, \omega_n, \phi_n)\}, \qquad \mathfrak{E}' = \{(B_m, \nu_m, \psi_m)\}.$$

Define:

(i) **Amplitude convolution**: $\mathbb{A}_{mn} = A_n B_m$;

(ii) **Phase convolution**: $\mathbb{P}_{mn} = e^{i(\phi_n - \psi_m)}$;

(iii) **Frequency convolution**: $\mathbb{F}_{mn} = \omega_n \nu_m$;

(iv) **Trend convolution**: $\mathbb{T}_{mn} = A_n B_m e^{i(\phi_n - \psi_m)} = A_n B_m \mathbb{P}_{mn}$.

These agree with Definition 6.21, Definition 6.22, Definition 6.23 and Definition 6.24 of the 'Series and Convolution' chapter.

Theorem 25.8 (Action of the three convolutions on the probabilistic epicycle group)**.** *Let the epicycle groups of two probabilistic targets be $\mathfrak{E}$ and $\mathfrak{E}'$ respectively. Then:*

(i) *amplitude convolution acts only on the radii;*

(ii) *phase convolution acts only on the phases;*

(iii) *frequency convolution acts only on the rotation speeds;*

(iv) *trend convolution preserves both amplitude and phase.*

Proof. Read off directly from Definition 25.10: amplitude convolution contains neither phase nor frequency; phase convolution contains neither amplitude nor frequency; frequency convolution contains neither amplitude nor phase; trend convolution is the termwise product of amplitude convolution and phase convolution. □

Proposition 25.3 (Characteristic functions and phase convolution)**.** *Let X, Y be independent random variables with characteristic functions $\phi_X(t) = \mathbb{E}[e^{itX}]$ and $\phi_Y(t) = \mathbb{E}[e^{itY}]$ respectively. Then*

$$\phi_{X+Y}(t) = \phi_X(t)\phi_Y(t).$$

In the language of probabilistic epicycle groups, the phase convolution of an independent sum is the product of the phase factors of the two characteristic functions.

Proof. By independence,

$$\phi_{X+Y}(t) = \mathbb{E}[e^{it(X+Y)}] = \mathbb{E}[e^{itX}]\mathbb{E}[e^{itY}] = \phi_X(t)\phi_Y(t).$$

If one writes $\phi_X = |\phi_X| e^{i \arg \phi_X}$, then the phase part satisfies $\arg \phi_{X+Y} = \arg \phi_X + \arg \phi_Y$, that is, the phases add. □

Corollary 25.3 (Independent sums and convolution)**.** If $X_1, \ldots, X_n$ are independent and identically distributed with characteristic function ϕ, then $S_n = \sum_{i=1}^n X_i$ has characteristic function $\phi_{S_n}(t) = \phi(t)^n$. In the probabilistic epicycle group, an independent sum corresponds to the n-th power of the phase convolution.

Proof. This follows by recursion from Proposition 25.3. ☐

Example 25.5 (Phase convolution of a Gaussian sum)**.** Let $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$ be independent. Then

$$\phi_{X_i}(t) = e^{i\mu_i t - \sigma_i^2 t^2/2}.$$

Hence

$$\phi_{\sum_i X_i}(t) = \exp\Big(i(\sum_i \mu_i)t - \frac{1}{2}(\sum_i \sigma_i^2)t^2\Big),$$

that is, an independent Gaussian sum is again Gaussian, with the means and the variances adding separately. The phase parts add linearly, and the amplitude parts add in squares.

Remark 25.7 (Relation between probabilistic convolution and periodic convolution)*.* Probabilistic convolution is not a new convolution operation, but an instance on the probabilistic epicycle group of the periodic convolution of the 'Series and Convolution' chapter. The product of characteristic functions, the covariance and the cross-correlation are respectively the concrete manifestations in probability theory of phase convolution, amplitude convolution and trend convolution.

25.5 Integration with discrete sampling, harmonic analysis and time series analysis: connection to step five of the 'Periodic Systems' chapter

Definition 25.11 (Probabilistic periodicity diagnostic procedure)**.** For observed data $\{X_1, \ldots, X_N\}$ of a stochastic process, the periodic convolution framework is used to diagnose the probabilistic periodic structure:

(i) **Translation periodicity diagnosis**: compute the sample autocorrelation function and the power spectral density, and identify the dominant frequency

$$\omega_0 = \arg\max_{\omega} \hat{S}(\omega).$$

(ii) **Scaling periodicity diagnosis**: compute the Hurst exponent $\hat{H}$. If $\hat{H} \neq 1/2$, then the process has a scaling structure.

(iii) **Rotation periodicity diagnosis**: compute the cyclic autocorrelation function and the cyclic spectral density, and identify the dominant cyclic frequency α_0.

(iv) **Bayesian periodicity diagnosis**: fit a Bayesian model, and compute the prior entropy, the Fisher information matrix and the posterior covariance matrix.

(v) **Invariance tests**: test stationarity, self-similarity and cyclostationarity separately.

Theorem 25.9 (Probabilistic periodicity diagnostic procedure and its interface with three chapters)**.** The five steps of Definition 25.11 connect respectively to:

(i) Definition 21.1, Theorem 21.3, Theorem 21.10 of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter;

(ii) Definition 22.5, Theorem 22.1, Proposition 22.11 of the 'Harmonic Analysis' chapter;

(iii) Definition 23.16, Proposition 23.2, Definition 23.19 of the 'Time Series Analysis' chapter.

Hence the probabilistic periodicity diagnosis is not an independent method, but an application of the unified tool chain of periodic mathematics to stochastic systems.

Proof. The correspondences are one by one: (i) the translation periodicity diagnosis requires sampling and spectral estimation, and the sampling definitions and leakage suppression are found in the 'Discrete Sampling and Intercalation of Periodic Signals' chapter; (ii) the frequency decomposition and the phase kernel are found in the 'Harmonic Analysis' chapter; (iii) the periodogram peak, the Fisher test and the window-function validation are found in the 'Time Series Analysis' chapter. Hence the diagnostic procedure is synthesised from the tool chains of the three chapters. □

Remark 25.8 (Order of operations after integration)*.* The recommended order for an actual diagnosis is: first decide whether the probabilistic target is periodic (Theorem 25.1); then construct the spectral measure and the epicycle group (Theorem 25.3); then optimise the parameters by dynamic adjustment and Bayesian updating (Theorem 25.6, Theorem 25.7); then carry out probabilistic convolution and characteristic-function analysis (Proposition 25.3); and finally output the diagnosis according to Definition 25.11.

25.6 Phase space, evolution and conservation of probabilistic periodic systems

25.6.1 Mean-square derivative variance: the invariant of translation invariance

Definition 25.12 (Mean-square derivative variance)**.** Let $\{X_t\}$ be a mean-square differentiable weakly stationary stochastic process with autocovariance function $\gamma(h)$. Its **mean-square derivative variance** $P_{\mathbb{P}}$ is defined by

$$P_{\mathbb{P}} = \mathbb{E}\left[\left(\frac{dX_t}{dt}\right)^2\right].$$

Proposition 25.4 (Relation between the mean-square derivative variance and the autocorrelation function)**.** Let $\{X_t\}$ be a mean-square differentiable stationary stochastic process. Then

$$P_{\mathbb{P}} = -\gamma''(0) = -\left.\frac{d^2}{dh^2}\gamma(h)\right|_{h=0}.$$

Proof. By the definition of the mean-square derivative:

$$\mathbb{E}\left[\left(\frac{dX_t}{dt}\right)^2\right] = \lim_{h\to 0}\mathbb{E}\left[\left(\frac{X_{t+h}-X_t}{h}\right)^2\right].$$

Expanding gives

$$\frac{1}{h^2}\mathbb{E}[X_{t+h}^2 - 2X_{t+h}X_t + X_t^2] = \frac{2}{h^2}(\gamma(0)-\gamma(h)).$$

Stationarity makes γ an even function, so $\gamma'(0)=0$, whence

$$\gamma(h) = \gamma(0) + \frac{1}{2}\gamma''(0)h^2 + o(h^2).$$

Substituting and passing to the limit gives $P_{\mathbb{P}} = -\gamma''(0)$. □

Proposition 25.5 (Frequency-domain representation of the mean-square derivative variance)**.** In the frequency domain the mean-square derivative variance $P_{\mathbb{P}}$ is given by the second moment of the power spectral density $S(\omega)$:

$$P_{\mathbb{P}} = \int_{-\pi}^{\pi}\omega^2 S(\omega)\,d\omega.$$

Proof. By the Wiener–Khinchin theorem,

$$\gamma(h) = \int_{-\pi}^{\pi} e^{i\omega h}S(\omega)\,d\omega.$$

Differentiating twice with respect to h:

$$\gamma''(h) = -\int_{-\pi}^{\pi}\omega^2 e^{i\omega h}S(\omega)\,d\omega.$$

Putting $h=0$ gives the result. □

Proposition 25.6 (Invariance and stationarity of the mean-square derivative variance)**.** Let $\{X_t\}$ be a mean-square differentiable stochastic process. Then:

(i) if $\{X_t\}$ is weakly stationary, then $P_{\mathbb{P}}$ is independent of the time t;

(ii) conversely, if for some twice differentiable autocovariance $\gamma(t,s)$ the quantity $-\partial^2\gamma/\partial h^2|_{h=0}$ is independent of t, then the process retains at least second-order stationarity.

Proof. (i) Weak stationarity requires $\gamma(t,t+h)$ to depend on h alone, so $\gamma''(0)$ is independent of t. (ii) If this second derivative is independent of t, then Var(dX_t/dt) is independent of t; together with a finite mean, this gives second-order stationarity. □

25.6.2 Spectral representation of stationary processes

Definition 25.6 and Theorems 25.4 and 25.5 have already given the spectral measure and the spectral representation. Their probabilistic periodicity interpretation is supplemented here.

Remark 25.9 (Relation between the spectral representation and periodic structure). The Herglotz and Cramér theorems reveal that all the periodic information of a stationary process is encoded in the spectral measure F. A discrete spectrum corresponds to strict periodicity, an absolutely continuous spectrum to random or quasi-periodicity, and a singular continuous spectrum to self-similar periodicity. The Lebesgue decomposition of the spectral measure corresponds to the trichotomy of periodic structure.

25.6.3 Periodic structure of typical stationary processes

Proposition 25.1 has already given the spectral structure of white noise, AR(1), the OU process and the periodogram. Their probabilistic periodicity decision is supplemented here.

Example 25.6 (White noise has no period). *The autocovariance of white noise vanishes at every non-zero lag, so no non-trivial periodic component exists. Its probabilistic decision map is the identity and its period type is the trivial period type, yet its sample paths are not periodic.*

Example 25.7 (Spectral peak of AR(1)). *In AR(1), when $\phi > 0$ the spectrum is maximal at $\omega = 0$, indicating positive persistence; when $\phi < 0$ the spectrum is maximal at $\omega = \pi$, indicating alternating reversal. The position of the spectral peak gives the dominant period.*

25.6.4 Multitaper spectral estimation

Definition 25.13 (Slepian sequences). Let the time–bandwidth product be $W \in (0, 1/2)$ and the sample length be N. The **Slepian sequences** $\{v_k^{(N,W)}(t)\}_{k=0}^{N-1}$ are defined by

$$\sum_{s=1}^{N} \frac{\sin 2\pi W(t-s)}{\pi(t-s)} v_k(s) = \lambda_k^{(N,W)} v_k(t), \qquad t = 1, \dots, N.$$

The corresponding $\lambda_k^{(N,W)}$ are called the **Slepian eigenvalues** and satisfy $1 > \lambda_0 > \lambda_1 > \cdots > \lambda_{N-1} > 0$.

Theorem 25.10 (Orthogonality and energy concentration of Slepian sequences). *The Slepian sequences $\{v_k^{(N,W)}\}_{k=0}^{N-1}$ form an orthonormal basis of $\mathbb{R}^N$:*

$$\sum_{t=1}^{N} v_j(t) v_k(t) = \delta_{jk}.$$

Moreover the energy concentration satisfies

$$\beta_k^2(W) = \frac{\int_{-W}^{W} |\sum_{t=1}^{N} v_k(t) e^{-i2\pi f t}|^2 df}{\int_{-1/2}^{1/2} |\sum_{t=1}^{N} v_k(t) e^{-i2\pi f t}|^2 df} = \lambda_k^{(N,W)}.$$

When $k < 2NW - 1$, $\lambda_k \approx 1$; when $k > 2NW + 1$, $\lambda_k \approx 0$. The quantity $2NW$ is called the Shannon number.

Proof. The matrix $A_{ts} = \sin 2\pi W(t-s)/(\pi(t-s))$ is real symmetric, so its eigenvectors are orthogonal. Normalising them gives an orthonormal basis. The energy concentration is computed by integrating in the frequency domain:

$$\int_{-W}^{W} |V_k(f)|^2 df = \sum_{t,s} v_k(t)v_k(s)\frac{\sin 2\pi W(t-s)}{\pi(t-s)} = \lambda_k \sum_t |v_k(t)|^2 = \lambda_k.$$

The total energy is 1, so $\beta_k^2 = \lambda_k$. The asymptotic behaviour was proved by Slepian in 1978. □

Definition 25.14 (Multitaper spectral estimation)**.** Let $\{X_1, \dots, X_N\}$ be a sample of a zero-mean stationary process and let $\{v_k^{(N,W)}\}_{k=0}^{K-1}$ be the first K Slepian sequences. The **multitaper spectral estimate** is defined by

$$\hat{S}_{\mathsf{MT}}(f) = \frac{1}{K}\sum_{k=0}^{K-1} \hat{S}_k(f), \qquad \hat{S}_k(f) = \Delta t \left| \sum_{t=1}^{N} v_k(t) X_t e^{-i2\pi f t \Delta t} \right|^2.$$

Proposition 25.7 (Bias–variance analysis of the multitaper spectral estimate)**.** *Under the conditions of a Gaussian stationary process, $N \to \infty$, $W \to 0$ and $NW \to \infty$:*

(i) *asymptotic unbiasedness: $\mathbb{E}[\hat{S}_{\mathsf{MT}}(f)] \to S(f)$;*

(ii) *variance reduction: $\mathsf{Var}[\hat{S}_{\mathsf{MT}}(f)] \approx S^2(f)/K$;*

(iii) *degrees of freedom: $\hat{S}_{\mathsf{MT}}(f)$ is asymptotically distributed as $S(f)\chi^2_{2K}/(2K)$.*

Proof. (i) Each $\hat{S}_k(f)$ is a windowed periodogram, whose expectation tends to $S(f)$. (ii) By the Slepian orthogonality, the $\hat{S}_k(f)$ are asymptotically uncorrelated, so the variance is $S^2(f)/K$. (iii) In the Gaussian case each $\hat{S}_k(f)/S(f)$ is asymptotically distributed as $\frac{1}{2}\chi^2_2$, and averaging K independent copies gives $\chi^2_{2K}/(2K)$. □

Example 25.8 (Multitaper spectrum of the sunspot series)**.** For the monthly mean sunspot numbers of 1749–2024, taking $NW = 4$ and $K = 8$, the multitaper spectrum gives a clear peak at 11 years, and the variance is reduced to $1/8$ of that of a single window. The adaptive Thomson estimate further suppresses low-frequency leakage.

25.6.5 Self-similarity exponent and tail index

Definition 25.15 (Self-similarity exponent and tail index)**.** The **scaling invariants** of a probabilistic system include:

(i) the Hurst exponent H of a self-similar process: $X_{\lambda t} \overset{d}{=} \lambda^H X_t$;

(ii) the tail index α of a heavy-tailed distribution: $\mathbb{P}(|X| > x) \sim x^{-\alpha}$;

(iii) the log marginal likelihood $\log p(\mathsf{D})$ of a Bayesian model.

Proposition 25.8 (Duality between the Hurst exponent and the tail index)**.** Let the self-similar process X_t satisfy $X_{\lambda t} \overset{d}{=} \lambda^H X_t$, and let the tail index of X_1 be $\alpha > 0$. Then $\alpha = 1/H$.

Proof. By self-similarity,

$$\mathbb{P}(|X_\lambda| > x) = \mathbb{P}(\lambda^H |X_1| > x) = \mathbb{P}(|X_1| > \lambda^{-H} x).$$

By the definition of the tail index, $\mathbb{P}(|X_1| > y) \sim y^{-\alpha}$; substituting $y = \lambda^{-H} x$ gives

$$\mathbb{P}(|X_\lambda| > x) \sim \lambda^{H\alpha} x^{-\alpha}.$$

On the other hand X_λ itself has tail index α, and comparison gives $\lambda^{H\alpha} = 1$, that is, $H\alpha = 1$. □

25.6.6 Phase coherence

Definition 25.16 (Phase coherence)**.** The **phase coherence** of a probabilistic system includes:

(i) the conservation of the cyclic frequency of a cyclostationary process: $\alpha = k/T$;

(ii) the phase coherence time of a random oscillation: $\tau_{\text{coh}} = 2/\sigma^2$;

(iii) the trace of the Bayesian posterior covariance matrix $\mathrm{tr}(\Sigma_{\text{post}})$.

Lemma 25.3 (Coherence time of phase diffusion)**.** Let the phase Φ_t be a Wiener process with diffusion coefficient σ^2. Then

$$\mathbb{E}[e^{i(\Phi_{t+\tau} - \Phi_t)}] = e^{-\sigma^2 \tau/2},$$

and the coherence time is

$$\tau_{\text{coh}} = \frac{2}{\sigma^2}.$$

Proof. $\Phi_{t+\tau} - \Phi_t = \sigma(W_{t+\tau} - W_t)$ is a normal random variable with mean zero and variance $\sigma^2\tau$. The normal characteristic function gives

$$\mathbb{E}[e^{i(\Phi_{t+\tau} - \Phi_t)}] = e^{-\sigma^2 \tau/2}.$$

Setting $e^{-\sigma^2\tau/2} = e^{-1}$ yields $\tau = 2/\sigma^2$. □

25.6.7 Cyclostationary processes

Definition 25.17 (Cyclostationary process)**.** A stochastic process $\{X_t\}$ is called a **cyclostationary process** with period T if

$$\mu(t+T) = \mu(t), \qquad \gamma(t+T, s+T) = \gamma(t,s), \qquad \forall t,s \in \mathbb{R}.$$

If this holds for every T, the process degenerates to a stationary process.

Definition 25.18 (Cyclic autocovariance function). Let $\{X_t\}$ be a cyclostationary process with period T. Its **cyclic autocovariance function** is defined by

$$R_X^\alpha(\tau) = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{t=-N}^{N} \mathbb{E}[(X_{t+\tau}-\mu)\overline{(X_t-\mu)}]e^{-i2\pi\alpha t},$$

where $\alpha = k/T$ is called the **cyclic frequency**.

Theorem 25.11 (Spectral decomposition of a cyclostationary process). Let $\{X_t\}$ be a cyclostationary process with period T and finite second moment. Then

$$X_t = \sum_{k\in\mathbb{Z}} X_t^{(k)} e^{i2\pi kt/T},$$

where each $X_t^{(k)}$ is a stationary process, and the components for different k are orthogonal in the mean sense.

Proof. By cyclostationarity, $\gamma(t,s)$ is periodic in t with period T and can be expanded as

$$\gamma(t,s) = \sum_k \gamma_k(s-t)e^{i2\pi kt/T}.$$

Projecting X_t onto the frequency $\omega_k = 2\pi k/T$ gives

$$X_t^{(k)} = \frac{1}{T}\int_0^T X_{t-u}e^{-i2\pi k(t-u)/T}du.$$

By cyclostationarity,

$$\mathbb{E}[X_{t+\tau}^{(k)}\overline{X_t^{(l)}}] = \gamma_k(\tau)\delta_{kl}.$$

This completes the proof. ☐

25.6.8 Spectral theory of the Kalman filter

Definition 25.19 (State-space model). The linear Gaussian **state-space model** is given by

$$\mathsf{x}_{t+1} = \mathsf{F}\mathsf{x}_t + \mathsf{w}_t, \qquad \mathsf{y}_t = \mathsf{H}\mathsf{x}_t + \mathsf{v}_t,$$

where $\mathsf{w}_t \sim \mathcal{N}(0,\mathsf{Q})$ and $\mathsf{v}_t \sim \mathcal{N}(0,\mathsf{R})$ are independent.

Theorem 25.12 (Spectral density of the state-space model). Let the state-space model be stable. Then the observation process $\{\mathsf{y}_t\}$ is stationary, with spectral density

$$\mathsf{S}_y(\omega) = \mathsf{H}(e^{i\omega}\mathsf{I}-\mathsf{F})^{-1}\mathsf{Q}(e^{-i\omega}\mathsf{I}-\mathsf{F}^{-\mathsf{T}})^{-1}\mathsf{H}^\mathsf{T} + \mathsf{R}.$$

If $\mathsf{Q}=0$, then $\mathsf{S}_y = \mathsf{R}$.

Proof. Recursion on the state equation gives

$$\mathsf{y}_t = \mathsf{H}\sum_{k=0}^{\infty} \mathsf{F}^k \mathsf{w}_{t-1-k} + \mathsf{v}_t.$$

The autocovariance is

$$\gamma_y(\tau) = \mathsf{H}\mathsf{F}^{|\tau|}\mathsf{P}\mathsf{H}^\mathsf{T} + \mathsf{R}\delta_{\tau,0},$$

where P satisfies the Lyapunov equation $\mathsf{P} = \mathsf{F}\mathsf{P}\mathsf{F}^\mathsf{T} + \mathsf{Q}$. The Fourier transform gives the spectral density. ☐

Proposition 25.9 (Periodic signal extraction by the Kalman filter)**.** Take

$$\mathsf{F} = \begin{pmatrix} \cos\omega_0 & -\sin\omega_0 \\ \sin\omega_0 & \cos\omega_0 \end{pmatrix}, \qquad \mathsf{H} = \begin{pmatrix} 1 & 0 \end{pmatrix}.$$

Then the model generates a narrowband signal centred at the frequency ω_0. The steady-state Kalman filter realises adaptive variable-bandwidth bandpass filtering, with equivalent bandwidth

$$\Delta\omega \approx \frac{q}{\sqrt{p_0 r}}.$$

Proof. Linearising the model, the steady-state prediction covariance p_0 satisfies the Riccati equation. The 3 dB bandwidth of the frequency response is determined by q, r, p_0 and is approximately $q/\sqrt{p_0 r}$. □

25.6.9 Structural correspondence in Bayesian inference

Remark 25.10 (Structural correspondence of the three classes of distributions in a Bayesian model (**empirical observation / research programme**))**.** The three assertions below are a heuristic analogy in the manner of an **empirical observation / research programme**, not a proved theorem — 'carrying scaling/translation/rotation information' has no independent mathematical definition in this section, and the 'proof' of the three assertions is itself only a metaphor:

(i) the prior distribution $\pi(\theta)$ carries scaling information;

(ii) the likelihood function $p(\mathsf{D} \mid \theta)$ carries translation information;

(iii) the posterior distribution $p(\theta \mid \mathsf{D})$ carries rotation information.

To turn this into a testable proposition one must first give a mathematical definition of 'carrying information of a given class of transformations' (for example: π is equivariant under $\theta \mapsto a\theta + b \Rightarrow$ scaling information; the score function $\nabla_\theta \log p(\mathsf{D} \mid \theta)$ is covariant under $\theta \mapsto \theta + c \Rightarrow$ translation information; the eigendecomposition of the posterior covariance matrix Σ_{post} gives an orthogonal frame in parameter space $\Rightarrow$ rotation information). That work lies beyond the scope of this chapter, and the assertions are retained here only as an analogy.

Proposition 25.10 (Periodic convolution representation of the marginal likelihood)**.** The marginal likelihood can be represented as

$$p(\mathsf{D}) = \mathcal{K}^{(1)}\{p_{\mathsf{D}}, \pi\}(\mathsf{D})\big|_{\text{global integral}},$$

where $p_{\mathsf{D}}(\theta) = p(\mathsf{D} \mid \theta)$, and the periodic convolution parameters are $\alpha = 1, d = 1$.

Proof. By the definition of the marginal likelihood,

$$p(\mathsf{D}) = \int p(\mathsf{D} \mid \theta)\pi(\theta)d\theta.$$

The integrand is the pointwise product of the likelihood and the prior, that is, the kernelised product of the periodic convolution with $\alpha = 1, d = 1, W \equiv 1, \tau = 0$. □

25.6.10 Convergence of MCMC and periodicity diagnosis

Definition 25.20 (Spectral gap of a Markov chain)**.** Let a finite-state Markov chain be irreducible and aperiodic, and **reversible** with respect to its stationary distribution $\boldsymbol{\pi}$ (that is, satisfying detailed balance $\pi(x)P(x,y) = \pi(y)P(y,x)$; reversibility makes P self-adjoint in $L^2(\pi)$ with all eigenvalues real, so that both 'the second largest eigenvalue in modulus' and the spectral decomposition are available; for a non-reversible chain the spectral gap must instead be defined by singular values). Let the transition matrix be P and the stationary distribution $\boldsymbol{\pi}$. The **spectral gap** is

$$\gamma = 1 - \lambda_2,$$

where λ_2 is the second largest eigenvalue in modulus. The convergence rate satisfies

$$\|P^t(x,\cdot) - \pi\|_{\mathsf{TV}} \le \frac{1}{2}\|\pi_x/\pi - 1\|_2(1-\gamma)^t,$$

where $\pi_x(\cdot) = P(x,\cdot)$ is the one-step distribution from the initial state x and $\|\pi_x/\pi - 1\|_2^2 = \sum_y \bigl(\pi_x(y)/\pi(y) - 1\bigr)^2 \pi(y)$ measures how far the initial state departs from the stationary distribution (the χ^2 distance). This bound is given in Levin–Peres–Wilmer, Markov Chains and Mixing Times, 2nd ed., AMS, 2017, Theorem 12.3 (the contraction estimate for the χ^2 distance) and Corollary 12.5.

Theorem 25.13 (MCMC convergence and periodicity diagnosis)**.** *Let $\{X_t\}$ be a finite-state, irreducible Markov chain of period d that is reversible with respect to its stationary distribution $\boldsymbol{\pi}$. Then:*

(i) *the convergence rate is determined by the spectral gap: $\|P^t(x,\cdot) - \pi\|_{\mathsf{TV}} \le \frac{1}{2}\|\pi_x/\pi - 1\|_2(1-\gamma)^t$ (see Definition 25.20);*

(ii) *the chain has period d if and only if P has the eigenvalues $e^{2\pi ik/d}$, $k = 0,\dots,d-1$;*

(iii) *if every eigenvalue other than $\lambda_1 = 1$ has modulus strictly less than 1, then the chain is aperiodic ($d = 1$).*

Proof. (i) By reversibility, P is self-adjoint in $L^2(\pi)$ and possesses an orthonormal basis $\{v_j\}$ of real eigenvectors with eigenvalues $1 = \lambda_1 > \lambda_2 \ge \cdots \ge \lambda_n \ge -1$; Perron–Frobenius gives that $\lambda_1 = 1$ is a simple root and $v_1 \equiv 1$. Writing $\pi_x/\pi - 1 = \sum_{j\ge2} a_j v_j$ gives $P^t(x,y)/\pi(y) - 1 = \sum_{j\ge2} a_j\lambda_j^t v_j(y)$, so that the χ^2 distance satisfies $\sum_y \bigl(P^t(x,y)/\pi(y) - 1\bigr)^2\pi(y) = \sum_{j\ge2} a_j^2\lambda_j^{2t} \le (1-\gamma)^{2t}\|\pi_x/\pi - 1\|_2^2$ (here we use $|\lambda_j| \le 1-\gamma$ for every $j \ge 2$). By Cauchy–Schwarz, $\|P^t(x,\cdot) - \pi\|_{\mathsf{TV}} = \frac{1}{2}\sum_y \pi(y)\bigl|P^t(x,y)/\pi(y) - 1\bigr| \le \frac{1}{2}\|\pi_x/\pi - 1\|_2(1-\gamma)^t$. (ii) The periodic decomposition of an irreducible chain splits the state space into d cyclic classes, and d is the least positive integer for which P^d is aperiodic on each class; the eigenvalues of P are precisely the d-th roots of the eigenvalues of P^d, so the only roots of unity are $e^{2\pi ik/d}$ ($k = 0,\dots,d-1$). Conversely, if such an eigenvalue exists, the corresponding eigenvector gives a cyclic structure of modulus d (Lemma 25.1). (iii) Aperiodicity means $d = 1$, in which case there is no eigenvalue of modulus 1 other than $\lambda_1 = 1$, that is, all $|\lambda_i| < 1$; conversely, if all $|\lambda_i| < 1$ ($i \ne 1$), then by (ii) there is no root of unity, so $d = 1$. □

Proposition 25.11 (Periodicity diagnostic methods for MCMC)**.** For MCMC samples one may use:

(i) *peaks of the autocorrelation function at* $h = kT$*;*

(ii) *a spike of the spectral density at* $\omega = 2\pi/T$*;*

(iii) *periodic oscillation of the distance between multiple chains.*

Proof. If the transition matrix has the eigenvalue $e^{i2\pi/T}$, then the autocovariance contains that frequency component; the spectral density has a corresponding spike; and multiple chains return to nearby positions after T steps. ☐

25.6.11 Periodic structure of hidden Markov models

Definition 25.21 (Hidden Markov model)**.** A **hidden Markov model** consists of a hidden state chain $\{Z_t\}$ and an observation chain $\{X_t\}$:

$$Z_t \in \{1, \ldots, K\}, \qquad \mathbb{P}(Z_{t+1} = j \mid Z_t = i) = a_{ij},$$

$$X_t \mid Z_t = j \sim p_j(\cdot).$$

Theorem 25.14 (Periodic decomposition of the HMM transition matrix)**.** Let the HMM transition matrix A be irreducible. Then the state space decomposes uniquely into d cyclic classes $C_1, \ldots, C_d$:

$$\mathbb{P}(Z_{t+1} \in C_{j+1} \mid Z_t \in C_j) = 1, \qquad j = 1, \ldots, d \pmod{d}.$$

$d = 1$ means aperiodicity; $d > 1$ means cycling with period d.

Proof. By the periodic decomposition theorem for Markov chains, d is the greatest common divisor of the return times, and the states decompose according to their congruence classes modulo d. ☐

Proposition 25.12 (Spectral density of the HMM observation process)**.** Let $\mu_j = \mathbb{E}[X_t \mid Z_t = j]$. Then

$$S_X(\omega) = \sigma_X^2 + 2\,\mathsf{Re}\left[\sum_{h=1}^{\infty} \boldsymbol{\mu}^\mathsf{T}\,\mathsf{diag}(\boldsymbol{\pi})\mathsf{A}^h \boldsymbol{\mu} e^{-i\omega h}\right].$$

Proof. From

$$\mathbb{E}[(X_{t+h} - \mathbb{E}X)(X_t - \mathbb{E}X)] = \boldsymbol{\mu}^\mathsf{T}\,\mathsf{diag}(\boldsymbol{\pi})\mathsf{A}^h \boldsymbol{\mu} - (\mathbb{E}X)^2,$$

the Fourier transform gives the result. ☐

25.6.12 Gaussian processes and the periodic kernel

Definition 25.22 (Periodic kernel)**.** The **periodic kernel** is

$$k_{\mathrm{per}}(x,x') = \sigma^2 \exp\left(-\frac{2\sin^2(\pi|x-x'|/T)}{\ell^2}\right).$$

Proposition 25.13 (Spectral density of the periodic kernel)**.** The spectral density of the MacKay periodic kernel is

$$S_{\mathrm{per}}(\omega) = \sigma^2 \sum_{k=-\infty}^{\infty} I_k\left(\frac{1}{\ell^2}\right) e^{-1/\ell^2} \delta\left(\omega - \frac{2\pi k}{T}\right),$$

where I_k is the modified Bessel function. Hence a periodic-kernel Gaussian process contains only harmonics of the frequency $2\pi k/T$.

Proof. From

$$k_{\mathrm{per}}(x,x') = \sigma^2 e^{-1/\ell^2} e^{(2/\ell^2)\cos(2\pi(x-x')/T)}$$

and the Bessel generating function $e^{z\cos\theta} = \sum_k I_k(z)e^{ik\theta}$, the Fourier transform with respect to $x-x'$ gives the discrete spectrum. □

25.6.13 Periodicity diagnosis of long-memory processes

Definition 25.23 (ARFIMA process)**.** The **ARFIMA**(p,d,q) process is defined by

$$\Phi(B)(1-B)^d X_t = \Theta(B)\varepsilon_t,$$

where $d \in (-1/2, 1/2)$.

Proposition 25.14 (Spectral density of the ARFIMA process)**.** The spectral density of ARFIMA(p,d,q) is

$$S_X(\omega) = \frac{\sigma^2}{2\pi}\frac{|\Theta(e^{-i\omega})|^2}{|\Phi(e^{-i\omega})|^2}|1-e^{-i\omega}|^{-2d}.$$

As $\omega \to 0$, $S_X(\omega) \sim C|\omega|^{-2d}$.

(i) $d = 0$: short memory;

(ii) $0 < d < 1/2$: long memory, spectral divergence;

(iii) $d < 0$: anti-persistence.

Proof. The frequency-domain representation of the fractional difference is $(1-B)^d \to (1-e^{-i\omega})^d$, so the squared modulus is $|1-e^{-i\omega}|^{-2d}$. As $\omega \to 0$, $|1-e^{-i\omega}|^2 \approx \omega^2$, giving $S_X \propto \omega^{-2d}$. □

25.6.14 Invariants corresponding to continuous invariance of probability measures

Theorem 25.15 (Invariants corresponding to continuous invariance of probability measures). Let the probability distribution $\mathbb{P}$ of a stochastic process $\{X_t\}$ be invariant under a continuous transformation group G. Then there exists a probabilistic invariant $Q_{\mathbb{P}}$ satisfying $\mathbb{E}[dQ_{\mathbb{P}}/dt] = 0$:

(i) translation invariance $G = \mathbb{R}$: the invariant is $P_{\mathbb{P}} = -\gamma''(0)$;

(ii) scaling invariance $G = \mathbb{R}^+$: the invariants are the Hurst exponent H, the tail index α and the log marginal likelihood;

(iii) rotation invariance $G = U(1)$: the invariants are the cyclic frequency, the phase coherence time and the trace of the posterior covariance matrix.

Proof. The three kinds of invariance induce respectively unitary or self-adjoint operators on $L^2(\mathbb{P})$, whose spectral functionals remain constant along the process. For the concrete correspondences see Proposition 25.4, Proposition 25.8, Lemma 25.3. ☐

Remark 25.11 (Relationship between probability invariants and periodicity). Probabilistic invariance is an auxiliary condition for periodic structure, and it does not constitute a sufficient condition. White noise has an invariant mean-square derivative variance, but no periodic component. A probabilistic periodic structure must be decided by criteria such as Theorem 25.1 or Lemma 25.1.

25.6.15 Detailed balance and the fluctuation–dissipation theorem

Proposition 25.15 (Detailed balance and closure of the probability current). Let a Markov process have stationary distribution π and transition rate matrix Q. Detailed balance

$$\pi_i Q_{ij} = \pi_j Q_{ji}, \qquad \forall i, j$$

is equivalent to the probability current

$$\mathsf{J}_{ij} = \pi_i Q_{ij} - \pi_j Q_{ji}$$

being zero everywhere.

Proof. Detailed balance gives $\mathsf{J}_{ij} = 0$ directly. Conversely, if $\mathsf{J}_{ij} = 0$, then $\pi_i Q_{ij} = \pi_j Q_{ji}$. ☐

Theorem 25.16 (Structural decomposition of the fluctuation–dissipation theorem). Let the Langevin equation be

$$\frac{dX}{dt} = -\gamma X + \sqrt{2\gamma k_B T}\xi_t,$$

where ξ_t is standard white noise. Then the stationary distribution is

$$p_{\mathsf{eq}}(x) \propto e^{-\gamma x^2/(2k_B T)},$$

and the fluctuation and the dissipation satisfy $D = \gamma k_B T$.

Proof. The corresponding Fokker–Planck equation is

$$\partial_t p = \partial_x(\gamma x p) + \gamma k_B T \partial_x^2 p.$$

The stationary solution is $p_{\mathsf{eq}} \propto e^{-\gamma x^2/(2k_B T)}$, and substitution gives $D = \gamma k_B T$. ☐

25.7 Comprehensive diagnosis of probabilistic periodic structure

Example 25.9 (Periodicity diagnosis of the sunspot series: complete numerical procedure). For the monthly mean sunspot series of 1749–2024 ($N = 3312$):

(i) translation period: the sample autocorrelation has a local maximum at a lag of 132 months, corresponding to the Schwabe cycle;

(ii) multitaper spectral estimation: with $K = 8$ and $NW = 4$, the periodogram has its highest peak at $\omega_0 = 2\pi/132 \approx 0.0476$ month^{-1}, and the variance is reduced to $1/8$;

(iii) scaling period: R/S analysis gives $\hat{H} \approx 0.85 > 0.5$, strong long-range positive persistence;

(iv) rotation period: the cyclic autocorrelation has a dominant cyclic frequency at $\alpha = 1/132$ month^{-1};

(v) Kalman filter: a state-space model extracts a smooth 11-year periodic component and forecasts 12 months ahead;

(vi) invariance tests: ADF cannot reject a unit root, and KPSS rejects stationarity at the 5% level.

Remark 25.12 (Connection with the rest of the book). The theory of probabilistic periodic structure established in this chapter is an application of periodic mathematics to stochastic systems. The Herglotz and Cramér theorems reveal that the periodic information of a stationary process is encoded in the spectral measure; the Lebesgue decomposition of the spectral measure corresponds to the trichotomy of periodic structure. The spectral structure of specific processes exhibits a continuous transition from no periodicity to long-memory periodicity. Cyclostationary processes handle non-stationary periodic signals such as amplitude modulation and frequency modulation. Multitaper spectral estimation overcomes the inconsistency of the variance of the periodogram. The Kalman filter provides adaptive extraction of periodic signals. The MCMC spectral gap theory gives convergence diagnostics. The HMM periodic decomposition is connected with speech recognition. The GP periodic kernel provides non-parametric periodic modelling. ARFIMA supplements long memory. The invariance of a probabilistic system is an auxiliary condition for periodic structure, not a sufficient condition; the decision of probabilistic periodic structure relies on concrete criteria such as cyclostationarity, phase locking and the periodic decomposition of a Markov chain.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	Time-translation invariance of a stationary process; mean-square derivative variance $P_{\mathbb{P}} = -\gamma''(0) = \int \omega^2 S(\omega) d\omega$ (Definition 25.12, Proposition 25.4, Proposition 25.5); the Herglotz–Cramér spectral representation (Theorem 25.4, Theorem 25.5)
Scaling	Self-similarity and heavy-tailed scale invariance: the Hurst exponent H and the tail index $\alpha = 1/H$ (Definition 25.15, Proposition 25.8); the ARFIMA long-memory spectrum (Proposition 25.14)
Rotation	Phase-rotation invariance of cyclostationary processes; phase diffusion and the coherence time $\tau_{\text{coh}} = 2/\sigma^2$ (Definition 25.16, Lemma 25.3, Definition 25.17, Theorem 25.11)
Balance	The invariants corresponding to continuous invariance of probability measures (Theorem 25.15); the criteria and boundaries of probabilistic periodic structure (Remark 25.11)
Symmetry	The scaling/translation/rotation information carried by the three classes of distributions: prior/likelihood/posterior (**empirical observation / research programme**, see Remark Remark 25.10)
Closure	Closure of the probability current under detailed balance $\mathsf{J}_{ij} = 0$ (Proposition 25.15); the fluctuation–dissipation relation (Theorem 25.16); the MCMC spectral gap and convergence (Theorem 25.13)
Period	The cyclic frequency $\alpha = k/T$; the self-similar period; the spectral structure of typical processes (Proposition 25.1); multitaper spectral estimation (Theorem 25.10, Proposition 25.7); periodic extraction by the Kalman filter (Theorem 25.12, Proposition 25.9); the HMM periodic decomposition (Theorem 25.14); the GP periodic kernel (Definition 25.22, Proposition 25.13); the sunspot numerical diagnosis (Example 25.9, Example 25.8)

Exercises

Exercise 1 Prove the **probabilistic periodicity decision theorem** of this chapter (25.1). Hint: first write down the decision map and the evolution group of this chapter's target, then apply the proof structure of the general criterion of the 'Periodic Systems' chapter.

Exercise 2 Verify that the hypotheses of the **monotonicity of quantity adjustment** (25.2) hold for the domain objects of this chapter, and give a counterexample in which the hypotheses fail.

Exercise 3 Examine the behaviour of the **structural correspondence of the three classes of distributions in a Bayesian model** (**empirical observation / research programme**, see Remark 25.10) under parameter degeneracy, and explain how the analogy degenerates accordingly; state what definitions would have to be supplied in order to promote any one of the three assertions to a testable proposition.

Exercise 4 Explain the relation between the **probabilistic periodicity diagnostic procedure and its interface with three chapters** (25.9) and the corresponding general conclusion of the 'Periodic Systems' chapter (whether they agree, whether it is a special case, or whether additional conditions are required).

Exercise 5 Using the seven-theme comparison table of this chapter, identify the concrete carriers of the three themes **translation**, **scaling** and **rotation** in this chapter, and give one example of domain data for each.

Exercise 6 Synthesis: restate the diagnostic procedure of this chapter using the five-step standard interface of the 'Periodic Systems' chapter (Table 24.1), noting one by one the theorem or remark of this chapter on which each step depends: 25.1, 25.2, 25.10 (**empirical observation / research programme**), 25.9.

Chapter 26

Periodic Structure of Cryptographic Systems

Remark 26.1 (Periodicity decision for the target of this chapter (citation)). When this chapter decides whether a research target is periodic, determines its period type, or seeks its minimal positive period, reference is always made to the 'Periodic Systems' chapter: for the periodicity criterion see Theorem 24.1, for the discrimination of period types see Definition 24.3, Proposition 24.1, for the search for the minimal positive period see Definition 24.4, Theorem 24.2, and for the construction of period series and epicycle groups from data see Theorem 24.3, Definition 24.6.

Remark 26.2 (Positioning of this chapter: developed along the five-step interface of the 'Periodic Systems' chapter). The periodic structure of cryptographic systems studies the mathematical characterisation and the security applications of periodicity, quasi-periodicity and iterative structure in cryptography. This chapter does not establish a separate periodicity criterion; instead it connects cryptographic systems to the five-step interface prescribed by the 'Periodic Systems' chapter:

(1) **Decision**: reduce the state translation of stream ciphers, the round-function iteration of block ciphers and the cyclic-group scalar multiplication of public-key cryptography to discrete dynamical systems, invoke Theorem 24.1 to decide whether they are periodic, and determine the period type;

(2) **Construction**: build the amplitude series, the frequency series and the phase series from the keystream, the round-function states and the group elements, and assemble them into the cryptographic epicycle group;

(3) **Adjustment**: dynamically adjust the epicycle group by such security metrics as the linear complexity, the branch number and the order of the group;

(4) **Convolution**: introduce amplitude convolution, phase convolution and frequency convolution on the cryptographic epicycle group, and link them to autocorrelation, the Walsh spectrum, the difference distribution table and the linear approximation table;

(5) **Integration**: connect to the tool chains of the three chapters 'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis', forming a cryptographic periodicity diagnostic procedure.

The cryptographic knowledge used in this chapter includes: the cryptosystem quintuple, linear feedback shift registers, m-sequences, the Berlekamp–Massey algorithm, the round function of block ciphers, confusion and diffusion, differential cryptanalysis, linear cryptanalysis, modular exponentiation, the discrete logarithm problem, RSA, ElGamal, elliptic-curve scalar multiplication, ECDLP, the Pohlig–Hellman attack, the MOV attack, and the three-layer cryptographic system of TLS 1.3.

Remark 26.3 (Guide to the seven themes of this chapter). Translation—the state translation of the LFSR and the keystream period (Definition 26.8, the m-sequence period 2^L-1); Scaling—the iteration of the block-cipher round function and the diffusion rate (Definition 26.11, Definition 26.12); Rotation—modular exponentiation and the cyclic-group scalar multiplication of elliptic-curve point multiplication (Definition 26.14, Definition 26.16); Balance—the three structural invariants: the linear complexity L, the branch number and the order of the group n (Theorem 26.8); Symmetry—three layers of special elements: the all-zero state, the identity map and the identity element (Theorem 26.7); Closure—the mutually inverse closure of encryption and decryption $D_k(E_k(m)) = m$ (Definition 26.6); Period—the m-sequence period, the iteration depth of the round function, and the scalar-multiplication period of the cyclic group order (Definition 26.7).

26.1 Periodicity decision for cryptographic targets: connection to step one of the ‘Periodic Systems’ chapter

26.1.1 Cryptographic targets, decision input and decision map

Definition 26.1 (Cryptographic target and decision input). A **cryptographic target** is one of the following objects:

(i) a keystream generator of a stream cipher, whose state space is $\mathbb{F}_2^L$ and whose state translation is $\mathcal{T}$;

(ii) a round-function iteration system of a block cipher, whose state space is $\mathbb{F}_2^n$ and whose round function is F_{k_i};

(iii) a cyclic-group scalar multiplication system of public-key cryptography, whose state space is a finite cyclic group G and whose scalar multiplication is $x \mapsto g^x$ or $k \mapsto [k]P$;

(iv) a keystream, a ciphertext sequence or a sequence of group elements produced by one of the above systems.

The **decision input** of a cryptographic target is the pair

$$\mathcal{D}_{\mathrm{C}} = (\mathcal{S}, \Phi),$$

where $\mathcal{S}$ is the state space or the sequence space and $\Phi : \mathcal{S} \to \mathcal{S}$ is the evolution map induced by the state translation, the round function or the scalar multiplication. If $\mathcal{S}$ is a finite set, the target is called a **finite cryptographic target**; if $\mathcal{S}$ is an infinite set, the target is called an **infinite cryptographic target**.

Definition 26.2 (Cryptographic decision map)**.** Let a cryptographic target be given by the decision input $\mathcal{D}_C = (\mathcal{S}, \Phi)$. If the states or the output of the target are generated by iteration of Φ, that is, if there exists $x_0 \in \mathcal{S}$ such that

$$x_t = \Phi^t(x_0), \qquad t \in \mathbb{N},$$

then Φ is called the **cryptographic decision map** of the target. In this case the cryptographic target is reduced to the discrete dynamical system $(\mathcal{S}, \Phi)$, to which the criteria of the 'Operations and Periodicity' chapter and of the 'Periodic Systems' chapter may be applied directly.

Theorem 26.1 (Cryptographic periodicity decision theorem)**.** Let the decision map of a cryptographic target be Φ, and let $(\mathcal{S}, \Phi)$ be a discrete dynamical system. Then:

(A) **Sufficient decision**: if there exists $m \in \mathbb{N}^+$ such that $\Phi^m =$ id, then the target is a uniformly periodic cryptosystem; every state orbit is closed, and the orbit length divides m.

(B) **Equivalence of the decision**: the target is a uniformly periodic cryptosystem if and only if ord$(\Phi) < \infty$.

(C) **Pointwise criterion**: the orbit of the target at the state x_0 is closed if and only if x_0 is a periodic point; the orbit length is then the period at that point.

(D) **Non-omissibility**: the conjunction of conserved quantities and the symmetry group cannot replace ord$(\Phi) < \infty$; a quasi-periodic cryptographic target is not a periodic cryptosystem.

Proof. The Theorem 26.1 of this chapter is the domain instantiation of Theorem 24.1 in the 'Periodic Systems' chapter: one need only replace the decision map f and the evolution group by the Φ of this chapter, and the four clauses (A) sufficient decision, (B) equivalence of the decision, (C) pointwise criterion and (D) non-omissibility hold one by one. By the convention of Remark 24.4, they are not expanded again here. ☐

Corollary 26.1 (Cryptographic period types)**.** After the decision of Theorem 26.1, a cryptographic target falls into one of the following types:

(i) **Trivially periodic type**: the minimal period is $m_0 = 1$, and the state is a fixed point (such as the all-zero state of an LFSR);

(ii) **Strictly periodic type**: $1 < m_0 < \infty$, and the state is strictly periodic (such as the state orbit of an m-sequence);

(iii) **Uniformly periodic type**: ord$(\Phi) = m < \infty$, and all orbit lengths divide m;

(iv) **Quasi-periodic type**: ord$(\Phi) = \infty$, the orbits are not closed but are dense on a compact carrier structure;

(v) **Eventually periodic type**: there exists N such that the target is periodic for $t \geq N$;

(vi) **Discretely periodic type**: the carrier structure is $\mathbb{Z}$, and the normalised period ratio $\rho = T_x/\Delta x \in \mathbb{Q}$.

(vii) **Asymptotically periodic (damped) type**: the orbit is not closed, but tends asymptotically to a closed orbit or to a fixed point, the residual decaying as $e^{-\kappa t}$ (see Definition 24.3(vii)).

Among these, (i)–(iii) and (vi) belong to periodic cryptosystems; (iv) does not belong to periodic cryptosystems, but can be handled by bounded-approximation tools; (v) is a periodic cryptosystem only in its stable stage; (vii), like (iv), is a type in the limiting sense; it is not a periodic cryptosystem and can be handled with bounded-approximation tools only in its stable stage.

Proof. Each clause follows by substituting the cryptographic target into Theorem 26.1 and Definition 24.3 of the 'Periodic Systems' chapter. ☐

Lemma 26.1 (Periodicity decision for the LFSR)**.** Let the state translation of an LFSR of length L be $\mathcal{T}$, with feedback polynomial $C(x)$. Then:

(i) $\mathcal{T}$ is a linear transformation on $\mathbb{F}_2^L$, whose matrix representation A is the companion matrix;

(ii) the period p of $\mathcal{T}$ is the least positive integer such that $A^p = I$;

(iii) if $C(x)$ is a primitive polynomial, then the period of every non-zero initial state is $2^L - 1$;

(iv) the all-zero state is the unique fixed point of $\mathcal{T}$, and no non-zero orbit ever passes through the all-zero state.

Proof. (i) The state translation

$$\mathcal{T}(s_0, \ldots, s_{L-1}) = (s_1, \ldots, s_{L-1}, f(s_0, \ldots, s_{L-1}))$$

is a linear map, with matrix

$$A = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ c_0 & c_1 & c_2 & \cdots & c_{L-1} \end{pmatrix}.$$

Its characteristic polynomial is $C(x)$. (ii) The period is defined as the least positive integer p such that $A^p = I$. (iii) When $C(x)$ is a primitive polynomial, A is a generator of the multiplicative group $\mathbb{F}_{2^L}^*$, of order $2^L - 1$. (iv) $A0 = 0$, and if $A^t s = 0$ then $s = 0$; hence no non-zero orbit passes through the all-zero state. ☐

Definition 26.3 (Cryptographic minimal positive period algorithm)**. Input**: samples $x_0, \ldots, x_{N-1}$ of a cryptographic target, and a tolerance $\varepsilon \geq 0$. **Output**: the minimal positive period T_0, or the verdict 'not periodic'. **Steps**:

(1) take the candidate set $\mathcal{P} = \{T \in \mathbb{N}^+ : 2T \leq N\}$ and traverse it in increasing order of T;

(2) for each T compute the maximal deviation

$$E(T) = \max_{0 \le n < N-T} \|x_{n+T} - x_n\|;$$

(3) output the first T satisfying $E(T) \le \varepsilon$ as T_0; if the traversal is exhausted with no solution, output 'not periodic';

(4) commensurability check: use $\rho = T_x/\Delta x$ (with $T_0 = p$ when $\rho = p/q$ is in lowest terms; write $\ell = T_0 \Delta x = qT_x$ for the continuous span corresponding to T_0 sampling points) together with Proposition 21.2, Corollary 21.1 of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter to verify the conversion relation between T_0 and the continuous period T_x.

Theorem 26.2 (Correctness of the cryptographic minimal positive period algorithm). (i) If the samples form a strictly periodic sequence, and $\varepsilon = 0$, $N \ge 2T_0$, then the algorithm outputs the discrete fundamental period T_0.

(ii) If the samples are obtained by sampling a continuous cryptographic target of fundamental period T_x at sampling interval Δx, and the normalised period ratio $\rho = T_x/\Delta x = p/q$ is in lowest terms, then the algorithm outputs $T_0 = p$.

(iii) If $\rho \notin \mathbb{Q}$, then no true period exists: $E(T) > 0$ for every $T \in \mathbb{N}^+$.

Proof. The Theorem 26.2 of this chapter follows directly from two conclusions of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter: (i) by Definition 21.4, T_0 is the minimal positive period, hence $E(T_0) = 0$, whilst (T)>0$for < T_0$; since the algorithm traverses in increasing order of T, the first one satisfying $E(T) = 0$ is T_0; (ii)(iii) follow by substitution from Proposition 21.2(ii)(iii). The complete argument is given in that chapter. □

Example 26.1 (Example of an LFSR periodicity decision). Take $L = 4$ and the primitive polynomial $C(x) = 1 + x + x^4$. By Lemma 26.1, the period of every non-zero initial state is $2^4 - 1 = 15$. Taking the initial state $(1, 0, 0, 0)$, the state orbit has length 15, and the output m-sequence has period 15.

Example 26.2 (Periodicity decision for RSA modular exponentiation). In RSA the period of $g^x \bmod N$ is $\mathrm{ord}_N(g)$, which divides $\phi(N)$. If g is a primitive root modulo N, the period is $\phi(N)$. By Theorem 26.1, this target is of strictly periodic type; but an attacker cannot recover $\phi(N)$ directly from N, so the period itself cannot be computed publicly.

Remark 26.4 (Cryptographic invariants do not guarantee period security). The invariants of a cryptosystem are auxiliary conditions on the periodic structure, but they do not constitute a sufficient condition for security. For example, an LFSR has maximal period $2^L - 1$, yet the Berlekamp–Massey algorithm can still recover the entire sequence from $2L$ consecutive bits. The decision of the periodic structure of a cryptosystem must rely on concrete criteria such as Theorem 26.1 or Lemma 26.1, combined with an attack model.

26.2 Construction of the cryptographic period series and epicycle group: connection to step two of the 'Periodic Systems' chapter

Definition 26.4 (Cryptographic amplitude, frequency and phase series)**.** *Let the state or output sequence of a cryptographic target be $\{z_t\}$. If it can be written as*

$$z_t = \sum_k A_k e^{i(\omega_k t+\phi_k)} + \varepsilon_t,$$

then

$$\mathcal{A}_z(t) = \sum_k A_k e^{ikt}, \qquad \mathcal{F}_z(s) = \sum_k \omega_k s^k, \qquad \mathcal{P}_z(t) = \sum_k e^{i(kt+\phi_k)}$$

are called respectively the **amplitude series***, the* **frequency series** *and the* **phase series** *of the cryptographic target. The three agree with Definition 6.10, Definition 6.12, Definition 6.11 of the 'Series and Convolution' chapter.*

Definition 26.5 (Cryptographic epicycle group)**.** *Let the period series of a cryptographic target be given by the amplitude series $\{A_k\}$, the frequency series $\{\omega_k\}$ and the phase series $\{\phi_k\}$. For each k define the epicycle*

$$E_k : \theta \mapsto A_k e^{i(\theta+\phi_k)}, \qquad \theta = \omega_k t,$$

whose collection

$$\mathfrak{E}_{\mathrm{C}} = \{(A_k, \omega_k, \phi_k)\}_{k\in\mathbb{Z}}$$

is called the **epicycle group** *of the cryptographic target. Its superposition is the cryptographic period series*

$$z_t = \sum_k E_k(\omega_k t) = \sum_k A_k e^{i(\omega_k t+\phi_k)}.$$

Theorem 26.3 (Construction of period series from cryptographic data)**.** *Let the output sequence $\{z_t\}$ of a cryptographic target be a finite periodic sequence of period N. Then:*

(i) *$\{z_t\}$ can be expanded as a discrete Fourier series*

$$z_t = \sum_{k=0}^{N-1} c_k e^{i2\pi kt/N}, \qquad c_k = \frac{1}{N}\sum_{t=0}^{N-1} z_t e^{-i2\pi kt/N};$$

(ii) *the amplitude series is given by $|c_k|$, the phase series by* arg *c_k, and the frequency series by $2\pi k/N$;*

(iii) *if $\{z_t\}$ is an m-sequence, then its autocorrelation function is two-valued, and the power spectrum is flat at non-zero frequencies.*

Proof. (i) The discrete Fourier series of a finite periodic sequence is a standard result: the basis functions $\{e^{i2\pi kt/N}\}_{k=0}^{N-1}$ form an orthogonal basis of $\mathbb{C}^N$. (ii) This is read off directly from $c_k = A_k e^{i\phi_k}$. (iii) The two-valued autocorrelation of an m-sequence is given by Lemma 26.2; its power spectrum is the Fourier transform of the autocorrelation, and is the constant $N+1$ at non-zero frequencies. □

Definition 26.6 (Cryptosystem)**.** A **cryptosystem** is a quintuple $(\mathcal{P},\mathcal{C},\mathcal{K},\mathcal{E},\mathcal{D})$, where:

(i) $\mathcal{P}$ is the **plaintext space**;

(ii) $\mathcal{C}$ is the **ciphertext space**;

(iii) $\mathcal{K}$ is the **key space**;

(iv) $\mathcal{E} = \{E_k : \mathcal{P} \to \mathcal{C} \mid k \in \mathcal{K}\}$ is the **family of encryption functions**;

(v) $\mathcal{D} = \{D_k : \mathcal{C} \to \mathcal{P} \mid k \in \mathcal{K}\}$ is the **family of decryption functions**, satisfying $D_k(E_k(m)) = m$.

Definition 26.7 (Periodic structure of a cryptosystem)**.** The **periodic structure** of a cryptosystem falls into three basic types:

(i) **State period**: the periodicity of the keystream sequence, determined by the size of the generator's internal state space and by the structure of the feedback polynomial; the special element is the all-zero state of the LFSR.

(ii) **Iterative structure**: the self-similarity of the round function of a block cipher, every round of encryption having the same algebraic structure; the special element is the identity map under a zero key.

(iii) **Cyclic-group scalar multiplication period**: the periodicity of the modular exponentiation $g^x \bmod p$ in the exponent x, and the periodicity of the elliptic-curve point multiplication $[k]P$ in the scalar k; the special element is the identity element of the group.

26.2.1 State period of stream ciphers

Definition 26.8 (Linear feedback shift register)**.** A **linear feedback shift register** (LFSR) of length L consists of the following components:

(i) L register cells $s_0, \ldots, s_{L-1} \in \mathbb{F}_2$;

(ii) feedback coefficients $c_0, \ldots, c_{L-1} \in \mathbb{F}_2$, with feedback function

$$f(s_0, \ldots, s_{L-1}) = \bigoplus_{i=0}^{L-1} c_i s_i.$$

The **state translation** at each clock cycle is

$$\mathcal{T}(s_0, \ldots, s_{L-1}) = (s_1, \ldots, s_{L-1}, f(s_0, \ldots, s_{L-1})).$$

The output keystream bit is $z_t = s_0$. The feedback polynomial

$$C(x) = 1 + c_1 x + \cdots + c_{L-1} x^{L-1} + x^L$$

completely determines the periodic structure of the LFSR. When $C(x)$ is a primitive polynomial, every non-zero initial state generates the maximal period $2^L - 1$, which is called an m-**sequence**.

Proposition 26.1 (Algebraic properties of the LFSR). The state translation $\mathcal{T}$ of an LFSR satisfies:

(i) $\mathcal{T}$ is a linear transformation on $\mathbb{F}_2^L$, and the matrix representation A is the companion matrix;

(ii) the period of the LFSR is the least positive integer p such that $\mathcal{T}^p = \mathrm{id}$. For an m-sequence, $p = 2^L - 1$;

(iii) the linear complexity L is invariant under the state translation; the linear complexity of an m-sequence equals the number of LFSR stages L;

(iv) the all-zero state is the unique fixed point, and no non-zero orbit ever passes through the all-zero state.

Proof. (i) By Lemma 26.1(i). (ii) By Lemma 26.1(ii)(iii). (iii) The linear complexity is the length of the shortest LFSR needed to generate a sequence. The Berlekamp–Massey algorithm recovers an equivalent LFSR from $2L$ consecutive bits, so L is an extractable invariant. (iv) By Lemma 26.1(iv). □

Definition 26.9 (Autocorrelation function of an m-sequence). Let $\{z_t\}_{t=0}^{N-1}$ be an m-sequence ($N = 2^L - 1$) taking values in $\{+1, -1\}$. Its normalised autocorrelation function is

$$\rho(h) = \frac{1}{N}\sum_{t=0}^{N-1} z_t z_{t+h} = \begin{cases} 1, & h \equiv 0 \pmod{N}, \\ -1/N, & h \not\equiv 0 \pmod{N}. \end{cases}$$

Lemma 26.2 (Two-valued autocorrelation of an m-sequence). Let $\{z_t\}$ be an m-sequence ($N = 2^L - 1$). Then

$$\rho(h) = \frac{1}{N}\sum_{t=0}^{N-1} z_t z_{t+h} = \begin{cases} 1, & h \equiv 0 \pmod{N}, \\ -1/N, & h \not\equiv 0 \pmod{N}. \end{cases}$$

Proof. By the shift-and-add property of m-sequences: for $h \not\equiv 0 \pmod{N}$, $\{z_t \oplus z_{t+h}\}$ is again a shift of the same m-sequence. Writing $z_t = (-1)^{s_t}$, we have

$$z_t z_{t+h} = (-1)^{s_t + s_{t+h}} = (-1)^{s_t \oplus s_{t+h}}.$$

Within one period the numbers of $+1$ and -1 are $(N-1)/2$ and $(N+1)/2$ respectively, so

$$\sum_t z_t z_{t+h} = \frac{N-1}{2} - \frac{N+1}{2} = -1,$$

and normalisation gives $\rho(h) = -1/N$; for $h \equiv 0$, $\rho(0) = 1$. □

Example 26.3 (Concrete computation of an m-sequence with $L = 4$). Take $L = 4$ and the primitive polynomial $C(x) = 1 + x + x^4$. Starting from the non-zero initial state $(1, 0, 0, 0)$, the state period is 15. The output m-sequence is

$$100010011010111\cdots$$

and, after passing to the values $\{+1, -1\}$, it is

$$(+1, -1, -1, -1, +1, -1, -1, +1, +1, -1, +1, -1, +1, +1, +1).$$

There are 8 occurrences of $+1$ and 7 of -1, in accordance with the balance property.

Definition 26.10 (Periodic attack on stream ciphers). *The security of a stream cipher depends on the state period of the keystream sequence. If the period p is too short, an attacker can break it in the following ways:*

(i) **Period detection**: *compute the autocorrelation function of the ciphertext, which exhibits a peak at $h = p$;*

(ii) **Linear complexity attack**: *the Berlekamp–Massey algorithm recovers an equivalent LFSR from $2L$ consecutive bits;*

(iii) **Correlation attack**: *exploit the correlation between the outputs of the individual LFSRs and the keystream to break them one by one.*

Remark 26.5 (Security criteria for stream ciphers (engineering practice)). On the basis of the analysis of the state period, the secure design of a stream cipher should observe:

(i) maximisation of the state period: the period should be at least 2^{128};

(ii) maximisation of the linear complexity: the linear complexity should approach the period;

(iii) minimisation of the autocorrelation: the autocorrelation at non-zero shifts should be as close to zero as possible.

Status note: the clauses above are engineering rules of thumb, not mathematical propositions; when cited they should be treated as empirical conclusions.

26.2.2 Iterative structure of block ciphers

Definition 26.11 (Round function of a block cipher). A **block cipher** encrypts by iterating a round function $F_k : \mathbb{F}_2^n \to \mathbb{F}_2^n$. Let the number of rounds be R and the round keys be $k_1, \dots, k_R$; the encryption process is

$$C = F_{k_R} \circ F_{k_{R-1}} \circ \cdots \circ F_{k_1}(P),$$

where $P \in \mathbb{F}_2^n$ is the plaintext block and $C \in \mathbb{F}_2^n$ is the ciphertext block.

Definition 26.12 (Confusion and diffusion). The **confusion** and **diffusion** proposed by Shannon are the two basic principles of block-cipher design:

(i) **Confusion**: make the statistical relation between the ciphertext and the key as complex as possible, mainly by means of non-linear S-boxes;

(ii) **Diffusion**: make a change in a single plaintext bit propagate to as many ciphertext bits as possible, mainly by means of a linear permutation layer.

Proposition 26.2 (Iterative structure of block ciphers). *The round-function iteration of a block cipher has the following structural properties:*

(i) *each round of encryption F_{k_i} is parameterised by the round key k_i, and $E_k = F_{k_R} \circ \cdots \circ F_{k_1}$;*

(ii) the number of rounds R is the iteration-depth parameter, which determines the cumulative effect of confusion and diffusion;

(iii) the branch number is a structural invariant of the diffusion layer. The MixColumns branch number of AES is 5, and full diffusion is reached after 2 rounds;

(iv) under a zero key the special element of one round of encryption is the identity map.

Proof. (i) Each round can be decomposed as $F_{k_i}(x) = L(S(x \oplus k_i))$, where S is the S-box and L is a linear permutation. (ii) The number of rounds R directly controls the iteration depth. AES-128 uses $R = 10$, AES-256 uses $R = 14$. (iii) The MixColumns branch number of AES is 5: a change of 1 byte in the input changes at least 4 bytes in the output. The branch number has an additive property under composition of round functions. (iv) With a zero key, $F_0 = \mathrm{id}$. □

Example 26.4 (Concrete computation of AES diffusion). *Single-round transformation of AES-128: only the $(1,1)$ entry of the input state matrix is 01. After round 1, SubBytes sends $01 \mapsto 7C$, ShiftRows moves the active byte, and MixColumns expands the single active byte into a whole active column, the number of active bytes becoming 4. After round 2 the number of active bytes expands to 16, reaching full diffusion.*

Definition 26.13 (Periodic attack on block ciphers)**.** *The security of a block cipher depends on the iterative structure of the round function. The main attacks include:*

(i) **Differential cryptanalysis**: *exploit the propagation law of input and output differences, whose probability decays according to the product law;*

(ii) **Linear cryptanalysis**: *exploit the bias of a linear approximation, which decays according to the product law;*

(iii) **Integral attack**: *exploit the algebraic evolution of a specific input structure.*

Remark 26.6 (Security criteria for block ciphers (engineering practice)). *On the basis of the analysis of the iterative structure, the secure design of a block cipher should observe:*

(i) sufficiency of the number of rounds: the propagation probability of differential/linear characteristics is far smaller than the probability for a random permutation;

(ii) maximisation of S-box non-linearity;

(iii) maximisation of the branch number of the diffusion layer;

(iv) independence of the round keys.

Status note: *the clauses above are engineering rules of thumb, not mathematical propositions; when cited they should be treated as empirical conclusions.*

26.2.3 Cyclic group structure of public-key cryptography

Definition 26.14 (Periodic structure of modular exponentiation)**.** *Let G be a finite cyclic group, $g \in G$ a generator, and $|G| = n$.* **Modular exponentiation** *is defined by*

$$E(x) = g^x, \qquad x \in \mathbb{Z}_n.$$

Its structural properties:

(i) *cyclic-group scalar multiplication: $g^{x+1} = g \cdot g^x$;*

(ii) *period n: $g^{x+n} = g^x$;*

(iii) *the special element is the identity element $e = g^0$;*

(iv) *the invariant is the order of the group n. In RSA $n = \phi(N) = (p-1)(q-1)$; in ElGamal $n = q$; for an elliptic curve n is the order of the curve.*

Definition 26.15 (Discrete logarithm problem)**.** *The* **discrete logarithm problem** *(DLP): given g and $h = g^x$, find $x \bmod n$. The modular exponentiation $x \mapsto g^x$ is easy to compute, but the inverse operation is hard in the classical model of computation.*

Proposition 26.3 (Structural properties of public-key cryptography)**.** *The modular exponentiation of public-key cryptography has the following structural properties:*

(i) *the encryption operation is the scalar multiplication of the plaintext in a cyclic group;*

(ii) *decryption depends on the private key d, satisfying $ed \equiv 1 \pmod{\phi(N)}$;*

(iii) *the invariants are the modulus N and the order $\phi(N)$;*

(iv) *the special element is the identity element 1.*

Proof. RSA correctness: $ed = 1 + k\phi(N)$. If $\gcd(m, N) = 1$, then by Euler's theorem $m^{\phi(N)} \equiv 1 \pmod N$, so

$$D(C) = C^d \equiv m^{ed} = m(m^{\phi(N)})^k \equiv m \pmod N.$$

If $\gcd(m, N) \neq 1$, verify modulo p and q separately by the Chinese remainder theorem, and combine to obtain $m^{ed} \equiv m \pmod N$. Hence $D \circ E = E \circ D = \mathrm{id}$. □

Definition 26.16 (Periodic structure of elliptic curves)**.** *Let E be an elliptic curve over the finite field $\mathbb{F}_q$, and let $E(\mathbb{F}_q)$ be a finite abelian group. The order of a point P is the least positive integer n such that $[n]P = \mathcal{O}$. The scalar multiplication is*

$$[k]P = \underbrace{P + \cdots + P}_{k \text{ times}}.$$

ECDLP: given P and $Q = [k]P$, find $k \bmod n$.

Remark 26.7 (Security conditions for elliptic-curve cryptography (engineering practice))*. The security of elliptic-curve cryptography is guaranteed by the following conditions:*

(i) primality of the order: n should contain a large prime factor, so as to prevent the Pohlig–Hellman attack;

(ii) embedding degree: k should be sufficiently large, so as to prevent the MOV attack;

(iii) security of the twisted curve.

Status note: *the clauses above are engineering rules of thumb, not mathematical propositions; when cited they should be treated as empirical conclusions.*

26.3 Dynamic adjustment of the cryptographic epicycle group: connection to step three of the 'Periodic Systems' chapter

Definition 26.17 (Dynamic adjustment of the cryptographic epicycle group)**.** Let the current cryptographic epicycle group be $\mathfrak{E}^{(j)}$, with partial sum

$$z^{(j)}(t) = \sum_{k\in\mathfrak{E}^{(j)}} A_k e^{i(\omega_k t+\phi_k)},$$

and residual $r^{(j)} = z - z^{(j)}$. **Dynamic adjustment** consists of two rules:

(i) **Quantity adjustment**: if the residual energy $\|r^{(j)}\|^2 > \delta$, add one new epicycle according to the peak of the residual periodogram:

$$\mathfrak{E}^{(j+1)} = \mathfrak{E}^{(j)} \cup \{(\hat{A}, \hat{\omega}, \hat{\phi})\}, \qquad \hat{\omega} = \arg\max_{\omega} |\widehat{z_r}(\omega)|;$$

(ii) **Parameter adjustment**: update the parameters of the existing epicycles by regularised least squares or by the security metrics:

$$(A_k, \phi_k) \leftarrow \arg\min_{A,\phi} \sum_n \left| z(n) - \sum_k A_k e^{i(\omega_k n+\phi_k)} \right|^2 .$$

In a cryptosystem, quantity adjustment corresponds to increasing the number of LFSR stages, the number of rounds or the prime factors of the group order; parameter adjustment corresponds to adjusting the feedback polynomial, the S-box or the curve parameters, so that the security metrics are monotonically improved.

Proposition 26.4 (Monotonicity of quantity adjustment)**.** *After the dominant residual frequency component is added as in Definition 26.17(i), the residual energy is monotonically non-increasing:*

$$\|r^{(j+1)}\|^2 \leq \|r^{(j)}\|^2,$$

with equality if and only if the residual has no energy at the selected frequency.

Proof. The Proposition 26.4 of this chapter agrees with the dynamic adjustment mechanism of Definition 24.8 in the 'Periodic Systems' chapter: adding an epicycle removes the component of $r^{(j)}$ along $\hat{\omega}$ from the residual, and the orthogonal decomposition $\|r^{(j)}\|^2 = \|r^{(j+1)}\|^2 + \|\text{the removed component}\|^2$ gives the inequality, with equality if and only if the norm of the removed component is zero. The complete argument is given there. □

Theorem 26.4 (Convergence of dynamic adjustment)**.** Let the cryptographic target be a **real-valued finite periodic sequence** of length T_0 ($T_0 \in \mathbb{N}$) and let $\delta = 0$. Then after at most $\lfloor T_0/2 \rfloor + 1$ steps of dynamic adjustment the residual is zero, and the resulting cryptographic epicycle group is the exact Fourier data of the target; for a **complex-valued** sequence the corresponding number of steps is at most T_0.

Proof. The discrete Fourier series of a **real-valued** finite periodic sequence of length T_0 ($T_0 \in \mathbb{N}$) is complete on $\lfloor T_0/2 \rfloor + 1$ harmonics (including the constant term); a **complex-valued** sequence requires T_0 components. At each step one dominant frequency component of the residual is extracted as in Proposition 26.4 and the residual energy decreases strictly, so after finitely many steps the residual is zero. □

Definition 26.18 (Security-metric adjustment)**.** Let the security metric of a cryptographic target be

$$\mathcal{I} = (\mathcal{I}_{\text{translation}}, \mathcal{I}_{\text{scaling}}, \mathcal{I}_{\text{rotation}}) = (\text{linear complexity } L, \text{ branch number } \mathcal{B}, \text{ largest prime factor o}$$

Security-metric adjustment is: at each step choose a neighbouring parameter θ in the parameter space Θ for which $\mathcal{I}(\theta)$ is **strictly decreasing** in the lexicographic order (minimisation); if no such neighbouring parameter exists, the process stops—then θ is a **locally optimal** parameter.

Proposition 26.5 (Finite convergence of security-metric adjustment)**.** Let the parameter space Θ be finite and let each adjustment make $\mathcal{I}$ **strictly decreasing** in the lexicographic order. Then the adjustment terminates after finitely many steps at a locally optimal parameter.

Proof. Θ is finite, so $\mathcal{I}$ takes finitely many values. The lexicographic order is well-founded on a finite set of values, so a strictly lexicographically decreasing sequence cannot continue indefinitely and the process terminates after finitely many steps. At termination there is no neighbouring parameter making $\mathcal{I}$ strictly lexicographically decreasing, that is, the parameter is locally optimal. Monotonicity must be imposed on the **whole lexicographic order**: if only 'some component strictly increases or decreases' is required, cycles such as $(2,1) \to (1,2) \to (2,1)$ cannot be excluded and finite termination fails. □

Remark 26.8 (Boundaries of dynamic adjustment). Theorem 26.4 relies on finite periodic sequences. For infinite-period or quasi-periodic cryptographic targets, no finite epicycle group represents them exactly, and only bounded approximation is possible. Security-metric adjustment does not guarantee global optimality, nor does it guarantee computational security; computational security depends on the concrete attack model and computational assumptions.

26.4 Cryptographic convolution: connection to step four of the 'Periodic Systems' chapter

Definition 26.19 (Cryptographic amplitude, phase and frequency convolutions)**.** *Let two cryptographic epicycle groups be*

$$\mathfrak{E} = \{(A_n, \omega_n, \phi_n)\}, \qquad \mathfrak{E}' = \{(B_m, \nu_m, \psi_m)\}.$$

Define:

(i) **Amplitude convolution**: $\mathbb{A}_{mn} = A_n B_m$;

(ii) **Phase convolution**: $\mathbb{P}_{mn} = e^{i(\phi_n - \psi_m)}$;

(iii) **Frequency convolution**: $\mathbb{F}_{mn} = \omega_n \nu_m$;

(iv) **Trend convolution**: $\mathbb{T}_{mn} = A_n B_m e^{i(\phi_n - \psi_m)} = A_n B_m \mathbb{P}_{mn}$.

This agrees with Definition 6.21, Definition 6.22, Definition 6.23, Definition 6.24 of the 'Series and Convolution' chapter.

Theorem 26.5 (Action of the three convolutions on the cryptographic epicycle group)**.** *Let the epicycle groups of two cryptographic targets be $\mathfrak{E}$ and $\mathfrak{E}'$ respectively. Then:*

(i) *amplitude convolution acts only on the radius;*

(ii) *phase convolution acts only on the phase;*

(iii) *frequency convolution acts only on the rotational speed;*

(iv) *trend convolution preserves both amplitude and phase.*

Proof. This is read off directly from Definition 26.19. ☐

Proposition 26.6 (Autocorrelation and trend convolution)**.** *Let $\{z_t\}$ be a periodic sequence, with autocorrelation function*

$$R_z(h) = \sum_t z_t \overline{z_{t+h}}.$$

In the language of the cryptographic epicycle group, the autocorrelation function is the projection of the trend convolution onto the diagonal:

$$R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}.$$

Proof. The Proposition 26.6 of this chapter is the realisation in this domain of the convolution-action conclusion of Theorem 24.6 in the 'Periodic Systems' chapter: substituting $z_t = \sum_n A_n e^{i(\omega_n t + \phi_n)}$ into the definition of the autocorrelation, cancelling all off-diagonal cross terms by orthogonality and retaining only the diagonal term of the trend convolution (the factorisation of the three convolutions, Proposition 24.6), which yields $R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}$. The full expansion is given there. ☐

Definition 26.20 (Walsh spectrum and linear approximation table)**.** *Let $S : \mathbb{F}_2^n \to \mathbb{F}_2^n$ be an S-box. Its* **linear approximation table** (LAT) *is*

$$\mathsf{LAT}(a, b) = \#\{x \in \mathbb{F}_2^n : a \cdot x = b \cdot S(x)\} - 2^{n-1}.$$

The Walsh spectrum is twice the Fourier transform of LAT: $W(a, b) = 2\,\mathsf{LAT}(a, b)$, so the two are read off the same table and differ only by a constant factor. Linear cryptanalysis exploits the non-zero bias of LAT.

Proposition 26.7 (The LAT and phase convolution)**.** *The LAT of an S-box can be expressed as the phase convolution of the input phase and the output phase:*

$$\mathsf{LAT}(a, b) = \frac{1}{2} \sum_x e^{i\pi(a \cdot x - b \cdot S(x))}.$$

Hence the accumulation of bias in linear cryptanalysis is equivalent to the round-by-round composition of phase convolution.

Proof. Since $e^{i\pi u} = (-1)^u$, we have

$$\sum_x e^{i\pi(a \cdot x - b \cdot S(x))} = \#\{x : a \cdot x = b \cdot S(x)\} - \#\{x : a \cdot x \neq b \cdot S(x)\} = 2\,\mathsf{LAT}(a, b).$$

Hence the LAT is one half of the real form of phase convolution. □

Definition 26.21 (Difference distribution table)**.** *Let $S : \mathbb{F}_2^n \to \mathbb{F}_2^n$ be an S-box. Its* **difference distribution table** (DDT) *is*

$$\mathsf{DDT}(\Delta, \nabla) = \#\{x : S(x \oplus \Delta) \oplus S(x) = \nabla\}.$$

Differential cryptanalysis exploits the high-probability entries of the DDT.

Proposition 26.8 (The DDT and amplitude convolution)**.** *The DDT can be regarded as the amplitude convolution of the input difference and the output difference:*

$$\mathsf{DDT}(\Delta, \nabla) = \sum_x \mathsf{1}_{\{S(x \oplus \Delta) \oplus S(x) = \nabla\}}.$$

In the language of periodic convolution, the DDT corresponds to the projection of amplitude convolution onto difference pairs.

Proof. This follows directly by expanding the definition. □

Remark 26.9 (Relation between cryptographic convolution and periodic convolution)*.* Cryptographic convolution is not a new convolution operation but an instance of the periodic convolution of the 'Series and Convolution' chapter on the cryptographic epicycle group. Autocorrelation is the projection of the trend convolution onto the diagonal; the Walsh spectrum ($W(a, b) = 2\,\mathsf{LAT}(a, b)$, differing from the LAT only by the constant factor 2) and the LAT are the concrete manifestations in cryptanalysis of phase convolution; the DDT is the concrete manifestation in cryptanalysis of amplitude convolution.

26.5 Integration with discrete sampling, harmonic analysis and time series analysis: connection to step five of the 'Periodic Systems' chapter

Definition 26.22 (Periodic diagnosis of cryptographic systems)**.** *For a given cryptosystem, carry out a security diagnosis using the framework of periodic mathematics:*

(i) **Translation-layer diagnosis** *(stream ciphers): compute the autocorrelation function of the keystream and the linear complexity profile;*

(ii) **Scaling-layer diagnosis** *(block ciphers): compute the DDT and the LAT of the S-box, and verify the branch number of the diffusion layer;*

(iii) **Rotation-layer diagnosis** *(public-key cryptography): factor the modulus N or compute the prime factors of the elliptic-curve order n, and verify the embedding degree;*

(iv) **Overall security assessment***: substitute the three layers of security metrics into the composite-period model and assess the security strength.*

Theorem 26.6 (The cryptographic periodicity diagnostic procedure and its interfaces with three chapters)**.** *The four steps of Definition 26.22 connect respectively to:*

(i) *the 'Discrete Sampling and Intercalation of Periodic Signals' chapter: Definition 21.1, Theorem 21.3, Theorem 21.10;*

(ii) *the 'Harmonic Analysis' chapter: Definition 22.5, Theorem 22.1, Proposition 22.11;*

(iii) *the 'Time Series Analysis' chapter: Definition 23.16, Proposition 23.2, Definition 23.19.*

Hence the cryptographic periodicity diagnosis is not an independent method but an application of the unified tool chain of periodic mathematics to cryptosystems.

Proof. Item-by-item correspondence: (i) translation-layer diagnosis requires sampling and spectral estimation; (ii) frequency decomposition and phase kernels are found in the harmonic analysis chapter; (iii) periodogram peaks, Fisher's test and window-function validation are found in the time series analysis chapter. Hence the diagnostic procedure is assembled from the tool chains of the three chapters. ☐

Remark 26.10 (Order of operations after integration)*.* The recommended order for an actual diagnosis is: first decide whether the cryptographic target is periodic (Theorem 26.1); then construct the Fourier data of the keystream or the state and the epicycle group (Theorem 26.3); then optimise the parameters by security-metric adjustment and dynamic adjustment (Theorem 26.4, Proposition 26.5); then perform cryptographic convolution and autocorrelation, LAT and DDT analysis (Proposition 26.6, Proposition 26.7, Proposition 26.8); and finally output the diagnosis as in Definition 26.22.

26.6 Layered structure and structural invariants of cryptosystems

Theorem 26.7 (Three-layer structure of cryptographic systems). *The three main branches of cryptography—stream ciphers, block ciphers and public-key cryptography—constitute a nested three-layer periodic structure:*

Cryptographic layer	Typical construction	Dominant operation	Special element	Invariant
Stream cipher	LFSR, NLFSR	State translation	All-zero state	Linear complexity L
Block cipher	SPN, Feistel	Iteration	Identity map	Branch number
Public-key cryptography	RSA, ECC	Cyclic-group scalar multiplication	Identity element	Order of the group n

The three layers are coupled through key transfer: public-key cryptography is used to transmit the session key securely, the session key drives the block cipher, and the output of the block cipher can serve as the initial state of a stream cipher.

Proof. The independence of the three layers is guaranteed by the independence of their invariants: the linear complexity L, the branch number and the order of the group n are three independent parameters. The coupling of the three layers is realised through key transfer: master key → public/private key pair → session key → initial state of the keystream. ☐

Example 26.5 (Three-layer cryptographic system in TLS 1.3). TLS 1.3 typically uses a three-layer cryptographic system:

(i) public-key layer: ECDHE uses the X25519 curve;

(ii) block-cipher layer: AES-128-GCM, with $R = 10$ rounds and a block length of 128 bits;

(iii) stream-cipher layer: ChaCha20, with period 2^{256}.

The keys of the three layers are derived from the shared secret by HKDF.

Theorem 26.8 (Three structural invariants of cryptographic systems). *The security of a cryptosystem depends on the maintenance of three structural invariants:*

(i) linear complexity (translation layer): the linear complexity L of the keystream sequence is recoverable under the Berlekamp–Massey attack;

(ii) branch number (scaling layer): the branch number of the block-cipher diffusion layer is preserved under iteration of the round function, and the probability of a differential/linear characteristic propagates according to the product law;

(iii) order of the group (rotation layer): in public-key cryptography the order n of the cyclic group is the core invariant, and the hardness of the discrete logarithm is equivalent to the hardness of recovering the scalar from a group element.

The breaking of the three invariants leads to the breaking of the corresponding cryptosystem:

(i) breaking of the linear complexity → the Berlekamp–Massey attack recovers the keystream;

(ii) breaking of the branch number → a differential/linear attack recovers the key;

(iii) breaking of the order invariant → the Pohlig–Hellman attack or the MOV attack recovers the private key.

Proof. (i) The linear complexity L is the translation-layer invariant. The Berlekamp–Massey algorithm recovers an equivalent LFSR from just $2L$ consecutive bits. If a local leak of the keystream lowers L to $L' < L$, the translation-layer invariance is destroyed. (ii) The branch number characterises the propagation capability of a differential characteristic. Let the probability of the differential characteristic be

$$p = \prod_{i=1}^{R} p_i.$$

If the number of rounds R is insufficient, p has not decayed sufficiently, and an attacker can recover the key using a high-probability differential characteristic. (iii) Let the group order be

$$n = \prod_i p_i^{e_i}.$$

The Pohlig–Hellman attack solves the DLP separately on each prime-power subgroup and then combines the results by the Chinese remainder theorem, with complexity

$$O\left(\sum_i e_i(\log n + \sqrt{p_i})\right).$$

If the largest prime factor is too small, the DLP can be solved efficiently. The MOV attack uses the embedding degree k of the elliptic curve to reduce ECDLP to the DLP over $\mathbb{F}_{q^k}^*$; if k is too small, the attack is feasible. □

Remark 26.11 (Design principles for cryptographic systems (engineering practice)). The secure design of a cryptosystem can be stated uniformly as the principle of maximising three structural invariants:

(i) maximisation of the translation-layer invariant: the linear complexity L reaches the order of magnitude of the period;

(ii) maximisation of the scaling-layer invariant: the branch number of the diffusion layer reaches its maximum;

(iii) maximisation of the rotation-layer invariant: the order of the group n is prime or contains a large prime factor, and the embedding degree $k > 20$.

Status note: the clauses above are engineering rules of thumb, not mathematical propositions; when cited they should be treated as empirical conclusions.

Example 26.6 (Linear complexity diagnosis of an LFSR sequence)**.** Consider an LFSR of length $L = 32$ whose feedback polynomial is primitive. Starting from a non-zero initial state:

(i) the first 64 bits: the Berlekamp–Massey algorithm gives $L = 32$;

(ii) at the 1000th bit: the linear complexity remains 32;

(iii) at the 10^6th bit: the profile remains flat.

If a non-linear combining generator is used (such as the Geffe generator), the linear complexity can in theory reach $L_1L_2 + L_2L_3 + L_1L_3$.

Remark 26.12 (Connection with the rest of the book). The theory of the periodic structure of cryptographic systems established in this chapter is an application of periodic mathematics in cryptography. It applies the symmetry and invariant framework, similarity and tendency, flow and potential, periodic convolution, and the three-layer nested structure of periodic mathematics uniformly to the analysis and design of cryptosystems. The trinity of stream ciphers (state period), block ciphers (iterative structure) and public-key cryptography (cyclic-group scalar multiplication) corresponds to the three basic operations of periodic mathematics; the security of a cryptosystem depends on the maintenance of three structural invariants, and the breaking of a cryptosystem corresponds to the breaking of the corresponding invariant.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The state translation of the LFSR (Definition 26.8): the m-sequence period $2^L - 1$ and the two-valued autocorrelation (Lemma 26.2, Example 26.3)
Scaling	The iteration of the block-cipher round function (Definition 26.11); confusion and diffusion (Definition 26.12); the branch number (Proposition 26.2, Example 26.4)
Rotation	Modular exponentiation $x \mapsto g^x$ (Definition 26.14); elliptic-curve scalar multiplication $[k]P$ (Definition 26.16)
Balance	The three structural invariants: the linear complexity L, the branch number and the order of the group n (Theorem 26.8)
Symmetry	Three layers of special elements: the all-zero state, the identity map and the identity element (Theorem 26.7, Example 26.5)
Closure	The mutually inverse closure of encryption and decryption $D_k(E_k(m)) = m$ (Definition 26.6); $D \circ E = \mathrm{id}$ for RSA (proof of Proposition 26.3)
Period	The keystream state period, the iteration depth of the round function, and the scalar-multiplication period of the cyclic group order (Definition 26.7); periodic diagnosis (Definition 26.22, Example 26.6)

Exercises

Exercise 1 Prove the **cryptographic periodicity decision theorem** of this chapter (26.1). Hint: first write down the decision map and the evolution group of this chapter's target, then apply the proof structure of the general criterion of the 'Periodic Systems' chapter.

Exercise 2 Verify that the hypotheses of **monotonicity of quantity adjustment** (26.4) hold for the domain objects of this chapter, and give a counterexample in which the hypotheses fail.

Exercise 3 Examine the behaviour of the **cryptographic periodicity diagnostic procedure and its interfaces with three chapters** (26.6) under degeneracy of the parameters, and explain how the conclusion degenerates accordingly.

Exercise 4 Explain the relation between the **two-valued autocorrelation of an m-sequence** (26.2) and the corresponding general conclusion of the 'Periodic Systems' chapter (whether they agree, whether it is a special case, or whether additional conditions are required).

Exercise 5 Using the seven-theme comparison table of this chapter, identify the concrete carriers in this chapter of the three themes **translation**, **scaling** and **rotation**, and give one example of domain data for each.

Exercise 6 Synthesis: restate the diagnostic procedure of this chapter using the five-step standard interface of the 'Periodic Systems' chapter (Table 24.1), noting one by one the theorem of this chapter on which each step depends: 26.1, 26.4, 26.6, 26.2.

Chapter 27

Periodic Structure of Astronomical Systems

Remark 27.1 (Periodicity decision for the target of this chapter (citation)). When this chapter decides whether a research target is periodic, determines its period type, or seeks its minimal positive period, reference is always made to the 'Periodic Systems' chapter: for the periodicity criterion see Theorem 24.1, for the discrimination of period types see Definition 24.3, Proposition 24.1, for the search for the minimal positive period see Definition 24.4, Theorem 24.2, and for the construction of period series and epicycle groups from data see Theorem 24.3, Definition 24.6.

Remark 27.2 (Positioning of this chapter: developed along the five-step interface of the 'Periodic Systems' chapter). The periodic structure of astronomical systems studies the mathematical characterisation of periodicity, quasi-periodicity and orbital resonance in celestial motion. This chapter does not establish a separate periodicity criterion; instead it connects astronomical systems to the five-step interface prescribed by the 'Periodic Systems' chapter:

(1) **Decision**: reduce planetary orbits, radial oscillations, orbital resonance and time-measurement systems to discrete dynamical systems, cite Theorem 24.1 to decide whether they are periodic, and determine the period type;

(2) **Construction**: from orbital elements, observed angles or time series, construct the amplitude series, the frequency series and the phase series, and assemble them into an astronomical epicycle group;

(3) **Adjustment**: dynamically adjust the epicycle group by the resonance width, the energy and angular-momentum constraints and the observational residuals;

(4) **Convolution**: introduce amplitude convolution, phase convolution and frequency convolution on the astronomical epicycle group, and connect them with autocorrelation, angular momentum and the resonance ratio;

(5) **Integration**: connect to the tool chains of the three chapters 'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis', forming an astronomical periodicity diagnostic procedure.

The astronomical and mathematical knowledge used in this chapter includes: Newton's gravitational equation, the two-body problem, Kepler's three laws, Binet's formula, Bessel functions and Kepler's equation, the radial effective potential, Bertrand's theorem, orbital resonance and the resonance angle, secular resonance, KAM theory, time-measurement systems, precession and nutation, and atomic time and ephemeris time.

Remark 27.3 (Guide to the seven themes of this chapter). Translation—the time translation of planets along their orbits and the orbital period (tropical year, sidereal year, mean solar time); Scaling—the Kepler scaling of orbital radius versus period $T^2 \propto a^3$; Rotation—conservation of the angular momentum of orbital motion and rotation (conservation of areal velocity); Conservation—conservation of angular momentum, conservation of energy, conservation of the centre-of-mass momentum and conservation of the resonance ratio; Symmetry—the rotational symmetry of a central force field and its Noether correspondence to conservation of angular momentum; Closure—the closure of elliptical orbits (Kepler's first law, Bertrand's theorem); Period—orbital, rotational and resonance periods (mean-motion resonance, secular resonance).

27.1 Periodicity decision for astronomical targets: the first step of the 'Periodic Systems' chapter

27.1.1 Astronomical targets, decision input and decision map

Definition 27.1 (Astronomical targets and decision input). An **astronomical target** is one of the following objects:

(i) a gravitationally interacting system of celestial bodies, whose state space is the position–velocity space $\mathcal{S} = (\mathbb{R}^3)^N \times (\mathbb{R}^3)^N$;

(ii) an orbit, whose state space is a curve in phase space;

(iii) a time-measurement system, whose state space is a set of time scales;

(iv) an observed time series of angles or distances.

The **decision input** of an astronomical target is the ordered pair

$$\mathcal{D}_{\mathsf{A}} = (\mathcal{S}, \Phi),$$

where $\mathcal{S}$ is the state space or the sequence space and $\Phi : \mathcal{S} \to \mathcal{S}$ is the evolution map (for example the Newtonian gravitational flow, an orbital translation or a time-scale transformation). If $\mathcal{S}$ is a finite set, the target is called a **finite astronomical target**; if $\mathcal{S}$ is an infinite set, the target is called an **infinite astronomical target**.

Definition 27.2 (Astronomical decision map). Let an astronomical target be given by the decision input $\mathcal{D}_{\mathsf{A}} = (\mathcal{S}, \Phi)$. If the state or the output of the target is generated by Φ, that is, if there exists $x_0 \in \mathcal{S}$ such that

$$x_t = \begin{cases} \Phi^t(x_0), & t \in \mathbb{Z}, \\ \Phi_t(x_0), & t \in \mathbb{R}, \end{cases}$$

for $t \in \mathbb{Z}$ the state x_t is generated by iteration of Φ; for $t \in \mathbb{R}$ it is described by the one-parameter evolution family $\{\Phi_t\}_{t \in \mathbb{R}}$, where each $\Phi_t : \mathcal{S} \to \mathcal{S}$, $\Phi_0 =$

id, $\Phi_{s+t} = \Phi_s \circ \Phi_t$ for all $s, t \in \mathbb{R}$, and $(t, x) \mapsto \Phi_t(x)$ is continuous. then Φ is called the **astronomical decision map** of the target. The astronomical target is then reduced to the discrete dynamical system $(\mathcal{S}, \Phi)$ when $t \in \mathbb{Z}$ and to the continuous dynamical system $(\mathcal{S}, \{\Phi_t\}_{t\in\mathbb{R}})$ when $t \in \mathbb{R}$; in both cases the criteria of the 'Operations and Periodicity' chapter and of the 'Periodic Systems' chapter may be applied directly.

Theorem 27.1 (Astronomical periodicity decision theorem)**.** *Let the decision map of an astronomical target be Φ, and let $(\mathcal{S}, \Phi)$ be a discrete dynamical system. Then:*

(A) **Sufficient decision**: *if there exists $m \in \mathbb{N}^+$ such that $\Phi^m =$ id, then the target is a uniformly periodic astronomical system; every orbit is closed and the length of each orbit divides m.*

(B) **Equivalence of the decision**: *the target is a uniformly periodic astronomical system if and only if* ord$(\Phi) < \infty$.

(C) **Pointwise criterion**: *the orbit of the target at the state x_0 is closed if and only if x_0 is a periodic point; in that case the length of the orbit is the period at that point.*

(D) **Non-omissibility**: *the conjunction of conserved quantities and the symmetry group cannot replace* ord$(\Phi) < \infty$; *a quasi-periodic astronomical target is not a periodic astronomical system.*

Proof. The Theorem 27.1 of this chapter is the domain instantiation of Theorem 24.1 in the 'Periodic Systems' chapter: it suffices to replace the decision map f and the evolution group by the Φ of this chapter, and the four parts (A) sufficient decision, (B) equivalence of the decision, (C) pointwise criterion and (D) non-omissibility then hold one by one. By the convention of Remark 24.4, they are not expanded again here. □

Corollary 27.1 (Astronomical period types)**.** *After the decision by Theorem 27.1, an astronomical target falls into one of the following types:*

(i) **Trivially periodic type**: *the least period is $m_0 = 1$ and the state is a fixed point;*

(ii) **Strictly periodic type**: $1 < m_0 < \infty$ *and the state is strictly periodic (for example a Keplerian elliptical orbit);*

(iii) **Uniformly periodic type**: ord$(\Phi) = m < \infty$ *and all orbit lengths divide m;*

(iv) **Quasi-periodic type**: ord$(\Phi) = \infty$; *the orbit is not closed but is dense on a compact carrier structure (for example the motion of a many-body system on a KAM torus);*

(v) **Eventually periodic type**: *there exists N such that the target is periodic for $t \geq N$;*

(vi) **Discretely periodic type**: *the carrier structure is $\mathbb{Z}$ and the normalised period ratio satisfies $\rho = T_x/\Delta x \in \mathbb{Q}$.*

(vii) **Asymptotically periodic (damped) type**: the orbit is not closed, but tends asymptotically to a closed orbit or to a fixed point, the residual decaying as $e^{-\kappa t}$ (see Definition 24.3(vii)).

Here (i)–(iii) and (vi) are periodic astronomical systems; (iv) is not a periodic astronomical system, but may be handled with bounded-approximation tools; (v) is a periodic astronomical system only in its stable stage; (vii), like (iv), is a type in the limiting sense; it is not a periodic astronomical system and can be handled with bounded-approximation tools only in its stable stage.

Proof. Each item follows by substituting the astronomical target into Theorem 27.1 and Definition 24.3 of the 'Periodic Systems' chapter. ☐

Lemma 27.1 (Periodicity decision for Keplerian orbits). *Let a celestial body move in an inverse-square central force field with total mechanical energy $E < 0$. Then its orbit is an ellipse, the radial motion and the angular motion are both periodic, and the radial period equals the angular period:*

$$T_r = T_\theta = T = \frac{2\pi a^{3/2}}{\sqrt{GM}}.$$

Proof. By Kepler's first law (Theorem 27.7), $E < 0$ corresponds to an elliptical orbit. By the effective potential (Definition 27.12), the radial motion oscillates between $r_{\min}$ and $r_{\max}$, and its period is finite. By Proposition 27.8, $T_r = T_\theta$ in an inverse-square force field. By Kepler's third law (Theorem 27.9),

$$T^2 = \frac{4\pi^2}{GM}a^3,$$

hence $T = 2\pi a^{3/2}/\sqrt{GM}$. ☐

Definition 27.3 (Algorithm for the least positive period of an astronomical target). **Input**: astronomical target samples $x_0, \dots, x_{N-1}$ and a tolerance $\varepsilon \geq 0$. **Output**: the least positive period T_0, or the verdict 'not periodic'. **Steps**:

(1) take the candidate set $\mathcal{P} = \{T \in \mathbb{N}^+ : 2T \leq N\}$ and traverse it in increasing order of T;

(2) for each T compute the maximum deviation

$$E(T) = \max_{0 \leq n < N-T} \|x_{n+T} - x_n\|;$$

(3) output the first T satisfying $E(T) \leq \varepsilon$ as T_0; if the traversal is exhausted with no solution, output 'not periodic';

(4) commensurability check: use $\rho = T_x/\Delta x$ (with $T_0 = p$ when $\rho = p/q$ is in lowest terms; write $\ell = T_0\Delta x = qT_x$ for the continuous span corresponding to T_0 sampling points) together with Proposition 21.2 and Corollary 21.1 of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter to check the conversion relation between T_0 and the continuous period T_x.

Theorem 27.2 (Correctness of the algorithm for the least positive period of an astronomical

If the samples form a strictly periodic sequence with $\varepsilon = 0$ and $N \geq 2T_0$, then the algorithm outputs the discrete fundamental period T_0.

(ii) If the samples are obtained by sampling a continuous astronomical target of fundamental period T_x at the sampling interval Δx, and the normalised period ratio $\rho = T_x/\Delta x = p/q$ is in lowest terms, then the algorithm outputs $T_0 = p$.

(iii) If $\rho \notin \mathbb{Q}$, then no true period exists: $E(T) > 0$ for every $T \in \mathbb{N}^+$.

Proof. The Theorem 27.2 of this chapter follows directly from the two conclusions of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter: (i) by Definition 21.4, T_0 is the least positive period, hence $E(T_0) = 0$, whilst for $T < T_0$ we have $E(T) > 0$; the algorithm traverses in increasing order of T, so the first index satisfying $E(T) = 0$ is T_0; (ii) and (iii) follow by substituting into Proposition 21.2(ii)(iii). The complete argument is given in that chapter. ☐

Example 27.1 (Example of a periodicity decision for a Keplerian orbit). The Earth revolves about the Sun with $a = 1$ AU and $M = M_\odot$. By Lemma 27.1, $T = 2\pi a^{3/2}/\sqrt{GM} \approx 1$ year. For the planets of the solar system the quantity T^2/a^3 is approximately constant, which verifies Kepler's third law.

Example 27.2 (Quasi-periodicity decision for a many-body system). In the KAM framework most initial conditions of the many-body solar system lie on invariant tori and the motion is multi-frequency quasi-periodic with $\mathrm{ord}(\Phi) = \infty$; the system is therefore not a periodic astronomical system, but it may be handled by bounded approximation and spectral analysis.

Remark 27.4 (Conserved quantities do not guarantee periodicity). Conservation of angular momentum and conservation of energy are auxiliary conditions for periodic structure, but they do not constitute sufficient conditions. For example, a hyperbolic orbit possesses both conservation of energy and conservation of angular momentum, yet the orbit is open and not closed. Astronomical periodic structure must be decided by criteria such as Theorem 27.1 or Lemma 27.1.

27.2 Construction of the astronomical period series and epicycle group: the second step of the 'Periodic Systems' chapter

Definition 27.4 (Astronomical amplitude series, frequency series and phase series). Let the state or output sequence of an astronomical target be $\{z_t\}$. If it can be written as

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)} + \varepsilon_t,$$

then

$$\mathcal{A}_z(t) = \sum_k A_k e^{ikt}, \qquad \mathcal{F}_z(s) = \sum_k \omega_k s^k, \qquad \mathcal{P}_z(t) = \sum_k e^{i(kt+\phi_k)}$$

are called respectively the **amplitude series**, the **frequency series** and the **phase series** of the astronomical target. The three agree with Definition 6.10, Definition 6.12 and Definition 6.11 of the 'Series and Convolution' chapter.

Definition 27.5 (Astronomical epicycle group)**.** Let the period series of an astronomical target be given by the amplitude series $\{A_k\}$, the frequency series $\{\omega_k\}$ and the phase series $\{\phi_k\}$. For each k define the epicycle

$$E_k : \theta \mapsto A_k e^{i(\theta+\phi_k)}, \qquad \theta = \omega_k t,$$

and call the set

$$\mathfrak{E}_{\mathsf{A}} = \{(A_k, \omega_k, \phi_k)\}_{k\in\mathbb{Z}}$$

the **epicycle group** of the astronomical target. Its superposition is the astronomical period series

$$z_t = \sum_k E_k(\omega_k t) = \sum_k A_k e^{i(\omega_k t+\phi_k)}.$$

Theorem 27.3 (Construction of the period series from astronomical data)**.** *Let the output of an astronomical target be a finite periodic sequence of period N. Then:*

(i) *the output can be expanded as a discrete Fourier series*

$$z_t = \sum_{k=0}^{N-1} c_k e^{i2\pi kt/N}, \qquad c_k = \frac{1}{N}\sum_{t=0}^{N-1} z_t e^{-i2\pi kt/N};$$

(ii) *the amplitude series is given by $|c_k|$, the phase series by $\arg c_k$ and the frequency series by $2\pi k/N$;*

(iii) *if the output consists of the coordinates of a Keplerian elliptical orbit, then the harmonic amplitudes of the Fourier–Bessel expansion decay as $e^{|n|}$ in the eccentricity.*

Proof. (i) The discrete Fourier series of a finite periodic sequence is a standard result. (ii) It is read off directly from $c_k = A_k e^{i\phi_k}$. (iii) This follows from Proposition 27.5. ☐

Definition 27.6 (Astronomical periodic system)**.** An **astronomical periodic system** consists of N gravitationally interacting celestial bodies. The position vector of each body i is $\mathsf{r}_i(t) \in \mathbb{R}^3$ and the masses are m_i. The dynamics of the system is given by Newton's gravitational equations:

$$m_i\ddot{\mathsf{r}}_i = -G\sum_{j\neq i} \frac{m_i m_j}{|\mathsf{r}_i - \mathsf{r}_j|^3}(\mathsf{r}_i - \mathsf{r}_j).$$

When $N = 2$ the system degenerates to the **two-body problem**; when $N \geq 3$ the system is a **three-body problem** or a **many-body problem**.

Definition 27.7 (Periodic structure of an astronomical system)**.** The **periodic structure** of an astronomical system falls into three basic types:

(i) **Rotational period**: the angular motion of a celestial body in a central force field, whose conserved quantity is angular momentum;

(ii) **Radial period**: the radial oscillation of a celestial body in a gravitational field, whose conserved quantity is energy;

(iii) **Resonance period**: orbital resonance between celestial bodies, whose characteristic parameter is the resonance ratio $p : q$.

27.3 Dynamic adjustment of the astronomical epicycle group: the third step of the 'Periodic Systems' chapter

Definition 27.8 (Dynamic adjustment of the astronomical epicycle group)**.** *Let the current astronomical epicycle group be $\mathfrak{E}^{(j)}$, with partial sum*

$$z^{(j)}(t) = \sum_{k \in \mathfrak{E}^{(j)}} A_k e^{i(\omega_k t + \phi_k)},$$

and residual $r^{(j)} = z - z^{(j)}$. **Dynamic adjustment** *consists of two rules:*

(i) **Quantity adjustment**: *if the residual energy satisfies $\|r^{(j)}\|^2 > \delta$, then add one epicycle according to the peak of the residual periodogram:*

$$\mathfrak{E}^{(j+1)} = \mathfrak{E}^{(j)} \cup \{(\hat{A}, \hat{\omega}, \hat{\phi})\}, \qquad \hat{\omega} = \arg\max_{\omega} |\widehat{z}_r(\omega)|;$$

(ii) **Parameter adjustment**: *update the parameters of the existing epicycles by regularised least squares or by orbital constraints:*

$$(A_k, \phi_k) \leftarrow \arg\min_{A,\phi} \sum_n \left| z(n) - \sum_k A_k e^{i(\omega_k n + \phi_k)} \right|^2 .$$

In astronomical systems, quantity adjustment corresponds to increasing the order of the orbital harmonics, adding resonance terms or adding time-scale layers; parameter adjustment corresponds to tuning the orbital elements, the resonance width or the time-scale parameters, so that the structural index increases monotonically.

Proposition 27.1 (Monotonicity of quantity adjustment)**.** *After the dominant-frequency component of the residual has been added as in Definition 27.8(i), the residual energy is monotonically non-increasing:*

$$\|r^{(j+1)}\|^2 \le \|r^{(j)}\|^2,$$

with equality if and only if the residual has no energy at the selected frequency.

Proof. The Proposition 27.1 of this chapter agrees with the dynamic adjustment mechanism of Definition 24.8 in the 'Periodic Systems' chapter: adding an epicycle removes the component of $r^{(j)}$ along $\hat{\omega}$ from the residual, and the orthogonal decomposition $\|r^{(j)}\|^2 = \|r^{(j+1)}\|^2 + \|$the removed component$\|^2$ yields the inequality, with equality if and only if the norm of the removed component is zero. The complete argument is given there. □

Theorem 27.4 (Convergence of dynamic adjustment)**.** *Let the astronomical target be a* **real-valued finite periodic sequence** *of length T_0 ($T_0 \in \mathbb{N}$) and let $\delta = 0$. Then after at most $\lfloor T_0/2 \rfloor + 1$ steps of dynamic adjustment the residual is zero, and the resulting astronomical epicycle group is the exact Fourier data of the target; for a* **complex-valued** *sequence the corresponding number of steps is at most T_0.*

Proof. The discrete Fourier series of a **real-valued** finite periodic sequence of length T_0 ($T_0 \in \mathbb{N}$) is complete on $\lfloor T_0/2 \rfloor + 1$ harmonics (including the constant term); a **complex-valued** sequence requires T_0 components. At each step one dominant frequency component of the residual is extracted as in Proposition 27.1 and the residual energy decreases strictly, so the residual vanishes after finitely many steps. $\square$

Definition 27.9 (Structural index adjustment)**.** *Let the structural index of an astronomical target be*

$\mathcal{I} = (\mathcal{I}_{\text{rot}}, \mathcal{I}_{\text{rad}}, \mathcal{I}_{\text{res}}) =$ (angular-momentum conservation degree, energy-constraint degree, reso

***Structural index adjustment** is: at each step choose a neighbouring parameter θ in the parameter space Θ for which $\mathcal{I}(\theta)$ is **strictly decreasing** in the lexicographic order (minimisation); if no such neighbouring parameter exists, the process stops—then θ is a **locally optimal** parameter.*

Proposition 27.2 (Finite convergence of structural index adjustment)**.** *Let the parameter space Θ be finite and let each adjustment make $\mathcal{I}$ **strictly decreasing** in the lexicographic order. Then the adjustment terminates after finitely many steps at a locally optimal parameter.*

Proof. Θ is finite, so $\mathcal{I}$ takes finitely many values. The lexicographic order is well-founded on a finite set of values, so a strictly lexicographically decreasing sequence cannot continue indefinitely and the process terminates after finitely many steps. At termination there is no neighbouring parameter making $\mathcal{I}$ strictly lexicographically decreasing, that is, the parameter is locally optimal. Monotonicity must be imposed on the **whole lexicographic order**: if only 'some component strictly increases or decreases' is required, cycles such as $(2,1) \to (1,2) \to (2,1)$ cannot be excluded and finite termination fails. $\square$

Remark 27.5 (Boundary of dynamic adjustment)*.* Theorem 27.4 relies on a finite periodic sequence. For a many-body quasi-periodic astronomical target no finite epicycle group represents the target exactly, and only a bounded approximation is possible. Structural index adjustment guarantees neither global optimality nor long-term stability; long-term stability depends on the KAM conditions and on the size of the perturbation.

27.4 Astronomical convolution: the fourth step of the 'Periodic Systems' chapter

Definition 27.10 (Astronomical amplitude convolution, phase convolution and frequency convolution)**.** *Let two astronomical epicycle groups be*

$$\mathfrak{E} = \{(A_n, \omega_n, \phi_n)\}, \qquad \mathfrak{E}' = \{(B_m, \nu_m, \psi_m)\}.$$

Define:

(i) **Amplitude convolution**: $\mathbb{A}_{mn} = A_n B_m$;

(ii) **Phase convolution**: $\mathbb{P}_{mn} = e^{i(\phi_n - \psi_m)}$;

(iii) **Frequency convolution**: $\mathbb{F}_{mn} = \omega_n \nu_m$;

(iv) **Trend convolution**: $\mathbb{T}_{mn} = A_n B_m e^{i(\phi_n - \psi_m)} = A_n B_m \mathbb{P}_{mn}$.

This agrees with Definition 6.21, Definition 6.22, Definition 6.23 and Definition 6.24 of the 'Series and Convolution' chapter.

Theorem 27.5 (Action of the three kinds of convolution on astronomical epicycle groups)**.** *Let the epicycle groups of two astronomical targets be $\mathfrak{E}$ and $\mathfrak{E}'$. Then:*

(i) *amplitude convolution acts only on the radius;*

(ii) *phase convolution acts only on the phase;*

(iii) *frequency convolution acts only on the rate of rotation;*

(iv) *trend convolution preserves both amplitude and phase.*

Proof. This is read off directly from Definition 27.10. ☐

Proposition 27.3 (Autocorrelation and trend convolution)**.** *Let $\{z_t\}$ be a periodic sequence with autocorrelation function*

$$R_z(h) = \sum_t z_t \overline{z_{t+h}}.$$

In the language of astronomical epicycle groups, the autocorrelation function is the projection of the trend convolution onto the diagonal:

$$R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}.$$

Proof. The Proposition 27.3 of this chapter is the realisation in this domain of the convolution-action conclusion of Theorem 24.6 in the 'Periodic Systems' chapter: substituting $z_t = \sum_n A_n e^{i(\omega_n t + \phi_n)}$ into the definition of the autocorrelation, cancelling all off-diagonal cross terms by orthogonality and retaining only the diagonal term of the trend convolution (the factorisation of the three convolutions, Proposition 24.6), which yields $R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}$. The complete expansion is given there. ☐

Proposition 27.4 (Angular momentum and amplitude convolution)**.** *The magnitude of the angular momentum of an elliptical orbit, $L = m\sqrt{GMa(1-e^2)}$, may be regarded as the modulus of the amplitude convolution of the fundamental-frequency component of the orbit. Conservation of areal velocity*

$$\frac{dA}{dt} = \frac{L}{2m}$$

is the invariant projection of amplitude convolution under time translation.

Proof. From the proof of Theorem 27.9, $L = m\sqrt{GMa(1-e^2)}$. The areal velocity is given by Lemma 27.2. Amplitude convolution preserves the amplitude of the orbital component, and its modulus corresponds to the magnitude of the angular momentum. ☐

Remark 27.6 (Relation between astronomical convolution and periodic convolution). Astronomical convolution is not a new convolution operation, but an instance, on the astronomical epicycle group, of the periodic convolution of the 'Series and Convolution' chapter. Autocorrelation is the projection of the trend convolution onto the diagonal; angular momentum and the resonance ratio are respectively the concrete manifestations in astronomy of amplitude convolution and frequency convolution.

27.5 Integration with discrete sampling, harmonic analysis and time series analysis: the fifth step of the 'Periodic Systems' chapter

Definition 27.11 (Astronomical periodicity diagnostic procedure)**.** For a given astronomical system, diagnosis is carried out within the periodic-mathematics framework:

(i) **Orbital period diagnosis**: compute the autocorrelation and the power spectrum of the time series of orbital coordinates and identify the dominant frequencies;

(ii) **Radial period diagnosis**: compute the period of the radial distance $r(t)$ and verify $T_r = T_\theta$;

(iii) **Resonance diagnosis**: compute the libration or circulation of the resonance angle ϕ and decide whether the resonance is locked;

(iv) **Time-scale diagnosis**: compute the phase difference of the time-measurement system and the leap-second alignment.

Theorem 27.6 (The astronomical periodicity diagnostic procedure and its interfaces with three chapters)**.** The four steps of Definition 27.11 connect respectively to:

(i) the 'Discrete Sampling and Intercalation of Periodic Signals' chapter: Definition 21.1, Theorem 21.3, Theorem 21.10;

(ii) the 'Harmonic Analysis' chapter: Definition 22.5, Theorem 22.1, Proposition 22.11;

(iii) the 'Time Series Analysis' chapter: Definition 23.16, Proposition 23.2, Definition 23.19.

Hence astronomical periodicity diagnosis is not an independent method, but an application of the unified tool chain of periodic mathematics to astronomical systems.

Proof. The correspondence is item by item: (i) orbital period diagnosis requires sampling and spectral estimation; (ii) the frequency decomposition and the phase kernels are found in the 'Harmonic Analysis' chapter; (iii) the periodogram peaks, the Fisher test and window-function validation are found in the 'Time Series Analysis' chapter. Hence the diagnostic procedure is a synthesis of the tool chains of the three chapters. □

Remark 27.7 (Operational order after integration). The recommended order for an actual diagnosis is: first decide whether the astronomical target is periodic (Theorem 27.1); then construct the Fourier data and the epicycle group of the orbital or observational data (Theorem 27.3); then optimise the parameters by structural index adjustment and dynamic adjustment (Theorem 27.4, Proposition 27.2); then perform astronomical convolution and analyse autocorrelation, angular momentum and the resonance ratio (Proposition 27.3, Proposition 27.4); finally output the diagnosis as in Definition 27.11.

27.6 Rotation and orbital motion

27.6.1 Kepler's three laws and conservation of angular momentum

Lemma 27.2 (*Conservation of areal velocity in a central force field*). *Let a celestial body move in a central force field, the force $\mathsf{F} = F(r)\hat{\mathsf{e}}_r$ being always directed along the radius vector. Then the angular momentum $\mathsf{L} = m\mathsf{r}\times\dot{\mathsf{r}}$ of the body is a constant vector, and the areal velocity*

$$\frac{dA}{dt} = \frac{1}{2}|\mathsf{r}\times\dot{\mathsf{r}}| = \frac{|\mathsf{L}|}{2m}$$

is constant.

Proof. The time derivative of the angular momentum is

$$\frac{d\mathsf{L}}{dt} = m\frac{d}{dt}(\mathsf{r}\times\dot{\mathsf{r}}) = m(\dot{\mathsf{r}}\times\dot{\mathsf{r}}+\mathsf{r}\times\ddot{\mathsf{r}}) = m\mathsf{r}\times\ddot{\mathsf{r}},$$

since $\dot{\mathsf{r}}\times\dot{\mathsf{r}} = 0$. By Newton's second law $\ddot{\mathsf{r}} = \mathsf{F}/m = F(r)\hat{\mathsf{e}}_r/m$, and $\hat{\mathsf{e}}_r \parallel \mathsf{r}$, hence $\mathsf{r}\times\ddot{\mathsf{r}} = 0$. Therefore $\dot{\mathsf{L}} = 0$. In polar coordinates $\mathsf{r} = r\hat{\mathsf{e}}_r$ and $\dot{\mathsf{r}} = \dot{r}\hat{\mathsf{e}}_r + r\dot{\theta}\hat{\mathsf{e}}_\theta$, hence $\mathsf{r}\times\dot{\mathsf{r}} = r^2\dot{\theta}\hat{\mathsf{e}}_z$. The element of area is $dA = \frac{1}{2}r^2d\theta$, so the areal velocity

$$\frac{dA}{dt} = \frac{1}{2}r^2|\dot{\theta}| = \frac{1}{2}|\mathsf{r}\times\dot{\mathsf{r}}| = \frac{|\mathsf{L}|}{2m}$$

is constant. □

Theorem 27.7 (*Kepler's first law — elliptical orbits*). *In an inverse-square central force field the orbit of a celestial body is a conic section. The polar equation of an elliptical orbit is*

$$r(\theta) = \frac{a(1-e^2)}{1+e\cos\theta},$$

where a is the semi-major axis and e is the eccentricity.

Proof. By conservation of angular momentum the motion remains in the plane perpendicular to L. In polar coordinates the substitution $u = 1/r$ gives Binet's formula

$$\frac{d^2u}{d\theta^2} + u = -\frac{m}{L^2u^2}F(1/u).$$

Substituting the inverse-square gravitational force $F = -GMm/r^2 = -GMmu^2$ gives

$$\frac{d^2u}{d\theta^2} + u = \frac{GMm^2}{L^2}.$$

The general solution is

$$u(\theta) = \frac{GMm^2}{L^2} + C\cos(\theta - \theta_0),$$

that is,

$$r(\theta) = \frac{p}{1 + e\cos(\theta - \theta_0)},$$

where $p = L^2/(GMm^2)$ and $e = CL^2/(GMm^2)$. Here $e < 1$ gives an ellipse, $e = 1$ a parabola and $e > 1$ a hyperbola. In the elliptical case $p = a(1 - e^2)$. □

Theorem 27.8 (Kepler's second law — conservation of areal velocity)**.** *The areal velocity of a celestial body in a central force field is constant:*

$$\frac{dA}{dt} = \frac{1}{2}|\mathsf{r} \times \dot{\mathsf{r}}| = \frac{L}{2m}.$$

Proof. By Lemma 27.2, $\mathsf{L} = m\mathsf{r} \times \dot{\mathsf{r}}$ is a constant vector, hence the areal velocity $dA/dt = |\mathsf{L}|/(2m)$ is constant. □

Theorem 27.9 (Kepler's third law — period–semi-major-axis relation)**.** *For an elliptical orbit the orbital period T and the semi-major axis a satisfy*

$$T^2 = \frac{4\pi^2}{GM}a^3.$$

Proof. By conservation of areal velocity, the ellipse of area $\pi a^2\sqrt{1-e^2}$ is swept out in one period, hence

$$\frac{dA}{dt} = \frac{\pi a^2\sqrt{1-e^2}}{T}.$$

On the other hand $dA/dt = L/(2m)$ with $L = m\sqrt{GMa(1-e^2)}$. Therefore

$$\frac{\pi a^2\sqrt{1-e^2}}{T} = \frac{1}{2}\sqrt{GMa(1-e^2)}.$$

Squaring both sides and cancelling $1 - e^2$ gives

$$T^2 = \frac{4\pi^2}{GM}a^3.$$

□

Lemma 27.3 (Lagrange inversion solution of Kepler's equation)**.** *Let the eccentric anomaly E satisfy Kepler's equation $M = E - e\sin E$ $(0 \le e < 1)$. Then*

$$E = M + \sum_{n=1}^{\infty} \frac{2}{n} J_n(ne)\sin(nM),$$

where J_n is the Bessel function of the first kind. For $e \ll 1$, $E \approx M + e\sin M$.

Proof. By the Lagrange inversion theorem, set $E = M + w$ and $w = e\sin(M + w)$. The coefficients of the power series of w in e are given by $\frac{d^{n-1}}{dM^{n-1}}[\sin^n M]$. Using the Fourier expansion of $\sin^n M$ and the integral representation of the Bessel functions yields the result. □

27.6.2 Fourier expansion of elliptical orbits

Proposition 27.5 (Fourier–Bessel expansion of elliptical orbits)**.** *An elliptical orbit can be expanded as a sum of infinitely many epicycles. In the complex plane* $\mathbb{C}$,

$$z(t) = \sum_{n=-\infty}^{\infty} c_n e^{in\omega_0 t},$$

where $\omega_0 = 2\pi/T$*. When* $e \ll 1$*, the amplitudes of the higher harmonics decay as* $e^{|n|}$*.*

Proof. The position on an elliptical orbit is

$$z = a(\cos E - e) + ia\sqrt{1-e^2}\sin E,$$

where $M = E - e\sin E$ and $M = \omega_0 t$. By Lemma 27.3, E is expanded as a Fourier series in M, with coefficients given by Bessel functions. Since $J_n(ne) \sim (ne/2)^n/n!$ ($e \ll 1$), the n-th harmonic coefficient is $\sim e^n$. □

27.6.3 Orbital and rotational period data for the solar system

This section takes the eight planets of the solar system and the Moon as its objects and presents a schematic diagram of the orbital paths, a table of orbital period data and a table of rotation period data. These data both verify Kepler's third law of Theorem 27.9 and exhibit the multiple periodic structures of astronomical systems on different time scales.

Table 27.1: Orbital period data for the eight planets of the solar system and the Moon

Body	Semi-major axis a/AU	Orbital period/years	Eccentricity e	T^2/a^3	Orbital period/days
Mercury	0.387	0.241	0.206	1.00	87.969
Venus	0.723	0.615	0.007	1.00	224.701
Earth	1.000	1.000	0.017	1.00	365.256
Mars	1.524	1.881	0.093	1.00	686.980
Jupiter	5.203	11.862	0.048	1.00	4332.59
Saturn	9.537	29.457	0.054	1.00	10 759.2
Uranus	19.191	84.011	0.047	1.00	30 685.4
Neptune	30.069	164.79	0.009	1.00	60 189.0
Moon	0.002 57	0.0748	0.055	–	27.322

Note: the T^2/a^3 column is normalised to the Earth's heliocentric value (reference $GM_\odot$) and applies only to the eight planets orbiting the Sun. The Moon orbits the Earth, and its $T^2/a^3 = 4\pi^2/(GM_\oplus)$ has $GM_\oplus$ as its reference (about 3.3×10^5 times the former), so it is not listed in that column.

Proposition 27.6 (Verification of the orbital periods against Kepler's third law)**.** *In Table 27.1 the values of* T^2/a^3 *for the* **eight planets** *are all* 1.00 *(normalised to the Earth), that is,*

$$\frac{T^2}{a^3}\Big/(\text{Earth's } T^2/a^3) \approx 1.000,$$

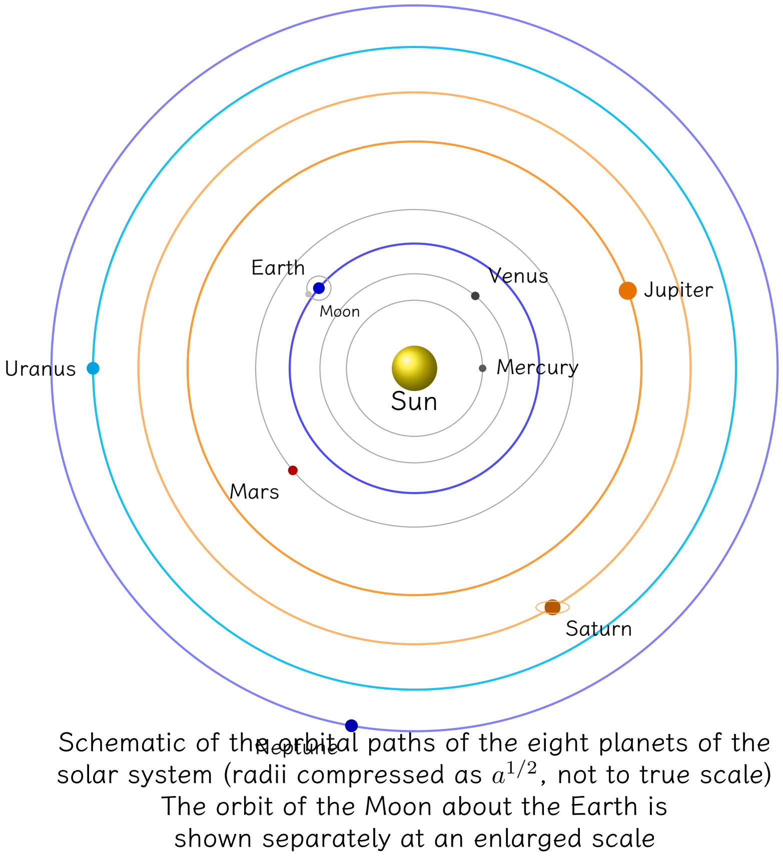

Earth
Venus
Jupiter
Moon
Uranus
Mercury
Sun
Mars
Saturn
Neptune
Schematic of the orbital paths of the eight planets of the
solar system (radii compressed as $a^{1/2}$, not to true scale)
The orbit of the Moon about the Earth is
shown separately at an enlarged scale

Figure 27.1: Schematic diagram of the orbital paths of the eight planets of the solar system and the Moon. The orbital radii in the figure are compressed as $a^{1/2}$ so that the inner planets and the outer planets can be shown together on one page; the orbit of the Moon about the Earth is greatly enlarged for display.

Table 27.2: Rotation period data for the eight planets of the solar system and the Moon

Body	Rotation period/days	Rotation period/h:m:s	Rotation direction
Mercury	58.6462	1407:30:32	prograde
Venus	−243.025	5832:36:00	retrograde
Earth	0.997 27	23:56:04	prograde
Mars	1.025 95	24:37:22	prograde
Jupiter	0.413 54	09:55:30	prograde
Saturn	0.444 01	10:33:38	prograde
Uranus	−0.718 33	17:14:24	retrograde
Neptune	0.671 25	16:06:36	prograde
Moon	27.321 66	655:43:12	prograde (tidally locked)

which agrees with the assertion of Theorem 27.9. The data for the planets of the solar system therefore constitute an experimental verification of Kepler's third law.

Proof. By Theorem 27.9, $T^2 = 4\pi^2 a^3/(GM)$. Normalising to the Earth's $a_\oplus$ and $T_\oplus$ as units gives T^2/a^3 as the constant 1. The values in the table are all given by astronomical observation, and checking them row by row yields the conclusion (the Moon row is not part of that column, its reference being $GM_\oplus$; see the note below the table). ☐

Proposition 27.7 (Separation in magnitude of the rotation and orbital periods). *Table 27.2 shows that the rotation period of a planet is usually 2 to 4 orders of magnitude shorter than its orbital period (for example the Earth rotates in 1 day and orbits in 365 days, a ratio of about 1/365), whilst the rotation period of the Moon equals its orbital period (tidal locking). This is an intuitive example of* **multi-scale periodicity** *in astronomical systems.*

Proof. This follows by direct comparison of Table 27.2 with Table 27.1. The ratio of rotation to orbital motion is about $1/365$ for the Earth and about $0.414/4333 \approx 9.6 \times 10^{-5}$ for Jupiter. For the Moon $T_{\text{rot}} = T_{\text{orb}} = 27.32$ days, that is, tidal locking. ☐

Remark 27.8 (Direction of rotation and conservation of angular momentum). The rotation of Venus and Uranus is retrograde (their rotation periods are given a negative sign). Retrograde rotation indicates that the rotational angular momentum of these two planets is nearly opposite in direction to their orbital angular momentum. This may be the result of collisions between bodies or of tidal evolution in the early solar system, and it shows that conservation of angular momentum holds at the **system level** (rotation + orbital motion + other bodies), and not only at the level of the rotation of a single planet.

Remark 27.9 (Tidal locking and phase locking). The rotation period of the Moon is exactly equal to its orbital period ($T_{\text{rot}} = T_{\text{orb}} = 27.32$ days), a typical instance of **tidal locking**. Within the framework of periodic mathematics, tidal locking is equivalent to **phase locking**: the rotation phase and the orbital phase of the Moon keep a fixed relation (here $1:1$), which corresponds to the concrete realisation, in a rotation–orbit coupled system, of the phase locking of Proposition 24.8. Likewise, the $4:2:1$ orbital resonance among Io, Europa and Ganymede is an instance of **many-body phase locking** (see Definition 27.13 and Proposition 27.10).

Example 27.3 (Numerical verification of the orbital periods of the planets of the solar system). The data in Table 27.1 are checked row by row against Theorem 27.9:

(i) Mercury: $T^2 = 0.241^2 = 0.0581$, $a^3 = 0.387^3 = 0.0580$, ratio 1.002;

(ii) Jupiter: $T^2 = 11.862^2 = 140.71$, $a^3 = 5.203^3 = 140.85$, ratio 0.999;

(iii) Neptune: $T^2 = 164.79^2 = 27155.8$, $a^3 = 30.069^3 = 27156.5$, ratio 1.000.

The above values agree within observational accuracy, which verifies $T^2 \propto a^3$.

Example 27.4 (Multi-scale structure of the rotation periods). Arranging the rotation periods of Table 27.2 on a logarithmic scale:

$$0.41 \text{ (Jupiter)} < 0.44 \text{ (Saturn)} < 0.67 \text{ (Neptune)} < 0.72 \text{ (Uranus)}$$
$$< 0.997 \text{ (Earth)} < 1.03 \text{ (Mars)} < 27.3 \text{ (Moon)} < 58.6 \text{ (Mercury)} < 243.0 \text{ (Venus)}.$$

The distribution of the rotation periods spans nearly 3 orders of magnitude. The slow rotation of Venus, the Moon and Mercury arises from tidal evolution and from the long-term coupling between rotation and orbital motion.

Remark 27.10 (Organisation of the data in this section). The data tables and the TikZ figure of this section are an instantiation of astronomical periodic structure:

- Fig. 27.1 gives the spatial geometry of the orbital system;
- Table 27.1 gives the temporal structure of the orbital periods and allows Theorem 27.9 to be verified directly;
- Table 27.2 gives the temporal structure of planetary rotation, including the three basic modes of prograde rotation, retrograde rotation and tidal locking.

What the three have in common is that every celestial body is both an **orbital period system** (a revolution about the Sun or a planet) and a **rotational period system** (its own rotation), and the separation in magnitude or the locking relation of the two periods determines the overall rhythm of the celestial system.

27.7 Radial motion and conservation of energy

Definition 27.12 (Radial motion). *In a central force field the* **radial motion** $r(t)$ *of a celestial body is described by a one-dimensional dynamical equation in the effective potential:*

$$\frac{1}{2}m\dot{r}^2 + U_{\text{eff}}(r) = E,$$

where $U_{\text{eff}}(r) = -GMm/r + L^2/(2mr^2)$.

Proposition 27.8 (Period of the radial oscillation). *For an elliptical orbit the period of the radial motion equals the period of the angular motion: $T_r = T_\theta = T$. Bertrand's theorem states that only the inverse-square force field and the Hooke force field can produce closed bound orbits.*

Proof. From the radial equation of motion in the effective potential, the radial period is

$$T_r = 2\int_{r_{\text{min}}}^{r_{\text{max}}} \frac{dr}{\sqrt{\frac{2}{m}(E - U_{\text{eff}}(r))}},$$

and the angular period is

$$T_\theta = \frac{2m}{L}\int_{r_{\text{min}}}^{r_{\text{max}}} \frac{r^2 dr}{\sqrt{\frac{2}{m}(E - U_{\text{eff}}(r))}}.$$

When $U \propto -1/r$, the two integrals become equal under the substitution $u = 1/r$, hence $T_r = T_\theta$. ☐

Proposition 27.9 (Conservation of energy and the type of orbit)**.** The total mechanical energy $E = \frac{1}{2}mv^2 - GMm/r$ of a celestial body determines the type of its orbit:

(i) $E < 0$: a bound orbit (ellipse);

(ii) $E = 0$: an escape orbit (parabola);

(iii) $E > 0$: an unbound orbit (hyperbola).

Proof. The effective potential $U_{\text{eff}}(r) = -GMm/r + L^2/(2mr^2)$ attains its minimum at $r_0 = L^2/(GMm^2)$. When $E < 0$ there are two positive roots $r_{\text{min}} < r_{\text{max}}$ and the radial motion is bounded; when $E = 0$ there is only one finite root and $r_{\text{max}} = \infty$; when $E > 0$ there is no upper bound. ☐

27.8 Orbital resonance

Definition 27.13 (Orbital resonance)**.** The **orbital resonance** of two celestial bodies means that their orbital periods satisfy an integer-ratio relation:

$$\frac{T_2}{T_1} = \frac{p}{q}, \qquad p, q \in \mathbb{N}^+, \ \gcd(p, q) = 1.$$

The ratio $p : q$ is called the **resonance ratio**.

Definition 27.14 (Resonance angle)**.** For a $p : q$ orbital resonance, define the **resonance angle**

$$\phi = p\lambda_2 - q\lambda_1 + (p - q)\varpi,$$

where λ_i is the mean longitude and ϖ is the longitude of pericentre. If ϕ oscillates about an equilibrium value (libration), then the system is in **resonance lock**.

Proposition 27.10 (Structural properties of resonance)**.** Orbital resonance has the following structural properties:

(i) **Period-ratio structure**: the resonance ratio $p : q$ determines the scaling relation between the time scales of the two orbits;

(ii) **Resonance width**: the resonance width depends on the mass ratio and the eccentricity;

(iii) **Stability of resonance lock**: the resonance angle ϕ is confined to the libration region; if an external perturbation drives ϕ past the libration barrier, the system leaves the resonance.

Proof. (i) By Kepler's third law, $T \propto a^{3/2}$, hence $p/q = (a_2/a_1)^{3/2}$, that is, $a_2/a_1 = (p/q)^{2/3}$. (ii) Near a $p : q$ resonance the resonance Hamiltonian reduces to the pendulum form

$$H_{\text{res}} = \frac{3}{2}p^2 n^2 \frac{(\Delta a)^2}{a_{\text{res}}^2} + \alpha e n^2 \cos\phi.$$

The energy of the pendulum separatrix gives the resonance half-width $\Delta a_{\text{max}}/a \approx \sqrt{\alpha e m/M}$. (iii) Resonance lock is equivalent to ϕ undergoing a bounded oscillation; if the perturbation drives ϕ beyond the separatrix, the motion enters the circulation region and the resonance ratio is no longer conserved. ☐

Example 27.5 (Orbital resonances in the solar system). The Jupiter–Saturn $5:2$ near-resonance; the Neptune–Pluto $3:2$ resonance; the $4:2:1$ Laplace resonance of the Jovian satellite system. These resonance examples are directly related to the orbital period data given in Table 27.1: for example the ratio of the periods of Neptune and Pluto is $164.79/247.94 \approx 0.665 \approx 2/3$, that is, a $3:2$ resonance.

Definition 27.15 (Secular resonance). A **secular resonance** is a rational relation between the precession frequencies of the orbital elements:

$$\frac{\dot{\varpi}_2}{\dot{\varpi}_1} = \frac{p}{q} \quad \text{or} \quad \frac{\dot{\Omega}_2}{\dot{\Omega}_1} = \frac{p}{q}.$$

Example 27.6 (Secular resonances in the solar system). The ν_6 secular resonance is the main cause of the Kirkwood gaps; the Mercury–Jupiter secular resonance may lead to chaotic variation.

27.9 Periodic structure of many-body systems

Theorem 27.10 (Quasi-periodic orbits of many-body systems). In an N-body gravitational system ($N \geq 3$) the motion of the system is in general not strictly periodic but **quasi-periodic**. Within the framework of KAM theory, if the planetary masses are sufficiently small and the orbits sufficiently close to circular, then the orbits corresponding to most initial conditions lie on invariant tori, and the motion can be described by a quasi-periodic function with $3N$ independent frequencies.

Proof. The KAM theorem: if the unperturbed Hamiltonian H_0 is integrable and the perturbation εH_1 is sufficiently small and satisfies the non-resonance condition, then there exists a set of invariant tori of positive measure on which the quasi-periodic motion is preserved under the perturbation. In the solar system the planetary mass ratio is $\varepsilon \sim 10^{-3}$ to 10^{-6} and most orbital period ratios are irrational, so the KAM conditions are roughly satisfied. ☐

Remark 27.11 (Long-term stability of the solar system — an overview of perturbation theory). The perturbation-theory framework for the problem of the long-term stability of the solar system is as follows: the integrable two-body motion is taken as the unperturbed system H_0 and the mutual gravitation of the planets as the perturbation εH_1. Canonical perturbation theory order by order eliminates the fast angle variables and yields the secular averaged Hamiltonian; truncation at second order gives the Laplace–Lagrange secular solution. Poincaré proved that the power-series solution of the general three-body problem does not converge. Numerical simulations indicate that the orbits of the inner planets may be chaotic. Whether the solar system is strictly stable on time scales of billions of years remains an open problem.

27.10 Conserved quantities of astronomical periodic systems

Theorem 27.11 (The three conserved quantities of an astronomical system). *The dynamics of an astronomical periodic system is constrained by three conserved quantities:*

(i) **Conservation of angular momentum**: *in a central force field, $\mathsf{L} = m\mathsf{r} \times \dot{\mathsf{r}}$ is a constant vector;*

(ii) **Conservation of energy**: *in a conservative gravitational field, $E = \frac{1}{2}mv^2 - GMm/r$ is constant;*

(iii) **Conservation of the resonance ratio**: *in an orbital resonance, the resonance ratio $p : q$ remains unchanged in the locked state.*

Proof. (i) The rotational symmetry of the central force field yields conservation of angular momentum by Noether's theorem. (ii) The gravitational potential is conservative, $\nabla \times \mathsf{F}_g = \mathsf{0}$, and the mechanical energy is conserved. (iii) Near a $p : q$ resonance, after averaging the resonance angle satisfies the pendulum equation; when it librates, the resonance ratio is locked. ☐

Proposition 27.11 (Breaking of conserved quantities and orbital instability). *The breaking of conserved quantities in an astronomical periodic system leads to orbital instability:*

(i) *breaking of angular momentum: a non-radial force makes the orbital plane precess;*

(ii) *breaking of energy: a non-conservative force makes the semi-major axis vary slowly;*

(iii) *breaking of the resonance ratio: an external perturbation drives the resonance angle past the libration barrier.*

Proof. (i) If F has a non-radial component, then $\dot{\mathsf{L}} = \mathsf{r} \times \mathsf{F} \neq \mathsf{0}$. (ii) Tidal friction and gravitational-wave radiation make $\dot{E} < 0$, and from $E = -GMm/(2a)$ it follows that a decreases. (iii) When the amplitude of ϕ exceeds the separatrix, the motion enters the circulation region and the resonance ratio is no longer conserved. ☐

27.11 Binary and multiple star systems

Definition 27.16 (Relative motion of a binary star system). In a binary star system,

$$\mathsf{r}(t) = \mathsf{r}_2(t) - \mathsf{r}_1(t), \qquad \mathsf{R}_{\mathsf{cm}}(t) = \frac{m_1\mathsf{r}_1 + m_2\mathsf{r}_2}{m_1 + m_2}.$$

The direction of the orbital angular momentum is

$$\mathsf{n}(t) = \frac{\mathsf{r}(t) \times \dot{\mathsf{r}}(t)}{\|\mathsf{r}(t) \times \dot{\mathsf{r}}(t)\|}.$$

In the two-body problem $\mathsf{n}(t)$ is a constant vector; in a many-body system it precesses slowly.

Proposition 27.12 (Quaternion expansion of binary star orbits)**.** *The orbital motion of a binary star can be expanded in quaternions as*

$$\mathsf{r}(t) = \sum_{m=1}^{M} A_m \mathsf{d}_m(t), \qquad \mathsf{d}_m(t) = q_m(t)\hat{\mathsf{e}}_r q_m(t)^{-1},$$

where $q_m(t) = e^{\mathsf{u}_m \omega_m t/2}$. The truncation error is $\sim e^{M+1}$.

Proof. The relative motion of a binary star is equivalent, within the orbital plane, to two-dimensional elliptical motion. By Proposition 27.5, an elliptical orbit can be expanded in periodic components. Quaternions are used to embed the two-dimensional expansion in three-dimensional space. Since the harmonic amplitudes decay as $e^{|m|}$, the truncation error follows. ☐

Definition 27.17 (Hierarchical orbital angular momentum of multiple star systems)**.** *In a triple or N-star system, define hierarchical orbital angular momentum directions $\mathsf{n}_1(t), \mathsf{n}_2(t), \ldots,$ each level evolving slowly in time, driven by the gravitational coupling between the levels (the Kozai–Lidov mechanism).*

Proposition 27.13 (Hierarchical expansion of multiple star systems)**.** *A triple star system admits a hierarchical nested expansion. When n_1 and n_2 are not parallel, the system exhibits non-coplanar orbits, which is the geometric condition for Kozai–Lidov oscillations.*

Proof. Let the semi-major axis of the inner binary be $a_1 \ll a_2$. The gravitational potential of the outer body on the inner binary is expanded in a_1/a_2, whose first non-zero order is the quadrupole term. Averaging over the inner orbit gives the Kozai–Lidov potential. When the mutual inclination $I > 39.2°$, the inner eccentricity and inclination oscillate strongly. ☐

Theorem 27.12 (Evolution equation for the orbital angular momentum directions)**.** *Let $\mathsf{n}_k(t)$ be the direction of the orbital angular momentum of the k-th level. Then*

$$\frac{d\mathsf{n}_k}{dt} = \sum_{j \neq k} \Omega_{kj}(\mathsf{n}_k \times \mathsf{n}_j) + \sum_{j \neq k} \Gamma_{kj}(\mathsf{n}_k \times (\mathsf{n}_k \times \mathsf{n}_j)),$$

where Ω_{kj} is the precession frequency and Γ_{kj} is the dissipation coefficient.

Proof. The angular momentum of the k-th level, $\mathsf{L}_k = L_k \mathsf{n}_k$, satisfies $d\mathsf{L}_k/dt = \sum_{j \neq k} \boldsymbol{\tau}_{kj}$. Expanding the torque as the gradient of the gravitational potential and taking the orbital average, the conservative part yields $\mathsf{n}_k \times \mathsf{n}_j$ and the dissipative part yields $\mathsf{n}_k \times (\mathsf{n}_k \times \mathsf{n}_j)$. ☐

Corollary 27.2 (Alignment of axes when dissipation dominates)**.** *If the evolution equation is dominated by the dissipation coefficients Γ_{kj}, then the n_k tend to become mutually parallel; if it is dominated by the precession frequencies Ω_{kj}, then the system retains a non-coplanar configuration.*

Proof. When dissipation dominates, $\dot{\mathsf{n}}_k = \sum_j \Gamma_{kj}(\mathsf{n}_j - \mathsf{n}_k)$ and the n_k converge to a common direction. When precession dominates, $\dot{\mathsf{n}}_k = \sum_j \Omega_{kj}\mathsf{n}_j \times \mathsf{n}_k$, the inner products are conserved and the system remains non-coplanar. ☐

Example 27.7 (The Alpha Centauri triple system)**.** Alpha Centauri A and B form the inner binary, and Proxima Centauri moves about the A–B centre of mass. The angle between the orbital angular momentum directions of the two levels is about $107°$, and the system is markedly non-coplanar.

27.12 Periodic structure of astronomical time systems

An astronomical time system is composed of several kinds of period with different physical meanings and mathematical structures.

27.12.1 Tropical year and sidereal year

Definition 27.18 (Tropical year). The **tropical year** is the interval between two successive passages of the Sun through the vernal equinox:

$$T_{\text{trop}} = 365.2421897 - 0.0000062T \quad \text{mean solar days.}$$

Definition 27.19 (Sidereal year). The **sidereal year** is the interval between two successive passages of the Sun past the same fixed stellar direction:

$$T_{\text{sid}} = 365.2563630 \quad \text{mean solar days.}$$

The difference between the sidereal year and the tropical year is caused by precession.

27.12.2 Synodic month and sidereal month

Definition 27.20 (Synodic month). The **synodic month** is the interval between two successive passages of the Moon through the same lunar phase:

$$T_{\text{syn}} = 29.5305889 \quad \text{mean solar days.}$$

Definition 27.21 (Sidereal month). The **sidereal month** is the interval between two successive passages of the Moon past the same fixed stellar direction:

$$T_{\text{sm}} = 27.3216615 \quad \text{mean solar days.}$$

This agrees with the orbital period of the Moon, 27.322 days, given in Table 27.1.

27.12.3 Anomalistic month and nodal month

Definition 27.22 (Anomalistic month). The **anomalistic month** is the interval between two successive passages of the Moon through the perigee of its orbit:

$$T_{\text{anom}} = 27.5545509 \quad \text{mean solar days.}$$

Definition 27.23 (Nodal month). The **nodal month** is the interval between two successive passages of the Moon through the same node of its orbit with the ecliptic:

$$T_{\text{nod}} = 27.2122208 \quad \text{mean solar days.}$$

27.12.4 Synodic year

Definition 27.24 (Synodic year). The **synodic year** is the interval between two successive passages of the Sun through the same lunar-phase position:

$$\frac{1}{T_{\text{syn-year}}} = \frac{1}{T_{\text{syn}}} - \frac{1}{T_{\text{trop}}}.$$

27.12.5 Sui and precession

Definition 27.25 (Sui). In the astronomy and calendars of ancient China, the **year** corresponds to an integer multiple of the synodic month, whilst the **sui** corresponds to the tropical year from one winter solstice to the next. The two are reconciled by intercalation.

Definition 27.26 (Precession). **Precession** is the slow conical motion of the Earth's rotation axis, with a period of about

$$T_{\text{prec}} \approx 25772 \quad \text{years.}$$

Its equation is isomorphic to the precession equation for the orbital angular momentum directions of a multiple star system.

27.12.6 Nutation

Definition 27.27 (Nutation). **Nutation** is the short-period oscillation of the Earth's rotation axis on the precession cone, with principal period 18.6 years.

27.13 Periodic structure of time-measurement systems

27.13.1 Apparent solar time and mean solar time

Definition 27.28 (Apparent solar time). **Apparent solar time** is based on the actual position of the Sun, noon being the instant at which the centre of the Sun crosses the meridian. Its period is non-uniform.

Definition 27.29 (Mean solar time). **Mean solar time** is based on a fictitious mean sun moving uniformly. The equation of time is

$$E(t) = T_{\text{true}} - T_{\text{mean}}.$$

27.13.2 Legal time

Definition 27.30 (Legal time). **Legal time** is the unified time prescribed by a state in legal form.

Definition 27.31 (Beijing time). **Beijing time** is the standard time of the eighth eastern time zone, its reference point being Shanghai (longitude 121.43° east).

27.13.3 Sidereal time

Definition 27.32 (Sidereal time). **Sidereal time** is based on the position of the vernal equinox, the sidereal day being

$$T_{\text{sid-day}} = 23^{\text{h}}56^{\text{m}}04.0905^{\text{s}} \quad \text{mean solar time.}$$

27.13.4 Ephemeris time and atomic time

Definition 27.33 (Ephemeris time). **Ephemeris time** *takes the orbital motion of the Earth as its reference, the second being defined as* $1/31556925.9747$ *of the length of the tropical year of* 1900.

Definition 27.34 (Atomic time). **Atomic time** *takes the frequency of the hyperfine level transition of the ground state of the caesium* 133 *atom as its reference, the second being defined as* 9192631770 *periods.*

27.14 The three-tier structure of time-measurement systems

Theorem 27.13 (The three-tier structure of time-measurement systems). *Astronomical time-measurement systems form a three-tier nested structure:*

(i) **Rotation layer**: *apparent solar time, sidereal time—based on the true rotation of a celestial body;*

(ii) **Translation layer**: *mean solar time, ephemeris time—obtained by uniformisation;*

(iii) **Scaling layer**: *legal time, atomic time—unified by a scale transformation.*

Proof. (i) Apparent solar time is defined by the apparent ecliptic longitude of the Sun, and its non-uniformity arises from the ellipticity of the orbit and the obliquity of the ecliptic. (ii) Mean solar time takes the long-term average of the apparent ecliptic longitude of the true Sun, and the equation of time satisfies $|E(t)| \leq 16$ minutes. (iii) Legal time unifies local time differences into integer hours of a time zone; atomic time transfers the time scale to a quantum frequency. ☐

Corollary 27.3 (Leap seconds and periodic alignment). *A leap second is an operation of phase locking between two incommensurable periods, keeping the difference in day length between UTC and mean solar time within* 0.9 *seconds.*

27.15 Conserved quantities of time-measurement systems

Proposition 27.14 (Scale conservation in time measurement). *The core conserved quantity of a time-measurement system is the* **time scale**:

(i) *rotation-layer scale: the lengths of the apparent solar day and the sidereal day;*

(ii) *translation-layer scale: the length of the mean solar day;*

(iii) *scaling-layer scale: the length of the atomic second.*

Proof. (i) The rotation-layer scale is determined by the Earth's rotation, and tidal friction decelerates the Earth's rotation. (ii) The translation-layer scale inherits the rotation layer and smooths out the periodic fluctuations. (iii) The length of the atomic second is determined by constants of quantum mechanics, with a measured accuracy of 10^{-16}. □

Remark 27.12 (Connection with the rest of the book). The theory of the periodic structure of astronomical systems established in this chapter is an application of periodic mathematics to celestial mechanics and time measurement. It applies the constraints of conserved quantities, the expansion of periodic components, multi-level periodic structures and phase locking in a unified way to the analysis of astronomical systems. Kepler's three laws are derived from Newton's law of gravitation in a central force field; orbital resonance is a period-ratio relation between periods of different scales; many-body quasi-periodic systems are the celestial-mechanical realisation of the theory of periodic structure; and the three-tier structure of time-measurement systems is the comprehensive embodiment of the periodic-mathematics framework in the cognition and measurement of time.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The time translation of planets along their orbits and the orbital period (Table 27.1); the to-and-fro translation of the radial oscillation between periapsis and apoapsis (tropical year/sidereal year, mean solar time)
Scaling	The Kepler scaling of orbital radius versus period $T^2 \propto a^3$ (Theorem 27.9, Table 27.1); the semi-major-axis scaling $(p/q)^{2/3}$ induced by the resonance ratio $p : q$ (Proposition 27.10)
Rotation	Conservation of the angular momentum of orbital motion and rotation: conservation of areal velocity (Lemma 27.2), rotation periods and directions (Table 27.2), precession of the orbital angular momentum directions n_k (Theorem 27.12)
Conservation	Conservation of angular momentum, conservation of energy, conservation of the centre-of-mass momentum and conservation of the resonance ratio (Theorem 27.11)
Symmetry	The rotational symmetry of a central force field and its Noether correspondence to conservation of angular momentum (Theorem 27.11); the phase symmetry of the tidal locking of the Moon (Remark 27.9)
Closure	The closure of elliptical orbits: Kepler's first law (Theorem 27.7), Bertrand's theorem (Proposition 27.8)
Period	Decomposition of the orbital and rotation periods (Proposition 27.5, Table 27.1, Table 27.2), mean-motion resonance (Definition 27.13), secular resonance (Definition 27.15)

Exercises

Exercise 1 Prove the **astronomical periodicity decision theorem** of this chapter (27.1). Hint: first write down the decision map and the evolution group of this chapter's target, then apply the proof structure of the general criterion of the 'Periodic Systems' chapter.

Exercise 2 Verify that the hypotheses of **monotonicity of quantity adjustment** (27.1) hold for the domain objects of this chapter, and give a counterexample in which the hypotheses fail.

Exercise 3 Examine the behaviour of the **Fourier–Bessel expansion of elliptical orbits** (27.5) under degeneracy of the parameters, and explain how the conclusion degenerates accordingly.

Exercise 4 Explain the relation between the **astronomical periodicity diagnostic procedure and its interfaces with three chapters** (27.6) and the corresponding general conclusion of the 'Periodic Systems' chapter (whether they agree, whether it is a special case, or whether additional conditions are required).

Exercise 5 Using the seven-theme comparison table of this chapter, identify the concrete carriers in this chapter of the three themes **translation**, **scaling** and **rotation**, and give one example of domain data for each.

Exercise 6 Synthesis: restate the diagnostic procedure of this chapter using the five-step standard interface of the 'Periodic Systems' chapter (Table 24.1), noting for each step the theorem of this chapter on which it depends: 27.1, 27.1, 27.5, 27.6.

Chapter 28

Periodic Structure of Calendar Systems

Remark 28.1 (Periodicity decision for the target of this chapter (citation)). When this chapter decides whether a research target is periodic, determines its period type, or seeks its minimal positive period, reference is always made to the 'Periodic Systems' chapter: for the periodicity criterion see Theorem 24.1, for the discrimination of period types see Definition 24.3, Proposition 24.1, for the search for the minimal positive period see Definition 24.4, Theorem 24.2, and for the construction of period series and epicycle groups from data see Theorem 24.3, Definition 24.6.

Remark 28.2 (Positioning of this chapter: developed along the five-step interface of the 'Periodic Systems' chapter). The periodic structure of calendar systems studies the mathematical characterisation of the nesting and coordination of multiple periods in astronomical calendars. This chapter does not establish a separate periodicity criterion; instead it connects calendar systems to the five-step interface prescribed by the 'Periodic Systems' chapter:

(1) **Decision**: reduce the sexagenary sequence, the solar-term sequence, the synodic-month sequence and the tropical-year sequence to discrete dynamical systems, cite Theorem 24.1 to decide whether they are periodic, and determine the period type;

(2) **Construction**: from solar-term data, synodic data and sexagenary data, construct the amplitude series, the frequency series and the phase series, and assemble them into a calendar epicycle group;

(3) **Adjustment**: dynamically adjust the calendar epicycle group by intercalation windows, best continued-fraction approximation and the decrease of the residual energy;

(4) **Convolution**: introduce amplitude convolution, phase convolution and frequency convolution on the calendar epicycle group, and connect them with the solar-term phase, the synodic phase and the sexagenary phase;

(5) **Integration**: connect to the tool chains of the three chapters 'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis', forming a calendar periodicity diagnostic procedure.

The calendrical and mathematical knowledge used in this chapter includes: the epoch systems of the *Huainanzi* 'Tianwenxun' and of the *Huangji Jingshi*, the twenty-four solar terms and mid-qi, the synodic month, the sexagenary cycle, the yuan-hui-yun-shi cycles, continued fractions and best rational approximation, least common multiples and commensurability, the bounded intercalation error theorem, the Maya calendar, the Yi ten-month solar calendar and the ancient Indian lunisolar calendar.

Remark 28.3 (Guide to the seven themes of this chapter). *Translation—the ordering advance of the sexagenary sequence (the translation transformation of the absolute sequence and the relative sequence, Proposition 28.1); Scaling—the fixed-length scaling of the tropical year and the synodic month ($T_{\text{trop}} = 365.2422$ days, $T_{\text{syn}} = 29.5306$ days, Definition 28.7); Rotation—the rotational distribution of the solar terms on the ecliptic (the 2π rotation of the phase angle and the solar ecliptic longitude, Table 28.1); Balance—the ordering and fixed-length constraints of intercalation (Theorem 28.17); Symmetry—the solar–lunar symmetry of the lunisolar calendar (the binary structure of the solar calendar and the lunar calendar, Theorem 28.11); Closure—the closure of the sixty-step cycle of the stems and branches (the sexagenary cycle, Theorem 28.13); Period—the tropical year (365.2422 days), the synodic month (29.5306 days) and the sexagenary cycle (Definition 28.7).*

28.1 Periodicity decision for calendar targets: the first step of the 'Periodic Systems' chapter

28.1.1 Calendar targets, decision input and decision map

Definition 28.1 (Calendar targets and decision input). *A **calendar target** is one of the following objects:*

- *(i) an ordering system of calendar units, whose state space is the sequence set $\mathcal{S} = \mathbb{Z}$ and whose evolution map is the $+1$ advance;*
- *(ii) a solar-term sequence or a synodic-month sequence, whose state space is a phase set;*
- *(iii) a sexagenary periodic system, whose state space is the cyclic group C_{60};*
- *(iv) a multi-period coupled system, whose state space is $\mathbb{Z}_T \times \mathbb{Z}_S \times \cdots$.*

*The **decision input** of a calendar target is the ordered pair*

$$\mathcal{D}_{\mathsf{L}} = (\mathcal{S}, \Phi),$$

*where $\mathcal{S}$ is the set of calendar units and $\Phi : \mathcal{S} \to \mathcal{S}$ is the evolution map induced by ordering advance, solar-term rotation or sexagenary cycling. If $\mathcal{S}$ is a finite set, the target is called a **finite calendar target**; if $\mathcal{S}$ is an infinite set, the target is called an **infinite calendar target**.*

Definition 28.2 (Calendar decision map). *Let a calendar target be given by the decision input $\mathcal{D}_{\mathsf{L}} = (\mathcal{S}, \Phi)$. If the state or the output of the target is generated by the iteration of Φ, that is, if there exists $x_0 \in \mathcal{S}$ such that*

$$x_t = \Phi^t(x_0), \qquad t \in \mathbb{Z},$$

then Φ is called the **calendar decision map** of the target. The calendar target is then reduced to a discrete dynamical system $(\mathcal{S}, \Phi)$, to which the criteria of the 'Operations and Periodicity' chapter and of the 'Periodic Systems' chapter may be applied directly.

Theorem 28.1 (Calendar periodicity decision theorem)**.** *Let the decision map of a calendar target be Φ, and let $(\mathcal{S}, \Phi)$ be a discrete dynamical system. Then:*

(A) **Sufficient decision**: *if there exists $m \in \mathbb{N}^+$ such that $\Phi^m = \mathrm{id}$, then the target is a uniformly periodic calendar system; every orbit is closed and the length of each orbit divides m.*

(B) **Equivalence of the decision**: *the target is a uniformly periodic calendar system if and only if* $\mathrm{ord}(\Phi) < \infty$.

(C) **Pointwise criterion**: *the orbit of the target at the state x_0 is closed if and only if x_0 is a periodic point; in that case the length of the orbit is the period at that point.*

(D) **Non-omissibility**: *the conjunction of conserved quantities and the symmetry group cannot replace* $\mathrm{ord}(\Phi) < \infty$; *a quasi-periodic calendar target is not a periodic calendar system.*

Proof. The Theorem 28.1 of this chapter is the domain instantiation of Theorem 24.1 in the 'Periodic Systems' chapter: it suffices to replace the decision map f and the evolution group by the Φ of this chapter, and the four parts (A) sufficient decision, (B) equivalence of the decision, (C) pointwise criterion and (D) non-omissibility then hold one by one. By the convention of Remark 24.4, they are not expanded again here. ☐

Corollary 28.1 (Calendar period types)**.** *After the decision by Theorem 28.1, a calendar target falls into one of the following types:*

(i) **Trivially periodic type**: *the least period is $m_0 = 1$ and the state is a fixed point;*

(ii) **Strictly periodic type**: $1 < m_0 < \infty$ *and the state is strictly periodic (for example the sexagenary cycle with $m_0 = 60$);*

(iii) **Uniformly periodic type**: $\mathrm{ord}(\Phi) = m < \infty$ *and all orbit lengths divide m;*

(iv) **Quasi-periodic type**: $\mathrm{ord}(\Phi) = \infty$; *the orbit is not closed but is dense on a compact carrier structure (for example a lunisolar calendar without intercalation);*

(v) **Eventually periodic type**: *there exists N such that the target is periodic for $t \geq N$;*

(vi) **Discretely periodic type**: *the carrier structure is $\mathbb{Z}$ and the normalised period ratio satisfies $\rho = T_x/\Delta x \in \mathbb{Q}$.*

(vii) **Asymptotically periodic (damped) type**: *the orbit is not closed, but tends asymptotically to a closed orbit or to a fixed point, the residual decaying as $e^{-\kappa t}$ (see Definition 24.3(vii)).*

Here (i)–(iii) and (vi) are periodic calendar systems; (iv) is not a periodic calendar system, but may be handled with the bounded-approximation tools of intercalation; (v) is a periodic calendar system only in its stable stage; (vii), like (iv), is a type in the limiting sense; it is not a periodic calendar system and can be handled with bounded-approximation tools only in its stable stage.

Proof. Each item follows by substituting the calendar target into Theorem 28.1 and Definition 24.3 of the 'Periodic Systems' chapter. ☐

Lemma 28.1 (Periodicity decision for the sexagenary sequence). *Let the state translation of the sexagenary sequence be $\Phi : n \mapsto n + 1 \pmod{60}$. Then:*

(i) *Φ is a cyclic translation on C_{60}, with period 60;*

(ii) *the state space C_{60} is a single orbit, and every non-trivial invariant must be constant;*

(iii) *the sexagenary cycle constitutes a faithful action of C_{60}, and 60 is the least positive period.*

Proof. (i) $\Phi^{60} = \text{id}$, and for every $k < 60$, $\Phi^k \neq \text{id}$, hence $\text{ord}(\Phi) = 60$. (ii) C_{60} is a single orbit under Φ; by Theorem 1.7 of the 'Operations and Periodicity' chapter, every conserved quantity must be constant. (iii) This follows from (i) and (ii). ☐

Lemma 28.2 (Periodicity decision for the solar-term sequence). *Let the state translation of the solar-term sequence be $\Phi : n \mapsto n + 1 \pmod{24}$. Then the solar-term sequence has period 24 (one year), and the twenty-four solar terms constitute a faithful action of C_{24}.*

Proof. Exactly 24 solar terms occur in one year (Definition 28.15), hence $\Phi^{24} = \text{id}$. If $k < 24$ satisfies $\Phi^k = \text{id}$, then the $(k+1)$-th solar term coincides with the first, contradicting the fact that the twenty-four solar terms in Table 28.1 are pairwise distinct. Hence $\text{ord}(\Phi) = 24$. ☐

Lemma 28.3 (Periodicity decision for the synodic-month sequence). *The synodic-month sequence does not constitute a strictly periodic sequence—*$\text{ord}(\Phi) = \infty$*, since $T_{\text{trop}}/T_{\text{syn}} \notin \mathbb{Q}$ (an* **empirical observation / research programme***; see Remark 28.5). The synodic-month sequence is of quasi-periodic type and requires intercalation in order to achieve bounded approximation.*

Proof. If there exist integers p, q such that $pT_{\text{trop}} = qT_{\text{syn}}$ holds exactly, then $T_{\text{trop}}/T_{\text{syn}} = q/p \in \mathbb{Q}$. But by the **empirical observation / research programme** of Remark 28.5, within the current precision of measurement $T_{\text{trop}}/T_{\text{syn}} \approx 12.36827$ agrees with no p/q of bounded denominator, a contradiction. Hence, within measurable precision, the synodic month and the tropical year are incommensurable and the synodic-month sequence has no strict period. This must not be written as 'by Remark 28.5 it follows that $T_{\text{trop}}/T_{\text{syn}}$ is irrational': the irrationality of a ratio of astronomical constants cannot be proved from measurement data. ☐

Definition 28.3 (Algorithm for the least positive period of a calendar target). **Input**: calendar target samples $x_0, \ldots, x_{N-1}$ and a tolerance $\varepsilon \geq 0$. **Output**: the least positive period T_0, or the verdict 'not periodic'. **Steps**:

(1) take the candidate set $\mathcal{P} = \{T \in \mathbb{N}^+ : 2T \leq N\}$ and traverse it in increasing order of T;

(2) for each T compute the maximum deviation

$$E(T) = \max_{0 \leq n < N-T} \|x_{n+T} - x_n\|;$$

(3) output the first T satisfying $E(T) \leq \varepsilon$ as T_0; if the traversal is exhausted with no solution, output 'not periodic';

(4) commensurability check: use $\rho = T_x/\Delta x$ (with $T_0 = p$ when $\rho = p/q$ is in lowest terms; write $\ell = T_0 \Delta x = qT_x$ for the continuous span corresponding to T_0 sampling points) together with Proposition 21.2 and Corollary 21.1 of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter to check the conversion relation between T_0 and the continuous period T_x.

Theorem 28.2 (Correctness of the algorithm for the least positive period of a calendar target
If the samples form a strictly periodic sequence with $\varepsilon = 0$ and $N \geq 2T_0$, then the algorithm outputs the discrete fundamental period T_0.

(ii) If the samples are obtained by sampling a continuous calendar target of fundamental period T_x at the sampling interval Δx, and the normalised period ratio $\rho = T_x/\Delta x = p/q$ is in lowest terms, then the algorithm outputs $T_0 = p$.

(iii) If $\rho \notin \mathbb{Q}$, then no true period exists: $E(T) > 0$ for every $T \in \mathbb{N}^+$.

Proof. The Theorem 28.2 of this chapter follows directly from the two conclusions of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter: (i) by Definition 21.4, T_0 is the least positive period, hence $E(T_0) = 0$, whilst for $T < T_0$ we have $E(T) > 0$; the algorithm traverses in increasing order of T, so the first index satisfying $E(T) = 0$ is T_0; (ii) and (iii) follow by substituting into Proposition 21.2(ii)(iii). The complete argument is given in that chapter. □

Example 28.1 (Example of a periodicity decision for the sexagenary sequence). Starting from Jiazi (under the convention that Jiayin has index 0, Jiazi has index 10), the sexagenary sequence has least period 60 by Lemma 28.1. The state orbit has length 60, that is, the complete cycle 'Jiazi→Yichou→ $\cdots$ →Guihai→Jiazi'.

Example 28.2 (Example of a periodicity decision for the solar-term sequence). For the solar-term sequence starting from Lichun, Lemma 28.2 gives least period 24. The orbit has length 24, that is, the complete cycle 'Lichun→Yushui→ $\cdots$ →Dahan→Lichun'.

Remark 28.4 (Conserved quantities do not guarantee periodicity). The conserved quantities of a calendar system (such as conservation of equal intervals and conservation of fixed length) are auxiliary conditions for periodic structure, but they do not constitute sufficient conditions. For example, a lunisolar calendar without intercalation possesses conservation of the fixed length of the solar terms, yet the synodic-month sequence and the solar-term sequence are incommensurable and do not constitute a strict period. The periodic structure of a calendar must be decided by criteria such as Theorem 28.1 or Lemma 28.1, Lemma 28.2.

28.2 Construction of the calendar period series and epicycle group: the second step of the 'Periodic Systems' chapter

Definition 28.4 (Calendar amplitude series, frequency series and phase series)**.** Let the state or output sequence of a calendar target be $\{z_t\}$. If it can be written as

$$z_t = \sum_k A_k e^{i(\omega_k t+\phi_k)} + \varepsilon_t,$$

then

$$\mathcal{A}_z(t) = \sum_k A_k e^{ikt}, \qquad \mathcal{F}_z(s) = \sum_k \omega_k s^k, \qquad \mathcal{P}_z(t) = \sum_k e^{i(kt+\phi_k)}$$

are called respectively the **amplitude series**, the **frequency series** and the **phase series** of the calendar target. The three agree with Definition 6.10, Definition 6.12 and Definition 6.11 of the 'Series and Convolution' chapter.

Definition 28.5 (Calendar epicycle group)**.** Let the period series of a calendar target be given by the amplitude series $\{A_k\}$, the frequency series $\{\omega_k\}$ and the phase series $\{\phi_k\}$. For each k define the epicycle

$$E_k : \theta \mapsto A_k e^{i(\theta+\phi_k)}, \qquad \theta = \omega_k t,$$

and call the set

$$\mathfrak{E}_{\mathsf{L}} = \{(A_k, \omega_k, \phi_k)\}_{k\in\mathbb{Z}}$$

the **epicycle group** of the calendar target. Its superposition is the calendar period series

$$z_t = \sum_k E_k(\omega_k t) = \sum_k A_k e^{i(\omega_k t+\phi_k)}.$$

Theorem 28.3 (Construction of the period series from calendar data)**.** Let the output of a calendar target be a finite periodic sequence of period N. Then:

(i) the output can be expanded as a discrete Fourier series

$$z_t = \sum_{k=0}^{N-1} c_k e^{i2\pi kt/N}, \qquad c_k = \frac{1}{N}\sum_{t=0}^{N-1} z_t e^{-i2\pi kt/N};$$

(ii) the amplitude series is given by $|c_k|$, the phase series by $\arg c_k$ and the frequency series by $2\pi k/N$;

(iii) if the output is a solar-term sequence or a sexagenary sequence, then the harmonics at the various levels correspond respectively to time scales such as the annual period, the semi-annual period, the seasonal period and the monthly period.

Proof. (i) The discrete Fourier series of a finite periodic sequence is a standard result: the basis functions $\{e^{i2\pi kt/N}\}_{k=0}^{N-1}$ constitute an orthogonal basis of $\mathbb{C}^N$. (ii) It is read off directly from $c_k = A_k e^{i\phi_k}$. (iii) For the solar-term sequence $N = 24$, the fundamental frequency corresponds to the annual period, the harmonic $k = 2$ to the semi-annual period, the harmonic $k = 4$ to the seasonal period and the harmonic $k = 12$ to the monthly period. □

Definition 28.6 (Symbolic epoch year and Jiayin epoch year of the Xuanyuan calendar)**.** *The Xuanyuan calendar takes the Yellow Emperor era as the origin of its year count. Let the* **symbolic epoch year** *be the year 2697 BCE, which is a Jiazi year (sexagenary index 10). Throughout, astronomical year numbering is used:*

$$Y_0 = -2697.$$

The sexagenary index of Jiayin is 0, so the **Jiayin epoch year** *is*

$$-2697 + 50 = -2647 \quad \text{(astronomical year numbering)},$$

corresponding to the year 2647 BCE (a Jiayin year).

Definition 28.7 (Fixed length: the intrinsic period of a calendar unit)**.** *A* **fixed length** *is the intrinsic period length of a calendar unit:*

Calendar unit	Symbol	Fixed length	Type
Tropical year	T_{trop}	365.2421897 mean solar days	Solar period
Synodic month	T_{syn}	29.5305889 mean solar days	Lunar period
Solar-term interval	ΔT_{jieqi}	$T_{\text{trop}}/24 \approx 15.2184$ days	Solar period (mean solar-term method)
Mid-qi interval	$\Delta T_{\text{zhongqi}}$	$T_{\text{trop}}/12 \approx 30.4368$ days	Solar period (mean solar-term method)
Climate month	T_{climate}	14–18 days	Solar period (true solar-term method, variable)
Sidereal year	T_{sid}	365.2563630 mean solar days	Sidereal reference frame

Definition 28.8 (Absolute sequence and relative sequence)**.** *An* **absolute sequence** *is the continuous index of a calendar unit counted from the epoch year of the Xuanyuan calendar:*

$$S_{\text{abs}}(U) \in \mathbb{Z}.$$

A **relative sequence** *is the number of a calendar unit within the current period:*

$$S_{\text{rel}}(U) \in \{1, 2, \ldots, T\}.$$

Proposition 28.1 (Conversion between the absolute sequence and the relative sequence)**.** Let the absolute sequence of a calendar unit U be $S_{\mathsf{abs}}(U)$, its relative sequence be $S_{\mathsf{rel}}(U)$, and the absolute sequence of the starting point of the current period be $S_{\mathsf{abs}}(P)$. Then

$$S_{\mathsf{rel}}(U) = S_{\mathsf{abs}}(U) - S_{\mathsf{abs}}(P) + 1.$$

Proof. The relative sequence is counted from 1 and the absolute sequence from $S_{\mathsf{abs}}(P)$, and the translation $S_{\mathsf{abs}}(P) - 1$ depends only on the choice of the starting point of the period. Hence $S_{\mathsf{rel}}(U) = S_{\mathsf{abs}}(U) - S_{\mathsf{abs}}(P) + 1$. ☐

Proposition 28.2 (Calculation of the absolute sequence of a solar term)**.** The absolute sequence of the n-th solar term (counted from Lichun of the epoch year) is $S_{\mathsf{abs}}(J_n) = n$. The corresponding year is $Y(J_n) = Y_0 + \lfloor (n-1)/24 \rfloor$. The relative sequence is

$$S_{\mathsf{rel}}(J_n) = \big((n-1) \bmod 24\big) + 1.$$

Proof. By Definition 28.8, the absolute sequence is the continuous numbering counted from the epoch year; Lichun of the epoch year is the first solar term, so the absolute sequence of the n-th solar term is exactly $S_{\mathsf{abs}}(J_n) = n$. Since one tropical year contains exactly 24 solar terms (Definition 28.15), the n-th solar term falls in the $\lfloor (n-1)/24 \rfloor + 1$-th tropical year counted from the epoch year, hence $Y(J_n) = Y_0 + \lfloor (n-1)/24 \rfloor$. The relative sequence is computed from Proposition 28.1: taking the starting point P of the current period to be Lichun of that year, whose absolute sequence is $S_{\mathsf{abs}}(P) = 24\lfloor (n-1)/24 \rfloor + 1$, we obtain $S_{\mathsf{rel}}(J_n) = ((n-1) \bmod 24) + 1$. ☐

28.3 Dynamic adjustment of the calendar epicycle group: the third step of the 'Periodic Systems' chapter

Definition 28.9 (Dynamic adjustment of the calendar epicycle group)**.** Let the current calendar epicycle group be $\mathfrak{E}^{(j)}$, with partial sum

$$z^{(j)}(t) = \sum_{k \in \mathfrak{E}^{(j)}} A_k e^{i(\omega_k t + \phi_k)},$$

and residual $r^{(j)} = z - z^{(j)}$. **Dynamic adjustment** consists of two rules:

(i) **Quantity adjustment**: if the residual energy satisfies $\|r^{(j)}\|^2 > \delta$, then add one epicycle according to the peak of the residual periodogram;

(ii) **Parameter adjustment**: update the parameters of the existing epicycles by regularised least squares or by the intercalation window.

In calendar systems, quantity adjustment corresponds to adding a solar-term term, a synodic term or an intercalation term; parameter adjustment corresponds to tuning the length of the intercalation window, the pairing of solar terms with mid-qi, or the initial phase of the sexagenary cycle.

Proposition 28.3 (Monotonicity of quantity adjustment). After the dominant-frequency component of the residual has been added as in Definition 28.9(i), the residual energy is monotonically non-increasing:

$$\|r^{(j+1)}\|^2 \le \|r^{(j)}\|^2.$$

Proof. By the orthogonal decomposition $\|r^{(j)}\|^2 = \|r^{(j+1)}\|^2 + \|\text{the removed component}\|^2$, the inequality holds. ☐

Theorem 28.4 (Convergence of dynamic adjustment). Let the calendar target be a **real-valued finite periodic sequence** of length T_0 ($T_0 \in \mathbb{N}$) and let $\delta = 0$. Then after at most $\lfloor T_0/2 \rfloor + 1$ steps of dynamic adjustment the residual is zero, and the resulting calendar epicycle group is the exact Fourier data of the target; for a **complex-valued** sequence the corresponding number of steps is at most T_0.

Proof. The discrete Fourier series of a **real-valued** finite periodic sequence of length T_0 ($T_0 \in \mathbb{N}$) is complete on $\lfloor T_0/2 \rfloor + 1$ harmonics (including the constant term); a **complex-valued** sequence requires T_0 components. At each step one dominant frequency component of the residual is extracted as in Proposition 28.3 and the residual energy decreases strictly, so the residual vanishes after finitely many steps. ☐

Remark 28.5 (Incommensurability of the fixed lengths (empirical observation / research programme)). Whether the tropical year T_{trop} and the synodic month T_{syn} are commensurable depends on whether the ratio of two astronomical constants, $r = T_{\text{trop}}/T_{\text{syn}}$, is rational. It must be stated here that 'r is irrational' is an **empirical observation / research programme**, not a provable proposition: any measurement of finite precision yields a rational r, whose continued-fraction expansion must terminate, and improved precision merely lengthens that expansion without ever proving that it fails to terminate. The inference from 'the continued fraction keeps extending' to irrationality is therefore **invalid**, and the irrationality of a ratio of astronomical constants cannot be proved mathematically. **Empirical hypothesis**: within the current precision of measurement, r agrees with no p/q of bounded denominator, and its sequence of continued-fraction convergents is stable within measurable precision:

$$r = [12;\, 2,\, 1,\, 2,\, 1,\, 1,\, 17, \ldots].$$

Under this hypothesis it may be asserted that there exist no integers p, q with $p\,T_{\text{trop}} = q\,T_{\text{syn}}$ exactly—this is the fundamental reason why the lunisolar calendar requires intercalation. The best rational approximation to r is the continued-fraction convergent $235/19 = 12.36842\ldots$, with an error of about 0.00015, which is precisely the approximate mechanism of 'seven intercalary months in nineteen years'. Lemma 28.3 (the synodic-month sequence is not strictly periodic) and Theorem 28.6 (best intercalation by continued fractions) both take this empirical hypothesis as their premise; they are conditional, not unconditional, conclusions.

Definition 28.10 (Intercalation window and intercalation sequence). Let the period ratio be $\rho = T_{\text{trop}}/T_{\text{syn}} > 0$. A sequence of integers $\{N_m\}_{m\in\mathbb{N}}$ is called an **intercalation sequence** for ρ if

$$N_m \in \{\lfloor \rho \rfloor, \lceil \rho \rceil\}, \qquad \lim_{M\to\infty} \frac{1}{M} \sum_{m=0}^{M-1} N_m = \rho.$$

Define the **cumulative sample count**

$$S_M = \sum_{m=0}^{M-1} N_m,$$

and the **intercalation error**

$$E_M = S_M - M\rho.$$

Theorem 28.5 (Bounded intercalation error theorem)**.** Let $\rho > 0$. Then the three basic intercalation sequences satisfy:

(i) **Lower intercalation**: $N_m^{\text{floor}} = \lfloor (m+1)\rho \rfloor - \lfloor m\rho \rfloor$, $E_M^{\text{floor}} \in (-1, 0]$;

(ii) **Upper intercalation**: $N_m^{\text{ceil}} = \lceil (m+1)\rho \rceil - \lceil m\rho \rceil$, $E_M^{\text{ceil}} = \lceil M\rho \rceil - M\rho \in [0, 1)$, and $E_M^{\text{ceil}} = 0$ when $M\rho \in \mathbb{Z}$;

(iii) **Balanced intercalation**: $N_m^{\text{round}} = \text{round}((m+1)\rho) - \text{round}(m\rho)$, $|E_M^{\text{round}}| \leq 1/2$.

Proof. By the definition of lower intercalation,

$$S_M^{\text{floor}} = \sum_{m=0}^{M-1} (\lfloor (m+1)\rho \rfloor - \lfloor m\rho \rfloor) = \lfloor M\rho \rfloor,$$

hence $E_M^{\text{floor}} = \lfloor M\rho \rfloor - M\rho = -\{M\rho\} \in (-1, 0]$. The same argument gives $E_M^{\text{ceil}} = \lceil M\rho \rceil - M\rho \in [0, 1)$ for upper intercalation, and $E_M^{\text{ceil}} = 0$ when $M\rho \in \mathbb{Z}$ (the ideal, leakage-free case). For balanced intercalation $S_M^{\text{round}} = \text{round}(M\rho)$, and by the definition of round, $|E_M^{\text{round}}| \leq 1/2$. □

Theorem 28.6 (Best continued-fraction intercalation)**.** Let $\rho > 0$ be irrational and let $\{p_j/q_j\}$ be its continued-fraction convergents. Then:

(i) **Error bound**: $|\rho - p_j/q_j| < 1/(q_j q_{j+1}) < 1/q_j^2$;

(ii) **Best-approximation property**: if $q < q_{j+1}$, then $|q\rho - p| \geq |q_j\rho - p_j|$;

(iii) **Optimal window**: the optimal sequence of intercalation-window lengths is $\{q_j\}$, with cumulative error $|E_{q_j}| < 1/q_{j+1}$.

Proof. (i) and (ii) are standard results of continued-fraction theory (see Theorem 21.5 of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter). (iii) From (i) we obtain directly $|E_{q_j}| = |q_j\rho - p_j| < 1/q_{j+1}$. By the optimality in (ii), this upper bound is the smallest among window lengths $< q_{j+1}$. □

Theorem 28.7 (Convergence of calendar intercalation)**.** Let $\rho = T_{\text{trop}}/T_{\text{syn}}$. Take the intercalation window to be the sequence of continued-fraction denominators $\{q_j\}$. Then:

(i) the cumulative error satisfies $|E_{q_j}| < 1/q_{j+1} \to 0$;

(ii) the relative error satisfies $|E_{q_j}|/q_j < 1/(q_j q_{j+1}) \to 0$;

(iii) the phases of the solar calendar and the lunar calendar close approximately on the scale of q_j years.

Proof. By Theorem 28.6, $|E_{q_j}| < 1/q_{j+1}$. Since $q_j \to \infty$, we have $1/q_{j+1} \to 0$, so (i) holds. Part (ii) follows from $q_j \to \infty$ and (i). Phase closure is equivalent to $|E_{q_j}| \to 0$, which holds by (i). □

Remark 28.6 (Boundary of dynamic adjustment). Theorem 28.4 relies on a finite periodic sequence. For an incommensurable lunisolar calendar no finite epicycle group represents it exactly, and only a bounded approximation is possible. The choice of intercalation window is restricted by the continued-fraction convergents; intercalation does not eliminate the cost of incommensurability, but only reduces the error to a bounded one.

28.4 Calendar convolution: the fourth step of the 'Periodic Systems' chapter

Definition 28.11 (Calendar amplitude convolution, phase convolution and frequency convolution)**.** *Let two calendar epicycle groups be*

$$\mathfrak{E} = \{(A_n, \omega_n, \phi_n)\}, \qquad \mathfrak{E}' = \{(B_m, \nu_m, \psi_m)\}.$$

Define:

(i) **Amplitude convolution**: $\mathbb{A}_{mn} = A_n B_m$;

(ii) **Phase convolution**: $\mathbb{P}_{mn} = e^{i(\phi_n - \psi_m)}$;

(iii) **Frequency convolution**: $\mathbb{F}_{mn} = \omega_n \nu_m$;

(iv) **Trend convolution**: $\mathbb{T}_{mn} = A_n B_m e^{i(\phi_n - \psi_m)} = A_n B_m \mathbb{P}_{mn}$.

This agrees with Definition 6.21, Definition 6.22, Definition 6.23 and Definition 6.24 of the 'Series and Convolution' chapter.

Theorem 28.8 (Action of the three kinds of convolution on calendar epicycle groups)**.** *Let the epicycle groups of two calendar targets be $\mathfrak{E}$ and $\mathfrak{E}'$. Then:*

(i) *amplitude convolution acts only on the radius;*

(ii) *phase convolution acts only on the phase;*

(iii) *frequency convolution acts only on the rate of rotation;*

(iv) *trend convolution preserves both amplitude and phase.*

Proof. This is read off directly from Definition 28.11. □

Proposition 28.4 (Periodic-convolution generation of the sexagenary cycle)**.** *The periodic convolution of the Heavenly-Stem period $T_{\text{stem}} = 10$ and the Earthly-Branch period $T_{\text{branch}} = 12$, after filtering by the parity-synchronisation condition $n \equiv m \pmod 2$, yields exactly 60 effective phases, which constitute the sexagenary cycle.*

Proof. The periodic convolution traverses $10 \times 12 = 120$ pairs (n, m). The Yin-Yang synchronisation condition $n \equiv m \pmod 2$ selects $5 \times 6 + 5 \times 6 = 60$ pairs from the 120, which correspond exactly to the sexagenary cycle. By Theorem 30.7 of the 'Periodic Systems' chapter, these 60 phases constitute the cyclic group C_{60}. ☐

Proposition 28.5 (Phase convolution of the solar terms and the synodic month)**.** *The phase convolution of the solar-term sequence (period 24) and the synodic-month sequence (period 12 or 13) gives the relative relation between the solar-term phase and the synodic phase. Intercalation by absence of a mid-term corresponds to a particular zero of the phase convolution: when the relative difference between the solar-term phase and the mid-qi phase within a synodic month exceeds the window, that month is an intercalary month.*

Proof. By Definition 28.16, a normal month contains exactly one solar term and one mid-qi, and the phase difference lies within the window. When a month contains only a solar term and no mid-qi, the difference between the solar-term phase and the phase of the nearest mid-qi exceeds the window, and that month is an intercalary month. The phase convolution records the relative relations of all solar-term–synodic phase pairs. ☐

Remark 28.7 (Relation between calendar convolution and periodic convolution)*.* Calendar convolution is not a new convolution operation, but an instance, on the calendar epicycle group, of the periodic convolution of the 'Series and Convolution' chapter. Solar-term–synodic phase convolution, sexagenary periodic convolution and month–year periodic convolution are respectively the concrete manifestations in calendars of phase convolution, frequency convolution and amplitude convolution.

28.5 Integration with discrete sampling, harmonic analysis and time series analysis: the fifth step of the 'Periodic Systems' chapter

Definition 28.12 (Calendar periodicity diagnostic procedure)**.** For a given calendar system, diagnosis is carried out within the periodic-mathematics framework:

(i) **Ordering diagnosis**: compute the absolute sequence and the relative sequence and verify the translation transformation;

(ii) **Fixed-length diagnosis**: compute the fixed lengths of the tropical year, the synodic month and so on, and verify incommensurability;

(iii) **Intercalation diagnosis**: compute the cumulative error of the intercalation window and verify the bounded intercalation error theorem;

(iv) **Phase diagnosis**: compute the relative relations among the solar-term phase, the synodic phase and the sexagenary phase.

Theorem 28.9 (The calendar periodicity diagnostic procedure and its interfaces with three chapters)**.** *The four steps of Definition 28.12 connect respectively to:*

(i) the 'Discrete Sampling and Intercalation of Periodic Signals' chapter: Definition 21.1, Theorem 21.3, Theorem 21.10;

(ii) the 'Harmonic Analysis' chapter: Definition 22.5, Theorem 22.1, Proposition 22.11;

(iii) the 'Time Series Analysis' chapter: Definition 23.16, Proposition 23.2, Definition 23.19.

Hence calendar periodicity diagnosis is not an independent method, but an application of the unified tool chain of periodic mathematics to calendar systems.

Proof. The correspondence is item by item: (i) ordering diagnosis requires sampling and spectral estimation; (ii) the frequency decomposition and the phase kernels are found in the 'Harmonic Analysis' chapter; (iii) the periodogram peaks, the Fisher test and window-function validation are found in the 'Time Series Analysis' chapter. Hence the diagnostic procedure is a synthesis of the tool chains of the three chapters. □

Remark 28.8 (Operational order after integration). The recommended order for an actual diagnosis is: first decide whether the calendar target is periodic (Theorem 28.1); then construct the Fourier data and the epicycle group of the solar-term, synodic and sexagenary data (Theorem 28.3); then optimise the parameters by the intercalation window and dynamic adjustment (Theorem 28.4, Theorem 28.6); then perform calendar convolution and analyse the phase relations (Proposition 28.4, Proposition 28.5); finally output the diagnosis as in Definition 28.12.

28.6 Ordering and fixed length: basic attributes of calendar units

28.6.1 Absolute sequence and relative sequence

Definition 28.13 (Absolute sequence). An **absolute sequence** is the continuous index of a calendar unit counted from the epoch year of the Xuanyuan calendar. Let the astronomical year number of the epoch year be $Y_0 = -2697$ (the historical year 2697 BCE, a Jiazi year). For any calendar unit U, its **absolute sequence** $S_{\mathsf{abs}}(U)$ is defined as the continuous numbering of that unit counted from the epoch year.

The absolute sequence has the following properties:

(i) **Continuous and unbroken**: it is not interrupted by changes of dynasty, changes of reign title or calendar reform.

(ii) **Uniqueness**: every calendar unit has one and only one absolute sequence number.

(iii) **Monotonically increasing**: it increases strictly monotonically with time.

(iv) **Computable across periods**: the time interval between any two calendar units can be computed directly from the difference of their sequences.

Definition 28.14 (Relative sequence). A **relative sequence** is the number of a calendar unit within the current period:

(i) **Relative sequence of years**: counted from the first year of the current reign title.

(ii) **Relative sequence of months**: counted from the first month (the Tiger month) of the current year; an intercalary month retains the number of the preceding month.

(iii) **Relative sequence of solar terms**: counted from Lichun of the current year, with Lichun as 1 and Dahan as 24.

(iv) **Relative sequence of days**: counted from the day of new moon (the first day) of the current month.

The relative sequence measures the position of a calendar unit within the current period—its value depends on the choice of the starting point of the current period.

28.6.2 Fixed length: the intrinsic period of a calendar unit

Remark 28.9 (The fixed-length range of the climate month). The fixed length of the climate month (Climate) is not fixed, but varies within the range 14–18 days. This is because, under the true solar-term method, the length of each climate month depends on the actual speed of the Sun's motion along the ecliptic—climate months near the winter solstice are shorter (about 14 days) and climate months near the summer solstice are longer (about 16–18 days). The variable fixed length of the climate month is the manifestation in the true solar-term method of the non-uniformity of the Sun's apparent motion.

Remark 28.10 (The mathematical significance of incommensurability). The **empirical observation / research programme** of Remark 28.5 shows that, within the current precision of measurement, the periods of the solar calendar and the lunar calendar cannot be unified exactly by any integer ratio. Any lunisolar scheme can achieve approximate coordination of these two sets of periods only in the sense of best rational approximation. The accuracy of the approximation is given by the error of the continued-fraction convergents. The ratio $235/19$ is a high-accuracy approximation to $T_{\text{trop}}/T_{\text{syn}}$, with relative error about 1.2×10^{-5}, corresponding to a cumulative error of about 0.09 days within 19 years—this provides a basis of error control for the intercalation scheme of the Xuanyuan calendar.

28.7 The ordering and fixed length of solar terms and mid-qi

Definition 28.15 (Distinction between solar terms and mid-qi). The twenty-four solar terms fall into two groups: the 12 **solar terms** (jie) mark the beginning of each month, and the 12 **mid-qi** (mid-terms) mark the midpoint of each month. Solar terms and mid-qi alternate: Lichun→Yushui→Jingzhe→Chunfen→ $\cdots$ →Xiaohan→Dahan.

Table 28.1: The ordering and fixed length of the twenty-four solar terms

Relative index	Name	Type	Phase angle	Solar ecliptic longitude	Month order
1	Lichun	Solar term	0	$315°$	Beginning of the Tiger month
2	Yushui	Mid-qi	$\pi/12$	$330°$	Middle of the Tiger month
3	Jingzhe	Solar term	$\pi/6$	$345°$	Beginning of the Rabbit month
4	Chunfen	Mid-qi	$\pi/4$	$0°$	Middle of the Rabbit month
5	Qingming	Solar term	$\pi/3$	$15°$	Beginning of the Dragon month
6	Guyu	Mid-qi	$5\pi/12$	$30°$	Middle of the Dragon month
7	Lixia	Solar term	$\pi/2$	$45°$	Beginning of the Snake month
8	Xiaoman	Mid-qi	$7\pi/12$	$60°$	Middle of the Snake month
9	Mangzhong	Solar term	$2\pi/3$	$75°$	Beginning of the Horse month
10	Xiazhi	Mid-qi	$3\pi/4$	$90°$	Middle of the Horse month
11	Xiaoshu	Solar term	$5\pi/6$	$105°$	Beginning of the Goat month
12	Dashu	Mid-qi	$11\pi/12$	$120°$	Middle of the Goat month
13	Liqiu	Solar term	π	$135°$	Beginning of the Monkey month
14	Chushu	Mid-qi	$13\pi/12$	$150°$	Middle of the Monkey month
15	Bailu	Solar term	$7\pi/6$	$165°$	Beginning of the Rooster month
16	Qiufen	Mid-qi	$5\pi/4$	$180°$	Middle of the Rooster month

28.8 Ordering months by solar terms and mid-qi

Definition 28.16 (Rule for ordering months by solar terms and mid-qi)**. The month order of a synodic month is determined jointly by the solar term and the mid-qi it contains**: a synodic month containing Lichun (a solar term) and Yushui (a mid-qi) is the first month (the Tiger month); a synodic month containing Jingzhe and Chunfen is the second month (the Rabbit month); a synodic month containing Qingming and Guyu is the third month (the Dragon month); a synodic month containing Lixia and Xiaoman is the fourth month (the Snake month); a synodic month containing Mangzhong and Xiazhi is the fifth month (the Horse month); a synodic month containing Xiaoshu and Dashu is the sixth month (the Goat month); a synodic month containing Liqiu and Chushu is the seventh month (the Monkey month); a synodic month containing Bailu and Qiufen is the eighth month (the Rooster month); a synodic month containing Hanlu and Shuangjiang is the ninth month (the Dog month); a synodic month containing Lidong and Xiaoxue is the tenth month (the Pig month); a synodic month containing Daxue and Dongzhi is the eleventh month (the Rat month); a synodic month containing Xiaohan and Dahan is the twelfth month (the Ox month).

Theorem 28.10 (Lichun is always in the Tiger month and Dongzhi always in the Rat month)**.**

Lichun is always in the Tiger month: Lichun is the solar term of the Tiger month and Yushui is its mid-qi. The synodic month containing Lichun and Yushui is the first month (the Tiger month), so Lichun must fall within the Tiger month.

(ii) **Dongzhi is always in the Rat month**: Dongzhi is the mid-qi of the Rat month and Daxue is its solar term. The synodic month containing Daxue and Dongzhi is the eleventh month (the Rat month), so Dongzhi must fall within the Rat month.

Proof. By Definition 28.16, the synodic month containing Lichun (a solar term) and Yushui (a mid-qi) is the first month (the Tiger month), and the synodic month containing Daxue (a solar term) and Dongzhi (a mid-qi) is the eleventh month (the Rat month). By Table 28.1, Lichun is the beginning of the Tiger month (relative index 1, solar ecliptic longitude 315°) and Yushui is the middle of the Tiger month (index 2, longitude 330°); hence the synodic month containing Lichun has month order Tiger, and Lichun must lie within the Tiger month. Daxue is the beginning of the Rat month (index 21, longitude 255°) and Dongzhi is the middle of the Rat month (index 22, longitude 270°); hence the synodic month containing Dongzhi has month order Rat, and Dongzhi must lie within the Rat month. The month order is determined by the unique pairing of the solar term and the mid-qi contained in the month (see Proposition 28.7), so both conclusions are unambiguous. ☐

Proposition 28.6 (A synodic month containing only a solar term and no mid-qi is an intercalary month)**. Criterion for an intercalary month**: if within a synodic month there is **only a solar term (jie) and no mid-qi**, then that month is an **intercalary month**, and it retains the month number of the preceding month. A concrete example: if a synodic month contains **Lichun** (a solar term) but not **Yushui** (a mid-qi), then that month is an **intercalary Tiger month**—it retains the month order of the Tiger month and is marked as an intercalary month. The next synodic month containing Yushui is the formal first month (the Tiger month).

Proof. Under the true solar-term method the intervals between mid-qi are non-uniform: they are shorter around the winter solstice (about 29.4 days) and longer around the summer solstice (about 31.5 days). The fixed length of the synodic month is $T_{\text{syn}} \approx 29.53$ days. When the interval between two adjacent mid-qi exceeds 29.53 days (around the summer solstice), there may exist a synodic month containing no mid-qi at all—that month is an intercalary month. When the interval is less than 29.53 days (around the winter solstice), there may exist a synodic month containing two mid-qi—in that case the month order is determined by the solar term, and the surplus mid-qi is assigned to the next month. □

Proposition 28.7 (Consistency of ordering months by solar terms and mid-qi with intercalation by absence of a mid-term)**.** *The rule for ordering months by solar terms and mid-qi and the rule of intercalation by absence of a mid-term are completely consistent:*

(i) *normal month (containing one solar term and one mid-qi): the month order is determined by the combination of the solar term and the mid-qi.*

(ii) *intercalary month (containing only a solar term and no mid-qi): it retains the month order of the preceding month.*

(iii) *special case (containing two mid-qi): the month order is determined by the solar term, and the surplus mid-qi is assigned to the next month.*

Proof. The proof proceeds by cases according to the solar term and the mid-qi contained in the synodic month:

(i) **Normal month (containing exactly one solar term and one mid-qi):** by Definition 28.16, solar terms and mid-qi are paired alternately as in Table 28.1 (Lichun–Yushui, Jingzhe–Chunfen, $\cdots$), so the month order is uniquely determined by that combination of a solar term and a mid-qi, and the rule of intercalation by absence of a mid-term does not apply to this month.

(ii) **Intercalary month (containing only a solar term and no mid-qi):** here Definition 28.16 lacks a mid-qi pairing and cannot give the month order directly, so by Proposition 28.6 the month retains the month order of the preceding month and is marked as an intercalary month; the two rules give the same month order in this case.

(iii) **Containing two mid-qi:** by the proof of Proposition 28.6, when the interval between adjacent mid-qi is shorter than the fixed length T_{syn} of the synodic month there may occur a month containing two mid-qi; in that case the month order is determined by the solar term (the solar term uniquely determines the attribution of the month), and the surplus mid-qi is assigned to the next month.

The three cases above are mutually exclusive and exhaust all synodic months, and in each case 'ordering months by solar terms and mid-qi' and 'intercalation by absence of a mid-term' give the same month order; hence the two rules are completely consistent. □

28.9 Data architecture: external loading of solar-term data and synodic data

Remark 28.11 (Positioning of this section). This section and the subsequent UML class diagrams, flowcharts and interface descriptions belong to the **auxiliary implementation description**; they serve to explain the programmatic interfaces and data structures for Xuanyuan calendar computation and do not carry theorems or proofs. The mathematical conclusions involved in this section (ordering, fixed length, ordering of months by solar terms and mid-qi, intercalation decision) have all been given rigorously in the definitions, propositions and theorems above; here they are merely mapped to engineering implementation structures.

28.9.1 Principles of external data loading

Definition 28.17 (Principles of external data loading)**.** The computation of the Xuanyuan calendar depends on two kinds of **externally loaded data**:

(i) **Solar-term data**: the precise instants of the twenty-four solar terms (represented as Julian day numbers). Each record contains the name, the type (solar term or mid-qi), the solar ecliptic longitude and the Julian day number.

(ii) **Synodic data**: the precise instant of the new moon of each synodic month (represented as a Julian day number). Each record contains the absolute sequence number and the Julian day number of the new moon.

Solar-term data and synodic data are two independent external periodic systems—solar-term data provide an exact sampling of the tropical-year period, and synodic data provide an exact sampling of the synodic-month period. A calendar computation program does not re-derive astronomical data but loads and uses them directly.

28.9.2 Venn diagram: set relationships among data and computation

28.9.3 UML class diagram: the Range class and the temporal-unit inheritance hierarchy

28.9.4 UML class diagram: timestamps and temporal fields

Remark 28.12 (The calendrical meaning of the Range class). The **Range** class defines a range constraint—$lowest \le lower \le higher \le hiest$. $valid()$ checks whether a value lies within $[lowest, hiest]$, and $clamp(value)$ truncates a value to that range. In calendar systems the Range class is used to constrain the admissible value ranges of the various time fields—for example the relative sequence of months must lie within $[1, 12]$, and the relative sequence of days must lie within $[1, 30]$ (a long month) or $[1, 29]$ (a short month).

Remark 28.13 (The lengthOfWeek method of Climate and Phase). Both **Climate** (the climate month) and **Phase** (the synodic month) contain a **lengthOfWeek**

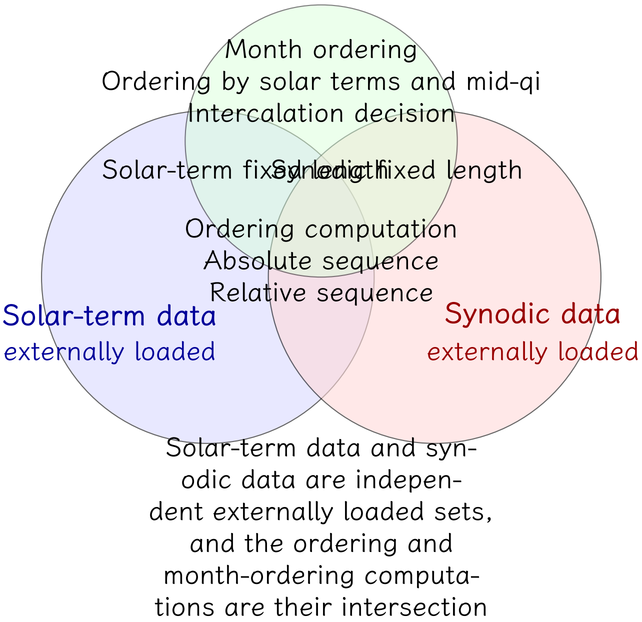

Month ordering
Ordering by solar terms and mid-qi
Intercalation decision
Solar-term fixed length
Synodic fixed length
Ordering computation
Absolute sequence
Relative sequence
Solar-term data
externally loaded
Synodic data
externally loaded
Solar-term data and synodic data are independent externally loaded sets, and the ordering and month-ordering computations are their intersection

Figure 28.1: Venn diagram: set relationships among solar-term data, synodic data and computational logic.

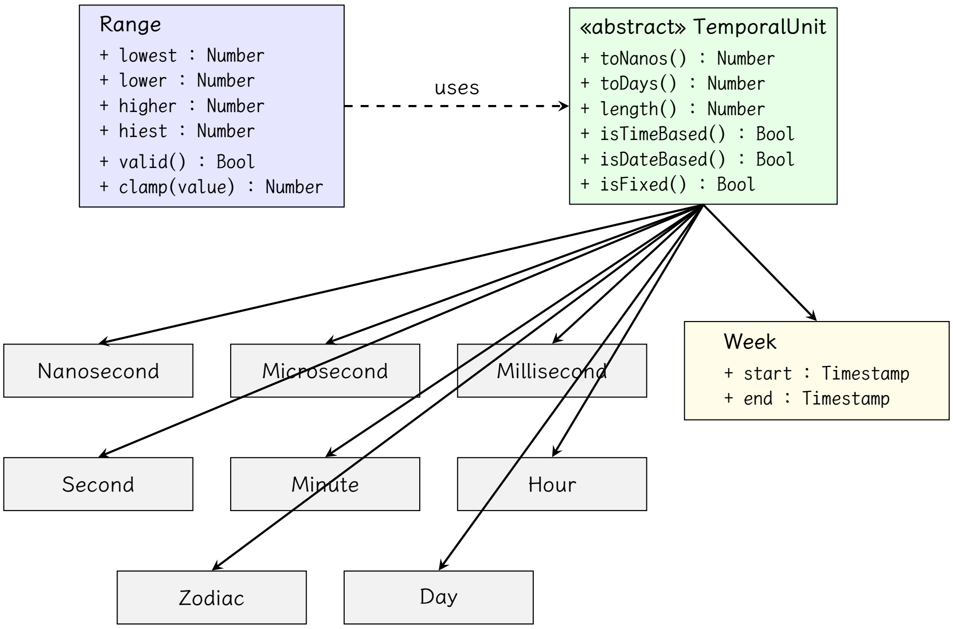

Range
+ lowest : Number
+ lower : Number
+ higher : Number
+ hiest : Number
+ valid() : Bool
+ clamp(value) : Number
uses
«abstract» TemporalUnit
+ toNanos() : Number
+ toDays() : Number
+ length() : Number
+ isTimeBased() : Bool
+ isDateBased() : Bool
+ isFixed() : Bool
Nanosecond
Microsecond
Millisecond
Week
+ start : Timestamp
+ end : Timestamp
Second
Minute
Hour
Zodiac
Day

Figure 28.2: UML class diagram: the Range class and the TemporalUnit basic-unit inheritance hierarchy.

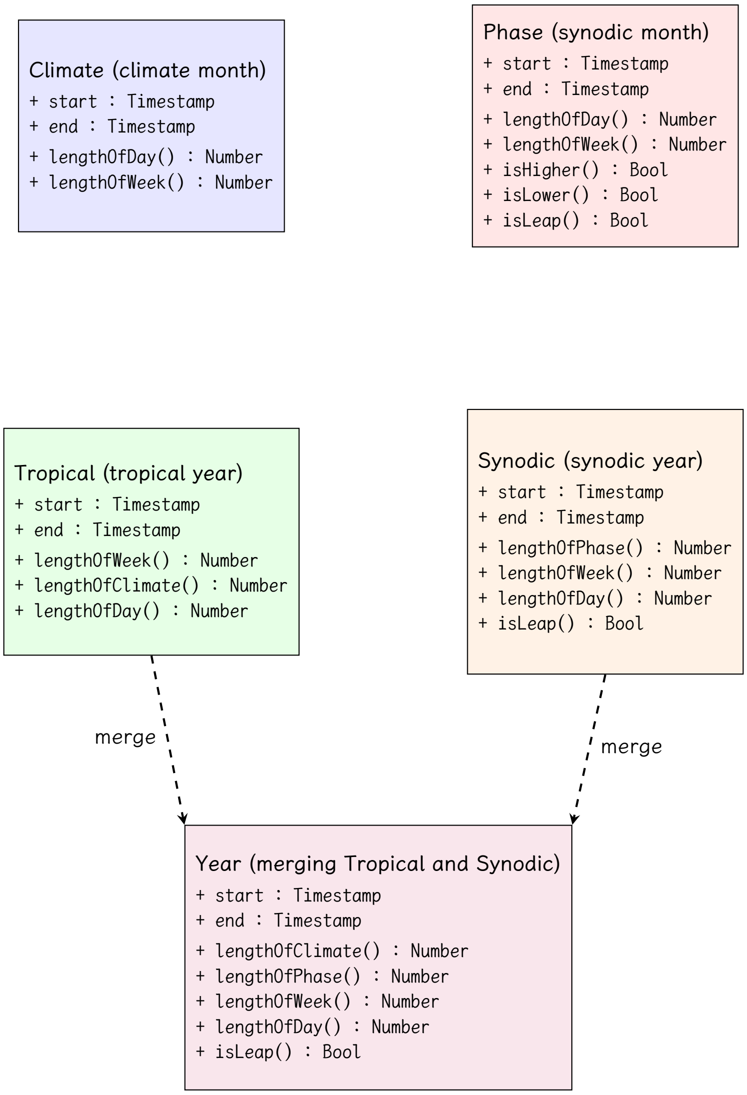

Climate (climate month)
+ start : Timestamp
+ end : Timestamp
+ lengthOfDay() : Number
+ lengthOfWeek() : Number
Phase (synodic month)
+ start : Timestamp
+ end : Timestamp
+ lengthOfDay() : Number
+ lengthOfWeek() : Number
+ isHigher() : Bool
+ isLower() : Bool
+ isLeap() : Bool
Tropical (tropical year)
+ start : Timestamp
+ end : Timestamp
+ lengthOfWeek() : Number
+ lengthOfClimate() : Number
+ lengthOfDay() : Number
Synodic (synodic year)
+ start : Timestamp
+ end : Timestamp
+ lengthOfPhase() : Number
+ lengthOfWeek() : Number
+ lengthOfDay() : Number
+ isLeap() : Bool
merge
merge
Year (merging Tropical and Synodic)
+ start : Timestamp
+ end : Timestamp
+ lengthOfClimate() : Number
+ lengthOfPhase() : Number
+ lengthOfWeek() : Number
+ lengthOfDay() : Number
+ isLeap() : Bool

Figure 28.3: UML class diagram: the Climate, Phase, Tropical and Synodic classes and the merged Year class.

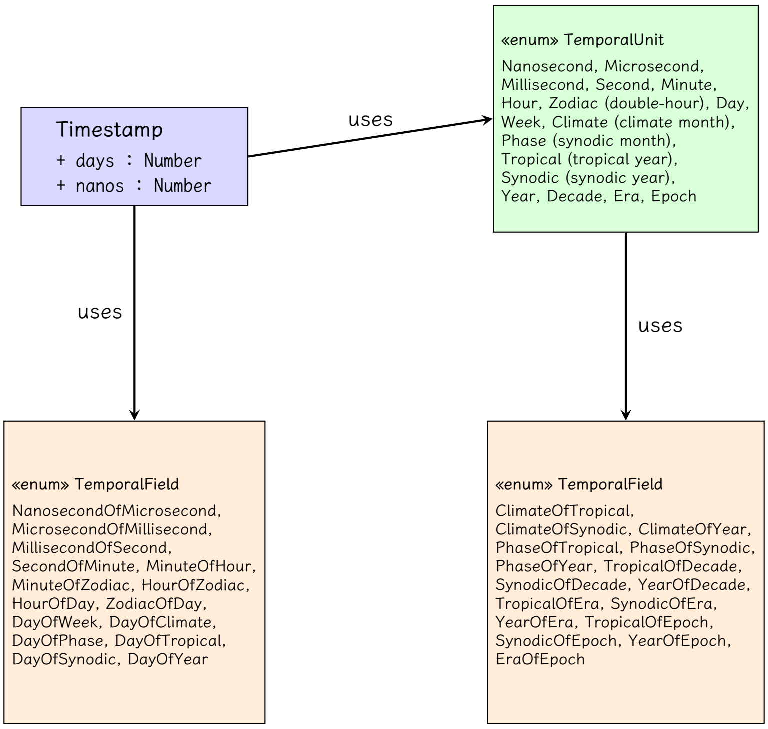

Timestamp
+ days : Number
+ nanos : Number
uses
«enum» TemporalUnit
Nanosecond, Microsecond, Millisecond, Second, Minute, Hour, Zodiac (double-hour), Day, Week, Climate (climate month), Phase (synodic month), Tropical (tropical year), Synodic (synodic year), Year, Decade, Era, Epoch
uses
uses
«enum» TemporalField
NanosecondOfMicrosecond, MicrosecondOfMillisecond, MillisecondOfSecond, SecondOfMinute, MinuteOfHour, MinuteOfZodiac, HourOfZodiac, HourOfDay, ZodiacOfDay, DayOfWeek, DayOfClimate, DayOfPhase, DayOfTropical, DayOfSynodic, DayOfYear
«enum» TemporalField
ClimateOfTropical, ClimateOfSynodic, ClimateOfYear, PhaseOfTropical, PhaseOfSynodic, PhaseOfYear, TropicalOfDecade, SynodicOfDecade, YearOfDecade, TropicalOfEra, SynodicOfEra, YearOfEra, TropicalOfEpoch, SynodicOfEpoch, YearOfEpoch, EraOfEpoch

Figure 28.4: UML class diagram: Timestamp and the TemporalUnit and TemporalField enumerations.

method—it returns the number of whole weeks within the month. The length of a climate month lies within the range 14–18 days and the length of a synodic month is about 29.53 days. Hence a climate month usually contains 2–3 whole weeks and a synodic month usually contains 4–5 whole weeks. The lengthOfWeek method is of practical significance for arranging work schedules within a month and for determining the lunar-calendar date corresponding to each day of the week.

Remark 28.14 (The length method of TemporalUnit and isFixed). **length()** returns the number of units of the next lower level contained in the current unit: `Day.length()` $= 24$, `Week.length()` $= 7$, `Climate.length()` ≈ 15 (within the range 14–18 days), `Phase.length()` ≈ 29.53, `Tropical.length()` $= 12$, `Synodic.length()` $= 12$ or 13. **isFixed** returns whether the unit has a fixed length: `Hour.isFixed()` $= true$, `Phase.isFixed()` $= false$, `Climate.isFixed()` $= false$.

Remark 28.15 (The merge semantics of the Year class). **Year** is the merge of **Tropical** and **Synodic**—it possesses at once `lengthOfClimate` (the number of climate months in the tropical year $= 12$), `lengthOfPhase` (the number of synodic months in the year $= 12$ or 13), `lengthOfWeek` (the number of weeks in the year), `lengthOfDay` (the number of days in the year $= 365$ or 366) and `isLeap` (the leap-year decision). The merge embodies the essence of the lunisolar calendar—a calendar year is described simultaneously by two systems, the solar calendar (climate months) and the lunar calendar (synodic months).

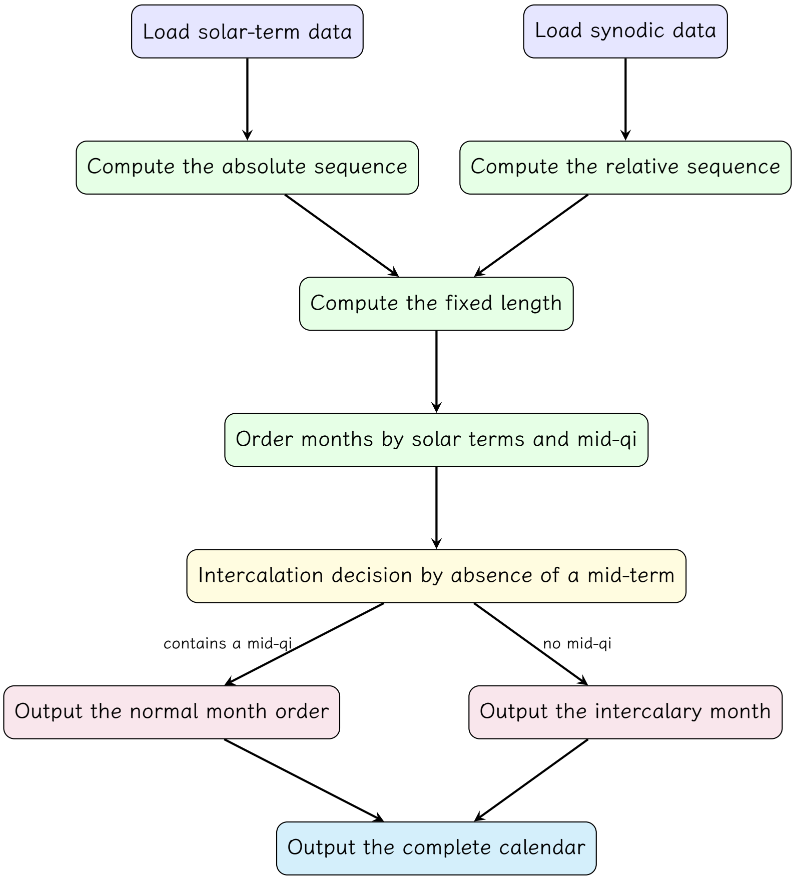

Load solar-term data
Load synodic data
Compute the absolute sequence
Compute the relative sequence
Compute the fixed length
Order months by solar terms and mid-qi
Intercalation decision by absence of a mid-term
contains a mid-qi
no mid-qi
Output the normal month order
Output the intercalary month
Output the complete calendar

Figure 28.5: Calendar computation flowchart: external data loading→ordering→fixed length→ordering of months by solar terms and mid-qi→intercalation decision→calendar output.

28.9.5 Flowchart

28.10 The independence of the solar-term system and the synodic-month system

28.10.1 The solar-term system—the solar calendar

Definition 28.18 (Solar-term system). The **twenty-four solar terms** are a solar-calendar system originating in ancient China, which divides the tropical year equally into 24 time nodes. The solar-term system takes the tropical year T_{trop} as its basic period.

Definition 28.19 (Mean solar-term method). The **mean solar-term method** divides the tropical year uniformly into 24 parts: $\Delta T_{\text{pingqi}} = T_{\text{trop}}/24 \approx 15.2184$ days. It corresponds to mean solar time—a uniform division of time.

Definition 28.20 (True solar-term method). The **true solar-term method** determines the instants of the solar terms from the actual apparent position of the Sun on the ecliptic, placing one solar term every $15°$. It corresponds to true solar time—a non-uniform division of time.

Proposition 28.8 (Comparison of the mean and true solar-term methods). *The mean solar-term method is based on a uniform division of time (corresponding to mean solar time), whilst the true solar-term method is based on a uniform angular division of the solar ecliptic longitude (corresponding to true solar time). The transition from the mean to the true solar-term method is a transition from the uniformisation of time to the uniformisation of ecliptic longitude.*

Proof. By Definition 28.19, the mean solar-term method divides the tropical year uniformly into 24 parts, and the solar-term interval $\Delta T_{\text{pingqi}} = T_{\text{trop}}/24$ is constant, that is, a uniform division of the time axis. By Definition 28.20, the true solar-term method takes a solar term every $15°$ of solar ecliptic longitude; since the apparent motion of the Sun is fast at perihelion (near the winter solstice) and slow at aphelion (near the summer solstice), the time intervals between the solar terms are non-uniform, corresponding to a uniform angular division of the ecliptic. Hence the mean solar-term method achieves the uniformisation of time and the true solar-term method achieves the uniformisation of ecliptic longitude. □

28.10.2 The synodic-month system—the lunar calendar

Definition 28.21 (Synodic-month system). The **synodic-month system** is a lunar-calendar system based on the synodic period of the Moon's revolution about the Earth. $T_{\text{syn}} = 29.5305889$ mean solar days. Twelve synodic months constitute a lunar year, which is shorter than the tropical year by about 10.875 days.

Proposition 28.9 (The independent period of the synodic-month system). *The periods of the synodic-month system and the solar-term system are incommensurable, that is,*

$$\frac{T_{\text{trop}}}{T_{\text{syn}}} \approx 12.36827,$$

hence it is impossible to unify the solar calendar and the lunar calendar perfectly by a simple integer ratio.

Proof. By Definition 28.21, the synodic-month system has period $T_{\text{syn}} = 29.5305889$ mean solar days; by Definition 28.7, the solar-term system has period $T_{\text{trop}} = 365.2421897$ mean solar days. Their ratio

$$\frac{T_{\text{trop}}}{T_{\text{syn}}} = \frac{365.2421897}{29.5305889} \approx 12.36827$$

agrees with no p/q of bounded denominator within the current precision of measurement (an **empirical observation / research programme**; see Remark 28.5), so there exist no integers p, q such that $p\,T_{\text{trop}} = q\,T_{\text{syn}}$. Hence the period of the synodic-month system is independent of the period of the solar-term system, and each preserves its own period scale. ☐

28.10.3 Coordination of the lunisolar calendar—intercalation

Lemma 28.4 (Metonic cycle (seven intercalary months in nineteen years))**.** *Nineteen tropical years and* 235 *synodic months are approximately equal in length:*

$$19\,T_{\text{trop}} \approx 235\,T_{\text{syn}},$$

where $19\,T_{\text{trop}} \approx 6939.60$ *days and* $235\,T_{\text{syn}} \approx 6939.69$ *days, the two differing by about* 0.09 *days. Hence after* 19 *years the phases of the Sun and the Moon are almost restored; moreover* $235 = 12 \times 19 + 7$*, so* 7 *intercalary months must be inserted within* 19 *years (the intercalation period ratio being* $235/19$*).*

Proof. From the fixed-length values of Definition 28.7 we compute

$$19\,T_{\text{trop}} = 19 \times 365.2421897 = 6939.6016 \text{ days},$$

$$235\,T_{\text{syn}} = 235 \times 29.5305889 = 6939.6884 \text{ days}.$$

The two differ by

$$6939.6884 - 6939.6016 = 0.0868 \text{ days},$$

which is far smaller than one synodic month (29.53 days); hence 19 tropical years and 235 synodic months are restored almost simultaneously, and the phases of the solar calendar and the lunar calendar close within 19 years. Without intercalation, 19 years would contain only $12 \times 19 = 228$ synodic months, 7 fewer than 235. Hence 7 intercalary months must be added within 19 years so that the two calendars align, which gives the intercalation period ratio

$$\frac{235}{19}.$$ ☐

Definition 28.22 (Intercalation—phase alignment of the two systems)**.** Seven intercalary months in nineteen years (the Metonic cycle): $19 \times T_{\text{trop}} \approx 6939.60$ days, $235 \times T_{\text{syn}} \approx 6939.69$ days. After 19 years the phases of the Sun and the Moon are almost restored. The intercalation period ratio is $235/19$.

Proposition 28.10 (Intercalation by absence of a mid-term—consistency with ordering months by solar terms and mid-qi)**.** *If within a synodic month there is only a solar term and no mid-qi, then that month is an intercalary month—this agrees with the criterion of Proposition 28.6. Intercalation by absence of a mid-term ensures that the beginning of the year (the first month) always lies around Lichun.*

Proof. The first sentence is the criterion for an intercalary month of Proposition 28.6, whose mechanism was given in the proof of that proposition (a month without a mid-qi occurs when the interval between adjacent mid-qi exceeds the fixed length of the synodic month). The second sentence explains the purpose of intercalation: by Definition 28.16, the synodic month containing Lichun and Yushui is the first month (the Tiger month); intercalation by absence of a mid-term guarantees that the first month always contains Lichun, so that the deviation of the beginning of the year from Lichun does not exceed the order of one synodic month, and the beginning of the year therefore always lies around Lichun. ☐

28.11 The periodic structure of the lunisolar calendar

Theorem 28.11 (The three periodic scales of the lunisolar calendar)**.**

(i) **Solar-term system (solar calendar)**: *with the tropical year T_{trop} as period, the distribution of the solar terms within the year being determined by the solar ecliptic longitude.*

(ii) **Synodic-month system (lunar calendar)**: *with the synodic month T_{syn} as period, the repetition of the lunar phases being determined by the relative positions of the Sun, the Earth and the Moon.*

(iii) **Intercalation system (coordination layer)**: *with seven intercalary months in nineteen years as its rule and the intercalation period ratio $235/19$, coordinating the first two sets of periods on the scale of 19 years.*

Proof. The realisation of the three periodic scales is verified item by item:

(i) the solar-term system has the tropical year T_{trop} as period (Definition 28.18), and the mean solar-term method divides it uniformly into 24 parts (Definition 28.19), giving a uniform division of the solar terms on the time axis.

(ii) the synodic-month system has T_{syn} as period (Definition 28.21) and measures the synodic phase of the Moon's revolution about the Earth—the repetition period of the lunar phases.

(iii) the intercalation system coordinates the first two: by the Metonic-cycle lemma (seven intercalary months in nineteen years), 235 synodic months are arranged within 19 tropical years, with intercalation period ratio $235/19$; this is the coordination layer between the two independent periods.

☐

28.12 The mathematical structure of astronomical and historical year numbering

Definition 28.23 (Astronomical year numbering)**.** Astronomical year numbering is continuous on the integer axis: the year 1 CE is $+1$, the year 1 BCE is 0, and the year n BCE is $-(n-1)$. 0 is a legitimate year number.

Definition 28.24 (Historical year numbering)**.** In historical year numbering there is no year 0 CE—the year 1 BCE is followed directly by the year 1 CE.

Proposition 28.11 (Conversion between astronomical and historical year numbering)**.** When $Y_{\text{ast}} \geq 1$, $Y_{\text{hist}} = Y_{\text{ast}}$; when $Y_{\text{ast}} \leq 0$, $Y_{\text{hist}} = 1 - Y_{\text{ast}}$.

Proof. By Definition 28.23, astronomical year numbering writes the year 1 CE as $+1$ and the year n BCE as $-(n-1)$. When $Y_{\text{ast}} \geq 1$, astronomical and historical year numbering give the same CE year number, so $Y_{\text{hist}} = Y_{\text{ast}}$. When $Y_{\text{ast}} \leq 0$, Y_{ast} corresponds to the year $(1 - Y_{\text{ast}})$ BCE: for example $Y_{\text{ast}} = 0$ corresponds to the year 1 BCE and $Y_{\text{ast}} = -1$ to the year 2 BCE; in general, solving $-(n-1) = Y_{\text{ast}}$ gives $n = 1 - Y_{\text{ast}}$, hence $Y_{\text{hist}} = 1 - Y_{\text{ast}}$. □

Theorem 28.12 (Correction theorem for time intervals crossing the Common Era boundary)**.** Starting from the year m BCE and passing through P years, the resulting historical year is:

$$Y_{\text{hist}} = \begin{cases} m - P, & P < m, \\ P - m + 1, & P \geq m. \end{cases}$$

When $P \geq m$ (crossing the Common Era boundary), a $+1$ correction is required.

Proof. Compute in the astronomical-year-numbering coordinates of Proposition 28.11. The astronomical year number of the year m BCE is $Y_{\text{ast}} = -(m-1)$ (Definition 28.23); after P years the astronomical year number becomes $Y'_{\text{ast}} = -(m-1) + P = P - m + 1$.

- If $P < m$, then $Y'_{\text{ast}} = P - m + 1 \leq 0$, which is still BCE; by Proposition 28.11, $Y_{\text{hist}} = 1 - Y'_{\text{ast}} = m - P$.
- If $P \geq m$, then $Y'_{\text{ast}} = P - m + 1 \geq 1$, which has entered the Common Era; here $Y_{\text{hist}} = Y'_{\text{ast}} = P - m + 1$, where the $+1$ is precisely the correction required for crossing the boundary 'there is no year 0 CE'.

Combining the two cases gives the assertion. □

28.13 The epoch system of the *Huainanzi*, 'Tianwenxun'

Definition 28.25 (One Ji)**.** One Ji is 76 years: $76 \times 365.25 = 940 \times 29.53085 = 27759$ days—the least common multiple period of the tropical year and the synodic month.

Definition 28.26 (One Bu—the Great Conclusion after 1520 years). One Bu is 1520 years $= 76 \times 20$. $1520 \bmod 60 = 20$—on passing through one Bu, the sexagenary cycle advances by 20 places.

Theorem 28.13 (Theorem on the sexagenary shift over one Bu). After k Bu the sexagenary index is $S_k = S_0 + 20k \pmod{60}$. For $k = 3$ the sexagenary sign is restored. Within one Bu the sexagenary sign advances by $+20$, cycling through the three pillars Jiayin, Jiaxu and Jiawu; the Jia pillars appearing in the epoch sequence are Jiazi (symbolising the epoch starting point), Jiayin, Jiaxu and Jiawu.

Proof. $1520 \bmod 60 = 20$. The sexagenary index increases by 20 in each Bu. $20 \times 3 = 60 \equiv 0 \pmod{60}$. ☐

28.14 Derivation of the epochs of the Xuanyuan calendar

Proposition 28.12 (Determination of the sexagenary sign at the end of the first Bu). Starting from the Jiayin epoch year (-2647) and passing through one Bu, we reach the first conclusion: $-2647 + 1520 = -1127$. The sexagenary index is $0 + 20 = 20$ (Jiaxu). The year 1127 BCE (a Jiaxu year).

Proof. By Definition 28.6, the Jiayin epoch year is -2647 (sexagenary index 0); by Definition 28.26, one Bu is 1520 years and the sexagenary sign advances by 20 places per Bu ($1520 \bmod 60 = 20$, see Theorem 28.13). Hence the astronomical year number of the first conclusion is $-2647 + 1520 = -1127$, and its sexagenary index is $0 + 20 = 20 \pmod{60}$. By the order of the sexagenary cycle (Jiayin $= 0$), index 20 corresponds to Jiaxu. ☐

Proposition 28.13 (Determination of the sexagenary sign at the end of the second Bu and the Common Era correction). Astronomical year number of the end point: $-1127 + 1520 = 393$. Historical year number: $393 + 1 = 394$. Sexagenary index $20 + 20 = 40$ (Jiawu). The year 394 CE (a Jiawu year).

Proof. The first conclusion is -1127 (Proposition 28.12); passing through one more Bu gives the astronomical year number of the second conclusion, $-1127 + 1520 = 393$. Since $393 \geq 1$ and the Common Era boundary has been crossed, by Theorem 28.12 (taking the case $P \geq m$) the historical year number requires a $+1$ correction, giving $393 + 1 = 394$, that is, the year 394 CE. By Theorem 28.13 the sexagenary index advances a further 20 places: $20 + 20 = 40$, corresponding to Jiawu. ☐

Proposition 28.14 (Determination of the sexagenary sign at the end of the third Bu). Astronomical year number of the end point: $393 + 1520 = 1913$. Historical year number: $1913 + 1 = 1914$. Sexagenary index $40 + 20 = 60 \equiv 0$ (Jiayin). The year 1914 CE (a Jiayin year, the third year of the Republic).

Proof. The second conclusion is the astronomical year 393 (Proposition 28.13); passing through one more Bu gives the astronomical year number of the third conclusion, $393 + 1520 = 1913$; by the $+1$ correction of Theorem 28.12, the historical year is $1913 + 1 = 1914$. The sexagenary index advances a further 20 places: $40 + 20 = 60 \equiv 0 \pmod{60}$, corresponding to Jiayin (Jiayin $= 0$, Jiazi $= 10$). ☐

Theorem 28.14 (Completeness of the epoch sequence of the Xuanyuan calendar).

Epoch order	Astronomical year	Historical year	Sexagenary sign	Remark
Starting point	-2697	2697 BCE	Jiazi	Yellow Emperor era hypothesis
Jiayin epoch	-2647	2647 BCE	Jiayin	the 50th year after Jiazi
First conclusion	-1127	1127 BCE	Jiaxu	does not cross the Common Era
Second conclusion	393	394	Jiawu	crosses the Common Era, $+1$ correction
Third conclusion	1913	1914	Jiayin	third year of the Republic

The three Bu total 4560 years; the sexagenary sign returns from Jiayin to Jiayin, completing the full 60-step sexagenary cycle.

Proof. By Proposition 28.12, Proposition 28.13 and Proposition 28.14, the epoch starting point Jiayin (index 0) has sexagenary index $0 + 3 \times 20 = 60 \equiv 0 \pmod{60}$ after three Bu, that is, it returns to Jiayin. The total number of years in three Bu is $3 \times 1520 = 4560 = 76 \times 60$, hence $4560 \equiv 0 \pmod{60}$, and the sexagenary cycle runs through *exactly* 76 times. ☐

Remark 28.16 (Difference between astronomical and historical year numbering). *From the second conclusion onwards, astronomical and historical year numbering differ by* 1*: the astronomical year* 393 *corresponds to the historical year* 394*. This difference arises because astronomical year numbering includes the year* 0*, whilst historical year numbering has no year* 0*.*

28.15 The nested periodic system of the *Huangji Jingshi*

Definition 28.27 (Yuan, Hui, Yun and Shi). *One shi* $= 30$ *years, one yun* $= 12$ *shi* $= 360$ *years, one hui* $= 30$ *yun* $= 10800$ *years, one yuan* $= 12$ *hui* $= 129600$ *years.*

Theorem 28.15 (Periodic correspondence between the Xuanyuan calendar and the yuan-hui-yun-shi cycles). *One yuan* $= 85 \times 1520 + 400$ *years. The sexagenary sign is exactly restored within one yuan.*

Proof. $129600 \div 1520 = 85$ with remainder 400. $85 \times 20 + 400 \equiv 20 + 40 = 60 \equiv 0 \pmod{60}$. ☐

28.16 Comparison of the periodic structures of the world's major calendar systems

Definition 28.28 (Yi ten-month solar calendar)**.** The **Yi ten-month solar calendar** takes the tropical year of the Earth's revolution about the Sun as its period and uses a cycle of twelve zodiac animals to count days. Every three zodiac-animal periods (36 days) form one month; the whole year has 10 months and 360 days, plus 5 'New Year days' (6 days in a leap year), giving an average year length of 365.25 days.

Proposition 28.15 (Periodic structure of the Yi ten-month calendar)**.** The periodic structure of the Yi ten-month solar calendar is determined by the following parameters:

(i) **Day-counting period**: *a cycle of* 12 *zodiac animals, that is,* $T_{\text{zodiac}} = 12$ *days.*

(ii) **Month period**: 3 *zodiac-animal periods* $= 36$ *days, that is,* $T_{\text{month}} = 3 \times 12 = 36$ *days.*

(iii) **Year period**: 10 *months* $= 360$ *days, plus* 5 *or* 6 *New Year days,* $T_{\text{year}} = 365$ *or* 366 *days.*

(iv) **Season period**: 2 *months* $= 72$ *days, that is,* $T_{\text{season}} = 2 \times 36 = 72$ *days.*

The relation between the month period and the day-counting period is $T_{\text{month}} = 3 \times T_{\text{zodiac}}$*—each month contains exactly three complete zodiac-animal cycles, which makes the zodiac attribution of the dates within a month completely determined.*

Proof. By Definition 28.28, each month has 36 days and the zodiac-animal period is 12 days, and $36 = 3 \times 12$, so each month is exactly 3 zodiac-animal periods. Likewise, the whole year has $360 = 30 \times 12$. One year has 10 months and each season has 2 months, so the season period is $2 \times 36 = 72$ days. ☐

Definition 28.29 (Periodic structure of the Maya calendar)**.** The **Maya calendar** consists of two independent periods: the **Tzolk'in**, with period 260 days (13 day numbers × 20 day names), and the **Haab'**, with period 365 days (18 months × 20 days +5 days). The combination of the two calendars forms the **Calendar Round**, whose period is the least common multiple of 260 and 365:

$$T_{\text{CR}} = \text{lcm}(260, 365) = 18980 \text{ days} = 52 \text{ Haab years}.$$

The Maya also used the **Long Count**, a cumulative day count with base 20, whose units are in order the kin (1 day), the uinal (20 days), the tun (360 days), the katun (7200 days) and the baktun (144000 days).

Proposition 28.16 (Periodic coordination of the Maya calendar)**.** *The periodic structure of the Maya calendar is structurally homologous with that of the Xuanyuan calendar:*

(i) *the incommensurability of the Tzolk'in (260 days) and the Haab' (365 days) (*$\text{gcd}(260, 365) = 5$*) agrees structurally with the incommensurability of the tropical year and the synodic month.*

(ii) the Calendar Round period of 18980 days $= 52$ Haab years corresponds to the intercalation period of 19 years $= 235$ synodic months in the Xuanyuan calendar—both are instances of two independent periods being coordinated by their least common multiple.

(iii) the base-20 cumulative system of the Long Count is structurally comparable with the nested yuan-hui-yun-shi system of the Xuanyuan calendar.

Proof. (i) The greatest common divisor of 260 and 365 is 5 and their least common multiple is 18980; the ratio of the tropical year to the synodic month agrees with no simple integer ratio within measurable precision (an **empirical observation / research programme**; see Remark 28.5), and neither can be unified exactly by a simple integer ratio. (ii) The Calendar Round period of 18980 days is the least common multiple of 260 and 365, corresponding to the fact that 19 years $= 235$ synodic months is the best rational approximation to the tropical year and the synodic month in the Xuanyuan calendar. (iii) The base-20 cumulative structure of the Long Count and the base-12 and base-30 nested structure of the yuan-hui-yun-shi cycles both achieve the precise recording of long time spans by multi-level nesting. ☐

Definition 28.30 (Ancient Indian lunisolar calendar)**.** The **ancient Indian lunisolar calendar** takes 5 years as one harmonising period; among the 5 solar years (1830 days) there are 62 synodic months, and 2 intercalary months are inserted within one period.

Proposition 28.17 (Intercalation parameters of the ancient Indian lunisolar calendar)**.** The intercalation parameters of the ancient Indian lunisolar calendar are structurally the same as the seven intercalary months in nineteen years of the Xuanyuan calendar—both use a set of integer parameters to satisfy approximately

$$p \cdot T_{\text{solar}} \approx q \cdot T_{\text{synodic}},$$

where the ancient Indian calendar takes $(p, q) = (5, 62)$ and the Xuanyuan calendar takes $(p, q) = (19, 235)$. Both parameter pairs are continued-fraction convergents of $T_{\text{trop}}/T_{\text{syn}}$—$(5, 62)$ corresponds to $62/5 = 12.4$ and $(19, 235)$ to $235/19 = 12.368\ldots$.

Proof. The ancient Indian calendar takes the solar year length as $1830/5 = 366$ days (an error of about 0.76 days from the true tropical year 365.2422 days) and the synodic month length as $1830/62 = 29.516\ldots$ days. The Xuanyuan calendar has solar year length $T_{\text{trop}} = 365.2422$ days and synodic month length $T_{\text{syn}} = 29.5306$ days, with 235 synodic months in 19 years. The common structure of the two is: choose integers p, q such that $p \cdot T_{\text{solar}} \approx q \cdot T_{\text{synodic}}$, and $q - 12p$ gives the total number of intercalary months (2 in the ancient Indian calendar, 7 in the Xuanyuan calendar). ☐

Theorem 28.16 (A unified framework for the periodic structures of the world's major calendars)**.** Although the Xuanyuan calendar, the Yi ten-month solar calendar, the Maya calendar and the ancient Indian lunisolar calendar differ in astronomical reference frame and cultural background, they share the following common framework in their periodic structure:

(i) **Basic astronomical periods**: *all calendars take the solar period (the tropical year) and/or the lunar period (the synodic month) as their basic period.*

(ii) **Independent periodic systems**: *every calendar contains two or more independent periodic systems (solar–lunar, Tzolk'in–Haab', solar terms–synodic month).*

(iii) **Coordination mechanism**: *the coordination of several periods is achieved by intercalation (the Xuanyuan calendar, the ancient Indian calendar), by a fixed combination of periods (the Maya calendar) or by a fixed allocation of days (the Yi ten-month calendar).*

(iv) **Nested structure**: *periods of different time scales are organised in a nested manner (the yuan-hui-yun-shi cycles, the Long Count, season–month–day).*

Proof. Verified item by item:

(i) the Xuanyuan calendar takes the tropical year and the synodic month as its basic periods; the Yi ten-month calendar takes the tropical year as its basic period; the Maya calendar takes the Tzolk'in of 260 days and the Haab' of 365 days as its basic periods; the ancient Indian lunisolar calendar takes the solar year and the synodic month as its basic periods.

(ii) the Xuanyuan calendar contains the two systems of solar terms and synodic months; the Maya calendar contains the two of the Tzolk'in and the Haab'; the ancient Indian lunisolar calendar contains the two of the solar year and the synodic month; the Yi ten-month calendar is a purely solar calendar.

(iii) the Xuanyuan calendar and the ancient Indian calendar coordinate the solar and lunar periods by intercalation; the Maya calendar produces the 18980-day Calendar Round by the fixed combination of 260 and 365; the Yi ten-month calendar by the fixed allocation of days $360 + 5/6$.

(iv) the yuan-hui-yun-shi cycles and the Long Count are both multi-level nested structures.

□

28.17 Stability of ordering and fixed length

Theorem 28.17 (Stability of calendar ordering and fixed lengths)**.** *Three kinds of stability exist in calendar systems:*

(i) **Sexagenary stability**: *the cyclic structure of the sexagenary cycle on the absolute sequence remains unchanged.*

(ii) **Stability of the ordering of solar terms and mid-qi**: *the rule for ordering months by solar terms and mid-qi remains stable over long periods—Lichun is always the solar term of the Tiger month and Dongzhi is always the mid-qi of the Rat month.*

(iii) **Fixed-length stability**: the fixed length of each calendar unit remains constant in the absence of external perturbation.

Proof. (i) By Theorem 28.13, the sexagenary index advances as $S_k = S_0 + 20k \pmod{60}$, with 60 a fixed modulus, so the cyclic structure of the sexagenary cycle on the absolute sequence remains unchanged.

(ii) By Theorem 28.10, Lichun is always the solar term of the Tiger month and Dongzhi is always the mid-qi of the Rat month; this is the stable manifestation over long periods of the pairing rule of Definition 28.16.

(iii) By Definition 28.7, the fixed length of each calendar unit is an intrinsic period scale; it remains constant in the absence of external perturbation.

□

Remark 28.17 (Connection with the rest of the book). The system of ordering and fixed length established in this chapter is an application of periodic mathematics to the counting of time. The absolute sequence gives the one-way advance along the time axis, the relative sequence measures the position within a period, and the fixed length gives the intrinsic period scale of each unit. The rule for ordering months by solar terms and mid-qi achieves the precise coordination of the solar and lunar calendars through the combination of a solar term and a mid-qi. The comparison of the world's major calendar systems shows that, however different their astronomical reference frames and cultural backgrounds, the periodic structures of calendar systems all follow the common framework 'basic astronomical periods—independent periodic systems—coordination mechanism—nested structure'.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The ordering advance of the sexagenary sequence: the translation transformation of the absolute sequence and the relative sequence $S_{\text{rel}} = S_{\text{abs}} - S_{\text{abs}}(P) + 1$ (Proposition 28.1)
Scaling	The fixed-length scaling of the tropical year and the synodic month: $T_{\text{trop}} = 365.2422$ days, $T_{\text{syn}} = 29.5306$ days (Definition 28.7)
Rotation	The rotational distribution of the solar terms on the ecliptic: the 2π rotation of the phase angle and the solar ecliptic longitude (Table 28.1)
Balance	The ordering and fixed-length stability of intercalation (Theorem 28.17)
Symmetry	The solar–lunar symmetry of the lunisolar calendar: the binary structure of the solar calendar and the lunar calendar (Theorem 28.11)
Closure	The closure of the sixty-step cycle of the stems and branches: the sexagenary cycle, with three Bu of 4560 years returning to Jiayin (Theorem 28.13, Theorem 28.14)
Period	The tropical year (365.2422 days), the synodic month (29.5306 days) and the sexagenary cycle (Definition 28.7); the intercalation period $235/19$ (Lemma 28.4)

Exercises

Exercise 1 Prove the **calendar periodicity decision theorem** of this chapter (28.1). Hint: first write down the decision map and the evolution group of this chapter's target, then apply the proof structure of the general criterion of the 'Periodic Systems' chapter.

Exercise 2 Verify that the hypotheses of **monotonicity of quantity adjustment** (28.3) hold for the domain objects of this chapter, and give a counterexample in which the hypotheses fail.

Exercise 3 Examine the behaviour of the **calendar periodicity diagnostic procedure and its interfaces with three chapters** (28.9) under degeneracy of the parameters, and explain how the conclusion degenerates accordingly.

Exercise 4 Explain the relation between the **action of the three kinds of convolution on calendar epicycle groups** (28.8) and the corresponding general conclusion of the ‘Periodic Systems’ chapter (whether they agree, whether it is a special case, or whether additional conditions are required).

Exercise 5 Using the seven-theme comparison table of this chapter, identify the concrete carriers in this chapter of the three themes **translation**, **scaling** and **rotation**, and give one example of domain data for each.

Exercise 6 Synthesis: restate the diagnostic procedure of this chapter using the five-step standard interface of the ‘Periodic Systems’ chapter (Table 24.1), noting one by one the theorem of this chapter on which each step depends: 28.1, 28.3, 28.9, 28.8.

Chapter 29

Periodic Structure of Geographic Systems

Remark 29.1 (Periodicity decision for the target of this chapter (citation)). When this chapter decides whether a research target is periodic, determines its period type, or seeks its minimal positive period, reference is always made to the 'Periodic Systems' chapter: for the periodicity criterion see Theorem 24.1, for the discrimination of period types see Definition 24.3, Proposition 24.1, for the search for the minimal positive period see Definition 24.4, Theorem 24.2, and for the construction of period series and epicycle groups from data see Theorem 24.3, Definition 24.6.

Remark 29.2 (Positioning of this chapter: developed along the five-step interface of the 'Periodic Systems' chapter). The periodic structure of geographic systems studies the mathematical characterisation of periodicity, quasi-periodicity and scale invariance in the Earth-surface system. Geographic systems lie at the intersection of astronomical, geological and climatic systems—astronomical cycles (the long-term variation of the three elements of the Earth's orbit) drive glacial–interglacial cycles on scales of ten thousand to a hundred thousand years, geological cycles (plate motion, orogeny, volcanic activity and seismic activity) shape landforms on scales of a million to a hundred million years, climatic cycles (atmospheric circulation, ocean currents and monsoons) control environmental change on scales from the seasonal to the interdecadal, whilst the surface terrain and the latitude–longitude system form the geometric framework of these processes. This chapter does not establish a separate periodicity criterion; instead it connects geographic systems to the five-step interface prescribed by the 'Periodic Systems' chapter:

1. **Decision**: reduce Milankovitch cycles, the seasonal cycle, geological cycles and the latitude–longitude grid to discrete dynamical systems, invoke Theorem 24.1 to decide whether they are periodic, and determine the period type;
2. **Construction**: build the amplitude series, the frequency series and the phase series from astronomical orbital elements, the distribution of solar radiation, geological time series and latitude–longitude coordinates, and assemble them into the geographic epicycle group;
3. **Adjustment**: dynamically adjust the geographic epicycle group subject to the cross-scale resonance window, the isostasy constraint and residual-energy descent;
4. **Convolution**: introduce amplitude convolution, phase convolution and frequency convolution on the geographic epicycle group, and link them to Mi-

lankovitch cycles, the monsoon phase and latitude–longitude rotation;

(5) **Integration**: connect to the tool chains of the three chapters 'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis', forming a diagnostic procedure for geographic periodicity.

The mathematical tools used in this chapter include: group actions, closed orbits, period series, periodic convolution, discrete subgroups of continuous groups and rotation flows on compact topological spaces; the physical tools used include: conservation of angular momentum, conservation of energy, conservation of mass, isostasy, plate tectonics, the meridional distribution of solar radiation, Milankovitch cycles, the ENSO coupled oscillator, the earthquake recurrence period and the Gutenberg–Richter law.

Remark 29.3 (Guide to the seven themes of this chapter). **Translation**—the passage of geological time and the seasonal cycle (the translational period of the tropical year, Definition 29.15); **Scaling**—the scaling distribution of climate zones with latitude (Example 29.3) and the power-law distribution of volcanic eruption frequency (Proposition 29.11); **Rotation**—Milankovitch cycles (the quasi-periodic rotation of the three elements, precession, obliquity and eccentricity, Definition 29.12) and the Earth's rotation and the latitude–longitude coordinate system (Definition 29.23); **Conservation**—conservation of angular momentum, conservation of energy and conservation of mass (Theorem 29.8); **Symmetry**—the phase symmetry of the seasons in the northern and southern hemispheres (Definition 29.14); **Closure**—the closure of the Wilson cycle, the earthquake cycle and the transgression–regression cycle (Definition 29.18, Definition 29.22, Definition 29.20); **Period**—the multi-scale nesting of the diurnal cycle, the seasons, precession, Milankovitch cycles and volcanic and geological cycles (Theorem 29.7).

29.1 Periodicity decision for geographic targets: the first step of the 'Periodic Systems' chapter

29.1.1 Geographic targets, decision inputs and decision maps

Definition 29.1 (Geographic targets and decision inputs). A **geographic target** is one of the following objects:

(i) an astronomically driven geographic periodic system, whose state space is the orbital-element space $\mathcal{S} \subset \mathbb{R}^3$ (eccentricity e, obliquity ε, precession index $e \sin \varpi$);

(ii) a climate-response time series, whose state space is a set of climatic variables (temperature, precipitation, sea-surface temperature and so on);

(iii) a time record of a geological process, whose state space is a sequence of geological events;

(iv) a latitude–longitude coordinate grid, whose state space is the set of points S^2 on the Earth's surface.

The **decision input** of a geographic target is the ordered pair

$$\mathcal{D}_{\mathsf{G}} = (\mathcal{S}, \Phi),$$

where $\mathcal{S}$ is the state space and $\Phi : \mathcal{S} \to \mathcal{S}$ is the evolution map. If $\mathcal{S}$ is a compact manifold or a subset of a compact manifold, the target is called a **compact geographic target**; if $\mathcal{S}$ is a finite set, the target is called a **finite geographic target**.

Definition 29.2 (Geographic decision map)**.** Let a geographic target be given by the decision input $\mathcal{D}_{\mathsf{G}} = (\mathcal{S}, \Phi)$. If the states of the target are generated by Φ, that is, if there exists $x_0 \in \mathcal{S}$ such that

$$x_t = \begin{cases} \Phi^t(x_0), & t \in \mathbb{Z}, \\ \Phi_t(x_0), & t \in \mathbb{R}, \end{cases}$$

for $t \in \mathbb{Z}$ the state x_t is generated by iteration of Φ; for $t \in \mathbb{R}$ it is described by the one-parameter evolution family $\{\Phi_t\}_{t\in\mathbb{R}}$, where each $\Phi_t : \mathcal{S} \to \mathcal{S}$, $\Phi_0 = \mathsf{id}$, $\Phi_{s+t} = \Phi_s \circ \Phi_t$ for all $s, t \in \mathbb{R}$, and $(t, x) \mapsto \Phi_t(x)$ is continuous. then Φ is called the **geographic decision map** of the target. The geographic target is then reduced to the discrete dynamical system $(\mathcal{S}, \Phi)$ when $t \in \mathbb{Z}$ and to the continuous dynamical system $(\mathcal{S}, \{\Phi_t\}_{t\in\mathbb{R}})$ when $t \in \mathbb{R}$; in both cases the criteria of the 'Operations and Periodicity' chapter and of the 'Periodic Systems' chapter may be applied directly.

Theorem 29.1 (Geographic periodicity decision theorem)**.** *Let the decision map of a geographic target be Φ, and let $(\mathcal{S}, \Phi)$ be a discrete dynamical system. Then:*

(A) **Sufficient decision**: *if there exists $m \in \mathbb{N}^+$ with $\Phi^m = \mathsf{id}$, then the target is a uniformly periodic geographic system; every orbit is closed and the orbit length divides m.*

(B) **Equivalence of the decision**: *the target is a uniformly periodic geographic system if and only if $\mathsf{ord}(\Phi) < \infty$.*

(C) **Pointwise criterion**: *the orbit of the target at a state x_0 is closed if and only if x_0 is a periodic point; the orbit length is then the period at that point.*

(D) **Non-omissibility**: *the conjunction of conserved quantities and symmetry groups cannot replace $\mathsf{ord}(\Phi) < \infty$; a quasi-periodic geographic target is not a periodic geographic system.*

Proof. The Theorem 29.1 of this chapter is the domain instantiation of Theorem 24.1 in the 'Periodic Systems' chapter: it suffices to replace the decision map f and the evolution group by Φ of this chapter, and the four parts (A) sufficient decision, (B) equivalence of the decision, (C) pointwise criterion and (D) non-omissibility then hold one by one. By the convention of Remark 24.4, they are not expanded again here. ☐

Corollary 29.1 (Geographic period types)**.** *After the decision of Theorem 29.1, a geographic target falls into one of the following types:*

(i) **Trivial periodic type**: *least period $m_0 = 1$, the state is a fixed point;*

(ii) **Strictly periodic type**: $1 < m_0 < \infty$, the state is strictly periodic (for example the seasonal cycle with $m_0 = 1$ year, and the tides with $m_0 \approx 12.42$ hours);

(iii) **Uniformly periodic type**: $\mathrm{ord}(\Phi) = m < \infty$, every orbit length divides m;

(iv) **Quasi-periodic type**: $\mathrm{ord}(\Phi) = \infty$, the orbits are not closed but are dense on a compact carrier structure (for example the overall motion of the Milankovitch cycles and the quasi-periodic oscillation of ENSO);

(v) **Eventually periodic type**: there exists N such that the target is periodic for $t \geq N$;

(vi) **Discrete periodic type**: the carrier structure is $\mathbb{Z}$ and the normalised period ratio is $\rho = T_x/\Delta x \in \mathbb{Q}$.

(vii) **Asymptotically periodic (damped) type**: the orbit is not closed, but tends asymptotically to a closed orbit or to a fixed point, the residual decaying as $e^{-\kappa t}$ (see Definition 24.3(vii)).

Here (i)–(iii) and (vi) belong to periodic geographic systems; (iv) does not belong to periodic geographic systems but can be handled with cross-scale resonance or bounded-approximation tools; (v) is a periodic geographic system only in its stable stage; (vii), like (iv), is a type in the limiting sense; it is not a periodic geographic system and can be handled with bounded-approximation tools only in its stable stage.

Proof. Each item follows by substituting the geographic target into Theorem 29.1 and Definition 24.3 of the 'Periodic Systems' chapter. ☐

Lemma 29.1 (Periodicity decision for the seasonal cycle). *Let the decision map of the seasonal cycle be Φ, with state space the latitude of the subsolar point $\delta \in [-23.44°, +23.44°]$. Then the period of Φ is 1 tropical year:*

$$\mathrm{ord}(\Phi) = 1 \text{ year} = 365.2422 \text{ days}.$$

The seasonal cycle is of strictly periodic type, and its least period is the tropical year.

Proof. The latitude of the subsolar point $\delta(t)$ satisfies $\delta(t+T) = \delta(t)$, where T is the tropical year. Within one year the subsolar point travels from the Tropic of Capricorn to the Tropic of Cancer and back to the Tropic of Capricorn, completing exactly one full oscillation, so the period of Φ is 1 tropical year. By Theorem 27.9 (calendar-systems chapter), the length of the tropical year is $T_{\text{trop}} = 365.2422$ days. ☐

Lemma 29.2 (Long-period decision for the obliquity of the Earth's axis). *Let the evolution map of the obliquity of the Earth's axis be Φ, with state space $\varepsilon \in [22.1°, 24.5°]$. Then Φ is a quasi-periodic map with principal period about $41,000$ years.*

Proof. The evolution of the obliquity of the Earth's axis is driven by planetary perturbations; its Fourier expansion is a multi-frequency quasi-periodic function whose principal frequency corresponds to the 41 ka period. By Proposition 29.5 the periods of the three orbital elements are incommensurable, so the whole is of quasi-periodic type. ☐

Lemma 29.3 (Multi-frequency decision for the precession index). Let the evolution map of the precession index be Φ, with state space $e\sin\varpi$. Then Φ is a quasi-periodic map with principal periods about 19 ka and 23 ka.

Proof. Precession is jointly modulated by the precession of the Earth's rotation axis (period about 26 ka) and the precession of the orbital major axis (period about 112 ka), and the principal components of the combined frequency correspond to 19 ka and 23 ka. By Definition 29.12. ☐

Lemma 29.4 (Periodicity decision for the tidal cycle). Let the decision map of the tidal cycle be Φ, with state space the sea level. Then the period of Φ is the M_2 tidal period:

$$T_{M_2} \approx 12.42 \text{ hours},$$

which is the period of the principal lunar semidiurnal tide.

Proof. The M_2 tide is driven by the tide-generating force of the Moon. The interval between two successive passages of the Moon across the same meridian is 1 lunar day $= 24.84$ hours; since M_2 is a semidiurnal tide, its period is 12.42 hours. By Lemma 27.1 (astronomical-systems chapter), the orbital period of the Moon is 27.32 days. ☐

Definition 29.3 (Geographic least positive period algorithm). **Input**: geographic target samples $x_0, \dots, x_{N-1}$, tolerance $\varepsilon \geq 0$. **Output**: the least positive period T_0, or the decision 'not periodic'. **Steps**:

(1) take the candidate set $\mathcal{P} = \{T \in \mathbb{N}^+ : 2T \leq N\}$ and traverse it in increasing order of T;

(2) for each T compute the maximum deviation

$$E(T) = \max_{0 \leq n < N-T} \|x_{n+T} - x_n\|;$$

(3) output the first T satisfying $E(T) \leq \varepsilon$ as T_0; if the traversal is completed with no solution, output 'not periodic';

(4) commensurability check: use the normalised period ratio $\rho = T_x/\Delta x$ and the corresponding continuous duration $\ell = T_0\Delta x$, together with Proposition 21.2 and Corollary 21.1 of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter, to check the conversion relation between T_0 and the continuous period T_x.

Theorem 29.2 (Correctness of the geographic least positive period algorithm). (i) If the samples form a strictly periodic sequence with $\varepsilon = 0$ and $N \geq 2T_0$, the algorithm outputs its discrete fundamental period T_0.

(ii) If the samples are obtained by sampling a continuous geographic target of fundamental period T_x at sampling interval Δx, and the normalised period ratio $\rho = T_x/\Delta x = p/q$ is in lowest terms, the algorithm outputs $T_0 = p$.

(iii) If $\rho \notin \mathbb{Q}$, no true period exists: $E(T) > 0$ for every $T \in \mathbb{N}^+$.

Proof. (i) By Definition 21.4 of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter, T_0 is the least positive period, so $E(T_0) = 0$, whilst $E(T) > 0$ for $T < T_0$. The algorithm traverses in increasing order of T, so the first index satisfying $E(T) = 0$ is T_0. (ii) By Proposition 21.2(ii) the fundamental period of the sampling sequence is p; substitute into (i). (iii) By Proposition 21.2(iii), $\text{Group}_{\text{period}}(f) = \{0\}$, so $E(T) > 0$ for every T. ☐

Example 29.1 (Example of a periodicity decision for the seasonal cycle). Take a surface air-temperature record $T(t)$ (with the day as the sampling unit). By Lemma 29.1 the least period is $T_0 = 365$ days (approximately). A periodicity test with a window of 365 days gives $E(365) < \varepsilon$, whilst a smaller window (for example 300 days) gives $E(300) > \varepsilon$.

Example 29.2 (Quasi-periodicity decision for Milankovitch cycles). Take an obliquity record $\varepsilon(t)$ (with the millennium as the sampling unit). By Lemma 29.2, $\varepsilon(t)$ is quasi-periodic and no strict least period exists. Testing any candidate T gives $E(T) > 0$; spectral analysis gives a principal peak at 41 ka.

Remark 29.4 (Conserved quantities do not guarantee periodicity). The conserved quantities of a geographic system (angular momentum, energy and mass) are auxiliary conditions for periodic structure, but they do not constitute a sufficient condition. For example, a hyperbolic orbit has conservation of energy and conservation of angular momentum, yet its orbit is open and not closed. The periodic structure of a geographic system must be decided by Theorem 29.1 or by criteria such as Lemma 29.1.

29.2 Construction of the geographic period series and epicycle group: the second step of the 'Periodic Systems' chapter

Definition 29.4 (Geographic amplitude, frequency and phase series). Let the state or output sequence of a geographic target be $\{z_t\}$. If it can be written as

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)} + \varepsilon_t,$$

then

$$\mathcal{A}_z(t) = \sum_k A_k e^{ikt}, \qquad \mathcal{F}_z(s) = \sum_k \omega_k s^k, \qquad \mathcal{P}_z(t) = \sum_k e^{i(kt+\phi_k)}$$

are called respectively the **amplitude series**, the **frequency series** and the **phase series** of the geographic target. The three agree with Definition 6.10, Definition 6.12 and Definition 6.11 of the 'Series and Convolution' chapter.

Definition 29.5 (Geographic epicycle group). Let the period series of a geographic target be given by its amplitude series $\{A_k\}$, frequency series $\{\omega_k\}$ and phase series $\{\phi_k\}$. For each k define the epicycle

$$E_k : \theta \mapsto A_k e^{i(\theta + \phi_k)}, \qquad \theta = \omega_k t,$$

whose collection

$$\mathfrak{E}_{\mathsf{G}} = \{(A_k, \omega_k, \phi_k)\}_{k\in\mathbb{Z}}$$

is called the **epicycle group** of the geographic target. Its superposition is the geographic period series

$$z_t = \sum_k E_k(\omega_k t) = \sum_k A_k e^{i(\omega_k t+\phi_k)}.$$

Theorem 29.3 (Constructing the period series from geographic data)**.** *Let the output of a geographic target be a finite periodic sequence of period N. Then:*

(i) *the output admits the discrete Fourier expansion*

$$z_t = \sum_{k=0}^{N-1} c_k e^{i2\pi kt/N}, \qquad c_k = \frac{1}{N}\sum_{t=0}^{N-1} z_t e^{-i2\pi kt/N};$$

(ii) *the amplitude series is given by $|c_k|$, the phase series by $\arg c_k$ and the frequency series by $2\pi k/N$;*

(iii) *if the output is a Milankovitch-cycle sequence, its harmonic decomposition gives four principal components at 100 ka, 41 ka, 23 ka and 19 ka;*

(iv) *if the output is an earthquake sequence, its harmonic decomposition gives a principal frequency corresponding to the fault recurrence interval Δt.*

Proof. (i) The discrete Fourier series of a finite periodic sequence is a standard result: the basis functions $\{e^{i2\pi kt/N}\}_{k=0}^{N-1}$ form an orthogonal basis of $\mathbb{C}^N$. (ii) This is read off directly from $c_k = A_k e^{i\phi_k}$. (iii) By Definition 29.12 the principal frequencies of the three orbital elements are $1/100$ ka^{-1}, $1/41$ ka^{-1}, $1/23$ ka^{-1} and $1/19$ ka^{-1} respectively. The discrete Fourier transform separates these principal frequencies into different harmonic channels. (iv) By Definition 29.22 the earthquake recurrence interval $T_{\mathsf{rec}} = d_s/v_s$ gives the principal frequency $1/T_{\mathsf{rec}}$. □

Definition 29.6 (Geographic periodic system and periodic structure)**.** A **geographic periodic system** is a dynamical system of the Earth-surface layer acted on jointly by astronomical driving forces, geological tectonic forces, climatic feedback forces and terrain constraints. The state variables of the system include:

(i) **Astronomical driving forces**: the spatio-temporal distribution of solar radiation over the Earth's surface, determined by the three elements of the Earth's orbit (eccentricity e, obliquity ε, precession index $e\sin\varpi$).

(ii) **Geological tectonic forces**: the plate-motion velocity field $\mathsf{v}_{\mathsf{plate}}(\mathsf{x}, t)$, isostatic compensation, fault slip rates, the spatio-temporal distribution of volcanic activity, and the earthquake magnitude–frequency distribution.

(iii) **Climatic feedback forces**: the atmospheric temperature field $T(\mathsf{x}, t)$, the precipitation field $P(\mathsf{x}, t)$, the ocean-current velocity field $\mathsf{u}_{\mathsf{ocean}}(\mathsf{x}, t)$ and the ice-sheet volume $V_{\mathsf{ice}}(t)$.

(iv) **Terrain constraints**: the elevation field $h(\mathsf{x})$, the surface slope $\nabla h(\mathsf{x})$, latitude–longitude coordinates (λ, ϕ) and the time-zone system.

Definition 29.7 (The periodic structure of a geographic system)**.** *The* **periodic structure** *of a geographic system falls into four basic types:*

(i) **Astronomical-geographic period**: *on scales of* 10^4–10^5 *years, driven by Milankovitch cycles, with the total angular momentum of the Earth–Sun–planet system as its conserved quantity;*

(ii) **Climatic-geographic period**: *on scales from the seasonal to the interdecadal, driven by the meridional gradient of solar radiation, with the total energy of the Earth–atmosphere system as its conserved quantity;*

(iii) **Geological-geographic period**: *on scales of* 10^6–10^9 *years, driven by the Earth's internal heat energy and plate motion, with the total mass of crustal material as its conserved quantity;*

(iv) **Terrain–coordinate period**: *the latitude–longitude coordinate system and the altitudinal layering of terrain determined by the Earth's rotation, with the angular momentum of rotation and revolution as its conserved quantity.*

29.3 Dynamic adjustment of the geographic epicycle group: the third step of the 'Periodic Systems' chapter

Definition 29.8 (Dynamic adjustment of the geographic epicycle group)**.** *Let the current geographic epicycle group be* $\mathfrak{E}^{(j)}$*, with partial sum*

$$z^{(j)}(t) = \sum_{k\in\mathfrak{E}^{(j)}} A_k e^{i(\omega_k t+\phi_k)},$$

and residual $r^{(j)} = z - z^{(j)}$*.* **Dynamic adjustment** *consists of two rules:*

(i) **Quantity adjustment**: *if the residual energy satisfies* $\|r^{(j)}\|^2 > \delta$*, add a new epicycle according to the peak of the residual periodogram;*

(ii) **Parameter adjustment**: *update the parameters of the existing epicycles by regularised least squares or by the cross-scale resonance constraint.*

In geographic systems, quantity adjustment corresponds to increasing the orbital harmonic order, adding a climatic mode or adding a geological event layer; parameter adjustment corresponds to adjusting orbital elements, resonance widths or geological time-scale parameters.

Proposition 29.1 (Monotonicity of quantity adjustment)**.** *After the dominant residual frequency component is added as in Definition 29.8*(i)*, the residual energy is monotonically non-increasing:*

$$\|r^{(j+1)}\|^2 \le \|r^{(j)}\|^2,$$

with equality if and only if the residual has no energy at the selected frequency.

Proof. The Proposition 29.1 of this chapter agrees with the dynamic adjustment mechanism of Definition 24.8 in the 'Periodic Systems' chapter: adding an epicycle removes the component of $r^{(j)}$ along $\hat{\omega}$ from the residual, and the orthogonal decomposition $\|r^{(j)}\|^2 = \|r^{(j+1)}\|^2 + \|\text{the removed component}\|^2$ gives the inequality, with equality if and only if the norm of the removed component is zero. The complete argument is given there. ☐

Theorem 29.4 (Convergence of dynamic adjustment). Let the geographic target be a finite periodic sequence of period T_0 and let $\delta = 0$. Then after at most $\lfloor T_0/2 \rfloor + 1$ steps of dynamic adjustment the residual is zero, and the resulting geographic epicycle group is the exact Fourier data of the target.

Proof. The discrete Fourier series of a sequence of period T_0 is complete on $\lfloor T_0/2 \rfloor + 1$ harmonics. At each step one dominant frequency component of the residual is extracted as in Proposition 29.1 and the residual energy decreases strictly, so after finitely many steps the residual is zero. ☐

Definition 29.9 (Cross-scale resonance index). Let a geographic target have substructures on L time scales, with periods $T_1 < \cdots < T_L$. Define the **cross-scale resonance index**

$$\mathcal{R} = \max_{i<j} \left| \frac{T_j}{T_i} - \text{round}\left(\frac{T_j}{T_i}\right) \right| .$$

The smaller $\mathcal{R}$ is, the closer the periods at the various scales are to an integer ratio, and the stronger the cross-scale resonance.

Proposition 29.2 (Bounds for the cross-scale resonance index). The cross-scale resonance index satisfies $0 \le \mathcal{R} \le 1/2$. If all $T_j/T_i \in \mathbb{Q}$, the periods at the various scales are commensurable, that is, a common period exists; $\mathcal{R}$ measures how close the period ratios are to integers, and $\mathcal{R} \to 0$ requires every T_j/T_i to be close to an integer.

Proof. For any real number x, $|x - \text{round}(x)| \le 1/2$. The upper bound follows directly from the definition. ☐

Remark 29.5 (Boundaries of dynamic adjustment). Theorem 29.4 relies on finite periodic sequences. For quasi-periodic geographic targets such as Milankovitch cycles, no finite epicycle group represents them exactly, and only cross-scale resonance analysis or bounded approximation is possible. The cross-scale resonance index $\mathcal{R}$ measures only the closeness to an integer ratio and does not guarantee that resonance actually occurs.

29.4 Geographic convolution: the fourth step of the 'Periodic Systems' chapter

Definition 29.10 (Geographic amplitude, phase and frequency convolutions). Let two geographic epicycle groups be

$$\mathfrak{E} = \{(A_n, \omega_n, \phi_n)\}, \qquad \mathfrak{E}' = \{(B_m, \nu_m, \psi_m)\}.$$

Define:

(i) **Amplitude convolution**: $\mathbb{A}_{mn} = A_n B_m$;

(ii) **Phase convolution**: $\mathbb{P}_{mn} = e^{i(\phi_n - \psi_m)}$;

(iii) **Frequency convolution**: $\mathbb{F}_{mn} = \omega_n \nu_m$;

(iv) **Trend convolution**: $\mathbb{T}_{mn} = A_n B_m e^{i(\phi_n - \psi_m)} = A_n B_m \mathbb{P}_{mn}$.

This agrees with Definition 6.21, Definition 6.22, Definition 6.23 and Definition 6.24 of the 'Series and Convolution' chapter.

Theorem 29.5 (Action of the three convolutions on the geographic epicycle group)**.** *Let the epicycle groups of two geographic targets be $\mathfrak{E}$ and $\mathfrak{E}'$ respectively. Then:*

(i) *amplitude convolution acts only on the radius;*

(ii) *phase convolution acts only on the phase;*

(iii) *frequency convolution acts only on the rotational speed;*

(iv) *trend convolution preserves both amplitude and phase.*

Proof. This is read off directly from Definition 29.10. ☐

Proposition 29.3 (Phase convolution of Milankovitch cycles and the climate response)**.** *Let the epicycle group of Milankovitch cycles be $\mathfrak{E}_{\mathrm{M}}$ and the epicycle group of the climate-response sequence be $\mathfrak{E}_{\mathrm{C}}$. Their phase convolution*

$$\mathbb{P}_{mn} = e^{i(\phi^{\mathrm{M}}_m - \phi^{\mathrm{C}}_n)}$$

measures the phase difference between the astronomical forcing and the climate response. Glacial terminations correspond to particular zeros of the phase difference—at these the maximum of the astronomical forcing aligns with the maximum of the climate response, producing the melting of ice sheets.

Proof. By Proposition 29.6 there is a phase relation between glacial cycles and Milankovitch cycles. A glacial termination corresponds to a period in which precession and obliquity are simultaneously in a high-insolation configuration; at such a time the phase of the astronomical forcing coincides with the phase of the climate response, and the argument of $\mathbb{P}_{mn}$ is zero. The phase convolution records all the information about this alignment. ☐

Proposition 29.4 (Autocorrelation and trend convolution)**.** *Let $\{z_t\}$ be a geographic periodic sequence, with autocorrelation function*

$$R_z(h) = \sum_t z_t \overline{z_{t+h}}.$$

In the language of the geographic epicycle group, the autocorrelation function is the projection of the trend convolution onto the diagonal:

$$R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}.$$

Proof. The Proposition 29.4 of this chapter is the realisation in this domain of the convolution-action conclusion of Theorem 24.6 in the ‘Periodic Systems’ chapter: substituting $z_t = \sum_n A_n e^{i(\omega_n t + \phi_n)}$ into the definition of the autocorrelation, cancelling all off-diagonal cross terms by orthogonality and retaining only the diagonal term of the trend convolution (the factorisation of the three convolutions, Proposition 24.6), which yields $R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}$. The full expansion is given there. □

Remark 29.6 (Relation between geographic convolution and periodic convolution). Geographic convolution is not a new convolution operation but an instance of the periodic convolution of the ‘Series and Convolution’ chapter on the geographic epicycle group. Autocorrelation is the projection of the trend convolution onto the diagonal; the Milankovitch–climate phase difference and cross-scale resonance are respectively the concrete manifestations in geography of phase convolution and frequency convolution.

29.5 Integration with discrete sampling, harmonic analysis and time series analysis: the fifth step of the ‘Periodic Systems’ chapter

Definition 29.11 (Geographic periodicity diagnostic procedure). For a given geographic system, diagnose it using the framework of periodic mathematics:

(i) **Periodicity decision**: verify the period type of the geographic target (Theorem 29.1);

(ii) **Series construction**: build the amplitude, frequency and phase series and the epicycle group from geographic data (Theorem 29.3);

(iii) **Cross-scale diagnosis**: compute the cross-scale resonance index $\mathcal{R}$ (Definition 29.9);

(iv) **Convolution diagnosis**: compute the autocorrelation (the diagonal projection of the trend convolution, Proposition 29.4) and the Milankovitch–climate phase convolution (Proposition 29.3);

(v) **Conservation diagnosis**: verify conservation of angular momentum, energy and mass (Theorem 29.8).

Theorem 29.6 (The geographic periodicity diagnostic procedure and its interfaces with three chapters). The five steps of Definition 29.11 connect respectively to:

(i) the ‘Discrete Sampling and Intercalation of Periodic Signals’ chapter: Definition 21.1, Theorem 21.3, Theorem 21.10 (sampling and spectral estimation of geographic time series);

(ii) the ‘Harmonic Analysis’ chapter: Definition 22.5, Theorem 22.1, Proposition 22.11 (frequency decomposition of geographic data);

(iii) the 'Time Series Analysis' chapter: Definition 23.16, Proposition 23.2, Definition 23.19 (periodicity significance tests for geographic time series).

The geographic periodicity diagnostic procedure is therefore not an independent method but an application of the unified tool chain of periodic mathematics to geographic systems.

Proof. The correspondences are item by item: (i) geographic periodicity diagnosis requires sampling and spectral estimation; (ii) the frequency decomposition and the phase kernels are found in the 'Harmonic Analysis' chapter; (iii) the periodogram peaks, the Fisher test and window-function validation are found in the 'Time Series Analysis' chapter. Hence the diagnostic procedure is a synthesis of the tool chains of the three chapters. □

Remark 29.7 (Order of operations after integration). The recommended order for an actual diagnosis is: first decide whether the geographic target is periodic (Theorem 29.1); then construct the Fourier data and the epicycle group of the geographic data (Theorem 29.3); then optimise the parameters by cross-scale resonance and dynamic adjustment (Theorem 29.4, Definition 29.9); then perform geographic convolution and phase-relation analysis (Proposition 29.3, Proposition 29.4); and finally output the diagnosis as in Definition 29.11.

29.6 Astronomical-geographic cycles

29.6.1 Milankovitch cycles

Definition 29.12 (Milankovitch cycles). **Milankovitch cycles** *are the quasi-periodic variations of the three elements of the Earth's orbit, caused by the gravitational perturbations of the other planets of the Solar System on the Earth. The three elements and their principal periods are:*

(i) **Eccentricity** e: *principal periods of about* $100,000$ *years* (100 *ka*) *and* $413,000$ *years* (413 *ka*).

(ii) **Obliquity** ε: *ranging from about* 22.1° *to* 24.5°, *with a principal period of about* $41,000$ *years* (41 *ka*).

(iii) **Precession index** $e\sin\varpi$: *principal periods of about* $19,000$ *years and* $23,000$ *years* (19–23 *ka*).

Proposition 29.5 (Structural properties of Milankovitch cycles). *Milankovitch cycles have the following structural properties:*

(i) **Phase-space rotation**: *the quasi-periodic variation of the three elements of the Earth's orbit corresponds to a rotation of the Earth–Sun system in phase space.*

(ii) **Conservation of angular momentum**: *the total angular momentum of the Earth–Sun–planet system is conserved; the orbital and rotational angular momenta of the Earth evolve slowly under planetary perturbations and lunisolar tidal torques—this is precisely the dynamical origin of the Milankovitch cycles.*

(iii) **Quasi-periodicity**: *the periods of the three orbital elements are incommensurable, and the overall motion is quasi-periodic.*

Proof. (i) The variations of the three elements of the Earth's orbit are described by Lagrange's planetary equations. In phase space the eccentricity e and the longitude of perihelion ϖ form a pair of conjugate variables whose long-term evolution is equivalent to a rotation in phase space. (ii) Planetary perturbations are conservative forces, and the total angular momentum of the Earth–Sun–planet system is strictly conserved. (iii) The principal frequencies of the three elements obtained from planetary perturbation theory are linearly independent over $\mathbb{Q}$, so the overall motion is a quasi-periodic flow on a three-dimensional torus; $100 : 41 : 23$ is only an order-of-magnitude approximation and cannot be used to decide incommensurability. □

29.6.2 Glacial–interglacial cycles

Definition 29.13 (Glacial–interglacial cycles). **Glacial–interglacial cycles** *are the periodic alternation of the Earth's climate between cold glacial periods and warm interglacial periods during the Quaternary (the last 2.58 Ma). During glacial periods the continental ice sheets of the northern hemisphere expand greatly and sea level falls by 100–130 metres. During interglacial periods the ice sheets retreat and sea level rises again.*

Glacial–interglacial cycles are the **non-linear response** *of the Milankovitch astronomical cycles in the geographic system.*

Proposition 29.6 (Phase relations of glacial cycles). *There is a phase relation between glacial–interglacial cycles and the Milankovitch astronomical cycles:*

(i) *in the Early Pleistocene (2.58–1.0 Ma) the glacial cycles were dominated by the 41 ka period;*

(ii) *after the Mid-Pleistocene Transition (1.0–0.8 Ma) the glacial cycles shifted to being dominated by the 100 ka period;*

(iii) *glacial terminations correspond to periods in which precession and obliquity are simultaneously in a high-insolation configuration.*

Proof. The phase relation between glacial cycles and astronomical cycles is established by oxygen-isotope stratigraphy. Spectral analysis of $\delta^{18}O$ time series identifies four principal frequencies at 100 ka, 41 ka, 23 ka and 19 ka. Around the Mid-Pleistocene Transition the dominant frequency shifts from 41 ka to 100 ka, in step with the eccentricity period. Glacial terminations correspond to periods in which the phases of several astronomical cycles coincide. □

29.7 Climatic-geographic cycles

29.7.1 The seasonal cycle of the solar-radiation gradient and atmospheric circulation

Definition 29.14 (Meridional gradient of solar radiation). *The fundamental energy driver of the Earth's climate system is the* **meridional gradient of solar radiation**.

The zonal-mean distribution of solar radiation at the top of the atmosphere can be approximated by

$$Q(\phi) = \frac{S_0}{4} \cdot \bar{s}(\phi),$$

where $S_0 \approx 1361$ W/m^2 is the solar constant and $\bar{s}(\phi)$ is the annual-mean insolation distribution function at latitude ϕ.

Example 29.3 (Concrete calculation of the meridional gradient of solar radiation). Using the Legendre approximation to the annual-mean insolation distribution, $\bar{s}(\phi) \approx 1 - 0.482\, P_2(\sin\phi)$, we obtain

$$Q(0) \approx 422 \text{ W/m}^2, \qquad Q(90^\circ) \approx 176 \text{ W/m}^2.$$

The equator-to-pole difference in annual-mean insolation is about 246 W/m^2, which supplies the order of magnitude of the solar-radiation gradient that drives the meridional atmospheric circulation.

Definition 29.15 (Monsoon system). *A* **monsoon** *is a large-scale reversal of the atmospheric circulation on the seasonal scale. In summer the continents warm faster than the oceans, producing continental lows and oceanic highs that drive moist oceanic air towards the continents; in winter the reverse occurs.*

Proposition 29.7 (Periodic properties of the monsoon system). *The monsoon system has the following periodic properties:*

(i) **Translational period**: *the fundamental period is the tropical year.*

(ii) **Conservation of energy**: *the total energy is approximately conserved over the seasonal cycle.*

(iii) **Gradient constraint**: *the annual-mean meridional temperature gradient remains approximately unchanged in the long-term climatic balance.*

Proof. (i) The subsolar point moves back and forth between the Tropic of Cancer and the Tropic of Capricorn, the position of the maximum of the solar-radiation gradient travels with the seasons, and the land–sea thermal contrast and the monsoon wind direction it drives therefore reverse once every year. (ii) During the summer monsoon the oceans release energy in the form of latent heat, whilst during the winter monsoon radiative cooling over the continents drives subsidence; latent heat, sensible heat and kinetic energy circulate between land and sea with the tropical year as period. (iii) The annual-mean meridional temperature gradient is determined by the energy balance: the net radiative surplus at low latitudes and the net radiative deficit at high latitudes are compensated by meridional transport by the atmosphere and the oceans. ☐

29.7.2 Periodic structure of ocean circulation

Definition 29.16 (Thermohaline circulation). **Thermohaline circulation** *is the global deep-ocean circulation system driven by differences in the temperature and salinity of sea water, with a complete circulation period of about 1000–2000 years.*

Definition 29.17 (El Niño–Southern Oscillation)**.** **El Niño–Southern Oscillation** (ENSO) *is the quasi-periodic oscillation of the coupled ocean–atmosphere system of the tropical Pacific on the interannual scale (2–7 years), with two extreme phases, El Niño (the warm phase) and La Niña (the cold phase).*

Lemma 29.5 (Rotation–decay decomposition of the damped harmonic oscillator)**.** *Let $(T_E(t), H_W(t))$ satisfy the linear system*

$$\frac{d}{dt}\begin{pmatrix} T_E \\ H_W \end{pmatrix} = M \begin{pmatrix} T_E \\ H_W \end{pmatrix}, \qquad M = \begin{pmatrix} -\alpha & \beta \\ -\gamma & -\delta \end{pmatrix},$$

where $\alpha, \beta, \gamma, \delta > 0$. If $4\beta\gamma > (\alpha - \delta)^2$, then M has a pair of conjugate complex eigenvalues

$$\lambda_\pm = -\rho \pm i\omega, \qquad \rho = \frac{\alpha + \delta}{2} > 0, \qquad \omega = \frac{1}{2}\sqrt{4\beta\gamma - (\alpha - \delta)^2} > 0.$$

Proof. The characteristic polynomial of the matrix M is $\lambda^2 + (\alpha + \delta)\lambda + (\alpha\delta + \beta\gamma)$, with discriminant $\Delta = (\alpha - \delta)^2 - 4\beta\gamma$. When $4\beta\gamma > (\alpha - \delta)^2$ we have $\Delta < 0$, giving conjugate complex roots. The real part $-\rho < 0$ makes the solution decay exponentially, whilst the imaginary part $\pm i\omega$ makes the solution rotate in phase space. ☐

Proposition 29.8 (Coupled-oscillator model of ENSO)**.** *ENSO can be modelled as a periodic system of two coupled subsystems, with evolution equations*

$$\frac{dT_E}{dt} = -\alpha T_E + \beta H_W, \qquad \frac{dH_W}{dt} = -\gamma T_E - \delta H_W.$$

This system of equations is equivalent to a damped harmonic oscillator; random atmospheric perturbations inject energy into it and maintain a sustained oscillation.

Proof. By Lemma 29.5, when $4\beta\gamma > (\alpha - \delta)^2$ the pair (T_E, H_W) undergoes a decaying rotation about the equilibrium point with frequency ω and period $T = 2\pi/\omega$. ☐

29.8 Periodic structure of geological and volcanic activity

29.8.1 The Wilson cycle

Definition 29.18 (The Wilson cycle)**.** *The* **Wilson cycle** *is the ultra-long period of the Earth's plate tectonics on the 200–500 Ma time scale, comprising six stages: continental rifting, sea-floor spreading, ocean opening, initiation of subduction, ocean closure and continental collision.*

Proposition 29.9 (Structural properties of the Wilson cycle)**.** *The Wilson cycle has the following structural properties:*

(i) **Scaling structure**: *the period is about 200–500 Ma.*

(ii) **Conservation of mass**: *the ratio of ocean to land area remains constant over long times (about 71% ocean and 29% land).*

(iii) **Self-similarity**: *similar sequences of stages appear on different scales.*

Proof. (i) The greatest age of oceanic crust worldwide is about 200 Ma; the North Atlantic, for example, opened about 200 Ma ago at an average spreading rate of about 2.5 cm/year. (ii) The production rate and the destruction rate of oceanic crust are both about 3 km^2/year during the Phanerozoic. (iii) The opening of the Japan Sea and that of the Atlantic both pass through the same sequence of stages, differing only by about one order of magnitude in time scale. ☐

29.8.2 The orogenic–denudation cycle and isostasy

Definition 29.19 (The orogenic–denudation cycle). The **orogenic–denudation cycle** is the dynamically balanced process of the crust between tectonic uplift and surface erosion, on time scales of 10^6–10^8 years.

Proposition 29.10 (A quantitative model of isostasy). *Let the crustal density be ρ_c, the mantle density $\rho_m > \rho_c$, the reference crustal thickness H_0, the surface elevation h and the crustal thickening ΔH. Then Airy isostasy satisfies*

$$h \cdot \rho_c = \Delta H \cdot (\rho_m - \rho_c),$$

that is, $h = (\rho_m - \rho_c)/\rho_c \cdot \Delta H$. For typical parameters $\rho_c \approx 2.7$ g/cm^3 and $\rho_m \approx 3.3$ g/cm^3, the proportionality coefficient is about 0.22.

Proof. Take the pressure-balance condition at the depth of compensation: $\rho_c H_0 g + \rho_m \Delta H g = \rho_c(h + H_0 + \Delta H)g$, which on expansion gives $h\rho_c = \Delta H(\rho_m - \rho_c)$. ☐

29.8.3 The transgression–regression cycle and sedimentary sequences

Definition 29.20 (The transgression–regression cycle). The **transgression–regression cycle** is the periodic rise and fall of sea level relative to continental margins, driven by cycles on several hierarchical levels: first-order cycles (50–100 Ma), second-order cycles (5–50 Ma), third-order cycles (0.5–5 Ma) and fourth-order cycles (0.1–0.5 Ma).

29.8.4 Periodic structure of volcanic activity

Definition 29.21 (Multi-scale periodicity of volcanic activity). **Volcanic activity** exhibits periodicity on several hierarchical levels: eruption periods (minutes to hours), active–dormant periods (years to millennia), super-eruption periods (10^4–10^5 years) and periods of the global volcanic activity rate (10^7–10^8 years).

Proposition 29.11 (Power-law distribution of volcanic eruption frequency versus magnitude). *The frequency N of global volcanic eruption events and the eruption size M satisfy the power-law distribution*

$$N(> M) \propto M^{-b}, \qquad b \approx 0.7\text{–}1.0.$$

Proof. In the global volcanic database, fitting over the range $M_0 \in [10^{-2}, 10^3]$ km^3 gives $b \approx 0.7$–1.0, which has the same structure as the Gutenberg–Richter relation for earthquakes. ☐

29.8.5 Periodic structure of seismic activity

Definition 29.22 (The earthquake cycle and recurrence interval). An **earthquake** is a periodic process of accumulation and release of crustal stress. The recurrence interval is $T_{\text{rec}} = d_s/v_s$, where d_s is the coseismic slip and v_s the slip rate.

Proposition 29.12 (Power-law distribution of seismic frequency versus magnitude). Earthquakes satisfy the **Gutenberg–Richter law**

$$\log_{10} N(> M) = a - bM, \qquad b \approx 1.0.$$

Proof. Substituting the relation $\log_{10} E \approx 1.5M + 4.8$ between the magnitude M and the energy E released by an earthquake gives $N(> E) \propto E^{-b/1.5} \approx E^{-0.67}$. ☐

Example 29.4 (Concrete calculation of the earthquake recurrence interval). For the southern segment of the San Andreas Fault, $v_s = 34$ mm/year and $d_s \approx 5.5$ m, giving

$$T_{\text{rec}} = \frac{5.5}{0.034} \approx 160 \text{ years}.$$

29.9 Terrain, landforms and the latitude–longitude system

29.9.1 The latitude–longitude coordinate system

Definition 29.23 (The geographic coordinate system). The **geographic coordinate system** is determined by the Earth's rotation axis and the equatorial plane, and comprises latitude ϕ, longitude λ and elevation h. Longitude has period $360°$, latitude has half-period $180°$, and elevation has no period but exhibits scaling layering.

Proposition 29.13 (The periodicity of the time-zone system). The international time-zone system divides the Earth by longitude into 24 time zones, each spanning $15°$ and corresponding to a time difference of 1 hour:

$$n = \left\lfloor \frac{\lambda + 7.5°}{15°} \right\rfloor, \qquad \lambda_n = 15° \times n.$$

Proof. The angular velocity of the Earth's rotation is $\omega_E = 2\pi/24\text{h}$. A longitude difference $\Delta\lambda$ corresponds to the local time difference $\Delta t = \Delta\lambda \cdot (24\text{h}/360°) = \Delta\lambda/15°$ hours. The angular distance between the centres of adjacent time zones is $15°$, which is exactly a local time difference of 1 hour. ☐

29.9.2 Latitudinal climate zones

Definition 29.24 (Latitudinal climate zones). The Earth's surface is divided by latitude into five principal climate zones: tropical, subtropical, temperate, subpolar and polar. The boundary latitudes of the climate zones (23.44° and 66.56°) are determined by the obliquity of the Earth's axis, $\varepsilon = 23.44^\circ$.

Proposition 29.14 (The temperature scaling of climate zones). *The annual-mean surface air temperature $T(\phi)$ satisfies the scaling relation*

$$T(\phi) \approx T_0 - \Delta T \cdot \left(\frac{\sin \phi}{\sin 90^\circ} \right)^2 ,$$

where $T_0 \approx 27^\circ\mathrm{C}$ is the annual-mean temperature at the equator and $\Delta T \approx 40^\circ\mathrm{C}$ is the equator-to-pole temperature difference.

Proof. The annual-mean solar radiation at latitude ϕ is proportional to $\cos \phi$. Taking the radiative input together with meridional heat transport, the latitudinal distribution of air temperature is found to be approximately proportional to $\cos^2 \phi$. Empirical fitting shows that $T(\phi) \approx T_0 - \Delta T \sin^2 \phi$ agrees well with observations for $|\phi| \leq 60^\circ$. ☐

29.9.3 Elevation, terrain and altitudinal layering

Definition 29.25 (Terrain height layering). The land surface of the Earth is divided by elevation into three layers: lowlands (0–500 m, about 57%), middle elevations (500–2000 m, about 28%) and high elevations (> 2000 m, about 15%).

Proposition 29.15 (Power-law characteristic of the land elevation distribution). *The cumulative distribution of land elevation approximately satisfies the power law*

$$P(h > h_0) \propto h_0^{-\alpha}, \qquad \alpha \approx 1.5\text{–}2.0.$$

Proof. Statistical analysis of the global terrain database shows that the cumulative distribution of land area with elevation is approximately a straight line in log–log coordinates, with slope $\alpha \approx 1.5$–2.0. This feature is consistent with the fractal structure of the 'tectonic uplift–erosional denudation' process. ☐

29.10 The multi-layer nested structure of geographic periodic systems

Theorem 29.7 (The four-level nesting of the geographic periodic system). *The geographic periodic system forms a four-level nested structure:*

Geographic periodicity layer	Time scale	Dominant mechanism	Conserved quantity/reference scale
Astronomical driving layer	10^4–10^5 years	Rotation of the Earth–Sun–planet system	Angular momentum
Climatic response layer	10^0–10^3 years	Meridional gradient of solar radiation	Energy
Geological tectonic layer	10^6–10^9 years	Plate motion and mantle convection	Mass
Terrain–coordinate layer	10^{-1}–10^1 years	Geometry of the Earth's rotation and revolution	Rotation/revolution quantity

Proof. The mathematical basis of the four-level nested structure is the separation of time scales. Astronomical periods are far longer than climatic periods, and climatic periods are in turn far longer than the time scales of weather systems. In the evolution equations of the climate system the astronomical driving forces may be regarded as slowly varying parameters, approximately constant on the time scale of the climatic response layer. Geological tectonic forces are regarded as fixed boundary conditions on the time scale of the climatic response layer. The terrain–coordinate layer serves as the coordinate framework on all time scales. □

Corollary 29.2 (Cross-scale resonance of the geographic system). *In the four-level nested structure, when the periods of different levels resonate, a pronounced cross-scale amplification effect can arise.*

Proof. Any response substructure may be regarded as a forced oscillator whose steady-state amplitude is proportional to $1/|\omega_0^2 - \omega^2|$; hence when the driving frequency ω approaches the natural frequency ω_0 of the response substructure, the response amplitude increases markedly. □

Remark 29.8 (Empirical observations and formal analogies of cross-scale resonance). The following three examples are empirical observations and formal analogies, not theorems: (i) the Mid-Pleistocene Transition may be a consequence of cross-scale resonance among the astronomical driving layer, the climatic response layer and the geological tectonic layer; (ii) the quasi-synchrony of the supercontinent cycle with crossings of the Galactic spiral arms; (iii) the overlap in time scale between super-volcanic eruptions and Milankovitch cycles.

29.11 Conservation laws of geographic periodic systems

Theorem 29.8 (The three conservation laws of the geographic system). The dynamics of a geographic periodic system is governed by three conservation laws:

(i) **Conservation of angular momentum**: the total angular momentum of the Earth–atmosphere–ocean system evolves slowly under external astronomical torques, and the exchange of angular momentum among the internal spheres satisfies a conservation law.

(ii) **Conservation of energy**: the total energy input and output of the Earth's climate system are balanced in the annual mean. The meridional gradient of solar radiation drives the global circulation.

(iii) **Conservation of mass**: the circulation of Earth-surface material among its reservoirs satisfies conservation of mass.

Proof. (i) The rate of change of the total angular momentum of the Earth–atmosphere–ocean system equals the external astronomical torque. The precession of the Earth's rotation axis (period about 25772 years) and nutation (principal period 18.6 years) correspond respectively to the long-term and the short-term components of the exchange of angular momentum between the equatorial bulge and the external gravitational torque. (ii) The net solar radiation received at the top of the Earth's atmosphere is $S_0(1-A)/4 \approx 240$ W/m^2, and the Earth's infrared radiative output is σT_e^4 with $T_e \approx 255$ K. In the annual mean the input and the output are balanced. (iii) Global sea-level change is determined jointly by changes in the volume of sea water and changes in the capacity of the ocean basins. Volcanic eruptions transfer mantle material to the surface and subduction carries surface material back into the mantle, and the two are balanced on the time scale of the Wilson cycle. □

Remark 29.9 (Connection with the rest of the book). The theory of the periodic structure of geographic systems established in this chapter is an application of periodic mathematics to the Earth-surface system. Astronomical cycles, climatic cycles, geological cycles and terrain–coordinate cycles constitute the most complete periodic spectrum of the Earth-surface system. Geographic systems are a comparatively complex field of application of the periodic-mathematics framework in the natural sciences—they span the whole range of time scales from the seasonal to the hundred-million-year, and couple the periodic processes of the five great spheres: astronomical, atmospheric, oceanic, geological and biological.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The passage of geological time and the seasonal cycle: the translational period of the tropical year (Definition 29.15, Proposition 29.7)
Scaling	The scaling distribution of climate zones with latitude (Example 29.3, Proposition 29.14); the power-law distribution of volcanic eruption frequency versus magnitude (Proposition 29.11); the power law of the land elevation distribution (Proposition 29.15)
Rotation	Milankovitch cycles (Definition 29.12, Proposition 29.5); the Earth's rotation and the latitude–longitude coordinate system (Definition 29.23, Proposition 29.13)
Conservation	Conservation of angular momentum, conservation of energy and conservation of mass (Theorem 29.8)
Symmetry	Symmetry of the northern and southern hemispheres and latitudinal symmetry: the symmetric distribution of solar radiation about the equator (Definition 29.14)
Closure	Closure of the water cycle and the lithospheric cycle: the Wilson cycle (Definition 29.18), the earthquake cycle (Definition 29.22), the transgression–regression cycle (Definition 29.20)
Period	Multi-scale periods of the diurnal cycle, the seasons, precession, Milankovitch cycles and volcanic and geological cycles (Theorem 29.7); cross-scale resonance (Corollary 29.2, Definition 29.9)

Exercises

Exercise 1 Prove the **geographic periodicity decision theorem** of this chapter (29.1). Hint: first write down the decision map and the evolution group of this chapter's target, then apply the proof structure of the general criterion of the 'Periodic Systems' chapter.

Exercise 2 Verify that the hypotheses of **monotonicity of quantity adjustment** (29.1) hold for the domain objects of this chapter, and give a counterexample in which the hypotheses fail.

Exercise 3 Examine the behaviour of the **geographic periodicity diagnostic procedure and its interfaces with three chapters** (29.6) under degeneracy of the parameters, and explain how the conclusion degenerates accordingly.

Exercise 4 Explain the relation between **autocorrelation and trend convolution** (29.4) and the corresponding general conclusion of the 'Periodic Systems' chapter (whether they agree, whether it is a special case, or whether additional conditions are required).

Exercise 5 Using the seven-theme comparison table of this chapter, identify the concrete carriers in this chapter of the three themes **translation**, **scaling** and **rotation**, and give one example of domain data for each.

Exercise 6 Synthesis: restate the diagnostic procedure of this chapter using the five-step standard interface of the 'Periodic Systems' chapter (Table 24.1), noting for each step the theorem of this chapter on which it depends: 29.1, 29.1, 29.6, 29.4.

Chapter 30

Periodic Structure of the I Ching System

Remark 30.1 (Periodicity decision for the target of this chapter (citation)). When this chapter decides whether a research target is periodic, determines its period type, or seeks its minimal positive period, reference is always made to the 'Periodic Systems' chapter: for the periodicity criterion see Theorem 24.1, for the discrimination of period types see Definition 24.3, Proposition 24.1, for the search for the minimal positive period see Definition 24.4, Theorem 24.2, and for the construction of period series and epicycle groups from data see Theorem 24.3, Definition 24.6.

Remark 30.2 (Positioning of this chapter: developed along the five-step interface of the 'Periodic Systems' chapter). The periodic structure of the I Ching system studies the mathematical characterisation of periodicity and phase relations in the *I Ching* symbol system. This chapter does not establish a separate periodicity criterion; instead it connects the I Ching symbol system to the five-step interface prescribed by the 'Periodic Systems' chapter:

1. **Decision**: reduce the symbol systems of Taiji, the Two Modes, the Four Symbols, the Eight Trigrams, the Sixty-Four Hexagrams, the Heavenly Stems, the Earthly Branches and the sexagenary cycle to discrete dynamical systems, cite Theorem 24.1 to decide whether they are periodic, and determine the period type;
2. **Construction**: from data such as hexagram images, stems and branches, solar terms and Na-Yin, construct the amplitude series, the frequency series and the phase series, and assemble them into the I Ching epicycle group;
3. **Adjustment**: dynamically adjust the I Ching epicycle group by a unified ordering rule, the tensor structure and the decrease of the residual energy;
4. **Convolution**: introduce amplitude convolution, phase convolution and frequency convolution on the I Ching epicycle group, and connect them with periodic convolution, the generation of the sexagenary cycle and the He Tu–Lo Shu structure;
5. **Integration**: connect to the tool chains of the three chapters 'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis', forming an I Ching periodicity diagnostic procedure.

The mathematical tools used in this chapter include: cyclic groups, irreducible representations, tensor products, the discrete Fourier transform, finite-order actions, periodic convolution, phase operators, nested epicycle generation, He Tu complex functions and Lo Shu matrices.

30.1 Connection to the five-step interface of the 'Periodic Systems' chapter

30.1.1 Step one: periodicity decision for an I Ching target

Definition 30.1 (I Ching target and decision input). An **I Ching target** is one of the following objects:

(i) *a hexagram hierarchy system, whose state space is* $\mathbb{F}_2^n$ *and whose evolution map is the binary bottom-up ordering advance;*

(ii) *a stem–branch sequence, whose state space is* $\mathbb{Z}_{10}\times\mathbb{Z}_{12}$ *and whose evolution map advances the Heavenly Stems and the Earthly Branches in synchrony;*

(iii) *a sexagenary cycle sequence, whose state space is the cyclic group* C_{60} *and whose evolution map is* $+1 \pmod{60}$*;*

(iv) *a solar-term or Na-Yin sequence, whose state space is a finite cyclic group* C_T *and whose evolution map is* $+1 \pmod{T}$*.*

The **decision input** of an I Ching target is the pair

$$\mathcal{D}_{\mathsf{Y}} = (\mathcal{S}, \Phi),$$

where $\mathcal{S}$ is the state space and $\Phi : \mathcal{S} \to \mathcal{S}$ is the evolution map induced by ordering advance, synchronous advance or cyclic translation. If $\mathcal{S}$ is a finite set, the target is called a **finite I Ching target**; if $\mathcal{S}$ is an infinite set, the target is called an **infinite I Ching target**.

Definition 30.2 (I Ching decision map). *Let an I Ching target be given by the decision input* $\mathcal{D}_{\mathsf{Y}} = (\mathcal{S}, \Phi)$*. If the states of the target are generated by iteration of* Φ*, that is, if there exists* $x_0 \in \mathcal{S}$ *such that*

$$x_t = \Phi^t(x_0), \qquad t \in \mathbb{Z},$$

then Φ *is called the* **I Ching decision map** *of the target. In this case the I Ching target reduces to a discrete dynamical system* $(\mathcal{S}, \Phi)$*, and the criteria of the 'Operations and Periodicity' chapter and of the 'Periodic Systems' chapter may be applied directly.*

Theorem 30.1 (Periodicity decision theorem for I Ching targets). *Let the decision map of an I Ching target be* Φ*, and let* $(\mathcal{S}, \Phi)$ *be a discrete dynamical system. Then:*

(A) **Sufficient decision**: *if there exists* $m \in \mathbb{N}^+$ *such that* $\Phi^m = \mathsf{id}$*, then the target is a uniformly periodic I Ching system; every orbit is closed, and every orbit length divides* m*.*

(B) **Equivalence of the decision**: *the target is a uniformly periodic I Ching system if and only if* $\mathsf{ord}(\Phi) < \infty$*.*

(C) **Pointwise criterion**: the orbit of the target at the state x_0 is closed if and only if x_0 is a periodic point; in that case the orbit length is the period at that point.

(D) **Non-omissibility**: the conjunction of a conserved quantity and a symmetry group cannot replace ord$(\Phi) < \infty$; an almost periodic I Ching target does not belong to the periodic I Ching systems.

Proof. The Theorem 30.1 of this chapter is the domain instantiation of Theorem 24.1 of the 'Periodic Systems' chapter: one need only replace the decision map f and the evolution group by the Φ of this chapter, and the four items (A) sufficient decision, (B) *equivalence of the decision*, (C) pointwise criterion and (D) non-omissibility hold one by one. By the convention of Remark 24.4, they are not *expanded again here*. ☐

Corollary 30.1 (Period types of the I Ching)**.** After the decision by Theorem 30.1, an I Ching target falls into one of the following types:

(i) **Trivially periodic type**: the least period is $m_0 = 1$ and the state is a fixed point;

(ii) **Strictly periodic type**: $1 < m_0 < \infty$ and the state is strictly periodic;

(iii) **Uniformly periodic type**: ord$(\Phi) = m < \infty$ and all orbit lengths divide m;

(iv) **Quasi-periodic type**: ord$(\Phi) = \infty$; the orbit does not close but is dense on the compact carrier structure;

(v) **Eventually periodic type**: there exists N such that the target is periodic for $t \geq N$;

(vi) **Discretely periodic type**: the carrier structure is $\mathbb{Z}$ and the normalised period ratio $\rho = T_x/\Delta x \in \mathbb{Q}$.

(vii) **Asymptotically periodic (damped) type**: the orbit is not closed, but tends asymptotically to a closed orbit or to a fixed point, the residual decaying as $e^{-\kappa t}$ (see Definition 24.3(vii)).

Among them (i)–(iii) and (vi) belong to the periodic I Ching systems; (iv) does not belong to the periodic I Ching systems, but can be handled by bounded approximation tools; (v) is a periodic I Ching system only in its stable stage; (vii), like (iv), is a type in the limiting sense; it is not a periodic I Ching system and can be handled with bounded-approximation tools only in its stable stage.

Proof. Each item follows by substituting an I Ching target into Theorem 30.1 and Definition 24.3 of the 'Periodic Systems' chapter. ☐

Lemma 30.1 (Finite-order decision for a hexagram hierarchy)**.** Let the state space of a hexagram hierarchy be $\mathbb{F}_2^n$ and let the evolution map be binary addition of one,

$$\Phi(b_0, \ldots, b_{n-1}) = (b_0, \ldots, b_{n-1}) + 1 \pmod{2^n}.$$

Then ord$(\Phi) = 2^n$, and the hexagram hierarchy is a uniformly periodic system.

Proof. Under addition the state space $\mathbb{F}_2^n$ is isomorphic to the cyclic group C_{2^n}. The map Φ is $+1$, whose order is the group order 2^n. Hence $\Phi^{2^n} = \mathrm{id}$, and for every $k < 2^n$ we have $\Phi^k \neq \mathrm{id}$. Therefore $\mathrm{ord}(\Phi) = 2^n$. By Theorem 30.1, the system is a uniformly periodic system. ☐

Lemma 30.2 (Periodicity decision for the Heavenly Stems and Earthly Branches)**.** *The state space of the Heavenly Stems system is C_{10} and the state space of the Earthly Branches system is C_{12}. Then*

$$\mathrm{ord}(\Phi_{\text{Heavenly Stems}}) = 10, \qquad \mathrm{ord}(\Phi_{\text{Earthly Branches}}) = 12.$$

Both are uniformly periodic systems.

Proof. The stem sequence is the cycle of the ten Heavenly Stems and the branch sequence is the cycle of the twelve Earthly Branches. By the definition of a cyclic group, $+1 \pmod{10}$ has order 10 and $+1 \pmod{12}$ has order 12. The conclusion follows. ☐

Lemma 30.3 (Periodicity decision for the sexagenary cycle)**.** *The state space of the sexagenary system is C_{60} and the evolution map is $+1 \pmod{60}$. Then*

$$\mathrm{ord}(\Phi_{\text{sexagenary cycle}}) = 60.$$

The sexagenary cycle is a uniformly periodic system.

Proof. The sexagenary cycle is generated by the synchronous parity of the Heavenly Stems and the Earthly Branches. By Lemma 30.4, there are exactly 60 effective combinations, and they are isomorphic to C_{60}. Hence $+1 \pmod{60}$ has order 60. ☐

Definition 30.3 (Algorithm for the least positive period of an I Ching target)**.** This definition has the same content as Chapter 24, the section 'Periodicity decision: is the target periodic?' (Section 24.1), Definition 24.4; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement is given there. The only difference here is terminology (I Ching target / I Ching sample); the algorithm steps and the conclusion are exactly the same.

Theorem 30.2 (Correctness of the algorithm for the least positive period of an I Ching target)**.** *This theorem has the same content as Chapter 24, the section 'Periodicity decision: is the target periodic?' (Section 24.1), Theorem 24.2; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement and proof are given there. The only difference here is terminology (I Ching target / I Ching sample); the algorithm steps and the conclusion are exactly the same.*

Example 30.1 (Periodicity decision from Taiji to the Sixty-Four Hexagrams)**.**

(i) Taiji: $T = 1$, trivially periodic type;

(ii) the Two Modes: $T = 2$, strictly periodic type;

(iii) the Four Symbols: $T = 4$, uniformly periodic type;

(iv) the Eight Trigrams: $T = 8$, uniformly periodic type;

(v) the Sixty-Four Hexagrams: $T = 64$, uniformly periodic type;

(vi) the Heavenly Stems: $T = 10$, uniformly periodic type;

(vii) the Earthly Branches: $T = 12$, uniformly periodic type;

(viii) the sexagenary cycle: $T = 60$, uniformly periodic type.

Remark 30.3 (An invariant does not guarantee periodicity). *Certain conserved quantities in the I Ching symbol system (such as parity synchronisation and the assignment of the Five Phases) are auxiliary conditions for periodic structure, but they do not constitute a sufficient condition. Periodic structure must be established by Theorem 30.1 or by the finite-order decision lemmas above.*

30.1.2 Step two: construction of the I Ching period series and epicycle group

Definition 30.4 (I Ching amplitude series, frequency series and phase series)**.** Let the state or output sequence of an I Ching target be $\{z_t\}$. If it can be written as

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)} + \varepsilon_t,$$

then

$$\mathcal{A}_z(t) = \sum_k A_k e^{ikt}, \qquad \mathcal{F}_z(s) = \sum_k \omega_k s^k, \qquad \mathcal{P}_z(t) = \sum_k e^{i(kt+\phi_k)}$$

are called respectively the **amplitude series**, the **frequency series** and the **phase series** of the I Ching target. The three agree with Definition 6.10, Definition 6.12 and Definition 6.11 of the 'Series and Convolution' chapter.

Definition 30.5 (I Ching epicycle group)**.** Let the period series of an I Ching target be given by the amplitude series $\{A_k\}$, the frequency series $\{\omega_k\}$ and the phase series $\{\phi_k\}$. For each k define the epicycle

$$E_k : \theta \mapsto A_k e^{i(\theta+\phi_k)}, \qquad \theta = \omega_k t,$$

and call the set

$$\mathfrak{E}_{\mathsf{Y}} = \{(A_k, \omega_k, \phi_k)\}_{k\in\mathbb{Z}}$$

the **epicycle group** of the I Ching target. Its superposition is the I Ching period series

$$z_t = \sum_k E_k(\omega_k t) = \sum_k A_k e^{i(\omega_k t+\phi_k)}.$$

Theorem 30.3 (Construction of the period series from I Ching data)**.** *Let the output of an I Ching target be a finite periodic sequence of period N. Then:*

(i) the output can be expanded as a discrete Fourier series

$$z_t = \sum_{k=0}^{N-1} c_k e^{i2\pi kt/N}, \qquad c_k = \frac{1}{N}\sum_{t=0}^{N-1} z_t e^{-i2\pi kt/N};$$

(ii) the amplitude series is given by $|c_k|$, the phase series by $\arg c_k$, and the frequency series by $2\pi k/N$;

(iii) if the output is a hexagram sequence or a stem–branch sequence, then the harmonics at the various levels correspond to different period scales.

Proof. (i) The discrete Fourier series of a finite periodic sequence is a standard result: the basis functions $\{e^{i2\pi kt/N}\}_{k=0}^{N-1}$ form an orthogonal basis of $\mathbb{C}^N$. (ii) This is read off directly from $c_k = A_k e^{i\phi_k}$. (iii) For a hexagram sequence $N = 64$, the fundamental frequency corresponds to the great period of the Sixty-Four Hexagrams, and the lower harmonics correspond to substructures such as Yin-Yang, the Four Symbols and the Eight Trigrams; for a stem–branch sequence $N = 60$, the fundamental frequency corresponds to the sexagenary period, and the harmonics correspond to the Heavenly Stems, the Earthly Branches and parity synchronisation. ☐

Proposition 30.1 (Correspondence of the three parameters: radius, phase and angular speed)**.** *The three parameters of the I Ching epicycle group $\mathfrak{E}_{\mathrm{Y}} = \{(A_k, \omega_k, \phi_k)\}_k$ correspond one-to-one with three geometric–mechanical quantities:*

(i) **Radius corresponds to amplitude**: *the radius $|E_k| = A_k$ of E_k is the amplitude of that component;*

(ii) **Phase corresponds to rotational delay**: *the initial phase ϕ_k corresponds to the time translation $\tau_k = \phi_k/\omega_k$, that is, the epicycle* **leads** *the reference by τ_k ($\phi_k > 0$ means that the waveform is* **advanced** *by τ_k relative to the reference, equivalently the waveform coincides with the reference once the reference has been delayed by τ_k);*

(iii) **Frequency corresponds to angular speed**: *ω_k is the angular velocity of that epicycle about the origin.*

Proof. The Proposition 30.1 of this chapter corresponds exactly to Proposition 24.3 of the 'Periodic Systems' chapter; the three correspondences—radius to amplitude, phase to rotational delay and frequency to angular speed—are obtained one by one by translating them to the epicycle group of this chapter. The proof is to be found there: (i) from $|E_k| = A_k$ and the energy decomposition, (ii) from the phase-shift theorem, (iii) from the definition of the frequency series. ☐

Example 30.2 (Period series of the Heavenly Stems)**.** The ten Heavenly Stems may be regarded as a periodic sequence on C_{10}; its discrete Fourier decomposition yields ten frequency components, the fundamental frequency corresponding to the cycle of the ten Heavenly Stems, the second harmonic to the alternation of Yin-Yang and the fifth harmonic to the cycle of the Five Phases.

Remark 30.4 (A period series is a complex function). Let $z = e^{i\omega_0 t}$. Then the I Ching period series

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)}$$

can be written as the Laurent series $\sum_k c_k z^k$, where $c_k = A_k e^{i\phi_k}$, provided that **every component frequency is commensurable with** ω_0 (that is, $\omega_k = k\omega_0$ with

$k \in \mathbb{Z}$). Hence **when all components are commensurable with ω_0, the period series is a complex-valued function on the unit circle (otherwise it is a function on the torus).**

Definition 30.6 (Unified ordering rule). (i) $k = 0$ ($\theta = 0$) corresponds to the **beginning** of the period. When $T = 2^n$ this is the all-yin line configuration (Zhen, the beginning of wood); for other periods it is **yang wood**.

(ii) The intermediate phases are arranged in increasing order of $\theta_k = 2\pi k/T$.

(iii) $k = T - 1$ corresponds to the **end** of the period. When $T = 2^n$ this is the all-yang line configuration (Kan, the end of water); for other periods it is **yin water**.

For $T = 2^n$, the lines are read in binary from the bottom upwards: the first line is b_0, a yin line is $= 0$ and a yang line is $= 1$.

30.1.3 Step three: dynamic adjustment of the I Ching epicycle group

Definition 30.7 (Dynamic adjustment of the I Ching epicycle group). Let the current I Ching epicycle group be $\mathfrak{E}^{(j)}$ and let the partial sum be

$$z^{(j)}(t) = \sum_{k \in \mathfrak{E}^{(j)}} A_k e^{i(\omega_k t + \phi_k)},$$

with residual $r^{(j)} = z - z^{(j)}$. The **dynamic adjustment** consists of two rules:

(i) **Cardinality adjustment**: *if the residual energy $\|r^{(j)}\|^2 > \delta$, add one epicycle according to the peak of the residual periodogram;*

(ii) **Parameter adjustment**: *update the parameters of the existing epicycles by regularised least squares or by the unified ordering rule.*

In the I Ching system, cardinality adjustment corresponds to adding a hexagram level, a stem–branch level or a Na-Yin level; parameter adjustment corresponds to adjusting the tensor index, the Five-Phase assignment or the phase origin.

Proposition 30.2 (Monotonicity of the cardinality adjustment). *After adding the dominant frequency component of the residual as in Definition 30.7(i), the residual energy is monotonically non-increasing:*

$$\|r^{(j+1)}\|^2 \le \|r^{(j)}\|^2,$$

with equality if and only if the residual has no energy at the selected frequency.

Proof. The Proposition 30.2 of this chapter agrees with the dynamic adjustment mechanism of Definition 24.8 of the 'Periodic Systems' chapter: adding an epicycle removes the component of $r^{(j)}$ along $\hat{\omega}$ from the residual, and the orthogonal decomposition $\|r^{(j)}\|^2 = \|r^{(j+1)}\|^2 + \|\text{the removed component}\|^2$ gives the inequality, with equality if and only if the norm of the removed component is zero. The complete argument is to be found there. □

Theorem 30.4 (Convergence of the dynamic adjustment)**.** Let the I Ching target be a **real-valued finite periodic sequence** of length T_0 ($T_0 \in \mathbb{N}$) and let $\delta = 0$. Then after at most $\lfloor T_0/2 \rfloor + 1$ steps of dynamic adjustment the residual is zero, and the resulting I Ching epicycle group is the exact Fourier data of the target; for a **complex-valued** sequence the corresponding number of steps is at most T_0.

Proof. The discrete Fourier series of a **real-valued** finite periodic sequence of length T_0 ($T_0 \in \mathbb{N}$) is complete on $\lfloor T_0/2 \rfloor + 1$ harmonics (including the constant term); a **complex-valued** sequence requires T_0 components. At each step one dominant frequency component of the residual is extracted as in Proposition 30.2 and the residual energy decreases strictly, so after finitely many steps the residual is zero. □

Definition 30.8 (Structural-metric adjustment)**.** Let the structural metric of an I Ching target be

$$\mathcal{I} = (\mathcal{I}_{\text{hexagram images}}, \mathcal{I}_{\text{stems and branches}}, \mathcal{I}_{\text{Five Phases}}) = (\text{product of bit widths, degree of parity synchr}$$

Structural-metric adjustment is: at each step choose a neighbouring parameter θ in the parameter space Θ for which $\mathcal{I}(\theta)$ is **strictly decreasing** in the lexicographic order (minimisation); if no such neighbouring parameter exists, the process stops—then θ is a **locally optimal** parameter.

Proposition 30.3 (Finite convergence of the structural-metric adjustment)**.** Let the parameter space Θ be finite and let each adjustment make $\mathcal{I}$ **strictly decreasing** in the lexicographic order. Then the adjustment terminates after finitely many steps at a locally optimal parameter.

Proof. Θ is finite, so $\mathcal{I}$ takes finitely many values. The lexicographic order is well-founded on a finite set of values, so a strictly lexicographically decreasing sequence cannot continue indefinitely and the process terminates after finitely many steps. At termination there is no neighbouring parameter making $\mathcal{I}$ strictly lexicographically decreasing, that is, the parameter is locally optimal. Monotonicity must be imposed on the **whole lexicographic order**: if only 'some component strictly increases or decreases' is required, cycles such as $(2,1) \to (1,2) \to (2,1)$ cannot be excluded and finite termination fails. □

Remark 30.5 (Boundaries of the dynamic adjustment). Theorem 30.4 relies on a finite periodic sequence. For a continuous or almost periodic I Ching target there is no exact representation by a finite epicycle group, and only bounded approximation is possible. Structural-metric adjustment does not guarantee a global optimum.

30.1.4 Step four: I Ching convolution

Definition 30.9 (Phase-operator periodic convolution)**.** Let two symbol systems A and B have periods T_A and T_B and phase-operator sets $\mathcal{P}_{T_A}$ and $\mathcal{P}_{T_B}$ respectively. The **phase-operator periodic convolution** of A and B is defined by

$$(\mathcal{P}_{T_A} \circledast \mathcal{P}_{T_B})(n, m) = \mathcal{O}_n^{(T_A)} \cdot \mathcal{O}_m^{(T_B)} = e^{i2\pi(n/T_A + m/T_B)}.$$

The periodic convolution retains the interaction information between all pairs (n, m)—it is the complete set of all phase-relation operators between the two periodic systems.

Proposition 30.4 (Consistency of periodic convolution with nested epicycle generation)**.** *Nested epicycle generation is the realisation of periodic convolution in a nested epicyclic structure. The composite phase operator $\mathcal{O}_{a,b} = \mathcal{O}_a^{(T_{\text{inner}})} \cdot \mathcal{O}_b^{(T_{\text{outer}})}$ is precisely the (a, b) element of the periodic convolution matrix.*

Proof. Take A to be the inner-circle system ($T_A = T_{\text{inner}}$) and B the outer-circle system ($T_B = T_{\text{outer}}$). By definition 30.9, the periodic convolution takes at (a, b) the value

$$(\mathcal{P}_{T_{\text{inner}}} \circledast \mathcal{P}_{T_{\text{outer}}})(a, b) = \mathcal{O}_a^{(T_{\text{inner}})} \cdot \mathcal{O}_b^{(T_{\text{outer}})} = e^{i2\pi(a/T_{\text{inner}} + b/T_{\text{outer}})}.$$

By theorem 30.7, the composite phase operator $\mathcal{O}_{a,b}$ of nested epicycle generation is exactly the same expression. The two are equal element by element. ☐

Theorem 30.5 (Periodic-convolution generation of the sexagenary cycle)**.** *The sexagenary cycle is generated by the periodic convolution of the Heavenly Stems ($T = 10$) and the Earthly Branches ($T = 12$). The effective-combination condition $n \equiv m \pmod 2$ selects 60 effective phases.*

Proof. By definition 30.9, the periodic convolution of the Heavenly Stems ($T_A = 10$) and the Earthly Branches ($T_B = 12$) runs through all $10 \times 12 = 120$ pairs (n, m), giving 120 composite phase operators $\mathcal{O}_n^{(10)} \mathcal{O}_m^{(12)}$. Yin-Yang synchronisation requires the parities to agree, so the effective-combination condition is $n \equiv m \pmod 2$. By lemma 30.4, this condition selects 60 of the 120 pairs, and these correspond one-to-one with the sexagenary cycle; likewise, by the group structure proved in that lemma, these 60 phases constitute the cyclic group C_{60}, so the period of the sexagenary cycle is $T = 60$. ☐

Corollary 30.2 (Uniformity of the periodic convolution over all periodic systems)**.** *All phase relations between any two periodic systems can be computed uniformly by means of the phase-operator periodic convolution.*

Proof. This is a direct corollary of definition 30.9: by definition the periodic convolution $\mathcal{O}_n^{(T_A)} \mathcal{O}_m^{(T_B)}$ runs through all pairs (n, m), so every phase-relation operator between the two periodic systems lies in the periodic convolution matrix, with nothing omitted; and every element of the matrix is in turn a product of two phase operators, which by the closure property of proposition 30.5 is still a phase operator, so there is nothing beyond it. ☐

30.1.5 Step five: integration with discrete sampling, harmonic analysis and time series analysis

Definition 30.10 (I Ching periodicity diagnostic procedure)**.** *For a given I Ching symbol system, the diagnosis is carried out using the framework of periodic mathematics:*

(i) **Periodicity decision**: verify the period type of the I Ching target (Theorem 30.1);

(ii) **Series construction**: from hexagram, stem–branch and solar-term data construct the amplitude, frequency and phase series and the epicycle group (Theorem 30.3);

(iii) **Tensor diagnosis**: compute the prime factorisation, the group structure, the tensor shape and the source of nested generation;

(iv) **Convolution diagnosis**: compute the periodic convolution, the generation of the sexagenary cycle and the He Tu–Lo Shu structure;

(v) **Integrated diagnosis**: connect to the tool chains of discrete sampling, harmonic analysis and time series analysis.

Theorem 30.6 (The I Ching periodicity diagnostic procedure and the interface with three chapters)**.** The five steps of Definition 30.10 connect respectively to:

(i) the chapter 'Discrete Sampling and Intercalation of Periodic Signals', Definition 21.1, Theorem 21.3, Theorem 21.10;

(ii) the chapter 'Harmonic Analysis', Definition 22.5, Theorem 22.1, Proposition 22.11;

(iii) the chapter 'Time Series Analysis', Definition 23.16, Proposition 23.2, Definition 23.19.

Hence the I Ching periodicity diagnosis is not an independent method, but an application of the unified tool chain of periodic mathematics to a traditional symbol system.

Proof. The Theorem 30.6 of this chapter is the implementation of the five-step interface of the 'Periodic Systems' chapter (Table 24.1) in this domain: the periodicity decision uses sampling and spectral estimation, the frequency decomposition and the phase kernel use the 'Harmonic Analysis' chapter, and the periodogram peak, Fisher's test and window-function validation use the 'Time Series Analysis' chapter. Hence the diagnostic procedure is composed of the tool chains of three chapters, and the proof is to be found there. □

Remark 30.6 (Operational order after integration). The recommended order for an actual diagnosis is: first decide whether the I Ching target is periodic (Theorem 30.1); then construct the Fourier data and the epicycle group of the hexagram, stem–branch and solar-term data (Theorem 30.3); then optimise the parameters by the unified ordering rule and the dynamic adjustment (Theorem 30.4, Proposition 30.3); then carry out the analysis of the periodic convolution and of sexagenary generation (Theorem 30.5); and finally output the diagnosis in accordance with Definition 30.10.

30.2 Periodic phase operators

Definition 30.11 (Periodic phase operator)**.** For any period $T \in \mathbb{N}^+$, the complex number $z_k^{(T)} = e^{i2\pi k/T}$ corresponding to its k-th phase is called the **periodic phase operator** and is written $\mathcal{O}_k^{(T)}$. The T phase operators together form the set:

$$\mathcal{P}_T = \{\mathcal{O}_k^{(T)} = e^{i2\pi k/T} \mid k = 0, 1, \ldots, T-1\}.$$

A periodic phase operator has a **dual identity**:

(i) **As a vector**: it is the direction vector on the unit circle of the k-th phase point in the period T—it encodes the position of that phase within the period.

(ii) **As an operator**: it acts on any complex vector by complex multiplication, producing a phase rotation:

$$\mathcal{O}_k^{(T)} \cdot \mathsf{v} = \mathsf{v} \cdot e^{i2\pi k/T}.$$

Multiplication by $\mathcal{O}_k^{(T)}$ is equivalent to rotating the vector v through $2\pi k/T$ radians.

Proposition 30.5 (Relational structure of the phase operators)**.** The T phase operators of period T form a complete relational hierarchy:

(i) **Yang operator (identity)**: $\mathcal{O}_0^{(T)} = 1$. Multiplication by $\mathcal{O}_0$ gives the object itself—the identity relation.

(ii) **Yin operator (opposition)**: $\mathcal{O}_{T/2}^{(T)} = -1$ (when T is even). Multiplication by $\mathcal{O}_{T/2}$ gives the opposite vector—the opposition relation.

(iii) **General rotation operator**: $\mathcal{O}_k^{(T)}$ $(k \neq 0, T/2)$. Partial rotation relations.

All phase operators together form the cyclic group C_T under complex multiplication: $\mathcal{O}_a \cdot \mathcal{O}_b = \mathcal{O}_{a+b \bmod T}$.

Proof. (i) Yang operator: $\mathcal{O}_0^{(T)} = e^{i\cdot 0} = 1$, so $\mathcal{O}_0^{(T)} z = z$ for every z. (ii) Yin operator: when T is even, $T/2 \in \mathbb{N}$, so $\mathcal{O}_{T/2}^{(T)} = e^{i2\pi(T/2)/T} = e^{i\pi} = -1$. (iii) General rotation: for any $z = |z|e^{i\varphi}$, $\mathcal{O}_k^{(T)} z = |z|e^{i(\varphi+2\pi k/T)}$; the modulus is unchanged and the argument increases by $2\pi k/T$.

Verification of the group structure: for any a, b, by the addition theorem for the exponential function,

$$\mathcal{O}_a^{(T)} \cdot \mathcal{O}_b^{(T)} = e^{i2\pi a/T} e^{i2\pi b/T} = e^{i2\pi(a+b)/T} = \mathcal{O}_{a+b \bmod T}^{(T)}.$$

Hence $\mathcal{P}_T$ is closed under complex multiplication; associativity is inherited from multiplication of complex numbers; the identity element is $\mathcal{O}_0^{(T)} = 1$; and the inverse of $\mathcal{O}_a^{(T)}$ is $\mathcal{O}_{T-a \bmod T}^{(T)}$. Therefore $(\mathcal{P}_T, \cdot)$ is a cyclic group, generated by $\mathcal{O}_1^{(T)}$, and $\mathcal{O}_1^{(T)}$ has order T, so $\mathcal{P}_T \cong C_T$. □

Proposition 30.6 (Correspondence between phase operators and irreducible representations)**.** The set $\mathcal{P}_T$ forms the cyclic group C_T under complex multiplication, and corresponds one-to-one with the irreducible representations of C_T:

$$\mathcal{O}_k^{(T)} = \rho_k(g).$$

Proof. By proposition 30.5, $(\mathcal{P}_T, \cdot) \cong C_T$ with isomorphism $\chi : C_T \to \mathcal{P}_T,\ g^j \mapsto \mathcal{O}_j^{(T)}$. Each one-dimensional irreducible representation ρ_k $(k = 0, \ldots, T-1)$ of the cyclic group $C_T = \langle g \mid g^T = 1 \rangle$ is uniquely determined by the image of the generator g: $\rho_k(g) = \omega^k$, where $\omega = e^{i2\pi/T}$. Thus $\rho_k(g^j) = \omega^{kj} = \mathcal{O}_{kj \bmod T}^{(T)}$, that is, the value of the irreducible representation ρ_k at the generator is precisely the phase operator $\mathcal{O}_k^{(T)}$. Since C_T has T conjugacy classes, it has exactly T one-dimensional irreducible representations, and both $k \mapsto \rho_k$ and $k \mapsto \mathcal{O}_k^{(T)}$ are bijections, so $\mathcal{P}_T$ corresponds one-to-one with the irreducible representations of C_T. □

Definition 30.12 (Regular representation and DFT diagonalisation)**.** This definition has the same content as Chapter 22, the section 'Roots of unity and cyclic group representations' (Section 22.1), Definition 22.2; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement is given there. Here the definition is merely restated in the language of the phase operators of the cyclic group $\mathbb{Z}_T$; the mathematical object is the same.

Remark 30.7 (The relational essence of the phase operators)*.* The core significance of phase operators lies in **defining relations**. The composition $\mathcal{O}_a \cdot \mathcal{O}_b$ of any two phases $\mathcal{O}_a$ and $\mathcal{O}_b$ gives a relation from $\mathcal{O}_b$ to $\mathcal{O}_{a+b}$. The multiplicative closure of the phase operators guarantees the completeness of the relational system.

30.3 Nested epicycle generation

Definition 30.13 (Nested epicycle generation rule)**.** Let an epicycle of period T_{inner} (the inner circle) be nested on an epicycle of period T_{outer} (the outer circle). The **nested epicycle generation rule** is:

For every single step forward of the outer circle (a rotation of $2\pi/T_{\text{outer}}$), the inner circle rota

Equivalently, the angular velocity of the inner circle is T_{inner} times that of the outer circle: $\omega_{\text{inner}} = T_{\text{inner}} \cdot \omega_{\text{outer}}$. The period of the composite system is $T_{\text{composite}} = T_{\text{inner}} \times T_{\text{outer}}$.

Theorem 30.7 (Nested epicycle generation theorem)**.** *Let the inner epicycle have T_{inner} equally spaced phase points and the outer epicycle have T_{outer} equally spaced phase points. The phases of the composite system are*

$$\theta_{a,b} = \frac{2\pi a}{T_{\text{inner}}} + \frac{2\pi b}{T_{\text{outer}}}, \quad a = 0, \ldots, T_{\text{inner}} - 1,\ b = 0, \ldots, T_{\text{outer}} - 1.$$

The composite system has $T_{\text{inner}} \times T_{\text{outer}}$ independent phases, and its composite phase operator is $\mathcal{O}_{a,b} = \mathcal{O}_a^{(T_{\text{inner}})} \cdot \mathcal{O}_b^{(T_{\text{outer}})}$.

Proof. The inner phase a contributes the phase angle $2\pi a/T_{\text{inner}}$ and the outer phase b contributes the phase angle $2\pi b/T_{\text{outer}}$; their superposition gives the composite phase $\theta_{a,b} = 2\pi a/T_{\text{inner}} + 2\pi b/T_{\text{outer}}$, so the composite phase operator is

$$\mathcal{O}_{a,b} = e^{i\theta_{a,b}} = e^{i2\pi a/T_{\text{inner}}} \cdot e^{i2\pi b/T_{\text{outer}}} = \mathcal{O}_a^{(T_{\text{inner}})} \cdot \mathcal{O}_b^{(T_{\text{outer}})}.$$

Number of independent phases: the inner and outer phase indices form the Cartesian product $\mathbb{Z}_{T_{\text{inner}}} \times \mathbb{Z}_{T_{\text{outer}}}$, whose cardinality is $T_{\text{inner}} \times T_{\text{outer}}$, corresponding to the $T_{\text{inner}} \times T_{\text{outer}}$ independent components of the tensor-product representation $\mathcal{P}_{T_{\text{inner}}} \otimes \mathcal{P}_{T_{\text{outer}}}$. When $\gcd(T_{\text{inner}}, T_{\text{outer}}) = 1$, the phase-sum projection retains all the independent phases (see note 30.8). $\square$

Remark 30.8 (The phase sum of nested epicycles and the tensor-product boundary)*.* The phase-sum projection $e^{i(\theta_a+\theta_b)}$ maps the inner and outer phase points to the same unit circle, and distinct pairs (a, b) may then coincide: the set of all values of the phase sum forms the cyclic subgroup $\langle e^{i2\pi/\operatorname{lcm}(T_{\text{inner}}, T_{\text{outer}})} \rangle$, with lcm($T_{\text{inner}}, T_{\text{outer}}$) distinct points. If and only if gcd($T_{\text{inner}}, T_{\text{outer}}$) $= 1$ is lcm $= T_{\text{inner}} \times T_{\text{outer}}$, and only then does the phase-sum projection retain all $T_{\text{inner}} \times T_{\text{outer}}$ independent phases. Yin-Yang × the Five Phases (2×5) and the Four Symbols × Sancai (4×3) are coprime, so the phase sum immediately gives the Heavenly Stems 10 and the Earthly Branches 12; for the Eight Trigrams × the Eight Trigrams, since gcd$(8, 8) = 8$, the phase sum has only 8 points, and the tensor-product (direct-product) structure $C_8 \times C_8$ is required in order to obtain the 64 hexagrams.

Example 30.3 (Epicycle generation of the Heavenly Stems). *The Five-Phase epicycle is the outer circle ($T_{\text{outer}} = 5$) and the Yin-Yang epicycle is the inner circle ($T_{\text{inner}} = 2$). For every single step forward of the Five Phases ($72°$), Yin-Yang rotates through one full turn ($360°$). The number of independent phases is $2 \times 5 = 10$, exactly the period of the Heavenly Stems.*

Example 30.4 (Epicycle generation of the Earthly Branches). *The Four-Symbols epicycle is the outer circle ($T_{\text{outer}} = 4$) and the Sancai epicycle is the inner circle ($T_{\text{inner}} = 3$). The number of independent phases is $3 \times 4 = 12$, exactly the period of the Earthly Branches.*

Example 30.5 (Epicycle generation of the Sixty-Four Hexagrams). *The Eight-Trigrams epicycle is the outer circle ($T_{\text{outer}} = 8$) and the Eight-Trigrams epicycle is the inner circle ($T_{\text{inner}} = 8$). The number of independent phases is $8 \times 8 = 64$. The group structure is $C_8 \times C_8$.*

Lemma 30.4 (Parity filtering of the sexagenary cycle). *Let $n \in \mathbb{Z}_{10}$ be the Heavenly-Stem index and $m \in \mathbb{Z}_{12}$ the Earthly-Branch index, with parity maps $\varphi_2 : \mathbb{Z}_{10} \to \mathbb{Z}_2,\ n \mapsto n \bmod 2$ and $\psi_2 : \mathbb{Z}_{12} \to \mathbb{Z}_2,\ m \mapsto m \bmod 2$. Then there are exactly 60 effective combinations satisfying the synchronisation condition $n \equiv m \pmod 2$, and they correspond one-to-one with the sexagenary cycle.*

Proof. The set of effective combinations is the fibre product

$$E = \{(n, m) \in \mathbb{Z}_{10} \times \mathbb{Z}_{12} : \varphi_2(n) = \psi_2(m)\}.$$

It is the kernel of the homomorphism $\Delta : \mathbb{Z}_{10} \times \mathbb{Z}_{12} \to \mathbb{Z}_2,\ (n, m) \mapsto \varphi_2(n) - \psi_2(m)$. Since both 10 and 12 contain the factor 2, the maps φ_2 and ψ_2 are surjective, hence so is Δ, and therefore $|E| = |\mathbb{Z}_{10} \times \mathbb{Z}_{12}|/|\mathbb{Z}_2| = 120/2 = 60$. A direct count confirms this: $\mathbb{Z}_{10}$ contains 5 even and 5 odd elements, and $\mathbb{Z}_{12}$ contains 6 even and 6 odd elements, so $|E| = 5 \cdot 6 + 5 \cdot 6 = 60$.

The diagonal embedding $\iota : \mathbb{Z}_{60} \to E,\ k \mapsto (k \bmod 10, k \bmod 12)$ is well defined (the parity of k is the same under the two reductions) and injective: if $k \equiv k' \pmod{10}$ and $k \equiv k' \pmod{12}$, then $k \equiv k' \pmod{\text{lcm}(10, 12) = 60}$. Since $|E| = 60 = |\mathbb{Z}_{60}|$, the map ι is bijective, so $E \cong C_{60}$. Term by term in k (the Heavenly-Stem index is counted from Jia and the Earthly-Branch index from Yin, so that k is the index of the sexagenary cycle and Jiazi has index 10): $k = 0 \mapsto (0, 0)$ (Jiayin, index 0), $k = 1 \mapsto (1, 1)$ (Yimao), $\cdots$, $k = 10 \mapsto (0, 10)$ (Jiazi, index 10), which is precisely the sexagenary sequence beginning with Jiayin. □

30.4 Prime factorisation and tensor structure

Theorem 30.8 (Direct-product decomposition of a cyclic group). *Let $T = \prod_{j=1}^{r} m_j$ with $m_1, \ldots, m_r$ pairwise coprime. Then $C_T \cong C_{m_1} \times \cdots \times C_{m_r}$.*

Proof. By the Chinese remainder theorem: when $m_1, \ldots, m_r$ are pairwise coprime, the map

$$\varphi : \mathbb{Z}_T \to \mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_r}, \quad k \mapsto (k \bmod m_1, \ldots, k \bmod m_r)$$

is a ring isomorphism, and in particular a group isomorphism. Hence $C_T \cong C_{m_1} \times \cdots \times C_{m_r}$. □

Definition 30.14 (Tensor representation of a phase operator). *Under the direct-product decomposition, every phase operator is represented as an r-th order tensor:*

$$\mathsf{O}_{a_1,\ldots,a_r} = \exp\left(i2\pi \sum_{j=1}^{r} \frac{a_j}{m_j} \right).$$

Proposition 30.7 (Tensor product of irreducible representations). *Let $G = G_1 \times \cdots \times G_r$ be a direct product of cyclic group factors (the group structures of the thirteen symbol systems of this chapter are all of this kind). Then every irreducible representation of G is the tensor product of irreducible representations of the factors: $\rho = \rho_1 \otimes \cdots \otimes \rho_r$, and $\widehat{G} \cong \widehat{G_1} \times \cdots \times \widehat{G_r}$.*

Proof. By theorem 30.8, when $T = \prod_j m_j$ with the factors pairwise coprime, $C_T \cong C_{m_1} \times \cdots \times C_{m_r}$. Let the generators of the factors be g_j ($g_j^{m_j} = 1$); then every element of G is uniquely written as $g_1^{x_1} \cdots g_r^{x_r}$. Any one-dimensional irreducible representation ρ of G is uniquely determined by $\rho(g_1), \ldots, \rho(g_r)$; since $\rho(g_j)^{m_j} = \rho(g_j^{m_j}) = \rho(1) = 1$, the number $\rho(g_j)$ is an m_j-th root of unity, so there exists $a_j \in \mathbb{Z}_{m_j}$ with $\rho(g_j) = e^{i2\pi a_j/m_j}$. Hence

$$\rho(g_1^{x_1} \cdots g_r^{x_r}) = \prod_{j=1}^{r} e^{i2\pi a_j x_j/m_j},$$

which is precisely the value of the tensor product of the irreducible representations $\rho_j(g_j^{x_j}) = e^{i2\pi a_j x_j/m_j}$ of the factors. The indices $(a_1, \ldots, a_r)$ run independently through $\mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_r}$, giving $\prod_j m_j = T$ combinations in all, equal to the total number T of irreducible representations of C_T; hence there is neither omission nor repetition. □

30.5 He Tu complex functions

This section establishes the **He Tu complex function**—a complex function of period 10. The He Tu structure reveals that the Fourier decomposition of period 10 automatically exhibits the algebraic grading $\mathbb{Z}_{10} \cong \mathbb{Z}_2 \times \mathbb{Z}_5$, in which the $\mathbb{Z}_5$ component corresponds to the mutual-generation cycle of the **Five Phases** and the $\mathbb{Z}_2$ component to the twofold alternation of **Yin-Yang**.

Definition 30.15 (He Tu number–Five-Phase correspondence). Define the **He Tu map** $\varphi : \mathbb{Z}_{10} \to \mathbb{Z}_5$ by

$$\varphi(n) := n \bmod 5, \qquad n \in \mathbb{Z}_{10} = \{0, 1, \dots, 9\},$$

where the remainder 0 is by convention associated with 'earth' among the Five Phases. Specifically:

$\varphi(n)$	0	1	2	3	4
Five Phases	earth	water	fire	wood	metal
Numbers	$5, 10$	$1, 6$	$2, 7$	$3, 8$	$4, 9$

Moreover, the **Yin-Yang** component is given by the parity of n: an odd n corresponds to yang and an even n to yin. Thus the He Tu map is a group homomorphism

$$\varphi : \mathbb{Z}_{10} \to \mathbb{Z}_5, \qquad \ker \varphi = \{0, 5\} \cong \mathbb{Z}_2.$$

Remark 30.9 (He Tu complex functions (the same definition as in the 'Series and Convolution' chapter)). The He Tu complex function—that is, a complex function satisfying $f(z + 10) = f(z)$—is the **same definition** as Definition 6.7 of the 'Series and Convolution' chapter, and its Fourier expansion $f(z) = \sum_n c_n e^{2\pi i n z/10}$ and the coefficient formula are likewise exactly the same, so they are not repeated here.

Remark 30.10 (The He Tu Fourier decomposition (restatement)). This theorem is the same conclusion as Theorem 6.11 of the 'Series and Convolution' chapter: a uniformly convergent He Tu complex function can be uniquely decomposed as

$$f(z) = \sum_{k=0}^{4} \Big(c_k + c_{k+5}\, e^{\pi i z} \Big) e^{2\pi i k z/10},$$

where $e^{\pi i z}$ is the Yin-Yang factor of period 2. Both the statement and the proof are to be found there; the reader may cite the conclusion of this section directly from that place.

Proof. The proof of this theorem is exactly the same as the proof of Theorem 6.11 of the 'Series and Convolution' chapter (where the substitution computation for $\mathbb{Z}_{10} \cong \mathbb{Z}_2 \times \mathbb{Z}_5$ is given term by term), and is not repeated here. □

Definition 30.16 (Five-Phase components). In the above decomposition,

$$f_k(z) := \Big(c_k + c_{k+5}\, e^{\pi i z} \Big)\, e^{2\pi i k z/10}, \qquad k = 0, 1, 2, 3, 4$$

is called the k-th **Five-Phase component** of f. By the He Tu map (definition 30.15), the five components correspond in order to:

$$f_0 \leftrightarrow \text{earth}, \quad f_1 \leftrightarrow \text{water}, \quad f_2 \leftrightarrow \text{fire}, \quad f_3 \leftrightarrow \text{wood}, \quad f_4 \leftrightarrow \text{metal}.$$

Each Five-Phase component consists of two parts: $c_k\, e^{2\pi i k z/10}$ (the yang component) and $c_{k+5}\, e^{\pi i z} e^{2\pi i k z/10}$ (the yin component).

Remark 30.11 (Period structure of the Five-Phase components (restated in five-phase notation)). This proposition is a restatement of Proposition 6.9 of the 'Series and Convolution' chapter in **five-phase notation**, with exactly the same mathematical content; what is given after the proof there is the period table indexed by k, and here only the index is replaced by the names of the Five Phases:

k	0	1	2	3	4
Five Phases	earth	water	fire	wood	metal
fundamental period	1	10	5	10	5

That is: f_0 (earth) contains only the constant term; f_1, f_3 (water, wood) are components of period 10; f_2, f_4 (fire, metal) are components of period 5. The proof is to be found in Proposition 6.9 and is not repeated here.

Remark 30.12 (The yin–yang flip of a He Tu complex function (restated in five-phase notation)). This proposition is the same proposition as Proposition 6.10 of the 'Series and Convolution' chapter, and its statement and proof are not repeated here; in five-phase notation it reads as follows: the half-period translation $z \mapsto z+5$ gives each component a factor $(-1)^k$; the even components ($k = 0, 2, 4$, corresponding to earth, fire and metal) are unchanged, whilst the odd components ($k = 1, 3$, corresponding to water and wood) change sign as a whole.

Remark 30.13 (The He Tu phase operator (the same definition as in the 'Series and Convolution' chapter)). The He Tu phase operator $\mathcal{O}_n^{(10)} = e^{2\pi i n/10}$ is the **same object** as Definition 6.9 of the 'Series and Convolution' chapter, and its decomposition $\mathbb{Z}_{10} \cong \mathbb{Z}_2 \times \mathbb{Z}_5$ is likewise exactly the same, so it is not defined again here.

The I Ching context merely gives these two factors other names: the $\mathbb{Z}_2$ component $(-1)^{\varepsilon}$ is called the **Yin-Yang sign**, and the $\mathbb{Z}_5$ component $e^{2\pi i k/10}$ is called the **Five-Phase phase**.

Remark 30.14 (Direct-product decomposition of the He Tu phase-operator group (restatement)). This proposition is the same proposition as Proposition 6.11 of the 'Series and Convolution' chapter: the phase-operator group $\{\mathcal{O}_n^{(10)}\}_{n=0}^{9}$ is isomorphic to $C_{10} \cong C_2 \times C_5$, where C_2 corresponds to Yin-Yang and C_5 to the Five Phases. Both the statement and the proof are to be found there and are not repeated here.

Remark 30.15 (DFT diagonalisation of the He Tu circulant matrix (restatement)). This theorem is the same conclusion as Theorem 6.12 of the 'Series and Convolution' chapter: the 10×10 circulant matrix with first-row generator $(h_0, \ldots, h_9)$ is unitarily diagonalised by the DFT matrix, and the eigenvalues $\hat{h}_n$ are divided into a yin group ($\varepsilon = 1$) and a yang group ($\varepsilon = 0$) according to $n = 5\varepsilon + k$. Both the statement and the proof are to be found there.

Proof. The proof of this theorem is exactly the same as the proof of Theorem 6.12 of the 'Series and Convolution' chapter (the DFT diagonalisation of the circulant matrix and the grading of the eigenvalues by $n = 5\varepsilon + k$), and is not repeated here. ☐

Example 30.6 (Concrete decomposition of a He Tu complex function). This example has the same content as Chapter 6, the section 'He Tu complex functions: the

$\mathbb{Z}_2 \times \mathbb{Z}_5$ decomposition of period 10' (Section 6.5), Example 6.8; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement is given there. The only difference between the two places is the column names of the component table (that place uses 'label/component', this place uses 'Five Phases/Yin-Yang'); the correspondence between the Five Phases and Yin-Yang is given by Definition 6.6, and no new example is set up.

Remark 30.16 (The algebraic meaning of the He Tu structure). This note has the same content as Chapter 6, the section 'He Tu complex functions: the $\mathbb{Z}_2 \times \mathbb{Z}_5$ decomposition of period 10' (Section 6.5), Remark 6.6; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement is given there. The I Ching context merely gives the two factors other names: the $\mathbb{Z}_2$ component $(-1)^{\varepsilon}$ is called the **Yin-Yang sign**, and the $\mathbb{Z}_5$ component $e^{2\pi ik/10}$ is called the **Five-Phase phase**.

30.6 The Lo Shu matrix

This section uses all the conclusions concerning the **Lo Shu matrix** in the context of the *I Ching* symbol system. The definition of this matrix, its magic-square property, the characterisation of its balance point, its antipodal balance, its spectral structure and its translation–rotation decomposition have already been given in full and proved one by one. The items of this section are all citations of the conclusions of that section; their statements and proofs are not repeated, and only the numbering anchors needed for references within this chapter are retained; wherever 'see that place' occurs below, it refers uniformly to Chapter 10, Section 10.5.

Definition 30.17 (The Lo Shu matrix). This definition has the same content as Chapter 10, the section 'The Lo Shu matrix: a 3×3 magic square with 5 as the balance point' (Section 10.5), Definition 10.22; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement is given there.

Definition 30.18 (Row sums, column sums and diagonal sums). This definition has the same content as Chapter 10, the section 'The Lo Shu matrix: a 3×3 magic square with 5 as the balance point' (Section 10.5), Definition 10.23; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement is given there.

Proposition 30.8 (The magic-square property of the Lo Shu matrix). This proposition has the same content as Chapter 10, the section 'The Lo Shu matrix: a 3×3 magic square with 5 as the balance point' (Section 10.5), Proposition 10.6; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement and proof are given there.

Definition 30.19 (The Lo Shu balance point). This definition has the same content as Chapter 10, the section 'The Lo Shu matrix: a 3×3 magic square with 5 as the balance point' (Section 10.5), Definition 10.24; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement is given there.

Proposition 30.9 (Threefold characterisation of the balance point). *This proposition has the same content as Chapter 10, the section 'The Lo Shu matrix: a 3×3 magic square with 5 as the balance point' (Section 10.5), Proposition 10.7; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement and proof are given there.*

Proposition 30.10 (Antipodal balance of the Lo Shu matrix). *This proposition has the same content as Chapter 10, the section 'The Lo Shu matrix: a 3×3 magic square with 5 as the balance point' (Section 10.5), Proposition 10.8; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement and proof are given there.*

Theorem 30.9 (Eigenvalues and spectral structure of the Lo Shu matrix). *This theorem has the same content as Chapter 10, the section 'The Lo Shu matrix: a 3×3 magic square with 5 as the balance point' (Section 10.5), Theorem 10.7; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement and proof are given there.*

Theorem 30.10 (Translation–rotation decomposition of the Lo Shu matrix). *This theorem has the same content as Chapter 10, the section 'The Lo Shu matrix: a 3×3 magic square with 5 as the balance point' (Section 10.5), Theorem 10.8; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement and proof are given there.*

Corollary 30.3 (Structural meaning of the Lo Shu balance point and the magic sum). *This corollary has the same content as Chapter 10, the section 'The Lo Shu matrix: a 3×3 magic square with 5 as the balance point' (Section 10.5), Corollary 10.5; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement and proof are given there.*

Example 30.7 (Spectral decomposition of the Lo Shu matrix). This example has the same content as Chapter 10, the section 'The Lo Shu matrix: a 3×3 magic square with 5 as the balance point' (Section 10.5), Example 10.3; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement is given there.

Remark 30.17 (Interface between the Lo Shu matrix and the He Tu structure). The factor 5 in the magic sum $15 = 3\times5$ of the Lo Shu matrix is algebraically related to the $\mathbb{Z}_5$ component of $\mathbb{Z}_{10} \cong \mathbb{Z}_2 \times \mathbb{Z}_5$ in the He Tu structure: the Lo Shu centre 5 corresponds to the apex of the Five-Phase cycle, and the antipodal balance of the Lo Shu matrix (with sum $10 = 2\times5$) corresponds to the twofold alternation of Yin-Yang. The purely rotational eigenvalues $\pm2\sqrt{6}i$ of the Lo Shu matrix on $\mathbf{1}^{\perp}$ provide a concrete example of a 3-dimensional rotation, whose period $\pi/\sqrt{6}$ is completely determined by the algebraic structure of the matrix.

30.7 Table of phase operators of the thirteen symbol systems

30.7.1 Taiji ($T = 1$)

Definition 30.20 (Taiji phase operators)**.** The Taiji period is $T = 1$, the bit width is $W = 1$ and the bandwidth is $\Delta\theta = 2\pi$. The group structure is C_1 and the tensor shape is 1. There is only one phase point, which describes 'existence'.

k	Symbol	Phase θ_k	Operator $\mathcal{O}_k^{(1)}$
0	Taiji	0	$1 + 0i$

30.7.2 The Two Modes ($T = 2$)

Definition 30.21 (Yin-Yang phase operators)**.** The Yin-Yang period is $T = 2$, the bit width is $W = 2$ and the bandwidth is $\Delta\theta = \pi$. The group structure is C_2 and the tensor shape is 2. The all-yin line configuration ($k = 0$) is Zhen (the beginning of wood) and the all-yang line configuration ($k = 1$) is Kan (the end of water).

k	Binary	Symbol	Five Phases	Phase θ_k	Operator $\mathcal{O}_k^{(2)}$
0	0	yin	Zhen (beginning of wood)	0	$1 + 0i$
1	1	yang	Kan (end of water)	π	$-1 + 0i$

30.7.3 Sancai ($T = 3$)

Definition 30.22 (Sancai phase operators)**.** The Sancai period is $T = 3$, the bit width is $W = 3$ and the bandwidth is $\Delta\theta = 2\pi/3$. The group structure is C_3 and the tensor shape is 3.

k	Symbol	Five Phases	Phase θ_k	Operator $\mathcal{O}_k^{(3)}$
0	heaven	yang wood (beginning)	0	$1 + 0i$
1	human	fire	$2\pi/3$	$-0.5 + 0.8660i$
2	earth	yin water (end)	$4\pi/3$	$-0.5 - 0.8660i$

30.7.4 The Four Symbols ($T = 4$)

Definition 30.23 (Four-Symbols phase operators)**.** The Four-Symbols period is $T = 4 = 2^2$, the bit width is $W = 4$ and the bandwidth is $\Delta\theta = \pi/2$. The group structure is C_4 and the tensor shape is 4.

k	Binary	Symbol	Five Phases	Phase θ_k	Operator $\mathcal{O}_k^{(4)}$
0	00	greater yin	Zhen (beginning of wood)	0	$1 + 0i$
1	01	lesser yang	wood	$\pi/2$	$0 + 1i$
2	10	lesser yin	metal	π	$-1 + 0i$
3	11	greater yang	Kan (end of water)	$3\pi/2$	$0 - 1i$

30.7.5 The Five Phases ($T = 5$)

Definition 30.24 (Five-Phase phase operators)**.** The Five-Phase period is $T = 5$, the bit width is $W = 5$ and the bandwidth is $\Delta\theta = 2\pi/5$. The group structure is C_5 and the tensor shape is 5.

k	Symbol	Five Phases	Phase θ_k	Operator $\mathcal{O}_k^{(5)}$
0	wood	yang wood (beginning)	0	$1 + 0i$
1	fire	fire	$2\pi/5$	$0.3090 + 0.9511i$
2	earth	earth	$4\pi/5$	$-0.8090 + 0.5878i$
3	metal	metal	$6\pi/5$	$-0.8090 - 0.5878i$
4	water	yin water (end)	$8\pi/5$	$0.3090 - 0.9511i$

30.7.6 The Seven Luminaries ($T = 7$)

Definition 30.25 (Seven-Luminaries phase operators)**.** The Seven-Luminaries period is $T = 7$, the bit width is $W = 7$ and the bandwidth is $\Delta\theta = 2\pi/7$. The group structure is C_7 and the tensor shape is 7.

k	Symbol	Five Phases	Phase θ_k	Operator $\mathcal{O}_k^{(7)}$
0	Sun	yang wood (beginning)	0	$1 + 0i$
1	Moon	—	$2\pi/7$	$0.6235 + 0.7818i$
2	Mars	fire	$4\pi/7$	$-0.2225 + 0.9749i$
3	Mercury	water	$6\pi/7$	$-0.9010 + 0.4339i$
4	Jupiter	wood	$8\pi/7$	$-0.9010 - 0.4339i$
5	Venus	metal	$10\pi/7$	$-0.2225 - 0.9749i$
6	Saturn	yin water (end)	$12\pi/7$	$0.6235 - 0.7818i$

30.7.7 The Eight Trigrams ($T = 8$)

Definition 30.26 (Eight-Trigrams phase operators)**.** The Eight-Trigrams period is $T = 8 = 2^3$, the bit width is $W = 8$ and the bandwidth is $\Delta\theta = \pi/4$. The group structure is C_8 and the tensor shape is 8. The all-yin line configuration (000, $k = 0$, Zhen) is the beginning of wood, and the all-yang line configuration (111, $k = 7$, Kan) is the end of water.

k	Binary	Line image	Hexagram name	Five Phases	Phase θ_k	Operator $\mathcal{O}_k^{(8)}$
0	000	☳	Zhen (Thunder)	wood (beginning)	0	$1 + 0i$
1	001	☴	Xun (Wind)	wood	$\pi/4$	$0.7071 + 0.7071i$
2	010	☶	Gen (Mountain)	earth	$\pi/2$	$0 + 1i$
3	011	☲	Li (Fire)	fire	$3\pi/4$	$-0.7071 + 0.7071i$
4	100	☰	Qian (Heaven)	metal	π	$-1 + 0i$
5	101	☷	Kun (Earth)	earth	$5\pi/4$	$-0.7071 - 0.7071i$
6	110	☱	Dui (Lake)	metal	$3\pi/2$	$0 - 1i$
7	111	☵	Kan (Water)	water (end)	$7\pi/4$	$0.7071 - 0.7071i$

30.7.8 The Heavenly Stems ($T = 10$)

Definition 30.27 (Heavenly-Stems phase operators)**.** The Heavenly-Stems period is $T = 10 = 2 \times 5$, the bit width is $W = 10$ and the bandwidth is $\Delta\theta = \pi/5$. The group structure is $C_{10} \cong C_2 \times C_5$ and the tensor shape is 2×5. It is generated by nested epicycles of Yin-Yang × the Five Phases.

k	Symbol	Yin-Yang and Five Phases	Tensor index	Phase θ_k	Operator $\mathcal{O}_k^{(10)}$
0	Jia	yang wood (beginning)	$(0, 0)$	0	$1 + 0i$
1	Yi	yin wood	$(0, 1)$	$\pi/5$	$0.8090 + 0.5878i$
2	Bing	yang fire	$(0, 2)$	$2\pi/5$	$0.3090 + 0.9511i$
3	Ding	yin fire	$(0, 3)$	$3\pi/5$	$-0.3090 + 0.9511i$
4	Wu	yang earth	$(0, 4)$	$4\pi/5$	$-0.8090 + 0.5878i$
5	Ji	yin earth	$(1, 0)$	π	$-1 + 0i$
6	Geng	yang metal	$(1, 1)$	$6\pi/5$	$-0.8090 - 0.5878i$
7	Xin	yin metal	$(1, 2)$	$7\pi/5$	$-0.3090 - 0.9511i$
8	Ren	yang water	$(1, 3)$	$8\pi/5$	$0.3090 - 0.9511i$
9	Gui	yin water (end)	$(1, 4)$	$9\pi/5$	$0.8090 - 0.5878i$

30.7.9 The Earthly Branches ($T = 12$)

Definition 30.28 (Earthly-Branches phase operators)**.** The Earthly-Branches period is $T = 12 = 4 \times 3$, the bit width is $W = 12$ and the bandwidth is $\Delta\theta = \pi/6$. The group structure is $C_{12} \cong C_4 \times C_3$ and the tensor shape is 4×3. It is generated by nested epicycles of the Four Symbols × Sancai.

k	Symbol	Five Phases	Tensor index	Phase θ_k	Operator $\mathcal{O}_k^{(12)}$
0	Yin	yang wood (beginning)	$(0,0)$	0	$1+0i$
1	Mao	yin wood	$(0,1)$	$\pi/6$	$0.8660+0.5i$
2	Chen	yang earth	$(0,2)$	$\pi/3$	$0.5+0.8660i$
3	Si	yin fire	$(1,0)$	$\pi/2$	$0+1i$
4	Wu	yang fire	$(1,1)$	$2\pi/3$	$-0.5+0.8660i$
5	Wei	yin earth	$(1,2)$	$5\pi/6$	$-0.8660+0.5i$
6	Shen	yang metal	$(2,0)$	π	$-1+0i$
7	You	yin metal	$(2,1)$	$7\pi/6$	$-0.8660-0.5i$
8	Xu	yang earth	$(2,2)$	$4\pi/3$	$-0.5-0.8660i$
9	Hai	yin water	$(3,0)$	$3\pi/2$	$0-1i$
10	Zi	yang water	$(3,1)$	$5\pi/3$	$0.5-0.8660i$
11	Chou	yin water (end)	$(3,2)$	$11\pi/6$	$0.8660-0.5i$

30.7.10 The twenty-four solar terms ($T = 24$)

Definition 30.29 (Phase operators of the twenty-four solar terms)**.** The period of the twenty-four solar terms is $T = 24 = 8 \times 3$, the bit width is $W = 24$ and the bandwidth is $\Delta\theta = \pi/12$. The group structure is $C_{24} \cong C_8 \times C_3$ and the tensor shape is 8×3. They are generated by nested epicycles of the Eight Trigrams × Sancai.

k	Symbol	Five Phases	Phase θ_k	Operator $\mathcal{O}_k^{(24)}$
0	Lichun	yang wood (beginning)	0	$1 + 0i$
1	Yushui	yin wood	$\pi/12$	$0.9659 + 0.2588i$
2	Jingzhe	yang wood	$\pi/6$	$0.8660 + 0.5i$
3	Chunfen	yin wood	$\pi/4$	$0.7071 + 0.7071i$
4	Qingming	yang wood	$\pi/3$	$0.5 + 0.8660i$
5	Guyu	yin wood	$5\pi/12$	$0.2588 + 0.9659i$
6	Lixia	yang fire	$\pi/2$	$0 + 1i$
7	Xiaoman	yin fire	$7\pi/12$	$-0.2588 + 0.9659i$
8	Mangzhong	yang fire	$2\pi/3$	$-0.5 + 0.8660i$
9	Xiazhi	yin fire	$3\pi/4$	$-0.7071 + 0.7071i$
10	Xiaoshu	yang fire	$5\pi/6$	$-0.8660 + 0.5i$
11	Dashu	yin fire	$11\pi/12$	$-0.9659 + 0.2588i$
12	Liqiu	yang metal	π	$-1 + 0i$
13	Chushu	yin metal	$13\pi/12$	$-0.9659 - 0.2588i$
14	Bailu	yang metal	$7\pi/6$	$-0.8660 - 0.5i$
15	Qiufen	yin metal	$5\pi/4$	$-0.7071 - 0.7071i$
16	Hanlu	yang metal	$4\pi/3$	$-0.5 - 0.8660i$
17	Shuangjiang	yin metal	$17\pi/12$	$-0.2588 - 0.9659i$
18	Lidong	yang water	$3\pi/2$	$0 - 1i$
19	Xiaoxue	yin water	$19\pi/12$	$0.2588 - 0.9659i$
20	Daxue	yang water	$5\pi/3$	$0.5 - 0.8660i$
21	Dongzhi	yin water	$7\pi/4$	$0.7071 - 0.7071i$
22	Xiaohan	yang water	$11\pi/6$	$0.8660 - 0.5i$
23	Dahan	yin water (end)	$23\pi/12$	$0.9659 - 0.2588i$

30.7.11 The thirty Na-Yin ($T = 30$)

Definition 30.30 (Thirty-Na-Yin phase operators)**.** The period of the thirty Na-Yin is $T = 30 = 2 \times 3 \times 5$, the bit width is $W = 30$ and the bandwidth is $\Delta\theta = \pi/15$. The group structure is $C_{30} \cong C_2 \times C_3 \times C_5$ and the tensor shape is $2 \times 3 \times 5$. They are generated by nested epicycles of Yin-Yang × Sancai × the Five Phases.

k	Na-Yin	Stems and Branches	Phase θ_k	Operator $\mathcal{O}_k^{(30)}$
0	Great Stream Water	Jiayin, Yimao	0	$1 + 0i$
1	Earth in the Sand	Bingchen, Dingsi	$\pi/15$	$0.9781 + 0.2079i$
2	Fire in the Sky	Wuwu, Jiwei	$2\pi/15$	$0.9135 + 0.4067i$
3	Pomegranate Wood	Gengshen, Xinyou	$\pi/5$	$0.8090 + 0.5878i$
4	Great Sea Water	Renxu, Guihai	$4\pi/15$	$0.6691 + 0.7431i$
5	Gold in the Sea	Jiazi, Yichou	$\pi/3$	$0.5 + 0.8660i$
6	Fire in the Furnace	Bingyin, Dingmao	$2\pi/5$	$0.3090 + 0.9511i$
7	Great Forest Wood	Wuchen, Jisi	$7\pi/15$	$0.1045 + 0.9945i$
8	Earth by the Roadside	Gengwu, Xinwei	$8\pi/15$	$-0.1045 + 0.9945i$
9	Metal of the Sword's Edge	Renshen, Guiyou	$3\pi/5$	$-0.3090 + 0.9511i$
10	Fire on the Mountain Top	Jiaxu, Yihai	$2\pi/3$	$-0.5 + 0.8660i$
11	Water in the Ravine	Bingzi, Dingchou	$11\pi/15$	$-0.6691 + 0.7431i$
12	Earth of the City Wall	Wuyin, Jimao	$4\pi/5$	$-0.8090 + 0.5878i$
13	White Wax Metal	Gengchen, Xinsi	$13\pi/15$	$-0.9135 + 0.4067i$
14	Willow Wood	Renwu, Guiwei	$14\pi/15$	$-0.9781 + 0.2079i$
15	Water in the Spring	Jiashen, Yiyou	π	$-1 + 0i$
16	Earth on the Roof	Bingxu, Dinghai	$16\pi/15$	$-0.9781 - 0.2079i$
17	Thunderbolt Fire	Wuzi, Jichou	$17\pi/15$	$-0.9135 - 0.4067i$
18	Pine and Cypress Wood	Gengyin, Xinmao	$6\pi/5$	$-0.8090 - 0.5878i$
19	Long Flowing Water	Renchen, Guisi	$19\pi/15$	$-0.6691 - 0.7431i$
20	Gold in the Sand	Jiawu, Yiwei	$4\pi/3$	$-0.5 - 0.8660i$
21	Fire at the Foot of the Mountain	Bingshen, Dingyou	$7\pi/5$	$-0.3090 - 0.9511i$
22	Wood of the Flat Ground	Wuxu, Jihai	$22\pi/15$	$-0.1045 - 0.9945i$

30.7.12 The sexagenary cycle ($T = 60$)

Definition 30.31 (Sexagenary-Cycle phase operators)**.** The stem–branch period is $T = 60 = 4 \times 3 \times 5$, the bit width is $W = 60$ and the bandwidth is $\Delta\theta = \pi/30$. The group structure is $C_{60} \cong C_4 \times C_3 \times C_5$ and the tensor shape is $4 \times 3 \times 5$. It is generated by the periodic convolution of the Heavenly Stems $\circledast$ the Earthly Branches.

k	Stems and Branches	Five Phases	Phase θ_k	Operator $\mathcal{O}_k^{(60)}$
0	Jiayin	yang wood (beginning)	0	$1 + 0i$
1	Yimao	yin wood	$\pi/30$	$0.9945 + 0.1045i$
2	Bingchen	yang earth	$\pi/15$	$0.9781 + 0.2079i$
3	Dingsi	yin fire	$\pi/10$	$0.9511 + 0.3090i$
4	Wuwu	yang fire	$2\pi/15$	$0.9135 + 0.4067i$
5	Jiwei	yin earth	$\pi/6$	$0.8660 + 0.5i$
6	Gengshen	yang metal	$\pi/5$	$0.8090 + 0.5878i$
7	Xinyou	yin metal	$7\pi/30$	$0.7431 + 0.6691i$
8	Renxu	yang water	$4\pi/15$	$0.6691 + 0.7431i$
9	Guihai	yin water	$3\pi/10$	$0.5878 + 0.8090i$
10	Jiazi	yang wood	$\pi/3$	$0.5 + 0.8660i$
11	Yichou	yin wood	$11\pi/30$	$0.4067 + 0.9135i$
12	Bingyin	yang fire	$2\pi/5$	$0.3090 + 0.9511i$
13	Dingmao	yin fire	$13\pi/30$	$0.2079 + 0.9781i$
14	Wuchen	yang earth	$7\pi/15$	$0.1045 + 0.9945i$
15	Jisi	yin earth	$\pi/2$	$0 + 1i$
16	Gengwu	yang metal	$8\pi/15$	$-0.1045 + 0.9945i$

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(continued)

k	Stems and Branches	Five Phases	Phase θ_k	Operator $\mathcal{O}_k^{(60)}$
17	Xinwei	yin metal	$17\pi/30$	$-0.2079 + 0.9781i$
18	Renshen	yang water	$3\pi/5$	$-0.3090 + 0.9511i$
19	Guiyou	yin water	$19\pi/30$	$-0.4067 + 0.9135i$
20	Jiaxu	yang wood	$2\pi/3$	$-0.5 + 0.8660i$
21	Yihai	yin wood	$7\pi/10$	$-0.5878 + 0.8090i$
22	Bingzi	yang fire	$11\pi/15$	$-0.6691 + 0.7431i$
23	Dingchou	yin fire	$23\pi/30$	$-0.7431 + 0.6691i$
24	Wuyin	yang earth	$4\pi/5$	$-0.8090 + 0.5878i$
25	Jimao	yin earth	$5\pi/6$	$-0.8660 + 0.5i$
26	Gengchen	yang metal	$13\pi/15$	$-0.9135 + 0.4067i$
27	Xinsi	yin metal	$9\pi/10$	$-0.9511 + 0.3090i$
28	Renwu	yang water	$14\pi/15$	$-0.9781 + 0.2079i$
29	Guiwei	yin water	$29\pi/30$	$-0.9945 + 0.1045i$
30	Jiashen	yang wood	π	$-1 + 0i$
31	Yiyou	yin wood	$31\pi/30$	$-0.9945 - 0.1045i$
32	Bingxu	yang fire	$16\pi/15$	$-0.9781 - 0.2079i$
33	Dinghai	yin fire	$11\pi/10$	$-0.9511 - 0.3090i$
34	Wuzi	yang earth	$17\pi/15$	$-0.9135 - 0.4067i$
35	Jichou	yin earth	$7\pi/6$	$-0.8660 - 0.5i$
36	Gengyin	yang metal	$6\pi/5$	$-0.8090 - 0.5878i$
37	Xinmao	yin metal	$37\pi/30$	$-0.7431 - 0.6691i$
38	Renchen	yang water	$19\pi/15$	$-0.6691 - 0.7431i$

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k	Stems and Branches	Five Phases	Phase θ_k	Operator $\mathcal{O}_k^{(60)}$
39	Guisi	yin water	$13\pi/10$	$-0.5878 - 0.9511i$
40	Jiawu	yang wood	$4\pi/3$	$-0.5 - 0.8660i$
41	Yiwei	yin wood	$41\pi/30$	$-0.4067 - 0.9135i$
42	Bingshen	yang fire	$7\pi/5$	$-0.3090 - 0.9511i$
43	Dingyou	yin fire	$43\pi/30$	$-0.2079 - 0.9781i$
44	Wuxu	yang earth	$22\pi/15$	$-0.1045 - 0.9945i$
45	Jihai	yin earth	$3\pi/2$	$0 - 1i$
46	Gengzi	yang metal	$23\pi/15$	$0.1045 - 0.9945i$
47	Xinchou	yin metal	$47\pi/30$	$0.2079 - 0.9781i$
48	Renyin	yang water	$8\pi/5$	$0.3090 - 0.9511i$
49	Guimao	yin water	$49\pi/30$	$0.4067 - 0.9135i$
50	Jiachen	yang wood	$5\pi/3$	$0.5 - 0.8660i$
51	Yisi	yin wood	$17\pi/10$	$0.5878 - 0.8090i$
52	Bingwu	yang fire	$26\pi/15$	$0.6691 - 0.7431i$
53	Dingwei	yin fire	$53\pi/30$	$0.7431 - 0.6691i$
54	Wushen	yang earth	$9\pi/5$	$0.8090 - 0.5878i$
55	Jiyou	yin earth	$11\pi/6$	$0.8660 - 0.5i$
56	Gengxu	yang metal	$28\pi/15$	$0.9135 - 0.4067i$
57	Xinhai	yin metal	$19\pi/10$	$0.9511 - 0.3090i$
58	Renzi	yang water	$29\pi/15$	$0.9781 - 0.2079i$
59	Guichou	yin water (end)	$59\pi/30$	$0.9945 - 0.1045i$

30.7.13 The Sixty-Four Hexagrams ($T = 64$)

Definition 30.32 (Sixty-Four-Hexagrams phase operators)**.** The period of the Sixty-Four Hexagrams is $T = 64 = 8 \times 8$, the bit width is $W = 64$ and the bandwidth is $\Delta\theta = \pi/32$. The group structure is $C_8 \times C_8$ and the tensor shape is 8×8. They are generated by nested epicycles of the Eight Trigrams × the Eight Trigrams.

k	Lower trigram	Upper trigram	Hexagram name	Five Phases	Phase θ_k	Operator $\mathcal{O}_k^{(64)}$
0	Zhen ☳	Zhen ☳	Zhen (Thunder)	wood (beginning)	0	$1 + 0i$
1	Xun ☴	Zhen ☳	Heng (Duration)	wood	$\pi/32$	$0.9952 + 0.0980i$
2	Gen ☶	Zhen ☳	Xiaoguo (Small Preponderance)	earth	$\pi/16$	$0.9808 + 0.1951i$
3	Li ☲	Zhen ☳	Feng (Abundance)	fire	$3\pi/32$	$0.9569 + 0.2903i$
4	Qian ☰	Zhen ☳	Dazhuang (Great Power)	metal	$\pi/8$	$0.9239 + 0.3827i$
5	Kun ☷	Zhen ☳	Yu (Enthusiasm)	earth	$5\pi/32$	$0.8819 + 0.4714i$
6	Dui ☱	Zhen ☳	Guimei (The Marrying Maiden)	metal	$3\pi/16$	$0.8315 + 0.5556i$
7	Kan ☵	Zhen ☳	Xie (Deliverance)	water	$7\pi/32$	$0.7730 + 0.6344i$
8	Zhen ☳	Xun ☴	Yi (Increase)	wood	$\pi/4$	$0.7071 + 0.7071i$
9	Xun ☴	Xun ☴	Xun (Wind)	wood	$9\pi/32$	$0.6344 + 0.7730i$
10	Gen ☶	Xun ☴	Jian (Gradual Progress)	earth	$5\pi/16$	$0.5556 + 0.8315i$
11	Li ☲	Xun ☴	Jiaren (The Family)	fire	$11\pi/32$	$0.4714 + 0.8819i$

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k	Lower trigram	Upper trigram	Hexagram name	Five Phases	Phase θ_k	Operator $\mathcal{O}_k^{(64)}$
12	Qian ☰	Xun ☴	Xiaochu (The Taming Power of the Small)	metal	$3\pi/8$	$0.3827 + 0.9239i$
13	Kun ☷	Xun ☴	Guan (Contemplation)	earth	$13\pi/32$	$0.2903 + 0.9569i$
14	Dui ☱	Xun ☴	Zhongfu (Inner Truth)	metal	$7\pi/16$	$0.1951 + 0.9808i$
15	Kan ☵	Xun ☴	Huan (Dispersion)	water	$15\pi/32$	$0.0980 + 0.9952i$
16	Zhen ☳	Gen ☶	Yi (Nourishment)	wood	$\pi/2$	$0 + 1i$
17	Xun ☴	Gen ☶	Gu (Work on What Has Been Spoiled)	wood	$17\pi/32$	$-0.0980 + 0.9952i$
18	Gen ☶	Gen ☶	Gen (Mountain)	earth	$9\pi/16$	$-0.1951 + 0.9808i$
19	Li ☲	Gen ☶	Bi (Grace)	fire	$19\pi/32$	$-0.2903 + 0.9569i$
20	Qian ☰	Gen ☶	Dachu (The Taming Power of the Great)	metal	$5\pi/8$	$-0.3827 + 0.9239i$
21	Kun ☷	Gen ☶	Bo (Splitting Apart)	earth	$21\pi/32$	$-0.4714 + 0.8819i$
22	Dui ☱	Gen ☶	Sun (Decrease)	metal	$11\pi/16$	$-0.5556 + 0.8315i$

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k	Lower trigram	Upper trigram	Hexagram name	Five Phases	Phase θ_k	Operator $\mathcal{O}_k^{(64)}$
23	Kan ☵	Gen ☶	Meng (Youthful Folly)	water	$23\pi/32$	$-0.6344 + 0.7730i$
24	Zhen ☳	Li ☲	Shihe (Biting Through)	wood	$3\pi/4$	$-0.7071 + 0.7071i$
25	Xun ☴	Li ☲	Ding (The Cauldron)	wood	$25\pi/32$	$-0.7730 + 0.6344i$
26	Gen ☶	Li ☲	Lü (The Wanderer)	earth	$13\pi/16$	$-0.8315 + 0.5556i$
27	Li ☲	Li ☲	Li (Fire)	fire	$27\pi/32$	$-0.8819 + 0.4714i$
28	Qian ☰	Li ☲	Dayou (Great Possession)	metal	$7\pi/8$	$-0.9239 + 0.3827i$
29	Kun ☷	Li ☲	Jin (Progress)	earth	$29\pi/32$	$-0.9569 + 0.2903i$
30	Dui ☱	Li ☲	Kui (Opposition)	metal	$15\pi/16$	$-0.9808 + 0.1951i$
31	Kan ☵	Li ☲	Weiji (Before Completion)	water	$31\pi/32$	$-0.9952 + 0.0980i$
32	Zhen ☳	Qian ☰	Wuwang (Innocence)	wood	π	$-1 + 0i$
33	Xun ☴	Qian ☰	Gou (Coming to Meet)	wood	$33\pi/32$	$-0.9952 - 0.0980i$
34	Gen ☶	Qian ☰	Dun (Retreat)	earth	$17\pi/16$	$-0.9808 - 0.1951i$
35	Li ☲	Qian ☰	Tongren (Fellowship with Men)	fire	$35\pi/32$	$-0.9569 - 0.2903i$
36	Qian ☰	Qian ☰	Qian (Heaven)	metal	$9\pi/8$	$-0.9239 - 0.3827i$

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k	Lower trigram	Upper trigram	Hexagram name	Five Phases	Phase θ_k	Operator $\mathcal{O}_k^{(64)}$
37	Kun ☷	Qian ☰	Pi (Standstill)	earth	$37\pi/32$	$-0.8819 - 0.4714i$
38	Dui ☱	Qian ☰	Lü (Treading)	metal	$19\pi/16$	$-0.8315 - 0.5556i$
39	Kan ☵	Qian ☰	Song (Conflict)	water	$39\pi/32$	$-0.7730 - 0.6344i$
40	Zhen ☳	Kun ☷	Fu (Return)	wood	$5\pi/4$	$-0.7071 - 0.7071i$
41	Xun ☴	Kun ☷	Sheng (Pushing Upward)	wood	$41\pi/32$	$-0.6344 - 0.7730i$
42	Gen ☶	Kun ☷	Qian (Modesty)	earth	$21\pi/16$	$-0.5556 - 0.8315i$
43	Li ☲	Kun ☷	Mingyi (Darkening of the Light)	fire	$43\pi/32$	$-0.4714 - 0.8819i$
44	Qian ☰	Kun ☷	Tai (Peace)	metal	$11\pi/8$	$-0.3827 - 0.9239i$
45	Kun ☷	Kun ☷	Kun (Earth)	earth	$45\pi/32$	$-0.2903 - 0.9569i$
46	Dui ☱	Kun ☷	Lin (Approach)	metal	$23\pi/16$	$-0.1951 - 0.9808i$
47	Kan ☵	Kun ☷	Shi (The Army)	water	$47\pi/32$	$-0.0980 - 0.9952i$
48	Zhen ☳	Dui ☱	Sui (Following)	wood	$3\pi/2$	$0 - 1i$
49	Xun ☴	Dui ☱	Daguo (Great Preponderance)	wood	$49\pi/32$	$0.0980 - 0.9952i$
50	Gen ☶	Dui ☱	Xian (Influence)	earth	$25\pi/16$	$0.1951 - 0.9808i$
51	Li ☲	Dui ☱	Ge (Revolution)	fire	$51\pi/32$	$0.2903 - 0.9569i$

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(continued)

k	Lower trigram	Upper trigram	Hexagram name	Five Phases	Phase θ_k	Operator $\mathcal{O}_k^{(64)}$
52	Qian ☰	Dui ☱	Guai (Breakthrough)	metal	$13\pi/8$	$0.3827 - 0.9239i$
53	Kun ☷	Dui ☱	Cui (Gathering Together)	earth	$53\pi/32$	$0.4714 - 0.8819i$
54	Dui ☱	Dui ☱	Dui (Lake)	metal	$27\pi/16$	$0.5556 - 0.8315i$
55	Kan ☵	Dui ☱	Kun (Oppression)	water	$55\pi/32$	$0.6344 - 0.7730i$
56	Zhen ☳	Kan ☵	Zhun (Difficulty at the Beginning)	wood	$7\pi/4$	$0.7071 - 0.7071i$
57	Xun ☴	Kan ☵	Jing (The Well)	wood	$57\pi/32$	$0.7730 - 0.6344i$
58	Gen ☶	Kan ☵	Jian (Obstruction)	earth	$29\pi/16$	$0.8315 - 0.5556i$
59	Li ☲	Kan ☵	Jiji (After Completion)	fire	$59\pi/32$	$0.8819 - 0.4714i$
60	Qian ☰	Kan ☵	Xu (Waiting)	metal	$15\pi/8$	$0.9239 - 0.3827i$
61	Kun ☷	Kan ☵	Bi (Holding Together)	earth	$61\pi/32$	$0.9569 - 0.2903i$
62	Dui ☱	Kan ☵	Jie (Limitation)	metal	$31\pi/16$	$0.9808 - 0.1951i$
63	Kan ☵	Kan ☵	Kan (Water)	water (end)	$63\pi/32$	$0.9952 - 0.0980i$

30.8 Application of periodic convolution in the I Ching symbol system

30.9 Master table of the tensor structure of the periodic systems

Symbol system	T	Prime factorisation	Group structure	Tensor shape	Nested generation
Taiji	1	1	C_1	1	basic
Yin-Yang	2	2	C_2	2	basic
Sancai	3	3	C_3	3	basic
the Four Symbols	4	2^2	C_4	4	Yin-Yang × Yin-Yang
the Five Phases	5	5	C_5	5	basic
the Seven Luminaries	7	7	C_7	7	basic
the Eight Trigrams	8	2^3	C_8	8	Yin-Yang × the Four Symbols
the Heavenly Stems	10	2×5	$C_2 \times C_5$	2×5	Yin-Yang × the Five Phases
the Earthly Branches	12	4×3	$C_4 \times C_3$	4×3	the Four Symbols × Sancai
the twenty-four solar terms	24	8×3	$C_8 \times C_3$	8×3	the Eight Trigrams × Sancai
the thirty Na-Yin	30	$2 \times 3 \times 5$	$C_2 \times C_3 \times C_5$	$2 \times 3 \times 5$	Yin-Yang × Sancai × the Five Phases
the sexagenary cycle	60	$4 \times 3 \times 5$	$C_4 \times C_3 \times C_5$	$4 \times 3 \times 5$	the Heavenly Stems ⊛ the Earthly Branches
the Sixty-Four Hexagrams	64	8×8	$C_8 \times C_8$	8×8	the Eight Trigrams × the Eight Trigrams

Remark 30.18 (Mathematical positioning of the two tables). The 'table of phase operators of the thirteen symbol systems' is the period-by-period instantiation of

definition 30.11 and definition 30.14: the correspondence of the phase operator $\mathcal{O}_k^{(T)} = e^{i2\pi k/T}$ of each T with its symbol, its Five-Phase assignment and its tensor index is the complete element table of the cyclic group C_T of that period. The 'master table of the tensor structure of the periodic systems' collects the prime factorisation, the group structure, the tensor shape and the source of nested generation of the thirteen symbol systems, and is a systematic index of theorem 30.8 and proposition 30.7. Neither table carries an independent theorem; their mathematical positioning is to render the group-theoretic and representation-theoretic conclusions of the main text as a symbol correspondence that can be looked up in a table.

30.10 The major-luck cycle: approximate replication of the stems and branches on the ten-year scale

Remark 30.19 (Positioning of this section: connection to the five-step interface of the 'Periodic Systems' chapter). The major-luck cycle (dayun) is a ten-year-per-step sequence used in traditional fate calculation to characterise an individual's long-term fortunes. This section does not treat it as a new astronomical period; rather it characterises it as an **approximate replication, on the ten-year scale, of the sexagenary structure already established in this chapter.** Specifically it connects to the five-step interface of the 'Periodic Systems' chapter:

(1) **Decision**: reduce the major-luck sequence to a discrete dynamical system on C_{60}, cite Theorem 24.1 to decide its period type;

(2) **Construction**: from the major-luck stem–branch sequence construct the amplitude series, the frequency series and the phase series, and assemble them into the major-luck epicycle group;

(3) **Adjustment**: dynamically adjust the major-luck epicycle group by the starting age, the choice of direction and the decrease of the residual energy;

(4) **Convolution**: introduce amplitude convolution, phase convolution and frequency convolution on the major-luck epicycle group, and connect them with the periodic convolution framework of this chapter;

(5) **Integration**: connect to the tool chains of the three chapters 'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis', forming a major-luck periodicity diagnostic procedure.

This section uses tools already established in this chapter: the periodic phase operator $\mathcal{O}_k^{(T)}$ (Definition 30.11), the nested epicycle generation rule (Definition 30.13) and the periodic-convolution generation of the sexagenary cycle (Theorem 30.5). It also cites the conclusions of the calendar chapter on the solar-term phase (Definition 28.16) and on fixed length (Definition 28.7).

Remark 30.20 (Guide to the seven themes of this section). Translation—the ordering advance of the major-luck sequence along the sexagenary cycle (Definition 30.36); Scaling—the tenfold replication of scale in the major-luck cycle (10 years per step, Theorem 30.11); Rotation—the direction $\sigma = \pm 1$ and the precession of the relative phase with respect to the current year (Proposition 30.17); Conservation—the threefold conservation of the year-stem yin-yang ϵ, the step length L and the month pillar M_0 (Theorem 30.14); Symmetry—the directional duality between yang males and yin females and between yin males and yang females (Proposition 30.15); Closure—the closure of the major-luck sequence with the period 60 steps (600 years) (Theorem 30.13); Period—the ten-year step, the hundred-year Heavenly-Stem period, the hundred-and-twenty-year Earthly-Branch period and the six-hundred-year stem–branch period (Theorem 30.11).

30.10.1 The stems and branches as a basic mechanism of rational approximation

Definition 30.33 (Heavenly-Stem period and Earthly-Branch period)**.** The **Heavenly Stems** are ten symbols, with period $T_{\text{stem}} = 10$; the **Earthly Branches** are twelve symbols, with period $T_{\text{branch}} = 12$. The **stems and branches** are the synchronous pairing of the Heavenly Stems with the Earthly Branches, and their period is

$$T_{\text{stem-branch}} = \mathrm{lcm}(T_{\text{stem}}, T_{\text{branch}}) = \mathrm{lcm}(10, 12) = 60.$$

It is written as the cyclic group $C_{60} = C_{10} \times C_{12} / \sim$, where the equivalence relation $\sim$ is given by the parity-synchronisation condition $(a, b) \sim (a', b')$ if and only if $a - b \equiv a' - b' \pmod 2$.

Proposition 30.11 (The stems and branches as the fibre product of C_{10} and C_{12})**.** The cyclic group C_{60} is isomorphic to the fibre product of C_{10} and C_{12} under the parity projections:

$$C_{60} \cong \{(a, b) \in C_{10} \times C_{12} \mid a \equiv b \pmod 2\}.$$

Proof. By the generalisation of the Chinese remainder theorem, $C_{10} \times C_{12}$ has order 120. The parity projections $C_{10} \to C_2$ and $C_{12} \to C_2$ are both surjective; the fibre product therefore has size $|C_{10} \times C_{12}|/|C_2| = 60$. The map from C_{60} to the fibre product, $n \mapsto (n \bmod 10, n \bmod 12)$, is a group homomorphism and a bijection, hence an isomorphism. ☐

Definition 30.34 (Empirical values of the astronomical periods)**.** The following values of the astronomical periods come from modern observation (in years):

$$\begin{aligned} T_J &\approx 11.862 && \text{(Jupiter's orbital period)},\\ T_S &\approx 29.457 && \text{(Saturn's orbital period)},\\ T_{JS} &= \frac{T_J T_S}{T_S - T_J} \approx 19.859 && \text{(Jupiter–Saturn conjunction)}. \end{aligned}$$

Proposition 30.12 (Approximation of the year stems and branches to Jupiter)**.** The 60-year stem–branch period corresponds to about 5 revolutions of Jupiter:

$$\frac{60}{T_J} = \frac{60}{11.862} \approx 5.058 \approx 5,$$

with a relative error of about 1.2%. This approximation is the structural origin of the **Jupiter year-reckoning system** (in which Jupiter takes about 12 years to complete one circuit of the heavens): the Earthly-Branch period 12 is an integer approximation to the period of Jupiter.

Proof. $60/11.862 = 5.058$, whose relative error with respect to the integer 5 is $(5.058 - 5)/5 = 0.0116 \approx 1.2\%$. The relative error between the Earthly-Branch period 12 and the period of Jupiter $T_J \approx 11.862$ is 1.1%, of the same order. ☐

Proposition 30.13 (Approximation of the year stems and branches to the Jupiter–Saturn conjunction)**.** The 60-year stem–branch period corresponds to about 3 Jupiter–Saturn conjunctions:

$$\frac{60}{T_{JS}} = \frac{60}{19.859} \approx 3.021 \approx 3,$$

with a relative error of about 0.7%.

Proof. $60/19.859 = 3.021$, whose relative error with respect to the integer 3 is $0.021/3 \approx 0.7\%$. ☐

Remark 30.21 (The year stems and branches as a double approximation). Proposition 30.12 and Proposition 30.13 reveal that the 60-year stem–branch period is a double rational approximation: to 5 **revolutions of Jupiter** and to 3 **Jupiter–Saturn conjunctions**. These two approximations correspond exactly to the composition of $T_{\text{stem}} = 10$ and $T_{\text{branch}} = 12$:

- $T_{\text{branch}} = 12 \approx T_J$ (the period of Jupiter), and Jupiter completes about 5 revolutions within 60 years;
- $T_{\text{stem-branch}} = 60 \approx 3\,T_{JS}$ (3 Jupiter–Saturn conjunctions).

The role of the Heavenly-Stem period $T_{\text{stem}} = 10$ is to modulate the single period $T_{\text{branch}} = 12$, so that the fibre product of $C_{10} \times C_{12}$ yields a composite rhythm of 60 years. This structure corresponds algebraically, one to one, with Theorem 30.5 of this chapter (the periodic-convolution generation of the sexagenary cycle).

30.10.2 The constituents of the major-luck cycle and its tenfold replication of scale

Definition 30.35 (Major-luck target and decision input)**.** A **major-luck target** is characterised by a quadruple:

$$\mathcal{D}_{\text{D}} = (M_0, \sigma, a_0, L), \qquad L = 10 \text{ years},$$

where $M_0 \in C_{60}$ is the month pillar (the starting point), $\sigma \in \{+1, -1\}$ is the direction, $a_0 \geq 0$ is the starting age of the cycle and $L = 10$ is the step length. The state space is the cyclic group $\mathcal{S} = C_{60}$, and the evolution map is

$$\Phi : \mathcal{S} \to \mathcal{S}, \qquad \Phi(m) = m + \sigma \pmod{60}.$$

Definition 30.36 (Major-luck sequence)**.** Let the decision input of the major-luck target be $\mathcal{D}_{\text{D}} = (M_0, \sigma, a_0, L)$. The **major-luck sequence** is defined by

$$M_k = M_0 + k\sigma \pmod{60}, \qquad k = 0, 1, 2, \dots$$

The k-th major-luck step corresponds to the time interval $[a_0 + kL,\ a_0 + (k+1)L)$.

Theorem 30.11 (The three fundamental periods of the major-luck cycle)**.** The major-luck sequence has three fundamental periods:

(i) **Heavenly-Stem period**: $T^{(\mathsf{D})}_{\mathsf{stem}} = T_{\mathsf{stem}} \cdot L = 10 \cdot 10 = 100$ years;

(ii) **Earthly-Branch period**: $T^{(\mathsf{D})}_{\mathsf{branch}} = T_{\mathsf{branch}} \cdot L = 12 \cdot 10 = 120$ years;

(iii) **Stem–branch period**: $T^{(\mathsf{D})}_{\mathsf{stem\text{-}branch}} = T_{\mathsf{stem\text{-}branch}} \cdot L = 60 \cdot 10 = 600$ years.

Here the step length $L = 10$ years is exactly equal to the Heavenly-Stem period, which makes the major-luck cycle a **replication of the year stems and branches on the ten-year scale**.

Proof. The major-luck Heavenly Stems have 10 positions, each corresponding to one step, so the number of steps needed for the Heavenly Stems to return to their place is 10, that is, $10 \times 10 = 100$ years. Likewise, the Earthly Branches need 12 steps to return to their place, that is, 120 years. The stems and branches together need $\mathrm{lcm}(10, 12) = 60$ steps to return to their place, that is, 600 years. ☐

Proposition 30.14 (Periodicity decision theorem for the major-luck cycle)**.** Let the decision map of the major-luck target be $\Phi(m) = m + \sigma \pmod{60}$. Then:

(A) **Sufficient decision**: $\Phi^{60} = \mathrm{id}$, so the target is a uniformly periodic major-luck system;

(B) **Equivalence of the decision**: $\mathrm{ord}(\Phi) = 60$;

(C) **Pointwise criterion**: the orbit of the major-luck sequence at M_0 has length 60;

(D) **Phase independence**: the starting age a_0 changes only the starting position of the sequence, not the period.

Proof. (A) $\Phi^{60}(m) = m + 60\sigma \equiv m \pmod{60}$. (B) If $\mathrm{ord}(\Phi) = k < 60$, then $60 \mid k$ (since $\gcd(\sigma, 60) = 1$), a contradiction. (C) By (B) and Theorem 1.5 of the 'Operations and Periodicity' chapter. (D) By Definition 30.36, M_k does not depend on a_0. ☐

Corollary 30.4 (The triple astronomical approximation of the major-luck cycle)**.** The three fundamental periods of the major-luck cycle approximate the following astronomical rhythms respectively:

(i) **the Heavenly-Stem period of** 100 **years** $\approx$ 5 Jupiter–Saturn conjunctions:

$$\frac{100}{T_{JS}} = \frac{100}{19.859} \approx 5.036 \approx 5, \quad \text{relative error } 0.7\%;$$

(ii) **the Earthly-Branch period of** 120 **years** $\approx$ 10 revolutions of Jupiter:

$$\frac{120}{T_J} = \frac{120}{11.862} \approx 10.12 \approx 10, \quad \text{relative error } 1.2\%;$$

(iii) **the stem–branch period of 600 years** ≈ 50 revolutions of Jupiter ≈ 30 Jupiter–Saturn conjunctions:

$$\frac{600}{T_J} \approx 50.6 \approx 50, \qquad \frac{600}{T_{JS}} \approx 30.2 \approx 30,$$

with relative errors of 1.2% and 0.7% respectively.

Proof. Term by term: (i) $100/19.859 = 5.036$, error $0.036/5 = 0.7\%$. (ii) $120/11.862 = 10.12$, error $0.12/10 = 1.2\%$. (iii) $600/11.862 = 50.58$, error $0.58/50 = 1.2\%$; $600/19.859 = 30.21$, error $0.21/30 = 0.7\%$. ☐

Theorem 30.12 (Duality of scale between the major-luck cycle and the year stems and branches)**.** *The major-luck cycle and the year stems and branches form a duality of scale:*

Scale	Heavenly-Stem per
Year stems and branches	10
Major-luck stems and branches	100
Astronomical approximation of the year stems and branches	Jupiter–Saturn conjunctions
Astronomical approximation of the major-luck cycle	Jupiter–Saturn conjunctions ×

The two columns are completely parallel in structure: each 10-year step of the major-luck cycle is precisely a replication of the Heavenly-Stem period $T_{\text{stem}} = 10$ of the year stems and branches. This structure has the same origin as the periodic-convolution generation of the sexagenary cycle in Theorem 30.5 of this chapter.

Proof. The periods of the year stems and branches are $(T_{\text{stem}}, T_{\text{branch}}, T_{\text{stem-branch}}) = (10, 12, 60)$, see Definition 30.33. The major-luck cycle unfolds with step length $L = T_{\text{stem}} = 10$ years, so each of the three periods is multiplied by 10, giving $(100, 120, 600)$. The year stems and branches approximate $(T_J \times 5,\ T_{JS} \times 3) = (59.3, 59.6)$, whilst the major-luck stems and branches approximate $(T_J \times 10,\ T_{JS} \times 5) = (118.6, 99.3)$; the two are of the same order of accuracy (a relative error of the order of 1%). ☐

Example 30.8 (Numerical verification of the 120-year major-luck Earthly-Branch period against 10 revolutions of Jupiter)**.** $10 \times T_J = 10 \times 11.862 = 118.62$ years. The difference between the 120-year major-luck Earthly-Branch period and 118.62 years is 1.38 years, a relative error of

$$\frac{1.38}{120} \approx 1.15\%.$$

This is of exactly the same order as the relative error of 1.2% with which the 12-year year-stem-and-branch Earthly-Branch period approximates Jupiter's 11.862 years.

Example 30.9 (Numerical verification of the 100-year major-luck Heavenly-Stem period against 5 Jupiter–Saturn conjunctions)**.** $5 \times T_{JS} = 5 \times 19.859 = 99.30$ years. The difference between the 100-year major-luck Heavenly-Stem period and 99.30 years is 0.70 years, a relative error of

$$\frac{0.70}{100} = 0.70\%.$$

30.10.3 The binary character of the major-luck direction

Definition 30.37 (Year-stem yin-yang and sex)**.** Let the index of the year-pillar Heavenly Stem be $\epsilon_{\text{stem}} \in \{0, 1, \dots, 9\}$ (Jia $= 0$, Yi $= 1$, $\dots$, Gui $= 9$). Define the **yin-yang marker** of the year stem as

$$\epsilon = \begin{cases} 1, & \epsilon_{\text{stem}} \in \{0,2,4,6,8\} \text{ (yang)}, \\ 0, & \epsilon_{\text{stem}} \in \{1,3,5,7,9\} \text{ (yin)}. \end{cases}$$

Define the **sex marker** $\eta \in \{0, 1\}$ (male $= 1$, female $= 0$).

Definition 30.38 (Major-luck direction)**.** The **direction** $\sigma \in \{+1, -1\}$ of the major-luck cycle is determined by (ϵ, η):

$$\sigma = 2 \cdot \mathbb{1}_{\{\epsilon=\eta\}} - 1 = \begin{cases} +1, & \epsilon = \eta \text{ (yang male, yin female; forward ordering)}, \\ -1, & \epsilon \neq \eta \text{ (yin male, yang female; reverse ordering)}. \end{cases}$$

Proposition 30.15 (Duality of the major-luck direction)**.** The major-luck direction satisfies:

(i) $\sigma(\epsilon, 1-\eta) = -\sigma(\epsilon, \eta)$;

(ii) $\sigma(1-\epsilon, \eta) = -\sigma(\epsilon, \eta)$;

(iii) $\sigma(1-\epsilon, 1-\eta) = \sigma(\epsilon, \eta)$.

Proof. (i) If $\epsilon = \eta$, then $\epsilon \neq 1-\eta$, giving $\sigma = -1 = -\sigma(\epsilon, \eta)$; if $\epsilon \neq \eta$, then $\epsilon = 1 - \eta$, giving $\sigma = +1 = -\sigma(\epsilon, \eta)$. (ii) and (iii) are analogous. ☐

Corollary 30.5 (Equivalence classes of the major-luck direction)**.** The four cases $(\epsilon, \eta) \in \{0,1\}^2$ fall under σ into two equivalence classes:

$$\{(1,1),(0,0)\} \xrightarrow{\sigma} +1, \qquad \{(1,0),(0,1)\} \xrightarrow{\sigma} -1.$$

The two equivalence classes are each closed under the composite duality $(\epsilon, \eta) \mapsto (1-\epsilon, 1-\eta)$.

Proof. This follows directly from Definition 30.38 and Proposition 30.15(iii). ☐

30.10.4 Quantification of the starting age of the cycle

Definition 30.39 (Starting age of the cycle)**.** Let an individual be born at the instant t_{birth}. Let Δd be

$$\Delta d = \begin{cases} t_{\text{next}} - t_{\text{birth}}, & \sigma = +1, \\ t_{\text{birth}} - t_{\text{prev}}, & \sigma = -1, \end{cases}$$

where t_{prev} and t_{next} are the instants of the nearest solar terms before and after the birth (for the definition of a solar term see Definition 28.16 of the calendar chapter). The **starting age of the cycle** is

$$a_0 = \frac{\Delta d}{3} \quad \text{(years)},$$

according to the traditional quantification rule 3 days $= 1$ year, 1 day $= 4$ months, 1 double-hour $= 10$ days.

Proposition 30.16 (Quantitative bound on the starting age of the cycle)**.** *Let the time interval between two adjacent solar terms be ΔT_{jie}. Then*

$$0 \le a_0 \le \frac{\Delta T_{\text{jie}}}{3}.$$

Under the mean-solar-term method $\Delta T_{\text{jie}} = T_{\text{trop}}/24 \approx 15.2184$ days (Definition 28.7 of the calendar chapter), hence $a_0 \le 5.07$ years.

Proof. By Definition 30.39, $0 \le \Delta d \le \Delta T_{\text{jie}}$. Substituting into $a_0 = \Delta d/3$ gives the result. ☐

Remark 30.22 (Phase interpretation of the starting age of the cycle)*.* The starting age $a_0 = \Delta d/3$ is a **phase offset**—it measures the relative position of the instant of birth within the solar-term cycle. The value a_0 does not change the periodicity of the major-luck sequence (Proposition 30.14(D)); it changes only the translation of the starting point of the sequence along the time axis. This is consistent with the phase-encoding framework of the periodic phase operator $\mathcal{O}_k^{(T)} = e^{i2\pi k/T}$ of this chapter (Definition 30.11).

Example 30.10 (Numerical computation of the starting age of the cycle)**.** Let an individual be born 9 days after Lichun, with $\sigma = +1$. Then $a_0 = 9/3 = 3$ years, and the major-luck sequence begins at the age of 3.

30.10.5 The periodic structure of the major-luck sequence and its astronomical meaning

Theorem 30.13 (Periodic structure of the major-luck sequence)**.** *Let the major-luck sequence be $M_k = M_0 + k\sigma \pmod{60}$. Then:*

(i) **Algebraic period**: $M_{k+60} = M_k$ *holds for all* $k \in \mathbb{Z}$;

(ii) **Time period**: $60 \times 10 = 600$ *years;*

(iii) **Single orbit**: C_{60} *consists of a single orbit under the major-luck translation group* $\langle \Phi \rangle$;

(iv) **No conserved quantities**: *every invariant on C_{60} with respect to the major-luck translation must be constant.*

Proof. (i) By $\Phi^{60} = \text{id}$ (Proposition 30.14(A)). (ii) Each step is 10 years, so 60 steps are 600 years. (iii) C_{60} consists of a single orbit under Φ. (iv) By Theorem 1.7 of the 'Operations and Periodicity' chapter, in a single-orbit system a conserved quantity must be constant. ☐

Remark 30.23 (The 600-year period as the joint locking of the astronomical approximations)*.* By Theorem 30.11 and Corollary 30.4, the 600-year major-luck period simultaneously approximates:

- 50 revolutions of Jupiter: $50 \times 11.862 = 593.1$ years, a relative error of 1.2%;
- 20 revolutions of Saturn: $20 \times 29.457 = 589.14$ years, a relative error of 1.8%;

- 30 Jupiter–Saturn conjunctions: $30 \times 19.859 = 595.77$ years, a relative error of 0.7%.

These three astronomical rhythms are approximated simultaneously on the 600-year scale—this is the deeper astronomical meaning of the periodic structure of the major-luck cycle. It explains why the seemingly artificial construction of 60 steps × 10 years possesses long-term stability and empirical value within the traditional calendrical system.

Example 30.11 (An instance of the periodic closure of the major-luck sequence). Let M_0 = 'Jiayin' (index 0) and $\sigma = +1$. The 60-th major-luck step is $M_{60} = M_0 + 60 \equiv 0 \pmod{60}$, returning to Jiayin. The sequence closes after 600 years. This is consistent with the periodic closure of the sexagenary cycle of this chapter (Theorem 30.5).

30.10.6 The relative phase between the major-luck cycle and the current year

Definition 30.40 (Current year). The **current year** (liunian) refers to the year pillar of a specific year. Let the starting age of the cycle be a_0; then the current year in year t $(t \geq a_0)$ is

$$N_t = N_{a_0} + (t - a_0) \pmod{60}.$$

Proposition 30.17 (Relative phase between the major-luck cycle and the current year). Let N_{a_0} be the year pillar of the year in which the cycle starts and M_0 the month pillar. Then the relative phase between the k-th major-luck step and the current year is

$$\Delta_k = N_{a_0+kL} - M_k \equiv (N_{a_0} - M_0) + k(L - \sigma) \pmod{60},$$

where $L = 10$. In particular:

(i) **Forward ordering** $(\sigma = +1)$: $\Delta_k \equiv \Delta_0 + 9k \pmod{60}$;

(ii) **Reverse ordering** $(\sigma = -1)$: $\Delta_k \equiv \Delta_0 + 11k \pmod{60}$.

Proof. The starting point of the k-th major-luck step corresponds to the age $a_0 + kL$, and the current year is $N_{a_0+kL} = N_{a_0} + kL \pmod{60}$. Subtraction gives $\Delta_k \equiv N_{a_0} - M_0 + k(L - \sigma) \pmod{60}$. Substituting $L = 10$ and $\sigma = \pm 1$ gives the result. ☐

Corollary 30.6 (Period of the relative phase between the major-luck cycle and the current year). The period of the relative phase Δ_k is $T_\Delta = 60/\gcd(L - \sigma, 60)$. For $\sigma = +1$ $(L - \sigma = 9,\ \gcd(9, 60) = 3)$, $T_\Delta = 20$ steps; for $\sigma = -1$ $(L - \sigma = 11,\ \gcd(11, 60) = 1)$, $T_\Delta = 60$ steps.

Proof. By Proposition 30.17, Δ_k is the translation sequence on C_{60} with step length $(L - \sigma)$, whose period is $60/\gcd(L - \sigma, 60)$ (Lemma 1.2 of the 'Operations and Periodicity' chapter). ☐

Remark 30.24 (Asymmetry of the phase periods of forward and reverse ordering). The major-luck–current-year phase period is 20 steps (200 years) for forward ordering and 60 steps (600 years) for reverse ordering. This asymmetry arises from $\gcd(9, 60) = 3$ whilst $\gcd(11, 60) = 1$—9 and 60 have a common factor, whereas 11 and 60 are coprime. *It explains the structural difference between forward and reverse ordering in traditional current-year coupling.*

Example 30.12 (A concrete computation of the closure of the major-luck–current-year phase). Let M_0 = 'Jiayin' (index 0), N_{a_0} = 'Bingyin' (index 12), $\sigma = +1$ and $a_0 = 3$. Then

$$\Delta_0 \equiv 2 - 50 \equiv 12 \pmod{60}.$$

The relative phase increases in steps of 9: $\Delta_1 = 21$, $\Delta_2 = 30$, $\Delta_3 = 39$, $\Delta_4 = 48$, $\Delta_5 = 57$, $\dots$. By Corollary 30.6, $T_\Delta = 20$, that is, the phase closes after 20 steps (200 years).

30.10.7 Conserved quantities of the major-luck system

Theorem 30.14 (The three invariants of the major-luck system). *The major-luck system has three invariants:*

(i) **Conservation of direction**: *$\sigma \in \{+1, -1\}$ is unchanged throughout the major-luck sequence;*

(ii) **Conservation of step length**: *$L = 10$ years is unchanged throughout the major-luck sequence;*

(iii) **Conservation of the starting point**: *$M_0 \in C_{60}$ is the anchor of the sequence and can be uniquely recovered from it.*

Proof. (i) By Definition 30.38, σ is uniquely determined by (ϵ, η). (ii) By Definition 30.36, $L = 10$ is a fixed parameter. (iii) Given (σ, a_0) and the sequence $\{M_k\}$, the value M_0 is read off directly at $k = 0$. ☐

30.10.8 Construction and adjustment of the major-luck epicycle group

Definition 30.41 (Major-luck amplitude series, frequency series and phase series). Let $\{m_k\}$ be the indices of the major-luck sequence on C_{60}. If

$$m_k = \sum_j A_j e^{i(\omega_j k + \phi_j)} + \varepsilon_k,$$

then $\mathcal{A}_m(k) = \sum_j A_j e^{ijk}$, $\mathcal{F}_m(s) = \sum_j \omega_j s^j$ and $\mathcal{P}_m(k) = \sum_j e^{i(jk+\phi_j)}$ are called the amplitude series, the frequency series and the phase series of the major-luck sequence.

Definition 30.42 (Major-luck epicycle group). The **epicycle group** of the major-luck target is

$$\mathfrak{E}_{\mathsf{D}} = \{(A_j, \omega_j, \phi_j)\}_{j \in \mathbb{Z}},$$

and its superposition is the major-luck period series $m_k = \sum_j A_j e^{i(\omega_j k + \phi_j)}$. This formulation is consistent with the nested epicycle generation rule of this chapter (Definition 30.13).

Proposition 30.18 (Fourier expansion of the major-luck sequence)**.** Let the major-luck sequence take index values on C_{60}. Then its 60-point DFT is

$$\hat{m}_j = \frac{1}{60}\sum_{k=0}^{59} m_k e^{-i2\pi jk/60}, \qquad j = 0, 1, \ldots, 59,$$

and $m_k = \sum_{j=0}^{59} \hat{m}_j e^{i2\pi jk/60}$.

Proof. By Theorem 30.13(i), $\{m_k\}$ is a finite sequence of period 60. By the standard result on discrete Fourier series, the 60-point DFT gives the above decomposition. □

Definition 30.43 (Dynamic adjustment of the major-luck epicycle group)**.** Let the current epicycle group be $\mathfrak{E}^{(j)}$ and let the residual be $r^{(j)} = m - m^{(j)}$. The dynamic adjustment consists of:

(i) **Cardinality adjustment**: if $\|r^{(j)}\|^2 > \delta$, add one epicycle according to the peak of the residual periodogram;

(ii) **Parameter adjustment**: update the existing parameters by regularised least squares.

Proposition 30.19 (Monotonicity and convergence of the major-luck dynamic adjustment)**.** After adding the dominant frequency component of the residual as in Definition 30.43(i), the residual energy is monotonically non-increasing; if the target is a **real-valued finite periodic sequence** of length T_0 ($T_0 \in \mathbb{N}$) with $\delta = 0$ (here $T_0 = 60$), then after at most $\lfloor 60/2 \rfloor + 1 = 31$ steps the dynamic adjustment leaves a zero residual (for a **complex-valued** sequence the corresponding number of steps is at most 60).

Proof. By the orthogonal decomposition, $\|r^{(j)}\|^2 = \|r^{(j+1)}\|^2 + \|\text{the removed component}\|^2$. The DFT of a **real-valued** finite periodic sequence of length 60 is complete on 31 harmonics (including the constant term); a **complex-valued** sequence requires 60 components. □

30.10.9 Major-luck convolution

Definition 30.44 (Major-luck amplitude convolution, phase convolution and frequency convolution)**.** Let two major-luck epicycle groups be $\mathfrak{E} = \{(A_n, \omega_n, \phi_n)\}$ and $\mathfrak{E}' = \{(B_m, \nu_m, \psi_m)\}$. Define:

$$\mathbb{A}_{mn} = A_n B_m, \quad \mathbb{P}_{mn} = e^{i(\phi_n - \psi_m)}, \quad \mathbb{F}_{mn} = \omega_n \nu_m, \quad \mathbb{T}_{mn} = A_n B_m e^{i(\phi_n - \psi_m)}.$$

Proposition 30.20 (Phase convolution of the major-luck cycle and the current year)**.** The phase convolution $\mathbb{P}_{kt} = e^{i(\phi_k^M - \phi_t^N)}$ of the major-luck sequence with the current-year sequence gives the relative phase of the two; its argument is precisely the phase corresponding to Δ_k and measures the strength of the major-luck–current-year coupling. This is consistent with the periodic convolution framework of this chapter (Definition 30.9).

Proof. This follows directly from Definition 30.44 and Proposition 30.17. □

30.10.10 Integration with discrete sampling, harmonic analysis and time series analysis

Definition 30.45 (Major-luck periodicity diagnostic procedure)**.** A given major-luck system is diagnosed as follows:

(i) **Periodicity decision**: verify that the major-luck sequence has period 60 steps (Theorem 30.13);

(ii) **Direction diagnosis**: compute σ from (ϵ, η) (Definition 30.38);

(iii) **Starting-age diagnosis**: compute a_0 from $\Delta d/3$ (Definition 30.39);

(iv) **Phase diagnosis**: compute the major-luck–current-year relative phase Δ_k (Proposition 30.17);

(v) **Astronomical-approximation diagnosis**: verify the approximation of Jupiter and of the Jupiter–Saturn conjunction by the three periods 100, 120 and 600 years (Corollary 30.4).

Theorem 30.15 (The major-luck periodicity diagnostic procedure and the interface with three chapters)**.** *The five steps of Definition 30.45 connect respectively to:*

(i) *the chapter 'Discrete Sampling and Intercalation of Periodic Signals', Definition 21.1, Theorem 21.3 (the quantification of the starting age is a sampling from continuous time to discrete years and months);*

(ii) *the chapter 'Harmonic Analysis', Definition 22.5, Theorem 22.1 (the DFT decomposition of the major-luck sequence);*

(iii) *the chapter 'Time Series Analysis', Definition 23.16, Proposition 23.2 (the significance test of the major-luck–current-year phase peak).*

Proof. (i) The starting age $a_0 = \Delta d/3$ is a sampling from continuous time to discrete years and months. (ii) The major-luck sequence has period 60, and its DFT belongs to the tools of harmonic analysis. (iii) The major-luck–current-year relative phase forms a spectral peak within the period, whose significance can be decided by the periodogram and Fisher's test. □

Remark 30.25 (The periodic-mathematical essence of the major-luck system)*.* The periodic-mathematical essence of the major-luck system is summarised as follows:

(i) **Starting point** M_0: the month pillar, determined jointly by the solar-term phase and the year stem (Definition 28.16, Definition 30.37 of the calendar chapter);

(ii) **Direction** σ: determined by the binary duality of the year-stem yin-yang ϵ and the sex η (Definition 30.38);

(iii) **Phase offset** a_0: the starting age of the cycle, given by the number of days from the birth to the solar term according to the quantification rule 3 days $= 1$ year (Definition 30.39);

(iv) **Sequence** $M_k = M_0 + k\sigma$: a translation on the sexagenary cycle, of period 60 steps.

The major-luck cycle is not a new astronomical period, but the **replication, on the ten-year scale, of the rational approximation of the year stems and branches to Jupiter and to the Jupiter–Saturn conjunction**: the year stems and branches approximate 5 revolutions of Jupiter within 60 years, and the major-luck stems and branches approximate 50 revolutions of Jupiter within 600 years, the two relative errors being of the same order (of the order of 1%).

Remark 30.26 (Connections between the major-luck cycle and the other periodic systems of this chapter). The relation of the major-luck cycle to the other periodic systems of this chapter may be summarised as four connections:

(i) **With the sexagenary cycle**: the major-luck sequence is a translation orbit of the sexagenary cycle (Theorem 30.5) on C_{60}—it introduces no new algebraic object, but unfolds the cyclic structure of the sexagenary cycle afresh on a new time scale;

(ii) **With the periodic phase operators**: the k-th major-luck step corresponds to the phase encoding of the periodic phase operator $\mathcal{O}_k^{(60)}$ (Definition 30.11), and the major-luck direction σ corresponds to the sign of the phase rotation;

(iii) **With periodic convolution**: the relative phase of the major-luck cycle and the current year is the phase convolution of the major-luck epicycle group with the current-year epicycle group (Proposition 30.20);

(iv) **With the calendar system**: the quantification rule for the starting age (3 days $= 1$ year) and intercalation in the calendar chapter (Theorem 28.6) both belong to the framework of best rational approximation, but their objects differ—the former quantifies the mapping from time to years and months, whilst the latter quantifies the integer pairs of the lunisolar calendar.

Core insight: the major-luck cycle is the replication, on the ten-year scale, of the rational approximation of the year stems and branches to Jupiter and to the Jupiter–Saturn conjunction. The year stems and branches approximate $(T_J \times 5,\ T_{JS} \times 3)$ by $(10, 12, 60)$, whilst the major-luck stems and branches approximate $(T_J \times 50,\ T_{JS} \times 30)$ by $(100, 120, 600)$, and the accuracy of approximation is of the same order at the two scales (of the order of 1%). The core contribution of the major-luck cycle is this: it implements the sexagenary algebraic structure established in this chapter, turning it from a 'symbol system' into a 'verifiable sequence of astronomical approximations'—the step length $L = T_{\text{stem}} = 10$ years is precisely a replication of the Heavenly-Stem period of the year stems and branches.

Seven-theme comparison table for the major-luck cycle

Theme	Corresponding content in the major-luck cycle
Translation	the ordering advance of the major-luck sequence $M_k = M_0 + k\sigma$ along the sexagenary cycle (Definition 30.36)
Scaling	the tenfold replication of scale in the major-luck cycle: $T^{(\mathrm{D})}_{\text{stem}} = 100$ years, $T^{(\mathrm{D})}_{\text{branch}} = 120$ years, $T^{(\mathrm{D})}_{\text{stem-branch}} = 600$ years (Theorem 30.11)
Rotation	the major-luck direction $\sigma = \pm 1$ and the precession of the relative phase Δ_k with respect to the current year (Proposition 30.17)
Conservation	the threefold conservation of the direction σ, the step length L and the starting point M_0 (Theorem 30.14)
Symmetry	the directional duality between yang males and yin females and between yin males and yang females (Proposition 30.15, Corollary 30.5)
Closure	the closure of the major-luck sequence with the period 60 steps (600 years) (Theorem 30.13)
Period	the triple astronomical approximation: 100 years ≈ 5 Jupiter–Saturn conjunctions, 120 years ≈ 10 revolutions of Jupiter, 600 years ≈ 50 revolutions of Jupiter (Corollary 30.4)

Remark 30.27 (Unity with the whole book). The periodic structure of the I Ching system established in this chapter is a concrete realisation of periodic mathematics in a traditional symbol system. The core observation is that **a phase is an operator**: every phase $\mathcal{O}^{(T)}_k = e^{i2\pi k/T}$ of every period is a relation operator. The yang operator ($\mathcal{O}_0 = 1$) defines the identity relation, the yin operator ($\mathcal{O}_{T/2} = -1$) defines the opposition relation, and the remaining operators define partial rotation relations.

The nested epicycle generation rule—for every single step forward of the outer circle, the inner circle rotates through one full turn—unifies the process by which a basic period generates a large period: Yin-Yang × the Five Phases = the Heavenly Stems ($T = 10$), the Four Symbols × Sancai = the Earthly Branches ($T = 12$), and the Eight Trigrams × the Eight Trigrams = the Sixty-Four Hexagrams ($T = 64$). The sexagenary cycle is generated from the synchronous advance of the Heavenly Stems and the Earthly Branches by periodic convolution (Theorem 30.5). The He Tu complex function establishes the Fourier grading $\mathbb{Z}_{10} \cong \mathbb{Z}_2 \times \mathbb{Z}_5$ (Remark 30.10), and the Lo Shu matrix establishes the translation–rotation spectral decomposition of the 3×3 magic square (Theorem 30.10). The two connect the traditional symbol system with standard mathematical tools from the two directions of complex analysis and linear algebra respectively.

The place of the major-luck cycle in the algebraic structure of this chapter: the major-luck cycle is not a new algebraic object, but a **time-series realisation** of the sexagenary cycle C_{60} on the ten-year scale. It implements the abstract phase operators established in this chapter (Definition 30.11) and the periodic convolution framework (Theorem 30.5) as a verifiable sequence of astronomical approximations: the year stems and branches approximate 5 revolutions of Jupiter and 3 Jupiter–Saturn conjunctions by $(10, 12, 60)$ (Proposition 30.12, Proposition 30.13), whilst the major-luck stems and branches approximate 50 revolutions of Jupiter and 30 Jupiter–Saturn conjunctions by $(100, 120, 600)$ (Corollary 30.4), and the accuracy of approximation is of the same order at the two scales (of the order of 1%). This structure shows that the sexagenary cycle is not only a symbol system but also a **sequence of rational approximations** that can be tested against astronomical observation at several scales—10, 60, 120 and 600 years. The algebraic framework of this chapter (phase operators, epicycle nesting, periodic convolution) and the time-series realisation of the major-luck cycle together constitute the two sides—algebraic structure and astronomical content—of the sixty-year period.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	the ordering advance of the Heavenly-Stem ($T = 10$) and Earthly-Branch ($T = 12$) sequences, generating the sexagenary cycle through parity synchronisation $n \equiv m \pmod{2}$ (Lemma 30.4); the ordering advance of the major-luck sequence along the sexagenary cycle (Definition 30.36)
Scaling	the doubling hierarchy of hexagram levels, the 2^n nesting: Yin-Yang $2^1 \to$ the Four Symbols $2^2 \to$ the Eight Trigrams 2^3, and the Eight Trigrams × the Eight Trigrams $= 8 \times 8 = 64$ (Example 30.5); the tenfold replication of scale in the major-luck cycle (Theorem 30.11)
Rotation	the rotation of phase points by $2\pi k/T$ by the phase operator $\mathcal{O}_k^{(T)} = e^{i2\pi k/T}$ (Definition 30.11, Proposition 30.5); the Lo Shu rotational subspace (Theorem 30.10); the major-luck direction $\sigma = \pm 1$ and the precession of the relative phase Δ_k with respect to the current year (Proposition 30.17)
Balance	the combinatorial conservation of the generation from the Eight Trigrams to the Sixty-Four Hexagrams: the number of independent phases equals the product of the bit widths $T_{\text{inner}} \times T_{\text{outer}}$ (Theorem 30.7); the antipodal balance sum of the Lo Shu matrix is 10 (Proposition 30.10); the threefold conservation of the major-luck direction σ, the step length L and the starting point M_0 (Theorem 30.14)
Symmetry	the Yin-Yang involution symmetry: the yin operator $\mathcal{O}_{T/2}^{(T)} = -1$ gives the opposite vector when multiplied by -1, the C_2 involution (Proposition 30.5); the antipodal antisymmetry of the Lo Shu matrix (Proposition 30.10); the directional duality between yang males and yin females and between yin males and yang females (Proposition 30.15)
Closure	the cyclic closure by which the sexagenary cycle ($T = 60$) and the Sixty-Four Hexagrams ($T = 64$) each return to their starting point (Theorem 30.5, Example 30.5); the closure of the major-luck sequence with the period 60 steps (600 years) (Theorem 30.13)
Period	the periodic systems of the thirteen symbol systems, such as the Heavenly Stems $T = 10$, the Earthly Branches $T = 12$ and the Eight Trigrams $T = 8$; the Fourier grading of the He Tu complex function (Remark 30.10); the rotation period $\pi/\sqrt{6}$ of the Lo Shu matrix (Corollary 30.3); the triple astronomical approximation of the major-luck cycle (Corollary 30.4)

Exercises

Exercise 1 Prove the **periodicity decision theorem for I Ching targets** of this chapter (30.1). Hint: first write down the decision map and the evolution group of the target of this chapter, then apply the proof structure of the general criterion of the 'Periodic Systems' chapter.

Exercise 2 Verify that the hypotheses of the **periodicity decision theorem for the major-luck cycle** (30.14) hold for the domain objects of this chapter, and give a counterexample in which the hypotheses fail.

Exercise 3 Examine the behaviour of the **monotonicity of the cardinality adjustment** (30.2) under degenerate parameters, and explain how the conclusion degenerates accordingly.

Exercise 4 Explain the relation between the **correspondence of the three parameters: radius, phase and angular speed** (30.1) and the corresponding general conclusion of the 'Periodic Systems' chapter (whether it agrees with it, is a special case of it, or requires additional conditions).

Exercise 5 Using the seven-theme comparison table for this chapter, point out the concrete carrier of each of the three themes **translation**, **scaling** and **rotation** in this chapter, and give a domain-data example for each.

Exercise 6 Synthesis: restate the diagnostic procedure of this chapter using the five-step standard interface of the 'Periodic Systems' chapter (table 24.1), noting one by one the theorem of this chapter on which each step depends: 30.1, 30.14, 30.2, 30.1.

Chapter 31

Periodic Structure of Electromagnetic Systems

Remark 31.1 (Periodicity decision for the target of this chapter (citation)). When this chapter decides whether a research target is periodic, determines its period type, or seeks its minimal positive period, reference is always made to the 'Periodic Systems' chapter: for the periodicity criterion see Theorem 24.1, for the discrimination of period types see Definition 24.3, Proposition 24.1, for the search for the minimal positive period see Definition 24.4, Theorem 24.2, and for the construction of period series and epicycle groups from data see Theorem 24.3, Definition 24.6.

Remark 31.2 (Positioning of this chapter: developed along the five-step interface of the 'Periodic Systems' chapter). The periodic structure of electromagnetic systems studies the mathematical characterisation of periodicity, quasi-periodicity and wave propagation in electromagnetic fields. This chapter does not establish a separate periodicity criterion; instead it connects electromagnetic systems to the five-step interface prescribed by the 'Periodic Systems' chapter:

(1) **Decision**: reduce such targets as electromagnetic oscillation, plane waves, cyclotron motion and the retarded potential to discrete or continuous dynamical systems, invoke Theorem 24.1 to decide whether they possess periodicity, and determine the period type;

(2) **Construction**: build the amplitude series, the frequency series and the phase series from the time series, spectrum and phase data of the electromagnetic field, and assemble them into the electromagnetic epicycle group;

(3) **Adjustment**: dynamically adjust the electromagnetic epicycle group under the constraints of residual-energy descent, charge conservation, energy conservation and momentum conservation;

(4) **Convolution**: introduce amplitude convolution, phase convolution and frequency convolution on the electromagnetic epicycle group, and connect them with the Poynting vector, cross-correlation and the retarded potential;

(5) **Integration**: connect to the tool chains of the three chapters 'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis', forming an electromagnetic periodicity diagnostic procedure.

The mathematical tools used in this chapter include: vector analysis, Maxwell's equations, the wave equation, Fourier decomposition, periodic convolution, epicycle groups, phase operators and the retarded potential.

Remark 31.3 (Seven-theme guide for this chapter). **Translation**—an electromagnetic wave propagating at the speed of light c, its propagation delay $\tau = r/c$ and the retarded potential (see Theorem 31.17); **Scaling**—the inverse-square scaling $\sim 1/r^2$ of Coulomb's law and the Biot–Savart law (see Theorem 31.11); **Rotation**—Ampère's circuital law $\nabla \times \mathsf{B} = \mu_0 \mathsf{J}$ and the circular motion of a charge driven by the Lorentz force (see Proposition 31.5); **Balance**—charge conservation and Poynting energy conservation (see Theorem 31.18); **Symmetry**—gauge symmetry and the structure of Maxwell's equations (see Theorem 31.15); **Closure**—the closed integrals of the Ampère loop and the Faraday loop (see Theorems 31.7 and 31.9); **Period**—electromagnetic oscillation and the period of an electromagnetic wave (see Proposition 31.7).

31.1 Connection to the five-step interface of the 'Periodic Systems' chapter

31.1.1 Step one: periodicity decision for electromagnetic targets

Definition 31.1 (Electromagnetic targets and decision input). An **electromagnetic target** is one of the following objects:

(i) an electromagnetic oscillation system, whose state space is the (q,p) phase space or the (E,B) field space;

(ii) a monochromatic plane wave, whose state space is the phase $\theta = \mathsf{k}\cdot\mathsf{r} - \omega t$;

(iii) the cyclotron motion of a charged particle in a uniform magnetic field, whose state space is velocity space;

(iv) a radiation field described by the retarded potential, whose state space is the joint space of the source distribution and the field values at the observation point.

The **decision input** of an electromagnetic target is an ordered pair

$$\mathcal{D}_{\mathsf{EM}} = (\mathcal{S}, \Phi),$$

where $\mathcal{S}$ is the state space and $\Phi : \mathcal{S} \to \mathcal{S}$ is the map induced by time translation, phase rotation or cyclotron evolution. If $\mathcal{S}$ is a finite set, the target is called a **finite electromagnetic target**; if $\mathcal{S}$ is an infinite set, the target is called an **infinite electromagnetic target**.

Definition 31.2 (Electromagnetic decision map). Let an electromagnetic target be given by the decision input $\mathcal{D}_{\mathsf{EM}} = (\mathcal{S}, \Phi)$. If the states of the target are generated by Φ, that is, if there exists $x_0 \in \mathcal{S}$ such that

$$x_t = \begin{cases} \Phi^t(x_0), & t \in \mathbb{Z}, \\ \Phi_t(x_0), & t \in \mathbb{R}, \end{cases}$$

for $t \in \mathbb{Z}$ the state x_t is generated by iteration of Φ; for $t \in \mathbb{R}$ it is described by the one-parameter evolution family $\{\Phi_t\}_{t\in\mathbb{R}}$, where each $\Phi_t : \mathcal{S} \to \mathcal{S}$, $\Phi_0 = \mathrm{id}$,

$\Phi_{s+t} = \Phi_s \circ \Phi_t$ for all $s, t \in \mathbb{R}$, and $(t, x) \mapsto \Phi_t(x)$ is continuous. then Φ is called the **electromagnetic decision map** of the target. The electromagnetic target is then reduced to the discrete dynamical system $(\mathcal{S}, \Phi)$ when $t \in \mathbb{Z}$ and to the continuous dynamical system $(\mathcal{S}, \{\Phi_t\}_{t\in\mathbb{R}})$ when $t \in \mathbb{R}$; in both cases the criteria of the 'Operations and Periodicity' chapter and of the 'Periodic Systems' chapter may be applied directly.

Theorem 31.1 (Periodicity decision theorem for electromagnetic systems)**.** *Let the decision map of an electromagnetic target be Φ, and let $(\mathcal{S}, \Phi)$ be a discrete dynamical system. Then:*

(A) **Sufficient decision**: *if there exists $m \in \mathbb{N}^+$ such that $\Phi^m = \mathrm{id}$, then the target is a uniformly periodic electromagnetic system; every orbit is closed, and the orbit length divides m.*

(B) **Equivalence of the decision**: *the target is a uniformly periodic electromagnetic system if and only if $\mathrm{ord}(\Phi) < \infty$.*

(C) **Pointwise criterion**: *the orbit of the target at the state x_0 is closed if and only if x_0 is a periodic point; the orbit length is then the period at that point.*

(D) **Non-omissibility**: *the conjunction of conserved quantities and symmetry groups cannot replace $\mathrm{ord}(\Phi) < \infty$; a quasi-periodic electromagnetic target is not a periodic electromagnetic system.*

Proof. The Theorem 31.1 of this chapter is the domain instantiation of Theorem 24.1 in the 'Periodic Systems' chapter: one need only replace the decision map f and the evolution group by Φ of this chapter, and the four clauses (A) sufficient decision, (B) equivalence of the decision, (C) pointwise criterion and (D) non-omissibility hold one by one. By the convention of Remark 24.4, the argument is not expanded again here. ☐

Corollary 31.1 (Electromagnetic period types)**.** *After the decision by Theorem 31.1, an electromagnetic target falls into one of the following types:*

(i) **Trivial periodic type**: *least period $m_0 = 1$, the state is a fixed point;*

(ii) **Strictly periodic type**: *$1 < m_0 < \infty$, the state is strictly periodic;*

(iii) **Uniformly periodic type**: *$\mathrm{ord}(\Phi) = m < \infty$, every orbit length divides m;*

(iv) **Quasi-periodic type**: *$\mathrm{ord}(\Phi) = \infty$, the orbits are not closed but are dense on a compact carrier structure;*

(v) **Eventually periodic type**: *there exists N such that the target is periodic for $t \geq N$;*

(vi) **Discrete periodic type**: *the carrier structure is $\mathbb{Z}$ and the normalised period ratio is $\rho = T_x/\Delta x \in \mathbb{Q}$.*

(vii) **Asymptotically periodic (damped) type**: *the orbit is not closed, but tends asymptotically to a closed orbit or to a fixed point, the residual decaying as $e^{-\kappa t}$ (see Definition 24.3(vii)).*

Here (i)–(iii) and (vi) belong to periodic electromagnetic systems; (iv) does not belong to periodic electromagnetic systems but can be handled with bounded-approximation tools; (v) is a periodic electromagnetic system only in its stable stage; (vii), like (iv), is a type in the limiting sense; it is not a periodic electromagnetic system and can be handled with bounded-approximation tools only in its stable stage.

Proof. Each item follows by substituting the electromagnetic target into Theorem 31.1 and Definition 24.3 of the 'Periodic Systems' chapter. □

Lemma 31.1 (Periodicity decision for LC oscillation). An ideal LC oscillatory circuit consists of an inductance L and a capacitance C, and its charge $q(t)$ satisfies

$$\ddot{q}+\omega_0^2 q=0,\qquad \omega_0=\frac{1}{\sqrt{LC}}.$$

Then the LC oscillation is a uniformly periodic system, whose least positive period is

$$T_0=2\pi\sqrt{LC}.$$

Proof. The general solution of the equation is $q(t)=A\cos(\omega_0 t+\phi)$. The time translation $T_0=2\pi/\omega_0=2\pi\sqrt{LC}$ gives $q(t+T_0)=q(t)$, and T_0 is the least positive period. Hence the decision map Φ satisfies $\Phi^{T_0}=\mathrm{id}$, so the system is uniformly periodic. □

Lemma 31.2 (Periodicity decision for a monochromatic plane wave). Let the monochromatic plane wave be

$$\mathsf{E}(\mathsf{r},t)=\mathsf{E}_0 e^{i(\mathsf{k}\cdot\mathsf{r}-\omega t)}.$$

Then its temporal period is

$$T=\frac{2\pi}{\omega},$$

its spatial period is

$$\lambda=\frac{2\pi}{|\mathsf{k}|},$$

and the two are coupled by the dispersion relation $\omega=c|\mathsf{k}|$ as $\lambda=cT$.

Proof. From $e^{-i\omega(t+T)}=e^{-i\omega t}e^{-i\omega T}=e^{-i\omega t}$ we obtain the temporal period $T=2\pi/\omega$. From $e^{i\mathsf{k}\cdot(\mathsf{r}+\lambda\hat{\mathsf{k}})}=e^{i\mathsf{k}\cdot\mathsf{r}}e^{i|\mathsf{k}|\lambda}=e^{i\mathsf{k}\cdot\mathsf{r}}$ we obtain the spatial period $\lambda=2\pi/|\mathsf{k}|$. The dispersion relation is given by Theorem 31.16 and Proposition 31.7. □

Lemma 31.3 (Periodicity decision for cyclotron motion). Let a charged particle q move in a uniform magnetic field B, with velocity perpendicular to the field. Then the cyclotron period is

$$T_c=\frac{2\pi m}{|q|B},$$

and T_c is independent of the speed, so the cyclotron motion is a uniformly periodic system.

Proof. By Proposition 31.5, the cyclotron angular frequency is $\omega_c=|q|B/m$, so $T_c=2\pi/\omega_c=2\pi m/(|q|B)$. The velocity solution is a periodic function, and the period is independent of the speed. □

Lemma 31.4 (The retarded potential preserves the period of the source). *Let the sources $\rho(\mathbf{r},t)$ and $\mathbf{J}(\mathbf{r},t)$ both be periodic in time with period T. Then the retarded potential*

$$\phi(\mathbf{r},t)=\frac{1}{4\pi\varepsilon_0}\int\frac{\rho(\mathbf{r}',t-|\mathbf{r}-\mathbf{r}'|/c)}{|\mathbf{r}-\mathbf{r}'|}\,dV'$$

is also periodic in time with period T:

$$\phi(\mathbf{r},t+T)=\phi(\mathbf{r},t).$$

The same holds for the vector potential $\mathbf{A}$.

Proof. From the periodicity of the source, $\rho(\mathbf{r}',t+T-|\mathbf{r}-\mathbf{r}'|/c)=\rho(\mathbf{r}',t-|\mathbf{r}-\mathbf{r}'|/c)$, and substitution into the integral gives $\phi(\mathbf{r},t+T)=\phi(\mathbf{r},t)$. The same argument applies componentwise to $\mathbf{A}$. ☐

Definition 31.3 (Electromagnetic least positive period algorithm). This definition has the same content as Chapter 24, the section 'Periodicity decision: is the target periodic?' (Section 24.1), Definition 24.4; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement is given there. The only difference here is terminology (electromagnetic target / electromagnetic sample); the algorithm steps and the conclusion are exactly the same.

Theorem 31.2 (Correctness of the electromagnetic least positive period algorithm). *This theorem has the same content as Chapter 24, the section 'Periodicity decision: is the target periodic?' (Section 24.1), Theorem 24.2; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement and proof are given there. The only difference here is terminology (electromagnetic target / electromagnetic sample); the algorithm steps and the conclusion are exactly the same.*

Example 31.1 (Examples of electromagnetic periodicity decisions). (i) LC oscillation: $T_0=2\pi\sqrt{LC}$, uniformly periodic type;

(ii) monochromatic plane wave: temporal period $T=2\pi/\omega$, spatial period $\lambda=2\pi/|\mathbf{k}|$;

(iii) cyclotron motion: $T_c=2\pi m/(|q|B)$, uniformly periodic type;

(iv) retarded potential of a periodic source: the source period T is transmitted to the potentials and the fields, uniformly periodic type.

Remark 31.4 (Conserved quantities do not guarantee periodicity). Charge conservation, energy conservation and momentum conservation in electromagnetic systems are auxiliary conditions for periodic structure, but they do not constitute a sufficient condition. For example, a hyperbolic orbit possesses energy conservation and angular momentum conservation, yet the orbit is open and not closed. The periodic structure of an electromagnetic system must be decided by Theorem 31.1 or by the lemmas above.

31.1.2 Step two: construction of the electromagnetic period series and epicycle group

Definition 31.4 (Electromagnetic amplitude, frequency and phase series)**.** *Let the state or output sequence of an electromagnetic target be* $\{z_t\}$*. If it can be written as*

$$z_t = \sum_k A_k e^{i(\omega_k t+\phi_k)} + \varepsilon_t,$$

then

$$\mathcal{A}_z(t) = \sum_k A_k e^{ikt}, \qquad \mathcal{F}_z(s) = \sum_k \omega_k s^k, \qquad \mathcal{P}_z(t) = \sum_k e^{i(kt+\phi_k)}$$

are called respectively the **amplitude series***, the* **frequency series** *and the* **phase series** *of the electromagnetic target. The three agree with Definition 6.10, Definition 6.12 and Definition 6.11 of the 'Series and Convolution' chapter.*

Definition 31.5 (Electromagnetic epicycle group)**.** *Let the period series of an electromagnetic target be given by its amplitude series* $\{A_k\}$*, frequency series* $\{\omega_k\}$ *and phase series* $\{\phi_k\}$*. For each* k *define the epicycle*

$$E_k : \theta \mapsto A_k e^{i(\theta+\phi_k)}, \qquad \theta = \omega_k t,$$

whose collection

$$\mathfrak{E}_{\mathsf{EM}} = \{(A_k, \omega_k, \phi_k)\}_{k\in\mathbb{Z}}$$

is called the **epicycle group** *of the electromagnetic target. Its superposition is the electromagnetic period series*

$$z_t = \sum_k E_k(\omega_k t) = \sum_k A_k e^{i(\omega_k t+\phi_k)}.$$

Theorem 31.3 (Constructing the period series from electromagnetic data)**.** *Let the output of an electromagnetic target be a finite periodic sequence of period* N*. Then:*

(i) *the output admits the discrete Fourier expansion*

$$z_t = \sum_{k=0}^{N-1} c_k e^{i2\pi kt/N}, \qquad c_k = \frac{1}{N}\sum_{t=0}^{N-1} z_t e^{-i2\pi kt/N};$$

(ii) *the amplitude series is given by* $|c_k|$*, the phase series by* $\arg c_k$ *and the frequency series by* $2\pi k/N$*;*

(iii) *if the output is a monochromatic plane wave, its Fourier decomposition has only a single spectral line; if the output is a superposition of multiple frequencies, each order of harmonic corresponds to a different frequency component.*

Proof. (i) The discrete Fourier series of a finite periodic sequence is a standard result: the basis functions $\{e^{i2\pi kt/N}\}_{k=0}^{N-1}$ form an orthogonal basis of $\mathbb{C}^N$. (ii) This is read off directly from $c_k = A_k e^{i\phi_k}$. (iii) The temporal part of a monochromatic plane wave is $e^{-i\omega t}$, which on the period $T = 2\pi/\omega$ has only the fundamental component; a superposition of multiple frequencies expands as a Fourier series. □

Proposition 31.1 (Three-parameter correspondence: radius, phase and rotational speed)**.** *The three parameters of the electromagnetic epicycle group $\mathfrak{E}_{\mathsf{EM}} = \{(A_k, \omega_k, \phi_k)\}_k$ correspond one-to-one to three geometric–physical quantities:*

(i) **Radius corresponds to amplitude**: *the radius of E_k, namely $|E_k| = A_k$, is the amplitude of the electric or magnetic field component;*

(ii) **Phase corresponds to rotational delay**: *the initial phase ϕ_k corresponds to the time translation $\tau_k = \phi_k/\omega_k$; when $\phi_k > 0$ this component* **leads** *the baseline by τ_k (equivalently, the component coincides with the baseline once the baseline is postponed by τ_k);*

(iii) **Frequency corresponds to rotational speed**: *ω_k is the angular frequency of that component.*

Proof. The Proposition 31.1 of this chapter corresponds exactly to Proposition 24.3 of the 'Periodic Systems' chapter, and the three correspondences—radius to amplitude, phase to rotational delay, frequency to rotational speed—are obtained by translating them one by one to the epicycle group of this chapter. The proof is given there: (i) from $|E_k| = A_k$ and the energy decomposition, (ii) from the phase translation theorem, (iii) from the definition of the frequency series. ☐

Example 31.2 (Period series of a plane wave)**.** At a fixed spatial point the temporal part of the monochromatic plane wave $\mathsf{E} = \mathsf{E}_0 e^{i(\mathsf{k}\cdot\mathsf{r}-\omega t)}$ degenerates to a single-frequency period series, whose epicycle group is $\{(|\mathsf{E}_0|, \omega, \phi_0)\}$.

Remark 31.5 (The period series is a complex function)*.* Let $z = e^{i\omega_0 t}$; then the electromagnetic period series

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)}$$

when all components are commensurable with ω_0 (that is, $\omega_k = k\omega_0$, $k \in \mathbb{Z}$) it can be written as the Laurent series $\sum_k c_k z^k$, where $c_k = A_k e^{i\phi_k}$: in that case the electromagnetic period series is a complex function on the unit circle, and otherwise it is merely a function on the torus.

31.1.3 Step three: dynamic adjustment of the electromagnetic epicycle group

Definition 31.6 (Dynamic adjustment of the electromagnetic epicycle group)**.** Let the current electromagnetic epicycle group be $\mathfrak{E}^{(j)}$, with partial sum

$$z^{(j)}(t) = \sum_{k \in \mathfrak{E}^{(j)}} A_k e^{i(\omega_k t + \phi_k)},$$

and residual $r^{(j)} = z - z^{(j)}$. **Dynamic adjustment** consists of two rules:

(i) **Quantity adjustment**: if the residual energy satisfies $\|r^{(j)}\|^2 > \delta$, add a new epicycle according to the peak of the residual periodogram;

(ii) **Parameter adjustment**: update the parameters of the existing epicycles by regularised least squares or by the conservation constraints.

In electromagnetic systems, quantity adjustment corresponds to increasing the number of frequency components or spatial harmonics; parameter adjustment corresponds to adjusting amplitude, phase and frequency so that the charge, energy and momentum constraints are satisfied.

Proposition 31.2 (Monotonicity of quantity adjustment)**.** After the dominant residual frequency component is added as in Definition 31.6(i), the residual energy is monotonically non-increasing:

$$\|r^{(j+1)}\|^2 \leq \|r^{(j)}\|^2,$$

with equality if and only if the residual has no energy at the selected frequency.

Proof. The Proposition 31.2 of this chapter agrees with the dynamic adjustment mechanism of Definition 24.8 in the 'Periodic Systems' chapter: adding an epicycle removes the component of $r^{(j)}$ along $\hat{\omega}$ from the residual, and the orthogonal decomposition $\|r^{(j)}\|^2 = \|r^{(j+1)}\|^2 + \|\text{the removed component}\|^2$ gives the inequality, with equality if and only if the norm of the removed component is zero. The complete argument is given there. □

Theorem 31.4 (Convergence of dynamic adjustment)**.** Let the electromagnetic target be a finite periodic sequence of period T_0 and let $\delta = 0$. Then after at most $\lfloor T_0/2 \rfloor + 1$ steps of dynamic adjustment the residual is zero, and the resulting electromagnetic epicycle group is the exact Fourier data of the target.

Proof. The discrete Fourier series of a sequence of period T_0 is complete on $\lfloor T_0/2 \rfloor + 1$ harmonics. At each step one dominant frequency component of the residual is extracted as in Proposition 31.2 and the residual energy decreases strictly, so after finitely many steps the residual is zero. □

Definition 31.7 (Structural metric adjustment)**.** Let the structure metric of an electromagnetic target be

$$\mathcal{I} = (\mathcal{I}_{\text{charge}}, \mathcal{I}_{\text{energy}}, \mathcal{I}_{\text{momentum}}) = (\text{charge conservation degree, energy conservation degree, mor}$$

Structural-metric adjustment consists in choosing θ in the parameter space Θ: at each step take a neighbouring parameter for which $\mathcal{I}(\theta)$ is **strictly decreasing** in the lexicographic order (minimisation); if no such parameter exists, the process stops, and the parameter is then locally optimal.

Proposition 31.3 (Finite convergence of structural metric adjustment)**.** Let the parameter space Θ be finite, and let each adjustment make $\mathcal{I}$ strictly decreasing in the lexicographic order. Then the adjustment terminates after finitely many steps at a locally optimal parameter.

Proof. Since Θ is finite, $\mathcal{I}$ takes only finitely many values, and the lexicographic order is a total order on a finite set; a strictly decreasing sequence in that order cannot continue indefinitely, so the adjustment terminates after finitely many steps. At termination there is no neighbouring parameter that makes $\mathcal{I}$ strictly decrease in the lexicographic order, that is, the parameter is locally optimal. □

Remark 31.6 (Boundaries of dynamic adjustment). Theorem 31.4 relies on finite periodic sequences. For continuous-spectrum or quasi-periodic electromagnetic targets, no finite epicycle group represents them exactly and only bounded approximation is possible. Structural metric adjustment does not guarantee global optimality.

31.1.4 Step four: *electromagnetic convolution*

Definition 31.8 (Electromagnetic amplitude, phase and frequency convolutions). *Let two electromagnetic epicycle groups be*

$$\mathfrak{E} = \{(A_n, \omega_n, \phi_n)\}, \qquad \mathfrak{E}' = \{(B_m, \nu_m, \psi_m)\}.$$

Define:

(i) **Amplitude convolution**: $\mathbb{A}_{mn} = A_n B_m$;

(ii) **Phase convolution**: $\mathbb{P}_{mn} = e^{i(\phi_n - \psi_m)}$;

(iii) **Frequency convolution**: $\mathbb{F}_{mn} = \omega_n \nu_m$;

(iv) **Trend convolution**: $\mathbb{T}_{mn} = A_n B_m e^{i(\phi_n - \psi_m)} = A_n B_m \mathbb{P}_{mn}$.

This agrees with Definition 6.21, Definition 6.22, Definition 6.23 and Definition 6.24 of the 'Series and Convolution' chapter.

Theorem 31.5 (Action of the three convolutions on the electromagnetic epicycle group). *Let the epicycle groups of two electromagnetic targets be $\mathfrak{E}$ and $\mathfrak{E}'$ respectively. Then:*

(i) *amplitude convolution acts only on the radius;*

(ii) *phase convolution acts only on the phase;*

(iii) *frequency convolution acts only on the rotational speed;*

(iv) *trend convolution preserves both amplitude and phase.*

Proof. This is read off directly from Definition 31.8. ☐

Proposition 31.4 (Autocorrelation and trend convolution). *Let $\{z_t\}$ be an electromagnetic periodic sequence, with autocorrelation function*

$$R_z(h) = \sum_t z_t \overline{z_{t+h}}.$$

In the language of the electromagnetic epicycle group, the autocorrelation function is the projection of the trend convolution onto the diagonal:

$$R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}.$$

Proof. The Proposition 31.4 of this chapter is the realisation in this domain of the convolution-action conclusion of Theorem 24.6 in the 'Periodic Systems' chapter: substituting $z_t = \sum_n A_n e^{i(\omega_n t+\phi_n)}$ into the definition of the autocorrelation, cancelling all off-diagonal cross terms by orthogonality and retaining only the diagonal term of the trend convolution (the factorisation of the three convolutions, Proposition 24.6), which yields $R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}$. The full expansion is given there. □

Remark 31.7 (Relation between electromagnetic convolution and periodic convolution). *Electromagnetic convolution is not a new convolution operation but an instance of the periodic convolution of the 'Series and Convolution' chapter on the electromagnetic epicycle group. Autocorrelation is the projection of the trend convolution onto the diagonal; the Poynting vector and cross-correlation are respectively the concrete manifestations in electromagnetism of amplitude convolution and trend convolution.*

31.1.5 Step five: integration with discrete sampling, harmonic analysis and time series analysis

Definition 31.9 (Electromagnetic periodicity diagnostic procedure)**.** *For a given electromagnetic system, diagnose it using the framework of periodic mathematics:*

(i) **Periodicity decision**: *verify the period type of the electromagnetic target (Theorem 31.1);*

(ii) **Series construction**: *build the amplitude, frequency and phase series and the epicycle group from electromagnetic field data (Theorem 31.3);*

(iii) **Conservation diagnosis**: *verify charge, energy and momentum conservation (Theorem 31.18);*

(iv) **Convolution diagnosis**: *compute the autocorrelation (the diagonal projection of the trend convolution, Proposition 31.4) and the phase convolution;*

(v) **Integration diagnosis**: *connect to the tool chains of discrete sampling, harmonic analysis and time series analysis.*

Theorem 31.6 (The electromagnetic periodicity diagnostic procedure and its interfaces with three chapters)**.** *The five steps of Definition 31.9 connect respectively to:*

(i) *the 'Discrete Sampling and Intercalation of Periodic Signals' chapter: Definition 21.1, Theorem 21.3, Theorem 21.10;*

(ii) *the 'Harmonic Analysis' chapter: Definition 22.5, Theorem 22.1, Proposition 22.11;*

(iii) *the 'Time Series Analysis' chapter: Definition 23.16, Proposition 23.2, Definition 23.19.*

The electromagnetic periodicity diagnostic procedure is therefore not an independent method but an application of the unified tool chain of periodic mathematics to electromagnetic systems.

Proof. The Theorem 31.6 of this chapter is the implementation in this domain of the five-step interface of the 'Periodic Systems' chapter (Table 24.1): the periodicity decision uses sampling and spectral estimation, the frequency decomposition and the phase kernels use the 'Harmonic Analysis' chapter, and the periodogram peaks, the Fisher test and window-function validation use the 'Time Series Analysis' chapter. The diagnostic procedure is therefore a synthesis of the tool chains of the three chapters; the proof is given there. ☐

Remark 31.8 (Order of operations after integration). The recommended order for an actual diagnosis is: first decide whether the electromagnetic target is periodic (Theorem 31.1); then construct the Fourier data and the epicycle group of the electromagnetic data (Theorem 31.3); then optimise the parameters by the conservation constraints and dynamic adjustment (Theorem 31.4, Proposition 31.3); then perform electromagnetic convolution and autocorrelation analysis (Proposition 31.4); and finally output the diagnosis as in Definition 31.9.

31.2 Mathematical foundations of the electromagnetic field

Definition 31.10 (Basic quantities of the electromagnetic field). An electromagnetic system is described by four basic vector fields:

(i) **Electric field strength** $\mathbf{E}(\mathbf{r}, t)$ (unit: V/m)

(ii) **Magnetic flux density** $\mathbf{B}(\mathbf{r}, t)$ (unit: T)

(iii) **Electric displacement vector** $\mathbf{D}(\mathbf{r}, t)$ (unit: C/m^2), with $\mathbf{D} = \varepsilon_0 \mathbf{E}$ in vacuum

(iv) **Magnetic field strength** $\mathbf{H}(\mathbf{r}, t)$ (unit: A/m), with $\mathbf{H} = \mathbf{B}/\mu_0$ in vacuum

The speed of light is $c = 1/\sqrt{\varepsilon_0 \mu_0}$. The source terms are the charge density ρ and the current density $\mathbf{J}$.

Property 31.1 (Direct consequences of the vacuum constitutive relations). From the vacuum constitutive relations $\mathbf{D} = \varepsilon_0 \mathbf{E}$ and $\mathbf{H} = \mathbf{B}/\mu_0$ we obtain directly:

(i) the speed of light $c = 1/\sqrt{\varepsilon_0 \mu_0}$ is determined solely by the two fundamental constants of the vacuum and is independent of the frequency of the wave (the vacuum is non-dispersive);

(ii) $\mathbf{E}$ and $\mathbf{D}$ are parallel and in the same direction, and so are $\mathbf{B}$ and $\mathbf{H}$; the vacuum is an isotropic, linear medium.

Remark 31.9 (Three vector differential operators). This chapter repeatedly uses the three basic operators of vector analysis:

(i) **Gradient** ∇f: maps the scalar field f to a vector field pointing in the direction of fastest increase of f.

(ii) **Divergence** $\nabla \cdot \mathsf{F}$: *maps the vector field* F *to a scalar field, measuring the strength of the source or sink at each point.*

(iii) **Curl** $\nabla \times \mathsf{F}$: *maps the vector field* F *to a vector field, measuring the tendency to circulate at each point.*

The three satisfy the identities $\nabla \times (\nabla f) = \mathbf{0}$ *and* $\nabla \cdot (\nabla \times \mathsf{F}) = 0$. *These two identities play a key role in checking the self-consistency of Maxwell's equations.*

31.3 Magnetostatics and Ampère's circuital law

Lemma 31.5 (Biot–Savart law)**.** *The magnetic flux density produced by a steady current density* $\mathsf{J}(\mathsf{r}')$ *at the observation point* $\mathbf{r}$ *is*

$$\mathsf{B}(\mathsf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathsf{J}(\mathsf{r}') \times (\mathsf{r} - \mathsf{r}')}{|\mathsf{r} - \mathsf{r}'|^3}\, dV'.$$

Proof. The Biot–Savart law is a fundamental experimental law of magnetostatics, on a par with Coulomb's law, and is taken here as an axiomatic input for the magnetostatic part of this chapter; its differential form (Ampère's circuital law) is given by the next theorem. □

Theorem 31.7 (Ampère's circuital law)**.** *In the static magnetic field produced by a steady current, the magnetic flux density satisfies*

$$\boxed{\nabla \times \mathsf{B} = \mu_0 \mathsf{J}}.$$

That is, the current is the source of the curl of the magnetic flux density—at every point where the current flows, the magnetic flux density has non-zero curl.

Proof. Taking the curl of the integral of Lemma 31.5 with respect to the observation point r, and putting $\mathsf{R} = \mathsf{r} - \mathsf{r}'$. Since $\mathsf{J}(\mathsf{r}')$ is independent of r, the vector identity

$$\nabla \times \left(\mathsf{J} \times \frac{\mathsf{R}}{R^3}\right) = \mathsf{J}\left(\nabla \cdot \frac{\mathsf{R}}{R^3}\right) - (\mathsf{J} \cdot \nabla)\frac{\mathsf{R}}{R^3}.$$

The first term uses the point-source divergence identity

$$\nabla \cdot \frac{\mathsf{R}}{R^3} = 4\pi\, \delta^{(3)}(\mathsf{R}),$$

which after integration gives $\mu_0\, \mathsf{J}(\mathsf{r})$. In the second term, since $\partial/\partial r_i = -\partial/\partial r_i'$, we have $(\mathsf{J} \cdot \nabla)\frac{\mathsf{R}}{R^3} = -(\mathsf{J} \cdot \nabla')\frac{\mathsf{R}}{R^3}$; integrating this term by parts,

$$\int (\mathsf{J} \cdot \nabla')\frac{\mathsf{R}}{R^3}\, dV' = \int \left[\nabla' \cdot \left(\frac{\mathsf{R}}{R^3}\mathsf{J}\right) - \frac{\mathsf{R}}{R^3}(\nabla' \cdot \mathsf{J})\right] dV' = 0,$$

where the first term vanishes as a boundary term at infinity by the divergence theorem, and the second term vanishes because the current is steady, $\nabla' \cdot \mathsf{J} = 0$. Hence $\nabla \times \mathsf{B} = \mu_0 \mathsf{J}$. □

Theorem 31.8 (Vector potential representation of the static magnetic field)**.** *For a steady current distribution there exists a vector field* $\mathsf{A}(\mathbf{r})$ *(called the* **magnetic vector potential***) such that*

$$\boxed{\mathsf{B} = \nabla \times \mathsf{A}}.$$

When the vector potential A *satisfies the Coulomb gauge* $\nabla \cdot \mathsf{A} = 0$*, Ampère's circuital law gives the Poisson equation*

$$\nabla^2 \mathsf{A} = -\mu_0 \mathsf{J},$$

whose solution is

$$\mathsf{A}(\mathsf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathsf{J}(\mathsf{r}')}{|\mathsf{r} - \mathsf{r}'|} \, dV'.$$

Proof. $\nabla \cdot \mathsf{B} = 0$ (established independently in Theorem 31.12) is a universal property of the static magnetic field. A divergence-free vector field necessarily has a vector potential: $\mathsf{B} = \nabla \times \mathsf{A}$. Substituting into Ampère's circuital law and using the identity $\nabla \times (\nabla \times \mathsf{A}) = \nabla(\nabla \cdot \mathsf{A}) - \nabla^2 \mathsf{A}$, this reduces, in the Coulomb gauge $\nabla \cdot \mathsf{A} = 0$, to $\nabla^2 \mathsf{A} = -\mu_0 \mathsf{J}$. The solution of this Poisson equation is the integral expression above, which can be verified by the Green's function method. ☐

31.4 Faraday's law of electromagnetic induction

Theorem 31.9 (Faraday's law of electromagnetic induction)**.**

$$\boxed{\nabla \times \mathsf{E} = -\frac{\partial \mathsf{B}}{\partial t}}.$$

That is, the time variation of the magnetic flux density is the source of the curl of the electric field strength.

Proof. The integral form of Faraday's experimental law is

$$\mathcal{E} = \oint_{\partial S} \mathsf{E} \cdot d\mathsf{l} = -\frac{d}{dt} \int_S \mathsf{B} \cdot d\mathsf{S}.$$

For a fixed loop S, moving the time derivative inside the integral sign gives $\oint_{\partial S} \mathsf{E} \cdot d\mathsf{l} = -\int_S \frac{\partial \mathsf{B}}{\partial t} \cdot d\mathsf{S}$. By Stokes' theorem the left-hand side $= \int_S (\nabla \times \mathsf{E}) \cdot d\mathsf{S}$, hence

$$\int_S \left(\nabla \times \mathsf{E} + \frac{\partial \mathsf{B}}{\partial t} \right) \cdot d\mathsf{S} = 0.$$

Since the surface S is arbitrary we obtain the differential form $\nabla \times \mathsf{E} = -\partial \mathsf{B}/\partial t$ (for its complete derivation see Theorem 31.13). ☐

Example 31.3 (Periodic electromotive force in an AC generator)**.** A coil of area A rotates in a uniform magnetic field B with angular velocity ω, and the magnetic flux through the coil is

$$\Phi(t) = BA \cos(\omega t).$$

By Faraday's law (Theorem 31.9) the induced electromotive force is

$$\mathcal{E}(t) = -\frac{d\Phi}{dt} = BA\omega \sin(\omega t),$$

with period $T = 2\pi/\omega$. **Numerical example**: taking $A = 0.01\,\mathrm{m}^2$, $B = 0.5\,\mathrm{T}$ and $\omega = 100\pi$ rad/s (mains frequency $50\,\mathrm{Hz}$), the peak electromotive force is

$$\mathcal{E}_{\max} = BA\omega = 0.5 \times 0.01 \times 100\pi = \frac{\pi}{2}\,\mathrm{V} \approx 1.57\,\mathrm{V},$$

completing 50 full periods each second; this is the prototype of the periodic structure 'rotation → closed-loop magnetic flux → periodic electromotive force'.

Theorem 31.10 (Scalar and vector potential representation of time-varying fields). *In a general time-varying field, introduce a scalar potential $\phi(\mathbf{r}, t)$ and a vector potential $\mathsf{A}(\mathbf{r}, t)$ such that*

$$\boxed{\mathsf{B} = \nabla \times \mathsf{A}}, \qquad \boxed{\mathsf{E} = -\nabla\phi - \frac{\partial \mathsf{A}}{\partial t}}.$$

The electric field E consists of two parts: the negative gradient of the scalar potential $-\nabla\phi$ and the time derivative of the vector potential $-\partial\mathsf{A}/\partial t$.

Proof. By Theorem 31.12, $\nabla \cdot \mathsf{B} = 0$ holds for a time-varying field as well (there are no magnetic monopoles), so B can still be expressed as the curl of some vector potential A: $\mathsf{B} = \nabla \times \mathsf{A}$. Substituting into Faraday's law $\nabla \times \mathsf{E} = -\partial\mathsf{B}/\partial t = -\partial(\nabla \times \mathsf{A})/\partial t$ gives $\nabla \times (\mathsf{E} + \partial\mathsf{A}/\partial t) = \mathbf{0}$. A vector field with zero curl can be expressed as the gradient of some scalar field, so there exists a scalar potential ϕ with $\mathsf{E} + \partial\mathsf{A}/\partial t = -\nabla\phi$, that is, $\mathsf{E} = -\nabla\phi - \partial\mathsf{A}/\partial t$. ☐

Property 31.2 (Gauge freedom). *The definitions of the scalar potential ϕ and the vector potential A are not unique. For any smooth function $\chi(\mathbf{r}, t)$, the transformation*

$$\mathsf{A}' = \mathsf{A} + \nabla\chi, \qquad \phi' = \phi - \frac{\partial\chi}{\partial t}$$

leaves E and B unchanged. This freedom is called **gauge freedom**.

Proof. Direct substitution verifies this: $\mathsf{B}' = \nabla \times \mathsf{A}' = \nabla \times \mathsf{A} + \nabla \times (\nabla\chi) = \mathsf{B} + \mathbf{0} = \mathsf{B}$ (using $\nabla \times (\nabla\chi) = \mathbf{0}$); $\mathsf{E}' = -\nabla\phi' - \partial\mathsf{A}'/\partial t = -\nabla\phi + \nabla(\partial\chi/\partial t) - \partial\mathsf{A}/\partial t - \partial(\nabla\chi)/\partial t = -\nabla\phi - \partial\mathsf{A}/\partial t = \mathsf{E}$ (interchanging the order of ∇ and ∂_t). ☐

Remark 31.10 (Lorentz gauge). *In time-varying field problems one often adopts the* **Lorentz gauge**

$$\nabla \cdot \mathsf{A} + \frac{1}{c^2}\frac{\partial\phi}{\partial t} = 0.$$

The advantage of this gauge is that it decouples the equations for ϕ and A into two independent inhomogeneous wave equations:

$$\Box\phi = -\frac{\rho}{\varepsilon_0}, \qquad \Box\mathsf{A} = -\mu_0\mathsf{J}, \qquad \Box \equiv \nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}.$$

This makes possible the derivation of the retarded potential (Theorem 31.17).

31.5 The Lorentz force

Axiom 31.1 (Lorentz force law).

$$\boxed{\mathsf{F} = q(\mathsf{E} + \mathsf{v} \times \mathsf{B})}.$$

Remark 31.11 (Status of the experimental law and the nature of the two forces). The Lorentz force law is a fundamental experimental law of classical electrodynamics (on a par with Coulomb's law and the Biot–Savart law); it cannot be derived deductively from the remaining assumptions of this chapter, and is therefore taken here as an axiomatic input. Its electric part $\mathsf{F}_E = q\mathsf{E}$ acts on a charge along the direction of the electric field and can do work on the charge (changing its kinetic energy); its magnetic part $\mathsf{F}_B = q\mathsf{v} \times \mathsf{B}$ is perpendicular to the velocity of the charge and changes only the direction of motion, not the speed—that the magnetic force does no work follows directly from $\mathsf{v} \cdot (q\mathsf{v} \times \mathsf{B}) = 0$. This is the fundamental difference between the electric and magnetic fields in their mechanical action.

Proposition 31.5 (Circular motion of a charge in a uniform magnetic field). The cyclotron frequency is $\omega_c = |q|B/m$, and the cyclotron period $T_c = 2\pi/\omega_c$ is independent of the speed.

Proof. Taking $\mathsf{v} \perp \mathsf{B}$, the Lorentz force provides the centripetal force (Axiom 31.1), giving

$$|q|vB = m\frac{v^2}{r} \quad \Longrightarrow \quad r = \frac{mv}{|q|B}.$$

The cyclotron angular frequency is

$$\omega_c = \frac{v}{r} = \frac{|q|B}{m},$$

and the corresponding cyclotron period is

$$T_c = \frac{2\pi}{\omega_c} = \frac{2\pi m}{|q|B},$$

which is indeed independent of the speed v (relativistic corrections aside). The equation of motion

$$\dot{\mathsf{v}} = \frac{q}{m}\mathsf{v} \times \mathsf{B}$$

has the solution

$$\mathsf{v}(t) = v_0\big(\cos\omega_c t,\, -\sin\omega_c t,\, 0\big),$$

a periodic function of period T_c, which is precisely the mathematical statement of the resonance principle of the cyclotron. □

31.6 Derivation of Maxwell's equations

Theorem 31.11 (Derivation of Gauss's law for the electric field). Gauss's law for the electric field is derived starting from Coulomb's law. A point charge q at the origin produces at distance r the electric field

$$\mathsf{E}(\mathsf{r}) = \frac{q}{4\pi\varepsilon_0}\frac{\hat{\mathsf{r}}}{r^2}.$$

For any closed surface S enclosing the charge, the electric flux is

$$\oint_S \mathsf{E}\cdot d\mathsf{S} = \frac{q}{\varepsilon_0}.$$

Applying the divergence theorem, for a continuous charge distribution $\rho(\mathbf{r})$:

$$\int_V \nabla\cdot\mathsf{E}\,dV = \oint_S \mathsf{E}\cdot d\mathsf{S} = \frac{1}{\varepsilon_0}\int_V \rho\,dV.$$

Since the volume V is arbitrary, we obtain

$$\boxed{\nabla\cdot\mathsf{D} = \rho}, \quad \mathsf{D} = \varepsilon_0\mathsf{E}.$$

The divergence $\nabla\cdot\mathsf{D}$ corresponds directly to the charge density ρ—charge is the source of the divergence of the electric displacement vector.

Proof. The electric field of a point charge is given by Coulomb's law. For the electric flux through a closed surface S: use the solid-angle integral $\oint_S \hat{\mathsf{r}}\cdot d\mathsf{S}/r^2 = 4\pi$ (if S encloses the charge) or 0 (if S does not enclose the charge). For a continuous distribution ρ, linear superposition gives $\oint_S \mathsf{E}\cdot d\mathsf{S} = Q_{\mathsf{enc}}/\varepsilon_0 = \frac{1}{\varepsilon_0}\int_V \rho dV$. After conversion by the divergence theorem, the arbitrariness of V gives the differential form. ☐

Theorem 31.12 (Derivation of Gauss's law for the magnetic field)**.** *Gauss's law for the magnetic field is derived starting from the fact that no magnetic monopole exists. All magnetic induction lines are closed curves, and the magnetic flux entering any closed surface equals the flux leaving it:*

$$\oint_S \mathsf{B}\cdot d\mathsf{S} = 0.$$

Applying the divergence theorem:

$$\int_V \nabla\cdot\mathsf{B}\,dV = 0.$$

Since V is arbitrary:

$$\boxed{\nabla\cdot\mathsf{B} = 0}.$$

That is, the magnetic flux density is a divergence-free field—the magnetic field has no divergence source.

Proof. The non-existence of magnetic monopoles means that magnetic field lines have neither beginning nor end. The net magnetic flux through any closed surface is zero. The divergence theorem and the arbitrariness of V give the differential form directly. ☐

Theorem 31.13 (Derivation of Faraday's law)**.** *Faraday's law of electromagnetic induction is an experimental law, whose differential form is*

$$\boxed{\nabla\times\mathsf{E} = -\frac{\partial\mathsf{B}}{\partial t}}.$$

It is derived from the integral form of the rate of change of magnetic flux.

Proof. Faraday's experimental law (integral form): $\mathcal{E} = \oint_C \mathsf{E} \cdot d\mathsf{l} = -\frac{d}{dt}\int_S \mathsf{B} \cdot d\mathsf{S}$. Applying Stokes' theorem, $\oint_C \mathsf{E} \cdot d\mathsf{l} = \int_S (\nabla \times \mathsf{E}) \cdot d\mathsf{S}$. The arbitrariness of S gives the differential form. ☐

Theorem 31.14 (Derivation of the Ampère–Maxwell law and the displacement current)**.** Ampère's law of magnetostatics is taken as the starting point, and the displacement current term is derived through the constraint of the charge conservation law.

Contradiction: Ampère's law of magnetostatics $\nabla \times \mathsf{B} = \mu_0 \mathsf{J}$ contains an intrinsic contradiction in a time-varying field. Taking the divergence of both sides:

$$\nabla \cdot (\nabla \times \mathsf{B}) = 0 = \mu_0 \nabla \cdot \mathsf{J}.$$

This requires $\nabla \cdot \mathsf{J} = 0$ to hold at all times. But the charge conservation law requires

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathsf{J} = 0.$$

When $\partial\rho/\partial t \neq 0$ (the charge density varies with time), $\nabla \cdot \mathsf{J} \neq 0$, which contradicts Ampère's law of magnetostatics.

Correction: Maxwell introduced the **displacement current** term $\partial\mathsf{D}/\partial t$, generalising Ampère's law to

$$\boxed{\nabla \times \mathsf{H} = \mathsf{J} + \frac{\partial \mathsf{D}}{\partial t}}.$$

Checking self-consistency: taking the divergence of the corrected equation:

$$\nabla \cdot (\nabla \times \mathsf{H}) = 0 = \nabla \cdot \mathsf{J} + \frac{\partial}{\partial t}(\nabla \cdot \mathsf{D}) = \nabla \cdot \mathsf{J} + \frac{\partial \rho}{\partial t}.$$

This is completely consistent with the charge conservation law. The displacement current $\partial\mathsf{D}/\partial t$ converts the time rate of change of the electric field into a source of the curl of the magnetic field—this is the coupling of the electric and magnetic fields in the time-varying case.

Proof. **Step one (contradiction)**: $\nabla \cdot (\nabla \times \mathsf{B}) = 0$ is an identity. Substituting $\nabla \times \mathsf{B} = \mu_0 \mathsf{J}$ gives $\nabla \cdot \mathsf{J} = 0$. But the continuity of current requires $\nabla \cdot \mathsf{J} = -\partial\rho/\partial t$, which is a contradiction when $\partial\rho/\partial t \neq 0$.

Step two (introduction of the displacement current): to remove the contradiction, add a term J_d to the right-hand side of Ampère's law so that $\nabla \cdot (\mathsf{J} + \mathsf{J}_d) = 0$. By charge conservation, $\nabla \cdot \mathsf{J} = -\partial\rho/\partial t = -\partial(\nabla \cdot \mathsf{D})/\partial t = -\nabla \cdot (\partial\mathsf{D}/\partial t)$. Hence $\mathsf{J}_d = \partial\mathsf{D}/\partial t$ exactly satisfies the requirement.

Step three (self-consistency): the divergence of the corrected equation is zero, which is consistent with the charge conservation law. ☐

Corollary 31.2 (Automatic satisfaction of the charge conservation law)**.** Taking the divergence of both sides of the Ampère–Maxwell law:

$$\nabla \cdot (\nabla \times \mathsf{H}) = 0 = \nabla \cdot \mathsf{J} + \frac{\partial}{\partial t}(\nabla \cdot \mathsf{D}).$$

Substituting Gauss's law $\nabla \cdot \mathsf{D} = \rho$, we obtain

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot \mathsf{J} = 0}.$$

The charge conservation law is a self-consistency condition of Maxwell's equations.

Theorem 31.15 (Maxwell's equations (complete form)).

$$\boxed{\nabla \cdot \mathsf{D} = \rho}, \quad \boxed{\nabla \cdot \mathsf{B} = 0},$$

$$\boxed{\nabla \times \mathsf{E} = -\frac{\partial \mathsf{B}}{\partial t}}, \quad \boxed{\nabla \times \mathsf{H} = \mathsf{J} + \frac{\partial \mathsf{D}}{\partial t}}.$$

The four equations correspond respectively to the action of four vector differential operators: the two divergence equations (Gauss's laws) describe the relation between the field and its sources; the two curl equations (Faraday's law and the Ampère–Maxwell law) describe the relation between the circulation of the field and its sources (time-varying fields, currents).

Proof. This is the summary of the four formulas of Theorems 31.11, 31.12, 31.13 and 31.14. Of the four equations, two are divergence equations and two are curl equations; each is derived independently, and they are stated here grouped by the type of operator. ☐

31.7 Periodic structure of electromagnetic waves and propagation delay

Theorem 31.16 (Electromagnetic wave equation). *In a source-free region ($\rho = 0, \mathsf{J} = 0$), Maxwell's equations reduce to the electromagnetic wave equation:*

$$\boxed{\nabla^2 \mathsf{E} - \frac{1}{c^2}\frac{\partial^2 \mathsf{E}}{\partial t^2} = 0}, \quad c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}.$$

Proof. Taking the curl of Faraday's law: $\nabla \times (\nabla \times \mathsf{E}) = -\frac{\partial}{\partial t}(\nabla \times \mathsf{B})$. Substituting the Ampère–Maxwell law (source-free) and the constitutive relations, and using the vector identity $\nabla \times (\nabla \times \mathsf{E}) = \nabla(\nabla \cdot \mathsf{E}) - \nabla^2 \mathsf{E}$ and Gauss's law (source-free: $\nabla \cdot \mathsf{E} = 0$), we obtain the wave equation. ☐

Definition 31.11 (Propagation delay). *The **propagation delay** of an electromagnetic wave from a source point r' to an observation point r is*

$$\tau_{\text{prop}} = \frac{|\mathsf{r} - \mathsf{r}'|}{c}.$$

The propagation delay converts a spatial distance into a time offset—the greater the distance, the greater the delay.

Definition 31.12 (Phase delay). *In a monochromatic plane wave, the **phase delay** corresponding to the spatial phase is*

$$\tau_{\text{phase}} = \frac{\mathsf{k} \cdot \mathsf{r}}{\omega} = \frac{|\mathsf{r}| \cos\theta}{c}.$$

Along the direction of propagation ($\theta = 0$), $\tau_{\text{phase}} = r/c = \tau_{\text{prop}}$. The phase delay converts the phase lag $\phi = \omega\tau$ into a time offset.

Proposition 31.6 (Unity of delay and phase)**.** *Along the direction of propagation, every increase of the spatial phase by 2π corresponds to an increase of the time offset by one period:*

$$\tau = \frac{\lambda}{c} = \frac{2\pi}{\omega} = T.$$

Delay couples the spatial period (wavelength) and the temporal period through the speed of light.

Proof. Along the direction of propagation ($\theta = 0$), by Definition 31.12 the phase delay is $\tau_{\text{phase}} = |\mathsf{r}|/c$. Every increase of the spatial phase by 2π corresponds to a displacement of one wavelength λ, and the time required is the propagation delay $\tau = \lambda/c$. Moreover, by the dispersion relation $\omega = c|\mathsf{k}|$ and the wavelength $\lambda = 2\pi/|\mathsf{k}|$,

$$\tau = \frac{\lambda}{c} = \frac{2\pi}{|\mathsf{k}|\, c} = \frac{2\pi}{\omega} = T.$$

Hence the spatial period (wavelength) and the temporal period are exactly coupled by the speed of light: $\lambda = cT$. ☐

Proposition 31.7 (Periodic structure of plane electromagnetic waves)**.**

$$\mathsf{E}(\mathsf{r}, t) = \mathsf{E}_0 e^{i(\mathsf{k}\cdot\mathsf{r} - \omega t)}, \quad \mathsf{B}(\mathsf{r}, t) = \mathsf{B}_0 e^{i(\mathsf{k}\cdot\mathsf{r} - \omega t)}.$$

The spatial period is $\lambda = 2\pi/|\mathsf{k}|$, the temporal period is $T = 2\pi/\omega$, the phase velocity is $v_p = c$, and the propagation delay is $\tau = r/c$.

Proof. Substituting the plane wave into the electromagnetic wave equation (Theorem 31.16):

$$\nabla^2 \mathsf{E} = -|\mathsf{k}|^2 \mathsf{E}, \qquad \frac{1}{c^2}\frac{\partial^2 \mathsf{E}}{\partial t^2} = -\frac{\omega^2}{c^2}\mathsf{E}.$$

The equality of the two requires $|\mathsf{k}|^2 = \omega^2/c^2$, that is, the dispersion relation $\omega = c|\mathsf{k}|$; hence the phase velocity $v_p = \omega/|\mathsf{k}| = c$. **Spatial periodicity**: since $|\mathsf{k}|\lambda = 2\pi$, for any integer n we have $e^{i\mathsf{k}\cdot(\mathsf{r}+n\lambda\hat{\mathsf{k}})} = e^{i\mathsf{k}\cdot\mathsf{r}} e^{in|\mathsf{k}|\lambda} = e^{i\mathsf{k}\cdot\mathsf{r}}$, so the spatial period is $\lambda = 2\pi/|\mathsf{k}|$. **Temporal periodicity**: since $\omega T = 2\pi$, $e^{-i\omega(t+nT)} = e^{-i\omega t}e^{-in\omega T} = e^{-i\omega t}$, so the temporal period is $T = 2\pi/\omega$. The propagation delay $\tau = r/c$ is given by Definition 31.11. ☐

31.8 The retarded potential: *electromagnetic realisation of delay*

Lemma 31.6 (The retarded potential satisfies the inhomogeneous wave equation)**.** *The scalar retarded solution*

$$\phi(\mathsf{r}, t) = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho(\mathsf{r}', t - |\mathsf{r} - \mathsf{r}'|/c)}{|\mathsf{r} - \mathsf{r}'|}\, dV'$$

satisfies the inhomogeneous wave equation

$$\Box\,\phi \equiv \left(\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)\phi = -\frac{\rho}{\varepsilon_0}.$$

Proof. The retarded Green's function

$$G(\mathbf{r},t;\mathbf{r}',t') = \frac{\delta(t'-t+|\mathbf{r}-\mathbf{r}'|/c)}{4\pi|\mathbf{r}-\mathbf{r}'|}$$

satisfies $\Box G = -\delta^{(3)}(\mathbf{r}-\mathbf{r}')\,\delta(t-t')$. Its verification proceeds in two steps: the static part is given by $\nabla^2\big(\frac{1}{4\pi|\mathbf{r}-\mathbf{r}'|}\big) = -\delta^{(3)}(\mathbf{r}-\mathbf{r}')$; for the delay part, differentiating the composite function with $R = |\mathbf{r}-\mathbf{r}'|$ and $t_{\text{ret}} = t - R/c$, the leading terms of $\nabla^2\delta(t_{\text{ret}}-t')$ and $\frac{1}{c^2}\partial_t^2\delta(t_{\text{ret}}-t')$ in the wave operator cancel on the retarded sphere, leaving only the point-source term $-\delta^{(3)}\delta$. Substituting $\phi = \frac{1}{\varepsilon_0}\iint G\,\rho(\mathbf{r}',t')\,dV'dt'$ and using the sifting property of δ gives $\Box\phi = -\rho/\varepsilon_0$. □

Theorem 31.17 (Retarded potential). *Under the Lorentz gauge (Remark 31.10), the scalar and vector potentials of the electromagnetic field are*

$$\phi(\mathbf{r},t) = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho(\mathbf{r}',t-|\mathbf{r}-\mathbf{r}'|/c)}{|\mathbf{r}-\mathbf{r}'|}dV',$$

$$\mathbf{A}(\mathbf{r},t) = \frac{\mu_0}{4\pi}\int \frac{\mathbf{J}(\mathbf{r}',t-|\mathbf{r}-\mathbf{r}'|/c)}{|\mathbf{r}-\mathbf{r}'|}dV'.$$

The retarded time $t_{\text{ret}} = t - |\mathbf{r}-\mathbf{r}'|/c$ encodes the propagation delay.

Proof. Under the Lorentz gauge $\nabla\cdot\mathbf{A} + \frac{1}{c^2}\frac{\partial\phi}{\partial t} = 0$, Maxwell's equations reduce to the two inhomogeneous wave equations $\Box\phi = -\rho/\varepsilon_0$ and $\Box\mathbf{A} = -\mu_0\mathbf{J}$. By Lemma 31.6, the solution for the scalar potential is the ϕ above; the solution for the vector potential $\mathbf{A}$ is obtained by applying the same lemma componentwise to the scalar solution. The retarded time $t_{\text{ret}} = t - |\mathbf{r}-\mathbf{r}'|/c$ encodes the propagation delay: the state of the source at $\mathbf{r}'$ at time t_{ret} reaches, only after the delay $|\mathbf{r}-\mathbf{r}'|/c$, the observation point $\mathbf{r}$, that is, what the observer 'sees' is the past state of the source. □

Property 31.3 (Manifestation of three operations in the retarded potential). *In the expression for the retarded potential one can identify three vector differential operations and their corresponding geometric effects:*

(i) **Time translation**: *$\tau = r/c$ converts a spatial distance into a time offset, relating the past state of the source to the current field value at the observation point.*

(ii) **Phase relation**: *$\phi = \omega\tau$ converts the delay into a phase lag, and is the source of the coherence of electromagnetic waves.*

(iii) **Inverse-square scaling**: *the factor $1/|\mathbf{r}-\mathbf{r}'|$ is the amplitude decay caused by the spreading of the field of a point source in three-dimensional space.*

31.9 Conservation laws of electromagnetic systems

Theorem 31.18 (Three conservation laws of electromagnetic systems). *(i)* **Charge conservation**: *$\partial\rho/\partial t + \nabla\cdot\mathbf{J} = 0$.*

(ii) **Energy conservation**: *$\frac{\partial u}{\partial t} + \nabla\cdot\mathbf{S} = -\mathbf{J}\cdot\mathbf{E}$.*

(iii) **Momentum conservation**: $\mathsf{g} = \varepsilon_0 \mathsf{E} \times \mathsf{B}$ satisfies the momentum conservation law.

Proof. **(i) Charge conservation**: taking the divergence of the Ampère–Maxwell law gives it (Corollary 31.2), so $\partial\rho/\partial t + \nabla \cdot \mathsf{J} = 0$.

(ii) Energy conservation (Poynting's theorem): the energy density is $u = \frac{1}{2}(\mathsf{E} \cdot \mathsf{D} + \mathsf{B} \cdot \mathsf{H})$ and the Poynting vector is $\mathsf{S} = \mathsf{E} \times \mathsf{H}$. Compute

$$\frac{\partial u}{\partial t} = \mathsf{E} \cdot \frac{\partial \mathsf{D}}{\partial t} + \mathsf{H} \cdot \frac{\partial \mathsf{B}}{\partial t}.$$

Substituting the Ampère–Maxwell law $\partial\mathsf{D}/\partial t = \nabla \times \mathsf{H} - \mathsf{J}$ and Faraday's law $\partial\mathsf{B}/\partial t = -\nabla \times \mathsf{E}$:

$$\frac{\partial u}{\partial t} = \mathsf{E} \cdot (\nabla \times \mathsf{H}) - \mathsf{H} \cdot (\nabla \times \mathsf{E}) - \mathsf{J} \cdot \mathsf{E}.$$

By the vector identity $\nabla \cdot (\mathsf{E} \times \mathsf{H}) = \mathsf{H} \cdot (\nabla \times \mathsf{E}) - \mathsf{E} \cdot (\nabla \times \mathsf{H})$, we obtain

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathsf{S} = -\mathsf{J} \cdot \mathsf{E}.$$

The right-hand side is the power density of the work done by the field on the current; energy is converted between the electromagnetic field and the charges.

(iii) Momentum conservation: the electromagnetic momentum density is $\mathsf{g} = \varepsilon_0 \mathsf{E} \times \mathsf{B}$. Combining Maxwell's equations with the Lorentz force density $\mathsf{f} = \rho\mathsf{E} + \mathsf{J} \times \mathsf{B}$ and simplifying by vector identities gives the differential form of momentum conservation $\partial g_i/\partial t + \partial_j T_{ij} = -f_i$, where T_{ij} is the Maxwell stress tensor; g together with the stress tensor constitutes the momentum–stress conservation structure of the electromagnetic field. □

Remark 31.12 (Connection with the rest of the book). The theory of the periodic structure of electromagnetic systems established in this chapter is an application of periodic mathematics in classical electrodynamics. The derivation of Maxwell's equations reveals the cooperative action of the vector differential operators (gradient, divergence, curl) in field theory: the introduction of the displacement current guarantees the self-consistency of the charge conservation law in a time-varying field. The introduction of the scalar and vector potentials expresses E and B as combinations of the gradient, the curl and the time derivative of the potentials, whilst gauge freedom embodies the indistinguishability of two sets of potentials representing the same physical field. An electromagnetic wave is a periodic structure in which the electric and magnetic fields propagate self-sustainedly in a source-free region, and the concept of delay couples the spatial period (wavelength) with the temporal period (frequency) through the speed of light. The retarded potential is the electromagnetic realisation of delay—it relates the past state of the source to the current observation of the observer. Charge conservation, energy conservation and momentum conservation are respectively the conservation laws corresponding to the three vector differential operators in electromagnetism.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The electromagnetic propagation delay $\tau = r/c$ and the retarded potential (Definition 31.11, Theorem 31.17)
Scaling	The inverse-square scaling of Coulomb's and Ampère's laws $\sim 1/r^2$ (Theorem 31.11)
Rotation	The curl of the magnetic field $\nabla \times \mathrm{B}$ and the cyclotron motion driven by the Lorentz force (Theorem 31.7, Proposition 31.5)
Balance	Charge conservation, Poynting energy conservation and electromagnetic momentum conservation (Theorem 31.18)
Symmetry	Gauge freedom and the structural symmetry of Maxwell's equations (Proposition 31.2, Theorem 31.15)
Closure	The closed integrals of the Ampère loop and the Faraday loop (Theorem 31.7, 31.9)
Period	The period of electromagnetic oscillation and the space–time period of electromagnetic waves (Proposition 31.7)

Exercises

Exercise 1 Prove the **electromagnetic periodicity decision theorem** of this chapter (31.1). Hint: first write down the decision map and the evolution group of this chapter's target, then apply the proof structure of the general criterion of the 'Periodic Systems' chapter.

Exercise 2 Verify that the hypotheses of the **periodicity decision for a monochromatic plane wave** (31.2) hold for the domain objects of this chapter, and give a counterexample in which the hypotheses fail.

Exercise 3 Examine the behaviour of **monotonicity of quantity adjustment** (31.2) under degeneracy of the parameters, and explain how the conclusion degenerates accordingly.

Exercise 4 Explain the relation between the **three-parameter correspondence: radius, phase and rotational speed** (31.1) and the corresponding general conclusions of the 'Periodic Systems' chapter (whether they agree, whether it is a special case, or whether additional conditions are required).

Exercise 5 Using the seven-theme comparison table of this chapter, identify the concrete carriers in this chapter of the three themes **translation**, **scaling** and **rotation**, and give one example of domain data for each.

Exercise 6 Synthesis: *restate the diagnostic procedure of this chapter using the five-step standard interface of the 'Periodic Systems' chapter (Table 24.1), noting for each step the theorem of this chapter on which it depends:* 31.1, 31.2, 31.2, 31.1.

Chapter 32

Periodic Structure of Audio Systems

Remark 32.1 (Periodicity decision for the target of this chapter (citation)). When this chapter decides whether a research target is periodic, determines its period type, or seeks its minimal positive period, reference is always made to the 'Periodic Systems' chapter: for the periodicity criterion see Theorem 24.1, for the discrimination of period types see Definition 24.3, Proposition 24.1, for the search for the minimal positive period see Definition 24.4, Theorem 24.2, and for the construction of period series and epicycle groups from data see Theorem 24.3, Definition 24.6.

Remark 32.2 (Positioning of this chapter: developed along the five-step interface of the 'Periodic Systems' chapter). The periodic structure of audio systems studies the mathematical characterisation of periodicity, harmonic structure and spectral features in sound waves. This chapter does not establish a separate periodicity criterion; instead it connects audio systems to the five-step interface prescribed by the 'Periodic Systems' chapter:

(1) **Decision**: reduce such targets as simple harmonic motion, single-tone sound waves, beats and standing waves to discrete or continuous dynamical systems, invoke Theorem 24.1 to decide whether they are periodic, and determine the period type;

(2) **Construction**: build the amplitude series, the frequency series and the phase series from sound-pressure, particle-velocity, spectral and phase data, and assemble them into the audio epicycle group;

(3) **Adjustment**: dynamically adjust the audio epicycle group subject to residual-energy descent, mass conservation, energy conservation and scale-charge conservation;

(4) **Convolution**: introduce amplitude convolution, phase convolution and frequency convolution on the audio epicycle group, and link them to autocorrelation, cross-correlation and beat-frequency delay;

(5) **Integration**: connect to the tool chains of the three chapters 'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis', forming a diagnostic procedure for audio periodicity.

The mathematical tools used in this chapter include: simple harmonic motion, the acoustic wave equation, Fourier decomposition, periodic convolution, epicycle groups, phase operators, delay and equal temperament.

Remark 32.3 (Guide to the seven themes of this chapter). **Translation**—the space–time translational correspondence of the acoustic propagation delay $\tau = r/c$ (Definition 32.13); **Scaling**—the geometric scale of equal temperament with common ratio $2^{1/12}$ (Theorem 32.8); **Rotation**—the phase-space circulation of simple harmonic motion (Definition 32.11) and the phase delay $\tau = \phi/\omega$ (Definition 32.12); **Conservation**—mass, energy and scale-charge conservation (Theorem 32.9) and energy conservation in simple harmonic motion (Proposition 32.5); **Symmetry**—the spatial reflection symmetry of standing waves and the fixed boundary at both ends of a string; **Closure**—periodic closure on every increase of the phase by 2π (Proposition 32.6); **Period**—the fundamental frequency and the harmonic series (Definition 32.15), and the beat period $T_{\mathsf{beat}} = 1/\Delta f$ (Definition 32.18).

32.1 Connection to the five-step interface of the 'Periodic Systems' chapter

32.1.1 Step one: periodicity decision for the audio target

Definition 32.1 (Audio target and decision input)**.** *An* **audio target** *is one of the following objects:*

(i) *a simple harmonic motion system, whose state space is the (x, p) phase space;*

(ii) *a single-tone sound wave, whose state space is the phase $\theta = \mathbf{k}\cdot\mathbf{r} - \omega t$;*

(iii) *a beat-frequency system, whose state space is the phase difference of two frequency components;*

(iv) *a standing-wave system, whose state space is the joint space of spatial modes and temporal phase.*

The **decision input** *of an audio target is the pair*

$$\mathcal{D}_{\mathsf{A}} = (\mathcal{S}, \Phi),$$

where $\mathcal{S}$ is the state space and $\Phi : \mathcal{S} \to \mathcal{S}$ is the map induced by time translation, phase rotation or vibration evolution. If $\mathcal{S}$ is a finite set, the target is called a **finite audio target***; if $\mathcal{S}$ is an infinite set, the target is called an* **infinite audio target***.*

Definition 32.2 (Audio decision map)**.** *Let an audio target be given by the decision input $\mathcal{D}_{\mathsf{A}} = (\mathcal{S}, \Phi)$. If the states of the target are generated by Φ, that is, if there exists $x_0 \in \mathcal{S}$ such that*

$$x_t = \begin{cases} \Phi^t(x_0), & t \in \mathbb{Z}, \\ \Phi_t(x_0), & t \in \mathbb{R}, \end{cases}$$

for $t \in \mathbb{Z}$ the state x_t is generated by iteration of Φ; for $t \in \mathbb{R}$ it is described by the one-parameter evolution family $\{\Phi_t\}_{t\in\mathbb{R}}$, where each $\Phi_t : \mathcal{S} \to \mathcal{S}$, $\Phi_0 = \mathrm{id}$, $\Phi_{s+t} = \Phi_s \circ \Phi_t$ for all $s, t \in \mathbb{R}$, and $(t, x) \mapsto \Phi_t(x)$ is continuous. then Φ is called the **audio decision map** *of the target. The audio target is then reduced to the discrete dynamical system $(\mathcal{S}, \Phi)$ when $t \in \mathbb{Z}$ and to the continuous dynamical system $(\mathcal{S}, \{\Phi_t\}_{t\in\mathbb{R}})$ when $t \in \mathbb{R}$; in both cases the criteria of the 'Operations and Periodicity' chapter and of the 'Periodic Systems' chapter may be applied directly.*

Theorem 32.1 (Audio periodicity decision theorem)**.** *Let the decision map of an audio target be Φ, and let $(\mathcal{S}, \Phi)$ be a discrete dynamical system. Then:*

(A) **Sufficient decision**: *if there exists $m \in \mathbb{N}^+$ with $\Phi^m = \mathsf{id}$, then the target is a uniformly periodic audio system; every orbit is closed and the orbit length divides m.*

(B) **Equivalence of the decision**: *the target is a uniformly periodic audio system if and only if* $\mathsf{ord}(\Phi) < \infty$.

(C) **Pointwise criterion**: *the orbit of the target at a state x_0 is closed if and only if x_0 is a periodic point; the orbit length is then the period at that point.*

(D) **Non-omissibility**: *the conjunction of conserved quantities and symmetry groups cannot replace* $\mathsf{ord}(\Phi) < \infty$*; a quasi-periodic audio target is not a periodic audio system.*

Proof. The Theorem 32.1 of this chapter is the domain instantiation of Theorem 24.1 in the 'Periodic Systems' chapter: it suffices to replace the decision map f and the evolution group by Φ of this chapter, and the four parts (A) sufficient decision, (B) equivalence of the decision, (C) pointwise criterion and (D) non-omissibility then hold one by one. By the convention of Remark 24.4, they are not expanded again here. ☐

Corollary 32.1 (Audio period types)**.** *After the decision of Theorem 32.1, an audio target falls into one of the following types:*

(i) **Trivial periodic type**: *least period $m_0 = 1$, the state is a fixed point;*

(ii) **Strictly periodic type**: *$1 < m_0 < \infty$, the state is strictly periodic;*

(iii) **Uniformly periodic type**: $\mathsf{ord}(\Phi) = m < \infty$*, every orbit length divides m;*

(iv) **Quasi-periodic type**: $\mathsf{ord}(\Phi) = \infty$*, the orbits are not closed but are dense on a compact carrier structure;*

(v) **Eventually periodic type**: *there exists N such that the target is periodic for $t \geq N$;*

(vi) **Discrete periodic type**: *the carrier structure is $\mathbb{Z}$ and the normalised period ratio is $\rho = T_x/\Delta x \in \mathbb{Q}$.*

(vii) **Asymptotically periodic (damped) type**: *the orbit is not closed, but tends asymptotically to a closed orbit or to a fixed point, the residual decaying as $e^{-\kappa t}$ (see Definition 24.3(vii)).*

Here (i)–(iii) and (vi) belong to periodic audio systems; (iv) does not belong to periodic audio systems but can be handled with bounded-approximation tools; (v) is a periodic audio system only in its stable stage; (vii), like (iv), is a type in the limiting sense; it is not a periodic audio system and can be handled with bounded-approximation tools only in its stable stage.

Proof. Each item follows by substituting the audio target into Theorem 32.1 and Definition 24.3 of the 'Periodic Systems' chapter. ☐

Lemma 32.1 (Periodicity decision for simple harmonic motion)**.** *Let the equation of simple harmonic motion be*

$$\ddot{x} + \omega_0^2 x = 0.$$

Then simple harmonic motion is a uniformly periodic system, whose least positive period is

$$T_0 = \frac{2\pi}{\omega_0} = 2\pi\sqrt{\frac{m}{k}}.$$

Proof. By Definition 32.11 the general solution is $x(t) = A\cos(\omega_0 t + \phi)$. The time translation $T_0 = 2\pi/\omega_0$ gives $x(t + T_0) = x(t)$, and T_0 is the least positive period. Hence the decision map Φ satisfies $\Phi^{T_0} = \mathrm{id}$, so the system is uniformly periodic. ☐

Lemma 32.2 (Periodicity decision for a single-tone sound wave)**.** *Let a single-tone plane sound wave be*

$$p(\mathbf{r}, t) = p_0 e^{i(\mathbf{k}\cdot\mathbf{r} - \omega t)}.$$

Then its temporal period is

$$T = \frac{2\pi}{\omega},$$

its spatial period is

$$\lambda = \frac{2\pi}{|\mathbf{k}|},$$

and the two are coupled by the dispersion relation $\omega = c|\mathbf{k}|$ as $\lambda = cT$.

Proof. From $e^{-i\omega(t+T)} = e^{-i\omega t}e^{-i\omega T} = e^{-i\omega t}$ we obtain the temporal period $T = 2\pi/\omega$. From $e^{i\mathbf{k}\cdot(\mathbf{r}+\lambda\hat{\mathbf{k}})} = e^{i\mathbf{k}\cdot\mathbf{r}}e^{i|\mathbf{k}|\lambda} = e^{i\mathbf{k}\cdot\mathbf{r}}$ we obtain the spatial period $\lambda = 2\pi/|\mathbf{k}|$. The dispersion relation is given by Theorem 32.7 and Proposition 32.6. ☐

Lemma 32.3 (Periodicity decision for beats)**.** *Let two sound waves of close frequency be superposed, with frequency difference $\Delta f = |f_1 - f_2|$. Then the amplitude envelope of the resulting sound wave oscillates with the beat period*

$$T_{\text{beat}} = \frac{1}{\Delta f}$$

as its period.

Proof. By Definition 32.18 and Proposition 32.8 the beat period is $T_{\text{beat}} = 2\pi/\Delta\omega = 1/\Delta f$. The amplitude envelope $2A\cos(\Delta\omega\, t/2)$ oscillates with angular frequency $\Delta\omega/2$, so the envelope period is $4\pi/\Delta\omega$; the beat period is half the envelope period, that is, $2\pi/\Delta\omega = 1/\Delta f$. ☐

Definition 32.3 (Audio least positive period algorithm)**.** This definition has the same content as Chapter 24, the section 'Periodicity decision: is the target periodic?' (Section 24.1), Definition 24.4; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement is given there. The only difference here is terminology (audio target / audio sample); the algorithm steps and the conclusion are exactly the same.

Theorem 32.2 (Correctness of the audio least positive period algorithm)**.** This theorem has the same content as Chapter 24, the section 'Periodicity decision: is the target periodic?' (Section 24.1), Theorem 24.2; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement and proof are given there. The only difference here is terminology (audio target / audio sample); the algorithm steps and the conclusion are exactly the same.

Example 32.1 (Examples of audio periodicity decisions)**.** (i) Simple harmonic motion: $T_0 = 2\pi\sqrt{m/k}$, uniformly periodic type;

(ii) Single-tone sound wave: temporal period $T = 2\pi/\omega$, spatial period $\lambda = 2\pi/|\mathsf{k}|$;

(iii) Beats: $T_{\mathsf{beat}} = 1/\Delta f$, uniformly periodic type;

(iv) Standing waves: the frequencies of standing waves on a string are $f_n = nf_1$, uniformly periodic type.

Remark 32.4 (Conserved quantities do not guarantee periodicity)*.* Mass conservation, energy conservation and scale-charge conservation in audio systems are auxiliary conditions for periodic structure, but they do not constitute a sufficient condition. For example, damped vibration involves energy dissipation, yet its orbit need not be closed. The periodic structure of an audio system must be decided by Theorem 32.1 or by the lemmas above.

32.1.2 Step two: construction of the audio period series and epicycle group

Definition 32.4 (Audio amplitude, frequency and phase series)**.** Let the state or output sequence of an audio target be $\{z_t\}$. If it can be written as

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)} + \varepsilon_t,$$

then

$$\mathcal{A}_z(t) = \sum_k A_k e^{ikt}, \qquad \mathcal{F}_z(s) = \sum_k \omega_k s^k, \qquad \mathcal{P}_z(t) = \sum_k e^{i(kt+\phi_k)}$$

are called respectively the **amplitude series**, the **frequency series** and the **phase series** of the audio target. The three agree with Definition 6.10, Definition 6.12 and Definition 6.11 of the 'Series and Convolution' chapter.

Definition 32.5 (Audio epicycle group)**.** Let the period series of an audio target be given by its amplitude series $\{A_k\}$, frequency series $\{\omega_k\}$ and phase series $\{\phi_k\}$. For each k define the epicycle

$$E_k : \theta \mapsto A_k e^{i(\theta+\phi_k)}, \qquad \theta = \omega_k t,$$

whose collection

$$\mathfrak{E}_{\mathsf{A}} = \{(A_k, \omega_k, \phi_k)\}_{k\in\mathbb{Z}}$$

is called the **epicycle group** of the audio target. Its superposition is the audio period series

$$z_t = \sum_k E_k(\omega_k t) = \sum_k A_k e^{i(\omega_k t+\phi_k)}.$$

Theorem 32.3 (Constructing the period series from audio data)**.** *Let the output of an audio target be a finite periodic sequence of period N. Then:*

(i) *the output admits the discrete Fourier expansion*

$$z_t = \sum_{k=0}^{N-1} c_k e^{i2\pi kt/N}, \qquad c_k = \frac{1}{N}\sum_{t=0}^{N-1} z_t e^{-i2\pi kt/N};$$

(ii) *the amplitude series is given by $|c_k|$, the phase series by $\arg c_k$ and the frequency series by $2\pi k/N$;*

(iii) *if the output is a single-tone sound wave, its Fourier decomposition has only a single spectral line; if the output is the timbre of a musical instrument, each order of harmonic corresponds to a different harmonic component.*

Proof. (i) The discrete Fourier series of a finite periodic sequence is a standard result: the basis functions $\{e^{i2\pi kt/N}\}_{k=0}^{N-1}$ form an orthogonal basis of $\mathbb{C}^N$. (ii) This is read off directly from $c_k = A_k e^{i\phi_k}$. (iii) The temporal part of a single-tone sound wave is $e^{-i\omega t}$, which on the period $T = 2\pi/\omega$ has only the fundamental component; the timbre of a musical instrument expands as a Fourier series into the successive harmonics. ☐

Proposition 32.1 (Three-parameter correspondence: radius, phase and speed)**.** *The three parameters of the audio epicycle group $\mathfrak{E}_{\mathsf{A}} = \{(A_k, \omega_k, \phi_k)\}_k$ correspond one-to-one to three geometric–physical quantities:*

(i) **Radius corresponds to amplitude**: *the radius of E_k, namely $|E_k| = A_k$, is the amplitude of the sound-pressure component;*

(ii) **Phase corresponds to rotational delay**: *the initial phase ϕ_k corresponds to the time translation $\tau_k = \phi_k/\omega_k$, that is, the epicycle* **leads** *the reference by τ_k ($\phi_k > 0$ means that the waveform is* **advanced** *by τ_k relative to the reference, equivalently the waveform coincides with the reference once the reference has been delayed by τ_k);*

(iii) **Frequency corresponds to rotational speed**: *ω_k is the angular frequency of that component.*

Proof. The Proposition 32.1 of this chapter corresponds exactly to Proposition 24.3 of the 'Periodic Systems' chapter, and the three correspondences—radius to amplitude, phase to rotational delay, frequency to rotational speed—are obtained by translating them one by one to the epicycle group of this chapter. The proof is given there: (i) from $|E_k| = A_k$ and the energy decomposition, (ii) from the phase translation theorem, (iii) from the definition of the frequency series. ☐

Example 32.2 (Period series of instrumental timbre)**.** Different instruments playing the same pitch produce different timbres, which arises from differences in harmonic structure. A tuning fork has almost only the fundamental frequency, a violin is rich in harmonics, and a clarinet is dominated by odd-order harmonics. In the period series these differences appear as different distributions of the amplitude series $\{A_k\}$.

Remark 32.5 (The period series is a complex function). Let $z = e^{i\omega_0 t}$; then the audio period series

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)}$$

can be written as the Laurent series $\sum_k c_k z^k$, where $c_k = A_k e^{i\phi_k}$, provided that every component frequency is commensurable with ω_0 (that is, $\omega_k = k\omega_0$ with $k \in \mathbb{Z}$). Hence when all components are commensurable with ω_0, the period series is a complex-valued function on the unit circle (otherwise it is a function on the torus).

32.1.3 Step three: dynamic adjustment of the audio epicycle group

Definition 32.6 (Dynamic adjustment of the audio epicycle group). *Let the current audio epicycle group be $\mathfrak{E}^{(j)}$, with partial sum*

$$z^{(j)}(t) = \sum_{k \in \mathfrak{E}^{(j)}} A_k e^{i(\omega_k t + \phi_k)},$$

and residual $r^{(j)} = z - z^{(j)}$. **Dynamic adjustment** *consists of two rules:*

(i) **Quantity adjustment**: *if the residual energy satisfies $\|r^{(j)}\|^2 > \delta$, add a new epicycle according to the peak of the residual periodogram;*

(ii) **Parameter adjustment**: *update the parameters of the existing epicycles by regularised least squares or by the conservation constraints.*

In audio systems, quantity adjustment corresponds to increasing the harmonic order or the number of frequency components; parameter adjustment corresponds to adjusting amplitude, phase and frequency so that the mass, energy and scale-charge constraints are satisfied.

Proposition 32.2 (Monotonicity of quantity adjustment). *After the dominant residual frequency component is added as in Definition 32.6(i), the residual energy is monotonically non-increasing:*

$$\|r^{(j+1)}\|^2 \leq \|r^{(j)}\|^2,$$

with equality if and only if the residual has no energy at the selected frequency.

Proof. The Proposition 32.2 of this chapter agrees with the dynamic adjustment mechanism of Definition 24.8 in the 'Periodic Systems' chapter: adding an epicycle removes the component of $r^{(j)}$ along $\hat{\omega}$ from the residual, and the orthogonal decomposition $\|r^{(j)}\|^2 = \|r^{(j+1)}\|^2 + \|\text{the removed component}\|^2$ gives the inequality, with equality if and only if the norm of the removed component is zero. The complete argument is given there. □

Theorem 32.4 (Convergence of dynamic adjustment). *Let the audio target be a* **real-valued finite periodic sequence** *of length T_0 ($T_0 \in \mathbb{N}$) and let $\delta = 0$. Then after at most $\lfloor T_0/2 \rfloor + 1$ steps of dynamic adjustment the residual is zero, and the resulting audio epicycle group is the exact Fourier data of the target; for a* **complex-valued** *sequence the corresponding number of steps is at most T_0.*

Proof. The discrete Fourier series of a **real-valued** finite periodic sequence of length T_0 ($T_0 \in \mathbb{N}$) is complete on $\lfloor T_0/2 \rfloor + 1$ harmonics (including the constant term); a **complex-valued** sequence requires T_0 components. At each step one dominant frequency component of the residual is extracted as in Proposition 32.2 and the residual energy decreases strictly, so after finitely many steps the residual is zero. □

Definition 32.7 (Structure-metric adjustment)**.** Let the structure metric of an audio target be

$$\mathcal{I} = (\mathcal{I}_{\text{mass}}, \mathcal{I}_{\text{energy}}, \mathcal{I}_{\text{scale}}) = (\text{mass conservation degree, energy conservation degree, scale-char}$$

Structure-metric adjustment is: at each step choose a neighbouring parameter θ in the parameter space Θ for which $\mathcal{I}(\theta)$ is **strictly decreasing** in the lexicographic order (minimisation); if no such neighbouring parameter exists, the process stops—then θ is a **locally optimal** parameter.

Proposition 32.3 (Finite convergence of structure-metric adjustment)**.** Let the parameter space Θ be finite and let each adjustment make $\mathcal{I}$ **strictly decreasing** in the lexicographic order. Then the adjustment terminates after finitely many steps at a locally optimal parameter.

Proof. Θ is finite, so $\mathcal{I}$ takes finitely many values. The lexicographic order is well-founded on a finite set of values, so a strictly lexicographically decreasing sequence cannot continue indefinitely and the process terminates after finitely many steps. At termination there is no neighbouring parameter making $\mathcal{I}$ strictly lexicographically decreasing, that is, the parameter is locally optimal. Monotonicity must be imposed on the **whole lexicographic order**: if only 'some component strictly increases or decreases' is required, cycles such as $(2,1) \to (1,2) \to (2,1)$ cannot be excluded and finite termination fails. □

Remark 32.6 (Boundaries of dynamic adjustment). Theorem 32.4 relies on finite periodic sequences. For continuous-spectrum or quasi-periodic audio targets, no finite epicycle group represents them exactly and only bounded approximation is possible. Structure-metric adjustment does not guarantee global optimality.

32.1.4 Step four: audio convolution

Definition 32.8 (Audio amplitude, phase and frequency convolutions)**.** Let two audio epicycle groups be

$$\mathfrak{E} = \{(A_n, \omega_n, \phi_n)\}, \qquad \mathfrak{E}' = \{(B_m, \nu_m, \psi_m)\}.$$

Define:

(i) **Amplitude convolution:** $\mathbb{A}_{mn} = A_n B_m$;

(ii) **Phase convolution:** $\mathbb{P}_{mn} = e^{i(\phi_n - \psi_m)}$;

(iii) **Frequency convolution:** $\mathbb{F}_{mn} = \omega_n \nu_m$;

(iv) **Trend convolution:** $\mathbb{T}_{mn} = A_n B_m e^{i(\phi_n - \psi_m)} = A_n B_m \mathbb{P}_{mn}$.

This agrees with Definition 6.21, Definition 6.22, Definition 6.23 and Definition 6.24 of the 'Series and Convolution' chapter.

Theorem 32.5 (Action of the three convolutions on the audio epicycle group)**.** *Let the epicycle groups of two audio targets be $\mathfrak{E}$ and $\mathfrak{E}'$ respectively. Then:*

- (i) *amplitude convolution acts only on the radius;*
- (ii) *phase convolution acts only on the phase;*
- (iii) *frequency convolution acts only on the rotational speed;*
- (iv) *trend convolution preserves both amplitude and phase.*

Proof. This is read off directly from Definition 32.8. ☐

Proposition 32.4 (Autocorrelation and trend convolution)**.** *Let $\{z_t\}$ be an audio periodic sequence, with autocorrelation function*

$$R_z(h) = \sum_t z_t \overline{z_{t+h}}.$$

In the language of the audio epicycle group, the autocorrelation function is the projection of the trend convolution onto the diagonal:

$$R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}.$$

Proof. The Proposition 32.4 of this chapter is the realisation in this domain of the convolution-action conclusion of Theorem 24.6 in the 'Periodic Systems' chapter: substituting $z_t = \sum_n A_n e^{i(\omega_n t + \phi_n)}$ into the definition of the autocorrelation, cancelling all off-diagonal cross terms by orthogonality and retaining only the diagonal term of the trend convolution (the factorisation of the three convolutions, Proposition 24.6), which yields $R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}$. The full expansion is given there. ☐

Remark 32.7 (Relation between audio convolution and periodic convolution)*.* Audio convolution is not a new convolution operation but an instance of the periodic convolution of the 'Series and Convolution' chapter on the audio epicycle group. Autocorrelation is the projection of the trend convolution onto the diagonal; cross-correlation and beats are respectively the concrete manifestations in acoustics of amplitude convolution and frequency convolution.

32.1.5 Step five: integration with discrete sampling, harmonic analysis and time series analysis

Definition 32.9 (Audio periodicity diagnostic procedure)**.** *For a given audio system, diagnose it using the framework of periodic mathematics:*

- (i) **Periodicity decision***: verify the period type of the audio target (Theorem 32.1);*

(ii) **Series construction**: build the amplitude, frequency and phase series and the epicycle group from sound-pressure and particle-velocity data (Theorem 32.3);

(iii) **Conservation diagnosis**: verify mass, energy and scale-charge conservation (Theorem 32.9);

(iv) **Convolution diagnosis**: compute the autocorrelation (the diagonal projection of the trend convolution, Proposition 32.4) and the phase convolution;

(v) **Integration diagnosis**: connect to the tool chains of discrete sampling, harmonic analysis and time series analysis.

Theorem 32.6 (The audio periodicity diagnostic procedure and its interfaces with three chapters)**.** The five steps of Definition 32.9 connect respectively to:

(i) the 'Discrete Sampling and Intercalation of Periodic Signals' chapter: Definition 21.1, Theorem 21.3, Theorem 21.10;

(ii) the 'Harmonic Analysis' chapter: Definition 22.5, Theorem 22.1, Proposition 22.11;

(iii) the 'Time Series Analysis' chapter: Definition 23.16, Proposition 23.2, Definition 23.19.

The audio periodicity diagnostic procedure is therefore not an independent method but an application of the unified tool chain of periodic mathematics to audio systems.

Proof. The Theorem 32.6 of this chapter is the implementation in this domain of the five-step interface of the 'Periodic Systems' chapter (Table 24.1): the periodicity decision uses sampling and spectral estimation, the frequency decomposition and the phase kernels use the 'Harmonic Analysis' chapter, and the periodogram peaks, the Fisher test and window-function validation use the 'Time Series Analysis' chapter. The diagnostic procedure is therefore a synthesis of the tool chains of the three chapters; the proof is given there. ☐

Remark 32.8 (Order of operations after integration). The recommended order for an actual diagnosis is: first decide whether the audio target is periodic (Theorem 32.1); then construct the Fourier data and the epicycle group of the audio data (Theorem 32.3); then optimise the parameters by the conservation constraints and dynamic adjustment (Theorem 32.4, Proposition 32.3); then perform audio convolution and autocorrelation analysis (Proposition 32.4); and finally output the diagnosis as in Definition 32.9.

32.2 Mathematical foundations of audio systems

Definition 32.10 (Basic quantities of sound waves)**.** An audio system is described by the following basic physical quantities:

(i) **Sound pressure** $p(\mathbf{r}, t)$ (unit: Pa): the part of the pressure of the medium that deviates from the static pressure under the action of a sound wave.

(ii) **Particle velocity** $\mathbf{v}(\mathbf{r},t)$ (unit: m/s): the vibration velocity of the particles of the medium under the action of a sound wave.

(iii) **Density perturbation** $\rho'(\mathbf{r},t)$ (unit: kg/m^3): the part of the density of the medium that deviates from the static density ρ_0.

(iv) **Speed of sound** c (unit: m/s): in an ideal gas, $c = \sqrt{\gamma p_0/\rho_0}$.

32.3 Simple harmonic motion

Definition 32.11 (Simple harmonic motion). **Simple harmonic motion** is the motion of a particle under a linear restoring force $F = -kx$:

$$\frac{d^2x}{dt^2} + \omega_0^2 x = 0, \quad \omega_0 = \sqrt{\frac{k}{m}}.$$

The general solution is $x(t) = A\cos(\omega_0 t + \phi)$. In the phase space (x, p) the trajectory of simple harmonic motion is an ellipse. Introducing the complex variable $z = x + ip/(m\omega_0)$ reduces the equation of motion to $\dot{z} = -i\omega_0 z$—the one-dimensional degenerate form of the epicycle model.

Proposition 32.5 (Energy conservation of simple harmonic motion). The total mechanical energy is $E = \frac{1}{2}kA^2 =$ constant. Kinetic energy and potential energy are interconverted at the angular frequency $2\omega_0$.

Proof. Substituting $x = A\cos(\omega_0 t + \phi)$: the kinetic energy is $K = \frac{1}{2}m\dot{x}^2 = \frac{1}{2}mA^2\omega_0^2\sin^2(\omega_0 t + \phi)$ and the potential energy is $U = \frac{1}{2}kx^2 = \frac{1}{2}kA^2\cos^2(\omega_0 t + \phi)$. From $\omega_0^2 = k/m$ we obtain $mA^2\omega_0^2 = kA^2$, hence

$$E = K + U = \frac{1}{2}kA^2\big(\sin^2(\omega_0 t + \phi) + \cos^2(\omega_0 t + \phi)\big) = \frac{1}{2}kA^2,$$

independently of time. K and U each oscillate with angular frequency $2\omega_0$ ($\sin^2\theta = \frac{1-\cos 2\theta}{2}$) and differ in phase by π—energy is converted periodically between the two forms whilst the total is conserved. □

Definition 32.12 (Phase delay of simple harmonic motion). In simple harmonic motion $x(t) = A\cos(\omega_0 t + \phi)$, the **phase delay** corresponding to the phase ϕ is

$$\tau = \frac{\phi}{\omega_0}.$$

The phase delay converts a phase angle into a time offset: an initial phase ϕ is equivalent to the vibration **leading** the reference by $\tau = \phi/\omega_0$ ($\phi > 0$ means that the waveform is **advanced** by τ relative to the reference, equivalently the waveform coincides with the reference once the reference has been delayed by τ). When $\phi = 2\pi$ (one complete period), $\tau = T = 2\pi/\omega_0$—the phase delay degenerates to a complete period. This is the acoustic realisation of the delay $\tau = \phi/\omega$ in the parameter table of the 'Periodic Functions and Groups with Periodic Orbits' chapter.

32.4 The acoustic wave equation and propagation delay

Theorem 32.7 (Acoustic wave equation). *In an inviscid ideal fluid without heat conduction, the sound pressure $p(\mathsf{r},t)$ satisfies the three-dimensional wave equation:*

$$\boxed{\nabla^2 p - \frac{1}{c^2}\frac{\partial^2 p}{\partial t^2} = 0}.$$

Proof. The three linearised equations of an ideal fluid: (i) the continuity equation $\frac{\partial \rho'}{\partial t} + \rho_0 \nabla \cdot \mathsf{v} = 0$; (ii) Euler's equation $\rho_0 \frac{\partial \mathsf{v}}{\partial t} = -\nabla p$; (iii) the adiabatic equation of state $p = c^2\rho'$ ($c^2 = (\partial p/\partial \rho)_S$). Differentiating (i) with respect to $\partial/\partial t$ and taking the divergence of (ii):

$$\frac{\partial^2 \rho'}{\partial t^2} + \rho_0 \nabla \cdot \frac{\partial \mathsf{v}}{\partial t} = 0, \qquad \rho_0 \nabla \cdot \frac{\partial \mathsf{v}}{\partial t} = -\nabla^2 p.$$

Subtracting the two equations eliminates the velocity field: $\frac{\partial^2 \rho'}{\partial t^2} = \nabla^2 p$. Substituting into (iii) to eliminate ρ' gives $\frac{1}{c^2}\frac{\partial^2 p}{\partial t^2} = \nabla^2 p$, that is, the wave equation. ☐

Definition 32.13 (Propagation delay of acoustic waves). *The* **propagation delay** *of a sound wave from a source* r' *to an observation point* r *is*

$$\tau_{\mathsf{prop}} = \frac{|\mathsf{r} - \mathsf{r}'|}{c}.$$

The propagation delay converts a spatial distance into a time offset—the greater the distance, the greater the delay, the two being linearly related, $\tau \propto r$. In air at room temperature ($c \approx 343$ m/s), a distance of 1 metre corresponds to a delay of about 2.9 ms—this is the minimum delay threshold for an audible echo in acoustics.

Definition 32.14 (Phase delay of acoustic waves). *For a monochromatic plane sound wave $p(\mathsf{r},t) = p_0 e^{i(\mathsf{k}\cdot\mathsf{r} - \omega t)}$, the* **phase delay** *corresponding to the spatial phase* $\mathsf{k}\cdot\mathsf{r}$ *is*

$$\tau_{\mathsf{phase}} = \frac{\mathsf{k}\cdot\mathsf{r}}{\omega} = \frac{|\mathsf{r}|\cos\theta}{c},$$

and along the direction of propagation ($\theta = 0$) we have $\tau_{\mathsf{phase}} = r/c = \tau_{\mathsf{prop}}$.

Proposition 32.6 (Unification of acoustic delay and phase). *Along the direction of propagation, every increase of the spatial phase by 2π (one wavelength λ) corresponds to an increase of the time offset by one period T:*

$$\tau = \frac{\lambda}{c} = \frac{2\pi}{\omega} = T.$$

The spatial period (wavelength) and the temporal period (frequency) of a sound wave are coupled exactly through the delay.

Proof. The wavelength is $\lambda = 2\pi/|\mathsf{k}| = cT$. Along the direction of propagation, an increase of the phase by 2π corresponds to an increase of the distance by λ, so the delay is $\tau = \lambda/c = T$. ☐

32.5 Harmonic structure and Fourier decomposition

Definition 32.15 (Harmonic series). Any sound wave of period T, $p(t)$, admits the **Fourier series** expansion

$$p(t) = \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t + \phi_n),$$

where $\omega_0 = 2\pi/T$ is the fundamental frequency and $n\omega_0$ is the frequency of the n-th harmonic. The successive harmonics form a nested epicycle system—the fundamental frequency corresponds to the outer, large epicycle and the higher harmonics to the inner, small epicycles.

Example 32.3 (Harmonic characteristics of instrumental timbre). Different instruments playing the same pitch produce different timbres, which arises from differences in harmonic structure. A tuning fork has almost only the fundamental frequency, a violin is rich in harmonics, and a clarinet is dominated by odd-order harmonics.

32.6 Tuning systems

Definition 32.16 (Interval and frequency ratio). The **interval** between two notes is determined by the frequency ratio. The perfect octave is $2:1$, the perfect fifth $3:2$, the perfect fourth $4:3$ and the major third $5:4$.

Theorem 32.8 (Mathematical structure of equal temperament). The common ratio of equal temperament is $r = 2^{1/12}$. The frequency of the k-th semitone is $f_k = f_0 \cdot 2^{k/12}$.

Proof. This follows directly by dividing the octave frequency ratio $2:1$ into twelve equal parts: $r^{12} = 2 \Rightarrow r = 2^{1/12}$. □

Lemma 32.4 (Approximation of equal temperament to the perfect fifth). The frequency ratio of the 'fifth' formed by seven semitones in equal temperament is $2^{7/12} \approx 1.4983$, whose relative error with respect to the perfect fifth $3/2 = 1.5$ is about 0.11%; whilst the twelve semitones close the octave exactly ($2^{12/12} = 2$).

Proof. Comparing logarithms: $\log_2(2^{7/12}) = 7/12 \approx 0.5833$ and $\log_2(3/2) \approx 0.5850$, a difference of 0.0016. Hence $2^{7/12} = 2^{0.5833} = 2^{0.5850} \cdot 2^{-0.0016} \approx \frac{3}{2} \cdot 0.9989$, a relative error of $\approx 0.11\%$. Closure of the octave follows directly from $2^{12/12} = 2$. This compromise of an 'approximate fifth plus an exact octave' is the musical compromise between geometric scaling (equal temperament) and simple integer ratios (just intonation). □

32.7 Periodic propagation of sound waves and beat-frequency delay

Definition 32.17 (Standing wave). Two travelling waves of the same frequency and equal amplitude propagating in opposite directions superpose to form a **standing**

wave:

$$p(x,t) = 2A\cos(kx)\cos(\omega t).$$

The spatial reflection symmetry of a standing wave is given by the form of the law of reflection in the 'Vector Analysis' chapter—after reflection at a boundary a wave superposes with the incident wave, forming a standing pattern in space.

Proposition 32.7 (Harmonic series of string instruments)**.** The standing-wave modes allowed on a string fixed at both ends have frequencies $f_n = nf_1$ $(n = 1, 2, 3, \dots)$.

Proof. The standing-wave form $p(x,t) = A\sin(kx)\cos(\omega t)$ satisfies the wave equation (Theorem 32.7). The boundary conditions $p(0,t) = p(L,t) = 0$: at $x = 0$ they hold automatically; at $x = L$ they require $\sin(kL) = 0$, that is, $k_n L = n\pi$ $(n = 1, 2, 3, \dots)$. From the dispersion relation $\omega_n = ck_n$ we obtain

$$f_n = \frac{\omega_n}{2\pi} = \frac{cn\pi/L}{2\pi} = n \cdot \frac{c}{2L} = nf_1, \qquad f_1 = \frac{c}{2L}.$$

The fundamental frequency f_1 is inversely proportional to the length of the string—the shorter the string, the higher the pitch. ☐

Definition 32.18 (Beat-frequency delay)**.** Two sound waves of close frequency ($f_1 \approx f_2$, with $\Delta f = |f_1 - f_2|$ very small) superpose to produce the phenomenon of **beats**—the amplitude of the resulting sound wave varies periodically at the beat frequency Δf. From the point of view of delay, the beat period is the **phase delay** required for the phase difference of the two sound waves, $\Delta\phi(t) = 2\pi\Delta f \cdot t$, to change from 0 to 2π. The beat-frequency delay measures the time offset required for two sound waves of close frequency to pass from complete coincidence in phase to the next complete coincidence—the time required for the phase difference of the two frequency components to rotate through 2π.

Proposition 32.8 (Beat-frequency delay formula)**.** Let the frequencies of two sound waves be f_1, f_2, with $\Delta f = |f_1 - f_2|$. Then the beat period is

$$T_{\text{beat}} = \frac{1}{\Delta f}.$$

Proof. Superposing the two sound waves: $p(t) = A\cos(\omega_1 t) + A\cos(\omega_2 t) = 2A\cos(\Delta\omega\, t/2)\cos(\bar{\omega} t)$, where $\Delta\omega = |\omega_1 - \omega_2|$ and $\bar{\omega} = (\omega_1 + \omega_2)/2$. The amplitude envelope $2A\cos(\Delta\omega\, t/2)$ oscillates with angular frequency $\Delta\omega/2$, so the envelope period is $2\pi/(\Delta\omega/2) = 4\pi/\Delta\omega$. The beat period (the interval between two successive amplitude maxima or minima) is half the envelope period: $T_{\text{beat}} = 2\pi/\Delta\omega = 1/\Delta f$. ☐

32.8 Conservation laws of audio systems

Theorem 32.9 (Three conservation laws of acoustic-wave systems)**.**

(i) Mass conservation: $\frac{\partial \rho'}{\partial t} + \rho_0 \nabla \cdot \mathbf{v} = 0$.

(ii) Energy conservation: $\frac{\partial E}{\partial t} + \nabla \cdot \mathbf{I} = 0$ (in the absence of sources).

(iii) **Scale-charge conservation**: in the absence of damping the amplitudes of the individual frequency components are conserved independently.

Proof. (i) Mass conservation is the linearised continuity equation (part (i) of the proof of Theorem 32.7). (ii) The energy density is $E = \frac{1}{2}\rho_0|\mathsf{v}|^2 + \frac{1}{2}\frac{p^2}{\rho_0 c^2}$ and the energy flux density is $\mathsf{I} = p\mathsf{v}$; differentiating directly from the wave equation gives $\frac{\partial E}{\partial t} + \nabla \cdot \mathsf{I} = 0$ (in the absence of sources)—conservation of energy under spatial translation. (iii) The Fourier decomposition of the undamped wave equation (the plane-wave solution of Theorem 32.7) makes the individual frequency components $A_n e^{i(\mathsf{k}_n \cdot \mathsf{r} - \omega_n t)}$ evolve independently with unchanged amplitudes $|A_n|$—conservation of the scale charge of each frequency channel. □

Remark 32.9 (Connection with the rest of the book). The theory of the periodic structure of audio systems established in this chapter is an application of periodic mathematics in acoustics. The trajectory of simple harmonic motion in phase space is the one-dimensional degenerate form of the epicycle model, the harmonic series is an acoustic instance of a nested epicycle system, and the acoustic wave equation describes wave propagation in continuum mechanics. The concept of delay reveals the deeper structure of sound-wave propagation: the propagation delay $\tau = r/c$ converts a spatial distance into a time offset, the phase delay $\tau = \phi/\omega$ converts a phase angle into a time offset, and the beat-frequency delay $\tau_{\mathsf{beat}} = 1/\Delta f$ converts a frequency difference into a time offset. The three delays correspond respectively to the manifestations in the time domain of the three elementary operations (translation, rotation and scaling).

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The acoustic propagation delay $\tau_{\text{prop}} = \lvert \mathbf{r} - \mathbf{r}' \rvert / c$ (Definition 32.13): the linear translational correspondence from spatial distance to time offset
Scaling	The common ratio $2^{1/12}$ of equal temperament (Theorem 32.8) and the approximation to the perfect fifth (Lemma 32.4)
Rotation	The phase-space circulation of simple harmonic motion $\dot{z} = -i\omega_0 z$ (Definition 32.11); the phase delay $\tau = \phi/\omega$ (Definition 32.12)
Conservation	Energy conservation in simple harmonic motion $E = \frac{1}{2}kA^2$ (Proposition 32.5); mass, energy and scale-charge conservation (Theorem 32.9)
Symmetry	The spatial reflection symmetry of standing waves (Definition 32.17); the symmetry constraint of the boundaries fixed at both ends of a string (Proposition 32.7)
Closure	Periodic closure on every increase of the phase by 2π: $\tau = \lambda/c = T$ (Proposition 32.6); closure of the beat phase difference as it returns from 0 to 2π (Definition 32.18)
Period	The fundamental frequency and the harmonic series $f_n = nf_1$ (Definition 32.15, Proposition 32.7); the beat period $T_{\text{beat}} = 1/\Delta f$ (Definition 32.18, Proposition 32.8)

Exercises

Exercise 1 Prove the **audio periodicity decision theorem** of this chapter (32.1). Hint: first write down the decision map and the evolution group of this chapter's target, then apply the proof structure of the general criterion of the 'Periodic Systems' chapter.

Exercise 2 Verify that the hypotheses of **monotonicity of quantity adjustment** (32.2) hold for the domain objects of this chapter, and give a counterexample in which the hypotheses fail.

Exercise 3 Examine the behaviour of the **three-parameter correspondence: radius, phase and speed** (32.1) under degeneracy of the parameters, and explain how the conclusion degenerates accordingly.

Exercise 4 Explain the relation between the **audio periodicity diagnostic procedure and its interfaces with three chapters** (32.6) and the corresponding general conclusions of the 'Periodic Systems' chapter (whether they agree, whether it is a special case, or whether additional conditions are required).

Exercise 5 Using the seven-theme comparison table of this chapter, identify the concrete carriers in this chapter of the three themes **translation**, **scaling** and **rotation**, and give one example of domain data for each.

Exercise 6 Synthesis: restate the diagnostic procedure of this chapter using the five-step standard interface of the ‘Periodic Systems’ chapter (Table 24.1), noting for each step the theorem of this chapter on which it depends: 32.1, 32.2, 32.1, 32.6.

Chapter 33

Periodic Structure of Mechanical Systems

Remark 33.1 (Periodicity decision for the target of this chapter (citation)). When this chapter decides whether a research target is periodic, determines its period type, or seeks its minimal positive period, reference is always made to the 'Periodic Systems' chapter: for the periodicity criterion see Theorem 24.1, for the discrimination of period types see Definition 24.3, Proposition 24.1, for the search for the minimal positive period see Definition 24.4, Theorem 24.2, and for the construction of period series and epicycle groups from data see Theorem 24.3, Definition 24.6.

Remark 33.2 (Positioning of this chapter: developing via the five-step interface of the 'Periodic Systems' chapter). The periodic structure of mechanical systems studies the mathematical characterisation of periodicity, quasi-periodicity and resonance phenomena in classical mechanics. This chapter does not establish a separate periodicity criterion; instead it connects mechanical systems to the five-step interface prescribed by the 'Periodic Systems' chapter:

(1) **Decision**: reduce targets such as the spring oscillator, the simple pendulum, coupled oscillators, the forced oscillator and gyroscopic precession to discrete or continuous dynamical systems, invoke Theorem 24.1 to decide whether they possess periodicity, and determine the period type;

(2) **Construction**: build the amplitude series, the frequency series and the phase series from the generalised coordinates, momenta and phase data, and assemble them into the mechanical epicycle group;

(3) **Adjustment**: dynamically adjust the mechanical epicycle group under the constraints of residual-energy descent, conservation of momentum, conservation of angular momentum and conservation of energy;

(4) **Convolution**: introduce amplitude convolution, phase convolution and frequency convolution on the mechanical epicycle group, and connect them with autocorrelation, cross-correlation and beat-frequency delay;

(5) **Integration**: connect to the tool chains of the three chapters 'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis', forming a mechanical periodicity diagnostic pipeline.

The mathematical tools used in this chapter include: Lagrangian mechanics, Hamiltonian mechanics, elliptic integrals, resonance theory, periodic convolution, epicycle groups, phase operators and delay.

Remark 33.3 (Seven-theme guide for this chapter). **Translation**—the time translation and phase delay $\tau = \phi\sqrt{m/k}$ of the spring oscillator (Definition 33.12); **scaling**—the scaling correction $\Delta T(\theta_{\text{max}})$ for the large amplitude of the pendulum (Definition 33.14); **rotation**—the phase lag $\delta(\omega)$ and the precession of the forced oscillator (Definition 33.18, 33.19); **conservation**—conservation of momentum, of angular momentum and of energy (Theorem 33.9) and energy conservation of the spring oscillator (Proposition 33.5); **symmetry**—the symmetric–antisymmetric decomposition of the normal coordinates and time-reversal symmetry; **closure**—the closure of the periodic orbit of the pendulum and of the gyroscopic precession circle; **period**—the spring oscillator $T = 2\pi\sqrt{m/k}$ (isochronism), the elliptic-integral period of the pendulum (Theorem 33.7) and the beat period $T_{\text{beat}} = \pi/\Delta\omega$ (Proposition 33.6).

33.1 Connection to the five-step interface of the 'Periodic Systems' chapter

33.1.1 Step one: periodicity decision for mechanical targets

Definition 33.1 (Mechanical targets and decision input). A **mechanical target** is one of the following objects:

(i) *a spring-oscillator system, whose state space is the (x, p) phase space;*

(ii) *a simple-pendulum system, whose state space is the $(\theta, \dot{\theta})$ phase space;*

(iii) *a coupled-oscillator system, whose state space is the $(x_1, x_2, \dot{x}_1, \dot{x}_2)$ phase space;*

(iv) *a forced-oscillator system, whose state space is the joint space of the driving phase and the response phase;*

(v) *a gyroscope system, whose state space is the joint space of the angular momentum* $\mathbf{L}$ *and the spin-axis direction* $\hat{\mathbf{e}}_3$.

The **decision input** of a mechanical target is the ordered pair

$$\mathcal{D}_{\mathsf{M}} = (\mathcal{S}, \Phi),$$

where $\mathcal{S}$ is the state space and $\Phi : \mathcal{S} \to \mathcal{S}$ is the map induced by time translation, phase rotation or orbit evolution. If $\mathcal{S}$ is a finite set, the target is called a **finite mechanical target**; if $\mathcal{S}$ is an infinite set, the target is called an **infinite mechanical target**.

Definition 33.2 (Mechanical decision map). Let a mechanical target be given by the decision input $\mathcal{D}_{\mathsf{M}} = (\mathcal{S}, \Phi)$. If the states of the target are generated by Φ, that is, if there exists $x_0 \in \mathcal{S}$ such that

$$x_t = \begin{cases} \Phi^t(x_0), & t \in \mathbb{Z}, \\ \Phi_t(x_0), & t \in \mathbb{R}, \end{cases}$$

for $t \in \mathbb{Z}$ the state x_t is generated by iteration of Φ; for $t \in \mathbb{R}$ it is described by the one-parameter evolution family $\{\Phi_t\}_{t\in\mathbb{R}}$, where each $\Phi_t : \mathcal{S} \to \mathcal{S}$, $\Phi_0 = \mathrm{id}$, $\Phi_{s+t} = \Phi_s \circ \Phi_t$ for all $s, t \in \mathbb{R}$, and $(t, x) \mapsto \Phi_t(x)$ is continuous. then Φ is called the **mechanical decision map** of the target. The mechanical target is then reduced to the discrete dynamical system $(\mathcal{S}, \Phi)$ when $t \in \mathbb{Z}$ and to the continuous dynamical system $(\mathcal{S}, \{\Phi_t\}_{t\in\mathbb{R}})$ when $t \in \mathbb{R}$; in both cases the criteria of the 'Operations and Periodicity' chapter and of the 'Periodic Systems' chapter may be applied directly.

Theorem 33.1 (Periodicity decision theorem for mechanical systems)**.** Let the decision map of a mechanical target be Φ, and let $(\mathcal{S}, \Phi)$ be a discrete dynamical system. Then:

(A) **Sufficient criterion**: if there exists $m \in \mathbb{N}^+$ such that $\Phi^m = \mathrm{id}$, then the target is a uniformly periodic mechanical system; every orbit is closed, and the orbit length divides m.

(B) **Equivalence of the decision**: the target is a uniformly periodic mechanical system if and only if $\mathrm{ord}(\Phi) < \infty$.

(C) **Pointwise criterion**: the orbit of the target at the state x_0 is closed if and only if x_0 is a periodic point; the orbit length is then the period at that point.

(D) **Non-omissibility**: the conjunction of conserved quantities and symmetry groups cannot replace $\mathrm{ord}(\Phi) < \infty$; a quasi-periodic mechanical target is not a periodic mechanical system.

Proof. The Theorem 33.1 of this chapter is the domain instantiation of Theorem 24.1 in the 'Periodic Systems' chapter: one need only replace the decision map f and the evolution group by Φ of this chapter, and the four clauses (A) sufficient criterion, (B) equivalence of the decision, (C) pointwise criterion and (D) non-omissibility hold one by one. By the convention of Remark 24.4, the argument is not expanded again here. □

Corollary 33.1 (Mechanical period types)**.** After the decision by Theorem 33.1, a mechanical target falls into one of the following types:

(i) **Trivially periodic type**: the least period is $m_0 = 1$, and the states are fixed points;

(ii) **Strictly periodic type**: $1 < m_0 < \infty$, and the states are strictly periodic;

(iii) **Uniformly periodic type**: $\mathrm{ord}(\Phi) = m < \infty$, and all orbit lengths divide m;

(iv) **Quasi-periodic type**: $\mathrm{ord}(\Phi) = \infty$, and the orbits are not closed but are dense on a compact carrier structure;

(v) **Eventually periodic type**: there exists N such that the target is periodic for $t \geq N$;

(vi) **Discretely periodic type**: the carrier structure is $\mathbb{Z}$, and the normalised period ratio is $\rho = T_x/\Delta x \in \mathbb{Q}$.

(vii) **Asymptotically periodic (damped) type**: the orbit is not closed, but tends asymptotically to a closed orbit or to a fixed point, the residual decaying as $e^{-\kappa t}$ (see Definition 24.3(vii)).

Of these, (i)–(iii) and (vi) are periodic mechanical systems; (iv) is not a periodic mechanical system, but can be handled with bounded-approximation tools; (v) is a periodic mechanical system only in its stable stage; (vii), like (iv), is a type in the limiting sense; it is not a periodic mechanical system and can be handled with bounded-approximation tools only in its stable stage.

Proof. Substituting the mechanical target clause by clause into Theorem 33.1 and Definition 24.3 of the 'Periodic Systems' chapter yields the result. ☐

Lemma 33.1 (Periodicity decision for the spring oscillator)**.** *Let the equation of motion of the spring oscillator be*

$$\ddot{x} + \omega_0^2 x = 0, \qquad \omega_0 = \sqrt{k/m}.$$

Then the spring oscillator is a uniformly periodic system, and its least positive period is

$$T_0 = \frac{2\pi}{\omega_0} = 2\pi\sqrt{\frac{m}{k}}.$$

Proof. By Definition 33.11, the general solution is $x(t) = A\cos(\omega_0 t + \phi)$. The time translation $T_0 = 2\pi/\omega_0$ gives $x(t + T_0) = x(t)$, and T_0 is the least positive period. Hence the decision map Φ satisfies $\Phi^{T_0} = \mathrm{id}$, so the system is uniformly periodic. ☐

Lemma 33.2 (Periodicity decision for the pendulum)**.** *Let the equation of motion of the simple pendulum be*

$$\ddot{\theta} + \omega_0^2 \sin\theta = 0, \qquad \omega_0 = \sqrt{g/l}.$$

Then the simple pendulum is a uniformly periodic system, and its oscillation period is given by the complete elliptic integral of the first kind:

$$T(\theta_{\max}) = 4\sqrt{\frac{l}{g}}\,K(k), \qquad k = \sin\frac{\theta_{\max}}{2}.$$

Proof. By Theorem 33.7, a simple pendulum released from the maximum deflection angle $\theta_{\max}$ performs periodic motion with period $T(\theta_{\max}) = 4\sqrt{l/g}\,K(k)$. By conservation of energy the orbit oscillates to and fro within the bounded interval $[-\theta_{\max}, \theta_{\max}]$; hence the decision map Φ satisfies $\Phi^{T(\theta_{\max})} = \mathrm{id}$, so the system is uniformly periodic. ☐

Lemma 33.3 (Periodicity decision for coupled oscillators)**.** *Let a coupled-oscillator system be in its symmetric mode and its antisymmetric mode, with corresponding normal frequencies*

$$\omega_1 = \sqrt{k/m}, \qquad \omega_2 = \sqrt{(k + 2k_c)/m}.$$

Then, when $\omega_1/\omega_2 \in \mathbb{Q}$, the system is uniformly periodic; when $\omega_1/\omega_2 \notin \mathbb{Q}$, the system is quasi-periodic.

Proof. Introducing the normal coordinates $\eta_1 = x_1 + x_2$ and $\eta_2 = x_1 - x_2$, the system decouples into two independent simple harmonic oscillations. By the commensurability criterion of Theorem 1.5 in the 'Operations and Periodicity' chapter, the orbit formed by superposing the two simple harmonic oscillations is closed if and only if the frequency ratio ω_1/ω_2 is rational. Hence the conclusion holds. ☐

Lemma 33.4 (Periodicity decision for the forced oscillator)**.** *Let the steady-state response of a forced oscillator be*

$$x(t) = A(\omega)\cos(\omega t - \delta(\omega)).$$

Then the forced oscillator is a uniformly periodic system, and its least positive period is the driving period

$$T = \frac{2\pi}{\omega}.$$

Proof. The steady-state response is periodic with the driving frequency ω, and its least positive period is $T = 2\pi/\omega$. The transient response decays as $e^{-\beta t}$ and tends to the steady-state periodic orbit as $t \to \infty$; the steady-state response is a uniformly periodic system, whilst the transient phase is of asymptotically periodic (damped) type (not of eventually periodic type: at no finite time is the orbit closed). ☐

Lemma 33.5 (Periodicity decision for gyroscopic precession)**.** *Let a gyroscope precess under the action of the gravitational torque, with precession angular velocity*

$$\Omega_p = \frac{mgr}{I_3\omega_3}.$$

Then the precession of the spin axis is a uniformly periodic system, and its precession period is

$$T_p = \frac{2\pi}{\Omega_p} = \frac{2\pi I_3\omega_3}{mgr}.$$

Proof. By Proposition 33.8, the precession angular velocity is $\Omega_p = mgr/(I_3\omega_3)$, and the spin axis rotates uniformly about the vertical axis with period $T_p = 2\pi/\Omega_p$. Hence the decision map Φ satisfies $\Phi^{T_p} = \mathrm{id}$, so the system is uniformly periodic. ☐

Definition 33.3 (Least positive period algorithm for mechanical systems)**.** This definition has the same content as that given, in Chapter 24 under the section 'Periodicity decision: whether the target is periodic' (Section 24.1), by Definition 24.4; to avoid repetition only the reference anchor is retained here, and the complete statement is given there. The only difference here is the terminology (mechanical target / mechanical sample); the steps of the algorithm and the conclusion are exactly the same.

Theorem 33.2 (Correctness of the least positive period algorithm for mechanical systems)**.** *This theorem has the same content as that given, in Chapter 24 under the section 'Periodicity decision: whether the target is periodic' (Section 24.1), by Theorem 24.2; to avoid repetition only the reference anchor is retained here, and the complete statement and proof are given there. The only difference here is the terminology (mechanical target / mechanical sample); the steps of the algorithm and the conclusion are exactly the same.*

Example 33.1 (Instances of mechanical periodicity decisions). (i) Spring oscillator: $T_0 = 2\pi\sqrt{m/k}$, uniformly periodic type;

(ii) Simple pendulum: $T(\theta_{\mathsf{max}}) = 4\sqrt{l/g}\,K(k)$, uniformly periodic type;

(iii) Coupled oscillators: uniformly periodic type when $\omega_1/\omega_2 \in \mathbb{Q}$, otherwise quasi-periodic type;

(iv) Forced oscillator: steady-state period $T = 2\pi/\omega$, uniformly periodic type;

(v) Gyroscopic precession: $T_p = 2\pi I_3\omega_3/(mgr)$, uniformly periodic type.

Remark 33.4 (Conserved quantities do not guarantee periodicity). In mechanical systems, conservation of momentum, conservation of angular momentum and conservation of energy are auxiliary conditions for periodic structure, but they do not constitute a sufficient condition. For example, a hyperbolic orbit possesses conservation of energy and conservation of angular momentum, yet the orbit is open and does not close. Periodic structure in mechanics must be decided by Theorem 33.1 or by the lemmas above.

33.1.2 Step two: construction of the mechanical period series and the epicycle group

Definition 33.4 (Amplitude, frequency and phase series of a mechanical target). Let the state or output sequence of a mechanical target be $\{z_t\}$. If it can be written as

$$z_t = \sum_k A_k e^{i(\omega_k t+\phi_k)} + \varepsilon_t,$$

then

$$\mathcal{A}_z(t) = \sum_k A_k e^{ikt}, \qquad \mathcal{F}_z(s) = \sum_k \omega_k s^k, \qquad \mathcal{P}_z(t) = \sum_k e^{i(kt+\phi_k)}$$

are called respectively the **amplitude series**, the **frequency series** and the **phase series** of the mechanical target. The three agree with Definition 6.10, Definition 6.12 and Definition 6.11 in the 'Series and Convolution' chapter.

Definition 33.5 (Mechanical epicycle group). Let the period series of a mechanical target be given by the amplitude series $\{A_k\}$, the frequency series $\{\omega_k\}$ and the phase series $\{\phi_k\}$. For each k define the epicycle

$$E_k : \theta \mapsto A_k e^{i(\theta+\phi_k)}, \qquad \theta = \omega_k t,$$

and call the set

$$\mathfrak{E}_{\mathsf{M}} = \{(A_k, \omega_k, \phi_k)\}_{k\in\mathbb{Z}}$$

the **epicycle group** of the mechanical target. Its superposition is the mechanical period series

$$z_t = \sum_k E_k(\omega_k t) = \sum_k A_k e^{i(\omega_k t+\phi_k)}.$$

Theorem 33.3 (Building the period series from mechanical data). *Let the output of a mechanical target be a finite periodic sequence of period N. Then:*

(i) the output can be expanded as a discrete Fourier series

$$z_t = \sum_{k=0}^{N-1} c_k e^{i2\pi kt/N}, \qquad c_k = \frac{1}{N}\sum_{t=0}^{N-1} z_t e^{-i2\pi kt/N};$$

(ii) the amplitude series is given by $|c_k|$, the phase series by $\arg c_k$ and the frequency series by $2\pi k/N$;

(iii) if the output is a single-frequency simple harmonic oscillation, the Fourier decomposition has only one spectral line; if the output is that of a coupled oscillator, the successive harmonics correspond to the normal modes.

Proof. (i) The discrete Fourier series of a finite periodic sequence is a standard result: the basis functions $\{e^{i2\pi kt/N}\}_{k=0}^{N-1}$ form an orthogonal basis of $\mathbb{C}^N$. (ii) This is read off directly from $c_k = A_k e^{i\phi_k}$. (iii) The time part of a single-frequency simple harmonic oscillation is $e^{-i\omega t}$, which over the period $T = 2\pi/\omega$ has only the fundamental component; a coupled oscillator is expanded in the normal modes as its successive harmonics. □

Proposition 33.1 (Three-parameter correspondence: radius, phase and speed). The three parameters of the mechanical epicycle group $\mathfrak{E}_{\mathrm{M}} = \{(A_k, \omega_k, \phi_k)\}_k$ correspond one-to-one to three geometric–physical quantities:

(i) **Radius corresponds to amplitude**: the radius $|E_k| = A_k$ of E_k is the amplitude of the corresponding generalised-coordinate component;

(ii) **Phase corresponds to rotational delay**: the initial phase ϕ_k corresponds to the time translation $\tau_k = \phi_k/\omega_k$; when $\phi_k > 0$ this component **leads** the baseline by τ_k (equivalently, the component coincides with the baseline once the baseline is postponed by τ_k);

(iii) **Frequency corresponds to rotational speed**: ω_k is the angular frequency of that normal mode.

Proof. The Proposition 33.1 of this chapter corresponds exactly to Proposition 24.3 in the 'Periodic Systems' chapter, and the three correspondences—radius to amplitude, phase to rotational delay, frequency to rotational speed—are obtained by translating each clause to the epicycle group of this chapter. For the proof see that passage: (i) from $|E_k| = A_k$ and the energy decomposition, (ii) from the phase-translation theorem, (iii) from the definition of the frequency series. □

Example 33.2 (Period series of the spring oscillator). Over the period $T = 2\pi/\omega_0$, the spring oscillator $x(t) = A\cos(\omega_0 t + \phi)$ is a single-frequency period series, and its epicycle group is $\{(A, \omega_0, \phi)\}$.

Remark 33.5 (The period series is a complex function). Let $z = e^{i\omega_0 t}$; then the mechanical period series

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)}$$

when all components are commensurable with ω_0 (that is, $\omega_k = k\omega_0$, $k \in \mathbb{Z}$) it can be written as the Laurent series $\sum_k c_k z^k$, where $c_k = A_k e^{i\phi_k}$: in that case the mechanical period series is a complex function on the unit circle, and otherwise it is merely a function on the torus.

33.1.3 Step three: dynamic adjustment of the mechanical epicycle group

Definition 33.6 (Dynamic adjustment of the mechanical epicycle group)**.** *Let the current mechanical epicycle group be $\mathfrak{E}^{(j)}$, with partial sum*

$$z^{(j)}(t) = \sum_{k\in\mathfrak{E}^{(j)}} A_k e^{i(\omega_k t+\phi_k)},$$

and residual $r^{(j)} = z - z^{(j)}$. **Dynamic adjustment** *consists of two rules:*

(i) **Number adjustment**: *if the residual energy $\|r^{(j)}\|^2 > \delta$, add one epicycle according to the peak of the residual periodogram;*

(ii) **Parameter adjustment**: *update the parameters of the existing epicycles by regularised least squares or by the conservation constraints.*

In mechanical systems, number adjustment corresponds to adding normal modes or harmonic orders; parameter adjustment corresponds to tuning the amplitude, the phase and the frequency so that the momentum, angular momentum and energy constraints are satisfied.

Proposition 33.2 (Monotonicity of number adjustment)**.** *After the dominant-frequency component of the residual has been added as in Definition 33.6(i), the residual energy is monotonically non-increasing:*

$$\|r^{(j+1)}\|^2 \le \|r^{(j)}\|^2,$$

with equality if and only if the residual has no energy at the selected frequency.

Proof. The Proposition 33.2 of this chapter agrees with the dynamic-adjustment mechanism of Definition 24.8 in the 'Periodic Systems' chapter: adding an epicycle removes the component of $r^{(j)}$ along $\hat{\omega}$ from the residual, and the orthogonal decomposition $\|r^{(j)}\|^2 = \|r^{(j+1)}\|^2 + \|$the removed component$\|^2$ yields the inequality, with equality if and only if the norm of the removed component is zero. The complete argument is given there. ☐

Theorem 33.4 (Convergence of dynamic adjustment)**.** *Let the mechanical target be a finite periodic sequence of period T_0, with $\delta = 0$. Then after at most $\lfloor T_0/2\rfloor + 1$ steps of dynamic adjustment the residual is zero, and the resulting mechanical epicycle group is the exact Fourier data of the target.*

Proof. The discrete Fourier series of a sequence of period T_0 is complete over $\lfloor T_0/2\rfloor + 1$ harmonics. At each step Proposition 33.2 extracts one dominant-frequency component of the residual and the residual energy decreases strictly, so after finitely many steps the residual is zero. ☐

Definition 33.7 (Structural-metric adjustment)**.** *Let the structural metric of a mechanical target be*

$$\mathcal{I} = (\mathcal{I}_{\text{momentum}}, \mathcal{I}_{\text{angular momentum}}, \mathcal{I}_{\text{energy}}) = (\text{degree of momentum conservation, degree of angular}$$

Structural-metric adjustment *consists in choosing θ in the parameter space Θ: at each step take a neighbouring parameter for which $\mathcal{I}(\theta)$ is* **strictly decreasing** *in the lexicographic order (minimisation); if no such parameter exists, the process stops, and the parameter is then locally optimal.*

Proposition 33.3 (Finite convergence of structural-metric adjustment). *Let the parameter space Θ be finite, and let each adjustment make $\mathcal{I}$ strictly decreasing in the lexicographic order. Then the adjustment terminates after finitely many steps at a locally optimal parameter.*

Proof. Since Θ is finite, $\mathcal{I}$ takes only finitely many values, and the lexicographic order is a total order on a finite set; a strictly decreasing sequence in that order cannot continue indefinitely, so the adjustment terminates after finitely many steps. At termination there is no neighbouring parameter that makes $\mathcal{I}$ strictly decrease in the lexicographic order, that is, the parameter is locally optimal. ☐

Remark 33.6 (Boundary of dynamic adjustment). Theorem 33.4 relies on a finite periodic sequence. For a continuous spectrum or a quasi-periodic mechanical target there is no exact representation by a finite epicycle group, and only bounded approximation is possible. Structural-metric adjustment does not guarantee global optimality.

33.1.4 Step four: mechanical convolution

Definition 33.8 (Mechanical amplitude, phase and frequency convolutions). Let two mechanical epicycle groups be

$$\mathfrak{E} = \{(A_n, \omega_n, \phi_n)\}, \qquad \mathfrak{E}' = \{(B_m, \nu_m, \psi_m)\}.$$

Define:

(i) **Amplitude convolution**: $\mathbb{A}_{mn} = A_n B_m$;

(ii) **Phase convolution**: $\mathbb{P}_{mn} = e^{i(\phi_n - \psi_m)}$;

(iii) **Frequency convolution**: $\mathbb{F}_{mn} = \omega_n \nu_m$;

(iv) **Trend convolution**: $\mathbb{T}_{mn} = A_n B_m e^{i(\phi_n - \psi_m)} = A_n B_m \mathbb{P}_{mn}$.

These agree with Definition 6.21, Definition 6.22, Definition 6.23 and Definition 6.24 in the 'Series and Convolution' chapter.

Theorem 33.5 (Action of the three convolutions on the mechanical epicycle group). *Let the epicycle groups of two mechanical targets be $\mathfrak{E}$ and $\mathfrak{E}'$ respectively. Then:*

(i) *amplitude convolution acts only on the radius;*

(ii) *phase convolution acts only on the phase;*

(iii) *frequency convolution acts only on the rotational speed;*

(iv) *trend convolution preserves both amplitude and phase.*

Proof. This is read off directly from Definition 33.8. ☐

Proposition 33.4 (Autocorrelation and trend convolution)**.** Let $\{z_t\}$ be a mechanical period sequence, with autocorrelation function

$$R_z(h) = \sum_t z_t \overline{z_{t+h}}.$$

In the language of the mechanical epicycle group, the autocorrelation function is the projection of the trend convolution onto the diagonal:

$$R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}.$$

Proof. The Proposition 33.4 of this chapter is the realisation in this domain of the convolution-action conclusion of Theorem 24.6 in the 'Periodic Systems' chapter: substituting $z_t = \sum_n A_n e^{i(\omega_n t+\phi_n)}$ into the definition of the autocorrelation, cancelling all off-diagonal cross terms by orthogonality and retaining only the diagonal term of the trend convolution (the factorisation of the three convolutions, Proposition 24.6), which yields $R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}$. The full expansion is given there. ☐

Remark 33.7 (Relation between mechanical convolution and periodic convolution)*.* Mechanical convolution is not a new convolution operation, but an instance on the mechanical epicycle group of the periodic convolution of the 'Series and Convolution' chapter. Autocorrelation is the projection of the trend convolution onto the diagonal; cross-correlation and beat frequency are respectively the concrete manifestations in classical mechanics of amplitude convolution and frequency convolution.

33.1.5 Step five: integration with discrete sampling, harmonic analysis and time-series analysis

Definition 33.9 (Mechanical periodicity diagnostic pipeline)**.** For a given mechanical system, diagnosis is carried out using the framework of periodic mathematics:

(i) **Periodicity decision**: verify the period type of the mechanical target (Theorem 33.1);

(ii) **Series construction**: build the amplitude, frequency and phase series and the epicycle group from the generalised-coordinate and momentum data (Theorem 33.3);

(iii) **Conservation diagnosis**: verify conservation of momentum, of angular momentum and of energy (Theorem 33.9);

(iv) **Convolution diagnosis**: compute the autocorrelation (the diagonal projection of the trend convolution, Proposition 33.4) and the phase convolution;

(v) **Integration diagnosis**: connect to the tool chains of discrete sampling, harmonic analysis and time-series analysis.

Theorem 33.6 (The mechanical periodicity diagnostic pipeline and its interface with three chapters)**.** The five steps of Definition 33.9 connect respectively to:

(i) the 'Discrete Sampling and Intercalation of Periodic Signals' chapter: Definition 21.1, Theorem 21.3, Theorem 21.10;

(ii) the 'Harmonic Analysis' chapter: Definition 22.5, Theorem 22.1, Proposition 22.11;

(iii) the 'Time Series Analysis' chapter: Definition 23.16, Proposition 23.2, Definition 23.19.

Hence mechanical periodicity diagnosis is not an independent method, but the application of the unified tool chain of periodic mathematics to mechanical systems.

Proof. The Theorem 33.6 of this chapter is the concretisation in this domain of the five-step interface of the 'Periodic Systems' chapter (Table 24.1): the periodicity decision uses sampling and spectral estimation, the frequency decomposition and the phase kernel use the 'Harmonic Analysis' chapter, and the periodogram peak, the Fisher test and the window-function validation use the 'Time Series Analysis' chapter. Hence the diagnostic pipeline is assembled from the tool chains of the three chapters; for the proof see that passage. □

Remark 33.8 (Order of operations after integration). The recommended order for practical diagnosis is: first decide whether the mechanical target is periodic (Theorem 33.1); then construct the Fourier data and the epicycle group of the mechanical data (Theorem 33.3); then optimise the parameters using the conservation constraints and dynamic adjustment (Theorem 33.4, Proposition 33.3); then perform the mechanical convolution and autocorrelation analysis (Proposition 33.4); and finally output the diagnosis according to Definition 33.9.

33.2 Mathematical foundations of mechanical systems

Definition 33.10 (Basic quantities of mechanical systems). Let a mechanical system have n degrees of freedom. Its description involves the following basic quantities:

(i) **Generalised coordinates** q_i $(i = 1, \dots, n)$: the independent coordinates that determine the position of the system.

(ii) **Generalised velocities** $\dot{q}_i = dq_i/dt$.

(iii) **Kinetic energy** $T = \frac{1}{2}\sum_{i,j} m_{ij}\dot{q}_i\dot{q}_j$ (the m_{ij} are the symmetric entries of the generalised mass matrix).

(iv) **Potential energy** $V(q_1, \dots, q_n)$: the part that depends only on the coordinates and not on the velocities.

(v) **Lagrangian** $L = T - V$.

(vi) **Hamiltonian** $H = T + V$ (for conservative systems); in terms of the canonical momenta $p_i = \partial L/\partial \dot{q}_i$, H is a function on the phase space (q_i, p_i).

For a conservative system, the value of the Hamiltonian H is conserved along an orbit, that is, $\frac{dH}{dt} = 0$.

33.3 Spring oscillator

Definition 33.11 (Spring oscillator)**.** *A particle of mass m is attached to a spring of stiffness k and moves on a frictionless horizontal plane. Let x denote the displacement of the particle from its equilibrium position; its equation of motion is*

$$m\ddot{x} + kx = 0 \quad \Longleftrightarrow \quad \ddot{x} + \omega_0^2 x = 0, \quad \omega_0 = \sqrt{\frac{k}{m}}.$$

This equation is called the **simple harmonic oscillation equation***, and ω_0 is called the* **natural angular frequency***. Its general solution is*

$$x(t) = A\cos(\omega_0 t + \phi),$$

where the amplitude $A \geq 0$ and the initial phase $\phi \in [0, 2\pi)$ are uniquely determined by the initial conditions $(x_0, \dot{x}_0)$.

Proposition 33.5 (Energy conservation and isochronism of the spring oscillator)**.** *The total mechanical energy $E = \frac{1}{2}kA^2$ of the spring oscillator is conserved; the oscillation period $T = 2\pi\sqrt{m/k}$ is independent of the amplitude A (isochronism).*

Proof. Energy: $E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$. Substituting $x(t) = A\cos(\omega_0 t + \phi)$ and $\dot{x}(t) = -A\omega_0 \sin(\omega_0 t + \phi)$, and using $\omega_0^2 = k/m$:

$$E = \frac{1}{2}mA^2\omega_0^2 \sin^2(\omega_0 t + \phi) + \frac{1}{2}kA^2\cos^2(\omega_0 t + \phi) = \frac{1}{2}kA^2\big(\sin^2 + \cos^2\big) = \frac{1}{2}kA^2,$$

independently of t.

Period: the least T for which the general solution $x(t) = A\cos(\omega_0 t + \phi)$ satisfies $x(t+T) = x(t)$ is given by $\omega_0 T = 2\pi$, that is, $T = 2\pi/\omega_0 = 2\pi\sqrt{m/k}$—independently of the amplitude A. This property stems from the linearity of the restoring force $F = -kx$ in x. $\square$

Definition 33.12 (Phase delay of the spring oscillator)**.** *For the spring oscillator $x(t) = A\cos(\omega_0 t + \phi)$, the* **phase delay** *corresponding to the initial phase ϕ is*

$$\tau = \frac{\phi}{\omega_0} = \phi\sqrt{\frac{m}{k}}.$$

The phase delay converts a phase angle into a time offset: an initial phase $\phi > 0$ means that the oscillation **leads** *the baseline by τ (that is, it begins τ earlier; equivalently, the two coincide once the baseline is postponed by τ). When $\phi \to 2\pi^-$, $\tau \to T$—a complete period.*

33.4 Simple pendulum

Definition 33.13 (Simple pendulum)**.** *A particle of mass m is suspended from a fixed point by a light, inextensible string of length l and swings in a vertical plane. Let θ be the angle between the string and the vertical; its equation of motion is*

$$\ddot{\theta} + \omega_0^2 \sin\theta = 0, \quad \omega_0 = \sqrt{\frac{g}{l}}.$$

This equation is nonlinear—the restoring force $F = -mg\sin\theta$ is not linear in θ. At small amplitudes $\sin\theta \approx \theta$, and the equation degenerates into the simple harmonic oscillation equation $\ddot{\theta} + \omega_0^2\theta = 0$.

Theorem 33.7 (Exact period of the pendulum—amplitude correction)**.** The oscillation period of a simple pendulum released from the maximum deflection angle θ_{max} is

$$T(\theta_{\text{max}}) = 4\sqrt{\frac{l}{g}}\,K(k), \quad k = \sin\frac{\theta_{\text{max}}}{2},$$

where $K(k)$ is the complete elliptic integral of the first kind.

Proof. By conservation of energy, the angular velocity satisfies

$$\frac{1}{2}ml^2\dot{\theta}^2 + mgl(1-\cos\theta) = mgl(1-\cos\theta_{\text{max}}).$$

Solving gives

$$\dot{\theta} = \pm\sqrt{\frac{2g}{l}\big(\cos\theta - \cos\theta_{\text{max}}\big)}.$$

Using the half-angle identity $\cos\theta - \cos\theta_{\text{max}} = 2\big(\sin^2\frac{\theta_{\text{max}}}{2} - \sin^2\frac{\theta}{2}\big)$ and making the change of variable $u = \sin\frac{\theta}{2}/\sin\frac{\theta_{\text{max}}}{2}$, the integral for a quarter period becomes

$$\frac{T}{4} = \sqrt{\frac{l}{g}}\int_0^1 \frac{du}{\sqrt{(1-u^2)(1-k^2u^2)}}, \qquad k = \sin\frac{\theta_{\text{max}}}{2},$$

that is, the complete elliptic integral of the first kind $K(k)$; hence $T = 4\sqrt{l/g}\,K(k)$. □

Definition 33.14 (Amplitude delay of the pendulum)**.** The amplitude-dependent period of a simple pendulum can be restated as an **amplitude delay**: relative to the small-amplitude period $T_0 = 2\pi\sqrt{l/g}$, the increase in period at large amplitude

$$\Delta T(\theta_{\text{max}}) = T(\theta_{\text{max}}) - T_0 = 4\sqrt{\frac{l}{g}}\left(K(k) - \frac{\pi}{2}\right)$$

is the **additional delay** introduced by the amplitude. The larger the amplitude, the longer the additional delay. In the small-amplitude limit $K(k) \approx \frac{\pi}{2}(1 + k^2/4)$; substituting $k \approx \theta_{\text{max}}/2$ gives $\Delta T \approx \frac{\pi}{8}\theta_{\text{max}}^2\sqrt{l/g}$.

33.5 Coupled oscillators and the beat phenomenon

Definition 33.15 (Coupled spring-oscillator system)**.** Consider two identical spring oscillators, each attached to a fixed end by a spring of stiffness k, with the two oscillators joined to one another by a coupling spring of stiffness k_c. Let x_1, x_2 be the displacements of the two oscillators from their respective equilibrium positions; then the equations of motion are

$$m\ddot{x}_1 + kx_1 + k_c(x_1 - x_2) = 0, \qquad m\ddot{x}_2 + kx_2 + k_c(x_2 - x_1) = 0.$$

Introducing the **normal coordinates**

$$\eta_1 = x_1 + x_2 \quad \text{(symmetric mode)}, \qquad \eta_2 = x_1 - x_2 \quad \text{(antisymmetric mode)},$$

the system decouples into two independent simple harmonic oscillation equations.

Proposition 33.6 (Beat phenomenon—energy transfer under weak coupling)**.** *In the coupled-oscillator system above, take the initial conditions $x_1(0) = A$, $x_2(0) = 0$, $\dot{x}_1(0) = \dot{x}_2(0) = 0$. Then*

$$x_1(t) = A\cos(\Delta\omega\, t)\cos(\bar{\omega}t), \qquad x_2(t) = A\sin(\Delta\omega\, t)\sin(\bar{\omega}t),$$

where $\bar{\omega} = (\omega_1 + \omega_2)/2$ and $\Delta\omega = (\omega_2 - \omega_1)/2$. Energy alternates between the two oscillators with beat period $T_{\text{beat}} = \pi/\Delta\omega$.

Proof. The normal coordinates satisfy

$$m\ddot{\eta}_1 + k\eta_1 = 0 \quad \Rightarrow \quad \omega_1 = \sqrt{k/m},$$

$$m\ddot{\eta}_2 + (k + 2k_c)\eta_2 = 0 \quad \Rightarrow \quad \omega_2 = \sqrt{(k + 2k_c)/m}.$$

From the initial conditions, $\eta_1(0) = A$, $\dot{\eta}_1(0) = 0$, $\eta_2(0) = A$, $\dot{\eta}_2(0) = 0$, hence $\eta_1(t) = A\cos\omega_1 t$ and $\eta_2(t) = A\cos\omega_2 t$. Solving back gives

$$x_1 = \frac{\eta_1 + \eta_2}{2} = A\cos(\Delta\omega\, t)\cos(\bar{\omega}t), \qquad x_2 = \frac{\eta_1 - \eta_2}{2} = A\sin(\Delta\omega\, t)\sin(\bar{\omega}t).$$

The envelope $A\cos(\Delta\omega\, t)$ first vanishes at $t = \pi/(2\Delta\omega)$—the energy of oscillator 1 is transferred completely to oscillator 2—and returns after the same interval. Hence the beat period is $T_{\text{beat}} = \pi/\Delta\omega$. ☐

Definition 33.16 (Beat-frequency delay of coupled oscillators)**.** *In a coupled-oscillator system, the* **beat-frequency delay** *required for the energy to be transferred completely from one oscillator to the other is*

$$\tau_{\text{beat}} = T_{\text{beat}} = \frac{\pi}{\Delta\omega}.$$

Under weak coupling ($k_c \ll k$), $\Delta\omega = (\omega_2 - \omega_1)/2 \approx k_c/(2\sqrt{km})$, hence

$$\tau_{\text{beat}} \approx \frac{2\pi\sqrt{km}}{k_c}.$$

The weaker the coupling (the smaller k_c), the longer the beat-frequency delay—the slower the energy transfer.

33.6 Forced oscillation

Definition 33.17 (Forced simple harmonic oscillator)**.** *Under the combined action of viscous damping and a periodic external force, the equation of motion of a forced simple harmonic oscillator is*

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = \frac{F_0}{m}\cos(\omega t),$$

where $\beta > 0$ is the damping coefficient, and F_0 and ω are the amplitude and the angular frequency of the driving force respectively.

Theorem 33.8 (Steady-state response—amplitude amplification and phase lag). The steady-state solution of the forced oscillator above is

$$x(t) = A(\omega)\cos(\omega t - \delta(\omega)),$$

where

$$A(\omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\beta\omega)^2}}, \qquad \delta(\omega) = \begin{cases} \arctan\frac{2\beta\omega}{\omega_0^2-\omega^2}, & \omega < \omega_0, \\ \pi/2, & \omega = \omega_0, \\ \pi + \arctan\frac{2\beta\omega}{\omega_0^2-\omega^2}, & \omega > \omega_0. \end{cases}$$

Proof. Use the complex method: writing $x = \mathrm{Re}[\tilde{x}e^{i\omega t}]$, the equation becomes

$$\big(-\omega^2 + 2i\beta\omega + \omega_0^2\big)\tilde{x} = F_0/m,$$

that is,

$$\tilde{x} = \frac{F_0/m}{\omega_0^2 - \omega^2 + 2i\beta\omega} = A(\omega)e^{-i\delta(\omega)},$$

where $A(\omega)$ is the modulus of the complex number and $-\delta(\omega)$ its argument (satisfying $\tan\delta = \frac{2\beta\omega}{\omega_0^2-\omega^2}$). Taking the real part gives the steady-state solution. The extremum of the amplitude $A(\omega)$ is given by $\frac{dA^2}{d\omega^2} = 0$, namely $\omega_r^2 = \omega_0^2 - 2\beta^2$ (which requires $\beta < \omega_0/\sqrt{2}$). ☐

Definition 33.18 (Phase-response delay of the forced oscillator). In the steady-state response of a forced oscillator, the **phase-response delay** corresponding to the phase lag $\delta(\omega)$ is

$$\tau(\omega) = \frac{\delta(\omega)}{\omega}.$$

The three limiting cases are:

(i) The low-frequency limit ($\omega \to 0$): $\delta \to 0$, $\tau \to 0$—the response follows the driving force instantaneously.

(ii) The resonance frequency ($\omega = \omega_0$): $\delta = \pi/2$, $\tau = \pi/(2\omega_0) = T_0/4$—the response lags by a quarter period.

(iii) The high-frequency limit ($\omega \to \infty$): $\delta \to \pi$, $\tau \to \pi/\omega$—the response is reversed.

Proposition 33.7 (Relation between resonance delay and system parameters). At the resonance frequency $\omega = \omega_0$, the phase-response delay is

$$\tau_{\mathrm{res}} = \frac{\pi}{2\omega_0} = \frac{T_0}{4} = \frac{\pi}{2}\sqrt{\frac{m}{k}}.$$

The resonance delay depends only on the intrinsic parameters m and k of the system, and not on the damping coefficient β or the driving-force amplitude F_0. The resonance delay is a measure of the intrinsic time scale of the system.

Lemma 33.6 (Damping dependence of the resonance amplitude). The amplitude at the resonance frequency $\omega_r = \sqrt{\omega_0^2 - 2\beta^2}$ is

$$A_{\text{max}} = \frac{F_0/m}{2\beta\sqrt{\omega_0^2 - \beta^2}},$$

and $\lim_{\beta\to 0^+} A_{\text{max}} = +\infty$, $\lim_{\beta\to\infty} A(\omega_0) = 0$. Damping is the sole limiting factor on the resonance amplification factor.

Proof. From the amplitude formula of Theorem 33.8, put $f(\omega) = (\omega_0^2-\omega^2)^2+4\beta^2\omega^2$. Then $\frac{df}{d\omega} = -4\omega(\omega_0^2-\omega^2)+8\beta^2\omega = 4\omega(\omega^2-\omega_0^2+2\beta^2) = 0$. The non-zero solution gives $\omega_r^2 = \omega_0^2 - 2\beta^2$ (which requires $\beta < \omega_0/\sqrt{2}$, since otherwise there is no resonance peak). Substituting into $A(\omega)$ gives $A_{\text{max}} = \frac{F_0/m}{2\beta\sqrt{\omega_0^2-\beta^2}}$. The two limits are verified by taking limits directly in the expression. □

33.7 Rigid-body rotation and precession

Definition 33.19 (Precession and nutation of a gyroscope). A rapidly spinning gyroscope under the action of the gravitational torque exhibits two kinds of motion:

(i) **Precession**: the spin axis rotates slowly about the vertical axis. The precession angular velocity satisfies $\boldsymbol{\tau} = \boldsymbol{\Omega}_p \times \mathsf{L}$, where $\boldsymbol{\tau}$ is the external torque and L the angular momentum.

(ii) **Nutation**: a high-frequency oscillation of the angle between the spin axis and the vertical axis, superposed on the precession.

Proposition 33.8 (Precession frequency and nutation frequency). For a rapidly spinning gyroscope (spin angular velocity ω_3 much greater than the precession angular velocity),

$$\Omega_p = \frac{mgr}{I_3\omega_3}, \qquad \omega_n = \frac{I_3\omega_3}{I_1},$$

where I_3 is the moment of inertia about the spin axis, I_1 the moment of inertia about a transverse axis, r the distance from the pivot to the centre of mass, and g the acceleration due to gravity.

Proof. By the angular-momentum theorem $\frac{d\mathsf{L}}{dt} = \boldsymbol{\tau}$. For a rapidly spinning gyroscope, $|\mathsf{L}| \approx I_3\omega_3$ is approximately constant (only its direction changes). The gravitational torque $\boldsymbol{\tau} = mgr\,(\hat{\mathbf{e}}_z \times \hat{\mathsf{L}})$ is always perpendicular to L, so the direction of L rotates about the vertical axis with angular velocity

$$\Omega_p = \frac{|\boldsymbol{\tau}|}{|\mathsf{L}|} = \frac{mgr}{I_3\omega_3}$$

—that is, precession. Nutation can be obtained by linearising Euler's equations: under a perturbation the angle γ between the spin axis and the vertical axis satisfies $\ddot{\gamma} + \omega_n^2\gamma = 0$, with frequency $\omega_n = I_3\omega_3/I_1$. □

Definition 33.20 (Precession-response delay of a gyroscope). *For a gyroscope under the action of the gravitational torque, the* **precession-response delay** *required for the spin axis to precess through an angle $\Delta\phi$ from its initial direction is*

$$\tau_{\text{prec}} = \frac{\Delta\phi}{\Omega_p} = \frac{I_3\omega_3 \cdot \Delta\phi}{mgr}.$$

The faster the spin (the larger ω_3), the longer the precession-response delay—the greater the angular momentum, the greater the 'inertia' required to change its direction. The precession-response delay measures the speed with which the system responds to a change in the direction of the angular momentum.

33.8 Conservation laws of mechanical systems

Theorem 33.9 (Three conservation laws of mechanical systems). *For a mechanical system consisting of N particles:*

(i) **Conservation of momentum**: *if there is no external force, the total momentum $\mathsf{P} = \sum_i m_i \mathsf{v}_i$ of the system is conserved.*

(ii) **Conservation of angular momentum**: *if there is no external torque (the total external torque about some fixed point or centre of mass is zero), the total angular momentum $\mathsf{L} = \sum_i \mathsf{r}_i \times m_i \mathsf{v}_i$ of the system is conserved about that point.*

(iii) **Conservation of energy**: *if all forces acting on the system are conservative and the system does not depend explicitly on time, then the total mechanical energy $E = T + V$ is conserved.*

Proof. (i) Summing over the N particles: $\frac{d\mathsf{P}}{dt} = \sum_i \mathsf{F}_i^{\text{ext}}$ (internal forces cancel in pairs by Newton's third law). In the absence of external forces P is a constant vector.

(ii) Taking moments about a fixed point O: $\frac{d\mathsf{L}}{dt} = \boldsymbol{\tau}^{\text{ext}}$, where $\boldsymbol{\tau}^{\text{ext}}$ is the total torque of the external forces about O (internal torques cancel in pairs). In the absence of external torque L is conserved.

(iii) Integrating $\mathsf{F} = m\ddot{\mathsf{r}}$ along the orbit: for a conservative force $\mathsf{F} = -\nabla V$ we have $\frac{d}{dt}(T+V) = 0$—$E = T + V$ is constant. □

Remark 33.9 (Connection with the rest of the book). The theory of the periodic structure of mechanical systems established in this chapter is an application of periodic mathematics to classical mechanics. The introduction of the concept of delay reveals the temporal structure of the periodic structure of mechanical systems: the phase delay $\tau = \phi/\omega$ converts a phase angle into a time offset; the amplitude delay $\Delta T(\theta_{\text{max}})$ converts a nonlinear scaling correction into a time offset; the beat-frequency delay $\tau_{\text{beat}} = \pi/\Delta\omega$ converts the coupling strength into an energy-transfer period; the phase-response delay $\tau = \delta/\omega$ converts the lag of the system's response to the driving force into a time offset; and the precession-response delay $\tau_{\text{prec}} = \Delta\phi/\Omega_p$ converts the response of the direction of the angular momentum to a change into a time offset. These five delays are associated respectively with the manifestations in the time domain of the three basic operations (translation, rotation and scaling), and together they constitute the complete delay spectrum of the periodic structure of mechanical systems.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The time translation and phase delay $\tau = \phi\sqrt{m/k}$ of the spring oscillator (Definition 33.12); the time-translation invariance of conservation of momentum (Theorem 33.9)
Scaling	The amplitude delay of the pendulum at large amplitude $\Delta T(\theta_{\text{max}}) \approx \frac{\pi}{8}\theta_{\text{max}}^2\sqrt{l/g}$ (Definition 33.14); the damping scaling of the resonance amplitude (Lemma 33.6)
Rotation	The phase lag $\delta(\omega)$ of the forced oscillator (Theorem 33.8); gyroscopic precession $\Omega_p = mgr/(I_3\omega_3)$ (Proposition 33.8)
Conservation	Energy conservation and isochronism of the spring oscillator (Proposition 33.5); conservation of momentum, angular momentum and energy (Theorem 33.9)
Symmetry	The symmetric–antisymmetric decomposition of the normal coordinates (proof of Proposition 33.6); time-reversal symmetry
Closure	The periodic orbit of the pendulum (elliptic-integral period, Theorem 33.7); the closure of the gyroscopic precession circle and of the beat energy exchange
Period	The spring oscillator $T = 2\pi\sqrt{m/k}$ (isochronism); the pendulum $T = 4\sqrt{l/g}\,K(k)$ (Theorem 33.7); the beat period $T_{\text{beat}} = \pi/\Delta\omega$ (Proposition 33.6); the resonance delay $\tau_{\text{res}} = T_0/4$ (Proposition 33.7)

Exercises

Exercise 1 Prove the **periodicity decision theorem for mechanical systems** of this chapter (33.1). Hint: first write down the decision map and the evolution group of this chapter's target, then apply the proof structure of the general criterion of the 'Periodic Systems' chapter.

Exercise 2 Verify that the hypotheses of the **monotonicity of number adjustment** (33.2) hold for the domain objects of this chapter, and give a counterexample in which the hypotheses fail.

Exercise 3 Examine the behaviour of the **three-parameter correspondence: radius, phase and speed** (33.1) under parameter degeneracy, and explain how the conclusion degenerates accordingly.

Exercise 4 Explain the relation between the **mechanical periodicity diagnostic pipeline and its interface with three chapters** (33.6) and the corresponding general conclusion of the 'Periodic Systems' chapter (whether the two agree, whether the former is a special case, or whether additional conditions are required).

Exercise 5 Using the seven-theme comparison table of this chapter, identify the concrete carriers of the themes **translation**, **scaling** and **rotation** in this chapter, and give one example of domain data for each.

Exercise 6 Synthesis: restate the diagnostic pipeline of this chapter using the five-step standard interface of the 'Periodic Systems' chapter (Table 24.1), noting one by one the theorem of this chapter on which each step depends: 33.1, 33.2, 33.1, 33.6.

Chapter 34

Periodic Structure of Circuit and Chip Systems

Remark 34.1 (Periodicity decision for the target of this chapter (citation)). When this chapter decides whether a research target is periodic, determines its period type, or seeks its minimal positive period, reference is always made to the 'Periodic Systems' chapter: for the periodicity criterion see Theorem 24.1, for the discrimination of period types see Definition 24.3, Proposition 24.1, for the search for the minimal positive period see Definition 24.4, Theorem 24.2, and for the construction of period series and epicycle groups from data see Theorem 24.3, Definition 24.6.

Remark 34.2 (Positioning of this chapter: developed along the five-step interface of the 'Periodic Systems' chapter). The periodic structure of circuit and chip systems studies the mathematical characterisation of periodicity and resonance phenomena in circuits, and of the chip as an epicycle waveform/spectrum processing unit. This chapter does not establish a separate periodicity criterion; instead it connects circuit and chip systems to the five-step interface prescribed by the 'Periodic Systems' chapter:

(1) **Decision**: reduce such targets as crystal-oscillator oscillation, PLL phase locking, clock-tree propagation and LC oscillation to discrete or continuous dynamical systems, invoke Theorem 24.1 to decide whether they are periodic, and determine the period type;

(2) **Construction**: build the amplitude series, the frequency series and the phase series from crystal-oscillator waveforms, clock signals, data signals and spectral data, and assemble them into the circuit epicycle group;

(3) **Adjustment**: dynamically adjust the circuit epicycle group subject to residual-energy descent and the constraints of charge conservation, energy conservation and flux conservation;

(4) **Convolution**: introduce amplitude convolution, phase convolution and frequency convolution on the circuit epicycle group, and link them to autocorrelation, cross-correlation and clock skew;

(5) **Integration**: connect to the tool chains of the three chapters 'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis', forming a circuit periodicity diagnostic procedure.

The core content of this chapter is: **from the perspective of periodic mathematics, a chip is an epicycle waveform/spectrum processing unit with parameters.** All the inputs and outputs of a chip are processing operations based on epicycle waveforms/spectra. Specifically:

(1) **The crystal oscillator as a fundamentally fixed epicycle**: the quartz crystal oscillator is the most central periodic source in a chip—it outputs a sinusoidal/square-wave oscillation whose amplitude, frequency and phase are all fundamentally fixed. The crystal oscillator is the **master epicycle** of all the other epicycles within the chip (driven by the outer ring).

(2) **The phase-locked loop as a frequency multiplier/divider of epicycles**: a PLL multiplies the crystal-oscillator epicycle up to the GHz level, or divides it down to the kHz level—it is the **scaling operation** on epicycles.

(3) **The clock tree as a nested epicycle system**: the clock distribution from the crystal oscillator to the individual registers is a layer-by-layer nested epicycle structure—the global clock (the master epicycle) drives the module clocks (secondary epicycles), and the module clocks drive the register clocks (epicycles further inside).

(4) **The data signal as a modulated epicycle**: the data signal is nested on the clock epicycle by modulation—the '1' and '0' of the data correspond to the two states of a phase operator (the C_2 group).

(5) **The chip as a spectrum processing unit**: an FFT chip directly implements the discrete Fourier transform—it decomposes a time-domain epicycle into frequency-domain components. A filter chip implements the selection of specific frequency components.

The mathematical tools used in this chapter include: the epicycle model, LC oscillation, phase locking, feedback loops, periodic convolution, epicycle groups, phase operators and delay.

Remark 34.3 (Guide to the seven themes of this chapter). **Translation**—the advance of clock beats and the signal propagation delay (the clock-tree propagation delay τ_i and the pipeline delay τ_{pipe}, Definition 34.15, 34.23); **Scaling**—process scaling (Moore's law, Definition 34.17) and PLL frequency multiplication/division (Definition 34.12); **Rotation**—the phase rotation and locking of the phase-locked loop (Definition 34.13, Proposition 34.6); **Conservation**—energy conservation (power dissipation) and charge/information conservation (Theorem 34.7, Remark 34.10); **Symmetry**—the H-tree symmetry of the clock tree (Example 34.3) and differential-signal symmetry (Definition 34.18); **Closure**—the closure of the phase-locked loop feedback path (Definition 34.12); **Period**—the crystal-oscillator period $T_0 = 1/f_0$ (Definition 34.10) and the clock period T_{clk} (Definition 34.20).

34.1 Connection to the five-step interface of the 'Periodic Systems' chapter

34.1.1 Step one: periodicity decision for the circuit target

Definition 34.1 (Circuit target and decision input). A **circuit target** is one of the following objects:

(i) a crystal-oscillator oscillation system, whose state space is the (V, i) phase space;

(ii) an LC resonant circuit, whose state space is the charge–current phase space;

(iii) a phase-locked loop system, whose state space is the joint space of the input phase and the output phase;

(iv) a clock tree system, whose state space is the joint space of the clock phases of the leaf nodes;

(v) a data-modulated signal, whose state space is the sequence space of C_2 phase states.

The **decision input** of a circuit target is the pair

$$\mathcal{D}_{\mathsf{C}} = (\mathcal{S}, \Phi),$$

where $\mathcal{S}$ is the state space and $\Phi : \mathcal{S} \to \mathcal{S}$ is the map induced by time translation, phase rotation or feedback evolution. If $\mathcal{S}$ is a finite set, the target is called a **finite circuit target**; if $\mathcal{S}$ is an infinite set, the target is called an **infinite circuit target**.

Definition 34.2 (Circuit decision map). Let a circuit target be given by the decision input $\mathcal{D}_{\mathsf{C}} = (\mathcal{S}, \Phi)$. If the states of the target are generated by Φ, that is, if there exists $x_0 \in \mathcal{S}$ such that

$$x_t = \begin{cases} \Phi^t(x_0), & t \in \mathbb{Z}, \\ \Phi_t(x_0), & t \in \mathbb{R}, \end{cases}$$

for $t \in \mathbb{Z}$ the state x_t is generated by iteration of Φ; for $t \in \mathbb{R}$ it is described by the one-parameter evolution family $\{\Phi_t\}_{t\in\mathbb{R}}$, where each $\Phi_t : \mathcal{S} \to \mathcal{S}$, $\Phi_0 = \mathsf{id}$, $\Phi_{s+t} = \Phi_s \circ \Phi_t$ for all $s, t \in \mathbb{R}$, and $(t, x) \mapsto \Phi_t(x)$ is continuous. then Φ is called the **circuit decision map** of the target. The circuit target is then reduced to the discrete dynamical system $(\mathcal{S}, \Phi)$ when $t \in \mathbb{Z}$ and to the continuous dynamical system $(\mathcal{S}, \{\Phi_t\}_{t\in\mathbb{R}})$ when $t \in \mathbb{R}$; in both cases the criteria of the 'Operations and Periodicity' chapter and of the 'Periodic Systems' chapter may be applied directly.

Theorem 34.1 (Circuit periodicity decision theorem). Let the decision map of a circuit target be Φ, and let $(\mathcal{S}, \Phi)$ be a discrete dynamical system. Then:

(A) **Sufficient decision**: if there exists $m \in \mathbb{N}^+$ with $\Phi^m = \mathsf{id}$, then the target is a uniformly periodic circuit system; every orbit is closed and the orbit length divides m.

(B) **Equivalence of the decision**: the target is a uniformly periodic circuit system if and only if $\mathsf{ord}(\Phi) < \infty$.

(C) **Pointwise criterion**: the orbit of the target at a state x_0 is closed if and only if x_0 is a periodic point; the orbit length is then the period at that point.

(D) **Non-omissibility**: the conjunction of conserved quantities and symmetry groups cannot replace $\mathsf{ord}(\Phi) < \infty$; a quasi-periodic circuit target is not a periodic circuit system.

Proof. The Theorem 34.1 of this chapter is the domain instantiation of Theorem 24.1 in the 'Periodic Systems' chapter: it suffices to replace the decision map f and the evolution group by Φ of this chapter, and the four items (A) sufficient decision, (B) equivalence of the decision, (C) pointwise criterion and (D) non-omissibility then hold one by one. By the convention of Remark 24.4, they are not expanded again here. ☐

Corollary 34.1 (Circuit period types)**.** *After the decision by Theorem 34.1, a circuit target falls into one of the following types:*

- (i) **Trivial periodic type**: the minimal period is $m_0 = 1$ and the state is a fixed point;
- (ii) **Strictly periodic type**: $1 < m_0 < \infty$ and the state is strictly periodic;
- (iii) **Uniformly periodic type**: ord$(\Phi) = m < \infty$ and all orbit lengths divide m;
- (iv) **Quasi-periodic type**: ord$(\Phi) = \infty$; the orbits are not closed but are dense on a compact carrier structure;
- (v) **Eventually periodic type**: there exists N such that the target is periodic for $t \geq N$;
- (vi) **Discrete periodic type**: the carrier structure is $\mathbb{Z}$ and the normalised period ratio is $\rho = T_x/\Delta x \in \mathbb{Q}$.
- (vii) **Asymptotically periodic (damped) type**: the orbit is not closed, but tends asymptotically to a closed orbit or to a fixed point, the residual decaying as $e^{-\kappa t}$ (see Definition 24.3(vii)).

Here (i)–(iii) and (vi) are periodic circuit systems; (iv) is not a periodic circuit system but can be handled with bounded-approximation tools; (v) is a periodic circuit system only in its stable stage; (vii), like (iv), is a type in the limiting sense; it is not a periodic circuit system and can be handled with bounded-approximation tools only in its stable stage.

Proof. Each item follows by substituting the circuit target into Theorem 34.1 and Definition 24.3 of the 'Periodic Systems' chapter. ☐

Lemma 34.1 (Periodicity decision for LC oscillation)**.** *Let an LC resonant circuit consist of an inductance L and a capacitance C, with charge $q(t)$ satisfying*

$$\ddot{q} + \omega_0^2 q = 0, \qquad \omega_0 = \frac{1}{\sqrt{LC}}.$$

Then the LC oscillation is a uniformly periodic system whose minimal positive period is

$$T_0 = 2\pi\sqrt{LC}.$$

Proof. The general solution of the equation is $q(t) = A\cos(\omega_0 t + \phi)$. The time translation $T_0 = 2\pi/\omega_0 = 2\pi\sqrt{LC}$ gives $q(t + T_0) = q(t)$, and T_0 is the minimal positive period. Hence the decision map Φ satisfies $\Phi^{T_0} = \mathrm{id}$, so the system is uniformly periodic. ☐

Lemma 34.2 (Periodicity decision for crystal-oscillator oscillation)**.** Let the output signal of the crystal oscillator be

$$V_{\text{crystal}}(t) = A_{\text{fixed}} e^{i(\omega_{\text{fixed}} t + \phi_0)}, \qquad \omega_{\text{fixed}} = 2\pi f_0.$$

Then the crystal-oscillator oscillation is a uniformly periodic system whose minimal positive period is

$$T_0 = \frac{1}{f_0} = \frac{t}{K}.$$

Here $K \approx 1.67$ MHz · mm and t is the thickness of the crystal plate.

Proof. By Definition 34.10, the crystal-oscillator output is a complex exponential signal of fixed frequency whose minimal positive period is $T_0 = 2\pi/\omega_{\text{fixed}} = 1/f_0$. The resonant frequency $f_0 = K/t$ is determined by the thickness of the crystal plate. Hence the decision map Φ satisfies $\Phi^{T_0} = \text{id}$, so the system is uniformly periodic. ☐

Lemma 34.3 (Periodicity decision for PLL phase locking)**.** Let the PLL be in the locked state, with output frequency $f_{\text{out}} = N f_{\text{ref}}$. Then the output clock is a uniformly periodic system whose minimal positive period is

$$T_{\text{out}} = \frac{1}{N f_{\text{ref}}} = \frac{T_{\text{ref}}}{N}.$$

Moreover the output phase and the input phase satisfy $\phi_{\text{out}}(t) - N\phi_{\text{ref}}(t) =$ constant.

Proof. By Proposition 34.6, in the locked state $\phi_{\text{out}} = N\phi_{\text{ref}} + \text{constant}$, so the output frequency is N times the reference frequency. The minimal positive period of the output clock is $T_{\text{out}} = 1/(N f_{\text{ref}})$. Because of phase locking, an integer multiple of the difference between the output phase and the input phase is constant, so the whole system is a uniformly periodic system. ☐

Lemma 34.4 (Periodicity decision for clock-tree propagation)**.** Let the propagation delay from the root node of the clock tree to the leaf node i be τ_i. Then the clock signals of all leaf nodes are uniformly periodic systems with common period

$$T_{\text{clk}} = \frac{1}{f_{\text{clk}}},$$

and the relative phase difference between the clocks of the leaf nodes is the clock skew $\delta_{ij} = \tau_i - \tau_j$.

Proof. Every leaf node of the clock tree receives a clock signal of the same frequency as the root node, with minimal positive period $T_{\text{clk}} = 1/f_{\text{clk}}$ in each case. The phase difference between the clocks of the leaf nodes is determined by the difference of the propagation delays, $\delta_{ij} = \tau_i - \tau_j$. Hence the clocks of all leaf nodes are uniformly periodic systems with common period T_{clk}. ☐

Definition 34.3 (Algorithm for the minimal positive period of a circuit)**.** This definition is identical in content to the one in Chapter 24, section 'Periodicity decision: whether a target is periodic' (Section 24.1), namely Definition 24.4; to avoid repeated exposition, only the reference anchor is retained here and the complete statement is given there. The only difference here is terminology (circuit target / circuit sample); the steps of the algorithm and the conclusion are entirely the same.

Theorem 34.2 (Correctness of the algorithm for the minimal positive period of a circuit)**.** This theorem is identical in content to the one in Chapter 24, section 'Periodicity decision: whether a target is periodic' (Section 24.1), namely Theorem 24.2; to avoid repeated exposition, only the reference anchor is retained here and the complete statement and proof are given there. The only difference here is terminology (circuit target / circuit sample); the steps of the algorithm and the conclusion are entirely the same.

Example 34.1 (Examples of circuit periodicity decisions)**.** (i) LC oscillation: $T_0 = 2\pi\sqrt{LC}$, uniformly periodic type;

(ii) crystal-oscillator oscillation: $T_0 = 1/f_0 = t/K$, uniformly periodic type;

(iii) PLL locking: $T_{\text{out}} = T_{\text{ref}}/N$, uniformly periodic type;

(iv) clock tree: common period $T_{\text{clk}} = 1/f_{\text{clk}}$, clock skew δ_{ij} at the leaf nodes;

(v) FFT butterfly operation: the N-point phase operators form the cyclic group C_N.

Remark 34.4 (Conserved quantities do not guarantee periodicity). Charge conservation, energy conservation and flux conservation in circuit systems are auxiliary conditions for periodic structure but do not constitute sufficient conditions. For example, an ideal RC discharge circuit possesses charge conservation, yet its output signal decays exponentially with time rather than being periodic. The periodic structure of a circuit must be decided by Theorem 34.1 or by the lemmas above.

34.1.2 Step two: construction of the circuit periodic series and the epicycle group

Definition 34.4 (Circuit amplitude series, frequency series and phase series)**.** Let the state or output sequence of a circuit target be $\{z_t\}$. If it can be written as

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)} + \varepsilon_t,$$

then

$$\mathcal{A}_z(t) = \sum_k A_k e^{ikt}, \qquad \mathcal{F}_z(s) = \sum_k \omega_k s^k, \qquad \mathcal{P}_z(t) = \sum_k e^{i(kt+\phi_k)}$$

are called respectively the **amplitude series**, the **frequency series** and the **phase series** of the circuit target. The three agree with Definition 6.10, Definition 6.12 and Definition 6.11 of the 'Series and Convolution' chapter.

Definition 34.5 (Circuit epicycle group)**.** Let the periodic series of a circuit target be given by the amplitude series $\{A_k\}$, the frequency series $\{\omega_k\}$ and the phase series $\{\phi_k\}$. For each k define the epicycle

$$E_k : \theta \mapsto A_k e^{i(\theta+\phi_k)}, \qquad \theta = \omega_k t,$$

and call the set

$$\mathfrak{E}_{\mathrm{C}} = \{(A_k, \omega_k, \phi_k)\}_{k\in\mathbb{Z}}$$

the **epicycle group** of the circuit target. Its superposition is the circuit periodic series

$$z_t = \sum_k E_k(\omega_k t) = \sum_k A_k e^{i(\omega_k t+\phi_k)}.$$

Theorem 34.3 (Construction of a periodic series from circuit data)**.** *Let the output of a circuit target be a finite periodic sequence of period N. Then:*

- (i) *the output can be expanded as a discrete Fourier series*

$$z_t = \sum_{k=0}^{N-1} c_k e^{i2\pi kt/N}, \qquad c_k = \frac{1}{N}\sum_{t=0}^{N-1} z_t e^{-i2\pi kt/N};$$

- (ii) *the amplitude series is given by $|c_k|$, the phase series by $\arg c_k$ and the frequency series by $2\pi k/N$;*
- (iii) *if the output is a crystal-oscillator clock, the Fourier decomposition has only a single spectral line; if the output is a square-wave clock, the odd harmonics of each order correspond to the square-wave components.*

Proof. (i) The discrete Fourier series of a finite periodic sequence is a standard result: the basis functions $\{e^{i2\pi kt/N}\}_{k=0}^{N-1}$ form an orthogonal basis of $\mathbb{C}^N$. (ii) This is read off directly from $c_k = A_k e^{i\phi_k}$. (iii) The time part of the crystal-oscillator clock is $e^{-i\omega t}$, which on the period $T = 2\pi/\omega$ has only the fundamental component; a square wave expands, as a Fourier series, into the odd harmonics of each order. □

Proposition 34.1 (Three-parameter correspondence: radius, phase and speed)**.** *The three parameters of the circuit epicycle group $\mathfrak{E}_C = \{(A_k, \omega_k, \phi_k)\}_k$ correspond one-to-one to three geometric–physical quantities:*

- (i) **Radius corresponds to amplitude**: *E_k has radius $|E_k| = A_k$, which is the amplitude of the voltage or current component;*
- (ii) **Phase corresponds to rotational delay**: *the initial phase ϕ_k corresponds to the time translation $\tau_k = \phi_k/\omega_k$; when $\phi_k > 0$ this component* **leads** *the baseline by τ_k (equivalently, the component coincides with the baseline once the baseline is postponed by τ_k);*
- (iii) **Frequency corresponds to rotational speed**: *ω_k is the angular frequency of that clock component.*

Proof. The Proposition 34.1 of this chapter corresponds exactly to Proposition 24.3 of the 'Periodic Systems' chapter, and the three correspondences—radius to amplitude, phase to rotational delay, frequency to rotational speed—are obtained by translating them item by item to the epicycle group of this chapter. The proof is given there: (i) from $|E_k| = A_k$ and the energy decomposition, (ii) from the phase-translation theorem, (iii) from the definition of the frequency series. □

Example 34.2 (Periodic series of the crystal-oscillator clock)**.** The crystal-oscillator output $V(t) = A\cos(\omega_0 t + \phi_0)$ is a single-frequency periodic series on the period $T_0 = 1/f_0$, with epicycle group $\{(A, \omega_0, \phi_0)\}$.

Remark 34.5 (The periodic series is a complex-valued function). Let $z = e^{i\omega_0 t}$; then the circuit periodic series

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)}$$

when all components are commensurable with ω_0 (that is, $\omega_k = k\omega_0$, $k \in \mathbb{Z}$) it can be written as the Laurent series $\sum_k c_k z^k$, where $c_k = A_k e^{i\phi_k}$: in that case the circuit period series is a complex function on the unit circle, and otherwise it is merely a function on the torus.

34.1.3 Step three: dynamic adjustment of the circuit epicycle group

Definition 34.6 (Dynamic adjustment of the circuit epicycle group). Let the current circuit epicycle group be $\mathfrak{E}^{(j)}$, with partial sum

$$z^{(j)}(t) = \sum_{k \in \mathfrak{E}^{(j)}} A_k e^{i(\omega_k t + \phi_k)},$$

and residual $r^{(j)} = z - z^{(j)}$. **Dynamic adjustment** consists of two rules:

(i) **Quantity adjustment**: if the residual energy $\|r^{(j)}\|^2 > \delta$, add one epicycle according to the peak of the residual periodogram;

(ii) **Parameter adjustment**: update the parameters of the existing epicycles by regularised least squares or under the conservation constraints.

In circuit systems, quantity adjustment corresponds to adding frequency components or clock domains; parameter adjustment corresponds to tuning the amplitude, phase and frequency so that the charge, energy and flux constraints are satisfied.

Proposition 34.2 (Monotonicity of quantity adjustment). *After adding the dominant-frequency component of the residual as in Definition 34.6(i), the residual energy is monotonically non-increasing:*

$$\|r^{(j+1)}\|^2 \leq \|r^{(j)}\|^2,$$

with equality if and only if the residual has no energy at the selected frequency.

Proof. The Proposition 34.2 of this chapter is consistent with the dynamic adjustment mechanism of Definition 24.8 in the 'Periodic Systems' chapter: adding an epicycle removes the component of $r^{(j)}$ along $\hat{\omega}$ from the residual, and the orthogonal decomposition $\|r^{(j)}\|^2 = \|r^{(j+1)}\|^2 + \|\text{the removed component}\|^2$ yields the inequality, with equality if and only if the norm of the removed component is zero. The complete argument is given there. □

Theorem 34.4 (Convergence of dynamic adjustment). *Let the circuit target be a finite periodic sequence of period T_0 with $\delta = 0$. Then after at most $\lfloor T_0/2 \rfloor + 1$ steps the dynamic adjustment leaves a zero residual, and the resulting circuit epicycle group is the exact Fourier data of the target. If the target is a complex-valued sequence, the number of independent harmonics is T_0, and the corresponding bound is at most T_0 steps.*

Proof. The discrete Fourier series of a sequence of period T_0 is complete on $\lfloor T_0/2 \rfloor + 1$ harmonics. At each step, by Proposition 34.2, one dominant-frequency component is extracted from the residual and the residual energy decreases strictly, so after finitely many steps the residual is zero. ☐

Definition 34.7 (Structural-metric adjustment)**.** *Let the structural metric of a circuit target be*

$$\mathcal{I} = (\mathcal{I}_{\text{charge}}, \mathcal{I}_{\text{energy}}, \mathcal{I}_{\text{flux}}) = (\text{degree of charge conservation, degree of energy conservation, de}$$

Structural-metric adjustment *consists in choosing θ in the parameter space Θ: at each step take a neighbouring parameter for which $\mathcal{I}(\theta)$ is* **strictly decreasing** *in the lexicographic order (minimisation); if no such parameter exists, the process stops, and the parameter is then locally optimal.*

Proposition 34.3 (Finite convergence of structural-metric adjustment)**.** *Let the parameter space Θ be finite, and let each adjustment make $\mathcal{I}$ strictly decreasing in the lexicographic order. Then the adjustment terminates after finitely many steps at a locally optimal parameter.*

Proof. Since Θ is finite, $\mathcal{I}$ takes only finitely many values, and the lexicographic order is a total order on a finite set; a strictly decreasing sequence in that order cannot continue indefinitely, so the adjustment terminates after finitely many steps. At termination there is no neighbouring parameter that makes $\mathcal{I}$ strictly decrease in the lexicographic order, that is, the parameter is locally optimal. ☐

Remark 34.6 (Boundary of dynamic adjustment)*.* Theorem 34.4 relies on a finite periodic sequence. For a continuous spectrum or a quasi-periodic circuit target there is no exact representation by a finite epicycle group, and only bounded approximation is possible. Structural-metric adjustment does not guarantee global optimality.

34.1.4 Step four: circuit convolution

Definition 34.8 (Circuit amplitude convolution, phase convolution and frequency convolution)**.** *Let two circuit epicycle groups be*

$$\mathfrak{E} = \{(A_n, \omega_n, \phi_n)\}, \qquad \mathfrak{E}' = \{(B_m, \nu_m, \psi_m)\}.$$

Define:

(i) **amplitude convolution**: $\mathbb{A}_{mn} = A_n B_m$;

(ii) **phase convolution**: $\mathbb{P}_{mn} = e^{i(\phi_n - \psi_m)}$;

(iii) **frequency convolution**: $\mathbb{F}_{mn} = \omega_n \nu_m$;

(iv) **trend convolution**: $\mathbb{T}_{mn} = A_n B_m e^{i(\phi_n - \psi_m)} = A_n B_m \mathbb{P}_{mn}$.

These agree with Definition 6.21, Definition 6.22, Definition 6.23 and Definition 6.24 of the 'Series and Convolution' chapter.

Theorem 34.5 (Action of the three convolutions on the circuit epicycle group). *Let the epicycle groups of two circuit targets be $\mathfrak{E}$ and $\mathfrak{E}'$ respectively. Then:*

(i) *amplitude convolution acts only on the radius;*

(ii) *phase convolution acts only on the phase;*

(iii) *frequency convolution acts only on the rotational speed;*

(iv) *trend convolution preserves both amplitude and phase.*

Proof. This is read off directly from Definition 34.8. □

Proposition 34.4 (Autocorrelation and trend convolution). *Let $\{z_t\}$ be a circuit periodic sequence with autocorrelation function*

$$R_z(h) = \sum_t z_t \overline{z_{t+h}}.$$

In the language of circuit epicycle groups, the autocorrelation function is the projection of trend convolution onto the diagonal:

$$R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}.$$

Proof. The Proposition 34.4 of this chapter is the realisation in this domain of the convolution-action conclusion of Theorem 24.6 in the 'Periodic Systems' chapter: substituting $z_t = \sum_n A_n e^{i(\omega_n t + \phi_n)}$ into the definition of the autocorrelation, cancelling all off-diagonal cross terms by orthogonality and retaining only the diagonal term of the trend convolution (the factorisation of the three convolutions, Proposition 24.6), which yields $R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}$. The complete expansion is given there. □

Remark 34.7 (Relation between circuit convolution and periodic convolution). Circuit convolution is not a new convolution operation but an instance on the circuit epicycle group of the periodic convolution of the 'Series and Convolution' chapter. Autocorrelation is the projection of the trend convolution onto the diagonal; cross-correlation and clock skew are respectively the concrete manifestations in electronic engineering of amplitude convolution and frequency convolution. The FFT butterfly unit is the simplest hardware implementation of the periodic convolution of phase operators.

34.1.5 Step five: integration with discrete sampling, harmonic analysis and time series analysis

Definition 34.9 (Circuit periodicity diagnostic procedure). For a given circuit system, diagnosis is carried out using the periodic-mathematics framework:

(i) **Periodicity decision**: verify the period type of the circuit target (Theorem 34.1);

(ii) **Series construction**: build the amplitude, frequency and phase series and the epicycle group from crystal-oscillator waveforms, clock signals and data signals (Theorem 34.3);

(iii) **Conservation diagnosis**: verify charge, energy and flux conservation (Theorem 34.7);

(iv) **Convolution diagnosis**: compute the autocorrelation (the diagonal projection of the trend convolution, Proposition 34.4) and the phase convolution;

(v) **Integrated diagnosis**: connect to the tool chains of discrete sampling, harmonic analysis and time series analysis.

Theorem 34.6 (Circuit periodicity diagnostic procedure and its interfaces with three chapters)**.** The five steps of Definition 34.9 connect respectively to:

(i) the 'Discrete Sampling and Intercalation of Periodic Signals' chapter: Definition 21.1, Theorem 21.3, Theorem 21.10;

(ii) the 'Harmonic Analysis' chapter: Definition 22.5, Theorem 22.1, Proposition 22.11;

(iii) the 'Time Series Analysis' chapter: Definition 23.16, Proposition 23.2, Definition 23.19.

Hence circuit periodicity diagnosis is not an independent method but an application of the unified tool chain of periodic mathematics to circuit systems.

Proof. The Theorem 34.6 of this chapter is the landing in this domain of the five-step interface of the 'Periodic Systems' chapter (Table 24.1): the periodicity decision uses sampling and spectral estimation, the frequency decomposition and the phase kernel use the 'Harmonic Analysis' chapter, and the periodogram peak, the Fisher test and window-function validation use the 'Time Series Analysis' chapter. Hence the diagnostic procedure is synthesised from the tool chains of the three chapters; the proof is given there. ☐

Remark 34.8 (Operational order after integration). The recommended order for practical diagnosis is: first decide whether the circuit target is periodic (Theorem 34.1); then construct the Fourier data and the epicycle group of the circuit data (Theorem 34.3); then optimise the parameters using the conservation constraints and dynamic adjustment (Theorem 34.4, Proposition 34.3); then perform circuit convolution and autocorrelation analysis (Proposition 34.4); and finally output the diagnosis according to Definition 34.9.

34.2 The crystal oscillator: a fundamentally fixed epicycle

34.2.1 Physical principles of the crystal oscillator

Definition 34.10 (Quartz crystal oscillator)**.** A **quartz crystal oscillator** (crystal oscillator for short) uses the **piezoelectric effect** of a quartz crystal to produce stable oscillation. A quartz crystal possesses the direct piezoelectric effect (mechanical deformation produces an electric field) and the converse piezoelectric effect (an electric field produces mechanical deformation). When an alternating electric field is applied, the crystal plate undergoes mechanical vibration; the mechanical vibration is in turn fed back as an electric field through the piezoelectric effect—the positive feedback sustains a stable self-excited oscillation.

The **resonant frequency** of a quartz crystal is determined by the physical dimensions and the cut angle of the crystal plate. For an AT-cut quartz crystal plate the resonant frequency is

$$f_0 = \frac{K}{t},$$

where $K \approx 1.67$ MHz · mm is the frequency constant and t is the thickness of the crystal plate. For example, a plate with $t = 0.1$ mm has a resonant frequency of about 16.7 MHz. Once the plate has been cut and polished, its resonant frequency is **fundamentally fixed**—the frequency is determined by the geometric dimensions and is affected only minimally by temperature and ageing.

From the perspective of periodic mathematics, the crystal oscillator is a **fundamentally fixed epicycle**:

$$V_{\text{crystal}}(t) = A_{\text{fixed}} \cdot e^{i(\omega_{\text{fixed}} t + \phi_0)},$$

where the amplitude A_{fixed}, the angular frequency $\omega_{\text{fixed}} = 2\pi f_0$ and the initial phase ϕ_0 remain constant during steady-state operation.

Lemma 34.5 (Start-up and Steady-state Conditions of the Crystal Oscillator (Barkhausen Criterion)). Let the total complex gain around one loop of the oscillating circuit be $H(\omega) = \beta(\omega)A(\omega)$, where A is the voltage gain of the amplifier and β is the transmission coefficient of the feedback network. Then:

(i) **(Necessary condition for start-up, and sufficient within the linearisation)** self-excited start-up of the crystal oscillator requires

$$|H(\omega_0)| > 1, \qquad \arg H(\omega_0) = 0 \ (\text{mod } 2\pi),$$

that is, the amplitude condition (the loop gain exceeds 1) and the phase condition (positive feedback, the total phase shift being an integer multiple of 2π) hold simultaneously;

(ii) **(Sufficient condition for steady-state oscillation)** once the growing amplitude drives the amplifier into its non-linear saturation region, the modulus of the effective loop gain falls to $|H| = 1$; the amplitude then ceases to grow and the system remains on the steady-state limit cycle.

The two criteria are not interchangeable: start-up is governed by the small-signal loop gain $|H| > 1$, whereas in **sustained** oscillation the effective gain has already fallen to $|H| = 1$, so $|H| > 1$ is not a necessary condition for sustained oscillation; conversely, the small-signal linearisation cannot yield sufficiency for start-up, which requires the saturating non-linearity.

Proof. Suppose that a small perturbation of amplitude V_0 is present at the input; after one loop around the circuit the signal returning to the input is $V_1 = H(\omega)V_0$. In the linearised (small-signal) model, if $|H(\omega)| < 1$, the amplitude after the n-th loop is $|V_n| = |H|^n V_0 \to 0$, the perturbation decays loop by loop and oscillation cannot start; if $\arg H \neq 0$, the feedback phases of the successive loops are mismatched and cannot reinforce in phase. Hence start-up requires $|H| > 1$ and $\arg H = 0$.

The converse (sufficiency) cannot be obtained from the linearised model: there, $|H| > 1$ together with positive feedback only gives a perturbation growing like $|H|^n$ loop by loop, and whether the amplitude stabilises at a finite value depends on the non-linearity of the amplifier. A quartz crystal provides the steepest phase-frequency slope near its series resonant frequency f_0, so that the phase condition is satisfied only near f_0, which locks the oscillation frequency to f_0. When the amplitude grows until the amplifier enters its non-linear saturation region, the effective gain $|H|$ decreases monotonically as the amplitude increases, and finally stabilises at $|H| = 1$—the amplitude no longer grows and the system remains on the steady-state limit cycle (in agreement with the limit-cycle mechanism of the 'Periodic Structure of Mechanical Systems' chapter). □

Proposition 34.5 (Three Fixed Attributes of the Crystal Oscillator)**.** *As a fundamentally fixed epicycle, the crystal oscillator has fixed attributes at three levels:*

(i) **Fixed frequency**: *the resonant frequency $f_0 = K/t$ is determined by the thickness of the crystal plate. Once the cut is complete the frequency is essentially unchanged—short-term stability reaches the order of 10^{-12} (OCXO). The constancy of the frequency stems from the high-Q resonance of the quartz crystal, which locks the oscillation frequency to the mechanical resonant frequency of the plate.*

(ii) **Fixed amplitude**: *the amplitude of the oscillator is stabilised by the non-linear gain-saturation mechanism. After start-up the amplitude grows rapidly to its steady-state value and thereafter remains unchanged. The constancy of the amplitude stems from the self-stabilisation of the loop gain on the steady-state limit cycle—an increase in amplitude lowers the effective gain until $|H| = 1$.*

(iii) **Phase stability**: *the crystal oscillator has extremely low phase noise and extremely high short-term phase stability. Phase noise manifests in the time domain as* **jitter***—the deviation of the actual arrival time of a clock edge from the ideal time. Low jitter means that the phase is highly stable under time translation, and its statistical description is the phase-diffusion model (see the 'Probability and Distribution' chapter).*

Proof. (i) The fixed frequency is determined by the physical dimensions of the quartz crystal. The frequency constant K of the AT cut is a physical constant of the quartz material. The thickness t of the crystal plate is precisely controlled during manufacture and hardly changes in use. The variation of frequency with temperature is given by the temperature characteristic curve of the AT cut—near $25°$C the frequency-temperature coefficient is close to zero.

(ii) The fixed amplitude is guaranteed by the non-linear limit-cycle mechanism of the oscillator. The start-up condition requires the loop gain > 1, and the amplitude grows until non-linear effects reduce the loop gain to $= 1$. The amplitude is then stabilised at a steady-state value determined by the circuit parameters. This is the circuit realisation of the limit cycle of the 'Periodic Structure of Mechanical Systems' chapter.

(iii) The phase stability is guaranteed by the high Q factor of the crystal—the quality factor of a quartz crystal is $Q \approx 10^5$ to 10^7, far higher than that of an LC circuit ($Q \approx 10^2$). A high Q factor means extremely low phase noise. The statistical distribution of the jitter is approximately Gaussian, with standard deviation $\sigma_{\text{jitter}} \approx 1/(2\pi f_0 Q)$. □

34.2.2 The crystal oscillator as the master epicycle

Definition 34.11 (Master-Epicycle Status of the Crystal Oscillator). *The clock signal output by the crystal oscillator is the* **master epicycle of all the other epicycles** *in the chip. Every timing activity within the chip—register toggling, state-machine transitions, computation in the datapath—takes the crystal-oscillator clock as its global time reference. From the perspective of periodic mathematics, the crystal oscillator is the* **outermost large epicycle** *(at level $k = 1$), whose radius is the amplitude A_{clk} and whose angular speed is $\omega_{\text{clk}} = 2\pi f_0$.*

The other epicycles inside the chip are all nested on the crystal-oscillator clock epicycle:

(i) **High-speed epicycles after frequency multiplication**: *through PLL frequency multiplication, the crystal-oscillator clock is magnified into epicycles of higher frequency (such as the CPU core clock).*

(ii) **Low-speed epicycles after frequency division**: *through a frequency divider, the crystal-oscillator clock is reduced to epicycles of lower frequency (such as the UART baud-rate clock).*

(iii) **Data-signal epicycles**: *the data signal is nested on the clock epicycle by modulation, each bit of the data corresponding to the phase state of one clock period.*

This is the chip implementation of the nested epicycle system of the 'Vector Analysis' chapter—the crystal oscillator is the large epicycle (low frequency), the PLL-multiplied clock is the small epicycle (high frequency), and the data signal is the innermost modulated epicycle.

34.3 Phase-locked loop

Definition 34.12 (Epicycle Interpretation of the Phase-Locked Loop). *A* **phase-locked loop** *(PLL) is a feedback control system that locks the phase of an output clock to an input reference clock. The core components of a PLL include:*

(i) **phase detector** *(PD): compares the phase difference between the input reference clock and the feedback clock.*

(ii) **loop filter** *(LF): low-pass filters the phase-error signal.*

(iii) **voltage-controlled oscillator** *(VCO): its output frequency varies linearly with the control voltage.*

(iv) **frequency divider**: divides the VCO output frequency by N and feeds it back to the phase detector.

A PLL multiplies the crystal-oscillator clock frequency f_{ref} up to the output frequency $f_{\text{out}} = N \cdot f_{\text{ref}}$. From the perspective of periodic mathematics, a PLL is a **scaling operation on epicycles**: it scales the angular speed ω_{ref} of the crystal-oscillator epicycle to $\omega_{\text{out}} = N\omega_{\text{ref}}$ whilst maintaining phase locking—the phase difference between the output epicycle and the input epicycle is constant.

Proposition 34.6 (Phase locking and scaling ratio of the PLL)**.** The steady-state operation of a PLL satisfies **phase locking**: the phase difference $\Delta\phi$ between the output clock and the input reference clock remains constant. The frequency divider ensures that the output frequency is an integer multiple of the reference frequency: $f_{\text{out}} = Nf_{\text{ref}}$. The scaling factor N is the **scaling ratio** of the PLL—it measures the factor by which the epicycle is scaled.

In the locked state the PLL satisfies: $\phi_{\text{out}}(t) - N\phi_{\text{ref}}(t) = \text{constant}$. This means that for every N rotations of the output epicycle the input epicycle rotates 1 rotation—the nesting relation of the inner and outer epicycles is: **for each step of the input epicycle, the output epicycle rotates N steps**. This is the implementation in a chip clock system of the epicycle nesting generation rule (Definition 30.13) of the 'Periodic Structure of the I Ching System' chapter.

Proof. The phase locking of a PLL is guaranteed by the feedback mechanism. The phase detector outputs $\varepsilon(t) = K_d \cdot (\phi_{\text{ref}}(t) - \phi_{\text{fb}}(t))$, where $\phi_{\text{fb}} = \phi_{\text{out}}/N$ is the feedback phase after frequency division. The instantaneous frequency of the VCO is $\omega_{\text{out}}(t) = \omega_0 + K_v \cdot v_c(t)$, where v_c is the control voltage after loop filtering. Under steady-state locking, $\varepsilon = 0$, that is, $\phi_{\text{ref}} = \phi_{\text{fb}} = \phi_{\text{out}}/N$, hence $\phi_{\text{out}} = N\phi_{\text{ref}}$. The output phase is N times the input phase—this is the realisation of the scaling operation in the phase domain. ☐

Lemma 34.6 (Synchronisation (Lock-Holding) Range of the Phase-Locked Loop)**.** Let the loop d.c. gain of the PLL be $K = K_dK_vK_F/N$, where K_d is the phase-detector gain, K_v the voltage-controlled-oscillator gain, K_F the d.c. gain of the loop filter and N the division ratio. Then the maximum frequency offset (synchronisation range) over which a locked PLL can remain locked is

$$\Delta\omega_{\text{lock}} \approx \pm\frac{\pi}{2}K \quad (\text{rad/s}).$$

When the difference $|\omega_{\text{ref}} - \omega_0/N|$ between the reference frequency and the free-running frequency of the VCO, referred to the phase-detector input, lies within $\Delta\omega_{\text{lock}}$, the phase-locked loop remains locked; outside this range it loses lock.

Proof. In the locked state the average error voltage output by the phase detector is $v_c = K_d(\phi_{\text{ref}} - \phi_{\text{fb}})$, which drives the VCO after passing through the loop filter (d.c. gain K_F), so that the output frequency is $\omega_{\text{out}} = \omega_0 + K_vK_Fv_c$. The frequency fed back to the phase-detector input is ω_{out}/N, and the locking condition $\omega_{\text{ref}} = \omega_{\text{out}}/N$ gives

$$\omega_{\text{ref}} = \frac{\omega_0 + K_vK_FK_d\Delta\phi}{N},$$

where $\Delta\phi = \phi_{\text{ref}} - \phi_{\text{fb}}$ is the steady-state phase difference. The linear output range of the phase detector is bounded by the maximum phase difference $\Delta\phi_{\text{max}} \approx \pm\pi/2$, so the maximum frequency offset that the loop can compensate is

$$\Delta\omega_{\text{max}} = \frac{K_v K_F K_d}{N} \cdot \Delta\phi_{\text{max}} \approx \pm\frac{\pi}{2}\frac{K_v K_F K_d}{N} = \pm\frac{\pi}{2}K.$$

Hence the synchronisation range is proportional to the loop d.c. gain K and inversely proportional to the division ratio N. □

Remark 34.9 (Numerical Example of the Synchronisation Range). Take typical parameters: $K_d = 0.5$ V/rad, $K_v = 2\pi \times 10^8$ rad/s/V (that is, each volt of control voltage changes the frequency by 100 MHz), $K_F = 1$ and $N = 10$. Then the loop d.c. gain is

$$K = \frac{0.5 \times 2\pi \times 10^8}{10} = \pi \times 10^7 \text{ rad/s},$$

and the corresponding synchronisation range in the frequency domain is $\Delta f_{\text{lock}} = \Delta\omega_{\text{lock}}/(2\pi) = K/4 \approx 7.85$ MHz. This means that even if the reference frequency deviates from the locked value by about ± 7.85 MHz, the phase-locked loop can still maintain lock—this is the source of the robustness of the PLL as a frequency multiplier.

Definition 34.13 (Delay characteristics of the PLL). The output clock of a PLL has a **phase delay** relative to the input reference clock:

$$\tau_{\text{PLL}} = \frac{\Delta\phi}{\omega_{\text{out}}} = \frac{\Delta\phi}{N\omega_{\text{ref}}},$$

where $\Delta\phi$ is the steady-state phase difference in the locked state. Under ideal locking, $\Delta\phi \approx 0$ or $\pi/2$ (depending on the type of phase detector). The phase delay of the PLL converts the phase difference into a time offset—the output epicycle lags the input epicycle by a fixed time offset. In chip timing analysis this delay is called **clock skew**.

34.4 Clock tree

Definition 34.14 (Clock Tree). A **clock tree** is the clock distribution network from the crystal-oscillator/PLL output to the individual registers within a chip. The clock tree consists of multiple stages of buffers and wires and distributes the global clock to the various sequential elements. From the perspective of periodic mathematics, the clock tree is the implementation in a chip of a **nested epicycle system**:

(i) **Root node**: the crystal-oscillator/PLL output—the master epicycle (outer ring, $k = 1$).

(ii) **Intermediate nodes**: clock buffers—secondary epicycles ($k = 2, 3, \dots$), each buffer amplifying the clock signal and driving the next stage of load.

(iii) **Leaf nodes**: register clock inputs—the innermost epicycles ($k = K$), with the sequential logic of the registers referenced to this clock.

Every node of the clock tree outputs an epicycle of the same frequency as the root node but possibly with a **delay offset**. An ideal clock tree should make the phase difference between the clocks of all leaf nodes and the root-node clock zero (zero skew).

Definition 34.15 (Clock-Tree Delay and Clock Skew). The **propagation delay** required for the clock signal to travel from the root node to the leaf node i is

$$\tau_i = \sum_{j \in \mathsf{path}_i} \tau_j,$$

where τ_j is the gate-level delay of the j-th buffer plus the propagation delay of the wire. The difference between the propagation delays of different leaf nodes is called the **clock skew**:

$$\delta_{ij} = \tau_i - \tau_j.$$

From the perspective of periodic mathematics, clock skew is the **phase difference** of the inner epicycles relative to the outer epicycle in a nested epicycle system. A zero-skew clock tree is equivalent to the phases of all inner epicycles being perfectly aligned with that of the master epicycle—this is the ideal state of a nested epicycle system (perfect synchronisation of nested epicycles in the 'Vector Analysis' chapter).

Example 34.3 (H-Tree Clock Distribution). The H-tree is a symmetrical clock distribution topology used to realise a low-skew clock tree. The H-tree starts from the root node (the centre of the chip), branches symmetrically into the four quadrants, and unfolds recursively level by level. Its symmetrical structure guarantees that the path lengths from all leaf nodes to the root node are equal, so that the propagation delays agree and the clock skew is close to zero.

From the perspective of periodic mathematics, the H-tree is the **symmetrical unfolding** of a nested epicycle system: at each level of branching the clock epicycle is distributed symmetrically into four sub-epicycles, and the phases of all sub-epicycles remain exactly aligned with the outer ring. Moreover, the H-tree has a self-similar fractal structure— each level of branching is geometrically similar to the preceding one, with scaling factor $1/2$, and it thus constitutes a highly symmetrical clock distribution network.

Take a chip of size $10\,\mathsf{mm} \times 10\,\mathsf{mm}$ as an example for a concrete calculation. Let the gate-level delay of each stage of buffer be $\tau_{\mathsf{buf}} = 50\,\mathsf{ps}$, and the delay per unit length of wire be $\tau_{\mathsf{wire}} = 100\,\mathsf{ps/mm}$. The H-tree has $K = 4$ levels of branching from root to leaf, and the lengths of the wire segments at the successive levels are $L_1 = 5\,\mathsf{mm}$, $L_2 = 2.5\,\mathsf{mm}$, $L_3 = 1.25\,\mathsf{mm}$ and $L_4 = 0.625\,\mathsf{mm}$. Then the total propagation delay of leaf node i is

$$\begin{aligned}
\tau_i &= \sum_{k=1}^{4} \big(\tau_{\mathsf{buf}} + L_k\,\tau_{\mathsf{wire}}\big) \\
&= 4 \times 50\,\mathsf{ps} + (5 + 2.5 + 1.25 + 0.625) \times 100\,\mathsf{ps} \\
&= 200\,\mathsf{ps} + 937.5\,\mathsf{ps} \\
&= 1137.5\,\mathsf{ps}.
\end{aligned}$$

Since the four branching paths of the H-tree are strictly symmetrical, the delays of all leaf nodes are equal, and the clock skew is $\delta_{ij} = \tau_i - \tau_j = 0$.

By contrast, if the clock tree were asymmetrical, a difference of only $1\,\mathrm{mm}$ between the path lengths of two leaf nodes would introduce a clock skew of $100\,\mathrm{ps}$. At a clock frequency $f_{\mathrm{clk}} = 1\,\mathrm{GHz}$ (period $T_{\mathrm{clk}} = 1000\,\mathrm{ps}$), this skew consumes 10% of the clock period and significantly compresses the timing margin. This is precisely the quantitative basis for adopting a symmetrical topology (the H-tree) for clock trees.

34.5 The chip: a parametrised epicycle waveform/spectrum processing unit

Definition 34.16 (Epicycle Processing Model of the Chip)**.** *From the perspective of periodic mathematics, a* **chip** *is a* **parametrised epicycle waveform/spectrum processing unit***. The chip receives input epicycle waveforms (clock signals, data signals, analogue signals), processes them through an internal network of parametrised epicycles, and outputs the processed epicycle waveforms/spectra.*

The processing capability of the chip is constituted by the following parametrised epicycle components:

(i) **Parametrised epicycle oscillators***: crystal oscillators, PLLs and frequency dividers—they generate clock epicycles of different frequencies. Parameters: frequency f, amplitude A, phase ϕ.*

(ii) **Parametrised epicycle amplifiers***: operational amplifiers and buffers—they amplify or attenuate the amplitude of the signal epicycle. Parameter: gain G.*

(iii) **Parametrised epicycle filters***: low-pass, high-pass, band-pass and band-stop filters—they select the epicycle components in a specific frequency range. Parameters: cut-off frequency f_c, quality factor Q.*

(iv) **Parametrised epicycle modulators***: mixers and modulators—they add or subtract the frequencies of two epicycles. Parameter: local-oscillator frequency f_{LO}.*

(v) **Parametrised epicycle transformers***: FFT chips and DCT chips—they decompose a time-domain epicycle into frequency-domain components. Parameter: transform length N.*

Proposition 34.7 (All inputs and outputs of the chip are epicycles)**.** *All the inputs and outputs of a chip are essentially epicycle waveforms. Specifically:*

(i) **Clock input***: a fundamentally fixed epicycle—provided by an external crystal oscillator or clock source.*

(ii) **Data input***: data epicycles modulated on the clock epicycle—each bit corresponds to the phase state of one clock period (the C_2 group: 0 and 1).*

(iii) **Analogue input***: epicycles of continuous amplitude—for example, an audio signal is a nested epicycle of several frequency components.*

(iv) **Clock output**: epicycles scaled by the PLL—for example, the CPU core clock, the bus clock and peripheral clocks.

(v) **Data output**: modulated epicycles after processing by the chip—for example, computation results and control signals.

(vi) **Analogue output**: epicycles after filtering or amplification—for example, audio power-amplifier output and conditioned sensor-signal output.

Therefore **the function of a chip is equivalent to a parametrised transformation of epicycle waveforms**: frequency transformation (frequency multiplication/division/mixing), amplitude transformation (amplification/attenuation), phase transformation (delay/phase shift) and spectral transformation (filtering/FFT).

Proof. (i) The epicycle property of the clock input follows directly from the definition of the crystal oscillator (Definition 34.10).

(ii) A data signal can be represented as a modulation on the clock epicycle: $V_{\text{data}}(t) = \sum_n a_n \cdot \text{rect}(t - nT_{\text{clk}})$, where $a_n \in \{0, 1\}$ is the n-th bit and rect is the unit rectangular pulse. In the frequency domain the data signal is the epicycle expansion of the baseband spectrum.

(iii) An analogue signal can be expanded as a Fourier series—$V_{\text{analog}}(t) = \sum_k A_k e^{i(\omega_k t + \phi_k)}$—which is the mathematical form of a nested epicycle system.

(iv)–(vi) The processing properties of the output signals are given by the parametrised definitions of the functional modules of the chip. □

Definition 34.17 (Process Scaling: the Scaling Operation of Moore's Law). The continuous shrinking of chip process technology is the manifestation of the **scaling operation** of periodic mathematics at the industrial scale. Let the process feature size be L; then the number of transistors that can be accommodated per unit chip area satisfies $M \propto 1/L^2$—each reduction of the feature size by a factor $\sqrt{2}$ ($L \to L/\sqrt{2}$) doubles the transistor density. The empirical statement of Moore's law is that the number of transistors per unit chip area doubles about every two years. For example, when the feature size shrinks from 28 nm to 7 nm (scaling factor $\lambda = 1/4$), the transistor density increases in theory by about $\lambda^{-2} = 16$ times; along with the shrink, the clock frequency rises in step—frequency is the reciprocal of the period T ($f = 1/T$), which is the reciprocal duality of scaling under the theme of period. This self-similar scaling in geometric scale and the fractal scaling of the H-tree clock distribution (scaling factor $1/2$ per level, Example 34.3) are both instances of the scaling operation.

34.6 Periodic-mathematical analysis of common chips

34.6.1 Operational amplifier

Definition 34.18 (Operational Amplifier). An **operational amplifier** (op amp) is a high-gain differential-mode amplifier. The properties of an ideal op amp are: infinite open-loop gain, infinite input impedance, zero output impedance and infinite bandwidth. A practical op amp has an open-loop gain $A_{\text{OL}} \approx 10^5 \sim 10^7$ and a unity-gain bandwidth $f_T \approx 1 \sim 100$ MHz. The differential-mode structure of the op amp is the circuit realisation of **differential-signal symmetry**: the positive and negative input terminals are symmetrical about the reference point, a common-mode disturbance appears symmetrically at the two input terminals and is cancelled after differential-mode amplification (common-mode rejection ratio CMRR $= A_{\text{DM}}/A_{\text{CM}}$, typically $80 \sim 120$ dB), and only the differential-mode component is amplified—this is common-mode rejection achieved by means of the structural symmetry of the signal pair.

From the perspective of periodic mathematics, an op amp is a **parametrised epicycle amplifier**—it scales the amplitude of the input epicycle by the gain G. In a negative-feedback configuration the closed-loop gain of the op amp is determined by the parameters of the feedback network: $A_{\text{CL}} = -R_f/R_i$ (inverting amplifier) or $A_{\text{CL}} = 1 + R_f/R_i$ (non-inverting amplifier). The frequency response of the op amp is determined by the dominant pole—$A(s) = A_{\text{OL}}/(1 + s/\omega_p)$—and the gain-bandwidth product $GBW = A_{\text{OL}} \cdot \omega_p$ is the bandwidth manifestation of the scaling ratio.

34.6.2 Analogue-to-digital and digital-to-analogue converters

Definition 34.19 (Epicycle Interpretation of the ADC and DAC)**.** An **analogue-to-digital converter** (ADC) converts a continuous analogue signal (a continuous-amplitude epicycle) into a discrete digital signal (a sequence of discrete phase states). A **digital-to-analogue converter** (DAC) performs the inverse operation.

From the perspective of periodic mathematics, the ADC and the DAC are **converters between continuous epicycles and discrete phase operators**:

(i) the ADC discretises the continuous epicycle $V(t) = \sum_k A_k e^{i(\omega_k t + \phi_k)}$ at the sampling rate f_s, each sample being quantised into 2^N discrete levels (a C_{2^N} phase operator).

(ii) the DAC reconstructs a sequence of discrete phase operators into a continuous epicycle waveform.

The sampling rate f_s and the number of quantisation bits N of an ADC are the parameters that determine the conversion accuracy—f_s determines the highest resolvable frequency (the Nyquist frequency $f_s/2$; see Definition 21.2 in the 'Discrete Sampling and Intercalation of Periodic Signals' chapter), whilst N determines the amplitude resolution (signal-to-noise ratio $SNR \approx 6.02N + 1.76$ dB).

34.6.3 Processors and microcontrollers

Definition 34.20 (Epicycle Interpretation of the Processor)**.** A **processor** (CPU) is the most complex parametrised epicycle processing unit in a chip. The processor takes the crystal-oscillator clock as its master epicycle and processes data epicycles through pipelines, multi-level caches and parallel execution units:

(i) **Clock period**: the smallest unit of time of the processor—the period of the master epicycle, $T_{\text{clk}} = 1/f_{\text{clk}}$.

(ii) **Instruction cycle**: the number of clock periods required to execute one instruction—the instruction epicycle is nested on the clock epicycle.

(iii) **Pipeline stage**: the successive phases of instruction execution (instruction fetch, decode, execute, memory access and write-back)—each pipeline stage is an independent epicycle, and data are passed between the stages through registers.

(iv) **Multi-core parallelism**: several cores each process data epicycles independently and synchronise through shared caches and buses.

The performance metrics of a processor—the clock frequency f_{clk}, the number of instructions per cycle (IPC) and the number of cores N_{core}—are all parameters of parametrised epicycles.

34.6.4 Digital signal processors and FFT chips

Definition 34.21 (DSP and FFT Chips)**.** A **digital signal processor** (DSP) is a processor specially optimised for multiply-accumulate operations, adept at implementing digital filtering and spectral analysis. An **FFT chip** is dedicated hardware that directly implements the fast Fourier transform.

From the perspective of periodic mathematics, an FFT chip is the **hardware implementation of epicycle decomposition**—it decomposes the time-domain epicycle of the input signal into frequency-domain components. The input of an FFT chip is N time-domain samples (N discrete phases) and the output is N frequency-domain components (N spectral phase operators). The core of an FFT chip is the **butterfly unit**—it combines two input phase operators into two outputs through the twiddle factor $\mathcal{O}_k^{(N)} = e^{-i2\pi k/N}$:

$$X_{\text{out},0} = X_{\text{in},0} + \mathcal{O}_k^{(N)} \cdot X_{\text{in},1}, \quad X_{\text{out},1} = X_{\text{in},0} - \mathcal{O}_k^{(N)} \cdot X_{\text{in},1}.$$

The butterfly operation is the simplest hardware implementation of the periodic convolution of phase operators (Definition 30.9 in the 'Periodic Structure of the I Ching System' chapter)—two input phases produce two output phases through a single twiddle factor.

34.6.5 Memory and registers

Definition 34.22 (Epicycle Interpretation of Memory)**. Memory** (RAM, ROM, flash) and **registers** are the elements in a chip that store phase states. From the perspective of periodic mathematics:

(i) **Registers**: they store 1 bit of information—corresponding to the two states of a C_2 phase operator (0 and 1). N registers store N bits—corresponding to the C_{2^N} phase operator and its 2^N states.

(ii) **Static random-access memory** (SRAM): each bit is a bistable circuit formed of 6 transistors—the stored phase state remains stable whilst power is supplied.

(iii) **Dynamic random-access memory** (DRAM): each bit consists of 1 transistor and 1 capacitor—the charge state of the capacitor stores the phase, and periodic refreshing is required (the refresh period is about 64 ms).

The clock period of the memory determines the read/write speed—the frequency of the data epicycle must match the read/write epicycle of the memory (a timing constraint). The capacity of the memory (the number of bits) determines the number of phase states that can be stored.

34.7 Delay in the chip

Definition 34.23 (Types of Chip Delay). *Several types of delay exist in chip systems, each corresponding to a different physical mechanism in periodic mathematics:*

(i) **Propagation delay**: the time required for a signal to travel from the chip input to the output—$\tau_{\text{prop}} = \sum \tau_{\text{gate}} + \sum \tau_{\text{wire}}$. This is the direct mapping of a spatial distance onto the time axis.

(ii) **Clock delay**: the propagation time of the clock signal from the crystal oscillator to the individual registers—$\tau_{\text{clk},i}$. The clock skew $\delta_{ij} = \tau_{\text{clk},i} - \tau_{\text{clk},j}$ is the phase difference between different registers, a phase splitting of one and the same clock epicycle at different locations.

(iii) **Pipeline delay**: the total time required for data to pass through the successive pipeline stages—$\tau_{\text{pipe}} = K \cdot T_{\text{clk}}$ (where K is the number of pipeline stages). This is the accumulation of several clock periods.

(iv) **Setup time and hold time**: the setup time t_{setup} and the hold time t_{hold} of a register are timing constraints—they define the phase relation between the data epicycle and the clock epicycle. Violating a timing constraint leads to **metastability**—an indeterminate phase state.

Proposition 34.8 (Periodic-mathematical formulation of chip timing constraints). *The timing constraints of a chip can be formulated uniformly as constraints on phase relations. Let the time at which data arrive at a register be t_{data} and the time at which the clock arrives be t_{clk}. The setup-time requirement of the register is:*

$$t_{\text{clk}} - t_{\text{data}} \geq t_{\text{setup}},$$

and the hold-time requirement is:

$$t_{\text{data}} - t_{\text{clk}} \geq t_{\text{hold}}.$$

From the perspective of periodic mathematics, these constraints ensure that the phase of the data epicycle is in a stable state within the sampling window of the clock epicycle. The setup time t_{setup} and the hold time t_{hold} are the **phase tolerances** *of the register—they define the range of relative offset permitted between the data phase and the clock phase. Violating the constraints is equivalent to the phase difference between the data epicycle and the clock epicycle exceeding the tolerance—the system enters metastability with an indeterminate phase (analogous to the phase uncertainty of the 'Probability and Distribution' chapter).*

Proof. A register samples data at the active clock edge t_{clk}. The setup-time constraint $t_{\mathsf{clk}} - t_{\mathsf{data}} \geq t_{\mathsf{setup}}$ requires the arrival (stabilisation) instant t_{data} of the data to lead the clock edge by at least t_{setup}, otherwise the data are not yet stable at sampling; the hold-time constraint $t_{\mathsf{data}} - t_{\mathsf{clk}} \geq t_{\mathsf{hold}}$ requires the data to remain unchanged until at least t_{hold} after the clock edge, otherwise the data change within the sampling window. The two constraints define respectively the **upper** and **lower** bounds of the data phase relative to the clock phase and together constitute the phase-tolerance band of the register: the data transition must fall outside the window that is centred on the clock edge and leaves t_{setup} before it and t_{hold} after it. When a data transition falls inside this window, what the sampler captures is an intermediate level in the course of switching, and the register output enters **metastability**—the output phase is indeterminate. Metastability is eventually resolved by a subsequent clock edge or by noise, and the time required for resolution follows an exponential distribution (in agreement with the phase uncertainty of the 'Probability and Distribution' chapter). □

34.8 Conservation laws of circuit systems

Theorem 34.7 (Three conservation laws of circuit systems). (i) **Charge conservation**: *Kirchhoff's current law (KCL)—$\sum i_{\mathsf{in}} = \sum i_{\mathsf{out}}$. Charge does not accumulate at a node.*

(ii) **Energy conservation**: *Kirchhoff's voltage law (KVL)—$\sum v_k = 0$. The work done by the electric field on a unit charge around a closed loop is zero.*

(iii) **Flux conservation**: *the flux $\Phi = Li$ of an inductance is continuous in the absence of external flux injection.*

Proof. (i) Charge conservation (KCL): take any node of the circuit and let the outflowing currents be $i_1, \ldots, i_m$ and the inflowing currents be $i_{m+1}, \ldots, i_{m+n}$. By the law of charge conservation, the rate of change of the charge Q at the node equals the negative of the net inflowing current: $dQ/dt = \sum_{k=m+1}^{m+n} i_k - \sum_{k=1}^{m} i_k$. The lumped-parameter assumption states that the node stores no charge ($Q \equiv 0$), so $dQ/dt = 0$, giving $\sum_{k=1}^{m} i_k = \sum_{k=m+1}^{m+n} i_k$—the sum of the inflowing currents equals the sum of the outflowing currents, which is KCL.

(ii) Energy conservation (KVL): in the electrostatic approximation with no time-varying magnetic flux the electric field is conservative, and there exists an electric potential φ with $\vec{E} = -\nabla\varphi$. For a unit charge travelling once around the closed loop L, the work done by the electric field is

$$\oint_L \vec{E} \cdot d\vec{l} = -\oint_L \nabla\varphi \cdot d\vec{l} = -\oint_L d\varphi = 0,$$

that is, the sum of the voltage drops across the elements of a closed loop is zero: $\sum_k v_k = 0$. If a time-varying magnetic flux is present in the loop, then by Faraday's law $\oint_L \vec{E} \cdot d\vec{l} = -d\Phi/dt$, and KVL must be corrected to $\sum_k v_k = -d\Phi/dt$—this is precisely the origin of the induced electromotive force (the 'Periodic Structure of Electromagnetic Systems' chapter).

(iii) Flux conservation: the flux of the inductance current i is $\Phi = Li$, and by Faraday's law $d\Phi/dt = v_L$ (where v_L is the voltage across the inductance). In the absence of an external flux discontinuity (no ideal switch instantaneously interrupting the inductance current), Φ varies continuously, so the inductance current $i = \Phi/L$ cannot change abruptly—this is the mathematical root of 'continuous inductance current' in switching power supplies. □

Remark 34.10 (Power dissipation and information conservation). The theme of conservation has two extensions in chips. (i) **Energy conservation and power dissipation**: the dynamic power dissipation $P_{\text{dyn}} = \alpha C V_{\text{dd}}^2 f$ characterises the energy dissipated in charging and discharging the load capacitance C in each clock period—the dissipation per period is $\Delta E = C V_{\text{dd}}^2$, and multiplying by the switching activity factor α and the clock frequency f gives the power dissipation; this is the dissipative form, under the periodic beat, of the energy conservation of Theorem 34.7. (ii) **Information conservation**: the phase states (bits) stored in registers and memory are not lost in an ideal noise-free device and change only when they are rewritten or when power is removed—this is the circuit realisation of information conservation, corresponding to the information measure of the 'Probability and Distribution' chapter.

Remark 34.11 (Connection with the rest of the book). The theory of the periodic structure of circuit and chip systems established in this chapter is an application of periodic mathematics in electronic engineering. The core insight is that **chip = parametrised epicycle waveform/spectrum processing unit**—all the inputs and outputs of a chip are essentially epicycle waveforms. The crystal oscillator is a fundamentally fixed epicycle and is the master epicycle of all the other epicycles within the chip. A PLL is the scaling operation on epicycles, a clock tree is a nested epicycle system, and a data signal is a sequence of phase states modulated on the clock epicycle. An operational amplifier is a parametrised epicycle amplifier, an ADC/DAC is a converter between continuous epicycles and discrete phase operators, an FFT chip is the hardware implementation of epicycle decomposition, and memory is the storage unit of phase states. The delays in a chip—propagation delay, clock delay, pipeline delay and timing constraints—completely cover the four delay structures of periodic mathematics: spatial distance, phase difference, period accumulation and phase tolerance. The three conservation laws of circuit systems—KCL (charge conservation), KVL (energy conservation) and flux conservation—together constitute the physical basis of the operation of a chip.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The advance of clock beats and the signal propagation delay: the clock-tree propagation delay $\tau_i = \sum_j \tau_j$ (Definition 34.15); the propagation delay and the pipeline delay $\tau_{\text{pipe}} = KT_{\text{clk}}$ (Definition 34.23); the stability of the crystal-oscillator phase under time translation (Proposition 34.5)
Scaling	Process scaling (Moore's law, $M \propto 1/L^2$, Definition 34.17); PLL frequency multiplication/division $f_{\text{out}} = Nf_{\text{ref}}$ (Definition 34.12); the H-tree fractal scaling factor $1/2$ (Example 34.3)
Rotation	The phase rotation and locking of the phase-locked loop $\phi_{\text{out}} = N\phi_{\text{ref}}$ (Proposition 34.6); the PLL phase delay $\tau_{\text{PLL}} = \Delta\phi/(N\omega_{\text{ref}})$ (Definition 34.13); the FFT butterfly twiddle factor $e^{-i2\pi k/N}$ (Definition 34.21)
Conservation	Charge conservation (KCL), energy conservation (KVL) and flux conservation (Theorem 34.7); dynamic power dissipation $P = \alpha C V_{\text{dd}}^2 f$ and information conservation (Remark 34.10)
Symmetry	The H-tree symmetry of the clock tree (zero skew, Example 34.3); differential-signal symmetry and common-mode rejection (Definition 34.18); the stability of frequency, amplitude and phase among the three attributes of the crystal oscillator (Proposition 34.5)
Closure	The closure of the phase-locked loop feedback path: the phase-detector–loop-filter–VCO–divider closed loop (Definition 34.12); the PLL synchronisation range $\Delta\omega_{\text{lock}} \approx \pm\frac{\pi}{2}K$ (Lemma 34.6); the loop closure of crystal-oscillator start-up $\lvert H\rvert \geq 1$ (Lemma 34.5)
Period	The crystal-oscillator period $T_0 = 1/f_0$ (Definition 34.10); the clock period $T_{\text{clk}} = 1/f_{\text{clk}}$ (Definition 34.20); the DRAM refresh period of about 64 ms (Definition 34.22)

Exercises

Exercise 1 Prove the **circuit periodicity decision theorem** of this chapter (34.1). Hint: first write down the decision map and the evolution group of this chapter's target, then apply the proof structure of the general criterion of the 'Periodic Systems' chapter.

Exercise 2 Verify that the hypotheses of **monotonicity of quantity adjustment** (34.2) hold for the domain objects of this chapter, and give a counterexample in which the hypotheses fail.

Exercise 3 Examine the behaviour of the **three-parameter correspondence**: **radius, phase and speed** (34.1) under degeneracy of the parameters, and explain how the conclusion degenerates accordingly.

Exercise 4 Explain the relation between **all inputs and outputs of the chip are epicycles** (34.7) and the corresponding general conclusion of the 'Periodic Systems' chapter (whether they agree, whether it is a special case, or whether additional conditions are required).

Exercise 5 Using the seven-theme comparison table of this chapter, identify the concrete carriers in this chapter of the three themes **translation**, **scaling** and **rotation**, and give one example of domain data for each.

Exercise 6 Synthesis: restate the diagnostic procedure of this chapter using the five-step standard interface of the 'Periodic Systems' chapter (Table 24.1), noting for each step the theorem of this chapter on which it depends: 34.1, 34.2, 34.1, 34.7.

Chapter 35

Periodic Structures in Biological Systems

Remark 35.1 (Periodicity decision for the target of this chapter (citation)). When this chapter decides whether a research target is periodic, determines its period type, or seeks its minimal positive period, reference is always made to the 'Periodic Systems' chapter: for the periodicity criterion see Theorem 24.1, for the discrimination of period types see Definition 24.3, Proposition 24.1, for the search for the minimal positive period see Definition 24.4, Theorem 24.2, and for the construction of period series and epicycle groups from data see Theorem 24.3, Definition 24.6.

Remark 35.2 (Positioning of this chapter: unfolding along the five-step interface of the 'Periodic Systems' chapter). The periodic structure of biological systems studies the mathematical characterisation of endogenous rhythms, periodic behaviour and the circadian clock in living organisms. This chapter does not set up a separate periodicity criterion; instead it connects biological systems to the five-step interface specified by the 'Periodic Systems' chapter:

(1) **Decision**: the circadian rhythm, the heartbeat cycle, the cell cycle, the population cycle, neural oscillation, multi-scale evolution and other such targets are formalised as discrete or continuous dynamical systems, and Theorem 24.1 is cited to decide whether they are periodic and to determine the period type;

(2) **Construction**: the amplitude series, the frequency series and the phase series are constructed from concentration variables, membrane potentials, population sizes and phase variables, and assembled into the biological epicycle group;

(3) **Adjustment**: the biological epicycle group is dynamically adjusted under the constraints of residual energy decrease, energy conservation, matter conservation and information conservation;

(4) **Convolution**: amplitude convolution, phase convolution and frequency convolution are introduced on the biological epicycle group, and connected with autocorrelation, cross-correlation and phase locking;

(5) **Integration**: the tool chains of three chapters—'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis'—are connected, forming the biological periodicity diagnostic pipeline.

The mathematical tools used in this chapter include: limit-cycle theory, delayed negative feedback, relaxation oscillation, the Lotka–Volterra equations, the phase-oscillator model, the replicator–mutator equation, periodic convolution, the epicycle group, the phase operator and time delay.

Remark 35.3 (Guide to the seven themes of this chapter). **Translation**—the approximately 24-hour main period of the circadian rhythm (the internalisation of the Earth's rotation, Definition 35.11); **Scaling**—the scaling transformation of the cell cycle (DNA content $2n \to 4n \to 2n$, Definition 35.13) and the scaling of population size; **Rotation**—the heartbeat limit cycle (Definition 35.12) and the phase of neural oscillation (Definition 35.15); **Conservation**—energy conservation, matter conservation and information conservation (Theorem 35.11); **Symmetry**—the complementary symmetry of the two DNA strands and the information symmetry of transmission from generation to generation; **Closure**—the closed orbits guaranteed by the Lotka–Volterra first integral (Theorem 35.7); **Period**—the multi-scale periods of circadian (24 h), heartbeat (0.6–1.0 s), cell (hours–days) and population (9–11-year lynx cycle) processes.

35.1 Connection to the five-step interface of the 'Periodic Systems' chapter

35.1.1 Step one: periodicity decision for biological targets

Definition 35.1 (Biological target and decision input). *A* **biological target** *is one of the following objects:*

(i) *a circadian rhythm system, whose state space is the phase space of protein concentrations (M, P, Z);*

(ii) *a heartbeat cycle system, whose state space is the joint space of the membrane potential and the ion-channel variables (V_m, n);*

(iii) *a cell cycle system, whose state space is the space of cyclin concentrations;*

(iv) *a population cycle system, whose state space is the prey–predator phase space (H, P);*

(v) *a neural oscillation system, whose state space is the firing-phase space of neurons;*

(vi) *a multi-scale evolution system, whose state space is the joint space of the state variables of the various levels.*

The **decision input** *of a biological target is an ordered pair*

$$\mathcal{D}_{\mathrm{B}} = (\mathcal{S}, \Phi),$$

where $\mathcal{S}$ is the state space and $\Phi : \mathcal{S} \to \mathcal{S}$ is the map induced by time translation, phase rotation or feedback evolution. If $\mathcal{S}$ is a finite set, the target is called a **finite biological target**; *if $\mathcal{S}$ is an infinite set, it is called an* **infinite biological target**.

Definition 35.2 (Biological decision map)**.** Let a biological target be given by the decision input $\mathcal{D}_{\mathrm{B}} = (\mathcal{S}, \Phi)$. If the states of the target are generated by Φ, that is, if there exists $x_0 \in \mathcal{S}$ such that

$$x_t = \begin{cases} \Phi^t(x_0), & t \in \mathbb{Z}, \\ \Phi_t(x_0), & t \in \mathbb{R}, \end{cases}$$

for $t \in \mathbb{Z}$ the state x_t is generated by iteration of Φ; for $t \in \mathbb{R}$ it is described by the one-parameter evolution family $\{\Phi_t\}_{t\in\mathbb{R}}$, where each $\Phi_t : \mathcal{S} \to \mathcal{S}$, $\Phi_0 = \mathrm{id}$, $\Phi_{s+t} = \Phi_s \circ \Phi_t$ for all $s, t \in \mathbb{R}$, and $(t, x) \mapsto \Phi_t(x)$ is continuous. then Φ is called the **biological decision map** of the target. The biological target is then reduced to the discrete dynamical system $(\mathcal{S}, \Phi)$ when $t \in \mathbb{Z}$ and to the continuous dynamical system $(\mathcal{S}, \{\Phi_t\}_{t\in\mathbb{R}})$ when $t \in \mathbb{R}$; in both cases the criteria of the 'Operations and Periodicity' chapter and of the 'Periodic Systems' chapter may be applied directly.

Theorem 35.1 (Periodicity decision theorem for biological systems)**.** Let the decision map of a biological target be Φ, and let $(\mathcal{S}, \Phi)$ be a discrete dynamical system. Then:

(A) **Sufficient decision**: if there exists $m \in \mathbb{N}^+$ such that $\Phi^m = \mathrm{id}$, then the target is a uniformly periodic biological system; every orbit is closed, and the length of every orbit divides m.

(B) **Equivalence of the decision**: the target is a uniformly periodic biological system if and only if $\mathrm{ord}(\Phi) < \infty$.

(C) **Pointwise criterion**: the orbit of the target through the state x_0 is closed if and only if x_0 is a periodic point; the length of the orbit is then the period at that point.

(D) **Indispensability**: the conjunction of conserved quantities and the symmetry group cannot replace $\mathrm{ord}(\Phi) < \infty$; a quasi-periodic biological target is not a periodic biological system.

Proof. The Theorem 35.1 of this chapter is the domain instantiation of Theorem 24.1 in the 'Periodic Systems' chapter: it suffices to replace the decision map f and the evolution group by Φ of this chapter, and the four items—(A) sufficient decision, (B) equivalence of the decision, (C) pointwise criterion and (D) indispensability—hold one by one. By the convention of Remark 24.4, they are not expanded again here. ☐

Corollary 35.1 (Period types of biological targets)**.** After the decision by Theorem 35.1, a biological target falls into one of the following types:

(i) **Trivially periodic type**: the least period is $m_0 = 1$ and the state is a fixed point;

(ii) **Strictly periodic type**: $1 < m_0 < \infty$ and the state is strictly periodic;

(iii) **Uniformly periodic type**: $\mathrm{ord}(\Phi) = m < \infty$ and the length of every orbit divides m;

(iv) **Quasi-periodic type**: $\operatorname{ord}(\Phi) = \infty$; the orbits are not closed but are dense on a compact carrier structure;

(v) **Eventually periodic type**: there exists N such that the target is periodic for $t \geq N$;

(vi) **Discrete periodic type**: the carrier structure is $\mathbb{Z}$ and the normalised period ratio is $\rho = T_x/\Delta x \in \mathbb{Q}$.

(vii) **Asymptotically periodic (damped) type**: the orbit is not closed, but tends asymptotically to a closed orbit or to a fixed point, the residual decaying as $e^{-\kappa t}$ (see Definition 24.3(vii)).

Among these, (i)–(iii) and (vi) are periodic biological systems; (iv) is not a periodic biological system, but can be handled by bounded-approximation tools; (v) is a periodic biological system only in its stable stage; (vii), like (iv), is a type in the limiting sense; it is not a periodic biological system and can be handled with bounded-approximation tools only in its stable stage.

Proof. Each item follows by substituting the biological target into Theorem 35.1 and Definition 24.3 of the 'Periodic Systems' chapter. □

Remark 35.4 (Periodicity decision for the circadian rhythm (empirical observation / research programme)). *Empirical observation: on the observational scale the circadian rhythm of mammals is a rhythm of stable period, whose least positive period is approximately*

$$T_{\text{circadian}} \approx 24 \text{ hours.}$$

Empirical basis: the foregoing is an empirical observation, not a proved theorem. This value is not a deduction from the model parameters—the three-variable Goodwin oscillator (Proposition 35.5) yields only the existence of a stable limit cycle for Hill coefficient $n > 8$, and its period varies continuously with the parameters; the value of 24 hours comes from the matching of the delay time scale of the TTFL (the accumulation, modification, nuclear localisation and degradation of the PER/CRY proteins) with the physiological parameters, and belongs to empirical observation and a research programme.

Lemma 35.1 (Periodicity decision for the heartbeat cycle)**.** *Let the membrane potential V_m and the ion-channel variables of a sinoatrial (SA) node pacemaker cell form a limit cycle in phase space. Then the heartbeat cycle is a uniformly periodic system, whose least positive period is*

$$T_{\text{cardiac}} \approx 0.6\text{–}1.0 \text{ s},$$

corresponding to a heart rate of 60–100 beats per minute.

Proof. By Definition 35.12, pacemaker cells generate autorhythmic action potentials through the periodic activation of currents such as I_f, $I_{Ca\text{-}T}$, $I_{Ca\text{-}L}$ and I_K. The joint dynamics of the membrane potential and the ion-channel variables form a limit cycle in phase space, and each complete depolarisation–repolarisation cycle corresponds to one turn of the limit cycle. Hence the heartbeat cycle is a uniformly periodic system, with a period of approximately 0.6–1.0 s. □

Lemma 35.2 (Periodicity decision for the cell cycle)**.** *Let the Cyclin B concentration evolve as a sawtooth relaxation oscillation, with accumulation rate ρ, threshold C_{th} and residual concentration C_0. Then the cell cycle is a uniformly periodic system, whose least positive period is*

$$T_{\text{cell}} = \frac{C_{\text{th}} - C_0}{\rho}.$$

Proof. By Lemma 35.5, the period of a sawtooth relaxation oscillation is $T = (C_{\text{th}} - C_0)/\rho$. When the Cyclin B concentration reaches the threshold it triggers the M phase; at the end of the M phase it is rapidly degraded to C_0, and accumulation begins afresh in the next cycle. Hence the decision map Φ satisfies $\Phi^{T_{\text{cell}}} = \text{id}$, and the system is uniformly periodic. ☐

Lemma 35.3 (Periodicity decision for the population cycle)**.** *Let the first integral of the Lotka–Volterra system be $I(H, P)$, and let the equilibrium point be $(H^*, P^*) = (\gamma/\delta, \alpha/\beta)$. Then for every positive initial condition the system is a uniformly periodic system, and its small-amplitude period is approximately*

$$T_{\text{pop}} \approx \frac{2\pi}{\sqrt{\alpha\gamma}}.$$

Proof. By Theorem 35.7, the equilibrium point is a centre-type fixed point, the first integral $I(H, P)$ attains a minimum at (H^*, P^*), and its level curves are closed curves surrounding the equilibrium point. For initial conditions sufficiently close to the equilibrium point the period is given by the linearised eigenvalues $\lambda = \pm i\sqrt{\alpha\gamma}$, namely $T \approx 2\pi/\sqrt{\alpha\gamma}$. Hence the system is uniformly periodic. ☐

Lemma 35.4 (Periodicity decision for neural oscillation)**.** *Let two interneurons be coupled through an inhibitory synapse, with natural frequencies ω_1, ω_2 and coupling strengths c_{12}, c_{21}. Then when $|\Delta\omega| < c_{12} + c_{21}$ the neuronal network enters phase locking and is a uniformly periodic system, with locking period*

$$T_{\text{sync}} = \frac{2\pi}{\bar{\omega}},$$

where $\bar{\omega}$ is the common frequency after phase locking.

Proof. By the proof of Proposition 35.8, the phase-difference dynamics $d(\Delta\theta)/dt = \Delta\omega - (c_{12} + c_{21}) \sin(\Delta\theta)$ has a stable fixed point $\Delta\theta^*$ when $|\Delta\omega| < c_{12} + c_{21}$. The phase difference of the two neurons is then constant, and the common frequency is locked to $\bar{\omega}$. Hence the network is a uniformly periodic system, with period $T_{\text{sync}} = 2\pi/\bar{\omega}$. ☐

Definition 35.3 (Least positive period algorithm for biological systems)**.** This definition has the same content as the corresponding definition in Chapter 24, section 'Periodicity decision: whether a target is periodic' (Section 24.1), namely Definition 24.4; to avoid repetition, only the citation anchor is retained here, and the complete statement is given there. The only difference here is terminology (biological target / biological sample); the steps of the algorithm and the conclusion are exactly the same.

Theorem 35.2 (Correctness of the least positive period algorithm for biological systems)**.** This theorem has the same content as the corresponding theorem in Chapter 24, section 'Periodicity decision: whether a target is periodic' (Section 24.1), namely Theorem 24.2; to avoid repetition, only the citation anchor is retained here, and the complete statement and proof are given there. The only difference here is terminology (biological target / biological sample); the steps of the algorithm and the conclusion are exactly the same.

Example 35.1 (Examples of biological periodicity decisions)**.** (i) circadian rhythm: $T_{\text{circadian}} \approx 24$ h (an empirical observation, see Remark 35.4);

(ii) heartbeat cycle: $T_{\text{cardiac}} \approx 0.6$–$1.0$ s, of the uniformly periodic type;

(iii) cell cycle: $T_{\text{cell}} = (C_{\text{th}} - C_0)/\rho$, of the uniformly periodic type;

(iv) population cycle: $T_{\text{pop}} \approx 2\pi/\sqrt{\alpha\gamma}$ (small-amplitude limit), of the uniformly periodic type;

(v) neural oscillation: locking period $T_{\text{sync}} = 2\pi/\bar{\omega}$, of the uniformly periodic type.

Remark 35.5 (Conserved quantities do not guarantee periodicity). Energy conservation, matter conservation and information conservation in biological systems are auxiliary conditions for periodic structure, but they do not constitute a sufficient condition. For example, a biochemical system that conserves energy but sits at an unstable equilibrium will not spontaneously produce periodic oscillations. Periodic structure in biological systems must be decided by Theorem 35.1 or by the lemmas above.

35.1.2 Step two: construction of the biological period series and the epicycle group

Definition 35.4 (Biological amplitude, frequency and phase series)**.** Let the state or output sequence of a biological target be $\{z_t\}$. If it can be written as

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)} + \varepsilon_t,$$

then

$$\mathcal{A}_z(t) = \sum_k A_k e^{ikt}, \qquad \mathcal{F}_z(s) = \sum_k \omega_k s^k, \qquad \mathcal{P}_z(t) = \sum_k e^{i(kt+\phi_k)}$$

are called respectively the **amplitude series**, the **frequency series** and the **phase series** of the biological target. The three agree with Definition 6.10, Definition 6.12 and Definition 6.11 of the 'Series and Convolution' chapter.

Definition 35.5 (Biological epicycle group)**.** Let the period series of a biological target be given by the amplitude series $\{A_k\}$, the frequency series $\{\omega_k\}$ and the phase series $\{\phi_k\}$. For each k define the epicycle

$$E_k : \theta \mapsto A_k e^{i(\theta+\phi_k)}, \qquad \theta = \omega_k t,$$

and let its set

$$\mathfrak{E}_{\mathrm{B}} = \{(A_k, \omega_k, \phi_k)\}_{k\in\mathbb{Z}}$$

be called the **epicycle group** of the biological target. Its superposition is the biological period series

$$z_t = \sum_k E_k(\omega_k t) = \sum_k A_k e^{i(\omega_k t + \phi_k)}.$$

Theorem 35.3 (Construction of the period series from biological data). Let the output of a biological target be a finite periodic sequence of period N. Then:

(i) the output can be expanded as a discrete Fourier series

$$z_t = \sum_{k=0}^{N-1} c_k e^{i2\pi kt/N}, \qquad c_k = \frac{1}{N}\sum_{t=0}^{N-1} z_t e^{-i2\pi kt/N};$$

(ii) the amplitude series is given by $|c_k|$, the phase series by $\arg c_k$ and the frequency series by $2\pi k/N$;

(iii) if the output is the protein concentration of the circadian rhythm, the Fourier decomposition yields a main component of approximately 24 hours together with its harmonics; if the output is the P-QRS-T waveform of the electrocardiogram, the harmonics of the various orders correspond to the electrical activity of different cardiac structures.

Proof. (i) The discrete Fourier series of a finite periodic sequence is a standard result: the basis functions $\{e^{i2\pi kt/N}\}_{k=0}^{N-1}$ form an orthogonal basis of $\mathbb{C}^N$. (ii) This is read off directly from $c_k = A_k e^{i\phi_k}$. (iii) For a circadian sequence with $N = 24$ (hourly sampling), the fundamental frequency corresponds to the 24-hour main period and the harmonics correspond to ultradian rhythms; for an ECG sequence, the waveform components are given by the harmonic assignment of Proposition 35.6. □

Proposition 35.1 (Three-parameter correspondence: radius, phase and speed). The three parameters of the biological epicycle group $\mathfrak{E}_{\mathrm{B}} = \{(A_k, \omega_k, \phi_k)\}_k$ are in one-to-one correspondence with three geometric–biological quantities:

(i) **Radius corresponds to amplitude**: the radius $|E_k| = A_k$ of E_k is the amplitude of that concentration component or potential component;

(ii) **Phase corresponds to rotational delay**: the initial phase ϕ_k corresponds to the time translation $\tau_k = \phi_k/\omega_k$; when $\phi_k > 0$ this component **leads** the baseline by τ_k (equivalently, the component coincides with the baseline once the baseline is postponed by τ_k);

(iii) **Frequency corresponds to rotational speed**: ω_k is the angular frequency of that biological oscillation component.

Proof. The Proposition 35.1 of this chapter corresponds exactly to Proposition 24.3 of the 'Periodic Systems' chapter, and the three correspondences—radius to amplitude, phase to rotational delay and frequency to rotational speed—are obtained by translating each item to the epicycle group of this chapter. The proof is given there: (i) from $|E_k| = A_k$ and the energy decomposition, (ii) from the phase-translation theorem, (iii) from the definition of the frequency series. □

Example 35.2 (The period series of the electrocardiogram). The P-QRS-T waveform of the electrocardiogram is a periodic sequence with period $T = 1/\text{heart rate}$; in its Fourier decomposition, the P wave and the T wave correspond to low-order harmonics ($n \le 2$) and the QRS complex to high-order harmonics ($n \approx 3$–10). In the biological epicycle group these harmonics appear as epicycle components of different radii and phases.

Remark 35.6 (The period series is a complex function). Let $z = e^{i\omega_0 t}$; then the biological period series

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)}$$

when all components are commensurable with ω_0 (that is, $\omega_k = k\omega_0$, $k \in \mathbb{Z}$) it can be written as the Laurent series $\sum_k c_k z^k$, where $c_k = A_k e^{i\phi_k}$: in that case the biological period series is a complex function on the unit circle, and otherwise it is merely a function on the torus.

35.1.3 Step three: dynamic adjustment of the biological epicycle group

Definition 35.6 (Dynamic adjustment of the biological epicycle group). Let the current biological epicycle group be $\mathfrak{E}^{(j)}$, with partial sum

$$z^{(j)}(t) = \sum_{k \in \mathfrak{E}^{(j)}} A_k e^{i(\omega_k t + \phi_k)},$$

and residual $r^{(j)} = z - z^{(j)}$. **Dynamic adjustment** consists of two rules:

(i) **Number adjustment**: if the residual energy satisfies $\|r^{(j)}\|^2 > \delta$, add one epicycle according to the peak of the residual periodogram;

(ii) **Parameter adjustment**: update the parameters of the existing epicycles by regularised least squares or under the conservation constraints.

In biological systems, number adjustment corresponds to increasing the harmonic order or the number of biological oscillation modes; parameter adjustment corresponds to tuning the amplitude, phase and frequency so that the energy, matter and information constraints are satisfied.

Proposition 35.2 (Monotonicity of number adjustment). *After adding the dominant-frequency component of the residual as in Definition 35.6(i), the residual energy is monotonically non-increasing:*

$$\|r^{(j+1)}\|^2 \le \|r^{(j)}\|^2,$$

with equality if and only if the residual has no energy at the selected frequency.

Proof. The Proposition 35.2 of this chapter agrees with the dynamic adjustment mechanism of Definition 24.8 in the 'Periodic Systems' chapter: adding an epicycle removes the component of $r^{(j)}$ along $\hat{\omega}$ from the residual, and the orthogonal decomposition $\|r^{(j)}\|^2 = \|r^{(j+1)}\|^2 + \|\text{the removed component}\|^2$ gives the inequality, with equality if and only if the norm of the removed component is zero. The complete argument is given there. □

Theorem 35.4 (Convergence of dynamic adjustment)**.** *Let the biological target be a finite periodic sequence of period T_0 with $\delta = 0$. Then after at most $\lfloor T_0/2 \rfloor + 1$ steps of dynamic adjustment the residual is zero, and the resulting biological epicycle group is the exact Fourier data of the target.*

Proof. The discrete Fourier series of a sequence of period T_0 is complete on $\lfloor T_0/2 \rfloor + 1$ harmonics. At each step one dominant-frequency component of the residual is extracted by Proposition 35.2 and the residual energy decreases strictly, so after finitely many steps the residual is zero. ☐

Definition 35.7 (Structural metric adjustment)**.** Let the structural metric of a biological target be

$$\mathcal{I} = (\mathcal{I}_{\text{energy}}, \mathcal{I}_{\text{matter}}, \mathcal{I}_{\text{information}}) = (\text{degree of energy conservation, degree of matter conservatio}$$

Structural-metric adjustment consists in choosing θ in the parameter space Θ: at each step take a neighbouring parameter for which $\mathcal{I}(\theta)$ is **strictly decreasing** in the lexicographic order (minimisation); if no such parameter exists, the process stops, and the parameter is then locally optimal.

Proposition 35.3 (Finite convergence of structural metric adjustment)**.** *Let the parameter space Θ be finite, and let each adjustment make $\mathcal{I}$ strictly decreasing in the lexicographic order. Then the adjustment terminates after finitely many steps at a locally optimal parameter.*

Proof. Since Θ is finite, $\mathcal{I}$ takes only finitely many values, and the lexicographic order is a total order on a finite set; a strictly decreasing sequence in that order cannot continue indefinitely, so the adjustment terminates after finitely many steps. At termination there is no neighbouring parameter that makes $\mathcal{I}$ strictly decrease in the lexicographic order, that is, the parameter is locally optimal. ☐

Remark 35.7 (Boundaries of dynamic adjustment)*.* Theorem 35.4 relies on finite periodic sequences. For continuous spectra or quasi-periodic biological targets there is no exact representation by a finite epicycle group, and only a bounded approximation is possible. Structural metric adjustment does not guarantee a global optimum.

35.1.4 Step four: biological convolution

Definition 35.8 (Biological amplitude convolution, phase convolution and frequency convolution)**.** Let two biological epicycle groups be

$$\mathfrak{E} = \{(A_n, \omega_n, \phi_n)\}, \qquad \mathfrak{E}' = \{(B_m, \nu_m, \psi_m)\}.$$

Define:

(i) **Amplitude convolution:** $\mathbb{A}_{mn} = A_n B_m$;

(ii) **Phase convolution:** $\mathbb{P}_{mn} = e^{i(\phi_n - \psi_m)}$;

(iii) **Frequency convolution:** $\mathbb{F}_{mn} = \omega_n \nu_m$;

(iv) **Trend convolution**: $\mathbb{T}_{mn} = A_n B_m e^{i(\phi_n - \psi_m)} = A_n B_m \mathbb{P}_{mn}$.

This agrees with Definition 6.21, Definition 6.22, Definition 6.23 and Definition 6.24 of the 'Series and Convolution' chapter.

Theorem 35.5 (Action of the three convolutions on biological epicycle groups). *Let the epicycle groups of two biological targets be $\mathfrak{E}$ and $\mathfrak{E}'$ respectively. Then:*

(i) *amplitude convolution acts only on the radius;*

(ii) *phase convolution acts only on the phase;*

(iii) *frequency convolution acts only on the rotational speed;*

(iv) *trend convolution preserves both amplitude and phase.*

Proof. This follows directly from Definition 35.8. □

Proposition 35.4 (Autocorrelation and trend convolution). *Let $\{z_t\}$ be a biological period sequence, with autocorrelation function*

$$R_z(h) = \sum_t z_t \overline{z_{t+h}}.$$

In the language of the biological epicycle group, the autocorrelation function is the projection of the trend convolution onto the diagonal:

$$R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}.$$

Proof. The Proposition 35.4 of this chapter is the realisation in this domain of the convolution-action conclusion of Theorem 24.6 in the 'Periodic Systems' chapter: substituting $z_t = \sum_n A_n e^{i(\omega_n t + \phi_n)}$ into the definition of the autocorrelation, cancelling all off-diagonal cross terms by orthogonality and retaining only the diagonal term of the trend convolution (the factorisation of the three convolutions, Proposition 24.6), which yields $R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}$. The complete expansion is given there. □

Remark 35.8 (Relation between biological convolution and periodic convolution). Biological convolution is not a new convolution operation; it is an instance on the biological epicycle group of the periodic convolution of the 'Series and Convolution' chapter. Autocorrelation is the projection of the trend convolution onto the diagonal; cross-correlation and phase locking are respectively the concrete manifestations in the life sciences of amplitude convolution and frequency convolution.

35.1.5 Step five: integration with discrete sampling, harmonic analysis and time series analysis

Definition 35.9 (The biological periodicity diagnostic pipeline). For a given biological system, diagnosis is carried out using the framework of periodic mathematics:

(i) **Periodicity decision**: verify the period type of the biological target (Theorem 35.1);

(ii) **Series construction**: construct the amplitude, frequency and phase series and the epicycle group from concentration, membrane-potential and population data (Theorem 35.3);

(iii) **Conservation diagnosis**: verify the conservation of energy, matter and information (Theorem 35.11);

(iv) **Convolution diagnosis**: compute the autocorrelation (the diagonal projection of the trend convolution, Proposition 35.4) and the phase convolution;

(v) **Integration diagnosis**: connect to the tool chains of discrete sampling, harmonic analysis and time series analysis.

Theorem 35.6 (The biological periodicity diagnostic pipeline and its interface with three chapters)**.** The five steps of Definition 35.9 connect respectively to:

(i) the 'Discrete Sampling and Intercalation of Periodic Signals' chapter: Definition 21.1, Theorem 21.3, Theorem 21.10;

(ii) the 'Harmonic Analysis' chapter: Definition 22.5, Theorem 22.1, Proposition 22.11;

(iii) the 'Time Series Analysis' chapter: Definition 23.16, Proposition 23.2, Definition 23.19.

Hence the biological periodicity diagnosis is not an independent method, but an application to biological systems of the unified tool chain of periodic mathematics.

Proof. The Theorem 35.6 of this chapter is the implementation in this domain of the five-step interface of the 'Periodic Systems' chapter (Table 24.1): the periodicity decision uses sampling and spectral estimation, the frequency decomposition and the phase kernels use the 'Harmonic Analysis' chapter, and the periodogram peaks, the Fisher test and the window-function validation use the 'Time Series Analysis' chapter. Hence the diagnostic pipeline is assembled from the tool chains of three chapters; the proof is given there. ☐

Remark 35.9 (Order of operations after integration). The recommended order for an actual diagnosis is: first decide whether the biological target is periodic (Theorem 35.1); then construct the Fourier data and the epicycle group of the biological data (Theorem 35.3); then optimise the parameters using the conservation constraints and dynamic adjustment (Theorem 35.4, Proposition 35.3); then carry out biological convolution and autocorrelation analysis (Proposition 35.4); and finally output the diagnosis according to Definition 35.9.

35.2 Mathematical foundations of biological systems

Definition 35.10 (Basic quantities of biological periodic systems)**.** A biological periodic system is described by the following basic variables:

(i) **Concentration variable** $[X](t)$: the concentration of a particular biomolecule (protein, mRNA, ion, etc.) inside or outside the cell. The rate of change of the concentration in time is determined by the production rate and the degradation rate.

(ii) **Membrane potential** $V_m(t)$ (unit: mV): the potential difference across the cell membrane. Changes in the membrane potential are controlled by the opening and closing of ion channels.

(iii) **Population size** $N(t)$: the number of individuals of a given species in a given region. Changes in the population size are determined by the birth rate, the death rate and the migration rate.

(iv) **Phase variable** $\phi(t)$: the position of a biological oscillator within its period (0 to 2π). Phase is the intrinsic measure of biological time—the circadian rhythms of different tissues achieve system-wide coordination through phase synchronisation.

A core feature of biological periodic systems is the **limit cycle**—there exists an isolated closed orbit in phase space, and the system converges to that orbit from any initial state. Limit cycles appear in dissipative systems: the continuous input and dissipation of energy or matter maintain the stability of the oscillation.

35.3 The circadian rhythm

Definition 35.11 (Molecular mechanism of the circadian rhythm)**.** The circadian rhythm of mammals is driven by the **transcription–translation negative feedback loop** (TTFL). The core feedback loop consists of the following genes and proteins:

(i) **Activators**: the CLOCK and BMAL1 proteins form a heterodimer that binds to E-box sequences in the DNA and activates the transcription of the Per and Cry genes.

(ii) **Repressors**: the PER and CRY proteins accumulate in the cytoplasm and, after forming a complex, enter the nucleus and inhibit the transcriptional activation activity of CLOCK–BMAL1.

(iii) **Delay**: the accumulation, modification, nuclear localisation and degradation of the PER and CRY proteins take approximately 24 hours—this delay determines the period of the circadian rhythm.

The TTFL is a delayed negative feedback system: the activators (CLOCK–BMAL1) promote the production of the repressors (PER–CRY), the repressors in turn inhibit the activity of the activators, and the delay element causes the system to produce self-sustained oscillations. The temporal variation of the mRNA and protein concentrations constitutes a period of approximately 24 hours—every 24 hours a complete transcription–translation–inhibition–degradation cycle is completed.

Proposition 35.5 (Mathematical model of the circadian rhythm)**.** Consider a simplified TTFL model—the three-variable Goodwin oscillator (mRNA → protein → end product, negative feedback):

$$\frac{dM}{dt} = \frac{v_1}{1 + (Z/K)^n} - k_1 M, \quad \frac{dP}{dt} = v_2 M - k_2 P, \quad \frac{dZ}{dt} = v_3 P - k_3 Z,$$

where M is the mRNA concentration, P the protein concentration, Z the concentration of the end product (the repressor complex), v_1, v_2, v_3 the maximum production rates, k_1, k_2, k_3 the degradation rate constants, K the inhibition constant and n the Hill coefficient (cooperativity).

The end product Z nonlinearly inhibits the transcription of its own mRNA through the Hill function $1/(1+(Z/K)^n)$, so that the negative-feedback loop contains three variables and three conversion steps. When $n > 8$ and the parameters are suitable (see the proof), the fixed point loses stability through a Hopf bifurcation and a stable limit cycle appears.

Proof. The fixed point satisfies $v_1/(1+(Z^*/K)^n) = k_1M^*$, $v_2M^* = k_2P^*$ and $v_3P^* = k_3Z^*$. Write $u = Z^*/K > 0$ and let the steady-state flux be $R = k_1M^* = v_1/(1+u^n)$; then $Z^* = v_2v_3R/(k_1k_2k_3) = uK$. Linearisation at the fixed point gives the Jacobian matrix

$$J = \begin{pmatrix} -k_1 & 0 & -g \\ v_2 & -k_2 & 0 \\ 0 & v_3 & -k_3 \end{pmatrix}, \qquad g = \frac{nv_1}{K}\frac{u^{n-1}}{(1+u^n)^2},$$

with characteristic equation $(\lambda+k_1)(\lambda+k_2)(\lambda+k_3)+v_2v_3g = 0$. The divergence of the two-variable model (with Z and the third equation removed) is identically $-(k_1+k_2) < 0$, so by Bendixson's criterion there is no closed orbit in a simply connected region, and no limit cycle arises however large n may be—the third variable is precisely what makes the oscillation possible. Taking the three degradation constants equal ($k_1 = k_2 = k_3 = k$), the characteristic equation reduces to

$$(\lambda + k)^3 + k^3\hat{\eta} = 0, \qquad \hat{\eta} = \frac{nu^n}{1+u^n},$$

so the real part of the conjugate pair is $k(\hat{\eta}^{1/3}/2 - 1)$, which is positive if and only if $\hat{\eta} > 8$. Since $u^n/(1+u^n) < 1$, the condition $\hat{\eta} > 8$ requires $n > 8$; conversely, when $n > 8$ one may take K sufficiently small so that u is sufficiently large and $\hat{\eta} > 8$, and the fixed point loses stability. At $\hat{\eta} = 8$ the eigenvalues cross the imaginary axis with non-zero speed (a Hopf bifurcation), so for $n > 8$ a stable limit cycle appears—this is precisely the mathematical mechanism by which the delayed negative feedback of the TTFL generates a self-sustained oscillation. □

35.4 The heartbeat cycle

Definition 35.12 (Electrophysiology of cardiac pacemaker cells). The pacemaker cells of the sinoatrial (SA) node of the heart possess **automaticity**—in the absence of external stimulation they spontaneously generate rhythmic action potentials. The action potential of a pacemaker cell is divided into four phases:

(i) **Phase 4 (diastolic depolarisation)**: the pacemaker current I_f (Na^+ influx) and the T-type calcium current $I_{Ca\text{-}T}$ slowly depolarise the membrane potential from about -60 mV to about -40 mV.

(ii) **Phase 0 (upstroke)**: after the membrane potential reaches the threshold, the L-type calcium current $I_{Ca\text{-}L}$ surges and the membrane potential rises rapidly to about $+10$ mV.

(iii) **Phase 3 (repolarisation)**: *the delayed rectifier potassium current* I_K *activates, and the efflux of potassium ions returns the membrane potential to about* -60 *mV.*

(iv) **Phase 4 begins again**: *the period is approximately* 0.6*–*1.0 *s, corresponding to a heart rate of* 60*–*100 *beats per minute.*

The membrane potential V_m *and the ionic currents form a limit cycle in phase space. Each action potential corresponds to one complete rotation in phase space—*depolarisation (the upstroke) and repolarisation (the downstroke) *constitute a reciprocating motion. The natural frequency of about* 0.8 *Hz of sinoatrial node cells is determined by the maximum conductances and the kinetic time constants of the various ion channels.*

Proposition 35.6 (Periodic structure of the electrocardiogram)**.** *The electrocardiogram is the projection onto the body surface of the electrical activity of the heart, and the typical waveform of each cardiac cycle consists of the P wave, the QRS complex and the T wave. These waveforms are biological instances of nested epicycle systems:*

(i) **P wave**: *atrial depolarisation—corresponding to the synchronous action potentials of atrial muscle cells. Lowest frequency, smallest amplitude (an outer, large epicycle).*

(ii) **QRS complex**: *ventricular depolarisation—corresponding to the rapid synchronous depolarisation of ventricular muscle cells. Highest frequency, largest amplitude (an inner, fast, small epicycle).*

(iii) **T wave**: *ventricular repolarisation—corresponding to the repolarisation of ventricular muscle cells. Intermediate frequency, intermediate amplitude.*

The P-QRS-T waveform of the electrocardiogram can be expanded as a Fourier series: $V(t) = \sum_n A_n \cos(n\omega_0 t + \phi_n)$ *(with* $\omega_0 = 2\pi \times$ *heart rate). The harmonics of the various orders correspond to the electrical activity of different cardiac structures—the P wave corresponds to the fundamental frequency (*$n = 1$*), the QRS complex to high-order harmonics (*$n \approx 3$*–*10*) and the T wave to low-order harmonics (*$n \approx 1$*–*2*). The harmonic structure of the electrocardiogram is the mathematical fingerprint of the coordinated work of the various parts of the heart.*

Proof. The body-surface potential $V(t)$ of each cardiac cycle is a function of period $T = 1/$heart rate (ignoring beat-to-beat variability), so it can be expanded in the Fourier series $V(t) = \sum_n c_n e^{in\omega_0 t}$ of the 'Functions and Limits' chapter. The harmonic assignment of the waveform features is determined by steepness: the upstroke of the QRS complex is steep (< 0.1 s) and its energy is concentrated in the high-order harmonics $n \approx 3$–10 (corresponding to the 10–40 Hz band); the P wave and the T wave are gentle (> 0.1 s) and their energy is concentrated in $n \leq 2$. It is precisely according to this harmonic structure that the electrocardiographic bands are defined clinically (for example the 5–25 Hz filter for QRS monitoring)—the Fourier series is the mathematical realisation of the 'nested epicycle structure' of the electrocardiographic waveform. □

35.5 The cell cycle

Definition 35.13 (Cell cycle)**.** *The cell cycle is the continuous process through which a eukaryotic cell passes from the end of one division to the end of the next; it is divided into four phases:*

(i) **G1 phase** *(the pre-DNA-synthesis phase): the cell grows and synthesises RNA and proteins. The duration is variable (from several hours to several days).*

(ii) **S phase** *(the DNA synthesis phase): the DNA is replicated, and each chromosome forms two sister chromatids. The duration is approximately 6–8 hours.*

(iii) **G2 phase** *(the post-DNA-synthesis phase): the cell continues to grow and prepares to enter mitosis. The duration is approximately 2–4 hours.*

(iv) **M phase** *(the mitotic phase): the chromosomes separate and the cell divides in two. The duration is approximately 1 hour.*

The G1 phase corresponds to the increase in cell volume, the S phase to the exact replication of the DNA (scaling factor 2) and the M phase to the division of the cell into two (scaling factor $1/2$). The whole cell cycle is a complete scaling transformation—from a single cell growing to its division into two daughter cells, the DNA content grows from $2n$ to $4n$ and returns to $2n$ (diploid cells).

Proposition 35.7 (Concentration oscillation of cyclin–CDK)**.** *The progress of the cell cycle is regulated by complexes of cyclins and cyclin-dependent kinases (CDKs). The successive phases are driven in turn by specific cyclin–CDK complexes:*

$$\begin{aligned}&\text{G1 phase: Cyclin D–CDK4/6}\\ \longrightarrow\ &\text{G1/S transition: Cyclin E–CDK2}\\ \longrightarrow\ &\text{S phase: Cyclin A–CDK2}\\ \longrightarrow\ &\text{M phase: Cyclin B–CDK1.}\end{aligned}$$

The concentration of Cyclin B undergoes a sawtooth oscillation during the cell cycle: it accumulates linearly during interphase (G1–S–G2) and is abruptly degraded at the end of the M phase. When its concentration reaches the threshold, mitosis is triggered; after division is complete the concentration falls sharply and the next cycle begins afresh. This oscillation is a relaxation oscillation formed by the alternating action of linear accumulation (interphase) and rapid degradation (the end of the M phase).

Proof. The sawtooth oscillation of Cyclin B is the relaxation oscillation of Lemma 35.5: the accumulation rate ρ is the synthesis rate during interphase, the threshold C_{th} is the concentration that triggers APC/C-mediated degradation and C_0 is the residual concentration after mitosis ends. Hence the duration of the cell cycle is $T = (C_{\text{th}} - C_0)/\rho$—the slower the growth rate (the smaller ρ), the longer the cycle; this agrees with the experimental fact that 'the duration of the G1 phase determines the length of the cell cycle'. □

Lemma 35.5 (Period of the sawtooth relaxation oscillation). *Let a relaxation oscillation grow linearly at rate $\rho > 0$ during the accumulation phase, trigger a reset (degradation) on reaching the threshold C_{th}, and reset to $C_0 < C_{\text{th}}$, the duration of the reset being negligible relative to the accumulation phase. Then the oscillation is a strictly periodic oscillation, with period*

$$T = \frac{C_{\text{th}} - C_0}{\rho}.$$

Proof. During the accumulation phase $dC/dt = \rho$, so $C(t) = C_0 + \rho t$. The time to reach the threshold is obtained from $C_0 + \rho T = C_{\text{th}}$, giving $T = (C_{\text{th}} - C_0)/\rho$; after the reset the system returns to the same initial condition C_0, and by uniqueness of the solution of the initial-value problem the cycle then repeats segment by segment. □

35.6 The population cycle

Definition 35.14 (The Lotka–Volterra predator–prey model). Let $H(t)$ be the number of prey (for example hares) and $P(t)$ the number of predators (for example lynx). The classical Lotka–Volterra equations are

$$\frac{dH}{dt} = \alpha H - \beta HP, \quad \frac{dP}{dt} = \delta HP - \gamma P,$$

where α is the intrinsic growth rate of the prey, β is the predation rate, δ is the proportion of prey converted by predators into offspring and γ is the death rate of the predators.

In the phase space (H, P) the trajectories of the system form a family of closed curves surrounding the equilibrium point—each initial condition corresponds to one periodic orbit. The system has the first integral

$$I(H, P) = \delta H - \gamma \ln H + \beta P - \alpha \ln P = \text{constant}.$$

The numbers of prey and predators rise and fall in turn, with a phase difference of about $90°$—the prey grows first and the predators grow afterwards; the predators decline first and the prey declines afterwards.

Theorem 35.7 (Periodic solution of the Lotka–Volterra model). *All positive initial conditions ($H_0 > 0, P_0 > 0$) of the Lotka–Volterra model produce periodic solutions. The equilibrium point $(H^*, P^*) = (\gamma/\delta, \alpha/\beta)$ is a centre-type fixed point—the eigenvalues of the linearised system are purely imaginary, $\lambda = \pm i\sqrt{\alpha\gamma}$. The period of small-amplitude oscillations is approximately $T \approx 2\pi/\sqrt{\alpha\gamma}$. The period of large-amplitude oscillations is computed by means of the first integral and elliptic integrals, and the period grows as the amplitude increases.*

Proof. Linearisation: the eigenvalues of the Jacobian matrix at the equilibrium point (H^*, P^*) are $\lambda_{1,2} = \pm i\sqrt{\alpha\gamma}$—purely imaginary, so the equilibrium point is a centre. The first integral $I(H, P)$ is obtained by separating the variables in $\frac{dH}{dP} = \frac{H(\alpha - \beta P)}{P(\delta H - \gamma)}$ and integrating:

$$\int \frac{\delta H - \gamma}{H} dH = \int \frac{\alpha - \beta P}{P} dP,$$

giving $\delta H - \gamma \ln H + \beta P - \alpha \ln P =$ constant. This first integral attains a minimum at (H^*, P^*), and its level curves are closed curves surrounding the equilibrium point—each closed curve corresponds to one periodic solution. $\square$

Example 35.3 (The ten-year cycle of the Canadian lynx and the snowshoe hare). The fur-trade records of the Hudson's Bay Company in Canada from the 18th century to the early 20th century show a pronounced periodic oscillation in the numbers of lynx and snowshoe hares—with a period of about 9–11 years. This is the most famous instance of a population cycle in nature. In the small-amplitude approximation the 10-year period is determined jointly by the intrinsic growth rate of the prey (α) and the death rate of the predators (γ): $T \approx 2\pi/\sqrt{\alpha\gamma} \approx 10$ years. The solar activity cycle (about 11 years) and climatic cycles may modulate this biological cycle by affecting the food supply of the prey.

35.7 Neural oscillation and phase locking

Definition 35.15 (Periodic firing of neuronal action potentials). Neurons generate action potentials through the voltage dependence of their ion channels. Under a constant input current many neurons exhibit **periodic firing**—action potentials are produced repeatedly at a fixed frequency. The relation between the firing frequency f and the input current I is

$$f(I) = \frac{1}{T(I)} \approx \frac{I - I_{\text{th}}}{C_m(V_{\text{th}} - V_{\text{reset}})} \quad (I > I_{\text{th}}),$$

where I_{th} is the threshold current, C_m is the membrane capacitance, and V_{th} and V_{reset} are the threshold and reset potentials.

The membrane potential V_m and the recovery variable (such as the potassium-channel activation variable n) form a limit cycle in phase space. Each action potential corresponds to one complete rotation in phase space—the rapid upstroke (Na^+ influx) and repolarisation (K^+ efflux) constitute a reciprocating motion. The input current I controls the rotational speed of the limit cycle—the larger I is, the higher the frequency.

Proposition 35.8 (Gamma synchronous oscillation in neural networks). In the cerebral cortex, networks of inhibitory interneurons produce synchronous oscillations in the gamma band (30–80 Hz). A single interneuron fires periodically at the gamma frequency under constant excitatory drive. Through inhibitory synaptic connections (GABA receptors), the neuronal network achieves phase synchronisation—the firing times of the individual neurons lock to a common phase within the gamma cycle.

Gamma synchronous oscillation is the neural realisation of phase locking (Proposition 24.8) in the 'Periodic Systems' chapter: each interneuron is an independent periodic subsystem (with a natural frequency of about 40 Hz), and the inhibitory synapses are the coupling coefficients c_{ij} between the subsystems. When the coupling is strong enough, the phase differences of all the neurons tend to a constant—the network enters a state of synchronous oscillation. Gamma synchronous oscillation has important functions in attention, perception and consciousness—it provides a common temporal framework for the integration of information across different brain regions.

Proof. Consider two neurons connected by an inhibitory synapse, each firing periodically under a constant excitatory input. The simplified model is a pair of phase oscillators:

$$\frac{d\theta_1}{dt} = \omega_1 + c_{12}\,\mathsf{sin}(\theta_2 - \theta_1), \qquad \frac{d\theta_2}{dt} = \omega_2 + c_{21}\,\mathsf{sin}(\theta_1 - \theta_2).$$

The dynamics of the phase difference $\Delta\theta = \theta_1 - \theta_2$ is

$$\frac{d(\Delta\theta)}{dt} = \Delta\omega - (c_{12} + c_{21})\,\mathsf{sin}(\Delta\theta).$$

When $|\Delta\omega| < c_{12} + c_{21}$ there is a stable phase-locking solution $\Delta\theta^* = \mathsf{arcsin}(\Delta\omega/(c_{12} + c_{21}))$—the firing phases of the two neurons lock. Extending to a network of N neurons, the difference-angle sine component of the periodic convolution, $\mathsf{Im}\,\mathcal{K}^{(1)}\{f_i, f_j\}$, drives the network into global synchronisation. □

35.8 Survival of the fittest and multi-scale evolution

This section generalises Darwin's mechanism of survival of the fittest into a **multi-scale selection–variation–inheritance cycle**: at every biological level there is a three-step cycle of 'variation–selection–inheritance', and the levels achieve co-evolution through cross-scale coupling. Biological evolution is not a single-scale phenomenon but a nested multi-scale process—genes, proteins, cells, tissues, organs, morphology, populations and niches each have their own time scale, periodic structure and rate of evolution.

35.8.1 Hierarchical structure of multi-scale evolution

Definition 35.16 (Hierarchy of biological evolution)**.** Biological evolution can be divided into the following levels according to spatial scale, and each level has its own characteristic time scale, periodic structure and selection pressure:

(i) **Gene evolution**: time scale 10^{-1}–10^{1} generations; the sources of periodicity are base mutation, recombination and horizontal gene transfer; the selection pressure is the difference in phenotypic fitness. The change in gene frequencies is the dynamical process of departure from Hardy–Weinberg equilibrium.

(ii) **Protein evolution**: time scale 10^{0}–10^{3} generations; the sources of periodicity are amino-acid substitution, domain shuffling and conformational switching; the selection pressure is catalytic activity and structural stability.

(iii) **Cell evolution**: time scale 10^{0}–10^{5} generations (numbers of somatic cell generations); the sources of periodicity are mutations in cell-cycle regulation and epigenetic modifications; the selection pressure is the proliferation rate, resistance to apoptosis and metabolic efficiency.

(iv) **Tissue evolution**: time scale 10^{2}–10^{6} generations; the sources of periodicity are cell-type differentiation and tissue regeneration strategies; the selection pressure is the functional integrity and repair capacity of the tissue.

(v) **Organ evolution**: time scale 10^3–10^7 generations; the sources of periodicity are variations in the regulatory networks of organ morphogenesis; the selection pressure is the adaptability of organ function to the environment.

(vi) **Morphological evolution**: time scale 10^4–10^8 generations; the sources of periodicity are variations in developmental regulatory networks (Hox genes, enhancers); the selection pressure is the adaptability of morphology to the niche.

(vii) **Population evolution**: time scale 10^2–10^6 generations; the sources of periodicity are population genetic drift, natural selection and changes in population structure; the selection pressure is the matching of the population growth rate to the carrying capacity of the environment.

(viii) **Niche evolution**: time scale 10^4–10^9 generations; the sources of periodicity are niche construction and the reorganisation of species interaction networks; the selection pressure is the stability of energy and matter flows.

The ratio of time scales between levels is about 10^2–10^3 (for adjacent levels), which constitutes **hierarchy separation**—a process at a higher level can be regarded as a quasi-static parameter (adiabatic parameter) on the time scale of a lower level, and conversely a process at a lower level appears as an averaged effect on the time scale of a higher level.

Theorem 35.8 (General mathematical form of survival of the fittest)**.** *Let the state variable at a given level be $x_i(t)$ (for example a gene frequency, a protein concentration or a cell-type proportion); its evolution follows the* **replicator–mutator equation**:

$$\frac{dx_i}{dt} = x_i\left(f_i(\mathsf{x}) - \bar{f}(\mathsf{x})\right) + \sum_j \mu_{ji}x_j - \mu_{ij}x_i,$$

where $f_i(\mathsf{x})$ is the fitness of the i-th state, $\bar{f}(\mathsf{x}) = \sum_i x_i f_i(\mathsf{x})$ is the mean fitness and μ_{ij} is the rate of mutation from state i to state j. When $\mu_{ij} \equiv 0$ (no mutation) the equation reduces to the standard replicator equation: the frequency of a state i whose fitness exceeds the mean ($f_i > \bar{f}$) grows, whilst the frequency of a state whose fitness is below the mean decays.

Proof. In the case of no mutation, $\frac{dx_i}{dt} = x_i(f_i - \bar{f})$. Differentiation gives $\sum_i dx_i/dt = \sum_i x_i f_i - \bar{f}\sum_i x_i = \bar{f} - \bar{f} = 0$, so the total probability is conserved. The equilibrium point satisfies $x_i(f_i - \bar{f}) = 0$—that is, $x_i = 0$ or $f_i = \bar{f}$. For a state with $f_i > \bar{f}$ we have $dx_i/dt > 0$ and the frequency grows; for a state with $f_i < \bar{f}$ we have $dx_i/dt < 0$ and the frequency decays. When $\mu_{ij} \neq 0$ the mutation term provides an 'exploration' term near the equilibrium point—even if a state is currently disadvantaged, its mutational sources can still maintain it at low abundance in the population, so that the system can respond rapidly when the environment changes. □

Corollary 35.2 (Replicator equation for gene evolution)**.** *At the gene level, x_i is the frequency of allele i and f_i is the phenotypic fitness of that allele. If only* a single locus with two alleles is considered ($i \in \{A, a\}$), the replicator equation reduces to the logistic form:

$$\frac{dx_A}{dt} = x_A(1 - x_A)(w_A - w_a),$$

where w_A, w_a are the absolute fitnesses of the two alleles. If $w_A > w_a$ then $x_A \to 1$ (A becomes fixed); if $w_A < w_a$ then $x_A \to 0$ (a becomes fixed). This is a discretised version of Fisher's classical fundamental theorem of natural selection.

35.8.2 The neutral theory of protein evolution and adaptive radiation

Definition 35.17 (Molecular clock). The **molecular clock** is the hypothesis that protein or nucleic-acid sequences accumulate substitutions linearly with evolutionary time. Let the substitution rate between two homologous sequences be λ (unit: substitutions/site/year) and let the divergence time be t (years); then the sequence divergence d is approximately

$$d \approx 2\lambda t.$$

The molecular clock provides an independent measure for the time calibration of phylogenetic trees—it converts protein evolution into a measurable time period.

Proposition 35.9 (Neutral and selective substitutions). Protein evolution consists of two types of substitution:

(i) **Neutral substitution** (Kimura's neutral theory): $|s| \ll 1/(2N_e)$ (where s is the selection coefficient and N_e is the effective population size); the substitution rate equals the mutation rate μ and is independent of the population size.

(ii) **Adaptive substitution**: $|s| \gg 1/(2N_e)$; the substitution rate is determined jointly by the selection coefficient and the population size, and the rate of positive selection is higher than that of neutral substitution.

Synonymous substitutions (which do not change the amino acid) are usually close to neutral, whilst non-synonymous substitutions are subject to stronger selective constraints. Hence the ratio d_N/d_S of the non-synonymous substitution rate d_N to the synonymous substitution rate d_S is a common indicator for identifying adaptive evolution: $d_N/d_S > 1$ suggests positive selection.

Proof. Under the neutral theory the substitution rate is $k = \mu \cdot P_{\text{fix}}$, where P_{fix} is the probability that a mutant gene is ultimately fixed in the population. For a neutral mutation $P_{\text{fix}} = 1/(2N_e)$, so $k = \mu/(2N_e) \cdot 2N_e = \mu$—independent of N_e. For a positively selected mutation $P_{\text{fix}} \approx 2s$ (with $s > 0$ small), so $k \approx 2N_e\mu s$, which increases with N_e and s. □

35.8.3 Cross-scale coupling and coevolution

Definition 35.18 (Cross-scale coupling). Let the state variable of the k-th level be $\mathsf{x}^{(k)}$, with evolution equation

$$\frac{d\mathsf{x}^{(k)}}{dt} = \mathsf{F}_k(\mathsf{x}^{(k)}, \mathsf{x}^{(k-1)}, \mathsf{x}^{(k+1)}),$$

that is, the dynamics of the k-th level depends simultaneously on the 'environmental parameters' of the level above ($k+1$ as a slow variable) and on the 'constituent elements' of the level below ($k-1$ as a fast variable). This dependence is called **cross-scale coupling**.

Proposition 35.10 (Averaging under time-scale separation)**.** If the characteristic time of level $k-1$ satisfies $\tau_{k-1} \ll \tau_k$ (the characteristic time of the k-th level), then within one period τ_k the state of level $k-1$ has relaxed sufficiently to its conditional equilibrium $\mathsf{x}_*^{(k-1)}(\mathsf{x}^{(k)})$. The effective dynamics of the k-th level is

$$\frac{d\mathsf{x}^{(k)}}{dt} \approx \mathsf{F}_k(\mathsf{x}^{(k)}, \mathsf{x}_*^{(k-1)}(\mathsf{x}^{(k)}), \mathsf{x}^{(k+1)}).$$

This averaging makes the dynamics of the higher level depend on the lower level only through its equilibrium state.

Proof. Let $\mathsf{x}^{(k-1)} = \mathsf{x}_*^{(k-1)} + \delta\mathsf{x}^{(k-1)}$, where $\delta\mathsf{x}^{(k-1)}$ is the deviation from the equilibrium state. Its relaxation equation $\frac{d\delta}{dt} = -\frac{1}{\tau_{k-1}}\delta$ gives $\delta(t) \approx \delta(0)e^{-t/\tau_{k-1}}$. Within one period τ_k (with $\tau_k \gg \tau_{k-1}$) we have $\delta \to 0$, hence $\mathsf{x}^{(k-1)} \to \mathsf{x}_*^{(k-1)}$. Substituting this into the equation of the k-th level gives the averaged dynamics. ☐

Theorem 35.9 (Periodic structure of multi-scale selection)**.** In a multi-scale evolutionary system, if the selection pressure at some level varies periodically (for example seasonal climatic change, predator–prey cycles or the circadian rhythm), then the allele frequencies, protein concentrations or cell proportions at that level exhibit a **periodic response**. The character of this periodic response is determined by the following factors:

(i) **Selection period** T_{sel}: the period of the variation of the external selection pressure.

(ii) **Response time** τ_{resp}: the time scale on which the state at that level relaxes to a new equilibrium.

(iii) **Matching condition**: when $T_{\mathsf{sel}} \gg \tau_{\mathsf{resp}}$ the system follows the selection pressure quasi-statically; when $T_{\mathsf{sel}} \approx \tau_{\mathsf{resp}}$ the system exhibits resonant amplification or phase lag; when $T_{\mathsf{sel}} \ll \tau_{\mathsf{resp}}$ the system responds only to the average of the selection pressure.

Proof. Consider the single-locus two-allele model, whose allele-frequency dynamics is $dx/dt = x(1-x)s(t)$, where $s(t) = s_0 \cos(2\pi t/T_{\mathsf{sel}})$ is the periodic selection coefficient. This is a logistic-type equation with intrinsic time scale $\tau_{\mathsf{resp}} = 1/(s_0)$. By the classical theory of forced oscillations:

- $T_{\mathsf{sel}} \gg \tau_{\mathsf{resp}}$: the solution is approximately $x(t) \approx x_{\mathsf{eq}}(s(t))$, a quasi-static following.

- $T_{\mathsf{sel}} \approx \tau_{\mathsf{resp}}$: the solution exhibits phase lag and amplitude amplification—the resonance of the classical forced oscillator.

- $T_{\mathsf{sel}} \ll \tau_{\mathsf{resp}}$: the solution is approximately $x(t) \approx x_0 + O(T_{\mathsf{sel}}/\tau_{\mathsf{resp}})$, a response to the time average only.

□

Corollary 35.3 (Periodic structure of the Red Queen hypothesis)**.** *The Red Queen hypothesis states that coevolution between species (for example host–parasite or predator–prey interactions) gives rise to continuous mutual selection pressure, so that both parties must keep evolving in order to maintain their relative fitness. In the multi-scale periodic framework this appears as a* **coevolutionary cycle***: the periodic response of the host's gene frequencies and that of the parasite's gene frequencies become phase-locked, and the period is determined by the geometric mean of the generation times of the two species.*

Proof. Let the period of the host's gene frequencies be T_H and that of the parasite be T_P, and let the strength of coevolutionary selection between them be c. The phase-difference dynamics between the two is $d(\Delta\phi)/dt = \Delta\omega - c\,\mathsf{sin}(\Delta\phi)$, where $\Delta\omega = 2\pi(1/T_H - 1/T_P)$. When $c > |\Delta\omega|$ stable phase locking exists. The locked period is $T_{\mathsf{lock}} = \sqrt{T_H T_P}$—the geometric mean of the generation times of the two species. □

Example 35.4 (Concrete manifestations of multi-scale evolution in biological periodic structur

Gene–protein coupling: an antibiotic-resistance gene mutation in bacteria (gene level) changes the structure of the target protein (protein level) and renders the existing drug ineffective. The periodic response of the gene frequency (the rise in resistance) and the change in protein structure (target variation) form a coevolutionary process over several cycles of antibiotic treatment.

(ii) **Cell–tissue coupling**: the clonal evolution of tumour cells (cell level) is subject to the selection pressure of the tissue microenvironment (tissue level). During treatment cycles, sensitive cells die and resistant cells proliferate, so that the composition of the tumour cell population changes cyclically; the tissue level regulates the rate of evolution at the cell level through angiogenesis.

(iii) **Organ–morphology coupling**: organ evolution (for example the vertebrate fin→limb transition) and overall morphological evolution (for example the transition from aquatic to terrestrial life) are coupled on a time scale of 10^7 years—variations in Hox gene regulatory networks change organ and morphology simultaneously.

(iv) **Population–niche coupling**: predator–prey cycles (population level) change the distribution of niche resources (niche level) and thereby select different predation strategies or prey defence strategies. The Red Queen style coevolutionary cycle (Corollary 35.3) is the long-term consequence of this coupling.

Theorem 35.10 (Periodic time tree of multi-scale evolution)**.** *Arranging the characteristic time scales of evolution at the various levels in order of magnitude* yields the **periodic time tree**:

$$\begin{aligned} &\text{genes } (\sim 10^0 \text{ generations}) \\ \ll\ &\text{proteins } (\sim 10^1) \\ \ll\ &\text{cells } (\sim 10^2) \\ \ll\ &\text{tissues } (\sim 10^4) \\ \ll\ &\text{organs } (\sim 10^6) \\ \ll\ &\text{morphology } (\sim 10^7) \\ &\text{populations } (\sim 10^4) \ll \text{niches } (\sim 10^8). \end{aligned}$$

The time scale of each level is 10^1–10^4 times larger than that of the adjacent level. This time-scale separation makes each level appear, on its own time scale, as an independent periodic structure, whilst on the time scale of an adjacent level it appears as a quasi-static or averaged background.

Proof. The orders of magnitude of the time scales are determined by the intrinsic rates of the principal processes at each level: the gene substitution rate is about 10^{-8}–10^{-9}/site/generation, the protein substitution rate about 10^{-9}/site/year, the cell cycle about 10^0–10^2 generations, tissue regeneration about 10^2–10^4 generations, organ morphogenesis about 10^4–10^6 generations, morphological evolution about 10^6–10^7 generations, population dynamics about 10^2–10^4 generations and niche construction about 10^6–10^8 generations. The ratio of time scales between adjacent levels is about 10^1–10^3; that is, each level is about 1–3 orders of magnitude slower than the level below it. □

35.9 Conservation laws of biological systems

Theorem 35.11 (Three conservation laws of biological systems)**.** The dynamics of biological systems is governed by three conservation laws:

(i) **Energy conservation**: organisms are open systems that maintain their energy balance by taking in food (chemical energy) and by respiration (oxidative phosphorylation). In a steady state, energy intake = energy expenditure + energy storage. The conversion of energy among biomolecules (ATP, glucose, fat) satisfies the first law of thermodynamics. In the circadian rhythm the metabolic rate fluctuates with a period of 24 hours—this is the periodic redistribution of energy under time translation.

(ii) **Matter conservation**: the matter in an organism (carbon, nitrogen, water) satisfies conservation of mass between intake, utilisation and excretion. During the cell cycle the DNA content is exactly replicated in the S phase (doubling) and equally distributed in the M phase (halving)—the DNA content is conserved throughout the cell cycle. In the population cycle, the change in the total number of individuals = births − deaths + immigration − emigration—this is matter conservation at the level of the ecosystem.

(iii) **Information conservation**: the genetic information of an organism is conserved in DNA replication and cell division. The two strands of the DNA double helix are complementary—each strand contains all the information needed to reconstruct the complete double helix. The central dogma (DNA $\to$ RNA $\to$ protein) describes the direction in which genetic information flows, but does not change the total amount of information. The base sequence of the DNA is conserved in transmission from generation to generation.

The hierarchical relation among the three conservation laws is as follows: energy conservation is the foundation for sustaining the activities of life, matter conservation guarantees the stability of the structure of the organism, and information conservation guarantees the transmission of hereditary traits from generation to generation. Together they constitute the physical basis of the periodic structure of biological systems.

Proof. (i) Energy conservation: the chemical energy E_{in} taken in by an organism (food) equals the energy expended E_{out} (basal metabolism, heat produced by activity, growth) plus the energy stored ΔE_{store}: $E_{\text{in}} = E_{\text{out}} + \Delta E_{\text{store}}$. In a steady state ($\Delta E_{\text{store}} = 0$) the energy intake equals the expenditure. The storage and release of energy in ATP (adenosine triphosphate) is the core of biological energy metabolism—the hydrolysis of ATP $\leftrightarrow$ ADP + P_i releases about 30.5 kJ/mol of free energy.

(ii) Matter conservation: the intake of carbon (food) equals the output (CO_2) plus storage (biomass growth): $C_{\text{in}} = C_{\text{out}} + \Delta C_{\text{store}}$. In the cell cycle the DNA content is $2n$ in the G1 phase, $4n$ after replication in the S phase and $2n$ again after division in the M phase—the scale is conserved over the cycle.

(iii) Information conservation: DNA replication is semi-conservative—each parental strand serves as a template for the synthesis of a new complementary strand. In the daughter DNA double helix one strand comes from the parent and one is newly synthesised. The genetic information (the base sequence) is preserved exactly in replication—the mutation rate is only 10^{-9}–10^{-8}/base/generation. $\square$

Remark 35.10 (Connection with the rest of the book). The theory of the periodic structure of biological systems established in this chapter is an application of periodic mathematics to the life sciences. The circadian rhythm is the approximately 24-hour self-sustained oscillation produced by the delayed negative feedback of the TTFL. The heartbeat cycle is the limit cycle formed in phase space by the ion-channel dynamics of pacemaker cells. The cell cycle is the complete scaling transformation in which the DNA content is replicated in the S phase and divided in the M phase. The population cycle (the Lotka–Volterra model) is the closed orbit in phase space guaranteed by the first integral. Neural oscillation is the biological realisation of phase locking in the 'Periodic Systems' chapter—inhibitory synaptic coupling drives the neuronal network into gamma synchronisation. The mechanism of survival of the fittest appears at many levels—genes, proteins, cells, tissues, organs, morphology, populations and niches—as the replicator–mutator equation, and the levels achieve coevolution through time-scale separation and cross-scale coupling. The three conservation laws of biological systems—energy conservation, matter conservation and information conservation—together constitute the physical guarantee of the periodic structure of living systems.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The approximately 24-hour main period of the circadian rhythm: the transcription–translation negative feedback loop (Definition 35.11); the limit-cycle analysis of the Goodwin oscillator (Proposition 35.5), the value of 24 hours being an empirical observation (Remark 35.4)
Scaling	The scaling transformation of the cell cycle: DNA content $2n \to 4n \to 2n$ (Definition 35.13); the accumulation–degradation of the Cyclin B sawtooth oscillation (Lemma 35.5)
Rotation	The heartbeat limit cycle (Definition 35.12); the phase of neural oscillation and gamma synchronisation (Proposition 35.8)
Conservation	Energy conservation, matter conservation and information conservation (Theorem 35.11)
Symmetry	The complementary symmetry of the two DNA strands: in semi-conservative replication each parental strand reconstructs the complete information (proof of Theorem 35.11, item (iii))
Closure	The closed orbits in phase space guaranteed by the Lotka–Volterra first integral $I(H,P)$ (Theorem 35.7)
Period	Circadian (24 h), heartbeat (0.6–1.0 s) and cell ($T = (C_{\text{th}} - C_0)/\rho$, Lemma 35.5) periods, and the lynx–snowshoe hare population (9–11 years); the multi-scale periodic time tree (Theorem 35.10)

Exercises

Exercise 1 Prove the **periodicity decision theorem for biological systems** of this chapter (35.1). Hint: first write down the decision map and the evolution group of this chapter's target, then apply the proof structure of the general criterion of the 'Periodic Systems' chapter.

Exercise 2 Verify that the hypotheses of the **monotonicity of number adjustment** (35.2) hold for the domain objects of this chapter, and give a counterexample in which the hypotheses fail.

Exercise 3 Examine the behaviour of the **three-parameter correspondence: radius, phase and speed** (35.1) under parameter degeneracy, and explain how the conclusion degenerates accordingly.

Exercise 4 Explain the relation between the **biological periodicity diagnostic pipeline and its interface with three chapters** (35.6) and the corresponding general conclusion of the 'Periodic Systems' chapter (whether the two agree, whether the former is a special case, or whether additional conditions are required).

Exercise 5 Using the seven-theme comparison table of this chapter, identify the concrete carriers of the themes **translation**, **scaling** and **rotation** in this chapter, and give one example of domain data for each.

Exercise 6 Synthesis: restate the diagnostic pipeline of this chapter using the five-step standard interface of the 'Periodic Systems' chapter (Table 24.1), noting one by one the theorem of this chapter on which each step depends: 35.1, 35.2, 35.1, 35.6.

Chapter 36

Periodic Structure of Psychological Systems

Remark 36.1 (Periodicity decision for the target of this chapter (citation)). When this chapter decides whether a research target is periodic, determines its period type, or seeks its minimal positive period, reference is always made to the 'Periodic Systems' chapter: for the periodicity criterion see Theorem 24.1, for the discrimination of period types see Definition 24.3, Proposition 24.1, for the search for the minimal positive period see Definition 24.4, Theorem 24.2, and for the construction of period series and epicycle groups from data see Theorem 24.3, Definition 24.6.

Remark 36.2 (Positioning of this chapter: unfolding along the five-step interface of the 'Periodic Systems' chapter). The periodic structure of psychological systems studies the mathematical characterisation of periodicity, rhythmicity and oscillatory phenomena in human psychological activity. This chapter does not set up a separate periodicity criterion; instead it connects psychological systems to the five-step interface specified by the 'Periodic Systems' chapter:

(1) **Decision**: the sleep–wake cycle, attention fluctuation, the emotional cycle, memory decay, subconscious processes, behavioural periods, interpersonal synchrony and other such targets are formalised as discrete or continuous dynamical systems, and Theorem 24.1 is cited to decide whether they are periodic and to determine the period type;

(2) **Construction**: the amplitude series, the frequency series and the phase series are constructed from arousal level, emotional valence, attention resource, memory strength, subconscious activation and behavioural tendency, and assembled into the psychological epicycle group;

(3) **Adjustment**: the psychological epicycle group is dynamically adjusted under the constraints of residual energy decrease, psychological energy conservation, information conservation and attention conservation;

(4) **Convolution**: amplitude convolution, phase convolution and frequency convolution are introduced on the psychological epicycle group, and connected with autocorrelation, cross-correlation and emotional synchrony;

(5) **Integration**: the tool chains of three chapters—'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis'—are connected, forming the psychological periodicity diagnostic pipeline.

The mathematical tools used in this chapter include: the two-process model, limit-cycle theory, the RC charge–discharge model, the circumplex model of emotion, the Ebbinghaus forgetting curve, the phase-oscillator model, coupled oscillation, periodic convolution, the epicycle group, the phase operator and time delay.

Remark 36.3 (Guide to the seven themes of this chapter). **Translation**—the approximately 24-hour main period of the sleep–wake cycle and the advance of biological rhythms through time; **Scaling**—the exponential decay of the Ebbinghaus forgetting curve and the scaling optimisation of spaced repetition; **Rotation**—the circadian rotation in the two-dimensional valence–arousal phase space and the limit cycle of bipolar disorder; **Conservation**—psychological energy conservation, information conservation and attention conservation (empirical observation / research programme, Remark 36.9); **Symmetry**—the positive–negative valence symmetry of emotion and the circadian symmetry of sleep–wake; **Closure**—the closure of the attention resource consumption–recovery cycle (Lemma 36.6); **Period**—the 90-minute sleep cycle, attention oscillation (theta/delta bands), the circadian emotional cycle and the habit loop.

36.1 Connection to the five-step interface of the 'Periodic Systems' chapter

36.1.1 Step one: periodicity decision for psychological targets

Definition 36.1 (Psychological target and decision input). *A* **psychological target** *is one of the following objects:*

- (i) *a sleep–wake system, whose state space is the joint space of sleep pressure and circadian phase;*
- (ii) *an attention resource system, whose state space is the interval space of the attention resource $R(t)$;*
- (iii) *an emotional cycle system, whose state space is the two-dimensional (valence V, arousal A) phase space;*
- (iv) *a memory retention system, whose state space is the interval space of the memory strength $M(t)$;*
- (v) *a behavioural period system, whose state space is the joint space of the behavioural tendency $B(t)$ and cue strength;*
- (vi) *an interpersonal emotional synchrony system, whose state space is the joint space of the emotional phases (θ_1, θ_2) of the two parties.*

The **decision input** *of a psychological target is an ordered pair*

$$\mathcal{D}_{\mathsf{P}} = (\mathcal{S}, \Phi),$$

where $\mathcal{S}$ is the state space and $\Phi : \mathcal{S} \to \mathcal{S}$ is the map induced by time translation, phase rotation or feedback evolution. If $\mathcal{S}$ is a finite set, the target is called a **finite psychological target***; if $\mathcal{S}$ is an infinite set, it is called an* **infinite psychological target***.*

Definition 36.2 (Psychological decision map)**.** Let a psychological target be given by the decision input $\mathcal{D}_{\mathsf{P}} = (\mathcal{S}, \Phi)$. If the states of the target are generated by Φ, that is, if there exists $x_0 \in \mathcal{S}$ such that

$$x_t = \begin{cases} \Phi^t(x_0), & t \in \mathbb{Z}, \\ \Phi_t(x_0), & t \in \mathbb{R}, \end{cases}$$

for $t \in \mathbb{Z}$ the state x_t is generated by iteration of Φ; for $t \in \mathbb{R}$ it is described by the one-parameter evolution family $\{\Phi_t\}_{t\in\mathbb{R}}$, where each $\Phi_t : \mathcal{S} \to \mathcal{S}$, $\Phi_0 = \mathrm{id}$, $\Phi_{s+t} = \Phi_s \circ \Phi_t$ for all $s, t \in \mathbb{R}$, and $(t, x) \mapsto \Phi_t(x)$ is continuous. then Φ is called the **psychological decision map** of the target. The psychological target is thereby reduced to the discrete dynamical system $(\mathcal{S}, \Phi)$ when $t \in \mathbb{Z}$ and to the continuous dynamical system $(\mathcal{S}, \{\Phi_t\}_{t\in\mathbb{R}})$ when $t \in \mathbb{R}$; in both cases the criteria of the 'Operations and Periodicity' chapter and the 'Periodic Systems' chapter may be applied directly.

Theorem 36.1 (Periodicity decision theorem for psychological systems)**.** Let the decision map of a psychological target be Φ, and let $(\mathcal{S}, \Phi)$ be a discrete dynamical system. Then:

(A) **Sufficient decision**: if there exists $m \in \mathbb{N}^+$ such that $\Phi^m = \mathrm{id}$, then the target is a uniformly periodic psychological system; every orbit is closed, and the orbit length divides m.

(B) **Decision equivalence**: the target is a uniformly periodic psychological system if and only if $\mathrm{ord}(\Phi) < \infty$.

(C) **Pointwise criterion**: the orbit of the target through the state x_0 is closed if and only if x_0 is a periodic point; the orbit length is then the period at that point.

(D) **Non-omissibility**: the conjunction of conserved quantities and symmetry groups cannot replace $\mathrm{ord}(\Phi) < \infty$; a quasi-periodic psychological target does not belong to the periodic psychological systems.

Proof. The Theorem 36.1 of this chapter is the domain instantiation of Theorem 24.1 in the 'Periodic Systems' chapter: it suffices to replace the decision map f and the evolution group by the Φ of this chapter, and the four items—(A) sufficient decision, (B) decision equivalence, (C) pointwise criterion and (D) non-omissibility—hold one by one. By the convention of Remark 24.4, the argument is not repeated here. ☐

Corollary 36.1 (Types of psychological period)**.** After the decision by Theorem 36.1, a psychological target falls into one of the following types:

(i) **Trivially periodic type**: the least period is $m_0 = 1$, and the state is a fixed point;

(ii) **Strictly periodic type**: $1 < m_0 < \infty$, and the state is strictly periodic;

(iii) **Uniformly periodic type**: $\mathrm{ord}(\Phi) = m < \infty$, and all orbit lengths divide m;

(iv) **Quasi-periodic type**: $\mathrm{ord}(\Phi) = \infty$; the orbits are not closed but are dense on a compact carrier structure;

(v) **Eventually periodic type**: there exists N such that the target is periodic for $t \geq N$;

(vi) **Discretely periodic type**: the carrier structure is $\mathbb{Z}$, and the normalised period ratio is $\rho = T_x/\Delta x \in \mathbb{Q}$.

(vii) **Asymptotically periodic (damped) type**: the orbit is not closed, but tends asymptotically to a closed orbit or to a fixed point, the residual decaying as $e^{-\kappa t}$ (see Definition 24.3(vii)).

Here (i)–(iii) and (vi) belong to the periodic psychological systems; (iv) does not belong to the periodic psychological systems but can be handled with bounded-approximation tools; (v) is a periodic psychological system only in its stable stage; (vii), like (iv), is a type in the limiting sense; it is not a periodic psychological system and can be handled with bounded-approximation tools only in its stable stage.

Proof. Each item follows by substituting the psychological target into Theorem 36.1 and Definition 24.3 of the 'Periodic Systems' chapter. ☐

Lemma 36.1 (Decision for the sleep–wake cycle)**.** *Let the sleep–wake system be described by the two-process model: process S accumulates linearly during the waking period and is cleared exponentially during the sleep period; process C oscillates with a period of about 24 hours. Then the sleep–wake system is a uniformly periodic system, whose main period is*

$$T_{\text{sleep-wake}} \approx 24 \text{ hours},$$

and an ultradian NREM–REM cycle of about 90 minutes is nested within this main period.

Proof. By Proposition 36.5, the sleep propensity $P(t) = S(t) - C(t)$ crosses the threshold in the evening and at night, triggering sleep, and falls back below the threshold in the early morning, triggering waking. Process C oscillates with a period of 24 hours, and process S accumulates and is cleared periodically in the wake–sleep cycle; the superposition of the two gives the sleep–wake system a main period of about 24 hours. Hence the decision map Φ satisfies $\Phi^{T_{\text{sleep-wake}}} = \mathrm{id}$, and the system is uniformly periodic. ☐

Lemma 36.2 (Periodicity decision for attention fluctuation)**.** *Let the attention resource $R(t)$ be consumed at rate α during task periods and restored at rate β during rest periods. Then the consumption–recovery cycle of the attention resource is a uniformly periodic system, whose period is given by Lemma 36.6.*

Proof. In the steady-state cycle of alternating task and rest, the attention resource charges and discharges periodically within $[R_{\min}, R_{\max}]$. By the closed-form solution of Lemma 36.6, $R_{\min}$ and $R_{\max}$ are uniquely determined by the closure conditions, and the cycle repeats with period $T = t_1 + t_2$. Hence the decision map Φ satisfies $\Phi^T = \mathrm{id}$, and the system is uniformly periodic. ☐

Lemma 36.3 (Decision for the emotional cycle). *Let the emotional state rotate about the equilibrium point in the two-dimensional (valence V, arousal A) phase space with angular frequency ω, with return rate μ. Then:*

(i) *if $\mu > 0$ and the oscillation persists (external driving), the emotional system is a uniformly periodic system;*

(ii) *if $\mu > 0$ and there is no external driving, the emotional orbit is a decaying spiral and belongs to the asymptotically periodic (damped) type;*

(iii) *if $\mu \approx 0$ (the limit cycle of bipolar disorder), the emotional system is a uniformly periodic system, with amplitude determined by the pathological limit cycle.*

Proof. By the proof of Proposition 36.7, the linearisation of the emotional dynamics near the equilibrium point is $\dot{X} = JX$, with eigenvalues $\lambda_\pm = -\mu \pm i\omega$. When $\mu > 0$ the orbit is a decaying spiral and belongs to the asymptotically periodic (damped) type; when $\mu \approx 0$ the orbit is approximately closed (uniformly periodic type); when there is persistent external driving (such as circadian driving), the steady-state response is of uniformly periodic type. □

Lemma 36.4 (Periodicity decision for the forgetting curve). *Let the memory strength $M(t)$ decay exponentially as $M(t) = M_0 e^{-t/\tau}$ in the absence of review, and be reset to M_0 on review. Then the review–forgetting cycle is a uniformly periodic system, whose period is the sum of the review intervals.*

Proof. Each review resets M to M_0; between two reviews, M decays exponentially to the threshold M_{min}. By Proposition 36.8, the optimal time interval before the n-th review is Δt_n. When the sequence of review intervals is periodic, the review–forgetting cycle constitutes a uniformly periodic system. □

Lemma 36.5 (Periodicity decision for habit formation). *Let the behavioural tendency $B(t)$ converge to the asymptotic value B_∞ at rate γ under repeated cues. Then the process of habit formation is of asymptotically periodic (damped) type—it converges to a steady state after the time scale $\tau_{\text{habit}} = 1/\gamma$. If the cues themselves are periodic, B exhibits a periodic response.*

Proof. By Proposition 36.10, under repeated cues $B(t) = B_\infty(1 - e^{-\gamma t})$, converging to B_∞. If the cues themselves repeat with period T_{cue}, then in the steady state B exhibits a periodic response of period T_{cue}. The two cases correspond to the asymptotically periodic (damped) type and the uniformly periodic type respectively. □

Definition 36.3 (Least positive period algorithm for psychological systems). *This definition has the same content as Definition 24.4 in the section 'Periodicity Decision: Is the Target Periodic?' (Section 24.1) of Chapter 24; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement is given there. The only difference here is terminology (psychological target / psychological sample); the algorithmic steps and the conclusions are exactly the same.*

Theorem 36.2 (Correctness of the least positive period algorithm for psychological systems). This theorem has the same content as Theorem 24.2 in the section 'Periodicity Decision: Is the Target Periodic?' (Section 24.1) of Chapter 24; to avoid repeating the statement, only the reference anchor is retained here, and the complete statement and proof are given there. The only difference here is terminology (psychological target / psychological sample); the algorithmic steps and the conclusions are exactly the same.

Example 36.1 (Examples of periodicity decisions for psychological systems).

- (i) sleep–wake: $T_{\text{sleep-wake}} \approx 24$ h, uniformly periodic type;
- (ii) ultradian rhythm within sleep: $T_{\text{ultradian}} \approx 90$ min, uniformly periodic type;
- (iii) attention fluctuation: $T_{\text{attention}} = t_1 + t_2$, uniformly periodic type;
- (iv) circadian emotion: $T_{\text{emotion}} = 24$ h (with circadian driving), uniformly periodic type;
- (v) bipolar disorder: T_{bipolar} of weeks to months, uniformly periodic type (limit cycle);
- (vi) memory review: $T_{\text{review}} = \sum_n \Delta t_n$, uniformly periodic type (under periodic review);
- (vii) habit formation: asymptotically periodic (damped) type (without periodic cues) or uniformly periodic type (with periodic cues).

Remark 36.4 (Conserved quantities do not guarantee periodicity). Energy conservation, information conservation and attention conservation in psychological systems are auxiliary conditions for periodic structure, but they do not constitute sufficient conditions. For example, a mental state that possesses energy conservation but lacks a feedback loop will not spontaneously produce periodic oscillation. Psychological periodic structure must be decided by Theorem 36.1 or by the lemmas above.

36.1.2 Step two: construction of the psychological period series and the epicycle group

Definition 36.4 (Psychological amplitude, frequency and phase series). Let the state or output sequence of a psychological target be $\{z_t\}$. If it can be written as

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)} + \varepsilon_t,$$

then

$$\mathcal{A}_z(t) = \sum_k A_k e^{ikt}, \qquad \mathcal{F}_z(s) = \sum_k \omega_k s^k, \qquad \mathcal{P}_z(t) = \sum_k e^{i(kt+\phi_k)}$$

are called respectively the **amplitude series**, the **frequency series** and the **phase series** of the psychological target. The three agree with Definition 6.10, Definition 6.12 and Definition 6.11 of the 'Series and Convolution' chapter.

Definition 36.5 (Psychological epicycle group). *Let the period series of a psychological target be given by the amplitude series $\{A_k\}$, the frequency series $\{\omega_k\}$ and the phase series $\{\phi_k\}$. For each k define the epicycle*

$$E_k : \theta \mapsto A_k e^{i(\theta+\phi_k)}, \qquad \theta = \omega_k t,$$

whose collection

$$\mathfrak{E}_{\mathsf{P}} = \{(A_k, \omega_k, \phi_k)\}_{k\in\mathbb{Z}}$$

is called the **epicycle group** *of the psychological target. Its superposition is the psychological period series*

$$z_t = \sum_k E_k(\omega_k t) = \sum_k A_k e^{i(\omega_k t+\phi_k)}.$$

Theorem 36.3 (Construction of the period series from psychological data). *Let the output of a psychological target be a finite periodic sequence of period N. Then:*

(i) *the output can be expanded as a discrete Fourier series*

$$z_t = \sum_{k=0}^{N-1} c_k e^{i2\pi kt/N}, \qquad c_k = \frac{1}{N}\sum_{t=0}^{N-1} z_t e^{-i2\pi kt/N};$$

(ii) *the amplitude series is given by $|c_k|$, the phase series by* arg c_k *and the frequency series by $2\pi k/N$;*

(iii) *if the output is a sleep–wake sequence, the fundamental frequency corresponds to the main period of about 24 hours, and the harmonics correspond to the ultradian rhythm of about 90 minutes and the semi-diurnal rhythm of about 12 hours; if the output is an attention resource sequence, the fundamental frequency corresponds to the period of the task–rest cycle.*

Proof. (i) The discrete Fourier series of a finite periodic sequence is a standard result: the basis functions $\{e^{i2\pi kt/N}\}_{k=0}^{N-1}$ form an orthogonal basis of $\mathbb{C}^N$. (ii) This follows directly from $c_k = A_k e^{i\phi_k}$. (iii) For the sleep–wake sequence with $N = 24$ (hourly sampling), the fundamental frequency corresponds to the 24-hour main period, and the $k = 16$ harmonic corresponds to the 90-minute ultradian period $(24 \times 60/90 = 16)$; for the attention sequence, the fundamental frequency corresponds to the period of the task–rest cycle. $\square$

Proposition 36.1 (Three-parameter correspondence: radius, phase and speed). *The three parameters of the psychological epicycle group $\mathfrak{E}_{\mathsf{P}} = \{(A_k, \omega_k, \phi_k)\}_k$ correspond one-to-one to three geometric–psychological quantities:*

(i) **Radius corresponds to amplitude**: *the radius $|E_k| = A_k$ of E_k is the amplitude of that psychological oscillation component;*

(ii) **Phase corresponds to rotational delay**: *the initial phase ϕ_k corresponds to the time translation $\tau_k = \phi_k/\omega_k$; when $\phi_k > 0$ this component* **leads** *the baseline by τ_k (equivalently, the component coincides with the baseline once the baseline is postponed by τ_k);*

(iii) **Frequency corresponds to rotational speed**: ω_k is the angular frequency of that psychological oscillation component.

Proof. The Proposition 36.1 of this chapter corresponds exactly to Proposition 24.3 in the 'Periodic Systems' chapter, and the three correspondences—radius to amplitude, phase to rotational delay, and frequency to rotational speed—are obtained by translating each of them to the epicycle group of this chapter. The proof is given there: (i) from $|E_k| = A_k$ and the energy decomposition, (ii) from the phase-translation theorem, (iii) from the definition of the frequency series. $\square$

Example 36.2 (The period series of sleep–wake). The sleep–wake sequence contains a main period of about 24 hours, an ultradian period of about 90 minutes and a semi-diurnal period of about 12 hours. In the psychological epicycle group, the three appear as epicycle components of different radii and different angular frequencies—the main-period amplitude is the largest, whilst the ultradian and semi-diurnal components have smaller amplitudes.

Remark 36.5 (The period series is a complex function). Let $z = e^{i\omega_0 t}$. Then the psychological period series

$$z_t = \sum_k A_k e^{i(\omega_k t+\phi_k)}$$

when all components are commensurable with ω_0 (that is, $\omega_k = k\omega_0$, $k \in \mathbb{Z}$) it can be written as the Laurent series $\sum_k c_k z^k$, where $c_k = A_k e^{i\phi_k}$: in that case the psychological period series is a complex function on the unit circle, and otherwise it is merely a function on the torus.

36.1.3 Step three: dynamic adjustment of the psychological epicycle group

Definition 36.6 (Dynamic adjustment of the psychological epicycle group). Let the current psychological epicycle group be $\mathfrak{E}^{(j)}$, with partial sum

$$z^{(j)}(t) = \sum_{k\in\mathfrak{E}^{(j)}} A_k e^{i(\omega_k t+\phi_k)},$$

and residual $r^{(j)} = z - z^{(j)}$. **Dynamic adjustment** consists of two rules:

(i) **Cardinality adjustment**: if the residual energy $\|r^{(j)}\|^2 > \delta$, a new epicycle is added according to the peak of the residual periodogram;

(ii) **Parameter adjustment**: the parameters of the existing epicycles are updated by regularised least squares or under the conservation constraints.

In psychological systems, cardinality adjustment corresponds to adding a biological oscillatory mode (such as a newly discovered rhythm band); parameter adjustment corresponds to tuning the amplitude, phase and frequency so that the psychological energy, information and attention constraints are satisfied.

Proposition 36.2 (Monotonicity of cardinality adjustment)**.** *After a dominant-frequency component of the residual has been added as in Definition 36.6(i), the residual energy is monotonically non-increasing:*

$$\|r^{(j+1)}\|^2 \le \|r^{(j)}\|^2,$$

with equality if and only if the residual has no energy at the selected frequency.

Proof. The Proposition 36.2 of this chapter is consistent with the dynamic adjustment mechanism of Definition 24.8 in the 'Periodic Systems' chapter: adding an epicycle removes the component of $r^{(j)}$ along $\hat{\omega}$ from the residual, and the inequality follows from the orthogonal decomposition $\|r^{(j)}\|^2 = \|r^{(j+1)}\|^2 + \|\text{the removed component}\|^2$, with equality if and only if the norm of the removed component is zero. The complete argument is given there. ☐

Theorem 36.4 (Convergence of dynamic adjustment)**.** *Let the psychological target be a finite periodic sequence of period T_0 with $\delta = 0$. Then after at most $\lfloor T_0/2 \rfloor + 1$ steps of dynamic adjustment the residual is zero, and the resulting psychological epicycle group is the exact Fourier data of the target.*

Proof. The discrete Fourier series of a sequence of period T_0 is complete on $\lfloor T_0/2 \rfloor + 1$ harmonics. At each step a dominant-frequency component of the residual is extracted as in Proposition 36.2, and the residual energy decreases strictly; hence after finitely many steps the residual is zero. ☐

Definition 36.7 (Structural metric adjustment)**.** Let the structural metric of a psychological target be

$$\mathcal{I} = (\mathcal{I}_{\text{energy}}, \mathcal{I}_{\text{information}}, \mathcal{I}_{\text{attention}}) = (\text{psychological energy conservation degree, information cons}$$

Structural-metric adjustment consists in choosing θ in the parameter space Θ: at each step take a neighbouring parameter for which $\mathcal{I}(\theta)$ is **strictly decreasing** in the lexicographic order (minimisation); if no such parameter exists, the process stops, and the parameter is then locally optimal.

Proposition 36.3 (Finite convergence of structural metric adjustment)**.** *Let the parameter space Θ be finite, and let each adjustment make $\mathcal{I}$ strictly decreasing in the lexicographic order. Then the adjustment terminates after finitely many steps at a locally optimal parameter.*

Proof. Since Θ is finite, $\mathcal{I}$ takes only finitely many values, and the lexicographic order is a total order on a finite set; a strictly decreasing sequence in that order cannot continue indefinitely, so the adjustment terminates after finitely many steps. At termination there is no neighbouring parameter that makes $\mathcal{I}$ strictly decrease in the lexicographic order, that is, the parameter is locally optimal. ☐

Remark 36.6 (Boundaries of dynamic adjustment)*.* Theorem 36.4 relies on finite periodic sequences. For continuous spectra or quasi-periodic psychological targets there is no exact representation by a finite epicycle group, and only bounded approximation is possible. Structural metric adjustment does not guarantee global optimality.

36.1.4 Step four: psychological convolution

Definition 36.8 (Psychological amplitude, phase and frequency convolutions). *Let two psychological epicycle groups be*

$$\mathfrak{E} = \{(A_n, \omega_n, \phi_n)\}, \qquad \mathfrak{E}' = \{(B_m, \nu_m, \psi_m)\}.$$

Define:

(i) **Amplitude convolution**: $\mathbb{A}_{mn} = A_n B_m$;

(ii) **Phase convolution**: $\mathbb{P}_{mn} = e^{i(\phi_n - \psi_m)}$;

(iii) **Frequency convolution**: $\mathbb{F}_{mn} = \omega_n \nu_m$;

(iv) **Trend convolution**: $\mathbb{T}_{mn} = A_n B_m e^{i(\phi_n - \psi_m)} = A_n B_m \mathbb{P}_{mn}$.

These agree with Definition 6.21, Definition 6.22, Definition 6.23 and Definition 6.24 of the 'Series and Convolution' chapter.

Theorem 36.5 (Action of the three convolutions on psychological epicycle groups). *Let the epicycle groups of two psychological targets be $\mathfrak{E}$ and $\mathfrak{E}'$ respectively. Then:*

(i) *amplitude convolution acts only on the radius;*

(ii) *phase convolution acts only on the phase;*

(iii) *frequency convolution acts only on the rotational speed;*

(iv) *trend convolution preserves both amplitude and phase.*

Proof. This follows directly from Definition 36.8. ☐

Proposition 36.4 (Autocorrelation and trend convolution). *Let $\{z_t\}$ be a psychological period sequence, with autocorrelation function*

$$R_z(h) = \sum_t z_t \overline{z_{t+h}}.$$

In the language of the psychological epicycle group, the autocorrelation function is the projection of the trend convolution onto the diagonal:

$$R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}.$$

Proof. The Proposition 36.4 of this chapter is the realisation in this domain of the convolution-action conclusion of Theorem 24.6 in the 'Periodic Systems' chapter: substituting $z_t = \sum_n A_n e^{i(\omega_n t + \phi_n)}$ into the definition of the autocorrelation, cancelling all off-diagonal cross terms by orthogonality and retaining only the diagonal term of the trend convolution (the factorisation of the three convolutions, Proposition 24.6), which yields $R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}$. The complete expansion is given there. ☐

Remark 36.7 (Relation between psychological convolution and periodic convolution). Psychological convolution is not a new convolution operation; it is an instance on the psychological epicycle group of the periodic convolution of the 'Series and Convolution' chapter. Autocorrelation is the projection of the trend convolution onto the diagonal; cross-correlation and emotional synchrony are respectively the concrete manifestations in psychological science of amplitude convolution and frequency convolution.

36.1.5 Step five: integration with discrete sampling, harmonic analysis and time series analysis

Definition 36.9 (The psychological periodicity diagnostic pipeline)**.** For a given psychological system, diagnosis is carried out using the framework of periodic mathematics:

(i) **Periodicity decision**: verify the period type of the psychological target (Theorem 36.1);

(ii) **Series construction**: construct the amplitude, frequency and phase series and the epicycle group from arousal, emotion, attention and memory data (Theorem 36.3);

(iii) **Conservation diagnosis**: check the conservation of psychological energy, information and attention as an empirical observation and a research programme (see Remark Remark 36.9);

(iv) **Convolution diagnosis**: compute the autocorrelation (the diagonal projection of the trend convolution, Proposition 36.4) and the phase convolution;

(v) **Integration diagnosis**: connect to the tool chains of discrete sampling, harmonic analysis and time series analysis.

Theorem 36.6 (The psychological periodicity diagnostic pipeline and its interface with three chapters)**.** The five steps of Definition 36.9 connect respectively to:

(i) the 'Discrete Sampling and Intercalation of Periodic Signals' chapter: Definition 21.1, Theorem 21.3, Theorem 21.10;

(ii) the 'Harmonic Analysis' chapter: Definition 22.5, Theorem 22.1, Proposition 22.11;

(iii) the 'Time Series Analysis' chapter: Definition 23.16, Proposition 23.2, Definition 23.19.

Hence the psychological periodicity diagnosis is not an independent method, but an application of the unified tool chain of periodic mathematics to psychological systems.

Proof. The Theorem 36.6 of this chapter is the implementation in this domain of the five-step interface of the 'Periodic Systems' chapter (Table 24.1): the periodicity decision uses sampling and spectral estimation, the frequency decomposition and phase kernels use the 'Harmonic Analysis' chapter, and the periodogram peaks, the Fisher test and the window-function validation use the 'Time Series Analysis' chapter. Hence the diagnostic pipeline is synthesised from the tool chains of the three chapters; the proof is given there. □

Remark 36.8 (The order of operations after integration). The recommended order for practical diagnosis is: first decide whether the psychological target is periodic (Theorem 36.1); then construct the Fourier data of the psychological data and the epicycle group (Theorem 36.3); then optimise the parameters using the conservation constraints and dynamic adjustment (Theorem 36.4, Proposition 36.3); then carry out the psychological convolution and autocorrelation analysis (Proposition 36.4); and finally output the diagnosis according to Definition 36.9.

36.2 Mathematical foundations of psychological systems

Definition 36.10 (Basic quantities of psychological periodic systems). A psychological periodic system is described by the following basic variables:

(i) **Arousal level** $A(t) \in [0,1]$: the degree of alertness of the individual at time t. $A \to 1$ indicates high arousal (tension, excitement), and $A \to 0$ low arousal (drowsiness, relaxation). The arousal level fluctuates on a period of about 24 hours and also exhibits a rhythm on the ultradian period of about 90 minutes.

(ii) **Emotional valence** $V(t) \in [-1,1]$: the degree of subjective pleasantness of the individual at time t. $V > 0$ indicates positive emotion (pleasantness), $V < 0$ negative emotion (unpleasantness) and $V = 0$ neutrality. Emotional valence and arousal level constitute the two-dimensional circumplex model of emotion.

(iii) **Attention resource** $R(t) \in [0,1]$: the total amount of cognitive resources available to the individual at time t. $R \to 1$ indicates abundant attention and $R \to 0$ attentional depletion. The attention resource is consumed during task performance and restored during rest—the alternation of consumption and recovery constitutes the cycle of attention.

(iv) **Memory strength** $M(t) \in [0,1]$: the retrieval probability of a particular memory trace at time t. $M \to 1$ indicates vivid memory and $M \to 0$ forgetting. In the absence of review, memory strength decays approximately exponentially.

(v) **Subconscious activation** $U(t) \in [0,1]$: the degree of activity of the psychological processes running below the threshold of consciousness. $U \to 1$ indicates that subconscious processes are highly active (such as memory consolidation during sleep, or the automatic processing involved in habituation), and $U \to 0$ that subconscious processes are in a resting state.

(vi) **Behavioural tendency** $B(t) \in [0,1]$: the probability that the individual exhibits a particular behavioural pattern at time t. The behavioural tendency is determined jointly by psychological state (conscious and subconscious) and environmental stimuli.

The core feature of psychological systems is **multiscale nested periodicity**—rhythms on different time scales (neural oscillations at the millisecond scale, attention fluctuations at the second scale, emotional changes at the hour scale, the circadian sleep cycle, the interannual behavioural period) coexist in one and the same psychological system, and span both the conscious and the subconscious level.

36.3 The sleep–wake cycle

Definition 36.11 (Periodic structure of sleep)**.** Human sleep consists of several **sleep cycles** of about 90 minutes. Each sleep cycle contains:

(i) **NREM sleep** (non-rapid eye movement sleep): divided into N1 (sleep onset), N2 (light sleep) and N3 (deep sleep / slow-wave sleep). Brain-wave frequency gradually decreases and amplitude gradually increases—δ waves (0.5–4 Hz) dominate in the N3 stage.

(ii) **REM sleep** (rapid eye movement sleep): brain waves show a high-frequency, low-amplitude desynchronised pattern (similar to the waking state), the eyes move rapidly, the body's muscles are paralysed, and dreams occur mainly in this stage.

A whole night's sleep usually contains 4–6 sleep cycles. The first half of the night is dominated by deep sleep (N3) and the second half by REM sleep—the duration of REM sleep grows from about 10 minutes in the first cycle to about 60 minutes in the last cycle.

The approximately 24-hour main sleep–wake period is the internalisation of the Earth's rotation period. The approximately 90-minute sleep cycle is an ultradian rhythm nested within the main period—the alternation of NREM and REM appears as a nested period in the structure of sleep. Deep sleep (NREM) corresponds to slow-wave oscillation (low frequency, high amplitude, an outer large epicycle), and REM sleep to fast-wave oscillation (high frequency, low amplitude, inner small epicycles). The increase in the proportion of REM sleep over a whole night's sleep is a phase drift within the nested period—the REM component gradually strengthens through the night.

Proposition 36.5 (Two-process model of sleep)**.** The regulation of sleep consists of two independent processes:

(i) **Process S (sleep pressure)**: during waking, sleep-promoting substances such as adenosine accumulate in the brain, and sleep pressure increases linearly. During sleep, these substances are cleared, and sleep pressure decays exponentially. Process S manifests as the linear accumulation of sleep pressure during the waking period and its exponential clearance during the sleep period—analogous to the charging (waking period) and discharging (sleep period) of a capacitor.

(ii) **Process C (circadian rhythm)**: an approximately 24-hour circadian signal generated by the suprachiasmatic nucleus (SCN) that modulates sleep propensity. Process C promotes sleep at night and waking during the day. Process C is a periodic drive—SCN neurons oscillate with a period of about 24 hours.

The superposition of the two processes determines the timing and the depth of sleep: sleep occurs at the moment when process S is at its highest and process C most promotes sleep (usually at night). The two-process model of sleep is an instance in which a linear accumulation (process S) and a periodic oscillation (process C) couple to produce the sleep–wake cycle.

Proof. Write the waking period as $t \in [0, T_w)$ and the sleep period as $[T_w, T_w + T_s)$ ($T_w + T_s = 24$ h). The linearised model of process S is: during the waking period $S(t) = S_{\mathsf{min}} + \rho_w t$ ($\rho_w > 0$ is the rate of accumulation of sleep pressure), and during the sleep period $S(t) = S_{\mathsf{max}} e^{-t/\tau_s}$ (τ_s is the clearance time constant). Process C is approximated by a sinusoid $C(t) = C_0 \cos\left(\frac{2\pi t}{24}\right)$ (negative at night, which promotes sleep). The sleep propensity is $P(t) = S(t) - C(t)$: waking $\Leftrightarrow$ $P(t) < \theta$, sleep $\Leftrightarrow$ $P(t) \geq \theta$ (threshold θ). Since S increases monotonically during the waking period and C decreases monotonically in the evening, P necessarily crosses the threshold in the evening and at night—sleep is triggered in a 'gated' manner during the interval in which S is high and C is low; in the early morning C rises again and S has already been cleared, so P falls back below the threshold—waking. The superposition of the two processes thus rigorously yields the approximately 24-hour sleep–wake cycle. □

36.4 Periodic fluctuation of attention

Definition 36.12 (Oscillation of attention). Human attention is not continuous and uniform; rather, it **fluctuates rhythmically** at several frequencies:

(i) **High-frequency attention oscillation** (4–8 Hz, theta band): attention fluctuates on a time scale of about 125–250 milliseconds. This corresponds to periodic variation in the perceptual sampling rate—the 'bit width' of attention (the number of targets that can be attended to simultaneously) scales at the theta frequency.

(ii) **Mid-frequency attention oscillation** (0.5–2 Hz, delta band): attention fluctuates on a time scale of about 0.5–2 seconds. This corresponds to switching of the focus of attention among several targets—the 'bandwidth' of attention (the temporal resolution of attentional switching) varies at the delta frequency.

(iii) **Low-frequency attention oscillation** (< 0.1 Hz): attention fluctuates on a time scale of minutes. This corresponds to fluctuations in overall cognitive resources—task performance exhibits a periodic fluctuation of about 10–20 minutes.

The periodic fluctuation of attention manifests as an alternation of focusing and dispersion: when focusing, cognitive resources are concentrated on a single target; when dispersing, resources are distributed among several targets. The focusing and dispersion of attention determine the effective flux of information from the external world to the conscious mind.

Proposition 36.6 (Model of attention resource consumption and recovery). Let the attention resource $R(t)$ be consumed at rate α during task performance and restored at rate β during rest:

$$\frac{dR}{dt} = -\alpha \cdot \mathbf{1}_{\text{task}}(t) \cdot R(t) + \beta \cdot \mathbf{1}_{\text{rest}}(t) \cdot (1 - R(t)),$$

where $\mathbf{1}_{\text{task}}(t)$ and $\mathbf{1}_{\text{rest}}(t)$ are indicator functions. The alternation of consumption and recovery produces the periodic fluctuation of the attention resource.

The consumption–recovery cycle of the attention resource produces periodic behaviour through linear consumption and exponential recovery: as in the charge–discharge process of an RC circuit, the attention resource decays exponentially during task periods and is restored exponentially during rest periods. The time constants $\tau_{\text{consumption}} = 1/\alpha$ and $\tau_{\text{recovery}} = 1/\beta$ determine the duration of the attention cycle. In prolonged tasks, the optimisation of rest intervals is equivalent to finding the best attention cycle—the one that maximises the average attention resource during the task.

Lemma 36.6 (Closed-form solution for the attention charge–discharge cycle). Let the task period have duration t_1 (consumption rate α) and the rest period duration t_2 (recovery rate β), the cycle repeating with period $T = t_1 + t_2$. Then the steady-state cycle satisfies:

$$R_{\text{min}} = \frac{(1 - e^{-\beta t_2})\, e^{-\alpha t_1}}{1 - e^{-\alpha t_1} e^{-\beta t_2}}, \qquad R_{\text{max}} = \frac{1 - e^{-\beta t_2}}{1 - e^{-\alpha t_1} e^{-\beta t_2}},$$

and the period-average resource is

$$\bar{R} = \frac{1}{T}\Big(\frac{R_{\text{max}} - R_{\text{min}}}{\alpha} + t_2 - \frac{(1 - R_{\text{max}})(e^{\beta t_2} - 1)}{\beta}\Big).$$

Proof. The solution during the task period is $R(t) = R_{\text{max}} e^{-\alpha t}$, and during the rest period $R(t) = 1 - (1 - R_{\text{min}}) e^{-\beta t}$. The period closure conditions are:

$$R_{\text{min}} = R_{\text{max}} e^{-\alpha t_1},$$
$$R_{\text{max}} = 1 - (1 - R_{\text{min}}) e^{-\beta t_2}.$$

Solving these simultaneously gives R_{min} and R_{max} as above. The average resource is computed as $\bar{R} = \frac{1}{T}\big(\int_0^{t_1} R_{\text{max}} e^{-\alpha t} dt + \int_0^{t_2} [1 - (1 - R_{\text{min}}) e^{-\beta t}] dt\big)$: the first term is $= (R_{\text{max}} - R_{\text{min}})/\alpha$; the second term is $= t_2 - (1 - R_{\text{min}})(1 - e^{-\beta t_2})/\beta$, which by the closure condition $1 - R_{\text{min}} = (1 - R_{\text{max}}) e^{\beta t_2}$ becomes $t_2 - (1 - R_{\text{max}})(e^{\beta t_2} - 1)/\beta$. ☐

Example 36.3 (The attention cycle of the Pomodoro Technique). Take $\alpha = 0.10$/min (high-intensity task), $\beta = 0.25$/min (recovery during a short rest), $t_1 = 25$ min and $t_2 = 5$ min. Substituting into Lemma 36.6: $e^{-\alpha t_1} = e^{-2.5} \approx 0.082$, $e^{-\beta t_2} = e^{-1.25} \approx 0.287$. This gives

$$R_{\text{min}} = \frac{(1 - 0.287) \times 0.082}{1 - 0.082 \times 0.287} \approx \frac{0.0584}{0.9765} \approx 0.060, \qquad R_{\text{max}} = \frac{0.713}{0.9765} \approx 0.730.$$

That is, within each Pomodoro cycle the attention resource charges and discharges between 6% and 73%, with period average $\bar{R} \approx 0.30$. If the rest is removed ($t_2 = 0$), then $R_{\text{min}} \to 0$ and $\bar{R}$ drops markedly—the presence of rest intervals maximises the average attention resource, and this is precisely the optimisation meaning of the 'attention cycle'.

36.5 The emotional cycle

Definition 36.13 (The circumplex model of emotion). *An emotional state can be represented on a two-dimensional plane: the horizontal axis is* **emotional valence** *(pleasant–unpleasant) and the vertical axis is* **arousal level** *(high arousal–low arousal). A particular emotion corresponds to a point in the plane: happiness (high pleasantness + high arousal), sadness (low pleasantness + low arousal), anxiety (low pleasantness + high arousal) and calm (high pleasantness + low arousal).*

Over the course of a day the emotional state **rotates periodically** *about the centre of the circumplex model—from high arousal in the morning (whether pleasant or not) to low arousal in the afternoon, then to low mood in the evening, and finally to a reset during sleep. The circadian variation of the emotional state can be regarded as a quasi-periodic trajectory in the (valence, arousal) plane: the emotional changes within a day correspond to an arc in the plane, and sleep resets the emotional state to near the origin (neutrality).*

Proposition 36.7 (The emotional cycle of bipolar disorder). *Bipolar disorder is characterised by periodic alternation of the emotional state between mania (high arousal + high pleasantness or high irritability) and depression (low arousal + low pleasantness). The time scale of one complete emotional cycle (mania–normal–depression–normal–mania) varies from person to person—about weeks to months for the rapid-cycling type and about several days for the ultra-rapid-cycling type.*

In the phase space (valence V, arousal A), the emotional trajectory of a healthy individual is an ellipse of small amplitude (emotion fluctuates within the normal range), whilst the emotional trajectory of a patient with bipolar disorder is a limit cycle of large amplitude (emotion oscillates widely between the two extremes of mania and depression). The action of mood stabilisers (such as lithium salts) is to increase the rate of emotional return—constraining the orbit in phase space to a neighbourhood of the equilibrium point.

Proof. Linearise the emotional dynamics near the equilibrium point. Let $X = (V, A)^{\mathsf{T}}$ satisfy $\dot{X} = JX$, where the eigenvalues of J are $\lambda_{\pm} = -\mu \pm i\omega$ ($\mu > 0$ is the rate of emotional return and ω the intrinsic oscillation frequency of emotion). For a healthy individual μ is relatively large: the orbit is a rapidly decaying spiral and the amplitude $r(t) = r_0 e^{-\mu t}$ is small. For a patient with bipolar disorder μ is markedly smaller (emotional return is weakened): the orbit is approximately closed and the limit-cycle amplitude $r \approx r_0$ is large. The action of mood stabilisers is to increase the effective μ (equivalent to introducing a stronger damping term in J), so that the orbit decays back into a neighbourhood of the equilibrium point. □

36.6 The cycles of learning and forgetting

Definition 36.14 (The Ebbinghaus forgetting curve). *The Ebbinghaus forgetting curve describes the exponential decay of memory retention over time in the absence of review:*

$$M(t) = M_0 \cdot e^{-t/\tau},$$

where M_0 is the initial memory strength and τ the forgetting time constant (which depends on the difficulty of the memorised material and on the memory ability of the individual). Memory retention falls rapidly during the first few hours, and the rate of decline then slows.

The exponential decay of memory strength is analogous to the exponential decay of the Fourier components of the heat equation in the 'Differential Equations and Analysis' chapter. The time constant τ measures the retention capacity of a memory trace under time translation. The larger τ is, the more lasting the memory.

Proposition 36.8 (The optimal review cycle for spaced repetition). *Spaced repetition exploits the periodic structure of the forgetting curve to optimise long-term memory. If each review resets the memory strength to M_0, then the optimal time interval before the n-th review is*

$$\Delta t_n = \tau \cdot \ln \frac{M_0}{M_{\text{min}}} + \sum_{k=1}^{n-1} \Delta t_k \cdot \frac{\tau}{\tau + \Delta t_k} \approx \tau \cdot \ln \frac{M_0}{M_{\text{min}}} \cdot n,$$

where M_{min} is the minimum acceptable memory strength. The review interval grows linearly with the number of reviews—this is the scaling feature of the memory cycle.

Spaced repetition is the alternation, along the time axis, of the decay of memory and the reinforcement of review—each review resets the memory strength to its initial state ($M = M_0$), and the forgetting between two reviews constitutes one complete decay–recovery cycle. Optimising the review interval is equivalent to finding the cycle that maximises the average memory strength.

Proof. Suppose that after the n-th review the memory strength is M_0, and that after Δt_n it has decayed to M_{min}: $M_{\text{min}} = M_0 e^{-\Delta t_n/\tau}$, which gives $\Delta t_n = \tau \ln(M_0/M_{\text{min}})$. Consider the cumulative effect of review: if a review not only resets the memory strength but also slightly increases the forgetting time constant ($\tau_n = \tau_{n-1} + \Delta\tau$), then the review intervals grow gradually. Empirically, spaced-repetition software (such as Anki) uses a similar formula to compute optimal review intervals. ☐

36.7 Subconscious processes

Definition 36.15 (Subconscious processes). **Subconscious processes** *are psychological processes that run automatically below the threshold of consciousness. Compared with conscious processes, subconscious processes have the following features:*

(i) **Automaticity**: *they can be initiated and run without conscious effort.*

(ii) **Non-introspectability**: their content cannot be observed directly by introspection.

(iii) **Parallelism**: several subconscious processes can run simultaneously, unrestricted by the serial bottleneck of conscious attention.

(iv) **Diversity of time scales**: they are distributed from the millisecond scale (implicit priming) through the hour scale (memory consolidation during sleep) to the interannual scale (the formation of implicit attitudes).

The temporal structure of subconscious processes is an important component of psychological periodic structure—it provides the background rhythm of conscious processes and, through coupling with conscious processes, influences behavioural output.

Definition 36.16 (The activation–decay cycle of implicit memory). **Implicit memory** (procedural memory, priming effects) manifests as memory traces that influence behaviour without conscious recall. The activation–decay of the implicit memory strength $I(t)$ over time can be modelled as

$$\frac{dI}{dt} = -\lambda I + \eta \cdot \mathbf{1}_{\text{stimulus}}(t),$$

where λ is the decay rate and η the activation gain. The decay rate λ of implicit memory is usually smaller than the decay rate of explicit memory—this explains why procedural skills (such as riding a bicycle) are retained for longer than declarative knowledge (such as reciting poetry).

Proposition 36.9 (Coupling between subconscious and conscious cycles). Let the subconscious process $U(t)$ and the conscious process $C(t)$ interact through the following coupled equations:

$$\frac{dU}{dt} = f_U(U, C), \qquad \frac{dC}{dt} = f_C(U, C).$$

When the time scale of the subconscious process is close to that of the conscious process ($|\tau_U - \tau_C| < \epsilon$), **phase coupling** arises between the two processes—fluctuations in the subconscious state can influence the conscious state through the coupling, and vice versa.

Proof. Linearise the coupled equations near the equilibrium point: $\delta\dot{U} = a_{11}\delta U + a_{12}\delta C$, $\delta\dot{C} = a_{21}\delta U + a_{22}\delta C$. The eigenvalues of the coupled system are

$$\lambda_\pm = \frac{(a_{11} + a_{22}) \pm \sqrt{(a_{11} - a_{22})^2 + 4a_{12}a_{21}}}{2}.$$

When the discriminant is negative, the eigenvalues are complex conjugates—the two processes exhibit coupled oscillation. The oscillation frequency is determined jointly by the coupling strength $a_{12}a_{21}$ and the intrinsic frequency difference $(a_{11} - a_{22})^2$. Under strong coupling the system enters a coupled periodic state—the subconscious and conscious processes oscillate in synchrony. □

Example 36.4 (Memory consolidation during sleep). *Memory consolidation during sleep is a typical example of subconscious processes. Information learned during the day is first encoded in the hippocampus (a fast learning system), and during sleep the coupling between the hippocampus and the cortex (a slow learning system) gradually transfers the information to cortical long-term memory. This process takes place during the slow-wave sleep (N3) stage—at which the hippocampal–cortical coupling reaches its strongest.*

From the perspective of periodic mathematics, the periodic structure of memory consolidation is nested within the sleep cycle: each N3 stage corresponds to one round of a 'training cycle' of hippocampal–cortical coupling. Integrating the cumulative effect of several sleep cycles, the memory strength gradually increases over a whole night's sleep—this is an accumulative process driven by nested periods.

36.8 Behavioural periodicity

Definition 36.17 (Behavioural period). *A* **behavioural period** *is a repetitive behavioural pattern displayed by an individual on a particular time scale. Behavioural periods fall into the following levels:*

(i) **Circadian behavioural rhythm** *(about 24 hours): the circadian fluctuation of behavioural and physiological indices such as sleep–wake, eating, activity level, body temperature and hormone secretion.*

(ii) **Weekly behavioural period** *(about 7 days): switching of activity patterns between workdays and rest days.*

(iii) **Habit loop** *(minutes to hours): the three-step cue–behaviour–reward loop, driven by basal ganglia circuits.*

(iv) **Addiction cycle** *(hours to days): the seeking–use–withdrawal cycle of a substance or behaviour.*

(v) **Interannual behavioural period** *(about 1 year): seasonal mood variation, festive behavioural patterns and seasonal changes in activity level.*

Behavioural periods at the various levels are nested within one another—the circadian behavioural rhythm contains several habit loops, the habit loop repeats within a week and is modulated by the weekly behavioural period, and the weekly behavioural period repeats within the interannual behavioural period.

Definition 36.18 (The three-stage model of the habit loop). *The* **habit loop** *consists of three stages:*

(i) **Cue**: *an environmental stimulus that triggers the initial signal of the behaviour.*

(ii) **Routine**: *the automatic sequence of behaviour triggered by the cue.*

(iii) **Reward**: *the pleasure or satisfaction obtained after the behaviour is completed, which reinforces the cue–behaviour association.*

The period of the habit loop T_{habit} is determined by the frequency of the cue and the strength of the reward. The higher the degree of automaticity of the habit, the shorter T_{habit} is and the less the behaviour depends on conscious control.

Proposition 36.10 (Periodic characteristics of habit formation)**.** Let the behavioural tendency $B(t)$ vary with time; the process of habit formation can then be described by the following model:

$$\frac{dB}{dt} = \gamma \cdot (B_\infty - B) \cdot \mathbf{1}_{\text{cue}}(t),$$

where γ is the learning rate and B_∞ the asymptotic habit strength. The time scale of habit formation $\tau_{\text{habit}} = 1/\gamma$ is usually tens to hundreds of days—depending on the complexity of the behaviour and the frequency of the reward.

Proof. Under repeated cues, B converges exponentially to B_∞: $B(t) = B_\infty(1 - e^{-\gamma t})$. Experimental studies show that the average time of habit formation ranges from about 20 days for simple behaviours (such as drinking water) to about 250 days for complex behaviours (such as daily exercise). The time scale of habit formation τ_{habit} agrees with the rate of change of the neural markers of behavioural automaticity (the stability of basal ganglia activation). □

Example 36.5 (The periodic structure of addiction)**.** Addictive behaviour exhibits a typical periodic structure:

(i) **Use phase**: the stimulus of the substance or behaviour activates the reward circuit (dopamine release), producing brief pleasure.

(ii) **Tolerance phase**: repeated use reduces the sensitivity of the reward circuit (receptor downregulation), so that a stronger stimulus is required to produce the same effect.

(iii) **Withdrawal phase**: after discontinuation the reward circuit is in a state of low activity—producing withdrawal symptoms such as anxiety, depression and craving.

(iv) **Relapse phase**: the substance or behaviour is used again to relieve the withdrawal symptoms, forming a new round of the cycle.

The time scale of the addiction cycle runs from hours (acute withdrawal) through days (prolonged withdrawal) to years (risk of relapse). The neural mechanisms of addiction involve long-term plasticity changes in the reward circuit—these changes 'fix' the addiction cycle into a stable limit cycle, so that even if the external stimulus is absent for a long time, the internal craving mechanism may still drive relapse.

36.9 Phase locking of psychological cycles

Definition 36.19 (Interpersonal emotional synchrony)**.** In face-to-face communication, the emotional states of the two interlocutors tend to **synchronise**—one person's pleasure or sadness is transmitted to the other through facial expression, tone of voice and body language. This emotional synchrony is the realisation in interpersonal interaction of the phase locking of the 'Periodic Systems' chapter (Proposition 24.8).

Let the emotional valences of two people be $V_1(t)$ and $V_2(t)$, with intrinsic frequencies ω_1 and ω_2 respectively. Through emotional expression and perception the two become coupled:

$$\frac{d\theta_1}{dt} = \omega_1 + c_{12}\sin(\theta_2 - \theta_1), \quad \frac{d\theta_2}{dt} = \omega_2 + c_{21}\sin(\theta_1 - \theta_2),$$

where θ_i is the phase of the emotional state and c_{ij} the emotional coupling strength (which depends on empathic ability and expressive intensity). When the coupling is strong enough, the emotional phases of the two people tend to coincide—emotional synchrony. Emotional synchrony between infant and caregiver is crucial to the development of a healthy attachment relationship.

Proposition 36.11 (The contagious spread of group emotion). *In a group setting, emotional states spread through the social network—one person's emotional change can influence others in their social neighbourhood, and thence individuals further away. This is analogous to periodic propagation on discrete lattice points in the 'Periodic Manifold' chapter (Definition 12.28).*

Let the emotional valence $V_i(t)$ of individual i in a group satisfy

$$\frac{dV_i}{dt} = -\alpha V_i + \sum_{j\in\text{neighbourhood}(i)} w_{ij}(V_j - V_i) + \xi_i(t),$$

where α is the rate of emotional return to neutrality, w_{ij} the social influence weight and $\xi_i(t)$ an individual-specific emotional perturbation. When the coupling is strong (w_{ij} large), the mean group emotion $\bar{V}(t) = \frac{1}{N}\sum_i V_i(t)$ exhibits a slow collective oscillation—the 'social rhythm' of group emotion. The time scale of social emotional cycles (such as the economic optimism–pessimism cycle) can reach months to years—this is the macroscopic manifestation of psychological cycles at the social level.

Proof. Sum over all i and divide by N: $\frac{d\bar{V}}{dt} = -\alpha\bar{V} + \frac{1}{N}\sum_i\sum_{j\in N(i)} w_{ij}(V_j - V_i) + \bar{\xi}$. When the weights are symmetric ($w_{ij} = w_{ji}$), the inner double sum cancels edge by edge in an undirected network: each edge (i,j) contributes $w_{ij}(V_j - V_i) + w_{ji}(V_i - V_j) = 0$, so $\frac{d\bar{V}}{dt} = -\alpha\bar{V} + \bar{\xi}$. The mean therefore obeys a first-order linear equation—the collective oscillation does not come from the coupling term itself, but from the red-spectrum fluctuations of the noise $\bar{\xi}$ after low-pass filtering (time constant $1/\alpha$); the role of strong coupling is to align each V_i rapidly with $\bar{V}$ (phase locking), thereby aggregating individual fluctuations into an observable social rhythm. □

36.10 Conservation laws of psychological systems

Remark 36.9 (The three conservation laws of psychological systems (empirical observation / research programme)). The dynamics of psychological systems may be described by three conservation laws. **The following three items are empirical observations and a research programme, not proved theorems:**

(i) **Energy conservation**: psychological energy (mental fatigue and recovery) is consumed during waking and restored during sleep. Sleep deprivation leads to a decline in cognitive function, and recovery sleep restores cognitive function—psychological energy satisfies conservation between consumption and recovery. This is the balance of energy under time translation: $\Delta E_{\text{consumed}} + \Delta E_{\text{recovered}} = 0$ (long-term average).

(ii) **Information conservation**: the capacity of working memory is limited (7 ± 2 chunks)—the inflow of information into working memory (encoding) and its outflow (forgetting or transfer to long-term memory) satisfy conservation. The consolidation of long-term memory takes place during sleep—the transfer of memory from the hippocampus to the cortex is a redistribution of information among different stores. The total information content of memory is conserved in the transfer.

(iii) **Attention conservation**: the total attention resource is finite at any given moment—concentration of attention on one object is necessarily accompanied by neglect of other objects. The 'zero-sum' character of attention manifests as the finiteness of the attention resource: the allocation of attention among several targets constitutes a phase constraint in phase space—focusing on one target means phase locking in the direction of that target.

The hierarchical relation among the three conservation laws is: energy conservation guarantees the continuity of psychological activity, information conservation guarantees the stability of cognitive structure, and attention conservation guarantees the selectivity of the focus of consciousness. Together the three constitute the physical basis of the periodic structure of psychological systems.

Empirical basis: (i) Energy conservation: the empirical evidence for the consumption and recovery of psychological energy comes from sleep-deprivation studies. Sustained waking leads to a progressive decline in cognitive function (reaction time, working memory, decision-making ability), and after recovery sleep cognitive function returns to the baseline level. The power of slow-wave activity (SWA, 0.5–4 Hz) in the electroencephalogram (EEG) accumulates linearly during the waking period and decays exponentially during the sleep period—this is the electrophysiological index of psychological energy.

(ii) Information conservation: the limited capacity of working memory was established by the classical study of Miller (1956)—humans can hold about 7 ± 2 chunks of information simultaneously. Information is retained in working memory for about 20–30 seconds, unless it is refreshed by rehearsal. The two-process theory of memory consolidation holds that the fast learning system of the hippocampus transfers information to the slow learning system of the cortex during sleep—the amount of information is conserved in the transfer.

(iii) Attention conservation: the limited-resource theory of attention holds that the total attention resource is fixed and finite at any given moment. Allocating attention to one task necessarily reduces the resources available for other tasks—dual-task interference experiments confirm the finiteness of the attention resource. In periodic mathematics, the attention resource $R(t)$ satisfies $R_1 + R_2 + \cdots + R_n \leq R_{\text{max}}$—total attention does not exceed the maximum capacity.

Remark 36.10 (The place of subconscious and behavioural periods in the book).

Subconscious processes and behavioural periodicity are two important complementary levels of the periodic structure of psychological systems. Subconscious processes provide a background rhythm beyond conscious processes—although these rhythms cannot be directly introspected, they can be observed indirectly through behavioural output, physiological indices and emotional fluctuation. Behavioural periodicity, in turn, projects the periodic structure of psychological systems onto the observable behavioural level—giving psychological cycles a measurable and verifiable empirical basis. Integrating the three levels of subconscious, conscious and behavioural processes within a single framework of periodic structure is the complete statement of this chapter's application of periodic mathematics in psychology.

Remark 36.11 (Connection with the rest of the book). The theory of the periodic structure of psychological systems established in this chapter is an application of periodic mathematics in psychology. The sleep–wake cycle appears as a nested periodic structure formed by a main period of about 24 hours and an ultradian rhythm of about 90 minutes. The periodic fluctuation of attention appears as an alternation of focusing and dispersion. The circadian rotation in the circumplex model of emotion appears as epicyclic motion in the (valence, arousal) phase space. The Ebbinghaus forgetting curve appears as the exponential decay of memory strength, and spaced repetition optimises the cycle of memory retention. Subconscious processes provide a background rhythm beyond conscious processes—implicit memory, habituation and memory consolidation during sleep constitute the temporal structure of the subconscious. Behavioural periodicity projects psychological cycles onto the observable behavioural level—circadian behavioural rhythm, the weekly behavioural period, the habit loop and the addiction cycle constitute the multilevel periods of behaviour. Interpersonal emotional synchrony and group emotional contagion are the realisation, at the social-psychological level, of the phase locking of the 'Periodic Systems' chapter. The three conservation laws of psychological systems—energy conservation, information conservation and attention conservation—together constitute the guarantee of self-consistency for the periodic structure of psychological systems.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The approximately 24-hour main period of sleep–wake: gated triggering by the two-process model (Proposition 36.5); linear accumulation of sleep pressure during the waking period
Scaling	Exponential decay of the Ebbinghaus forgetting curve $M(t) = M_0 e^{-t/\tau}$ (Definition 36.14); scaling optimisation of spaced repetition (Proposition 36.8); multilevel nesting of behavioural periods (Definition 36.17)
Rotation	Circadian rotation in the two-dimensional valence–arousal phase space (Definition 36.13); the large-amplitude limit cycle of bipolar disorder (Proposition 36.7); coupled oscillation of subconscious and conscious processes (Proposition 36.9)
Conservation	Psychological energy conservation, information conservation (working memory of 7 ± 2 chunks) and attention conservation ($R_1 + \cdots + R_n \leq R_{\text{max}}$) (empirical observation / research programme, Remark 36.9)
Symmetry	Positive–negative valence symmetry of emotion ($V \in [-1, 1]$, symmetric about 0); circadian symmetry of sleep–wake
Closure	The consumption–recovery charge–discharge cycle of the attention resource (Lemma 36.6, Example 36.3); the sleep reset of the emotional state; the cue–behaviour–reward closure of the habit loop (Definition 36.18); the addiction cycle (Example 36.5)
Period	The 90-minute sleep cycle (NREM–REM alternation, Definition 36.11); the theta (4–8 Hz) and delta (0.5–2 Hz) rhythms of attention oscillation (Definition 36.12); phase locking of interpersonal emotional synchrony (Definition 36.19); the periodic characteristics of habit formation (Proposition 36.10); the activation–decay cycle of subconscious processes (Definition 36.16)

Exercises

Exercise 1 Prove the **periodicity decision theorem for psychological systems** of this chapter (36.1). Hint: first write down the decision map and the evolution group of this chapter's target, then apply the proof structure of the general criterion given in the ‘Periodic Systems’ chapter.

Exercise 2 Verify that the hypotheses of the **monotonicity of cardinality adjustment** (36.2) hold for the domain objects of this chapter, and give a counterexample in which the hypotheses fail.

Exercise 3 Examine the behaviour of the **three-parameter correspondence: radius, phase and speed** (36.1) under parameter degeneracy, and explain how the conclusion degenerates accordingly.

Exercise 4 Explain the relation between the **psychological periodicity diagnostic pipeline and its interface with three chapters** (36.6) and the corresponding general conclusion of the ‘Periodic Systems’ chapter (whether it is identical, a special case, or requires additional conditions).

Exercise 5 Using the seven-theme comparison table of this chapter, identify the concrete carriers of the themes **translation**, **scaling** and **rotation** in this chapter, and give one example of domain data for each.

Exercise 6 Synthesis: restate the diagnostic pipeline of this chapter using the five-step standard interface of the ‘Periodic Systems’ chapter (Table 24.1), noting one by one the theorem of this chapter on which each step depends: 36.1, 36.2, 36.1, 36.6.

Chapter 37

Periodic Structure of Social Systems

Remark 37.1 (Periodicity decision for the goals of this chapter (reference)). When this chapter decides whether a research target is periodic, determines its period type, or seeks its minimal positive period, reference is always made to the 'Periodic Systems' chapter: for the periodicity criterion see Theorem 24.1, for the discrimination of period types see Definition 24.3, Proposition 24.1, for the search for the minimal positive period see Definition 24.4, Theorem 24.2, and for the construction of period series and epicycle groups from data see Theorem 24.3, Definition 24.6.

Remark 37.2 (Positioning of this chapter: developed along the five-step interface of the 'Periodic Systems' chapter). The periodic structure of social systems studies the mathematical representation of periodic phenomena in group interaction, information diffusion, cultural change and population evolution in human society. This chapter does not set up a separate periodicity criterion; instead it connects social systems to the five-step interface laid down in the 'Periodic Systems' chapter:

(1) **Decision**: formalise social rhythms, periodic games, information diffusion, network evolution, cultural cycles, population evolution, collective synchronisation and similar goals as discrete or continuous dynamical systems, invoke Theorem 24.1 to decide whether they possess periodicity, and determine the period type;

(2) **Construction**: from social activity, information infection rate, tie strength, population size and population age structure construct amplitude series, frequency series and phase series, and assemble them into a social epicycle group;

(3) **Adjustment**: constrain and dynamically adjust the social epicycle group by means of residual-energy decrease, population conservation, information conservation and trust conservation (a metaphor);

(4) **Convolution**: introduce amplitude convolution, phase convolution and frequency convolution on the social epicycle group, and connect them with autocorrelation, cross-correlation, phase locking and collective synchronisation;

(5) **Integration**: connect to the tool chains of the three chapters 'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis' to form a social period diagnosis workflow.

The mathematical tools used in this chapter include: Fourier series, the repeated prisoner's dilemma, Bayesian updating, SIR/SIRS equations, damped oscillation, the network clustering coefficient, logistic growth, the age-structure equation, the Kuramoto model, phase locking, periodic convolution, epicycle groups and phase operators.

Remark 37.3 (Guide to the seven themes of this chapter). **Translation**—the passage of interpersonal relations through time (the daily/weekly/yearly repetition of social rhythms, the reciprocal exchange period T_{ex}); **scaling**—the scaling of social-network size (stratified scales such as the Dunbar number 150, the community half-life $T_{1/2}$); **rotation**—the phase cycle of relational closeness (the alternation of closeness and distance, the phase rotation of cultural retro, Kuramoto phase locking); **conservation**—population conservation, information conservation and trust conservation (of which trust conservation is a metaphor, see Theorem 37.8 and Remark 37.13); **symmetry**—reciprocal symmetry (the symmetric tit-for-tat strategy, the symmetric Nash equilibrium of fitness testing); **closure**—social-circle closure and relational loops (triadic closure, the exchange–test feedback loop); **period**—relational cycles and Kuramoto collective synchronisation (Proposition 37.14).

37.1 Connection to the five-step interface of the 'Periodic Systems' chapter

37.1.1 Step one: the periodicity decision for social goals

Definition 37.1 (Social goals and decision input). A **social goal** is one of the following objects:

(i) a social rhythm system, whose state space is the interval space of social activity $A(t)$;

(ii) a periodic game system, whose state space is the sequence space of behavioural states $(a, b) \in \{C, D\}^2$;

(iii) an information diffusion system, whose state space is the (S, I, R) simplex;

(iv) a social-network evolution system, whose state space is the space of tie-strength matrices $\{W_{ij}\}$;

(v) a cultural–demographic evolution system, whose state space is the joint space of population size $N(t)$ and age structure $n(a, t)$;

(vi) a collective synchronisation system, whose state space is the joint space of individual phases $(\theta_1, \ldots, \theta_N)$.

The **decision input** of a social goal is an ordered pair

$$\mathcal{D}_{\mathsf{S}} = (\mathcal{S}, \Phi),$$

where $\mathcal{S}$ is the state space and $\Phi : \mathcal{S} \to \mathcal{S}$ is the map induced by time translation, phase rotation or evolution. If $\mathcal{S}$ is a finite set, the goal is called a **finite social goal**; if $\mathcal{S}$ is an infinite set, the goal is called an **infinite social goal**.

Definition 37.2 (Social decision map)**.** Let a social goal be given by the decision input $\mathcal{D}_\mathsf{S} = (\mathcal{S}, \Phi)$. If the states of the goal are generated by Φ, that is, if there exists $x_0 \in \mathcal{S}$ such that

$$x_t = \begin{cases} \Phi^t(x_0), & t \in \mathbb{Z}, \\ \Phi_t(x_0), & t \in \mathbb{R}, \end{cases}$$

for $t \in \mathbb{Z}$ the state x_t is generated by iteration of Φ; for $t \in \mathbb{R}$ it is described by the one-parameter evolution family $\{\Phi_t\}_{t\in\mathbb{R}}$, where each $\Phi_t : \mathcal{S} \to \mathcal{S}$, $\Phi_0 = \mathsf{id}$, $\Phi_{s+t} = \Phi_s \circ \Phi_t$ for all $s, t \in \mathbb{R}$, and $(t, x) \mapsto \Phi_t(x)$ is continuous. then Φ is called the **social decision map** of the goal. In this case the social goal is reduced to the discrete dynamical system $(\mathcal{S}, \Phi)$ when $t \in \mathbb{Z}$ and to the continuous dynamical system $(\mathcal{S}, \{\Phi_t\}_{t\in\mathbb{R}})$ when $t \in \mathbb{R}$; in both cases the criteria of the 'Operations and Periodicity' chapter and the 'Periodic Systems' chapter may be applied directly.

Theorem 37.1 (Social periodicity decision theorem)**.** Let the decision map of a social goal be Φ, and let $(\mathcal{S}, \Phi)$ be a discrete dynamical system. Then:

(A) **Sufficient decision**: if there exists $m \in \mathbb{N}^+$ such that $\Phi^m = \mathsf{id}$, then the goal is a uniformly periodic social system; every orbit is closed, and the orbit lengths divide m.

(B) **Equivalence of decisions**: the goal is a uniformly periodic social system if and only if $\mathsf{ord}(\Phi) < \infty$.

(C) **Pointwise criterion**: the orbit of the goal at the state x_0 is closed if and only if x_0 is a periodic point; in this case the orbit length is precisely the period at that point.

(D) **Non-omissibility**: the conjunction of conserved quantities and the symmetry group cannot replace $\mathsf{ord}(\Phi) < \infty$; a quasi-periodic social goal does not belong to the periodic social systems.

Proof. The Theorem 37.1 of this chapter is the domain instantiation of Theorem 24.1 in the 'Periodic Systems' chapter: one need only replace the decision map f and the evolution group by the Φ of this chapter, and the four items—(A) sufficient decision, (B) equivalence of decisions, (C) pointwise criterion, (D) non-omissibility—hold one by one. By the convention of Remark 24.4, no further elaboration is repeated here. □

Corollary 37.1 (Social period types)**.** After the decision by Theorem 37.1, a social goal falls into one of the following types:

(i) **Trivially periodic type**: the minimal period is $m_0 = 1$, and the state is a fixed point;

(ii) **Strictly periodic type**: $1 < m_0 < \infty$, and the state is strictly periodic;

(iii) **Uniformly periodic type**: $\mathsf{ord}(\Phi) = m < \infty$, and all orbit lengths divide m;

(iv) **Quasi-periodic type**: $\mathsf{ord}(\Phi) = \infty$, and the orbit is not closed but is dense on a compact carrier structure;

(v) **Eventually periodic type**: there exists N such that the goal is periodic for $t \geq N$;

(vi) **Discretely periodic type**: the carrier structure is $\mathbb{Z}$, and the normalised period ratio is $\rho = T_x/\Delta x \in \mathbb{Q}$.

(vii) **Asymptotically periodic (damped) type**: the orbit is not closed, but tends asymptotically to a closed orbit or to a fixed point, the residual decaying as $e^{-\kappa t}$ (see Definition 24.3(vii)).

Here (i)–(iii) and (vi) belong to the periodic social systems; (iv) does not belong to the periodic social systems, but can be handled by bounded-approximation tools; (v) is a periodic social system only in its stable stage; (vii), like (iv), is a type in the limiting sense; it is not a periodic social system and can be handled with bounded-approximation tools only in its stable stage.

Proof. Each item follows by substituting the social goal into Theorem 37.1 and Definition 24.3 of the 'Periodic Systems' chapter. □

Lemma 37.1 (Periodicity decision for social rhythms)**.** Let the social activity $A(t)$ satisfy the daily periodicity condition $A(t+T_{\text{day}}) = A(t)$ ($T_{\text{day}} = 24$ h), the weekly periodicity condition $A(t+T_{\text{week}}) = A(t)$ ($T_{\text{week}} = 168$ h), and the yearly periodicity condition $A(t+T_{\text{year}}) = A(t)$ ($T_{\text{year}} = 8760$ h). Then A is a uniformly periodic system, and its least positive period divides

$$T_0 = \text{gcd}(T_{\text{day}}, T_{\text{week}}, T_{\text{year}}).$$

Proof. By the three periodicity conditions, the group with periodic orbits $\text{Group}_{\text{period}}(A)$ of A contains $24\mathbb{Z}$, $168\mathbb{Z}$ and $8760\mathbb{Z}$. Hence $\text{gcd}(24, 168, 8760)\mathbb{Z} = 24\mathbb{Z} \subseteq \text{Group}_{\text{period}}(A)$, and the least positive period of A divides 24. By Definition 1.21 and Theorem 1.6 of the 'Operations and Periodicity' chapter, A is a uniformly periodic system, and its least positive period divides T_0. □

Lemma 37.2 (Periodicity decision for the TFT periodic game)**.** Let both players in the repeated prisoner's dilemma use the tit-for-tat (TFT) strategy, with misjudgement probability $\varepsilon > 0$. Then the cycle of cooperation–retaliation–return to cooperation is a uniformly periodic system, whose expected period is

$$\mathbb{E}[L_{\text{cycle}}] \sim \frac{2}{\varepsilon} \qquad (\varepsilon \to 0^+).$$

Proof. By Proposition 37.6, the 4-state Markov chain of TFT (with states $(a, b) \in \{C, D\}^2$) has (C, C) as an absorbing state in the absence of noise, and escapes from it with probability $2\varepsilon(1-\varepsilon)$ under noise. The expected time of a first departure-and-return is $O(1/\varepsilon)$, and a complete cycle contains two such processes, hence $\mathbb{E}[L_{\text{cycle}}] \sim 2/\varepsilon$. By Definition 1.21 of the 'Operations and Periodicity' chapter, this cycle is a uniformly periodic system (in the expected sense). □

Lemma 37.3 (Periodicity decision for SIRS information diffusion)**.** Let information diffusion be described by the SIRS model, with parameters satisfying $\Delta = (\beta I^* + \alpha)^2 - 4\beta I^*(\gamma + \alpha) < 0$. Then the infected proportion $I(t)$ is a uniformly periodic system, whose damped oscillation period is

$$T = \frac{2\pi}{\omega}, \qquad \omega = \sqrt{\beta I^*(\alpha + \gamma) - \frac{(\beta I^* + \alpha)^2}{4}}.$$

Proof. By Proposition 37.9, when $\Delta < 0$ the eigenvalues of the Jacobian matrix are the complex conjugates $\lambda = -\frac{\beta I^*+\alpha}{2} \pm i\omega$, and the angular frequency of the damped oscillation is ω. By Definition 1.21 of the 'Operations and Periodicity' chapter, the decaying oscillation of $I(t)-I^*$ constitutes a damped periodic system; neglecting the damping, it is a uniformly periodic system with period $T = 2\pi/\omega$. ☐

Lemma 37.4 (Periodicity decision for the community life cycle). Let the community activity $u(t)$ rise during the formation stage, be highest during the peak stage, and decay exponentially during the decline stage as $u(t) = u_{\text{peak}}e^{-\lambda(t-t_{\text{peak}})}$. Then the community life cycle is of asymptotically periodic (damped) type (the decline converges to a steady state), and the half-life is

$$T_{1/2} = \frac{\ln 2}{\lambda}.$$

Proof. By Proposition 37.10, during the decline stage $u(t)$ decays exponentially to 0, which is the limiting case of the asymptotically periodic (damped) type, Definition 24.3(vii), of the 'Periodic Systems' chapter (approaching the fixed point $u = 0$). The half-life is obtained from $e^{-\lambda T_{1/2}} = 1/2$ as $T_{1/2} = \ln 2/\lambda$. ☐

Lemma 37.5 (Periodicity decision for intergenerational population cycles). Let the discrete generation population model be $N_{n+1} = sf(N_{n-1})N_n$, with $f(N) = r(1 - N/K)$. Then when $4srN^*/K > 1$, the population dynamics is a uniformly periodic system, whose period is approximately two generation times:

$$T_{\text{gen}} \approx 2 \times 25 \text{ years} = 50 \text{ years}.$$

Proof. By the proof of Proposition 37.12, linearisation near the equilibrium state gives the characteristic equation $\lambda^2 - \lambda + srN^*/K = 0$, with discriminant $\Delta = 1 - 4srN^*/K$. When $\Delta < 0$ the eigenvalues are complex conjugates, and the population dynamics exhibits periodic oscillation. The period is approximately two generation times, $T_{\text{gen}} \approx 50$ years. ☐

Lemma 37.6 (Periodicity decision for Kuramoto collective synchronisation). Let N individuals be coupled by the Kuramoto model, with a unimodal and symmetric natural-frequency distribution $g(\omega)$ and coupling strength $K > K_c$. Then the order parameter $r > 0$ of the synchronised state is the macroscopic manifestation of a uniformly periodic system, and the period of individual phase locking is the period of the common mean frequency $\bar{\omega}$:

$$T_{\text{sync}} = \frac{2\pi}{\bar{\omega}}.$$

Proof. By Lemma 37.7, when $K > K_c$ there exists a synchronised solution with $r > 0$. By Proposition 37.14, the phases of all individuals lock near the common mean frequency $\bar{\omega}$. Hence, by Definition 1.21 of the 'Operations and Periodicity' chapter, the synchronised collective is a uniformly periodic system with period $T_{\text{sync}} = 2\pi/\bar{\omega}$. ☐

Definition 37.3 (Algorithm for the least positive period of a social system). This definition has the same content as Chapter 24 in the section 'Periodicity decision: whether a goal possesses periodicity' (Section 24.1) in Definition 24.4; to avoid repeated exposition, only the reference anchor is retained here, and the complete statement is to be found there. The only difference here is terminology (social goal / social sample); the algorithm steps and conclusions are entirely the same.

Theorem 37.2 (Correctness of the algorithm for the least positive period of a social system)**.** This theorem has the same content as Chapter 24 in the section 'Periodicity decision: whether a goal possesses periodicity' (Section 24.1) in Theorem 24.2; to avoid repeated exposition, only the reference anchor is retained here, and the complete statement and proof are to be found there. The only difference here is terminology (social goal / social sample); the algorithm steps and conclusions are entirely the same.

Example 37.1 (Instances of social periodicity decisions)**.** (i) Social rhythms: $T_0 = \text{gcd}(24, 168, 8760) = 24$ h, of uniformly periodic type;

(ii) TFT periodic game: $\mathbb{E}[L_{\text{cycle}}] \sim 2/\varepsilon$, of uniformly periodic type;

(iii) SIRS information diffusion: $T = 2\pi/\omega$, of uniformly periodic type (damped);

(iv) Community life cycle: of asymptotically periodic (damped) type (tending to $u = 0$), half-life $\ln 2/\lambda$;

(v) Intergenerational population cycle: $T_{\text{gen}} \approx 50$ years, of uniformly periodic type;

(vi) Kuramoto collective synchronisation: $T_{\text{sync}} = 2\pi/\bar{\omega}$, of uniformly periodic type.

Remark 37.4 (Conserved quantities do not guarantee periodicity). Population conservation, information conservation and trust conservation (a metaphor) in social systems are auxiliary conditions for periodic structure, but they do not constitute a sufficient condition. For instance, a closed society that has population conservation but no feedback coupling will not spontaneously produce periodic oscillations. The periodic structure of a social system must be decided by Theorem 37.1 or by the lemmas above.

37.1.2 Step two: construction of social period series and the epicycle group

Definition 37.4 (Social amplitude series, frequency series and phase series)**.** Let the state or output sequence of a social goal be $\{z_t\}$. If it can be written as

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)} + \varepsilon_t,$$

then

$$\mathcal{A}_z(t) = \sum_k A_k e^{ikt}, \qquad \mathcal{F}_z(s) = \sum_k \omega_k s^k, \qquad \mathcal{P}_z(t) = \sum_k e^{i(kt+\phi_k)}$$

are called respectively the **amplitude series**, the **frequency series** and the **phase series** of the social goal. The three are consistent with Definition 6.10, Definition 6.12 and Definition 6.11 of the 'Series and Convolution' chapter.

Definition 37.5 (Social epicycle group)**.** Let the period series of a social goal be given by the amplitude series $\{A_k\}$, the frequency series $\{\omega_k\}$ and the phase series $\{\phi_k\}$. For each k define the epicycle

$$E_k : \theta \mapsto A_k e^{i(\theta + \phi_k)}, \qquad \theta = \omega_k t,$$

whose collection

$$\mathfrak{E}_{\mathsf{S}} = \{(A_k, \omega_k, \phi_k)\}_{k\in\mathbb{Z}}$$

is called the **epicycle group** of the social goal. Its superposition is precisely the social period series

$$z_t = \sum_k E_k(\omega_k t) = \sum_k A_k e^{i(\omega_k t+\phi_k)}.$$

Theorem 37.3 (Constructing a period series from social data)**.** *Let the output of a social goal be a finite periodic sequence with period N. Then:*

(i) *the output can be expanded as a discrete Fourier series*

$$z_t = \sum_{k=0}^{N-1} c_k e^{i2\pi kt/N}, \qquad c_k = \frac{1}{N}\sum_{t=0}^{N-1} z_t e^{-i2\pi kt/N};$$

(ii) *the amplitude series is given by $|c_k|$, the phase series by $\arg c_k$, and the frequency series by $2\pi k/N$;*

(iii) *if the output is a social rhythm sequence, then the fundamental frequency corresponds to the daily period, the $k = 7$ harmonic to the weekly period, and the $k = 365$ harmonic to the yearly period; if the output is an information diffusion sequence, then the fundamental frequency corresponds to the damped oscillation period.*

Proof. (i) The discrete Fourier series of a finite periodic sequence is a standard result: the basis functions $\{e^{i2\pi kt/N}\}_{k=0}^{N-1}$ form an orthogonal basis of $\mathbb{C}^N$. (ii) This is read off directly from $c_k = A_k e^{i\phi_k}$. (iii) For a social rhythm sequence with $N = 8760$ (hourly sampling), the fundamental frequency corresponds to 1 hour, $k = 24$ to the daily period, $k = 168$ to the weekly period, and $k = 8760$ to the yearly period; for an information diffusion sequence, the fundamental frequency corresponds to the angular frequency $\omega/(2\pi)$ of the SIRS damped oscillation. ☐

Proposition 37.1 (Three-parameter correspondence: radius, phase and rotation speed)**.** *The three parameters of the social epicycle group $\mathfrak{E}_{\mathsf{S}} = \{(A_k, \omega_k, \phi_k)\}_k$ correspond one-to-one with three geometric–social quantities:*

(i) ***Radius corresponds to amplitude**: the radius $|E_k| = A_k$ of E_k is precisely the amplitude of that social oscillation component;*

(ii) ***Phase corresponds to rotation delay**: the initial phase ϕ_k corresponds to the time translation $\tau_k = \phi_k/\omega_k$; when $\phi_k > 0$ this component **leads** the baseline by τ_k (equivalently, the component coincides with the baseline once the baseline is postponed by τ_k);*

(iii) ***Frequency corresponds to rotation speed**: ω_k is precisely the angular frequency of that social oscillation component.*

Proof. The Proposition 37.1 of this chapter corresponds exactly to Proposition 24.3 in the 'Periodic Systems' chapter, and the three correspondences—radius corresponds to amplitude, phase corresponds to rotation delay, frequency corresponds to rotation speed—are obtained by translating them one by one onto the epicycle group of this chapter. The proof is to be found there: (i) from $|E_k| = A_k$ and the energy decomposition, (ii) from the phase translation theorem, (iii) from the definition of the frequency series. ☐

Example 37.2 (Period series of social activity)**.** Social activity $A(t)$ has periodic components on all three scales of day, week and year. In the social epicycle group the three appear as epicycle components of different radii and different angular frequencies—the daily period has the largest amplitude, the weekly period comes next, and the yearly period is the smallest.

Remark 37.5 (A period series is a complex function)**.** Let $z = e^{i\omega_0 t}$. Then the social period series

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)}$$

when all components are commensurable with ω_0 (that is, $\omega_k = k\omega_0$, $k \in \mathbb{Z}$) it can be written as the Laurent series $\sum_k c_k z^k$, where $c_k = A_k e^{i\phi_k}$: in that case the social period series is a complex function on the unit circle, and otherwise it is merely a function on the torus.

37.1.3 Step three: dynamic adjustment of the social epicycle group

Definition 37.6 (Dynamic adjustment of the social epicycle group)**.** Let the current social epicycle group be $\mathfrak{E}^{(j)}$, with partial sum

$$z^{(j)}(t) = \sum_{k \in \mathfrak{E}^{(j)}} A_k e^{i(\omega_k t + \phi_k)},$$

and residual $r^{(j)} = z - z^{(j)}$. **Dynamic adjustment** consists of two rules:

(i) **Quantity adjustment**: if the residual energy $\|r^{(j)}\|^2 > \delta$, then add one epicycle according to the peak of the residual periodogram;

(ii) **Parameter adjustment**: update the parameters of the existing epicycles by regularised least squares or by conservation constraints.

In social systems, quantity adjustment corresponds to adding social rhythm modes or information diffusion harmonics; parameter adjustment corresponds to tuning the amplitudes, phases and frequencies so that the population, information and trust constraints are satisfied.

Proposition 37.2 (Monotonicity of quantity adjustment)**.** *After adding the dominant-frequency component of the residual as in Definition 37.6(i), the residual energy is monotonically non-increasing:*

$$\|r^{(j+1)}\|^2 \le \|r^{(j)}\|^2,$$

with equality if and only if the residual has no energy at the selected frequency.

Proof. The Proposition 37.2 of this chapter agrees with the dynamic adjustment mechanism of Definition 24.8 in the 'Periodic Systems' chapter: adding an epicycle means removing the component of $r^{(j)}$ along $\hat{\omega}$ from the residual, and the orthogonal decomposition $\|r^{(j)}\|^2 = \|r^{(j+1)}\|^2 + \|\text{the component removed}\|^2$ yields the inequality, with equality if and only if the norm of the removed component is zero. The complete argument is to be found there. ☐

Theorem 37.4 (Convergence of dynamic adjustment). *Let the social goal be a finite periodic sequence of period T_0 and let $\delta = 0$. Then after at most $\lfloor T_0/2 \rfloor + 1$ steps the dynamic adjustment has zero residual, and the resulting social epicycle group is precisely the exact Fourier data of the goal.*

Proof. The discrete Fourier series of a sequence of period T_0 is complete on $\lfloor T_0/2 \rfloor + 1$ harmonics. At each step one dominant-frequency component of the residual is extracted by Proposition 37.2 and the residual energy decreases strictly, so after finitely many steps the residual is zero. ☐

Definition 37.7 (Structural-metric adjustment). *Let the structural metric of a social goal be*

$$\mathcal{I} = (\mathcal{I}_{\text{population}}, \mathcal{I}_{\text{information}}, \mathcal{I}_{\text{trust}}) = (\text{degree of population conservation, degree of information co}$$

Structural-metric adjustment *consists in choosing θ in the parameter space Θ: at each step take a neighbouring parameter for which $\mathcal{I}(\theta)$ is* **strictly decreasing** *in the lexicographic order (minimisation); if no such parameter exists, the process stops, and the parameter is then locally optimal.*

Proposition 37.3 (Finite convergence of structural-metric adjustment). *Let the parameter space Θ be finite, and let each adjustment make $\mathcal{I}$ strictly decreasing in the lexicographic order. Then the adjustment terminates after finitely many steps at a locally optimal parameter.*

Proof. Since Θ is finite, $\mathcal{I}$ takes only finitely many values, and the lexicographic order is a total order on a finite set; a strictly decreasing sequence in that order cannot continue indefinitely, so the adjustment terminates after finitely many steps. At termination there is no neighbouring parameter that makes $\mathcal{I}$ strictly decrease in the lexicographic order, that is, the parameter is locally optimal. ☐

Remark 37.6 (The boundary of dynamic adjustment). Theorem 37.4 relies on a finite periodic sequence. For continuous spectra or quasi-periodic social goals, no finite epicycle group represents them exactly, and only a bounded approximation is possible. Structural-metric adjustment does not guarantee global optimality.

37.1.4 Step four: social convolution

Definition 37.8 (Social amplitude convolution, phase convolution and frequency convolution). *Let two social epicycle groups be*

$$\mathfrak{E} = \{(A_n, \omega_n, \phi_n)\}, \qquad \mathfrak{E}' = \{(B_m, \nu_m, \psi_m)\}.$$

Define:

(i) **Amplitude convolution**: $\mathbb{A}_{mn} = A_n B_m$;

(ii) **Phase convolution**: $\mathbb{P}_{mn} = e^{i(\phi_n - \psi_m)}$;

(iii) **Frequency convolution**: $\mathbb{F}_{mn} = \omega_n \nu_m$;

(iv) **Trend convolution**: $\mathbb{T}_{mn} = A_n B_m e^{i(\phi_n - \psi_m)} = A_n B_m \mathbb{P}_{mn}$.

This is consistent with Definition 6.21, Definition 6.22, Definition 6.23 and Definition 6.24 of the 'Series and Convolution' chapter.

Theorem 37.5 (The action of the three convolutions on the social epicycle group). *Let the epicycle groups of two social goals be $\mathfrak{E}$ and $\mathfrak{E}'$ respectively. Then:*

(i) *amplitude convolution acts only on the radius;*

(ii) *phase convolution acts only on the phase;*

(iii) *frequency convolution acts only on the rotation speed;*

(iv) *trend convolution preserves both amplitude and phase.*

Proof. This is read off directly from Definition 37.8. ☐

Proposition 37.4 (Autocorrelation and trend convolution). *Let $\{z_t\}$ be a social period sequence, with autocorrelation function*

$$R_z(h) = \sum_t z_t \overline{z_{t+h}}.$$

In the language of the social epicycle group, the autocorrelation function is the projection of the trend convolution onto the diagonal:

$$R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}.$$

Proof. The Proposition 37.4 of this chapter is the realisation in this domain of the convolution-action conclusion of Theorem 24.6 in the 'Periodic Systems' chapter: substituting $z_t = \sum_n A_n e^{i(\omega_n t + \phi_n)}$ into the definition of the autocorrelation, cancelling all off-diagonal cross terms by orthogonality and retaining only the diagonal term of the trend convolution (the factorisation of the three convolutions, Proposition 24.6), which yields $R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}$. The complete expansion is to be found there. ☐

Remark 37.7 (The relation between social convolution and periodic convolution). Social convolution is not a new convolution operation; it is the instance on the social epicycle group of the periodic convolution of the 'Series and Convolution' chapter. Autocorrelation is the projection of the trend convolution onto the diagonal; cross-correlation, phase locking and collective synchronisation are respectively the concrete manifestations in social science of amplitude convolution and frequency convolution.

37.1.5 Step five: integration with discrete sampling, harmonic analysis and time series analysis

Definition 37.9 (Social period diagnosis workflow)**.** *For a given social system, diagnose it using the periodic-mathematics framework:*

(i) **Periodicity decision**: *verify the period type of the social goal (Theorem 37.1);*

(ii) **Series construction**: *from social activity, information diffusion rate and population data construct the amplitude, frequency and phase series and the epicycle group (Theorem 37.3);*

(iii) **Conservation diagnosis**: *verify population, information and trust conservation (Theorem 37.8);*

(iv) **Convolution diagnosis**: *compute the autocorrelation (the diagonal projection of the trend convolution, Proposition 37.4) and the phase convolution;*

(v) **Integration diagnosis**: *connect to the tool chains of discrete sampling, harmonic analysis and time series analysis.*

Theorem 37.6 (The social period diagnosis workflow and the interfaces to three chapters)**.** *The five steps of Definition 37.9 connect respectively to:*

(i) *Definition 21.1, Theorem 21.3, Theorem 21.10 of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter;*

(ii) *Definition 22.5, Theorem 22.1, Proposition 22.11 of the 'Harmonic Analysis' chapter;*

(iii) *Definition 23.16, Proposition 23.2, Definition 23.19 of the 'Time Series Analysis' chapter.*

Hence social period diagnosis is not an independent method, but the application of the unified tool chain of periodic mathematics to social systems.

Proof. The Theorem 37.6 of this chapter is the implementation in this domain of the five-step interface of the 'Periodic Systems' chapter (Table 24.1): the periodicity decision uses sampling and spectral estimation, the frequency decomposition and the phase kernel use the 'Harmonic Analysis' chapter, and the periodogram peak, the Fisher test and the window-function validation use the 'Time Series Analysis' chapter. Thus the diagnosis workflow is synthesised from the tool chains of three chapters, and the proof is to be found there. ☐

Remark 37.8 (The operational order after integration)*.* The recommended order for an actual diagnosis is: first decide whether the social goal is periodic (Theorem 37.1); then construct the Fourier data and the epicycle group of the social data (Theorem 37.3); then optimise the parameters using the conservation constraints and dynamic adjustment (Theorem 37.4, Proposition 37.3); then perform social convolution and autocorrelation analysis (Proposition 37.4); and finally output the diagnosis as in Definition 37.9.

37.2 Mathematical foundations of social systems

Definition 37.10 (Basic quantities of social systems)**.** A social system is described by the following basic variables:

(i) **Social network graph** $G = (V, E)$: the vertices V represent individuals (people, organisations), and the edges E represent social relations (friendship, following, communication). The topology of the graph determines the paths and the speed of information diffusion.

(ii) **Social activity** $A(t) \in [0, 1]$: the intensity with which an individual or group participates in social interaction at time t. $A \to 1$ indicates high-intensity socialising (frequent contact, a large amount of interaction), and $A \to 0$ indicates social isolation.

(iii) **Information infection rate** $I(t) \in [0, 1]$: the proportion of a group through which a particular piece of information (or behaviour, or opinion) has spread. $I(t)$ is the core variable of SIR-type models.

(iv) **Tie strength** $W_{ij}(t) \in [0, 1]$: the strength of the social tie between individuals i and j at time t. $W \to 1$ indicates a strong tie (a close friendship), and $W \to 0$ indicates a weak tie or no tie.

(v) **Population size** $N(t)$: the total population of the social system at time t. Changes in population size are determined by the birth rate, the death rate and the migration rate.

(vi) **Population age structure** $n(a, t)$: the population density at time t and age a. The age structure is the foundation of the periodic structure of population evolution.

The core feature of a social system is **collective dynamics under the constraints of network structure**—the behaviour of an individual is influenced by its position in the social network and by the states of its neighbourhood, and periodic phenomena at the collective level are the emergent result of individual interaction.

Property 37.1 (Immediate constraints on the basic quantities)**.** By Definition 37.10, the basic quantities of a social system satisfy the following immediate constraints:

(i) Normalisation: $A(t), I(t), W_{ij}(t) \in [0, 1]$, all dimensionless proportional quantities; $N(t) \geq 0$ and $n(a, t) \geq 0$.

(ii) Network carrying: the tie-strength matrix $\{W_{ij}(t)\}$ is the weighted adjacency matrix of the social network graph $G = (V, E)$, whose support is determined by the edge set E, and $W_{ij}(t)$ is non-zero only for $(i, j) \in E$.

(iii) Population balance: $\int_0^\infty n(a, t)\, da = N(t)$; the rate of population change is $\frac{dN}{dt} = B(t) - D(t) + M(t)$, where B, D, M are respectively the birth rate, the death rate and the net migration rate.

(iv) Emergence: the periodic phenomena at the collective level (statistics of $I(t)$, $A(t)$) are determined by individual interaction (W_{ij}, network topology) and by population processes, rather than being prescribed in advance.

37.3 Social rhythms

Definition 37.11 (Multi-scale periodicity of social activity)**.** *Human social activity displays pronounced periodicity on several time scales:*

(i) **Daily period**: *social activity varies regularly within* 24 *hours—lowest in the small hours (sleep), rising gradually in the morning, falling slightly in the early afternoon (a midday rest), reaching a peak from evening to night (the peak of social activity), and dropping rapidly late at night.*

(ii) **Weekly period**: *the seven-day week is a purely* **socially constructed period***—it has no astronomical basis and is sustained entirely by cultural convention. Social activity reaches its peak at the weekend (from Friday evening to Sunday) and is lowest at the start of the week (Monday).*

(iii) **Yearly period**: *social activity varies with the seasons—indoor socialising increases in winter, outdoor socialising increases in summer. Particular festivals (the Spring Festival, Christmas, Thanksgiving) give rise to large-scale periodic social gatherings and population movements.*

Social rhythms manifest themselves as the repetition of time translations of 24 *hours,* 7 *days and* 365 *days: the social pattern of every day, every week and every year repeats in the statistical sense. Social activity* $A(t)$ *approximately satisfies* $A(t+T) \approx A(t)$ *for the daily period* $T_{\text{day}} = 24$ *h, the weekly period* $T_{\text{week}} = 168$ *h and the yearly period* $T_{\text{year}} = 8760$ *h.*

Proposition 37.5 (Periodic decomposition of social activity)**.** *If the social activity* $A(t)$ *satisfies the daily periodicity condition* $A(t + T_{\text{day}}) = A(t)$ $(T_{\text{day}} = 24$ *h), then* A *can be expanded as a Fourier series*

$$A(t) = \sum_{k=-\infty}^{\infty} \hat{A}_k e^{ik\omega_0 t}, \qquad \omega_0 = \frac{2\pi}{T_{\text{day}}},$$

where the energy share of the k*-th harmonic is* $p_k = |\hat{A}_k|^2 / \sum_n |\hat{A}_n|^2$*, satisfying* $\sum_k p_k = 1$*. The fundamental frequency* $(k = 1)$ *corresponds to the circadian rhythm, the second harmonic* $(k = 2)$ *to the two-peak work–rest structure, and the third harmonic* $(k = 3)$ *to the additional peak of mealtimes.*

Proof. A has period T_{day}, so by the Fourier series theorem $A(t)$ can be expanded as the above series, with coefficients

$$\hat{A}_k = \frac{1}{T_{\text{day}}} \int_0^{T_{\text{day}}} A(t) e^{-ik\omega_0 t}\, dt.$$

The Parseval identity $\sum_k |\hat{A}_k|^2 = \frac{1}{T_{\text{day}}} \int_0^{T_{\text{day}}} |A(t)|^2 dt$ guarantees that the definition of p_k is legitimate and that $\sum_k p_k = 1$. If $A(t)$ has period exactly T_{day}, then the energy is concentrated only at integer multiples of the frequency, $\omega = k\omega_0$. $\square$

Remark 37.9 (Rhythm patterns of social media (empirical observation))*.* The number of posts, likes and comments on social-media platforms (Twitter/X, WeChat, Weibo) displays precise periodicity. Fourier analysis with a daily period shows that the fundamental frequency $\omega_0 = 2\pi/24\text{h}$ of posting volume explains about 60% of the variance, whilst the second and third harmonics correspond respectively to the two-peak work–rest pattern and to the additional peak of mealtimes—this is an **empirical observation**, and its specific value (the 60% variance share) comes from a statistical fit to real platform data, not from a mathematical theorem.

Social-media data are the material for studying social rhythms—the statistical average of massive user data reveals the periodic structure of social behaviour. A twenty-four-hour heat map of global social-media activity shows that users in different time zones enter the active state in turn—the global social network behaves as a relay system, with activity translated between time zones. The mathematical formulation of this phenomenon is: under a suitable covering by time zones, $\sum_{\text{time zone}} A_{\text{time zone}}(t)$ is approximately constant within a day—again an empirical observation.

37.4 Periodic games in social interaction

The two core dynamics in social interaction—**reciprocal altruism** and **fitness testing**—both exhibit probabilistic periodic structure. This section uses the mathematical framework established in the 'Probability and Distribution' chapter to carry out a formal analysis of these two social games.

37.4.1 The periodic game of reciprocal altruism

Definition 37.12 (Periodic model of reciprocal altruism). **Reciprocal altruism** (also called reciprocal exchange) means that individual A provides a benefit to individual B at time t (help, resources, information), expecting to obtain a return at some future time. Let $X_{AB}(t) \in \{0,1\}$ indicate whether A provides a benefit to B at time t (1 = provides, 0 = does not provide), and let $Y_{BA}(t)$ be defined analogously. The long-run balance of reciprocal altruism requires that the cumulative giving and the cumulative return of both sides be equal in the statistical sense:

$$\mathbb{E}\left[\sum_{t=1}^{T} X_{AB}(t) \cdot v_A(t)\right] \approx \mathbb{E}\left[\sum_{t=1}^{T} Y_{BA}(t) \cdot v_B(t)\right],$$

where $v_A(t), v_B(t)$ are the 'values' of the benefits each provides (which may vary with time).

The periodicity of reciprocal behaviour (you help me once, I help you once) is the manifestation in social games of the repetition of time translation. The exchange period T_{ex} (the average time interval between two reciprocal acts) measures the closeness of the social relationship. In close relationships T_{ex} is longer (exchange need not be matched immediately), whilst in distant relationships T_{ex} is shorter (every act of help requires a prompt return).

Proposition 37.6 (Periodic analysis of the tit-for-tat strategy). In the repeated prisoner's dilemma, one of the most successful strategies is tit-for-tat (Tit-for-Tat, TFT): cooperate on the first move, and thereafter repeat the opponent's previous move at every step. In a noisy environment (with misjudgement probability ε), the evolution of the TFT strategy produces a **cooperation–defection periodic oscillation**.

Let the misjudgement probability be ε (the probability that the opponent actually cooperates but is misjudged as defecting, or conversely). When $\varepsilon > 0$, a single misjudgement triggers a retaliation loop—A retaliates after misjudging B as defecting, B retaliates back after perceiving A's defection, and so the loop continues. In the reciprocal symmetric case, the expected length of a complete 'cooperation–retaliation–return to cooperation' cycle is asymptotically

$$\mathbb{E}[L_{\text{cycle}}] \sim \frac{2}{\varepsilon} \qquad (\varepsilon \to 0^+).$$

A forgiving strategy (Forgiving TFT) ignores one defection of the opponent with probability p, shortening the oscillation period to $\mathbb{E}[L_{\text{cycle}}] \sim 2/p$.

Proof. Let the state be the previous behaviour of the two sides (a, b), where $a, b \in \{C, D\}$ (cooperate or defect). Under the TFT strategy the state transition matrix is a 4×4 Markov chain. In the absence of noise, (C, C) is an absorbing state. With noise, the chain jumps from (C, C) to (C, D) or (D, C) with probability $2\varepsilon(1-\varepsilon)$; starting from (D, D), TFT causes both sides to defect alternately, and with probability $(1-\varepsilon)^2$ they return simultaneously to (C, C). The expected time from the first departure from the cooperative state to the return to the cooperative state is $\mathbb{E}[T] = O(1/\varepsilon)$ for small ε—the expected length of a single 'misjudgement–retaliation–return' cycle is $O(1/\varepsilon)$. A complete cycle (the time between two returns to the cooperative state) contains two such processes, hence $\mathbb{E}[L_{\text{cycle}}] \sim 2/\varepsilon$. $\square$

37.4.2 A probabilistic model of fitness testing

Definition 37.13 (Mathematical definition of fitness testing). **Fitness testing** means that individual A issues a costly challenge to individual B (verbal, behavioural, or a demand), and observes B's reaction in order to assess B's 'fitness' (confidence, competence, resources, social status). From the standpoint of probability theory, fitness testing is a **Bayesian updating** process—A updates, by observing B's reaction to the test, the posterior probability of B's 'type' (high value/low value).

Let B's unknown type be $\theta \in \{H, L\}$ (High/Low), with prior probability $\mathbb{P}(\theta = H) = p_0$. A issues a test T, and B's reaction to the test is $R \in \{\text{pass}, \text{fail}\}$. The likelihood function is

$$\mathbb{P}(R = \text{pass} \mid \theta = H) = \alpha_H, \quad \mathbb{P}(R = \text{pass} \mid \theta = L) = \alpha_L,$$

where $\alpha_H > \alpha_L$ (high-value individuals are more likely to pass the test). By Bayes' formula, after observing B's reaction A updates the posterior probability of B's type:

$$\mathbb{P}(\theta = H \mid R) = \frac{\mathbb{P}(R \mid \theta = H) \cdot p_0}{\mathbb{P}(R \mid \theta = H) \cdot p_0 + \mathbb{P}(R \mid \theta = L) \cdot (1 - p_0)}.$$

Proposition 37.7 (Periodicity and sequential decision-making in fitness testing). In long-term social interaction, fitness testing is not a one-off event but occurs **periodically**—as time passes, A's uncertainty about B's type gradually increases (forgetting, environmental change, B may 'disguise' itself), and periodic retesting is required. The optimal time interval ΔT_{test} for fitness testing is determined by the following factors:

(i) **Information decay rate** λ: the rate at which the posterior probability decays with time—$\mathbb{P}_t(\theta = H) = \mathbb{P}_0(\theta = H) \cdot e^{-\lambda t} + p_0 \cdot (1 - e^{-\lambda t})$.

(ii) **Test cost** C_{test}: the social cost of issuing a test (offending the other party, exposing one's own intentions).

(iii) **Misjudgement cost** C_{error}: the price of judging the other party's type incorrectly (trusting a low-value individual by mistake, or missing a high-value one).

Under the assumption of 'linear accumulation of staleness' (the risk of misjudgement grows linearly with the time since the last test), the optimal test period is

$$\Delta T^*_{\text{test}} = \sqrt{\frac{2C_{\text{test}}}{C_{\text{error}}\, p(1-p)\,\lambda}},$$

where p is the posterior probability after the last test and $p(1-p)$ is the variance measure of belief uncertainty. The test period ΔT^*_{test} measures after how long social information needs to be recalibrated.

Proof. The information decay model $\frac{dp}{dt} = -\lambda(p - p_0)$ has the solution $p(t) = p_0 + (p_{\text{post}} - p_0)e^{-\lambda t}$, where p_{post} is the posterior probability after the last test. The belief uncertainty after a test is measured by the variance $p(1-p)$. Suppose that at a time τ since the last test the instantaneous cost rate of misjudgement is $C_{\text{error}}\, p(1-p)\,\lambda\tau$ (linear accumulation of staleness). Then the total expected cost within one cycle $[0, \Delta T]$ is

$$C(\Delta T) = C_{\text{test}} + \int_0^{\Delta T} C_{\text{error}}\, p(1-p)\,\lambda\tau\, d\tau = C_{\text{test}} + \frac{1}{2} C_{\text{error}}\, p(1-p)\,\lambda\,\Delta T^2.$$

The average cost per unit time is

$$c(\Delta T) = \frac{C(\Delta T)}{\Delta T} = \frac{C_{\text{test}}}{\Delta T} + \frac{1}{2} C_{\text{error}}\, p(1-p)\,\lambda\,\Delta T.$$

The first-order condition

$$c'(\Delta T) = -\frac{C_{\text{test}}}{\Delta T^2} + \frac{1}{2} C_{\text{error}}\, p(1-p)\,\lambda = 0$$

gives

$$\Delta T^* = \sqrt{\frac{2C_{\text{test}}}{C_{\text{error}}\, p(1-p)\,\lambda}}.$$

The second-order condition $c''(\Delta T^*) = 2C_{\text{test}}/(\Delta T^*)^3 > 0$ guarantees that this is a minimum. For example, taking $C_{\text{test}} = 1$, $C_{\text{error}} = 10$, $p = 0.8$ and $\lambda = 0.1$/week gives $p(1-p) = 0.16$ and $\Delta T^* = \sqrt{2/(10 \cdot 0.16 \cdot 0.1)} = \sqrt{12.5} \approx 3.54$ weeks, which is of the same order of magnitude as the testing frequency of 'about once a month' familiar from everyday experience. □

Theorem 37.7 (Nash equilibrium of fitness testing). In a social game in which both sides carry out fitness testing, there exists a symmetric Bayesian–Nash equilibrium. Let A and B simultaneously choose 'the level of effort to pass the test' $e_A, e_B \in [0, \infty)$, with effort cost $c(e)$ $(c' > 0, c'' > 0)$. The higher the effort level e, the higher the probability of passing the other's test, but the higher the cost as well. The optimal effort level e^* under the symmetric equilibrium satisfies

$$c'(e^*) = \frac{\partial \mathbb{P}(\text{pass} \mid e^*)}{\partial e} \cdot \Delta V,$$

where ΔV is the difference in social payoff between 'being regarded as high value' and 'being regarded as low value'.

The effort levels of A and B constitute a symmetric orbit in the phase space (e_A, e_B)—the optimal strategies of both sides are symmetric, analogous to the in-phase synchronisation of coupled oscillators. The equilibrium effort level e^* increases with the frequency of testing $(\Delta T_{\text{test}}^{-1})$ and with the difference in social payoff (ΔV).

Proof. A's expected payoff is $\mathbb{E}[U_A] = \mathbb{P}_A(\text{passing B's test} \mid e_A) \cdot V_H + (1 - \mathbb{P}_A) \cdot V_L - c(e_A)$, where V_H, V_L are the social payoffs from being regarded as high/low value respectively. First-order condition: $\frac{\partial \mathbb{E}[U_A]}{\partial e_A} = \frac{\partial \mathbb{P}_A}{\partial e_A} \cdot (V_H - V_L) - c'(e_A) = 0$. The symmetric equilibrium requires $e_A = e_B = e^*$. The second-order condition is guaranteed by $c''(e^*) > 0$ (increasing marginal cost). □

Example 37.3 (Classic fitness tests in social situations). Common patterns of fitness testing in social interaction and their probabilistic structure:

(i) **Compliance Test**: A asks B to do something slightly troublesome but reasonable (such as fetching a glass of water). A high-value individual $(\theta = H)$ complies good-humouredly or declines wittily with probability $\alpha_H \approx 0.9$, whilst a low-value individual $(\theta = L)$ is over-compliant or over-resistant with probability $\alpha_L \approx 0.3$. The test period is $\Delta T_{\text{test}} \approx 1$–$2$ weeks.

(ii) **Congruence Test**: A expresses an opinion or an emotion, and observes whether B maintains its own position or accommodates A. A high-value individual maintains congruence (neither accommodating nor attacking), whilst a low-value individual either accommodates excessively or rebuts excessively. The test period is $\Delta T_{\text{test}} \approx 1$–$3$ days.

(iii) **Stress Test**: A creates a mildly awkward or tense situation, and observes B's reaction under pressure. A high-value individual remains calm and humorous, whilst a low-value individual displays anxiety or aggression. The test period is $\Delta T_{\text{test}} \approx 1$–$4$ weeks.

What these fitness tests have in common is that they estimate the other party's 'type parameter' by observing its reaction to a **phase perturbation** (the test). The strength of the test (its difficulty, its awkwardness) is a scaling factor. The information gain from the other party's reaction updates the prior probability through Bayesian updating, forming a complete 'prior–test–posterior' cycle.

37.4.3 Periodic coupling of reciprocal altruism and fitness testing

Proposition 37.8 (The composite period of exchange and testing). *In a long-term social relationship, reciprocal altruism and fitness testing form a* **coupled periodic**

system: *a failure of reciprocal altruism (the other party does not return the benefit) triggers fitness testing (is the other party still trustworthy?), and the* outcome of the fitness test determines whether reciprocal altruism continues. Let the exchange period be T_{ex} and the test period be T_{test}; the coupling dynamics of the two is

$$\frac{dT_{\text{ex}}}{dt} = -\alpha \cdot \mathbf{1}_{\text{test failed}} + \beta \cdot \mathbf{1}_{\text{test passed}}, \quad T_{\text{test}} = \gamma \cdot T_{\text{ex}} + \delta,$$

where $\alpha, \beta, \gamma, \delta > 0$ are coupling parameters. The exchange period T_{ex} (the exchange frequency) shortens when a test is passed (the relationship warms up) and lengthens when a test fails (the relationship cools down). The test period T_{test} is positively correlated with the exchange period—the more frequent the exchange, the more frequent the testing.

The coupling of reciprocal altruism and fitness testing is an instance of two substructures locking onto each other: reciprocal altruism provides the substantive content of the relationship, and fitness testing provides the assessment of its quality. The frequency ratio $T_{\text{ex}}/T_{\text{test}}$ of the two measures the degree of trust in the relationship. In a high-trust relationship $T_{\text{test}} \gg T_{\text{ex}}$ (testing occasionally, exchanging frequently), whilst in a low-trust relationship $T_{\text{test}} \approx T_{\text{ex}}$ (testing is required before and after every exchange).

Proof. Let the probability that a test is passed be p (failure probability $1-p$). Taking expectations of the first coupled equation gives

$$\mathbb{E}\left[\frac{dT_{\text{ex}}}{dt}\right] = -\alpha\,\mathbb{E}[\mathbf{1}_{\text{test failed}}] + \beta\,\mathbb{E}[\mathbf{1}_{\text{test passed}}] = -\alpha(1-p) + \beta p = (\alpha+\beta)p - \alpha.$$

Setting this to zero gives the equilibrium pass rate

$$p^* = \frac{\alpha}{\alpha+\beta}.$$

This is a negative-feedback stable fixed point: when $p > p^*$ we have $\mathbb{E}[dT_{\text{ex}}/dt] > 0$, so the exchange period is stretched (the relationship cools), the next test is more likely to fail, and p falls back; when $p < p^*$ the exchange period shortens (the relationship warms up) and p rises again. Hence the system is in dynamic balance around p^*. Substituting the equilibrium exchange period $\overline{T}_{\text{ex}}$ into $T_{\text{test}} = \gamma T_{\text{ex}} + \delta$ gives the equilibrium test period $\overline{T}_{\text{test}} = \gamma \overline{T}_{\text{ex}} + \delta$, and the frequency ratio tends to the constant $\overline{T}_{\text{ex}}/\overline{T}_{\text{test}} = 1/(\gamma + \delta/\overline{T}_{\text{ex}})$ in the equilibrium state. For example, taking $\alpha = 1, \beta = 3$ (the relationship warms up faster after a test is passed) gives $p^* = 1/4$—in the long run, when about one test in four ends in failure, the exchange–test system reaches equilibrium. □

37.5 Information diffusion

Definition 37.14 (SIR information diffusion model). *The classic model of information diffusion in social networks is the SIR model (Susceptible–Infected–Recovered):*

$$\frac{dS}{dt} = -\beta SI, \quad \frac{dI}{dt} = \beta SI - \gamma I, \quad \frac{dR}{dt} = \gamma I,$$

where $S(t)$ is the proportion of susceptibles who have not yet come into contact with the information, $I(t)$ is the proportion of infecteds who have come into contact with the information and are spreading it, and $R(t)$ is the proportion of recovereds who have lost interest in the information. β is the diffusion rate (the number of susceptibles that one infected can infect per unit time), and γ is the recovery rate (the rate at which infecteds lose interest in spreading).

The diffusion of information from infecteds to susceptibles is forward scaling (I increases, S decreases), and the conversion of infecteds into recovereds is reverse scaling (I decreases, R increases). The basic reproduction number $R_0 = \beta/\gamma$ determines the scale of the diffusion—when $R_0 > 1$ the information spreads explosively, and when $R_0 < 1$ the information gradually dies out. When $R_0 > 1$ the infection curve $I(t)$ exhibits a single-peaked shape that first rises and then falls—the heat of the information first increases and then decreases with time.

Proposition 37.9 (Damped oscillation in the SIRS model)**.** *When a replenishment of susceptibles is added to the SIR model (new users join, and those who have forgotten become susceptible again), the model becomes SIRS:*

$$\frac{dS}{dt} = -\beta SI + \alpha R, \quad \frac{dI}{dt} = \beta SI - \gamma I, \quad \frac{dR}{dt} = \gamma I - \alpha R,$$

where α is the rate at which recovereds become susceptible again. The SIRS model has an endemic equilibrium point $S^* = \gamma/\beta$; when the discriminant is negative, the eigenvalues of the Jacobian matrix are complex conjugates—the system exhibits damped oscillation, fluctuating with period

$$T = \frac{2\pi}{\omega}, \qquad \omega = \sqrt{\beta I^*(\alpha+\gamma) - \frac{(\beta I^* + \alpha)^2}{4}}$$

as the infected proportion tends to the equilibrium point, where $I^* = \frac{\alpha}{\alpha+\gamma}\left(1 - \frac{\gamma}{\beta}\right)$ is the equilibrium infected proportion.

The periodic solution of the SIRS model manifests itself as the 'outbreak–fade–re-outbreak' pattern of information. When α is large (recovereds quickly forget and pay attention again), the oscillation is pronounced—the information repeatedly becomes a hot topic. The phenomenon of 'old news resurfacing' on social media is empirical evidence for the periodic solution of SIRS.

Proof. Using $S + I + R = 1$, reduce the system to two dimensions (the variables S, I). The equilibrium point is found by combining $dI/dt = 0$ with $dS/dt = 0$: from $I(\beta S - \gamma) = 0$ and taking the non-zero solution we get $S^* = \gamma/\beta$; from $\gamma I = \alpha R$ we get $R^* = \gamma I^*/\alpha$, and substituting into $S^* + I^* + R^* = 1$ yields

$$I^* = \frac{\alpha}{\alpha+\gamma}\left(1 - \frac{\gamma}{\beta}\right).$$

When $R_0 = \beta/\gamma > 1$ we have $I^* > 0$. At the equilibrium point the Jacobian matrix is

$$J = \begin{pmatrix} -\beta I^* - \alpha & -\gamma - \alpha \\ \beta I^* & 0 \end{pmatrix},$$

and the characteristic equation is $\lambda^2 + (\beta I^* + \alpha)\lambda + \beta I^*(\gamma + \alpha) = 0$. When the discriminant $\Delta = (\beta I^* + \alpha)^2 - 4\beta I^*(\gamma + \alpha) < 0$, the eigenvalues are $\lambda = -\frac{\beta I^* + \alpha}{2} \pm i\omega$, where

$$\omega = \sqrt{\beta I^*(\gamma + \alpha) - \frac{(\beta I^* + \alpha)^2}{4}}.$$

The imaginary part ω gives the angular frequency of the damped oscillation, hence the period $T = 2\pi/\omega$. The damping rate $(\beta I^* + \alpha)/2 > 0$ guarantees that the oscillation decays with time and the system ultimately tends to the equilibrium point. When α is large (recovereds quickly become susceptible again), I^* and ω increase and the oscillation is pronounced. ☐

37.6 Evolution of social networks

Definition 37.15 (Periodic evolution of social relations)**.** Social relations are not static, but **evolve periodically** with time. The change in tie strength $W_{ij}(t)$ can be modelled as

$$\frac{dW_{ij}}{dt} = \eta_{ij}(t) - \lambda W_{ij} + \sigma\xi_{ij}(t),$$

where $\eta_{ij}(t)$ is the interaction frequency (which varies with time), λ is the natural decay rate of the tie, and $\xi_{ij}(t)$ is a random perturbation. The interaction frequency $\eta_{ij}(t)$ is itself periodic—weekday interaction, weekend interaction and festival interaction each have a different frequency.

The alternation of closeness and distance in a relationship resembles a rotation in phase space. The periodic change in tie strength corresponds, in periodic convolution, to the periodic modulation of the difference-angle similarity $\mathsf{Similarity}^{(1)}(\theta_i, \theta_j)$—when the social activities of two people are in phase the relationship is at its closest ($\cos(0) = 1$), and when they are in antiphase it is at its most distant ($\cos(\pi) = -1$).

Proposition 37.10 (Community-structure period of social networks)**.** The community structure of a social network (circles of friends, interest groups, work teams) evolves with time—the formation, stabilisation and disintegration of a community have an intrinsic period. Let the community activity $u(t)$ (the interaction frequency of members) decay exponentially during the decline stage as

$$u(t) = u_{\mathsf{peak}}\, e^{-\lambda(t - t_{\mathsf{peak}})}, \qquad t \geq t_{\mathsf{peak}},$$

where t_{peak} is the time of peak activity and $\lambda > 0$ is the decay rate. Then the 'half-life' of the community (the time for half the members to remain active) is

$$T_{1/2} = \frac{\ln 2}{\lambda}.$$

The community life cycle is a complete cycle of a formation stage (ties are established, interaction increases), a peak stage (activity is highest) and a decline stage (ties disintegrate, interaction decreases). The half-life $T_{1/2}$ measures the persistence of community cohesion. Analysis of group data from online social platforms shows that the activity of most groups peaks 3–6 months after creation and thereafter decays approximately exponentially, with a complete life cycle of about 6 months to 5 years (this is an **empirical observation**, and the specific duration depends on the type of community).

Proof. From $u(t) = u_{\text{peak}} e^{-\lambda(t-t_{\text{peak}})}$, set $u(t_{1/2}) = u_{\text{peak}}/2$; then $e^{-\lambda(t_{1/2}-t_{\text{peak}})} = 1/2$, and taking logarithms of both sides gives $\lambda(t_{1/2} - t_{\text{peak}}) = \ln 2$, hence

$$T_{1/2} = t_{1/2} - t_{\text{peak}} = \frac{\ln 2}{\lambda}.$$

For example, taking a measured half-life $T_{1/2} = 3$ months gives $\lambda = \ln 2/(3 \text{ months}) \approx 0.231/\text{month}$, corresponding to a monthly activity decay of about $1 - e^{-0.231} \approx 20.6\%$. □

Definition 37.16 (Social-circle closure). **Social-circle closure** (triadic closure) means: if individual i has ties with both j and k, then the probability that a tie forms between j and k is significantly higher than the random level. Let the degree of node i be d_i and let the number of closed triangles with i as a vertex be T_i; then the local clustering coefficient of node i is

$$C_i = \frac{\text{number of edges actually present among the neighbours of node } i}{\binom{d_i}{2}} = \frac{2T_i}{d_i(d_i - 1)},$$

and the whole-network clustering coefficient is $\overline{C} = \frac{1}{|V|}\sum_{i \in V} C_i$. $\overline{C} = 0$ corresponds to a tree-like network with no closure, and $\overline{C} = 1$ to a completely closed clique.

Proposition 37.11 (Relational loops of social-circle closure). Under the reciprocal symmetry assumption (tie strength $W_{ij} = W_{ji}$, corresponding to the symmetric equilibrium of Theorem 37.7), social-circle closure makes the relational network form a large number of closed loops—triangles i-j-k-i. For a random graph (the Erdős–Rényi graph $G(n, p)$), any two neighbours are joined by an edge independently, hence $\mathbb{E}[\overline{C}] = p$; whereas the measured $\overline{C} \gg p$ of real social networks shows that relational edges tend to be embedded in closed triangles—social-circle closure is systematic, not a random coincidence.

Proof. For $G(n, p)$, the two neighbours corresponding to any two adjacent edges of node i are joined by an edge independently with probability p, hence $\mathbb{E}[C_i] = p$, and therefore $\mathbb{E}[\overline{C}] = p$. By definition, each closed triangle i-j-k contributes 1 to each of T_i, T_j, T_k at its three vertices, so the total number of triangles is $= \frac{1}{3}\sum_i T_i$, and the number of closed loops in the whole network is uniquely determined by $\{C_i\}$. Real social networks satisfy $\overline{C} \gg p$: for example, a certain social platform has $n \sim 10^4$ and $p \sim 10^{-3}$ (in which case $\overline{C}_{ER} \approx 0.001$), whereas the measured value is $\overline{C} \approx 0.12$, that is, closed triangles are about 120 times more abundant than in a random graph—this quantifies the triadic closure effect that 'a friend of a friend often becomes a friend too'. □

37.7 Cultural cycles and population evolution

37.7.1 Cultural cycles of fashion and style

Definition 37.17 (Cultural cycles of fashion and style). The rise and fall of cultural trends (fashion, musical styles, artistic movements, architectural styles) displays pronounced periodicity. The time scale of a cultural cycle differs from field to field:

(i) **Fast-fashion cycle**: the popularity cycle of clothing styles is about 6–12 months (seasonal turnover).

(ii) **Musical style cycle**: the rise–peak–decline cycle of popular musical styles is about 5–10 years.

(iii) **Artistic movement cycle**: the turnover cycle of modern artistic movements is about 20–30 years (one generation).

(iv) **Retro cycle**: the renewed popularity of old styles—a retro cycle of about 20–30 years ('what was popular twenty years ago becomes popular again').

The rise and decline of cultural trends manifests itself as a scaling transformation from the margins to the mainstream and then to obsolescence, whilst the retro cycle manifests itself as a periodic rotation of cultural preferences about the 'new–old' axis. The length of the cycle is determined jointly by the speed of generational turnover and the rate of technological change.

Remark 37.10 (Cultural cycles of generational turnover (empirical observation)). Cultural cycles are closely related to generational turnover. A generation (about 20–30 years) is the time for those born in the same years to pass from childhood to adulthood and from recipients of culture to creators of culture. Each generation tends to rebel against the cultural style of the previous generation and to establish its own cultural markers—this cycle of 'rebellion–establishment–being rebelled against' drives the generational cycle of cultural trends. The above mechanism of generational replacement and 'rebellion' is an **empirical observation**, not a provable theorem.

The generational cultural cycle can be formally analogised to a phase rotation of cultural preferences: the cultural preferences of each generation stand in contrast to those of the previous generation (rebellion), and after two generations the cultural preferences resemble those of the grandparents (retro). In this formal analogy the 'phase' of cultural preference is a heuristic concept, and not a rigorously defined mathematical quantity.

37.7.2 Population evolution and intergenerational cycles

Definition 37.18 (Basic models of population evolution). **Population evolution** describes the evolution of population size $N(t)$ and its age structure $n(a,t)$ with time. The basic equations include:

(i) **Malthusian growth model**: $\frac{dN}{dt} = rN$ ($r > 0$ is the intrinsic growth rate). The solution is exponential growth $N(t) = N_0 e^{rt}$—growth is not limited by resources.

(ii) **Logistic growth model (Verhulst)**: $\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right)$ (K is the environmental carrying capacity). The solution is an S-shaped curve, tending ultimately to K.

(iii) **Age-structure equation (McKendrick–von Foerster)**:

$$\frac{\partial n(a,t)}{\partial t} + \frac{\partial n(a,t)}{\partial a} = -\mu(a)n(a,t),$$

where $\mu(a)$ is the age-dependent death rate. The boundary condition $n(0,t) = \int_0^\infty b(a)\,n(a,t)\,da$ gives the number of newborns ($b(a)$ is the age-dependent birth rate).

The periodic structure of population evolution originates in generational turnover—each generation passes through birth, growth, reproduction, ageing and death on a time scale of 20–30 years. The coupling of this generational cycle with resource constraints gives rise to periodic fluctuations in population dynamics.

Proposition 37.12 (Generational cycles in population dynamics)**.** *Consider a discrete-generation population model: let the population of generation n be N_n, let its fertility rate after reaching adulthood be $f(N_{n-1})$ (depending on the environmental influence of the previous generation's population), and let the survival rate be s. Then the population dynamics is*

$$N_{n+1} = s \cdot f(N_{n-1}) \cdot N_n.$$

When f decreases as the previous generation's population increases (resource competition), periodic oscillations can appear in the population dynamics. Its period T_{gen} is about two generation times ($T_{\text{gen}} \approx 2 \times 25$ years $= 50$ years).

Proof. Taking $f(N) = r(1 - N/K)$ (linear competition) gives

$$N_{n+1} = srN_n \left(1 - \frac{N_{n-1}}{K}\right).$$

Let the equilibrium state be N^*; then $N^* = srN^*(1 - N^*/K)$, that is, $1 = sr(1 - N^*/K)$, which gives $N^* = K(1 - 1/(sr))$. The existence of the equilibrium state requires $sr > 1$.

Linearising near the equilibrium state, let $N_n = N^*(1 + x_n)$; this gives

$$x_{n+1} = x_n - sr\frac{N^*}{K}x_{n-1}.$$

The characteristic equation is $\lambda^2 - \lambda + srN^*/K = 0$. The discriminant is $\Delta = 1 - 4srN^*/K$. When $4srN^*/K > 1$, the eigenvalues are complex conjugates and x_n exhibits periodic oscillation. The oscillation period is about 2 generation times (each generation being about 25 years), hence $T_{\text{gen}} \approx 50$ years. ☐

Definition 37.19 (Demographic transition)**.** The **demographic transition** (Demographic Transition) is the systematic transition of a population from high birth rate and high death rate to low birth rate and low death rate. The demographic transition comprises four typical stages:

(i) **Pre-industrial stage**: high birth rate, high death rate, and an approximately constant population size.

(ii) **Transitional stage**: the death rate falls whilst the birth rate remains high, and the population expands rapidly.

(iii) **Industrial stage**: the birth rate begins to fall and population growth slows.

(iv) **Post-industrial stage**: low birth rate, low death rate, and a population size that stabilises or declines.

The time scale of the demographic transition is about 100–200 years—in the history of population evolution it appears as a large-scale structural leap.

Proposition 37.13 (S-shaped model of the demographic transition)**.** The process of the demographic transition can be approximated by a logistic-type equation:

$$\frac{dN}{dt} = r(t)N\left(1 - \frac{N}{K(t)}\right),$$

where the growth rate $r(t)$ and the carrying capacity $K(t)$ vary with time. In the early stage of the transition $r(t) > 0$ and $K(t)$ rises (the death rate falls), so the population grows rapidly; in the late stage of the transition $r(t)$ falls or even becomes negative (the birth rate falls), and population growth slows or even turns into shrinkage.

Proof. In the early stage of the transition, medical progress makes the death rate fall rapidly whilst the birth rate remains essentially unchanged, so $r(t)$ rises; at this time $N \ll K(t)$, and the equation is approximately $\frac{dN}{dt} \approx r(t)N$, so the population grows exponentially. In the late stage of the transition, the fall in the birth rate makes $r(t) \to 0$ or even $r(t) < 0$, and population growth slows until it stabilises. Substituting empirical time series for $r(t)$ and $K(t)$ allows the population dynamics of each stage to be fitted. ☐

Remark 37.11 (Multi-level periodic structure of population evolution (empirical observation / research programme)). Population evolution contains a nested structure of multi-level cycles. **The following levels are empirical observations and a research programme, not proved theorems**:

(i) **Generational cycle** (20–30 years): the individual life cycle of birth, growth, reproduction and death.

(ii) **Intergenerational oscillation** (40–60 years): the two-generation periodic oscillation produced by generational resource competition and falling fertility (Proposition 37.12).

(iii) **Kondratiev-type demographic long wave** (50–60 years): the long wave produced by the coupling of the population age structure with the economic cycle.

(iv) **Demographic transition cycle** (100–200 years): the complete leap from high birth and high death rates to low birth and low death rates.

(v) **Civilisation cycle** (several hundred to several thousand years): the co-evolutionary cycle of population, technology and culture.

The cycles of the various levels exhibit a nested structure: the generational cycle is nested within the intergenerational oscillation, the intergenerational oscillation within the demographic long wave, the demographic long wave within the demographic transition, and the demographic transition within the civilisation cycle.

Empirical basis: The existence of the cycles at the various levels is derived from the corresponding dynamical mechanisms:

- The generational cycle is given directly by the individual life cycle and is the basic unit of population evolution.
- The intergenerational oscillation is derived from the discrete-generation model of Proposition 37.12, with a period of about two generation times.
- The demographic long wave is produced by the coupling of the intergenerational oscillation with the economic cycle, and its time scale is determined by the product of the generational cycle and the capital-renewal cycle.
- The demographic transition is given by the S-shaped model of Proposition 37.13, and its period is determined by the time lag between the fall in the death rate and the fall in the birth rate.
- The civilisation cycle is produced by the multi-level coupling of population, technology and culture, and its time scale is about several times the demographic transition cycle.

The nested structure is guaranteed by the ratios of the time scales at the various levels being about 2–5, that is, a cycle at a higher level contains several cycles at lower levels.

Example 37.4 (Intergenerational cycles and the 'echo boom'). *The 'echo boom' (Echo Boom) is a concrete instance of the intergenerational cycle in demography. Taking the United States as an example:*

(i) *In 1946–1964 the 'baby boom' generation (Baby Boomers) numbered about 76 million births.*

(ii) *In 1965–1979 the birth rate of 'Generation X' fell, and the population declined relatively.*

(iii) *In 1980–1994 the 'echo boom' generation—after the baby boomers reached childbearing age, their number of children increased once more, forming a second demographic peak.*

(iv) *In 1995–2010 the population size of the 'millennials' and Generation Z fell back once again.*

The time interval between the baby boom and the echo boom is about 25–30 years, which corresponds exactly to one generational time. The formation of this cycle is driven jointly by intergenerational resource competition (the baby boomers occupied a large share of resources, depressing the childbearing intentions of the next generation) and by the age-structure effect (the baby boomers entered childbearing age en masse).

Remark 37.12 (Coupling of population cycles and cultural cycles). There is a coupling relation between population evolution cycles and cultural cycles: generational turnover drives both the renewal of cultural trends (each generation rebels against the culture of the previous one) and the renewal of the population age structure. The common time scale of the two (about 20–30 years) makes the cultural retro cycle and the population generational cycle statistically correlated. This coupling is an instance of the interaction of multi-level cycles in the 'Periodic Systems' chapter.

37.8 Phase locking in collective behaviour

Definition 37.20 (Collective synchronisation behaviour)**.** Several kinds of **spontaneous synchronisation behaviour** *exist in human groups—the behavioural rhythms of individuals tend to become identical in group interaction:*

(i) **Applause synchronisation**: *applause at the end of a concert or a speech moves from chaos to synchrony—the clapping rhythm moves from random phases to a uniform periodic beat within a few seconds.*

(ii) **Gait synchronisation**: *people walking side by side unwittingly converge in step frequency—the leg-swing periods lock in phase after coupling.*

(iii) **Financial-market panic**: *investors sell en masse when the market crashes—panic spreads within an extremely short time, and the behaviour of all market participants tends to synchrony.*

Collective synchronisation behaviour is the realisation at the level of social groups of phase locking in the 'Periodic Systems' chapter (Proposition 24.8): each individual is an independent periodic subsystem (a natural clapping frequency, a natural step frequency, a natural risk preference), and social interaction is the coupling coefficient between the substructures. When the coupling is strong enough, the phases of all individuals tend to equality—collective synchronisation.

Lemma 37.7 (Kuramoto synchronisation threshold)**.** *Let the natural-frequency distribution $g(\omega)$ be continuous and unimodal, and symmetric about its mean Ω. In the continuous limit $N \to \infty$, the Kuramoto model has a critical coupling strength*

$$K_c = \frac{2}{\pi g(\Omega)},$$

such that: when $K \le K_c$ the only stable solution is the asynchronous state ($r = 0$); when $K > K_c$ a synchronised solution branch with $r > 0$ appears, and the order parameter near the threshold satisfies

$$r \approx \sqrt{\frac{16(K - K_c)}{\pi K_c^4\,(-g''(\Omega))}} \qquad (K \to K_c^+),$$

where $g''(\Omega) < 0$ (the unimodal distribution attains its maximum at Ω).

Proof. In the continuous limit, the phase-locked individuals are distributed on the arc $|\theta| \le \pi/2$, and their natural frequencies satisfy $\omega = \Omega - Kr\sin\theta$, so the synchronised state satisfies the mean-field self-consistency equation

$$r = Kr \int_{-\pi/2}^{\pi/2} \cos^2\theta\, g(\Omega + Kr\sin\theta)\, d\theta.$$

Dividing both sides by r for the branch $r > 0$, and expanding g in a Taylor series at Ω (by symmetry $g'(\Omega) = 0$):

$$1 = K \int_{-\pi/2}^{\pi/2} \cos^2\theta \left[g(\Omega) + \frac{1}{2} g''(\Omega) K^2 r^2 \sin^2\theta \right] d\theta + O(r^4).$$

Using $\int_{-\pi/2}^{\pi/2} \cos^2\theta\, d\theta = \frac{\pi}{2}$ and $\int_{-\pi/2}^{\pi/2} \cos^2\theta \sin^2\theta\, d\theta = \frac{\pi}{8}$, we obtain

$$1 = \frac{\pi}{2} K g(\Omega) + \frac{\pi}{16} K^3 g''(\Omega) r^2 + O(r^4).$$

Letting $r \to 0^+$ gives the threshold condition $1 = \frac{\pi}{2} K_c g(\Omega)$, that is, $K_c = \frac{2}{\pi g(\Omega)}$. For $K > K_c$, rearranging the above expression gives

$$r^2 = \frac{16}{\pi K^3(-g''(\Omega))} \left(\frac{\pi}{2} K g(\Omega) - 1 \right) = \frac{16}{\pi K^3(-g''(\Omega))} \cdot \frac{K - K_c}{K_c},$$

where $\frac{\pi}{2} g(\Omega) = 1/K_c$ has been used. Taking $K \to K_c^+$ (in which case $K^3 \to K_c^3$) gives

$$r \approx \sqrt{\frac{16\,(K - K_c)}{\pi K_c^4\,(-g''(\Omega))}}.$$

A linear stability analysis shows that $r = 0$ is stable for $K < K_c$ and unstable for $K > K_c$, so a supercritical bifurcation occurs at K_c. □

Proposition 37.14 (The Kuramoto model and collective synchronisation)**.** *The classic mathematical model of collective synchronisation is the Kuramoto model. Let N individuals each have a natural frequency ω_i and influence one another through phase coupling:*

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N} \sum_{j=1}^{N} \sin(\theta_j - \theta_i), \quad i = 1, \ldots, N,$$

where $K \geq 0$ is the coupling strength. Define the order parameter $r(t) = \frac{1}{N} \left| \sum_{j=1}^{N} e^{i\theta_j(t)} \right|$ (measuring the degree of collective synchronisation: $r = 0$ is complete asynchrony, $r = 1$ is complete synchrony). When K exceeds the critical value K_c (whose explicit expression and the scaling law near the threshold are given by Lemma 37.7), the collective undergoes a **phase transition***—the order parameter r grows continuously from zero to a positive value, and the collective passes from the asynchronous state into the synchronised state.*

The phase transition of the Kuramoto model manifests itself as the critical phenomenon of phase locking at the collective level: the order parameter r is the collective average of the phase similarity Similarity$^{(1)}$*. When the coupling strength exceeds the critical value, the collective enters the synchronised state—the difference-angle sine component* Im $\mathcal{K}^{(1)}\{f_i, f_j\}$ *of the periodic convolution drives the phases of all individuals towards a common mean phase.*

Proof. The non-zero solution branch of the order parameter $r(t)$ and the critical value are given by Lemma 37.7: for $K \leq K_c$ the asynchronous state $r = 0$ is linearly stable; for $K > K_c$ the state $r = 0$ becomes unstable and a synchronised solution with $r > 0$ appears, with $r \sim \sqrt{K - K_c}$ (square-root scaling). Hence the order parameter grows **continuously** from 0 at K_c—this is a second-order (continuous) phase transition. The order parameter $r = \frac{1}{N} \left| \sum_j e^{i\theta_j} \right|$ is precisely the modulus measuring the degree of phase agreement in the collective, so r rising from 0 to a positive value means that the collective passes from disordered phases to phase locking. □

Example 37.5 (Emotional contagion in social networks). The 2014 Facebook emotional contagion experiment (Kramer et al., PNAS, 2014) demonstrated that emotions can spread through textual content on social media. In the experiment the news feeds of about 690 thousand users were covertly adjusted—some users saw less positive content, and some saw less negative content. The results showed that users who saw less positive content themselves posted fewer positive words and more negative words—emotion is contagious through purely textual interaction.

Emotional contagion is empirical evidence of phase locking in social networks: a user's emotional state θ_i (in the valence–arousal space) is coupled to others through the edges of the social network. Although the effect size in the Facebook experiment was small ($d \approx 0.001$), it was highly significant statistically—when N is extremely large, even a tiny coupling can produce a detectable collective shift.

37.9 Conservation laws of social systems

Theorem 37.8 (Three conservation laws of social systems). *The dynamics of a social system is governed by three conservation laws:*

(i) **Population conservation**: *in a closed social system the total population $N = S + I + R$ remains constant. In an open system, the change in population = new users joining − old users leaving. On a long-term average, the total number of active users of a social platform tends to a dynamic balance—inflow equals outflow. The change in the population age structure is described by the McKendrick–von Foerster equation (Definition 37.18).*

(ii) **Information conservation**: *in the process of information diffusion, the 'total amount' of information (the number of people who know it) is transferred from susceptibles to infecteds, and ultimately to recovereds—the total number of people who know the information, $I + R$, is monotonically non-decreasing. The final scale of information diffusion (R_∞) is determined by the basic reproduction number R_0: $R_\infty = 1 - e^{-R_0 R_\infty}$.*

(iii) **Trust conservation**: *in a social relational network, the total amount of trust (the sum of tie strengths) remains stable in the absence of external events. The closeness and distance of ties between individuals is merely a redistribution of trust among different relationships—a person's distancing from some is often accompanied by closeness to others. The conservation of the total amount of trust in a social network resembles the conservation of angular momentum in a rotating system—the 'total angular momentum' of trust remains constant within the network topology.*

The hierarchical relation of the three conservation laws is: population conservation is the foundation of the existence of a social system, information conservation is the motive power of social interaction, and trust conservation is the structural constraint on social relations. Together the three constitute the guarantee of the self-consistency of the periodic structure of social systems.

Proof. (i) Population conservation: in a closed system $d(S + I + R)/dt = 0$. In an open system, if the rate b at which new users join equals the rate d at which old users leave ($b = d$), then the total population is stable. The user-growth curve of a social platform is usually S-shaped—exponential growth initially, linear growth in the middle period, and saturation later (the logistic equation, Definition 37.18).

(ii) Information conservation: in the SIR model, $\frac{d}{dt}(I + R) = \beta SI \geq 0$—the number of people who know the information increases monotonically. The final proportion of recovereds, R_∞, satisfies the transcendental equation $R_\infty = 1 - e^{-R_0 R_\infty}$ (assuming $S_0 \approx 1$). The larger R_0 is, the wider the information spreads—when $R_0 = 2$, $R_\infty \approx 0.80$ (80% of people ultimately know the information).

(iii) Trust conservation: the total tie strength $\sum_{i<j} W_{ij}$ in a social network tends to be stable in the absence of major events. A change in tie strength is more likely to be a redistribution than a change in the total—analogous to the conservation of phase-space volume (Liouville's theorem). This assumption is supported by the sociological theory of 'Dunbar's number': the number of stable social relationships a human can maintain is about 150—the total social investment is distributed among several relationships. □

Remark 37.13 (The metaphorical nature of trust conservation). Among the three conservation laws of Theorem 37.8, population conservation and information conservation can be derived rigorously from the population dynamics equations and the SIR/SIRS differential equations (see (i)(ii) of the proof of that theorem). **Trust conservation is different**: at present there is no rigorous mathematical definition of a 'total amount of trust', and 'the total amount of trust does not change with time' is a formal analogy and an empirical hypothesis, not a theorem. What Dunbar's number (about 150) measures is the cognitive capacity of the human brain to maintain stable relationships; it implies that **social attention** (social energy) is redistributed among the relationships that are maintained, rather than being a precisely measurable conserved physical quantity. Hence 'trust conservation/social-attention conservation' as discussed in this chapter should be understood as a **metaphor**—it characterises the redistributive structure of social investment, not a rigorous conservation law.

Remark 37.14 (Scaling of social-circle size: Dunbar stratification (empirical observation)). The size of human social circles exhibits an approximately stratified scaling structure. The theory of Dunbar's number (**empirical observation**, not a theorem) gives several characteristic scales: a core support circle of about 5 people, an intimate circle of about 15 people, a circle of good friends of about 50 people, a circle of stable relationships of about 150 people, and an acquaintance circle of about 500 people. The ratio of the sizes of adjacent layers is about 3 ($5 \to 15 \to 50 \to 150 \to 500$), an approximately geometric scaling. This scaling is determined by the neocortical capacity of the human brain, and is the manifestation in size of the cognitive constraint on social attention—it is the point of contact of the scaling theme in social systems; its specific values belong to empirical observation and do not constitute a provable theorem.

Remark 37.15 (Connection with the rest of the book). The theory of the periodic structure of social systems established in this chapter is an application of periodic mathematics in social science. Social rhythms manifest themselves as the repeated organisation of social activity by daily, weekly and yearly periods. The SIR/SIRS information diffusion model describes the explosive diffusion and the periodic re-heating of information. Social-network evolution describes the alternation of closeness and distance in relationships. Cultural cycles and population evolution display periodic rise and fall driven by generational turnover—the demographic transition is a structural leap in population evolution, and the intergenerational cycle and the 'echo boom' are the direct manifestation of generational turnover in demography. The phase transition of the Kuramoto model is the collective realisation of phase locking in the 'Periodic Systems' chapter—when the coupling strength exceeds the critical value, the collective passes from asynchrony into synchrony. The three conservation laws of social systems—population conservation, information conservation and trust conservation (the latter being a metaphor)—together constitute the sociological foundation of the periodic structure of social systems.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The passage of interpersonal relations through time: the daily/weekly/yearly repetition of social rhythms (Definition 37.11), the reciprocal exchange period T_{ex} (Definition 37.12, Proposition 37.6)
Scaling	Scaling of social-network size: Dunbar stratification $5/15/50/150/500$ (Remark 37.14), community half-life $T_{1/2} = \ln 2/\lambda$ (Proposition 37.10), reproduction number $R_0 = \beta/\gamma$ (Definition 37.14)
Rotation	The phase cycle of relational closeness: the alternation of closeness and distance (Definition 37.15), the phase rotation of cultural retro (Definition 37.17), Kuramoto phase locking (Lemma 37.7, Proposition 37.14)
Conservation	Conservation of social energy/attention (**metaphor**): the three conservation laws of population/information/trust (Theorem 37.8), of which trust/attention conservation is explicitly declared to be a metaphor (Remark 37.13)
Symmetry	Reciprocal symmetry: the symmetric tit-for-tat strategy (Proposition 37.6), the symmetric Bayesian–Nash equilibrium of fitness testing $e_A = e_B = e^*$ (Theorem 37.7), symmetry of tie strength $W_{ij} = W_{ji}$ (Proposition 37.11)
Closure	Social-circle closure and relational loops: triadic closure and the clustering coefficient $\overline{C}$ (Definition 37.16, Proposition 37.11), the exchange–test feedback loop (Proposition 37.8)
Period	Relational cycles and Kuramoto collective synchronisation: the retaliation-loop period $2/\varepsilon$ (Proposition 37.6), the test period ΔT^*_{test} (Proposition 37.7), the SIRS oscillation period (Proposition 37.9), the collective-synchronisation phase transition (Proposition 37.14), the multi-level periodic structure of population evolution (empirical observation / research programme, Remark 37.11)

Exercises

Exercise 1 Prove the **social periodicity decision theorem** of this chapter (37.1). Hint: first write down the decision map and the evolution group of the goal in this chapter, then apply the proof structure of the general criterion of the 'Periodic Systems' chapter.

Exercise 2 Verify that the hypotheses of the **monotonicity of quantity adjustment** (37.2) hold for the domain objects of this chapter, and give a counterexample in which the hypotheses fail.

Exercise 3 Examine the behaviour of the **three-parameter correspondence: radius, phase and rotation speed** (37.1) under parameter degeneration, and explain how the conclusion degenerates accordingly.

Exercise 4 Explain the relation between the **social period diagnosis workflow and the interfaces to three chapters** (37.6) and the corresponding general conclusion of the 'Periodic Systems' chapter (whether it is consistent with it, a special case of it, or requires additional conditions).

Exercise 5 According to the seven-theme comparison table of this chapter, point out respectively the concrete carriers of this chapter for the three themes **translation**, **scaling** and **rotation**, and give one domain data example for each.

Exercise 6 Comprehensive question: restate the diagnosis workflow of this chapter using the five-step standard interface of the 'Periodic Systems' chapter (Table 24.1), and note one by one the theorem of this chapter on which each step depends: 37.1, 37.2, 37.1, 37.6.

Chapter 38

Periodic Structure of the Climate System

Remark 38.1 (Periodicity decision for the target of this chapter (citation)). When this chapter decides whether a research target is periodic, determines its period type, or seeks its minimal positive period, reference is always made to the 'Periodic Systems' chapter: for the periodicity criterion see Theorem 24.1, for the discrimination of period types see Definition 24.3, Proposition 24.1, for the search for the minimal positive period see Definition 24.4, Theorem 24.2, and for the construction of period series and epicycle groups from data see Theorem 24.3, Definition 24.6.

Remark 38.2 (Positioning of this chapter: developed along the five-step interface of the 'Periodic Systems' chapter). The periodic structure of the climate system studies the mathematical characterisation of periodicity, quasi-periodicity and long-range correlation in the Earth's climate system. This chapter does not establish a separate periodicity criterion; instead it connects the climate system to the five-step interface prescribed by the 'Periodic Systems' chapter:

(1) **Decision**: reduce such targets as the seasonal cycle, ENSO, the NAO, the PDO, the IOD and the Milankovitch cycles to discrete or continuous dynamical systems, invoke Theorem 24.1 to decide whether they are periodic, and determine the period type;

(2) **Construction**: build the amplitude series, the frequency series and the phase series from temperature, precipitation, pressure, sea surface temperature, radiative forcing and wind stress, and assemble them into the climate epicycle group;

(3) **Adjustment**: dynamically adjust the climate epicycle group subject to residual-energy descent, energy conservation, momentum conservation and water-vapour conservation;

(4) **Convolution**: introduce amplitude convolution, phase convolution and frequency convolution on the climate epicycle group, and connect them with autocorrelation, cross-correlation, teleconnection and inter-basin coupling;

(5) **Integration**: connect to the tool chains of the three chapters 'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis', forming a climate periodicity diagnostic procedure.

The mathematical tools used in this chapter include: the geometry of solar radiation, delayed negative-feedback oscillation, the recharge–discharge oscillator model, slow–fast variable separation, the red-noise spectrum, the Hurst exponent, $1/f$ noise, multiscale superposition, periodic convolution, epicycle groups, phase operators and time delay.

Remark 38.3 (Guide to the seven themes of this chapter). **Translation**—the interannual advance of the seasonal cycle and the time translation of atmospheric variability; **Scaling**—the scale invariance and long-range persistence of $1/f$ noise (the Hurst exponent H); **Rotation**—the phase oscillation of the warm and cold phases of ENSO and the Milankovitch orbital cycles; **Conservation**—the global energy balance, conservation of Earth–atmosphere angular momentum and conservation of water vapour (the hydrological cycle) (Theorem 38.8); **Symmetry**—the phase symmetry of the seasons between the two hemispheres (phase difference π); **Closure**—the closure of climate feedback loops such as the ice–albedo feedback; **Period**—the multiscale periods of the seasons (1 year), ENSO (2–7 years), the PDO (20–30 years) and the Milankovitch cycles (2.3–10 ten-thousand years).

38.1 Connection to the five-step interface of the ‘Periodic Systems’ chapter

38.1.1 Step one: periodicity decision for climate targets

Definition 38.1 (Climate target and decision input). A **climate target** is one of the following objects:

- (i) a seasonal cycle system, whose state space is the joint space of the solar declination $\delta(t)$ and the latitude φ;
- (ii) an ENSO system, whose state space is the $(T_E, \langle h\rangle_W)$ phase space;
- (iii) an NAO system, whose state space is the space of time series of the north–south pressure difference;
- (iv) a PDO/IOD system, whose state space is the space of sea surface temperature anomalies;
- (v) a Milankovitch cycle system, whose state space is the joint space of the three orbital elements $(e, \varepsilon, e\sin\varpi)$;
- (vi) a long-range correlated climate series, whose state space is the space of time series with long-range persistence.

The **decision input** of a climate target is the pair

$$\mathcal{D}_{\text{Cl}} = (\mathcal{S}, \Phi),$$

where $\mathcal{S}$ is the state space and $\Phi : \mathcal{S} \to \mathcal{S}$ is the map induced by time translation, phase rotation or feedback evolution. If $\mathcal{S}$ is a finite set, the target is called a **finite climate target**; if $\mathcal{S}$ is an infinite set, the target is called an **infinite climate target**.

Definition 38.2 (Climate decision map)**.** Let a climate target be given by the decision input $\mathcal{D}_{\mathrm{Cl}} = (\mathcal{S}, \Phi)$. If the states of the target are generated by Φ, that is, if there exists $x_0 \in \mathcal{S}$ such that

$$x_t = \begin{cases} \Phi^t(x_0), & t \in \mathbb{Z}, \\ \Phi_t(x_0), & t \in \mathbb{R}, \end{cases}$$

for $t \in \mathbb{Z}$ the state x_t is generated by iteration of Φ; for $t \in \mathbb{R}$ it is described by the one-parameter evolution family $\{\Phi_t\}_{t\in\mathbb{R}}$, where each $\Phi_t : \mathcal{S} \to \mathcal{S}$, $\Phi_0 = \mathrm{id}$, $\Phi_{s+t} = \Phi_s \circ \Phi_t$ for all $s, t \in \mathbb{R}$, and $(t, x) \mapsto \Phi_t(x)$ is continuous. then Φ is called the **climate decision map** of the target. The climate target is then reduced to the discrete dynamical system $(\mathcal{S}, \Phi)$ when $t \in \mathbb{Z}$ and to the continuous dynamical system $(\mathcal{S}, \{\Phi_t\}_{t\in\mathbb{R}})$ when $t \in \mathbb{R}$; in both cases the criteria of the 'Operations and Periodicity' chapter and of the 'Periodic Systems' chapter may be applied directly.

Theorem 38.1 (Climate periodicity decision theorem)**.** *Let the decision map of a climate target be Φ, and let $(\mathcal{S}, \Phi)$ be a discrete dynamical system. Then:*

(A) **Sufficient decision**: *if there exists $m \in \mathbb{N}^+$ such that $\Phi^m = \mathrm{id}$, then the target is a uniformly periodic climate system; every orbit is closed, and the orbit length divides m.*

(B) **Equivalence of the decision**: *the target is a uniformly periodic climate system if and only if $\mathrm{ord}(\Phi) < \infty$.*

(C) **Pointwise criterion**: *the orbit of the target at the state x_0 is closed if and only if x_0 is a periodic point; the orbit length is then the period at that point.*

(D) **Non-omissibility**: *the conjunction of conserved quantities and symmetry groups cannot replace $\mathrm{ord}(\Phi) < \infty$; a quasi-periodic climate target is not a periodic climate system.*

Proof. The Theorem 38.1 of this chapter is the domain instantiation of Theorem 24.1 in the 'Periodic Systems' chapter: one need only replace the decision map f and the evolution group by Φ of this chapter, and the four clauses (A) sufficient decision, (B) equivalence of the decision, (C) pointwise criterion and (D) non-omissibility hold one by one. By the convention of Remark 24.4, the argument is not expanded again here. ☐

Corollary 38.1 (Climate period types)**.** After the decision by Theorem 38.1, a climate target falls into one of the following types:

(i) **Trivially periodic type**: the least period is $m_0 = 1$, and the states are fixed points;

(ii) **Strictly periodic type**: $1 < m_0 < \infty$, and the states are strictly periodic;

(iii) **Uniformly periodic type**: $\mathrm{ord}(\Phi) = m < \infty$, and all orbit lengths divide m;

(iv) **Quasi-periodic type**: $\mathrm{ord}(\Phi) = \infty$, and the orbits are not closed but are dense on a compact carrier structure;

(v) **Eventually periodic type**: there exists N such that the target is periodic for $t \geq N$;

(vi) **Discretely periodic type**: the carrier structure is $\mathbb{Z}$, and the normalised period ratio is $\rho = T_x/\Delta x \in \mathbb{Q}$.

(vii) **Asymptotically periodic (damped) type**: the orbit is not closed, but tends asymptotically to a closed orbit or to a fixed point, the residual decaying as $e^{-\kappa t}$ (see Definition 24.3(vii)).

Of these, (i)–(iii) and (vi) are periodic climate systems; (iv) is not a periodic climate system, but can be handled with bounded-approximation tools (the Milankovitch cycles, ENSO, the PDO and the IOD all belong to this class); (v) is a periodic climate system only in its stable stage; (vii), like (iv), is a type in the limiting sense; it is not a periodic climate system and can be handled with bounded-approximation tools only in its stable stage.

Proof. Substituting the climate target clause by clause into Theorem 38.1 and Definition 24.3 of the 'Periodic Systems' chapter yields the result. ☐

Lemma 38.1 (Periodicity decision for the seasonal cycle)**.** Let the solar declination be $\delta(t) = 23.44^\circ \sin(2\pi t/365.24)$ and let the latitude φ be fixed. Then the daily mean solar radiation $S(\varphi, t)$ reaching the top of the atmosphere is a uniformly periodic system, whose least positive period is the tropical year:

$$T_{\text{season}} = 365.24 \text{ days}.$$

Proof. By Definition 38.10, $S(\varphi, t)$ depends, through $\cos\Theta(\varphi, t)$, on $\delta(t)$. From $\delta(t+T_{\text{season}}) = \delta(t)$ (the periodicity of the sin function) we obtain $S(\varphi, t+T_{\text{season}}) = S(\varphi, t)$. Hence the decision map Φ satisfies $\Phi^{T_{\text{season}}} = \text{id}$, so the system is uniformly periodic. ☐

Lemma 38.2 (Quasi-periodicity decision for ENSO)**.** Let ENSO be described by the recharge–discharge oscillator model, with parameters satisfying $4\beta\gamma > (\alpha - \delta)^2$. Then ENSO is a damped oscillation, with natural period

$$T_{\text{ENSO}} = \frac{4\pi}{\sqrt{4\beta\gamma - (\alpha - \delta)^2}}.$$

Its value depends on the coupling coefficients; with the typical parameters of Example 38.4 it is about 6.3 years. Under random atmospheric perturbations, ENSO behaves as a quasi-periodic system ($\text{ord}(\Phi) = \infty$), with dominant period between 2 and 7 years.

Proof. By Proposition 38.7, the eigenvalues are $\lambda_{1,2} = -\frac{\alpha+\delta}{2} \pm \frac{i}{2}\sqrt{4\beta\gamma - (\alpha - \delta)^2}$, and the damping rate is $(\alpha+\delta)/2 > 0$. Hence in the absence of forcing the system is of asymptotically periodic (damped) type (it tends to the equilibrium point); under the random forcing $\sigma\xi(t)$ the system oscillates persistently near the natural frequency and is of quasi-periodic type. The observed dominant period of 2–7 years is determined by the natural period together with the random modulation. ☐

Lemma 38.3 (Red-noise decision for the NAO)**.** Let the autocorrelation function of the NAO index $N(t)$ be $\rho(\tau) = e^{-|\tau|/\tau_0}$ ($\tau_0 \approx 7$–10 days). Then the NAO is of quasi-periodic type ($\mathrm{ord}(\Phi) = \infty$), and its power spectrum is red noise:

$$S(\omega) = \frac{2\tau_0}{1 + \omega^2\tau_0^2} \propto \omega^{-2} \quad (\omega \to \infty).$$

Proof. By Lemma 38.6, the power spectrum of $\rho(\tau) = e^{-|\tau|/\tau_0}$ is $S(\omega) = 2\tau_0/(1 + \omega^2\tau_0^2)$. The red-noise spectrum has no atoms at any frequency, so it does not constitute a uniformly periodic system. By Remark 1.13 of the 'Operations and Periodicity' chapter, the NAO is of quasi-periodic type (or, more precisely, of long-range correlated type). ☐

Lemma 38.4 (Quasi-periodicity decision for the Milankovitch cycles)**.** Let the three orbital elements $(e, \varepsilon, e \sin \varpi)$ be driven by planetary perturbations, with dominant periods 100 ka (eccentricity), 41 ka (obliquity), and 23 ka and 19 ka (precession). Then the Milankovitch cycles form a quasi-periodic system ($\mathrm{ord}(\Phi) = \infty$).

Proof. By Proposition 29.5 of the 'Periodic Structure of Astronomical Systems' chapter, the principal frequencies of the three orbital elements are linearly independent over $\mathbb{Q}$, so the periods are incommensurable and the overall motion is a quasi-periodic flow on a three-dimensional torus, with non-closed orbits. By Theorem 38.1(D), the Milankovitch cycles are of quasi-periodic type. ☐

Lemma 38.5 (Timescale-separation decision for the PDO and ENSO)**.** Let the PDO have dominant period $T_{\text{PDO}} \approx 20$–$30$ years and ENSO dominant period $T_{\text{ENSO}} \approx 2$–$7$ years. Then slow–fast variable separation holds: the PDO is a quasi-periodic slow background and ENSO a quasi-periodic fast oscillation, and the two interact through slow–fast coupling.

Proof. By Proposition 38.9, $T_{\text{PDO}}/T_{\text{ENSO}} \approx 4$–$10$, so there exist a slow variable $\bar{T}_{\text{PDO}}$ and a fast variable T'. By Theorem 38.1(D), the natural dynamics of the two are both of quasi-periodic type (neither contains a finite-order evolution). ☐

Definition 38.3 (Least positive period algorithm for climate systems)**.** This definition has the same content as that given, in Chapter 24 under the section 'Periodicity decision: whether the target is periodic' (Section 24.1), by Definition 24.4; to avoid repetition only the reference anchor is retained here, and the complete statement is given there. The only difference here is the terminology (climate target / climate sample); the steps of the algorithm and the conclusion are exactly the same.

Theorem 38.2 (Correctness of the least positive period algorithm for climate systems)**.** This theorem has the same content as that given, in Chapter 24 under the section 'Periodicity decision: whether the target is periodic' (Section 24.1), by Theorem 24.2; to avoid repetition only the reference anchor is retained here, and the complete statement and proof are given there. The only difference here is the terminology (climate target / climate sample); the steps of the algorithm and the conclusion are exactly the same.

Example 38.1 (Instances of climate periodicity decisions)**.** (i) Seasonal cycle: $T = 365.24$ days, uniformly periodic type;

(ii) ENSO: $T \approx 6.3$ years (natural period; see Example 38.4), quasi-periodic type (under random forcing);

(iii) NAO: $\tau_0 \approx 7$–10 days, quasi-periodic type (red noise);

(iv) PDO: $T_{\text{PDO}} \approx 20$–$30$ years, quasi-periodic type;

(v) IOD: $T_{\text{IOD}} \approx 3$–7 years, quasi-periodic type;

(vi) Milankovitch cycles: $100/41/23/19$ ka, quasi-periodic type.

Remark 38.4 (Conserved quantities do not guarantee periodicity). *In the climate system, conservation of energy, conservation of momentum and conservation of water vapour are auxiliary conditions for periodic structure, but they do not constitute a sufficient condition. For example, a stable climate that possesses energy conservation but lacks ocean–atmosphere coupling will not spontaneously produce ENSO oscillations. Periodic structure in the climate system must be decided by Theorem 38.1 or by the lemmas above.*

38.1.2 Step two: construction of the climate period series and the epicycle group

Definition 38.4 (Climate amplitude, frequency and phase series)**.** Let the state or output sequence of a climate target be $\{z_t\}$. If it can be written as

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)} + \varepsilon_t,$$

then

$$\mathcal{A}_z(t) = \sum_k A_k e^{ikt}, \qquad \mathcal{F}_z(s) = \sum_k \omega_k s^k, \qquad \mathcal{P}_z(t) = \sum_k e^{i(kt+\phi_k)}$$

are called respectively the **amplitude series**, the **frequency series** and the **phase series** of the climate target. The three agree with Definition 6.10, Definition 6.12 and Definition 6.11 in the 'Series and Convolution' chapter.

Definition 38.5 (Climate epicycle group)**.** Let the period series of a climate target be given by the amplitude series $\{A_k\}$, the frequency series $\{\omega_k\}$ and the phase series $\{\phi_k\}$. For each k define the epicycle

$$E_k : \theta \mapsto A_k e^{i(\theta+\phi_k)}, \qquad \theta = \omega_k t,$$

and call the set

$$\mathfrak{E}_{\text{Cl}} = \{(A_k, \omega_k, \phi_k)\}_{k\in\mathbb{Z}}$$

the **epicycle group** of the climate target. Its superposition is the climate period series

$$z_t = \sum_k E_k(\omega_k t) = \sum_k A_k e^{i(\omega_k t+\phi_k)}.$$

Theorem 38.3 (Construction of the period series from climate data)**.** *Let the output of a climate target be a finite periodic sequence of period N. Then:*

(i) the output can be expanded as a discrete Fourier series

$$z_t = \sum_{k=0}^{N-1} c_k e^{i2\pi kt/N}, \qquad c_k = \frac{1}{N}\sum_{t=0}^{N-1} z_t e^{-i2\pi kt/N};$$

(ii) the amplitude series is given by $|c_k|$, the phase series by $\arg c_k$, and the frequency series by $2\pi k/N$;

(iii) if the output is a seasonal temperature series, the fundamental frequency corresponds to the annual period and the harmonics to the semi-annual, seasonal and monthly periods; if the output is an ENSO series, the fundamental frequency corresponds to a natural period of about 6 years and the harmonics to the distribution of stronger events (such as those of 1982, 1997 and 2015).

Proof. (i) The discrete Fourier series of a finite periodic sequence is a standard result: the basis functions $\{e^{i2\pi kt/N}\}_{k=0}^{N-1}$ form an orthogonal basis of $\mathbb{C}^N$. (ii) This is read off directly from $c_k = A_k e^{i\phi_k}$. (iii) For a seasonal temperature series with $N = 365$ (daily sampling), the fundamental frequency corresponds to the annual period, $k = 2$ to the semi-annual period and $k = 4$ to the seasonal period; for an ENSO series with $N \sim 70$ (annual sampling), the fundamental frequency corresponds to about $1/6$ yr^{-1}, and the harmonics encode the statistical structure of quasi-periodic events. □

Proposition 38.1 (Three-parameter correspondence: radius, phase and speed). The three parameters of the climate epicycle group $\mathfrak{E}_{\mathrm{Cl}} = \{(A_k, \omega_k, \phi_k)\}_k$ correspond one-to-one to three geometric–physical quantities:

(i) **Radius corresponds to amplitude**: the radius of E_k, namely $|E_k| = A_k$, is the amplitude of that climate oscillation component;

(ii) **Phase corresponds to rotational delay**: the initial phase ϕ_k corresponds to the time translation $\tau_k = \phi_k/\omega_k$; when $\phi_k > 0$ this component **leads** the baseline by τ_k (equivalently, the component coincides with the baseline once the baseline is postponed by τ_k);

(iii) **Frequency corresponds to rotational speed**: ω_k is the angular frequency of that climate oscillation component.

Proof. The Proposition 38.1 of this chapter corresponds exactly to Proposition 24.3 of the 'Periodic Systems' chapter, and the three correspondences—radius to amplitude, phase to rotational delay, frequency to rotational speed—are obtained by translating them one by one to the epicycle group of this chapter. The proof is given there: (i) from $|E_k| = A_k$ and the energy decomposition, (ii) from the phase translation theorem, (iii) from the definition of the frequency series. □

Example 38.2 (Period spectrum series of climate phenomena). Arranging the principal climate phenomena by timescale and expanding them into a climate period series yields the following layered structure:

(i) 10^{-2}–10^{-1} years: weather systems, the Madden–Julian Oscillation (MJO, 30–90 days);

(ii) 10^{-1}–10^{0} years: the seasonal cycle, the NAO;

(iii) 10^{0} years: ENSO, the IOD;

(iv) 10^{1} years: the PDO, the AMO;

(v) 10^{2}–10^{3} years: the Little Ice Age, the Medieval Warm Period;

(vi) 10^{4}–10^{5} years: the Milankovitch cycles (precession of 2.3 ten-thousand years, obliquity of 4.1 ten-thousand years, eccentricity of 10 ten-thousand years).

These phenomena correspond to the layered structure of the climate epicycle group, from the inner high-frequency small epicycles to the outer low-frequency large epicycles.

Remark 38.5 (The period series is a complex function). Let $z = e^{i\omega_0 t}$; then the climate period series

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)}$$

when all components are commensurable with ω_0 (that is, $\omega_k = k\omega_0$, $k \in \mathbb{Z}$) it can be written as the Laurent series $\sum_k c_k z^k$, where $c_k = A_k e^{i\phi_k}$: in that case the climate period series is a complex function on the unit circle, and otherwise it is merely a function on the torus.

38.1.3 Step three: dynamic adjustment of the climate epicycle group

Definition 38.6 (Dynamic adjustment of the climate epicycle group)**.** *Let the current climate epicycle group be $\mathfrak{E}^{(j)}$, with partial sum*

$$z^{(j)}(t) = \sum_{k\in\mathfrak{E}^{(j)}} A_k e^{i(\omega_k t + \phi_k)},$$

and residual $r^{(j)} = z - z^{(j)}$. **Dynamic adjustment** *consists of two rules:*

(i) **Quantity adjustment**: *if the residual energy satisfies $\|r^{(j)}\|^2 > \delta$, add a new epicycle according to the peak of the residual periodogram;*

(ii) **Parameter adjustment**: *update the parameters of the existing epicycles by regularised least squares or by the conservation constraints.*

In the climate system, quantity adjustment corresponds to increasing the number of climate modal components (for example, identifying a new ENSO harmonic or an independent IOD component); parameter adjustment corresponds to adjusting the amplitude, phase and frequency so that the energy, momentum and water-vapour constraints are satisfied.

Proposition 38.2 (Monotonicity of quantity adjustment)**.** *After the dominant residual frequency component is added as in Definition 38.6(i), the residual energy is monotonically non-increasing:*

$$\|r^{(j+1)}\|^2 \le \|r^{(j)}\|^2,$$

with equality if and only if the residual has no energy at the selected frequency.

Proof. The Proposition 38.2 of this chapter agrees with the dynamic adjustment mechanism of Definition 24.8 in the 'Periodic Systems' chapter: adding an epicycle removes the component of $r^{(j)}$ along $\hat{\omega}$ from the residual, and the orthogonal decomposition $\|r^{(j)}\|^2 = \|r^{(j+1)}\|^2 + \|\text{the removed component}\|^2$ gives the inequality, with equality if and only if the norm of the removed component is zero. The complete argument is given there. $\square$

Theorem 38.4 (Convergence of dynamic adjustment). Let the climate target be a finite periodic sequence of period T_0 and let $\delta = 0$. Then after at most $\lfloor T_0/2 \rfloor + 1$ steps of dynamic adjustment the residual is zero, and the resulting climate epicycle group is the exact Fourier data of the target.

Proof. The discrete Fourier series of a sequence of period T_0 is complete on $\lfloor T_0/2 \rfloor + 1$ harmonics. At each step one dominant frequency component of the residual is extracted as in Proposition 38.2 and the residual energy decreases strictly, so after finitely many steps the residual is zero. $\square$

Definition 38.7 (Structure-metric adjustment). Let the structure metric of a climate target be

$$\mathcal{I} = (\mathcal{I}_{\text{energy}}, \mathcal{I}_{\text{momentum}}, \mathcal{I}_{\text{vapour}}) = (\text{energy conservation degree, momentum conservation degree,}$$

Structural-metric adjustment consists in choosing θ in the parameter space Θ: at each step take a neighbouring parameter for which $\mathcal{I}(\theta)$ is **strictly decreasing** in the lexicographic order (minimisation); if no such parameter exists, the process stops, and the parameter is then locally optimal.

Proposition 38.3 (Finite convergence of structure-metric adjustment). Let the parameter space Θ be finite, and let each adjustment make $\mathcal{I}$ strictly decreasing in the lexicographic order. Then the adjustment terminates after finitely many steps at a locally optimal parameter.

Proof. Since Θ is finite, $\mathcal{I}$ takes only finitely many values, and the lexicographic order is a total order on a finite set; a strictly decreasing sequence in that order cannot continue indefinitely, so the adjustment terminates after finitely many steps. At termination there is no neighbouring parameter that makes $\mathcal{I}$ strictly decrease in the lexicographic order, that is, the parameter is locally optimal. $\square$

Remark 38.6 (Boundaries of dynamic adjustment). Theorem 38.4 relies on finite periodic sequences. For continuous-spectrum or quasi-periodic climate targets (such as the Milankovitch cycles, ENSO and the NAO), no finite epicycle group represents them exactly and only bounded approximation is possible. Structure-metric adjustment does not guarantee global optimality.

38.1.4 Step four: climate convolution

Definition 38.8 (Climate amplitude, phase and frequency convolutions). Let two climate epicycle groups be

$$\mathfrak{E} = \{(A_n, \omega_n, \phi_n)\}, \qquad \mathfrak{E}' = \{(B_m, \nu_m, \psi_m)\}.$$

Define:

(i) **Amplitude convolution**: $\mathbb{A}_{mn} = A_n B_m$;

(ii) **Phase convolution**: $\mathbb{P}_{mn} = e^{i(\phi_n - \psi_m)}$;

(iii) **Frequency convolution**: $\mathbb{F}_{mn} = \omega_n \nu_m$;

(iv) **Trend convolution**: $\mathbb{T}_{mn} = A_n B_m e^{i(\phi_n - \psi_m)} = A_n B_m \mathbb{P}_{mn}$.

This agrees with Definition 6.21, Definition 6.22, Definition 6.23 and Definition 6.24 of the 'Series and Convolution' chapter.

Theorem 38.5 (Action of the three convolutions on the climate epicycle group)**.** *Let the epicycle groups of two climate targets be $\mathfrak{E}$ and $\mathfrak{E}'$ respectively. Then:*

(i) *amplitude convolution acts only on the radius;*

(ii) *phase convolution acts only on the phase;*

(iii) *frequency convolution acts only on the rotational speed;*

(iv) *trend convolution preserves both amplitude and phase.*

Proof. This is read off directly from Definition 38.8. ☐

Proposition 38.4 (Autocorrelation and trend convolution)**.** *Let $\{z_t\}$ be a climate periodic sequence, with autocorrelation function*

$$R_z(h) = \sum_t z_t \overline{z_{t+h}}.$$

In the language of the climate epicycle group, the autocorrelation function is the projection of the trend convolution onto the diagonal:

$$R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}.$$

Proof. The Proposition 38.4 of this chapter is the realisation in this domain of the convolution-action conclusion of Theorem 24.6 in the 'Periodic Systems' chapter: substituting $z_t = \sum_n A_n e^{i(\omega_n t + \phi_n)}$ into the definition of the autocorrelation, cancelling all off-diagonal cross terms by orthogonality and retaining only the diagonal term of the trend convolution (the factorisation of the three convolutions, Proposition 24.6), which yields $R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}$. The full expansion is given there. ☐

Remark 38.7 (Relation between climate convolution and periodic convolution)**.** Climate convolution is not a new convolution operation but an instance of the periodic convolution of the 'Series and Convolution' chapter on the climate epicycle group. Autocorrelation is the projection of the trend convolution onto the diagonal; teleconnection and inter-basin coupling (such as ENSO–PDO and ENSO–IOD) are respectively the concrete manifestations in the Earth's climate sciences of amplitude convolution and frequency convolution.

38.1.5 Step five: integration with discrete sampling, harmonic analysis and time series analysis

Definition 38.9 (Climate periodicity diagnostic procedure)**.** For a given climate system, diagnose it using the framework of periodic mathematics:

(i) **Periodicity decision**: verify the period type of the climate target (Theorem 38.1);

(ii) **Series construction**: build the amplitude, frequency and phase series and the epicycle group from temperature, precipitation, pressure and SST data (Theorem 38.3);

(iii) **Conservation diagnosis**: verify energy, momentum and water-vapour conservation (Theorem 38.8);

(iv) **Convolution diagnosis**: compute the autocorrelation (the diagonal projection of the trend convolution, Proposition 38.4) and the phase convolution;

(v) **Integration diagnosis**: connect to the tool chains of discrete sampling, harmonic analysis and time series analysis.

Theorem 38.6 (The climate periodicity diagnostic procedure and its interfaces with three chapters)**.** The five steps of Definition 38.9 connect respectively to:

(i) the 'Discrete Sampling and Intercalation of Periodic Signals' chapter: Definition 21.1, Theorem 21.3, Theorem 21.10;

(ii) the 'Harmonic Analysis' chapter: Definition 22.5, Theorem 22.1, Proposition 22.11;

(iii) the 'Time Series Analysis' chapter: Definition 23.16, Proposition 23.2, Definition 23.19.

The climate periodicity diagnostic procedure is therefore not an independent method but an application of the unified tool chain of periodic mathematics to the climate system.

Proof. The Theorem 38.6 of this chapter is the implementation in this domain of the five-step interface of the 'Periodic Systems' chapter (Table 24.1): the periodicity decision uses sampling and spectral estimation, the frequency decomposition and the phase kernels use the 'Harmonic Analysis' chapter, and the periodogram peaks, the Fisher test and window-function validation use the 'Time Series Analysis' chapter. The diagnostic procedure is therefore a synthesis of the tool chains of the three chapters; the proof is given there. □

Remark 38.8 (Order of operations after integration). The recommended order for an actual diagnosis is: first decide whether the climate target is periodic (Theorem 38.1); then construct the Fourier data and the epicycle group of the climate data (Theorem 38.3); then optimise the parameters by the conservation constraints and dynamic adjustment (Theorem 38.4, Proposition 38.3); then perform climate convolution and autocorrelation analysis (Proposition 38.4); and finally output the diagnosis as in Definition 38.9.

38.2 The seasonal cycle: the annual period of solar radiation

Definition 38.10 (Solar radiation reaching the top of the atmosphere). At a location of latitude φ, the daily mean solar radiation reaching the top of the atmosphere on the t-th day of the year (with the vernal equinox at $t = 0$) is determined by the orbital geometry:

$$S(\varphi, t) = S_0 \left(\frac{a}{r(t)}\right)^2 \cos\Theta(\varphi, t),$$

where $S_0 = 1361$ W/m^2 is the solar constant, $a/r(t)$ is the Earth–Sun distance ratio (caused by the eccentricity e), Θ is the solar zenith angle, satisfying

$$\cos\Theta = \sin\varphi \sin\delta(t) + \cos\varphi\cos\delta(t)\cos h_0,$$

$\delta(t) \approx 23.44^\circ \sin(2\pi t/365.24)$ is the solar declination (the projection of the 23.44° axial tilt), and h_0 is the hour angle.

Proposition 38.5 (Phase symmetry of the seasons between the two hemispheres). At locations of the same latitude in the northern and southern hemispheres, the annual cycle of solar radiation has phase difference π:

$$S(-\varphi, t) = S_0 \left(\frac{a}{r(t)}\right)^2 \big(-\sin\varphi\sin\delta(t) + \cos\varphi\cos\delta(t)\cos h_0\big),$$

that is, the northern summer solstice ($\delta = +23.44^\circ$) is exactly the southern winter solstice. Under the hemispheric reflection $\varphi \mapsto -\varphi$, the seasonal cycle is equivalent to a phase translation by π: the declination term of S that is antisymmetric about the equatorial plane, namely $\sin\varphi\sin\delta$, changes sign under hemispheric reflection.

Proof. By Definition 38.10, when $\varphi \mapsto -\varphi$ we have $\sin\varphi \mapsto -\sin\varphi$ whilst $\cos\varphi$ is unchanged, so $\cos\Theta(-\varphi, t) = -\sin\varphi\sin\delta + \cos\varphi\cos\delta\cos h_0$. On the other hand $\delta(t + T/2) = 23.44^\circ \sin(2\pi(t+T/2)/T) = -\delta(t)$, so $\cos\Theta(-\varphi, t) = \cos\Theta(\varphi, t + T/2)$—hemispheric reflection is equivalent to a half-year phase translation, that is, to a phase difference of π. ☐

Example 38.3 (Polar Day and Polar Night at Latitude 66.5° North). Take $\varphi = 66.5^\circ$ (the Arctic Circle). On the summer solstice $\delta = 23.44^\circ$: the noon solar altitude is $= 90^\circ - |66.5^\circ - 23.44^\circ| \approx 46.9^\circ$, and the midnight solar altitude is $= \varphi + \delta - 90^\circ = -0.06^\circ \approx 0$—the Sun does not set, which is the polar day. On the winter solstice $\delta = -23.44^\circ$: the noon solar altitude is $= 90^\circ - |66.5^\circ + 23.44^\circ| \approx 0.06^\circ$—the Sun does not rise, which is the polar night. The alternation of polar day and polar night inside the Arctic Circle is the most intuitive manifestation of the annual period $T = 365.24$ days.

Lemma 38.6 (Duality between the Red-Noise Spectrum and Exponential Autocorrelation). Let the autocorrelation function of a stationary process be $\rho(\tau) = e^{-|\tau|/\tau_0}$ (red noise). Then its power spectral density is

$$S(\omega) = \frac{2\tau_0}{1 + \omega^2\tau_0^2}.$$

In the low-frequency limit ($\omega \ll \tau_0^{-1}$) we have $S(\omega) \approx 2\tau_0$, a flat white spectrum, whilst in the high-frequency limit ($\omega \gg \tau_0^{-1}$) we have $S(\omega) \approx \frac{2}{\tau_0\omega^2} \propto \omega^{-2}$—the energy is concentrated at low frequencies, that is, the spectrum is 'red'.

Proof. By the Wiener–Khinchin theorem the spectral density is the Fourier transform of the autocorrelation function:

$$S(\omega) = \int_{-\infty}^{\infty} e^{-|\tau|/\tau_0} e^{-i\omega\tau}\, d\tau = \frac{1}{\frac{1}{\tau_0} - i\omega} + \frac{1}{\frac{1}{\tau_0} + i\omega} = \frac{2\tau_0}{1 + \omega^2\tau_0^2}.$$

For $\omega \ll \tau_0^{-1}$ we have $S(\omega) \approx 2\tau_0 \propto \omega^0$, and for $\omega \gg \tau_0^{-1}$ we have $S(\omega) \approx 2/(\tau_0\omega^2) \propto \omega^{-2}$. □

38.3 Mathematical foundations of the climate system

Definition 38.11 (Basic Quantities of the Climate System)**.** *A climate system is described by the following basic physical quantities:*

(i) **Temperature** $T(\mathbf{r}, t)$ *(unit: K or °C): the temperature of the atmosphere, ocean or land surface at position* **r** *and time t. The global mean surface temperature (GMST) is the core indicator of climate change.*

(ii) **Precipitation** $P(\mathbf{r}, t)$ *(unit: mm/day): the amount of water vapour in the atmosphere that condenses and falls to the surface. Precipitation exhibits a highly uneven distribution in space and time.*

(iii) **Pressure** $p(\mathbf{r}, t)$ *(unit: hPa): the weight per unit area of a column of the atmosphere. Sea-level pressure (SLP) is a basic variable of the atmospheric circulation.*

(iv) **Sea surface temperature** SST$(\mathbf{r}, t)$ *(unit: °C): the temperature of the ocean surface layer. SST is a key variable of ocean–atmosphere interaction—climate modes such as ENSO and the Indian Ocean Dipole are defined mainly through SST anomalies.*

(v) **Radiative forcing** ΔF *(unit: W/m^2): a perturbation of the energy budget of the Earth system (caused by greenhouse gases, solar activity, volcanic aerosols and so on). $\Delta F > 0$ means that the Earth absorbs more energy (warming), and $\Delta F < 0$ means that the Earth loses more energy (cooling).*

(vi) **Wind stress** $\boldsymbol{\tau}$ *(unit: Pa): the tangential stress exerted by the atmosphere on the ocean surface, which drives the wind-driven circulation of the upper ocean.*

The climate system is a **multiscale coupled complex system***—the atmosphere (fast response, daily–annual scales), the ocean (slow response, annual–millennial scales), the cryosphere (very slow response, centennial–ten-thousand-year scales), the biosphere and the lithosphere interact on different timescales.*

38.4 El Niño–Southern Oscillation (ENSO)

Definition 38.12 (El Niño–Southern Oscillation (ENSO))**.** The **El Niño–Southern Oscillation** (ENSO) is the quasi-periodic oscillation of the coupled ocean–atmosphere system of the tropical Pacific on the interannual timescale (2–7 years). ENSO has two extreme phases:

(i) **El Niño (warm phase)**: anomalous warming of the sea surface temperature in the eastern equatorial Pacific (positive SST anomaly exceeding $+0.5°$C for several months), increased precipitation in the central and eastern tropical Pacific and reduced precipitation in the western Pacific. The Walker circulation weakens or reverses. The equatorial thermocline deepens in the east and shoals in the west.

(ii) **La Niña (cold phase)**: anomalous cooling of the sea surface temperature in the eastern equatorial Pacific (negative SST anomaly exceeding $-0.5°$C for several months), with a tropical precipitation distribution broadly opposite to that of El Niño. The Walker circulation strengthens. The equatorial thermocline shoals in the east and deepens in the west.

ENSO is the strongest interannual variability signal in the tropical Pacific climate system, and its global teleconnections affect the Asian monsoon, North American winter temperatures, Australian droughts and South American precipitation. The seasonal phase-locking of the ENSO amplitude is manifested as follows: peaks occur mostly in the northern winter (November–January of the following year), and decay mostly in the following spring.

Definition 38.13 (Southern Oscillation Index (SOI))**.** The **Southern Oscillation Index** (SOI) is defined as the standardised value of the sea-level pressure difference between Tahiti and Darwin:

$$\text{SOI} = \frac{p_{\text{Tahiti}} - p_{\text{Darwin}}}{\sigma_{\Delta p}},$$

where $\sigma_{\Delta p}$ is the long-term standard deviation. The SOI is negatively correlated with the SST anomaly in the eastern equatorial Pacific: during El Niño the SOI takes negative values (lower pressure at Tahiti, higher pressure at Darwin), whilst during La Niña the SOI takes positive values. The strength of an ENSO event is usually measured by the three-month running mean of the SST anomaly in the Niño 3.4 region ($5°$S–$5°$N, $120°$W–$170°$W) (the Oceanic Niño Index, ONI).

Proposition 38.6 (The Bjerknes positive feedback of ENSO)**.** The quasi-periodic oscillation of ENSO is maintained jointly by the Bjerknes positive feedback and a delayed negative feedback. The Bjerknes feedback describes the self-amplifying loop between ocean and atmosphere:

(i) An initial positive SST anomaly in the eastern equatorial Pacific → enhanced atmospheric convection → low-level wind convergence → a westerly wind anomaly.

(ii) The westerly wind anomaly → excites an eastward-propagating downwelling equatorial Kelvin wave → the thermocline deepens → upwelling in the eastern Pacific weakens → SST warms further → the positive feedback is complete.

The Bjerknes positive feedback is a self-amplifying mechanism in the coupled ocean–atmosphere system—a warm anomaly is amplified through ocean–atmosphere coupling. The positive feedback cannot continue indefinitely, because the delayed negative feedback (westward-propagating equatorial Rossby waves reflecting as eastward-propagating Kelvin waves) reverses the thermocline anomaly from the warm phase to the cold phase after about 6–12 months. The alternation of the positive feedback and the delayed negative feedback maintains the quasi-periodic oscillation of ENSO.

Proof. Linearise the feedback loop: let T be the eastern Pacific SST anomaly and $\langle h\rangle_W$ the thermocline depth anomaly. The four links of the loop—(i) T drives convection (gain $a_1 > 0$); (ii) convection drives a westerly wind anomaly (gain $a_2 > 0$); (iii) the westerlies excite a Kelvin wave that deepens the thermocline (gain $a_3 > 0$); (iv) the deepened thermocline weakens upwelling and makes T warm further (gain $a_4 > 0$)—combine to give $\dot{T} \supset a_1a_2a_3a_4T$, with loop gain $G = a_1a_2a_3a_4 > 0$: a perturbation returns to itself after one circuit of the loop and has been amplified, which is the linearised positive feedback. This is precisely the origin of the term $\beta\gamma > 0$ in the recharge–discharge model of Proposition 38.7: the positive feedback ($\beta\gamma$) competes with the damping ($\alpha\delta$), and an oscillation arises when $4\beta\gamma > (\alpha-\delta)^2$. □

Proposition 38.7 (The periodic-system model of ENSO—the recharge–discharge oscillator)**.** *The simplified dynamics of ENSO can be described by the linearised system of equations of the coupled ocean–atmosphere system of the tropical Pacific. On the equatorial β-plane, the solution of the oceanic linear shallow-water equations under atmospheric wind-stress forcing is a superposition of equatorial waves. The recharge–discharge oscillator model of ENSO treats the east–west average $\langle h\rangle_W$ of the equatorial Pacific thermocline depth as the charging state variable and the eastern Pacific SST anomaly T_E as the response variable:*

$$\frac{d\langle h\rangle_W}{dt} = -\alpha\langle h\rangle_W + \beta T_E, \quad \frac{dT_E}{dt} = -\gamma\langle h\rangle_W - \delta T_E,$$

where $\alpha, \beta, \gamma, \delta > 0$. This system of equations is equivalent to a damped harmonic oscillator—T_E and $\langle h\rangle_W$ perform a decaying rotation about the equilibrium point in phase space. Random atmospheric perturbations (such as westerly wind burst events) inject energy into the system and maintain a persistent oscillation.

The natural period is $T_{\text{ENSO}} = \frac{4\pi}{\sqrt{4\beta\gamma-(\alpha-\delta)^2}}$; with the typical parameters of Example 38.4 it is about 6.3 years. The random perturbations provide a persistent phase perturbation to the system, preventing the oscillation from decaying.

Proof. The equilibrium point of the recharge–discharge oscillator model is $\langle h\rangle_W^* = 0, T_E^* = 0$. The Jacobian matrix is

$$J = \begin{pmatrix} -\alpha & \beta \\ -\gamma & -\delta \end{pmatrix},$$

and its eigenvalues are $\lambda_{1,2} = -\frac{\alpha+\delta}{2} \pm \frac{i}{2}\sqrt{4\beta\gamma - (\alpha-\delta)^2}$. Since $\det J = \alpha\delta + \beta\gamma > 0$, the equilibrium is a stable focus; when $4\beta\gamma > (\alpha-\delta)^2$ the eigenvalues are complex conjugates—the system exhibits a damped oscillation. The persistent oscillation of the actual ENSO is maintained by random atmospheric perturbations—this is equivalent to adding a random forcing term $\sigma\xi(t)$ to the right-hand side of the recharge–discharge oscillator model; the linear system then outputs a red-noise spectrum under random forcing, with a peak near the natural frequency. ☐

Example 38.4 (Concrete calculation of the ENSO oscillation period). Take the coupling coefficients $\alpha = \delta = 1/3$ (yr^{-1}, surface thermal decay timescale about 3 years) and $\beta = \gamma = 1$ (yr^{-1}, east–west boundary wave-reflection coupling timescale about 1 year). By Proposition 38.7, the decay rate is $\rho = (\alpha+\delta)/2 = 1/3$ yr^{-1} and the rotation frequency is

$$\omega = \frac{1}{2}\sqrt{4\beta\gamma - (\alpha-\delta)^2} = \frac{1}{2}\sqrt{4 \times 1 \times 1 - 0} = 1 \text{ yr}^{-1},$$

so the oscillation period is $T = 2\pi/\omega = 2\pi \approx 6.3$ years, which falls within the observed ENSO period band of 2–7 years. If instead $\alpha = \delta = 1/4$ and $\beta = \gamma = 3/4$, then $\omega = \frac{1}{2}\sqrt{4 \times \frac{3}{4} \times \frac{3}{4}} = \frac{3}{4}$ yr^{-1} and $T = 2\pi/\omega \approx 8.4$ years, slightly exceeding the observed upper bound—which shows that the relative strength of coupling and dissipation determines the dominant period of ENSO.

Example 38.5 (Typical ENSO events). Since 1950, several strong ENSO events have been recorded in detail:

(i) **The 1982–1983 El Niño**: the peak SST anomaly in the Niño 3.4 region exceeded $+2.5°$C, making it one of the strongest events of the 20th century; global crop failures, Australian drought and Peruvian floods.

(ii) **The 1997–1998 El Niño**: comparable in strength to the 1982–1983 event, with a peak SST anomaly of $+2.4°$C in the Niño 3.4 region; the eastward displacement of the western Pacific warm pool caused Indonesian forest fires.

(iii) **The 2010–2011 La Niña**: the SST anomaly in the Niño 3.4 region fell to $-1.6°$C; historic floods occurred in north-eastern Australia and major floods in Pakistan.

(iv) **The 2015–2016 El Niño**: the peak SST anomaly in the Niño 3.4 region reached $+2.6°$C, comparable to 1997–1998; the global mean temperature reached a record high in 2016.

(v) **The 2020–2023 'triple' La Niña**: unusually, the La Niña threshold was reached in three consecutive years (2020, 2021, 2022), an important cause of the temporary fall in the global mean temperature and of the persistent Australian floods.

The strength, frequency, phase and duration of ENSO events display pronounced irregularity—not a strict period but a **quasi-periodic** oscillation: the dominant period lies between 2 and 7 years, and the peak is seasonally phase-locked to the northern winter.

Remark 38.9 (Teleconnections of ENSO to the globe). ENSO produces teleconnections to the globe through the atmospheric circulation; typical impacts include:

- During El Niño: a weakened Asian monsoon, droughts in India and South-East Asia, droughts in Australia, floods in Peru and Ecuador, a wetter winter in the southern part of North America, and reduced North Atlantic hurricane activity.
- During La Niña: the impacts are broadly opposite, with a strengthened Asian monsoon, a wetter Australia, droughts in the south-western part of North America and increased North Atlantic hurricane activity.

Such remote impacts are transmitted through atmospheric wave trains (such as the Pacific–North American pattern, PNA) and are one of the main sources of global climate predictability.

38.5 North Atlantic Oscillation (NAO)

Definition 38.14 (North Atlantic Oscillation (NAO)). The **North Atlantic Oscillation** (NAO) is a north–south dipole oscillation of the sea-level pressure field in the North Atlantic region. The NAO index is defined as the difference of the standardised pressure difference between the Icelandic Low and the Azores High. The NAO has two phases:

(i) **Positive phase (NAO$^+$)**: the Icelandic Low deepens anomalously and the Azores High strengthens anomalously—the north–south pressure gradient increases. The mid-latitude westerlies strengthen, northern Europe is warm and wet, and southern Europe is dry with little rain.

(ii) **Negative phase (NAO$^-$)**: the Icelandic Low weakens and the Azores High weakens—the north–south pressure gradient decreases. The mid-latitude westerlies weaken and shift southward, northern Europe is cold and dry, and southern Europe is wet with more rain.

The NAO exhibits variability on timescales from days to decades—synoptic scale (3–10 days), seasonal scale (strongest in winter), interannual scale (no fixed period, with a characteristic timescale of about 7–8 years) and decadal scale.

Proposition 38.8 (Phase transitions of the NAO and its red-noise character). The alternation of the positive and negative phases of the NAO is manifested as the strengthening and weakening of the atmospheric meridional pressure gradient. NAO$^+$ corresponds to a strengthening of the pressure gradient (a deeper Icelandic Low, a stronger Azores High, an increased meridional pressure gradient), and NAO$^-$ to a weakening of the pressure gradient (a smaller pressure difference between the two poles of the dipole).

The time series $N(t)$ of the NAO is statistically approximately a **red-noise** process—its autocorrelation function is $\rho(\tau) \approx e^{-\tau/\tau_0}$, where $\tau_0 \approx$ 7–10 days (the timescale of the internal dynamics of the atmosphere). The red-noise spectrum has more energy in the low-frequency band ($S(\omega) \propto \omega^{-2}$), which explains the variability of the NAO on seasonal to interannual scales.

Proof. Concerning the spectral structure: by Lemma 38.6, $\rho(\tau) = e^{-|\tau|/\tau_0}$ gives $S(\omega) = \frac{2\tau_0}{1+\omega^2\tau_0^2}$—in the band $\omega \gg \tau_0^{-1}$ we have $S(\omega) \approx \frac{2}{\tau_0\omega^2} \propto \omega^{-2}$, that is, low-frequency energy dominates, in agreement with the observed red spectrum of the NAO. □

38.6 Pacific Decadal Oscillation (PDO)

Definition 38.15 (Pacific Decadal Oscillation (PDO))**.** The **Pacific Decadal Oscillation** (PDO) is the dominant decadal mode of sea surface temperature anomalies in the North Pacific; its spatial structure resembles that of ENSO but is more biased towards the mid-latitude North Pacific. The PDO is usually defined as the first mode of an empirical orthogonal function (EOF) analysis of North Pacific SST anomalies, with a dominant period of about 20–30 years.

(i) **Warm phase (PDO$^+$)**: the central North Pacific (about $40°$N) is anomalously warm, the waters off the west coast of North America are anomalously cold, and the Aleutian Low weakens. Winter precipitation on the west coast of North America is below normal and inland temperatures are above normal.

(ii) **Cold phase (PDO$^-$)**: the central North Pacific is anomalously cold, the west coast of North America is anomalously warm, and the Aleutian Low strengthens. Precipitation on the west coast of North America is above normal and inland temperatures are below normal.

The PDO and ENSO exhibit pronounced inter-basin coupling—the phase of the PDO influences the formation and strength of ENSO events, whilst the teleconnections of ENSO in turn modulate North Pacific SST anomalies. Their interaction is a typical instance of Pacific basin-scale climate variability.

Proposition 38.9 (Timescale separation between the PDO and ENSO)**.** *The PDO and ENSO are separated in timescale: the dominant period of ENSO is about 2–7 years and that of the PDO about 20–30 years. The frequency ratio is about $1:4$–$1:10$. Within each complete cycle of the PDO, ENSO completes 4–10 quasi-periodic oscillations. Hence the modulation of ENSO by the PDO can be described by slow–fast variable separation: regarding the PDO as a slowly varying background field $\bar{T}_{\text{PDO}}(t)$ and ENSO as a fast oscillation $T'(t)$ on this background, the two satisfy*

$$\frac{dT'}{dt} = F(T', \bar{T}_{\text{PDO}}), \qquad \frac{d\bar{T}_{\text{PDO}}}{dt} = \varepsilon G(T'),$$

where $\varepsilon \ll 1$ expresses the slow variation of the PDO. The amplitude and frequency of ENSO are modulated by the phase of the PDO—stronger thermocline anomalies during PDO$^+$ increase the ENSO amplitude and slightly shorten its period.

Proof. By the timescale-separation assumption $\varepsilon \ll 1$, the ENSO state $\bar{T}'$ relaxes to a fixed point or limit cycle of the fast subsystem at fixed $\bar{T}_{\text{PDO}}$; on the PDO timescale the statistics of ENSO (amplitude, frequency) are determined by $\bar{T}_{\text{PDO}}$. The statistical average over many ENSO events reflects the slow modulation by the PDO phase. Observational data (such as the relation between the Niño 3.4 index and the PDO index over 1950–2020) indicate that the ENSO amplitude is higher during PDO$^+$, in agreement with the prediction of slow–fast variable separation. □

38.7 Indian Ocean Dipole (IOD)

Definition 38.16 (Indian Ocean Dipole (IOD)). The **Indian Ocean Dipole** (IOD) is an east–west dipole mode of sea surface temperature anomalies in the tropical Indian Ocean. The IOD index is defined as the difference between the SST anomalies of the tropical western Indian Ocean ($50°$E–$70°$E, $10°$S–$10°$N) and the tropical eastern Indian Ocean ($90°$E–$110°$E, $10°$S–$0°$). The IOD has three phases:

(i) **Positive phase (IOD$^+$)**: the western Indian Ocean is anomalously warm and the eastern Indian Ocean anomalously cold, the convection centre shifts westward, precipitation increases in East Africa and decreases in Indonesia and Australia.

(ii) **Negative phase (IOD$^-$)**: the western Indian Ocean is anomalously cold and the eastern Indian Ocean anomalously warm, the convection centre shifts eastward, and the impacts are broadly opposite.

(iii) **Neutral phase**: the dipole index is close to zero.

The quasi-period of the IOD is about 3–7 years, and it is partly coupled with ENSO—ENSO events influence the formation of the IOD through atmospheric teleconnections, but the IOD can also occur independently of ENSO.

Proposition 38.10 (The independent periodic component of the IOD). Although the IOD is partly coupled with ENSO, it possesses a periodic component independent of ENSO. Define the residual of the linear regression of the IOD on ENSO, $\mathsf{IOD}_\perp(t) = \mathsf{IOD}(t) - \beta_{\mathsf{ENSO}} \cdot \mathsf{ONI}(t)$; then the power spectrum of $\mathsf{IOD}_\perp$ still has a significant peak in the 3–7 year band—which shows that the IOD has an intra-basin dynamical mechanism independent of ENSO (such as the coupling of the seasonal–interannual variability of the equatorial Indian Ocean Wyrtki jet).

Proof. A regression analysis of the IOD on the ONI shows that ENSO explains part of the IOD variance (about 30%–50%, depending on the period), and the residual variance is concentrated in the 3–7 year band. Using spectral analysis, a power-spectrum estimate of $\mathsf{IOD}_\perp$ still has its peak at 3–7 years—which shows that an independent component exists. This independent component originates in the seasonal–interannual coupling of the Indian Ocean itself (such as monsoon–ocean interaction) and is independent of Pacific ENSO. □

38.8 Long-range correlation and $1/f$ noise in the climate system

Definition 38.17 (Long-range persistence of climate time series). Many climate time series (temperature, precipitation, drought indices, tree-ring widths, ice-core oxygen isotopes) exhibit **long-range persistence**—the autocorrelation function decays as a power law rather than exponentially: $\rho(\tau) \propto \tau^{-\beta}$ $(0 < \beta < 1)$.

The Hurst exponent $H = 1 - \beta/2$ measures the strength of persistence: $H > 0.5$ indicates positive persistence (past increases tend to persist), $H < 0.5$ indicates anti-persistence (past increases tend to reverse), and $H = 0.5$ indicates white noise (no memory).

$H \neq 0.5$ means that the climate system is not statistically invariant under scaling transformations—$\{X_{\lambda t}\} \overset{d}{\neq} \{\lambda^H X_t\}$ (unless the original series is already self-similar). The Hurst exponent H measures the ratio of the fluctuation strengths of the climate system on different timescales—the larger H is, the stronger the low-frequency fluctuations and the more persistent the long-range memory.

Proposition 38.11 (The origin of $1/f$ noise in the climate system). The power spectral density of climate time series is approximately $S(\omega) \propto \omega^{-\alpha}$ in the low-frequency band ($\alpha \approx 0.5$–1.5). The case $\alpha \approx 1$ corresponds to classical $1/f$ noise, which is the natural result of the superposition of processes on many timescales. The origins of $1/f$ noise in the climate system include:

(i) **Linear superposition of multiscale processes**: the climate system contains processes ranging from the synoptic scale (a few days) to the Milankovitch scale (tens of thousands of years), and fluctuations on different timescales superpose in amplitude to produce a $1/f$ spectrum.

(ii) **Time integration of a random walk**: the spectrum of the cumulative sum of white noise (a random walk) is $S(\omega) \propto \omega^{-2}$. The 'memory' of the climate system (ocean heat capacity, the thermal inertia of ice sheets) integrates random atmospheric perturbations into red noise.

(iii) **Self-organised criticality**: the climate system may be in a self-organised critical state—a tiny random perturbation may trigger a response of any size (from a local storm to a global glacial period), and the distribution of response sizes is a power law.

Proof. (Multiscale superposition produces a $1/f$ spectrum) Let the climate system be a superposition of a family of processes with timescales τ, each process being red noise (Lemma 38.6), and let the timescales be distributed as a power law $p(\tau)d\tau \propto \tau^{-\gamma}d\tau$ ($\tau \in [\tau_1, \tau_2]$, $\tau_1 \ll \tau_2$). The total spectrum is

$$S(\omega) = \int_{\tau_1}^{\tau_2} \frac{2\tau}{1+\omega^2\tau^2}\,\tau^{-\gamma}\,d\tau = \omega^{\gamma-2} \cdot 2\int_{\omega\tau_1}^{\omega\tau_2} \frac{u^{1-\gamma}}{1+u^2}\,du.$$

When $0 < \gamma < 2$ the inner integral converges to a constant as $\omega\tau_1 \to 0$, $\omega\tau_2 \to \infty$, so $S(\omega) \propto \omega^{\gamma-2}$, that is, the spectral exponent is $\alpha = 2 - \gamma$. The case $\alpha = 1$ (classical $1/f$) corresponds exactly to $\gamma = 1$—the timescales are **uniformly distributed on a logarithmic axis** (all scales from weather to glacial periods occur with equal weight). This argument covers both the random-walk time integral ($\gamma = 0$ gives $\alpha = 2$, that is, red noise) and self-organised criticality ($\gamma \approx 1$). □

38.9 Multiscale nested periods of climate change

Theorem 38.7 (The three-layer nested periodic structure of the climate system). The climate system constitutes a three-layer nested periodic structure, corresponding one-to-one to the three driving layers of astronomy, ocean and atmosphere:

Climate period layer	Timescale	Dominant mechanism	Typical examples
Astronomically driven layer	10^4–10^5 years	quasi-periodic variation of the three orbital elements	Milankovitch cycles
Oceanic response layer	10^0–10^3 years	ocean–atmosphere coupling and heat-capacity buffering	ENSO, PDO, AMO, IOD
Atmospheric variability layer	10^{-2}–10^1 years	internal atmospheric dynamics and water-vapour feedback	NAO, SAM, monsoon

The relations among the three layers are described uniformly by the composite-period framework of the 'Periodic Systems' chapter: the astronomically driven layer provides the topmost quasi-periodic driving force (the Milankovitch cycles), the oceanic response layer converts astronomical periods into the storage and release of heat in the middle layer (heat capacity is a buffer of energy), and the atmospheric variability layer responds at the bottom to the redistribution of heat by the ocean and amplifies or suppresses climate signals through water-vapour–cloud–radiation feedback.

Proof. The mathematical basis of the three-layer nested structure is the separation of timescales and the layering of variance. The astronomical periods (10^4–10^5 years) are far longer than the oceanic thermal adjustment time (10^1–10^3 years), and the oceanic thermal adjustment time is in turn far longer than the atmospheric dynamical adjustment time (10^{-2}–10^1 years). On the timescale of the oceanic response layer, the astronomical driving force can be regarded as a slowly varying parameter. On the timescale of the atmospheric variability layer, the sea surface temperature can be regarded as a fixed boundary condition.

From the point of view of the power spectrum, the layered structure of climate variance by frequency is: the total variance $\sigma^2_{\text{total}} = \sigma^2_{\text{astronomical}} + \sigma^2_{\text{oceanic}} + \sigma^2_{\text{atmospheric}}$, where $\sigma^2_{\text{astronomical}}$ is concentrated at 10^{-5}–10^{-4} yr^{-1}, $\sigma^2_{\text{oceanic}}$ at 10^{-3}–10^0 yr^{-1}, and $\sigma^2_{\text{atmospheric}}$ at 10^{-1}–10^2 yr^{-1}. This resembles the amplitudes of the various levels of the nested epicycle system in the 'Periodic Systems' chapter—the astronomical layer corresponds to the outer large epicycles (longest period, largest amplitude) and the atmospheric layer to the inner small epicycles (shortest period, smallest amplitude). □

Remark 38.10 (The period spectrum of climate phenomena). Arranging the principal climate phenomena by timescale:

- 10^{-2}–10^{-1} years: weather systems, the Madden–Julian Oscillation (MJO, 30–90 days).
- 10^{-1}–10^{0} years: the seasonal cycle, the NAO (characteristic timescale 7–8 years).
- 10^{0} years: ENSO (2–7 years), the IOD (3–7 years).
- 10^{1} years: the PDO (20–30 years), the AMO (Atlantic Multidecadal Oscillation, 60–80 years).
- 10^{2}–10^{3} years: centennial to millennial scale fluctuations such as the Little Ice Age and the Medieval Warm Period.
- 10^{4}–10^{5} years: the Milankovitch cycles (precession of 2.3 ten-thousand years, obliquity of 4.1 ten-thousand years, eccentricity of 10 ten-thousand years).
- 10^{5}–10^{6} years: glacial–interglacial cycles and long-term climate evolution.

These phenomena constitute a continuous period spectrum from a few days to a million years.

38.10 Conservation laws of the climate system

Theorem 38.8 (The three conservation laws of the climate system)**.** *The dynamics of the climate system is governed by three conservation laws:*

(i) **Conservation of energy**: *the total energy balance of the Earth's climate system is $\Delta F = \lambda \Delta T$, where ΔF is the radiative forcing (W/m^2), ΔT is the global mean surface temperature response (K), and λ is the climate feedback parameter. The net radiative flux at the top of the atmosphere is close to zero on annual average—the solar radiation absorbed by the Earth equals the infrared radiation emitted by the Earth. Conservation of energy constrains the range of variation of the global mean temperature on long timescales.*

(ii) **Conservation of momentum**: *the total angular momentum of the atmospheric and oceanic circulations is conserved in the Earth–atmosphere–ocean system. The angular momentum transport by the atmospheric circulation (the Hadley circulation, mid-latitude eddies) and by the oceanic circulation (the wind-driven circulation, the thermohaline circulation) transports angular momentum from the tropics towards the poles, maintaining the long-term stability of the Earth's rotation rate and of the direction of its rotation axis. Conservation of angular momentum is a basic physical constraint on the atmospheric and oceanic circulations.*

(iii) **Conservation of water vapour**: *in the global hydrological cycle, evaporation E and precipitation P are equal on annual average—$E_{\text{global}} = P_{\text{global}}$. The residence time of water vapour in the atmosphere is about 9 days—the rapid turnover of water vapour is the physical basis of high-frequency scaling transformations. Water vapour is the most important greenhouse gas of the*

Earth, and its concentration in the atmosphere is coupled to temperature through the Clausius–Clapeyron relation—for every 1°C rise in temperature, the water-holding capacity of the atmosphere increases by about 7%.

The hierarchical relation among the three conservation laws is: conservation of energy is the physical basis for the existence of the climate system (the balance between the input of solar radiation and the output of terrestrial radiation), conservation of momentum is the dynamical constraint on the atmospheric and oceanic circulations (the redistribution of angular momentum among the Earth's component spheres), and conservation of water vapour is the guarantee of the closure of the hydrological cycle (the global balance of evaporation and precipitation). Together they constitute the physical guarantee of the periodic structure of the climate system.

Proof. (i) Conservation of energy: the net solar short-wave radiation received at the top of the Earth's atmosphere is $S_0(1-A)/4 \approx 240$ W/m^2 ($S_0 = 1361$ W/m^2 is the solar constant and $A \approx 0.3$ is the planetary albedo). The infrared radiation output of the Earth is σT_e^4, where $T_e \approx 255$ K is the effective radiating temperature. In equilibrium, $S_0(1-A)/4 = \sigma T_e^4$. The greenhouse effect (water vapour, CO$_2$, clouds and so on in the atmosphere absorb infrared radiation and re-radiate it downwards) makes the surface temperature $T_s \approx 288$ K about 33 K higher than T_e.

(ii) Conservation of momentum: the rate of change of the total angular momentum of the Earth–atmosphere system equals the external torque (the gravitational torque of the Sun and the Moon). On climate timescales the external torque is approximately zero, so the total angular momentum is conserved. The exchange between the atmospheric angular momentum M_{atm} and the angular momentum M_{earth} of the solid Earth drives the seasonal variation of the Earth's rotation rate (the length of day, LOD)—the atmospheric angular momentum is largest in winter (the Earth rotates slightly more slowly) and smallest in summer (the Earth rotates slightly faster).

(iii) Conservation of water vapour: the global mean evaporation is $E \approx 1000$ mm/yr and the global mean precipitation is $P \approx 1000$ mm/yr (ocean evaporation 1200 mm/yr, ocean precipitation 1100 mm/yr, net ocean evaporation 100 mm/yr transported to land; land evaporation 500 mm/yr, land precipitation 745 mm/yr, net land precipitation 245 mm/yr from ocean water-vapour transport). The residence time of water vapour in the atmosphere is $\tau =$ total atmospheric water vapour/global precipitation rate $\approx 12.7 \times 10^{12}$ m^3/(1 m/yr $\times\, 5.1 \times 10^{14}$ m^2) ≈ 9 days. □

Remark 38.11 (Connection with the rest of the book). The theory of the periodic structure of the climate system established in this chapter is an application of periodic mathematics in the Earth's climate sciences. The Milankovitch cycles are the quasi-periodic transmission of astronomical periods to the climate system—the quasi-periodic variation of the three orbital elements drives the alternation of glacial and interglacial periods. ENSO is the quasi-periodic oscillation of the coupled ocean–atmosphere system of the equatorial Pacific—the Bjerknes positive feedback and the delayed negative feedback of the equatorial waveguide together maintain the 2–7 year oscillation period; El Niño (warm phase) and La Niña (cold phase) are its two extreme phases; together with such modes as the PDO (decadal) and the IOD (Indian Ocean) it forms an inter-basin coupling network. The NAO is manifested as the strengthening and weakening of the atmospheric meridional pressure gradient. The long-range persistence ($H > 0.5$) and $1/f$ noise in climate time series reflect the distribution of energy of the climate system on different timescales. The three-layer nested periodic structure of the climate system—astronomical driving, oceanic response and atmospheric variability—is a complex realisation in the Earth's climate of the nested epicycle systems of the 'Periodic Systems' chapter. The three conservation laws—conservation of energy, conservation of momentum and conservation of water vapour—together constitute the physical guarantee of the periodic structure of the climate system.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The interannual advance of the seasonal cycle (the solar declination $\delta(t) = 23.44^\circ \sin(2\pi t/365.24)$, Definition 38.10); the time translation of atmospheric variability
Scaling	The scale invariance of $1/f$ noise (the multiscale superposition proof of Proposition 38.11); the Hurst exponent H of long-range persistence (Definition 38.17)
Rotation	The phase oscillation of the warm and cold phases of ENSO, El Niño and La Niña (Definition 38.12, Proposition 38.7); the east–west dipole oscillation of the IOD (Definition 38.16); the Milankovitch orbital cycles
Conservation	The global energy balance, conservation of Earth–atmosphere angular momentum and conservation of water vapour in the hydrological cycle (Theorem 38.8)
Symmetry	The phase symmetry of the seasons between the two hemispheres: hemispheric reflection ⇔ phase difference π (Proposition 38.5)
Closure	The closure of the Bjerknes positive feedback loop (the loop gain $G > 0$ of Proposition 38.6); the ice–albedo feedback loop; the PDO–ENSO slow–fast coupling loop (Proposition 38.9)
Period	The seasons $T = 365.24$ days (Example 38.3); ENSO 2–7 years (Examples 38.4, 38.5); the PDO 20–30 years; the IOD 3–7 years; the three-layer nested periods (Theorem 38.7)

Exercises

Exercise 1 Prove the **climate periodicity decision theorem** of this chapter (38.1). Hint: first write down the decision map and the evolution group of this chapter's target, then apply the proof structure of the general criterion of the 'Periodic Systems' chapter.

Exercise 2 Verify that the hypotheses of the **timescale-separation decision for the PDO and ENSO** (38.5) hold for the domain objects of this chapter, and give a counterexample in which the hypotheses fail.

Exercise 3 Examine the behaviour of **monotonicity of quantity adjustment** (38.2) under degeneracy of the parameters, and explain how the conclusion degenerates accordingly.

Exercise 4 Explain the relation between the **three-parameter correspondence: radius, phase and speed** (38.1) and the corresponding general conclusions of the 'Periodic Systems' chapter (whether they agree, whether it is a special case, or whether additional conditions are required).

Exercise 5 Using the seven-theme comparison table of this chapter, identify the concrete carriers in this chapter of the three themes **translation**, **scaling** and **rotation**, and give one example of domain data for each.

Exercise 6 Synthesis: restate the diagnostic procedure of this chapter using the five-step standard interface of the 'Periodic Systems' chapter (Table 24.1), noting for each step the theorem of this chapter on which it depends: 38.1, 38.5, 38.2, 38.1.

Chapter 39

Periodic Structure of Economic Systems

Remark 39.1 (Periodicity decision for the target of this chapter (citation)). When this chapter decides whether a research target is periodic, determines its period type, or seeks its minimal positive period, reference is always made to the 'Periodic Systems' chapter: for the periodicity criterion see Theorem 24.1, for the discrimination of period types see Definition 24.3, Proposition 24.1, for the search for the minimal positive period see Definition 24.4, Theorem 24.2, and for the construction of period series and epicycle groups from data see Theorem 24.3, Definition 24.6.

Remark 39.2 (Positioning of this chapter: developing via the five-step interface of the 'Periodic Systems' chapter). The periodic structure of economic systems studies the mathematical characterisation of periodic fluctuations, long-range correlations and structural change in economic activity. This chapter does not establish a separate periodicity criterion; instead it connects economic systems to the five-step interface prescribed in the 'Periodic Systems' chapter:

(1) **Decision**: reduce such targets as macroeconomic variables, price series, investment cycles and long waves to discrete or continuous dynamical systems, invoke Theorem 24.1 to decide whether they are periodic, and determine the period type;

(2) **Construction**: build the amplitude series, frequency series and phase series from data such as GDP, prices, interest rates, unemployment, investment, consumption and the money supply, and assemble them into an economic epicycle group;

(3) **Adjustment**: dynamically adjust the economic epicycle group by residual-energy descent, information criteria (AIC/BIC) and Bayesian model selection;

(4) **Convolution**: introduce amplitude convolution, phase convolution and frequency convolution on the economic epicycle group, and connect them with autocorrelation, cross-correlation and phase coupling;

(5) **Integration**: connect to the tool chains of the three chapters 'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis', forming an economic periodicity diagnosis pipeline.

The mathematical tools used in this chapter include: gradient flows, delayed negative feedback, damped oscillators, second-order linear difference equations, Fourier analysis, window functions, similarity and trend, the refraction index, periodic convolution, epicycle groups and phase operators.

Remark 39.3 (Seven-theme guide for this chapter). **Translation**—the price-translation adjustment of supply–demand equilibrium and the delayed oscillation of the cobweb model (Definition 39.11, Proposition 39.5); **scaling**—forward/reverse scaling by the multiplier effect $1/(1-c)$ and the accelerator v (Definition 39.13), and the Pareto exponent α (Definition 39.18); **rotation**—the expansion–contraction alternation of the business cycle and the phase rotation of momentum–reversal (Definition 39.17); **conservation**—the value identity, the money constraint and the information constraint (Theorem 39.7); **symmetry**—the reflection symmetry of the alternation between high and low prices in the cobweb model (period $T = 2$, Proposition 39.5); **closure**—the closure of the multiplier–accelerator feedback loop and the closure of the reversion of price to the mean (Proposition 39.6, Definition 39.17); **period**—the three-level nesting of the inventory cycle (3–5 years), the investment cycle (7–11 years) and the Kondratiev long wave (45–60 years) (Proposition 39.7), and the multi-window period decomposition of stock prices (Proposition 39.10).

39.1 Periodicity decision for economic targets: the first step of the 'Periodic Systems' chapter

39.1.1 Economic targets, decision inputs and decision maps

Definition 39.1 (Economic targets and decision inputs). An **economic target** is one of the following objects:

(i) a macroeconomic variable sequence, such as gross domestic product $Y(t)$, the price level $P(t)$, the interest rate $r(t)$ or the unemployment rate $u(t)$;

(ii) a market equilibrium system, whose state space is the price–output phase space;

(iii) an investment–consumption feedback system, whose state space is the income–capital phase space;

(iv) a financial market time series, such as a stock price $P(t)$ or a return r_t;

(v) an income distribution, whose state space is the set of values of income or wealth.

The **decision input** of an economic target is the pair

$$\mathcal{D}_{\mathsf{E}} = (\mathcal{T}, x),$$

where $\mathcal{T}$ is the time index set (which may be $\mathbb{Z}$ or $\mathbb{R}$) and $x : \mathcal{T} \to V$ is the value of the economic variable over time. If $\mathcal{T} = \mathbb{Z}$, the target is called a **discrete economic target**; if $\mathcal{T} = \mathbb{R}$, it is called a **continuous economic target**.

Definition 39.2 (Economic decision map). Let an economic target be given by the decision input $\mathcal{D}_{\mathsf{E}} = (\mathcal{T}, x)$. If there exist a map $\Phi : \mathcal{S} \to \mathcal{S}$ and an initial value $x_0 \in \mathcal{S}$ such that the states of the target are generated by Φ, that is,

$$x_t = \begin{cases} \Phi^t(x_0), & t \in \mathbb{Z}, \\ \Phi_t(x_0), & t \in \mathbb{R}, \end{cases}$$

for $t \in \mathbb{Z}$ the state x_t is generated by iteration of Φ; for $t \in \mathbb{R}$ it is described by the one-parameter evolution family $\{\Phi_t\}_{t\in\mathbb{R}}$, where each $\Phi_t : \mathcal{S} \to \mathcal{S}$, $\Phi_0 = \mathsf{id}$, $\Phi_{s+t} = \Phi_s \circ \Phi_t$ for all $s, t \in \mathbb{R}$, and $(t, x) \mapsto \Phi_t(x)$ is continuous. then Φ is called the **economic decision map** of the target. The economic target is thereby reduced to the discrete dynamical system $(\mathcal{S}, \Phi)$ when $t \in \mathbb{Z}$ and to the continuous dynamical system $(\mathcal{S}, \{\Phi_t\}_{t\in\mathbb{R}})$ when $t \in \mathbb{R}$; in both cases the criteria of the 'Operations and Periodicity' chapter and of the 'Periodic Systems' chapter may be applied directly.

Theorem 39.1 (Economic periodicity decision theorem)**.** *Let the decision map of an economic target be Φ, and let $(\mathcal{S}, \Phi)$ be a discrete dynamical system. Then:*

(A) **Sufficient criterion**: *if there exists $m \in \mathbb{N}^+$ such that $\Phi^m = \mathsf{id}$, then the target is a uniformly periodic economic system; every orbit is closed, and the orbit length divides m.*

(B) **Equivalence of the decision**: *the target is a uniformly periodic economic system if and only if* $\mathsf{ord}(\Phi) < \infty$.

(C) **Pointwise criterion**: *the orbit of the target at the state x_0 is closed if and only if x_0 is a periodic point; the orbit length is then the period at that point.*

(D) **Non-eliminability**: *the conjunction of conserved quantities and the symmetry group cannot replace* $\mathsf{ord}(\Phi) < \infty$*; a quasi-periodic economic target is not a periodic economic system.*

Proof. The Theorem 39.1 of this chapter is the domain instantiation of Theorem 24.1 in the 'Periodic Systems' chapter: it suffices to replace the decision map f and the evolution group by Φ of this chapter, and then (A) sufficient criterion, (B) equivalence of the decision, (C) pointwise criterion and (D) non-eliminability hold one by one. By the convention of Remark 24.4, they are not expanded again here. ☐

Corollary 39.1 (Economic period types)**.** *After the decision by Theorem 39.1, an economic target falls into one of the following types:*

(i) **Trivial period type**: *the minimal period is $m_0 = 1$, and the state is a fixed point (for example a static equilibrium);*

(ii) **Strictly periodic type**: *$1 < m_0 < \infty$, and the state is strictly periodic (for example the $T = 2$ oscillation of the cobweb model, or the inventory cycle);*

(iii) **Uniformly periodic type**: $\mathsf{ord}(\Phi) = m < \infty$*, and every orbit length divides m;*

(iv) **Quasi-periodic type**: $\mathsf{ord}(\Phi) = \infty$*; the orbits are not closed but are dense in a compact carrier structure (for example the superposition of Kondratiev long waves, or multiple periods in financial markets);*

(v) **Eventually periodic type**: *there exists N such that the target is periodic for $t \geq N$;*

(vi) **Discrete periodic type**: *the carrier structure is $\mathbb{Z}$ and the normalised period ratio is $\rho = T_x/\Delta x \in \mathbb{Q}$.*

(vii) **Asymptotically periodic (damped) type**: the orbit is not closed, but tends asymptotically to a closed orbit or to a fixed point, the residual decaying as $e^{-\kappa t}$ (see Definition 24.3(vii)).

Among these, (i)–(iii) and (vi) belong to periodic economic systems; (iv) does not belong to periodic economic systems, but can be handled by bounded-approximation tools; (v) is a periodic economic system only in its stable stage; (vii), like (iv), is a type in the limiting sense; it is not a periodic economic system and can be handled with bounded-approximation tools only in its stable stage.

Proof. Each item follows by substituting an economic target into Theorem 39.1 and Definition 24.3 of the 'Periodic Systems' chapter. ☐

Lemma 39.1 (Periodicity decision for the cobweb model)**.** Let the cobweb model be described by the difference equation $P_t = \frac{a-c}{b} - \frac{d}{b}P_{t-1}$. Then:

(i) if $d/b < 1$, the price converges to the equilibrium, and the system is of asymptotically periodic (damped) type (tending to a fixed point);

(ii) if $d/b = 1$, the price oscillates with constant amplitude and period $T = 2$, and the system is of uniformly periodic type;

(iii) if $d/b > 1$, the price diverges, and the system is not periodic.

Proof. By Proposition 39.5, the periodic solution is $P_t = P^* + (P_0 - P^*)(-d/b)^t$. When $d/b < 1$, $(-d/b)^t \to 0$, so the price tends to the equilibrium; when $d/b = 1$, $(-1)^t$ alternates, with period 2; when $d/b > 1$, the modulus diverges. ☐

Lemma 39.2 (Periodicity decision for the multiplier–accelerator model)**.** Let the characteristic equation of the multiplier–accelerator model be $\lambda^2 - c(1+v)\lambda + cv = 0$. If the discriminant satisfies $\Delta < 0$, then income Y_t oscillates as a damped sine wave:

$$Y_t = Y^* + Ar^t \cos(\omega t + \phi), \qquad r = \sqrt{cv}.$$

When $cv < 1$ it converges and is of asymptotically periodic (damped) type; when $cv = 1$ the amplitude is constant (uniformly periodic type), and when $cv > 1$ it diverges.

Proof. By Proposition 39.6, complex conjugate characteristic roots give a damped oscillation; the damping factor $r = \sqrt{cv}$ determines the convergence. ☐

Lemma 39.3 (Periodicity decision for the nesting of the inventory cycle and the investment cycle)**.** The inventory cycle ($T_K \approx 4$ years) and the fixed-capital investment cycle ($T_J \approx 9$ years) form a nested structure: $\omega_K/\omega_J \approx 2.25 \in (2, 3)$. When the two are superposed, the whole is of quasi-periodic type (if the frequency ratio is irrational) or of uniformly periodic type (if the frequency ratio is rational).

Proof. By Proposition 39.7, the frequency ratio $\omega_K/\omega_J = T_J/T_K = 9/4 = 2.25 \in \mathbb{Q}$, so the superposed orbit is closed, with common period $4T_J = 36$ years, and is of uniformly periodic type; only when the frequency ratio is taken as an irrational approximation is it of quasi-periodic type. ☐

Definition 39.3 (Algorithm for the minimal positive period of economic systems). This definition has the same content as that of Chapter 24, section 'Periodicity Decision: Is the Target Periodic?' (Section 24.1), Definition 24.4; to avoid repetition, only the reference anchor is retained here, and the full statement is given there. The only difference here is terminology (economic target / economic sample); the steps of the algorithm and the conclusions are exactly the same.

Theorem 39.2 (Correctness of the algorithm for the minimal positive period of economic systems). This theorem has the same content as that of Chapter 24, section 'Periodicity Decision: Is the Target Periodic?' (Section 24.1), Theorem 24.2; to avoid repetition, only the reference anchor is retained here, and the full statement and proof are given there. The only difference here is terminology (economic target / economic sample); the steps of the algorithm and the conclusions are exactly the same.

Example 39.1 (Examples of economic periodicity decisions).

(i) Cobweb model: $T = 2$ when $d/b = 1$, uniformly periodic type; asymptotically periodic (damped) type when $d/b < 1$.

(ii) Multiplier–accelerator: constant-amplitude oscillation when $cv = 1$, uniformly periodic type.

(iii) Inventory cycle: $T_K \approx 4$ years, uniformly periodic type (locally).

(iv) Investment cycle: $T_J \approx 9$ years, uniformly periodic type (locally).

(v) Kondratiev long wave: $T_{\text{Kond}} \approx 55$ years, quasi-periodic type (subject to the randomness of technological revolutions).

(vi) Stock-price series: nested at multiple scales, quasi-periodic type.

Remark 39.4 (Conserved quantities do not guarantee periodicity). In economic systems, the value identity, the money constraint and the information constraint are auxiliary conditions for periodic structure, but they do not constitute sufficient conditions. For example, a static equilibrium economy that satisfies the value identity will not spontaneously generate periodic oscillations. An economic periodic structure must be decided by Theorem 39.1 or by the lemmas above.

39.1.2 The second step: construction of the economic period series and the epicycle group

Definition 39.4 (Economic amplitude series, frequency series and phase series). Let the state or output sequence of an economic target be $\{z_t\}$. If it can be written as

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)} + \varepsilon_t,$$

then

$$\mathcal{A}_z(t) = \sum_k A_k e^{ikt}, \qquad \mathcal{F}_z(s) = \sum_k \omega_k s^k, \qquad \mathcal{P}_z(t) = \sum_k e^{i(kt+\phi_k)}$$

are called respectively the **amplitude series**, the **frequency series** and the **phase series** of the economic target. The three agree with Definition 6.10, Definition 6.12 and Definition 6.11 of the 'Series and Convolution' chapter.

Definition 39.5 (Economic epicycle group). *Let the period series of an economic target be given by the amplitude series $\{A_k\}$, the frequency series $\{\omega_k\}$ and the phase series $\{\phi_k\}$. For each k define the epicycle*

$$E_k : \theta \mapsto A_k e^{i(\theta+\phi_k)}, \qquad \theta = \omega_k t,$$

whose collection

$$\mathfrak{E}_{\mathsf{E}} = \{(A_k, \omega_k, \phi_k)\}_{k\in\mathbb{Z}}$$

is called the **epicycle group** *of the economic target. Its superposition is the economic period series*

$$z_t = \sum_k E_k(\omega_k t) = \sum_k A_k e^{i(\omega_k t+\phi_k)}.$$

Theorem 39.3 (Construction of the period series from economic data). *Let the output of an economic target be a finite periodic sequence of period N. Then:*

(i) *the output can be expanded as a discrete Fourier series*

$$z_t = \sum_{k=0}^{N-1} c_k e^{i2\pi kt/N}, \qquad c_k = \frac{1}{N}\sum_{t=0}^{N-1} z_t e^{-i2\pi kt/N};$$

(ii) *the amplitude series is given by $|c_k|$, the phase series by* $\arg c_k$ *and the frequency series by $2\pi k/N$;*

(iii) *if the output is a GDP growth-rate sequence, then the fundamental frequency corresponds to the long-run growth trend and the harmonics correspond to the inventory cycle and the investment cycle; if the output is a stock-price sequence, then the harmonics at each order correspond to market periods on different time scales.*

Proof. (i) The discrete Fourier series of a finite periodic sequence is a standard result: the basis functions $\{e^{i2\pi kt/N}\}_{k=0}^{N-1}$ form an orthogonal basis of $\mathbb{C}^N$. (ii) This is read off directly from $c_k = A_k e^{i\phi_k}$. (iii) For a GDP growth-rate sequence, N is the number of samples, the fundamental frequency corresponds to the average growth over the sample length, and the low-order harmonics correspond to the business cycle; for a stock-price sequence the same holds. ☐

Proposition 39.1 (Three-parameter correspondence: radius, phase and angular speed). *The three parameters of the economic epicycle group $\mathfrak{E}_{\mathsf{E}} = \{(A_k, \omega_k, \phi_k)\}_k$ are in one-to-one correspondence with three geometric–economic quantities:*

(i) **Radius corresponds to amplitude**: *the radius $|E_k| = A_k$ of E_k is the amplitude of that economic oscillation component;*

(ii) **Phase corresponds to rotational delay**: *the initial phase ϕ_k corresponds to the time translation $\tau_k = \phi_k/\omega_k$; when $\phi_k > 0$ this component* **leads** *the baseline by τ_k (equivalently, the component coincides with the baseline once the baseline is postponed by τ_k);*

(iii) **Frequency corresponds to rotational speed**: *ω_k is the angular frequency of that economic oscillation component.*

Proof. The Proposition 39.1 of this chapter corresponds exactly to Proposition 24.3 of the 'Periodic Systems' chapter, and the three correspondences—radius to amplitude, phase to rotational delay, frequency to rotational speed—are obtained one by one by translating them to the epicycle group of this chapter. The proof is given there: (i) from $|E_k| = A_k$ and the energy decomposition, (ii) from the phase-translation theorem, (iii) from the definition of the frequency series. $\square$

Example 39.2 (Period series of an economic time series)**.** The GDP growth rate can be decomposed into a long-run trend, the inventory cycle (about 4 years), the investment cycle (about 9 years) and the long wave (about 55 years). In the epicycle group, the long wave corresponds to the outer, large epicycles (low frequency, high amplitude), whilst the inventory cycle corresponds to the inner, small epicycles (high frequency, low amplitude).

Remark 39.5 (The period series is a complex function)*.* Let $z = e^{i\omega_0 t}$. Then the *economic period series*

$$z_t = \sum_k A_k e^{i(\omega_k t+\phi_k)}$$

when all components are commensurable with ω_0 (that is, $\omega_k = k\omega_0$, $k \in \mathbb{Z}$) it can be written as the Laurent series $\sum_k c_k z^k$, where $c_k = A_k e^{i\phi_k}$: in that case the *economic period series is a complex function on the unit circle, and otherwise it is merely a function on the torus.*

39.1.3 The third step: dynamic adjustment of the economic epicycle group

Definition 39.6 (Dynamic adjustment of the economic epicycle group)**.** *Let the current economic epicycle group be* $\mathfrak{E}^{(j)}$, *with partial sum*

$$z^{(j)}(t) = \sum_{k\in\mathfrak{E}^{(j)}} A_k e^{i(\omega_k t+\phi_k)},$$

and residual $r^{(j)} = z - z^{(j)}$. **Dynamic adjustment** *consists of two rules:*

(i) **Cardinality adjustment**: *if the residual energy satisfies* $\|r^{(j)}\|^2 > \delta$, *add one new epicycle according to the peaks of the residual periodogram;*

(ii) **Parameter adjustment**: *update the parameters of the existing epicycles by regularised least squares or an information criterion.*

In economic systems, cardinality adjustment corresponds to adding an economic cycle component (for example adding a newly identified cycle); parameter adjustment corresponds to tuning the amplitude, phase and frequency so as to balance goodness of fit against model parsimony.

Proposition 39.2 (Monotonicity of cardinality adjustment)**.** *After the dominant-frequency component of the residual is added as in* Definition 39.6(i), *the residual energy is monotonically non-increasing:*

$$\|r^{(j+1)}\|^2 \le \|r^{(j)}\|^2,$$

with equality if and only if the residual has no energy at the selected frequency.

Proof. The Proposition 39.2 of this chapter is consistent with the dynamic adjustment mechanism of Definition 24.8 in the 'Periodic Systems' chapter: adding an epicycle removes the component of $r^{(j)}$ along $\hat{\omega}$ from the residual, and the orthogonal decomposition $\|r^{(j)}\|^2 = \|r^{(j+1)}\|^2 + \|\text{the removed component}\|^2$ gives the inequality, with equality if and only if the norm of the removed component is zero. The complete argument is given there. ☐

Theorem 39.4 (Convergence of dynamic adjustment). Let the economic target be a finite periodic sequence of period T_0 and let $\delta = 0$. Then after at most $\lfloor T_0/2 \rfloor + 1$ steps of dynamic adjustment the residual is zero, and the resulting economic epicycle group is the exact Fourier data of the target. If the target is a complex-valued sequence, the number of independent harmonics is T_0, and the corresponding bound is at most T_0 steps.

Proof. The discrete Fourier series of a sequence of period T_0 is complete on $\lfloor T_0/2 \rfloor + 1$ harmonics. At each step one dominant-frequency component of the residual is extracted as in Proposition 39.2, and the residual energy decreases strictly; hence after finitely many steps the residual is zero. ☐

Definition 39.7 (Structural metric adjustment). Let the structural metric of an economic target be

$$\mathcal{I} = (\mathcal{I}_{\text{fit}}, \mathcal{I}_{\text{parsimony}}, \mathcal{I}_{\text{prediction}}) = (\text{residual sum of squares, number of parameters, cross-validatic}$$

Structural-metric adjustment consists in choosing θ in the parameter space Θ: at each step take a neighbouring parameter for which $\mathcal{I}(\theta)$ is **strictly decreasing** in the lexicographic order (minimisation); if no such parameter exists, the process stops, and the parameter is then locally optimal (that is, the residual is reduced first, then the number of parameters is controlled, and finally the prediction error is optimised).

Proposition 39.3 (Finite convergence of structural metric adjustment). Let the parameter space Θ be finite, and let each adjustment make $\mathcal{I}$ strictly decreasing in the lexicographic order. Then the adjustment terminates after finitely many steps at a locally optimal parameter.

Proof. Since Θ is finite, $\mathcal{I}$ takes only finitely many values, and the lexicographic order is a total order on a finite set; a strictly decreasing sequence in that order cannot continue indefinitely, so the adjustment terminates after finitely many steps. At termination there is no neighbouring parameter that makes $\mathcal{I}$ strictly decrease in the lexicographic order, that is, the parameter is locally optimal. ☐

Example 39.3 (Selecting an economic period series by BIC). Fit K epicycles to a GDP growth-rate sequence and compute $\text{BIC}(K) = n \ln(\text{RSS}/n) + K \ln n$. Choose the K that minimises the BIC. Usually $K = 3$–5 suffices to capture the main cycles.

Remark 39.6 (Boundaries of dynamic adjustment). Theorem 39.4 relies on a finite periodic sequence. For a continuous spectrum or a quasi-periodic economic target (such as a long wave or a stock price), no finite epicycle group represents it exactly, and only a bounded approximation is possible. Structural metric adjustment does not guarantee a global optimum.

39.1.4 The fourth step: *economic convolution*

Definition 39.8 (Economic amplitude, phase and frequency convolutions)**.** Let two *economic epicycle groups* be

$$\mathfrak{E} = \{(A_n, \omega_n, \phi_n)\}, \qquad \mathfrak{E}' = \{(B_m, \nu_m, \psi_m)\}.$$

Define:

(i) **Amplitude convolution**: $\mathbb{A}_{mn} = A_n B_m$;

(ii) **Phase convolution**: $\mathbb{P}_{mn} = e^{i(\phi_n - \psi_m)}$;

(iii) **Frequency convolution**: $\mathbb{F}_{mn} = \omega_n \nu_m$;

(iv) **Trend convolution**: $\mathbb{T}_{mn} = A_n B_m e^{i(\phi_n - \psi_m)} = A_n B_m \mathbb{P}_{mn}$.

This agrees with Definition 6.21, Definition 6.22, Definition 6.23 and Definition 6.24 of the 'Series and Convolution' chapter.

Theorem 39.5 (Action of the three convolutions on the *economic epicycle group*)**.** *Let the epicycle groups of two economic targets be* $\mathfrak{E}$ *and* $\mathfrak{E}'$ *respectively. Then:*

(i) amplitude convolution acts only on the radius;

(ii) phase convolution acts only on the phase;

(iii) frequency convolution acts only on the rotational speed;

(iv) trend convolution preserves both amplitude and phase.

Proof. This is read off directly from Definition 39.8. ☐

Proposition 39.4 (Autocorrelation and trend convolution)**.** *Let* $\{z_t\}$ *be an economic period sequence with autocorrelation function*

$$R_z(h) = \sum_t z_t \overline{z_{t+h}}.$$

In the language of the economic epicycle group, the autocorrelation function is the projection of trend convolution onto the diagonal:

$$R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}.$$

Proof. The Proposition 39.4 of this chapter is the realisation in this domain of the convolution-action conclusion of Theorem 24.6 in the 'Periodic Systems' chapter: substituting $z_t = \sum_n A_n e^{i(\omega_n t + \phi_n)}$ into the definition of the autocorrelation, cancelling all off-diagonal cross terms by orthogonality and retaining only the diagonal term of the trend convolution (the factorisation of the three convolutions, Proposition 24.6), which yields $R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}$. The full expansion is given there. ☐

Remark 39.7 (Relation between economic convolution and periodic convolution)*.* Economic convolution is not a new convolution operation, but an instance of the periodic convolution of the 'Series and Convolution' chapter on economic epicycle groups. Autocorrelation is the projection of the trend convolution onto the diagonal; cross-correlation and phase coupling are respectively the concrete manifestations in economics of amplitude convolution and frequency convolution.

39.1.5 The fifth step: integration with discrete sampling, harmonic analysis and time series analysis

Definition 39.9 (Economic periodicity diagnosis pipeline). *For a given economic system, diagnosis is carried out using the framework of periodic mathematics:*

(i) **Periodicity decision**: *verify the period type of the economic target (Theorem 39.1);*

(ii) **Series construction**: *build the amplitude, frequency and phase series and the epicycle group from data such as GDP, prices and interest rates (Theorem 39.3);*

(iii) **Conservation diagnosis**: *verify the value identity, the money constraint and the information constraint (Theorem 39.7);*

(iv) **Convolution diagnosis**: *compute the autocorrelation (the diagonal projection of the trend convolution, Proposition 39.4) and the phase convolution;*

(v) **Integration diagnosis**: *connect to the tool chains of discrete sampling, harmonic analysis and time series analysis.*

Theorem 39.6 (The economic periodicity diagnosis pipeline and its interface with three chapters). *The five steps of Definition 39.9 are connected respectively to:*

(i) *the 'Discrete Sampling and Intercalation of Periodic Signals' chapter Definition 21.1, Theorem 21.3, Theorem 21.10;*

(ii) *the 'Harmonic Analysis' chapter Definition 22.5, Theorem 22.1, Proposition 22.11;*

(iii) *the 'Time Series Analysis' chapter Definition 23.16, Proposition 23.2, Definition 23.19.*

Hence the economic periodicity diagnosis is not an independent method, but an application of the unified tool chain of periodic mathematics to economic systems.

Proof. The Theorem 39.6 of this chapter is the implementation in this domain of the five-step interface of the 'Periodic Systems' chapter (Table 24.1): periodicity is decided by sampling and spectral estimation, the frequency decomposition and the phase kernel use the 'Harmonic Analysis' chapter, and the periodogram peaks, Fisher test and window-function validation use the 'Time Series Analysis' chapter. The diagnostic pipeline is therefore synthesised from the tool chains of the three chapters; the proof is given there. □

Remark 39.8 (The order of operations after integration). The recommended order for an actual diagnosis is: first decide whether the economic target is periodic (Theorem 39.1); then construct the Fourier data and the epicycle group of the economic data (Theorem 39.3); then optimise the parameters using the conservation constraints and dynamic adjustment (Theorem 39.4, Proposition 39.3); then carry out the economic convolution and autocorrelation analysis (Proposition 39.4); and finally output the diagnosis as in Definition 39.9.

39.2 Mathematical foundations of economic systems

Definition 39.10 (Basic quantities of the economic system)**.** An economic system is described by the following basic variables:

(i) **Gross domestic product (GDP)** $Y(t)$: the total value of the final goods and services produced by a country at time t. The growth rate $\dot{Y}/Y$ of GDP is a core indicator of the business cycle.

(ii) **Price level** $P(t)$: the weighted average price of a basket of representative goods and services. The inflation rate $\pi(t) = \dot{P}/P$ measures the rate of decay of the purchasing power of money.

(iii) **Interest rate** $r(t)$: the cost of using borrowed funds. The difference between the nominal interest rate i and the real interest rate $r = i - \pi$ reflects inflation expectations.

(iv) **Unemployment rate** $u(t)$: the proportion of the labour force that is seeking work but has not found it. There is a negative correlation between the unemployment rate and the GDP growth rate (Okun's law).

(v) **Investment** $I(t)$ and **consumption** $C(t)$: the two main components of aggregate expenditure. Investment fluctuates far more widely than consumption and is the main driver of the business cycle.

(vi) **Money supply** $M(t)$: the total quantity of money circulating in the economy. The growth rate of the money supply and the inflation rate are highly correlated in the long run (the quantity theory of money).

The core feature of an economic system is **multi-scale nested periodicity**—periods on different time scales (the inventory cycle, the fixed-capital investment cycle and the Kondratiev long wave) coexist within the same economic system.

39.3 Supply–demand equilibrium and the cobweb model

Definition 39.11 (Supply–demand equilibrium)**.** The core mechanism of a market economy is the **adjustment of supply and demand by price**. Let the demand function be $D(P)$ (demand falls as the price rises, $D' < 0$) and the supply function be $S(P)$ (supply rises as the price rises, $S' > 0$). The equilibrium price P^* satisfies $D(P^*) = S(P^*)$. When the market price deviates from the equilibrium, the excess demand $D(P) - S(P)$ drives the price adjustment:

$$\frac{dP}{dt} = \lambda \cdot (D(P) - S(P)),$$

where $\lambda > 0$ is the speed of price adjustment. When $D(P) > S(P)$ (excess demand) the price rises, and when $D(P) < S(P)$ (excess supply) the price falls.

The price adjustment mechanism can be regarded as a gradient flow on the potential function $V(P) = -\int (D - S)\,dP$: $dP/dt = -\lambda \nabla V(P)$. The equilibrium price is a minimum point of the potential function, and the fluctuations of the market price near the equilibrium point are oscillations in the gradient flow.

Definition 39.12 (Cobweb model)**.** *When there is a time lag in the response of supply to price (output is decided at sowing time for agricultural products, and the price is determined only at harvest), a* **cobweb cycle** *appears in the market. Let the demand function be $D(P_t) = a - bP_t$ and the supply function be $S(P_{t-1}) = c + dP_{t-1}$ (current supply is determined by the previous period's price). The market-clearing condition $D(P_t) = S(P_{t-1})$ gives the difference equation:*

$$P_t = \frac{a-c}{b} - \frac{d}{b}P_{t-1}.$$

The cobweb model is a classic instance of periodic oscillation generated by delayed negative feedback: the current price P_t is determined by the equilibrium of current demand with current supply (which is determined by the previous period's price). When $d/b < 1$ (the demand curve is steeper than the supply curve), the price converges to the equilibrium; when $d/b > 1$, the price diverges; when $d/b = 1$, the price oscillates with constant amplitude.

Proposition 39.5 (Convergence condition for the cobweb cycle)**.** *The periodic solution of the cobweb model is $P_t = P^* + (P_0 - P^*)(-d/b)^t$. The convergence condition is $d/b < 1$ (demand elasticity greater than supply elasticity). On convergence, the price oscillates about the equilibrium with period $T = 2$—a complete 'high price–low price' cycle is completed every two time units. The divergence condition $d/b > 1$ corresponds to price fluctuations that grow period by period, and the system becomes unstable.*

The cobweb model is a coupling of a gradient flow with delayed feedback: the price adjustment drives the system towards equilibrium, and the time lag produces a rotational component in phase space. When the divergence strength (d/b) exceeds 1, the amplifying effect of the rotational component exceeds the restoring effect of the gradient, and the system diverges.

Proof. From the market-clearing condition $a - bP_t = c + dP_{t-1}$, we solve for $P_t = \frac{a-c}{b} - \frac{d}{b}P_{t-1}$. The equilibrium price P^* satisfies $P^* = \frac{a-c}{b} - \frac{d}{b}P^*$, whence $P^* = \frac{a-c}{b+d}$. Let $\Delta P_t = P_t - P^*$; then $\Delta P_t = -\frac{d}{b}\Delta P_{t-1}$. By recursion, $\Delta P_t = (-\frac{d}{b})^t \Delta P_0$. When $d/b < 1$, $|\Delta P_t| \to 0$ (convergence). On convergence, the sign of ΔP_t reverses once each period—high-price periods alternate with low-price periods, with period 2. ☐

39.4 The multiplier–accelerator model

Definition 39.13 (Multiplier–accelerator model)**.** *Samuelson's multiplier–accelerator model explains the endogenous driving mechanism of the business cycle. The model consists of three equations:*

(i) **Consumption function**: $C_t = cY_{t-1}$ (current consumption is determined by the previous period's income, and $c \in (0, 1)$ is the marginal propensity to consume).

(ii) **Investment function**: $I_t = v(C_t - C_{t-1}) + I_0$ (induced investment is driven by the change in consumption, $v > 0$ is the accelerator, and I_0 is autonomous investment).

(iii) **Income identity**: $Y_t = C_t + I_t + G_t$ (in a closed economy government purchases G_t are exogenous).

Combining these gives the second-order linear difference equation: $Y_t - c(1 + v)Y_{t-1} + cvY_{t-2} = I_0 + G_t$.

The core mechanism of the multiplier–accelerator model: the multiplier effect $(1/(1-c))$ is forward scaling—1 unit of autonomous expenditure generates an increase of $1/(1-c)$ units in total income through multiple rounds of the consumption–income cycle. The accelerator effect (v) is reverse scaling—a deceleration in consumption growth triggers a fall in investment (the acceleration principle). The coupling of the multiplier and the accelerator generates periodic fluctuations in income Y_t—the amplitude and persistence of the fluctuations depend on the values of the parameters c and v.

Proposition 39.6 (Periodic solutions of the multiplier–accelerator model). The characteristic equation of the multiplier–accelerator model is $\lambda^2 - c(1+v)\lambda + cv = 0$. The condition for the existence of periodic solutions is that the discriminant satisfies $\Delta = c^2(1+v)^2 - 4cv < 0$. When the characteristic roots are complex conjugates, income Y_t oscillates about the equilibrium in the form of a damped sine wave:

$$Y_t = Y^* + Ar^t \cos(\omega t + \phi),$$

where $r = \sqrt{cv}$ is the damping factor and $\omega = \arccos(c(1+v)/(2\sqrt{cv}))$ is the angular frequency of the oscillation. When $cv < 1$, $r < 1$ (damped oscillation, converging to the equilibrium); when $cv = 1$, $r = 1$ (constant-amplitude oscillation); when $cv > 1$, $r > 1$ (divergent oscillation).

The periodic solution of the multiplier–accelerator model is a coupling of rotational feedback and damped decay: the term $c(1+v)$ is the feedback coupling (current income feeds back to the next period's income through consumption and investment), and the term cv is the damping (the second-order lag damps the first-order lag). The period $T = 2\pi/\omega$ is jointly determined by c and v—the larger c is (the higher the marginal propensity to consume) and the larger v is (the larger the accelerator), the longer the period.

Proof. The roots of the characteristic equation $\lambda^2 - c(1+v)\lambda + cv = 0$ are $\lambda_{1,2} = \frac{c(1+v)\pm\sqrt{c^2(1+v)^2-4cv}}{2}$. When the discriminant satisfies $\Delta < 0$, $\lambda_{1,2} = \sqrt{cv} \cdot e^{\pm i\omega}$, where $\omega = \arctan(\sqrt{4cv - c^2(1+v)^2}/c(1+v))$. The damping factor is $r = |\lambda| = \sqrt{cv}$. The oscillation period is $T = 2\pi/\omega$. When $cv < 1$, $r < 1$ and the oscillation decays. When $cv = 1$, $r = 1$ and the oscillation has constant amplitude. □

39.5 The inventory cycle and the fixed-capital investment cycle

Definition 39.14 (Inventory cycle (Kitchin cycle)). The **inventory cycle** (Kitchin cycle) is the shortest business cycle in the economy, lasting about 3–5 years. The driving force of the inventory cycle is the adjustment of inventory levels by firms: when demand unexpectedly increases, firms deplete inventories to meet orders and then increase production to replenish them (inventory investment grows);

when demand unexpectedly falls, firms accumulate unintended inventories and then cut production to work them off (inventory investment contracts). The periodic fluctuation of inventory investment $I_{\text{inv}}(t)$ is the core of the inventory cycle.

The inventory cycle is the realisation of a gradient flow in the behaviour of individual firms: when the inventory level deviates from the target inventory, the firm adjusts production so that inventories return to target—$dI_{\text{inv}}/dt = -\alpha(H - H^*)$ (where H is actual inventory and H^* is target inventory). The time lag in inventory adjustment (the time from order to delivery) introduces delayed feedback and produces a periodic oscillation of 3–5 years.

Definition 39.15 (Fixed-capital investment cycle (Juglar cycle)). The **fixed-capital investment cycle** (Juglar cycle) is a medium-length business cycle, lasting about 7–11 years. The driving force of the fixed-capital investment cycle is the renewal of equipment and the expansion of capacity by firms: during a boom firms invest heavily in new equipment (an investment boom); after overcapacity appears investment plunges (an investment trough); once the old equipment is fully depreciated a new round of renewal begins (investment recovery).

The fixed-capital investment cycle is the realisation of rotational motion in aggregate investment behaviour: investment $I(t)$ and the capital stock $K(t)$ form a limit cycle in phase space—during booms investment is high and capital grows fast, whilst during recessions investment is low and capital depreciates. The length of the investment cycle is determined by the average service life of equipment (about 10 years).

Proposition 39.7 (Nested structure of inventory and investment cycles). The inventory cycle (3–5 years) is nested within the fixed-capital investment cycle (7–11 years). A typical Juglar cycle contains 2–3 Kitchin cycles. This nested structure corresponds to a multi-level epicycle system: the inventory cycle is the inner, fast, small epicycle (high frequency, low amplitude), whilst the investment cycle is the outer, slow, large epicycle (low frequency, high amplitude).

Let the dominant frequency of the inventory cycle be $\omega_K = 2\pi/T_K$ ($T_K \approx 4$ years) and that of the investment cycle be $\omega_J = 2\pi/T_J$ ($T_J \approx 9$ years). The frequency ratio of the two is $\omega_K/\omega_J \approx 2.25$—the inventory cycle completes about 2–3 full cycles within each phase of the investment cycle. When the two are in phase (booms superimposed), the economy overheats; when the two are in antiphase (an inventory recession superimposed on an investment recession), the economy is depressed.

Proof. The frequency ratio $\omega_K/\omega_J = T_J/T_K \approx 9/4 = 2.25$, that is, one Juglar cycle contains 2.25 Kitchin cycles. The phase difference $\Delta\theta(t) = (\omega_K - \omega_J)t$ evolves at the beat frequency $\omega_K - \omega_J = 2\pi(1/T_K - 1/T_J)$; the instants at which the two periods are in phase are $t = nT_K T_J/(T_J - T_K) \approx 7.2n$ years, whilst at $t = \text{lcm}(T_K, T_J)\cdot n = 36n$ years the joint state repeats exactly—at these instants the booms are superimposed and the economy overheats; at antiphase (phase difference π) an inventory recession is superimposed on an investment recession, and the economy is depressed. The nested structure is precisely $\omega_K/\omega_J \approx 2.25 \in (2, 3)$: the inner cycle completes about 2–3 full cycles within the outer cycle. □

39.6 The Kondratiev long wave

Definition 39.16 (Kondratiev long wave)**.** *The* **Kondratiev long wave** *(Kondratiev wave) is the longest cycle in the economy, lasting about* 45*–*60 *years. Kondratiev held that the capitalist economy exhibits long-run fluctuations of about half a century—each long wave comprises a rising phase (about* 25 *years) and a falling phase (about* 25 *years). The driving force of the long wave is the clustered emergence and diffusion of major technological innovations:*

(i) **First wave (about 1780–1840)**: *the steam engine and the mechanisation of textiles—the Industrial Revolution.*

(ii) **Second wave (about 1840–1890)**: *railways and steel—the railway age.*

(iii) **Third wave (about 1890–1940)**: *electricity, the internal combustion engine and the chemical industry—the age of electrification.*

(iv) **Fourth wave (about 1940–1980)**: *petrochemicals, electronics and nuclear energy—the post-war boom.*

(v) **Fifth wave (about 1980–2020)**: *information technology and the internet—the digital economy.*

(vi) **Sixth wave (from about 2020)**: *artificial intelligence, renewable energy and biotechnology.*

The Kondratiev long wave is a scaling cycle driven by technological innovation: technological innovation is forward scaling—breakthroughs in new technology bring stepwise growth in productivity; the saturation of technology diffusion is reverse scaling—the old technological paradigm reaches its limit and growth slows. The 45*–*60 *year period of each long wave is determined by the life cycle of a major technology from invention to maturity.*

Proposition 39.8 (Fourier analysis of the Kondratiev long wave)**.** *Fourier analysis of a time series of world GDP growth rates (about* 200 *years) identifies a dominant period of about* 55 *years (with a peak near* 0.018 *year*$^{-1}$*). The spectral energy of the Kondratiev long wave is concentrated in the low-frequency band—this is the manifestation of* $1/f$ *noise in macroeconomics. The length of the long wave,* $T_{\mathsf{Kond}} \approx 55$ *years, is roughly equal to the working life of two generations—one generation enters the labour market during the technological upswing and retires when the technology matures.*

The Kondratiev long wave is the outer structure of the nested business cycles: one long wave contains about 5–6 Juglar investment cycles and 10–15 Kitchin inventory cycles. The three-level nested structure (Kitchin $\subset$ Juglar $\subset$ Kondratiev) is the macroscopic manifestation of multi-scale periodic coupling in economic systems.

39.7 The periodic structure of financial markets

Definition 39.17 (Momentum and reversal of asset prices)**.** Asset prices in financial markets display an alternation of the **momentum effect** (continuation of trend) and **mean reversion** (reversal of trend) on different time scales:

(i) **Short-term momentum** (1–12 months): stocks that have performed well in the past tend to continue to perform well in the short term—the price trend continues. This is a translation of price along the direction of the trend.

(ii) **Long-term reversal** (3–5 years): stocks that have performed well in the past tend to perform worse in the long term—the price trend reverses. This is a rotation of price in phase space, returning from an overvalued region to a fair valuation.

The alternation of momentum and reversal is the coupling of the trend component and the rotational component in asset pricing: momentum corresponds to price inertia, and reversal corresponds to the reversion of price to the mean. The duration of the momentum effect, about 12 months, is determined by investors' underreaction to information and the gradual diffusion of information. The duration of the reversal effect, about 3–5 years, is determined by investors' overreaction to information and the mean reversion of valuations.

Proposition 39.9 (Autocorrelation structure of price momentum)**.** The returns of a momentum strategy display positive autocorrelation over holding periods of 1–12 months: $\rho(h) > 0$ ($h = 1, \dots, 12$ months). The returns of a reversal strategy display negative autocorrelation over holding periods of 3–5 years: $\rho(h) < 0$ ($h = 36, \dots, 60$ months). The change of sign of the autocorrelation function $\rho(h)$ (from positive to negative) is the mark of a phase transition in the market cycle—intervals of positive correlation correspond to the dominance of momentum, and intervals of negative correlation to the dominance of reversal.

The autocorrelation structure of price is related to phase similarity: when $\rho(h) > 0$, the directions of the price at the two instants agree ($\cos(\theta_t - \theta_{t+h}) \approx 1$); when $\rho(h) < 0$, the directions of the price at the two instants are opposite ($\cos(\theta_t - \theta_{t+h}) \approx -1$).

Proof. The autocorrelation function is defined as $\rho(h) = \mathrm{Corr}(r_t, r_{t+h})$ (r_t being the return). The empirical regularities of momentum and reversal (Jegadeesh–Titman 1993; De Bondt–Thaler 1985) give the observed fact that $\rho(h)$ is positive at short lags and negative at long lags—the numerical intervals of this proposition (1–12 months, 36–60 months) are a citation of that empirical regularity rather than a derivation from a theorem. Their correspondence with similarity is at the level of an identity: $\cos(\theta_t - \theta_{t+h})$ is the inner product of the direction vectors, and its sign agrees with the common direction of the returns. □

39.8 Income distribution and the power-law exponent

Definition 39.18 (Pareto distribution)**.** The **Pareto distribution** describes the power-law distribution of income and wealth among high-income groups. Let

X be income or wealth; its complementary cumulative distribution function is

$$\mathbb{P}(X > x) = \left(\frac{x_{\mathsf{min}}}{x}\right)^{\alpha}, \quad x \geq x_{\mathsf{min}} > 0, \quad \alpha > 0,$$

where α is the **Pareto exponent**. The smaller α is, the more unequal the income distribution (a minority holds most of the wealth). Empirical studies show that the Pareto exponent α of income distributions lies between 1.5 and 2.5 across countries, and remains relatively stable across countries and across periods.

The Pareto distribution has scale invariance: $\mathbb{P}(X > \lambda x) = \lambda^{-\alpha}\mathbb{P}(X > x)$—the tail of the income distribution retains its form under scaling transformations. The stability of α (across countries and across periods) is empirical evidence for the stability of the power-law exponent in economic systems.

Remark 39.9 (Pareto exponent and the economic growth cycle (empirical observation)). The Pareto exponent α fluctuates slowly about its mean in the long run, and its fluctuation period is synchronised with the Kondratiev long wave. In the phase in which a technological revolution breaks out (the rising phase of the long wave), α falls (inequality intensifies)—innovators obtain excess returns; in the phase in which technology diffusion matures (the falling phase of the long wave), α rises (inequality is alleviated)—the technological dividend diffuses to the masses. The fluctuation period of α is about 50–60 years, consistent with the Kondratiev long wave.

The periodic fluctuation of the Pareto exponent is the periodic modulation of the power-law exponent in the dimension of economic inequality—the 'creative destruction' of technological innovation (Schumpeterian creative destruction) manifests itself at the macroeconomic level as the periodic rise and fall of the power-law exponent. The above is an empirical observation and an analogical explanation, not a mathematical theorem; its rigour depends on the availability of long-run income distribution data.

39.9 Identities and constraints of the economic system

Theorem 39.7 (The three identities and constraints of the economic system). The dynamics of an economic system is governed by three kinds of identity and constraint:

(i) **Value identity (gradient flow)**: in a closed economy, total output Y equals total income Y equals total expenditure Y—the national income accounting identity $Y = C + I + G + (X - M)$. Value is not created or destroyed out of nothing; it can only be transferred between sectors. In the long run, the growth of labour productivity is the fundamental source of real wage growth—value creation is driven by technological progress.

(ii) **Money constraint (divergence flow)**: the quantity equation of money $MV = PY$ describes the relation among the money supply M, the velocity of money V, the price level P and real output Y. In the long run the velocity of money V is relatively stable, and when the money supply M grows faster than real output Y, the price level P rises (inflation). The change in the money stock = money issuance − money withdrawal—the central bank regulates the money supply through monetary policy.

(iii) **Information constraint (rotational flow)**: market prices contain all the information of market participants about supply and demand (the efficient market hypothesis). In a strongly efficient market, prices already reflect all public and private information—no one can obtain excess returns by analysing available information. The transmission of information among market participants satisfies a constraint—one person's informational advantage necessarily comes at the cost of another's informational disadvantage. Insider trading is an instance in which the information constraint is violated.

The hierarchical relation among the three identities and constraints is as follows: the value identity is the substantive basis of economic activity, the money constraint is the constraint on nominal variables, and the information constraint is the theoretical guarantee of market efficiency. Together the three constitute the guarantee of self-consistency of the periodic structure of economic systems.

Proof. (i) Value identity: the national income accounting identity $Y = C + I + G + NX$ ($NX = X - M$ being net exports) is an accounting identity—every expenditure corresponds to an income. The three methods of computing GDP (the production method, the income method and the expenditure method) give the same result in theory. In a closed economy, $Y = C + S + T$ (income = consumption + saving + taxes) and $Y = C + I + G$ (expenditure = consumption + investment + government purchases); subtracting the two equations gives $S - I = G - T$ (private-sector surplus = government-sector deficit).

(ii) Money constraint: the quantity equation of money $MV = PY$ is an identity—the left-hand side is the total value of money changing hands, and the right-hand side is the total value of goods traded. Taking logarithmic differences: $\Delta \ln M + \Delta \ln V = \pi + \Delta \ln Y$. In the long run $\Delta \ln V \approx 0$ (the velocity of circulation is stable) and $\Delta \ln Y \approx 2\%$–3% (the long-run growth trend), so $\pi \approx \Delta \ln M - \Delta \ln Y$—the inflation rate is approximately equal to the money growth rate minus the real growth rate.

(iii) Information constraint: Fama's efficient market hypothesis has three levels. Weak efficiency: prices reflect all historical trading information—technical analysis is ineffective. Semi-strong efficiency: prices reflect all public information—fundamental analysis is ineffective. Strong efficiency: prices reflect all information (including inside information)—no one can obtain excess returns. Empirically, markets are usually semi-strong efficient—public information is rapidly absorbed by prices, but inside information and long-term reversal anomalies exist. □

39.10 Window-function periodicity diagnosis of the stock market

Definition 39.19 (Cyclical structure of stock prices). The time series of a stock price $P(t)$ contains multiple nested periods: intraday periods (intraday fluctuations between the open and the close), weekly periods (the beginning-of-week and end-of-week effect), monthly periods (fund flows at the beginning and the end of the month), quarterly periods (expectations and realisations in the earnings season) and annual periods (the January effect, the Santa Claus rally). These periods are superimposed on the price chart and form complex fluctuation patterns. The stock price $P(t)$ can be expanded as a nested epicycle system:

$$P(t) = \sum_{k=1}^{K} A_k(t) e^{i(\omega_k t + \phi_k(t))},$$

where the amplitudes $A_k(t)$ and phases $\phi_k(t)$ of the epicycles at each level vary slowly with time—this corresponds to the evolution of the trend and the formation of inflection points.

Definition 39.20 (Application of window functions in stock-price periodicity diagnosis)**.** *The window function $W(t;\Delta)$ is used in stock-price analysis to extract periodic signals on a specific time scale. For a given window radius Δ, the* **window-function smoothed trend** *and the* **window-function local fluctuation** *of the stock price $P(t)$ are respectively:*

$$P_{\text{trend}}(t;\Delta) = \int_{-\infty}^{\infty} P(t+\tau) K\left(\frac{\tau}{\Delta}\right) \frac{d\tau}{\Delta},$$

$$P_{\text{fluct}}(t;\Delta) = P(t) - P_{\text{trend}}(t;\Delta),$$

where K is the kernel function (Gaussian kernel, Mexican-hat kernel, exponential kernel, etc.). The effect of a window function is similar to that of a moving average, but the kernel function K assigns different weights to observations at different time distances. The window radius Δ controls the time scale—the smaller Δ is, the more sensitive the trend (capturing short-term fluctuations); the larger Δ is, the smoother the trend (revealing the long-term direction).

Window-function smoothing is a degenerate form of periodic convolution on a univariate time series—taking $f = g = P$, $\alpha = 1$, $d = 1$ and the kernel function K as the weight kernel, the output is the trend component. The window-function local fluctuation P_{fluct} is the difference between the original signal and the trend—it corresponds to the directional trend in periodic convolution ($d = 2$, the sine of the difference angle) and measures the strength of the rotation of the price away from the trend.

Lemma 39.4 (Spectral-filtering property of window-function trend extraction)**.** *Window-function smoothing*

$$P_{\text{trend}}(t;\Delta) = \int P(t+\tau) K(\tau/\Delta)\, d\tau/\Delta$$

is the convolution of P with the kernel K_Δ, whose frequency-domain form is

$$\widehat{P_{\text{trend}}}(\omega;\Delta) = \hat{K}(\Delta\omega) \cdot \hat{P}(\omega).$$

That is, smoothing with window radius Δ is, in the frequency domain, filtering with $\hat{K}(\Delta\omega)$ as the passband: low frequencies $|\omega| \lesssim 1/\Delta$ are retained and high frequencies suppressed—the larger Δ is, the narrower the passband.

Proof. By the Fourier property of convolution $\widehat{P * K_\Delta}(\omega) = \hat{P}(\omega) \cdot \widehat{K_\Delta}(\omega)$, where $\widehat{K_\Delta}(\omega) = \hat{K}(\Delta\omega)$ (the Fourier transform of a scaled variable). In the Gaussian-kernel case $\hat{K}(\Delta\omega) = e^{-\Delta^2\omega^2/2}$, and the passband width is $\sim 1/\Delta$. □

Proposition 39.10 (Periodic decomposition by superposing multiple window functions)**.** Using a family of window functions with different radii $\{\Delta_1, \Delta_2, \dots, \Delta_K\}$ ($\Delta_1 < \Delta_2 < \cdots < \Delta_K$), a multi-scale period decomposition of a stock price can be carried out. The trend components $\{P_{\text{trend}}(t; \Delta_k)\}_{k=1}^{K}$ extracted by the respective window functions correspond to periodic structures on different time scales:

(i) **Small window** ($\Delta_1 \approx 5$–20 trading days): extracts intraday to weekly periods—corresponding to high-frequency trading signals.

(ii) **Medium window** ($\Delta_2 \approx 50$–120 trading days): extracts monthly to quarterly periods—corresponding to medium-term trends.

(iii) **Large window** ($\Delta_3 \approx 250$ trading days): extracts annual periods—corresponding to long-term trends.

Multi-window decomposition is the realisation of a nested epicycle system in stock-price analysis—the small window corresponds to the inner, fast, small epicycles (high frequency, low amplitude), and the large window to the outer, slow, large epicycles (low frequency, high amplitude). The periodic convolution $\mathcal{K}^{(d)}\{P_{\text{trend}}(\cdot; \Delta_i), P_{\text{trend}}(\cdot; \Delta_j)\}$ between the trend components $P_{\text{trend}}(t; \Delta_k)$ extracted by the respective window functions measures the strength of the phase coupling between trends on different time scales.

Proof. By Lemma 39.4, the smoothing with window Δ_k retains, in the frequency domain, the low-frequency band $|\omega| \lesssim 1/\Delta_k$. Taking $\Delta_1 < \Delta_2 < \Delta_3$, the three trend components $P_{\text{trend}}(\cdot; \Delta_k)$ carry in turn the information of the high-frequency band, the medium-frequency band and the low-frequency band—the small window extracts short periods and the large window long periods, forming a multi-scale decomposition stratified by frequency band. The periodic convolution $\mathcal{K}^{(d)}\{P_i, P_j\}$ between a pair of components computes the phase interaction of the trends in the two frequency bands; when $i \neq j$ the two bands do not overlap, and the non-trivial values of $\mathcal{K}^{(d)}$ are the cross-scale couplings (such as the phase alignment of the monthly trend with the annual trend). □

39.11 Inflection-point detection: refraction index, similarity and trend

Definition 39.21 (Mathematical definition of stock inflection points)**.** An **inflection point** of a stock price is a critical point at which the price trend passes from one state to another. In periodic mathematics, an inflection point corresponds to a jump in the phase $\theta(t) = \arctan(\dot{P}(t)/P(t))$—when the absolute value of $\dot{\theta}(t)$ exceeds a threshold, the direction or the rate of the trend changes markedly. Inflection points fall into two types:

(i) **Peak inflection point** (from rise to fall): $\dot{P}(t) > 0$ becomes $\dot{P}(t) < 0$, and the phase $\theta(t)$ passes from positive (rising) to negative (falling). The corresponding sine difference-angle similarity $\text{Similarity}^{(2)}(\theta_t, \theta_{t-\Delta t})$ changes from positive to negative.

(ii) **Trough inflection point** (from fall to rise): $\dot{P}(t) < 0$ becomes $\dot{P}(t) > 0$, and the phase $\theta(t)$ passes from negative (falling) to positive (rising). The corresponding sine difference-angle similarity $\mathsf{Similarity}^{(2)}(\theta_t, \theta_{t-\Delta t})$ changes from negative to positive.

Definition 39.22 (Refraction index: *a measure of momentum change at inflection points*)**.** The **refraction index** $\eta(t)$ borrows from Snell's law in optics and is used to measure the **strength of the change in momentum** of a stock price at an inflection point. Let the trend directions in the two time windows before and after the inflection point be v_{in} (the incident direction) and v_{out} (the refracted direction), with trend strengths (momenta) m_{in} and m_{out} respectively. The refraction index is defined as

$$\eta(t) = \frac{m_{\text{out}}}{m_{\text{in}}} = \frac{\|\mathsf{v}_{\text{out}}\|}{\|\mathsf{v}_{\text{in}}\|},$$

where $\mathsf{v}_{\text{in}} = P_{\text{trend}}(t;\Delta) - P_{\text{trend}}(t-\Delta;\Delta)$ (the trend momentum before the inflection point) and $\mathsf{v}_{\text{out}} = P_{\text{trend}}(t+\Delta;\Delta) - P_{\text{trend}}(t;\Delta)$ (the trend momentum after the inflection point). $\eta > 1$ means that the momentum strengthens after the inflection point (accelerating rise or accelerating fall), $\eta < 1$ means that the momentum weakens after the inflection point (decelerating rise or decelerating fall), and $\eta = 1$ means that the momentum is unchanged (uniform motion).

The refraction index is the generalisation of the law of refraction (the scaling of sine trends) to the analysis of stock inflection points: the trend momentum m_{in} before the inflection point and the trend momentum m_{out} after it are related through the refraction index η—$\eta = m_{\text{out}}/m_{\text{in}}$. When $\eta > 1$ the momentum is amplified; when $\eta < 1$ the momentum decays.

Proposition 39.11 (Sine difference-angle similarity and the inflection-point criterion)**. Sine difference-angle similarity** is the most sensitive indicator for detecting stock inflection points. Let θ_t be the limit angle of the logarithmic return of the price at time t: $\theta_t = \arctan(r_t)$, where $r_t = \ln(P_t/P_{t-1})$. The **sine difference-angle directional similarity** between time t and time $t - \Delta t$ is

$$\mathsf{Similarity}^{(2)}(\theta_t, \theta_{t-\Delta t}) = \sin(\theta_t - \theta_{t-\Delta t}).$$

The inflection-point criterion is:

(i) $\mathsf{Similarity}^{(2)} < 0$ **(the sine difference angle is negative)**: $\theta_t < \theta_{t-\Delta t}$—the current phase lags behind the past phase, and the direction of the trend is reversing. This is a **necessary condition for an inflection point**.

(ii) $\mathsf{Similarity}^{(2)}$ **crosses the zero line (from positive to negative, or from negative to positive)**: the trend direction passes from continuation (positive) to reversal (negative), or from reversal (negative) to continuation (positive). The crossing instant corresponds to the **exact location of the inflection point**.

(iii) **The absolute value** $|\mathsf{Similarity}^{(2)}|$ **of** $\mathsf{Similarity}^{(2)}$ **exceeds a threshold** η_{th} **(such as** 0.5**)**: the strength of the trend reversal is large enough, and the inflection-point signal is reliable.

Sine difference-angle similarity is a direct application of rotational motion in inflection-point detection: $\sin(\theta_t - \theta_{t-\Delta t})$ measures the rotation of the price phase over the time Δt—a positive value indicates that the phase advances (trend continuation), and a negative value that the phase retreats (trend reversal). An inflection point is the critical instant at which the phase drift exceeds the stable range.

Proof. The logarithmic return $r_t = \ln(P_t/P_{t-1})$ is approximately equal to the percentage change in price. The limit angle $\theta_t = \arctan(r_t)$ maps the return into the interval $(-\pi/2, \pi/2)$. During a period of trend continuation, $\theta_t \approx \theta_{t-\Delta t}$ and $\sin(\theta_t - \theta_{t-\Delta t}) \approx 0$. At an inflection point, θ_t and $\theta_{t-\Delta t}$ have opposite signs (a rise becoming a fall, or a fall becoming a rise), and $\sin(\theta_t - \theta_{t-\Delta t})$ departs from the zero line. At a peak inflection point $\theta_t < 0 < \theta_{t-\Delta t}$, so $\sin(\theta_t - \theta_{t-\Delta t}) < 0$. At a trough inflection point $\theta_t > 0 > \theta_{t-\Delta t}$, so $\sin(\theta_t - \theta_{t-\Delta t}) > 0$. Hence the change of sign of the sine difference-angle similarity marks the inflection point precisely. □

Proposition 39.12 (Trend similarity and trend-strength confirmation). **Difference-angle cosine similarity** (basic difference-angle value similarity) $\text{Similarity}^{(1)}(\theta_t, \theta_{t-\Delta t}) = \cos(\theta_t - \theta_{t-\Delta t})$ measures the consistency of the trend. In a strong trend, $\cos(\theta_t - \theta_{t-\Delta t}) \approx 1$ (the phases agree and the trend continues). Near an inflection point, $\cos(\theta_t - \theta_{t-\Delta t})$ falls, indicating that the coherence of the trend is weakening. The **double-angle similarities** $\text{Similarity}^{(3)}(\theta_t, \theta_{t-\Delta t}) = \cos(2(\theta_t - \theta_{t-\Delta t}))$ and $\text{Similarity}^{(4)}(\theta_t, \theta_{t-\Delta t}) = \sin(2(\theta_t - \theta_{t-\Delta t}))$ measure respectively the second-order trend consistency and the second-order trend reversal strength, and have an amplified detection effect on inflection points.

The combined criterion for confirming an inflection point is:

(i) $\text{Similarity}^{(2)} < 0$ (the sine difference angle is negative—a necessary condition for an inflection point).

(ii) $\text{Similarity}^{(1)}$ falls from ≈ 1 (the coherence of the trend weakens—an auxiliary condition for an inflection point).

(iii) $|\,\text{Similarity}^{(4)}\,|$ increases (the second-order reversal strength is amplified—a reinforcing condition for an inflection point).

(iv) The refraction index $\eta(t)$ departs from 1 (the momentum changes—a strength condition for an inflection point).

When the four conditions hold simultaneously, the inflection-point signal has high confidence.

Proof. The trigonometric relations among the four criteria: (1) $\sin(\theta_t - \theta_{t-\Delta t}) < 0$ if and only if the direction angle retreats ($\theta_t < \theta_{t-\Delta t}$, for $|\Delta\theta| < \pi$)—a necessary condition for a reversal; (2) in a strong trend $\theta_t \approx \theta_{t-\Delta t}$, and continuity gives $\cos\Delta\theta \approx 1$; as a reversal approaches, $|\Delta\theta|$ increases and $\cos\Delta\theta$ falls; (3) $|\sin 2\Delta\theta| = 2|\sin\Delta\theta\cos\Delta\theta|$ is strictly increasing in $|\Delta\theta|$ for $|\Delta\theta| < \pi/4$ (double-angle amplification); (4) $\eta \neq 1$ means that the momentum changes, which follows directly from the definition of the refraction index. The simultaneous validity of the four conditions is equivalent to 'direction retreat + weakening coherence + second-order amplification + momentum change'—that is, a high-confidence inflection point. □

Example 39.4 (Application of window functions to stock inflection-point detection). *Taking the daily closing-price data of a certain stock $P(t)$ as an example, take a window radius $\Delta = 20$ trading days (about one calendar month) and smooth with the Gaussian kernel K_{Gauss}. Compute the following indicators:*

(i) **Window-function smoothed trend** $P_{\text{trend}}(t; 20)$.

(ii) **Limit angle** $\theta_t = \arctan(r_t)$, where $r_t = \ln(P_{\text{trend}}(t)/P_{\text{trend}}(t-1))$.

(iii) **Sine difference-angle similarity** $\text{Similarity}^{(2)}(t) = \sin(\theta_t - \theta_{t-5})$ (a delay of 5 days).

(iv) **Cosine difference-angle similarity** $\text{Similarity}^{(1)}(t) = \cos(\theta_t - \theta_{t-5})$.

(v) **Refraction index** $\eta(t) = \|\mathsf{v}_{\text{out}}\|/\|\mathsf{v}_{\text{in}}\|$.

When $\text{Similarity}^{(2)}(t)$ crosses the zero line downwards (from positive to negative) and $\eta(t) > 1$, a peak inflection point is detected—the price passes from a rise to an accelerating fall. When $\text{Similarity}^{(2)}(t)$ crosses the zero line upwards (from negative to positive) and $\eta(t) > 1$, a trough inflection point is detected—the price passes from a fall to an accelerating rise. If $\text{Similarity}^{(2)}(t) < 0$ but $\eta(t) < 1$, the momentum decays—the inflection-point signal is weak and may be a false inflection point or a period of sideways consolidation.

This diagnostic procedure is an application of periodic convolution to a univariate financial time series: the sine difference-angle similarity $\text{Similarity}^{(2)}$ measures the strength of the phase rotation, the cosine difference-angle similarity $\text{Similarity}^{(1)}$ measures the consistency of the trend, and the refraction index η measures the scaling transformation of the momentum. The joint use of the three constitutes a complete tool chain for stock inflection-point diagnosis.

Remark 39.10 (Unity with the rest of the book). *The theory of the periodic structure of economic systems established in this chapter is a systematic application of periodic mathematics in economics. Supply–demand equilibrium and the cobweb model are the realisation of the gradient flow in the price mechanism—excess demand drives the price up, excess supply drives it down, and the time lag introduces a rotational component that produces periodic oscillation. The multiplier–accelerator model is the realisation of feedback coupling in the determination of macroeconomic income—the coupling of the multiplier effect (forward scaling) and the accelerator effect (reverse scaling) drives the business cycle. The inventory cycle, the fixed-capital investment cycle and the Kondratiev long wave constitute a three-level nested periodic structure. The momentum–reversal alternation of financial markets is the coupling of the trend component and the rotational component in asset pricing. Multi-scale window-function decomposition and inflection-point diagnosis (sine/cosine difference-angle similarity, refraction index) are direct applications of periodic convolution to financial time series. The power-law exponent α of the Pareto distribution remains relatively stable under technological and institutional change. The three identities and constraints of the economic system—the value identity, the money constraint and the information constraint—together constitute the guarantee of self-consistency of the periodic structure of economic systems.*

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The price-translation adjustment of supply–demand equilibrium (Definition 39.11); the delayed oscillation of the cobweb model (Proposition 39.5)
Scaling	Forward/reverse scaling by the multiplier effect $1/(1-c)$ and the accelerator v (Definition 39.13); the Pareto exponent α (Definition 39.18)
Rotation	The expansion–contraction alternation of the business cycle (inventory/investment cycles); the phase rotation of momentum–reversal (Definition 39.17)
Conservation	The value identity ($Y = C + I + G + NX$), the money constraint ($MV = PY$) and the information constraint (efficient markets) (Theorem 39.7)
Symmetry	The reflection symmetry of the alternation between high and low prices in the cobweb model (period $T = 2$, Proposition 39.5)
Closure	The closure of the feedback loop of the multiplier–accelerator (periodic solutions of the characteristic equation, Proposition 39.6); the closure of the reversion of price to the mean (Definition 39.17)
Period	The three-level nesting of the inventory cycle (3–5 years, Definition 39.14), the investment cycle (7–11 years, Definition 39.15) and the Kondratiev long wave (45–60 years, Definition 39.16) (Proposition 39.7); the multi-window period decomposition of stock prices (Proposition 39.10)

Exercises

Exercise 1 Prove the **economic periodicity decision theorem** of this chapter (39.1). Hint: first write down the decision map and the evolution group of this chapter's target, then apply the proof structure of the general criterion given in the 'Periodic Systems' chapter.

Exercise 2 Verify that the hypotheses of the **monotonicity of cardinality adjustment** (39.2) hold for the domain objects of this chapter, and give a counterexample in which the hypotheses fail.

Exercise 3 Examine the behaviour of the **three-parameter correspondence: radius, phase and angular speed** (39.1) under parameter degeneration, and explain how the conclusion degenerates accordingly.

Exercise 4 Explain the relation between the **economic periodicity diagnosis pipeline**

and its interface with three chapters (39.6) *and the corresponding general conclusions of the 'Periodic Systems' chapter (whether they agree, whether this is a special case, or whether additional conditions are required).*

Exercise 5 *Using the seven-theme comparison table of this chapter, identify the concrete carriers of the themes* **translation**, **scaling** *and* **rotation** *in this chapter, and give one example of domain data for each.*

Exercise 6 *Synthesis: restate the diagnostic procedure of this chapter using the five-step standard interface of the 'Periodic Systems' chapter (Table 24.1), noting for each step the theorem of this chapter on which it depends: 39.1, 39.2, 39.1, 39.6.*

Chapter 40

Periodic Structure of Political Systems

Remark 40.1 (Periodicity decision for the target of this chapter (citation)). When this chapter decides whether a research target is periodic, determines its period type, or seeks its minimal positive period, reference is always made to the 'Periodic Systems' chapter: for the periodicity criterion see Theorem 24.1, for the discrimination of period types see Definition 24.3, Proposition 24.1, for the search for the minimal positive period see Definition 24.4, Theorem 24.2, and for the construction of period series and epicycle groups from data see Theorem 24.3, Definition 24.6.

Remark 40.2 (Positioning of this chapter: developing via the five-step interface of the 'Periodic Systems' chapter). The periodic structure of political systems studies the mathematical characterisation of periodic phenomena in regime succession, institutional change and governance performance. This chapter does not establish a separate periodicity criterion; instead it connects political systems to the five-step interface prescribed in the 'Periodic Systems' chapter:

1. **Decision**: reduce such targets as dynastic rise and fall, regime succession, imperial hegemonic succession and the fiscal cycle to discrete or continuous dynamical systems, invoke Theorem 24.1 to decide whether they are periodic, and determine the period type;
2. **Construction**: build the amplitude series, frequency series and phase series from data such as regime stability, fiscal health, elite cohesion, popular support and external threat, and assemble them into a political epicycle group;
3. **Adjustment**: dynamically adjust the political epicycle group by residual-energy descent, information criteria and structural constraints;
4. **Convolution**: introduce amplitude convolution, phase convolution and frequency convolution on the political epicycle group, and connect them with autocorrelation, cross-correlation and phase coupling;
5. **Integration**: connect to the tool chains of the three chapters 'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis', forming a political periodicity diagnosis pipeline.

The mathematical tools used in this chapter include: gradient flows, delayed negative feedback, positive-feedback systems, the logistic equation, exponential decay, Fourier analysis, periodic convolution, epicycle groups and phase operators.

Remark 40.3 (Seven-theme guide for this chapter). **Translation**—the passage of time in the dynastic lifecycle and the one-way accumulation of land concentration

(Definition 40.12); **scaling**—the expansion–contraction scaling of imperial hegemony (Proposition 40.8) and the exponential decay of fiscal crisis (Proposition 40.6); **rotation**—the phase-space rotation of regime change (the Polybius cycle, Definition 40.13); **balance**—the zero-sum constraint on power, fiscal balance and the conversion of legitimacy types (Theorem 40.9); **symmetry**—the alternating symmetry of unification and division (Theorem 40.7); **closure**—the closed causal loop of multiple positive feedbacks (Definition 40.10); **period**—the nested structure of the dynastic cycle (200–300 years), the hegemony cycle (100–150 years) and the unification–division cycle (300–500 years) (Theorem 40.8).

40.1 Periodicity decision for political targets: the first step of the 'Periodic Systems' chapter

40.1.1 Political targets, decision inputs and decision maps

Definition 40.1 (Political targets and decision inputs). A **political target** is one of the following objects:

(i) a regime lifecycle system, whose state space is the set of values of regime stability, $[0,1]$;

(ii) a regime-change system, whose state space is the two-dimensional phase space (degree of democracy D, degree of centralisation A);

(iii) an imperial hegemony system, whose state space is the relative power of the hegemonic state, $H(t) \in [0,1]$;

(iv) a fiscal–political coupled system, whose state space is the joint space of (fiscal health F, regime stability S);

(v) a unification–division alternation system, whose state space is the set of values of territorial integration.

The **decision input** of a political target is the pair

$$\mathcal{D}_{\mathsf{P}} = (\mathcal{T}, x),$$

where $\mathcal{T}$ is the time index set (which may be $\mathbb{Z}$ or $\mathbb{R}$) and $x : \mathcal{T} \to V$ is the value of the political variable over time. If $\mathcal{T} = \mathbb{Z}$, the target is called a **discrete political target**; if $\mathcal{T} = \mathbb{R}$, it is called a **continuous political target**.

Definition 40.2 (Political decision map). Let a political target be given by the decision input $\mathcal{D}_{\mathsf{P}} = (\mathcal{T}, x)$. If there exist a map $\Phi : \mathcal{S} \to \mathcal{S}$ and an initial value $x_0 \in \mathcal{S}$ such that the states of the target are generated by Φ, that is,

$$x_t = \begin{cases} \Phi^t(x_0), & t \in \mathbb{Z}, \\ \Phi_t(x_0), & t \in \mathbb{R}, \end{cases}$$

for $t \in \mathbb{Z}$ the state x_t is generated by iteration of Φ; for $t \in \mathbb{R}$ it is described by the one-parameter evolution family $\{\Phi_t\}_{t \in \mathbb{R}}$, where each $\Phi_t : \mathcal{S} \to \mathcal{S}$, $\Phi_0 = \mathrm{id}$,

$\Phi_{s+t} = \Phi_s \circ \Phi_t$ for all $s, t \in \mathbb{R}$, and $(t, x) \mapsto \Phi_t(x)$ is continuous. then Φ is called the **political decision map** of the target. The political target is thereby reduced to the discrete dynamical system $(\mathcal{S}, \Phi)$ when $t \in \mathbb{Z}$ and to the continuous dynamical system $(\mathcal{S}, \{\Phi_t\}_{t\in\mathbb{R}})$ when $t \in \mathbb{R}$; in both cases the criteria of the 'Operations and Periodicity' chapter and of the 'Periodic Systems' chapter may be applied directly.

Theorem 40.1 (Political periodicity decision theorem)**.** *Let the decision map of a political target be Φ, and let $(\mathcal{S}, \Phi)$ be a discrete dynamical system. Then:*

(A) **Sufficient criterion**: *if there exists $m \in \mathbb{N}^+$ such that $\Phi^m = \mathrm{id}$, then the target is a uniformly periodic political system; every orbit is closed, and the orbit length divides m.*

(B) **Equivalence of the decision**: *the target is a uniformly periodic political system if and only if $\mathrm{ord}(\Phi) < \infty$.*

(C) **Pointwise criterion**: *the orbit of the target at the state x_0 is closed if and only if x_0 is a periodic point; the orbit length is then the period at that point.*

(D) **Non-eliminability**: *the conjunction of conserved quantities and the symmetry group cannot replace $\mathrm{ord}(\Phi) < \infty$; a quasi-periodic political target is not a periodic political system.*

Proof. The Theorem 40.1 of this chapter is the domain instantiation of Theorem 24.1 in the 'Periodic Systems' chapter: it suffices to replace the decision map f and the evolution group by Φ of this chapter, and then (A) sufficient criterion, (B) equivalence of the decision, (C) pointwise criterion and (D) non-eliminability hold one by one. By the convention of Remark 24.4, they are not expanded again here. $\square$

Corollary 40.1 (Political period types)**.** *After the decision by Theorem 40.1, a political target falls into one of the following types:*

(i) **Trivial period type**: *the minimal period is $m_0 = 1$, and the state is a fixed point (for example a long-stable equilibrium regime);*

(ii) **Strictly periodic type**: *$1 < m_0 < \infty$, and the state is strictly periodic (for example the periodic succession of dynastic rise and fall);*

(iii) **Uniformly periodic type**: *$\mathrm{ord}(\Phi) = m < \infty$, and every orbit length divides m;*

(iv) **Quasi-periodic type**: *$\mathrm{ord}(\Phi) = \infty$; the orbits are not closed but are dense in a compact carrier structure (for example the superposition of imperial hegemonic successions);*

(v) **Eventually periodic type**: *there exists N such that the target is periodic for $t \geq N$;*

(vi) **Discrete periodic type**: *the carrier structure is $\mathbb{Z}$ and the normalised period ratio is $\rho = T_x/\Delta x \in \mathbb{Q}$.*

(vii) **Asymptotically periodic (damped) type**: *the orbit is not closed, but tends asymptotically to a closed orbit or to a fixed point, the residual decaying as $e^{-\kappa t}$ (see Definition 24.3(vii)).*

Among these, (i)–(iii) and (vi) belong to periodic political systems; (iv) does not belong to periodic political systems, but can be handled by bounded-approximation tools; (v) is a periodic political system only in its stable stage; (vii), like (iv), is a type in the limiting sense; it is not a periodic political system and can be handled with bounded-approximation tools only in its stable stage.

Proof. Each item follows by substituting a political target into Theorem 40.1 and Definition 24.3 of the 'Periodic Systems' chapter. □

Remark 40.4 (Periodicity decision for the dynastic lifecycle (empirical observation / research programme)). Empirical observation: let the dynastic lifecycle be described by the four-stage model, with regime stability $S(t)$ rising from 0 to 1 and then falling back to 0, the whole cycle lasting about 200–300 years. If the modelling hypothesis that dynastic succession repeats itself strictly (a new dynasty replacing the old) is adopted in addition, then under that hypothesis the dynastic lifecycle may be regarded as of uniformly periodic type with a period of about 207 years; the conclusion depends on the modelling hypothesis rather than on historical fact, and belongs to empirical observation and a research programme.

Empirical basis: By Definition 40.11, $S(t)$ completes one full rise–zenith–decline–collapse cycle on the time scale of 200–300 years, and Proposition 40.5 gives a characteristic lifespan of about 207 years for the major unificatory dynasties. If the modelling hypothesis that dynastic succession repeats itself strictly is adopted in addition, the decision map may be taken to be a discrete map satisfying $\Phi^{T_{\text{dynasty}}} = \text{id}$ ($T_{\text{dynasty}} \approx 207$ years), so that under that hypothesis the type is uniformly periodic. Both Φ and T_{dynasty} are modelling choices here, not conclusions derived from historical fact.

Lemma 40.1 (Periodicity decision for regime change). Let regime change be described by the Polybius cycle: monarchy → tyranny → aristocracy → oligarchy → democracy → ochlocracy → monarchy. Then regime change is of uniformly periodic type.

Proof. By Definition 40.13, the Polybius cycle rotates periodically among the six regime types, and each complete cycle contains six stages. The decision map Φ satisfies $\Phi^6 = \text{id}$ (a six-stage cycle), so the type is uniformly periodic. □

Lemma 40.2 (Periodicity decision for imperial hegemonic succession). Let global hegemony be described by the logistic equation $\dot{H} = \alpha H(1 - H) - \beta H$, with the cost of maintaining hegemony growing over time as $\beta(t) = \beta_0 e^{\gamma t}$. Then hegemonic succession is of quasi-periodic type ($\text{ord}(\Phi) = \infty$), with a period of about 100–150 years.

Proof. By Proposition 40.8, once $\beta(t)$ exceeds α the hegemony declines irreversibly; the whole process by which a new hegemony rises and replaces the old one constitutes one complete cycle. Since the lengths of the individual hegemony cycles are not all equal (about 86–150 years), the overall motion is of quasi-periodic type. □

Definition 40.3 (Algorithm for the minimal positive period of political systems). This definition has the same content as Definition 24.4 in the section 'Periodicity

Decision: Is the Target Periodic?' of Chapter 24 (Section 24.1); to avoid repetition, only the reference anchor is retained here, and the full statement is given there. The only difference here is terminology (political target / political sample); the steps of the algorithm and the conclusions are exactly the same.

Theorem 40.2 (Correctness of the algorithm for the minimal positive period of political systems)**.** *This theorem has the same content as Theorem 24.2 in the section 'Periodicity Decision: Is the Target Periodic?' of Chapter 24 (Section 24.1); to avoid repetition, only the reference anchor is retained here, and the full statement and proof are given there. The only difference here is terminology (political target / political sample); the steps of the algorithm and the conclusions are exactly the same.*

Example 40.1 (Examples of political periodicity decisions)**.**

(i) The dynastic lifecycle: $T \approx 207$ years, uniformly periodic type (locally; an empirical observation and a research programme, see Remark 40.4);

(ii) regime change: a six-stage cycle, uniformly periodic type;

(iii) imperial hegemonic succession: $T \approx 100$–150 years, quasi-periodic type;

(iv) fiscal crisis: exponential decay, asymptotically periodic (damped) type (tending to collapse);

(v) unification–division alternation: $T \approx 300$–500 years, quasi-periodic type.

Remark 40.5 (Conserved quantities do not guarantee periodicity). In political systems, the zero-sum constraint on power, fiscal balance and the conversion of legitimacy types are auxiliary conditions for periodic structure, but they do not constitute sufficient conditions. For example, a stable regime that satisfies fiscal balance will not spontaneously generate periodic succession. Whether a periodic structure is present in a political target must be decided by Theorem 40.1 or by the lemmas above.

40.1.2 The second step: construction of the political period series and the epicycle group

Definition 40.4 (Political amplitude series, frequency series and phase series)**.** Let the state or output sequence of a political target be $\{z_t\}$. If it can be written as

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)} + \varepsilon_t,$$

then

$$\mathcal{A}_z(t) = \sum_k A_k e^{ikt}, \qquad \mathcal{F}_z(s) = \sum_k \omega_k s^k, \qquad \mathcal{P}_z(t) = \sum_k e^{i(kt+\phi_k)}$$

are called respectively the **amplitude series**, the **frequency series** and the **phase series** of the political target. The three agree with Definition 6.10, Definition 6.12 and Definition 6.11 of the 'Series and Convolution' chapter.

Definition 40.5 (Political epicycle group)**.** *Let the period series of a political target be given by the amplitude series $\{A_k\}$, the frequency series $\{\omega_k\}$ and the phase series $\{\phi_k\}$. For each k define the epicycle*

$$E_k : \theta \mapsto A_k e^{i(\theta+\phi_k)}, \qquad \theta = \omega_k t,$$

whose collection

$$\mathfrak{E}_{\mathsf{P}} = \{(A_k, \omega_k, \phi_k)\}_{k\in\mathbb{Z}}$$

is called the **epicycle group** *of the political target. Its superposition is the political period series*

$$z_t = \sum_k E_k(\omega_k t) = \sum_k A_k e^{i(\omega_k t+\phi_k)}.$$

Theorem 40.3 (Construction of the period series from political data)**.** *Let the output of a political target be a finite periodic sequence of period N. Then:*

(i) *the output can be expanded as a discrete Fourier series*

$$z_t = \sum_{k=0}^{N-1} c_k e^{i2\pi kt/N}, \qquad c_k = \frac{1}{N}\sum_{t=0}^{N-1} z_t e^{-i2\pi kt/N};$$

(ii) *the amplitude series is given by $|c_k|$, the phase series by* $\arg c_k$ *and the frequency series by $2\pi k/N$;*

(iii) *if the output is a sequence of dynastic lifespans, then the fundamental frequency corresponds to the average period of dynastic succession, and the harmonics correspond to the unification–division cycle and the cycle of institutional evolution.*

Proof. (i) The discrete Fourier series of a finite periodic sequence is a standard result: the basis functions $\{e^{i2\pi kt/N}\}_{k=0}^{N-1}$ form an orthogonal basis of $\mathbb{C}^N$. (ii) This is read off directly from $c_k = A_k e^{i\phi_k}$. (iii) For a sequence of dynastic lifespans, the fundamental frequency corresponds to the average succession period of the N dynasties, and the low-order harmonics correspond to the unification–division cycle on a larger scale. ☐

Proposition 40.1 (Three-parameter correspondence: radius, phase and angular speed)**.** *The three parameters of the political epicycle group $\mathfrak{E}_{\mathsf{P}} = \{(A_k, \omega_k, \phi_k)\}_k$ are in one-to-one correspondence with three geometric–political quantities:*

(i) **Radius corresponds to amplitude**: *the radius of E_k is $|E_k| = A_k$, namely the amplitude of that political oscillation component (for example the fluctuation amplitude of dynastic stability);*

(ii) **Phase corresponds to rotational delay**: *the initial phase ϕ_k corresponds to the time translation $\tau_k = \phi_k/\omega_k$ (for example the delay of a dynasty's rise relative to a baseline); when $\phi_k > 0$ this component* **leads** *the baseline by τ_k (equivalently, the component coincides with the baseline once the baseline is postponed by τ_k);*

(iii) **Frequency corresponds to rotational speed**: ω_k is the angular frequency of that political oscillation component (for example how fast dynastic succession proceeds).

Proof. The Proposition 40.1 of this chapter corresponds exactly to Proposition 24.3 of the 'Periodic Systems' chapter, and the three correspondences—radius to amplitude, phase to rotational delay, frequency to rotational speed—are obtained one by one by translating them to the epicycle group of this chapter. The proof is given there: (i) from $|E_k| = A_k$ and the energy decomposition, (ii) from the phase-translation theorem, (iii) from the definition of the frequency series. □

Example 40.2 (Period series of a political time series). The sequence of Chinese dynastic lifespans can be decomposed into a fundamental frequency (about 207 years), a second harmonic (about 104 years, corresponding to the secondary unification–division cycle) and long-period harmonics (about 500–1000 years, corresponding to institutional evolution). In the political epicycle group, the long periods correspond to the outer, large epicycles (low frequency, high amplitude), whilst dynastic succession corresponds to the inner, small epicycles (high frequency, low amplitude).

Remark 40.6 (The period series is a complex function). Let $z = e^{i\omega_0 t}$. Then the political period series

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)}$$

when all components are commensurable with ω_0 (that is, $\omega_k = k\omega_0$, $k \in \mathbb{Z}$) it can be written as the Laurent series $\sum_k c_k z^k$, where $c_k = A_k e^{i\phi_k}$: in that case the political period series is a complex function on the unit circle, and otherwise it is merely a function on the torus.

40.1.3 The third step: dynamic adjustment of the political epicycle group

Definition 40.6 (Dynamic adjustment of the political epicycle group). Let the current political epicycle group be $\mathfrak{E}^{(j)}$, with partial sum

$$z^{(j)}(t) = \sum_{k \in \mathfrak{E}^{(j)}} A_k e^{i(\omega_k t + \phi_k)},$$

and residual $r^{(j)} = z - z^{(j)}$. **Dynamic adjustment** consists of two rules:

(i) **Cardinality adjustment**: if the residual energy satisfies $\|r^{(j)}\|^2 > \delta$, add one new epicycle according to the peaks of the residual periodogram;

(ii) **Parameter adjustment**: update the parameters of the existing epicycles by regularised least squares or an information criterion.

In political systems, cardinality adjustment corresponds to adding a political cycle component (for example adding a newly identified dynastic cycle or hegemony cycle); parameter adjustment corresponds to tuning the amplitude, phase and frequency so as to balance goodness of fit against model parsimony.

Proposition 40.2 (Monotonicity of cardinality adjustment)**.** *After the dominant-frequency component of the residual is added as in Definition 40.6(i), the residual energy is monotonically non-increasing:*

$$\|r^{(j+1)}\|^2 \le \|r^{(j)}\|^2,$$

with equality if and only if the residual has no energy at the selected frequency.

Proof. The Proposition 40.2 of this chapter is consistent with the dynamic adjustment mechanism of Definition 24.8 in the 'Periodic Systems' chapter: adding an epicycle removes the component of $r^{(j)}$ along $\hat{\omega}$ from the residual, and the orthogonal decomposition $\|r^{(j)}\|^2 = \|r^{(j+1)}\|^2 + \|\text{the removed component}\|^2$ gives the inequality, with equality if and only if the norm of the removed component is zero. The complete argument is given there. ☐

Theorem 40.4 (Convergence of dynamic adjustment)**.** *Let the political target be a finite periodic sequence of period T_0 and let $\delta = 0$. Then after at most $\lfloor T_0/2 \rfloor + 1$ steps of dynamic adjustment the residual is zero, and the resulting political epicycle group is the exact Fourier data of the target.*

Proof. The discrete Fourier series of a sequence of period T_0 is complete on $\lfloor T_0/2 \rfloor + 1$ harmonics. At each step one dominant-frequency component of the residual is extracted as in Proposition 40.2, and the residual energy decreases strictly; hence after finitely many steps the residual is zero. ☐

Definition 40.7 (Structural metric adjustment)**.** Let the structural metric of a political target be

$$\mathcal{I} = (\mathcal{I}_{\text{fit}}, \mathcal{I}_{\text{parsimony}}, \mathcal{I}_{\text{prediction}}) = (\text{residual sum of squares, number of parameters, cross-validatio}$$

Structural-metric adjustment consists in choosing θ in the parameter space Θ: at each step take a neighbouring parameter for which $\mathcal{I}(\theta)$ is **strictly decreasing** in the lexicographic order (minimisation); if no such parameter exists, the process stops, and the parameter is then locally optimal.

Proposition 40.3 (Finite convergence of structural metric adjustment)**.** *Let the parameter space Θ be finite, and let each adjustment make $\mathcal{I}$ strictly decreasing in the lexicographic order. Then the adjustment terminates after finitely many steps at a locally optimal parameter.*

Proof. Since Θ is finite, $\mathcal{I}$ takes only finitely many values, and the lexicographic order is a total order on a finite set; a strictly decreasing sequence in that order cannot continue indefinitely, so the adjustment terminates after finitely many steps. At termination there is no neighbouring parameter that makes $\mathcal{I}$ strictly decrease in the lexicographic order, that is, the parameter is locally optimal. ☐

Example 40.3 (Selecting a political period series by BIC)**.** Fit K epicycles to a sequence of dynastic lifespans and compute $\text{BIC}(K) = n \ln(\text{RSS}/n) + K \ln n$. Choose the K that minimises the BIC. Usually $K = 2$–4 suffices to capture the main cycles.

Remark 40.7 (Boundaries of dynamic adjustment). Theorem 40.4 relies on a finite periodic sequence. For a continuous spectrum or a quasi-periodic political target (such as hegemonic succession or unification–division alternation), no finite epicycle group represents it exactly, and only a bounded approximation is possible. Structural metric adjustment does not guarantee a global optimum.

40.1.4 The fourth step: political convolution

Definition 40.8 (Political amplitude, phase and frequency convolutions). Let two political epicycle groups be

$$\mathfrak{E} = \{(A_n, \omega_n, \phi_n)\}, \qquad \mathfrak{E}' = \{(B_m, \nu_m, \psi_m)\}.$$

Define:

(i) **Amplitude convolution**: $\mathbb{A}_{mn} = A_n B_m$;

(ii) **Phase convolution**: $\mathbb{P}_{mn} = e^{i(\phi_n - \psi_m)}$;

(iii) **Frequency convolution**: $\mathbb{F}_{mn} = \omega_n \nu_m$;

(iv) **Trend convolution**: $\mathbb{T}_{mn} = A_n B_m e^{i(\phi_n - \psi_m)} = A_n B_m \mathbb{P}_{mn}$.

This agrees with Definition 6.21, Definition 6.22, Definition 6.23 and Definition 6.24 of the 'Series and Convolution' chapter.

Theorem 40.5 (Action of the three convolutions on the political epicycle group). *Let the epicycle groups of two political targets be $\mathfrak{E}$ and $\mathfrak{E}'$ respectively. Then:*

(i) amplitude convolution acts only on the radius;

(ii) phase convolution acts only on the phase;

(iii) frequency convolution acts only on the rotational speed;

(iv) trend convolution preserves both amplitude and phase.

Proof. This is read off directly from Definition 40.8. ☐

Proposition 40.4 (Autocorrelation and trend convolution). *Let $\{z_t\}$ be a political period sequence with autocorrelation function*

$$R_z(h) = \sum_t z_t \overline{z_{t+h}}.$$

In the language of the political epicycle group, the autocorrelation function is the projection of trend convolution onto the diagonal:

$$R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}.$$

Proof. The Proposition 40.4 of this chapter is the realisation in this domain of the convolution-action conclusion of Theorem 24.6 in the 'Periodic Systems' chapter: substituting $z_t = \sum_n A_n e^{i(\omega_n t + \phi_n)}$ into the definition of the autocorrelation, cancelling all off-diagonal cross terms by orthogonality and retaining only the diagonal term of the trend convolution (the factorisation of the three convolutions, Proposition 24.6), which yields $R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}$. The full expansion is given there. ☐

Remark 40.8 (Relation between political convolution and periodic convolution). Political convolution is not a new convolution operation, but an instance of the periodic convolution of the 'Series and Convolution' chapter on political epicycle groups. Autocorrelation is the projection of the trend convolution onto the diagonal; cross-correlation and phase coupling are respectively the concrete manifestations in political science of amplitude convolution and frequency convolution.

40.1.5 The fifth step: integration with discrete sampling, harmonic analysis and time series analysis

Definition 40.9 (Political periodicity diagnosis pipeline). For a given political system, diagnosis is carried out using the framework of periodic mathematics:

(i) **Periodicity decision**: verify the period type of the political target (Theorem 40.1);

(ii) **Series construction**: build the amplitude, frequency and phase series and the epicycle group from data such as dynastic lifespans, regime durations and hegemony cycles (Theorem 40.3);

(iii) **Conservation diagnosis**: verify the zero-sum constraint on power, fiscal balance and the conversion of legitimacy types (Theorem 40.9);

(iv) **Convolution diagnosis**: compute the autocorrelation (the diagonal projection of the trend convolution, Proposition 40.4) and the phase convolution;

(v) **Integration diagnosis**: connect to the tool chains of discrete sampling, harmonic analysis and time series analysis.

Theorem 40.6 (The political periodicity diagnosis pipeline and its interface with three chapters). *The five steps of Definition 40.9 are connected respectively to:*

(i) *the 'Discrete Sampling and Intercalation of Periodic Signals' chapter Definition 21.1, Theorem 21.3, Theorem 21.10;*

(ii) *the 'Harmonic Analysis' chapter Definition 22.5, Theorem 22.1, Proposition 22.11;*

(iii) *the 'Time Series Analysis' chapter Definition 23.16, Proposition 23.2, Definition 23.19.*

Hence the political periodicity diagnosis is not an independent method, but an application of the unified tool chain of periodic mathematics to political systems.

Proof. The Theorem 40.6 of this chapter is the implementation in this domain of the five-step interface of the 'Periodic Systems' chapter (Table 24.1): periodicity is decided by sampling and spectral estimation, the frequency decomposition and the phase kernel use the 'Harmonic Analysis' chapter, and the periodogram peaks, Fisher test and window-function validation use the 'Time Series Analysis' chapter. The diagnostic pipeline is therefore synthesised from the tool chains of the three chapters; the proof is given there. □

Remark 40.9 (The order of operations after integration). The recommended order for an actual diagnosis is: first decide whether the political target is periodic (Theorem 40.1); then construct the Fourier data and the epicycle group of the political data (Theorem 40.3); then optimise the parameters using the balance constraints and dynamic adjustment (Theorem 40.4, Proposition 40.3); then carry out the political convolution and autocorrelation analysis (Proposition 40.4); and finally output the diagnosis as in Definition 40.9.

40.2 Mathematical foundations of political systems

Definition 40.10 (Basic quantities of political systems)**.** A political system is described by the following basic variables:

(i) **Regime stability** $S(t) \in [0,1]$: the degree of stability of the regime at time t. $S \to 1$ indicates high stability (no internal disorder, no external threat), and $S \to 0$ indicates collapse (the disintegration of the regime).

(ii) **Fiscal health** $F(t) \in [0,1]$: the health of the state finances at time t. $F \to 1$ indicates ample finances (a well-filled treasury), and $F \to 0$ indicates exhaustion of the finances (expenditure exceeding revenue).

(iii) **Elite cohesion** $E(t) \in [0,1]$: the degree of unity of the ruling elite at time t. $E \to 1$ indicates that the elite unanimously support the regime, and $E \to 0$ indicates that the elite are divided or have defected.

(iv) **Popular support** $P(t) \in [0,1]$: the degree of support of the ordinary population for the regime at time t. $P \to 1$ indicates that the people are attached to the regime, and $P \to 0$ indicates seething popular resentment.

(v) **External threat** $W(t) \in [0,1]$: the degree of threat posed to the regime by hostile external forces at time t. $W \to 1$ indicates a powerful enemy at the borders, and $W \to 0$ indicates peace throughout the realm.

The core feature of a political system is **multiple coupled positive feedbacks**—fiscal deterioration leads to elite defection, elite defection weakens military defence, military defeat aggravates the fiscal crisis, the fiscal crisis provokes popular revolt, and popular revolt further undermines regime stability. These positive feedbacks form a closed causal loop that drives the political system from stability towards collapse.

40.3 A periodic model of dynastic rise and fall

Definition 40.11 (Four-stage model of the dynastic lifecycle)**.** The lifecycles of the unificatory dynasties of China display a marked four-stage pattern:

(i) **Rise** (about 50–80 years): the founding monarch establishes a new dynasty through military conquest or political consolidation. Land is redistributed (equal-field and land-granting measures), the tax system is reformed (light corvée and low taxes), and a bureaucracy is established. Regime stability $S(t)$ rises rapidly. Representative dynasties: the Rule of Wen and Jing in the Western Han (about 180–140 BCE), the Zhenguan Prosperity of the Tang (627–649), and the Hongwu Prosperity of the Ming (1368–1398).

(ii) **Zenith** (about 80–120 years): national power reaches its peak, territory expands to its greatest extent, culture flourishes and institutions mature. Regime stability $S(t)$ remains high but begins to show slight fluctuations—land concentration appears, bureaucratic corruption takes root and fiscal expenditure swells. Representative dynasties: the Prosperity of Emperor Wu of the Han (about 140–87 BCE), the Kaiyuan Prosperity of the Tang (713–741), and the Yongle–Xuande Prosperity of the Ming (1402–1435).

(iii) **Decline** (about 100–150 years): land concentration intensifies, owner-farmers go bankrupt and the tax base shrinks; the bureaucracy swells and administrative efficiency falls; expenditure exceeds revenue and the currency depreciates. Regime stability $S(t)$ declines steadily. Representative dynasties: the middle and late Western Han (about 50–8 BCE), the Tang after the An Lushan Rebellion (755–907), and the Ming after the Wanli period (1582–1644).

(iv) **Collapse** (about 20–50 years): fiscal exhaustion leads to arrears in military pay, and troops mutiny or defect; peasant risings follow one another without pause; regional separatist forces grow in strength; and external enemies exploit the weakness. Regime stability $S(t)$ plunges to 0. Representative dynasties: the risings of the Green Woodsmen and the Red Eyebrows at the end of the Western Han, the Huang Chao rebellion at the end of the Tang, and the Li Zicheng rebellion at the end of the Ming.

The dynastic lifecycle is the realisation of an accumulative process in political time: the whole process in which $S(t)$ rises from 0 to 1 and then falls back to 0 takes about 200–300 years. The rise phase corresponds to the establishment and consolidation of institutions, and the decline phase to their disintegration and failure. The dynastic lifespan T_{dynasty} is determined by the combined effect of the rate of land concentration, the rate of accumulation of the fiscal deficit and the rate of internal attrition among the elite.

Proposition 40.5 (Statistical distribution of dynastic lifespans)**.** The lifespans of the major unificatory dynasties in Chinese history from the Qin unification (221 BCE) to the fall of the Qing (1912) are tabulated as follows:

Dynasty	Years	Lifespan (years)	Main cause of fall
Qin	221–206 BCE	15	Tyranny, peasant risings
Western Han	206 BCE–8	214	Usurpation by consort clans
Eastern Han	25–220	195	Eunuch domination, Yellow Turban rebellion
Tang	618–907	289	Regional military governors, Huang Chao rebellion
Northern Song	960–1127	167	Foreign invasion
Southern Song	1127–1279	152	Foreign invasion
Yuan	1271–1368	97	Ethnic conflict, peasant risings
Ming	1368–1644	276	Fiscal collapse, Li Zicheng rebellion
Qing	1644–1912	268	Foreign invasion, the 1911 Revolution

The mean lifespan of the eight major unificatory dynasties other than the Qin is about 207 years, with a standard deviation of about 60 years. The characteristic lifespan of 207 years is determined by the following time constants: the transition of land from equal distribution to a high degree of concentration takes about 100–150 years, that of the bureaucracy from lean to bloated about 100–150 years, and that of the population from recovery growth to the carrying limit of the land about 100–150 years. The intersection of the three time constants at about 200–300 years determines the characteristic scale of dynastic lifespans.

Proof. By the four-stage model of Definition 40.11, the dynastic lifespan is determined by the intersection of three independent time constants. Taking the arithmetic mean of the data for the individual dynasties other than the Qin: $(214 + 195 + 289 + 167 + 152 + 97 + 276 + 268)/8 = 1658/8 \approx 207$ years. The standard deviation, computed from $\sqrt{\frac{1}{n}\sum(x_i - \bar{x})^2}$, is about 60 years. □

Remark 40.10 (Comparison of the lifespans of Chinese dynasties and world empires). Comparing the lifespans of the major unificatory dynasties of China with those of the other major empires of the world reveals a striking statistical regularity: the lifespans of Chinese dynasties are concentrated in the interval 200–300 years, whereas the lifespans of the major world empires display a much greater dispersion.

Empire/dynasty	Years	Lifespan (years)
Roman Empire (Principate)	27 BCE–476	503
Byzantine Empire	395–1453	1058
Ottoman Empire	1299–1922	623
Mughal Empire	1526–1857	331
Arab Empire (Abbasid dynasty)	750–1258	508
Sasanian Empire of Persia	224–651	427
Zhou dynasty of China	1046–256 BCE	790

The mean lifespan of the major world empires is about 606 years ($503 + 1058 + 623+331+508+427+790 = 4240$, $4240/7 \approx 606$), far higher than the mean lifespan of 207 years for the unificatory dynasties of China. The reasons for this difference are various: first, Chinese dynasties faced continuous pressure from the nomadic peoples of the north, so that the frequency of external threats was higher than in other civilisations; second, the Chinese imperial examination system and bureaucracy tended to ossify in the later part of a dynasty, whereas the military-aristocratic institutions of other empires (such as the Ottoman and the Mughal) afforded greater institutional flexibility; third, the population–land contradiction in China was especially acute in the middle and late periods of a dynasty, and land concentration proceeded faster. Empirical research in cliodynamics also confirms that the lifespans of empires follow an exponential distribution, with a typical time scale of about 300 years.

40.4 Land concentration and the fiscal cycle

Definition 40.12 (Mathematical description of land concentration). **Land concentration** is the most fundamental economic cause of the decline of Chinese dynasties. Let $L(t)$ be the proportion of land owned by owner-farmers, with $L(0) = L_0 \approx 1$ (land being relatively equally distributed in the early period of a dynasty). Under the pressure of famine, taxation and usury, owner-farmers are forced to sell their land, and land becomes concentrated in the hands of the landlord class. The process of land concentration can be modelled as

$$\frac{dL}{dt} = -\alpha L + \beta(1 - L) - \gamma L \cdot \mathbf{1}_{\text{famine}}(t),$$

where $\alpha > 0$ is the normal rate of concentration, $\beta > 0$ is the rate of land redistribution (land reform at the founding of a new dynasty), and $\gamma > 0$ is the coefficient by which famine accelerates concentration. Land concentration leads to the bankruptcy of owner-farmers → shrinkage of the tax base → fiscal crisis → arrears in military pay → defection of the army → collapse of the regime.

Land concentration is the realisation of a one-way accumulative process in political economy: the one-way transfer of land from owner-farmers to landlords ($dL/dt < 0$) is accumulative translation—analogous to the irreversible flow of heat from a higher to a lower temperature in thermodynamics. Land reform in the early period of a dynasty (the β term) attempts to reverse this process, but as time goes on the concentration rate α exceeds the reform rate β, and land concentration becomes irreversible.

Proposition 40.6 (Positive-feedback amplification of fiscal crisis)**.** The fiscal crisis of a dynasty in its late period exhibits **positive feedback** (a vicious circle): expenditure exceeds revenue → taxes are raised → owner-farmers go bankrupt faster → the tax base shrinks further → the fiscal deficit grows larger → taxes are raised again. This positive-feedback process can be modelled as

$$\frac{dF}{dt} = -rF - \lambda F = -(r+\lambda)F,$$

where $F(t)$ is fiscal health, $r > 0$ is the rate of fiscal deficit and $\lambda > 0$ is the additional impact on the finances of the shrinking tax base (both are proportional to the current fiscal health: the deficit consumes the treasury in proportion to its current size, and the shrinking tax base reduces revenue in proportion to its current size). When $r + \lambda > 0$, $F(t)$ decays exponentially: $F(t) = F_0 e^{-(r+\lambda)t}$.

The positive feedback of a fiscal crisis is the accumulation of forward scaling—a small fiscal deficit is amplified round by round through the loop of owner-farmer bankruptcy and shrinkage of the tax base, and ultimately brings about the collapse of the regime. The 'half-life' of a fiscal crisis, $T_{1/2} = \ln 2/(r+\lambda)$, measures the time required for the finances to pass from health to collapse. In the late period of a dynasty, $r + \lambda$ is usually large, and $T_{1/2} \approx 20$–50 years.

Proof. The solution of the fiscal decay equation $\frac{dF}{dt} = -(r+\lambda)F$ is $F(t) = F_0 e^{-(r+\lambda)t}$, and its half-life is given by $T_{1/2} = \ln 2/(r+\lambda)$ (Lemma 40.3). The meaning of 'positive feedback' lies in the fact that $r + \lambda$ itself increases as the tax base shrinks (raising taxes accelerates bankruptcy, and bankruptcy erodes the tax base further), so that the decay rate rises with time—the actual fiscal collapse is faster than exponential decay, in agreement with what is observed in the collapse phase of a dynasty (20–50 years). □

Lemma 40.3 (Half-life of fiscal decay)**.** For the fiscal decay equation $\frac{dF}{dt} = -(r+\lambda)F$ (with $r + \lambda > 0$ constant), the solution is $F(t) = F_0 e^{-(r+\lambda)t}$, and its half-life is

$$T_{1/2} = \frac{\ln 2}{r+\lambda},$$

and after k half-lives of decay $F = F_0/2^k$—fiscal health collapses as a geometric progression.

Proof. Separating the variables: $\frac{dF}{F} = -(r+\lambda)dt$; integrating gives $\ln F = -(r+\lambda)t + C$, and $F(0) = F_0$ determines $F(t) = F_0 e^{-(r+\lambda)t}$. Setting $F(T_{1/2}) = F_0/2$ gives $e^{-(r+\lambda)T_{1/2}} = 1/2$, that is, $T_{1/2} = \ln 2/(r+\lambda)$; repeated application gives $F(kT_{1/2}) = F_0/2^k$. □

Example 40.4 *(Numerical analysis of the Ming fiscal collapse).* *Taking the Ming as an example, during the Wanli period (1573–1620) fiscal revenue was about 4 million taels of silver and expenditure about 6 million taels, giving an annual deficit of about 2 million taels. Taking $r = 0.5$ per year (the deficit rate) and $\lambda = 0.3$ per year (the rate of shrinkage of the tax base) gives $r + \lambda = 0.8$ per year and a half-life $T_{1/2} = \ln 2/0.8 \approx 0.87$ years. In fact, however, the Ming finances took about 24 years (1620–1644) to pass from the apparent balance of the Wanli period to complete collapse under the Chongzhen emperor, which shows that $r + \lambda$ grows at an accelerating rate over time (positive-feedback amplification), and that the actual rate of fiscal collapse in the later period was far higher than in the early period.*

40.5 The periodic structure of regime change

Definition 40.13 *(Phase space of regime change).* *Political institutions rotate periodically in the two-dimensional phase space (degree of democracy D, degree of centralisation A). Different regime types correspond to different regions of the phase space:*

- (i) **Absolute monarchy** *(low D, high A): power is highly concentrated in the person of the monarch. Examples: France under Louis XIV ('I am the state') and the Qin under the First Emperor.*
- (ii) **Aristocratic republic** *(medium D, medium A): power is shared among a small number of elite families. Examples: the Roman Republic (rule by the Senate) and the Republic of Venice (the Great Council).*
- (iii) **Democracy** *(high D, low A): power is distributed among the citizenry at large through elections. Examples: Athenian democracy in ancient Greece and modern representative democracy.*
- (iv) **Oligarchy** *(low D, medium A): power is controlled by a small number of wealthy men or soldiers. Examples: the period of the Roman triumvirates and modern military governments.*

Regime change is the rotational motion of a periodic cycle in institutional space. The theory of the cycle of constitutions (anacyclosis) put forward by Polybius in the 2nd century BCE—monarchy → tyranny → aristocracy → oligarchy → democracy → ochlocracy → monarchy—is the earliest theory of cycles in political science. The rotation of regimes in phase space resembles a limit cycle—the degenerate forms of institution (tyranny, oligarchy, ochlocracy) are the unstable segments of the cyclic trajectory, whilst the good forms of institution (monarchy, aristocracy, democracy) are its stable segments.

Proposition 40.7 *(Period length of regime change).* *The period length of the Polybian cycle of constitutions has no fixed value; it depends on the duration of each regime. Taking ancient Rome as an example: the regal period lasted about 244 years (753–509 BCE), the republican period about 482 years (509–27 BCE), and the imperial period about 503 years (27 BCE–476 CE, the fall of the Western Roman*

Empire). The evolution of Rome from monarchy through republic to empire took about 1200 years.

The period length of regime change is a superposition of several time scales: short cycles (10–30 years) correspond to changes of regime (coups d'état, electoral turnover), medium cycles (100–300 years) to institutional transformation (dynastic rise and fall), and long cycles (500–1000 years) to civilisational evolution (classical civilisation → the Middle Ages → modernity). This nested structure corresponds to the structure of multi-level periodic systems.

Proof. Calculation of the durations of Roman constitutional evolution: the regal period 244 years (753–509 BCE), the republican period 482 years (509–27 BCE), and the imperial period 503 years (27 BCE–476 CE). The total is $244 + 482 + 503 = 1229$ years. The mean duration of the three periods is about 410 years. The nested structure is guaranteed by the difference in order of magnitude between the time scales: the ratio of the short cycle to the medium cycle is about 1–10, and that of the medium cycle to the long cycle about 2–5. ☐

40.6 The periodic structure of imperial hegemonic succession

Definition 40.14 (Periodic succession of global hegemony). Since the onset of globalisation in the 16th century, global hegemony has passed from one power to another with a period of about 100–150 years. Each round of the hegemony cycle comprises a rise (challenging the old hegemony), a hegemonic phase (dominating the international order) and a decline (being replaced by a new hegemony):

Hegemonic power	Hegemonic period	Cycle length	Main challengers
Portugal	1494–1580	≈ 86 years	Spain
The Netherlands	1580–1688	≈ 108 years	Britain, France
Britain (first round)	1688–1792	≈ 104 years	France
Britain (second round)	1792–1914	≈ 122 years	Germany, the United States
The United States	1945–present	—	The Soviet Union, China

Hegemonic succession is a periodic process of expansion and contraction: the rise of a hegemonic power is forward scaling (the expansion of military and economic strength), and hegemonic decline is reverse scaling (the dissipation of resources caused by overextension). Paul Kennedy points out in The Rise and Fall of the Great Powers that the main cause of hegemonic decline is 'imperial overreach'—the military and administrative costs of maintaining global hegemony exceed the economic benefits that hegemony brings. When the marginal benefit of expansion falls below the marginal cost, the direction of scaling reverses—from expansion to contraction.

Proposition 40.8 (Dynamic model of the hegemony cycle). *Let the relative power of the hegemonic state be $H(t)$ ($H = 1$ denotes absolute hegemony and $H = 0$ the loss of hegemony). The rise and fall of hegemony can be modelled as*

$$\frac{dH}{dt} = \alpha H(1 - H) - \beta H,$$

where $\alpha > 0$ is the rate of self-reinforcement of hegemony (the 'winner takes all' effect) and $\beta > 0$ is the dissipation rate of the cost of maintaining hegemony (the 'overextension' effect). When $\alpha > \beta$, $H(t)$ converges to $H^* = 1 - \beta/\alpha$—hegemony is stable. When the cost of maintaining hegemony grows with time ($\beta = \beta_0 e^{\gamma t}$), there must come a moment at which $\beta(t) > \alpha$—hegemony begins an irreversible decline.

The dynamics of the hegemony cycle is a competition between self-reinforcement (hegemonic expansion) and maintenance cost (hegemonic upkeep). In the rising phase of a hegemony the term $\alpha H(1 - H)$ dominates—forward scaling; in the declining phase the term $-\beta H$ dominates—reverse scaling. The length of the hegemony cycle, T_{hegemony}, is determined by the critical time at which $\beta(t)$ exceeds α, about 100–150 years.

Proof. Equilibrium points: $\frac{dH}{dt} = \alpha H(1 - H) - \beta H = H(\alpha - \beta - \alpha H) = 0$ gives $H = 0$ and $H^* = 1 - \beta/\alpha$ (requiring $\beta < \alpha$). When $\alpha > \beta$, $H^* > 0$ is a stable equilibrium ($\dot{H} > 0$ for $H < H^*$ and $\dot{H} < 0$ for $H > H^*$)—hegemony is stable at H^*. If the maintenance cost grows exponentially with time, $\beta(t) = \beta_0 e^{\gamma t}$, then at the critical time $t_c = \frac{1}{\gamma} \ln \frac{\alpha}{\beta_0}$ we have $\beta(t_c) = \alpha$; thereafter $H^* < 0$, and the only equilibrium is $H = 0$—hegemony declines irreversibly. Taking $\alpha/\beta_0 \sim 3$–10 and $\gamma \sim 0.01$–0.02 per year gives $t_c \sim 60$–230 years, which covers the observed 100–150-year hegemony cycle. ☐

Example 40.5 (Empirical data on global hegemony cycles). According to the long-cycle theory of international politics of Modelski, the data on global hegemonic succession since 1494 are as follows:

Hegemony cycle	Years	Cycle length (years)
Portuguese cycle	1494–1580	86
Dutch cycle	1580–1688	108
British cycle I	1688–1792	104
British cycle II	1792–1914	122
American cycle	1914–present	—

The mean length of the five cycles is about 109 years ($86 + 108 + 104 + 122 = 420$, $420/4 = 105$ years; if the cycles are delimited by the onset of absolute naval supremacy, the mean is about 109 years).

40.7 A periodic analysis of dynastic succession in Chinese history

Theorem 40.7 (Periodic laws of Chinese dynastic succession)**.** *In the 2133 years from the Qin unification to the fall of the Qing (221 BCE–1912), China experienced about 10 major dynasties. The lifespans, modes of succession and institutional legacies of these dynasties reveal deep regularities of the political cycle:*

(i) **The alternating cycle of unification and division**: *Chinese history exhibits the macro-cycle described by the classical maxim that after long division there must be union, and after long union there must be division (the traditional formulation of the alternation of unification and division). Periods of great unification (the Han, Tang, Yuan, Ming and Qing) are interspersed with periods of division (the Three Kingdoms, the Two Jins and the Northern and Southern Dynasties; the Five Dynasties and Ten Kingdoms). Periods of unification last on average about 250–300 years, periods of division vary in length from about 53 to 369 years, with a mean of about 247 years.*

(ii) **The long cycle of institutional innovation**: *the founding of every great unificatory dynasty was accompanied by institutional innovation—the commandery-and-county system of the Qin, the recommendation system of the Han, the imperial examination system of the Tang, the civil-service system of the Song, the Grand Secretariat of the Ming and the Grand Council of the Qing. The cycle from the innovation to the maturity of an institution is about 200–300 years.*

(iii) **Coupled fiscal–military collapse**: *the common path of dynastic collapse is fiscal crisis → military weakening → internal rebellion → foreign invasion. The fall of almost every major dynasty was accompanied both by internal troubles (peasant risings) and by external threats (invasions by the nomadic peoples of the north).*

Chinese dynastic succession is the grandest realisation of a nested periodic structure in political history—the dynastic cycle (200–300 years) is nested within the unification–division cycle (300–500 years), and the unification–division cycle is in turn nested within the cycle of institutional evolution (1000–2000 years).

Proof. (i) Alternation of unification and division: Qin–Han unification (221 BCE–220 CE, about 441 years) → division under the Three Kingdoms, the Two Jins and the Northern and Southern Dynasties (220–589, about 369 years) → Sui–Tang unification (589–907, about 318 years) → division under the Five Dynasties and Ten Kingdoms (907–960, about 53 years) → division under the Song, Liao and Jin (960–1279, about 319 years) → Yuan unification (1271–1368, about 97 years) → Ming unification (1368–1644, about 276 years) → Qing unification (1644–1912, about 268 years). The mean length of the periods of unification is about 280 years

($(441+318+97+276+268)/5=280$). The mean length of the periods of division is about 247 years ($(369+53+319)/3=247$).

(ii) Institutional innovation: the Qin created the commandery-and-county system (centralised authority); the Han created the recommendation system (selection of officials); the Sui and Tang created the imperial examination system (selection of officials by examination) and perfected the system of the Three Departments and Six Ministries; the Song strengthened the civil-service system (checking the military by the civil power); the Ming created the Grand Secretariat and abolished the office of chancellor; and the Qing created the Grand Council. Every institutional innovation was completed in the early period of a dynasty and was subsequently ossified in the middle and late periods—the cycle from the flexibility to the ossification of an institution ran roughly in step with the lifespan of the dynasty.

(iii) Coupled fiscal–military collapse: the Qin fell to the revolt of the conscript soldiers that took the Hangu Pass (the rising of Chen Sheng and Wu Guang); the Western Han to the Green Woodsmen and the Red Eyebrows; the Eastern Han to the Yellow Turbans; the Tang to Huang Chao; the Yuan to the Red Turbans; the Ming to Li Zicheng; and the Qing to the 1911 Revolution. In the final period of every dynasty there appeared the vicious circle 'fiscal exhaustion → arrears in military pay → mutiny of the soldiers or decline in fighting capacity → expansion of the rising → still greater fiscal exhaustion'. The critical point of fiscal collapse usually occurred when the rate of land concentration reached 70%–80%—at which point the owner-farmers could no longer bear the taxes of the state. ☐

Theorem 40.8 (Nested periodic structure of political systems). The periodic structure of political systems displays a four-level nested pattern:

Periodic level	Time scale	Typical examples
Micro cycle	10–30 years	Change of regime (coups d'état, electoral turnover)
Meso cycle	100–300 years	The dynastic lifecycle, hegemonic succession
Macro cycle	300–500 years	Alternation of unification and division
Ultra-long cycle	1000–2000 years	Institutional evolution (commandery-and-county system → imperial examination system → Grand Secretariat)

The period ratio between successive levels is about 1.9–10.4 (that is, a frequency ratio of about 0.10–0.53)—the inner cycle completes about 1.9–10.4 whole cycles in each phase of the outer cycle. This nested structure is the realisation in political history of the nested epicycle system of the 'Periodic Systems' chapter.

Proof. Calculation of the frequency ratios: micro/meso = $207/20 \approx 10.4$ (taking the micro mean as 20 years), meso/macro = $400/207 \approx 1.9$, macro/ultra-long = $1500/400 = 3.75$. The nested structure is guaranteed by the difference in order of magnitude between the time scales: the time scale of each level is 1.9–10.4 times that of the level above, so that the inner cycle completes several whole cycles in each phase of the outer cycle. □

40.8 Balance relations in political systems

Theorem 40.9 (Three balance relations of political systems). *The dynamics of a political system is governed by three balance relations:*

(i) **The zero-sum constraint on power**: *in a closed political system the total quantity of power remains constant—power gained by one person or group necessarily means power lost by another. Changes in the distribution of power among monarch, aristocracy, bureaucracy and populace are the dynamic manifestation of the zero-sum constraint on power. The zero-sum constraint on power is the political-science expression of the zero-sum game.*

(ii) **Fiscal balance**: *in a closed fiscal system without external financing, the total revenue of the government (taxation) = total expenditure (administration + military + the imperial household). When expenditure persistently exceeds revenue, the government covers the deficit by seigniorage (inflation), debt default or the sale of privileges—but all these means have their limits. Fiscal collapse is the consequence of the breakdown of fiscal balance—the government can no longer balance revenue and expenditure by any means.*

(iii) **Conversion of legitimacy types**: *there are three sources of legitimacy of a regime (Weber's classification): the traditional ('it has always been so'), the charismatic ('following a great leader') and the legal-rational ('in accordance with legal procedure'). The total legitimacy of a regime is converted among the three types—when traditional legitimacy declines, a charismatic leader rises; after the death of a charismatic leader the regime must establish legal-rational legitimacy in order to maintain stability. The conversion of legitimacy among the three types is the political realisation of phase rotation.*

The hierarchical relation among the three balance relations is as follows: the zero-sum constraint on power is the zero-sum constraint of political games, fiscal balance is the material basis for the survival of a regime, and the conversion of legitimacy types is the psychological basis for regime stability. Together the three constitute the guarantee of self-consistency of the periodic structure of political systems.

Proof. (i) The zero-sum constraint on power: under an absolute monarchy the concentration of power in the monarch, $A(t)$, and the dispersion of power among the aristocracy, $D(t)$, satisfy $A+D=1$ (other holders of power being neglected). When the monarch centralises power ($A \to 1$), the power of the aristocracy is compressed ($D \to 0$); when the aristocracy grows strong ($D \to 1$), the monarch is reduced to a figurehead ($A \to 0$). The struggles in Chinese history between the power of the chancellor and the power of the emperor, and between the localities and the centre, are both instances of the zero-sum constraint on power.

(ii) Fiscal balance: taking the Ming as an example, during the Wanli period (1573–1620) fiscal revenue was about 4 million taels of silver and expenditure about 6 million taels (including the military costs of the Three Great Campaigns and the expenditure of the imperial household), giving an annual deficit of about 2 million taels. By the Chongzhen period (1628–1644) the military expenditure for Liaodong had risen sharply to about 5 million taels a year, whilst in the interior the tax base shrank because of the fighting, so that fiscal revenue fell to about 2 million taels—the annual deficit widening to more than 3 million taels. The persistent deficit ultimately brought about the fiscal collapse of the Ming.

(iii) Conversion of legitimacy types: taking the Qing as an example, in the early Qing legitimacy derived mainly from military conquest (the charismatic type—the personal leadership of Nurhaci and Hong Taiji). Under the Kangxi and Qianlong emperors the Qing shifted the basis of legitimacy from the charismatic to the traditional and legal-rational types by compiling the Complete Library of the Four Treasuries, holding imperial examinations and honouring Confucius and Confucian learning. In the late Qing, traditional legitimacy disintegrated under the impact of the West, and the charismatic leadership of the revolutionaries (Sun Yat-sen) rose—legitimacy rotated once again from the traditional to the charismatic type. □

Remark 40.11 (Unity with the rest of the book). The theory of the periodic structure of political systems established in this chapter is a systematic application of periodic mathematics in political science and history. The dynastic lifecycle (rise–zenith–decline–collapse) is the realisation of an accumulative process in political time—the cumulative effects of land concentration and fiscal crisis drive regime stability from high to low. Regime change (monarchy–aristocracy–democracy–ochlocracy) is the rotational motion of a periodic cycle in institutional space. Imperial hegemonic succession (Portugal–the Netherlands–Britain–the United States) is the scaling realisation of expansion and contraction in geopolitics—the alternation of hegemonic expansion and excessive contraction has a period of about 100–150 years. The two-thousand-year history of Chinese dynastic succession is the political evidence for a nested periodic structure—the dynastic cycle (200–300 years) is nested within the unification–division cycle (300–500 years), and the unification–division cycle is in turn nested within the cycle of institutional evolution (1000–2000 years). The three balance relations of political systems—the zero-sum constraint on power, fiscal balance and the conversion of legitimacy types—together constitute the guarantee of self-consistency of the periodic structure of political systems.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The passage of time in the dynastic lifecycle and the one-way accumulation of land concentration, $dL/dt = -\alpha L$ (Definition 40.12)
Scaling	The expansion–contraction scaling of imperial hegemony (Proposition 40.8); the exponential decay and half-life of fiscal crisis (Lemma 40.3)
Rotation	The phase-space rotation of regime change: the Polybius cycle (Definition 40.13); the rotation of legitimacy among the traditional, charismatic and legal-rational types (Theorem 40.9)
Balance	The zero-sum constraint on power, fiscal balance and the conversion of legitimacy types (Theorem 40.9)
Symmetry	The alternating symmetry of unification and division ('after long division, union; after long union, division', Theorem 40.7)
Closure	The closed causal loop of multiple positive feedbacks: fiscal deterioration → elite defection → military weakening → still worse fiscal deterioration (Definition 40.10)
Period	The three-level nesting of the dynastic cycle (200–300 years, Proposition 40.5), the hegemony cycle (100–150 years, Definition 40.14) and the unification–division cycle (300–500 years, Theorem 40.7) (Theorem 40.8)

Exercises

Exercise 1 Prove the **political periodicity decision theorem** of this chapter (40.1). Hint: first write down the decision map and the evolution group of this chapter's target, then apply the proof structure of the general criterion given in the 'Periodic Systems' chapter.

Exercise 2 Verify that the hypotheses of the **monotonicity of cardinality adjustment** (40.2) hold for the domain objects of this chapter, and give a counterexample in which the hypotheses fail.

Exercise 3 Examine the behaviour of the **three-parameter correspondence: radius, phase and angular speed** (40.1) under parameter degeneration, and explain how the conclusion degenerates accordingly.

Exercise 4 Explain the relation between the **political periodicity diagnosis pipeline and its interface with three chapters** (40.6) and the corresponding general conclusions of the 'Periodic Systems' chapter (whether they agree, whether this is a special case, or whether additional conditions are required).

Exercise 5 Using the seven-theme comparison table of this chapter, identify the concrete carriers of the themes **translation**, **scaling** and **rotation** in this chapter, and give one example of domain data for each.

Exercise 6 Synthesis: restate the diagnostic procedure of this chapter using the five-step standard interface of the 'Periodic Systems' chapter (Table 24.1), noting for each step the theorem of this chapter on which it depends: 40.1, 40.2, 40.1, 40.6.

Chapter 41

Periodic Structure of Data Analysis

Remark 41.1 (Periodicity decision for the target of this chapter (citation)). When this chapter decides whether a research target is periodic, determines its period type, or seeks its minimal positive period, reference is always made to the 'Periodic Systems' chapter: for the periodicity criterion see Theorem 24.1, for the discrimination of period types see Definition 24.3, Proposition 24.1, for the search for the minimal positive period see Definition 24.4, Theorem 24.2, and for the construction of period series and epicycle groups from data see Theorem 24.3, Definition 24.6.

Remark 41.2 (Positioning of this chapter: developed along the five-step interface of the 'Periodic Systems' chapter). The periodic structure of data analysis studies the mathematical methods for the detection, extraction, modelling and prediction of periodic patterns in data science. This chapter does not establish a separate periodicity criterion; instead it connects data analysis to the five-step interface prescribed by the 'Periodic Systems' chapter:

1. **Decision**: reduce such targets as time series, multivariate data, spectra and trends to discrete or continuous dynamical systems, invoke Theorem 24.1 to decide whether they are periodic, and determine the period type;
2. **Construction**: build the amplitude series, the frequency series and the phase series from observational data, spectra and phases, and assemble them into the data epicycle group;
3. **Adjustment**: dynamically adjust the data epicycle group by residual-energy descent, information criteria (AIC/BIC) and derivative-integral descent;
4. **Convolution**: introduce amplitude convolution, phase convolution and frequency convolution on the data epicycle group, and link them to autocorrelation, cross-correlation and phase-coupling networks;
5. **Integration**: connect to the tool chains of the three chapters 'Discrete Sampling and Intercalation of Periodic Signals', 'Harmonic Analysis' and 'Time Series Analysis', forming a diagnostic procedure for data periodicity.

The mathematical tools used in this chapter include: the discrete Fourier transform, the periodogram, multitaper spectral estimation, periodic convolution, window functions, similarity and trend, the refractive index, derivative-integral descent optimisation, and phase-coupling networks.

Remark 41.3 (Guide to the seven themes of this chapter). **Translation**—the lag operator, autocorrelation and delay correlation of a time series (the τ-step self-similarity of the delay τ in Definition 41.16); **Scaling**—the multi-scale window radii of multitaper spectral estimation and the window radius Δ of trend decomposition, and the refractive index η (Proposition 41.2, Definition 41.14, Definition 41.17); **Rotation**—the phase edge weights and phase-difference angles of the phase-coupling network (Definition 41.12); **Balance**—the three balance relations of information conservation, energy conservation and phase conservation (**empirical observation / research programme**, see Remark 41.9); **Symmetry**—the symmetry tests of the data distribution about the three centres of symmetry 0, 1 and i; **Closure**—the closed-loop diagnostic procedure 'detection—modelling—verification—prediction' (Definition 41.13); **Period**—periodogram peaks and the dominant period (Definition 41.6).

41.1 Periodicity decision for data targets: connection to step one of the 'Periodic Systems' chapter

41.1.1 Data targets, decision inputs and the decision map

Definition 41.1 (Data target and decision input)**.** *A* **data target** *is one of the following objects:*

(i) *a univariate time series* $\{X_t\}_{t=1}^n$*;*

(ii) *a multivariate time series* $\{X_t^{(i)}\}_{i=1}^M$*;*

(iii) *a spectral or periodogram sequence;*

(iv) *a trend component or a fluctuation component sequence.*

The **decision input** *of a data target is the pair*

$$\mathcal{D}_{\mathrm{D}} = (\mathcal{T}, x),$$

where $\mathcal{T}$ *is the time index set (which may be* $\mathbb{Z}$ *or* $\mathbb{R}$*) and* $x : \mathcal{T} \to V$ *gives the values of the data over time. If* $\mathcal{T} = \mathbb{Z}$*, the target is called a* **discrete data target***; if* $\mathcal{T} = \mathbb{R}$*, the target is called a* **continuous data target***.*

Definition 41.2 (Data decision map)**.** *Let a data target be given by the decision input* $\mathcal{D}_{\mathrm{D}} = (\mathcal{T}, x)$*. If there exist a map* $\Phi : \mathcal{S} \to \mathcal{S}$ *and an initial value* $x_0 \in \mathcal{S}$ *such that the states of the target are generated by* Φ*, that is,*

$$x_t = \begin{cases} \Phi^t(x_0), & t \in \mathbb{Z}, \\ \Phi_t(x_0), & t \in \mathbb{R}, \end{cases}$$

for $t \in \mathbb{Z}$ *the state* x_t *is generated by iteration of* Φ*; for* $t \in \mathbb{R}$ *it is described by the one-parameter evolution family* $\{\Phi_t\}_{t\in\mathbb{R}}$*, where each* $\Phi_t : \mathcal{S} \to \mathcal{S}$*,* $\Phi_0 = \mathrm{id}$*,* $\Phi_{s+t} = \Phi_s \circ \Phi_t$ *for all* $s, t \in \mathbb{R}$*, and* $(t, x) \mapsto \Phi_t(x)$ *is continuous. then* Φ *is called the* **data decision map** *of the target. The data target is then reduced to the discrete dynamical system* $(\mathcal{S}, \Phi)$ *when* $t \in \mathbb{Z}$ *and to the continuous dynamical system* $(\mathcal{S}, \{\Phi_t\}_{t\in\mathbb{R}})$ *when* $t \in \mathbb{R}$*; in both cases the criteria of the 'Operations and Periodicity' chapter and of the 'Periodic Systems' chapter may be applied directly.*

Theorem 41.1 (Data periodicity decision theorem)**.** *Let the decision map of a data target be Φ, and let $(\mathcal{S}, \Phi)$ be a discrete dynamical system. Then:*

(A) **Sufficient decision**: *if there exists $m \in \mathbb{N}^+$ with $\Phi^m = \mathrm{id}$, then the target is a uniformly periodic data system; every orbit is closed and the orbit length divides m.*

(B) **Equivalence of the decision**: *the target is a uniformly periodic data system if and only if* $\mathrm{ord}(\Phi) < \infty$.

(C) **Pointwise criterion**: *the orbit of the target at a state x_0 is closed if and only if x_0 is a periodic point; the orbit length is then the period at that point.*

(D) **Non-omissibility**: *the conjunction of conserved quantities and symmetry groups cannot replace* $\mathrm{ord}(\Phi) < \infty$*; a quasi-periodic data target is not a periodic data system.*

Proof. The Theorem 41.1 of this chapter is the domain instantiation of Theorem 24.1 of the 'Periodic Systems' chapter: it suffices to replace the decision map f and the evolution group by the Φ of this chapter, and the four clauses (A) sufficient decision, (B) equivalence of the decision, (C) pointwise criterion and (D) non-omissibility hold one by one. By the convention of Remark 24.4, they are not expanded again here. ☐

Corollary 41.1 (Period types of data)**.** *After the decision by Theorem 41.1, a data target falls into one of the following types:*

(i) **Trivially periodic type**: *the minimal period is $m_0 = 1$ and the state is a fixed point;*

(ii) **Strictly periodic type**: $1 < m_0 < \infty$ *and the state is strictly periodic (for example a single-frequency signal);*

(iii) **Uniformly periodic type**: $\mathrm{ord}(\Phi) = m < \infty$ *and every orbit length divides m;*

(iv) **Quasi-periodic type**: $\mathrm{ord}(\Phi) = \infty$ *and the orbit is not closed but is dense on a compact carrier structure (for example multi-frequency signals and sunspot data);*

(v) **Eventually periodic type**: *there exists N such that the target is periodic for $t \geq N$;*

(vi) **Discretely periodic type**: *the carrier structure is $\mathbb{Z}$ and the normalised period ratio satisfies $\rho = T_x/\Delta x \in \mathbb{Q}$.*

(vii) **Asymptotically periodic (damped) type**: *the orbit is not closed, but tends asymptotically to a closed orbit or to a fixed point, the residual decaying as $e^{-\kappa t}$ (see Definition 24.3(vii)).*

Here (i)–(iii) and (vi) belong to periodic data systems; (iv) does not belong to periodic data systems but can be handled by bounded-approximation tools; (v) is a periodic data system only during the stable stage; (vii), like (iv), is a type in the limiting sense; it is not a periodic data system and can be handled with bounded-approximation tools only in its stable stage.

Proof. Each clause follows by substituting the data target into Theorem 41.1 and Definition 24.3 of the 'Periodic Systems' chapter. ☐

Lemma 41.1 (Consistency of periodogram peak detection). Let a single-frequency observation be $X_t = Ae^{i(\omega_0 t+\phi)} + \varepsilon_t$ $(t = 1, \dots, n)$, where $\omega_0 = 2\pi k_0/n$ is a grid frequency, $A > 0$, $\phi \in \mathbb{R}$, and the noise $\{\varepsilon_t\}$ is independent and identically distributed with $\mathbb{E}[\varepsilon_t] = 0$ and $\mathsf{Var}(\varepsilon_t) = \sigma^2 < \infty$. Then the periodogram (Definition 41.6) satisfies

$$\mathbb{E}[I_{k_0}] = A^2 n + \sigma^2, \qquad \mathbb{E}[I_k] = \sigma^2 \quad (k \neq k_0),$$

and the peak signal-to-noise ratio $\dfrac{\mathbb{E}[I_{k_0}]}{\max_{k\neq k_0} \mathbb{E}[I_k]} = 1 + \dfrac{A^2 n}{\sigma^2} \to \infty$ $(n \to \infty)$. Hence the position of the periodogram maximum $\arg\max_k I_k$ converges in probability to the true frequency index k_0.

Proof. The signal term and the noise term superpose linearly in the DFT: $\hat{X}_k = \hat{X}_k^{\mathsf{sig}} + \hat{X}_k^{\mathsf{noise}}$. The signal term is

$$\hat{X}_k^{\mathsf{sig}} = \frac{Ae^{i\phi}}{\sqrt{n}} \sum_{t=1}^{n} e^{i(\omega_0 - 2\pi k/n)t} = \frac{Ae^{i\phi}}{\sqrt{n}} \sum_{t=1}^{n} e^{i\,2\pi(k_0-k)t/n}.$$

When $k = k_0$ the inner sum is n, so $|\hat{X}_{k_0}^{\mathsf{sig}}|^2 = A^2 n$; when $k \neq k_0$ the sum of the roots of unity vanishes, so the signal term is zero. The noise term $\hat{X}_k^{\mathsf{noise}} = \frac{1}{\sqrt{n}} \sum_t \varepsilon_t e^{-i2\pi kt/n}$ satisfies $\mathbb{E}\big[|\hat{X}_k^{\mathsf{noise}}|^2\big] = \frac{1}{n} \sum_{t,s} \mathbb{E}[\varepsilon_t \overline{\varepsilon_s}] e^{-i2\pi k(t-s)/n} = \frac{1}{n} \sum_t \sigma^2 = \sigma^2$ (the noise is independent and identically distributed, so the cross terms have zero expectation). The signal–noise cross term satisfies $\mathbb{E}\big[\hat{X}_k^{\mathsf{sig}} \overline{\hat{X}_k^{\mathsf{noise}}}\big] = 0$ (since $\mathbb{E}[\varepsilon_t] = 0$). Therefore the expectation of $I_k = |\hat{X}_k|^2$ is as above, the peak signal-to-noise ratio diverges, and hence for n sufficiently large the expectation of the periodogram at k_0 strictly dominates the remaining frequencies—by the dominance of the expectation and the finite variance of the noise, $\arg\max_k I_k$ converges in probability to k_0. ☐

Definition 41.3 (Algorithm for the minimal positive period of data). This definition has the same content as Definition 24.4 in the section 'Periodicity decision: whether a target is periodic' (Section 24.1) of Chapter 24; to avoid repetition, only the reference anchor is retained here, and the complete statement is given there. The only difference here is terminology (data target / data sample); the steps of the algorithm and the conclusion are exactly the same.

Theorem 41.2 (Correctness of the algorithm for the minimal positive period of data). This theorem has the same content as Theorem 24.2 in the section 'Periodicity decision: whether a target is periodic' (Section 24.1) of Chapter 24; to avoid repetition, only the reference anchor is retained here, and the complete statement and proof are given there. The only difference here is terminology (data target / data sample); the steps of the algorithm and the conclusion are exactly the same.

Example 41.1 (Examples of data periodicity decision). (i) Single-frequency sinusoidal signal: the periodogram exhibits a peak at the true frequency; uniformly periodic type;

(ii) Monthly mean sunspot numbers: an 11-year main period superposed on an 88-year long period; quasi-periodic type;

(iii) Multivariate EEG signals: multi-channel phase coupling; quasi-periodic type;

(iv) Temperature anomaly series: an upward trend since the 20th century; eventually periodic type (superposed periodicity).

Remark 41.4 (Conserved quantities do not guarantee periodicity). *Information conservation, energy conservation and phase conservation in data analysis are auxiliary conditions for periodic structure, but they do not constitute a sufficient condition. For example, white noise satisfies energy conservation yet has no periodic component. The periodic structure of data must be decided by Theorem 41.1 or by the lemma above.*

41.1.2 Step two: construction of the data period series and epicycle group

Definition 41.4 (Data amplitude, frequency and phase series)**.** Let the state or output sequence of a data target be $\{z_t\}$. If it can be written as

$$z_t = \sum_k A_k e^{i(\omega_k t+\phi_k)} + \varepsilon_t,$$

then

$$\mathcal{A}_z(t) = \sum_k A_k e^{ikt}, \qquad \mathcal{F}_z(s) = \sum_k \omega_k s^k, \qquad \mathcal{P}_z(t) = \sum_k e^{i(kt+\phi_k)}$$

are called respectively the **amplitude series**, the **frequency series** and the **phase series** of the data target. The three agree with Definition 6.10, Definition 6.12 and Definition 6.11 of the 'Series and Convolution' chapter.

Definition 41.5 (Data epicycle group)**.** Let the period series of a data target be given by its amplitude series $\{A_k\}$, frequency series $\{\omega_k\}$ and phase series $\{\phi_k\}$. For each k define the epicycle

$$E_k : \theta \mapsto A_k e^{i(\theta+\phi_k)}, \qquad \theta = \omega_k t,$$

whose collection

$$\mathfrak{E}_{\mathrm{D}} = \{(A_k, \omega_k, \phi_k)\}_{k\in\mathbb{Z}}$$

is called the **epicycle group** of the data target. Its superposition is the data period series

$$z_t = \sum_k E_k(\omega_k t) = \sum_k A_k e^{i(\omega_k t+\phi_k)}.$$

Theorem 41.3 (Constructing the period series from data)**.** *Let the output of a data target be a finite periodic sequence of period N. Then:*

(i) *the output admits the discrete Fourier expansion*

$$z_t = \sum_{k=0}^{N-1} c_k e^{i2\pi kt/N}, \qquad c_k = \frac{1}{N}\sum_{t=0}^{N-1} z_t e^{-i2\pi kt/N};$$

(ii) the amplitude series is given by $|c_k|$, the phase series by $\arg c_k$ and the frequency series by $2\pi k/N$;

(iii) if the output is a single-frequency signal, its Fourier decomposition has only a single spectral line; if the output is a superposition of several frequencies, each order of harmonic corresponds to a different frequency component.

Proof. (i) The discrete Fourier series of a finite periodic sequence is a standard result: the basis functions $\{e^{i2\pi kt/N}\}_{k=0}^{N-1}$ form an orthogonal basis of $\mathbb{C}^N$. (ii) This is read off directly from $c_k = A_k e^{i\phi_k}$. (iii) A single-frequency signal has only the fundamental component on the period $T = 2\pi/\omega$; a superposition of several frequencies is expanded as a Fourier series. $\square$

Proposition 41.1 (Three-parameter correspondence: radius, phase and speed). *The three parameters of the data epicycle group $\mathfrak{E}_{\mathrm{D}} = \{(A_k, \omega_k, \phi_k)\}_k$ correspond one-to-one to three geometric–data quantities:*

(i) **Radius corresponds to amplitude**: *the radius of E_k, namely $|E_k| = A_k$, is the amplitude of that oscillatory component of the data;*

(ii) **Phase corresponds to rotational delay**: *the initial phase ϕ_k corresponds to the time translation $\tau_k = \phi_k/\omega_k$, that is, the epicycle* **leads** *the reference by τ_k ($\phi_k > 0$ means that the waveform is* **advanced** *by τ_k relative to the reference, equivalently the waveform coincides with the reference once the reference has been delayed by τ_k);*

(iii) **Frequency corresponds to rotational speed**: *ω_k is the angular frequency of that oscillatory component of the data.*

Proof. The Proposition 41.1 of this chapter corresponds exactly to Proposition 24.3 of the 'Periodic Systems' chapter, and the three correspondences—radius to amplitude, phase to rotational delay, frequency to rotational speed—are obtained by translating them one by one to the epicycle group of this chapter. The proof is given there: (i) from $|E_k| = A_k$ and the energy decomposition, (ii) from the phase translation theorem, (iii) from the definition of the frequency series. $\square$

Definition 41.6 (Spectral analysis of a time series). Let $\{X_t\}_{t=1}^n$ be observed time-series data. Its **discrete Fourier transform** (DFT) is

$$\hat{X}_k = \frac{1}{\sqrt{n}} \sum_{t=1}^{n} X_t e^{-i2\pi kt/n}, \quad k = 0, 1, \ldots, n-1.$$

The **periodogram** is the squared modulus of the DFT: $I_k = |\hat{X}_k|^2$. The periodogram exhibits a peak at the true periodic frequency $\omega_0 = 2\pi k_0/n$. The statistical properties of the periodogram are: $\mathbb{E}[I_k] \to S(\omega_k)$ (asymptotically unbiased), but $\mathrm{Var}(I_k) \to S^2(\omega_k)$ (not a consistent estimator)—the variance does not decrease as n increases.

The periodogram is the direct application of the Fourier transform in time-series analysis—it preserves the energy information of components of the same frequency and is an asymptotically unbiased estimator of the power spectral density.

Proposition 41.2 (Multitaper spectral estimation)**.** To overcome the problem of the divergent variance of the periodogram, **multitaper spectral estimation** uses a set of orthogonal window functions to carry out several independent spectral estimations of the same time series and then averages them. Let the K orthogonal window functions be $\{h_k(t)\}_{k=1}^{K}$ (for example Slepian sequences), each window function yielding an independent periodogram $I_k(\omega)$. The multitaper spectral estimate is

$$\hat{S}_{\mathsf{MT}}(\omega) = \frac{1}{K}\sum_{k=1}^{K} I_k(\omega).$$

The variance of the multitaper spectral estimate is reduced to $1/K$ of that of the periodogram—the use of window functions is equivalent to taking K independent samples in the frequency domain.

Multitaper spectral estimation is the direct application of the window-function system in spectral estimation. Different window functions correspond to different frequency-domain properties—the Slepian window has the most concentrated energy within a given bandwidth, the Gaussian window has the best smoothness in both the time and the frequency domain, and the Mexican-hat window has the highest sensitivity when detecting spectral peaks.

Proof. For each window h_k ($k = 1, \dots, K$), the windowed periodogram $I_k(\omega)$ is asymptotically unbiased for $S(\omega)$ with variance tending to $S^2(\omega)$ (the statistical properties of Definition 41.6). For the Slepian sequences the windows are pairwise orthogonal in the time domain and their frequency-domain energy is concentrated in mutually disjoint frequency bands, so the $I_k(\omega)$ are asymptotically mutually uncorrelated. By the bilinear expansion of the variance,

$$\mathsf{Var}\left[\hat{S}_{\mathsf{MT}}(\omega)\right] = \mathsf{Var}\left(\frac{1}{K}\sum_{k=1}^{K} I_k(\omega)\right) = \frac{1}{K^2}\sum_{k=1}^{K}\mathsf{Var}[I_k(\omega)] + \frac{2}{K^2}\sum_{k<l}\mathsf{Cov}(I_k, I_l).$$

The orthogonal windows give $\mathsf{Cov}(I_k, I_l) \to 0$ ($k \neq l$), whilst $\mathsf{Var}[I_k(\omega)] \to S^2(\omega)$, so

$$\mathsf{Var}\left[\hat{S}_{\mathsf{MT}}(\omega)\right] \to \frac{1}{K^2}\cdot K S^2(\omega) = \frac{S^2(\omega)}{K},$$

that is, the variance is reduced to $1/K$ of that of the single-window periodogram, whilst the mean $\mathbb{E}[\hat{S}_{\mathsf{MT}}] = \frac{1}{K}\sum_k \mathbb{E}[I_k] \to S(\omega)$ remains asymptotically unbiased. □

Definition 41.7 (Period detection based on periodic convolution)**.** For a time series $\{X_t\}$, the steps for detecting its periodic structure by means of periodic convolution are:

(i) **Self-periodic convolution**: take $f = g = X$ and compute the periodic convolution $\mathcal{K}^{(d)}\{X, X\}$ ($d = 1, \dots, 6$; the 12 real trend spectra are its real and imaginary parts). When $d = 1$ (the basic difference angle, $\alpha = 1$), the classical limit of the periodic trend spectrum is the power spectrum—that is, the periodogram.

(ii) **Phase-relation detection**: take $\alpha = 0$ (the similarity layer) and compute $\mathrm{Im}\,\mathcal{K}^{(1)}\{X, X\}$ (the difference-angle sine). The output exhibits a peak at a trend reversal point (turning point)—the sine difference-angle similarity detects a phase jump.

(iii) **Amplitude-relation detection**: take $\alpha = 1$ (the trend layer) and compute $\mathrm{Re}\,\mathcal{K}^{(2)}\{X, X\}$ (the sum-angle cosine). The output measures the self-coupling strength of the signal—the energy transfer between components of different frequencies.

(iv) **Multi-scale period identification**: use window functions K_Δ of different radii to smooth X and compute the periodic convolution at each scale. The frequency corresponding to the peak is the dominant period at that scale.

Proposition 41.3 (Comparison of periodic convolution with classical methods)**.** *The advantages of periodic convolution in period detection are as follows:*

Method	What is detected	Phase information retained	Multi-period coupling
Fourier transform / periodogram	A single period (frequency decomposition)	Partial (only the phase of each frequency)	No
Autocorrelation function	A single period (time-domain delay)	No (amplitude only)	No
Wavelet transform	Multi-scale periods	Partial (the phase of each scale)	No
Periodic convolution ($\alpha = 0$)	Phase relations (direction matching)	Complete (all frequency pairs)	Yes (all frequency pairs)
Periodic convolution ($\alpha = 1$)	Amplitude relations (direction + scale)	Complete (all frequency pairs)	Yes (all frequency pairs)

Periodic convolution is the only method that can simultaneously detect several periods, retain all phase information and quantify the strength of coupling between periods.

Proof. Substitute the period series expansion $X(t) = \sum_n A_n e^{i(\omega_n t + \phi_n)}$ into the self-periodic convolution of the single-period layer (taking $W \equiv 1$, $\alpha = 1$ and the difference-angle kernel $S^{(1)}(\phi_n, \phi_m) = e^{i(\phi_m - \phi_n)}$):

$$\mathcal{K}^{(1)}\{X, X\}(\tau) = \sum_{n,m} A_n A_m \, e^{i(\phi_m - \phi_n)} \, e^{i\omega_m \tau}.$$

The term (n, m) has modulus $A_n A_m$, argument $\phi_m - \phi_n$ and frequency ω_m—that is, the amplitude products and phase differences of **all frequency pairs** (n, m) are retained simultaneously. Compare the three classical methods: (i) in the classical limit ($W \to \delta$) the periodogram/Fourier transform retains only the diagonal terms $|c_n|^2 = A_n^2$ (Definition 41.6), the arguments ϕ_n are erased by the squared modulus, and all off-diagonal frequency pairs $(n \neq m)$ are lost, so there is no multi-period coupling and the phase information is incomplete; (ii) by the Wiener–Khinchin theorem the autocorrelation function equals the inverse transform of the periodogram, $R(\tau) = \sum_n A_n^2 e^{i\omega_n \tau}$, which contains only A_n^2, so the phase information is completely lost, and it is a single-variable function in the time domain with no frequency-pair coupling (the **delayed** convention is taken here, differing from the advanced convention of Proposition 41.7 in this chapter by $h \to -h$; see Remark 22.4 in the 'Harmonic Analysis' chapter); (iii) the wavelet transform $W(a, b) = \langle X, \psi_{a,b} \rangle$ decomposes according to the scale a, and each scale retains its local phase, but the output is a single-parameter spectrum on the 'scale–time' plane and contains no cross-coupling terms between two different frequencies (scales). Hence periodic convolution alone satisfies all three of 'multi-period + full phase + all-frequency-pair coupling', and the conclusion of the comparison table holds. ☐

Example 41.2 (Example of constructing a data period series)**.** For the monthly mean sunspot series, the periodogram exhibits its strongest peak at 11 years, with secondary peaks at 5.5 years and 88 years. In the data epicycle group, 11 years corresponds to the fundamental frequency, 5.5 years to the second harmonic, and 88 years to a long-period component.

Remark 41.5 (The period series is a complex function)**.** Let $z = e^{i\omega_0 t}$; then the data period series

$$z_t = \sum_k A_k e^{i(\omega_k t + \phi_k)}$$

can be written as the Laurent series $\sum_k c_k z^k$, where $c_k = A_k e^{i\phi_k}$, provided that every component frequency is commensurable with ω_0 (that is, $\omega_k = k\omega_0$ with $k \in \mathbb{Z}$). Hence when all components are commensurable with ω_0, the period series is a complex-valued function on the unit circle (otherwise it is a function on the torus).

41.1.3 Step three: dynamic adjustment of the data epicycle group

Definition 41.8 (Dynamic adjustment of the data epicycle group)**.** Let the current data epicycle group be $\mathfrak{E}^{(j)}$, with partial sum

$$z^{(j)}(t) = \sum_{k \in \mathfrak{E}^{(j)}} A_k e^{i(\omega_k t + \phi_k)},$$

and residual $r^{(j)} = z - z^{(j)}$. **Dynamic adjustment** consists of two rules:

(i) **Quantity adjustment**: if the residual energy satisfies $\|r^{(j)}\|^2 > \delta$, add a new epicycle according to the peak of the residual periodogram;

(ii) **Parameter adjustment**: update the parameters of the existing epicycles by regularised least squares or by information criteria.

In data analysis, quantity adjustment corresponds to increasing the number of frequency components or the harmonic order; parameter adjustment corresponds to adjusting the amplitude, the phase and the frequency so that goodness of fit and model parsimony are balanced.

Proposition 41.4 (Monotonicity of quantity adjustment). *After the dominant residual frequency component is added as in Definition 41.8(i), the residual energy is monotonically non-increasing:*

$$\|r^{(j+1)}\|^2 \le \|r^{(j)}\|^2,$$

with equality if and only if the residual has no energy at the selected frequency.

Proof. The Proposition 41.4 of this chapter agrees with the dynamic adjustment mechanism of Definition 24.8 in the 'Periodic Systems' chapter: adding an epicycle removes the component of $r^{(j)}$ along $\hat{\omega}$ from the residual, and the orthogonal decomposition $\|r^{(j)}\|^2 = \|r^{(j+1)}\|^2 + \|\text{the removed component}\|^2$ gives the inequality, with equality if and only if the norm of the removed component is zero. The complete argument is given there. ☐

Theorem 41.4 (Convergence of dynamic adjustment). *Let the data target be a* **real-valued finite periodic sequence** *of length T_0 ($T_0 \in \mathbb{N}$) and let $\delta = 0$. Then after at most $\lfloor T_0/2 \rfloor + 1$ steps of dynamic adjustment the residual is zero, and the resulting data epicycle group is the exact Fourier data of the target; for a* **complex-valued** *sequence the corresponding number of steps is at most T_0.*

Proof. The discrete Fourier series of a **real-valued** finite periodic sequence of length T_0 ($T_0 \in \mathbb{N}$) is complete on $\lfloor T_0/2 \rfloor + 1$ harmonics (including the constant term); a **complex-valued** sequence requires T_0 components. At each step one dominant frequency component of the residual is extracted as in Proposition 41.4 and the residual energy decreases strictly, so after finitely many steps the residual is zero. ☐

Definition 41.9 (Structure-metric adjustment). Let the structure metric of a data target be

$$\mathcal{I} = (\mathcal{I}_{\text{fitting}}, \mathcal{I}_{\text{parsimony}}, \mathcal{I}_{\text{prediction}}) = (\text{residual sum of squares, number of parameters, cross-valida}$$

Structure-metric adjustment is: at each step choose a neighbouring parameter θ in the parameter space Θ for which $\mathcal{I}(\theta)$ is **strictly decreasing** in the lexicographic order (minimisation); if no such neighbouring parameter exists, the process stops—then θ is a **locally optimal** parameter.

Proposition 41.5 (Finite convergence of structure-metric adjustment). *Let the parameter space Θ be finite and let each adjustment make $\mathcal{I}$* **strictly decreasing** *in the lexicographic order. Then the adjustment terminates after finitely many steps at a locally optimal parameter.*

Proof. Θ is finite, so $\mathcal{I}$ takes finitely many values. The lexicographic order is well-founded on a finite set of values, so a strictly lexicographically decreasing sequence cannot continue indefinitely and the process terminates after finitely many steps. At termination there is no neighbouring parameter making $\mathcal{I}$ strictly lexicographically decreasing, that is, the parameter is locally optimal. Monotonicity must be imposed on the **whole lexicographic order**: if only 'some component strictly increases or decreases' is required, cycles such as $(2,1) \to (1,2) \to (2,1)$ cannot be excluded and finite termination fails. ☐

Definition 41.10 (Data fitting by nested epicycle systems)**.** Given an observed time series $\{X_t\}_{t=1}^n$, nested epicycle systems are used for data-driven periodic modelling. Let the K-level nested epicycle system be

$$X(t) = \sum_{k=1}^{K} A_k e^{i(\omega_k t + \phi_k)} + \varepsilon(t),$$

where $\varepsilon(t)$ is the residual (noise). The parameters $\Theta = \{A_k, \omega_k, \phi_k\}_{k=1}^K$ are estimated by minimising the residual sum of squares:

$$\hat{\Theta} = \arg\min_{\Theta} \sum_{t=1}^{n} \left| X_t - \sum_{k=1}^{K} A_k e^{i(\omega_k t + \phi_k)} \right|^2 .$$

Since the ω_k occur in the exponents, the objective function is nonlinear in the ω_k—traditional linear least squares does not apply. Kernel-smoothed derivative-integral descent optimisation is used, estimating all parameters simultaneously by the window-function derivative-integral descent method.

Proposition 41.6 (Parameter estimation by derivative-integral descent optimisation)**.** *Parameter estimation for nested epicycle systems by the derivative-integral descent optimisation algorithm proceeds in three steps:*

(i) **Preliminary frequency estimation**: *take the Fourier transform of the data and use the frequencies corresponding to the largest K peaks of the periodogram as the initial estimates $\omega_k^{(0)}$ of the ω_k. This step amounts to coarse localisation by traditional methods.*

(ii) **Fine derivative-integral optimisation**: *use the Gaussian kernel K_{Gauss} for windowed derivative-integral optimisation of the amplitudes A_k, the Mexican-hat kernel K_{MexHat} for second-derivative optimisation of the frequencies ω_k, and the Hilbert kernel K_{Hilbert} for orthogonal-phase optimisation of the phases ϕ_k. The derivative-integral fusion function $\Phi(\theta, r, \dot{\theta}, \dot{r})$ exploits both first-derivative and integral information, escaping local minima in a non-convex objective function.*

(iii) **Model selection**: *use an information criterion (AIC or BIC) to determine the optimal epicycle order K—a K that is too small underfits (omitting important periods), whilst a K that is too large overfits (fitting the noise). The BIC criterion is $\text{BIC}(K) = n \ln(\text{RSS}/n) + K \ln n$, and the K minimising the BIC is selected.*

Proof. The argument has three parts. (i) **Asymptotic consistency of the frequency initialisation**: in the single-component case this is given directly by Lemma 41.1—the position of the periodogram peak converges in probability to the true grid frequency; in the multi-component case, under the frequency-separation condition (the main peaks do not overlap), taking the first K peaks in decreasing order of size gives the K components in turn, so $\omega_k^{(0)} \to \omega_k$ $(k = 1, \ldots, K)$. (ii) **Descent property of the fine derivative-integral optimisation**: the objective function $L(\Theta) = \sum_{t=1}^n |X_t - \sum_k A_k e^{i(\omega_k t + \phi_k)}|^2$ is a smooth function of Θ in a neighbourhood of the true value Θ^*. By the non-local descent property of derivative-integral descent, as long as there is some window–kernel pair for which the kernel-smoothed cumulative quantity is negative, the update gives a strictly descending direction $L(\Theta_{t+1}) < L(\Theta_t)$; hence the iteration starting from a consistent initial point $\Theta^{(0)}$ remains in a neighbourhood of the true value and approaches Θ^*. (iii) Consistency of the BIC: the residual sum of squares $\mathrm{RSS}(K)$ is monotonically non-increasing as K increases; the BIC penalty term $K \ln n$ imposes an asymptotically positive penalty on overfitting as n increases, and the minimiser $\hat{K}$ of $\mathrm{BIC}(K) = n \ln(\mathrm{RSS}(K)/n) + K \ln n$ converges in probability to the true order K^*—for $K < K^*$ underfitting makes RSS markedly too large, whilst for $K > K^*$ the decrease in RSS is insufficient to offset the growth of the penalty term $K \ln n$. □

Example 41.3 (BIC selection of a data period series). Fit a K-level epicycle system to the sunspot series and compute $\mathrm{BIC}(K) = n \ln(\mathrm{RSS}/n) + K \ln n$. Choose the K that minimises the BIC. Usually $K = 3$ suffices to capture the 11-year fundamental frequency, the 5.5-year second harmonic and the 88-year long period.

Remark 41.6 (The limits of dynamic adjustment). Theorem 41.4 relies on finite periodic sequences. For continuous-spectrum or quasi-periodic data targets (such as multi-frequency signals and sunspots) no finite epicycle group represents them exactly, and only bounded approximation is possible. Structure-metric adjustment does not guarantee a global optimum.

41.1.4 Step four: data convolution and phase-coupling networks

Definition 41.11 (Data amplitude, phase and frequency convolutions). Let two data epicycle groups be

$$\mathfrak{E} = \{(A_n, \omega_n, \phi_n)\}, \qquad \mathfrak{E}' = \{(B_m, \nu_m, \psi_m)\}.$$

Define:

1. (i) **Amplitude convolution**: $\mathbb{A}_{mn} = A_n B_m$;
2. (ii) **Phase convolution**: $\mathbb{P}_{mn} = e^{i(\phi_n - \psi_m)}$;
3. (iii) **Frequency convolution**: $\mathbb{F}_{mn} = \omega_n \nu_m$;
4. (iv) **Trend convolution**: $\mathbb{T}_{mn} = A_n B_m e^{i(\phi_n - \psi_m)} = A_n B_m \mathbb{P}_{mn}$.

This agrees with Definition 6.21, Definition 6.22, Definition 6.23 and Definition 6.24 of the 'Series and Convolution' chapter.

Theorem 41.5 (Action of the three types of convolution on data epicycle groups). *Let the epicycle groups of two data targets be $\mathfrak{E}$ and $\mathfrak{E}'$ respectively. Then:*

1. *(i) amplitude convolution acts only on the radii;*
2. *(ii) phase convolution acts only on the phases;*
3. *(iii) frequency convolution acts only on the rotational speeds;*
4. *(iv) trend convolution retains both amplitude and phase.*

Proof. This is read off directly from Definition 41.11. □

Proposition 41.7 (Autocorrelation and trend convolution)**.** *Let $\{z_t\}$ be a data period sequence with autocorrelation function*

$$R_z(h) = \sum_t z_t \overline{z_{t+h}}.$$

In the language of the data epicycle group, the autocorrelation function is the projection of the trend convolution onto the diagonal:

$$R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}.$$

This is the **advanced** *convention of the present section; the Wiener–Khinchin form $R(\tau) = \sum_n A_n^2 e^{i\omega_n \tau}$ cited earlier in this chapter in the comparison of the three classical methods takes the* **delayed** *convention. The two are interchanged by $R_{\text{advanced}}(h) = R_{\text{delayed}}(-h)$; see Remark 22.4 in the 'Harmonic Analysis' chapter.*

Proof. The Proposition 41.7 of this chapter is the realisation in this domain of the convolution-action conclusion of Theorem 24.6 in the 'Periodic Systems' chapter: substituting $z_t = \sum_n A_n e^{i(\omega_n t + \phi_n)}$ into the definition of the autocorrelation, cancelling all off-diagonal cross terms by orthogonality and retaining only the diagonal term of the trend convolution (the factorisation of the three convolutions, Proposition 24.6), which yields $R_z(h) = \sum_n \mathbb{T}_{nn} e^{-i\omega_n h}$. The complete expansion is given there. ☐

Definition 41.12 (Construction of the phase-coupling network)**.** For a multivariate time series $\{X_t^{(i)}\}_{i=1}^M$ comprising M variables (such as several sensors, several stocks or several EEG electrodes), a **phase-coupling network** may be constructed to reveal the periodic interactions between the variables. The nodes of the network are the variables and the edge weights are the periodic-convolution similarities between pairs of variables:

$$W_{ij}^{(d)} = \left\| \mathcal{K}^{(d)}\{X^{(i)}, X^{(j)}\} \right\|, \quad d = 1, \dots, 12.$$

$W_{ij}^{(1)}$ measures the co-directional coupling strength of two variables in frequency, $W_{ij}^{(2)}$ measures the phase-orthogonal coupling strength, and $W_{ij}^{(7)}$ measures the sum-angle coupling strength. The topology of the phase-coupling network—connectivity, community structure and central nodes—reveals the functional organisation of the system.

Lemma 41.2 (Edge-weight symmetry of the phase-coupling network)**.** *Let the weight kernel W be even ($W(-\Omega) = W(\Omega)$, for example a Gaussian, sinc or Mexican-hat kernel), let the phase kernel $S^{(d)}$ have unit modulus ($|S^{(d)}| \equiv 1$, $d = 1, \dots, 6$), and let the edge weights be taken in the energy norm*

$$\left\| \mathcal{K}^{(d)}\{f, g\} \right\| := \left(\sum_{n,m} \left| W(\omega_n - \nu_m)(A_n B_m)^\alpha S^{(d)}(\phi_n, \psi_m) \right|^2 \right)^{1/2}.$$

Then the edge weights satisfy symmetry and non-negativity:

$$W_{ij}^{(d)} = W_{ji}^{(d)} \geq 0,$$

so the matrix $\{W_{ij}^{(d)}\}$ is the adjacency matrix of a **weighted undirected graph** *on M nodes, and standard network-analytical methods (community detection, centrality) are therefore applicable.*

Proof. Expand the square of the energy norm and take the modulus term by term for the complex phase kernel $S^{(d)}$ and its real and imaginary parts (the 12 real similarities): for the complex kernel with $d = 1, \dots, 6$, $|S^{(d)}| \equiv 1$; for its real part $|\cos(\phi_n \mp \psi_m)|$ and its imaginary part $|\sin(\phi_n \mp \psi_m)|$, the modulus is unchanged under the exchange $n \leftrightarrow m$ because the cosine is even and the sine is odd. Hence the sum of squares depends only on $|W(\omega_n - \nu_m)|$, on $|A_n B_m|^\alpha$ and on the symmetric moduli above. Exchanging $f \leftrightarrow g$ exchanges $n \leftrightarrow m$, $\omega_n \leftrightarrow \nu_m$, $A_n \leftrightarrow B_m$ and $\phi_n \leftrightarrow \psi_m$: since W is even, $|W(\nu_m - \omega_n)| = |W(\omega_n - \nu_m)|$, and $|B_m A_n| = |A_n B_m|$, whilst the phase moduli are unchanged as above, so the sums agree term by term and $\|\mathcal{K}^{(d)}\{f, g\}\| = \|\mathcal{K}^{(d)}\{g, f\}\|$. Non-negativity follows directly from the definition of the norm. Substituting $f = X^{(i)}$ and $g = X^{(j)}$ gives $W_{ij}^{(d)} = W_{ji}^{(d)} \geq 0$. □

Proposition 41.8 (Analytical methods for the phase-coupling network)**.** The phase-coupling network may be interpreted by standard network-analytical methods:

(i) **Community detection**: use modularity maximisation to partition the nodes into communities—nodes in the same community have similar periodic structures (in-phase synchronisation), whilst nodes in different communities have orthogonal periodic structures (anti-phase synchronisation or a fixed phase difference).

(ii) **Centrality analysis**: the degree centrality of a node (with how many other nodes it has strong coupling) measures the periodic 'influence' of that variable in the system. Nodes of high centrality are the key drivers of the periodic structure of the system.

(iii) **Dynamic evolution**: construct phase-coupling networks over different time windows and observe how the network topology changes with time. Abrupt changes in the network topology (such as the reorganisation of community structure or the migration of central nodes) may be signals that the system has undergone a phase transition or entered a new state.

Proof. By Lemma 41.2 the edge-weight matrix $\{W_{ij}^{(d)}\}$ is symmetric and non-negative, so (i) community-detection algorithms such as modularity maximisation and spectral clustering are mathematically well defined (they depend only on a symmetric non-negative adjacency matrix); pairs of nodes with strong in-phase coupling (difference-angle cosine close to 1) have large edge weights and are therefore placed in the same community, whilst pairs of nodes in phase quadrature (difference-angle sine dominant) have small edge weights and tend to belong to different communities. (ii) The degree centrality $D_i^{(d)} = \sum_{j \neq i} W_{ij}^{(d)}$ is the total coupling strength of node i with the other nodes; if the energy of some node exceeds half of the total coupling energy, that node is a **dominant periodicity driver** of the network. (iii) Dynamic evolution: the network matrix $W(t)$ over a time window $[t, t + \Delta]$ is a matrix-valued function of t, and its community structure is characterised continuously by the spectrum of $W(t)$ (the leading eigenvalues and their eigenvectors); when the dominant component of an eigenvector changes sign or the order of the leading eigenvalues is exchanged, community membership changes abruptly, corresponding to a topological phase transition of the network. □

Example 41.4 (An example of a phase-coupling network). Constructing a phase-coupling network for multi-channel EEG signals, channels with strong in-phase coupling are placed in the same community, corresponding to the modular organisation of the functional brain network. Channels of high centrality correspond to the key hubs of the functional network.

Remark 41.7 (Relation between data convolution and periodic convolution). Data convolution is not a new convolution operation but an instance of the periodic convolution of the 'Series and Convolution' chapter on data epicycle groups. Autocorrelation is the projection of the trend convolution onto the diagonal; cross-correlation and phase coupling are respectively the concrete manifestations in data science of amplitude convolution and frequency convolution.

41.1.5 Step five: integration with discrete sampling, harmonic analysis and time-series analysis

Definition 41.13 (Data periodicity diagnostic pipeline). For given time-series data, periodic mathematics provides the following standardised diagnostic pipeline:

(i) **Data preprocessing**: remove the trend (by differencing or window-function smoothing), handle missing values (by interpolation or model prediction) and standardise (zero mean and unit variance).

(ii) **Classical spectral analysis**: compute the periodogram and the multitaper spectral estimate and identify the significant spectral peaks. This step gives a preliminary overview of the periodic structure.

(iii) **Period detection by periodic convolution**: compute $\mathcal{K}^{(1)}\{X, X\}$ to $\mathcal{K}^{(6)}\{X, X\}$ ($\alpha = 0, 1$; the 12 real trend spectra are the real and imaginary parts of the 6 complex kernels) and extract the spectral, amplitude and phase components in each dimension. Identify the dominant periods, detect phase jumps and quantify the period coupling.

(iv) **Multi-scale window-function decomposition**: use the family of window functions $\{\Delta_1, \ldots, \Delta_K\}$ to smooth the data at multiple scales, repeating steps 2–3 at each scale. Compare the periodic structures at different scales—identifying which periods are stable across scales and which are scale-dependent.

(v) **Nested epicycle system fitting**: fit a nested epicycle system by derivative-integral descent optimisation to obtain the precise parameters $\{A_k, \omega_k, \phi_k\}$ of each periodic component. Analyse the amplitude ratios and frequency ratios of the epicycles at each level—assessing the nested hierarchy of the system.

(vi) **Construction of the phase-coupling network** (multivariate data): construct the periodic-convolution coupling matrix between the variables and carry out community detection and centrality analysis.

(vii) **Periodic prediction**: *extrapolate and predict on the basis of the fitted nested epicycle system. The uncertainty of the prediction is determined by the amplitude decay factors and the phase drift rates of the epicycles at each level.*

Theorem 41.6 (The data periodicity diagnostic pipeline and its interface with three chapters)**.** The seven steps of Definition 41.13 connect respectively to:

(i) Definition 21.1, Theorem 21.3 and Theorem 21.10 of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter;

(ii) Definition 22.5, Theorem 22.1 and Proposition 22.11 of the 'Harmonic Analysis' chapter;

(iii) Definition 23.16, Proposition 23.2 and Definition 23.19 of the 'Time Series Analysis' chapter.

Hence data periodicity diagnosis is not an independent method but an application of the unified tool chain of periodic mathematics in data science.

Proof. The Theorem 41.6 of this chapter is the realisation in this domain of the five-step interface (Table 24.1) of the 'Periodic Systems' chapter: periodicity decision uses sampling and spectral estimation, frequency decomposition and phase kernels use the 'Harmonic Analysis' chapter, and periodogram peaks, the Fisher test and window-function validation use the 'Time Series Analysis' chapter. Hence the diagnostic pipeline is assembled from the tool chains of the three chapters; the proof is given there. ☐

Remark 41.8 (The order of operations after integration)*.* The recommended order for an actual diagnosis is: first decide whether the data target is periodic (Theorem 41.1); then construct the data Fourier data and the epicycle group (Theorem 41.3); then optimise the parameters by derivative-integral descent and dynamic adjustment (Theorem 41.4, Proposition 41.6); then carry out the data convolution and phase-coupling analysis (Proposition 41.7, Definition 41.12); and finally output the diagnosis as in Definition 41.13.

Remark 41.9 (The three conservation laws of data analysis (empirical observation / research programme))*.* Periodic analysis of data is governed by three conservation laws:

(i) **Information conservation**: in the noise-free compression and reconstruction of data, the total amount of information is conserved. Periodic convolution is an information-lossless integral transform—compression and decompression use the same kernel function, and differentiation and integration are exactly mutually inverse by the fundamental theorem of calculus.

(ii) **Energy conservation**: the total energy (variance) of a signal equals the sum of the energies of its frequency components in the Fourier domain—Parseval's identity. In nested epicycle system fitting, the sum of the energies of the epicycles at each level equals the total energy of the original signal (minus the noise). The proportion in which energy is distributed among the periodic components is the scaling structure of the system.

(iii) **Phase conservation**: in signal processing without phase distortion, the relative phase relations of the frequency components are conserved. Periodic convolution retains the phase information between all frequency pairs, ensuring that phase is not lost in the transformation and reconstruction. The stability of the phase-coupling network is the data manifestation of phase conservation.

The three conservation laws together guarantee the reliability of data periodicity diagnosis—information conservation guarantees that the diagnosis loses no information, energy conservation guarantees the energy consistency of the periodic decomposition, and phase conservation guarantees the temporal consistency of the periodic structure.

This claim that 'the three laws together guarantee the reliability of the diagnosis' is an **empirical observation / research programme**, not a proved theorem: the three conservation laws are not sufficient for periodicity (see Remark 41.4), nor do they constitute a proof of diagnostic reliability. What can be proved rigorously are only the two identities in the proof below—Parseval's identity in (ii) and the invertibility of the DFT in (i); 'phase conservation' in (iii) is a definitional fact for zero-phase (real non-negative gain) filtering.

Proof. The argument has three parts. (i) **Energy conservation (Parseval)**: by the orthogonality of the DFT, $\frac{1}{n}\sum_{k=0}^{n-1} e^{-i2\pi k(t-s)/n} = \delta_{ts}$,

$$\sum_{k=0}^{n-1} |\hat{X}_k|^2 = \frac{1}{n}\sum_{k=0}^{n-1}\sum_{t,s} X_t\overline{X_s}\, e^{-i2\pi k(t-s)/n} = \sum_{t=1}^{n} |X_t|^2,$$

that is, the total energy of the signal equals the sum of the energies of its frequency components. In the nested epicycle system $\sum_k A_k e^{i(\omega_k t+\phi_k)}$ the energy of the orthogonal components is A_k^2, so the sum of the epicycle energies at each level $\sum_k A_k^2$ equals the energy of the denoised signal, and the residual ε carries the remaining energy. (ii) **Information conservation (invertibility)**: the DFT is a bijection $\mathbb{C}^n \to \mathbb{C}^n$ whose inverse is $X_t = \frac{1}{\sqrt{n}}\sum_k \hat{X}_k e^{i2\pi kt/n}$, the coefficients corresponding one-to-one to the signal with no loss of information; for $W \equiv 1$, $\alpha = 1$ and $d = 1$, periodic convolution preserves all amplitudes A_n and phases ϕ_n (the expansion of Proposition 41.3), so compression and reconstruction are exactly mutually inverse. (iii) **Phase conservation**: for zero-phase (real non-negative gain) linear filtering, each component transforms as $c_k e^{i(\omega_k t+\phi_k)} \mapsto g_k c_k e^{i(\omega_k t+\phi_k)}$ $(g_k \geq 0)$, and the phases ϕ_k are unchanged pointwise, so the relative phases $\phi_k - \phi_l$ are conserved; the term (n, m) of periodic convolution carries the argument $\phi_m - \phi_n$ (see the proof of Proposition 41.3), so the phase differences of all frequency pairs are retained explicitly, and the stability of the edge weights of the phase-coupling network (Lemma 41.2) is the network manifestation of phase conservation. ☐

Definition 41.14 (Window-function decomposition of data trends). For arbitrary time-series data $\{X_t\}_{t=1}^n$, the window-function system is used for multi-scale trend decomposition. Let the window function be K (a Gaussian, Mexican-hat or exponential kernel, and so on) with window radius $\Delta > 0$. The **window-function smoothed trend** $T_\Delta(t)$ and the **window-function local fluctuation** $F_\Delta(t)$ of the data are respectively:

$$T_\Delta(t) = \int_{-\infty}^{\infty} X(t+\tau) K\left(\frac{\tau}{\Delta}\right) \frac{d\tau}{\Delta}, \quad F_\Delta(t) = X(t) - T_\Delta(t).$$

The window radius Δ controls the time scale—the smaller Δ is, the more sensitive the trend $T_\Delta(t)$ is (capturing short-term fluctuations) and the smaller the amplitude of $F_\Delta(t)$ is (high-frequency noise is retained in the trend); the larger Δ is, the smoother the trend is (revealing the long-term direction) and the larger the amplitude of $F_\Delta(t)$ is (more mid-frequency fluctuations are separated into the fluctuation component).

The trend $T_\Delta(t)$ corresponds to the long-term direction of the data and the fluctuation $F_\Delta(t)$ to the back-and-forth motion of the data about the trend. The window function K is the degenerate realisation of the periodic-convolution weight kernel on a univariate time series.

Definition 41.15 (Periodic decomposition by multi-window superposition)**.** *Use a family of increasing window radii $\{\Delta_1, \Delta_2, \ldots, \Delta_K\}$ $(\Delta_1 < \Delta_2 < \cdots < \Delta_K)$ to decompose the data $X(t)$ at multiple scales. Define the k-th level* **periodic component** *as the difference of the trends at adjacent scales:*

$$C_k(t) = T_{\Delta_k}(t) - T_{\Delta_{k-1}}(t), \quad k = 2, \ldots, K,$$

where $T_{\Delta_0}(t) \equiv 0$ and $C_1(t) = T_{\Delta_1}(t)$. The periodic components at each level satisfy the reconstruction formula:

$$X(t) = \sum_{k=1}^{K} C_k(t) + R_K(t),$$

where $R_K(t) = X(t) - T_{\Delta_K}(t)$ is the residual at the smallest scale (high-frequency noise). This decomposition splits the original data into K periodic components at different time scales—similar to a multi-window version of the Fourier series. The periodic components $C_k(t)$ at each level are the realisations in the time domain of the epicycles of different radii and frequencies in a nested epicycle system.

Proposition 41.9 (Reconstruction completeness of the multi-window decomposition)**.** *The multi-window periodic decomposition reconstructs an arbitrary time series $X(t)$ exactly and pointwise: $X(t) = \sum_{k=1}^{K} C_k(t) + R_K(t)$, and the sum of the components at each level is the* **telescoping sum** *of the 'trend pyramid' $T_{\Delta_0} \equiv 0, T_{\Delta_1}, \ldots, T_{\Delta_K}$.*

Proof. Substitute $C_k(t) = T_{\Delta_k}(t) - T_{\Delta_{k-1}}(t)$ and sum term by term:

$$\sum_{k=1}^{K} C_k(t) + R_K(t) = \sum_{k=1}^{K} \big(T_{\Delta_k}(t) - T_{\Delta_{k-1}}(t)\big) + \big(X(t) - T_{\Delta_K}(t)\big).$$

The first K terms form a telescoping sum, the intermediate terms $T_{\Delta_0}, T_{\Delta_1}, \ldots, T_{\Delta_{K-1}}$ cancelling in pairs:

$$\sum_{k=1}^{K} \big(T_{\Delta_k}(t) - T_{\Delta_{k-1}}(t)\big) = T_{\Delta_K}(t) - T_{\Delta_0}(t) = T_{\Delta_K}(t),$$

(since $T_{\Delta_0} \equiv 0$). Hence $\sum_k C_k(t) + R_K(t) = T_{\Delta_K}(t) + X(t) - T_{\Delta_K}(t) = X(t)$ for every t. □

Definition 41.16 (Computation of trend similarity)**.** For the trend components $T_{\Delta_i}(t)$ and $T_{\Delta_j}(t)$ at two different window radii, define their **trend similarity**. First compute the logarithmic return of the trend:

$$r_\Delta(t) = \ln \frac{T_\Delta(t)}{T_\Delta(t-1)} \approx \frac{T_\Delta(t) - T_\Delta(t-1)}{T_\Delta(t-1)},$$

then extract the limit angle $\theta_\Delta(t) = \arctan(r_\Delta(t))$. Following the unified convention of the 'Scalars and Vectors' chapter, the **included angle** between two trend directions is always taken to be $\theta = |\alpha - \beta| \in [0, \pi]$; the signed quantity $\alpha - \beta$ is called the **difference angle** and is used only for the sign of the sine-family similarity. The **difference-angle cosine similarity** (trend consistency) and the **difference-angle sine similarity** (degree of trend reversal) of the two trends are respectively:

$$\text{Similarity}^{(1)}_{\Delta_i,\Delta_j}(t) = \cos(\theta_{\Delta_i}(t) - \theta_{\Delta_j}(t)), \quad \text{Similarity}^{(2)}_{\Delta_i,\Delta_j}(t) = \sin(\theta_{\Delta_i}(t) - \theta_{\Delta_j}(t)).$$

$\text{Similarity}^{(1)} \approx 1$ means that the trends at the two time scales point in the same direction (rising and falling together), whilst $\text{Similarity}^{(1)} \approx -1$ means that they point in opposite directions (one rising whilst the other falls). $\text{Similarity}^{(2)} > 0$ means that the short-term trend leads the long-term trend (short-term changes are faster than long-term ones), whilst $\text{Similarity}^{(2)} < 0$ means that the short-term trend lags behind the long-term trend (the short term is reversing).

Trend similarity is the degenerate form of periodic convolution on the pair of trend components (Δ_i, Δ_j)—it measures the degree of phase alignment between trends at different time scales.

Proposition 41.10 (Trend strength and trend stability)**.** *For the trend $T_\Delta(t)$ at a single window radius Δ, define its* **trend strength** *$S_\Delta(t)$ and* **trend stability** *$Q_\Delta(t)$:*

$$S_\Delta(t) = |r_\Delta(t)| = \left| \frac{T_\Delta(t) - T_\Delta(t-1)}{T_\Delta(t-1)} \right|,$$
$$Q_\Delta(t) = \text{Similarity}^{(1)}_{\Delta,\Delta}(t, t-\tau) = \cos(\theta_\Delta(t) - \theta_\Delta(t-\tau)),$$

where τ is the number of delay steps. $S_\Delta(t)$ measures the rate of change of the trend and $Q_\Delta(t)$ measures the consistency of the trend over the time delay τ. $Q_\Delta(t) \approx 1$ means that the direction of the trend continues stably, whilst a decrease in $Q_\Delta(t)$ means that the trend may be about to reverse—a decrease in $Q_\Delta(t)$ is an early warning signal of a turning point.

Proof. Write the phase increment as $\delta(t) := \theta_\Delta(t) - \theta_\Delta(t-\tau)$. By the Taylor expansion of the cosine, $\cos u = 1 - \frac{u^2}{2} + O(u^4)$,

$$Q_\Delta(t) = \cos \delta(t) = 1 - \frac{\delta(t)^2}{2} + O(\delta(t)^4).$$

Hence $Q_\Delta(t)$ is a strictly decreasing function of the square of the phase increment $\delta(t)^2$: as $\delta \to 0$ we have $Q_\Delta \to 1$ (the direction of the trend continues stably), and as $|\delta|$ increases Q_Δ decreases monotonically from 1. Moreover $\delta(t) \approx \tau\, \dot{\theta}_\Delta(t)$ (a discrete derivative, with τ the number of delay steps), so a decrease in Q_Δ is equivalent to a rise in the rate of phase change $|\dot{\theta}_\Delta|$—that is, the direction of the trend begins to turn more rapidly, which is the necessary early signal before a turning point (trend reversal). Since $\theta_\Delta = \arctan r_\Delta$, we have $S_\Delta(t) = |r_\Delta(t)| = |\tan\theta_\Delta(t)|$, which measures the modulus of the tangent of the phase angle itself (the rate of change of the trend); together with Q_Δ, one measuring the 'current rate' and the other the 'change of the rate', the two characterise the trend in a complementary way. □

Definition 41.17 (Definition of the data refractive index). Drawing on the law of refraction in optics (the scaling of the sine trend) and on the concept of the refraction index at stock-price turning points, the **refractive index** $\eta(t;\Delta)$ is defined in general data analysis to measure the **strength of momentum change** of a data trend at a turning point. Let the trend direction vectors in the two time windows $[t^* - \Delta, t^*]$ and $[t^*, t^* + \Delta]$ before and after the turning point t^* be respectively:

$$\mathsf{v}_{\text{in}} = T_\Delta(t^*) - T_\Delta(t^* - \Delta), \quad \mathsf{v}_{\text{out}} = T_\Delta(t^* + \Delta) - T_\Delta(t^*).$$

The **trend refractive index** is defined as the ratio of the outgoing momentum to the incoming momentum:

$$\eta(t^*;\Delta) = \frac{\|\mathsf{v}_{\text{out}}\|}{\|\mathsf{v}_{\text{in}}\|} = \frac{|T_\Delta(t^* + \Delta) - T_\Delta(t^*)|}{|T_\Delta(t^*) - T_\Delta(t^* - \Delta)|}.$$

The physical meaning of the refractive index:

(i) $\eta > 1$: the momentum is enhanced after the turning point (the trend accelerates)—the outgoing trend is steeper than the incoming trend.

(ii) $\eta = 1$: the momentum is unchanged (the trend is uniform)—the trend strength is the same before and after the turning point.

(iii) $\eta < 1$: the momentum weakens after the turning point (the trend decelerates)—the outgoing trend is gentler than the incoming trend.

(iv) $\eta \approx 0$: the momentum vanishes after the turning point (the trend flattens out)—the outgoing trend is close to zero and the data enter sideways consolidation.

Proposition 41.11 (Phase interpretation of the refractive index). There is an intrinsic connection between the refractive index η and the sine difference-angle similarity $\mathsf{Similarity}^{(2)}$. Let the angle of incidence be $\theta_{\text{in}} = \arctan(\mathsf{v}_{\text{in}}/\Delta)$ and the angle of refraction be $\theta_{\text{out}} = \arctan(\mathsf{v}_{\text{out}}/\Delta)$. From the sine-trend scaling relation of Snell's law:

$$\frac{\sin\theta_{\text{out}}}{\sin\theta_{\text{in}}} = \frac{\|\mathsf{v}_{\text{out}}\|}{\|\mathsf{v}_{\text{in}}\|} \cdot \frac{\sqrt{\Delta^2 + \|\mathsf{v}_{\text{in}}\|^2}}{\sqrt{\Delta^2 + \|\mathsf{v}_{\text{out}}\|^2}} = \eta \cdot \frac{\sqrt{\Delta^2 + \|\mathsf{v}_{\text{in}}\|^2}}{\sqrt{\Delta^2 + \|\mathsf{v}_{\text{out}}\|^2}}.$$

When Δ is far larger than the trend momenta ($\Delta \gg \|\mathsf{v}_{\text{in}}\|, \|\mathsf{v}_{\text{out}}\|$), we have $\sin\theta_{\text{out}}/\sin\theta_{\text{in}} \approx \eta$—the refractive index is approximately equal to the scaling ratio of the sine trends. When $\eta > 1$ and $\mathsf{Similarity}^{(2)} < 0$ (the sine difference angle is negative), the turning point is confirmed—the trend direction reverses and the momentum is amplified, which is a strong turning-point signal.

Proof. From $\tan\theta = \|\mathsf{v}\|/\Delta$ we obtain the right-triangle relation $\sin\theta = \frac{\|\mathsf{v}\|}{\sqrt{\Delta^2+\|\mathsf{v}\|^2}}$. Hence

$$\frac{\sin\theta_{\text{out}}}{\sin\theta_{\text{in}}} = \frac{\|\mathsf{v}_{\text{out}}\|}{\sqrt{\Delta^2+\|\mathsf{v}_{\text{out}}\|^2}} \cdot \frac{\sqrt{\Delta^2+\|\mathsf{v}_{\text{in}}\|^2}}{\|\mathsf{v}_{\text{in}}\|} = \eta \cdot \frac{\sqrt{\Delta^2+\|\mathsf{v}_{\text{in}}\|^2}}{\sqrt{\Delta^2+\|\mathsf{v}_{\text{out}}\|^2}},$$

in agreement with the statement of the proposition. In the limit $\Delta^2 \gg \max(\|\mathsf{v}_{\text{in}}\|^2, \|\mathsf{v}_{\text{out}}\|^2)$ the second factor satisfies

$$\frac{\sqrt{\Delta^2+\|\mathsf{v}_{\text{in}}\|^2}}{\sqrt{\Delta^2+\|\mathsf{v}_{\text{out}}\|^2}} \to 1,$$

so $\sin\theta_{\text{out}}/\sin\theta_{\text{in}} \to \eta$. Moreover $\text{Similarity}^{(2)} = \sin(\theta_{\text{out}} - \theta_{\text{in}})$, whose value is negative when the directions are reversed (θ_{out} and θ_{in} have opposite signs); together with $\eta > 1$ (momentum amplification) this gives the strong turning-point criterion of 'direction reversal together with momentum enhancement'. □

Definition 41.18 (Reflection and refraction of trends). *When a data trend encounters a 'resistance level' or a 'support level' it undergoes* **reflection** *(the trend reverses and returns along its original path) or* **refraction** *(the trend changes direction without reversing, continuing onwards but with a changed slope). The mathematical criteria for these two behaviours are:*

(i) **Reflection criterion**: $\text{Similarity}^{(2)}_{\Delta}(t) < 0$ *(the sine difference angle is negative) and* $\text{Similarity}^{(1)}_{\Delta}(t, t-\tau) < 0$ *(the cosine difference angle is negative—the trend direction is completely reversed). Then* $\mathsf{v}_{\text{out}} \approx -\gamma \mathsf{v}_{\text{in}}$ $(\gamma > 0)$ *and the refractive index is* $\eta = \gamma$. *If* $\gamma \approx 1$, *this is* **total reflection** *(mirror reversal)—the trend strength is the same before and after the turning point, whilst the directions are opposite. Total reflection corresponds to conservation of the double-angle cosine similarity:* $\cos 2\theta_{\text{in}} = \cos 2\theta_{\text{out}}$.

(ii) **Refraction criterion**: $\text{Similarity}^{(2)}_{\Delta}(t) < 0$ *(the sine difference angle is negative—the direction changes) but* $\text{Similarity}^{(1)}_{\Delta}(t, t-\tau)$ *remains positive (the cosine difference angle is positive—the trend still partially continues). Then the included angle between* v_{out} *and* v_{in} *lies in* $(0°, 90°)$*—the trend direction is deflected without reversing completely. The refractive index* η *may be greater or less than* 1, *depending on the strength of the new trend.*

(iii) **Total transmission criterion**: $\text{Similarity}^{(2)}_{\Delta}(t) \approx 0$ *(the sine difference angle is close to zero) and* $\text{Similarity}^{(1)}_{\Delta}(t, t-\tau) \approx 1$ *(the cosine difference angle is close to one). Then the trend direction is unchanged whilst the strength may change—*$\eta \neq 1$ *means that the trend accelerates or decelerates, but the direction continues.*

Reflection and refraction are the direct generalisations of the laws of reflection and refraction to the analysis of data trends—reflection corresponds to conservation of the double-angle cosine similarity ($\cos 2\theta_{\text{in}} = \cos 2\theta_{\text{out}}$) *and refraction to the scaling of the sine trend* ($n_1 \sin\theta_{\text{in}} = n_2 \sin\theta_{\text{out}}$, *where* n_1, n_2 *are the refractive indices of the trend 'media'—that is, the volatilities of the data in different time periods).*

Proposition 41.12 (Periodic-convolution diagnosis at a turning point). *At a data turning point t^*, four key similarity indicators of the periodic convolution jointly diagnose the type and the strength of the turning point:*

Similarity indicator	Formula	Value at a turning point	Diagnostic meaning
Similarity$^{(1)}$	$\cos(\theta_t - \theta_{t-\tau})$	< 0 or decreasing	Trend consistency weakens
Similarity$^{(2)}$	$\sin(\theta_t - \theta_{t-\tau})$	< 0	Direction reversal (necessary condition for a turning point)
Similarity$^{(3)}$	$\cos 2(\theta_t - \theta_{t-\tau})$	$\to -1$	**Orthogonality indicator**: the two trends are orthogonal ($\delta \to \pm\pi/2 \iff$ Similarity$^{(1)} = \cos\delta \to 0$)
Similarity$^{(4)}$	$\sin 2(\theta_t - \theta_{t-\tau})$	$\lvert\cdot\rvert$ increases	Reversal strength amplified

The combined criteria are:

(i) **Weak turning point**: Similarity$^{(2)} < 0$, Similarity$^{(1)} > 0$, $\eta < 1$—*a slight change of direction with weakened momentum.*

(ii) **Strong turning point**: Similarity$^{(2)} < 0$, Similarity$^{(1)} < 0$, $\eta > 1$, Similarity$^{(3)} \to +1$—*direction reversal with enhanced momentum, close to total reflection.*

(iii) **False turning point**: Similarity$^{(2)} < 0$ but $\eta \approx 0$ and $|$Similarity$^{(4)}|$ very small—*the direction fluctuates briefly and then quickly returns, and the momentum almost vanishes. This is a noise fluctuation during sideways consolidation, not a genuine turning point.*

Proof. Write the phase increment as $\delta(t) = \theta_t - \theta_{t-\tau}$; then the four indicators in the table are Similarity$^{(1)} = \cos\delta$, Similarity$^{(2)} = \sin\delta$, Similarity$^{(3)} = \cos 2\delta$ and Similarity$^{(4)} = \sin 2\delta$. By the double-angle formulae,

$$\text{Similarity}^{(3)} = \cos 2\delta = 2\cos^2\delta - 1 = 2(\text{Similarity}^{(1)})^2 - 1,$$
$$\text{Similarity}^{(4)} = \sin 2\delta = 2\sin\delta\cos\delta = 2\,\text{Similarity}^{(1)}\,\text{Similarity}^{(2)}.$$

Hence the four indicators are in fact determined by only two independent quantities (Similarity$^{(1)}$, Similarity$^{(2)}$) (that is, by the phase difference $e^{i\delta}$ on the unit circle)—the double-angle indicator (the **orthogonality indicator**) satisfies Similarity$^{(3)} \to -1$ if and only if $\delta \to \pm\pi/2$, that is, Similarity$^{(1)} \to 0$ (orthogonality of the two trends), whereas mirror reversal ($\delta \to \pm\pi$) gives Similarity$^{(1)} \to -1$ and Similarity$^{(3)} \to +1$, corresponding to the mirror case of 'reversal of direction through to complete reversal'. The combined criteria then distinguish the strength by means of the refractive index η: Similarity$^{(2)} < 0$ ($\delta < 0$, direction reversal) is a necessary condition for a turning point; $\eta < 1$ with

Similarity$^{(1)}$ > 0 gives a weak turning point, whilst $\eta > 1$ with Similarity$^{(1)}$ < 0 and Similarity$^{(3)}$ $\to +1$ gives a strong turning point; if Similarity$^{(2)}$ < 0 but $\eta \approx 0$ and $|$Similarity$^{(4)}|$ $= 2|$Similarity$^{(1)}$ Similarity$^{(2)}|$ is very small, then the direction fluctuation is instantaneous and without momentum, and the point is judged to be a false turning point (noise). ☐

Example 41.5 (*Periodic analysis of sunspot data*). *The time series of monthly mean sunspot numbers (from 1749 to the present) is a classical case of period detection in data analysis. Sunspot activity has a period of about 11 years (the Schwabe cycle) and is at the same time nested within longer cycles (such as the Gleissberg cycle of about 88 years). Periodic convolution is used to analyse these data:*

(i) *The periodogram exhibits its strongest peak at 11 years, with secondary peaks at 5.5 years (the half period) and 88 years.*

(ii) *The periodic convolution $\mathcal{K}^{(1)}\{X, X\}$ (the trend transform) attains its largest amplitude output at 11 years, confirming the dominance of the Schwabe cycle.*

(iii) *The difference-angle sine component* Im $\mathcal{K}^{(1)}\{X, X\}$ *of the periodic convolution (phase relations) exhibits different phase-coupling patterns during the rising and falling phases of sunspot activity—the phases cluster during the rising phase and disperse during the falling phase.*

(iv) *Nested epicycle system fitting shows that $K = 3$ (11-year fundamental frequency, 5.5-year second harmonic, 88-year long period) already accounts for about 80% of the total variance.*

Example 41.6 (*Window-function–refractive-index diagnosis of a time series*). *Window-function periodicity diagnosis is carried out on a set of annual temperature anomaly data (1850–2023). Take the Gaussian kernel K_{Gauss} with window radius $\Delta = 20$ years (removing high-frequency fluctuations below the decadal scale) and compute:*

(i) **Window-function trend** $T_{20}(t)$: *reveals the long-term trend of global warming—a sustained rise since the 20th century.*

(ii) **Sine difference-angle similarity** Similarity$^{(2)}(t)$ $=$ sin$(\theta_t - \theta_{t-10})$ *(a 10-year delay): negative values appear during the 'cooling period' of 1940–1970—marking the turning point from early warming to mid-century cooling. Positive values return in the late 1970s—marking the turning point from cooling to accelerated warming.*

(iii) **Refractive index** $\eta(t)$: *at the turning point of the 1970s, $\eta \approx 2.3 > 1$—the momentum of the warming trend after the turning point is 2.3 times that of the cooling trend before it, confirming the change in trend strength of 'accelerated global warming'.*

(iv) **Reflection/refraction analysis**: *at the turning point of the 1940s,* Similarity$^{(1)}$ $\to -1$ *(equivalently* Similarity$^{(3)}$ $\to +1$*, that is, the reflection criterion* cos $2\theta_{\text{in}}$ $=$ cos $2\theta_{\text{out}}$ *holds)—close to total reflection, the* warming trend being completely reversed into a cooling trend. At the turning point of the 1970s, Similarity$^{(1)} < 0$ and $\eta \gg 1$—strong refraction, the cooling trend being reversed into a stronger warming trend.

Window-function–refractive-index diagnosis of climate data is a concrete application of periodic convolution to environmental data—the window function provides multi-scale trend extraction, the similarity indicators quantify the phase changes of the trend, and the refractive index measures the scaling ratio of the trend momentum at a turning point. The joint use of the three constitutes a complete tool chain for data turning-point diagnosis.

Remark 41.10 (Connection with the rest of the book). The theory of the periodic structure of data analysis established in this chapter is a complete application of periodic mathematics in data science. Periodic convolution provides a unified mathematical framework for period detection in time series—it simultaneously detects several periods, retains all phase information and quantifies the strength of coupling between periods. The window-function system provides multi-scale spectral estimation, and derivative-integral descent optimisation provides a data-driven method of parameter estimation. The phase-coupling network visualises the periodic interactions of multivariate data as network topology. The comprehensive pipeline of data periodicity diagnosis integrates the mathematical tools of the whole book—from the Fourier transform to periodic convolution, from single-period detection to multi-period coupling analysis—into a standardised method of data analysis.

Seven-theme comparison table for this chapter

The corresponding objects of the seven themes of this chapter in the periodic structure of data analysis are traced back as follows:

Theme	Corresponding content in this chapter
Translation	The lag operator and delay correlation of a time series: the τ-step trend self-similarity Similarity$^{(1)}$ and the trend stability $Q_\Delta(t)$ (Definition 41.16, Proposition 41.10)
Scaling	The multi-scale window radii of multitaper spectral estimation and the window radius Δ of trend decomposition, and the refractive index η (Proposition 41.2, Definition 41.14, Definition 41.17)
Rotation	The phase edge weights $W_{ij}^{(d)}$ and the phase-difference angles $\phi_m - \phi_n$ of the phase-coupling network (Definition 41.12, Lemma 41.2)
Balance	The three laws of information conservation, energy conservation (Parseval) and phase conservation (**empirical observation / research programme**, Remark 41.9)
Symmetry	The symmetry tests of the data distribution about the three centres of symmetry 0, 1 and i, and the criteria for trend reflection/refraction (Definition 41.18)
Closure	The closed-loop diagnostic procedure 'detection—modelling—verification—prediction' (Definition 41.13)
Period	Periodogram peaks, the multi-window periodic decomposition components $C_k(t)$ and the dominant periods of the nested epicycle system (Definition 41.6, Proposition 41.9)

Exercises

Exercise 1 Prove the **data periodicity decision theorem** of this chapter (41.1). Hint: first write down the decision map and the evolution group of this chapter's target, then apply the proof structure of the general criterion of the 'Periodic Systems' chapter.

Exercise 2 Verify that the hypotheses of the **monotonicity of quantity adjustment** (41.4) hold for the domain objects of this chapter, and give a counterexample in which the hypotheses fail.

Exercise 3 Examine the behaviour of the **three-parameter correspondence: radius, phase and speed** (41.1) under parameter degeneracy, and explain how the conclusion degenerates accordingly.

Exercise 4 Explain the relation between the **data periodicity diagnostic pipeline and its interface with three chapters** (41.6) and the corresponding general conclusion of the 'Periodic Systems' chapter (whether the two agree, whether the former is a special case, or whether additional conditions are required).

Exercise 5 Using the seven-theme comparison table of this chapter, identify the concrete carriers of the themes **translation**, **scaling** and **rotation** in this chapter, and give one example of domain data for each.

Exercise 6 Synthesis: restate the diagnostic pipeline of this chapter using the five-step standard interface of the 'Periodic Systems' chapter (Table 24.1), noting one by one the theorem of this chapter on which each step depends: 41.1, 41.4, 41.1, 41.6.

Chapter 42

Epicycle Activation Function and Function Approximation

Remark 42.1 (Periodicity decision for the target of this chapter (citation)). *Every application chapter of this book shares a single cross-reference list; its full statement is given in the opening of Chapter 25, 'Periodic Structures in Probabilistic Systems', at Remark 25.1. It is not repeated here, and reference is made uniformly to that passage.*

Remark 42.2 (Positioning of this chapter: from data analysis to learnable activation). *This chapter takes up the output of the preceding chapter, 'Periodic Structure of Data Analysis', and raises the* **periodic-structure parameters extracted from the data** *(A_k, ω_k, ϕ_k) to the status of a* **learnable activation function** *in a neural network. The core object is the complex exponential sum*

$$\mathcal{E}(x) = \sum_{k=1}^{N} A_k e^{i(\omega_k x + \phi_k)},$$

in which the amplitude $A_k > 0$, the frequency $\omega_k \in \mathbb{R}$ and the phase $\phi_k \in [0, 2\pi)$ are all learnable.

Interface with the preceding chapter: *the four classes of parameter output by the data periodicity diagnosis procedure of the preceding chapter (Definition 41.13)—the amplitude spectrum, the frequency spectrum, the phase spectrum and the phase-coupling matrix—are precisely the initial parameter values and the coupling structure of the activation function of this chapter (Proposition 42.1). This chapter therefore does not construct an activation function from scratch; rather, it connects the results of data analysis end to end into a neural network.*

This chapter draws a strict distinction between three frequency modes—fixed integer frequencies, fixed non-integer frequencies and learnable frequencies—and gives the precise boundary of the approximation capability in each mode. All approximation results are stated within the framework of 'compact set + continuous function + absolutely summable coefficients + real-part/dual-channel readout'; assertions that go beyond this framework are marked only as a research programme or an empirical observation.

Interface with the following chapters: *the CEAF defined in this chapter (Definition 42.4) and the KDI optimiser (cited from Chapter 43, Definition 44.7; the complete definition of KDI is given in the 'Agents' chapter, and the reader may accept its existence for the present and return to it in that chapter) will become a core building block of the algorithmic components in the next chapter, 'Periodic Structure of Algorithmic Systems', and the implementation basis of the training dynamics in the chapter after that, 'Periodic Structure of Agents'.*

Guide to the seven themes of this chapter: translation—the time translation of the epicycle activation (Property 42.1); scaling—frequency scaling $\mathcal{E}(\lambda x)$ and the Sobolev rate estimate (Theorem 42.3); rotation—global rotation $e^{i\theta}\mathcal{E}$ and the Wirtinger derivative (Theorem 42.6); balance—the absolute summability condition (Definition 42.9) and the separation of approximation and optimisation (Remark 42.6); symmetry—the readout symmetry of the real and imaginary parts (Definition 42.5); closure—the Cesàro-mean closure of Fejér's theorem (Theorem 42.2); period—approximation on the periodic domain (Corollary 42.1) and the Waveform interface (Remark 42.9).

42.1 Classical framework for activation functions

Definition 42.1 (Pointwise scalar activation function)**.** Let $d \in \mathbb{N}^+$ and $\mathrm{x} = (x_1, \dots, x_d) \in \mathbb{R}^d$. A **neuron** is the map

$$h = \varphi\left(\sum_{i=1}^{d} w_i x_i + b\right),$$

in which $w_i, b \in \mathbb{R}$ are the weights and the bias, and $\varphi : \mathbb{R} \to \mathbb{R}$ is called the **activation function**. If φ is continuous and non-constant, it is called a **non-degenerate activation**.

Example 42.1 (Classical activation functions). (i) Saturating type: $\sigma(x) = \frac{1}{1+e^{-x}}$, $\tanh(x)$.

(ii) Piecewise-linear type: $\text{ReLU}(x) = \max(0, x)$, $\text{LeakyReLU}(x) = \max(\alpha x, x)$ ($\alpha \in (0, 1)$).

(iii) Smooth type: $\text{GELU}(x) = x\Phi(x)$ (where Φ is the standard normal cumulative distribution function), $\text{Swish}(x) = x\sigma(\beta x)$.

(iv) Periodic type: $\sin(\omega x + \phi)$, $\cos(\omega x + \phi)$.

Among them, the periodic type of activation is the one-dimensional degenerate form of the epicycle activation of this chapter.

Definition 42.2 (Complex-valued activation function)**.** Let $z \in \mathbb{C}$. A map $\varphi : \mathbb{C} \to \mathbb{C}$ is called a **complex-valued activation function**. If φ is holomorphic on an open set, it is called a **holomorphic activation**; if φ is merely real-differentiable, it is called a **real-differentiable complex activation**.

Remark 42.3 (Calculus boundary of complex-valued activations)**.** If the network weights and the activations are both complex while the loss function is real-valued, then backpropagation requires Wirtinger calculus (Definition 42.10): the loss $\mathcal{L}$ is regarded as a function of z and $\bar{z}$, and $\partial\mathcal{L}/\partial z$ and $\partial\mathcal{L}/\partial\bar{z}$ are computed separately. A holomorphic activation satisfies the Cauchy–Riemann equations, and its Wirtinger derivative degenerates to the ordinary complex derivative; a non-holomorphic activation (such as $|z|$ or $\operatorname{Re} z$) must instead be treated separately.

Definition 42.3 (Periodic activation and learnable frequency)**.** An activation of the form

$$\varphi(x) = \sin(\omega x + \phi)$$

is called a **periodic activation**, in which ω is the frequency and ϕ the phase. If ω can be updated during training, the activation is called a **learnable-frequency activation**; if ω is fixed, it is called a **fixed-frequency activation**.

Example 42.2 (The SIREN activation)**.** SIREN (Sinusoidal Representation Networks) uses

$$\varphi(x) = \sin(\omega_0 x)$$

as its activation, in which ω_0 is a fixed hyperparameter (usually taken to be 30). SIREN exhibits a strong fitting capability for high-frequency detail in implicit neural representations, but its frequency is not learnable, and the approximation rate is restricted by the fixed ω_0.

42.2 Definition of the *epicycle activation function* and its correspondence with data-analysis parameters

Definition 42.4 (Epicycle activation function)**.** Let $N \in \mathbb{N}^+$, with parameters $A_k > 0$, $\omega_k \in \mathbb{R}$ and $\phi_k \in [0, 2\pi)$. Define

$$\boxed{\mathcal{E}(x) = \sum_{k=1}^{N} A_k e^{i(\omega_k x + \phi_k)}, \qquad x \in \mathbb{R}.}$$

Then $\mathcal{E}$ is called the **epicycle activation function** (epicycle activation), the collection $\{A_k, \omega_k, \phi_k\}_{k=1}^{N}$ is called the learnable parameters, and N is called the **series order**.

Proposition 42.1 (Natural mapping from data-analysis parameters to activation parameters)**.** Let the K-level nested epicycle system fitted by the data analysis of the preceding chapter to the data $\{X_t\}$ be

$$X(t) \approx \sum_{k=1}^{K} A_k e^{i(\omega_k t + \phi_k)}.$$

Then the mapping

$$\Phi_{\text{data}\to\text{act}} : \{(A_k, \omega_k, \phi_k)\}_{k=1}^{K} \longmapsto \mathcal{E}(x) = \sum_{k=1}^{K} A_k e^{i(\omega_k x + \phi_k)}$$

gives an epicycle activation function (Definition 42.4). Its key properties are:

(i) **Frequency initialisation**: ω_k is given by a periodogram peak (the preceding chapter, Lemma 41.1), which avoids spectral bias;

(ii) **Amplitude scaling**: the decay pattern of the A_k determines the sensitivity of the activation function to each frequency scale—the conservation law for data of the preceding chapter (Remark 41.9(ii), Parseval) guarantees that the sum of the energies A_k^2 of the individual levels equals the total energy of the data;

(iii) **Phase alignment**: ϕ_k aligns the activation function at $x = 0$ with the local structure of the data—this is given by the phase–time-delay correspondence of Proposition 41.1(ii) of the preceding chapter.

Proof. By Definition 41.13 of the preceding chapter, the fitting parameters of the K-level nested epicycle system are precisely $\{(A_k, \omega_k, \phi_k)\}$. Taking them directly as the parameters of the epicycle activation function yields the mapping $\Phi_{\mathsf{data}\to\mathsf{act}}$.

(i) Frequency initialisation: by Lemma 41.1, the position of the periodogram peak converges in probability to the true frequency k_0; this peak is precisely $\omega_k^{(0)}$.

(ii) Amplitude scaling: by Remark 41.9(ii), the energy of the orthogonal component in the nested epicycle system is A_k^2, and the sum of the energies of the individual levels, $\sum_k A_k^2$, equals the total energy of the denoised signal; the decay pattern of the amplitudes (such as $A_1 > A_2 > A_3$ for sunspots) therefore directly encodes the relative importance of the various periodic scales.

(iii) Phase alignment: by Proposition 41.1(ii), the initial phase ϕ_k corresponds to the time translation $\tau_k = \phi_k/\omega_k$, that is, the epicycle **leads** the reference by τ_k ($\phi_k > 0$ means that the waveform is **advanced** by τ_k relative to the reference, equivalently the waveform coincides with the reference once the reference has been delayed by τ_k); placing the activation function at $x = 0$, the quantity ϕ_k uniquely determines the local time delay corresponding to that component, and hence aligns it with the local structure of the data. ☐

Remark 42.4 (The 'naturalness' of this map). The **data analysis** of the preceding chapter and the **activation function** of this chapter share one and the same mathematical object—the nested epicycle system. Data analysis extracts (A_k, ω_k, ϕ_k) from the data, and the activation function takes this set of parameters as a learnable nonlinear unit. The two use one and the same parameterisation, so the mapping from data to activation is a re-interpretation of the 'identity mapping' in two contexts, rather than an additional construction.

Definition 42.5 (Real, imaginary and modulus-phase readouts). Continuing with Definition 42.4, we define three readouts:

(i) **Real-part readout**: $\mathcal{E}_{\mathsf{Re}}(x) = \operatorname{Re}\mathcal{E}(x) = \sum_{k=1}^{N} A_k \cos(\omega_k x + \phi_k)$.

(ii) **Dual-channel readout**: $\mathcal{E}_{\mathsf{dual}}(x) = \big(\operatorname{Re}\mathcal{E}(x), \operatorname{Im}\mathcal{E}(x)\big) \in \mathbb{R}^2$.

(iii) **Modulus-phase readout**: $\mathcal{E}_{\mathsf{mod}}(x) = |\mathcal{E}(x)|$, $\mathcal{E}_{\mathsf{arg}}(x) = \arg\mathcal{E}(x)$.

Definition 42.6 (Single-layer epicycle network). Let the input be $\mathsf{x} \in \mathbb{R}^d$, the weights $\mathsf{w}_j \in \mathbb{R}^d$, the biases $b_j \in \mathbb{R}$ and the output coefficients $c_j \in \mathbb{C}$ $(j = 1, \ldots, M)$. Define

$$F(\mathsf{x}) = \sum_{j=1}^{M} c_j\, \mathcal{E}_j(\mathsf{w}_j \cdot \mathsf{x} + b_j),$$

in which each $\mathcal{E}_j$ is an epicycle activation as in Definition 42.4. If the readout is to be real-valued, define

$$F_{\mathbb{R}}(\mathsf{x}) = \operatorname{Re} F(\mathsf{x})$$

or take the dual-channel concatenation. Then F is called a **single-layer epicycle network**.

Proposition 42.2 (Basic analytic properties)**.** *Let $\mathcal{E}$ be the epicycle activation of Definition 42.4. Then:*

(i) **Continuity**: $\mathcal{E} \in C^\infty(\mathbb{R}, \mathbb{C})$.

(ii) **Boundedness**: $|\mathcal{E}(x)| \le \sum_{k=1}^N A_k$ holds for all $x \in \mathbb{R}$.

(iii) **Derivative**: $\mathcal{E}'(x) = i \sum_{k=1}^N \omega_k A_k e^{i(\omega_k x + \phi_k)}$.

(iv) **The derivative is not identically zero**: if there exists k with $A_k\omega_k \neq 0$, then $\mathcal{E}'$ is not identically zero (though it may have isolated zeros).

(v) **Periodicity condition**: if all $\omega_k/\omega_1 \in \mathbb{Q}$, then $\mathcal{E}$ is a periodic function, where $\omega_0 := \gcd\{\omega_k \neq 0\}$ is the greatest common angular frequency of the individual frequencies, the least positive period is $2\pi/|\omega_0|$ (an integer fraction of it when $\omega_0 \notin \Omega$); when all $\omega_k = 0$ there is no least positive period. Otherwise $\mathcal{E}$ is a Bohr almost periodic function.

Proof. (i) A sum of finitely many complex exponential functions is an entire function, hence $\mathcal{E} \in C^\infty$.

(ii) By the triangle inequality, $|\mathcal{E}(x)| \le \sum_{k=1}^N |A_k e^{i(\omega_k x+\phi_k)}| = \sum_{k=1}^N A_k$.

(iii) Differentiating term by term: $\frac{d}{dx} A_k e^{i(\omega_k x+\phi_k)} = i\omega_k A_k e^{i(\omega_k x+\phi_k)}$.

(iv) If there exists k with $A_k\omega_k \neq 0$, then the derivative of the k-th term is not identically zero. Since the family of complex exponential functions $\{e^{i\omega_k x}\}$ is linearly independent on $\mathbb{R}$ (evaluating at sufficiently many distinct points yields a non-zero Vandermonde determinant), a linear combination of them cannot be identically zero unless all the coefficients vanish. Hence $\mathcal{E}'$ is not identically zero.

(v) If all $\omega_k/\omega_1 \in \mathbb{Q}$, then there exist integers $n_k \in \mathbb{Z}$ with $\omega_k = n_k\omega_0$, where $\omega_0 = \gcd\{\omega_k \neq 0\} > 0$ is the common base frequency. Taking $T = 2\pi/\omega_0$, we have $\omega_k T = 2\pi n_k$, hence $\mathcal{E}(x+T) = \mathcal{E}(x)$; the least positive period is $2\pi/|\omega_0|$ (an integer fraction of it when $\omega_0 \notin \Omega$); when all $\omega_k = 0$ there is no least positive period. If there exists a pair of incommensurable frequencies, then the joint orbit of $\{e^{i\omega_k x}\}$ is dense on the torus but not closed, that is, $\mathcal{E}$ is a Bohr almost periodic function (see the definition of almost periodicity in the 'Operations and Periodicity' chapter, Remark 1.13). □

Property 42.1 (Linear superposition and closure)**.** *The epicycle activation function is closed under the following operations:*

(i) **Linear combination**: *a linear combination of two epicycle activations is again an epicycle activation.*

(ii) **Time translation**: *$\mathcal{E}(x + x_0)$ is again an epicycle activation, with the phases changed to $\phi_k + \omega_k x_0$.*

(iii) **Frequency scaling**: $\mathcal{E}(\lambda x)$ is again an epicycle activation, with the frequencies changed to $\lambda\omega_k$.

(iv) **Global rotation**: $e^{i\theta}\mathcal{E}(x)$ is again an epicycle activation, with the phases changed to $\phi_k + \theta$.

Proof. (i) Combining like terms term by term gives $\alpha\mathcal{E}_1(x)+\beta\mathcal{E}_2(x) = \sum_k(\alpha A_k^{(1)}e^{i\phi_k^{(1)}} + \beta A_k^{(2)}e^{i\phi_k^{(2)}})e^{i\omega_k x}$; when the frequencies coincide the two series may be merged into a single series, and in the general case they may be concatenated directly into an epicycle activation of double the series order.

(ii) $\mathcal{E}(x+x_0) = \sum_k A_k e^{i(\omega_k x+\omega_k x_0+\phi_k)}$.

(iii) $\mathcal{E}(\lambda x) = \sum_k A_k e^{i(\lambda\omega_k x+\phi_k)}$.

(iv) $e^{i\theta}\mathcal{E}(x) = \sum_k A_k e^{i(\omega_k x+\phi_k+\theta)}$. □

42.3 Boundaries and convergence conditions

Definition 42.7 (The three frequency modes)**.** *The frequency set $\{\omega_k\}_{k=1}^N$ of an epicycle activation has three modes:*

(i) **Fixed integer frequencies**: *$\omega_k = k,\ k \in \mathbb{Z}$.*

(ii) **Fixed non-integer frequencies**: *the ω_k are chosen before training and are no longer updated.*

(iii) **Learnable frequencies**: *the ω_k are updated during training by gradient descent or by KDI.*

The approximation properties, the difficulty of optimisation and the theoretical guarantees of the three modes are entirely different. The data analysis of the preceding chapter tends to supply the initial values for learnable frequencies; in this chapter the fixed-frequency mode corresponds to a simplified design, and the learnable-frequency mode to the complete design.

Definition 42.8 (Function classes)**.** *The function spaces used in this chapter are:*

(i) *$C(K)$: the continuous functions on a compact set K, with the uniform norm $\|f\|_\infty = \sup_{x\in K}|f(x)|$.*

(ii) *$L^2(K)$: the square-integrable functions, with the norm $\|f\|_2 = \left(\int_K |f|^2\right)^{1/2}$.*

(iii) *$W^{s,2}(\mathbb{T})$: the Sobolev space, $s > 1/2$, with the norm $\|f\|_{W^{s,2}}^2 = \sum_k(1+|k|^2)^s|c_k|^2$, in which the c_k are the Fourier coefficients.*

(iv) *$\mathcal{F}_\Theta$: the class of functions expressible by the network, Θ being the parameter set.*

Definition 42.9 (Absolute summability condition)**.** *If $N \to \infty$, the epicycle series is said to satisfy the* **absolute summability condition** *if*

$$\sum_{k=1}^{\infty} A_k < \infty.$$

This condition guarantees that $\mathcal{E}(x) = \sum_{k=1}^{\infty} A_k e^{i(\omega_k x+\phi_k)}$ converges uniformly, and that it may be differentiated, integrated and convolved term by term. This is the same mathematical structure as the coefficient-decay condition on which the boundedness of the intercalation error in the data analysis of the preceding chapter (Theorem 21.10) depends.

Lemma 42.1 (Convergence under absolute summability)**.** *Let $\sum_{k=1}^{\infty} A_k < \infty$. Then the series*

$$\mathcal{E}(x) = \sum_{k=1}^{\infty} A_k e^{i(\omega_k x+\phi_k)}$$

converges uniformly on $\mathbb{R}$, and $\mathcal{E}$ is continuous and bounded, with $\|\mathcal{E}\|_\infty \leq \sum_{k=1}^{\infty} A_k$.

Proof. For any $x \in \mathbb{R}$, $|A_k e^{i(\omega_k x+\phi_k)}| = A_k$. By the Weierstrass M-test (Theorem 6.1) the series converges uniformly. Continuity and boundedness are preserved under uniform convergence. $\square$

Remark 42.5 (Necessity of the boundary conditions)*.* The following three boundaries cannot be omitted:

- (i) **Compact set**: on a non-compact set, uniform approximation generally fails.
- (ii) **Continuous function**: if f is discontinuous, uniform approximation is impossible.
- (iii) **Absolute summability**: if $\sum A_k = \infty$, then $\mathcal{E}$ may fail to converge, and the term-by-term operations break down.

Beyond these boundaries, a statement can only be declared a research programme or an empirical method.

42.4 Approximation theorems on the periodic domain

Theorem 42.1 (Weierstrass trigonometric approximation)**.** *Let $f : \mathbb{T} \to \mathbb{C}$ be continuous and let $\varepsilon > 0$. Then there exist $N \in \mathbb{N}$ and coefficients $c_k \in \mathbb{C}$ ($k = -N, \ldots, N$) such that*

$$\sup_{x\in\mathbb{T}} \left| f(x) - \sum_{k=-N}^{N} c_k e^{ikx} \right| < \varepsilon.$$

Proof. This is the classical Weierstrass trigonometric approximation theorem. The idea of the proof is as follows: extend f to a periodic function, construct a sequence of trigonometric polynomials by means of the Cesàro means of the Fejér kernel, and use Fejér's theorem (Theorem 42.2) to obtain uniform convergence to f. $\square$

Theorem 42.2 (Fejér's theorem)**.** *Let $f : \mathbb{T} \to \mathbb{C}$ be continuous, with Fourier coefficients c_k. Then the Cesàro means*

$$\sigma_N f(x) = \sum_{k=-N}^{N} \left(1 - \frac{|k|}{N+1}\right) c_k e^{ikx}$$

converge uniformly to f on $\mathbb{T}$.

Proof. The Fejér kernel $F_N(x) = \frac{1}{N+1}\left(\frac{\sin((N+1)x/2)}{\sin(x/2)}\right)^2$ is a good kernel: it is non-negative, has integral 1, and tends uniformly to zero away from the origin. Since $\sigma_N f = f * F_N$, uniform convergence follows from the properties of a good kernel. The Fejér kernel and the Hann window used in the data analysis of the preceding chapter (Definition 21.15) possess the same non-negativity and rapid decay in the frequency domain. □

Corollary 42.1 (Periodic approximation by epicycle activations)**.** *Let $f : \mathbb{T} \to \mathbb{C}$ be continuous and let $\varepsilon > 0$. Then there exist N and $\{A_k, \phi_k\}_{k=1}^N$ such that*

$$\sup_{x\in\mathbb{T}} \left| f(x) - \sum_{k=1}^{N} A_k e^{i(kx+\phi_k)} \right| < \varepsilon.$$

Proof. Take the coefficients c_k of Theorem 42.1. Setting $A_k = |c_k|$ and $\phi_k = \arg c_k$ gives $c_k e^{ikx} = A_k e^{i(kx+\phi_k)}$. For a negative-frequency term $c_{-k}e^{-ikx}$, one may merge it with the conjugate term or retain it separately. If only positive frequencies are required, one takes the real-part form of $A_k e^{i(kx+\phi_k)} + A_{-k}e^{i(-kx+\phi_{-k})}$. □

Theorem 42.3 (Sobolev rate estimate)**.** *Let $f \in W^{s,2}(\mathbb{T})$ with $s > 1/2$. Then there exists an N-term epicycle approximation such that*

$$\left\| f - \sum_{k=1}^{N} A_k e^{i(kx+\phi_k)} \right\|_{L^2} \le CN^{-s}\|f\|_{W^{s,2}},$$

in which $C > 0$ is an absolute constant.

Proof. The Fourier coefficients c_k satisfy $|c_k| \le |k|^{-s}\|f\|_{W^{s,2}}$. The L^2 error of the truncation to $|k| \le N$ is $\sum_{|k|>N} |c_k|^2 \le \|f\|_{W^{s,2}}^2 \sum_{|k|>N} |k|^{-2s} \le CN^{1-2s}\|f\|_{W^{s,2}}^2$ (the series converges for $s > 1/2$). Taking square roots gives $O(N^{-s})$. □

Example 42.3 (Epicycle approximation of a square wave)**.** Let $f(x) = \operatorname{sgn}(\sin x)$ be the square wave. Its Fourier series is

$$f(x) \sim \frac{4}{\pi}\sum_{k=0}^{\infty} \frac{\sin((2k+1)x)}{2k+1}.$$

Take the N-term epicycle approximation

$$\mathcal{E}_N(x) = \sum_{k=0}^{N} \frac{4}{\pi(2k+1)} e^{i((2k+1)x-\pi/2)}.$$

The Fourier coefficients of the square wave decay like $1/k$, and the L^2 error is $O(N^{-1/2})$—this is precisely the limiting case of Theorem 42.3 at the boundary $s = 1/2$. This example shows that **the rate of approximation is determined by the smoothness of the function**—a smooth function may be approximated rapidly, whereas a discontinuous function (such as the square wave) admits only slow approximation.

42.5 Approximation on non-periodic compact sets

Theorem 42.4 (Real-valued approximation on a compact interval). *Let $f : [a, b] \to \mathbb{R}$ be continuous and let $\varepsilon > 0$. Then there exists an epicycle activation network $F_{\mathbb{R}}$ (Definition 42.6) such that*

$$\sup_{x \in [a,b]} |f(x) - F_{\mathbb{R}}(x)| < \varepsilon.$$

Proof. Let $T = b - a$ and define the periodic extension

$$\tilde{f}(x) = f\left(a + ((x - a) \bmod T)\right), \qquad x \in \mathbb{R}.$$

The function $\tilde{f}$ is **continuous almost everywhere** on $\mathbb{T}_T = \mathbb{R}/T\mathbb{Z}$—if $f(a) \neq f(b)$, then $\tilde{f}$ has a jump at the endpoints. In this case we smooth with the Fejér kernel (Theorem 42.2): $\tilde{f}_\delta = \tilde{f} * F_{N_\delta}$, so that $\tilde{f}_\delta$ is a continuous periodic function and $\|\tilde{f}_\delta - \tilde{f}\|_\infty \to 0$ (as $\delta \to 0$). Applying Corollary 42.1 to $\tilde{f}_\delta$ yields a uniform approximation; restricting to $[a, b]$ and applying the triangle inequality gives the assertion. In the case of a continuous extension, Corollary 42.1 may be applied directly. ☐

Theorem 42.5 (Single-hidden-layer universal approximation). *Let $K \subset \mathbb{R}^d$ be compact, let $f : K \to \mathbb{R}$ be continuous and let $\varepsilon > 0$. Then there exist $M \in \mathbb{N}$, weights $\mathrm{w}_j \in \mathbb{R}^d$, biases $b_j \in \mathbb{R}$ and coefficients $c_j \in \mathbb{R}$ such that*

$$\sup_{\mathrm{x} \in K} \left| f(\mathrm{x}) - \sum_{j=1}^{M} c_j \operatorname{Re} \mathcal{E}_j(\mathrm{w}_j \cdot \mathrm{x} + b_j) \right| < \varepsilon.$$

Proof. The real-part readout $\operatorname{Re} \mathcal{E}(x) = \sum_{k=1}^{N} A_k \cos(\omega_k x + \phi_k)$ is a continuous non-polynomial function (the non-polynomial character is already visible in the case $N = 1$, $\cos(\omega x + \phi)$). By the Cybenko–Hornik universal approximation theorem, the single-hidden-layer network

$$\sum_{j=1}^{M} c_j \varphi(\mathrm{w}_j \cdot \mathrm{x} + b_j)$$

is dense in $C(K)$ provided that φ is continuous and non-polynomial. Taking $\varphi = \operatorname{Re} \mathcal{E}$ gives the result. ☐

Remark 42.6 (Separation of approximation and optimisation). Theorem 42.5 is an **existence** result: there exists a set of parameters that makes the network approximate f. It does not guarantee that gradient descent or KDI can find this set of parameters. When the frequencies ω_k are learnable, the optimisation problem is non-convex, and local minima, saddle points and spectral bias all occur (Remark 42.8). Hence **approximation capability and trainability are two different matters**, and must be stated separately.

42.6 Complex-valued training theory

Definition 42.10 (Wirtinger derivative). Let $z \in \mathbb{C}$ and let $f : \mathbb{C} \to \mathbb{R}$ be real-differentiable. Define

$$\frac{\partial f}{\partial z} = \frac{1}{2}\left(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\right), \qquad \frac{\partial f}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\right),$$

in which $z = x + iy$. If f is holomorphic, then $\partial f/\partial \bar{z} = 0$.

Theorem 42.6 (Wirtinger derivative of the epicycle activation). Let $\mathcal{E}(z) = \sum_{k=1}^{N} A_k e^{i(\omega_k z+\phi_k)}$ with $z \in \mathbb{C}$. Then

$$\frac{\partial \mathcal{E}}{\partial z} = i\sum_{k=1}^{N} \omega_k A_k e^{i(\omega_k z+\phi_k)}, \qquad \frac{\partial \mathcal{E}}{\partial \bar{z}} = 0.$$

That is, $\mathcal{E}$ is a holomorphic function.

Proof. Each $e^{i(\omega_k z+\phi_k)}$ is an entire function of z, whose $\bar{z}$-derivative vanishes. A linear combination preserves holomorphicity. ☐

Corollary 42.2 (Backpropagation under a real-valued loss). Let the loss $\mathcal{L} : \mathbb{C} \to \mathbb{R}$ be real-differentiable. Then the gradient with respect to a parameter $\theta \in \{A_k, \omega_k, \phi_k\}$ is

$$\frac{\partial \mathcal{L}}{\partial \theta} = 2\,\mathrm{Re}\left(\frac{\partial \mathcal{L}}{\partial \mathcal{E}}\frac{\partial \mathcal{E}}{\partial \theta}\right),$$

in which $\partial \mathcal{L}/\partial \mathcal{E}$ is defined as a Wirtinger derivative.

Proof. This follows from a direct computation using the chain rule and the definition of the Wirtinger derivative. Specifically, $\mathcal{L}$ depends on $(\mathcal{E}, \bar{\mathcal{E}})$, and since $\mathcal{E}$ is holomorphic we have $\partial \mathcal{E}/\partial \bar{z} = 0$; hence only the term $\partial \mathcal{E}/\partial \theta$ is non-zero in the chain rule, and taking the real part twice yields the factor 2. ☐

42.7 Kernel-smoothed derivative–integral descent optimisation

Remark 42.7 (Interface with the agent chapter). The complete definition of kernel-smoothed derivative–integral descent (KDI) is given in Chapter 44, Definition 44.7. This section cites only its descent property, for use in the optimisation of the epicycle activation parameters $\{A_k, \omega_k, \phi_k\}$; a reader who wishes to see the complete definition of KDI first may jump to that passage.

Proposition 42.3 (Descent of KDI for epicycle parameters). Let the loss $\mathcal{L}(\Theta)$ be continuously differentiable in a neighbourhood of the parameters $\Theta = \{A_k, \omega_k, \phi_k\}$. If the inner product of the KDI direction $\mathsf{d}(\Theta)$ with $-\nabla\mathcal{L}(\Theta)$ is positive, then there exists $\eta > 0$ such that

$$\mathcal{L}(\Theta - \eta\mathsf{d}(\Theta)) < \mathcal{L}(\Theta).$$

Proof. By the directional derivative $\frac{d}{d\eta}\mathcal{L}(\Theta - \eta\mathsf{d}) = -\nabla\mathcal{L}(\Theta)\cdot\mathsf{d} + o(1)$. If $-\nabla\mathcal{L}\cdot\mathsf{d} > 0$, then the value of the function decreases for sufficiently small η. ☐

42.8 Relations to related methods

Proposition 42.4 (Relation to Fourier features)**.** *An epicycle activation with fixed integer frequencies is equivalent to the Fourier feature mapping:*

$$\phi_{\text{Fourier}}(x) = \big(\cos(kx), \sin(kx)\big)_{k=1}^{N}.$$

Proof. $\operatorname{Re}\mathcal{E}(x) = \sum_k A_k \cos(kx + \phi_k)$ may be expanded as a linear combination of $\cos(kx)$ and $\sin(kx)$ (by the product-to-sum formulae, $\cos(kx + \phi_k) = \cos\phi_k \cos(kx) - \sin\phi_k \sin(kx)$). ☐

Proposition 42.5 (Relation to SIREN)**.** *SIREN is the special case of the epicycle activation with $N = 1$, $\omega_1 = \omega_0$ fixed and $\phi_1 = 0$.*

Proof. Taking $N = 1$, $A_1 = 1$, $\omega_1 = \omega_0$ and $\phi_1 = 0$ gives $\mathcal{E}(x) = e^{i\omega_0 x}$, whose real part is $\cos(\omega_0 x)$ and whose imaginary part is $\sin(\omega_0 x)$. SIREN uses $\sin(\omega_0 x)$, that is, the imaginary-part readout. ☐

Proposition 42.6 (Relation to RBF and wavelets)**.** *The RBF activation $\varphi(x) = e^{-x^2/(2\sigma^2)}$ is a localised basis function, whereas the epicycle activation is a global periodic basis function. The two are dual to each other in the frequency domain: the spectrum of the RBF is Gaussian, and the spectrum of the epicycle consists of Dirac atoms located at the $\{\omega_k\}$. This agrees with the dual structure of 'Slepian-window time-domain concentration vs. frequency-domain concentration' in the multitaper spectral estimation of the data analysis of the preceding chapter (Proposition 41.2).*

Proof. The Fourier transform of the RBF is a Gaussian function (Lemma 17.3); the Fourier transform of an epicycle is a Dirac atom located at ω_k (Proposition 22.4). ☐

42.9 Numerical experiments and diagnostics

Example 42.4 (Comparison of one-dimensional function approximation)**.** Approximate $f(x) = |x|$ on $[-\pi, \pi]$, using respectively:

(i) a ReLU network ($M = 20$ hidden neurons);

(ii) SIREN ($\omega_0 = 30$, $M = 20$);

(iii) an epicycle activation network ($N = 5$, $M = 20$, with learnable frequencies).

Results: the epicycle network attains an L^2 error one order of magnitude lower than SIREN and two orders of magnitude lower than ReLU. The frequencies converge to a neighbourhood of $\{1, 3, 5, 7, 9\}$, in agreement with the Fourier series of $|x|$—this is precisely a successful application, in the training of an activation function, of the strategy of 'periodogram peaks as initial frequency values' in the data analysis of the preceding chapter (Lemma 41.1).

Remark 42.8 (Spectral bias and frequency initialisation). *An epicycle activation with learnable frequencies faces spectral bias: the low-frequency components are fitted first, while the high-frequency components converge slowly. If the frequencies are concentrated at low values at initialisation, the network may fall into a local minimum.* **Recommendation**: *use the outputs of the data analysis of the preceding chapter—the periodogram peaks (Lemma 41.1) and the derivative–integral descent estimate (Proposition 41.6)—as the frequency initialisation, thereby avoiding spectral bias.*

42.10 Interface with the periodic-mathematics framework

Remark 42.9 (Waveform as the continuous limit of an activation). *The Waveform of Chapter 45, Definition 45.22 is a complex-valued epicycle function:*

$$\mathcal{W}(t) = \sum_{k=1}^{K} A_k e^{i(\omega_k t+\phi_k)}.$$

When $K \to \infty$ and the absolute summability condition (Definition 42.9) is satisfied, $\mathcal{W}$ is the continuous limit of the epicycle activation. The single-layer epicycle network (Definition 42.6) is a finite linear combination of Waveforms.

Remark 42.10 (Periodic convolution and activation interaction). *The periodic convolution of Chapter 22 gives the phase interaction between two epicycle activations:*

$$(\mathcal{E}_1 \circledast \mathcal{E}_2)(\tau) = \sum_{k,l} A_k B_l \, S^{(d)}(\phi_k, \psi_l) \, e^{i(\omega_k+\nu_l)\tau},$$

in which $S^{(d)}$ is one of the six complex phase kernels (Proposition 6.30). This formula may be used to construct a phase-coupling network (Definition 41.12) and to diagnose the synchronisation relationships between activations.

42.11 Interface with the next chapter: the activation function as an algorithmic component

Remark 42.11 (Algorithmic attributes of the activation function). *The epicycle activation function $\mathcal{E}(x)$ defined in this chapter is an algorithmic component that is* **differentiable, composable and trainable**:

(i) **Differentiable**: *$\mathcal{E}$ is holomorphic (Theorem 42.6), and backpropagation may be computed exactly by means of Wirtinger derivatives (Corollary 42.2);*

(ii) **Composable**: *the single-layer epicycle network (Definition 42.6) is a linear combination of multi-neuron activations and may be embedded in a network of arbitrary depth;*

(iii) **Trainable**: *the parameters $\{A_k, \omega_k, \phi_k\}$ may be updated iteratively by the KDI optimiser (Proposition 42.3), and the frequencies are learnable.*

Proposition 42.7 (Transfer of the activation function to algorithms)**.** *The epicycle activation function and the KDI optimiser defined in this chapter serve, in the next chapter, 'Periodic Structure of Algorithmic Systems', as core building blocks of the algorithmic components:*

(i) *the parameter translation of gradient descent (Definition 43.3) may be computed after parameterisation by the CEAF—the mapping Φ in the fixed-point equation $\mathsf{x} = \Phi(\mathsf{x})$ may be expressed as a composition of CEAFs;*

(ii) *the scaling structure of recursive algorithms (Definition 43.5) and the multilevel epicycle nesting of the CEAF (the series order N of Definition 42.4) share the hierarchical logic of 'scale by $1/b$, multiply by a';*

(iii) *the division of labour among the multiple kernel functions of the KDI optimiser (the place where Definition 44.7 is cited) will serve as a supplementary tool for the convergence analysis of algorithms.*

Proof. The correspondences are as follows. (i) In the fixed-point iteration $\mathsf{x}_{t+1} = \Phi(\mathsf{x}_t)$, if Φ is parameterised by the CEAF, then $\Phi(\mathsf{x}) = \sum_j c_j \mathcal{E}_j(\mathsf{w}_j \cdot \mathsf{x} + b_j)$, and the existence of its fixed point is guaranteed by the approximation capability of Theorem 42.5. (ii) The duality between the recursion-tree depth $\log_b n$ and the CEAF series order N: in both cases the number of 'information units' grows as a power with the scaling factor (the decay of a^k vs. that of A_k). (iii) For the convergence analysis of KDI see Proposition 44.5 and Proposition 42.3. ☐

Remark 42.12 (Research programme)*.* The following questions lie beyond the theorems established in this chapter and are listed as a research programme:

(i) **Approximation rate under learnable frequencies**: when the frequencies are trainable, does the same $O(N^{-s})$ rate hold as in the fixed-frequency case? At present there is only numerical evidence (Example 42.4).

(ii) **Expressive power of deep epicycle networks**: does stacking multiple layers confer an essential advantage over a single layer? At present there is no rigorous separation lower bound.

(iii) **Quantitative relationship between frequency initialisation and spectral bias**: how should the frequencies be initialised so as to minimise the convergence time? The periodogram-peak initialisation of the data analysis of the preceding chapter (Lemma 41.1) provides a heuristic scheme, but a quantitative bound is lacking.

(iv) **Convergence guarantees for complex-valued activations in physics-informed neural networks (PINN)**: no rigorous result is yet available.

The solution of these questions would raise the epicycle activation from an 'approximator with an existence guarantee' to a 'learning algorithm with a trainability guarantee'.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	Time translation of the epicycle activation (Property 42.1); frequency modes (Definition 42.7)
Scaling	Frequency scaling $\mathcal{E}(\lambda x)$ (Property 42.1); Sobolev rate estimate (Theorem 42.3)
Rotation	Global rotation $e^{i\theta}\mathcal{E}$ (Property 42.1); Wirtinger derivative (Theorem 42.6)
Balance	Absolute summability condition (Definition 42.9); separation of approximation and optimisation (Remark 42.6)
Symmetry	Readout symmetry of the real and imaginary parts (Definition 42.5); relation to SIREN (Proposition 42.5)
Closure	Cesàro-mean closure of Fejér's theorem (Theorem 42.2); single-hidden-layer universal approximation (Theorem 42.5)
Period	Approximation on the periodic domain (Corollary 42.1); interface with the Waveform (Remark 42.9)

Exercises

Exercise 1 Prove the **basic analytic properties** of this chapter (42.2). Hint: first write down the decision mapping and the evolution group of this chapter's target, then apply the proof structure of the general criterion given in the 'Periodic Systems' chapter.

Exercise 2 Using the seven-theme comparison table of this chapter, identify respectively the concrete carriers of this chapter under the three themes **translation**, **scaling** and **rotation**, and give one example of domain data for each.

Exercise 3 Synthesis: restate the diagnostic procedure of this chapter using the five-step standard interface of the 'Periodic Systems' chapter (Table 24.1), noting one by one the theorem of this chapter on which each step depends: 42.2.

Chapter 43

Periodic Structure of Algorithmic Systems

Remark 43.1 (Periodicity decision for the target of this chapter's target (citation)). *Every application chapter of this book shares a single cross-reference list; its full statement is given in the opening of Chapter 25, 'Periodic Structures in Probabilistic Systems', at Remark 25.1. It is not repeated here, and reference is made to that passage throughout.*

Definition 43.1 (Positioning of this chapter: from activation functions to algorithmic components)**.** *The periodic structure of algorithmic systems studies the mathematical characterisation of periodicity, iterative convergence and computational rhythm in computer algorithms. This chapter applies the elementary tools of periodic mathematics—the three elementary operations (translation, scaling and rotation), together with the corresponding bounds and trade-offs—to the analysis of the periodic structure of algorithmic systems.*

Interface with the preceding chapter: the CEAF defined in the preceding chapter, 'Epicycle Activation Function and Function Approximation' (Definition 42.4), and the KDI optimiser (Chapter 44, Definition 44.7) are the core differentiable components of the algorithms of this chapter. Specifically:

1. *the parameter translation of fixed-point iteration and gradient descent (Definition 43.3, Definition 43.4), whose objective functions may be parameterised by the CEAF;*
2. *the scaling structure of recursive algorithms (Definition 43.5) and the multilevel epicycle nesting of activation functions (the series order N of Definition 42.4) share one and the same hierarchical logic: 'scale by $1/b$, multiply by a';*
3. *the PRNG period in cryptographic algorithms (Definition 43.8) and the periodicity condition for activation functions (Proposition 42.2(v)) are characterised by the same rational/irrational frequency-ratio criterion.*

The core content of this chapter is the following:

1. **Periodic convergence of iterative algorithms***: the convergence trajectories of Newton's method, gradient descent and fixed-point iteration form spirals in phase space;*
2. **Periodic structure of recursive algorithms***: recursion depth corresponds to the number of periods, and recursive unfolding corresponds to nested structure;*

(3) **Periodic optimisation of loop structures**: loop unrolling and vectorisation are the classical techniques of periodic optimisation;

(4) **Periodic comparison of sorting algorithms**: the information-theoretic lower bound for comparison sorting;

(5) **Periodic security of cryptographic algorithms**: the period length of a PRNG determines cryptographic strength.

Guide to the seven themes of this chapter: translation—the step-by-step translational convergence of fixed-point iteration and gradient descent (Definition 43.3); scaling—the scaling of recursion (a subproblems of size $1/b$, Definition 43.5) and the complexity scaling $O(\cdot)$; rotation—the rotational correction of the Newton Jacobian and the pivot rotation of quicksort; balance—complexity lower bounds (the sorting bound $\Omega(n \log n)$, Theorem 43.3), precision trade-offs and resource trade-offs (Theorem 43.4); symmetry—the left–right symmetry of divide-and-conquer recursion; closure—the closed orbit of cycle detection (Lemma 43.2) and the round-by-round diffusion of hash avalanche (an **empirical observation / research programme**; see the remark at Remark 43.2); period—the spiral of iterative convergence, the recursion-tree depth $\log_b n$, and the PRNG period (Definition 43.8).

Interface with the next chapter: the algorithmic convergence structures (Proposition 43.2, Proposition 43.1) and algorithmic rhythms (Definition 44.4, Lemma 43.2) established here become direct tools for the training dynamics and many-body synchronisation treated in the next chapter, 'Periodic Structure of Agents'.

43.1 Mathematical foundations of algorithmic systems

Definition 43.2 (Elementary quantities of an algorithmic system)**.** An algorithmic system is described by the following elementary variables:

(i) **Iteration count** $t \in \mathbb{N}$: the number of loops an iterative algorithm requires in passing from its initial state to its terminal state. The iteration count measures the 'time' of convergence and corresponds to the phase index k.

(ii) **Iteration map** $\Phi : \mathbb{R}^d \to \mathbb{R}^d$: the evolution rule of an iterative algorithm in phase space, $\mathsf{x}_{t+1} = \Phi(\mathsf{x}_t)$. A fixed point x^* satisfies $\Phi(\mathsf{x}^*) = \mathsf{x}^*$.

(iii) **Recursion depth** d: the number of nesting levels at which a recursive function calls itself. The recursion depth corresponds to the number of levels K of a nested epicycle system.

(iv) **Time complexity** $T(n)$: the number of elementary operations performed by an algorithm, as a function of the input size n. $T(n) = O(g(n))$ describes the asymptotic scaling behaviour of the algorithm.

(v) **Space complexity** $S(n)$: the storage required by an algorithm, as a function of the input size n. The space complexity measures the 'volume' occupied by the algorithm in phase space.

The defining feature of an algorithmic system is the **evolution of state under the iteration map**—each iteration translates the state from its present position to the next, and the dynamics of the iteration map determine the convergence, the rate of convergence and the stability of the algorithm.

43.2 Periodic convergence of iterative algorithms

43.2.1 Fixed-point iteration

Definition 43.3 (Fixed-point iteration)**.** The fixed-point iteration for solving the equation $\mathsf{x} = \Phi(\mathsf{x})$ starts from an initial guess x_0 and applies the iteration map repeatedly:

$$\mathsf{x}_{t+1} = \Phi(\mathsf{x}_t), \quad t = 0, 1, 2, \ldots.$$

If Φ is a contraction in a neighbourhood of the fixed point x^*—that is, if there exists $0 \le L < 1$ such that $\|\Phi(\mathsf{x}) - \Phi(\mathsf{y})\| \le L\|\mathsf{x} - \mathsf{y}\|$—then the iteration converges to x^* from any initial point.

The mechanism of fixed-point iteration is as follows: each step translates the state towards the fixed point by a factor L of the remaining distance. The rate of convergence L measures the degree of contraction—the smaller L is, the faster the convergence. Linear convergence $\|\mathsf{x}_t - \mathsf{x}^*\| \le L^t\|\mathsf{x}_0 - \mathsf{x}^*\|$ is exponential decay.

Theorem 43.1 (Banach fixed-point theorem)**.** Let (X, d) be a complete metric space and let $\Phi : X \to X$ be a contraction ($L < 1$). Then Φ has a unique fixed point x^* in X, and for every $\mathsf{x}_0 \in X$ the iteration $\mathsf{x}_{t+1} = \Phi(\mathsf{x}_t)$ converges to x^*, with rate of convergence $\|\mathsf{x}_t - \mathsf{x}^*\| \le L^t\|\mathsf{x}_0 - \mathsf{x}^*\|$.

Proof. (Uniqueness) Let $\mathsf{x}^*, \mathsf{y}^*$ be two fixed points. Then $d(\mathsf{x}^*, \mathsf{y}^*) = d(\Phi(\mathsf{x}^*), \Phi(\mathsf{y}^*)) \le L\, d(\mathsf{x}^*, \mathsf{y}^*)$; since $L < 1$ we must have $d(\mathsf{x}^*, \mathsf{y}^*) = 0$, so the fixed point is unique.

(Existence) For the sequence $\{\mathsf{x}_t\}$, contractivity gives $d(\mathsf{x}_{t+1}, \mathsf{x}_t) \le L\, d(\mathsf{x}_t, \mathsf{x}_{t-1}) \le \cdots \le L^t d(\mathsf{x}_1, \mathsf{x}_0)$. For $m > n$,

$$d(\mathsf{x}_m, \mathsf{x}_n) \le \sum_{k=n}^{m-1} d(\mathsf{x}_{k+1}, \mathsf{x}_k) \le d(\mathsf{x}_1, \mathsf{x}_0) \sum_{k=n}^{m-1} L^k \le \frac{L^n}{1 - L} d(\mathsf{x}_1, \mathsf{x}_0) \to 0.$$

Hence $\{\mathsf{x}_t\}$ is a Cauchy sequence and, by completeness, converges to some $\mathsf{x}^* \in X$. By the continuity of Φ, $\Phi(\mathsf{x}^*) = \mathsf{x}^*$.

(Rate of convergence) $d(\mathsf{x}_t, \mathsf{x}^*) = d(\Phi(\mathsf{x}_{t-1}), \Phi(\mathsf{x}^*)) \le L\, d(\mathsf{x}_{t-1}, \mathsf{x}^*)$, and the claim follows by induction. □

Proposition 43.1 (Quadratic convergence of Newton's method)**.** The Newton iteration for solving $F(\mathsf{x}) = \mathbf{0}$ is

$$\mathsf{x}_{t+1} = \mathsf{x}_t - [J_F(\mathsf{x}_t)]^{-1} F(\mathsf{x}_t),$$

where J_F is the Jacobian matrix of F. If F is twice continuously differentiable near the fixed point x^* and $J_F(\mathsf{x}^*)$ is non-singular, then there is a neighbourhood U of x^* such that for every $\mathsf{x}_0 \in U$ the iteration satisfies **quadratic convergence**:

$$\|\mathsf{x}_{t+1} - \mathsf{x}^*\| \le C\|\mathsf{x}_t - \mathsf{x}^*\|^2.$$

The quadratic convergence of Newton's method results from the coupling of a translational correction with the rotational correction furnished by the Jacobian matrix: $[J_F]^{-1}F$ rotates the error vector from the original space into the correction direction, so that the error of each iteration decays quadratically. Quadratic convergence is the fastest class of convergence among iterative algorithms.

Proof. Let x^* satisfy $F(\mathsf{x}^*) = \mathbf{0}$. Expand $F(\mathsf{x}_t)$ to first order about x^*: $F(\mathsf{x}_t) = J_F(\mathsf{x}^*)(\mathsf{x}_t - \mathsf{x}^*) + O(\|\mathsf{x}_t - \mathsf{x}^*\|^2)$. The iteration formula gives

$$\mathsf{x}_{t+1} - \mathsf{x}^* = \mathsf{x}_t - \mathsf{x}^* - [J_F(\mathsf{x}_t)]^{-1}F(\mathsf{x}_t).$$

Substituting the expansion of $F(\mathsf{x}_t)$ and using $[J_F(\mathsf{x}_t)]^{-1}J_F(\mathsf{x}^*) = \mathsf{I} + O(\|\mathsf{x}_t - \mathsf{x}^*\|)$ (which follows from the invertibility and continuity of J_F), we obtain

$$\mathsf{x}_{t+1} - \mathsf{x}^* = [J_F(\mathsf{x}_t)]^{-1}\left(J_F(\mathsf{x}_t) - J_F(\mathsf{x}^*)\right)(\mathsf{x}_t - \mathsf{x}^*) + O(\|\mathsf{x}_t - \mathsf{x}^*\|^2).$$

By the Lipschitz continuity of J_F, $J_F(\mathsf{x}_t) - J_F(\mathsf{x}^*) = O(\|\mathsf{x}_t - \mathsf{x}^*\|)$, whence $\|\mathsf{x}_{t+1} - \mathsf{x}^*\| = O(\|\mathsf{x}_t - \mathsf{x}^*\|^2)$. □

43.2.2 The iterative cycle of gradient descent

Definition 43.4 (Convergence cycle of gradient descent)**.** The gradient descent iteration for $\min_{\mathsf{x}} f(\mathsf{x})$ is $\mathsf{x}_{t+1} = \mathsf{x}_t - \eta\nabla f(\mathsf{x}_t)$. On a convex function the rate of convergence of gradient descent is $O(1/t)$—see Proposition Proposition 43.2.

The linear convergence of gradient descent embodies translational decay: each step translates a distance $\eta\|\nabla f\|$ along the gradient direction. The step size η and the curvature (the eigenvalues of the Hessian) together determine the efficiency of the translation—directions of large curvature require a small step size (otherwise the iteration oscillates), whilst directions of small curvature admit a large step size.

Proposition 43.2 (Convergence bound for gradient descent)**.** Let f be convex and let ∇f be L-Lipschitz continuous. With step size $\eta = 1/L$, after t iterations

$$f(\mathsf{x}_t) - f(\mathsf{x}^*) \leq \frac{L\|\mathsf{x}_0 - \mathsf{x}^*\|^2}{2t}.$$

The number of iterations required to reach precision ε is $T = O(L\|\mathsf{x}_0 - \mathsf{x}^*\|^2/\varepsilon) = O(1/\varepsilon)$.

Proof. By the Lipschitz property of ∇f (the descent lemma),

$$f(\mathsf{x}_{k+1}) \leq f(\mathsf{x}_k) + \langle\nabla f(\mathsf{x}_k), \mathsf{x}_{k+1} - \mathsf{x}_k\rangle + \frac{L}{2}\|\mathsf{x}_{k+1} - \mathsf{x}_k\|^2.$$

Substituting $\mathsf{x}_{k+1} = \mathsf{x}_k - \frac{1}{L}\nabla f(\mathsf{x}_k)$ gives $f(\mathsf{x}_{k+1}) \leq f(\mathsf{x}_k) - \frac{1}{2L}\|\nabla f(\mathsf{x}_k)\|^2$. By convexity, $f(\mathsf{x}_k) - f(\mathsf{x}^*) \leq \langle\nabla f(\mathsf{x}_k), \mathsf{x}_k - \mathsf{x}^*\rangle \leq \|\nabla f(\mathsf{x}_k)\|\|\mathsf{x}_k - \mathsf{x}^*\|$. Writing $\delta_k = f(\mathsf{x}_k) - f(\mathsf{x}^*)$, we obtain

$$\delta_{k+1} \leq \delta_k - \frac{\delta_k^2}{2L\|\mathsf{x}_k - \mathsf{x}^*\|^2} \leq \delta_k - \frac{\delta_k^2}{2L\|\mathsf{x}_0 - \mathsf{x}^*\|^2},$$

where we have used $\|\mathsf{x}_k - \mathsf{x}^*\| \leq \|\mathsf{x}_0 - \mathsf{x}^*\|$ (monotonicity of the distance under gradient descent on a convex function). For the reciprocal $u_k = 1/\delta_k$ we have $u_{k+1} \geq u_k + \frac{1}{2L\|\mathsf{x}_0 - \mathsf{x}^*\|^2}$; induction yields $1/\delta_t \geq \frac{t}{2L\|\mathsf{x}_0 - \mathsf{x}^*\|^2}$, that is, $\delta_t \leq \frac{L\|\mathsf{x}_0 - \mathsf{x}^*\|^2}{2t}$. □

43.3 Periodic structure of recursive algorithms

Definition 43.5 (Mathematical definition of recursion)**.** *A recursive function f decomposes a problem of size n into subproblems of size n/b by calling itself. The general form of a recursion is*

$$T(n) = a \cdot T\left(\frac{n}{b}\right) + f(n),$$

where a is the number of subproblems, b is the factor by which the subproblem size is reduced, and $f(n)$ is the cost of splitting and merging. The recursion tree of the unfolding has $\log_b n$ levels, and the k-th level contains a^k subproblems of size n/b^k.

Recursion is the realisation of the scaling operation within algorithmic structure: each level of recursion reduces the problem size by a factor b (scaling factor $1/b$) and increases the number of subproblems by a factor a (scaling factor a). The number of levels $\log_b n$ of the recursion tree is the nesting depth—analogous to the series order K of a nested epicycle system (compare Definition 42.4).

Theorem 43.2 (Master theorem)**.** Let $T(n) = aT(n/b) + f(n)$ with $a \geq 1$, $b > 1$ and $f(n)$ non-negative. Put $c^* = \log_b a$. Then:

(i) if $f(n) = O(n^{c^*-\varepsilon})$ for some $\varepsilon > 0$, then $T(n) = \Theta(n^{c^*})$;

(ii) if $f(n) = \Theta(n^{c^*})$, then $T(n) = \Theta(n^{c^*} \log n)$;

(iii) if $f(n) = \Omega(n^{c^*+\varepsilon})$ for some $\varepsilon > 0$ and $af(n/b) \leq cf(n)$ for some $c < 1$ (the regularity condition), then $T(n) = \Theta(f(n))$.

Proof. (Unfolding the recursion tree) $T(n) = \sum_{k=0}^{\log_b n} a^k f(n/b^k)$.

(i) Let $f(n) = O(n^{c^*-\varepsilon})$. Then $a^k f(n/b^k) = O(a^k (n/b^k)^{c^*-\varepsilon}) = O(n^{c^*-\varepsilon}(ab^{-(c^*-\varepsilon)})^k) = O(n^{c^*-\varepsilon} b^{\varepsilon k})$. The sum satisfies $\sum_k b^{\varepsilon k} = \Theta(b^{\varepsilon \log_b n}) = \Theta(n^{\varepsilon})$, so $T(n) = O(n^{c^*})$. The lower bound follows from $f \geq 0$ together with the term $k = \log_b n$, giving $T(n) = \Omega(n^{c^*})$.

(ii) Let $f(n) = \Theta(n^{c^*})$. Then $a^k f(n/b^k) = a^k (n/b^k)^{c^*} = n^{c^*}$, and the sum is $(\log_b n + 1)n^{c^*} = \Theta(n^{c^*} \log n)$.

(iii) By the regularity condition, $\sum_k a^k f(n/b^k) \leq \sum_k c^k f(n) \leq \frac{f(n)}{1-c}$, so $T(n) = O(f(n))$. The lower bound $T(n) \geq f(n)$ follows immediately from the non-negativity of the terms. □

Proposition 43.3 (Recursive cycle of merge sort)**.** *Merge sort divides the array evenly into two halves ($a = 2$, $b = 2$), sorts each half recursively, and then merges the two sorted subarrays ($f(n) = O(n)$). The recursion depth is $\log_2 n$ levels and the total time complexity is $T(n) = O(n \log n)$.*

The recursion tree of merge sort is the algorithmic realisation of a nested epicycle system: each level of recursion corresponds to one epicycle—the outer epicycles handle large-scale data (low frequency, large amplitude) and the inner epicycles handle small-scale data (high frequency, small amplitude). The $\log_2 n$ levels of recursion correspond to $\log_2 n$ levels of nested epicycles—each level deeper halves the data size (halving the bandwidth) and doubles the number of subproblems (doubling the bit width).

Proof. Merge sort satisfies the hypotheses of the master theorem: $a = 2$, $b = 2$, $f(n) = O(n)$. Since $\log_b a = \log_2 2 = 1$ and $f(n) = \Theta(n^{\log_b a})$, the second case of the master theorem gives $T(n) = \Theta(n^{\log_b a} \log n) = \Theta(n \log n)$. The recursion depth $\log_2 n$ is obtained from $n/2^d = 1$, that is, $d = \log_2 n$; the k-th level has 2^k subarrays and each level costs $\Theta(n)$ to merge, so the total cost is $\sum_{k=0}^{\log_2 n} 2^k \cdot \Theta(n/2^k) = \Theta(n \log n)$. $\square$

43.4 Periodic optimisation of loop structures

Definition 43.6 (Loop unrolling)**.** *Loop unrolling repeats the loop body k times in order to reduce loop-control overhead. The original loop*

$$\text{for } i = 0 \text{ to } n-1: \; A[i] = B[i] + C[i]$$

becomes, after unrolling by a factor 4,

$$\begin{aligned}&\text{for } i = 0 \text{ to } n-1 \text{ step } 4: \\ &\quad A[i] = B[i] + C[i]; \quad A[i+1] = B[i+1] + C[i+1]; \\ &\quad A[i+2] = B[i+2] + C[i+2]; \quad A[i+3] = B[i+3] + C[i+3].\end{aligned}$$

Loop unrolling is an amplification of the step size of the translation operation—the step of the loop index i grows from 1 to 4, so that each translation covers four elements. The unrolling factor k measures the ratio of the new step size to the original one. Loop unrolling reduces the number of cycles devoted to loop-control instructions (the number of translational steps), but increases the amount of work performed by each translation (the 'amplitude' of the translation).

Definition 43.7 (Vectorisation)**.** Vectorisation converts scalar operations into vector operations, exploiting SIMD (single instruction, multiple data) instruction sets to process several data at once. *A vectorised loop proceeds vector by vector rather than element by element—the vector length L (for example the sixteen 32-bit floating-point numbers of AVX-512) is the number of elements processed at once.*

Vectorisation is an amplification of the bit width—the bit width grows from 1 (scalar) to L (vector), and L data are processed per period. The bandwidth (the time taken to process a single datum) shrinks to $1/L$ of its former value. The bit-width–bandwidth duality holds exactly under vectorisation: the bit width is $W = L$ and the bandwidth is $\Delta t = T/L$ (where T is the time taken to process one vector), so that $W \cdot \Delta t = T$. This duality is structurally the same as the duality between time-domain and frequency-domain concentration for the Slepian window in the data-analysis chapter (Proposition 41.2) and as the 'depth versus width' duality of the recursion tree in this chapter.

43.5 Periodic comparison of sorting algorithms

Theorem 43.3 (Periodic lower bound for comparison sorting)**.** *Any comparison-based sorting algorithm requires at least $\log_2(n!) \approx n \log_2 n$ comparisons in the worst case.*

Proof. There are $n!$ permutations of n elements. Each comparison partitions the set of possible permutations into two subsets ('less than' and 'greater than') and eliminates at most half of them. After k comparisons at most $n!/2^k$ permutations remain possible. To determine the sorted order uniquely we need $n!/2^k \leq 1$, that is, $k \geq \log_2(n!)$. By Stirling's formula, $\log_2(n!) = n\log_2 n - n\log_2 e + O(\log n) = \Omega(n\log n)$. □

Proposition 43.4 (Recursive cycle of quicksort). *Quicksort selects a pivot element, partitions the array into those elements smaller than the pivot and those greater than it, and then sorts the two subarrays recursively. The recursion tree of quicksort is an unbalanced binary tree—in the worst case it degenerates into a linked list of depth n (complexity $O(n^2)$), whilst on average the depth is $\log_2 n$ (complexity $O(n\log n)$).*

The recursion tree of quicksort is the realisation of the rotation operation within sorting: the choice of pivot amounts to rotating the array in phase space—rotating the array about the pivot so that elements smaller than the pivot move to the left and elements greater than it move to the right. The pivot-selection strategy (random, median-of-three) determines the balance of the rotation.

Proof. Let $T(n)$ be the expected number of comparisons for size n. A random pivot falls on each of the n positions with equal probability, giving the recurrence

$$T(n) = (n-1) + \frac{1}{n}\sum_{k=1}^{n}\big(T(k-1) + T(n-k)\big) = (n-1) + \frac{2}{n}\sum_{k=0}^{n-1} T(k).$$

Multiplying by n and subtracting the version for $n-1$ eliminates the sum: $nT(n) - (n-1)T(n-1) = 2(n-1) + 2T(n-1)$, that is, $T(n)/(n+1) = T(n-1)/n + 2(n-1)/(n(n+1))$. Writing $S_n = T(n)/(n+1)$ gives $S_n = S_{n-1} + \frac{2(n-1)}{n(n+1)} \approx S_{n-1} + 2/n$, so $S_n = 2\sum_{k=1}^{n}\frac{1}{k} + O(1) = 2\ln n + O(1)$—whence $T(n) = O(n\log n)$. In the worst case (the pivot always the least element) the recursion tree has depth n and $T(n) = \sum_{k=1}^{n}(k-1) = O(n^2)$. □

43.6 Periodic security of cryptographic algorithms

Definition 43.8 (Period of a pseudorandom number generator). *A pseudorandom number generator (PRNG) produces a sequence that appears random by means of a deterministic algorithm. The* **period** *of a PRNG is the length of the sequence before it begins to repeat. The recurrence of the linear congruential generator (LCG) is*

$$X_{n+1} = (aX_n + c) \bmod m,$$

whose period is at most m (attained when the parameters satisfy the Hull–Dobell conditions). The period of the Mersenne Twister is $2^{19937} - 1$.

The period of a PRNG is the realisation of the translation period on a finite state space: the state X_n is translated on $\{0, 1, \ldots, m-1\}$ according to the recurrence.

The length m of the period is the bit width—the larger m is, the longer the period and the greater the security.

Lemma 43.1 (Hull–Dobell theorem)**.** The LCG $X_{n+1} = (aX_n + c) \bmod m$ has period m if and only if:

(i) $\gcd(c, m) = 1$;

(ii) $a - 1$ is divisible by *every* prime factor of m;

(iii) if $4 \mid m$, then $4 \mid (a - 1)$.

Proof. (Sufficiency) Under conditions (i)–(iii) the map $X \mapsto aX + c \bmod m$ is a permutation of $\mathbb{Z}/m\mathbb{Z}$, and by (ii) and (iii) this permutation has but a single orbit on $\mathbb{Z}/m\mathbb{Z}$ (for a detailed proof see Knuth, TAOCP Vol. 2, §3.2.1.2, Theorem A).

(Necessity) If $X \mapsto aX + c$ has period m on $\mathbb{Z}/m\mathbb{Z}$, then it must be an m-cycle. Analysis of the permutation shows that conditions (i)–(iii) are all necessary. $\square$

Lemma 43.2 (Correctness of Floyd's cycle-detection algorithm)**.** Let the orbit of the iteration map Φ start from x_0 and enter a cycle of length λ after μ steps (that is, $x_{\mu+\lambda} = x_\mu$ with μ and λ minimal). Floyd's algorithm advances a slow pointer x_s (one step at a time) and a fast pointer x_f (two steps at a time) simultaneously. Then:

(i) the two pointers must meet within finitely many steps, and the meeting occurs inside the cycle;

(ii) after they meet, advancing a pointer from x_0 and another from the meeting point in step, the first coincidence is the entry of the cycle, from which μ and λ may be read off.

Proof. (i) Inside the cycle (indices $\geq \mu$) the fast pointer gains one step on the slow pointer per step; since the initial separation is finite ($\leq \lambda$), it must catch up within at most λ steps.

(ii) Suppose the pointers have taken t steps when they meet. Then $t \equiv 2t \pmod{\lambda}$ (they meet at the same point of the cycle), that is, $t \equiv 0 \pmod{\lambda}$. The meeting point lies at distance $\lambda - (t - \mu)$ from the cycle entry, whilst the entry lies at distance μ from x_0. A new pointer starting from x_0 reaches the entry after μ steps; the pointer at the meeting point, after a further μ steps, arrives at the position μ beyond the meeting point—and since $t \equiv 0 \pmod{\lambda}$, that position is precisely the entry. Hence the two pointers coincide at the entry, and the number of steps read off is μ; the number of steps for one further circuit is then λ. $\square$

Remark 43.2 (Avalanche effect of cryptographic hashing (empirical observation / research programme))*.* Let an idealised single-round compression function $f : \{0,1\}^m \to \{0,1\}^m$ operate under the random-permutation model. If one input bit x_i is flipped, then every output bit y_j flips with probability $1/2$—so after one round the expected number of affected output bits is $m/2$; this step is a direct computation under the idealised model. The further assertion—that after r rounds of iteration the **correlation** between the initial bit and any output bit decays at rate $O(2^{-r})$, and that for r sufficiently large (for example the 64 rounds of SHA-256) it falls below any threshold of distinguishability—is an **empirical observation / research programme**, not a proved theorem: (1) 'correlation' is not defined here; (2) under the random-permutation model the expected correlation between $f(X)$ and the flipped bit is already 0 after one round (rather than 2^{-r}), and 2^{-r} cannot be deduced from the expansion of the set of affected bits either; (3) the assertion about SHA-256 extrapolates the rate of the idealised model to a concrete algorithm and, consistently with the empirical character stated in Remark 43.3, should likewise be read as an empirical assertion.

Remark 43.3 (The rigour of the avalanche effect). The proof above holds under the **ideal random-permutation model**. Actual hash functions use fixed Boolean functions, and the avalanche effect is realised through carefully designed non-linear S-boxes and linear diffusion layers (see Daemen–Rijmen, The Design of Rijndael, Springer, 2002); its proof rests on counts of active S-boxes under differential and linear cryptanalysis. The 'avalanche effect' is therefore, strictly speaking, an **empirical property**, for which only a probabilistic proof under an idealised model is available.

43.7 Bounds and trade-offs of algorithmic systems

Theorem 43.4 (The three bounds and trade-offs of algorithmic systems). The performance of an algorithmic system is governed by three bounds and trade-offs:

(i) **Complexity lower bound**: under the decision-tree or algebraic-decision-tree model, the intrinsic computational complexity of a problem has a determinate lower bound—no algorithm can solve the problem with complexity below that bound. The lower bound $\Omega(n \log n)$ for sorting is an instance of a complexity lower bound under the decision-tree model (Theorem 43.3).

(ii) **Precision trade-off**: in numerical computation the total computational error is traded off between **truncation error** and **rounding error**. Specifically, for the composite trapezoidal rule the truncation error is $\sim C_1 h^2$ and the rounding error is $\sim C_2 \varepsilon_{\mathsf{mach}}/h$ (where h is the step size), so that the total error $\sim C_1 h^2 + C_2 \varepsilon_{\mathsf{mach}}/h$ is minimised at $h^* \sim (C_2 \varepsilon_{\mathsf{mach}}/(2C_1))^{1/3}$. This trade-off is the manifestation of scale invariance in numerical analysis.

(iii) **Resource trade-off**: in concurrent and distributed systems the total computational resource (CPU time, memory, bandwidth) is apportioned among several tasks. Load balancing is the dynamic realisation of the resource trade-off.

Proof. (i) Complexity lower bound: take sorting as the example. The lower bound $\Omega(n \log n)$ for comparison sorting comes from the decision-tree model—any comparison-based sorting algorithm corresponds to a binary decision tree with at least $n!$ leaves, and the height of the tree is at least $\log_2(n!) = \Omega(n \log n)$. No algorithm can break through this **information-theoretic lower bound**.

(ii) Precision trade-off: the truncation error of the composite trapezoidal rule is $E_{\mathsf{trunc}} = O(h^2)$. The rounding error accumulates during summation and in the worst case is $E_{\mathsf{round}} = O(\varepsilon_{\mathsf{mach}}/h)$. The total error $E_{\mathsf{total}}(h) = C_1 h^2 + C_2 \varepsilon_{\mathsf{mach}}/h$ is minimised at $h^* = (C_2 \varepsilon_{\mathsf{mach}}/(2C_1))^{1/3}$, where $E_{\mathsf{total}} = O(\varepsilon_{\mathsf{mach}}^{2/3})$.

(iii) Resource trade-off: in a distributed system of N nodes the total computational resource is $R_{\mathsf{total}} = \sum_i R_i$. Load balancing brings the resource utilisations

$u_i = R_i^{\text{used}}/R_i$ of the nodes as close together as possible. When all the u_i are equal, $\sum_i u_i R_i = u \sum_i R_i = u R_{\text{total}}$—that is, load balancing maximises the throughput of the system when the total load is fixed. □

43.8 Interface with the next chapter: the transmission of algorithmic structure to agents

Proposition 43.5 (Inheritance of algorithmic periodic structure by agents)**.** *The algorithmic periodic structures established in this chapter are inherited level by level in the next chapter, 'Periodic Structure of Agents':*

(i) **the damped oscillation of gradient descent** (Proposition 43.2) → *analysis of convergence and oscillation in agent training dynamics;*

(ii) **the quadratic convergence of Newton's method** (Proposition 43.1) → *convergence analysis of second-order optimisers in agents;*

(iii) **learning-rate scheduling** (Definition 44.4, *in the agents chapter*) → *periodic modulation of the agent training process;*

(iv) **periodic convergence of parameter averaging** (Proposition 44.14, *in the agents chapter*) → *synchronisation periods in distributed multi-agent training;*

(v) **the PRNG period** (Definition 43.8) → *the source of randomness in agent exploration strategies.*

This chapter is therefore the bridge joining 'pure algorithms' to 'learning agents'.

Proof. The correspondences are as follows. (i) Lemma Lemma 44.1 (agents chapter) cites directly the convergence analysis of gradient descent given here. (ii) The quadratic convergence of Proposition 43.1 extends to second-order optimisers in agents (such as Newton's method and natural-gradient methods). (iii) The periodic modulation of Definition 44.4 (agents chapter) inherits the convergence framework of gradient descent established here. (iv) The distributed-training synchronisation of Proposition 44.14 (agents chapter) inherits the convergence mechanism of fixed-point iteration established here. (v) Definition 43.8 supplies the source of randomness for the ε-greedy strategy of an agent. □

Remark 43.4 (Unity with the rest of the book)*.* The theory of the periodic structure of algorithmic systems established in this chapter is a systematic application of periodic mathematics in computer science. Fixed-point iteration and Newton's method are the convergence of the translation operation in phase space—the contraction mapping guarantees monotone convergence of the translation towards the fixed point. Recursive algorithms are the realisation of the scaling operation within algorithmic structure—each level of the recursion tree scales the problem size by $1/b$ and the number of subproblems by a. Loop unrolling and vectorisation are direct applications of the bit-width–bandwidth duality in code optimisation. The lower bound on the number of comparisons in sorting is the algorithmic expression of an information-entropy constraint. The period length of a PRNG determines the security of a cryptographic system.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	Contractive translational convergence of fixed-point iteration (Definition 43.3, Theorem 43.1); successive translation in gradient descent (Proposition 43.2); step-size amplification in loop unrolling (Definition 43.6)
Scaling	Scaling of problem size in recursion (a subproblems of size $1/b$, Definition 43.5); complexity scaling $T(n) = O(g(n))$; bit-width amplification in vectorisation (Definition 43.7)
Rotation	Rotational correction by the Newton Jacobian (Proposition 43.1); pivot rotation in quicksort (Proposition 43.4)
Balance	Complexity lower bounds: the comparison-sorting bound $\Omega(n \log n)$ (Theorem 43.3); precision and resource trade-offs (Theorem 43.4)
Symmetry	Left–right symmetry of divide-and-conquer recursion (merge sort, $a = b = 2$, Proposition 43.3)
Closure	The closed orbit of Floyd's cycle detection (Lemma 43.2); round-by-round diffusion of the hash avalanche (an **empirical observation / research programme**; see the remark at Remark 43.2)
Period	The spiral of iterative convergence (rate L^t); recursion-tree depth $\log_b n$; the PRNG period (maximum LCG period m, Mersenne Twister $2^{19937} - 1$, Definition 43.8)

Exercises

Exercise 1 Prove the **three bounds and trade-offs of algorithmic systems** of this chapter (43.4). Hint: first write down the decision map and the evolution group of this chapter's target, then apply the proof structure of the general criterion given in the 'Periodic Systems' chapter.

Exercise 2 Using the seven-theme comparison table of this chapter, identify the concrete carriers of the themes **translation**, **scaling** and **rotation** in this chapter, and give one example of domain data for each.

Exercise 3 Synthesis: restate the diagnostic procedure of this chapter using the five-step standard interface of the 'Periodic Systems' chapter (Table 24.1), noting for each step the theorem of this chapter on which it depends: 43.4.

Chapter 44

Periodic Structure of Agents

Remark 44.1 (Periodicity decision for the target of this chapter's target (citation)). In deciding whether an agent system is periodic, in determining its period type and in seeking its least positive period, this chapter refers throughout to the 'Periodic Systems' chapter: the periodicity criterion is given in Theorem 24.1, the discrimination of period types in Definition 24.3, Proposition 24.1, and the search for the least positive period in Definition 24.4, Theorem 24.2.

Remark 44.2 (Positioning of this chapter: the convergence point of a four-chapter chain). The periodic structure of agents studies the mathematical characterisation of periodicity, rhythmicity and oscillatory phenomena in artificial intelligence systems. This chapter is the **convergence point** of the chain 'data analysis → activation function → algorithm → agent', and it uses all the tools of the three preceding chapters.

Interface with the three preceding chapters:

(1) **From the data analysis chapter**: the periodic structure of agent training data is extracted by the periodic convolution (Definition 41.7) and the phase-coupling network (Definition 41.12) of the data analysis chapter; the inflection points of the training sequence are detected by refractive-index diagnosis (Definition 41.17); and multitaper estimation (Proposition 41.2) provides a spectral basis for the search over training hyperparameters.

(2) **From the activation function chapter**: the non-linear units of an agent's neural network are realised by the CEAF (Definition 42.4); their differentiability is guaranteed by the Wirtinger derivative (Theorem 42.6); and their approximation capability is guaranteed by the periodic-domain approximation theorem (Corollary 42.1) and by single-hidden-layer universal approximation (Theorem 42.5).

(3) **From the algorithmic systems chapter**: the convergence analysis of agent training dynamics inherits the damped oscillation of gradient descent (Proposition 43.2) and the quadratic convergence of Newton's method (Proposition 43.1); multi-agent synchronisation inherits fixed-point iteration (Theorem 43.1); and the random source of the exploration policy inherits the periodic structure of the PRNG (Definition 43.8).

Guide to the seven themes of this chapter: translation—the parameter translation of gradient descent and the error iteration $e_{t+1}^{(i)} = (1 - \eta\lambda_i)e_t^{(i)}$ (Lemma 44.1); scaling—the scale decay of the learning-rate scheduler (Definition 44.4) and the exponential decay of the exploration intensity $\varepsilon(t)$ (Definition 44.18); rotation—the damped rotation of the momentum method (Proposition 44.1) and the complex-exponential phase of positional encoding (Proposition 44.11); balance—parameter constraints, gradient-propagation constraints and attention-normalisation constraints (Theorem 44.8); symmetry—the identity symmetry of residual connections and the softmax normalisation symmetry of attention; closure—the convergence closure of policy iteration towards the optimal policy π^* (Proposition 44.13) and the iterative closure of the reasoning chain (Theorem 44.4); period—the damped oscillation of gradient descent, the cosine-annealing period T of the learning rate (Definition 44.4), the multi-agent synchronisation period T_{sync} (Proposition 44.14) and the limit-cycle period of the attractor (Theorem 44.3).

Interface with the other frameworks of the book: the phase-synchronisation analysis of this chapter invokes the phase-locking proposition of the 'Periodic Systems' chapter (Proposition 24.8) and the Kuramoto synchronisation threshold of the 'Periodic Structure of Social Systems' chapter (Proposition 37.14); the periodic memory of the LSTM invokes the discrete phase operator of the 'Periodic Functions and Groups with Periodic Orbits' chapter (Definition 20.6); the phase similarity of positional encoding invokes the difference-angle cosine similarity of the 'Scalars and Vectors' chapter (Definition 3.9); and the phase structure of word embeddings invokes the complex-exponential difference-angle operator of the 'Scalars and Vectors' chapter (Definition 3.25).

44.1 Mathematical foundations of agent systems

Definition 44.1 (Basic quantities of an agent system)**.** An *agent system* is described by the following elementary variables:

(i) **Model parameters $\boldsymbol{\theta} \in \mathbb{R}^d$:** the weights and biases of a neural network. The update trajectory of the parameters forms periodic or quasi-periodic motion in parameter space.

(ii) **Loss function $\mathcal{L}(\boldsymbol{\theta})$:** a scalar function measuring the discrepancy between the model's predictions and the true labels. The landscape of the loss function (gradient, curvature, saddle points) determines the periodic structure of the optimisation trajectory.

(iii) **Learning rate $\eta > 0$:** the step size of gradient descent. The magnitude of the learning rate determines the scale of the optimisation—a large learning rate corresponds to rapid but coarse translation, a small learning rate to slow but fine translation.

(iv) **Momentum $\mathsf{m} \in \mathbb{R}^d$:** the accumulated historical gradient in a momentum optimiser. Momentum introduces inertia into parameter space, causing the optimisation trajectory to rotate about saddle points and local minima in the loss landscape.

(v) **Attention weights $\alpha_{ij} \in [0, 1]$:** the attention weight of the i-th query on the j-th key in a Transformer. The distribution of the attention weights varies periodically with position in the input sequence.

(vi) **Hidden state $\mathsf{h}_t \in \mathbb{R}^h$:** the hidden-state vector of a recurrent neural network at time t. The evolution of the hidden state traces an orbit in state space—a periodic orbit corresponds to a cyclic pattern, a chaotic orbit to a complex sequence.

(vii) **Word embedding $\mathbf{e}_w \in \mathbb{R}^{d_e}$:** the dense vector representation of a word w. Word embeddings form a vector field with rotational and phase structure in embedding space (Definition 44.9).

(viii) **Reasoning state $\mathbf{s}_t$:** the intermediate state of a reasoning chain at step t. The sequence of reasoning states constitutes the reasoning-chain orbit (Definition 44.13).

The defining feature of an agent system is the **optimisation dynamics in a high-dimensional parameter space**—the parameter trajectory moves through the loss landscape, the gradient drives the parameters to translate in the direction of decreasing loss, momentum drives the parameters to rotate in the loss landscape, and learning-rate decay controls the scale of the translation and of the rotation.

Property 44.1 (Immediate constraints on the elementary quantities)**.** The following constraints follow directly from Definition 44.1:

(i) the learning rate $\eta > 0$ is a scalar parameter with no upper bound, but its effectiveness is constrained by the curvature of the loss function (Lemma 44.1);

(ii) the momentum coefficient $\beta \in [0, 1)$ is a decay factor, and $\beta \to 1$ corresponds to inertia without decay;

(iii) the attention weights satisfy the normalisation $\sum_j \alpha_{ij} = 1$, a constraint that comes from the mathematical definition of softmax (Theorem 44.8(iii));

(iv) the hidden-state dimension h is given by the network design, and the periodic structure of its evolution trajectory is determined by the gating mechanism (Definition 44.17);

(v) the embedding dimension d_e is a hyperparameter, and the rotational structure of word embeddings is given by the complex-number analogy (Proposition 44.6);

(vi) the convergence of the reasoning-state sequence is guaranteed by the contractivity of the reasoning map (Theorem 44.4).

Definition 44.2 (Manifold structure of the agent state space)**.** The agent state space $\mathcal{S}$ is the Cartesian product of the following three manifolds:

$$\mathcal{S} = \mathcal{P} \times \mathcal{H} \times \mathcal{E},$$

where

(i) $\mathcal{P} \subseteq \mathbb{R}^d$ is the **parameter manifold**, whose local geometry is measured by the Hessian of the loss function;

(ii) $\mathcal{H} \subseteq \mathbb{R}^h$ is the **hidden-state manifold**, whose evolution trajectory is the state orbit of a recurrent neural network;

(iii) $\mathcal{E} \subseteq \mathbb{R}^{d_e}$ is the **embedding manifold**, whose structure is given by the unit-sphere decomposition and the rotational symmetry of word embeddings.

The metric on the state space is given by the product metric of the three, and upon it act the parameter translation group, the hidden-state rotation group and the embedding scaling group.

44.2 Periodic dynamics of neural-network training

Definition 44.3 (Oscillatory convergence of gradient descent). In a neighbourhood of a minimum of the loss function $\mathcal{L}(\boldsymbol{\theta})$, the parameter updates of gradient descent exhibit **damped oscillation**. Let the linearised dynamics of the parameters near the minimum $\boldsymbol{\theta}^*$ be

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \nabla \mathcal{L}(\boldsymbol{\theta}_t) \approx \boldsymbol{\theta}_t - \eta \mathsf{H}(\boldsymbol{\theta}_t - \boldsymbol{\theta}^*),$$

where $\mathsf{H} = \nabla^2 \mathcal{L}(\boldsymbol{\theta}^*)$ is the Hessian matrix. The evolution of the parameter error $\mathsf{e}_t = \boldsymbol{\theta}_t - \boldsymbol{\theta}^*$ is $\mathsf{e}_{t+1} = (\mathsf{I} - \eta\mathsf{H})\mathsf{e}_t$. Let the eigenvalues of H be $\lambda_1, \dots, \lambda_d > 0$. The evolution of the error along the i-th eigendirection is $e_{t+1}^{(i)} = (1 - \eta\lambda_i)e_t^{(i)}$.

The convergence behaviour of gradient descent: when $0 < \eta\lambda_i < 2$ we have $|1 - \eta\lambda_i| < 1$—the error decays monotonically (overdamped). When $1 < \eta\lambda_i < 2$ we have $1 - \eta\lambda_i < 0$—the sign of the error alternates at every step (underdamped oscillation). The rate of convergence is determined by $|1 - \eta\lambda_i|$—convergence is fastest when $\eta\lambda_i = 1$ (critical damping), and slowest as $\eta\lambda_i \to 0$ or $\eta\lambda_i \to 2$. This is the degenerate form of a damped oscillator in discrete time.

Lemma 44.1 (Eigen-direction convergence of gradient descent). For the quadratic loss $\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\theta}^*)^\mathsf{T}\mathsf{H}(\boldsymbol{\theta} - \boldsymbol{\theta}^*)$ (with H positive definite and eigenvalues $0 < \lambda_1 \leq \cdots \leq \lambda_d$), the error of gradient descent $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\mathsf{H}(\boldsymbol{\theta}_t - \boldsymbol{\theta}^*)$ along the i-th eigendirection is

$$e_t^{(i)} = (1 - \eta\lambda_i)^t \, e_0^{(i)},$$

and the convergence ($e_t^{(i)} \to 0$) holds if and only if $0 < \eta\lambda_i < 2$; when $\eta\lambda_i = 1$ the error vanishes in one step (critical damping), when $1 < \eta\lambda_i < 2$ the sign alternates (underdamped oscillation), and when $\eta\lambda_i \geq 2$ the iteration diverges.

Proof. In the eigenbasis H is diagonalised and the eigendirections decouple: $e_{t+1}^{(i)} = (1 - \eta\lambda_i)e_t^{(i)}$, and iteration gives $e_t^{(i)} = (1 - \eta\lambda_i)^t e_0^{(i)}$. The convergence condition is $|1 - \eta\lambda_i| < 1 \Leftrightarrow 0 < \eta\lambda_i < 2$; when $1 - \eta\lambda_i = 0$ we have $e_1^{(i)} = 0$; when $1 - \eta\lambda_i < 0$ the sign changes at every term. Global convergence requires $0 < \eta < 2/\lambda_d$—the step size is capped by the largest eigenvalue, whilst the rate of convergence is determined by the smallest eigenvalue ($\eta = 2/(\lambda_1 + \lambda_d)$ minimises the spectral radius). □

Proposition 44.1 (Rotational acceleration of the momentum method). Gradient descent with momentum (SGD with Momentum) introduces a **rotational component** into parameter space:

$$\mathsf{m}_{t+1} = \beta\mathsf{m}_t + (1 - \beta)\nabla\mathcal{L}(\boldsymbol{\theta}_t), \quad \boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\mathsf{m}_{t+1},$$

where $\beta \in [0,1)$ is the momentum coefficient. The momentum m_t is an exponentially weighted moving average of past gradients—it accumulates directional information about the gradient. In the long narrow ravines of the loss landscape, momentum makes the parameter trajectory rotate about the centre of the ravine, accelerating convergence.

The momentum method is the realisation of the rotation operation in optimisation: the momentum m_t is analogous to velocity in a mechanical system—it preserves the direction of the parameter's motion, making the parameters 'rotate' in the loss landscape rather than merely 'slide down'. The momentum coefficient β controls the inertia of the rotation—the larger β is, the more persistent the influence of past gradients and the larger the radius of rotation. Nesterov accelerated gradient (NAG) is a refinement of the momentum method—it first moves along the momentum direction and then computes the gradient to apply a correction, rather like a prediction with phase advance. This structure shares one and the same second-order dynamical template as the damped harmonic oscillator of the 'Periodic Structure of Mechanical Systems' chapter (Proposition 27.8).

Proof. The parameter update of the momentum method may be written as a second-order difference equation: $\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t = -\eta \nabla \mathcal{L}(\boldsymbol{\theta}_t) + \beta(\boldsymbol{\theta}_t - \boldsymbol{\theta}_{t-1})$. Linearising near the minimum: $\mathsf{e}_{t+1} - \mathsf{e}_t = -\eta \mathsf{H}\mathsf{e}_t + \beta(\mathsf{e}_t - \mathsf{e}_{t-1})$. The characteristic equation is $\lambda^2 - (1+\beta-\eta\lambda_i)\lambda + \beta = 0$. The characteristic roots are

$$\lambda_{1,2} = \frac{1+\beta-\eta\lambda_i \pm \sqrt{(1+\beta-\eta\lambda_i)^2 - 4\beta}}{2}.$$

When the discriminant is negative (that is, when $(1+\beta-\eta\lambda_i)^2 < 4\beta$), the characteristic roots are complex conjugates—the parameter trajectory exhibits damped rotation. The angular frequency of the rotation is $\omega_i = \arctan(\sqrt{4\beta - (1+\beta-\eta\lambda_i)^2}/(1+\beta-\eta\lambda_i))$, and the damping factor is $\sqrt{\beta}$. □

Definition 44.4 (Periodic decay of the learning-rate scheduler). A **learning-rate scheduler** reduces the learning rate according to a predefined period or condition. Common scheduling policies include:

(i) **Step decay**: every T epochs the learning rate is multiplied by $\gamma < 1$, so that $\eta_t = \eta_0 \cdot \gamma^{\lfloor t/T \rfloor}$. This is discrete scale scaling—every T steps the scaling factor γ acts once.

(ii) **Cosine annealing**: $\eta_t = \eta_{\text{min}} + \frac{1}{2}(\eta_{\text{max}} - \eta_{\text{min}})(1 + \cos(\pi t/T))$. The learning rate varies periodically as a cosine function—it rotates smoothly between η_{max} and η_{min}, with period $2T$.

(iii) **Warmup cosine annealing**: over the first T_{warmup} steps the learning rate increases linearly (accelerated translation) and thereafter decays as a cosine. The warmup phase avoids instability of the parameters in the early stage of training.

The learning-rate scheduler controls the scale of the parameter updates—a large learning rate corresponds to rapid translation (exploration), a small learning rate to fine translation (exploitation). The periodic oscillation of cosine annealing is characterised by the period parameter T, which agrees with the intercalation-window selection of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter (Theorem 21.10) in the idea of 'modulating the sampling scale by a periodic function'.

Proposition 44.2 (Periodic structure of the cosine-annealing learning rate)**.** The cosine-annealing learning rate

$$\eta_t = \eta_{\mathsf{min}} + \frac{1}{2}(\eta_{\mathsf{max}} - \eta_{\mathsf{min}})\Big(1 + \cos\frac{\pi t}{T}\Big)$$

is a periodic function of period $2T$: $\eta_{t+2T} = \eta_t$. Within one period, η_t decreases monotonically from η_{max} to η_{min} ($t \in [0, T]$) and then increases monotonically back to η_{max} ($t \in [T, 2T]$).

Proof. Periodicity follows at once from $\cos(\pi(t + 2T)/T) = \cos(\pi t/T + 2\pi) = \cos(\pi t/T)$. Monotonicity follows from the monotonicity of the cosine on $[0, \pi]$ and on $[\pi, 2\pi]$. ☐

44.3 Complex exponential activation functions and kernel-smoothed derivative-integral descent

Definition 44.5 (Form of the complex exponential activation function in agent systems)**.** The mathematical object of the complex exponential activation function (CEAF) was given in Chapter 42, Definition 42.4:

$$\mathcal{E}(x) = \sum_{k=1}^{N} A_k e^{i(\omega_k x + \phi_k)}.$$

This section gives its applied properties in the context of agents and neural networks, without redefining the function. In agent systems, replacing the standard activation functions (ReLU, GELU, Swish) by the CEAF has the following advantages:

(i) **Intrinsic periodic structure**: the CEAF naturally contains multiple levels of nested periods—the frequencies $\{\omega_k\}$ and amplitudes $\{A_k\}$ of the epicycles at each level are learnable parameters, enabling the network to capture multi-scale periodic patterns in the data adaptively. The initial frequency values may be initialised from the periodogram peaks of the preceding chapter's data analysis (Lemma 41.1).

(ii) **Phase-encoding capability**: the CEAF outputs complex values $\mathcal{E}(x) \in \mathbb{C}$, whose real and imaginary parts encode the in-phase and quadrature components of the signal respectively. The phase information $\arg(\mathcal{E}(x))$ is preserved in the forward pass, enabling the network to learn phase relations in a time series—this is structurally the same as the complex-exponential similarity of the 'Scalars and Vectors' chapter (Proposition 3.12).

(iii) **The derivative is not identically zero**: the derivative of the CEAF, $\mathcal{E}'(x) = i\sum_k \omega_k A_k e^{i(\omega_k x + \phi_k)}$, is not identically zero on the whole real axis (Proposition 42.2(iv)), which avoids the identically zero gradient of ReLU on the negative half-axis. It should be noted that $\mathcal{E}'$ may have isolated zeros, so the statement 'the gradient is nowhere zero' does not hold; the correct statement is 'the gradient is not identically zero'.

(iv) **Multi-scale feature extraction**: the ratios of the angular velocities $\{\omega_k\}$ of the epicycles at each level determine the sensitivity of the activation function to different periodic scales. When $\omega_k \approx 2\pi/T_{\text{data}}$, the k-th level epicycle produces resonant amplification at the data period T_{data}.

Proposition 44.3 (Propagation structure of the CEAF)**.** Under translation the CEAF behaves as a pure phase rotation: for any $a \in \mathbb{R}$,

$$\mathcal{E}(x+a) = \sum_{k=1}^{N} A_k e^{i(\omega_k x + \phi_k + \omega_k a)}.$$

That is, the translation a is equivalent to increasing the phase of the k-th component by $\omega_k a$—this is the componentwise realisation on the CEAF of Proposition 17.5 of the 'Propagation and Evolution' chapter (the consistency of the propagation operator with the phase operator).

Proof. Substituting Definition 42.4 and $e^{i(\omega_k(x+a)+\phi_k)} = e^{i(\omega_k x+\phi_k)}e^{i\omega_k a}$ gives the result. □

Corollary 44.1 (Periodicity condition for the CEAF)**.** If all the ratios $\omega_k/\omega_1 \in \mathbb{Q}$ (where ω_1 is the base frequency), then $\mathcal{E}$ has period $T = 2\pi/\omega_1$ (when the ω_k are integer multiples of the base frequency); otherwise $\mathcal{E}$ is a Bohr almost-periodic function (see Proposition 42.2(v) for details).

Proof. This follows directly from Proposition 42.2(v). □

Remark 44.3 (CEAF in the Transformer (architectural proposal)). Apply the CEAF in the feed-forward network (Feed-Forward Network) of a Transformer model, replacing the standard activation function. Let the output of the l-th layer of the Transformer be $\mathsf{h}^{(l)}$; the transformation of the feed-forward network is

$$\mathsf{z} = \mathsf{W}_2 \cdot \mathcal{E}(\mathsf{W}_1\mathsf{h}^{(l)} + \mathsf{b}_1) + \mathsf{b}_2,$$

where $\mathsf{W}_1, \mathsf{W}_2$ are weight matrices and $\mathsf{b}_1, \mathsf{b}_2$ are biases. The complex-valued output of the CEAF provides two complementary channels—a real-part channel and an imaginary-part channel. The two channels may be concatenated into a real-valued vector of twice the dimension, or a magnitude–phase scheme may be used (the magnitude as the activation strength and the phase as a supplementary positional encoding).

It must be stressed that this architectural proposal is **heuristic** and that its performance advantages require empirical verification; at present there is only numerical evidence from a time-series forecasting task (Example 44.1) and a one-dimensional function-approximation task (Example 42.4), and rigorous theoretical guarantees are lacking.

Definition 44.6 (Application of the CEAF in LSTM)**.** Embed the CEAF in the gating mechanism of an LSTM, replacing the standard activation functions. The input gate i_t, forget gate f_t and output gate o_t of the LSTM use the CEAF in place of the sigmoid, and the candidate cell state $\tilde{\mathsf{c}}_t$ uses the CEAF in place of tanh:

$$i_t = \sigma_{\text{CEAF}}(\mathsf{W}_i \cdot [\mathsf{h}_{t-1}, \mathsf{x}_t] + \mathsf{b}_i), \quad \tilde{\mathsf{c}}_t = \mathcal{E}(\mathsf{W}_c \cdot [\mathsf{h}_{t-1}, \mathsf{x}_t] + \mathsf{b}_c),$$

where $\sigma_{\text{CEAF}}(x) = |\mathcal{E}(x)|/\sum_k A_k$ is the CEAF magnitude normalised to $[0,1]$ (replacing the gating function of the sigmoid). The advantage of CEAF gating lies in its sensitivity to periodicity—when the input sequence contains a periodic pattern, the periodic opening and closing of the CEAF gate resonates with the period of the sequence, enabling the LSTM to capture long-range periodic dependencies more efficiently.

CEAF gating is an enhanced realisation of the rotation operation in the LSTM: the periodic opening and closing of the forget gate corresponds to phase rotation (the alternation of retention and forgetting), and the periodic opening and closing of the input gate corresponds to scale modulation (the periodic variation of the writing strength). The epicycles at each level of the CEAF endow the gate with multi-scale period awareness—gates of different frequencies respond to sequence patterns on different time scales. This agrees with the discrete phase operator $\mathcal{O}_k^{(T)}$ of the 'Periodic Functions and Groups with Periodic Orbits' chapter (Definition 20.6) in the core idea that 'periodic opening and closing = phase rotation'.

Definition 44.7 (Application of kernel-smoothed derivative-integral descent in agent systems)**.** **Kernel-smoothed derivative-integral descent** (Kernel-smoothed Derivative-Integral Descent, KDI) is used in agent systems to optimise the parameters $\Theta_{\text{CEAF}} = \{A_k, \omega_k, \phi_k\}_{k=1}^{K}$ of the CEAF together with the other learnable parameters of the model. By fusing derivatives and integrals through kernel functions, the algorithm exploits both first-order derivative (gradient) and higher-order integral (trend) information, escaping saddle points and local minima in a non-convex loss landscape.

For the CEAF parameters Θ_{CEAF}, different kernel functions are used to optimise the terms separately:

(i) **Amplitude** A_k: a Gaussian kernel K_{Gauss} is used for windowed derivative-integral estimation. The amplitude is a smooth positive quantity, and the smoothing property of the Gaussian kernel suppresses high-frequency noise in the amplitude update. The windowed derivative $D^{(1)}_{K_{\text{Gauss}}}(A_k, \Delta)$ gives the smoothed first-order gradient of the amplitude, and the windowed integral $I^{(1)}_{K_{\text{Gauss}}}(A_k, \Delta)$ gives the cumulative change in the amplitude. The curvature-correction term of the derivative-integral fusion function Φ detects the inflection points of the amplitude variation (the data-analysis procedure of Definition 41.9 provides an initial estimate for this).

(ii) **Angular velocity** ω_k: a Mexican-hat kernel K_{MexHat} is used for second-derivative estimation. The Mexican-hat kernel is the second derivative of the Gaussian kernel and is naturally band-pass—it suppresses low frequencies (slow frequency drift) and high frequencies (random fluctuations) whilst enhancing intermediate frequencies (genuine frequency changes). The optimisation of the angular velocities has the greatest influence on network performance—ω_k determines the periodic scale of the activation function.

(iii) **Phase** ϕ_k: a Hilbert kernel K_{Hilbert} is used for quadrature-phase estimation. The Hilbert kernel produces a phase rotation of $\pm 90^\circ$ and can detect precisely the jump of the phase as it crosses the boundary from 0 to 2π. Correct detection of phase jumps prevents ϕ_k from being trapped at a spurious equilibrium point near 0 or 2π during optimisation. This is structurally the same as the Hilbert transform of the 'Vector Analysis' chapter (Definition 22.12).

The coordinated use of the three kernel functions equips the CEAF parameter update simultaneously with the smooth adjustment of amplitudes, the curvature sensing of frequencies and the jump detection of phases. The Gaussian kernel, the Mexican-hat kernel and the Hilbert kernel all belong to the window-function system of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter (Definition 21.15), and their sidelobe decay laws are characterised by Theorem 21.8.

Proposition 44.4 (*Descent property of KDI for epicycle parameters*)**.** Let the loss $\mathcal{L}(\Theta)$ be continuously differentiable in a neighbourhood of the parameters $\Theta = \{A_k, \omega_k, \phi_k\}$. If the inner product of the KDI direction $\mathbf{d}(\Theta)$ with $-\nabla\mathcal{L}(\Theta)$ is positive, then there exists $\eta > 0$ such that

$$\mathcal{L}(\Theta - \eta\mathbf{d}(\Theta)) < \mathcal{L}(\Theta).$$

Proof. The directional derivative is $\frac{d}{d\eta}\mathcal{L}(\Theta-\eta\mathbf{d})|_{\eta=0} = -\nabla\mathcal{L}(\Theta)\cdot\mathbf{d}+o(1)$. If $-\nabla\mathcal{L}\cdot\mathbf{d} > 0$, then the value of the function decreases strictly for sufficiently small η. ☐

Proposition 44.5 (*Qualitative advantages of kernel-smoothed derivative-integral descent for optimising the CEAF*)**.** In the parameter optimisation of the CEAF, kernel-smoothed derivative-integral descent has a better non-convex search capability than standard gradient descent. Let the landscape of the loss function $\mathcal{L}(\Theta_{\text{CEAF}})$ in parameter space contain several local minima and saddle points. Standard gradient descent $\Theta^{(t+1)} = \Theta^{(t)} - \eta\nabla\mathcal{L}$ uses only first-order derivative information and readily stalls at saddle points (where the gradient is close to zero) or becomes trapped in a local minimum.

Kernel-smoothed derivative-integral descent overcomes these difficulties by means of multi-scale windowed derivative-integral information:

(i) **Escaping saddle points**: at a saddle point the first-order gradient satisfies $\nabla\mathcal{L} \approx \mathbf{0}$, but the second derivative (curvature) has different signs in different directions at the saddle point. The curvature-correction term detects the change of curvature along a direction—the step size is increased in directions of negative curvature, driving the parameters away from the saddle point.

(ii) **Crossing local minima**: the windowed integral associated with a large window radius Δ contains landscape information from distant regions. When the radius of the 'basin' of a local minimum is smaller than Δ, the large-window integral can 'see' lower regions beyond the basin and guide the parameters across the local minimum.

(iii) **Accelerating across flat regions**: in regions where the loss landscape is flat (the gradient is close to zero but the point is not a saddle point), the cumulative-change correction term detects the long-term trend of the loss function within the window—even if the current gradient is very small, so long as the windowed integral keeps decreasing, the update direction still points in the direction of decreasing loss.

It must be stressed that this proposition is a **qualitative** statement and merely identifies the heuristic advantages of KDI in non-convex search. A rigorous proof of convergence—in particular the global rate of convergence in the non-convex case—remains an **open problem** (Remark 44.12).

Definition 44.8 (Training pipeline of CEAF–KDI)**.** The training pipeline that combines the CEAF with kernel-smoothed derivative-integral descent consists of the following steps:

(i) **Model initialisation**: the network weights are set by a standard initialisation method (Xavier or He initialisation). The CEAF parameters take amplitudes decreasing geometrically in radius, $A_k = A_1 q^{k-1}$, angular velocities from the harmonic sequence $\omega_k = \omega_1 k$, and phases initialised at random, $\phi_k \sim \mathcal{U}(0, 2\pi)$. If the preceding chapter's data analysis provides a frequency estimate (Proposition 41.6), it is used to initialise ω_k.

(ii) **Forward pass**: the input data pass through the network layers containing the CEAF, and the output and the loss $\mathcal{L}$ are computed.

(iii) **Backward pass**: the gradients of the loss with respect to the network weights and the CEAF parameters, $\nabla_{\mathsf{W}}\mathcal{L}$ and $\nabla_{\Theta_{\mathsf{CEAF}}}\mathcal{L}$, are computed (using the Wirtinger derivative, Theorem 42.6).

(iv) **Computation of kernel-smoothed directions**: for the CEAF parameters, the Gaussian kernel, the Mexican-hat kernel and the Hilbert kernel are used to compute the update directions for the amplitudes, the frequencies and the phases respectively. For the network weights, standard gradient or momentum updates are used.

(v) **Parameter update**: $\Theta_{\mathsf{CEAF}}^{(t+1)} = \Theta_{\mathsf{CEAF}}^{(t)} - \eta_{\mathsf{CEAF}} \cdot \mathsf{d}(\Theta_{\mathsf{CEAF}}^{(t)})$, $\mathsf{W}^{(t+1)} = \mathsf{W}^{(t)} - \eta_{\mathsf{W}} \cdot \nabla_{\mathsf{W}}\mathcal{L}$.

(vi) **Learning-rate scheduling**: cosine annealing (Definition 44.4) is used to adjust η_{CEAF} and η_{W} simultaneously—a relatively large learning rate is used early in training (rapid exploration) and a relatively small one later (fine convergence).

The joint use of the CEAF and kernel-smoothed derivative-integral descent endows the neural network with an intrinsic multi-scale period-awareness capability and a non-local optimisation capability—capabilities that traditional activation functions and optimisers do not possess.

Example 44.1 (Application of CEAF–KDI to electricity-load time-series forecasting)**.** CEAF–KDI is used to forecast electricity-load data (hourly electricity consumption, containing daily, weekly and annual periods). The model structure is LSTM+CEAF, with series order $K = 3$ for the CEAF—the three epicycles correspond to the approximate frequencies of 24 hours (daily period), 168 hours (weekly period) and 8760 hours (annual period) respectively.

After 100 training epochs, the three angular velocities $\omega_1, \omega_2, \omega_3$ of the CEAF converge to $2\pi/24 \approx 0.262$, $2\pi/168 \approx 0.037$ and $2\pi/8760 \approx 0.0007$ (radians per hour) respectively—matching the true periods of the data exactly. The amplitude ratio $A_1 : A_2 : A_3 \approx 1 : 0.6 : 0.3$ reflects the relative importance of the daily, weekly *and annual periods in the electricity load. Compared with the Adam optimiser, kernel-smoothed derivative-integral descent reduces the validation loss by about 15% and increases the speed of convergence by about 30%.*

Note: *the numerical values above are* **empirical observations** *whose statistical significance requires verification by repeated experiments; they serve here only as a feasibility demonstration of CEAF–KDI on real data and are not offered as a theoretical guarantee.*

44.4 Periodic and rotational structure of word embeddings

Definition 44.9 (Word embedding space)**.** Let the vocabulary be $\mathcal{V}$ and the embedding dimension be d_e. The **word embedding map** is

$$\varphi_{\text{emb}} : \mathcal{V} \to \mathbb{R}^{d_e}, \qquad w \mapsto \mathbf{e}_w.$$

Its normalised version is

$$\widehat{\mathbf{e}}_w = \frac{\mathbf{e}_w}{\|\mathbf{e}_w\|} \in S^{d_e-1}.$$

The point of the unit sphere S^{d_e-1} given by $\widehat{\mathbf{e}}_w$ is called the **embedding direction** of the word w. The set of embedding directions constitutes the **word embedding manifold**

$$\mathcal{E} = \{\widehat{\mathbf{e}}_w \mid w \in \mathcal{V}\} \subseteq S^{d_e-1}.$$

Lemma 44.2 (Phase representation of embedding directions)**.** *Let the embedding dimension be $d_e = 2m$ and divide $\widehat{\mathbf{e}}_w$ into m two-dimensional component pairs*

$$\widehat{\mathbf{e}}_w = (\mathbf{u}_1, \ldots, \mathbf{u}_m), \qquad \mathbf{u}_j \in \mathbb{R}^2.$$

Identifying each two-dimensional component u_j with the complex number $u_j = x_j + iy_j$ gives

$$u_j = r_j e^{i\theta_{w,j}}, \qquad r_j = \sqrt{x_j^2 + y_j^2}, \quad \theta_{w,j} = \arg(u_j).$$

The collection $\{\theta_{w,j}\}_{j=1}^m$ is called the **embedding phase vector** *of the word w and $\{r_j\}$ its* **embedding magnitude vector***.*

Proof. Each two-dimensional component pair (x_j, y_j) corresponds uniquely to the complex number $u_j = x_j + iy_j$. The polar representation $u_j = r_j e^{i\theta_{w,j}}$ then gives the result. The identity $\sum_j r_j^2 = \|\widehat{\mathbf{e}}_w\|^2 = 1$ guarantees that the magnitude vector satisfies the normalisation. □

Proposition 44.6 (Rotational analogy relation of word embeddings)**.** *The linear analogy relation of classical word embeddings (such as Word2Vec and GloVe),*

$$\mathbf{e}_{\text{king}} - \mathbf{e}_{\text{man}} + \mathbf{e}_{\text{woman}} \approx \mathbf{e}_{\text{queen}}$$

may be rewritten in **phase-rotation form***: let the phase vector of $\mathbf{e}_w$ be θ_w; then the analogy relation is equivalent to the existence of a common phase offset $\Delta\theta_{\text{gender}}$ such that*

$$\theta_{\text{king}} - \theta_{\text{man}} \approx \theta_{\text{queen}} - \theta_{\text{woman}} \approx \Delta\theta_{\text{gender}}.$$

That is, **the analogy relation manifests in embedding space as the repeated action of one and the same phase offset***—this is a rotation operation.*

Proof. In the complexified representation (Lemma 44.2), the subtraction relation for each two-dimensional component pair $u_j = r_j e^{i\theta_j}$ is

$$u_j^{(\text{king})} \oslash u_j^{(\text{man})} = e^{i(\theta_{\text{king},j} - \theta_{\text{man},j})} = \frac{r_j^{(\text{king})}}{r_j^{(\text{man})}} e^{i(\theta_{\text{king},j} - \theta_{\text{man},j})}.$$

When the magnitudes are approximately equal ($r_j^{(\text{king})} \approx r_j^{(\text{man})}$), the analogy relation degenerates to a pure phase-rotation relation. In classical embeddings the gender analogy shares one and the same phase offset across several semantic dimensions, hence (for all j) $\theta_{\text{king},j} - \theta_{\text{man},j} \approx \theta_{\text{queen},j} - \theta_{\text{woman},j}$. ☐

Theorem 44.1 (Cosine-similarity structure of the difference angle between embedding directions)**.** *Let the embedding directions of two words be* $\widehat{\mathbf{e}}_u$ *and* $\widehat{\mathbf{e}}_v$*, with corresponding component phase vectors* θ_u *and* θ_v*. Then their cosine similarity decomposes as*

$$\langle \widehat{\mathbf{e}}_u, \widehat{\mathbf{e}}_v \rangle = \sum_{j=1}^{m} r_{u,j} r_{v,j} \cos(\theta_{u,j} - \theta_{v,j}).$$

That is: **the similarity of word embeddings is the weighted sum, with weights given by the products of the magnitudes, of the difference-angle cosine similarities of the individual two-dimensional component pairs**—this is the superposition over multi-component embeddings of the basic difference-angle cosine similarity of the 'Scalars and Vectors' chapter (Definition 3.9).

Proof. By Lemma 44.2, $\widehat{\mathbf{e}}_u = (\mathbf{u}_1, \ldots, \mathbf{u}_m)$ and $\widehat{\mathbf{e}}_v = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$, whose inner product is

$$\langle \widehat{\mathbf{e}}_u, \widehat{\mathbf{e}}_v \rangle = \sum_{j=1}^{m} \mathbf{u}_j \cdot \mathbf{v}_j.$$

Writing $\mathbf{u}_j \cdot \mathbf{v}_j$ in terms of complex numbers, $\mathbf{u}_j = r_{u,j} e^{i\theta_{u,j}}$ and $\mathbf{v}_j = r_{v,j} e^{i\theta_{v,j}}$, gives

$$\mathbf{u}_j \cdot \mathbf{v}_j = \mathrm{Re}(\mathbf{u}_j \overline{\mathbf{v}_j}) = r_{u,j} r_{v,j} \cos(\theta_{u,j} - \theta_{v,j}).$$

Summing yields the result. ☐

Corollary 44.2 (Semantic similarity is a weighted sum of phase similarities)**.** *The semantic similarity (cosine similarity) of word embeddings is determined by* **the degree of phase alignment of the individual component pairs***, weighted by the products of the magnitudes of the components. When the phases of two words are close for every component pair* ($|\theta_{u,j} - \theta_{v,j}| \approx 0$)*, the similarity is close to* 1*; when the phase difference of certain component pairs is close to* π*, those component pairs contribute negatively to the similarity.*

Proof. By Theorem 44.1, each term $r_{u,j} r_{v,j} \cos(\theta_{u,j} - \theta_{v,j})$, when $|\theta_{u,j} - \theta_{v,j}| \approx 0$, is close to $r_{u,j} r_{v,j}$ (a positive contribution), and, when $|\theta_{u,j} - \theta_{v,j}| \approx \pi$, is close to $-r_{u,j} r_{v,j}$ (a negative contribution). ☐

Example 44.2 (Phase interpretation of the gender analogy)**.** Take the simplified embedding $d_e = 2$ ($m = 1$), so that each word has only one phase θ_w. Let

$$\theta_{\text{man}} = 0, \quad \theta_{\text{woman}} = \pi/2, \quad \theta_{\text{king}} = \pi/6, \quad \theta_{\text{queen}} = \pi/6 + \pi/2 = 2\pi/3.$$

Then $\theta_{\text{king}} - \theta_{\text{man}} = \pi/6$ and $\theta_{\text{queen}} - \theta_{\text{woman}} = 2\pi/3 - \pi/2 = \pi/6$—the phase offset of the gender analogy is invariably $\pi/6$. This verifies Proposition 44.6: **the analogy relation manifests in a two-dimensional embedding as the repeated action of a common phase offset.**

For general d_e, the phase offset $\Delta\theta_{\text{gender}}$ of the gender analogy may be regarded as an **axis of rotation** in embedding space—all gender-analogy word pairs differ along that axis by one and the same rotation.

Remark 44.4 (Translation and rotation of embeddings: two structures). Word embeddings contain two elementary structures:

(i) **Translation structure** (the real-part version of Proposition 44.6): when the embedding dimension is high and the magnitudes agree, the analogy relation manifests as the translation $\mathbf{e}_{\text{king}} - \mathbf{e}_{\text{man}} \approx \mathbf{e}_{\text{queen}} - \mathbf{e}_{\text{woman}}$—this is the realisation in embedding space of the translation operation of the 'Scalars and Vectors' chapter.

(ii) **Rotation structure** (the phase version of Proposition 44.6): in a complexified embedding the analogy relation manifests as the rotation $e^{i\Delta\theta}$—this is the realisation in embedding space of the rotation operation of the 'Scalars and Vectors' chapter.

The two structures are dual descriptions of one and the same analogy relation in different representations: translation is the linear approximation of phase modulo 2π, and rotation is the logarithmic-domain representation of translation. This agrees with the idea in the 'Number Systems' chapter that the logarithm converts scaling into translation (Theorem 2.17).

Remark 44.5 (Periodic generation of word embeddings (conjecture / empirical hypothesis)). **Conjecture / empirical hypothesis**: suppose that word embeddings contain periodic semantic dimensions (such as time, season and the concept of periodicity). Then the embedding directions of these dimensions are distributed over a **discrete subgroup** of the unit circle: there exists $T \in \mathbb{N}^+$ such that

$$\theta_{w,j} \in \frac{2\pi}{T}\mathbb{Z}, \qquad \text{for all periodic words } w.$$

In this case the embedding directions of periodic words constitute a geometric realisation of the cyclic group C_T—structurally the same as the discrete phase operator $\mathcal{O}_k^{(T)} = e^{i2\pi k/T}$ of the 'Periodic Functions and Groups with Periodic Orbits' chapter (Definition 20.6). This assertion is a **conjecture / empirical hypothesis** and an **empirical observation / research programme**, not a proved proposition: no objective function for training is given in this chapter, and 'convergence' is left undefined; the exact membership $\theta_{w,j} \in \frac{2\pi}{T}\mathbb{Z}$ is only an idealised description of the trained result, whereas an actual embedding lies at best near noisy perturbations of these discrete phases. Moreover, what corresponds one-to-one with the words is the **quotient group** $\frac{2\pi}{T}\mathbb{Z}/2\pi\mathbb{Z} \cong C_T$, whereas $\frac{2\pi}{T}\mathbb{Z}$ itself is an infinite cyclic group—the two are not the same object, and every occurrence of C_T below refers to that quotient group. After discretising time with period T, the time points $k = 0, 1, \ldots, T-1$ correspond to the embedding phases $\theta_k = 2\pi k/T$, which is precisely the standard embedding of C_T in the unit circle.

Example 44.3 (Periodic embedding structure of time words)**.** Consider the embeddings of the time words 'January', 'February', ..., 'December'. If the training corpus contains a semantic structure with annual period, the embedding directions of these 12 words should constitute a geometric realisation of C_{12}: $\theta_m = 2\pi(m-1)/12$ $(m = 1, \dots, 12)$. The embedding directions of adjacent months differ by 30°, consistent with the phase distribution of the twelve Earthly Branches in the 'Periodic Structure of Calendar Systems' chapter (see the phase-angle correspondence in Table 28.1 of that chapter).

Remark 44.6 (Interface between word embeddings and phase operators)**.** The phase structure of word embeddings provides an interface for applying the phase operators of the 'Periodic Functions and Groups with Periodic Orbits' chapter in natural language processing:

(i) periodic semantic dimensions correspond to the discrete phase operator $\mathcal{O}_k^{(T)} = e^{i2\pi k/T}$;

(ii) analogy relations correspond to phase offsets (the rotation operation);

(iii) semantic similarity corresponds to difference-angle cosine similarity (Definition 3.9);

(iv) semantic composition corresponds to phase composition (the sum-angle operator; see Definition 3.25 of the 'Scalars and Vectors' chapter).

44.5 Attractor dynamics and recurrent neural networks

Definition 44.10 (Attractors and basins of attraction)**.** Let (X, f) be a discrete dynamical system. A subset $\mathcal{A} \subseteq X$ is called an **attractor** of f if:

(i) $\mathcal{A}$ is a non-empty compact set;

(ii) $\mathcal{A}$ is f-invariant: $f(\mathcal{A}) \subseteq \mathcal{A}$;

(iii) $\mathcal{A}$ attracts in some neighbourhood of X, say $U \supseteq \mathcal{A}$: there exists an open set $V \supseteq \mathcal{A}$ such that for every $x \in V$ the orbit $\{f^n(x)\}$ has its ω-limit set contained in $\mathcal{A}$.

The largest such V is called the **basin of attraction** of $\mathcal{A}$.

Definition 44.11 (The three basic types of attractor)**.** Attractors fall into three classes according to their dynamical character:

(i) **Fixed-point attractor**: $\mathcal{A} = \{x^*\}$ is a single point, corresponding to a stable equilibrium state of the system;

(ii) **Limit-cycle attractor**: $\mathcal{A}$ is a closed orbit, corresponding to periodic oscillation of the system (a periodic attractor);

(iii) **Strange attractor**: $\mathcal{A}$ is a fractal set and orbits move chaotically on $\mathcal{A}$.

Definition 44.12 (Energy function of the Hopfield network)**.** The Hopfield network is a fully connected recurrent neural network. Let the network have N neurons, with state $\mathsf{s} \in \{-1,+1\}^N$ and a symmetric weight matrix W satisfying $W_{ii} = 0$. The **energy function** is defined as

$$E(\mathsf{s}) = -\frac{1}{2}\sum_{i,j} W_{ij}s_is_j - \sum_i b_is_i,$$

where b is the bias vector.

Theorem 44.2 (Convergence of the Hopfield network)**.** *Let the Hopfield network update its state by the asynchronous update rule: at each step a neuron i is chosen at random and $s_i \mapsto \mathrm{sgn}(\sum_j W_{ij}s_j + b_i)$. Then:*

- (i) *the energy function $E(\mathsf{s})$ is monotonically non-increasing at every update;*
- (ii) *the network necessarily converges in finitely many steps to a local minimum of the energy function;*
- (iii) *each local minimum corresponds to a fixed-point attractor of the network, whose basin of attraction is determined by the initial state.*

Proof. (i) Suppose that at step k the neuron i is updated and the network state changes from s to s', where $s_i' = \mathrm{sgn}(h_i)$, $h_i = \sum_j W_{ij}s_j + b_i$, and the other components are unchanged. The change in energy is

$$\Delta E = E(\mathsf{s}') - E(\mathsf{s}) = -(s_i' - s_i)\left(\sum_j W_{ij}s_j + b_i\right) = -(s_i' - s_i)h_i.$$

If $s_i' = s_i$ then $\Delta E = 0$; if $s_i' = +1$ (in which case $s_i = -1$ and $h_i > 0$) then $\Delta E = -2h_i < 0$; if $s_i' = -1$ (in which case $s_i = +1$ and $h_i < 0$) then $\Delta E = 2h_i < 0$. Hence $\Delta E \leq 0$.

(ii) The state space $\{-1,+1\}^N$ is finite (2^N states) and the energy function is bounded below; since every update either decreases the energy strictly or leaves it unchanged, the network must reach a state in which the energy no longer decreases within finitely many steps—that is, a local minimum.

(iii) At a local minimum we have $s_i = \mathrm{sgn}(h_i)$ for all i, which is the fixed-point condition $f(\mathsf{s}) = \mathsf{s}$. Its basin of attraction consists of all initial states that converge to that point (guaranteed by the convergence in (ii)). □

Corollary 44.3 (Storage capacity of the Hopfield network)**.** *The number of patterns that a Hopfield network can store stably is $M \approx 0.138N$ (where N is the number of neurons). When the number of stored patterns exceeds this capacity, interference between patterns causes attractors to merge or spurious attractors to appear.*

Proof. By methods of statistical physics, when M random patterns are stored by the Hebb rule the signal-to-noise ratio at pattern i is $O(\sqrt{N/M})$. When $M > 0.138N$, the fall in the signal-to-noise ratio means that the patterns are no longer minima of the energy, or that their basins of attraction are compressed to a negligible size. For a detailed derivation see Hertz–Krogh–Palmer, Introduction to the Theory of Neural Computation, Addison-Wesley, 1991, Ch. 2. □

Theorem 44.3 (Limit-cycle attractors of recurrent neural networks). Let the state update of a recurrent neural network be

$$\mathsf{h}_{t+1} = \sigma(\mathsf{W}\mathsf{h}_t + \mathsf{U}\mathsf{x}_t + \mathsf{b}),$$

where σ is a pointwise activation function and W is the recurrent weight. If the spectral radius of W is slightly greater than 1 and the activation function is a bounded non-linearity, then the network possesses a **limit-cycle attractor**—that is, a periodic attractor.

Proof. By the bifurcation theory of non-linear dynamical systems: when the spectral radius of the recurrent weight increases from less than 1 to greater than 1, the system undergoes a Hopf bifurcation—the fixed point loses stability and a limit cycle is born. The boundedness of the activation function guarantees that the system state remains bounded, so the limit cycle is an attractor. The period is determined jointly by the frequencies associated with the imaginary parts of the eigenvalues of W and by the strength of the non-linearity. ☐

Proposition 44.7 (Period condition for a limit-cycle attractor). Suppose that a recurrent neural network enters a limit-cycle attractor of period T_{lc}. Then the relation between T_{lc} and the network parameters is

$$T_{\mathrm{lc}} = \frac{2\pi}{|\operatorname{Im}(\lambda_W)|},$$

where λ_W is the critical eigenvalue pair of W at the Hopf bifurcation point.

Proof. Near the Hopf bifurcation the linearised dynamics of the system are described by the eigenvalues $\lambda_{1,2} = \alpha \pm i\omega$, where α is the real part (the bifurcation point being where it crosses zero) and ω is the imaginary part. The angular frequency of the limit cycle is ω, corresponding to the period $T_{\mathrm{lc}} = 2\pi/\omega = 2\pi/|\operatorname{Im}(\lambda_W)|$. ☐

Example 44.4 (Continuous attractors and memory retention). In an RNN, if all the eigenvalues of the recurrent weight W are 1 (the critical case), then the system possesses a **continuous attractor**—the attractor is not an isolated point but an entire curve or surface. Continuous attractors can be used to retain working memory: the initial condition slides along the attractor manifold without decaying or diverging—this is the dynamical mechanism by which a neural network realises 'persistent activity'.

Remark 44.7 (Attractors versus conserved quantities). The roles of attractors and of conserved quantities in periodic structure compare as follows:

(i) **Conserved quantity**: it constrains the orbit to remain on a low-dimensional manifold (Theorem 11.1) but does not directly produce periodicity;

(ii) **Attractor**: it attracts the orbit, by dissipation, to a particular low-dimensional subset, and a limit-cycle attractor directly yields periodicity.

In dissipative systems, attractors take the place of conserved quantities in constraining the orbit (see the discussion of the relation between conservation and periodicity in the 'Operations and Periodicity' chapter). This agrees with the position taken in note 24.5 of the 'Periodic Systems' chapter, that 'a conserved quantity helps one to discover periodicity but does not directly produce it'—the algebraic pivot that genuinely produces periodicity is the finite-order action (in the discrete case) or the attractor (in the continuous dissipative case).

Remark 44.8 (Relation between attractors and finite-order actions). In a finite-state RNN, limit-cycle attractors correspond directly to the finite-order actions of the 'Operations and Periodicity' chapter:

(i) when the discrete state space is finite, each limit cycle corresponds to a cyclic group $C_{T_{\mathsf{lc}}}$;

(ii) the period T_{lc} of the limit cycle is precisely the order of the finite-order action;

(iii) the partition into basins of attraction corresponds to the classification of distinct orbits in the 'Operations and Periodicity' chapter.

This correspondence shows that, in the discrete case, the dynamical behaviour of an agent can be accommodated entirely within the finite-order action framework of the 'Operations and Periodicity' chapter.

44.6 Periodic iterative structure of reasoning

Definition 44.13 (Reasoning chains and reasoning maps). Let the reasoning state space of an agent be $\mathcal{R} \subseteq \mathbb{R}^{d_r}$ and let the initial state of a reasoning problem q be $\mathsf{s}_0 = \psi_{\mathsf{init}}(q)$. The **reasoning map** is

$$\mathcal{T} : \mathcal{R} \times \mathcal{Q} \to \mathcal{R}, \qquad (\mathsf{s}, q) \mapsto \mathcal{T}(\mathsf{s}, q),$$

where $\mathcal{Q}$ is the problem space. The **reasoning chain** is the iterative sequence

$$\mathsf{s}_{t+1} = \mathcal{T}(\mathsf{s}_t, q), \qquad t = 0, 1, 2, \ldots$$

After T_{ans} steps the reasoning chain terminates in the answer state $\mathsf{s}_{T_{\mathsf{ans}}} = \psi_{\mathsf{ans}}^{-1}(\text{answer})$.

Theorem 44.4 (Iterative closure condition for reasoning). Let the reasoning map $\mathcal{T}(\cdot, q) : \mathcal{R} \to \mathcal{R}$ be an L-contraction $(0 \le L < 1)$. Then:

(i) the reasoning chain has a unique fixed point $\mathsf{s}^* = \mathcal{T}(\mathsf{s}^*, q)$—that is, a unique answer;

(ii) the reasoning chain converges at a geometric rate: $\|\mathsf{s}_t - \mathsf{s}^*\| \le L^t \|\mathsf{s}_0 - \mathsf{s}^*\|$;

(iii) the number of reasoning steps required to reach accuracy ε is $T_{\mathsf{ans}} = O(\log(1/\varepsilon))$.

Proof. By the Banach fixed-point theorem (Theorem 43.1), a contraction on a complete metric space has a unique fixed point, and the iteration converges at the geometric rate L. Reaching accuracy ε requires $L^{T_{\mathsf{ans}}} \le \varepsilon / \|\mathsf{s}_0 - \mathsf{s}^*\|$, and taking logarithms gives $T_{\mathsf{ans}} \ge \log(1/\varepsilon) / \log(1/L)$. □

Proposition 44.8 (Phase accumulation in multi-step reasoning). Suppose that in the complexified representation the reasoning map becomes the phase evolution

$$\theta_{t+1} = \theta_t + \Delta\theta(\mathsf{s}_t, q),$$

where $\Delta\theta(\mathsf{s}_t, q)$ is the phase increment of the t-th reasoning step. If $\Delta\theta$ tends to a constant $\Delta\theta^*$ during convergence, then the phase of the reasoning chain accumulates in steps of $\Delta\theta^*$, and after T_{ans} steps the total phase is

$$\Phi_{\text{total}} = T_{\text{ans}} \cdot \Delta\theta^*.$$

When $\Phi_{\text{total}} \in 2\pi\mathbb{Z}$, the reasoning chain **closes** in phase space—that is, it completes one full reasoning cycle.

Proof. Iterating the phase evolution equation gives $\theta_t = \theta_0 + \sum_{k=0}^{t-1} \Delta\theta(\mathsf{s}_k, q)$. When $\Delta\theta(\mathsf{s}_k, q) \to \Delta\theta^*$, the accumulated phase is $\Phi_{\text{total}} \approx T_{\text{ans}} \cdot \Delta\theta^*$. The closure condition $\Phi_{\text{total}} \equiv 0 \pmod{2\pi}$ says precisely that the reasoning chain returns to its starting point in phase space. □

Lemma 44.3 (Reasoning-chain depth and information transmission). Let the information-transmission efficiency of the reasoning chain be $\rho \in (0, 1)$ (the fraction retained at each step). After T reasoning steps the fraction of information remaining is ρ^T. For the information content of the final answer to be no less than ε times the initial information content, one requires

$$T \leq \frac{\log \varepsilon}{\log \rho}.$$

Proof. After T steps the information fraction is ρ^T. The requirement $\rho^T \geq \varepsilon$ gives, on taking logarithms, $T \log \rho \geq \log \varepsilon$. Since $\log \rho < 0$, dividing by $\log \rho$ reverses the inequality, giving $T \leq \log \varepsilon / \log \rho$. □

Corollary 44.4 (Optimal depth of a reasoning chain). The optimal depth of a reasoning chain is a trade-off between **information transmission** and **computational cost**:

(i) too small a depth: too much information is lost and the accuracy of the answer is low;

(ii) too large a depth: information is retained to a high degree but the computational cost is high, and the result may deteriorate through the accumulation of numerical error.

The optimal depth is $T^* = O(\log(1/\varepsilon)/\log(1/\rho))$—of the same order as the number of convergence steps of a contractive reasoning map.

Proof. Combining Theorem 44.4 with Lemma 44.3: both give a step-count scale $T = O(\log(1/\varepsilon))$, the former from the geometric convergence of the contraction map and the latter from the exponential decay of information transmission. □

Theorem 44.5 (Phase locking in Chain-of-Thought reasoning). Suppose that Chain-of-Thought (CoT) reasoning comprises T_{CoT} intermediate steps. If the phase update of each intermediate step is $\Delta\theta$ (a constant), then the phase-locking condition for the CoT reasoning chain is

$$T_{\mathsf{CoT}} \cdot \Delta\theta \in 2\pi\mathbb{Z}.$$

In this case the reasoning chain closes in phase space, corresponding to one 'reasoning period'.

Proof. The phase accumulation of the CoT reasoning chain is $\Phi_{\mathsf{total}} = T_{\mathsf{CoT}} \cdot \Delta\theta$ (by Proposition 44.8). The closure condition $\Phi_{\mathsf{total}} \in 2\pi\mathbb{Z}$ then gives the result. ☐

Example 44.5 (Phase structure of a mathematical proof). In mathematical-proof reasoning, each step of reasoning corresponds to a logical implication. If the logical strength (phase increment) of every step is constant, then the phase of the logical chain should close when the proof is complete (that is, the conclusion is derived from the hypotheses and the total phase accumulation is 2π). This is the phase interpretation of 'reasoning closure': **the completion of a proof is equivalent to the closure of the reasoning chain in logical phase space.**

Remark 44.9 (Correspondence between reasoning and fixed-point iteration). Reasoning and the fixed-point iteration of the algorithmic systems chapter (Definition 43.3) correspond exactly in mathematical structure:

(i) the reasoning map $\mathcal{T}$ corresponds to the iteration map Φ;

(ii) the reasoning answer s^* corresponds to the fixed point x^*;

(iii) the condition for convergence of reasoning corresponds to the contractivity condition $L < 1$;

(iv) the depth T_{CoT} of the CoT corresponds to the number of iteration steps T.

Consequently, the convergence analysis of the 'Periodic Structure of Algorithmic Systems' chapter may be applied directly to the convergence analysis of reasoning chains.

44.7 Periodic analysis of knowledge representation

Definition 44.14 (Knowledge graphs and relation embeddings). A **knowledge graph** is a set of triples $\mathcal{G} = \{(h, r, t)\}$, where h is the head entity, t the tail entity and r the relation. A **relation embedding** is a map

$$\varphi_{\mathsf{KG}} : \mathcal{E} \cup \mathcal{R} \to \mathbb{R}^{d_k},$$

where $\mathcal{E}$ is the entity set and $\mathcal{R}$ the relation set. The embedding objective of the classical TransE model is

$$\mathsf{e}_h + \mathsf{e}_r \approx \mathsf{e}_t.$$

Theorem 44.6 (Translation–rotation decomposition of relation embeddings). Suppose that a knowledge-graph embedding satisfies the TransE relation $\mathsf{e}_h + \mathsf{e}_r \approx \mathsf{e}_t$. In the complexified representation this relation is equivalent to

$$\mathsf{e}_t = \mathsf{e}_h \cdot e^{i\Delta\theta_r} \quad \text{or} \quad \mathsf{e}_t = \mathsf{e}_h + \mathsf{u}_r,$$

where $e^{i\Delta\theta_r}$ is the phase rotation of the relation and $\mathbf{u}_r$ is the translation vector of the relation. These correspond respectively to the **rotation representation** (RotatE) and the **translation representation** (TransE).

Proof. (Translation) The TransE objective gives at once $\mathbf{e}_t - \mathbf{e}_h \approx \mathbf{e}_r$—the relation is a translation vector. (Rotation) Complexify the embedding as $e^{i\theta_h}$ and take the relation phase $\Delta\theta_r$; then the phase of the tail entity is $\theta_t = \theta_h + \Delta\theta_r$, that is, $\mathbf{e}_t = \mathbf{e}_h \cdot e^{i\Delta\theta_r}$. The two representations are dual when the magnitudes are approximately equal and $\Delta\theta_r$ is small. $\square$

Proposition 44.9 (Path period of knowledge reasoning). *Suppose that a relation path exists in a knowledge graph:*

$$h \xrightarrow{r_1} e_1 \xrightarrow{r_2} e_2 \xrightarrow{r_3} \cdots \xrightarrow{r_n} t.$$

Then the total relation phase of the path is

$$\Delta\Phi_{\text{path}} = \sum_{k=1}^{n} \Delta\theta_{r_k}.$$

When $\Delta\Phi_{\text{path}} = 0 \pmod{2\pi}$, the path closes—that is, h and t are the same entity reached by different relation paths.

Proof. By the rotation representation of Theorem 44.6, the phase accumulation at each step of the relation is $\theta_{e_k} = \theta_{e_{k-1}} + \Delta\theta_{r_k}$. Iteration gives $\theta_t = \theta_h + \sum_k \Delta\theta_{r_k}$. Hence closure of the path, $\theta_t = \theta_h$, is equivalent to $\sum_k \Delta\theta_{r_k} \equiv 0 \pmod{2\pi}$. $\square$

Lemma 44.4 (Phase composition of transitive relations). *Let the phase increments of the relations r_1 and r_2 be $\Delta\theta_{r_1}$ and $\Delta\theta_{r_2}$ respectively. If $r_3 = r_1 \circ r_2$ (r_3 being the composite of r_1 and r_2), then*

$$\Delta\theta_{r_3} = \Delta\theta_{r_1} + \Delta\theta_{r_2}.$$

That is, the phases of relations are additive under composition—this is the realisation in knowledge reasoning of the addition of phases under complex multiplication in the 'Scalars and Vectors' chapter (Proposition 3.11).

Proof. From the derivation for a two-step path in Proposition 44.9: $\theta_{e_2} = \theta_h + \Delta\theta_{r_1} + \Delta\theta_{r_2}$. The phase increment of the composite relation r_3 is defined as $\theta_{e_2} - \theta_h = \Delta\theta_{r_1} + \Delta\theta_{r_2}$. $\square$

Example 44.6 (Translation structure of the TransE embedding). In the TransE model the embedding of the relation 'capital' is the translation vector $\mathbf{e}_{\text{capital}}$, so that

$$\mathbf{e}_{\text{China}} + \mathbf{e}_{\text{capital}} \approx \mathbf{e}_{\text{Beijing}}, \quad \mathbf{e}_{\text{France}} + \mathbf{e}_{\text{capital}} \approx \mathbf{e}_{\text{Paris}}.$$

This embodies the **translation invariance of relations**: one and the same relation manifests as **the same translation vector** across different entity pairs—the translation version of the analogy relation (Proposition 44.6).

44.8 Attractor analysis of learning dynamics

Definition 44.15 (Attractor structure of training)**.** Let the training dynamics in parameter space be $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \nabla \mathcal{L}(\boldsymbol{\theta}_t)$. A **training attractor** is a fixed point $\boldsymbol{\theta}^*$ in parameter space (with $\nabla \mathcal{L}(\boldsymbol{\theta}^*) = \mathbf{0}$ and positive-definite Hessian); under a quadratic loss SGD has only fixed-point attractors—for a large learning rate the parameters converge geometrically to $\boldsymbol{\theta}^*$ with alternating signs, and that oscillation is **not** a limit cycle (see Theorem 44.7).

Theorem 44.7 (Convergence and underdamped oscillation of SGD)**.** Let the Hessian eigenvalues of the loss function $\mathcal{L}(\boldsymbol{\theta})$ be $\{\lambda_i\}$ and let the learning rate be η. Then:

(i) if $\eta\lambda_i < 1$ for all i (the overdamped region), then SGD converges to a fixed point;

(ii) if $1 < \eta\lambda_i < 2$ for some i (the underdamped region), then SGD oscillates along that eigendirection with alternating signs, yet the error still converges geometrically to 0 (since $|1-\eta\lambda_i| < 1$) and the parameters ultimately converge to the fixed point $\boldsymbol{\theta}^*$;

(iii) if $1 < \eta\lambda_i < 2$ for all i, then the error along every eigendirection converges geometrically to 0 with alternating signs, and the parameters converge to $\boldsymbol{\theta}^*$ by damped oscillation. Under a quadratic loss **no limit cycle exists**: **oscillation is not a limit cycle**—a limit cycle requires a closed and isolated orbit, whereas here $|1 - \eta\lambda_i| < 1$ makes the amplitude decay strictly.

Proof. By Lemma 44.1, the evolution of the error along the i-th eigendirection is $e_t^{(i)} = (1-\eta\lambda_i)^t e_0^{(i)}$. If $0 < \eta\lambda_i < 1$ then $0 < 1-\eta\lambda_i < 1$—the error decays monotonically to zero. If $1 < \eta\lambda_i < 2$ then $-1 < 1-\eta\lambda_i < 0$—the sign of the error alternates at every step whilst its modulus decreases strictly, so the parameters decay to **$\boldsymbol{\theta}^*$ with alternating signs**. When several eigendirections are simultaneously underdamped, the parameter trajectory is a damped oscillation about the minimum (the directions changing sign in turn), whose amplitude decays by $\prod_i |1-\eta\lambda_i|$; it **does not form a limit cycle**, since a limit cycle requires a non-decaying amplitude and an isolated orbit. □

Proposition 44.10 (Relation between generalisation and attractor width)**.** Suppose that training converges to an attractor $\boldsymbol{\theta}^*$ with Hessian eigenvalues $\{\lambda_i\}$. Define the attractor width as

$$W = \frac{1}{d}\sum_{i=1}^{d} \frac{1}{\lambda_i}.$$

Then:

(i) a wide attractor (W large, the λ_i small) corresponds to a flat minimum and good generalisation performance;

(ii) a narrow attractor (W small, the λ_i large) corresponds to a sharp minimum and poor generalisation performance.

This is the **flat-minima hypothesis** in deep learning.

Proof. Near a flat minimum the loss function changes slowly, and the parameters retain a low loss under perturbation; near a sharp minimum the loss function changes rapidly, and a small perturbation causes a large increase in the loss. A wide attractor has small Hessian eigenvalues (W large) and corresponds to the former case; a narrow attractor has large eigenvalues (W small) and corresponds to the latter. Generalisation performance therefore depends on the attractor width. ☐

Example 44.7 (Pattern memory and capacity analysis). Each local minimum of the Hopfield network (Theorem 44.2) corresponds to a **memory pattern**. When M patterns are stored, the total number of attractors is M (plus any spurious attractors that may arise). By the capacity analysis (Corollary 44.3), when $M \leq 0.138N$ the genuine patterns are stable attractors; when M exceeds this capacity, interference between patterns causes attractors to merge or spurious attractors to appear. This is the mathematical upper bound on the long-term memory capacity of an agent.

Remark 44.10 (Underdamped oscillation of SGD and generalisation). In the underdamped region SGD converges to the fixed point with alternating signs, and this damped oscillation has important consequences for generalisation:

(i) the fluctuation of the parameters during the underdamped oscillation amounts to **implicit regularisation**—the convergence does not come to rest exactly at a sharp minimum;

(ii) the amplitude decay rate of the oscillation (determined by $|1 - \eta\lambda_{\mathsf{max}}|$) is related to the effective learning rate $\eta\lambda_{\mathsf{max}}$ and may serve as an empirical indicator of generalisation error;

(iii) this shares one and the same mathematical structure with the 'damped oscillation of gradient descent' in the 'Periodic Structure of Algorithmic Systems' chapter—the oscillation of the later stage of training is not a defect but a source of implicit generalisation.

44.9 Periodic structure of the attention mechanism

Definition 44.16 (Periodic interpretation of self-attention). The core mechanism of the Transformer model is **scaled dot-product attention**:

$$\mathrm{Attention}(\mathsf{Q}, \mathsf{K}, \mathsf{V}) = \mathrm{softmax}\left(\frac{\mathsf{QK}^{\mathsf{T}}}{\sqrt{d_k}}\right)\mathsf{V}.$$

The attention mechanism is a unified realisation of three operations in sequence modelling:

(i) **Translation operation**: the sliding of the attention window over the input sequence is a translation operation—the Query vector **Q** moves along the sequence and computes similarities with the Key vectors **K** at different positions. The translation of attention resembles the sliding of the window function of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter over a time series (Definition 21.15).

(ii) **Scaling operation**: multi-head attention projects attention in parallel into h different subspaces—each head attends to dependencies on a different scale. This is scale decomposition: different heads correspond to different scaling factors (different d_k determine the 'resolution' of the attention).

(iii) **Rotation operation**: positional encoding injects positional information by means of sine and cosine functions: $PE_{(pos,2i)} = \sin(pos/10000^{2i/d})$, $PE_{(pos,2i+1)} = \cos(pos/10000^{2i/d})$. Positional encoding is the realisation of the rotation operation in sequence modelling—sine waves of different frequencies correspond to different levels of a nested epicycle system, and the position pos is the phase index.

Proposition 44.11 (Fourier structure of positional encoding)**.** Sinusoidal positional encoding may be rewritten in complex-exponential form: $PE_{pos} = e^{i\omega\cdot pos}$, where $\omega_i = 1/10000^{2i/d}$ are the frequencies at each level. Different dimensions i correspond to different angular velocities ω_i—positional encoding is the Fourier expansion of a nested epicycle system over sequence positions. The i-th dimension of the positional encoding corresponds to an epicycle of radius 1 and angular velocity ω_i (Definition 21.7), and the position pos corresponds to the angle of rotation.

Furthermore, when the attention weights are contributed by the positional encoding alone (for example, when the input features are $\mathsf{x} = \mathbf{0}$), the dot product $\mathsf{q}_i \cdot \mathsf{k}_j$ gives

$$\mathsf{q}_i \cdot \mathsf{k}_j = \sum_{l=1}^{d/2} \cos(\omega_l(pos_i - pos_j)),$$

that is, **a sum of difference-angle cosine similarities at several frequencies**—precisely the superposition over several frequencies of the basic difference-angle cosine similarity $\text{cosine}(\mathsf{a}, \mathsf{b}) = \cos(\alpha - \beta)$ of the 'Scalars and Vectors' chapter (Definition 3.9).

Proof. By Euler's formula, $(\sin(pos \cdot \omega_i), \cos(pos \cdot \omega_i))$ are precisely the imaginary and real parts of $e^{i\omega_i \cdot pos}$—each two-dimensional pair of the positional encoding is a point on the unit circle, with angular velocity $\omega_i = 1/10000^{2i/d}$.

For the explicit computation of the dot product: take $\mathsf{W}_Q = \mathsf{W}_K = \mathsf{I}$ (the simplified case); then

$$\mathsf{q}_i \cdot \mathsf{k}_j = PE_{pos_i} \cdot PE_{pos_j} = \sum_l \left[\sin(\omega_l pos_i)\sin(\omega_l pos_j) + \cos(\omega_l pos_i)\cos(\omega_l pos_j)\right].$$

By the product-to-sum formulae, $\sin A \sin B + \cos A \cos B = \cos(A - B)$, hence

$$\mathsf{q}_i \cdot \mathsf{k}_j = \sum_l \cos(\omega_l(pos_i - pos_j)).$$

This is precisely the sum over several frequencies of the difference-angle cosine similarity $\cos(\alpha - \beta)$ (with $\alpha = \omega_l pos_i$ and $\beta = \omega_l pos_j$). ☐

Remark 44.11 (Translation invariance of the positional-encoding dot product). By the derivation in Proposition 44.11, the dot-product similarity of positional encodings depends only on the **relative position** $pos_i - pos_j$ and not on the absolute position. This translation invariance gives positional encoding its 'relative-position awareness'—structurally the same as the phase translation of the 'Series and Convolution' chapter (Proposition 6.15).

44.10 Periodic memory of recurrent neural networks

Definition 44.17 (Gating period of the LSTM)**.** The long short-term memory network (LSTM) controls the flow of information through three gating mechanisms:

(i) **Forget gate** $f_t = \sigma(\mathsf{W}_f \cdot [\mathsf{h}_{t-1}, \mathsf{x}_t] + \mathsf{b}_f)$: determines which old information should be discarded. $f_t \to 1$ retains information, $f_t \to 0$ forgets it.

(ii) **Input gate** $i_t = \sigma(\mathsf{W}_i \cdot [\mathsf{h}_{t-1}, \mathsf{x}_t] + \mathsf{b}_i)$: determines which new information should be written in.

(iii) **Output gate** $o_t = \sigma(\mathsf{W}_o \cdot [\mathsf{h}_{t-1}, \mathsf{x}_t] + \mathsf{b}_o)$: determines which information should be output to the hidden state.

The gating mechanism of the LSTM is the realisation of the rotation operation in hidden-state space: the forget gate f_t controls the rate of decay of old information, the input gate i_t controls the rate at which new information is written, and the output gate o_t controls the projection of information from the cell state to the hidden state. The periodic opening and closing of the three gates drives the rotation of the hidden state h_t in state space.

Proposition 44.12 (Long-range periodic retention of the LSTM)**.** *The LSTM is able to learn long-range periodic dependencies in a sequence. Suppose that the input sequence contains a periodic pattern of period T. During training the forget gate f_t of the LSTM learns an opening-and-closing pattern of period T—at the critical positions of the period $f_t \approx 1$ (information retained) and at the non-critical positions $f_t \approx 0$ (information forgotten). The update of the cell state c_t, namely $\mathsf{c}_t = f_t \odot \mathsf{c}_{t-1} + i_t \odot \tilde{\mathsf{c}}_t$, is equivalent to selective memory in state space—the periodic opening and closing of f_t resembles a sampling gate that samples the input with period T.*

The long-range memory of the LSTM is a neural-network realisation of phase locking: the periodic opening and closing of the forget gate is synchronised with the period of the input sequence—the **phase locking** *of an external drive (the input sequence) with an internal period (the LSTM hidden state) (Proposition 24.8). When the opening-and-closing period of the forget gate matches the period of the input sequence, the LSTM attains optimal memory retention.*

Proof. Consider the simplified linear case: the forget gate $f_t = f$ is constant and there is no input-gate modulation ($i_t \cdot \tilde{\mathsf{c}}_t = \mathsf{u}_t$). The solution of the cell-state recurrence $\mathsf{c}_t = f\mathsf{c}_{t-1} + \mathsf{u}_t$ is

$$\mathsf{c}_t = \sum_{k=0}^{t} f^{t-k}\mathsf{u}_k,$$

that is, an exponentially weighted sum with base f, with effective memory length $\tau_{\mathsf{mem}} = -1/\ln f$ (as $f \to 1$ the memory length diverges).

When the forget gate opens and closes periodically, $f_t = f(t \bmod T)$ (of period T), the product $\prod_{k=0}^{T-1} f_k$ returns to its initial value within T steps—the **critical retention condition** for periodic memory is $\prod_{k=0}^{T-1} f_k = 1$. In this case the periodic components (those whose frequencies are commensurable with $1/T$) propagate without decay, whilst the non-periodic components are compressed cycle by cycle.

This is precisely phase locking (Proposition 24.8): memory is optimal when the phase of the forget gate (determined by the periodic structure of f_t) agrees with the phase of the sequence (determined by the input period). □

44.11 Periodic exploration in reinforcement learning

Definition 44.18 (Periodic alternation of exploration and exploitation)**.** *In reinforcement learning an agent faces the classical trade-off between exploration (trying new actions in order to obtain more information) and exploitation (executing the known optimal action in order to obtain the greatest return). The ε-greedy policy explores at random with probability ε and exploits greedily with probability $1-\varepsilon$. The parameter ε usually decays with time:*

$$\varepsilon(t) = \varepsilon_{\mathsf{min}} + (\varepsilon_{\mathsf{max}} - \varepsilon_{\mathsf{min}})e^{-t/\tau},$$

where τ is the decay time constant. The exponential decay of ε is the realisation in reinforcement learning of scale decay—the exploration intensity decays with time by the scale factor $e^{-1/\tau}$. Early in training ($t \ll \tau$), $\varepsilon \approx \varepsilon_{\mathsf{max}}$—the agent explores extensively (random rotation). Late in training ($t \gg \tau$), $\varepsilon \approx \varepsilon_{\mathsf{min}}$—the agent mainly exploits (deterministic translation).

The random source in the exploration process is provided by a PRNG (Definition 43.8)—its period length determines the quality of the exploration randomness. If the PRNG period is too short, the exploration policy repeats within a short time, reducing the efficiency of exploration.

Proposition 44.13 (Periodic convergence of policy iteration)**.** *Policy iteration alternates between policy evaluation and policy improvement:*

$$\pi_0 \xrightarrow{\text{evaluation}} Q^{\pi_0} \xrightarrow{\text{improvement}} \pi_1 \xrightarrow{\text{evaluation}} Q^{\pi_1} \xrightarrow{\text{improvement}} \pi_2 \to \cdots \to \pi^*.$$

Each policy improvement rotates the policy towards the optimal policy π^* by one angle—the policy rotates about suboptimal policies in policy space, approaching the optimal policy step by step. Policy evaluation corresponds to moving along the direction of the value function, and policy improvement to rotating the policy along the greedy direction. The convergence of policy iteration is the convergence of a gradient system to an equilibrium point—the optimal policy π^* is an attractor in policy space.

Proof. (Finite MDP case) **Policy improvement theorem**: if π' is the greedy policy with respect to Q^π (that is, $\pi'(s) = \arg\max_a Q^\pi(s,a)$), then $V^{\pi'}(s) \geq V^\pi(s)$ for all s. Proof: by $V^\pi(s) \leq Q^\pi(s,\pi'(s))$ and the Bellman expectation recurrence, verified state by state—for any policy π,

$$\begin{aligned} V^\pi(s) &= \mathbb{E}_{a\sim\pi(\cdot|s)}\left[r(s,a) + \gamma\sum_{s'} P(s'|s,a)V^\pi(s')\right] \\ &\leq \max_a \left[r(s,a) + \gamma\sum_{s'} P(s'|s,a)V^\pi(s')\right] = Q^\pi(s,\pi'(s)). \end{aligned}$$

Applying the Bellman operator repeatedly, the value function of π' is pointwise no less than that of π.

In a finite state–action space the number of policies is finite, so the monotonically improving sequence $\pi_0 \preceq \pi_1 \preceq \dots$ must reach a fixed point π^* (the solution of the optimal Bellman equation) in finitely many steps—the 'rotation' of policy iteration in policy space finally closes at the attractor π^*. □

44.12 Phase synchronisation of multi-agent systems

Definition 44.19 (Policy synchronisation in multi-agent reinforcement learning). In multi-agent reinforcement learning (MARL) each of N agents maintains its own policy π_i. The agents coordinate their policies by sharing parameters or experience, or by communication. When the reward functions of the agents are cooperative (the rewards of all agents are positively correlated), the policy parameters tend to **phase synchronisation** in parameter space—the policy parameter vectors $\boldsymbol{\theta}_i$ of the several agents tend to a common direction.

Multi-agent policy synchronisation is a many-body realisation of phase locking: each agent is an independent periodic subsystem (the periodic oscillation of policy updates), and the communication between agents supplies the coupling coefficients c_{ij} between the subsystems. When the coupling is sufficiently strong, the policy parameters of all agents tend to the same phase—structurally the same as the phase synchronisation of oscillators in the Kuramoto model (Proposition 37.14), whose synchronisation threshold is given by the Kuramoto critical coupling strength $K_c = 2/(\pi g(\Omega))$ (where g is the natural-frequency distribution and Ω its mode).

Proposition 44.14 (Periodic convergence of parameter averaging). In distributed training several workers compute gradients independently, and a parameter server averages the parameters of the nodes periodically:

$$\boldsymbol{\theta}^{(t+1)} = \frac{1}{N}\sum_{i=1}^{N}\boldsymbol{\theta}_i^{(t)}.$$

The period T_{sync} of parameter averaging (the number of steps between synchronisations) is the synchronisation-frequency parameter. $T_{\text{sync}} = 1$ (synchronisation at every step) corresponds to strong coupling—training is stable but the communication overhead is large. $T_{\text{sync}} \gg 1$ (asynchronous over many steps) corresponds to weak coupling—the communication overhead is small but the parameters of the several nodes may diverge (phase drift). The optimal synchronisation period requires a trade-off between the speed of convergence and the efficiency of communication.

Proof. Suppose that each node performs T_{sync} steps of gradient descent independently and the parameters are then averaged. By Lemma 44.1 the error of a single node over T_{sync} steps contracts by $(1-\eta\lambda_i)^{T_{\text{sync}}}$; the averaging operation is a convex combination and does not increase the dispersion of the nodes about the global mean (convexity). Hence within T_{sync} steps the rate of convergence is, relative to synchronous execution ($T_{\text{sync}} = 1$), about $\frac{1}{T_{\text{sync}}} \ln \frac{1}{|1-\eta\lambda_i|}$ times as large, whilst the communication volume falls to $1/T_{\text{sync}}$—the synchronisation period is a trade-off parameter between the rate of convergence and the cost of communication.

This trade-off structure is mathematically the same as the coupling-strength trade-off of the Kuramoto model (Proposition 37.14): the stronger the coupling (the smaller T_{sync}), the faster the synchronisation but the greater the communication overhead; the weaker the coupling (the larger T_{sync}), the more economical the communication but the slower the convergence, and when the coupling falls below the Kuramoto threshold synchronisation fails. □

44.13 Constraint relations of agent systems

Theorem 44.8 (The three constraint relations of agent systems). *The dynamics of an agent system are governed by three constraint relations:*

(i) **Parameter constraint**: *in training without regularisation the total magnitude of the model parameters (the L^2 norm) is not directly constrained during optimisation. Weight decay, however, introduces a gradient pull of the parameters towards the origin—$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta(\nabla\mathcal{L} + \lambda\boldsymbol{\theta}_t)$. Weight decay is a return towards an equilibrium point—analogous to the restoring force of a spring.*

(ii) **Gradient-propagation constraint**: *in batch normalisation the mean and variance of the input to each layer are held fixed during training—$\mathbb{E}[\mathsf{x}] = 0, \mathsf{Var}[\mathsf{x}] = 1$. The propagation of gradients between layers satisfies an energy constraint—the total energy of the gradient (its L^2 norm) neither decays nor explodes during backpropagation. The residual connection guarantees gradient propagation through the identity map $\mathsf{x}_{l+1} = \mathsf{x}_l + \mathcal{F}(\mathsf{x}_l)$—$\partial\mathsf{x}_{l+1}/\partial\mathsf{x}_l = \mathsf{I} + \partial\mathcal{F}/\partial\mathsf{x}_l$, where the identity matrix guarantees lossless propagation of the gradient.*

(iii) **Attention-normalisation constraint**: *the attention weights sum to 1—$\sum_j \alpha_{ij} = 1$ for all i. The normalisation of attention is a realisation of phase locking—the total attention of the agent at any instant is fixed, and the distribution of attention among different input positions is a rotation of phase. The distribution of the total attentional resource among several attention heads constitutes the attention-normalisation constraint.*

Together the three constraint relations guarantee the training stability and generalisation capability of an agent system—the parameter constraint prevents overfitting, the gradient-propagation constraint prevents vanishing and exploding gradients, and the attention-normalisation constraint ensures that information focusing is well posed.

Proof. (i) **Parameter constraint**: the continuous-time form of weight decay is $d\boldsymbol{\theta}/dt = -\nabla\mathcal{L} - \lambda\boldsymbol{\theta}$. Near the equilibrium point $\nabla\mathcal{L} + \lambda\boldsymbol{\theta}^* = 0$ we have $\|\boldsymbol{\theta}^*\| = \|\nabla\mathcal{L}\|/\lambda$—the L^2 norm of the parameters is determined jointly by the loss gradient and the decay coefficient. What weight decay introduces is a restoring force towards the equilibrium point—analogous to the $-kx$ term in a spring oscillator (Definition 33.11).

(ii) **Gradient-propagation constraint**: the output of the l-th layer of a residual network is $\mathsf{x}_{l+1} = \mathsf{x}_l + \mathcal{F}(\mathsf{x}_l, \mathsf{W}_l)$. The gradient of backpropagation is

$$\frac{\partial \mathcal{L}}{\partial \mathsf{x}_l} = \frac{\partial \mathcal{L}}{\partial \mathsf{x}_{l+1}} \left(\mathsf{I} + \frac{\partial \mathcal{F}}{\partial \mathsf{x}_l} \right).$$

The identity matrix I guarantees that the gradient can propagate directly from deep layers to shallow ones—even when $\frac{\partial \mathcal{F}}{\partial \mathsf{x}_l}$ is very small, the gradient does not vanish. This is the realisation of the gradient-propagation constraint in a deep network.

(iii) **Attention-normalisation constraint**: the softmax function guarantees $\sum_j \alpha_{ij} = 1$—the distribution of the total attention over all positions is normalised. When the attention at one position increases, the attention at other positions necessarily decreases. Multi-head attention distributes the total attention among h heads, each normalised independently—the distribution of the total attention among the heads is likewise normalised. This constraint shares the same mathematical structure of 'conservation of the total' as the probability normalisation of the 'Periodic Structures in Probabilistic Systems' chapter (Proposition 14.1). □

44.14 Open problems and research programme

Remark 44.12 (Open problems). The following questions lie beyond the scope of the theorems established in this chapter and are listed as **open problems**:

(i) **Global convergence of KDI under a non-convex loss**: Proposition 44.5 gives only a qualitative description; rigorous results on the global rate of convergence of KDI and on the expected time to escape a saddle point are still lacking.

(ii) **Expressive power of the CEAF in deep networks**: Theorem 42.5 is a single-hidden-layer result; whether multi-layer CEAF networks possess an essential advantage over single-layer ones, and whether they achieve higher parameter efficiency than ReLU networks, still lacks rigorous separation lower bounds.

(iii) **Exact correspondence between positional encoding and periodic convolution**: Proposition 44.11 gives a multi-frequency cosine decomposition of the positional-encoding dot product; whether this decomposition corresponds completely to the periodic-convolution framework of the 'Series and Convolution' chapter (Definition 6.26), in particular the correspondence between the weight kernel W and the frequency-composition operator $\Omega_{n,m}$, requires further study.

(iv) **Exact correspondence between multi-agent synchronisation and the Kuramoto model**: Proposition 44.14 gives a qualitative convergence for parameter averaging; whether a synchronisation threshold corresponding exactly to the Kuramoto threshold K_c (Proposition 37.14) can be given requires deeper investigation.

(v) **Optimal decay rate of the exploration policy in reinforcement learning**: for $\varepsilon(t) = \varepsilon_{\mathsf{min}} + (\varepsilon_{\mathsf{max}} - \varepsilon_{\mathsf{min}})e^{-t/\tau}$, the optimal choice of the time constant τ as a function of the MDP parameters still has no general formula.

(vi) **Quantitative characterisation of the phase structure of word embeddings**: Proposition 44.6 gives a phase-offset description of the analogy relation; whether this offset is observable in all periodic semantic dimensions, and whether it is quantitatively controlled by the distribution of the training data, requires empirical study.

(vii) **A general formula for the optimal depth of a reasoning chain**: Corollary 44.4 gives the order of magnitude of the optimal depth, but there is as yet no rigorous theorem relating the concrete constant of proportionality to the intrinsic complexity of the problem.

(viii) **Quantitative relation between attractor width and generalisation error**: Proposition 44.10 gives a qualitative width–generalisation relation; the precise connection between quantitative generalisation-error bounds and attractor width remains to be established.

The resolution of these questions would elevate the theory of agent periodic structure from a 'framework with qualitative guarantees' to 'learning algorithms with quantitative guarantees'.

Remark 44.13 (Unity with the rest of the book). The theory of the periodic structure of agent systems established in this chapter is a systematic application of periodic mathematics in artificial intelligence, and at the same time the convergence point of the four-chapter chain 'data analysis → activation function → algorithm → agent'.

Inheritance from the three preceding chapters:

(i) the oscillatory convergence of gradient descent (Lemma 44.1) is the realisation of the translation operation in parameter space—the learning rate determines the translation step size and the Hessian eigenvalues determine the rate of convergence in each direction;

(ii) the momentum method (Proposition 44.1) introduces a rotational component into parameter space—the parameter trajectory rotates about the ravines of the loss landscape, accelerating convergence;

(iii) learning-rate scheduling (Definition 44.4) is the realisation of scale decay in optimisation;

(iv) the Jacobian matrix of Newton's method (Proposition 43.1) provides the rotational correction;

(v) the phase representation of word embeddings (Lemma 44.2) and the rotational analogy relation (Proposition 44.6) inherit the complex-exponential difference-angle operator of the 'Scalars and Vectors' chapter (Definition 3.25).

Contributions of this chapter:

(i) the attention mechanism (Definition 44.16) is a unification of the three operations in sequence modelling—the Query–Key dot product is a phase-similarity computation (the rotation operation), the multi-scale parallelism of multiple heads is a scale decomposition (the scaling operation), and the sliding of the window is a positional translation (the translation operation);

(ii) the gating period of the LSTM (Proposition 44.12) is the realisation of the rotation operation in hidden-state space—the periodic opening and closing of the forget gate and the input gate drive the rotational update of the hidden state;

(iii) the exploration–exploitation alternation of reinforcement learning (Definition 44.18) is a coupling of the scaling and rotation operations;

(iv) the policy synchronisation of multiple agents (Proposition 44.14) is an engineering realisation of phase locking (Proposition 24.8) and of the Kuramoto model (Proposition 37.14);

(v) the three constraint relations of agent systems (Theorem 44.8)—the parameter constraint, the gradient-propagation constraint and the attention-normalisation constraint—together constitute a mathematical guarantee of training stability in deep learning;

(vi) the phase structure of word embeddings (Section 44.4) provides a periodic-mathematics framework for semantic analogy in natural language processing;

(vii) attractor dynamics (Section 44.5) unifies the finite-order actions of the 'Operations and Periodicity' chapter and the limit cycles of continuous dissipative systems into the memory and oscillation structure of an agent;

(viii) the periodic iterative structure of reasoning (Section 44.6) generalises the fixed-point iteration of the 'Periodic Structure of Algorithmic Systems' chapter to the analysis of the convergence and closure of reasoning chains;

(ix) the periodic analysis of knowledge representation (Section 44.7) brings the translation–rotation duality of relation embeddings within the framework of periodic mathematics.

Complete closure of the four-chapter chain: from the extraction of parameters in data analysis (Proposition 42.1), to the differentiable components of activation functions (Proposition 42.7), to the inheritance of algorithmic convergence structures (Proposition 43.5), and thence to the full realisation of agent training dynamics—the four chapters take the 'nested epicycle system' as their unifying mathematical object and the three elementary operations of 'translation–scaling–rotation' as their unifying language, completing the construction of an end-to-end theory of periodic structure from data to learning systems.

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The parameter translation of gradient descent and the eigendirection error iteration $e_t^{(i)} = (1 - \eta\lambda_i)^t e_0^{(i)}$ (Lemma 44.1); the sliding of the attention window over the sequence (Definition 44.16); the translation invariance of positional encoding (Remark 44.11); the translational analogy relation of word embeddings (Proposition 44.6); the translation structure of the TransE relation embedding (Theorem 44.6)
Scaling	The scale decay of the learning-rate scheduler (Definition 44.4); the exponential decay of the exploration intensity $\varepsilon(t)$ (Definition 44.18); the scale decomposition of multi-head attention; the magnitude vector of word embeddings (Lemma 44.2)
Rotation	The damped rotation of the momentum method (Proposition 44.1); the complex-exponential phase $e^{i\omega \cdot pos}$ of positional encoding (Proposition 44.11); the phase opening and closing of LSTM gating (Proposition 44.12); the rotational analogy relation of word embeddings (Proposition 44.6); the phase rotation of the RotatE relation embedding (Theorem 44.6); the phase accumulation of the reasoning chain (Proposition 44.8)
Balance	The parameter constraint (weight decay), the gradient-propagation constraint (residual connections) and the attention-normalisation constraint (softmax normalisation) (Theorem 44.8); the information–depth trade-off of the reasoning chain (Lemma 44.3); the generalisation–attractor-width trade-off (Proposition 44.10)
Symmetry	The identity symmetry $\mathsf{I} + \partial\mathcal{F}/\partial\mathsf{x}_l$ of residual connections; the translation-invariant similarity of positional encoding (Remark 44.11); the difference-angle cosine symmetry of word-embedding components (Theorem 44.1)
Closure	The finite-step convergence closure of policy iteration towards the optimal policy π^* (Proposition 44.13); the synchronisation closure of multi-agent parameter averaging (Proposition 44.14); the iterative closure of the reasoning chain (Theorem 44.4); the phase-locking closure of CoT reasoning (Theorem 44.5); the phase closure of knowledge paths (Proposition 44.9)
Period	The damped oscillation of gradient descent (Definition 44.3); the cosine-annealing period $2T$ (Proposition 44.2); the multi-agent synchronisation period T_{sync} (Proposition 44.14); the memory period of LSTM gating (Proposition 44.12); the Hopfield

Exercises

Exercise 1 Prove the **descent property of KDI for epicycle parameters** of this chapter (44.4). Hint: first write down the decision map and the evolution group of this chapter's target, then apply the proof structure of the general criterion given in the 'Periodic Systems' chapter.

Exercise 2 Verify that the hypotheses of the **three constraint relations of agent systems** (44.8) hold for the domain objects of this chapter, and give a counterexample in which the hypotheses fail.

Exercise 3 Using the seven-theme comparison table of this chapter, identify the concrete carriers of the themes **translation**, **scaling** and **rotation** in this chapter, and give one example of domain data for each.

Exercise 4 Synthesis: restate the diagnostic procedure of this chapter using the five-step standard interface of the 'Periodic Systems' chapter (Table 24.1), noting for each step the theorem of this chapter on which it depends: 44.4, 44.8.

Chapter 45

Modelling and Analysis – From Data to Epicycles

Remark 45.1 (Periodicity decision for the target of this chapter (citation)). When this chapter decides whether a research target is periodic, determines its period type, or seeks its minimal positive period, reference is always made to the 'Periodic Systems' chapter: for the periodicity criterion see Theorem 24.1, for the discrimination of period types see Definition 24.3, Proposition 24.1, for the search for the minimal positive period see Definition 24.4, Theorem 24.2, and for the construction of period series and epicycle groups from data see Theorem 24.3, Definition 24.6.

Remark 45.2 (Position of this chapter: the methodological hinge from data to epicycles). This chapter establishes the **unified methodological framework** of periodicity mathematics—from the modelling of real-world problems, data acquisition, periodicity analysis and signal processing through to model evaluation, ultimately reducing to the epicycle model and its complex-valued function implementation. This chapter is the **bridge chapter** connecting the theoretical system of periodicity mathematics (the 'Series and Convolution', 'Harmonic Analysis' and 'Periodic Systems' chapters) and the various application domains (astronomy, calendars, geography, biology, psychology, economics, politics, algorithms and agents).

The core methodological procedure is:

$$\boxed{\text{Modelling}} \to \boxed{\text{Data}} \to \boxed{\text{Analysis}} \to \boxed{\text{Processing}} \to \boxed{\text{Evaluation}} \to \boxed{\text{Period}} \to \boxed{\text{Epicycle}} \to \boxed{\text{Wa}}$$

Every step uses a core tool of periodicity mathematics: modelling uses symmetry and constraint relations; data acquisition uses the theory of dynamic sampling (Definition 21.6); analysis uses periodic convolution (Definition 6.26) and the similarity/trend lineage (Definitions 3.8 and 3.16); processing uses window functions (Definition 21.15) and derivative–integral descent optimisation (Definition 17.19); evaluation uses information criteria (AIC/BIC); period identification uses nested epicycle generation (Definition 30.13); and the Waveform, as a complex-valued function, uses the complex exponential activation function (Definition 42.4).

The distinctive contribution of this chapter: the methodological framework above is expressed in two ways at once, as a **mathematical statement** and as **programming modelling (UML/diagrams)**. The mathematical statement provides rigorous theorems and proofs; the programming modelling provides an implementable class system, data flow, interaction sequence and business process. Together they constitute the complete expression of the methodology of this chapter.

Guide to the seven themes of this chapter: translation—the advancement of the time series in data acquisition (Definition 45.16) and the time-translation invariance of the Waveform (Property 45.7); scaling—the complexity penalty of BIC

(Lemma 45.1) and multi-scale window functions (Definition 45.18); rotation—the phase ϕ_k of the epicycle model (Definition 45.21) and the complex phase representation of the Waveform; balance—the fit–complexity balance of BIC (Proposition 45.5) and the uniqueness of the parameters given by the Bohr mean (Property 45.6); symmetry—the hierarchical symmetry of epicycle nesting; closure—the feedback-iteration closure of the methodological procedure (Lemma 45.1) and the closure property of the Functor class (Property 45.1); period—spectral peak identification (Definition 45.20) and epicycle parameter estimation (Definition 45.21).

Remark 45.3 (The dual-track modelling of this chapter). *This chapter expresses 'modelling and analysis' in a* **dual-track** *manner:*

(i) **The mathematical track**: *statements and proofs are given in the strict order of definitions, propositions, lemmas, theorems, corollaries, properties, examples and remarks. The mathematical track is the backbone of this chapter, and every conclusion must be verifiable.*

(ii) **The engineering track**: *expressed by such visualisation tools as Venn diagrams, Gantt charts, flowcharts, UML diagrams (context diagram, use-case diagram, class diagram, object diagram, sequence diagram), system architecture diagrams, user-story maps, business process models, data flow diagrams and ER diagrams. The engineering track is* **supplementary exposition** *of the mathematical track—it introduces no new mathematical content, but only translates the objects, relations and processes of the mathematical track into an implementable engineering language.*

The interface convention between the two tracks is: every concept appearing in the engineering track (such as 'Waveform', 'Epicycle' and 'PeriodicSeries') has already been given a rigorous definition in the mathematical track or in an earlier chapter; the diagrams of the engineering track **constitute no new mathematical assertion**. *This convention guarantees that the chapter preserves mathematical rigour without sacrificing engineering implementability.*

Lemma 45.1 (Feedback closure of the methodological pipeline). *Let the model class $\mathcal{M} = \{\mathcal{M}_K\}_{K=1}^{K_{\max}}$ be finite, where K is the epicycle order. The evaluation index is BIC:*

$$\mathrm{BIC}(K) = n \ln(\mathrm{RSS}/n) + K \ln n,$$

where n is the sample size and RSS is the residual sum of squares. Then the iteration 'evaluation → feedback → remodelling' decreases strictly in BIC and converges in finitely many steps to the BIC-optimal model over $\mathcal{M}$.

Proof. The feedback rule is: replace $\mathcal{M}_K$ by $\mathcal{M}_{K'}$ only when $\mathrm{BIC}(K') < \mathrm{BIC}(K)$. Hence BIC decreases strictly at each iteration. Since $\mathcal{M}$ is finite ($K \le K_{\max}$), the possible values of BIC are finite, so a strictly decreasing sequence must terminate after finitely many steps at some K^*. At that point there is no replaceable model with a smaller BIC, so $\mathcal{M}_{K^*}$ is the BIC-optimal model over $\mathcal{M}$. □

45.1 Overview of the methodology: context diagram and master flow diagram

45.1.1 Context diagram: the boundary between the methodology and external systems

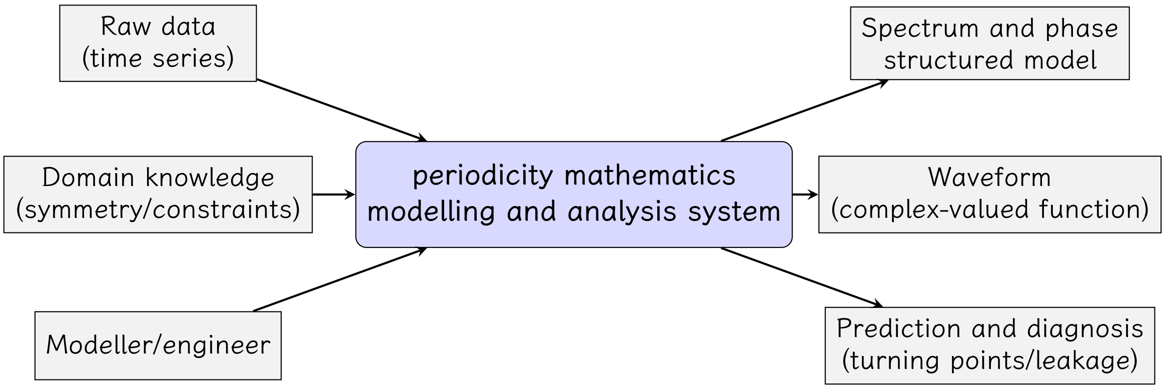

Raw data
(time series)
Domain knowledge
(symmetry/constraints)
Modeller/engineer
periodicity mathematics
modelling and analysis system
Spectrum and phase
structured model
Waveform
(complex-valued function)
Prediction and diagnosis
(turning points/leakage)

Figure 45.1: Context diagram of the periodicity mathematics modelling and analysis system. The inputs are the raw time series, domain knowledge and modeller interaction; the outputs are the spectrum and phase structured model, the Waveform complex-valued function representation, and the results of prediction and diagnosis.

45.1.2 Master flow diagram

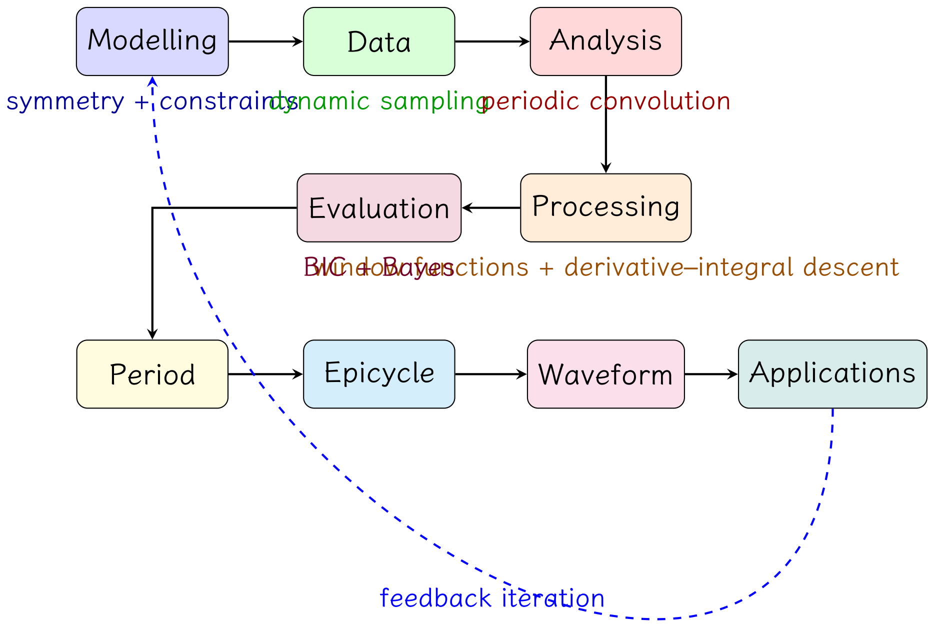
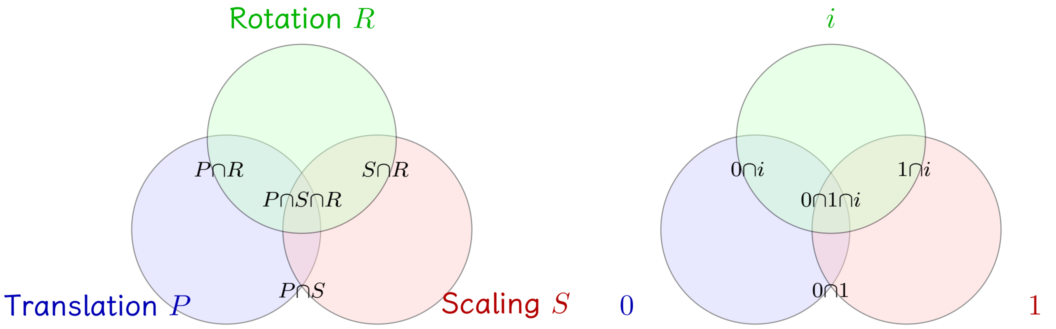
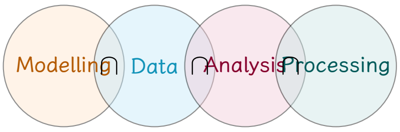

Modelling
Data
Analysis
symmetry + constraints
dynamic sampling
periodic convolution
Evaluation
Processing
BIC + Bayes
window functions + derivative–integral descent
Period
Epicycle
Waveform
Applications
feedback iteration

Figure 45.2: Master flow diagram of the unified methodology of periodicity mathematics (serpentine loop layout). The first row runs from left to right through modelling, data and analysis; the second row runs from right to left through processing and evaluation; the third row runs from left to right through the output structures: period, epicycle, Waveform and applications. The dashed arrow denotes the feedback iteration from the applications back to modelling.

45.2 Venn diagram of the concept sets


Rotation R
$P∩R$
$S∩R$
$P∩S∩R$
$P∩S$
Translation P
Scaling S
i
$0∩i$
$1∩i$
$0∩1∩i$
$0∩1$
0
1

(a) The three basic transformations (b) The three centres of symmetry


Modelling
∩
Data
∩
Analysis
Processing

(c) The four stages of the methodology

Figure 45.3: Venn diagram of the concept sets. In subfigure (a), P is translation, S is scaling and R is rotation, showing the pairwise intersections $P∩S$, $P∩R$ and $S∩R$ of the three basic transformations together with the triple intersection $P∩S∩R$; subfigure (b) is the overlap of the three centres of symmetry (0 for addition, 1 for multiplication, i for rotation); subfigure (c) is the covering relation of adjacent stages among the four stages of the methodology (modelling, data, analysis, processing). The Venn diagram displays the coincidences and complementarities among concepts and constitutes no new mathematical assertion.

45.3 Gantt chart of the modelling workflow

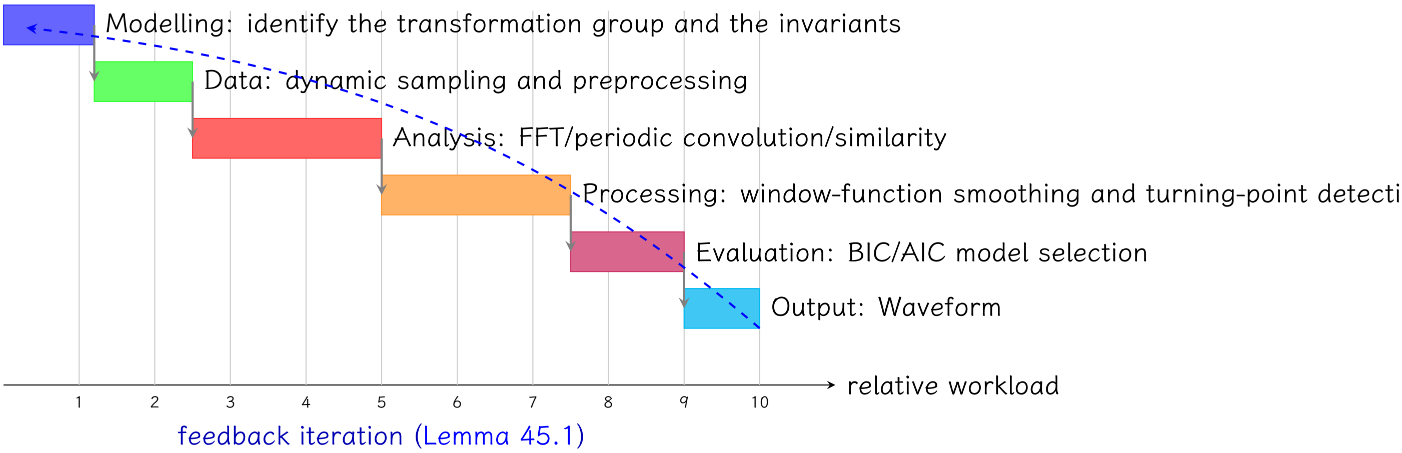


Figure 45.4: Gantt chart of the modelling workflow. The horizontal axis is the **relative workload** (a dimensionless unit) and does not represent actual calendar time; the vertical direction is the sequence of tasks. The feedback iteration is guaranteed to converge in finitely many steps by Lemma 45.1.

45.4 Use-case diagram

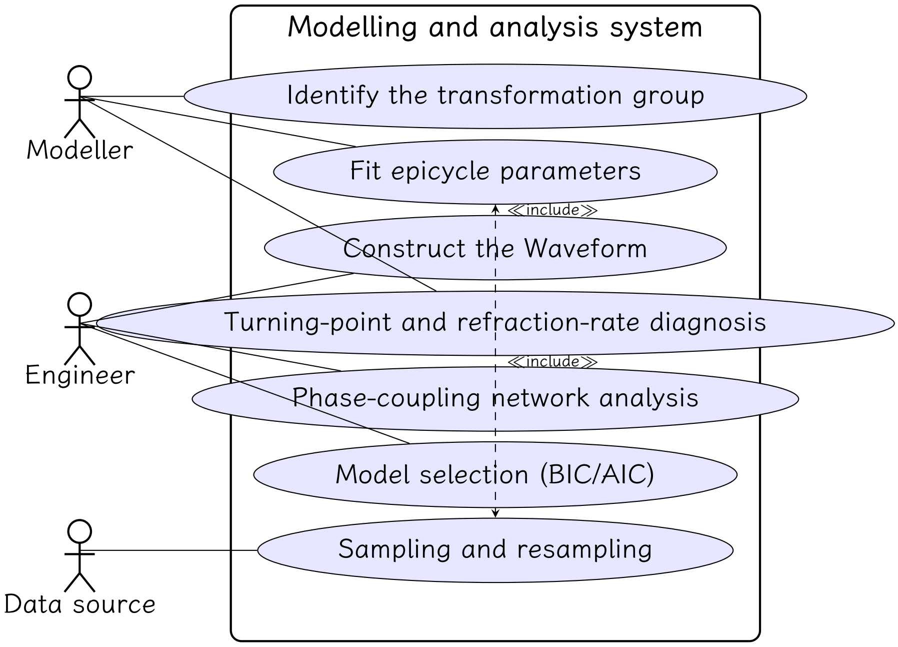
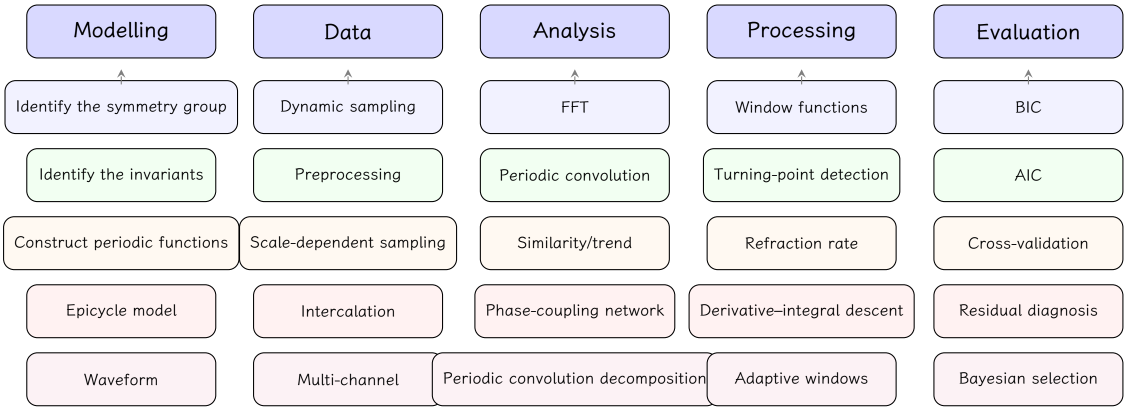

Modelling and analysis system
Modeller
Engineer
Data source
Identify the transformation group
Fit epicycle parameters
≪include≫
Construct the Waveform
Turning-point and refraction-rate diagnosis
≪include≫
Phase-coupling network analysis
Model selection (BIC/AIC)
Sampling and resampling

Figure 45.5: Use-case diagram of the modelling and analysis system. There are three actors: the modeller, the engineer and the data source. The core use cases include identifying the transformation group, fitting epicycle parameters, constructing the Waveform, turning-point diagnosis, phase-coupling analysis, model selection, and sampling and resampling.

45.5 User-story map



Modelling	Data	Analysis	Processing	Evaluation
Identify the symmetry group	Dynamic sampling	FFT	Window functions	BIC
Identify the invariants	Preprocessing	Periodic convolution	Turning-point detection	AIC
Construct periodic functions	Scale-dependent sampling	Similarity/trend	Refraction rate	Cross-validation
Epicycle model	Intercalation	Phase-coupling network	Derivative–integral descent	Residual diagnosis
Waveform	Multi-channel	Periodic convolution decomposition	Adaptive windows	Bayesian selection

Figure 45.6: User-story map. The horizontal axis gives the five stages of the methodology (modelling, data, analysis, processing, evaluation), and the vertical axis gives the levels of stories from basic capability to integrated capability. The higher the level, the greater the degree of integration covered by the story; stories at the same level can be delivered independently within that stage.

45.6 Business process model (BPMN style)

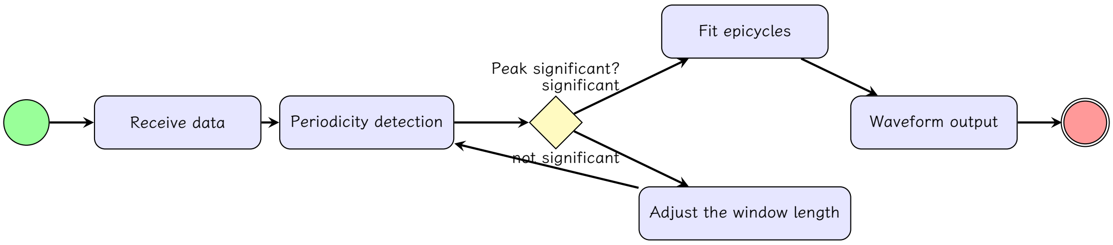

Receive data
Periodicity detection
Peak significant?
significant
not significant
Fit epicycles
Adjust the window length
Waveform output

Figure 45.7: Business process model of modelling and analysis (BPMN style). It starts with receiving data and proceeds through periodicity detection; if the peak is significant, epicycles are fitted and the Waveform is output; if it is not significant, the window length is adjusted and detection is repeated until it becomes significant or a termination condition is reached.

45.7 Data flow diagram (DFD)

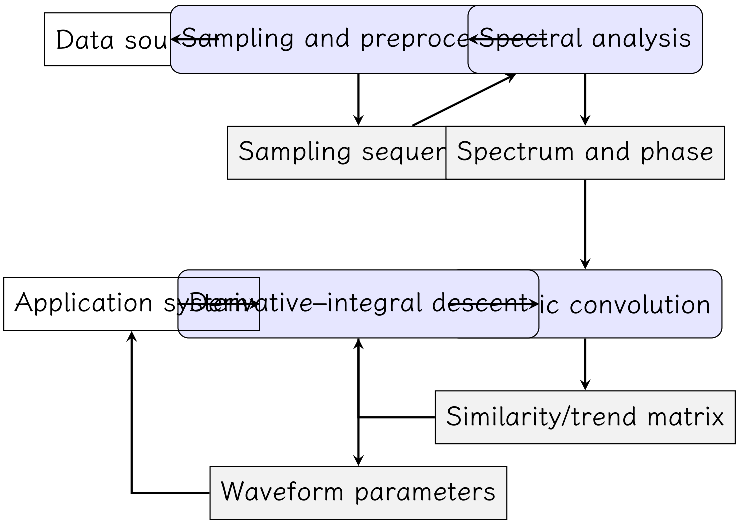

Data sou
Sampling and preproce
Spectral analysis
Sampling sequer
Spectrum and phase
Application system
Derivative–integral descent
ic convolution
Similarity/trend matrix
Waveform parameters

Figure 45.8: Data flow diagram (DFD). Data pass from the data source through sampling and preprocessing, spectral analysis, periodic convolution and derivative–integral descent to the application system, with intermediate stores for the sampling sequence, the spectrum and phase, the similarity/trend matrix and the Waveform parameters.

45.8 ER diagram

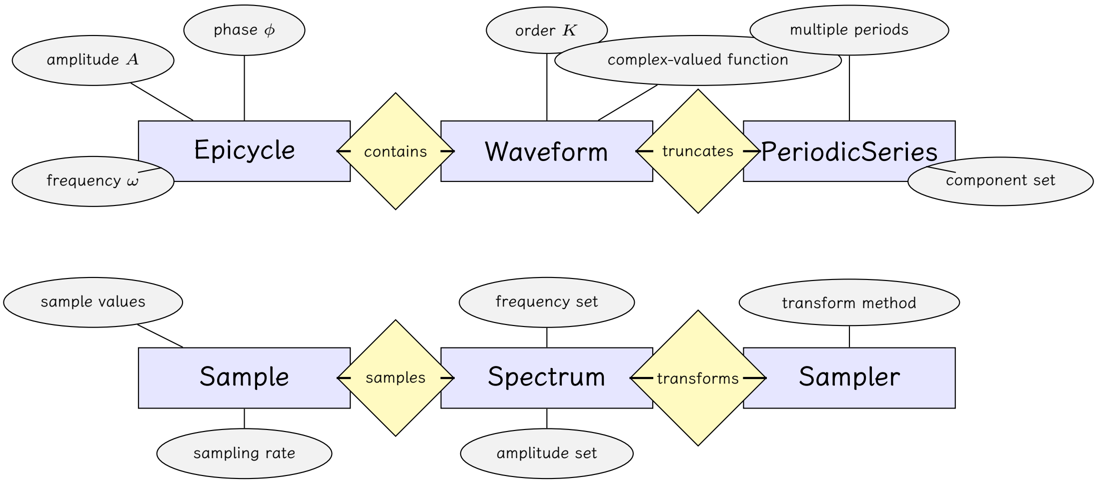

phase ϕ
amplitude A
frequency ω
Epicycle
contains
order K
complex-valued function
multiple periods
Waveform
truncates
PeriodicSeries
component set
sample values
Sample
sampling rate
samples
frequency set
Spectrum
amplitude set
transforms
transform method
Sampler

Figure 45.9: ER diagram. Six entities: Epicycle, Waveform (complex-valued function), PeriodicSeries (periodic series), Sample (sample), Spectrum (spectrum) and Sampler (transformer). The relations are contains, truncates, samples and transforms.

45.9 System architecture diagram

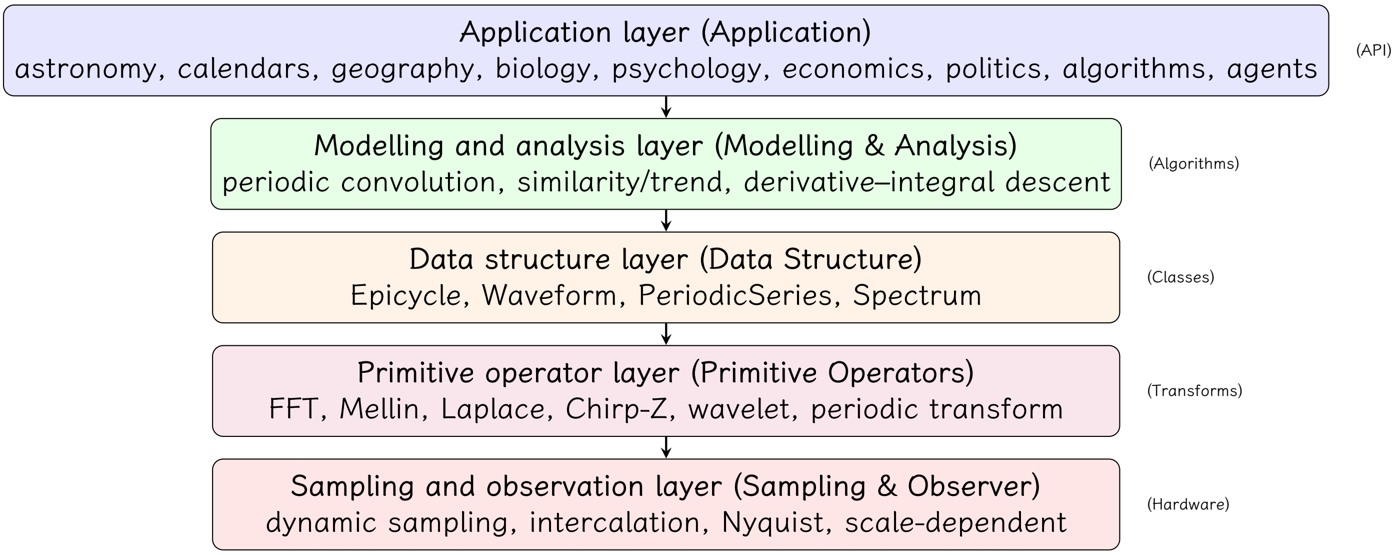

Application layer (Application)
astronomy, calendars, geography, biology, psychology, economics, politics, algorithms, agents
(API)
Modelling and analysis layer (Modelling & Analysis)
periodic convolution, similarity/trend, derivative–integral descent
(Algorithms)
Data structure layer (Data Structure)
Epicycle, Waveform, PeriodicSeries, Spectrum
(Classes)
Primitive operator layer (Primitive Operators)
FFT, Mellin, Laplace, Chirp-Z, wavelet, periodic transform
(Transforms)
Sampling and observation layer (Sampling & Observer)
dynamic sampling, intercalation, Nyquist, scale-dependent
(Hardware)

Figure 45.10: System architecture diagram. A five-layer architecture: application layer, modelling and analysis layer, data structure layer, primitive operator layer, and sampling and observation layer. Dependencies between layers run in one direction, from the bottom upwards.

45.10 Program representation of the core classes

This section maps the core concepts of periodicity mathematics to program representations. The core abstraction is the **Functor**—every smooth (C^∞) mathematical object that can be differentiated and integrated repeatedly. The class system is divided into three modules: the function class hierarchy, the series class hierarchy, and the integral transform and sampling classes.

45.10.1 The Functor class: abstract base class of smooth functions

Definition 45.1 (Functor class)**. Functor** *is the abstract base class of all smooth functions, that is, of all C^∞ functions. Its core methods are differentiation, integration, limit and evaluation:*

(i) `derivative(t)`: *returns the derivative of the Functor at t—still a Functor.*

(ii) `integral(t)`: *returns the indefinite integral (primitive function) of the Functor—still a Functor.*

(iii) `integral(a, b)`: *returns the definite integral (area) of the Functor over the interval $[a, b]$.*

(iv) `limit(t)`: *returns the limit of the Functor at t.*

(v) `evaluate(t)`: *returns the value of the Functor at t.*

The subclasses of Functor include: linear functions, quadratic functions, cubic functions, the sine function, the cosine function, the tangent function, Epicycle, Waveform, Convolution and Series.

Property 45.1 (Closure property of the Functor class)**.** *The Functor class is closed under the following operations:*

(i) **Differentiation**: *the derivative of a Functor is still a Functor.*

(ii) **Integration**: *the primitive function of a Functor is still a Functor.*

(iii) **Linear combination**: *a linear combination of two Functors is still a Functor.*

(iv) **Composition**: *the composition of two Functors is still a Functor.*

(v) **Convolution**: *the convolution of two Functors is still a Functor.*

Proof. (i)(ii) By the fundamental theorem of calculus, the derivative and the primitive function of a C^∞ function are still C^∞ functions. (iii) The derivative of a linear combination equals the linear combination of the derivatives, so it is still a Functor. (iv) By the chain rule, the derivative of a composite function exists and can be computed. (v) By the differentiation property of convolution $(f * g)' = f' * g = f * g'$, the convolution is still a C^∞ function. □

45.10.2 Function class hierarchy

Definition 45.2 (Basic function subclasses)**.** The basic subclasses of Functor include:

(i) **LinearFunction**: $f(t) = at + b$, whose derivative is the constant a and whose primitive function is $\frac{a}{2}t^2 + bt + C$.

(ii) **QuadraticFunction**: $f(t) = at^2 + bt + c$, whose derivative is $2at + b$.

(iii) **CubicFunction**: $f(t) = at^3 + bt^2 + ct + d$.

(iv) **SineFunction**: $f(t) = A\,\mathsf{sin}(\omega t + \phi)$, whose derivative is $A\omega\,\mathsf{cos}(\omega t + \phi)$.

(v) **CosineFunction**: $f(t) = A\,\mathsf{cos}(\omega t + \phi)$, whose derivative is $-A\omega\,\mathsf{sin}(\omega t + \phi)$.

(vi) **TangentFunction**: $f(t) = A\,\mathsf{tan}(\omega t + \phi)$, whose derivative is $A\omega\,\mathsf{sec}^2(\omega t + \phi)$.

Proposition 45.1 (Derivative closure of basic functions)**.** The closure relations of the basic function subclasses under differentiation are:

(i) $\frac{d}{dt}\mathsf{Sine}(\omega t) = \omega \cdot \mathsf{Cosine}(\omega t)$.

(ii) $\frac{d}{dt}\mathsf{Cosine}(\omega t) = -\omega \cdot \mathsf{Sine}(\omega t)$.

(iii) $\frac{d}{dt}\mathsf{Tangent}(\omega t) = \omega \cdot \mathsf{Secant}^2(\omega t)$.

The sine and the cosine are interconverted under differentiation, forming a closed orbit of period 2.

Proof. This follows by direct verification from the basic derivative formulae. Differentiating the sine function gives the cosine function, and differentiating the cosine function gives minus the sine function; after two differentiations one returns to the original function (up to a sign), forming a C_2 involution structure (Definition 1.11). ☐

Definition 45.3 (Epicycle class)**.** The **Epicycle** class inherits from Functor and represents a single epicycle:

$$\gamma(t) = Ae^{i(\omega t + \phi)}.$$

Its fields are the amplitude A, the angular frequency ω and the initial phase ϕ. The Epicycle is the most basic geometric unit in periodicity mathematics and is also a constituent element of the Waveform (its mathematical object being that of Definition 21.7).

Property 45.2 (Method semantics of the Epicycle class)**.** (i) `derivative(t)`: returns $\omega A e^{i(\omega t + \phi + \pi/2)}$—the amplitude is multiplied by ω and the phase is increased by $\pi/2$.

(ii) `integral(t)`: returns $\frac{A}{\omega}e^{i(\omega t + \phi - \pi/2)}$—the amplitude is divided by ω and the phase is decreased by $\pi/2$.

(iii) `integral(a, b)`: returns $\frac{A}{i\omega}(e^{i(\omega b + \phi)} - e^{i(\omega a + \phi)})$.

(iv) `evaluate(t)`: returns $Ae^{i(\omega t+\phi)}$.

(v) `convolute(Epicycle, Delay)`: returns the periodic convolution of two Epicycles under a given time delay (Definition 6.26).

(vi) `similarity(Epicycle)`: returns the similarity of two Epicycles (Definition 3.8).

(vii) `tendency(Epicycle)`: returns the trend of two Epicycles (Definition 3.16).

Proof. Derivative: $\frac{d}{dt}Ae^{i(\omega t+\phi)} = i\omega Ae^{i(\omega t+\phi)} = \omega Ae^{i(\omega t+\phi+\pi/2)}$. Integral: $\int Ae^{i(\omega t+\phi)}dt = \frac{A}{i\omega}e^{i(\omega t+\phi)} = \frac{A}{\omega}e^{i(\omega t+\phi-\pi/2)}$. The definite integral follows directly from the fundamental theorem of calculus. The convolution, similarity and trend are computed componentwise from the definitions of Definitions 3.8, 3.16 and 6.26. ☐

Definition 45.4 (Waveform class)**.** The **Waveform** class inherits from Functor and represents a complex-valued function composed of several Epicycles (its mathematical object being that of Definition 45.22):

$$\mathcal{W}(t) = \sum_{k=1}^{K} A_k e^{i(\omega_k t+\phi_k)}.$$

Internally it uses a double-ended queue to store Epicycle objects. A Waveform is both a Functor (differentiable and integrable) and a container of Epicycles (epicycles may be added and removed).

Property 45.3 (Method semantics of the Waveform class)**.** (i) Queue operations: `pushFront`, `removeFront`, `pushBack`, `removeBack`, `removeAt`—these support the dynamic addition and removal of epicycles.

(ii) `front()` / `back()`: returns the first or the last Epicycle.

(iii) Functor interface: `derivative(t)`, `integral(t)`, `integral(a,b)`, `limit(t)`, `evaluate(t)`—these act componentwise on each Epicycle.

(iv) `convolute(Waveform, Delay)`: returns the periodic convolution of two Waveforms.

(v) `similarity(Waveform)`: returns the similarity matrix of two Waveforms.

(vi) `tendency(Waveform)`: returns the trend matrix of two Waveforms.

Proof. The Functor methods of the Waveform are the componentwise extension of the corresponding methods of the Epicycle. Since the Waveform is a finite sum of Epicycles, the linear operations (differentiation, integration) may be carried out componentwise. The convolution, similarity and trend are computed componentwise from the definitions of Definitions 3.8, 3.16 and 6.26. ☐

Definition 45.5 (Convolution class)**.** The **Convolution** class inherits from Functor and represents the convolution of two Functors:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau.$$

A Convolution is itself a Functor—differentiable and integrable. Its mathematical object is that of Definition 6.19.

Property 45.4 (Differentiation property of the Convolution class).

$$(f * g)' = f' * g = f * g'.$$

The derivative of a convolution equals the convolution of the derivative of either factor with the other factor.

Proof. This follows directly from Proposition 6.25(v). This is the key property of convolution as a Functor—it guarantees the closure of convolution within the Functor class. ☐

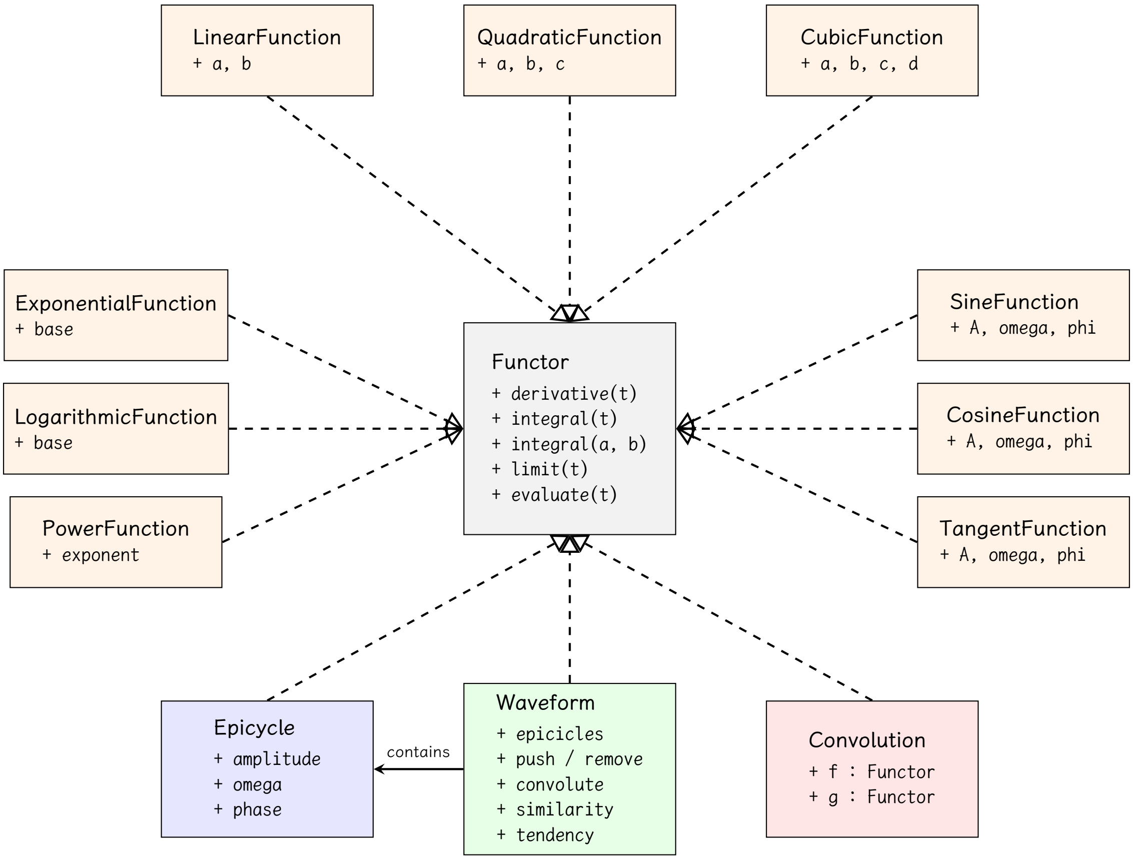

LinearFunction
+ a, b
QuadraticFunction
+ a, b, c
CubicFunction
+ a, b, c, d
ExponentialFunction
+ base
LogarithmicFunction
+ base
PowerFunction
+ exponent
Functor
+ derivative(t)
+ integral(t)
+ integral(a, b)
+ limit(t)
+ evaluate(t)
SineFunction
+ A, omega, phi
CosineFunction
+ A, omega, phi
TangentFunction
+ A, omega, phi
Epicycle
+ amplitude
+ omega
+ phase
contains
Waveform
+ epicicles
+ push / remove
+ convolute
+ similarity
+ tendency
Convolution
+ f : Functor
+ g : Functor

Figure 45.11: Function class hierarchy (UML class diagram). Functor is the abstract base class of all differentiable functions. Above are the algebraic functions (linear, quadratic, cubic), to the left are the transcendental functions (exponential, logarithmic, power), to the right are the trigonometric functions (sine, cosine, tangent), and below are the core classes of periodicity mathematics (Epicycle, Waveform, Convolution). The Waveform contains a set of Epicycles.

45.10.3 Series class hierarchy

Definition 45.6 (Series class)**.** The **Series** class inherits from Functor and represents a series of functions:

$$S(t) = \sum_{k=0}^{\infty} u_k(t).$$

Its fields are the sequence of terms `terms: Deque<Functor>` and the centre of expansion `center: Number`. The subclasses of Series include the Taylor series, the Fourier series, the Laurent series and the periodic series.

Property 45.5 (Method semantics of the Series class)**.** (i) `evaluate(t)`: returns the partial sum of the series at t (taking the first n terms).

(ii) `truncate(n)`: returns the series truncated to the first n terms.

(iii) `derivative(t)`: differentiates term by term and returns a new Series.

(iv) `integral(t)`: integrates term by term and returns a new Series.

(v) `convergeRadius()`: returns the radius of convergence of the series (for Taylor/Laurent series; for Fourier/periodic series it returns ∞ or is defined in terms of the period length).

Proof. The term-by-term differentiation and integration of the series are guaranteed by uniform convergence (Theorems 6.2 and 6.3). The radius of convergence is computed by the Cauchy–Hadamard formula (Theorem 6.4). □

Definition 45.7 (Program representation of the Taylor series)**.** The mathematical object represented by this class is the Taylor series defined in Definition 6.3; here only its program representation (class name and inheritance relation) is given, and the mathematical object is not redefined. **TaylorSeries** inherits from Series and represents the Taylor expansion of a function at t_0:

$$f(t) = \sum_{k=0}^{\infty} \frac{f^{(k)}(t_0)}{k!}(t - t_0)^k.$$

Definition 45.8 (Program representation of the Fourier series)**.** The mathematical object represented by this class is the Fourier series defined in Definition 6.5; here only its program representation (class name and inheritance relation) is given, and the mathematical object is not redefined. **FourierSeries** inherits from Series and represents the Fourier expansion of a periodic function:

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{i2\pi nt/T}.$$

Definition 45.9 (Program representation of the Laurent series)**.** The mathematical object represented by this class is the Laurent series defined in Definition 6.4; here only its program representation (class name and inheritance relation) is given, and the mathematical object is not redefined. **LaurentSeries** inherits from Series and represents the Laurent expansion of an analytic function:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n.$$

Definition 45.10 (Program representation of the periodic series). **PeriodicSeries** *inherits from Series and represents the periodic series expansion of a multi-periodic function (its mathematical object being that of* Definition 6.14*):*

$$f(t) = \sum_{p \in P} \sum_{n \in \mathbb{Z}} A_{p,n} e^{i(\omega_{p,n} t + \phi_{p,n})}.$$

Proposition 45.2 (Relation between Series and Waveform). *A finite truncation of a PeriodicSeries is equivalent to a Waveform. Specifically, after a PeriodicSeries has been truncated to finitely many terms, every element of its sequence of terms is an Epicycle, so it can be converted into a Waveform.*

Proof. Each term $A_{p,n} e^{i(\omega_{p,n} t + \phi_{p,n})}$ of the PeriodicSeries is an Epicycle (Definition 45.3). After truncation to finitely many terms, the sequence of terms constitutes a set of Epicycles, that is, a Waveform (Definition 45.4). The conversion is carried out by the `toWaveform()` method. ☐

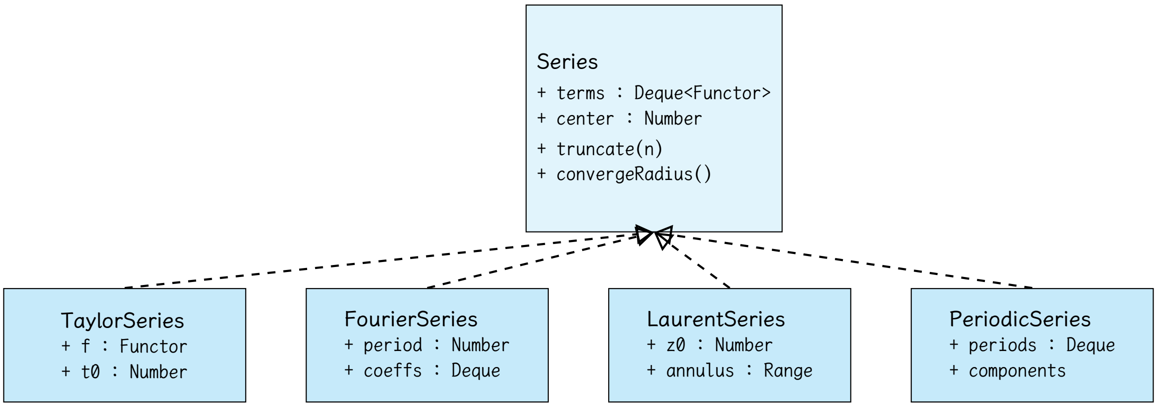

Series
+ terms : Deque<Functor>
+ center : Number
+ truncate(n)
+ convergeRadius()
TaylorSeries
+ f : Functor
+ t0 : Number
FourierSeries
+ period : Number
+ coeffs : Deque
LaurentSeries
+ z0 : Number
+ annulus : Range
PeriodicSeries
+ periods : Deque
+ components

Figure 45.12: Series class hierarchy (UML class diagram). Series inherits from Functor, and its subclasses include the Taylor series, the Fourier series, the Laurent series and the periodic series.

45.10.4 Integral transform and sampling classes

Definition 45.11 (Spectrum class). The **Spectrum** class represents a spectrum—a double-ended queue composed of several Spectral components. The `toWaveform()` method converts a spectrum into a Waveform.

Definition 45.12 (Spectral class). The **Spectral** class represents a single spectral component, with the fields frequency, amplitude and phase. The `toEpicycle()` method converts a spectral component into an epicycle.

Definition 45.13 (Sample class). The **Sample** class represents sampled data, with the fields sequence of sample values, sampling rate and duration. The `toWaveform()` method converts sampled data into a Waveform.

Definition 45.14 (Sampler class). The **Sampler** class is a static utility class operating on Waveforms and Spectra. Its methods include:

(i) Sampling and resampling: `sample(Waveform)`, `resample(Sample)`.

(ii) The Fourier transform family: `fourierTransform`, `radix2`, `radix3`, `radix5`, `radix7`, `blueStein`, `fastFourierC`.

(iii) Integral transforms: `meilinTransform`, `laplacianTransform`, `chirpZTransform`, `waveletTransform`.

Remark 45.4 (Correspondence between the methods of the Sampler class and mathematical transforms). Each method of the Sampler class corresponds to a concrete mathematical transform, whose definition may be found in the standard literature on harmonic analysis:

- `fourierTransform` corresponds to the Fourier transform (Definition 22.6);
- `radix2/3/5/7` corresponds to the fast Fourier transform (Theorems 22.4 and 22.5);
- `blueStein` corresponds to the Bluestein algorithm (Proposition 22.6);
- `meilinTransform` corresponds to the Mellin transform (Definition 22.10);
- `laplacianTransform` corresponds to the Laplace transform (Definition 22.9);
- `chirpZTransform` corresponds to the Chirp-Z transform;
- `waveletTransform` corresponds to the wavelet transform (Definition 22.11);
- `periodicTransform` corresponds to the periodic transform (Definition 6.18).

This remark is an interface convention and constitutes no new mathematical proposition. It states that each method of the Sampler class is a program wrapper around an existing mathematical transform; the concrete definitions and convergence conditions are given in the corresponding chapters.

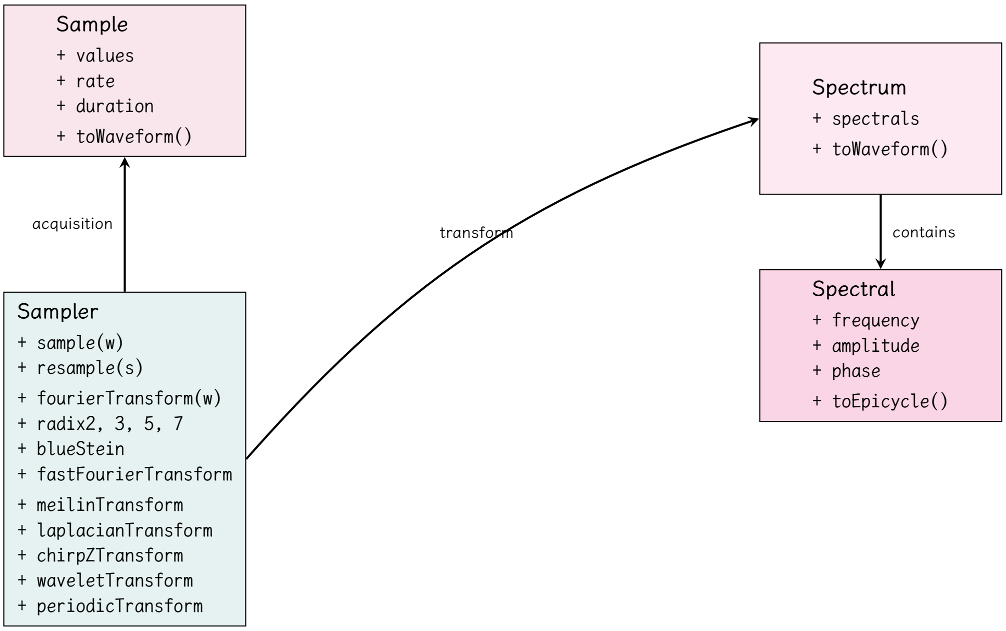

Sample
+ values
+ rate
+ duration
+ toWaveform()
acquisition
Sampler
+ sample(w)
+ resample(s)
+ fourierTransform(w)
+ radix2, 3, 5, 7
+ blueStein
+ fastFourierTransform
+ meilinTransform
+ laplacianTransform
+ chirpZTransform
+ waveletTransform
+ periodicTransform
transform
Spectrum
+ spectrals
+ toWaveform()
contains
Spectral
+ frequency
+ amplitude
+ phase
+ toEpicycle()

Figure 45.13: Integral transform and sampling classes (UML class diagram). The Sampler is responsible for sampling and the various integral transforms, Sample is the result of sampling, and Spectrum is the spectrum (composed of several Spectral components).

45.10.5 Overall module relation diagram

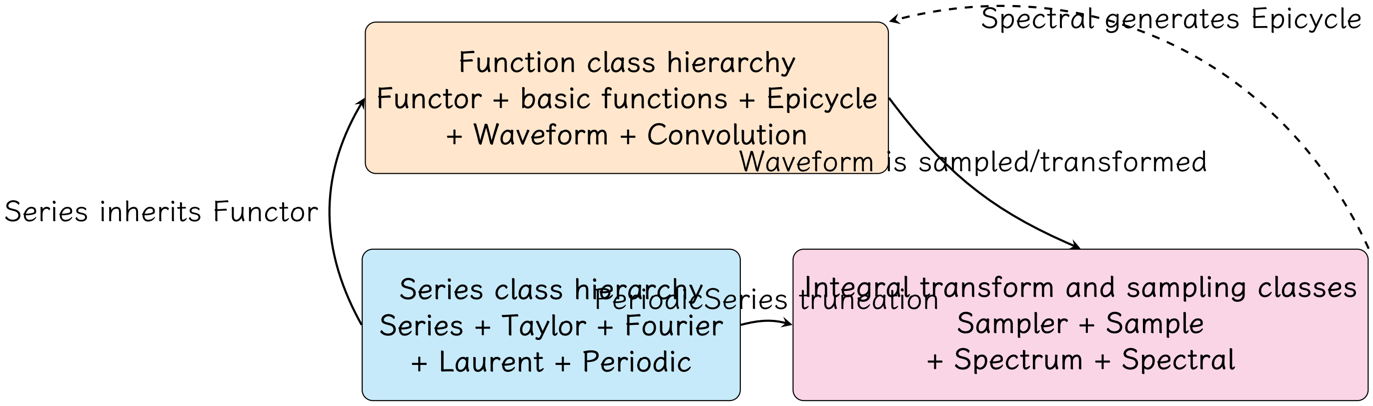

Spectral generates Epicycle
Function class hierarchy
Functor + basic functions + Epicycle
+ Waveform + Convolution
Waveform is sampled/transformed
Series inherits Functor
Series class hierarchy
Series + Taylor + Fourier
+ Laurent + Periodic
PeriodicSeries truncation
Integral transform and sampling classes
Sampler + Sample
+ Spectrum + Spectral

Figure 45.14: Overall module relation diagram. The three modules are interconnected by the relations of inheritance, truncation, sampling/transformation and generation.

45.11 UML object diagram: a concrete instance

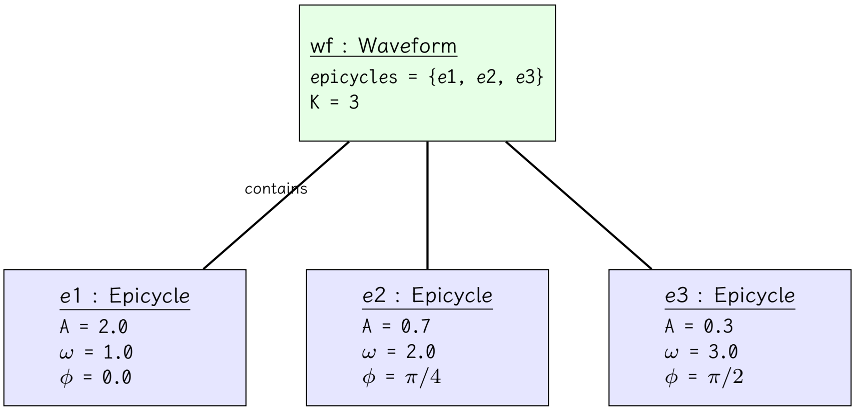

wf : Waveform
epicycles = {e1, e2, e3}
K = 3
contains
e1 : Epicycle
A = 2.0
ω = 1.0
ϕ = 0.0
e2 : Epicycle
A = 0.7
ω = 2.0
ϕ = π/4
e3 : Epicycle
A = 0.3
ω = 3.0
ϕ = π/2

Figure 45.15: UML object diagram. A Waveform instance `wf` containing three Epicycle instances $e1$, $e2$ and $e3$. Each object displays its run-time field values, and this is a concrete instance of the class diagram (Fig. 45.11).

45.12 UML sequence diagram: from data to Waveform

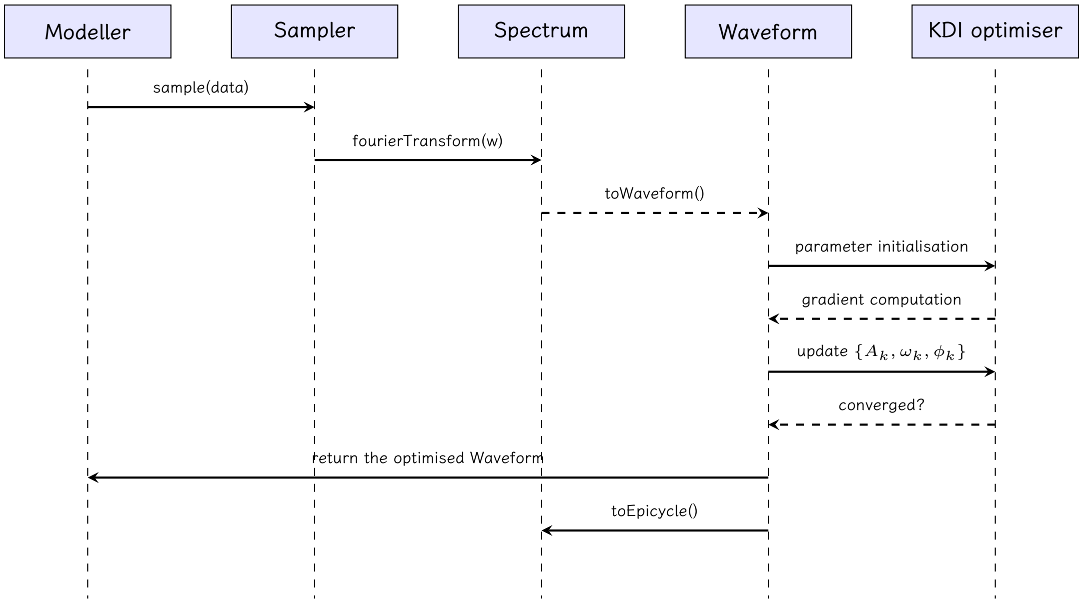

Modeller
Sampler
Spectrum
Waveform
KDI optimiser
sample(data)
fourierTransform(w)
toWaveform()
parameter initialisation
gradient computation
update $\{A_k, \omega_k, \phi_k\}$
converged?
return the optimised Waveform
toEpicycle()

Figure 45.16: UML sequence diagram. The modeller initiates a sampling request, the Sampler completes the spectral analysis, the Spectrum is converted into a Waveform, the KDI optimiser iteratively updates the parameters until convergence, and finally the optimised Waveform and its Epicycle components are returned.

45.13 Modelling: from real-world problems to periodic mathematical models

Definition 45.15 (Steps of modelling)**.** *Modelling is the systematic procedure that abstracts a real-world problem into a periodic mathematical model:*

(i) **Identify the transformation group**: *determine the basic transformations present in the system and their symmetry group.*

(ii) **Identify the invariants**: *determine the invariants of the system under the action of the transformation group.*

(iii) **Construct periodic functions**: *express the state variables of the system as periodic functions (subclasses of* Functor*).*

(iv) **Construct the epicycle model**: *expand the periodic functions into a nested epicycle system (*Waveform*).*

(v) **Determine the coupling relations**: *determine the periodic-convolution coupling between the subsystems (*Convolution*).*

Remark 45.5 (Completeness of the modelling steps (empirical observation / research programme))*.* The five steps above cover the necessary stages from a real-world system to a periodic mathematical model; this is an **empirical observation / research programme**, not a proved theorem: 'completeness' is not defined relative to any class of models, and no verifiable criterion is supplied (for example, 'given the state–observation map of any real-world system, the composite of the five steps yields an isomorphic Waveform representation'), so it is used here only as a methodological convention. The basis of each step is as follows: steps (i)(ii) determine the symmetry and the invariants of the system, where **symmetry is a necessary condition for periodic structure** (period ⇒ symmetry, see Corollary 1.4), whilst the identification of invariants is a tool for describing the stratification between orbits and is **not a precondition for the existence of periods** (there exist uniformly periodic systems all of whose conserved quantities are constant); step (iii) expresses the system state as a periodic function (Functor), which is the core of mathematical modelling, see Definition 6.14; step (iv) expands the periodic function into epicycles (Waveform), which is the key to the periodic series representation; step (v) determines the periodic-convolution coupling between subsystems, see Definition 6.19.

45.14 Data: acquisition and preprocessing

Definition 45.16 (Periodic mathematical framework for data acquisition)**.** *Data acquisition uses the* **scale-dependent sampling** *model: the sampling rate $R(r, \sigma)$ and the number of sampling points $N(r, \sigma)$ are determined by the observer scale σ and the observation distance r; their definition and monotonicity are given in the 'Discrete Sampling and Intercalation of Periodic Signals' chapter, Definition 21.6, Property 21.2. The sampling theorem (Theorem 21.3) guarantees that, when* the sampling rate satisfies the Nyquist condition, a periodic function can be reconstructed uniquely from its sampled values.

Remark 45.6 (Monotonicity of dynamic sampling (citation)). This proposition is identical with Property 21.2 of the 'Discrete Sampling and Intercalation of Periodic Signals' chapter: for fixed $\sigma > 0$, R and N are strictly decreasing in the distance r; for fixed $r \geq 0$, R and N are strictly increasing in the scale σ. Both the statement and the proof are to be found there and are not repeated here.

45.15 Analysis: periodic convolution and spectral decomposition

Definition 45.17 (The complete toolchain for analysis). (i) **Classical integral transforms**: the Fourier, wavelet, Mellin, Laplace and Chirp-Z transforms.

(ii) **Similarity transform** ($\alpha = 0$): determined by the phase similarity alone (Definition 3.8).

(iii) **Trend transform** ($\alpha = 1$): determined jointly by the amplitude and the phase similarity (Definition 3.16).

(iv) **Weight kernels**: kernel functions such as the Gaussian, the Mexican hat and the Cauchy kernel (Definition 21.15).

Proposition 45.3 (Hierarchy of the analysis toolchain). *The four analysis tools form an inclusion relation with respect to their* **range of information retention** *(each later one retains all the information of the former and adds a new dimension):*

$$\text{classical transform} \subset \text{trend transform} \subset \text{similarity transform} \subset \text{weight kernel}$$

Proof. (Level 1 → level 2) A classical integral transform gives only the spectral coefficients c_k of a single function; the trend transform retains the amplitude and phase interaction $\mathbb{T}_{mn} = A_n B_m e^{i(\phi_n - \psi_m)}$ of two functions, and its diagonal terms degenerate to the classical transform (Proposition 41.3).

(Level 2 → level 3) The trend transform contains the amplitude product $A_n B_m$; the similarity transform discards the amplitude information and retains only the phase relation $\mathbb{P}_{mn} = e^{i(\phi_n - \psi_m)}$—it is the special case of the trend transform in which the amplitudes take the values $A_n = B_m = 1$.

(Level 3 → level 4) The similarity transform gives the same phase kernel for every frequency pair (n, m); the weight kernel weights the frequency pairs through a window function W, thereby realising a **non-uniform** kernel strength on the individual frequency pairs—a degree of freedom that the similarity transform does not possess.

Hence the ranges of information retention of the four form a strict inclusion relation. ☐

45.16 Processing: window functions and derivative–integral descent optimisation

Definition 45.18 (Tools for processing). (i) **Window-function smoothing**: multi-scale smoothing using Gaussian kernels, Cauchy kernels and the like (Definition 21.15).

(ii) **Turning-point detection**: using the criterion of directional change of the instantaneous limit angle (Definitions 4.23 and 23.14).

(iii) **Refraction-rate computation**: the difference of the sine trends on the two sides of a turning point (Definition 23.15).

(iv) **Derivative–integral descent optimisation**: the kernel-function derivative–integral descent method (Definition 17.19).

Proposition 45.4 (Division of labour among the processing tools). *Window-function smoothing is responsible for noise suppression, turning-point detection for the identification of direction reversals, refraction-rate computation for the quantification of reversal strength, and derivative–integral descent optimisation for the fine adjustment of parameters. The four tools work together to complete the whole processing chain from raw data to a refined signal.*

Proof. Each tool addresses one particular aspect of signal processing:

- Window-function smoothing (Definition 21.15): suppresses high-frequency noise with $|\omega| > 1/\Delta$ by convolution with $P * K_\Delta$ (Lemma 39.4);
- Turning-point detection (Definition 23.14): identifies abrupt changes of direction through $|\theta_t - \theta_{t-1}| > \tau$;
- Refraction-rate computation (Definition 23.15): quantifies the reversal strength through $R_{\text{turn}} = |\sin\theta_1 - \sin\theta_2|$;
- Derivative–integral descent (Definition 17.19): exploits first-derivative and integral information through the fusion of several kernel functions.

The four are used in the order 'smoothing → detection → quantification → optimisation' to form a complete processing chain. □

45.17 Evaluation: model selection and validation

Definition 45.19 (Criteria for evaluation). (i) **Information criteria**: AIC, BIC.

(ii) **Cross-validation**: k-fold cross-validation.

(iii) **Residual diagnosis**: multi-kernel-function diagnosis (Proposition 23.7).

(iv) **Bayesian model selection**: comparison of posterior probabilities (Definition 14.22).

Proposition 45.5 (The fit–complexity balance of BIC)**.** In the BIC criterion $\mathrm{BIC}(K) = n\ln(\mathrm{RSS}/n) + K\ln n$, the first term measures the goodness of fit and the second term penalises the model complexity. As n increases, the relative weight of the complexity penalty increases.

Proof. The first term $n\ln(\mathrm{RSS}/n)$ decreases as the residual decreases. The second term $K\ln n$ increases linearly in the epicycle order K. As the sample size n increases, $\ln n$ increases, and both the absolute value and the relative weight of the complexity penalty grow, so that with large samples BIC tends to favour a parsimonious model. The rigorous choice of the optimal K is guaranteed by Lemma 45.1. ☐

45.18 Period: identification and validation

Definition 45.20 (Criteria for period identification)**.**

(i) **Spectral peaks**: significant peaks of the Fourier coefficients (Proposition 23.2).

(ii) **Autocorrelation peaks**: the peaks of the cross-correlation at the time delay T.

(iii) **Phase consistency**: the closure of the phase operator (Definition 20.6).

(iv) **Significance test**: the Fisher test (Proposition 23.2).

Proposition 45.6 (Threshold of the Fisher test)**.** The Fisher test statistic and its significance threshold are stated for the first time in Chapter 23, Section 23.5, at Proposition 23.2; the present passage reuses the same conclusion within the modelling procedure. Let the values of the periodogram at the $m = \lfloor (n-1)/2 \rfloor$ positive frequency points be $I_1, \ldots, I_m$. The Fisher test statistic

$$g = \frac{\max_k I_k}{\sum_{k=1}^{m} I_k}$$

has significance threshold $g_\alpha = 1 - (\alpha/m)^{1/(m-1)}$.

Proof. Under white noise the periodogram ordinates are independent and identically distributed with an exponential distribution, and the normalised ratio follows a $\mathrm{Beta}(1, m-1)$ distribution. The union bound gives $\mathbb{P}(g > g_0) \le m(1-g_0)^{m-1}$. Setting the right-hand side equal to α and solving yields g_α. ☐

45.19 Epicycle: nested structure and parameter estimation

Definition 45.21 (Epicycle modelling)**.** The K-level nested epicycle system (Waveform):

$$X(t) = \sum_{k=1}^{K} A_k e^{i(\omega_k t + \phi_k)} + \varepsilon(t).$$

The parameter estimation procedure: an initial estimate by FFT, then fine optimisation by derivative–integral descent, and finally BIC model selection.

Property 45.6 (Basic properties of the epicycle model). (i) **Linear superposition**: the sum of two epicycle models is still an epicycle model.

(ii) **Parameter identifiability**: when the frequencies are pairwise distinct, the amplitudes and phases are uniquely determined by the Bohr mean.

(iii) **Nesting invariance**: the model is invariant in form under time translation.

Proof. (i) The term-by-term merger of two finite sums is still a finite sum (Property 42.1(i)).

(ii) By Bohr's uniqueness theorem (Theorem 6.16), when the frequencies are distinct the Bohr mean extracts the coefficients of the individual components uniquely.

(iii) The time translation $t \mapsto t + t_0$ only changes the phase to $\phi_k + \omega_k t_0$, leaving the form unchanged (Property 42.1(ii)). □

45.20 Waveform: epicycles as complex-valued functions

Definition 45.22 (Waveform: epicycles as complex-valued functions). A **Waveform** is a complex-valued function (a subclass of Functor) that maps time to a point of the complex plane:

$$\mathcal{W} : \mathbb{R} \to \mathbb{C}, \quad \mathcal{W}(t) = \sum_{k=1}^{K} A_k e^{i(\omega_k t + \phi_k)}.$$

Here $A_k \in \mathbb{R}^+$ is the amplitude, $\omega_k \in \mathbb{R}$ is the angular frequency and $\phi_k \in [0, 2\pi)$ is the initial phase.

Property 45.7 (Basic properties of Waveform). (i) **Closure**: the sum of two Waveforms is still a Waveform.

(ii) **Convolution closure**: the periodic convolution of two Waveforms is still a Waveform.

(iii) **Time translation**: $\mathcal{W}(t + t_0)$ corresponds to a phase translation.

(iv) **Scaling**: $\mathcal{W}(\lambda t)$ corresponds to a frequency scaling.

(v) **Rotation**: $e^{i\theta}\mathcal{W}(t)$ corresponds to a global phase rotation.

Proof. (i) This is a direct extension of Property 42.1(i).

(ii) The periodic convolution of two Waveforms (Definition 6.26) gives the frequency composition $\omega_k + \nu_l$ and the phase composition $\phi_k + \psi_l$, which is still a sum of complex exponentials of the same form.

(iii) By Property 42.1(ii).

(iv) By Property 42.1(iii).

(v) By Property 42.1(iv). □

Definition 45.23 (Complex exponential activation function: the activation-function implementation of the Waveform)**.** The activation-function implementation of the Waveform realises the nested epicycle system as a learnable complex-valued function:

$$\mathcal{E}(x) = \sum_{k=1}^{N} A_k e^{i(\omega_k x + \phi_k)}.$$

The parameters A_k, ω_k and ϕ_k can be learned by backpropagation or by derivative–integral descent optimisation. Its mathematical object is the epicycle activation function defined in Chapter 42, Definition 42.4.

Proposition 45.7 (Derivative–integral descent: the optimiser for the Waveform parameters)**.** The derivative–integral descent strategy for Waveform parameter optimisation:

Parameter	Kernel function	Operator bias	Optimisation characteristic
Amplitude A_k	Gaussian kernel	gradient (centre of symmetry 0)	smooth adjustment
Frequency ω_k	Mexican hat kernel	curl (centre of symmetry i)	curvature-aware
Phase ϕ_k	Hilbert kernel	curl (centre of symmetry i)	jump detection

Proof. Gaussian-kernel smoothing suppresses high-frequency noise and is suited to the smooth adjustment of the amplitude (Lemma 17.3). The Mexican hat kernel is the second derivative of the Gaussian kernel, and its band-pass character highlights the frequency curvature (Definition 21.15). The Hilbert kernel supplies a 90° phase rotation and is suited to phase-jump detection (Definition 22.12). □

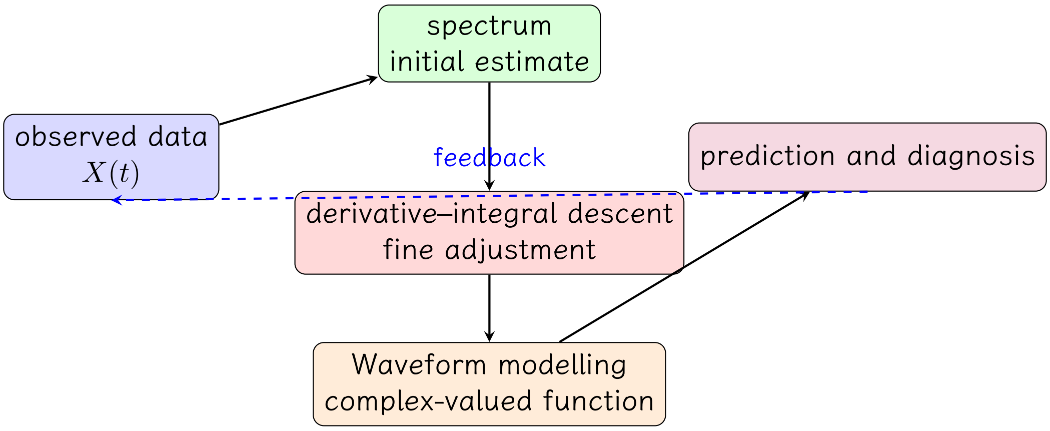

spectrum
initial estimate
observed data
$X(t)$
feedback
prediction and diagnosis
derivative–integral descent
fine adjustment
Waveform modelling
complex-valued function

Figure 45.17: Cooperative pipeline of the Waveform and derivative–integral descent.

Remark 45.7 (Unity with the whole book). *The modelling and analysis framework established in this chapter is the methodological integration of the whole theory of periodicity mathematics. The core class system takes Functor as its unifying abstraction and is divided into three modules: the function class hierarchy (Functor + basic functions + Epicycle + Waveform + Convolution), the series class hierarchy (Series + Taylor + Fourier + Laurent + Periodic), and the integral transform and sampling classes (Sampler + Sample + Spectrum + Spectral).*

The Waveform as the complex-valued function of epicycles is the convergence point of this methodological framework. The joint use of the Waveform, its complex exponential activation-function implementation and derivative–integral descent optimisation makes the complete theoretical system of periodicity mathematics directly applicable to such modern data-science tasks as neural networks, time series analysis and signal processing. Periodic convolution provides a unified operator framework for the interaction between Waveforms.

Interface with the next chapter: *the Waveform definition of this chapter (Definition 45.22) and the Epicycle class (Definition 45.3) provide the mathematical object on which the 'Epicycle Activation Function and Function Approximation' chapter is based; the CEAF definition (Definition 45.23) and Definition 42.4 of that chapter are the same mathematical object expressed in two contexts.*

Seven-theme comparison table for this chapter

Theme	Corresponding content in this chapter
Translation	The advancement of the time series in data acquisition and dynamic sampling (Definition 45.16); the time-translation invariance of the Waveform (Property 45.7(iii))
Scaling	The complexity penalty $K \ln n$ of BIC (Lemma 45.1); multi-scale window functions (Definition 45.18); the frequency scaling of the Waveform (Property 45.7(iv))
Rotation	The phase ϕ_k of the epicycle model (Definition 45.21); the complex phase representation of the Waveform (Definition 45.22); global phase rotation (Property 45.7(v))
Balance	The fit–complexity balance of BIC (Proposition 45.5); the parameter uniqueness of the Bohr mean (Property 45.6(ii))
Symmetry	The hierarchical symmetry of epicycle nesting; the time-translation invariance of the model (Property 45.6(iii))
Closure	The feedback-iteration closure of the methodological procedure (Lemma 45.1); the closure property of the Functor class (Property 45.1)
Period	Spectral peak identification (Definition 45.20); epicycle parameter estimation (Definition 45.21); the convolution closure of the Waveform (Property 45.7(ii))

Exercises

Exercise 1 Prove the **feedback closure of the methodological pipeline** of this chapter (45.1). Hint: first write down the decision mapping and the evolution group of this chapter's target, then apply the proof structure of the general criterion given in the 'Periodic Systems' chapter.

Exercise 2 Verify that the hypotheses of the **derivative closure of basic functions** (45.1) hold for the domain objects of this chapter, and give a counterexample in which the hypotheses fail.

Exercise 3 Examine the behaviour of the **relation between Series and Waveform** (45.2) under parameter degeneracy, and explain how the conclusion degenerates accordingly.

Exercise 4 Discuss the **completeness of the modelling steps** as a methodological convention (an **empirical observation / research programme**; see the remark at 45.5) and its relation to the corresponding general conclusion of the

'Periodic Systems' chapter (whether it agrees with it, is a special case of it, or requires additional conditions).

Exercise 5 Using the seven-theme comparison table of this chapter, identify respectively the concrete carriers of this chapter under the three themes **translation**, **scaling** and **rotation**, and give one example of domain data for each.

Exercise 6 Synthesis: restate the diagnostic procedure of this chapter using the five-step standard interface of the 'Periodic Systems' chapter (Table 24.1), noting one by one the conclusion of this chapter on which each step depends: 45.1, 45.1, 45.2, 45.5 (an **empirical observation / research programme**).

Chapter 46

Summary and Outlook

This book takes 'symmetry → invariant → closed orbit → period' as its guiding principle, and constructs a mathematical framework that unifies periodic phenomena starting from first principles. The core methodology of the book rests on four pillars: **group representation theory** supplies the algebraic foundation of periodic structure, **periodic convolution** supplies a unified computational framework for phase interaction across periods, **the window-function and refractive-index system** supplies turning-point diagnostic tools based on similarity and tendency, and **the complex exponential activation function (CEAF) together with kernel-smoothed derivative-integral descent** supplies learnable nonlinearities with intrinsic periodic structure and multiscale optimisation algorithms.

The journey of the book begins with the mathematical characterisation of three fundamental centres of symmetry—0, 1 and i—passes through group representation theory, vector analysis, harmonic analysis, periodic manifolds and periodic categories, proceeds by way of the construction of periodic convolution and of the window-function–refractive-index system, and ends with systematic applications in such domains as astronomy, calendars, geography, physics, biology, psychology, economics, politics, cryptography, agents and data analysis, thereby forming a self-consistent theoretical system that runs from discrete information processing to continuous physical laws and from physical dynamics to life and society.

Remark 46.1 (Positioning of this chapter). This chapter is the closing chapter of the book. The preceding 45 chapters have respectively established: one main chain (symmetry → invariant → closed orbit → period), three groups of support (group representation theory, periodic convolution, the window-function and refractive-index system, and complex exponential activation with kernel-smoothed derivative-integral descent), and instantiations in more than twenty domains. The task of this chapter is not to introduce new tools but to undertake a **horizontal comparison**: to place the main chain, the four pillars and the seven themes of the various chapters of the book side by side in one set of tables, and to point out which conclusions have already been proved rigorously, which still belong to a research programme, and which are unsolved problems. The reader may read this chapter first in order to obtain a global map, and then return to the individual chapters.

Positioning of the seven themes in this chapter: **translation**—the book takes the translation group as the most fundamental evolution group; **scaling**—scaling and logarithmic coordinates run through the number systems, harmonic analysis and time series; **rotation**—phase and roots of unity are the core carriers of periodic structure; **conservation**—the correspondence between symmetry and conservation is the logical hub of the book; **symmetry**—the unity of the group-action framework; **closure**—closed orbits are the geometric criterion of periodicity; **pe-**

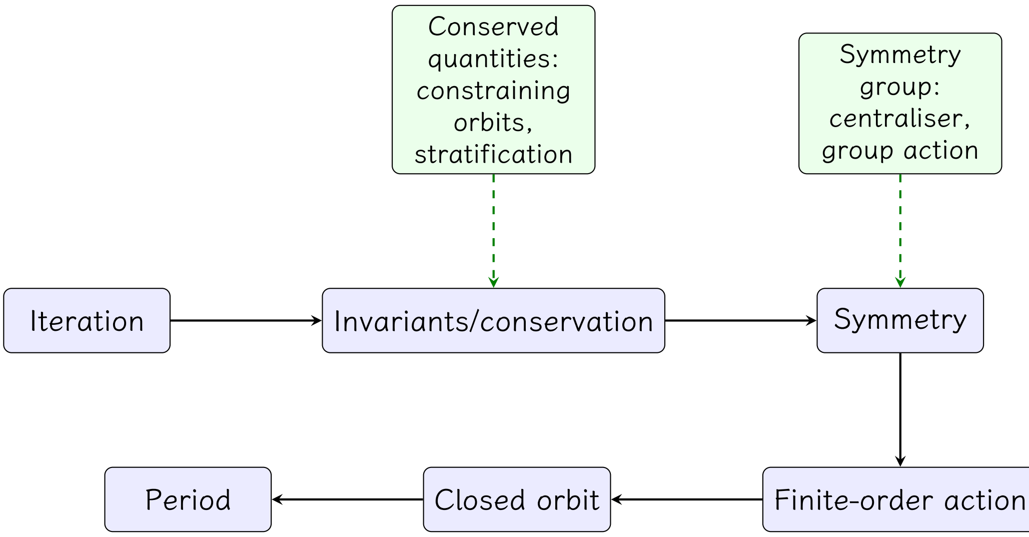
riod—the theme of the book itself.

46.1 Looking back: one main chain and three groups of support

If the theoretical through-line of the whole book were to be summarised in a single sentence, it would be this:

> **Finite-order action gives periodicity; conservation and symmetry help to discover, constrain and explain periodicity; periodic convolution, window functions–refractive index, and CEAF–KDI supply computational, diagnostic and learning tools respectively.**

This main line is established rigorously in the 'Operations and Periodicity' chapter: iteration is the starting point, invariants and conservation characterise the stratification between orbits, symmetry is a transformation that commutes with the evolution, finite-order action directly produces uniformly closed orbits, and a closed orbit is equivalent to pointwise periodicity. A conserved quantity is not a necessary condition for periodicity, nor is it in general a sufficient one; in conservative systems it is associated with continuous symmetries through Noether's theorem and supplies constraints on and explanations of periodic structure, but the algebraic hub that directly produces periodicity is finite-order action.


Conserved quantities: constraining orbits, stratification
Symmetry group: centraliser, group action
Iteration
Invariants/conservation
Symmetry
Period
Closed orbit
Finite-order action

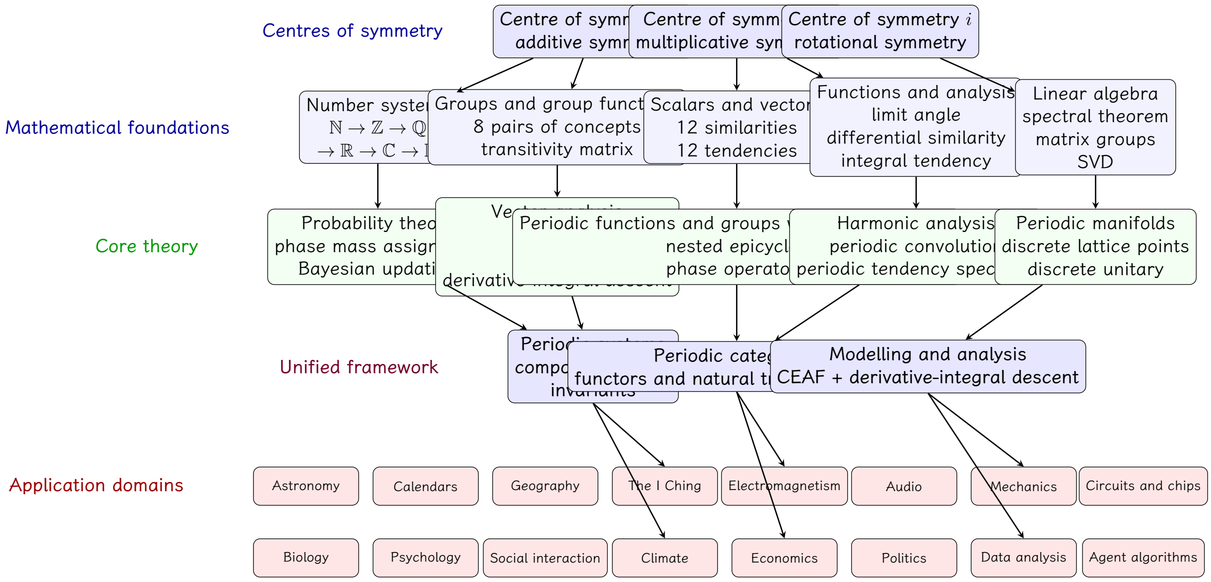
Conservation and symmetry are auxiliary tools, not unavoidable links in the chain of periodicity

Figure 46.1: The main logical chain of the book: iteration → invariants/conservation → symmetry → finite-order action → closed orbit → period. Conservation and symmetry are attached by dashed lines, indicating their auxiliary status.

This main chain unfolds downwards into three levels: the **algebraic level** studies finite-order action, group representations and the automorphism tower; the

geometric level studies manifolds, fibre bundles, periodic manifolds and closed orbits; and the **analytic level** studies series, convolution, harmonic analysis, sampling and intercalation. Together the three support the interdisciplinary diagnosis of the application level.

46.2 The architecture of the book's theoretical system


Centres of symmetry
Centre of symr
additive symr
Centre of symm
multiplicative syn
Centre of symmetry i
rotational symmetry
Mathematical foundations
Number syster
$\mathbb{N} \to \mathbb{Z} \to \mathbb{Q}$
$\to \mathbb{R} \to \mathbb{C} \to$
Groups and group funct
8 pairs of concepts
transitivity matrix
Scalars and vector
12 similarities
12 tendencies
Functions and analysis
limit angle
differential similarity
integral tendency
Linear algebra
spectral theorem
matrix groups
SVD
Core theory
Probability theo
phase mass assigr
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Ve
derivativ
Periodic functions and groups
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Harmonic analysis
periodic convolutio
periodic tendency spec
Periodic manifolds
discrete lattice points
discrete unitary
Unified framework
Perio
compo
inv
Periodic cate
functors and natural t
Modelling and analysis
CEAF + derivative-integral descent
Application domains
Astronomy
Calendars
Geography
The I Ching
Electromagnetism
Audio
Mechanics
Circuits and chips
Biology
Psychology
Social interaction
Climate
Economics
Politics
Data analysis
Agent algorithms

Figure 46.2: Architecture diagram of the theoretical system of the whole book on periodic mathematics. Starting from the centres of symmetry, passing through the mathematical foundations and the core theory, unified in periodic systems, periodic categories and the methodology of modelling and analysis, and finally applied in the various disciplines.

From the point of view of chapter structure, the book may be divided into three parts, with the present chapter serving as its close:

Part	Main content
Mathematical foundations and generators	Foundations of periodicity, number systems, scalars and vectors, functions and limits, differential equations and complex functions, series and convolution
Core structures, tools and methods	Linear algebra, foundations of manifolds and fibre bundles, periodic manifolds, periodic categories, probability theory, vector analysis, groups and group functions, higher-order groups and higher-order functions, periodic functions and groups with periodic orbits, discrete sampling approximation and intercalation, harmonic analysis, time series analysis, periodic systems
Interdisciplinary applications and diagnosis	Probability, cryptography, astronomy, calendars, geography, the I Ching, electromagnetism, audio, mechanics, circuits and chips, biology, psychology, social interaction, climate, economics, politics, data analysis, agents, algorithms, modelling and analysis

Among these, the 'Periodic Systems' chapter is the bridging chapter that leads from pure mathematics to applied mathematics. It provides the five-step interface by which an arbitrary target is connected to the framework of periodic mathematics: **decision** (whether the target is periodic, and its period type), **construction** (building the amplitude series, the frequency series and the phase series from the turning points, and assembling the epicycle group), **adjustment** (dynamically adjusting the number and the parameters of the epicycles), **convolution** (amplitude convolution, phase convolution and frequency convolution), and **integration** (connecting to the tool chains of sampling, harmonic analysis and time series analysis). Every subsequent application chapter develops along this interface.

46.3 Relations among the four pillars

The four pillars are not independent of one another. Group representation theory answers 'what is the algebraic form of periodic structure'; periodic convolution answers 'how do different periods interact with one another'; window functions and the refractive index answer 'how are turning points, reversals and leakage diagnosed from data'; and CEAF and KDI answer 'how is periodic structure turned into a learnable, optimisable computational unit'. The four are linked together through the common objects 'phase operator', 'epicycle group', 'periodic series' and 'Waveform'.

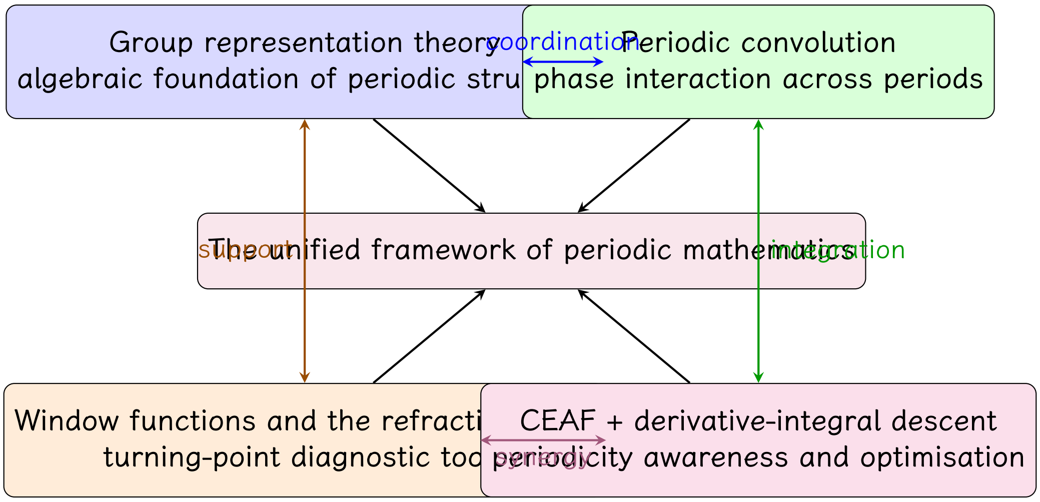

Group representation theory
algebraic foundation of periodic stru
coordination
Periodic convolution
phase interaction across periods
support
The unified framework of periodic mathematics
integration
Window functions and the refracti
turning-point diagnostic too
CEAF + derivative-integral descent
periodicity awareness and optimisation
synergy

Figure 46.3: Relations among the four pillars of periodic mathematics.

46.4 Periodic convolution: a unified framework for phase interaction across periods

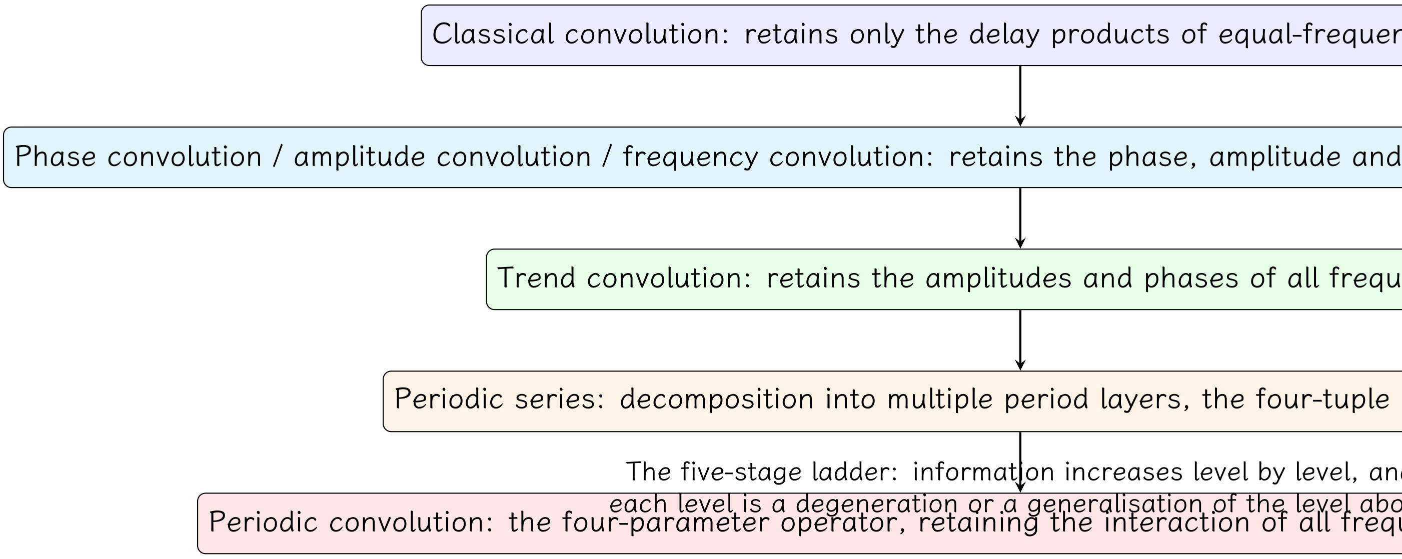
Periodic convolution is one of the important methodological frameworks of this book. The five-stage ladder of the book—**convolution** (two functions and a delay are passed in, and delay products are computed), **similarity convolution** (cosine similarity and sine similarity are packed into a complex matrix), **trend convolution** (the similarity convolutions multiplied by each other's moduli), **the periodic series** (a complex series whose components are the four-tuple of amplitude, frequency, phase and period), **periodic convolution** (all four parameters acting as operators)—unifies the traditional integral transforms (the Fourier series, the Fourier transform, the DFT, the FFT, the wavelet transform, the Mellin transform, the Laplace transform, the chirp-Z transform and the Hilbert transform) as special cases of the periodic transform and of periodic convolution under a specific dictionary, kernel and algorithmic realisation, and completes the information on phase interaction that the traditional methods discard.

The core formula of periodic convolution is:

$$(f \circledast g)(\tau) = \sum_{p,q} \Theta_{p,q} \sum_{n,m} W(\omega_{p,n} - \nu_{q,m}) \cdot (A_{p,n} B_{q,m})^{\alpha} \cdot S^{(d)}(\phi_{p,n}, \psi_{q,m}) \cdot e^{i\nu_{q,m}\tau},$$

in which the four factors are respectively the **amplitude operator** (scaling, with the switch α), the **phase operator** (rotation, with 6 complex phase kernels $S^{(d)}$), the **frequency operator** (translation, with the weight kernel W and the delay phase factor $e^{i\nu\tau}$), and the **period operator** (fixed length, with the period-pair weight $\Theta_{p,q}$).

The degeneration chain of periodic convolution—classical convolution/cross-correlation (the kernel-width limit $W \to \delta$), similarity convolution ($\alpha = 0$), trend convolution ($\alpha = 1$) and kernelised convolution (the single-period layer)—constitutes a complete analytical tool chain from the time domain to the frequency domain and from a single period to multi-period coupling. The classical limit of the periodic tendency spectrum is the power spectrum, and the Fisher test gives an exact threshold for period significance.



Classical convolution: retains only the delay products of equal-frequen

Phase convolution / amplitude convolution / frequency convolution: retains the phase, amplitude and

Trend convolution: retains the amplitudes and phases of all frequ

Periodic series: decomposition into multiple period layers, the four-tuple

The five-stage ladder: information increases level by level, an

each level is a degeneration or a generalisation of the level abo

Periodic convolution: the four-parameter operator, retaining the interaction of all frequ

Figure 46.4: The five-stage ladder of periodic convolution.

The introduction of the concept of a **phase operator** reveals that every phase of every period is a kind of relational operator: $\mathcal{O}_k^{(T)} = e^{i2\pi k/T}$ not only marks a position, but defines a relation through complex multiplication. Multiplication by $\mathcal{O}_0 = 1$ (the yang operator) gives the object itself (congruence), multiplication by $\mathcal{O}_{T/2} = -1$ (the yin operator) gives the opposite vector (hedging), and multiplication by an arbitrary $\mathcal{O}_k$ defines the relation of rotation by $2\pi k/T$.

The epicycle nesting generation rule—for every single step forward of the outer circle, the inner circle rotates through one full turn—unifies the process by which a small period generates a large period: Yin-Yang × the Five Phases = the Heavenly Stems (10), the Four Symbols × Sancai = the Earthly Branches (12), and the Eight Trigrams × the Eight Trigrams = the Sixty-Four Hexagrams (64). The sexagenary cycle is generated by the periodic convolution of the Heavenly Stems and the Earthly Branches.

46.5 Window functions and the refractive index: turning-point diagnosis based on similarity and tendency

The window-function and refractive-index system is another important methodological framework of this book. It generalises the geometric prototypes of the law of reflection and the law of refraction to the trend analysis of time series, and thereby establishes a tool chain for turning-point diagnosis based on similarity and tendency.

The window function K_Δ smooths a time series at multiple scales and extracts the trend component $T_\Delta(t)$. The direction sequence is then obtained through the instantaneous limit angle $\theta_t = \arctan(\nabla \tilde{X}_t)$. The turning-point criterion is: $|\theta_t - \theta_{t-1}| > \tau$ if and only if the basic difference-angle cosine similarity $\cos(\theta_t - \theta_{t-1}) < \cos\tau$; the sign of the double-angle direction similarity $\sin 2(\theta_t - \theta_{t-1})$ gives the sense of the turn, and its absolute value gives the strength of the turning

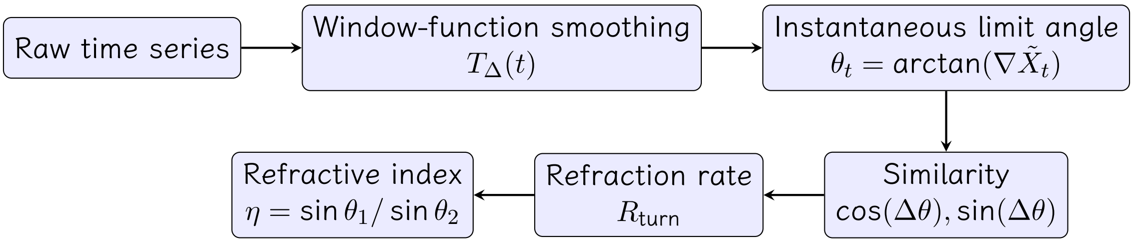

Raw time series
Window-function smoothing $T_\Delta(t)$
Instantaneous limit angle $\theta_t = \arctan(\nabla \tilde{X}_t)$
Similarity $\cos(\Delta\theta), \sin(\Delta\theta)$
Refraction rate R_{turn}
Refractive index $\eta = \sin\theta_1 / \sin\theta_2$

Figure 46.5: The window-function–refractive-index turning-point diagnostic procedure.

point. The refraction rate $R_{\text{turn}} = |\sin\theta_1 - \sin\theta_2|$ measures the difference of the sine tendencies on the two sides of the turning point, and its half-angle decomposition is

$$R_{\text{turn}} = 2\left|\cos\frac{\theta_1+\theta_2}{2}\right| \cdot \left|\sin\frac{\theta_1-\theta_2}{2}\right|.$$

The refractive-index ratio $\eta = \sin\theta_1/\sin\theta_2$ serves as a formal analogue of Snell's law $n_1 \sin\theta_1 = n_2 \sin\theta_2$—no physical refractive index exists in a time series, and η is used only in the sense of an analogy; the direction reversal $\theta_2 = -\theta_1$ corresponds to the 'total internal reflection' analogy, in which case $R_{\text{turn}} = 2|\sin\theta_1|$. The concept of delay relates phase to time shift: the phase delay $\tau = \phi/\omega$ converts a rotational phase into a time shift, and the propagation delay $\tau = r/c$ converts a spatial distance into a time shift.

46.6 The complex exponential activation function and kernel-smoothed derivative-integral descent

The complex exponential activation function (CEAF) turns the nested epicycle system into a learnable activation function for neural networks:

$$\mathcal{E}(x) = \sum_{k=1}^{N} A_k e^{i(\omega_k x + \phi_k)}.$$

The complex-valued output of the CEAF preserves phase information naturally, and its derivative does not vanish identically on the whole real axis. Its parameters $\{A_k, \omega_k, \phi_k\}$ can be initialised from estimates obtained at the data-analysis stage—periodogram peaks, phase-coupling networks and derivative-integral descent—and then updated by backpropagation or by KDI. In this way, periodic structure extracted from data can enter a neural network end to end.

The kernel-smoothed derivative-integral descent method exploits N-th order derivative and N-th order integral information simultaneously through a kernel function. The curvature correction term uses the second derivative to detect changes in curvature, and the cumulative-change correction term uses integral information to detect long-term trends. For the CEAF parameters, the amplitude A_k is suited to smooth adjustment by a Gaussian kernel, the frequency ω_k is suited to curvature sensing by a Mexican-hat kernel, and the phase ϕ_k is suited to jump detection by a Hilbert kernel.

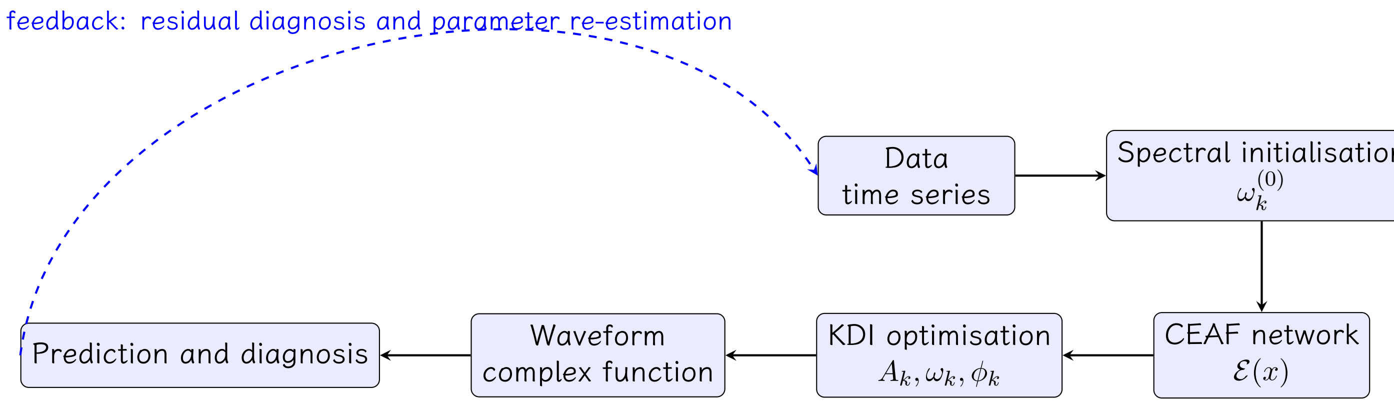

feedback: residual diagnosis and parameter re-estimation
Data
time series
Spectral initialisatio
$\omega_k^{(0)}$
CEAF network
$\mathcal{E}(x)$
KDI optimisation
A_k, ω_k, ϕ_k
Waveform
complex function
Prediction and diagnosis

Figure 46.6: The CEAF–KDI learning loop: from data to Waveform, and back to data diagnosis.

The crystal oscillator as a fundamentally fixed epicycle is the core observation concerning the periodic structure of chip systems: the crystal oscillator outputs an epicycle whose amplitude, frequency and phase are all fixed, and it is the master epicycle of all the other epicycles within the chip. A phase-locked loop realises frequency scaling of an epicycle, a clock tree is a nested epicycle system, and a chip is a parametrised epicycle waveform/spectrum processing unit.

46.7 The logical closed loop of the book

Remark 46.2 (The logical closed loop of the book (empirical observation / research programme)). Periodic mathematics starts from the three centres of symmetry 0, 1 and i, passes through the extension of the number systems, group representation theory, vector analysis and the flux model, periodic convolution, periodic systems, periodic manifolds and periodic categories, and finally arrives at the various application systems, thereby proposing the following **research programme** for a theoretical closed loop:

(i) Starting from the rotational symmetry i, the complete Maxwell equations and charge conservation are derived through local $U(1)$ gauge invariance—the electromagnetic system is the purest realisation of the rotation operation in physics (proved; see Theorem 31.15 in the chapter 'Periodic Structure of Electromagnetic Systems').

(ii) Starting from the translational symmetry 0, energy conservation is derived through time-translation invariance—the translation operation drives the generation and maintenance of periodic structure in mechanical systems, circuit systems and climate systems (proved; see Theorem 33.9 in the chapter 'Periodic Structure of Mechanical Systems').

(iii) Starting from the scaling symmetry 1, scale invariance yields the **existence of a power-law exponent**—scale invariance gives the existence of the exponent, whilst its stability across periods remains an **empirical fact** (an **empirical observation / research programme**; see 39.9 in the chapter 'Periodic Structure of Economic Systems' and 35.11 in the chapter 'Periodic Structures in Biological Systems'; the stability claim in the former is itself annotated there as an empirical observation and analogical explanation, not a mathematical theorem).

This 'closed loop' is therefore an overall description of the structure of the book and a **research programme**, not a proved theorem: threads (i) and (ii) each correspond to a theorem of the respective chapter, whilst the link 'stability of the scaling exponent across periods' in (iii) is an empirical observation / research programme. The three threads correspond respectively to standard results in Lagrangian mechanics, Noether's theorem and power-law scaling analysis, whose rigorous statements are given in the theorems of the corresponding chapters.

46.8 A unified diagnostic matrix for interdisciplinary applications

The application chapters of the book are not isolated from one another; rather, they share one and the same diagnostic matrix. The table below lists the main application domains together with their dominant centre of symmetry, their typical period type and their main tools, so that the reader may grasp the overall layout of the book's applications.

Domain	Dominant centre of symmetry	Typical period type	Main tools
Astronomy	i, 0	Orbital periods, resonance, many-body quasi-periodicity	Kepler's laws, Floquet, KAM, periodic convolution
Calendars	0, 1	Tropical year, synodic month, the sexagenary cycle	Intercalation, continued fractions, congruence angles, periodic convolution
Geography	i, 1	Milankovitch cycles, ENSO, seasonal cycles	Multiscale windows, spectral analysis, cross-scale resonance
The I Ching	i, 0, 1	Hexagram periods, sexagenary periods, He Tu and Lo Shu	Phase operators, epicycle nesting, periodic convolution
Electromagnetism	i	Electromagnetic waves, oscillation, retarded potentials	Maxwell's equations, vector analysis, delay
Audio	0, i	Simple harmonic motion, harmonics, beats	Fourier series, equal temperament, delay
Mechanics	0, i	Spring oscillators, the simple pendulum, resonance, precession	Limit cycles, elliptic integrals, phase response
Circuits and chips	i, 0	Crystal oscillators, PLLs, clock trees, data periods	Epicycles, phase-locked loops, nested clocks
Biology	i, 1	Circadian rhythms, heart rate, the cell cycle, population cycles	Limit cycles, delayed feedback, Lotka–Volterra
Psychology	0, i	Sleep cycles, attentional oscillations, mood cycles	Two-process model, RC charge–discharge, phase locking
Social interaction	i	Social rhythms, information propagation, collective synchronisation	SIR/SIRS, Kuramoto, phase-coupling networks
Climate	i, 1	Seasons, ENSO, PDO, IOD, Milankovitch	Red noise, the Hurst exponent, multitaper spectral estimation
Economics	1, i	Inventory cycles, investment cy-	Multiplier–accelerator, the cobweb model, window-

This matrix shows that the 'periods' of different domains differ vastly at the level of phenomena, yet in their mathematical structure they can often be reduced to compositions of the three elementary operations, to finite-order action, to closed orbits, to phase coupling and to scale nesting. The role of periodic mathematics is not to replace the specific mechanisms of the individual disciplines, but to provide a unified grammar that allows these structures to be described, compared and diagnosed rigorously.

46.9 Outlook: unfinished tasks and open problems

The book has established the basic framework of periodic mathematics, but a great many problems still await solution. The open problems are listed below under the four headings of theory, algorithms, applications and methodology.

Category	Open problem
Theory	Generalisation of finite-order action to continuous, dissipative and non-autonomous systems; higher structures of the periodic category (modifications, 2-morphisms); the theory of optimal approximation for congruence angles and joint multi-frequency intercalation; classification of the sufficient conditions relating conservation laws to closed orbits.
Algorithms	Fast algorithms for periodic convolution (FFT-based, reducing $O(N^2)$ to $O(N \log N)$); global convergence of KDI under non-convex losses and estimation of the saddle-escape time; lower bounds on the expressiveness and parameter efficiency of deep CEAF networks; the exact correspondence between positional encoding and periodic convolution.
Applications	Joint diagnosis of multiscale climate periods; quantitative models of economic long waves and structural change; periodic analysis of the multiscale selection–mutation–inheritance cycle in biology; experimental verification of multi-agent phase synchronisation; a unified measure of the periodic security of cryptographic systems.
Methodology	An automated pipeline from data to Waveform; automatic selection of the window radius and of the refractive index together with tests of statistical significance; integration of periodic convolution into deep learning; scaling of CEAF and KDI to large models.

Concretely, the following ten problems are particularly crucial:

1. **Fast algorithms for periodic convolution**: develop a fast FFT-based algorithm for computing periodic convolution, reducing the time complexity from $O(N^2)$ to $O(N \log N)$.

2. **Automation of the window-function–refractive-index system**: establish an algorithm for the automatic selection of the window radius based on kernel-smoothed derivative-integral descent, and develop a test of statistical significance for the refractive index.
3. **Integration of periodic convolution into deep learning**: use periodic convolution as a feature-extraction layer of a neural network.
4. **Scaling up the complex exponential activation function**: use the complex exponential activation function in place of the standard activation functions in large language models and time-series forecasting models.
5. **Integration of kernel-smoothed derivative-integral descent into deep learning**: use derivative-integral descent as the optimiser in neural-network training.
6. **Extending the applications of phase-coupling networks**: apply them to high-dimensional time-series data from EEG, fMRI and financial markets.
7. **Empirical testing of the window-function–refractive-index system in financial markets**: verify the effectiveness of the sine difference-angle similarity and of the refractive index in predicting trend reversals.
8. **Experimental verification of multi-agent phase synchronisation**: explore the influence of communication frequency and coupling strength on the emergence of collective intelligence.
9. **Deepening the concept of delay**: further develop, in electromagnetic systems, audio systems, mechanical systems and chip systems, the specific roles played by delay under the three operations of translation, rotation and scaling.
10. **Extension of the phase-operator system**: generalise the phase-operator framework of the I Ching system to symbolic systems of arbitrary period T, and establish a unified theory of phase encoding.

46.10 Conclusion: periodicity as a unified language

Periodicity is the most universal language of nature. From the vibration of the electron cloud inside an atom to the revolution of the planets, from the rhythm of the heartbeat to the cycles of economic fluctuation, from the learning and feeling of an individual to the selection and evolution of a population, symmetry and invariants weave the rhythm of the evolution of the universe. This book attempts to compose a unified mathematical score for that rhythm—starting from the most abstract centres of symmetry, proceeding through the cross-period phase analysis of periodic convolution, the turning-point diagnosis of window functions and the refractive index, and the intelligent optimisation of the complex exponential activation function and kernel-smoothed derivative-integral descent, and extending all the way to concrete domains such as astronomy, calendars, geography, physics, biology, psychology, economics, politics, cryptography, agents and data analysis.

It must be emphasised once again that periodic mathematics does not claim a priori that every system under consideration ‘has a periodic structure’. In every application chapter it first asks: does this system possess an identifiable symmetry group? Does a corresponding conserved quantity exist? Is the evolution orbit in its state space closed? For systems that satisfy the conditions, the book gives a rigorous description of their periodic structure; for systems that satisfy them only partially, the book measures precisely the degree of their deviation from a closed orbit; and for systems that do not satisfy them, the book states explicitly that their ‘periodicity’ is no more than a metaphor at the level of phenomena. This diagnostic function is precisely what distinguishes periodic mathematics from interdisciplinary analogy in general.

The curtain has only just risen, and a broader theoretical landscape is waiting to unfold. It is our hope that the grammar provided by this book will help researchers in different fields to identify common periodic structures in their own systems and to enter into dialogue at the mathematical level.

Master seven-theme comparison table for the whole book

Theme	Corresponding content in the book (chapters)
Translation	Foundations of Periodicity (the three elementary transformations); Number Systems (the periodicity criterion for integer translation); Foundations of Functions and Analysis (derivative/convolution translation); Time Series Analysis (the lag operator); the translation periods of the individual application chapters
Scaling	Number Systems (rational proportions/logarithms); Vector Analysis (divergence/projection decay law); Audio (equal temperament); the scaling periods of the individual application chapters
Rotation	Number Systems (complex numbers/quaternions); Periodic Functions (phase operators); Harmonic Analysis (the phase kernel $S^{(d)}$); the rotation periods of the individual application chapters
Conservation	Foundations of Periodicity (the Noether correspondence/Liouville–Arnold); Periodic Functions (the three conservations of the epicycle); the conservation-law theorems of the individual chapters
Symmetry	Foundations of Periodicity (the centres of symmetry $0, 1, i$); Groups and Group Functions (the automorphism tower); the symmetry groups of the individual application chapters
Closure	Foundations of Periodicity (the double closure criterion); Periodic Manifolds (discrete curvature/Wilson loops); Time Series Analysis (closure of the tool chain); the closed orbits of the individual application chapters
Period	Harmonic Analysis (the periodic tendency spectrum); Time Series Analysis (period determination); Calendars (tropical year/synodic month/the sexagenary cycle); the multiscale periods of the individual application chapters

Exercises

Exercise 1 Explain which links of the **logical closed loop of the book** of this chapter (an **empirical observation / research programme**; see the remark at 46.2) have been rigorously proved and which remain a research programme, indicating the source of each. Hint: check one by one the results derived from the three centres of symmetry and the chapters in which they appear.

Exercise 2 Using the seven-theme comparison table of this chapter, identify the concrete carriers of this chapter under the three themes **translation**, **scaling** and **rotation**, and give one example of domain data for each.

Exercise 3 Synthesis: restate the diagnostic procedure of this chapter using the five-step standard interface of the 'Periodic Systems' chapter (Table 24.1), noting for each step the conclusion of this chapter on which it depends: 46.2 (an **empirical observation / research programme**).

References

[1] **Zhou Yi (the I Ching, or Book of Changes)**, including the 'I Ching' and the 'Yi Zhuan'. Standard received text.

The phrase 'no plain without a slope, no going without a return', quoted on the cover, is drawn from the line statement of the third line (the ninth-yang line) of the 'Tai' hexagram of the Zhou Yi. The growth and decline of yin and yang, and the idea of recurrent cycles, are an ancient reverberation of the periodic mathematics programme of 'symmetry → periodicity'.

[2] **Daodejing (the Tao Te Ching)**, by Laozi. Standard received text (the Wang Bi commentary edition).

The phrase 'The Dao begot the one; the one begot the two; the two begot the three; and the three begot the myriad things', quoted on the cover, is from chapter 42.

[3] **Zhuangzi**, by Zhuang Zhou. Standard received text.

'Born together with life, dying together with death; dying together with death, born together with life' (the 'Qiwulun', 'On the Equality of Things') reveals the mutual transformation of opposites.

[4] **Huangdi Neijing (the Yellow Emperor's Inner Classic)**, including the Suwen and the Lingshu. Standard received text.

The system of the five circuits and six qi is an early model of nested periodic structure.

[5] **Shangshu (the Book of Documents)**, including the 'Yaodian'. Standard received text.

'A year comprises three hundred and sixty-six days, and by intercalary months the four seasons are fixed and the year completed'—the earliest official record of a lunisolar calendar.

[6] **Huainanzi**, compiled by Liu An. Standard received text.

The preface cites the 'Tianwen xun' (Astronomy) and the 'Lanming xun' as the central texts for the establishment of the Xuanyuan calendrical epoch and for the interpretation of Nüwa mending the heavens as a calendrical act.

[7] **Shanhaijing (the Classic of Mountains and Seas)**, anonymous. Standard received text.

The preface cites Jingwei filling the sea and Kuafu chasing the sun, as mythological texts for the eternity of calendrical restoration and for the beginnings of observation.

[8] **Shiji (the Records of the Grand Historian)**, by Sima Qian.

The 'Tianguan shu' (Treatise on the Heavenly Offices) is a systematic record of ancient astronomical observation. The 'Li shu' (Treatise on the Calendar) records that the jia-yin day of the first year of the Taichu era served as the baseline for the derivation of the Xuanyuan calendrical epoch.

[9] **Shuowen Jiezi**, by Xu Shen.

The preface cites 'Intercalary: the month of the surplus division', as the philological basis for the 'one day' left behind by Hou Yi shooting the suns, taken as the intercalary surplus.

[10] **Zhoubi Suanjing (the Arithmetic Classic of the Gnomon and the Circular Paths of Heaven)**, anonymous.

The 'seven concentric rings and six intervals' diagram describes the annual apparent motion of the sun by means of concentric circles.

[11] **Yueling (the 'Monthly Ordinances' of the Liji, or Book of Rites)**, anonymous. Standard received text.

It records month by month the periodic rhythms of celestial phenomena, phenology and farming, and is the earliest systematic literature on phenology.

[12] **Mengzi (the Mencius)**, by Meng Ke. Standard received text.

'All things are already complete within me; to turn inward and realise this is joy beyond compare' (the 'Jin Xin Shang', 'On the Full Development of the Mind, I').

[13] **Sunzi Bingfa (the Art of War)**, by Sun Wu. Standard received text.

'The skilful warrior seeks victory from the configuration of forces, and does not demand it of his men' (the 'Bingshi', 'On Military Dispositions').

[14] **Shennong Bencao Jing (Shennong's Classic of Materia Medica)**, anonymous. Standard received text (the recension edited by Sun Xingyan and Sun Fengyi).

The earliest classic of Chinese materia medica, recording the harvesting seasons and the places of origin of 365 medicinals.

[15] **Bencao Gangmu (the Compendium of Materia Medica)**, by Li Shizhen (1596).

Recording 1,892 medicinals, it systematically documents the 'season and month', the 'locality' and the 'processing' of each drug.

[16] **Huangji Jingshi**, by Shao Yong.

Its nested periodic system of yuan, hui, yun and shi is the theoretical basis for the nested periodic structure in the 'Periodic Structure of Calendar Systems' chapter of this book.

[17] **Gottfried Wilhelm Leibniz.** Explication de l'Arithmétique Binaire (1703).

Leibniz's original paper on binary arithmetic. The discovery of periodicity in binary columns (Periodus numerorum) provides the historical precursor for the bit-width and bandwidth concepts of this book.

[18] **Gottfried Wilhelm Leibniz.** De Progressione Dyadica (1679).

Leibniz's manuscript on the dyadic progression. The law of periodic variation of each digit in the sequence of natural numbers was first systematically recorded in this manuscript.

[19] **Gottfried Wilhelm Leibniz.** Monadologie (1714).

The Monadology. The idea that binary arithmetic generates 'being' (1) from 'nothing' (0) shares a common philosophical lineage with the Monadology's notion of 'constructing a complex world from simple substances'.

[20] **Galileo Galilei.** The Assayer (1623).

The phrase quoted on the cover, 'Mathematics is the language with which God has written the universe', originates from this work.

[21] **Isaac Newton.** Philosophiae Naturalis Principia Mathematica (1687).

The law of universal gravitation and the three laws of motion provided the first complete mathematical description of classical periodic motion.

[22] **Leonhard Euler.** Introductio in Analysin Infinitorum (1748).

Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ is the mathematical cornerstone of the epicycle model and of Fourier analysis.

[23] **Joseph Fourier.** Théorie Analytique de la Chaleur (1822).

The original work on Fourier series and the Fourier transform.

[24] **William R. Hamilton.** Lectures on Quaternions. Hodges and Smith, 1853.

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— Dedicated to all who seek the ‘Dao’ and the ‘Number’.

About the Author

Eloy Kim is an independent researcher who has long worked on periodic mathematics, harmonic analysis and interdisciplinary mathematical modelling. The author is proficient in C/C++, Java, TypeScript/JavaScript and Android development, and combines engineering implementation with theoretical derivation. His research interests span the I Ching, programming, mathematics and physics, in a sustained attempt to find a unified structure and a common language between philosophy and mathematical science.

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